

Flet Documentation

Source: <https://flet.dev/docs/>

Generated: 2026-06-23

Introduction

Flet is a framework that allows building web, desktop and mobile applications in Python without prior experience in frontend development.

Flet app example

Below is a simple "Counter" app, with a text field and two buttons to increment and decrement the counter value:

counter.py

```
import flet as ft

def main(page: ft.Page):
    page.title = "Flet counter example"
    page.vertical_alignment = ft.MainAxisAlignment.CENTER

    input = ft.TextField(value="0", text_align=ft.TextAlign.RIGHT, width=100)

    def minus_click(e):
        input.value = str(int(input.value) - 1)

    def plus_click(e):
        input.value = str(int(input.value) + 1)

    page.add(
        ft.Row(
            alignment=ft.MainAxisAlignment.CENTER,
            controls=[
                ft.IconButton(ft.Icons.REMOVE, on_click=minus_click),
                input,
                ft.IconButton(ft.Icons.ADD, on_click=plus_click),
            ],
        )
    )

ft.run(main)
```

To run the app, install flet:

```
pip install 'flet[all]'
```

then launch the app:

```
flet run counter.py
```

This will open the app in a native OS window - what a nice alternative to Electron! ☐

Image: Image - <https://flet.dev/docs/assets/getting-started/counter-app/macOS.png>

To run the same app as a web app use --web option with flet run command:

```
flet run --web counter.py
```

Image: Image - <https://flet.dev/docs/assets/getting-started/counter-app/safari.png>

Flet CLI

Flet command-line interface (CLI) provides an easy way to create, run, test and package Flet apps.

Usage

```
flet [-h] [--version]
      {create,run,build,debug,pack,publish,serve,emulators,devices,doctor} ...
```

Options

--help

Show this help message and exit.

Aliases: -h

--version

Show program's version number and exit.

Aliases: -V

Subcommands

- **build:** Build a Flet Python app into a platform-specific executable or.
- **create:** Create a new Flet project using a predefined template.
- **debug:** Run a Flet Python app in debug mode on a specified platform (desktop, web, mobile).
- **devices:** List all connected iOS and Android devices.
- **doctor:** Get information about the system and environment setup.
- **emulators:** List, create, and launch available emulators.
- **pack:** Package a Flet application into a standalone desktop executable or app bundle.
- **publish:** Compile and package a Flet app as a standalone static web application.
- **run:** Run a Flet application in hot reload mode.
- **serve:** Serve static files from a directory with a lightweight web server,.

FastAPI

FastAPI wrapper that integrates Flet app-manager startup/shutdown lifecycle.

It mirrors `fastapi.FastAPI` constructor options while wiring a custom lifespan context that starts and stops `flet_web.fastapi.app_manager`.

Inherits: `fastapi.FastAPI`

Publishing a Flet app

Flet CLI provides the `flet build` command to package a Flet app into a standalone executable or installable package for distribution.

Prerequisites

Platform matrix

Use the following matrix to determine on which OS flet build can be run in order to target each platform:

Run on			
macOS	☐	☐	☐
Windows	☐		
Linux	☐		

apk/aabipa/ios-simulatormacoslinuxwindowsweb

Flutter SDK

Flutter is required to build Flet apps for any platform.

If the minimum required version of the Flutter SDK is not already available in the system PATH, it will be automatically downloaded and installed (in the `$HOME/flutter/{version}` directory) during the first build process.

Tip

The recommended (minimum required) Flutter SDK version depends on the Flet version installed or in use.

It can be viewed by running one of the following commands:

```
flet --version
uv run python -c "import flet.version; print(flet.version.flutter_version)"
```

or the below Python code snippet:

```
import flet.version
print(flet.version.flutter_version)
```

Project structure

The `flet build` command assumes the following minimal Flet project structure:

```
README.md
pyproject.toml
src
  assets
    icon.png
  main.py
```

Tip

To quickly set up a project with the correct structure, use the `flet create` command:

```
flet create <project-name>
```

Where `<project-name>` is the name for your project directory.

Using `requirements.txt` instead of `pyproject.toml`

Instead of a `pyproject.toml` file, you can also use a `requirements.txt` file to specify dependencies.

In this case, two things to keep in mind:

- if both files are present, `flet build` will ignore `requirements.txt`.
- don't use `pip freeze > requirements.txt` to generate this file or fill it with dependencies, as it may include packages incompatible with the target platform. Instead, hand-pick and include only the direct dependencies required by your app, including `flet`.

How it works

When you run `flet build <target_platform>`, the pipeline is:

- Create a Flutter project in {flet_app_directory}/build/flutter from the template. The Flutter app embeds your packaged Python app in its assets and uses flet and serious_python to run the app and render the UI. The project is cached and reused across builds for rapid iterations; use --clear-cache to force a rebuild.

Copy custom icons and splash images from assets into the Flutter project, then generate:

- Icons for all platforms via flutter_launcher_icons.
- Splash screens for web, iOS, and Android via flutter_native_splash.

Package the Python app using serious_python package:

- Install app dependencies from pypi.org and pypi.flet.dev for the selected platform, as configured in App dependencies and Source packages.
- If configured, compile .py files to .pyc.
- Add all project files, except those excluded, to the app asset.
- Run flutter build to produce the executable or installable package.
- Copy build outputs from Step 4 into the output directory.

Configuration options

Placeholders

Throughout this documentation, the following placeholders are used:

- <target_platform> - one of: apk, aab, ipa, ios-simulator, web, macos, windows, linux.
- <PLATFORM> - the config namespace under [tool.flet.<PLATFORM>]; one of: android (for apk and aab targets), ios (for ipa and ios-simulator targets), web, macos, windows, linux.
- <python_app_path> - the path passed to flet build (defaults to the current directory).
- <flet_app_directory> - the resolved project root for <python_app_path>; pyproject.toml and requirements.txt are read from here.
- <flet_version> - the version of Flet in use. You can check with flet --version or uv run python -c "import flet; print(flet.__version__)".

Understanding pyproject.toml structure

Flet loads pyproject.toml as a nested dictionary and looks up settings using dot-separated paths (for example, tool.flet.web.base_url).

The two forms below are equivalent and resolve to the same key-value pair:

Form 1 (will be used/preferred throughout this documentation)

```
[tool.flet.section]
key = "value"
```

Form 2

```
[tool.flet]
section.key = "value"
```

But they are different or should not be confused with the below ("quoted keys" are literals and do not create nesting):

```
[tool.flet]
"section.key" = "value"
```

App path

Defines the root directory of your Python app within <python_app_path>. Flet looks for the entry point, the assets directory, and exclude paths relative to this directory.

Resolution order

Its value is determined in the following order of precedence:

- [tool.flet.app].path
- <python_app_path>

path is resolved relative to <python_app_path>.

Example

- pyproject.toml

```
[tool.flet.app]
path = "src"
```

Entry point

This is the Python module that starts your app and contains the call to `flet.run()` or `flet.render()`. Flet uses the module stem and looks for `<module>.py` in your app path.

Resolution order

Its value is determined in the following order of precedence:

- `--module-name`
- `[tool.flet.app].module`
- "main" (entry file `main.py`)

Its value can either be `<module>` or `<module>.py`; both resolve to the same Python module.

Example

- flet build

- pyproject.toml

```
flet build <target_platform> --module-name app.py

[tool.flet.app]
module = "app.py"
```

Project name

The project name is the base identifier for bundle IDs and other internal names. The source value is normalized to a safe identifier: lowercased, punctuation and spaces removed or collapsed, and hyphens converted to underscores (for example, `My App` or `my-app` becomes `my_app`).

Resolution order

Its value is determined in the following order of precedence:

- `--project`
- `[project].name`
- project/app directory name

Example

- flet build

- pyproject.toml

```
flet build <target_platform> --project my_app

[project]
name = "my_app"
```

Product name

The display (user-facing) name shown in window titles, launcher labels, and about dialogs.

It does not control the on-disk executable or bundle name. Use the artifact name for artifact naming.

Resolution order

Its value is determined in the following order of precedence:

- `--product`
- `[tool.flet].product`
- `--project`
- `[project].name`
- project/app directory name

Example

- flet build

- pyproject.toml

```
flet build <target_platform> --product "My Awesome App"

[tool.flet]
product = "My Awesome App"
```

Artifact name

The on-disk name for executables and/or app bundles. For example, on Windows it determines the name of the .exe file, and on macOS it sets the name of the .app bundle.

It does not affect bundle IDs or package identifiers.

It can contain spaces or accents, but keep file system restrictions in mind on your target platforms.

Resolution order

Its value is determined in the following order of precedence:

- --artifact
- [tool.flet.<PLATFORM>].artifact
- --project
- [project].name
- project/app directory name

Example

- flet build

- pyproject.toml

```
flet build <target_platform> --artifact "My Awesome App"

[tool.flet]
artifact = "My Awesome App"
```

Organization name

Platform support

Android, iOS, macOS, and Linux only.

The organization name in reverse domain name notation, typically in the form com.mycompany. It is used as the prefix for the bundle ID and for package identifiers on mobile and desktop targets.

Resolution order

Its value is determined in the following order of precedence:

- --org
- [tool.flet.<PLATFORM>].org
- [tool.flet].org
- "com.flet"

Example

- flet build

- pyproject.toml

```
flet build <target_platform> --org com.mycompany

[tool.flet]      # or [tool.flet.<PLATFORM>]
org = "com.mycompany"
```

Bundle ID

Platform support

Android, iOS, macOS, and Linux only.

The bundle ID for the application, typically in the form "com.mycompany.my_app".

If not explicitly specified, it is derived from the organization name and the project name used by the build template.

Resolution order

Its value is determined in the following order of precedence:

- --bundle-id
- [tool.flet.<PLATFORM>].bundle_id
- [tool.flet].bundle_id

Example

- flet build
 - pyproject.toml
- ```
flet build <target_platform> --bundle-id com.mycompany.my_app

[tool.flet] # or [tool.flet.<PLATFORM>]
bundle_id = "com.mycompany.my_app"
```

## Company Name

Platform support

Windows and macOS only.

The company name displayed in about app dialogs and metadata (notably on desktop builds).

## Resolution order

Its value is determined in the following order of precedence:

- --company
- [tool.flet].company
- Build template default (see Template Source)

## Example

- flet build
  - pyproject.toml
- ```
flet build <target_platform> --company "My Company Inc."

[tool.flet]
company = "My Company Inc."
```

Copyright

Platform support

Windows and macOS only.

Copyright text displayed in about app dialogs and metadata.

Resolution order

Its value is determined in the following order of precedence:

- --copyright
- [tool.flet].copyright
- Build template default (see Template Source)

Example

- flet build
 - pyproject.toml
- ```
flet build <target_platform> --copyright "Copyright © 2026 My Company Inc."

[tool.flet]
copyright = "Copyright © 2026 My Company Inc."
```

## Versioning

### Build Number

An integer identifier used internally to distinguish one build from another.

Each new build must have a unique, incrementing number; higher numbers indicate more recent builds.

#### Resolution order

Its value is determined in the following order of precedence:

- `--build-number`
- `[tool.flet].build_number`
- Otherwise, the build number from the generated `pubspec.yaml` (see Template Source) will be used.

#### Example

- `flet build`
- `pyproject.toml`

```
flet build <target_platform> --build-number 1

[tool.flet]
build_number = 1
```

### Build Version

A user-facing version string in x.y.z format. Increment this for each new release to differentiate it from previous versions.

#### Resolution order

Its value is determined in the following order of precedence:

- `--build-version`
- `[project].version`
- `[tool.poetry].version`
- Otherwise, the build version from the generated `pubspec.yaml` (see Template Source) will be used.

#### Example

- `flet build`
- `pyproject.toml`

```
flet build <target_platform> --build-version 1.0.0

[project]
version = "1.0.0"
```

### Output directory

The directory where the build output is saved. If the directory already exists, it is deleted and recreated on each build.

For web builds, the app's assets directory is copied into the output directory.

#### Resolution order

Its value is determined in the following order of precedence:

- `--output (or -o)`
- `<python_app_path>/build/<target_platform>`

#### Example

- `flet build`

```
flet build <target_platform> --output <path-to-output-dir>
```

### App dependencies

These are the Python packages that your Flet app depends on to function correctly.

#### Resolution order

Its value is determined in the following order of precedence:

- [tool.poetry].dependencies if present; otherwise [project].dependencies (PEP 621).
- If [tool.flet.<PLATFORM>].dependencies is set (where <PLATFORM> corresponds to <target\_platform>), its values are appended to the list above.
- If the result of all above is empty and requirements.txt exists in <python\_app\_path>, it is used.
- If the result of all the above is empty, flet==<flet\_version> is used.

To use a local development version of a dependency during builds, configure [tool.flet].dev\_packages or [tool.flet.<PLATFORM>].dev\_packages with a package name to path mapping.

If your app uses Flet extensions (third-party packages), list them in your Python dependencies so they are packaged with the app. Examples of extensions can be found in Built-in extensions.

## Example

- pyproject.toml

```
[project]
dependencies = [
 "flet",
 "requests",
 "flet-extension1",
 "flet-extension2 @ git+https://github.com/account/flet-extension2.git", # git repo
 "flet-extension3 @ file:///path/to/flet-extension3", # local package
]

[tool.flet.<PLATFORM>] # will be used/appended only if <PLATFORM> corresponds to <target_platform>
dependencies = [
 "dep1",
 "dep2",
]
```

## Source packages

### Platform support

Android, iOS, and Web only.

By default, packaging for mobile and web only installs binary wheels. Use source packages to allow specific dependencies to be installed from source distributions (sdist).

This can be useful for installing - pure Python - dependencies that do not have pre-built wheels for the target platform or an all-platform wheel (\*.py3-none-any.whl), but instead provide a source distribution (\*.tar.gz).

For more information on pure vs non-pure Python packages, see our blog post on the topic.

On desktop targets, source installs are already allowed, so this setting is mainly useful for Android, iOS, and Web (flet build web only — not flet publish).

## Resolution order

Its value is determined in the following order of precedence:

- --source-packages
- [tool.flet.<PLATFORM>].source\_packages
- [tool.flet].source\_packages

## Example

- flet build

- pyproject.toml

```
flet build <target_platform> --source-packages package1 package2

[tool.flet]
source_packages = ["package1", "package2"]
```

## Icons

### Platform support

Android, iOS, macOS, Windows and Web only.

You can customize app icons for all platforms (except Linux) using image files placed in the assets directory of your Flet app.

If a platform-specific icon (as in the table below) is not provided, icon.png (or any supported format like .bmp, .jpg, or .webp) will be used as fallback. For the iOS platform, transparency (alpha channel) will be automatically removed, if present.

| Platform | File Name | Recommended Size | Notes                                                                                       |
|----------|-----------|------------------|---------------------------------------------------------------------------------------------|
| iOS      |           | ≥ 1024×1024 px   | Transparency (alpha channel) is not supported and will be automatically removed if present. |
| Android  |           | ≥ 192×192 px     |                                                                                             |
| Web      |           | ≥ 512×512 px     |                                                                                             |
| Windows  | or        | 256×256 px       | file will be internally converted to a 256×256 px icon.                                     |
| macOS    |           | ≥ 1024×1024 px   |                                                                                             |

icon\_ios.pngicon\_android.pngicon\_web.pngicon\_windows.icoicon\_windows.png.pngicoicon\_macos.png

## Splash screen

Platform support

Android, iOS, and Web only.

A splash screen is a visual element displayed when an app is launching, typically showing a logo or image while the app loads.

You can customize splash screens for iOS, Android, and Web platforms by placing image files in the assets directory of your Flet app.

If platform-specific splash images are not provided, Flet will fall back to splash.png. If that is also missing, it will use icon.png or any supported format such as .bmp, .jpg, or .webp.

## Splash images

| Platform | Dark Fallback Order | Light Fallback Order |
|----------|---------------------|----------------------|
| iOS      | → → → →             | → →                  |
| Android  | → → → →             | → →                  |
| Web      | → → → →             | → →                  |

splash\_dark\_ios.pngsplash\_dark.pngsplash\_ios.pngsplash.pngicon.pngsplash\_ios.pngsplash.pngicon.pngsplash\_dark\_android.pngsplash\_dark.pngsplash\_android.pngsplash.pngicon.pngsplash\_android.pngsplash.pngicon.pngsplash\_dark\_web.pngsplash\_dark.pngsplash\_web.pngsplash.pngicon.pngsplash\_web.pngsplash.pngicon.png

## Splash Background Colors

You can customize splash background colors using the following options:

- Splash Color: Background color for light mode splash screens.
- Splash Dark Color: Background color for dark mode splash screens.

Resolution order

Their values are respectively determined in the following order of precedence:

- --splash-color / --splash-dark-color
- [tool.flet.<PLATFORM>.splash].color / [tool.flet.<PLATFORM>.splash].dark\_color
- [tool.flet.splash].color / [tool.flet.splash].dark\_color
- Build template defaults

Example

- flet build

- pyproject.toml

```
flet build <target_platform> --splash-color #ffffff --splash-dark-color #333333
```

```
[tool.flet.splash]
color = "#ffffff"
dark_color = "#333333"
```

## Disabling Splash Screens

Splash screens are enabled by default but can be disabled.

Resolution order

Its value is determined in the following order of precedence:

on Android:

- `--no-android-splash`
- `[tool.flet.splash].android`

on iOS:

- `--no-ios-splash`
- `[tool.flet.splash].ios`

on Web:

- `--no-web-splash`
- `[tool.flet.splash].web`

Example

- `flet build`
- `pyproject.toml`

```
flet build apk --no-android-splash
flet build ipa --no-ios-splash
flet build ios-simulator --no-ios-splash
flet build web --no-web-splash

[tool.flet.splash]
android = false
ios = false
web = false
```

## Boot screen

Platform support

Windows, macOS, Linux, Android, and iOS only.

The boot screen is shown while the packaged app archive (app.zip) is extracted to the app data directory (typically on first launch or after the app bundle changes). It appears after the splash screen and before the startup screen.

It is not shown by default. Enable it, for example, when then extraction time is noticeable.

## Example

- `pyproject.toml`

```
[tool.flet.app.boot_screen] # or [tool.flet.<PLATFORM>.app.boot_screen]
show = true
message = "Preparing the app for its first launch..."
```

## Startup screen

Platform support

Windows, macOS, Linux, Android, and iOS only.

The startup screen is shown while the Python runtime and your app are starting. On mobile targets this can include preparing packaged dependencies. It appears after the boot screen.

It is not shown by default.

## Example

- `pyproject.toml`

```
[tool.flet.app.startup_screen] # or [tool.flet.<PLATFORM>.app.startup_screen]
show = true
message = "Starting up the app..."
```

## Hidden app window on startup

### Platform support

Windows, macOS, and Linux only.

A Flet desktop app (Windows, macOS, or Linux) can start with its window hidden. This lets your app perform initial setup (for example, add content, resize or position the window) before showing it to the user.

See this code example.

### Resolution order

Its value is determined in the following order of precedence:

- [tool.flet.<PLATFORM>.app].hide\_window\_on\_start, where <PLATFORM> can be windows, macos or linux
- [tool.flet.app].hide\_window\_on\_start
- FLET\_HIDE\_WINDOW\_ON\_START

### Example

- pyproject.toml

```
[tool.flet.app] # or [tool.flet.<PLATFORM>.app]
hide_window_on_start = true
```

## Deep linking

### Platform support

Android and iOS only.

Deep linking allows users to navigate directly to specific content within a mobile app using a URI (Uniform Resource Identifier). Instead of opening the app's homepage, deep links direct users to a specific page, feature, or content within the app, enhancing user experience and engagement.

- Scheme: deep linking URL scheme, e.g. "https" or "myapp".
- Host: deep linking URL host.

See also:

- Flutter deep linking
- Android intents and intent filters
- Defining a custom URL scheme for your app
- Universal Links

### Resolution order

Its value is determined in the following order of precedence:

- --deep-linking-scheme and --deep-linking-host (only when both are provided)
- [tool.flet.<PLATFORM>.deep\_linking].scheme / [tool.flet.<PLATFORM>.deep\_linking].host, where <PLATFORM> can be android or ios
- [tool.flet.deep\_linking].scheme / [tool.flet.deep\_linking].host

Both scheme and host are required; if either is missing, the deep-linking entries are not added.

### Example

- flet build
- pyproject.toml

```
flet build <target_platform> \
--deep-linking-scheme "https" \
--deep-linking-host "mydomain.com"

[tool.flet.deep_linking] # or [tool.flet.<PLATFORM>.deep_linking]
scheme = "https"
host = "mydomain.com"
```

### Template translation

In the Android AndroidManifest.xml, the pyproject.toml example above will be translated accordingly into this:

```
<meta-data android:name="flutter_deeplinking_enabled" android:value="true" />
<intent-filter>
 <action android:name="android.intent.action.VIEW" />
 <category android:name="android.intent.category.DEFAULT" />
 <category android:name="android.intent.category.BROWSABLE" />
 <data android:scheme="https" android:host="mydomain.com" />
</intent-filter>
```

In the iOS ios/Runner/Info.plist, the pyproject.toml example above will be translated accordingly into this:

```
<key>FlutterDeepLinkingEnabled</key>
<true />
<key>CFBundleURLTypes</key>
<array>
 <dict>
 <key>CFBundleTypeRole</key>
 <string>Editor</string>
 <key>CFBundleURLName</key>
 <string>mydomain.com</string>
 <key>CFBundleURLSchemes</key>
 <array>
 <string>https</string>
 </array>
 </dict>
</array>
```

## Target Architecture

### Platform support

Android and macOS only.

A target platform can have different CPU architectures, which in turn support different instruction sets.

It is possible to build your app for specific CPU architectures. This is useful for reducing the size of the resulting binary or package, or for targeting specific devices.

For more/complementary information, see the specific platform guides: Android, macOS.

## Resolution order

Its value is determined in the following order of precedence:

- --arch
- [tool.flet.<PLATFORM>].target\_arch, where <PLATFORM> can be android or macos
- [tool.flet].target\_arch
- Platform defaults for the <target\_platform>

## Example

- flet build
- pyproject.toml

```
flet build macos --arch arm64 x86_64

[tool.flet.macos] # or [tool.flet]
target_arch = ["arm64", "x86_64"]
```

## Excluding files and directories

Files and/or directories can be excluded from the build process. This can be useful for reducing the size of the resulting binary or package.

## Resolution order

Its value is determined in the following order of precedence:

- --exclude (can be used multiple times)
- [tool.flet.<PLATFORM>].app.exclude (type: list of strings)
- [tool.flet.app].exclude (type: list of strings)

The files and/or directories specified should be provided as relative paths to the app path directory. Paths are matched exactly (no globs), and directories are excluded recursively.

By default, the build directory is always excluded. Additionally, when the `target_platform` is web, the assets directory is always excluded.

## Example

- `flet build`

- `pyproject.toml`

```
flet build <target_platform> --exclude .git .venv
[tool.flet.app] # or [tool.flet.<PLATFORM>.app]
exclude = [".git", ".venv"]
```

## Compilation and cleanup

Flet can compile your app's `.py` files and/or installed packages' `.py` files into `.pyc` files during the packaging process (via `python -m compileall -b`). Cleanup removes known junk files and any additional globs you specify.

Compilation:

- `compile-app`: compile app's `.py` files
- `compile-packages`: compile site/installed packages' `.py` files

Cleanup:

- `cleanup-app`: remove junk files from the app directory
- `cleanup-app-files`: additional globs to delete from the app directory (implies `cleanup-app`)
- `cleanup-package-files`: additional globs to delete from site-packages (implies `cleanup-packages`)
- `cleanup-packages`: remove junk files from site-packages (defaults to true)

By default, Flet does not compile your app files during packaging. This allows the build process to complete even if there are syntax errors, which can be useful for debugging or rapid iteration.

## Resolution order

The values of `compile-app` and `cleanup-app` are respectively determined in the following order of precedence:

- `--compile-app` / `--cleanup-app`
- `[tool.flet.<PLATFORM>.compile].app` / `[tool.flet.<PLATFORM>.cleanup].app`
- `[tool.flet.compile].app` / `[tool.flet.cleanup].app`
- empty list / empty list

The values of `compile-packages` and `cleanup-packages` are respectively determined in the following order of precedence:

- `--compile-packages` / `--cleanup-packages`
- `[tool.flet.<PLATFORM>.compile].packages` / `[tool.flet.<PLATFORM>.cleanup].packages`
- `[tool.flet.compile].packages` / `[tool.flet.cleanup].packages`
- `False` / `True`

The values of `cleanup-app-files` and `cleanup-package-files` are respectively determined in the following order of precedence:

- `--cleanup-app-files` / `--cleanup-package-files`
- `[tool.flet.<PLATFORM>.cleanup].app_files` / `[tool.flet.<PLATFORM>.cleanup].package_files`
- `[tool.flet.cleanup].app_files` / `[tool.flet.cleanup].package_files`
- `False` / `False`

## Example

- `flet build`

- `pyproject.toml`

```
flet build <target_platform> \
 --compile-app --compile-packages \
 --cleanup-app-files "**/*.c" "**/*.h" --cleanup-package-files "**/*.pyi"
[tool.flet.compile] # or [tool.flet.<PLATFORM>.compile]
```



```

app = true
packages = true

[tool.flet.cleanup] # or [tool.flet.<PLATFORM>.cleanup]
app = true
packages = true
app_files = ["**/*.c", "**/*.h"]
package_files = ["**/*.pyi"]

```

## Permissions

### Platform support

Android, iOS, and macOS only.

flet build allows granular control over permissions, features, and entitlements embedded into AndroidManifest.xml, Info.plist and .entitlements files.

See platform guides for setting specific iOS, Android and macOS permissions.

### Predefined cross-platform permission bundles

Cross-platform permissions are named and predefined bundles that apply a baseline set of platform-specific entries required for a feature. Each bundle expands into the corresponding platform-specific equivalents. This is especially useful for beginners who may be unfamiliar with the underlying platform APIs or prefer not to interact with them directly.

Only the bundles you list are applied. If you need different wording or extra entries, set the platform-specific tables directly; those values are merged on top and can override the bundle defaults. The examples below show the exact pyproject.toml equivalents for each bundle.

Below is a list of available bundles:

#### location

##### pyproject.toml equivalent

```

iOS
[tool.flet.ios.info]
NSLocationWhenInUseUsageDescription = "This app uses location service when in use."
NSLocationAlwaysAndWhenInUseUsageDescription = "This app uses location service."

macOS
[tool.flet.macos.info]
NSLocationUsageDescription = "This app needs access to your location."

[tool.flet.macos.entitlement]
"com.apple.security.personal-information.location" = true

Android
[tool.flet.android.permission]
"android.permission.ACCESS_FINE_LOCATION" = true
"android.permission.ACCESS_COARSE_LOCATION" = true
"android.permission.ACCESS_BACKGROUND_LOCATION" = true

[tool.flet.android.feature]
"android.hardware.location.network" = false
"android.hardware.location.gps" = false

```

#### camera

##### pyproject.toml equivalent

```

iOS
[tool.flet.ios.info]
NSCameraUsageDescription = "This app uses the camera to capture photos and videos."

macOS
[tool.flet.macos.info]
NSCameraUsageDescription = "This app uses the camera to capture photos and videos."

[tool.flet.macos.entitlement]
"com.apple.security.device.camera" = true

Android
[tool.flet.android.permission]
"android.permission.CAMERA" = true

[tool.flet.android.feature]
"android.hardware.camera" = false
"android.hardware.camera.any" = false
"android.hardware.camera.front" = false
"android.hardware.camera.external" = false
"android.hardware.camera.autofocus" = false

```

## microphone

### pyproject.toml equivalent

```
iOS
[tool.flet.ios.info]
NSMicrophoneUsageDescription = "This app uses microphone to record sounds."

macOS
[tool.flet.macos.info]
NSMicrophoneUsageDescription = "This app uses microphone to record sounds."

[tool.flet.macos.entitlement]
"com.apple.security.device.audio-input" = true

Android
[tool.flet.android.permission]
"android.permission.RECORD_AUDIO" = true
"android.permission.WRITE_EXTERNAL_STORAGE" = true
"android.permission.READ_EXTERNAL_STORAGE" = true
```

## photo\_library

### pyproject.toml equivalent

```
iOS
[tool.flet.ios.info]
NSPhotoLibraryUsageDescription = "This app saves photos and videos to the photo library."

macOS
[tool.flet.macos.info]
NSPhotoLibraryUsageDescription = "This app saves photos and videos to the photo library."

[tool.flet.macos.entitlement]
"com.apple.security.personal-information.photos-library" = true

Android
[tool.flet.android.permission]
"android.permission.READ_MEDIA_VISUAL_USER_SELECTED" = true
```

## Resolution order

Its value is determined in the following order of precedence:

- `--permissions`
- `[tool.flet].permissions` (type: list of strings)
- `[]`

## Example

- `flet build`
  - `pyproject.toml`
- ```
flet build <target_platform> --permissions location microphone

[tool.flet]
permissions = ["location", "microphone"]
```

Build template

`flet build` creates (and reuses) a Flutter project under `<app_root>/build/flutter` using a cookiecutter template. By default, the template is downloaded as a zip artifact from the matching Flet GitHub Release. The version of the template used is determined by the installed Flet version.

The cached project is refreshed when template inputs change or when you pass `--clear-cache`.

Template Source

Defines the location of the cookiecutter build-template to be used.

Supported values include:

- A GitHub repository using the `gh:` prefix (e.g., `gh:org/template`)
- A full Git URL (e.g., `https://github.com/org/template.git`)
- A zip URL (e.g., `https://github.com/flet-dev/flet/releases/download/v0.83.0/flet-build-template.zip`)
- A local directory path

Resolution order

Its value is determined in the following order of precedence:

- `--template`
- `[tool.flet.template].url`
- The default zip URL from the Flet GitHub Release matching the installed version

Example

- `flet build`
- `pyproject.toml`

```
flet build apk --template gh:my-org/my-custom-template

[tool.flet.template]
url = "gh:my-org/my-custom-template"
```

Template Reference

Defines the branch, tag, or commit to check out from the template source.

Resolution order

Its value is determined in the following order of precedence:

- `--template-ref`
- `[tool.flet.template].ref`
- `<flet_version>`

Example

- `flet build`
- `pyproject.toml`

```
flet build <target_platform> --template-ref main

[tool.flet.template]
ref = "main"
```

Template Directory

Defines the relative path to the cookiecutter template. If template source is set, the path is treated as a subdirectory within its root; otherwise, it is relative to the template root.

Resolution order

Its value is determined in the following order of precedence:

- `--template-dir`
- `[tool.flet.template].dir`
- root of the template source

Example

- `flet build`
- `pyproject.toml`

```
flet build <target_platform> --template gh:org/template --template-dir sub/directory

[tool.flet.template]
url = "gh:org/template"
dir = "sub/directory"
```

Additional flutter build Arguments

During the flet build process, flutter build command gets called internally to package your app for the specified platform. However, not all flutter build arguments are exposed or usable through the flet build command directly.

For possible flutter build arguments, see Flutter docs guide. For most targets, run `flutter build <target_platform> --help`; for `ios-simulator`, run `flutter build ios --simulator --help`.

Important

Passing additional flutter build arguments might cause unexpected behavior. Use at your own risk, and only if you fully know what you're doing!

Resolution order

Its value is determined in the following order of precedence:

- `--flutter-build-args` (can be used multiple times)
- `[tool.flet.<PLATFORM>.flutter].build_args`
- `[tool.flet.flutter].build_args`

Example

- `flet build`
- `pyproject.toml`

```
flet build apk \  
  --flutter-build-args=--obfuscate \  
  --flutter-build-args=--export-method=development \  
  --flutter-build-args=--dart-define=API_URL=https://api.example.com  
  
[tool.flet.flutter]      # or [tool.flet.<PLATFORM>.flutter]  
build_args = [  
  "--obfuscate",  
  "--export-method=development",  
  "--dart-define=API_URL=https://api.example.com",  
]
```

Flutter dependencies

When you run `flet build`, Flet generates a Flutter shell project and then updates its `pubspec.yaml` using values from `pyproject.toml`.

Use:

- `[tool.flet.flutter.pubspec.dependencies]` for normal package declarations. (Dart docs)
- `[tool.flet.flutter.pubspec.dependency_overrides]` when you must force a version or source, for example, a local path or Git fork. (Dart docs)

Values follow standard Pub dependency syntax, expressed in TOML.

Note

- Important: In most cases, you usually do not need to add/override Flutter dependencies. We recommend doing it only if you fully know what you are doing, as it can lead to unexpected behavior.
- If the same package appears in both `pyproject.toml` and the resulting `pubspec.yaml`, the value from `pyproject.toml` wins.
- If you use `{ path = "..."} under [tool.flet.flutter.pubspec.dependencies] or [tool.flet.flutter.pubspec.dependency_overrides], that path is resolved by Flutter from the generated pubspec.yaml location: <flet_app_directory>/build/flutter/pubspec.yaml. This means relative paths are not resolved from your pyproject.toml file.`

Example

```
[tool.flet.flutter.pubspec.dependencies]      # or [tool.flet.flutter.pubspec.dependency_overrides]  
# Version  
pkg_1 = "^1.2.3"  
  
# Local path  
pkg_2 = { path = "../pkg_2" }  
  
# Git (short form)  
pkg_3 = { git = "https://github.com/org/pkg_3.git" }  
  
# Git (expanded form: URL + ref + subdirectory)  
pkg_4 = { git = { url = "https://github.com/org/mono_repo.git", ref = "main", path = "packages/pkg_4" } }  
  
# Hosted source  
pkg_5 = { hosted = { name = "pkg_5", url = "https://pub.dev" }, version = "^1.0.0" }  
  
# SDK package (dependencies only; typically not used in dependency_overrides)  
flutter_test = { sdk = "flutter" }
```

Verbose logging

The `-v` (or `--verbose`) and `-vv` flags enable detailed output from all commands during the flet build process.

Use `-v` for standard/basic verbose logging, or `-vv` for even more detailed output (higher verbosity level). If you need support, we may ask you to share this verbose log.

Console output

In packaged apps (flet build output), all output from your Python code such as `print()` statements, `sys.stdout.write()` calls, and messages from the Python logging module is redirected to a `console.log` file. The full path to this file is available via `StoragePaths.get_console_log_filename()` or the `FLET_APP_CONSOLE` environment variable.

Note: `FLET_APP_CONSOLE` is only set in production builds; in development runs, output stays in your terminal.

The log file is written in an unbuffered manner, allowing you to read it at any point in your Python program using:

```
import os
import flet as ft

async def main(page: ft.Page):
    log_file = await ft.StoragePaths().get_console_log_filename()
    # or
    # log_file = os.getenv("FLET_APP_CONSOLE")

    with open(log_file, "r") as f:
        logs = f.read()
        page.add(ft.Text(logs)) # display on UI

ft.run(main)
```

If your program calls `sys.exit(100)`, the complete log will automatically be shown in a scrollable window. This is a special "magic" exit code for debugging purposes:

```
import sys
sys.exit(100)
```

Calling `sys.exit()` with any other code will terminate the app without displaying the log.

Continuous Integration/Continuous Deployment (CI/CD)

You can use flet build command in your CI/CD pipelines to automate the build and release process of your Flet apps.

GitHub Actions

You can use GitHub Actions to build your Flet app automatically on every push, pull request, or manual run.

```
name: Build Flet App

on:
  push:
  pull_request:
  workflow_dispatch:

env:
  UV_PYTHON: 3.12
  PYTHONUTF8: 1

  # https://flet.dev/docs/reference/environment-variables
  FLET_CLI_NO_RICH_OUTPUT: 1

jobs:
  build:
    name: Build ${{ matrix.name }}
    runs-on: ${{ matrix.runner }}
    strategy:
      fail-fast: false
    matrix:
      include:
        # ----- Desktop -----
        - name: linux
          runner: ubuntu-latest
          build_cmd: "flet build linux"
          artifact_path: build/linux
          needs_linux_deps: true

        - name: macos
          runner: macos-26
          build_cmd: "flet build macos"
          artifact_path: build/macos
          needs_linux_deps: false

        - name: windows
          runner: windows-latest
          build_cmd: "flet build windows"
```

```

    artifact_path: build/windows
    needs_linux_deps: false

# ----- Android -----
- name: aab
  runner: ubuntu-latest
  build_cmd: "flet build aab"
  artifact_path: build/aab
  needs_linux_deps: false

- name: apk
  runner: ubuntu-latest
  build_cmd: "flet build apk"
  artifact_path: build/apk
  needs_linux_deps: false

# ----- iOS -----
- name: ipa
  runner: macos-26
  build_cmd: "flet build ipa"
  artifact_path: build/ipa
  needs_linux_deps: false

- name: ios-simulator
  runner: macos-26
  build_cmd: "flet build ios-simulator"
  artifact_path: build/ios-simulator
  needs_linux_deps: false

# ----- Web -----
- name: web
  runner: ubuntu-latest
  build_cmd: "flet build web"
  artifact_path: build/web
  needs_linux_deps: false

steps:
- name: Checkout repository
  uses: actions/checkout@v4

- name: Setup uv
  uses: astral-sh/setup-uv@v6

- name: Install Linux dependencies
  if: matrix.needs_linux_deps
  shell: bash
  run: |
    sudo apt update --allow-releaseinfo-change
    LINUX_DEPS="$(uv run python -c 'from flet.utils.linux_deps import linux_dependencies; print(" ".join(linux_dependencies))')
    sudo apt-get install -y --no-install-recommends $LINUX_DEPS
    sudo apt-get clean

- name: Build app
  shell: bash
  run: |
    uv run ${matrix.build_cmd} --yes --verbose

- name: Upload Artifact
  uses: actions/upload-artifact@v4
  with:
    name: ${matrix.name}-build-artifact
    path: ${matrix.artifact_path}
    if-no-files-found: error
    overwrite: false

```

The workflow file above builds for all major targets and uploads each build output as an artifact. You can further customize the workflow for your specific needs, for example, restricting the build targets or adding additional steps.

See it in action [here](#).

Troubleshooting

Prerelease compatibility

If you are using a prerelease version of the Flet Python package (for example, 0.80.6.devNNNN) to build an app, the build template may still resolve the latest stable flet Flutter package, which can lead to version incompatibility issues.

Why?: Under normal circumstances, each prerelease of the Flet Python package would require a matching prerelease of the Flutter Flet package to guarantee compatibility. However, we don't publish prerelease versions of the Flutter package to pub.dev. Because of this, the build template resolves the latest stable Flutter flet release instead.

This creates a version mismatch/incompatibility for apps packaged with flet build:

- Your Python code may depend on newly introduced controls or features.
- The packaged Flutter shell may still be using an older stable flet version.
- At runtime, the app fails because the Flutter layer does not recognize the new controls/features in your prerelease flet package, leading to errors like Unknown control: <ControlName>.

Note: this issue does not affect the development workflows (ex: running an app with flet run), as the flet Flutter dependency is only resolved during the flet build process.

Solution

The rule-of-thumb is, if you are using a prerelease Flet Python package, always ensure the Flutter flet dependency is aligned with the same development version before building your app:

- Override the Flutter flet dependency to point to the corresponding development Git reference.
- pyproject.toml

```
[tool.flet.flutter.pubspec.dependency_overrides]
flet = { git = { url = "https://github.com/flet-dev/flet.git", ref = "main", path = "packages/flet" } }
```

- Rebuild the app with the build cache cleared (use --clear-cache; or manually delete build/flutter)

To ensure reproducible builds (ex: in production or CI), prefer using a specific commit SHA, instead of a branch or tag ref.

API Reference

- Controls - UI building blocks with properties, events, and usage examples.
- Services - Device and platform capabilities such as sensors, storage, and permissions.
- CLI - flet commands for creating, running, packaging, and debugging apps.
- Types - Core types, enums, events, exceptions, and utilities shared across the SDK.
- Built-in binary Python packages for Android and iOS - Prebuilt wheels available from <https://pypi.flet.dev> for mobile targets.
- Environment Variables - Runtime configuration toggles for apps and the CLI.

Flet Studio

Flet Studio is a browser-based editor for writing, running, saving and sharing Flet apps. Your Python code runs entirely in the browser via Pyodide — there is no Flet account required to try it, and no local install required to run something someone else built and shared with you.

Why it exists

Flet Studio started as a tool we wanted for ourselves and kept being useful for things we hadn't planned. In rough order:

- A playground that saves your work. Unlike a transient REPL or a paste-and-run sandbox, projects in Studio persist across sessions and across devices once you sign in.
- A way to share small apps via a link. Send a colleague or someone on the Flet team a URL and they can open, run, and read the code without setting anything up. Handy for bug reports, design sketches, and "look at this control".
- A gallery of Flet examples that you can actually edit. Browse examples, open one, tweak it, fork it into your own project — same editor, same runtime.
- A zero-commitment way to try Flet. Before `pip install flet`, before reading the getting-started docs, you can write a few lines and see them run.
- A dogfooding project. Studio's UI is itself a Flet app. Building it surfaces gaps in Flet and ideas for improvements, which feed back into the framework.
- The base UI for future Flet services. Cloud packaging and publishing are on the roadmap — Studio is the surface those will plug into. (Today: editor only.)

How it works

- Your code runs in the browser. Flet Studio uses Pyodide — CPython compiled to WebAssembly — to execute your Python directly in the browser tab. Nothing you write is sent to a server to be run.
- There is a Flet Studio backend, but it's only for the IDE itself — file storage, authentication, accounts, project metadata. Your app's Python code never executes on our servers; it executes client-side, in your browser, in Pyodide.
- The rendering engine is the same one `flet publish` and `flet build web` produce. What you see in Studio is what you'd get from a published Flet static site — same controls, same layout, same web runtime.

Limitations

Studio inherits the constraints of running Python in the browser. Most of these are covered in more detail in the static website publishing docs:

- Pure-Python packages and Pyodide-built wheels only. Most of the Python ecosystem works, but packages with C/Rust extensions need a Pyodide build. See the Pyodide built-in packages list. Source-only sdist (no wheel) cannot be installed.
- Single browser thread. No real threads. Long blocking work freezes the UI — prefer async I/O.
- Browser sandbox. No access to your local filesystem, no shell, no access to local hardware that isn't exposed by the browser.
- Performance. Pyodide is roughly 3×–5× slower than CPython for CPU-bound code.
- Cold start. The first load fetches the Python runtime and any wheels you import, so initial load is slower than a native install. Subsequent loads are cached.

Does it replace local Flet development?

No.

Studio is for trying Flet, prototyping small apps, sharing snippets, learning, and showing off ideas. For everything else — desktop and mobile builds, native packages, real multi-file projects, your favorite IDE and debugger, the full Python ecosystem without the Pyodide constraints above — install Flet locally and use `flet run` and `flet build`. See installation when you're ready to make the jump.

Types

Explore the shared building blocks that power Flet apps: aliases, base controls, reusable classes, enums, events, and exceptions.

- Aliases - shorthand names and conveniences for common patterns.
- Base Controls - foundational control types to extend and compose.
- Classes - core data structures (layout, animation, theming, platform info, and more).
- Decorators - helpers for defining components, controls, and observables.
- Enums - constant sets that define options for controls and services.
- Events - callback payloads emitted by controls and services.
- Exceptions - Flet-specific error types for diagnostics and handling.
- Methods - utilities for lifecycle, hooks, and component state.

Stay up to date

Use this section to track Flet releases, breaking changes, deprecations, and compatibility policy.

- Release notes - release announcements and changelog links.
- Breaking changes and deprecations - migration guides for changes that can require code or configuration updates.
- Compatibility policy - how Flet handles breaking changes and deprecations.

flet-controls

User interface is made of Controls (aka widgets).

To make controls visible to a user they must be added to a Page (the top-most control in the tree) or inside other controls. Nesting controls into each other could be represented as a tree with Page as a root.

Controls are just regular Python classes. Create control instances via constructors with parameters matching their properties, for example:

```
ft.Text(value="Hello, world!", color="green")
```

To display control on a page add it to controls list of a Page and call `page.update()` to send page changes to a browser or desktop client:

```
import flet as ft

def main(page: ft.Page):
    t = ft.Text(value="Hello, world!", color="green")
    page.controls.append(t)
    page.update()

ft.run(main)
```

Image:  - <https://flet.dev/img/docs/getting-started/controls-text.png>

note

In the following examples we will be showing just the contents of main function.

You can modify control properties and the UI will be updated on the next `page.update()`:

```
t = ft.Text()
page.add(t) # it's a shortcut for page.controls.append(t) and then page.update()

for i in range(10):
    t.value = f"Step {i}"
    page.update()
    time.sleep(1)
```

Some controls are "container" controls (like Page) which could contain other controls. For example, Row control allows arranging other controls in a row one-by-one:

```
page.add(
    ft.Row(
        controls=[
            ft.Text("A"),
            ft.Text("B"),
            ft.Text("C")
        ]
    )
)
```

or TextField and Button next to it:

```
page.add(
    ft.Row(
        controls=[
            ft.TextField(label="Your name"),
            ft.Button(text="Say my name!")
        ]
    )
)
```

`page.update()` is smart enough to send only the changes made since its last call, so you can add a couple of new controls to the page, remove some of them, change other controls' properties and then call `page.update()` to do a batched update, for example:

```
for i in range(10):
    page.controls.append(ft.Text(f"Line {i}"))
    if i > 4:
        page.controls.pop(0)
    page.update()
    time.sleep(0.3)
```

Some controls, like buttons, could have event handlers reacting on a user input, for example `Button.on_click`:

```
def button_clicked(e):
    page.add(ft.Text("Clicked!"))

page.add(ft.Button(text="Click me", on_click=button_clicked))
```

and more advanced example for a simple To-Do:

```
import flet as ft

def main(page):
    async def add_clicked(e):
        page.add(ft.Checkbox(label=new_task.value))
        new_task.value = ""
        await new_task.focus()
        new_task.update()

    new_task = ft.TextField(hint_text="What's needs to be done?", width=300)
    page.add(ft.Row([new_task, ft.Button("Add", on_click=add_clicked)]))

ft.run(main)
```

Image: Image - <https://flet.dev/img/docs/getting-started/simple-ToDo.png>

info

Flet implements imperative UI model where you "manually" build application UI with stateful controls and then mutate it by updating control properties. Flutter implements declarative model where UI is automatically re-built on application data changes. Managing application state in modern frontend applications is inherently complex task and Flet's "old-school" approach could be more attractive to programmers without frontend experience.

disabled property

Every control has disabled property which is false by default - control and all its children are enabled. disabled property is mostly used with data entry controls like TextField, Dropdown, Checkbox, buttons. However, disabled could be set to a parent control and its value will be propagated down to all children recursively.

For example, if you have a form with multiple entry control you can set disabled property for each control individually:

```
first_name = ft.TextField()
last_name = ft.TextField()
first_name.disabled = True
last_name.disabled = True
page.add(first_name, last_name)
```

or you can put form controls into container, e.g. Column and then set disabled for the column:

```
first_name = ft.TextField()
last_name = ft.TextField()
c = ft.Column(controls=[
    first_name,
    last_name
])
c.disabled = True
page.add(c)
```

Buttons

Button is the most essential input control which generates click event when pressed:

```
btn = ft.Button("Click me!")
page.add(btn)
```

Image: Image - <https://flet.dev/img/docs/getting-started/getting-user-input-elevated-button.png>

All events generated by controls on a web page are continuously sent back to your script, so how do you respond to a button click?

Event handlers

Buttons with events in "Counter" app:

```
import flet as ft

def main(page: ft.Page):
    page.title = "Flet counter example"
    page.vertical_alignment = ft.MainAxisAlignment.CENTER

    txt_number = ft.TextField(value="0", text_align="right", width=100)

    def minus_click(e):
        txt_number.value = str(int(txt_number.value) - 1)
        page.update()

    def plus_click(e):
        txt_number.value = str(int(txt_number.value) + 1)
        page.update()

    page.add(
        ft.Row(
            [
```

```

        ft.IconButton(ft.Icons.REMOVE, on_click=minus_click),
        txt_number,
        ft.IconButton(ft.Icons.ADD, on_click=plus_click),
    ],
    alignment=ft.MainAxisAlignment.CENTER,
)
)

```

```
ft.run(main)
```

Image - <https://flet.dev/img/docs/getting-started/getting-user-input-event-handlers.png>

Textbox

Flet provides a number of controls for building forms: TextField, Checkbox, Dropdown, Button.

Let's ask a user for a name:

```

import flet as ft

def main(page):
    def btn_click(e):
        if not txt_name.value:
            txt_name.error_text = "Please enter your name"
            page.update()
        else:
            name = txt_name.value
            page.clean()
            page.add(ft.Text(f"Hello, {name}!"))

    txt_name = ft.TextField(label="Your name")

    page.add(txt_name, ft.Button("Say hello!", on_click=btn_click))

ft.run(main)

```

Image - <https://flet.dev/img/docs/getting-started/getting-user-input-textbox.png>

Checkbox

The Checkbox control provides you with various properties and events emmiters for ease of use.

Let's create a one checkbox ToDo:

```

import flet as ft

def main(page):
    def checkbox_changed(e):
        output_text.value = (
            f"You have learned how to ski : {todo_check.value}."
        )
        page.update()

    output_text = ft.Text()
    todo_check = ft.Checkbox(label="ToDo: Learn how to use ski", value=False, on_change=checkbox_changed)
    page.add(todo_check, output_text)

ft.run(main)

```

Image - <https://flet.dev/img/docs/getting-started/getting-user-input-checkbox.png>

Dropdown

```

import flet as ft

def main(page: ft.Page):
    def button_clicked(e):
        output_text.value = f"Dropdown value is: {color_dropdown.value}"
        page.update()

    output_text = ft.Text()
    submit_btn = ft.Button(text="Submit", on_click=button_clicked)
    color_dropdown = ft.Dropdown(
        width=100,
        options=[
            ft.dropdown.Option("Red"),
            ft.dropdown.Option("Green"),
            ft.dropdown.Option("Blue"),
        ],
    )
    page.add(color_dropdown, submit_btn, output_text)

ft.run(main)

```

Image - <https://flet.dev/img/docs/getting-started/getting-user-input-dropdown.png>

Mobile support

This article covers Flet's vision for going mobile and provides a roadmap.

Server-Driven UI

Flet is a Server-driven UI (SDUI) framework. SDUI is an emerging technology which is the best described in Technology Radar post:

Server-driven UI separates the rendering into a generic container in the mobile app while the structure and data for each view is provided by the server. This means that changes that once required a round trip to an app store can now be accomplished via simple changes to the responses the server sends.

Companies like DoorDash, Airbnb, Lyft and others have been successfully implementing Server-driven UI in their mobile apps to reduce time-to-market.

Flet approach

Flet is going to implement Server-Driven UI approach where program written in Python or other language is running on the server and only a thin client - either standalone Flutter app (.apk or .ipa package) in app store or a Flutter widget as a part of another app - is delivered to a mobile:

Missing image omitted: Flet highlevel diagram

Once SDUI experience is ready we'll start working on a standalone mobile package.

Roadmap

To provide the best experience for Flet apps on mobile platforms, we plan to release the following items by the end of this year:

Flet widget for Flutter

The first step we are going to do is to separate Flet client into a Flutter widget and publish the package at <https://pub.dev>. Flet widget could be then integrated by mobile developers into existing or new Flutter apps for adding dynamic server-driven UI experiences to the core app functionality. A new Flutter app could be also created with a single Flet widget just for the purpose of hosting a complete Flet app.

Developers will follow Flutter guide for packaging, signing and distributing their apps to Android, iOS, Linux, macOS or Windows platforms.

Flet team will provide sample CI pipelines to automate packaging, signing and publishing of Flutter apps.

Flet Studio for iOS and Android

The next step is a standalone "Flet Studio" app (the name is not final) in App Store and Google Play for "testing mobile experiences developed with Flet framework". Developers or beta testers will be able to "register" URL of their hosted Flet app within Flet Studio and instantly see how it performs on a mobile device.

White-labeled Flet mobile app

We are going to provide a guide and CI pipeline for automatic publishing of white-labeled Flet app to a user App Store or Google Play account. This app will be "pinned" to a specific app URL and could additionally bundle app assets (media, fonts) to minimize network usage.

Standalone mobile package for Flet app

We are going to investigate the way and develop a prototype for bundling together Flet framework, user program, language runtime and all dependencies into a standalone mobile package (.apk or .ipa package), so Flet program does not require a web server.

Embedding Flet into native apps

We are going to provide a guide, sample apps and CI pipeline to integrate Flet widget into existing native Android and iOS apps (not developed with Flutter) using Flutter Add-to-App feature. Put Flutter to work article gives a real-world example on how to integrate Flutter into existing mobile app.

packaging-desktop-app-with-pyinstaller

A better way to package is here

You can now use flet build command to package your Flet app into executable or install bundle for macOS, Windows and Linux.

flet build does not longer rely on PyInstaller like flet pack does, but uses Flutter SDK to produce a fast, offline, fully customizable (your own icons, about dialog and metadata) executable for Windows, Linux and macOS with Python runtime embedded into executable and running in-process.

Follow this guide for desktop packaging

Flet Python app and all its dependencies can be packaged into an executable and user can run it on their computer without installing a Python interpreter or any modules.

Flet wraps PyInstaller API to package Flet Python app and all its dependencies into a single package for Windows, macOS and Linux. To create Windows package, PyInstaller must be run on Windows; to build Linux app, it must be run on Linux; and to build macOS app - on macOS.

Start from installing PyInstaller:

```
pip install pyinstaller
```

Navigate to the directory where your .py file is located and build your app with the following command:

```
flet pack your_program.py
```

Your bundled Flet app should now be available in dist folder. Try running the program to see if it works.

On macOS:

```
open dist/your_program.app
```

on Windows:

```
dist\your_program.exe
```

on Linux:

```
dist/your_program
```

Now you can just zip the contents of dist folder and distribute to your users! They don't need Python or Flet installed to run your packaged program - what a great alternative to Electron!

By default, an executable/bundle has the same name as a Python script. You can change it with --name argument:

```
flet pack your_program.py --name bundle_name
```

Customizing package icon

Default bundle app icon is diskette which might be confusing for younger developers missed those ancient times when floppy disks were used to store computer data.

You can replace the icon with your own by adding --icon argument:

```
flet pack your_program.py --icon <your-image.png>
```

PyInstaller will convert provided PNG to a platform specific format (.ico for Windows and .icns for macOS), but you need to install Pillow module for that:

```
pip install pillow
```

Packaging assets

Your Flet app can include assets. Provided app assets are in assets folder next to your_program.py they can be added to an application package with --add-data argument, on macOS/Linux:

```
flet pack your_program.py --add-data "assets:assets"
```

On Windows assets;assets must be delimited with ;:

```
flet pack your_program.py --add-data "assets;assets"
```

Customizing macOS bundle

macOS bundle details can be customized with the following flet pack macOS-specific arguments:

- `--product-name` - display name of macOS bundle, shown in Dock, Activity Monitor, About dialog.
- `--product-version` - bundle version shown in "About" dialog.
- `--copyright` - copyright notice shown in "About" dialog.
- `--bundle-id` unique bundle ID.

Image: Flet app bundle about - <https://flet.dev/docs/assets/getting-started/package-desktop/flet-app-bundle-about.png>

Customizing Windows executable metadata

Windows executable "Details" properties dialog can be customized with the following flet pack arguments:

- `--product-name` - "Product name" field.
- `--product-version` - "Product version" field.
- `--file-version` - "File version" field.
- `--file-description` - "File description" field, also program display name in Task Manager.
- `--copyright` - "Copyright" field.

Using CI for multi-platform packaging

To create an app package with PyInstaller for specific OS it must be run on that OS.

If you don't have an access to Mac or PC you can bundle your app for all three platforms with AppVeyor - Continuous Integration service for Windows, Linux and macOS. In short, Continuous Integration (CI) is an automated process of building, testing and deploying (Continuous Delivery - CD) application on every push to a repository.

AppVeyor is free for open source projects hosted on GitHub, GitLab and Bitbucket. To use AppVeyor, push your app to a repository within one of those source-control providers.

note

AppVeyor is the company behind Flet.

To get started with AppVeyor sign up for a free account.

Click "New project" button, authorize AppVeyor to access your GitHub, GitLab or Bitbucket account, choose a repository with your program and create a new project.

Now, to configure packaging of your app for Windows, Linux and macOS, add file with the following contents into the root of your repository `appveyor.yml`. `appveyor.yml` is a build configuration file, or CI workflow, describing build, test, packaging and deploy commands that must be run on every commit.

note

You can just fork `flet-dev/python-ci-example` repository and customize it to your needs.

When you push any changes to GitHub repository, AppVeyor will automatically start a new build:

Missing image omitted: AppVeyor CI Flet Python project

What that CI workflow does on every push to the repository:

- Clones the repository to a clean virtual machine.
- Installs app dependencies using `pip`.
- Runs flet pack to package Python app into a bundle for Windows, macOS and Ubuntu.
- Zip/Tar app bundles and uploads them to "Artifacts".
- Uploads app bundles to GitHub releases when a new tag is pushed. Just push a new tag to make a release!

GITHUB_TOKEN

GITHUB_TOKEN in `appveyor.yml` is a GitHub Personal Access Token (PAT) used by AppVeyor to publish created packages to repository "Releases". You need to generate your own token and replace it in `appveyor.yml`. Login to your GitHub account and navigate to Personal access token page. Click "Generate new token" and select "public_repo" or "repo" scope for public or private repository respectively. Copy generated token to a clipboard and return to AppVeyor Portal. Navigate to Encrypt configuration data page and paste token to "Value to encrypt" field, click "Encrypt" button. Put encrypted value under GITHUB_TOKEN in your `appveyor.yml`.

Configure AppVeyor for your Python project, push a new tag to a repository and "automagically" get desktop bundle for all three platforms in GitHub releases! ☐

Missing image omitted: AppVeyor CI Flet GitHub releases

In addition to GitHub Releases, you can also configure releasing of artifacts to Amazon S3 bucket or Azure Blob storage.

flet build

Build a Flet Python app into a platform-specific executable or installable bundle. It supports building for desktop (macOS, Linux, Windows), web, Android (APK/AAB), and iOS (IPA and simulator .app), with a wide range of customization options for metadata, assets, splash screens, and signing.

Detailed guide with usage examples: <https://flet.dev/docs/publish>.

Usage

```
flet build [-h] [-v] [-o OUTPUT_DIR]
            [--arch TARGET_ARCH [TARGET_ARCH ...]]
            [--exclude EXCLUDE [EXCLUDE ...]] [--clear-cache]
            [--project PROJECT_NAME] [--artifact ARTIFACT_NAME]
            [--description DESCRIPTION] [--product PRODUCT_NAME]
            [--org ORG_NAME] [--bundle-id BUNDLE_ID]
            [--company COMPANY_NAME] [--copyright COPYRIGHT]
            [--android-adaptive-icon-background ANDROID_ADAPTIVE_ICON_BACKGROUND]
            [--splash-color SPLASH_COLOR]
            [--splash-dark-color SPLASH_DARK_COLOR] [--no-web-splash]
            [--no-ios-splash] [--no-android-splash]
            [--ios-team-id IOS_TEAM_ID]
            [--ios-export-method IOS_EXPORT_METHOD]
            [--ios-provisioning-profile IOS_PROVISIONING_PROFILE]
            [--ios-signing-certificate IOS_SIGNING_CERTIFICATE]
            [--base-url BASE_URL] [--web-renderer {auto,canvaskit,skwasm}]
            [--route-url-strategy {path,hash}]
            [--pwa-background-color PWA_BACKGROUND_COLOR]
            [--pwa-theme-color PWA_THEME_COLOR] [--no-wasm] [--no-cdn]
            [--split-per-abi] [--compile-app] [--compile-packages]
            [--cleanup-app]
            [--cleanup-app-files CLEANUP_APP_FILES [CLEANUP_APP_FILES ...]]
            [--cleanup-packages]
            [--cleanup-package-files CLEANUP_PACKAGE_FILES [CLEANUP_PACKAGE_FILES ...]]
            [--flutter-build-args [FLUTTER_BUILD_ARGS ...]]
            [--source-packages SOURCE_PACKAGES [SOURCE_PACKAGES ...]]
            [--info-plist INFO_PLIST [INFO_PLIST ...]]
            [--macos-entitlements MACOS_ENTITLEMENTS [MACOS_ENTITLEMENTS ...]]
            [--android-features ANDROID_FEATURES [ANDROID_FEATURES ...]]
            [--android-permissions ANDROID_PERMISSIONS [ANDROID_PERMISSIONS ...]]
            [--android-meta-data ANDROID_META_DATA [ANDROID_META_DATA ...]]
            [--permissions {location,camera,microphone,photo_library} [{location,camera,microphone,photo_library} ...]]
            [--deep-linking-scheme DEEP_LINKING_SCHEME]
            [--deep-linking-host DEEP_LINKING_HOST]
            [--android-signing-key-store ANDROID_SIGNING_KEY_STORE]
            [--android-signing-key-store-password ANDROID_SIGNING_KEY_STORE_PASSWORD]
            [--android-signing-key-password ANDROID_SIGNING_KEY_PASSWORD]
            [--android-signing-key-alias ANDROID_SIGNING_KEY_ALIAS]
            [--build-number BUILD_NUMBER] [--build-version BUILD_VERSION]
            [--module-name MODULE_NAME] [--template TEMPLATE]
            [--template-dir TEMPLATE_DIR] [--template-ref TEMPLATE_REF]
            [--show-platform-matrix] [--no-rich-output] [--yes]
            [--skip-flutter-doctor]
            {macos,linux,windows,web,apk,aab,ipa,ios-simulator}
            [python_app_path]
```

Positional arguments

target_platform

The target platform or type of package to build.

Possible values: aab, apk, ios-simulator, ipa, linux, macos, web, windows

Required: true

python_app_path

Path to a directory with a Flet Python program.

Default: .

Options

--android-adaptive-icon-background

The color to be used to fill out the background of Android adaptive icons.

--android-features

The list of <feature_name>=true|false features to add to AndroidManifest.xml (android only); can be used multiple times.

Value: ANDROID_FEATURES+

--android-meta-data

The list of <name>=<value> app meta-data entries to add to AndroidManifest.xml (android only); can be used multiple times.

Value: ANDROID_META_DATA+

--android-permissions

The list of <permission_name>=true|false permissions to add to AndroidManifest.xml (android only); can be used multiple times.

Value: ANDROID_PERMISSIONS+

--android-signing-key-alias

Android signing key alias.

Environment variable: FLET_ANDROID_SIGNING_KEY_ALIAS

--android-signing-key-password

Android signing key password.

Environment variable: FLET_ANDROID_SIGNING_KEY_PASSWORD

--android-signing-key-store

Path to an upload keystore .jks file for Android apps.

Environment variable: FLET_ANDROID_SIGNING_KEY_STORE

--android-signing-key-store-password

Android signing store password.

Environment variable: FLET_ANDROID_SIGNING_KEY_STORE_PASSWORD

--arch

Build for specific CPU architectures (used in macOS and Android builds only). Example: --arch arm64 x64.

Value: TARGET_ARCH+

--artifact

Executable or bundle name on disk.

--base-url

Base URL from which the app is served (web only).

--build-number

Build number - an identifier used as an internal version number.

Value: BUILD_NUMBER <int>

--build-version

Build version - a x.y.z string used as the version number shown to users.

--bundle-id

Bundle ID for the application, e.g. com.mycompany.app-name. It is used as an iOS, Android, macOS and Linux bundle ID.

--cleanup-app

Remove unnecessary app files upon packaging.

--cleanup-app-files

The list of globs to delete extra app files and directories.

Value: CLEANUP_APP_FILES+

--cleanup-package-files

The list of globs to delete extra package files and directories.

Value: CLEANUP_PACKAGE_FILES+

--cleanup-packages

Remove unnecessary package files upon packaging.

--clear-cache

Remove any existing build cache before starting the build process.

--company

Company name to display in about app dialogs.

--compile-app

Pre-compile app's .py files to .pyc.

--compile-packages

Pre-compile site packages' .py files to .pyc.

--copyright

Copyright text to display in about app dialogs.

--deep-linking-host

Deep linking URL host for iOS and Android builds.

--deep-linking-scheme

Deep linking URL scheme to configure for iOS and Android builds, i.g. https or myapp.

--description

Short description of the application.

--exclude

Files and/or directories to exclude from the package; can be used multiple times.

Value: EXCLUDE+

--flutter-build-args

Additional arguments for flutter build command.

Value: FLUTTER_BUILD_ARGS*

--help

Show this help message and exit.

Aliases: -h

--info-plist

The list of <key>=<value> pairs to add to Info.plist. Values can be booleans, strings, numbers, TOML arrays, or TOML inline tables (macos, ipa and ios-simulator only); can be used multiple times.

Value: INFO_PLIST+

--ios-export-method

Export method for iOS app bundle (default: debugging).

--ios-provisioning-profile

Provisioning profile name or UUID that should be used to sign and export iOS app bundle.

--ios-signing-certificate

Signing certificate name, SHA-1 hash, or automatic selector to use for signing iOS app bundle.

--ios-team-id

Apple developer team ID for signing iOS app bundle (ipa only).

--macos-entitlements

The list of <key>=<value> entitlements. Values can be booleans, strings, numbers, TOML arrays, or TOML inline tables (macos only); can be used multiple times.

Value: MACOS_ENTITLEMENTS+

--module-name

Python module name with an app entry point.

--no-android-splash

Disable splash screen on Android platform.

--no-cdn

Disable loading of CanvasKit, Pyodide and fonts from CDN.

Environment variable: FLET_WEB_NO_CDN

--no-ios-splash

Disable splash screen on iOS platform.

--no-rich-output

Disable rich output and prefer plain text. Useful on Windows builds.

Environment variable: FLET_CLI_NO_RICH_OUTPUT

--no-wasm

Disable WASM target for web build (web only).

--no-web-splash

Disable splash screen on web platform.

--org

Organization name in reverse domain name notation, e.g. com.mycompany, combined with project name and used in bundle IDs and signing.

--output

Output directory for the final executable/bundle (default: <python_app_path>/build/<target_platform>).

Aliases: -o

--permissions

The list of pre-defined cross-platform permissions for iOS, Android and macOS builds.

Possible values: camera, location, microphone, photo_library

--product

Display name shown in app launchers, window titles, and about dialogs.

--project

Project name for bundle IDs and identifiers; used as the default for artifact and product names.

--pwa-background-color

Initial background color for your web app (web only).

--pwa-theme-color

Default color for your web app's user interface (web only).

--route-url-strategy

Base URL path to serve the app from. Useful if the app is hosted in a subdirectory (web only).

Possible values: hash, path

Environment variable: FLET_WEB_ROUTE_URL_STRATEGY

--show-platform-matrix

Display the build platform matrix in a table, then exit.

--skip-flutter-doctor

Skip running Flutter doctor upon failed builds.

Environment variable: FLET_CLI_SKIP_FLUTTER_DOCTOR

--source-packages

The list of Python packages to install from source distributions.

Value: SOURCE_PACKAGES+

--splash-color

Background color of app splash screen on iOS, Android and web.

--splash-dark-color

Background color in dark mode of app splash screen on iOS, Android and web.

--split-per-abi

Split the APKs per ABIs (Android only).

--template

Directory containing Flutter bootstrap template, or a URL to a git repository template.

--template-dir

Relative path to a Flutter bootstrap template in a repository.

--template-ref

The branch, tag or commit ID to checkout after cloning the repository with Flutter bootstrap template.

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

--web-renderer

Flutter web renderer to use (web only).

Possible values: auto, canvaskit, skwasm

Environment variable: FLET_WEB_RENDERER

--yes

Answer yes to all prompts (install dependencies without confirmation).

flet create

Create a new Flet project using a predefined template. It sets up the initial directory structure, metadata, and required files to help you get started quickly.

Usage

```
flet create [-h] [-v] [--project-name PROJECT_NAME]
            [--description DESCRIPTION] [--template {app,extension}]
            [--template-ref TEMPLATE_REF]
            [output_directory]
```

Positional arguments

output_directory

Directory where the new Flet project will be created. If omitted, the project is created in the current directory.

Default: .

Options

--description

Short description of the new Flet project. This will appear in generated metadata.

--help

Show this help message and exit.

Aliases: -h

--project-name

Name of the new Flet project. This will be used in metadata files such as pyproject.toml.

--template

The template to use (or type of project to create) for new Flet project.

Possible values: app, extension

Default: app

--template-ref

Git reference (branch, tag, or commit ID) of the Flet template repository (flet-dev/flet-app-templates) to use. Useful when using a custom or development version of templates.

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

flet debug

Run a Flet Python app in debug mode on a specified platform (desktop, web, mobile).

Usage

```
flet debug [-h] [-v] [--device-id DEVICE_ID] [--show-devices] [--release]
            [--route ROUTE] [--arch TARGET_ARCH [TARGET_ARCH ...]]
            [--exclude EXCLUDE [EXCLUDE ...]] [--clear-cache]
            [--project PROJECT_NAME] [--artifact ARTIFACT_NAME]
            [--description DESCRIPTION] [--product PRODUCT_NAME]
            [--org ORG_NAME] [--bundle-id BUNDLE_ID]
            [--company COMPANY_NAME] [--copyright COPYRIGHT]
            [--android-adaptive-icon-background ANDROID_ADAPTIVE_ICON_BACKGROUND]
            [--splash-color SPLASH_COLOR]
            [--splash-dark-color SPLASH_DARK_COLOR] [--no-web-splash]
            [--no-ios-splash] [--no-android-splash]
            [--ios-team-id IOS_TEAM_ID]
            [--ios-export-method IOS_EXPORT_METHOD]
            [--ios-provisioning-profile IOS_PROVISIONING_PROFILE]
            [--ios-signing-certificate IOS_SIGNING_CERTIFICATE]
            [--base-url BASE_URL] [--web-renderer {auto,canvaskit,skwasm}]
            [--route-url-strategy {path,hash}]
            [--pwa-background-color PWA_BACKGROUND_COLOR]
            [--pwa-theme-color PWA_THEME_COLOR] [--no-wasm] [--no-cdn]
            [--split-per-abi] [--compile-app] [--compile-packages]
            [--cleanup-app]
            [--cleanup-app-files CLEANUP_APP_FILES [CLEANUP_APP_FILES ...]]
            [--cleanup-packages]
            [--cleanup-package-files CLEANUP_PACKAGE_FILES [CLEANUP_PACKAGE_FILES ...]]
            [--flutter-build-args [FLUTTER_BUILD_ARGS ...]]
            [--source-packages SOURCE_PACKAGES [SOURCE_PACKAGES ...]]
            [--info-plist INFO_PLIST [INFO_PLIST ...]]
            [--macos-entitlements MACOS_ENTITLEMENTS [MACOS_ENTITLEMENTS ...]]
            [--android-features ANDROID_FEATURES [ANDROID_FEATURES ...]]
            [--android-permissions ANDROID_PERMISSIONS [ANDROID_PERMISSIONS ...]]
            [--android-meta-data ANDROID_META_DATA [ANDROID_META_DATA ...]]
            [--permissions {location,camera,microphone,photo_library} [{location,camera,microphone,photo_library} ...]]
            [--deep-linking-scheme DEEP_LINKING_SCHEME]
            [--deep-linking-host DEEP_LINKING_HOST]
            [--android-signing-key-store ANDROID_SIGNING_KEY_STORE]
            [--android-signing-key-store-password ANDROID_SIGNING_KEY_STORE_PASSWORD]
            [--android-signing-key-password ANDROID_SIGNING_KEY_PASSWORD]
            [--android-signing-key-alias ANDROID_SIGNING_KEY_ALIAS]
            [--build-number BUILD_NUMBER] [--build-version BUILD_VERSION]
            [--module-name MODULE_NAME] [--template TEMPLATE]
            [--template-dir TEMPLATE_DIR] [--template-ref TEMPLATE_REF]
            [--show-platform-matrix] [--no-rich-output] [--yes]
            [--skip-flutter-doctor]
            [{macos,linux,windows,web,ios,android}] [python_app_path]
```

Positional arguments

platform

The target platform to run the app on.

Possible values: android, ios, linux, macos, web, windows

Required: false

python_app_path

Path to a directory with a Flet Python program.

Default: .

Options

--android-adaptive-icon-background

The color to be used to fill out the background of Android adaptive icons.

--android-features

The list of <feature_name>=true|false features to add to AndroidManifest.xml (android only); can be used multiple times.

Value: ANDROID_FEATURES+

--android-meta-data

The list of <name>=<value> app meta-data entries to add to AndroidManifest.xml (android only); can be used multiple times.

Value: ANDROID_META_DATA+

--android-permissions

The list of <permission_name>=true|false permissions to add to AndroidManifest.xml (android only); can be used multiple times.

Value: ANDROID_PERMISSIONS+

--android-signing-key-alias

Android signing key alias.

Environment variable: FLET_ANDROID_SIGNING_KEY_ALIAS

--android-signing-key-password

Android signing key password.

Environment variable: FLET_ANDROID_SIGNING_KEY_PASSWORD

--android-signing-key-store

Path to an upload keystore .jks file for Android apps.

Environment variable: FLET_ANDROID_SIGNING_KEY_STORE

--android-signing-key-store-password

Android signing store password.

Environment variable: FLET_ANDROID_SIGNING_KEY_STORE_PASSWORD

--arch

Build for specific CPU architectures (used in macOS and Android builds only). Example: --arch arm64 x64.

Value: TARGET_ARCH+

--artifact

Executable or bundle name on disk.

--base-url

Base URL from which the app is served (web only).

--build-number

Build number - an identifier used as an internal version number.

Value: BUILD_NUMBER <int>

--build-version

Build version - a x.y.z string used as the version number shown to users.

--bundle-id

Bundle ID for the application, e.g. com.mycompany.app-name. It is used as an iOS, Android, macOS and Linux bundle ID.

--cleanup-app

Remove unnecessary app files upon packaging.

--cleanup-app-files

The list of globs to delete extra app files and directories.

Value: CLEANUP_APP_FILES+

--cleanup-package-files

The list of globs to delete extra package files and directories.

Value: CLEANUP_PACKAGE_FILES+

--cleanup-packages

Remove unnecessary package files upon packaging.

--clear-cache

Remove any existing build cache before starting the build process.

--company

Company name to display in about app dialogs.

--compile-app

Pre-compile app's .py files to .pyc.

--compile-packages

Pre-compile site packages' .py files to .pyc.

--copyright

Copyright text to display in about app dialogs.

--deep-linking-host

Deep linking URL host for iOS and Android builds.

--deep-linking-scheme

Deep linking URL scheme to configure for iOS and Android builds, i.g. https or myapp.

--description

Short description of the application.

--device-id

Device ID to run the app on for iOS and Android builds.

Aliases: -d

--exclude

Files and/or directories to exclude from the package; can be used multiple times.

Value: EXCLUDE+

--flutter-build-args

Additional arguments for flutter build command.

Value: FLUTTER_BUILD_ARGS*

--help

Show this help message and exit.

Aliases: -h

--info-plist

The list of <key>=<value> pairs to add to Info.plist. Values can be booleans, strings, numbers, TOML arrays, or TOML inline tables (macos, ipa and ios-simulator only); can be used multiple times.

Value: INFO_PLIST+

--ios-export-method

Export method for iOS app bundle (default: debugging).

--ios-provisioning-profile

Provisioning profile name or UUID that should be used to sign and export iOS app bundle.

--ios-signing-certificate

Signing certificate name, SHA-1 hash, or automatic selector to use for signing iOS app bundle.

--ios-team-id

Apple developer team ID for signing iOS app bundle (ipa only).

--macos-entitlements

The list of <key>=<value> entitlements. Values can be booleans, strings, numbers, TOML arrays, or TOML inline tables (macos only); can be used multiple times.

Value: MACOS_ENTITLEMENTS+

--module-name

Python module name with an app entry point.

--no-android-splash

Disable splash screen on Android platform.

--no-cdn

Disable loading of CanvasKit, Pyodide and fonts from CDN.

Environment variable: FLET_WEB_NO_CDN

--no-ios-splash

Disable splash screen on iOS platform.

--no-rich-output

Disable rich output and prefer plain text. Useful on Windows builds.

Environment variable: FLET_CLI_NO_RICH_OUTPUT

--no-wasm

Disable WASM target for web build (web only).

--no-web-splash

Disable splash screen on web platform.

--org

Organization name in reverse domain name notation, e.g. com.mycompany, combined with project name and used in bundle IDs and signing.

--permissions

The list of pre-defined cross-platform permissions for iOS, Android and macOS builds.

Possible values: camera, location, microphone, photo_library

--product

Display name shown in app launchers, window titles, and about dialogs.

--project

Project name for bundle IDs and identifiers; used as the default for artifact and product names.

--pwa-background-color

Initial background color for your web app (web only).

--pwa-theme-color

Default color for your web app's user interface (web only).

--release

Build the app in release mode.

--route

Route to open the app on for web, iOS and Android builds.

--route-url-strategy

Base URL path to serve the app from. Useful if the app is hosted in a subdirectory (web only).

Possible values: hash, path

Environment variable: FLET_WEB_ROUTE_URL_STRATEGY

--show-devices

Show connected devices for iOS and Android builds.

--show-platform-matrix

Display the build platform matrix in a table, then exit.

--skip-flutter-doctor

Skip running Flutter doctor upon failed builds.

Environment variable: FLET_CLI_SKIP_FLUTTER_DOCTOR

--source-packages

The list of Python packages to install from source distributions.

Value: SOURCE_PACKAGES+

--splash-color

Background color of app splash screen on iOS, Android and web.

--splash-dark-color

Background color in dark mode of app splash screen on iOS, Android and web.

--split-per-abi

Split the APKs per ABIs (Android only).

--template

Directory containing Flutter bootstrap template, or a URL to a git repository template.

--template-dir

Relative path to a Flutter bootstrap template in a repository.

--template-ref

The branch, tag or commit ID to checkout after cloning the repository with Flutter bootstrap template.

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

--web-renderer

Flutter web renderer to use (web only).

Possible values: auto, canvaskit, skwasm

Environment variable: FLET_WEB_RENDERER

--yes

Answer yes to all prompts (install dependencies without confirmation).

flet devices

List all connected iOS and Android devices.

Usage

```
flet devices [-h] [-v] [--device-timeout DEVICE_TIMEOUT]
               [--device-connection {both,attached,wireless}]
               [--no-rich-output] [--yes] [--skip-flutter-doctor]
               [{ios,android}]
```

Positional arguments

platform

The target platform to list devices for. If not specified, lists all platforms.

Possible values: android, ios

Required: false

Options

--device-connection

Filter devices by connection type: attached (USB) or wireless.

Possible values: attached, both, wireless

Default: both

--device-timeout

Time (in seconds) to wait for devices to attach.

Value: DEVICE_TIMEOUT <int>

Default: 10

--help

Show this help message and exit.

Aliases: -h

--no-rich-output

Disable rich output and prefer plain text. Useful on Windows builds.

Environment variable: FLET_CLI_NO_RICH_OUTPUT

--skip-flutter-doctor

Skip running Flutter doctor upon failed builds.

Environment variable: FLET_CLI_SKIP_FLUTTER_DOCTOR

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

--yes

Answer yes to all prompts (install dependencies without confirmation).

flet doctor

Get information about the system and environment setup.

Usage

```
flet doctor [-h] [-v]
```

Options

--help

Show this help message and exit.

Aliases: -h

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

flet emulators

List, create, and launch available emulators.

Usage

```
flet emulators [-h] [-v] [--cold] [--no-rich-output] [--yes]
                [--skip-flutter-doctor]
                [{start,create,delete}] [emulator]
```

Positional arguments

action

Action to perform: start an emulator, create a new one, or delete it.

Possible values: create, delete, start

Required: false

emulator

Emulator ID or name (required for start, create, and delete).

Required: false

Options

--cold

Cold boot the emulator when starting.

--help

Show this help message and exit.

Aliases: -h

--no-rich-output

Disable rich output and prefer plain text. Useful on Windows builds.

Environment variable: FLET_CLI_NO_RICH_OUTPUT

--skip-flutter-doctor

Skip running Flutter doctor upon failed builds.

Environment variable: FLET_CLI_SKIP_FLUTTER_DOCTOR

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

--yes

Answer yes to all prompts (install dependencies without confirmation).

flet pack

Package a Flet application into a standalone desktop executable or app bundle using PyInstaller.

Usage

```
flet pack [-h] [-v] [-i ICON] [-n NAME] [-D] [--distpath DISTPATH]
          [--add-data [ADD_DATA ...]] [--add-binary [ADD_BINARY ...]]
          [--hidden-import [HIDDEN_IMPORT ...]]
          [--product-name PRODUCT_NAME]
          [--file-description FILE_DESCRIPTION]
          [--product-version PRODUCT_VERSION]
          [--file-version FILE_VERSION] [--company-name COMPANY_NAME]
          [--copyright COPYRIGHT] [--codesign-identity CODESIGN_IDENTITY]
          [--bundle-id BUNDLE_ID] [--debug-console DEBUG_CONSOLE]
          [--uac-admin]
          [--pyinstaller-build-args [PYINSTALLER_BUILD_ARGS ...]] [-y]
script
```

Positional arguments

script

Path to the Python script that launches your Flet app.

Required: true

Options

--add-binary

Additional binary files to be added to the executable.Format: source:destination[:platform].

Value: ADD_BINARY*

--add-data

Add additional non-binary files or folders to the bundle. Accepts one or more arguments in the form source:destination.

Value: ADD_DATA*

--bundle-id

Bundle identifier used for macOS app packaging.

--codesign-identity

Code signing identity to sign the app bundle (macOS only).

--company-name

Company name metadata for the Windows executable.

--copyright

Copyright string embedded in the executable (Windows) or bundle (macOS).

--debug-console

Show python debug console window (ensure correct DEBUG level). Useful for troubleshooting runtime errors.

--distpath

Directory where the packaged app will be placed.

Default: dist

--file-description

File description to embed in the executable (Windows).

--file-version

File version for the executable in n.n.n.n format (Windows only).

--help

Show this help message and exit.

Aliases: -h

--hidden-import

Add Python modules that are dynamically imported and not detected by static analysis.

Value: HIDDEN_IMPORT*

--icon

Path to an icon file for your executable or app bundle. Supported formats: .ico (Windows), .png (Linux) and .icns (macOS).

Aliases: -i

--name

Name for the generated executable (Windows) or app bundle (macOS).

Aliases: -n

--onedir

Create a one-folder bundle instead of a single-file executable (Windows only).

Aliases: -D

--product-name

Product name to be embedded in the executable (Windows) or bundle (macOS).

--product-version

Product version for the executable (Windows) or bundle (macOS).

--pyinstaller-build-args

Additional raw arguments to the underlying pyinstaller build command.

Value: PYINSTALLER_BUILD_ARGS*

--uac-admin

Request elevated (admin) permissions on application start (Windows only). Adds a UAC manifest to the executable.

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

--yes

Enable non-interactive mode. All prompts will be skipped.

Aliases: -y

flet publish

Compile and package a Flet app as a standalone static web application.

Usage

```
flet publish [-h] [-v] [--pre] [-a ASSETS_DIR] [--distpath DISTPATH]
              [--app-name APP_NAME] [--app-short-name APP_SHORT_NAME]
              [--app-description APP_DESCRIPTION] [--base-url BASE_URL]
              [--web-renderer {auto,canvaskit,skwasm}]
              [--route-url-strategy {path,hash}]
              [--pwa-background-color PWA_BACKGROUND_COLOR]
              [--pwa-theme-color PWA_THEME_COLOR] [--no-cdn]
              [script]
```

Positional arguments

script

Path to the Python script that starts your Flet app.

Default: .

Options

--app-description

Short description of the application. Used in PWA manifests and metadata.

--app-name

Full name of the application. This is used in PWA metadata and may appear in the install prompt.

--app-short-name

A shorter version of the application name, often used in homescreen icons or install prompts.

--assets

Path to a directory containing static assets used by the app (e.g., images, fonts, icons).

Environment variable: FLET_ASSETS_DIR

Aliases: -a

--base-url

Base URL path to serve the app from. Useful if the app is hosted in a subdirectory.

--distpath

Directory where the published web app should be placed.

Default: dist

--help

Show this help message and exit.

Aliases: -h

--no-cdn

Disable loading of CanvasKit, Pyodide, and fonts from CDNs. Use this for full offline deployments or air-gapped environments.

--pre

Allow micropip to install pre-release Python packages. Use this if your app depends on a prerelease version of a package.

--pwa-background-color

Initial background color of your web app during the loading phase (used in splash screens).

--pwa-theme-color

Default color of the browser UI (e.g., address bar) when your app is installed as a PWA.

--route-url-strategy

Controls how routes are handled in the browser.

Possible values: hash, path

Default: path

Environment variable: FLET_WEB_ROUTE_URL_STRATEGY

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

--web-renderer

Flutter web renderer to use.

Possible values: auto, canvaskit, skwasm

Default: auto

Environment variable: FLET_WEB_RENDERER

flet run

Run a Flet application in hot reload mode.

Usage

```
flet run [-h] [-v] [-p PORT] [--host HOST] [--name APP_NAME] [-m] [-d]
          [-r] [-n] [-w] [--ios] [--android] [-a ASSETS_DIR]
          [--ignore-dirs IGNORE_DIRS]
          [script]
```

Positional arguments

script

Path to the Python script that starts your Flet app.

Default: .

Options

--android

Launch the app on an Android device.

--assets

Path to a directory containing static assets used by the app (e.g. images, fonts).

Default: assets

Aliases: -a

--directory

Watch the directory of the script for changes and hot reload the app accordingly.

Aliases: -d

--help

Show this help message and exit.

Aliases: -h

--hidden

Start the application with the window hidden.

Aliases: -n

--host

The host to run Flet web app on. Use "*" to listen on all IPs.

--ignore-dirs

Comma-separated list of directory names to ignore when watching for file changes.

--ios

Launch the app on an iOS device.

--module

Treat the script as a python module path instead of a file path. Example: flet run -m my_app.main.

Aliases: -m

--name

A unique name for your app. Useful when running multiple apps on the same port.

--port

Custom TCP or HTTP (if --web option is used) port to run the Flet app on. If not specified, a random port will be chosen.

Value: PORT <int>

Aliases: -p

--recursive

Watch script directory and all its sub-directories recursively for file changes.

Aliases: -r

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

--web

Launch the Flet app as a dynamic website and automatically open it in your web browser after startup.

Aliases: -w

flet serve

Serve static files from a directory with a lightweight web server, optionally adding WebAssembly-related headers for Flet web apps.

Usage

```
flet serve [-h] [-v] [-p PORT] [web_root]
```

Positional arguments

web_root

Directory to serve.

Default: ./build/web

Options

--help

Show this help message and exit.

Aliases: -h

--port

Port number to serve the files on. Use this to customize the port if the default is already in use or needs to be changed.

Value: PORT <int>

Default: 8000

Aliases: -p

--verbose

Enable verbose output. Use -v for standard verbose logging and -vv for more detailed output.

Aliases: -v

AdaptiveControl

Base class for controls that support adaptive behavior, which allows them to adjust their appearance and behavior based on the target platform (ex: Material design on Android/Windows/Linux, Cupertino design on iOS/macOS).

The adaptive property is applicable in two common scenarios:

- Platform-adaptive controls: These controls have a corresponding version on both Material and Cupertino platforms. When adaptive is set to True, the control renders the appropriate platform-specific implementation.
- Container controls: Controls that contain children (ex: Row, Column) can pass the adaptive value down to their children that do not explicitly define it themselves. This enables nested adaptive behavior in complex layouts.

Extension developers can use this base class to create their own adaptive controls by checking the adaptive flag at runtime and rendering accordingly.

Note

This class does not implement any platform-specific rendering itself. It is up to the control inheriting from it to interpret the adaptive flag and render accordingly.

Inherits: Control

Properties

- adaptive - Enables platform-specific rendering or inheritance of adaptiveness from parent controls.

Properties

buildadaptiveclass-attributeinstance-attribute

adaptive: bool | None = None

Enables platform-specific rendering or inheritance of adaptiveness from parent controls.

Ads

Displaying Google Ads in Flet apps.

It is powered by the `google_mobile_ads` Flutter package.

Platform Support

| Platform | Windows | macOS | Linux |
|-----------|--------------------------|--------------------------|--------------------------|
| Supported | ☐ | ☐ | ☐ |

Usage

To use ads controls add `flet-ads` package to your project dependencies:

- `uv`
- `pip`
`uv add flet-ads`
`pip install flet-ads`

Requirements

The below sections show the required configurations for each platform.

Android

A valid AdMob app ID is required to be specified, otherwise the app might crash on launch or behave unexpectedly.

- `flet build`
- `pyproject.toml`

```
flet build apk \
  --android-meta-data com.google.android.gms.ads.APPLICATION_ID="ca-app-pub-xxxxxxxxxxxxxxxx~yyyyyyyyyy"

[tool.flet.android.meta_data]
"com.google.android.gms.ads.APPLICATION_ID" = "ca-app-pub-xxxxxxxxxxxxxxxx~yyyyyyyyyy"
```

See also:

- [setting Android permissions](#)
- [AdMob app ID for testing purposes](#)

iOS

A valid AdMob app ID is required to be specified, otherwise the app might crash on launch or behave unexpectedly.

- `flet build`
- `pyproject.toml`

```
flet build ipa \
  --info-plist GADApplicationIdentifier="ca-app-pub-xxxxxxxxxxxxxxxx~yyyyyyyyyy"

[tool.flet.ios.info]
GADApplicationIdentifier = "ca-app-pub-xxxxxxxxxxxxxxxx~yyyyyyyyyy"
```

See also:

- [setting iOS permissions](#)
- [AdMob app ID for testing purposes](#)

Test Values

AdMob provides app and ad unit IDs for testing purposes.

- AdMob app ID: `"ca-app-pub-3940256099942544~3347511713"`

`BannerAd.unit_id`

- Android: `"ca-app-pub-3940256099942544/9214589741"`

- iOS: "ca-app-pub-3940256099942544/2435281174"

InterstitialAd.unit_id

- Android: "ca-app-pub-3940256099942544/1033173712"
- iOS: "ca-app-pub-3940256099942544/4411468910"

They are not meant to be used in production.

Example

BannerAd and InterstitialAd

```
import flet as ft
import flet_ads as fta

# Test ad unit IDs
ids = {
    ft.PagePlatform.ANDROID: {
        "banner": "ca-app-pub-3940256099942544/6300978111",
        "interstitial": "ca-app-pub-3940256099942544/1033173712",
    },
    ft.PagePlatform.IOS: {
        "banner": "ca-app-pub-3940256099942544/2934735716",
        "interstitial": "ca-app-pub-3940256099942544/4411468910",
    },
}

def main(page: ft.Page):
    page.appbar = ft.AppBar(
        adaptive=True,
        title="Mobile Ads Playground",
        bgcolor=ft.Colors.LIGHT_BLUE_300,
    )
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.scroll = ft.ScrollMode.AUTO

    def show_new_interstitial_ad():
        async def show_iad(e: ft.Event[fta.InterstitialAd]):
            await iad.show()

        iad = fta.InterstitialAd(
            unit_id=ids[page.platform]["interstitial"],
            on_load=show_iad,
            on_error=lambda e: print("InterstitialAd error", e.data),
            on_open=lambda e: print("InterstitialAd opened"),
            on_close=lambda e: print("InterstitialAd closed"),
            on_impression=lambda e: print("InterstitialAd impression"),
            on_click=lambda e: print("InterstitialAd clicked"),
        )

    def get_new_banner_ad() -> fta.BannerAd:
        return fta.BannerAd(
            unit_id=ids[page.platform]["banner"],
            width=320,
            height=50,
            on_click=lambda e: print("BannerAd clicked"),
            on_load=lambda e: print("BannerAd loaded"),
            on_error=lambda e: print("BannerAd error", e.data),
            on_open=lambda e: print("BannerAd opened"),
            on_close=lambda e: print("BannerAd closed"),
            on_impression=lambda e: print("BannerAd impression"),
            on_will_dismiss=lambda e: print("BannerAd will dismiss"),
        )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.OutlinedButton(
                        content="Show InterstitialAd",
                        on_click=show_new_interstitial_ad,
                        disabled=page.web
                        or not page.platform.is_mobile(), # mobile only
                    ),
                    ft.OutlinedButton(
                        content="Show BannerAd",
                        on_click=lambda e: page.add(get_new_banner_ad()),
                        disabled=page.web
                        or not page.platform.is_mobile(), # mobile only
                    ),
                ],
            ),
        )
    )
```

```
)    ),
```

```
if __name__ == "__main__":  
    ft.run(main)
```

Image: Image - https://flet.dev/docs/examples/extensions/ads/media/example_1.gif

AlertDialog

Can be used to inform the user about situations that require acknowledgement.

It has an optional title and an optional list of actions . The title is displayed above the content and the actions are displayed below the content.

```
ft.AlertDialog(  
    title=ft.Text("Session expired"),  
    content=ft.Text("Please sign in again to continue."),  
    actions=[ft.TextButton("Dismiss")],  
    open=True,  
)
```

Image: AlertDialog - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/alert_dialog/image_for_docs.png

AlertDialog

Inherits: DialogControl

Properties

- `action_button_padding` - The padding that surrounds each button in actions.
- `actions` - A set of actions that are displayed at the bottom of this dialog.
- `actions_alignment` - Defines the horizontal layout of the actions.
- `actions_overflow_button_spacing` - The spacing between actions when the OverflowBar switches to a column layout because the actions don't fit horizontally.
- `actions_padding` - Padding around the set of actions at the bottom of this dialog.
- `alignment` - How to align this dialog.
- `barrier_color` - The color of the modal barrier below this dialog.
- `bgcolor` - The background color of this dialog's surface.
- `clip_behavior` - Defines how the contents of this dialog are clipped (or not) to the given shape.
- `content` - The content of this dialog is displayed in the center of this dialog in a lighter font.
- `content_padding` - Padding around the content.
- `content_text_style` - The style for the text in the content of this dialog.
- `elevation` - Defines the elevation (z-coordinate) at which this dialog should appear.
- `icon` - A control that is displayed at the top of this dialog.
- `icon_color` - The color for the Icon in the icon of this dialog.
- `icon_padding` - Padding around the icon.
- `inset_padding` - Padding around this dialog itself.
- `modal` - Whether dialog can be dismissed/closed by clicking the area outside of it.
- `scrollable` - Determines whether the title and content controls are wrapped in a scrollable.
- `semantics_label` - The semantic label of this dialog used by accessibility frameworks to announce screen transitions when this dialog is opened and closed.
- `shadow_color` - The color used to paint a drop shadow under this dialog, which reflects this dialog's elevation.
- `shape` - The shape of this dialog.
- `title` - The title of this dialog is displayed in a large font at its top.
- `title_padding` - Padding around the title.
- `title_text_style` - TBD

Examples

Modal and non-modal dialogs

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "AlertDialog examples"

    dialog = ft.AlertDialog(
        title=ft.Text("Hello"),
        content=ft.Text("You are notified!"),
        alignment=ft.Alignment.CENTER,
        on_dismiss=lambda e: print("Dialog dismissed!"),
        title_padding=ft.Padding.all(25),
    )

    modal_dialog = ft.AlertDialog(
        modal=True,
        title=ft.Text("Please confirm"),
        content=ft.Text("Do you really want to delete all those files?"),
        actions=[
            ft.TextButton("Yes", on_click=lambda e: page.pop_dialog()),
            ft.TextButton("No", on_click=lambda e: page.pop_dialog()),
        ],
        actions_alignment=ft.MainAxisAlignment.END,
        on_dismiss=lambda e: print("Modal dialog dismissed!"),
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button(
                        content="Open dialog",
                        on_click=lambda e: page.show_dialog(dialog),
                    ),
                    ft.Button(
                        content="Open modal dialog",
                        on_click=lambda e: page.show_dialog(modal_dialog),
                    ),
                ]
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: Modal and non-modal dialogs - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/alert_dialog/alert_dialog_flow.gif

Modal and non-modal dialogs

Properties

buildaction_button_paddingclass-attributeinstance-attribute

action_button_padding: PaddingValue | None = None

The padding that surrounds each button in actions.

buildactionsclass-attributeinstance-attribute

actions: list[Control] = field(default_factory=list)

A set of actions that are displayed at the bottom of this dialog.

Typically this is a list of TextButton controls.

Raises:

- ValueError - If none of title, content, or actions are provided, as the dialog would have nothing to display.

buildactions_alignmentclass-attributeinstance-attribute

actions_alignment: MainAxisAlignment | None = None

Defines the horizontal layout of the actions.

Internally defaults to MainAxisAlignment.END.

buildactions_overflow_button_spacingclass-attributeinstance-attribute

actions_overflow_button_spacing: Number | None = None

The spacing between actions when the OverflowBar switches to a column layout because the actions don't fit horizontally.

If the controls in actions do not fit into a single row, they are arranged into a column. This parameter provides additional vertical space between buttons when it does overflow.

buildactions_paddingclass-attributeinstance-attribute

`actions_padding: PaddingValue | None = None`

Padding around the set of actions at the bottom of this dialog.

Typically used to provide padding to the button bar between the button bar and the edges of this dialog.

If are no actions, then no padding will be included. The padding around the button bar defaults to zero.

buildalignmentclass-attributeinstance-attribute

`alignment: Alignment | None = None`

How to align this dialog.

If None, then DialogTheme.alignment is used. If that is also None, the default is Alignment.CENTER.

buildbarrier_colorclass-attributeinstance-attribute

`barrier_color: ColorValue | None = None`

The color of the modal barrier below this dialog.

If None, then DialogTheme.barrier_color is used. If that is also None, the default is Colors.BLACK_54.

buildbgcolorclass-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

The background color of this dialog's surface.

buildclip_behaviorclass-attributeinstance-attribute

`clip_behavior: ClipBehavior = ClipBehavior.NONE`

Defines how the contents of this dialog are clipped (or not) to the given shape.

buildcontentclass-attributeinstance-attribute

`content: Control | None = None`

The content of this dialog is displayed in the center of this dialog in a lighter font.

Typically this is a Column that contains this dialog's Text message.

buildcontent_paddingclass-attributeinstance-attribute

`content_padding: PaddingValue | None = None`

Padding around the content.

If there is no content, no padding will be provided. Otherwise, padding of 20 pixels is provided above the content to separate the content from the title, and padding of 24 pixels is provided on the left, right, and bottom to separate the content from the other edges of this dialog.

buildcontent_text_styleclass-attributeinstance-attribute

`content_text_style: TextStyle | None = None`

The style for the text in the content of this dialog.

If None, DialogTheme.content_text_style is used. If that's is also None, defaults to TextTheme.body_medium (if Theme.use_material3 is True; TextTheme.title_medium otherwise) of Theme.text_theme.

buildelevationclass-attributeinstance-attribute

`elevation: Number | None = None`

Defines the elevation (z-coordinate) at which this dialog should appear.

buildiconclass-attributeinstance-attribute

`icon: Control | None = None`

A control that is displayed at the top of this dialog.

Typically a Icon control.

buildicon_colorclass-attributeinstance-attribute

icon_color: ColorValue | None = None

The color for the Icon in the icon of this dialog.

If None, DialogTheme.icon_color is used. If that is null, defaults to color scheme's ColorScheme.secondary if Theme.use_material3 is True, Colors.BLACK otherwise.

buildicon_paddingclass-attributeinstance-attribute

icon_padding: PaddingValue | None = None

Padding around the icon.

buildinset_paddingclass-attributeinstance-attribute

inset_padding: PaddingValue | None = None

Padding around this dialog itself.

Defaults to Padding.symmetric(vertical=40, horizontal=24) - 40 pixels horizontally and 24 pixels vertically outside of this dialog box.

buildmodalclass-attributeinstance-attribute

modal: bool = False

Whether dialog can be dismissed/closed by clicking the area outside of it.

buildscrollableclass-attributeinstance-attribute

scrollable: bool = False

Determines whether the title and content controls are wrapped in a scrollable.

This configuration is used when the title and content are expected to overflow. Both title and content are wrapped in a scroll view, allowing all overflowed content to be visible while still showing the button bar.

buildsemantics_labelclass-attributeinstance-attribute

semantics_label: str | None = None

The semantic label of this dialog used by accessibility frameworks to announce screen transitions when this dialog is opened and closed.

On iOS, if this label is not provided, a semantic label will be inferred from the title if it is not None.

buildshadow_colorclass-attributeinstance-attribute

shadow_color: ColorValue | None = None

The color used to paint a drop shadow under this dialog, which reflects this dialog's elevation.

buildshapeclass-attributeinstance-attribute

shape: OutlinedBorder | None = None

The shape of this dialog.

If None, defaults to DialogTheme.shape. If it is also None, it defaults to RoundedRectangleBorder(radius=4.0).

buildtitleclass-attributeinstance-attribute

title: StrOrControl | None = None

The title of this dialog is displayed in a large font at its top.

Typically a Text control.

buildtitle_paddingclass-attributeinstance-attribute

title_padding: PaddingValue | None = None

Padding around the title.

If there is no title, no padding will be provided. Otherwise, this padding is used.

Defaults to 24 pixels on the top, left, and right of the title. If the content is not None, then no bottom padding is provided (see content_padding). If it is not set, then an extra 20 pixels of bottom padding is added to separate the title from the actions.

buildtitle_text_styleclass-attributeinstance-attribute

title_text_style: TextStyle | None = None

TBD

AnimatedSwitcher

Used to switch between controls with an animation.

Inherits: `LayoutControl`

Properties

- `content` - The content to display.
- `duration` - The duration of the transition from the old content to the new one.
- `reverse_duration` - The duration of the transition from the new content to the old one.
- `switch_in_curve` - The animation curve to use when transitioning in a new content.
- `switch_out_curve` - The animation curve to use when transitioning an old content out.
- `transition` - An animation type to transition between new and old content.

Examples

Scale, fade, and rotate effects

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    c1 = ft.Container(
        content=ft.Text("Hello!", theme_style=ft.TextThemeStyle.HEADLINE_MEDIUM),
        alignment=ft.Alignment.CENTER,
        width=200,
        height=200,
        bgcolor=ft.Colors.GREEN,
    )
    c2 = ft.Container(
        content=ft.Text("Bye!", size=50),
        alignment=ft.Alignment.CENTER,
        width=200,
        height=200,
        bgcolor=ft.Colors.YELLOW,
    )
    switcher = ft.AnimatedSwitcher(
        content=c1,
        transition=ft.AnimatedSwitcherTransition.SCALE,
        duration=500,
        reverse_duration=100,
        switch_in_curve=ft.AnimationCurve.BOUNCE_OUT,
        switch_out_curve=ft.AnimationCurve.BOUNCE_IN,
    )

    def scale(e):
        switcher.content = c2 if switcher.content == c1 else c1
        switcher.transition = ft.AnimatedSwitcherTransition.SCALE
        switcher.update()

    def fade(e):
        switcher.content = c2 if switcher.content == c1 else c1
        switcher.transition = ft.AnimatedSwitcherTransition.FADE
        switcher.update()

    def rotate(e):
        switcher.content = c2 if switcher.content == c1 else c1
        switcher.transition = ft.AnimatedSwitcherTransition.ROTATION
        switcher.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    switcher,
                    ft.Button("Scale", on_click=scale),
                    ft.Button("Fade", on_click=fade),
                    ft.Button("Rotate", on_click=rotate),
                ]
            ),
        ),
    )
```

```
if __name__ == "__main__":
    ft.run(main)
```

Image: scale-effect - https://flet.dev/docs/examples/controls/core/animated_switcher/media/scale_effect.gif

Animate image switch

play_arrowTry Online

```
import time

import flet as ft

def main(page: ft.Page):
    def animate(e: ft.Event[ft.Button]):
        switcher.content = ft.Image(
            src=f"https://picsum.photos/200/300?{time.time()}",
            width=200,
            height=300,
        )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    switcher := ft.AnimatedSwitcher(
                        content=ft.Image(
                            src="https://picsum.photos/200/300",
                            width=200,
                            height=300,
                        ),
                        transition=ft.AnimatedSwitcherTransition.SCALE,
                        duration=500,
                        reverse_duration=100,
                        switch_in_curve=ft.AnimationCurve.BOUNCE_OUT,
                        switch_out_curve=ft.AnimationCurve.BOUNCE_IN,
                    ),
                    ft.Button("Animate!", on_click=animate),
                ]
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Animate image switch buffered

play_arrowTry Online

```
import base64
import time
from dataclasses import field

import httpx

import flet as ft

@ft.control
class BufferingSwitcher(ft.AnimatedSwitcher):
    content: ft.Image | None = None
    image_queue: list[str] = field(default_factory=list)

    def init(self):
        self.transition = ft.AnimatedSwitcherTransition.SCALE
        self.duration = 500
        self.reverse_duration = 100
        self.switch_in_curve = ft.AnimationCurve.EASE_IN
        self.switch_out_curve = ft.AnimationCurve.EASE_OUT
        if self.content and self.content.src:
            self.image_queue.append(self.content.src)

    def animate(self, e):
        self.content = ft.Image(
            src=self.image_queue.pop(),
            width=200,
            height=300,
            gapless_playback=True,
        )
        self.update()

    async def fill_queue(self):
        while len(self.image_queue) < 10:
            image_base64 = await self.image_to_base64(
```

```

        f"https://picsum.photos/200/300?{time.time()}")
    )
    if image_base64:
        self.image_queue.append(image_base64)

    async def image_to_base64(self, url):
        response = await httpx.AsyncClient(follow_redirects=True).get(url)
        if response.status_code == 200:
            base64_str = (
                base64.standard_b64encode(response.content).decode("utf-8").strip()
            )
            return base64_str
        else:
            print(f"Image request failed with {response.status_code}")
            return None

    def before_update(self):
        self.page.run_task(self.fill_queue)

def main(page: ft.Page):
    switcher = BufferingSwitcher(
        content=ft.Image(
            src=f"https://picsum.photos/200/300?{time.time()}",
            width=200,
            height=300,
        )
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    switcher,
                    ft.Button("Animate!", on_click=switcher.animate),
                ]
            )
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Properties

buildcontentinstance-attribute

content: Control

The content to display.

When it changes, this switcher will animate the transition from the old/previous content to the new one.

Raises:

- ValueError - If it is not visible.

bulldurationclass-attributeinstance-attribute

```

duration: DurationValue = field(
    default_factory=lambda: Duration(seconds=1)
)

```

The duration of the transition from the old content to the new one.

buildreverse_durationclass-attributeinstance-attribute

```

reverse_duration: DurationValue = field(
    default_factory=lambda: Duration(seconds=1)
)

```

The duration of the transition from the new content to the old one.

buildswitch_in_curveclass-attributeinstance-attribute

```

switch_in_curve: AnimationCurve = AnimationCurve.LINEAR

```

The animation curve to use when transitioning in a new content.

buildswitch_out_curveclass-attributeinstance-attribute

```

switch_out_curve: AnimationCurve = AnimationCurve.LINEAR

```

The animation curve to use when transitioning an old content out.

buildtransitionclass-attributeinstance-attribute

```
transition: AnimatedSwitcherTransition = (  
    AnimatedSwitcherTransition.FADE  
)
```

An animation type to transition between new and old content.

AppBar

A material design app bar.

Example:

```
ft.AppBar(  
    leading=ft.Icon(ft.Icons.MENU),  
    title=ft.Text("Dashboard"),  
    bgcolor=ft.Colors.SURFACE_CONTAINER,  
    actions=[  
        ft.IconButton(ft.Icons.SEARCH),  
        ft.IconButton(ft.Icons.MORE_VERT),  
    ],  
)
```

Image: AppBar - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/app_bar/image_for_docs.png

AppBar

Inherits: `AdaptiveControl`

Properties

- `actions` - A list of Controls to display in a row after the title control.
- `actions_padding` - The padding between the actions and the end of this app bar.
- `automatically_imply_leading` - Whether we should try to imply the leading control if it is None.
- `bgcolor` - The fill color to use for this app bar.
- `center_title` - Whether the title should be centered.
- `clip_behavior` - The content will be clipped (or not) according to this option.
- `color` - The default color for Text and Icon controls within this app bar.
- `elevation` - The app bar's elevation.
- `elevation_on_scroll` - The elevation to be used if this app bar has something scrolled underneath it.
- `exclude_header_semantics` - Whether the title should be wrapped with header Semantics.
- `force_material_transparency` - Forces this app bar to be transparent (instead of Material's default type).
- `leading` - A control to display before the toolbar's title.
- `leading_width` - Defines the width of the leading control.
- `secondary` - Whether this app bar is not being displayed at the top of the screen.
- `shadow_color` - The color of the shadow below this app bar.
- `shape` - The shape of this app bar's Material as well as its shadow.
- `title` - The primary Control displayed in this app bar.
- `title_spacing` - The spacing around title on the horizontal axis.
- `title_text_style` - The style to be used for the Text controls in the title.
- `toolbar_height` - Defines the height of the toolbar component of this app bar.
- `toolbar_opacity` - The opacity of the toolbar.
- `toolbar_text_style` - The style to be used for the Text controls in the app bar's leading and actions.

Examples

Actions and popup menu

`play_arrow` Try Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.title = "AppBar Example"
```

```

def handle_checked_item_click(e: ft.Event[ft.PopupMenuItem]):
    e.control.checked = not e.control.checked

page.appbar = ft.AppBar(
    leading=ft.Icon(ft.Icons.PALETTE),
    leading_width=40,
    title=ft.Text("AppBar Example"),
    center_title=False,
    bgcolor=ft.Colors.BLUE_GREY_400,
    actions=[
        ft.IconButton(ft.Icons.WB_SUNNY_OUTLINED),
        ft.IconButton(ft.Icons.FILTER_3),
        ft.PopupMenuButton(
            key="popup",
            items=[
                ft.PopupMenuItem(content="Item 1"),
                ft.PopupMenuItem(), # divider
                ft.PopupMenuItem(
                    content="Checked item",
                    checked=False,
                    on_click=handle_checked_item_click,
                ),
            ],
        ),
    ],
)

page.add(ft.SafeArea(content=ft.Column(controls=[ft.Text("Body!")])))

if __name__ == "__main__":
    ft.run(main)

```

Image: actions-and-popup-menu - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/app_bar/app_bar_flow.gif

Theme mode toggle

play_arrowTry Online

```

import flet as ft

LIGHT_SEED_COLOR = ft.Colors.DEEP_ORANGE
DARK_SEED_COLOR = ft.Colors.DEEP_PURPLE

def main(page: ft.Page):
    page.title = "AppBar Example"

    def handle_checked_item_click(e: ft.Event[ft.PopupMenuItem]):
        e.control.checked = not e.control.checked

    page.theme_mode = ft.ThemeMode.LIGHT
    page.theme = ft.Theme(color_scheme_seed=LIGHT_SEED_COLOR)
    page.dark_theme = ft.Theme(color_scheme_seed=DARK_SEED_COLOR)

    def toggle_theme_mode(e: ft.Event[ft.IconButton]):
        page.theme_mode = (
            ft.ThemeMode.DARK
            if page.theme_mode == ft.ThemeMode.LIGHT
            else ft.ThemeMode.LIGHT
        )
        theme_mode_toggle.icon = (
            ft.Icons.WB_SUNNY_OUTLINED
            if page.theme_mode == ft.ThemeMode.LIGHT
            else ft.Icons.WB_SUNNY
        )

    theme_mode_toggle = ft.IconButton(
        key="theme_mode_toggle",
        icon=(
            ft.Icons.WB_SUNNY_OUTLINED
            if page.theme_mode == ft.ThemeMode.LIGHT
            else ft.Icons.WB_SUNNY
        ),
        on_click=toggle_theme_mode,
    )

    page.padding = 50
    page.appbar = ft.AppBar(
        # toolbar_height=100,
        bgcolor=ft.Colors.SECONDARY_CONTAINER,
        leading=ft.Icon(ft.Icons.PALETTE),
        leading_width=40,
        title=ft.Text("AppBar Example"),
        center_title=False,
        actions=[
            theme_mode_toggle,
            ft.PopupMenuButton(

```

```

items=[
    ft.PopupMenuItem(content="Item 1"),
    ft.PopupMenuItem(icon=ft.Icons.POWER_INPUT, content="Check power"),
    ft.PopupMenuItem(
        on_click=lambda _: print(
            "Button with a custom content clicked!"
        ),
        content=ft.Row(
            controls=[
                ft.Icon(ft.Icons.HOURLASS_TOP_OUTLINED),
                ft.Text("Item with a custom content"),
            ]
        ),
    ),
    ft.PopupMenuItem(), # divider
    ft.PopupMenuItem(
        content="Checked item",
        checked=False,
        on_click=handle_checked_item_click,
    ),
]
),
],
)
page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                ft.Text(
                    value="Flet is a framework that allows building web, desktop "
                    "and mobile applications in Python without prior experience "
                    "in frontend development.You can build a UI for your program "
                    "with Flet controls which are based on Flutter by Google. "
                    "Flet goes beyond merely wrapping Flutter widgets. It adds "
                    "its own touch by combining smaller widgets, simplifying "
                    "complexities, implementing UI best practices, and applying "
                    "sensible defaults. This ensures that your applications look "
                    "stylish and polished without requiring additional design "
                    "efforts on your part.",
                    text_align=ft.TextAlign.END,
                ),
                ft.Button("Click me!"),
            ]
        )
    )
)

if __name__ == "__main__":
    ft.run(main)

```

Image: theme-and-material-mode-toggles -

https://flet.dev/docs/test-images/examples/controls/material/golden/macos/app_bar/theme_mode_toggle_flow.gif

Properties

buildactionsclass-attributeinstance-attribute

actions: list[Control] | None = None

A list of Controls to display in a row after the title control.

Typically, these controls are IconButton representing common operations. For less common operations, consider using a PopupMenuButton as the last action.

Info

If AdaptiveControl.adaptive is True and this app is opened on an iOS or macOS device, these actions will be automatically placed in a Row. This is because CupertinoAppBar.trailing (which is the counterpart property of actions) takes only a single Control.

buildactions_paddingclass-attributeinstance-attribute

actions_padding: PaddingValue | None = None

The padding between the actions and the end of this app bar.

buildautomatically_imply_leadingclass-attributeinstance-attribute

automatically_imply_leading: bool = True

Whether we should try to imply the leading control if it is None.

- If True and leading is None, this app bar will automatically determine an appropriate leading control.

- If False and leading is None, the space is allocated to the title.
- If a leading control is provided, this parameter has no effect.

buildbgcolorclass-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

The fill color to use for this app bar.

Default color is defined by `AppBarTheme.bgcolor`

buildcenter_titleclass-attributeinstance-attribute

`center_title: bool | None = None`

Whether the title should be centered.

Default value is defined by `AppBarTheme.center_title`

buildclip_behaviorclass-attributeinstance-attribute

`clip_behavior: ClipBehavior | None = None`

The content will be clipped (or not) according to this option.

buildcolorclass-attributeinstance-attribute

`color: ColorValue | None = None`

The default color for Text and Icon controls within this app bar.

Default color is defined by `AppBarTheme.color`

buildelevationclass-attributeinstance-attribute

`elevation: Number | None = None`

The app bar's elevation.

Note

This effect is only visible when using the Material 2 design (when `Theme.use_material3` is False).

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildelevation_on_scrollclass-attributeinstance-attribute

`elevation_on_scroll: Number | None = None`

The elevation to be used if this app bar has something scrolled underneath it.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildexclude_header_semanticsclass-attributeinstance-attribute

`exclude_header_semantics: bool = False`

Whether the title should be wrapped with header Semantics.

buildforce_material_transparencyclass-attributeinstance-attribute

`force_material_transparency: bool = False`

Forces this app bar to be transparent (instead of Material's default type).

This will also remove the visual display of `bgcolor` and `elevation`, and affect other characteristics of this app bar.

buildleadingclass-attributeinstance-attribute

`leading: Control | None = None`

A control to display before the toolbar's title.

Typically an `Icon` or `IconButton` control.

buildleading_widthclass-attributeinstance-attribute

`leading_width: Number | None = None`

Defines the width of the leading control.

buildsecondaryclass-attributeinstance-attribute

secondary: bool = False

Whether this app bar is not being displayed at the top of the screen.

buildshadow_colorclass-attributeinstance-attribute

shadow_color: ColorValue | None = None

The color of the shadow below this app bar.

A shadow is only visible and displayed if the elevation is greater than zero.

buildshapeclass-attributeinstance-attribute

shape: OutlinedBorder | None = None

The shape of this app bar's Material as well as its shadow.

buildtitleclass-attributeinstance-attribute

title: StrOrControl | None = None

The primary Control displayed in this app bar.

Typically a Text control that contains a description of the current contents of this app.

Note

If AdaptiveControl.adaptive and this app is opened on an iOS or macOS device, this title control will be automatically centered, independent of the value of center_title.

buildtitle_spacingclass-attributeinstance-attribute

title_spacing: Number | None = None

The spacing around title on the horizontal axis. It is applied even if there are no leading or actions controls.

Tip

If you want title to take all the space available, set title_spacing to 0.0.

buildtitle_text_styleclass-attributeinstance-attribute

title_text_style: TextStyle | None = None

The style to be used for the Text controls in the title.

buildtoolbar_heightclass-attributeinstance-attribute

toolbar_height: Number | None = None

Defines the height of the toolbar component of this app bar.

buildtoolbar_opacityclass-attributeinstance-attribute

toolbar_opacity: Number = 1.0

The opacity of the toolbar.

- 0.0: transparent
- 1.0: fully opaque

Raises:

- ValueError - If it is not between 0.0 and 1.0, inclusive.

buildtoolbar_text_styleclass-attributeinstance-attribute

toolbar_text_style: TextStyle | None = None

The style to be used for the Text controls in the app bar's leading and actions.

AutoComplete

Helps the user make a selection by entering some text and choosing from among a list of displayed options.

Inherits: `LayoutControl`

Properties

- `selected_index` - The index of the (last) selected suggestion.
- `suggestions` - A list of `AutoCompleteSuggestion` controls representing the suggestions to be displayed.
- `suggestions_max_height` - The maximum - visual - height of the suggestions list.
- `value` - Current text displayed in the input field.

Events

- `on_change` - Called when the input text changes.
- `on_select` - Called when a suggestion is selected.

Examples

AutoComplete

[play_arrow](#) Try Online

```
import flet as ft

numbers = [
    ("one 1", "One"),
    ("two 2", "Two"),
    ("three 3", "Three"),
    ("four 4", "Four"),
    ("five 5", "Five"),
    ("six 6", "Six"),
    ("seven 7", "Seven"),
    ("eight 8", "Eight"),
    ("nine 9", "Nine"),
    ("ten 10", "Ten"),
]

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.vertical_alignment = ft.MainAxisAlignment.CENTER

    def handle_change(e: ft.Event[ft.AutoComplete]):
        info.value = f"Current input ■: {e.data!r} \n"

    def handle_select(e: ft.AutoCompleteSelectEvent):
        info.value = (
            f"Current input ■: {e.control.value!r} \n"
            f"Your selection: {e.selection.value}"
        )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.AutoComplete(
                        value="One",
                        width=200,
                        on_change=handle_change,
                        on_select=handle_select,
                        suggestions=[
                            ft.AutoCompleteSuggestion(key=key, value=value)
                            for key, value in numbers
                        ],
                    ),
                    info := ft.Text(
                        "Enter a number (in words or digits) to get suggestions."
                    ),
                ]
            ),
        ),
    )
```

```
if __name__ == "__main__":
    ft.run(main)
```

Image: auto-complete-basic-example -

https://flet.dev/docs/test-images/examples/controls/material/golden/macros/auto_complete/auto_complete_basic_flow.gif

Properties

buildselected_indexproperty

```
selected_index: int | None
```

The index of the (last) selected suggestion.

It is None until a suggestion has been selected from the UI.

Note

This property is read-only.

buildsuggestionsclass-attributeinstance-attribute

```
suggestions: list[AutoCompleteSuggestion] = field(
    default_factory=list
)
```

A list of AutoCompleteSuggestion controls representing the suggestions to be displayed.

Note

- A valid AutoCompleteSuggestion must have at least a key or value specified, else it will be ignored. If only key is provided, value will be set to key as fallback and vice versa.
- The internal filtration process of the suggestions (based on their keys) with respect to the user's input is case-insensitive because the comparison is done in lowercase.

buildsuggestions_max_heightclass-attributeinstance-attribute

```
suggestions_max_height: Number = 200
```

The maximum - visual - height of the suggestions list.

buildvalueclass-attributeinstance-attribute

```
value: str = ''
```

Current text displayed in the input field.

This value reflects user input even if it does not match any provided suggestion.

Events

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[AutoComplete] | None = None
```

Called when the input text changes.

bolton_selectclass-attributeinstance-attribute

```
on_select: EventHandler[AutoCompleteSelectEvent] | None = (
    None
)
```

Called when a suggestion is selected.

AutofillGroup

Used to group autofill controls together.

Inherits: Control

Properties

- content - The content of this group.
- dispose_action - The action to be run when this group is the topmost and it's being disposed, in order to clean up the current autofill context.

Examples

AutofillGroup

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.AutofillGroup(
                content=ft.Column(
                    controls=[
                        ft.TextField(
                            label="Name",
                            autofill_hints=ft.AutofillHint.NAME,
                        ),
                        ft.TextField(
                            label="Email",
                            autofill_hints=[ft.AutofillHint.EMAIL],
                        ),
                        ft.TextField(
                            label="Phone Number",
                            autofill_hints=[ft.AutofillHint.TELEPHONE_NUMBER],
                        ),
                        ft.TextField(
                            label="Street Address",
                            autofill_hints=ft.AutofillHint.FULL_STREET_ADDRESS,
                        ),
                        ft.TextField(
                            label="Postal Code",
                            autofill_hints=ft.AutofillHint.POSTAL_CODE,
                        ),
                    ],
                ),
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: basic - https://flet.dev/docs/examples/controls/material/autofill_group/media/basic.gif

Properties

buildcontentinstance-attribute

content: Control

The content of this group.

Raises:

- ValueError - If it is not visible.

builddispose_actionclass-attributeinstance-attribute

```
dispose_action: AutofillGroupDisposeAction = (
    AutofillGroupDisposeAction.COMMIT
)
```

The action to be run when this group is the topmost and it's being disposed, in order to clean up the current autofill context.

Banner

A banner displays an important, succinct message, and provides actions for users to address (or dismiss the banner). A user action is required for it to be dismissed.

Banners are displayed at the top of the screen, below a top app bar. They are persistent and non-modal, allowing the user to either ignore them or interact with them at any time.

Example:

```
banner = ft.Banner(  
    leading=ft.Icon(ft.Icons.INFO_OUTLINED, color=ft.Colors.PRIMARY),  
    content=ft.Text("Backup completed successfully."),  
    actions=[ft.TextButton("Dismiss")],  
    bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,  
    open=True,  
)  
page.show_dialog(banner)
```

Image: Banner - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/banner/image_for_docs.png

Banner

Inherits: DialogControl

Properties

- **actions** - The set of actions that are displayed at the bottom or trailing side of this banner.
- **bgcolor** - The color of the surface of this banner.
- **content** - The content of this banner.
- **content_padding** - The amount of space by which to inset the content.
- **content_text_style** - The style to be used for the Text controls in the content.
- **divider_color** - The color of the divider.
- **elevation** - The elevation of this banner.
- **force_actions_below** - An override to force the actions to be below the content regardless of how many there are.
- **leading** - The leading Control of this banner.
- **leading_padding** - The amount of space by which to inset the leading control.
- **margin** - The amount of space surrounding this banner.
- **min_action_bar_height** - The minimum action bar height.
- **shadow_color** - The color of the shadow below this banner.

Events

- **on_visible** - Called when this banner is shown or made visible for the first time.

Examples

Banner

play_arrowTry Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER  
  
    def handle_banner_close(e: ft.Event[ft.TextButton]):  
        page.pop_dialog()  
        page.add(ft.Text("Action clicked: " + e.control.data))  
  
    action_button_style = ft.ButtonStyle(color=ft.Colors.BLUE)  
    banner = ft.Banner(  
        bgcolor=ft.Colors.AMBER_100,  
        leading=ft.Icon(ft.Icons.WARNING_AMBER_ROUNDED, color=ft.Colors.AMBER, size=40),  
        content=ft.Text(  
            value="Oops, there were some errors while trying to delete the file. "  

```

```

        "What would you like to do?",
        color=ft.Colors.BLACK,
    ),
    actions=[
        ft.TextButton(
            content="Retry",
            style=action_button_style,
            on_click=handle_banner_close,
            data="retry",
        ),
        ft.TextButton(
            content="Ignore",
            style=action_button_style,
            on_click=handle_banner_close,
            data="ignore",
        ),
        ft.TextButton(
            content="Cancel",
            style=action_button_style,
            on_click=handle_banner_close,
            data="cancel",
        ),
    ],
)

page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                ft.Button(
                    "Show Banner", on_click=lambda e: page.show_dialog(banner)
                )
            ]
        )
    )
)

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/banner/banner_flow.gif

Properties

buildactionsinstance-attribute

actions: list[Control]

The set of actions that are displayed at the bottom or trailing side of this banner.

Typically this is a list of TextButton controls.

Raises:

- ValueError - If it does not contain at least one visible Control.

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The color of the surface of this banner.

buildcontentinstance-attribute

content: StrOrControl

The content of this banner.

Typically a Text control.

Raises:

- ValueError - If content is not visible.

buildcontent_paddingclass-attributeinstance-attribute

content_padding: PaddingValue | None = None

The amount of space by which to inset the content.

If the actions are below the content, this defaults to Padding.only(left=16.0, top=24.0, right=16.0, bottom=4.0).

If the actions are trailing the content, this defaults to Padding.only(left=16.0, top=2.0).

buildcontent_text_styleclass-attributeinstance-attribute

`content_text_style: TextStyle | None = None`

The style to be used for the Text controls in the content.

builddivider_colorclass-attributeinstance-attribute

`divider_color: ColorValue | None = None`

The color of the divider.

buildelevationclass-attributeinstance-attribute

`elevation: Number | None = None`

The elevation of this banner.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildforce_actions_belowclass-attributeinstance-attribute

`force_actions_below: bool = False`

An override to force the actions to be below the content regardless of how many there are.

If this is `True`, the actions will be placed below the content. If this is `False`, the actions will be placed on the trailing side of the content if actions length is 1 and below the content if greater than 1.

buildleadingclass-attributeinstance-attribute

`leading: IconDataOrControl | None = None`

The leading Control of this banner.

Typically an Icon control.

buildleading_paddingclass-attributeinstance-attribute

`leading_padding: PaddingValue | None = None`

The amount of space by which to inset the leading control.

Defaults to `BannerTheme.leading_padding`, or if that is `None`, falls back to `Padding.only(end=16)`.

buildmarginclass-attributeinstance-attribute

`margin: MarginValue | None = None`

The amount of space surrounding this banner.

buildmin_action_bar_heightclass-attributeinstance-attribute

`min_action_bar_height: Number = 52.0`

The minimum action bar height.

buildshadow_colorclass-attributeinstance-attribute

`shadow_color: ColorValue | None = None`

The color of the shadow below this banner.

Events

bolton_visibleclass-attributeinstance-attribute

`on_visible: ControlEventHandler[Banner] | None = None`

Called when this banner is shown or made visible for the first time.

BaseControl

Base class for all Flet controls and services.

Properties

- `data` - Arbitrary data of any type.
- `key`
- `page` - The page to which this control belongs to.
- `parent` - The direct ancestor(parent) of this control.

Methods

- `before_event` - Intercept an event before its handler is executed.
- `build` - Called once during control initialization to define its child controls.
- `did_mount` - Called after the control is mounted into the page tree.
- `is_isolated` - Return whether this control is marked as isolated.
- `update` - Request a UI update for this control.
- `will_unmount` - Called before the control is removed from the page tree.

Properties

`builddataclass-attributeinstance-attribute`

`data: Any = skip_field()`

Arbitrary data of any type.

`buildkeyclass-attributeinstance-attribute`

`key: KeyVal | None = None`

`buildpageproperty`

`page: Page | BasePage`

The page to which this control belongs to.

`buildparentproperty`

`parent: BaseControl | None`

The direct ancestor(parent) of this control.

It defaults to `None` and will only have a value when this control is mounted (added to the page tree).

The `Page` control (which is the root of the tree) is an exception - it always has `parent=None`.

Methods

`deployed_codebefore_event`

`before_event(e: ControlEvent)`

Intercept an event before its handler is executed.

Return `False` to cancel dispatch. Return `True` or `None` to continue normal event processing.

`deployed_codebuild`

`build()`

Called once during control initialization to define its child controls. `page` property is available/usable in this method.

`deployed_codedid_mount`

`did_mount()`

Called after the control is mounted into the page tree.

Override to start resources that require an attached page, for example subscriptions, timers, or service listeners.

deployed_codeis_isolated

`is_isolated()`

Return whether this control is marked as isolated.

Isolated controls are excluded from parent-driven update traversal and are expected to manage their own update boundaries.

deployed_codeupdate

`update() -> None`

Request a UI update for this control.

Call after changing control state or properties. The control must be attached to a page and not marked as frozen.

deployed_codewill_unmount

`will_unmount()`

Called before the control is removed from the page tree.

Override to dispose resources created in `did_mount()`, such as subscriptions, timers, or external handles.

BasePage

A visual container representing a top-level view in a Flet application.

BasePage serves as the base class for Page and MultiView, and provides a unified surface for rendering application content, app bars, navigation elements, dialogs, overlays, and more. It manages one or more View instances and exposes high-level layout, scrolling, and theming properties.

Unlike lower-level layout controls (e.g., Column, Container), BasePage represents an entire logical view or screen of the app. It provides direct access to view-level controls such as AppBar, NavigationBar, FloatingActionButton, and supports system-level events like window resizing and media changes.

This class is not intended to be used directly in most apps; instead, use Page or MultiView, which extend this base functionality.

Inherits: AdaptiveControl

Properties

- `appbar` - Gets or sets the top application bar (AppBar or CupertinoAppBar) for the view.
- `auto_scroll` - Whether root view should auto-scroll to the end on content changes.
- `bgcolor` - Background color of the root view.
- `bottom_appbar` - Bottom app bar displayed in the root view.
- `controls` - Root view content controls displayed by this page.
- `dark_theme` - Customizes the theme of the application when in dark theme mode.
- `decoration` - Background decoration of the root view container.
- `drawer` - Navigation drawer opened from the leading edge.
- `enable_screenshots` - Enable taking screenshots of the entire page with `take_screenshot` method.
- `end_drawer` - Navigation drawer opened from the trailing edge.
- `floating_action_button` - Floating action button shown for the root view.
- `floating_action_button_location` - Placement of the floating action button in the root view.
- `foreground_decoration` - Foreground decoration painted above root view content.
- `height` - Page height in logical pixels.
- `horizontal_alignment` - Horizontal alignment of root view child controls.
- `locale_configuration` - Configures supported locales and the current locale.
- `media` - The current environmental metrics of the page or window.
- `navigation_bar` - Bottom navigation bar for the root view.
- `overlay` - The list of overlay controls rendered above page content.
- `padding` - Inner padding for the root view content.
- `scroll` - Scroll behavior mode for root view content.
- `services` - Service instances attached to the root view lifecycle.
- `show_semantics_debugger` - Whether to turn on an overlay that shows the accessibility information reported by the framework.
- `spacing` - Default spacing between root view child controls.
- `theme` - Customizes the theme of the application when in light theme mode.
- `theme_animation_style` - Overrides the default animation style used when the application's theme is changed (for example, when `theme_mode` switches between light and dark).
- `theme_mode` - The page's theme mode.
- `title` - Page or window title.

- `vertical_alignment` - Vertical alignment of root view child controls.
- `views` - A list of views managed by the page.
- `width` - Page width in logical pixels.

Events

- `on_media_change` - Called when media has changed.
- `on_resize` - Called when a user resizes a browser or native OS window containing Flet app

Methods

- `add` - Adds controls to the page.
- `clean` - Remove all root view controls and send update to the client.
- `close_drawer` - Close the drawer.
- `close_end_drawer` - Close the end drawer.
- `insert` - Inserts controls at specific index of `page.controls` list.
- `pop_dialog` - Closes the most recently opened dialog.
- `remove` - Removes specific controls from `page.controls` list.
- `remove_at` - Remove controls from `page.controls` list at specific index.
- `scroll_to` - Moves scroll position to either absolute offset, relative delta or jump to the control with specified `scroll_key`.
- `show_dialog` - Displays a dialog and manages its dismissal lifecycle.
- `show_drawer` - Show the drawer.
- `show_end_drawer` - Show the end drawer.
- `take_animation` - Captures an animated sequence of page screenshots in a single round-trip.
- `take_screenshot` - Captures a screenshot of the entire page with overlays.
- `update` - Push pending updates to the connected client.

Properties

`buildappbarpropertywritable`

`appbar`: `AppBar` | `CupertinoAppBar` | `None`

Gets or sets the top application bar (`AppBar` or `CupertinoAppBar`) for the view.

The app bar typically displays the page title and optional actions such as navigation icons, menus, or other interactive elements.

`buildauto_scrollpropertywritable`

`auto_scroll`: `bool`

Whether root view should auto-scroll to the end on content changes.

`buildbgcolorpropertywritable`

`bgcolor`: `ColorValue` | `None`

Background color of the root view.

`buildbottom_appbarpropertywritable`

`bottom_appbar`: `BottomAppBar` | `None`

Bottom app bar displayed in the root view.

`buildcontrolspropertywritable`

`controls`: `list[BaseControl]`

Root view content controls displayed by this page.

`bulddark_themeclass-attributeinstance-attribute`

`dark_theme`: `Theme` | `None` = `None`

Customizes the theme of the application when in dark theme mode.

builddecorationpropertywritable

decoration: BoxDecoration | None

Background decoration of the root view container.

builddrawerpropertywritable

drawer: NavigationDrawer | None

Navigation drawer opened from the leading edge.

buildenable_screenshotsclass-attributeinstance-attribute

enable_screenshots: bool = False

Enable taking screenshots of the entire page with take_screenshot method.

buildend_drawerpropertywritable

end_drawer: NavigationDrawer | None

Navigation drawer opened from the trailing edge.

buildfloating_action_buttonpropertywritable

floating_action_button: FloatingActionButton | None

Floating action button shown for the root view.

buildfloating_action_button_locationpropertywritable

```
floating_action_button_location: (  
    FloatingActionButtonLocation | OffsetValue | None  
)
```

Placement of the floating action button in the root view.

buildforeground_decorationpropertywritable

foreground_decoration: BoxDecoration | None

Foreground decoration painted above root view content.

buildheightclass-attributeinstance-attribute

height: Number | None = None

Page height in logical pixels.

Note

- This property is read-only.
- To get or set the full window height including window chrome (e.g., title bar and borders) when running a Flet app on desktop, use the height property of Page.window instead.

buildhorizontal_alignmentpropertywritable

horizontal_alignment: CrossAxisAlignment

Horizontal alignment of root view child controls.

buildlocale_configurationclass-attributeinstance-attribute

locale_configuration: LocaleConfiguration | None = None

Configures supported locales and the current locale.

buildmediaclass-attributeinstance-attribute

```
media: PageMediaData = field(  
    default_factory=lambda: PageMediaData(  
        padding=Padding.zero(),  
        view_padding=Padding.zero(),  
        view_insets=Padding.zero(),  
        device_pixel_ratio=0,  
        orientation=Orientation.PORTRAIT,  
        always_use_24_hour_format=False,  
    )  
)
```

The current environmental metrics of the page or window.

This data is updated whenever the platform window or layout changes, such as when rotating a device, resizing a browser window, or adjusting system UI elements like the keyboard or safe areas.

buildnavigation_barpropertywritable

```
navigation_bar: (  
  NavigationBar | CupertinoNavigationBar | None  
)
```

Bottom navigation bar for the root view.

buildoverlayproperty

```
overlay: list[BaseControl]
```

The list of overlay controls rendered above page content.

buildpaddingpropertywritable

```
padding: PaddingValue | None
```

Inner padding for the root view content.

buildscrollpropertywritable

```
scroll: ScrollMode | Scrollbar | None
```

Scroll behavior mode for root view content.

buildservicespropertywritable

```
services: list[Service]
```

Service instances attached to the root view lifecycle.

buildshow_semantics_debuggerclass-attributeinstance-attribute

```
show_semantics_debugger: bool | None = None
```

Whether to turn on an overlay that shows the accessibility information reported by the framework.

buildspacingpropertywritable

```
spacing: Number
```

Default spacing between root view child controls.

buildthemeclass-attributeinstance-attribute

```
theme: Theme | None = None
```

Customizes the theme of the application when in light theme mode. Currently, a theme can only be automatically generated from a "seed" color. For example, to generate light theme from a green color.

buildtheme_animation_styleclass-attributeinstance-attribute

```
theme_animation_style: AnimationStyle | None = None
```

Overrides the default animation style used when the application's theme is changed (for example, when `theme_mode` switches between light and dark).

Tip

Use `AnimationStyle.no_animation` to disable the theme transition entirely.

buildtheme_modeclass-attributeinstance-attribute

```
theme_mode: ThemeMode | None = ThemeMode.SYSTEM
```

The page's theme mode.

buildtitleclass-attributeinstance-attribute

```
title: str | None = None
```

Page or window title.

buildvertical_alignmentpropertywritable

```
vertical_alignment: MainAxisAlignment
```

Vertical alignment of root view child controls.

buildviewsclass-attributeinstance-attribute

```
views: list[View] = field(default_factory=lambda: [View()])
```

A list of views managed by the page.

Each View represents a distinct navigation state or screen in the application.

The first view in the list is considered the active one by default.

buildwidthclass-attributeinstance-attribute

```
width: Number | None = None
```

Page width in logical pixels.

Note

- This property is read-only.
- To get or set the full window height including window chrome (e.g., title bar and borders) when running a Flet app on desktop, use the width property of Page.window instead.

Events

bolton_media_change

```
on_media_change: EventHandler[PageMediaData] | None = None
```

Called when media has changed.

bolton_resize

```
on_resize: EventHandler[PageResizeEvent] | None = None
```

Called when a user resizes a browser or native OS window containing Flet app

Example

```
def main(page: ft.Page):
    def handle_page_size(e):
        print("New page size:", page.window.width, page.window_height)

    page.on_resize = handle_page_size
```

Methods

deployed_codeadd

```
add(*controls: Control = ()) -> None
```

Adds controls to the page.

Example:

```
page.add(ft.Text("Hello!"), ft.FilledButton("Button"))
```

deployed_codeclean

```
clean() -> None
```

Remove all root view controls and send update to the client.

deployed_codeclose_drawer

```
close_drawer()
```

Close the drawer.

deployed_codeclose_end_drawer

```
close_end_drawer()
```

Close the end drawer.

deployed_codeinsert

```
insert(at: int, *controls: Control = ()) -> None
```

Inserts controls at specific index of page.controls list.

deployed_codepop_dialog

```
pop_dialog() -> DialogControl | None
```

Closes the most recently opened dialog.

This method searches the active dialog stack for the topmost dialog that is currently open, marks it as closed, updates its state, and returns the closed dialog.

Returns:

- DialogControl | None - The closed dialog instance if one was found, otherwise None.

deployed_coderemove

```
remove(*controls: Control = ()) -> None
```

Removes specific controls from page.controls list.

deployed_coderemove_at

```
remove_at(index: int) -> None
```

Remove controls from page.controls list at specific index.

deployed_codescroll_toasync

```
scroll_to(
    offset: Number | None = None,
    delta: Number | None = None,
    scroll_key: ScrollKey
    | str
    | int
    | float
    | bool
    | None = None,
    duration: DurationValue | None = None,
    curve: AnimationCurve | None = None,
) -> None
```

Moves scroll position to either absolute offset, relative delta or jump to the control with specified scroll_key.

See Column.scroll_to for method details and examples.

deployed_codeshow_dialog

```
show_dialog(dialog: DialogControl) -> None
```

Displays a dialog and manages its dismissal lifecycle.

This method adds the specified dialog to the active dialog stack and renders it on the page. The on_dismiss handler of the dialog is temporarily wrapped to ensure the dialog is removed from the stack and its dismissal event is triggered appropriately.

Parameters:

- dialog (DialogControl) - The dialog instance to display. Must not yet be open.

Raises:

- RuntimeError - If the specified dialog is already open.

deployed_codeshow_drawerasync

```
show_drawer()
```

Show the drawer.

Raises:

- ValueError - If no drawer is defined.

deployed_codeshow_end_drawerasync

```
show_end_drawer()
```

Show the end drawer.

Raises:

- ValueError - If no end_drawer is defined.

deployed_codetake_animationasync

```
take_animation(
    name: str,
    frame_delays_ms: list[int],
    pixel_ratio: Number | None = None,
) -> list[bytes]
```


Captures an animated sequence of page screenshots in a single round-trip. Each entry in `frame_delays_ms` is the delay in milliseconds to wait before capturing that frame. The wait and capture loop runs entirely on the Flutter side, so animation timing isn't distorted by Python-Flutter RPC latency the way it is with a Python-driven `take_screenshot` loop.

Requires `enable_screenshots = True`.

Parameters:

- `name` (str) - Name prefix for the captured frames (for debugging).
- `frame_delays_ms` (list[int]) - Per-frame delays in milliseconds. The list length determines the number of frames captured.
- `pixel_ratio` (Number | None, default: None) - A pixel ratio of the captured frames. If None, device-specific pixel ratio will be used.

Returns:

- list[bytes] - List of PNG-encoded frames, one per entry in `frame_delays_ms`.

`deployed_codetake_screenshotasync`

```
take_screenshot(  
    pixel_ratio: Number | None = None,  
    delay: DurationValue | None = None,  
) -> bytes
```

Captures a screenshot of the entire page with overlays.

Parameters:

- `pixel_ratio` (Number | None, default: None) - A pixel ratio of the captured screenshot. If None, device-specific pixel ratio will be used.
- `delay` (DurationValue | None, default: None) - A delay before taking a screenshot. The delay will be 20 milliseconds if not specified.

Returns:

- bytes - Screenshot in PNG format.

`deployed_codeupdate`

```
update(*controls: Control = ()) -> None
```

Push pending updates to the connected client.

Parameters:

- `*controls` (default: ()) - Specific controls to update. When omitted, updates this page object.

BottomAppBar

A material design bottom app bar.

```
ft.BottomAppBar(  
    bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,  
    content=ft.Row(  
        alignment=ft.MainAxisAlignment.SPACE_AROUND,  
        controls=[  
            ft.IconButton(ft.Icons.MENU),  
            ft.IconButton(ft.Icons.SEARCH),  
            ft.IconButton(ft.Icons.SETTINGS),  
        ],  
    ),  
)
```

Image: BottomAppBar - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/bottom_app_bar/image_for_docs.png

BottomAppBar

Inherits: `LayoutControl`

Properties

- `bgcolor` - The fill color to use for this app bar.
- `border_radius` - The border radius to apply when clipping and painting this app bar.
- `clip_behavior` - Defines how the content of this app bar should be clipped.
- `content` - The content of this bottom app bar.
- `elevation` - The z-coordinate at which to place this bottom app bar relative to its parent.
- `notch_margin` - The margin between the `FloatingActionButton` and this app bar's notch.
- `padding` - Empty space to inscribe inside a container decoration (background, border).
- `shadow_color` - The color of the shadow below this app bar.
- `shape` - The notch that is made for the floating action button.

Examples

Notched FloatingActionButton

`play_arrow` Try Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.vertical_alignment = ft.MainAxisAlignment.CENTER  
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER  
  
    page.floating_action_button = ft.FloatingActionButton(  
        icon=ft.Icons.ADD,  
        shape=ft.CircleBorder(),  
    )  
    page.floating_action_button_location = ft.FloatingActionButtonLocation.CENTER_DOCKED  
  
    page.bottom_appbar = ft.BottomAppBar(  
        bgcolor=ft.Colors.BLUE,  
        shape=ft.CircularRectangleNotchShape(),  
        content=ft.Row(  
            controls=[  
                ft.IconButton(icon=ft.Icons.MENU, icon_color=ft.Colors.WHITE),  
                ft.Container(expand=True),  
                ft.IconButton(icon=ft.Icons.SEARCH, icon_color=ft.Colors.WHITE),  
                ft.IconButton(icon=ft.Icons.FAVORITE, icon_color=ft.Colors.WHITE),  
            ],  
        ),  
    )  
  
    page.add(ft.SafeArea(content=ft.Column(controls=[ft.Text("Content goes here...")]))))  
  
if __name__ == "__main__":  
    ft.run(main)
```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/bottom_app_bar/notched_fab.png

BottomAppBar border radius

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    page.bottom_appbar = ft.BottomAppBar(
        border_radius=ft.BorderRadius.all(20),
        bgcolor=ft.Colors.BLUE,
        content=ft.Row(
            controls=[
                ft.IconButton(icon=ft.Icons.MENU, icon_color=ft.Colors.WHITE),
                ft.Container(expand=True),
                ft.IconButton(icon=ft.Icons.SEARCH, icon_color=ft.Colors.WHITE),
                ft.IconButton(icon=ft.Icons.FAVORITE, icon_color=ft.Colors.WHITE),
            ],
        ),
    )

    page.add(ft.SafeArea(content=ft.Column(controls=[ft.Text("Content goes here...")]))))

if __name__ == "__main__":
    ft.run(main)
```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/bottom_app_bar/border_radius.png

Properties

buildbgcolorclass-attributeinstance-attribute

bgcolor: `ColorValue` | `None` = `None`

The fill color to use for this app bar.

If `None`, `BottomAppBarTheme.bgcolor` is used; if that is also `None`, then defaults to `ColorScheme.surface`.

buildborder_radiusclass-attributeinstance-attribute

border_radius: `BorderRadiusValue` | `None` = `None`

The border radius to apply when clipping and painting this app bar.

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: `ClipBehavior` | `None` = `None`

Defines how the content of this app bar should be clipped.

If `None`, defaults to:

- `ClipBehavior.ANTI_ALIAS` if `border_radius` is set and not equal to `BorderRadius.all`;
- Else `ClipBehavior.NONE`.

buildcontentclass-attributeinstance-attribute

content: `Control` | `None` = `None`

The content of this bottom app bar.

buildelevationclass-attributeinstance-attribute

elevation: `Number` | `None` = `None`

The z-coordinate at which to place this bottom app bar relative to its parent. It controls the size of the shadow below this app bar.

If `None`, `BottomAppBarTheme.elevation` is used; if that is also `None`, then defaults to 3.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildnotch_marginclass-attributeinstance-attribute

notch_margin: `Number` = 4.0

The margin between the `FloatingActionButton` and this app bar's notch.

Note

Has effect only if shape is not None.

buildpaddingclass-attributeinstance-attribute

`padding: PaddingValue | None = None`

Empty space to inscribe inside a container decoration (background, border).

If None, BottomAppBarTheme.padding is used; if that is also None, then defaults to `Padding.symmetric(vertical=12.0, horizontal=16.0)`.

buildshadow_colorclass-attributeinstance-attribute

`shadow_color: ColorValue | None = None`

The color of the shadow below this app bar.

If None, BottomAppBarTheme.shadow_color is used; if that is also None, then defaults to `Colors.TRANSPARENT`.

buildshapeclass-attributeinstance-attribute

`shape: NotchShape | None = None`

The notch that is made for the floating action button.

If None, BottomAppBarTheme.shape is used; if that is also None, then the shape will be rectangular with no notch.

BottomSheet

A modal bottom sheet.

Displays a temporary surface anchored to the bottom of the screen that presents supplemental content or actions. Prevents interaction with the underlying app while visible.

Example:

```
sheet = ft.BottomSheet(
    content=ft.Column(
        width=150,
        alignment=ft.MainAxisAlignment.CENTER,
        horizontal_alignment=ft.CrossAxisAlignment.CENTER,
        controls=[
            ft.Text("Choose an option"),
            ft.TextButton("Dismiss"),
        ],
    )
)
page.show_dialog(sheet)
```

Image: BottomSheet - https://flet.dev/docs/test-images/examples/controls/material/golden/macos/bottom_sheet/image_for_docs.png

BottomSheet

Inherits: DialogControl

Properties

- `animation_style` - The animation style of this bottom sheet.
- `barrier_color` - The color of the scrim that obscures content behind this bottom sheet.
- `bgcolor` - The background color of this bottom sheet.
- `clip_behavior` - Defines how the content of this bottom sheet should be clipped.
- `content` - The content of this bottom sheet.
- `dismissible` - Specifies whether this bottom sheet will be dismissed when user taps on the scrim.
- `draggable` - Specifies whether this bottom sheet can be dragged up and down and dismissed by swiping downwards.
- `elevation` - Defines the size of the shadow below this bottom sheet.
- `fullscreen` - Expands the sheet to fill the window/page height.
- `maintain_bottom_view_insets_padding` - Adds a padding at the bottom to avoid obstructing this bottom sheet's content with on-screen keyboard or other system elements.
- `scrollable` - Removes the half-height cap so the sheet can grow with its content.
- `shape` - Defines the shape of this bottom sheet.
- `show_drag_handle` - Whether to display drag handle at the top of sheet or not.
- `size_constraints` - The size constraints to apply to this bottom sheet.
- `use_safe_area` - Specifies whether the sheet will avoid system intrusions on the top, left, and right.

Examples

BottomSheet

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "BottomSheet Example"
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def handle_sheet_dismissal(e: ft.Event[ft.DialogControl]):
        page.add(ft.Text("Bottom sheet dismissed"))

    sheet = ft.BottomSheet(
        on_dismiss=handle_sheet_dismissal,
```

```

        content=ft.Container(
            padding=50,
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                tight=True,
                controls=[
                    ft.Text("Here is a bottom sheet!"),
                    ft.Button("Dismiss", on_click=lambda _: page.pop_dialog()),
                ],
            ),
        ),
    ),
)

page.add(
    ft.SafeArea(
        # avoid_intrusions_left=False,
        # avoid_intrusions_top=False,
        # avoid_intrusions_right=False,
        # avoid_intrusions_bottom=False,
        content=ft.Column(
            controls=[
                ft.Button(
                    content="Display bottom sheet",
                    on_click=lambda e: page.show_dialog(sheet),
                ),
            ],
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/bottom_sheet/basic.gif

BottomSheet fullscreen

play_arrowTry Online

import flet as ft

```

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def handle_switch_change(e: ft.Event[ft.Switch]):
        sheet.fullscreen = e.control.value

    sheet = ft.BottomSheet(
        fullscreen=True,
        show_drag_handle=True,
        content=ft.Container(
            padding=ft.Padding.all(10),
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    ft.Text("This is bottom sheet's content!"),
                    ft.Button("Close bottom sheet", on_click=lambda: page.pop_dialog()),
                ],
            ),
        ),
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.START,
                controls=[
                    ft.Button(
                        content="Display bottom sheet",
                        on_click=lambda e: page.show_dialog(sheet),
                    ),
                    ft.Switch(
                        value=True,
                        label="Fullscreen",
                        on_change=handle_switch_change,
                    ),
                ],
            ),
        ),
    )

    if __name__ == "__main__":
        ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/bottom_sheet/fullscreen.gif

Properties

buildanimation_styleclass-attributeinstance-attribute

`animation_style: AnimationStyle | None = None`

The animation style of this bottom sheet.

buildbarrier_colorclass-attributeinstance-attribute

`barrier_color: ColorValue | None = None`

The color of the scrim that obscures content behind this bottom sheet.

buildbgcolorclass-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

The background color of this bottom sheet.

buildclip_behaviorclass-attributeinstance-attribute

`clip_behavior: ClipBehavior | None = None`

Defines how the content of this bottom sheet should be clipped.

buildcontentinstance-attribute

`content: Control`

The content of this bottom sheet.

Tip

Set `scrollable` `True` if this content is or contains scrollable controls (e.g., `ListView`, `GridView`) or you plan to expand the content or give it a custom height, else the bottom sheet might ignore the custom height and stop around mid-screen.

bulddismissibleclass-attributeinstance-attribute

`dismissible: bool = True`

Specifies whether this bottom sheet will be dismissed when user taps on the scrim.

bulddraggableclass-attributeinstance-attribute

`draggable: bool = False`

Specifies whether this bottom sheet can be dragged up and down and dismissed by swiping downwards.

buldelevationclass-attributeinstance-attribute

`elevation: Number | None = None`

Defines the size of the shadow below this bottom sheet.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildfullscreenclass-attributeinstance-attribute

`fullscreen: bool = False`

Expands the sheet to fill the window/page height.

If set to `True`, `scrollable` is internally set to `True` equally, so the sheet can grow freely to fill the page.

buildmaintain_bottom_view_insets_paddingclass-attributeinstance-attribute

`maintain_bottom_view_insets_padding: bool = True`

Adds a padding at the bottom to avoid obstructing this bottom sheet's content with on-screen keyboard or other system elements.

buildscrollableclass-attributeinstance-attribute

`scrollable: bool = False`

Removes the half-height cap so the sheet can grow with its content.

Set this to `True` whenever the sheet body contains scrollable controls (e.g., `ListView`, `GridView`) or you plan to expand the content or give it a custom height, else the bottom sheet might ignore the custom height and stop around mid-screen.

buildshapeclass-attributeinstance-attribute

`shape: OutlinedBorder | None = None`

Defines the shape of this bottom sheet.

buildshow_drag_handleclass-attributeinstance-attribute

`show_drag_handle: bool = False`

Whether to display drag handle at the top of sheet or not.

buildsize_constraintsclass-attributeinstance-attribute

`size_constraints: BoxConstraints | None = None`

The size constraints to apply to this bottom sheet.

builduse_safe_areaclass-attributeinstance-attribute

`use_safe_area: bool = True`

Specifies whether the sheet will avoid system intrusions on the top, left, and right.

Button

A material button.

It supports various styles, colors, event handlers for user interaction, and can be used to display text, icons, etc.

Example:

```
ft.Button(content="Enabled button")
ft.Button(content="Disabled button", disabled=True)
```

Image: Button - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/image_for_docs.png

Enabled and disabled buttons

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `autofocus` - Whether this button should be focused initially.
- `bgcolor` - The button's background color.
- `clip_behavior` - The button's clip behavior.
- `color` - The button's foreground color.
- `content` - The button's label.
- `elevation` - The button's elevation.
- `icon` - The icon to display inside the button.
- `icon_color` - The color of the icon.
- `style` - The button's style.
- `url` - The URL to open when the button is clicked.

Events

- `on_blur` - Called when the button loses focus.
- `on_click` - Called when the button is clicked.
- `on_focus` - Called when the button is focused.
- `on_hover` - Called when the button is hovered.
- `on_long_press` - Called when the button is long-pressed.

Methods

- `focus` - Requests focus for this control.

Examples

Button

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button(content="Enabled button"),
                    ft.Button(content="Disabled button", disabled=True),
                ]
            )
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: Basic button - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/basic.png>

Buttons with icons

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button(content="Button with icon", icon=ft.Icons.WAVES_ROUNDED),
                    ft.Button(
                        content="Button with colorful icon",
                        icon=ft.Icons.PARK_ROUNDED,
                        icon_color=ft.Colors.GREEN_400,
                    ),
                ]
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: Basic button - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/icons.png>

Handling clicks

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    def button_clicked(e: ft.Event[ft.Button]):
        button.data += 1
        message.value = f"Button clicked {button.data} time(s)"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    button := ft.Button(
                        content="Button with 'click' event",
                        data=0,
                        on_click=button_clicked,
                    ),
                    message := ft.Text(),
                ]
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: Handling clicks - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/handling_clicks.png

Custom content

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button(
                        width=150,
                        content=ft.Row(
                            alignment=ft.MainAxisAlignment.SPACE_AROUND,
                            controls=[
                                ft.Icon(ft.Icons.FAVORITE, color=ft.Colors.PINK),
                                ft.Icon(ft.Icons.AUDIOTRACK, color=ft.Colors.GREEN),
                                ft.Icon(ft.Icons.BEACH_ACCESS, color=ft.Colors.BLUE),
                            ],
                        ),
                    ),
                ],
            ),
        ),
    )
```

```

        ft.Button(
            content=ft.Container(
                padding=ft.Padding.all(10),
                content=ft.Column(
                    alignment=ft.MainAxisAlignment.CENTER,
                    spacing=5,
                    controls=[
                        ft.Text(value="Compound button", size=20),
                        ft.Text(value="This is secondary text"),
                    ],
                ),
            ),
        ),
    ],
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Buttons with custom content - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/custom_content.png

Button shapes

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    page.padding = 30
    page.spacing = 30

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button(
                        content="Stadium",
                        style=ft.ButtonStyle(shape=ft.StadiumBorder()),
                    ),
                    ft.Button(
                        content="Rounded rectangle",
                        style=ft.ButtonStyle(
                            shape=ft.RoundedRectangleBorder(radius=10)
                        ),
                    ),
                    ft.Button(
                        content="Continuous rectangle",
                        style=ft.ButtonStyle(
                            shape=ft.ContinuousRectangleBorder(radius=30)
                        ),
                    ),
                    ft.Button(
                        content="Beveled rectangle",
                        style=ft.ButtonStyle(
                            shape=ft.BeveledRectangleBorder(radius=10)
                        ),
                    ),
                    ft.Button(
                        content="Circle",
                        style=ft.ButtonStyle(shape=ft.CircleBorder(), padding=30),
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: Buttons with different shapes - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/button_shapes.png

Button styling

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(

```

```

controls=[
    ft.Button(
        content="Styled button 1",
        style=ft.ButtonStyle(
            color={
                ft.ControlState.HOVERED: ft.Colors.WHITE,
                ft.ControlState.FOCUSED: ft.Colors.BLUE,
                ft.ControlState.DEFAULT: ft.Colors.BLACK,
            },
            bgcolor={
                ft.ControlState.FOCUSED: ft.Colors.PINK_200,
                ft.ControlState.DEFAULT: ft.Colors.YELLOW,
            },
            padding={ft.ControlState.HOVERED: 20},
            overlay_color=ft.Colors.TRANSPARENT,
            elevation={
                ft.ControlState.DEFAULT: 0,
                ft.ControlState.HOVERED: 5,
                ft.ControlState.PRESSED: 10,
            },
            animation_duration=500,
            side={
                ft.ControlState.DEFAULT: ft.BorderSide(
                    1, color=ft.Colors.BLUE_100
                ),
                ft.ControlState.HOVERED: ft.BorderSide(
                    3, color=ft.Colors.BLUE_400
                ),
                ft.ControlState.PRESSED: ft.BorderSide(
                    6, color=ft.Colors.BLUE_600
                ),
            },
            shape={
                ft.ControlState.HOVERED: ft.RoundedRectangleBorder(
                    radius=20
                ),
                ft.ControlState.DEFAULT: ft.RoundedRectangleBorder(
                    radius=2
                ),
            },
        ),
    ],
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: Styled button - default state - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/styled_initial.png

Default state

Image: Styled button - hovered state - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/styled_hovered.png

Hovered state

Animate on hover

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    def animate(e: ft.Event[ft.Button]):
        e.control.rotate = 0.1 if e.data else 0

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button(
                        content="Hover over me, I'm animated!",
                        rotate=0,
                        animate_rotation=100,
                        on_hover=animate,
                        on_click=lambda e: page.add(
                            ft.Text("Clicked! Try a long press!")
                        ),
                        on_long_press=lambda e: page.add(
                            ft.Text("I knew you could do it!")
                        ),
                    ),
                ],
            ),
        ),
    )

```

```
)
    )
)
```

```
if __name__ == "__main__":
    ft.run(main)
```

Image: Unhovered button - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/animate_on_hover_initial.png

Normal button

Image: Hovered button - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/button/animate_on_hover_hovered.png

Hovered button

Properties

buildautofocusclass-attributeinstance-attribute

```
autofocus: bool | None = None
```

Whether this button should be focused initially.

buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue | None = field(
    default=None, metadata={"skip": True}
)
```

The button's background color. If not specified, defaults to the theme's primary color.

buildclip_behaviorclass-attributeinstance-attribute

```
clip_behavior: ClipBehavior | None = None
```

The button's clip behavior.

buildcolorclass-attributeinstance-attribute

```
color: ColorValue | None = field(
    default=None, metadata={"skip": True}
)
```

The button's foreground color. If not specified, defaults to the theme's primary color.

buildcontentclass-attributeinstance-attribute

```
content: StrOrControl | None = None
```

The button's label. Typically a Text control or a string. If a string is provided, it will be wrapped in a Text control.

Raises:

- ValueError - If neither content nor icon (string or visible control) is provided.

buildelevationclass-attributeinstance-attribute

```
elevation: Number = field(
    default=1, metadata={"skip": True}
)
```

The button's elevation. If not specified, defaults to 1.

buildiconclass-attributeinstance-attribute

```
icon: IconDataOrControl | None = None
```

The icon to display inside the button. Typically an Icon control or an IconData. If an IconData is provided, it will be wrapped in an Icon control.

Raises:

- ValueError - If neither icon nor content (string or visible control) is provided.

buildicon_colorclass-attributeinstance-attribute

```
icon_color: ColorValue | None = None
```

The color of the icon. If not specified, defaults to the current foreground color.

buildstyleclass-attributeinstance-attribute

```
style: ButtonStyle | None = field(
    default=None, metadata={"skip": True}
)
```

)

The button's style.

buildurlclass-attributeinstance-attribute

`url: str | Url | None = None`

The URL to open when the button is clicked.

Events

bolton_blurclass-attributeinstance-attribute

`on_blur: ControlEventHandler[Button] | None = None`

Called when the button loses focus.

bolton_clickclass-attributeinstance-attribute

`on_click: ControlEventHandler[Button] | None = None`

Called when the button is clicked.

bolton_focusclass-attributeinstance-attribute

`on_focus: ControlEventHandler[Button] | None = None`

Called when the button is focused.

bolton_hoverclass-attributeinstance-attribute

`on_hover: ControlEventHandler[Button] | None = None`

Called when the button is hovered.

bolton_long_pressclass-attributeinstance-attribute

`on_long_press: ControlEventHandler[Button] | None = None`

Called when the button is long-pressed.

Methods

deployed_codefocusasync

`focus()`

Requests focus for this control.

Camera

Display a live camera preview, capture photos and videos, and stream camera frames directly in your Flet apps.

Powered by the camera Flutter package.

Platform Support

| Platform | iOS | Android | Web |
|-----------|--------------------------|--------------------------|--------------------------|
| Supported | ☐ | ☐ | ☐ |

Usage

Add the flet-camera package to your project dependencies:

- uv
 - pip
- ```
uv add flet-camera
pip install flet-camera
```

### Permissions

Request camera (and microphone if recording video with audio) permissions on mobile platforms before initializing the control. You can use `PermissionHandler` to prompt the user. You can also use predefined cross-platform permission bundles for camera and microphone permissions.

## Requirements

The below sections show the required configurations for each platform.

### Android

Configuration to be made to access the camera and optionally the microphone:

- `android.permission.CAMERA`: Allows camera usage.
- `android.permission.RECORD_AUDIO` (optional): Allows video recording with audio.
- `flet build`
- `pyproject.toml`

```
flet build apk \
 --android-permissions android.permission.CAMERA=true \
 --android-permissions android.permission.RECORD_AUDIO=true

[tool.flet.android.permission]
"android.permission.CAMERA" = true
"android.permission.RECORD_AUDIO" = true
```

See also:

- [setting Android permissions](#)

### iOS

Configuration to be made to access the camera and optionally the microphone:

- `NSCameraUsageDescription`: Required for camera usage.
- `NSMicrophoneUsageDescription` (optional): Required only for video recording with audio. For example, when `enable_audio` parameter of `Camera.initialize` is set to `True` (default).
- `flet build`
- `pyproject.toml`

```
flet build ipa \
 --info-plist NSCameraUsageDescription="Some message to describe why you need this permission..." \
 --info-plist NSMicrophoneUsageDescription="Some message to describe why you need this permission..."

[tool.flet.ios.info]
NSCameraUsageDescription = "Some message to describe why you need this permission..."
```

```
NSMicrophoneUsageDescription = "Some message to describe why you need this permission..."
```

See also:

- [setting iOS permissions](#)

## Cross-platform

Additionally/alternatively, you can make use of our predefined cross-platform camera (and optionally microphone) permission bundles:

- [flet build](#)
- [pyproject.toml](#)

```
flet build <target_platform> --permissions camera microphone

[tool.flet]
permissions = ["camera", "microphone"]
```

## Example

### Camera playground

[play\\_arrow](#)Try Online

```
import base64
import logging
from dataclasses import dataclass, field
from datetime import datetime
from math import cos, pi, sin

import flet as ft
import flet_camera as fc

@dataclass
class State:
 cameras: list[fc.CameraDescription] = field(default_factory=list)
 selected_camera: fc.CameraDescription | None = None
 camera_labels: dict[str, str] = field(default_factory=dict)
 is_streaming: bool = False
 is_streaming_supported: bool = False
 is_initialized: bool = False
 is_preview_paused: bool = False
 is_recording: bool = False
 is_recording_paused: bool = False
 device_orientation: ft.DeviceOrientation | None = None
 last_frame_width: int | None = None
 last_frame_height: int | None = None

async def main(page: ft.Page):
 image_frame_width = 280
 image_frame_height = 200
 page.title = "Camera control"
 page.padding = 16
 page.scroll = ft.ScrollMode.AUTO
 page.horizontal_alignment = ft.CrossAxisAlignment.STRETCH

 state = State()

 preview = fc.Camera(
 expand=True,
 preview_enabled=True,
 content=ft.Container(
 alignment=ft.Alignment.CENTER,
 content=ft.Icon(
 ft.Icons.CENTER_FOCUS_STRONG,
 color=ft.Colors.WHITE_70,
 size=48,
),
),
),

 status = ft.Text(value="Select a camera", size=12)
 last_photo_label = ft.Text("Last photo", visible=False)
 last_image = ft.Image(
 src=base64.b64decode(
 "iVBORw0KGgoAAAANSUHEUgAAAAEAAAABCAQAAAC1HAWCAAAAC0lEQVR42mP8/wIAAgMBAp0YVwAAAABJRUErkJggg=="
),
 width=image_frame_width,
 height=image_frame_height,
 fit=ft.BoxFit.CONTAIN,
 gapless_playback=True,
```



```

)
last_image_frame = ft.Container(
 width=image_frame_width,
 height=image_frame_height,
 alignment=ft.Alignment.CENTER,
 content=last_image,
 visible=False,
)
selector = ft.Dropdown(label="Camera", options=[])
recorded_video_path = ft.Text(value="Recorded video: not saved yet", size=12)
take_photo_btn = ft.FilledIconButton(
 icon=ft.Icons.PHOTO_CAMERA,
 tooltip="Take photo",
 disabled=True,
)
record_btn = ft.FilledTonalIconButton(
 icon=ft.Icons.VIDEOCAM,
 selected_icon=ft.Icons.STOP,
 selected=False,
 tooltip="Start / stop recording",
 disabled=True,
)
pause_recording_btn = ft.OutlinedIconButton(
 icon=ft.Icons.PAUSE,
 selected_icon=ft.Icons.PLAY_ARROW,
 selected=False,
 tooltip="Pause / resume recording",
 disabled=True,
)
stream_btn = ft.OutlinedIconButton(
 icon=ft.Icons.PLAY_ARROW,
 selected_icon=ft.Icons.STOP,
 selected=False,
 tooltip="Start / stop image stream",
 disabled=True,
)
preview_btn = ft.OutlinedIconButton(
 icon=ft.Icons.VISIBILITY_OFF,
 selected_icon=ft.Icons.VISIBILITY,
 selected=True,
 tooltip="Pause / resume preview",
 disabled=True,
)

def has_human_readable_name(camera: fc.CameraDescription) -> bool:
 name = camera.name.strip()
 if not name:
 return False
 if name.startswith("com.apple.avfoundation."):
 return False
 return not (":" in name and "." in name)

def camera_label(camera: fc.CameraDescription) -> str:
 if has_human_readable_name(camera):
 return camera.name

 direction = camera.lens_direction.value.capitalize()
 lens_map = {
 "wide": "Wide",
 "telephoto": "Telephoto",
 "ultrawide": "Ultra Wide",
 "unknown": "Unknown",
 }
 lens_type = lens_map.get(camera.lens_type.value, camera.lens_type.value)
 return f"{direction} ({lens_type})"

def device_orientation_degrees(orientation: ft.DeviceOrientation | None) -> int:
 if orientation == ft.DeviceOrientation.PORTRAIT_UP:
 return 0
 if orientation == ft.DeviceOrientation.LANDSCAPE_RIGHT:
 return 90
 if orientation == ft.DeviceOrientation.PORTRAIT_DOWN:
 return 180
 if orientation == ft.DeviceOrientation.LANDSCAPE_LEFT:
 return 270
 return 0

def image_rotation_radians() -> float:
 camera = state.selected_camera
 if camera is None:
 return 0.0
 sensor_orientation = camera.sensor_orientation
 device_degrees = device_orientation_degrees(state.device_orientation)
 if camera.lens_direction == fc.CameraLensDirection.FRONT:
 rotation_degrees = (sensor_orientation + device_degrees) % 360
 else:
 rotation_degrees = (sensor_orientation - device_degrees + 360) % 360

```

```

 return rotation_degrees * pi / 180

def apply_last_image_transform(src_width: int | None, src_height: int | None):
 _ = (src_width, src_height)
 last_image.width = image_frame_width
 last_image.height = image_frame_height

 if page.platform != ft.PagePlatform.ANDROID:
 last_image.offset = None
 last_image.rotate = None
 return

 angle = image_rotation_radians()
 width = float(image_frame_width)
 height = float(image_frame_height)
 calc_angle = -angle

 # Flet uses clockwise radians for Rotate.angle.
 def rotate_point(x: float, y: float) -> tuple[float, float]:
 return (
 x * cos(calc_angle) + y * sin(calc_angle),
 -x * sin(calc_angle) + y * cos(calc_angle),
)

 corners = [
 rotate_point(0.0, 0.0),
 rotate_point(width, 0.0),
 rotate_point(0.0, height),
 rotate_point(width, height),
]
 min_x = min(p[0] for p in corners)
 min_y = min(p[1] for p in corners)

 # Control.offset is normalized by control width/height.
 last_image.offset = ft.Offset(
 x=(-min_x / width) if width else 0.0,
 y=(-min_y / height) if height else 0.0,
)
 last_image.rotate = ft.Rotate(angle=angle, alignment=ft.Alignment.TOP_LEFT)

async def get_cameras():
 state.cameras = await preview.get_available_cameras()
 state.camera_labels.clear()
 seen_labels: dict[str, int] = {}
 for camera in state.cameras:
 label = camera_label(camera)
 seen_labels[label] = seen_labels.get(label, 0) + 1
 if seen_labels[label] > 1:
 label = f"{label} {seen_labels[label]}"
 state.camera_labels[camera.name] = label

 selector.options = [
 ft.DropdownOption(key=c.name, text=state.camera_labels[c.name])
 for c in state.cameras
]
 if selector.value and selector.value not in state.camera_labels:
 selector.value = None
 state.selected_camera = None
 state.is_initialized = False
 state.is_streaming = False
 state.is_recording = False
 state.is_recording_paused = False
 state.is_preview_paused = False
 state.is_streaming_supported = False
 state.device_orientation = None
 sync_action_buttons()
 status.value = "Select a camera"
 page.update()

def sync_action_buttons():
 take_photo_btn.disabled = not state.is_initialized
 record_btn.disabled = not state.is_initialized
 record_btn.selected = state.is_recording
 pause_recording_btn.selected = state.is_recording_paused
 pause_recording_btn.disabled = not state.is_recording
 stream_btn.selected = state.is_streaming
 stream_btn.disabled = not (
 state.is_initialized and state.is_streaming_supported
)
 preview_btn.selected = not state.is_preview_paused
 preview_btn.disabled = not state.is_initialized

async def init_camera(e=None):
 state.selected_camera = next(

```

## Description

A control that provides camera preview and capture capabilities.

Inherits: `LayoutControl`

## Properties

- `content` - Optional child to overlay on top of the camera preview.
- `preview_enabled` - Whether the preview surface is shown.

## Events

- `on_state_change` - Fires when the camera controller state changes.
- `on_stream_image` - Fires when an image frame is available while streaming.

## Methods

- `get_available_cameras` - Lists the available camera devices on the current platform.
- `get_exposure_offset_step_size` - Returns the smallest increment supported for exposure offset changes.
- `get_max_exposure_offset` - Maximum exposure offset supported by the current camera.
- `get_max_zoom_level` - Maximum zoom level supported by the current camera.
- `get_min_exposure_offset` - Minimum exposure offset supported by the current camera.
- `get_min_zoom_level` - Minimum zoom level supported by the current camera.
- `initialize` - Initializes a new camera controller for the given description.
- `lock_capture_orientation` - Locks capture orientation to the specified device orientation.
- `pause_preview` - Pauses the camera preview.
- `pause_video_recording` - Pauses an active video recording.
- `prepare_for_video_recording` - Prepares the capture session for video recording.
- `resume_preview` - Resumes the camera preview.
- `resume_video_recording` - Resumes a paused video recording.
- `set_description` - Switches to another camera description.
- `set_exposure_mode` - Changes the exposure mode.
- `set_exposure_offset` - Sets exposure offset in EV units.
- `set_exposure_point` - Sets the exposure metering point.
- `set_flash_mode` - Changes the flash mode.
- `set_focus_mode` - Changes the focus mode.
- `set_focus_point` - Sets the focus metering point.
- `set_zoom_level` - Applies the provided zoom level.
- `start_image_stream` - Begins streaming camera image frames.
- `start_video_recording` - Starts capturing video.
- `stop_image_stream` - Stops streaming camera image frames.
- `stop_video_recording` - Stops video recording and returns the recorded bytes.
- `supports_image_streaming` - Indicates whether image streaming is supported on the current platform.
- `take_picture` - Captures a still image and returns the encoded bytes.
- `unlock_capture_orientation` - Unlocks the capture orientation.

## Properties

`buildcontentclass-attributeinstance-attribute`

`content: Control | None = None`

Optional child to overlay on top of the camera preview.

### **buildpreview\_enabled**class-attributeinstance-attribute

```
preview_enabled: bool = True
```

Whether the preview surface is shown.

## **Events**

### **bolton\_state\_change**class-attributeinstance-attribute

```
on_state_change: (
 EventHandler[CameraStateEvent] | None
) = None
```

Fires when the camera controller state changes.

### **bolton\_stream\_image**class-attributeinstance-attribute

```
on_stream_image: (
 EventHandler[CameraImageEvent] | None
) = None
```

Fires when an image frame is available while streaming.

## **Methods**

### **deployed\_code**get\_available\_camerasasync

```
get_available_cameras() -> list[CameraDescription]
```

Lists the available camera devices on the current platform.

Returns:

- list[CameraDescription] - A list of available camera descriptions.

### **deployed\_code**get\_exposure\_offset\_step\_sizeasync

```
get_exposure_offset_step_size() -> float
```

Returns the smallest increment supported for exposure offset changes.

### **deployed\_code**get\_max\_exposure\_offsetasync

```
get_max_exposure_offset() -> float
```

Maximum exposure offset supported by the current camera.

### **deployed\_code**get\_max\_zoom\_levelasync

```
get_max_zoom_level() -> float
```

Maximum zoom level supported by the current camera.

### **deployed\_code**get\_min\_exposure\_offsetasync

```
get_min_exposure_offset() -> float
```

Minimum exposure offset supported by the current camera.

### **deployed\_code**get\_min\_zoom\_levelasync

```
get_min_zoom_level() -> float
```

Minimum zoom level supported by the current camera.

### **deployed\_code**initializeasync

```
initialize(
 description: CameraDescription,
 resolution_preset: ResolutionPreset,
 enable_audio: bool = True,
 fps: int | None = None,
 video_bitrate: int | None = None,
 audio_bitrate: int | None = None,
 image_format_group: ImageFormatGroup | None = None,
)
```

Initializes a new camera controller for the given description.

Parameters:

- `description` (`CameraDescription`) - Camera device to bind to.
- `resolution_preset` (`ResolutionPreset`) - Desired resolution preset.
- `enable_audio` (bool, default: True) - Whether audio is enabled for recordings.
- `fps` (int | None, default: None) - Optional target frames per second.
- `video_bitrate` (int | None, default: None) - Optional video bitrate.
- `audio_bitrate` (int | None, default: None) - Optional audio bitrate.
- `image_format_group` (`ImageFormatGroup` | None, default: None) - Optional image format group override.

### `deployed_codelock_capture_orientationasync`

```
lock_capture_orientation(
 orientation: DeviceOrientation | None = None,
)
```

Locks capture orientation to the specified device orientation.

Parameters:

- `orientation` (`DeviceOrientation` | None, default: None) - Specific orientation to lock, or current device orientation if None.

### `deployed_codepause_previewasync`

```
pause_preview()
```

Pauses the camera preview.

### `deployed_codepause_video_recordingasync`

```
pause_video_recording()
```

Pauses an active video recording.

### `deployed_codeprepare_for_video_recordingasync`

```
prepare_for_video_recording()
```

Prepares the capture session for video recording.

### `deployed_coderesume_previewasync`

```
resume_preview()
```

Resumes the camera preview.

### `deployed_coderesume_video_recordingasync`

```
resume_video_recording()
```

Resumes a paused video recording.

### `deployed_codeset_descriptionasync`

```
set_description(description: CameraDescription)
```

Switches to another camera description.

Parameters:

- `description` (`CameraDescription`) - Camera to switch to.

### `deployed_codeset_exposure_modeasync`

```
set_exposure_mode(mode: ExposureMode)
```

Changes the exposure mode.

Parameters:

- `mode` (`ExposureMode`) - Exposure mode to apply.

### `deployed_codeset_exposure_offsetasync`

```
set_exposure_offset(offset: float) -> float
```

Sets exposure offset in EV units.

Parameters:

- `offset` (float) - Exposure offset to apply.

Returns:

- float - The offset value that was set.

### **deployed\_codeset\_exposure\_pointasync**

```
set_exposure_point(point: OffsetValue | None)
```

Sets the exposure metering point.

Parameters:

- point (OffsetValue | None) - Normalized offset (0..1) or None to reset.

### **deployed\_codeset\_flash\_modeasync**

```
set_flash_mode(mode: FlashMode)
```

Changes the flash mode.

Parameters:

- mode (FlashMode) - Flash mode to apply.

### **deployed\_codeset\_focus\_modeasync**

```
set_focus_mode(mode: FocusMode)
```

Changes the focus mode.

Parameters:

- mode (FocusMode) - Focus mode to apply.

### **deployed\_codeset\_focus\_pointasync**

```
set_focus_point(point: OffsetValue | None)
```

Sets the focus metering point.

Parameters:

- point (OffsetValue | None) - Normalized offset (0..1) or None to reset.

### **deployed\_codeset\_zoom\_levelasync**

```
set_zoom_level(zoom: float)
```

Applies the provided zoom level.

Parameters:

- zoom (float) - Zoom level to set.

### **deployed\_codestart\_image\_streamasync**

```
start_image_stream()
```

Begins streaming camera image frames.

### **deployed\_codestart\_video\_recordingasync**

```
start_video_recording()
```

Starts capturing video.

### **deployed\_codestop\_image\_streamasync**

```
stop_image_stream()
```

Stops streaming camera image frames.

### **deployed\_codestop\_video\_recordingasync**

```
stop_video_recording() -> bytes
```

Stops video recording and returns the recorded bytes.

Returns:

- bytes - Encoded video file bytes.

### **deployed\_codesupports\_image\_streamingasync**

```
supports_image_streaming() -> bool
```

Indicates whether image streaming is supported on the current platform.

### **deployed\_codetake\_pictureasync**

`take_picture()` -> bytes

Captures a still image and returns the encoded bytes.

Returns:

- bytes - Encoded image bytes.

### **deployed\_codeunlock\_capture\_orientationasync**

`unlock_capture_orientation()`

Unlocks the capture orientation.

# Canvas

Canvas is a control for drawing arbitrary graphics using a set of primitives or "shapes" such as line, arc, path and text.

Example:

```
cv.Canvas(
 width=160,
 height=160,
 shapes=[
 cv.Rect(
 0,
 0,
 160,
 160,
 paint=ft.Paint(
 color=ft.Colors.BLUE_100,
 style=ft.PaintingStyle.FILL,
),
),
 cv.Circle(
 80,
 80,
 50,
 paint=ft.Paint(
 color=ft.Colors.BLUE_400,
 style=ft.PaintingStyle.FILL,
),
),
],
)
```

Image: Canvas - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/canvas/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/canvas/image_for_docs.png)

## Canvas

Inherits: `LayoutControl`

## Properties

- `content` - The content of this canvas.
- `resize_interval` - Sampling interval in milliseconds for `on_resize` event.
- `shapes` - A list of shapes to draw on this canvas.

## Events

- `on_resize` - Called when the size of this canvas has changed.

## Methods

- `capture` - Captures the current visual state of the canvas.
- `clear_capture` - Clears the previously captured canvas image.
- `get_capture` - Retrieves the most recent canvas capture as PNG bytes.

# Examples

## Smiling face

`play_arrow`Try Online

```
import math

import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 stroke_paint = ft.Paint(stroke_width=2, style=ft.PaintingStyle.STROKE)
 fill_paint = ft.Paint(style=ft.PaintingStyle.FILL)

 page.add(
 ft.SafeArea(
 content=cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
```



```

 cv.Circle(x=100, y=100, radius=50, paint=stroke_paint),
 cv.Circle(x=80, y=90, radius=10, paint=stroke_paint),
 cv.Circle(x=84, y=87, radius=5, paint=fill_paint),
 cv.Circle(x=120, y=90, radius=10, paint=stroke_paint),
 cv.Circle(x=124, y=87, radius=5, paint=fill_paint),
 cv.Arc(
 x=70,
 y=95,
 width=60,
 height=40,
 start_angle=0,
 sweep_angle=math.pi,
 paint=stroke_paint,
),
],
)
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: smiling-face - [https://flet.dev/docs/examples/controls/core/canvas/media/smiling\\_face.png](https://flet.dev/docs/examples/controls/core/canvas/media/smiling_face.png)

## Flet logo

play\_arrowTry Online

```

import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT

 page.add(
 ft.SafeArea(
 content=cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Path(
 elements=[
 cv.Path.MoveTo(x=25, y=125),
 cv.Path.QuadraticTo(cp1x=50, cp1y=25, x=135, y=35, w=0.35),
 cv.Path.QuadraticTo(cp1x=75, cp1y=115, x=135, y=215, w=0.6),
 cv.Path.QuadraticTo(cp1x=50, cp1y=225, x=25, y=125, w=0.35),
],
 paint=ft.Paint(
 stroke_width=2,
 style=ft.PaintingStyle.FILL,
 color=ft.Colors.PINK_400,
),
),
 cv.Path(
 elements=[
 cv.Path.MoveTo(x=85, y=125),
 cv.Path.QuadraticTo(cp1x=120, cp1y=85, x=165, y=75, w=0.5),
 cv.Path.QuadraticTo(
 cp1x=120, cp1y=115, x=165, y=175, w=0.3
),
 cv.Path.QuadraticTo(cp1x=120, cp1y=165, x=85, y=125, w=0.5),
],
 paint=ft.Paint(
 stroke_width=2,
 style=ft.PaintingStyle.FILL,
 color=ft.Colors.with_opacity(0.5, ft.Colors.BLUE_400),
),
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: flet-logo - [https://flet.dev/docs/examples/controls/core/canvas/media/flet\\_logo.png](https://flet.dev/docs/examples/controls/core/canvas/media/flet_logo.png)

## Triangles

play\_arrowTry Online

```

import flet as ft
import flet.canvas as cv

```

```

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Path(
 paint=ft.Paint(style=ft.PaintingStyle.FILL),
 elements=[
 cv.Path.MoveTo(x=25, y=25),
 cv.Path.LineTo(x=105, y=25),
 cv.Path.LineTo(x=25, y=105),
],
),
 cv.Path(
 paint=ft.Paint(stroke_width=2, style=ft.PaintingStyle.STROKE),
 elements=[
 cv.Path.MoveTo(x=125, y=125),
 cv.Path.LineTo(x=125, y=45),
 cv.Path.LineTo(x=45, y=125),
 cv.Path.Close(),
],
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: triangles - <https://flet.dev/docs/examples/controls/core/canvas/media/triangles.png>

## Bezier curves

play\_arrowTry Online

```

import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 cp = cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Path(
 paint=ft.Paint(stroke_width=2, style=ft.PaintingStyle.STROKE),
 elements=[
 cv.Path.MoveTo(x=75, y=25),
 cv.Path.QuadraticTo(cp1x=25, cp1y=25, x=25, y=62.5),
 cv.Path.QuadraticTo(cp1x=25, cp1y=100, x=50, y=100),
 cv.Path.QuadraticTo(cp1x=50, cp1y=120, x=30, y=125),
 cv.Path.QuadraticTo(cp1x=60, cp1y=120, x=65, y=100),
 cv.Path.QuadraticTo(cp1x=125, cp1y=100, x=125, y=62.5),
 cv.Path.QuadraticTo(cp1x=125, cp1y=25, x=75, y=25),
],
),
 cv.Path(
 elements=[
 cv.Path.SubPath(
 x=100,
 y=100,
 elements=[
 cv.Path.MoveTo(x=75, y=40),
 cv.Path.CubicTo(
 cp1x=75, cp1y=37, cp2x=70, cp2y=25, x=50, y=25
),
 cv.Path.CubicTo(
 cp1x=20, cp1y=25, cp2x=20, cp2y=62.5, x=20, y=62.5
),
 cv.Path.CubicTo(
 cp1x=20, cp1y=80, cp2x=40, cp2y=102, x=75, y=120
),
 cv.Path.CubicTo(
 cp1x=110, cp1y=102, cp2x=130, cp2y=80, x=130, y=62.5
),
 cv.Path.CubicTo(
 cp1x=130, cp1y=62.5, cp2x=130, cp2y=25, x=100, y=25
),
 cv.Path.CubicTo(
 cp1x=85, cp1y=25, cp2x=75, cp2y=37, x=75, y=40
),
],
),
],
),
],
)

```

```

],
paint=ft.Paint(
 style=ft.PaintingStyle.FILL,
 gradient=ft.PaintRadialGradient(
 center=ft.Offset(150, 150),
 radius=50,
 colors=[ft.Colors.PINK_100, ft.Colors.PINK],
),
),
),
],
)

```

```
page.add(ft.SafeArea(content=cp))
```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: bezier-curves - [https://flet.dev/docs/examples/controls/core/canvas/media/bezier\\_curves.png](https://flet.dev/docs/examples/controls/core/canvas/media/bezier_curves.png)

## Canvas text

play\_arrowTry Online

```

import math

import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Text(x=0, y=0, value="Just a text"),
 cv.Circle(
 x=200, y=100, radius=2, paint=ft.Paint(color=ft.Colors.RED)
),
 cv.Text(
 x=200,
 y=100,
 style=ft.TextStyle(weight=ft.FontWeight.BOLD, size=30),
 alignment=ft.Alignment.TOP_CENTER,
 rotate=math.pi * 0.15,
 value="Rotated",
 spans=[
 ft.TextSpan(
 text="around top_center",
 style=ft.TextStyle(
 italic=True, color=ft.Colors.GREEN, size=20
),
),
],
),
 cv.Circle(
 x=400, y=100, radius=2, paint=ft.Paint(color=ft.Colors.RED)
),
 cv.Text(
 x=400,
 y=100,
 value="Rotated around top_left",
 style=ft.TextStyle(size=20),
 alignment=ft.Alignment.TOP_LEFT,
 rotate=math.pi * -0.15,
),
 cv.Circle(
 x=600, y=200, radius=2, paint=ft.Paint(color=ft.Colors.RED)
),
 cv.Text(
 x=600,
 y=200,
 value="Rotated around center",
 style=ft.TextStyle(size=20),
 alignment=ft.Alignment.CENTER,
 rotate=math.pi / 2,
),
 cv.Text(
 x=300,
 y=400,
 value=(
 "Limited to max_width and right-aligned.\n"
 "Lorem ipsum dolor sit amet, consectetur adipiscing elit, "
 "sed do eiusmod tempor incididunt ut labore et dolore "
)
),
],
),
),
)

```

```

 "magna aliqua. Ut enim ad minim veniam, quis nostrud "
 "exercitation "
 "ullamco laboris nisi ut aliquip ex ea commodo consequat."
),
 text_align=ft.TextAlign.RIGHT,
 max_width=400,
),
cv.Text(
 x=200,
 y=200,
 value="WOW!",
 style=ft.TextStyle(
 weight=ft.FontWeight.BOLD,
 size=100,
 foreground=ft.Paint(
 color=ft.Colors.PINK,
 stroke_width=6,
 style=ft.PaintingStyle.STROKE,
 stroke_join=ft.StrokeJoin.ROUND,
 stroke_cap=ft.StrokeCap.ROUND,
),
),
),
cv.Text(
 x=200,
 y=200,
 value="WOW!",
 style=ft.TextStyle(
 weight=ft.FontWeight.BOLD,
 size=100,
 foreground=ft.Paint(
 gradient=ft.PaintLinearGradient(
 begin=(200, 200),
 end=(300, 300),
 colors=[ft.Colors.YELLOW, ft.Colors.RED],
),
 stroke_join=ft.StrokeJoin.ROUND,
 stroke_cap=ft.StrokeCap.ROUND,
),
),
),
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: text - <https://flet.dev/docs/examples/controls/core/canvas/media/text.png>

## Brush

play\_arrowTry Online

```

from dataclasses import dataclass

import flet as ft
import flet.canvas as cv

MAX_SHAPES_PER_CAPTURE = 30

@dataclass
class State:
 x: float = 0
 y: float = 0
 shapes_count: int = 1

state = State()

def main(page: ft.Page):
 page.title = "Canvas Example"

 file_picker = ft.FilePicker()

 def handle_pan_start(e: ft.DragStartEvent):
 state.x = e.local_position.x
 state.y = e.local_position.y

 async def handle_pan_update(e: ft.DragUpdateEvent):
 # ft.context.disable_auto_update()
 canvas.shapes.append(
 cv.Line(

```

```

 x1=state.x,
 y1=state.y,
 x2=e.local_position.x,
 y2=e.local_position.y,
 paint=ft.Paint(stroke_width=3),
)
)
canvas.update()
state.shapes_count += 1

if state.shapes_count == MAX_SHAPES_PER_CAPTURE:
 await canvas.capture()
 canvas.shapes.clear()
 canvas.update()
 state.shapes_count = 0

state.x = e.local_position.x
state.y = e.local_position.y

canvas = cv.Canvas(
 expand=False,
 shapes=[
 cv.Fill(ft.Paint(color=ft.Colors.WHITE)),
],
 content=ft.GestureDetector(
 on_pan_start=handle_pan_start,
 on_pan_update=handle_pan_update,
 drag_interval=10,
),
)

async def save_image():
 await canvas.capture()
 capture = await canvas.get_capture()
 if capture:
 file_path = await file_picker.save_file(
 file_name="flet_picture.png", src_bytes=capture
)
 if file_path and page.platform in [
 ft.PagePlatform.MACOS,
 ft.PagePlatform.WINDOWS,
 ft.PagePlatform.LINUX,
]:
 with open(file_path, "wb") as f:
 f.write(capture)

page.add(
 ft.SafeArea(
 expand=True,
 content=ft.Column(
 controls=[
 ft.Button("Save image", on_click=save_image),
 ft.Container(
 border_radius=5,
 border=ft.Border.all(2),
 bgcolor=ft.Colors.WHITE,
 width=float("inf"),
 expand=True,
 content=canvas,
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Gradients

play\_arrowTry Online

```

import math

import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Rect(

```

```

 x=10,
 y=10,
 width=100,
 height=100,
 border_radius=5,
 paint=ft.Paint(
 style=ft.PaintingStyle.FILL,
 gradient=ft.PaintLinearGradient(
 begin=(0, 10),
 end=(100, 50),
 colors=[ft.Colors.BLUE, ft.Colors.YELLOW],
),
),
),
 cv.Circle(
 x=60,
 y=170,
 radius=50,
 paint=ft.Paint(
 style=ft.PaintingStyle.FILL,
 gradient=ft.PaintRadialGradient(
 radius=50,
 center=(60, 170),
 colors=[ft.Colors.YELLOW, ft.Colors.BLUE],
),
),
),
 cv.Path(
 elements=[
 cv.Path.Arc(
 x=10,
 y=230,
 width=100,
 height=100,
 start_angle=3 * math.pi / 4,
 sweep_angle=3 * math.pi / 2,
),
],
 paint=ft.Paint(
 stroke_width=15,
 stroke_join=ft.StrokeJoin.ROUND,
 style=ft.PaintingStyle.STROKE,
 gradient=ft.PaintSweepGradient(
 start_angle=0,
 end_angle=math.pi * 2,
 rotation=3 * math.pi / 4,
 center=(60, 280),
 colors=[ft.Colors.YELLOW, ft.Colors.PURPLE],
 color_stops=[0.0, 1.0],
),
),
),
],
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildcontentclass-attributeinstance-attribute

content: Control | None = None

The content of this canvas.

### buildresize\_intervalclass-attributeinstance-attribute

resize\_interval: Number = 10

Sampling interval in milliseconds for on\_resize event.

Setting to 0 calls on\_resize immediately on every change.

### buildshapesclass-attributeinstance-attribute

shapes: list[Shape] = field(default\_factory=list)

A list of shapes to draw on this canvas.

## Events

## **bolton\_resizeclass-attributeinstance-attribute**

`on_resize: EventHandler[CanvasResizeEvent] | None = None`

Called when the size of this canvas has changed.

## **Methods**

### **deployed\_codecaptureasync**

`capture(pixel_ratio: Number | None=None)`

Captures the current visual state of the canvas.

The captured image is stored internally and will be rendered as a background beneath all subsequently drawn shapes.

Parameters:

- `pixel_ratio` (Number | None, default: None) - The pixel density multiplier to use when rendering the capture. 1.0 means 1 device pixel per logical pixel (no scaling). Values greater than 1.0 produce higher-resolution captures. If None, the device's default pixel ratio is used.

### **deployed\_codeclear\_captureasync**

`clear_capture()`

Clears the previously captured canvas image.

After clearing, no background will be rendered from a prior capture.

### **deployed\_codeget\_captureasync**

`get_capture() -> bytes`

Retrieves the most recent canvas capture as PNG bytes.

Returns:

- `bytes` (bytes) - The captured image in PNG format, or an empty result
- `bytes` - if no capture has been made.

# Card

A material design card: a panel with slightly rounded corners and an elevation shadow.

```
ft.Card(
 shadow_color=ft.Colors.ON_SURFACE_VARIANT,
 content=ft.Container(
 width=400,
 padding=10,
 content=ft.ListTile(
 bgcolor=ft.Colors.GREY_400,
 leading=ft.Icon(ft.Icons.FOREST),
 title=ft.Text("Card Name"),
),
),
)
```

Image: Card - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/card/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/card/image_for_docs.png)

## Card

Inherits: `LayoutControl`, `AdaptiveControl`

## Properties

- `bgcolor` - The card's background color.
- `clip_behavior` - Defines how the content will be clipped.
- `content` - The Control to display inside the card.
- `elevation` - The z-coordinate at which to place this card.
- `semantic_container` - Whether this card represents a single semantic container, or if it instead represents a collection of individual semantic nodes (different types of content).
- `shadow_color` - The color to paint the shadow below this card.
- `shape` - The shape of this card.
- `show_border_on_foreground` - Whether the shape of the border should be painted in front of the content or behind.
- `variant` - Defines the card variant to be used.

## Examples

### Music info

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Card Example"
 page.theme_mode = ft.ThemeMode.LIGHT

 page.add(
 ft.SafeArea(
 content=ft.Card(
 shadow_color=ft.Colors.ON_SURFACE_VARIANT,
 content=ft.Container(
 width=400,
 padding=10,
 content=ft.Column(
 controls=[
 ft.ListTile(
 bgcolor=ft.Colors.GREY_400,
 leading=ft.Icon(ft.Icons.ALBUM),
 title=ft.Text("The Enchanted Nightingale"),
 subtitle=ft.Text(
 "Music by Julie Gable. Lyrics by Sidney Stein."
),
),
],
),
 ft.Row(
 alignment=ft.MainAxisAlignment.END,
 controls=[
 ft.TextButton("Buy tickets"),
 ft.TextButton("Listen"),
],
),
),
),
),
)
```



```

),
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: music-info - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/card/music\\_info.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/card/music_info.png)

## Properties

### buildbgcolorclass-attributeinstance-attribute

bgcolor: `ColorValue` | `None` = `None`

The card's background color.

### buildclip\_behaviorclass-attributeinstance-attribute

clip\_behavior: `ClipBehavior` | `None` = `None`

Defines how the content will be clipped.

Defaults to `CardTheme.clip_behavior`, or if that is `None`, falls back to `ClipBehavior.NONE`.

### buildcontentclass-attributeinstance-attribute

content: `Control` | `None` = `None`

The `Control` to display inside the card.

Tip

To display multiple children, wrap them in a control like `Row`, `Column`, or `Stack`, which accept a controls list.

### buildelevationclass-attributeinstance-attribute

elevation: `Number` | `None` = `None`

The z-coordinate at which to place this card. Defines the size of the shadow below the card.

Defaults to `CardTheme.elevation`, or if that is `None`, falls back to `1.0`.

### buildsemantic\_containerclass-attributeinstance-attribute

semantic\_container: `bool` = `True`

Whether this card represents a single semantic container, or if it instead represents a collection of individual semantic nodes (different types of content).

### buildshadow\_colorclass-attributeinstance-attribute

shadow\_color: `ColorValue` | `None` = `None`

The color to paint the shadow below this card.

Defaults to `CardTheme.shadow_color`

### buildshapeclass-attributeinstance-attribute

shape: `OutlinedBorder` | `None` = `None`

The shape of this card.

Defaults to `CardTheme.shape`, or if that is `None`, falls back to `RoundedRectangleBorder(radius=12.0)`.

### buildshow\_border\_on\_foregroundclass-attributeinstance-attribute

show\_border\_on\_foreground: `bool` = `True`

Whether the shape of the border should be painted in front of the content or behind.

### buildvariantclass-attributeinstance-attribute

variant: `CardVariant` = `CardVariant.ELEVATED`

Defines the card variant to be used.

# Charts

Interactive chart controls powered by flet-charts let you display data as bar, line, pie, scatter and plotly visualisations directly in your Flet apps.

It is built on top of the fl\_chart Flutter package and ships with helper classes for axis labels, tooltips and more.

## Platform Support

Platform	Windows	macOS	Linux
Supported	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>

## Usage

Add flet-charts to your project dependencies:

- `uv`
- `pip`  
`uv add flet-charts`  
`pip install flet-charts`

## Available Charts

- `BarChart`
- `CandlestickChart`
- `LineChart`
- `MatplotlibChart`
- `PieChart`
- `PlotlyChart`
- `RadarChart`
- `ScatterChart`

# Checkbox

Checkbox allows to select one or more items from a group, or switch between two mutually exclusive options (checked or unchecked, on or off).

```
ft.Checkbox()
ft.Checkbox(label="Checked", value=True)
ft.Checkbox(label="Disabled", disabled=True)
```

Image: Checkbox - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/checkbox/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/checkbox/image_for_docs.png)

Checkbox

Inherits: `LayoutControl`, `AdaptiveControl`

## Properties

- `active_color` - The color to use when this checkbox is checked.
- `autofocus` - True if the control will be selected as the initial focus.
- `border_side` - The color and width of this checkbox's border in all or specific `ControlStates`.
- `check_color` - The color to use for the check icon when this checkbox is checked.
- `error` - Whether this checkbox wants to show an error state.
- `fill_color` - The color that fills this checkbox in all or specific `ControlStates`.
- `focus_color` - The color for the checkbox's Material when it has the input focus.
- `hover_color` - The color to use when this checkbox is hovered.
- `label` - The clickable label to display on the right of a checkbox.
- `label_position` - Defines on which side of the checkbox the label should be shown.
- `label_style` - The label's text style.
- `mouse_cursor` - The cursor to be displayed when a mouse pointer enters or is hovering over this checkbox.
- `overlay_color` - The color of this checkbox's overlay in various `ControlState` states.
- `semantics_label` - The semantic label for the checkbox that is not shown in the UI, but will be announced by screen readers in accessibility modes (e.g TalkBack/VoiceOver).
- `shape` - The shape of the checkbox.
- `splash_radius` - The radius of the circular Material ink response (ripple) in logical pixels.
- `tristate` - If True the checkbox's value can be True, False, or None.
- `value` - The value of this checkbox.
- `visual_density` - Defines how compact the checkbox's layout will be.

## Events

- `on_blur` - Called when this checkbox has lost focus.
- `on_change` - Called when the state of this checkbox is changed.
- `on_focus` - Called when this checkbox has received focus.

## Examples

### Checkbox

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 def handle_button_click(e: ft.Event[ft.Button]):
 message.value = (
 f"Checkboxes values are: {c1.value}, {c2.value}, {c3.value}, "
 f"{c4.value}, {c5.value}."
)
 page.add(ft.Button(label="Click", on_click=handle_button_click))
 page.add(ft.Checkbox(label="c1", value=True))
 page.add(ft.Checkbox(label="c2", value=False))
 page.add(ft.Checkbox(label="c3", value=None))
 page.add(ft.Checkbox(label="c4", value=False))
 page.add(ft.Checkbox(label="c5", value=True))
 page.add(message)
```

```

)
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 c1 := ft.Checkbox(
 label="Unchecked by default checkbox", value=False
),
 c2 := ft.Checkbox(
 label="Undefined by default tristate checkbox", tristate=True
),
 c3 := ft.Checkbox(label="Checked by default checkbox", value=True),
 c4 := ft.Checkbox(label="Disabled checkbox", disabled=True),
 c5 := ft.Checkbox(
 label="Checkbox with LEFT label_position",
 label_position=ft.LabelPosition.LEFT,
),
 ft.Button(content="Submit", on_click=handle_button_click),
 message := ft.Text(),
]
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: basic - <https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/checkbox/basic.png>

After clicking Submit

## Checkbox events

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 events = ft.Column()

 def handle_checkbox_change(e: ft.Event[ft.Checkbox]):
 events.controls.append(ft.Text(f"Checkbox value changed to {e.control.value}"))

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Checkbox(
 label="Checkbox with 'change' event",
 on_change=handle_checkbox_change,
),
 events,
]
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: handling-events - [https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/checkbox/handling\\_events.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/checkbox/handling_events.png)

After three clicks

## Styled checkboxes

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Checkbox(label="Checkbox with default style"),
 ft.Checkbox(
 label="Checkbox with constant fill color",
 fill_color=ft.Colors.RED,
 check_color=ft.Colors.YELLOW,
),
 ft.Row(

```

```

 controls=[
 ft.Checkbox(
 key="dynamic_fill_checkbox",
 fill_color={
 ft.ControlState.HOVERED: ft.Colors.BLUE,
 ft.ControlState.SELECTED: ft.Colors.GREEN,
 ft.ControlState.DEFAULT: ft.Colors.RED,
 },
),
 ft.Text("Checkbox with dynamic fill color"),
],
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Styled checkboxes - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/checkbox/styled\\_checkboxes\\_selected.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/checkbox/styled_checkboxes_selected.png)

## Properties

### buildactive\_colorclass-attributeinstance-attribute

active\_color: ColorValue | None = None

The color to use when this checkbox is checked.

### buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

True if the control will be selected as the initial focus. If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

### buildborder\_sideclass-attributeinstance-attribute

border\_side: ControlStateValue[BorderSide] | None = None

The color and width of this checkbox's border in all or specific ControlStates.

Defaults to CheckboxTheme.border\_side, or if that is None, falls back to BorderSide with a width of 2.0.

#### Note

Supported states: ControlState.SELECTED, ControlState.HOVERED, ControlState.DISABLED, ControlState.FOCUSED, ControlState.PRESSED, ControlState.ERROR, and ControlState.DEFAULT.

### buildcheck\_colorclass-attributeinstance-attribute

check\_color: ColorValue | None = None

The color to use for the check icon when this checkbox is checked.

### builderrorclass-attributeinstance-attribute

error: bool = False

Whether this checkbox wants to show an error state.

If True this checkbox will have a different default container color and check color.

### buildfill\_colorclass-attributeinstance-attribute

fill\_color: ControlStateValue[ColorValue] | None = None

The color that fills this checkbox in all or specific ControlStates.

#### Note

Supported states: ControlState.SELECTED, ControlState.HOVERED, ControlState.DISABLED, ControlState.FOCUSED, and ControlState.DEFAULT.

### buildfocus\_colorclass-attributeinstance-attribute

focus\_color: ColorValue | None = None

The color for the checkbox's Material when it has the input focus. If overlay\_color returns a non-None color in the ControlState.FOCUSED state, it will be used instead.

Defaults to `CheckboxTheme.overlay_color` in the `ControlState.FOCUSED` state, or if that is `None`, falls back to `Theme.focus_color`.

### **buildhover\_colorclass-attributeinstance-attribute**

`hover_color: ColorValue | None = None`

The color to use when this checkbox is hovered.

### **buildlabelclass-attributeinstance-attribute**

`label: StrOrControl | None = None`

The clickable label to display on the right of a checkbox.

### **buildlabel\_positionclass-attributeinstance-attribute**

`label_position: LabelPosition = LabelPosition.RIGHT`

Defines on which side of the checkbox the label should be shown.

### **buildlabel\_styleclass-attributeinstance-attribute**

`label_style: TextStyle | None = None`

The label's text style.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: MouseCursor | None = None`

The cursor to be displayed when a mouse pointer enters or is hovering over this checkbox.

Defaults to `CheckboxTheme.mouse_cursor`, or if that is `None`, falls back to `MouseCursor.CLICK`.

### **buildoverlay\_colorclass-attributeinstance-attribute**

`overlay_color: ControlStateValue[ColorValue] | None = None`

The color of this checkbox's overlay in various `ControlState` states.

Note

Supported states: `ControlState.PRESSED`, `ControlState.SELECTED`, `ControlState.HOVERED`, `ControlState.FOCUSED`, and `ControlState.DEFAULT`.

### **buildsemantics\_labelclass-attributeinstance-attribute**

`semantics_label: str | None = None`

The semantic label for the checkbox that is not shown in the UI, but will be announced by screen readers in accessibility modes (e.g `TalkBack/VoiceOver`).

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

The shape of the checkbox.

Defaults to `CheckboxTheme.shape`, or if that is `None`, falls back to `RoundedRectangleBorder(radius=2)`.

### **buildsplash\_radiusclass-attributeinstance-attribute**

`splash_radius: Number | None = None`

The radius of the circular Material ink response (ripple) in logical pixels.

Defaults to `CheckboxTheme.splash_radius`, or if that is `None`, falls back to `20.0`.

### **buildtristateclass-attributeinstance-attribute**

`tristate: bool = False`

If `True` the checkbox's value can be `True`, `False`, or `None`.

### **buildvalueclass-attributeinstance-attribute**

`value: bool | None = False`

The value of this checkbox.

- If `True`, this checkbox is checked.
- If `False`, this checkbox is unchecked.

- If None and tristate is True, this checkbox is indeterminate (displayed as a dash).

## **buildvisual\_densityclass-attributeinstance-attribute**

`visual_density: VisualDensity | None = None`

Defines how compact the checkbox's layout will be.

## **Events**

### **bolton\_blurclass-attributeinstance-attribute**

`on_blur: ControlEventHandler[Checkbox] | None = None`

Called when this checkbox has lost focus.

### **bolton\_changeclass-attributeinstance-attribute**

`on_change: ControlEventHandler[Checkbox] | None = None`

Called when the state of this checkbox is changed.

### **bolton\_focusclass-attributeinstance-attribute**

`on_focus: ControlEventHandler[Checkbox] | None = None`

Called when this checkbox has received focus.

# Chip

Chips are compact elements that represent an attribute, text, entity, or action.

Example:

```
ft.Chip(
 label="Explore topics",
 leading=ft.Icon(ft.Icons.EXPLORE_OUTLINED),
)
```

Image: Chip - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/chip/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/chip/image_for_docs.png)

Chip

Inherits: `LayoutControl`

## Properties

- `autofocus` - Whether this chip will be selected as the initial focus.
- `bgcolor` - Color to be used for the unselected, enabled chip's background.
- `border_side` - Defines the color and weight of this chip's outline.
- `check_color` - The color of this chip's check mark when a check mark is visible.
- `clip_behavior` - The content will be clipped (or not) according to this option.
- `color` - The color that fills this chip in various `ControlState`.
- `delete_drawer_animation_style` - The animation style for the `delete_icon`'s animations.
- `delete_icon` - A `Control` to display to the right of this chip's label in case `on_delete` event is specified.
- `delete_icon_color` - The color of the `delete_icon`.
- `delete_icon_size_constraints` - The size constraints for the `delete_icon` control.
- `delete_icon_tooltip` - The text to be used for this chip's `delete_icon` tooltip.
- `disabled_color` - The color used for this chip's background if it is disabled.
- `elevation` - A non-negative value which defines the size of the shadow below this chip.
- `elevation_on_click` - The elevation to be applied on this chip relative to its parent during the press motion.
- `enable_animation_style` - The animation style for the enable and disable animations.
- `label` - The primary content of this chip.
- `label_padding` - The padding around the label.
- `label_text_style` - The style to be applied to this chip's label.
- `leading` - A `Control` to display to the left of this chip's label.
- `leading_drawer_animation_style` - The animation style for the leading control's animations.
- `leading_size_constraints` - The size constraints for the leading control.
- `padding` - The padding between the label and the outside shape.
- `select_animation_style` - The animation style for the select and unselect animations.
- `selected` - If `on_select` event is specified, `selected` property is used to determine whether this chip is selected or not.
- `selected_color` - The color used for this chip's background when it is selected.
- `selected_shadow_color` - The color used for this chip's background when the elevation is greater than 0 and this chip is selected.
- `shadow_color` - The color used for this chip's background when the elevation is greater than 0 and this chip is not selected.
- `shape` - The shape of the border around this chip.



- `show_checkmark` - If `on_select` event is specified and chip is selected, `show_checkmark` is used to determine whether or not to show a checkmark.
- `visual_density` - TBD

## Events

- `on_blur` - Called when this chip has lost focus.
- `on_click` - Called when the user clicks on this chip.
- `on_delete` - Called when the user clicks on the `delete_icon`.
- `on_focus` - Called when this chip has received focus.
- `on_select` - Called when the user clicks on this chip.

## Examples

### Assist chips

Assist chips are chips with leading icon and `on_click` event specified.

They represent smart or automated actions that appear dynamically and contextually in a UI.

An alternative to assist chips are buttons, which should appear persistently and consistently.

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 url_launcher = ft.UrlLauncher()
 page.services.append(url_launcher)

 def handle_chip1_click(e: ft.Event[ft.Chip]):
 e.control.label.value = "Saved to favorites"
 e.control.leading = ft.Icon(ft.Icons.FAVORITE_OUTLINED)
 e.control.disabled = True

 async def handle_chip2_click(e: ft.Event[ft.Chip]):
 await url_launcher.launch_url("https://maps.google.com")

 page.add(
 ft.SafeArea(
 content=ft.Row(
 controls=[
 ft.Chip(
 label=ft.Text("Save to favourites"),
 leading=ft.Icon(ft.Icons.FAVORITE_BORDER_OUTLINED),
 bgcolor=ft.Colors.GREEN_200,
 disabled_color=ft.Colors.GREEN_100,
 autofocus=True,
 on_click=handle_chip1_click,
),
 ft.Chip(
 label=ft.Text("9 min walk"),
 leading=ft.Icon(ft.Icons.MAP_SHARP),
 bgcolor=ft.Colors.GREEN_200,
 on_click=handle_chip2_click,
),
]
)
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: assist-chips - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/chip/assist\\_chips.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/chip/assist_chips.png)

### Filter chips

Filter chips are chips with `on_select` event specified.

They use tags or descriptive words provided in the label to filter content. They can be a good alternative to switches or checkboxes.

[play\\_arrowTry Online](#)

```
import flet as ft
```

```
def main(page: ft.Page):
 def handle_amenity_selection(e: ft.Event[ft.Chip]):
 print("Amenity selected:", e.control.label.value)

 amenities = ["Washer / Dryer", "Ramp access", "Dogs OK", "Cats OK", "Smoke-free"]

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Row(
 controls=[
 ft.Icon(ft.Icons.HOTEL_CLASS),
 ft.Text("Amenities"),
]
),
 ft.Row(
 controls=[
 ft.Chip(
 label=ft.Text(amenity),
 bgcolor=ft.Colors.GREEN_200,
 disabled_color=ft.Colors.GREEN_100,
 autofocus=True,
 on_select=handle_amenity_selection,
)
 for amenity in amenities
]
),
]
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: filter-chips - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/chip/filter\\_chips.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/chip/filter_chips.png)

## Properties

### buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

Whether this chip will be selected as the initial focus.

If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

### buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

Color to be used for the unselected, enabled chip's background.

### buildborder\_sideclass-attributeinstance-attribute

border\_side: BorderSide | None = None

Defines the color and weight of this chip's outline.

### buildcheck\_colorclass-attributeinstance-attribute

check\_color: ColorValue | None = None

The color of this chip's check mark when a check mark is visible.

### buildclip\_behaviorclass-attributeinstance-attribute

clip\_behavior: ClipBehavior = ClipBehavior.NONE

The content will be clipped (or not) according to this option.

### buildcolorclass-attributeinstance-attribute

color: ControlStateValue[ColorValue] | None = None

The color that fills this chip in various ControlState.

### builddelete\_drawer\_animation\_styleclass-attributeinstance-attribute

delete\_drawer\_animation\_style: AnimationStyle | None = None

The animation style for the delete\_icon's animations.

## **builddelete\_iconclass-attributeinstance-attribute**

`delete_icon: Control | None = None`

A Control to display to the right of this chip's label in case on\_delete event is specified.

## **builddelete\_icon\_colorclass-attributeinstance-attribute**

`delete_icon_color: ColorValue | None = None`

The color of the delete\_icon.

## **builddelete\_icon\_size\_constraintsclass-attributeinstance-attribute**

`delete_icon_size_constraints: BoxConstraints | None = None`

The size constraints for the delete\_icon control.

If None, it defaults to a minimum size of chip height or label height (whichever is greater) and a padding of 8.0 pixels on all sides.

## **builddelete\_icon\_tooltipclass-attributeinstance-attribute**

`delete_icon_tooltip: str | None = None`

The text to be used for this chip's delete\_icon tooltip. If not provided or provided with an empty string, the tooltip of the delete icon will not be displayed.

## **builddisabled\_colorclass-attributeinstance-attribute**

`disabled_color: ColorValue | None = None`

The color used for this chip's background if it is disabled.

## **buildelevationclass-attributeinstance-attribute**

`elevation: Number | None = None`

A non-negative value which defines the size of the shadow below this chip.

Defaults to 0.

Raises:

- ValueError - If elevation is negative.

## **buildelevation\_on\_clickclass-attributeinstance-attribute**

`elevation_on_click: Number | None = None`

The elevation to be applied on this chip relative to its parent during the press motion. This controls the size of the shadow below this chip.

Defaults to 8.0.

Raises:

- ValueError - If it is not greater than or equal to 0.0.

## **buildenable\_animation\_styleclass-attributeinstance-attribute**

`enable_animation_style: AnimationStyle | None = None`

The animation style for the enable and disable animations.

## **buildlabelinstance-attribute**

`label: StrOrControl`

The primary content of this chip.

Typically a Text control.

## **buildlabel\_paddingclass-attributeinstance-attribute**

`label_padding: PaddingValue | None = None`

The padding around the label.

By default, this is 4 logical pixels at the beginning and the end of the label, and zero on top and bottom.

## **buildlabel\_text\_styleclass-attributeinstance-attribute**

`label_text_style: TextStyle | None = None`

The style to be applied to this chip's label.

### **buildleadingclass-attributeinstance-attribute**

leading: Control | None = None

A Control to display to the left of this chip's label.

Typically the leading control is an Icon or a CircleAvatar.

### **buildleading\_drawer\_animation\_styleclass-attributeinstance-attribute**

leading\_drawer\_animation\_style: AnimationStyle | None = None

The animation style for the leading control's animations.

### **buildleading\_size\_constraintsclass-attributeinstance-attribute**

leading\_size\_constraints: BoxConstraints | None = None

The size constraints for the leading control.

If None, it defaults to a minimum size of chip height or label height (whichever is greater) and a padding of 8.0 pixels on all sides.

### **buildpaddingclass-attributeinstance-attribute**

padding: PaddingValue | None = None

The padding between the label and the outside shape.

Defaults to 8 logical pixels on all sides.

### **buildselect\_animation\_styleclass-attributeinstance-attribute**

select\_animation\_style: AnimationStyle | None = None

The animation style for the select and unselect animations.

### **buildselectedclass-attributeinstance-attribute**

selected: bool = False

If on\_select event is specified, selected property is used to determine whether this chip is selected or not.

### **buildselected\_colorclass-attributeinstance-attribute**

selected\_color: ColorValue | None = None

The color used for this chip's background when it is selected.

### **buildselected\_shadow\_colorclass-attributeinstance-attribute**

selected\_shadow\_color: ColorValue | None = None

The color used for this chip's background when the elevation is greater than 0 and this chip is selected.

### **buildshadow\_colorclass-attributeinstance-attribute**

shadow\_color: ColorValue | None = None

The color used for this chip's background when the elevation is greater than 0 and this chip is not selected.

### **buildshapeclass-attributeinstance-attribute**

shape: OutlinedBorder | None = None

The shape of the border around this chip.

Defaults to ChipTheme.shape, or if that is resolves to None, falls back to RoundedRectangleBorder(radius=8).

### **buildshow\_checkmarkclass-attributeinstance-attribute**

show\_checkmark: bool = True

If on\_select event is specified and chip is selected, show\_checkmark is used to determine whether or not to show a checkmark.

### **buildvisual\_densityclass-attributeinstance-attribute**

visual\_density: VisualDensity | None = None

TBD

## Events

### **bolton\_blurclass-attributeinstance-attribute**

`on_blur: ControlEventHandler[Chip] | None = None`

Called when this chip has lost focus.

### **bolton\_clickclass-attributeinstance-attribute**

`on_click: ControlEventHandler[Chip] | None = None`

Called when the user clicks on this chip.

Raises:

- `ValueError` - If specified together with `on_select`.

### **bolton\_deleteclass-attributeinstance-attribute**

`on_delete: ControlEventHandler[Chip] | None = None`

Called when the user clicks on the `delete_icon`.

### **bolton\_focusclass-attributeinstance-attribute**

`on_focus: ControlEventHandler[Chip] | None = None`

Called when this chip has received focus.

### **bolton\_selectclass-attributeinstance-attribute**

`on_select: ControlEventHandler[Chip] | None = None`

Called when the user clicks on this chip.

It internally changes `selected` property to the opposite value.

Raises:

- `ValueError` - If specified together with `on_click`.

# CircleAvatar

A circle that represents a user.

If `foreground_image_src` fails then `background_image_src` is used, and if this also fails, then `bgcolor` is used.

Example:

```
ft.CircleAvatar(
 content=ft.Text("AB"),
 bgcolor=ft.Colors.PRIMARY,
 color=ft.Colors.ON_PRIMARY,
)
```

Image: CircleAvatar - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/circle\\_avatar/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/circle_avatar/image_for_docs.png)

CircleAvatar

Inherits: `LayoutControl`

## Properties

- `background_image_src` - The source (local asset file or URL) of the background image in the circle.
- `bgcolor` - The color with which to fill the circle.
- `color` - The default color for text in this avatar.
- `content` - The content of this avatar.
- `foreground_image_src` - The source (local asset file or URL) of the foreground image in the circle.
- `max_radius` - The maximum size of the avatar, expressed as the radius (half the diameter).
- `min_radius` - The minimum size of the avatar, expressed as the radius (half the diameter).
- `radius` - The size of the avatar, expressed as the radius (half the diameter).

## Events

- `on_image_error` - Called when an error occurs while loading the `background_image_src` or `foreground_image_src`.

## Examples

### User avatars

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 # a "normal" avatar with background image
 ft.CircleAvatar(
 foreground_image_src="https://avatars.githubusercontent.com/u/5041459?s=88&v=4",
 content=ft.Text("FF"),
),
 # avatar with failing foreground image and fallback text
 ft.CircleAvatar(
 foreground_image_src="https://avatars.githubusercontent.com/u/_5041459?s=88&v=4",
 content=ft.Text("FF"),
),
 # avatar with icon, aka icon with inverse background
 ft.CircleAvatar(content=ft.Icon(ft.Icons.ABC)),
 # avatar with icon and custom colors
 ft.CircleAvatar(
 content=ft.Icon(ft.Icons.WARNING_ROUNDED),
 color=ft.Colors.YELLOW_200,
 bgcolor=ft.Colors.AMBER_700,
),
 # avatar with online status
 ft.Stack(
 width=40,
 height=40,
 controls=[
 ft.CircleAvatar(

```

```

 foreground_image_src="https://avatars.githubusercontent.com/u/5041459?s=88&v=4"
),
 ft.Container(
 alignment=ft.Alignment.BOTTOM_LEFT,
 content=ft.CircleAvatar(
 bgcolor=ft.Colors.GREEN, radius=5
),
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: user-avatars - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/circle\\_avatar/user\\_avatars.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/circle_avatar/user_avatars.png)

## Properties

### buildbackground\_image\_srcclass-attributeinstance-attribute

`background_image_src: str | bytes | None = None`

The source (local asset file or URL) of the background image in the circle. Changing the background image will cause the avatar to animate to the new image.

If this avatar is to have the user's initials, use `content` instead.

Typically used as a fallback image for `foreground_image_src`.

### buildbgcolorclass-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

The color with which to fill the circle.

Changing the background color will cause this avatar to animate to the new color.

### buildcolorclass-attributeinstance-attribute

`color: ColorValue | None = None`

The default color for text in this avatar.

Defaults to the primary text theme color if no `bgcolor` is specified.

### buildcontentclass-attributeinstance-attribute

`content: StrOrControl | None = None`

The content of this avatar.

Typically a `Text` control.

Tip

If this avatar is to have an image, use `background_image_src` instead.

### buildforeground\_image\_srcclass-attributeinstance-attribute

`foreground_image_src: str | bytes | None = None`

The source (local asset file or URL) of the foreground image in the circle.

Fallbacks to `background_image_src`.

Typically used as profile image.

### buildmax\_radiusclass-attributeinstance-attribute

`max_radius: Number | None = None`

The maximum size of the avatar, expressed as the radius (half the diameter). If set to a non `None` value, then radius must be `None` (default).

Defaults to `float('inf')` i.e. infinity.

Raises:

- `ValueError` - If it is not greater than or equal to 0.0.

- ValueError - If it is set while radius is set.

### **buildmin\_radiusclass-attributeinstance-attribute**

`min_radius: Number | None = None`

The minimum size of the avatar, expressed as the radius (half the diameter). If set to a non None value, then radius must be None (default).

Defaults to 0.0.

Raises:

- ValueError - If it is not greater than or equal to 0.0.
- ValueError - If it is set while radius is set.

### **buildradiusclass-attributeinstance-attribute**

`radius: Number | None = None`

The size of the avatar, expressed as the radius (half the diameter).

If set to a non None value, then neither min\_radius nor max\_radius may be specified.

Raises:

- ValueError - If it is not greater than or equal to 0.
- ValueError - If it is set while min\_radius or max\_radius is set.

## **Events**

### **bolton\_image\_errorclass-attributeinstance-attribute**

`on_image_error: ControlEventHandler[CircleAvatar] | None = (None)`

Called when an error occurs while loading the background\_image\_src or foreground\_image\_src.

The data property of the event handler argument is a string whose value is either "background" or "foreground" indicating the error's origin.



# CodeEditor

Edit and highlight source code.

Image: CodeEditor - [https://flet.dev/docs/test-images/examples/extensions/code\\_editor/golden/macros/code\\_editor/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/extensions/code_editor/golden/macros/code_editor/image_for_docs.png)

CodeEditor

Inherits: `LayoutControl`

## Properties

- `autocomplete` - Whether autocomplete is enabled.
- `autocomplete_words` - Words offered by autocomplete.
- `autofocus` - Whether this editor should focus itself if nothing else is focused.
- `code_theme` - Syntax highlighting theme.
- `gutter_style` - Gutter styling.
- `issues` - List of code analysis issues to display in the editor gutter.
- `language` - Syntax highlighting language.
- `padding` - Padding around the editor.
- `read_only` - Whether the editor is read-only.
- `selection` - Represents the current text selection or caret position in the editor.
- `text_style` - Text style for the editor content.
- `value` - Full text including folded sections and service comments.

## Events

- `on_blur` - Called when the editor loses focus.
- `on_change` - Called when the editor text changes.
- `on_focus` - Called when the editor receives focus.
- `on_selection_change` - Called when the text selection or caret position changes.

## Methods

- `focus` - Request focus for this editor.
- `fold_at` - Fold the block starting at the given line number.
- `fold_comment_at_line_zero` - Fold the comment block at line 0.
- `fold_imports` - Fold import sections.

## Usage

Add flet-code-editor to your project dependencies:

- `uv`
- `pip`  
`uv add flet-code-editor`  
`pip install flet-code-editor`

## Examples

### CodeEditor

`play_arrow`Try Online

```
import flet_code_editor as fce

import flet as ft
```

```

CODE = """import flet as ft

def main(page: ft.Page):
 counter = ft.Text("0", size=50, data=0)

 def btn_click(e):
 counter.data += 1
 counter.value = str(counter.data)
 counter.update()

 page.floating_action_button = ft.FloatingActionButton(
 icon=ft.Icons.ADD, on_click=btn_click
)
 page.add(
 ft.SafeArea(
 ft.Container(
 counter,
 alignment=ft.Alignment.CENTER,
 expand=True,
),
 expand=True,
)
)

ft.run(main)
"""

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 expand=True,
 content=fce.CodeEditor(
 language=fce.CodeLanguage.PYTHON,
 code_theme=fce.CodeTheme.ATOM_ONE_LIGHT,
 value=CODE,
 expand=True,
 on_change=lambda e: print("Changed:", e.data),
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: code-editor-example-1 - [https://flet.dev/docs/test-images/examples/extensions/code\\_editor/golden/macros/code\\_editor/example\\_1.png](https://flet.dev/docs/test-images/examples/extensions/code_editor/golden/macros/code_editor/example_1.png)

## Selection handling

play\_arrowTry Online

```

import flet_code_editor as fce

import flet as ft

CODE = """import flet as ft

def main(page: ft.Page):
 counter = ft.Text("0", size=50, data=0)

 def btn_click(e):
 counter.data += 1
 counter.value = str(counter.data)
 counter.update()

 page.floating_action_button = ft.FloatingActionButton(
 icon=ft.Icons.ADD, on_click=btn_click
)
 page.add(
 ft.SafeArea(
 ft.Container(
 counter,
 alignment=ft.Alignment.CENTER,
 expand=True,
),
 expand=True,
)
)

ft.run(main)
"""

font_families = [
 # Apple platforms
 "SF Mono",
 "Menlo",

```

```

Android
"Roboto Mono",
Windows
"Consolas",
Linux
"Ubuntu Mono",
Universal fallbacks
"Courier New",
]

def main(page: ft.Page):
 page.title = "CodeEditor selection"
 max_selection_preview = 80

 theme = fce.CustomCodeTheme(
 keyword=ft.TextStyle(color=ft.Colors.INDIGO_600, weight=ft.FontWeight.W_600),
 string=ft.TextStyle(color=ft.Colors.RED_700),
 comment=ft.TextStyle(color=ft.Colors.GREY_600, italic=True),
)

 text_style = ft.TextStyle(
 font_family="monospace", font_family_fallback=font_families, size=12
)

 gutter_style = fce.GutterStyle(
 text_style=ft.TextStyle(
 font_family="monospace", font_family_fallback=font_families, size=12
),
 show_line_numbers=True,
 show_folding_handles=True,
 width=60,
)

 def handle_selection_change(e: ft.TextSelectionChangeEvent[fce.CodeEditor]):
 if e.selected_text:
 normalized = " ".join(e.selected_text.split())
 suffix = "..." if len(normalized) > max_selection_preview else ""
 preview = normalized[:max_selection_preview]
 selection.value = (
 f"Selection ({len(e.selected_text)} chars): '{preview}{suffix}'"
)
 else:
 selection.value = "No selection."
 selection_details.value = f"start={e.selection.start}, end={e.selection.end}"
 caret.value = f"Caret position: {e.selection.end}"

 async def select_all(e: ft.Event[ft.Button]):
 await editor.focus()
 editor.selection = ft.TextSelection(
 base_offset=0,
 extent_offset=len(editor.value or ""),
)

 async def move_caret_to_start(e: ft.Event[ft.Button]):
 await editor.focus()
 editor.selection = ft.TextSelection(base_offset=0, extent_offset=0)

 page.padding = 0
 page.spacing = 0
 page.add(
 ft.SafeArea(
 expand=True,
 content=ft.Column(
 spacing=10,
 expand=True,
 controls=[
 editor := fce.CodeEditor(
 language=fce.CodeLanguage.PYTHON,
 code_theme=theme,
 autocomplete=True,
 autocomplete_words=[
 "Container",
 "Button",
 "Text",
 "Row",
 "Column",
],
 value=CODE,
 text_style=text_style,
 gutter_style=gutter_style,
 padding=10,
 on_selection_change=handle_selection_change,
 expand=True,
),
 ft.Column(
 margin=10,

```

```

 controls=[
 selection := ft.Text("Select some text from the editor."),
 selection_details := ft.Text(),
 caret := ft.Text("Caret position: -"),
 ft.Row(
 controls=[
 ft.Button("Select all text", on_click=select_all),
 ft.Button(
 "Move caret to start",
 on_click=move_caret_to_start,
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: code-editor-example-2 - [https://flet.dev/docs/test-images/examples/extensions/code\\_editor/golden/macros/code\\_editor/example\\_2.png](https://flet.dev/docs/test-images/examples/extensions/code_editor/golden/macros/code_editor/example_2.png)

## Folding and initial selection

play\_arrowTry Online

```

import flet_code_editor as fce

import flet as ft

CODE = """# 1
2
3
import json
import textwrap

print("Folding demo")
"""

def main(page: ft.Page):
 editor = fce.CodeEditor(
 language=fce.CodeLanguage.PYTHON,
 value=CODE,
 selection=ft.TextSelection(base_offset=41, extent_offset=62),
 autofocus=True,
 expand=True,
 on_selection_change=lambda e: print("Selection:", e),
)

 async def fold_imports():
 await editor.fold_imports()

 async def fold_comment():
 await editor.fold_comment_at_line_zero()

 page.add(
 ft.SafeArea(
 expand=True,
 content=ft.Column(
 controls=[
 ft.Row(
 controls=[
 ft.Button("Fold imports", on_click=fold_imports),
 ft.Button("Fold comment", on_click=fold_comment),
],
),
 editor,
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: code-editor-example-3 - [https://flet.dev/docs/test-images/examples/extensions/code\\_editor/golden/macros/code\\_editor/example\\_3.png](https://flet.dev/docs/test-images/examples/extensions/code_editor/golden/macros/code_editor/example_3.png)

## Properties

buildautocompleteclass-attributeinstance-attribute

`autocomplete: bool | None = False`

Whether autocomplete is enabled.

### **buildautocomplete\_wordsclass-attributeinstance-attribute**

`autocomplete_words: list[str] | None = None`

Words offered by autocomplete.

### **buildautofocusclass-attributeinstance-attribute**

`autofocus: bool | None = False`

Whether this editor should focus itself if nothing else is focused.

### **buildcode\_themeclass-attributeinstance-attribute**

`code_theme: CodeTheme | CustomCodeTheme | None = None`

Syntax highlighting theme.

### **buildgutter\_styleclass-attributeinstance-attribute**

`gutter_style: GutterStyle | None = None`

Gutter styling.

### **buildissuesclass-attributeinstance-attribute**

`issues: list[Issue] = field(default_factory=list)`

List of code analysis issues to display in the editor gutter.

### **buildlanguageclass-attributeinstance-attribute**

`language: CodeLanguage | None = None`

Syntax highlighting language.

### **buildpaddingclass-attributeinstance-attribute**

`padding: PaddingValue | None = None`

Padding around the editor.

### **buildread\_onlyclass-attributeinstance-attribute**

`read_only: bool | None = False`

Whether the editor is read-only.

### **buildselectionclass-attributeinstance-attribute**

`selection: TextSelection | None = None`

Represents the current text selection or caret position in the editor.

Setting this property updates the editor selection and may trigger `on_selection_change` when the editor is focused.

### **buildtext\_styleclass-attributeinstance-attribute**

`text_style: TextStyle | None = None`

Text style for the editor content.

### **buildvalueclass-attributeinstance-attribute**

`value: str | None = None`

Full text including folded sections and service comments.

## **Events**

### **bolton\_blurclass-attributeinstance-attribute**

`on_blur: ControlEventHandler[CodeEditor] | None = None`

Called when the editor loses focus.

### **bolton\_changeclass-attributeinstance-attribute**

`on_change: ControlEventHandler[CodeEditor] | None = None`

Called when the editor text changes.

### **bolton\_focusclass-attributeinstance-attribute**

```
on_focus: ControlEventHandler[CodeEditor] | None = None
```

Called when the editor receives focus.

### **bolton\_selection\_changeclass-attributeinstance-attribute**

```
on_selection_change: (
 EventHandler[TextSelectionChangeEvent[CodeEditor]]
 | None
) = None
```

Called when the text selection or caret position changes.

## **Methods**

### **deployed\_codefocusasync**

```
focus()
```

Request focus for this editor.

### **deployed\_codefold\_atasync**

```
fold_at(line_number: int)
```

Fold the block starting at the given line number.

### **deployed\_codefold\_comment\_at\_line\_zeroasync**

```
fold_comment_at_line_zero()
```

Fold the comment block at line 0.

### **deployed\_codefold\_importsasync**

```
fold_imports()
```

Fold import sections.

# Color pickers

Multiple color picker controls built on Flutter's `flutter_colorpicker` package.

## Usage

Add `flet-color-pickers` to your project dependencies:

- `uv`

- `pip`

```
uv add flet-color-pickers
```

```
pip install flet-color-pickers
```

## Available color pickers

- `ColorPicker`
- `BlockPicker`
- `HueRingPicker`
- `MaterialPicker`
- `MultipleChoiceBlockPicker`
- `SlidePicker`

# Column

Arranges child controls vertically, optionally aligning and spacing them within the available space.

```
ft.Column(
 width=220,
 height=120,
 spacing=12,
 controls=[
 ft.Text("Daily planning", size=20, weight=ft.FontWeight.W_600),
 ft.Text("Review pull requests"),
 ft.Text("Ship release"),
],
)
```

Image: Column - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/column/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/column/image_for_docs.png)

## Column

Inherits: `LayoutControl`, `ScrollableControl`, `AdaptiveControl`

## Properties

- `alignment` - How the child Controls should be placed vertically.
- `controls` - A list of controls to display.
- `horizontal_alignment` - Defines how the controls should be placed horizontally.
- `intrinsic_width` - If True, the Column will be as wide as the widest child control.
- `run_alignment` - How the runs should be placed in the cross-axis when wrap is True.
- `run_spacing` - The spacing between runs when wrap is True.
- `spacing` - Spacing between the controls.
- `tight` - Determines how vertical space is allocated.
- `wrap` - Whether the controls should wrap into additional columns (runs) when they don't fit in a single vertical column.

## Examples

### Column spacing

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 def generate_items(count: int):
 """Generates a list of custom Containers with length `count`."""
 return [
 ft.Container(
 content=ft.Text(value=str(i)),
 alignment=ft.Alignment.CENTER,
 width=50,
 height=50,
 bgcolor=ft.Colors.AMBER,
 border_radius=ft.BorderRadius.all(5),
)
 for i in range(1, count + 1)
]

 def handle_slider_change(e: ft.Event[ft.Slider]):
 """Updates the spacing between items based on slider value."""
 column.spacing = int(e.control.value)
 column.update()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Column(
 controls=[
 ft.Text("Spacing between items"),
 ft.Slider(
 min=0,
 max=100,
 divisions=10,

```



```

 value=0,
 label="{value}",
 width=500,
 on_change=handle_slider_change,
),
]
),
column := ft.Column(spacing=0, controls=generate_items(5)),
]
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: spacing - <https://flet.dev/docs/examples/controls/core/column/media/spacing.gif>

## Column wrapping

play\_arrow Try Online

```

import flet as ft

HEIGHT = 400

def main(page: ft.Page):
 def items(count: int):
 return [
 ft.Container(
 content=ft.Text(value=str(i)),
 alignment=ft.Alignment.CENTER,
 width=30,
 height=30,
 bgcolor=ft.Colors.AMBER,
 border_radius=ft.BorderRadius.all(5),
)
 for i in range(1, count + 1)
]

 def handle_slider_change(e: ft.Event[ft.Slider]):
 col.height = float(e.control.value)
 col.update()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Column(
 controls=[
 ft.Text(
 "Change the column height to see how child items "
 "wrap onto multiple columns:"
),
 ft.Slider(
 min=0,
 max=HEIGHT,
 divisions=20,
 value=HEIGHT,
 label="{value}",
 width=500,
 on_change=handle_slider_change,
),
],
),
 ft.Container(
 bgcolor=ft.Colors.TRANSPARENT,
 content=(
 col := ft.Column(
 wrap=True,
 spacing=10,
 run_spacing=10,
 controls=items(10),
 height=HEIGHT,
)
),
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Column vertical alignment

[play\\_arrowTry Online](#)

```
import flet as ft

@ft.control
class ColumnFromVerticalAlignment(ft.Column):
 alignment: ft.MainAxisAlignment = ft.MainAxisAlignment.START

 def init(self):
 self.controls = [
 ft.Text(str(self.alignment), size=10),
 ft.Container(
 content=ft.Column(self.generate_items(3), alignment=self.alignment),
 bgcolor=ft.Colors.AMBER_100,
 height=400,
),
]

 @staticmethod
 def generate_items(count: int):
 """Generates a list of custom Containers with length `count`."""
 return [
 ft.Container(
 content=ft.Text(value=str(i)),
 alignment=ft.Alignment.CENTER,
 width=50,
 height=50,
 bgcolor=ft.Colors.AMBER_500,
)
 for i in range(1, count + 1)
]

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Row(
 spacing=30,
 alignment=ft.MainAxisAlignment.START,
 scroll=ft.ScrollMode.AUTO,
 controls=[
 ColumnFromVerticalAlignment(alignment=ft.MainAxisAlignment.START),
 ColumnFromVerticalAlignment(alignment=ft.MainAxisAlignment.CENTER),
 ColumnFromVerticalAlignment(alignment=ft.MainAxisAlignment.END),
 ColumnFromVerticalAlignment(
 alignment=ft.MainAxisAlignment.SPACE_BETWEEN
),
 ColumnFromVerticalAlignment(
 alignment=ft.MainAxisAlignment.SPACE_AROUND
),
 ColumnFromVerticalAlignment(
 alignment=ft.MainAxisAlignment.SPACE_EVENLY
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: alignment - <https://flet.dev/docs/examples/controls/core/column/media/alignment.png>

## Column horizontal alignment

[play\\_arrowTry Online](#)

```
import flet as ft

@ft.control
class ColumnFromHorizontalAlignment(ft.Column):
 alignment: ft.CrossAxisAlignment = ft.CrossAxisAlignment.START

 def init(self):
 self.controls = [
 ft.Text(str(self.alignment), size=16),
 ft.Container(
 bgcolor=ft.Colors.AMBER_100,
 width=100,
 content=ft.Column(
```

```

 controls=self.generate_items(3),
 alignment=ft.MainAxisAlignment.START,
 horizontal_alignment=self.alignment,
),
),
]

@staticmethod
def generate_items(count: int):
 """Generates a list of custom Containers with length `count`."""
 return [
 ft.Container(
 content=ft.Text(value=str(i)),
 alignment=ft.Alignment.CENTER,
 width=50,
 height=50,
 bgcolor=ft.Colors.AMBER_500,
)
 for i in range(1, count + 1)
]

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Row(
 spacing=30,
 alignment=ft.MainAxisAlignment.START,
 controls=[
 ColumnFromHorizontalAlignment(
 alignment=ft.CrossAxisAlignment.START
),
 ColumnFromHorizontalAlignment(
 alignment=ft.CrossAxisAlignment.CENTER
),
 ColumnFromHorizontalAlignment(alignment=ft.CrossAxisAlignment.END),
],
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: horizontal-alignment - [https://flet.dev/docs/examples/controls/core/column/media/horizontal\\_alignment.png](https://flet.dev/docs/examples/controls/core/column/media/horizontal_alignment.png)

## Infinite scrolling

This example demonstrates adding of list items on-the-fly, as user scroll to the bottom, creating the illusion of infinite list:

[play\\_arrow](#)Try Online

```

import threading

import flet as ft

class State:
 i = 0

s = State()
sem = threading.Semaphore()

def main(page: ft.Page):
 def on_scroll(e: ft.OnScrollEvent):
 if e.pixels >= e.max_scroll_extent - 100 and sem.acquire(blocking=False):
 try:
 for _i in range(0, 10):
 cl.controls.append(ft.Text(f"Text line {s.i}", key=str(s.i)))
 s.i += 1
 cl.update()
 finally:
 sem.release()

 cl = ft.Column(
 spacing=10,
 height=200,
 width=200,
 scroll=ft.ScrollMode.ALWAYS,
 scroll_interval=0,
 on_scroll=on_scroll,
)
 for _i in range(0, 50):
 cl.controls.append(ft.Text(f"Text line {s.i}", key=str(s.i)))

```

```

 s.i += 1

page.add(ft.SafeArea(content=ft.Container(cl, border=ft.Border.all(1))))

if __name__ == "__main__":
 ft.run(main)

```

## Scrolling programmatically

This example shows how to use `scroll_to()` to programmatically scroll a column:

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 column = ft.Column(
 spacing=10,
 height=200,
 width=float("inf"),
 scroll=ft.ScrollMode.ALWAYS,
 controls=[
 ft.Text(f"Text line {i}", key=ft.ScrollKey(i)) for i in range(0, 100)
],
)

 async def scroll_to_offset(e):
 await column.scroll_to(offset=500, duration=1000)

 async def scroll_to_start(e):
 await column.scroll_to(offset=0, duration=1000)

 async def scroll_to_end(e):
 await column.scroll_to(
 offset=-1, duration=2000, curve=ft.AnimationCurve.EASE_IN_OUT
)

 async def scroll_to_key(e):
 await column.scroll_to(scroll_key="20", duration=1000)

 async def scroll_to_delta(e):
 await column.scroll_to(delta=100, duration=200)

 async def scroll_to_minus_delta(e):
 await column.scroll_to(delta=-100, duration=200)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Container(content=column, border=ft.Border.all(1)),
 ft.Button("Scroll to offset 500", on_click=scroll_to_offset),
 ft.Row(
 controls=[
 ft.Button("Scroll -100", on_click=scroll_to_minus_delta),
 ft.Button("Scroll +100", on_click=scroll_to_delta),
]
),
 ft.Button("Scroll to key '20'", on_click=scroll_to_key),
 ft.Row(
 controls=[
 ft.Button("Scroll to start", on_click=scroll_to_start),
 ft.Button("Scroll to end", on_click=scroll_to_end),
]
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: programmatic-scroll - [https://flet.dev/docs/examples/controls/core/column/media/programmatic\\_scroll.png](https://flet.dev/docs/examples/controls/core/column/media/programmatic_scroll.png)

## Properties

buildalignmentclass-attributeinstance-attribute

```
alignment: MainAxisAlignment = MainAxisAlignment.START
```

How the child Controls should be placed vertically.

### **buildcontrolsclass-attributeinstance-attribute**

```
controls: list[Control] = field(default_factory=list)
```

A list of controls to display.

### **buildhorizontal\_alignmentclass-attributeinstance-attribute**

```
horizontal_alignment: CrossAxisAlignment = (
 CrossAxisAlignment.START
)
```

Defines how the controls should be placed horizontally.

### **buildintrinsic\_widthclass-attributeinstance-attribute**

```
intrinsic_width: bool = False
```

If True, the Column will be as wide as the widest child control.

### **buildrun\_alignmentclass-attributeinstance-attribute**

```
run_alignment: MainAxisAlignment = MainAxisAlignment.START
```

How the runs should be placed in the cross-axis when wrap is True.

### **buildrun\_spacingclass-attributeinstance-attribute**

```
run_spacing: Number = 10
```

The spacing between runs when wrap is True.

### **buildspacingclass-attributeinstance-attribute**

```
spacing: Number = 10
```

Spacing between the controls.

It is applied only when alignment is MainAxisAlignment.START, MainAxisAlignment.END or MainAxisAlignment.CENTER.

### **buildtightclass-attributeinstance-attribute**

```
tight: bool = False
```

Determines how vertical space is allocated.

If False (default), children expand to fill the available vertical space. If True, only the minimum vertical space required by the children is used.

### **buildwrapclass-attributeinstance-attribute**

```
wrap: bool = False
```

Whether the controls should wrap into additional columns (runs) when they don't fit in a single vertical column.

# Container

Allows to decorate a control with background color and border and position it with padding, margin and alignment.

Image: Container - [https://flet.dev/docs/examples/controls/core/container/media/overview\\_padding\\_margin\\_border.png](https://flet.dev/docs/examples/controls/core/container/media/overview_padding_margin_border.png)

Container explained

Inherits: `LayoutControl`, `AdaptiveControl`

## Properties

- `alignment` - Defines the alignment of the content inside this container.
- `animate` - Enables container "implicit" animation that gradually changes its values over a period of time.
- `bgcolor` - Defines the background color of this container.
- `blend_mode` - The blend mode applied to the color or gradient background of the container.
- `blur` - Defines how Gaussian blur effect should be applied under this container.
- `border` - A border to draw above the background color.
- `border_radius` - The border radius of this container.
- `clip_behavior` - Defines how the content of this container is clipped.
- `color_filter` - Applies a color filter to this container.
- `content` - The content of this container.
- `dark_theme` - Allows setting a nested theme to be used when in dark theme mode for all controls inside the container and down its tree.
- `foreground_decoration` - The foreground decoration of this container.
- `gradient` - Defines the gradient background of this container.
- `ignore_interactions` - Whether to ignore all interactions with this container and its descendants.
- `image` - An image to paint above the bgcolor or gradient.
- `ink` - True to produce ink ripples effect when user clicks this container.
- `ink_color` - The splash color of the ink response.
- `padding` - Empty space to inscribe inside a container decoration (background, border).
- `shadow` - The shadow(s) below this container.
- `shape` - Sets the shape of this container.
- `theme` - Allows setting a nested theme for all controls inside this container and down its tree.
- `theme_mode` - "Resets" parent theme and creates a new, unique scheme for all controls inside the container.
- `url` - The URL to open when this container is clicked.

## Events

- `on_click` - Called when a user clicks the container.
- `on_hover` - Called when a mouse pointer enters or exists the container area.
- `on_long_press` - Called when this container is long-pressed.
- `on_tap_down` - Called when a user clicks the container with or without a long press.

## Examples

### Clickable

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
```

```

page.title = "Container Example"
page.theme_mode = ft.ThemeMode.LIGHT
page.vertical_alignment = ft.MainAxisAlignment.CENTER
page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

page.add(
 ft.SafeArea(
 content=ft.Row(
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 ft.Container(
 margin=10,
 padding=10,
 alignment=ft.Alignment.CENTER,
 bgcolor=ft.Colors.AMBER,
 width=150,
 height=150,
 border_radius=10,
 content=ft.Text("Non clickable"),
),
 ft.Container(
 margin=10,
 padding=10,
 alignment=ft.Alignment.CENTER,
 bgcolor=ft.Colors.GREEN_200,
 width=150,
 height=150,
 border_radius=10,
 on_click=lambda e: print("Clickable without Ink clicked!"),
 content=ft.Text("Clickable without Ink"),
),
 ft.Container(
 margin=10,
 padding=10,
 alignment=ft.Alignment.CENTER,
 bgcolor=ft.Colors.CYAN_200,
 width=150,
 height=150,
 border_radius=10,
 ink=True,
 on_click=lambda e: print("Clickable with Ink clicked!"),
 content=ft.Text("Clickable with Ink"),
),
 ft.Container(
 margin=10,
 padding=10,
 alignment=ft.Alignment.CENTER,
 width=150,
 height=150,
 border_radius=10,
 ink=True,
 on_click=lambda e: print(
 "Clickable transparent with Ink clicked!"
),
 content=ft.Text("Clickable transparent with Ink"),
),
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image - <https://flet.dev/docs/examples/controls/core/container/media/clickable.gif>

## Handling clicks

play\_arrow Try Online

```
import flet as ft
```

```

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT
 page.vertical_alignment = ft.MainAxisAlignment.CENTER
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 lp_counter = 0
 cl_counter = 0
 td_counter = 0

 def on_click(e):
 nonlocal cl_counter
 cl_counter += 1
 t1.spans[-1] = ft.TextSpan(
 text=f" {cl_counter} ",

```

```

 style=ft.TextStyle(size=16, bgcolor=ft.Colors.TEAL_300),
)
 t1.update()

def on_long_press(e):
 nonlocal lp_counter
 lp_counter += 1
 t3.spans[-1] = ft.TextSpan(
 text=f" {lp_counter} ",
 style=ft.TextStyle(size=16, bgcolor=ft.Colors.TEAL_300),
)
 t3.update()

def on_tap_down(e):
 nonlocal td_counter
 td_counter += 1
 t2.spans[-1] = ft.TextSpan(
 text=f" {td_counter} ",
 style=ft.TextStyle(size=16, bgcolor=ft.Colors.TEAL_300),
)
 t2.update()

c = ft.Container(
 bgcolor=ft.Colors.PINK_900,
 alignment=ft.Alignment.CENTER,
 padding=ft.Padding.all(10),
 height=150,
 width=150,
 on_click=on_click,
 on_long_press=on_long_press,
 on_tap_down=on_tap_down,
 content=ft.Text(
 "Press Me!",
 text_align=ft.TextAlign.CENTER,
 style=ft.TextStyle(
 size=30,
 # weight=ft.FontWeight.BOLD,
 foreground=ft.Paint(
 color=ft.Colors.BLUE_700,
 stroke_cap=ft.StrokeCap.BUTT,
 stroke_width=2,
 stroke_join=ft.StrokeJoin.BEVEL,
 style=ft.PaintingStyle.STROKE,
),
),
 theme_style=ft.TextThemeStyle.DISPLAY_MEDIUM,
),
)

t1 = ft.Text(
 spans=[
 ft.TextSpan(
 text="On Click", style=ft.TextStyle(size=16, weight=ft.FontWeight.BOLD)
),
 ft.TextSpan(text=" counter: ", style=ft.TextStyle(size=16, italic=True)),
 ft.TextSpan(
 text=f" {cl_counter} ",
 style=ft.TextStyle(size=16, bgcolor=ft.Colors.TEAL_300),
),
],
)

t2 = ft.Text(
 spans=[
 ft.TextSpan(
 text="Tap Down", style=ft.TextStyle(size=16, weight=ft.FontWeight.BOLD)
),
 ft.TextSpan(text=" counter: ", style=ft.TextStyle(size=16, italic=True)),
 ft.TextSpan(
 text=f" {td_counter} ",
 style=ft.TextStyle(size=16, bgcolor=ft.Colors.TEAL_300),
),
],
)

t3 = ft.Text(
 spans=[
 ft.TextSpan(
 text="Long Press",
 style=ft.TextStyle(size=16, weight=ft.FontWeight.BOLD),
),
 ft.TextSpan(text=" counter: ", style=ft.TextStyle(size=16, italic=True)),
 ft.TextSpan(
 text=f" {lp_counter} ",
 style=ft.TextStyle(size=16, bgcolor=ft.Colors.TEAL_300),
),
],
)

page.add(ft.SafeArea(content=ft.Column(controls=[c, t1, t3, t2])))

```



```
if __name__ == "__main__":
 ft.run(main)
```

Image: [https://flet.dev/docs/examples/controls/core/container/media/handling\\_clicks.gif](https://flet.dev/docs/examples/controls/core/container/media/handling_clicks.gif)

## Handling hovers

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 def handle_hover(e: ft.Event[ft.Container]):
 e.control.bgcolor = ft.Colors.BLUE if e.data else ft.Colors.RED
 e.control.update()

 page.add(
 ft.SafeArea(
 content=ft.Container(
 width=200,
 height=200,
 bgcolor=ft.Colors.RED,
 ink=False,
 on_hover=handle_hover,
)
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: [https://flet.dev/docs/examples/controls/core/container/media/handling\\_hovers.gif](https://flet.dev/docs/examples/controls/core/container/media/handling_hovers.gif)

## Animate size and color

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 def animate_container(e: ft.Event[ft.Button]):
 container.width = 100 if container.width == 150 else 150
 container.height = 50 if container.height == 150 else 150
 container.bgcolor = (
 ft.Colors.BLUE if container.bgcolor == ft.Colors.RED else ft.Colors.RED
)
 container.update()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 container := ft.Container(
 width=150,
 height=150,
 bgcolor=ft.Colors.RED,
 animate=ft.Animation(
 duration=1000, curve=ft.AnimationCurve.BOUNCE_OUT
),
),
 ft.Button("Animate container", on_click=animate_container),
]
)
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: [https://flet.dev/docs/examples/controls/core/container/media/animate\\_1.gif](https://flet.dev/docs/examples/controls/core/container/media/animate_1.gif)

## Animate gradient and shape

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 gradient1 = ft.LinearGradient(
 begin=ft.Alignment.TOP_CENTER,
 end=ft.Alignment.BOTTOM_CENTER,
 colors=[ft.Colors.GREEN, ft.Colors.BLUE],
 stops=[0.5, 1.0],
```

```

)
gradient2 = ft.RadialGradient(
 center=ft.Alignment.TOP_LEFT,
 radius=1.0,
 colors=[ft.Colors.YELLOW, ft.Colors.DEEP_ORANGE_900],
 tile_mode=ft.GradientTileMode.CLAMP,
)

message = ft.Text("Animate me!")

def animate_container(e: ft.Event[ft.Button]):
 message.value = (
 "Animate me back!" if message.value == "Animate me!" else "Animate me!"
)
 container.width = 150 if container.width == 250 else 250
 container.height = 150 if container.height == 250 else 250
 container.gradient = gradient2 if container.gradient == gradient1 else gradient1
 if container.alignment == ft.Alignment.TOP_LEFT:
 container.alignment = ft.Alignment.BOTTOM_RIGHT
 else:
 container.alignment = ft.Alignment.TOP_LEFT
 container.border_radius = 30 if container.border_radius == 10 else 10
 container.border = (
 ft.Border.all(width=2, color=ft.Colors.BLACK)
 if container.border == ft.Border.all(width=2, color=ft.Colors.BLUE)
 else ft.Border.all(width=2, color=ft.Colors.BLUE)
)
 container.update()

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 container := ft.Container(
 width=250,
 height=250,
 gradient=gradient2,
 alignment=ft.Alignment.TOP_LEFT,
 animate=ft.Animation(
 duration=1000, curve=ft.AnimationCurve.BOUNCE_OUT
),
 border=ft.Border.all(width=2, color=ft.Colors.BLUE),
 border_radius=10,
 padding=10,
 content=message,
),
 ft.Button("Animate container", on_click=animate_container),
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

## Animated slide-in menu

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 def show_menu(e: ft.Event[ft.Button]):
 container.offset = ft.Offset(0, 0)
 container.update()

 def hide_menu(e: ft.Event[ft.IconButton]):
 container.offset = ft.Offset(-2, 0)
 container.update()

 page.overlay.append(
 container := ft.Container(
 left=10,
 top=10,
 width=200,
 height=300,
 bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
 border_radius=5,
 offset=ft.Offset(-2, 0),
 animate_offset=ft.Animation(300, ft.AnimationCurve.EASE_IN),
 content=ft.Column(
 controls=[
 ft.Row(
 alignment=ft.MainAxisAlignment.END,
 controls=[

```

```

 ft.IconButton(icon=ft.Icons.CLOSE, on_click=hide_menu)
],
),
ft.ListTile(
 title=ft.Text("Menu A"),
 on_click=lambda _: print("Menu A clicked"),
),
ft.ListTile(
 title=ft.Text("Menu B"),
 on_click=lambda _: print("Menu B clicked"),
),
],
),
),
)

page.add(ft.SafeArea(content=ft.Button("Show menu", on_click=show_menu)))

if __name__ == "__main__":
 ft.run(main)

```

## Inherited and overridden theme

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 # Yellow page theme with SYSTEM (default) mode
 page.theme = ft.Theme(
 color_scheme_seed=ft.Colors.YELLOW,
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 # Page theme
 ft.Container(
 bgcolor=ft.Colors.SURFACE_TINT,
 padding=20,
 width=300,
 content=ft.Button("Page theme button"),
),
 # Inherited theme with primary color overridden
 ft.Container(
 theme=ft.Theme(
 color_scheme=ft.ColorScheme(primary=ft.Colors.PINK)
),
 bgcolor=ft.Colors.SURFACE_TINT,
 padding=20,
 width=300,
 content=ft.Button("Inherited theme button"),
),
 # Unique always DARK theme
 ft.Container(
 theme=ft.Theme(color_scheme_seed=ft.Colors.INDIGO),
 theme_mode=ft.ThemeMode.DARK,
 bgcolor=ft.Colors.SURFACE_TINT,
 padding=20,
 width=300,
 content=ft.Button("Unique theme button"),
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/container/nested\\_themes\\_1.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/container/nested_themes_1.png)

## Page, dark and light themes

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 # page.theme = ft.Theme(
 # color_scheme_seed=ft.Colors.YELLOW,
 # color_scheme=ft.ColorScheme(

```

```

primary=ft.Colors.GREEN, primary_container=ft.Colors.GREEN_200
),
)

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Row(
 controls=[
 ft.Button("Page theme"),
 ft.TextButton("Page theme text button"),
 ft.Text(
 "Text in primary container color",
 color=ft.Colors.PRIMARY_CONTAINER,
),
],
),
 ft.Container(
 height=100,
 theme=ft.Theme(
 color_scheme=ft.ColorScheme(primary=ft.Colors.PINK)
),
 content=ft.Row(
 controls=[
 ft.Button(
 "Inherited theme with primary color overridden"
),
 ft.TextButton("Button 2"),
],
),
),
 ft.Container(
 padding=20,
 bgcolor=ft.Colors.SURFACE_TINT,
 theme_mode=ft.ThemeMode.DARK,
 theme=ft.Theme(
 color_scheme_seed=ft.Colors.GREEN,
 color_scheme=ft.ColorScheme(
 primary_container=ft.Colors.BLUE
),
),
 content=ft.Row(
 controls=[
 ft.Button("Always DARK theme"),
 ft.TextButton("Text button"),
 ft.Text(
 "Text in primary container color",
 color=ft.Colors.PRIMARY_CONTAINER,
),
],
),
),
 ft.Container(
 padding=20,
 bgcolor=ft.Colors.SURFACE_TINT,
 border=ft.Border.all(3, ft.Colors.OUTLINE),
 theme_mode=ft.ThemeMode.LIGHT,
 theme=ft.Theme(),
 content=ft.Row(
 controls=[
 ft.Button("Always LIGHT theme"),
 ft.TextButton("Text button"),
 ft.Text(
 "Text in primary container color",
 color=ft.Colors.PRIMARY_CONTAINER,
),
],
),
),
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/container/nested\\_themes\\_2.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/container/nested_themes_2.png)

## Theme mode toggle

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):

```

```

page.theme_mode = ft.ThemeMode.DARK

def handle_switch_change(e: ft.Event[ft.Switch]):
 if page.theme_mode == ft.ThemeMode.DARK:
 page.theme_mode = ft.ThemeMode.LIGHT
 switch.thumb_icon = ft.Icons.LIGHT_MODE
 else:
 switch.thumb_icon = ft.Icons.DARK_MODE
 page.theme_mode = ft.ThemeMode.DARK
 page.update()

Yellow page theme with SYSTEM (default) mode
page.theme = ft.Theme(color_scheme_seed=ft.Colors.YELLOW)

switch = ft.Switch(thumb_icon=ft.Icons.DARK_MODE, on_change=handle_switch_change)

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 # Page theme
 ft.Row(
 alignment=ft.MainAxisAlignment.SPACE_BETWEEN,
 controls=[
 ft.Container(
 bgcolor=ft.Colors.SURFACE_TINT,
 padding=20,
 width=300,
 content=ft.Button("Page theme button"),
),
 ft.Container(
 padding=ft.Padding.only(bottom=50),
 alignment=ft.Alignment.TOP_RIGHT,
 content=switch,
),
],
),
 # Inherited theme with primary color overridden
 ft.Container(
 theme=ft.Theme(
 color_scheme=ft.ColorScheme(primary=ft.Colors.PINK)
),
 bgcolor=ft.Colors.SURFACE_TINT,
 padding=20,
 width=300,
 content=ft.Button("Inherited theme button"),
),
 # Unique always DARK theme
 ft.Container(
 theme=ft.Theme(color_scheme_seed=ft.Colors.INDIGO),
 theme_mode=ft.ThemeMode.DARK,
 bgcolor=ft.Colors.SURFACE_TINT,
 padding=20,
 width=300,
 content=ft.Button("Unique theme button"),
),
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image - [https://flet.dev/docs/examples/controls/core/container/media/nested\\_themes\\_3.gif](https://flet.dev/docs/examples/controls/core/container/media/nested_themes_3.gif)

## Size aware

### play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 def handle_size_change(e: ft.LayoutSizeChangeEvent[ft.Container]):
 e.control.content.value = f"{int(e.width)} x {int(e.height)}"

 page.add(
 ft.SafeArea(
 expand=True,
 content=ft.Container(
 expand=True,
 alignment=ft.Alignment.CENTER,
 bgcolor=ft.Colors.BLUE_ACCENT,
 size_change_interval=100,
 on_size_change=handle_size_change,
),
),
)

```

```

 content=ft.Text(color=ft.Colors.WHITE, weight=ft.FontWeight.BOLD),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/container/size\\_aware.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/container/size_aware.png)

## Properties

### buildalignmentclass-attributeinstance-attribute

alignment: Alignment | None = None

Defines the alignment of the content inside this container.

#### Note

If alignment is non-None, this container may expand to fill the available space from its parent (before positioning its content within itself according to the given alignment) instead of shrinking to its content. If you need this container to keep a fixed size, give it container an explicit width and/or height values, or constrain it via its parent.

### buildanimateclass-attributeinstance-attribute

animate: AnimationValue | None = None

Enables container "implicit" animation that gradually changes its values over a period of time.

### buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

Defines the background color of this container.

### buildblend\_modeclass-attributeinstance-attribute

blend\_mode: BlendMode | None = None

The blend mode applied to the color or gradient background of the container.

Defaults to BlendMode.MODULATE.

### buildblurclass-attributeinstance-attribute

blur: BlurValue | None = None

Defines how Gaussian blur effect should be applied under this container.

#### Example

```

ft.Stack(
 controls=[
 ft.Container(
 content=ft.Text("Hello"),
 image_src="https://picsum.photos/100/100",
 width=100,
 height=100,
),
 ft.Container(
 width=50,
 height=50,
 blur=10,
 bgcolor="#44CCCC00",
),
 ft.Container(
 width=50,
 height=50,
 left=10,
 top=60,
 blur=(0, 10),
),
 ft.Container(
 top=10,
 left=60,
 blur=ft.Blur(10, 0, ft.BlurTileMode.MIRROR),
 width=50,
 height=50,
 bgcolor="#44CCCC00",
 border=ft.Border.all(2, ft.Colors.BLACK),
),
],
)

```

### **buildborderclass-attributeinstance-attribute**

`border: Border | None = None`

A border to draw above the background color.

### **buildborder\_radiusclass-attributeinstance-attribute**

`border_radius: BorderRadiusValue | None = None`

The border radius of this container.

### **buildclip\_behaviorclass-attributeinstance-attribute**

`clip_behavior: ClipBehavior | None = None`

Defines how the content of this container is clipped.

Defaults to ClipBehavior.ANTI\_ALIAS if border\_radius is not None; otherwise ClipBehavior.NONE.

### **buildcolor\_filterclass-attributeinstance-attribute**

`color_filter: ColorFilter | None = None`

Applies a color filter to this container.

### **buildcontentclass-attributeinstance-attribute**

`content: Control | None = None`

The content of this container.

### **bulddark\_themeclass-attributeinstance-attribute**

`dark_theme: Theme | None = None`

Allows setting a nested theme to be used when in dark theme mode for all controls inside the container and down its tree.

### **buildforeground\_decorationclass-attributeinstance-attribute**

`foreground_decoration: BoxDecoration | None = None`

The foreground decoration of this container.

### **buildgradientclass-attributeinstance-attribute**

`gradient: Gradient | None = None`

Defines the gradient background of this container.

### **buildignore\_interactionsclass-attributeinstance-attribute**

`ignore_interactions: bool = False`

Whether to ignore all interactions with this container and its descendants.

### **buildimageclass-attributeinstance-attribute**

`image: DecorationImage | None = None`

An image to paint above the bgcolor or gradient. If shape=BoxShape.CIRCLE then this image is clipped to the circle's boundary; if border\_radius is not None then the image is clipped to the given radii.

### **buildinkclass-attributeinstance-attribute**

`ink: bool = False`

True to produce ink ripples effect when user clicks this container.

### **buildink\_colorclass-attributeinstance-attribute**

`ink_color: ColorValue | None = None`

The splash color of the ink response.

### **buildpaddingclass-attributeinstance-attribute**

`padding: PaddingValue | None = None`

Empty space to inscribe inside a container decoration (background, border). The child control is placed inside this padding.

### **buildshadowclass-attributeinstance-attribute**

`shadow: BoxShadowValue | None = None`

The shadow(s) below this container.

### **buildshapeclass-attributeinstance-attribute**

`shape: BoxShape = BoxShape.RECTANGLE`

Sets the shape of this container.

### **buildthemeclass-attributeinstance-attribute**

`theme: Theme | None = None`

Allows setting a nested theme for all controls inside this container and down its tree.

### **buildtheme\_modeclass-attributeinstance-attribute**

`theme_mode: ThemeMode | None = None`

"Resets" parent theme and creates a new, unique scheme for all controls inside the container. Otherwise the styles defined in container's theme property override corresponding styles from the parent, inherited theme.

Defaults to ThemeMode.SYSTEM.

### **buildurlclass-attributeinstance-attribute**

`url: str | Url | None = None`

The URL to open when this container is clicked.

Additionally, if an `on_click` callback is provided, it is fired after that.

## **Events**

### **bolton\_clickclass-attributeinstance-attribute**

`on_click: ControlEventHandler[Container] | None = None`

Called when a user clicks the container.

It will not be called if this container is long pressed.

### **bolton\_hoverclass-attributeinstance-attribute**

`on_hover: ControlEventHandler[Container] | None = None`

Called when a mouse pointer enters or exists the container area.

The data property of the event handler argument is a boolean: True when the cursor enters and False when it exits this container.

### **bolton\_long\_pressclass-attributeinstance-attribute**

`on_long_press: ControlEventHandler[Container] | None = None`

Called when this container is long-pressed.

### **bolton\_tap\_downclass-attributeinstance-attribute**

`on_tap_down: EventHandler[TapEvent[Container]] | None = None`

Called when a user clicks the container with or without a long press.



# ContextMenu

Wraps its content and displays contextual menus for specific mouse events.

## Tip

On web, call `disable` method of `Page.browser_context_menu` to suppress the default browser context menu before relying on custom menus.

Image: ContextMenu - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/context\\_menu/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/context_menu/image_for_docs.png)

## ContextMenu

Inherits: `LayoutControl`

## Properties

- `content` - The child control that listens for mouse interaction.
- `items` - A list of menu items to display in the context menu, when open is called.
- `primary_items` - A list of menu items to display in the context menu, for primary (usually left) mouse button actions.
- `primary_trigger` - Defines a trigger mode for the display of `primary_items`.
- `secondary_items` - A list of menu items to display in the context menu for secondary (usually right) mouse button actions.
- `secondary_trigger` - Defines a trigger mode for the display of `secondary_items`.
- `tertiary_items` - A list of menu items to display in the context menu for tertiary (usually middle) mouse button actions.
- `tertiary_trigger` - Defines a trigger mode for the display of `tertiary_items`.

## Events

- `on_dismiss` - Fires when the menu is dismissed without a selection, or when an attempt is made to open the menu but no items are available.
- `on_select` - Fires when a context menu item is selected.

## Methods

- `open` - Opens the context menu programmatically, and displays items.

# Examples

## Triggers

`play_arrow` Try Online

```
import flet as ft
```

```
async def main(page: ft.Page):
 # on web, disable default browser context menu
 if page.web:
 await page.browser_context_menu.disable()

 def handle_item_click(e: ft.Event[ft.PopupMenuItem]):
 action = e.control.content
 page.show_dialog(ft.SnackBar(content=f"Item '{action}' selected."))

 page.add(
 ft.SafeArea(
 expand=True,
 content=ft.ContextMenu(
 primary_items=[
 ft.PopupMenuItem(content="Primary 1", on_click=handle_item_click),
 ft.PopupMenuItem(content="Primary 2", on_click=handle_item_click),
],
 primary_trigger=ft.ContextMenuTrigger.DOWN,
 secondary_items=[
 ft.PopupMenuItem(content="Secondary 1", on_click=handle_item_click),
 ft.PopupMenuItem(content="Secondary 2", on_click=handle_item_click),
],
 secondary_trigger=ft.ContextMenuTrigger.DOWN,
 tertiary_items=[
```

```

 ft.PopupMenuItem(content="Tertiary 1", on_click=handle_item_click),
 ft.PopupMenuItem(content="Tertiary 2", on_click=handle_item_click),
],
 tertiary_trigger=ft.ContextMenuTrigger.DOWN,
 on_select=lambda e: print(f"Selected item: {e.item.content}"),
 on_dismiss=lambda e: print("Menu dismissed"),
 expand=True,
 content=ft.Container(
 key="context_menu_trigger_area",
 expand=True,
 bgcolor=ft.Colors.BLUE,
 alignment=ft.Alignment.CENTER,
 border_radius=ft.BorderRadius.all(12),
 content=ft.Text("Left/middle/right click to open a context menu."),
),
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/context\\_menu/triggers\\_flow.gif](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/context_menu/triggers_flow.gif)

## Programmatic open

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.vertical_alignment = ft.MainAxisAlignment.CENTER

 def handle_select(e: ft.ContextMenuSelectEvent):
 action = e.item.content
 page.show_dialog(ft.SnackBar(f"Item '{action}' selected."))

 async def open_menu(e: ft.Event[ft.Button]):
 await menu.open()

 menu = ft.ContextMenu(
 on_select=handle_select,
 items=[
 ft.PopupMenuItem(
 content="Item 1",
 on_click=lambda e: print(f"{e.control.content}"),
),
 ft.PopupMenuItem(
 content="Item 2",
 on_click=lambda e: print(f"{e.control.content}"),
),
 ft.PopupMenuItem(
 content="Item 3",
 on_click=lambda e: print(f"{e.control.content}"),
),
],
 content=ft.Button("Click to open menu", on_click=open_menu),
)

 page.add(
 ft.SafeArea(
 content=menu,
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/context\\_menu/programmatic\\_open.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/context_menu/programmatic_open.png)

## Custom trigger

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 async def open_menu(e: ft.TapEvent[ft.GestureDetector]):
 await menu.open(
 local_position=e.local_position,
 global_position=e.global_position,
)

```

```

menu = ft.ContextMenu(
 expand=True,
 items=[
 ft.PopupMenuItem(
 content="Cut",
 on_click=lambda e: print(f"{e.control.content}"),
),
 ft.PopupMenuItem(
 content="Copy",
 on_click=lambda e: print(f"{e.control.content}"),
),
 ft.PopupMenuItem(
 content="Paste",
 on_click=lambda e: print(f"{e.control.content}"),
),
],
 content=ft.GestureDetector(
 expand=True,
 on_double_tap_down=open_menu,
 content=ft.Container(
 key="context_menu_custom_trigger_area",
 bgcolor=ft.Colors.BLUE,
 alignment=ft.Alignment.CENTER,
 content=ft.Text("Double-click to open the context menu."),
),
),
)

page.add(
 ft.SafeArea(
 expand=True,
 content=menu,
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/context\\_menu/custom\\_trigger\\_flow.gif](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/context_menu/custom_trigger_flow.gif)

## Properties

### buildcontentinstance-attribute

content: Control

The child control that listens for mouse interaction.

Raises:

- ValueError - If not visible.

### builditemsclass-attributeinstance-attribute

items: list[PopupMenuItem] = field(default\_factory=list)

A list of menu items to display in the context menu, when open is called.

### buildprimary\_itemsclass-attributeinstance-attribute

primary\_items: list[PopupMenuItem] = field(
 default\_factory=list
)

A list of menu items to display in the context menu, for primary (usually left) mouse button actions.

These items are displayed when the corresponding primary\_trigger is activated.

### buildprimary\_triggerclass-attributeinstance-attribute

primary\_trigger: ContextMenuTrigger | None = None

Defines a trigger mode for the display of primary\_items.

If set to None, the trigger is disabled.

### buildsecondary\_itemsclass-attributeinstance-attribute

secondary\_items: list[PopupMenuItem] = field(
 default\_factory=list
)

A list of menu items to display in the context menu for secondary (usually right) mouse button actions.

These items are displayed when the corresponding `secondary_trigger` is activated.

### **buildsecondary\_triggerclass-attributeinstance-attribute**

```
secondary_trigger: ContextMenuTrigger | None = (
 ContextMenuTrigger.DOWN
)
```

Defines a trigger mode for the display of `secondary_items`.

If set to `None`, the trigger is disabled.

### **buildtertiary\_itemsclass-attributeinstance-attribute**

```
tertiary_items: list[PopupMenuItem] = field(
 default_factory=list
)
```

A list of menu items to display in the context menu for tertiary (usually middle) mouse button actions.

These items are displayed when the corresponding `tertiary_trigger` is activated.

### **buildtertiary\_triggerclass-attributeinstance-attribute**

```
tertiary_trigger: ContextMenuTrigger | None = (
 ContextMenuTrigger.DOWN
)
```

Defines a trigger mode for the display of `tertiary_items`.

If set to `None`, the trigger is disabled.

## **Events**

### **bolton\_dismissclass-attributeinstance-attribute**

```
on_dismiss: EventHandler[ContextMenuDismissEvent] | None = (
 None
)
```

Fires when the menu is dismissed without a selection, or when an attempt is made to open the menu but no items are available.

### **bolton\_selectclass-attributeinstance-attribute**

```
on_select: EventHandler[ContextMenuSelectEvent] | None = (
 None
)
```

Fires when a context menu item is selected.

## **Methods**

### **deployed\_codeopenasync**

```
open(
 global_position: OffsetValue | None = None,
 local_position: OffsetValue | None = None,
) -> None
```

Opens the context menu programmatically, and displays items.

Parameters:

- `global_position` (`OffsetValue` | `None`, default: `None`) - A global coordinate describing where the menu should appear. If omitted, `local_position` or the center of the content is used.
- `local_position` (`OffsetValue` | `None`, default: `None`) - A local coordinate relative to the content. When provided without `global_position`, the coordinate is translated to global space automatically.

# Control

Base class for controls.

Not meant to be used directly.

Inherits: BaseControl

## Properties

- **badge** - A badge to show on top of this control.
- **col** - If a parent of this control is a `ResponsiveRow`, this property is used to determine how many virtual columns of a screen this control will span.
- **disabled** - Every control has disabled property which is `False` by default - control and all its children are enabled.
- **expand** - Specifies whether/how this control should expand to fill available space in its parent layout.
- **expand\_loose** - Allows the control to expand along the main axis if space is available, but does not require it to fill all available space.
- **opacity** - Defines the transparency of the control.
- **rtl** - Whether the text direction of the control should be right-to-left (RTL).
- **tooltip** - The tooltip to show when this control is hovered over.
- **visible** - Every control has visible property which is `True` by default - control is rendered on the page.

## Properties

### **buildbadgeclass-attributeinstance-attribute**

`badge: BadgeValue | None = None`

A badge to show on top of this control.

### **buildcolclass-attributeinstance-attribute**

`col: ResponsiveNumber = 12`

If a parent of this control is a `ResponsiveRow`, this property is used to determine how many virtual columns of a screen this control will span.

Can be a number or a dictionary configured to have a different value for specific breakpoints, for example `col={"sm": 6}`.

A value of 0 hides the control for that breakpoint, so it does not occupy any columns in the parent `ResponsiveRow`.

This control spans the 12 virtual columns by default.

Breakpoint	Dimension
xs	<576px
sm	≥576px
md	≥768px
lg	≥992px
xl	≥1200px
xxl	≥1400px

### **builddisabledclass-attributeinstance-attribute**

`disabled: bool = False`

Every control has disabled property which is `False` by default - control and all its children are enabled.

Note

The value of this property will be propagated down to all children controls recursively.

Example

For example, if you have a form with multiple entry controls you can disable them all together by disabling container:

```
ft.Column(
 disabled = True,
 controls=[
 ft.TextField(),
 ft.TextField()
]
)
```

## buildexpandclass-attributeinstance-attribute

expand: bool | int | None = None

Specifies whether/how this control should expand to fill available space in its parent layout.

More information here.

### Note

Has effect only if the direct parent of this control is one of the following controls, or their subclasses: Column, Row, View, Page.

Raises:

- ValueError - If it is not of type bool or int.

## buildexpand\_looseclass-attributeinstance-attribute

expand\_loose: bool = False

Allows the control to expand along the main axis if space is available, but does not require it to fill all available space.

More information here.

### Note

If expand\_loose is True, it will have effect only if:

- expand is not None and
- the direct parent of this control is one of the following controls, or their subclasses: Column, Row, View, Page.

## buildopacityclass-attributeinstance-attribute

opacity: Number = 1.0

Defines the transparency of the control.

Value ranges from 0.0 (completely transparent) to 1.0 (completely opaque without any transparency).

Raises:

- ValueError - If it is not between 0.0 and 1.0, inclusive.

## buildrtlclass-attributeinstance-attribute

rtl: bool = False

Whether the text direction of the control should be right-to-left (RTL).

## buildtooltipclass-attributeinstance-attribute

tooltip: TooltipValue | None = None

The tooltip to show when this control is hovered over.

## buildvisibleclass-attributeinstance-attribute

visible: bool = True

Every control has visible property which is True by default - control is rendered on the page. Setting visible to False completely prevents control (and all its children if any) from rendering on a page canvas. Hidden controls cannot be focused or selected with a keyboard or mouse and they do not emit any events.

# CupertinoActionSheet

An iOS-style action sheet.

Action sheets are generally used to give the user a choice between two or more choices for the current context.

Example:

```
sheet = ft.CupertinoActionSheet(
 title=ft.Text("Choose an option"),
 message=ft.Text("Select what you would like to do"),
 cancel=ft.CupertinoActionSheetAction(content=ft.Text("Cancel")),
 actions=[
 ft.CupertinoActionSheetAction(content=ft.Text("Save")),
 ft.CupertinoActionSheetAction(
 content=ft.Text("Delete"), destructive=True
),
],
)
page.show_dialog(ft.CupertinoBottomSheet(sheet))
```

Image: CupertinoActionSheet -

[https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_action\\_sheet/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_action_sheet/image_for_docs.png)

## CupertinoActionSheet

Inherits: `LayoutControl`

## Properties

- `actions` - A list of action buttons to be shown in the sheet.
- `cancel` - An optional control to be shown below the actions but grouped separately from them.
- `message` - A control containing a descriptive message that provides more details about the reason for the alert.
- `title` - A control containing the title of the action sheet.

## Examples

### CupertinoActionSheet

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 messages = ft.Column(tight=True)

 def handle_click(e: ft.Event[ft.CupertinoActionSheetAction]):
 messages.controls.append(ft.Text(f"Action clicked: {e.control.content.value}"))
 page.pop_dialog()

 action_sheet = ft.CupertinoActionSheet(
 title=ft.Row(
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[ft.Text("Title"), ft.Icon(ft.Icons.BEDTIME)],
),
 message=ft.Row(
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[ft.Text("Description"), ft.Icon(ft.Icons.AUTO_AWESOME)],
),
 cancel=ft.CupertinoActionSheetAction(
 on_click=handle_click,
 content=ft.Text("Cancel"),
),
 actions=[
 ft.CupertinoActionSheetAction(
 default=True,
 on_click=handle_click,
 content=ft.Text("Default Action"),
),
 ft.CupertinoActionSheetAction(
 on_click=handle_click,
 content=ft.Text("Normal Action"),
),
 ft.CupertinoActionSheetAction(
 destructive=True,
```

```

 on_click=handle_click,
 content=ft.Text("Destructive Action"),
),
],
)

bottom_sheet = ft.CupertinoBottomSheet(action_sheet)

page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.CupertinoFilledButton(
 on_click=lambda _: page.show_dialog(bottom_sheet),
 content="Open CupertinoBottomSheet",
),
 messages,
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_action\\_sheet/media/basic.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_action_sheet/media/basic.png)

## Properties

### buildactionsclass-attributeinstance-attribute

actions: list[Control] | None = None

A list of action buttons to be shown in the sheet.

These actions are typically CupertinoActionSheetActions.

Raises:

- ValueError - If none of actions, title, message, or cancel are provided.

### buildcancelclass-attributeinstance-attribute

cancel: Control | None = None

An optional control to be shown below the actions but grouped separately from them.

Typically a CupertinoActionSheetAction button.

### buildmessageclass-attributeinstance-attribute

message: StrOrControl | None = None

A control containing a descriptive message that provides more details about the reason for the alert.

Typically a Text control.

### buildtitleclass-attributeinstance-attribute

title: StrOrControl | None = None

A control containing the title of the action sheet.

Typically a Text control.



# CupertinoActionSheetAction

An action button typically used in a CupertinoActionSheet.

Inherits: `LayoutControl`

## Properties

- `content` - The child control to be shown in this action button.
- `default` - Whether this action should receive the style of an emphasized, default action.
- `destructive` - Whether this action should receive the style of a destructive action.
- `mouse_cursor` - Defines the mouse cursor for this action button.

## Events

- `on_click` - Called when this action button is clicked.

## Properties

### `buildcontentinstance-attribute`

`content: StrOrControl`

The child control to be shown in this action button.

Raises:

- `ValueError` - If it is neither a string nor a visible `Control`.

### `builddefaultclass-attributeinstance-attribute`

`default: bool = False`

Whether this action should receive the style of an emphasized, default action.

### `builddestructiveclass-attributeinstance-attribute`

`destructive: bool = False`

Whether this action should receive the style of a destructive action.

### `buildmouse_cursorclass-attributeinstance-attribute`

`mouse_cursor: MouseCursor | None = None`

Defines the mouse cursor for this action button.

## Events

### `bolton_clickclass-attributeinstance-attribute`

`on_click: (ControlEventHandler[CupertinoActionSheetAction] | None) = None`

Called when this action button is clicked.

# CupertinoActivityIndicator

An iOS-style activity indicator that spins clockwise.

```
ft.CupertinoActivityIndicator(
 radius=30,
 color=ft.CupertinoColors.DARK_BACKGROUND_GRAY,
)
```

Image: CupertinoActivityIndicator -

[https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_activity\\_indicator/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_activity_indicator/image_for_docs.png)

CupertinoActivityIndicator

Inherits: LayoutControl

## Properties

- **animating** - Whether this indicator is running its animation.
- **color** - Defines the color of this indicator.
- **progress** - Determines the percentage of spinner ticks that will be shown.
- **radius** - The radius of this indicator.

## Examples

### CupertinoActivityIndicator

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.vertical_alignment = ft.MainAxisAlignment.CENTER

 page.add(
 ft.SafeArea(
 content=ft.CupertinoActivityIndicator(
 animating=True,
 color=ft.Colors.RED,
 radius=50,
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_activity\\_indicator/media/basic.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_activity_indicator/media/basic.png)

## Properties

### buildanimatingclass-attributeinstance-attribute

animating: bool = True

Whether this indicator is running its animation.

Note

Has no effect if progress is not None.

### buildcolorclass-attributeinstance-attribute

color: ColorValue | None = None

Defines the color of this indicator.

### buildprogressclass-attributeinstance-attribute

progress: Number | None = None

Determines the percentage of spinner ticks that will be shown.

Typical usage would display all ticks, however, this allows for more fine-grained control such as during pull-to-refresh when the drag-down action shows one tick at a time as the user continues to drag down.

Note

If not None, then animating will be ignored.

Raises:

- ValueError - If it is not between 0.0 and 1.0, inclusive.

**buildradiusclass-attributeinstance-attribute**

`radius: Number = 10`

The radius of this indicator.

Raises:

- ValueError - If it is not strictly greater than 0.

# CupertinoAlertDialog

An iOS-style alert dialog.

An alert dialog informs the user about situations that require acknowledgement. An alert dialog has an optional title and an optional list of actions. The title is displayed above the content and the actions are displayed below the content.

Inherits: DialogControl

## Properties

- `actions` - A set of actions that are displayed at the bottom of the dialog.
- `barrier_color` - The color of the modal barrier below this dialog.
- `content` - The content of this dialog, displayed in a light font at the center of this dialog.
- `inset_animation` - The animation style to be used when the system keyboard intrudes into the space that the dialog is placed in.
- `modal` - Whether this dialog cannot be dismissed by clicking the area outside of it.
- `title` - The title of this dialog, displayed in a large font at the top of this dialog.

## Examples

### File deletion confirmation

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 messages = ft.Column(tight=True)

 def handle_dialog_dismissal(_: ft.Event[ft.DialogControl]):
 messages.controls.append(ft.Text("Dialog dismissed"))

 def handle_action_click(e: ft.Event[ft.CupertinoDialogAction]):
 messages.controls.append(ft.Text(f"Action clicked: {e.control.content}"))
 page.pop_dialog()

 cupertino_alert_dialog = ft.CupertinoAlertDialog(
 title=ft.Text("Cupertino Alert Dialog"),
 content=ft.Text("Do you want to delete this file?"),
 on_dismiss=handle_dialog_dismissal,
 actions=[
 ft.CupertinoDialogAction(
 destructive=True,
 on_click=handle_action_click,
 content="Yes",
),
 ft.CupertinoDialogAction(
 default=True,
 on_click=handle_action_click,
 content="No",
),
],
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.CupertinoFilledButton(
 on_click=lambda _: page.show_dialog(cupertino_alert_dialog),
 content="Open CupertinoAlertDialog",
),
 messages,
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Cupertino, material and adaptive

play\_arrowTry Online

```
from typing import Union

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.scroll = ft.ScrollMode.AUTO

 messages = ft.Column(tight=True)

 def handle_action_click(
 e: ft.Event[Union[ft.TextButton, ft.CupertinoDialogAction]],
):
 messages.controls.append(ft.Text(f"Action clicked: {e.control.content}"))
 page.pop_dialog()

 cupertino_actions = [
 ft.CupertinoDialogAction(
 destructive=True,
 on_click=handle_action_click,
 content="Yes",
),
 ft.CupertinoDialogAction(
 default=False,
 on_click=handle_action_click,
 content="No",
),
]

 material_actions = [
 ft.TextButton(on_click=handle_action_click, content="Yes"),
 ft.TextButton(on_click=handle_action_click, content="No"),
]

 page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.FilledButton(
 on_click=lambda _: page.show_dialog(
 ft.AlertDialog(
 title=ft.Text("Material Alert Dialog"),
 content=ft.Text("Do you want to delete this file?"),
 actions=material_actions,
)
),
 content="Open Material Dialog",
),
 ft.CupertinoFilledButton(
 on_click=lambda _: page.show_dialog(
 ft.CupertinoAlertDialog(
 title=ft.Text("Cupertino Alert Dialog"),
 content=ft.Text("Do you want to delete this file?"),
 actions=cupertino_actions,
)
),
 content="Open Cupertino Dialog",
),
 ft.FilledButton(
 adaptive=True,
 bgcolor=ft.Colors.BLUE_ACCENT,
 on_click=lambda _: page.show_dialog(
 ft.AlertDialog(
 adaptive=True,
 title=ft.Text("Adaptive Alert Dialog"),
 content=ft.Text("Do you want to delete this file?"),
 actions=(
 cupertino_actions
 if page.platform.is_apple()
 else material_actions
),
)
),
 content="Open Adaptive Dialog",
),
],
),
),
)
```

```
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### buildactionsclass-attributeinstance-attribute

```
actions: list[Control] = field(default_factory=list)
```

A set of actions that are displayed at the bottom of the dialog.

Typically this is a list of CupertinoDialogAction controls.

### buildbarrier\_colorclass-attributeinstance-attribute

```
barrier_color: ColorValue | None = None
```

The color of the modal barrier below this dialog.

If None, then DialogTheme.barrier\_color is used. If that is also None, the default is Colors.BLACK\_54.

### buildcontentclass-attributeinstance-attribute

```
content: Control | None = None
```

The content of this dialog, displayed in a light font at the center of this dialog.

Typically a Column that contains the dialog's Text message.

### buildinset\_animationclass-attributeinstance-attribute

```
inset_animation: Animation = field(
 default_factory=lambda: Animation(
 curve=AnimationCurve.DECELERATE,
 duration=Duration(milliseconds=100),
)
)
```

The animation style to be used when the system keyboard intrudes into the space that the dialog is placed in.

### buildmodalclass-attributeinstance-attribute

```
modal: bool = False
```

Whether this dialog cannot be dismissed by clicking the area outside of it.

### buildtitleclass-attributeinstance-attribute

```
title: StrOrControl | None = None
```

The title of this dialog, displayed in a large font at the top of this dialog.

Typically a Text control.

# CupertinoAppBar

An iOS-styled app bar.

## Note

The alignment of the title depends on whether this app bar is large or not. If it is True, the title is left-aligned and if it is False (the default), the title is centered.

Inherits: Control

## Properties

- `automatic_background_visibility` - Whether the navigation bar should appear transparent when content is scrolled under it.
- `automatically_imply_leading` - Whether we should try to imply the leading control if None.
- `automatically_imply_title` - Whether we should try to imply the title control if None.
- `bgcolor` - The fill color to use for this app bar.
- `border` - The border of the app bar.
- `brightness` - The brightness of the specified bgcolor.
- `enable_background_filter_blur` - Whether to have a blur effect when a non-opaque bgcolor is used.
- `large` - Whether to use a large app bar layout.
- `leading` - A control to display at the start of this app bar.
- `padding` - Defines the padding for the contents of the app bar.
- `previous_page_title` - Manually specify the previous route's title when automatically implying the leading back button.
- `title` - A string or a control to display in the middle of this app bar.
- `trailing` - A Control to place at the end of the app bar.
- `transition_between_routes` - Determines whether the app bar transitions between routes.

## Examples

### CupertinoAppBar

`play_arrow`Try Online

```
import flet as ft

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT

 page.appbar = ft.CupertinoAppBar(
 leading=ft.Icon(ft.Icons.PALETTE, color=ft.Colors.ON_SECONDARY),
 title=ft.Text("CupertinoAppBar Example"),
 trailing=ft.Icon(ft.Icons.WB_SUNNY_OUTLINED, color=ft.Colors.ON_SECONDARY),
 automatic_background_visibility=False,
 bgcolor=ft.Colors.SECONDARY,
 brightness=ft.Brightness.LIGHT,
)

 page.add(
 ft.SafeArea(
 content=ft.Text("Body!"),
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_app\\_bar/media/basic.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_app_bar/media/basic.png)

### Theme mode toggle

`play_arrow`Try Online

```

import flet as ft

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT

 def handle_theme_mode_toggle(_: ft.Event[ft.IconButton]):
 page.theme_mode = (
 ft.ThemeMode.DARK
 if page.theme_mode == ft.ThemeMode.LIGHT
 else ft.ThemeMode.LIGHT
)
 theme_mode_button.icon = (
 ft.Icons.WB_SUNNY_OUTLINED
 if page.theme_mode == ft.ThemeMode.LIGHT
 else ft.Icons.WB_SUNNY
)

 theme_mode_button = ft.IconButton(
 icon=(
 ft.Icons.WB_SUNNY_OUTLINED
 if page.theme_mode == ft.ThemeMode.LIGHT
 else ft.Icons.WB_SUNNY
),
 icon_color=ft.Colors.ON_INVERSE_SURFACE,
 on_click=handle_theme_mode_toggle,
)

 page.appbar = ft.CupertinoAppBar(
 automatic_background_visibility=False,
 leading=ft.Icon(ft.Icons.PALETTE, color=ft.Colors.ON_INVERSE_SURFACE),
 bgcolor=ft.Colors.INVERSE_SURFACE,
 trailing=theme_mode_button,
 title=ft.Text("CupertinoAppBar Example", color=ft.Colors.ON_INVERSE_SURFACE),
)

 page.add(
 ft.SafeArea(
 content=ft.Text("Body!"),
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildautomatic\_background\_visibilityclass-attributeinstance-attribute

automatic\_background\_visibility: bool | None = None

Whether the navigation bar should appear transparent when content is scrolled under it.

If False, the navigation bar will display its bgcolor.

### buildautomatically\_imply\_leadingclass-attributeinstance-attribute

automatically\_imply\_leading: bool | None = None

Whether we should try to imply the leading control if None.

- If True and leading is None, the app bar will automatically determine an appropriate leading control.
- If False and leading is None, the space is allocated to the title.
- If a leading control is provided, this parameter has no effect.

### buildautomatically\_imply\_titleclass-attributeinstance-attribute

automatically\_imply\_title: bool | None = None

Whether we should try to imply the title control if None.

- If True and title is None, a Text control containing the current route's title will be automatically filled in.
- If the title is not None, this parameter has no effect.

### buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The fill color to use for this app bar.

Default color is defined by current theme.



## buildborderclass-attributeinstance-attribute

`border: Border | None = None`

The border of the app bar. By default, a single pixel bottom border side is rendered.

## buildbrightnessclass-attributeinstance-attribute

`brightness: Brightness | None = None`

The brightness of the specified bgcolor.

Setting this value changes the style of the system status bar. It is typically used to increase the contrast ratio of the system status bar over bgcolor.

If None (the default), its value will be inferred from the relative luminance of the bgcolor.

## buildenable\_background\_filter\_blurclass-attributeinstance-attribute

`enable_background_filter_blur: bool | None = None`

Whether to have a blur effect when a non-opaque bgcolor is used.

This will only be respected when `automatic_background_visibility` is False or until content scrolls under the navigation bar.

## buildlargeclass-attributeinstance-attribute

`large: bool = False`

Whether to use a large app bar layout.

If True, the title is left-aligned; if False, the title is centered.

## buildleadingclass-attributeinstance-attribute

`leading: Control | None = None`

A control to display at the start of this app bar.

Typically the leading control is an Icon or an IconButton .

If it is None and `automatically_imply_leading` is True, an appropriate button will be automatically created.

## buildpaddingclass-attributeinstance-attribute

`padding: PaddingValue | None = None`

Defines the padding for the contents of the app bar.

If None, the app bar will adopt the following defaults:

- vertically, contents will be sized to the same height as the app bar itself minus the status bar.
- horizontally, padding will be 16 pixels according to iOS specifications unless the leading widget is an automatically inserted back button, in which case the padding will be 0.

Note

Vertical padding (top and bottom) won't change the height of this app bar.

## buildprevious\_page\_titleclass-attributeinstance-attribute

`previous_page_title: str | None = None`

Manually specify the previous route's title when automatically implying the leading back button.

Overrides the text shown with the back chevron instead of automatically showing the previous route's title when `automatically_imply_leading` is True.

Note

Has no effect if leading is not None or if `automatically_imply_leading` is False.

## buildtitleclass-attributeinstance-attribute

`title: StrOrControl | None = None`

A string or a control to display in the middle of this app bar.

Typically a Text.

## buildtrailingclass-attributeinstance-attribute

`trailing: Control | None = None`

A Control to place at the end of the app bar.

Typically used for actions such as searching or editing.

## **buildtransition\_between\_routesclass-attributeinstance-attribute**

`transition_between_routes: bool = True`

Determines whether the app bar transitions between routes.

If True, this app bar will animate on top of route transitions when the destination route also contains a CupertinoAppBar or CupertinoSliverAppBar with `transition_between_routes` set to True.

This transition also occurs during edge back swipe gestures, mimicking native iOS behavior.

### **Note**

When enabled, only one app bar can be present per route unless a `hero_tag` is specified.

# CupertinoBottomSheet

A Cupertino bottom sheet.

Inherits: DialogControl

## Properties

- bgcolor - The background color of this bottom sheet.
- content - The control to be displayed in this bottom sheet.
- height - The height of this bottom sheet.
- modal - Whether this bottom sheet can be dismissed/closed by clicking the area outside of it.
- padding - The sheet's padding.

## Examples

### CupertinoActionSheet

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 messages = ft.Column(tight=True)

 def handle_click(e: ft.Event[ft.CupertinoActionSheetAction]):
 messages.controls.append(ft.Text(f"Action clicked: {e.control.content.value}"))
 page.pop_dialog()

 action_sheet = ft.CupertinoActionSheet(
 title=ft.Row(
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[ft.Text("Title"), ft.Icon(ft.Icons.BEDTIME)],
),
 message=ft.Row(
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[ft.Text("Description"), ft.Icon(ft.Icons.AUTO_AWESOME)],
),
 cancel=ft.CupertinoActionSheetAction(
 on_click=handle_click,
 content=ft.Text("Cancel"),
),
 actions=[
 ft.CupertinoActionSheetAction(
 default=True,
 on_click=handle_click,
 content=ft.Text("Default Action"),
),
 ft.CupertinoActionSheetAction(
 on_click=handle_click,
 content=ft.Text("Normal Action"),
),
 ft.CupertinoActionSheetAction(
 destructive=True,
 on_click=handle_click,
 content=ft.Text("Destructive Action"),
),
],
),
)

bottom_sheet = ft.CupertinoBottomSheet(action_sheet)

page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.CupertinoFilledButton(
 on_click=lambda _: page.show_dialog(bottom_sheet),
 content="Open CupertinoBottomSheet",
),
],
 messages,
),
),
)
```

```
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

Image: cupertinoactionsheet - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_action\\_sheet/media/basic.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_action_sheet/media/basic.png)

## Properties

### buildbgcolorclass-attributeinstance-attribute

bgcolor: `ColorValue` | `None` = `None`

The background color of this bottom sheet.

### buildcontentinstance-attribute

content: `Control`

The control to be displayed in this bottom sheet.

### buildheightclass-attributeinstance-attribute

height: `Number` | `None` = `None`

The height of this bottom sheet.

### buildmodalclass-attributeinstance-attribute

modal: `bool` = `False`

Whether this bottom sheet can be dismissed/closed by clicking the area outside of it.

### buildpaddingclass-attributeinstance-attribute

padding: `PaddingValue` | `None` = `None`

The sheet's padding.

# CupertinoButton

An iOS-style button.

Example:

```
ft.CupertinoButton("Tap me")
```

Image: CupertinoButton - [https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_button/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_button/image_for_docs.png)

CupertinoButton

Inherits: LayoutControl

## Properties

- alignment - The alignment of this button's content.
- autofocus - Whether this button should be selected as the initial focus when no other node in its scope is currently focused.
- bgcolor - The background color of this button.
- border\_radius - The radius of the button's corners when it has a background color.
- color - The color of this button's text.
- content - The content of this button.
- disabled\_bgcolor - The background color of this button when disabled.
- focus\_color - The color to use for the focus highlight for keyboard interactions.
- icon - An icon shown in this button.
- icon\_color - The foreground color of the icon.
- min\_size - The minimum size of this button.
- mouse\_cursor - The cursor for a mouse pointer when it enters or is hovering over this button.
- opacity\_on\_click - Defines the opacity of the button when it is clicked.
- padding - The amount of space to surround the content control inside the bounds of the button.
- size - The size of this button.
- url - The URL to open when this button is clicked.

## Events

- on\_blur - Called when this button loses focus.
- on\_click - Called when a user clicks this button.
- on\_focus - Called when this button receives focus.
- on\_long\_press - Called when a user long-presses this button.

## Methods

- focus - Requests focus for this control.

## Examples

### CupertinoButton

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.CupertinoButton(
```

```

 bgcolor=ft.CupertinoColors.LIGHT_BACKGROUND_GRAY,
 opacity_on_click=0.3,
 on_click=lambda _: print("Normal CupertinoButton clicked!"),
 content=ft.Text(
 value="Normal CupertinoButton",
 color=ft.CupertinoColors.DESTRUCTIVE_RED,
),
),
 ft.CupertinoButton(
 bgcolor=ft.Colors.PRIMARY,
 alignment=ft.Alignment.TOP_LEFT,
 border_radius=ft.BorderRadius.all(15),
 opacity_on_click=0.5,
 on_click=lambda _: print("Filled CupertinoButton clicked!"),
 content=ft.Text(
 "Filled CupertinoButton",
 color=ft.Colors.YELLOW,
),
),
 ft.CupertinoButton(
 bgcolor=ft.Colors.PRIMARY,
 disabled=True,
 alignment=ft.Alignment.TOP_LEFT,
 opacity_on_click=0.5,
 content=ft.Text("Disabled CupertinoButton"),
),
 ft.Button(
 adaptive=True,
 bgcolor=ft.CupertinoColors.SYSTEM_TEAL,
 content=ft.Row(
 tight=True,
 controls=[
 ft.Icon(ft.Icons.FAVORITE, color="pink"),
 ft.Text("Button+adaptive"),
],
),
),
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_button/media/basic.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_button/media/basic.png)

## Properties

### buildalignmentclass-attributeinstance-attribute

```

alignment: Alignment | None = field(
 default_factory=lambda: Alignment.CENTER
)

```

The alignment of this button's content.

Typically buttons are sized to be just big enough to contain the child and its padding. If the button's size is constrained to a fixed size, this property defines how the child is aligned within the available space.

### buildautofocusclass-attributeinstance-attribute

```

autofocus: bool = False

```

Whether this button should be selected as the initial focus when no other node in its scope is currently focused.

### buildbgcolorclass-attributeinstance-attribute

```

bgcolor: ColorValue | None = None

```

The background color of this button.

### buildborder\_radiusclass-attributeinstance-attribute

```

border_radius: BorderRadiusValue = field(
 default_factory=lambda: BorderRadius.all(8.0)
)

```

The radius of the button's corners when it has a background color.

### buildcolorclass-attributeinstance-attribute

```

color: ColorValue | None = None

```

The color of this button's text.

### **buildcontentclass-attributeinstance-attribute**

`content: StrOrControl | None = None`

The content of this button.

### **builddisabled\_bgcolorclass-attributeinstance-attribute**

`disabled_bgcolor: ColorValue | None = None`

The background color of this button when disabled.

### **buildfocus\_colorclass-attributeinstance-attribute**

`focus_color: ColorValue | None = None`

The color to use for the focus highlight for keyboard interactions.

Defaults to a slightly transparent bgcolor. If bgcolor is None, defaults to a slightly transparent CupertinoColors.ACTIVE\_BLUE. 'Slightly transparent' in this context means the color is used with an opacity of 0.80, a brightness of 0.69 and a saturation of 0.835.

### **buildiconclass-attributeinstance-attribute**

`icon: IconDataOrControl | None = None`

An icon shown in this button.

### **buildicon\_colorclass-attributeinstance-attribute**

`icon_color: ColorValue | None = None`

The foreground color of the icon.

### **buildmin\_sizeclass-attributeinstance-attribute**

`min_size: Size | None = None`

The minimum size of this button.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: MouseCursor | None = None`

The cursor for a mouse pointer when it enters or is hovering over this button.

### **buildopacity\_on\_clickclass-attributeinstance-attribute**

`opacity_on_click: Number = 0.4`

Defines the opacity of the button when it is clicked.

When not pressed, the button has an opacity of 1.0.

Raises:

- `ValueError` - If it is not between 0.0 and 1.0, inclusive.

### **buildpaddingclass-attributeinstance-attribute**

`padding: PaddingValue | None = None`

The amount of space to surround the content control inside the bounds of the button.

### **buildsizeclass-attributeinstance-attribute**

`size: CupertinoButtonSize = CupertinoButtonSize.LARGE`

The size of this button.

### **buildurlclass-attributeinstance-attribute**

`url: str | Url | None = None`

The URL to open when this button is clicked.

Additionally, if on\_click event callback is provided, it is fired after that.

## **Events**

### **bolton\_blurclass-attributeinstance-attribute**

```
on_blur: ControlEventHandler[CupertinoButton] | None = None
```

Called when this button loses focus.

### **bolton\_clickclass-attributeinstance-attribute**

```
on_click: ControlEventHandler[CupertinoButton] | None = None
```

Called when a user clicks this button.

### **bolton\_focusclass-attributeinstance-attribute**

```
on_focus: ControlEventHandler[CupertinoButton] | None = None
```

Called when this button receives focus.

### **bolton\_long\_pressclass-attributeinstance-attribute**

```
on_long_press: (
 ControlEventHandler[CupertinoButton] | None
) = None
```

Called when a user long-presses this button.

## **Methods**

### **deployed\_codefocusasync**

```
focus()
```

Requests focus for this control.



# CupertinoCheckbox

A macOS style checkbox.

Checkbox allows to select one or more items from a group, or switch between two mutually exclusive options (checked or unchecked, on or off).

```
ft.Column(
 intrinsic_width=True,
 controls=[
 ft.CupertinoCheckbox(),
 ft.CupertinoCheckbox(label="Checked", value=True),
 ft.CupertinoCheckbox(label="Disabled", disabled=True),
],
)
```

Image: CupertinoCheckbox - [https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_checkbox/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_checkbox/image_for_docs.png)

CupertinoCheckbox

Inherits: LayoutControl

## Properties

- `active_color` - The color used to fill checkbox when it is checked/selected.
- `autofocus` - Whether this checkbox will be selected as the initial focus.
- `border_side` - Defines the checkbox's border sides in all or specific `ControlState` states.
- `check_color` - The color to use for the check icon when this checkbox is checked.
- `fill_color` - The color used to fill this checkbox in all or specific `ControlState` states.
- `focus_color` - The color used for this checkbox's border shadow when it has the input focus.
- `label` - A clickable label to display on the right of this checkbox.
- `label_position` - Defines on which side of this checkbox the label should be shown.
- `mouse_cursor` - The cursor for a mouse pointer entering or hovering over this checkbox.
- `semantics_label` - The semantic label for this checkbox that will be announced by screen readers.
- `shape` - The shape of this checkbox.
- `spacing` - The space between this checkbox and the label.
- `tristate` - If True, this checkbox's value can be True, False, or None.
- `value` - The value of this checkbox.

## Events

- `on_blur` - Called when this checkbox has lost focus.
- `on_change` - Called when the state of this checkbox is changed.
- `on_focus` - Called when this checkbox has received focus.

## Examples

### Cupertino, material and adaptive

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.CupertinoCheckbox(label="Cupertino Checkbox", value=True),
 ft.Checkbox(label="Material Checkbox", value=True),
 ft.Container(height=20),
 ft.Text(
 value=
```

```

 "Adaptive Checkbox shows as CupertinoCheckbox on macOS "
 "and iOS and as Checkbox on other platforms:"
),
),
ft.Checkbox(
 adaptive=True,
 label="Adaptive Checkbox",
 value=True,
),
],
),
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: cupertino-material-and-adaptive -

[https://flet.dev/docs/examples/controls/cupertino/cupertino\\_checkbox/media/cupertino\\_material\\_and\\_adaptive.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_checkbox/media/cupertino_material_and_adaptive.png)

## Styled

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.CupertinoCheckbox(
 label="Cupertino Checkbox tristate",
 value=True,
 tristate=True,
 check_color=ft.Colors.GREY_900,
 fill_color={
 ft.ControlState.HOVERED: ft.Colors.PINK_200,
 ft.ControlState.PRESSED: ft.Colors.LIME_ACCENT_200,
 ft.ControlState.SELECTED: ft.Colors.DEEP_ORANGE_200,
 ft.ControlState.DEFAULT: ft.Colors.TEAL_200,
 },
),
 ft.CupertinoCheckbox(
 label="Cupertino Checkbox circle border",
 value=True,
 shape=ft.CircleBorder(),
),
 ft.CupertinoCheckbox(
 label="Cupertino Checkbox border states",
 value=True,
),
 ft.CupertinoCheckbox(
 label="Cupertino Checkbox label position",
 value=True,
 label_position=ft.LabelPosition.LEFT,
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

**buildactive\_color**class-attributeinstance-attribute

```

active_color: ColorValue | None = (
 CupertinoColors.ACTIVE_BLUE
)

```

The color used to fill checkbox when it is checked/selected.

If fill\_color returns a non-null color in the ControlState.SELECTED state, it will be used instead of this color.

**buildautofocus**class-attributeinstance-attribute

```

autofocus: bool = False

```

Whether this checkbox will be selected as the initial focus. If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

### **buildborder\_sideclass-attributeinstance-attribute**

`border_side: ControlStateValue[BorderSide] | None = None`

Defines the checkbox's border sides in all or specific ControlState states.

Note

Supported states: ControlState.SELECTED, ControlState.HOVERED, ControlState.DISABLED, ControlState.FOCUSED, ControlState.PRESSED, ControlState.ERROR, and ControlState.DEFAULT.

### **buildcheck\_colorclass-attributeinstance-attribute**

`check_color: ColorValue | None = None`

The color to use for the check icon when this checkbox is checked.

### **buildfill\_colorclass-attributeinstance-attribute**

`fill_color: ControlStateValue[ColorValue] | None = None`

The color used to fill this checkbox in all or specific ControlState states.

`active_color` is used as fallback color when the checkbox is in the SELECTED state, CupertinoColors.WHITE at 50% opacity is used as fallback color when this checkbox is in the DISABLED state, and CupertinoColors.WHITE otherwise.

Note

Supported states: ControlState.SELECTED, ControlState.HOVERED, ControlState.DISABLED, ControlState.FOCUSED, and ControlState.DEFAULT.

### **buildfocus\_colorclass-attributeinstance-attribute**

`focus_color: ColorValue | None = None`

The color used for this checkbox's border shadow when it has the input focus.

### **buildlabelclass-attributeinstance-attribute**

`label: str | None = None`

A clickable label to display on the right of this checkbox.

### **buildlabel\_positionclass-attributeinstance-attribute**

`label_position: LabelPosition = LabelPosition.RIGHT`

Defines on which side of this checkbox the label should be shown.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: MouseCursor | None = None`

The cursor for a mouse pointer entering or hovering over this checkbox.

### **buildsemantics\_labelclass-attributeinstance-attribute**

`semantics_label: str | None = None`

The semantic label for this checkbox that will be announced by screen readers.

This is announced by assistive technologies (e.g TalkBack/VoiceOver) and not shown on the UI.

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

The shape of this checkbox.

Internally defaults to `RoundedRectangleBorder(radius=4)`.

### **buildspacingclass-attributeinstance-attribute**

`spacing: Number | None = 10`

The space between this checkbox and the label.

### **buildtristateclass-attributeinstance-attribute**

`tristate: bool = False`

If True, this checkbox's value can be True, False, or None.

### **buildvalueclass-attributeinstance-attribute**

```
value: bool | None = False
```

The value of this checkbox.

- If True, this checkbox is checked.
- If False, this checkbox is unchecked.
- If None and tristate is True, this checkbox is indeterminate (displayed as a dash).

## **Events**

### **bolton\_blurclass-attributeinstance-attribute**

```
on_blur: ControlEventHandler[CupertinoCheckbox] | None = (
 None
)
```

Called when this checkbox has lost focus.

### **bolton\_changeclass-attributeinstance-attribute**

```
on_change: ControlEventHandler[CupertinoCheckbox] | None = (
 None
)
```

Called when the state of this checkbox is changed.

### **bolton\_focusclass-attributeinstance-attribute**

```
on_focus: ControlEventHandler[CupertinoCheckbox] | None = (
 None
)
```

Called when this checkbox has received focus.

# CupertinoContextMenu

A full-screen modal route that opens up when the content is long-pressed.

Inherits: `AdaptiveControl`

## Properties

- `actions` - A list of action buttons to be shown in the menu.
- `content` - The content of this context menu.
- `enable_haptic_feedback` - Whether a click on the actions should produce haptic feedback.

## Examples

### CupertinoContextMenu

[play\\_arrow](#) [Try Online](#)

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.vertical_alignment = ft.MainAxisAlignment.CENTER

 page.add(
 ft.SafeArea(
 content=ft.CupertinoContextMenu(
 enable_haptic_feedback=True,
 actions=[
 ft.CupertinoContextMenuAction(
 default=True,
 trailing_icon=ft.Icons.CHECK,
 on_click=lambda _: print("Action 1"),
 content="Action 1",
),
 ft.CupertinoContextMenuAction(
 trailing_icon=ft.Icons.MORE,
 on_click=lambda _: print("Action 2"),
 content="Action 2",
),
 ft.CupertinoContextMenuAction(
 destructive=True,
 trailing_icon=ft.Icons.CANCEL,
 on_click=lambda _: print("Action 3"),
 content="Action 3",
),
],
 content=ft.Image("https://picsum.photos/200/200"),
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_context\\_menu/media/basic.gif](https://flet.dev/docs/examples/controls/cupertino/cupertino_context_menu/media/basic.gif)

## Properties

### `buildactionsinstance-attribute`

`actions: list[Control]`

A list of action buttons to be shown in the menu.

Typically `CupertinoContextMenuActions`.

Raises:

- `ValueError` - If it does not contain at least one visible `Control`.

### `buildcontentinstance-attribute`

`content: Control`

The content of this context menu.

#### Info

When this context menu is long-pressed, the menu will open and this control will be moved to the new route and be expanded. This allows the content to resize to fit in its place in the new route, if it doesn't size itself.

Raises:

- `ValueError` - If it is not visible.

#### `buildenable_haptic_feedbackclass-attributeinstance-attribute`

`enable_haptic_feedback: bool = True`

Whether a click on the actions should produce haptic feedback.

# CupertinoContextMenuAction

A cupertino context menu action.

Typically used as a child of CupertinoContextMenu.actions.

Inherits: AdaptiveControl

## Properties

- `content` - The content of this action button.
- `default` - Whether this action should receive the style of an emphasized, default action.
- `destructive` - Whether this action should receive the style of a destructive action.
- `trailing_icon` - An icon to display at the right of the content control.

## Events

- `on_click` - Called when this action button is clicked.

## Properties

### `buildcontentinstance-attribute`

`content: StrOrControl`

The content of this action button.

Raises:

- `ValueError` - If it is neither a string nor a visible Control.

### `builddefaultclass-attributeinstance-attribute`

`default: bool = False`

Whether this action should receive the style of an emphasized, default action.

### `builddestructiveclass-attributeinstance-attribute`

`destructive: bool = False`

Whether this action should receive the style of a destructive action.

### `buildtrailing_iconclass-attributeinstance-attribute`

`trailing_icon: IconData | None = None`

An icon to display at the right of the content control.

## Events

### `bolton_clickclass-attributeinstance-attribute`

```
on_click: (
 ControlEventHandler[CupertinoContextMenuAction] | None
) = None
```

Called when this action button is clicked.

# CupertinoDatePicker

An iOS-styled date picker.

Inherits: `LayoutControl`

## Properties

- `bgcolor` - The background color of this date picker.
- `date_order` - The order in which the columns inside this picker are displayed.
- `date_picker_mode` - The mode of the date picker.
- `first_date` - The earliest allowable date that the user can select.
- `item_extent` - The uniform height of all children.
- `last_date` - The latest allowable date that the user can select.
- `locale` - The locale for this date picker.
- `maximum_year` - Maximum year to which the picker can be scrolled when in `CupertinoDatePickerMode.DATE` mode.
- `minimum_year` - Minimum year to which the picker can be scrolled when in `CupertinoDatePickerMode.DATE` mode.
- `minute_interval` - The granularity of the minutes spinner, if it is shown in the current `date_picker_mode`.
- `show_day_of_week` - Whether to show day of week alongside day.
- `use_24h_format` - Whether to use the 24-hour time format.
- `value` - The initial date and/or time of the picker.

## Events

- `on_change` - Called when the selected date and/or time changes.

## Examples

### CupertinoDatePicker

`play_arrow` Try Online

```
from datetime import datetime

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 message = ft.Text("Chosen Time:")

 def handle_date_change(e: ft.Event[ft.CupertinoDatePicker]):
 message.value = f"Chosen Date: {e.control.value.strftime('%Y-%m-%d %H:%M %p')}"

 cupertino_date_picker = ft.CupertinoDatePicker(
 value=datetime.now(),
 date_picker_mode=ft.CupertinoDatePickerMode.DATE_AND_TIME,
 on_change=handle_date_change,
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.CupertinoFilledButton(
 on_click=lambda _: page.show_dialog(
 ft.CupertinoBottomSheet(
 height=216,
 padding=ft.Padding.only(top=6),
 content=cupertino_date_picker,
)
),
),
 content="Open CupertinoDatePicker",
],
),
 message,
```



```

),
],
)

if __name__ == "__main__":
 ft.run(main)

```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_date\\_picker/media/basic.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_date_picker/media/basic.png)

## Properties

### buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue | None = None
```

The background color of this date picker.

### builddate\_orderclass-attributeinstance-attribute

```
date_order: CupertinoDatePickerDateOrder | None = None
```

The order in which the columns inside this picker are displayed.

#### Note

The final order in which the columns are displayed is also influenced by the `date_picker_mode`. For example, if `date_picker_mode` is `CupertinoDatePickerMode.MONTH_YEAR` both `CupertinoDatePickerDateOrder.DAY_MONTH_YEAR` and `CupertinoDatePickerDateOrder.MONTH_DAY_YEAR` will result in the month|year order.

### builddate\_picker\_modeclass-attributeinstance-attribute

```
date_picker_mode: CupertinoDatePickerMode = (
 CupertinoDatePickerMode.DATE_AND_TIME
)
```

The mode of the date picker.

### buildfirst\_dateclass-attributeinstance-attribute

```
first_date: DateTimeValue | None = None
```

The earliest allowable date that the user can select.

- If set to `None` (the default), there is no lower date limit.
- When not `None`, one can still scroll the picker to dates earlier than `first_date`, with the exception that the `on_change` will not be called. Once let go, the picker will scroll back to `first_date`.

#### Note

In `CupertinoDatePickerMode.TIME` mode, a time becomes unselectable if the datetime produced by combining that particular time and the date part of value is earlier than `last_date`. So typically, `first_date` needs to be set to a datetime that is on the same date as value.

### builditem\_extentclass-attributeinstance-attribute

```
item_extent: Number = 32.0
```

The uniform height of all children.

Raises:

- `ValueError` - If it is not strictly greater than 0.

### buildlast\_dateclass-attributeinstance-attribute

```
last_date: DateTimeValue | None = None
```

The latest allowable date that the user can select.

- If set to `None` (the default), there is no upper date limit.
- When not `None`, one can still scroll the picker to dates later than `last_date`, with the exception that the `on_change` will not be called. Once let go, the picker will scroll back to `last_date`.

#### Note

In CupertinoDatePickerMode.TIME mode, a time becomes unselectable if the datetime produced by combining that particular time and the date part of value is later than last\_date. So typically, last\_date needs to be set to a datetime that is on the same date as value.

### buildlocaleclass-attributeinstance-attribute

locale: Locale | None = None

The locale for this date picker. It is intended for rare cases where this control should be localized differently from the rest of the page.

#### Notes

- The locale must be supported by Flutter's global localization delegates; otherwise the override is ignored and the control uses the page or system locale.
- If None (the default), the page or system locale is used.

### buildmaximum\_yearclass-attributeinstance-attribute

maximum\_year: int | None = None

Maximum year to which the picker can be scrolled when in CupertinoDatePickerMode.DATE mode.

Defaults to None - no limit.

#### Raises:

- ValueError - If it is less than value year.

### buildminimum\_yearclass-attributeinstance-attribute

minimum\_year: int = 1

Minimum year to which the picker can be scrolled when in CupertinoDatePickerMode.DATE mode.

#### Raises:

- ValueError - If it is greater than value year.

### buildminute\_intervalclass-attributeinstance-attribute

minute\_interval: int = 1

The granularity of the minutes spinner, if it is shown in the current date\_picker\_mode.

#### Raises:

- ValueError - If it is not strictly greater than 0.
- ValueError - If it is not a factor of 60.

### buildshow\_day\_of\_weekclass-attributeinstance-attribute

show\_day\_of\_week: bool = False

Whether to show day of week alongside day.

#### Raises:

- ValueError - If it is set when date\_picker\_mode is not CupertinoDatePickerMode.DATE.

### builduse\_24h\_formatclass-attributeinstance-attribute

use\_24h\_format: bool = False

Whether to use the 24-hour time format.

If False, the 12-hour time format is used.

### buildvalueclass-attributeinstance-attribute

```
value: DateTimeValue = field(
 default_factory=lambda: datetime.now()
)
```

The initial date and/or time of the picker.

Defaults to the present date and time.

#### Raises:

- ValueError - If it is not greater than or equal to first\_date, when first\_date is set.

- ValueError - If it is not less than or equal to `last_date`, when `last_date` is set.
- ValueError - If its year is not greater than or equal to `minimum_year`, when `date_picker_mode` is `CupertinoDatePickerMode.DATE` or `CupertinoDatePickerMode.MONTH_YEAR`.
- ValueError - If its year is not less than or equal to `maximum_year`, when `maximum_year` is set.
- ValueError - If its minute is not a multiple of `minute_interval`.

## Events

### `bolton_change` class-attribute instance-attribute

```
on_change: (
 ControlEventHandler[CupertinoDatePicker] | None
) = None
```

Called when the selected date and/or time changes.

Will not be called if the new selected value is not valid, or is not in the range of `first_date` and `last_date`.

# CupertinoDialogAction

## Examples

### File deletion confirmation

play\_arrow Try Online

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 messages = ft.Column(tight=True)

 def handle_dialog_dismissal(_: ft.Event[ft.DialogControl]):
 messages.controls.append(ft.Text("Dialog dismissed"))

 def handle_action_click(e: ft.Event[ft.CupertinoDialogAction]):
 messages.controls.append(ft.Text(f"Action clicked: {e.control.content}"))
 page.pop_dialog()

 cupertino_alert_dialog = ft.CupertinoAlertDialog(
 title=ft.Text("Cupertino Alert Dialog"),
 content=ft.Text("Do you want to delete this file?"),
 on_dismiss=handle_dialog_dismissal,
 actions=[
 ft.CupertinoDialogAction(
 destructive=True,
 on_click=handle_action_click,
 content="Yes",
),
 ft.CupertinoDialogAction(
 default=True,
 on_click=handle_action_click,
 content="No",
),
],
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.CupertinoFilledButton(
 on_click=lambda _: page.show_dialog(cupertino_alert_dialog),
 content="Open CupertinoAlertDialog",
),
 messages,
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: file-deletion-confirmation - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_alert\\_dialog/media/file\\_deletion\\_confirmation.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_alert_dialog/media/file_deletion_confirmation.png)

## Cupertino, material and adaptive

play\_arrow Try Online

```
from typing import Union

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.scroll = ft.ScrollMode.AUTO

 messages = ft.Column(tight=True)

 def handle_action_click(
 e: ft.Event[Union[ft.TextButton, ft.CupertinoDialogAction]],
):
 messages.controls.append(ft.Text(f"Action clicked: {e.control.content}"))
 page.pop_dialog()
```

```

cupertino_actions = [
 ft.CupertinoDialogAction(
 destructive=True,
 on_click=handle_action_click,
 content="Yes",
),
 ft.CupertinoDialogAction(
 default=False,
 on_click=handle_action_click,
 content="No",
),
]

material_actions = [
 ft.TextButton(on_click=handle_action_click, content="Yes"),
 ft.TextButton(on_click=handle_action_click, content="No"),
]

page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.FilledButton(
 on_click=lambda _: page.show_dialog(
 ft.AlertDialog(
 title=ft.Text("Material Alert Dialog"),
 content=ft.Text("Do you want to delete this file?"),
 actions=material_actions,
)
),
 content="Open Material Dialog",
),
 ft.CupertinoFilledButton(
 on_click=lambda _: page.show_dialog(
 ft.CupertinoAlertDialog(
 title=ft.Text("Cupertino Alert Dialog"),
 content=ft.Text("Do you want to delete this file?"),
 actions=cupertino_actions,
)
),
 content="Open Cupertino Dialog",
),
 ft.FilledButton(
 adaptive=True,
 bgcolor=ft.Colors.BLUE_ACCENT,
 on_click=lambda _: page.show_dialog(
 ft.AlertDialog(
 adaptive=True,
 title=ft.Text("Adaptive Alert Dialog"),
 content=ft.Text("Do you want to delete this file?"),
 actions=(
 /cupertino_actions
 if page.platform.is_apple()
 else material_actions
),
)
),
 content="Open Adaptive Dialog",
),
 messages,
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

A dialog action button.

Typically used as a child of `CupertinoAlertDialog.actions`.

Inherits: `Control`

## Properties

- `content` - The content of this action button.
- `default` - Whether this action is a default action.
- `destructive` - If set to `True`, this button's text color will be red.
- `text_style` - The text style to use for text in this button.

## Events

- `on_click` - Called when a user clicks this button.

## Properties

### `buildcontentinstance-attribute`

`content: StrOrControl`

The content of this action button.

Raises:

- `ValueError` - If it is neither a string nor a visible `Control`.

### `builddefaultclass-attributeinstance-attribute`

`default: bool = False`

Whether this action is a default action. In this case, the button will have bold text.

Info

Multiple actions can have this property set to `True` in a `CupertinoAlertDialog`.

### `builddestructiveclass-attributeinstance-attribute`

`destructive: bool = False`

If set to `True`, this button's text color will be red.

Typically used for actions that destroy objects, such as an delete that deletes an email etc.

### `buildtext_styleclass-attributeinstance-attribute`

`text_style: TextStyle | None = None`

The text style to use for text in this button.

Can be useful when content is a string.

## Events

### `bolton_clickclass-attributeinstance-attribute`

```
on_click: (
 ControlEventHandler[CupertinoDialogAction] | None
) = None
```

Called when a user clicks this button.

# CupertinoFilledButton

An iOS-style button filled with default background color.

```
ft.CupertinoFilledButton("Tap me")
```

Image: CupertinoFilledButton -

[https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_filled\\_button/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_filled_button/image_for_docs.png)

CupertinoFilledButton

Inherits: CupertinoButton

## Examples

### CupertinoFilledButton

play\_arrowTry Online

```
import flet as ft
```

```
def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.CupertinoFilledButton(
 opacity_on_click=0.3,
 on_click=lambda _: print("CupertinoFilledButton clicked!"),
 content=ft.Text("CupertinoFilledButton"),
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_filled\\_button/media/basic.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_filled_button/media/basic.png)

# CupertinoListTile

An iOS-style list tile.

The CupertinoListTile is a Cupertino equivalent of the Material ListTile.

```
ft.CupertinoListTile(
 title="Notifications",
 subtitle="Enabled",
 width=400,
 leading=ft.Icon(ft.Icons.NOTIFICATIONS_OUTLINED),
 trailing=ft.Icon(ft.Icons.CHEVRON_RIGHT),
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
)
```

Image: CupertinoListTile - [https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_list\\_tile/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_list_tile/image_for_docs.png)

## CupertinoListTile

Inherits: LayoutControl

## Properties

- `additional_info` - A Control to display on the right of the list tile, before trailing.
- `bgcolor` - The background color of this list tile.
- `bgcolor_activated` - The background color of this list tile after it was tapped.
- `leading` - A control to display before the title.
- `leading_size` - Used to constrain the width and height of leading control.
- `leading_to_title` - The horizontal space between leading and title.
- `notched` - Whether this list tile should be created in an "Inset Grouped" form, known from either iOS Notes or Reminders app.
- `padding` - The tile's internal padding.
- `subtitle` - Additional content displayed below the title.
- `title` - The primary content of this list tile.
- `toggle_inputs` - Whether clicking on this tile should toggle the state of (visible) toggleable controls like Radio, Checkbox or Switch inside it.
- `trailing` - A control to display after the title.
- `url` - The URL to open when this button is clicked.

## Events

- `on_click` - Called when a user clicks/taps the list tile.

## Examples

### Notched

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 def handle_tile_click(_: ft.Event[ft.CupertinoListTile]):
 print("Tile clicked")

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.CupertinoListTile(
 additional_info=ft.Text("Wed Jan 24"),
 bgcolor_activated=ft.Colors.AMBER_ACCENT,
 leading=ft.Icon(ft.CupertinoIcons.GAME_CONTROLLER),
 title=ft.Text("CupertinoListTile: notched = False"),
 subtitle=ft.Text("Subtitle"),
 trailing=ft.Icon(ft.CupertinoIcons.ALARM),
)
]
)
)
)
```



```

 on_click=handle_tile_click,
),
 ft.CupertinoListTile(
 notched=True,
 additional_info=ft.Text("Thu Jan 25"),
 leading=ft.Icon(ft.CupertinoIcons.GAME_CONTROLLER),
 title=ft.Text("CupertinoListTile: notched = True"),
 subtitle=ft.Text("Subtitle"),
 trailing=ft.Icon(ft.CupertinoIcons.ALARM),
 on_click=handle_tile_click,
),
 1,
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: notched - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_list\\_tile/media/notched.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_list_tile/media/notched.png)

## Properties

### buildadditional\_infoclass-attributeinstance-attribute

`additional_info: StrOrControl | None = None`

A Control to display on the right of the list tile, before trailing.

Similar to subtitle, an `additional_info` is used to display additional information.

Typically a Text control.

### buildbgcolorclass-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

The background color of this list tile.

### buildbgcolor\_activatedclass-attributeinstance-attribute

`bgcolor_activated: ColorValue | None = None`

The background color of this list tile after it was tapped.

### buildleadingclass-attributeinstance-attribute

`leading: IconDataOrControl | None = None`

A control to display before the title.

### buildleading\_sizeclass-attributeinstance-attribute

`leading_size: Number | None = None`

Used to constrain the width and height of leading control.

Defaults to 30.0, if notched is True, else 28.0.

### buildleading\_to\_titleclass-attributeinstance-attribute

`leading_to_title: Number | None = None`

The horizontal space between leading and title.

Defaults to 12.0, if notched is True, else 16.0.

### buildnotchedclass-attributeinstance-attribute

`notched: bool = False`

Whether this list tile should be created in an "Inset Grouped" form, known from either iOS Notes or Reminders app.

### buildpaddingclass-attributeinstance-attribute

`padding: PaddingValue | None = None`

The tile's internal padding. Insets a CupertinoListTile's contents: its leading, title, subtitle, `additional_info` and trailing controls.

### buildsubtitleclass-attributeinstance-attribute

`subtitle: StrOrControl | None = None`

Additional content displayed below the title.

Typically a Text control.

### **buildtitleinstance-attribute**

```
title: StrOrControl
```

The primary content of this list tile.

Typically a Text control.

Raises:

- ValueError - If it is neither a string nor a visible Control.

### **buildtoggle\_inputsclass-attributeinstance-attribute**

```
toggle_inputs: bool = False
```

Whether clicking on this tile should toggle the state of (visible) toggleable controls like Radio, Checkbox or Switch inside it.

### **buildtrailingclass-attributeinstance-attribute**

```
trailing: IconDataOrControl | None = None
```

A control to display after the title.

Typically an Icon control.

### **buildurlclass-attributeinstance-attribute**

```
url: str | Url | None = None
```

The URL to open when this button is clicked.

Additionally, if on\_click event callback is provided, it is fired after that.

## **Events**

### **bolton\_clickclass-attributeinstance-attribute**

```
on_click: ControlEventHandler[CupertinoListTile] | None = (
 None
)
```

Called when a user clicks/taps the list tile.

# CupertinoNavigationBar

An iOS-styled bottom navigation tab bar.

Navigation bars offer a persistent and convenient way to switch between primary destinations in an app.

Inherits: `LayoutControl`

## Properties

- `active_color` - The foreground color of the icon and title of the selected destinations.
- `bgcolor` - The color of the navigation bar itself.
- `border` - Defines the border of this navigation bar.
- `destinations` - The destinations of this navigation bar.
- `icon_size` - The size of all destination icons.
- `inactive_color` - The foreground color of the icon and title of the unselected destinations.
- `selected_index` - The index into destinations for the currently selected `NavigationBarDestination`.

## Events

- `on_change` - Called when selected destination changed.

## Examples

### CupertinoNavigationBar

[play\\_arrow](#)Try Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "CupertinoNavigationBar Example"

 page.navigation_bar = ft.CupertinoNavigationBar(
 bgcolor=ft.Colors.AMBER_100,
 inactive_color=ft.Colors.GREY,
 active_color=ft.Colors.BLACK,
 on_change=lambda e: print("Selected tab:", e.control.selected_index),
 destinations=[
 ft.NavigationBarDestination(
 icon=ft.Icons.EXPLORE_OUTLINED,
 selected_icon=ft.Icons.EXPLORE,
 label="Explore",
),
 ft.NavigationBarDestination(
 icon=ft.Icons.COMMUTE_OUTLINED,
 selected_icon=ft.Icons.COMMUTE,
 label="Commute",
),
 ft.NavigationBarDestination(
 icon=ft.Icons.BOOKMARK_BORDER,
 selected_icon=ft.Icons.BOOKMARK,
 label="Favorites",
),
],
)

 page.add(
 ft.SafeArea(
 content=ft.Text("Body!"),
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_navigation\\_bar/media/basic.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_navigation_bar/media/basic.png)

## Wired

[play\\_arrow](#)Try Online

```

import flet as ft

def main(page: ft.Page):
 page.title = "CupertinoNavigationBar Example"

 body_text = ft.Text("Explore!")

 def handle_nav_destination_change(e: ft.Event[ft.CupertinoNavigationBar]):
 if e.control.selected_index == 0:
 body_text.value = "Explore!"
 elif e.control.selected_index == 1:
 body_text.value = "Find Your Way!"
 else:
 body_text.value = "Your Favorites!"

 page.navigation_bar = ft.CupertinoNavigationBar(
 bgcolor=ft.Colors.AMBER_100,
 inactive_color=ft.Colors.GREY,
 active_color=ft.Colors.BLACK,
 on_change=handle_nav_destination_change,
 destinations=[
 ft.NavigationBarDestination(
 icon=ft.Icons.EXPLORE_OUTLINED,
 selected_icon=ft.Icons.EXPLORE,
 label="Explore",
),
 ft.NavigationBarDestination(
 icon=ft.Icons.COMMUTE_OUTLINED,
 selected_icon=ft.Icons.COMMUTE,
 label="Commute",
),
 ft.NavigationBarDestination(
 icon=ft.Icons.BOOKMARK_BORDER,
 selected_icon=ft.Icons.BOOKMARK,
 label="Favorites",
),
],
)

 page.add(
 ft.SafeArea(
 content=body_text,
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildactive\_colorclass-attributeinstance-attribute

active\_color: ColorValue | None = None

The foreground color of the icon and title of the selected destinations.

### buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The color of the navigation bar itself.

### buildborderclass-attributeinstance-attribute

border: Border | None = None

Defines the border of this navigation bar.

### builddestinationsinstance-attribute

destinations: list[NavigationBarDestination]

The destinations of this navigation bar.

Raises:

- ValueError - If it does not contain at least two visible NavigationBarDestinations.

### buildicon\_sizeclass-attributeinstance-attribute

icon\_size: Number = 30

The size of all destination icons.

### **buildinactive\_color**class-attributeinstance-attribute

```
inactive_color: ColorValue = CupertinoColors.INACTIVE_GRAY
```

The foreground color of the icon and title of the unselected destinations.

### **buildselected\_index**class-attributeinstance-attribute

```
selected_index: int = 0
```

The index into destinations for the currently selected `NavigationBarDestination`.

Raises:

- `IndexError` - If it is not greater than or equal to 0.
- `IndexError` - If it is not less than the length of visible destinations.

## **Events**

### **bolton\_change**class-attributeinstance-attribute

```
on_change: (
 ControlEventHandler[CupertinoNavigationBar] | None
) = None
```

Called when selected destination changed.

# CupertinoPicker

An iOS-styled picker.

Inherits: `LayoutControl`

## Properties

- `bgcolor` - The background color of this timer picker.
- `controls` - A list of controls representing items in this picker.
- `default_selection_overlay_bgcolor` - The default background color of the `selection_overlay`.
- `diameter_ratio` - Relative ratio between this picker's height and the simulated cylinder's diameter.
- `item_extent` - The uniform height of all controls.
- `looping` - If `True`, children on a wheel can be scrolled in a loop.
- `magnification` - The zoomed-in or magnification rate of the magnifier.
- `off_axis_fraction` - How much the wheel is horizontally off-center, as a fraction of its width.
- `selected_index` - The index (starting from 0) of the selected item in the controls list.
- `selection_overlay` - A control overlaid on the picker to highlight the selected entry, centered and matching the height of the center row.
- `squeeze` - The angular compactness of the children on the wheel.
- `use_magnifier` - Whether to use the magnifier for the center item of this picker's wheel.

## Events

- `on_change` - Called when the selection changes.

## Examples

### Fruit selection

`play_arrow` Try Online

```
import flet as ft

FRUITS = [
 "Apple",
 "Mango",
 "Banana",
 "Orange",
 "Pineapple",
 "Strawberry",
]

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 selected_fruit_text = ft.Text(value=FRUITS[3], size=23)

 def handle_selection_change(e: ft.Event[ft.CupertinoPicker]):
 selected_fruit_text.value = FRUITS[int(e.data)]

 cupertino_picker = ft.CupertinoPicker(
 selected_index=3,
 magnification=1.22,
 squeeze=1.2,
 use_magnifier=True,
 on_change=handle_selection_change,
 controls=[ft.Text(value=f) for f in FRUITS],
)

 page.add(
 ft.SafeArea(
 content=ft.Row(
 tight=True,
 controls=[
 ft.Text("Selected Fruit:", size=23),
 ft.TextButton(
```

```

 style=ft.ButtonStyle(color=ft.Colors.BLUE),
 on_click=lambda _: page.show_dialog(
 ft.CupertinoBottomSheet(
 height=216,
 padding=ft.Padding.only(top=6),
 content=cupertino_picker,
)
),
 content=selected_fruit_text,
),
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: fruit-selection - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_picker/media/fruit\\_selection.gif](https://flet.dev/docs/examples/controls/cupertino/cupertino_picker/media/fruit_selection.gif)

## Properties

### buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The background color of this timer picker.

### buildcontrolsclass-attributeinstance-attribute

controls: list[Control] = field(default\_factory=list)

A list of controls representing items in this picker.

### builddefault\_selection\_overlay\_bgcolorclass-attributeinstance-attribute

```

default_selection_overlay_bgcolor: ColorValue = (
 CupertinoColors.TERTIARY_SYSTEM_FILL
)

```

The default background color of the selection\_overlay.

### bulddiameter\_ratiocontainer-attributeinstance-attribute

diameter\_ratio: Number = 1.07

Relative ratio between this picker's height and the simulated cylinder's diameter.

Smaller values create more pronounced curvatures in the scrollable wheel.

### builditem\_extentclass-attributeinstance-attribute

item\_extent: Number = 32.0

The uniform height of all controls.

Raises:

- ValueError - If it is not strictly greater than 0.0.

### buildloopingclass-attributeinstance-attribute

looping: bool = False

If True, children on a wheel can be scrolled in a loop.

### buildmagnificationclass-attributeinstance-attribute

magnification: Number = 1.0

The zoomed-in or magnification rate of the magnifier.

If the value is greater than 1.0, the item in the center will be zoomed in by that rate, and it will also be rendered as flat, not cylindrical like the rest of the list. The item will be zoomed-out if magnification is less than 1.0.

Note

Has effect only if use\_magnifier is True.

Raises:

- ValueError - If it is not strictly greater than 0.0.

### **buildoff\_axis\_fractionclass-attributeinstance-attribute**

`off_axis_fraction: Number = 0.0`

How much the wheel is horizontally off-center, as a fraction of its width.

### **buildselected\_indexclass-attributeinstance-attribute**

`selected_index: int = 0`

The index (starting from 0) of the selected item in the controls list.

### **buildselection\_overlayclass-attributeinstance-attribute**

`selection_overlay: Control | None = None`

A control overlaid on the picker to highlight the selected entry, centered and matching the height of the center row.

Defaults to a rounded rectangle in iOS 14 style with `default_selection_overlay_bgcolor` as background color.

### **buildsqueezecclass-attributeinstance-attribute**

`squeeze: Number = 1.45`

The angular compactness of the children on the wheel.

Raises:

- `ValueError` - If it is not strictly greater than 0.0.

### **builduse\_magnifierclass-attributeinstance-attribute**

`use_magnifier: bool = False`

Whether to use the magnifier for the center item of this picker's wheel.

## **Events**

### **bolton\_changecclass-attributeinstance-attribute**

```
on_change: ControlEventHandler[CupertinoPicker] | None = (
 None
)
```

Called when the selection changes.



# CupertinoRadio

A macOS-styled radio button, allowing the user to select a single option from two or more choices.

```
ft.RadioGroup(
 value="option_2",
 content=ft.Column(
 intrinsic_width=True,
 controls=[
 ft.CupertinoRadio(value="option_1", label="Option 1"),
 ft.CupertinoRadio(value="option_2", label="Option 2"),
 ft.CupertinoRadio(value="option_3", label="Option 3"),
],
),
)
```

Image: CupertinoRadio - [https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_radio/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_radio/image_for_docs.png)

## CupertinoRadio

Inherits: LayoutControl

## Properties

- `active_color` - The color used to fill this radio when it is selected.
- `autofocus` - Whether this radio will be selected as the initial focus.
- `fill_color` - The color that fills this radio.
- `focus_color` - The color for the radio's border when it has the input focus.
- `inactive_color` - The color used to fill this radio when it is not selected.
- `label` - The clickable label to display on the right of this radio.
- `label_position` - The position of the label relative to this radio.
- `mouse_cursor` - The cursor for a mouse pointer when it enters or is hovering over this radio.
- `toggleable` - Whether this radio button can return to an indeterminate state by selecting it again when already selected.
- `use_checkmark_style` - Whether the radio displays in a checkbox style instead of the default radio style.
- `value` - The value to set to RadioGroup ancestor/parent when this radio is selected.

## Events

- `on_blur` - Called when this radio has lost focus.
- `on_focus` - Called when this radio has received focus.

## Examples

### Cupertino, material and adaptive

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 message = ft.Text()

 group = ft.RadioGroup(
 content=ft.Column(
 controls=[
 ft.CupertinoRadio(
 value="red",
 label="Red - Cupertino Radio",
 active_color=ft.Colors.RED,
 inactive_color=ft.Colors.RED,
),
 ft.Radio(
 value="green",
 label="Green - Material Radio",
 fill_color=ft.Colors.GREEN,
),
 ft.Radio(
 value="blue",
```

```

 label="Blue - Adaptive Radio",
 adaptive=True,
 active_color=ft.Colors.BLUE,
),
],
)
)

def handle_button_click(_: ft.Event[ft.Button]):
 message.value = f"Your favorite color is: {group.value}"

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Select your favorite color:"),
 group,
 ft.Button(content="Submit", on_click=handle_button_click),
 message,
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: cupertino-material-and-adaptive -

[https://flet.dev/docs/examples/controls/cupertino/cupertino\\_radio/media/cupertino\\_material\\_and\\_adaptive.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_radio/media/cupertino_material_and_adaptive.png)

## Properties

### buildactive\_colorclass-attributeinstance-attribute

active\_color: ColorValue | None = Colors.PRIMARY

The color used to fill this radio when it is selected.

### buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

Whether this radio will be selected as the initial focus.

If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

### buildfill\_colorclass-attributeinstance-attribute

fill\_color: ColorValue | None = None

The color that fills this radio.

### buildfocus\_colorclass-attributeinstance-attribute

focus\_color: ColorValue | None = None

The color for the radio's border when it has the input focus.

### buildinactive\_colorclass-attributeinstance-attribute

inactive\_color: ColorValue | None = None

The color used to fill this radio when it is not selected.

### buildlabelclass-attributeinstance-attribute

label: str | None = None

The clickable label to display on the right of this radio.

### buildlabel\_positionclass-attributeinstance-attribute

label\_position: LabelPosition = LabelPosition.RIGHT

The position of the label relative to this radio.

### buildmouse\_cursorclass-attributeinstance-attribute

mouse\_cursor: MouseCursor | None = None

The cursor for a mouse pointer when it enters or is hovering over this radio.

### buildtoggleableclass-attributeinstance-attribute

toggleable: bool = False

Whether this radio button can return to an indeterminate state by selecting it again when already selected.

### **builduse\_checkmark\_styleclass-attributeinstance-attribute**

```
use_checkmark_style: bool = False
```

Whether the radio displays in a checkbox style instead of the default radio style.

### **buildvalueclass-attributeinstance-attribute**

```
value: str = ''
```

The value to set to RadioGroup ancestor/parent when this radio is selected.

## **Events**

### **bolton\_blurclass-attributeinstance-attribute**

```
on_blur: ControlEventHandler[CupertinoRadio] | None = None
```

Called when this radio has lost focus.

### **bolton\_focusclass-attributeinstance-attribute**

```
on_focus: ControlEventHandler[CupertinoRadio] | None = None
```

Called when this radio has received focus.

# CupertinoSegmentedButton

An iOS-style segmented button.

Example:

```
ft.CupertinoSegmentedButton(
 selected_index=1,
 controls=[
 ft.Text("One"),
 ft.Text("Two"),
 ft.Text("Three"),
],
)
```

Image: CupertinoSegmentedButton -

[https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macos/cupertino\\_segmented\\_button/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macos/cupertino_segmented_button/image_for_docs.png)

## CupertinoSegmentedButton

Inherits: `LayoutControl`

## Properties

- `border_color` - The color of this button's border.
- `click_color` - The color used to fill the background of this control when temporarily interacting with through a long press or drag.
- `controls` - The list of segments to be displayed.
- `disabled_color` - The color used to fill the background of the segment when it is disabled.
- `disabled_text_color` - The color used for the text of the segment when it is disabled.
- `padding` - This button's padding.
- `selected_color` - The color of this button when it is selected.
- `selected_index` - The index (starting from 0) of the selected segment in the controls list.
- `unselected_color` - The color of this button when it is not selected.

## Events

- `on_change` - Called when the state of the button is changed - when one of the controls is clicked.

## Examples

### CupertinoSegmentedButton

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT

 page.add(
 ft.SafeArea(
 content=ft.CupertinoSegmentedButton(
 selected_index=1,
 selected_color=ft.Colors.RED_400,
 on_change=lambda e: print(f"selected_index: {e.data}"),
 padding=ft.Padding.symmetric(vertical=20, horizontal=50),
 controls=[
 ft.Text("One"),
 ft.Container(
 padding=ft.Padding.symmetric(vertical=10, horizontal=30),
 content=ft.Text("Two"),
),
 ft.Container(
 padding=ft.Padding.symmetric(vertical=5, horizontal=10),
 content=ft.Text("Three"),
),
],
),
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_segmented\\_button/media/basic.gif](https://flet.dev/docs/examples/controls/cupertino/cupertino_segmented_button/media/basic.gif)

## Segments padding

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT

 segmented_button = ft.CupertinoSegmentedButton(
 selected_index=1,
 selected_color=ft.Colors.RED_400,
 unselected_color=ft.Colors.GREY_400,
 on_change=lambda e: print(f"selected_index: {e.data}"),
 controls=[
 ft.Text("All"),
 ft.Container(
 padding=ft.Padding.symmetric(vertical=30, horizontal=0),
 content=ft.Text("None"),
),
 ft.Container(
 padding=ft.Padding.symmetric(vertical=0, horizontal=30),
 content=ft.Text("Some"),
),
],
)

 def handle_vertical_change(e: ft.Event[ft.Slider]):
 segmented_button.controls[1].padding = ft.Padding.only(
 top=e.control.value,
 bottom=e.control.value,
)

 def handle_horizontal_change(e: ft.Event[ft.Slider]):
 segmented_button.controls[2].padding = ft.Padding.only(
 left=e.control.value,
 right=e.control.value,
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 segmented_button,
 ft.Text("Vertical padding button 1:"),
 ft.Slider(
 label="{value}",
 min=0,
 max=50,
 divisions=50,
 value=30,
 on_change=handle_vertical_change,
),
 ft.Text("Horizontal padding button 2:"),
 ft.Slider(
 label="{value}",
 min=0,
 max=50,
 divisions=50,
 value=30,
 on_change=handle_horizontal_change,
),
 ft.Text(
 value=(
 "**note that padding changes to one segment can effect "
 "padding on other segments*"
),
 theme_style=ft.TextThemeStyle.LABEL_MEDIUM,
 color=ft.Colors.ORANGE_ACCENT,
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildborder\_colorclass-attributeinstance-attribute**

`border_color: ColorValue | None = None`

The color of this button's border.

### **buildclick\_colorclass-attributeinstance-attribute**

`click_color: ColorValue | None = None`

The color used to fill the background of this control when temporarily interacting with through a long press or drag.

Defaults to the `selected_color` with 20% opacity.

### **buildcontrolsinstance-attribute**

`controls: list[Control]`

The list of segments to be displayed.

Raises:

- `ValueError` - If it does not contain at least two visible Controls.

### **builddisabled\_colorclass-attributeinstance-attribute**

`disabled_color: ColorValue | None = None`

The color used to fill the background of the segment when it is disabled.

If `None`, this color will be 50% opacity of the `selected_color` when the segment is selected. If the segment is unselected, this color will be set to the `unselected_color`.

### **builddisabled\_text\_colorclass-attributeinstance-attribute**

`disabled_text_color: ColorValue | None = None`

The color used for the text of the segment when it is disabled.

### **buildpaddingclass-attributeinstance-attribute**

`padding: PaddingValue | None = None`

This button's padding.

### **buildselected\_colorclass-attributeinstance-attribute**

`selected_color: ColorValue | None = None`

The color of this button when it is selected.

### **buildselected\_indexclass-attributeinstance-attribute**

`selected_index: int = 0`

The index (starting from 0) of the selected segment in the controls list.

Raises:

- `IndexError` - If `selected_index` is out of range relative to the visible controls.

### **buildunselected\_colorclass-attributeinstance-attribute**

`unselected_color: ColorValue | None = None`

The color of this button when it is not selected.

## **Events**

### **bolton\_changeclass-attributeinstance-attribute**

`on_change: (ControlEventHandler[CupertinoSegmentedButton] | None) = None`

Called when the state of the button is changed - when one of the controls is clicked.

# CupertinoSlider

An iOS-type slider.

It provides a visual indication of adjustable content, as well as the current setting in the total range of content.

Use a slider when you want people to set defined values (such as volume or brightness), or when people would benefit from instant feedback on the effect of setting changes.

Example:

```
ft.CupertinoSlider(value=0.6)
```

Image: CupertinoSlider - [https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_slider/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_slider/image_for_docs.png)

CupertinoSlider

Inherits: LayoutControl

## Properties

- `active_color` - The color to use for the portion of the slider track that is active.
- `divisions` - The number of discrete divisions.
- `max` - The maximum value the user can select.
- `min` - The minimum value the user can select.
- `thumb_color` - The color of this slider's thumb.
- `value` - The currently selected value for this slider.

## Events

- `on_blur` - Called when this slider has lost focus.
- `on_change` - Called when the state of this slider changed.
- `on_change_end` - Called when the user is done selecting a new value for this slider.
- `on_change_start` - Called when the user starts selecting a new value for this slider.
- `on_focus` - Called when this slider has received focus.

## Examples

### Handling events

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.vertical_alignment = ft.MainAxisAlignment.CENTER
 page.theme_mode = ft.ThemeMode.LIGHT

 slider_value = ft.Text("0.0")
 slider_status = ft.Text()

 def handle_change_start(_: ft.Event[ft.CupertinoSlider]):
 slider_status.value = "Sliding"

 def handle_change(e: ft.Event[ft.CupertinoSlider]):
 slider_value.value = str(e.control.value)

 def handle_change_end(_: ft.Event[ft.CupertinoSlider]):
 slider_status.value = "Finished sliding"

 page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 slider_value,
 ft.CupertinoSlider(
 divisions=20,
```

```

 min=0,
 max=100,
 active_color=ft.Colors.PURPLE,
 thumb_color=ft.Colors.PURPLE,
 on_change_start=handle_change_start,
 on_change_end=handle_change_end,
 on_change=handle_change,
),
 slider_status,
 1,
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: handling-events - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_slider/media/handling\\_events.gif](https://flet.dev/docs/examples/controls/cupertino/cupertino_slider/media/handling_events.gif)

## Properties

### buildactive\_colorclass-attributeinstance-attribute

active\_color: ColorValue | None = None

The color to use for the portion of the slider track that is active.

The "active" side of the slider is the side between the thumb and the minimum value.

### bulldivisionsclass-attributeinstance-attribute

divisions: int | None = None

The number of discrete divisions.

If None, the slider is continuous.

### buildmaxclass-attributeinstance-attribute

max: Number = 1.0

The maximum value the user can select.

If min is equal to the max, then this slider is disabled.

Raises:

- ValueError - If it is not greater than or equal to min.
- ValueError - If it is not greater than or equal to value, when value is set.

### buildminclass-attributeinstance-attribute

min: Number = 0.0

The minimum value the user can select.

If max is equal to the min, then this slider is disabled.

Raises:

- ValueError - If it is not less than or equal to max.
- ValueError - If it is not less than or equal to value, when value is set.

### buildthumb\_colorclass-attributeinstance-attribute

thumb\_color: ColorValue | None = None

The color of this slider's thumb.

### buildvalueclass-attributeinstance-attribute

value: Number | None = None

The currently selected value for this slider.

The slider's thumb is drawn at a position that corresponds to this value.

Raises:

- ValueError - If it is not greater than or equal to min.



- ValueError - If it is not less than or equal to max.

## Events

### **bolton\_blurclass-attributeinstance-attribute**

```
on_blur: ControlEventHandler[CupertinoSlider] | None = None
```

Called when this slider has lost focus.

### **bolton\_changeclass-attributeinstance-attribute**

```
on_change: ControlEventHandler[CupertinoSlider] | None = (
 None
)
```

Called when the state of this slider changed.

### **bolton\_change\_endclass-attributeinstance-attribute**

```
on_change_end: (
 ControlEventHandler[CupertinoSlider] | None
) = None
```

Called when the user is done selecting a new value for this slider.

### **bolton\_change\_startclass-attributeinstance-attribute**

```
on_change_start: (
 ControlEventHandler[CupertinoSlider] | None
) = None
```

Called when the user starts selecting a new value for this slider.

### **bolton\_focusclass-attributeinstance-attribute**

```
on_focus: ControlEventHandler[CupertinoSlider] | None = None
```

Called when this slider has received focus.

# CupertinoSlidingSegmentedButton

A cupertino sliding segmented button.

Example:

```
ft.CupertinoSlidingSegmentedButton(
 selected_index=1,
 controls=[
 ft.Text("One"),
 ft.Text("Two"),
 ft.Text("Three"),
],
)
```

Image: CupertinoSlidingSegmentedButton -

[https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macos/cupertino\\_sliding\\_segmented\\_button/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macos/cupertino_sliding_segmented_button/image_for_docs.png)

## CupertinoSlidingSegmentedButton

Inherits: `LayoutControl`

## Properties

- `bgcolor` - The background color of this button.
- `controls` - The list of segments to be displayed.
- `padding` - The amount of space by which to inset the controls.
- `proportional_width` - Determine whether segments have proportional widths based on their content.
- `selected_index` - The index (starting from 0) of the selected segment in the controls list.
- `thumb_color` - The color of this button when it is not selected.

## Events

- `on_change` - Called when the state of the button is changed - when one of the controls is clicked.

## Examples

### CupertinoSlidingSegmentedButton

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "CupertinoSlidingSegmentedButton Example"
 page.theme_mode = ft.ThemeMode.LIGHT

 def handle_selection_change(e: ft.Event[ft.CupertinoSlidingSegmentedButton]):
 page.show_dialog(
 ft.SnackBar(ft.Text(f"Segment {e.control.selected_index + 1} was chosen!"))
)

 page.add(
 ft.SafeArea(
 content=ft.CupertinoSlidingSegmentedButton(
 selected_index=1,
 thumb_color=ft.Colors.BLUE_400,
 on_change=handle_selection_change,
 controls=[
 ft.Text("One"),
 ft.Text("Two"),
 ft.Text("Three"),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_sliding\\_segmented\\_button/media/basic.gif](https://flet.dev/docs/examples/controls/cupertino/cupertino_sliding_segmented_button/media/basic.gif)

## Properties

### buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue = CupertinoColors.TERTIARY_SYSTEM_FILL
```

The background color of this button.

### buildcontrolsinstance-attribute

```
controls: list[Control]
```

The list of segments to be displayed.

Raises:

- `ValueError` - If it does not contain at least two visible Controls.

### buildpaddingclass-attributeinstance-attribute

```
padding: PaddingValue = field(
 default_factory=lambda: Padding.symmetric(
 vertical=2, horizontal=3
)
)
```

The amount of space by which to inset the controls.

### buildproportional\_widthclass-attributeinstance-attribute

```
proportional_width: bool = False
```

Determine whether segments have proportional widths based on their content.

If `False`, all segments will have the same width, determined by the longest segment. If `True`, each segment's width will be determined by its individual content.

If the max width of parent constraints is smaller than the width that this control needs, the segment widths will scale down proportionally to ensure this control fits within the boundaries; similarly, if the min width of parent constraints is larger, the segment width will scale up to meet the min width requirement.

### buildselected\_indexclass-attributeinstance-attribute

```
selected_index: int = 0
```

The index (starting from 0) of the selected segment in the controls list.

Raises:

- `IndexError` - If it is not between 0 and the length of visible controls, inclusive.

### buildthumb\_colorclass-attributeinstance-attribute

```
thumb_color: ColorValue | None = None
```

The color of this button when it is not selected.

## Events

### bolton\_changeclass-attributeinstance-attribute

```
on_change: (
 ControlEventHandler[CupertinoSlidingSegmentedButton]
 | None
) = None
```

Called when the state of the button is changed - when one of the controls is clicked.

# CupertinoSwitch

An iOS-style switch.

Used to toggle the on/off state of a single setting.

Example:

```
ft.CupertinoSwitch(value=True)
```

Image: CupertinoSwitch - [https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_switch/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_switch/image_for_docs.png)

CupertinoSwitch

Inherits: `LayoutControl`

## Properties

- `active_thumb_image_src` - An image to use on the thumb of this switch when the switch is on.
- `active_track_color` - The color to use on the track when this switch is on.
- `autofocus` - Whether this switch will be selected as the initial focus.
- `focus_color` - The color to use for the focus highlight for keyboard interactions.
- `inactive_thumb_color` - The color to use on the thumb when this switch is off.
- `inactive_thumb_image_src` - An image to use on the thumb of this switch when the switch is off.
- `inactive_track_color` - The color to use on the track when this switch is off.
- `label` - The clickable label to display on the right of this switch.
- `label_position` - The position of the label relative to this switch.
- `off_label_color` - The color to use for the accessibility label when the switch is off.
- `thumb_color` - The color of this switch's thumb.
- `thumb_icon` - The icon of this Switch's thumb in various `ControlStates`.
- `track_outline_color` - The outline color of this switch's track in various `ControlStates`.
- `track_outline_width` - The outline width of this switch's track in all or specific `ControlStates`.
- `value` - The current value of this switch.

## Events

- `on_blur` - Called when this switch has lost focus.
- `on_change` - Called when the state of this switch is changed.
- `on_focus` - Called when this switch has received focus.
- `on_image_error` - Called when `active_thumb_image_src` or `inactive_thumb_image_src` fails to load.
- `on_label_color` - The color to use for the accessibility label when the switch is on.

## Examples

### Cupertino, material and adaptive

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.CupertinoSwitch(label="Cupertino Switch", value=True),
 ft.Switch(
 label="Material Switch",
 value=True,
 thumb_color={ft.ControlState.SELECTED: ft.Colors.BLUE},
```

```

 track_color=ft.Colors.YELLOW,
 focus_color=ft.Colors.PURPLE,
),
 ft.Container(height=20),
 ft.Text(
 value=(
 "Adaptive Switch shows as CupertinoSwitch on macOS and "
 "iOS and as Switch on other platforms:"
)
),
 ft.Switch(adaptive=True, label="Adaptive Switch", value=True),
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: cupertino-material-and-adaptive -

[https://flet.dev/docs/examples/controls/cupertino/cupertino\\_switch/media/cupertino\\_material\\_and\\_adaptive.gif](https://flet.dev/docs/examples/controls/cupertino/cupertino_switch/media/cupertino_material_and_adaptive.gif)

## Properties

### buildactive\_thumb\_image\_srcclass-attributeinstance-attribute

active\_thumb\_image\_src: str | bytes | None = None

An image to use on the thumb of this switch when the switch is on.

It can be one of the following:

- A URL or local asset file path;
- A base64 string;
- Raw bytes.

### buildactive\_track\_colorclass-attributeinstance-attribute

active\_track\_color: ColorValue | None = None

The color to use on the track when this switch is on.

### buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

Whether this switch will be selected as the initial focus.

If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

### buildfocus\_colorclass-attributeinstance-attribute

focus\_color: ColorValue | None = None

The color to use for the focus highlight for keyboard interactions.

### buildinactive\_thumb\_colorclass-attributeinstance-attribute

inactive\_thumb\_color: ColorValue | None = None

The color to use on the thumb when this switch is off.

If None, defaults to thumb\_color, and if this is also None, defaults to CupertinoColors.WHITE.

### buildinactive\_thumb\_image\_srcclass-attributeinstance-attribute

inactive\_thumb\_image\_src: str | bytes | None = None

An image to use on the thumb of this switch when the switch is off.

It can be one of the following:

- A URL or local asset file path;
- A base64 string;
- Raw bytes.

### buildinactive\_track\_colorclass-attributeinstance-attribute

inactive\_track\_color: ColorValue | None = None

The color to use on the track when this switch is off.

### **buildlabelclass-attributeinstance-attribute**

```
label: str | None = None
```

The clickable label to display on the right of this switch.

### **buildlabel\_positionclass-attributeinstance-attribute**

```
label_position: LabelPosition = LabelPosition.RIGHT
```

The position of the label relative to this switch.

### **buildoff\_label\_colorclass-attributeinstance-attribute**

```
off_label_color: ColorValue | None = None
```

The color to use for the accessibility label when the switch is off.

### **buildthumb\_colorclass-attributeinstance-attribute**

```
thumb_color: ColorValue | None = None
```

The color of this switch's thumb.

### **buildthumb\_iconclass-attributeinstance-attribute**

```
thumb_icon: ControlStateValue[IconData] | None = None
```

The icon of this Switch's thumb in various ControlStates.

Supported states: ControlState.SELECTED, ControlState.HOVERED, ControlState.DISABLED, ControlState.FOCUSED, and ControlState.DEFAULT.

### **buildtrack\_outline\_colorclass-attributeinstance-attribute**

```
track_outline_color: (
 ControlStateValue[ColorValue] | None
) = None
```

The outline color of this switch's track in various ControlStates.

Supported states: ControlState.SELECTED, ControlState.HOVERED, ControlState.DISABLED, ControlState.FOCUSED, and ControlState.DEFAULT.

### **buildtrack\_outline\_widthclass-attributeinstance-attribute**

```
track_outline_width: (
 ControlStateValue[Number | None] | None
) = None
```

The outline width of this switch's track in all or specific ControlStates.

Supported states: ControlState.SELECTED, ControlState.HOVERED, ControlState.DISABLED, ControlState.FOCUSED, and ControlState.DEFAULT.

### **buildvalueclass-attributeinstance-attribute**

```
value: bool = False
```

The current value of this switch.

## **Events**

### **bolton\_blurclass-attributeinstance-attribute**

```
on_blur: ControlEventHandler[CupertinoSwitch] | None = None
```

Called when this switch has lost focus.

### **bolton\_changeclass-attributeinstance-attribute**

```
on_change: ControlEventHandler[CupertinoSwitch] | None = (
 None
)
```

Called when the state of this switch is changed.

### **bolton\_focusclass-attributeinstance-attribute**

```
on_focus: ControlEventHandler[CupertinoSwitch] | None = None
```

Called when this switch has received focus.

### **bolton\_image\_error**class-attributeinstance-attribute

```
on_image_error: (
 ControlEventHandler[CupertinoSwitch] | None
) = None
```

Called when active\_thumb\_image\_src or inactive\_thumb\_image\_src fails to load.

### **bolton\_label\_color**class-attributeinstance-attribute

```
on_label_color: ColorValue | None = None
```

The color to use for the accessibility label when the switch is on.

# CupertinoTextField

An iOS-style text field.

```
ft.CupertinoTextField(placeholder_text="Search")
```

Image: CupertinoTextField - [https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_textfield/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_textfield/image_for_docs.png)

CupertinoTextField

Inherits: TextField

## Properties

- `blend_mode` - The blend mode applied to the bgcolor or gradient background.
- `clear_button_semantics_label` - The semantic label for the clear button used by screen readers.
- `clear_button_visibility_mode` - Defines the visibility of the clear button based on the state of text entry.
- `gradient` - Configures the gradient background.
- `image` - An image to paint above the bgcolor or gradient background.
- `padding` - The padding around the text entry area between the prefix and suffix or the clear button when `clear_button_visibility_mode` is not `OverlayVisibilityMode.NEVER`.
- `placeholder_style` - The TextStyle to use for placeholder\_text.
- `placeholder_text` - A lighter colored placeholder hint that appears on the first line of the text field when the text entry is empty.
- `prefix_visibility_mode` - Defines the visibility of the prefix control based on the state of text entry.
- `shadows` - A list of shadows behind this text field.
- `suffix_visibility_mode` - Defines the visibility of the suffix control based on the state of text entry.

## Examples

### Cupertino, material and adaptive

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.TextField(
 label="Material text field",
 label_style=ft.TextStyle(color=ft.Colors.GREY_400),
),
 ft.CupertinoTextField(
 placeholder_text="Cupertino text field",
 placeholder_style=ft.TextStyle(color=ft.Colors.GREY_400),
),
 ft.TextField(
 adaptive=True,
 label="Adaptive text field",
 label_style=ft.TextStyle(color=ft.Colors.GREY_400),
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: cupertino-material-and-adaptive -

[https://flet.dev/docs/examples/controls/cupertino/cupertino\\_text\\_field/media/cupertino\\_material\\_and\\_adaptive.png](https://flet.dev/docs/examples/controls/cupertino/cupertino_text_field/media/cupertino_material_and_adaptive.png)

### Selection change



## play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Text selection"

 selection = ft.Text("Select some text from the field.")
 selection_details = ft.Text()
 caret = ft.Text("Caret position: -")

 def handle_selection_change(e: ft.TextSelectionChangeEvent[ft.CupertinoTextField]):
 selection.value = (
 f"Selection: '{e.selected_text}'" if e.selected_text else "No selection."
)
 selection_details.value = f"start={e.selection.start}, end={e.selection.end}"
 caret.value = f"Caret position: {e.selection.end}"

 field = ft.CupertinoTextField(
 value="Lorem ipsum dolor sit amet, consectetur adipiscing elit.",
 multiline=True,
 min_lines=3,
 autofocus=True,
 on_selection_change=handle_selection_change,
)

 async def select_characters(_: ft.Event[ft.Button]):
 await field.focus()
 field.selection = ft.TextSelection(
 base_offset=0,
 extent_offset=len(field.value),
)

 async def move_caret(_: ft.Event[ft.Button]):
 await field.focus()
 field.selection = ft.TextSelection(base_offset=0, extent_offset=0)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 spacing=10,
 controls=[
 field,
 selection,
 selection_details,
 caret,
 ft.Button(content="Select all text", on_click=select_characters),
 ft.Button(content="Move caret to start", on_click=move_caret),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Background image

### play\_arrowTry Online

```
import flet as ft

async def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT

 page.add(
 ft.SafeArea(
 content=ft.CupertinoTextField(
 label="Textfield Label",
 label_style=ft.TextStyle(italic=True, weight=ft.FontWeight.BOLD),
 bgcolor=ft.Colors.BLUE_GREY,
 image=ft.DecorationImage(src="https://picsum.photos/1000/260"),
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

buildblend\_modeclass-attributeinstance-attribute

```
blend_mode: BlendMode | None = None
```

The blend mode applied to the bgcolor or gradient background.

### **buildclear\_button\_semantics\_labelclass-attributeinstance-attribute**

```
clear_button_semantics_label: str | None = 'Clear'
```

The semantic label for the clear button used by screen readers.

This will be used by screen reading software to identify the clear button widget.

### **buildclear\_button\_visibility\_modeclass-attributeinstance-attribute**

```
clear_button_visibility_mode: OverlayVisibilityMode = (
 OverlayVisibilityMode.NEVER
)
```

Defines the visibility of the clear button based on the state of text entry.

Will appear only if no suffix is provided.

### **buildgradientclass-attributeinstance-attribute**

```
gradient: Gradient | None = None
```

Configures the gradient background.

### **buildimageclass-attributeinstance-attribute**

```
image: DecorationImage | None = None
```

An image to paint above the bgcolor or gradient background.

### **buildpaddingclass-attributeinstance-attribute**

```
padding: PaddingValue = field(
 default_factory=lambda: Padding.all(7)
)
```

The padding around the text entry area between the prefix and suffix or the clear button when `clear_button_visibility_mode` is not `OverlayVisibilityMode.NEVER`.

### **buildplaceholder\_styleclass-attributeinstance-attribute**

```
placeholder_style: TextStyle | None = None
```

The `TextStyle` to use for `placeholder_text`.

### **buildplaceholder\_textclass-attributeinstance-attribute**

```
placeholder_text: str | None = None
```

A lighter colored placeholder hint that appears on the first line of the text field when the text entry is empty.

Defaults to an empty string.

### **buildprefix\_visibility\_modeclass-attributeinstance-attribute**

```
prefix_visibility_mode: OverlayVisibilityMode = (
 OverlayVisibilityMode.ALWAYS
)
```

Defines the visibility of the prefix control based on the state of text entry.

Note

Has no effect if prefix is not specified.

### **buildshadowsclass-attributeinstance-attribute**

```
shadows: BoxShadowValue | None = None
```

A list of shadows behind this text field.

### **buildsuffix\_visibility\_modeclass-attributeinstance-attribute**

```
suffix_visibility_mode: OverlayVisibilityMode = (
 OverlayVisibilityMode.ALWAYS
)
```

Defines the visibility of the suffix control based on the state of text entry.

Note

Has no effect if suffix is not specified.

# CupertinoTimerPicker

A countdown timer picker in iOS style.

It can show a countdown duration with hour, minute and second spinners. The duration is bound between 0 and 23 hours 59 minutes 59 seconds.

Example:

```
ft.CupertinoTimerPicker(value=1000)
```

Image: CupertinoTimerPicker -

[https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_timer\\_picker/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_timer_picker/image_for_docs.png)

Simple Cupertino timer picker

Inherits: `LayoutControl`

## Properties

- `alignment` - Defines how this picker should be positioned within its parent.
- `bgcolor` - The background color of this picker.
- `item_extent` - The uniform height of all children.
- `minute_interval` - The granularity of the minute spinner.
- `mode` - The mode of this picker.
- `second_interval` - The granularity of the second spinner.
- `value` - The initial duration of the countdown timer.

## Events

- `on_change` - Called when the timer's duration changes.

## Examples

### CupertinoTimerPicker

`play_arrow` Try Online

```
import time

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 timer_value_text = ft.Text(
 value="00:01:10",
 size=23,
 color=ft.CupertinoColors.DESTRUCTIVE_RED,
)

 def handle_timer_picker_change(e: ft.Event[ft.CupertinoTimerPicker]):
 timer_value_text.value = time.strftime("%H:%M:%S", time.gmtime(e.data))

 timer_picker = ft.CupertinoTimerPicker(
 value=300,
 second_interval=10,
 minute_interval=1,
 mode=ft.CupertinoTimerPickerMode.HOUR_MINUTE_SECONDS,
 on_change=handle_timer_picker_change,
)

 page.add(
 ft.SafeArea(
 content=ft.Row(
 tight=True,
 controls=[
 ft.Text("TimerPicker Value:", size=23),
 ft.CupertinoButton(
 on_click=lambda _: page.show_dialog(
 ft.CupertinoBottomSheet(
 height=216,
 padding=ft.Padding.only(top=6),
```

```

),
 content=timer_picker,
),
 content=timer_value_text,
),
],
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: basic - [https://flet.dev/docs/examples/controls/cupertino/cupertino\\_timer\\_picker/media/basic.gif](https://flet.dev/docs/examples/controls/cupertino/cupertino_timer_picker/media/basic.gif)

## Properties

### buildalignmentclass-attributeinstance-attribute

```

alignment: Alignment = field(
 default_factory=lambda: Alignment.CENTER
)

```

Defines how this picker should be positioned within its parent.

### buildbgcolorclass-attributeinstance-attribute

```

bgcolor: ColorValue | None = None

```

The background color of this picker.

### builditem\_extentclass-attributeinstance-attribute

```

item_extent: Number = 32.0

```

The uniform height of all children.

Raises:

- ValueError - If it is not strictly greater than 0.0.

### buildminute\_intervalclass-attributeinstance-attribute

```

minute_interval: int = 1

```

The granularity of the minute spinner.

Raises:

- ValueError - If it is not strictly greater than 0.
- ValueError - If it is not a factor of 60.

### buildmodeclass-attributeinstance-attribute

```

mode: CupertinoTimerPickerMode = (
 CupertinoTimerPickerMode.HOUR_MINUTE_SECONDS
)

```

The mode of this picker.

### buildsecond\_intervalclass-attributeinstance-attribute

```

second_interval: int = 1

```

The granularity of the second spinner.

Raises:

- ValueError - If it is not strictly greater than 0.
- ValueError - If it is not a factor of 60.

### buildvalueclass-attributeinstance-attribute

```

value: DurationValue = field(
 default_factory=lambda: Duration()
)

```

The initial duration of the countdown timer.

If specified as an integer, it will be assumed to be in seconds.

Raises:

- ValueError - If it is not greater than or equal to 0.
- ValueError - If it is not strictly less than 24 hours.
- ValueError - If it is not a multiple of `minute_interval`.
- ValueError - If it is not a multiple of `second_interval`.

## Events

`bolton_change` class-attribute instance-attribute

```
on_change: (
 ControlEventHandler[CupertinoTimerPicker] | None
) = None
```

Called when the timer's duration changes.

The data property of the event handler contains the new duration value. Its type matches value: if value is a `Duration`, then data is also a `Duration`; otherwise, it is an `int` (seconds).

# CupertinoTintedButton

An iOS-style button filled with default background color.

```
ft.CupertinoTintedButton("Tap me")
```

Image: CupertinoTintedButton -

[https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino\\_tinted\\_button/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/cupertino/golden/macOS/cupertino_tinted_button/image_for_docs.png)

CupertinoTintedButton

Inherits: CupertinoButton

# DataCell

The data for a cell of a DataTable.

Inherits: Control

## Properties

- `content` - The content of this cell.
- `placeholder` - Whether the content is actually a placeholder.
- `show_edit_icon` - Whether to show an edit icon at the end of this cell.

## Events

- `on_double_tap` - Called when this cell is double tapped.
- `on_long_press` - Called if this cell is long-pressed.
- `on_tap` - Called if this cell is tapped.
- `on_tap_cancel` - Called if the user cancels a tap was started on cell.
- `on_tap_down` - Called if this cell is tapped down.

## Properties

### `buildcontentinstance-attribute`

`content: StrOrControl`

The content of this cell.

Typically a Text control or a Dropdown control.

If the cell has no data, then a Text control with placeholder text should be provided instead, and `placeholder` should be set to `True`.

#### Tip

To lay out multiple children, set the content to a container-like control such as Row, Column, or Stack, which have a `controls` property.

Raises:

- `ValueError` - If it is neither a string nor a visible Control.

### `buildplaceholderclass-attributeinstance-attribute`

`placeholder: bool = False`

Whether the content is actually a placeholder.

If this is `True`, the default text style for the cell is changed to be appropriate for placeholder text.

### `buildshow_edit_iconclass-attributeinstance-attribute`

`show_edit_icon: bool = False`

Whether to show an edit icon at the end of this cell.

This does not make the cell actually editable; the caller must implement editing behavior if desired (initiated from the `on_tap` callback).

#### Note

If this is set, `on_tap` should also be set, otherwise tapping the icon will have no effect.

## Events

### `bolton_double_tapclass-attributeinstance-attribute`

`on_double_tap: ControlEventHandler[DataCell] | None = None`

Called when this cell is double tapped.



#### Note

If this is None (including on\_tap, on\_long\_press, on\_tap\_cancel, on\_tap\_down), tapping this cell will attempt to select its row (if DataRow.on\_select\_change is provided).

#### **bolton\_long\_press**class-attributeinstance-attribute

on\_long\_press: ControlEventHandler[DataCell] | None = None

Called if this cell is long-pressed.

#### Note

If this is None (including on\_tap, on\_double\_tap, on\_tap\_cancel, on\_tap\_down), tapping this cell will attempt to select its row (if DataRow.on\_select\_change is provided).

#### **bolton\_tap**class-attributeinstance-attribute

on\_tap: ControlEventHandler[DataCell] | None = None

Called if this cell is tapped.

#### Note

If this is None (including on\_double\_tap, on\_long\_press, on\_tap\_cancel, on\_tap\_down), tapping this cell will attempt to select its row (if DataRow.on\_select\_change is provided).

#### **bolton\_tap\_cancel**class-attributeinstance-attribute

on\_tap\_cancel: ControlEventHandler[DataCell] | None = None

Called if the user cancels a tap was started on cell.

#### Note

If this is None (including on\_tap, on\_double\_tap, on\_long\_press, on\_tap\_down), tapping this cell will attempt to select its row (if DataRow.on\_select\_change is provided).

#### **bolton\_tap\_down**class-attributeinstance-attribute

on\_tap\_down: EventHandler[TapEvent[DataCell]] | None = None

Called if this cell is tapped down.

#### Note

If this is None (including on\_tap, on\_double\_tap, on\_long\_press, on\_tap\_cancel), tapping this cell will attempt to select its row (if DataRow.on\_select\_change is provided).

# DataColumn

Column configuration for a DataTable.

Inherits: Control

## Properties

- `heading_row_alignment` - Defines the horizontal layout of the label and sort indicator in the heading row.
- `label` - The column heading.
- `numeric` - Whether this column represents numeric data or not.
- `tooltip` - The column heading's tooltip.

## Events

- `on_sort` - Called when the user asks to sort the table using this column.

## Properties

**buildheading\_row\_alignmentclass-attributeinstance-attribute**

`heading_row_alignment: MainAxisAlignment | None = None`

Defines the horizontal layout of the label and sort indicator in the heading row.

**buildlabelinstance-attribute**

`label: StrOrControl`

The column heading.

Typically, this will be a Text control. It could also be an Icon (typically using size 18), or a combination of both in a Row.

Raises:

- `ValueError` - If it is neither a string nor a visible Control.

**buildnumericclass-attributeinstance-attribute**

`numeric: bool = False`

Whether this column represents numeric data or not.

The contents of cells of columns containing numeric data are right-aligned.

**buildtooltipclass-attributeinstance-attribute**

`tooltip: str | None = field(default=None, kw_only=True)`

The column heading's tooltip.

This is a longer description of the column heading, for cases where the heading might have been abbreviated to keep the column width to a reasonable size.

## Events

**bolton\_sortclass-attributeinstance-attribute**

`on_sort: EventHandler[DataColumnSortEvent] | None = None`

Called when the user asks to sort the table using this column.

If not set, the column will not be considered sortable.

# DataRow

Row configuration and cell data for a DataTable.

One row configuration must be provided for each row to display in the table.

The data for this row of the table is provided in the cells property.

Inherits: Control

## Properties

- cells - The data for this row: a list of DataCell controls.
- color - The color of this row.
- selected - Whether the row is selected.

## Events

- on\_long\_press - Called when this row is long-pressed.
- on\_select\_change - Called when the user selects or unselects a selectable row.

## Properties

### buildcellsclass-attributeinstance-attribute

```
cells: list[DataCell] = field(default_factory=list)
```

The data for this row: a list of DataCell controls.

Note

There must be exactly as many cells as there are visible columns in the table.

Raises:

- ValueError - If it does not contain at least one visible DataCell.

### buildcolorclass-attributeinstance-attribute

```
color: ControlStateValue[ColorValue] | None = None
```

The color of this row.

By default, the color is transparent unless selected. Selected rows has a grey translucent color.

The effective color can depend on the ControlState state, if the row is selected, pressed, hovered, focused, disabled or enabled. The color is painted as an overlay to the row. To make sure that the row's InkWell is visible (when pressed, hovered and focused), it is recommended to use a translucent color.

### buildselectedclass-attributeinstance-attribute

```
selected: bool = False
```

Whether the row is selected.

If on\_select\_change is non-null for any row in the table, then a checkbox is shown at the start of each row. If the row is selected (True), the checkbox will be checked and the row will be highlighted.

Otherwise, the checkbox, if present, will not be checked.

## Events

### bolton\_long\_pressclass-attributeinstance-attribute

```
on_long_press: ControlEventHandler[DataRow] | None = None
```

Called when this row is long-pressed.

If a DataCell in the row has its DataCell.on\_tap, DataCell.on\_double\_tap, DataCell.on\_long\_press, DataCell.on\_tap\_cancel or DataCell.on\_tap\_down callback defined, that callback behavior overrides the gesture behavior of the row for that particular cell.

## bolton\_select\_change

on\_select\_change: ControlEventHandler[DataRow] | None = None

Called when the user selects or unselects a selectable row.

If this is not null, then this row is selectable. The current selection state of this row is given by selected.

If any row is selectable, then the table's heading row will have a checkbox that can be checked to select all selectable rows (and which is checked if all the rows are selected), and each subsequent row will have a checkbox to toggle just that row.

A row whose on\_select\_change callback is null is ignored for the purposes of determining the state of the "all" checkbox, and its checkbox is disabled.

If a DataCell in the row has its DataCell.on\_tap callback defined, that callback behavior overrides the gesture behavior of the row for that particular cell.

# DataTable

A Material Design data table.

Example:

```
ft.DataTable(
 columns=[
 ft.DataColumn(label=ft.Text("Name")),
 ft.DataColumn(label=ft.Text("Role")),
],
 rows=[
 ft.DataRow(
 cells=[
 ft.DataCell(ft.Text("Alice")),
 ft.DataCell(ft.Text("Engineer")),
]
),
 ft.DataRow(
 cells=[
 ft.DataCell(ft.Text("Bob")),
 ft.DataCell(ft.Text("Designer")),
]
),
],
)
```

Tip

Outgrown this table? Its beefier cousin, `DataTable2` (from the `flet-datatable2` extension package), adds sticky headers, fixed rows/columns, per-column widths (fixed or relative S/M/L), and a few other goodies — handy for large or wide datasets.

Image: `DataTable` - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/image_for_docs.png)

`DataTable`

Inherits: `LayoutControl`

## Properties

- `bgcolor` - The background color for this table.
- `border` - The border around the table.
- `border_radius` - Border corners.
- `checkbox_horizontal_margin` - Horizontal margin around the checkbox, if it is displayed.
- `clip_behavior` - Defines how the contents of this table are clipped.
- `column_spacing` - The horizontal margin between the contents of each data column.
- `columns` - A list of `DataColumn` controls describing table columns.
- `data_row_color` - The background color for the data rows.
- `data_row_max_height` - The maximum height of each row (excluding the row that contains column headings).
- `data_row_min_height` - The minimum height of each row (excluding the row that contains column headings).
- `data_text_style` - The text style of the data rows.
- `divider_thickness` - The width of the divider that appears between rows.
- `gradient` - The background gradient of this table.
- `heading_row_color` - The background color for the heading row.
- `heading_row_height` - The height of the heading row.
- `heading_text_style` - The text style for the heading row.
- `horizontal_lines` - Set the color and width of horizontal lines between rows.
- `horizontal_margin` - The horizontal margin between the edges of this table and the content in the first and last cells of each row.
- `rows` - A list of `DataRow` controls defining table rows.

- `show_bottom_border` - Whether a border at the bottom of the table is displayed.
- `show_checkbox_column` - Whether the control should display checkboxes for selectable rows.
- `sort_ascending` - Whether the column mentioned in `sort_column_index`, if any, is sorted in ascending order.
- `sort_column_index` - The current primary sort key's column.
- `vertical_lines` - Set the color and width of vertical lines between columns.

## Events

- `on_select_all` - Invoked when the user selects or unselects every row, using the checkbox in the heading row.

## Examples

### DataTable

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.DataTable(
 expand=True,
 columns=[
 ft.DataColumn(label=ft.Text("First name")),
 ft.DataColumn(label=ft.Text("Last name")),
 ft.DataColumn(label=ft.Text("Age"), numeric=True),
],
 rows=[
 ft.DataRow(
 cells=[
 ft.DataCell(ft.Text("John")),
 ft.DataCell(ft.Text("Smith")),
 ft.DataCell(ft.Text("43")),
],
),
 ft.DataRow(
 cells=[
 ft.DataCell(ft.Text("Jack")),
 ft.DataCell(ft.Text("Brown")),
 ft.DataCell(ft.Text("19")),
],
),
 ft.DataRow(
 cells=[
 ft.DataCell(ft.Text("Alice")),
 ft.DataCell(ft.Text("Wong")),
 ft.DataCell(ft.Text("25")),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/basic.png>

### Horizontal margin and column spacing

Use `horizontal_margin` to control the left and right edge spacing of the first and last columns. Use `column_spacing` to control spacing between columns.

play\_arrowTry Online

```
import flet as ft

def _cell(label: str, color: str = ft.Colors.SURFACE_CONTAINER_HIGHEST) -> ft.DataCell:
 return ft.DataCell(
 ft.Container(
 width=90,
 height=32,
 alignment=ft.Alignment.CENTER,
 bgcolor=color,
 border=ft.Border.all(1, ft.Colors.BLACK_26),
```

```

 content=ft.Text(label, size=12, weight=ft.FontWeight.W_600),
)
)

def main(page: ft.Page):
 page.scroll = ft.ScrollMode.AUTO
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.theme_mode = ft.ThemeMode.DARK

 def update_spacing() -> None:
 table.horizontal_margin = horizontal_margin_slider.value
 table.column_spacing = column_spacing_slider.value
 table.update()

 def handle_spacing_change(_: ft.Event[ft.Slider]) -> None:
 update_spacing()

 def set_preset(horizontal_margin: float, column_spacing: float) -> None:
 horizontal_margin_slider.value = horizontal_margin
 column_spacing_slider.value = column_spacing
 horizontal_margin_slider.update()
 column_spacing_slider.update()
 update_spacing()

 horizontal_margin_slider = ft.Slider(
 key="horizontal_margin_slider",
 min=0,
 max=40,
 divisions=40,
 value=16,
 label="{value}",
 on_change=handle_spacing_change,
)
 column_spacing_slider = ft.Slider(
 key="column_spacing_slider",
 min=0,
 max=40,
 divisions=40,
 value=16,
 label="{value}",
 on_change=handle_spacing_change,
)

 table = ft.DataTable(
 border=ft.Border.all(1, ft.Colors.ON_SURFACE_VARIANT),
 horizontal_margin=horizontal_margin_slider.value,
 column_spacing=column_spacing_slider.value,
 horizontal_lines=ft.BorderSide(1, ft.Colors.ON_SURFACE_VARIANT),
 vertical_lines=ft.BorderSide(1, ft.Colors.ON_SURFACE_VARIANT),
 heading_row_height=40,
 data_row_min_height=40,
 data_row_max_height=40,
 columns=[
 ft.DataColumn(label="Col A"),
 ft.DataColumn(label="Col B"),
 ft.DataColumn(label="Col C"),
],
 rows=[
 ft.DataRow(cells=[_cell("A1"), _cell("B1"), _cell("C1")]),
 ft.DataRow(cells=[_cell("A2"), _cell("B2"), _cell("C2")]),
],
)

 page.appbar = ft.AppBar(title="DataTable spacing")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Container(
 width=520,
 padding=12,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 border_radius=8,
 content=ft.Column(
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Text("horizontal_margin (outer edges)"),
 horizontal_margin_slider,
 ft.Text("column_spacing (between columns)"),
 column_spacing_slider,
 ft.Row(
 wrap=True,
 controls=[
 ft.FilledButton(
 "Reset",

```

```

 on_click=lambda _: set_preset(16, 16),
),
 ft.OutlinedButton(
 "Compact preset",
 on_click=lambda _: set_preset(0, 0),
),
 ft.OutlinedButton(
 "Spacious preset",
 on_click=lambda _: set_preset(24, 32),
),
],
),
],
),
table,
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/spacing\\_flow.gif](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/spacing_flow.gif)

## Adaptive row heights

Setting `data_row_max_height` to `float('inf')` (infinity) will cause the `DataTable` to let each individual row adapt its height to its respective content, instead of all rows having the same height.

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.DataTable(
 width=560,
 data_row_min_height=48,
 data_row_max_height=float("inf"),
 columns=[
 ft.DataColumn(label="Description"),
 ft.DataColumn(label="Notes"),
],
 rows=[
 ft.DataRow(
 cells=[
 ft.DataCell("TWO lines visible without overflow"),
 ft.DataCell("Line 1\nLine 2"),
],
),
 ft.DataRow(
 cells=[
 ft.DataCell("FOUR lines visible without overflow"),
 ft.DataCell("Line 1\nLine 2\nLine 3\nLine 4"),
],
),
 ft.DataRow(
 cells=[
 ft.DataCell("FIVE lines visible without overflow"),
 ft.DataCell("Line 1\nLine 2\nLine 3\nLine 4\nLine 5"),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/adaptive\\_row\\_heights.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/adaptive_row_heights.png)

## Sortable columns and selectable rows

This example demonstrates row selection (including select-all), sortable string and numeric columns, and stable selection across sorts and refreshes.

play\_arrowTry Online

```

import flet as ft

```



```

def main(page: ft.Page):
 inventory_items = [
 {"id": 1, "name": "Alpha", "qty": 4},
 {"id": 2, "name": "Bravo", "qty": 9},
 {"id": 3, "name": "Charlie", "qty": 2},
 {"id": 4, "name": "Delta", "qty": 6},
 {"id": 5, "name": "Echo", "qty": 3},
 {"id": 6, "name": "Foxtrot", "qty": 8},
 {"id": 7, "name": "Golf", "qty": 1},
 {"id": 8, "name": "Hotel", "qty": 7},
 {"id": 9, "name": "India", "qty": 5},
 {"id": 10, "name": "Juliet", "qty": 10},
]
 displayed_items = list(inventory_items)
 selected_item_ids: set[int] = {1, 3, 5}

 sort_key_for_column = {
 0: lambda item: str(item["name"]).lower(),
 1: lambda item: int(item["qty"]),
 }

 def build_rows(items: list[dict[str, int | str]]) -> list[ft.DataRow]:
 return [
 ft.DataRow(
 selected=item["id"] in selected_item_ids,
 on_select_change=handle_row_selection_change,
 data=item["id"],
 cells=[
 ft.DataCell(ft.Text(item["name"])),
 ft.DataCell(ft.Text(str(item["qty"]))),
],
)
 for item in items
]

 def refresh_table_rows() -> None:
 table.rows = build_rows(displayed_items)
 table.update()

 def handle_row_selection_change(e: ft.Event[ft.DataRow]) -> None:
 row = e.control
 item_id = row.data
 is_selected = e.data

 if is_selected:
 selected_item_ids.add(item_id)
 else:
 selected_item_ids.discard(item_id)

 row.selected = is_selected
 row.update()

 def handle_select_all(e: ft.Event[ft.DataTable]) -> None:
 if e.data:
 selected_item_ids.update(int(item["id"]) for item in displayed_items)
 else:
 selected_item_ids.clear()

 refresh_table_rows()

 def handle_column_sort(e: ft.DataColumnSortEvent) -> None:
 displayed_items.sort(
 key=sort_key_for_column[e.column_index],
 reverse=not e.ascending,
)

 table.sort_column_index = e.column_index
 table.sort_ascending = e.ascending
 refresh_table_rows()

 table = ft.DataTable(
 width=700,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 border_radius=10,
 vertical_lines=ft.border.BorderSide(1, ft.Colors.OUTLINE_VARIANT),
 horizontal_lines=ft.border.BorderSide(1, ft.Colors.OUTLINE_VARIANT),
 sort_column_index=0,
 sort_ascending=True,
 heading_row_color=ft.Colors.SURFACE_CONTAINER_HIGHEST,
 heading_row_height=100,
 data_row_color={
 ft.ControlState.HOVERED: ft.Colors.with_opacity(0.08, ft.Colors.PRIMARY),
 ft.ControlState.SELECTED: ft.Colors.with_opacity(0.14, ft.Colors.PRIMARY),
 },
 show_checkbox_column=True,
 on_select_all=handle_select_all,

```

```

divider_thickness=1,
column_spacing=200,
columns=[
 ft.DataColumn(
 label=ft.Text("Item"),
 on_sort=handle_column_sort,
),
 ft.DataColumn(
 label=ft.Text("Quantity"),
 tooltip="Numeric quantity",
 numeric=True,
 on_sort=handle_column_sort,
),
],
rows=build_rows(displayed_items),
)

page.add(
 ft.SafeArea(
 content=table,
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/sortable\\_and\\_selectable.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/sortable_and_selectable.png)

## Handling events

### play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 def handle_row_selection_change(e: ft.Event[ft.DataRow]) -> None:
 if e.control.data == 1:
 row1.selected = not row1.selected
 elif e.control.data == 2:
 row2.selected = not row2.selected
 elif e.control.data == 3:
 row3.selected = not row3.selected

 table.update()

 def handle_column_sort(e: ft.DataColumnSortEvent) -> None:
 if e.control.data in [1, 2]:
 print(f"{e.column_index}, {e.ascending}")
 table.sort_ascending = e.ascending
 table.update()

 table = ft.DataTable(
 width=700,
 bgcolor=ft.Colors.TEAL_ACCENT_200,
 border=ft.Border.all(2, ft.Colors.RED_ACCENT_200),
 border_radius=10,
 vertical_lines=ft.border.BorderSide(3, ft.Colors.BLUE_600),
 horizontal_lines=ft.border.BorderSide(1, ft.Colors.GREEN_600),
 sort_column_index=0,
 sort_ascending=True,
 heading_row_color=ft.Colors.BLACK_12,
 heading_row_height=100,
 data_row_color={ft.ControlState.HOVERED: "0x30FF0000"},
 show_checkbox_column=True,
 divider_thickness=0,
 column_spacing=200,
 columns=[
 ft.DataColumn(
 label=ft.Text("Column 1"),
 tooltip="This is the first column",
 data=1,
 on_sort=handle_column_sort,
),
 ft.DataColumn(
 label=ft.Text("Column 2"),
 tooltip="This is a second column",
 numeric=True,
 data=2,
 on_sort=handle_column_sort,
),
],
 rows=[
 row1 := ft.DataRow(
 cells=[ft.DataCell(ft.Text("A")), ft.DataCell(ft.Text("1"))],
 selected=True,
),

```

```

 on_select_change=handle_row_selection_change,
 data=1,
),
 row2 := ft.DataRow(
 cells=[ft.DataCell(ft.Text("B")), ft.DataCell(ft.Text("2"))],
 selected=False,
 on_select_change=handle_row_selection_change,
 data=2,
),
 row3 := ft.DataRow(
 cells=[ft.DataCell(ft.Text("C")), ft.DataCell(ft.Text("3"))],
 selected=False,
 on_select_change=handle_row_selection_change,
 data=3,
),
],
)

page.add(
 ft.SafeArea(
 content=table,
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/handling\\_events\\_flow.gif](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/datatable/handling_events_flow.gif)

## Properties

### buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The background color for this table.

### buildborderclass-attributeinstance-attribute

border: Border | None = None

The border around the table.

### buildborder\_radiusclass-attributeinstance-attribute

border\_radius: BorderRadiusValue | None = None

Border corners.

### buildcheckbox\_horizontal\_marginclass-attributeinstance-attribute

checkbox\_horizontal\_margin: Number | None = None

Horizontal margin around the checkbox, if it is displayed.

### buildclip\_behaviorclass-attributeinstance-attribute

clip\_behavior: ClipBehavior = ClipBehavior.NONE

Defines how the contents of this table are clipped.

### buildcolumn\_spacingclass-attributeinstance-attribute

column\_spacing: Number | None = None

The horizontal margin between the contents of each data column.

### buildcolumnsinstance-attribute

columns: list[DataColumn]

A list of DataColumn controls describing table columns.

Raises:

- ValueError - If there are no visible columns.

### bulddata\_row\_colorclass-attributeinstance-attribute

data\_row\_color: ControlStateValue[ColorValue] | None = None

The background color for the data rows.

The effective background color can be made to depend on the `ControlState` state, i.e. if the row is selected, pressed, hovered, focused, disabled or enabled. The color is painted as an overlay to the row. To make sure that the row's `InkWell` is visible (when pressed, hovered and focused), it is recommended to use a translucent background color.

### **builddata\_row\_max\_heightclass-attributeinstance-attribute**

`data_row_max_height: Number | None = None`

The maximum height of each row (excluding the row that contains column headings). Set to `float("inf")` for the height of each row to adjust automatically with its content.

Defaults to 48.0.

Raises:

- `ValueError` - If it is not greater than or equal to `data_row_min_height`.

### **builddata\_row\_min\_heightclass-attributeinstance-attribute**

`data_row_min_height: Number | None = None`

The minimum height of each row (excluding the row that contains column headings).

Defaults to 48.0.

Raises:

- `ValueError` - If it is not less than or equal to `data_row_max_height`.

### **builddata\_text\_styleclass-attributeinstance-attribute**

`data_text_style: TextStyle | None = None`

The text style of the data rows.

### **builddivider\_thicknessclass-attributeinstance-attribute**

`divider_thickness: Number = 1.0`

The width of the divider that appears between rows.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

### **buildgradientclass-attributeinstance-attribute**

`gradient: Gradient | None = None`

The background gradient of this table.

### **buildheading\_row\_colorclass-attributeinstance-attribute**

`heading_row_color: ControlStateValue[ColorValue] | None = (None)`

The background color for the heading row.

The effective background color can be made to depend on the `ControlState` state, i.e. if the row is pressed, hovered, focused when sorted. The color is painted as an overlay to the row. To make sure that the row's `InkWell` is visible (when pressed, hovered and focused), it is recommended to use a translucent color.

### **buildheading\_row\_heightclass-attributeinstance-attribute**

`heading_row_height: Number | None = None`

The height of the heading row.

### **buildheading\_text\_styleclass-attributeinstance-attribute**

`heading_text_style: TextStyle | None = None`

The text style for the heading row.

### **buildhorizontal\_linesclass-attributeinstance-attribute**

`horizontal_lines: BorderSide | None = None`

Set the color and width of horizontal lines between rows.

### **buildhorizontal\_marginclass-attributeinstance-attribute**

`horizontal_margin: Number | None = None`

The horizontal margin between the edges of this table and the content in the first and last cells of each row.

When a checkbox is displayed, it is also the margin between the checkbox the content in the first data column.

### **buildrowsclass-attributeinstance-attribute**

`rows: list[DataRow] = field(default_factory=list)`

A list of DataRow controls defining table rows.

Raises:

- **ValueError** - If any visible row does not contain exactly as many visible DataCells as there are visible columns.

### **buildshow\_bottom\_borderclass-attributeinstance-attribute**

`show_bottom_border: bool = False`

Whether a border at the bottom of the table is displayed.

By default, a border is not shown at the bottom to allow for a border around the table defined by decoration.

### **buildshow\_checkbox\_columnclass-attributeinstance-attribute**

`show_checkbox_column: bool = False`

Whether the control should display checkboxes for selectable rows.

If True, a checkbox will be placed at the beginning of each row that is selectable. However, if DataRow.on\_select\_change is not set for any row, checkboxes will not be placed, even if this value is True.

If False, all rows will not display a checkbox.

### **buildsort\_ascendingclass-attributeinstance-attribute**

`sort_ascending: bool = False`

Whether the column mentioned in sort\_column\_index, if any, is sorted in ascending order.

If True, the order is ascending (meaning the rows with the smallest values for the current sort column are first in the table).

If False, the order is descending (meaning the rows with the smallest values for the current sort column are last in the table).

### **buildsort\_column\_indexclass-attributeinstance-attribute**

`sort_column_index: int | None = None`

The current primary sort key's column.

If specified, indicates that the indicated column is the column by which the data is sorted. The number must correspond to the index of the relevant column in columns.

Setting this will cause the relevant column to have a sort indicator displayed.

When this is None, it implies that the table's sort order does not correspond to any of the columns.

Raises:

- **ValueError** - If it is out of range relative to the visible columns.

### **buildvertical\_linesclass-attributeinstance-attribute**

`vertical_lines: BorderSide | None = None`

Set the color and width of vertical lines between columns.

## **Events**

### **bolton\_select\_allclass-attributeinstance-attribute**

`on_select_all: ControlEventHandler[DataTable] | None = None`

Invoked when the user selects or unselects every row, using the checkbox in the heading row.

If this is None, then the DataRow.on\_select\_change callback of every rows of this table is invoked appropriately instead.

Tip

To control whether a particular row is selectable or not, see `DataRow.on_select_change`. This callback is only relevant if any row is selectable.

# DataTable2

Enhanced data table for Flet that adds sticky headers, fixed rows/columns, and other UX improvements via the flet-datatable2 extension.

It wraps the Flutter data\_table\_2 package.

## Platform Support

Platform	Windows	macOS	Linux
Supported	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>

## Usage

Add flet-datatable2 to your project dependencies:

- uv
  - pip
- ```
uv add flet-datatable2
pip install flet-datatable2
```

Examples

Empty state

[play_arrowTry Online](#)

```
import flet as ft
import flet_datatable2 as fdt

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=fdt.DataTable2(
                expand=True,
                empty=ft.Text("This table is empty."),
                columns=[
                    fdt.DataColumn2(label=ft.Text("First name")),
                    fdt.DataColumn2(label=ft.Text("Last name")),
                    fdt.DataColumn2(label=ft.Text("Age"), numeric=True),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: [Image - https://flet.dev/docs/test-images/examples/extensions/datatable2/golden/macos/datatable2/empty_state.png](https://flet.dev/docs/test-images/examples/extensions/datatable2/golden/macos/datatable2/empty_state.png)

Sortable and selectable

[play_arrowTry Online](#)

```
from dataclasses import dataclass

import flet as ft
import flet_datatable2 as ftd

@dataclass
class Dessert:
    name: str
    calories: float
    fat: float
    carbs: float
    protein: float
    sodium: float
    calcium: float
    iron: float

desserts = [
```

```

Dessert("Frozen Yogurt", 159, 6.0, 24, 4.0, 87, 14, 1),
Dessert("Ice Cream Sandwich", 237, 9.0, 37, 4.3, 129, 8, 1),
Dessert("Eclair", 262, 16.0, 24, 6.0, 337, 6, 7),
Dessert("Cupcake", 305, 3.7, 67, 4.3, 413, 3, 8),
Dessert("Gingerbread", 356, 16.0, 49, 3.9, 327, 7, 16),
Dessert("Jelly Bean", 375, 0.0, 94, 0.0, 50, 0, 0),
Dessert("Lollipop", 392, 0.2, 98, 0.0, 38, 0, 2),
Dessert("Honeycomb", 408, 3.2, 87, 6.5, 562, 0, 45),
Dessert("Donut", 452, 25.0, 51, 4.9, 326, 2, 22),
Dessert("Apple Pie", 518, 26.0, 65, 7.0, 54, 12, 6),
Dessert("Frozen Yougurt with sugar", 168, 6.0, 26, 4.0, 87, 14, 1),
Dessert("Ice Cream Sandwich with sugar", 246, 9.0, 39, 4.3, 129, 8, 1),
Dessert("Eclair with sugar", 271, 16.0, 26, 6.0, 337, 6, 7),
Dessert("Cupcake with sugar", 314, 3.7, 69, 4.3, 413, 3, 8),
Dessert("Gingerbread with sugar", 345, 16.0, 51, 3.9, 327, 7, 16),
Dessert("Jelly Bean with sugar", 364, 0.0, 96, 0.0, 50, 0, 0),
Dessert("Lollipop with sugar", 401, 0.2, 100, 0.0, 38, 0, 2),
Dessert("Honeycomb with sugar", 417, 3.2, 89, 6.5, 562, 0, 45),
Dessert("Donut with sugar", 461, 25.0, 53, 4.9, 326, 2, 22),
Dessert("Apple pie with sugar", 527, 26.0, 67, 7.0, 54, 12, 6),
Dessert("Frozen yougurt with honey", 223, 6.0, 36, 4.0, 87, 14, 1),
Dessert("Ice Cream Sandwich with honey", 301, 9.0, 49, 4.3, 129, 8, 1),
Dessert("Eclair with honey", 326, 16.0, 36, 6.0, 337, 6, 7),
Dessert("Cupcake with honey", 369, 3.7, 79, 4.3, 413, 3, 8),
Dessert("Gingerbread with honey", 420, 16.0, 61, 3.9, 327, 7, 16),
Dessert("Jelly Bean with honey", 439, 0.0, 106, 0.0, 50, 0, 0),
Dessert("Lollipop with honey", 456, 0.2, 110, 0.0, 38, 0, 2),
Dessert("Honeycomb with honey", 472, 3.2, 99, 6.5, 562, 0, 45),
Dessert("Donut with honey", 516, 25.0, 63, 4.9, 326, 2, 22),
Dessert("Apple pie with honey", 582, 26.0, 77, 7.0, 54, 12, 6),
]

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    sorted_desserts = list(desserts)

    def handle_row_selection_change(e: ft.Event[ftd.DataRow2]) -> None:
        e.control.selected = not e.control.selected
        e.control.update()

    def sort_column(e: ft.DataColumnSortEvent) -> None:
        sorters = [
            lambda d: d.name.lower(),
            lambda d: d.calories,
            lambda d: d.fat,
            lambda d: d.carbs,
            lambda d: d.protein,
            lambda d: d.sodium,
            lambda d: d.calcium,
            lambda d: d.iron,
        ]
        sorted_desserts.sort(key=sorters[e.column_index], reverse=not e.ascending)
        table.rows = get_data_rows(sorted_desserts)
        table.sort_column_index = e.column_index
        table.sort_ascending = e.ascending
        table.update()

    def get_data_columns() -> list[ftd.DataColumn2]:
        return [
            ftd.DataColumn2(
                label=ft.Text("Name"),
                size=ftd.DataColumnSize.L,
                on_sort=sort_column,
                heading_row_alignment=ft.MainAxisAlignment.START,
            ),
            ftd.DataColumn2(
                label=ft.Text("Calories"),
                on_sort=sort_column,
                numeric=True,
                heading_row_alignment=ft.MainAxisAlignment.END,
            ),
            ftd.DataColumn2(label=ft.Text("Fat"), on_sort=sort_column, numeric=True),
            ftd.DataColumn2(label=ft.Text("Carbs"), on_sort=sort_column, numeric=True),
            ftd.DataColumn2(
                label=ft.Text("Protein"),
                on_sort=sort_column,
                numeric=True,
            ),
            ftd.DataColumn2(label=ft.Text("Sodium"), on_sort=sort_column, numeric=True),
            ftd.DataColumn2(
                label=ft.Text("Calcium"),
                on_sort=sort_column,
                numeric=True,
            ),
        ],

```



```

        ftd.DataColumn2(label=ft.Text("Iron"), on_sort=sort_column, numeric=True),
    ]

def get_data_rows(items: list) -> list[ft.DataRow2]:
    return [
        ftd.DataRow2(
            specific_row_height=50,
            on_select_change=handle_row_selection_change,
            cells=[
                ft.DataCell(content=ft.Text(dessert.name)),
                ft.DataCell(content=ft.Text(dessert.calories)),
                ft.DataCell(content=ft.Text(dessert.fat)),
                ft.DataCell(content=ft.Text(dessert.carbs)),
                ft.DataCell(content=ft.Text(dessert.protein)),
                ft.DataCell(content=ft.Text(dessert.sodium)),
                ft.DataCell(content=ft.Text(dessert.calcium)),
                ft.DataCell(content=ft.Text(dessert.iron)),
            ],
        )
        for dessert in items
    ]

table = ftd.DataTable2(
    expand=True,
    show_checkbox_column=True,
    column_spacing=0,
    heading_row_color=ft.Colors.SECONDARY_CONTAINER,
    horizontal_margin=12,
    sort_ascending=True,
    bottom_margin=10,
    min_width=600,
    on_select_all=lambda _: print("All selected"),
    columns=get_data_columns(),
    rows=get_data_rows(sorted_desserts),
)

page.add(ft.SafeArea(expand=True, content=table))

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: https://flet.dev/docs/examples/extensions/datatable2/media/sortable_and_selectable.gif

Column widths

play_arrowTry Online

```

import flet as ft
import flet_datatable2 as fdt

def main(page: ft.Page):
    def cell_text(value: str) -> ft.Text:
        """A helper to truncate any overflowing cell text with an ellipsis."""
        return ft.Text(value, overflow=ft.TextOverflow.ELLIPSIS, max_lines=1)

    page.add(
        ft.SafeArea(
            expand=True,
            content=fdt.DataTable2(
                expand=True,
                min_width=600,
                columns=[
                    # Absolute pixel width – best for predictable, short fields.
                    fdt.DataColumn2(label="Name", fixed_width=140),
                    # Relative size S – compact, auto-fits the remaining space.
                    fdt.DataColumn2(label="Role", size=fdt.DataColumnSize.S),
                    # Relative size L – takes the lion's share of what's left.
                    fdt.DataColumn2(label="Recent work", size=fdt.DataColumnSize.L),
                ],
                rows=[
                    ft.DataRow(
                        cells=[
                            ft.DataCell(cell_text("Alice Nakamura")),
                            ft.DataCell(cell_text("Engineer")),
                            ft.DataCell(
                                cell_text(
                                    "Led the migration of our checkout service "
                                    "to a set of composable workers, cutting "
                                    "p99 latency in half."
                                )
                            ),
                        ],
                    ),
                    ft.DataRow(
                        cells=[

```

```

        # Longer than 140px - shows ellipsis in a fixed column.
        ft.DataCell(cell_text("Bartholomew Laurent-Fitzgerald")),
        ft.DataCell(cell_text("Designer")),
        ft.DataCell(
            cell_text(
                "Rebuilt the onboarding flow and maintains "
                "the internal design-system token registry."
            )
        ),
    ],
),
ft.DataRow(
    cells=[
        ft.DataCell(cell_text("Chen")),
        ft.DataCell(cell_text("PM")),
        ft.DataCell(
            cell_text("Owns the Platform Reliability roadmap.")
        ),
    ],
),
],
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/extensions/datatable2/golden/macros/datatable2/column_widths.png

Description

Provides sticky header row, scrollable data rows, and additional layout flexibility with DataColumn2 and DataRow2.

Note

DataTable2 doesn't support DataTable.data_row_min_height and DataTable.data_row_max_height properties present in the parent DataTable. Use data_row_height instead.

Inherits: DataTable

Properties

- bottom_margin - Adds space after the last row if set.
- checkbox_alignment - Alignment of the checkbox.
- columns - A list of table columns.
- data_row_checkbox_theme - Overrides theme of checkboxes in each data row.
- data_row_height - Height of each data row.
- data_row_max_height
- data_row_min_height
- empty - Placeholder control shown when there are no data rows.
- fixed_columns_color - Background color for sticky left columns.
- fixed_corner_color - Background color of the fixed top-left corner cell.
- fixed_left_columns - Number of sticky columns on the left.
- fixed_top_rows - Number of sticky rows from the top.
- heading_checkbox_theme - Overrides theme of the heading checkbox.
- lm_ratio - Ratio of Large column width to Medium.
- min_width - Minimum table width before horizontal scrolling kicks in.
- rows - A list of table rows.
- show_heading_checkbox - Controls visibility of the heading checkbox.
- sm_ratio - Ratio of Small column width to Medium.
- sort_arrow_animation_duration - Duration of sort arrow animation.

- `sort_arrow_icon` - Icon shown when sorting is applied.
- `sort_arrow_icon_color` - When set always overrides/preceeds default arrow icon color.
- `visible_horizontal_scroll_bar` - Determines visibility of the horizontal scrollbar.
- `visible_vertical_scroll_bar` - Determines visibility of the vertical scrollbar.

Properties

`buildbottom_marginclass-attributeinstance-attribute`

`bottom_margin: Number | None = None`

Adds space after the last row if set.

`buildcheckbox_alignmentclass-attributeinstance-attribute`

```
checkbox_alignment: Alignment = field(
    default_factory=lambda: Alignment.CENTER
)
```

Alignment of the checkbox.

`buildcolumnsinstance-attribute`

`columns: list[DataColumn2 | DataColumn]`

A list of table columns.

`bulddata_row_checkbox_themeclass-attributeinstance-attribute`

`data_row_checkbox_theme: CheckboxTheme | None = None`

Overrides theme of checkboxes in each data row.

`bulddata_row_heightclass-attributeinstance-attribute`

`data_row_height: Number | None = None`

Height of each data row.

`bulddata_row_max_heightclass-attributeinstance-attribute`

```
data_row_max_height: None = field(
    init=False,
    repr=False,
    compare=False,
    metadata={"skip": True},
)
```

`bulddata_row_min_heightclass-attributeinstance-attribute`

```
data_row_min_height: None = field(
    init=False,
    repr=False,
    compare=False,
    metadata={"skip": True},
)
```

`buildeptyclass-attributeinstance-attribute`

`empty: Control | None = None`

Placeholder control shown when there are no data rows.

`buildfixed_columns_colorclass-attributeinstance-attribute`

`fixed_columns_color: ColorValue | None = None`

Background color for sticky left columns.

`buildfixed_corner_colorclass-attributeinstance-attribute`

`fixed_corner_color: ColorValue | None = None`

Background color of the fixed top-left corner cell.

`buildfixed_left_columnsclass-attributeinstance-attribute`

`fixed_left_columns: int = 0`

Number of sticky columns on the left. Includes checkbox column, if present.

buildfixed_top_rowsclass-attributeinstance-attribute

```
fixed_top_rows: int = 1
```

Number of sticky rows from the top. Includes heading row by default.

buildheading_checkbox_themeclass-attributeinstance-attribute

```
heading_checkbox_theme: CheckboxTheme | None = None
```

Overrides theme of the heading checkbox.

buildlm_ratiorclass-attributeinstance-attribute

```
lm_ratio: Number = 1.2
```

Ratio of Large column width to Medium.

buildmin_widthclass-attributeinstance-attribute

```
min_width: Number | None = None
```

Minimum table width before horizontal scrolling kicks in.

buildrowsclass-attributeinstance-attribute

```
rows: list[DataRow | DataRow2] = field(
    default_factory=list
)
```

A list of table rows.

buildshow_heading_checkboxclass-attributeinstance-attribute

```
show_heading_checkbox: bool = True
```

Controls visibility of the heading checkbox.

buildsm_ratiorclass-attributeinstance-attribute

```
sm_ratio: Number = 0.67
```

Ratio of Small column width to Medium.

buildsort_arrow_animation_durationclass-attributeinstance-attribute

```
sort_arrow_animation_duration: DurationValue = field(
    default_factory=lambda: Duration(milliseconds=150)
)
```

Duration of sort arrow animation.

buildsort_arrow_iconclass-attributeinstance-attribute

```
sort_arrow_icon: IconData = Icons.ARROW_UPWARD
```

Icon shown when sorting is applied.

buildsort_arrow_icon_colorclass-attributeinstance-attribute

```
sort_arrow_icon_color: ColorValue | None = None
```

When set always overrides/preceeds default arrow icon color.

buildvisible_horizontal_scroll_barclass-attributeinstance-attribute

```
visible_horizontal_scroll_bar: bool | None = None
```

Determines visibility of the horizontal scrollbar.

buildvisible_vertical_scroll_barclass-attributeinstance-attribute

```
visible_vertical_scroll_bar: bool | None = None
```

Determines visibility of the vertical scrollbar.

DatePicker

A Material-style date picker dialog.

It can be opened by calling `Page.show_dialog` method.

Depending on the `entry_mode`, it will show either a Calendar or an Input (TextField) for picking a date.

```
picker = ft.DatePicker()

ft.Button(
    "Pick date",
    icon=ft.Icons.CALENDAR_MONTH,
    on_click=lambda _: page.show_dialog(picker),
)
```

Image: DatePicker - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/date_picker/image_for_docs.png

Date picker

Inherits: `DialogControl`

Properties

- `barrier_color` - The color of the modal barrier that darkens everything below this picker's dialog.
- `cancel_text` - The text that is displayed on the cancel button.
- `confirm_text` - The text that is displayed on the confirm button.
- `current_date` - The date representing today.
- `date_picker_mode` - Initial display mode of this picker.
- `entry_mode` - The initial mode of date entry method for the date picker dialog.
- `error_format_text` - The error message displayed below the text field if the entered date is not in the correct format.
- `error_invalid_text` - The error message displayed below the text field if the date is earlier than `first_date` or later than `last_date`.
- `field_hint_text` - The hint text displayed in the text field.
- `field_label_text` - The label text displayed in the TextField.
- `first_date` - The earliest allowable date that the user can select.
- `help_text` - The text that is displayed at the top of the header.
- `inset_padding` - The amount of padding added to `view_insets` of the `Page.media` on the outside of this picker's dialog.
- `keyboard_type` - The type of keyboard to use for editing the text.
- `last_date` - The latest allowable date that the user can select.
- `locale` - The locale for this date picker dialog.
- `modal` - Whether this date picker cannot be dismissed by clicking the area outside of it.
- `switch_to_calendar_icon` - The icon displayed in the corner of this picker's dialog when `entry_mode` is `DatePickerEntryMode.INPUT`.
- `switch_to_input_icon` - The icon displayed in the corner of this picker's dialog when `entry_mode` is `DatePickerEntryMode.CALENDAR`.
- `value` - The selected date that the picker should display.

Events

- `on_change` - Called when user clicks confirm button.
- `on_entry_mode_change` - Called when the `entry_mode` is changed from the user interface.

Examples

DatePicker

play_arrowTry Online

```
import datetime

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    messages = ft.Column(tight=True)

    def handle_change(e: ft.Event[ft.DatePicker]):
        messages.controls.append(
            ft.Text(f"Date changed: {e.control.value.strftime('%m/%d/%Y')}")
        )

    def handle_dismissal(_: ft.Event[ft.DialogControl]):
        messages.controls.append(ft.Text("DatePicker dismissed"))

    today = datetime.datetime(year=2025, month=4, day=15)

    picker = ft.DatePicker(
        first_date=datetime.datetime(year=today.year - 1, month=1, day=1),
        last_date=datetime.datetime(year=today.year + 1, month=today.month, day=20),
        current_date=today,
        on_change=handle_change,
        on_dismiss=handle_dismissal,
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    ft.Button(
                        icon=ft.Icons.CALENDAR_MONTH,
                        on_click=lambda _: page.show_dialog(picker),
                        content="Pick date",
                    ),
                    messages,
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/date_picker/basic.png

Custom locale

play_arrowTry Online

```
import datetime

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    current_date = datetime.datetime(year=2025, month=4, day=15)

    page.add(
        ft.SafeArea(
            content=ft.Button(
                icon=ft.Icons.CALENDAR_MONTH,
                on_click=lambda _: page.show_dialog(
                    ft.DatePicker(
                        locale=ft.Locale("zh", "Hans"),
                        first_date=datetime.datetime(year=2024, month=1, day=1),
                        last_date=datetime.datetime(year=2026, month=12, day=31),
                        current_date=current_date,
                    )
                ),
                content="Pick date (zh_Hans locale)",
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: custom-locale - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/date_picker/custom_locale.png

Properties

buildbarrier_colorclass-attributeinstance-attribute

`barrier_color: ColorValue | None = None`

The color of the modal barrier that darkens everything below this picker's dialog.

If None, the `DialogTheme.barrier_color` is used. If it is also None, then `Colors.BLACK_54` is used.

buildcancel_textclass-attributeinstance-attribute

`cancel_text: str | None = None`

The text that is displayed on the cancel button.

Defaults to "Cancel".

buildconfirm_textclass-attributeinstance-attribute

`confirm_text: str | None = None`

The text that is displayed on the confirm button.

Defaults to "OK".

buildcurrent_dateclass-attributeinstance-attribute

```
current_date: DateTimeValue = field(
    default_factory=lambda: datetime.now()
)
```

The date representing today. It will be highlighted in the day grid.

bulddate_picker_modeclass-attributeinstance-attribute

`date_picker_mode: DatePickerMode = DatePickerMode.DAY`

Initial display mode of this picker.

buildentry_modeclass-attributeinstance-attribute

```
entry_mode: DatePickerEntryMode = (
    DatePickerEntryMode.CALENDAR
)
```

The initial mode of date entry method for the date picker dialog.

builderror_format_textclass-attributeinstance-attribute

`error_format_text: str | None = None`

The error message displayed below the text field if the entered date is not in the correct format.

Defaults to "Invalid format".

builderror_invalid_textclass-attributeinstance-attribute

`error_invalid_text: str | None = None`

The error message displayed below the text field if the date is earlier than `first_date` or later than `last_date`.

Defaults to "Out of range".

buildfield_hint_textclass-attributeinstance-attribute

`field_hint_text: str | None = None`

The hint text displayed in the text field.

The default value is the date format string that depends on your locale. For example, 'mm/dd/yyyy' for en_US.

buildfield_label_textclass-attributeinstance-attribute

`field_label_text: str | None = None`

The label text displayed in the TextField.

If None, defaults to the words representing the date format string. For example, 'Month, Day, Year' for en_US.

Defaults to "Enter Date".

buildfirst_dateclass-attributeinstance-attribute

```

first_date: DateTimeValue = field(
    default_factory=lambda: datetime(
        year=1900, month=1, day=1
    )
)

```

The earliest allowable date that the user can select.

Defaults to January 1, 1900.

buildhelp_textclass-attributeinstance-attribute

```

help_text: str | None = None

```

The text that is displayed at the top of the header.

This is used to indicate to the user what they are selecting a date for.

Defaults to "Select date".

buildinset_paddingclass-attributeinstance-attribute

```

inset_padding: PaddingValue = field(
    default_factory=lambda: Padding.symmetric(
        horizontal=16.0, vertical=24.0
    )
)

```

The amount of padding added to view_insets of the Page.media on the outside of this picker's dialog.

This defines the minimum space between the screen's edges and the dialog.

buildkeyboard_typeclass-attributeinstance-attribute

```

keyboard_type: KeyboardType = KeyboardType.DATETIME

```

The type of keyboard to use for editing the text.

buildlast_dateclass-attributeinstance-attribute

```

last_date: DateTimeValue = field(
    default_factory=lambda: datetime(
        year=2050, month=1, day=1
    )
)

```

The latest allowable date that the user can select.

Defaults to January 1, 2050.

buildlocaleclass-attributeinstance-attribute

```

locale: Locale | None = None

```

The locale for this date picker dialog. It is intended for (rare) cases where this dialog should be localized differently from the rest of the page.

It overrides the locale used by the page (see Page.locale_configuration), but does not participate in page-level locale resolution.

If set to None (the default) or an inexistent/unsupported locale, the current_locale of the Page.locale_configuration is used as fallback.

buildmodalclass-attributeinstance-attribute

```

modal: bool = False

```

Whether this date picker cannot be dismissed by clicking the area outside of it.

buildswitch_to_calendar_iconclass-attributeinstance-attribute

```

switch_to_calendar_icon: IconData | None = None

```

The icon displayed in the corner of this picker's dialog when entry_mode is DatePickerEntryMode.INPUT.

Clicking on this icon changes the entry_mode to DatePickerEntryMode.CALENDAR.

If None, defaults to Icons.CALENDAR_TODAY.

buildswitch_to_input_iconclass-attributeinstance-attribute

```

switch_to_input_icon: IconData | None = None

```

The icon displayed in the corner of this picker's dialog when entry_mode is DatePickerEntryMode.CALENDAR.

Clicking on icon changes the entry_mode to DatePickerEntryMode.INPUT.

If None, defaults to Icons.EDIT_OUTLINED.

buildvalueclass-attributeinstance-attribute

value: DateTimeValue | None = None

The selected date that the picker should display.

Defaults to current_date.

Events

bolton_changeclass-attributeinstance-attribute

on_change: ControlEventHandler[DatePicker] | None = None

Called when user clicks confirm button. value is updated with selected date.

The data property of the event handler argument contains the selected date.

bolton_entry_mode_changeclass-attributeinstance-attribute

on_entry_mode_change: (
 EventHandler[DatePickerEntryModeChangeEvent] | None
) = None

Called when the entry_mode is changed from the user interface.

DateRangePicker

A Material-style date range picker dialog.

It can be opened by calling `Page.show_dialog` method.

Depending on the `entry_mode`, it will show either a Calendar or an Input (text field) for picking a date range.

```
picker = ft.DateRangePicker()

ft.Button(
    "Pick date range",
    icon=ft.Icons.DATE_RANGE,
    on_click=lambda _: page.show_dialog(picker),
)
```

Image: DateRangePicker - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/date_range_picker/image_for_docs.png

Date range picker

Inherits: `DialogControl`

Properties

- `barrier_color` - The color of the modal barrier that darkens everything below the date picker.
- `cancel_text` - The text that is displayed on the cancel button.
- `confirm_text` - The text that is displayed on the confirm button.
- `current_date` - The date representing today.
- `end_value` - The selected end date that the picker should display.
- `entry_mode` - The initial mode of date entry method for the date picker dialog.
- `error_format_text` - The error message displayed below the `TextField` if the entered date is not in the correct format.
- `error_invalid_range_text` - The message used when the date range is invalid (e.g.
- `error_invalid_text` - The error message displayed below the `TextField` if the date is earlier than `first_date` or later than `last_date`.
- `field_end_hint_text` - The text used to prompt the user when no text has been entered in the end field.
- `field_end_label_text` - The label for the end date text input field.
- `field_start_hint_text` - The text used to prompt the user when no text has been entered in the start field.
- `field_start_label_text` - The label for the start date text input field.
- `first_date` - The earliest allowable date on the date range.
- `help_text` - The text that is displayed at the top of the header.
- `keyboard_type` - The type of keyboard to use for editing the text.
- `last_date` - The latest allowable date on the date range.
- `locale` - The locale for this date picker dialog.
- `modal` - Whether this date picker cannot be dismissed by clicking the area outside of it.
- `save_text` - The label on the save button for the fullscreen calendar mode.
- `start_value` - The selected start date that the picker should display.
- `switch_to_calendar_icon` - The name of the icon displayed in the corner of the dialog when `entry_mode` is `DatePickerEntryMode.INPUT`.
- `switch_to_input_icon` - The name of the icon displayed in the corner of the dialog when `entry_mode` is `DatePickerEntryMode.CALENDAR`.

Events

- `on_change` - Called when user clicks confirm button.

Examples

DateRangePicker

[play_arrowTry Online](#)

```
import datetime

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    messages = ft.Column(tight=True)

    def handle_change(e: ft.Event[ft.DateRangePicker]):
        messages.controls.extend(
            [
                ft.Text(
                    f"Start Date changed: {e.control.start_value.strftime('%m/%d/%Y')}"
                ),
                ft.Text(
                    f"End Date changed: {e.control.end_value.strftime('%m/%d/%Y')}"
                ),
            ]
        )

    def handle_dismissal(_: ft.Event[ft.DialogControl]):
        messages.controls.append(ft.Text("DateRangePicker dismissed"))

    today = datetime.datetime(year=2025, month=4, day=15)

    picker = ft.DateRangePicker(
        start_value=datetime.datetime(year=today.year, month=today.month, day=1),
        end_value=datetime.datetime(year=today.year, month=today.month, day=15),
        first_date=datetime.datetime(year=2024, month=1, day=1),
        last_date=datetime.datetime(year=2026, month=12, day=31),
        current_date=today,
        on_change=handle_change,
        on_dismiss=handle_dismissal,
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    ft.Button(
                        icon=ft.Icons.DATE_RANGE,
                        on_click=lambda _: page.show_dialog(picker),
                        content="Pick date range",
                    ),
                    messages,
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/date_range_picker/basic.png

Custom locale

[play_arrowTry Online](#)

```
import datetime

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    current_date = datetime.datetime(year=2025, month=4, day=15)

    page.add(
        ft.SafeArea(
            content=ft.Button(
                icon=ft.Icons.CALENDAR_MONTH,
                on_click=lambda _: page.show_dialog(
                    ft.DateRangePicker(
                        locale=ft.Locale("zh", "Hans"),

```

```

        start_value=datetime.datetime(year=2025, month=4, day=10),
        end_value=datetime.datetime(year=2025, month=4, day=20),
        first_date=datetime.datetime(year=2024, month=1, day=1),
        last_date=datetime.datetime(year=2026, month=12, day=31),
        current_date=current_date,
    )
),
content="Pick dates (zh_Hans locale)",
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: custom-locale - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/date_range_picker/custom_locale.png

Properties

buildbarrier_colorclass-attributeinstance-attribute

```
barrier_color: ColorValue | None = None
```

The color of the modal barrier that darkens everything below the date picker.

If None, the DialogTheme.barrier_color is used. If it is also None, then Colors.BLACK_54 is used.

buildcancel_textclass-attributeinstance-attribute

```
cancel_text: str | None = None
```

The text that is displayed on the cancel button.

buildconfirm_textclass-attributeinstance-attribute

```
confirm_text: str | None = None
```

The text that is displayed on the confirm button.

buildcurrent_dateclass-attributeinstance-attribute

```
current_date: DateTimeValue = field(
    default_factory=lambda: datetime.now()
)
```

The date representing today. It will be highlighted in the day grid.

buildend_valueclass-attributeinstance-attribute

```
end_value: DateTimeValue | None = None
```

The selected end date that the picker should display.

Defaults to current_date.

buildentry_modeclass-attributeinstance-attribute

```
entry_mode: DatePickerEntryMode = (
    DatePickerEntryMode.CALENDAR
)
```

The initial mode of date entry method for the date picker dialog.

builderror_format_textclass-attributeinstance-attribute

```
error_format_text: str | None = None
```

The error message displayed below the TextField if the entered date is not in the correct format.

builderror_invalid_range_textclass-attributeinstance-attribute

```
error_invalid_range_text: str | None = None
```

The message used when the date range is invalid (e.g. start date is after end date).

builderror_invalid_textclass-attributeinstance-attribute

```
error_invalid_text: str | None = None
```

The error message displayed below the TextField if the date is earlier than first_date or later than last_date.

buildfield_end_hint_textclass-attributeinstance-attribute

```
field_end_hint_text: str | None = None
```

The text used to prompt the user when no text has been entered in the end field.

buildfield_end_label_textclass-attributeinstance-attribute

```
field_end_label_text: str | None = None
```

The label for the end date text input field.

buildfield_start_hint_textclass-attributeinstance-attribute

```
field_start_hint_text: str | None = None
```

The text used to prompt the user when no text has been entered in the start field.

buildfield_start_label_textclass-attributeinstance-attribute

```
field_start_label_text: str | None = None
```

The label for the start date text input field.

buildfirst_dateclass-attributeinstance-attribute

```
first_date: DateTimeValue = field(
    default_factory=lambda: datetime(
        year=1900, month=1, day=1
    )
)
```

The earliest allowable date on the date range.

buildhelp_textclass-attributeinstance-attribute

```
help_text: str | None = None
```

The text that is displayed at the top of the header.

This is used to indicate to the user what they are selecting a date for.

buildkeyboard_typeclass-attributeinstance-attribute

```
keyboard_type: KeyboardType = KeyboardType.DATETIME
```

The type of keyboard to use for editing the text.

buildlast_dateclass-attributeinstance-attribute

```
last_date: DateTimeValue = field(
    default_factory=lambda: datetime(
        year=2050, month=1, day=1
    )
)
```

The latest allowable date on the date range.

buildlocaleclass-attributeinstance-attribute

```
locale: Locale | None = None
```

The locale for this date picker dialog. It is intended for (rare) cases where this dialog should be localized differently from the rest of the page.

It overrides the locale used by the page (see `Page.locale_configuration`), but does not participate in page-level locale resolution.

If set to `None` (the default) or an inexistent/unsupported locale, the `current_locale` of the `Page.locale_configuration` is used as fallback.

buildmodalclass-attributeinstance-attribute

```
modal: bool = False
```

Whether this date picker cannot be dismissed by clicking the area outside of it.

buildsave_textclass-attributeinstance-attribute

```
save_text: str | None = None
```

The label on the save button for the fullscreen calendar mode.

buildstart_valueclass-attributeinstance-attribute

```
start_value: DateTimeValue | None = None
```

The selected start date that the picker should display.

Defaults to `current_date`.

buildswitch_to_calendar_iconclass-attributeinstance-attribute

```
switch_to_calendar_icon: IconData | None = None
```

The name of the icon displayed in the corner of the dialog when `entry_mode` is `DatePickerEntryMode.INPUT`.

Clicking on this icon changes the `entry_mode` to `DatePickerEntryMode.CALENDAR`.

If `None`, `Icons.CALENDAR_TODAY` is used.

buildswitch_to_input_iconclass-attributeinstance-attribute

```
switch_to_input_icon: IconData | None = None
```

The name of the icon displayed in the corner of the dialog when `entry_mode` is `DatePickerEntryMode.CALENDAR`.

Clicking on this icon changes the `entry_mode` to `DatePickerEntryMode.INPUT`.

If `None`, `Icons.EDIT_OUTLINED` is used.

Events

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[DateRangePicker] | None = (
    None
)
```

Called when user clicks confirm button.

`start_value` and `end_value` are updated with selected dates.

The `data` property of the event handler argument contains the selected dates.

DialogControl

A base class for dialog controls.

Inherits: `AdaptiveControl`

Properties

- `open` - Set to `True` to display this dialog.

Events

- `on_dismiss` - Called when dialog is dismissed.

Properties

buildopenclass-attributeinstance-attribute

`open: bool = False`

Set to `True` to display this dialog.

Events

bolton_dismissclass-attributeinstance-attribute

`on_dismiss: ControlEventHandler[DialogControl] | None = None`

Called when dialog is dismissed.

Dismissible

A control that can be dismissed by dragging in the indicated `dismiss_direction`. When dragged or flung in the specified `dismiss_direction`, its content smoothly slides out of view.

After completing the sliding animation, if a `resize_duration` is provided, this control further animates its height (or width, depending on what is perpendicular to the `dismiss_direction`), gradually reducing it to zero over the specified `resize_duration`.

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `background` - A control that is stacked behind the content.
- `content` - The control that is being dismissed.
- `cross_axis_end_offset` - Specifies the end offset along the main axis once the content has been dismissed.
- `dismiss_direction` - The direction in which the control can be dismissed.
- `dismiss_thresholds` - The offset threshold the item has to be dragged in order to be considered as dismissed.
- `movement_duration` - The duration for content to dismiss or to come back to original position if not dismissed.
- `resize_duration` - The amount of time the control will spend contracting before `on_dismiss` is called.
- `secondary_background` - A control that is stacked behind the content and is exposed when it has been dragged up or to the left.

Events

- `on_confirm_dismiss` - Gives the app an opportunity to confirm or veto a pending dismissal.
- `on_dismiss` - Called when this control has been dismissed, after finishing resizing.
- `on_resize` - Called when this dismissible changes size, for example, when contracting before being dismissed.
- `on_update` - Called when this control has been dragged.

Methods

- `confirm_dismiss` - Resolve a pending dismissal decision triggered by `on_confirm_dismiss`.

Examples

Dismissible ListTiles

[play_arrow](#) [Try Online](#)

```
import flet as ft

def main(page: ft.Page):
    async def handle_dialog_action_click(e: ft.Event[ft.TextButton]):
        page.pop_dialog()
        await dialog.data.confirm_dismiss(e.control.data)

    dialog = ft.AlertDialog(
        modal=True,
        title=ft.Text("Please confirm"),
        content=ft.Text("Do you really want to delete this item?"),
        actions=[
            ft.TextButton("Yes", data=True, on_click=handle_dialog_action_click),
            ft.TextButton("No", data=False, on_click=handle_dialog_action_click),
        ],
        actions_alignment=ft.MainAxisAlignment.CENTER,
    )

    async def handle_confirm_dismiss(e: ft.DismissibleDismissEvent):
        if e.direction == ft.DismissDirection.END_TO_START:
            dialog.data = e.control
            page.show_dialog(dialog)
        else:
            await e.control.confirm_dismiss(True)

    def handle_dismiss(e: ft.Event[ft.Dismissible]):
```



```

        e.control.parent.controls.remove(e.control)
        e.control.parent.update()

def handle_update(e: ft.DismissibleUpdateEvent):
    print(e)

page.add(
    ft.SafeArea(
        content=ft.ListView(
            expand=True,
            controls=[
                ft.Dismissible(
                    dismiss_direction=ft.DismissDirection.HORIZONTAL,
                    background=ft.Container(bgcolor=ft.Colors.GREEN),
                    secondary_background=ft.Container(bgcolor=ft.Colors.RED),
                    on_dismiss=handle_dismiss,
                    on_update=handle_update,
                    on_confirm_dismiss=handle_confirm_dismiss,
                    dismiss_thresholds={
                        ft.DismissDirection.END_TO_START: 0.2,
                        ft.DismissDirection.START_TO_END: 0.2,
                    },
                    content=ft.ListTile(title=ft.Text(f"Item {i}")),
                )
                for i in range(10)
            ],
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: dismissible-list-tiles - https://flet.dev/docs/examples/controls/core/dismissible/media/dismissible_list_tiles.gif

Remove Dismissible on_dismiss inside component

Important

Always specify a key for Dismissible when using inside Flet component.

The issue you may encounter here is specific to the Dismissible control used inside Flet component (declarative UI).

When a user swipes (dismisses) an item, that widget is marked as "dismissed" on the Flutter side and effectively removed from the UI. However, when Flet recalculates the UI diff on the Python side, it may attempt to reuse widgets in the list based on their order rather than their identity.

If no key is provided, Flet's diffing algorithm can't tell that a particular Dismissible corresponds to a specific item — so it assumes the items have merely shifted. That leads to update commands like:

"Update text in items 0...N-1, then delete the last item (N)."

On Flutter's side, though, the already-dismissed Dismissible widget in the middle of the list can't be updated — it's gone — causing runtime errors.

Always assign a stable, unique key to each Dismissible, typically based on the item's identifier or index.

play_arrowTry Online

```

import flet as ft

@ft.component
def App():
    items, set_items = ft.use_state(list(range(5)))

    return ft.SafeArea(
        content=ft.ListView(
            controls=[
                ft.Dismissible(
                    key=i,
                    dismiss_direction=ft.DismissDirection.HORIZONTAL,
                    background=ft.Container(bgcolor=ft.Colors.GREEN),
                    secondary_background=ft.Container(bgcolor=ft.Colors.RED),
                    on_dismiss=lambda _, index=i: set_items(
                        [item for item in items if item != index]
                    ),
                    dismiss_thresholds={
                        ft.DismissDirection.HORIZONTAL: 0.1,
                        ft.DismissDirection.START_TO_END: 0.1,
                    },
                    content=ft.ListTile(title=ft.Text(f"Item {i}")),
                )
                for i in items
            ],
        ),
    )

```

```

    ],
    ),
)

def main(page: ft.Page):
    page.render(App)

if __name__ == "__main__":
    ft.run(main)

```

Properties

buildbackgroundclass-attributeinstance-attribute

```
background: Control | None = None
```

A control that is stacked behind the content.

If secondary_background is also specified, then this control only appears when the content has been dragged down or to the right.

buildcontentinstance-attribute

```
content: Control
```

The control that is being dismissed.

Raises:

- ValueError - If it is not visible.

buildcross_axis_end_offsetclass-attributeinstance-attribute

```
cross_axis_end_offset: Number = 0.0
```

Specifies the end offset along the main axis once the content has been dismissed.

If set to a non-zero value, then this dismissible moves in cross direction depending on whether it is positive or negative.

bulddismiss_directionclass-attributeinstance-attribute

```
dismiss_direction: DismissDirection = (
    DismissDirection.HORIZONTAL
)
```

The direction in which the control can be dismissed.

bulddismiss_thresholdsclass-attributeinstance-attribute

```
dismiss_thresholds: dict[
    DismissDirection, Number | None
] = field(default_factory=dict)
```

The offset threshold the item has to be dragged in order to be considered as dismissed. This is specified as a dictionary where the key is of type DismissDirection and the value is the threshold (a fractional/decimal value between 0.0 and 1.0, inclusive).

Example

```
ft.Dismissible(
    # ...
    dismiss_thresholds={
        ft.DismissDirection.VERTICAL: 0.1,
        ft.DismissDirection.START_TO_END: 0.7
    }
)
```

buildmovement_durationclass-attributeinstance-attribute

```
movement_duration: DurationValue = field(
    default_factory=lambda: Duration(milliseconds=200)
)
```

The duration for content to dismiss or to come back to original position if not dismissed.

buildresize_durationclass-attributeinstance-attribute

```
resize_duration: DurationValue = field(
    default_factory=lambda: Duration(milliseconds=300)
)
```

The amount of time the control will spend contracting before on_dismiss is called.

buildsecondary_backgroundclass-attributeinstance-attribute

```
secondary_background: Control | None = None
```

A control that is stacked behind the content and is exposed when it has been dragged up or to the left.

Raises:

- ValueError - If it is provided and visible but the background is not provided and visible.

Events

bolton_confirm_dismissclass-attributeinstance-attribute

```
on_confirm_dismiss: (
    EventHandler[DismissibleDismissEvent] | None
) = None
```

Gives the app an opportunity to confirm or veto a pending dismissal. This dismissible cannot be dragged again until this pending dismissal is resolved.

To resolve the pending dismissal, call the confirm_dismiss method passing it a boolean representing the decision. If True, then the control will be dismissed, otherwise it will be moved back to its original location.

bolton_dismissclass-attributeinstance-attribute

```
on_dismiss: EventHandler[DismissibleDismissEvent] | None = (
    None
)
```

Called when this control has been dismissed, after finishing resizing.

bolton_resizeclass-attributeinstance-attribute

```
on_resize: ControlEventHandler[Dismissible] | None = None
```

Called when this dismissible changes size, for example, when contracting before being dismissed.

bolton_updateclass-attributeinstance-attribute

```
on_update: EventHandler[DismissibleUpdateEvent] | None = (
    None
)
```

Called when this control has been dragged.

Methods

deployed_codeconfirm_dismissasync

```
confirm_dismiss(dismiss: bool)
```

Resolve a pending dismissal decision triggered by on_confirm_dismiss.

Call this method from your confirmation flow after handling on_confirm_dismiss.

Parameters:

- dismiss (bool) - True to continue dismissing the control, False to cancel and return it to the original position.

Divider

A thin horizontal line (divider), with padding on either side.

```
ft.Column(
    width=240,
    spacing=10,
    controls=[
        ft.Text("Section A", weight=ft.FontWeight.W_600),
        ft.Divider(),
        ft.Text("Section B"),
    ],
)
```

Image: Divider - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/divider/image_for_docs.png

Divider

Inherits: Control

Properties

- color - The color to use when painting the line.
- height - The divider's height extent.
- leading_indent - The amount of empty space to the leading edge of the divider.
- radius - The border radius of the divider.
- thickness - The thickness of the line drawn within the divider.
- trailing_indent - The amount of empty space to the trailing edge of the divider.

Examples

Divider

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                spacing=0,
                expand=True,
                controls=[
                    ft.Container(
                        expand=True,
                        bgcolor=ft.Colors.AMBER,
                        alignment=ft.Alignment.CENTER,
                    ),
                    ft.Divider(),
                    ft.Container(
                        expand=True,
                        bgcolor=ft.Colors.PINK,
                        alignment=ft.Alignment.CENTER,
                    ),
                    ft.Divider(height=1, color=ft.Colors.WHITE),
                    ft.Container(
                        expand=True,
                        bgcolor=ft.Colors.BLUE_300,
                        alignment=ft.Alignment.CENTER,
                    ),
                    ft.Divider(height=9, thickness=3),
                    ft.Container(
                        expand=True,
                        bgcolor=ft.Colors.DEEP_PURPLE_200,
                        alignment=ft.Alignment.CENTER,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
```

```
ft.run(main)
```

Image: basic - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/divider/basic.png>

Properties

buildcolorclass-attributeinstance-attribute

```
color: ColorValue | None = None
```

The color to use when painting the line.

If None, DividerTheme.color is used.

buildheightclass-attributeinstance-attribute

```
height: Number | None = None
```

The divider's height extent. The divider itself is always drawn as a horizontal line that is centered within the height specified by this value.

If None, DividerTheme.space is used. If that's is also None, defaults to 16.0.

Raises:

- ValueError - If height is negative.

buildleading_indentclass-attributeinstance-attribute

```
leading_indent: Number | None = None
```

The amount of empty space to the leading edge of the divider.

If None, DividerTheme.leading_indent is used. If that's is also None, defaults to 0.0.

Raises:

- ValueError - If it is not greater than or equal to 0.

buildradiusclass-attributeinstance-attribute

```
radius: BorderRadiusValue | None = None
```

The border radius of the divider.

buildthicknessclass-attributeinstance-attribute

```
thickness: Number | None = None
```

The thickness of the line drawn within the divider.

A divider with a thickness of 0.0 is always drawn as a line with a height of exactly one device pixel.

If None, DividerTheme.thickness is used. If that is also None, defaults to 0.0.

Raises:

- ValueError - If thickness is negative.

buildtrailing_indentclass-attributeinstance-attribute

```
trailing_indent: Number | None = None
```

The amount of empty space to the trailing edge of the divider.

If None, DividerTheme.trailing_indent is used. If that is also None, defaults to 0.0.

Raises:

- ValueError - If it is not greater than or equal to 0.

Draggable

A control that can be dragged from to a DragTarget.

When a draggable control recognizes the start of a drag gesture, it displays the `content_feedback` control that tracks the user's finger across the screen. If the user lifts their finger while on top of a `DragTarget`, this target is given the opportunity to complete drag-and-drop flow.

Inherits: `Control`

Properties

- `affinity` - Specifies the axis along which this control competes with other gestures to initiate a drag.
- `axis` - Restricts the draggable's movement to a specific axis.
- `content` - The control to display when the draggable is not being dragged.
- `content_feedback` - The control to show under the pointer when a drag is under way.
- `content_when_dragging` - The control to display instead of content when this draggable is being dragged.
- `group` - The group this draggable belongs to.
- `max_simultaneous_drag`s - Specifies how many simultaneous drag operations are allowed for this draggable.

Events

- `on_drag_complete` - Called when this draggable is dropped and accepted by a `DragTarget`.
- `on_drag_start` - Called when this draggable starts being dragged.

Examples

Drag and drop containers

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    def handle_drag_will_accept(e: ft.DragWillAcceptEvent):
        e.control.content.border = ft.Border.all(
            2,
            ft.Colors.BLACK_45 if e.accept else ft.Colors.RED,
        )
        e.control.update()

    def handle_drag_accept(e: ft.DragTargetEvent):
        e.control.content.bgcolor = e.src.content.bgcolor
        e.control.content.border = None
        e.control.update()

    def handle_drag_leave(e: ft.DragTargetLeaveEvent):
        e.control.content.border = None
        e.control.update()

    page.add(
        ft.SafeArea(
            content=ft.Row(
                controls=[
                    ft.Column(
                        controls=[
                            ft.Draggable(
                                group="color",
                                content=ft.Container(
                                    width=50,
                                    height=50,
                                    bgcolor=ft.Colors.CYAN,
                                    border_radius=5,
                                ),
                                content_feedback=ft.Container(
                                    width=20,
                                    height=20,
                                    bgcolor=ft.Colors.CYAN,
                                    border_radius=3,
                                ),
                            ),
                        ],
                    ),
                ],
            ),
        ),
    ),
```

```

        ft.Draggable(
            group="color",
            content=ft.Container(
                width=50,
                height=50,
                bgcolor=ft.Colors.YELLOW,
                border_radius=5,
            ),
        ),
        ft.Draggable(
            group="color",
            content=ft.Container(
                width=50,
                height=50,
                bgcolor=ft.Colors.GREEN,
                border_radius=5,
            ),
        ),
    ],
),
ft.Container(width=100),
ft.DragTarget(
    group="color",
    on_will_accept=handle_drag_will_accept,
    on_accept=handle_drag_accept,
    on_leave=handle_drag_leave,
    content=ft.Container(
        width=50,
        height=50,
        bgcolor=ft.Colors.BLUE_GREY_100,
        border_radius=5,
    ),
),
],
),
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: drag-and-drop-containers - https://flet.dev/docs/examples/controls/core/drag_target_and_draggable/media/drag_and_drop_containers.gif

Drag and drop containers declarative

play_arrowTry Online

```

from dataclasses import dataclass

import flet as ft

@dataclass
@ft.observable
class TargetState:
    bgcolor: ft.Colors = ft.Colors.BLUE_GREY_100
    is_drag_over: bool = False

@ft.component
def App():
    target, _ = ft.use_state(lambda: TargetState())

    def on_will_accept(_: ft.DragWillAcceptEvent):
        target.is_drag_over = True

    def on_accept(e: ft.DragTargetEvent):
        target.bgcolor = e.src.data
        target.is_drag_over = False

    def on_leave(_: ft.DragTargetLeaveEvent):
        target.is_drag_over = False

    return ft.SafeArea(
        content=ft.Row(
            controls=[
                ft.Column(
                    controls=[
                        ft.Draggable(
                            group="color",
                            data=ft.Colors.CYAN,
                            content=ft.Container(
                                width=50,
                                height=50,
                                bgcolor=ft.Colors.CYAN,
                                border_radius=5,

```

```

        ),
        content_feedback=ft.Container(
            width=20,
            height=20,
            bgcolor=ft.Colors.CYAN,
            border_radius=3,
        ),
    ),
    ft.Draggable(
        group="color",
        data=ft.Colors.YELLOW,
        content=ft.Container(
            width=50,
            height=50,
            bgcolor=ft.Colors.YELLOW,
            border_radius=5,
        ),
    ),
    ft.Draggable(
        group="color",
        data=ft.Colors.GREEN,
        content=ft.Container(
            width=50,
            height=50,
            bgcolor=ft.Colors.GREEN,
            border_radius=5,
        ),
    ),
],
),
ft.Container(width=100),
ft.DragTarget(
    group="color",
    on_will_accept=on_will_accept,
    on_accept=on_accept,
    on_leave=on_leave,
    content=ft.Container(
        width=50,
        height=50,
        bgcolor=target.bgcolor,
        border=(
            ft.Border.all(2, ft.Colors.BLACK_45)
            if target.is_drag_over
            else None
        ),
        border_radius=5,
    ),
),
),
],
),
)

```

```

def main(page: ft.Page):
    page.render(App)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Properties

buildaffinityclass-attributeinstance-attribute

affinity: Axis | None = None

Specifies the axis along which this control competes with other gestures to initiate a drag.

- If None, the drag starts as soon as a tap down gesture is recognized, regardless of direction.
- If set to Axis.HORIZONTAL or Axis.VERTICAL, the control will only initiate a drag when the gesture matches the specified axis, allowing it to compete with other gestures in that direction.

buildaxisclass-attributeinstance-attribute

axis: Axis | None = None

Restricts the draggable's movement to a specific axis.

- Axis.HORIZONTAL: Only allows horizontal dragging.
- Axis.VERTICAL: Only allows vertical dragging.
- None: Allows dragging in any direction.

buildcontentinstance-attribute

`content: Control`

The control to display when the draggable is not being dragged.

If the draggable is being dragged, the `content_when_dragging` is displayed instead.

Raises:

- `ValueError` - If it is not visible.

buildcontent_feedbackclass-attributeinstance-attribute

`content_feedback: Control | None = None`

The control to show under the pointer when a drag is under way.

buildcontent_when_draggingclass-attributeinstance-attribute

`content_when_dragging: Control | None = None`

The control to display instead of content when this draggable is being dragged.

If set, this control visually replaces content during an active drag operation, allowing you to show a different appearance or an "empty" placeholder. If `None`, the original content remains visible while dragging.

buildgroupclass-attributeinstance-attribute

`group: str = 'default'`

The group this draggable belongs to.

Note

For a `DragTarget` to accept an incoming drop from a `Draggable`, they must both be in the same group.

buildmax_simultaneous_dragsclass-attributeinstance-attribute

`max_simultaneous_draggs: int | None = None`

Specifies how many simultaneous drag operations are allowed for this draggable.

- 0 - disables dragging entirely.
- 1 - allows only one drag at a time. For a better user experience, you may want to provide an "empty" widget for `content_when_dragging` to visually indicate the item is being moved.
- any other positive integer - allows that many concurrent drags.
- `None` - no limit on the number of simultaneous drags.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

Events

bolton_drag_completeclass-attributeinstance-attribute

`on_drag_complete: ControlEventHandler[Draggable] | None = (None)`

Called when this draggable is dropped and accepted by a `DragTarget`.

bolton_drag_startclass-attributeinstance-attribute

`on_drag_start: ControlEventHandler[Draggable] | None = None`

Called when this draggable starts being dragged.

DragTarget

A control that completes drag operation when a Draggable control is dropped.

When a Draggable is dragged on top of a DragTarget, the DragTarget is asked whether it will accept the data the Draggable is carrying. The DragTarget will accept incoming drag if it belongs to the same group as Draggable. If the user does drop the Draggable on top of the DragTarget (and the DragTarget has indicated that it will accept the Draggable's data), then the DragTarget is asked to accept the Draggable's data.

Inherits: Control

Properties

- content - The content of this control.
- group - The group this target belongs to.

Events

- on_accept - Called when the user does drop an acceptable (same group) Draggable on this target.
- on_leave - Called when a Draggable leaves this target.
- on_move - Called when a Draggable moves within this target.
- on_will_accept - Called when a Draggable is dragged on this target.

Examples

Drag and drop containers

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    def handle_drag_will_accept(e: ft.DragWillAcceptEvent):
        e.control.content.border = ft.Border.all(
            2,
            ft.Colors.BLACK_45 if e.accept else ft.Colors.RED,
        )
        e.control.update()

    def handle_drag_accept(e: ft.DragTargetEvent):
        e.control.content.bgcolor = e.src.content.bgcolor
        e.control.content.border = None
        e.control.update()

    def handle_drag_leave(e: ft.DragTargetLeaveEvent):
        e.control.content.border = None
        e.control.update()

    page.add(
        ft.SafeArea(
            content=ft.Row(
                controls=[
                    ft.Column(
                        controls=[
                            ft.Draggable(
                                group="color",
                                content=ft.Container(
                                    width=50,
                                    height=50,
                                    bgcolor=ft.Colors.CYAN,
                                    border_radius=5,
                                ),
                                content_feedback=ft.Container(
                                    width=20,
                                    height=20,
                                    bgcolor=ft.Colors.CYAN,
                                    border_radius=3,
                                ),
                            ),
                            ft.Draggable(
                                group="color",
                                content=ft.Container(
                                    width=50,
                                    height=50,
```

```

        bgcolor=ft.Colors.YELLOW,
        border_radius=5,
    ),
),
ft.Draggable(
    group="color",
    content=ft.Container(
        width=50,
        height=50,
        bgcolor=ft.Colors.GREEN,
        border_radius=5,
    ),
),
],
),
ft.Container(width=100),
ft.DragTarget(
    group="color",
    on_will_accept=handle_drag_will_accept,
    on_accept=handle_drag_accept,
    on_leave=handle_drag_leave,
    content=ft.Container(
        width=50,
        height=50,
        bgcolor=ft.Colors.BLUE_GREY_100,
        border_radius=5,
    ),
),
],
),
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: drag-and-drop-containers - https://flet.dev/docs/examples/controls/core/drag_target_and_draggable/media/drag_and_drop_containers.gif

Drag and drop containers declarative

play_arrowTry Online

```
from dataclasses import dataclass
```

```
import flet as ft
```

```

@dataclass
@ft.observable
class TargetState:
    bgcolor: ft.Colors = ft.Colors.BLUE_GREY_100
    is_drag_over: bool = False

@ft.component
def App():
    target, _ = ft.use_state(lambda: TargetState())

    def on_will_accept(_: ft.DragWillAcceptEvent):
        target.is_drag_over = True

    def on_accept(e: ft.DragTargetEvent):
        target.bgcolor = e.src.data
        target.is_drag_over = False

    def on_leave(_: ft.DragTargetLeaveEvent):
        target.is_drag_over = False

    return ft.SafeArea(
        content=ft.Row(
            controls=[
                ft.Column(
                    controls=[
                        ft.Draggable(
                            group="color",
                            data=ft.Colors.CYAN,
                            content=ft.Container(
                                width=50,
                                height=50,
                                bgcolor=ft.Colors.CYAN,
                                border_radius=5,
                            ),
                            content_feedback=ft.Container(
                                width=20,
                                height=20,
                                bgcolor=ft.Colors.CYAN,

```

```

        border_radius=3,
    ),
),
ft.Draggable(
    group="color",
    data=ft.Colors.YELLOW,
    content=ft.Container(
        width=50,
        height=50,
        bgcolor=ft.Colors.YELLOW,
        border_radius=5,
    ),
),
ft.Draggable(
    group="color",
    data=ft.Colors.GREEN,
    content=ft.Container(
        width=50,
        height=50,
        bgcolor=ft.Colors.GREEN,
        border_radius=5,
    ),
),
],
),
ft.Container(width=100),
ft.DragTarget(
    group="color",
    on_will_accept=on_will_accept,
    on_accept=on_accept,
    on_leave=on_leave,
    content=ft.Container(
        width=50,
        height=50,
        bgcolor=target.bgcolor,
        border=(
            ft.Border.all(2, ft.Colors.BLACK_45)
            if target.is_drag_over
            else None
        ),
        border_radius=5,
    ),
),
],
),
),
)

```

```

def main(page: ft.Page):
    page.render(App)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Properties

buildcontentinstance-attribute

content: Control

The content of this control.

Raises:

- ValueError - If it is not visible.

buildgroupclass-attributeinstance-attribute

group: str = 'default'

The group this target belongs to.

Note

For a DragTarget to accept an incoming drop from a Draggable, they must both be in the same group.

Events

bolton_acceptclass-attributeinstance-attribute

on_accept: EventHandler[DragTargetEvent] | None = None

Called when the user does drop an acceptable (same group) Draggable on this target.

bolton_leaveclass-attributeinstance-attribute

`on_leave: EventHandler[DragTargetLeaveEvent] | None = None`

Called when a Draggable leaves this target.

bolton_moveclass-attributeinstance-attribute

`on_move: EventHandler[DragTargetEvent] | None = None`

Called when a Draggable moves within this target.

bolton_will_acceptclass-attributeinstance-attribute

`on_will_accept: EventHandler[DragWillAcceptEvent] | None = (`
 `None`
`)`

Called when a Draggable is dragged on this target.

Dropdown

A dropdown control that allows users to select a single option from a list of options.

Example:

```
ft.Dropdown(  
    width=220,  
    value="alice",  
    options=[  
        ft.DropdownOption(key="alice", text="Alice"),  
        ft.DropdownOption(key="bob", text="Bob"),  
    ],  
)
```

Image: Dropdown - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/dropdown/image_for_docs.png

Dropdown

Inherits: `LayoutControl`

Properties

- `autofocus` - Whether the control will be selected as the initial focus.
- `bgcolor` - The background color of the dropdown menu in various `ControlState` states.
- `border` - Border around input.
- `border_color` - Border color.
- `border_radius` - The border radius applied to the corners of the dropdown input field.
- `border_width` - The width of the border in virtual pixels.
- `capitalization` - Configures how the text input should be capitalized.
- `color` - Text color.
- `content_padding` - The padding for the input decoration's container.
- `dense` - Whether the `TextField` is part of a dense form (i.e., uses less vertical space).
- `editable` - Whether the dropdown allows editing of the text input field.
- `elevation` - The dropdown's menu elevation in various `ControlState` states.
- `enable_filter` - Determine if the menu list can be filtered by the text input.
- `enable_search` - Determine if the first item that matches the text input can be highlighted.
- `error_style` - The `TextStyle` to use for `error_text`.
- `error_text` - Text that appears below the input border.
- `expanded_insets` - The insets for the expanded dropdown menu.
- `fill_color` - Background color of the dropdown input text field.
- `filled` - Whether the decoration's container is filled with theme `fill_color`.
- `focused_border_color` - Border color in focused state.
- `focused_border_width` - Border width in focused state.
- `helper_style` - The `TextStyle` to use for `helper_text`.
- `helper_text` - Text that provides context about the input's value, such as how the value will be used.
- `hint_style` - The `TextStyle` to use for `hint_text`.
- `hint_text` - Text that suggests what sort of input the field accepts.
- `hover_color` - The color of the dropdown input text field when hovered.
- `input_filter` - A filter to apply to the text input field.
- `label` - Optional text that describes the input field.

- `label_style` - The label's text style.
- `leading_icon` - An optional Icon at the front of the text input field inside the decoration box.
- `menu_height` - The height of the dropdown menu.
- `menu_style` - The menu style that defines the visual attributes of the menu.
- `menu_width` - The width of the dropdown menu.
- `options` - A list of options to display in the dropdown.
- `selected_suffix` - A control to display after the selected item in the dropdown.
- `selected_trailing_icon` - An optional icon at the end of the text field to indicate that the text field is pressed.
- `text` - The text entered in the text field.
- `text_align` - The text align for the TextField of the Dropdown.
- `text_size` - Text size in virtual pixels.
- `text_style` - The TextStyle to use for text in input text field.
- `trailing_icon` - An icon to display at the end of the text field.
- `value` - The key of the dropdown options corresponding to the selected option.

Events

- `on_blur` - Called when the control has lost focus.
- `on_focus` - Called when the control has received focus.
- `on_select` - Called when the selected item of this dropdown has changed.
- `on_text_change` - Called when the text input of this dropdown has changed.

Methods

- `focus` - Requests focus for this control.

Examples

Color selection with filtering

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT

    def get_options() -> list[ft.DropdownOption]:
        colors = [
            ft.Colors.RED,
            ft.Colors.BLUE,
            ft.Colors.YELLOW,
            ft.Colors.PURPLE,
            ft.Colors.LIME,
        ]
        return [
            ft.DropdownOption(
                key=color.value,
                content=ft.Text(value=color.value, color=color),
            )
            for color in colors
        ]

    def handle_dropdown_select(e: ft.Event[ft.Dropdown]):
        e.control.color = e.control.value

    page.add(
        ft.SafeArea(
            content=ft.Dropdown(
                key="color_dropdown",
                editable=True,
                label="Color",
                options=get_options(),
                on_select=handle_dropdown_select,
            ),
        ),
    ),
```

```
)
)
```

```
if __name__ == "__main__":
    ft.run(main)
```

Image: color-selection-with-filtering -

https://flet.dev/docs/test-images/examples/controls/material/golden/macros/dropdown/color_selection_with_filtering_flow.gif

Icon selection

play_arrowTry Online

```
import flet as ft
```

```
def main(page: ft.Page):
    def get_options() -> list[ft.DropdownOption]:
        icons = [
            {"name": "Smile", "icon": ft.Icons.SENTIMENT_SATISFIED_OUTLINED},
            {"name": "Cloud", "icon": ft.Icons.CLOUD_OUTLINED},
            {"name": "Brush", "icon": ft.Icons.BRUSH_OUTLINED},
            {"name": "Heart", "icon": ft.Icons.FAVORITE},
        ]
        return [
            ft.DropdownOption(key=icon["name"], leading_icon=icon["icon"])
            for icon in icons
        ]

    page.add(
        ft.SafeArea(
            content=ft.Dropdown(
                key="icon_dropdown",
                border=ft.InputBorder.UNDERLINE,
                enable_filter=True,
                editable=True,
                leading_icon=ft.Icons.SEARCH,
                label="Icon",
                options=get_options(),
            ),
        ),
    )
```

```
if __name__ == "__main__":
    ft.run(main)
```

Image: icon-selection - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/dropdown/icon_selection_flow.gif

Declarative

play_arrowTry Online

```
from dataclasses import dataclass
from typing import cast
```

```
import flet as ft
```

```
@dataclass
@ft.observable
class Form:
    color: str = "red"

    def change_color(self, new_color: str):
        print("New color:", new_color)
        self.color = new_color
```

```
@ft.component
def App():
    form, _ = ft.use_state(lambda: Form())

    return ft.SafeArea(
        content=ft.Column(
            controls=cast(
                list[ft.Control],
                [
                    ft.Text(f"Selected color: {form.color}"),
                    ft.Column(
                        controls=[
                            ft.Dropdown(
                                key="declarative_dropdown",
                                editable=True,
                                label="Color",
                                value=form.color,
```



```

        on_select=lambda e: form.change_color(
            cast(str, e.control.value)
        ),
        options=[
            ft.DropdownOption(key="red", text="Red"),
            ft.DropdownOption(key="green", text="Green"),
            ft.DropdownOption(key="blue", text="Blue"),
        ],
    ),
],
),
),
),
def main(page: ft.Page):
    page.render(App)

if __name__ == "__main__":
    ft.run(main)
```

Select and change events

```
import flet as ft
```

```
ft.run(main)
```

Image: select-and-change-events -

https://flet.dev/docs/test-images/examples/controls/material/golden/macros/dropdown/select_and_change_events_flow.gif

Styled

play_arrowTry Online

```
import flet as ft
```

```
def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Dropdown(
                        key="styled_dropdown_1",
                        text_size=20,
                        content_padding=10,
                        color=ft.Colors.PURPLE_200,
                        bgcolor=ft.Colors.BLUE_200,
                        filled=True,
                        border_radius=30,
                        border_color=ft.Colors.GREEN_800,
                        focused_border_color=ft.Colors.GREEN_ACCENT_400,
                        focused_border_width=5,
                        options=[
                            ft.DropdownOption("a", "Style 1A"),
                            ft.DropdownOption("b", "Style 1B"),
                            ft.DropdownOption("c", "Style 1C"),
                        ],
                    ),
                    ft.Dropdown(
                        key="styled_dropdown_2",
                        border_radius=30,
                        filled=True,
                        fill_color=ft.Colors.RED_400,
                        border_color=ft.Colors.TRANSPARENT,
                        bgcolor=ft.Colors.RED_200,
                        color=ft.Colors.CYAN_400,
                        focused_border_color=ft.Colors.PINK_300,
                        focused_border_width=20,
                        options=[
                            ft.DropdownOption("a", "Style 2A"),
                            ft.DropdownOption("b", "Style 2B"),
                            ft.DropdownOption("c", "Style 2C"),
                        ],
                    ),
                    ft.Dropdown(
                        key="styled_dropdown_3",
                        border_color=ft.Colors.PINK_ACCENT,
                        focused_border_color=ft.Colors.GREEN_ACCENT_400,
                        focused_border_width=25,
                        border_radius=30,
                        width=150,
                        border_width=5,
                        options=[
                            ft.DropdownOption("a", "Style 3A"),
                            ft.DropdownOption("b", "Style 3B"),
                            ft.DropdownOption("c", "Style 3C"),
                        ],
                    ),
                    ft.Container(
                        padding=ft.Padding.only(bottom=20),
                        content=ft.Dropdown(
                            key="styled_dropdown_4",
                            text_size=30,
                            color=ft.Colors.ORANGE_ACCENT,
                            border_radius=20,
                            filled=True,
                            border_width=0,
                            autofocus=True,
                            focused_border_color=ft.Colors.GREEN_100,
                            focused_border_width=10,
                            width=200,
                            height=50,
                            options=[
                                ft.dropdown.Option("a", "Style 4A"),
                                ft.dropdown.Option("b", "Style 4B"),
                                ft.dropdown.Option("c", "Style 4C"),
                            ],
                        ),
                    ),
                    ft.Dropdown(
                        key="styled_dropdown_5",
```

```

text_size=30,
border_radius=20,
filled=True,
border_width=0,
focused_border_color=ft.Colors.GREEN_100,
focused_border_width=10,
content_padding=20,
width=200,
options=[
    ft.DropdownOption(
        key="a",
        text="Style 5A",
        style=ft.ButtonStyle(
            shape=ft.BeveledRectangleBorder(radius=15),
            color={
                ft.ControlState.HOVERED: ft.Colors.WHITE,
                ft.ControlState.FOCUSED: ft.Colors.BLUE,
                ft.ControlState.DEFAULT: ft.Colors.BLACK,
            },
        ),
    ),
    ft.DropdownOption(
        key="b",
        text="Style 5B",
        style=ft.ButtonStyle(
            shape=ft.BeveledRectangleBorder(radius=15),
            color={
                ft.ControlState.HOVERED: ft.Colors.WHITE,
                ft.ControlState.FOCUSED: ft.Colors.BLUE,
                ft.ControlState.DEFAULT: ft.Colors.BLACK,
            },
        ),
    ),
    ft.DropdownOption(
        key="c",
        text="Style 5C",
        style=ft.ButtonStyle(
            shape=ft.BeveledRectangleBorder(radius=15),
            color={
                ft.ControlState.HOVERED: ft.Colors.WHITE,
                ft.ControlState.FOCUSED: ft.Colors.BLUE,
                ft.ControlState.DEFAULT: ft.Colors.BLACK,
            },
        ),
    ),
],
),
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: styled-dropdowns - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/dropdown/styled_flow.gif

Properties

buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

Whether the control will be selected as the initial focus. If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

buildbgcolorclass-attributeinstance-attribute

bgcolor: ControlStateValue[ColorValue] | None = None

The background color of the dropdown menu in various ControlState states.

buildborderclass-attributeinstance-attribute

border: InputBorder | None = None

Border around input.

Defaults to InputBorder.OUTLINE.

buildborder_colorclass-attributeinstance-attribute

border_color: ColorValue | None = None

Border color.

Tip

Set to `Colors.TRANSPARENT` to hide the border.

buildborder_radiusclass-attributeinstance-attribute

`border_radius: BorderRadiusValue | None = None`

The border radius applied to the corners of the dropdown input field. Accepts a value in virtual pixels or a `BorderRadiusValue` object. If set to `None`, the default border radius defined by the theme or system is used.

buildborder_widthclass-attributeinstance-attribute

`border_width: Number = 1`

The width of the border in virtual pixels.

Tip

Set to 0 to completely remove the border.

buildcapitalizationclass-attributeinstance-attribute

`capitalization: TextCapitalization | None = None`

Configures how the text input should be capitalized.

buildcolorclass-attributeinstance-attribute

`color: ColorValue | None = None`

Text color.

buildcontent_paddingclass-attributeinstance-attribute

`content_padding: PaddingValue | None = None`

The padding for the input decoration's container.

builddenseclass-attributeinstance-attribute

`dense: bool = False`

Whether the `TextField` is part of a dense form (i.e., uses less vertical space).

buildditableclass-attributeinstance-attribute

`editable: bool = False`

Whether the dropdown allows editing of the text input field.

buildelevationclass-attributeinstance-attribute

`elevation: ControlStateValue[Number | None] | None = 8`

The dropdown's menu elevation in various `ControlState` states.

buildenable_filterclass-attributeinstance-attribute

`enable_filter: bool = False`

Determine if the menu list can be filtered by the text input. Defaults to false.

If set to true, dropdown menu will show a filtered list. The filtered list will contain items that match the text provided by the input field, with a case-insensitive comparison.

buildenable_searchclass-attributeinstance-attribute

`enable_search: bool = True`

Determine if the first item that matches the text input can be highlighted.

builderror_styleclass-attributeinstance-attribute

`error_style: TextStyle | None = None`

The `TextStyle` to use for `error_text`.

builderror_textclass-attributeinstance-attribute

`error_text: str | None = None`

Text that appears below the input border.

If non-null, the border's color animates to red and the helper_text is not shown.

buildexpanded_insetsclass-attributeinstance-attribute

expanded_insets: PaddingValue | None = None

The insets for the expanded dropdown menu.

buildfill_colorclass-attributeinstance-attribute

fill_color: ColorValue | None = None

Background color of the dropdown input text field.

Note

Will not be visible if filled=False.

buildfilledclass-attributeinstance-attribute

filled: bool = False

Whether the decoration's container is filled with theme fill_color.

buildfocused_border_colorclass-attributeinstance-attribute

focused_border_color: ColorValue | None = None

Border color in focused state.

buildfocused_border_widthclass-attributeinstance-attribute

focused_border_width: Number | None = None

Border width in focused state.

buildhelper_styleclass-attributeinstance-attribute

helper_style: TextStyle | None = None

The TextStyle to use for helper_text.

buildhelper_textclass-attributeinstance-attribute

helper_text: str | None = None

Text that provides context about the input's value, such as how the value will be used.

If non-null, the text is displayed below the input decorator, in the same location as error_text. If a non-null error_text value is specified then the helper text is not shown.

buildhint_styleclass-attributeinstance-attribute

hint_style: TextStyle | None = None

The TextStyle to use for hint_text.

buildhint_textclass-attributeinstance-attribute

hint_text: str | None = None

Text that suggests what sort of input the field accepts.

Displayed on top of the input when it's empty and either (a) label is null or (b) the input has the focus.

buildhover_colorclass-attributeinstance-attribute

hover_color: ColorValue | None = None

The color of the dropdown input text field when hovered.

buildinput_filterclass-attributeinstance-attribute

input_filter: InputFilter | None = None

A filter to apply to the text input field.

buildlabelclass-attributeinstance-attribute

label: StrOrControl | None = None

Optional text that describes the input field.

When the input field is empty and unfocused, the label is displayed on top of the input field (i.e., at the same location on the screen where text may be entered in the input field). When the input field receives focus (or if the field is non-empty) the

label moves above, either vertically adjacent to, or to the center of the input field.

buildlabel_styleclass-attributeinstance-attribute

```
label_style: TextStyle | None = None
```

The label's text style.

buildleading_iconclass-attributeinstance-attribute

```
leading_icon: IconDataOrControl | None = None
```

An optional Icon at the front of the text input field inside the decoration box.

If this is not null, the menu items will have extra paddings to be aligned with the text in the text field.

buildmenu_heightclass-attributeinstance-attribute

```
menu_height: Number | None = None
```

The height of the dropdown menu.

If this is None, the menu will display as many items as possible on the screen.

buildmenu_styleclass-attributeinstance-attribute

```
menu_style: MenuStyle | None = None
```

The menu style that defines the visual attributes of the menu.

The default width of the menu is set to the width of the text field.

buildmenu_widthclass-attributeinstance-attribute

```
menu_width: Number | None = None
```

The width of the dropdown menu.

If this is None, the menu width will be the same as input textfield width.

builddropdownoptionsclass-attributeinstance-attribute

```
options: list[DropdownOption] = field(default_factory=list)
```

A list of options to display in the dropdown.

buildselected_suffixclass-attributeinstance-attribute

```
selected_suffix: Control | None = None
```

A control to display after the selected item in the dropdown.

buildselected_trailing_iconclass-attributeinstance-attribute

```
selected_trailing_icon: IconDataOrControl | None = None
```

An optional icon at the end of the text field to indicate that the text field is pressed.

buildtextclass-attributeinstance-attribute

```
text: str | None = None
```

The text entered in the text field.

buildtext_alignclass-attributeinstance-attribute

```
text_align: TextAlign = TextAlign.START
```

The text align for the TextField of the Dropdown.

buildtext_sizeclass-attributeinstance-attribute

```
text_size: Number | None = None
```

Text size in virtual pixels.

buildtext_styleclass-attributeinstance-attribute

```
text_style: TextStyle | None = None
```

The TextStyle to use for text in input text field.

buildtrailing_iconclass-attributeinstance-attribute

```
trailing_icon: IconDataOrControl | None = None
```

An icon to display at the end of the text field.

buildvalueclass-attributeinstance-attribute

`value: str | None = None`

The key of the dropdown options corresponding to the selected option.

Events

bolton_blurclass-attributeinstance-attribute

`on_blur: ControlEventHandler[Dropdown] | None = None`

Called when the control has lost focus.

bolton_focusclass-attributeinstance-attribute

`on_focus: ControlEventHandler[Dropdown] | None = None`

Called when the control has received focus.

bolton_selectclass-attributeinstance-attribute

`on_select: ControlEventHandler[Dropdown] | None = None`

Called when the selected item of this dropdown has changed.

bolton_text_changeclass-attributeinstance-attribute

`on_text_change: ControlEventHandler[Dropdown] | None = None`

Called when the text input of this dropdown has changed.

Methods

deployed_codefocusasync

`focus()`

Requests focus for this control.

DropdownOption

Represents an item in a dropdown.

Inherits: Control

Properties

- `content` - A Control to display in this option.
- `key` - Option's key.
- `leading_icon` - An optional icon to display before the content or text.
- `style` - Customizes this menu item's appearance.
- `text` - Option's display text.
- `trailing_icon` - An optional icon to display after the content or text.

Properties

`buildcontentclass-attributeinstance-attribute`

`content: Control | None = None`

A Control to display in this option. If not specified, text will be used as fallback, else text will be ignored.

`buildkeyclass-attributeinstance-attribute`

`key: str | None = None`

Option's key.

If not specified text will be used as fallback.

Raises:

- `ValueError` - If neither key nor text is provided.

`buildleading_iconclass-attributeinstance-attribute`

`leading_icon: IconDataOrControl | None = None`

An optional icon to display before the content or text.

`buildstyleclass-attributeinstance-attribute`

`style: ButtonStyle | None = None`

Customizes this menu item's appearance.

`buildtextclass-attributeinstance-attribute`

`text: str | None = None`

Option's display text.

If not specified key will be used as fallback.

Raises:

- `ValueError` - If neither key nor text is provided.

`buildtrailing_iconclass-attributeinstance-attribute`

`trailing_icon: IconDataOrControl | None = None`

An optional icon to display after the content or text.

ExpansionPanel

A material expansion panel. It can either be expanded or collapsed. Its body is only visible when it is expanded.

Example:

```
ft.ExpansionPanelList(  
    width=400,  
    controls=[  
        ft.ExpansionPanel(  
            header=ft.Text("Shipping address"),  
            content=ft.Text("123 Market Street, Springfield"),  
            expanded=True,  
        ),  
        ft.ExpansionPanel(  
            header=ft.Text("Billing address"),  
            content=ft.Text("Same as shipping"),  
        ),  
    ],  
)
```

Image: ExpansionPanel - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_panel/image_for_docs.png

ExpansionPanel

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `bgcolor` - The background color of this panel.
- `can_tap_header` - Whether tapping on this panel's header will expand or collapse it.
- `content` - The control to be found in the body of this panel.
- `expanded` - Whether this panel is in expanded (True) or collapsed (False) state.
- `header` - The control to be found in the header of this panel.
- `highlight_color` - Defines the highlight color of this panel if `can_tap_header` is True, or the highlight color of the expand/collapse `IconButton` if `can_tap_header` is False.
- `splash_color` - Defines the splash color of this panel if `can_tap_header` is True, or the splash color of the expand/collapse `IconButton` if `can_tap_header` is False.

Properties

`buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

The background color of this panel.

`buildcan_tap_headerclass-attributeinstance-attribute`

`can_tap_header: bool = False`

Whether tapping on this panel's header will expand or collapse it.

`buildcontentclass-attributeinstance-attribute`

`content: Control | None = None`

The control to be found in the body of this panel.

It is displayed below the header when this panel is expanded.

If this property is None, this panel will have a placeholder Text as content.

`buildexpandedclass-attributeinstance-attribute`

`expanded: bool = False`

Whether this panel is in expanded (True) or collapsed (False) state.

`buildheaderclass-attributeinstance-attribute`

`header: Control | None = None`

The control to be found in the header of this panel.

It is always visible, regardless of whether this panel is expanded or collapsed. If `can_tap_header` is `True`, tapping on this header will expand or collapse this panel.

If this property is `None`, this panel will have a placeholder `Text` as header.

buildhighlight_colorclass-attributeinstance-attribute

`highlight_color: ColorValue | None = None`

Defines the highlight color of this panel if `can_tap_header` is `True`, or the highlight color of the expand/collapse `IconButton` if `can_tap_header` is `False`.

If this is `None`, then the icon button will use its default highlight color `Theme.highlight_color`, and this panel will use its default highlight color `Theme.highlight_color` (if `can_tap_header` is `True`).

buildsplash_colorclass-attributeinstance-attribute

`splash_color: ColorValue | None = None`

Defines the splash color of this panel if `can_tap_header` is `True`, or the splash color of the expand/collapse `IconButton` if `can_tap_header` is `False`.

If `can_tap_header` is `False`, and `Theme.use_material3` is `True`, this field will be ignored, as `IconButton.splash_color` will be ignored, and you should use `highlight_color` instead.

If this is `None`, then the icon button will use its default splash color `Theme.splash_color`, and this panel will use its default splash color `Theme.splash_color` (if `can_tap_header` is `True`).

ExpansionPanellist

A material expansion panel list that lays out its children and animates expansions.

Example:

```
ft.ExpansionPanellist(
    width=400,
    controls=[
        ft.ExpansionPanel(
            header=ft.Text("Details"),
            content=ft.Text("More information here"),
            expanded=True,
        ),
        ft.ExpansionPanel(
            header=ft.Text("History"),
            content=ft.Text("View previous updates"),
        ),
    ],
)
```

Image: ExpansionPanellist - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_panel_list/image_for_docs.png

ExpansionPanellist

Inherits: LayoutControl, ScrollableControl

Properties

- `controls` - A list of panels to display.
- `divider_color` - The color of the divider when `ExpansionPanel.expanded` is `False`.
- `elevation` - Defines the elevation of the controls, when expanded.
- `expand_icon_color` - The color of the icon.
- `expanded_header_padding` - Defines the padding around the header when expanded.
- `spacing` - The size of the gap between the controls when expanded.

Events

- `on_change` - Called when an item of controls is expanded or collapsed.

Examples

ExpansionPanellist

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT

    def handle_change(e: ft.Event[ft.ExpansionPanellist]):
        print(f"change on panel with index {e.data}")

    def handle_delete(e: ft.Event[ft.IconButton]):
        icon_button = e.control
        tile = icon_button.parent
        panel = tile.parent

        panel_list.controls.remove(panel)
        panel_list.update()

    panel_list = ft.ExpansionPanellist(
        expand_icon_color=ft.Colors.AMBER,
        elevation=8,
        divider_color=ft.Colors.AMBER,
        on_change=handle_change,
        controls=[
            ft.ExpansionPanel(
                bgcolor=ft.Colors.BLUE_400,
                expanded=True,
            ),
        ],
    )
```

```

colors = [
    ft.Colors.GREEN_500,
    ft.Colors.BLUE_800,
    ft.Colors.RED_800,
]

for i, bgcolor in enumerate(colors):
    panel_list.controls.append(
        ft.ExpansionPanel(
            bgcolor=bgcolor,
            header=ft.ListTile(title=ft.Text(f"Panel {i}"), bgcolor=bgcolor),
            content=ft.ListTile(
                bgcolor=bgcolor,
                title=ft.Text(f"This is in Panel {i}"),
                subtitle=ft.Text(f"Press the icon to delete panel {i}"),
                trailing=ft.IconButton(
                    icon=ft.Icons.DELETE,
                    on_click=handle_delete,
                ),
            ),
        ),
    )

page.add(
    ft.SafeArea(
        content=panel_list,
    )
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_panel_list/basic_flow.gif

Scrolling

ExpansionPanelList supports scrolling through its scroll property.

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.ExpansionPanelList(
                expand=True,
                scroll=ft.ScrollMode.ALWAYS,
                spacing=8,
                controls=[
                    ft.ExpansionPanel(
                        can_tap_header=True,
                        header=ft.ListTile(title=ft.Text(f"Panel {i}")),
                        content=ft.ListTile(
                            title=ft.Text(f"Details for panel {i}"),
                            subtitle=ft.Text(
                                "This content can be expanded or collapsed."
                            ),
                        ),
                    ),
                ],
            ),
        ),
    )

    for i in range(1, 41)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_panel_list/scrollable.png

Properties

buildcontrolsclass-attributeinstance-attribute

controls: list[ExpansionPanel] = field(default_factory=list)

A list of panels to display.

builddivider_colorclass-attributeinstance-attribute

divider_color: ColorValue | None = None

The color of the divider when `ExpansionPanel.expanded` is `False`.

buildelevationclass-attributeinstance-attribute

```
elevation: Number = 2
```

Defines the elevation of the controls, when expanded.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildexpand_icon_colorclass-attributeinstance-attribute

```
expand_icon_color: ColorValue | None = None
```

The color of the icon.

Defaults to `Colors.BLACK_54` in light theme mode and `Colors.WHITE_60` in dark theme mode.

buildexpanded_header_paddingclass-attributeinstance-attribute

```
expanded_header_padding: PaddingValue = field(
    default_factory=lambda: Padding.symmetric(vertical=16.0)
)
```

Defines the padding around the header when expanded.

buildspacingclass-attributeinstance-attribute

```
spacing: Number | None = None
```

The size of the gap between the controls when expanded.

Events

bolton_change class-attributeinstance-attribute

```
on_change: (
    EventHandler[ExpansionPanelListChangeEvent] | None
) = None
```

Called when an item of controls is expanded or collapsed.

ExpansionTile

A single-line ListTile with an expansion arrow icon that expands or collapses the tile to reveal or hide its controls.

Example:

```
ft.ExpansionTile(  
    width=400,  
    title="Account",  
    subtitle="Manage profile and security",  
    expanded=True,  
    controls=[  
        ft.ListTile(title=ft.Text("Profile")),  
        ft.ListTile(title=ft.Text("Security")),  
    ],  
)
```

Image: ExpansionTile - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_tile/image_for_docs.png

ExpansionTile

Inherits: LayoutControl, AdaptiveControl

Properties

- **affinity** - Typically used to force the expansion arrow icon to the tile's leading or trailing edge.
- **animation_style** - Defines the animation style (curve and duration) for this tile's expansion and collapse.
- **bgcolor** - The color to display behind the sublist when expanded.
- **clip_behavior** - Defines how the content of this tile is clipped.
- **collapsed_bgcolor** - Defines the background color of this tile when the sublist is collapsed (expanded is False).
- **collapsed_icon_color** - The icon color of this tile's expansion arrow icon when the sublist is collapsed (expanded is False).
- **collapsed_shape** - The tile's border shape when the sublist is collapsed.
- **collapsed_text_color** - The color of this tile's titles when the sublist is collapsed (expanded is False).
- **controls** - The controls to be displayed when this tile expanded.
- **controls_padding** - Defines the padding around the controls.
- **dense** - Whether this list tile is part of a vertically dense list.
- **enable_feedback** - Whether detected gestures should provide acoustic and/or haptic feedback.
- **expanded** - The expansion state of this tile.
- **expanded_alignment** - Defines the alignment of controls, which are arranged in a column when the tile is expanded.
- **expanded_cross_axis_alignment** - Defines the alignment of each child control within controls when the tile is expanded.
- **icon_color** - The icon color of this tile's expansion arrow icon when the sublist is expanded.
- **leading** - A Control to display before the title.
- **maintain_state** - A boolean value which defines whether the state of the controls is maintained when this tile expanded and collapses.
- **min_tile_height** - The minimum height of this tile.
- **shape** - The border shape of this tile when the sublist is expanded.
- **show_trailing_icon** - Whether this tile should build/show a default trailing icon, if trailing is None.
- **subtitle** - Additional content displayed below the title.
- **text_color** - The color of this tile's titles when the sublist is expanded.
- **tile_padding** - Defines the tile's padding.
- **title** - A Control to display as primary content of this tile.
- **trailing** - A Control to display after the title.

- `visual_density` - Defines how compact this tile's layout will be.

Events

- `on_change` - Called when a user clicks or taps the list tile.

Examples

ExpansionTile

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT
    page.spacing = 0
    page.padding = 0

    def handle_tile_change(e: ft.Event[ft.ExpansionTile]):
        page.show_dialog(
            ft.SnackBar(
                duration=1000,
                content=ft.Text(
                    value=(
                        f"ExpansionTile was "
                        f"{'expanded' if e.data == 'true' else 'collapsed'}"
                    )
                ),
            ),
        )
        if e.control.trailing:
            e.control.trailing.icon = (
                ft.Icons.ARROW_DROP_DOWN
                if e.control.trailing.icon == ft.Icons.ARROW_DROP_DOWN_CIRCLE
                else ft.Icons.ARROW_DROP_DOWN_CIRCLE
            )
            e.control.trailing.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                spacing=0,
                controls=[
                    ft.ExpansionTile(
                        expanded=True,
                        title=ft.Text("ExpansionTile 1"),
                        subtitle=ft.Text("Trailing expansion arrow icon"),
                        affinity=ft.TileAffinity.PLATFORM,
                        maintain_state=True,
                        collapsed_text_color=ft.Colors.RED,
                        text_color=ft.Colors.RED,
                        controls=[
                            ft.ListTile(title=ft.Text("This is sub-tile number 1.1")),
                            ft.ListTile(title=ft.Text("This is sub-tile number 1.2")),
                        ],
                    ),
                    ft.ExpansionTile(
                        expanded=True,
                        title=ft.Text("ExpansionTile 2"),
                        subtitle=ft.Text("Custom expansion arrow icon"),
                        trailing=ft.Icon(ft.Icons.ARROW_DROP_DOWN),
                        collapsed_text_color=ft.Colors.GREEN,
                        text_color=ft.Colors.GREEN,
                        on_change=handle_tile_change,
                        controls=[
                            ft.ListTile(title=ft.Text("This is sub-tile number 2.1")),
                            ft.ListTile(title=ft.Text("This is sub-tile number 2.2")),
                        ],
                    ),
                    ft.ExpansionTile(
                        expanded=True,
                        title=ft.Text("ExpansionTile 3"),
                        subtitle=ft.Text("Leading expansion arrow icon"),
                        affinity=ft.TileAffinity.LEADING,
                        collapsed_text_color=ft.Colors.BLUE_800,
                        text_color=ft.Colors.BLUE_200,
                        controls=[
                            ft.ListTile(title=ft.Text("This is sub-tile number 3.1")),
                            ft.ListTile(title=ft.Text("This is sub-tile number 3.2")),
                        ],
                    ),
                ],
            ),
        ),
    )
```

```
)
),
```

```
if __name__ == "__main__":
    ft.run(main)
```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_tile/basic.png

Programmatic expansion

play_arrow Try Online

```
import flet as ft
```

```
def main(page: ft.Page):
    page.spacing = 20

    tile = ft.ExpansionTile(
        title=ft.Text("I am the title of this tile.", weight=ft.FontWeight.BOLD),
        subtitle=ft.Text("This is the subtitle."),
        affinity=ft.TileAffinity.LEADING,
        controls=[ft.Text("■", size=80)],
        expanded=True,
        on_change=lambda e: print(f"Tile was {'expanded' if e.data else 'collapsed'}"),
    )

    def expand_tile(_: ft.Event[ft.FilledButton]):
        tile.expanded = True
        tile.update()

    def collapse_tile(_: ft.Event[ft.OutlinedButton]):
        tile.expanded = False
        tile.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Row(
                        alignment=ft.MainAxisAlignment.CENTER,
                        controls=[
                            ft.FilledButton("Expand Tile", on_click=expand_tile),
                            ft.OutlinedButton("Collapse Tile", on_click=collapse_tile),
                        ],
                    ),
                    tile,
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_tile/programmatic_expansion_flow.gif

Custom animations

play_arrow Try Online

```
import flet as ft
```

```
def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.spacing = 20

    tile = ft.ExpansionTile(
        expanded=True,
        title=ft.Text("Expand/Collapse me while being attentive to the animations!"),
        controls=[
            ft.ListTile(title=ft.Text("Sub-item 1")),
            ft.ListTile(title=ft.Text("Sub-item 2")),
            ft.ListTile(title=ft.Text("Sub-item 3")),
        ],
    )

    def switch_animation(e: ft.Event[ft.CupertinoSlidingSegmentedButton]):
        if e.control.selected_index == 0:
            tile.animation_style = None
        elif e.control.selected_index == 1:
            tile.animation_style = ft.AnimationStyle(
                curve=ft.AnimationCurve.BOUNCE_OUT,
                duration=ft.Duration(seconds=5),
            )
```



```

    )
    else:
        tile.animation_style = ft.AnimationStyle.no_animation()
        tile.update()

page.add(
    ft.SafeArea(
        content=ft.Column(
            horizontal_alignment=ft.CrossAxisAlignment.CENTER,
            controls=[
                ft.CupertinoSlidingSegmentedButton(
                    selected_index=0,
                    thumb_color=ft.Colors.BLUE_400,
                    on_change=switch_animation,
                    controls=[
                        ft.Text("Default animation"),
                        ft.Text("Custom animation"),
                        ft.Text("No animation"),
                    ],
                ),
                tile,
            ],
        ),
    ),
)
)
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: [Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_tile/custom_animations_default.gif](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_tile/custom_animations_default.gif)

Theme mode toggle

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    page.spacing = 0
    page.padding = 0
    page.theme_mode = ft.ThemeMode.LIGHT

    def handle_switch_change(e: ft.Event[ft.Switch]):
        page.theme_mode = ft.ThemeMode.DARK if e.control.value else ft.ThemeMode.LIGHT
        e.control.thumb_icon = (
            ft.Icons.DARK_MODE
            if page.theme_mode == ft.ThemeMode.DARK
            else ft.Icons.LIGHT_MODE
        )
        page.update()

    def handle_expansion_tile_change(e: ft.Event[ft.ExpansionTile]):
        page.show_dialog(
            ft.SnackBar(
                duration=1000,
                content=ft.Text(
                    value=(
                        f"ExpansionTile was "
                        f"{'expanded' if e.data == 'true' else 'collapsed'}"
                    )
                ),
            ),
        )
        if e.control.trailing:
            e.control.trailing.icon = (
                ft.Icons.ARROW_DROP_DOWN
                if e.control.trailing.icon == ft.Icons.ARROW_DROP_DOWN_CIRCLE
                else ft.Icons.ARROW_DROP_DOWN_CIRCLE
            )
            e.control.trailing.update()

    switch = ft.Switch(
        thumb_icon=ft.Icons.DARK_MODE,
        on_change=handle_switch_change,
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                spacing=0,
                controls=[
                    ft.ExpansionTile(
                        key="theme_mode_toggle_top_panel",
                        title=ft.Text("ExpansionTile 1"),
                        subtitle=ft.Text("Trailing expansion arrow icon"),

```

```

        bgcolor=ft.Colors.BLUE_GREY_200,
        collapsed_bgcolor=ft.Colors.BLUE_GREY_200,
        affinity=ft.TileAffinity.PLATFORM,
        maintain_state=True,
        collapsed_text_color=ft.Colors.RED,
        text_color=ft.Colors.RED,
        controls=[
            ft.ListTile(
                title=ft.Text("This is sub-tile number 1"),
                bgcolor=ft.Colors.BLUE_GREY_200,
            )
        ],
    ),
    ft.ExpansionTile(
        title=ft.Text("ExpansionTile 2"),
        subtitle=ft.Text("Custom expansion arrow icon"),
        trailing=ft.Icon(ft.Icons.ARROW_DROP_DOWN),
        collapsed_text_color=ft.Colors.GREEN,
        text_color=ft.Colors.GREEN,
        on_change=handle_expansion_tile_change,
        controls=[
            ft.ListTile(title=ft.Text("This is sub-tile number 2"))
        ],
    ),
    ft.ExpansionTile(
        title=ft.Text("ExpansionTile 3"),
        subtitle=ft.Text("Leading expansion arrow icon"),
        affinity=ft.TileAffinity.LEADING,
        expanded=True,
        collapsed_text_color=ft.Colors.BLUE_800,
        text_color=ft.Colors.BLUE_200,
        controls=[
            ft.ListTile(title=ft.Text("This is sub-tile number 3")),
            ft.ListTile(title=ft.Text("This is sub-tile number 4")),
            ft.ListTile(title=ft.Text("This is sub-tile number 5")),
        ],
    ),
    ft.Row(
        expand=True,
        alignment=ft.MainAxisAlignment.END,
        controls=[
            ft.Container(
                padding=ft.Padding.only(bottom=50),
                alignment=ft.Alignment.BOTTOM_RIGHT,
                expand=True,
                content=switch,
            ),
        ],
    ),
),
),
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Borders

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT

    page.add(
        ft.SafeArea(
            content=ft.ExpansionTile(
                title=ft.Text(
                    value="Expansion Tile with changing borders",
                    text_align=ft.TextAlign.CENTER,
                ),
                subtitle=ft.Text(
                    value="Tile border changes when expanded",
                    text_align=ft.TextAlign.CENTER,
                ),
                bgcolor=ft.Colors.BLUE_GREY_200,
                controls_padding=ft.Padding.symmetric(horizontal=10),
                collapsed_bgcolor=ft.Colors.BLUE_GREY_200,
                affinity=ft.TileAffinity.PLATFORM,
                maintain_state=True,
                shape=ft.RoundedRectangleBorder(radius=20),
                collapsed_shape=ft.StadiumBorder(side=ft.BorderSide(width=2)),
                collapsed_text_color=ft.Colors.GREY_800,
            ),
        ),
    )

```

```

        text_color=ft.Colors.GREY_800,
        controls=[
            ft.ListTile(
                title=ft.Text("A sub-tile"),
                bgcolor=ft.Colors.BLUE_GREY_200,
                shape=ft.RoundedRectangleBorder(radius=20),
            ),
            ft.ListTile(
                title=ft.Text("Another sub-tile"),
                bgcolor=ft.Colors.BLUE_GREY_200,
                shape=ft.RoundedRectangleBorder(radius=20),
            ),
        ],
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/expansion_tile/borders_flow.gif

Properties

buildaffinityclass-attributeinstance-attribute

affinity: TileAffinity | None = None

Typically used to force the expansion arrow icon to the tile's leading or trailing edge.

If None, ListTileTheme.affinity is used; if that is also None, then defaults to TileAffinity.TRAILING (the expansion arrow icon appears on the tile's trailing edge).

buildanimation_styleclass-attributeinstance-attribute

animation_style: AnimationStyle | None = None

Defines the animation style (curve and duration) for this tile's expansion and collapse.

If AnimationStyle.duration is provided, it will be used to override the expansion animation duration. If it is None, then AnimationStyle.duration from the ExpansionTileTheme.animation_style will be used. If that is also None, Duration(milliseconds=200) will be used as default.

If AnimationStyle.curve is provided, it will be used to override the expansion animation curve. If it is None, then AnimationStyle.curve from the ExpansionTileTheme.animation_style will be used. If that is also None, AnimationCurve.EASE_IN will be used as default.

If AnimationStyle.reverse_curve is provided, it will be used to override the collapse animation curve. If it is None, then AnimationStyle.reverse_curve from the ExpansionTileTheme.animation_style will be used. If that is also None, the expansion curve will be used as default.

Tip

To disable the animations, use AnimationStyle.no_animation.

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The color to display behind the sublist when expanded.

If None, ExpansionTileTheme.bgcolor is used; if that is also None, then defaults to Colors.TRANSPARENT.

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: ClipBehavior | None = None

Defines how the content of this tile is clipped.

If set and a custom collapsed or expanded shape is provided, this value determines how this tile is clipped.

If None, ExpansionTileTheme.clip_behavior is used; if that is also None, then defaults to ClipBehavior.ANTI_ALIAS.

buildcollapsed_bgcolorclass-attributeinstance-attribute

collapsed_bgcolor: ColorValue | None = None

Defines the background color of this tile when the sublist is collapsed (expanded is False).

If None, ExpansionTileTheme.collapsed_bgcolor is used; if that is also None, then defaults to Colors.TRANSPARENT.

buildcollapsed_icon_colorclass-attributeinstance-attribute

`collapsed_icon_color: ColorValue | None = None`

The icon color of this tile's expansion arrow icon when the sublist is collapsed (expanded is False).

If None, ExpansionTileTheme.collapsed_icon_color is used; if that is also None, then defaults to ColorScheme.on_surface of the Page.theme.

buildcollapsed_shapeclass-attributeinstance-attribute

`collapsed_shape: OutlinedBorder | None = None`

The tile's border shape when the sublist is collapsed.

If None, ExpansionTileTheme.shape is used; if that is also None, then defaults to a Border with vertical sides of color Colors.TRANSPARENT.

buildcollapsed_text_colorclass-attributeinstance-attribute

`collapsed_text_color: ColorValue | None = None`

The color of this tile's titles when the sublist is collapsed (expanded is False).

If None, ExpansionTileTheme.collapsed_text_color is used; if that is also None, then defaults to body_large of the Theme.text_theme.

buildcontrolsclass-attributeinstance-attribute

`controls: List[Control] | None = None`

The controls to be displayed when this tile expanded.

Typically a list of ListTile controls.

buildcontrols_paddingclass-attributeinstance-attribute

`controls_padding: PaddingValue | None = None`

Defines the padding around the controls.

If None, ExpansionTileTheme.controls_padding is used; if that is also None, then defaults to Padding.all(0).

builddenseclass-attributeinstance-attribute

`dense: bool | None = None`

Whether this list tile is part of a vertically dense list.

Dense tiles default to having a smaller height.

It is not recommended to set this property to True when in Material3.

If None, then its value is based on ListTileTheme.dense.

buildenable_feedbackclass-attributeinstance-attribute

`enable_feedback: bool = True`

Whether detected gestures should provide acoustic and/or haptic feedback. For example, on Android a tap will produce a clicking sound and a long-press will produce a short vibration, when feedback is enabled.

buildexpandedclass-attributeinstance-attribute

`expanded: bool = False`

The expansion state of this tile.

True - expanded, False - collapsed.

buildexpanded_alignmentclass-attributeinstance-attribute

`expanded_alignment: Alignment | None = None`

Defines the alignment of controls, which are arranged in a column when the tile is expanded.

If None, ExpansionTileTheme.expanded_alignment is used; if that is also None, then defaults to Alignment.CENTER.

buildexpanded_cross_axis_alignmentclass-attributeinstance-attribute

`expanded_cross_axis_alignment: CrossAxisAlignment = (
 CrossAxisAlignment.CENTER
)`

Defines the alignment of each child control within controls when the tile is expanded.

Raises:

- `ValueError` - If set to `CrossAxisAlignment.BASELINE`.

buildicon_colorclass-attributeinstance-attribute

`icon_color: ColorValue | None = None`

The icon color of this tile's expansion arrow icon when the sublist is expanded.

If `None`, `ExpansionTileTheme.icon_color` is used; if that is also `None`, then defaults to `ColorScheme.primary` of the `Page.theme`.

buildleadingclass-attributeinstance-attribute

`leading: IconDataOrControl | None = None`

A `Control` to display before the title.

Typically a `CircleAvatar` control.

Depending on the value of `affinity`, this control may replace the rotating expansion arrow icon.

buildmaintain_stateclass-attributeinstance-attribute

`maintain_state: bool = False`

A boolean value which defines whether the state of the controls is maintained when this tile expanded and collapses.

When `True`, the children are kept in the tree while the tile is collapsed. When `False` (default), the controls are removed from the tree when the tile is collapsed and recreated upon expansion.

buildmin_tile_heightclass-attributeinstance-attribute

`min_tile_height: Number | None = None`

The minimum height of this tile.

If `None`, the default tile heights are 56.0, 72.0, and 88.0 for one, two, and three lines of text respectively. If `dense` is `True`, these defaults are changed to 48.0, 64.0, and 76.0. A visual density value or a large title will also adjust the default tile heights.

buildshapeclass-attributeinstance-attribute

`shape: OutlinedBorder | None = None`

The border shape of this tile when the sublist is expanded.

If `None`, `ExpansionTileTheme.shape` is used; if that is also `None`, then defaults to a `Border` with vertical sides of color `Theme.divider_color`.

buildshow_trailing_iconclass-attributeinstance-attribute

`show_trailing_icon: bool = True`

Whether this tile should build/show a default trailing icon, if trailing is `None`.

buildsubtitleclass-attributeinstance-attribute

`subtitle: StrOrControl | None = None`

Additional content displayed below the title.

Typically a `Text` control.

buildtext_colorclass-attributeinstance-attribute

`text_color: ColorValue | None = None`

The color of this tile's titles when the sublist is expanded.

If `None`, `ExpansionTileTheme.text_color` is used; if that is also `None`, then defaults to `body_large` of the `Theme.text_theme`.

buildtile_paddingclass-attributeinstance-attribute

`tile_padding: PaddingValue | None = None`

Defines the tile's padding.

Analogous to `ListTile.content_padding`, this property defines the insets for the leading, title, subtitle and trailing controls. It does not inset the expanded controls widgets.

If `None`, `ExpansionTileTheme.tile_padding` is used; if that is also `None`, then defaults to `Padding.symmetric(horizontal=16.0)`.

buildtitleinstance-attribute

`title: StrOrControl`

A Control to display as primary content of this tile.

Typically a Text control.

Raises:

- `ValueError` - If it is neither a string nor a visible Control.

buildtrailingclass-attributeinstance-attribute

`trailing: IconDataOrControl | None = None`

A Control to display after the title.

Typically an Icon control.

Depending on the value of `affinity`, this control may replace the rotating expansion arrow icon.

buildvisual_densityclass-attributeinstance-attribute

`visual_density: VisualDensity | None = None`

Defines how compact this tile's layout will be.

Events

bolton_changeclass-attributeinstance-attribute

`on_change: ControlEventHandler[ExpansionTile] | None = None`

Called when a user clicks or taps the list tile.

The data property of the event handler argument is a boolean representing the expanded state of the tile after the change.

FilledButton

Filled buttons have the most visual impact after the `FloatingActionButton`, and is typically used for important, final actions that complete a flow, like "Save", "Join now", or "Confirm".

```
ft.FilledButton(content="Tap me")
```

Image: FilledButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/filled_button/image_for_docs.png

FilledButton

Inherits: Button

Examples

FilledButton

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    page.title = "FilledButton Example"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.FilledButton(content="Filled button"),
                    ft.FilledButton(content="Disabled button", disabled=True),
                    ft.FilledButton(
                        content="Button with icon",
                        icon=ft.Icons.ADD_OUTLINED,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/filled_button/basic.png

FilledIconButton

A filled variant of IconButton.

Filled icon buttons have higher visual impact and should be used for high emphasis actions, such as turning off a microphone or camera.

```
ft.FilledIconButton(icon=ft.Icons.CHECK)
```

Image: FilledIconButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/filled_icon_button/image_for_docs.png

FilledIconButton

Inherits: IconButton

FilledTonalButton

A filled tonal button is an alternative middle ground between `FilledButton` and `OutlinedButton` buttons. They're useful in contexts where a lower-priority button requires slightly more emphasis than an outline would give, such as "Next" in an onboarding flow. Tonal buttons use the secondary color mapping.

```
ft.FilledTonalButton(content="Tap me")
```

Image: FilledTonalButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/filled_tonal_button/image_for_docs.png

FilledTonalButton

Inherits: `Button`

Examples

FilledTonalButton

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "FilledTonalButton Example"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.FilledTonalButton(content="Filled tonal button"),
                    ft.FilledTonalButton(content="Disabled button", disabled=True),
                    ft.FilledTonalButton(
                        content="Button with icon",
                        icon=ft.Icons.ADD_OUTLINED,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/filled_tonal_button/basic.png

FilledTonalIconButton

A filled tonal variant of IconButton.

Filled tonal icon buttons are a middle ground between filled and outlined icon buttons. They're useful in contexts where the button requires slightly more emphasis than an outline would give, such as a secondary action paired with a high emphasis action.

```
ft.FilledTonalIconButton(icon=ft.Icons.CHECK)
```

Image: FilledTonalIconButton -

https://flet.dev/docs/test-images/examples/controls/material/golden/macros/filled_tonal_icon_button/image_for_docs.png

FilledTonalIconButton

Inherits: **IconButton**

FletApp

Renders another Flet app in the current app, similar to HTML IFrame, but for Flet.

Inherits: `LayoutControl`

Properties

- `app_error_message` - Template message to display when the app fails to load.
- `app_startup_screen_message` - Message to display on the app startup screen.
- `args` - Optional dictionary of arguments to pass to the Flet app.
- `assets_dir` - Base location for assets referenced by the embedded app.
- `force_pyodide` - Whether to force the use of Pyodide.
- `reconnect_interval_ms` - Delay, in milliseconds, between reconnection attempts.
- `reconnect_timeout_ms` - Total time to try reconnecting.
- `show_app_startup_screen` - Whether to show the app startup screen.
- `url` - Flet app URL, e.g.

Events

- `on_error` - Called when a connection or any unhandled error occurs.
- `on_python_output` - Fires once per stdout/stderr write inside the embedded Pyodide app.

Properties

`buildapp_error_messageclass-attributeinstance-attribute`

`app_error_message: str | None = None`

Template message to display when the app fails to load. Use `{message}` placeholder to include the error message and `{details}` to include error details.

`buildapp_startup_screen_messageclass-attributeinstance-attribute`

`app_startup_screen_message: str | None = None`

Message to display on the app startup screen.

`buildargsclass-attributeinstance-attribute`

`args: dict[str, Any] | None = None`

Optional dictionary of arguments to pass to the Flet app.

`buildassets_dirclass-attributeinstance-attribute`

`assets_dir: str | None = None`

Base location for assets referenced by the embedded app. On web this is a URL prefix joined with relative src values (e.g. on Image/Lottie/Markdown); on desktop it is a filesystem path.

`buildforce_pyodideclass-attributeinstance-attribute`

`force_pyodide: bool = False`

Whether to force the use of Pyodide.

`buildreconnect_interval_msclass-attributeinstance-attribute`

`reconnect_interval_ms: int | None = None`

Delay, in milliseconds, between reconnection attempts.

`buildreconnect_timeout_msclass-attributeinstance-attribute`

`reconnect_timeout_ms: int | None = None`

Total time to try reconnecting.

buildshow_app_startup_screenclass-attributeinstance-attribute

`show_app_startup_screen: bool = False`

Whether to show the app startup screen.

buildurlclass-attributeinstance-attribute

`url: str | None = None`

Flet app URL, e.g. `http://localhost:8550` or `flet.sock`.

Events

bolton_errorclass-attributeinstance-attribute

`on_error: ControlEventHandler[FletApp] | None = None`

Called when a connection or any unhandled error occurs.

bolton_python_outputclass-attributeinstance-attribute

`on_python_output: (EventHandler[FletAppOutputEvent] | None) = None`

Fires once per stdout/stderr write inside the embedded Pyodide app. Pyodide line-buffers by default, so each event is typically one `print(...)` call. Only fires for embedded FletApps with `force_pyodide=True`; root-level Pyodide pages have nowhere to bubble the event.

FloatingActionButton

A floating action button is a circular icon button that hovers over content to promote a primary action in the application. Floating action button is usually set to `page.floating_action_button`, but can also be added as a regular control at any place on a page.

Example:

```
ft.FloatingActionButton(icon=ft.Icons.ADD)
```

Image: FloatingActionButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/floating_action_button/image_for_docs.png

FloatingActionButton

Inherits: `LayoutControl`

Properties

- `autofocus` - True if the control will be selected as the initial focus.
- `bgcolor` - Button background color.
- `clip_behavior` - Defines how the content is clipped.
- `content` - The content of this button.
- `disabled_elevation` - The elevation of this button when disabled.
- `elevation` - The elevation of this button.
- `enable_feedback` - Whether detected gestures should provide acoustic and/or haptic feedback.
- `focus_color` - The color to use for filling this button when it has input focus.
- `focus_elevation` - The elevation of this button when it has input focus.
- `foreground_color` - The default foreground color for icons and text within this button.
- `highlight_elevation` - The elevation of this button when it is highlighted.
- `hover_color` - The color to use for filling this button when hovered.
- `hover_elevation` - The elevation of this button it is enabled and being hovered over.
- `icon` - Icon shown in this button.
- `mini` - Controls the size of this button.
- `mouse_cursor` - The cursor to be displayed when a mouse pointer enters or is hovering over this control.
- `shape` - The shape of the FAB's border.
- `splash_color` - The color to use for the ink splash.
- `url` - The URL to open when this button is clicked.

Events

- `on_click` - Called when a user clicks this button.

Examples

Handling clicks

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "Floating Action Button"
    page.theme_mode = ft.ThemeMode.LIGHT
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.padding = 0
    page.scroll = ft.ScrollMode.HIDDEN

    count = 1
```

```

def handle_fab_click(e: ft.Event[ft.FloatingActionButton]):
    nonlocal count
    page.add(
        ft.ListTile(
            title=ft.Text(f"Tile {count}"),
            bgcolor=ft.Colors.TEAL_300,
            leading=ft.Icon(
                ft.Icons.CIRCLE_OUTLINED,
                color=ft.Colors.DEEP_ORANGE_300,
            ),
            on_click=lambda x: print(x.control.title.value + " was clicked!"),
        )
    )
    page.show_dialog(ft.SnackBar(ft.Text("Tile was added successfully!")))
    count += 1

page.floating_action_button = ft.FloatingActionButton(
    key="handling_clicks_fab",
    icon=ft.Icons.ADD,
    on_click=handle_fab_click,
    bgcolor=ft.Colors.LIME_300,
)

page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                ft.Container(
                    bgcolor=ft.Colors.BLUE,
                    padding=ft.Padding.all(20),
                    content=ft.Row(
                        alignment=ft.MainAxisAlignment.CENTER,
                        controls=[
                            ft.Text(
                                value="Floating Action Button Example",
                                style=ft.TextStyle(
                                    size=20,
                                    weight=ft.FontWeight.W_500,
                                ),
                            ),
                        ],
                    ),
                ),
                ft.Text("Press the FAB to add a tile!"),
            ],
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: handling-clicks - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/floating_action_button/handling_clicks.gif

Properties

buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

True if the control will be selected as the initial focus. If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

Button background color.

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: ClipBehavior = ClipBehavior.NONE

Defines how the content is clipped.

buildcontentclass-attributeinstance-attribute

content: StrOrControl | None = None

The content of this button.

Raises:

- ValueError - If neither icon nor a valid content (string or visible Control) is provided.

builddisabled_elevationclass-attributeinstance-attribute

`disabled_elevation: Number | None = None`

The elevation of this button when disabled.

Defaults to elevation.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildelevationclass-attributeinstance-attribute

`elevation: Number | None = None`

The elevation of this button.

Defaults to 6.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildenable_feedbackclass-attributeinstance-attribute

`enable_feedback: bool | None = None`

Whether detected gestures should provide acoustic and/or haptic feedback. On Android, for example, setting this to `True` will produce a click sound and a long-press will produce a short vibration.

buildfocus_colorclass-attributeinstance-attribute

`focus_color: ColorValue | None = None`

The color to use for filling this button when it has input focus.

buildfocus_elevationclass-attributeinstance-attribute

`focus_elevation: Number | None = None`

The elevation of this button when it has input focus.

Defaults to 8.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildforeground_colorclass-attributeinstance-attribute

`foreground_color: ColorValue | None = None`

The default foreground color for icons and text within this button.

buildhighlight_elevationclass-attributeinstance-attribute

`highlight_elevation: Number | None = None`

The elevation of this button when it is highlighted.

Defaults to 12.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildhover_colorclass-attributeinstance-attribute

`hover_color: ColorValue | None = None`

The color to use for filling this button when hovered.

buildhover_elevationclass-attributeinstance-attribute

`hover_elevation: Number | None = None`

The elevation of this button it is enabled and being hovered over.

Defaults to 8.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildiconclass-attributeinstance-attribute

`icon: IconDataOrControl | None = None`

Icon shown in this button.

buildminiclass-attributeinstance-attribute

`mini: bool = False`

Controls the size of this button.

By default, floating action buttons are non-mini and have a height and width of 56.0 logical pixels. Mini floating action buttons have a height and width of 40.0 logical pixels with a layout width and height of 48.0 logical pixels.

buildmouse_cursorclass-attributeinstance-attribute

`mouse_cursor: MouseCursor | None = None`

The cursor to be displayed when a mouse pointer enters or is hovering over this control.

buildshapeclass-attributeinstance-attribute

`shape: OutlinedBorder | None = None`

The shape of the FAB's border.

buildsplash_colorclass-attributeinstance-attribute

`splash_color: ColorValue | None = None`

The color to use for the ink splash.

buildurlclass-attributeinstance-attribute

`url: str | Url | None = None`

The URL to open when this button is clicked.

Additionally, if `on_click` event callback is provided, it is fired after that.

Events

bolton_clickclass-attributeinstance-attribute

```
on_click: (
    ControlEventHandler[FloatingActionButton] | None
) = None
```

Called when a user clicks this button.

FormFieldControl

Inherits: `LayoutControl`

Properties

- `align_label_with_hint` - TBD
- `bgcolor` - TextField background color.
- `border` - Border around input.
- `border_color` - The border color.
- `border_radius`
- `border_width` - The width of the border in virtual pixels.
- `collapsed` - TBD
- `color` - Text color.
- `content_padding` - The padding for the input decoration's container.
- `counter` - A Control to place below the line as a character count.
- `counter_style` - The text style to use for counter.
- `dense` - Whether this control is part of a dense form (ie, uses less vertical space).
- `error` - Text that appears below the input border.
- `error_max_lines` - TBD
- `error_style` - The text style to use for error.
- `fill_color` - Background color of TextField.
- `filled` - If True the decoration's container is filled with theme `fill_color`.
- `fit_parent_size` - TBD
- `focus_color` - TBD
- `focused_bgcolor` - Background color in focused state.
- `focused_border_color` - Border color in focused state.
- `focused_border_width` - Border width in focused state.
- `focused_color` - The text's color when focused.
- `helper` - Text that provides context about the input's value, such as how the value will be used.
- `helper_max_lines` - TBD
- `helper_style` - The text style to use for helper.
- `hint_fade_duration` - TBD
- `hint_max_lines` - TBD
- `hint_style` - The text style to use for `hint_text`.
- `hint_text` - Text that suggests what sort of input the field accepts.
- `hover_color` - Background color of TextField when hovered.
- `icon` - The icon to show before the input field and outside of the decoration's container.
- `label` - Optional text that describes the input field.
- `label_style` - The text style to use for label.
- `prefix` - A Control to place on the line before the input.
- `prefix_icon` - An icon that appears before the editable part of the text field, within the decoration's container.

- `prefix_icon_size_constraints` - TBD
- `prefix_style` - The text style to use for prefix.
- `size_constraints` - TBD
- `suffix` - A Control to place on the line after the input.
- `suffix_icon` - An icon that appears after the editable part of the text field and after the suffix, within the decoration's container.
- `suffix_icon_size_constraints` - TBD
- `suffix_style` - The text style to use for suffix.
- `text_size` - Text size in virtual pixels.
- `text_style` - The TextStyle to use for the text being edited.
- `text_vertical_align` - Defines how the text should be aligned vertically.

Methods

- `focus` - Request focus for this control.

Properties

`buildalign_label_with_hint`class-attributeinstance-attribute

`align_label_with_hint`: bool | None = None

TBD

`buildbgcolor`class-attributeinstance-attribute

`bgcolor`: ColorValue | None = None

TextField background color.

Note

Will not be visible if `filled` is False.

`buildborder`class-attributeinstance-attribute

`border`: InputBorder = InputBorder.OUTLINE

Border around input.

`buildborder_color`class-attributeinstance-attribute

`border_color`: ColorValue | None = None

The border color.

Tip

Set to `Colors.TRANSPARENT` to invisible/hide the border.

`buildborder_radius`class-attributeinstance-attribute

`border_radius`: BorderRadiusValue | None = None

`buildborder_width`class-attributeinstance-attribute

`border_width`: Number | None = None

The width of the border in virtual pixels.

Defaults to 1.

Tip

Set to 0 to completely remove the border.

`buildcollapsed`class-attributeinstance-attribute

`collapsed`: bool | None = None

TBD

`buildcolor`class-attributeinstance-attribute

`color: ColorValue | None = None`

Text color.

buildcontent_paddingclass-attributeinstance-attribute

`content_padding: PaddingValue | None = None`

The padding for the input decoration's container.

buildcounterclass-attributeinstance-attribute

`counter: StrOrControl | None = None`

A Control to place below the line as a character count.

If None or an empty string then nothing will appear in the counter's location.

buildcounter_styleclass-attributeinstance-attribute

`counter_style: TextStyle | None = None`

The text style to use for counter.

builddenseclass-attributeinstance-attribute

`dense: bool | None = None`

Whether this control is part of a dense form (ie, uses less vertical space).

builderrorclass-attributeinstance-attribute

`error: StrOrControl | None = None`

Text that appears below the input border.

If non-null, the border's color animates to red and the helper is not shown.

builderror_max_linesclass-attributeinstance-attribute

`error_max_lines: int | None = None`

TBD

builderror_styleclass-attributeinstance-attribute

`error_style: TextStyle | None = None`

The text style to use for error.

buildfill_colorclass-attributeinstance-attribute

`fill_color: ColorValue | None = None`

Background color of TextField.

Note

Will not be visible if filled is False.

buildfilledclass-attributeinstance-attribute

`filled: bool | None = None`

If True the decoration's container is filled with theme fill_color.

If filled=None (the default), then it is implicitly set to True when at least one of the following is not None: fill_color, focused_bgcolor, hover_color and bgcolor.

buildfit_parent_sizeclass-attributeinstance-attribute

`fit_parent_size: bool | None = None`

TBD

buildfocus_colorclass-attributeinstance-attribute

`focus_color: ColorValue | None = None`

TBD

buildfocused_bgcolorclass-attributeinstance-attribute

`focused_bgcolor: ColorValue | None = None`

Background color in focused state.

Note

Will not be visible if filled is False.

buildfocused_border_colorclass-attributeinstance-attribute

focused_border_color: ColorValue | None = None

Border color in focused state.

buildfocused_border_widthclass-attributeinstance-attribute

focused_border_width: Number | None = None

Border width in focused state.

buildfocused_colorclass-attributeinstance-attribute

focused_color: ColorValue | None = None

The text's color when focused.

buildhelperclass-attributeinstance-attribute

helper: StrOrControl | None = None

Text that provides context about the input's value, such as how the value will be used.

If non-null, the text is displayed below the input decorator, in the same location as error. If a non-null error value is specified then the helper text is not shown.

buildhelper_max_linesclass-attributeinstance-attribute

helper_max_lines: int | None = None

TBD

buildhelper_styleclass-attributeinstance-attribute

helper_style: TextStyle | None = None

The text style to use for helper.

buildhint_fade_durationclass-attributeinstance-attribute

hint_fade_duration: DurationValue | None = None

TBD

buildhint_max_linesclass-attributeinstance-attribute

hint_max_lines: int | None = None

TBD

buildhint_styleclass-attributeinstance-attribute

hint_style: TextStyle | None = None

The text style to use for hint_text.

buildhint_textclass-attributeinstance-attribute

hint_text: str | None = None

Text that suggests what sort of input the field accepts.

Displayed on top of the input when the it's empty and either (a) label is None or (b) the input has the focus.

buildhover_colorclass-attributeinstance-attribute

hover_color: ColorValue | None = None

Background color of TextField when hovered.

Note

Will not be visible if filled is False.

buildiconclass-attributeinstance-attribute

icon: IconDataOrControl | None = None

The icon to show before the input field and outside of the decoration's container.

buildlabelclass-attributeinstance-attribute

`label: StrOrControl | None = None`

Optional text that describes the input field.

When the input field is empty and unfocused, the label is displayed on top of the input field (i.e., at the same location on the screen where text may be entered in the input field). When the input field receives focus (or if the field is non-empty) the label moves above, either vertically adjacent to, or to the center of the input field.

buildlabel_styleclass-attributeinstance-attribute

`label_style: TextStyle | None = None`

The text style to use for label.

buildprefixclass-attributeinstance-attribute

`prefix: StrOrControl | None = None`

A Control to place on the line before the input.

It appears after the `prefix_icon`, if both are specified.

This can be used, for example, to add some padding to text that would otherwise be specified using `prefix`, or to add a custom control in front of the input. The control's baseline is lined up with the input baseline.

buildprefix_iconclass-attributeinstance-attribute

`prefix_icon: IconDataOrControl | None = None`

An icon that appears before the editable part of the text field, within the decoration's container.

If `prefix` is specified and visible, this icon will appear to its left.

buildprefix_icon_size_constraintsclass-attributeinstance-attribute

`prefix_icon_size_constraints: BoxConstraints | None = None`

TBD

buildprefix_styleclass-attributeinstance-attribute

`prefix_style: TextStyle | None = None`

The text style to use for prefix.

buildsize_constraintsclass-attributeinstance-attribute

`size_constraints: BoxConstraints | None = None`

TBD

buildsuffixclass-attributeinstance-attribute

`suffix: StrOrControl | None = None`

A Control to place on the line after the input.

It appears before the `suffix_icon`, if both are specified.

This can be used, for example, to add some padding to the text that would otherwise be specified using `suffix`, or to add a custom control after the input. The control's baseline is lined up with the input baseline.

buildsuffix_iconclass-attributeinstance-attribute

`suffix_icon: IconDataOrControl | None = None`

An icon that appears after the editable part of the text field and after the suffix, within the decoration's container.

buildsuffix_icon_size_constraintsclass-attributeinstance-attribute

`suffix_icon_size_constraints: BoxConstraints | None = None`

TBD

buildsuffix_styleclass-attributeinstance-attribute

`suffix_style: TextStyle | None = None`

The text style to use for suffix.

buildtext_sizeclass-attributeinstance-attribute

```
text_size: Number | None = None
```

Text size in virtual pixels.

buildtext_styleclass-attributeinstance-attribute

```
text_style: TextStyle | None = None
```

The TextStyle to use for the text being edited.

buildtext_vertical_alignclass-attributeinstance-attribute

```
text_vertical_align: VerticalAlignment | Number | None = (
    None
)
```

Defines how the text should be aligned vertically.

Value can either be a number ranging from -1.0 (topmost location) to 1.0 (bottommost location) or of type VerticalAlignment Defaults to VerticalAlignment.CENTER.

Methods

deployed_codefocusasync

```
focus()
```

Request focus for this control.

Example

```
async def main(page: ft.Page):
    page.add(ctf := ft.TextField())
    await ctf.focus()
```

GestureDetector

A control that detects gestures.

Attempts to recognize gestures that correspond to its non-None callbacks.

If this control has a content, it defers to that child control for its sizing behavior, else it grows to fit the parent instead.

Inherits: LayoutControl, AdaptiveControl

Properties

- `allowed_devices` - TBD
- `content` - A child Control contained by the gesture detector.
- `drag_interval` - Throttling in milliseconds for horizontal drag, vertical drag and pan update events.
- `exclude_from_semantics` - TBD
- `hover_interval` - Throttling in milliseconds for `on_hover` event.
- `mouse_cursor` - The mouse cursor for mouse pointers that are hovering over the control.
- `multi_tap_touches` - The minimum number of pointers to trigger `on_multi_tap` event.
- `trackpad_scroll_causes_scale` - TBD

Events

- `on_double_tap` - The user has tapped the screen with a primary button at the same location twice in quick succession.
- `on_double_tap_cancel` - The pointer sequence that was expected to cause a double tap will not do so.
- `on_double_tap_down` - Called when a pointer that might cause a double tap has contacted the screen at a particular location.
- `on_enter` - Called when a mouse pointer has entered this control.
- `on_exit` - Called when a mouse pointer has exited this control.
- `on_force_press_end` - Called when the pointer that triggered a force press is no longer in contact with the screen.
- `on_force_press_peak` - Called when a pointer has pressed with a force exceeding the peak pressure.
- `on_force_press_start` - Called when a pointer has pressed with a force exceeding the start pressure.
- `on_force_press_update` - Called for each update after a force press has started.
- `on_horizontal_drag_cancel` - The pointer that previously triggered `on_horizontal_drag_down` will not end up causing a horizontal drag.
- `on_horizontal_drag_down` - Called when a pointer has contacted the screen and might begin to move horizontally.
- `on_horizontal_drag_end` - Called when a pointer moving horizontally is no longer in contact and was moving at a specific velocity.
- `on_horizontal_drag_start` - Called when a pointer has contacted the screen with a primary button and has begun to move horizontally.
- `on_horizontal_drag_update` - Called when a pointer that is in contact with the screen and moving horizontally has moved in the horizontal direction.
- `on_hover` - Called when a mouse pointer has entered this control.
- `on_long_press` - Called when a long press gesture with a primary button has been recognized.
- `on_long_press_cancel` - The pointer that previously triggered `on_long_press_down` will not end up causing a long-press.
- `on_long_press_down` - Called when a pointer that might cause a long press with a primary button has contacted the screen.
- `on_long_press_end` - Called when a pointer that has triggered a long-press with a primary button has stopped contacting the screen.

- `on_long_press_move_update` - Called when, after a long press has been accepted, the pointer moves.
- `on_long_press_start` - Triggered when a pointer has remained in contact with the screen at the same location for a long period of time.
- `on_long_press_up` - Called when a pointer that has triggered a long press with a primary button is no longer in contact with the screen.
- `on_multi_long_press` - Called when a long press gesture with multiple pointers has been recognized.
- `on_multi_tap` - Called when multiple pointers contacted the screen.
- `on_pan_cancel` - The pointer that previously triggered `on_pan_down` will not end up causing a pan gesture.
- `on_pan_down` - Called when a pointer has contacted the screen and might begin to move.
- `on_pan_end` - Called when a pointer is no longer in contact and was moving at a specific velocity.
- `on_pan_start` - Called when a pointer has contacted the screen and has begun to move.
- `on_pan_update` - Called when a pointer that is in contact with the screen and moving has moved again.
- `on_right_pan_end` - A pointer with secondary button pressed is no longer in contact and was moving at a specific velocity.
- `on_right_pan_start` - Pointer has contacted the screen while secondary button pressed and has begun to move.
- `on_right_pan_update` - A pointer that is in contact with the screen, secondary button pressed and moving has moved again.
- `on_scale_end` - TBD
- `on_scale_start` - Called when the pointers in contact with the screen have established a focal point and initial scale of 1.0.
- `on_scale_update` - TBD
- `on_scroll` - TBD
- `on_secondary_long_press` - Called when a long press gesture with a secondary button has been recognized.
- `on_secondary_long_press_cancel` - The pointer that previously triggered `on_secondary_long_press_down` not end up causing a long-press.
- `on_secondary_long_press_down` - Called when a pointer that might cause a long press with a secondary button has contacted the screen.
- `on_secondary_long_press_end` - Called when a pointer that has triggered a long-press with a secondary button has stopped contacting the screen.
- `on_secondary_long_press_move_update` - Called when, after a secondary long press has been accepted, the pointer moves.
- `on_secondary_long_press_start` - Triggered when a pointer has remained in contact with the screen at the same location for a long period of time.
- `on_secondary_long_press_up` - Called when a pointer that has triggered a long press with a secondary button is no longer in contact with the screen.
- `on_secondary_tap` - A tap with a secondary button has occurred.
- `on_secondary_tap_cancel` - The pointer that previously triggered `on_secondary_tap_down` will not end up causing a tap.
- `on_secondary_tap_down` - Called when a pointer that might cause a tap with a secondary button has contacted the screen at a particular location.
- `on_secondary_tap_up` - Called when a pointer that will trigger a tap with a secondary button has stopped contacting the screen at a particular location.
- `on_tap` - Called when a tap with a primary button has occurred.
- `on_tap_cancel` - The pointer that previously triggered `on_tap_down` will not end up causing a tap.
- `on_tap_down` - Called when a pointer that might cause a tap with a primary button has contacted the screen at a particular location.
- `on_tap_move` - Called when a pointer that triggered a tap has moved.

- `on_tap_up` - Called when a pointer that will trigger a tap with a primary button has stopped contacting the screen at a particular location.
- `on_tertiary_long_press` - Called when a long press gesture with a tertiary button has been recognized.
- `on_tertiary_long_press_cancel` - The pointer that previously triggered `on_tertiary_long_press_down` will not end up causing a long-press.
- `on_tertiary_long_press_down` - Called when a pointer that might cause a long press with a tertiary button has contacted the screen.
- `on_tertiary_long_press_end` - Called when a pointer that has triggered a long-press with a tertiary button has stopped contacting the screen.
- `on_tertiary_long_press_move_update` - Called when, after a tertiary long press has been accepted, the pointer moves.
- `on_tertiary_long_press_start` - Triggered when a pointer has remained in contact with the screen at the same location for a long period of time.
- `on_tertiary_long_press_up` - Called when a pointer that has triggered a long press with a tertiary button is no longer in contact with the screen.
- `on_tertiary_tap_cancel` - The pointer that previously triggered `on_tertiary_tap_down` will not end up causing a tap.
- `on_tertiary_tap_down` - Called when a pointer that might cause a tap with a tertiary button has contacted the screen at a particular location.
- `on_tertiary_tap_up` - Called when a pointer that will trigger a tap with a tertiary button has stopped contacting the screen at a particular location.
- `on_vertical_drag_cancel` - The pointer that previously triggered `on_vertical_drag_down` will not end up causing a vertical drag.
- `on_vertical_drag_down` - Called when a pointer has contacted the screen and might begin to move vertically.
- `on_vertical_drag_end` - Called when a pointer moving vertically is no longer in contact and was moving at a specific velocity.
- `on_vertical_drag_start` - Called when a pointer has contacted the screen and has begun to move vertically.
- `on_vertical_drag_update` - A pointer moving vertically has moved in the vertical direction.

Examples

Solitaire game tutorial

Handling events

[play_arrowTry Online](#)

```
import flet as ft
```

```
def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.GestureDetector(
                hover_interval=50,
                on_tap=lambda e: print(e),
                on_tap_down=lambda e: print(e),
                on_tap_up=lambda e: print(e),
                on_secondary_tap=lambda e: print(e),
                on_secondary_tap_down=lambda e: print(e),
                on_secondary_tap_up=lambda e: print(e),
                on_long_press_start=lambda e: print(e),
                on_long_press_end=lambda e: print(e),
                on_secondary_long_press_start=lambda e: print(e),
                on_secondary_long_press_end=lambda e: print(e),
                on_double_tap=lambda e: print(e),
                on_double_tap_down=lambda e: print(e),
                on_pan_start=lambda e: print(e),
                on_pan_update=lambda e: print(e),
                on_pan_end=lambda e: print(e),
                on_hover=lambda e: print(e),
                on_enter=lambda e: print(e),
                on_exit=lambda e: print(e),
                content=ft.Container(
                    bgcolor=ft.Colors.GREEN,
                    width=200,
```

```

        height=200,
    ),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Draggable containers

The following example demonstrates how a control can be freely dragged inside a Stack.

The sample also shows that GestureDetector can have a child control (blue container) as well as be nested inside another control (yellow container) giving the same results.

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    def handle_pan_update1(e: ft.DragUpdateEvent[ft.GestureDetector]):
        container = e.control.parent
        container.top = max(0.0, container.top + e.local_delta.y)
        container.left = max(0.0, container.left + e.local_delta.x)
        container.update()

    def handle_pan_update2(e: ft.DragUpdateEvent[ft.GestureDetector]):
        e.control.top = max(0.0, e.control.top + e.local_delta.y)
        e.control.left = max(0.0, e.control.left + e.local_delta.x)
        e.control.update()

    page.add(
        ft.SafeArea(
            content=ft.Stack(
                width=1000,
                height=500,
                controls=[
                    ft.Container(
                        bgcolor=ft.Colors.AMBER,
                        width=50,
                        height=50,
                        left=0,
                        top=0,
                        content=ft.GestureDetector(
                            mouse_cursor=ft.MouseCursor.MOVE,
                            drag_interval=10,
                            on_pan_update=handle_pan_update1,
                        ),
                    ),
                    ft.GestureDetector(
                        mouse_cursor=ft.MouseCursor.MOVE,
                        drag_interval=10,
                        on_vertical_drag_update=handle_pan_update2,
                        left=100,
                        top=100,
                        content=ft.Container(
                            bgcolor=ft.Colors.BLUE,
                            width=50,
                            height=50,
                        ),
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: draggable-containers - https://flet.dev/docs/examples/controls/core/gesture_detector/media/draggable_containers.gif

Window drag area

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    def on_pan_update(e: ft.DragUpdateEvent[ft.GestureDetector]):
        page.window.left += e.global_delta.x
        page.window.top += e.global_delta.y

```

```

page.add(
    ft.SafeArea(
        content=ft.Stack(
            width=1000,
            height=500,
            controls=[
                ft.GestureDetector(
                    mouse_cursor=ft.MouseCursor.MOVE,
                    on_pan_update=on_pan_update,
                    left=200,
                    top=200,
                    content=ft.Container(
                        bgcolor=ft.Colors.PINK,
                        width=50,
                        height=50,
                    ),
                ),
            ],
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Mouse cursors

play_arrowTry Online

```

import random

import flet as ft

def main(page: ft.Page):
    def on_pan_update(event: ft.DragUpdateEvent[ft.GestureDetector]):
        container.top = max(0.0, container.top + event.local_delta.y)
        container.left = max(0.0, container.left + event.local_delta.x)
        container.update()

    gesture_detector = ft.GestureDetector(
        mouse_cursor=ft.MouseCursor.BASIC,
        drag_interval=50,
        on_pan_update=on_pan_update,
    )
    container = ft.Container(
        bgcolor=ft.Colors.AMBER,
        width=150,
        height=150,
        left=0,
        top=0,
        content=gesture_detector,
    )

    def handle_button_click(e: ft.Event[ft.Button]):
        gesture_detector.mouse_cursor = random.choice(list(ft.MouseCursor))
        text.value = f"Mouse Cursor: {gesture_detector.mouse_cursor}"
        gesture_detector.update()
        text.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Stack(
                        width=1000,
                        height=500,
                        controls=[container],
                    ),
                    ft.Button("Change mouse Cursor", on_click=handle_button_click),
                    text := ft.Text(f"Mouse Cursor: {gesture_detector.mouse_cursor}"),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Properties

buildallowed_deviceclass-attributeinstance-attribute

allowed_devices: list[PointerType] | None = None

TBD

buildcontentclass-attributeinstance-attribute

```
content: Control | None = None
```

A child Control contained by the gesture detector.

builddrag_intervalclass-attributeinstance-attribute

```
drag_interval: int = 0
```

Throttling in milliseconds for horizontal drag, vertical drag and pan update events.

When a user moves a pointer a lot of events are being generated to do precise tracking. `drag_interval` allows sending drag update events to a Flet program every X milliseconds, thus preserving the bandwidth (web and mobile apps).

0 means no throttling: all events are sent to a Flet program, resulting in very smooth tracking.

buildexclude_from_semanticsclass-attributeinstance-attribute

```
exclude_from_semantics: bool = False
```

TBD

buildhover_intervalclass-attributeinstance-attribute

```
hover_interval: int = 0
```

Throttling in milliseconds for `on_hover` event.

buildmouse_cursorclass-attributeinstance-attribute

```
mouse_cursor: MouseCursor | None = None
```

The mouse cursor for mouse pointers that are hovering over the control.

buildmulti_tap_touchesclass-attributeinstance-attribute

```
multi_tap_touches: int = 0
```

The minimum number of pointers to trigger `on_multi_tap` event.

buildtrackpad_scroll_causes_scaleclass-attributeinstance-attribute

```
trackpad_scroll_causes_scale: bool = False
```

TBD

Events

bolton_double_tapclass-attributeinstance-attribute

```
on_double_tap: (
    ControlEventHandler[GestureDetector] | None
) = None
```

The user has tapped the screen with a primary button at the same location twice in quick succession.

bolton_double_tap_cancelclass-attributeinstance-attribute

```
on_double_tap_cancel: (
    ControlEventHandler[GestureDetector] | None
) = None
```

The pointer sequence that was expected to cause a double tap will not do so.

bolton_double_tap_downclass-attributeinstance-attribute

```
on_double_tap_down: (
    EventHandler[TapEvent[GestureDetector]] | None
) = None
```

Called when a pointer that might cause a double tap has contacted the screen at a particular location.

Triggered immediately after the down event of the second tap.

bolton_enterclass-attributeinstance-attribute

```
on_enter: (
    EventHandler[HoverEvent[GestureDetector]] | None
) = None
```

Called when a mouse pointer has entered this control.

bolton_exitclass-attributeinstance-attribute

```
on_exit: (  
    EventHandler[HoverEvent[GestureDetector]] | None  
) = None
```

Called when a mouse pointer has exited this control.

bolton_force_press_endclass-attributeinstance-attribute

```
on_force_press_end: (  
    EventHandler[ForcePressEvent[GestureDetector]] | None  
) = None
```

Called when the pointer that triggered a force press is no longer in contact with the screen.

bolton_force_press_peakclass-attributeinstance-attribute

```
on_force_press_peak: (  
    EventHandler[ForcePressEvent[GestureDetector]] | None  
) = None
```

Called when a pointer has pressed with a force exceeding the peak pressure.

bolton_force_press_startclass-attributeinstance-attribute

```
on_force_press_start: (  
    EventHandler[ForcePressEvent[GestureDetector]] | None  
) = None
```

Called when a pointer has pressed with a force exceeding the start pressure.

bolton_force_press_updateclass-attributeinstance-attribute

```
on_force_press_update: (  
    EventHandler[ForcePressEvent[GestureDetector]] | None  
) = None
```

Called for each update after a force press has started.

bolton_horizontal_drag_cancelclass-attributeinstance-attribute

```
on_horizontal_drag_cancel: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

The pointer that previously triggered on_horizontal_drag_down will not end up causing a horizontal drag.

bolton_horizontal_drag_downclass-attributeinstance-attribute

```
on_horizontal_drag_down: (  
    EventHandler[DragDownEvent[GestureDetector]] | None  
) = None
```

Called when a pointer has contacted the screen and might begin to move horizontally.

bolton_horizontal_drag_endclass-attributeinstance-attribute

```
on_horizontal_drag_end: (  
    EventHandler[DragEndEvent[GestureDetector]] | None  
) = None
```

Called when a pointer moving horizontally is no longer in contact and was moving at a specific velocity.

bolton_horizontal_drag_startclass-attributeinstance-attribute

```
on_horizontal_drag_start: (  
    EventHandler[DragStartEvent[GestureDetector]] | None  
) = None
```

Called when a pointer has contacted the screen with a primary button and has begun to move horizontally.

bolton_horizontal_drag_updateclass-attributeinstance-attribute

```
on_horizontal_drag_update: (  
    EventHandler[DragUpdateEvent[GestureDetector]] | None  
) = None
```

Called when a pointer that is in contact with the screen and moving horizontally has moved in the horizontal direction.

bolton_hoverclass-attributeinstance-attribute

```
on_hover: (  
    EventHandler[HoverEvent[GestureDetector]] | None  
) = None
```

Called when a mouse pointer has entered this control.

bolton_long_pressclass-attributeinstance-attribute

```
on_long_press: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

Called when a long press gesture with a primary button has been recognized.

bolton_long_press_cancelclass-attributeinstance-attribute

```
on_long_press_cancel: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

The pointer that previously triggered on_long_press_down will not end up causing a long-press.

bolton_long_press_downclass-attributeinstance-attribute

```
on_long_press_down: (  
    EventHandler[LongPressDownEvent[GestureDetector]] | None  
) = None
```

Called when a pointer that might cause a long press with a primary button has contacted the screen.

bolton_long_press_endclass-attributeinstance-attribute

```
on_long_press_end: (  
    EventHandler[LongPressEndEvent[GestureDetector]] | None  
) = None
```

Called when a pointer that has triggered a long-press with a primary button has stopped contacting the screen.

bolton_long_press_move_updateclass-attributeinstance-attribute

```
on_long_press_move_update: (  
    EventHandler[LongPressMoveUpdateEvent[GestureDetector]]  
    | None  
) = None
```

Called when, after a long press has been accepted, the pointer moves.

bolton_long_press_startclass-attributeinstance-attribute

```
on_long_press_start: (  
    EventHandler[LongPressStartEvent[GestureDetector]]  
    | None  
) = None
```

Triggered when a pointer has remained in contact with the screen at the same location for a long period of time.

bolton_long_press_upclass-attributeinstance-attribute

```
on_long_press_up: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

Called when a pointer that has triggered a long press with a primary button is no longer in contact with the screen.

bolton_multi_long_pressclass-attributeinstance-attribute

```
on_multi_long_press: (  
    EventHandler[LongPressEndEvent[GestureDetector]] | None  
) = None
```

Called when a long press gesture with multiple pointers has been recognized.

bolton_multi_tapclass-attributeinstance-attribute

```
on_multi_tap: (  
    EventHandler[TapEvent[GestureDetector]] | None  
) = None
```

Called when multiple pointers contacted the screen.

bolton_pan_cancelclass-attributeinstance-attribute

```
on_pan_cancel: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

The pointer that previously triggered on_pan_down will not end up causing a pan gesture.

bolton_pan_downclass-attributeinstance-attribute

```
on_pan_down: (
    EventHandler[DragDownEvent[GestureDetector]] | None
) = None
```

Called when a pointer has contacted the screen and might begin to move.

bolton_pan_endclass-attributeinstance-attribute

```
on_pan_end: (
    EventHandler[DragEndEvent[GestureDetector]] | None
) = None
```

Called when a pointer is no longer in contact and was moving at a specific velocity.

bolton_pan_startclass-attributeinstance-attribute

```
on_pan_start: (
    EventHandler[DragStartEvent[GestureDetector]] | None
) = None
```

Called when a pointer has contacted the screen and has begun to move.

bolton_pan_updateclass-attributeinstance-attribute

```
on_pan_update: (
    EventHandler[DragUpdateEvent[GestureDetector]] | None
) = None
```

Called when a pointer that is in contact with the screen and moving has moved again.

bolton_right_pan_endclass-attributeinstance-attribute

```
on_right_pan_end: (
    EventHandler[PointerEvent[GestureDetector]] | None
) = None
```

A pointer with secondary button pressed is no longer in contact and was moving at a specific velocity.

bolton_right_pan_startclass-attributeinstance-attribute

```
on_right_pan_start: (
    EventHandler[PointerEvent[GestureDetector]] | None
) = None
```

Pointer has contacted the screen while secondary button pressed and has begun to move.

bolton_right_pan_updateclass-attributeinstance-attribute

```
on_right_pan_update: (
    EventHandler[PointerEvent[GestureDetector]] | None
) = None
```

A pointer that is in contact with the screen, secondary button pressed and moving has moved again.

bolton_scale_endclass-attributeinstance-attribute

```
on_scale_end: (
    EventHandler[ScaleEndEvent[GestureDetector]] | None
) = None
```

TBD

bolton_scale_startclass-attributeinstance-attribute

```
on_scale_start: (
    EventHandler[ScaleStartEvent[GestureDetector]] | None
) = None
```

Called when the pointers in contact with the screen have established a focal point and initial scale of 1.0.

bolton_scale_updateclass-attributeinstance-attribute

```
on_scale_update: (
    EventHandler[ScaleUpdateEvent[GestureDetector]] | None
) = None
```

TBD

bolton_scrollclass-attributeinstance-attribute

```
on_scroll: (
    EventHandler[ScrollEvent[GestureDetector]] | None
) = None
```

TBD

bolton_secondary_long_pressclass-attributeinstance-attribute

```
on_secondary_long_press: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

Called when a long press gesture with a secondary button has been recognized.

bolton_secondary_long_press_cancelclass-attributeinstance-attribute

```
on_secondary_long_press_cancel: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

The pointer that previously triggered on_secondary_long_press_down not end up causing a long-press.

bolton_secondary_long_press_downclass-attributeinstance-attribute

```
on_secondary_long_press_down: (  
    EventHandler[LongPressDownEvent[GestureDetector]] | None  
) = None
```

Called when a pointer that might cause a long press with a secondary button has contacted the screen.

bolton_secondary_long_press_endclass-attributeinstance-attribute

```
on_secondary_long_press_end: (  
    EventHandler[LongPressEndEvent[GestureDetector]] | None  
) = None
```

Called when a pointer that has triggered a long-press with a secondary button has stopped contacting the screen.

bolton_secondary_long_press_move_updateclass-attributeinstance-attribute

```
on_secondary_long_press_move_update: (  
    EventHandler[LongPressMoveUpdateEvent[GestureDetector]]  
    | None  
) = None
```

Called when, after a secondary long press has been accepted, the pointer moves.

bolton_secondary_long_press_startclass-attributeinstance-attribute

```
on_secondary_long_press_start: (  
    EventHandler[LongPressStartEvent[GestureDetector]]  
    | None  
) = None
```

Triggered when a pointer has remained in contact with the screen at the same location for a long period of time.

bolton_secondary_long_press_upclass-attributeinstance-attribute

```
on_secondary_long_press_up: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

Called when a pointer that has triggered a long press with a secondary button is no longer in contact with the screen.

bolton_secondary_tapclass-attributeinstance-attribute

```
on_secondary_tap: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

A tap with a secondary button has occurred.

bolton_secondary_tap_cancelclass-attributeinstance-attribute

```
on_secondary_tap_cancel: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

The pointer that previously triggered on_secondary_tap_down will not end up causing a tap.

bolton_secondary_tap_downclass-attributeinstance-attribute

```
on_secondary_tap_down: (  
    EventHandler[TapEvent[GestureDetector]] | None  
) = None
```

Called when a pointer that might cause a tap with a secondary button has contacted the screen at a particular location.

bolton_secondary_tap_upclass-attributeinstance-attribute

```
on_secondary_tap_up: (  
    EventHandler[TapEvent[GestureDetector]] | None
```



```
) = None
```

Called when a pointer that will trigger a tap with a secondary button has stopped contacting the screen at a particular location.

bolton_tapclass-attributeinstance-attribute

```
on_tap: EventHandler[TapEvent[GestureDetector]] | None = (
    None
)
```

Called when a tap with a primary button has occurred.

bolton_tap_cancelclass-attributeinstance-attribute

```
on_tap_cancel: (
    ControlEventHandler[GestureDetector] | None
) = None
```

The pointer that previously triggered on_tap_down will not end up causing a tap.

bolton_tap_downclass-attributeinstance-attribute

```
on_tap_down: (
    EventHandler[TapEvent[GestureDetector]] | None
) = None
```

Called when a pointer that might cause a tap with a primary button has contacted the screen at a particular location.

bolton_tap_moveclass-attributeinstance-attribute

```
on_tap_move: (
    EventHandler[TapMoveEvent[GestureDetector]] | None
) = None
```

Called when a pointer that triggered a tap has moved.

bolton_tap_upclass-attributeinstance-attribute

```
on_tap_up: (
    EventHandler[TapEvent[GestureDetector]] | None
) = None
```

Called when a pointer that will trigger a tap with a primary button has stopped contacting the screen at a particular location.

bolton_tertiary_long_pressclass-attributeinstance-attribute

```
on_tertiary_long_press: (
    ControlEventHandler[GestureDetector] | None
) = None
```

Called when a long press gesture with a tertiary button has been recognized.

bolton_tertiary_long_press_cancelclass-attributeinstance-attribute

```
on_tertiary_long_press_cancel: (
    ControlEventHandler[GestureDetector] | None
) = None
```

The pointer that previously triggered on_tertiary_long_press_down will not end up causing a long-press.

bolton_tertiary_long_press_downclass-attributeinstance-attribute

```
on_tertiary_long_press_down: (
    EventHandler[LongPressDownEvent[GestureDetector]] | None
) = None
```

Called when a pointer that might cause a long press with a tertiary button has contacted the screen.

bolton_tertiary_long_press_endclass-attributeinstance-attribute

```
on_tertiary_long_press_end: (
    EventHandler[LongPressEndEvent[GestureDetector]] | None
) = None
```

Called when a pointer that has triggered a long-press with a tertiary button has stopped contacting the screen.

bolton_tertiary_long_press_move_updateclass-attributeinstance-attribute

```
on_tertiary_long_press_move_update: (
    EventHandler[LongPressMoveUpdateEvent[GestureDetector]]
    | None
) = None
```

Called when, after a tertiary long press has been accepted, the pointer moves.

bolton_tertiary_long_press_startclass-attributeinstance-attribute

```
on_tertiary_long_press_start: (  
    EventHandler[LongPressStartEvent[GestureDetector]]  
    | None  
) = None
```

Triggered when a pointer has remained in contact with the screen at the same location for a long period of time.

bolton_tertiary_long_press_upclass-attributeinstance-attribute

```
on_tertiary_long_press_up: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

Called when a pointer that has triggered a long press with a tertiary button is no longer in contact with the screen.

bolton_tertiary_tap_cancelclass-attributeinstance-attribute

```
on_tertiary_tap_cancel: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

The pointer that previously triggered on_tertiary_tap_down will not end up causing a tap.

bolton_tertiary_tap_downclass-attributeinstance-attribute

```
on_tertiary_tap_down: (  
    EventHandler[TapEvent[GestureDetector]] | None  
) = None
```

Called when a pointer that might cause a tap with a tertiary button has contacted the screen at a particular location.

bolton_tertiary_tap_upclass-attributeinstance-attribute

```
on_tertiary_tap_up: (  
    EventHandler[TapEvent[GestureDetector]] | None  
) = None
```

Called when a pointer that will trigger a tap with a tertiary button has stopped contacting the screen at a particular location.

bolton_vertical_drag_cancelclass-attributeinstance-attribute

```
on_vertical_drag_cancel: (  
    ControlEventHandler[GestureDetector] | None  
) = None
```

The pointer that previously triggered on_vertical_drag_down will not end up causing a vertical drag.

bolton_vertical_drag_downclass-attributeinstance-attribute

```
on_vertical_drag_down: (  
    EventHandler[DragDownEvent[GestureDetector]] | None  
) = None
```

Called when a pointer has contacted the screen and might begin to move vertically.

bolton_vertical_drag_endclass-attributeinstance-attribute

```
on_vertical_drag_end: (  
    EventHandler[DragEndEvent[GestureDetector]] | None  
) = None
```

Called when a pointer moving vertically is no longer in contact and was moving at a specific velocity.

bolton_vertical_drag_startclass-attributeinstance-attribute

```
on_vertical_drag_start: (  
    EventHandler[DragStartEvent[GestureDetector]] | None  
) = None
```

Called when a pointer has contacted the screen and has begun to move vertically.

bolton_vertical_drag_updateclass-attributeinstance-attribute

```
on_vertical_drag_update: (  
    EventHandler[DragUpdateEvent[GestureDetector]] | None  
) = None
```

A pointer moving vertically has moved in the vertical direction.

GridView

A scrollable, 2D array of controls.

It is very effective for large lists (thousands of items). Prefer it over wrapping Columns or Rows for smooth scrolling.

```
ft.GridView(  
    width=180,  
    runs_count=2,  
    spacing=8,  
    controls=[  
        ft.Container(  
            width=50, height=50, bgcolor=ft.Colors.PRIMARY, border_radius=8  
        ),  
        ft.Container(  
            width=50, height=50, bgcolor=ft.Colors.SECONDARY, border_radius=8  
        ),  
        ft.Container(  
            width=50, height=50, bgcolor=ft.Colors.TERTIARY, border_radius=8  
        ),  
        ft.Container(  
            width=50, height=50, bgcolor=ft.Colors.ERROR, border_radius=8  
        ),  
    ],  
)
```

Image: GridView - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/grid_view/image_for_docs.png

GridView

Inherits: LayoutControl, ScrollableControl, AdaptiveControl

Properties

- build_controls_on_demand - TBD
- cache_extent - Items that fall in the cache area (area before or after the visible area that are about to become visible when the user scrolls) are laid out even though they are not (yet) visible on screen.
- child_aspect_ratio - The ratio of the cross-axis to the main-axis extent of each child.
- clip_behavior - The content will be clipped (or not) according to this option.
- controls - A list of controls to display inside grid.
- horizontal - Whether to layout the grid items horizontally.
- max_extent - The maximum width or height of the grid item.
- padding - The amount of space by which to inset the children.
- reverse - Whether the scroll view scrolls in the reading direction.
- run_spacing - The number of logical pixels between each child along the cross axis.
- runs_count - The number of children in the cross axis.
- semantic_child_count - The number of children that will contribute semantic information.
- spacing - The number of logical pixels between each child along the main axis.

Examples

Photo gallery

play_arrowTry Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.title = "GridView Example"  
    page.theme_mode = ft.ThemeMode.DARK  
    page.padding = 50  
  
    page.add(  
        ft.SafeArea(  
            expand=True,  
            content=ft.GridView(  
                expand=True,
```

```

        runs_count=5,
        max_extent=150,
        child_aspect_ratio=1.0,
        spacing=5,
        run_spacing=5,
        controls=[
            ft.Image(
                src=f"https://picsum.photos/150/150?{i}",
                fit=ft.BoxFit.NONE,
                repeat=ft.ImageRepeat.NO_REPEAT,
                border_radius=ft.BorderRadius.all(10),
            )
            for i in range(0, 60)
        ],
    ),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: photo-gallery - https://flet.dev/docs/examples/controls/core/grid_view/media/photo_gallery.png

Properties

buildbuild_controls_on_demandclass-attributeinstance-attribute

build_controls_on_demand: bool = True

TBD

buildcache_extentclass-attributeinstance-attribute

cache_extent: Number | None = None

Items that fall in the cache area (area before or after the visible area that are about to become visible when the user scrolls) are laid out even though they are not (yet) visible on screen.

The cacheExtent describes how many pixels the cache area extends before the leading edge and after the trailing edge of the viewport.

The total extent, which the viewport will try to cover with controls, is cache_extent before the leading edge + extent of the main axis + cache_extent after the trailing edge.

buildchild_aspect_ratioclass-attributeinstance-attribute

child_aspect_ratio: Number = 1.0

The ratio of the cross-axis to the main-axis extent of each child.

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: ClipBehavior | None = None

The content will be clipped (or not) according to this option.

buildcontrolsclass-attributeinstance-attribute

controls: list[Control] = field(default_factory=list)

A list of controls to display inside grid.

buildhorizontalclass-attributeinstance-attribute

horizontal: bool = False

Whether to layout the grid items horizontally.

buildmax_extentclass-attributeinstance-attribute

max_extent: int | None = None

The maximum width or height of the grid item.

buildpaddingclass-attributeinstance-attribute

padding: PaddingValue | None = None

The amount of space by which to inset the children.

buildreverseclass-attributeinstance-attribute

reverse: bool = False

Whether the scroll view scrolls in the reading direction.

For example, if the reading direction is left-to-right and horizontal is True, then the scroll view scrolls from left to right when reverse is False and from right to left when reverse is True.

Similarly, if horizontal is False, then the scroll view scrolls from top to bottom when reverse is False and from bottom to top when reverse is True.

buildrun_spacingclass-attributeinstance-attribute

`run_spacing: Number = 10`

The number of logical pixels between each child along the cross axis.

buildruns_countclass-attributeinstance-attribute

`runs_count: int = 1`

The number of children in the cross axis.

buildsemantic_child_countclass-attributeinstance-attribute

`semantic_child_count: int | None = None`

The number of children that will contribute semantic information.

buildspacingclass-attributeinstance-attribute

`spacing: Number = 10`

The number of logical pixels between each child along the main axis.

Hero

Marks the content as a shared element for route transitions.

Place a Hero with the same tag in both source and destination views to animate that control between routes.

Inherits: `LayoutControl`

Properties

- `content` - The control to display and animate.
- `tag` - A unique identifier used to match source and destination Hero controls.
- `transition_on_user_gestures` - Whether to animate when the route is transitioned by a user gesture (for example, iOS back swipe).

Examples

Hero

[play_arrow](#) Try Online

```
import flet as ft

def main(page: ft.Page):
    hero_tag = "demo-hero-card"

    def build_card(size: int, label: str) -> ft.Container:
        icon_size = max(24, int(size * 0.28))
        return ft.Container(
            width=size,
            height=size,
            border_radius=ft.BorderRadius.all(20),
            gradient=ft.LinearGradient(
                begin=ft.Alignment.TOP_LEFT,
                end=ft.Alignment.BOTTOM_RIGHT,
                colors=[ft.Colors.BLUE_700, ft.Colors.CYAN_400],
            ),
            alignment=ft.Alignment.CENTER,
            content=ft.Stack(
                fit=ft.StackFit.EXPAND,
                controls=[
                    ft.Container(
                        alignment=ft.Alignment.CENTER,
                        content=ft.Icon(
                            ft.Icons.ROCKET_LAUNCH,
                            size=icon_size,
                            color=ft.Colors.WHITE,
                        ),
                    ),
                    ft.Container(
                        alignment=ft.Alignment.BOTTOM_CENTER,
                        padding=ft.Padding.only(left=8, right=8, bottom=10),
                        content=ft.Text(
                            label,
                            color=ft.Colors.WHITE,
                            weight=ft.FontWeight.BOLD,
                            size=12,
                            no_wrap=True,
                            max_lines=1,
                            overflow=ft.TextOverflow.ELLIPSIS,
                            text_align=ft.TextAlign.CENTER,
                        ),
                    ),
                ],
            ),
        )

    async def go_to_details(_):
        await page.push_route("/details")

    async def go_home(_):
        await page.push_route("/")

    def build_home_view() -> ft.View:
        return ft.View(
            route="/",
            appbar=ft.AppBar(title=ft.Text("Hero animation")),
```

```

        controls=[
            ft.SafeArea(
                expand=True,
                content=ft.Container(
                    expand=True,
                    alignment=ft.Alignment.CENTER,
                    content=ft.Column(
                        alignment=ft.MainAxisAlignment.CENTER,
                        horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                        spacing=20,
                        controls=[
                            ft.Text("Tap the card to navigate"),
                            ft.GestureDetector(
                                mouse_cursor=ft.MouseCursor.CLICK,
                                on_tap=go_to_details,
                                content=ft.Hero(
                                    tag=hero_tag,
                                    content=build_card(130, "Open details"),
                                ),
                            ),
                        ],
                    ),
                ),
            ],
        )

def build_details_view() -> ft.View:
    return ft.View(
        route="/details",
        appbar=ft.AppBar(title=ft.Text("Details")),
        controls=[
            ft.SafeArea(
                expand=True,
                content=ft.Container(
                    expand=True,
                    alignment=ft.Alignment.CENTER,
                    content=ft.Column(
                        alignment=ft.MainAxisAlignment.CENTER,
                        horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                        spacing=20,
                        controls=[
                            ft.Hero(
                                tag=hero_tag,
                                transition_on_user_gestures=True,
                                content=build_card(280, "Details"),
                            ),
                            ft.Button("Back", on_click=go_home),
                        ],
                    ),
                ),
            ],
        )

def route_change(e: ft.RouteChangeEvent = None):
    page.views.clear()
    page.views.append(build_home_view())
    if page.route == "/details":
        page.views.append(build_details_view())
    page.update()

async def view_pop(e: ft.ViewPopEvent):
    if e.view is not None:
        page.views.remove(e.view)
        await page.push_route(page.views[-1].route)

page.on_route_change = route_change
page.on_view_pop = view_pop
route_change()

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - <https://flet.dev/docs/test-images/examples/controls/core/golden/macros/hero/basic.gif>

Gallery

play_arrowTry Online

```

import flet as ft

PRODUCTS = [
    {
        "id": 1,
        "name": "Comet",

```

```

        "subtitle": "Fast launch profile",
        "description": "Balanced setup for quick iteration and lightweight payloads.",
        "icon": ft.Icons.ROCKET_LAUNCH,
        "start_color": ft.Colors.BLUE_700,
        "end_color": ft.Colors.CYAN_400,
    },
    {
        "id": 2,
        "name": "Nebula",
        "subtitle": "Visual pipeline",
        "description": "Optimized for rich UI transitions and animated data surfaces.",
        "icon": ft.Icons.AUTO_AWESOME,
        "start_color": ft.Colors.DEEP_PURPLE_700,
        "end_color": ft.Colors.PINK_400,
    },
    {
        "id": 3,
        "name": "Orbit",
        "subtitle": "Reliable baseline",
        "description": "Stable profile focused on consistency and performance.",
        "icon": ft.Icons.PUBLIC,
        "start_color": ft.Colors.GREEN_700,
        "end_color": ft.Colors.LIGHT_GREEN_400,
    },
    {
        "id": 4,
        "name": "Aurora",
        "subtitle": "Analytics focus",
        "description": "Configured for telemetry-heavy apps and live metrics.",
        "icon": ft.Icons.ANALYTICS,
        "start_color": ft.Colors.ORANGE_700,
        "end_color": ft.Colors.AMBER_400,
    },
    {
        "id": 5,
        "name": "Pulse",
        "subtitle": "Real-time profile",
        "description": "Tuned for frequent updates, messaging, and event streams.",
        "icon": ft.Icons.BOLT,
        "start_color": ft.Colors.RED_700,
        "end_color": ft.Colors.DEEP_ORANGE_400,
    },
]

```

```

def hero_tag(product_id: int) -> str:
    return f"product-hero-{product_id}"

```

```

def main(page: ft.Page):
    def product_by_id(product_id: int):
        for product in PRODUCTS:
            if product["id"] == product_id:
                return product
        return None

    def route_product_id(route: str):
        segments = [segment for segment in route.split("/") if segment]
        if len(segments) != 2 or segments[0] != "product":
            return None
        try:
            return int(segments[1])
        except ValueError:
            return None

    async def open_product(product_id: int):
        await page.push_route(f"/product/{product_id}")

    async def back_to_gallery(e: ft.Event[ft.Button]):
        await page.push_route("/")

    def build_hero_tile(product: dict, size: int) -> ft.Hero:
        icon_size = max(24, int(size * 0.4))
        return ft.Hero(
            tag=hero_tag(product["id"]),
            transition_on_user_gestures=True,
            content=ft.Container(
                width=size,
                height=size,
                border_radius=ft.BorderRadius.all(20),
                gradient=ft.LinearGradient(
                    begin=ft.Alignment.TOP_LEFT,
                    end=ft.Alignment.BOTTOM_RIGHT,
                    colors=[product["start_color"], product["end_color"]],
                ),
                alignment=ft.Alignment.CENTER,
                content=ft.Icon(product["icon"], color=ft.Colors.WHITE, size=icon_size),
            ),
        )

```



```

    ),
)

def build_home_view() -> ft.View:
    rows: list[ft.Control] = []
    for product in PRODUCTS:
        rows.append(
            ft.GestureDetector(
                mouse_cursor=ft.MouseCursor.CLICK,
                on_tap=lambda _, pid=product["id"]: page.run_task(
                    open_product, pid
                ),
                content=ft.Card(
                    content=ft.Container(
                        padding=12,
                        content=ft.Row(
                            vertical_alignment=ft.CrossAxisAlignment.CENTER,
                            spacing=12,
                            controls=[
                                build_hero_tile(product, 78),
                                ft.Column(
                                    spacing=4,
                                    controls=[
                                        ft.Text(
                                            product["name"],
                                            size=18,
                                            weight=ft.FontWeight.W_600,
                                        ),
                                        ft.Text(
                                            product["subtitle"],
                                            color=ft.Colors.ON_SURFACE_VARIANT,
                                        ),
                                        ft.Text(
                                            "Tap to view details",
                                            size=12,
                                            color=ft.Colors.PRIMARY,
                                        ),
                                    ],
                                ),
                            ],
                        ),
                    ),
                ),
            ),
        )
    return ft.View(
        route="/",
        appbar=ft.AppBar(title=ft.Text("Hero gallery")),
        controls=[
            ft.SafeArea(
                expand=True,
                content=ft.Column(
                    expand=True,
                    spacing=10,
                    scroll=ft.ScrollMode.AUTO,
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    controls=[
                        ft.Text(
                            "Each card has a unique Hero tag. "
                            "Open any item to see a matched transition.",
                            color=ft.Colors.ON_SURFACE_VARIANT,
                        ),
                        *rows,
                    ],
                ),
            ),
        ],
    )

def build_details_view(product: dict) -> ft.View:
    return ft.View(
        route=f"/product/{product['id']}",
        appbar=ft.AppBar(title=ft.Text(product["name"])),
        controls=[
            ft.SafeArea(
                expand=True,
                content=ft.Column(
                    expand=True,
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    spacing=16,
                    scroll=ft.ScrollMode.AUTO,
                    controls=[
                        ft.Container(height=8,
                                    build_hero_tile(product, 220),
                                    ft.Text(
                                        product["subtitle"],

```

```

        size=20,
        weight=ft.FontWeight.W_600,
        text_align=ft.TextAlign.CENTER,
    ),
    ft.Container(
        width=520,
        content=ft.Text(
            product["description"],
            text_align=ft.TextAlign.CENTER,
            color=ft.Colors.ON_SURFACE_VARIANT,
        ),
    ),
    ft.Button("Back to gallery", on_click=back_to_gallery),
],
),
),
],
)

def route_change(e: ft.RouteChangeEvent = None):
    page.views.clear()
    page.views.append(build_home_view())

    pid = route_product_id(page.route)
    if pid is not None:
        product = product_by_id(pid)
        if product is not None:
            page.views.append(build_details_view(product))

    page.update()

async def view_pop(e: ft.ViewPopEvent):
    if e.view is not None:
        page.views.remove(e.view)
        await page.push_route(page.views[-1].route

```

Properties

buildcontentinstance-attribute

content: Control

The control to display and animate.

Raises:

- ValueError - If it is not visible.

buildtaginstance-attribute

tag: str

A unique identifier used to match source and destination Hero controls.

Raises:

- ValueError - If it is not of type str.

buildtransition_on_user_gesturesclass-attributeinstance-attribute

transition_on_user_gestures: bool = False

Whether to animate when the route is transitioned by a user gesture (for example, iOS back swipe).

Icon

A control that displays an icon from a built-in or custom icon set.

Icons can be customized in color, size, and visual style using various parameters such as stroke weight, fill level, and shadows.

Example:

```
ft.Icon(ft.Icons.FAVORITE, color=ft.Colors.PRIMARY, size=40)
```

Image: Icon - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/icon/image_for_docs.png

Icon

Inherits: `LayoutControl`

Properties

- `apply_text_scaling` - Whether to scale the icon based on the system or user's preferred text size.
- `blend_mode` - The blend mode used when rendering the icon.
- `color` - The color to use when drawing the icon.
- `fill` - The fill amount of the icon, between 0.0 (outline) and 1.0 (solid).
- `grade` - A fine-tuning adjustment for the stroke thickness of the icon.
- `icon` - The icon to display, selected from a predefined icon set.
- `optical_size` - Adjusts the icon's visual style for different sizes to maintain clarity and balance.
- `semantics_label` - An accessibility label for the icon.
- `shadows` - A list of shadows to apply beneath the icon.
- `size` - The size (width and height) of the square area the icon will occupy.
- `weight` - The stroke weight (thickness) of the icon's lines.

Examples

To browse and visualize all available icons, visit our icons browser

Icon

`play_arrow` Try Online

```
from typing import cast

import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    # material
                    ft.Row(
                        controls=[
                            ft.Icon(ft.Icons.FAVORITE, color=ft.Colors.PINK),
                            ft.Icon(
                                ft.Icons.AUDIOTRACK,
                                color=ft.Colors.GREEN_400,
                                size=30,
                            ),
                            ft.Icon(
                                ft.Icons.BEACH_ACCESS,
                                color=ft.Colors.BLUE,
                                size=50,
                            ),
                            ft.Icon(ft.Icons.SETTINGS, color="#c1c1c1"),
                        ],
                    ),
                    # cupertino
                    ft.Row(
                        controls=[
                            ft.Icon(
```

Image: basic - <https://flet.dev/docs/examples/controls/core/icon/media/basic.png>

```
buildapply_text_scalingclass-attributeinstance-attribute
```

Whether to scale the icon based on the system or user's preferred text size.

buildblend_modeclass-attributeinstance-attribute

The blend mode used when rendering the icon.

buildcolorclass-attributeinstance-attribute

The color to use when drawing the icon.

The fill amount of the icon, between 0.0 (outline) and 1.0 (solid).

This feature requires the icon's font to support fill variation. It can be used to indicate state transitions or selection visually.

Raises:

- buildgradeclass-attributeinstance-attribute

A fine-tuning adjustment for the stroke thickness of the icon.

340

buildiconinstance-attribute

`icon: IconData`

The icon to display, selected from a predefined icon set.

You can explore available icons using the Flet Icons Browser.

buildoptical_sizeclass-attributeinstance-attribute

`optical_size: Number | None = None`

Adjusts the icon's visual style for different sizes to maintain clarity and balance.

This requires the icon font to support optical sizing.

Raises:

- `ValueError` - If it is not strictly greater than 0.0.

buildsemantics_labelclass-attributeinstance-attribute

`semantics_label: str | None = None`

An accessibility label for the icon.

This text is not displayed visually but may be announced by screen readers or other assistive technologies.

buildshadowsclass-attributeinstance-attribute

`shadows: BoxShadowValue | None = None`

A list of shadows to apply beneath the icon.

Use multiple shadows to simulate complex lighting effects. The order of shadows matters for how transparency is blended.

buildsizeclass-attributeinstance-attribute

`size: Number | None = None`

The size (width and height) of the square area the icon will occupy.

If not set, a default size will be used. When placing this icon inside other controls (such as buttons), those controls may also affect sizing.

buildweightclass-attributeinstance-attribute

`weight: Number | None = None`

The stroke weight (thickness) of the icon's lines.

This requires the icon font to support weight variation.

Raises:

- `ValueError` - If it is not strictly greater than 0.0.

IconButton

An icon button is a round button with an icon in the middle that reacts to touches by filling with color (ink).

Icon buttons are commonly used in the toolbars, but they can be used in many other places as well.

```
ft.IconButton(icon=ft.Icons.FAVORITE, icon_color=ft.Colors.PRIMARY)
```

Image: IconButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/icon_button/image_for_docs.png

IconButton

Inherits: LayoutControl, AdaptiveControl

Properties

- alignment - Defines how the icon is positioned within this button.
- autofocus - Whether this control will be provided initial focus.
- bgcolor - The background color of this button.
- disabled_color - The color to use for the icon inside the button when disabled.
- enable_feedback - Whether detected gestures should provide acoustic and/or haptic feedback.
- focus_color - The color of this button when in focus.
- highlight_color - The color of this button when it is pressed.
- hover_color - The color of this button when hovered.
- icon - An icon to be shown in this button.
- icon_color - The foreground color of the icon.
- icon_size - The icon's size in virtual pixels.
- mouse_cursor - The cursor to be displayed when a mouse pointer enters or is hovering over this control.
- padding - Defines the padding around this button.
- selected - The optional selection state of this button.
- selected_icon - The icon to be shown in this button for the 'selected' state.
- selected_icon_color - The icon color for the 'selected' state of this button.
- size_constraints - Size constraints for this button.
- splash_color - The primary color of this button when it is in the pressed state.
- splash_radius - The splash radius.
- style - Customizes this button's appearance.
- url - The URL to open when this button is clicked.
- visual_density - Defines how compact this button's layout will be.

Events

- on_blur - Called when this button has lost focus.
- on_click - Called when a user clicks this button.
- on_focus - Called when this button has received focus.
- on_hover - Called when the button is hovered.
- on_long_press - Called when the button is long-pressed.

Methods

- focus - Moves focus to this button.

Examples

Handling clicks

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "IconButton Example"

    def button_clicked(e: ft.Event[ft.IconButton]):
        button.data += 1
        message.value = f"Button clicked {button.data} time(s)"
        message.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    button := ft.IconButton(
                        key="handling_clicks_icon_button",
                        icon=ft.Icons.PLAY_CIRCLE_FILL_OUTLINED,
                        data=0,
                        on_click=button_clicked,
                    ),
                    message := ft.Text(),
                ],
            ),
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: handling-clicks - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/icon_button/handling_clicks.gif

Selected icon

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "IconButton Example"
    page.padding = 10
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.vertical_alignment = ft.MainAxisAlignment.CENTER

    def handle_click(e: ft.Event[ft.IconButton]):
        e.control.selected = not e.control.selected
        e.control.update()

    page.add(
        ft.SafeArea(
            content=ft.IconButton(
                key="selected_icon_button",
                icon=ft.Icons.BATTERY_1_BAR,
                selected_icon=ft.Icons.BATTERY_FULL,
                scale=5,
                on_click=handle_click,
                selected=False,
                style=ft.ButtonStyle(
                    color={
                        ft.ControlState.SELECTED: ft.Colors.GREEN,
                        ft.ControlState.DEFAULT: ft.Colors.RED,
                    }
                ),
            ),
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: selected-icon - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/icon_button/selected_icon.gif

Variants

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
```

```

page.title = "IconButton variants"
page.padding = 10
page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
page.vertical_alignment = ft.MainAxisAlignment.CENTER

def toggle_icon_button(e):
    e.control.selected = not e.control.selected

page.add(
    ft.SafeArea(
        content=ft.Row(
            alignment=ft.MainAxisAlignment.CENTER,
            spacing=50,
            controls=[
                # Normal buttons column (enabled only)
                ft.Column(
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    spacing=20,
                    controls=[
                        ft.Text(
                            "Normal",
                            theme_style=ft.TextThemeStyle.BODY_MEDIUM,
                        ),
                        ft.IconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                        ),
                        ft.FilledIconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                        ),
                        ft.FilledTonalIconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                        ),
                        ft.OutlinedIconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                        ),
                    ],
                ),
                # Disabled buttons column
                ft.Column(
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    spacing=20,
                    controls=[
                        ft.Text(
                            "Disabled",
                            theme_style=ft.TextThemeStyle.BODY_MEDIUM,
                        ),
                        ft.IconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                            disabled=True,
                        ),
                        ft.FilledIconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                            disabled=True,
                        ),
                        ft.FilledTonalIconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                            disabled=True,
                        ),
                        ft.OutlinedIconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                            disabled=True,
                        ),
                    ],
                ),
                # Toggle buttons column
                ft.Column(
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    spacing=20,
                    controls=[
                        ft.Text(
                            "Toggle",
                            theme_style=ft.TextThemeStyle.BODY_MEDIUM,
                        ),
                        ft.IconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                            selected=False,
                            on_click=toggle_icon_button,
                        ),
                        ft.FilledIconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                            selected=False,
                            on_click=toggle_icon_button,
                        ),
                        ft.FilledTonalIconButton(
                            icon=ft.Icons.FILTER_DRAMA,
                            selected=False,
                            on_click=toggle_icon_button,

```


Image: variants - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/icon_button/variants.png

The foreground color of the icon.

buildicon_sizeclass-attributeinstance-attribute

```
icon_size: Number | None = None
```

The icon's size in virtual pixels.

Defaults to 24.

buildmouse_cursorclass-attributeinstance-attribute

```
mouse_cursor: MouseCursor | None = field(
    default=None, metadata={"skip": True}
)
```

The cursor to be displayed when a mouse pointer enters or is hovering over this control.

buildpaddingclass-attributeinstance-attribute

```
padding: PaddingValue | None = None
```

Defines the padding around this button. The entire padded icon will react to input gestures.

Defaults to `Padding.all(8)`.

buildselectedclass-attributeinstance-attribute

```
selected: bool | None = None
```

The optional selection state of this button.

If this property is not set, the button will behave as a normal push button, otherwise, the button will toggle between showing icon (when `False`), and `selected_icon` (when `True`).

buildselected_iconclass-attributeinstance-attribute

```
selected_icon: IconDataOrControl | None = None
```

The icon to be shown in this button for the 'selected' state.

buildselected_icon_colorclass-attributeinstance-attribute

```
selected_icon_color: ColorValue | None = None
```

The icon color for the 'selected' state of this button.

buildsize_constraintsclass-attributeinstance-attribute

```
size_constraints: BoxConstraints | None = None
```

Size constraints for this button.

buildsplash_colorclass-attributeinstance-attribute

```
splash_color: ColorValue | None = None
```

The primary color of this button when it is in the pressed state.

buildsplash_radiusclass-attributeinstance-attribute

```
splash_radius: Number | None = None
```

The splash radius.

Note

This value is honoured only when in Material 2 (`Theme.use_material3` is `False`).

Raises:

- `ValueError` - If it is not strictly greater than 0.

buildstyleclass-attributeinstance-attribute

```
style: ButtonStyle | None = None
```

Customizes this button's appearance.

Note

- Only honoured in Material 3 design (`Theme.use_material3` is `True`).

- If `Theme.use_material3` is `True`, any parameters defined in `style` will be overridden by the corresponding parameters in this button. For example, if icon button `visual_density` is set to `VisualDensity.STANDARD` and `style's visual_density` is set to `VisualDensity.COMPACT`, `VisualDensity.STANDARD` will be used.

buildurlclass-attributeinstance-attribute

```
url: str | Url | None = None
```

The URL to open when this button is clicked.

Additionally, if `on_click` event callback is provided, it is fired after that.

buildvisual_densityclass-attributeinstance-attribute

```
visual_density: VisualDensity | None = field(
    default=None, metadata={"skip": True}
)
```

Defines how compact this button's layout will be.

Events

bolton_blurclass-attributeinstance-attribute

```
on_blur: ControlEventHandler[IconButton] | None = None
```

Called when this button has lost focus.

bolton_clickclass-attributeinstance-attribute

```
on_click: ControlEventHandler[IconButton] | None = None
```

Called when a user clicks this button.

bolton_focusclass-attributeinstance-attribute

```
on_focus: ControlEventHandler[IconButton] | None = None
```

Called when this button has received focus.

bolton_hoverclass-attributeinstance-attribute

```
on_hover: ControlEventHandler[IconButton] | None = None
```

Called when the button is hovered.

bolton_long_pressclass-attributeinstance-attribute

```
on_long_press: ControlEventHandler[IconButton] | None = None
```

Called when the button is long-pressed.

Methods

deployed_codefocusasync

```
focus()
```

Moves focus to this button.

Image

Displays an image.

The following popular formats are supported: JPEG, PNG, SVG, GIF, Animated GIF, WebP, Animated WebP, BMP, and WBMP.

Example:

```
ft.Image(  
    src="https://flet.dev/img/logo.svg",  
    width=100,  
    height=100,  
)
```

Image: Image - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/image/image_for_docs.png

Image

Inherits: `LayoutControl`

Properties

- `anti_alias` - Whether to paint the image with anti-aliasing.
- `border_radius` - Clips this image to have rounded corners.
- `cache_height` - The size at which this image should be decoded.
- `cache_width` - The size at which this image should be decoded.
- `color` - If set, this color is blended with each image pixel using `color_blend_mode`.
- `color_blend_mode` - Used to combine color with the image.
- `error_content` - Fallback control to display if the image cannot be loaded from the provided source.
- `exclude_from_semantics` - Whether to exclude this image from semantics.
- `fade_in_animation` - Fade-in animation of the `src` image as it appears after loading, replacing the `placeholder_src`.
- `filter_quality` - The rendering quality of the image.
- `fit` - Defines how to inscribe this image into the space allocated during layout.
- `gapless_playback` - Whether to continue showing the old image (True), or briefly show nothing (False), when the image provider changes.
- `placeholder_fade_out_animation` - Fade-out animation for the `placeholder_src`, after the `src` loads.
- `placeholder_fit` - Defines how to inscribe the placeholder into its space.
- `placeholder_src` - A placeholder displayed while the image is loading.
- `repeat` - How to paint any portions of the layout bounds not covered by this image.
- `semantics_label` - A semantic description of this image.
- `src` - The image source.

Examples

Image gallery

`play_arrow` Try Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.title = "Image Example"  
    page.theme_mode = ft.ThemeMode.LIGHT  
    page.padding = 50  
  
    gallery = ft.Row(expand=True, wrap=False, scroll=ft.ScrollMode.ALWAYS)  
  
    for i in range(0, 30):  
        gallery.controls.append(  
            ft.Image(  
                src="https://flet.dev/img/logo.svg",  
                width=100,  
                height=100,  
            )  
        )
```

Image: Image - <https://flet.dev/docs/examples/controls/core/image/media/gallery.gif>

```

page.add(
  ft.SafeArea(
    content=ft.Row(
      controls=[
        ft.Image(
          src=base64_src,
          width=100,
          height=100,
        ),
        ft.Image(
          src=bytes_src,
          width=100,
          height=100,
        ),
      ],
    ),
  ),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Displaying a static SVG image

play_arrowTry Online

```
import flet as ft
```

```

def main(page: ft.Page):
    svg_image = """
<svg version="1.1" id="Layer_1"
xmlns="http://www.w3.org/2000/svg"
xmlns:xlink="http://www.w3.org/1999/xlink"
x="0px" y="0px"
viewBox="0 0 30 30" style="enable-background:new 0 0 30 30;" xml:space="preserve">
<g>
<path d="M14.43339,18.88085c2.15923,0,3.91203-1.7476,3.91203-3.91226
c0-2.15805-1.7528-3.90566-3.91723-3.90566
c-0.74107,0-1.43221,0.21538-2.02332,0.57576c0.18598-0.05526,0.37562-0.09493,0.58084-0.09493
c1.09756,0,1.98365,0.88609,1.98365,1.98235c0,1.09248-0.88608,1.98376-1.98365,1.98376
c-1.09627,0-1.98235-0.88608-1.98235-1.98376c0-0.20499,0.03979-0.39486,0.09493-0.58096
c-0.36027,0.59135-0.57576,1.28236-0.57576,2.02344c10.52254,17.13325,12.27026,18.88085,14.43339,18.88085z"/>
<path d="M24.55235,1.00006H5.44812C2.99534,1.00006,
1,2.99563,1,5.44747v19.10459c0,2.45278,1.99533,4.44788,4.44812,4.44788
h19.10423c2.45243,0,4.44765-1.9951,4.44765-4.44788V5.44747C29,2.99563,27.00479,1.00006,24.55235,1.00006z
M27.32951,24.55206
c0,1.53175-1.246,2.77774-2.77715,2.77774H5.44812c-1.53151,0-2.77751-1.246-2.77751-2.77774v-8.83719h4.08065
c0.37644,3.90943,3.67563,6.97482,7.67705,6.97482c4.00709,0,7.3065-3.06539,7.68295-6.97482h5.21824V24.55206z
M8.36366,14.96859
c0-3.34831,2.7161-6.06418,6.06453-6.06418c3.35469,0,6.07079,2.71587,6.07079,6.06418c0,3.35492-2.7161,6.07079-6.07079,6.07079
C11.07977,21.03938,8.36366,18.32351,8.36366,14.96859z
M27.32951,13.75472h-5.28579
c-0.58568-3.67705-3.77116-6.5006-7.61541-6.5006c-3.83882,0-7.02383,2.82356-7.60951,6.5006H2.67061V5.44747
c0-1.53128,1.246-2.7768,2.77751-2.7768h19.10423c1.53116,0,2.77715,1.24552,2.77715,2.7768V13.75472z"/>
<path d="M24.06196,5.11637h-1.75185c-0.48236,0-0.87593,0.39345-0.87593,
0.87569v1.75186
c0,0.48225,0.39356,0.87616,0.87593,0.87616h1.75186c0.47776,0,0.87038-0.39392,0.87038-0.87616V5.99205
C24.93234,5.50982,24.53972,5.11637,24.06196,5.11637z"/>
</g>
</svg>"""

page.add(
  ft.SafeArea(
    content=ft.Column(
      controls=[
        ft.Image(
          src="https://raw.githubusercontent.com/dnfield/flutter_svg/master/packages/flutter_svg/example/assets/wik
          width=200,
          height=200,
        ),
        ft.Image(
          src=svg_image,
          width=100,
          height=100,
          color=ft.Colors.RED,
        ),
        ft.Image(
          src=svg_image,
          width=100,
          height=100,
          color=ft.Colors.BLUE,
        ),
        ft.Container(
          bgcolor=ft.Colors.BLACK_87,
          border_radius=5,

```

```

        content=ft.Image(
            src=svg_image,
            width=100,
            height=100,
            color=ft.Colors.WHITE,
        ),
    ],
),
)

if __name__ == "__main__":
    ft.run(main)

```

Displaying a dynamic SVG image

play_arrowTry Online

```

import asyncio

import flet as ft

async def main(page: ft.Page):
    svg_image = """
<svg width="400" height="400" xmlns="http://www.w3.org/2000/svg">
  <g>
    <ellipse ry="{}" rx="{}" id="svg_1" cy="200" cx="200" stroke="#000" fill="#fff"/>
  </g>
</svg>
"""

    img = ft.Image(src=svg_image.format(0, 0))
    page.add(ft.SafeArea(content=img))

    for _ in range(0, 10):
        for i in range(0, 10):
            img.src = svg_image.format(i * 10, i * 10)
            img.update()
            await asyncio.sleep(0.1)

if __name__ == "__main__":
    ft.run(main)

```

Displaying a Lucide icon

play_arrowTry Online

```

import flet as ft

"""
- Browse Lucide icons: https://lucide.dev/
- Copy SVG and use it as value for `Image.src`
"""

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Row(
                controls=[
                    ft.Image(
                        src='<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="pink" color=ft.Colors.PINK,
                    ),
                    ft.Image(
                        src='<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="green" color=ft.Colors.GREEN,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Gapless playback when changing image sources

This example updates both images to a new network URL on each click. With `gapless_playback` set to `True`, the previous frame remains visible while the next image loads. With `gapless_playback` set to `False`, the image area can briefly be empty, causing a flicker/blink effect.

play_arrowTry Online

```
import random

import flet as ft

def get_new_image_src() -> str:
    random_id = random.randint(1, 85)
    return f"https://picsum.photos/id/{random_id}/1280/720"

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    initial_src = get_new_image_src()
    image_with_gapless = ft.Image(
        src=initial_src,
        width=340,
        height=220,
        fit=ft.BoxFit.COVER,
        gapless_playback=True,
    )
    image_without_gapless = ft.Image(
        src=initial_src,
        width=340,
        height=220,
        fit=ft.BoxFit.COVER,
        gapless_playback=False,
    )

    def image_panel(title: str, image: ft.Image):
        return ft.Column(
            spacing=8,
            width=340,
            horizontal_alignment=ft.CrossAxisAlignment.CENTER,
            controls=[
                ft.Text(title, weight=ft.FontWeight.BOLD),
                ft.Container(
                    content=image,
                    bgcolor=ft.Colors.BLUE_GREY_900,
                    border_radius=ft.BorderRadius.all(12),
                    clip_behavior=ft.ClipBehavior.ANTI_ALIAS,
                ),
            ],
        )

    def next_photo(e: ft.Event[ft.Button]):
        new_src = get_new_image_src()
        image_with_gapless.src = new_src
        image_without_gapless.src = new_src
        image_with_gapless.update()
        image_without_gapless.update()

    page.appbar = ft.AppBar(title="Gapless Playback Showcase")
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        "Click 'Load next photo' to switch both images to a new URL.\n"
                        "The left/top image keeps showing the previous frame while "
                        "loading "
                        "the next one. This is referred to as gapless playback.",
                        text_align=ft.TextAlign.CENTER,
                    ),
                    ft.Row(
                        wrap=True,
                        spacing=20,
                        run_spacing=20,
                        alignment=ft.MainAxisAlignment.CENTER,
                        controls=[
                            image_panel("gapless_playback=True", image_with_gapless),
                            image_panel(
                                "gapless_playback=False",
                                image_without_gapless,
                            ),
                        ],
                    ),
                    ft.FilledButton("Load next photo", on_click=next_photo),
                ],
            ),
        ),
    )

if __name__ == "__main__":
```



```
ft.run(main)
```

Properties

buildanti_aliasclass-attributeinstance-attribute

```
anti_alias: bool = False
```

Whether to paint the image with anti-aliasing.

Anti-aliasing alleviates the sawtooth artifact when this image is rotated.

buildborder_radiusclass-attributeinstance-attribute

```
border_radius: BorderRadiusValue | None = None
```

Clips this image to have rounded corners.

buildcache_heightclass-attributeinstance-attribute

```
cache_height: int | None = None
```

The size at which this image should be decoded.

The image will, however, be rendered to the constraints of the layout regardless of this parameter.

buildcache_widthclass-attributeinstance-attribute

```
cache_width: int | None = None
```

The size at which this image should be decoded.

The image will, however, be rendered to the constraints of the layout regardless of this parameter.

buildcolorclass-attributeinstance-attribute

```
color: ColorValue | None = None
```

If set, this color is blended with each image pixel using `color_blend_mode`.

buildcolor_blend_modeclass-attributeinstance-attribute

```
color_blend_mode: BlendMode | None = None
```

Used to combine color with the image.

In terms of the blend mode, color is the source and this image is the destination.

builderror_contentclass-attributeinstance-attribute

```
error_content: Control | None = None
```

Fallback control to display if the image cannot be loaded from the provided source.

buildexclude_from_semanticsclass-attributeinstance-attribute

```
exclude_from_semantics: bool = False
```

Whether to exclude this image from semantics.

buildfade_in_animationclass-attributeinstance-attribute

```
fade_in_animation: Animation | None = None
```

Fade-in animation of the src image as it appears after loading, replacing the `placeholder_src`.

If `None`, defaults to `Animation(Duration(milliseconds=250), AnimationCurve.EASE_IN_OUT)`, if `placeholder_src` or `placeholder_fade_out_animation` is not `None`.

buildfilter_qualityclass-attributeinstance-attribute

```
filter_quality: FilterQuality = FilterQuality.MEDIUM
```

The rendering quality of the image.

buildfitclass-attributeinstance-attribute

```
fit: BoxFit | None = None
```

Defines how to inscribe this image into the space allocated during layout.

buildgapless_playbackclass-attributeinstance-attribute

```
gapless_playback: bool = False
```

Whether to continue showing the old image (True), or briefly show nothing (False), when the image provider changes.

Has no effect on svg images.

buildplaceholder_fade_out_animationclass-attributeinstance-attribute

`placeholder_fade_out_animation: Animation | None = None`

Fade-out animation for the placeholder_src, after the src loads.

If None, defaults to `Animation(Duration(milliseconds=150), AnimationCurve.EASE_OUT)`, if placeholder_src or fade_in_animation is not None.

buildplaceholder_fitclass-attributeinstance-attribute

`placeholder_fit: BoxFit | None = None`

Defines how to inscribe the placeholder into its space.

If None, default to fit.

buildplaceholder_srcclass-attributeinstance-attribute

`placeholder_src: str | bytes | None = None`

A placeholder displayed while the image is loading.

It can be one of the following:

- A URL or local asset file path;
- A base64 string;
- Raw bytes.

If None, no placeholder is shown while loading; the src simply appears (or fades in if placeholder_src or placeholder_fade_out_animation is not None).

Note

SVG sources are currently not supported as placeholders. If provided, this property will be ignored and the src will be displayed directly instead.

buildrepeatclass-attributeinstance-attribute

`repeat: ImageRepeat = ImageRepeat.NO_REPEAT`

How to paint any portions of the layout bounds not covered by this image.

buildsemantics_labelclass-attributeinstance-attribute

`semantics_label: str | None = None`

A semantic description of this image.

Used to provide a description of the image to TalkBack on Android, and VoiceOver on iOS.

buildsrcinstance-attribute

`src: str | bytes`

The image source.

It can be one of the following:

- A URL or local asset file path;
- A base64 string;
- Raw bytes.

InteractiveViewer

Allows you to pan, zoom, and rotate its content.

Inherits: `LayoutControl`

Properties

- `alignment` - The alignment of the content within this viewer.
- `boundary_margin` - A margin for the visible boundaries of the content.
- `clip_behavior` - Defines how to clip the content.
- `constrained` - Whether the normal size constraints at this point in the control tree are applied to the content.
- `content` - The `Control` to be transformed.
- `interaction_end_friction_coefficient` - Changes the deceleration behavior after a gesture.
- `interaction_update_interval` - The interval (in milliseconds) at which the `on_interaction_update` event is fired.
- `max_scale` - The maximum allowed scale.
- `min_scale` - The minimum allowed scale.
- `pan_enabled` - Whether panning is enabled.
- `scale_enabled` - Whether scaling is enabled.
- `scale_factor` - The amount of scale to be performed per pointer scroll.
- `trackpad_scroll_causes_scale` - Whether scrolling up/down on a trackpad should cause scaling instead of panning.

Events

- `on_interaction_end` - Called when the user ends a pan or scale gesture.
- `on_interaction_start` - Called when the user begins a pan or scale gesture.
- `on_interaction_update` - Called when the user updates a pan or scale gesture.

Methods

- `pan` - Translates the current transform matrix.
- `reset` - Resets the current transform matrix to identity.
- `restore_state` - Restores the transform matrix previously captured by `save_state`.
- `save_state` - Saves a snapshot of the current transform matrix.
- `zoom` - Applies multiplicative zoom to the current transform.

Examples

Handling events

[play_arrow](#) Try Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.InteractiveViewer(
                min_scale=0.1,
                max_scale=15,
                boundary_margin=ft.Margin.all(20),
                on_interaction_start=lambda e: print(e),
                on_interaction_end=lambda e: print(e),
                on_interaction_update=lambda e: print(e),
                content=ft.Image(src="https://picsum.photos/500/500"),
            ),
        ),
    )
```

```
if __name__ == "__main__":
    ft.run(main)
```

Programmatic transformations

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.vertical_alignment = ft.MainAxisAlignment.CENTER

    async def handle_zoom_in(e: ft.Event[ft.Button]):
        await i.zoom(1.2)

    async def handle_zoom_out(e: ft.Event[ft.Button]):
        await i.zoom(0.8)

    async def handle_pan(e: ft.Event[ft.Button]):
        await i.pan(dx=50, dy=50)

    async def handle_reset(e: ft.Event[ft.Button]):
        await i.reset()

    async def handle_reset_slow(e: ft.Event[ft.Button]):
        await i.reset(animation_duration=ft.Duration(seconds=2))

    async def handle_save_state(e: ft.Event[ft.Button]):
        await i.save_state()

    async def handle_restore_state(e: ft.Event[ft.Button]):
        await i.restore_state()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    i := ft.InteractiveViewer(
                        min_scale=0.1,
                        max_scale=5,
                        boundary_margin=ft.Margin.all(20),
                        content=ft.Image(src="https://picsum.photos/500/500"),
                    ),
                    ft.Row(
                        wrap=True,
                        controls=[
                            ft.Button("Zoom In", on_click=handle_zoom_in),
                            ft.Button("Zoom Out", on_click=handle_zoom_out),
                            ft.Button("Pan", on_click=handle_pan),
                            ft.Button("Save State", on_click=handle_save_state),
                            ft.Button("Restore State", on_click=handle_restore_state),
                            ft.Button("Reset (instant)", on_click=handle_reset),
                            ft.Button("Reset (slow)", on_click=handle_reset_slow),
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

buildalignmentclass-attributeinstance-attribute

alignment: Alignment | None = None

The alignment of the content within this viewer.

buildboundary_marginclass-attributeinstance-attribute

```
boundary_margin: MarginValue = field(
    default_factory=lambda: Margin.all(0)
)
```

A margin for the visible boundaries of the content.

Any transformation that results in the viewport being able to view outside of the boundaries will be stopped at the boundary. The boundaries do not rotate with the rest of the scene, so they are always aligned with the viewport.

To produce no boundaries at all, pass an infinite value.

Defaults to `Margin.all(0)`, which results in boundaries that are the exact same size and position as the content.

buildclip_behaviorclass-attributeinstance-attribute

```
clip_behavior: ClipBehavior = ClipBehavior.HARD_EDGE
```

Defines how to clip the content.

If set to `ClipBehavior.NONE`, the content can visually overflow the bounds of this `InteractiveViewer`, but gesture events (such as pan or zoom) will only be recognized within the viewer's area. Ensure this `InteractiveViewer` is sized appropriately when using `ClipBehavior.NONE`.

buildconstrainedclass-attributeinstance-attribute

```
constrained: bool = True
```

Whether the normal size constraints at this point in the control tree are applied to the content.

If set to `False`, then the content will be given infinite constraints. This is often useful when a content should be bigger than this `InteractiveViewer`.

For example, for a content which is bigger than the viewport but can be panned to reveal parts that were initially offscreen, `constrained` must be set to `False` to allow it to size itself properly. If `constrained` is `True` and the content can only size itself to the viewport, then areas initially outside of the viewport will not be able to receive user interaction events. If experiencing regions of the content that are not receptive to user gestures, make sure `constrained` is `False` and the content is sized properly.

buildcontentinstance-attribute

```
content: Control
```

The `Control` to be transformed.

Raises:

- `ValueError` - If it is not visible.

buildinteraction_end_friction_coefficientclass-attributeinstance-attribute

```
interaction_end_friction_coefficient: Number = 1.35e-05
```

Changes the deceleration behavior after a gesture.

Raises:

- `ValueError` - If it is not strictly greater than 0.

buildinteraction_update_intervalclass-attributeinstance-attribute

```
interaction_update_interval: int = 200
```

The interval (in milliseconds) at which the `on_interaction_update` event is fired.

buildmax_scaleclass-attributeinstance-attribute

```
max_scale: Number = 2.5
```

The maximum allowed scale.

Raises:

- `ValueError` - If it is not strictly greater than 0.
- `ValueError` - If it is not greater than or equal to `min_scale`.

buildmin_scaleclass-attributeinstance-attribute

```
min_scale: Number = 0.8
```

The minimum allowed scale.

The effective scale is limited by the value of `boundary_margin`. If scaling would cause the content to be displayed outside the defined boundary, it is prevented. By default, `boundary_margin` is set to `Margin.all(0)`, so scaling below 1.0 is typically not possible unless you increase the `boundary_margin` value.

Raises:

- `ValueError` - If it is not strictly greater than 0.

- `ValueError` - If it is not less than or equal to `max_scale`.

`buildpan_enabled`class-attributeinstance-attribute

```
pan_enabled: bool = True
```

Whether panning is enabled.

`buildscale_enabled`class-attributeinstance-attribute

```
scale_enabled: bool = True
```

Whether scaling is enabled.

`buildscale_factor`class-attributeinstance-attribute

```
scale_factor: Number = 200
```

The amount of scale to be performed per pointer scroll.

Increasing this value above the default causes scaling to feel slower, while decreasing it causes scaling to feel faster.

Note

Has effect only on pointer device scrolling, not pinch to zoom.

`buildtrackpad_scroll_causes_scale`class-attributeinstance-attribute

```
trackpad_scroll_causes_scale: bool = False
```

Whether scrolling up/down on a trackpad should cause scaling instead of panning.

Events

`bolton_interaction_end`class-attributeinstance-attribute

```
on_interaction_end: (
    EventHandler[ScaleEndEvent[InteractiveViewer]] | None
) = None
```

Called when the user ends a pan or scale gesture.

`bolton_interaction_start`class-attributeinstance-attribute

```
on_interaction_start: (
    EventHandler[ScaleStartEvent[InteractiveViewer]] | None
) = None
```

Called when the user begins a pan or scale gesture.

`bolton_interaction_update`class-attributeinstance-attribute

```
on_interaction_update: (
    EventHandler[ScaleUpdateEvent[InteractiveViewer]] | None
) = None
```

Called when the user updates a pan or scale gesture.

Methods

`deployed_codepanasync`

```
pan(dx: Number, dy: Number=0, dz: Number=0)
```

Translates the current transform matrix.

Parameters:

- `dx` (Number) - Horizontal translation delta in logical pixels.
- `dy` (Number, default: 0) - Vertical translation delta in logical pixels.
- `dz` (Number, default: 0) - Z-axis translation delta applied to the transform matrix.

Note

XY translation is clamped to the same interaction boundaries used for gesture-driven panning. `dz` is applied directly to the matrix and defaults to 0.

`deployed_coderesetasync`

```
reset(animation_duration: DurationValue | None = None)
```

Resets the current transform matrix to identity.

Parameters:

- `animation_duration` (`DurationValue` | `None`, default: `None`) - Optional animation duration for the reset transition. If `None`, the reset is applied immediately.

Notes

When `animation_duration` is provided, Flutter interpolates from the current transform to identity using a matrix tween.

`deployed_coderestore_stateasync`

```
restore_state()
```

Restores the transform matrix previously captured by `save_state`.

If no state has been saved yet, this method has no effect.

`deployed_codesave_stateasync`

```
save_state()
```

Saves a snapshot of the current transform matrix.

The saved state can later be restored using `restore_state`. Calling this method again overwrites the previously saved snapshot.

`deployed_codezoomasync`

```
zoom(factor: Number)
```

Applies multiplicative zoom to the current transform.

Parameters:

- `factor` (`Number`) - Scale multiplier relative to the current scale. Values greater than 1 zoom in, values between 0 and 1 zoom out.

Note

The resulting scale is clamped to `min_scale` and `max_scale`. Additional boundary clamping is applied so the visible viewport remains within `boundary_margin` limits.

KeyboardListener

A control that calls a callback whenever the user presses or releases a key on a keyboard.

Inherits: Control

Properties

- autofocus - True if this control will be selected as the initial focus when no other node in its scope is currently focused.
- content - The content control of the keyboard listener.
- include_semantics - Include semantics information in this control.

Events

- on_key_down - Fires when a keyboard key is pressed.
- on_key_repeat - Fires when a keyboard key is being hold, causing repeated events.
- on_key_up - Fires when a keyboard key is released.

Methods

- focus - Requests focus for this control.

Examples

Press any keys

play_arrowTry Online

```
import flet as ft

async def main(page: ft.Page):
    pressed_keys = set()

    def key_down(e: ft.KeyDownEvent):
        pressed_keys.add(e.key)
        keys.controls = [ft.Text(k, size=20) for k in pressed_keys]
        keys.update()

    def key_up(e: ft.KeyUpEvent):
        pressed_keys.remove(e.key)
        keys.controls = [ft.Text(k, size=20) for k in pressed_keys]
        keys.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text("Press any keys..."),
                    ft.KeyboardListener(
                        autofocus=True,
                        on_key_down=key_down,
                        on_key_up=key_up,
                        content=(keys := ft.Row()),
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

True if this control will be selected as the initial focus when no other node in its scope is currently focused.

buildcontentinstance-attribute

`content: Control`

The content control of the keyboard listener.

buildinclude_semanticsclass-attributeinstance-attribute

`include_semantics: bool = True`

Include semantics information in this control.

Events

bolton_key_downclass-attributeinstance-attribute

`on_key_down: EventHandler[KeyDownEvent] | None = None`

Fires when a keyboard key is pressed.

bolton_key_repeatclass-attributeinstance-attribute

`on_key_repeat: EventHandler[KeyRepeatEvent] | None = None`

Fires when a keyboard key is being hold, causing repeated events.

bolton_key_upclass-attributeinstance-attribute

`on_key_up: EventHandler[KeyUpEvent] | None = None`

Fires when a keyboard key is released.

Methods

deployed_codefocusasync

`focus()`

Requests focus for this control.

LayoutControl

Base class for visual controls that participate in page layout.

LayoutControl extends Control with common visual layout capabilities, including:

- explicit sizing (width, height, aspect_ratio);
- absolute positioning (left, top, right, bottom);
- parent-space placement (align, margin);
- 2D transforms (rotate, scale, offset, flip, transform);
- implicit animations for those properties (animate_*);
- layout/animation lifecycle events (on_size_change, on_animation_end).

Use LayoutControl as the base for custom visual controls rendered on the page surface. For popup controls, use DialogControl; for non-visual integrations, use Service.

Inherits: Control

Properties

- align - Alignment of the control within its parent.
- animate_align - Enables implicit animation of the align property.
- animate_margin - Enables implicit animation of the margin property.
- animate_offset - Enables implicit animation of the offset property.
- animate_opacity - Enables implicit animation of the opacity property.
- animate_position - Enables implicit animation of the positioning properties (left, right, top and bottom).
- animate_rotation - Enables implicit animation of the rotate property.
- animate_scale - Enables implicit animation of the scale property.
- animate_size - TBD
- aspect_ratio - The aspect ratio of the control.
- bottom - The distance that the child's bottom edge is inset from the bottom of the stack.
- flip - Flips this control horizontally and/or vertically.
- height - Imposed Control height in virtual pixels.
- left - The distance that the child's left edge is inset from the left of the stack.
- margin - Sets the margin of the control.
- offset - Applies a translation transformation before painting the control.
- right - The distance that the child's right edge is inset from the right of the stack.
- rotate - Transforms this control using a rotation around its center.
- scale - Scales this control along the 2D plane.
- size_change_interval - Sampling interval in milliseconds for on_size_change event.
- top - The distance that the child's top edge is inset from the top of the stack.
- transform - Applies a generic matrix transform to this control.
- width - Imposed Control width in virtual pixels.

Events

- on_animation_end - Called when animation completes.
- on_size_change - Called when the size of this control changes.

Examples

Flip

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.spacing = 20

    def toggle_x(e: ft.Event[ft.Button]):
        card.flip.flip_x = not card.flip.flip_x
        card.update()

    def toggle_y(e: ft.Event[ft.Button]):
        card.flip.flip_y = not card.flip.flip_y
        card.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                spacing=20,
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    card := ft.Container(
                        width=220,
                        height=120,
                        bgcolor=ft.Colors.BLUE_300,
                        border_radius=16,
                        alignment=ft.Alignment.CENTER,
                        content=ft.Text("Flip me", size=24, weight=ft.FontWeight.BOLD),
                        flip=ft.Flip(
                            flip_x=False,
                            flip_y=False,
                            filter_quality=ft.FilterQuality.MEDIUM,
                        ),
                    ),
                    ft.Row(
                        alignment=ft.MainAxisAlignment.CENTER,
                        controls=[
                            ft.Button("Toggle X", on_click=toggle_x),
                            ft.Button("Toggle Y", on_click=toggle_y),
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: Image - https://flet.dev/docs/test-images/controls/core/golden/macros/layout_control/flip.png

Rotate

play_arrowTry Online

```
from math import pi

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.add(
        ft.SafeArea(
            content=ft.Container(
                width=220,
                height=120,
                bgcolor=ft.Colors.BLUE_300,
                border_radius=16,
                alignment=ft.Alignment.CENTER,
                content=ft.Text("Rotate", size=28, weight=ft.FontWeight.BOLD),
                rotate=ft.Rotate(
                    angle=pi / 10,
                    alignment=ft.Alignment.CENTER,
                    filter_quality=ft.FilterQuality.MEDIUM,
                ),
            ),
        ),
    )
```

```
)
```

```
if __name__ == "__main__":  
    ft.run(main)
```

Image - https://flet.dev/docs/test-images/controls/core/golden/macros/layout_control/rotate.png

RotatedBox

play_arrowTry Online

```
import flet as ft
```

```
def _demo_control(content: ft.Control) -> ft.Container:  
    return ft.Container(  
        padding=10,  
        border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),  
        border_radius=8,  
        content=content,  
    )  
  
def _lane(title: str, controls: list[ft.Control]) -> ft.Container:  
    return ft.Container(  
        width=540,  
        padding=12,  
        border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),  
        border_radius=12,  
        content=ft.Column(  
            spacing=8,  
            controls=[  
                ft.Text(title, size=16, weight=ft.FontWeight.BOLD),  
                ft.Divider(height=1),  
                ft.Row(  
                    spacing=14,  
                    vertical_alignment=ft.CrossAxisAlignment.START,  
                    controls=controls,  
                ),  
            ],  
        ),  
    )  
  
def main(page: ft.Page):  
    page.padding = 24  
    page.scroll = ft.ScrollMode.AUTO  
    page.add(  
        ft.SafeArea(  
            content=ft.Column(  
                controls=[  
                    ft.Text(  
                        "RotatedBox rotates before layout. Compare occupied "  
                        "space below:",  
                        size=16,  
                        weight=ft.FontWeight.W_500,  
                    ),  
                    ft.Column(  
                        spacing=16,  
                        controls=[  
                            _lane(  
                                "Normal controls",  
                                [  
                                    _demo_control(ft.Text("Text", size=26)),  
                                    _demo_control(  
                                        ft.ProgressBar(  
                                            width=170, value=0.65, color=ft.Colors.GREEN  
                                        )  
                                    ),  
                                    _demo_control(ft.Button("Button")),  
                                ],  
                            ),  
                            _lane(  
                                "RotatedBox quarter_turns=1",  
                                [  
                                    _demo_control(  
                                        ft.RotatedBox(  
                                            quarter_turns=1,  
                                            content=ft.Text("Text", size=26),  
                                        )  
                                    ),  
                                    _demo_control(  
                                        ft.RotatedBox(  
                                            quarter_turns=1,  
                                            content=ft.ProgressBar(  
                                                width=170,  

```

```

        value=0.65,
        color=ft.Colors.GREEN,
    ),
),
_demo_control(
    ft.RotatedBox(
        quarter_turns=1,
        content=ft.Button("Button"),
    )
),
1,
),
1,
),
1
)
)
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/controls/core/golden/macros/rotated_box/rotated_box.png

Scale

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.add(
        ft.SafeArea(
            content=ft.Container(
                width=220,
                height=120,
                bgcolor=ft.Colors.GREEN_300,
                border_radius=16,
                alignment=ft.Alignment.CENTER,
                content=ft.Text("Scale", size=28, weight=ft.FontWeight.BOLD),
                scale=ft.Scale(
                    scale_x=1.18,
                    scale_y=0.82,
                    alignment=ft.Alignment.CENTER,
                    filter_quality=ft.FilterQuality.MEDIUM,
                ),
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/controls/core/golden/macros/layout_control/scale.png

Offset

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.add(
        ft.SafeArea(
            content=ft.Stack(
                width=460,
                height=260,
                controls=[
                    ft.Text(
                        "Offset translates by control size.",
                        left=12,
                        top=8,
                        size=16,
                        color=ft.Colors.ON_SURFACE_VARIANT,
                    ),
                    ft.Container(
                        left=30,
                        top=70,

```

```

        width=170,
        height=90,
        border_radius=16,
        bgcolor=ft.Colors.BLUE_100,
        border=ft.Border.all(2, ft.Colors.BLUE_GREY_400),
        alignment=ft.Alignment.CENTER,
        content=ft.Text(
            "Original",
            size=20,
            color=ft.Colors.BLUE_GREY_700,
        ),
    ),
    ft.Container(
        left=30,
        top=70,
        width=170,
        height=90,
        border_radius=16,
        bgcolor=ft.Colors.AMBER_300,
        alignment=ft.Alignment.CENTER,
        content=ft.Text("Offset", size=26, weight=ft.FontWeight.BOLD),
        offset=ft.Offset(
            x=1.05,
            y=0.55,
            filter_quality=ft.FilterQuality.MEDIUM,
        ),
    ),
    ft.Icon(
        ft.Icons.ARROW_RIGHT_ALT_ROUNDED,
        left=212,
        top=82,
        size=44,
        color=ft.Colors.BLUE_GREY_600,
    ),
    ft.Text(
        "offset = Offset(1.05, 0.55)",
        left=194,
        top=222,
        size=14,
        color=ft.Colors.ON_SURFACE_VARIANT,
    ),
),
],
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/controls/core/golden/macros/layout_control/offset.png

Matrix4 Transform

play_arrowTry Online

```

from math import pi

import flet as ft

def card(title: str, color: str, matrix: ft.Matrix4) -> ft.Container:
    return ft.Container(
        width=220,
        height=130,
        border_radius=18,
        bgcolor=color,
        padding=12,
        content=ft.Text(title, size=18, weight=ft.FontWeight.BOLD),
        transform=ft.Transform(
            matrix=matrix,
            alignment=ft.Alignment.CENTER,
            filter_quality=ft.FilterQuality.MEDIUM,
        ),
    )

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.scroll = ft.ScrollMode.AUTO
    page.spacing = 20

    perspective_tilt = (
        ft.Matrix4.identity()
        .set_entry(3, 2, 0.0018)
        .rotate_x(-0.35)
        .rotate_y(0.45)
    )

```

```

        .translate(0, -10, 0)
    )

    skew_and_rotate = ft.Matrix4.skew_y(0.28).rotate_z(-pi / 14)

    mirrored_spin = ft.Matrix4.diagonal3_values(-1, 1, 1).rotate_z(pi / 10)

    mix = ft.Matrix4.translation_values(24, -8, 0).multiply(
        ft.Matrix4.rotation_z(pi / 16).scale(0.9, 0.9)
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                spacing=20,
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    ft.Text("Matrix4 transform recording + replay", size=24),
                    ft.ResponsiveRow(
                        controls=[
                            ft.Container(
                                col={"sm": 6, "md": 3},
                                content=card(
                                    "Perspective tilt",
                                    ft.Colors.CYAN_300,
                                    perspective_tilt,
                                ),
                            ),
                            ft.Container(
                                col={"sm": 6, "md": 3},
                                content=card(
                                    "Skew + rotate",
                                    ft.Colors.AMBER_300,
                                    skew_and_rotate,
                                ),
                            ),
                            ft.Container(
                                col={"sm": 6, "md": 3},
                                content=card(
                                    "Mirror + spin",
                                    ft.Colors.PINK_200,
                                    mirrored_spin,
                                ),
                            ),
                            ft.Container(
                                col={"sm": 6, "md": 3},
                                content=card(
                                    "Multiply chain",
                                    ft.Colors.LIGHT_GREEN_300,
                                    mix,
                                ),
                            ),
                        ],
                    ),
                ],
            ),
        ),
    )
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/layout_control/matrix4_transform.png

Properties ☐ ☐

buildalignclass-attributeinstance-attribute

align: Alignment | None = None

Alignment of the control within its parent.

buildanimate_alignclass-attributeinstance-attribute

animate_align: AnimationValue | None = None

Enables implicit animation of the align property.

More information here.

buildanimate_marginclass-attributeinstance-attribute

animate_margin: AnimationValue | None = None

Enables implicit animation of the margin property.

More information here.

buildanimate_offsetclass-attributeinstance-attribute

`animate_offset: AnimationValue | None = None`

Enables implicit animation of the offset property.

More information here.

buildanimate_opacityclass-attributeinstance-attribute

`animate_opacity: AnimationValue | None = None`

Enables implicit animation of the opacity property.

More information here.

buildanimate_positionclass-attributeinstance-attribute

`animate_position: AnimationValue | None = None`

Enables implicit animation of the positioning properties (left, right, top and bottom).

More information here.

buildanimate_rotationclass-attributeinstance-attribute

`animate_rotation: AnimationValue | None = None`

Enables implicit animation of the rotate property.

More information here.

buildanimate_scaleclass-attributeinstance-attribute

`animate_scale: AnimationValue | None = None`

Enables implicit animation of the scale property.

More information here.

buildanimate_sizeclass-attributeinstance-attribute

`animate_size: AnimationValue | None = None`

TBD

buildaspect_raticlass-attributeinstance-attribute

`aspect_ratio: Number | None = None`

The aspect ratio of the control. It is defined as the ratio of width to height.

Note

In current implementation, if `aspect_ratio` is set, width and height on the same control are ignored for final rendered size.

buildbottomclass-attributeinstance-attribute

`bottom: Number | None = None`

The distance that the child's bottom edge is inset from the bottom of the stack.

Note

Effective only if this control is a descendant of one of the following: Stack control, `Page.overlay` list.

buildflipclass-attributeinstance-attribute

`flip: Flip | None = None`

Flips this control horizontally and/or vertically.

Set to an instance of `Flip` to mirror across x-axis, y-axis, or both.

buildheightclass-attributeinstance-attribute

`height: Number | None = None`

Imposed Control height in virtual pixels.

buildleftclass-attributeinstance-attribute

`left: Number | None = None`

The distance that the child's left edge is inset from the left of the stack.

Note

Effective only if this control is a descendant of one of the following: Stack control, Page.overlay list.

buildmarginclass-attributeinstance-attribute

margin: MarginValue | None = None

Sets the margin of the control.

buildoffsetclass-attributeinstance-attribute

offset: OffsetValue | None = None

Applies a translation transformation before painting the control.

The translation is expressed as an Offset scaled to the control's size. So, Offset(x=0.25, y=0), for example, will result in a horizontal translation of one quarter the width of this control.

Example The following example displays container at 0, 0 top left corner of a stack as transform applies -1 * 100, -1 * 100 (offset * control's size) horizontal and vertical translations to the control:

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.Stack(
            width=1000,
            height=1000,
            controls=[
                ft.Container(
                    bgcolor=ft.Colors.RED,
                    width=100,
                    height=100,
                    left=100,
                    top=100,
                    offset=ft.Offset(-1, -1),
                ),
            ],
        ),
    )

ft.run(main)
```

buildrightclass-attributeinstance-attribute

right: Number | None = None

The distance that the child's right edge is inset from the right of the stack.

Note

Effective only if this control is a descendant of one of the following: Stack control, Page.overlay list.

buildrotateclass-attributeinstance-attribute

rotate: RotateValue | None = None

Transforms this control using a rotation around its center.

The value of rotate property could be one of the following types:

- number - a rotation in clockwise radians. Full circle 360° is math.pi * 2 radians, 90° is pi / 2, 45° is pi / 4, etc.
- Rotate - allows to specify rotation angle as well as alignment - the location of rotation center.

Example

```
ft.Image(
    src="https://picsum.photos/100/100",
    width=100,
    height=100,
    border_radius=5,
    rotate=Rotate(angle=0.25 * pi, alignment=ft.Alignment.CENTER_LEFT)
)
```

buildscaleclass-attributeinstance-attribute

scale: ScaleValue | None = None

Scales this control along the 2D plane. Default scale factor is 1.0, meaning no-scale.

Setting this property to 0.5, for example, makes this control twice smaller, while 2.0 makes it twice larger.

Different scale multipliers can be specified for x and y axis, by setting `Control.scale` property to an instance of `Scale` class. Either `scale` or `scale_x` and `scale_y` could be specified, but not all of them.

Example

```
ft.Image(  
    src="https://picsum.photos/100/100",  
    width=100,  
    height=100,  
    border_radius=5,  
    scale=ft.Scale(scale_x=2, scale_y=0.5)  
)
```

`buildsize_change_interval`class-attributeinstance-attribute

`size_change_interval: int = 10`

Sampling interval in milliseconds for `on_size_change` event.

Setting to 0 calls `on_size_change` immediately on every change.

`buildtop`class-attributeinstance-attribute

`top: Number | None = None`

The distance that the child's top edge is inset from the top of the stack.

Note

Effective only if this control is a descendant of one of the following: `Stack` control, `Page.overlay` list.

`buildtransform`class-attributeinstance-attribute

`transform: Transform | None = None`

Applies a generic matrix transform to this control.

Set to an instance of `Transform` with a recorded `Matrix4` to describe arbitrary transform sequences.

`buildwidth`class-attributeinstance-attribute

`width: Number | None = None`

Imposed `Control` width in virtual pixels.

Events

`bolton_animation_end`class-attributeinstance-attribute

```
on_animation_end: (  
    ControlEventHandler[LayoutControl] | None  
) = None
```

Called when animation completes.

Can be used to chain multiple animations.

The data property of the event handler argument contains the name of the animation.

More information [here](#).

`bolton_size_change`class-attributeinstance-attribute

```
on_size_change: (  
    EventHandler[LayoutSizeChangeEvent[LayoutControl]]  
    | None  
) = None
```

Called when the size of this control changes.

`size_change_interval` defines how often this event is called.

ListTile

A single fixed-height row that typically contains some text as well as a leading or trailing icon.

Example:

```
ft.ListTile(  
    width=400,  
    leading=ft.Icon(ft.Icons.ACCOUNT_CIRCLE),  
    title="Jane Doe",  
    subtitle="Product Manager",  
    trailing=ft.Icon(ft.Icons.CHEVRON_RIGHT),  
    bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,  
)
```

Image: ListTile - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/list_tile/image_for_docs.png

ListTile

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `autofocus` - True if the control will be selected as the initial focus.
- `bgcolor` - The list tile's background color.
- `content_padding` - The tile's internal padding.
- `dense` - Whether this list tile is part of a vertically dense list.
- `enable_feedback` - Whether detected gestures should provide acoustic and/or haptic feedback.
- `horizontal_spacing` - The horizontal gap between the title and the leading and trailing controls.
- `hover_color` - The tile's color when hovered.
- `icon_color` - Defines the default color for the icons present in leading and trailing.
- `is_three_line` - Whether this list tile is intended to display three lines of text.
- `leading` - A control to display before the title.
- `leading_and_trailing_text_style` - The `TextStyle` for the leading and trailing controls.
- `min_height` - The minimum height allocated for this control.
- `min_leading_width` - The minimum width allocated for the leading control.
- `min_vertical_padding` - The minimum padding on the top and bottom of the title and subtitle controls.
- `mouse_cursor` - The cursor to be displayed when a mouse pointer enters or is hovering over this control.
- `selected` - If this tile is also enabled then icons and text are rendered with the same color.
- `selected_color` - Defines the color used for icons and text when `selected=True`.
- `selected_tile_color` - Defines the background color of `ListTile` when `selected=True`.
- `shape` - The tile's shape.
- `splash_color` - The list tile's splash color after the control has been tapped.
- `style` - Defines the font used for the title.
- `subtitle` - Additional content displayed below the title.
- `subtitle_text_style` - The `TextStyle` for the subtitle control.
- `text_color` - The color used for texts in title, subtitle, leading, and trailing.
- `title` - A control to display as primary content of the list tile.
- `title_alignment` - Defines how leading and trailing are vertically aligned relative to the titles (title and subtitle).
- `title_text_style` - The `TextStyle` for the title control.
- `toggle_inputs` - Whether clicking on a list tile should toggle the state of `Radio`, `Checkbox` or `Switch` inside this tile.

- trailing - A control to display after the title.
- url - The URL to open when this button is clicked.
- visual_density - Defines how compact the control's layout will be.

Events

- on_click - Called when a user clicks or taps the list tile.
- on_long_press - Called when the user long-presses on this list tile.

Examples

ListTile

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "ListTile Example"

    page.add(
        ft.SafeArea(
            content=ft.Card(
                content=ft.Container(
                    width=500,
                    padding=ft.Padding.symmetric(vertical=10),
                    content=ft.Column(
                        spacing=0,
                        controls=[
                            ft.ListTile(title=ft.Text("One-line list tile")),
                            ft.ListTile(
                                title=ft.Text("One-line dense list tile"),
                                dense=True,
                            ),
                            ft.ListTile(
                                leading=ft.Icon(ft.Icons.SETTINGS),
                                title=ft.Text("One-line selected list tile"),
                                selected=True,
                            ),
                            ft.ListTile(
                                leading=ft.Image(
                                    src="assets/icon-192.png",
                                    fit=ft.BoxFit.CONTAIN,
                                ),
                                title=ft.Text("One-line with leading control"),
                            ),
                            ft.ListTile(
                                title=ft.Text("One-line with trailing control"),
                                trailing=ft.PopupMenuButton(
                                    icon=ft.Icons.MORE_VERT,
                                    items=[
                                        ft.PopupMenuItem(content="Item 1"),
                                        ft.PopupMenuItem(content="Item 2"),
                                    ],
                                ),
                            ),
                            ft.ListTile(
                                leading=ft.Icon(ft.Icons.ALBUM),
                                title=ft.Text(
                                    value=(
                                        "One-line with leading and trailing controls"
                                    )
                                ),
                                trailing=ft.PopupMenuButton(
                                    icon=ft.Icons.MORE_VERT,
                                    items=[
                                        ft.PopupMenuItem(content="Item 1"),
                                        ft.PopupMenuItem(content="Item 2"),
                                    ],
                                ),
                            ),
                            ft.ListTile(
                                leading=ft.Icon(ft.Icons.SNOOZE),
                                title=ft.Text(
                                    value=(
                                        "Two-line with leading and trailing controls"
                                    )
                                ),
                                subtitle=ft.Text("Here is a second title."),
                                trailing=ft.PopupMenuButton(
                                    icon=ft.Icons.MORE_VERT,
```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/list_tile/basic.png

When using a Text control for title and subtitle, you can enforce line limits using `Text.max_lines`.

buildleadingclass-attributeinstance-attribute

`leading: IconDataOrControl | None = None`

A control to display before the title.

buildleading_and_trailing_text_styleclass-attributeinstance-attribute

`leading_and_trailing_text_style: TextStyle | None = None`

The `TextStyle` for the leading and trailing controls.

buildmin_heightclass-attributeinstance-attribute

`min_height: Number | None = None`

The minimum height allocated for this control.

If `None` or not set, default tile heights are 56.0, 72.0, and 88.0 for one, two, and three lines of text respectively. If `dense` is `True`, these defaults are changed to 48.0, 64.0, and 76.0.

Note that, a visual density value or a large title will also adjust the default tile heights.

buildmin_leading_widthclass-attributeinstance-attribute

`min_leading_width: Number | None = None`

The minimum width allocated for the leading control.

buildmin_vertical_paddingclass-attributeinstance-attribute

`min_vertical_padding: Number | None = None`

The minimum padding on the top and bottom of the title and subtitle controls.

buildmouse_cursorclass-attributeinstance-attribute

`mouse_cursor: MouseCursor | None = None`

The cursor to be displayed when a mouse pointer enters or is hovering over this control. The value is `MouseCursor` enum.

buildselectedclass-attributeinstance-attribute

`selected: bool = False`

If this tile is also enabled then icons and text are rendered with the same color. By default the selected color is the theme's primary color.

buildselected_colorclass-attributeinstance-attribute

`selected_color: ColorValue | None = None`

Defines the color used for icons and text when `selected=True`.

buildselected_tile_colorclass-attributeinstance-attribute

`selected_tile_color: ColorValue | None = None`

Defines the background color of `ListTile` when `selected=True`.

buildshapeclass-attributeinstance-attribute

`shape: OutlinedBorder | None = None`

The tile's shape.

buildsplash_colorclass-attributeinstance-attribute

`splash_color: ColorValue | None = None`

The list tile's splash color after the control has been tapped.

buildstyleclass-attributeinstance-attribute

`style: ListTileStyle | None = None`

Defines the font used for the title.

Defaults to `ListTileStyle.LIST`.

buildsubtitleclass-attributeinstance-attribute

`subtitle: StrOrControl | None = None`

Additional content displayed below the title.

If `is_three_line` is `False`, this should not wrap. If `is_three_line` is `True`, this should be configured to take a maximum of two lines. For example, you can use `Text.max_lines` to enforce the number of lines.

Typically a `Text` control.

buildsubtitle_text_styleclass-attributeinstance-attribute

`subtitle_text_style: TextStyle | None = None`

The `TextStyle` for the subtitle control.

buildtext_colorclass-attributeinstance-attribute

`text_color: ColorValue | None = None`

The color used for texts in title, subtitle, leading, and trailing.

buildtitleclass-attributeinstance-attribute

`title: StrOrControl | None = None`

A control to display as primary content of the list tile.

Typically a `Text` control. This should not wrap. To enforce the single line limit, use `Text.max_lines`.

buildtitle_alignmentclass-attributeinstance-attribute

`title_alignment: ListTileTitleAlignment | None = None`

Defines how leading and trailing are vertically aligned relative to the titles (title and subtitle).

Defaults to `ListTileTitleAlignment.THREE_LINE` in Material 3 or `ListTileTitleAlignment.TITLE_HEIGHT` in Material 2.

buildtitle_text_styleclass-attributeinstance-attribute

`title_text_style: TextStyle | None = None`

The `TextStyle` for the title control.

buildtoggle_inputsclass-attributeinstance-attribute

`toggle_inputs: bool = False`

Whether clicking on a list tile should toggle the state of `Radio`, `Checkbox` or `Switch` inside this tile.

buildtrailingclass-attributeinstance-attribute

`trailing: IconDataOrControl | None = None`

A control to display after the title.

Typically an `Icon` control.

buildurlclass-attributeinstance-attribute

`url: str | Url | None = None`

The URL to open when this button is clicked.

Additionally, if `on_click` event callback is provided, it is fired after that.

buildvisual_densityclass-attributeinstance-attribute

`visual_density: VisualDensity | None = None`

Defines how compact the control's layout will be.

Events

bolton_clickclass-attributeinstance-attribute

`on_click: ControlEventHandler[ListTile] | None = None`

Called when a user clicks or taps the list tile.

bolton_long_pressclass-attributeinstance-attribute

`on_long_press: ControlEventHandler[ListTile] | None = None`

Called when the user long-presses on this list tile.

ListView

A scrollable list of controls arranged linearly.

ListView is the most commonly used scrolling control. It displays its children one after another in the scroll direction. In the cross axis, the children are required to fill the ListView.

```
ft.ListView(  
    controls=[ft.Text(f"Item {i}") for i in range(1, 6)],  
)
```

Image: ListView - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/list_view/image_for_docs.png

ListView

Inherits: LayoutControl, ScrollableControl, AdaptiveControl

Properties

- `build_controls_on_demand` - Whether the controls should be built lazily/on-demand, i.e.
- `cache_extent` - The viewport has an area before and after the visible area to cache items that are about to become visible when the user scrolls.
- `clip_behavior` - Defines how to clip the controls.
- `controls` - A list of Controls to display inside ListView.
- `divider_thickness` - If greater than 0 then Divider is used as a spacing between list view items.
- `first_item_prototype` - Whether the dimensions of the first item should be used as a "prototype" for all other items.
- `horizontal` - Whether the controls should be laid out horizontally.
- `item_extent` - A fixed height or width (when horizontal is True) of an item to optimize rendering.
- `padding` - The amount of space by which to inset the controls.
- `prototype_item` - A control to be used as a "prototype" for all items, i.e.
- `reverse` - Whether the scroll view scrolls in the reading direction.
- `semantic_child_count` - The number of children that will contribute semantic information.
- `spacing` - The height of divider between the controls.

Examples

Auto-scrolling and dynamical items addition

`play_arrow` Try Online

```
import asyncio  
  
import flet as ft  
  
async def main(page: ft.Page):  
    def handle_switch_change(e: ft.Event[ft.Switch]):  
        lv.auto_scroll = not lv.auto_scroll  
        lv.update()  
  
    lv = ft.ListView(  
        spacing=10,  
        padding=20,  
        width=150,  
        auto_scroll=True,  
        controls=[  
            ft.Text(f"Line {i}", color=ft.Colors.ON_SECONDARY) for i in range(0, 60)  
        ],  
    )  
  
    page.add(  
        ft.SafeArea(  
            expand=True,  
            content=ft.Row(  
                expand=True,  
                vertical_alignment=ft.CrossAxisAlignment.START,  
                controls=[
```



```

        ft.Container(
            bgcolor=ft.Colors.GREY_500,
            content=lv,
        ),
        ft.Switch(
            thumb_icon=ft.Icons.LIST_OUTLINED,
            value=True,
            label="Auto-scroll",
            label_position=ft.LabelPosition.RIGHT,
            on_change=handle_switch_change,
        ),
    ],
),
)

# Add a new item to the ListView every second.
for i in range(len(lv.controls), 120):
    await asyncio.sleep(1)
    lv.controls.append(ft.Text(f"Line {i}", color=ft.Colors.ON_SECONDARY))
    lv.update()

if __name__ == "__main__":
    ft.run(main)

```

Image: autoscroll-and-dynamic-items - https://flet.dev/docs/examples/controls/core/list_view/media/autoscroll_and_dynamic_items.gif

Properties

buildcontrols_on_demandclass-attributeinstance-attribute

buildcontrols_on_demand: bool = True

Whether the controls should be built lazily/on-demand, i.e. only when they are about to become visible.

This is particularly useful when dealing with a large number of controls.

buildcache_extentclass-attributeinstance-attribute

cache_extent: Number | None = None

The viewport has an area before and after the visible area to cache items that are about to become visible when the user scrolls.

Items that fall in this cache area are laid out even though they are not (yet) visible on screen. The cache_extent describes how many pixels the cache area extends before the leading edge and after the trailing edge of the viewport.

The total extent, which the viewport will try to cover with children, is cache_extent before the leading edge + extent of the main axis + cache_extent after the trailing edge.

The cache area is also used to implement implicit accessibility scrolling on iOS: When the accessibility focus moves from an item in the visible viewport to an invisible item in the cache area, the framework will bring that item into view with an (implicit) scroll action.

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: ClipBehavior = ClipBehavior.HARD_EDGE

Defines how to clip the controls.

buildcontrolsclass-attributeinstance-attribute

controls: list[Control] = field(default_factory=list)

A list of Controls to display inside ListView.

bulddivider_thicknessclass-attributeinstance-attribute

divider_thickness: Number = 0

If greater than 0 then Divider is used as a spacing between list view items.

buildfirst_item_prototypeclass-attributeinstance-attribute

first_item_prototype: bool = False

Whether the dimensions of the first item should be used as a "prototype" for all other items.

If True, their height or width will be the same as the first item.

buildhorizontalclass-attributeinstance-attribute

```
horizontal: bool = False
```

Whether the controls should be laid out horizontally.

builditem_extentclass-attributeinstance-attribute

```
item_extent: Number | None = None
```

A fixed height or width (when horizontal is True) of an item to optimize rendering.

Note

This property has effect only when build_controls_on_demand is True or spacing is 0.

buildpaddingclass-attributeinstance-attribute

```
padding: PaddingValue | None = None
```

The amount of space by which to inset the controls.

buildprototype_itemclass-attributeinstance-attribute

```
prototype_item: Control | None = None
```

A control to be used as a "prototype" for all items, i.e. their height or width will be the same as the prototype_item.

Note

This property has effect only when build_controls_on_demand is True or spacing is 0.

buildreverseclass-attributeinstance-attribute

```
reverse: bool = False
```

Whether the scroll view scrolls in the reading direction.

For example, if the reading direction is left-to-right and horizontal is True, then the scroll view scrolls from left to right when reverse is False and from right to left when reverse is True.

Similarly, if horizontal is False, then the scroll view scrolls from top to bottom when reverse is False and from bottom to top when reverse is True.

buildsemantic_child_countclass-attributeinstance-attribute

```
semantic_child_count: int | None = None
```

The number of children that will contribute semantic information.

buildspacingclass-attributeinstance-attribute

```
spacing: Number = 0
```

The height of divider between the controls.

Lottie

Render rich Lottie animations inside your Flet apps with a simple control.

It is backed by the lottie Flutter package.

Platform Support

| Platform | Windows | macOS | Linux |
|-----------|--------------------------|--------------------------|--------------------------|
| Supported | ☐ | ☐ | ☐ |

Usage

Add the flet-lottie package to your project dependencies:

- uv
 - pip
- ```
uv add flet-lottie
pip install flet-lottie
```

## Example

### Lottie animations

play\_arrowTry Online

```
import flet as ft
import flet_lottie as ftl

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ftl.Lottie(
 src="https://raw.githubusercontent.com/xvrh/lottie-flutter/master/example/assets/Mobilo/A.json",
 reverse=False,
 animate=True,
 error_content=ft.Placeholder(ft.Text("Error loading Lottie")),
 on_error=lambda e: print(f"Error loading Lottie: {e.data}"),
),
 ftl.Lottie(
 src="sample.json",
 reverse=False,
 animate=True,
 enable_merge_paths=True,
 enable_layers_opacity=True,
 error_content=ft.Placeholder(ft.Text("Error loading Lottie")),
 on_error=lambda e: print(f"Error loading Lottie: {e.data}"),
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Description

Displays lottie animations.

Note

- Layer effects are currently not supported. See [airbnb/lottie-android#1964](#) and [xvrh/lottie-flutter#189](#) for details.

Inherits: LayoutControl

## Properties

- animate - Whether the animation should be played automatically.

- `background_loading` - Whether the animation should be loaded in the background.
- `enable_layers_opacity` - Whether to enable layer-level opacity.
- `enable_merge_paths` - Whether to enable merge path support.
- `error_content` - A control to display when an error occurs while loading the Lottie animation.
- `filter_quality` - The quality of the image layer.
- `fit` - Defines how to inscribe the Lottie composition into the space allocated during layout.
- `headers` - Headers for network requests.
- `repeat` - Whether the animation should repeat in a loop.
- `reverse` - Whether the animation should be played in reverse (from start to end and then continuously from end to start).
- `src` - The lottie animation source.

## Events

- `on_error` - Fires when an error occurs while loading the Lottie animation.

## Properties

### `buildanimateclass-attributeinstance-attribute`

`animate: bool = True`

Whether the animation should be played automatically.

### `buildbackground_loadingclass-attributeinstance-attribute`

`background_loading: bool | None = None`

Whether the animation should be loaded in the background.

### `buildenable_layers_opacityclass-attributeinstance-attribute`

`enable_layers_opacity: bool = False`

Whether to enable layer-level opacity.

### `buildenable_merge_pathsclass-attributeinstance-attribute`

`enable_merge_paths: bool = False`

Whether to enable merge path support.

### `builderror_contentclass-attributeinstance-attribute`

`error_content: Control | None = None`

A control to display when an error occurs while loading the Lottie animation.

For more information on the error, see `on_error`.

### `buildfilter_qualityclass-attributeinstance-attribute`

`filter_quality: FilterQuality = FilterQuality.LOW`

The quality of the image layer.

### `buildfitclass-attributeinstance-attribute`

`fit: BoxFit | None = None`

Defines how to inscribe the Lottie composition into the space allocated during layout.

### `buildheadersclass-attributeinstance-attribute`

`headers: dict[str, str] | None = None`

Headers for network requests.

### `buildrepeatclass-attributeinstance-attribute`

`repeat: bool = True`

Whether the animation should repeat in a loop.

Note

Has no effect if animate is False.

### **buildreverseclass-attributeinstance-attribute**

`reverse: bool = False`

Whether the animation should be played in reverse (from start to end and then continuously from end to start).

Note

Has no effect if animate or repeat is False.

### **buildsrcinstance-attribute**

`src: str | bytes`

The lottie animation source.

It can be one of the following:

- A URL or local asset file path;
- A base64 string;
- Raw bytes.

## **Events**

### **bolton\_errorclass-attributeinstance-attribute**

`on_error: ControlEventHandler[Lottie] | None = None`

Fires when an error occurs while loading the Lottie animation.

The data property of the event handler argument contains information on the error.

# Overview

Display interactive maps in your Flet apps with markers, overlays, and rich attributions provided by the flet-map extension. The control is built on top of flutter\_map and supports multiple tile providers and layers.

## Platform Support

Platform	Windows	macOS	Linux
Supported	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>

## Usage

Add the flet-map package to your project dependencies:

- uv
  - pip
- ```
uv add flet-map
pip install flet-map
```

Important

Different tile providers have their own usage policies. Make sure you fully comply with their requirements (ex: attribution, rate limits) when using them in your app, to avoid being blocked or facing legal issues. More details here.

Examples

Map

play_arrowTry Online

```
import flet as ft
import flet_map as ftm

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ftm.Map(
                expand=True,
                layers=[
                    ftm.TileLayer(
                        url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
                        user_agent_package_name="flet-map-examples/1.0",
                        on_image_error=lambda e: print(f"TileLayer Error: {e.data}"),
                    ),
                    ftm.SimpleAttribution(text="OpenStreetMap contributors"),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Attribution

play_arrowTry Online

```
import flet as ft
import flet_map as ftm

def main(page: ft.Page):
    def launch_url(url: str):
        page.run_task(ft.UrlLauncher().launch_url, url)

    page.add(
        ft.SafeArea(
            expand=True,
            content=ftm.Map(
                expand=True,
                initial_zoom=11,
```

```

initial_center=ftm.MapLatitudeLongitude(
    latitude=40.7128, longitude=-74.0060
),
layers=[
    ftm.TileLayer(
        url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
        user_agent_package_name="flet-map-examples/1.0",
        on_image_error=lambda e: print(f"TileLayer Error: {e.data}"),
    ),
    ftm.SimpleAttribution(
        text="OpenStreetMap contributors",
        text_style=ft.TextStyle(
            color=ft.Colors.WHITE, weight=ft.FontWeight.BOLD, size=15
        ),
        alignment=ft.Alignment.TOP_RIGHT,
        bgcolor=ft.Colors.BLUE,
        on_click=lambda: launch_url(
            "https://www.openstreetmap.org/copyright"
        ),
    ),
    ftm.RichAttribution(
        alignment=ftm.AttributionAlignment.BOTTOM_RIGHT,
        popup_bgcolor=ft.Colors.WHITE,
        popup_border_radius=8,
        popup_initial_display_duration=4000,
        permanent_height=28,
        attributions=[
            ftm.ImageSourceAttribution(
                image=ft.Image(
                    src="iVBORw0KGgoAAAANSUhEUgAAABkAAAAGCAYAAADnnNMGAAAACXBIXMAAAOAAAADkQFnq8zdAAAAGXRFWHRTb2Z0
                    width=74,
                    height=28,
                    fit=ft.BoxFit.CONTAIN,
                ),
                height=28,
                tooltip="Flet Logo",
                on_click=lambda: launch_url("https://flet.dev"),
            ),
            ftm.TextSourceAttribution(
                text="OpenStreetMap contributors",
                text_style=ft.TextStyle(size=12),
                on_click=lambda: launch_url(
                    "https://www.openstreetmap.org/copyright"
                ),
            ),
            ftm.TextSourceAttribution(
                text="flet-map",
                text_style=ft.TextStyle(
                    color=ft.Colors.RED,
                    size=15,
                    weight=ft.FontWeight.BOLD,
                    decoration=ft.TextDecoration.UNDERLINE,
                ),
                prepend_copyright=False,
                on_click=lambda: launch_url(
                    "https://flet.dev/docs/controls/map/"
                ),
            ),
        ],
    ),
],
),
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Camera Controls

play_arrowTry Online

```

import flet as ft
import flet_map as ftm

```

```

def main(page: ft.Page):
    page.padding = 16

    async def update_camera_status(trigger: str):
        camera = await my_map.get_camera()
        camera_status.value = (
            f"Camera [{trigger}]: "
            f"center={({camera.center.latitude:.5f}, {camera.center.longitude:.5f}), "
            f"zoom={camera.zoom:.2f}, rotation={camera.rotation:.1f}"
        )
    )

```

```

page.update()

async def zoom_in(e: ft.Event[ft.Button]):
    await my_map.zoom_in()
    await update_camera_status("zoom_in")

async def zoom_out(e: ft.Event[ft.Button]):
    await my_map.zoom_out()
    await update_camera_status("zoom_out")

async def rotate_plus_15(e: ft.Event[ft.Button]):
    await my_map.rotate_from(15)
    await update_camera_status("rotate_from(+15)")

async def reset_rotation(e: ft.Event[ft.Button]):
    await my_map.reset_rotation()
    await update_camera_status("reset_rotation")

async def center_berlin(e: ft.Event[ft.Button]):
    await my_map.center_on(point=ftm.MapLatitudeLongitude(52.52, 13.405), zoom=12)
    await update_camera_status("center_on(berlin)")

async def move_tokyo(e: ft.Event[ft.Button]):
    await my_map.move_to(
        destination=ftm.MapLatitudeLongitude(35.6762, 139.6503), zoom=11
    )
    await update_camera_status("move_to(tokyo)")

async def set_world_zoom(e: ft.Event[ft.Button]):
    await my_map.zoom_to(3)
    await update_camera_status("zoom_to(3)")

page.appbar = ft.AppBar(title="Camera controls")
page.add(
    ft.SafeArea(
        expand=True,
        content=ft.Column(
            expand=True,
            controls=[
                ft.Text(
                    "Use buttons to control map camera programmatically.",
                    size=12,
                ),
                ft.Row(
                    wrap=True,
                    spacing=8,
                    run_spacing=8,
                    controls=[
                        ft.Button("Zoom in", on_click=zoom_in),
                        ft.Button("Zoom out", on_click=zoom_out),
                        ft.Button("Rotate +15°", on_click=rotate_plus_15),
                        ft.Button("Reset rotation", on_click=reset_rotation),
                        ft.Button("Center Berlin", on_click=center_berlin),
                        ft.Button("Move to Tokyo", on_click=move_tokyo),
                        ft.Button("World zoom (3)", on_click=set_world_zoom),
                    ],
                ),
            ],
        ),
        camera_status := ft.Text(selectable=True, font_family="monospace"),
        my_map := ftm.Map(
            expand=True,
            initial_center=ftm.MapLatitudeLongitude(52.52, 13.405),
            initial_zoom=5,
            layers=[
                ftm.TileLayer(
                    url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
                    user_agent_package_name="flet-map-examples/1.0",
                ),
                ftm.SimpleAttribution(
                    text="OpenStreetMap contributors",
                    on_click=lambda e: e.page.launch_url(
                        "https://www.openstreetmap.org/copyright"
                    ),
                ),
                ftm.MarkerLayer(
                    markers=[
                        ftm.Marker(
                            coordinates=ftm.MapLatitudeLongitude(
                                52.52, 13.405
                            ),
                            content=ft.Icon(ft.Icons.LOCATION_ON),
                        ),
                        ftm.Marker(
                            coordinates=ftm.MapLatitudeLongitude(
                                35.6762, 139.6503
                            ),
                            content=ft.Icon(ft.Icons.LOCATION_ON),
                        ),
                    ],
                ),
            ],
        ),
    ),
)

```



```
if __name__ == "__main__":  
    ft.run(main)
```

```

        on_click=lambda e: e.page.launch_url(
            "https://www.openstreetmap.org/copyright"
        ),
    ),
    ],
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Interaction Flags

play_arrowTry Online

```

import flet as ft
import flet_map as ftm

FLAG_OPTIONS: list[tuple[str, ftm.InteractionFlag]] = [
    ("Drag", ftm.InteractionFlag.DRAG),
    ("Fling animation", ftm.InteractionFlag.FLING_ANIMATION),
    ("Pinch move", ftm.InteractionFlag.PINCH_MOVE),
    ("Pinch zoom", ftm.InteractionFlag.PINCH_ZOOM),
    ("Double tap zoom", ftm.InteractionFlag.DOUBLE_TAP_ZOOM),
    ("Double tap drag zoom", ftm.InteractionFlag.DOUBLE_TAP_DRAG_ZOOM),
    ("Scroll wheel zoom", ftm.InteractionFlag.SCROLL_WHEEL_ZOOM),
    ("Rotate", ftm.InteractionFlag.ROTATE),
]

def main(page: ft.Page):
    page.padding = 16

    def get_selected_flags() -> ftm.InteractionFlag:
        flags = ftm.InteractionFlag.NONE
        for checkbox in checkboxes:
            if checkbox.value:
                flags |= checkbox.data
        return flags

    def update_interaction_flags(e: ft.Event[ft.Checkbox] = None):
        flags = get_selected_flags()
        my_map.interaction_configuration = ftm.InteractionConfiguration(flags=flags)
        page.update()

    def handle_map_event(e: ftm.MapEvent):
        event_type = e.event_type.value if e.event_type else "-"
        last_event.value = (
            "Last event: "
            f"type={event_type}, source={e.source.value}, zoom={e.camera.zoom:.2f}"
        )
        page.update()

    checkboxes = [
        ft.Checkbox(
            label=label,
            value=True,
            data=flag,
            on_change=update_interaction_flags,
        )
        for label, flag in FLAG_OPTIONS
    ]

    my_map = ftm.Map(
        expand=True,
        initial_center=ftm.MapLatitudeLongitude(latitude=52.52, longitude=13.405),
        initial_zoom=11,
        on_event=handle_map_event,
        interaction_configuration=ftm.InteractionConfiguration(
            flags=ftm.InteractionFlag.ALL
        ),
        layers=[
            ftm.TileLayer(
                url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
                user_agent_package_name="flet-map-examples/1.0",
                on_image_error=lambda e: print(f"TileLayer Error: {e.data}"),
            ),
            ftm.SimpleAttribution(
                text="OpenStreetMap contributors",
                on_click=lambda e: e.page.launch_url(
                    "https://www.openstreetmap.org/copyright"
                ),
            ),
        ],
    ),

```

```

    ],
)

page.appbar = ft.AppBar(title="Interaction flags")
page.add(
    ft.SafeArea(
        expand=True,
        content=ft.Column(
            expand=True,
            controls=[
                ft.Text(
                    (
                        "Toggle flags and try dragging, zooming, rotating, "
                        "and scrolling."
                    ),
                    size=12,
                ),
                ft.ResponsiveRow(
                    controls=[
                        ft.Container(col={"sm": 6, "md": 4}, content=c)
                        for c in checkboxes
                    ]
                ),
                last_event := ft.Text(
                    "Last event: -", selectable=True, font_family="monospace"
                ),
                ft.Container(expand=True, content=my_map),
            ],
        ),
    ),
)

update_interaction_flags()

if __name__ == "__main__":
    ft.run(main)

```

Multiple Layers

play_arrowTry Online

```

import random

import flet as ft
import flet_map as ftm

def main(page: ft.Page):
    def handle_tap(e: ftm.MapTapEvent):
        if e.name == "tap":
            marker_layer.markers.append(
                ftm.Marker(
                    content=ft.Icon(
                        ft.Icons.LOCATION_ON, color=ft.CupertinoColors.DESTRUCTIVE_RED
                    ),
                    coordinates=e.coordinates,
                )
            )
        elif e.name == "secondary_tap":
            circle_layer.circles.append(
                ftm.CircleMarker(
                    radius=random.randint(5, 10),
                    coordinates=e.coordinates,
                    color=ft.Colors.random(),
                    border_color=ft.Colors.random(),
                    border_stroke_width=4,
                )
            )
        page.update()

    page.appbar = ft.AppBar(title="Multiple Layers")
    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                expand=True,
                controls=[
                    ft.Text(
                        "Click anywhere to add a Marker, right-click to add a "
                        "CircleMarker."
                    ),
                    ftm.Map(
                        expand=True,
                        initial_center=ftm.MapLatitudeLongitude(15, 10),
                        initial_zoom=4.2,
                        interaction_configuration=ftm.InteractionConfiguration(

```



```

    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Overlay Images

play_arrowTry Online

```

import flet as ft
import flet_map as ftm

```

```
base64_image = "iVBORw0KGgoAAAANSUhEUgAAABkAAAgCAYAAADnnNMGAAACXBIWXMMAA0RAAADkQFnq8zdAAAAGXRFWHRTb2Z0d2FyZQB3d3cuaW5rc2NhcnGUub"

```

```

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ftm.Map(
                expand=True,
                initial_center=ftm.MapLatitudeLongitude(51.5, -0.09),
                initial_zoom=6,
                layers=[
                    ftm.TileLayer(
                        url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
                        user_agent_package_name="flet-map-examples/1.0",
                    ),
                    ftm.OverlayImageLayer(
                        overlay_images=[
                            ftm.OverlayImage(
                                src=base64_image,
                                bounds=ftm.MapLatitudeLongitudeBounds(
                                    corner_1=ftm.MapLatitudeLongitude(51.5, -0.09),
                                    corner_2=ftm.MapLatitudeLongitude(48.8566, 2.3522),
                                ),
                                opacity=0.8,
                            ),
                            ftm.RotatedOverlayImage(
                                src=base64_image,
                                top_left_corner=ftm.MapLatitudeLongitude(
                                    53.377, -2.999
                                ),
                                bottom_left_corner=ftm.MapLatitudeLongitude(
                                    52.503, -1.868
                                ),
                                bottom_right_corner=ftm.MapLatitudeLongitude(
                                    53.475, 0.275
                                ),
                                opacity=0.8,
                            ),
                        ],
                    ),
                    ftm.SimpleAttribution(text="OpenStreetMap contributors"),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Reference

Map

- Layers: TileLayer, MarkerLayer, OverlayImageLayer, CircleLayer, PolygonLayer, PolylineLayer
- Markers and overlays: Marker, CircleMarker, PolygonMarker, PolylineMarker, OverlayImage, RotatedOverlayImage
- Attributions: SimpleAttribution, RichAttribution, SourceAttribution

See the types section for additional configuration helpers.

Markdown

Renders text in markdown format.

```
ft.Markdown(  
    value="# Welcome\n\nThis is **Markdown** rendered in Flet.",  
    width=260,  
)
```

Image: Markdown - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/markdown/image_for_docs.png

Markdown

Inherits: `LayoutControl`

Properties

- `auto_follow_links` - Automatically open URLs in the document.
- `auto_follow_links_target` - Where to open URL in the web mode.
- `code_style_sheet` - The styles to use when displaying the code blocks.
- `code_theme` - A syntax highlighting theme for code blocks.
- `extension_set` - The extensions to use when rendering the markdown content.
- `fit_content` - Whether to allow the widget to fit the child content.
- `image_error_content` - The control to display when an image fails to load.
- `latex_scale_factor` - The scale factor for LaTeX formulas rendering.
- `latex_style` - The text style to apply to LaTeX formulas.
- `md_style_sheet` - The styles to use when displaying the markdown.
- `selectable` - Whether rendered text is selectable or not.
- `shrink_wrap` - Whether the extent of the scroll view in the scroll direction should be determined by the contents being viewed.
- `soft_line_break` - The soft line break is used to identify the spaces at the end of a line of text and the leading spaces in the immediately following the line of text.
- `value` - Markdown content to render.

Events

- `on_selection_change` - Called when the text selection changes.
- `on_tap_link` - Called when a link within Markdown document is clicked/tapped.
- `on_tap_text` - Called when some text is clicked/tapped.

Examples

Markdown

`play_arrow` Try Online

```
import flet as ft  
  
sample = """  
# Markdown Example  
Markdown allows you to easily include formatted text, images, and even formatted Dart  
code in your app.  
  
## Titles  
  
Setext-style  
  
This is an H1  
=====
```

```
This is an H2  
-----  
  
Atx-style
```

```

# This is an H1

## This is an H2

##### This is an H6

Select the valid headers:

- [x] `# hello`
- [ ] `#hello`

## Links

[inline-style](https://www.google.com)

## Images

![Image from Flet assets](/icons/icon-192.png)

![Test image](https://picsum.photos/200/300)

## Tables

Syntax	Result
`*italic 1*`	*italic 1*
`_italic 2_`	_italic 2_
`**bold 1**`	**bold 1**
`__bold 2__`	__bold 2__
`This is a ~~strikethrough~~`	This is a ~~strikethrough~~
`***italic bold 1***`	***italic bold 1***
`__italic bold 2__`	__italic bold 2__
`***~~italic bold strikethrough 1~~***`	***~~italic bold strikethrough 1~~***
`~~***italic bold strikethrough 2***~~`	~~***italic bold strikethrough 2***~~

## Styling

Style text as _italic_, __bold__, ~~strikethrough~~, or `inline code`.

- Use bulleted lists
- To better clarify
- Your points

## Code blocks

Formatted Dart code looks really pretty too:

~~~dart
void main() {
  runApp(MaterialApp(
    home: Scaffold(
      body: ft.Markdown(data: markdownData),
    ),
  ));
}
~~~
"""

def main(page: ft.Page):
    page.scroll = ft.ScrollMode.AUTO

    async def handle_link_tap(e: ft.Event[ft.Markdown]):
        await page.launch_url(e.data)

    page.add(
        ft.SafeArea(
            content=ft.Markdown(
                value=sample,
                selectable=True,
                extension_set=ft.MarkdownExtensionSet.GITHUB_WEB,
                on_tap_link=handle_link_tap,
            ),
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - <https://flet.dev/docs/examples/controls/core/markdown/media/basic.gif>

Code syntax highlight

play_arrowTry Online

```
import flet as ft
```

```
sample = ""
# Flet
```

```

```

Flet is a framework for adding server-driven UI (SDUI) experiences to existing Flutter apps or building standalone web, mobile and desktop apps with Flutter UI.

Add an interactive `FletApp` widget to your Flutter app whose content is controlled by a remote Python script.

It is an ideal solution for building non-core or frequently changing functionality such as product catalog, feedback form, in-app

On the server side Flet provides an easy to learn programming model that enables Python developers without prior Flutter (or even

```
## Getting started with Flet
```

```
### Install `flet` Python module
```

Flet requires Python 3.7 or above. To start with Flet, you need to install flet module first:

```
~~~bash
pip install flet
~~~
```

```
### Create Python program
```

Create a new Python program using Flet which will be driving the content of `FletApp` widget.

Let's do a simple `counter.py` app similar to a Flutter new project template:

```
~~~python
import flet
from flet import IconButton, Page, Row, TextField, icons

def main(page: Page):
    page.title = "Flet counter example"
    page.vertical_alignment = "center"

    txt_number = TextField(value="0", text_align="right", width=100)

    def minus_click(e):
        txt_number.value = int(txt_number.value) - 1
        page.update()

    def plus_click(e):
        txt_number.value = int(txt_number.value) + 1
        page.update()

    page.add(
        Row(
            [
                IconButton(icons.REMOVE, on_click=minus_click),
                txt_number,
                IconButton(icons.ADD, on_click=plus_click),
            ],
            alignment="center",
        )
    )

flet.app(main, port=8550)
~~~
```

Run the app:

```
~~~bash
python counter.py
~~~
```

You should see the app running in a native OS window.

There is a web server (Fletd) running in the background on a fixed port `8550`. Fletd web server is a "bridge" between Python and

`FletApp` widget in your Flutter application will be communicating with Fletd web server via WebSockets to receive UI updates and

For production use Python app along with Fletd could be [deployed to a public web host](https://flet.dev/docs/publish/web/dynamic

```
### Add Flet widget to a Flutter app
```

Create a new or open existing Flutter project.

Install Flutter `flet` package:

```
~~~bash
flutter pub add flet
~~~
```

For a new project replace `main.dart` with the following:

```
~~~dart
```



```

import 'package:flet/flet.dart';
import 'package:flutter/material.dart';

void main() async {
  await setupDesktop();
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      title: 'Flet Flutter Demo',
      home: FletApp(pageUrl: "http://localhost:8550"),
    );
  }
}
~~~

```

In the app above `FletApp` widget is hosted inside `MaterialApp` widget.

If Flet app must be able to handle page route change events (web browser URL changes, mobile app deep linking) it must be the top

```

~~~dart
import 'package:flet/flet.dart';
import 'package:flutter/material.dart';

void main() async {
  await setupDesktop();
  runApp(const FletApp(pageUrl: "http://localhost:8550"));
}
~~~

```

Run the program and see Flet app running inside a Flutter app.

When adding `FletApp` widget to the existing desktop Flutter app make sure `setupDesktop()` is called before `runApp()` to initia

Flet learning resources

```

* [Getting started for Python](https://flet.dev/docs/getting-started/installation/)
* [Controls reference](https://flet.dev/docs/controls)
* [Tutorials](https://flet.dev/docs/tutorials)
* [Examples](https://github.com/flet-dev/examples/tree/main/python)

```

Flet community

```

* [Discussions](https://github.com/flet-dev/flet/discussions)
* [Discord](https://discord.gg/dzWXP8SHG8)
* [Twitter](https://twitter.com/fletdev)
* [Email](mailto:[email protected])

```

FAQ

Coming soon.

Adding custom Flutter widgets

Coming soon.

""" # noqa: E501

```

def main(page: ft.Page):
    page.scroll = ft.ScrollMode.AUTO

    page.fonts = {
        "Roboto Mono": "RobotoMono-VariableFont_wght.ttf",
    }

    async def navigate_md_link(e: ft.Event[ft.Markdown]):
        await page.launch_url(e.data)

    page.add(
        ft.SafeArea(
            content=ft.Markdown(
                value=sample,
                selectable=True,
                extension_set=ft.MarkdownExtensionSet.GITHUB_WEB,
                code_theme=ft.MarkdownCodeTheme.ATOM_ONE_DARK,
                code_style_sheet=ft.MarkdownStyleSheet(
                    code_text_style=ft.TextStyle(font_family="Roboto Mono")
                ),
                on_tap_link=navigate_md_link,
            ),
        ),
    )

```

```
if __name__ == "__main__":
    ft.run(main)
```

Image: code-syntax-highlight - https://flet.dev/docs/examples/controls/core/markdown/media/code_syntax_highlight.png

Custom text theme

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.DARK

    def change_theme_mode(e: ft.Event[ft.Switch]):
        if page.theme_mode == ft.ThemeMode.DARK:
            page.theme_mode = ft.ThemeMode.LIGHT
            switch.thumb_icon = ft.Icons.LIGHT_MODE
        else:
            switch.thumb_icon = ft.Icons.DARK_MODE
            page.theme_mode = ft.ThemeMode.DARK
        page.update()

    switch = ft.Switch(thumb_icon=ft.Icons.DARK_MODE, on_change=change_theme_mode)

    page.add(
        ft.SafeArea(
            content=ft.Row(
                alignment=ft.MainAxisAlignment.SPACE_BETWEEN,
                controls=[
                    ft.Container(
                        bgcolor="#550000",
                        padding=20,
                        theme=ft.Theme(
                            text_theme=ft.TextTheme(
                                body_medium=ft.TextStyle(color=ft.Colors.WHITE),
                                body_large=ft.TextStyle(color=ft.Colors.WHITE),
                                body_small=ft.TextStyle(color=ft.Colors.WHITE),
                            )
                        ),
                        content=ft.Markdown("I can read this!"),
                    ),
                    ft.Container(
                        padding=ft.Padding.only(bottom=50),
                        alignment=ft.Alignment.TOP_RIGHT,
                        content=switch,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

buildauto_follow_linksclass-attributeinstance-attribute

auto_follow_links: bool = False

Automatically open URLs in the document.

If registered, on_tap_link event is fired after that.

buildauto_follow_links_targetclass-attributeinstance-attribute

auto_follow_links_target: UrlTarget | None = None

Where to open URL in the web mode.

buildcode_style_sheetclass-attributeinstance-attribute

code_style_sheet: MarkdownStyleSheet | None = None

The styles to use when displaying the code blocks.

buildcode_themeclass-attributeinstance-attribute

code_theme: (
 MarkdownCodeTheme | MarkdownCustomCodeTheme | None
) = None

A syntax highlighting theme for code blocks.

Defaults to MarkdownCodeTheme.GITHUB.

buildextension_setclass-attributeinstance-attribute

```
extension_set: MarkdownExtensionSet = (  
    MarkdownExtensionSet.NONE  
)
```

The extensions to use when rendering the markdown content.

buildfit_contentclass-attributeinstance-attribute

```
fit_content: bool = True
```

Whether to allow the widget to fit the child content.

buildimage_error_contentclass-attributeinstance-attribute

```
image_error_content: Control | None = None
```

The control to display when an image fails to load.

buildlatex_scale_factorclass-attributeinstance-attribute

```
latex_scale_factor: float | None = None
```

The scale factor for LaTeX formulas rendering. Controls the size of rendered mathematical expressions.

buildlatex_styleclass-attributeinstance-attribute

```
latex_style: TextStyle | None = None
```

The text style to apply to LaTeX formulas. Allows customization of font, color, and other text properties for mathematical expressions.

buildmd_style_sheetclass-attributeinstance-attribute

```
md_style_sheet: MarkdownStyleSheet | None = None
```

The styles to use when displaying the markdown.

buildselectableclass-attributeinstance-attribute

```
selectable: bool = False
```

Whether rendered text is selectable or not.

buildshrink_wrapclass-attributeinstance-attribute

```
shrink_wrap: bool = True
```

Whether the extent of the scroll view in the scroll direction should be determined by the contents being viewed.

buildsoft_line_breakclass-attributeinstance-attribute

```
soft_line_break: bool = False
```

The soft line break is used to identify the spaces at the end of a line of text and the leading spaces in the immediately following the line of text.

buildvalueclass-attributeinstance-attribute

```
value: str = ''
```

Markdown content to render.

Events

bolton_selection_changeclass-attributeinstance-attribute

```
on_selection_change: (  
    EventHandler[TextSelectionChangeEvent[Markdown]] | None  
) = None
```

Called when the text selection changes.

bolton_tap_linkclass-attributeinstance-attribute

```
on_tap_link: ControlEventHandler[Markdown] | None = None
```

Called when a link within Markdown document is clicked/tapped.

The data property of the event handler argument contains the clickedURL.

Example: <https://github.com/flet-dev/examples/blob/main/python/controls/information-displays/markdown/markdown-event-example.py>

bolton_tap_textclass-attributeinstance-attribute

```
on_tap_text: ControlEventHandler[Markdown] | None = None
```

Called when some text is clicked/tapped.

MenuBar

A menu bar that manages cascading child menus.

It could be placed anywhere but typically resides above the main body of the application and defines a menu system for invoking callbacks in response to user selection of a menu item.

```
ft.MenuBar(
    controls=[
        ft.SubmenuButton(
            content=ft.Text("Submenu"),
            controls=[
                ft.MenuItemButton(content=ft.Text("Item 1")),
                ft.MenuItemButton(content=ft.Text("Item 2")),
                ft.MenuItemButton(content=ft.Text("Item 3")),
            ],
        ),
    ],
)
```

Image: MenuBar - https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/menu_bar/image_for_docs.png

MenuBar

Inherits: Control

Properties

- `clip_behavior` - Whether to clip the content of this control or not.
- `controls` - A list of top-level menu controls to display in this menu bar.
- `style` - The menu bar style.

Examples

MenuBar with nested submenus

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    appbar_text_ref = ft.Ref[ft.Text]()

    def handle_menu_item_click(e: ft.Event[ft.MenuItemButton]):
        text = e.control.content.value
        page.show_dialog(ft.SnackBar(ft.Text(f"{text} was clicked!")))
        appbar_text_ref.current.value = text
        appbar_text_ref.current.update()

    def handle_submenu_open(e: ft.Event[ft.SubmenuButton]):
        print(f"{e.control.content.value}.on_open")

    def handle_submenu_close(e: ft.Event[ft.SubmenuButton]):
        print(f"{e.control.content.value}.on_close")

    def handle_submenu_hover(e: ft.Event[ft.SubmenuButton]):
        print(f"{e.control.content.value}.on_hover")

    page.appbar = ft.AppBar(
        title=ft.Text("Menus", ref=appbar_text_ref),
        center_title=True,
        bgcolor=ft.Colors.BLUE,
    )

    page.add(
        ft.SafeArea(
            content=ft.Row(
                controls=[
                    ft.MenuBar(
                        expand=True,
                        style=ft.MenuStyle(
                            alignment=ft.Alignment.TOP_LEFT,
                            bgcolor=ft.Colors.RED_300,
                            mouse_cursor={
                                ft.ControlState.HOVERED: ft.MouseCursor.WAIT,
                                ft.ControlState.DEFAULT: ft.MouseCursor.ZOOM_OUT,
                            },
                        ),
                    ),
                ],
            ),
        ),
    )
```

```

controls=[
    ft.SubmenuButton(
        content=ft.Text("File"),
        on_open=handle_submenu_open,
        on_close=handle_submenu_close,
        on_hover=handle_submenu_hover,
        controls=[
            ft.MenuItemButton(
                content=ft.Text("About"),
                leading=ft.Icon(ft.Icons.INFO),
                on_click=handle_menu_item_click,
                style=ft.ButtonStyle(
                    bgcolor={
                        ft.ControlState.HOVERED: (
                            ft.Colors.GREEN_100
                        )
                    }
                ),
            ),
            ft.MenuItemButton(
                content=ft.Text("Save"),
                leading=ft.Icon(ft.Icons.SAVE),
                on_click=handle_menu_item_click,
                style=ft.ButtonStyle(
                    bgcolor={
                        ft.ControlState.HOVERED: (
                            ft.Colors.GREEN_100
                        )
                    }
                ),
            ),
            ft.MenuItemButton(
                content=ft.Text("Quit"),
                leading=ft.Icon(ft.Icons.CLOSE),
                on_click=handle_menu_item_click,
                style=ft.ButtonStyle(
                    bgcolor={
                        ft.ControlState.HOVERED: (
                            ft.Colors.GREEN_100
                        )
                    }
                ),
            ),
        ],
    ),
    ft.SubmenuButton(
        content=ft.Text("View"),
        on_open=handle_submenu_open,
        on_close=handle_submenu_close,
        on_hover=handle_submenu_hover,
        controls=[
            ft.SubmenuButton(
                content=ft.Text("Zoom"),
                controls=[
                    ft.MenuItemButton(
                        content=ft.Text("Magnify"),
                        leading=ft.Icon(ft.Icons.ZOOM_IN),
                        close_on_click=False,
                        on_click=handle_menu_item_click,
                        style=ft.ButtonStyle(
                            bgcolor={
                                ft.ControlState.HOVERED: (
                                    ft.Colors.PURPLE_200
                                )
                            }
                        ),
                    ),
                    ft.MenuItemButton(
                        content=ft.Text("Minify"),
                        leading=ft.Icon(ft.Icons.ZOOM_OUT),
                        close_on_click=False,
                        on_click=handle_menu_item_click,
                        style=ft.ButtonStyle(
                            bgcolor={
                                ft.ControlState.HOVERED: (
                                    ft.Colors.PURPLE_200
                                )
                            }
                        ),
                    ),
                ],
            ),
        ],
    ),
],
)

```

```

    ),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/menu_bar/nested_submenus.gif

Properties

buildclip_behaviorclass-attributeinstance-attribute

```
clip_behavior: ClipBehavior = ClipBehavior.NONE
```

Whether to clip the content of this control or not.

buildcontrolsclass-attributeinstance-attribute

```
controls: list[Control] = field(default_factory=list)
```

A list of top-level menu controls to display in this menu bar.

Raises:

- `ValueError` - If it does not contain at least one visible `Control`.

buildstyleclass-attributeinstance-attribute

```
style: MenuStyle | None = None
```

The menu bar style.

MenuItemButton

A button for use in a MenuBar or on its own, that can be activated by click or keyboard navigation.

```
ft.Row(
    controls=[
        ft.MenuItemButton(
            content=ft.Text("Yes"),
            on_click=lambda e: print("yes"),
        ),
        ft.MenuItemButton(
            content=ft.Text("No"),
            on_click=lambda e: print("no"),
        ),
        ft.MenuItemButton(
            content=ft.Text("Maybe"),
            on_click=lambda e: print("maybe"),
        ),
    ],
)
```

Image: MenuItemButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/menu_item_button/image_for_docs.png

Menu item buttons outside of menubar

Inherits: LayoutControl

Properties

- autofocus - Whether this button should automatically request focus.
- clip_behavior - Whether to clip the content of this control or not.
- close_on_click - Defines if the menu will be closed when the MenuItemButton is clicked.
- content - The child control or text to be displayed in the center of this button.
- focus_on_hover - Determine if hovering can request focus.
- leading - An optional control to display before the content.
- overflow_axis - The direction in which the menu item expands.
- semantic_label - A string that describes the button's action to assistive technologies.
- style - Customizes this button's appearance.
- trailing - An optional control to display after the content.

Events

- on_blur - Called when this button loses focus.
- on_click - Called when the button is clicked.If not defined the button will be disabled.
- on_focus - Called when the button receives focus.
- on_hover - Called when the button is hovered.

Examples

MenuItemButton

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.padding = 0
    page.spacing = 0
    page.theme_mode = ft.ThemeMode.LIGHT

    def handle_color_click(e: ft.Event[ft.MenuItemButton]):
        color = e.control.content.value
        background_container.content.value = f"{color} background color"
        background_container.bgcolor = color.lower()

    def handle_on_hover(e: ft.Event[ft.MenuItemButton]):
        print(e)
```



```

menubar = ft.MenuBar(
    expand=True,
    controls=[
        ft.SubmenuButton(
            content=ft.Text("BgColors"),
            controls=[
                ft.MenuItemButton(
                    content=ft.Text("Blue"),
                    leading=ft.Icon(ft.Icons.COLORIZE),
                    style=ft.ButtonStyle(
                        bgcolor={ft.ControlState.HOVERED: ft.Colors.BLUE}
                    ),
                    on_click=handle_color_click,
                    on_hover=handle_on_hover,
                ),
                ft.MenuItemButton(
                    content=ft.Text("Green"),
                    leading=ft.Icon(ft.Icons.COLORIZE),
                    style=ft.ButtonStyle(
                        bgcolor={ft.ControlState.HOVERED: ft.Colors.GREEN}
                    ),
                    on_click=handle_color_click,
                    on_hover=handle_on_hover,
                ),
                ft.MenuItemButton(
                    content=ft.Text("Red"),
                    leading=ft.Icon(ft.Icons.COLORIZE),
                    style=ft.ButtonStyle(
                        bgcolor={ft.ControlState.HOVERED: ft.Colors.RED}
                    ),
                    on_click=handle_color_click,
                    on_hover=handle_on_hover,
                ),
            ],
        ),
    ],
)

page.add(
    ft.SafeArea(
        expand=True,
        avoid_intrusions_left=False,
        avoid_intrusions_top=False,
        avoid_intrusions_right=False,
        avoid_intrusions_bottom=False,
        content=ft.Column(
            expand=True,
            spacing=0,
            controls=[
                ft.Row(controls=[menubar]),
                background_container := ft.Container(
                    expand=True,
                    bgcolor=ft.Colors.WHITE,
                    alignment=ft.Alignment.CENTER,
                    content=ft.Text(
                        value="Choose a bgcolor from the menu",
                        style=ft.TextStyle(weight=ft.FontWeight.W_500),
                    ),
                ),
            ],
        ),
    ),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/menu_item_button/basic.gif

Properties

buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

Whether this button should automatically request focus.

Defaults to False.

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: ClipBehavior = ClipBehavior.NONE

Whether to clip the content of this control or not.

buildclose_on_clickclass-attributeinstance-attribute

`close_on_click: bool = True`

Defines if the menu will be closed when the MenuItemButton is clicked.

buildcontentclass-attributeinstance-attribute

`content: StrOrControl | None = None`

The child control or text to be displayed in the center of this button.

Typically this is the button's label, using a Text control.

buildfocus_on_hoverclass-attributeinstance-attribute

`focus_on_hover: bool = True`

Determine if hovering can request focus.

buildleadingclass-attributeinstance-attribute

`leading: Control | None = None`

An optional control to display before the content.

Typically an Icon control.

buildoverflow_axisclass-attributeinstance-attribute

`overflow_axis: Axis = Axis.HORIZONTAL`

The direction in which the menu item expands.

If the menu item button is a descendent of MenuBar, then this property is ignored.

buildsemantic_labelclass-attributeinstance-attribute

`semantic_label: str | None = None`

A string that describes the button's action to assistive technologies.

buildstyleclass-attributeinstance-attribute

`style: ButtonStyle | None = None`

Customizes this button's appearance.

buildtrailingclass-attributeinstance-attribute

`trailing: Control | None = None`

An optional control to display after the content.

Typically an Icon control.

Events

bolton_blurclass-attributeinstance-attribute

`on_blur: ControlEventHandler[MenuItemButton] | None = None`

Called when this button loses focus.

bolton_clickclass-attributeinstance-attribute

`on_click: ControlEventHandler[MenuItemButton] | None = None`

Called when the button is clicked. If not defined the button will be disabled.

bolton_focusclass-attributeinstance-attribute

`on_focus: ControlEventHandler[MenuItemButton] | None = None`

Called when the button receives focus.

bolton_hoverclass-attributeinstance-attribute

`on_hover: ControlEventHandler[MenuItemButton] | None = None`

Called when the button is hovered.

MergeSemantics

A control that merges the semantics of its descendants.

Causes all the semantics of the subtree rooted at this node to be merged into one node in the semantics tree.

Used by accessibility tools, search engines, and other semantic analysis software to determine the meaning of the application.

Inherits: Control

Properties

- content - The Control to annotate.

Properties

buildcontentclass-attributeinstance-attribute

content: Control | None = None

The Control to annotate.

MultiView

TBD

Inherits: BasePage

Properties

- `initial_data` - TBD
- `view_id` - TBD

Properties

buildinitial_datainstance-attribute

`initial_data: dict[str, Any]`

TBD

buildview_idinstance-attribute

`view_id: int`

TBD

NavigationBar

Material 3 Navigation Bar component.

Navigation bars offer a persistent and convenient way to switch between primary destinations in an app.

```
ft.NavigationBar(  
    destinations=[  
        ft.NavigationBarDestination(icon=ft.Icons.CIRCLE, label="Item 1"),  
        ft.NavigationBarDestination(icon=ft.Icons.SQUARE, label="Item 2"),  
        ft.NavigationBarDestination(icon=ft.Icons.HEXAGON, label="Item 3"),  
    ],  
)
```

Image: NavigationBar - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_bar/image_for_docs.png

Simple navigation bar

Inherits: LayoutControl, AdaptiveControl

Properties

- `animation_duration` - The transition time for each destination as it goes between selected and unselected.
- `bgcolor` - The color of the navigation bar itself.
- `border` - TBD
- `destinations` - Defines the appearance of the button items that are arrayed within the navigation bar.
- `elevation` - The elevation of the navigation bar itself.
- `indicator_color` - The color of the selected destination indicator.
- `indicator_shape` - The shape of the selected destination indicator.
- `label_behavior` - Defines how the destinations' labels will be laid out and when they'll be displayed.
- `label_padding` - The padding around the `NavigationBarDestination.label`.
- `overlay_color` - The highlight color of the `NavigationBarDestination` in various `ControlState` states.
- `selected_index` - The index into destinations for the current selected `NavigationBarDestination` or `None` if no destination is selected.
- `shadow_color` - The color used for the drop shadow to indicate elevation.

Events

- `on_change` - Called when selected destination changed.

Examples

NavigationBar

[play_arrow](#)Try Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.title = "NavigationBar Example"  
  
    page.navigation_bar = ft.NavigationBar(  
        destinations=[  
            ft.NavigationBarDestination(icon=ft.Icons.EXPLORE, label="Explore"),  
            ft.NavigationBarDestination(icon=ft.Icons.COMMUTE, label="Commute"),  
            ft.NavigationBarDestination(  
                icon=ft.Icons.BOOKMARK_BORDER,  
                selected_icon=ft.Icons.BOOKMARK,  
                label="Favorites",  
            ),  
        ],  
    )  
  
    page.add(  
        ft.SafeArea(  
            content=ft.Text("Body!"),  
        )  
    )
```

)

```
if __name__ == "__main__":  
    ft.run(main)
```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_bar/basic.png

Adaptive Navigation

This example switches between a `NavigationBar` on narrow layouts and a `NavigationRail` with an end `NavigationDrawer` on wider layouts.

play_arrow Try Online

```
import flet as ft

DESTINATIONS = [
    ("Messages", ft.Icons.WIDGETS_OUTLINED, ft.Icons.WIDGETS),
    ("Profile", ft.Icons.FORMAT_PAINT_OUTLINED, ft.Icons.FORMAT_PAINT),
    ("Settings", ft.Icons.SETTINGS_OUTLINED, ft.Icons.SETTINGS),
]

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT
    page.padding = 0

    screen_index = 0

    def build_page_index_text() -> ft.Text:
        return ft.Text(f"Page Index = {screen_index}", size=24)

    def set_screen(index: int):
        nonlocal screen_index
        screen_index = index
        render()

    def build_navigation_bar() -> ft.NavigationBar:
        def handle_nav_bar_change(e: ft.Event[ft.NavigationBar]):
            set_screen(e.control.selected_index)

        return ft.NavigationBar(
            selected_index=screen_index,
            on_change=handle_nav_bar_change,
            destinations=[
                ft.NavigationBarDestination(
                    label=label,
                    icon=icon,
                    selected_icon=selected_icon,
                )
                for label, icon, selected_icon in DESTINATIONS
            ],
        )

    def build_navigation_rail() -> ft.NavigationRail:
        def handle_nav_rail_change(e: ft.Event[ft.NavigationRail]):
            if e.control.selected_index is not None:
                set_screen(e.control.selected_index)

        return ft.NavigationRail(
            min_width=50,
            selected_index=screen_index,
            use_indicator=True,
            on_change=handle_nav_rail_change,
            destinations=[
                ft.NavigationRailDestination(
                    label=label,
                    icon=icon,
                    selected_icon=selected_icon,
                )
                for label, icon, selected_icon in DESTINATIONS
            ],
        )

    def build_end_drawer() -> ft.NavigationDrawer:
        async def handle_drawer_change(e: ft.Event[ft.NavigationDrawer]):
            set_screen(e.control.selected_index)
            await page.close_end_drawer()

        return ft.NavigationDrawer(
            selected_index=screen_index,
            on_change=handle_drawer_change,
            controls=[
                ft.Container(
                    padding=ft.Padding.only(left=28, top=16, right=16, bottom=10),
                    content=ft.Text(
```

```

        "Header", theme_style=ft.TextThemeStyle.TITLE_SMALL
    ),
),
*[
    ft.NavigationDrawerDestination(
        label=label,
        icon=icon,
        selected_icon=selected_icon,
    )
    for label, icon, selected_icon in DESTINATIONS
],
],
)

def build_bottom_bar_layout() -> ft.SafeArea:
    return ft.SafeArea(
        expand=True,
        content=ft.Container(
            expand=True,
            alignment=ft.Alignment.CENTER,
            content=build_page_index_text(),
        ),
    )

def build_drawer_layout() -> ft.SafeArea:
    async def open_drawer(e: ft.Event[ft.Button]):
        await page.show_end_drawer()

    return ft.SafeArea(
        expand=True,
        avoid_intrusions_top=False,
        avoid_intrusions_bottom=False,
        content=ft.Row(
            expand=True,
            controls=[
                ft.Container(
                    padding=ft.Padding.symmetric(horizontal=5),
                    content=build_navigation_rail(),
                ),
                ft.VerticalDivider(thickness=1, width=1),
                ft.Column(
                    expand=True,
                    alignment=ft.MainAxisAlignment.SPACE_EVENLY,
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    controls=[
                        build_page_index_text(),
                        ft.Button("Open Drawer", on_click=open_drawer),
                    ],
                ),
            ],
        ),
    )

def render():
    page.clean()
    if (page.width or page.window.width) >= 450: # wide layout
        page.navigation_bar = None
        page.end_drawer = build_end_drawer()
        page.add(build_drawer_layout())
    else: # narrow layout
        page.end_drawer = None
        page.navigation_bar = build_navigation_bar()
        page.add(build_bottom_bar_layout())
    page.update()

page.on_resize = lambda e: render()
render()

if __name__ == "__main__":
    ft.run(main)

```

Image: Adaptive navigation example switching between a navigation bar and a navigation rail with an end drawer - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_drawer/adaptive_navigation.gif

Properties

buildanimation_durationclass-attributeinstance-attribute

animation_duration: DurationValue | None = None

The transition time for each destination as it goes between selected and unselected.

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The color of the navigation bar itself.

buildborderclass-attributeinstance-attribute

```
border: Border | None = None
```

TBD

builddestinationsclass-attributeinstance-attribute

```
destinations: list[NavigationBarDestination] = field(
    default_factory=list
)
```

Defines the appearance of the button items that are arrayed within the navigation bar.

Raises:

- **ValueError** - If it does not contain at least two visible destinations.

buildelevationclass-attributeinstance-attribute

```
elevation: Number | None = None
```

The elevation of the navigation bar itself.

buildindicator_colorclass-attributeinstance-attribute

```
indicator_color: ColorValue | None = None
```

The color of the selected destination indicator.

buildindicator_shapeclass-attributeinstance-attribute

```
indicator_shape: OutlinedBorder | None = None
```

The shape of the selected destination indicator.

buildlabel_behaviorclass-attributeinstance-attribute

```
label_behavior: NavigationBarLabelBehavior | None = None
```

Defines how the destinations' labels will be laid out and when they'll be displayed.

Can be used to show all labels, show only the selected label, or hide all labels.

Defaults to `NavigationBarLabelBehavior.ALWAYS_SHOW`.

buildlabel_paddingclass-attributeinstance-attribute

```
label_padding: PaddingValue | None = None
```

The padding around the `NavigationBarDestination.label`.

buildoverlay_colorclass-attributeinstance-attribute

```
overlay_color: ControlStateValue[ColorValue] | None = None
```

The highlight color of the `NavigationBarDestination` in various `ControlState` states.

The following `ControlState` values are supported: `PRESSED`, `HOVERED` and `FOCUSED`.

buildselected_indexclass-attributeinstance-attribute

```
selected_index: int = 0
```

The index into destinations for the current selected `NavigationBarDestination` or `None` if no destination is selected.

buildshadow_colorclass-attributeinstance-attribute

```
shadow_color: ColorValue | None = None
```

The color used for the drop shadow to indicate elevation.

Events

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[NavigationBar] | None = None
```

Called when selected destination changed.

NavigationBarDestination

Defines the appearance of the button items that are arrayed within the navigation bar.

The value must be a list of two or more NavigationBarDestination instances.

Inherits: AdaptiveControl

Properties

- bgcolor - The color of this destination.
- icon - The name of the icon or Control of the destination.
- label - The text label that appears below the icon of this NavigationBarDestination.
- selected_icon - The name of alternative icon or Control displayed when this destination is selected.

Properties

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The color of this destination.

buildiconinstance-attribute

icon: IconDataOrControl

The name of the icon or Control of the destination.

Example with icon name:

```
icon=ft.Icons.BOOKMARK
```

Example with Control:

```
icon=ft.Icon(ft.Icons.BOOKMARK)
```

If selected_icon is provided, this will only be displayed when the destination is not selected.

To make the NavigationBar more accessible, consider choosing an icon with a stroked and filled version, such as ft.Icons.CLOUD and ft.Icons.CLOUD_QUEUE. The icon should be set to the stroked version and selected_icon to the filled version.

buildlabelclass-attributeinstance-attribute

label: str | None = None

The text label that appears below the icon of this NavigationBarDestination.

buildselected_iconclass-attributeinstance-attribute

selected_icon: IconDataOrControl | None = None

The name of alternative icon or Control displayed when this destination is selected.

Example with icon name:

```
selected_icon=ft.Icons.BOOKMARK
```

Example with Control:

```
selected_icon=ft.Icon(ft.Icons.BOOKMARK)
```

If this icon is not provided, the NavigationBar will display icon in either state.

NavigationDrawer

Material Design Navigation Drawer component.

Navigation Drawer is a panel that slides in horizontally from the left or right edge of the view to show primary destinations in an app.

Example:

```
ft.NavigationDrawer(  
    tile_padding=ft.Padding(top=10),  
    controls=[  
        ft.NavigationDrawerDestination(  
            icon=ft.Icons.HOME_OUTLINED,  
            label="Item 1",  
        ),  
        ft.NavigationDrawerDestination(  
            icon=ft.Icons.MAIL_OUTLINED,  
            label="Item 2",  
        ),  
        ft.NavigationDrawerDestination(  
            icon=ft.Icons.SETTINGS_OUTLINED,  
            label="Item 3",  
        ),  
    ],  
)
```

Image: NavigationDrawer - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_drawer/image_for_docs.png

Navigation drawer extended

Inherits: `AdaptiveControl`

Properties

- `bgcolor` - The color of this navigation drawer.
- `controls` - The list of items to display in this drawer.
- `elevation` - The elevation of this navigation drawer.
- `indicator_color` - The color of the selected destination indicator.
- `indicator_shape` - The shape of the selected destination indicator.
- `selected_index` - The index for the currently selected `NavigationDrawerDestination`.
- `shadow_color` - The color used for the drop shadow to indicate elevation.
- `tile_padding` - Defines the padding around destination tiles inside the drawer.
- `width` - Width of the drawer in logical pixels.

Events

- `on_change` - Called when the selected destination changed.
- `on_dismiss` - Called when this drawer is dismissed.

Examples

Start Position

`play_arrow` Try Online

```
import flet as ft  
  
def main(page: ft.Page):  
    async def handle_show_drawer():  
        await page.show_drawer()  
  
    def handle_dismissal(e: ft.Event[ft.NavigationDrawer]):  
        print("Drawer dismissed!")  
  
    async def handle_change(e: ft.Event[ft.NavigationDrawer]):  
        print(f"Selected Index changed: {e.control.selected_index}")  
        await page.close_drawer()  
  
    page.drawer = ft.NavigationDrawer(  
        controls=[  
            ft.NavigationDrawerDestination(  
                icon=ft.Icons.HOME_OUTLINED,  
                label="Item 1",  
            ),  
            ft.NavigationDrawerDestination(  
                icon=ft.Icons.MAIL_OUTLINED,  
                label="Item 2",  
            ),  
            ft.NavigationDrawerDestination(  
                icon=ft.Icons.SETTINGS_OUTLINED,  
                label="Item 3",  
            ),  
        ],  
        on_change=handle_change,  
        on_dismiss=handle_dismissal,  
    )
```

```

on_dismiss=handle_dismissal,
on_change=handle_change,
controls=[
    ft.Container(height=12),
    ft.NavigationDrawerDestination(
        label="Item 1",
        icon=ft.Icons.DOOR_BACK_DOOR_OUTLINED,
        selected_icon=ft.Icon(ft.Icons.DOOR_BACK_DOOR),
    ),
    ft.Divider(thickness=2),
    ft.NavigationDrawerDestination(
        icon=ft.Icon(ft.Icons.MAIL_OUTLINED),
        label="Item 2",
        selected_icon=ft.Icons.MAIL,
    ),
    ft.NavigationDrawerDestination(
        icon=ft.Icon(ft.Icons.PHONE_OUTLINED),
        label="Item 3",
        selected_icon=ft.Icons.PHONE,
    ),
],
)

page.add(
    ft.SafeArea(
        content=ft.Button(
            content="Show drawer",
            on_click=handle_show_drawer,
        )
    )
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Start-aligned navigation drawer example -

https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_drawer/position_start.gif

End Position

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    async def handle_show_drawer():
        await page.show_end_drawer()

    def handle_dismissal(e: ft.Event[ft.NavigationDrawer]):
        print("Drawer dismissed!")

    async def handle_change(e: ft.Event[ft.NavigationDrawer]):
        print(f"Selected Index changed: {e.control.selected_index}")
        await page.close_end_drawer()

    page.end_drawer = ft.NavigationDrawer(
        on_dismiss=handle_dismissal,
        on_change=handle_change,
        controls=[
            ft.NavigationDrawerDestination(
                icon=ft.Icons.ADD_TO_HOME_SCREEN_SHARP,
                label="Item 1",
            ),
            ft.NavigationDrawerDestination(
                icon=ft.Icon(ft.Icons.ADD_COMMENT),
                label="Item 2",
            ),
        ],
    )

    page.add(
        ft.SafeArea(
            content=ft.Button(
                content="Show end drawer",
                on_click=handle_show_drawer,
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: End-aligned navigation drawer example -

https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_drawer/position_end.gif

Theming

play_arrow Try Online

```
import flet as ft

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT
    page.theme = ft.Theme(
        color_scheme_seed=ft.Colors.INDIGO,
        navigation_drawer_theme=ft.NavigationDrawerTheme(
            bgcolor=ft.Colors.RED_300,
            indicator_color=ft.Colors.INDIGO_100,
            icon_theme={
                ft.ControlState.DEFAULT: ft.IconTheme(
                    color=ft.Colors.INDIGO_900,
                    size=22,
                ),
                ft.ControlState.SELECTED: ft.IconTheme(
                    color=ft.Colors.YELLOW,
                    size=28,
                ),
            },
        ),
        label_text_style={
            ft.ControlState.DEFAULT: ft.TextStyle(
                color=ft.Colors.YELLOW,
                size=11,
                italic=True,
            ),
            ft.ControlState.SELECTED: ft.TextStyle(
                color=ft.Colors.INDIGO_900,
                size=15,
                weight=ft.FontWeight.BOLD,
            ),
        },
    ),
)

page.drawer = ft.NavigationDrawer(
    controls=[
        ft.Container(
            padding=ft.Padding.only(left=28, top=20, bottom=12),
            content=ft.Text(
                "Workspace", theme_style=ft.TextThemeStyle.TITLE_MEDIUM
            ),
        ),
        ft.NavigationDrawerDestination(
            icon=ft.Icons.DASHBOARD_OUTLINED,
            selected_icon=ft.Icons.DASHBOARD,
            label="Dashboard",
        ),
        ft.NavigationDrawerDestination(
            icon=ft.Icons.NOTIFICATIONS_OUTLINED,
            selected_icon=ft.Icons.NOTIFICATIONS,
            label="Notifications",
        ),
        ft.NavigationDrawerDestination(
            icon=ft.Icons.SETTINGS_OUTLINED,
            selected_icon=ft.Icons.SETTINGS,
            label="Settings",
        ),
    ],
)

async def handle_show_drawer(e: ft.Event[ft.Button]):
    await page.show_drawer()

page.add(
    ft.SafeArea(
        content=ft.Button(
            "Show drawer",
            icon=ft.Icons.MENU,
            on_click=handle_show_drawer,
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)
```

Image: Themed navigation drawer example - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_drawer/theming.gif

Adaptive Navigation

This example switches between a `NavigationBar` on narrow layouts and a `NavigationRail` with an end `NavigationDrawer` on wider layouts.

play_arrowTry Online

```
import flet as ft

DESTINATIONS = [
    ("Messages", ft.Icons.WIDGETS_OUTLINED, ft.Icons.WIDGETS),
    ("Profile", ft.Icons.FORMAT_PAINT_OUTLINED, ft.Icons.FORMAT_PAINT),
    ("Settings", ft.Icons.SETTINGS_OUTLINED, ft.Icons.SETTINGS),
]

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT
    page.padding = 0

    screen_index = 0

    def build_page_index_text() -> ft.Text:
        return ft.Text(f"Page Index = {screen_index}", size=24)

    def set_screen(index: int):
        nonlocal screen_index
        screen_index = index
        render()

    def build_navigation_bar() -> ft.NavigationBar:
        def handle_nav_bar_change(e: ft.Event[ft.NavigationBar]):
            set_screen(e.control.selected_index)

        return ft.NavigationBar(
            selected_index=screen_index,
            on_change=handle_nav_bar_change,
            destinations=[
                ft.NavigationBarDestination(
                    label=label,
                    icon=icon,
                    selected_icon=selected_icon,
                )
                for label, icon, selected_icon in DESTINATIONS
            ],
        )

    def build_navigation_rail() -> ft.NavigationRail:
        def handle_nav_rail_change(e: ft.Event[ft.NavigationRail]):
            if e.control.selected_index is not None:
                set_screen(e.control.selected_index)

        return ft.NavigationRail(
            min_width=50,
            selected_index=screen_index,
            use_indicator=True,
            on_change=handle_nav_rail_change,
            destinations=[
                ft.NavigationRailDestination(
                    label=label,
                    icon=icon,
                    selected_icon=selected_icon,
                )
                for label, icon, selected_icon in DESTINATIONS
            ],
        )

    def build_end_drawer() -> ft.NavigationDrawer:
        async def handle_drawer_change(e: ft.Event[ft.NavigationDrawer]):
            set_screen(e.control.selected_index)
            await page.close_end_drawer()

        return ft.NavigationDrawer(
            selected_index=screen_index,
            on_change=handle_drawer_change,
            controls=[
                ft.Container(
                    padding=ft.Padding.only(left=28, top=16, right=16, bottom=10),
                    content=ft.Text(
                        "Header", theme_style=ft.TextThemeStyle.TITLE_SMALL
                    ),
                ),
                *[
                    ft.NavigationDrawerDestination(
                        label=label,
                        icon=icon,
                        selected_icon=selected_icon,
                    )
                    for label, icon, selected_icon in DESTINATIONS
                ]
            ]
        )
```

```

        ],
    ),
)

def build_bottom_bar_layout() -> ft.SafeArea:
    return ft.SafeArea(
        expand=True,
        content=ft.Container(
            expand=True,
            alignment=ft.Alignment.CENTER,
            content=build_page_index_text(),
        ),
    )

def build_drawer_layout() -> ft.SafeArea:
    async def open_drawer(e: ft.Event[ft.Button]):
        await page.show_end_drawer()

    return ft.SafeArea(
        expand=True,
        avoid_intrusions_top=False,
        avoid_intrusions_bottom=False,
        content=ft.Row(
            expand=True,
            controls=[
                ft.Container(
                    padding=ft.Padding.symmetric(horizontal=5),
                    content=build_navigation_rail(),
                ),
                ft.VerticalDivider(thickness=1, width=1),
                ft.Column(
                    expand=True,
                    alignment=ft.MainAxisAlignment.SPACE_EVENLY,
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    controls=[
                        build_page_index_text(),
                        ft.Button("Open Drawer", on_click=open_drawer),
                    ],
                ),
            ],
        ),
    )

def render():
    page.clean()
    if (page.width or page.window.width) >= 450: # wide layout
        page.navigation_bar = None
        page.end_drawer = build_end_drawer()
        page.add(build_drawer_layout())
    else: # narrow layout
        page.end_drawer = None
        page.navigation_bar = build_navigation_bar()
        page.add(build_bottom_bar_layout())
    page.update()

page.on_resize = lambda e: render()
render()

if __name__ == "__main__":
    ft.run(main)

```

Image: Adaptive navigation example switching between a navigation bar and a navigation rail with an end drawer - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_drawer/adaptive_navigation.gif

Properties

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The color of this navigation drawer.

buildcontrolsclass-attributeinstance-attribute

controls: list[Control] = field(default_factory=list)

The list of items to display in this drawer.

The list typically contains NavigationDrawerDestination items and/or other controls such as headlines, padding containers, and dividers.

buildelevationclass-attributeinstance-attribute

elevation: Number | None = None

The elevation of this navigation drawer.

buildindicator_colorclass-attributeinstance-attribute

```
indicator_color: ColorValue | None = None
```

The color of the selected destination indicator.

buildindicator_shapeclass-attributeinstance-attribute

```
indicator_shape: OutlinedBorder | None = None
```

The shape of the selected destination indicator.

buildselected_indexclass-attributeinstance-attribute

```
selected_index: int = 0
```

The index for the currently selected `NavigationDrawerDestination`.

A valid `selected_index` is an integer between 0 and the number of destinations minus 1. For an invalid value, for example -1, all destinations will appear unselected.

buildshadow_colorclass-attributeinstance-attribute

```
shadow_color: ColorValue | None = None
```

The color used for the drop shadow to indicate elevation.

buildtile_paddingclass-attributeinstance-attribute

```
tile_padding: PaddingValue = field(
    default_factory=lambda: Padding.symmetric(horizontal=12)
)
```

Defines the padding around destination tiles inside the drawer.

buildwidthclass-attributeinstance-attribute

```
width: Number | None = None
```

Width of the drawer in logical pixels. When `None`, falls back to the Material default (~304dp).

Events

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[NavigationDrawer] | None = (
    None
)
```

Called when the selected destination changed.

bolton_dismissclass-attributeinstance-attribute

```
on_dismiss: ControlEventHandler[NavigationDrawer] | None = (
    None
)
```

Called when this drawer is dismissed.

NavigationDrawerDestination

A NavigationDrawer destination.

Inherits: Control

Properties

- bgcolor - The background color of the whole rectangular area behind this drawer destination.
- icon - An icon name (or Control) that's displayed for this destination.
- label - The label that appears alongside the icon of this destination.
- selected_icon - An icon name (or Control) that's displayed for this destination when selected.

Properties

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The background color of the whole rectangular area behind this drawer destination.

To customize only the indicator color consider using NavigationDrawer.indicator_color.

If it is None, no background color is set for this destination and bgcolor will be visible.

buildiconinstance-attribute

icon: IconDataOrControl

An icon name (or Control) that's displayed for this destination.

If selected_icon is provided, icon will only be displayed when this destination is not selected.

The icon will use NavigationDrawerTheme.icon_theme. If this is None, the default IconTheme would use a size of 24.0 and ColorScheme.on_surface_variant.

buildlabelinstance-attribute

label: StrOrControl

The label that appears alongside the icon of this destination.

buildselected_iconclass-attributeinstance-attribute

selected_icon: IconDataOrControl | None = None

An icon name (or Control) that's displayed for this destination when selected.

If provided, this destination will fade from icon to selected_icon when this destination goes from unselected to selected state.

If not provided, icon will be displayed for both selected and unselected states.

The icon will use NavigationDrawerTheme.icon_theme with ControlState.SELECTED. If this is None, the default IconTheme would use a size of 24.0 and ColorScheme.on_secondary_container.

NavigationRail

A material widget that is meant to be displayed at the left or right of an app to navigate between a small number of views, typically between three and five.

Example:

```
ft.NavigationRail(  
    selected_index=0,  
    height=200,  
    width=100,  
    destinations=[  
        ft.NavigationRailDestination(icon=ft.Icons.STAR, label="Star"),  
        ft.NavigationRailDestination(icon=ft.Icon(ft.Icons.ADD), label="Add"),  
        ft.NavigationRailDestination(icon=ft.Icons.DELETE, label=ft.Text("Delete"))  
    ],  
)
```

Image: NavigationRail - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_rail/image_for_docs.png

Navigation rail extended

Inherits: `LayoutControl`

Properties

- `bgcolor` - Sets the color of the Container that holds all of this `NavigationRail`'s contents.
- `destinations` - Defines the appearance of the button items that are arrayed within the navigation rail.
- `elevation` - Controls the size of the shadow below the `NavigationRail`.
- `extended` - Indicates that the `NavigationRail` should be in the extended state.
- `group_alignment` - The vertical alignment for the group of destinations within the rail.
- `indicator_color` - The color of the navigation rail's indicator.
- `indicator_shape` - The shape of the navigation rail's indicator.
- `label_type` - Defines the layout and behavior of the labels for the default, unextended navigation rail.
- `leading` - An optional leading control in the rail that is placed above the destinations.
- `min_extended_width` - The final width when the animation is complete for setting extended to `True`.
- `min_width` - The smallest possible width for the rail regardless of the destination's icon or label size.
- `pin_leading_to_top` - Whether to pin the leading control to the top of the rail.
- `pin_trailing_to_bottom` - Whether to pin the trailing control to the bottom of the rail.
- `scrollable` - Whether the main group of destinations should become scrollable when vertical space is insufficient.
- `selected_index` - The index into destinations for the current selected `NavigationRailDestination` or `None` if no destination is selected.
- `selected_label_text_style` - The `TextStyle` of a destination's label when it is selected.
- `trailing` - An optional trailing control in the rail that is placed below the destinations.
- `unselected_label_text_style` - The `TextStyle` of a destination's label when it is not selected.
- `use_indicator` - Whether to add a rounded navigation indicator behind the selected destination's icon.

Events

- `on_change` - Called when selected destination changed.

Examples

NavigationRail

`play_arrow` Try Online

```
import flet as ft  
def main(page: ft.Page):
```

```

rail = ft.NavigationRail(
    selected_index=0,
    label_type=ft.NavigationRailLabelType.ALL,
    min_width=100,
    min_extended_width=400,
    group_alignment=-0.9,
    on_change=lambda e: print("Selected destination:", e.control.selected_index),
    leading=ft.FloatingActionButton(
        icon=ft.Icons.CREATE,
        content="Add",
        on_click=lambda e: print("FAB clicked!"),
    ),
    destinations=[
        ft.NavigationRailDestination(
            icon=ft.Icons.FAVORITE_BORDER,
            selected_icon=ft.Icons.FAVORITE,
            label="First",
        ),
        ft.NavigationRailDestination(
            icon=ft.Icon(ft.Icons.BOOKMARK_BORDER),
            selected_icon=ft.Icon(ft.Icons.BOOKMARK),
            label="Second",
        ),
        ft.NavigationRailDestination(
            icon=ft.Icons.SETTINGS_OUTLINED,
            selected_icon=ft.Icon(ft.Icons.SETTINGS),
            label=ft.Text("Settings"),
        ),
    ],
)

page.add(
    ft.SafeArea(
        expand=True,
        content=ft.Row(
            expand=True,
            controls=[
                ft.SelectionArea(content=rail),
                ft.VerticalDivider(width=1),
                ft.Column(
                    alignment=ft.MainAxisAlignment.START,
                    expand=True,
                    controls=[ft.Text("Body!")],
                ),
            ],
        ),
    ),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_rail/basic.png

Adaptive Navigation

This example switches between a `NavigationBar` on narrow layouts and a `NavigationRail` with an end `NavigationDrawer` on wider layouts.

`play_arrow` Try Online

```

import flet as ft

DESTINATIONS = [
    ("Messages", ft.Icons.WIDGETS_OUTLINED, ft.Icons.WIDGETS),
    ("Profile", ft.Icons.FORMAT_PAINT_OUTLINED, ft.Icons.FORMAT_PAINT),
    ("Settings", ft.Icons.SETTINGS_OUTLINED, ft.Icons.SETTINGS),
]

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT
    page.padding = 0

    screen_index = 0

    def build_page_index_text() -> ft.Text:
        return ft.Text(f"Page Index = {screen_index}", size=24)

    def set_screen(index: int):
        nonlocal screen_index
        screen_index = index
        render()

    def build_navigation_bar() -> ft.NavigationBar:
        def handle_nav_bar_change(e: ft.Event[ft.NavigationBar]):

```

```

        set_screen(e.control.selected_index)

    return ft.NavigationBar(
        selected_index=screen_index,
        on_change=handle_nav_bar_change,
        destinations=[
            ft.NavigationBarDestination(
                label=label,
                icon=icon,
                selected_icon=selected_icon,
            )
            for label, icon, selected_icon in DESTINATIONS
        ],
    )

def build_navigation_rail() -> ft.NavigationRail:
    def handle_nav_rail_change(e: ft.Event[ft.NavigationRail]):
        if e.control.selected_index is not None:
            set_screen(e.control.selected_index)

    return ft.NavigationRail(
        min_width=50,
        selected_index=screen_index,
        use_indicator=True,
        on_change=handle_nav_rail_change,
        destinations=[
            ft.NavigationRailDestination(
                label=label,
                icon=icon,
                selected_icon=selected_icon,
            )
            for label, icon, selected_icon in DESTINATIONS
        ],
    )

def build_end_drawer() -> ft.NavigationDrawer:
    async def handle_drawer_change(e: ft.Event[ft.NavigationDrawer]):
        set_screen(e.control.selected_index)
        await page.close_end_drawer()

    return ft.NavigationDrawer(
        selected_index=screen_index,
        on_change=handle_drawer_change,
        controls=[
            ft.Container(
                padding=ft.Padding.only(left=28, top=16, right=16, bottom=10),
                content=ft.Text(
                    "Header", theme_style=ft.TextThemeStyle.TITLE_SMALL
                ),
            ),
            *[
                ft.NavigationDrawerDestination(
                    label=label,
                    icon=icon,
                    selected_icon=selected_icon,
                )
                for label, icon, selected_icon in DESTINATIONS
            ],
        ],
    )

def build_bottom_bar_layout() -> ft.SafeArea:
    return ft.SafeArea(
        expand=True,
        content=ft.Container(
            expand=True,
            alignment=ft.Alignment.CENTER,
            content=build_page_index_text(),
        ),
    )

def build_drawer_layout() -> ft.SafeArea:
    async def open_drawer(e: ft.Event[ft.Button]):
        await page.show_end_drawer()

    return ft.SafeArea(
        expand=True,
        avoid_intrusions_top=False,
        avoid_intrusions_bottom=False,
        content=ft.Row(
            expand=True,
            controls=[
                ft.Container(
                    padding=ft.Padding.symmetric(horizontal=5),
                    content=build_navigation_rail(),
                ),
                ft.VerticalDivider(thickness=1, width=1),
            ],
        ),
    )

```

```

        ft.Column(
            expand=True,
            alignment=ft.MainAxisAlignment.SPACE_EVENLY,
            horizontal_alignment=ft.CrossAxisAlignment.CENTER,
            controls=[
                build_page_index_text(),
                ft.Button("Open Drawer", on_click=open_drawer),
            ],
        ),
    ],
),
)

def render():
    page.clean()
    if (page.width or page.window.width) >= 450: # wide layout
        page.navigation_bar = None
        page.end_drawer = build_end_drawer()
        page.add(build_drawer_layout())
    else: # narrow layout
        page.end_drawer = None
        page.navigation_bar = build_navigation_bar()
        page.add(build_bottom_bar_layout())
    page.update()

page.on_resize = lambda e: render()
render()

if __name__ == "__main__":
    ft.run(main)

```

Image: Adaptive navigation example switching between a navigation bar and a navigation rail with an end drawer - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/navigation_drawer/adaptive_navigation.gif

Properties

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

Sets the color of the Container that holds all of this NavigationRail's contents.

builddestinationsclass-attributeinstance-attribute

```

destinations: list[NavigationRailDestination] = field(
    default_factory=list
)

```

Defines the appearance of the button items that are arrayed within the navigation rail.

The value must be a list of two or more NavigationRailDestination instances.

buildelevationclass-attributeinstance-attribute□□

elevation: Number | None = None

Controls the size of the shadow below the NavigationRail.

Defaults to 0.0.

Raises:

- ValueError - If it is not greater than or equal to 0.

buildextendedclass-attributeinstance-attribute

extended: bool = False

Indicates that the NavigationRail should be in the extended state.

The extended state has a wider rail container, and the labels are positioned next to the icons. min_extended_width can be used to set the minimum width of the rail when it is in this state.

The rail will implicitly animate between the extended and normal state.

If the rail is going to be in the extended state, then label_type should be set to NavigationRailLabelType.NONE

buildgroup_alignmentclass-attributeinstance-attribute

group_alignment: Number | None = None

The vertical alignment for the group of destinations within the rail.

The `NavigationRailDestinations` are grouped together with the trailing widget, between the leading widget and the bottom of the rail.

The value must be between -1.0 and 1.0.

If `group_alignment` is -1.0, then the items are aligned to the top. If `group_alignment` is 0.0, then the items are aligned to the center. If `group_alignment` is 1.0, then the items are aligned to the bottom.

Defaults to -1.0.

buildindicator_colorclass-attributeinstance-attribute

`indicator_color: ColorValue | None = None`

The color of the navigation rail's indicator.

buildindicator_shapeclass-attributeinstance-attribute

`indicator_shape: OutlinedBorder | None = None`

The shape of the navigation rail's indicator.

Defaults to `StadiumBorder`.

buildlabel_typeclass-attributeinstance-attribute

`label_type: NavigationRailLabelType | None = None`

Defines the layout and behavior of the labels for the default, unextended navigation rail.

When a navigation rail is extended, the labels are always shown.

Defaults to `None` - no labels are shown.

buildleadingclass-attributeinstance-attribute

`leading: Control | None = None`

An optional leading control in the rail that is placed above the destinations.

Its location is not affected by `group_alignment`.

Typically a `FloatingActionButton`, but may also be a non-button, such as a logo.

buildmin_extended_widthclass-attributeinstance-attribute

`min_extended_width: Number | None = None`

The final width when the animation is complete for setting `extended` to `True`.

Defaults to 256.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildmin_widthclass-attributeinstance-attribute

`min_width: Number | None = None`

The smallest possible width for the rail regardless of the destination's icon or label size.

Defaults to 72.

This value also defines the min width and min height of the destinations.

To make a compact rail, set this to 56 and set `label_type` to `NavigationRailLabelType.NONE`

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildpin_leading_to_topclass-attributeinstance-attribute

`pin_leading_to_top: bool = True`

Whether to pin the leading control to the top of the rail.

If `False`, the leading control becomes part of the main group together with the destinations and participates in scrolling/alignment.

buildpin_trailing_to_bottomclass-attributeinstance-attribute

`pin_trailing_to_bottom: bool = False`

Whether to pin the trailing control to the bottom of the rail.

If False, the trailing control becomes part of the main group together with the destinations and participates in scrolling/alignment.

buildscrollableclass-attributeinstance-attribute

`scrollable: bool = False`

Whether the main group of destinations should become scrollable when vertical space is insufficient.

When `pin_leading_to_top` or `pin_trailing_to_bottom` are False, the respective controls also become part of the scrollable group.

buildselected_indexclass-attributeinstance-attribute

`selected_index: int | None = None`

The index into destinations for the current selected `NavigationRailDestination` or None if no destination is selected.

buildselected_label_text_styleclass-attributeinstance-attribute

`selected_label_text_style: TextStyle | None = None`

The `TextStyle` of a destination's label when it is selected.

When a destination is not selected, `unselected_label_text_style` will instead be used.

buildtrailingclass-attributeinstance-attribute

`trailing: Control | None = None`

An optional trailing control in the rail that is placed below the destinations.

Its location is affected by `group_alignment`.

This is commonly a list of additional options or destinations that is usually only rendered when `extended=True`.

buildunselected_label_text_styleclass-attributeinstance-attribute

`unselected_label_text_style: TextStyle | None = None`

The `TextStyle` of a destination's label when it is not selected.

When a destination is selected, `selected_label_text_style` will instead be used.

builduse_indicatorclass-attributeinstance-attribute

`use_indicator: bool | None = None`

Whether to add a rounded navigation indicator behind the selected destination's icon.

The indicator's shape will be circular if `label_type` is `NavigationRailLabelType.NONE`, or a `StadiumBorder` if `label_type` is `NavigationRailLabelType.ALL` or `NavigationRailLabelType.SELECTED`.

If None, defaults to `NavigationRailTheme.use_indicator`. If that is also None, defaults to `Theme.use_material3`.

Events

bolton_changeclass-attributeinstance-attribute

`on_change: ControlEventHandler[NavigationRail] | None = None`

Called when selected destination changed.

NavigationRailDestination

Represents a destination in a NavigationRail.

Inherits: Control

Properties

- `icon` - The name of the icon or Control of the destination.
- `indicator_color` - The color of the `indicator_shape` when this destination is selected.
- `indicator_shape` - The shape of the selection indicator.
- `label` - A string or Control representing the destination's label.
- `padding` - The amount of space to inset the destination item.
- `selected_icon` - The name of alternative icon or Control displayed when this destination is selected.

Properties

`buildiconinstance-attribute`

`icon: IconDataOrControl`

The name of the icon or Control of the destination.

If `selected_icon` is provided, this will only be displayed when the destination is not selected.

To make the NavigationRail more accessible, consider choosing an icon with a stroked and filled version, such as `Icons.CLOUD` and `Icons.CLOUD_QUEUE`. The icon should be set to the stroked version and `selected_icon` to the filled version.

`buildindicator_colorclass-attributeinstance-attribute`

`indicator_color: ColorValue | None = None`

The color of the `indicator_shape` when this destination is selected.

`buildindicator_shapeclass-attributeinstance-attribute`

`indicator_shape: OutlinedBorder | None = None`

The shape of the selection indicator.

`buildlabelclass-attributeinstance-attribute`

`label: StrOrControl | None = None`

A string or Control representing the destination's label.

`buildpaddingclass-attributeinstance-attribute`

`padding: PaddingValue | None = None`

The amount of space to inset the destination item.

`buildselected_iconclass-attributeinstance-attribute`

`selected_icon: IconDataOrControl | None = None`

The name of alternative icon or Control displayed when this destination is selected.

If this icon is not provided, the NavigationRail will display icon in either state.

OutlinedButton

Outlined buttons are medium-emphasis buttons. They contain actions that are important, but aren't the primary action in an app. Outlined buttons pair well with filled buttons to indicate an alternative, secondary action.

Example:

```
ft.OutlinedButton(content="Outlined button")
```

Image: OutlinedButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/outlined_button/image_for_docs.png

Simple Outlined Button

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `autofocus` - True if the control will be selected as the initial focus.
- `clip_behavior` - The content will be clipped (or not) according to this option.
- `content` - A Control representing custom button content.
- `icon` - An icon to display in this button.
- `icon_color` - Icon color.
- `style`
- `url` - The URL to open when this button is clicked.

Events

- `on_blur` - Called when this button has lost focus.
- `on_click` - Called when a user clicks this button.
- `on_focus` - Called when this button has received focus.
- `on_hover` - Called when a mouse pointer enters or exists this button's response area.
- `on_long_press` - Called when this button is long-pressed.

Methods

- `focus` - Requests focus for this control.

Examples

OutlinedButton

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "OutlinedButton Example"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.OutlinedButton(content="Outlined button"),
                    ft.OutlinedButton(content="Disabled button", disabled=True),
                ]
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/outlined_button/basic.png

Handling clicks

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "OutlinedButton Example"
    page.theme_mode = ft.ThemeMode.LIGHT

    def handle_button_click(e: ft.Event[ft.OutlinedButton]):
        button.data += 1
        message.value = f"Button clicked {button.data} time(s)"
        page.update()

    button = ft.OutlinedButton(
        content="Button with 'click' event",
        data=0,
        on_click=handle_button_click,
    )
    message = ft.Text()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[button, message],
                alignment=ft.MainAxisAlignment.CENTER,
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                expand=True,
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: handling-clicks - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/outlined_button/handling_clicks.gif

Icons

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "OutlinedButton Example"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.OutlinedButton(
                        content="Button with icon", icon=ft.Icons.CHAIR_OUTLINED
                    ),
                    ft.OutlinedButton(
                        content="Button with colorful icon",
                        icon=ft.Icons.PARK_ROUNDED,
                        icon_color=ft.Colors.GREEN_400,
                    ),
                ]
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: icons - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/outlined_button/icons.png

Custom content

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "OutlinedButton Example"
    page.theme_mode = ft.ThemeMode.LIGHT

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.OutlinedButton(
                        width=150,
```

Image: custom-content - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/outlined_button/custom_content.png

bolton_blurclass-attributeinstance-attribute

`on_blur: ControlEventHandler[OutlinedButton] | None = None`

Called when this button has lost focus.

bolton_clickclass-attributeinstance-attribute

`on_click: ControlEventHandler[OutlinedButton] | None = None`

Called when a user clicks this button.

bolton_focusclass-attributeinstance-attribute

`on_focus: ControlEventHandler[OutlinedButton] | None = None`

Called when this button has received focus.

bolton_hoverclass-attributeinstance-attribute

`on_hover: ControlEventHandler[OutlinedButton] | None = None`

Called when a mouse pointer enters or exists this button's response area.

data property of event object contains true (string) when cursor enters and false when it exits.

bolton_long_pressclass-attributeinstance-attribute

`on_long_press: (ControlEventHandler[OutlinedButton] | None) = None`

Called when this button is long-pressed.

Methods

deployed_codefocusasync

`focus()`

Requests focus for this control.

OutlinedIconButton

An outlined variant of IconButton.

Outlined icon buttons are medium-emphasis buttons. They're useful when an icon button needs more emphasis than a standard icon button but less than a filled or filled tonal icon button.

Inherits: IconButton

Page

Page is a container for View controls.

A page instance and the root view are automatically created when a new user session started.

Inherits: `BasePage`

Properties

- `auth` - The current authorization context, or `None` if the user is not authorized.
- `browser_context_menu` deprecated - The `BrowserContextMenu` service for the current page.
- `client_ip` - IP address of the connected user.
- `client_user_agent` - Browser details of the connected user.
- `clipboard` deprecated - The `Clipboard` service for the current page.
- `debug` - True if Flutter client of Flet app is running in debug mode.
- `executor` - The executor for the current page.
- `fonts` - Defines the custom fonts to be used in the application.
- `loop` - The event loop for the current page.
- `multi_view` - True if the application is running with multi-view support.
- `multi_views` - The list of multi-views associated with this page.
- `name` - The name of the current page.
- `platform` - The operating system the application is running on.
- `platform_brightness` - The current brightness mode of the host platform.
- `pubsub` - The `PubSub` client for the current page.
- `pwa` - True if the application is running as Progressive Web App (PWA).
- `pyodide` - True if the application is running in Pyodide (WebAssembly) mode.
- `query` - The query parameters of the current page.
- `route` - Gets current app route.
- `session` - The session that this page belongs to.
- `shared_preferences` deprecated - The `SharedPreferences` service for the current page.
- `storage_paths` deprecated - The `StoragePaths` service for the current page.
- `test` - True if the application is running with test mode.
- `url` - The URL of the current page.
- `url_launcher` deprecated - The `UrlLauncher` service for the current page.
- `wasm` - True if the application is running in WebAssembly (WASM) mode.
- `web` - True if the application is running in the web browser.
- `window` - Provides properties/methods/events to monitor and control the app's native OS window.

Events

- `on_app_lifecycle_state_change` - Called when app lifecycle state changes.
- `on_close` - Called when a session has expired after configured amount of time (60 minutes by default).
- `on_connect` - Called when a web user (re-)connects to a page session.
- `on_disconnect` - Called when a web user disconnects from a page session, i.e.
- `on_error` - Called when unhandled exception occurs.

- `on_keyboard_event` - Called when a keyboard key is pressed.
- `on_locale_change` - Called when the locale preferences/settings of the host platform have changed.
- `on_login` - Called upon successful or failed OAuth authorization flow.
- `on_logout` - Called after `page.logout()` call.
- `on_multi_view_add` - TBD
- `on_multi_view_remove` - TBD
- `on_platform_brightness_change` - Called when brightness of app host platform has changed.
- `on_route_change` - Called when page route changes either programmatically, by editing application URL or using browser Back/Forward buttons.
- `on_view_pop` - Called when the user clicks automatic "Back" button in AppBar control.
- `on_views_pop_until` - Called when `pop_views_until` reaches the destination view.

Methods

- `before_event`
- `can_launch_url` deprecated - Checks whether the specified URL can be handled by some app installed on the device.
- `close_in_app_web_view` deprecated - Closes in-app web view opened with `launch_url()`.
- `error` - Report an application error to the current session/client.
- `get_control` - Get a control by its id.
- `get_device_info` - Returns device information.
- `get_upload_url` - Generates presigned upload URL for built-in upload storage:
- `go` deprecated - A helper method that updates `page.route`, calls `page.on_route_change` event handler to update views and finally calls `page.update()`.
- `launch_url` deprecated - Opens a web browser or popup window to a given url.
- `login` - Starts OAuth flow.
- `logout` - Clears current authentication context.
- `navigate` - Navigate to a new route (sync convenience wrapper).
- `pop_views_until` - Pops views from the navigation stack until a view with the given route is found, then delivers result via the `on_views_pop_until` event.
- `push_route` - Pushes a new navigation route to the browser history stack.
- `render` - Render a component tree into controls of the root view.
- `render_views` - Render a component tree as the full list of page views.
- `run_task` - Run handler coroutine as a new Task in the event loop associated with the current page.
- `run_thread` - Run handler function as a new Thread in the executor associated with the current page.
- `schedule_update` - Queue this page for a deferred batched update.
- `set_allowed_device_orientations` - Constrains the allowed orientations for the app when running on a mobile device.
- `update` - Push pending state changes to the client.

Examples

Keyboard events

`play_arrow`Try Online

```
import flet as ft

class ButtonControl(ft.Container):
    def __init__(self, text):
        super().__init__()
        self.content: ft.Text = ft.Text(text)
```

```

        self.border = ft.Border.all(1, ft.Colors.BLACK_54)
        self.border_radius = 3
        self.bgcolor = "0x09000000"
        self.padding = 10
        self.visible = False

def main(page: ft.Page):
    page.spacing = 50
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def on_keyboard(e: ft.KeyboardEvent):
        key.content.value = e.key
        key.visible = True
        shift.visible = e.shift
        ctrl.visible = e.ctrl
        alt.visible = e.alt
        meta.visible = e.meta
        page.update()

    page.on_keyboard_event = on_keyboard

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        "Press any key with a combination of CTRL, ALT, SHIFT "
                        "and META keys..."
                    ),
                    ft.Row(
                        controls=[
                            key := ButtonControl(""),
                            shift := ButtonControl("Shift"),
                            ctrl := ButtonControl("Control"),
                            alt := ButtonControl("Alt"),
                            meta := ButtonControl("Meta"),
                        ],
                        alignment=ft.MainAxisAlignment.CENTER,
                    ),
                ]
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Mobile device orientation configuration

Shows how to lock your app to specific device orientations (e.g., portrait up, landscape right) and listen for orientation changes on mobile devices.

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page) -> None:
    page.title = "Device orientation lock"
    page.appbar = ft.AppBar(
        title=ft.Text("Device orientation Playground"),
        center_title=True,
        bgcolor=ft.Colors.BLUE,
    )

    def handle_media_change(e: ft.PageMediaData) -> None:
        page.show_dialog(
            ft.SnackBar(
                f"I see you rotated the device to {e.orientation.name} orientation. ■",
                action="Haha!",
                duration=ft.Duration(seconds=3),
            )
        )

    page.on_media_change = handle_media_change

    async def on_checkbox_change(e: ft.Event[ft.Checkbox]) -> None:
        # get selection
        selected = [o for o, checkbox in checkboxes.items() if checkbox.value]
        # apply selection
        await page.set_allowed_device_orientations(selected)

    checkboxes: dict[ft.DeviceOrientation, ft.Checkbox] = {
        orientation: ft.Checkbox(

```

```

        label=orientation.name,
        value=True,
        on_change=on_checkbox_change,
        disabled=not page.platform.is_mobile(), # disabled on non-mobile platforms
    )
    for orientation in list(ft.DeviceOrientation)
}

page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                ft.Text(
                    spans=[
                        # shown only on mobile platforms
                        ft.TextSpan(
                            "Select the orientations that should remain enabled "
                            "for the app. "
                            "If no orientation is selected, "
                            "the system defaults will be used.",
                            visible=page.platform.is_mobile(),
                        ),
                        # shown only on non-mobile platforms
                        ft.TextSpan(
                            "Please open this example on a mobile device instead.",
                            visible=not page.platform.is_mobile(),
                            style=ft.TextStyle(weight=ft.FontWeight.BOLD),
                        ),
                    ],
                ),
                ft.Column(controls=list(checkboxes.values())),
            ]
        )
    )
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

App exit confirm dialog

```

import flet as ft

def main(page: ft.Page):
    def window_event(e: ft.WindowEvent):
        if e.type == ft.WindowEventType.CLOSE:
            page.show_dialog(confirm_dialog)
            page.update()

    page.window.prevent_close = True
    page.window.on_event = window_event

    async def handle_yes_click(e: ft.Event[ft.Button]):
        await page.window.destroy()

    def handle_no_click(e: ft.Event[ft.OutlinedButton]):
        page.pop_dialog()
        page.update()

    confirm_dialog = ft.AlertDialog(
        modal=True,
        title=ft.Text("Please confirm"),
        content=ft.Text("Do you really want to exit this app?"),
        actions=[
            ft.Button(content="Yes", on_click=handle_yes_click),
            ft.OutlinedButton(content="No", on_click=handle_no_click),
        ],
        actions_alignment=ft.MainAxisAlignment.END,
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        'Try exiting this app by clicking window\'s "Close" button!'
                    )
                ]
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)

```


Window hidden on start

A Flet desktop app (Windows, macOS, or Linux) can start with its window hidden. This lets your app perform initial setup (for example, add content, resize or position the window) before showing it to the user.

In the example below, the window is resized and centered before becoming visible:

```
import asyncio

import flet as ft

async def main(page: ft.Page):
    print("Window is hidden on start. Will show after 3 seconds...")
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text("Hello!"),
                ]
            )
        )
    )

    # some configuration that we want to do before showing the window
    page.window.width = 300
    page.window.height = 200
    page.update()
    await page.window.center()

    # wait for 3 seconds before showing the window
    await asyncio.sleep(3)

    # show the window
    page.window.visible = True
    page.update()

if __name__ == "__main__":
    ft.run(main, view=ft.AppView.FLET_APP_HIDDEN)
```

If you need this feature when packaging a desktop app using flet build, see this.

Semantics debugger

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def on_keyboard(e: ft.KeyboardEvent):
        if e.shift and e.key == "S":
            page.show_semantics_debugger = not page.show_semantics_debugger
            page.update()

    page.on_keyboard_event = on_keyboard

    def button_click(e: ft.Event[ft.Button]):
        counter.value = str(int(counter.value) + 1)
        page.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    counter := ft.Text("0", size=40),
                    ft.Text("Press Shift+S to toggle semantics debugger"),
                    ft.Button(
                        content="Increment number",
                        icon=ft.Icons.ADD,
                        on_click=button_click,
                    ),
                ]
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Device locale

play_arrowTry Online

```
import flet as ft

async def main(page: ft.Page):
    def format_locales(locales: list[ft.Locale]) -> str:
        """Format locale list for display."""
        return ", ".join(str(loc) for loc in locales)

    def handle_locale_change(e: ft.LocaleChangeEvent):
        page.add(ft.Text(f"Locales changed: {format_locales(e.locales)}"))

    page.on_locale_change = handle_locale_change
    page.scroll = ft.ScrollMode.AUTO
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    initial_locales = (await page.get_device_info()).locales
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(f"Initial locales: {format_locales(initial_locales)}"),
                    ft.Text(
                        "Change your system or browser language to trigger "
                        "on_locale_change.",
                        color=ft.Colors.BLUE,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

buildauthproperty

auth: Authorization | None

The current authorization context, or None if the user is not authorized.

buildbrowser_context_menudeprecatedproperty

Deprecated

Deprecated since version 0.80.0 and scheduled for removal in 0.90.0. Use BrowserContextMenu instead.

The BrowserContextMenu service for the current page.

buildclient_ipclass-attributeinstance-attribute

client_ip: str | None = None

IP address of the connected user.

Note

This property is web- and read-only only.

buildclient_user_agentclass-attributeinstance-attribute

client_user_agent: str | None = None

Browser details of the connected user.

Note

This property is web- and read-only only.

buildclipboarddeprecatedproperty

Deprecated

Deprecated since version 0.80.0 and scheduled for removal in 0.90.0. Use Clipboard instead.

The Clipboard service for the current page.

builddebugclass-attributeinstance-attribute

debug: bool = False

True if Flutter client of Flet app is running in debug mode.

Note

This property is read-only.

buildexecutorproperty

executor: `ThreadPoolExecutor` | `None`

The executor for the current page.

buildfontsclass-attributeinstance-attribute

fonts: `dict[str, str]` | `None` = `None`

Defines the custom fonts to be used in the application.

Value is a dictionary, in which the keys represent the font family name used for reference and the values:

- Key: The font family name used for reference.
- Value: The font source, either an absolute URL or a relative path to a local asset. The following font file formats are supported .ttc, .ttf and .otf.

Usage example here.

buildloopproperty

loop: `asyncio.AbstractEventLoop`

The event loop for the current page.

buildmulti_viewclass-attributeinstance-attribute

multi_view: `bool` = `False`

True if the application is running with multi-view support.

Note

This property is read-only.

buildmulti_viewsclass-attributeinstance-attribute

multi_views: `list[MultiView]` = `field(default_factory=list)`

The list of multi-views associated with this page.

buildnameproperty

name: `str`

The name of the current page.

buildplatformclass-attributeinstance-attribute

platform: `PagePlatform` | `None` = `None`

The operating system the application is running on.

buildplatform_brightnessclass-attributeinstance-attribute

platform_brightness: `Brightness` | `None` = `None`

The current brightness mode of the host platform.

Note

This property is read-only.

buildpubsubproperty

pubsub: `PubSubClient`

The PubSub client for the current page.

buildpwaclass-attributeinstance-attribute

pwa: `bool` = `False`

True if the application is running as Progressive Web App (PWA).

Note

This property is read-only.

buildpyodideclass-attributeinstance-attribute

`pyodide: bool = False`

True if the application is running in Pyodide (WebAssembly) mode.

Note

This property is read-only.

buildqueryproperty

`query: QueryString`

The query parameters of the current page.

buildrouteclass-attributeinstance-attribute

`route: str = '/'`

Gets current app route.

Note

This property is read-only.

buildsessionproperty

`session: Session`

The session that this page belongs to.

buildshared_preferencesdeprecatedproperty

Deprecated

Deprecated since version 0.80.0 and scheduled for removal in 0.90.0. Use `SharedPreferences` instead.

The `SharedPreferences` service for the current page.

buildstorage_pathsdeprecatedproperty

Deprecated

Deprecated since version 0.80.0 and scheduled for removal in 0.90.0. Use `StoragePaths` instead.

The `StoragePaths` service for the current page.

buildtestclass-attributeinstance-attribute

`test: bool = False`

True if the application is running with test mode.

Note

This property is read-only.

buildurlproperty

`url: str | None`

The URL of the current page.

buildurl_launcherdeprecatedproperty

`url_launcher: UrlLauncher`

Deprecated

Deprecated since version 0.80.0 and scheduled for removal in 0.90.0. Use `UrlLauncher` instead.

The `UrlLauncher` service for the current page.

buildwasmclass-attributeinstance-attribute

`wasm: bool = False`

True if the application is running in WebAssembly (WASM) mode.

Note

This property is read-only.

buildwebclass-attributeinstance-attribute

```
web: bool = False
```

True if the application is running in the web browser.

Note

This property is read-only.

buildwindowclass-attributeinstance-attribute

```
window: Window = field(default_factory=lambda: Window())
```

Provides properties/methods/events to monitor and control the app's native OS window.

Events

bolton_app_lifecycle_state_changeclass-attributeinstance-attribute

```
on_app_lifecycle_state_change: (
    EventHandler[AppLifecycleStateChangeEvent] | None
) = None
```

Called when app lifecycle state changes.

bolton_closeclass-attributeinstance-attribute

```
on_close: ControlEventHandler[Page] | None = None
```

Called when a session has expired after configured amount of time (60 minutes by default).

bolton_connectclass-attributeinstance-attribute

```
on_connect: ControlEventHandler[Page] | None = None
```

Called when a web user (re-)connects to a page session.

It is not triggered when an app page is first opened, but is triggered when the page is refreshed, or Flet web client has re-connected after computer was unlocked. This event could be used to detect when a web user becomes "online".

bolton_disconnectclass-attributeinstance-attribute

```
on_disconnect: ControlEventHandler[Page] | None = None
```

Called when a web user disconnects from a page session, i.e. closes browser tab/window.

bolton_errorclass-attributeinstance-attribute

```
on_error: ControlEventHandler[Page] | None = None
```

Called when unhandled exception occurs.

bolton_keyboard_eventclass-attributeinstance-attribute

```
on_keyboard_event: EventHandler[KeyboardEvent] | None = None
```

Called when a keyboard key is pressed.

bolton_locale_changeclass-attributeinstance-attribute

```
on_locale_change: EventHandler[LocaleChangeEvent] | None = (
    None
)
```

Called when the locale preferences/settings of the host platform have changed.

For example, when the user updates device language settings or browser preferred languages.

bolton_loginclass-attributeinstance-attribute

```
on_login: EventHandler[LoginEvent] | None = None
```

Called upon successful or failed OAuth authorization flow.

See Authentication guide for more information and examples.

bolton_logoutclass-attributeinstance-attribute

```
on_logout: ControlEventHandler[Page] | None = None
```

Called after `page.logout()` call.

bolton_multi_view_addclass-attributeinstance-attribute

```
on_multi_view_add: (
    EventHandler[MultiViewAddEvent] | None
) = None
```

TBD

bolton_multi_view_removeclass-attributeinstance-attribute

```
on_multi_view_remove: (
    EventHandler[MultiViewRemoveEvent] | None
) = None
```

TBD

bolton_platform_brightness_change-attributeinstance-attribute

```
on_platform_brightness_change: (
    EventHandler[PlatformBrightnessChangeEvent] | None
) = None
```

Called when brightness of app host platform has changed.

bolton_route_change-attributeinstance-attribute

```
on_route_change: EventHandler[RouteChangeEvent] | None = (
    None
)
```

Called when page route changes either programmatically, by editing application URL or using browser Back/Forward buttons.

bolton_view_pop-attributeinstance-attribute

```
on_view_pop: EventHandler[ViewPopEvent] | None = None
```

Called when the user clicks automatic "Back" button in AppBar control.

bolton_views_pop_until-attributeinstance-attribute

```
on_views_pop_until: (
    EventHandler[ViewsPopUntilEvent] | None
) = None
```

Called when pop_views_until reaches the destination view.

The event carries the result value passed by the caller.

Methods

deployed_codebefore_event

```
before_event(e: ControlEvent)
```

deployed_codecan_launch_urlasynccdeprecated

```
can_launch_url(url: str) -> bool
```

Deprecated

Deprecated since version 0.90.0. Use `UrlLauncher().can_launch_url()` instead.

Checks whether the specified URL can be handled by some app installed on the device.

Parameters:

- url (str) - The URL to check.

Returns:

- bool - True if it is possible to verify that there is a handler available. False if there is no handler available, or the application does not have permission to check. For example: - On recent versions of Android and iOS, this will always return False unless the application has been configuration to allow querying the system for launch support. - In web mode, this will always return False except for a few specific schemes that are always assumed to be supported (such as http(s)), as web pages are never allowed to query installed applications.

deployed_codeclose_in_app_web_viewasynccdeprecated

```
close_in_app_web_view() -> None
```

Deprecated

Deprecated since version 0.90.0. Use `UrlLauncher().close_in_app_web_view()` instead.

Closes in-app web view opened with `launch_url()`.

□ Mobile only.

`deployed_codeerror`

`error(message: str) -> None`

Report an application error to the current session/client.

Parameters:

- `message (str)` - Error message to send.

`deployed_codeget_control`

`get_control(id: int) -> BaseControl | None`

Get a control by its id.

Example

```
def main(page: ft.Page):
    x = ft.IconButton(ft.Icons.ADD)
    page.add(x)
    print(type(page.get_control(x._i)))
```

`deployed_codeget_device_infoasync`

`get_device_info() -> DeviceInfo | None`

Returns device information.

Returns:

- `DeviceInfo | None` - The device information object for the current platform, or `None` if unavailable.

`deployed_codeget_upload_url`

`get_upload_url(file_name: str, expires: int) -> str`

Generates presigned upload URL for built-in upload storage:

- `file_name` - a relative to upload storage path.
- `expires` - a URL time-to-live in seconds.

Example

```
upload_url = page.get_upload_url("dir/filename.ext", 60)
```

To enable built-in upload storage, provide the `upload_dir` argument to `ft.run()` call:

```
ft.run(main, upload_dir="uploads")
```

`deployed_codecodedeprecated`

`go(route: str, skip_route_change_event: bool = False, **kwargs: Any = {}) -> None`

Deprecated

Deprecated since version 0.80.0 and scheduled for removal in 0.90.0. Use `push_route()` instead.

A helper method that updates `page.route`, calls `page.on_route_change` event handler to update views and finally calls `page.update()`.

`deployed_codelaunch_urlasynccodeprecated`

```
launch_url(
    url: str | Url,
    web_popup_window_name: str | UrlTarget | None = None,
    web_popup_window: bool = False,
    web_popup_window_width: int | None = None,
    web_popup_window_height: int | None = None,
) -> None
```

Deprecated

Deprecated since version 0.90.0. Use `UrlLauncher().launch_url()` instead.

Opens a web browser or popup window to a given url.

Parameters:

- `url (str | Url)` - The URL to open.
- `web_popup_window_name (str | UriTarget | None, default: None)` - Window tab/name to open URL in. Use `UriTarget.SELF` for the same browser tab, `UriTarget.BLANK` for a new browser tab (or in external application on a mobile device), or a custom name for a named tab.
- `web_popup_window (bool, default: False)` - Display the URL in a browser popup window.
- `web_popup_window_width (int | None, default: None)` - Popup window width.
- `web_popup_window_height (int | None, default: None)` - Popup window height.

deployed_codeloginasync

```
login(
    provider: OAuthProvider,
    fetch_user: bool = True,
    fetch_groups: bool = False,
    scope: list[str] | None = None,
    saved_token: str | None = None,
    on_open_authorization_url: Callable[
        [str], Coroutine[Any, Any, None]
    ] | None = None,
    complete_page_html: str | None = None,
    redirect_to_page: bool | None = False,
    authorization: type[AT] = AuthorizationImpl,
) -> AT
```

Starts OAuth flow.

See Authentication guide for more information and examples.

deployed_codelogout

```
logout() -> None
```

Clears current authentication context. See Authentication guide for more information and examples.

deployed_codenavigate

```
navigate(route: str, **kwargs: Any = {}) -> None
```

Navigate to a new route (sync convenience wrapper).

Equivalent to `asyncio.create_task(page.push_route(route, **kwargs))`. Use this in synchronous callbacks (e.g. `on_click`) where awaiting is not possible.

Parameters:

- `route (str)` - New navigation route.
- `**kwargs (default: {})` - Additional query string parameters to be added to the route.

deployed_codepop_views_untilasync

```
pop_views_until(route: str, result: Any = None) -> None
```

Pops views from the navigation stack until a view with the given route is found, then delivers result via the `on_views_pop_until` event.

Example

```
import flet as ft

def main(page: ft.Page):
    def on_pop_result(e: ft.ViewsPopUntilEvent):
        page.show_dialog(ft.SnackBar(ft.Text(f"Result: {e.result}")))

    page.on_views_pop_until = on_pop_result

    # ... later, from a deeply nested view:
    async def go_back(ev):
        await page.pop_views_until("/", result="Done!")
```

Parameters:

- `route (str)` - Target route to navigate back to. Must match the route of an existing View in views.
- `result (Any, default: None)` - Optional value delivered to `on_views_pop_until` on the destination view.

Raises:

- `ValueError` - If no view with the given route exists in views.

deployed_codepush_routeasync

`push_route(route: str, **kwargs: Any = {})` -> None

Pushes a new navigation route to the browser history stack. Changing route will fire `page.on_route_change` event handler.

Example

```
import asyncio

import flet as ft

def main(page: ft.Page):
    page.title = "Routes Example"

    def route_change():
        page.views.clear()
        page.views.append(
            ft.View(
                route="/",
                controls=[
                    ft.AppBar(
                        title=ft.Text("Flet app"),
                    ),
                    ft.Button(
                        "Visit Store",
                        on_click=lambda: asyncio.create_task(
                            page.push_route("/store")
                        ),
                    ),
                ],
            )
        )
        if page.route == "/store":
            page.views.append(
                ft.View(
                    route="/store",
                    controls=[
                        ft.AppBar(
                            title=ft.Text("Store"),
                        ),
                        ft.Button(
                            "Go Home",
                            on_click=lambda: asyncio.create_task(
                                page.push_route("/")
                            ),
                        ),
                    ],
                )
            )
        page.update()

    async def view_pop(e):
        if e.view is not None:
            print("View pop:", e.view)
            page.views.remove(e.view)
            top_view = page.views[-1]
            await page.push_route(top_view.route)

    page.on_route_change = route_change
    page.on_view_pop = view_pop

    route_change()

if __name__ == "__main__":
    ft.run(main)
```

Parameters:

- `route (str)` - New navigation route.
- `**kwargs (default: {})` - Additional query string parameters to be added to the route.

deployed_coderender

`render(component: Callable[..., Any], *args: Any = (), **kwargs: Any = {})`

Render a component tree into controls of the root view.

The rendered result replaces `page.views[0].controls`, then triggers initial page update and component update scheduler startup.

Parameters:

- component (Callable[..., Any]) - Component function to render.
- *args (default: ()) - Positional arguments passed to component.
- **kwargs (default: {}) - Keyword arguments passed to component.

deployed_coderender_views

```
render_views(component: Callable[..., Any], *args: Any = (), **kwargs: Any = {})
```

Render a component tree as the full list of page views.

The rendered result replaces page.views, then triggers initial page update and component update scheduler startup.

Parameters:

- component (Callable[..., Any]) - Component function to render.
- *args (default: ()) - Positional arguments passed to component.
- **kwargs (default: {}) - Keyword arguments passed to component.

deployed_coderun_task

```
run_task(handler: Callable[InputT, Awaitable[RetT]], *args: InputT.args = (), **kwargs: InputT.kwargs = {}) -> Future[RetT]
```

Run handler coroutine as a new Task in the event loop associated with the current page.

deployed_coderun_thread

```
run_thread(handler: Callable[InputT, Any], *args: InputT.args = (), **kwargs: InputT.kwargs = {}) -> None
```

Run handler function as a new Thread in the executor associated with the current page.

deployed_codeschedule_update

```
schedule_update()
```

Queue this page for a deferred batched update.

deployed_codeset_allowed_device_orientationsasync

```
set_allowed_device_orientations(  
    orientations: list[DeviceOrientation],  
) -> None
```

Constrains the allowed orientations for the app when running on a mobile device.

Parameters:

- orientations (list[DeviceOrientation]) - A list of allowed device orientations. Set to an empty list to use the system default behavior.

Raises:

- FletUnsupportedPlatformException - If the method is called on a non-mobile platform.

Limitations:

Android: On Android 16 (API 36) or later, this method won't be able to change the orientation of devices with a display width ≥ 600 dp cannot change orientation. For more details see Android 16 docs [here](#). Also, Android limits the orientations to the following combinations:

- [] $\rightarrow$ unspecified
- [PORTRAIT_UP] $\rightarrow$ portrait
- [LANDSCAPE_LEFT] $\rightarrow$ landscape
- [PORTRAIT_DOWN] $\rightarrow$ reversePortrait
- [PORTRAIT_UP, PORTRAIT_DOWN] $\rightarrow$ userPortrait
- [LANDSCAPE_RIGHT] $\rightarrow$ reverseLandscape
- [LANDSCAPE_LEFT, LANDSCAPE_RIGHT] $\rightarrow$ userLandscape
- [PORTRAIT_UP, LANDSCAPE_LEFT, LANDSCAPE_RIGHT] $\rightarrow$ user
- [PORTRAIT_UP, PORTRAIT_DOWN, LANDSCAPE_LEFT, LANDSCAPE_RIGHT] $\rightarrow$ fullUser

- iOS: This setting will only be respected on iPad if multitasking is disabled. You can decide to opt out of multitasking on iPad, then this will work but your app will not support Slide Over and Split View multitasking anymore. Should you decide to opt out of multitasking you can do this by setting "Requires full screen" to true in the Xcode Deployment Info.

deployed_codeupdate

update(*controls: Control = ()) -> None

Push pending state changes to the client.

Parameters:

- *controls (default: ()) - Specific controls to patch. When omitted, patches the whole page state.

Pagelet

Implements the basic Material Design visual layout structure.

Use it for projects that require a "page within a page" layouts with its own AppBar, BottomAppBar, NavigationDrawer, such as demos and galleries.

Image: Pagelet - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/pagelet/image_for_docs.png

Pagelet

Inherits: LayoutControl, AdaptiveControl

Properties

- `appbar` - An AppBar control to display at the top of the Pagelet.
- `bgcolor` - Background color of this Pagelet.
- `bottom_appbar` - A BottomAppBar control to display at the bottom of this Pagelet.
- `bottom_sheet` - The persistent bottom sheet to show information that supplements the primary content of this Pagelet.
- `content` - A child Control contained by this Pagelet.
- `drawer` - A NavigationDrawer control to display as a panel sliding from the start edge of the page.
- `end_drawer` - A NavigationDrawer control to display as a panel sliding from the end edge of the page.
- `floating_action_button` - A FloatingActionButton control to display on top of this Pagelet's content.
- `floating_action_button_location` - Defines the position of the `floating_action_button`.
- `navigation_bar` - A navigation bar (NavigationBar or CupertinoNavigationBar) control to display at the bottom of the Pagelet.

Methods

- `close_drawer` - Close the drawer.
- `close_end_drawer` - Close the end drawer.
- `show_drawer` - Show the drawer.
- `show_end_drawer` - Show the end drawer.

Examples

Pagelet

`play_arrow` Try Online

```
import asyncio

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.MainAxisAlignment.CENTER
    page.vertical_alignment = ft.CrossAxisAlignment.CENTER

    async def handle_show_drawer(e: ft.Event[ft.FloatingActionButton]):
        await pagelet.show_drawer()

    async def handle_show_end_drawer(e: ft.Event[ft.Button]):
        await pagelet.show_end_drawer()
        await asyncio.sleep(3)
        await pagelet.close_end_drawer()

    page.add(
        ft.SafeArea(
            content=(
                pagelet := ft.Pagelet(
                    width=500,
                    height=500,
                    appbar=ft.AppBar(
                        title=ft.Text("Pagelet AppBar"),
                        center_title=True,
```

```

        bgcolor=ft.Colors.RED_500,
    ),
    content=ft.Text("Pagelet Body"),
    bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
    bottom_appbar=ft.BottomAppBar(
        bgcolor=ft.Colors.BLUE,
        shape=ft.CircularRectangleNotchShape(),
        content=ft.Row(
            controls=[
                ft.IconButton(
                    icon=ft.Icons.MENU, icon_color=ft.Colors.WHITE
                ),
                ft.Container(expand=True),
                ft.IconButton(
                    icon=ft.Icons.SEARCH, icon_color=ft.Colors.WHITE
                ),
                ft.IconButton(
                    icon=ft.Icons.FAVORITE, icon_color=ft.Colors.WHITE
                ),
            ],
        ),
    ),
    drawer=ft.NavigationDrawer(
        on_dismiss=lambda e: print("Drawer dismissed"),
        controls=[
            ft.NavigationDrawerDestination(
                icon=ft.Icons.ADD_TO_HOME_SCREEN_SHARP,
                label="Item 1",
            ),
            ft.NavigationDrawerDestination(
                icon=ft.Icons.ADD_COMMENT,
                label="Item 2",
            ),
        ],
    ),
    end_drawer=ft.NavigationDrawer(
        on_dismiss=lambda e: print("End Drawer dismissed"),
        controls=[
            ft.NavigationDrawerDestination(
                icon=ft.Icons.SLOW_MOTION_VIDEO,
                label="Item 3",
            ),
            ft.NavigationDrawerDestination(
                icon=ft.Icons.INSERT_CHART,
                label="Item 4",
            ),
        ],
    ),
    floating_action_button=ft.FloatingActionButton(
        icon=ft.Icons.ADD,
        shape=ft.CircleBorder(),
    ),
    floating_action_button_location=ft.FloatingActionButtonLocation.CENTER_DOCKED,
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - <https://flet.dev/docs/test-images/examples/controls/core/golden/macros/pagelet/basic.png>

Properties

buildappbarclass-attributeinstance-attribute

appbar: AppBar | CupertinoAppBar | None = None

An AppBar control to display at the top of the Pagelet.

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

Background color of this Pagelet.

buildbottom_appbarclass-attributeinstance-attribute

bottom_appbar: BottomAppBar | None = None

A BottomAppBar control to display at the bottom of this Pagelet.

Note

If both the `bottom_appbar` and `navigation_bar` properties are specified, `bottom_appbar` takes precedence and will be displayed.

buildbottom_sheetclass-attributeinstance-attribute

`bottom_sheet: Control | None = None`

The persistent bottom sheet to show information that supplements the primary content of this Pagelet.

buildcontentinstance-attribute

`content: Control`

A child Control contained by this Pagelet.

The control in the content of the Pagelet is positioned at the top-left of the available space between the app bar and the bottom of the Pagelet.

Raises:

- `ValueError` - If it is not visible.

builddrawerclass-attributeinstance-attribute

`drawer: NavigationDrawer | None = None`

A `NavigationDrawer` control to display as a panel sliding from the start edge of the page.

buildend_drawerclass-attributeinstance-attribute

`end_drawer: NavigationDrawer | None = None`

A `NavigationDrawer` control to display as a panel sliding from the end edge of the page.

buildfloating_action_buttonclass-attributeinstance-attribute

`floating_action_button: Control | None = None`

A `FloatingActionButton` control to display on top of this Pagelet's content.

buildfloating_action_button_locationclass-attributeinstance-attribute

`floating_action_button_location: (FloatingActionButtonLocation | OffsetValue | None) = FloatingActionButtonLocation.END_FLOAT`

Defines the position of the `floating_action_button`.

buildnavigation_barclass-attributeinstance-attribute

`navigation_bar: (NavigationBar | CupertinoNavigationBar | None) = None`

A navigation bar (`NavigationBar` or `CupertinoNavigationBar`) control to display at the bottom of the Pagelet.

Note

If both the `navigation_bar` and `bottom_appbar` properties are specified, `navigation_bar` takes precedence and will be displayed.

Methods

deployed_codeclose_drawerasync

`close_drawer()`

Close the drawer.

deployed_codeclose_end_drawerasync

`close_end_drawer()`

Close the end drawer.

deployed_codeshow_drawerasync

`show_drawer()`

Show the drawer.

Raises:

- `ValueError` - If no drawer is defined.

`deployed_codeshow_end_drawerasync`

`show_end_drawer()`

Show the end drawer.

Raises:

- `ValueError` - If no `end_drawer` is defined.

PageView

Displays its child controls one page at a time and lets users swipe between them, similar to a carousel.

It is helpful for onboarding flows, photo carousels, or any scenario where content is divided into discrete full-width pages.

Inherits: `LayoutControl`

Properties

- `clip_behavior` - Defines how pages are clipped if they overflow their bounds.
- `controls` - A list of controls to display, one per page, in the order they should appear.
- `horizontal` - Whether the pages should be arranged and scrolled horizontally.
- `implicit_scrolling` - Whether to allow adjacent pages to render partially before the user scrolls to them, enabling smoother transitions and improved accessibility by allowing focus to move seamlessly between pages during navigation.
- `keep_page` - Whether this page view should restore the most recently viewed page when rebuilt.
- `pad_ends` - Whether to add padding to both ends of the list.
- `reverse` - Whether to reverse the order in which pages are read and swiped.
- `selected_index` - The zero-based index of the currently visible page.
- `snap` - Whether the view should snap to exact page boundaries after a drag.
- `viewport_fraction` - The fraction of the viewport that each page should occupy in the scrolling direction (see `horizontal`).

Events

- `on_change` - Fired when the visible page changes.

Methods

- `go_to_page` - Animates to the page at index.
- `jump_to` - Immediately sets the scroll position to the given value, without animation and without validating whether the value is within bounds.
- `jump_to_page` - Jumps immediately to the page at index without animation, moving the page position from its current value to the given value without animation nor range checking.
- `next_page` - Animates to the next page.
- `previous_page` - Animates to the previous page.

Examples

PageView

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.padding = 0

    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.PageView(
                expand=True,
                viewport_fraction=0.9,
                on_change=lambda e: print(f"Currently viewing page {int(e.data)}"),
                selected_index=1,
                horizontal=True,
                controls=[
                    ft.Container(
                        bgcolor=ft.Colors.INDIGO_400,
                        content=ft.Column(
                            alignment=ft.MainAxisAlignment.CENTER,
                            horizontal_alignment=ft.CrossAxisAlignment.CENTER,
```



```

        controls=[
            ft.Text("Welcome", size=40, weight=ft.FontWeight.BOLD),
            ft.Text("Swipe to see what PageView can do."),
        ],
    ),
),
ft.Container(
    bgcolor=ft.Colors.PINK_300,
    content=ft.Column(
        alignment=ft.MainAxisAlignment.CENTER,
        horizontal_alignment=ft.CrossAxisAlignment.CENTER,
        controls=[
            ft.Icon(ft.Icons.ANIMATION, size=72),
            ft.Text(
                "Viewport fraction lets you peek at the next slide."
            ),
        ],
    ),
),
ft.Container(
    bgcolor=ft.Colors.TEAL_300,
    content=ft.Column(
        alignment=ft.MainAxisAlignment.CENTER,
        horizontal_alignment=ft.CrossAxisAlignment.CENTER,
        controls=[
            ft.Icon(ft.Icons.TOUCH_APP, size=72),
            ft.Text("Use on_change to respond to page swipes."),
        ],
    ),
),
],
),
),
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Programmatic Swipe

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    page.padding = 0

    async def show_previous_page(e: ft.Event[ft.FloatingActionButton]):
        await view.previous_page(
            animation_curve=ft.AnimationCurve.BOUNCE_OUT,
            animation_duration=ft.Duration(seconds=1),
        )

    async def show_next_page(e: ft.Event[ft.FloatingActionButton]):
        await view.next_page(
            animation_curve=ft.AnimationCurve.BOUNCE_OUT,
            animation_duration=ft.Duration(seconds=1),
        )

    page.floating_action_button = ft.Row(
        alignment=ft.MainAxisAlignment.CENTER,
        wrap=True,
        controls=[
            ft.FloatingActionButton(
                icon=ft.Icons.SWIPE_LEFT,
                on_click=show_previous_page,
                tooltip="Previous Page",
            ),
            ft.FloatingActionButton(
                icon=ft.Icons.SWIPE_RIGHT,
                on_click=show_next_page,
                tooltip="Next Page",
            ),
        ],
    )

    page.add(
        ft.SafeArea(
            expand=True,
            content=(
                view := ft.PageView(
                    expand=True,
                    viewport_fraction=0.9,
                    selected_index=1,
                    horizontal=True,

```

```

        controls=[
            ft.Container(
                bgcolor=bgcolor,
                content=ft.Column(
                    alignment=ft.MainAxisAlignment.CENTER,
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    controls=[
                        ft.Text(
                            f"Page {idx}",
                            size=55,
                            weight=ft.FontWeight.BOLD,
                        ),
                    ],
                ),
            ),
        ]
        for idx, bgcolor in enumerate(
            [
                ft.Colors.RED_800,
                ft.Colors.BLUE_800,
                ft.Colors.GREEN_800,
                ft.Colors.ORANGE_800,
                ft.Colors.PURPLE_800,
                ft.Colors.PINK_800,
            ]
        ):
            page.controls.extend(controls)
        page.update()
    ),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Properties

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: ClipBehavior = ClipBehavior.HARD_EDGE

Defines how pages are clipped if they overflow their bounds.

buildcontrolsclass-attributeinstance-attribute

controls: list[Control] = field(default_factory=list)

A list of controls to display, one per page, in the order they should appear.

buildhorizontalclass-attributeinstance-attribute

horizontal: bool = True

Whether the pages should be arranged and scrolled horizontally.

False implies vertical arrangement and scrolling.

buildimplicit_scrollingclass-attributeinstance-attribute

implicit_scrolling: bool = False

Whether to allow adjacent pages to render partially before the user scrolls to them, enabling smoother transitions and improved accessibility by allowing focus to move seamlessly between pages during navigation.

With this flag set to False, when accessibility focus reaches the end of the current page and the user attempts to move it to the next element, the focus will traverse to the next widget outside of the page view.

With this flag set to True, when accessibility focus reaches the end of the current page and user attempts to move it to the next element, focus will traverse to the next page in the page view.

buildkeep_pageclass-attributeinstance-attribute

keep_page: bool = True

Whether this page view should restore the most recently viewed page when rebuilt.

buildpad_endsclass-attributeinstance-attribute

pad_ends: bool = True

Whether to add padding to both ends of the list.

If this is set to True and `viewport_fraction < 1.0`, padding will be added before the first page and after the last page so they snap to the center of the viewport when scrolled all the way to the start or end.

Note

If `viewport_fraction >= 1.0`, this property has no effect.

buildreverseclass-attributeinstance-attribute

```
reverse: bool = False
```

Whether to reverse the order in which pages are read and swiped.

For example, if the reading direction is left-to-right and `horizontal` is True, then this page view scrolls from left to right when `reverse` is False and from right to left when `reverse` is True.

Similarly, if `horizontal` is False, then this page view scrolls from top to bottom when `reverse` is False and from bottom to top when `reverse` is True.

buildselected_indexclass-attributeinstance-attribute

```
selected_index: int = 0
```

The zero-based index of the currently visible page.

Changing it later on (followed by `update`) jumps to the specified page without animation.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildsnapclass-attributeinstance-attribute

```
snap: bool = True
```

Whether the view should snap to exact page boundaries after a drag.

If the `pad_ends` is False and `viewport_fraction < 1.0`, the page will snap to the beginning of the viewport; otherwise, the page will snap to the center of the viewport.

buildviewport_fractionclass-attributeinstance-attribute

```
viewport_fraction: Number = 1.0
```

The fraction of the viewport that each page should occupy in the scrolling direction (see `horizontal`).

For example, 1.0 (default), means each page fills the viewport.

Raises:

- `ValueError` - If it is not strictly greater than 0.

Events

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[PageView] | None = None
```

Fired when the visible page changes.

The data property of the event argument contains the index of the new page.

Methods

deployed_codego_to_pageasync

```
go_to_page(
    index: int,
    animation_duration: DurationValue = DEFAULT_ANIMATION_DURATION,
    animation_curve: AnimationCurve = DEFAULT_ANIMATION_CURVE,
)
```

Animates to the page at index.

Parameters:

- `index` (int) - The index of the page to show.
- `animation_duration` (DurationValue, default: `DEFAULT_ANIMATION_DURATION`) - Length of the animation.

- `animation_curve` (`AnimationCurve`, default: `DEFAULT_ANIMATION_CURVE`) - The easing curve of the animation.

Raises:

- `ValueError` - If index is negative.

`deployed_codejump_toasync`

```
jump_to(value: Number)
```

Immediately sets the scroll position to the given value, without animation and without validating whether the value is within bounds.

Any active scrolling or animation is canceled.

If the scroll position changes, a start/update/end sequence of scroll notifications is dispatched. This method does not generate overscroll notifications.

After the jump, a ballistic activity is initiated if the value is outside the valid scroll range.

Parameters:

- `value` (`Number`) - The new scroll position.

`deployed_codejump_to_pageasync`

```
jump_to_page(index: int)
```

Jumps immediately to the page at index without animation, moving the page position from its current value to the given value without animation nor range checking.

Parameters:

- `index` (`int`) - The index of the page to show.

Raises:

- `ValueError` - If index is negative.

`deployed_codenext_pageasync`

```
next_page(
    animation_duration: DurationValue = DEFAULT_ANIMATION_DURATION,
    animation_curve: AnimationCurve = DEFAULT_ANIMATION_CURVE,
)
```

Animates to the next page. Same as calling `go_to_page` with `selected_index + 1`.

Parameters:

- `animation_duration` (`DurationValue`, default: `DEFAULT_ANIMATION_DURATION`) - Length of the animation.
- `animation_curve` (`AnimationCurve`, default: `DEFAULT_ANIMATION_CURVE`) - The easing curve of the animation.

`deployed_codeprevious_pageasync`

```
previous_page(
    animation_duration: DurationValue = DEFAULT_ANIMATION_DURATION,
    animation_curve: AnimationCurve = DEFAULT_ANIMATION_CURVE,
)
```

Animates to the previous page. Same as calling `go_to_page` with `selected_index - 1`.

Parameters:

- `animation_duration` (`DurationValue`, default: `DEFAULT_ANIMATION_DURATION`) - Length of the animation.
- `animation_curve` (`AnimationCurve`, default: `DEFAULT_ANIMATION_CURVE`) - The easing curve of the animation.

Placeholder

A placeholder box.

```
ft.Placeholder(  
    expand=True,  
    color=ft.Colors.RED_500,  
)
```

Image: Placeholder - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/placeholder/image_for_docs.png

Placeholder

Inherits: `LayoutControl`

Properties

- `color` - The color of the placeholder box.
- `content` - An optional `Control` to display inside the placeholder.
- `fallback_height` - The height to use when the placeholder is in a situation with an unbounded height.
- `fallback_width` - The width to use when the placeholder is in a situation with an unbounded width.
- `stroke_width` - The width of the lines in the placeholder box.

Examples

Placeholder

`play_arrow` Try Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.add(  
        ft.SafeArea(  
            expand=True,  
            content=ft.Placeholder(  
                expand=True,  
                color=ft.Colors.GREEN_ACCENT,  
                fallback_height=200,  
                fallback_width=300,  
                stroke_width=20,  
            ),  
        ),  
    )  
  
if __name__ == "__main__":  
    ft.run(main)
```

Image: basic - <https://flet.dev/docs/test-images/examples/controls/core/golden/macros/placeholder/basic.png>

Properties

`buildcolor` class-attribute instance-attribute

`color: ColorValue = Colors.BLUE_GREY_700`

The color of the placeholder box.

`buildcontent` class-attribute instance-attribute

`content: Control | None = None`

An optional `Control` to display inside the placeholder.

`buildfallback_height` class-attribute instance-attribute

`fallback_height: Number = 400.0`

The height to use when the placeholder is in a situation with an unbounded height.

`buildfallback_width` class-attribute instance-attribute

`fallback_width: Number = 400.0`

The width to use when the placeholder is in a situation with an unbounded width.

buildstroke_widthclass-attributeinstance-attribute

stroke_width: Number | None = 2.0

The width of the lines in the placeholder box.

PopupMenuButton

An icon button which displays a menu when clicked.

```
ft.PopupMenuButton(  
    items=[  
        ft.PopupMenuItem(content="Sm"),  
        ft.PopupMenuItem(content="Med"),  
        ft.PopupMenuItem(content="Lg"),  
    ],  
    menu_position=ft.PopupMenuPosition.UNDER,  
)
```

Image: PopupMenuButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/popup_menu_button/image_for_docs.png

Opened popup menu under button

Inherits: LayoutControl

Properties

- bgcolor - The menu's background color.
- clip_behavior - The content will be clipped (or not) according to this option.
- content - A Control that will be displayed instead of "more" icon.
- elevation - The menu's elevation when opened.
- enable_feedback - Whether detected gestures should provide acoustic and/or haptic feedback.
- icon - If provided, an icon to draw on the button.
- icon_color - The icon's color.
- icon_size - The icon's size.
- items - A collection of PopupMenuItem controls to display in a dropdown menu.
- menu_padding - TBD
- menu_position - Defines position of the popup menu relative to the button.
- padding - TBD
- popup_animation_style - TBD
- shadow_color - The color used to paint the shadow below the menu.
- shape - The menu's shape.
- size_constraints - TBD
- splash_radius - The splash radius.
- style - TBD

Events

- on_cancel - Called when the user dismisses/cancels the popup menu without selecting an item.
- on_open - Called when the popup menu is shown.
- on_select - TBD

Examples

PopupMenuButton

play_arrowTry Online

```
import flet as ft  
  
def main(page: ft.Page):  
    def handle_check_item_click(e: ft.Event[ft.PopupMenuItem]):  
        e.control.checked = not e.control.checked  
  
    page.add(  

```

```

ft.SafeArea(
  content=ft.PopupMenuButton(
    key="popup",
    items=[
      ft.PopupMenuItem(content="Item 1"),
      ft.PopupMenuItem(icon=ft.Icons.POWER_INPUT, content="Check power"),
      ft.PopupMenuItem(
        content=ft.Row(
          controls=[
            ft.Icon(ft.Icons.HOURLASS_TOP_OUTLINED),
            ft.Text("Item with a custom content"),
          ],
        ),
        on_click=lambda _: print("Button with custom content clicked!"),
      ),
      ft.PopupMenuItem(), # divider
      ft.PopupMenuItem(
        content="Checked item",
        checked=False,
        on_click=handle_check_item_click,
      ),
    ],
  ),
)

if __name__ == "__main__":
  ft.run(main)

```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/popup_menu_button/basic.gif

Properties

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

The menu's background color.

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: ClipBehavior = ClipBehavior.NONE

The content will be clipped (or not) according to this option.

buildcontentclass-attributeinstance-attribute

content: StrOrControl | None = None

A Control that will be displayed instead of "more" icon.

buildelevationclass-attributeinstance-attribute

elevation: Number | None = None

The menu's elevation when opened.

Defaults to 8.

buildenable_feedbackclass-attributeinstance-attribute

enable_feedback: bool | None = None

Whether detected gestures should provide acoustic and/or haptic feedback.

On Android, for example, setting this to True produce a click sound and a long-press will produce a short vibration.

Defaults to True.

buildiconclass-attributeinstance-attribute

icon: IconDataOrControl | None = None

If provided, an icon to draw on the button.

buildicon_colorclass-attributeinstance-attribute

icon_color: ColorValue | None = None

The icon's color.

buildicon_sizeclass-attributeinstance-attribute

icon_size: Number | None = None

The icon's size.

builditemsclass-attributeinstance-attribute

```
items: list[PopupMenuItem] = field(default_factory=list)
```

A collection of PopupMenuItem controls to display in a dropdown menu.

buildmenu_paddingclass-attributeinstance-attribute

```
menu_padding: PaddingValue | None = None
```

TBD

buildmenu_positionclass-attributeinstance-attribute

```
menu_position: PopupMenuPosition | None = None
```

Defines position of the popup menu relative to the button.

Defaults to PopupMenuPosition.OVER.

buildpaddingclass-attributeinstance-attribute

```
padding: PaddingValue = 8
```

TBD

buildpopup_animation_styleclass-attributeinstance-attribute

```
popup_animation_style: AnimationStyle | None = None
```

TBD

buildshadow_colorclass-attributeinstance-attribute

```
shadow_color: ColorValue | None = None
```

The color used to paint the shadow below the menu.

buildshapeclass-attributeinstance-attribute

```
shape: OutlinedBorder | None = None
```

The menu's shape.

Defaults to CircleBorder(radius=10.0).

buildsize_constraintsclass-attributeinstance-attribute

```
size_constraints: BoxConstraints | None = None
```

TBD

buildsplash_radiusclass-attributeinstance-attribute

```
splash_radius: Number | None = None
```

The splash radius.

buildstyleclass-attributeinstance-attribute

```
style: ButtonStyle | None = None
```

TBD

Events

bolton_cancelclass-attributeinstance-attribute

```
on_cancel: ControlEventHandler[PopupMenuButton] | None = (
    None
)
```

Called when the user dismisses/cancels the popup menu without selecting an item.

bolton_openclass-attributeinstance-attribute

```
on_open: ControlEventHandler[PopupMenuButton] | None = None
```

Called when the popup menu is shown.

bolton_selectclass-attributeinstance-attribute

```
on_select: ControlEventHandler[PopupMenuButton] | None = (
```

```
) None
```

TBD

A popup menu item.

Inherits: Control

Properties

- `checked` - Whether this menu item is checked.
- `content` - A Control representing custom content of this menu item.
- `height` - The minimum height of this menu item.
- `icon` - An icon to draw before the text label of this menu item.
- `label_text_style` - The text style of the label of this menu item.
- `mouse_cursor` - The cursor to be displayed when a mouse pointer enters or is hovering over this item.
- `padding` - The padding of this menu item.

Events

- `on_click` - Called when a user clicks on this menu item.

Properties

`buildcheckedclass-attributeinstance-attribute`

```
checked: bool | None = None
```

Whether this menu item is checked.

If set to True, a checkmark will be shown on the left of the content.

`buildcontentclass-attributeinstance-attribute`

```
content: StrOrControl | None = None
```

A Control representing custom content of this menu item.

`buildheightclass-attributeinstance-attribute`

```
height: Number = 48.0
```

The minimum height of this menu item.

`buildiconclass-attributeinstance-attribute`

```
icon: IconDataOrControl | None = None
```

An icon to draw before the text label of this menu item.

`buildlabel_text_styleclass-attributeinstance-attribute`

```
label_text_style: TextStyle | None = None
```

The text style of the label of this menu item.

`buildmouse_cursorclass-attributeinstance-attribute`

```
mouse_cursor: MouseCursor | None = None
```

The cursor to be displayed when a mouse pointer enters or is hovering over this item.

`buildpaddingclass-attributeinstance-attribute`

```
padding: PaddingValue | None = None
```

The padding of this menu item.

Defaults to `Padding.symmetric(horizontal=12)`.

Note

The height value of this menu item may influence the applied padding.

For example, if a height greater than the height of the sum of the padding and a content is provided, then the padding's effect will not be visible.

Events

`bolton_click`class-attributeinstance-attribute

`on_click`: `ControlEventHandler[PopupMenuItem]` | `None` = `None`

Called when a user clicks on this menu item.

ProgressBar

A material design linear progress indicator, also known as a progress bar.

A control that shows progress along a line.

Example:

```
ft.ProgressBar(width=400, value=0.8)
```

Image: ProgressBar - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/progress_bar/image_for_docs.png

Fixed progress bar

Inherits: LayoutControl

Properties

- `bar_height` - The minimum height of the line used to draw the linear indicator.
- `bgcolor` - Color of the track being filled by the linear indicator.
- `border_radius` - The border radius of both the indicator and the track.
- `color` - The progress indicator's color.
- `semantics_label` - The semantics label for this progress indicator.
- `semantics_value` - The semantics label for this progress indicator.
- `stop_indicator_color` - The color of the stop indicator.
- `stop_indicator_radius` - The radius of the stop indicator.
- `track_gap` - The gap between the indicator and the track.
- `value` - The value of this progress indicator.
- `year_2023` - If this is set to `False`, the `ProgressBar` will use the latest Material Design 3 appearance, which was introduced in December 2023.

Examples

Determinate and Indeterminate

play_arrowTry Online

```
import asyncio

import flet as ft

async def main(page: ft.Page):
    determinate_bar = ft.ProgressBar(width=400)
    determinate_message = ft.Text("Doing something...")

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        value="Linear progress indicator",
                        theme_style=ft.TextThemeStyle.HEADLINE_SMALL,
                    ),
                    ft.Column(controls=[determinate_message, determinate_bar]),
                    ft.Text(
                        value="Indeterminate progress bar",
                        theme_style=ft.TextThemeStyle.HEADLINE_SMALL,
                    ),
                    ft.ProgressBar(width=400, color=ft.Colors.AMBER),
                ]
            )
        )
    )

    for i in range(0, 101):
        determinate_bar.value = i * 0.01
        await asyncio.sleep(0.1)
        if i == 100:
            determinate_message.value = "Finished!"
```

```
page.update()
```

```
if __name__ == "__main__":  
    ft.run(main)
```

Image: determinate-and-indeterminate - https://flint.dev/docs/examples/controls/material/progress_bar/media/determinate_and_indeterminate.gif

Properties

buildbar_heightclass-attributeinstance-attribute

```
bar_height: Number | None = None
```

The minimum height of the line used to draw the linear indicator.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue | None = None
```

Color of the track being filled by the linear indicator.

buildborder_radiusclass-attributeinstance-attribute

```
border_radius: BorderRadiusValue | None = None
```

The border radius of both the indicator and the track.

Defaults to `BorderRadius.all(0)` - rectangular shape.

buildcolorclass-attributeinstance-attribute

```
color: ColorValue | None = None
```

The progress indicator's color.

buildsemantics_labelclass-attributeinstance-attribute

```
semantics_label: str | None = None
```

The semantics label for this progress indicator.

buildsemantics_valueclass-attributeinstance-attribute

```
semantics_value: Number | None = None
```

The semantics label for this progress indicator.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildstop_indicator_colorclass-attributeinstance-attribute

```
stop_indicator_color: ColorValue | None = None
```

The color of the stop indicator.

If `ProgressBar.year_2023` is `True` or `Theme.use_material3` is `False`, then no stop indicator will be drawn.

If not set, then the `ProgressIndicatorTheme.stop_indicator_color` will be used. If that is not set, then the `ColorScheme.primary` will be used.

buildstop_indicator_radiusclass-attributeinstance-attribute

```
stop_indicator_radius: Number | None = None
```

The radius of the stop indicator.

If `ProgressBar.year_2023` is `True` or `Theme.use_material3` is `False`, then no stop indicator will be drawn.

Set `stop_indicator_radius` to 0 to hide the stop indicator.

If not set, then the `ProgressIndicatorTheme.stop_indicator_radius` will be used. If that is not set, then defaults to 2.

buildtrack_gapclass-attributeinstance-attribute

```
track_gap: Number | None = None
```

The gap between the indicator and the track.

If `ProgressBar.year_2023` is `True` or `Theme.use_material3` is `False`, then no track gap will be drawn.

If not set, then the `ProgressIndicatorTheme.track_gap` will be used. If that is not set, then defaults to 4.

Tip

Set `track_gap` to 0 to hide the track gap.

buildvalueclass-attributeinstance-attribute

`value: Number | None = None`

The value of this progress indicator.

A value of 0.0 means no progress and 1.0 means that progress is complete. The value will be clamped to be in the range 0.0 - 1.0.

Defaults to `None`, meaning that this progress indicator is indeterminate - displays a predetermined animation that does not indicate how much actual progress is being made.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildyear_2023class-attributeinstance-attribute

`year_2023: bool | None = None`

If this is set to `False`, the `ProgressBar` will use the latest Material Design 3 appearance, which was introduced in December 2023.

When `True`, the `ProgressBar` will use the 2023 Material Design 3 appearance.

If not set, then the `ProgressIndicatorTheme.year_2023` will be used, which is `False` by default.

If `Theme.use_material3` is `False`, then this property is ignored.

ProgressRing

A material design circular progress indicator, which spins to indicate that the application is busy.

A control that shows progress along a circle.

```
ft.ProgressRing(value=0.4, padding=ft.Padding.all(10))
```

Image: ProgressRing - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/progress_ring/image_for_docs.png

Fixed progress ring

Inherits: LayoutControl

Properties

- `bgcolor` - Color of the circular track being filled by the circular indicator.
- `color` - The progress indicator's color.
- `padding` - The padding around the indicator track.
- `semantics_label` - Used to identify the purpose of this progress bar for screen reading software.
- `semantics_value` - Used for determinate progress indicators to indicate how much progress has been made.
- `size_constraints` - Defines the minimum and maximum size of the progress indicator.
- `stroke_align` - The relative position of the stroke.
- `stroke_cap` - The progress indicator's line ending.
- `stroke_width` - The width of the line used to draw the circle.
- `track_gap` - The gap between the active indicator and the background track.
- `value` - The value of this progress indicator.
- `year_2023` - If this is set to `False`, the `ProgressRing` will use the latest Material Design 3 appearance, which was introduced in December 2023.

Examples

Determinate and Indeterminate

play_arrowTry Online

```
import asyncio

import flet as ft

async def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        value="Circular progress indicator",
                        theme_style=ft.TextThemeStyle.HEADLINE_SMALL,
                    ),
                    ft.Row(
                        controls=[
                            determinate_ring := ft.ProgressRing(
                                width=16, height=16, stroke_width=2
                            ),
                            determinate_message := ft.Text(
                                "Wait for the completion..."
                            ),
                        ],
                    ),
                    ft.Text(
                        value="Indeterminate circular progress",
                        theme_style=ft.TextThemeStyle.HEADLINE_SMALL,
                    ),
                    ft.Column(
                        horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                        controls=[
                            ft.ProgressRing(),
                            ft.Text("I'm going to run for ages..."),
                        ],
                    ),
                ],
            ),
        ),
    )
```

```

        ],
    ),
)

for i in range(0, 101):
    determinate_ring.value = i * 0.01
    await asyncio.sleep(0.1)
    if i == 100:
        determinate_message.value = "Finished!"
        page.update()

if __name__ == "__main__":
    ft.run(main)

```

Image: determinate-and-indeterminate - https://flet.dev/docs/examples/controls/material/progress_ring/media/determinate_and_indeterminate.gif

Gauge with Progress

play_arrowTry Online

import flet as ft

```

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Stack(
                width=100,
                height=100,
                controls=[
                    ft.Container(content=ft.Text("60%"), alignment=ft.Alignment.CENTER),
                    ft.ProgressRing(value=0.6, width=100, height=100),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: determinate-and-indeterminate - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/progress_ring/gauge_with_progress.png

Properties

buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

Color of the circular track being filled by the circular indicator.

buildcolorclass-attributeinstance-attribute

color: ColorValue | None = None

The progress indicator's color.

buildpaddingclass-attributeinstance-attribute

padding: PaddingValue | None = None

The padding around the indicator track.

If None, ProgressIndicatorTheme.circular_track_padding is used.

If that's is also None and year_2023 is False, defaults to Padding.all(4.0).

Otherwise, defaults to Padding.all(0.0).

buildsemantics_labelclass-attributeinstance-attribute

semantics_label: str | None = None

Used to identify the purpose of this progress bar for screen reading software.

buildsemantics_valueclass-attributeinstance-attribute

semantics_value: Number | None = None

Used for determinate progress indicators to indicate how much progress has been made.

buildsize_constraintsclass-attributeinstance-attribute

`size_constraints: BoxConstraints | None = None`

Defines the minimum and maximum size of the progress indicator.

If `None`, `ProgressIndicatorTheme.size_constraints` is used.

If that's is also `None`, defaults to a minimum width and height of 36.

buildstroke_alignclass-attributeinstance-attribute

`stroke_align: Number | None = None`

The relative position of the stroke.

Value typically ranges be -1.0 (inside stroke) and 1.0 (outside stroke).

A value of 0 (center stroke) will center the border on the edge of the control.

If `ProgressRing.year_2023` is `True`, then the default value is 0. Otherwise, the default value is -1.

buildstroke_capclass-attributeinstance-attribute

`stroke_cap: StrokeCap | None = None`

The progress indicator's line ending.

buildstroke_widthclass-attributeinstance-attribute

`stroke_width: Number | None = None`

The width of the line used to draw the circle.

buildtrack_gapclass-attributeinstance-attribute

`track_gap: Number | None = None`

The gap between the active indicator and the background track.

If `year_2023` is `True` or `Theme.use_material3` is `False`, then no track gap will be drawn.

Set `track_gap` to 0 to hide this track gap.

If `None`, `ProgressIndicatorTheme.track_gap` is used.

If that's is also `None`, defaults to 4.0.

buildvalueclass-attributeinstance-attribute

`value: Number | None = None`

The value of this progress indicator.

A value of 0.0 means no progress and 1.0 means that progress is complete. The value will be clamped to be in the range 0.0 - 1.0. If `None`, this progress indicator is indeterminate, which means the indicator displays a predetermined animation that does not indicate how much actual progress is being made.

buildyear_2023class-attributeinstance-attribute

`year_2023: bool | None = None`

If this is set to `False`, the `ProgressRing` will use the latest Material Design 3 appearance, which was introduced in December 2023.

When `True`, the `ProgressRing` will use the 2023 Material Design 3 appearance.

If not set, then the `ProgressIndicatorTheme.year_2023` will be used, which is `False` by default.

If `Theme.use_material3` is `False`, then this property is ignored.

Radio

Radio buttons let people select a single option from two or more choices.

Example:

```
ft.RadioGroup(  
    content=ft.Row(  
        controls=[ft.Radio(label=f"{i}") for i in range(1, 4)],  
        alignment=ft.MainAxisAlignment.CENTER,  
    )  
)
```

Image: Radio - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/radio/image_for_docs.png

Simple radio buttons

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `active_color` - The color used to fill this radio when it is selected.
- `autofocus` - True if the control will be selected as the initial focus.
- `fill_color` - The color that fills the radio, in all or specific `ControlState` states.
- `focus_color` - The color of this radio when it has the input focus.
- `hover_color` - The color of this radio when it is hovered.
- `label` - The clickable label to display on the right of a Radio.
- `label_position` - Defaults to `LabelPosition.RIGHT`.
- `label_style` - The label's style.
- `mouse_cursor` - The cursor for a mouse pointer entering or hovering over this control.
- `overlay_color` - The overlay color of this radio in all or specific `ControlState` states.
- `splash_radius` - The splash radius of the circular Material ink response.
- `toggleable` - Set to True if this radio button is allowed to be returned to an indeterminate state by selecting it again when selected.
- `value` - The value to set to containing `RadioGroup` when the radio is selected.
- `visual_density` - Defines how compact the radio's layout will be.

Events

- `on_blur` - Called when the control has lost focus.
- `on_focus` - Called when the control has received focus.

Examples

Radio

[play_arrow](#) [Try Online](#)

```
import flet as ft  
  
def main(page: ft.Page):  
    def handle_button_click(e: ft.Event[ft.Button]):  
        message.value = f"Your favorite color is: {group.value}"  
  
    page.add(  
        ft.SafeArea(  
            content=ft.Column(  
                controls=[  
                    ft.Text("Select your favorite color:"),  
                    group := ft.RadioGroup(  
                        content=ft.Column(  
                            controls=[  
                                ft.Radio(  
                                    key="basic_radio_red", value="red", label="Red"                                ]  
                        )  
                    ]  
                )  
            )  
        )  
    )
```

```

        ),
        ft.Radio(value="green", label="Green"),
        ft.Radio(value="blue", label="Blue"),
    ]
)
),
ft.Button(
    key="basic_submit_button",
    content="Submit",
    on_click=handle_button_click,
),
message := ft.Text(),
]
)
)
)

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/radio/basic.gif>

Handling Selection Changes

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    def handle_selection_change(e: ft.Event[ft.RadioGroup]):
        message.value = f"Your favorite color is: {e.control.value}"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text("Select your favorite color:"),
                    ft.RadioGroup(
                        on_change=handle_selection_change,
                        content=ft.Column(
                            controls=[
                                ft.Radio(value="red", label="Red"),
                                ft.Radio(value="green", label="Green"),
                                ft.Radio(value="blue", label="Blue"),
                            ]
                        ),
                    ),
                    message := ft.Text(),
                ]
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: handling-selection-changes - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/radio/handling_selection_changes.gif

Styled

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.RadioGroup(
                content=ft.Column(
                    controls=[
                        ft.Radio(
                            key="styled_radio_default",
                            label="Radio with default style",
                            value="1",
                        ),
                        ft.Radio(
                            key="styled_radio_constant",
                            label="Radio with constant fill color",
                            value="2",
                            fill_color=ft.Colors.RED,
                        ),
                        ft.Radio(
                            key="styled_radio_dynamic",

```

```

        label="Radio with dynamic fill color",
        value="3",
        fill_color={
            ft.ControlState.HOVERED: ft.Colors.BLUE,
            ft.ControlState.SELECTED: ft.Colors.GREEN,
            ft.ControlState.DEFAULT: ft.Colors.RED,
        },
    ),
]

)
)
)

if __name__ == "__main__":
    ft.run(main)

```

Image: styled - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/radio/styled.gif>

Properties

buildactive_colorclass-attributeinstance-attribute

active_color: ColorValue | None = None

The color used to fill this radio when it is selected.

buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

True if the control will be selected as the initial focus.

If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

buildfill_colorclass-attributeinstance-attribute

fill_color: ControlStateValue[ColorValue] | None = None

The color that fills the radio, in all or specific ControlState states.

buildfocus_colorclass-attributeinstance-attribute

focus_color: ColorValue | None = None

The color of this radio when it has the input focus.

buildhover_colorclass-attributeinstance-attribute

hover_color: ColorValue | None = None

The color of this radio when it is hovered.

buildlabelclass-attributeinstance-attribute

label: str = ''

The clickable label to display on the right of a Radio.

buildlabel_positionclass-attributeinstance-attribute

label_position: LabelPosition = LabelPosition.RIGHT

Defaults to LabelPosition.RIGHT.

buildlabel_styleclass-attributeinstance-attribute

label_style: TextStyle | None = None

The label's style.

buildmouse_cursorclass-attributeinstance-attribute

mouse_cursor: MouseCursor | None = None

The cursor for a mouse pointer entering or hovering over this control.

buildoverlay_colorclass-attributeinstance-attribute

overlay_color: ControlStateValue[ColorValue] | None = None

The overlay color of this radio in all or specific ControlState states.

buildsplash_radiusclass-attributeinstance-attribute

`splash_radius: Number | None = None`

The splash radius of the circular Material ink response.

buildtoggleableclass-attributeinstance-attribute

`toggleable: bool = False`

Set to True if this radio button is allowed to be returned to an indeterminate state by selecting it again when selected.

buildvalueclass-attributeinstance-attribute

`value: str | None = None`

The value to set to containing RadioGroup when the radio is selected.

buildvisual_densityclass-attributeinstance-attribute

`visual_density: VisualDensity | None = None`

Defines how compact the radio's layout will be.

Events

bolton_blurclass-attributeinstance-attribute

`on_blur: ControlEventHandler[Radio] | None = None`

Called when the control has lost focus.

bolton_focusclass-attributeinstance-attribute

`on_focus: ControlEventHandler[Radio] | None = None`

Called when the control has received focus.

RadioGroup

Radio buttons let people select a single option from two or more choices.

Inherits: `Control`

Properties

- `content` - The content of the `RadioGroup`.
- `value` - Current value of the `RadioGroup`.

Events

- `on_change` - Called when the state of the `RadioGroup` is changed.

Properties

`buildcontentinstance-attribute`

`content: Control`

The content of the `RadioGroup`.

Typically a list of `Radio` controls nested in a container control, e.g. `Column`, `Row`.

Raises:

- `ValueError` - If content is not visible.

`buildvalueclass-attributeinstance-attribute`

`value: str | None = None`

Current value of the `RadioGroup`.

Events

`bolton_change-attributeinstance-attribute`

`on_change: ControlEventHandler[RadioGroup] | None = None`

Called when the state of the `RadioGroup` is changed.

RangeSlider

A Material Design range slider. Used to select a range from a range of values.

A range slider can be used to select from either a continuous or a discrete set of values. The default is to use a continuous range of values from min to max.

Example:

```
ft.RangeSlider(  
    min=0,  
    max=10,  
    start_value=2,  
    divisions=10,  
    end_value=7,  
)
```

Image: RangeSlider - https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/range_slider/image_for_docs.png

Range Slider

Inherits: `LayoutControl`

Properties

- `active_color` - The color to use for the portion of the slider track that is active.
- `divisions` - The number of discrete divisions.
- `end_value` - The currently selected end value of this slider.
- `inactive_color` - The color for the inactive portions of the slider track.
- `label` - A label to show above the slider thumbs when the slider is active.
- `max` - The maximum value the user can select.
- `min` - The minimum value the user can select.
- `mouse_cursor` - The cursor for a mouse pointer entering or hovering over this control.
- `overlay_color` - The highlight color that's typically used to indicate that the range slider thumb is in `ControlState.HOVERED` or `ControlState.DRAGGED` state.
- `round` - The number of decimals displayed on the label containing {value}.
- `start_value` - The currently selected start value for this slider.

Events

- `on_change` - Called when the state of this slider is changed.
- `on_change_end` - Called when the user is done selecting a new value for this slider.
- `on_change_start` - Called when the user starts selecting a new value for this slider.

Examples

RangeSlider

`play_arrow` Try Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.add(  
        ft.SafeArea(  
            content=ft.Column(  
                controls=[  
                    ft.Text(  
                        value="Range slider with divisions and labels",  
                        size=20,  
                        weight=ft.FontWeight.BOLD,  
                    ),  
                    ft.Container(height=30),  
                    ft.RangeSlider(  
                        min=0,  
                        max=50,  
                    ),  
                ],  
            ),  
        ),  
    )
```

```

        start_value=10,
        divisions=10,
        end_value=20,
        inactive_color=ft.Colors.GREEN_300,
        active_color=ft.Colors.GREEN_700,
        overlay_color=ft.Colors.GREEN_100,
        label="{value}",
    ),
],
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/range_slider/basic.gif

Handling Change Events

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    page.scroll = ft.ScrollMode.AUTO

    def handle_slider_change_start(e: ft.Event[ft.RangeSlider]):
        print(f"on_change_start: {e.control.start_value}, {e.control.end_value}")

    def handle_slider_change(e: ft.Event[ft.RangeSlider]):
        print(f"on_change: {e.control.start_value}, {e.control.end_value}")

    def handle_slider_change_end(e: ft.Event[ft.RangeSlider]):
        print(f"on_change_end: {e.control.start_value}, {e.control.end_value}")
        message.value = f"on_change_end: {e.control.start_value}, {e.control.end_value}"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        value="Range slider with events",
                        size=20,
                        weight=ft.FontWeight.BOLD,
                    ),
                    ft.Container(height=30),
                    ft.RangeSlider(
                        divisions=100,
                        min=0,
                        max=100,
                        start_value=10,
                        end_value=20,
                        on_change_start=handle_slider_change_start,
                        on_change=handle_slider_change,
                        on_change_end=handle_slider_change_end,
                        label="{value}%",
                    ),
                    message := ft.Text(),
                ],
            ),
        ),
    )

    if __name__ == "__main__":
        ft.run(main)

```

Image: handling_events - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/range_slider/handling_events.gif

Properties

buildactive_colorclass-attributeinstance-attribute

active_color: ColorValue | None = None

The color to use for the portion of the slider track that is active.

The "active" segment of the range slider is the span between the thumbs.

bulldivisionsclass-attributeinstance-attribute

divisions: int | None = None

The number of discrete divisions.

Typically used with label to show the current discrete values.

If not set, this slider is continuous and label is not displayed.

Raises:

- ValueError - If it is not strictly greater than 0.

buildend_valueinstance-attribute

`end_value: Number`

The currently selected end value of this slider.

The right thumb of this slider is drawn at a position that corresponds to this value. Use label to change the label displayed on the thumb.

Raises:

- ValueError - If it is not less than or equal to max.
- ValueError - If it is not greater than or equal to start_value.

buildinactive_colorclass-attributeinstance-attribute

`inactive_color: ColorValue | None = None`

The color for the inactive portions of the slider track.

The "inactive" segments of the slider are the span of tracks between the min and the start thumb, and the end thumb and the max.

buildlabelclass-attributeinstance-attribute

`label: str | None = None`

A label to show above the slider thumbs when the slider is active.

It may contain {value} which will be replaced with realtime values of start_value and end_value, in the corresponding slider thumbs.

If not set, then the labels will not be displayed. If divisions is not set, this slider is continuous and labels are not displayed.

buildmaxclass-attributeinstance-attribute

`max: Number = 1.0`

The maximum value the user can select.

If the max is equal to the min, then the slider is disabled.

Raises:

- ValueError - If it is not greater than or equal to end_value.
- ValueError - If it is not greater than or equal to min.

buildminclass-attributeinstance-attribute

`min: Number = 0.0`

The minimum value the user can select.

If the max is equal to the min, then the slider is disabled.

Raises:

- ValueError - If it is not less than or equal to start_value.
- ValueError - If it is not less than or equal to max.

buildmouse_cursorclass-attributeinstance-attribute

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

The cursor for a mouse pointer entering or hovering over this control.

buildoverlay_colorclass-attributeinstance-attribute

`overlay_color: ControlStateValue[ColorValue] | None = None`

The highlight color that's typically used to indicate that the range slider thumb is in `ControlState.HOVERED` or `ControlState.DRAGGED` state.

buildroundclass-attributeinstance-attribute

```
round: int = 0
```

The number of decimals displayed on the label containing {value}.

Defaults to 0 - value rounded to the nearest integer.

Raises:

- `ValueError` - If it is not between 0 and 20, inclusive.

buildstart_valueinstance-attribute

```
start_value: Number
```

The currently selected start value for this slider.

The left thumb of this slider is drawn at a position that corresponds to this value. Use `label` to change the label displayed on the thumb.

Raises:

- `ValueError` - If it is not greater than or equal to `min`.
- `ValueError` - If it is not less than or equal to `end_value`.

Events

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[RangeSlider] | None = None
```

Called when the state of this slider is changed.

bolton_change_endclass-attributeinstance-attribute

```
on_change_end: ControlEventHandler[RangeSlider] | None = (
    None
)
```

Called when the user is done selecting a new value for this slider.

bolton_change_startclass-attributeinstance-attribute

```
on_change_start: ControlEventHandler[RangeSlider] | None = (
    None
)
```

Called when the user starts selecting a new value for this slider.

ReorderableDragHandle

Used to drag an item in a ReorderableListView.

It creates a listener for a drag immediately following a pointer down event over the given content control.

Example:

```
ft.ReorderableListView(
    show_default_drag_handles=False,
    controls=[
        ft.ListTile(
            title=ft.Text(f"Draggable Item {i}", color=ft.Colors.BLACK),
            leading=ft.ReorderableDragHandle(
                content=ft.Icon(ft.Icons.DRAG_INDICATOR, color=ft.Colors.RED),
                mouse_cursor=ft.MouseCursor.GRAB,
            ),
        ),
    ],
    for i in range(10)
)
```

Image: ReorderableDragHandle -

https://flet.dev/docs/test-images/examples/controls/material/golden/macros/reorderable_drag_handle/image_for_docs.png

Reorderable list with custom drag handles

Inherits: LayoutControl, AdaptiveControl

Properties

- `content` - The control for which the application would like to respond to a tap and drag gesture by starting a reordering drag on a reorderable list.
- `mouse_cursor` - The mouse cursor for mouse pointers that are hovering over the control.

Examples

Custom drag handle

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    def on_reorder(e: ft.OnReorderEvent):
        # Reorder controls list to match the UI change
        e.control.controls.insert(e.new_index, e.control.controls.pop(e.old_index))

    page.add(
        ft.SafeArea(
            content=ft.ReorderableListView(
                expand=True,
                show_default_drag_handles=False,
                on_reorder=on_reorder,
                controls=[
                    ft.ListTile(
                        title=ft.Text(f"Draggable Item {i}", color=ft.Colors.BLACK),
                        leading=ft.ReorderableDragHandle(
                            key=f"drag_handle_{i}",
                            content=ft.Icon(
                                ft.Icons.DRAG_INDICATOR, color=ft.Colors.RED
                            ),
                            mouse_cursor=ft.MouseCursor.GRAB,
                        ),
                        bgcolor=ft.Colors.ERROR
                        if i % 2 == 0
                        else ft.Colors.ON_ERROR_CONTAINER,
                    ),
                ],
                for i in range(10)
            ),
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: basic - https://flet.dev/docs/examples/controls/material/reorderable_drag_handle/media/basic.gif

Properties

buildcontentinstance-attribute

`content: Control`

The control for which the application would like to respond to a tap and drag gesture by starting a reordering drag on a reorderable list.

Raises:

- `ValueError` - If it is not visible.

buildmouse_cursorclass-attributeinstance-attribute

`mouse_cursor: MouseCursor | None = None`

The mouse cursor for mouse pointers that are hovering over the control.

ReorderableListView

A scrollable list of controls that can be reordered.

Tip

By default, each child control (from controls) is draggable using an automatically created drag handle (see `show_default_drag_handles`). To customize the draggable area, use `ReorderableDragHandle` to define your own drag handle or region.

Example:

```
ft.ReorderableListView(
    controls=[
        ft.ListTile(
            title=ft.Text(f"Item {i}"),
            bgcolor=ft.Colors.BLUE_GREY_300,
        )
        for i in range(1, 6)
    ],
)
```

Image: ReorderableListView - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/reorderable_list_view/image_for_docs.png

Reorderable list view

Inherits: `ListView`

Properties

- `anchor` - The relative position of the zero scroll offset.
- `auto_scroller_velocity_scalar` - The velocity scalar per pixel over scroll.
- `controls` - The controls displayed by this `ReorderableListView`.
- `footer` - A non-reorderable footer item to show after controls.
- `header` - A non-reorderable header item to show before controls.
- `mouse_cursor` - The cursor for a mouse pointer when it enters or is hovering over the drag handle.
- `show_default_drag_handles` - Whether to show default drag handles for each controls item.

Events

- `on_reorder` - Called when a controls item has been dragged to a new location/position.
- `on_reorder_end` - Called when the dragged controls item is dropped.
- `on_reorder_start` - Called when a controls item drag has started.

Examples

Horizontal and vertical

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    # the primary color is the color of the reorder handle
    page.theme = page.dark_theme = ft.Theme(
        color_scheme=ft.ColorScheme(primary=ft.Colors.BLUE)
    )

    def handle_reorder(e: ft.OnReorderEvent):
        # Reorder controls list to match the UI change
        e.control.controls.insert(e.new_index, e.control.controls.pop(e.old_index))

    def get_color(i):
        return ft.Colors.ERROR if i % 2 == 0 else ft.Colors.ON_ERROR_CONTAINER

    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                [ # horizontal
```

```

ft.ReorderableListView(
    expand=True,
    horizontal=True,
    on_reorder=handle_reorder,
    controls=[
        ft.Container(
            content=ft.Text(f"Item {i}", color=ft.Colors.BLACK),
            bgcolor=get_color(i),
            margin=ft.Margin.symmetric(horizontal=5, vertical=10),
            width=100,
            alignment=ft.Alignment.CENTER,
        )
        for i in range(10)
    ],
),
# vertical
ft.ReorderableListView(
    expand=True,
    on_reorder=handle_reorder,
    controls=[
        ft.ListTile(
            title=ft.Text(f"Item {i}", color=ft.Colors.BLACK),
            leading=ft.Icon(ft.Icons.CHECK, color=ft.Colors.RED),
            bgcolor=get_color(i),
        )
        for i in range(10)
    ],
),
],
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: horizontal-and-vertical - https://flet.dev/docs/examples/controls/material/reorderable_list_view/media/horizontal_and_vertical.gif

Custom drag handle

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    def on_reorder(e: ft.OnReorderEvent):
        # Reorder controls list to match the UI change
        e.control.controls.insert(e.new_index, e.control.controls.pop(e.old_index))

    page.add(
        ft.SafeArea(
            content=ft.ReorderableListView(
                expand=True,
                show_default_drag_handles=False,
                on_reorder=on_reorder,
                controls=[
                    ft.ListTile(
                        title=ft.Text(f"Draggable Item {i}", color=ft.Colors.BLACK),
                        leading=ft.ReorderableDragHandle(
                            key=f"drag_handle_{i}",
                            content=ft.Icon(
                                ft.Icons.DRAG_INDICATOR, color=ft.Colors.RED
                            ),
                            mouse_cursor=ft.MouseCursor.GRAB,
                        ),
                        bgcolor=ft.Colors.ERROR
                        if i % 2 == 0
                        else ft.Colors.ON_ERROR_CONTAINER,
                    )
                    for i in range(10)
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: custom-drag-handle - https://flet.dev/docs/examples/controls/material/reorderable_drag_handle/media/basic.gif

Properties

buildanchorclass-attributeinstance-attribute

anchor: Number = 0.0

The relative position of the zero scroll offset.

buildauto_scroller_velocity_scalarclass-attributeinstance-attribute

auto_scroller_velocity_scalar: Number | None = None

The velocity scalar per pixel over scroll.

It represents how the velocity scale with the over scroll distance. The auto-scroll velocity = (distance of overscroll) * velocity scalar.

buildcontrolsclass-attributeinstance-attribute

controls: list[Control] = field(default_factory=list)

The controls displayed by this ReorderableListView.

Note

When an item of this list gets reordered, the on_reorder event gets fired, but it doesn't reorder the controls list automatically. So, to keep controls list in sync with the UI, reorder this list inside your on_reorder event handler. See on_reorder for an example.

buildfooterclass-attributeinstance-attribute

footer: Control | None = None

A non-reorderable footer item to show after controls.

buildheaderclass-attributeinstance-attribute

header: Control | None = None

A non-reorderable header item to show before controls.

buildmouse_cursorclass-attributeinstance-attribute

mouse_cursor: MouseCursor | None = None

The cursor for a mouse pointer when it enters or is hovering over the drag handle.

buildshow_default_drag_handlesclass-attributeinstance-attribute

show_default_drag_handles: bool = True

Whether to show default drag handles for each controls item.

If True: on desktop platforms, a drag handle is stacked over the center of each item's trailing edge; on mobile platforms, a long press anywhere on the item starts a drag.

The default desktop drag handle is just an Icons.DRAG_HANDLE wrapped by a ReorderableDragHandle. On mobile platforms, the entire item is wrapped with a ReorderableDragHandle.

To customize the appearance or layout of drag handles, wrap each controls item, or a control within each of them, with a ReorderableDragHandle. For full control over the drag handles, you might want to set show_default_drag_handles to False.

Example

```
ft.ReorderableListView(
    show_default_drag_handles=False,
    controls=[
        ft.ListTile(
            title=ft.Text(f"Draggable Item {i}", color=ft.Colors.BLACK),
            leading=ft.ReorderableDragHandle(
                content=ft.Icon(ft.Icons.DRAG_INDICATOR, color=ft.Colors.RED),
                mouse_cursor=ft.MouseCursor.GRAB,
            ),
        ),
    ],
    for i in range(10)
)
```

Events

bolton_reorderclass-attributeinstance-attribute

on_reorder: EventHandler[OnReorderEvent] | None = None

Called when a controls item has been dragged to a new location/position.

Note

This event does not reorder controls automatically. So, to keep controls list in sync with the UI, reorder it accordingly inside your `on_reorder` event handler.

Example:

```
def handle_reorder(e: ft.OnReorderEvent):
    rlv = e.control
    moved_item = rlv.controls.pop(e.old_index) # Remove the reordered item from its old position
    rlv.controls.insert(e.new_index, moved_item) # Insert the reordered item into its new position
    rlv.update()
```

`bolton_reorder_end`class-attributeinstance-attribute

`on_reorder_end`: `EventHandler[OnReorderEvent]` | `None` = `None`

Called when the dragged controls item is dropped.

`bolton_reorder_start`class-attributeinstance-attribute

`on_reorder_start`: `EventHandler[OnReorderEvent]` | `None` = `None`

Called when a controls item drag has started.

ResponsiveRow

Allows aligning child controls to virtual columns.

By default, a virtual grid has 12 columns, but that can be customized with columns property.

Similar to expand property, every control has col property which allows specifying how many columns a control should span.

Use scroll to enable vertical scrolling when the responsive content is taller than the available height.

Example:

```
ft.ResponsiveRow(
    controls=[
        ft.Button(
            f"Button {i}",
            color=ft.Colors.BLUE_GREY_300,
            col={
                ft.ResponsiveRowBreakpoint.XS: 12,
                ft.ResponsiveRowBreakpoint.MD: 6,
                ft.ResponsiveRowBreakpoint.LG: 3,
            },
        ),
        for i in range(1, 6)
    ],
)
```

Image: ResponsiveRow - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/responsive_row/image_for_docs.png

Responsive row

Inherits: LayoutControl, ScrollableControl, AdaptiveControl

Properties

- alignment - Defines how the child controls should be placed horizontally.
- breakpoints - Defines the minimum widths (in px) for each breakpoint key used by responsive properties such as col, spacing, and run_spacing.
- columns - The number of virtual columns to layout children.
- controls - A list of Controls to display.
- run_spacing - The spacing between runs.
- spacing - The spacing between controls in a row in virtual pixels.
- vertical_alignment - Defines how the child controls should be placed vertically.

Examples

ResponsiveRow

play_arrow Try Online

```
import flet as ft

def main(page: ft.Page):
    def handle_page_resize(e: ft.PageResizeEvent):
        pw.value = f"{page.width} px"
        pw.update()

    page.on_resize = handle_page_resize

    pw = ft.Text(text_align=ft.TextAlign.END, style=ft.TextTheme.display_small)

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.ResponsiveRow(
                        controls=[
                            ft.Container(
                                content=ft.Text("Column 1"),
                                padding=5,
                                bgcolor=ft.Colors.YELLOW,
                                col={
```

```

        ft.ResponsiveRowBreakpoint.XS: 12,
        ft.ResponsiveRowBreakpoint.MD: 6,
        ft.ResponsiveRowBreakpoint.LG: 3,
    },
),
ft.Container(
    content=ft.Text("Column 2"),
    padding=5,
    bgcolor=ft.Colors.GREEN,
    col={
        ft.ResponsiveRowBreakpoint.XS: 12,
        ft.ResponsiveRowBreakpoint.MD: 6,
        ft.ResponsiveRowBreakpoint.LG: 3,
    },
),
ft.Container(
    content=ft.Text("Column 3"),
    padding=5,
    bgcolor=ft.Colors.BLUE,
    col={
        ft.ResponsiveRowBreakpoint.XS: 12,
        ft.ResponsiveRowBreakpoint.MD: 6,
        ft.ResponsiveRowBreakpoint.LG: 3,
    },
),
ft.Container(
    content=ft.Text("Column 4"),
    padding=5,
    bgcolor=ft.Colors.PINK_300,
    col={
        ft.ResponsiveRowBreakpoint.XS: 12,
        ft.ResponsiveRowBreakpoint.MD: 6,
        ft.ResponsiveRowBreakpoint.LG: 3,
    },
),
],
),
ft.ResponsiveRow(
    run_spacing={ft.ResponsiveRowBreakpoint.XS: 10},
    controls=[
        ft.TextField(
            label="TextField 1",
            col={ft.ResponsiveRowBreakpoint.MD: 4},
        ),
        ft.TextField(
            label="TextField 2",
            col={ft.ResponsiveRowBreakpoint.MD: 4},
        ),
        ft.TextField(
            label="TextField 3",
            col={ft.ResponsiveRowBreakpoint.MD: 4},
        ),
    ],
),
),
pw,
]
)
)
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: [Image - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/responsive_row/basic.gif](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/responsive_row/basic.gif)

Custom Breakpoints

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    page.title = "ResponsiveRow with custom breakpoints"
    page.padding = 16

    breakpoints = {
        "phone": 0,
        "tablet": 540,
        "desktop": 800,
    }

    sorted_breakpoints = sorted(breakpoints.items(), key=lambda item: item[1])

    breakpoint_labels = {
        name: ft.Text(f"{name}: >= {value}px", weight=ft.FontWeight.W_500)
    }

```

```

    for name, value in sorted_breakpoints
}

width_label = ft.Text()
breakpoint_label = ft.Text()

def update_status(_=None):
    width = (
        (page.window.width if page.window and page.window.width else None)
        or page.width
        or 0
    )
    width_label.value = f"Page width: {width:.0f}px"
    active_breakpoint = max(
        (bp for bp, min_width in breakpoints.items() if width >= min_width),
        key=lambda bp: breakpoints[bp],
        default="phone",
    )
    breakpoint_label.value = f"Active breakpoint: {active_breakpoint}"
    for name, label in breakpoint_labels.items():
        is_active = name == active_breakpoint
        label.color = ft.Colors.BLUE_700 if is_active else None
        label.weight = ft.FontWeight.W_700 if is_active else ft.FontWeight.W_400
        label.update()
    width_label.update()
    breakpoint_label.update()

page.on_resize = update_status

page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                ft.Text("Resize the window to see custom breakpoints in action."),
                ft.Text(
                    "Cards switch column spans at phone, tablet, and "
                    "desktop widths."
                ),
                ft.Column(
                    [
                        ft.Text(
                            "Custom breakpoints (min widths):",
                            weight=ft.FontWeight.W_600,
                        ),
                        ft.Column(list(breakpoint_labels.values()), spacing=2),
                    ],
                    spacing=6,
                ),
                ft.ResponsiveRow(
                    breakpoints=breakpoints,
                    columns={"phone": 4, "tablet": 8, "desktop": 12},
                    spacing=10,
                    run_spacing=10,
                    controls=[
                        ft.Container(
                            content=ft.Text(
                                "Card 1", size=16, weight=ft.FontWeight.W_600
                            ),
                            alignment=ft.Alignment.CENTER,
                            bgcolor=ft.Colors.AMBER_200,
                            height=60,
                            col={"phone": 4, "tablet": 4, "desktop": 3},
                        ),
                        ft.Container(
                            content=ft.Text(
                                "Card 2", size=16, weight=ft.FontWeight.W_600
                            ),
                            alignment=ft.Alignment.CENTER,
                            bgcolor=ft.Colors.GREEN_200,
                            height=60,
                            col={"phone": 4, "tablet": 4, "desktop": 3},
                        ),
                        ft.Container(
                            content=ft.Text(
                                "Card 3", size=16, weight=ft.FontWeight.W_600
                            ),
                            alignment=ft.Alignment.CENTER,
                            bgcolor=ft.Colors.BLUE_200,
                            height=60,
                            col={"phone": 4, "tablet": 4, "desktop": 3},
                        ),
                        ft.Container(
                            content=ft.Text(
                                "Card 4", size=16, weight=ft.FontWeight.W_600
                            ),
                            alignment=ft.Alignment.CENTER,
                            bgcolor=ft.Colors.PINK_200,

```

```

        height=60,
        col={"phone": 4, "tablet": 4, "desktop": 3},
    ),
],
),
width_label,
breakpoint_label,
]
)
)
)
update_status()

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/responsive_row/custom_breakpoint.gif

Scrollable

play_arrow Try Online

```

import flet as ft

def build_card(index: int, color: ft.Colors) -> ft.Container:
    return ft.Container(
        bgcolor=color,
        border_radius=8,
        padding=16,
        content=ft.Column(
            spacing=10,
            controls=[
                ft.Text(f"Card {index}", size=22, weight=ft.FontWeight.BOLD),
                ft.Text("Line 1"),
                ft.Text("Line 2"),
                ft.Text("Line 3"),
                ft.Text("Line 4"),
                ft.Text("Line 5"),
            ],
        ),
    )

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                expand=True,
                spacing=12,
                controls=[
                    ft.Text(
                        "Resize the window height. The header stays visible while "
                        "the responsive content scrolls."
                    ),
                    ft.ResponsiveRow(
                        expand=True,
                        scroll=ft.ScrollMode.AUTO,
                        spacing=16,
                        run_spacing=16,
                        controls=[
                            ft.Container(
                                col={
                                    ft.ResponsiveRowBreakpoint.XS: 12,
                                    ft.ResponsiveRowBreakpoint.MD: 6,
                                    ft.ResponsiveRowBreakpoint.LG: 4,
                                },
                                content=build_card(
                                    i,
                                    [
                                        ft.Colors.BLUE_50,
                                        ft.Colors.GREEN_50,
                                        ft.Colors.AMBER_50,
                                    ][i % 3],
                                ),
                            )
                            for i in range(1, 10)
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Properties

buildalignmentclass-attributeinstance-attribute

```
alignment: MainAxisAlignment = MainAxisAlignment.START
```

Defines how the child controls should be placed horizontally.

buildbreakpointsclass-attributeinstance-attribute

```
breakpoints: dict[ResponsiveRowBreakpoint | str, Number] = (
    field(
        default_factory=lambda: {
            ResponsiveRowBreakpoint.XS: 0,
            ResponsiveRowBreakpoint.SM: 576,
            ResponsiveRowBreakpoint.MD: 768,
            ResponsiveRowBreakpoint.LG: 992,
            ResponsiveRowBreakpoint.XL: 1200,
            ResponsiveRowBreakpoint.XXL: 1400,
        }
    )
)
```

Defines the minimum widths (in px) for each breakpoint key used by responsive properties such as col, spacing, and run_spacing.

Keys can be ResponsiveRowBreakpoint values or custom strings. Breakpoint names in responsive values must match the names used here.

The default mirrors Bootstrap breakpoints.

buildcolumnsclass-attributeinstance-attribute

```
columns: ResponsiveNumber = 12
```

The number of virtual columns to layout children.

Raises:

- ValueError - If it is not strictly greater than 0.
- ValueError - If any breakpoint-specific value is not strictly greater than 0.

buildcontrolsclass-attributeinstance-attribute

```
controls: list[Control] = field(default_factory=list)
```

A list of Controls to display.

buildrun_spacingclass-attributeinstance-attribute

```
run_spacing: ResponsiveNumber = 10
```

The spacing between runs.

buildspacingclass-attributeinstance-attribute

```
spacing: ResponsiveNumber = 10
```

The spacing between controls in a row in virtual pixels.

Note

Has effect only when alignment is set to MainAxisAlignment.START, MainAxisAlignment.END, or MainAxisAlignment.CENTER.

buildvertical_alignmentclass-attributeinstance-attribute

```
vertical_alignment: CrossAxisAlignment = (
    CrossAxisAlignment.START
)
```

Defines how the child controls should be placed vertically.

Rive

Render Rive animations in your Flet app with the flet-rive extension.

Platform Support

| Platform | Windows | macOS | Linux |
|-----------|--------------------------|--------------------------|-------------------------------------|
| Supported | ☐ | ☐ | ☐ (x64 only) |

Usage

Add flet-rive to your project dependencies:

- uv
 - pip
- ```
uv add flet-rive
pip install flet-rive
```

### Hosting Rive files

Host .riv files locally or load them from a CDN. Use placeholder to keep layouts responsive while animations load.

## Example

### Rive animations

play\_arrowTry Online

```
import flet as ft
import flet_rive as ftr

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ftr.Rive(
 src="https://cdn.rive.app/animations/vehicles.riv",
 placeholder=ft.ProgressBar(),
 width=300,
 height=200,
),
 ftr.Rive(
 src="vehicles.riv",
 placeholder=ft.ProgressBar(),
 width=300,
 height=200,
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Description

Displays rive animations.

Inherits: LayoutControl

### Properties

- alignment - Alignment for the animation in the Rive control.
- animations - List of animations to play; default animation is played if empty.
- artboard - The name of the artboard to use.
- clip\_rect - Clip the artboard to this rect.

- `enable_antialiasing` - Whether to enable anti-aliasing when rendering.
- `fit` - The animation's fit.
- `headers` - Headers for network requests.
- `placeholder` - Control displayed while the Rive is loading.
- `speed_multiplier` - A multiplier for controlling the speed of the Rive animation playback.
- `src` - The source of your rive animation.
- `state_machines` - List of state machines to play; none will play if empty.
- `use_artboard_size` - Determines whether to use the inherent size of the artboard, i.e.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

`alignment: Alignment | None = None`

Alignment for the animation in the Rive control.

### `buildanimationsclass-attributeinstance-attribute`

`animations: list[str] = field(default_factory=list)`

List of animations to play; default animation is played if empty.

### `buildartboardclass-attributeinstance-attribute`

`artboard: str | None = None`

The name of the artboard to use. If not specified, the default artboard of the provided src is used.

### `buildclip_rectclass-attributeinstance-attribute`

`clip_rect: Rect | None = None`

Clip the artboard to this rect.

If not supplied it'll default to the constraint size provided by the parent control. Unless the Artboard has clipping disabled, then no clip will be applied.

### `buildenable_antialiasingclass-attributeinstance-attribute`

`enable_antialiasing: bool = True`

Whether to enable anti-aliasing when rendering.

### `buildfitclass-attributeinstance-attribute`

`fit: BoxFit | None = None`

The animation's fit.

### `buildheadersclass-attributeinstance-attribute`

`headers: dict[str, str] | None = None`

Headers for network requests.

### `buildplaceholderclass-attributeinstance-attribute`

`placeholder: Control | None = None`

Control displayed while the Rive is loading.

### `buildspeed_multiplierclass-attributeinstance-attribute`

`speed_multiplier: Number = 1.0`

A multiplier for controlling the speed of the Rive animation playback.

### `buildsrcinstance-attribute`

`src: str`

The source of your rive animation.

Can either be a URL or a path to a local asset file.

### **buildstate\_machinesclass-attributeinstance-attribute**

```
state_machines: list[str] = field(default_factory=list)
```

List of state machines to play; none will play if empty.

### **builduse\_artboard\_sizeclass-attributeinstance-attribute**

```
use_artboard_size: bool = False
```

Determines whether to use the inherent size of the artboard, i.e. the absolute size defined by the artboard, or size the control based on the available constraints only (sized by parent).



# RotatedBox

Rotates its content by an integral number of quarter turns.

Unlike `LayoutControl.rotate` (which uses `Transform.rotate` and applies the rotation only at paint time), `RotatedBox` applies the rotation before layout. This means the control's rotated dimensions participate in layout and can affect surrounding controls.

Inherits: `LayoutControl`

## Properties

- `content` - The Control to rotate.
- `quarter_turns` - The number of clockwise quarter turns.

## Example

### RotatedBox

[play\\_arrow](#)Try Online

```
import flet as ft

def _demo_control(content: ft.Control) -> ft.Container:
 return ft.Container(
 padding=10,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 border_radius=8,
 content=content,
)

def _lane(title: str, controls: list[ft.Control]) -> ft.Container:
 return ft.Container(
 width=540,
 padding=12,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 border_radius=12,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(title, size=16, weight=ft.FontWeight.BOLD),
 ft.Divider(height=1),
 ft.Row(
 spacing=14,
 vertical_alignment=ft.CrossAxisAlignment.START,
 controls=controls,
),
],
),
)

def main(page: ft.Page):
 page.padding = 24
 page.scroll = ft.ScrollMode.AUTO
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "RotatedBox rotates before layout. Compare occupied "
 "space below:",
 size=16,
 weight=ft.FontWeight.W_500,
),
 ft.Column(
 spacing=16,
 controls=[
 _lane(
 "Normal controls",
 [
 _demo_control(ft.Text("Text", size=26)),
 _demo_control(
 ft.ProgressBar(
 width=170, value=0.65, color=ft.Colors.GREEN
)
),
],
),
],
),
],
),
),
)
```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/rotated\\_box/rotated\\_box.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/rotated_box/rotated_box.png)

# Router

Top-level router component that matches the current page route against a tree of Route definitions and renders the matched component chain.

The Router subscribes to `on_route_change` and re-renders automatically when the route changes.

Navigation is done via `push_route` or `navigate`.

When `manage_views` is `True`, the Router returns a list of View objects (one per path level) instead of a single component tree. This enables swipe-back gestures, system back button, and AppBar implicit back button on mobile. Must be used with `render_views`.

Parameters:

- `routes` (list[Route]) - List of top-level Route definitions.
- `not_found` (Callable | None, default: None) - Optional component to render when no route matches (404).
- `manage_views` (bool, default: False) - When `True`, produce a list of View objects (one per path level) instead of a single component tree. Route components should return View instances with route and appbar set. Use with `render_views`.

Example

```
@ft.component
def App():
 return ft.Router(
 [
 ft.Route(index=True, component=Home),
 ft.Route(path="about", component=About),
 ft.Route(path="products/:pid", component=ProductDetails),
],
 manage_views=True,
)
```

Router matches the current page route against a tree of Route definitions and renders the matched component chain with nested outlet contexts.

Navigation is done via `page.navigate()` or `page.push_route()`.

## Examples

### Routing

`play_arrow` Try Online

```
"""Basic Router – simplest example with flat routes."""

import flet as ft

@ft.component
def Home():
 return ft.Text("Home page", size=24)

@ft.component
def About():
 return ft.Text("About page", size=24)

@ft.component
def App():
 return ft.SafeArea(
 content=ft.Column(
 [
 ft.Row(
 [
 ft.Button(
 "Home",
 on_click=lambda: ft.context.page.navigate("/"),
),
 ft.Button(
 "About",
 on_click=lambda: ft.context.page.navigate("/about"),
),
],
),
],
),
 ft.Router(
```

```

 [
 ft.Route(index=True, component=Home),
 ft.Route(path="about", component=About),
]
),
]
)

if __name__ == "__main__":
 ft.run(lambda page: page.render(App))

```

## Layout outlet

### play\_arrowTry Online

```

"""Layout routes with Outlet – shared layout wrapping child routes."""

import flet as ft

@ft.component
def Home():
 return ft.Text("Welcome home!", size=24)

@ft.component
def About():
 return ft.Text("About us", size=24)

@ft.component
def Contact():
 return ft.Text("Contact page", size=24)

@ft.component
def AppLayout():
 outlet = ft.use_route_outlet()
 return ft.Column(
 [
 ft.Container(
 content=ft.Row(
 [
 ft.Text("My App", size=20, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Home",
 on_click=lambda: ft.context.page.navigate("/"),
),
 ft.Button(
 "About",
 on_click=lambda: ft.context.page.navigate("/about"),
),
 ft.Button(
 "Contact",
 on_click=lambda: ft.context.page.navigate("/contact"),
),
]
),
 bgcolor=ft.Colors.SURFACE_BRIGHT,
 padding=10,
),
 ft.Container(content=outlet, padding=20),
 ft.Text("Footer - (c) 2026", text_align=ft.TextAlign.CENTER),
],
)

@ft.component
def App():
 return ft.SafeArea(
 content=ft.Router(
 [
 ft.Route(
 component=AppLayout,
 children=[
 ft.Route(index=True, component=Home),
 ft.Route(path="about", component=About),
 ft.Route(path="contact", component=Contact),
],
),
],
)
)

if __name__ == "__main__":

```

```
ft.run(lambda page: page.render(App))
```

## Dynamic segments

### play\_arrowTry Online

```
"""Dynamic segments - :param paths with use_route_params()."""
```

```
import flet as ft
```

```
@ft.component
def UserProfile():
 params = ft.use_route_params()
 return ft.Column(
 [
 ft.Text(f"User: {params['userId']}", size=24),
 ft.Text(f"All params: {params}"),
 ft.Button(
 "View post #10",
 on_click=lambda: ft.context.page.navigate(
 f"/users/{params['userId']}/posts/10"
),
),
 ft.Button(
 "Back to users",
 on_click=lambda: ft.context.page.navigate("/"),
),
]
)
```

```
@ft.component
def UserPost():
 params = ft.use_route_params()
 return ft.Column(
 [
 ft.Text(f"User: {params['userId']}, Post: {params['postId']}", size=24),
 ft.Text(f"All params: {params}"),
 ft.Button(
 "Back to user",
 on_click=lambda: ft.context.page.navigate(f"/users/{params['userId']}"),
),
]
)
```

```
@ft.component
def UserList():
 return ft.Column(
 [
 ft.Text("Users", size=24),
 ft.Button(
 "User alice",
 on_click=lambda: ft.context.page.navigate("/users/alice"),
),
 ft.Button(
 "User bob",
 on_click=lambda: ft.context.page.navigate("/users/bob"),
),
]
)
```

```
@ft.component
def App():
 return ft.SafeArea(
 content=ft.Router(
 [
 ft.Route(index=True, component=UserList),
 ft.Route(
 path="users/:userId",
 children=[
 ft.Route(index=True, component=UserProfile),
 ft.Route(path="posts/:postId", component=UserPost),
],
),
]
)
)
```

```
if __name__ == "__main__":
 ft.run(lambda page: page.render(App))
```

## Route loaders

## play\_arrowTry Online

```
"""Loaders – data loading with loader and use_route_loader_data()."""

import flet as ft

Simulated data loaders
def home_loader(params):
 return {"title": "Home", "message": "Welcome to the app!"}

def products_loader(params):
 return {
 "products": [
 {"id": 1, "name": "Widget A", "price": 9.99},
 {"id": 2, "name": "Gadget B", "price": 24.99},
 {"id": 3, "name": "Doohickey C", "price": 4.99},
]
 }

def product_detail_loader(params):
 pid = params.get("pid", "?")
 # In a real app, you'd fetch from a database or API
 products = {
 "1": {"id": 1, "name": "Widget A", "price": 9.99, "stock": 42},
 "2": {"id": 2, "name": "Gadget B", "price": 24.99, "stock": 7},
 "3": {"id": 3, "name": "Doohickey C", "price": 4.99, "stock": 100},
 }
 return products.get(pid, {"id": pid, "name": "Unknown", "price": 0, "stock": 0})

@ft.component
def Home():
 data = ft.use_route_loader_data()
 return ft.Column(
 [
 ft.Text(data["title"], size=24),
 ft.Text(data["message"]),
]
)

@ft.component
def ProductsList():
 data = ft.use_route_loader_data()
 return ft.Column(
 [
 ft.Text("Products", size=24),
 *[
 ft.ListTile(
 title=ft.Text(p["name"]),
 subtitle=ft.Text(f"${p['price']:.2f}"),
 on_click=lambda _, pid=p["id"]: ft.context.page.navigate(
 f"/products/{pid}"
),
)
 for p in data["products"]
],
]
)

@ft.component
def ProductDetails():
 data = ft.use_route_loader_data()
 params = ft.use_route_params()
 return ft.Column(
 [
 ft.Text(data["name"], size=24),
 ft.Text(f"Price: ${data['price']:.2f}"),
 ft.Text(f"In stock: {data['stock']}"),
 ft.Text(f"Product ID (from params): {params['pid']}"),
 ft.Button(
 "Back to products",
 on_click=lambda: ft.context.page.navigate("/products"),
),
]
)

@ft.component
def App():
 return ft.SafeArea(
 content=ft.Column(
 [

```

```

 ft.Row(
 [
 ft.Button(
 "Home",
 on_click=lambda: ft.context.page.navigate("/"),
),
 ft.Button(
 "Products",
 on_click=lambda: ft.context.page.navigate("/products"),
),
]
),
 ft.Divider(),
 ft.Router(
 [
 ft.Route(index=True, component=Home, loader=home_loader),
 ft.Route(
 path="products",
 children=[
 ft.Route(
 index=True,
 component=ProductsList,
 loader=products_loader,
),
 ft.Route(
 path=":pid",
 component=ProductDetails,
 loader=product_detail_loader,
),
],
),
],
),
],
),
)
)

```

```

if __name__ == "__main__":
 ft.run(lambda page: page.render(App))

```

## Active links

play\_arrowTry Online

```

"""Active links – nav highlighting with is_route_active()."""

```

```

import flet as ft

```

```

@ft.component
def Home():
 return ft.Text("Home page", size=24)

```

```

@ft.component
def Products():
 return ft.Text("Products page", size=24)

```

```

@ft.component
def Settings():
 return ft.Text("Settings page", size=24)

```

```

@ft.component
def NavLink(label, path):
 """A navigation link that highlights when its path is active."""
 active = ft.is_route_active(path)
 return ft.Container(
 content=ft.Text(
 label,
 weight=ft.FontWeight.BOLD if active else ft.FontWeight.NORMAL,
 color=ft.Colors.BLUE if active else ft.Colors.ON_SURFACE,
),
 bgcolor=ft.Colors.BLUE_50 if active else None,
 padding=ft.Padding.symmetric(horizontal=16, vertical=8),
 border_radius=8,
 on_click=lambda: ft.context.page.navigate(path),
)

```

```

@ft.component
def AppLayout():
 """Layout route – NavLink must be inside Router to use is_route_active()."""
 outlet = ft.use_route_outlet()
 return ft.Column(

```

```

 [
 ft.Row(
 [
 NavLink("Home", "/"),
 NavLink("Products", "/products"),
 NavLink("Settings", "/settings"),
]
),
 ft.Divider(),
 outlet,
]
)

@ft.component
def App():
 return ft.SafeArea(
 content=ft.Router(
 [
 ft.Route(
 component=AppLayout,
 children=[
 ft.Route(index=True, component=Home),
 ft.Route(path="products", component=Products),
 ft.Route(path="settings", component=Settings),
],
),
]
)
)

if __name__ == "__main__":
 ft.run(lambda page: page.render(App))

```

## Router featured app (declarative)

[play\\_arrowTry Online](#)

```

"""Featured Router example – layout, nav, nested routes, params, loaders, auth."""

from dataclasses import dataclass

import flet as ft

Auth context

@ft.observable
@dataclass
class AuthState:
 is_authenticated: bool = False
 username: str = ""
 is_admin: bool = False

 def login(self, username, admin=False):
 self.username = username
 self.is_authenticated = True
 self.is_admin = admin

 def logout(self):
 self.username = ""
 self.is_authenticated = False
 self.is_admin = False

AuthContext: ft.ContextProvider[AuthState | None] = ft.create_context(None)

Data loaders

def home_loader(params):
 return {"greeting": "Welcome to the Router Demo!", "featured_count": 3}

def projects_loader(params):
 return [
 {"id": 1, "name": "Flet", "status": "Active"},
 {"id": 2, "name": "Flutter", "status": "Active"},
 {"id": 3, "name": "FastAPI", "status": "Maintenance"},
]

```



```

def project_detail_loader(params):
 projects = {
 "1": {"id": 1, "name": "Flet", "description": "Build apps in Python"},
 "2": {"id": 2, "name": "Flutter", "description": "UI toolkit by Google"},
 "3": {"id": 3, "name": "FastAPI", "description": "Modern Python web framework"},
 }
 return projects.get(params.get("projectId"), {"name": "Unknown", "description": ""})

def settings_loader(params):
 return {"sections": ["Profile", "Security", "Notifications"]}

Auth components

@ft.component
def LoginPage():
 auth = ft.use_context(AuthContext)
 username_ref = ft.use_ref(None)

 def handle_login():
 auth.login(username_ref.current.value or "user")
 ft.context.page.navigate("/")

 def handle_admin_login():
 auth.login(username_ref.current.value or "admin", admin=True)
 ft.context.page.navigate("/")

 return ft.Column(
 [
 ft.Text("Sign In", size=28),
 ft.TextField(label="Username", value="admin", ref=username_ref),
 ft.Row(
 [
 ft.Button("Login", on_click=handle_login),
 ft.Button("Login as Admin", on_click=handle_admin_login),
]
),
],
 width=300,
)

@ft.component
def ProtectedRoute():
 auth = ft.use_context(AuthContext)
 outlet = ft.use_route_outlet()

 if not auth.is_authenticated:
 ft.context.page.navigate("/login")
 return ft.ProgressRing()

 return outlet

Navigation

@ft.component
def NavLink(label, path):
 active = ft.is_route_active(path)
 return ft.Container(
 content=ft.Text(
 label,
 weight=ft.FontWeight.BOLD if active else ft.FontWeight.NORMAL,
 color=ft.Colors.PRIMARY if active else ft.Colors.ON_SURFACE,
),
 bgcolor=ft.Colors.PRIMARY_CONTAINER if active else None,
 padding=ft.Padding.symmetric(horizontal=16, vertical=8),
 border_radius=8,
 on_click=lambda: ft.context.page.navigate(path),
)

@ft.component
def AppLayout():
 auth = ft.use_context(AuthContext)
 outlet = ft.use_route_outlet()

 if auth is None:
 return ft.ProgressRing()

 return ft.Column(

```

```

[
 # Header
 ft.Container(
 content=ft.Row(
 [
 ft.Text("Router Demo", size=20, weight=ft.FontWeight.BOLD),
 NavLink("Home", "/"),
 NavLink("Projects", "/projects"),
 NavLink("Settings", "/settings"),
 ft.Text(f"Hi, {auth.username}"),
 ft.IconButton(
 ft.Icons.LOGOUT,
 on_click=lambda: (
 auth.logout(),
 ft.context.page.navigate("/login"),
),
),
],
),
 padding=10,
 bgcolor=ft.Colors.SURFACE_BRIGHT,
),
 ft.Divider(height=1),
 # Content
 ft.Container(content=outlet, padding=20),
],
)

Page components

@ft.component
def Home():
 data = ft.use_route_loader_data()
 return ft.Column(
 [
 ft.Text(data["greeting"], size=24),
 ft.Text(f"{data['featured_count']} featured projects available"),
 ft.Button(
 "Browse projects",
 on_click=lambda: ft.context.page.navigate("/projects"),
),
],
)

@ft.component
def ProjectsList():
 data = ft.use_route_loader_data()
 return ft.Column(
 [
 ft.Text("Projects", size=24),
 *[
 ft.ListTile(
 title=ft.Text(p["name"]),
 subtitle=ft.Text(p["status"]),
 on_click=lambda _, pid=p["id"]: ft.context.page.navigate(
 f"/projects/{pid}"
),
)
 for p in data
],
],
)

@ft.component
def ProjectDetails():
 data = ft.use_route_loader_data()
 params = ft.use_route_params()
 location = ft.use_route_location()

 return ft.Column(
 [
 ft.Text(data["name"], size=24),
 ft.Text(data["description"]),
 ft.Text(f"Project ID: {params['projectId']}", italic=True),
 ft.Text(f"Location: {location}", italic=True, size=12),
 ft.Button(
 "Back to projects",
 on_click=lambda: ft.context.page.navigate("/projects"),
),
],
)

```

```

@ft.component
def ProjectsLayout():
 outlet = ft.use_route_outlet()
 return ft.Column(
 [
 ft.Container(
 content=ft.Text(
 "PROJECTS",
 weight=ft.FontWeight.BOLD,
 color=ft.Colors.ON_SECONDARY_CONTAINER,
),
 bgcolor=ft.Colors.SECONDARY_CONTAINER,
 padding=ft.Padding.symmetric(horizontal=12, vertical=4),
 border_radius=4,
),
 outlet,
]
)

@ft.component
def SettingsHome():
 data = ft.use_route_loader_data()
 return ft.Column(
 [
 ft.Text("Settings", size=24),
 ft.Text("Available sections:"),
 *[
 ft.ListTile(
 title=ft.Text(section),
 on_click=lambda _, s=section: ft.context.page.navigate(
 f"/settings/{s.lower()}"
),
)
 for section in data["sections"]
],
]
)

@ft.component
def SettingsSection():
 params = ft.use_route_params()
 return ft.Column(
 [
 ft.Text(f"{params['section'].title()} Settings", size=24),
 ft.Text(f"Configure your {params['section']} preferences here."),
 ft.Button(
 "Back to settings",
 on_click=lambda: ft.context.page.navigate("/settings"),
),
]
)

```

# -----

## Managed views — nested routes

Each route component returns a View with its own AppBar. Navigating deeper pushes views onto the stack; swipe-back and AppBar back button pop them.

play\_arrowTry Online

```

import flet as ft

@ft.component
def Home():
 return ft.View(
 route="/",
 can_pop=False,
 appbar=ft.AppBar(title=ft.Text("Home")),
 controls=[
 ft.Text("Home", size=24),
 ft.Button(
 "Go to Products",
 on_click=lambda: ft.context.page.navigate("/products"),
),
],
)

@ft.component
def ProductsList():
 return ft.View(
 route="/products",

```

```

 appbar=ft.AppBar(title=ft.Text("Products")),
 controls=[
 ft.Text("All Products", size=24),
 ft.Button(
 "View Product #1",
 on_click=lambda: ft.context.page.navigate("/products/1"),
),
 ft.Button(
 "View Product #2",
 on_click=lambda: ft.context.page.navigate("/products/2"),
),
],
),
)

@ft.component
def ProductDetails():
 params = ft.use_route_params()
 return ft.View(
 route=f"/products/{params['pid']}",
 appbar=ft.AppBar(title=ft.Text(f"Product #{params['pid']}")),
 controls=[
 ft.Text(f"Details for product #{params['pid']}", size=24),
],
)

@ft.component
def App():
 return ft.Router(
 [
 ft.Route(
 component=Home,
 children=[
 ft.Route(
 path="products",
 component=ProductsList,
 children=[
 ft.Route(path=":pid", component=ProductDetails),
],
),
],
),
],
 manage_views=True,
)

if __name__ == "__main__":
 ft.run(lambda page: page.render_views(App))

```

## Managed views — shared layout with outlet

A layout route with `outlet=True` wraps child routes in a shared View. Leaf components return regular controls; the layout provides the View.

[play\\_arrowTry Online](#)

```

import flet as ft

@ft.component
def Home():
 return ft.View(
 route="/",
 can_pop=False,
 appbar=ft.AppBar(title=ft.Text("Home")),
 controls=[
 ft.Text("Welcome!", size=24),
 ft.Button(
 "Browse Products",
 on_click=lambda: ft.context.page.navigate("/products"),
),
],
)

@ft.component
def ProductsList():
 """Returns a regular control – the layout provides the View."""
 return ft.Column(
 [
 ft.Text("All Products", size=24),
 ft.Button(
 "View Product #1",
 on_click=lambda: ft.context.page.navigate("/products/1"),
),
],
)

```

```

 ft.Button(
 "View Product #2",
 on_click=lambda: ft.context.page.navigate("/products/2"),
),
]
)

@ft.component
def ProductDetails():
 """Returns a regular control – the layout provides the View."""
 params = ft.use_route_params()
 return ft.Text(f"Product #{params['pid']}", size=24)

@ft.component
def ProductsLayout():
 """Shared layout – wraps each child route in a View with AppBar + footer."""
 outlet = ft.use_route_outlet()
 return ft.View(
 # use_view_path() returns the per-view resolved URL (unique per level)
 # – required for Flutter Navigator keying when the layout wraps
 # multiple child routes.
 route=ft.use_view_path(),
 appbar=ft.AppBar(
 title=ft.Text("Products"),
 bgcolor=ft.Colors.SURFACE_BRIGHT,
),
 controls=[
 ft.Container(content=outlet, padding=20, expand=True),
 ft.Divider(),
 ft.Text("Products Footer", text_align=ft.TextAlign.CENTER),
],
)

@ft.component
def App():
 return ft.Router(
 [
 ft.Route(
 component=Home,
 children=[
 ft.Route(
 path="products",
 component=ProductsLayout,
 outlet=True,
 children=[
 ft.Route(
 component=ProductsList,
 children=[
 ft.Route(path=":pid", component=ProductDetails),
],
),
],
),
],
),
],
 manage_views=True,
)

if __name__ == "__main__":
 ft.run(lambda page: page.render_views(App))

```

## Managed views — full app with NavigationRail

Complete app with NavigationRail, stacked project views, and tabbed settings — all using `manage_views=True`.

[play\\_arrow Try Online](#)

```

import flet as ft

Home

@ft.component
def HomeContent():
 return ft.Column(
 [
 ft.Text("Home", size=28, weight=ft.FontWeight.BOLD),
 ft.Text("Welcome to the app!", size=16),
],
)

```

```

Projects

PROJECTS = {
 "1": "Landing Page Redesign",
 "2": "Mobile App v2",
 "3": "API Migration",
}

@ft.component
def ProjectsList():
 return ft.Column(
 [
 ft.Text("All Projects", size=24),
 *[
 ft.Button(
 name,
 on_click=lambda _, pid=pid: ft.context.page.navigate(
 f"/projects/{pid}"
),
)
 for pid, name in PROJECTS.items()
],
],
)

@ft.component
def ProjectDetails():
 params = ft.use_route_params()
 pid = params.get("pid", "?")
 name = PROJECTS.get(pid, "Unknown")
 return ft.Column(
 [
 ft.Text(name, size=24, weight=ft.FontWeight.BOLD),
 ft.Text(f"Project ID: {pid}", size=16),
 ft.Text(
 "This is where project details, tasks, and team info would go.",
 size=14,
),
],
)

Settings

@ft.component
def GeneralSettings():
 return ft.Column(
 [
 ft.Text("General Settings", size=24),
 ft.Switch(label="Dark mode"),
 ft.Switch(label="Notifications"),
 ft.Switch(label="Auto-save"),
],
)

@ft.component
def AccountSettings():
 return ft.Column(
 [
 ft.Text("Account Settings", size=24),
 ft.TextField(label="Display name", value="Feodor"),
 ft.TextField(label="Email", value=""),
 ft.Button("Save changes"),
],
)

@ft.component
def SettingsLayout():
 """Tab navigation for settings – returns controls, not a View."""
 outlet = ft.use_route_outlet()
 return ft.Column(
 [
 ft.Text("Settings", size=28, weight=ft.FontWeight.BOLD),
 ft.Row(
 [
 ft.Button(
 "General",
 style=ft.ButtonStyle(

```

```

 bgcolor=ft.Colors.PRIMARY_CONTAINER
 if ft.is_route_active("/settings/general", exact=True)
 else None,
),
 on_click=lambda: ft.context.page.navigate("/settings/general"),
),
ft.Button(
 "Account",
 style=ft.ButtonStyle(
 bgcolor=ft.Colors.PRIMARY_CONTAINER
 if ft.is_route_active("/settings/account", exact=True)
 else None,
),
 on_click=lambda: ft.context.page.navigate("/settings/account"),
),
],
),
ft.Divider(),
ft.Container(content=outlet, expand=True),
],
expand=True,
)

```

```

Root layout with NavigationRail

```

```
NAV_ROUTES = ["/", "/projects", "/settings/general"]
```

```

@ft.component
def RootLayout():
 """Root layout – returns a View with NavigationRail + outlet."""
 outlet = ft.use_route_outlet()

 selected = 0
 if ft.is_route_active("/projects"):
 selected = 1
 elif ft.is_route_active("/settings"):
 selected = 2

 def on_nav_change(e):
 ft.context.page.navigate(NAV_ROUTES[e.control.selected_index])

 return ft.View(
 route="/", # fixed key – no transition animation between top-level pages
 can_pop=False,
 controls=[
 ft.Row(
 [
 ft.NavigationRail(
 selected_index=selected,
 label_type=ft.NavigationRailLabelType.ALL,
 destinations=[
 ft.NavigationRailDestination(
 icon=ft.Icons.HOME_OUTLINED,
 selected_icon=ft.Icons.HOME,
 label="Home",
),
 ft.NavigationRailDestination(
 icon=ft.Icons.FOLDER_OUTLINED,
 selected_icon=ft.Icons.FOLDER,
 label="Projects",
),
 ft.NavigationRailDestination(
 icon=ft.Icons.SETTINGS_OUTLINED,
 selected_icon=ft.Icons.SETTINGS,
 label="Settings",
),
],
 on_change=on_nav_change,
),
 ft.VerticalDivider(width=1),
 ft.Container(content=outlet, expand=True, padding=20),
],
 expand=True,
),
],
)

```

```

App

```

```

@ft.component
def App():

```

```

return ft.Router(
 [
 ft.Route(
 component=RootLayout,
 outlet=True,
 children=[
 ft.Route(index=True, component=HomeContent),
 ft.Route(
 path="projects",
 component=ProjectsList,
 children=[
 ft.Route(path=":pid", component=ProjectDetails),
],
),
 ft.Route(
 path="settings",
 component=SettingsLayout,
 outlet=True,
 children=[
 ft.Route(path="general", component=GeneralSettings),
 ft.Route(path="account", component=AccountSettings),
],
),
],
),
],
 manage_views=True,
)

```

```

if __name__ == "__main__":
 ft.run(lambda page: page.render_views(App))

```

## Modal routes

Routes marked `modal=True` are rendered as a fullscreen-dialog overlay on top of the previous (non-modal) location's view stack. A *global* modal is declared at the top level (the URL works from anywhere); a *local* modal is declared as a child of a non-modal parent (the URL embeds the parent's segment, so deep-link works without any state).

### play\_arrowTry Online

"""Modal routes – fullscreen-dialog overlays that don't unmount the underlying stack.

A route marked `modal=True` is rendered on top of the previous location's view stack. Closing the modal pops it without rebuilding the views underneath – they stay mounted.

Two flavours, distinguished by placement in the route tree:

\* **Global modal** – declared at the top level. Reachable from anywhere with a single URL; the base stack comes from the last non-modal location the Router saw. Demonstrated by `/settings` here.

\* **Local modal** – declared as a child of a non-modal parent. The URL embeds the parent's path, so deep-link works without any state. Demonstrated by `/items/<id>/edit` here.

"""

```
import flet as ft
```

```

Pages

```

```

@ft.component
def Home():
 page = ft.context.page
 return ft.View(
 route="/",
 appbar=ft.AppBar(title=ft.Text("Home")),
 controls=[
 ft.Text("Welcome!", size=22, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open Items",
 on_click=lambda: page.navigate("/items"),
),
 ft.Button(
 "Open Settings (global modal)",
 on_click=lambda: page.navigate("/settings"),
),
],
)

```

```
ITEMS = {"1": "Apples", "2": "Bananas", "3": "Cherries"}
```



```

@ft.component
def ItemList():
 page = ft.context.page
 return ft.View(
 route="/items",
 appbar=ft.AppBar(title=ft.Text("Items")),
 controls=[
 ft.Text("Pick an item:"),
 *[
 ft.Button(
 name,
 on_click=lambda _e, i=iid: page.navigate(f"/items/{i}"),
)
 for iid, name in ITEMS.items()
],
 ft.Button(
 "Open Settings (global modal)",
 on_click=lambda: page.navigate("/settings"),
),
],
)

```

```

@ft.component
def ItemDetails():
 page = ft.context.page
 params = ft.use_route_params()
 iid = params["id"]
 name = ITEMS.get(iid, "Unknown")
 return ft.View(
 route=ft.use_view_path(),
 appbar=ft.AppBar(title=ft.Text(f"Item: {name}")),
 controls=[
 ft.Text(f"Details for item {iid}: {name}"),
 ft.Button(
 "Edit (local modal)",
 on_click=lambda: page.navigate(f"/items/{iid}/edit"),
),
 ft.Button(
 "Open Settings (global modal)",
 on_click=lambda: page.navigate("/settings"),
),
],
)

```

```

Modals

```

```

@ft.component
def SettingsModal():
 """Global modal – `/settings` works from any base location."""
 return ft.View(
 route="/settings",
 fullscreen_dialog=True,
 appbar=ft.AppBar(title=ft.Text("Settings")),
 controls=[
 ft.Text("Account / System / About would live here."),
 ft.Text(
 "Close the modal – the underlying view (Home, Items, or "
 "Item details) is still mounted underneath, no rebuild.",
 size=12,
 color=ft.Colors.ON_SURFACE_VARIANT,
),
],
)

```

```

@ft.component
def EditItemModal():
 """Local modal ■– `/items/<id>/edit` is pinned to a specific item.

 Deep-link to it directly and the stack rebuilds from URL alone:
 `[ItemList, ItemDetails(id), EditItemModal]`.
 """
 params = ft.use_route_params()
 iid = params["id"]
 name = ITEMS.get(iid, "Unknown")
 return ft.View(
 route=ft.use_view_path(),
 fullscreen_dialog=True,
 appbar=ft.AppBar(title=ft.Text(f"Edit {name}")),
 controls=[
 ft.Text(f"Editing item {iid}"),
 ft.TextField(label="Name", value=name),

```

```

 ft.Text(
 "URL: " + ft.context.page.route,
 size=12,
 color=ft.Colors.ON_SURFACE_VARIANT,
),
],
)

App

@ft.component
def App():
 return ft.Router(
 [
 # Pathless parent so /items/<id> stacks on top of Home.
 ft.Route(
 component=Home,
 children=[
 ft.Route(
 path="items",
 component=ItemList,
 children=[
 ft.Route(
 path=":id",
 component=ItemDetails,
 children=[
 # Local modal – child of /items/:id.
 ft.Route(
 path="edit",
 component=EditItemModal,
 modal=True,
),
],
),
],
),
],
),
 # Global modal – top-level route, reachable from anywhere.
 ft.Route(path="settings", component=SettingsModal, modal=True),
],
 manage_views=True,
)

if __name__ == "__main__":
 ft.run(lambda page: page.render_views(App))

```

## Recursive routes

A route marked `recursive=True` can match itself as its own descendant — one View is emitted per consumed URL segment. Use this for tree-shaped URLs with unbounded depth (e.g. a file browser at `/folder/a/b/c`).

### play\_arrow Try Online

"""Recursive routes – unbounded URL depth, one View per segment.

A route marked ``recursive=True`` can match itself as its own descendant. The Router emits one ``_RouteMatch`` per consumed URL segment, which (in ``manage_views=True`` mode) turns into one View per segment in the navigation stack – so the back gesture walks one level at a time.

Sibling-wins-over-recursion: non-recursive children of a recursive route are tried BEFORE self-recursion at every depth. Here that means ``/folder/<anything>/search`` matches the ``Search`` page (not a folder named "search"), without needing to declare ``search`` at every depth manually.

Try this in the browser bar:

```

* `/folder` → folder root
* `/folder/a` → 1 level deep
* `/folder/a/b` → 2 levels deep
* `/folder/a/b/c/d/e` → 5 levels deep (no fixed limit)
* `/folder/a/b/search` → search page below `a/b`
"""

```

import flet as ft

```

@ft.component
def Home():

```

```

page = ft.context.page
return ft.View(
 route="/",
 appbar=ft.AppBar(title=ft.Text("Folder Browser")),
 controls=[
 ft.Text("A toy file browser.", size=18),
 ft.Button("Open /folder", on_click=lambda: page.navigate("/folder")),
],
)

@ft.component
def Folders():
 """Root of the folder tree (`/folder`)."""
 page = ft.context.page
 return ft.View(
 route="/folder",
 appbar=ft.AppBar(title=ft.Text("Folders")),
 controls=[
 ft.Text("Top-level folders:", weight=ft.FontWeight.BOLD),
 *[
 ft.Button(
 f"Open {name}",
 on_click=lambda _e, n=name: page.navigate(f"/folder/{n}"),
)
 for name in ("docs", "projects", "scratch")
],
 ft.Button(
 "Search at root",
 on_click=lambda: page.navigate("/folder/search"),
),
],
)

@ft.component
def Folder():
 """A folder at any depth. Re-used by the recursive route at every level.

 `use_route_params()` returns just the current segment (e.g.
 `{"name": "b"}` at depth 2). `use_view_path()` returns the
 accumulated URL up to this view, useful for showing the full path
 and for building child URLs.
 """
 page = ft.context.page
 params = ft.use_route_params()
 view_path = ft.use_view_path()
 segments = [s for s in view_path[len("/folder") :].split("/") if s]
 return ft.View(
 route=view_path,
 appbar=ft.AppBar(title=ft.Text(" / ".join(segments) or "Folders")),
 controls=[
 ft.Text(f"Current segment: {params['name']}"),
 ft.Text(f"Full path: {view_path}"),
 ft.Text("Open a sub-folder:", weight=ft.FontWeight.BOLD),
 *[
 ft.Button(
 f"Into {child}",
 on_click=lambda _e, c=child: page.navigate(f"{view_path}/{c}"),
)
 for child in ("alpha", "beta", "gamma")
],
 ft.Button(
 "Search here",
 on_click=lambda: page.navigate(f"{view_path}/search"),
),
],
)

@ft.component
def Search():
 """Non-recursive child of the recursive Folder route. Wins over
 self-recursion at every depth, so `/folder/<any-segments>/search`
 always lands here instead of being eaten as another folder named
 "search"."""
 page = ft.context.page
 view_path = ft.use_view_path()
 # Strip trailing "/search" to learn the folder we're searching in.
 folder = view_path[: -len("/search")] or "/folder"
 return ft.View(
 route=view_path,
 appbar=ft.AppBar(title=ft.Text("Search")),
 controls=[
 ft.Text(f"Searching in: {folder}"),
 ft.TextField(label="Query", autofocus=True),
 ft.Button("Back to folder", on_click=lambda: page.navigate(folder)),
],
)

```

```

],
)

@ft.component
def App():
 return ft.Router(
 [
 ft.Route(
 component=Home,
 children=[
 ft.Route(
 path="folder",
 component=Folders,
 children=[
 # Declared ONCE – the `search` sibling check
 # happens at every recursion depth.
 ft.Route(
 path=":name",
 component=Folder,
 recursive=True,
 children=[
 ft.Route(path="search", component=Search),
],
),
],
),
],
),
],
 manage_views=True,
)

if __name__ == "__main__":
 ft.run(lambda page: page.render_views(App))

```

# Row

Displays its children in a horizontal array.

To cause a child control to expand and fill the available horizontal space, set its `expand` property.

Example:

```
ft.Row(
 controls=[
 ft.Card(
 shape=ft.ContinuousRectangleBorder(radius=10),
 content=ft.Container(
 padding=5,
 border_radius=ft.BorderRadius.all(5),
 bgcolor=ft.Colors.AMBER_100,
 content=ft.Text(f"Control {i}"),
),
),
 for i in range(1, 6)
],
)
```

Image: Row - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/row/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/row/image_for_docs.png)

## Row

Inherits: `LayoutControl`, `ScrollableControl`, `AdaptiveControl`

## Properties

- `alignment` - Defines how the child controls should be placed horizontally.
- `controls` - A list of Controls to display.
- `intrinsic_height` - Whether this row should be as tall as the tallest child control in controls.
- `run_alignment` - How the runs should be placed in the cross-axis when `wrap` is `True`.
- `run_spacing` - The spacing between runs when `wrap` is `True`.
- `spacing` - The spacing between the child controls.
- `tight` - Whether this row should occupy all available horizontal space (`True`), or only as much as needed by its children controls (`False`).
- `vertical_alignment` - Defines how the child controls should be placed vertically.
- `wrap` - Whether this row should put child controls into additional rows (runs) if they don't fit in a single row.

## Examples

### Row spacing

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 def generate_items(count: int):
 return [
 ft.Container(
 content=ft.Text(value=str(i)),
 alignment=ft.Alignment.CENTER,
 width=50,
 height=50,
 bgcolor=ft.Colors.AMBER,
 border_radius=ft.BorderRadius.all(5),
)
 for i in range(1, count + 1)
]

 def handle_slider_change(e: ft.Event[ft.Slider]):
 row.spacing = int(e.control.value)
 row.update()

 row = ft.Row(spacing=0, scroll=ft.ScrollMode.AUTO, controls=generate_items(10))
 page.add(
```

```

ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Column(
 controls=[
 ft.Text("Spacing between items"),
 ft.Slider(
 key="slider",
 min=0,
 max=50,
 divisions=50,
 value=0,
 label="{value}",
 on_change=handle_slider_change,
),
],
),
],
),
 row,
]
)
)
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: spacing - [https://flet.dev/docs/test-images/examples/controls/core/golden/macos/row/row\\_spacing\\_adjustment.gif](https://flet.dev/docs/test-images/examples/controls/core/golden/macos/row/row_spacing_adjustment.gif)

## Row wrapping

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 def generate_items(count: int):
 return [
 ft.Container(
 content=ft.Text(value=str(i)),
 alignment=ft.Alignment.CENTER,
 width=50,
 height=50,
 bgcolor=ft.Colors.AMBER,
 border_radius=ft.BorderRadius.all(5),
)
 for i in range(1, count + 1)
]

 def handle_slider_change(e: ft.Event[ft.Slider]):
 row.width = float(e.control.value)
 row.update()

 row = ft.Row(
 wrap=True,
 spacing=10,
 run_spacing=10,
 width=page.window.width,
 controls=generate_items(30),
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Column(
 controls=[
 ft.Text(
 "Change the row width to see how child items wrap "
 "onto multiple rows:"
),
 ft.Slider(
 min=0,
 max=page.window.width,
 divisions=20,
 value=page.window.width,
 label="{value}",
 on_change=handle_slider_change,
),
],
),
],
),
 row,
],
)
)

```

```
if __name__ == "__main__":
 ft.run(main)
```

Image: wrap - [https://flet.dev/docs/test-images/examples/controls/core/golden/macos/row/wrap\\_adjustment.gif](https://flet.dev/docs/test-images/examples/controls/core/golden/macos/row/wrap_adjustment.gif)

## Row alignment

play\_arrow Try Online

```
import flet as ft

@ft.control
class RowWithAlignment(ft.Column):
 alignment: ft.MainAxisAlignment = ft.MainAxisAlignment.START

 def init(self):
 self.controls = [
 ft.Text(str(self.alignment), size=16),
 ft.Container(
 bgcolor=ft.Colors.AMBER_100,
 content=ft.Row(
 alignment=self.alignment,
 controls=self.generate_items(3),
),
),
]

 @staticmethod
 def generate_items(count: int):
 return [
 ft.Container(
 content=ft.Text(value=str(i)),
 alignment=ft.Alignment.CENTER,
 width=50,
 height=50,
 bgcolor=ft.Colors.AMBER_500,
)
 for i in range(1, count + 1)
]

def main(page: ft.Page):
 page.scroll = ft.ScrollMode.AUTO
 page.add(
 ft.SafeArea(
 content=ft.Column(
 scroll=ft.ScrollMode.AUTO,
 controls=[
 RowWithAlignment(alignment=ft.MainAxisAlignment.START),
 RowWithAlignment(alignment=ft.MainAxisAlignment.CENTER),
 RowWithAlignment(alignment=ft.MainAxisAlignment.END),
 RowWithAlignment(alignment=ft.MainAxisAlignment.SPACE_BETWEEN),
 RowWithAlignment(alignment=ft.MainAxisAlignment.SPACE_AROUND),
 RowWithAlignment(alignment=ft.MainAxisAlignment.SPACE_EVENLY),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: alignment - <https://flet.dev/docs/test-images/examples/controls/core/golden/macos/row/alignment.png>

## Row vertical alignment

play\_arrow Try Online

```
import flet as ft

@ft.control
class RowWithVerticalAlignment(ft.Column):
 alignment: ft.CrossAxisAlignment = ft.CrossAxisAlignment.START

 def init(self):
 self.controls = [
 ft.Text(str(self.alignment), size=16),
 ft.Container(
 bgcolor=ft.Colors.AMBER_100,
 height=150,
 content=ft.Row(
 vertical_alignment=self.alignment,
 controls=self.generate_items(3),
),
),
]
```

```

),
]

@staticmethod
def generate_items(count: int):
 return [
 ft.Container(
 content=ft.Text(value=str(i)),
 alignment=ft.Alignment.CENTER,
 width=50,
 height=50,
 bgcolor=ft.Colors.AMBER_500,
)
 for i in range(1, count + 1)
]

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 RowWithVerticalAlignment(alignment=ft.CrossAxisAlignment.START),
 RowWithVerticalAlignment(alignment=ft.CrossAxisAlignment.CENTER),
 RowWithVerticalAlignment(alignment=ft.CrossAxisAlignment.END),
]
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: vertical-alignment - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/row/vertical\\_alignment.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/row/vertical_alignment.png)

## Properties

### buildalignmentclass-attributeinstance-attribute

```
alignment: MainAxisAlignment = MainAxisAlignment.START
```

Defines how the child controls should be placed horizontally.

### buildcontrolsclass-attributeinstance-attribute

```
controls: list[Control] = field(default_factory=list)
```

A list of Controls to display.

### buildintrinsic\_heightclass-attributeinstance-attribute

```
intrinsic_height: bool = False
```

Whether this row should be as tall as the tallest child control in controls.

### buildrun\_alignmentclass-attributeinstance-attribute

```
run_alignment: MainAxisAlignment = MainAxisAlignment.START
```

How the runs should be placed in the cross-axis when wrap is True.

### buildrun\_spacingclass-attributeinstance-attribute

```
run_spacing: Number = 10
```

The spacing between runs when wrap is True.

### buildspacingclass-attributeinstance-attribute

```
spacing: Number = 10
```

The spacing between the child controls.

### Note

Has effect only when alignment is set to MainAxisAlignment.START, MainAxisAlignment.END, or MainAxisAlignment.CENTER.

### buildtightclass-attributeinstance-attribute

```
tight: bool = False
```

Whether this row should occupy all available horizontal space (True), or only as much as needed by its children controls (False).



#### Note

Has effect only when wrap is False.

#### **buildvertical\_alignment**class-attributeinstance-attribute

```
vertical_alignment: CrossAxisAlignment = (
 CrossAxisAlignment.CENTER
)
```

Defines how the child controls should be placed vertically.

#### Note

When wrap is True, this property doesn't support CrossAxisAlignment.STRETCH or CrossAxisAlignment.BASELINE. If either is used, CrossAxisAlignment.CENTER will be applied instead.

#### **buildwrap**class-attributeinstance-attribute

```
wrap: bool = False
```

Whether this row should put child controls into additional rows (runs) if they don't fit in a single row.

# SafeArea

A control that insets its content by sufficient padding to avoid intrusions by the operating system.

For example, this will indent the content by enough to avoid the status bar at the top of the screen.

It will also indent the content by the amount necessary to avoid the Notch on the iPhone X, or other similar creative physical features of the display.

When a `minimum_padding` is specified, the greater of the minimum padding or the safe area padding will be applied.

Inherits: `LayoutControl`, `AdaptiveControl`

## Properties

- `avoid_intrusions_bottom` - Whether to avoid system intrusions on the bottom side of the screen.
- `avoid_intrusions_left` - Whether to avoid system intrusions on the left.
- `avoid_intrusions_right` - Whether to avoid system intrusions on the right.
- `avoid_intrusions_top` - Whether to avoid system intrusions at the top of the screen, typically the system status bar.
- `content` - The control to display.
- `maintain_bottom_view_padding` - Specifies whether the `SafeArea` should maintain the bottom `MediaQueryData.viewPadding` instead of the bottom `MediaQueryData.padding`.
- `minimum_padding` - The minimum padding to apply.

## Example

### SafeArea

[play\\_arrow](#) Try Online

```
import flet as ft

class State:
 counter = 0

def main(page: ft.Page):
 state = State()
 message = ft.Text("0", size=50)

 def handle_button_click(e: ft.Event[ft.FloatingActionButton]):
 state.counter += 1
 message.value = str(state.counter)
 message.update()

 page.floating_action_button = ft.FloatingActionButton(
 icon=ft.Icons.ADD,
 on_click=handle_button_click,
)

 page.add(
 ft.SafeArea(
 expand=True,
 content=ft.Container(
 alignment=ft.Alignment.CENTER,
 content=message,
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### `avoid_intrusions_bottom`class-attributeinstance-attribute

`avoid_intrusions_bottom`: bool = True

Whether to avoid system intrusions on the bottom side of the screen.

### **buildavoid\_intrusions\_leftclass-attributeinstance-attribute**

`avoid_intrusions_left: bool = True`

Whether to avoid system intrusions on the left.

### **buildavoid\_intrusions\_rightclass-attributeinstance-attribute**

`avoid_intrusions_right: bool = True`

Whether to avoid system intrusions on the right.

### **buildavoid\_intrusions\_topclass-attributeinstance-attribute**

`avoid_intrusions_top: bool = True`

Whether to avoid system intrusions at the top of the screen, typically the system status bar.

### **buildcontentinstance-attribute**

`content: Control`

The control to display.

Raises:

- `ValueError` - If it is not visible.

### **buildmaintain\_bottom\_view\_paddingclass-attributeinstance-attribute**

`maintain_bottom_view_padding: bool = False`

Specifies whether the `SafeArea` should maintain the bottom `MediaQueryData.viewPadding` instead of the bottom `MediaQueryData.padding`.

This avoids layout shifts caused by keyboard overlays, useful when flexible controls are used.

### **buildminimum\_paddingclass-attributeinstance-attribute**

`minimum_padding: PaddingValue = 0`

The minimum padding to apply.

The greater of the minimum insets and the media padding will be applied.

# Screenshot

Takes a screenshot of containing control.

Inherits: Control

## Properties

- content - The control to be captured.

## Methods

- capture - Captures a screenshot of the enclosed content control.

## Examples

### Taking screenshot

play\_arrowTry Online

```
from pathlib import Path

import flet as ft
from flet.utils.files import get_current_script_dir

def main(page: ft.Page):
 async def take_screenshot(e: ft.Event[ft.Button]):
 image = await scr.capture()
 with open(Path(get_current_script_dir(), "screenshot.png"), "wb") as f:
 f.write(image)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 scr := ft.Screenshot(
 content=ft.Container(
 padding=10,
 content=ft.Button(
 "Hello, world!",
 bgcolor=ft.Colors.BLUE,
 elevation=10,
),
),
),
 ft.Button("Take screenshot", on_click=take_screenshot),
]
)
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### buildcontentinstance-attribute

content: Control

The control to be captured.

## Methods

### deployed\_codecaptureasync

```
capture(
 pixel_ratio: Number | None = None,
 delay: DurationValue | None = None,
) -> bytes
```

Captures a screenshot of the enclosed content control.

Parameters:

- pixel\_ratio (Number | None, default: None) - A pixel ratio of the captured screenshot. If None, device-specific pixel ratio will be used.
- delay (DurationValue | None, default: None) - A delay before taking a screenshot. The delay will be 20 milliseconds if not specified.

Returns:

- bytes - Screenshot in PNG format.

# ScrollableControl

Shared scroll behavior for controls that expose a scrollable viewport.

This mixin-style control is inherited by controls such as Column, Row, ResponsiveRow, View, ListView, and GridView. It provides a common API for:

- enabling/disabling scrolling and scrollbar visibility via `scroll`;
- receiving throttled scroll notifications via `on_scroll` and `scroll_interval`;
- imperatively changing position with `scroll_to`.

Inherits: Control

## Properties

- `auto_scroll` - Whether the scrollbar should automatically move its position to the end when children updated.
- `scroll` - Defines the scroll bar configuration of this control.
- `scroll_interval` - Throttling in milliseconds for `on_scroll` event.

## Events

- `on_scroll` - Called when scroll position is changed by a user.

## Methods

- `scroll_to` - Moves the scroll position.

## Properties

### `buildauto_scrollclass-attributeinstance-attribute`

`auto_scroll: bool = False`

Whether the scrollbar should automatically move its position to the end when children updated.

Note

Must be False for `scroll_to` method to work.

### `buildscrollclass-attributeinstance-attribute`

`scroll: ScrollMode | Scrollbar | None = None`

Defines the scroll bar configuration of this control.

Can be a Scrollbar instance for full control over the appearance of the scrollbar, or a ScrollMode value, for ready-made scrollbar behaviors.

### `buildscroll_intervalclass-attributeinstance-attribute`

`scroll_interval: Number = 10`

Throttling in milliseconds for `on_scroll` event.

## Events

### `bolton_scrollclass-attributeinstance-attribute`

`on_scroll: EventHandler[OnScrollEvent] | None = None`

Called when scroll position is changed by a user.

## Methods

### `deployed_codescroll_toasync`

```
scroll_to(
 offset: float | None = None,
 delta: float | None = None,
 scroll_key: ScrollKey
 | str
```

```

| int
| float
| bool
| None = None,
duration: DurationValue = 0,
curve: AnimationCurve = AnimationCurve.EASE,
)

```

Moves the scroll position.

Parameters:

- offset (float | None, default: None) - Absolute scroll target in pixels. A negative value is interpreted relative to the end (e.g. -1 to jump to the very end).
- delta (float | None, default: None) - Relative scroll change in pixels. Positive values scroll forward, negative values scroll backward.
- scroll\_key (ScrollKey | str | int | float | bool | None, default: None) - Key of the target control to scroll to.
- duration (DurationValue, default: 0) - The scroll animation duration.
- curve (AnimationCurve, default: AnimationCurve.EASE) - The scroll animation curve.

Notes

- Exactly one of offset, delta or scroll\_key should be provided.
- auto\_scroll must be False.
- This method is ineffective for controls (e.g. ListView, GridView) that build items dynamically.

# SearchBar

Manages a "search view" route that allows the user to select one of the suggested completions for a search query.

```
ft.SearchBar(bar_hint_text="Search...")
```

Image: SearchBar - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/search\\_bar/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/search_bar/image_for_docs.png)

SearchBar

Inherits: `LayoutControl`

## Properties

- `autofocus` - Whether the text field should focus itself if nothing else is already focused.
- `bar_bgcolor` - Defines the background color of the search bar in all or specific `ControlState` states.
- `bar_border_side` - The color and weight of the search bar's outline.
- `bar_elevation` - The elevation of the search bar.
- `bar_hint_text` - Defines the text to be shown in the search bar when it is empty and the search view is close.
- `bar_hint_text_style` - The style to use for the `SearchBar.bar_hint_text`.
- `bar_leading` - A control to display before the text input field when the search view is close.
- `bar_overlay_color` - Defines the highlight color that's typically used to indicate that the search bar is in `FOCUSED`, `HOVERED`, or `PRESSED` states.
- `bar_padding` - The padding between the search bar's boundary and its contents.
- `bar_scroll_padding` - Configures the padding around a `Scrollable` when the text field scrolls into view.
- `bar_shadow_color` - The shadow color of the search bar.
- `bar_shape` - The shape of the search bar.
- `bar_size_constraints` - Optional size constraints for the search bar.
- `bar_text_style` - The style to use for the text being edited.
- `bar_trailing` - A list of controls to display after the text input field when the search view is close.
- `capitalization` - Enables automatic on-the-fly capitalization of entered text.
- `controls` - The list of controls to be displayed below the search bar when in search view.
- `divider_color` - The color of the divider when in search view.
- `full_screen` - Defines whether the search view grows to fill the entire screen when the search bar is tapped.
- `keyboard_type` - The type of action button to use for the keyboard.
- `shrink_wrap` - Whether the search view should shrink-wrap its contents.
- `value` - The text in the search bar.
- `view_bar_padding` - The padding to use for the search view's search bar.
- `view_bgcolor` - Defines the background color of the search view.
- `view_elevation` - Defines the elevation of the search view.
- `view_header_height` - The height of the search field on the search view.
- `view_header_text_style` - Defines the text style of the text being edited on the search view.
- `view_hint_text` - Defines the text to be displayed when the search bar's input field is empty.
- `view_hint_text_style` - Defines the text style of `view_hint_text`.
- `view_leading` - A Control to display before the text input field when the search view is open.
- `view_padding` - The padding to use for the search view.
- `view_shape` - Defines the shape of the search view.



- `view_side` - Defines the color and weight of the search view's outline.
- `view_size_constraints` - Optional size constraints for the search view.
- `view_trailing` - A list of Controls to display after the text input field when the search view is open.

## Events

- `on_blur` - Fired when the search bar loses focus.
- `on_change` - Called when the typed input in the search bar has changed.
- `on_focus` - Fired when the search bar gains focus.
- `on_submit` - Called when user presses ENTER while focus is on SearchBar.
- `on_tap` - Called when the search bar is tapped.
- `on_tap_outside_bar` - Fired when the user taps outside the search bar while the search view is open.

## Methods

- `close_view` - Closes an opened search view.
- `focus` - Requests focus for this control.
- `open_view` - Opens the search view.

## Examples

### SearchBar

play\_arrowTry Online

```
import flet as ft

colors = [
 "Amber",
 "Blue Grey",
 "Brown",
 "Deep Orange",
 "Green",
 "Light Blue",
 "Orange",
 "Red",
]

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 def build_tiles(items: list[str]) -> list[ft.Control]:
 return [
 ft.ListTile(
 title=ft.Text(item),
 data=item,
 on_click=handle_tile_click,
)
 for item in items
]

 async def handle_tile_click(e: ft.Event[ft.ListTile]):
 await anchor.close_view()

 async def handle_change(e: ft.Event[ft.SearchBar]):
 query = e.control.value.strip().lower()
 matching = (
 [color for color in colors if query in color.lower()] if query else colors
)
 anchor.controls = build_tiles(matching)

 def handle_submit(e: ft.Event[ft.SearchBar]):
 print(f"Submit: {e.data}")

 async def handle_tap(e: ft.Event[ft.SearchBar]):
 await anchor.open_view()

 anchor = ft.SearchBar(
 view_elevation=4,
 divider_color=ft.Colors.AMBER,
 bar_hint_text="Search colors...",
 view_hint_text="Choose a color from the suggestions...",
 on_change=handle_change,
```

```

 on_submit=handle_submit,
 on_tap=handle_tap,
 controls=build_tiles(colors),
)

 page.add(ft.SafeArea(content=anchor))

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: basic - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/search\\_bar/basic.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/search_bar/basic.png)

## Properties

### buildautofocusclass-attributeinstance-attribute

```
autofocus: bool = False
```

Whether the text field should focus itself if nothing else is already focused.

### buildbar\_bgcolorclass-attributeinstance-attribute

```
bar_bgcolor: ControlStateValue[ColorValue] | None = None
```

Defines the background color of the search bar in all or specific ControlState states.

### buildbar\_border\_sideclass-attributeinstance-attribute

```
bar_border_side: ControlStateValue[BorderSide] | None = None
```

The color and weight of the search bar's outline.

This value is combined with bar\_shape to create a shape decorated with an outline.

### buildbar\_elevationclass-attributeinstance-attribute

```
bar_elevation: ControlStateValue[Number | None] | None = (
 None
)
```

The elevation of the search bar.

### buildbar\_hint\_textclass-attributeinstance-attribute

```
bar_hint_text: str | None = None
```

Defines the text to be shown in the search bar when it is empty and the search view is close.

Usually some text that suggests what sort of input the field accepts.

### buildbar\_hint\_text\_styleclass-attributeinstance-attribute

```
bar_hint_text_style: ControlStateValue[TextStyle] | None = (
 None
)
```

The style to use for the SearchBar.bar\_hint\_text.

### buildbar\_leadingclass-attributeinstance-attribute

```
bar_leading: Control | None = None
```

A control to display before the text input field when the search view is close.

Typically an Icon or an IconButton.

### buildbar\_overlay\_colorclass-attributeinstance-attribute

```
bar_overlay_color: ControlStateValue[ColorValue] | None = (
 None
)
```

Defines the highlight color that's typically used to indicate that the search bar is in FOCUSED, HOVERED, or PRESSED states.

### buildbar\_paddingclass-attributeinstance-attribute

```
bar_padding: ControlStateValue[PaddingValue] | None = None
```

The padding between the search bar's boundary and its contents.

### buildbar\_scroll\_paddingclass-attributeinstance-attribute

```
bar_scroll_padding: PaddingValue = 20
```

Configures the padding around a Scrollable when the text field scrolls into view.

If the bar's text field is partially off-screen or covered (e.g., by the keyboard), it scrolls into view, ensuring it is positioned at the specified distance from the Scrollable edges.

### **buildbar\_shadow\_color**class-attributeinstance-attribute

```
bar_shadow_color: ControlStateValue[ColorValue] | None = (
 None
)
```

The shadow color of the search bar.

### **buildbar\_shape**class-attributeinstance-attribute

```
bar_shape: ControlStateValue[OutlinedBorder] | None = None
```

The shape of the search bar.

This shape is combined with `bar_border_side` to create a shape decorated with an outline.

### **buildbar\_size\_constraints**class-attributeinstance-attribute

```
bar_size_constraints: BoxConstraints | None = None
```

Optional size constraints for the search bar.

### **buildbar\_text\_style**class-attributeinstance-attribute

```
bar_text_style: ControlStateValue[TextStyle] | None = None
```

The style to use for the text being edited.

### **buildbar\_trailing**class-attributeinstance-attribute

```
bar_trailing: list[Control] | None = None
```

A list of controls to display after the text input field when the search view is close.

These controls can represent additional modes of searching (e.g voice search), an avatar, or an overflow menu and are usually not more than two.

### **buildcapitalization**class-attributeinstance-attribute

```
capitalization: TextCapitalization | None = None
```

Enables automatic on-the-fly capitalization of entered text.

### **buildcontrols**class-attributeinstance-attribute

```
controls: list[Control] = field(default_factory=list)
```

The list of controls to be displayed below the search bar when in search view.

Typically `ListTiles` and will be displayed in a `ListView`.

### **bulddivider\_color**class-attributeinstance-attribute

```
divider_color: ColorValue | None = None
```

The color of the divider when in search view.

### **buildfull\_screen**class-attributeinstance-attribute

```
full_screen: bool = False
```

Defines whether the search view grows to fill the entire screen when the search bar is tapped.

### **buildkeyboard\_type**class-attributeinstance-attribute

```
keyboard_type: KeyboardType = KeyboardType.TEXT
```

The type of action button to use for the keyboard.

### **buildshrink\_wrap**class-attributeinstance-attribute

```
shrink_wrap: bool | None = None
```

Whether the search view should shrink-wrap its contents.

### **buildvalue**class-attributeinstance-attribute

```
value: str = ''
```

The text in the search bar.

### **buildview\_bar\_paddingclass-attributeinstance-attribute**

`view_bar_padding: PaddingValue | None = None`

The padding to use for the search view's search bar.

If null, then the default value is 8.0 horizontally.

### **buildview\_bgcolorclass-attributeinstance-attribute**

`view_bgcolor: ColorValue | None = None`

Defines the background color of the search view.

### **buildview\_elevationclass-attributeinstance-attribute**

`view_elevation: Number | None = None`

Defines the elevation of the search view.

### **buildview\_header\_heightclass-attributeinstance-attribute**

`view_header_height: Number | None = None`

The height of the search field on the search view.

### **buildview\_header\_text\_styleclass-attributeinstance-attribute**

`view_header_text_style: TextStyle | None = None`

Defines the text style of the text being edited on the search view.

### **buildview\_hint\_textclass-attributeinstance-attribute**

`view_hint_text: str | None = None`

Defines the text to be displayed when the search bar's input field is empty.

### **buildview\_hint\_text\_styleclass-attributeinstance-attribute**

`view_hint_text_style: TextStyle | None = None`

Defines the text style of `view_hint_text`.

### **buildview\_leadingclass-attributeinstance-attribute**

`view_leading: Control | None = None`

A Control to display before the text input field when the search view is open.

Typically an Icon or an IconButton.

Defaults to a back button which closes/pops the search view.

### **buildview\_paddingclass-attributeinstance-attribute**

`view_padding: PaddingValue | None = None`

The padding to use for the search view.

Has no effect if the search view is full-screen.

### **buildview\_shapeclass-attributeinstance-attribute**

`view_shape: OutlinedBorder | None = None`

Defines the shape of the search view.

### **buildview\_sideclass-attributeinstance-attribute**

`view_side: BorderSide | None = None`

Defines the color and weight of the search view's outline.

### **buildview\_size\_constraintsclass-attributeinstance-attribute**

`view_size_constraints: BoxConstraints | None = None`

Optional size constraints for the search view.

By default, the search view has the same width as the search bar and is 2/3 the height of the screen. If the width and height of the view are within the `view_size_constraints`, the view will show its default size. Otherwise, the size of the view will be constrained by this property.

## buildview\_trailingclass-attributeinstance-attribute

view\_trailing: list[Control] | None = None

A list of Controls to display after the text input field when the search view is open.

Defaults to a close button which closes/pops the search view.

## Events

### bolton\_blurclass-attributeinstance-attribute

on\_blur: ControlEventHandler[SearchBar] | None = None

Fired when the search bar loses focus.

### bolton\_changeclass-attributeinstance-attribute

on\_change: ControlEventHandler[SearchBar] | None = None

Called when the typed input in the search bar has changed.

### bolton\_focusclass-attributeinstance-attribute

on\_focus: ControlEventHandler[SearchBar] | None = None

Fired when the search bar gains focus.

### bolton\_submitclass-attributeinstance-attribute

on\_submit: ControlEventHandler[SearchBar] | None = None

Called when user presses ENTER while focus is on SearchBar.

### bolton\_tapclass-attributeinstance-attribute

on\_tap: ControlEventHandler[SearchBar] | None = None

Called when the search bar is tapped.

### bolton\_tap\_outside\_barclass-attributeinstance-attribute

```
on_tap_outside_bar: (
 ControlEventHandler[SearchBar] | None
) = None
```

Fired when the user taps outside the search bar while the search view is open.

## Methods

### deployed\_codeclose\_viewasync

close\_view(text: str | None=None)

Closes an opened search view.

Parameters:

- text (str | None, default: None) - The text to set in the search bar when closing the view. If not provided, the current value of the search bar will be used.

### deployed\_codefocusasync

focus()

Requests focus for this control.

### deployed\_codeopen\_viewasync

open\_view()

Opens the search view.

# Segment

A segment for a SegmentedButton.

Inherits: Control

## Properties

- icon - The icon to be displayed in the segment.
- label - The label (usually a Text) to be displayed in the segment.
- value - Used to identify this segment.

## Properties

### buildiconclass-attributeinstance-attribute

icon: IconDataOrControl | None = None

The icon to be displayed in the segment.

Typically an Icon.

Raises:

- ValueError - If neither it nor label is set and visible.

### buildlabelclass-attributeinstance-attribute

label: StrOrControl | None = None

The label (usually a Text) to be displayed in the segment.

Raises:

- ValueError - If neither it nor icon is set and visible.

### buildvalueinstance-attribute

value: str

Used to identify this segment.

# SegmentedButton

A segmented button control.

Image: SegmentedButton - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/segmented\\_button/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/segmented_button/image_for_docs.png)

SegmentedButton

Inherits: LayoutControl

## Properties

- `allow_empty_selection` - A boolean value that indicates if having no selected segments is allowed.
- `allow_multiple_selection` - A boolean value that indicates if multiple segments can be selected at one time.
- `direction` - The orientation of the button's segments.
- `padding` - Defines the button's size and padding.
- `segments` - The segments of this button.
- `selected` - A set of `Segment.values` that indicate which segments are selected.
- `selected_icon` - An `Icon` control that is used to indicate a segment is selected.
- `show_selected_icon` - A boolean value that indicates if the `selected_icon` is displayed on the selected segments.
- `style` - Customizes this button's appearance.

## Events

- `on_change` - Called when the selection changes.

## Examples

### SegmentedButton single and multiple selection

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 def handle_selection_change(e: ft.Event[ft.SegmentedButton]):
 print(e)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.SegmentedButton(
 on_change=handle_selection_change,
 selected_icon=ft.Icon(ft.Icons.CHECK_SHARP),
 selected=["1", "4"],
 allow_empty_selection=True,
 allow_multiple_selection=True,
 segments=[
 ft.Segment(
 value="1",
 label=ft.Text("One"),
 icon=ft.Icon(ft.Icons.LOOKS_ONE),
),
 ft.Segment(
 value="2",
 label=ft.Text("Two"),
 icon=ft.Icon(ft.Icons.LOOKS_TWO),
),
 ft.Segment(
 value="3",
 label=ft.Text("Three"),
 icon=ft.Icon(ft.Icons.LOOKS_3),
),
 ft.Segment(
 value="4",
 label=ft.Text("Four"),
 icon=ft.Icon(ft.Icons.LOOKS_4),
),
],
),
],
),
),
)
```

```

ft.SegmentedButton(
 on_change=handle_selection_change,
 selected_icon=ft.Icon(ft.Icons.CHECK_SHARP),
 selected=["2"],
 allow_multiple_selection=False,
 segments=[
 ft.Segment(
 value="1",
 label=ft.Text("One"),
 icon=ft.Icon(ft.Icons.LOOKS_ONE),
),
 ft.Segment(
 value="2",
 label=ft.Text("Two"),
 icon=ft.Icon(ft.Icons.LOOKS_TWO),
),
 ft.Segment(
 value="3",
 label=ft.Text("Three"),
 icon=ft.Icon(ft.Icons.LOOKS_3),
),
 ft.Segment(
 value="4",
 label=ft.Text("Four"),
 icon=ft.Icon(ft.Icons.LOOKS_4),
),
],
),
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/segmented\\_button/single\\_multiple\\_selection.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/segmented_button/single_multiple_selection.png)

## Properties

### buildallow\_empty\_selectionclass-attributeinstance-attribute

allow\_empty\_selection: bool = False

A boolean value that indicates if having no selected segments is allowed.

If True, then it is acceptable for none of the segments to be selected and also that selected can be empty.

If False (the default), there must be at least one segment selected. If the user taps on the only selected segment it will not be deselected, and on\_change will not be called.

Raises:

- ValueError - If selected is empty while allow\_empty\_selection is False.

### buildallow\_multiple\_selectionclass-attributeinstance-attribute

allow\_multiple\_selection: bool = False

A boolean value that indicates if multiple segments can be selected at one time.

If True, more than one segment can be selected. When selecting a segment, the other selected segments will stay selected. Selecting an already selected segment will unselect it.

If False (the default), only one segment may be selected at a time. When a segment is selected, any previously selected segment will be unselected.

Raises:

- ValueError - If selected has more than one item while allow\_multiple\_selection is False.

### builddirectionclass-attributeinstance-attribute

direction: Axis | None = None

The orientation of the button's segments.

Defaults to Axis.HORIZONTAL.

### buildpaddingclass-attributeinstance-attribute

padding: PaddingValue | None = None



Defines the button's size and padding. If specified, the button expands to fill its parent's space with this padding.

When None, the button adopts its intrinsic content size.

### **buildsegmentsinstance-attribute**

```
segments: list[Segment]
```

The segments of this button.

Raises:

- ValueError - If it does not contain at least one visible Segment.

### **buildselectedclass-attributeinstance-attribute**

```
selected: list[str] = field(default_factory=list)
```

A set of Segment.values that indicate which segments are selected. It is updated when the user (un)selects a segment.

Raises:

- ValueError - If selected violates the constraints defined by allow\_empty\_selection or allow\_multiple\_selection.

### **buildselected\_iconclass-attributeinstance-attribute**

```
selected_icon: IconDataOrControl | None = None
```

An Icon control that is used to indicate a segment is selected.

If show\_selected\_icon is True then for selected segments this icon will be shown before the Segment.label, replacing the Segment.icon if it is specified.

Defaults to an Icon with the CHECK icon.

### **buildshow\_selected\_iconclass-attributeinstance-attribute**

```
show_selected_icon: bool = True
```

A boolean value that indicates if the selected\_icon is displayed on the selected segments.

If True, the selected\_icon will be displayed at the start of the selected segments.

If False, then the selected\_icon is not used and will not be displayed on selected segments.

### **buildstyleclass-attributeinstance-attribute**

```
style: ButtonStyle | None = None
```

Customizes this button's appearance.

## **Events**

### **bolton\_changeclass-attributeinstance-attribute**

```
on_change: ControlEventHandler[SegmentedButton] | None = (
 None
)
```

Called when the selection changes.

The data property of the event handler argument contains a list of strings identifying the selected segments.

# SelectionArea

Flet controls are not selectable by default. SelectionArea is used to enable selection for its child control.

Image: SelectionArea - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/selection\\_area/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/selection_area/image_for_docs.png)

Selectable and unselectable text

Inherits: Control

## Properties

- content - The child control this selection area applies to.

## Events

- on\_change - Called when the selected content changes.

## Examples

### SelectionArea

play\_arrowTry Online

```
import flet as ft

TEXT_STYLE = ft.TextStyle(
 size=22,
 weight=ft.FontWeight.W_600,
 decoration=ft.TextDecoration(
 ft.TextDecoration.UNDERLINE | ft.TextDecoration.OVERLINE
),
 decoration_style=ft.TextDecorationStyle.WAVY,
)

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.SelectionArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Selectable text",
 color=ft.Colors.GREEN,
 style=TEXT_STYLE,
 key="selectable",
),
 ft.Text(
 "Also selectable",
 color=ft.Colors.GREEN,
 style=TEXT_STYLE,
),
]
),
),
 ft.Text(
 "Not selectable",
 color=ft.Colors.RED,
 style=TEXT_STYLE,
 key="non-selectable",
),
]
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/examples/controls/material/selection\\_area/media/basic.gif](https://flet.dev/docs/examples/controls/material/selection_area/media/basic.gif)

## Properties

### buildcontentinstance-attribute

content: Control

The child control this selection area applies to.

If you need to have multiple selectable controls, use container-like controls like Row or Column, which have a controls property for this purpose.

Raises:

- ValueError - If content is not visible.

## Events

**bolton\_change**class-attributeinstance-attribute

on\_change: ControlEventHandler[SelectionArea] | None = None

Called when the selected content changes.

# Semantics

Provides semantic annotations for the control tree, describing the meaning and purpose of controls.

These annotations are utilized by accessibility tools, search engines, and semantic analysis software to better understand the structure and functionality of the application.

Inherits: Control

## Properties

- badge - TBD
- button - Whether this subtree represents a button.
- checked - Whether this subtree represents a checkbox or similar widget with a "checked" state, and what its current state is.
- container - TBD
- content - The Control to annotate.
- current\_value\_length - The current number of characters that have been entered into an editable text field.
- decreased\_value - The value that the semantics node represents when it is decreased.
- exclude\_semantics - TBD
- expanded - Whether this subtree represents something that can be in an "expanded" or "collapsed" state.
- focus - Whether the node currently holds input focus.
- focusable - Whether the node is able to hold input focus.
- header - Whether this subtree represents a header.
- heading\_level - The heading level in the DOM document structure.
- hidden - Whether this subtree is currently hidden.
- hint\_text - A brief textual description of the result of an action performed on the content control.
- image - Whether the node represents an image.
- increased\_value - The value that the semantics node represents when it is increased.
- label - A textual description of the content.
- link - Whether this subtree represents a link.
- live\_region - Whether this subtree should be considered a live region.
- max\_value\_length - The maximum number of characters that can be entered into an editable text field.
- mixed - Whether this subtree represents a checkbox or similar control with a "half-checked" state or similar, and whether it is currently in this half-checked state.
- multiline - Whether the value is coming from a field that supports multiline text editing.
- obscured - Whether value should be obscured.
- read\_only - Whether this subtree is read only.
- selected - Whether this subtree represents something that can be in a selected or unselected state, and what its current state is.
- slider - Whether this subtree represents a slider.
- textfield - Whether this subtree represents a text field.
- toggled - Whether this subtree represents a toggle switch or similar widget with an "on" state, and what its current state is.
- tooltip - A textual description of the widget's tooltip.
- value - A textual description of the value of the content control.

## Events

- `on_copy` - Called when the current selection is copied to the clipboard.
- `on_cut` - Called when the current selection is cut to the clipboard.
- `on_decrease` - Called when the value represented by the semantics node is decreased.
- `on_did_gain_accessibility_focus` - Called when the node has gained accessibility focus.
- `on_did_lose_accessibility_focus` - Called when the node has lost accessibility focus.
- `on_dismiss` - Called when the node is dismissed.
- `on_double_tap` - TBD
- `on_increase` - Called when the value represented by the semantics node is increased.
- `on_long_press` - Called when the node is long-pressed (pressing and holding the screen with the finger for a few seconds without moving it).
- `on_long_press_hint_text` - TBD
- `on_move_cursor_backward_by_character` - Called when the cursor is moved backward by one character.
- `on_move_cursor_forward_by_character` - Called when the cursor is moved forward by one character.
- `on_paste` - Called when the current content of the clipboard is pasted.
- `on_scroll_down` - Called when a user moves their finger across the screen from top to bottom.
- `on_scroll_left` - Called when a user moves their finger across the screen from right to left.
- `on_scroll_right` - Called when a user moves their finger across the screen from left to right.
- `on_scroll_up` - Called when a user moves their finger across the screen from bottom to top.
- `on_set_text` - Called when a user wants to replace the current text in the text field with a new text.
- `on_tap` - Called when this control is tapped.
- `on_tap_hint_text` - TBD

## Examples

### Semantics

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 def handle_gain_accessibility_focus(e: ft.Event[ft.Semantics]):
 print("focus gained")

 def handle_lose_accessibility_focus(e: ft.Event[ft.Semantics]):
 print("focus lost")

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Semantics(
 label="Input your occupation",
 on_did_gain_accessibility_focus=handle_gain_accessibility_focus,
 on_did_lose_accessibility_focus=handle_lose_accessibility_focus,
 content=ft.TextField(
 label="Occupation",
 hint_text="Use 20 words or less",
 value="What is your occupation?",
),
),
 ft.Icon(ft.Icons.SETTINGS, color="#c1c1c1"),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

# Properties

## buildbadgeclass-attributeinstance-attribute

badge: BadgeValue | None = None

TBD

## buildbuttonclass-attributeinstance-attribute

button: bool | None = None

Whether this subtree represents a button.

## buildcheckedclass-attributeinstance-attribute

checked: bool | None = None

Whether this subtree represents a checkbox or similar widget with a "checked" state, and what its current state is.

## buildcontainerclass-attributeinstance-attribute

container: bool | None = None

TBD

## buildcontentclass-attributeinstance-attribute

content: Control | None = None

The Control to annotate.

## buildcurrent\_value\_lengthclass-attributeinstance-attribute

current\_value\_length: int | None = None

The current number of characters that have been entered into an editable text field.

## builddecreased\_valueclass-attributeinstance-attribute

decreased\_value: str | None = None

The value that the semantics node represents when it is decreased.

## buildexclude\_semanticsclass-attributeinstance-attribute

exclude\_semantics: bool = False

TBD

## buildexpandedclass-attributeinstance-attribute

expanded: bool | None = None

Whether this subtree represents something that can be in an "expanded" or "collapsed" state.

## buildfocusclass-attributeinstance-attribute

focus: bool | None = None

Whether the node currently holds input focus.

## buildfocusableclass-attributeinstance-attribute

focusable: bool | None = None

Whether the node is able to hold input focus.

## buildheaderclass-attributeinstance-attribute

header: bool | None = None

Whether this subtree represents a header.

## buildheading\_levelclass-attributeinstance-attribute

heading\_level: int | None = None

The heading level in the DOM document structure.

## buildhiddenclass-attributeinstance-attribute

hidden: bool | None = None

Whether this subtree is currently hidden.

### **buildhint\_textclass-attributeinstance-attribute**

`hint_text: str | None = None`

A brief textual description of the result of an action performed on the content control.

### **buildimageclass-attributeinstance-attribute**

`image: bool | None = None`

Whether the node represents an image.

### **buildincreased\_valueclass-attributeinstance-attribute**

`increased_value: str | None = None`

The value that the semantics node represents when it is increased.

### **buildlabelclass-attributeinstance-attribute**

`label: str | None = None`

A textual description of the content.

### **buildlinkclass-attributeinstance-attribute**

`link: bool | None = None`

Whether this subtree represents a link.

### **buildlive\_regionclass-attributeinstance-attribute**

`live_region: bool | None = None`

Whether this subtree should be considered a live region.

### **buildmax\_value\_lengthclass-attributeinstance-attribute**

`max_value_length: Number | None = None`

The maximum number of characters that can be entered into an editable text field.

### **buildmixedclass-attributeinstance-attribute**

`mixed: bool | None = None`

Whether this subtree represents a checkbox or similar control with a "half-checked" state or similar, and whether it is currently in this half-checked state.

### **buildmultilineclass-attributeinstance-attribute**

`multiline: bool | None = None`

Whether the value is coming from a field that supports multiline text editing.

### **buildobscuredclass-attributeinstance-attribute**

`obscured: bool | None = None`

Whether value should be obscured.

### **buildread\_onlyclass-attributeinstance-attribute**

`read_only: bool | None = None`

Whether this subtree is read only.

### **buildselectedclass-attributeinstance-attribute**

`selected: bool | None = None`

Whether this subtree represents something that can be in a selected or unselected state, and what its current state is.

### **buildsliderclass-attributeinstance-attribute**

`slider: bool | None = None`

Whether this subtree represents a slider.

### **buildtextfieldclass-attributeinstance-attribute**

`textfield: bool | None = None`

Whether this subtree represents a text field.

## buildtoggledclass-attributeinstance-attribute

toggled: bool | None = None

Whether this subtree represents a toggle switch or similar widget with an "on" state, and what its current state is.

## buildtooltipclass-attributeinstance-attribute

tooltip: str | None = None

A textual description of the widget's tooltip.

## buildvalueclass-attributeinstance-attribute

value: str | None = None

A textual description of the value of the content control.

## Events

### bolton\_copyclass-attributeinstance-attribute

on\_copy: ControlEventHandler[Semantics] | None = None

Called when the current selection is copied to the clipboard.

### bolton\_cutclass-attributeinstance-attribute

on\_cut: ControlEventHandler[Semantics] | None = None

Called when the current selection is cut to the clipboard.

### bolton\_decreaseclass-attributeinstance-attribute

on\_decrease: ControlEventHandler[Semantics] | None = None

Called when the value represented by the semantics node is decreased.

### bolton\_did\_gain\_accessibility\_focusclass-attributeinstance-attribute

```
on_did_gain_accessibility_focus: (
 ControlEventHandler[Semantics] | None
) = None
```

Called when the node has gained accessibility focus.

### bolton\_did\_lose\_accessibility\_focusclass-attributeinstance-attribute

```
on_did_lose_accessibility_focus: (
 ControlEventHandler[Semantics] | None
) = None
```

Called when the node has lost accessibility focus.

### bolton\_dismissclass-attributeinstance-attribute

on\_dismiss: ControlEventHandler[Semantics] | None = None

Called when the node is dismissed.

### bolton\_double\_tapclass-attributeinstance-attribute

on\_double\_tap: ControlEventHandler[Semantics] | None = None

TBD

### bolton\_increaseclass-attributeinstance-attribute

on\_increase: ControlEventHandler[Semantics] | None = None

Called when the value represented by the semantics node is increased.

### bolton\_long\_pressclass-attributeinstance-attribute

on\_long\_press: ControlEventHandler[Semantics] | None = None

Called when the node is long-pressed (pressing and holding the screen with the finger for a few seconds without moving it).

### bolton\_long\_press\_hint\_textclass-attributeinstance-attribute

on\_long\_press\_hint\_text: str | None = None

TBD



## **bolton\_move\_cursor\_backward\_by\_characterclass-attributeinstance-attribute**

```
on_move_cursor_backward_by_character: (
 ControlEventHandler[Semantics] | None
) = None
```

Called when the cursor is moved backward by one character.

## **bolton\_move\_cursor\_forward\_by\_characterclass-attributeinstance-attribute**

```
on_move_cursor_forward_by_character: (
 ControlEventHandler[Semantics] | None
) = None
```

Called when the cursor is moved forward by one character.

## **bolton\_paste\_ class-attributeinstance-attribute**

```
on_paste: ControlEventHandler[Semantics] | None = None
```

Called when the current content of the clipboard is pasted.

## **bolton\_scroll\_down\_ class-attributeinstance-attribute**

```
on_scroll_down: ControlEventHandler[Semantics] | None = None
```

Called when a user moves their finger across the screen from top to bottom.

## **bolton\_scroll\_left\_ class-attributeinstance-attribute**

```
on_scroll_left: ControlEventHandler[Semantics] | None = None
```

Called when a user moves their finger across the screen from right to left.

## **bolton\_scroll\_right\_ class-attributeinstance-attribute**

```
on_scroll_right: ControlEventHandler[Semantics] | None = (
 None
)
```

Called when a user moves their finger across the screen from left to right.

## **bolton\_scroll\_up\_ class-attributeinstance-attribute**

```
on_scroll_up: ControlEventHandler[Semantics] | None = None
```

Called when a user moves their finger across the screen from bottom to top.

## **bolton\_set\_text\_ class-attributeinstance-attribute**

```
on_set_text: ControlEventHandler[Semantics] | None = None
```

Called when a user wants to replace the current text in the text field with a new text.

Voice access users can trigger this handler by speaking type <text> to their Android devices.

## **bolton\_tap\_ class-attributeinstance-attribute**

```
on_tap: ControlEventHandler[Semantics] | None = None
```

Called when this control is tapped.

## **bolton\_tap\_hint\_text\_ class-attributeinstance-attribute**

```
on_tap_hint_text: str | None = None
```

TBD

# Service

Base class for user services.

Inherits: `BaseControl`

# ShaderMask

Applies a mask generated by a shader to its content.

For example, it can be used to gradually fade out the edge of a control by using a LinearGradient mask.

```
ft.ShaderMask(
 blend_mode=ft.BlendMode.MULTIPLY,
 shader=ft.LinearGradient(
 begin=ft.Alignment.CENTER_LEFT,
 end=ft.Alignment.CENTER_RIGHT,
 colors=[ft.Colors.WHITE, ft.Colors.BLACK],
 tile_mode=ft.GradientTileMode.CLAMP,
),
 content=ft.Image(
 src="https://picsum.photos/id/288/300/300",
 height=300,
 fit=ft.BoxFit.FILL,
)
)
```

Image: ShaderMask - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shader\\_mask/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shader_mask/image_for_docs.png)

Linear gradient mask

Inherits: LayoutControl

## Properties

- `blend_mode` - The blend mode to use when applying the shader to the content.
- `border_radius` - The radius of the mask.
- `content` - The Control to which the shader is applied.
- `shader` - Use gradient as a shader.

## Examples

### ShaderMask pink radial glow

[play\\_arrowTry Online](#)

```
import flet as ft
```

```
def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Row(
 controls=[
 ft.Image(
 src="/9j/4QDeRXhpZgAASUkqAAgAAAAAGABIBAwABAAAAQAABBoBBQABAAAAVgAAABsBBQABAAAAAXgAAACgBAwABAAAAAgAAABMCAwAB
```

Image: pink-radial-glow - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shader\\_mask/pink\\_radial\\_glow.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shader_mask/pink_radial_glow.png)

### ShaderMask fade out image bottom

[play\\_arrowTry Online](#)

```
import flet as ft
```

```
def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Row(
 controls=[
 ft.ShaderMask(
 content=ft.Image(
 src="/9j/4QDeRXhpZgAASUkqAAgAAAAAGABIBAwABAAAAQAABBoBBQABAAAAVgAAABsBBQABAAAAAXgAAACgBAwABAAAAAgAAABMCAwAB
```

Image: fade-out-image-bottom - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shader\\_mask/fade\\_out\\_image\\_bottom.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shader_mask/fade_out_image_bottom.png)

### ShaderMask linear and radial gradients

[play\\_arrowTry Online](#)

```
import flet as ft
```

```
def main(page: ft.Page):
 page.add(
 ft.ShaderMask(
 blend_mode=ft.BlendMode.MULTIPLY,
 shader=ft.LinearGradient(
 begin=ft.Alignment.CENTER_LEFT,
 end=ft.Alignment.CENTER_RIGHT,
 colors=[ft.Colors.WHITE, ft.Colors.BLACK],
 tile_mode=ft.GradientTileMode.CLAMP,
),
 content=ft.Image(
 src="https://picsum.photos/id/288/300/300",
 height=300,
 fit=ft.BoxFit.FILL,
)
)
)
```

```

ft.SafeArea(
 content=ft.Row(
 controls=[
 ft.ShaderMask(
 blend_mode=ft.BlendMode.COLOR_BURN,
 shader=ft.RadialGradient(
 center=ft.Alignment.TOP_LEFT,
 radius=1.0,
 colors=[ft.Colors.YELLOW, ft.Colors.DEEP_ORANGE_900],
 tile_mode=ft.GradientTileMode.CLAMP,
),
 content=ft.Image(
 src="/9j/4QDeRXhpZgAASUkqAAgAAAAGABIBAwABAAAAQAAABoBBQABAAAVgAAABsBBQABAAAAXgAAACgBAwABAAAAAgAAABMC

```

Image: fade-out-image-bottom - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shader\\_mask/linear\\_and\\_radial\\_gradients.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shader_mask/linear_and_radial_gradients.png)

## Properties

### buildblend\_modeclass-attributeinstance-attribute

blend\_mode: BlendMode = BlendMode.MODULATE

The blend mode to use when applying the shader to the content.

### buildborder\_radiusclass-attributeinstance-attribute

border\_radius: BorderRadiusValue | None = None

The radius of the mask.

### buildcontentclass-attributeinstance-attribute

content: Control | None = None

The Control to which the shader is applied.

### buildshaderinstance-attribute

shader: Gradient

Use gradient as a shader.

# Shimmer

Applies an animated shimmering effect to its content.

Use it to create lightweight loading placeholders or to add motion to otherwise static layouts.

```
ft.Shimmer(
 base_color=ft.Colors.with_opacity(0.3, ft.Colors.GREY_400),
 highlight_color=ft.Colors.WHITE,
 content=ft.Column(
 controls=[
 ft.Container(height=80, bgcolor=ft.Colors.GREY_300),
 ft.Container(height=80, bgcolor=ft.Colors.GREY_300),
 ft.Container(height=80, bgcolor=ft.Colors.GREY_300),
],
),
)
```

Image: Shimmer - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shimmer/image\\_for\\_docs.gif](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shimmer/image_for_docs.gif)

Shimmer

Inherits: `LayoutControl`

## Properties

- `base_color` - Base color used when no gradient is provided.
- `content` - The control to render with the shimmer effect.
- `direction` - Direction of the shimmering animation.
- `gradient` - Custom gradient that defines the shimmer colors.
- `highlight_color` - Highlight color used when no gradient is provided.
- `loop` - Number of times the animation should repeat.
- `period` - Duration of a shimmer cycle in milliseconds.

## Examples

### Shimmer

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Shimmer(
 base_color=ft.Colors.with_opacity(0.3, ft.Colors.GREY_400),
 highlight_color=ft.Colors.WHITE,
 content=ft.Column(
 controls=[
 ft.Container(height=80, bgcolor=ft.Colors.GREY_300),
 ft.Container(height=80, bgcolor=ft.Colors.GREY_300),
 ft.Container(height=80, bgcolor=ft.Colors.GREY_300),
],
),
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: custom-label - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shimmer/image\\_for\\_docs.gif](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shimmer/image_for_docs.gif)

### Shimmer basic placeholder

[play\\_arrowTry Online](#)

```
import flet as ft

def _line(width: int, height: int = 12) -> ft.Control:
 return ft.Container(
 width=width,
 height=height,
 bgcolor=ft.Colors.GREY_300,
 shimmer=ft.Shimmer(
 base_color=ft.Colors.GREY_400,
 highlight_color=ft.Colors.WHITE,
 content=ft.Container(
 width=width,
 height=height,
 bgcolor=ft.Colors.GREY_300,
),
),
)
```

```

 width=width,
 height=height,
 bgcolor=ft.Colors.GREY_400,
 border_radius=ft.BorderRadius.all(height),
)

def _placeholder_tile() -> ft.Control:
 return ft.Container(
 padding=ft.Padding.all(16),
 bgcolor=ft.Colors.with_opacity(0.3, ft.Colors.WHITE),
 border_radius=ft.BorderRadius.all(20),
 content=ft.Row(
 spacing=16,
 vertical_alignment=ft.CrossAxisAlignment.START,
 controls=[
 ft.Container(
 width=48,
 height=48,
 bgcolor=ft.Colors.with_opacity(0.5, ft.Colors.GREY_400),
 border_radius=ft.BorderRadius.all(24),
 content=ft.Icon(ft.Icons.PERSON, color=ft.Colors.GREY_500),
),
 ft.Column(
 expand=True,
 spacing=10,
 controls=[
 _line(160),
 _line(120),
 ft.Row(
 spacing=10,
 vertical_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[_line(70, 10), _line(90, 10)],
),
],
),
 ft.Container(
 width=32,
 height=32,
 bgcolor=ft.Colors.GREY_200,
 border_radius=ft.BorderRadius.all(16),
),
],
),
)

def main(page: ft.Page):
 page.title = "Shimmer - loading placeholders"

 page.add(
 ft.SafeArea(
 content=ft.Shimmer(
 base_color=ft.Colors.with_opacity(0.3, ft.Colors.GREY_400),
 highlight_color=ft.Colors.WHITE,
 content=ft.Column(
 controls=[_placeholder_tile() for _ in range(3)],
),
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: custom-label - [https://flet.dev/docs/test-images/examples/controls/core/golden/macOS/shimmer/basic\\_placeholder.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macOS/shimmer/basic_placeholder.png)

## Shimmer custom gradient

play\_arrowTry Online

```

import flet as ft

def _stat_block(title: str, subtitle: str) -> ft.Control:
 def metric(width: int, height: int = 14) -> ft.Control:
 return ft.Container(
 width=width,
 height=height,
 bgcolor=ft.Colors.WHITE,
 opacity=0.6,
 border_radius=ft.BorderRadius.all(height),
)

 return ft.Container(
 width=200,

```

```

padding=ft.Padding.all(20),
bgcolor=ft.Colors.with_opacity(0.1, ft.Colors.BLACK),
border_radius=ft.BorderRadius.all(24),
content=ft.Column(
 spacing=16,
 controls=[
 metric(140),
 ft.Row(spacing=10, controls=[metric(60, 10), metric(90, 10)]),
 ft.Container(
 border_radius=ft.BorderRadius.all(16),
 bgcolor=ft.Colors.WHITE,
 opacity=0.35,
),
 ft.Column(spacing=8, controls=[metric(120, 12), metric(160, 12)]),
 ft.Text(title, weight=ft.FontWeight.W_600),
 ft.Text(subtitle, size=12),
],
),
)

def main(page: ft.Page):
 page.title = "Shimmer - custom gradients"
 page.bgcolor = "#0e0e18"
 accent = ft.LinearGradient(
 begin=ft.Alignment(-1.0, -0.5),
 end=ft.Alignment(1.0, 0.5),
 colors=[
 ft.Colors.PURPLE,
 ft.Colors.PURPLE,
 ft.Colors.AMBER_200,
 ft.Colors.PURPLE,
 ft.Colors.PURPLE,
],
 stops=[0.0, 0.35, 0.5, 0.65, 1.0],
)

 cards = ft.Row(
 wrap=True,
 controls=[
 ft.Shimmer(
 gradient=accent,
 direction=ft.ShimmerDirection.TTB,
 period=2200,
 content=_stat_block("Recent activity", "Smooth top-to-bottom sweep"),
),
],
)

 page.add(ft.SafeArea(content=cards))

if __name__ == "__main__":
 ft.run(main)

```

Image: custom-label - [https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shimmer/custom\\_gradient.png](https://flet.dev/docs/test-images/examples/controls/core/golden/macros/shimmer/custom_gradient.png)

## Properties

### buildbase\_colorclass-attributeinstance-attribute

base\_color: ColorValue | None = None

Base color used when no gradient is provided.

### buildcontentinstance-attribute

content: Control

The control to render with the shimmer effect.

Raises:

- ValueError - If it is not visible.

### bulldirectionclass-attributeinstance-attribute

direction: ShimmerDirection = ShimmerDirection.LTR

Direction of the shimmering animation.

### buildgradientclass-attributeinstance-attribute

gradient: Gradient | None = None

Custom gradient that defines the shimmer colors.

### **buildhighlight\_colorclass-attributeinstance-attribute**

`highlight_color: ColorValue | None = None`

Highlight color used when no gradient is provided.

### **buildloopclass-attributeinstance-attribute**

`loop: int | None = 0`

Number of times the animation should repeat. 0 means infinite.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

### **buildperiodclass-attributeinstance-attribute**

`period: DurationValue = 1500`

Duration of a shimmer cycle in milliseconds.



# Slider

A slider provides a visual indication of adjustable content, as well as the current setting in the total range of content.

Use a slider when you want people to set defined values (such as volume or brightness), or when people would benefit from instant feedback on the effect of setting changes.

Example:

```
ft.Slider(label="Slider", value=0.3)
```

Image: Slider - [https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/slider/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/slider/image_for_docs.png)

Slider

Inherits: `LayoutControl`, `AdaptiveControl`

## Properties

- `active_color` - The color to use for the portion of the slider track that is active.
- `autofocus` - True if the control will be selected as the initial focus.
- `divisions` - The number of discrete divisions.
- `inactive_color` - The color for the inactive portion of the slider track.
- `interaction` - The allowed way for the user to interact with this slider.
- `label` - A label to show above the slider when the slider is active.
- `max` - The maximum value the user can select.
- `min` - The minimum value the user can select.
- `mouse_cursor` - The cursor to be displayed when a mouse pointer enters or is hovering over this control.
- `overlay_color` - The highlight color that's typically used to indicate that the range slider thumb is in `ControlState.HOVERED` or `ControlState.DRAGGED` states.
- `padding` - Determines the padding around this slider.
- `round` - The number of decimals displayed on the label containing value.
- `secondary_active_color` - The color to use for the portion of the slider track between the thumb and the `secondary_track_value`.
- `secondary_track_value` - The secondary track value for this slider.
- `thumb_color` - The color of the thumb.
- `value` - The currently selected value for this slider.
- `year_2023` - If this is set to `False`, this slider will use the latest Material Design 3 appearance, which was introduced in December 2023.

## Events

- `on_blur` - Called when this slider has lost focus.
- `on_change` - Called when the state of this slider is changed.
- `on_change_end` - Called when the user is done selecting a new value for this slider.
- `on_change_start` - Called when the user starts selecting a new value for this slider.
- `on_focus` - Called when this slider has received focus.

## Examples

### Slider

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
```

```

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Default slider:"),
 ft.Slider(),
 ft.Text("Default disabled slider:"),
 ft.Slider(disabled=True),
]
)
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: basic - <https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/slider/basic.png>

## Slider custom label

play\_arrowTry Online

```
import flet as ft
```

```

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Slider with value:"),
 ft.Slider(value=0.3),
 ft.Text("Slider with a custom range and label:"),
 ft.Slider(min=0, max=100, divisions=10, label="{value}%"),
]
)
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: custom-label - [https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/slider/custom\\_label.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/slider/custom_label.png)

## Slider handling events

play\_arrowTry Online

```
import flet as ft
```

```

def main(page: ft.Page):
 def slider_changed(e: ft.Event[ft.Slider]):
 message.value = f"Slider changed to {e.control.value}"
 message.update()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Slider with 'on_change' event:"),
 ft.Slider(
 key="slider",
 min=0,
 max=100,
 divisions=10,
 label="{value}%",
 on_change=slider_changed,
),
 message := ft.Text(),
]
)
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: handling-events - [https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/slider/handling\\_events.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/slider/handling_events.png)

## Slider random values

play\_arrowTry Online

```
import asyncio
```

```

import random

import flet as ft

async def main(page: ft.Page):
 def handle_slider_change(e: ft.Event[ft.Slider]):
 message.value = f"Slider.value changed to {e.control.value}"
 message.update()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Slider with 'on_change' event:"),
 slider := ft.Slider(
 label="{value}", on_change=handle_slider_change
),
 message := ft.Text(),
]
)
)
)

 while True:
 await asyncio.sleep(1)
 slider.value = random.random()
 event = ft.Event("_", slider, data=slider.value)
 handle_slider_change(event)
 slider.update()

if __name__ == "__main__":
 ft.run(main)

```

Image: random-values - [https://flet.dev/docs/examples/controls/material/slider/media/random\\_values.gif](https://flet.dev/docs/examples/controls/material/slider/media/random_values.gif)

## Properties

### buildactive\_colorclass-attributeinstance-attribute

active\_color: ColorValue | None = None

The color to use for the portion of the slider track that is active.

The "active" side of the slider is the side between the thumb and the minimum value.

### buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

True if the control will be selected as the initial focus. If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

### bulldivisionsclass-attributeinstance-attribute

divisions: int | None = None

The number of discrete divisions.

Typically used with label to show the current discrete value.

If None, this slider is continuous.

### buildinactive\_colorclass-attributeinstance-attribute

inactive\_color: ColorValue | None = None

The color for the inactive portion of the slider track.

The "inactive" side of the slider is the side between the thumb and the maximum value.

### buildinteractionclass-attributeinstance-attribute

interaction: SliderInteraction | None = None

The allowed way for the user to interact with this slider.

If None, SliderTheme.interaction is used. If that's is also None, defaults to SliderInteraction.TAP\_AND\_SLIDE.

### buildlabelclass-attributeinstance-attribute

label: str | None = None

A label to show above the slider when the slider is active. The value of label may contain {value} which will be dynamically replaced with a current slider value. For example, "Volume: {value}".

It is used to display the value of a discrete slider, and it is displayed as part of the value indicator shape.

If not set, then the value indicator will not be displayed.

### **buildmaxclass-attributeinstance-attribute**

`max: Number = 1.0`

The maximum value the user can select.

If the min is equal to the max, then this slider is disabled.

Raises:

- ValueError - If it is not greater than or equal to min.
- ValueError - If it is not greater than or equal to value, when value is set.

### **buildminclass-attributeinstance-attribute**

`min: Number = 0.0`

The minimum value the user can select.

If the max is equal to the min, then this slider is disabled.

Raises:

- ValueError - If it is not less than or equal to max.
- ValueError - If it is not less than or equal to value, when value is set.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: MouseCursor | None = None`

The cursor to be displayed when a mouse pointer enters or is hovering over this control.

### **buildoverlay\_colorclass-attributeinstance-attribute**

`overlay_color: ControlStateValue[ColorValue] | None = None`

The highlight color that's typically used to indicate that the range slider thumb is in `ControlState.HOVERED` or `ControlState.DRAGGED` states.

### **buildpaddingclass-attributeinstance-attribute**

`padding: PaddingValue | None = None`

Determines the padding around this slider.

### **buildroundclass-attributeinstance-attribute**

`round: int = 0`

The number of decimals displayed on the label containing value.

Defaults to 0, which displays value rounded to the nearest integer.

### **buildsecondary\_active\_colorclass-attributeinstance-attribute**

`secondary_active_color: ColorValue | None = None`

The color to use for the portion of the slider track between the thumb and the `secondary_track_value`.

### **buildsecondary\_track\_valueclass-attributeinstance-attribute**

`secondary_track_value: Number | None = None`

The secondary track value for this slider.

If not null, a secondary track using `secondary_active_color` is drawn between the thumb and this value, over the inactive track. If less than value, then the secondary track is not shown.

It can be ideal for media scenarios such as showing the buffering progress while the value shows the play progress.

### **buildthumb\_colorclass-attributeinstance-attribute**

`thumb_color: ColorValue | None = None`

The color of the thumb.

### **buildvalueclass-attributeinstance-attribute**

value: Number | None = None

The currently selected value for this slider.

The slider's thumb is drawn at a position that corresponds to this value.

Defaults to value of min.

Raises:

- ValueError - If it is not greater than or equal to min.
- ValueError - If it is not less than or equal to max.

### **buildyear\_2023class-attributeinstance-attribute**

year\_2023: bool | None = None

If this is set to False, this slider will use the latest Material Design 3 appearance, which was introduced in December 2023.

When True, the Slider will use the 2023 Material Design 3 appearance.

If not set, then the SliderTheme.year\_2023 will be used, which is False by default.

If Theme.use\_material3 is False, then this property is ignored.

## **Events**

### **bolton\_blurclass-attributeinstance-attribute**

on\_blur: ControlEventHandler[Slider] | None = None

Called when this slider has lost focus.

### **bolton\_changeclass-attributeinstance-attribute**

on\_change: ControlEventHandler[Slider] | None = None

Called when the state of this slider is changed.

### **bolton\_change\_endclass-attributeinstance-attribute**

on\_change\_end: ControlEventHandler[Slider] | None = None

Called when the user is done selecting a new value for this slider.

### **bolton\_change\_startclass-attributeinstance-attribute**

on\_change\_start: ControlEventHandler[Slider] | None = None

Called when the user starts selecting a new value for this slider.

### **bolton\_focusclass-attributeinstance-attribute**

on\_focus: ControlEventHandler[Slider] | None = None

Called when this slider has received focus.

# SnackBar

A lightweight message with an optional action which briefly displays at the bottom of the screen.

Example:

```
page.show_dialog(ft.SnackBar(ft.Text("Opened snack bar")))
```

Image: SnackBar - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/snack\\_bar/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/snack_bar/image_for_docs.png)

Opened snack bar

Inherits: DialogControl

## Properties

- **action** - An optional action that the user can take based on the snack bar.
- **action\_overflow\_threshold** - The percentage threshold for action's width before it overflows to a new line.
- **behavior** - This defines the behavior and location of the snack bar.
- **bgcolor** - SnackBar background color.
- **clip\_behavior** - The content will be clipped (or not) according to this option.
- **close\_icon\_color** - The color of the close icon, if **show\_close\_icon** is True.
- **content** - The primary content of the snack bar.
- **dismiss\_direction** - The direction in which the SnackBar can be dismissed.
- **duration** - The amount of time this snack bar should stay open for.
- **elevation** - The z-coordinate at which to place the snack bar.
- **margin** - Empty space to surround the snack bar.
- **padding** - The amount of padding to apply to the snack bar's content and optional action.
- **persist** - Whether the snack bar will stay or auto-dismiss after timeout.
- **shape** - The shape of this snack bar.
- **show\_close\_icon** - Whether to include a "close" icon widget.
- **width** - The width of the snack bar.

## Events

- **on\_action** - Called when action button is clicked.
- **on\_visible** - Called the first time that the snackbar is visible within the page.

## Examples

### SnackBar

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 def on_click(e: ft.Event[ft.Button]):
 page.show_dialog(ft.SnackBar(ft.Text("Hello, world!")))

 page.add(ft.SafeArea(content=ft.Button("Open SnackBar", on_click=on_click)))

if __name__ == "__main__":
 ft.run(main)
```

Image: basic - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/snack\\_bar/basic.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/snack_bar/basic.png)

### SnackBar counter

play\_arrowTry Online

```

import flet as ft

class Data:
 def __init__(self) -> None:
 self.counter = 0

 def increment(self):
 self.counter += 1

 def decrement(self):
 self.counter -= 1

data = Data()

def main(page: ft.Page):
 page.title = "SnackBar Example"

 snack_bar = ft.SnackBar(
 content=ft.Text("You did it!"),
 action="Undo it!",
 on_action=lambda e: data.decrement(),
)

 def handle_button_click(e: ft.Event[ft.Button]):
 data.increment()
 snack_bar.content.value = f"You did it x {data.counter}"
 if not snack_bar.open:
 page.show_dialog(snack_bar)
 page.update()

 page.add(
 ft.SafeArea(content=ft.Button("Open SnackBar", on_click=handle_button_click))
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Snack bar with counter - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/snack\\_bar/snack\\_bar\\_flow.gif](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/snack_bar/snack_bar_flow.gif)

Snack bar with counter

## SnackBar action

play\_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
 def open_simple_action(e: ft.Event[ft.Button]):
 page.show_dialog(
 ft.SnackBar(
 ft.Text("The file has been deleted."),
 action="Undo",
 on_action=lambda e: print("Simple Undo clicked"),
)
)

 def open_custom_action(e: ft.Event[ft.Button]):
 page.show_dialog(
 ft.SnackBar(
 ft.Text("The directory has been deleted."),
 persist=False,
 action=ft.SnackBarAction(
 label="Undo delete",
 text_color=ft.Colors.YELLOW,
 bgcolor=ft.Colors.BLUE,
 on_click=lambda e: print("Custom Undo clicked"),
),
)
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Button(
 "Open SnackBar with a Simple action",
 on_click=open_simple_action,
),
 ft.Button(
 "Open SnackBar with a Custom action",
 on_click=open_custom_action,
),
],
)
)
)

```

```

),
],
)
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: Snack bar with a simple action - [https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/snack\\_bar/action\\_simple.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/snack_bar/action_simple.png)

Snack bar with a simple action

Image: Snack bar with a custom action - [https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/snack\\_bar/action\\_custom.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/snack_bar/action_custom.png)

Snack bar with a custom action

## Properties

### buildactionclass-attributeinstance-attribute

action: str | SnackbarAction | None = None

An optional action that the user can take based on the snack bar.

For example, the snack bar might let the user undo the operation that prompted the snackbar. Snack bars can have at most one action.

The action should not be "dismiss" or "cancel".

### buildaction\_overflow\_thresholdclass-attributeinstance-attribute

action\_overflow\_threshold: Number | None = 0.25

The percentage threshold for action's width before it overflows to a new line.

If the width of the snackbar's content is greater than this percentage of the width of the snackbar minus the width of its action, then the action will appear below the content.

At a value of 0.0, the action will not overflow to a new line.

Raises:

- ValueError - If it is not between 0.0 and 1.0, inclusive.

### buildbehaviorclass-attributeinstance-attribute

behavior: SnackbarBehavior | None = None

This defines the behavior and location of the snack bar.

Defines where a Snackbar should appear within a page and how its location should be adjusted when the page also includes a FloatingActionButton or a NavigationBar.

If None, SnackbarTheme.behavior is used. If that's is also None, defaults to SnackbarBehavior.FIXED.

Note

- If behavior is SnackbarBehavior.FLOATING, the length of the bar is defined by either width and margin, and if both are specified, width takes precedence over margin.
- width and margin are ignored if behavior is not SnackbarBehavior.FLOATING.

### buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

Snackbar background color.

### buildclip\_behaviorclass-attributeinstance-attribute

clip\_behavior: ClipBehavior = ClipBehavior.HARD\_EDGE

The content will be clipped (or not) according to this option.

### buildclose\_icon\_colorclass-attributeinstance-attribute

close\_icon\_color: ColorValue | None = None

The color of the close icon, if show\_close\_icon is True.



## buildcontentinstance-attribute

content: StrOrControl

The primary content of the snack bar.

Typically a Text control.

Raises:

- ValueError - If it is neither a string nor a visible Control.

## builddismiss\_directionclass-attributeinstance-attribute

dismiss\_direction: DismissDirection | None = None

The direction in which the Snackbar can be dismissed.

If None, SnackbarTheme.dismiss\_direction is used. If that's is also None, defaults to DismissDirection.DOWN.

## builddurationclass-attributeinstance-attribute

```
duration: DurationValue = field(
 default_factory=lambda: Duration(milliseconds=4000)
)
```

The amount of time this snack bar should stay open for.

## buildelevationclass-attributeinstance-attribute

elevation: Number | None = None

The z-coordinate at which to place the snack bar. This controls the size of the shadow below the snack bar.

Raises:

- ValueError - If it is not greater than or equal to 0.

## buildmarginclass-attributeinstance-attribute

margin: MarginValue | None = None

Empty space to surround the snack bar.

Has effect only when behavior=SnackbarBehavior.FLOATING and will be ignored if width is specified.

## buildpaddingclass-attributeinstance-attribute

padding: PaddingValue | None = None

The amount of padding to apply to the snack bar's content and optional action.

## buildpersistclass-attributeinstance-attribute

persist: bool | None = None

Whether the snack bar will stay or auto-dismiss after timeout.

If True, the snack bar remains visible even after the timeout, until the user taps the action button or the close icon.

If False, the snack bar will be dismissed after the timeout.

If not provided, but the snackbar action is not null, the snackbar will persist as well.

## buildshapeclass-attributeinstance-attribute

shape: OutlinedBorder | None = None

The shape of this snack bar.

## buildshow\_close\_iconclass-attributeinstance-attribute

show\_close\_icon: bool = False

Whether to include a "close" icon widget.

Tapping the icon will close the snack bar.

## buildwidthclass-attributeinstance-attribute

width: Number | None = None

The width of the snack bar.

If width is specified, the snack bar will be centered horizontally in the available space.

Note

Has effect only when behavior is `SnackBarBehavior.FLOATING`. It can not be used if margin is specified.

## Events

### `bolton_action`class-attributeinstance-attribute

`on_action: ControlEventHandler[SnackBar] | None = None`

Called when action button is clicked.

### `bolton_visible`class-attributeinstance-attribute

`on_visible: ControlEventHandler[SnackBar] | None = None`

Called the first time that the snackbar is visible within the page.

A button that can be used as an action in a `SnackBar`.

An action button for a `SnackBar`.

Note

- Snack bar actions are always enabled. Instead of disabling a snack bar action, avoid including it in the snack bar in the first place.
- Snack bar actions will only respond to first click. Subsequent clicks/presses are ignored.

Inherits: `Control`

## Properties

- `bgcolor` - The button background fill color.
- `disabled_bgcolor` - The button disabled background color.
- `disabled_text_color` - The button disabled label color.
- `label` - The button's label.
- `text_color` - The button label color.

## Events

- `on_click` - Called when this action button is clicked.

## Properties

### `buildbgcolor`class-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

The button background fill color.

If None, `SnackBarTheme.action_bgcolor` is used.

### `builddisabled_bgcolor`class-attributeinstance-attribute

`disabled_bgcolor: ColorValue | None = None`

The button disabled background color. This color is shown after the action is dismissed.

If None, `SnackBarTheme.disabled_action_bgcolor` is used.

### `builddisabled_text_color`class-attributeinstance-attribute

`disabled_text_color: ColorValue | None = None`

The button disabled label color. This color is shown after the action is dismissed.

### `buildlabel`instance-attribute

`label: str`

The button's label.

### `buildtext_color`class-attributeinstance-attribute

`text_color: ColorValue | None = None`

The button label color.

If None, `SnackBarTheme.action_text_color` is used.

## Events

**bolton\_click**class-attributeinstance-attribute

`on_click: ControlEventHandler[SnackBarAction] | None = None`

Called when this action button is clicked.

# SpinKit

30 animated loading spinner controls built on Flutter's flutter\_spinkit package.

## Usage

Add flet-spinkit to your project dependencies:

- uv
  - pip
- ```
uv add flet-spinkit
pip install flet-spinkit
```

Then import the package in your Flet app:

```
import flet_spinkit as spins
```

Examples

SpinKit Showcase

play_arrowTry Online

```
import flet_spinkit as spins

import flet as ft

SPINNERS = [
    ("RotatingPlain", spins.RotatingPlain),
    ("DoubleBounce", spins.DoubleBounce),
    ("Wave", spins.Wave),
    ("WanderingCubes", spins.WanderingCubes),
    ("FadingFour", spins.FadingFour),
    ("FadingCube", spins.FadingCube),
    ("Pulse", spins.Pulse),
    ("ChasingDots", spins.ChasingDots),
    ("ThreeBounce", spins.ThreeBounce),
    ("Circle", spins.Circle),
    ("CubeGrid", spins.CubeGrid),
    ("FadingCircle", spins.FadingCircle),
    ("RotatingCircle", spins.RotatingCircle),
    ("FoldingCube", spins.FoldingCube),
    ("PumpingHeart", spins.PumpingHeart),
    ("HourGlass", spins.HourGlass),
    ("PouringHourGlass", spins.PouringHourGlass),
    ("PouringHourGlassRefined", spins.PouringHourGlassRefined),
    ("FadingGrid", spins.FadingGrid),
    ("Ring", spins.Ring),
    ("Ripple", spins.Ripple),
    ("DualRing", spins.DualRing),
    ("SpinningCircle", spins.SpinningCircle),
    ("SpinningLines", spins.SpinningLines),
    ("SquareCircle", spins.SquareCircle),
    ("ThreeInOut", spins.ThreeInOut),
    ("DancingSquare", spins.DancingSquare),
    ("PianoWave", spins.PianoWave),
    ("PulsingGrid", spins.PulsingGrid),
    ("WaveSpinner", spins.WaveSpinner),
]

def main(page: ft.Page):
    page.title = "SpinKit Showcase"
    page.scroll = ft.ScrollMode.AUTO
    page.padding = 20

    color = ft.Colors.BLUE

    page.add(
        ft.GridView(
            runs_count=4,
            max_extent=160,
            child_aspect_ratio=0.9,
            spacing=12,
            run_spacing=12,
            controls=[
                ft.Container(
                    content=ft.Column(
                        [
```

```

        ft.Container(
            content=cls(color=color, size=50),
            alignment=ft.Alignment.CENTER,
            expand=True,
        ),
        ft.Text(
            name,
            size=11,
            text_align=ft.TextAlign.CENTER,
        ),
    ],
    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
    spacing=8,
),
bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
border_radius=12,
padding=12,
)
for name, cls in SPINNERS
],
)
)

if __name__ == "__main__":
    ft.run(main)

```

SpinKit Properties

play_arrowTry Online

```

import flet_spinkit as spins

import flet as ft

def section(title: str, controls: list) -> ft.Column:
    return ft.Column(
        [
            ft.Text(title, size=14, weight=ft.FontWeight.BOLD),
            ft.Row(controls, spacing=24, wrap=True),
        ],
        spacing=12,
    )

def labeled(spinner, label: str) -> ft.Column:
    return ft.Column(
        [
            ft.Container(
                content=spinner,
                alignment=ft.Alignment.CENTER,
                width=80,
                height=80,
            ),
            ft.Text(
                label,
                size=10,
                text_align=ft.TextAlign.CENTER,
            ),
        ],
        horizontal_alignment=ft.CrossAxisAlignment.CENTER,
        spacing=4,
    )

def main(page: ft.Page):
    page.title = "SpinKit Properties"
    page.scroll = ft.ScrollMode.AUTO
    page.padding = 24

    page.add(
        ft.Text("SpinKit Properties", size=24, weight=ft.FontWeight.BOLD),
        ft.Divider(height=16, color=ft.Colors.TRANSPARENT),
        # Colors
        section(
            "color",
            [
                labeled(spins.RotatingCircle(color=ft.Colors.BLUE, size=50), "BLUE"),
                labeled(spins.RotatingCircle(color=ft.Colors.RED, size=50), "RED"),
                labeled(spins.RotatingCircle(color=ft.Colors.GREEN, size=50), "GREEN"),
                labeled(
                    spins.RotatingCircle(color=ft.Colors.ORANGE, size=50), "ORANGE"
                ),
                labeled(
                    spins.RotatingCircle(color=ft.Colors.PURPLE, size=50), "PURPLE"
                ),
            ],
        ),
    )

```

```

        labeled(spins.RotatingCircle(color=ft.Colors.TEAL, size=50), "TEAL"),
    ],
),
ft.Divider(height=8),
# Sizes
section(
    "size",
    [
        labeled(spins.ThreeBounce(color=ft.Colors.BLUE, size=20), "20"),
        labeled(spins.ThreeBounce(color=ft.Colors.BLUE, size=30), "30"),
        labeled(spins.ThreeBounce(color=ft.Colors.BLUE, size=40), "40"),
        labeled(spins.ThreeBounce(color=ft.Colors.BLUE, size=50), "50"),
        labeled(spins.ThreeBounce(color=ft.Colors.BLUE, size=70), "70"),
    ],
),
ft.Divider(height=8),
# Durations
section(
    "duration",
    [
        labeled(
            spins.Pulse(
                color=ft.Colors.BLUE,
                size=50,
                duration=ft.Duration(milliseconds=400),
            ),
            "400ms",
        ),
        labeled(
            spins.Pulse(
                color=ft.Colors.BLUE,
                size=50,
                duration=ft.Duration(milliseconds=700),
            ),
            "700ms",
        ),
        labeled(
            spins.Pulse(
                color=ft.Colors.BLUE,
                size=50,
                duration=ft.Duration(milliseconds=1000),
            ),
            "1000ms",
        ),
        labeled(
            spins.Pulse(
                color=ft.Colors.BLUE,
                size=50,
                duration=ft.Duration(milliseconds=2000),
            ),
            "2000ms",
        ),
        labeled(
            spins.Pulse(
                color=ft.Colors.BLUE,
                size=50,
                duration=ft.Duration(milliseconds=4000),
            ),
            "4000ms",
        ),
    ],
),
ft.Divider(height=8),
# Wave – wave_type
section(
    "Wave · wave_type",
    [
        labeled(
            spins.Wave(
                color=ft.Colors.BLUE, size=50, wave_type=spins.WaveType.START
            ),
            "START",
        ),
        labeled(
            spins.Wave(
                color=ft.Colors.BLUE, size=50, wave_type=spins.WaveType.CENTER
            ),
            "CENTER",
        ),
        labeled(
            spins.Wave(
                color=ft.Colors.BLUE, size=50, wave_type=spins.WaveType.END
            ),
            "END",
        ),
    ],
),
),

```

```

ft.Divider(height=8),
# Wave - item_count
section(
    "Wave · item_count",
    [
        labeled(spins.Wave(color=ft.Colors.BLUE, size=50, item_count=3), "3"),
        labeled(spins.Wave(color=ft.Colors.BLUE, size=50, item_count=5), "5"),
        labeled(spins.Wave(color=ft.Colors.BLUE, size=50, item_count=7), "7"),
        labeled(spins.Wave(color=ft.Colors.BLUE, size=50, item_count=10), "10"),
    ],
),
ft.Divider(height=8),
# Ring - line_width
section(
    "Ring · line_width",
    [
        labeled(spins.Ring(color=ft.Colors.BLUE, size=50, line_width=2), "2"),
        labeled(spins.Ring(color=ft.Colors.BLUE, size=50, line_width=5), "5"),
        labeled(spins.Ring(color=ft.Colors.BLUE, size=50, line_width=8), "8"),
        labeled(spins.Ring(color=ft.Colors.BLUE, size=50, line_width=12), "12"),
    ],
),
ft.Divider(height=8),
# DualRing - line_width
section(
    "DualRing · line_width",
    [
        labeled(
            spins.DualRing(color=ft.Colors.BLUE, size=50, line_width=2), "2"
        ),
        labeled(
            spins.DualRing(color=ft.Colors.BLUE, size=50, line_width=5), "5"
        ),
        labeled(
            spins.DualRing(color=ft.Colors.BLUE, size=50, line_width=8), "8"
        ),
        labeled(
            spins.DualRing(color=ft.Colors.BLUE, size=50, line_width=12), "12"
        ),
    ],
),
ft.Divider(height=8),
# Ripple - border_width
section(
    "Ripple · border_width",
    [
        labeled(
            spins.Ripple(color=ft.Colors.BLUE, size=50, border_width=2), "2"
        ),
        labeled(
            spins.Ripple(color=ft.Colors.BLUE, size=50, border_width=4), "4"
        ),
        labeled(
            spins.Ripple(color=ft.Colors.BLUE, size=50, border_width=8), "8"
        ),
        labeled(
            spins.Ripple(color=ft.Colors.BLUE, size=50, border_width=14), "14"
        ),
    ],
),
ft.Divider(height=8),
# SpinningLines - line_width
section(
    "SpinningLines · line_width",
    [
        labeled(
            spins.SpinningLines(color=ft.Colors.BLUE, size=50, line_width=1),
            "1",
        ),
        labeled(
            spins.SpinningLines(color=ft.Colors.BLUE, size=50, line_width=2),
            "2",
        ),
        labeled(
            spins.SpinningLines(color=ft.Colors.BLUE, size=50, line_width=4),
            "4",
        ),
        labeled(
            spins.SpinningLines(color=ft.Colors.BLUE, size=50, line_width=6),
            "6",
        ),
    ],
),
ft.Divider(height=8),
# PianoWave - item_count
section(
    "PianoWave · item_count",

```

```
[
    labeled(
        spins.PianoWave(color=ft.Colors.BLUE, size=50, item_count=3), "3"
    ),
    labeled(
        spins.Piano
```

Available controls

- ChasingDots
- Circle
- CubeGrid
- DancingSquare
- DoubleBounce
- DualRing
- FadingCircle
- FadingCube
- FadingFour
- FadingGrid
- FoldingCube
- HourGlass
- PianoWave
- PouringHourGlass
- PouringHourGlassRefined
- Pulse
- PulsingGrid
- PumpingHeart
- Ring
- Ripple
- RotatingCircle
- RotatingPlain
- SpinningCircle
- SpinningLines
- SquareCircle
- ThreeBounce
- ThreeInOut
- WanderingCubes
- Wave
- WaveSpinner

Stack

Positions its children on top of each other, following a LIFO (Last In First Out) order.

This control is useful if you want to overlap several children in a simple way. For example having some text and an image, overlaid with a gradient and a button attached to the bottom.

Stack is also useful if you want to implement implicit animations (<https://flet.dev/docs/cookbook/animations>) that require knowing absolute position of a target value.

Example:

```
ft.Stack(
    width=300,
    height=300,
    controls=[
        ft.Image(
            src="https://picsum.photos/300/300",
            width=300,
            height=300,
            fit=ft.BoxFit.CONTAIN,
        ),
        ft.Row(
            alignment=ft.MainAxisAlignment.CENTER,
            controls=[
                ft.Text(
                    value="Image title",
                    color=ft.Colors.SURFACE_TINT,
                    size=40,
                    weight=ft.FontWeight.BOLD,
                    opacity=0.5,
                ),
            ],
        ),
    ],
)
```

Image: Stack - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/stack/image_for_docs.png

Stack

Inherits: LayoutControl, AdaptiveControl

Properties

- alignment - Specifies the alignment for non-positioned (those without explicit alignment properties such as top or bottom) and partially-positioned controls.
- clip_behavior - The content will be clipped (or not) according to this option.
- controls - A list of Controls to display.
- fit - How to size the non-positioned controls.

Examples

Online avatar

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Stack(
                width=40,
                height=40,
                controls=[
                    ft.CircleAvatar(
                        foreground_image_src="https://avatars.githubusercontent.com/u/5041459?s=88&v=4"
                    ),
                    ft.Container(
                        alignment=ft.Alignment.BOTTOM_LEFT,
                        content=ft.CircleAvatar(bgcolor=ft.Colors.GREEN, radius=5),
                    ),
                ],
            ),
        ),
    )
```

```
)
```

```
if __name__ == "__main__":  
    ft.run(main)
```

Image: online-avatar - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/stack/online_avatar.png

Absolute positioning

play_arrowTry Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER  
    page.vertical_alignment = ft.MainAxisAlignment.CENTER  
  
    page.add(  
        ft.SafeArea(  
            content=ft.Container(  
                border_radius=8,  
                padding=5,  
                width=200,  
                height=200,  
                bgcolor=ft.Colors.BLACK,  
                content=ft.Stack(  
                    controls=[  
                        ft.Container(  
                            width=20,  
                            height=20,  
                            bgcolor=ft.Colors.RED,  
                            border_radius=5,  
                        ),  
                        ft.Container(  
                            width=20,  
                            height=20,  
                            bgcolor=ft.Colors.YELLOW,  
                            border_radius=5,  
                            right=0,  
                        ),  
                        ft.Container(  
                            width=20,  
                            height=20,  
                            bgcolor=ft.Colors.BLUE,  
                            border_radius=5,  
                            right=0,  
                            bottom=0,  
                        ),  
                        ft.Container(  
                            width=20,  
                            height=20,  
                            bgcolor=ft.Colors.GREEN,  
                            border_radius=5,  
                            left=0,  
                            bottom=0,  
                        ),  
                        ft.Column(  
                            left=85,  
                            top=85,  
                            controls=[  
                                ft.Container(  
                                    width=20,  
                                    height=20,  
                                    bgcolor=ft.Colors.PURPLE,  
                                    border_radius=5,  
                                ),  
                            ],  
                        ),  
                    ],  
                ),  
            ),  
        ),  
    )  
  
if __name__ == "__main__":  
    ft.run(main)
```

Image: absolute-positioning - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/stack/absolute_positioning.png

Properties

buildalignmentclass-attributeinstance-attribute

`alignment: Alignment | None = None`

Specifies the alignment for non-positioned (those without explicit alignment properties such as top or bottom) and partially-positioned controls.

buildclip_behaviorclass-attributeinstance-attribute

`clip_behavior: ClipBehavior = ClipBehavior.HARD_EDGE`

The content will be clipped (or not) according to this option.

buildcontrolsclass-attributeinstance-attribute

`controls: list[Control] = field(default_factory=list)`

A list of Controls to display.

For the display order, it follows the order of the list, so the last control in the list will be displayed on top (LIFO - Last In First Out).

buildfitclass-attributeinstance-attribute

`fit: StackFit = StackFit.LOOSE`

How to size the non-positioned controls.

SubmenuButton

A menu button that displays a cascading menu.

Typically used in a MenuBar control.

```
ft.SubmenuButton(
    content=ft.Text("Choose text style"),
    key="smbutton",
    expand=True,
    menu_style=ft.MenuStyle(
        alignment=ft.Alignment.BOTTOM_LEFT, side=ft.BorderSide(1)
    ),
    controls=[
        ft.MenuItemButton(
            content=ft.Text("Underlined"),
            on_click=lambda e: print(f"{e.control.content.value}.on_click"),
        ),
        ft.MenuItemButton(...),
        ...,
    ],
)
```

Image: SubmenuButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/submenu_button/image_for_docs.png

Activated submenu button

Inherits: LayoutControl

Properties

- `alignment_offset` - The offset of the menu relative to the alignment origin determined by `MenuStyle.alignment` on the style attribute.
- `clip_behavior` - Whether to clip the content of this control or not.
- `content` - The child control to be displayed in the middle portion of this button.
- `controls` - A list of controls that appear in the menu when it is opened.
- `leading` - An optional control to display before the content.
- `menu_style` - Customizes this menu's appearance.
- `style` - Customizes this button's appearance.
- `trailing` - An optional control to display after the content.

Events

- `on_blur` - Called when this button loses focus.
- `on_close` - Called when the menu is closed.
- `on_focus` - Called when the button receives focus.
- `on_hover` - Called when the button is hovered.
- `on_open` - Called when the menu is opened.

Examples

SubmenuButton

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.padding = 0
    page.spacing = 0

    def handle_color_click(e: ft.Event[ft.MenuItemButton]):
        color = e.control.content.value
        background_container.content.value = f"{color} background color"
        background_container.bgcolor = color.lower()

    def handle_alignment_click(e: ft.Event[ft.MenuItemButton]):
```

```

background_container.alignment = e.control.data

def handle_on_hover(e: ft.Event[ft.MenuItemButton]):
    print(f"{e.control.content.value}.on_hover")

menubar = ft.MenuBar(
    expand=True,
    controls=[
        ft.SubmenuButton(
            content=ft.Text("Change Body"),
            key="submenubutton",
            controls=[
                ft.SubmenuButton(
                    content=ft.Text("BG Color"),
                    leading=ft.Icon(ft.Icons.COLORIZE),
                    controls=[
                        ft.MenuItemButton(
                            content=ft.Text("Blue"),
                            on_click=handle_color_click,
                            on_hover=handle_on_hover,
                            style=ft.ButtonStyle(
                                bgcolor={ft.ControlState.HOVERED: ft.Colors.BLUE}
                            ),
                        ),
                        ft.MenuItemButton(
                            content=ft.Text("Green"),
                            on_click=handle_color_click,
                            on_hover=handle_on_hover,
                            style=ft.ButtonStyle(
                                bgcolor={ft.ControlState.HOVERED: ft.Colors.GREEN}
                            ),
                        ),
                        ft.MenuItemButton(
                            content=ft.Text("Red"),
                            on_click=handle_color_click,
                            on_hover=handle_on_hover,
                            style=ft.ButtonStyle(
                                bgcolor={ft.ControlState.HOVERED: ft.Colors.RED}
                            ),
                        ),
                    ],
                ),
                ft.SubmenuButton(
                    content=ft.Text("Text alignment"),
                    leading=ft.Icon(ft.Icons.LOCATION_PIN),
                    controls=[
                        ft.MenuItemButton(
                            content=ft.Text("bottom_center"),
                            data=ft.Alignment.BOTTOM_CENTER,
                            on_click=handle_alignment_click,
                            style=ft.ButtonStyle(
                                bgcolor={
                                    ft.ControlState.HOVERED: ft.Colors.GREY_100
                                }
                            ),
                        ),
                        ft.MenuItemButton(
                            content=ft.Text("bottom_left"),
                            data=ft.Alignment.BOTTOM_LEFT,
                            on_click=handle_alignment_click,
                            style=ft.ButtonStyle(
                                bgcolor={
                                    ft.ControlState.HOVERED: ft.Colors.GREY_100
                                }
                            ),
                        ),
                        ft.MenuItemButton(
                            content=ft.Text("top_center"),
                            data=ft.Alignment.TOP_CENTER,
                            on_click=handle_alignment_click,
                            style=ft.ButtonStyle(
                                bgcolor={
                                    ft.ControlState.HOVERED: ft.Colors.GREY_100
                                }
                            ),
                        ),
                        ft.MenuItemButton(
                            content=ft.Text("center_left"),
                            data=ft.Alignment.CENTER_LEFT,
                            on_click=handle_alignment_click,
                            style=ft.ButtonStyle(
                                bgcolor={
                                    ft.ControlState.HOVERED: ft.Colors.GREY_100
                                }
                            ),
                        ),
                        ft.MenuItemButton(

```

```

        content=ft.Text("center_right"),
        data=ft.Alignment.CENTER_RIGHT,
        on_click=handle_alignment_click,
        style=ft.ButtonStyle(
            bgcolor={
                ft.ControlState.HOVERED: ft.Colors.GREY_100
            }
        ),
    ),
    1,
),
1,
)
)
page.add(
    ft.SafeArea(
        expand=True,
        content=ft.Column(
            expand=True,
            spacing=0,
            controls=[
                ft.Row(controls=[menubar]),
                background_container := ft.Container(
                    expand=True,
                    bgcolor=ft.Colors.SURFACE_TINT,
                    alignment=ft.Alignment.CENTER,
                    content=ft.Text(
                        value="Choose a bgcolor from the menu",
                        style=ft.TextStyle(size=24, weight=ft.FontWeight.BOLD),
                    ),
                ),
            ],
        ),
    ),
)
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/submenu_button/basic.png

Standalone SubmenuButton

play_arrowTry Online

import flet as ft

```

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT

    smb = ft.SubmenuButton(
        expand=True,
        content=ft.Text("Text Styles"),
        key="smbutton",
        menu_style=ft.MenuStyle(
            alignment=ft.Alignment.CENTER_RIGHT, side=ft.BorderSide(1)
        ),
        controls=[
            ft.MenuItemButton(
                content=ft.Text("Underlined"),
                on_click=lambda e: print(f"{e.control.content.value}.on_click"),
                style=ft.ButtonStyle(
                    text_style={
                        ft.ControlState.HOVERED: ft.TextStyle(
                            decoration=ft.TextDecoration.UNDERLINE
                        )
                    }
                ),
            ),
            ft.MenuItemButton(
                content=ft.Text("Bold"),
                on_click=lambda e: print(f"{e.control.content.value}.on_click"),
                style=ft.ButtonStyle(
                    text_style={
                        ft.ControlState.HOVERED: ft.TextStyle(weight=ft.FontWeight.BOLD)
                    }
                ),
            ),
            ft.MenuItemButton(
                content=ft.Text("Italic"),
                on_click=lambda e: print(f"{e.control.content.value}.on_click"),
                style=ft.ButtonStyle(
                    text_style={ft.ControlState.HOVERED: ft.TextStyle(italic=True)}
                ),
            ),
        ],
    )

```

```

        ),
    ],
)

page.add(
    ft.SafeArea(
        content=ft.Row(controls=[smb]),
    )
)

if __name__ == "__main__":
    ft.run(main)

```

Image: standalone - https://flet.dev/docs/examples/controls/material/submenu_button/media/standalone.gif

Properties

buildalignment_offsetclass-attributeinstance-attribute

alignment_offset: OffsetValue | None = None

The offset of the menu relative to the alignment origin determined by MenuStyle.alignment on the style attribute.

buildclip_behaviorclass-attributeinstance-attribute

clip_behavior: ClipBehavior = ClipBehavior.HARD_EDGE

Whether to clip the content of this control or not.

buildcontentclass-attributeinstance-attribute

content: StrOrControl | None = None

The child control to be displayed in the middle portion of this button.

Typically this is the button's label, using a Text control.

buildcontrolsclass-attributeinstance-attribute

controls: list[Control] = field(default_factory=list)

A list of controls that appear in the menu when it is opened.

Typically either MenuItemButton or SubMenuButton controls.

If this list is empty, then the button for this menu item will be disabled.

buildleadingclass-attributeinstance-attribute

leading: Control | None = None

An optional control to display before the content.

Typically an Icon control.

buildmenu_styleclass-attributeinstance-attribute

menu_style: MenuStyle | None = None

Customizes this menu's appearance.

buildstyleclass-attributeinstance-attribute

style: ButtonStyle | None = None

Customizes this button's appearance.

buildtrailingclass-attributeinstance-attribute

trailing: Control | None = None

An optional control to display after the content.

Typically an Icon control.

Events

bolton_blurclass-attributeinstance-attribute

on_blur: ControlEventHandler[SubMenuButton] | None = None

Called when this button loses focus.

bolton_closeclass-attributeinstance-attribute

`on_close: ControlEventHandler[SubmenuButton] | None = None`

Called when the menu is closed.

bolton_focusclass-attributeinstance-attribute

`on_focus: ControlEventHandler[SubmenuButton] | None = None`

Called when the button receives focus.

bolton_hoverclass-attributeinstance-attribute

`on_hover: ControlEventHandler[SubmenuButton] | None = None`

Called when the button is hovered.

bolton_openclass-attributeinstance-attribute

`on_open: ControlEventHandler[SubmenuButton] | None = None`

Called when the menu is opened.

Switch

A toggle represents a physical switch that allows someone to choose between two mutually exclusive options.

For example, "On/Off", "Show/Hide".

```
ft.Switch(label="Unchecked switch", value=False)
ft.Switch(label="Disabled switch", disabled=True)
```

Image: Switch - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/switch/image_for_docs.png

Switch

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `active_color` - The color to use when this switch is on.
- `active_track_color` - The color to use on the track when this switch is on.
- `adaptive` - Whether an adaptive Switch should be created based on the target platform.
- `autofocus` - Whether this switch will be selected as the initial focus.
- `focus_color` - The color to use for the focus highlight for keyboard interactions.
- `hover_color` - The color to be used when it is being hovered over by the mouse pointer.
- `inactive_thumb_color` - The color to use on the thumb when this switch is off.
- `inactive_track_color` - The color to use on the track when this switch is off.
- `label` - The clickable label to display on the right of this switch.
- `label_position` - The position of the label, if provided.
- `label_text_style` - The label's text style, when it is a string.
- `mouse_cursor` - The cursor to be displayed when a mouse pointer enters or is hovering over this control.
- `overlay_color` - The color for the switch's Material in various `ControlState` states.
- `padding` - The amount of space to surround the child inside the bounds of the Switch.
- `splash_radius` - The radius of the splash effect when the switch is pressed.
- `thumb_color` - The color of this switch's thumb in various `ControlState` states.
- `thumb_icon` - The icon of this Switch's thumb in various `ControlState` states.
- `track_color` - The color of this switch's track in various `ControlState` states.
- `track_outline_color` - The outline color of this switch's track in various `ControlState` states.
- `track_outline_width` - The outline width of this switch's track in all or specific `ControlState` states.
- `value` - Current value of this switch.

Events

- `on_blur` - Called when the control has lost focus.
- `on_change` - Called when the state of the Switch is changed.
- `on_focus` - Called when the control has received focus.

Examples

Switch

`play_arrow` Try Online

```
import flet as ft
```

```
def main(page: ft.Page):
    def handle_button_click(e: ft.Event[ft.Button]):
```

```

        message.value = (
            f"Switch values are: {c1.value}, {c2.value}, {c3.value}, {c4.value}."
        )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    c1 := ft.Switch(label="Unchecked switch", value=False),
                    c2 := ft.Switch(label="Checked switch", value=True),
                    c3 := ft.Switch(label="Disabled switch", disabled=True),
                    c4 := ft.Switch(
                        label="Switch with rendered label_position='left'",
                        label_position=ft.LabelPosition.LEFT,
                    ),
                    ft.Button(content="Submit", on_click=handle_button_click),
                    message := ft.Text(),
                ],
            ),
        ),
    )
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/switch/basic.png>

Handling events

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    def handle_switch_change(e: ft.Event[ft.Switch]):
        page.theme_mode = ft.ThemeMode.DARK if e.control.value else ft.ThemeMode.LIGHT
        e.control.label = (
            "Light ThemeMode"
            if page.theme_mode == ft.ThemeMode.LIGHT
            else "Dark ThemeMode"
        )

    page.theme_mode = ft.ThemeMode.LIGHT
    page.add(
        ft.SafeArea(
            content=ft.Switch(
                key="theme_mode_switch",
                label="Light ThemeMode",
                on_change=handle_switch_change,
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: handling-events - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/switch/handling_events.gif

Properties

buildactive_colorclass-attributeinstance-attribute

active_color: ColorValue | None = None

The color to use when this switch is on.

buildactive_track_colorclass-attributeinstance-attribute

active_track_color: ColorValue | None = None

The color to use on the track when this switch is on.

If track_color returns a non-none color in the ControlState.SELECTED state, it will be used instead of this color.

buildadaptiveclass-attributeinstance-attribute

adaptive: bool | None = None

Whether an adaptive Switch should be created based on the target platform.

On iOS and macOS, a CupertinoSwitch is created, which has matching functionality and presentation as Switch, and the graphics as expected on iOS. On other platforms, a Material Switch is created.

Defaults to False. See the example of usage [here](#).

buildautofocusclass-attributeinstance-attribute

`autofocus: bool = False`

Whether this switch will be selected as the initial focus. If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

buildfocus_colorclass-attributeinstance-attribute

`focus_color: ColorValue | None = None`

The color to use for the focus highlight for keyboard interactions.

buildhover_colorclass-attributeinstance-attribute

`hover_color: ColorValue | None = None`

The color to be used when it is being hovered over by the mouse pointer.

buildinactive_thumb_colorclass-attributeinstance-attribute

`inactive_thumb_color: ColorValue | None = None`

The color to use on the thumb when this switch is off.

Defaults to colors defined in the material design specification.

If thumb_color returns a non-none color in the `ControlState.DEFAULT` state, it will be used instead of this color.

buildinactive_track_colorclass-attributeinstance-attribute

`inactive_track_color: ColorValue | None = None`

The color to use on the track when this switch is off.

Defaults to colors defined in the material design specification.

If track_color returns a non-none color in the `ControlState.DEFAULT` state, it will be used instead of this color.

buildlabelclass-attributeinstance-attribute

`label: StrOrControl | None = None`

The clickable label to display on the right of this switch.

buildlabel_positionclass-attributeinstance-attribute

`label_position: LabelPosition = LabelPosition.RIGHT`

The position of the label, if provided.

buildlabel_text_styleclass-attributeinstance-attribute

`label_text_style: TextStyle | None = None`

The label's text style, when it is a string.

buildmouse_cursorclass-attributeinstance-attribute

`mouse_cursor: MouseCursor | None = None`

The cursor to be displayed when a mouse pointer enters or is hovering over this control.

buildoverlay_colorclass-attributeinstance-attribute

`overlay_color: ControlStateValue[ColorValue] | None = None`

The color for the switch's Material in various `ControlState` states.

The following states are supported: `ControlState.PRESSED`, `ControlState.SELECTED`, `ControlState.HOVERED`, `ControlState.FOCUSED` and `ControlState.DEFAULT`.

buildpaddingclass-attributeinstance-attribute

`padding: PaddingValue | None = None`

The amount of space to surround the child inside the bounds of the Switch.

Defaults to horizontal padding of 4 pixels. If `Theme.use_material3` is false, then there is no padding by default.

buildsplash_radiusclass-attributeinstance-attribute

`splash_radius: Number | None = None`

The radius of the splash effect when the switch is pressed.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildthumb_colorclass-attributeinstance-attribute

```
thumb_color: ControlStateValue[ColorValue] | None = None
```

The color of this switch's thumb in various `ControlState` states.

The following states are supported: `ControlState.SELECTED`, `ControlState.HOVERED`, `ControlState.DISABLED`, `ControlState.FOCUSED` and `ControlState.DEFAULT` (fallback).

buildthumb_iconclass-attributeinstance-attribute

```
thumb_icon: ControlStateValue[IconData] | None = None
```

The icon of this Switch's thumb in various `ControlState` states.

The following states are supported: `ControlState.SELECTED`, `ControlState.HOVERED`, `ControlState.DISABLED`, `ControlState.FOCUSED` and `ControlState.DEFAULT` (fallback).

buildtrack_colorclass-attributeinstance-attribute

```
track_color: ControlStateValue[ColorValue] | None = None
```

The color of this switch's track in various `ControlState` states.

The following states are supported: `ControlState.SELECTED`, `ControlState.HOVERED`, `ControlState.DISABLED`, `ControlState.FOCUSED` and `ControlState.DEFAULT` (fallback).

buildtrack_outline_colorclass-attributeinstance-attribute

```
track_outline_color: (
    ControlStateValue[ColorValue] | None
) = None
```

The outline color of this switch's track in various `ControlState` states.

The following states are supported: `ControlState.SELECTED`, `ControlState.HOVERED`, `ControlState.DISABLED`, `ControlState.FOCUSED` and `ControlState.DEFAULT` (fallback).

buildtrack_outline_widthclass-attributeinstance-attribute

```
track_outline_width: (
    ControlStateValue[Number | None] | None
) = None
```

The outline width of this switch's track in all or specific `ControlState` states.

The following states are supported: `ControlState.SELECTED`, `ControlState.HOVERED`, `ControlState.DISABLED`, `ControlState.FOCUSED` and `ControlState.DEFAULT` (fallback).

buildvalueclass-attributeinstance-attribute

```
value: bool = False
```

Current value of this switch.

Events

bolton_blurclass-attributeinstance-attribute

```
on_blur: ControlEventHandler[Switch] | None = None
```

Called when the control has lost focus.

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[Switch] | None = None
```

Called when the state of the Switch is changed.

bolton_focusclass-attributeinstance-attribute

```
on_focus: ControlEventHandler[Switch] | None = None
```

Called when the control has received focus.

Tab

A Material Design TabBar tab.

Inherits: `AdaptiveControl`

Properties

- `height` - The height of the tab.
- `icon` - An icon to display on the left of Tab text.
- `icon_margin` - The margin added around the tab's icon.
- `label` - The tab's name.

Properties

`buildheightclass-attributeinstance-attribute`

`height: Number | None = None`

The height of the tab.

If `None`, it will be calculated based on the content of the Tab. When `icon` is not `None` along with `label`, the default height is 72.0 pixels. Without an icon, the height is 46.0 pixels.

Currently, the provided tab height cannot be lower than the default height.

`buildiconclass-attributeinstance-attribute`

`icon: IconDataOrControl | None = None`

An icon to display on the left of Tab text.

`buildicon_marginclass-attributeinstance-attribute`

`icon_margin: MarginValue | None = None`

The margin added around the tab's icon.

Only useful when used in combination with `icon`, and `label` is not `None`.

Defaults to 2 pixels of bottom margin. If `Theme.use_material3` is `False`, then defaults to 10 pixels of bottom margin.

`buildlabelclass-attributeinstance-attribute`

`label: StrOrControl | None = None`

The tab's name. Can be either a string or a control.

Raises:

- `ValueError` - If both `label` and `icon` are not set.

TabBar

Used for navigating frequently accessed, distinct content categories. Tabs allow for navigation between two or more content views and relies on text headers to articulate the different sections of content.

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `divider_color` - The color of the divider.
- `divider_height` - The height of the divider.
- `enable_feedback` - Whether detected gestures should provide acoustic and/or haptic feedback.
- `indicator` - Defines the appearance of the selected tab indicator.
- `indicator_animation` - Specifies the animation behavior of the tab indicator.
- `indicator_color` - The color of the indicator(line that appears below the selected tab).
- `indicator_size` - Defines how the selected tab indicator's size is computed.
- `indicator_thickness` - The thickness of the indicator.
- `label_color` - The color of selected tab labels.
- `label_padding` - The padding around the tab label.
- `label_text_style` - The text style of the tab labels.
- `mouse_cursor` - The cursor to be displayed when a mouse pointer enters or is hovering over this control.
- `overlay_color` - Defines the ink response focus, hover, and splash colors in various `ControlState` states.
- `padding` - The padding around the Tabs control.
- `scrollable` - Whether this tab bar can be scrolled horizontally.
- `secondary` - Whether to create a secondary/nested tab bar.
- `splash_border_radius` - Defines the clipping radius of splashes that extend outside the bounds of the tab.
- `tab_alignment` - Specifies the horizontal alignment of the tabs within this tab bar.
- `tabs` - A list of controls.
- `unselected_label_color` - The color of unselected tab labels.
- `unselected_label_text_style` - The text style of the unselected tab labels.

Events

- `on_click` - Called when a tab is clicked.
- `on_hover` - Called when a tab's (from tabs) hover state in the tab bar changes.

Properties

`builddivider_color`class-attributeinstance-attribute

`divider_color: ColorValue | None = None`

The color of the divider.

`builddivider_height`class-attributeinstance-attribute

`divider_height: Number | None = None`

The height of the divider.

If None, defaults to `TabBarTheme.divider_height`. If this is also None, 1.0 will be used.

`buildenable_feedback`class-attributeinstance-attribute

`enable_feedback: bool | None = None`

Whether detected gestures should provide acoustic and/or haptic feedback.

On Android, for example, setting this to True produce a click sound and a long-press will produce a short vibration.

Defaults to True.

buildindicatorclass-attributeinstance-attribute

`indicator: UnderlineTabIndicator | None = None`

Defines the appearance of the selected tab indicator.

If this or `TabBarTheme.indicator` is not None, `indicator_color` and `indicator_thickness` properties are ignored.

The indicator's size is based on the tab's bounds. If `indicator_size` is `TabBarIndicatorSize.TAB`, the tab's bounds are as wide as the space occupied by the tab in the tab bar. If `indicator_size` is `TabBarIndicatorSize.LABEL`, then the tab's bounds are only as wide as the tab control itself.

buildindicator_animationclass-attributeinstance-attribute

`indicator_animation: TabIndicatorAnimation | None = None`

Specifies the animation behavior of the tab indicator.

If this is None, then the value of `TabBarTheme.indicator_animation` is used. If that is also None, then the tab indicator will animate linearly if the `indicator_size` is `TabBarIndicatorSize.TAB`, otherwise it will animate with an elastic effect if the `indicator_size` is `TabBarIndicatorSize.LABEL`.

buildindicator_colorclass-attributeinstance-attribute

`indicator_color: ColorValue | None = None`

The color of the indicator(line that appears below the selected tab).

Note

Will be ignored if `indicator` or `TabBarTheme.indicator` is not None.

buildindicator_sizeclass-attributeinstance-attribute

`indicator_size: TabBarIndicatorSize | None = None`

Defines how the selected tab indicator's size is computed.

The size of the selected tab indicator is defined relative to the tab's overall bounds if `indicator_size` is `TabBarIndicatorSize.TAB` (the default) or relative to the bounds of the tab's control if `indicator_size` is `TabBarIndicatorSize.LABEL`.

The selected tab's location appearance can be refined further with the `indicator_color`, `indicator_thickness`, and `indicator` properties.

buildindicator_thicknessclass-attributeinstance-attribute

`indicator_thickness: Number = 2.0`

The thickness of the indicator. Value must be greater than zero.

Note

Will be ignored if `indicator` or `TabBarTheme.indicator` is not None.

Raises:

- `ValueError` - If `indicator` is None and `indicator_thickness` is not strictly greater than 0.

buildlabel_colorclass-attributeinstance-attribute

`label_color: ColorValue | None = None`

The color of selected tab labels.

buildlabel_paddingclass-attributeinstance-attribute

`label_padding: PaddingValue | None = None`

The padding around the tab label.

buildlabel_text_styleclass-attributeinstance-attribute

`label_text_style: TextStyle | None = None`

The text style of the tab labels.

buildmouse_cursorclass-attributeinstance-attribute

`mouse_cursor: MouseCursor | None = None`

The cursor to be displayed when a mouse pointer enters or is hovering over this control.

buildoverlay_colorclass-attributeinstance-attribute

`overlay_color: ControlStateValue[ColorValue] | None = None`

Defines the ink response focus, hover, and splash colors in various ControlState states.

The following states are supported: ControlState.PRESSED, ControlState.HOVERED and ControlState.FOCUSED.

buildpaddingclass-attributeinstance-attribute

`padding: PaddingValue | None = None`

The padding around the Tabs control.

buildscrollableclass-attributeinstance-attribute

`scrollable: bool = True`

Whether this tab bar can be scrolled horizontally.

If True, then each tab is as wide as needed for its label and the entire tab bar is scrollable. Otherwise each tab gets an equal share of the available space.

buildsecondaryclass-attributeinstance-attribute

`secondary: bool = False`

Whether to create a secondary/nested tab bar.

Secondary tabs are used within a content area to further separate related content and establish hierarchy.

buildsplash_border_radiusclass-attributeinstance-attribute

`splash_border_radius: BorderRadiusValue | None = None`

Defines the clipping radius of splashes that extend outside the bounds of the tab.

buildtab_alignmentclass-attributeinstance-attribute

`tab_alignment: TabAlignment | None = None`

Specifies the horizontal alignment of the tabs within this tab bar.

If this is None, then the value of TabBarTheme.tab_alignment is used. If TabBarTheme.tab_alignment is None and Theme.use_material3 is True, then TabAlignment.START_OFFSET is used if scrollable is True, otherwise TabAlignment.FILL is used.

If TabBarTheme.tab_alignment is None and Theme.use_material3 is False, then TabAlignment.CENTER is used if scrollable is True, otherwise TabAlignment.FILL is used.

Note

- If scrollable is False, then tab_alignment must be either TabAlignment.FILL or TabAlignment.CENTER.
- If scrollable is True, then tab_alignment must be TabAlignment.START, TabAlignment.START_OFFSET or TabAlignment.CENTER.

Raises:

- ValueError - If it is not valid for the current scrollable configuration.

buildtabsinstance-attribute

`tabs: list[Control]`

A list of controls.

Typically Tabs.

buildunselected_label_colorclass-attributeinstance-attribute

`unselected_label_color: ColorValue | None = None`

The color of unselected tab labels.

buildunselected_label_text_styleclass-attributeinstance-attribute

`unselected_label_text_style: TextStyle | None = None`

The text style of the unselected tab labels.

Events

`bolton_click`class-attributeinstance-attribute

`on_click: ControlEventHandler[TabBar] | None = None`

Called when a tab is clicked.

The data property of the event handler argument contains the index of the clicked tab.

`bolton_hover`class-attributeinstance-attribute

`on_hover: EventHandler[TabBarHoverEvent] | None = None`

Called when a tab's (from tabs) hover state in the tab bar changes.

When hover is moved from one tab directly to another, this will be called twice. First to represent hover exiting the initial tab, and then second for the pointer entering hover over the next tab.

TabBarView

A page view with one child per tab.

Note

The length of controls must be the same as the length property of the ancestor Tabs.

Inherits: LayoutControl, AdaptiveControl

Properties

- `clip_behavior` - Defines how the controls will be clipped.
- `controls` - A list of controls, where each control represents the content of a corresponding tab.
- `viewport_fraction` - The fraction of the viewport that each page should occupy.

Properties

`buildclip_behaviorclass-attributeinstance-attribute`

```
clip_behavior: ClipBehavior = ClipBehavior.HARD_EDGE
```

Defines how the controls will be clipped.

`buildcontrolsinstance-attribute`

```
controls: list[Control]
```

A list of controls, where each control represents the content of a corresponding tab. So, a control at index *i* in this list is displayed when the Tab at index *i* is selected.

Note

The length of this list must be equal to the number of tabs specified in an ancestor Tabs control.

`buildviewport_fractionclass-attributeinstance-attribute`

```
viewport_fraction: Number = 1.0
```

The fraction of the viewport that each page should occupy.

For example, 1.0 (the default), means each page fills the viewport in the scrolling direction.

Tabs

Used for navigating frequently accessed, distinct content categories. Tabs allow for navigation between two or more content views and relies on text headers to articulate the different sections of content.

Example:

```
ft.Tabs(
    length=2,
    expand=True,
    content=ft.Column(
        expand=True,
        controls=[
            ft.TabBar(
                tabs=[
                    ft.Tab(label="Overview"),
                    ft.Tab(label="Settings", icon=ft.Icons.SETTINGS),
                ]
            ),
            ft.TabBarView(
                expand=True,
                controls=[
                    ft.Container(
                        alignment=ft.Alignment.CENTER,
                        content=ft.Text("Overview content"),
                    ),
                    ft.Container(
                        alignment=ft.Alignment.CENTER,
                        content=ft.Text("Settings content"),
                    ),
                ],
            ),
        ],
    ),
)
```

Image: Tabs - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/tabs/image_for_docs.png

Tabs

Inherits: `LayoutControl`, `AdaptiveControl`

Properties

- `animation_duration` - The duration of tab animations.
- `content` - The content to display.
- `length` - The total number of tabs.
- `selected_index` - The index of the currently selected tab.

Events

- `on_change` - Called when `selected_index` changes.

Methods

- `move_to` - Selects the tab at the given index.

Examples

Tabs

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Tabs(
                selected_index=1,
                length=3,
                expand=True,
                content=ft.Column(
                    expand=True,
                    controls=[
```

```
ft.TabBar(
    tabs=[
        ft.Tab(label="Tab 1", icon=ft.Icons.SETTINGS_PHONE),
        ft.Tab(label="Tab 2", icon=ft.Icons.SETTINGS),
        ft.Tab(
            label=ft.CircleAvatar(
                foreground_image_src="https://avatars.githubusercontent.com/u/102273996?s=200&v=4",
            ),
        ),
    ],
),
ft.TabBarView(
    expand=True,
    controls=[
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("This is Tab 1"),
        ),
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("This is Tab 2"),
        ),
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("This is Tab 3"),
        ),
    ],
),
],
),
),
)

if __name__ == "__main__":
    ft.run(main)
```

Nested tabs

```
import flet as ft
```

```
if __name__ == "__main__":
    ft.run(main)
```

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Tabs(
                length=2,
                expand=True,
                content=ft.Column(
                    expand=True,
                    controls=[
                        ft.TabBar(
                            tab_alignment=ft.TabAlignment.START,
                            indicator_animation=ft.TabIndicatorAnimation.ELASTIC,
                            indicator_size=ft.TabBarIndicatorSize.LABEL,
                            indicator=ft.UnderlineTabIndicator(
                                border_side=ft.BorderSide(5, color=ft.Colors.RED),
                                border_radius=ft.BorderRadius.all(1),
                                insets=ft.Padding.only(bottom=5),
                            ),
                        ),
                        tabs=[
                            ft.Tab(label=ft.Text("Home")),
                            ft.Tab(label=ft.Text("My Account")),
                        ],
                    ),
                    ft.TabBarView(
                        expand=True,
                        controls=[
                            ft.Text("Home Tab Content"),
                            ft.Text("Profile Tab Content"),
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Programmatic tab switch

play_arrowTry Online

```
import random

import flet as ft

def main(page: ft.Page):
    async def handle_move_to_random(e: ft.Event[ft.FloatingActionButton]):
        # random index, excluding the current one
        i = random.choice([i for i in range(tabs.length) if i != tabs.selected_index])

        await tabs.move_to(
            index=i,
            animation_curve=ft.AnimationCurve.FAST_OUT_SLOWIN,
            animation_duration=ft.Duration(seconds=3),
        )

    page.floating_action_button = ft.FloatingActionButton(
        icon=ft.Icons.MOVE_UP,
        content="Move to a random tab",
        on_click=handle_move_to_random,
    )

    tabs = ft.Tabs(
        length=6,
        selected_index=5,
        expand=True,
        content=ft.Column(
            expand=True,
            controls=[
                ft.TabBar(
                    tab_alignment=ft.TabAlignment.CENTER,
                    tabs=[
                        ft.Tab(label=ft.Text("Tab 1")),
                        ft.Tab(label=ft.Text("Tab 2")),
                        ft.Tab(label=ft.Text("Tab 3")),
                        ft.Tab(label=ft.Text("Tab 4")),
                        ft.Tab(label=ft.Text("Tab 5")),
                    ],
                ),
            ],
        ),
    )
```

```

        ft.Tab(label=ft.Text("Tab 6")),
    ],
),
ft.TabBarView(
    expand=True,
    controls=[
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("Tab 1 content"),
        ),
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("Tab 2 content"),
        ),
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("Tab 3 content"),
        ),
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("Tab 4 content"),
        ),
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("Tab 5 content"),
        ),
        ft.Container(
            alignment=ft.Alignment.CENTER,
            content=ft.Text("Tab 6 content"),
        ),
    ],
),
],
),
)

page.add(ft.SafeArea(expand=True, content=tabs))

if __name__ == "__main__":
    ft.run(main)

```

Properties

buildanimation_durationclass-attributeinstance-attribute

```

animation_duration: DurationValue = field(
    default_factory=lambda: Duration(milliseconds=100)
)

```

The duration of tab animations. For example, the animation that occurs when the selected tab changes.

buildcontentinstance-attribute

```

content: Control

```

The content to display.

buildlengthinstance-attribute

```

length: int

```

The total number of tabs.

Typically greater than one.

Note

Must match the length of both `TabBar.tabs` and `TabBarView.controls`. Don't forget to update it when adding/removing tabs.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

buildselected_indexclass-attributeinstance-attribute

```

selected_index: int = 0

```

The index of the currently selected tab.

Supports Python-style negative indexing, where -1 represents the last tab, -2 the second to last, and so on.

Note

- Must be in range `[-length, length - 1]`.

- Changing the value of this property will internally trigger `move_to` with `animation_duration` and `AnimationCurve.EASE` animation curve. To specify a different animation curve or duration for this particular change, use `move_to` directly.

Raises:

- `IndexError` - If it is not in the range `[-length, length - 1]`.

Events

`bolton_changeclass-attributeinstance-attribute`

`on_change: ControlEventHandler[Tabs] | None = None`

Called when `selected_index` changes.

The data property of the event handler argument contains the index of the selected tab.

Methods

`deployed_codemove_toasync`

```
move_to(
    index: int,
    animation_curve: AnimationCurve = AnimationCurve.EASE_IN,
    animation_duration: DurationValue | None = None,
)
```

Selects the tab at the given index.

Additionally, it triggers `on_change` event and updates `selected_index`.

Note

If index is negative, it is interpreted as a Python-style negative index (e.g., `-1` refers to the last tab). If the resolved index is already the currently selected tab, the method returns immediately and does nothing.

Parameters:

- `index` (int) - The index of the tab to select. Must be between in range `[-length, length - 1]`.
- `animation_curve` (`AnimationCurve`, default: `AnimationCurve.EASE_IN`) - The curve to apply to the animation.
- `animation_duration` (`DurationValue` | `None`, default: `None`) - The duration of the animation. If `None` (the default), `Tabs.animation_duration` will be used.

Raises:

- `IndexError` - If the index is outside the range `[-length, length - 1]`.

Text

Display text.

It consists of two sources combined to produce the final text: value and spans.

Example:

```
ft.Text("Hello from Flet!", size=24, weight=ft.FontWeight.W_600)
```

Image: Text - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/text/image_for_docs.png

Text

Inherits: LayoutControl

Properties

- bgcolor - The text's background color.
- color - The text's foreground color.
- enable_interactive_selection - Whether to enable user interface affordances for changing the text selection.
- font_family - System or custom font family to render text with.
- font_family_fallback - Ordered fallback font families used when a glyph is unavailable in font_family.
- italic - Whether to use italic typeface.
- max_lines - An optional maximum number of lines for the text to span, wrapping if necessary.
- no_wrap - If False (default) the text should break at soft line breaks.
- overflow - Defines how the text overflows.
- selectable - Whether the text should be selectable.
- selection_cursor_color - The color of the cursor.
- selection_cursor_height - Defines how tall the cursor should be.
- selection_cursor_width - Defines how thick the cursor should be.
- semantics_label - An alternative semantics label for this text.
- show_selection_cursor - Whether to show cursor (blinking caret) when the text is selected.
- size - Text size in virtual pixels.
- spans - The list of TextSpan objects to build a rich text paragraph.
- style - The text's style.
- text_align - Text horizontal align.
- theme_style - Pre-defined text style.
- value - The text displayed.
- weight - Font weight.

Events

- on_selection_change - Called when the user changes the selection of text (including the cursor location).
- on_tap - Called when the user taps on this selectable text.

Examples

Custom text styles

play_arrowTry Online

```
import flet as ft
```

```
def main(page: ft.Page):
```

```

page.title = "Text custom styles"
page.scroll = ft.ScrollMode.ADAPTIVE

page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                ft.Text("Size 10", size=10),
                ft.Text(
                    "Size 30, Italic",
                    size=30,
                    color=ft.Colors.PINK_600,
                    italic=True,
                ),
                ft.Text(
                    value="Size 40, w100",
                    size=40,
                    color=ft.Colors.WHITE,
                    bgcolor=ft.Colors.BLUE_600,
                    weight=ft.FontWeight.W_100,
                ),
                ft.Text(
                    value="Size 50, Normal",
                    size=50,
                    color=ft.Colors.WHITE,
                    bgcolor=ft.Colors.ORANGE_800,
                    weight=ft.FontWeight.NORMAL,
                ),
                ft.Text(
                    value="Size 60, Bold, Italic",
                    size=50,
                    color=ft.Colors.WHITE,
                    bgcolor=ft.Colors.GREEN_700,
                    weight=ft.FontWeight.BOLD,
                    italic=True,
                ),
                ft.Text(
                    value="Size 70, w900, selectable",
                    size=70,
                    weight=ft.FontWeight.W_900,
                    selectable=True,
                ),
                ft.Text(
                    value="Limit long text to 1 line with ellipsis",
                    theme_style=ft.TextThemeStyle.HEADLINE_SMALL,
                ),
                ft.Text(
                    value=(
                        "Proin rutrum, purus sit amet elementum volutpat, nunc "
                        "lacus vulputate orci, cursus ultrices neque dui quis "
                        "purus. Ut ultricies purus nec nibh bibendum, eget "
                        "vestibulum metus varius. Duis convallis maximus justo, "
                        "eu rutrum libero maximus id. Donec ullamcorper arcu in "
                        "sapien molestie, non pellentesque tellus pellentesque. "
                        "Nulla nec tristique ex. Maecenas euismod nisl enim, a "
                        "convallis arcu laoreet at. Ut at tortor finibus, rutrum "
                        "massa sit amet, pulvinar velit. Phasellus diam lorem, "
                        "viverra vitae leo vitae, consequat suscipit lorem."
                    ),
                    max_lines=1,
                    overflow=ft.TextOverflow.ELLIPSIS,
                ),
                ft.Text(
                    value="Limit long text to 2 lines and fading",
                    theme_style=ft.TextThemeStyle.HEADLINE_SMALL,
                ),
                ft.Text(
                    value=(
                        "Lorem ipsum dolor sit amet, consectetur adipiscing "
                        "elit. Curabitur quis nibh vitae purus consectetur "
                        "facilisis sed vitae ipsum. Quisque faucibus sed nulla "
                        "placerat sagittis. Phasellus condimentum risus vitae "
                        "nulla vestibulum auctor. Curabitur scelerisque, nibh "
                        "eget imperdiet consequat, odio ante tempus diam, sed "
                        "volutpat nisl erat eget turpis. Sed viverra, diam sit "
                        "amet blandit vulputate, mi tellus dapibus lorem, vitae "
                        "vehicula diam mauris placerat diam. Morbi sit amet "
                        "pretium turpis, et consequat ligula. Nulla velit sem, "
                        "suscipit sit amet dictum non, tincidunt sed nulla. "
                        "Aenean pellentesque odio porttitor sagittis aliquam. "
                        "Name varius at metus vitae vulputate. Praesent "
                        "faucibus nibh lorem, eu pretium dolor dictum nec. "
                        "Phasellus eget dui laoreet, viverra magna vitae, "
                        "pellentesque diam."
                    ),
                    max_lines=2,
                ),
            ],
        ),
    ),
)

```

```

ft.Text(
    value="Limit the width and height of long text",
    theme_style=ft.TextThemeStyle.HEADLINE_SMALL,
),
ft.Text(
    value=(
        "Lorem ipsum dolor sit amet, consectetur adipiscing "
        "elit. Curabitur quis nibh vitae purus consectetur "
        "facilisis sed vitae ipsum. Quisque faucibus sed nulla "
        "placerat sagittis. Phasellus condimentum risus vitae "
        "nulla vestibulum auctor. Curabitur scelerisque, nibh "
        "eget imperdiet consequat, odio ante tempus diam, sed "
        "volutpat nisl erat eget turpis. Sed viverra, diam sit "
        "amet blandit vulputate, mi tellus dapibus lorem, vitae "
        "vehicula diam mauris placerat diam. Morbi sit amet "
        "pretium turpis, et consequat ligula. Nulla velit sem, "
        "suscipit sit amet dictum non, tincidunt sed nulla. "
        "Aenean pellentesque odio porttitor sagittis aliquam. "
        "Name various at metus vitae vulputate. Praesent "
        "faucibus nibh lorem, eu pretium dolor dictum nec. "
        "Phasellus eget dui laoreet, viverra magna vitae, "
        "pellentesque diam."
    ),
    width=700,
    height=100,
),
],
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/examples/controls/core/text/media/custom_styles.gif

Theme text styles

play_arrowTry Online

import flet as ft

```

def main(page: ft.Page):
    page.title = "Text theme styles"
    page.scroll = ft.ScrollMode.ADAPTIVE

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        "Display Large",
                        theme_style=ft.TextThemeStyle.DISPLAY_LARGE,
                    ),
                    ft.Text(
                        "Display Medium",
                        theme_style=ft.TextThemeStyle.DISPLAY_MEDIUM,
                    ),
                    ft.Text(
                        "Display Small",
                        theme_style=ft.TextThemeStyle.DISPLAY_SMALL,
                    ),
                    ft.Text(
                        "Headline Large",
                        theme_style=ft.TextThemeStyle.HEADLINE_LARGE,
                    ),
                    ft.Text(
                        "Headline Medium",
                        theme_style=ft.TextThemeStyle.HEADLINE_MEDIUM,
                    ),
                    ft.Text(
                        "Headline Small",
                        theme_style=ft.TextThemeStyle.HEADLINE_SMALL,
                    ),
                    ft.Text("Title Large", theme_style=ft.TextThemeStyle.TITLE_LARGE),
                    ft.Text("Title Medium", theme_style=ft.TextThemeStyle.TITLE_MEDIUM),
                    ft.Text("Title Small", theme_style=ft.TextThemeStyle.TITLE_SMALL),
                    ft.Text("Label Large", theme_style=ft.TextThemeStyle.LABEL_LARGE),
                    ft.Text(
                        "Label Medium",
                        theme_style=ft.TextThemeStyle.LABEL_MEDIUM,
                    ),
                    ft.Text("Label Small", theme_style=ft.TextThemeStyle.LABEL_SMALL),
                    ft.Text("Body Large", theme_style=ft.TextThemeStyle.BODY_LARGE),
                    ft.Text("Body Medium", theme_style=ft.TextThemeStyle.BODY_MEDIUM),

```

```

        ft.Text("Body Small", theme_style=ft.TextThemeStyle.BODY_SMALL),
    ],
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/examples/controls/core/text/media/text_theme_styles.png

Variable font weight

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    page.fonts = {
        "RobotoSlab": "https://github.com/google/fonts/raw/main/apache/robotoslab/RobotoSlab%5Bwght%5D.ttf"
    }

    def handle_slider_change(e):
        text.weight = f"w{int(e.control.value)}" # noqa
        text.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    text := ft.Text(
                        "This is rendered with Roboto Slab",
                        size=30,
                        font_family="RobotoSlab",
                        weight=ft.FontWeight.W_100,
                    ),
                    ft.Slider(
                        min=100,
                        max=900,
                        divisions=8,
                        label="Weight = {value}",
                        width=500,
                        on_change=handle_slider_change,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/examples/controls/core/text/media/variable_font_weight.gif

Rich text

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text("Plain text with default style"),
                    ft.Text(
                        "Selectable plain text with default style",
                        selectable=True,
                    ),
                    ft.Text(
                        value="Some text",
                        selectable=True,
                        size=30,
                        spans=[
                            ft.TextSpan(
                                text="here goes italic",
                                style=ft.TextStyle(
                                    italic=True,
                                    size=20,
                                    color=ft.Colors.GREEN,
                                ),
                                spans=[

```

```

        ft.TextSpan(
            text="bold and italic",
            style=ft.TextStyle(weight=ft.FontWeight.BOLD),
        ),
        ft.TextSpan(
            text="just italic",
            spans=[
                ft.TextSpan(
                    "smaller italic",
                    ft.TextStyle(size=15),
                )
            ],
        ),
    ],
),
),
ft.Text(
    disabled=False,
    spans=[
        ft.TextSpan(
            text="underlined and clickable",
            style=ft.TextStyle(
                decoration=ft.TextDecoration.UNDERLINE
            ),
            on_click=lambda e: print(f"Clicked span: {e.control}"),
            on_enter=lambda e: print(f"Entered span: {e.control}"),
            on_exit=lambda e: print(f"Exited span: {e.control}"),
        ),
        ft.TextSpan(text=" "),
        ft.TextSpan(
            text="underlined red wavy",
            style=ft.TextStyle(
                decoration=ft.TextDecoration.UNDERLINE,
                decoration_color=ft.Colors.RED,
                decoration_style=ft.TextDecorationStyle.WAVY,
            ),
            on_enter=lambda e: print(f"Entered span: {e.control}"),
            on_exit=lambda e: print(f"Exited span: {e.control}"),
        ),
        ft.TextSpan(text=" "),
        ft.TextSpan(
            text="overlined blue",
            style=ft.TextStyle(
                decoration=ft.TextDecoration.OVERLINE,
                decoration_color="blue",
            ),
        ),
        ft.TextSpan(text=" "),
        ft.TextSpan(
            text="overlined and underlined",
            style=ft.TextStyle(
                decoration=ft.TextDecoration.OVERLINE
                | ft.TextDecoration.UNDERLINE
            ),
        ),
        ft.TextSpan(text=" "),
        ft.TextSpan(
            text="line through thick",
            style=ft.TextStyle(
                decoration=ft.TextDecoration.LINE_THROUGH,
                decoration_thickness=3,
            ),
        ),
    ],
),
),
),
),
)

def handle_link_highlight(e: ft.Event[ft.TextSpan]):
    e.control.style.color = ft.Colors.BLUE
    e.control.update()

def handle_link_unhighlight(e: ft.Event[ft.TextSpan]):
    e.control.style.color = None
    e.control.update()

page.add(
    ft.Text(
        disabled=False,
        spans=[
            ft.TextSpan(
                text="Go to Google",
                style=ft.TextStyle(decoration=ft.TextDecoration.UNDERLINE),
                url="https://google.com",
            ),
        ],
    ),
)

```

```

        on_enter=handle_link_highlight,
        on_exit=handle_link_unhighlight,
    ),
],
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/examples/controls/core/text/media/rich_text_basic.png

Rich text border stroke

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Stack(
                controls=[
                    ft.Text(
                        spans=[
                            ft.TextSpan(
                                text="Greetings, planet!",
                                style=ft.TextStyle(
                                    size=40,
                                    weight=ft.FontWeight.BOLD,
                                    foreground=ft.Paint(
                                        color=ft.Colors.BLUE_700,
                                        stroke_width=6,
                                        style=ft.PaintingStyle.STROKE,
                                    ),
                                ),
                            ),
                        ],
                    ),
                    ft.Text(
                        spans=[
                            ft.TextSpan(
                                text="Greetings, planet!",
                                style=ft.TextStyle(
                                    size=40,
                                    weight=ft.FontWeight.BOLD,
                                    color=ft.Colors.GREY_300,
                                ),
                            ),
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/examples/controls/core/text/media/rich_text_border_stroke.png

Rich text gradient

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Text(
                spans=[
                    ft.TextSpan(
                        text="Greetings, planet!",
                        style=ft.TextStyle(
                            size=40,
                            weight=ft.FontWeight.BOLD,
                            foreground=ft.Paint(
                                gradient=ft.PaintLinearGradient(
                                    begin=(0, 20),
                                    end=(150, 20),
                                    colors=[ft.Colors.RED, ft.Colors.YELLOW],
                                ),
                            ),
                        ),
                    ),
                ],
            ),
        ),
    )

```

```
if __name__ == "__main__":
    ft.run(main)
```

Properties

bgcolor: ColorValue | None = None

```
color: ColorValue | None = None
```

```
enable_interactive_selection: bool = True
```

```
font_family: str | None = None
```

font_family_fallback: list[str] | None = None

```
italic: bool = False
```

```
max_lines: int | None = None
```

```
no_wrap: bool | None = None
```

```
overflow: TextOverflow = TextOverflow.CLIP
```

Defines how the text overflows.

buildselectableclass-attributeinstance-attribute

```
selectable: bool | None = None
```

Whether the text should be selectable.

Defaults to False.

buildselection_cursor_colorclass-attributeinstance-attribute

```
selection_cursor_color: ColorValue | None = None
```

The color of the cursor.

The cursor indicates the current text insertion point.

buildselection_cursor_heightclass-attributeinstance-attribute

```
selection_cursor_height: Number | None = None
```

Defines how tall the cursor should be.

buildselection_cursor_widthclass-attributeinstance-attribute

```
selection_cursor_width: Number = 2.0
```

Defines how thick the cursor should be.

The cursor will be drawn under the text. The cursor width will extend to the right of the boundary between characters for left-to-right text and to the left for right-to-left text. This corresponds to extending downstream relative to the selected position. Negative values may be used to reverse this behavior.

Note

Has effect only when selectable is True.

buildsemantics_labelclass-attributeinstance-attribute

```
semantics_label: str | None = None
```

An alternative semantics label for this text.

If present, the semantics of this control will contain this value instead of the actual text. This will overwrite any of the `TextSpan.semantics_labels`.

This is useful for replacing abbreviations or shorthands with the full text value:

```
ft.Text("$$", semantics_label="Double dollars")
```

buildshow_selection_cursorclass-attributeinstance-attribute

```
show_selection_cursor: bool = False
```

Whether to show cursor (blinking caret) when the text is selected.

Note

Has effect only when selectable is True.

buildsizeclass-attributeinstance-attribute

```
size: Number | None = None
```

Text size in virtual pixels.

Defaults to 14.

buildspansclass-attributeinstance-attribute

```
spans: list[TextSpan] | None = None
```

The list of `TextSpan` objects to build a rich text paragraph.

buildstyleclass-attributeinstance-attribute

```
style: TextStyle | None = None
```

The text's style.

buildtext_alignclass-attributeinstance-attribute

```
text_align: TextAlign = TextAlign.START
```


Text horizontal align.

buildtheme_styleclass-attributeinstance-attribute

```
theme_style: TextThemeStyle | None = None
```

Pre-defined text style.

buildvalueclass-attributeinstance-attribute

```
value: str = ''
```

The text displayed.

buildweightclass-attributeinstance-attribute

```
weight: FontWeight | None = None
```

Font weight.

Defaults to `FontWeight.NORMAL`.

Events

bolton_selection_changeclass-attributeinstance-attribute

```
on_selection_change: (
    EventHandler[TextSelectionChangeEvent[Text]] | None
) = None
```

Called when the user changes the selection of text (including the cursor location).

Note

Has effect only when `selectable` is `True`.

bolton_tapclass-attributeinstance-attribute

```
on_tap: ControlEventHandler[Text] | None = None
```

Called when the user taps on this selectable text.

Note

Has effect only when `selectable` is `True`.

TextButton

Text buttons are used for the lowest priority actions, especially when presenting multiple options. Text buttons can be placed on a variety of backgrounds. Until the button is interacted with, its container isn't visible.

```
ft.TextButton(  
    content="Text Button",  
    icon=ft.Icons.STAR_BORDER,  
    icon_color=ft.Colors.BLUE_300,  
)
```

Image: TextButton - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/text_button/image_for_docs.png

Simple text button

Inherits: LayoutControl, AdaptiveControl

Properties

- autofocus - True if the control will be selected as the initial focus.
- clip_behavior - Defines how the content of this button is clipped.
- content - A Control representing custom button content.
- icon - An icon to show in this button.
- icon_color - Icon color.
- style - Defines the style of this button.
- url - The URL to open when this button is clicked.

Events

- on_blur - Called when this button has lost focus.
- on_click - Called when a user clicks this button.
- on_focus - Called when this button has received focus.
- on_hover - Called when a mouse pointer enters or exists this button's response area.
- on_long_press - Called when this button is long-pressed.

Methods

- focus - Requests focus for this control.

Examples

TextButton

play_arrowTry Online

```
import flet as ft  
  
def main(page: ft.Page):  
    page.title = "Basic text buttons"  
  
    page.add(  
        ft.SafeArea(  
            content=ft.Column(  
                controls=[  
                    ft.TextButton(content="Text button"),  
                    ft.TextButton(content="Disabled button", disabled=True),  
                ],  
            ),  
        ),  
    )  
  
if __name__ == "__main__":  
    ft.run(main)
```

Image: basic - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/text_button/basic.png

TextButton icons

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "TextButtons with icons"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.TextButton(
                        content="Button with icon",
                        icon=ft.Icons.WAVES_OUTLINED,
                    ),
                    ft.TextButton(
                        content="Button with colorful icon",
                        icon=ft.Icons.PARK_ROUNDED,
                        icon_color=ft.Colors.GREEN_400,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: icons - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/text_button/icons.png

TextButton handling clicks

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "TextButton with 'click' event"

    def button_clicked(e):
        button.data += 1
        message_text.value = f"Button clicked {button.data} time(s)"

    message_text = ft.Text()
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    button := ft.TextButton(
                        key="TextButton",
                        content="Button with 'click' event",
                        data=0,
                        on_click=button_clicked,
                    ),
                    ft.Container(
                        padding=ft.Padding(left=12),
                        content=message_text,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: handling-clicks - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/text_button/handling_clicks.png

TextButton custom content

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "TextButtons with custom content"

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.TextButton(
                        width=150,
```

Image: custom-content - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/text_button/custom_content.png

`on_blur: ControlEventHandler[TextButton] | None = None`

Called when this button has lost focus.

bolton_clickclass-attributeinstance-attribute

`on_click: ControlEventHandler[TextButton] | None = None`

Called when a user clicks this button.

bolton_focusclass-attributeinstance-attribute

`on_focus: ControlEventHandler[TextButton] | None = None`

Called when this button has received focus.

bolton_hoverclass-attributeinstance-attribute

`on_hover: ControlEventHandler[TextButton] | None = None`

Called when a mouse pointer enters or exists this button's response area.

The data property of the event handler argument is True when cursor enters and False when it exits.

bolton_long_pressclass-attributeinstance-attribute

`on_long_press: ControlEventHandler[TextButton] | None = None`

Called when this button is long-pressed.

Methods

deployed_codefocusasync

`focus()`

Requests focus for this control.

TextField

A text field lets the user enter text, either with hardware keyboard or with an onscreen keyboard.

Example:

```
ft.TextField(label="Name", hint_text="Jane Doe")
```

Image: TextField - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/textfield/image_for_docs.png

TextField

Inherits: FormFieldControl, AdaptiveControl

Properties

- `always_call_on_tap` - TBD
- `animate_cursor_opacity` - TBD
- `autocorrect` - Whether to enable autocorrection.
- `autofill_hints` - Helps the autofill service identify the type of this text input.
- `autofocus` - True if the control will be selected as the initial focus.
- `can_request_focus` - TBD
- `can_reveal_password` - Displays a toggle icon button that allows revealing the entered password.
- `capitalization` - Enables automatic on-the-fly capitalization of entered text.
- `clip_behavior` - TBD
- `cursor_color` - The color of TextField cursor.
- `cursor_error_color` - TBD
- `cursor_height` - Sets cursor height.
- `cursor_radius` - Sets cursor radius.
- `cursor_width` - Sets cursor width.
- `enable_ime_personalized_learning` - TBD
- `enable_interactive_selection` - TBD
- `enable_stylus_handwriting` - TBD
- `enable_suggestions` - Whether to show input suggestions as the user types.
- `ignore_pointers` - TBD
- `ignore_up_down_keys` - Whether to intercept Arrow-Up and Arrow-Down when this field has focus.
- `input_filter` - Provides as-you-type filtering/validation.
- `keyboard_brightness` - TBD
- `keyboard_type` - The type of keyboard to use for editing the text.
- `max_length` - Limits a maximum number of characters that can be entered into TextField.
- `max_lines` - The maximum number of lines to show at one time, wrapping if necessary.
- `min_lines` - The minimum number of lines to occupy when the content spans fewer lines.
- `mouse_cursor` - TBD
- `multiline` - Whether this field can contain multiple lines of text.
- `obscuring_character` - TBD
- `password` - Whether to hide the text being edited.
- `read_only` - Whether the text can be changed.
- `scroll_padding` - TBD

- selection - Represents the current text selection or caret position in the field.
- selection_color - The color of TextField selection.
- shift_enter - Changes the behavior of Enter button in multiline textfield to be chat-like, i.e.
- show_cursor - Whether the field's cursor is to be shown.
- smart_dashes_type - Whether to allow the platform to automatically format dashes.
- smart_quotes_type - Whether to allow the platform to automatically format quotes.
- strut_style - TBD
- text_align - How the text should be aligned horizontally.
- value - Current value of this text field.

Events

- on_blur - Called when the control has lost focus.
- on_change - Called when the typed input for the TextField has changed.
- on_click - TBD
- on_focus - Called when the control has received focus.
- on_selection_change - Called when the text selection or caret position changes.
- on_submit - Called when user presses ENTER while focus is on TextField.
- on_tap_outside - TBD

Examples

TextField

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    def handle_button_click(e: ft.Event[ft.Button]):
        message.value = (
            f"Textboxes values are: '{tb1.value}', '{tb2.value}', "
            f"'{tb3.value}', '{tb4.value}', '{tb5.value}'."
        )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    tb1 := ft.TextField(label="Standard"),
                    tb2 := ft.TextField(
                        label="Disabled",
                        disabled=True,
                        value="First name",
                    ),
                    tb3 := ft.TextField(
                        label="Read-only",
                        read_only=True,
                        value="Last name",
                    ),
                    tb4 := ft.TextField(
                        label="With placeholder",
                        hint_text="Please enter text here",
                    ),
                    tb5 := ft.TextField(
                        label="With an icon",
                        icon=ft.Icons.EMOJI_EMOTIONS,
                    ),
                    ft.Button(content="Submit", on_click=handle_button_click),
                    message := ft.Text(),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Handling change events

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    def handle_field_change(e: ft.Event[ft.TextField]):
        message.value = e.control.value

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.TextField(
                        key="handling_change_textfield",
                        label="Textbox with 'change' event:",
                        on_change=handle_field_change,
                    ),
                    message := ft.Text(),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: handling-change-events - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/textfield/handling_change_events.gif

Handling selection changes

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "Text selection"

    def handle_selection_change(e: ft.TextSelectionChangeEvent[ft.TextField]):
        selection.value = (
            f"Selection: '{e.selected_text}'" if e.selected_text else "No selection."
        )
        selection_details.value = f"start={e.selection.start}, end={e.selection.end}"
        caret.value = f"Caret position: {e.selection.end}"

    async def select_characters(e: ft.Event[ft.Button]):
        await field.focus()
        field.selection = ft.TextSelection(
            base_offset=0, extent_offset=len(field.value)
        )

    async def move_caret(e: ft.Event[ft.Button]):
        await field.focus()
        field.selection = ft.TextSelection(base_offset=0, extent_offset=0)

    page.add(
        ft.SafeArea(
            content=ft.Column(
                spacing=10,
                controls=[
                    field := ft.TextField(
                        value=(
                            "Lorem ipsum dolor sit amet, consectetur adipiscing elit."
                        ),
                        multiline=True,
                        min_lines=3,
                        autofocus=True,
                        on_selection_change=handle_selection_change,
                    ),
                    selection := ft.Text("Select some text from the field."),
                    selection_details := ft.Text(),
                    caret := ft.Text("Caret position: -"),
                    ft.Button(
                        content="Select all text",
                        on_click=select_characters,
                    ),
                    ft.Button(
                        content="Move caret to start",
                        on_click=move_caret,
                    ),
                ],
            ),
        ),
    )
```



```
    ),
  )
)
```

```
if __name__ == "__main__":
    ft.run(main)
```

Image: selection-change - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/textfield/selection_change.gif

Password with reveal button

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.TextField(
                key="password_textfield",
                label="Password with reveal button",
                password=True,
                can_reveal_password=True,
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: password - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/textfield/password.gif>

Multiline fields

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.TextField(
                        key="multiline_standard",
                        label="Standard",
                        multiline=True,
                    ),
                    ft.TextField(
                        label="Disabled",
                        multiline=True,
                        disabled=True,
                        value="line1\nline2\nline3\nline4\nline5",
                    ),
                    ft.TextField(
                        key="multiline_auto_height",
                        label="Auto adjusted height with max lines",
                        multiline=True,
                        min_lines=1,
                        max_lines=3,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: multiline - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/textfield/multiline.gif>

Underlined and borderless TextFields

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
```

```

        ft.TextField(
            key="underlined_field",
            label="Underlined",
            border=ft.InputBorder.UNDERLINE,
            hint_text="Enter text here",
        ),
        ft.TextField(
            key="underlined_filled_field",
            label="Underlined filled",
            border=ft.InputBorder.UNDERLINE,
            filled=True,
            hint_text="Enter text here",
        ),
        ft.TextField(
            key="borderless_field",
            label="Borderless",
            border=ft.InputBorder.NONE,
            hint_text="Enter text here",
        ),
        ft.TextField(
            key="borderless_filled_field",
            label="Borderless filled",
            border=ft.InputBorder.NONE,
            filled=True,
            hint_text="Enter text here",
        ),
    ],
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: underlined-and-borderless - https://flet.dev/docs/test-images/examples/controls/material/golden/macOS/textfield/underlined_and_borderless.gif

Setting prefixes and suffixes

[play_arrowTry Online](#)

```

import flet as ft

def main(page: ft.Page):
    def handle_button_click(e: ft.Event[ft.Button]):
        message.value = (
            "Textboxes values are: "
            f"'{prefix_field.value}', "
            f"'{suffix_field.value}', "
            f"'{prefix_suffix_field.value}', "
            f"'{color_field.value}'."
        )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    prefix_field := ft.TextField(
                        key="prefix_field",
                        label="With prefix",
                        prefix="https://",
                    ),
                    suffix_field := ft.TextField(
                        key="suffix_field",
                        label="With suffix",
                        suffix=".com",
                    ),
                    prefix_suffix_field := ft.TextField(
                        key="prefix_suffix_field",
                        label="With prefix and suffix",
                        prefix="https://",
                        suffix=".com",
                        enable_interactive_selection=True,
                    ),
                    color_field := ft.TextField(
                        key="color_field",
                        label="My favorite color",
                        icon=ft.Icons.FORMAT_SIZE,
                        hint_text="Type your favorite color",
                        helper="You can type only one color",
                        counter="{value_length}/{max_length} chars used",
                        prefix_icon=ft.Icons.COLOR_LENS,
                        suffix="...is your color",
                        max_length=20,
                    ),
                ],
            ),
            ft.Button(

```

```

        key="submit_button",
        content="Submit",
        on_click=handle_button_click,
    ),
    message := ft.Text(),
],
),
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: prefix-and-suffix - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/textfield/prefix_and_suffix.gif

Styled TextField

play_arrowTry Online

```
import flet as ft
```

```

async def main(page: ft.Page):
    page.padding = 50

    tf = ft.TextField(
        key="styled_textfield",
        text_size=30,
        cursor_color=ft.Colors.RED,
        selection_color=ft.Colors.YELLOW,
        color=ft.Colors.PINK,
        bgcolor=ft.Colors.BLACK_26,
        filled=True,
        focused_color=ft.Colors.GREEN,
        focused_bgcolor=ft.Colors.CYAN_200,
        border_radius=30,
        border_color=ft.Colors.GREEN_800,
        focused_border_color=ft.Colors.GREEN_ACCENT_400,
        max_length=20,
        capitalization=ft.TextCapitalization.CHARACTERS,
    )

    page.add(
        ft.SafeArea(
            content=tf,
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: styled - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/textfield/styled.gif>

Custom label, hint, helper, and counter texts and styles

play_arrowTry Online

```
import flet as ft
```

```

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT

    def handle_field_change(e: ft.Event[ft.TextField]):
        message.value = e.control.value

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.TextField(
                        key="label_hint_helper_counter_textfield",
                        on_change=handle_field_change,
                        text_style=ft.TextStyle(
                            size=15,
                            italic=True,
                            color=ft.Colors.DEEP_ORANGE_600,
                            bgcolor=ft.Colors.LIME_ACCENT_200,
                        ),
                        label="Label",
                        label_style=ft.TextStyle(
                            size=17,
                            weight=ft.FontWeight.BOLD,
                            italic=True,
                            color=ft.Colors.BLUE,
                        ),
                    ),
                ],
            ),
        )
    )

```

```

        bgcolor=ft.Colors.RED_700,
    ),
    hint_text="Hint",
    hint_style=ft.TextStyle(
        size=15,
        weight=ft.FontWeight.BOLD,
        italic=True,
        color=ft.Colors.PINK_ACCENT,
        bgcolor=ft.Colors.BROWN_400,
    ),
    helper="Helper",
    helper_style=ft.TextStyle(
        size=14,
        weight=ft.FontWeight.BOLD,
        color=ft.Colors.DEEP_PURPLE,
        bgcolor=ft.Colors.BLUE_50,
    ),
    counter="Counter",
    counter_style=ft.TextStyle(
        size=14,
        italic=True,
        color=ft.Colors.YELLOW,
        bgcolor=ft.Colors.GREEN_500,
    ),
),
message := ft.Text(),
],
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: label-hint-helper-counter - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/textfield/label_hint_helper_counter.gif

Properties

buildalways_call_on_tapclass-attributeinstance-attribute

always_call_on_tap: bool = False

TBD

buildanimate_cursor_opacityclass-attributeinstance-attribute

animate_cursor_opacity: bool | None = None

TBD

buildautocorrectclass-attributeinstance-attribute

autocorrect: bool = True

Whether to enable autocorrection.

buildautofill_hintsclass-attributeinstance-attribute

autofill_hints: AutofillHint | list[AutofillHint] | None = (None)

Helps the autofill service identify the type of this text input.

More information here.

buildautofocusclass-attributeinstance-attribute

autofocus: bool = False

True if the control will be selected as the initial focus. If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

buildcan_request_focusclass-attributeinstance-attribute

can_request_focus: bool = True

TBD

buildcan_reveal_passwordclass-attributeinstance-attribute

can_reveal_password: bool = False

Displays a toggle icon button that allows revealing the entered password. Is shown if both password and `can_reveal_password` are `True`.

The icon is displayed in the same location as suffix and in case both `can_reveal_password/password` and suffix are provided, then the suffix won't be shown.

buildcapitalizationclass-attributeinstance-attribute

`capitalization: TextCapitalization | None = None`

Enables automatic on-the-fly capitalization of entered text.

Defaults to `TextCapitalization.NONE`.

buildclip_behaviorclass-attributeinstance-attribute

`clip_behavior: ClipBehavior = ClipBehavior.HARD_EDGE`

TBD

buildcursor_colorclass-attributeinstance-attribute

`cursor_color: ColorValue | None = None`

The color of `TextField` cursor.

buildcursor_error_colorclass-attributeinstance-attribute

`cursor_error_color: ColorValue | None = None`

TBD

buildcursor_heightclass-attributeinstance-attribute

`cursor_height: Number | None = None`

Sets cursor height.

buildcursor_radiusclass-attributeinstance-attribute

`cursor_radius: Number | None = None`

Sets cursor radius.

buildcursor_widthclass-attributeinstance-attribute

`cursor_width: Number = 2.0`

Sets cursor width.

buildenable_ime_personalized_learningclass-attributeinstance-attribute

`enable_ime_personalized_learning: bool = True`

TBD

buildenable_interactive_selectionclass-attributeinstance-attribute

`enable_interactive_selection: bool = True`

TBD

buildenable_stylus_handwritingclass-attributeinstance-attribute

`enable_stylus_handwriting: bool = True`

TBD

buildenable_suggestionsclass-attributeinstance-attribute

`enable_suggestions: bool = True`

Whether to show input suggestions as the user types.

This flag only affects Android. On iOS, suggestions are tied directly to autocorrect, so that suggestions are only shown when autocorrect is `True`. On Android autocorrection and suggestion are controlled separately.

buildignore_pointersclass-attributeinstance-attribute

`ignore_pointers: bool = False`

TBD

buildignore_up_down_keysclass-attributeinstance-attribute

```
ignore_up_down_keys: bool = False
```

Whether to intercept Arrow-Up and Arrow-Down when this field has focus.

When set to True, pressing those keys does not move the caret to the beginning or end of the text respectively.

buildinput_filterclass-attributeinstance-attribute

```
input_filter: InputFilter | None = None
```

Provides as-you-type filtering/validation.

Similar to the `on_change` callback, the input filters are not applied when the content of the field is changed programmatically.

buildkeyboard_brightnessclass-attributeinstance-attribute

```
keyboard_brightness: Brightness | None = None
```

TBD

buildkeyboard_typeclass-attributeinstance-attribute

```
keyboard_type: KeyboardType = KeyboardType.TEXT
```

The type of keyboard to use for editing the text.

buildmax_lengthclass-attributeinstance-attribute

```
max_length: int | None = None
```

Limits a maximum number of characters that can be entered into TextField.

Raises:

- `ValueError` - If it is not strictly greater than 0 or equal to -1.

buildmax_linesclass-attributeinstance-attribute

```
max_lines: int | None = None
```

The maximum number of lines to show at one time, wrapping if necessary.

This affects the height of the field itself and does not limit the number of lines that can be entered into the field.

If this is 1 (the default), the text will not wrap, but will scroll horizontally instead.

Raises:

- `ValueError` - If it is not strictly greater than 0.
- `ValueError` - If it is not greater than or equal to `min_lines` when both are set.

buildmin_linesclass-attributeinstance-attribute

```
min_lines: int | None = None
```

The minimum number of lines to occupy when the content spans fewer lines.

This affects the height of the field itself and does not limit the number of lines that can be entered into the field.

Defaults to 1.

Raises:

- `ValueError` - If it is not strictly greater than 0.
- `ValueError` - If it is not less than or equal to `max_lines` when both are set.

buildmouse_cursorclass-attributeinstance-attribute

```
mouse_cursor: MouseCursor | None = None
```

TBD

buildmultilineclass-attributeinstance-attribute

```
multiline: bool = False
```

Whether this field can contain multiple lines of text.

buildobscuring_characterclass-attributeinstance-attribute

```
obscuring_character: str = '•'
```

TBD

buildpasswordclass-attributeinstance-attribute

```
password: bool = False
```

Whether to hide the text being edited.

buildread_onlyclass-attributeinstance-attribute

```
read_only: bool = False
```

Whether the text can be changed.

When this is set to True, the text cannot be modified by any shortcut or keyboard operation. The text is still selectable.

buildscroll_paddingclass-attributeinstance-attribute

```
scroll_padding: PaddingValue = 20
```

TBD

buildselectionclass-attributeinstance-attribute

```
selection: TextSelection | None = None
```

Represents the current text selection or caret position in the field.

When the user selects text, this property is updated to reflect the selected range. If no text is selected, it contains an empty range indicating the caret position.

Setting this property visually updates the field's selection to match the given value, and hence leads to the `on_selection_change` event being triggered. To ensure the selection is visible and the event is fired, the text field must be focused. Call `focus` on the field before setting this property.

buildselection_colorclass-attributeinstance-attribute

```
selection_color: ColorValue | None = None
```

The color of TextField selection.

buildshift_enterclass-attributeinstance-attribute

```
shift_enter: bool = False
```

Changes the behavior of Enter button in multiline textfield to be chat-like, i.e. new line can be added with Shift+Enter and pressing just Enter fires `on_submit` event.

buildshow_cursorclass-attributeinstance-attribute

```
show_cursor: bool = True
```

Whether the field's cursor is to be shown.

buildsmart_dashes_typeclass-attributeinstance-attribute

```
smart_dashes_type: bool = True
```

Whether to allow the platform to automatically format dashes.

This flag only affects iOS versions 11 and above. As an example of what this does, two consecutive hyphen characters will be automatically replaced with one en dash, and three consecutive hyphens will become one em dash.

buildsmart_quotes_typeclass-attributeinstance-attribute

```
smart_quotes_type: bool = True
```

Whether to allow the platform to automatically format quotes.

This flag only affects iOS. As an example of what this does, a standard vertical double quote character will be automatically replaced by a left or right double quote depending on its position in a word.

buildstrut_styleclass-attributeinstance-attribute

```
strut_style: StrutStyle | None = None
```

TBD

buildtext_alignclass-attributeinstance-attribute

```
text_align: TextAlign | None = None
```

How the text should be aligned horizontally.

Defaults to TextAlign.LEFT.

buildvalueclass-attributeinstance-attribute

```
value: str = ''
```

Current value of this text field.

Events

bolton_blurclass-attributeinstance-attribute

```
on_blur: ControlEventHandler[TextField] | None = None
```

Called when the control has lost focus.

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[TextField] | None = None
```

Called when the typed input for the TextField has changed.

bolton_clickclass-attributeinstance-attribute

```
on_click: ControlEventHandler[TextField] | None = None
```

TBD

bolton_focusclass-attributeinstance-attribute

```
on_focus: ControlEventHandler[TextField] | None = None
```

Called when the control has received focus.

bolton_selection_changeclass-attributeinstance-attribute

```
on_selection_change: (
    EventHandler[TextSelectionChangeEvent[TextField]] | None
) = None
```

Called when the text selection or caret position changes.

This can be triggered either by user interaction (selecting text or moving the caret) or programmatically (through the selection property).

bolton_submitclass-attributeinstance-attribute

```
on_submit: ControlEventHandler[TextField] | None = None
```

Called when user presses ENTER while focus is on TextField.

bolton_tap_outsideclass-attributeinstance-attribute

```
on_tap_outside: ControlEventHandler[TextField] | None = None
```

TBD

TimePicker

A Material-style time picker dialog.

Can be opened by calling the `Page.show_dialog` method.

Depending on the `entry_mode`, it will show either a Dial or an Input (hour and minute text fields) for picking a time.

Example:

```
ft.TimePicker(  
    open=True,  
    value=time(1, 2),  
    entry_mode=ft.TimePickerEntryMode.INPUT_ONLY,  
)
```

Image: TimePicker - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/time_picker/image_for_docs.png

Time picker

Inherits: `DialogControl`

Properties

- `barrier_color` - The color of the modal barrier that darkens everything below this picker's dialog.
- `cancel_text` - The text that is displayed on the cancel button.
- `confirm_text` - The text that is displayed on the confirm button.
- `entry_mode` - The initial mode of time entry method for this picker.
- `error_invalid_text` - The error message displayed below the input text field if the input is not a valid hour/minute.
- `help_text` - The text that is displayed at the top of the header.
- `hour_format` - Defines the hour format of this time picker.
- `hour_label_text` - The text that is displayed below the hour input text field.
- `locale` - The locale for this time picker dialog.
- `minute_label_text` - The text that is displayed below the minute input text field.
- `modal` - Whether this picker cannot be dismissed by clicking the area outside of it.
- `orientation` - The orientation of the dialog when displayed.
- `switch_to_input_icon` - The icon displayed in the corner of this picker's dialog when `entry_mode` is `TimePickerEntryMode.DIAL`.
- `switch_to_timer_icon` - The icon displayed in the corner of this picker's dialog when `entry_mode` is `TimePickerEntryMode.INPUT`.
- `value` - The selected time that this picker should display.

Events

- `on_change` - Called when user clicks confirm button.
- `on_entry_mode_change` - Called when the `entry_mode` is changed through the time picker dialog.

Examples

TimePicker

`play_arrow` Try Online

```
from datetime import time  
  
import flet as ft  
  
def main(page: ft.Page):  
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER  
  
    def handle_change(e: ft.Event[ft.TimePicker]):  
        selection.value = f"Selection: {time_picker.value}"
```

```

        page.show_dialog(ft.SnackBar(f"TimePicker change: {time_picker.value}"))

def handle_dismissal(e: ft.Event[ft.DialogControl]):
    page.show_dialog(ft.SnackBar("TimePicker dismissed!"))

def handle_entry_mode_change(e: ft.TimePickerEntryModeChangeEvent):
    page.show_dialog(ft.SnackBar(f"Entry mode changed: {time_picker.entry_mode}"))

time_picker = ft.TimePicker(
    value=time(hour=19, minute=30),
    confirm_text="Confirm",
    error_invalid_text="Time out of range",
    help_text="Pick your time slot",
    entry_mode=ft.TimePickerEntryMode.DIAL,
    on_change=handle_change,
    on_dismiss=handle_dismissal,
    on_entry_mode_change=handle_entry_mode_change,
)

page.add(
    ft.SafeArea(
        content=ft.Column(
            horizontal_alignment=ft.CrossAxisAlignment.CENTER,
            controls=[
                ft.Button(
                    key="pick_time_button",
                    content="Pick time",
                    icon=ft.Icons.TIME_TO_LEAVE,
                    on_click=lambda: page.show_dialog(time_picker),
                ),
                selection := ft.Text(weight=ft.FontWeight.BOLD),
            ],
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/time_picker/basic.gif

Hour formats

play_arrowTry Online

```

from datetime import time

import flet as ft

```

```

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def get_system_hour_format():
        """Returns the current system's hour format."""
        return "24h" if page.media.always_use_24_hour_format else "12h"

    def format_time(value: time) -> str:
        """Returns a formatted time string based on TimePicker and system settings."""
        use_24h = time_picker.hour_format == ft.TimePickerHourFormat.H24 or (
            time_picker.hour_format == ft.TimePickerHourFormat.SYSTEM
            and page.media.always_use_24_hour_format
        )
        return value.strftime("%H:%M" if use_24h else "%I:%M %p")

    def handle_change(e: ft.Event[ft.TimePicker]):
        selection.value = f"Selection: {format_time(time_picker.value)}"

    time_picker = ft.TimePicker(
        value=time(hour=19, minute=30),
        help_text="Choose your meeting time",
        on_change=handle_change,
    )

    def open_picker(e: ft.Event[ft.Button]):
        choice = format_dropdown.value
        hour_format_map = {
            "system": ft.TimePickerHourFormat.SYSTEM,
            "12h": ft.TimePickerHourFormat.H12,
            "24h": ft.TimePickerHourFormat.H24,
        }
        time_picker.hour_format = hour_format_map[choice]
        page.show_dialog(time_picker)

    page.add(
        ft.SafeArea(

```

```

        content=ft.Column(
            horizontal_alignment=ft.CrossAxisAlignment.CENTER,
            controls=[
                ft.Row(
                    alignment=ft.MainAxisAlignment.CENTER,
                    controls=[
                        format_dropdown := ft.Dropdown(
                            label="Hour format",
                            value="system",
                            width=260,
                            key="dd",
                            options=[
                                ft.DropdownOption(
                                    key="system",
                                    text=(
                                        f"System default "
                                        f"({get_system_hour_format()})"
                                    ),
                                ),
                                ft.DropdownOption(key="12h", text="12-hour clock"),
                                ft.DropdownOption(key="24h", text="24-hour clock"),
                            ],
                        ),
                        ft.Button(
                            "Open TimePicker",
                            icon=ft.Icons.SCHEDULE,
                            on_click=open_picker,
                        ),
                    ],
                ),
                selection := ft.Text(weight=ft.FontWeight.BOLD),
            ],
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: [Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/time_picker/hour_formats.gif](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/time_picker/hour_formats.gif)

Custom locale

play_arrowTry Online

```

from datetime import time

import flet as ft

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    page.add(
        ft.SafeArea(
            content=ft.Button(
                key="custom_locale_button",
                content="Pick time (zh_Hans locale)",
                icon=ft.Icons.CALENDAR_MONTH,
                on_click=lambda e: page.show_dialog(
                    ft.TimePicker(
                        value=time(hour=19, minute=30),
                        locale=ft.Locale("zh", "Hans"),
                    )
                ),
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: [Image - https://flet.dev/docs/test-images/examples/controls/material/golden/macros/time_picker/custom_locale.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/time_picker/custom_locale.png)

Properties

buildbarrier_colorclass-attributeinstance-attribute

barrier_color: ColorValue | None = None

The color of the modal barrier that darkens everything below this picker's dialog.

If None, the DialogTheme.barrier_color is used. If it is also None, then Colors.BLACK_54 is used.

buildcancel_textclass-attributeinstance-attribute

`cancel_text: str | None = None`

The text that is displayed on the cancel button.

The default value is "Cancel".

buildconfirm_textclass-attributeinstance-attribute

`confirm_text: str | None = None`

The text that is displayed on the confirm button.

The default value is "OK".

buildentry_modeclass-attributeinstance-attribute

`entry_mode: TimePickerEntryMode = TimePickerEntryMode.DIAL`

The initial mode of time entry method for this picker.

Defaults to TimePickerEntryMode.DIAL.

builderror_invalid_textclass-attributeinstance-attribute

`error_invalid_text: str | None = None`

The error message displayed below the input text field if the input is not a valid hour/minute.

The default value is "Enter a valid time".

buildhelp_textclass-attributeinstance-attribute

`help_text: str | None = None`

The text that is displayed at the top of the header.

This is used to indicate to the user what they are selecting a time for. The default value is "Enter time".

buildhour_formatclass-attributeinstance-attribute

```
hour_format: TimePickerHourFormat = (  
    TimePickerHourFormat.SYSTEM  
)
```

Defines the hour format of this time picker.

buildhour_label_textclass-attributeinstance-attribute

`hour_label_text: str | None = None`

The text that is displayed below the hour input text field.

The default value is "Hour".

buildlocaleclass-attributeinstance-attribute

`locale: Locale | None = None`

The locale for this time picker dialog. It is intended for (rare) cases where this dialog should be localized differently from the rest of the page.

It overrides the locale used by the page (see Page.locale_configuration), but does not participate in page-level locale resolution.

If set to None (the default) or an inexistent/unsupported locale, the current_locale of the Page.locale_configuration is used as fallback.

buildminute_label_textclass-attributeinstance-attribute

`minute_label_text: str | None = None`

The text that is displayed below the minute input text field.

The default value is "Minute".

buildmodalclass-attributeinstance-attribute

`modal: bool = False`

Whether this picker cannot be dismissed by clicking the area outside of it.

buildorientationclass-attributeinstance-attribute

```
orientation: Orientation | None = None
```

The orientation of the dialog when displayed.

buildswitch_to_input_iconclass-attributeinstance-attribute

```
switch_to_input_icon: IconData | None = None
```

The icon displayed in the corner of this picker's dialog when entry_mode is TimePickerEntryMode.DIAL.

Clicking on icon changes the entry_mode to TimePickerEntryMode.INPUT.

If None, defaults to Icons.KEYBOARD_OUTLINED.

buildswitch_to_timer_iconclass-attributeinstance-attribute

```
switch_to_timer_icon: IconData | None = None
```

The icon displayed in the corner of this picker's dialog when entry_mode is TimePickerEntryMode.INPUT.

Clicking on this icon changes the entry_mode to TimePickerEntryMode.DIAL.

If None, defaults to Icons.ACCESS_TIME.

buildvalueclass-attributeinstance-attribute

```
value: time | None = field(
    default_factory=lambda: datetime.now().time()
)
```

The selected time that this picker should display.

The default value is equal to the current time.

Events

bolton_changeclass-attributeinstance-attribute

```
on_change: ControlEventHandler[TimePicker] | None = None
```

Called when user clicks confirm button.

value property is updated with selected time. Additionally, the data property of the event handler argument also contains the selected time.

bolton_entry_mode_changeclass-attributeinstance-attribute

```
on_entry_mode_change: (
    EventHandler[TimePickerEntryModeChangeEvent] | None
) = None
```

Called when the entry_mode is changed through the time picker dialog.

TransparentPointer

TransparentPointer is the solution to "How to pass through all gestures between two widgets in Stack" problem.

For example, if there is an Button inside Container with GestureDetector then tapping on a button won't be "visible" to a gesture detector behind it. With TransparentPointer a tapping event doesn't stop on a button, but goes up to the parent, similar to event bubbling in HTML/JS.

Inherits: LayoutControl

Properties

- content - The Control that should be displayed.

Example

Basic Example

TransparentPointer

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    def button_clicked(e):
        print("transparent pointer button clicked")

    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Stack(
                expand=True,
                controls=[
                    ft.GestureDetector(
                        on_tap=lambda _: print("TAP!"),
                        multi_tap_touches=3,
                        on_multi_tap=lambda e: print("MULTI TAP:", e.global_position),
                        on_multi_long_press=lambda _: print("Multi tap long press"),
                    ),
                    ft.TransparentPointer(
                        content=ft.Container(
                            padding=50,
                            content=ft.Button(
                                "Test button",
                                on_click=button_clicked,
                            ),
                        ),
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

buildcontentclass-attributeinstance-attribute

content: Control | None = None

The Control that should be displayed.

VerticalDivider

A thin vertical line, with padding on either side.

In the material design language, this represents a divider.

Example:

```
ft.Row(
    width=120,
    height=60,
    expand=True,
    spacing=0,
    controls=[
        ft.Container(
            bgcolor=ft.Colors.BLUE_GREY_200,
            alignment=ft.Alignment.CENTER,
            expand=True,
        ),
        ft.VerticalDivider(),
        ft.Container(
            bgcolor=ft.Colors.GREY_500,
            alignment=ft.Alignment.CENTER,
            expand=True,
        ),
    ],
)
```

Image: VerticalDivider - https://flet.dev/docs/test-images/examples/controls/core/golden/macos/vertical_divider/image_for_docs.png

Vertical divider

Inherits: Control

Properties

- color - The color to use when painting the line.
- leading_indent - The amount of empty space to the leading edge of the divider.
- radius - The border radius of the divider.
- thickness - The thickness of this divider.
- trailing_indent - The amount of empty space to the trailing edge of the divider.
- width - The divider's width.

Examples

VerticalDivider

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Row(
                width=180,
                height=100,
                spacing=0,
                controls=[
                    ft.Container(
                        bgcolor=ft.Colors.ORANGE_300,
                        alignment=ft.Alignment.CENTER,
                        expand=True,
                    ),
                    ft.VerticalDivider(),
                    ft.Container(
                        bgcolor=ft.Colors.BROWN_400,
                        alignment=ft.Alignment.CENTER,
                        expand=True,
                    ),
                    ft.VerticalDivider(width=1, color=ft.Colors.WHITE),
                    ft.Container(
                        bgcolor=ft.Colors.BLUE_300,
                        alignment=ft.Alignment.CENTER,
                        expand=True,
                    ),
                ],
            ),
        ),
    )
```

```

    ),
    ft.VerticalDivider(width=9, thickness=3),
    ft.Container(
        bgcolor=ft.Colors.GREEN_300,
        alignment=ft.Alignment.CENTER,
        expand=True,
    ),
],
),
)
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: basic - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/vertical_divider/basic.png

Properties

buildcolorclass-attributeinstance-attribute

color: ColorValue | None = None

The color to use when painting the line.

If None, DividerTheme.color is used.

buildleading_indentclass-attributeinstance-attribute

leading_indent: Number | None = None

The amount of empty space to the leading edge of the divider.

If None, DividerTheme.leading_indent is used. If that's is also None, defaults to 0.0.

Raises:

- ValueError - If it is not greater than or equal to 0.

buildradiusclass-attributeinstance-attribute

radius: BorderRadiusValue | None = None

The border radius of the divider.

buildthicknessclass-attributeinstance-attribute

thickness: Number | None = None

The thickness of this divider.

Note

A divider with a thickness of 0.0 is always drawn as a line with a width of exactly one device pixel.

If None, DividerTheme.thickness is used. If that's is also None, defaults to 0.0.

Raises:

- ValueError - If it is not greater than or equal to 0.

buildtrailing_indentclass-attributeinstance-attribute

trailing_indent: Number | None = None

The amount of empty space to the trailing edge of the divider.

If None, DividerTheme.trailing_indent is used. If that's is also None, defaults to 0.0.

Raises:

- ValueError - If it is not greater than or equal to 0.

buildwidthclass-attributeinstance-attribute

width: Number | None = None

The divider's width. The divider itself is always drawn as a vertical line that is centered within the width specified by this value.

If None, DividerTheme.space is used. If that's is also None, defaults to 16.0.

Raises:

- ValueError - If it is not greater than or equal to 0.

Video

Embed a full-featured video player in your Flet app with playlist support, hardware acceleration controls, and subtitle configuration.

It is powered by the `media_kit` Flutter package.

Platform Support

| Platform | Windows | macOS | Linux |
|-----------|--------------------------|--------------------------|--------------------------|
| Supported | ☐ | ☐ | ☐ |

Usage

Add the `flet-video` package to your project dependencies:

- `uv`
- `pip`
`uv add flet-video`
`pip install flet-video`

Requirements

The below sections show the required configurations for each platform.

Android

You may need to declare and request file-system/storage permissions, depending on your use case:

- `android.permission.READ_MEDIA_AUDIO` (optional): Allows to read audio files from external storage. Android 13 or higher.
- `android.permission.READ_MEDIA_VIDEO` (optional): Allows to read video files from external storage. Android 13 or higher.
- `android.permission.READ_EXTERNAL_STORAGE` (optional): Allows reading from external storage. Android 12 or lower.
- `android.permission.WRITE_EXTERNAL_STORAGE` (optional): Allows writing to external storage. Android 12 or lower.
- `flet build`
- `pyproject.toml`

```
flet build apk \
  --android-permissions android.permission.READ_MEDIA_AUDIO=true \
  --android-permissions android.permission.READ_MEDIA_VIDEO=true \
  --android-permissions android.permission.READ_EXTERNAL_STORAGE=true \
  --android-permissions android.permission.WRITE_EXTERNAL_STORAGE=true

[tool.flet.android.permission]
"android.permission.READ_MEDIA_AUDIO" = true
"android.permission.READ_MEDIA_VIDEO" = true
"android.permission.READ_EXTERNAL_STORAGE" = true
"android.permission.WRITE_EXTERNAL_STORAGE" = true
```

Use `PermissionHandler` to request permissions at runtime.

See also:

- `setting Android permissions`

Linux

`libmpv` libraries must be installed and present on the machine running the app.

On Ubuntu/Debian, this can be done with:

```
sudo apt install libmpv-dev mpv
```

If you encounter `libmpv.so.1` load errors, run:

```
sudo apt update
```

```
sudo apt install libmpv-dev libmpv2
sudo ln -s /usr/lib/x86_64-linux-gnu/libmpv.so /usr/lib/libmpv.so.1
```

Examples

Video

play_arrowTry Online

```
import flet as ft
import flet_video as ftv

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ftv.Video(
                expand=True,
                autoplay=True,
                playlist=[
                    ftv.VideoMedia(
                        "https://user-images.githubusercontent.com/28951144/229373720-14d69157-1a56-4a78-a2f4-d7a134d7c3e9.mp4"
                    ),
                ],
            ),
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Playback Controls

Drive playback programmatically with methods like `play()`, `pause()`, `stop()`, `seek()`, `next()`, and `previous()`, and inspect status with methods like `is_playing()`, `get_current_position()`, and `get_duration()`.

play_arrowTry Online

```
import flet as ft
import flet_video as ftv

def main(page: ft.Page):
    async def handle_play(e: ft.Event[ft.Button]):
        await video.play()

    async def handle_play_or_pause(e: ft.Event[ft.Button]):
        await video.play_or_pause()

    async def handle_pause(e: ft.Event[ft.Button]):
        await video.pause()

    async def handle_stop(e: ft.Event[ft.Button]):
        await video.stop()

    async def handle_seek(e: ft.Event[ft.Button]):
        await video.seek(ft.Duration(seconds=10))

    async def handle_next(e: ft.Event[ft.Button]):
        await video.next()

    async def handle_previous(e: ft.Event[ft.Button]):
        await video.previous()

    async def handle_status(e: ft.Event[ft.Button]):
        playing = await video.is_playing()
        position = (await video.get_current_position()).in_seconds
        duration = (await video.get_duration()).in_seconds
        page.show_dialog(
            ft.SnackBar(
                f"Playing?: {playing} | Duration: {duration}s | Position: {position}s"
            )
        )

    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                expand=True,
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    video := ftv.Video(
                        expand=True,
```

```

        playlist=[
            ftv.VideoMedia(
                "https://user-images.githubusercontent.com/28951144/229373720-14d69157-1a56-4a78-a2f4-d7a134d7c3e9"
            ),
            ftv.VideoMedia(
                "https://user-images.githubusercontent.com/28951144/229373718-86ce5e1d-d195-45d5-baa6-ef94041d0b99"
            ),
            ftv.VideoMedia(
                "https://user-images.githubusercontent.com/28951144/229373716-76da0a4e-225a-44e4-9ee7-3e9006dbc3e9"
            ),
        ],
    ),
    ft.Row(
        wrap=True,
        alignment=ft.MainAxisAlignment.CENTER,
        controls=[
            ft.Button("Play", on_click=handle_play),
            ft.Button("Play Or Pause", on_click=handle_play_or_pause),
            ft.Button("Pause", on_click=handle_pause),
            ft.Button("Stop", on_click=handle_stop),
            ft.Button("Seek 10s", on_click=handle_seek),
            ft.Button("Previous", on_click=handle_previous),
            ft.Button("Next", on_click=handle_next),
            ft.Button("Show Status", on_click=handle_status),
        ],
    ),
],
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Video Screenshot

Shows how to capture the current video frame with `take_screenshot()` and display it as an image.

play_arrowTry Online

```

import flet as ft
import flet_video as ftv

def main(page: ft.Page):
    async def handle_screenshot(e: ft.Event[ft.FloatingActionButton]):
        image = await video.take_screenshot(format="image/png")
        if image is None:
            status.value = "No video frame is available yet."
        else:
            preview.content = ft.Image(
                src=image,
                width=260,
                height=146,
                fit=ft.BoxFit.CONTAIN,
            )
            status.value = f"Captured {len(image)} bytes."
        page.update()

    page.floating_action_button = ft.FloatingActionButton(
        content="Take screenshot",
        icon=ft.Icons.CAMERA_ALT,
        on_click=handle_screenshot,
    )
    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                expand=True,
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    video := ftv.Video(
                        expand=True,
                        autoplay=True,
                        playlist=[
                            ftv.VideoMedia(
                                "https://user-images.githubusercontent.com/28951144/229373720-14d69157-1a56-4a78-a2f4-d7a134d7c3e9"
                            ),
                        ],
                    ),
                    preview := ft.Container(
                        width=260,
                        height=146,
                        alignment=ft.Alignment.CENTER,
                        bgcolor=ft.Colors.BLACK_12,
                        content=ft.Text("No screenshot yet"),

```

```

    ),
    status := ft.Text(),
  ],
),
)

if __name__ == "__main__":
    ft.run(main)

```

Playlist

Mutate playlist directly to add, remove, or replace items, and navigate between tracks.

play_arrowTry Online

```

import random

import flet as ft
import flet_video as ftv

media = [
    ftv.VideoMedia(
        "https://user-images.githubusercontent.com/28951144/229373720-14d69157-1a56-4a78-a2f4-d7a134d7c3e9.mp4"
    ),
    ftv.VideoMedia(
        "https://user-images.githubusercontent.com/28951144/229373718-86ce5e1d-d195-45d5-baa6-ef94041d0b90.mp4"
    ),
    ftv.VideoMedia(
        "https://user-images.githubusercontent.com/28951144/229373716-76da0a4e-225a-44e4-9ee7-3e9006dbc3e3.mp4"
    ),
    ftv.VideoMedia(
        "https://user-images.githubusercontent.com/28951144/229373695-22f88f13-d18f-4288-9bf1-c3e078d83722.mp4"
    ),
]

def main(page: ft.Page):
    def handle_add(e: ft.Event[ft.Button]):
        video.playlist.append(random.choice(media))

    def handle_remove(e: ft.Event[ft.Button]):
        if video.playlist:
            video.playlist.pop(random.randint(0, len(video.playlist) - 1))

    def handle_replace(e: ft.Event[ft.Button]):
        video.playlist = random.sample(media, 3)

    async def handle_next(e: ft.Event[ft.Button]):
        await video.next()

    async def handle_previous(e: ft.Event[ft.Button]):
        await video.previous()

    async def handle_jump(e: ft.Event[ft.Button]):
        await video.jump_to(0)

    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                expand=True,
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    video := ftv.Video(
                        expand=True,
                        autoplay=True,
                        playlist=media[:2],
                        playlist_mode=ftv.PlaylistMode.LOOP,
                    ),
                    ft.Row(
                        wrap=True,
                        alignment=ft.MainAxisAlignment.CENTER,
                        controls=[
                            ft.Button("Add Random", on_click=handle_add),
                            ft.Button("Remove Random", on_click=handle_remove),
                            ft.Button("Replace Playlist", on_click=handle_replace),
                            ft.Button("Next", on_click=handle_next),
                            ft.Button("Previous", on_click=handle_previous),
                            ft.Button("Jump to First", on_click=handle_jump),
                        ],
                    ),
                ],
            ),
        ),
    )

```

```
if __name__ == "__main__":
    ft.run(main)
```

Events

Listen for player events like `on_load`, `on_complete`, `on_track_change`, etc.

`play_arrow` [Try Online](#)

```
import flet as ft
import flet_video as ftv
```

```
def main(page: ft.Page):
    def handle_load(e: ft.Event[ftv.Video]):
        print("Video loaded")

    def handle_complete(e: ft.Event[ftv.Video]):
        print(f"Track completed: {e.data}")

    def handle_track_change(e: ft.Event[ftv.Video]):
        print(f"Track changed to index {e.data}")

    def handle_position_change(e: ft.Event[ftv.Video]):
        print(f"Position: {e.data.in_milliseconds / 1000:.3f}s")

    def handle_duration_change(e: ft.Event[ftv.Video]):
        print(f"Duration changed: {e.data.in_seconds}s")

    def handle_error(e: ft.Event[ftv.Video]):
        print(f"Error: {e.data}")

    def handle_enter_fullscreen(e: ft.Event[ftv.Video]):
        print("Entered fullscreen")

    def handle_exit_fullscreen(e: ft.Event[ftv.Video]):
        print("Exited fullscreen")

    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                expand=True,
                controls=[
                    ftv.Video(
                        expand=True,
                        autoplay=True,
                        on_load=handle_load,
                        on_complete=handle_complete,
                        on_track_change=handle_track_change,
                        on_position_change=handle_position_change,
                        on_duration_change=handle_duration_change,
                        on_error=handle_error,
                        on_enter_fullscreen=handle_enter_fullscreen,
                        on_exit_fullscreen=handle_exit_fullscreen,
                        playlist=[
                            ftv.VideoMedia(
                                "https://user-images.githubusercontent.com/28951144/229373720-14d69157-1a56-4a78-a2f4-d7a134d7c3e1.mp4"
                            ),
                            ftv.VideoMedia(
                                "https://user-images.githubusercontent.com/28951144/229373718-86ce5e1d-d195-45d5-baa6-ef94041d0b99.mp4"
                            ),
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Subtitles

Attach a `VideoSubtitleTrack` (here, raw VTT text) and customize its appearance with `VideoSubtitleConfiguration`.

`play_arrow` [Try Online](#)

```
import flet as ft
import flet_video as ftv

SAMPLE_VTT = """WEBVTT

00:00:00.000 --> 00:00:04.000
Welcome to flet-video!
```

```
00:00:04.000 --> 00:00:08.000
Subtitles are styled via VideoSubtitleConfiguration.
```

```
00:00:08.000 --> 00:00:14.000
This track is provided as raw VTT text.
"""
```

```
def main(page: ft.Page):
    page.spacing = 20
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    subtitle_track = ftv.VideoSubtitleTrack(
        src=SAMPLE_VTT,
        title="English",
        language="en",
    )

    def handle_change(e: ft.Event[ft.Switch]):
        if e.control.value:
            video.subtitle_track = subtitle_track
        else:
            video.subtitle_track = ftv.VideoSubtitleTrack.none()

    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                expand=True,
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    video := ftv.Video(
                        expand=True,
                        autoplay=False,
                        playlist=[
                            ftv.VideoMedia(
                                "https://user-images.githubusercontent.com/28951144/229373720-14d69157-1a56-4a78-a2f4-d7a134d7c3e9",
                            ),
                        ],
                        subtitle_track=subtitle_track,
                        subtitle_configuration=ftv.VideoSubtitleConfiguration(
                            text_align=ft.TextAlign.CENTER,
                            text_style=ft.TextStyle(
                                size=24,
                                color=ft.Colors.YELLOW,
                                weight=ft.FontWeight.BOLD,
                                bgcolor=ft.Colors.BLACK_54,
                            ),
                        ),
                    ),
                    ft.Switch(
                        value=True, label="Show Subtitles", on_change=handle_change
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Controls

Switch between `AdaptiveVideoControls`, `MaterialVideoControls`, `MaterialDesktopVideoControls`, custom, and hidden control sets at runtime.

play_arrowTry Online

```
import flet as ft
import flet_video as ftv

def main(page: ft.Page):
    def set_controls(
        value: ftv.VideoControls | ft.Control | None,
    ):
        video.controls = value

    page.add(
        ft.SafeArea(
            expand=True,
            content=ft.Column(
                expand=True,
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    video := ftv.Video(
```

```

expand=True,
playlist=[
    ftv.VideoMedia(
        "https://user-images.githubusercontent.com/28951144/229373720-14d69157-1a56-4a78-a2f4-d7a134d7c3e1",
    ),
],
controls=ftv.MaterialDesktopVideoControls(
    visible_on_mount=True,
    display_seek_bar=True,
    modify_volume_on_scroll=True,
    toggle_fullscreen_on_double_press=True,
    play_and_pause_on_tap=True,
    seek_bar_position_color=ft.Colors.BLUE,
    volume_bar_active_color=ft.Colors.YELLOW,
    controls_hover_duration=ft.Duration(seconds=5),
),
),
ft.Row(
    wrap=True,
    alignment=ft.MainAxisAlignment.CENTER,
    controls=[
        ft.Button(
            "Adaptive",
            on_click=lambda _: set_controls(
                ftv.AdaptiveVideoControls(
                    material=ftv.MaterialVideoControls(
                        visible_on_mount=True,
                        display_seek_bar=True,
                        volume_gesture=True,
                        brightness_gesture=True,
                        seek_gesture=True,
                        seek_on_double_tap=True,
                        speed_up_on_long_press=True,
                        seek_bar_position_color=ft.Colors.BLUE,
                        button_bar_button_color="#E0F7FA",
                    ),
                    material_desktop=ftv.MaterialDesktopVideoControls(
                        visible_on_mount=True,
                        display_seek_bar=True,
                        modify_volume_on_scroll=True,
                        toggle_fullscreen_on_double_press=True,
                        play_and_pause_on_tap=True,
                        hide_mouse_on_controls_removal=True,
                        seek_bar_position_color=ft.Colors.BLUE,
                        volume_bar_active_color=ft.Colors.YELLOW,
                    ),
                ),
            ),
        ),
        ft.Button(
            "Material",
            on_click=lambda _: set_controls(
                ftv.MaterialVideoControls(
                    visible_on_mount=True,
                    volume_gesture=True,
                    brightness_gesture=True,
                    seek_gesture=True,
                    seek_on_double_tap=True,
                    speed_up_on_long_press=True,
                    speed_up_factor=2.5,
                    controls_transition_duration=ft.Duration(
                        milliseconds=450
                    ),
                    seek_bar_position_color=ft.Colors.BLUE,
                    button_bar_button_color="#E0F7FA",
                ),
            ),
        ),
        ft.Button(
            "Material Desktop",
            on_click=lambda _: set_controls(
                ftv.MaterialDesktopVideoControls(
                    visible_on_mount=True,
                    display_seek_bar=True,
                    modify_volume_on_scroll=True,
                    toggle_fullscreen_on_double_press=True,
                    play_and_pause_on_tap=True,
                    hide_mouse_on_controls_removal=True,
                    controls_hover_duration=ft.Duration(seconds=5),
                    seek_bar_position_color=ft.Colors.BLUE,
                    seek_bar_hover_height=8,
                    volume_bar_active_color=ft.Colors.YELLOW,
                ),
            ),
        ),
        ft.Button(
            "Custom Control",

```



```

        on_click=lambda _: set_controls(
            ft.Container(
                alignment=ft.Alignment.BOTTOM_CENTER,
                padding=16,
                content=ft.Text("This is a custom control..."),
            )
        ),
    ),
    ft.Button(
        "No Controls",
        on_click=lambda _: set_controls(None),
    ),
],
),
),
)

if __name__ == "__main__":
    ft.run(main)

```

Mode-specific controls

Show different controls in normal vs. fullscreen mode by mapping each VideoControlsMode to its own controls value.

[play_arrowTry Online](#)

```

import flet as ft
import flet_video as ftv

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=ftv.Video(
                expand=True,
                playlist=[
                    ftv.VideoMedia(
                        "https://user-images.githubusercontent.com/28951144/229373720-14d69157-1a56-4a78-a2f4-d7a134d7c3e9.mp4"
                    ),
                ],
                controls={
                    ftv.VideoControlsMode.NORMAL: ftv.MaterialDesktopVideoControls(
                        visible_on_mount=True,
                        bottom_button_bar=[
                            ftv.VideoPlayOrPauseButton(),
                            ftv.VideoSpacer(),
                            ft.Text("NORMAL Mode", weight=ft.FontWeight.BOLD),
                            ftv.VideoFullscreenButton(),
                        ],
                    ),
                    ftv.VideoControlsMode.FULLSCREEN: ftv.MaterialDesktopVideoControls(
                        visible_on_mount=True,
                        bottom_button_bar=[
                            ftv.VideoFullscreenButton(),
                            ftv.VideoSpacer(),
                            ft.Text("FULLSCREEN Mode", weight=ft.FontWeight.BOLD),
                            ftv.VideoPlayOrPauseButton(),
                        ],
                    ),
                },
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Button bars

Customize the primary_button_bar, top_button_bar, and bottom_button_bar of MaterialDesktopVideoControls with built-in and custom items.

[play_arrowTry Online](#)

```

import flet as ft
import flet_video as ftv

def main(page: ft.Page):
    label_style = ft.TextStyle(
        size=12,

```

```

        color=ft.Colors.WHITE,
        weight=ft.FontWeight.BOLD,
    )

    page.add(
        ft.SafeArea(
            expand=True,
            content=ftv.Video(
                expand=True,
                playlist=[
                    ftv.VideoMedia("video-sample.mp4"),
                ],
                controls=ftv.MaterialDesktopVideoControls(
                    visible_on_mount=True,
                    primary_button_bar=[
                        ftv.VideoSkipPreviousButton(icon_color=ft.Colors.CYAN),
                        ftv.VideoPlayOrPauseButton(
                            icon_size=40,
                            icon_color=ft.Colors.CYAN,
                        ),
                        ftv.VideoSkipNextButton(
                            icon=ft.Icon(ft.Icons.FAST_FORWARD),
                            icon_color=ft.Colors.CYAN,
                        ),
                    ],
                    top_button_bar=[
                        ft.Text("Top button bar", style=label_style),
                        ftv.VideoSpacer(),
                        ftv.VideoFullscreenButton(icon_color=ft.Colors.AMBER),
                    ],
                    bottom_button_bar=[
                        ft.Text("Bottom button bar", style=label_style),
                        ftv.VideoSpacer(),
                        ftv.VideoPositionIndicator(
                            text_style=ft.TextStyle(
                                size=13,
                                color=ft.Colors.WHITE,
                            )
                        ),
                        ftv.VideoVolumeButton(
                            slider_width=96,
                            icon_color=ft.Colors.AMBER,
                        ),
                    ],
                ),
            ),
        ),
    )

)

if __name__ == "__main__":
    ft.run(main)

```

Description

A control that displays a video from a playlist.

Inherits: `LayoutControl`

Properties

- `alignment` - Defines the Alignment of the viewport.
- `autoplay` - Whether the video should start playing automatically.
- `configuration` - Additional configuration for the video player.
- `controls` - Controls displayed over the video.
- `fill_color` - Defines the color used to fill the video background.
- `filter_quality` - Filter quality of the texture used to render the video output.
- `fit` - The box fit to use for the video.
- `fullscreen` - Whether the video player is presented in fullscreen mode.
- `muted` - Defines whether the video player should be started in muted state.
- `pause_upon_entering_background_mode` - Whether to pause the video when application enters background mode.
- `pitch` - Defines the relative pitch of the video player.
- `playback_rate` - Defines the playback rate of the video player.

- `playlist` - A list of VideoMedias representing the video files to be played.
- `playlist_mode` - Represents the mode of playback for the playlist.
- `resume_upon_entering_foreground_mode` - Whether to resume the video when application enters foreground mode.
- `show_controls deprecated` - Whether to show the video player controls.
- `shuffle_playlist` - Defines whether the playlist should be shuffled.
- `subtitle_configuration` - Defines the subtitle configuration for the video player.
- `subtitle_track` - Defines the subtitle track for the video player.
- `title` - Defines the name of the underlying window & process for native backend.
- `volume` - Defines the volume of the video player.
- `wakelock` - Whether to acquire wake lock while playing the video.

Events

- `on_complete` - Fires when a video player completes.
- `on_duration_change` - Fires when the current media duration changes.
- `on_enter_fullscreen` - Fires when the video player enters fullscreen.
- `on_error` - Fires when an error occurs.
- `on_exit_fullscreen` - Fires when the video player exits fullscreen.
- `on_load` - Fires when the video player is initialized and ready for playback.
- `on_position_change` - Fires when the current playback position changes.
- `on_track_change` - Fires when a video track changes.

Methods

- `get_current_position` - Returns: The current position of the currently playing media.
- `get_duration` - Returns: The duration of the currently playing media.
- `is_completed` - Returns: True if video player has reached the end of the currently playing media, False otherwise.
- `is_playing` - Returns: True if the video player is currently playing, False otherwise.
- `jump_to` - Jumps to the VideoMedia at the specified `media_index` in the playlist.
- `next` - Jumps to the next VideoMedia in the playlist.
- `pause` - Pauses the video player.
- `play` - Starts playing the video.
- `play_or_pause` - Cycles between play and pause states of the video player, i.e., plays if paused and pauses if playing.
- `playlist_add deprecated` - Appends/Adds the provided media to the playlist.
- `playlist_remove deprecated` - Removes the provided media from the playlist.
- `previous` - Jumps to the previous VideoMedia in the playlist.
- `seek` - Seeks the currently playing VideoMedia from the playlist at the specified position.
- `stop` - Stops the video player.
- `take_screenshot` - Captures a screenshot of the current video frame.

Properties

`buildalignmentclass-attributeinstance-attribute`

```
alignment: Alignment = field(
    default_factory=lambda: Alignment.CENTER
)
```

Defines the Alignment of the viewport.

buildautoplayclass-attributeinstance-attribute

```
autoplay: bool = False
```

Whether the video should start playing automatically.

buildconfigurationclass-attributeinstance-attribute

```
configuration: VideoConfiguration = field(
    default_factory=lambda: VideoConfiguration()
)
```

Additional configuration for the video player.

buildcontrolsclass-attributeinstance-attribute

```
controls: (
    VideoControls
    | Control
    | dict[
        VideoControlsMode, VideoControls | Control | None
    ]
    | None
) = field(default_factory=lambda: AdaptiveVideoControls())
```

Controls displayed over the video.

Set to a VideoControls object to use built-in controls, a Control object to use custom Flet controls, or None to hide controls.

To use different controls outside and inside fullscreen, set this property to a dictionary keyed by VideoControlsMode. The VideoControlsMode.DEFAULT value is used when VideoControlsMode.NORMAL controls are not provided. If VideoControlsMode.FULLSCREEN controls are not provided, VideoControlsMode.NORMAL controls are reused before falling back to VideoControlsMode.DEFAULT. A mode value of None hides controls for that mode only.

Note

During the show_controls deprecation period, show_controls=False hides controls even when this property is set.

buildfill_colorclass-attributeinstance-attribute

```
fill_color: ColorValue = Colors.BLACK
```

Defines the color used to fill the video background.

buildfilter_qualityclass-attributeinstance-attribute

```
filter_quality: FilterQuality = FilterQuality.LOW
```

Filter quality of the texture used to render the video output.

Note

Android was reported to show blurry images when using FilterQuality.HIGH. Prefer the usage of FilterQuality.MEDIUM on this platform.

buildfitclass-attributeinstance-attribute

```
fit: BoxFit = BoxFit.CONTAIN
```

The box fit to use for the video.

buildfullscreenclass-attributeinstance-attribute

```
fullscreen: bool = False
```

Whether the video player is presented in fullscreen mode.

Set to True to enter fullscreen or False to exit fullscreen programmatically.

buildmutedclass-attributeinstance-attribute

```
muted: bool = False
```

Defines whether the video player should be started in muted state.

buildpause_upon_entering_background_modeclass-attributeinstance-attribute

```
pause_upon_entering_background_mode: bool = True
```

Whether to pause the video when application enters background mode.

buildpitchclass-attributeinstance-attribute

```
pitch: Number = 1.0
```

Defines the relative pitch of the video player.

buildplayback_rateclass-attributeinstance-attribute

```
playback_rate: Number = 1.0
```

Defines the playback rate of the video player.

buildplaylistclass-attributeinstance-attribute

```
playlist: list[VideoMedia] = field(default_factory=list)
```

A list of VideoMedias representing the video files to be played.

buildplaylist_modeclass-attributeinstance-attribute

```
playlist_mode: PlaylistMode | None = None
```

Represents the mode of playback for the playlist.

buildresume_upon_entering_foreground_modeclass-attributeinstance-attribute

```
resume_upon_entering_foreground_mode: bool = False
```

Whether to resume the video when application enters foreground mode. Has effect only if `pause_upon_entering_background_mode` is also set to `True`.

buildshow_controlsclass-attributedeprecatedinstance-attribute

```
show_controls: bool | None = None
```

Deprecated

Deprecated since version 0.85.0 and scheduled for removal in 0.88.0. To hide controls, instead set `controls` to `None`.

Whether to show the video player controls.

buildshuffle_playlistclass-attributeinstance-attribute

```
shuffle_playlist: bool = False
```

Defines whether the playlist should be shuffled.

buildsubtitle_configurationclass-attributeinstance-attribute

```
subtitle_configuration: VideoSubtitleConfiguration = field(
    default_factory=lambda: VideoSubtitleConfiguration()
)
```

Defines the subtitle configuration for the video player.

buildsubtitle_trackclass-attributeinstance-attribute

```
subtitle_track: VideoSubtitleTrack | None = None
```

Defines the subtitle track for the video player.

buildtitleclass-attributeinstance-attribute

```
title: str = 'flet-video'
```

Defines the name of the underlying window & process for native backend. This is visible inside the windows' volume mixer.

buildvolumeclass-attributeinstance-attribute

```
volume: Number = 100.0
```

Defines the volume of the video player.

Note

Its value ranges between 0.0 to 100.0 (inclusive), where 0.0 is muted and 100.0 is the maximum volume. An exception will be raised if the value is outside this range.

Raises:

- `ValueError` - If its value is not between 0.0 and 100.0 (inclusive).

buildwakeuplockclass-attributeinstance-attribute

```
wakeuplock: bool = True
```

Whether to acquire wake lock while playing the video. When `True`, device's display will not go to standby/sleep while the video is playing.

Events

bolton_completeclass-attributeinstance-attribute

```
on_complete: ControlEventHandler[Video] | None = None
```

Fires when a video player completes.

bolton_duration_changeclass-attributeinstance-attribute

```
on_duration_change: ControlEventHandler[Video] | None = (
    None
)
```

Fires when the current media duration changes.

Event handler argument's data property contains the current duration as a Duration.

bolton_enter_fullscreenclass-attributeinstance-attribute

```
on_enter_fullscreen: (
    ControlEventHandler[Video] | None
) = None
```

Fires when the video player enters fullscreen.

bolton_errorclass-attributeinstance-attribute

```
on_error: ControlEventHandler[Video] | None = None
```

Fires when an error occurs.

Event handler argument's data property contains information about the error.

bolton_exit_fullscreenclass-attributeinstance-attribute

```
on_exit_fullscreen: ControlEventHandler[Video] | None = (
    None
)
```

Fires when the video player exits fullscreen

bolton_loadclass-attributeinstance-attribute

```
on_load: ControlEventHandler[Video] | None = None
```

Fires when the video player is initialized and ready for playback.

bolton_position_changeclass-attributeinstance-attribute

```
on_position_change: ControlEventHandler[Video] | None = (
    None
)
```

Fires when the current playback position changes.

Event handler argument's data property contains the current position as a Duration.

bolton_track_changeclass-attributeinstance-attribute

```
on_track_change: ControlEventHandler[Video] | None = None
```

Fires when a video track changes.

Event handler argument's data property contains the index of the new track.

Methods

deployed_codeget_current_positionasync

```
get_current_position() -> Duration
```

Returns:

- Duration - The current position of the currently playing media.

deployed_codeget_durationasync

```
get_duration() -> Duration
```

Returns:

- Duration - The duration of the currently playing media.

deployed_codeis_completedasync

`is_completed() -> bool`

Returns:

- bool - True if video player has reached the end of the currently playing media, False otherwise.

deployed_codeis_playingasync

`is_playing() -> bool`

Returns:

- bool - True if the video player is currently playing, False otherwise.

deployed_codejump_toasync

`jump_to(media_index: int)`

Jumps to the VideoMedia at the specified media_index in the playlist.

deployed_codenextasync

`next()`

Jumps to the next VideoMedia in the playlist.

deployed_codepauseasync

`pause()`

Pauses the video player.

deployed_codeplayasync

`play()`

Starts playing the video.

deployed_codeplay_or_pauseasync

`play_or_pause()`

Cycles between play and pause states of the video player, i.e., plays if paused and pauses if playing.

deployed_codeplaylist_addasyncdeprecated

`playlist_add(media: VideoMedia)`

Deprecated

Deprecated since version 0.85.0 and scheduled for removal in 0.88.0. Use playlist directly, for example `video.playlist.append(media)`.

Appends/Adds the provided media to the playlist.

deployed_codeplaylist_removeasyncdeprecated

`playlist_remove(media_index: int)`

Deprecated

Deprecated since version 0.85.0 and scheduled for removal in 0.88.0. Use playlist directly, for example `video.playlist.pop(media_index)`.

Removes the provided media from the playlist.

deployed_codepreviousasync

`previous()`

Jumps to the previous VideoMedia in the playlist.

deployed_codeseekasync

`seek(position: DurationValue)`

Seeks the currently playing VideoMedia from the playlist at the specified position.

deployed_codestopasync

`stop()`

Stops the video player.

deployed_codetake_screenshotsync

```
take_screenshot(  
    format: str | None = "image/png",  
    include_libass_subtitles: bool = False,  
) -> bytes | None
```

Captures a screenshot of the current video frame.

Parameters:

- `format` (str | None, default: 'image/png') - The image format to return. Supported values are "image/png" (PNG encoded image), "image/jpeg" (JPEG encoded image), and None (raw BGRA pixel buffer on native backends).
- `include_libass_subtitles` (bool, default: False) - Whether to include libass subtitles in the screenshot on native backends. This requires libass support in the underlying player and is ignored on the web backend.

Returns:

- bytes | None - Encoded image bytes, or None if the current backend cannot capture a video frame.

Raises:

- `ValueError` - If format is not supported.

View

View is the top most container for all other controls.

A root view is automatically created when a new user session started. From layout perspective the View represents a Column control, so it has a similar behavior and shares same properties.

Inherits: ScrollableControl, LayoutControl

Properties

- `appbar` - An AppBar control to display at the top of the Page.
- `bgcolor` - Background color of the view.
- `bottom_appbar` - A BottomAppBar control to display at the bottom of the Page.
- `can_pop` - Whether the view can be popped.
- `controls` - A list of controls to display.
- `decoration` - The background decoration.
- `drawer` - A NavigationDrawer control to display as a panel sliding from the start edge of the view.
- `end_drawer` - A NavigationDrawer control to display as a panel sliding from the end edge of the view.
- `floating_action_button` - A FloatingActionButton control to display on top of Page content.
- `floating_action_button_location` - Describes position of `floating_action_button`
- `foreground_decoration` - The foreground decoration.
- `fullscreen_dialog` - If True, the view is a fullscreen modal dialog.
- `horizontal_alignment` - How the child Controls should be placed horizontally.
- `navigation_bar` - A navigation bar (NavigationBar or CupertinoNavigationBar) control to display at the bottom of the Page.
- `padding` - A space between page contents and its edges.
- `route` - View's route - not currently used by Flet framework, but can be used in a user program to update `Page.route` when a view popped.
- `services` - A list of Service controls associated with this view.
- `spacing` - The vertical spacing between controls on the Page.
- `vertical_alignment` - Defines how the child controls should be placed vertically.

Events

- `on_confirm_pop` - An event handler that is called when the view is about to be popped.

Methods

- `close_drawer` - Close the drawer.
- `close_end_drawer` - Close the end drawer.
- `confirm_pop` - Resolves a pending pop-confirmation request for this view.
- `show_drawer` - Show the drawer.
- `show_end_drawer` - Show the end drawer.

Properties

`buildappbarclass-attributeinstance-attribute`

`appbar`: AppBar | CupertinoAppBar | None = None

An AppBar control to display at the top of the Page.

buildbgcolorclass-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

Background color of the view.

buildbottom_appbarclass-attributeinstance-attribute

`bottom_appbar: BottomAppBar | None = None`

A BottomAppBar control to display at the bottom of the Page.

buildcan_popclass-attributeinstance-attribute

`can_pop: bool = True`

Whether the view can be popped.

buildcontrolsclass-attributeinstance-attribute

`controls: list[BaseControl] = field(default_factory=list)`

A list of controls to display.

builddecorationclass-attributeinstance-attribute

`decoration: BoxDecoration | None = None`

The background decoration.

builddrawerclass-attributeinstance-attribute

`drawer: NavigationDrawer | None = None`

A NavigationDrawer control to display as a panel sliding from the start edge of the view.

builndend_drawerclass-attributeinstance-attribute

`end_drawer: NavigationDrawer | None = None`

A NavigationDrawer control to display as a panel sliding from the end edge of the view.

buildfloating_action_buttonclass-attributeinstance-attribute

`floating_action_button: FloatingActionButton | None = None`

A FloatingActionButton control to display on top of Page content.

buildfloating_action_button_locationclass-attributeinstance-attribute

```
floating_action_button_location: (
    FloatingActionButtonLocation | OffsetValue | None
) = None
```

Describes position of floating_action_button

buildforeground_decorationclass-attributeinstance-attribute

`foreground_decoration: BoxDecoration | None = None`

The foreground decoration.

buildfullscreen_dialogclass-attributeinstance-attribute

`fullscreen_dialog: bool = False`

If True, the view is a fullscreen modal dialog.

buildhorizontal_alignmentclass-attributeinstance-attribute

```
horizontal_alignment: CrossAxisAlignment = (
    CrossAxisAlignment.START
)
```

How the child Controls should be placed horizontally.

buildnavigation_barclass-attributeinstance-attribute

```
navigation_bar: (
    NavigationBar | CupertinoNavigationBar | None
) = None
```

A navigation bar (NavigationBar or CupertinoNavigationBar) control to display at the bottom of the Page.

buildpaddingclass-attributeinstance-attribute

```
padding: PaddingValue | None = field(
    default_factory=lambda: Padding.all(10)
)
```

A space between page contents and its edges.

buildroute**class-attributeinstance-attribute**

```
route: str = field(default_factory=lambda: '/')
```

View's route - not currently used by Flet framework, but can be used in a user program to update Page.route when a view popped.

buildservices**class-attributeinstance-attribute**

```
services: list[Service] = field(
    default_factory=list, metadata={"skip": True}
)
```

A list of Service controls associated with this view.

buildspacing**class-attributeinstance-attribute**

```
spacing: Number = 10
```

The vertical spacing between controls on the Page.

Note

Has effect only when vertical_alignment is set to MainAxisAlignment.START, MainAxisAlignment.END, or MainAxisAlignment.CENTER.

buildvertical_alignment**class-attributeinstance-attribute**

```
vertical_alignment: MainAxisAlignment = (
    MainAxisAlignment.START
)
```

Defines how the child controls should be placed vertically.

Events

bolton_confirm_pop**class-attributeinstance-attribute**

```
on_confirm_pop: ControlEventHandler[View] | None = None
```

An event handler that is called when the view is about to be popped. You can use this event to confirm or cancel the pop action by calling confirm_pop method.

Methods

deployed_code**close_drawer**async

```
close_drawer()
```

Close the drawer.

deployed_code**close_end_drawer**async

```
close_end_drawer()
```

Close the end drawer.

deployed_code**confirm_pop**async

```
confirm_pop(should_pop: bool) -> None
```

Resolves a pending pop-confirmation request for this view.

Call this from on_confirm_pop to allow or cancel the current back-navigation attempt.

Parameters:

- should_pop (bool) - True to proceed with popping this view, False to keep the view on the navigation stack.

Notes

- This method only has effect while a pop confirmation is pending.
- If not called, the frontend confirmation wait times out and the pop is canceled.

deployed_code**show_drawer**async

```
show_drawer()
```

Show the drawer.

Raises:

- `ValueError` - If no drawer is defined.

`deployed_codeshow_end_drawerasync`

```
show_end_drawer()
```

Show the end drawer.

Raises:

- `ValueError` - If no end_drawer is defined.

WebView

Display web content in a WebView to be shown in your Flet apps.

It is powered by the `webview_flutter` and `webview_flutter_web` Flutter packages.

Platform Support

| Platform | Windows | macOS | Linux |
|-----------|--------------------------|--------------------------|--------------------------|
| Supported | ☐ | ☐ | ☐ |

Usage

Add flet-webview to your project dependencies:

- `uv`
`uv add flet-webview`
- `pip`
`pip install flet-webview`

Example

WebView

```
import flet as ft
import flet_webview as fwv

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            expand=True,
            content=fwv.WebView(
                url="https://flet.dev",
                on_page_started=lambda _: print("Page started"),
                on_page_ended=lambda _: print("Page ended"),
                on_web_resource_error=lambda e: print("WebView error:", e.data),
                expand=True,
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Troubleshooting

NET::ERR_CLEARTEXT_NOT_PERMITTED Error

If you run into the `NET::ERR_CLEARTEXT_NOT_PERMITTED` error in Android, then the app you're using is trying to access a web page that wants to transmit cleartext or unsecured information. Android blocks apps from doing this in order to avoid compromising user data.

For more details, see [this](#) and [this](#).

To fix it, your app's configuration (precisely, the manifest application attributes) needs to be modified as follows:

- `pyproject.toml`

```
[tool.flet.android.manifest_application]
usesCleartextTraffic = "true"
```

Description

Easily load webpages while allowing user interaction.

Note

Supported only on the following platforms: iOS, Android, macOS, and Web. Concerning Windows and Linux support, subscribe to [this issue](#).

Inherits: `LayoutControl`

Properties

- `bgcolor` - Defines the background color of the `WebView`.
- `prevent_links` - List of url-prefixes that should not be followed/loaded/downloaded.
- `url` - The URL of the web page to load.

Events

- `on_console_message` - Fires when a log message is written to the JavaScript console.
- `on_javascript_alert_dialog` - Fires when the web page attempts to display a JavaScript `alert()` dialog.
- `on_page_ended` - Fires when all the webview page loading processes are ended.
- `on_page_started` - Fires soon as the first loading process of the webview page is started.
- `on_progress` - Fires when the progress of the webview page loading is changed.
- `on_scroll` - Fires when the web page's scroll position changes.
- `on_url_change` - Fires when the URL of the webview page is changed.
- `on_web_resource_error` - Fires when there is error with loading a webview page resource.

Methods

- `can_go_back` - Whether there's a back history item.
- `can_go_forward` - Whether there's a forward history item.
- `clear_cache` - Clears all caches used by the `WebView`.
- `clear_local_storage` - Clears the local storage used by the `WebView`.
- `disable_zoom` - Disables zooming using the on-screen zoom controls and gestures.
- `enable_zoom` - Enables zooming using the on-screen zoom controls and gestures.
- `get_current_url` - Gets the current URL that the `WebView` is displaying or `None` if no URL was ever loaded.
- `get_title` - Get the title of the currently loaded page.
- `get_user_agent` - Get the value used for the HTTP User-Agent: request header.
- `go_back` - Goes back in the history of the webview, if `can_go_back()` is `True`.
- `go_forward` - Goes forward in the history of the webview, if `can_go_forward` is `True`.
- `load_file` - Loads the provided local file.
- `load_html` - Loads the provided HTML string.
- `load_request` - Makes an HTTP request and loads the response in the webview.
- `reload` - Reloads the current URL.
- `run_javascript` - Runs the given JavaScript in the context of the current page.
- `scroll_by` - Scrolls by the provided number of webview pixels.
- `scroll_to` - Scrolls to the provided position of webview pixels.
- `set_javascript_mode` - Sets the JavaScript mode of the `WebView`.

Properties

`buildbgcolorclass-attributeinstance-attribute`

`bgcolor`: `ColorValue` | `None` = `None`

Defines the background color of the `WebView`.

`buildprevent_linksclass-attributeinstance-attribute`

`prevent_links`: `list[str]` | `None` = `None`

List of url-prefixes that should not be followed/loaded/downloaded.

buildurlclass-attributeinstance-attribute

```
url: str | None = None
```

The URL of the web page to load.

Events

bolton_console_messageclass-attributeinstance-attribute

```
on_console_message: (
    EventHandler[WebViewConsoleMessageEvent] | None
) = None
```

Fires when a log message is written to the JavaScript console.

Note

Works only on the following platforms: iOS, Android and macOS.

bolton_javascript_alert_dialogclass-attributeinstance-attribute

```
on_javascript_alert_dialog: (
    EventHandler[WebViewJavaScriptEvent] | None
) = None
```

Fires when the web page attempts to display a JavaScript alert() dialog.

Note

Works only on the following platforms: iOS, Android and macOS.

bolton_page_endedclass-attributeinstance-attribute

```
on_page_ended: ControlEventHandler[WebView] | None = None
```

Fires when all the webview page loading processes are ended.

The data property of the event handler argument is of type str and contains the URL.

Note

Works only on the following platforms: iOS, Android and macOS.

bolton_page_startedclass-attributeinstance-attribute

```
on_page_started: ControlEventHandler[WebView] | None = (
    None
)
```

Fires soon as the first loading process of the webview page is started.

The data property of the event handler argument is of type str and contains the URL.

Note

Works only on the following platforms: iOS, Android and macOS.

bolton_progressclass-attributeinstance-attribute

```
on_progress: ControlEventHandler[WebView] | None = None
```

Fires when the progress of the webview page loading is changed.

The data property of the event handler argument is of type int and contains the progress value.

Note

Works only on the following platforms: iOS, Android and macOS.

bolton_scrollclass-attributeinstance-attribute

```
on_scroll: EventHandler[WebViewScrollEvent] | None = None
```

Fires when the web page's scroll position changes.

Note

Works only on the following platforms: iOS and Android.

bolton_url_changeclass-attributeinstance-attribute

```
on_url_change: ControlEventHandler[WebView] | None = None
```

Fires when the URL of the webview page is changed.

The data property of the event handler argument is of type str and contains the new URL.

Note

Works only on the following platforms: iOS, Android and macOS.

bolton_web_resource_errorclass-attributeinstance-attribute

```
on_web_resource_error: (
    ControlEventHandler[WebView] | None
) = None
```

Fires when there is error with loading a webview page resource.

The data property of the event handler argument is of type str and contains the error message.

Note

Works only on the following platforms: iOS, Android and macOS.

Methods

deployed_codecan_go_backasync

```
can_go_back() -> bool
```

Whether there's a back history item.

Note

Works only on the following platforms: iOS, Android, and macOS.

Returns:

- bool - True if there is a back history item, False otherwise.

deployed_codecan_go_forwardasync

```
can_go_forward() -> bool
```

Whether there's a forward history item.

Note

Works only on the following platforms: iOS, Android, and macOS.

Returns:

- bool - True if there is a forward history item, False otherwise.

deployed_codeclear_cacheasync

```
clear_cache()
```

Clears all caches used by the WebView.

The following caches are cleared:

- Browser HTTP Cache
- Cache API caches. Service workers tend to use this cache.
- Application cache

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_codeclear_local_storageasync

```
clear_local_storage()
```

Clears the local storage used by the WebView.

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_codeisable_zoomasync

`disable_zoom()`

Disables zooming using the on-screen zoom controls and gestures.

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_codeenable_zoomasync

`enable_zoom()`

Enables zooming using the on-screen zoom controls and gestures.

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_codeget_current_urlasync

`get_current_url() -> str | None`

Gets the current URL that the WebView is displaying or None if no URL was ever loaded.

Note

Works only on the following platforms: iOS, Android, and macOS.

Returns:

- `str | None` - The current URL that the WebView is displaying or None if no URL was ever loaded.

deployed_codeget_titleasync

`get_title() -> str | None`

Get the title of the currently loaded page.

Note

Works only on the following platforms: iOS, Android, and macOS.

Returns:

- `str | None` - The title of the currently loaded page.

deployed_codeget_user_agentasync

`get_user_agent() -> str | None`

Get the value used for the HTTP User-Agent: request header.

Note

Works only on the following platforms: iOS, Android, and macOS.

Returns:

- `str | None` - The value used for the HTTP User-Agent: request header.

deployed_codego_backasync

`go_back()`

Goes back in the history of the webview, if `can_go_back()` is True.

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_codego_forwardasync

`go_forward()`

Goes forward in the history of the webview, if `can_go_forward` is True.

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_codeload_fileasync

```
load_file(path: str)
```

Loads the provided local file.

Note

Works only on the following platforms: iOS, Android, and macOS.

Parameters:

- path (str) - The absolute path to the file.

deployed_codeload_htmlasync

```
load_html(
    value: str, base_url: str | None = None
) -> Self
```

Loads the provided HTML string.

Note

Works only on the following platforms: iOS, Android, and macOS.

Parameters:

- value (str) - The HTML string to load.
- base_url (str | None, default: None) - The base URL to use when resolving relative URLs within the value.

deployed_codeload_requestasync

```
load_request(
    url: str, method: RequestMethod = RequestMethod.GET
)
```

Makes an HTTP request and loads the response in the webview.

Parameters:

- url (str) - The URL to load.
- method (RequestMethod, default: RequestMethod.GET) - The HTTP method to use.

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_codereloadasync

```
reload()
```

Reloads the current URL.

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_coderun_javascriptasync

```
run_javascript(value: str)
```

Runs the given JavaScript in the context of the current page.

Parameters:

- value (str) - The JavaScript code to run.

Note

Works only on the following platforms: iOS, Android, and macOS.

deployed_codescroll_byasync

```
scroll_by(x: int, y: int)
```

Scrolls by the provided number of webview pixels.

Note

Works only on the following platforms: iOS, Android, and macOS.

Parameters:

- x (int) - The number of pixels to scroll by on the x-axis.
- y (int) - The number of pixels to scroll by on the y-axis.

deployed_codescroll_toasync

```
scroll_to(x: int, y: int)
```

Scrolls to the provided position of webview pixels.

Note

Works only on the following platforms: iOS, Android, and macOS.

Parameters:

- x (int) - The x-coordinate of the scroll position.
- y (int) - The y-coordinate of the scroll position.

deployed_codeset_javascript_modeasync

```
set_javascript_mode(mode: JavaScriptMode)
```

Sets the JavaScript mode of the WebView.

Note

- Works only on the following platforms: iOS, Android, and macOS.
- Disabling the JavaScript execution on the page may result to unexpected web page behaviour.

Parameters:

- mode (JavaScriptMode) - The JavaScript mode to set.

WindowDragArea

It mimics the behavior (drag, move, maximize, restore) of a native OS window title bar on the content control.

Image: WindowDragArea - https://flet.dev/docs/test-images/examples/controls/core/golden/macOS/window_drag_area/image_for_docs.png

Simple Window Drag Area

Inherits: `LayoutControl`

Properties

- `content` - The content of this drag area.
- `maximizable` - Whether double-clicking on the `WindowDragArea` should maximize/maximize the app's window.

Events

- `on_double_tap` - Called when the `WindowDragArea` is double-tapped and `maximizable=True`.
- `on_drag_end` - Called when a pointer that was previously in contact with the screen and moving/dragging is no longer in contact with the screen.
- `on_drag_start` - Called when a pointer has contacted the screen and has begun to move/drag.

Examples

No frame window

```
import flet as ft

def main(page: ft.Page):
    page.window.title_bar_hidden = True
    page.window.title_bar_buttons_hidden = True

    async def handle_window_close(e: ft.Event[ft.IconButton]):
        await page.window.close()

    page.add(
        ft.SafeArea(
            content=ft.Row(
                controls=[
                    ft.WindowDragArea(
                        expand=True,
                        content=ft.Container(
                            bgcolor=ft.Colors.AMBER_300,
                            padding=10,
                            content=ft.Text(
                                "Drag this area to move, maximize and "
                                "restore application window."
                            ),
                        ),
                    ),
                    ft.IconButton(ft.Icons.CLOSE, on_click=handle_window_close),
                ]
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: no-frame-window - https://flet.dev/docs/test-images/examples/controls/core/golden/macOS/window_drag_area/no_frame_window.png

Properties

`buildcontentinstance-attribute`

`content: Control`

The content of this drag area.

Raises:

- `ValueError` - If it is not visible.

buildmaximizableclass-attributeinstance-attribute

```
maximizable: bool = True
```

Whether double-clicking on the WindowDragArea should maximize/maximize the app's window.

Events

bolton_double_tapclass-attributeinstance-attribute

```
on_double_tap: (
    EventHandler[WindowEvent[WindowDragArea]] | None
) = None
```

Called when the WindowDragArea is double-tapped and maximizable=True.

Info

When a double-tap event is fired, the type property of the event handler argument can only be one of the following: WindowEventType.MAXIMIZE, WindowEventType.UNMAXIMIZE.

bolton_drag_endclass-attributeinstance-attribute

```
on_drag_end: (
    EventHandler[DragEndEvent[WindowDragArea]] | None
) = None
```

Called when a pointer that was previously in contact with the screen and moving/dragging is no longer in contact with the screen.

bolton_drag_startclass-attributeinstance-attribute

```
on_drag_start: (
    EventHandler[DragStartEvent[WindowDragArea]] | None
) = None
```

Called when a pointer has contacted the screen and has begun to move/drag.

Accessibility

Flet is based on Flutter which includes first-class framework support for accessibility in addition to that provided by the underlying operating system.

Screen readers

For mobile, screen readers (TalkBack, VoiceOver) enable visually impaired users to get spoken feedback about the contents of the screen and interact with the UI via gestures on mobile and keyboard shortcuts on desktop. Turn on VoiceOver or TalkBack on your mobile device and navigate around your app.

For web, the following screen readers are currently supported:

Mobile Browsers:

- iOS - VoiceOver
- Android - TalkBack

Desktop Browsers:

- MacOS - VoiceOver
- Windows - JAWs & NVDA

Screen Readers users on web will need to toggle "Enable accessibility" button to build the semantics tree.

Text

Use `Text.semantics_label` property to override default Text control semantics.

Buttons

All buttons with text on them generate proper semantics.

Use `tooltip` property to add screen reader semantics for `IconButton`, `FloatingActionButton` and `PopupMenuButton` buttons.

TextField and Dropdown

Use `TextField.label` and `Dropdown.label` properties to add screen reader semantics to those controls.

Custom semantics

For any specific requirements use `Semantics` control.

Debugging semantics

Set `Page.show_semantics_debugger` to `True` to show an overlay that shows the accessibility information reported by the framework.

You can implement a specific keyboard shortcut to conveniently toggle semantics debugger during app development:

Image:  <https://flet.dev/docs/assets/cookbook/accessibility/debug-accessibility-toggle.gif>

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def on_keyboard(e: ft.KeyboardEvent):
        if e.shift and e.key == "S":
            page.show_semantics_debugger = not page.show_semantics_debugger
            page.update()

    page.on_keyboard_event = on_keyboard

    def button_click(e: ft.Event[ft.Button]):
        counter.value = str(int(counter.value) + 1)
        page.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
```

```

        controls=[
            counter := ft.Text("0", size=40),
            ft.Text("Press Shift+S to toggle semantics debugger"),
            ft.Button(
                content="Increment number",
                icon=ft.Icons.ADD,
                on_click=button_click,
            ),
        ]
    )
)

if __name__ == "__main__":
    ft.run(main)

```

Adaptive apps

Flet framework allows you to develop adaptive apps, which means having a single codebase that will deliver a different look depending on the device's platform.

Below is an example of a very simple app that has a different look on iOS and Android platforms:

```
import flet as ft

def main(page):

    page.adaptive = True

    page.appbar = ft.AppBar(
        leading=ft.TextButton("New", style=ft.ButtonStyle(padding=0)),
        title=ft.Text("Adaptive AppBar"),
        actions=[
            ft.IconButton(ft.CupertinoIcons.ADD, style=ft.ButtonStyle(padding=0))
        ],
        bgcolor=ft.Colors.with_opacity(0.04, ft.CupertinoColors.SYSTEM_BACKGROUND),
    )

    page.navigation_bar = ft.NavigationBar(
        destinations=[
            ft.NavigationBarDestination(icon=ft.Icons.EXPLORE, label="Explore"),
            ft.NavigationBarDestination(icon=ft.Icons.COMMUTE, label="Commute"),
            ft.NavigationBarDestination(
                icon=ft.Icons.BOOKMARK_BORDER,
                selected_icon=ft.Icons.BOOKMARK,
                label="Bookmark",
            ),
        ],
        border=ft.Border(
            top=ft.BorderSide(color=ft.CupertinoColors.SYSTEM_GREY2, width=0)
        ),
    )

    page.add(
        ft.SafeArea(
            ft.Column(
                [
                    ft.Checkbox(value=False, label="Dark Mode"),
                    ft.Text("First field:"),
                    ft.TextField(keyboard_type=ft.KeyboardType.TEXT),
                    ft.Text("Second field:"),
                    ft.TextField(keyboard_type=ft.KeyboardType.TEXT),
                    ft.Switch(label="A switch"),
                    ft.FilledButton(content=ft.Text("Adaptive button")),
                    ft.Text("Text line 1"),
                    ft.Text("Text line 2"),
                    ft.Text("Text line 3"),
                ]
            )
        )
    )

ft.run(main)
```

By setting just `page.adaptive = True` you can make your app look awesome on both iOS and Android devices:

Image:  - <https://flet.dev/docs/assets/cookbook/adaptive-apps/iphone.png>

Image:  - <https://flet.dev/docs/assets/cookbook/adaptive-apps/android.png>

| iPhone | Android |
|---|--|
|  |  |

Material and Cupertino controls

Most of Flet controls are based on Material design.

There is also a number of iOS-style controls in Flet that are called Cupertino controls.

Cupertino controls usually have a matching Material control that has adaptive property which defaults to `False`. When using a Material control with adaptive property set to `True`, a different control will be created depending on the platform, for example:

```
ft.Checkbox(adaptive=True, value=True, label="Adaptive Checkbox")
```

Flet checks the value of `Page.platform` property and if it is `PagePlatform.IOS` or `ft.PagePlatform.MACOS`, Cupertino control will be created; in all other cases Material control will be created.

Note

adaptive property can be set for an individual control or a container-controls (ex: Row, Column) that has children controls. If a container-control is made adaptive, all its children will be adaptive too, unless adaptive property is explicitly set to False for a child control.

Material vs Cupertino

Below is the list of adaptive Material controls and their matching Cupertino controls:

AlertDialogCupertinoAlertDialog

Image: Image - https://flutter.dev/docs/examples/controls/material/alert_dialog/media/index.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_alert_dialog/media/adaptive.png

Dialog actionsCupertinoDialogAction

Image: Image - https://flutter.dev/docs/examples/controls/material/alert_dialog/media/adaptive_dialog_action.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_alert_dialog/media/adaptive_dialog_action.png

AppBarCupertinoAppBar

Image: Image - https://flutter.dev/docs/examples/controls/material/app_bar/media/index.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_app_bar/media/index.png

NavigationBarCupertinoNavigationBar

Image: Image - https://flutter.dev/docs/examples/controls/material/navigation_bar/media/adaptive.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_navigation_bar/media/adaptive.png

ListTileCupertinoListTile

Image: Image - https://flutter.dev/docs/examples/controls/material/list_tile/media/index.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_list_tile/media/index.png

TextFieldCupertinoTextField

Image: Image - https://flutter.dev/docs/examples/controls/material/text_field/media/index.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_text_field/media/index.png

CheckboxCupertinoCheckbox

Image: Image - <https://flutter.dev/docs/examples/controls/material/checkbox/media/index.png>

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_checkbox/media/index.png

SliderCupertinoSlider

Image: Image - https://flutter.dev/docs/test-images/examples/controls/material/golden/macOS/slider/image_for_docs.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_slider/media/index.png

SwitchCupertinoSwitch

Image: Image - <https://flutter.dev/docs/examples/controls/material/switch/media/index.png>

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_switch/media/index.png

RadioCupertinoRadio

Image: Image - <https://flutter.dev/docs/examples/controls/material/radio/media/index.png>

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_radio/media/index.png

FilledButtonCupertinoFilledButton

Image: Image - https://flutter.dev/docs/examples/controls/material/filled_button/media/adaptive.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_filled_button/media/adaptive.png

FilledTonalButtonCupertinoButton

Image: Image - https://flutter.dev/docs/examples/controls/material/filled_tonal_button/media/adaptive.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_button/media/adaptive_tonal_button.png

IconButtonCupertinoButton

Image: Image - https://flutter.dev/docs/examples/controls/material/icon_button/media/adaptive.png

Image: Image - https://flutter.dev/docs/examples/controls/cupertino/cupertino_button/media/adaptive_icon_button.png

ButtonOutlinedButtonTextButtonCupertinoButton


```
page.update()
```

```
ft.run(main)
```

Now the `NavigationBar` and icons within it will look like different on Android and iOS:

Image: Image - https://flet.dev/docs/examples/controls/material/navigation_bar/media/adaptive.png

Image: Image - https://flet.dev/docs/examples/controls/material/navigation_bar/media/adaptive.png

| iOS | Android |
|-----|---------|
| | |

Note

You may utilise reusable controls approach to adapt your app not only depending on the `Page.platform`, but also use `Page.web` property to have different UI depending on whether the app is running in a browser or not, or even combine the usage of both properties to have specific UI for your apps.

Animations

Implicit animations

With implicit animations, you can animate a control property by setting a target value; whenever that target value changes, the control animates the property from the old value to the new one.

Animation produces interpolated values between the old and the new value over the given duration.

By default, the animation is linearly increasing the animation value, however, a curve can be applied to the animation which changes the value according to the provided curve. For example, `AnimationCurve.EASE_OUT_CUBIC` curve increases the animation value quickly at the beginning of the animation and then slows down until the target value is reached:

`LayoutControl` (and its subclasses) provides a number of `animate_{something}` properties, described below, to enable implicit animation of its appearance:

- `animate_opacity`
- `animate_rotation`
- `animate_scale`
- `animate_offset`
- `animate_position`
- `animate` (Container)

`animate_*` properties could have one of the following values:

- Instance of `Animation` - allows configuring the duration and the curve of the
- `animation`, for example `animate_rotation=Animation(duration=300, curve=AnimationCurve.BOUNCE_OUT)`. See this [Flutter docs on animation curves](#) for possible values. Default is `AnimationCurve.LINEAR`.
- `int` value - enables animation with specified duration in milliseconds and `AnimationCurve.LINEAR` curve.
- `bool` value - enables animation with the duration of 1000 milliseconds and `AnimationCurve.LINEAR` curve.

Opacity animation

Setting control's `animate_opacity` to either `True`, number or an instance of `Animation` class (see above) enables implicit animation of `Control.opacity` property.

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    def animate_opacity(e: ft.Event[ft.Button]):
        container.opacity = 0 if container.opacity == 1 else 1
        container.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    container := ft.Container(
                        width=150,
                        height=150,
                        bgcolor=ft.Colors.BLUE,
                        border_radius=10,
                        animate_opacity=300,
                    ),
                    ft.Button(
                        content="Animate opacity",
                        on_click=animate_opacity,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
```

```
ft.run(main)
```

Image: animate-opacity - https://flet.dev/docs/examples/controls/core/layout_control/media/animate_opacity.gif

Rotation animation

Setting control's `animate_rotation` to either `True`, number or an instance of `Animation` class (see above) enables implicit animation of `LayoutControl.rotate` property.

play_arrowTry Online

```
from math import pi

import flet as ft

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.spacing = 30

    def animate(e: ft.Event[ft.Button]):
        container.rotate.angle += pi / 2
        container.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    container := ft.Container(
                        width=100,
                        height=70,
                        bgcolor=ft.Colors.BLUE,
                        border_radius=5,
                        rotate=ft.Rotate(angle=0, alignment=ft.Alignment.CENTER),
                        animate_rotation=ft.Animation(
                            duration=300,
                            curve=ft.AnimationCurve.BOUNCE_OUT,
                        ),
                    ),
                    ft.Button("Animate!", on_click=animate),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: animate-rotation - https://flet.dev/docs/examples/controls/core/layout_control/media/animate_rotation.gif

Scale animation

Setting control's `animate_scale` to either `True`, number or an instance of `Animation` class (see above) enables implicit animation of `LayoutControl.scale` property.

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.spacing = 30

    def animate(e: ft.Event[ft.Button]):
        container.scale = 2 if container.scale == 1 else 1
        container.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    container := ft.Container(
                        width=100,
                        height=100,
                        bgcolor=ft.Colors.BLUE,
                        border_radius=5,
                        scale=1,
                        animate_scale=ft.Animation(
                            duration=600,
                            curve=ft.AnimationCurve.BOUNCE_OUT,
                        ),
                    ),
                ],
            ),
        ),
    )
```

```

        ),
    ),
    ft.Button("Animate!", on_click=animate),
],
),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: animate-scale - https://flet.dev/docs/examples/controls/core/layout_control/media/animate_scale.gif

Offset animation

Setting control's `animate_offset` to either `True`, number or an instance of `Animation` class (see above) enables implicit animation of `LayoutControl.offset` property.

`offset` property is an instance of `Offset` class which specifies horizontal `x` and vertical `y` offset of a control scaled to control's size. For example, an offset `Offset(-0.25, 0)` will result in a horizontal translation of one quarter the width of the control.

Offset animation is used for various sliding effects:

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    def animate(e: ft.Event[ft.Button]):
        container.offset = ft.Offset(0, 0)
        container.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    container := ft.Container(
                        width=150,
                        height=150,
                        bgcolor=ft.Colors.BLUE,
                        border_radius=ft.BorderRadius.all(10),
                        offset=ft.Offset(x=-1.1, y=0),
                        animate_offset=ft.Animation(
                            duration=600,
                            curve=ft.AnimationCurve.BOUNCE_OUT,
                        ),
                    ),
                    ft.Button("Reveal!", on_click=animate),
                ],
            ),
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: animate-offset - https://flet.dev/docs/examples/controls/core/layout_control/media/animate_offset.gif

Position animation

Setting control's `animate_position` to either `True`, number or an instance of `Animation` class (see above) enables implicit animation of the following `LayoutControl` properties: `left`, `right`, `bottom`, `top`.

Note: Positioning is effective only if the control is a descendant of one of the following:

- Stack control
- Page.overlay list

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    def animate_container(e: ft.Event[ft.Button]):
        c1.top = 20
        c1.left = 200
        c2.top = 100
        c2.left = 40
        c3.top = 180
        c3.left = 100

```

```

        stack.update()

page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                stack := ft.Stack(
                    height=400,
                    controls=[
                        c1 := ft.Container(
                            width=50,
                            height=50,
                            bgcolor="red",
                            animate_position=1000,
                        ),
                        c2 := ft.Container(
                            width=50,
                            height=50,
                            bgcolor="green",
                            top=60,
                            left=0,
                            animate_position=500,
                        ),
                        c3 := ft.Container(
                            width=50,
                            height=50,
                            bgcolor="blue",
                            top=120,
                            left=0,
                            animate_position=1000,
                        ),
                    ],
                ),
                ft.Button("Animate!", on_click=animate_container),
            ],
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Image: animate-position - https://flet.dev/docs/examples/controls/core/layout_control/media/animate_position.gif

Animate

Setting `Container.animate` to `AnimationValue` enables implicit animation of container properties such as size, background color, border style, gradient.

play_arrowTry Online

```

import flet as ft

def main(page: ft.Page):
    def animate_container(e: ft.Event[ft.Button]):
        container.width = 100 if container.width == 150 else 150
        container.height = 50 if container.height == 150 else 150
        container.bgcolor = (
            ft.Colors.BLUE if container.bgcolor == ft.Colors.RED else ft.Colors.RED
        )
        container.update()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    container := ft.Container(
                        width=150,
                        height=150,
                        bgcolor=ft.Colors.RED,
                        animate=ft.Animation(
                            duration=1000, curve=ft.AnimationCurve.BOUNCE_OUT
                        ),
                    ),
                    ft.Button("Animate container", on_click=animate_container),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Image: animate - https://flet.dev/docs/examples/controls/core/container/media/animate_1.gif

Animated content switcher

AnimatedSwitcher allows animated transition between two controls ('new' and 'old').

```
import time

import flet as ft

def main(page: ft.Page):
    i = ft.Image(src="https://picsum.photos/150/150", width=150, height=150)

    def animate(e):
        sw.content = ft.Image(
            src=f"https://picsum.photos/150/150?{time.time()}", width=150, height=150
        )
        page.update()

    sw = ft.AnimatedSwitcher(
        i,
        transition=ft.AnimatedSwitcherTransition.SCALE,
        duration=500,
        reverse_duration=500,
        switch_in_curve=ft.AnimationCurve.BOUNCE_OUT,
        switch_out_curve=ft.AnimationCurve.BOUNCE_IN,
    )

    page.add(
        sw,
        ft.Button("Animate!", on_click=animate),
    )

ft.run(main)
```

Image: animated-switcher - <https://flet.dev/docs/assets/cookbook/animations/animated-switcher.gif>

Animation end callback

LayoutControl also has an `on_animation_end` event handler, which is called when an animation is complete. It can be used to chain multiple animations.

Event's data field/property contains the name of animation:

- "opacity"
- "rotation"
- "scale"
- "offset"
- "position"
- "container"

For example:

```
ft.Container(
    content=ft.Text("Animate me!"),
    animate=ft.Animation(1000, ft.AnimationCurve.BOUNCE_OUT),
    on_animation_end=lambda e: print("Container animation end:", e.data)
)
```


Assets

Flet apps can include both code and assets/resources. An asset is a file that is bundled and deployed with your app and is accessible at runtime. Common types of assets include static data (e.g., JSON files), configuration files, icons, images, videos, etc.

To use relative paths for your asset files, you need to provide a path to your assets directory when launching your app with the `ft.run()` function. The parameter for this is called `assets_dir`, which defaults to "assets". This parameter specifies the folder where local assets are stored and can be either an absolute path or a path relative to the app's entry point file, such as `main.py`.

Example: Displaying a Local Image

Suppose you have a folder named `assets` in the same directory as your `main.py` file, and this folder contains an image file named `sample.png` in a subfolder called `images`:

```
assets
  images
    sample.png
main.py
```

To display this image in your app, you can do the following:

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.Image(src="images/sample.png")
    )

ft.run(main, assets_dir="assets")
```

The same approach applies to other asset types like fonts, Lottie animations, Rive files, etc.

Accessing Asset Files in Production

For UI controls such as `Image`, you usually pass a path relative to `assets_dir`:

```
ft.Image(src="images/sample.png")
```

For non-UI file operations (ex: reading JSON, opening a bundled SQLite database, loading ML models, etc.), it is often more convenient to use an absolute path to your assets directory. For this reason, Flet sets the `FLET_ASSETS_DIR` environment variable (in production apps built with `flet build`, excluding web) to an absolute path to your assets directory.

You can use it with a local-development fallback, as shown in the below example (which assumes the asset folder is named "assets"):

```
import json
import os
from pathlib import Path

import flet as ft

def get_assets_dir() -> Path:
    default_assets_dir = Path(__file__).parent / "assets" # fallback for local runs
    return Path(os.environ.get("FLET_ASSETS_DIR", str(default_assets_dir))).resolve()

def main(page: ft.Page):
    assets_dir = get_assets_dir()

    # load a JSON file
    with (assets_dir / "data" / "some_config.json").open() as f:
        config = json.load(f)

    page.add(ft.Text(f"Loaded profile: {config['profile_name']}"))

ft.run(main, assets_dir="assets")
```

This gives you a single way to resolve bundled files across local runs and production builds. See `FLET_ASSETS_DIR` for details.

Async apps

Flet app can be written as an async app and use asyncio and other Python async libraries. Calling coroutines is naturally supported in Flet, so you don't need to wrap them to run synchronously.

By default, Flet executes control event handlers in separate threads, but sometimes that could be an ineffective usage of CPU or it does nothing while waiting for a HTTP response or executing `sleep()`.

Asyncio, on the other hand, allows implementing concurrency in a single thread by switching execution context between "coroutines". This is especially important for apps that are going to be published as static websites using Pyodide. Pyodide is a Python runtime built as a WebAssembly (WASM) and running in the browser. At the time of writing it doesn't support threading yet.

Getting started with async

You could mark `main()` method of Flet app as async and then use any asyncio API inside it:

```
import flet as ft

async def main(page: ft.Page):
    await asyncio.sleep(1)
    page.add(ft.Text("Hello, async world!"))

ft.run(main)
```

You can use `await ft.run_async(main)` if Flet app is part of a larger app and called from async code.

Control event handlers

Control event handlers could be both sync and async.

If a handler does not call any async methods it could be a regular sync method:

```
def page_resize(e):
    print("New page size:", page.window.width, page.window.height)

page.on_resize = page_resize
```

However, if a handler calls async logic it must be async too:

```
async def main(page: ft.Page):

    async def button_click(e):
        await some_async_method()
        page.add(ft.Text("Hello!"))

    page.add(ft.Button("Say hello!", on_click=button_click))

ft.run(main)
```

Async lambdas

There are no async lambdas in Python. It's perfectly fine to have a lambda event handler in async app for simple things:

```
page.on_error = lambda e: print("Page error:", e.data)
```

but you can't have an async lambda, so an async event handler must be used.

Sleeping

To delay code execution in async Flet app you should use `asyncio.sleep()` instead of `time.sleep()`, for example:

```
import asyncio
import flet as ft

def main(page: ft.Page):
    async def button_click(e):
        await asyncio.sleep(1)
        page.add(ft.Text("Hello!"))

    page.add(
        ft.Button("Say hello with delay!", on_click=button_click)
    )

ft.run(main)
```

Threading

To run something in the background use `page.run_task()`. For example, "Countdown" custom control which is self-updating on background could be implemented as following:

```
import asyncio
import flet as ft

class Countdown(ft.Text):
    def __init__(self, seconds):
        super().__init__()
        self.seconds = seconds

    def did_mount(self):
        self.running = True
        self.page.run_task(self.update_timer)

    def will_unmount(self):
        self.running = False

    async def update_timer(self):
        while self.seconds and self.running:
            mins, secs = divmod(self.seconds, 60)
            self.value = "{:02d}:{:02d}".format(mins, secs)
            self.update()
            await asyncio.sleep(1)
            self.seconds -= 1

def main(page: ft.Page):
    page.add(Countdown(120), Countdown(60))

ft.run(main)
```

Image: <https://flet.dev/img/docs/getting-started/user-control-countdown.gif>

Authentication

You can implement user authentication ("Login with X" buttons) in your Flet app using 3rd-party identity providers such as GitHub, Google, Azure, Auth0, LinkedIn and others.

Identity provider must support OAuth 2.0 Authorization Code Flow to retrieve API access token.

Built-in Flet login with credentials and user management is planned for future releases. If you have a requirement to create and manage user accounts in your app you can implement it by your own or use Auth0 identity provider which provides a generous free tier.

Flet authentication features:

- Works with Flet desktop, web and mobile apps.
- Using multiple authentication providers in one app.

Built-in OAuth providers with automatic user details fetching:

- GitHub
- Azure
- Google
- Auth0
- Optional groups fetching.
- Automatic token refresh.
- Login with a saved token ("Remember me").
- Custom OAuth providers.

Login process overview

- Configure OAuth provider (built-in or generic) with Client ID, Client secret, Redirect URL.
- Call `page.login(provider)` to initiate OAuth web flow.
- User is being redirected to OAuth provider website.
- On provider website user signs in and gives consent to access service API with requested scopes.
- Provider website redirects to Flet's OAuth callback URL with authorization code.
- Flet exchanges authorization code for a token and calls `page.on_login` event handler.
- Flet app can retrieve API token from `page.auth.token` property and user details from `page.auth.user`.

Configuring OAuth provider

Flet has the following built-in OAuth providers:

- GitHub
- Azure
- Google
- Auth0

Additionally, you can configure a generic OAuth provider and provide authorization, token and user info endpoints.

In this guide we will configure Flet login page with GitHub account.

To integrate Flet authentication with GitHub a new GitHub OAuth app should be registered first (Profile settings → Developer settings → OAuth Apps).

Authorization callback URL should be in the format:

`{application-url}/oauth_callback`

Image: GitHub new OAuth app - <https://flet.dev/docs/assets/cookbook/authentication/github-new-oauth-app.png>

On OAuth app details page click "Generate a new client secret" button. Copy "Client ID" and "Client secret" values to a safe place - you'll need them in a Flet app.

Image: GitHub OAuth app details - <https://flet.dev/docs/assets/cookbook/authentication/github-oauth-app-details.png>

Sign in with OAuth provider

```
import os

import flet as ft
from flet.auth.providers import GitHubOAuthProvider

GITHUB_CLIENT_ID = os.getenv("GITHUB_CLIENT_ID")
assert GITHUB_CLIENT_ID, "set GITHUB_CLIENT_ID environment variable"
GITHUB_CLIENT_SECRET = os.getenv("GITHUB_CLIENT_SECRET")
assert GITHUB_CLIENT_SECRET, "set GITHUB_CLIENT_SECRET environment variable"

def main(page: ft.Page):
    provider = GitHubOAuthProvider(
        client_id=GITHUB_CLIENT_ID,
        client_secret=GITHUB_CLIENT_SECRET,
        redirect_url="http://localhost:8550/oauth_callback",
    )

    def login_click(e):
        page.login(provider)

    def on_login(e):
        print("Login error:", e.error)
        print("Access token:", page.auth.token.access_token)
        print("User ID:", page.auth.user.id)

    page.on_login = on_login
    page.add(ft.Button("Login with GitHub", on_click=login_click))

ft.run(main, port=8550, view=ft.WEB_BROWSER)
```

Caution

Notice, we are fetching OAuth app client ID and client secret from an environment variables. Do not embed any secrets into source code to avoid accidental exposure to a public!

Before running the app set the secret environment variables in a command line:

```
$ export GITHUB_CLIENT_ID="<client_id>"
$ export GITHUB_CLIENT_SECRET="<client_secret>"
```

Run the program and click "Login with GitHub" button. GitHub authorize app page will be opened in:

- Desktop app - a new browser window or tab.
- Web app - a new popup window (make sure popup blocker is off).
- Mobile app - an in-app web browser.

Image: GitHub OAuth authorize - <https://flet.dev/docs/assets/cookbook/authentication/github-oauth-authorize.png>

Redirect URL

We used `http://localhost:8550/oauth_callback` as a redirect URL while registering GitHub OAuth app. Notice it has a fixed port 8550. To run your Flet app on a fixed port use port argument in `flet.app` call:

```
ft.run(main, port=8550)
```

Scope

Most of OAuth providers allows applications to request one or more scopes to limit application's access to a user's account.

Built-in Flet providers, by default, request scopes to access user profile, but you can request additional scopes in login method, like `public_repo` in the example above:

```
page.login(
    provider,
    scope=["public_repo"]
)
```

`page.login()` method has a number of arguments to control authentication process:

- `fetch_user` (bool) - whether to fetch user details into `page.auth.user`. Default is `True`.
- `fetch_groups` (bool) - whether to fetch user groups into `page.auth.user.groups`. Default is `False`.

- `scope` - a list of scopes to request.
- `saved_token` - a JSON snapshot of `page.auth.token` to restore authorization from. Token can be serialized with `page.auth.token.to_json()`, encrypted and saved in `page.client_storage`. See below.
- `on_open_authorization_url` - a callback to open a browser with authorization URL. See below.
- `complete_page_html` - a custom HTML contents of "You've been successfully authenticated. Close this page now" page.
- `redirect_to_page` (bool) - used with Flet web app only when authorization page is opened in the same browser tab.

The result of `page.login()` call is an instance of `Authorization` class with the following fields:

- `token` - OAuth token used to access provider's API. See below.
- `user` - user details with a mandatory `id` field and other fields specific to OAuth provider.
- `provider` - an instance of OAuth provider used for authorization.

A reference to the last authorization is saved in `page.auth` property.

If your app allows authorizations with multiple OAuth providers you can save authorizations in a session, for example:

```
page.session["github_auth"] = page.login(github_provider)
page.session["google_auth"] = page.login(google_provider)
```

Checking authentication results

Upon successful or failed authorization `page.on_login` event handler is called.

Event handler argument `e` is an instance of `LoginEvent` class with the following properties:

- `error` (str) - OAuth error.
- `error_description` (str) - OAuth error description.

Authorization was successful if `error` is an empty string.

You can use this event handler to toggle signed in/out UI, for example:

```
import os

import flet as ft
from flet.auth.providers import GitHubOAuthProvider

def main(page: ft.Page):
    provider = GitHubOAuthProvider(
        client_id=os.getenv("GITHUB_CLIENT_ID"),
        client_secret=os.getenv("GITHUB_CLIENT_SECRET"),
        redirect_url="http://localhost:8550/oauth_callback",
    )

    def login_button_click(e):
        page.login(provider, scope=["public_repo"])

    def on_login(e: ft.LoginEvent):
        if not e.error:
            toggle_login_buttons()

    def logout_button_click(e):
        page.logout()

    def on_logout(e):
        toggle_login_buttons()

    def toggle_login_buttons():
        login_button.visible = page.auth is None
        logout_button.visible = page.auth is not None
        page.update()

    login_button = ft.Button("Login with GitHub", on_click=login_button_click)
    logout_button = ft.Button("Logout", on_click=logout_button_click)
    toggle_login_buttons()

    page.on_login = on_login
    page.on_logout = on_logout

    page.add(login_button, logout_button)

ft.run(main, port=8550, view=ft.AppView.WEB_BROWSER)
```

Accessing user details

If `page.login()` method is called with `fetch_user=True` (default) a user profile will be assigned to `page.auth.user`.

All built-in OAuth providers implement `user.id` property - unique user identifier - which value depends on the provider (a number, Guid or email) and can be used in your app as a user key.

The rest of user profile properties depends on provider and can be accessed with an indexer. For example, to print some properties of GitHub user:

```
print("Name:", page.auth.user["name"])
print("Login:", page.auth.user["login"])
print("Email:", page.auth.user["email"])
```

Using OAuth token

Upon successful authorization `page.auth.token` will contain OAuth token that can be used to access providers's API. Token object has the following properties:

- `access_token` - access token used as an authorization token in API request header.
- `scope` - token's scope.
- `token_type` - access token type, e.g. Bearer.
- `expires_in` - optional number of seconds when access token expires.
- `expires_at` - optional time (`time.time()` + `expires_in`) when access token expires.
- `refresh_token` - optional refresh token which is used to get a new access token, when the old one expires.

Usually, only `page.auth.token.access_token` is needed to call provider's API, for example to list user's GitHub repositories:

```
import requests
headers = {"Authorization": "Bearer {}".format(page.auth.token.access_token)}
repos_resp = requests.get("https://api.github.com/user/repos", headers=headers)
user_repos = json.loads(repos_resp.text)
for repo in user_repos:
    print(repo["full_name"])
```

note

Do not save a reference to `page.auth.token` somewhere in your code, but rather call `page.auth.token` every time you need to grab access token. `page.auth.token` is a property which automatically refreshes OAuth token if/when it expires.

Correct code:

```
access_token = page.auth.token.access_token
```

Wrong code:

```
token = page.auth.token
# some other code
access_token = token.access_token # token could expire by this moment
```

Saving and restoring an auth token

To implement persistent login ("Remember me" checkbox on login page) you can save auth token in a client storage and use it to login next time a user opens your Flet app.

To serialize auth token to JSON:

```
jt = page.auth.token.to_json()
```

caution

Encrypt sensitive data before sending it to a client storage.

Flet includes utility methods for encrypting text data using symmetric algorithm (where the same key is used for encryption and decryption). They use Fernet implementation from cryptography package, which is AES 128 with some additional hardening, plus PBKDF2 to derive encryption key from a user passphrase.

To encrypt JSON token:

```
import os
from flet.security import encrypt, decrypt

secret_key = os.getenv("MY_APP_SECRET_KEY")
# returns base64-encoded string
ejt = encrypt(jt, secret_key)
```

caution

Notice, we are fetching a secret key (aka passphrase, password, etc.) from an environment variable. Do not embed any secrets into source code to avoid accidental exposure to a public!

Before running the app set the secret in a command line:

```
$ export MY_APP_SECRET_KEY="<secret>"
```

Now, encrypted value can be stored in a client storage:

```
await page.shared_preferences.set("myapp.auth_token", ejt)
```

Next time a user opens the app you can read encrypted token from a client storage and, if it exists, decrypt it and use in `page.login()` method:

```
ejt = await page.shared_preferences.get("myapp.auth_token")
if ejt:
    jt = decrypt(ejt, secret_key)
    page.login(provider, saved_token=jt)
```

See complete app example.

Signing out

Calling `page.logout()` resets `page.auth` reference and triggers `page.on_logout` event handlers.

You can remove saved token in logout method, for example:

```
async def logout_button_click(e):
    await page.shared_preferences.remove(AUTH_TOKEN_KEY)
    page.logout()
```

See complete app example.

Customizing authorization flow

By default, OAuth authorization flow is happening in either new browser window/tab (desktop app), browser popup window (web) or in-app web view (mobile).

Upon completion of authorization flow a user is redirected to Flet's OAuth callback page (`/oauth_callback`) which tries to close a browser window/tab with JavaScript and provides user with instructions to close the window manually if JavaScript close didn't work.

This section is applicable to Flet desktop and web apps only as in-app web view in mobile apps can be closed by Flet without relying on JavaScript.

You can customize the contents of "Authorization complete" page in `page.login()` method, for example:

```
complete_page_html = """
<!DOCTYPE html>
<html>
  <head>
    <title>Signed in to MyApp</title>
  </head>
  <body>
    <script type="text/javascript">
      window.close();
    </script>
    <p>You've been successfully signed in! You can close this tab or window now.</p>
  </body>
</html>
"""

page.login(
    provider,
    complete_page_html=complete_page_html,
)
```

You can also change web app to open provider's authorization page in the same tab which might be more familiar to your users and save them from dealing with popup blockers:

```
page.login(
    provider,
    on_open_authorization_url=lambda url: asyncio.create_task(page.launch_url(url, web_window_name="_self")),
    redirect_to_page=True
)
```

To open flow in a new tab (notice `_self` replaced with `_blank`):

```
page.login(
    provider,
    on_open_authorization_url=lambda url: asyncio.create_task(page.launch_url(url, web_window_name="_blank"))
)
```


Configuring a custom OAuth provider

You can configure any OAuth-compatible authentication provider in your app with `flet.auth.oauth_provider.OAuthProvider` class.

Following the instructions in LinkedIn Authorization Code Flow guide we are able to get all required parameters to configure LinkedIn OAuth provider to allow users of your Flet app to login with their LinkedIn accounts:

```
import os

import flet
from flet import Button, Page
from flet.auth import OAuthProvider

def main(page: Page):
    provider = OAuthProvider(
        client_id=os.getenv("LINKEDIN_CLIENT_ID"),
        client_secret=os.getenv("LINKEDIN_CLIENT_SECRET"),
        authorization_endpoint="https://www.linkedin.com/oauth/v2/authorization",
        token_endpoint="https://www.linkedin.com/oauth/v2/accessToken",
        user_endpoint="https://api.linkedin.com/v2/me",
        user_scopes=["r_liteprofile", "r_emailaddress"],
        user_id_fn=lambda u: u["id"],
        redirect_url="http://localhost:8550/oauth_callback",
    )

    def login_click(e):
        page.login(provider)

    def on_login(e):
        if e.error:
            raise RuntimeError(e.error)
        print("User ID:", page.auth.user.id)
        print("Access token:", page.auth.token.access_token)

    page.on_login = on_login
    page.add(Button("Login with LinkedIn", on_click=login_click))

flet.app(main, port=8550, view=flet.WEB_BROWSER)
```

Mandatory provider settings:

- `client_id`
- `client_secret`
- `authorization_endpoint`
- `token_endpoint`
- `redirect_url`

Similar to other examples client ID and client secret are fetched from environment variables.

Client Storage

Flet's client storage API allows storing key-value data on a client side in a persistent storage. Flet implementation uses `shared_preferences` Flutter package.

The actual storage mechanism depends on a platform where Flet app is running:

- Web - Local storage.
- Desktop - JSON file.
- iOS - `NSUserDefaults`.
- Android - `SharedPreferences`.

Writing data to the storage:

```
# strings
await page.shared_preferences.set("key", "value")

# numbers, booleans
await page.shared_preferences.set("number_setting", 12345)
await page.shared_preferences.set("bool_setting", True)

# lists
await page.shared_preferences.set("favorite_colors", ["red", "green", "blue"])
```

note

Each Flutter application using `shared_preferences` plugin has its own set of preferences. As the same Flet client (which is a Flutter app) is used to run UI for multiple Flet apps any values stored in one Flet application are visible/available to another Flet app running by the same user.

To distinguish one application settings from another it is recommended to use some unique prefix for all storage keys, for example `{company}.{product}`.. For example to store auth token in one app you could use `acme.one_app.auth_token` key and in another app use `acme.second_app.auth_token`.

Caution

It is responsibility of Flet app developer to encrypt sensitive data before sending it to a client storage, so it's not read/tampered by another app or an app user.

Reading data:

```
# The value is automatically converted back to the original type
value = await page.shared_preferences.get("key")

colors = await page.shared_preferences.get("favorite_colors")
# colors = ["red", "green", "blue"]
```

Check if a key exists:

```
await page.shared_preferences.contains_key("key") # True if the key exists
```

Get all keys:

```
await page.shared_preferences.get_keys("key-prefix.")
```

Remove a value:

```
await page.shared_preferences.remove("key")
```

Clear the storage:

```
await page.shared_preferences.clear()
```

Caution

`clear()` is a dangerous function that removes all preferences of all Flet apps ever run by the same user and serves as a heads-up that permanent application data shouldn't be stored in the client storage.

Working with Colors

Image:  - https://flet.dev/img/docs/colors/color_palettes.png

Flet supports two primary ways of defining colors:

- Hexadecimal values
- Named colors

Hexadecimal Values

Hex color values follow either of these formats:

- #aarrgbbb or 0xaarrgbbb
- #rrgbbb or 0xrrgbbb (opacity defaults to ff if aa is omitted)

```
>>> ft.Container(bgcolor="#ff0000")
```

Named Color

Flet uses Material Design theme colors and color palettes.

You can define named colors in two ways:

- As a string (e.g., "blue", "redAccent100")
- Using the Colors or CupertinoColors enums for better type safety and autocompletion

```
>>> ft.Container(bgcolor="yellow")
>>> ft.Container(bgcolor=ft.Colors.YELLOW)
>>> ft.Container(bgcolor=ft.CupertinoColors.DESTRUCTIVE_RED)
```

Theme Colors

Image:  - https://flet.dev/img/docs/colors/theme_colors.png

There are 30 named theme colors in Theme.color_scheme that are generated based on the Theme.color_scheme_seed, which defaults to Colors.BLUE.

```
>>> page.theme = ft.Theme(color_scheme_seed=ft.Colors.GREEN)
```

Any of the 30 theme scheme colors can be overridden, in which case they will have an absolute value that will no longer be dependent on the seed color.

```
page.theme = ft.Theme(
    color_scheme=ft.ColorScheme(
        primary=ft.Colors.GREEN,
        error=ft.Colors.RED,
    ),
)
```

Image:  - https://flet.dev/img/docs/colors/theme_colors_green.png

Theme colors serve as fallback values for most Flet controls.

Color Palettes

Image:  - https://flet.dev/img/docs/colors/color_palettes_2.png

Originally introduced by Material Design in 2014, color palettes are collections of coordinated colors designed to work together harmoniously.

Each color swatch (palette) includes multiple shades of a specific color, where lower numbers represent lighter shades and higher numbers represent darker ones. Most palettes range from 50 to 900, in increments of 100, while accent swatches (e.g., Colors.RED_ACCENT) have only 100, 200, 400, and 700.

In addition to color swatches, Flet provides named black and white variants with built-in opacities, such as:

- Colors.BLACK_54 → black at 54% opacity
- Colors.WHITE_70 → white at 70% opacity

These palette colors can be used:

- directly as values for control color properties (e.g., bgcolor, color)

- as seed colors for generating dynamic theme color schemes using `Theme.color_scheme_seed`

Color opacity

Flet provides multiple ways to set opacity:

Using `with_opacity`

This method is present in both `Colors` and `CupertinoColors` enums. It takes a color and an opacity, and returns a string in the format "color,opacity", understood by Flet.

```
>>> ft.Colors.with_opacity(0.5, ft.Colors.RED)
"red,0.5"
>>> ft.CupertinoColors.with_opacity(0.8, ft.CupertinoColors.LINK)
"link,0.8"
```

Embedding Opacity in Hex Code

For colors in hex format, you can specify the aa channel with values between 00 and ff.

```
>>> "#7fff6666"
```

Defining colors for Flet controls

Most Flet controls have default colors defined by the `Theme.color_scheme`, and these can be overridden at various levels.

Image:  - https://flet.dev/img/docs/colors/colors_fallback.svg

Control Level

If a color is provided as value for a control's color-like property, it will be used directly.

```
>>> ft.Card(bgcolor=ft.Colors.GREEN_200)
```

Note: Not every Flet control has a color property that can be set on the control level. For example, `FilledButton` always has a default "primary" color that is defined by the nearest ancestor's theme.

Control Theme Level

The `Theme` object has a lot of properties that can be used to override default colors for Flet controls.

For example, the nearest `Theme.card_bgcolor` will be used for the `Card` control.

Note: If you need to change theme for particular descendants, you can wrap them in a `Container`, for example, and customize its theme property, which will be applied to all its descendants.

Ancestor Theme Level

Flet searches upward in the widget tree to find the nearest ancestor with a defined theme, and will take color from its `Theme.color_scheme`. In the example below, the nearest ancestor for the `FilledButton` is `Container`, and the primary color that is used for the button will be taken from the `Container.theme`.

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.Container(
            width=200,
            height=200,
            border=ft.Border.all(1, ft.Colors.BLACK),
            content=ft.FilledButton("Primary color"),
            theme=ft.Theme(color_scheme=ft.ColorScheme(primary=ft.Colors.YELLOW))
        )
    )

ft.run(main)
```

If no theme is defined for a control, its parent, or its parent's ancestors, the control defaults to using the uppermost ancestor's theme, which is `Page.theme` (or `Page.dark_theme` in dark mode).

Control Refs

Flet controls are objects and to access their properties we need to keep references (variables) to those objects.

Consider the following example:

```
import flet as ft

def main(page):
    first_name = ft.TextField(label="First name", autofocus=True)
    last_name = ft.TextField(label="Last name")
    greetings = ft.Column()

    async def btn_click(e):
        greetings.controls.append(ft.Text(f"Hello, {first_name.value} {last_name.value}!"))
        first_name.value = ""
        last_name.value = ""
        page.update()
        await first_name.focus()

    page.add(
        first_name,
        last_name,
        ft.Button("Say hello!", on_click=btn_click),
        greetings,
    )

ft.run(main)
```

In the very beginning of `main()` method we create three controls which we are going to use in button's `on_click` handler: two `TextField` for first and last names and a `Column` - container for greeting messages. We create controls with all their properties set and in the end of `main()` method, in `page.add()` call, we use their references (variables).

When more and more controls and event handlers are added it becomes challenging to keep all control definitions in one place, so they become scattered across `main()` body. Glancing at `page.add()` parameters it's hard to imagine (without constant jumping to variable definitions in IDE) what would the end form look like:

```
page.add(
    first_name,
    last_name,
    ft.Button("Say hello!", on_click=btn_click),
    greetings,
)
```

Is `first_name` a `TextField`, does it have `autofocus` set? Is `greetings` a `Row` or a `Column`?

Ref class

Flet provides `Ref` utility class which allows to define a reference to the control, use that reference in event handlers and set the reference to a real control later, while building a tree. The idea comes from React.

To define a new typed control reference:

```
first_name = ft.Ref[ft.TextField]()
```

To access referenced control (control de-reference) use `Ref.current` property:

```
# empty first name
first_name.current.value = ""
```

To assign control to a reference set `Control.ref` property to a reference:

```
page.add(
    ft.TextField(ref=first_name, label="First name", autofocus=True)
)
```

note

All Flet controls have `ref` property.

We could re-write our program to use references:

```
import flet as ft

def main(page):
    first_name = ft.Ref[ft.TextField]()
    last_name = ft.Ref[ft.TextField]()
    greetings = ft.Ref[ft.Column]()

    async def btn_click(e):
        greetings.current.controls.append(
```

```

        ft.Text(f"Hello, {first_name.current.value} {last_name.current.value}!")
    )
    first_name.current.value = ""
    last_name.current.value = ""
    page.update()
    await first_name.current.focus()

page.add(
    ft.TextField(ref=first_name, label="First name", autofocus=True),
    ft.TextField(ref=last_name, label="Last name"),
    ft.Button("Say hello!", on_click=btn_click),
    ft.Column(ref=greetings),
)

ft.run(main)

```

Now we can clearly see in `page.add()` the structure of the page and all the controls it's built of.

Yes, the logic becomes a little bit more verbose as you need to add `.current.` to access ref's control, but it's a matter of personal preference :)

Custom Controls

While Flet provides 140+ built-in controls that can be used on their own, the real beauty of programming with Flet is that all those controls can be utilized for creating your own reusable UI components using Python object-oriented programming concepts.

You can create custom controls in Python by styling and/or combining existing Flet controls.

Styled controls

The most simple custom control you can create is a styled control, for example, a button of a certain color and behaviour that will be used multiple times throughout your app.

To create a styled control, you need to create a new dataclass in Python that inherits from the Flet control you are going to customize, Button in this case:

```
@ft.control
class MyButton(ft.Button):
    bgcolor: ft.Colors = ft.Colors.ORANGE_300
    color: ft.Colors = ft.Colors.GREEN_800
    style: ft.ButtonStyle = field(
        default_factory=lambda: ft.ButtonStyle(
            shape=ft.RoundedRectangleBorder(radius=10)
        )
    )
    expand: int = 1
```

You can define either `@dataclass` or `@ft.control` decorator on the inherited class - both methods works the same:

```
@dataclass
class MyButton(ft.Button):
    bgcolor: ft.Colors = ft.Colors.ORANGE_300
    color: ft.Colors = ft.Colors.GREEN_800
    style: ft.ButtonStyle = field(
        default_factory=lambda: ft.ButtonStyle(
            shape=ft.RoundedRectangleBorder(radius=10)
        )
    )
    expand: int = 1
```

You can also set properties in `init()` method:

```
@ft.control
class MyButton(ft.Button):
    def init(self):
        self.bgcolor = ft.Colors.ORANGE_300
        self.color = ft.Colors.GREEN_800
        self.style = ft.ButtonStyle(
            shape=ft.RoundedRectangleBorder(radius=10)
        )
        self.expand = 1
```

Rules

You should define either `@dataclass` or `@ft.control` decorator on the inherited class - both methods works the same.

A field, to be a class field, must have a type annotation, so `expand: int = 1` will override inherited property, but `expand = 1` won't. Not sure which type to use? Just Any will work, for example `expand: Any = 1`.

Use literal default value for simple data types, such as `int`, `bool`, `str` and `field(default_factory=lambda: <new_value>)` for immutable types such as `class`, `list`, `dict`.

You can set property values in `init()` method, but you won't be able to override if write `MyButton3(expand=False)`.

Now you can use your brand-new control in your app:

```
import flet as ft

@ft.control
class MyButton(ft.Button):
    bgcolor: ft.Colors = ft.Colors.ORANGE_300
    color: ft.Colors = ft.Colors.GREEN_800

def main(page: ft.Page):
    def ok_clicked(e):
        print("OK clicked")

    def cancel_clicked(e):
        print("Cancel clicked")

    page.add(
```

```

        MyButton(content="OK", on_click=ok_clicked),
        MyButton(content="Cancel", on_click=cancel_clicked),
    )

```

```
ft.run(main)
```

Image: Styled controls - <https://flet.dev/docs/assets/cookbook/custom-controls/styled-controls.png>

See example of using styled controls in Calculator App tutorial.

Composite controls

Composite custom controls inherit from container controls such as Column, Row, Stack or even View to combine multiple Flet controls. The example below is a Task control that can be used in a To-Do app:

```

import flet as ft

@ft.control
class Task(ft.Row):
    text: str = ""

    def init(self):
        self.text_view = ft.Text(value=self.text)
        self.text_edit = ft.TextField(value=self.text, visible=False)
        self.edit_button = ft.IconButton(icon=ft.Icons.EDIT, on_click=self.edit)
        self.save_button = ft.IconButton(
            visible=False, icon=ft.Icons.SAVE, on_click=self.save
        )
        self.controls = [
            ft.Checkbox(),
            self.text_view,
            self.text_edit,
            self.edit_button,
            self.save_button,
        ]

    def edit(self, e):
        self.edit_button.visible = False
        self.save_button.visible = True
        self.text_view.visible = False
        self.text_edit.visible = True
        self.update()

    def save(self, e):
        self.edit_button.visible = True
        self.save_button.visible = False
        self.text_view.visible = True
        self.text_edit.visible = False
        self.text_view.value = self.text_edit.value
        self.update()

def main(page: ft.Page):
    page.add(
        Task(text="Do laundry"),
        Task(text="Cook dinner"),
    )

ft.run(main)

```

Image: Composite controls - <https://flet.dev/docs/assets/cookbook/custom-controls/composite-controls.gif>

Life-cycle methods

Custom controls provide life-cycle "hook" methods that you may need to use for different use cases in your app.

build()

build() method is called when the control is being created and assigned its self.page.

Override build() method if you need to implement logic that cannot be executed in control's constructor because it requires access to the self.page. For example, choose the right icon depending on self.page.platform for your adaptive app.

did_mount()

did_mount() method is called after the control is added to the page and assigned transient uid.

Override did_mount() method if you need to implement logic that needs to be executed after the control was added to the page, for example Weather widget which calls Open Weather API every minute to update itself with the new weather conditions.

will_unmount()

`will_unmount()` method is called before the control is removed from the page.

Override `will_unmount()` method to execute clean-up code.

`before_update()`

`before_update()` method is called every time when the control is being updated.

Make sure not to call `update()` method within `before_update()`.

Isolated controls

Custom control has `is_isolated` property which defaults to `False`.

If you set `is_isolated` to `True`, your control will be isolated from outside layout, i.e. when `update()` method is called for the parent control, the control itself will be updated but any changes to the controls' children are not included into the update digest. Isolated controls should call `self.update()` to push its changes to a Flet page.

As a best practice, any custom control that calls `self.update()` inside its class methods should be isolated.

In the above examples, simple styled `MyButton` doesn't need to be isolated, but the `Task` should be:

```
class Task(ft.Row):
    def __init__(self, text):
        super().__init__()

    def is_isolated(self):
        return True
```

Declarative dialogs

`ft.use_dialog()` lets a component show and update dialogs declaratively. Instead of imperatively calling `page.show_dialog()` and later remembering to close or remove the dialog, you render a `DialogControl` from component state:

- pass a dialog instance to show it;
- pass `None` to hide it.

This keeps dialog logic in the same state flow as the rest of a declarative app:

- state decides whether the dialog is visible;
- dialog content updates when state changes;
- there is no `page.update()` call in the component.

Basic pattern

Call `ft.use_dialog()` on every render. When the dialog should be open, return a dialog control; otherwise return `None`.

```
import flet as ft

@ft.component
def App():
    show, set_show = ft.use_state(False)

    ft.use_dialog(
        ft.AlertDialog(
            modal=True,
            title=ft.Text("Delete report.pdf?"),
            content=ft.Text("This cannot be undone."),
            actions=[
                ft.TextButton("Delete", on_click=lambda: set_show(False)),
                ft.TextButton("Cancel", on_click=lambda: set_show(False)),
            ],
            on_dismiss=lambda: set_show(False),
        )
        if show
        else None
    )

    return ft.Column(
        controls=[
            ft.TextButton("Open dialog", on_click=lambda: set_show(True)),
        ]
    )

ft.run(lambda page: page.render(App))
```

The important part is that `show` is the source of truth. The dialog is not opened by mutating the page tree directly; it appears because the component renders it.

Updating dialog content from state

Because the dialog is declarative, its content can react to state changes while it is open. This is useful for confirmations, form validation, and async workflows:

```
import asyncio
import flet as ft

@ft.component
def App():
    show, set_show = ft.use_state(False)
    deleting, set_deleting = ft.use_state(False)

    async def handle_delete():
        set_deleting(True)
        await asyncio.sleep(2)
        set_deleting(False)
        set_show(False)

    ft.use_dialog(
        ft.AlertDialog(
            modal=True,
            title=ft.Text("Delete report.pdf?"),
```

```

        content=ft.Text(
            "Deleting, please wait..." if deleting else "This cannot be undone."
        ),
        actions=[
            ft.Button(
                "Deleting..." if deleting else "Delete",
                disabled=deleting,
                on_click=handle_delete,
            ),
            ft.TextButton(
                "Cancel",
                disabled=deleting,
                on_click=lambda: set_show(False),
            ),
        ],
        on_dismiss=lambda: set_show(False),
    )
    if show
    else None
)

return ft.TextButton("Delete file", on_click=lambda: set_show(True))

```

```
ft.run(lambda page: page.render(App))
```

This pattern works well with `asyncio` and other async APIs in Flet apps. For more background, see [Async apps](#).

Chaining dialogs

You can call `ft.use_dialog()` more than once in the same component. That makes follow-up flows straightforward, for example:

- confirmation dialog;
- async action;
- success dialog after the first dialog fully closes.

```
import flet as ft
```

```

@ft.component
def App():
    show_confirm, set_show_confirm = ft.use_state(False)
    show_success, set_show_success = ft.use_state(False)
    should_chain = ft.use_ref(False)

    def confirm_delete():
        should_chain.current = True
        set_show_confirm(False)

    def on_confirm_dismiss():
        if should_chain.current:
            should_chain.current = False
            set_show_success(True)

    ft.use_dialog(
        ft.AlertDialog(
            title=ft.Text("Delete file?"),
            actions=[
                ft.TextButton("Delete", on_click=confirm_delete),
                ft.TextButton("Cancel", on_click=lambda: set_show_confirm(False)),
            ],
            on_dismiss=on_confirm_dismiss,
        )
        if show_confirm
        else None
    )

    ft.use_dialog(
        ft.AlertDialog(
            title=ft.Text("Done"),
            content=ft.Text("The file was deleted."),
            actions=[
                ft.TextButton("OK", on_click=lambda: set_show_success(False)),
            ],
        )
        if show_success
        else None
    )

    return ft.TextButton("Open", on_click=lambda: set_show_confirm(True))

ft.run(lambda page: page.render(App))

```

`ft.use_ref()` is helpful here because the value survives re-renders without causing another render by itself.

on_dismiss timing

`DialogControl.on_dismiss` fires after the dialog close animation completes, not immediately when `open` changes to `False`. This makes it safe to start follow-up UI after the dialog has actually finished closing.

Use `on_dismiss` for logic that should happen after the dialog is fully gone, such as:

- opening the next dialog in a chain;
- resetting temporary dialog-local state;
- starting a follow-up animation or toast.

When to use `page.show_dialog()` instead

`ft.use_dialog()` is a better fit inside `@ft.component` functions and other declarative flows.

`page.show_dialog()` is still a good option when:

- the app is written imperatively;
- dialog lifecycle is handled outside the component tree;
- you need to trigger a dialog from existing page-level event code and do not want to convert that part of the app to declarative style yet.

In practice, the two APIs serve different styles:

- use declarative dialogs when UI should follow component state;
- use imperative dialogs when UI is managed by direct page mutation.

From Imperative to Declarative in Flet: Migrating a Simple CRUD “User Manager”

If you’ve been using Flet, you’ve probably built your apps the imperative way, maybe without even noticing. You flip visibility flags, set control values, update lists of controls and call `page.update()` - that is the imperative approach, meaning you change UI directly when handling events. Flet now supports a declarative style: stop mutating controls, change state instead, and Flet updates the UI automatically.

We’ll show the switch using a tiny CRUD “User Manager” app. First, the imperative version: UI-first, mutate controls, then update the page. Then the declarative rewrite: model-first, observable classes for data, components that return UI from state.

The behavior in both examples stays the same - in the app you can see the list of users, add user, inline edit with save/cancel buttons and delete. This is how this simple app looks in both examples:

Image: view - <https://flet.dev/docs/assets/cookbook/declarative-vs-imperative-crud-app/crud1.png>

After clicking inline Edit button:

Image: view - <https://flet.dev/docs/assets/cookbook/declarative-vs-imperative-crud-app/crud2.png>

Example 1 — Imperative

In the imperative version, you think UI-first: decide exactly how the screen should look, and how it should change on each button click. Event handlers directly toggle control properties (like visible, value), insert/remove controls, and then call `page.update()` to push those visual changes. Edit hides the read-only label and action buttons, shows inputs and Save/Cancel; Save copies text field values back into the label and restores the original view; Cancel just restores the original view; Delete removes the whole row from the page. In short, behavior is implemented by mutating controls and manually triggering re-renders, not by evolving a separate state model.

```
import flet as ft

class Item(ft.Row):
    def __init__(self, first_name, last_name):
        super().__init__()

        self.first_name_field = ft.TextField(first_name)
        self.last_name_field = ft.TextField(last_name)
        self.text = ft.Text(f"{first_name} {last_name}")
        self.edit_text = ft.Row(
            [
                self.first_name_field,
                self.last_name_field,
            ],
            visible=False,
        )
        self.edit_button = ft.Button("Edit", on_click=self.edit_item)
        self.delete_button = ft.Button("Delete", on_click=self.delete_item)
        self.save_button = ft.Button("Save", on_click=self.save_item, visible=False)
        self.cancel_button = ft.Button(
            "Cancel", on_click=self.cancel_item, visible=False
        )
        self.controls = [
            self.text,
            self.edit_text,
            self.edit_button,
            self.delete_button,
            self.save_button,
            self.cancel_button,
        ]

    def delete_item(self, e):
        self.page.controls.remove(self)
        self.page.update()

    def edit_item(self, e):
        print("edit_item")
        self.text.visible = False
        self.edit_button.visible = False
        self.delete_button.visible = False
        self.save_button.visible = True
        self.cancel_button.visible = True
        self.edit_text.visible = True
        self.page.update()

    def save_item(self, e):
```

```

        self.text.value = f"{self.first_name_field.value} {self.last_name_field.value}"
        self.text.visible = True
        self.edit_button.visible = True
        self.delete_button.visible = True
        self.save_button.visible = False
        self.cancel_button.visible = False
        self.edit_text.visible = False
        self.page.update()

    def cancel_item(self, e):
        self.text.visible = True
        self.edit_button.visible = True
        self.delete_button.visible = True
        self.save_button.visible = False
        self.cancel_button.visible = False
        self.edit_text.visible = False
        self.page.update()

def main(page: ft.Page):
    page.title = "CRUD Imperative Example"

    def add_item(e):
        item = Item(first_name.value, last_name=last_name.value)
        page.add(item)
        first_name.value = ""
        last_name.value = ""
        page.update()

    first_name = ft.TextField(label="First Name", width=200)
    last_name = ft.TextField(label="Last Name", width=200)

    page.add(
        ft.Row(
            [
                first_name,
                last_name,
                ft.Button("Add", on_click=add_item),
            ]
        )
    )

ft.run(main)

```

Example 2 — Declarative

In the declarative version, you think model-first: the model is a set of classes, and the data their objects hold is the single source of truth. In our CRUD app, the model consists of `User` (persisted fields `first_name`, `last_name`) and a top-level App that owns users: `list[User]` plus actions like `add_user(first, last)` and `delete_user(user)`. Both classes are marked `@ft.observable`, so assigning to their attributes (e.g., `user.update(...)`, `app.users.remove(user)`) triggers re-rendering — no `page.update()`.

The UI is composed as components marked with `@ft.component` that return a view of the current state. Each row decides whether to show a read-only view or an inline editor using its own short-lived, local values (hooks), while the durable data lives on the model objects. Event handlers update state only (e.g., modify a user or add/remove items), not the controls themselves; Flet detects those changes and re-renders the affected parts. In short: `UI = f(state)`, with `User` and `App` providing the authoritative data.

Image: diagram - <https://flet.dev/docs/assets/cookbook/declarative-vs-imperative-crud-app/crud-declarative.drawio.png>

```

from dataclasses import dataclass, field

import flet as ft

@ft.observable
@dataclass
class User:
    first_name: str
    last_name: str

    def update(self, first_name: str, last_name: str):
        self.first_name = first_name
        self.last_name = last_name

@ft.observable
@dataclass
class App:
    users: list[User] = field(default_factory=list)

    def add_user(self, first_name: str, last_name: str):
        if first_name.strip() or last_name.strip():
            self.users.append(User(first_name, last_name))

    def delete_user(self, user: User):

```

```

        self.users.remove(user)

@ft.component
def UserView(user: User, delete_user) -> ft.Control:
    # Local (transient) editing state—NOT in User
    is_editing, set_is_editing = ft.use_state(False)
    new_first_name, set_new_first_name = ft.use_state(user.first_name)
    new_last_name, set_new_last_name = ft.use_state(user.last_name)

    def start_edit():
        set_new_first_name(user.first_name)
        set_new_last_name(user.last_name)
        set_is_editing(True)

    def save():
        user.update(new_first_name, new_last_name)
        set_is_editing(False)

    def cancel():
        set_is_editing(False)

    if not is_editing:
        return ft.Row(
            [
                ft.Text(f"{user.first_name} {user.last_name}"),
                ft.Button("Edit", on_click=start_edit),
                ft.Button("Delete", on_click=lambda: delete_user(user)),
            ]
        )

    return ft.Row(
        [
            ft.TextField(
                label="First Name",
                value=new_first_name,
                on_change=lambda e: set_new_first_name(e.control.value),
                width=180,
            ),
            ft.TextField(
                label="Last Name",
                value=new_last_name,
                on_change=lambda e: set_new_last_name(e.control.value),
                width=180,
            ),
            ft.Button("Save", on_click=save),
            ft.Button("Cancel", on_click=cancel),
        ]
    )

@ft.component
def AddUserForm(add_user) -> ft.Control:
    # Uses local buffers; calls parent action on Add
    new_first_name, set_new_first_name = ft.use_state("")
    new_last_name, set_new_last_name = ft.use_state("")

    def add_user_and_clear():
        add_user(new_first_name, new_last_name)
        set_new_first_name("")
        set_new_last_name("")

    return ft.Row(
        controls=[
            ft.TextField(
                label="First Name",
                width=200,
                value=new_first_name,
                on_change=lambda e: set_new_first_name(e.control.value),
            ),
            ft.TextField(
                label="Last Name",
                width=200,
                value=new_last_name,
                on_change=lambda e: set_new_last_name(e.control.value),
            ),
            ft.Button("Add", on_click=add_user_and_clear),
        ]
    )

@ft.component
def AppView() -> list[ft.Control]:
    app, _ = ft.use_state(
        App(
            users=[
                User("John", "Doe"),
                User("Jane", "Doe"),
                User("Foo", "Bar"),
            ]
        )
    )

```

```

    )
)

return [
    AddUserForm(app.add_user),
    *[UIView(user, app.delete_user) for user in app.users],
]

ft.run(lambda page: page.render(AppView))

```

Mindset shift: UI = f(state)

The core idea is determinism: given the same state, your component should return the same UI. Think in two phases:

Handle event → update state Event handlers change data only (e.g., `set_is_editing(True)`, `user.update(...)`). They don't hide/show controls or call `page.update()`.

Render → derive UI from state The component returns controls based on the current state snapshot. Because models are `@ft.observable` and locals come from `ft.use_state`, Flet re-runs the component when state changes and re-renders the right subtree.

Declarative Building Blocks: Observables, Components, Hooks

Below are the key pieces of the Flet framework that make the declarative approach work:

Observables — your source of truth

`@ft.observable` marks a dataclass as reactive. When you assign to its fields (`user.first_name = "Ada"` or `app.users.append(user)`), Flet re-renders any components that read those fields - no `page.update()` calls. Use observables for persisted/domain data (things you actually save).

```

from dataclasses import dataclass, field
import flet as ft

@ft.observable
@dataclass
class User:
    first_name: str
    last_name: str

@ft.observable
@dataclass
class AppState:
    users: list[User] = field(default_factory=list)

```

Components — functions that return UI

`@ft.component` turns a function into a rendering unit. It takes props (regular args), may use hooks, and returns controls that describe the UI for the current state. Components do not imperatively mutate the page tree; they just return what the UI should look like now.

```

import flet as ft

@ft.component
def UserRow(user: User, on_delete) -> ft.Control:
    # returns a row for the current snapshot of `user`
    return ft.Row([
        ft.Text(f"{user.first_name} {user.last_name}"),
        ft.Button("Delete", on_click=lambda _: on_delete(user)),
    ])

```

Hooks — local, short-lived UI state

Why they are needed: components are functions that re-run on every render. Plain local variables get reinitialized each time, and changing them doesn't tell Flet to update the view. Hooks (e.g., `ft.use_state`) give a component a place to persist values across renders and a way to signal a re-render when those values change.

What hooks solve

- Persistence: locals reset on each render; hook state survives.
- Reactivity: modifying a local doesn't refresh the UI; a hook's setter schedules a re-render.
- Fresh values in handlers: event callbacks won't see stale locals; they read the latest hook state.

Use hooks for short-lived, view-only concerns (like an "is editing?" flag or current input text) that belong to a single component. Use observables for durable app/domain data shared across components.

Example


```
# Broken: local resets every render and doesn't trigger updates
@ft.component
def CounterBroken():
    count = 0
    return ft.Row([
        ft.Text(str(count)),
        ft.Button("+", on_click=lambda _: (count := count + 1)), # no re-render
    ])

# Correct: persists across renders and re-renders when updated
@ft.component
def Counter():
    count, set_count = ft.use_state(0)
    return ft.Row([
        ft.Text(str(count)),
        ft.Button("+", on_click=lambda _: set_count(count + 1)),
    ])
```

Rule of thumb: if a value must survive re-renders and updating it should change the UI, don't use a plain local — use hook state (for local UI) or an observable (for shared, persisted data).

Rewrite recipes (imperative → declarative)

1) Visibility toggles → Conditional rendering

```
# Imperative
self.text.visible = False
self.save_button.visible = True
self.page.update()

# Declarative
return (
    ft.Row([...read-only...])
    if not is_editing
    else ft.Row([...edit form...])
)
```

2) Direct control mutation → Model mutation

```
# Imperative
self.text.value = f"{first} {last}"

# Declarative
user.update(new_first_name, new_last_name)
```

3) page.update() everywhere → Nowhere

- Imperative handlers end with page.update().
- Declarative code updates observable fields or use_state values and lets Flet re-render.

4) Handlers manipulate state, not the view

```
# Declarative example
set_is_editing(True)
set_new_first_name(user.first_name)
```

5) Extract UI into components

- UserView = one row (read-only/editing)
- AddUserForm = small, reusable add form

Summary

The declarative style makes your UI a straightforward function of your data. It may not make a big difference for a very simple app, but as your screen grows, you'll add state and components, not scattered mutations of controls in different places. The result: code that's easier to understand, maintain, and change — without chasing visible flags or manual updates.

Drag and Drop

The mechanics of drag-and-drop in Flet is pretty simple - a user starts dragging Draggable control and "drops" it on DragTarget. If both draggable and drag target has the same group a drag target will call on_accept event handler and pass draggable control ID as event data. In this case draggable serves as a source "data" for drag-and-drop operation.

Let's take a look at the following example. In the program below you can drag left control displaying "1" on top of the right control displaying "0" and when drag operation completes left control is replaced with "0" and the right control becomes "1":

```
import flet as ft

def main(page: ft.Page):
    page.title = "Drag and Drop example"

    def drag_accept(e):
        # update text inside draggable (source) control
        e.src.content.content.value = "0"
        # update text inside drag target control
        e.control.content.content.value = "1"
        page.update()

    page.add(
        ft.Row(
            controls=[
                ft.Draggable(
                    group="number",
                    content=ft.Container(
                        width=50,
                        height=50,
                        bgcolor=ft.Colors.CYAN_200,
                        border_radius=5,
                        content=ft.Text("1", size=20),
                        alignment=ft.Alignment.CENTER,
                    ),
                ),
                ft.Container(width=100),
                ft.DragTarget(
                    group="number",
                    content=ft.Container(
                        width=50,
                        height=50,
                        bgcolor=ft.Colors.PINK_200,
                        border_radius=5,
                        content=ft.Text("0", size=20),
                        alignment=ft.Alignment.CENTER,
                    ),
                    on_accept=drag_accept,
                ),
            ],
        ),
    )

ft.run(main)
```

Image: Drag and drop number - <https://flet.dev/docs/assets/cookbook/drag-and-drop/image-1.gif>

So, it's developer's responsibility to determine what happens with "source" (draggable) and "destination" (drag target) controls when on_accept event occurs.

Try something

Change DragTarget's group property to number1 and note on_accept is not called any more when you drop "1" on the target.

There are additional properties and event handlers to make drag-and-drop operation even more interactive. For example, draggable has content_when_dragging property to display a different control instead of content when drag operation is under way. There is also content_feedback property to show a different control under the pointer. By default, the same content control, but with 50% opacity is displayed under cursor when dragging.

Let's modify Draggable in our example to display a "hole" in place of dragged control and just "1" under cursor while dragging:

```
...
    ft.Draggable(
        group="number",
        content=ft.Container(
            width=50,
            height=50,
            bgcolor=ft.Colors.CYAN_200,
            border_radius=5,
```

```

        content=ft.Text("1", size=20),
        alignment=ft.alignment.center,
    ),
    content_when_dragging=ft.Container(
        width=50,
        height=50,
        bgcolor=ft.Colors.BLUE_GREY_200,
        border_radius=5,
    ),
    content_feedback=ft.Text("1"),
),
...

```

Image: Drag and drop number 2 - <https://flet.dev/docs/assets/cookbook/drag-and-drop/image-2.gif>

Drag target control additionally has `on_will_accept` and `on_leave` event handlers which help better visualize when it's a good time to "drop" something on the target. Let's modify `DragTarget` in our example to draw a border around target control when it's ready to accept incoming drag:

```

import flet as ft

def main(page: ft.Page):
    page.title = "Drag and Drop example 2"

    def drag_accept(e):
        # update text inside draggable (source) control
        e.src.content.value = "0"
        # reset source group, so it cannot be dropped to a target anymore
        e.src.group = ""
        # update text inside drag target control
        e.control.content.value = "1"
        # reset border
        e.control.border = None
        page.update()

    def drag_will_accept(e):
        # black border when it's allowed to drop and red when it's not
        e.control.border = ft.Border.all(
            2, ft.Colors.BLACK_45 if e.data == "true" else ft.Colors.RED
        )
        e.control.update()

    def drag_leave(e):
        e.control.border = None
        e.control.update()

    page.add(
        ft.Row(
            controls=[
                ft.Draggable(
                    group="number",
                    content=ft.Container(
                        width=50,
                        height=50,
                        bgcolor=ft.Colors.CYAN_200,
                        border_radius=5,
                        content=ft.Text("1", size=20),
                        alignment=ft.Alignment.CENTER,
                    ),
                    content_when_dragging=ft.Container(
                        width=50,
                        height=50,
                        bgcolor=ft.Colors.BLUE_GREY_200,
                        border_radius=5,
                    ),
                    content_feedback=ft.Text("1"),
                ),
                ft.Container(width=100),
                ft.DragTarget(
                    group="number",
                    content=ft.Container(
                        width=50,
                        height=50,
                        bgcolor=ft.Colors.PINK_200,
                        border_radius=5,
                        content=ft.Text("0", size=20),
                        alignment=ft.Alignment.CENTER,
                    ),
                    on_accept=drag_accept,
                    on_will_accept=drag_will_accept,
                    on_leave=drag_leave,
                ),
            ],
        ),
    )

ft.run(main)

```

Image: Drag and drop number 3 - <https://flet.dev/docs/assets/cookbook/drag-and-drop/image-3.gif>

Encrypting sensitive data

Sensitive data such as tokens, keys, credit card numbers and other "secrets" must be stored at rest (database, files, client storage) in encrypted form to avoid data breaches.

Flet includes utility methods to encrypt and decrypt sensitive text data using symmetric algorithm (where the same key is used for encryption and decryption). They use Fernet implementation from cryptography package, which is AES 128 with some additional hardening, plus PBKDF2 to derive encryption key from a user passphrase.

Secret key

Encryption secret key (aka password, or passphrase) is an arbitrary password-like string configured by a user and used for encrypting and decrypting data. Crypto algorithm uses secret key to "derive" encryption key (32 bytes).

danger

Do not embed any secrets into the source code to avoid accidental exposure to the public!

You can provide a secret to your app via environment variable:

```
import os
secret_key = os.getenv("MY_APP_SECRET_KEY")
```

Before running the app set the secret in a command line:

```
$ export MY_APP_SECRET_KEY="<secret>"
```

note

While passing secrets via environment variables is a common practice amongst developers and service providers it does not fully prevent secrets leaking in some environments. Other mechanisms can be used to inject secrets to your application such as mounting secret files or vault services.

Encrypting data

Use `encrypt()` method to encrypt a string:

```
import os
from flet.security import encrypt, decrypt

secret_key = os.getenv("MY_APP_SECRET_KEY")
plain_text = "This is a secret message!"
encrypted_data = encrypt(plain_text, secret_key)
```

`encrypted_data` is a URL-safe base64-encoded string.

`encrypt` accepts strings only, so any objects must be serialized to JSON, XML or other text-based format before encryption.

Decrypting data

Use `decrypt()` method to decrypt the data:

```
import os
from flet.security import encrypt, decrypt

secret_key = os.getenv("MY_APP_SECRET_KEY")
encrypted_data = "6011lp2zpPp4QjBWe2c0wGdBQUFBQUJqTTFJbmgyWU5jb1Vp..."
plain_text = decrypt(encrypted_data, secret_key)
print(plain_text)
```

Expanding Controls

expand

When a child control is placed into a Row, Column, View or Page you can "expand" it to fill the available space.

When adding child controls to a Row, you can make them automatically grow to fill available horizontal space using the expand property, which all Flet controls (inheriting from Control) have.

It lets you control how they use free space inside a layout like Row or Column.

You can set expand to one of the following values:

- a boolean — Whether the control should take all the available space.
- an integer — Used to proportionally divide free space among multiple expanding controls (useful when you want more control over sizing).

Example 1

In this example, a TextField stretches to fill all remaining space in the row, while the Button stays sized to its content:

Expand textfield in row

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Container(
                width=480,
                padding=10,
                border=ft.Border.all(2, ft.Colors.BLUE_GREY_200),
                border_radius=10,
                content=ft.Row(
                    controls=[
                        ft.TextField(hint_text="Enter your name", expand=True),
                        ft.Button("Join chat"),
                    ]
                ),
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: expand textfield in row - https://flet.dev/docs/test-images/examples/controls/core/golden/macros/control/expand_textfield_in_row.png

Example 2

In this example, we create a Row with three Containers, distributed like 20% / 60% / 20%:

Expand row proportional 1 3 1

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Container(
                width=500,
                padding=10,
                border=ft.Border.all(2, ft.Colors.BLUE_GREY_200),
                border_radius=10,
                content=ft.Row(
                    spacing=8,
                    controls=[
                        ft.Container(
                            expand=1,
                            height=60,
                            bgcolor=ft.Colors.CYAN_300,
```

```

        alignment=ft.Alignment.CENTER,
        border_radius=8,
        content=ft.Text("1"),
    ),
    ft.Container(
        expand=3,
        height=60,
        bgcolor=ft.Colors.AMBER_300,
        alignment=ft.Alignment.CENTER,
        border_radius=8,
        content=ft.Text("3"),
    ),
    ft.Container(
        expand=1,
        height=60,
        bgcolor=ft.Colors.PINK_200,
        alignment=ft.Alignment.CENTER,
        border_radius=8,
        content=ft.Text("1"),
    ),
],
),
)

if __name__ == "__main__":
    ft.run(main)

```

Here, the available space is split into 5 total parts (1+3+1). The first and third containers get 1 part each (20%), and the middle one gets 3 parts (60%).

Image: expand row proportional 1 3 1 -
https://flet.dev/docs/test-images/examples/controls/core/golden/macros/control/expand_row_proportional_1_3_1.png

Example 3

This example demonstrates how two controls inside a Row can each expand to fill half of the available horizontal space using `expand=True`.

The layout uses a parent Container and a nested row, where both controls are expanded equally, resulting in a 50/50 split.

Expand row equal split

play_arrow Try Online

```

import flet as ft

def main(page: ft.Page):
    page.add(
        ft.SafeArea(
            content=ft.Container(
                width=500,
                height=180,
                padding=10,
                border=ft.Border.all(2, ft.Colors.BLUE_GREY_200),
                border_radius=10,
                content=ft.Row(
                    spacing=8,
                    controls=[
                        ft.Container(
                            expand=True,
                            bgcolor=ft.Colors.ORANGE_300,
                            border_radius=8,
                            alignment=ft.Alignment.CENTER,
                            content=ft.Text("Card 1"),
                        ),
                        ft.Container(
                            expand=True,
                            bgcolor=ft.Colors.GREEN_200,
                            border_radius=8,
                            alignment=ft.Alignment.CENTER,
                            content=ft.Text("Card 2"),
                        ),
                    ],
                ),
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

expand_loose

The `expand_loose` property allows a control to optionally expand and fill the available space along the main axis of its parent container (e.g., horizontally in a `Row`, or vertically in a `Column`).

Unlike `expand`, which forces the control to occupy all available space, `expand_loose` gives the control flexibility to grow if needed, but it's not required to fill the space.

Note

For this property to have effect:

- it must be used on child controls within the following layout containers, or any of their subclasses: `Row`, `Column`, `View`, `Page`
- the control using this property must have a non-`None` value for `expand`.

Example 1

In this example, Containers being placed in Rows with `expand_loose = True`:

Expand loose chat messages

[play_arrowTry Online](#)

```
from dataclasses import field

import flet as ft

@ft.control
class Message(ft.Container):
    author: str = ""
    body: str = ""
    border: ft.Border = field(default_factory=lambda: ft.Border.all(1, ft.Colors.BLACK))
    border_radius: ft.BorderRadius = field(
        default_factory=lambda: ft.BorderRadius.all(10)
    )
    bgcolor: ft.Colors = ft.Colors.GREEN_200
    padding: ft.PaddingValue = 10
    expand: bool = True
    expand_loose: bool = True

    def init(self):
        self.content = ft.Column(
            controls=[
                ft.Text(self.author, weight=ft.FontWeight.BOLD),
                ft.Text(self.body),
            ],
        )

def main(page: ft.Page):
    chat = ft.ListView(
        padding=10,
        spacing=10,
        controls=[
            ft.Row(
                alignment=ft.MainAxisAlignment.START,
                controls=[
                    Message(
                        author="John",
                        body="Hi, how are you?",
                    ),
                ],
            ),
            ft.Row(
                alignment=ft.MainAxisAlignment.END,
                controls=[
                    Message(
                        author="Jake",
                        body="Hi I am good thanks, how about you?",
                    ),
                ],
            ),
            ft.Row(
                alignment=ft.MainAxisAlignment.START,
                controls=[
                    Message(
                        author="John",
                        body="
```



```

        "Lorem Ipsum is simply dummy text of the printing and "
        "typesetting industry. Lorem Ipsum has been the industry's "
        "standard dummy text ever since the 1500s, when an unknown "
        "printer took a galley of type and scrambled it to make a "
        "type specimen book."
    ),
),
],
),
ft.Row(
    alignment=ft.MainAxisAlignment.END,
    controls=[
        Message(
            author="Jake",
            body="Thank you!",
        ),
    ],
),
],
),
)
page.add(
    ft.SafeArea(
        content=ft.Container(
            width=300,
            height=500,
            content=chat,
        ),
    ),
)

```

```

if __name__ == "__main__":
    ft.run(main)

```

Image: expand loose chat messages -
https://flet.dev/docs/test-images/examples/controls/core/golden/macros/control/expand_loose_chat_messages.png

Fonts

System fonts

You can use the (system) fonts installed on your computer, e.g. "Consolas", "Arial", "Verdana", "Tahoma", etc.

Limitation

System fonts cannot be used in Flet web apps rendered with canvas kit web renderer.

Usage Example

The following example demonstrates how to use the "Consolas" font in a Flet application.

```
import flet as ft

def main(page: ft.Page):
    page.add(
        ft.Text(
            value="This text is rendered with Consolas font",
            font_family="Consolas"
        )
    )

ft.run(main)
```

Importing Fonts

Font can be imported from external resource by providing an absolute URL or from application assets directory (see Assets Guide).

This is done by setting the page's fonts property.

To apply one of the imported fonts, you can:

- Use Theme.font_family to set a default/fallback app-wide font family.
- Specify a font for individual controls. For example, Text.font_family.

Usage Example

The example below loads the "Kanit" font from GitHub and "Open Sans" from local assets. "Kanit" is set as the default app font, while "Open Sans" is applied to a specific Text control.

```
import flet as ft

def main(page: ft.Page):
    page.fonts = {
        "Kanit": "https://raw.githubusercontent.com/google/fonts/master/ofl/kanit/Kanit-Bold.ttf",
        "Open Sans": "/fonts/OpenSans-Regular.ttf"
    }

    page.theme = Theme(font_family="Kanit") # Default app font

    page.add(
        ft.Text("This text uses the Kanit font"),
        ft.Text("This text uses the Open Sans font", font_family="Open Sans")
    )

ft.run(main, assets_dir="assets")
```

Static and Variable Fonts

Currently, only static fonts are supported. These fonts have a specific width, weight, or style combination (e.g., "Open Sans Regular").

Support for variable fonts is in progress.

However, to use variable fonts, you can create static instances at specific weights using fonttools, e.g.:

```
fonttools varLib.mutator ./YourVariableFont-VF.ttf wght=140 wdth=85
```

To explore available font features (e.g. possible options for wght) use Wakamai Fondue online tool.

Keyboard Shortcuts

A solid keyboard support is a key for user productivity while using your web and, especially, desktop app. Indeed, it could be really annoying to constantly switch between mouse and keyboard.

In addition to form controls' `.autofocus` property and `TextField.focus()` method Flet allows handling "global" keyboard events.

To capture all keystrokes implement `page.on_keyboard_event` handler. Event handler parameter `e` is an instance of `KeyboardEvent` class with the following properties:

- `key` - a textual representation of a pressed key, e.g. A, Enter or F5.
- `shift` - True if "Shift" key is pressed.
- `ctrl` - True if "Control" key is pressed.
- `alt` - True if "Alt" ("Option") key is pressed.
- `meta` - True if "Command" key is pressed.

This is a simple usage example:

```
import flet as ft

def main(page: ft.Page):
    def on_keyboard(e: ft.KeyboardEvent):
        page.add(
            ft.Text(
                f"Key: {e.key}, Shift: {e.shift}, Control: {e.ctrl}, Alt: {e.alt}, Meta: {e.meta}"
            )
        )

    page.on_keyboard_event = on_keyboard
    page.add(
        ft.Text("Press any key with a combination of CTRL, ALT, SHIFT and META keys...")
    )

ft.run(main)
```

Image: Keyboard shortcuts - <https://flet.dev/docs/assets/cookbook/keyboard/shortcuts.png>

Below is a more advanced example:

[play_arrowTry Online](#)

```
import flet as ft

class ButtonControl(ft.Container):
    def __init__(self, text):
        super().__init__()
        self.content: ft.Text = ft.Text(text)
        self.border = ft.Border.all(1, ft.Colors.BLACK_54)
        self.border_radius = 3
        self.bgcolor = "0x09000000"
        self.padding = 10
        self.visible = False

def main(page: ft.Page):
    page.spacing = 50
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def on_keyboard(e: ft.KeyboardEvent):
        key.content.value = e.key
        key.visible = True
        shift.visible = e.shift
        ctrl.visible = e.ctrl
        alt.visible = e.alt
        meta.visible = e.meta
        page.update()

    page.on_keyboard_event = on_keyboard

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        "Press any key with a combination of CTRL, ALT, SHIFT "
```

```

        "and META keys..."
    ),
    ft.Row(
        controls=[
            key := ButtonControl(""),
            shift := ButtonControl("Shift"),
            ctrl := ButtonControl("Control"),
            alt := ButtonControl("Alt"),
            meta := ButtonControl("Meta"),
        ],
        alignment=ft.MainAxisAlignment.CENTER,
    ),
]
)
)

if __name__ == "__main__":
    ft.run(main)

```

Large Lists

You can use Column and Row controls to display lists in the most cases, but if the list contains hundreds or thousands of items Column and Row will be ineffective with lagging UI as they render all items at once even they are not visible at the current scrolling position.

In the following example we are adding 5,000 text controls to a page. Page uses Column as a default layout container:

```
import flet as ft

def main(page: ft.Page):
    for i in range(5000):
        page.controls.append(ft.Text(f"Line {i}"))
    page.scroll = "always"
    page.update()

ft.run(main, view=ft.AppView.WEB_BROWSER)
```

Run the program and notice that it's not just it takes a couple of seconds to initially load and render all text lines on a page, but scrolling is slow and laggy too:

Image: Scroll column - <https://flet.dev/docs/assets/cookbook/large-lists/scroll-column.gif>

For displaying lists with a lot of items use ListView and GridView controls which render items on demand, visible at the current scrolling position only.

ListView

ListView could be either vertical (default) or horizontal. ListView items are displayed one after another in the scroll direction.

ListView already implements effective on demand rendering of its children, but scrolling performance could be further improved if you can set the same fixed height or width (for horizontal ListView) for all items ("extent"). This could be done by either setting absolute extent with item_extent property or making the extent of all children equal to the extent of the first child by setting first_item_prototype to True.

Let's output a list of 5,000 items using ListView control:

```
import flet as ft

def main(page: ft.Page):
    lv = ft.ListView(expand=True, spacing=10)
    for i in range(5000):
        lv.controls.append(ft.Text(f"Line {i}"))
    page.add(lv)

ft.run(main, view=ft.AppView.WEB_BROWSER)
```

Now the scrolling is smooth and fast enough to follow mouse movements:

Image: Scroll listview - <https://flet.dev/docs/assets/cookbook/large-lists/scroll-listview.gif>

note

We used expand=True in ListView constructor. In order to function properly, ListView must have a height (or width if horizontal) specified. You could set an absolute size, e.g. ListView(height=300, spacing=10), but in the example above we make ListView to take all available space on the page, i.e. expand. Read more about Control.expand property.

GridView

GridView allows arranging controls into a scrollable grid.

You can make a "grid" with ft.Column(wrap=True) or ft.Row(wrap=True), for example:

```
import os
import flet as ft

os.environ["FLET_WS_MAX_MESSAGE_SIZE"] = "8000000"

def main(page: ft.Page):
    r = ft.Row(wrap=True, scroll="always", expand=True)
    page.add(r)

    for i in range(5000):
        r.controls.append(
            ft.Container(
                ft.Text(f"Item {i}"),
```

```

        width=100,
        height=100,
        alignment=ft.Alignment.CENTER,
        bgcolor=ft.Colors.AMBER_100,
        border=ft.Border.all(1, ft.Colors.AMBER_400),
        border_radius=ft.BorderRadius.all(5),
    )
)
page.update()

ft.run(main, view=ft.AppView.WEB_BROWSER)

```

Image: Row wrap as grid - <https://flet.dev/docs/assets/cookbook/large-lists/row-wrap-as-grid.png>

Try scrolling and resizing the browser window - everything works, but very laggy.

note

At the start of the program we are setting the value of FLET_WS_MAX_MESSAGE_SIZE environment variable to 8000000 - this is the maximum size of WebSocket message in bytes that can be received by Flet Server rendering the page. Default size is 1 MB, but the size of JSON message describing 5,000 container controls would exceed 1 MB, so we are increasing allowed size to 8 MB.

Squeezing large messages through WebSocket channel is, generally, not a good idea, so use batched updates approach to control channel load.

GridView, similar to ListView, is very effective to render a lot of children. Let's implement the example above using GridView:

```

import os
import flet as ft

os.environ["FLET_WS_MAX_MESSAGE_SIZE"] = "8000000"

def main(page: ft.Page):
    gv = ft.GridView(expand=True, max_extent=150, child_aspect_ratio=1)
    page.add(gv)

    for i in range(5000):
        gv.controls.append(
            ft.Container(
                ft.Text(f"Item {i}"),
                alignment=ft.Alignment.CENTER,
                bgcolor=ft.Colors.AMBER_100,
                border=ft.Border.all(1, ft.Colors.AMBER_400),
                border_radius=ft.BorderRadius.all(5),
            )
        )
    page.update()

ft.run(main, view=ft.AppView.WEB_BROWSER)

```

Image: Grid view - <https://flet.dev/docs/assets/cookbook/large-lists/grid-view.png>

With GridView scrolling and window resizing are smooth and responsive!

You can specify either fixed number of rows or columns (runs) with runs_count property or the maximum size of a "tile" with max_extent property, so the number of runs can vary automatically. In our example we set the maximum tile size to 150 pixels and set its shape to "square" with child_aspect_ratio=1. child_aspect_ratio is the ratio of the cross-axis to the main-axis extent of each child. Try changing it to 0.5 or 2.

Batch updates

When page.update() is called a message is being sent to Flet server over WebSockets containing page updates since the last page.update(). Sending a large message with thousands of added controls could make a user waiting for a few seconds until the messages is fully received and controls rendered.

To increase usability of your program and present the results to a user as soon as possible you can send page updates in batches. For example, the following program adds 5,100 child controls to a ListView in batches of 500 items:

```

import flet as ft

def main(page: ft.Page):
    # add ListView to a page first
    lv = ft.ListView(expand=1, spacing=10, item_extent=50)
    page.add(lv)

    for i in range(5100):
        lv.controls.append(ft.Text(f"Line {i}"))
        # send page to a page
        if i % 500 == 0:

```

```
        page.update()  
        # send the rest to a page  
        page.update()  
  
ft.run(main, view=ft.AppView.WEB_BROWSER)
```

Image: Sending page updates in batches - <https://flet.dev/docs/assets/cookbook/large-lists/sending-page-updates-in-batches.png>

Logging

This article explains how to control level of the logs messages generated by Flet library and its underlying components. You may need to enable detailed logging to troubleshoot Flet library or when submitting a new Flet issue.

Python

Flet Python uses the following named loggers:

- `flet` - general framework and transport logging.
- `flet_object_patch` - detailed control tree diff/patch logging.
- `flet_components` - declarative component lifecycle logging.

For normal use, configure logging before calling `ft.run()`:

```
import logging

logging.basicConfig(level=logging.INFO)
```

This gives you the usual flet logs without too much noise.

To see more detail from the main flet logger, either raise the root logging level:

```
import logging

logging.basicConfig(level=logging.DEBUG)
```

or set the flet logger explicitly:

```
import logging

logging.basicConfig(level=logging.INFO)
logging.getLogger("flet").setLevel(logging.DEBUG)
```

`flet_object_patch` and `flet_components` set their own level to `INFO`, so they do not inherit the root `DEBUG` level unless you enable them explicitly.

To enable the most verbose diagnostics, set the corresponding loggers to `DEBUG`:

```
import logging

logging.basicConfig(level=logging.DEBUG)
logging.getLogger("flet_object_patch").setLevel(logging.DEBUG)
logging.getLogger("flet_components").setLevel(logging.DEBUG)
```

Use this level mostly for troubleshooting and issue reports.

Built-in Web Server

When running a web app, Flet starts its built-in web transport on top of FastAPI and Uvicorn.

The effective level of the flet Python logger is passed to the built-in web server, so configure flet logging before starting the app:

```
import logging

logging.basicConfig(level=logging.INFO)
logging.getLogger("flet").setLevel(logging.DEBUG)
```

If you host a Flet ASGI app with your own server process, such as `uvicorn` or `gunicorn`, configure that server's logging separately using its own options.

Navigation and Routing

Navigation and routing is the core of building multi-screen Flet apps. It lets you organize your UI into virtual pages (View objects), keep URL/history in sync, and support deep links to specific app states.

This page focuses on the current routing model and maintained examples.

Routing model in Flet

A Page is a container of views (page.views), where each view represents one route-level screen.

- page.route is the current route string (for example /, /store, /settings/mail).
- page.views is the active navigation stack.
- page.on_route_change rebuilds the stack when route changes.
- page.on_view_pop handles Back navigation (system Back, AppBar Back, browser Back).

A reliable setup uses a single source of truth: derive page.views from page.route.

Route basics

The default route is / when no route is provided.

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    page.add(ft.SafeArea(content=ft.Text(f"Initial route: {page.route}")))

if __name__ == "__main__":
    ft.run(main)
```

All routes should start with /, for example /store, /products/42, /settings/mail.

Handling route changes

Whenever route changes (URL edit, browser Back/Forward, or app navigation), page.on_route_change event is triggered. Use this event as the place where you decide which views must exist for the current route.

[play_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
    route_log = ft.Column(controls=[ft.Text(f"Initial route: {page.route}")]
    page.add(ft.SafeArea(content=route_log))

    def route_change(e):
        route_log.controls.append(ft.Text(f"New route: {e.route}"))
        page.update()

    page.on_route_change = route_change
    page.update()

if __name__ == "__main__":
    ft.run(main)
```

Building views from route

The pattern below is the baseline for most apps:

- Clear page.views.
- Add root view (/).
- Add extra views conditionally based on page.route.
- Handle Back in page.on_view_pop and navigate to the new top view.

Why is this pattern important?

- Keeps URL, history stack, and visible UI synchronized.
- Supports deep links and reloads naturally.
- Makes navigation deterministic and easier to debug.

play_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
    page.title = "Routes Example"

    print("Initial route:", page.route)

    async def open_mail_settings(e):
        await page.push_route("/settings/mail")

    async def open_settings(e):
        await page.push_route("/settings")

    def route_change():
        print("Route change:", page.route)
        page.views.clear()
        page.views.append(
            ft.View(
                route="/",
                controls=[
                    ft.SafeArea(
                        content=ft.Column(
                            controls=[
                                ft.AppBar(title=ft.Text("Flet app")),
                                ft.Button("Go to settings", on_click=open_settings),
                            ]
                        )
                    )
                ],
            )
        )
        if page.route == "/settings" or page.route == "/settings/mail":
            page.views.append(
                ft.View(
                    route="/settings",
                    controls=[
                        ft.SafeArea(
                            content=ft.Column(
                                controls=[
                                    ft.AppBar(
                                        title=ft.Text("Settings"),
                                        bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
                                    ),
                                    ft.Text(
                                        "Settings!",
                                        theme_style=ft.TextThemeStyle.BODY_MEDIUM,
                                    ),
                                    ft.Button(
                                        content="Go to mail settings",
                                        on_click=open_mail_settings,
                                    ),
                                ]
                            )
                        )
                    ],
                )
            )
        if page.route == "/settings/mail":
            page.views.append(
                ft.View(
                    route="/settings/mail",
                    controls=[
                        ft.SafeArea(
                            content=ft.Column(
                                controls=[
                                    ft.AppBar(
                                        title=ft.Text("Mail Settings"),
                                        bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
                                    ),
                                    ft.Text("Mail settings!"),
                                ]
                            )
                        )
                    ],
                )
            )
    page.on_route_change(route_change)

ft.app(main)
```

```

        page.update()

    async def view_pop(e):
        if e.view is not None:
            print("View pop:", e.view)
            page.views.remove(e.view)
            top_view = page.views[-1]
            await page.push_route(top_view.route)

    page.on_route_change = route_change
    page.on_view_pop = view_pop

    route_change()

if __name__ == "__main__":
    ft.run(main)

```

Programmatic navigation

Use `page.navigate()` to navigate from synchronous callbacks (e.g. `on_click`), or `page.push_route()` in async contexts:

```

# Sync (on_click, etc.)
page.navigate("/search", q="flet", page=2)

# Async
await page.push_route("/search", q="flet", page=2)

```

Back navigation and pop confirmation

When users go back, Flet triggers `page.on_view_pop`. For flows requiring confirmation (for example, unsaved changes), disable automatic pop and confirm manually with `View.can_pop` + `View.on_confirm_pop`.

`play_arrow` Try Online

```

import flet as ft

def main(page: ft.Page):
    page.title = "Routes Example"

    def route_change():
        page.views.clear()
        page.views.append(MainView("/"))
        if page.route == "/store":
            page.views.append(PermissionView("/store"))
        page.update()

    async def view_pop(e: ft.ViewPopEvent):
        if e.view is not None:
            print("View pop:", e.view)
            page.views.remove(e.view)
            top_view = page.views[-1]
            await page.push_route(top_view.route)

    page.on_route_change = route_change
    page.on_view_pop = view_pop

    route_change()

class MainView(ft.View):
    def __init__(self, path):
        super().__init__(
            route=path,
            controls=[
                ft.SafeArea(
                    content=ft.Column(
                        controls=[
                            ft.AppBar(title=ft.Text("Flet app")),
                            ft.Button("Go to store", on_click=self.open_store),
                        ]
                    )
                )
            ],
        )

    async def open_store(self, e):
        await self.page.push_route("/store")

class PermissionView(ft.View):
    def __init__(self, path):
        super().__init__(
            route=path,

```

```

        controls=[
            ft.SafeArea(
                content=ft.Column(
                    controls=[ft.AppBar(title=ft.Text(f"{path} View"))]
                )
            ),
        ],
        can_pop=False,
        on_confirm_pop=self.ask_pop_permission,
    )

    async def ask_pop_permission(self, e):
        async def on_dlg_yes(e):
            self.page.pop_dialog()
            await self.confirm_pop(True)

        async def on_dlg_no(e):
            self.page.pop_dialog()
            await self.confirm_pop(False)

        dlg_modal = ft.AlertDialog(
            title=ft.Text("Please confirm"),
            content=ft.Text("Go home?"),
            actions=[
                ft.TextButton(
                    "Yes",
                    on_click=on_dlg_yes,
                ),
                ft.TextButton(
                    "No",
                    on_click=on_dlg_no,
                ),
            ],
            actions_alignment=ft.MainAxisAlignment.END,
            on_dismiss=lambda e: print("Modal dialog dismissed!"),
        )

        self.page.show_dialog(dlg_modal)

if __name__ == "__main__":
    ft.run(main)

```

Pop multiple views with a result

When a multi-step flow finishes deep in the view stack, use `page.pop_views_until()` to jump back to a target route and deliver a result to the destination view via `page.on_views_pop_until`.

[play_arrowTry Online](#)

```

import asyncio

import flet as ft

def main(page: ft.Page):
    page.title = "Routes Example"

    result_text = ft.Text("No result yet", size=18)

    def route_change():
        page.views.clear()

        # Home View (/)
        page.views.append(
            ft.View(
                route="/",
                controls=[
                    ft.AppBar(title=ft.Text("Home"), bgcolor=ft.Colors.SURFACE_BRIGHT),
                    result_text,
                    ft.Button(
                        "Start flow",
                        on_click=lambda _: asyncio.create_task(
                            page.push_route("/step1")
                        ),
                    ),
                ],
            )
        )

    if page.route == "/step1" or page.route == "/step2" or page.route == "/step3":
        page.views.append(
            ft.View(
                route="/step1",
                controls=[
                    ft.AppBar(

```

```

        title=ft.Text("Step 1"),
        bgcolor=ft.Colors.SURFACE_BRIGHT,
    ),
    ft.Text("Step 1 of the flow"),
    ft.Button(
        "Go to Step 2",
        on_click=lambda _: asyncio.create_task(
            page.push_route("/step2")
        ),
    ),
],
)
)

if page.route == "/step2" or page.route == "/step3":
    page.views.append(
        ft.View(
            route="/step2",
            controls=[
                ft.AppBar(
                    title=ft.Text("Step 2"),
                    bgcolor=ft.Colors.SURFACE_BRIGHT,
                ),
                ft.Text("Step 2 of the flow"),
                ft.Button(
                    "Go to Step 3",
                    on_click=lambda _: asyncio.create_task(
                        page.push_route("/step3")
                    ),
                ),
            ],
        ),
    )

if page.route == "/step3":
    page.views.append(
        ft.View(
            route="/step3",
            controls=[
                ft.AppBar(
                    title=ft.Text("Step 3 (Final)"),
                    bgcolor=ft.Colors.SURFACE_BRIGHT,
                ),
                ft.Text("Flow complete!"),
                ft.Button(
                    "Finish and go Home",
                    on_click=lambda _: asyncio.create_task(
                        page.pop_views_until("/", result="Flow completed!")
                    ),
                ),
            ],
        ),
    )

page.update()

def on_pop_result(e: ft.ViewsPopUntilEvent):
    result_text.value = f"Result: {e.result}"
    page.show_dialog(ft.SnackBar(ft.Text(f"Got result: {e.result}")))
    page.update()

async def view_pop(e: ft.ViewPopEvent):
    if e.view is not None:
        page.views.remove(e.view)
        top_view = page.views[-1]
        await page.push_route(top_view.route)

page.on_route_change = route_change
page.on_view_pop = view_pop
page.on_views_pop_until = on_pop_result

route_change()

if __name__ == "__main__":
    ft.run(main)

```

Navigation UI patterns

Routing composes well with navigation controls such as drawer, rail, and tabs. This example shows route-driven drawer navigation with multiple top-level destinations:

[play_arrowTry Online](#)

```
import asyncio
```

```

import flet as ft

def main(page: ft.Page):
    page.title = "Drawer navigation"

    async def handle_change(e):
        if e.control.selected_index == 0:
            await page.push_route("/")
        elif e.control.selected_index == 1:
            await page.push_route("/store")
        elif e.control.selected_index == 2:
            await page.push_route("/about")

    def create_drawer(selected_index=0):
        return ft.NavigationDrawer(
            selected_index=selected_index,
            on_change=handle_change,
            controls=[
                ft.Container(height=12),
                ft.NavigationDrawerDestination(
                    label="Home",
                    icon=ft.Icons.HOME_OUTLINED,
                    selected_icon=ft.Icon(ft.Icons.HOME),
                ),
                ft.Divider(thickness=2),
                ft.NavigationDrawerDestination(
                    label="Store",
                    icon=ft.Icon(ft.Icons.STORE_OUTLINED),
                    selected_icon=ft.Icon(ft.Icons.STORE),
                ),
                ft.NavigationDrawerDestination(
                    label="About",
                    icon=ft.Icon(ft.Icons.PHONE_OUTLINED),
                    selected_icon=ft.Icons.PHONE,
                ),
            ],
        )

    async def show_drawer():
        await page.show_drawer()

    def route_change(route):
        page.views.clear()
        page.views.append(
            ft.View(
                route="/",
                controls=[
                    ft.SafeArea(
                        content=ft.Column(
                            controls=[
                                ft.AppBar(
                                    title=ft.Text("Home", expand=True),
                                    bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
                                    leading=ft.IconButton(
                                        ft.Icons.MENU, on_click=show_drawer
                                    ),
                                ),
                                ft.Text("Welcome to Home Page"),
                            ]
                        )
                    )
                ],
            ),
            drawer=create_drawer(selected_index=0) if page.route == "/" else None,
        )

    if page.route == "/store":
        page.views.append(
            ft.View(
                route="/store",
                controls=[
                    ft.SafeArea(
                        content=ft.Column(
                            controls=[
                                ft.AppBar(
                                    title=ft.Text("Store", expand=True),
                                    bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
                                    leading=ft.IconButton(
                                        ft.Icons.MENU, on_click=show_drawer
                                    ),
                                    automatically_imply_leading=False,
                                ),
                                ft.Text("Welcome to Store Page"),
                                ft.Button(
                                    "Go About",
                                    on_click=lambda _: asyncio.create_task(

```

```

        page.push_route("/about")
    ),
    ],
    ),
    ],
    drawer=create_drawer(selected_index=1),
)
)
)
if page.route == "/about":
    page.views.append(
        ft.View(
            route="/about",
            controls=[
                ft.SafeArea(
                    content=ft.Column(
                        controls=[
                            ft.AppBar(
                                title=ft.Text("About", expand=True),
                                bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
                                leading=ft.IconButton(
                                    ft.Icons.MENU, on_click=show_drawer
                                ),
                                automatically_imply_leading=False,
                            ),
                            ft.Text("Welcome to About Page"),
                            ft.Button(
                                "Go Store",
                                on_click=lambda _: asyncio.create_task(
                                    page.push_route("/store")
                                ),
                            ),
                        ],
                    ),
                ),
            ],
            drawer=create_drawer(selected_index=2),
        ),
    ),
)
page.update()

async def view_pop(view):
    page.views.pop()
    top_view = page.views[-1]
    await page.push_route(top_view.route)

page.on_route_change = route_change
page.on_view_pop = view_pop
route_change(page.route)

if __name__ == "__main__":
    ft.run(main)

```

Route templates (parameterized routes)

Use `TemplateRoute` to match and parse route parameters, for example `/books/:id`. Template syntax is provided by `repath`.

```

import flet as ft

troute = ft.TemplateRoute(page.route)

if troute.match("/books/:id"):
    print("Book ID:", troute.id)
elif troute.match("/account/:account_id/orders/:order_id"):
    print("Account:", troute.account_id, "Order:", troute.order_id)
else:
    print("Unknown route")

```

Web URL strategy

Flet web apps support two URL strategies :

- "path" (default): in the form `https://myapp.dev/store`
- "hash": `https://myapp.dev/#/store`

It can be set via `route_url_strategy` in `ft.run()`

```

import flet as ft
ft.run(main, route_url_strategy="hash")

```

For Flet server deployments, you can also set the `FLET_ROUTE_URL_STRATEGY` environment variable.

Declarative Router

For declarative apps using `@component`, Flet provides a Router system inspired by React Router. It handles nested routes, layout routes with outlets, dynamic segments, data loaders, and active link detection — all without manual route matching.

With `manage_views=True`, Router can also manage the view stack directly, enabling swipe-back gestures and AppBar back buttons on mobile. Route components return View objects with their own AppBar, and navigating deeper pushes views onto the stack.

See the Router cookbook for a complete guide.

Practical recommendations

- Always keep a root / view in `page.views`.
- Keep route handling centralized in `page.on_route_change`; avoid mutating `page.views` from many places.
- When adding new routes, test these cases: direct deep link, browser Back/Forward, app Back button, and reload.
- Use route templates for dynamic segments instead of manual string splitting.

PubSub

If you build a chat app using Flet you need somehow to pass user messages between sessions. When a user sends a message it should be broadcasted to all other app sessions and displayed on their pages.

Flet provides a simple built-in PubSub mechanism for asynchronous communication between page sessions.

Flet PubSub allows broadcasting messages to all app sessions or sending only to specific "topic" (or "channel") subscribers.

A typical PubSub usage would be:

- `subscribe()` to broadcast messages or `subscribe_topic()` on app session start.
- `send_all()` broadcast message or `send_all_on_topic()` on some event, like "Send" button click.
- `unsubscribe()` from broadcast messages or `unsubscribe_topic()` on some event, like "Leave" button click.
- `unsubscribe_all()` from everything on `page.on_close`.

This is an example of a simple chat application that uses PubSub:

```
import flet as ft

def main(page: ft.Page):
    page.title = "Flet Chat"

    # subscribe to broadcast messages
    def on_message(msg):
        messages.controls.append(ft.Text(msg))
        page.update()

    page.pubsub.subscribe(on_message)

    def send_click(e):
        page.pubsub.send_all(f"{user.value}: {message.value}")
        # clean up the form
        message.value = ""
        page.update()

    messages = ft.Column()
    user = ft.TextField(hint_text="Your name", width=150)
    message = ft.TextField(hint_text="Your message...", expand=True) # fill all the space
    send = ft.Button("Send", on_click=send_click)
    page.add(messages, ft.Row(controls=[user, message, send]))

ft.run(main, view=ft.AppView.WEB_BROWSER)
```

Image: Chat app - <https://flet.dev/docs/assets/cookbook/pub-sub/chat-app.gif>

Read and Write Files

In some cases, you need to read and write files to disk. For example, you might need to persist data across app launches, or download data from the internet and save it for later offline use.

Flet makes it easy to work with files and directories on the mobile/desktop device, as seen in the following example.

Storage Paths

Flet provides two directory paths for data storage, available as environment variables: `FLET_APP_STORAGE_DATA` and `FLET_APP_STORAGE_TEMP`.

Their values can be gotten as follows:

```
import os

app_data_path = os.getenv("FLET_APP_STORAGE_DATA")
app_temp_path = os.getenv("FLET_APP_STORAGE_TEMP")
```

Writing to a File

To write data to a new/existing file, you can use the built-in open function.

For example:

```
import os

app_data_path = os.getenv("FLET_APP_STORAGE_DATA")
my_file_path = os.path.join(app_data_path, "test_file.txt")

with open(my_file_path, "w") as f:
    f.write("Some file content...")
```

Reading from a File

To read data from an existing file, you can equally use the built-in open function.

For example:

```
import os

app_data_path = os.getenv("FLET_APP_STORAGE_DATA")
my_file_path = os.path.join(app_data_path, "test_file.txt")

with open(my_file_path, "r") as f:
    file_content = f.read()
    print(file_content)
```

Also, you can use the `os` module (or any other out there) to perform various file operations like renaming, deleting, listing files present in the directory, etc.

Example: Counter App

Below is an example that showcases a basic Counter application, whose value persists across app launches. This is made possible by writing the counter value to a file in the app's data storage directory and reading it when the app launches.

```
import os
from datetime import datetime
import flet as ft

# constants
FLET_APP_STORAGE_DATA = os.getenv("FLET_APP_STORAGE_DATA")
COUNTER_FILE_PATH = os.path.join(FLET_APP_STORAGE_DATA, "counter.txt")
FLET_APP_CONSOLE = os.getenv("FLET_APP_CONSOLE")

class Counter(ft.Text):
    def __init__(self, storage_path=COUNTER_FILE_PATH):
        super().__init__(theme_style=ft.TextThemeStyle.HEADLINE_LARGE)
        self.storage_path = storage_path
        self.count = self.__read_from_storage()

    def increment(self):
        """Increment the counter, store the new value, and return it."""
        self.count += 1
        self.update()
        self.__write_to_storage()

    def before_update(self):
        super().before_update()
        self.value = f"Button tapped {self.count} time{' ' if self.count == 1 else 's'}"
```

```

def __log(self, action: str, value: int = None):
    """Log executed action."""
    if value is None:
        value = self.count
    print(f"{datetime.now().strftime('%Y/%m/%d %H:%M:%S')} - {action} = {value}")

def __read_from_storage(self):
    """Read counter value. If an error occurs, use 0."""
    try:
        with open(self.storage_path, "r") as f:
            value = int(f.read().strip())
    except (FileNotFoundError, ValueError):
        # file does not exist or int parsing failed
        value = 0

    self.__log("READ", value)
    return value

def __write_to_storage(self):
    """Write current counter value to storage."""
    with open(self.storage_path, "w") as f:
        f.write(str(self.count))
    self.__log("WRITE")

def main(page: ft.Page):
    page.theme_mode = ft.ThemeMode.LIGHT
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def show_logs(e: ft.ControlEvent):
        if FLET_APP_CONSOLE is not None:
            with open(FLET_APP_CONSOLE, "r") as f:
                dlg = ft.AlertDialog(
                    title=ft.Text("App Logs"),
                    content=ft.Text(f.read()),
                    scrollable=True,
                )
            page.open(dlg)

    counter = Counter()
    page.appbar = ft.AppBar(
        title=ft.Text("Storage Playground", weight=ft.FontWeight.BOLD),
        center_title=True,
        bgcolor=ft.Colors.BLUE,
        color=ft.Colors.WHITE,
        adaptive=True,
        actions=[
            ft.IconButton(
                icon=ft.Icons.REMOVE_RED_EYE,
                tooltip="Show logs",
                visible=FLET_APP_CONSOLE is not None,
                on_click=show_logs,
            ),
        ],
    )
    page.floating_action_button = ft.FloatingActionButton(
        icon=ft.Icons.ADD,
        text="Increment Counter",
        foreground_color=ft.Colors.WHITE,
        bgcolor=ft.Colors.BLUE,
        on_click=lambda e: counter.increment(),
    )
    page.floating_action_button_location = ft.FloatingActionButtonLocation.CENTER_FLOAT

    page.add(ft.SafeArea(counter))

ft.run(main)

```

- Counter class is a custom control, which is a subclass/extension of the Text control. More information [here](#).
- FLET_APP_CONSOLE is an environment variable that points to the application's console log file (console.log) which contains the app's console output (ex: print() statements). Its value is set in production mode.
- If you have an android emulator or physical device, you can download and install this apk.
- Follow this [guide](#) to package your app for all platforms.

Router

Router is a declarative routing system for Flet apps, inspired by React Router. It handles nested route matching, layout routes with outlets, dynamic segments, data loading, view-stack navigation, and provides hooks for accessing route state.

Why Router?

Without Router, multi-page Flet apps require manual route matching, conditional rendering, and parameter extraction. Router automates all of this:

- Nested routes — define parent/child hierarchies
- Layout routes — share layouts across pages using `use_route_outlet()`
- Dynamic segments — `:paramName` in paths, accessed via `use_route_params()`
- Data loading — fetch data before rendering with `loader`
- Active link detection — highlight nav items with `is_route_active()`
- View-stack navigation — swipe-back and AppBar back button with `manage_views=True`

Quick start

```
@ft.component
def Home():
    return ft.Text("Welcome home!")

@ft.component
def About():
    return ft.Text("About us")

@ft.component
def App():
    return ft.Router([
        ft.Route(index=True, component=Home),
        ft.Route(path="about", component=About),
    ])

ft.run(lambda page: page.render(App))
```

See full example.

Defining routes

Routes are defined as a tree of Route objects.

Flat routes

All routes at the top level:

```
ft.Router([
    ft.Route(index=True, component=Home),
    ft.Route(path="about", component=About),
    ft.Route(path="contact", component=Contact),
])
```

Nested routes

Child routes are nested under parent routes. The parent's path is automatically prepended:

```
ft.Route(
    path="products",
    component=ProductsList,
    children=[
        ft.Route(path=":pid", component=ProductDetails),
    ],
)
# /products → ProductsList
# /products/42 → ProductDetails (params: {"pid": "42"})
```

See full example.

Index routes

An index route (`index=True`) matches when the parent path matches exactly, with no further segments:

```
ft.Route(path="settings", children=[
```

```

    ft.Route(index=True, component=SettingsHome),    # /settings
    ft.Route(path="profile", component=Profile),      # /settings/profile
  ])

```

Prefix routes

Pathless layout routes — group children under a shared layout without adding a path segment:

```

ft.Route(component=AdminLayout, children=[
    ft.Route(path="users", component=Users),        # /users
    ft.Route(path="settings", component=Settings),  # /settings
])

```

Path-only routes — add a path prefix without rendering a component:

```

ft.Route(path="api", children=[
    ft.Route(path="users", component=ApiUsers),      # /api/users
    ft.Route(path="products", component=ApiProducts), # /api/products
])

```

Layout routes and use_route_outlet()

A layout route has both a component and children. Its component renders shared UI (header, sidebar, footer) and calls `use_route_outlet()` to place the matched child:

```

@ft.component
def AppLayout():
    outlet = ft.use_route_outlet()
    return ft.Column([
        ft.AppBar(title=ft.Text("My App")),
        ft.Container(content=outlet, expand=True),
        ft.Text("Footer"),
    ])

ft.Router([
    ft.Route(component=AppLayout, children=[
        ft.Route(index=True, component=Home),
        ft.Route(path="about", component=About),
    ]),
])

```

Layout routes can be nested to any depth. Each level calls `use_route_outlet()` to get its immediate child. See full example.

Dynamic segments, optional segments, and splats

Route paths support Express-style patterns via the `repath` library.

Dynamic segments — `:paramName` matches a single path segment:

```

ft.Route(path="users/:userId", component=UserProfile)
# /users/42 → use_route_params() returns {"userId": "42"}

```

Optional segments — `:paramName?` may be absent:

```

ft.Route(path="users/:userId?", component=UserProfile)
# /users → {"userId": None}
# /users/42 → {"userId": "42"}

```

Splats (catch-all) — `:paramName*` matches zero or more segments:

```

ft.Route(path="files/:path*", component=FileBrowser)
# /files/a/b/c → {"path": "a/b/c"}

```

Custom regex — `:paramName(\\d+)` constrains a segment:

```

ft.Route(path="item/:id(\\d+)", component=ItemDetails)
# /item/42 → matches
# /item/abc → does NOT match

```

Navigation

Use `page.navigate()` from synchronous callbacks:

```

ft.Button("About", on_click=lambda: ft.context.page.navigate("/about"))

```

For async contexts, use `page.push_route()` with `await`. The Router automatically re-renders when the route changes.

Active links with is_route_active()

`is_route_active(path)` returns `True` if the given path matches the current location:

```

ft.Button(
    "Products",
    style=ft.ButtonStyle(
        bgcolor=ft.Colors.PRIMARY_CONTAINER
    )
)

```

```

        if ft.is_route_active("/products")
        else None,
    ),
    on_click=lambda: ft.context.page.navigate("/products"),
)

```

By default it uses prefix matching — `is_route_active("/products")` returns `True` for `/products/42`. Pass `exact=True` for exact matching. See full example.

Data loading

Routes can define a loader function. It runs when the route matches and its return value is available via `use_route_loader_data()`:

```

def load_user(params):
    return db.get_user(params["userId"])

ft.Route(path="users/:userId", component=UserProfile, loader=load_user)

@ft.component
def UserProfile():
    user = ft.use_route_loader_data()
    return ft.Text(f"Hello, {user.name}")

```

See full example.

Authentication

Auth is implemented using layout routes as guards — no special Router API needed.

Page redirect — redirect to `/login` if not authenticated:

```

@ft.component
def ProtectedRoute():
    outlet = ft.use_route_outlet()
    if not auth.is_authenticated:
        ft.context.page.navigate("/login")
        return ft.Text("Redirecting...")
    return outlet

```

Role-based access — nest guard routes for layered access control:

```

ft.Router([
    ft.Route(component=ProtectedRoute, children=[
        ft.Route(index=True, component=Home),
        ft.Route(component=AdminOnly, children=[
            ft.Route(path="admin", component=AdminPanel),
        ]),
    ]),
])

```

404 handling

Pass a `not_found` component to render when no route matches:

```

@ft.component
def NotFound():
    location = ft.use_route_location()
    return ft.Text(f"Page not found: {location}")

ft.Router(routes, not_found=NotFound)

```

Multi-view navigation

By default, Router renders all route content within a single View. With `manage_views=True`, the Router returns a list of Views — one per path level. This enables:

- Swipe-back gesture on mobile
- System back button
- AppBar implicit back button
- Slide transition between route levels

Use `page.render_views()` instead of `page.render()`.

Components return Views

Each route component returns a View with route, appbar, and controls:

```

@ft.component

```

```
def ProductDetails():
    params = ft.use_route_params()
    return ft.View(
        route=f"/products/{params['pid']}",
        appbar=ft.AppBar(title=ft.Text(f"Product #{params['pid']}")),
        controls=[
            ft.Text(f"Details for product #{params['pid']}"),
        ],
    )
```

Nested route hierarchy

Structure routes as a nested tree. A pathless root ensures it is always in the view stack:

```
ft.Router([
    ft.Route(component=Home, children=[
        ft.Route(path="products", component=ProductsList, children=[
            ft.Route(path=":pid", component=ProductDetails),
        ]),
    ]),
], manage_views=True)
```

View stack

- / — 1 view (Home)
- /products — 2 views (Home, Products) — back button to Home
- /products/1 — 3 views (Home, Products, Product Details) — back button to Products

See full example.

Shared layouts with outlet=True

Set `outlet=True` on a route to make it a layout that wraps child routes via `use_route_outlet()`. The layout returns a `View`; child components return regular controls.

Use `use_view_path()` for `View.route` — it returns the resolved URL for each child view level, so Flutter's Navigator gets a unique key per view in the stack:

```
@ft.component
def ProductsLayout():
    outlet = ft.use_route_outlet()
    return ft.View(
        route=ft.use_view_path(),
        appbar=ft.AppBar(title=ft.Text("Products")),
        controls=[
            ft.Container(content=outlet, expand=True),
            ft.Text("Footer"),
        ],
    )

ft.Route(path="products", component=ProductsLayout, outlet=True, children=[
    ft.Route(component=ProductsList, children=[
        ft.Route(path=":pid", component=ProductDetails),
    ]),
])
```

`use_route_location()` vs `use_view_path()`

`use_route_location()` always returns the full current URL. `use_view_path()` returns the URL for the current view level only — essential as `View.route` when a single layout component wraps multiple child views, so the Flutter Navigator can distinguish them.

The layout's shared UI appears on every child route's `View`, and each child is still a separate `View` in the stack — back navigation works between them. See full example.

Avoiding transition animation

When switching between top-level `NavigationRail` destinations, set a fixed route value on the root layout's `View` to prevent slide transitions:

```
@ft.component
def RootLayout():
    outlet = ft.use_route_outlet()
    return ft.View(
        route="/", # fixed key — no animation between top-level pages
        can_pop=False,
        controls=[
            ft.Row([
                ft.NavigationRail(...),
                ft.Container(content=outlet, expand=True),
            ], expand=True),
        ],
    )
```

```
    ],  
  )  
)
```

See full example.

Back navigation

When the user pops the topmost View (close X on a modal, AppBar back arrow, iOS swipe-back, system back button) the Router by default navigates to the parent of the current route in the route tree — i.e. `chain[-2].resolved_path` of the matched chain, not `views[-2].route`. This is robust against shared View.route values (the "avoid transition animation" trick above) — a deep-Gallery pop still goes to `/gallery` even if `RootLayout` emits `route="/"` for both the Apps and Gallery tab roots.

If you need fully custom pop behavior, install your own `page.on_view_pop` handler before `page.render_views(App)`. The Router detects a pre-existing handler and delegates to it instead of running the default.

Modal routes

Set `modal=True` on a route to render its View as a fullscreen-dialog overlay that sits on top of the previous (non-modal) location's view stack. Closing the modal pops it without rebuilding the underlying stack — the views underneath stay mounted, so there's no rebuild flash.

The modal's component still returns a View like any other route — set `fullscreen_dialog=True` on the View so Flutter renders the slide-up presentation and the close (X) leading icon.

Global modals — reachable from anywhere

Declared at the top level of the route tree. The base stack comes from the last non-modal location the Router saw (defaults to `/` on first render):

```
ft.Router([  
  ft.Route(component=Home),  
  ft.Route(component=Profile, path="profile"),  
  ft.Route(component=SettingsModal, path="settings", modal=True),  
], manage_views=True)
```

- `/` → stack [Home]
- `/profile` → stack [Profile]
- Navigate from `/profile` to `/settings` → stack [Profile, SettingsModal] (Profile still mounted underneath the modal)
- Close → back to `/profile`, stack [Profile] (Profile not rebuilt)
- Deep-link `/settings` directly → stack [Home, SettingsModal] (base defaults to `/`)

Local modals — pinned to a specific page

Declared as a child of a non-modal parent. The base stack is the chain above the modal in the route tree, so the URL itself encodes which item the modal applies to and deep-link works without any state:

```
ft.Router([  
  ft.Route(component=ItemList, path="items", children=[  
    ft.Route(component=ItemDetails, path=":id", children=[  
      ft.Route(component=EditModal, path="edit", modal=True),  
    ]),  
  ]),  
], manage_views=True)
```

- `/items/42` → stack [ItemList, ItemDetails(42)]
- `/items/42/edit` → stack [ItemList, ItemDetails(42), EditModal]
- Close → `/items/42`

See full example.

Recursive routes

A route marked `recursive=True` can match itself as its own descendant — one View is emitted per consumed URL segment. Use this for tree-shaped URLs with unbounded depth without writing nested duplicates:

```
ft.Router([  
  ft.Route(path="folder", component=Folders, children=[  
    ft.Route(  
      path=":name",  
      component=Folder,  
      recursive=True,  
    ),  
  ]),  
])
```



```

        children=[
            ft.Route(path="search", component=Search),
        ],
    ),
], manage_views=True)

```

- /folder/a/b/c → stack [Folders, Folder(a), Folder(a/b), Folder(a/b/c)]
- Back-swipe walks one segment at a time

Inside the component, `use_route_params()` returns just the current segment (e.g. {"name": "b"} at depth 2). Use `use_view_path()` for the accumulated URL up to this level — useful when the component needs to know the full path.

Sibling-wins-over-recursion rule

A non-recursive child of the recursive route is tried before self-recursion at every depth. In the example above, /folder/x/search matches as [Folders, Folder(x), Search] — the Search sibling wins over Folder("search"), so you only need to declare the special child once, not at every depth.

This makes the recursive route ideal for replacing N-level fan-outs like:

```

# Without recursive=True — declared N times for max depth N:
ft.Route(path=":s1", component=Folder, children=[
    ft.Route(path="search", component=Search),
    ft.Route(path=":s2", component=Folder, children=[
        ft.Route(path="search", component=Search),
        ft.Route(path=":s3", component=Folder, children=[
            ft.Route(path="search", component=Search),
            ft.Route(path=":s4", component=Folder, children=[...]),
        ]),
    ]),
]),

# With recursive=True — one declaration, unbounded depth:
ft.Route(path=":name", component=Folder, recursive=True, children=[
    ft.Route(path="search", component=Search),
]),

```

See full example.

Hooks reference

| Hook | Returns | Description |
|------|-----------|---|
| | | All dynamic segment values from the matched route chain |
| | | Current URL pathname |
| | | Resolved URL for the current view level (unique per view in mode) |
| | component | Matched child route component (for layout routes) |
| | | Return value of the current route's |
| | | Whether matches the current location |

`use_route_params()`dict[str, str]`use_route_location()`str`use_view_path()`str`manage_views`use_route_outlet()`use_route_loader_data()`Any`loader`is_route_active(path)boolpath

Session Storage

Flet provides an API for storing key-value data in user's session on a server side.

caution

In the current Flet implementation the data stored in a session store is transient and is not preserved between app restarts.

Write data

```
# strings
page.session.store.set("key", "value")

# numbers
page.session.store.set("number.setting", 12345)

# booleans
page.session.store.set("bool_setting", True)

# lists
page.session.store.set("favorite_colors", ["red", "green", "blue"])
```

Read data

```
# The value is automatically converted back to the original type
value = page.session.store.get("key")

colors = page.session.store.get("favorite_colors")
# colors = ["red", "green", "blue"]
```

Check key existence

```
page.session.store.contains_key("key") # True if the key exists
```

Get all keys

```
page.session.store.get_keys()
```

Remove data by key

```
page.session.store.remove("key")
```

Clear all data

```
page.session.store.clear()
```

Subprocess

In this cookbook recipe, you'll learn how to run external system commands from your Flet app using Python's built-in subprocess module.

This approach is useful when you need to interact with platform tools or system utilities that are not exposed through Flet APIs.

Note

Running external commands is not supported when your app runs in a browser.

Examples

Launching another Android app

Sometimes your Flet app may need to open another Android application. For example, you might want to open the Settings app so the user can adjust system or app settings.

On Android, this can be done using the `am start` command, which launches activities from the Android shell.

The command requires the component name (-n) of the target activity in the form: `package.name/.ActivityName`.

For example, the Android Settings app uses:

```
com.android.settings/.Settings
```

It is also recommended to pass the `--user` flag. Without it, the command may fail with an error such as:

```
Security exception: Permission Denial: startActivityAsUser asks to run as user -2 but is calling from uid u0a164; this requires android.permission.INTERACT_ACROSS_USERS_FULL
```

This happens because Android may attempt to launch the activity for a different user than the one your app runs under. Setting `--user 0` ensures the command runs for the primary user.

The full command therefore becomes:

```
am start -n com.android.settings/.Settings --user 0
```

And can be run from a Flet app like this:

```
import subprocess
import flet as ft

def main(page: ft.Page):
    def open_settings(e):
        result = subprocess.run(
            ["am", "start", "-n", "com.android.settings/.Settings", "--user", "0"],
            shell=False, # default
            capture_output=True,
            text=True,
        )

        print("STDOUT:", result.stdout)
        print("STDERR:", result.stderr)

    page.add(
        ft.SafeArea(
            content=ft.Button("Open Settings app", on_click=open_settings)
        )
    )

ft.run(main)
```

Running this app on an Android device, you should see the Settings app launch when the button is clicked.

Common Pitfall

When using `subprocess.run()`, be careful with the `shell` argument.

This will not work correctly:

```
subprocess.run(
    ["am", "start", "-n", "com.android.settings/.Settings", "--user", "0"],
    shell=True
)
```

Because when `shell=True` is used, the command must be passed as a single string, not a list.

Correct usage:

```
subprocess.run(  
    "am start -n com.android.settings/.Settings --user 0",  
    shell=True  
)
```

However, when possible, prefer `shell=False` with a list of arguments, as shown in the main example. This is generally safer and avoids shell parsing issues.

Theming

It is possible to configure your application and/or the containing controls to follow a particular themes.

App-wide themes

The Page control (uppermost control in the tree) has two useful properties for this: `theme` and `dark_theme` properties to configure the appearance/theme of the entire app in light and dark theme modes respectively.

Both of type `Theme`, they represent the default/fallback themes to be used app-wide, except explicitly modified/overridden in the tree.

```
page.theme = ft.Theme(color_scheme_seed=ft.Colors.GREEN)
page.dark_theme = ft.Theme(color_scheme_seed=ft.Colors.BLUE)
```

Nested themes

You can have a part of your app to use a different theme or override some theme styles for specific controls.

Some container-like controls have `theme` and `theme_mode` properties of type `Theme` and `ThemeMode` respectively.

Specifying `theme_mode` in the Container means you don't want to inherit parent theme mode, but want a completely new, unique scheme for all controls inside the container. However, if the container does not have `theme_mode` property set then the styles from its `theme` property will override the ones from the parent inherited theme:

```
import flet as ft

def main(page: ft.Page):
    # Yellow page theme with SYSTEM (default) mode
    page.theme = ft.Theme(
        color_scheme_seed=ft.Colors.YELLOW,
    )

    page.add(
        # Page theme
        ft.Container(
            content=ft.Button("Page theme button"),
            bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
            padding=20,
            width=300,
        ),

        # Inherited theme with primary color overridden
        ft.Container(
            theme=ft.Theme(color_scheme=ft.ColorScheme(primary=ft.Colors.PINK)),
            content=ft.Button("Inherited theme button"),
            bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
            padding=20,
            width=300,
        ),

        # Unique always DARK theme
        ft.Container(
            theme=ft.Theme(color_scheme_seed=ft.Colors.INDIGO),
            theme_mode=ft.ThemeMode.DARK,
            content=ft.Button("Unique theme button"),
            bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
            padding=20,
            width=300,
        ),
    )

ft.run(main)
```

Image: Nested themes - <https://flet.dev/docs/assets/cookbook/theming/nested-themes.png>

Built-in Extensions

Flet controls based on 3rd-party Flutter packages that used to be a part of Flet repository, now have been moved to separate repos and published on pypi:

- `flet-ads`
- `flet-audio`
- `flet-audio-recorder`
- `flet-camera`
- `flet-charts`
- `flet-datatable2`
- `flet-flashlight`
- `flet-geolocator`
- `flet-lottie`
- `flet-map`
- `flet-permission-handler`
- `flet-rive`
- `flet-secure-storage`
- `flet-video`
- `flet-webview`

To use a built-in Flet extension in your project, add it to the dependencies section of your `pyproject.toml` file, for example:

```
dependencies = [  
    "flet-audio",  
    "flet>=0.26.0",  
]
```

Creating an Extension

While Flet controls leverage many built-in Flutter widgets to enable the creation of complex applications, not all Flutter widgets or third-party packages can be directly supported by the Flet team or included in the core Flet framework. At the same time, the Flutter ecosystem is vast and offers developers a wide range of possibilities to extend functionality beyond the core.

To address this, the Flet framework provides an extensibility mechanism. This allows you to incorporate widgets and APIs from your own custom Flutter packages or third-party libraries directly into your Flet application.

In this guide, you will learn how to create Flet extension from template and then customize it to integrate 3rd-party Flutter package into your Flet app.

Prerequisites

To integrate custom Flutter package into Flet you need to have basic understanding of how to create Flutter apps and packages in Dart language and have Flutter development environment configured. See [Flutter Getting Started](#) for more information about Flutter and Dart.

Create Flet extension from template

Flet now makes it easy to create and build projects with your custom controls based on Flutter widgets or Flutter 3rd-party packages. In the example below, we will be creating a custom Flet extension based on the `flutter_spinkit` package.

Step 1. Create new virtual environment and install Flet there.

Step 2. Create new extension project from template.

```
flet create --template extension --project-name flet-spinkit
```

A project with new FletSpinkit control will be created. The control is just a Flutter Text widget with text property, which we will customize later.

Step 3. Build example app.

Flet project created from extension template has `examples/flet_spinkit_example` folder with the example app.

When in the folder where your `pyproject.toml` for the app is (`examples/flet_spinkit_example`), run `flet build` command, for example, for macos:

```
flet build macos -v
```

Open the app and see the new custom Flet Control:

```
open build/macos/flet-spinkit-example.app
```

Image:  - <https://flet.dev/assets/extensions/example.png>

Change Python files

Once the project was built for desktop once, you can make changes to your python files and run it without rebuilding.

First, if you are not using `uv`, install dependencies from `pyproject.toml`:

```
pip install .
```

or

```
poetry install
```

Now you can make changes to your example app `main.py`:

```
import flet as ft

from flet_spinkit import FletSpinkit

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    page.add(
        ft.Container(
            height=150,
            width=300,
            alignment=ft.Alignment.CENTER,
            bgcolor=ft.Colors.PINK_200,
            content=FletSpinkit(
                tooltip="My new PINK FletSpinkit Control tooltip",
```

```

        value="My new PINK FletSpinkit Flet Control",
    ),
),
)

ft.run(main)

```

and run:

```
flet run
```

Image: Image - <https://flet.dev/assets/extensions/example-pink.png>

Change Flutter package

When you make any changes to your flutter package, you need to rebuild:

```
flet build macos -v
```

If you need to debug, run this command:

```
build/macos/flet-spinkit-example.app/Contents/MacOS/flet-spinkit-example --debug
```

Integrate 3rd-party Flutter package

Let's integrate flutter_spinkit package into our Flet app.

Step 1. Add dependency

Go to src/flutter/flet_spinkit folder and run this command to add dependency to flutter_spinkit to pubspec.yaml:

```
flutter pub add flutter_spinkit
```

Read more information about using Flutter packages here.

Step 2. Modify dart file

In the src/flutter/flet_spinkit/lib/src/flet_spinkit.dart file, add import statement and replace Text widget with SpinKitRotatingCircle widget:

```

import 'package:flet/flet.dart';
import 'package:flutter/material.dart';
import 'package:flutter_spinkit/flutter_spinkit.dart';

class FletSpinkitControl extends StatelessWidget {
  final Control control;

  const FletSpinkitControl({
    super.key,
    required this.control,
  });

  @override
  Widget build(BuildContext context) {
    Widget myControl = SpinKitRotatingCircle(color: Colors.red, size: 100.0);

    return LayoutControl(control: control, child: myControl);
  }
}

```

Step 3. Rebuild example app

Go to examples/flet_spinkit_example, clear cache and rebuild your app:

```
flet build macos -v
```

Step 4. Run your app

Image: Image - <https://flet.dev/assets/extensions/spinkit1.gif>

Flet extension structure

After creating new Flet project from extension template, you will see the following folder structure:

```

■■■■ LICENSE
■■■■ mkdocs.yml
■■■■ README.md
■■■■ docs
■   ■■■■ index.md
■   ■■■■ FletSpinkit.md
■■■■ examples
■   ■■■■ flet_spinkit_example
■       ■■■■ README.md
■       ■■■■ pyproject.toml
■       ■■■■ src
■           ■■■■ main.py
■■■■ pyproject.toml

```



```

■■■ src
  ■■■ flet_spinkit
  ■   ■■■ __init__.py
  ■   ■■■ flet_spinkit.py
  ■■■ flutter
    ■■■ flet_spinkit
    ■■■ CHANGELOG.md
    ■■■ LICENSE
    ■■■ README.md
    ■■■ lib
    ■   ■■■ flet_spinkit.dart
    ■   ■■■ src
    ■       ■■■ create_control.dart
    ■       ■■■ flet_spinkit.dart
    ■■■ pubspec.yaml

```

Flet extension consists of:

- package, located in src folder
- example app, located in examples/flet-spinkit_example folder
- docs, located in docs folder

Package

Package is the component that will be used in your app. It consists of two parts: Python and Flutter.

Python

flet_spinkit.py

Defines the Python-side Flet control. FletSpinkit is registered with `@ft.control("flet_spinkit")` and inherits from `ft.LayoutControl`, which ties it to the Flutter `Control.type` handled in the extension. The class currently includes a `value: str` property and a placeholder docstring.

Flutter

pubspec.yaml

Flutter package manifest for the extension. Declares SDK constraints and dependencies. Notable deps:

- flet for Flet extension APIs
- flutter_spinkit for the spinner widgets used by the control

flet_spinkit.dart

Library entrypoint. Exports the public `Extension` class from `extension.dart`.

src/extension.dart

Registers the extension with Flet. `Extension.createWidget` maps `Control.type` to the Flutter widget; currently maps "flet_spinkit" to `FletSpinkitControl`.

src/flet_spinkit.dart

Flutter wrapper widget for the control. `FletSpinkitControl` builds a `SpinKitRotatingCircle` and wraps it with `LayoutControl` so layout/state from the Python control are applied.

Example app

src/main.py

Python program that uses Flet Python control.

pyproject.toml

Here you specify dependency to your package, which can be:

- Path dependency

Absolute path to your Flet extension folder, for example:

```

dependencies = [
    "flet-spinkit @ file:///Users/user-name/projects/flet-spinkit",
    "flet>=0.80.2",
]

```

- Git dependency

Link to git repository, for example:

```
dependencies = [
    "flet-ads @ git+https://github.com/flet-dev/flet-ads.git",
    "flet>=0.80.2",
]
```

- PyPi dependency

Name of the package published on pypi.org, for example:

```
dependencies = [
    "flet-ads",
    "flet>=0.80.2",
]
```

Docs

If you are planning to share your extension with community, you can easily generate documentation from your source code using mkdocs.

Flet extension comes with a docs folder containing initial files for your documentation and a mkdocs.yml file at the project root.

From the folder that contains mkdocs.yml, run the following command to see how your docs look locally:

```
mkdocs serve
```

Open <http://127.0.0.1:8000> in your browser:

Image:  - <https://flet.dev/assets/extensions/mkdocs.png>

Once your documentation is ready, if your package is hosted on GitHub, you can run the following command to host your documentation on GitHub pages:

```
mkdocs gh-deploy
```

You may find this guide helpful to get started with mkdocs.

Customize properties

In the example above, Spinkit control creates a hardcoded Flutter widget. Now let's customize its properties.

Common properties

Generally, there are three types of controls in Flet:

Visual controls that are added to the app/page surface, such as FletSpinkit.

Dialog and other popup controls (dialogs, pickers, panels, etc.) that are opened from the page (for example, `page.open(dlg)`).

Services (Clipboard, Battery, Video, Audio, etc.) that are created as standalone instances and automatically registered with the page.

When creating a visual control that should participate in layout (size, position, transforms, margin, etc.), define a dataclass-based control annotated with `@ft.control("control_name")` and inherit from `LayoutControl`. In its Dart counterpart (`src/flet_spinkit.dart`), wrap your Flutter widget with `LayoutControl(...)`.

When creating a dialog or other popup control (opened with `page.open(...)`), define a dataclass-based control annotated with `@ft.control("control_name")` and inherit from `DialogControl`. In its Dart counterpart, show the dialog/popup (for example, `showDialog` or `showModalBottomSheet`) and return a placeholder widget like `SizedBox.shrink()` instead of wrapping with `LayoutControl(...)` or `BaseControl(...)`.

When creating a service control (Clipboard, Battery, Video, Audio, etc.), define a dataclass-based control annotated with `@ft.control("control_name")` and inherit from `Service`. In its Dart counterpart, implement `FletService` and register it via `FletExtension.createService` (no widget wrapper).

You can use all `LayoutControl`, `DialogControl`, and `Service` properties inherited by your dataclass-based control without re-declaring them as fields (unless you want to override defaults or metadata).

If you have created your extension project from Flet extension template, your Python Control is already inherited from `LayoutControl` and you can use its properties in your example app:

```
import flet as ft

from flet_spinkit import FletSpinkit

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
```

```

page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

page.add(
    ft.Stack(
        [
            ft.Container(height=200, width=200, bgcolor=ft.Colors.BLUE_100),
            FletSpinkit(opacity=0.5, tooltip="Spinkit tooltip", top=0, left=0),
        ]
    )
)

ft.run(main)

```

Image: <https://flet.dev/assets/extensions/spinkit2.gif>

Control-specific properties

Now that you have taken full advantage of the properties Flet LayoutControl offer, let's define the properties that are specific to the new Control you are building.

In the FletSpinkit example, let's define its color and size.

In Python class, define new color and size properties:

```

from typing import Optional

import flet as ft

@ft.control("flet_spinkit")
class FletSpinkit(ft.LayoutControl):
    """
    FletSpinkit Control description.
    """

    color: Optional[ft.ColorValue] = None
    size: float = 100.00

```

In `src/flet_spinkit.dart` file, use helper methods `getColor` and `getDouble` to access color and size values:

```

import 'package:flet/flet.dart';
import 'package:flutter/material.dart';
import 'package:flutter_spinkit/flutter_spinkit.dart';

class FletSpinkitControl extends StatelessWidget {
  final Control control;

  const FletSpinkitControl({
    super.key,
    required this.control,
  });

  @override
  Widget build(BuildContext context) {
    Widget myControl = SpinkitRotatingCircle(
      color: control.getColor("color", context),
      size: control.getDouble("size") ?? 100.0,
    );

    return LayoutControl(control: control, child: myControl);
  }
}

```

Use color and size properties in your app:

```

import flet as ft

from flet_spinkit import FletSpinkit

def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    page.add(
        ft.Stack(
            controls=[
                ft.Container(height=200, width=200, bgcolor=ft.Colors.BLUE_100),
                FletSpinkit(
                    color=ft.Colors.YELLOW,
                    size=150,
                    opacity=0.5,
                    tooltip="Spinkit tooltip",
                    top=0,
                    left=0,
                ),
            ],
        )
    )

```

```
ft.run(main)
```

Rebuild and run:

Image: Image - <https://flet.dev/assets/extensions/spinkit3.gif>

Important: Default values must match on both sides

Properties with default values on the Python side are not sent to Flutter when the user hasn't changed them from the default. This means your Dart code must provide the same default value for every property that has one in Python.

For example, if your Python control declares:

```
size: float = 100.0
animate: bool = True
```

Then your Dart code must use matching defaults:

```
// Correct - defaults match Python
final size = control.getDouble("size", 100.0!);
final animate = control.getBool("animate", true!);

// Wrong - no default, will be null when user doesn't set the property
final size = control.getDouble("size"); // returns null!
final animate = control.getBool("animate"); // returns null!
```

This also applies to `@ft.value` types parsed with helper functions. If a value type field has a default, the corresponding `parseDouble()`, `parseBool()`, `parseDuration()`, etc. call on the Dart side must provide the same default.

Common pitfalls:

- Missing defaults: `control.getDouble("prop")` instead of `control.getDouble("prop", 0.0)!`
- Mismatched defaults: Python has `True` but Dart defaults to `false`
- Unit mismatches: Python uses `Duration(milliseconds=150)` but Dart uses `Duration(microseconds: 150)`
- Empty collections: `field(default_factory=list)` means an empty list won't be sent; Dart must handle null with `?? const []`

You can find source code for this example here.

Examples

Flet has controls that are implemented as built-in extensions and could serve as a starting point for your own controls.

app

Mount all Flet FastAPI handlers in one call.

Parameters:

- `main` (Callable) - Application entry point. It is called for newly connected users. Handler (function or coroutine) must have 1 parameter of instance `Page`.
- `before_main` (Callable | None, default: None) - Called after `Page` was created, but before calling `main`.
- `proxy_path` (str | None, default: None) - URL prefix when the app is mounted under a proxy.
- `assets_dir` (str | None, default: None) - an absolute path to app's assets directory.
- `app_name` (str | None, default: None) - PWA application name.
- `app_short_name` (str | None, default: None) - PWA application short name.
- `app_description` (str | None, default: None) - PWA application description.
- `web_renderer` (WebRenderer, default: WebRenderer.AUTO) - web renderer defaulting to WebRenderer.AUTO.
- `route_url_strategy` (RouteUrlStrategy, default: RouteUrlStrategy.PATH) - routing URL strategy: path (default) or hash.
- `no_cdn` (bool, default: False) - do not load resources from CDN.
- `upload_dir` (str | None, default: None) - an absolute path to a directory with uploaded files.
- `upload_endpoint_path` (str | None, default: None) - absolute URL of upload endpoint, e.g. `/upload`.
- `max_upload_size` (int | None, default: None) - maximum size of a single upload, bytes. Unlimited if None.
- `secret_key` (str | None, default: None) - secret key to sign and verify upload requests.
- `session_timeout_seconds` (int, default: DEFAULT_FLET_SESSION_TIMEOUT) - session lifetime, in seconds, after the user disconnected.
- `oauth_state_timeout_seconds` (int, default: DEFAULT_FLET_OAUTH_STATE_TIMEOUT) - OAuth state lifetime, in seconds, which is the maximum allowed time between starting OAuth flow and redirecting to OAuth callback URL.

https://flet.dev/docs/fastapi/app_manager

app_manager

`FletAppManager()`

FletApp

Handle Flet app WebSocket connections.

Parameters:

- `loop` (`asyncio.AbstractEventLoop`) - asyncio event loop (`asyncio.get_running_loop()`).
- `executor` (`ThreadPoolExecutor`) - Thread pool executor (`app_manager.executor`).
- `main` (`AppCallable`) - Application entry point - an async method called for newly connected user. Handler coroutine must have 1 parameter of instance `Page`.
- `before_main` (`AppCallable` | `None`) - Called before main.
- `session_timeout_seconds` (`int`, default: `DEFAULT_FLET_SESSION_TIMEOUT`) - Session lifetime, in seconds, after user disconnected.
- `oauth_state_timeout_seconds` (`int`, default: `DEFAULT_FLET_OAUTH_STATE_TIMEOUT`) - OAuth state lifetime, in seconds, which is a maximum allowed time between starting OAuth flow and redirecting to OAuth callback URL.
- `upload_endpoint_path` (`str` | `None`, default: `None`) - Absolute URL of upload endpoint, e.g. `/upload`.
- `secret_key` (`str` | `None`, default: `None`) - Secret key to sign upload requests.

Inherits: `flet.messaging.connection.Connection`

Properties

- `executor`
- `loop`

Methods

- `dispose` - Release app-level session reference during teardown.
- `get_upload_url` - Build signed upload URL for a file.
- `handle` - Handle WebSocket connection.
- `oauth_authorize` - Persist OAuth state metadata for a pending authorization flow.
- `send_message` - Serialize and enqueue a server message for transport to the client.

Properties

`buildexecutorinstance-attribute`

`buildloopinstance-attribute`

Methods

`deployed_codedispose`

`dispose()`

Release app-level session reference during teardown.

`deployed_codeget_upload_url`

`get_upload_url(file_name: str, expires: int) -> str`

Build signed upload URL for a file.

Parameters:

- `file_name` (`str`) - File name to be uploaded.
- `expires` (`int`) - URL lifetime in seconds.

Returns:

- `str` - Signed relative upload URL.

Raises:

- `RuntimeError` - If upload endpoint is not configured.

`deployed_codehandleasync`

`handle(websocket: WebSocket)`

Handle WebSocket connection.

Parameters:

- `websocket (WebSocket)` - WebSocket instance.

`deployed_codeoauth_authorize`

`oauth_authorize(attrs: dict[str, Any])`

Persist OAuth state metadata for a pending authorization flow.

Parameters:

- `attrs (dict[str, Any])` - OAuth attributes payload containing state and optional completion page data.

`deployed_codesend_message`

`send_message(message: ClientMessage)`

Serialize and enqueue a server message for transport to the client.

Parameters:

- `message (ClientMessage)` - Outbound protocol message.

FletOAuth

HTTP handler for OAuth callback.

Methods

- `handle` - Handle OAuth callback request.

Methods

`deployed_codehandleasync`

`handle(request: Request)`

Handle OAuth callback request.

Request must contain minimum code and state query parameters.

Returns either redirect to a Flet page or a HTML page with further instructions.

FletStaticFiles

Serve Flet app static files.

Parameters:

- `assets_dir` (str | None, default: None) - An absolute path to app's assets directory.
- `app_name` (str | None, default: None) - PWA application name.
- `app_short_name` (str | None, default: None) - PWA application short name.
- `app_description` (str | None, default: None) - PWA application description.
- `web_renderer` (WebRenderer, default: WebRenderer.AUTO) - Type of web renderer.
- `route_url_strategy` (RouteUrlStrategy, default: RouteUrlStrategy.PATH) - Routing URL strategy.
- `no_cdn` (bool, default: False) - Whether not load CanvasKit, Pyodide, and fonts from CDN.
- `websocket_endpoint_path` (str | None, default: None) - Absolute URL of Flet app WebSocket handler. Defaults to `/ws`.

Inherits: `fastapi.staticfiles.StaticFiles`

Properties

- `font_manifest_json`
- `index`
- `manifest_json`

Methods

- `lookup_path` - Resolve a static file path, with SPA fallback for client-side routes.

Properties

`buildfont_manifest_jsoninstance-attribute`

`buildindexinstance-attribute`

`buildmanifest_jsoninstance-attribute`

Methods

`deployed_codelookup_path`

```
lookup_path(  
    path: str,  
) -> tuple[str, os.stat_result | None]
```

Resolve a static file path, with SPA fallback for client-side routes.

Route-like paths (no extension, or `.html`) that don't match a file fall back to `index.html` so the Flutter client can handle routing. Asset-like paths (any other extension) resolve to a real 404.

Parameters:

- `path` (str) - Requested path, relative to the mounted static root (e.g. `"about"`, `"sample.json"`).

Returns:

- `tuple[str, os.stat_result | None]` - A `(full_path, stat_result)` tuple. When `stat_result` is `None`, Starlette responds with 404.

FletUpload

Flet app uploads handler.

Parameters:

- `upload_dir` (str) - An absolute path to a directory with uploaded files.
- `max_upload_size` (int | None, default: None) - maximum size of a single upload, bytes. Unlimited if None.
- `secret_key` (str | None, default: None) - secret key to sign and verify upload requests.

Methods

- `handle` - Handle file upload.

Methods

`deployed_codehandleasync`

`handle(request: Request)`

Handle file upload.

Upload must be an non-encoded (raw) file in the request body.

Creating a new Flet app

Create a new directory (or directory with pyproject.toml already exists if initialized with a project manager) and switch into it.

To create a new "minimal" Flet app run the following command:

- uv
 - pip
- ```
uv run flet create
flet create
```

## Important

Any existing README.md or pyproject.toml (for example, created by uv init) will be replaced by the one created by flet create command.

The command will create the following directory structure:

```
README.md
pyproject.toml
src
 assets
 icon.png
 main.py
storage
 data
 temp
```

You can find more information about flet create command [here](#).

## Auto-update

Flet automatically calls `page.update()` (or `.update()` on the nearest isolated ancestor) at the end of every event handler and `main()` function. This means you don't need to call `.update()` yourself in most cases:

```
import flet as ft

def main(page: ft.Page):
 def button_click(e):
 page.controls.append(ft.Text("Clicked!"))
 # no need to call page.update() - it happens automatically

 page.controls.append(ft.Button("Click me", on_click=button_click))
 # no need to call page.update() here either

ft.run(main)
```

## Note

If your event handler already calls `.update()` explicitly (e.g. code written for Flet 0.x), the automatic update is skipped to avoid a redundant double update.

## Disabling auto-update

You can disable auto-update for fine-grained control over when updates are sent to the client. Use `ft.context.disable_auto_update()` and `ft.context.enable_auto_update()` to toggle the behavior.

When called inside a handler, the setting applies to the current handler context only:

```
import flet as ft

def main(page: ft.Page):
 def add_many_items(e):
 ft.context.disable_auto_update()
 for i in range(100):
 page.controls.append(ft.Text(f"Item {i}"))
 page.update() # single update for all 100 items

 page.controls.append(ft.Button("Add items", on_click=add_many_items))

ft.run(main)
```

When called outside of event handlers (e.g. at the module level), it controls the global default for the entire app:

```
import flet as ft

disable auto-update globally
```

```
ft.context.disable_auto_update()

def main(page: ft.Page):
 def button_click(e):
 page.controls.append(ft.Text("Clicked!"))
 page.update() # must call explicitly since auto-update is off

 page.controls.append(ft.Button("Click me", on_click=button_click))
 page.update()

ft.run(main)
```

Now let's see Flet in action by running the app!

# Installation

## Prerequisites

### Python version

Flet requires Python 3.10 or later. (1)

- Check your Python version using `python --version`.

### Operating System

#### macOS

Flet supports macOS 12 (Monterey) or later.

#### Windows

Flet supports 64-bit version of Microsoft Windows 10 and Windows 11.

#### Linux

Flet supports Debian 10, 11 and 12 and Ubuntu 20.04, 22.04 and 24.04 LTS.

Desktop flavor (audio & video support)

On Linux, the Flet desktop client is available in two flavors: full and light.

The light flavor (default on Linux) does not include audio and video extensions, resulting in a smaller download. The full flavor bundles audio and video support out of the box.

To select the flavor, either set the `FLET_DESKTOP_FLAVOR` environment variable:

```
export FLET_DESKTOP_FLAVOR=full
```

or add the setting to your project's `pyproject.toml`:

```
[tool.flet]
desktop_flavor = "full"
```

If you use the light flavor and need audio or video, you will need to install the required libraries yourself — see the Audio and Video setup guides.

#### Windows Subsystem for Linux (WSL)

Flet apps can be run on WSL 2 (Windows Subsystem for Linux 2).

However, if you are getting cannot open display error follow this guide for troubleshooting.

## Creating a virtual environment (venv)

We recommend using a virtual environment for your Flet projects to keep dependencies isolated and avoid conflicts with your other Python projects.

First, create a new directory for your Flet project and switch into it:

```
mkdir my-app
cd my-app
```

Next, create and activate a virtual environment (we recommend using `uv` as package manager):

- `uv`
- `pip`

`uv` is "An extremely fast Python package and project manager, written in Rust".

Install `uv` if you haven't already, then run the following commands:

```
uv init --python='>=3.10'
uv venv
source .venv/bin/activate
```

Using Python's built-in `venv` module:

```
python -m venv .venv
source .venv/bin/activate
```

## Install Flet

To install Flet and add it to your project dependencies, do the following depending on your package manager:

- uv
- pip

```
uv add 'flet[all]'
pip install 'flet[all]'
```

## Verify installation

To make sure Flet has been installed correctly, we can check its version using the `--version` (or `-V`) flag or the `doctor` command:

- uv
- pip

```
uv run flet --version
or
uv run flet doctor

flet --version
or
flet doctor
```

Now you are ready to create your first Flet app.

## Upgrade Flet

To upgrade Flet to its latest version:

- uv
- pip

```
uv add 'flet[all]' --upgrade
pip install 'flet[all]' --upgrade
```

# Running a Flet app (Hot Reload)

Flet apps can be executed as either desktop or web applications using the flet run command. Doing so will start the app in a native OS window or a web browser, respectively, with hot reload enabled to view changes in real-time.

## Desktop app

To run Flet app as a desktop app, use the following command:

- uv
  - pip
- ```
uv run flet run
flet run
```

When you run the command without any arguments, main.py script in the current directory will be executed, by default.

If you need to provide a different path, use the following command:

- uv
 - pip
- ```
uv run flet run [script]
flet run [script]
```

Where [script] is a relative (ex: counter.py) or absolute (ex: /Users/john/projects/flet-app/main.py) path to the Python script you want to run.

The app will be started in a native OS window:

macOS

Image: macOS - <https://flet.dev/docs/assets/getting-started/counter-app/macOS.png>

Windows

Image: windows - <https://flet.dev/docs/assets/getting-started/counter-app/windows.png>

## Web app

To run Flet app as a web app, use the --web (or -w) option:

- uv
  - pip
- ```
uv run flet run --web [script]
flet run --web [script]
```

A new browser window/tab will be opened and the app will be using a random TCP port:

Image: Web - <https://flet.dev/docs/assets/getting-started/counter-app/safari.png>

Running Flet app as a web app

Watching for changes

By default, Flet will watch the script file that was run and reload the app whenever the contents of this file are modified+saved, but will not watch for changes in other files.

To modify this behavior, you can use one or more of these flet run options:

- -d or --directory to watch for changes in the [script]'s directory only
- -r or --recursive to watch for changes in the [script]'s directory and all sub-directories recursively

Example

- uv
 - pip
- ```
uv run flet run --recursive [script]
flet run --recursive [script]
```



# Testing on Mobile

Start building awesome mobile apps in Python using just your computer and mobile phone!

Important

- Make sure your computer fulfills these requirements.
- Your device and computer must be connected to the same Wi-Fi or local network.
- We recommend starting with the creation of a new virtual environment.
- iOS
- Android

Install the Flet iOS app on your iOS device. You will be using this app to see how your Flet project is working on iPhone or iPad.

Image: Get it on App Store - <https://flet.dev/docs/assets/getting-started/testing-on-mobile/ios/qr-code.jpg>

[View on App Store](<https://apps.apple.com/app/flet/id1624979699>)

Run the following command to start Flet development server with your app:

- uv
  - pip
- ```
uv run flet run --ios [script]
flet run --ios [script]
```

Install the Flet Android app on your Android device. You will be using this app to see how your Flet project is working on Android device.

Image: Get it on Google Play - <https://flet.dev/docs/assets/getting-started/testing-on-mobile/android/google-play-badge.png>

[View on PlayStore](<https://play.google.com/store/apps/details?id=com.appveyor.flet>)

Run the following command to start Flet development server with your app:

- uv
 - pip
- ```
uv run flet run --android [script]
flet run --android [script]
```

A QR code with encoded project URL will be displayed in the terminal:

Image: app-qr-code - <https://flet.dev/docs/assets/getting-started/testing-on-mobile/ios/app-qr-code.png>

Open Camera app on your Android device, point to QR code you got and click URL to open it in Flet app.

Try updating your [script] - the app will be instantly refreshed on your Android device.

To return to "Home" tab either:

- Long-press anywhere on the screen with 3 fingers or
- Shake your Android device.

You can also "manually" add a new project by clicking "+" floating action button in the app and typing in its URL.

Examples

Below is a URL to a "Counter" Flet app that we hosted for testing purposes:

<https://flet-counter-test-ios.fly.dev>

- The "Gallery" tab of the app has some more examples that you can try out.
- Explore Flet examples for even more examples.

# Packaging app for Android

Instructions for packaging a Flet app into an Android APK and Android App Bundle (AAB).

Info

This guide provides detailed Android-specific information. Complementary and more general information is available [here](#).

## Prerequisites

### Android SDK

The build process requires both Java (JDK 17) and the Android SDK.

If either component is missing or an incompatible version is detected, the required tools will be automatically installed during the first run of the flet build command.

- The JDK is installed in `$HOME/java/<version>` (for example, `17.0.13+11`).

If Android Studio is installed, Flet CLI will use its SDK:

- macOS: `~/Library/Android/sdk`
- Windows: `%LOCALAPPDATA%\Android\Sdk`
- Linux: `~/Android/Sdk`

Otherwise, a standalone Android SDK is installed in:

- macOS/Linux: `~/Android/sdk`
- Windows: `%USERPROFILE%\Android\sdk`

`ANDROID_HOME` and `ANDROID_SDK_ROOT` are also respected if set.

### Android wheels for binary Python packages

Binary Python packages (in contrast to "pure" Python packages written in Python only) are packages that are partially written in C, Rust, or other languages producing native code. Example packages are `numpy`, `cryptography`, or `pydantic-core`.

Make sure all non-pure (binary) packages used in your Flet app have pre-built wheels for Android.

## flet build apk

Note

This command can be run on a macOS, Linux, or Windows.

Builds a release Android APK.

Release builds are optimized for production, meaning they don't support debugging and are intended for publishing to app stores such as Google Play.

For Play Store deployment, it's recommended to:

- Use an Android App Bundle (AAB) for more efficient delivery and smaller install size
- Or split the APK by ABI to reduce the APK size

### Split APK per ABI

Android devices use different CPUs, so APKs can target different Application Binary Interfaces (ABIs).

By default, Flet builds a single "fat" APK that contains native binaries for all supported ABIs. This maximizes device compatibility but increases APK size.

Enabling ABI splits produces one APK per ABI, which reduces file size but requires distributing the correct APK for each device.

### Supported target architectures

The following target architectures are supported:

- arm64-v8a
- armeabi-v7a
- x86\_64
- x86

## Resolution order

Its value is determined in the following order of precedence:

- --split-per-abi
- [tool.flet.android].split\_per\_abi
- false

When enabled, 3 APKs are produced by default, one for each of the following ABIs: arm64-v8a, armeabi-v7a, and x86\_64. These can be customized by setting target architectures.

## Example

- flet build
  - pyproject.toml
- ```
flet build apk --split-per-abi
[tool.flet.android]
split_per_abi = true
```

flet build aab

Note

This command can be run on a macOS, Linux, or Windows.

Builds a release Android App Bundle (AAB) file.

Release builds are optimized for production, meaning they don't support debugging and are intended for publishing to app stores such as the Google Play Store.

It is recommended to use this AAB format (instead of APK) for publishing to the Google Play Store due to its optimized app size.

If you need to limit the ABIs included in the bundle, use --arch / [tool.flet.android].target_arch while split_per_abi is false.

Signing an Android bundle

Android requires that all APKs be digitally signed with a certificate before they are installed on a device or updated. When releasing using Android App Bundles, you need to sign your app bundle with an upload key before uploading it to the Play Console, and Play App Signing takes care of the rest. For apps distributing using APKs on the Play Store or on other stores, you must manually sign your APKs for upload.

For detailed information, see this guide.

To publish on the Play Store, you need to sign your app with a digital certificate.

Android uses two signing keys: upload and app signing.

- Developers upload an .aab or .apk file signed with an upload key to the Play Store.
- The end-users download the .apk file signed with an app signing key.

To create your app signing key, use Play App Signing as described in the official Play Store documentation.

To sign your app, use the following instructions.

Note

If you don't provide an upload keystore, release builds are signed with the debug key. This is fine for local testing but cannot be uploaded to the Play Store.

Create an upload keystore

If you have an existing keystore, skip to the next step. If not, create one using one of the following methods:

Follow the Android Studio key generation steps.

Run the following command at the command line: On macOS or Linux, use the following command:

```
keytool -genkey -v -keystore ~/upload-keystore.jks -keyalg RSA \
-keysize 2048 -validity 10000 -alias upload
```

On Windows, use the following command in PowerShell:

```
keytool -genkey -v -keystore $env:USERPROFILE\upload-keystore.jks `
-storetype JKS -keyalg RSA -keysize 2048 -validity 10000 `
-alias upload
```

You will be prompted for several details, such as a keystore password, a key alias, your names, and location. Remember the password and alias for use in the configuration steps below.

A file named upload-keystore.jks will be created in your home directory. If you want to store it elsewhere, change the argument passed to the -keystore parameter. The location of the keystore file is equally important for the key store step below.

Note

The keytool command might not be in your path—it's part of Java, which is installed as part of Android Studio. For the concrete path, run flutter doctor -v and locate the path printed after 'Java binary at:'. Then use that fully qualified path replacing java (at the end) with keytool. If your path includes space-separated names, such as Program Files, use platform-appropriate notation for the names. For example, on macOS and Linux use Program\ Files, and on Windows use "Program Files".

The -storetype JKS tag is only required for Java 9 or newer. As of the Java 9 release, the keystore type defaults to PKCS12.

Important

Keep your keystore file private; never check it into public source control!

Key alias

An alias name for the key within the keystore.

Resolution order

Its value is determined in the following order of precedence:

- --android-signing-key-alias
- FLET_ANDROID_SIGNING_KEY_ALIAS
- [tool.flet.android.signing].key_alias
- "upload"

Example

- flet build
 - pyproject.toml
 - .env
- ```
flet build aab --android-signing-key-alias value
[tool.flet.android.signing]
key_alias = "value"
FLET_ANDROID_SIGNING_KEY_ALIAS="value"
```

### Key store

The path to the keystore file (with extension .jks).

If you used the CLI commands above as-is, this file might be located at /Users/<user name>/upload-keystore.jks on macOS or C:\Users\<user name>\upload-keystore.jks on Windows.

### Resolution order

Its value is determined in the following order of precedence:

- --android-signing-key-store
- [tool.flet.android.signing].key\_store

- FLET\_ANDROID\_SIGNING\_KEY\_STORE

## Example

- flet build
  - pyproject.toml
  - .env
- ```
flet build aab --android-signing-key-store path/to/store.jks
[tool.flet.android.signing]
key_store = "path/to/store.jks"
FLET_ANDROID_SIGNING_KEY_STORE="path/to/store.jks"
```

Key store password

A password to unlock the keystore file (can contain multiple key entries).

Resolution order

Its value is determined in the following order of precedence:

- --android-signing-key-store-password
- FLET_ANDROID_SIGNING_KEY_STORE_PASSWORD
- key password

Example

- flet build
 - pyproject.toml
 - .env
- ```
flet build aab --android-signing-key-store-password value
```

For security reasons, the keystore password is not read from pyproject.toml to prevent accidental exposure in source control. See the other tabs for supported alternatives.

```
FLET_ANDROID_SIGNING_KEY_STORE_PASSWORD="value"
```

## Key password

A password used to access the private key inside the keystore.

## Resolution order

Its value is determined in the following order of precedence:

- --android-signing-key-password
- FLET\_ANDROID\_SIGNING\_KEY\_PASSWORD
- key store password

## Example

- flet build
  - pyproject.toml
  - .env
- ```
flet build aab --android-signing-key-password value
```

For security reasons, the keystore password is not read from pyproject.toml to prevent accidental exposure in source control. See the other tabs for supported alternatives.

```
FLET_ANDROID_SIGNING_KEY_PASSWORD="value"
```

Android Manifest

The Android Manifest describes essential information about your app to the Android build tools, the Android operating system, and Google Play. The file in which this information is written is AndroidManifest.xml, which gets populated with the information you provide.

Application attributes

You can add or override attributes on the `<application>` element of the `AndroidManifest.xml` file in the build template.

See also:

- `<application>` element

Resolution order

Its value is determined in the following order of precedence:

- `[tool.flet.android.manifest_application]`

Example

- `pyproject.toml`

```
[tool.flet.android.manifest_application]
usesCleartextTraffic = "true"
allowBackup = "false"
```

Template translation

In the `AndroidManifest.xml`, the `pyproject.toml` example above will be translated accordingly into this:

```
<application
  android:usesCleartextTraffic="true"
  android:allowBackup="false">
</application>
```

Meta-data

A name-value pair for an item of additional, arbitrary data that can be supplied to the parent component. More information [here](#).

A meta-data item is composed of:

- **name:** A unique name for the item, usually with a Java-style naming convention, for example `"com.sample.project.activity.fred"`.
- **value:** The value of the item. Android supports strings, integers, booleans, and floats. Flet writes values as strings, so pass the literal value you want Android to read (for example `"true"`, `"123"`, `"1.23"`).

See also:

- `<meta-data>` element

Resolution order

Its value is determined in the following order of precedence:

- `--android-meta-data`
- `[tool.flet.android.meta_data]`

Example

- `flet build`
- `pyproject.toml`

```
flet build apk \
  --android-meta-data firebase_analytics_collection_enabled=true \
  --android-meta-data default_timeout_seconds=30

[tool.flet.android.meta_data]
"firebase_analytics_collection_enabled" = "true"
"default_timeout_seconds" = "30"
```

Template translation

In the `AndroidManifest.xml`, the `pyproject.toml` example above will be translated accordingly into this:

```
<application>
  <meta-data android:name="firebase_analytics_collection_enabled" android:value="true" />
  <meta-data android:name="default_timeout_seconds" android:value="30" />
</application>
```

Providers

A content provider component that supplies app data to other apps or components. More information [here](#).

Each provider is declared as a TOML table whose key is the provider's `android:name` (the fully-qualified class name). The table's entries become extra `android:<key>=<value>` attributes on the generated `<provider>` element. A reserved `meta_data` sub-table emits nested `<meta-data>` children inside the provider.

See also:

- `<provider>` element

Resolution order

Its value is determined in the following order of precedence:

- `[tool.flet.android.provider]`

Supported value forms

The value for each provider must be a TOML inline table or sub-table of attributes. A value of `false` skips the entry entirely; `true` is not accepted (a `<provider>` with no attributes is meaningless). An empty table `{}` is also treated as `false`.

Attribute values must be strings, booleans, or numbers — they are written verbatim into the manifest. The `name` key is reserved (the `android:name` comes from the table key).

The reserved `meta_data` sub-table emits `<meta-data>` children. Each sub-entry's key is the `android:name`; the value can be either:

- a scalar (string/bool/number) — rendered as `android:value="..."`, or
- an inline table — its entries become `android:<key>=<value>` attributes on the `<meta-data>` element (useful for `android:resource="@xml/..."`).

Example

- `pyproject.toml`

```
[tool.flet.android.provider]
"rikka.shizuku.ShizukuProvider" = { authorities = "${applicationId}.shizuku", multiprocess = "false", enabled = "true", exported :

[tool.flet.android.provider."com.example.MyProvider"]
authorities = "${applicationId}.myprovider"
exported = false
grantUriPermissions = true

[tool.flet.android.provider."com.example.MyProvider".meta_data]
"android.support.FILE_PROVIDER_PATHS" = { resource = "@xml/file_paths" }
"some.other.key" = "some-value"
```

Template translation

In the `AndroidManifest.xml`, the `pyproject.toml` example above will be translated accordingly into this:

```
<application>
  <provider android:name="rikka.shizuku.ShizukuProvider"
    android:authorities="${applicationId}.shizuku"
    android:multiprocess="false"
    android:enabled="true"
    android:exported="true"
    android:permission="android.permission.INTERACT_ACROSS_USERS_FULL" />
  <provider android:name="com.example.MyProvider"
    android:authorities="${applicationId}.myprovider"
    android:exported="false"
    android:grantUriPermissions="true">
    <meta-data android:name="android.support.FILE_PROVIDER_PATHS" android:resource="@xml/file_paths" />
    <meta-data android:name="some.other.key" android:value="some-value" />
  </provider>
</application>
```

note

Flet already declares a built-in `androidx.core.content.FileProvider` with authorities `${applicationId}.provider`. Declaring another provider that uses the same authorities will cause the Android manifest merger to fail — pick a different `android:authorities` value for your custom provider.

Features

A hardware or software feature that is used by the application. More information [here](#).

- `name`: Specifies a single hardware or software feature used by the application as a descriptor string. Valid attribute values are listed in the [Hardware features](#) and [Software features](#) sections. These attribute values are case-sensitive.

- **required:** A boolean value (true or false) that indicates whether the application requires the feature specified by the name.

See also:

- `<uses-feature>` element
- Features reference

Resolution order

Its value is determined in the following order of precedence:

- `--android-features`
- `[tool.flet.android.feature]`
- Values injected by cross-platform permission bundles, if any.
- defaults: `android.software.leanback=false`, `android.hardware.touchscreen=false`

Supported value forms

- flet build
- pyproject.toml

Accepts repeated `<name>=<required>` entries. The `<required>` value can be true or false (case-insensitive).

Use boolean values. TOML booleans must be lowercase: true or false.

Example

- flet build
- pyproject.toml

```
flet build apk \
  --android-features android.hardware.camera=true \
  --android-features android.hardware.location.gps=false

[tool.flet.android.feature]
"android.hardware.camera" = true
"android.hardware.location.gps" = false
```

Template translation

In the `AndroidManifest.xml`, the example above will be translated accordingly into this:

```
<manifest>
  <uses-feature android:name="android.hardware.camera" android:required="true" />
  <uses-feature android:name="android.hardware.location.gps" android:required="false" />
</manifest>
```

Permissions

Use cross-platform permissions from Permissions when possible, and add Android-specific permissions or features here.

See also:

- Manifest.permission constants
- Request app permissions

Resolution order

Its value is determined in the following order of precedence:

- `--android-permissions`
- `[tool.flet.android.permission]`
- Values injected by cross-platform permission bundles, if any.
- defaults: `android.permission.INTERNET=true`

Supported value forms

- flet build
- pyproject.toml

Accepts repeated `<name>=<enabled>` entries. The `<enabled>` value can be true or false (case-insensitive). Permissions with false are omitted from the generated manifest.

Use boolean values. TOML booleans must be lowercase: true or false. Permissions set to false are omitted from the generated manifest.

A value can also be a TOML inline table whose entries become extra `android:<key>=<value>` attributes on the generated `<uses-permission>` element — useful for attributes like `maxSdkVersion` or `usesPermissionFlags`. A non-empty table is always emitted; an empty table `{}` is treated as false.

Example

- flet build
- pyproject.toml

```
flet build apk \
  --android-permissions android.permission.READ_EXTERNAL_STORAGE=true \
  --android-permissions android.permission.WRITE_EXTERNAL_STORAGE=true

[tool.flet.android.permission]
"android.permission.READ_EXTERNAL_STORAGE" = true
"android.permission.WRITE_EXTERNAL_STORAGE" = true
"android.permission.ACCESS_FINE_LOCATION" = { maxSdkVersion = "30" }
"android.permission.BLUETOOTH_SCAN" = { usesPermissionFlags = "neverForLocation" }
```

Template translation

In the `AndroidManifest.xml`, the `pyproject.toml` example above will be translated accordingly into this:

```
<manifest>
  <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
  <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
  <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" android:maxSdkVersion="30" />
  <uses-permission android:name="android.permission.BLUETOOTH_SCAN" android:usesPermissionFlags="neverForLocation" />
</manifest>
```

Minimum SDK version

The minimum Android API level your app can be installed on.

See also:

- `<uses-sdk>` element (`minSdkVersion`)
- Android API level reference

Resolution order

Its value is determined in the following order of precedence:

- `[tool.flet.android].min_sdk_version`
- Flutter default: `flutter.minSdkVersion`

Example

- pyproject.toml
- ```
[tool.flet.android]
min_sdk_version = 24
```

### Template translation

In the `android/app/build.gradle.kts`, the `pyproject.toml` example above will be translated accordingly into this:

```
defaultConfig {
 val resolvedMinSdk = 24
 minSdk = resolvedMinSdk
}
```

## Target SDK version

The Android API level your app targets for runtime behavior and compatibility.

See also:

- `<uses-sdk>` element (`targetSdkVersion`)
- Target API level requirements and behavior changes

## Resolution order

Its value is determined in the following order of precedence:

- [tool.flet.android].target\_sdk\_version
- Flutter default: flutter.targetSdkVersion

## Example

- pyproject.toml

```
[tool.flet.android]
target_sdk_version = 35
```

### Template translation

In the android/app/build.gradle.kts, the pyproject.toml example above will be translated accordingly into this:

```
defaultConfig {
 val resolvedTargetSdk = 35
 targetSdk = resolvedTargetSdk
}
```

## Adaptive icon background

The background color used for the Android adaptive launcher icon.

This value is applied when app icons are generated for Android.

## Resolution order

Its value is determined in the following order of precedence:

- --android-adaptive-icon-background
- [tool.flet.android].adaptive\_icon\_background
- Build template default: #ffffff

## Example

- flet build
- pyproject.toml

```
flet build apk --android-adaptive-icon-background "#0B6BFF"

[tool.flet.android]
adaptive_icon_background = "#0B6BFF"
```

## ADB Tips

Android Debug Bridge (adb) is a command-line tool included in the Android SDK that lets you communicate with Android devices and emulators.

If you installed Android Studio on macOS, adb is typically located at: ~/Library/Android/sdk/platform-tools/adb.

See this guide for help installing and using adb on different platforms.

To run interactive commands inside an Android simulator or device:

```
adb shell
```

To overcome "permissions denied" error while trying to browse file system in interactive Android shell:

```
su
```

To download a file from a device to your local computer:

```
adb pull <device-path> <local-path>
```

To install an APK on an Android device:

```
adb install <path-to-your.apk>
```

This works for both physical devices and emulators. If more than one device is connected, specify the target device:

```
adb -s <device> install <path-to-your.apk>
```

You can list available devices with:

```
adb devices
```

# Packaging app for iOS

Instructions for packaging a Flet app into an Xcode archive and, when signing is configured, an IPA for distribution.

Info

This guide provides detailed on iOS-specific information. Complementary and more general information is available [here](#).

## Prerequisites

### Rosetta 2

Some Flutter tooling and dependencies still ship as Intel binaries. Install Rosetta 2 on Apple Silicon if Flutter or CocoaPods prompts for it:

```
sudo softwareupdate --install-rosetta --agree-to-license
```

### Xcode

Xcode 15 or later to compile native Swift or Objective-C code. Open Xcode once after installation to accept the license and install additional components.

### CocoaPods

CocoaPods 1.16 or later to compile and enable Flutter plugins.

### iOS wheels for binary Python packages

Binary Python packages (vs "pure" Python packages written in Python only) are packages that are partially written in C, Rust, or other languages producing native code. Example packages are numpy, cryptography, or pydantic-core.

Make sure all non-pure (binary) packages used in your Flet app have pre-built wheels for iOS.

## flet build ipa

Supported host platforms

This command can be run on macOS only.

Builds an iOS app archive (.xcarchive) and, when signing is configured, exports an .ipa for testing or distribution.

To generate an .ipa for testing on your device or uploading to App Store Connect for distribution, you will need the following:

- Apple Developer Program subscription with access to App Store Connect
- Application Identifier
- Bundle ID configured in your Flet project (must match the App ID)
- Signing Certificate
- Provisioning Profile

## flet build ios-simulator

Supported host platforms

This command can be run on macOS only.

Builds an (unsigned) iOS Simulator .app bundle, intended for testing on iOS Simulator and does not require a signing certificate, provisioning profile, or Apple Developer Program setup.

## Configuration options

### Application Identifier (App ID)

A unique string that identifies your app within the Apple ecosystem. It is required to sign and distribute an iOS app and is used for various services like Push Notifications, App Groups, iCloud, and In-App Purchases.

It consists of two parts:

- Team ID: A unique 10-character string assigned by Apple to your developer account.
- Bundle ID: A reverse domain-style identifier for your app (e.g., com.example.myapp).

The Bundle ID in your Flet configuration must match the Bundle ID registered in the App ID.

## Creating a new App ID

- Visit Apple Developer Portal.
- Sign in with your Apple Developer Account.
- Click the "+" button to add a new identifier.
- Select "App IDs" and click Continue.
- Enter a Description – This is just for reference (e.g., "MyApp Identifier").

Choose the App ID Type:

- Select "App" for a standard iOS/macOS app.
- If you need an identifier for services like Apple Pay or Passbook, select the appropriate option.

Bundle ID – Choose:

- Explicit Bundle ID (com.example.myapp) – Recommended for most apps.
- Wildcard Bundle ID (com.example.\*) – Allows multiple apps to use the same identifier (rarely used).
- Enable App Services – Check the boxes for any services your app needs (e.g., Push Notifications, Sign in with Apple, etc.).
- Click Continue and Register.

Now you have Bundle ID and Team ID that will be used to identify your app.

## Team ID

The developer team ID to include in export options.

Resolution order

Its value is determined in the following order of precedence:

- --ios-team-id
- [tool.flet.ios].team\_id
- [tool.flet.ios.export\_methods."EXPORT\_METHOD"].team\_id

Example

- flet build
  - pyproject.toml
- ```
flet build ipa --ios-team-id ABCDEFE234

[tool.flet.ios]
team_id = "ABCDEFE234"
```

Signing Certificate

The certificate name, SHA-1 hash, or automatic selector to use for signing the iOS app bundle. Automatic selectors allow Xcode to pick the newest installed certificate of a particular type.

The available automatic selectors are:

- "Apple Development"
- "Apple Distribution"
- "Developer ID Application"
- "iOS Developer"
- "iOS Distribution"
- "Mac App Distribution"

- "Mac Developer"

Generating a Certificate Signing Request (CSR)

Before creating a development or distribution certificate, you need a CSR (Certificate Signing Request).

- Open Keychain Access on your Mac (++cmd+space++, then search for "Keychain Access").
- In the top menu, go to Keychain Access → Certificate Assistant → Request a Certificate From a Certificate Authority...

Fill in:

- User Email Address: Your Apple Developer email.
- Common Name: A descriptive name (e.g., "My App Distribution").
- CA Email Address: Leave this blank.
- Request is for: Select "Saved to disk".
- Click Continue, choose a location to save the .certSigningRequest file, and click Save.

Creating a Certificate in Apple Developer Portal

- Go to the Apple Developer Certificates Page.
- Click the "+" button to create a new certificate.
- Select "Apple Distribution" (for App Store & Ad Hoc) or "Apple Development" (for development) and click Continue.
- Upload the CSR file you created earlier and click Continue.
- Apple will generate the certificate. Click Download to get the .cer file.
- Double-click the downloaded .cer file to install it in Keychain Access.
- Open the Keychain Access app and ensure the certificate is installed under "login" keychain. The name of the development certificate usually starts with "Apple development:" and the name of the distribution certificate starts with "Apple distribution:".

Resolution order

Its value is determined in the following order of precedence:

- --ios-signing-certificate
- [tool.flet.ios].signing_certificate
- [tool.flet.ios.export_methods."EXPORT_METHOD"].signing_certificate

Example

- flet build
 - pyproject.toml
- ```
flet build ipa --ios-signing-certificate "Apple Distribution"

[tool.flet.ios]
signing_certificate = "Apple Distribution"
```

## Provisioning Profile

A Provisioning Profile is a file that allows an iOS app to run on physical devices and be distributed through the App Store or internally. It links your App ID, Developer/Distribution Certificate, and Registered Devices.

There are different types of provisioning profiles:

- Development Profile – Used for testing on physical devices.
- Ad Hoc Profile – Used for distributing an app outside the App Store to specific devices.
- App Store Profile – Used for submitting an app to the App Store.
- Enterprise Profile – Used for internal distribution within an organization.

## Creating a New Provisioning Profile

Follow these steps to create a provisioning profile via the Apple Developer Portal:

Step 1: Go to Apple Developer Portal

- Visit the Apple Developer Portal;
- Sign in with your Apple Developer Account

Step 2: Create a New Provisioning Profile

- Click the "+" button to add a new provisioning profile;

Choose the type of profile:

- iOS App Development – for testing on devices
- Ad Hoc – for distributing to specific devices
- App Store – for submitting an app to the App Store
- In-House – for internal distribution (Enterprise accounts only)
- Click Continue

Step 3: Select an App ID

- Choose the App ID that matches your app;
- Click Continue.

Step 4: Select a Distribution Certificate

Choose the appropriate certificate:

- iOS Development Certificate - for iOS App Development profile
- iOS Distribution Certificate - for Ad Hoc or App Store profiles
- Click Continue.

Step 5: Select Registered Devices (for Development & Ad Hoc)

- If you selected an iOS App Development or Ad Hoc profile, select the devices to include;
- Click Continue.

Step 6: Name and Generate the Profile

- Enter a Profile Name (e.g., MyApp Development Profile);
- Click Generate;
- Click Download to get the .mobileprovision file.

## Installing Provisioning Profile

Provisioning profiles are stored in ~/Library/MobileDevice/Provisioning Profiles directory.

To install a downloaded provisioning profile, copy it to ~/Library/MobileDevice/Provisioning Profiles directory with a new {UUID}.mobileprovision name.

Run the following command to get profile UUID:

```
profile_uuid=$(security cms -D -i ~/Downloads/{profile-name}.mobileprovision | xmllint --xpath "string(//key['=UUID']/following-sibling::text)" |
echo $profile_uuid
```

Run this command to copy profile to ~/Library/MobileDevice/Provisioning Profiles with a new name {UUID}.mobileprovision:

```
cp ~/Downloads/{profile-name}.mobileprovision ~/Library/MobileDevice/Provisioning Profiles/${profile_uuid}.mobileprovision
```

Note

If the copied profile disappears from the ~/Library/MobileDevice/Provisioning Profiles directory, ensure that the Xcode process is not running in the background.

Finally, you can use the command below to list all installed provisioning profiles, with their names and UUIDs:

```
for profile in ~/Library/MobileDevice/Provisioning Profiles/*.mobileprovision; do security cms -D -i "$profile" | grep -E -A1 '<
```

## Resolution order

Its value is determined in the following order of precedence:

- `--ios-provisioning-profile`
- `[tool.flet.ios].provisioning_profile`
- `[tool.flet.ios.export_methods."EXPORT_METHOD"].provisioning_profile`

The profile must match your Bundle ID.

## Example

- `flet build`
- `pyproject.toml`

```
flet build ipa --ios-provisioning-profile "release-testing com.mycompany.example-app"

[tool.flet.ios]
provisioning_profile = "release-testing com.mycompany.example-app"
```

## Export configuration

### Export options

Additional keys to include in the generated `exportOptions.plist` of the build template.

For a complete list of supported keys and values, run the following command:

```
xcodebuild -help
```

and find the section titled "Available keys for `-exportOptionsPlist`".

#### Resolution order

Its value is determined in the following order of precedence:

- `[tool.flet.ios].export_options` (if set, per-method export options are ignored)
- `[tool.flet.ios.export_methods."EXPORT_METHOD"].export_options` (see export methods)
- `{}` (no extra keys)

#### Supported value forms

- `pyproject.toml`

Both simple and complex structures are supported:

- string
- boolean
- dictionary (nested key-value object)
- array of strings
- array of booleans
- array of dictionaries (including dictionaries that contain arrays)

Numbers are not yet supported.

## Example

- `pyproject.toml`

```
[tool.flet.ios.export_options]
uploadSymbols = false
compileBitcode = false
thinning = "<none>"

[tool.flet.ios.export_options.manifest]
appURL = "https://example.com/app.ipa"
displayImageURL = "https://example.com/icon57.png"
fullSizeImageURL = "https://example.com/icon512.png"
```

#### Template translation

In the `ios/exportOptions.plist`, the example above will be translated accordingly into this:

```
<key>uploadSymbols</key>
<false/>
<key>compileBitcode</key>
<false/>
<key>thinning</key>
<string><none></string>
```

```
<key>manifest</key>
<dict>
 <key>appURL</key>
 <string>https://example.com/app.ipa</string>
 <key>displayImageURL</key>
 <string>https://example.com/icon57.png</string>
 <key>fullSizeImageURL</key>
 <string>https://example.com/icon512.png</string>
</dict>
```

## Export method

Defines how the app should be packaged when exporting the .ipa file.

Can be one of the following:

- debugging (or deprecated development): used for debugging and testing on development devices.
- release-testing (or deprecated ad-hoc): used for distributing the app outside the App Store to specific registered devices.
- app-store-connect (or deprecated app-store): used for submitting the app to the App Store.
- enterprise: used for internal distribution within an organization (requires an enterprise account).

To configure individual settings for one or more export methods, see export methods.

Resolution order

Its value is determined in the following order of precedence:

- --ios-export-method
- [tool.flet.ios].export\_method
- "debugging"

Example

- flet build
  - pyproject.toml
- ```
flet build ipa --ios-export-method debugging

[tool.flet.ios]
export_method = "debugging"
```

Export methods

Signing settings can be configured individually per export method.

Per-method values are used only when the corresponding top-level [tool.flet.ios] setting is not set. The method key must match the export_method value exactly.

Supported keys (same as the top-level settings):

- provisioning_profile
- signing_certificate
- export_options
- team_id

Example

- pyproject.toml
- ```
[tool.flet.ios.export_methods."debugging"]
provisioning_profile = "debugging com.mycompany.example-app"
signing_certificate = "Apple Development"

[tool.flet.ios.export_methods."release-testing"]
provisioning_profile = "release-testing com.mycompany.example-app"
team_id = "ABCDEFE234"
signing_certificate = "Apple Distribution"
export_options = { uploadSymbols = false }

[tool.flet.ios.export_methods."app-store-connect"]
provisioning_profile = "app-store-connect com.mycompany.example-app"
team_id = "ABCDEFE234"
signing_certificate = "Apple Distribution"
export_options = { uploadSymbols = true }
```



## Permissions

iOS permissions are declared through Info.plist privacy usage strings. You can also use the cross-platform permission bundles to inject common entries, then override or extend them with platform-specific values.

## Info.plist

Add or override Info.plist entries for iOS builds. These values are written to ios/Runner/Info.plist of the build project.

### Resolution order

Its value is determined in the following order of precedence:

- --info-plist
- [tool.flet.ios.info]
- Values injected by cross-platform permission bundles, if any.

## Supported value forms

- flet build
- pyproject.toml

Accepts repeated <key>=<value> entries. The <value> can be in one of the following forms:

- true or false (case-insensitive) for boolean values
- integer and real number literals, for example 32 or 0.5
- TOML array literals, for example ["myapp", "myapp-beta"]
- TOML inline tables, for example { NSAllowsArbitraryLoads = false }
- any other value is treated as a string

Both simple and complex structures are supported:

- string
- boolean
- integer
- real
- dictionary (nested key-value object)
- array of strings
- array of booleans
- array of integers
- array of reals
- array of dictionaries (including dictionaries that contain arrays)

### Example

- flet build
- pyproject.toml

```
flet build ipa \
 --info-plist NSCameraUsageDescription="This app needs camera access." \
 --info-plist UIFileSharingEnabled=true \
 --info-plist ExampleInteger=32 \
 --info-plist ExampleReal=0.5 \
 --info-plist 'LSApplicationQueriesSchemes=["myapp", "myapp-beta"]' \
 --info-plist 'FeatureFlags=[true, false]' \
 --info-plist 'RetryDelays=[1, 2, 3]' \
 --info-plist 'OpacitySteps=[0.25, 0.5, 0.75]' \
 --info-plist 'NSAppTransportSecurity={ NSAllowsArbitraryLoads = false }' \
 --info-plist 'CFBundleURLTypes=[{ CFBundleTypeRole = "Editor", CFBundleURLName = "example.com", CFBundleURLSchemes = ["myapp"] }]'

[tool.flet.ios.info]
NSCameraUsageDescription = "This app needs camera access."
UIFileSharingEnabled = true
ExampleInteger = 32
ExampleReal = 0.5
LSApplicationQueriesSchemes = ["myapp", "myapp-beta"]
```

```

FeatureFlags = [true, false]
RetryDelays = [1, 2, 3]
OpacitySteps = [0.25, 0.5, 0.75]
NSAppTransportSecurity = { NSAllowsArbitraryLoads = false }
CFBundleURLTypes = [
 { CFBundleTypeRole = "Editor", CFBundleURLName = "example.com", CFBundleURLSchemes = ["myapp"] },
 { CFBundleTypeRole = "Viewer", CFBundleURLName = "example.org", CFBundleURLSchemes = ["myapp-beta"] },
]

```

## Template translation

In the ios/Runner/Info.plist, the example above will be translated accordingly into this:

```

<plist version="1.0">
<dict>

 <key>NSCameraUsageDescription</key>
 <string>This app needs camera access.</string>

 <key>UIFileSharingEnabled</key>
 <true/>

 <key>ExampleInteger</key>
 <integer>32</integer>

 <key>ExampleReal</key>
 <real>0.5</real>

 <key>LSApplicationQueriesSchemes</key>
 <array>
 <string>myapp</string>
 <string>myapp-beta</string>
 </array>

 <key>FeatureFlags</key>
 <array>
 <true/>
 <false/>
 </array>

 <key>RetryDelays</key>
 <array>
 <integer>1</integer>
 <integer>2</integer>
 <integer>3</integer>
 </array>

 <key>OpacitySteps</key>
 <array>
 <real>0.25</real>
 <real>0.5</real>
 <real>0.75</real>
 </array>

 <key>NSAppTransportSecurity</key>
 <dict>
 <key>NSAllowsArbitraryLoads</key>
 <false/>
 </dict>

 <key>CFBundleURLTypes</key>
 <array>
 <dict>
 <key>CFBundleTypeRole</key>
 <string>Editor</string>
 <key>CFBundleURLName</key>
 <string>example.com</string>
 <key>CFBundleURLSchemes</key>
 <array>
 <string>myapp</string>
 </array>
 </dict>
 <dict>
 <key>CFBundleTypeRole</key>
 <string>Viewer</string>
 <key>CFBundleURLName</key>
 <string>example.org</string>
 <key>CFBundleURLSchemes</key>
 <array>
 <string>myapp-beta</string>
 </array>
 </dict>
 </array>

</dict>
</plist>

```

# Deploying an App to an Apple Device for Testing

You can deploy .ipa files directly to an iPhone or iPad on macOS—ideal for internal testing without publishing to the App Store.

Follow these steps:

## Install and Launch Apple Configurator

- Install Apple Configurator from App Store;
- Find and open it from your Applications folder.

## Connect Your iOS Device

- Connect your device (iPhone or iPad) to your Mac via USB;
- Unlock the connected device, and tap Trust This Computer, if prompted;
- Wait for this device to appear in the Apple Configurator.

## Add the .ipa File

- Either drag the .ipa file onto the connected device in Apple Configurator;
- Or click Add → Apps, then browse and select the .ipa file from your Mac.

## Install the App

- Click Prepare or Install to begin the deployment/installation;
- Apple Configurator will install the app on the connected device.

## Trust the Developer (for Ad Hoc or Enterprise apps)

If your app is signed with an Ad Hoc or Enterprise provisioning profile, you'll need to manually trust the developer:

- On the iOS device, go to Settings → General → VPN & Device Management;
- Under Developer App, tap your Developer Profile;
- Tap Trust [Your Developer Name], then confirm.

# Uploading app to App Store Connect for distribution

## Install and Sign in to Transporter

- Install and open Transporter from the Mac App Store.
- Sign in using your Apple Developer Account credentials (the same account used for App Store Connect).

## Prepare Your .ipa File

- Build your app and export an .ipa file using the app-store-connect export method. (release-testing is for Ad Hoc device distribution, not App Store Connect.)

## Upload the .ipa File in Transporter

- Drag and drop the .ipa file directly into the Transporter window, or click "Add App" and select your .ipa file from your Mac.
- Click the "..." button next to "Deliver", and select "Verify".
- Wait for Transporter to complete the verification process.
- After successful verification, click "Deliver" to upload your .ipa file to App Store Connect.

## Check Upload Status

- Transporter will display a success message upon completion.
- If errors occur, carefully review the details provided, correct the issues, and repeat the upload process.

## Confirm Upload in App Store Connect

- Go to App Store Connect.

- Navigate to Apps → Your App → TestFlight or App Store Version.
- Your newly uploaded build will initially appear under Processing (processing typically takes a few minutes to an hour).
- Once processing completes, your build will become available for submission. You can now submit the app for review.

# Packaging app for Linux

Instructions for packaging a Flet app into a Linux executable.

## Note

This guide provides detailed Linux-specific information. Complementary and more general information is available [here](#).

## Prerequisites

Flet uses Flutter to build Linux apps. Compiling the app and its native plugins links against GTK and a number of system libraries, so these must be installed before running `flet build linux`.

On Debian/Ubuntu-based distributions, install the required packages with `apt`:

```
sudo apt update
sudo apt install -y \
 binutils clang cmake llvm lld ninja-build pkg-config \
 libgtk-3-dev libsecret-1-0 libsecret-1-dev libunwind-dev \
 gstreamer1.0-alsa gstreamer1.0-gl gstreamer1.0-gtk3 gstreamer1.0-libav \
 gstreamer1.0-plugins-bad gstreamer1.0-plugins-base gstreamer1.0-plugins-good \
 gstreamer1.0-plugins-ugly gstreamer1.0-pulseaudio gstreamer1.0-qt5 \
 gstreamer1.0-tools gstreamer1.0-x \
 libasound2-dev libgstreamer1.0-dev \
 libgstreamer-plugins-base1.0-dev libgstreamer-plugins-bad1.0-dev \
 libmpv-dev mpv
```

This is the same set of packages Flet uses in its own build environment. A few notes on what they are for:

- Build toolchain — `clang`, `cmake`, `ninja-build`, `pkg-config`, `llvm`, `lld`, `binutils` and `libgtk-3-dev` are required to compile and link the app. In particular, the `lld` linker must be present — without it the build fails with a linker error.
- Secret storage — `libsecret-1-0` and `libsecret-1-dev` are used for secure storage / keyring access.
- Audio and video — the `gstreamer1.0-*`, `libgstreamer*-dev`, `libasound2-dev`, `libmpv-dev` and `mpv` packages are required by the Audio service and Video control. You can omit them if your app does not play media, but installing the full set above avoids surprises. See those pages for control-specific details.

## Other distributions

Package names differ on non-Debian distributions (e.g. Fedora, Arch). Install the equivalent GTK 3, GStreamer, mpv/libmpv, libsecret, clang/llvm, lld, cmake and ninja development packages for your distribution.

## flet build linux

### Note

This command can be run on Linux only (or WSL).

Builds a Linux executable.

## Window positioning on Wayland

On Linux the display server controls window placement, and this differs between X11 and Wayland:

- X11 lets applications set their own top-level window position.
- Wayland (the default session on modern GNOME/Ubuntu) does not — by design, a client cannot position its own top-level window; the compositor (e.g. Mutter) decides where windows are placed.

As a result, on a Wayland session the following have no effect (window sizing still works — only positioning is restricted):

- `Page.window.center()`
- setting `Page.window.left` / `Page.window.top`
- moving the window programmatically

This is a Wayland protocol limitation, not a Flet bug. The same code works as expected on Windows, macOS, Linux X11 sessions, and Wayland sessions running the app through XWayland.

To force the X11 backend (XWayland) on a Wayland session and re-enable programmatic positioning, run the app with the `GDK_BACKEND` environment variable:

```
GDK_BACKEND=x11 ./your_app
```

You can check the current session type with:

```
echo $XDG_SESSION_TYPE # "wayland" or "x11"
```

# Packaging app for macOS

Instructions for packaging a Flet app into a macOS application bundle.

Info

This guide provides detailed macOS-specific information. Complementary and more general information is available [here](#).

## Prerequisites

### Rosetta 2

Flutter, which we use for packaging, requires Rosetta 2 on Apple Silicon:

```
sudo softwareupdate --install-rosetta --agree-to-license
```

### Xcode

Xcode 15 or later is required to compile native Swift or Objective-C code.

### CocoaPods

CocoaPods 1.16 or later is required to install and compile Flutter plugins.

## flet build macos

Note

This command can be run on macOS only.

Builds a macOS application bundle from your Flet app.

## Target architecture

By default, `flet build macos` creates a universal bundle that runs on both Apple Silicon and Intel Macs. Packaging downloads Python wheels for both `arm64` and `x86_64` architectures.

To limit packaging to a specific architecture, see [this](#). This affects which Python wheels are bundled and, in turn, which CPU architectures the app will run on. You will then have to provide your users with the correct build for their Macs.

## Permissions

macOS permissions are declared through `Info.plist` privacy usage strings and app entitlements. You can also use the cross-platform permission bundles to inject common entries, then override or extend them with platform-specific values.

### Info.plist

Add or override `Info.plist` entries for macOS builds. These values are written to `macos/Runner/Info.plist` of the build project.

### Resolution order

Its value is determined in the following order of precedence:

- `--info-plist`
- `[tool.flet.macos.info]`
- Values injected by cross-platform permission bundles, if any.

### Supported value forms

- `flet build`
- `pyproject.toml`

Accepts repeated `<key>=<value>` entries. The `<value>` can be in one of the following forms:

- `true` or `false` (case-insensitive) for boolean values
- integer and real number literals, for example `32` or `0.5`
- TOML array literals, for example `["basic", "advanced"]`

- TOML inline tables, for example { NSAllowsArbitraryLoads = false }
- any other value is treated as a string

Both simple and complex structures are supported:

- string
- boolean
- integer
- real
- dictionary (nested key-value object)
- array of strings
- array of booleans
- array of integers
- array of reals
- array of dictionaries (including dictionaries that contain arrays)

## Example

- flet build
- pyproject.toml

```
flet build macos \
--info-plist LSAApplicationCategoryType="public.app-category.utilities" \
--info-plist NSSupportsSuddenTermination=true \
--info-plist ExampleInteger=32 \
--info-plist ExampleReal=0.5 \
--info-plist 'SupportedModes=["basic", "advanced"]' \
--info-plist 'FeatureFlags=[true, false]' \
--info-plist 'RetryDelays=[1, 2, 3]' \
--info-plist 'OpacitySteps=[0.25, 0.5, 0.75]' \
--info-plist 'NSAppTransportSecurity={ NSAllowsArbitraryLoads = false }' \
--info-plist 'CFBundleDocumentTypes=[{ CFBundleTypeName = "Data File", CFBundleTypeExtensions = ["dat"] }, { CFBundleTypeName =
[tool.flet.macos.info]
LSApplicationCategoryType = "public.app-category.utilities"
NSSupportsSuddenTermination = true
ExampleInteger = 32
ExampleReal = 0.5
SupportedModes = ["basic", "advanced"]
FeatureFlags = [true, false]
RetryDelays = [1, 2, 3]
OpacitySteps = [0.25, 0.5, 0.75]
NSAppTransportSecurity = { NSAllowsArbitraryLoads = false }
CFBundleDocumentTypes = [
 { CFBundleTypeName = "Data File", CFBundleTypeExtensions = ["dat"] },
 { CFBundleTypeName = "JSON File", CFBundleTypeExtensions = ["json"] },
]
```

## Template translation

In the macos/Runner/Info.plist, the example above will be translated accordingly into this:

```
<plist version="1.0">
<dict>

 <key>LSApplicationCategoryType</key>
 <string>public.app-category.utilities</string>

 <key>NSSupportsSuddenTermination</key>
 <true/>

 <key>ExampleInteger</key>
 <integer>32</integer>

 <key>ExampleReal</key>
 <real>0.5</real>

 <key>SupportedModes</key>
 <array>
 <string>basic</string>
 <string>advanced</string>
 </array>

 <key>FeatureFlags</key>
 <array>
```



```

 <true/>
 <false/>
 </array>

 <key>RetryDelays</key>
 <array>
 <integer>1</integer>
 <integer>2</integer>
 <integer>3</integer>
 </array>

 <key>OpacitySteps</key>
 <array>
 <real>0.25</real>
 <real>0.5</real>
 <real>0.75</real>
 </array>

 <key>NSAppTransportSecurity</key>
 <dict>
 <key>NSAllowsArbitraryLoads</key>
 <false/>
 </dict>

 <key>CFBundleDocumentTypes</key>
 <array>
 <dict>
 <key>CFBundleTypeName</key>
 <string>Data File</string>
 <key>CFBundleTypeExtensions</key>
 <array>
 <string>dat</string>
 </array>
 </dict>
 <dict>
 <key>CFBundleTypeName</key>
 <string>JSON File</string>
 <key>CFBundleTypeExtensions</key>
 <array>
 <string>json</string>
 </array>
 </dict>
 </array>
</dict>
</plist>

```

## Entitlements

Entitlements are property-list key-value pairs that grant an executable permission to use a service or technology. The supported value type depends on the entitlement key defined in the Apple Developer Entitlements Reference.

Entitlements are written to `macos/Runner/DebugProfile.entitlements` and `macos/Runner/Release.entitlements` in the build template.

## Resolution order

Its value is determined in the following order of precedence:

--macos-entitlements

[tool.flet.macos.entitlement]

Values injected by cross-platform permission bundles, if any.

Defaults:

```

[tool.flet.macos.entitlement]
"com.apple.security.app-sandbox" = false
"com.apple.security.cs.allow-jit" = true
"com.apple.security.network.client" = true
"com.apple.security.network.server" = true
"com.apple.security.files.user-selected.read-write" = true

```

## Supported value forms

- flet build
- pyproject.toml

Accepts repeated `<key>=<value>` entries. The `<value>` can be in one of the following forms:

- true or false (case-insensitive) for boolean values

- integer and real number literals, for example 32 or 0.5
- TOML array literals, for example ["group.example.one", "group.example.two"]
- TOML inline tables, for example { "com.apple.mail" = ["compose"] }
- any other value is treated as a string

Both simple and complex structures are supported:

- string
- boolean
- integer
- real
- dictionary (nested key-value object)
- array of strings
- array of booleans
- array of integers
- array of reals
- array of dictionaries (including dictionaries that contain arrays)

## Example

- flet build
- pyproject.toml

```
flet build macos \
--macos-entitlements com.apple.security.network.client=true \
--macos-entitlements com.apple.developer.ubiquity-kvstore-identifier=ABCDE12345.dev.example.myapp \
--macos-entitlements ExampleInteger=32 \
--macos-entitlements ExampleReal=0.5 \
--macos-entitlements 'com.apple.security.application-groups=["group.dev.example.myapp", "group.dev.example.shared"]' \
--macos-entitlements 'ExampleBooleanArray=[true, false]' \
--macos-entitlements 'com.apple.security.scripting-targets={ "com.apple.mail" = ["compose", "send"] }' \
--macos-entitlements 'ExampleArrayOfDictionaries=[{ Name = "alpha", Enabled = true }, { Name = "beta", Enabled = false }]'

[tool.flet.macos.entitlement]
"com.apple.security.network.client" = true
"com.apple.developer.ubiquity-kvstore-identifier" = "ABCDE12345.dev.example.myapp"
ExampleInteger = 32
ExampleReal = 0.5
"com.apple.security.application-groups" = [
 "group.dev.example.myapp",
 "group.dev.example.shared",
]
ExampleBooleanArray = [true, false]
"com.apple.security.scripting-targets" = { "com.apple.mail" = ["compose", "send"] }
ExampleArrayOfDictionaries = [
 { Name = "alpha", Enabled = true },
 { Name = "beta", Enabled = false },
]
```

## Template translation

In both macos/Runner/DebugProfile.entitlements and macos/Runner/Release.entitlements, the example above will be translated accordingly into this:

```
<key>com.apple.security.network.client</key>
<true />
<key>com.apple.developer.ubiquity-kvstore-identifier</key>
<string>ABCDE12345.dev.example.myapp</string>
<key>ExampleInteger</key>
<integer>32</integer>
<key>ExampleReal</key>
<real>0.5</real>
<key>com.apple.security.application-groups</key>
<array>
 <string>group.dev.example.myapp</string>
 <string>group.dev.example.shared</string>
</array>
<key>ExampleBooleanArray</key>
<array>
 <true />
 <false />
</array>
```

```
<key>com.apple.security.scripting-targets</key>
<dict>
 <key>com.apple.mail</key>
 <array>
 <string>compose</string>
 <string>send</string>
 </array>
</dict>
<key>ExampleArrayOfDictionaries</key>
<array>
 <dict>
 <key>Name</key>
 <string>alpha</string>
 <key>Enabled</key>
 <true />
 </dict>
 <dict>
 <key>Name</key>
 <string>beta</string>
 <key>Enabled</key>
 <false />
 </dict>
</array>
```

## Publish Flet web app

Flet allows publishing your app as a static or dynamic website.

Static website - content is delivered as-is and Python runs in the browser (Pyodide).

Dynamic website - content is generated per user and Python runs on the server.

Here is a table comparing Flet app running as a static vs dynamic website:

|  |                                                                                                                                                                                                                   |                                                                                                                                                                           |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <input type="checkbox"/> <input type="checkbox"/> Slower - Python runtime (Pyodide) along with app's Python code and all its dependencies must be loaded into the browser. Pyodide initialization takes time too. | <input type="checkbox"/> Faster - the app stays and runs on the server.                                                                                                   |
|  | <input type="checkbox"/> <input type="checkbox"/> Not every program that works with native Python                                                                                                                 | <input type="checkbox"/> Any Python package can be used.                                                                                                                  |
|  | <input type="checkbox"/> Zero latency between user-generated events (clicks, text field changes, drags) and page updates.                                                                                         | <input type="checkbox"/> <input type="checkbox"/> Non-zero latency - user-generated events are communicated to a server via WebSockets. UI updates are communicated back. |
|  | <input type="checkbox"/> <input type="checkbox"/> Slower - Pyodide is currently 3x-5x slower than native Python because of WASM                                                                                   | <input type="checkbox"/> Faster - the code is running on the server by native Python.                                                                                     |
|  | <input type="checkbox"/> <input type="checkbox"/> Low - app's code is loaded into a web browser and can be inspected by a user.                                                                                   | <input type="checkbox"/> High - the app is running on the server.                                                                                                         |
|  | <input type="checkbox"/> Cheap/Free - no code is running on the server and thus the app can be hosted anywhere: GitHub Pages, Cloudflare Pages, Replit, Vercel, a shared hosting or your own VPS.                 | <input type="checkbox"/> <input type="checkbox"/> Paid - the app requires Python code to run on the server and communicate with a web browser via WebSockets.             |

Static websiteDynamic websiteLoading timePython compatibilitycan be run with  
PyodideResponsivenessPerformanceCode protectionHosting

# Packaging app for Windows

Instructions for packaging a Flet app into a Windows application.

## Info

This guide provides detailed Windows-specific information. Complementary and more general information is available [here](#).

## Prerequisites

### Visual Studio

Visual Studio (2022 or 2026) is required with the Desktop development with C++ workload installed.

Follow this guide for instructions on downloading and installing correct Visual Studio components for Flutter desktop development.

## flet build windows

### Note

This command can be run on Windows only.

Builds a Windows application.

## Troubleshooting

### Developer mode

If you get the below error:

Building with plugins requires symlink support.

Please enable Developer Mode in your system settings. Run  
`start ms-settings:developers`  
to open settings.

Then, you need to enable Developer Mode as it indicates. Follow this guide on how to do that.

## Built-in binary Python packages for Android and iOS

Binary Python packages (as opposed to "pure" Python packages written entirely in Python) contain components written in C, Rust, or other languages that produce native code. Common examples include numpy, cryptography, and pydantic.

Flet provides an alternative package index, `pypi.flet.dev`, which hosts prebuilt Python binary wheels (.whl files used by pip) for iOS and Android platforms.

Work in progress

New packages are created by adding a recipe to the Mobile Forge project. Currently, we author these recipes for you; once the build process is fully automated, you'll be able to submit a PR and test the compiled package immediately.

If a package is not available on `pypi.flet.dev`, you can request it in Flet discussions - Packages. Please do not request "pure" Python packages. Check out this guide for more details on the difference between pure and binary packages.

Below is the list of packages hosted on `pypi.flet.dev` and the specific wheel versions currently built and supported:

| Package        | Versions                          |
|----------------|-----------------------------------|
|                | 3.14.0, 3.9.5                     |
|                | 3.53.2.0                          |
|                | 25.1.0, 21.2.0                    |
|                | 5.0.0, 4.2.0                      |
|                | 1.87                              |
|                | 3.8.1, 3.6.1, 2.9.2               |
|                | 1.3.3, 1.0.0                      |
|                | 1.2.0, 1.1.0                      |
|                | 2.0.0, 1.17.1                     |
|                | 1.3.3, 1.3.1                      |
|                | 7.2.0                             |
|                | 48.0.0, 43.0.1                    |
|                | 2.0.13                            |
|                | 0.4.2, 0.4.1, 0.4.0, 0.3.2, 0.3.1 |
|                | 1.10.1                            |
| (Android only) | 27.3.13750724, 27.2.12479018      |
|                | 1.1.2                             |
|                | 8.11                              |
|                | 1.4.27                            |
|                | 2.13.3                            |
|                | 3.13.1, 3.10                      |
|                | 3.13.1, 3.13                      |
|                | 3.0.90                            |
|                | 1.7.1                             |
|                | 5.46                              |
|                | 1.0.1, 0.99.8                     |
|                | 0.5                               |
|                | 1.6.43                            |
|                | 9.5                               |
|                | 0.21.5                            |
| (Android only) | 1.0.1                             |
|                | 1.0.20                            |
|                | 4.7                               |
|                | 2.15.3, 2.9.8                     |
|                | 1.1.45, 1.1.32                    |



paqueopencv-pythonorjsonpandaspendulumpillowpolarspreshedprimpprotobufpsutilpycryptodomepycryptodomexpydantic  
\_corepyjniuspymongopymssqlPyNaClpyobjuspyogriopyprojpysodiumpython-magicpyxirrPyYAMLpyzmqrasterioregexrpd-  
pyruamel.yaml.clibselectolaxshapelyspacySQLAlchemysqlitesrslythinctiktokentime-machinetokenizersujsonwebsocketsyar  
lzoze.interfacezstandardzstd



# Environment Variables

## Setting boolean values

To set a boolean True, use one of the following string values: "true", "1" or "yes". Any other value will be interpreted as False.

## FLET\_APP\_CONSOLE

The path to the application's console log file (console.log) in the temporary storage directory.

Its value is set in production mode.

## FLET\_APP\_STORAGE\_DATA

A directory for the storage of persistent application data that is preserved between app updates. It is already pre-created and its location depends on the platform the app is running on.

## FLET\_APP\_STORAGE\_TEMP

A directory for the storage of temporary application files, i.e. cache. It is already pre-created and its location depends on the platform the app is running on.

## FLET\_APP\_USER\_MODEL\_ID

Windows AppUserModelID used by the desktop client process for taskbar grouping and pinning.

For apps packaged with flet pack, this value is set automatically so taskbar pins point to the packaged app executable instead of the cached Flet client executable.

## FLET\_ASSETS\_DIR

Absolute path to the app's assets directory.

In production apps built with flet build, this environment-variable points to the bundled assets absolute location at runtime. Use it when your code needs a filesystem path to bundled files (for example, JSON configs, databases, or model files).

For local runs, it may be unset depending on how the app is started, so use a fallback:

```
import os
from pathlib import Path
import flet as ft

default_assets_dir = Path(__file__).parent / "assets"
assets_dir = Path(os.environ.get("FLET_ASSETS_DIR", str(default_assets_dir))).resolve()

def main(page: ft.Page):
 # ...

ft.run(main, assets_dir="assets")
```

For control properties like Image.src, continue using paths relative to the ft.run(assets\_dir=...), as described in the assets cookbook.

## FLET\_ANDROID\_SIGNING\_KEY\_ALIAS

Android signing key alias used by flet build for Android app signing.

It is used only when a keystore is configured.

## FLET\_ANDROID\_SIGNING\_KEY\_PASSWORD

Android signing key password used by flet build for Android app signing.

If FLET\_ANDROID\_SIGNING\_KEY\_STORE\_PASSWORD is set but this variable is not, the keystore password is reused as the key password.

## FLET\_ANDROID\_SIGNING\_KEY\_STORE

Path to the Android upload keystore (.jks) used by flet build for Android app signing.

## FLET\_ANDROID\_SIGNING\_KEY\_STORE\_PASSWORD

Android signing keystore password used by flet build for Android app signing.

If FLET\_ANDROID\_SIGNING\_KEY\_PASSWORD is set but this variable is not, the key password is reused as the keystore password.

## FLET\_CLI\_NO\_RICH\_OUTPUT

Whether to disable rich output in the console.

Defaults to "false".

## FLET\_PLATFORM

The platform on which the application is running. Its value is one of the following: "android", "ios", "linux", "macos", "windows" or "fuchsia".

## FLET\_CLI\_SKIP\_FLUTTER\_DOCTOR

Whether to skip running flutter doctor when a build fails.

Defaults to False.

## FLET\_HIDE\_WINDOW\_ON\_START

Set to true to start app with the main window hidden.

Defaults to False.

## FLET\_FORCE\_WEB\_SERVER

Set to true to force running app as a web app. Automatically set on headless Linux hosts.

## FLET\_OAUTH\_CALLBACK\_HANDLER\_ENDPOINT

Custom path for OAuth handler.

Defaults to "/oauth\_callback".

## FLET\_OAUTH\_STATE\_TIMEOUT

Maximum allowed time (in seconds) to complete OAuth web flow.

Defaults to 600.

## FLET\_MAX\_UPLOAD\_SIZE

Maximum allowed size (in bytes) of uploaded files.

Default is unlimited.

## FLET\_SECRET\_KEY

A secret key to sign temporary upload URLs.

## FLET\_SERVER\_IP

IP address to listen web app on, e.g. "127.0.0.1".

Defaults to 0.0.0.0 - bound to all server IPs.

## FLET\_SERVER\_PORT

TCP port to run app on.

8000 if the program is running on a Linux server or FLET\_FORCE\_WEB\_SERVER is set; otherwise random port.

## FLET\_SERVER\_UDS\_PATH

The Unix Domain Socket (UDS) path for the Flet server. It enables inter-process communication on Unix-based systems, with its value being a socket file path in the format flet\_<pid>.sock.

## FLET\_SESSION\_TIMEOUT

Session lifetime in seconds.

Defaults to 3600.

## FLET\_UPLOAD\_DIR

Absolute path to app "upload" directory.

## FLET\_UPLOAD\_HANDLER\_ENDPOINT

Custom path for upload handler.

Defaults to "/upload".

## FLET\_WEB\_APP\_PATH

A URL path after domain name to host web app under, e.g. "/apps/myapp".

Defaults to "/" - host app in the root.

## FLET\_WEB\_NO\_CDN

Set to true to avoid loading CanvasKit, Pyodide, and fonts from CDNs.

## FLET\_WEBSOCKET\_HANDLER\_ENDPOINT

Custom path for WebSocket handler.

Defaults to "/ws".

## FLET\_WEB\_RENDERER

Web rendering mode: "canvaskit" (default), "skwasm" or "auto".

## FLET\_WEB\_USE\_COLOR\_EMOJI

Set to True, true or 1 to load web font with colorful emojis.

## FLET\_WEB\_ROUTE\_URL\_STRATEGY

The URL strategy of the web application. Its value can be either "path" (default) or "hash".

# Accelerometer

Streams raw accelerometer `AccelerometerReadingEvent`, which describe the acceleration of the device, in  $\text{m/s}^2$ , including the effects of gravity.

Unlike `UserAccelerometer`, this service reports raw data from the accelerometer (physical sensor embedded in the mobile device) without any post-processing.

The accelerometer is unable to distinguish between the effect of an accelerated movement of the device and the effect of the surrounding gravitational field. This means that, at the surface of Earth, even if the device is completely still, the reading of `Accelerometer` is an acceleration of intensity 9.8 directed upwards (the opposite of the gravitational acceleration). This can be used to infer information about the position of the device (horizontal/vertical/tilted). `Accelerometer` reports zero acceleration if the device is free falling.

Note

- Supported platforms: Android, iOS and web.
- Web ignores requested sampling intervals.

Inherits: `Service`

## Properties

- `cancel_on_error` - Whether the stream subscription should cancel on the first sensor error.
- `enabled` - Whether the sensor should be sampled.
- `interval` - Desired sampling interval provided as a `Duration`.

## Events

- `on_error` - Fired when the platform reports a sensor error (for example when the device does not expose the accelerometer).
- `on_reading` - Fires when a new reading is available.

# Examples

## Accelerometer

```
import flet as ft

def main(page: ft.Page):
 def handle_reading(e: ft.AccelerometerReadingEvent):
 reading.value = f"x={e.x:.2f} m/s^2, y={e.y:.2f} m/s^2, z={e.z:.2f} m/s^2"

 def handle_error(e: ft.SensorErrorEvent):
 page.add(ft.Text(f"Accelerometer error: {e.message}"))

 page.services.append(
 ft.Accelerometer(
 on_reading=handle_reading,
 on_error=handle_error,
 interval=ft.Duration(milliseconds=100),
 cancel_on_error=False,
)
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Move your device to see accelerometer readings."),
 reading := ft.Text("Waiting for data..."),
]
)
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildcancel\_on\_errorclass-attributeinstance-attribute**

`cancel_on_error: bool = True`

Whether the stream subscription should cancel on the first sensor error.

### **buildenabledclass-attributeinstance-attribute**

`enabled: bool = True`

Whether the sensor should be sampled. Disable to stop streaming.

### **buildintervalclass-attributeinstance-attribute**

`interval: Duration | None = None`

Desired sampling interval provided as a Duration. Defaults to 200 ms.

Note that mobile platforms treat this value as a suggestion and the actual rate can differ depending on hardware and OS limitations.

## **Events**

### **bolton\_errorclass-attributeinstance-attribute**

`on_error: EventHandler[SensorErrorEvent] | None = None`

Fired when the platform reports a sensor error (for example when the device does not expose the accelerometer).  
`event.message` contains the error text.

### **bolton\_readingclass-attributeinstance-attribute**

`on_reading: (EventHandler[AccelerometerReadingEvent] | None) = None`

Fires when a new reading is available.

event exposes x, y, z acceleration values and timestamp (microseconds since epoch).

# Audio

Allows playing audio in Flet apps.

## Platform Support

| Platform  | Windows                  | macOS                    | Linux                    |
|-----------|--------------------------|--------------------------|--------------------------|
| Supported | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Usage

To use Audio control add flet-audio package to your project dependencies:

- uv
  - pip
- ```
uv add flet-audio
pip install flet-audio
```

Linux requirements

To play audio on Linux (or WSL) you need to install GStreamer library.

To install the minimal set of GStreamer libs on Ubuntu/Debian, run:

```
sudo apt install libgtk-3-dev libgstreamer1.0-dev libgstreamer-plugins-base1.0-dev
```

To install the full set:

```
sudo apt install \
  libgstreamer1.0-dev libgstreamer-plugins-base1.0-dev libgstreamer-plugins-bad1.0-dev \
  gstreamer1.0-plugins-base gstreamer1.0-plugins-good gstreamer1.0-plugins-bad \
  gstreamer1.0-plugins-ugly gstreamer1.0-libav gstreamer1.0-doc gstreamer1.0-tools \
  gstreamer1.0-x gstreamer1.0-alsa gstreamer1.0-gl gstreamer1.0-gtk3 \
  gstreamer1.0-qt5 gstreamer1.0-pulseaudio
```

If you receive error while loading shared libraries: libgstapp-1.0.so.0, it means GStreamer is not installed in your WSL environment. Install the full set of GStreamer libs, as shown above.

See this guide for installing on other Linux distributions.

Examples

Audio playback controls

play_arrowTry Online

```
import flet as ft
import flet_audio as fta

def main(page: ft.Page):
    url = "viper.mp3"

    async def play():
        await audio.play()

    async def pause():
        await audio.pause()

    async def resume():
        await audio.resume()

    async def release():
        await audio.release()

    def set_volume(value: float):
        audio.volume += value

    def set_balance(value: float):
        audio.balance += value

    async def seek_2s():
        await audio.seek(ft.Duration(seconds=2))

    async def get_duration():
```

```

        duration = await audio.get_duration()
        print("Duration:", duration)

    async def on_get_current_position():
        position = await audio.get_current_position()
        print("Current position:", position)

    audio = fta.Audio(
        src=url,
        autoplay=False,
        volume=1,
        balance=0,
        release_mode=fta.ReleaseMode.STOP,
        on_loaded=lambda _: print("Loaded"),
        on_duration_change=lambda e: print("Duration changed:", e.duration),
        on_position_change=lambda e: print("Position changed:", e.position),
        on_state_change=lambda e: print("State changed:", e.state),
        on_seek_complete=lambda _: print("Seek complete"),
    )
    page.services.append(audio)

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button("Play", on_click=play),
                    ft.Button("Pause", on_click=pause),
                    ft.Button("Resume", on_click=resume),
                    ft.Button("Release", on_click=release),
                    ft.Button("Seek 2s", on_click=seek_2s),
                    ft.Row(
                        controls=[
                            ft.Button(
                                "Volume down",
                                on_click=lambda _: set_volume(-0.1),
                            ),
                            ft.Button("Volume up", on_click=lambda _: set_volume(0.1)),
                        ]
                    ),
                    ft.Row(
                        controls=[
                            ft.Button(
                                "Balance left",
                                on_click=lambda _: set_balance(-0.1),
                            ),
                            ft.Button(
                                "Balance right",
                                on_click=lambda _: set_balance(0.1),
                            ),
                        ]
                    ),
                    ft.Button("Get duration", on_click=get_duration),
                    ft.Button(
                        "Get current position",
                        on_click=on_get_current_position,
                    ),
                ]
            )
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Description

A control to simultaneously play multiple audio sources.

Inherits: Service

Properties

- `autoplay` - Starts playing audio as soon as audio control is added to a page.
- `balance` - Defines the stereo balance.
- `playback_rate` - Defines the playback rate.
- `release_mode` - Defines the release mode.
- `src` - The audio source.
- `volume` - Sets the volume (amplitude).

Events

- `on_duration_change` - Fires as soon as audio duration is available (it might take a while to download or buffer it).
- `on_loaded` - Fires when an audio is loaded/buffered.
- `on_position_change` - Fires when audio position is changed.
- `on_seek_complete` - Fires on seek completions.
- `on_state_change` - Fires when audio player state changes.

Methods

- `get_current_position` - Get the current position of the audio playback.
- `get_duration` - Get audio duration of the audio playback.
- `pause` - Pauses the audio that is currently playing.
- `play` - Starts playing audio from the specified position.
- `release` - Releases the resources associated with this media player.
- `resume` - Resumes the audio that has been paused or stopped.
- `seek` - Moves the cursor to the desired position.

Properties

`buildautoplayclass-attributeinstance-attribute`

`autoplay: bool = False`

Starts playing audio as soon as audio control is added to a page.

Note

Autoplay works in desktop, mobile apps and Safari browser, but doesn't work in Chrome/Edge.

`buildbalanceclass-attributeinstance-attribute`

`balance: Number = 0.0`

Defines the stereo balance.

- `-1` - The left channel is at full volume; the right channel is silent.
- `1` - The right channel is at full volume; the left channel is silent.
- `0` - Both channels are at the same volume.

`buildplayback_rateclass-attributeinstance-attribute`

`playback_rate: Number = 1.0`

Defines the playback rate.

Should ideally be set when creating the constructor.

Note

- iOS and macOS have limits between 0.5x and 2x.
- Android SDK version should be 23 or higher.

`buildrelease_modeclass-attributeinstance-attribute`

`release_mode: ReleaseMode = ReleaseMode.RELEASE`

Defines the release mode.

`buildsrcclass-attributeinstance-attribute`

`src: str | bytes | None = None`

The audio source.

It can be one of the following:

- A URL or local asset file path;

- A base64 string;
- Raw bytes.

Note

Here is a list of supported audio formats.

buildvolumeinstance-attribute

```
volume: Number = 1.0
```

Sets the volume (amplitude). It's value ranges between 0.0 (mute) and 1.0 (maximum volume). Intermediate values are linearly interpolated.

Events

bolton_duration_changeinstance-attribute

```
on_duration_change: (
    EventHandler[AudioDurationChangeEvent] | None
) = None
```

Fires as soon as audio duration is available (it might take a while to download or buffer it).

bolton_loadedinstance-attribute

```
on_loaded: ControlEventHandler[Audio] | None = None
```

Fires when an audio is loaded/buffered.

bolton_position_changeinstance-attribute

```
on_position_change: (
    EventHandler[AudioPositionChangeEvent] | None
) = None
```

Fires when audio position is changed. Will continuously update the position of the playback every 1 second if the status is playing.

Can be used for a progress bar.

bolton_seek_completeinstance-attribute

```
on_seek_complete: ControlEventHandler[Audio] | None = (
    None
)
```

Fires on seek completions. An event is going to be sent as soon as the audio seek is finished.

bolton_state_changeinstance-attribute

```
on_state_change: (
    EventHandler[AudioStateChangeEvent] | None
) = None
```

Fires when audio player state changes.

Methods

deployed_codeget_current_positionasync

```
get_current_position() -> Duration | None
```

Get the current position of the audio playback.

Returns:

- Duration | None - The current position of the audio playback.

deployed_codeget_durationasync

```
get_duration() -> Duration | None
```

Get audio duration of the audio playback.

It will be available as soon as the audio duration is available (it might take a while to download or buffer it if file is not local).

Returns:

- Duration | None - The duration of audio playback.

deployed_codepauseasync

`pause()`

Pauses the audio that is currently playing.

If you call resume later, the audio will resume from the point that it has been paused.

deployed_codeplayasync

`play(position: DurationValue=0)`

Starts playing audio from the specified position.

Parameters:

- position (DurationValue, default: 0) - The position to start playback from. Defaults to the beginning of the audio.

deployed_codereleaseasync

`release()`

Releases the resources associated with this media player. These are going to be fetched or buffered again as soon as you change the source or call resume.

deployed_coderesumeasync

`resume()`

Resumes the audio that has been paused or stopped.

deployed_codeseekasync

`seek(position: DurationValue)`

Moves the cursor to the desired position.

Parameters:

- position (DurationValue) - The position to seek/move to.

Audio Recorder

Allows recording audio in Flet apps.

It is based on the record Flutter package.

Platform Support

Platform	Windows	macOS	Linux
Supported	☐	☐	☐

Usage

To use AudioRecorder service add flet-audio-recorder package to your project dependencies:

- uv
 - pip
- ```
uv add flet-audio-recorder
pip install flet-audio-recorder
```

## Requirements

The below sections show the required configurations for each platform.

### Android

Configuration to be made to access the microphone:

- android.permission.RECORD\_AUDIO: Allows audio recording.
- android.permission.WRITE\_EXTERNAL\_STORAGE (optional): Allows saving your recordings in public folders.
- android.permission.MODIFY\_AUDIO\_SETTINGS (optional): Allows using bluetooth telephony device like headset/earbuds.
- flet build
- pyproject.toml

```
flet build apk \
 --android-permissions android.permission.RECORD_AUDIO=true \
 --android-permissions android.permission.WRITE_EXTERNAL_STORAGE=true \
 --android-permissions android.permission.MODIFY_AUDIO_SETTINGS=true

[tool.flet.android.permission]
"android.permission.RECORD_AUDIO" = true
"android.permission.WRITE_EXTERNAL_STORAGE" = true
"android.permission.MODIFY_AUDIO_SETTINGS" = true
```

See also:

- setting Android permissions
- Audio formats sample rate hints

### iOS

Configuration to be made to access the microphone:

- NSMicrophoneUsageDescription: Required for recording audio.
- flet build
- pyproject.toml

```
flet build ipa \
 --info-plist NSMicrophoneUsageDescription="Some message to describe why you need this permission..."

[tool.flet.ios.info]
NSMicrophoneUsageDescription = "Some message to describe why you need this permission..."
```

See also:

- setting iOS permissions.

## macOS

Configuration to be made to access the microphone:

- NSMicrophoneUsageDescription: Allows recording audio.
- com.apple.security.device.audio-input: Allows recording audio using the built-in microphone and accessing audio input using Core Audio.
- flet build
- pyproject.toml

```
flet build macos \
 --info-plist NSMicrophoneUsageDescription="Some message to describe why you need this permission..." \
 --macos-entitlements com.apple.security.device.audio-input=true

[tool.flet.macos.info]
NSMicrophoneUsageDescription = "Some message to describe why you need this permission..."

[tool.flet.macos.entitlement]
"com.apple.security.device.audio-input" = true
```

See also:

- setting macOS permissions
- setting macOS entitlements

## Linux

The following dependencies are required (and widely available on your system):

- parecord: Used for audio input.
- pactl: Used for utility methods like getting available devices.
- ffmpeg: Used for encoding and output.

On Ubuntu 24.04.3 LTS, you can install them using:

```
sudo apt install pulseaudio-utils ffmpeg
```

## Cross-platform

Additionally/alternatively, you can make use of our predefined cross-platform microphone permission bundle:

- flet build
  - pyproject.toml
- ```
flet build <target_platform> --permissions microphone

[tool.flet]
permissions = ["microphone"]
```

Examples

Audio recorder

play_arrowTry Online

```
import flet as ft
import flet_audio_recorder as far

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.appbar = ft.AppBar(title=ft.Text("Audio Recorder"), center_title=True)

    path = "test-audio-file.wav"

    def show_snackbar(message):
        page.show_dialog(ft.SnackBar(ft.Text(message)))

    async def handle_recording_start(e: ft.Event[ft.Button]):
        show_snackbar("Starting recording...")
        await recorder.start_recording(output_path=path)

    async def handle_recording_stop(e: ft.Event[ft.Button]):
        output_path = await recorder.stop_recording()
        show_snackbar(f"Stopped recording. Output Path: {output_path}")
        if page.web and output_path is not None:
            await page.launch_url(output_path)
```

```

async def handle_list_devices(e: ft.Event[ft.Button]):
    o = await recorder.get_input_devices()
    show_snackbar(f"Input Devices: {' '.join([f'{d.id}:{d.label}' for d in o])}")

async def handle_has_permission(e: ft.Event[ft.Button]):
    try:
        status = await recorder.has_permission()
        show_snackbar(f"Audio Recording Permission status: {status}")
    except Exception as e:
        show_snackbar(f"Error checking permission: {e}")

async def handle_pause(e: ft.Event[ft.Button]):
    print(f"isRecording: {await recorder.is_recording()}")
    if await recorder.is_recording():
        await recorder.pause_recording()

async def handle_resume(e: ft.Event[ft.Button]):
    print(f"isPaused: {await recorder.is_paused()}")
    if await recorder.is_paused():
        await recorder.resume_recording()

async def handle_audio_encoder_test(e: ft.Event[ft.Button]):
    print(await recorder.is_supported_encoder(far.AudioEncoder.WAV))

recorder = far.AudioRecorder(
    configuration=far.AudioRecorderConfiguration(encoder=far.AudioEncoder.WAV),
    on_state_change=lambda e: print(f"State Changed: {e.data}"),
)

page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                ft.Button(
                    content="Start Audio Recorder", on_click=handle_recording_start
                ),
                ft.Button(
                    content="Stop Audio Recorder", on_click=handle_recording_stop
                ),
                ft.Button(content="List Devices", on_click=handle_list_devices),
                ft.Button(content="Pause Recording", on_click=handle_pause),
                ft.Button(content="Resume Recording", on_click=handle_resume),
                ft.Button(
                    content="WAV Encoder Support",
                    on_click=handle_audio_encoder_test,
                ),
                ft.Button(
                    content="Get Audio Recording Permission Status",
                    on_click=handle_has_permission,
                ),
            ],
        ),
    ),
)

if __name__ == "__main__":
    ft.run(main)

```

Stream chunks and save/download

On web, `AudioRecorder.stop_recording()` returns a browser-local Blob URL. Use streaming when your app needs access to the recorded bytes.

Set `AudioRecorderConfiguration.encoder` to `AudioEncoder.PCM16BITS` and handle `AudioRecorder.on_stream` to receive `AudioRecorderStreamEvent.chunk` bytes in Python.

`AudioEncoder.PCM16BITS` encoded streams are supported on all platforms. Stream chunks are raw PCM16 bytes and are not directly playable as an audio file. Wrap the bytes in a container such as WAV in Python when the destination needs a directly playable recording.

The example below collects the streamed chunks, wraps the PCM16 bytes in a WAV container, and passes the resulting bytes to `FilePicker.save_file()` so users can save or download the recording.

Audio recorder stream and download

play_arrowTry Online

```

import io
import time
import wave

import flet as ft
import flet_audio_recorder as far

```

```

SAMPLE_RATE = 44100
CHANNELS = 1
BYTES_PER_SAMPLE = 2

```

```

def pcm_to_wav_bytes(pcm_data: bytes) -> bytes:
    wav_buffer = io.BytesIO()
    with wave.open(wav_buffer, "wb") as wav:
        wav.setnchannels(CHANNELS)
        wav.setsampwidth(BYTES_PER_SAMPLE)
        wav.setframerate(SAMPLE_RATE)
        wav.writeframes(pcm_data)
    return wav_buffer.getvalue()

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    buffer = bytearray()

    def show_snackbar(message: str):
        page.show_dialog(ft.SnackBar(content=message, duration=ft.Duration(seconds=5)))

    def handle_stream(e: far.AudioRecorderStreamEvent):
        buffer.extend(e.chunk)
        status.value = (
            f"Streaming chunk {e.sequence}; {e.bytes_streamed} bytes collected."
        )

    async def handle_recording_start(e: ft.Event[ft.Button]):
        if not await recorder.has_permission():
            show_snackbar("Microphone permission is required.")
            return

        buffer.clear()
        status.value = "Recording..."
        await recorder.start_recording(
            configuration=far.AudioRecorderConfiguration(
                encoder=far.AudioEncoder.PCM16BITS,
                sample_rate=SAMPLE_RATE,
                channels=CHANNELS,
            ),
        )

    async def handle_recording_stop(e: ft.Event[ft.Button]):
        await recorder.stop_recording()
        if not buffer:
            show_snackbar("Nothing was recorded.")
            return

        wav_bytes = pcm_to_wav_bytes(bytes(buffer))
        file_name = f"recording-{int(time.time())}.wav"
        file_path = await ft.FilePicker().save_file(
            file_name=file_name,
            file_type=ft.FilePickerFileType.CUSTOM,
            allowed_extensions=["wav"],
            src_bytes=wav_bytes,
        )

        if file_path and not (page.web or page.platform.is_mobile()):
            with open(file_path, "wb") as output:
                output.write(wav_bytes)

        if page.web:
            status.value = f"Downloaded {file_name} ({len(wav_bytes)} bytes)."
        else:
            status.value = f"Saved to: {file_path}" if file_path else "Save cancelled."
            show_snackbar(status.value)

    recorder = far.AudioRecorder(on_stream=handle_stream)

    page.add(
        ft.SafeArea(
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    ft.Text(
                        "Stream PCM16 audio chunks and save or download "
                        "them as a WAV file."
                    ),
                    ft.Button("Start recording", on_click=handle_recording_start),
                    ft.Button("Stop and save", on_click=handle_recording_stop),
                    status := ft.Text(),
                ],
            ),
        ),
    )

```

```
if __name__ == "__main__":
    ft.run(main)
```

Streaming upload

Pass `AudioRecorderUploadSettings` to `AudioRecorder.start_recording()` to upload `AudioEncoder.PCM16BITS` recording bytes directly while recording.

The uploaded file contains raw PCM16 bytes, so a `.pcm` extension is intentional. See the stream chunks and save/download example for inspiration, if you need to create a playable WAV file.

note

Built-in upload URLs from `Page.get_upload_url()` require upload storage and a signing key. Run with `upload_dir` and set `FLET_SECRET_KEY`.

Audio recorder upload

play_arrowTry Online

```
import time

import flet as ft
import flet_audio_recorder as far

def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    def show_snackbar(message: str):
        page.show_dialog(ft.SnackBar(content=message, duration=ft.Duration(seconds=5)))

    def handle_upload(e: far.AudioRecorderUploadEvent):
        if e.error:
            status.value = f"Upload error: {e.error}"
        elif e.progress == 1:
            status.value = f"Upload complete: {e.bytes_uploaded or 0} bytes."
        else:
            status.value = f"Uploading: {e.bytes_uploaded or 0} bytes sent."

    async def handle_recording_start(e: ft.Event[ft.Button]):
        if not await recorder.has_permission():
            show_snackbar("Microphone permission is required.")
            return

        file_name = f"recordings/rec-{int(time.time())}.pcm"
        try:
            upload_url = page.get_upload_url(file_name=file_name, expires=600)
        except RuntimeError as ex:
            if "FLET_SECRET_KEY" not in str(ex):
                raise
            status.value = (
                "Uploads require a secret key. "
                "Set FLET_SECRET_KEY before running this app."
            )
            show_snackbar(status.value)
            return

        status.value = "Recording..."
        await recorder.start_recording(
            upload=far.AudioRecorderUploadSettings(
                upload_url=upload_url,
                file_name=file_name,
            ),
            configuration=far.AudioRecorderConfiguration(
                encoder=far.AudioEncoder.PCM16BITS,
                channels=1,
            ),
        )

    async def handle_recording_stop(e: ft.Event[ft.Button]):
        await recorder.stop_recording()
        show_snackbar(
            "Recording stopped. See 'uploads/recordings' folder for the recorded file."
        )

    recorder = far.AudioRecorder(on_upload=handle_upload)

    page.add(
        ft.SafeArea(
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    ft.Text("Record PCM16 audio and upload it as it streams."),
                    ft.Button("Start upload", on_click=handle_recording_start),
                ]
            )
        )
    )
```

```

        ft.Button("Stop recording", on_click=handle_recording_stop),
        status := ft.Text(),
    ],
),
)

if __name__ == "__main__":
    ft.run(
        main,
        upload_dir="uploads",
    )

```

Description

A control that allows you to record audio from your device.

This control can record audio using different audio encoders and also allows configuration of various audio recording parameters such as noise suppression, echo cancellation, and more.

Inherits: Service

Properties

- `configuration` - The default configuration of the audio recorder.

Events

- `on_state_change` - Called when recording state changes.
- `on_stream` - Called when a raw PCM16BITS recording chunk is available.
- `on_upload` - Called when streaming upload progress or errors are available.

Methods

- `cancel_recording` - Cancels the current audio recording.
- `get_input_devices` - Retrieves the available input devices for recording.
- `has_permission` - Checks if the app has permission to record audio, requesting it if needed.
- `is_paused` - Checks whether the audio recorder is currently paused.
- `is_recording` - Checks whether the audio recorder is currently recording.
- `is_supported_encoder` - Checks if the given audio encoder is supported by the recorder.
- `pause_recording` - Pauses the ongoing audio recording.
- `resume_recording` - Resumes a paused audio recording.
- `start_recording` - Starts recording audio and saves it to a file or streams it.
- `stop_recording` - Stops the audio recording and optionally returns the recording location.

Properties

`buildconfigurationclass-attributeinstance-attribute`

```

configuration: AudioRecorderConfiguration = field(
    default_factory=lambda: AudioRecorderConfiguration()
)

```

The default configuration of the audio recorder.

Events

`bolton_state_changeclass-attributeinstance-attribute`

```

on_state_change: (
    EventHandler[AudioRecorderStateChangeEvent] | None
) = None

```

Called when recording state changes.

`bolton_streamclass-attributeinstance-attribute`

```

on_stream: (

```



```
    EventHandler[AudioRecorderStreamEvent] | None  
  ) = None
```

Called when a raw PCM16BITS recording chunk is available.

bolton_uploadclass-attributeinstance-attribute

```
on_upload: (  
    EventHandler[AudioRecorderUploadEvent] | None  
  ) = None
```

Called when streaming upload progress or errors are available.

Methods

deployed_codecancel_recordingasync

```
cancel_recording()
```

Cancels the current audio recording.

deployed_codeget_input_devicesasync

```
get_input_devices() -> list[InputDevice]
```

Retrieves the available input devices for recording.

Returns:

- list[InputDevice] - A list of available input devices.

deployed_codehas_permissionasync

```
has_permission() -> bool
```

Checks if the app has permission to record audio, requesting it if needed.

Returns:

- bool - True if permission is already granted or granted after the request; False otherwise.

deployed_codeis_pausedasync

```
is_paused() -> bool
```

Checks whether the audio recorder is currently paused.

Returns:

- bool - True if the recorder is paused, False otherwise.

deployed_codeis_recordingasync

```
is_recording() -> bool
```

Checks whether the audio recorder is currently recording.

Returns:

- bool - True if the recorder is currently recording, False otherwise.

deployed_codeis_supported_encoderasync

```
is_supported_encoder(encoder: AudioEncoder) -> bool
```

Checks if the given audio encoder is supported by the recorder.

Parameters:

- encoder (AudioEncoder) - The audio encoder to check.

Returns:

- bool - True if the encoder is supported, False otherwise.

deployed_codepause_recordingasync

```
pause_recording()
```

Pauses the ongoing audio recording.

deployed_coderesume_recordingasync

```
resume_recording()
```

Resumes a paused audio recording.

deployed_codestart_recordingasync

```
start_recording(  
    output_path: str | None = None,  
    configuration: AudioRecorderConfiguration | None = None,  
    upload: AudioRecorderUploadSettings | None = None,  
) -> bool
```

Starts recording audio and saves it to a file or streams it.

If neither upload nor on_stream is used, output_path must be provided on platforms other than web.

When streaming, use PCM16BITS as the encoder. In that case, emitted or uploaded chunks contain raw PCM16 data. In some use cases, these chunks can be wrapped in a container such as WAV if the output must be directly playable as an audio file.

Parameters:

- output_path (str | None, default: None) - The file path where the audio will be saved. It must be specified if not on web.
- configuration (AudioRecorderConfiguration | None, default: None) - The configuration for the audio recorder. If None, the AudioRecorder.configuration will be used.
- upload (AudioRecorderUploadSettings | None, default: None) - Upload settings to stream recording bytes directly to a destination, for example a URL returned by Page.get_upload_url.

Returns:

- bool - True if recording was successfully started, False otherwise.

Raises:

- ValueError - If output_path is not provided on platforms other than web when neither streaming nor uploads are requested.
- ValueError - If streaming is requested with an encoder other than PCM16BITS.

deployed_codestop_recordingasync

```
stop_recording() -> str | None
```

Stops the audio recording and optionally returns the recording location.

Returns:

- str | None - The local file path where the audio was saved, a Blob URL on web, or None when streaming (i.e. when upload or on_stream is set).

Barometer

Streams barometer `BarometerReadingEvent` samples (atmospheric pressure in hPa). Useful for altitude calculations and weather-related experiences.

Note

- Supported platforms: Android, iOS.
- Barometer APIs are not exposed on the web or desktop platforms.
- iOS ignores custom sampling intervals.

On iOS you must also include a key called `NSMotionUsageDescription` in your app's `Info.plist` file. This key provides a message that tells the user why the app is requesting access to the device's motion data. Barometer service needs access to motion data to get barometer data.

For example, add the following to your `pyproject.toml` file:

```
[tool.flet.ios.info]
NSMotionUsageDescription = "This app requires access to the barometer to provide altitude information."
```

Adding `NSMotionUsageDescription` is a requirement and not doing so will crash your app when it attempts to access motion data.

Inherits: `Service`

Properties

- `cancel_on_error` - Whether the stream subscription should cancel on the first sensor error.
- `enabled` - Whether the sensor should be sampled.
- `interval` - Desired sampling interval provided as a `Duration`.

Events

- `on_error` - Fired when the platform reports a sensor error.
- `on_reading` - Fires when a new reading is available.

Examples

Barometer

```
import flet as ft

def main(page: ft.Page):
    def handle_reading(e: ft.BarometerReadingEvent):
        reading.value = f"{e.pressure:.2f} hPa"

    def handle_error(e: ft.SensorErrorEvent):
        page.add(ft.Text(f"Barometer error: {e.message}"))

    page.services.append(
        ft.Barometer(
            on_reading=handle_reading,
            on_error=handle_error,
            interval=ft.Duration(milliseconds=500),
        )
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text("Atmospheric pressure (hPa)."),
                    reading := ft.Text("Waiting for data..."),
                ],
            ),
        )
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

buildcancel_on_errorclass-attributeinstance-attribute

```
cancel_on_error: bool = True
```

Whether the stream subscription should cancel on the first sensor error.

buildenabledclass-attributeinstance-attribute

```
enabled: bool = True
```

Whether the sensor should be sampled. Disable to stop streaming.

buildintervalclass-attributeinstance-attribute

```
interval: Duration | None = None
```

Desired sampling interval provided as a Duration. Defaults to 200 ms, though some platforms (such as iOS) ignore custom sampling intervals.

Events

bolton_errorclass-attributeinstance-attribute

```
on_error: EventHandler[SensorErrorEvent] | None = None
```

Fired when the platform reports a sensor error. event.message is the error description.

bolton_readingclass-attributeinstance-attribute

```
on_reading: EventHandler[BarometerReadingEvent] | None = (
    None
)
```

Fires when a new reading is available.

event contains pressure (hPa) and timestamp (microseconds since epoch).

Battery

Provides access to device battery information and state changes.

Inherits: Service

Events

- `on_state_change` - Called when battery state changes (charging, discharging, full, unknown).

Methods

- `get_battery_level` - Gets the current battery level as a percentage, in the range 0 to 100.
- `get_battery_state` - Returns current battery state.
- `is_in_battery_save_mode` - Returns True if the device is currently in battery save mode.

Examples

Battery

```
import flet as ft

async def main(page: ft.Page):
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    async def refresh_info(e: ft.Event[ft.Button] = None):
        level = await battery.get_battery_level()
        state = await battery.get_battery_state()
        save_mode = await battery.is_in_battery_save_mode()
        info.value = (
            f"Battery level: {level}% \n"
            f"Battery state: {state.name} \n"
            f"Battery saver: {'ON' if save_mode else 'OFF'}"
        )

    async def on_state_change(e: ft.BatteryStateChangeEvent):
        print(f"State changed: {e.state}")
        await refresh_info()

    battery = ft.Battery(on_state_change=on_state_change)
    page.services.append(battery) # need to keep a reference to the service

    page.add(
        ft.SafeArea(
            content=ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    info := ft.Text(),
                    ft.Button("Refresh battery info", on_click=refresh_info),
                ],
            ),
        )
    )
    await refresh_info()

if __name__ == "__main__":
    ft.run(main)
```

Events

`bolton_state_change`class-attributeinstance-attribute

```
on_state_change: (
    EventHandler[BatteryStateChangeEvent] | None
) = None
```

Called when battery state changes (charging, discharging, full, unknown).

Methods

`deployed_code``get_battery_level`async

```
get_battery_level() -> int | None
```

Gets the current battery level as a percentage, in the range 0 to 100.

Returns:

- int | None - The battery level, or None when it is unavailable on the current
- int | None - environment.

deployed_codeget_battery_stateasync

`get_battery_state()` -> BatteryState

Returns current battery state.

deployed_codeis_in_battery_save_modeasync

`is_in_battery_save_mode()` -> bool

Returns True if the device is currently in battery save mode.

BrowserContextMenu

Controls the browser's context menu on the web platform.

The context menu is the menu that appears on right clicking or selecting text in the browser, for example.

On web, by default, the browser's context menu is enabled and Flet's context menus are hidden.

On all non-web platforms, this does nothing.

Inherits: Service

Properties

- disabled - Whether browser context menu disabling is currently requested.

Methods

- disable - Disable the browser's context menu.
- enable - Enable the browser's context menu.

Examples

Browser context menu

play_arrowTry Online

```
#
# Run this example as a web app using the command:
#
# flet run --web main.py
#

import flet as ft

async def main(page: ft.Page):
    bcm = ft.BrowserContextMenu()

    async def disable_context_menu():
        await bcm.disable()

    async def enable_context_menu():
        await bcm.enable()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button(
                        "Disable browser context menu",
                        on_click=disable_context_menu,
                    ),
                    ft.Button(
                        "Enable browser context menu",
                        on_click=enable_context_menu,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

builddisabledproperty

Whether browser context menu disabling is currently requested.

This flag is managed by this service instance:

- initialized to False;

- set to True after disable;
- set to False after enable.

Note

On non-web platforms, browser context menu control is not applicable, but this property still reflects the last requested state.

Methods

`deployed_code.disable_async`

`disable()`

Disable the browser's context menu.

By default, when the app starts, the browser's context menu is already enabled.

`deployed_code.enable_async`

`enable()`

Enable the browser's context menu.

By default, when the app starts, the browser's context menu is already enabled.

Clipboard

Provides access to the system clipboard.

Inherits: Service

Methods

- `get` - Retrieves data from the clipboard.
- `get_files` - Retrieves file references from the clipboard.
- `get_image` - Retrieves image data from the clipboard.
- `set` - Stores the given data on the clipboard.
- `set_files` - Stores file references on the clipboard.
- `set_image` - Stores image bytes on the clipboard.

Examples

Clipboard text

[play_arrowTry Online](#)

```
import flet as ft

async def main(page: ft.Page):
    async def set_to_clipboard():
        await ft.Clipboard().set(text_to_copy.value)
        text_to_copy.value = ""
        page.show_dialog(ft.SnackBar("Text copied to clipboard"))

    async def get_from_clipboard():
        contents = await ft.Clipboard().get()
        page.add(ft.Text(f"Clipboard contents: {contents}"))

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Column(
                        controls=[
                            text_to_copy := ft.TextField(label="Text to copy"),
                            ft.Button("Set to clipboard", on_click=set_to_clipboard),
                            ft.Button(
                                "Get from clipboard", on_click=get_from_clipboard
                            ),
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Clipboard images

[play_arrowTry Online](#)

```
import base64

import flet as ft

SAMPLE_PNG = base64.b64decode(
    "iVBORw0KGgoAAAANSUhEUgAAABkAAAAGCAYAAADnnMGAAAAACXBIWMAAA0RAAADkQFnq8zdAAAAGXRFWHRTb2Z0d2FyZQB3d3cuaw5rc2NhcGUub3Jnm+48GgAA"
)

async def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    async def set_image_to_clipboard(e: ft.Event[ft.Button]):
        await ft.Clipboard().set_image(SAMPLE_PNG)
        status.value = f"Sample image copied to clipboard ({len(SAMPLE_PNG)} bytes)."
```

```

        preview.content = ft.Image(src=SAMPLE_PNG)

    async def get_image_from_clipboard(e: ft.Event[ft.Button]):
        clipboard_image = await ft.Clipboard().get_image()
        if clipboard_image is None:
            status.value = "No image found in clipboard."
            preview.content = None
            return
        else:
            preview.content = ft.Image(src=clipboard_image)
            status.value = (
                f"Image loaded from clipboard ({len(clipboard_image)} bytes)."
            )

    page.add(
        ft.SafeArea(
            ft.Column(
                horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                controls=[
                    ft.Button(
                        "Set example image to clipboard",
                        on_click=set_image_to_clipboard,
                        disabled=not (page.web or page.platform.is_mobile()),
                        tooltip="Supported on mobile platforms only."
                        if not (page.web or page.platform.is_mobile())
                        else None,
                    ),
                    ft.Button(
                        "Get image from clipboard",
                        on_click=get_image_from_clipboard,
                    ),
                    status := ft.Text(),
                    preview := ft.Container(width=360, height=240),
                ],
            ),
        )
    )

if __name__ == "__main__":
    ft.run(main)

```

Clipboard files

play_arrowTry Online

```

import os
import subprocess
import sys
import tempfile
from pathlib import Path

import flet as ft

def open_file(_, path: str):
    """Open a desktop file in the default application."""
    if os.path.isfile(path):
        if sys.platform == "darwin": # macOS
            subprocess.run(["open", path])
        elif sys.platform == "win32": # Windows
            os.startfile(path)
        else: # Linux / BSD
            subprocess.run(["xdg-open", path])

async def main(page: ft.Page):
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    # supported-platform checkers
    supports_set_files = not page.web and page.platform.is_desktop()
    supports_get_files = not page.web and (
        page.platform.is_desktop() or page.platform == ft.PagePlatform.ANDROID
    )

    def get_sample_files() -> list[str]:
        # create temporary files
        sample_dir = Path(tempfile.gettempdir()) / "flet_clipboard_files_example"
        sample_dir.mkdir(parents=True, exist_ok=True)
        sample_files = [
            sample_dir / "sample_1.txt",
            sample_dir / "sample_2.txt",
        ]
        sample_files[0].write_text("Clipboard sample file #1\n", encoding="utf-8")
        sample_files[1].write_text("Clipboard sample file #2\n", encoding="utf-8")
        return [str(p) for p in sample_files]

```

```

async def set_files_to_clipboard(e: ft.Event[ft.Button]):
    files = get_sample_files()
    ok = await ft.Clipboard().set_files(files)
    status.value = (
        f"Set {len(files)} file references to clipboard (result: {ok})."
        if ok
        else "Failed to set file references to clipboard."
    )

async def get_files_from_clipboard(e: ft.Event[ft.Button]):
    files = await ft.Clipboard().get_files()
    if files:
        files_lv.controls = [
            ft.Row(
                alignment=ft.MainAxisAlignment.CENTER,
                controls=[
                    ft.Text(f, selectable=True),
                    ft.IconButton(
                        icon=ft.Icons.OPEN_IN_NEW,
                        icon_size=15,
                        on_click=lambda e, f=f: open_file(e, f),
                        tooltip="Open file",
                    ),
                ],
            )
            for f in files
        ]
    else:
        files_lv.controls = []
    status.value = f"Read {len(files)} file reference(s) from clipboard."

page.add(
    ft.SafeArea(
        ft.Column(
            horizontal_alignment=ft.CrossAxisAlignment.CENTER,
            controls=[
                ft.Button(
                    "Set example files to clipboard",
                    on_click=set_files_to_clipboard,
                    disabled=not supports_set_files,
                    tooltip="Supported on desktop platforms only."
                    if not supports_set_files
                    else None,
                ),
                ft.Button(
                    "Get files from clipboard",
                    on_click=get_files_from_clipboard,
                    disabled=not supports_get_files,
                    tooltip="Supported on Android and desktop platforms only."
                    if not supports_get_files
                    else None,
                ),
                status := ft.Text(),
                files_lv := ft.Column(
                    horizontal_alignment=ft.CrossAxisAlignment.CENTER,
                    spacing=0,
                    controls=[],
                ),
            ],
        ),
    )
)

if __name__ == "__main__":
    ft.run(main)

```

Methods

deployed_codeget_async

get() -> str | None

Retrieves data from the clipboard.

Returns:

- str | None - The string data retrieved from the clipboard, or None if the clipboard is empty or does not contain string data.

deployed_codeget_files_async

get_files() -> list[str]

Retrieves file references from the clipboard.

Returns:

- `list[str]` - A list of file references available in the clipboard. On Android these are typically content URIs.

Raises:

- `FletUnsupportedPlatformException` - If called on platforms other than the following: Android, macOS, Windows, Linux.

`deployed_code.get_image_async`

`get_image()` -> bytes | None

Retrieves image data from the clipboard.

Returns:

- bytes | None - The image data retrieved from the clipboard as bytes, or None if the clipboard is empty or does not contain image data.

`deployed_code.set_async`

`set(value: str)` -> None

Stores the given data on the clipboard.

Parameters:

- value (str) - The string data to be stored on the clipboard.

`deployed_code.set_files_async`

`set_files(files: list[str])` -> bool

Stores file references on the clipboard.

Parameters:

- files (list[str]) - A list of file paths.

Returns:

- bool - True if the operation succeeded, otherwise False.

Raises:

- `FletUnsupportedPlatformException` - If called on platforms other than the following: macOS, Windows, Linux.

`deployed_code.set_image_async`

`set_image(value: bytes)` -> None

Stores image bytes on the clipboard.

Parameters:

- value (bytes) - The image data (in bytes) to be stored on the clipboard.

Raises:

- `FletUnsupportedPlatformException` - If called on platforms other than the following: Android, iOS, Web.

Connectivity

Provides device connectivity status and change notifications.

Inherits: Service

Events

- `on_change` - Called when connectivity changes.

Methods

- `get_connectivity` - Returns the current connectivity type(s).

Examples

Connectivity

play_arrowTry Online

```
import flet as ft

async def main(page: ft.Page):
    connectivity = ft.Connectivity()

    status = ft.Text()
    changes = ft.Text()

    async def refresh(_=None):
        results = await connectivity.get_connectivity()
        status.value = "Current connectivity: " + ", ".join(r.value for r in results)

    async def on_change(e: ft.ConnectivityChangeEvent):
        changes.value = "Connectivity changed: " + ", ".join(
            r.value for r in e.connectivity
        )
        await refresh()

    connectivity.on_change = on_change

    await refresh()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Column(
                        [
                            status,
                            ft.Button("Refresh connectivity", on_click=refresh),
                            changes,
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Events

`bolton_change`class-attributeinstance-attribute

```
on_change: EventHandler[ConnectivityChangeEvent] | None = (
    None
)
```

Called when connectivity changes.

Methods

`deployed_code``get_connectivity`async

```
get_connectivity() -> list[ConnectivityType]
```

Returns the current connectivity type(s).

FilePicker

A control that allows you to use the native file explorer to pick single or multiple files, with extensions filtering support and upload.

Important

On Linux, this control requires Zenity when running Flet as a desktop app. It is not required when running Flet in a browser.

To install Zenity on Ubuntu/Debian run the following commands:

```
sudo apt-get install zenity
```

Inherits: Service

Events

- `on_upload` - Called when a file is uploaded via upload method.

Methods

- `get_directory_path` - Selects a directory and returns its absolute path.
- `pick_files` - Opens a pick file dialog.
- `save_file` - Opens a save file dialog which lets the user select a file path and a file name to save a file.
- `upload` - Uploads picked files to specified upload URLs.

Usage

Create an instance of FilePicker:

```
import flet as ft

file_picker = ft.FilePicker()
```

To open the file picker dialog call one of these three methods: `pick_files()`, `save_file()` or `get_directory_path()`, depending on the use case.

In most cases you can use a lambda function for that:

```
ft.Button(
    content="Pick files,
    on_click=lambda _: file_picker.pick_files(allow_multiple=True)
)
```

Uploading files

To upload one or more files, call `FilePicker.pick_files()` to let the user select files, then pass the returned list to `FilePicker.upload()` to perform the upload.

Separate uploads per user

If you need to separate uploads for each user you can specify a filename prepended with any number of directories in `page.get_upload_url()` call, for example:

```
upload_url = page.get_upload_url(f"/{username}/pictures/{f.name}", 600)
```

`/username/pictures` directories will be automatically created inside `upload_dir` if non-existent.

Upload storage

Notice the usage of `page.get_upload_url()` method – it generates a presigned upload URL for Flet's internal upload storage.

Use any storage for file uploads

You can generate presigned upload URL for AWS S3 storage using `boto3` library.

The same technique should work for Wasabi, Backblaze, MinIO and any other storage providers with S3-compatible API.

To enable Flet saving uploaded files to a directory provide a full or relative path to that directory in `flet.run()` call:

```
ft.run(main, upload_dir="uploads")
```

You can even put uploads inside assets directory, so uploaded files, e.g. pictures, docs, or other media, can be accessed from a Flet client right away:

```
ft.run(main, assets_dir="assets", upload_dir="assets/uploads")
```

and in your app you can display the uploaded picture with:

```
ft.Image(src="/uploads/<some-uploaded-picture.png>")
```

Examples

Pick, save, and browse directories

play_arrowTry Online

```
import flet as ft
```

```
def main(page: ft.Page):
    async def handle_pick_files(e: ft.Event[ft.Button]):
        files = await ft.FilePicker().pick_files(allow_multiple=True)
        selected_files.value = (
            ", ".join(map(lambda f: f.name, files)) if files else "Cancelled!"
        )

    async def handle_save_file(e: ft.Event[ft.Button]):
        save_file_path.value = await ft.FilePicker().save_file()

    async def handle_get_directory_path(e: ft.Event[ft.Button]):
        directory_path.value = await ft.FilePicker().get_directory_path()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Row(
                        controls=[
                            ft.Button(
                                content="Pick files",
                                icon=ft.Icons.UPLOAD_FILE,
                                on_click=handle_pick_files,
                            ),
                            selected_files := ft.Text(),
                        ],
                    ),
                    ft.Row(
                        controls=[
                            ft.Button(
                                content="Save file",
                                icon=ft.Icons.SAVE,
                                on_click=handle_save_file,
                                disabled=page.web, # disable this button in web mode
                            ),
                            save_file_path := ft.Text(),
                        ],
                    ),
                    ft.Row(
                        controls=[
                            ft.Button(
                                content="Open directory",
                                icon=ft.Icons.FOLDER_OPEN,
                                on_click=handle_get_directory_path,
                                disabled=page.web, # disable this button in web mode
                            ),
                            directory_path := ft.Text(),
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Image: Image - https://flet.dev/docs/examples/services/file_picker/media/pick_save_and_get_directory_path.png

Pick and upload files

The following example demonstrates multi-file pick and upload app.

Pick and upload files

play_arrowTry Online

```
#
# Example of picking and uploading files with progress indication
#
```



```

# Run this example with:
#   export FLET_SECRET_KEY=<some_secret_key>
#   uv run flet run --web examples/services/file_picker/pick_and_upload/main.py
#
from dataclasses import dataclass, field

import flet as ft

@dataclass
class State:
    file_picker: ft.FilePicker | None = None
    picked_files: list[ft.FilePickerFile] = field(default_factory=list)

state = State()

def main(page: ft.Page):
    if not page.web:
        page.add(
            ft.SafeArea(
                content=ft.Column(
                    controls=[
                        ft.Text(
                            "This example is only available in Flet Web mode.\n"
                            "\n"
                            "Run this example with:\n"
                            "    export FLET_SECRET_KEY=<some_secret_key>\n"
                            "    flet run --web "
                            "examples/services/file_picker/pick_and_upload/main.py",
                            color=ft.Colors.RED,
                            selectable=True,
                        ),
                    ],
                ),
            ),
        )
    return

prog_bars: dict[str, ft.ProgressRing] = {}

def on_upload_progress(e: ft.FilePickerUploadEvent):
    prog_bars[e.file_name].value = e.progress

async def handle_files_pick(e: ft.Event[ft.Button]):
    state.file_picker = ft.FilePicker(on_upload=on_upload_progress)
    files = await state.file_picker.pick_files(allow_multiple=True)
    print("Picked files:", files)
    state.picked_files = files

    # update progress bars
    upload_button.disabled = len(files) == 0
    prog_bars.clear()
    upload_progress.controls.clear()
    for f in files:
        prog = ft.ProgressRing(value=0, bgcolor="#eeeeee", width=20, height=20)
        prog_bars[f.name] = prog
        upload_progress.controls.append(ft.Row([prog, ft.Text(f.name)]))

async def handle_file_upload(e: ft.Event[ft.Button]):
    upload_button.disabled = True
    await state.file_picker.upload(
        files=[
            ft.FilePickerUploadFile(
                name=file.name,
                upload_url=page.get_upload_url(f"dir/{file.name}", 60),
            )
            for file in state.picked_files
        ]
    )

page.add(
    ft.SafeArea(
        content=ft.Column(
            controls=[
                ft.Button(
                    content="Select files...",
                    icon=ft.Icons.FOLDER_OPEN,
                    on_click=handle_files_pick,
                ),
                upload_progress := ft.Column(),
                upload_button := ft.Button(
                    content="Upload",
                    icon=ft.Icons.UPLOAD,
                    on_click=handle_file_upload,
                    disabled=True,

```

```

    ),
    ],
),
)

```

```

if __name__ == "__main__":
    ft.run(main, upload_dir="examples")

```

Image: Image - https://flet.dev/docs/examples/services/file_picker/media/pick_and_upload.png

Pick text content and save/download it

Use `pick_files()` with `with_data=True` when you need file contents directly, such as in web apps where `FilePickerFile.path` is not available.

Pick and save text content

play_arrow Try Online

```

#
# Example of picking and saving text content with FilePicker.
#
# Run this example with:
#   uv run flet run --web main.py
#
import flet as ft

def main(page: ft.Page):
    selected_file_name = ft.Text("No file selected")
    selected_file_content = ft.TextField(
        label="Selected file content",
        multiline=True,
        min_lines=8,
        max_lines=14,
    )
    save_status = ft.Text()

    async def pick_text_file(_: ft.Event[ft.Button]):
        files = await ft.FilePicker().pick_files(
            allow_multiple=False,
            with_data=True,
            file_type=ft.FilePickerFileType.CUSTOM,
            allowed_extensions=["txt", "md"],
        )
        if not files:
            selected_file_name.value = "Selection cancelled"
            selected_file_content.value = ""
            return

        selected = files[0]
        selected_file_name.value = f"Selected: {selected.name} ({selected.size} bytes)"
        selected_file_content.value = (
            selected.bytes.decode("utf-8", errors="replace") if selected.bytes else ""
        )
        save_status.value = ""

    async def save_text_file(_: ft.Event[ft.Button]):
        file_name = "flet_text_content.txt"
        file_path = await ft.FilePicker().save_file(
            file_name=file_name,
            file_type=ft.FilePickerFileType.CUSTOM,
            allowed_extensions=["txt"],
            src_bytes=selected_file_content.value.encode("utf-8"),
        )
        if page.web:
            save_status.value = f"Downloaded as {file_name}"
        else:
            save_status.value = (
                f"Saved to: {file_path}" if file_path else "Save cancelled"
            )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text(
                        "Pick a .txt/.md file and load its text from "
                        "FilePickerFile.bytes"
                    ),
                    ft.Button(
                        content="Pick text file",
                        icon=ft.Icons.UPLOAD_FILE,
                        on_click=pick_text_file,

```

```

        ),
        selected_file_name,
        selected_file_content,
        ft.Button(
            content="Save / Download text",
            icon=ft.Icons.DOWNLOAD,
            on_click=save_text_file,
        ),
        save_status,
    ],
),
)

if __name__ == "__main__":
    ft.run(main)

```

Events

bolton_uploadclass-attributeinstance-attribute

on_upload: EventHandler[FilePickerUploadEvent] | None = None

Called when a file is uploaded via upload method.

This callback is invoked at least twice for each uploaded file: once with 0.0 progress before the upload starts, and once with 1.0 progress when the upload completes.

For files larger than 1 MB, additional progress events are emitted at every 10% increment (for example, 0.1, 0.2, ...).

Methods

deployed_codeget_directory_pathasync

```

get_directory_path(
    dialog_title: str | None = None,
    initial_directory: str | None = None,
) -> str | None

```

Selects a directory and returns its absolute path.

Parameters:

- dialog_title (str | None, default: None) - The title of the dialog window. Defaults to [FilePicker].
- initial_directory (str | None, default: None) - The initial directory where the dialog should open.

Returns:

- str | None - The selected directory path or None if the dialog was cancelled.

Raises:

- FletUnsupportedPlatformException - If called in web mode.

deployed_codepick_filesasync

```

pick_files(
    dialog_title: str | None = None,
    initial_directory: str | None = None,
    file_type: FilePickerFileType = FilePickerFileType.ANY,
    allowed_extensions: list[str] | None = None,
    allow_multiple: bool = False,
    with_data: bool = False,
) -> list[FilePickerFile]

```

Opens a pick file dialog.

Tip

To upload the picked files, pass them to upload method, along with their upload URLs.

Parameters:

- dialog_title (str | None, default: None) - The title of the dialog window.
- initial_directory (str | None, default: None) - The initial directory where the dialog should open.
- file_type (FilePickerFileType, default: FilePickerFileType.ANY) - The file types allowed to be selected.
- allow_multiple (bool, default: False) - Allow the selection of multiple files at once.

- `with_data` (bool, default: False) - Read selected file contents into bytes.
- `allowed_extensions` (list[str] | None, default: None) - The allowed file extensions. Has effect only if `file_type` is `FilePickerFileType.CUSTOM`.

Returns:

- `list[FilePickerFile]` - A list of selected files.

deployed_codesave_fileasync

```
save_file(
    dialog_title: str | None = None,
    file_name: str | None = None,
    initial_directory: str | None = None,
    file_type: FilePickerFileType = FilePickerFileType.ANY,
    allowed_extensions: list[str] | None = None,
    src_bytes: bytes | None = None,
) -> str | None
```

Opens a save file dialog which lets the user select a file path and a file name to save a file.

Note

- On desktop this method only opens a dialog for the user to select a location and file name, and returns the chosen path. The file itself is not created or saved.

Parameters:

- `dialog_title` (str | None, default: None) - The title of the dialog window.
- `file_name` (str | None, default: None) - The default file name.
- `initial_directory` (str | None, default: None) - The initial directory where the dialog should open.
- `file_type` (`FilePickerFileType`, default: `FilePickerFileType.ANY`) - The file types allowed to be selected.
- `src_bytes` (bytes | None, default: None) - The contents of a file. Must be provided in web, iOS or Android modes.
- `allowed_extensions` (list[str] | None, default: None) - The allowed file extensions. Has effect only if `file_type` is `FilePickerFileType.CUSTOM`.

Raises:

- `ValueError` - If `src_bytes` is not provided, when called in web mode, on iOS or Android.
- `ValueError` - If `file_name` is not provided in web mode.

deployed_codeuploadasync

```
upload(files: list[FilePickerUploadFile])
```

Uploads picked files to specified upload URLs.

Before calling this method, `pick_files` first has to be called to ensure the internal file picker selection is not empty.

Once called, Flet asynchronously starts uploading selected files one-by-one and reports the progress via `on_upload` event.

Parameters:

- `files` (list[`FilePickerUploadFile`]) - A list of `FilePickerUploadFile`, where each item specifies which file to upload, and where (with PUT or POST).

Flashlight

Control the device torch/flashlight in your Flet app via the flet-flashlight extension, built on top of Flutter's flashlight package.

Platform Support

Platform	Windows	macOS	Linux
Supported	☐	☐	☐

Usage

Add flet-flashlight to your project dependencies:

- uv
 - pip
- ```
uv add flet-flashlight
pip install flet-flashlight
```

## Example

### Flashlight

```
import flet as ft
import flet_flashlight as ffl

def main(page: ft.Page):
 async def turn_on_flashlight():
 await ffl.Flashlight().on()

 async def turn_off_flashlight():
 await ffl.Flashlight().off()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Button("Turn On Flashlight", on_click=turn_on_flashlight),
 ft.Button("Turn Off Flashlight", on_click=turn_off_flashlight),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Description

A control to use FlashLight. Works on iOS and Android.

Inherits: Service

### Methods

- is\_available - Checks if the flashlight is available on the device.
- off - Turns the flashlight off.
- on - Turns the flashlight on.

## Methods

### deployed\_codeis\_availableasync

```
is_available()
```

Checks if the flashlight is available on the device.

**deployed\_codeoffasync**

`off()`

Turns the flashlight off.

**deployed\_codeonasync**

`on()`

Turns the flashlight on.

# Geolocator

Access device location services in your Flet app using the flet-geolocator extension. The control wraps Flutter's geolocator package and exposes async helpers for permission checks and position streams.

## Platform Support

| Platform  | Windows                  | macOS                    | Linux                    |
|-----------|--------------------------|--------------------------|--------------------------|
| Supported | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Usage

Add flet-geolocator to your project dependencies:

- uv
  - pip
- ```
uv add flet-geolocator
pip install flet-geolocator
```

Requirements

The below sections show the required configurations for each platform.

Android

Configuration to be made to access the device's location:

- ACCESS_FINE_LOCATION: Allows access precise location. Will be preferred over ACCESS_COARSE_LOCATION, if both are set.
- ACCESS_COARSE_LOCATION: Allows access approximate location.
- ACCESS_BACKGROUND_LOCATION (optional): Allows access to location even when the app is in the background. Effective as from Android 10 (API level 29).
- FOREGROUND_SERVICE_LOCATION (optional): Allows access to location even when the app is in the foreground. Effective as from Android 14 (API level 34).

Note

- At least one of ACCESS_FINE_LOCATION or ACCESS_COARSE_LOCATION permission is required to get location updates, with the former being preferred if both are set.
- Specifying the ACCESS_COARSE_LOCATION permission results in location updates with accuracy approximately equivalent to a city block. It might take a long time (minutes) before you will get your first locations fix as ACCESS_COARSE_LOCATION will only use the network services to calculate the position of the device. More information here.
- flet build
- pyproject.toml

```
flet build apk \
  --android-permissions android.permission.ACCESS_FINE_LOCATION=true \
  --android-permissions android.permission.ACCESS_COARSE_LOCATION=true \
  --android-permissions android.permission.ACCESS_BACKGROUND_LOCATION=true \
  --android-permissions android.permission.FOREGROUND_SERVICE_LOCATION=true

[tool.flet.android.permission]
"android.permission.ACCESS_FINE_LOCATION" = true
"android.permission.ACCESS_COARSE_LOCATION" = true
"android.permission.ACCESS_BACKGROUND_LOCATION" = true
"android.permission.FOREGROUND_SERVICE_LOCATION" = true
```

See also:

- setting Android permissions

iOS

Configuration to be made to access the device's location:

- flet build
- pyproject.toml

```
flet build ipa \
  --info-plist NSLocationWhenInUseUsageDescription="Some message to describe why you need this permission..."

[tool.flet.ios.info]
NSLocationWhenInUseUsageDescription = "Some message to describe why you need this permission..."
```

See also:

- setting iOS Info.plist entries

macOS

Configuration to be made to access the device's location:

- flet build
- pyproject.toml

```
flet build macos \
  --info-plist NSLocationUsageDescription="Some message to describe why you need this permission..." \
  --macos-entitlements com.apple.security.personal-information.location=true

[tool.flet.macos.info]
NSLocationUsageDescription = "Some message to describe why you need this permission..."

[tool.flet.macos.entitlement]
"com.apple.security.personal-information.location" = true
```

See also:

- macOS permissions
- macOS entitlements

Cross-platform

Additionally/alternatively, you can make use of our predefined cross-platform location permission bundle:

- flet build
- pyproject.toml

```
flet build <target_platform> --permissions location

[tool.flet]
permissions = ["location"]
```

Web: cached vs. fresh positions

On the web, `Geolocator.get_current_position` accepts a position from the browser's cache up to 5 minutes old by default, otherwise it asks the browser for a fresh fix and times out after 30 seconds if none arrives. This default exists because desktop browsers' on-demand location pipeline is slow and frequently fails on stationary devices.

If you need stricter freshness (or a longer cache window), pass a `GeolocatorWebConfiguration` with an explicit `maximum_age` and/or `time_limit`:

```
geo = ftg.Geolocator(
    configuration=ftg.GeolocatorWebConfiguration(
        accuracy=ftg.GeolocatorPositionAccuracy.HIGH,
        maximum_age=ft.Duration(seconds=10),
        time_limit=ft.Duration(seconds=15),
    ),
)
```

Troubleshooting

macOS: web browser returns POSITION_UNAVAILABLE or never resolves

When running a Flet app in `--web` mode on macOS, the browser may fail to obtain a position even though Location Services is on and the browser is allowed in System Settings → Privacy & Security → Location Services. Symptoms include:

- `getCurrentPosition` hanging or returning a Location request timed out. error.
- The position stream emitting Failed to obtain position: Position update is unavailable (this is the W3C POSITION_UNAVAILABLE / error code 2 from the browser).

This is a known issue with macOS' locationd daemon getting into a stuck state where it reports the browser as authorized but never delivers a fix. Restart it:

```
sudo killall locationd
```

The daemon respawns automatically. You can verify the browser side independently by visiting browserleaks.com/geo — if that page also fails to obtain a position, the problem is with the OS, not your Flet app.

The native (non-web) macOS Flet app is unaffected because it uses CoreLocation directly through the platform plugin rather than going through the browser's Geolocation API.

Example

Geolocator

play_arrowTry Online

```
from typing import Callable

import flet as ft
import flet_geolocator as ftg

async def main(page: ft.Page):
    page.scroll = ft.ScrollMode.ADAPTIVE
    page.appbar = ft.AppBar(title=ft.Text("Geolocator Tests"))

    def handle_position_change(e: ftg.GeolocatorPositionChangeEvent):
        page.add(ft.Text(f"New position: {e.position.latitude} {e.position.longitude}"))

    def get_dialog(handler: Callable):
        return ft.AlertDialog(
            adaptive=True,
            title="Opening Location Settings...",
            content=ft.Text(
                "You are about to be redirected to the location/app settings. "
                "Please locate this app and grant it location permissions."
            ),
            actions=[ft.TextButton("Take me there", on_click=handler)],
            actions_alignment=ft.MainAxisAlignment.CENTER,
        )

    def show_snackbar(message):
        page.show_dialog(ft.SnackBar(ft.Text(message)))

    async def handle_permission_request(e: ft.Event[ft.OutlinedButton]):
        try:
            p = await geo.request_permission()
            page.add(ft.Text(f"request_permission: {p}"))
            show_snackbar(f"Permission request sent: {p}")
        except RuntimeError as ex:
            show_snackbar(f"Could not request permission: {ex}")

    async def handle_get_permission_status(e: ft.Event[ft.OutlinedButton]):
        try:
            p = await geo.get_permission_status()
            show_snackbar(f"Permission status: {p}")
        except RuntimeError as ex:
            show_snackbar(f"Could not read permission status: {ex}")

    async def handle_get_current_position(e: ft.Event[ft.OutlinedButton]):
        try:
            p = await geo.get_current_position()
            show_snackbar(f"Current position: ({p.latitude}, {p.longitude})")
            print(f"Current position: ({p.latitude}, {p.longitude})")
        except RuntimeError as ex:
            show_snackbar(f"Could not get current position: {ex}")
            print(f"Could not get current position: {ex}")

    async def handle_get_last_known_position(e):
        try:
            p = await geo.get_last_known_position()
            if p is None:
                show_snackbar("No last known position available.")
            else:
                show_snackbar(f"Last known position: ({p.latitude}, {p.longitude})")
                print(f"Last known position: ({p.latitude}, {p.longitude})")
        except RuntimeError as ex:
            show_snackbar(f"Could not get last known position: {ex}")

    async def handle_location_service_enabled(e):
        try:
            p = await geo.is_location_service_enabled()
            show_snackbar(f"Location service enabled: {p}")
        except RuntimeError as ex:
```

```

        show_snackbar(f"Could not check location service: {ex}")

    async def handle_open_location_settings(e: ft.Event[ft.OutlinedButton]):
        try:
            p = await geo.open_location_settings()
            page.pop_dialog()
            if p is True:
                show_snackbar("Location settings opened successfully.")
            else:
                show_snackbar("Location settings could not be opened.")
        except RuntimeError as ex:
            page.pop_dialog()
            show_snackbar(f"Could not open location settings: {ex}")

    async def handle_open_app_settings(e: ft.Event[ft.OutlinedButton]):
        try:
            p = await geo.open_app_settings()
            page.pop_dialog()
            if p:
                show_snackbar("App settings opened successfully.")
            else:
                show_snackbar("App settings could not be opened.")
        except RuntimeError as ex:
            page.pop_dialog()
            show_snackbar(f"Could not open app settings: {ex}")

    location_settings_dlg = get_dialog(handle_open_location_settings)
    app_settings_dlg = get_dialog(handle_open_app_settings)

    geo = ftg.Geolocator(
        configuration=ftg.GeolocatorConfiguration(
            accuracy=ftg.GeolocatorPositionAccuracy.LOW
        ),
        on_position_change=handle_position_change,
        on_error=lambda e: page.add(ft.Text(f"Error: {e.data}")),
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Row(
                        wrap=True,
                        controls=[
                            ft.OutlinedButton(
                                content="Request Permission",
                                on_click=handle_permission_request,
                            ),
                            ft.OutlinedButton(
                                content="Get Permission Status",
                                on_click=handle_get_permission_status,
                            ),
                            ft.OutlinedButton(
                                content="Get Current Position",
                                on_click=handle_get_current_position,
                            ),
                            ft.OutlinedButton(
                                content="Get Last Known Position",
                                visible=not page.web,
                                on_click=handle_get_last_known_position,
                            ),
                            ft.OutlinedButton(
                                content="Is Location Service Enabled",
                                on_click=handle_location_service_enabled,
                            ),
                            ft.OutlinedButton(
                                content="Open Location Settings",
                                visible=not page.web, # (1)!
                                on_click=lambda e: page.show_dialog(
                                    location_settings_dlg
                                ),
                            ),
                            ft.OutlinedButton(
                                content="Open App Settings",
                                visible=not page.web, # (1)!
                                on_click=lambda e: page.show_dialog(app_settings_dlg),
                            ),
                        ],
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Description

A control that allows you to fetch GPS data from your device.

Inherits: Service

Properties

- `configuration` - Some additional configuration.
- `position` - The current position of the device.

Events

- `on_error` - Fires when an error occurs.
- `on_position_change` - Fires when the position of the device changes.

Methods

- `distance_between` - Calculates the distance between the supplied coordinates in meters.
- `get_current_position` - Gets the current position of the device with the desired accuracy and settings.
- `get_last_known_position` - Gets the last known position stored on the user's device.
- `get_permission_status` - Gets which permission the app has been granted to access the device's location.
- `is_location_service_enabled` - Checks if location service is enabled.
- `open_app_settings` - Attempts to open the app's settings.
- `open_location_settings` - Attempts to open the device's location settings.
- `request_permission` - Requests the device for access to the device's location.

Properties

`buildconfigurationclass-attributeinstance-attribute`

```
configuration: GeolocatorConfiguration | None = None
```

Some additional configuration.

`buildpositionclass-attributeinstance-attribute`

```
position: GeolocatorPosition | None = field(
    default=None, init=False
)
```

The current position of the device. (read-only)

Starts as None and will be updated when the position changes.

Events

`bolton_errorclass-attributeinstance-attribute`

```
on_error: ControlEventHandler[Geolocator] | None = None
```

Fires when an error occurs.

The data property of the event handler argument contains information on the error.

`bolton_position_change class-attributeinstance-attribute`

```
on_position_change: (
    EventHandler[GeolocatorPositionChangeEvent] | None
) = None
```

Fires when the position of the device changes.

Methods

`deployed_code distance_between async`

```
distance_between(
    start_latitude: Number,
    start_longitude: Number,
```

```

        end_latitude: Number,
        end_longitude: Number,
    ) -> Number

```

Calculates the distance between the supplied coordinates in meters.

The distance between the coordinates is calculated using the Haversine formula (see https://en.wikipedia.org/wiki/Haversine_formula).

Parameters:

- start_latitude (Number) - The latitude of the starting point, in degrees.
- start_longitude (Number) - The longitude of the starting point, in degrees.
- end_latitude (Number) - The latitude of the ending point, in degrees.
- end_longitude (Number) - The longitude of the ending point, in degrees.

Returns:

- Number - The distance between the coordinates in meters.

deployed_codeget_current_positionasync

```

get_current_position(
    configuration: GeolocatorConfiguration | None = None,
) -> GeolocatorPosition

```

Gets the current position of the device with the desired accuracy and settings.

Note

Depending on the availability of different location services, this can take several seconds. It is recommended to call the `get_last_known_position` method first to receive a known/cached position and update it with the result of the `get_current_position` method.

Parameters:

- configuration (GeolocatorConfiguration | None, default: None) - Additional configuration for the location request. If not specified, then the `Geolocator.configuration` property is used.

Returns:

- GeolocatorPosition - The current position of the device as a GeolocatorPosition.

deployed_codeget_last_known_positionasync

```

get_last_known_position() -> GeolocatorPosition | None

```

Gets the last known position stored on the user's device. The accuracy can be defined using the configuration property.

Returns:

- GeolocatorPosition | None - The last known position of the device as a GeolocatorPosition,
- GeolocatorPosition | None - or None if no last known position is available.

Raises:

- FletUnsupportedPlatformException - If invoked on a web platform.

deployed_codeget_permission_statusasync

```

get_permission_status() -> GeolocatorPermissionStatus

```

Gets which permission the app has been granted to access the device's location.

Returns:

- GeolocatorPermissionStatus - The status of the permission.

deployed_codeis_location_service_enabledasync

```

is_location_service_enabled() -> bool

```

Checks if location service is enabled.

Returns:

- bool - True if location service is enabled, False otherwise.

deployed_codeopen_app_settingsasync

`open_app_settings()` -> bool

Attempts to open the app's settings.

Returns:

- bool - True if the app's settings were opened successfully, False otherwise.

Raises:

- `FletUnsupportedPlatformException` - If invoked on a web platform.

deployed_codeopen_location_settingsasync

`open_location_settings()` -> bool

Attempts to open the device's location settings.

Returns:

- bool - True if the device's settings were opened successfully, False otherwise.

Raises:

- `FletUnsupportedPlatformException` - If invoked on a web platform.

deployed_coderequest_permissionasync

`request_permission()` -> `GeolocationPermissionStatus`

Requests the device for access to the device's location.

Returns:

- `GeolocationPermissionStatus` - The status of the permission request.

Gyroscope

Streams gyroscope `GyroscopeReadingEvent`, reporting device rotation rate around each axis in rad/s.

Note

- Supported platforms: Android, iOS and web.
- Web ignores requested sampling intervals.

Inherits: `Service`

Properties

- `cancel_on_error` - Whether the stream subscription should cancel on the first sensor error.
- `enabled` - Whether the sensor should be sampled.
- `interval` - Desired sampling interval provided as a `Duration`.

Events

- `on_error` - Fired when the platform reports a sensor error.
- `on_reading` - Fires when a new reading is available.

Examples

Gyroscope

```
import flet as ft

def main(page: ft.Page):
    def handle_reading(e: ft.GyroscopeReadingEvent):
        reading.value = f"x={e.x:.2f} rad/s, y={e.y:.2f} rad/s, z={e.z:.2f} rad/s"

    def handle_error(e: ft.SensorErrorEvent):
        page.add(ft.Text(f"Gyroscope error: {e.message}"))

    page.services.append(
        ft.Gyroscope(
            on_reading=handle_reading,
            on_error=handle_error,
            interval=ft.Duration(milliseconds=100),
        )
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text("Rotate your device to see gyroscope readings."),
                    reading := ft.Text("Waiting for data..."),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

`buildcancel_on_error`class-attributeinstance-attribute

`cancel_on_error: bool = True`

Whether the stream subscription should cancel on the first sensor error.

`buildenabled`class-attributeinstance-attribute

`enabled: bool = True`

Whether the sensor should be sampled. Disable to stop streaming.

`buildinterval`class-attributeinstance-attribute

`interval: Duration | None = None`

Desired sampling interval provided as a Duration. Defaults to 200 ms.

Events

`bolton_error`class-attributeinstance-attribute

`on_error: EventHandler[SensorErrorEvent] | None = None`

Fired when the platform reports a sensor error. `event.message` is the error description.

`bolton_reading`class-attributeinstance-attribute

`on_reading: EventHandler[GyroscopeReadingEvent] | None = (`
 `None`
`)`

Fires when a new reading is available.

event contains x, y, z rotation rates and timestamp (microseconds since epoch).

HapticFeedback

Allows access to the haptic feedback interface on the device.

Inherits: Service

Methods

- `heavy_impact` - Provides a haptic feedback corresponding a collision impact with a heavy mass.
- `light_impact` - Provides a haptic feedback corresponding a collision impact with a light mass.
- `medium_impact` - Provides a haptic feedback corresponding a collision impact with a medium mass.
- `selection_click` - TBD
- `vibrate` - Provides vibration haptic feedback to the user for a short duration.

Examples

Haptic feedback

```
import flet as ft

def main(page: ft.Page):
    hf = ft.HapticFeedback()

    async def heavy_impact():
        await hf.heavy_impact()

    async def medium_impact():
        await hf.medium_impact()

    async def light_impact():
        await hf.light_impact()

    async def vibrate():
        await hf.vibrate()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Button("Heavy impact", on_click=heavy_impact),
                    ft.Button("Medium impact", on_click=medium_impact),
                    ft.Button("Light impact", on_click=light_impact),
                    ft.Button("Vibrate", on_click=vibrate),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Methods

`deployed_codeheavy_impactasync`

```
heavy_impact()
```

Provides a haptic feedback corresponding a collision impact with a heavy mass.

`deployed_codelight_impactasync`

```
light_impact()
```

Provides a haptic feedback corresponding a collision impact with a light mass.

`deployed_codemedium_impactasync`

```
medium_impact()
```

Provides a haptic feedback corresponding a collision impact with a medium mass.

`deployed_codeselection_clickasync`


```
selection_click()
```

TBD

deployed_codevibrateasync

```
vibrate()
```

Provides vibration haptic feedback to the user for a short duration.

Magnetometer

Streams magnetometer `MagnetometerReadingEvent` reporting the ambient magnetic field (uT) per axis for compass-style use cases.

Note

- Supported platforms: Android, iOS.
- Magnetometer APIs are not available on web. or desktop, so always handle `on_error` to detect unsupported hardware.

Inherits: `Service`

Properties

- `cancel_on_error` - Whether the stream subscription should cancel on the first sensor error.
- `enabled` - Whether the sensor should be sampled.
- `interval` - Desired sampling interval provided as a `Duration`.

Events

- `on_error` - Fired when the platform reports a sensor error.
- `on_reading` - Fires when a new reading is available.

Examples

Magnetometer

```
import flet as ft

def main(page: ft.Page):
    def handle_reading(e: ft.MagnetometerReadingEvent):
        reading.value = f"x={e.x:.2f} uT, y={e.y:.2f} uT, z={e.z:.2f} uT"

    def handle_error(e: ft.SensorErrorEvent):
        page.add(ft.Text(f"Magnetometer error: {e.message}"))

    page.services.append(
        ft.Magnetometer(
            on_reading=handle_reading,
            on_error=handle_error,
            interval=ft.Duration(milliseconds=200),
        )
    )

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Text("Monitor the ambient magnetic field (uT)."),
                    reading := ft.Text("Waiting for data..."),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)
```

Properties

`buildcancel_on_error`class-attributeinstance-attribute

`cancel_on_error: bool = True`

Whether the stream subscription should cancel on the first sensor error.

`buildenabled`class-attributeinstance-attribute

`enabled: bool = True`

Whether the sensor should be sampled. Disable to stop streaming.

buildintervalclass-attributeinstance-attribute

`interval: Duration | None = None`

Desired sampling interval provided as a Duration. Defaults to 200 ms.

Events

bolton_errorclass-attributeinstance-attribute

`on_error: EventHandler[SensorErrorEvent] | None = None`

Fired when the platform reports a sensor error. `event.message` is the error description.

bolton_readingclass-attributeinstance-attribute

`on_reading: (EventHandler[MagnetometerReadingEvent] | None) = None`

Fires when a new reading is available.

event contains x, y, z magnetic field strengths (uT) and timestamp (microseconds since epoch).

PermissionHandler

Helps manage runtime permissions in your Flet apps.

It is powered by the Flutter `permission_handler` package.

Platform Support

Platform	Windows	macOS	Linux
Supported	☐	☐	☐

Usage

Add flet-permission-handler to your project dependencies:

- `uv`
- `pip`
`uv add flet-permission-handler`
`pip install flet-permission-handler`

Note

On mobile platforms you must also declare permissions in the native project files. See Flet publish docs.

Requirements

While the permissions are being requested during runtime, you'll still need to tell the OS which permissions your app might potentially use.

Android

See:

- full list of Android permissions
- Permission enum, which lists all the supported permissions
- setting Android permissions

iOS

See:

- Permission enum, which lists all the supported permissions and their Info.plist keys
- setting iOS permissions

Example

Permission Handler

play_arrowTry Online

```
import flet as ft
import flet_permission_handler as fph

def main(page: ft.Page):
    page.appbar = ft.AppBar(title="PermissionHandler Playground")

    def show_snackbar(message: str):
        page.show_dialog(ft.SnackBar(ft.Text(message)))

    async def get_permission_status(e: ft.Event[ft.OutlinedButton]):
        status = await ph.get_status(fph.Permission.MICROPHONE)
        show_snackbar(f"Microphone permission status: {status.name}")

    async def request_permission(e: ft.Event[ft.OutlinedButton]):
        status = await ph.request(fph.Permission.MICROPHONE)
        show_snackbar(f"Requested microphone permission: {status.name}")

    async def open_app_settings(e: ft.Event[ft.OutlinedButton]):
```

```

        show_snackbar("Opening app settings...")
        await ph.open_app_settings()

    ph = fph.PermissionHandler()

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.OutlinedButton("Open app settings", on_click=open_app_settings),
                    ft.OutlinedButton(
                        "Request Microphone permission", on_click=request_permission
                    ),
                    ft.OutlinedButton(
                        "Get Microphone permission status",
                        on_click=get_permission_status,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Description

Manages permissions for the application.

Platform support

Currently only supported on Android, iOS, Windows, and Web platforms.

Raises:

- `FletUnsupportedPlatformException` - If the platform is not supported.

Inherits: `Service`

Methods

- `get_status` - Gets the current status of the given permission.
- `open_app_settings` - Opens the app settings page.
- `request` - Request the user for access to the permission if access hasn't already been granted access before.

Methods

`deployed_code.get_status_async`

```

get_status(
    permission: Permission,
) -> PermissionStatus | None

```

Gets the current status of the given permission.

Parameters:

- `permission` (`Permission`) - The Permission to check the status for.

Returns:

- `PermissionStatus | None` - A `PermissionStatus` if the status is known, otherwise `None`.

`deployed_code.open_app_settings_async`

```

open_app_settings() -> bool

```

Opens the app settings page.

Returns:

- `bool` - True if the app settings page could be opened, otherwise `False`.

`deployed_code.request_async`

```

request(
    permission: Permission,
) -> PermissionStatus | None

```

Request the user for access to the permission if access hasn't already been granted access before.

Parameters:

- permission (Permission) - The Permission to request.

Returns:

- PermissionStatus | None - The new PermissionStatus after the request, or None if the request was not successful.

ScreenBrightness

Provides access to control and observe system and application screen brightness.

Note

- Supported platforms: Android, iOS.

To adjust the system brightness on Android, add the following permission in your pyproject.toml file:

```
[tool.flet.android.permission]
"android.permission.WRITE_SETTINGS" = true
```

Inherits: Service

Events

- `on_application_screen_brightness_change` - Called when the application screen brightness changes.
- `on_system_screen_brightness_change` - Called when the system screen brightness changes.

Methods

- `can_change_system_screen_brightness` - Returns whether the app is allowed to change the system screen brightness.
- `get_application_screen_brightness` - Returns the current application screen brightness, in range 0.0..1.0.
- `get_system_screen_brightness` - Returns the current system screen brightness, in range 0.0..1.0.
- `is_animate` - Returns True if brightness changes are animated (platform dependent).
- `is_auto_reset` - Returns True if brightness resets automatically on lifecycle changes.
- `reset_application_screen_brightness` - Resets the application screen brightness back to the system value.
- `set_animate` - Enables or disables animation for brightness changes.
- `set_application_screen_brightness` - Sets the application screen brightness.
- `set_auto_reset` - Enables or disables automatic reset to system brightness on lifecycle changes.
- `set_system_screen_brightness` - Sets the system screen brightness.

Examples

Screen brightness

```
import flet as ft

async def main(page: ft.Page):
    sb = ft.ScreenBrightness()
    page.services.append(sb)

    info = ft.Text()
    system_change = ft.Text()
    app_change = ft.Text()

    level = ft.Slider(min=0, max=1, divisions=20, value=0.5, label="Brightness")
    animate_switch = ft.Switch(
        label="Animate application changes", value=True, on_change=None
    )
    auto_reset_switch = ft.Switch(
        label="Auto-reset application brightness on lifecycle changes",
        value=True,
        on_change=None,
    )

    async def refresh_info():
        system_brightness = await sb.get_system_screen_brightness()
        app_brightness = await sb.get_application_screen_brightness()
        can_change_system = await sb.can_change_system_screen_brightness()

        animate_switch.value = await sb.is_animate()
        auto_reset_switch.value = await sb.is_auto_reset()

    info.value = (
        f"System: {system_brightness:.2f} | "
        f"Application: {app_brightness:.2f} | "
        f"Can change system: {can_change_system}"
    )
```

```

    )

    async def set_application_brightness(_):
        await sb.set_application_screen_brightness(level.value)
        await refresh_info()

    async def set_system_brightness(_):
        await sb.set_system_screen_brightness(level.value)
        await refresh_info()

    async def reset_application_brightness(_):
        await sb.reset_application_screen_brightness()
        await refresh_info()

    async def toggle_animate(e: ft.Event[ft.Switch]):
        await sb.set_animate(e.control.value)
        await refresh_info()

    async def toggle_auto_reset(e: ft.Event[ft.Switch]):
        await sb.set_auto_reset(e.control.value)
        await refresh_info()

    async def on_system_change(e: ft.ScreenBrightnessChangeEvent):
        system_change.value = f"System brightness changed: {e.brightness:.2f}"

    async def on_application_change(e: ft.ScreenBrightnessChangeEvent):
        app_change.value = f"Application brightness changed: {e.brightness:.2f}"

    sb.on_system_screen_brightness_change = on_system_change
    sb.on_application_screen_brightness_change = on_application_change

    await refresh_info()

    animate_switch.on_change = toggle_animate
    auto_reset_switch.on_change = toggle_auto_reset

    page.add(
        ft.SafeArea(
            content=ft.Column(
                controls=[
                    ft.Column(
                        [
                            info,
                            level,
                            ft.Row(
                                [
                                    ft.Button(
                                        "Set application",
                                        on_click=set_application_brightness,
                                    ),
                                    ft.Button(
                                        "Set system", on_click=set_system_brightness
                                    ),
                                    ft.TextButton(
                                        "Reset application",
                                        on_click=reset_application_brightness,
                                    ),
                                ],
                                wrap=True,
                            ),
                            animate_switch,
                            auto_reset_switch,
                            ft.Divider(),
                            system_change,
                            app_change,
                        ],
                        spacing=12,
                    ),
                ],
            ),
        ),
    )

if __name__ == "__main__":
    ft.run(main)

```

Events

bolton_application_screen_brightness_changeclass-attributeinstance-attribute

```

on_application_screen_brightness_change: (
    EventHandler[ScreenBrightnessChangeEvent] | None
) = None

```

Called when the application screen brightness changes.

bolton_system_screen_brightness_changedclass-attributeinstance-attribute

```
on_system_screen_brightness_change: (  
    EventHandler[ScreenBrightnessChangeEvent] | None  
) = None
```

Called when the system screen brightness changes.

Methods

deployed_codecan_change_system_screen_brightness

```
can_change_system_screen_brightness() -> bool
```

Returns whether the app is allowed to change the system screen brightness.

deployed_codeget_application_screen_brightness

```
get_application_screen_brightness() -> float
```

Returns the current application screen brightness, in range 0.0..1.0.

deployed_codeget_system_screen_brightness

```
get_system_screen_brightness() -> float
```

Returns the current system screen brightness, in range 0.0..1.0.

deployed_codeis_animate

```
is_animate() -> bool
```

Returns True if brightness changes are animated (platform dependent).

deployed_codeis_auto_reset

```
is_auto_reset() -> bool
```

Returns True if brightness resets automatically on lifecycle changes.

deployed_codereset_application_screen_brightness

```
reset_application_screen_brightness()
```

Resets the application screen brightness back to the system value.

deployed_codeset_animate

```
set_animate(animate: bool)
```

Enables or disables animation for brightness changes.

deployed_codeset_application_screen_brightness

```
set_application_screen_brightness(brightness: Number)
```

Sets the application screen brightness.

deployed_codeset_auto_reset

```
set_auto_reset(auto_reset: bool)
```

Enables or disables automatic reset to system brightness on lifecycle changes.

deployed_codeset_system_screen_brightness

```
set_system_screen_brightness(brightness: Number)
```

Sets the system screen brightness.

Secure Storage

A service for safely storing sensitive key-value data using the platform's native secure storage mechanisms—Keychain on iOS/macOS, Windows Credential Manager, libsecret on Linux, and Keystore on Android.

Powered by Flutter's flutter_secure_storage package.

note

You need libsecret-1-dev on your machine to build the project, and libsecret-1-0 to run the application (add it as a dependency after packaging your app). If you using snapcraft to build the project use the following.

Apart from libsecret you also need a keyring service, for that you need either gnome-keyring (for Gnome users) or kwalletmanager (for KDE users) or other light provider like secret-service.

```
sudo apt-get install libsecret-1-dev libsecret-1-0
```

Platform Support

Platform	Windows	macOS	Linux
Supported	☐	☐	☐

Usage

Add flet-secure-storage to your project dependencies:

- uv
 - pip
- ```
uv add flet-secure-storage
pip install flet-secure-storage
```

## Example

### Secure storage

play\_arrowTry Online

```
import base64
import os

import flet as ft
import flet_secure_storage as fss

def main(page: ft.Page):
 page.vertical_alignment = ft.MainAxisAlignment.CENTER
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 storage = fss.SecureStorage(
 web_options=fss.WebOptions(
 db_name="customstorage",
 public_key="publickey",
 wrap_key=base64.urlsafe_b64encode(os.urandom(32)).decode(),
 wrap_key_iv=base64.urlsafe_b64encode(os.urandom(16)).decode(),
),
 android_options=fss.AndroidOptions(
 reset_on_error=True,
 migrate_on_algorithm_change=True,
 enforce_biometrics=True,
 key_cipher_algorithm=fss.KeyCipherAlgorithm.AES_GCM_NO_PADDING,
 storage_cipher_algorithm=fss.StorageCipherAlgorithm.AES_GCM_NO_PADDING,
),
)

 key = ft.TextField(label="Key", value="example_key")
 value = ft.TextField(label="Value", value="secret_value")
 result = ft.Text(theme_style=ft.TextThemeStyle.TITLE_LARGE)

 async def set_value(e):
 await storage.set(key.value, value.value)
 result.value = "Value saved"

 async def get_value(e):
 result.value = await storage.get(key.value)

 async def remove_value(e):
```

```

 await storage.remove(key.value)
 result.value = "Value removed"

 async def clear_storage(e):
 await storage.clear()
 result.value = "Storage cleared"

 async def contains_key(e):
 exists = await storage.contains_key(key.value)
 result.value = f"Key exists: {exists}"

 async def get_availability(e):
 is_availability = await storage.get_availability()
 page.show_dialog(
 ft.SnackBar(
 content=ft.Text(
 value=f"Protected data available: {is_availability}"
 if is_availability
 else "Protected data available: None"
)
)
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Column(
 alignment=ft.MainAxisAlignment.CENTER,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 spacing=10,
 controls=[
 result,
 key,
 value,
 ft.Row(
 width=300,
 wrap=True,
 controls=[
 ft.Button("Set", on_click=set_value),
 ft.Button("Get", on_click=get_value),
 ft.Button("Contains key", on_click=contains_key),
 ft.Button("Remove", on_click=remove_value),
 ft.Button("Clear", on_click=clear_storage),
 ft.Button(
 "Check Data Availability",
 on_click=get_availability,
),
],
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Description

A class to manage secure storage in a Flet application across multiple platforms.

Inherits: Service

## Properties

- `android_options` - Android-specific configuration for secure storage.
- `ios_options` - iOS-specific configuration for secure storage.
- `macos_options` - macOS-specific configuration for secure storage.
- `web_options` - Web-specific configuration for secure storage.
- `windows_options` - Windows-specific configuration for secure storage.

## Events

- `on_change` - Fires when secure storage availability changes.

## Methods

- `clear` - Clears all key-value pairs from secure storage.
- `contains_key` - Checks whether the given key exists in secure storage.
- `get` - Retrieves the value stored under the given key in secure storage.
- `get_all` - Retrieves all key-value pairs from secure storage.
- `get_availability` - Gets the current availability status of secure storage.
- `remove` - Removes the value stored under the given key in secure storage.
- `set` - Stores a value in secure storage under the given key.

## Properties

### `buildandroid_optionsclass-attributeinstance-attribute`

```
android_options: AndroidOptions = field(
 default_factory=lambda: AndroidOptions()
)
```

Android-specific configuration for secure storage.

### `buildios_optionsclass-attributeinstance-attribute`

```
ios_options: IOSOptions = field(
 default_factory=lambda: IOSOptions()
)
```

iOS-specific configuration for secure storage.

### `buildmacos_optionsclass-attributeinstance-attribute`

```
macos_options: MacOSOptions = field(
 default_factory=lambda: MacOSOptions()
)
```

macOS-specific configuration for secure storage.

### `buildweb_optionsclass-attributeinstance-attribute`

```
web_options: WebOptions = field(
 default_factory=lambda: WebOptions()
)
```

Web-specific configuration for secure storage.

### `buildwindows_optionsclass-attributeinstance-attribute`

```
windows_options: WindowsOptions = field(
 default_factory=lambda: WindowsOptions()
)
```

Windows-specific configuration for secure storage.

## Events

### `bolton_change`class-attributeinstance-attribute

```
on_change: EventHandler[SecureStorageEvent] | None = None
```

Fires when secure storage availability changes.

iOS only feature. For unsupported platforms, this event will never fire. The payload is a `SecureStorageEvent` object with the available field.

## Methods

### `deployed_codeclearasync`

```
clear(
 web: WebOptions | None = None,
 ios: IOSOptions | None = None,
 macos: MacOSOptions | None = None,
 android: AndroidOptions | None = None,
 windows: WindowsOptions | None = None,
) -> None
```

Clears all key-value pairs from secure storage.

Parameters:

- web (WebOptions | None, default: None) - Optional web-specific configuration.
- ios (IOOptions | None, default: None) - Optional iOS-specific configuration.
- macos (MacOsOptions | None, default: None) - Optional macOS-specific configuration.
- android (AndroidOptions | None, default: None) - Optional Android-specific configuration.
- windows (WindowsOptions | None, default: None) - Optional Windows-specific configuration.

## deployed\_codecontains\_keyasync

```
contains_key(
 key: str,
 web: WebOptions | None = None,
 ios: IOOptions | None = None,
 macos: MacOsOptions | None = None,
 android: AndroidOptions | None = None,
 windows: WindowsOptions | None = None,
) -> bool
```

Checks whether the given key exists in secure storage.

Parameters:

- key (str) - The key to check.
- web (WebOptions | None, default: None) - Optional web-specific configuration.
- ios (IOOptions | None, default: None) - Optional iOS-specific configuration.
- macos (MacOsOptions | None, default: None) - Optional macOS-specific configuration.
- android (AndroidOptions | None, default: None) - Optional Android-specific configuration.
- windows (WindowsOptions | None, default: None) - Optional Windows-specific configuration.

Returns:

- bool - True if the key exists, False otherwise.

## deployed\_codegetasync

```
get(
 key: str,
 web: WebOptions | None = None,
 ios: IOOptions | None = None,
 macos: MacOsOptions | None = None,
 android: AndroidOptions | None = None,
 windows: WindowsOptions | None = None,
) -> str | None
```

Retrieves the value stored under the given key in secure storage.

Parameters:

- key (str) - The key to retrieve.
- web (WebOptions | None, default: None) - Optional web-specific configuration.
- ios (IOOptions | None, default: None) - Optional iOS-specific configuration.
- macos (MacOsOptions | None, default: None) - Optional macOS-specific configuration.
- android (AndroidOptions | None, default: None) - Optional Android-specific configuration.
- windows (WindowsOptions | None, default: None) - Optional Windows-specific configuration.

Returns:

- str | None - The stored string value, or None if the key does not exist.

## deployed\_codeget\_allasync

```
get_all(
 web: WebOptions | None = None,
 ios: IOOptions | None = None,
 macos: MacOsOptions | None = None,
 android: AndroidOptions | None = None,
 windows: WindowsOptions | None = None,
) -> dict[str, str]
```

Retrieves all key-value pairs from secure storage.

Parameters:

- web (WebOptions | None, default: None) - Optional web-specific configuration.
- ios (IOOptions | None, default: None) - Optional iOS-specific configuration.
- macos (MacOsOptions | None, default: None) - Optional macOS-specific configuration.
- android (AndroidOptions | None, default: None) - Optional Android-specific configuration.
- windows (WindowsOptions | None, default: None) - Optional Windows-specific configuration.

Returns:

- dict[str, str] - A dictionary with all stored key-value pairs.

## deployed\_codeget\_availabilityasync

```
get_availability() -> bool | None
```

Gets the current availability status of secure storage.

iOS and macOS only. On macOS, available only on macOS 12+. On older macOS versions, always returns True. On unsupported platforms, returns None.

Returns:

- bool | None - A boolean indicating storage availability, or None if unsupported.

## deployed\_coderemoveasync

```
remove(
 key: str,
 web: WebOptions | None = None,
 ios: IOOptions | None = None,
 macos: MacOsOptions | None = None,
 android: AndroidOptions | None = None,
 windows: WindowsOptions | None = None,
) -> None
```

Removes the value stored under the given key in secure storage.

Parameters:

- key (str) - The key to remove.
- web (WebOptions | None, default: None) - Optional web-specific configuration.
- ios (IOOptions | None, default: None) - Optional iOS-specific configuration.
- macos (MacOsOptions | None, default: None) - Optional macOS-specific configuration.
- android (AndroidOptions | None, default: None) - Optional Android-specific configuration.
- windows (WindowsOptions | None, default: None) - Optional Windows-specific configuration.

## deployed\_codesetasync

```
set(
 key: str,
 value: Any,
 web: WebOptions | None = None,
 ios: IOOptions | None = None,
 macos: MacOsOptions | None = None,
 android: AndroidOptions | None = None,
 windows: WindowsOptions | None = None,
) -> None
```

Stores a value in secure storage under the given key.

Parameters:

- key (str) - The key to store the value under.
- value (Any) - The value to store (cannot be None).
- web (WebOptions | None, default: None) - Optional web-specific configuration.
- ios (IOOptions | None, default: None) - Optional iOS-specific configuration.
- macos (MacOsOptions | None, default: None) - Optional macOS-specific configuration.

- android (AndroidOptions | None, default: None) - Optional Android-specific configuration.
- windows (WindowsOptions | None, default: None) - Optional Windows-specific configuration.

Raises:

- ValueError - If value is None.

# SemanticsService

Allows access to the platform's accessibility services.

Inherits: Service

## Methods

- `announce_message` - Sends a semantic announcement with the given message.
- `announce_tooltip` - Sends a semantic announcement of a tooltip.
- `get_accessibility_features` - Returns the current platform accessibility feature flags.

## Examples

### Accessibility features

`play_arrow`Try Online

```
import flet as ft

async def main(page: ft.Page):
 page.title = "SemanticsService - Accessibility features"
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.vertical_alignment = ft.MainAxisAlignment.CENTER

 async def get_features() -> str:
 features = await ft.SemanticsService().get_accessibility_features()
 return "\n".join(
 [
 f"Accessible navigation: {features.accessible_navigation}",
 f"Bold text: {features.bold_text}",
 f"Disable animations: {features.disable_animations}",
 f"High contrast: {features.high_contrast}",
 f"Invert colors: {features.invert_colors}",
 f"Reduce motion: {features.reduce_motion}",
 f"On/off switch labels: {features.on_off_switch_labels}",
 f"Supports announcements: {features.supports_announcements}",
]
)

 async def refresh_features(e: ft.Event[ft.Button]):
 info.value = await get_features()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 info := ft.Text(await get_features()),
 ft.Button("Refresh features", on_click=refresh_features),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Methods

### `deployed_code``announce_message``async`

```
announce_message(
 message: str,
 rtl: bool = False,
 assertiveness: Assertiveness = Assertiveness.POLITE,
)
```

Sends a semantic announcement with the given message.

Parameters:

- `message` (str) - The message to be announced.
- `rtl` (bool, default: False) - Indicates if the message text direction is right-to-left.



- `assertiveness` (`Assertiveness`, default: `Assertiveness.POLITE`) - The assertiveness level of the announcement. Only supported on web.

#### Notes

This method should be used for announcements that are not automatically handled by the system as a result of a UI state change.

#### `deployed_codeannounce_tooltipasync`

`announce_tooltip(message: str)`

Sends a semantic announcement of a tooltip.

#### Note

Only supported on Android.

#### `deployed_codeget_accessibility_featuresasync`

`get_accessibility_features()` -> `AccessibilityFeatures`

Returns the current platform accessibility feature flags.

Returns:

- `AccessibilityFeatures` - A snapshot of the platform's accessibility preferences at the time of invocation.

# ShakeDetector

Detects phone shakes.

Inherits: Service

## Properties

- `minimum_shake_count` - Number of shakes required before shake is triggered.
- `shake_count_reset_time_ms` - Time, in milliseconds, before shake count resets.
- `shake_slop_time_ms` - Minimum time between shakes, in milliseconds.
- `shake_threshold_gravity` - Shake detection threshold, in Gs.

## Events

- `on_shake` - Called when a shake is detected.

## Examples

### Shake detector

```
import flet as ft

def main(page: ft.Page):
 page.services.append(
 ft.ShakeDetector(
 minimum_shake_count=2,
 shake_slop_time_ms=300,
 shake_count_reset_time_ms=1000,
 on_shake=lambda _: page.add(ft.Text("Shake detected!")),
)
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[ft.Text("Shake your device!")],
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### `buildminimum_shake_count`class-attributeinstance-attribute

`minimum_shake_count: int = 1`

Number of shakes required before shake is triggered.

### `buildshake_count_reset_time_ms`class-attributeinstance-attribute

`shake_count_reset_time_ms: int = 3000`

Time, in milliseconds, before shake count resets.

### `buildshake_slop_time_ms`class-attributeinstance-attribute

`shake_slop_time_ms: int = 500`

Minimum time between shakes, in milliseconds.

### `buildshake_threshold_gravity`class-attributeinstance-attribute

`shake_threshold_gravity: Number = 2.7`

Shake detection threshold, in Gs.

## Events

## bolton\_shakeclass-attributeinstance-attribute

`on_shake: ControlEventHandler[ShakeDetector] | None = None`

Called when a shake is detected.

# Share

Shares text, links, or files using the platform share sheet.

Inherits: Service

## Methods

- `share_files` - Share one or more files with optional text/metadata.
- `share_text` - Share plain text with optional subject, title, and thumbnail.
- `share_uri` - Share a link/URI.

## Examples

### Share

`play_arrow` Try Online

```
import os

import flet as ft

async def main(page: ft.Page):
 share = ft.Share()

 status = ft.Text()
 result_raw = ft.Text()

 async def do_share_text():
 result = await share.share_text(
 "Hello from Flet!",
 subject="Greeting",
 title="Share greeting",
)
 status.value = f"Share status: {result.status}"
 result_raw.value = f"Raw: {result.raw}"

 async def do_share_uri():
 result = await share.share_uri("https://flet.dev")
 status.value = f"Share status: {result.status}"
 result_raw.value = f"Raw: {result.raw}"

 async def do_share_files_from_bytes():
 file = ft.ShareFile.from_bytes(
 b"Sample content from memory",
 mime_type="text/plain",
 name="sample.txt",
)
 result = await share.share_files(
 [file],
 text="Sharing a file from memory",
)
 status.value = f"Share status: {result.status}"
 result_raw.value = f"Raw: {result.raw}"

 async def do_share_files_from_paths():
 if page.web:
 status.value = "File sharing from paths is not supported on the web."
 return

 #
 temp_dir = await ft.StoragePaths().get_temporary_directory()
 file_path = os.path.join(temp_dir, "sample_from_path.txt")
 with open(file_path, "wb") as f:
 f.write(b"Sample content from file path")

 result = await share.share_files(
 [ft.ShareFile.from_path(file_path)],
 text="Sharing a file from memory",
)
 status.value = f"Share status: {result.status}"
 result_raw.value = f"Raw: {result.raw}"

 page.add(
 ft.SafeArea(
 ft.Column(
 [
 ft.Row(
 [
```

```

 ft.Button("Share text", on_click=do_share_text),
 ft.Button("Share link", on_click=do_share_uri),
 ft.Button(
 "Share file from bytes",
 on_click=do_share_files_from_bytes,
),
 ft.Button(
 "Share file from path",
 on_click=do_share_files_from_paths,
),
],
 wrap=True,
),
status,
result_raw,
],
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Methods

### deployed\_codeshare\_filesasync

```

share_files(
 files: list[ShareFile],
 title: str | None = None,
 text: str | None = None,
 subject: str | None = None,
 preview_thumbnail: ShareFile | None = None,
 share_position_origin: Offset | None = None,
 download_fallback_enabled: bool = True,
 mail_to_fallback_enabled: bool = True,
 excluded_cupertino_activities: Iterable[
 ShareCupertinoActivityType
]
 | None = None,
) -> ShareResult

```

Share one or more files with optional text/metadata.

Parameters:

- files (list[ShareFile]) - List of ShareFile instances to share.
- title (str | None, default: None) - Optional title for the share sheet.
- text (str | None, default: None) - Optional text to accompany the files.
- subject (str | None, default: None) - Optional subject for the shared files.
- preview\_thumbnail (ShareFile | None, default: None) - Optional thumbnail file to show in the share sheet.
- share\_position\_origin (Offset | None, default: None) - Optional position origin for the share sheet.
- download\_fallback\_enabled (bool, default: True) - Whether to enable download fallback.
- mail\_to\_fallback\_enabled (bool, default: True) - Whether to enable mailto fallback.
- excluded\_cupertino\_activities (Iterable[ShareCupertinoActivityType] | None, default: None) - Optional list of iOS/macOS activities to exclude.

### deployed\_codeshare\_textasync

```

share_text(
 text: str,
 title: str | None = None,
 subject: str | None = None,
 preview_thumbnail: ShareFile | None = None,
 share_position_origin: Offset | None = None,
 download_fallback_enabled: bool = True,
 mail_to_fallback_enabled: bool = True,
 excluded_cupertino_activities: Iterable[
 ShareCupertinoActivityType
]
 | None = None,
) -> ShareResult

```

Share plain text with optional subject, title, and thumbnail.

Parameters:

- text (str) - The text to share.
- title (str | None, default: None) - Optional title for the share sheet.
- subject (str | None, default: None) - Optional subject for the shared text.
- preview\_thumbnail (ShareFile | None, default: None) - Optional thumbnail file to show in the share sheet.
- share\_position\_origin (Offset | None, default: None) - Optional position origin for the share sheet.
- download\_fallback\_enabled (bool, default: True) - Whether to enable download fallback.
- mail\_to\_fallback\_enabled (bool, default: True) - Whether to enable mailto fallback.
- excluded\_cupertino\_activities (Iterable[ShareCupertinoActivityType] | None, default: None) - Optional list of iOS/macOS activities to exclude.

## deployed\_codeshare\_uriasync

```
share_uri(
 uri: str,
 share_position_origin: Offset | None = None,
 excluded_cupertino_activities: Iterable[
 ShareCupertinoActivityType
]
 | None = None,
) -> ShareResult
```

Share a link/URI.

Parameters:

- uri (str) - The URI to share.
- share\_position\_origin (Offset | None, default: None) - Optional position origin for the share sheet.
- excluded\_cupertino\_activities (Iterable[ShareCupertinoActivityType] | None, default: None) - Optional list of iOS/macOS activities to exclude.

# SharedPreferences

Provides access to persistent key-value storage.

Inherits: Service

## Methods

- `clear` - Clears all keys and values.
- `contains_key` - Checks if the given key exists.
- `get` - Gets the value for the given key.
- `get_keys` - Gets all keys with the given prefix.
- `remove` - Removes the value for the given key.
- `set` - Sets a value for the given key.

## Examples

### Shared preferences

play\_arrow Try Online

```
import flet as ft

async def main(page: ft.Page):
 async def set_value():
 await ft.SharedPreferences().set(store_key.value, store_value.value)
 get_key.value = store_key.value
 store_key.value = ""
 store_value.value = ""
 page.show_dialog(ft.SnackBar("Value saved to SharedPreferences"))

 async def get_value():
 contents = await ft.SharedPreferences().get(get_key.value)
 page.add(ft.Text(f"SharedPreferences contents: {contents}"))

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Column(
 [
 ft.Row(
 [
 store_key := ft.TextField(label="Key"),
 store_value := ft.TextField(label="Value"),
 ft.Button("Set", on_click=set_value),
]
),
 ft.Row(
 [
 get_key := ft.TextField(label="Key"),
 ft.Button("Get", on_click=get_value),
]
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Methods

### `deployed_code``clear``async`

`clear()` -> bool

Clears all keys and values.

Returns:

- bool - True if the preferences were cleared successfully, False otherwise.

### deployed\_codecontains\_keyasync

```
contains_key(key: str) -> bool
```

Checks if the given key exists.

Parameters:

- key (str) - The key to check for existence.

Returns:

- bool - True if the key exists, False otherwise.

### deployed\_codegetasync

```
get(key: str) -> SharedPreferencesValueType | None
```

Gets the value for the given key.

Parameters:

- key (str) - The key to retrieve the value for.

Returns:

- SharedPreferencesValueType | None - The value for the given key, or None if the key doesn't exist.

### deployed\_codeget\_keysasync

```
get_keys(key_prefix: str) -> list[str]
```

Gets all keys with the given prefix.

Parameters:

- key\_prefix (str) - The prefix to filter keys by.

Returns:

- list[str] - A list of keys that start with the given prefix.

### deployed\_coderemoveasync

```
remove(key: str) -> bool
```

Removes the value for the given key.

Parameters:

- key (str) - The key to remove.

Returns:

- bool - True if the key was removed, False if the key didn't exist.

### deployed\_codesetasync

```
set(
 key: str, value: SharedPreferencesValueType
) -> bool
```

Sets a value for the given key.

Note

Due to limitations on Android, it is not possible to set values that start with any of the following:

VGhpcyBpcyB0aGUgcHJlZml4IGZvciBhIGxpc3Qu, VGhpcyBpcyB0aGUgcHJlZml4IGZvciBCaWdJbnRlZ2Vy, and VGhpcyBpcyB0aGUgcHJlZml4IGZvciBEb3VibGUu.

Parameters:

- key (str) - The key to store the value under.
- value (SharedPreferencesValueType) - The value to store.

Returns:

- bool - True if the value was set successfully, False otherwise.



Raises:

- `ValueError` - If value is of an unsupported type (str, int, float, bool, and list[str]).

# StoragePaths

Provides access to commonly used storage paths on the device.

Note

Its methods are not supported in web mode.

Inherits: Service

## Methods

- `get_application_cache_directory` - Returns the path to the application-specific cache directory.
- `get_application_documents_directory` - Returns the path to a directory for user-generated data.
- `get_application_support_directory` - Returns the path to a directory for application support files.
- `get_console_log_filename` - Returns the path to a console.log file for debugging.
- `get_downloads_directory` - Returns the path to the downloads directory.
- `get_external_cache_directories` - Returns paths to external cache directories.
- `get_external_storage_directories` - Returns paths to external storage directories.
- `get_external_storage_directory` - Returns the path to the top-level external storage directory.
- `get_library_directory` - Returns the path to the library directory.
- `get_temporary_directory` - Returns the path to the temporary directory.

## Examples

### Storage paths

[play\\_arrow](#)Try Online

```
import flet as ft

async def main(page: ft.Page):
 storage_paths = ft.StoragePaths()

 items = []
 for label, method in [
 ("Application cache directory", storage_paths.get_application_cache_directory),
 (
 "Application documents directory",
 storage_paths.get_application_documents_directory,
),
 (
 "Application support directory",
 storage_paths.get_application_support_directory,
),
 ("Downloads directory", storage_paths.get_downloads_directory),
 ("External cache directories", storage_paths.get_external_cache_directories),
 (
 "External storage directories",
 storage_paths.get_external_storage_directories,
),
 ("Library directory", storage_paths.get_library_directory),
 ("External storage directory", storage_paths.get_external_storage_directory),
 ("Temporary directory", storage_paths.get_temporary_directory),
 ("Console log filename", storage_paths.get_console_log_filename),
]:
 try:
 value = await method()
 except ft.FletUnsupportedPlatformException as e:
 value = f"Not supported: {e}"
 except Exception as e:
 value = f"Error: {e}"
 else:
 if isinstance(value, list):
 value = ", ".join(value)
 elif value is None:
 value = "Unavailable"

 items.append(
 ft.Text(
```

```

 spans=[
 ft.TextSpan(
 f"{label}: ", style=ft.TextStyle(weight=ft.FontWeight.BOLD)
),
 ft.TextSpan(value),
]
)
)

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[ft.Column(items, spacing=5)],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Methods

### deployed\_code.get\_application\_cache\_directory\_async

```
get_application_cache_directory() -> str
```

Returns the path to the application-specific cache directory.

If this directory does not exist, it is created automatically.

Returns:

- str - The path to a directory where the application may place cache files.

Raises:

- `FletUnsupportedPlatformException` - If called on the web platform.

### deployed\_code.get\_application\_documents\_directory\_async

```
get_application_documents_directory() -> str
```

Returns the path to a directory for user-generated data.

This directory is intended for data that cannot be recreated by your application.

For non-user-generated data, consider using:

- `get_application_support_directory`
- `get_application_cache_directory`
- `get_external_storage_directory`

Raises:

- `FletUnsupportedPlatformException` - If called on the web platform.

Returns:

- str - The path to the application documents directory.

### deployed\_code.get\_application\_support\_directory\_async

```
get_application_support_directory() -> str
```

Returns the path to a directory for application support files.

This directory is created automatically if it does not exist. Use this for files not exposed to the user. Do not use for user data files.

Raises:

- `FletUnsupportedPlatformException` - If called on the web platform.

Returns:

- str - The path to the application support directory.

### deployed\_code.get\_console\_log\_filename\_async

```
get_console_log_filename() -> str
```

Returns the path to a console.log file for debugging.

This file is located in the `StoragePaths.get_application_cache_directory`.

Raises:

- `FletUnsupportedPlatformException` - If called on the web platform.

Returns:

- `str` - The path to the console log file.

### `deployed_code.get_downloads_directory_async`

```
get_downloads_directory() -> str | None
```

Returns the path to the downloads directory.

The returned directory may not exist; clients should verify and create it if necessary.

Raises:

- `FletUnsupportedPlatformException` - If called on the web platform.

Returns:

- `str | None` - The path to the downloads directory, or `None` if unavailable.

### `deployed_code.get_external_cache_directories_async`

```
get_external_cache_directories() -> list[str] | None
```

Returns paths to external cache directories.

These directories are typically on external storage (e.g., SD cards). Multiple directories may be available on some devices.

Raises:

- `FletUnsupportedPlatformException` - If called on the web or non-Android platforms.

Returns:

- `list[str] | None` - A List of external cache directory paths, or `None` if unavailable.

### `deployed_code.get_external_storage_directories_async`

```
get_external_storage_directories() -> list[str] | None
```

Returns paths to external storage directories.

These directories are typically on external storage (e.g., SD cards). Multiple directories may be available on some devices.

Raises:

- `FletUnsupportedPlatformException` - If called on the web or non-Android platforms.

Returns:

- `list[str] | None` - A List of external storage directory paths, or `None` if unavailable.

### `deployed_code.get_external_storage_directory_async`

```
get_external_storage_directory() -> str | None
```

Returns the path to the top-level external storage directory.

Raises:

- `FletUnsupportedPlatformException` - If called on the web or non-Android platforms.

Returns:

- `str | None` - The path to the external storage directory, or `None` if unavailable.

### `deployed_code.get_library_directory_async`

```
get_library_directory() -> str
```

Returns the path to the library directory.

This directory is for persistent, backed-up files not visible to the user (e.g., `sqlite.db`).

Raises:

- `FletUnsupportedPlatformException` - If called on the web or non-Apple platforms.

Returns:

- `str` - The path to the library directory.

## `deployed_code.get_temporary_directory_async`

`get_temporary_directory()` -> `str`

Returns the path to the temporary directory.

This directory is not backed up and is suitable for storing caches of downloaded files. Files may be cleared at any time. The caller is responsible for managing files within this directory.

Raises:

- `FletUnsupportedPlatformException` - If called on the web platform.

Returns:

- `str` - The path to the temporary directory.

# UrlLauncher

Provides access to URL launching capabilities.

Inherits: Service

## Methods

- `can_launch_url` - Checks whether the specified URL can be handled by some app installed on the device.
- `close_in_app_web_view` - Closes the in-app web view if it is currently open.
- `launch_url` - Opens a web browser or in-app view to a given url.
- `open_window` - Opens a popup browser window in web environments.
- `supports_close_for_launch_mode` - Checks whether `close_in_app_web_view` is supported for a launch mode.
- `supports_launch_mode` - Checks whether the specified launch mode is supported.

## Examples

### URL launcher

`play_arrow` Try Online

```
import flet as ft

async def main(page: ft.Page):
 url_launcher = ft.UrlLauncher()

 url = ft.TextField(label="URL to open", value="https://flet.dev", expand=True)
 status = ft.Text()

 async def can_launch():
 can = await url_launcher.can_launch_url(url.value)
 status.value = f"Can launch: {can}"

 async def launch_default():
 await url_launcher.launch_url(url.value)

 async def launch_in_app_webview():
 await url_launcher.launch_url(
 url.value,
 mode=ft.LaunchMode.IN_APP_WEB_VIEW,
 web_view_configuration=ft.WebViewConfiguration(
 enable_javascript=True, enable_dom_storage=True
),
)

 async def launch_in_app_browser_view():
 await url_launcher.launch_url(
 url.value,
 mode=ft.LaunchMode.IN_APP_BROWSER_VIEW,
 browser_configuration=ft.BrowserConfiguration(show_title=True),
)

 async def launch_external():
 await url_launcher.launch_url(
 url.value,
 mode=ft.LaunchMode.EXTERNAL_APPLICATION,
 web_only_window_name="_blank",
)

 async def launch_popup():
 await url_launcher.open_window(
 url.value, title="Flet popup", width=480, height=640
)

 async def close_webview():
 supported = await url_launcher.supports_close_for_launch_mode(
 ft.LaunchMode.IN_APP_WEB_VIEW
)
 if supported:
 await url_launcher.close_in_app_web_view()
 else:
 status.value = "Close in-app web view not supported on this platform"

 page.add(
```

```

ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Column(
 [
 url,
 ft.Row(
 [
 ft.Button("Launch URL", on_click=launch_default),
 ft.Button(
 "Launch in-app webview",
 on_click=launch_in_app_webview,
),
 ft.Button(
 "Launch in-app browser view",
 on_click=launch_in_app_browser_view,
),
 ft.Button(
 "Launch external/new tab",
 on_click=launch_external,
),
 ft.Button(
 "Open popup window (web)", on_click=launch_popup
),
 ft.Button("Can launch?", on_click=can_launch),
 ft.Button(
 "Close in-app webview", on_click=close_webview
),
],
 wrap=True,
),
 status,
],
),
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Methods

### deployed\_codecan\_launch\_urlasync

can\_launch\_url(url: str | Url) -> bool

Checks whether the specified URL can be handled by some app installed on the device.

Parameters:

- url (str | Url) - The URL to check.

Returns:

- bool - True if it is possible to verify that there is a handler available. False if there is no handler available, or the application does not have permission to check. For example: - On recent versions of Android and iOS, this will always return False unless the application has been configuration to allow querying the system for launch support. - On web, this will always return False except for a few specific schemes that are always assumed to be supported (such as http(s)), as web pages are never allowed to query installed applications.

### deployed\_codeclose\_in\_app\_web\_viewasync

close\_in\_app\_web\_view()

Closes the in-app web view if it is currently open.

### deployed\_codelaunch\_urlasync

```

launch_url(
 url: str | Url,
 mode: LaunchMode = LaunchMode.PLATFORM_DEFAULT,
 web_view_configuration: WebViewConfiguration
 | None = None,
 browser_configuration: BrowserConfiguration
 | None = None,
 web_only_window_name: str | None = None,
)

```

Opens a web browser or in-app view to a given url.

Parameters:

- url (str | Url) - The URL to open.
- mode (LaunchMode, default: LaunchMode.PLATFORM\_DEFAULT) - Preferred launch mode for opening the URL.
- web\_view\_configuration (WebViewConfiguration | None, default: None) - Optional configuration for in-app web views.
- browser\_configuration (BrowserConfiguration | None, default: None) - Optional configuration for in-app browser views.
- web\_only\_window\_name (str | None, default: None) - Window name for web-only launches.

## deployed\_codeopen\_windowasync

```
open_window(
 url: str | Url,
 title: str | None = None,
 width: float | None = None,
 height: float | None = None,
)
```

Opens a popup browser window in web environments.

Parameters:

- url (str | Url) - The URL to open in the popup window.
- title (str | None, default: None) - The popup window title.
- width (float | None, default: None) - Desired popup width in logical pixels.
- height (float | None, default: None) - Desired popup height in logical pixels.

## deployed\_codesupports\_close\_for\_launch\_modeasync

```
supports_close_for_launch_mode(
 mode: LaunchMode,
) -> bool
```

Checks whether close\_in\_app\_web\_view is supported for a launch mode.

Parameters:

- mode (LaunchMode) - Launch mode to verify close support for.

Returns:

- bool - True if closing an in-app web view is supported; otherwise False.

## deployed\_codesupports\_launch\_modeasync

```
supports_launch_mode(mode: LaunchMode) -> bool
```

Checks whether the specified launch mode is supported.

Parameters:

- mode (LaunchMode) - Launch mode to verify.

Returns:

- bool - True if the launch mode is supported by the platform; otherwise False.



# UserAccelerometer

Streams linear acceleration readings.

If the device is still, or is moving along a straight line at constant speed, the reported acceleration is zero. If the device is moving e.g. towards north and its speed is increasing, the reported acceleration is towards north; if it is slowing down, the reported acceleration is towards south; if it is turning right, the reported acceleration is towards east. The data of this stream is obtained by filtering out the effect of gravity from AccelerometerReadingEvent.

Note

- Supported platforms: Android, iOS and web.
- Web ignores requested sampling intervals.

Inherits: Service

## Properties

- `cancel_on_error` - Whether the stream subscription should cancel on the first sensor error.
- `enabled` - Whether the sensor should be sampled.
- `interval` - Desired sampling interval provided as a Duration.

## Events

- `on_error` - Fired when the platform reports a sensor error.
- `on_reading` - Fires when a new reading is available.

## Examples

### User accelerometer

```
import flet as ft

def main(page: ft.Page):
 def handle_reading(e: ft.UserAccelerometerReadingEvent):
 reading.value = f"x={e.x:.2f} m/s^2, y={e.y:.2f} m/s^2, z={e.z:.2f} m/s^2"

 def handle_error(e: ft.SensorErrorEvent):
 page.add(ft.Text(f"UserAccelerometer error: {e.message}"))

 page.services.append(
 ft.UserAccelerometer(
 on_reading=handle_reading,
 on_error=handle_error,
 interval=ft.Duration(milliseconds=100),
)
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Linear acceleration without gravity. "
 "Keep the app running on a device with motion sensors."
),
 reading := ft.Text("Waiting for data..."),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

`buildcancel_on_errorclass-attributeinstance-attribute`

```
cancel_on_error: bool = True
```

Whether the stream subscription should cancel on the first sensor error.

#### **buildenabledclass-attributeinstance-attribute**

```
enabled: bool = True
```

Whether the sensor should be sampled. Disable to stop streaming.

#### **buildintervalclass-attributeinstance-attribute**

```
interval: Duration | None = None
```

Desired sampling interval provided as a Duration. Defaults to 200 ms.

## **Events**

#### **bolton\_errorclass-attributeinstance-attribute**

```
on_error: EventHandler[SensorErrorEvent] | None = None
```

Fired when the platform reports a sensor error. event.message is the error description.

#### **bolton\_readingclass-attributeinstance-attribute**

```
on_reading: (
 EventHandler[UserAccelerometerReadingEvent] | None
) = None
```

Fires when a new reading is available.

event contains x, y, z acceleration values and timestamp (microseconds since epoch).

# Wakelock

Prevents the device from sleeping while enabled.

Inherits: Service

## Methods

- `disable` - Allows the device to sleep again.
- `enable` - Keeps the device awake.
- `is_enabled` - Returns True if the wakelock is currently enabled.

## Examples

### Wakelock

[play\\_arrow](#)Try Online

```
import flet as ft

async def main(page: ft.Page):
 wakelock = ft.Wakelock()

 status = ft.Text()

 async def update_status():
 enabled = await wakelock.is_enabled()
 status.value = f"Wakelock enabled: {enabled}"

 async def enable_lock():
 await wakelock.enable()
 await update_status()

 async def disable_lock():
 await wakelock.disable()
 await update_status()

 await update_status()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Column(
 [
 status,
 ft.Row(
 [
 ft.Button("Enable wakelock", on_click=enable_lock),
 ft.Button(
 "Disable wakelock", on_click=disable_lock
),
],
 wrap=True,
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Methods

`deployed_code``disable``async`

`disable()`

Allows the device to sleep again.

`deployed_code``enable``async`

`enable()`

Keeps the device awake.

**deployed\_codeis\_enabledasync**

`is_enabled()` -> bool

Returns True if the wakelock is currently enabled.

# Flet Studio Privacy Policy

Effective date: May 26, 2026

This Privacy Policy explains how Appveyor Systems Inc. ("AppVeyor", "we", "us", or "our") collects, uses, shares, and protects personal information in connection with the Flet Studio service ("Flet Studio", "the Service"). Flet Studio is a hosted browser-based development environment for building Python applications with the Flet framework.

This policy applies to information we collect from users of Flet Studio at <https://flet.app> (and any related domains we operate for the Service) and through the Flet Studio web application.

## 1. Introduction

### 1.1 Who we are

Appveyor Systems Inc. is the operator of Flet Studio. Appveyor Systems Inc. also operates the AppVeyor CI/CD service, which is governed by a separate privacy policy at <https://www.appveyor.com/privacy-policy/>. This document applies only to Flet Studio.

### 1.2 Our role under GDPR

With respect to the personal information described in this policy — your account profile, sign-in metadata, usage data, and the content you create or upload to Flet Studio — Appveyor Systems Inc. is the data controller. We determine the purposes and means by which this information is processed.

Separately, when you build a Flet application using Flet Studio and that application collects or processes personal information about your own users, you are the controller for that personal information and Appveyor Systems Inc. may act as a processor on your behalf. That relationship is governed by a Data Processing Agreement (DPA), not by this Privacy Policy. Contact us at [email protected] to discuss a DPA if you require one.

### 1.3 Scope

This policy covers personal information processed by Flet Studio as currently implemented. Some features described in our marketing materials or roadmap (for example, paid subscriptions, email notifications, and AI assistance) are not yet integrated. If we add features that materially change how personal information is processed, we will update this policy as described in Section 12.

## 2. Information We Collect

### 2.1 Information you provide

Account profile. When you create an account, we store the information needed to identify you and authenticate your sessions. Today, this is primarily provided through GitHub OAuth and includes:

- Your email address
- Your full name (if available from GitHub)
- A unique GitHub user identifier
- Your GitHub username
- Your profile avatar URL

Account and team information. For each account, we store the account name, a URL-safe slug, the account type (personal or team), the plan tier, and — for team accounts — the membership and role of each member.

Content you create or upload. When you use Flet Studio, we store the app source code, files, and versions that you create, edit, or upload to the Service, along with their metadata (file path, size, timestamps, app name, description, dependencies, tags). This content is stored in object storage hosted by our infrastructure provider, Railway.

### 2.2 Information collected automatically

Sign-in and security data. When you sign in, we record the timestamp of your most recent successful sign-in and the IP address from which it occurred. We also maintain an internal token version counter that is used to invalidate active sessions when you sign out or change credentials. We use this information to maintain account security, detect suspicious activity, and provide an audit trail.

Usage data. To enforce per-account quotas and prevent abuse, we maintain daily counters of certain actions (for example, how many gallery clones or archive uploads you have performed in the last 24 hours) and a running total of the storage your account is consuming. These counters are tied to your account, not to individual sign-in sessions.

Server-side telemetry. When telemetry is enabled, our server software emits operational traces and logs to a server-side observability tool (Pydantic Logfire) to help us diagnose errors, investigate incidents, and monitor performance. These traces may incidentally include account or request identifiers; we apply a PII-scrubbing rule to redact sensitive fields. Telemetry is configurable and is not enabled in all environments.

We do not use client-side analytics services. We do not load Google Analytics, Mixpanel, Segment, advertising pixels, or any similar product into the Flet Studio web application.

## 2.3 Information from third parties

When you authenticate via GitHub OAuth, we receive the profile fields listed in Section 2.1 from GitHub. We do not receive your GitHub password and we do not request access to your private repositories unless you explicitly grant a scope that permits it.

## 3. How We Use Information

We use personal information to:

- Provide the Service: authenticate you, render your accounts and projects, store and serve your content
- Enforce per-account limits and prevent abuse of the Service
- Maintain the security and integrity of the Service, including investigating fraud, abuse, or unauthorized access
- Diagnose, debug, and improve the Service through server-side telemetry
- Communicate with you about your account or service-related matters when you contact us
- Comply with legal obligations

We do not use your personal information for advertising, profiling, automated decision-making with legal effects, or training machine-learning models.

## 4. Legal Bases for Processing (GDPR)

If you are located in the European Economic Area, the United Kingdom, or Switzerland, we rely on the following legal bases under the GDPR (or its UK equivalent) when we process your personal information:

- Performance of a contract — to provide the Service you have signed up for. This covers account creation, authentication, storage of your content, and quota enforcement.
- Legitimate interests — to keep the Service secure, prevent abuse, debug and improve the Service, and maintain operational telemetry. We balance these interests against your rights and interests, and we apply data-minimization measures (such as PII scrubbing in telemetry) accordingly.
- Compliance with a legal obligation — when we are required to retain, disclose, or process personal information to comply with applicable law.
- Consent — where we ask for it explicitly. Today, our processing is not consent-based; if we introduce features that require consent (for example, optional marketing communications), we will request it separately and you will be able to withdraw it at any time.

## 5. How We Share Information

We do not sell your personal information. We do not share your personal information with third parties for advertising or for cross-context behavioral advertising of any kind.

We share personal information only with the following categories of recipients:

### 5.1 Service providers (subprocessors)

We rely on a small number of third-party service providers to operate Flet Studio. Each acts as a processor on our behalf and is bound by appropriate contractual obligations:

| Service provider | Purpose                                               | Data shared                                                              |
|------------------|-------------------------------------------------------|--------------------------------------------------------------------------|
|                  | OAuth identity provider used to authenticate accounts | Authentication exchange; we receive profile fields listed in Section 2.1 |

| Service provider | Purpose                                                                                                                       | Data shared                                                                        |
|------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|                  | Hosting for our application servers, our managed PostgreSQL database, and the object storage that holds user-uploaded content | All personal information stored by the Service is hosted on Railway infrastructure |
|                  | Server-side observability and tracing                                                                                         | Request traces, query traces, and error spans; PII-scrubbed before transmission    |

GitHubRailwayPydantic Logfire

## 5.2 Legal and safety disclosures

We may disclose personal information when we believe in good faith that disclosure is required by law, by a valid legal process (such as a subpoena or court order), or is necessary to protect the rights, property, or safety of Appveyor Systems Inc., our users, or the public.

## 5.3 Business transfers

If Appveyor Systems Inc. is involved in a merger, acquisition, reorganization, or sale of assets, personal information may be transferred as part of that transaction. We will notify you of any change in ownership or material change in the use of your personal information.

## 6. Cookies and Similar Technologies

Flet Studio uses only what is strictly necessary to keep you signed in. Specifically:

- A single authentication cookie named `flet_studio_auth`, transmitted only over HTTPS in production environments
- A JSON Web Token stored in your browser's `localStorage` (via the Flet framework's shared-preferences mechanism), used to maintain your session

We do not set any advertising cookies, analytics cookies, social-media cookies, or other tracking technologies. Because we use only strictly-necessary technologies, we do not display a cookie consent banner.

## 7. Data Retention

- Account data (profile fields, account/team records) is retained for as long as your account is active.
- User content (apps, files, versions) is retained for as long as your account is active or until you delete it.
- Sign-in metadata (last sign-in timestamp, last sign-in IP) is overwritten on each new sign-in; older values are not kept.
- Usage counters are kept per day and are not retained beyond what is needed for quota enforcement and operational analysis.
- Server-side telemetry (Logfire traces and logs) is retained for the period configured with our observability provider, which is short by default.
- Account deletion: when you request deletion of your account, we delete or anonymize your account profile, content, and associated records within 30 days, except where we are required to retain certain information to comply with legal obligations or to resolve disputes.

To request deletion, email [email protected] from the email address associated with your account.

## 8. Data Security

We protect personal information using a combination of technical and organizational measures:

- All traffic between your browser and Flet Studio is encrypted in transit using TLS.
- Personal information is stored on infrastructure operated by Railway, which provides encryption at rest for managed databases and object storage.
- Access to production systems is limited to personnel who require it for their role, authenticated using strong credentials.
- Authentication tokens carry a version claim that allows us to revoke all of your active sessions immediately when needed.

No method of transmission or storage is perfectly secure, but we work to maintain protections appropriate to the sensitivity of the information we handle.

## 9. International Data Transfers

Appveyor Systems Inc. and our infrastructure providers may store and process personal information in jurisdictions other than the one in which you reside. Where personal information of users located in the European Economic Area, the United Kingdom, or Switzerland is transferred to a country that has not been recognized as providing an adequate level of data protection, we rely on appropriate safeguards, including the European Commission's Standard Contractual Clauses or equivalent mechanisms.

## 10. Your Rights (EEA, UK, and Swiss Residents)

If you are located in the European Economic Area, the United Kingdom, or Switzerland, you have the following rights with respect to your personal information under applicable data-protection law:

- Right of access — to obtain confirmation as to whether we process personal information about you, and to receive a copy of that information
- Right to rectification — to have inaccurate personal information corrected and incomplete information completed
- Right to erasure ("right to be forgotten") — to have your personal information deleted in the circumstances described in the GDPR
- Right to restriction of processing — to have processing limited in certain circumstances
- Right to data portability — to receive your personal information in a structured, commonly-used, machine-readable format and to transmit it to another controller
- Right to object — to object to processing carried out on the basis of our legitimate interests
- Right to lodge a complaint — with the data-protection supervisory authority in the country where you live or work, or where the alleged infringement took place

To exercise any of these rights, email [email protected] from the email address associated with your account. We will respond to your request within the time limits prescribed by applicable law (generally one month). We will not discriminate against you for exercising any of your rights.

## 11. Children's Privacy

Flet Studio is not directed to children under the age of 16, and we do not knowingly collect personal information from children under 16. If we become aware that we have collected personal information from a child under 16 without verified parental consent, we will take steps to delete that information. If you believe a child has provided us with personal information, please contact us at [email protected].

## 12. Changes to This Policy

We may update this Privacy Policy from time to time to reflect changes to the Service or to our legal obligations. When we make material changes, we will update the "Effective date" at the top of this policy and, where appropriate, notify you through the Service or by email before the changes take effect. Your continued use of Flet Studio after a change takes effect constitutes acceptance of the revised policy.

## 13. Contact Us

If you have questions about this Privacy Policy or about how Flet Studio handles your personal information, please contact us:

Appveyor Systems Inc. Email: [email protected]



# Flet Studio Terms of Service

Effective date: May 26, 2026

These Terms of Service ("Terms") form a binding agreement between you and Appveyor Systems Inc. ("AppVeyor", "we", "us", or "our") and govern your access to and use of the Flet Studio service ("Flet Studio" or the "Service"). Flet Studio is a hosted browser-based development environment and runtime for building applications with the Flet framework.

Please read these Terms carefully. By accessing or using Flet Studio, you agree to be bound by these Terms and by our Privacy Policy. If you do not agree, you must not use the Service.

## 1. Acceptance of Terms

By creating an account, signing in, or otherwise using Flet Studio, you confirm that you have read, understood, and agree to be bound by these Terms. If you are accepting these Terms on behalf of a company, organization, or other legal entity, you represent and warrant that you have the authority to bind that entity to these Terms, in which case "you" refers to that entity.

We may update these Terms from time to time as described in Section 13.4. Your continued use of the Service after an update takes effect constitutes acceptance of the updated Terms.

## 2. The Service

Flet Studio provides a hosted environment that allows you to create, edit, store, run, and share applications written in Python using the Flet framework. The Service includes account management, project storage, code editing, application execution, and related features.

We may add, modify, or discontinue features at any time. Where a change materially and adversely affects the way you use the Service, we will use commercially reasonable efforts to provide advance notice through the Service or by email.

## 3. Accounts and Eligibility

### 3.1 Eligibility

You must be at least 16 years old to use Flet Studio. If you are using the Service on behalf of an entity, you must have authority to act on its behalf. The Service is not directed to children under 16, and we do not knowingly permit them to create accounts.

### 3.2 Account creation

You can create an account by signing in with a supported identity provider (currently GitHub) or, where available, by creating a local account with an email and password. You must provide accurate and complete information, and you must keep it accurate over time.

### 3.3 Account security

You are responsible for safeguarding your credentials and for all activity that occurs under your account. You must notify us promptly at [email protected] if you suspect unauthorized access to your account.

### 3.4 One account per person

You may not create multiple personal accounts to circumvent quotas, restrictions, or suspensions, except where we expressly permit it (for example, separate personal and team accounts).

### 3.5 Privacy

Information you provide when creating and using an account is processed in accordance with our Privacy Policy.

## 4. Acceptable Use

You agree not to use Flet Studio to do any of the following, or to permit or encourage others to do so:

- Upload, create, transmit, or distribute content that is unlawful, infringing, defamatory, harassing, threatening, hateful, obscene, sexually explicit, or that promotes violence
- Send spam, conduct phishing, distribute malware, or engage in fraud

- Violate the intellectual-property, privacy, publicity, or other rights of any person or entity
- Impersonate any person or entity or misrepresent your affiliation with one
- Probe, scan, or test the vulnerability of the Service or circumvent any authentication, access control, or rate-limiting mechanism
- Reverse engineer, decompile, or disassemble any part of the Service, except to the extent applicable law expressly permits
- Use automated means (bots, crawlers, scrapers) to access the Service in ways that are not expressly authorized, or to extract data at a scale or rate that interferes with normal operation
- Exceed your assigned quotas, evade rate limits, or otherwise consume disproportionate resources
- Run cryptocurrency miners, denial-of-service tooling, botnet clients, or any workload whose purpose is to abuse compute, storage, or network resources
- Use the Service in a manner that violates any applicable law, regulation, or third-party agreement, or that violates export-control or sanctions laws

We may, but are not obligated to, investigate suspected violations and may take any action we consider appropriate, including limiting features, removing content, or suspending or terminating your account.

## 5. User Content and License Grant

### 5.1 Ownership

"User Content" means the source code, files, project metadata, and any other materials you create, upload, or store using Flet Studio. You retain all ownership rights in your User Content. Nothing in these Terms transfers ownership of your User Content to us.

### 5.2 License to AppVeyor

You grant AppVeyor a worldwide, non-exclusive, royalty-free, fully paid license to host, store, transmit, display, copy, and execute your User Content solely as needed to operate and provide the Service to you, including making backups, performing maintenance, and enabling features you have requested. The license is limited to these operational purposes; it is not a license to publish, monetize, or repurpose your User Content.

This license terminates when you delete the relevant User Content or close your account, except that we may retain copies in routine backups for a limited period and as required by law, as described in our Privacy Policy.

### 5.3 No publication or sharing without your instruction

AppVeyor will not publish, share, or redistribute your User Content, or make it publicly available, except as you explicitly direct us to do (for example, by using a feature that publishes content at your request). This commitment applies regardless of any other clause in these Terms.

### 5.4 Your responsibility

You are solely responsible for your User Content, including its legality and your right to use it. You represent and warrant that you have all rights necessary to grant the license in Section 5.2, and that your User Content does not violate the Acceptable Use restrictions in Section 4.

### 5.5 Copyright complaints

If you believe content on Flet Studio infringes your copyright, send a notice to [email protected] that includes:

- A physical or electronic signature of the rights holder or authorized representative
- Identification of the work claimed to have been infringed
- Identification of the allegedly infringing material with enough detail to locate it
- Your contact information (address, telephone number, email)
- A statement that you have a good-faith belief that the use is not authorized
- A statement, under penalty of perjury, that the information in the notice is accurate and that you are authorized to act for the rights holder

We will respond to valid notices in a manner consistent with applicable copyright law and may remove allegedly infringing material and terminate the accounts of repeat infringers.

## 6. Plans, Pricing, and the Free Tier

Today, Flet Studio is offered as a Free plan at no charge. The Free plan includes limits on, but not limited to, the number of applications, private applications, storage, files per application, project versions, file sizes, archive sizes, and daily operation counts (such as gallery clones and archive uploads). The current Free-plan limits are documented within the Service and may be reviewed at any time.

We reserve the right to change Free-plan limits with reasonable advance notice. Where we materially reduce a limit, we will provide notice through the Service or by email.

We may introduce paid plans in the future. Paid plans, when offered, will be governed by additional terms presented at the time of purchase, including pricing, billing, refunds, and applicable taxes. Nothing in these Terms creates an obligation on AppVeyor to offer a paid plan or any particular feature.

## 7. Service Availability and Modifications

We strive to keep Flet Studio available and performant, but the Service is provided without any uptime or availability guarantee, particularly for users on the Free plan. We may perform scheduled or emergency maintenance, suspend access to all or part of the Service, or modify, deprecate, or discontinue features at any time.

Where a change materially and adversely affects how you use the Service, we will use commercially reasonable efforts to provide advance notice through the Service or by email.

## 8. Suspension and Termination

### 8.1 Termination by you

You may stop using Flet Studio at any time. You may delete your account by emailing [email protected] from the email address associated with your account.

### 8.2 Suspension or termination by AppVeyor

We may suspend or terminate your access to all or part of the Service:

- Immediately, if we reasonably believe you have breached these Terms, your account presents a security risk, you are abusing the Service, or we are required to do so by law
- With reasonable notice, for any other reason

### 8.3 Effect of termination

Upon termination, your right to use the Service ends immediately. We will handle your account data and User Content in accordance with our Privacy Policy, which provides that account data and User Content are deleted or anonymized within 30 days of a deletion request, subject to backups and legal retention requirements.

### 8.4 Survival

The following sections survive termination of these Terms: Section 5.1 (Ownership), Section 5.4 (Your responsibility), Section 9 (Disclaimers), Section 10 (Limitation of Liability), Section 11 (Indemnification), Section 12 (Governing Law and Dispute Resolution), and Section 13 (General Provisions), together with any other provision that by its nature should survive.

## 9. Disclaimers

THE SERVICE IS PROVIDED "AS IS" AND "AS AVAILABLE" WITHOUT WARRANTIES OF ANY KIND, WHETHER EXPRESS, IMPLIED, OR STATUTORY. TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, APPVEYOR DISCLAIMS ALL WARRANTIES, INCLUDING WITHOUT LIMITATION ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NON-INFRINGEMENT, ACCURACY, QUIET ENJOYMENT, AND ANY WARRANTIES ARISING FROM COURSE OF DEALING OR USAGE OF TRADE.

WE DO NOT WARRANT THAT THE SERVICE WILL BE UNINTERRUPTED, ERROR-FREE, SECURE, OR FREE OF HARMFUL COMPONENTS, OR THAT ANY CONTENT (INCLUDING YOUR USER CONTENT) WILL BE PRESERVED WITHOUT LOSS. YOU ARE SOLELY RESPONSIBLE FOR THE CODE AND APPLICATIONS YOU CREATE WITH FLET STUDIO AND FOR THE CONSEQUENCES OF RUNNING THEM.

Some jurisdictions do not allow the exclusion of certain warranties, so some of the above exclusions may not apply to you.

## 10. Limitation of Liability

TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW:

- No indirect damages. APPVEYOR WILL NOT BE LIABLE FOR ANY INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL, EXEMPLARY, OR PUNITIVE DAMAGES, OR FOR ANY LOSS OF PROFITS, REVENUE, GOODWILL, OR DATA, ARISING OUT OF OR IN CONNECTION WITH THESE TERMS OR THE SERVICE, EVEN IF APPVEYOR HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.
- Aggregate cap. APPVEYOR'S TOTAL AGGREGATE LIABILITY ARISING OUT OF OR IN CONNECTION WITH THESE TERMS OR THE SERVICE WILL NOT EXCEED THE GREATER OF (a) THE AMOUNTS YOU PAID TO APPVEYOR FOR THE SERVICE IN THE TWELVE (12) MONTHS PRECEDING THE EVENT GIVING RISE TO THE CLAIM, OR (b) ONE HUNDRED CANADIAN DOLLARS (CA\$100).
- Data-loss exclusion. APPVEYOR IS NOT LIABLE FOR ANY LOSS, ALTERATION, OR DESTRUCTION OF YOUR USER CONTENT. YOU ARE RESPONSIBLE FOR MAINTAINING YOUR OWN INDEPENDENT BACKUPS.

These limitations apply regardless of the legal theory on which a claim is based (contract, tort, statute, or otherwise) and apply even if a remedy fails of its essential purpose. Some jurisdictions do not allow the exclusion or limitation of certain damages; in those jurisdictions, the above limitations apply to the maximum extent permitted by law.

## 11. Indemnification

You agree to defend, indemnify, and hold harmless Appveyor Systems Inc., its affiliates, and their respective officers, directors, employees, and agents from and against any and all claims, damages, losses, liabilities, costs, and expenses (including reasonable legal fees) arising out of or related to:

- Your User Content
- Your use of or access to the Service
- Your breach of these Terms
- Your violation of any applicable law or any rights of a third party

We may, at our option, assume the exclusive defense and control of any matter subject to indemnification by you, in which case you agree to cooperate with our defense.

## 12. Governing Law and Dispute Resolution

These Terms, and any dispute or claim arising out of or in connection with them or their subject matter (including non-contractual disputes or claims), are governed by the laws of the Province of British Columbia and the federal laws of Canada applicable therein, without regard to conflict-of-laws principles.

You and AppVeyor each consent to the exclusive jurisdiction of the courts located in the Province of British Columbia, Canada for the resolution of any dispute arising out of or relating to these Terms or the Service, and you waive any objection based on inconvenient forum.

These Terms do not require mandatory arbitration, and nothing in these Terms is intended to waive your right to participate in a class action where applicable law would otherwise permit one.

## 13. General Provisions

### 13.1 Entire agreement

These Terms, together with the Privacy Policy and any additional terms you accept at the time of purchasing a paid feature, constitute the entire agreement between you and AppVeyor concerning the Service and supersede all prior or contemporaneous agreements on the same subject.

### 13.2 Severability

If any provision of these Terms is held to be invalid or unenforceable, that provision will be enforced to the maximum extent permissible, and the remaining provisions will remain in full force and effect.

### 13.3 No waiver

Our failure to enforce any right or provision of these Terms will not be deemed a waiver of that right or provision.

### 13.4 Changes to these Terms

We may modify these Terms from time to time. When we make material changes, we will update the "Effective date" above and, where appropriate, notify you through the Service or by email before the changes take effect. Your continued use of

the Service after the new Terms take effect constitutes acceptance.

### **13.5 Assignment**

You may not assign or transfer these Terms or your account without our prior written consent. We may assign these Terms to an affiliate or in connection with a merger, acquisition, reorganization, or sale of assets, on notice to you.

### **13.6 Relationship of the parties**

Nothing in these Terms creates a partnership, joint venture, agency, employment, or fiduciary relationship between you and AppVeyor.

### **13.7 Force majeure**

Neither party will be liable for failure or delay in performance caused by events beyond its reasonable control, including acts of God, war, terrorism, civil unrest, labor disputes, internet or telecommunications outages, governmental actions, or epidemics.

### **13.8 Notices**

We may give notices to you through the Service, by posting on our website, or by email to the address associated with your account. You must give notices to us in writing at [email protected].

### **13.9 Contact**

Appveyor Systems Inc. Email: [email protected]

## What's new in Flet Studio

New features, improvements, and bug fixes in Flet Studio, listed in reverse-chronological order. Flet Studio ships independently of the Flet SDK — see Flet release notes for SDK changes.

### May 29, 2026

- New sign in options: Google and Microsoft.

### May 26, 2026

- The initial public release of Flet Studio.

# Calculator Tutorial

In this tutorial you will learn, step-by-step, how to create a Calculator app in Python using Flet framework and publish it as a desktop, mobile or web app. The app is a simple console program, yet it is a multi-platform application with similar to iPhone calculator app UI:

Image: calc-app - <https://flet.dev/docs/examples/tutorials/calculator/media/app.png>

You can find a live demo [here](#).

In this tutorial, we will cover all of the basic concepts for creating a Flet app: building a page layout, adding controls, making reusable UI components, handling events, and publishing options.

## Getting started with Flet

To create a multi-platform app in Python with Flet, you don't need to know HTML, CSS or JavaScript, but you do need a basic knowledge of Python and object-oriented programming.

Before you can create your first Flet app, you need to setup your development environment, which requires Python 3.10 or above and flet package.

Once you have Flet installed, let's create a simple hello-world app.

Create hello.py with the following contents:

hello.py

```
import flet as ft

def main(page: ft.Page):
 page.add(ft.Text(value="Hello, world!"))

ft.run(main)
```

Run this app and you will see a new window with a greeting:

Image: hello-world - <https://flet.dev/docs/examples/tutorials/todo/media/hello-world.png>

## Adding page controls

Now you are ready to create a calculator app.

To start, you'll need a Text control for showing the result of calculation, and a few Buttons with all the numbers and actions on them.

Create calc.py with the following contents:

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Calc App"
 result = ft.Text(value="0")

 page.add(
 result,
 ft.Button("AC"),
 ft.Button("+/-"),
 ft.Button("%"),
 ft.Button("/"),
 ft.Button("7"),
 ft.Button("8"),
 ft.Button("9"),
 ft.Button("*"),
 ft.Button("4"),
 ft.Button("5"),
 ft.Button("6"),
 ft.Button("-"),
 ft.Button("1"),
 ft.Button("2"),
 ft.Button("3"),
 ft.Button("+"),
 ft.Button("0"),
 ft.Button("."),
 ft.Button("="),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

Run the app and you should see a page like this:

Image: calc1 - <https://flet.dev/docs/test-images/tutorials/golden/macros/calculator/calc1.png>

## Building page layout

Now let's arrange the text and buttons in 6 horizontal Rows.

Replace calc.py contents with the following:

play\_arrow Try Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Calc App"
 result = ft.Text(value="0")

 page.add(
 ft.Row(controls=[result]),
 ft.Row(
 controls=[
 ft.Button("AC"),
 ft.Button("+/-"),
 ft.Button("%"),
 ft.Button("/"),
]
),
 ft.Row(
 controls=[
 ft.Button("7"),
 ft.Button("8"),
 ft.Button("9"),
 ft.Button("*"),
]
),
 ft.Row(
 controls=[
 ft.Button("4"),
 ft.Button("5"),
 ft.Button("6"),
 ft.Button("-"),
]
),
 ft.Row(
 controls=[
 ft.Button("1"),
 ft.Button("2"),
 ft.Button("3"),
 ft.Button("+"),
]
),
 ft.Row(
 controls=[
 ft.Button("0"),
 ft.Button("."),
 ft.Button("="),
]
),
)

if __name__ == "__main__":
 ft.run(main)
```

Run the app and you should see a page like this:

Image: calc2 - <https://flet.dev/docs/test-images/tutorials/golden/macros/calculator/calc2.png>

## Using Container for decoration

To add a black background with rounded border around the calculator, we will be using Container control. Container may decorate only one control, so we will need to wrap all the 6 rows into a single vertical Column that will be used as the container's content:

Image: container-layout - <https://flet.dev/docs/examples/tutorials/calculator/media/container-layout.svg>

Here is the code for adding the container to the page:

```
page.add(
 ft.Container(
 width=350,
```



```

 bgcolor=ft.Colors.BLACK,
 border_radius=ft.BorderRadius.all(20),
 padding=20,
 content=ft.Column(
 controls= [],
)
)
)

```

## Styled Controls

To complete the UI portion of the program, we need to update style for result text and buttons to look similar to iPhone calculator app.

For the result text, let's specify its color and size properties:

```
result = ft.Text(value="0", color=ft.Colors.WHITE, size=20)
```

For the buttons, if we look again at the UI we are aiming to achieve, there are 3 types of buttons:

**Digit Buttons.** They have dark grey background color and white text, size is the same for all except 0 button which is twice as large.

**Action Buttons.** They have orange background color and white text, size is the same for all.

**Extra action buttons.** They have light grey background color and dark text, size is the same for all.

The buttons will be used multiple time in the program, so we will be creating custom Styled Controls to reuse the code.

Since all those types should inherit from Button class and have common text and expand properties, let's create a parent CalcButton class:

```
@ft.control
class CalcButton(ft.Button):
 expand: int = 1

```

Now let's create child classes for all three types of buttons:

```
@ft.control
class DigitButton(CalcButton):
 bgcolor: ft.Colors = ft.Colors.WHITE_24
 color: ft.Colors = ft.Colors.WHITE

@ft.control
class ActionButton(CalcButton):
 bgcolor: ft.Colors = ft.Colors.ORANGE
 color: ft.Colors = ft.Colors.WHITE

@ft.control
class ExtraActionButton(CalcButton):
 bgcolor: ft.Colors = ft.Colors.BLUE_GREY_100
 color: ft.Colors = ft.Colors.BLACK

```

We will be using these new classes now to create rows of buttons in the Container:

```
content = ft.Column(
 controls=[
 ft.Row(
 controls=[result],
 alignment=ft.MainAxisAlignment.END,
),
 ft.Row(
 controls=[
 ExtraActionButton(content="AC"),
 ExtraActionButton(content="+/-"),
 ExtraActionButton(content="%"),
 ActionButton(content="/"),
]
),
 ft.Row(
 controls=[
 DigitButton(content="7"),
 DigitButton(content="8"),
 DigitButton(content="9"),
 ActionButton(content="*"),
]
),
 ft.Row(
 controls=[
 DigitButton(content="4"),
 DigitButton(content="5"),
 DigitButton(content="6"),
 ActionButton(content="-"),
]
),
 ft.Row(

```

```

 controls=[
 DigitButton(content="1"),
 DigitButton(content="2"),
 DigitButton(content="3"),
 ActionButton(content="+"),
]
),
 ft.Row(
 controls=[
 DigitButton(content="0", expand=2),
 DigitButton(content="."),
 ActionButton(content="="),
]
),
]
)

```

Full code

open: true

play\_arrow Try Online

```

from dataclasses import field

import flet as ft

def main(page: ft.Page):
 page.title = "Calc App"
 result = ft.Text(value="0", color=ft.Colors.WHITE, size=20)

 @ft.control
 class CalcButton(ft.Button):
 expand: int = field(default_factory=lambda: 1)

 @ft.control
 class DigitButton(CalcButton):
 bgcolor: ft.Colors = ft.Colors.WHITE_24
 color: ft.Colors = ft.Colors.WHITE

 @ft.control
 class ActionButton(CalcButton):
 bgcolor: ft.Colors = ft.Colors.ORANGE
 color: ft.Colors = ft.Colors.WHITE

 @ft.control
 class ExtraActionButton(CalcButton):
 bgcolor: ft.Colors = ft.Colors.BLUE_GREY_100
 color: ft.Colors = ft.Colors.BLACK

 page.add(
 ft.Container(
 width=350,
 bgcolor=ft.Colors.BLACK,
 border_radius=ft.BorderRadius.all(20),
 padding=20,
 content=ft.Column(
 controls=[
 ft.Row(controls=[result], alignment=ft.MainAxisAlignment.END),
 ft.Row(
 controls=[
 ExtraActionButton(content="AC"),
 ExtraActionButton(content="+/-"),
 ExtraActionButton(content="%"),
 ActionButton(content="/"),
]
),
 ft.Row(
 controls=[
 DigitButton(content="7"),
 DigitButton(content="8"),
 DigitButton(content="9"),
 ActionButton(content="*"),
]
),
 ft.Row(
 controls=[
 DigitButton(content="4"),
 DigitButton(content="5"),
 DigitButton(content="6"),
 ActionButton(content="-"),
]
),
 ft.Row(
 controls=[
 DigitButton(content="1"),

```

```

 DigitButton(content="2"),
 DigitButton(content="3"),
 ActionButton(content="+"),
],
),
ft.Row(
 controls=[
 DigitButton(content="0", expand=2),
 DigitButton(content="."),
 ActionButton(content="="),
],
),
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: calc3 - <https://flet.dev/docs/test-images/tutorials/golden/macros/calculator/calc3.png>

Just what we wanted!

## Reusable UI components

While you can continue writing your app in the main function, the best practice would be to create a reusable UI component.

Imagine you are working on an app header, a side menu, or UI that will be a part of a larger project (for example, at Flet we will be using this Calculator app in a bigger "Gallery" app that will show all the examples for Flet framework).

Even if you can't think of such uses right now, we still recommend creating all your Flet apps with composability and reusability in mind.

To make a reusable Calc app component, we are going to encapsulate its state and presentation logic in a separate `CalculatorApp` class.

Copy the entire code for this step from here.

Try something

Try adding two `CalculatorApp` components to the page:

```

create application instance
calc1 = CalculatorApp()
calc2 = CalculatorApp()

add application's root control to the page
page.add(calc1, calc2)

```

## Handling events

Now let's make the calculator do its job. We will be using the same event handler for all the buttons and use content property to differentiate between the actions depending on the button clicked.

We will define `button_click` method in `CalculatorClass` and pass it to each button. Below is `on_click` event handler that will reset the Text value when "AC" button is clicked:

```

def button_clicked(self, e):
 data = e.control.content
 print(f"Button clicked with data = {data}")
 if data == "AC":
 self.result.value = "0"

```

With similar approach, `button_click` method will handle different calculator actions depending on content property for each button. Copy the entire code for this step from here.

Run the app and see it in the action:

Image: calc-app2 - <https://flet.dev/docs/examples/tutorials/calculator/media/app.gif>

## Publishing your app

Congratulations! You have created your Calculator app with Flet, and it looks awesome! Now it's time to share your app with the world!

Flet Python app and all its dependencies can be packaged into a standalone executable a package for distribution using `flet build` command.

Follow these instructions to package your Calculator app into a desktop executable, mobile app bundle or web app.

## Summary

In this tutorial you have learned how to:

- Create a simple Flet app;
- Work with Reusable UI components;
- Design UI layout using Column, Row and Container controls;
- Handle events;
- Publish your Flet app to multiple platforms;

For further reading you can explore controls and examples.

# Chat Tutorial

In this tutorial we are going to create a trivial in-memory Chat app that will help you understand Flet framework basics. This app could be a good starting point to creating your own more complex and useful projects.

In this tutorial you will learn how to:

- Create your first Flet app
- Add page controls and handle events
- Broadcast messages using built-in PubSub library
- Use AlertDialog control for accepting user name
- Enhance user interface with reusable controls
- Deploy the app as a web app

The complete application will look like this:

Image: chat - <https://flet.dev/docs/examples/tutorials/chat/media/chat.gif>

You can try the live demo [here](#).

## Getting started with Flet

It's a tradition to start with "Hello, world!" app!

To create a multi-platform app in Python with Flet, you don't need to know HTML, CSS or JavaScript, but you do need a basic knowledge of Python and object-oriented programming.

Before you can create your first Flet app, you need to setup your development environment.

Once you have Flet installed, let's create a hello-world app.

Create `hello.py` with the following contents:

`hello.py`

```
import flet as ft

def main(page: ft.Page):
 page.add(ft.Text(value="Hello, world!"))

ft.run(main)
```

Run this app you will see a new window with a greeting:

Image: hello-world - <https://flet.dev/docs/examples/tutorials/todo/media/hello-world.png>

## Adding page controls and handling events

To start, we want to be able to take user input (chat message) and show messages history on the screen. The layout for this step could look like this:

Image: chat-layout-1 - <https://flet.dev/docs/examples/tutorials/chat/media/layout-1.svg>

To implement this layout, we will be using these Flet controls:

- Column - a container to display chat messages (Text controls) vertically.
- Text - chat message displayed in the chat Column.
- Row - a container to display TextField and Button horizontally.
- TextField - input control used for taking new message input from the user.
- Button - "Send" button that will add new message to the chat Column.

Create `chat.py` with the following contents:

```
import flet as ft

def main(page: ft.Page):
 chat = ft.Column()
 new_message = ft.TextField()
```

```
def send_click(e):
 chat.controls.append(ft.Text(new_message.value))
 new_message.value = ""

page.add(
 chat,
 ft.Row(controls=[new_message, ft.Button("Send", on_click=send_click)]),
)

ft.run(main)
```

When user clicks on the "Send" button, it triggers on\_click event which calls send\_click method. send\_click then adds new Text control to the list of Column.controls and clears new\_message text field value.

#### Note

After any properties of a control are updated in event handler, there is no need to call an update() method as the update will be done automatically, but it should be called in case properties are updated outside of the event.

Chat app now looks like this:

Image: chat-1 - <https://flet.dev/docs/examples/tutorials/chat/media/chat-1.png>

## Broadcasting chat messages

In the previous step we have created a simple app that takes input from the user and displays chat messages on the screen.

If you open this app in two web browser tabs, it will create two app sessions. Each session will have its own list of messages.

#### Tip

To open your app in two web browser tabs locally, run the following command:

```
flet run --web <path_to_your_app>
```

Once it is opened, copy the url and paste it into a new tab.

To build a realtime chat app, you need to somehow pass the messages between chat app sessions. When a user sends a message, it should be broadcasted to all other app sessions and displayed on their pages.

Flet provides a simple built-in PubSub mechanism for asynchronous communication between page sessions.

First, we need subscribe the user to receive broadcast messages:

```
page.pubsub.subscribe(on_message)
```

pubsub.subscribe() method will add current app session to the list of subscribers. It accepts handler as an argument, that will later be called at the moment a publisher calls pubsub.send\_all() method.

In the handler we will be adding new message (Text) to the list of chat controls:

```
def on_message(message: Message):
 chat.controls.append(ft.Text(f"{message.user}: {message.text}"))
 page.update()
```

Finally, you need to call pubsub.send\_all() method when the user clicks on "Send" button:

```
def send_click(e):
 page.pubsub.send_all(Message(user=page.session.index, text=new_message.value))
 new_message.value = ""
 page.update()

page.add(chat, ft.Row([new_message, ft.Button("Send", on_click=send_click)]))
```

pubsub.send\_all() will call the on\_message() and pass on the Message object down to it.

Here is the full code for this step:

```
from dataclasses import dataclass

import flet as ft
```

```
@dataclass
class Message:
 user: str
 text: str
```

```
def main(page: ft.Page):
 chat = ft.Column()
```

```

new_message = ft.TextField()

def on_message(message: Message):
 chat.controls.append(ft.Text(f"{message.user}: {message.text}"))
 page.update()

page.pubsub.subscribe(on_message)

def send_click(e):
 page.pubsub.send_all(Message(user=page.session.id, text=new_message.value))
 new_message.value = ""

page.add(chat, ft.Row([new_message, ft.Button("Send", on_click=send_click)]))

```

ft.run(main)

Image: chat-2 - <https://flet.dev/docs/examples/tutorials/chat/media/chat-2.gif>

## User name dialog

Chat app that you have created in the previous step has basic functionality needed to exchange messages between user sessions. It is not very user-friendly though, since it shows session.id that sent a message, which doesn't tell much about who you are communicating with.

Let's improve our app to show user name instead of session.id for each message. To capture user name we will be using AlertDialog control. Let's add it to the page:

```

user_name = ft.TextField(label="Enter your name")

page.show_dialog(
 ft.AlertDialog(
 open=True,
 modal=True,
 title=ft.Text("Welcome!"),
 content=ft.Column([user_name], tight=True),
 actions=[ft.Button(content="Join chat", on_click=join_click)],
 actions_alignment=ft.MainAxisAlignment.END,
)
)

```

### Note

A dialog will be opened on the start of the program since we have set its open property to True.

Image: username-dialog - <https://flet.dev/docs/examples/tutorials/chat/media/username-dialog.png>

When the user clicks on "Join chat" button, it will call join\_click method that should send a message to all subscribers, informing them that the user has joined the chat. This message should look different from the regular chat message, for example, like this:

Image: chat-4 - <https://flet.dev/docs/examples/tutorials/chat/media/chat-4.png>

Let's add message\_type property to the Message class to differentiate between login and chat messages:

```

@dataclass
class Message:
 user: str
 text: str
 message_type: str

```

We will be checking message\_type in on\_message method:

```

def on_message(message: Message):
 if message.message_type == "chat_message":
 chat.controls.append(ft.Text(f"{message.user}: {message.text}"))
 elif message.message_type == "login_message":
 chat.controls.append(
 ft.Text(message.text, italic=True, color=ft.Colors.BLACK_45, size=12)
)
 page.update()

```

Messages of "login\_message" and "chat\_message" types will now be sent on two events: when user joins the chat and when user sends a message.

Let's create join\_click method:

```

def join_click(e):
 if not user_name.value:
 user_name.error = "Name cannot be blank!"
 user_name.update()
 else:
 page.session.store.set("user_name", user_name.value)
 page.pop_dialog()
 page.pubsub.send_all(Message(user=user_name.value, text=f"{user_name.value} has joined the chat.", message_type="login_message"))

```

We used page session storage to store user\_name for its future use in send\_click method to send chat messages.

## Note

User name dialog will close as soon as we call page.pop\_dialog() method.

Finally, let's update send\_click method to use user\_name that we previously saved using page.session.store:

```
def send_click(e):
 page.pubsub.send_all(Message(user=page.session.store.get('user_name'), text=new_message.value, message_type="chat_message"))
 new_message.value = ""
```

## Code

```
from dataclasses import dataclass

import flet as ft

@dataclass
class Message:
 user: str
 text: str
 message_type: str

def main(page: ft.Page):
 chat = ft.Column()
 new_message = ft.TextField()

 def on_message(message: Message):
 if message.message_type == "chat_message":
 chat.controls.append(ft.Text(f"{message.user}: {message.text}"))
 elif message.message_type == "login_message":
 chat.controls.append(
 ft.Text(message.text, italic=True, color=ft.Colors.BLACK_45, size=12)
)
 page.update()

 page.pubsub.subscribe(on_message)

 def send_click(e):
 page.pubsub.send_all(
 Message(
 user=page.session.store.get("user_name"),
 text=new_message.value,
 message_type="chat_message",
)
)
 new_message.value = ""

 user_name = ft.TextField(label="Enter your name")

 def join_click(e):
 if not user_name.value:
 user_name.error_text = "Name cannot be blank!"
 else:
 page.session.store.set("user_name", user_name.value)
 # page.dialog.open = False
 page.pop_dialog()
 page.pubsub.send_all(
 Message(
 user=user_name.value,
 text=f"{user_name.value} has joined the chat.",
 message_type="login_message",
)
)

 page.show_dialog(
 ft.AlertDialog(
 open=True,
 modal=True,
 title=ft.Text("Welcome!"),
 content=ft.Column([user_name], tight=True),
 actions=[ft.Button(content="Join chat", on_click=join_click)],
 actions_alignment=ft.MainAxisAlignment.END,
)
)

 page.add(chat, ft.Row([new_message, ft.Button("Send", on_click=send_click)]))

ft.run(main)
```

Image: chat-3 - <https://flet.dev/docs/examples/tutorials/chat/media/chat-3.gif>



## Enhancing user interface

Chat app that you have created in the previous step already serves its purpose of exchanging messages between users with basic login functionality.

Before moving on to deploying your app, we suggest adding some extra features to it that will improve user experience and make the app look more professional.

### Reusable user controls

You may want to show messages in a different format, like this:

Image: chat-layout-chatmessage - <https://flet.dev/docs/examples/tutorials/chat/media/layout-chatmessage.svg>

Chat message will now be a Row containing CircleAvatar with username initials and Column that contains two Text controls: user name and message text.

We will need to show quite a few chat messages in the chat app, so it makes sense to create your own reusable control. Lets create a new ChatMessage dataclass that will inherit from Row.

When creating an instance of ChatMessage class, we will pass a Message object as an argument and then ChatMessage will display itself based on message.user\_name and message.text:

```
@ft.control
class ChatMessage(ft.Row):
 def __init__(self, message: Message):
 super().__init__()
 self.message = message
 self.vertical_alignment = ft.CrossAxisAlignment.START
 self.controls = [
 ft.CircleAvatar(
 content=ft.Text(self.get_initials(self.message.user_name)),
 color=ft.Colors.WHITE,
 bgcolor=self.get_avatar_color(self.message.user_name),
),
 ft.Column(
 tight=True,
 spacing=5,
 controls=[
 ft.Text(self.message.user_name, weight=ft.FontWeight.BOLD),
 ft.Text(self.message.text, selectable=True),
],
),
]

 def get_initials(self, user_name: str):
 if user_name:
 return user_name[:1].capitalize()
 else:
 return "Unknown" # or any default value you prefer

 def get_avatar_color(self, user_name: str):
 colors_lookup = [
 ft.Colors.AMBER,
 ft.Colors.BLUE,
 ft.Colors.BROWN,
 ft.Colors.CYAN,
 ft.Colors.GREEN,
 ft.Colors.INDIGO,
 ft.Colors.LIME,
 ft.Colors.ORANGE,
 ft.Colors.PINK,
 ft.Colors.PURPLE,
 ft.Colors.RED,
 ft.Colors.TEAL,
 ft.Colors.YELLOW,
]
 return colors_lookup[hash(user_name) % len(colors_lookup)]
```

ChatMessage control extracts initials and algorithmically derives avatar color from a username. Later, if you decide to improve control layout or its logic, it won't affect the rest of the program - that's the power of encapsulation!

### Laying out controls

Now you can use your brand new ChatMessage to build a better layout for the chat app:

Image: chat-layout-2 - <https://flet.dev/docs/examples/tutorials/chat/media/layout-2.svg>

Instances of ChatMessage will be created instead of plain Text chat messages in on\_message method:

```
def on_message(message: Message):
 if message.message_type == "chat_message":
 m = ChatMessage(message)
```

```

elif message.message_type == "login_message":
 m = ft.Text(message.text, italic=True, color=ft.Colors.BLACK_45, size=12)
 chat.controls.append(m)
 page.update()

```

Other improvements suggested with the new layout are:

- ListView instead of Column for displaying messages, to be able to scroll through the messages later
- Container for displaying border around ListView
- IconButton instead of Button to send messages
- Use of expand property for controls to fill available space

Here is how you can implement this layout:

```

Chat messages
chat = ft.ListView(
 expand=True,
 spacing=10,
 auto_scroll=True,
)

A new message entry form
new_message = ft.TextField(
 hint_text="Write a message...",
 autofocus=True,
 shift_enter=True,
 min_lines=1,
 max_lines=5,
 filled=True,
 expand=True,
 on_submit=send_message_click,
)

Add everything to the page
page.add(
 ft.Container(
 content=chat,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=5,
 padding=10,
 expand=True,
),
 ft.Row(
 [
 new_message,
 ft.IconButton(
 icon=ft.Icons.SEND_ROUNDED,
 tooltip="Send message",
 on_click=send_message_click,
),
],
),
)

```

Full code

open: true

```
from dataclasses import dataclass
```

```
import flet as ft
```

```
@dataclass
```

```
class Message: # noqa: B903
 user_name: str
 text: str
 message_type: str
```

```
@ft.control
```

```
class ChatMessage(ft.Row):
 def __init__(self, message: Message):
 super().__init__()
 self.message = message
 self.vertical_alignment = ft.CrossAxisAlignment.START
 self.controls = [
 ft.CircleAvatar(
 content=ft.Text(self.get_initials(self.message.user_name)),
 color=ft.Colors.WHITE,
 bgcolor=self.get_avatar_color(self.message.user_name),
),
 ft.Column(
 tight=True,

```

```

 spacing=5,
 controls=[
 ft.Text(self.message.user_name, weight=ft.FontWeight.BOLD),
 ft.Text(self.message.text, selectable=True),
],
),
]

def get_initials(self, user_name: str):
 if user_name:
 return user_name[:1].capitalize()
 else:
 return "Unknown" # or any default value you prefer

def get_avatar_color(self, user_name: str):
 colors_lookup = [
 ft.Colors.AMBER,
 ft.Colors.BLUE,
 ft.Colors.BROWN,
 ft.Colors.CYAN,
 ft.Colors.GREEN,
 ft.Colors.INDIGO,
 ft.Colors.LIME,
 ft.Colors.ORANGE,
 ft.Colors.PINK,
 ft.Colors.PURPLE,
 ft.Colors.RED,
 ft.Colors.TEAL,
 ft.Colors.YELLOW,
]
 return colors_lookup[hash(user_name) % len(colors_lookup)]

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.STRETCH
 page.title = "Flet Chat"

 def join_chat_click(e):
 if not join_user_name.value:
 join_user_name.error_text = "Name cannot be blank!"
 join_user_name.update()
 else:
 page.session.store.set("user_name", join_user_name.value)
 welcome_dlg.open = False
 new_message.prefix = ft.Text(f"{join_user_name.value}: ")
 page.pubsub.send_all(
 Message(
 user_name=join_user_name.value,
 text=f"{join_user_name.value} has joined the chat.",
 message_type="login_message",
)
)

 async def send_message_click(e):
 if new_message.value != "":
 page.pubsub.send_all(
 Message(
 page.session.store.get("user_name"),
 new_message.value,
 message_type="chat_message",
)
)
 new_message.value = ""
 await new_message.focus()

 def on_message(message: Message):
 if message.message_type == "chat_message":
 m = ChatMessage(message)
 elif message.message_type == "login_message":
 m = ft.Text(
 message.text, italic=True, color=ft.Colors.ON_SURFACE_VARIANT, size=12
)
 chat.controls.append(m)
 page.update()

 page.pubsub.subscribe(on_message)

 # A dialog asking for a user display name
 join_user_name = ft.TextField(
 label="Enter your name to join the chat",
 autofocus=True,
 on_submit=join_chat_click,
)
 welcome_dlg = ft.AlertDialog(
 open=True,
 modal=True,
 title=ft.Text("Welcome!"),

```

```

 content=ft.Column([join_user_name], width=300, height=70, tight=True),
 actions=[ft.Button(content="Join chat", on_click=join_chat_click)],
 actions_alignment=ft.MainAxisAlignment.END,
)

 page.overlay.append(welcome_dlg)

 # Chat messages
 chat = ft.ListView(
 expand=True,
 spacing=10,
 auto_scroll=True,
)

 # A new message entry form
 new_message = ft.TextField(
 hint_text="Write a message...",
 autofocus=True,
 shift_enter=True,
 min_lines=1,
 max_lines=5,
 filled=True,
 expand=True,
 on_submit=send_message_click,
)

 # Add everything to the page
 page.add(
 ft.Container(
 content=chat,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=5,
 padding=10,
 expand=True,
),
 ft.Row(
 controls=[
 new_message,
 ft.IconButton(
 icon=ft.Icons.SEND_ROUNDED,
 tooltip="Send message",
 on_click=send_message_click,
),
],
),
)

ft.run(main)

```

This is the final version of the chat app for the purpose of this tutorial. Below you can read more about the enhancements that we have made.

## Keyboard support

### Focusing input controls

All data entry controls have autofocus property which when set to True moves initial focus to the control. If there is more than one control on a page with autofocus set, then the first one added to the page will get focus.

We set autofocus=True on a username TextField inside a dialog and then on a TextField for entering chat message to set initial focus on it when the dialog is closed.

When a user click "Send" button or presses Enter to submit a chat message, TextField loses focus. To programmatically set control focus we used TextField.focus() method.

### Submitting forms on Enter

It's so tempting to submit forms with just pushing Enter button on the keyboard! Type your name in the dialog, hit Enter, type a new message, hit Enter, type another, hit Enter - no mouse involved at all! ☐

Flet has support for that by providing TextField.on\_submit event handler which fires when a user press Enter button while the focus is on the TextField.

### Entering multiline messages

What about multiline TextFields where Enter must advance a cursor to the next line? We've got that covered too! TextField control has shift\_enter property which when set to True enables Discord-like behavior: to get to a new line user presses Shift+Enter while hitting just Enter submits a form.

### Animated scrolling to the last message

Noticed a nice animation of scrolling to the last message in a chat window? It could be enabled by setting `ListView.auto_scroll` property to `True`. The top most `Page` class, being a scrollable container itself, also supports `auto_scroll`.

## Page title

Final touch - page title that could be changed as simply as:

```
page.title = "Flet Chat"
page.update()
```

## Deploying the app

Congratulations! You have created your Chat app in Python with Flet, and it looks awesome!

Now it's time to share your app with the world!

Follow these instructions to deploy your Flet app as a web app to [Fly.io](#) or [Replit](#).

## What's next

There are plenty of features we could implement to improve this chat app:

- Disconnect, reconnect, session timeout
- Upload/download images
- Authentication, avatars
- Using database for the storage
- Chat channels, topics
- Full-text search
- Emojis, markdown
- Bots
- Mobile app

Please let us know if you would like to contribute to the app/tutorial and share it with other Flet developers.

## Summary

In this tutorial, you have learnt how to:

- Create a simple Flet app;
- Add page controls and handle events;
- Use built in PubSub library;
- User `AlertDialog` for entering user name;
- Build page layout with reusable controls;
- Deploy your Flet app to the web;

For further reading you can explore controls and examples.

# Solitaire Tutorial

In this tutorial we will show you step-by-step creation of a famous Klondike solitaire game in Python with Flet. For an inspiration, we looked at this online game: <https://www.solitr.com/>

This tutorial is aimed at beginner/intermediate level Python developers who have basic knowledge of Python and object oriented programming.

You can find a live demo here.

Image: solitaire - <https://flet.dev/docs/examples/tutorials/solitaire/media/part1-final.gif>

We have broken down the game implementation into the following steps:

- Getting started with Flet
- Proof of concept app for draggable cards
- Fanned card piles
- Solitaire setup
- Solitaire rules
- Winning the game
- Deploying the app

In the Part 2 (will be covered in the next tutorial) we'll be adding AppBar with options to start new game, view game rules and change game settings.

## Getting started with Flet

To create a multi-platform app in Python with Flet, you don't need to know HTML, CSS or JavaScript, but you do need a basic knowledge of Python and object-oriented programming.

Before you can create your first Flet app, you need to setup your development environment, which requires Python 3.9 or above and flet package.

Once you have Flet installed, let's create a simple hello-world app.

Create hello.py with the following contents:

hello.py

```
import flet as ft

def main(page: ft.Page):
 page.add(ft.Text(value="Hello, world!"))

ft.run(main)
```

Run this app and you will see a new window with a greeting:

Image: hello-world - <https://flet.dev/docs/examples/tutorials/todo/media/hello-world.png>

## Proof of concept app for draggable cards

For the proof of concept, we will only be using three types of controls:

- Stack - will be used as a parent control for absolute positioning of slots and cards
- GestureDetector - the card that will be moved within the Stack
- Container - the slot where the card will be dropped. Will also be used as content for the GestureDetector.

We have broken down the proof of concept app into four easy steps, so that after each step you have a complete short program to run and test.

### Step 1: Drag the card around

In this step we will create a Stack (Solitaire game field) and a GestureDetector (Solitaire card). The card will then be added to the list of the Stack.controls. top and left properties of the GestureDetector are used for absolute positioning of the card in the Stack.

```
import flet as ft
```

```
def main(page: ft.Page):
 def drag(e: ft.DragUpdateEvent):
 ...

 card = ft.GestureDetector(
 on_pan_update=drag,
 left=0,
 top=0,
 content=ft.Container(bgcolor=ft.Colors.GREEN, width=70, height=100),
)

 page.add(ft.Stack(controls=[card], width=1000, height=500))

ft.run(main)
```

Run the app to see the card added to the stack:

Image: drag\_and\_drop1 - <https://flet.dev/docs/examples/tutorials/solitaire/media/drag-and-drop1.png>

To be able to move the card, we'll update the drag method that is called in on\_pan\_update event of GestureDetector which happens every drag\_interval while the user drags the card with their mouse.

To show the card's movement, we'll be updating the card's top and left properties in the drag callback each time the on\_pan\_update event happens.

Below is the simplest code for dragging GestureDetector in Stack:

```
import flet as ft

Use of GestureDetector with on_pan_update event for dragging card
Absolute positioning of controls within stack

def main(page: ft.Page):
 def drag(e: ft.DragUpdateEvent):
 e.control.top = max(0, e.control.top + e.local_delta.y)
 e.control.left = max(0, e.control.left + e.local_delta.x)
 e.control.update()

 card = ft.GestureDetector(
 mouse_cursor=ft.MouseCursor.MOVE,
 drag_interval=5,
 on_pan_update=drag,
 left=0,
 top=0,
 content=ft.Container(bgcolor=ft.Colors.GREEN, width=70, height=100),
)

 page.add(ft.Stack(controls=[card], width=1000, height=500))

ft.run(main)
```

Now you can see the card moving:

Image: drag\_and\_drop2 - <https://flet.dev/docs/examples/tutorials/solitaire/media/drag-and-drop2.gif>

#### Note

After any properties of a control are updated, an update() method of the control (or its parent control) should be called for the update to take effect.

## Step 2: Drop the card in the slot or bounce it back

The goal of this step is to be able to drop a card into a slot if it is close enough and bounce it back if it's not.

Image: drag-and-drop3 - <https://flet.dev/docs/examples/tutorials/solitaire/media/drag-and-drop3.gif>

Let's create a Container control that will represent a slot to which we'll be dropping the card:

```
slot = ft.Container(
 width=70,
 height=100,
 left=200,
 top=0,
 border=ft.Border.all(1)
)
page.add(ft.Stack(controls = [slot, card], width=1000, height=500))
```

on\_pan\_end event of the card is called when the card is dropped:

```
card = ft.GestureDetector(
 mouse_cursor=ft.MouseCursor.MOVE,
 drag_interval=5,
 on_pan_update=drag,
 on_pan_end=drop,
 left=0,
 top=0,
 content=ft.Container(bgcolor=ft.Colors.GREEN, width=70, height=100),
```

```
)
```

On this event, we'll call drop method to check if the card is close enough to the slot (let's say it's closer than 20px to the slot), and place it there:

```
def drop(e: ft.DragEndEvent):
 if (
 abs(e.control.top - slot.top) < 20
 and abs(e.control.left - slot.left) < 20
):
 place(e.control, slot)
 e.control.update()

def place(card, slot):
 """place card to the slot"""
 card.top = slot.top
 card.left = slot.left
 page.update()
```

Now, if the card is not close enough, we need to bounce it back to its original position. Unfortunately, we don't know the original position coordinates, since the card's top and left properties were changed on on\_pan\_update event.

To solve this problem, let's create a Solitaire class object to keep track of the original position of the card when on\_pan\_start event of the card is called:

```
class Solitaire:
 def __init__(self):
 self.start_top = 0
 self.start_left = 0

solitaire = Solitaire()

def start_drag(e: ft.DragStartEvent):
 solitaire.start_top = e.control.top
 solitaire.start_left = e.control.left
 e.control.update()
```

Now let's update on\_pan\_end event with the option to bounce card back:

```
def bounce_back(game, card):
 """return card to its original position"""
 card.top = game.start_top
 card.left = game.start_left
 page.update()

def drop(e: ft.DragEndEvent):
 if (
 abs(e.control.top - slot.top) < 20
 and abs(e.control.left - slot.left) < 20
):
 place(e.control, slot)
 else:
 bounce_back(solitaire, e.control)

 e.control.update()
```

The full code for this step can be found [here](#).

### Step 3: Adding a second card

Eventually, we'll need 52 cards to play the game. For our proof of concept, let's add a second card:

```
card2 = ft.GestureDetector(
 mouse_cursor=ft.MouseCursor.MOVE,
 drag_interval=5,
 on_pan_start=start_drag,
 on_pan_update=drag,
 on_pan_end=drop,
 left=100,
 top=0,
 content=ft.Container(bgcolor=ft.Colors.YELLOW, width=70, height=100),
)

controls = [slot, card1, card2]
page.add(ft.Stack(controls=controls, width=1000, height=500))
```

Now, if you run the app with the two cards, you will notice that when you move the cards around, the yellow card (card2) is moving as expected but the green the card (card1) is moving under the yellow card.

Image: drag\_and\_drop4.gif - <https://flet.dev/docs/examples/tutorials/solitaire/media/drag-and-drop4.gif>

It happens because card2 is added to the list of stack's controls after card1. To fix this problem, we need to move the draggable card to the top of the list of controls on on\_pan\_start event:

```
def move_on_top(card, controls):
 """Moves draggable card to the top of the stack"""
 controls.remove(card)
```



```

controls.append(card)
page.update()

def start_drag(e: ft.DragStartEvent):
 move_on_top(e.control, controls)
 solitaire.start_top = e.control.top
 solitaire.start_left = e.control.left

```

Now the two cards can be dragged without issues:

Image: drag\_and\_drop5.gif - <https://flet.dev/docs/examples/tutorials/solitaire/media/drag-and-drop5.gif>

The full code for this step can be found [here](#).

## Step 4: Adding more slots

As a final step for the proof of concept app, let's create two more slots:

```

slot0 = ft.Container(
 width=70, height=100, left=0, top=0, border=ft.Border.all(1)
)

slot1 = ft.Container(
 width=70, height=100, left=200, top=0, border=ft.Border.all(1)
)

slot2 = ft.Container(
 width=70, height=100, left=300, top=0, border=ft.Border.all(1)
)

slots = [slot0, slot1, slot2]

```

When creating new cards, we will not specify their top and left position now, but instead, will place them to the slot0:

```

deal cards
place(card1, slot0)
place(card2, slot0)

```

on\_pan\_end event, where we check if a card is close to a slot, we will now go through the list of slots to find where the card should be dropped:

```

def drop(e: ft.DragEndEvent):
 for slot in slots:
 if (
 abs(e.control.top - slot.top) < 20
 and abs(e.control.left - slot.left) < 20
):
 place(e.control, slot)
 e.control.update()
 return

 bounce_back(solitaire, e.control)
 e.control.update()

```

As a result, the two cards can be dragged between the three slots:

Image: drag\_and\_drop6.gif - <https://flet.dev/docs/examples/tutorials/solitaire/media/drag-and-drop6.gif>

The full code for this step can be found [here](#).

Congratulations on completing the proof of concept app for the Solitaire game! Now you can work with GestureDetector to move cards inside Stack and place them to certain Containers, which is a great part of the game to begin with.

## Fanned card piles

In the proof of concept app you have accomplished the task of dropping a card into a slot in proximity or bounce it back. If there is already a card in that slot, the new card is placed on top of it, covering it completely.

In the actual Solitaire game, if there is already a card in a tableau slot, you want to place the draggable card a bit lower, so that you can see the previous card too, and if there are two cards, even lower. Those are called “fanned piles”.

Then, we want to be able to pick a card from the fanned pile that is not the top card of the pile and drag the card together with all the cards below it:

Image: fanned\_piles1.gif - <https://flet.dev/docs/examples/tutorials/solitaire/media/fanned-piles3.gif>

To be able to do that, it would be useful to have the information about the pile of cards in the slot from which the card is dragged, as well as in the slot to which it is being dropped. Let's restructure our program and get it ready for the implementation of the fanned piles.

## Slot, Card and Solitaire classes

A slot could have a pile property that would hold a list of cards that were placed there. Now the slot is a Container control object, and we can't add any new properties to it. Let's create a new Slot class that will inherit from Container and add a pile property to it:

```
SLOT_WIDTH = 70
SLOT_HEIGHT = 100

import flet as ft

class Slot(ft.Container):
 def __init__(self, top, left):
 super().__init__()
 self.pile=[]
 self.width=SLOT_WIDTH
 self.height=SLOT_HEIGHT
 self.left=left
 self.top=top
 self.border=ft.Border.all(1)
```

Similarly to Slot class, let's create a new Card class with slot property to remember in which slot it resides. It will inherit from GestureDetector and we'll move all card-related methods to it:

```
CARD_WIDTH = 70
CARD_HEIGHT = 100
DROP_PROXIMITY = 20

import flet as ft

class Card(ft.GestureDetector):
 def __init__(self, solitaire, color):
 super().__init__()
 self.slot = None
 self.mouse_cursor=ft.MouseCursor.MOVE
 self.drag_interval=5
 self.on_pan_start=self.start_drag
 self.on_pan_update=self.drag
 self.on_pan_end=self.drop
 self.left=None
 self.top=None
 self.solitaire = solitaire
 self.color = color
 self.content=ft.Container(bgcolor=self.color, width=CARD_WIDTH, height=CARD_HEIGHT)

 def move_on_top(self):
 """Moves draggable card to the top of the stack"""
 self.solitaire.controls.remove(self)
 self.solitaire.controls.append(self)
 self.solitaire.update()

 def bounce_back(self):
 """Returns card to its original position"""
 self.top = self.slot.top
 self.left = self.slot.left
 self.update()

 def place(self, slot):
 """Place card to the slot"""
 self.top = slot.top
 self.left = slot.left
 self.slot = slot

 def start_drag(self, e: ft.DragStartEvent):
 self.move_on_top()
 self.update()

 def drag(self, e: ft.DragUpdateEvent):
 self.top = max(0, self.top + e.local_delta.y)
 self.left = max(0, self.left + e.local_delta.x)
 self.update()

 def drop(self, e: ft.DragEndEvent):
 for slot in self.solitaire.slots:
 if (
 abs(self.top - slot.top) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
):
 self.place(slot)
 self.update()
 return

 self.bounce_back()
 self.update()
```

Note

Since each card has slot property now, there is no need to remember start\_left and start\_top position of the draggable card in Solitaire class anymore, because we can just bounce it back to its slot.

Let's update Solitaire class to inherit from Stack, and move the creation of cards and slots there:

```
SOLITAIRE_WIDTH = 1000
SOLITAIRE_HEIGHT = 500

import flet as ft
from slot import Slot
from card import Card

class Solitaire(ft.Stack):
 def __init__(self):
 super().__init__()
 self.controls = []
 self.slots = []
 self.cards = []
 self.width = SOLITAIRE_WIDTH
 self.height = SOLITAIRE_HEIGHT

 def did_mount(self):
 self.create_card_deck()
 self.create_slots()
 self.deal_cards()

 def create_card_deck(self):
 card1 = Card(self, color="GREEN")
 card2 = Card(self, color="YELLOW")
 self.cards = [card1, card2]

 def create_slots(self):
 self.slots.append(Slot(top=0, left=0))
 self.slots.append(Slot(top=0, left=200))
 self.slots.append(Slot(top=0, left=300))
 self.controls.extend(self.slots)
 self.update()

 def deal_cards(self):
 self.controls.extend(self.cards)
 for card in self.cards:
 card.place(self.slots[0])
 self.update()
```

## Note

If you try to call create\_slots() and create\_card\_deck() and deal\_cards() methods \_\_init\_\_() method of the Solitaire class, it will cause an error "Control must be added to the page first". To fix this, we create slots and cards inside the did\_mount() method, which happens immediately after the stack is added to the page.

Our main program will be very simple now:

```
import flet as ft
from solitaire import Solitaire

def main(page: ft.Page):
 solitaire = Solitaire()

 page.add(solitaire)

ft.run(main)
```

You can find the full source code for this step [here](#). It works exactly the same way as the proof of concept app, but re-written with the new classes to be ready for adding more complex functionality to it.

## Placing card with offset

When the card is being placed to a slot in the card.place() method, we need to do three things:

- Remove the card from its original slot, if it exists
- Change card's slot to the new slot
- Add the card to the new slot's pile

```
def place(self, slot):
 self.top = slot.top
 self.left = slot.left

 # remove card from it's original slot, if exists
 if self.slot is not None:
 self.slot.pile.remove(self)

 # change card's slot to a new slot
```

```

self.slot = slot

add card to the new slot's pile
slot.pile.append(self)

```

When updating card's top and left position, left should remain the same, but top will depend on the length of the new slot's pile and CARD\_OFFSET:

```

CARD_OFFSET = 20

...

def place(self, slot):
 self.top = slot.top + len(slot.pile) * CARD_OFFSET
 self.left = slot.left
 ...

```

Now the cards are placed with offset which gives us the fanned pile look:

Image: fanned\_piles1 - <https://flet.dev/docs/examples/tutorials/solitaire/media/fanned-piles1.png>

## Drag pile of cards

If you try to drag the card from the bottom of the pile now, it will look like this:

Image: drag\_pile - <https://flet.dev/docs/examples/tutorials/solitaire/media/fanned-piles2.gif>

To fix this problem, we need to update all the methods in card.py that work with the draggable card to work with the draggable pile instead.

Let's create get\_draggable\_pile() method that will get the list of cards that need to be dragged together, starting with the card you picked:

```

def get_draggable_pile(self):
 """returns list of cards that will be dragged together, starting with the current card"""
 if self.slot is not None:
 self.draggable_pile = self.slot.pile[self.slot.pile.index(self) :]
 else: # slot == None when the cards are dealt and need to be place in slot for the first time
 self.draggable_pile = [self]

```

Since self.draggable\_pile is a new Card attribute, Let's add it to \_\_init\_\_():

```

class Card(ft.GestureDetector):
 def __init__(self, solitaire, color):
 super().__init__()
 ...
 self.draggable_pile = [self]

```

Now we need to update start\_drag() method to get the draggable pile each time a drag starts:

```

def start_drag(self, e: ft.DragStartEvent):
 self.get_draggable_pile()
 self.move_on_top()
 self.solitaire.update()

```

Then, we'll update move\_on\_top() method:

```

def move_on_top(self):
 """Brings draggable card pile to the top of the stack"""

 # for card in self.get_draggable_pile():
 for card in self.draggable_pile:
 self.solitaire.controls.remove(card)
 self.solitaire.controls.append(card)
 self.solitaire.update()

```

Additionally, we need to update drag() method to go through the draggable pile and update positions of all the cards being dragged:

```

def drag(self, e: ft.DragUpdateEvent):
 for card in self.draggable_pile:
 card.top = (
 max(0, self.top + e.local_delta.y)
 + self.draggable_pile.index(card) * CARD_OFFSET
)
 card.left = max(0, self.left + e.local_delta.x)
 self.solitaire.update()

```

Also, we need to update place() method to place the draggable pile to the slot:

```

def place(self, slot):
 """Place draggable pile to the slot"""
 for card in self.draggable_pile:
 card.top = slot.top + len(slot.pile) * CARD_OFFSET
 card.left = slot.left

 # remove card from it's original slot, if it exists
 if card.slot is not None:

```

```

 card.slot.pile.remove(card)

 # change card's slot to a new slot
 card.slot = slot

 # add card to the new slot's pile
 slot.pile.append(card)

 self.solitaire.update()

```

Finally, if no slot in proximity is found, we need to bounce the whole pile back to its original position:

```

def bounce_back(self):
 """Returns draggable pile to its original position"""
 for card in self.draggable_pile:
 card.top = card.slot.top + card.slot.pile.index(card) * CARD_OFFSET
 card.left = card.slot.left
 self.solitaire.update()

```

The full source code of this step can be found [here](#). Now we can drag and drop cards in fanned piles, which means we are ready for the real deal!

## Solitaire setup

Let's take a look at the wikipedia article about Klondike (solitaire):

Klondike is played with a standard 52-card deck.

After shuffling, a tableau of seven fanned piles of cards is laid from left to right. From left to right, each pile contains one more card than the last. The first and left-most pile contains a single upturned card, the second pile contains two cards, the third pile contains three cards, the fourth pile contains four cards, the fifth pile contains five cards, the sixth pile contains six cards, and the seventh pile contains seven cards. The topmost card of each pile is turned face up. The remaining cards form the stock and are placed facedown at the upper left of the layout.

The four foundations (light rectangles in the upper right of the figure) are built up by suit from Ace (low in this game) to King, and the tableau piles can be built down by alternate colors.

Image: [game\\_setup\\_wiki - https://flet.dev/docs/examples/tutorials/solitaire/media/game-setup-wiki.png](https://flet.dev/docs/examples/tutorials/solitaire/media/game-setup-wiki.png)

We will now work on this setup step by step.

### Create card deck

The first step is to create a full deck of cards in Solitaire class. Each card should have a suit property (hearts, diamonds, clubs and spades) and a rank property (from Ace to King). For the suit, its color is important, because tableau piles are built by alternate colors.

For the rank, its value is important, because foundations are built from the lowest (Ace) to the highest (King) rank value.

In `solitaire.py`, create Suite and Rank classes:

```

class Suite:
 def __init__(self, suite_name, suite_color):
 self.name = suite_name
 self.color = suite_color

class Rank:
 def __init__(self, card_name, card_value):
 self.name = card_name
 self.value = card_value

```

Now, in the Card class, instead of accepting the color as an argument, we'll be accepting suite and rank in `__init__()`. Additionally, we'll add `face_up` property to the card and the Container will now has image of the back of the card as its content:

```

class Card(ft.GestureDetector):
 def __init__(self, solitaire, suite, rank):
 super().__init__()
 self.mouse_cursor=ft.MouseCursor.MOVE
 self.drag_interval=5
 self.on_pan_start=self.start_drag
 self.on_pan_update=self.drag
 self.on_pan_end=self.drop
 self.suite=suite
 self.rank=rank
 self.face_up=False
 self.top=None
 self.left=None
 self.solitaire = solitaire
 self.slot = None
 self.content=ft.Container(
 width=CARD_WIDTH,

```

```

height=CARD_HEIGHT,
border_radius = ft.BorderRadius.all(6),
content=ft.Image(src="images/card_back.png"))

```

All the images for the face up cards, as well as card back are stored in the “assets/images” folder in the same directory as main.py.

note

You could specify the assets folder, see Assets guide for more information and examples.

Finally, in `solitaire.create_card_deck()` we'll create lists of suites and ranks and then the 52-card deck:

```

def create_card_deck(self):
 suites = [
 Suite("hearts", "RED"),
 Suite("diamonds", "RED"),
 Suite("clubs", "BLACK"),
 Suite("spades", "BLACK"),
]
 ranks = [
 Rank("Ace", 1),
 Rank("2", 2),
 Rank("3", 3),
 Rank("4", 4),
 Rank("5", 5),
 Rank("6", 6),
 Rank("7", 7),
 Rank("8", 8),
 Rank("9", 9),
 Rank("10", 10),
 Rank("Jack", 11),
 Rank("Queen", 12),
 Rank("King", 13),
]

 self.cards = []

 for suite in suites:
 for rank in ranks:
 self.cards.append(Card(solitaire=self, suite=suite, rank=rank))

```

The card deck is ready to be dealt, and now we need to create the layout for it.

## Create slots

Klondike solitaire game layout should look like this:

Image: solitaire-layout - <https://flet.dev/docs/examples/tutorials/solitaire/media/solitaire-layout.svg>

Now, in the Slot class, we'll be accepting `solitaire`, `border` and `border_radius` as arguments in `__init__()`:

```

class Slot(ft.Container):
 def __init__(self, top, left, border):
 super().__init__()
 ...
 self.solitaire = solitaire
 self.border = border
 self.border_radius = ft.BorderRadius.all(6)

```

Let's create all those slots in `solitaire.create_slots()`:

```

def create_slots(self):

 self.stock = Slot(self, top=0, left=0, border=ft.Border.all(1))
 self.waste = Slot(self, top=0, left=100, border=None)

 self.foundations = []
 x = 300
 for i in range(4):
 self.foundations.append(Slot(self, top=0, left=x, border=ft.Border.all(1, "outline")))
 x += 100

 self.tableau = []
 x = 0
 for i in range(7):
 self.tableau.append(Slot(self, top=150, left=x, border=None))
 x += 100

 self.controls.append(self.stock)
 self.controls.append(self.waste)
 self.controls.extend(self.foundations)
 self.controls.extend(self.tableau)
 self.update()

```

Note

Note: some slots should have visible border and some shouldn't, so we added border to the list of arguments for the creation of Slot objects.

## Note

If you try to run the app, it will cause an error "IndexError: list index out of range". To fix this, update deal\_cards() method as described below.

## Deal cards

Let's start with shuffling the cards and adding them to the list of controls:

```
import random

...

def deal_cards(self):
 random.shuffle(self.cards)
 self.controls.extend(self.cards)
 self.update()
```

Then we'll deal the cards to the tableau piles from left to right so that each pile contains one more card than the last, and place the remaining cards to the stock pile:

```
def deal_cards(self):
 random.shuffle(self.cards)
 self.controls.extend(self.cards)

 # deal to tableau
 first_slot = 0
 remaining_cards = self.cards

 while first_slot < len(self.tableau):
 for slot in self.tableau[first_slot:]:
 top_card = remaining_cards[0]
 top_card.place(slot)
 remaining_cards.remove(top_card)
 first_slot += 1

 # place remaining cards to stock pile
 for card in remaining_cards:
 card.place(self.stock)

 self.update()
```

Let's run the program and see where we are at now:

Image: game-setup1 - <https://flet.dev/docs/examples/tutorials/solitaire/media/game-setup1.png>

Cards in stock were placed in a fanned pile in the same manner as to the tableau, but they should have been placed to a regular pile instead. To fix this problem, let's add this condition to the card.place() method:

```
def place(self, slot):
 """Place draggable pile to the slot"""
 for card in self.draggable_pile:
 if slot in self.solitaire.tableau:
 card.top = slot.top + len(slot.pile) * CARD_OFFSET
 else:
 card.top = slot.top
 self.left = slot.left
 ...
```

Now cards are only placed in fanned piles to tableau:

Image: game-setup2 - <https://flet.dev/docs/examples/tutorials/solitaire/media/game-setup2.png>

If you try moving the cards around now, the program won't work. The reason for this is that in the card.drop() method iterates through list of slots which we don't have now.

Let's update the method to go separately through foundations and tableau:

```
def drop(self, e: ft.DragEndEvent):
 for slot in self.solitaire.tableau:
 if (
 abs(self.top - (slot.top + len(slot.pile) * CARD_OFFSET)) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
):
 self.place(slot)
 self.solitaire.update()
 return

 for slot in self.solitaire.foundations:
 if (
 abs(self.top - slot.top) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
):
 self.place(slot)
 self.solitaire.update()
 return
```

```

 self.place(slot)
 self.solitaire.update()
 return

 self.bounce_back()

```

## Reveal top cards in tableau piles

Now we have the correct game setup and as a last touch we need to reveal the topmost cards in tableau piles.

In Slot class, create a `get_top_card()` method:

```

def get_top_card(self):
 if len(self.pile) > 0:
 return self.pile[-1]

```

In Card class, create `turn_face_up()` method:

```

def turn_face_up(self):
 self.face_up = True
 self.content.content.src=f"images/{self.rank.name}_{self.suite.name}.svg"
 self.solitaire.update()

```

Finally, reveal the topmost cards at the end of `solitaire.deal_cards()`:

```

def deal_cards(self):
 ...
 self.update()

 for slot in self.tableau:
 slot.get_top_card().turn_face_up()

 self.update()

```

Let's see how it looks now:

Image: game-setup3 - <https://flet.dev/docs/examples/tutorials/solitaire/media/game-setup3.png>

The full source code for this step can be found [here](#).

Congratulations on completing the Solitaire game setup! You've created a full 52-card deck, built layout with stock, waste, foundations and tableau piles, dealt the cards and revealed the top cards in tableau. Let's move on to the next item on our todo list, which is Solitaire Rules.

## Solitaire rules

If you run your current version of Solitaire, you'll notice that you can do some crazy things with your cards:

Image: game-rules1 - <https://flet.dev/docs/examples/tutorials/solitaire/media/game-rules1.gif>

Now it is time to implement some rules.

### General rules

Currently, we can move any card, but only face-up cards should be moved. Let's add this check in `start_drag`, `drag` and `drop` methods of the card:

```

def start_drag(self, e: ft.DragStartEvent):
 if self.face_up:
 self.move_on_top()
 self.solitaire.update()

def drag(self, e: ft.DragUpdateEvent):
 if self.face_up:
 for card in self.draggable_pile:
 card.top = max(0, self.top + e.local_delta.y) + self.draggable_pile.index(card) * CARD_OFFSET
 card.left = max(0, self.left + e.local_delta.x)
 card.solitaire.update()

def drop(self, e: ft.DragEndEvent):
 if self.face_up:
 for slot in self.solitaire.tableau:
 if (
 abs(self.top - (slot.top + len(slot.pile) * CARD_OFFSET)) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
):
 self.place(slot)
 return

 for slot in self.solitaire.foundations:
 if (
 abs(self.top - slot.top) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
):
 self.place(slot)

```



```

 return

 self.bounce_back()

```

Now let's specify click method for the on\_tap event of the card to reveal the card if you click on a faced-down top card in a tableau pile:

```

class Card(ft.GestureDetector):
 def __init__(self, solitaire, suite, rank):
 ...
 self.on_tap = self.click
 ...

 ...

 def click(self, e):
 if self.slot in self.solitaire.tableau:
 if not self.face_up and self == self.slot.get_top_card():
 self.turn_face_up()
 self.solitaire.update()

```

Let's check how it works:

Image: game\_rules2 - <https://flet.dev/docs/examples/tutorials/solitaire/media/game-rules2.gif>

## Foundations rules

At the moment we can place fanned piles to foundations, which shouldn't be allowed. Let's check the draggable pile length to fix it:

```

def drop(self, e: ft.DragEndEvent):
 if self.face_up:
 for slot in self.solitaire.tableau:
 if (
 abs(self.top - (slot.top + len(slot.pile) * CARD_OFFSET)) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
):
 self.place(slot)
 return

 if len(self.draggable_pile) == 1:
 for slot in self.solitaire.foundations:
 if (
 abs(self.top - slot.top) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
):
 self.place(slot)
 return

 self.bounce_back()

```

Then, of course, not any card can be placed to a foundation. According to the rules, a foundation should start with an Ace and then the cards of the same suite can be placed on top of it to build a pile from Ace to King.

Let's add this rule to Solitaire class:

```

def check_foundations_rules(self, card, slot):
 top_card = slot.get_top_card()
 if top_card is not None:
 return (
 card.suite.name == top_card.suite.name
 and card.rank.value - top_card.rank.value == 1
)
 else:
 return card.rank.name == "Ace"

```

We'll check this rule in drop() method of the card before placing a card to a foundation slot:

```

def drop(self, e: ft.DragEndEvent):
 if self.face_up:
 for slot in self.solitaire.tableau:
 if (
 abs(self.top - (slot.top + len(slot.pile) * CARD_OFFSET)) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
):
 self.place(slot)
 return

 if len(self.draggable_pile) == 1:
 for slot in self.solitaire.foundations:
 if (
 abs(self.top - slot.top) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
) and self.solitaire.check_foundations_rules(self, slot):
 self.place(slot)
 return

```

```
self.bounce_back()
```

As a final touch for foundations rules, let's implement doubleclick method for on\_double\_tap event of a card. It will be checking if the faced-up card fits into any of the foundations and place it there:

```
class Card(ft.GestureDetector):
 def __init__(self, solitaire, suite, rank):
 ...
 self.on_double_tap = self.doubleclick
 ...

 ...

 def doubleclick(self, e):
 self.get_draggable_pile()
 if self.face_up and len(self.draggable_pile) == 1:
 self.move_on_top()
 for slot in self.solitaire.foundations:
 if self.solitaire.check_foundations_rules(self, slot):
 self.place(slot)
 return
```

## Tableau rules

Finally, let's implement the rules to build tableau piles down from King to Ace by alternating suite color. Additionally, only King can be placed to an empty tableau slot.

Let's add these rules for Solitaire class:

```
def check_tableau_rules(self, card, slot):
 top_card = slot.get_top_card()
 if top_card is not None:
 return (
 card.suite.color != top_card.suite.color
 and top_card.rank.value - card.rank.value == 1
 and top_card.face_up
)
 else:
 return card.rank.name == "King"
```

Similarly to the foundations rules, we'll check tableau rules before placing a card to a tableau pile:

```
def drop(self, e: ft.DragEndEvent):
 if self.face_up:
 for slot in self.solitaire.tableau:
 if (
 abs(self.top - (slot.top + len(slot.pile) * CARD_OFFSET)) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
) and self.solitaire.check_tableau_rules(self, slot):
 self.place(slot)
 return

 if len(self.draggable_pile) == 1:
 for slot in self.solitaire.foundations:
 if (
 abs(self.top - slot.top) < DROP_PROXIMITY
 and abs(self.left - slot.left) < DROP_PROXIMITY
) and self.solitaire.check_foundations_rules(self, slot):
 self.place(slot)
 return

 self.bounce_back()
```

## Stock and waste

To properly play Solitaire game right now we are missing the remaining cards that are piled in the stock.

Let's update click() method of the card to go through the stock pile and place the cards to waste as we go:

```
def click(self, e):
 if self.slot in self.solitaire.tableau:
 if not self.face_up and self == self.slot.get_top_card():
 self.turn_face_up()
 elif self.slot == self.solitaire.stock:
 self.move_on_top()
 self.place(self.solitaire.waste)
 self.turn_face_up()
```

Now, in Card class, create turn\_face\_down() method:

```
def turn_face_down(self):
 """Hides card"""
 self.face_up = False
 self.content.content.src = "/images/card_back.png"
 self.solitaire.update()
```

That's it! Now you can properly play solitaire, but it's very difficult to win the game if you cannot pass through the waste again. Let's implement `click()` for `on_click` event of the stock Slot to go through the stock pile again:

```
class Slot(ft.Container):
 def __init__(self, solitaire, top, left, border):
 super().__init__()
 self.pile=[]
 self.width=SLOT_WIDTH
 self.height=SLOT_HEIGHT
 self.left=left
 self.top=top
 self.on_click=self.click
 self.solitaire=solitaire
 self.border=border
 self.border_radius = ft.BorderRadius.all(6)

 def click(self, e):
 if self == self.solitaire.stock:
 self.solitaire.restart_stock()
```

`restart_stock()` method in `Solitaire` class will place all cards from waste to stock again:

```
def restart_stock(self):
 while len(self.waste.pile) > 0:
 card = self.waste.get_top_card()
 card.turn_face_down()
 card.move_on_top()
 card.place(self.stock)
```

For `card.place()` method to work properly with cards from Stock and Waste, we've added a condition to `card.get_draggable_pile()`, so that it returns the top card only and not the whole pile:

```
def get_draggable_pile(self):
 """returns list of cards that will be dragged together, starting with the current card"""

 if (
 self.slot is not None
 and self.slot != self.solitaire.stock
 and self.slot != self.solitaire.waste
):
 self.draggable_pile = self.slot.pile[self.slot.pile.index(self) :]
 else: # slot == None when the cards are dealt and need to be placed in slot for the first time
 self.draggable_pile = [self]
```

All done! The full source code for this step can be found [here](#).

Let's move on to the last step of the game itself - detecting the situation when you have won.

## Winning the game

According to wikipedia, some suggest the chances of winning the game as being 1 in 30 games.

Knowing that the chances of winning are quite low, we should plan on showing the user something exciting when that finally happens.

First, let's add a check for the winning condition to `Solitaire` class. If all four foundations contain a total of 52 cards, then you have won:

```
def check_win(self):
 cards_num = 0
 for slot in self.foundations:
 cards_num += len(slot.pile)
 if cards_num == 52:
 return True
 return False
```

We'll be checking if this condition is true each time a card is placed to a foundation:

```
def place(self, slot):
 """Place draggable pile to the slot"""

 for card in self.draggable_pile:
 if slot in self.solitaire.tableau:
 card.top = slot.top + len(slot.pile) * CARD_OFFSET
 else:
 card.top = slot.top
 card.left = slot.left

 # remove card from its original slot, if exists
 if card.slot is not None:
 card.slot.pile.remove(card)

 # change card's slot to a new slot
 card.slot = slot

 # add card to the new slot's pile
```

```

 slot.pile.append(card)

 if self.solitaire.check_win():
 self.solitaire.winning_sequence()

 self.solitaire.update()

```

Finally, if the winning condition is met, it will trigger a winning sequence involving position animation:

```

def winning_sequence(self):
 for slot in self.foundations:
 for card in slot.pile:
 card.animate_position=1000
 card.move_on_top()
 card.top = random.randint(0, SOLITAIRE_HEIGHT)
 card.left = random.randint(0, SOLITAIRE_WIDTH)
 self.update()
 self.controls.append(ft.AlertDialog(title=ft.Text("Congratulations! You won!"), open=True))

```

Image: winning\_the\_game - <https://flet.dev/docs/examples/tutorials/solitaire/media/winning-the-game.gif>

Wow! We did it. You can find the full source code for the Solitaire game [here](#).

Now, as we have a decent desktop version of the game, let's deploy it as a web app to share with your friends and colleagues.

## Deploying the app

Congratulations! You have created your Solitaire game app in Python with Flet, and it looks awesome!

Now it's time to share your app with the world!

Follow these instructions to deploy your Flet app as a web app to Fly.io or Replit.

## Summary

In this tutorial, you have learnt how to:

- Create a simple Flet app;
- Drag and drop cards with GestureDetector;
- Create your own classes that inherit from Flet controls;
- Design UI layout using absolute positioning of controls in Stack;
- Implement implicit animations;
- Deploy your Flet app to the web;

For further reading you can explore controls and examples.

We would love to hear your feedback! Please drop us an email, join the discussion on Discord.

# ToDo Tutorial

In this tutorial we will show you, step-by-step, how to create a To-Do app in Python using Flet framework and then publish it as a desktop, mobile or web app. The app is a single-file console program of just 164 lines (formatted!) of Python code, yet it is a multi-platform application with rich, responsive UI:

Image: Image - <https://flet.dev/docs/examples/tutorials/todo/media/complete-demo-web.gif>

You can see the live demo here.

We chose a To-Do app for the tutorial, because it covers all of the basic concepts you would need to create a Flet app: building a page layout, adding controls, handling events, displaying and editing lists, making reusable UI components, and publishing options.

The tutorial consists of the following steps:

- Getting started with Flet
- Adding page controls and handling events
- View, edit and delete list items
- Filtering list items
- Final touches
- Publishing the app

## Getting started with Flet

To create a multi-platform app in Python with Flet, you don't need to know HTML, CSS or JavaScript, but you do need a basic knowledge of Python and object-oriented programming.

Before you can create your first Flet app, you need to setup your development environment, which requires Python 3.10 or above and flet package.

Once you have Flet installed, let's create a simple hello-world app.

Create hello.py with the following contents:

hello.py

```
import flet as ft

def main(page: ft.Page):
 page.add(ft.Text(value="Hello, world!"))

ft.run(main)
```

Run this app and you will see a new window with a greeting:

Image: Image - <https://flet.dev/docs/examples/tutorials/todo/media/hello-world.png>

## Adding page controls and handling events

To start, we'll need a TextField for entering a task name, and "+" FloatingActionButton with an event handler that will display a Checkbox with a new task.

Create todo.py with the following contents:

todo.py

```
import flet
from flet import Checkbox, FloatingActionButton, Icons, Page, TextField

def main(page: Page):
 def add_clicked(e):
 page.add(Checkbox(label=new_task.value))
 new_task.value = ""
 page.update()

 new_task = TextField(hint_text="Whats needs to be done?")

 page.add(new_task, FloatingActionButton(icon=Icons.ADD, on_click=add_clicked))
```

```
flet.run(main)
```

Run the app and you should see a page like this:

Image: Image - <https://flet.dev/docs/examples/tutorials/todo/media/app-1.png>

## Page layout

Now let's make the app look nice! We want the entire app to be at the top center of the page, taking up 600 px width. The TextField and the "+" button should be aligned horizontally, and take up full app width:

Image: Image - <https://flet.dev/docs/examples/tutorials/todo/media/diagram-1.svg>

Row is a control that is used to lay its children controls out horizontally on a page. Column is a control that is used to lay its children controls out vertically on a page.

Replace todo.py contents with the following:

hello.py

```
import flet as ft

def main(page: ft.Page):
 def add_clicked(e):
 tasks_view.controls.append(ft.Checkbox(label=new_task.value))
 new_task.value = ""
 view.update()

 new_task = ft.TextField(hint_text="What needs to be done?", expand=True)
 tasks_view = ft.Column()
 view = ft.Column(
 width=600,
 controls=[
 ft.Row(
 controls=[
 new_task,
 ft.FloatingActionButton(icon=ft.Icons.ADD, on_click=add_clicked),
],
),
 tasks_view,
],
)

 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.add(view)

ft.run(main)
```

Run the app and you should see a page like this:

Image: Image - <https://flet.dev/docs/examples/tutorials/todo/media/app-2.png>

## Reusable UI components

While we could continue writing our app in the main function, the best practice would be to create a reusable UI component. Imagine you are working on an app header, a side menu, or UI that will be a part of a larger project. Even if you can't think of such uses right now, we still recommend creating all your Flet apps with composability and reusability in mind.

To make a reusable To-Do app component, we are going to encapsulate its state and presentation logic in a separate class:

todo.py

```
import flet as ft

@ft.control
class TodoApp(ft.Column):
 # application's root control is a Column containing all other controls
 def init(self):
 self.new_task = ft.TextField(hint_text="What needs to be done?", expand=True)
 self.tasks_view = ft.Column()
 self.width = 600
 self.controls = [
 ft.Row(
 controls=[
 self.new_task,
 ft.FloatingActionButton(
 icon=ft.Icons.ADD, on_click=self.add_clicked
),
],
),
 self.tasks_view,
]
```

```

),
 self.tasks_view,
]

 def add_clicked(self, e):
 self.tasks_view.controls.append(ft.Checkbox(label=self.new_task.value))
 self.new_task.value = ""
 self.update()

def main(page: ft.Page):
 page.title = "To-Do App"
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.update()

 # create application instance
 todo = TodoApp()

 # add application's root control to the page
 page.add(todo)

ft.run(main)

```

Try this out!

Try adding two TodoApp components to the page:

```

create application instance
app1 = TodoApp()
app2 = TodoApp()

add application's root control to the page
page.add(app1, app2)

```

## View, edit and delete list items

In the previous step, we created a basic To-Do app with task items shown as checkboxes. Let's improve the app by adding "Edit" and "Delete" buttons next to a task name. The "Edit" button will switch a task item to edit mode.

Image: [Image - https://flet.dev/docs/examples/tutorials/todo/media/diagram-2.svg](https://flet.dev/docs/examples/tutorials/todo/media/diagram-2.svg)

Each task item is represented by two rows: display\_view row with Checkbox, "Edit" and "Delete" buttons and edit\_view row with TextField and "Save" button. view column serves as a container for both display\_view and edit\_view rows.

To encapsulate task item views and actions, we introduced a new Task class.

Additionally, we changed TodoApp class to create and hold Task instances when the "Add" button is clicked.

For "Delete" task operation, we implemented task\_delete() method in TodoApp class which accepts task control instance as a parameter.

Then, we passed a reference to task\_delete method into Task constructor and called it on "Delete" button event handler.

todo.py

```

from dataclasses import field
from typing import Callable

import flet as ft

@ft.control
class Task(ft.Column):
 task_name: str = ""
 on_task_delete: Callable[["Task"], None] = field(default=lambda task: None)

 def init(self):
 self.display_task = ft.Checkbox(value=False, label=self.task_name)
 self.edit_name = ft.TextField(expand=1)

 self.display_view = ft.Row(
 alignment=ft.MainAxisAlignment.SPACE_BETWEEN,
 vertical_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 self.display_task,
 ft.Row(
 spacing=0,
 controls=[
 ft.IconButton(
 icon=ft.Icons.CREATE_OUTLINED,
 tooltip="Edit To-Do",
 on_click=self.edit_clicked,
),
 ft.IconButton(

```

```

 ft.Icons.DELETE_OUTLINE,
 tooltip="Delete To-Do",
 on_click=self.delete_clicked,
),
],
),
],
)

self.edit_view = ft.Row(
 visible=False,
 alignment=ft.MainAxisAlignment.SPACE_BETWEEN,
 vertical_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 self.edit_name,
 ft.IconButton(
 icon=ft.Icons.DONE_OUTLINE_OUTLINED,
 icon_color=ft.Colors.GREEN,
 tooltip="Update To-Do",
 on_click=self.save_clicked,
),
],
)

self.controls = [self.display_view, self.edit_view]

def edit_clicked(self, e):
 self.edit_name.value = self.display_task.label
 self.display_view.visible = False
 self.edit_view.visible = True
 self.update()

def save_clicked(self, e):
 self.display_task.label = self.edit_name.value
 self.display_view.visible = True
 self.edit_view.visible = False
 self.update()

def delete_clicked(self, e):
 self.on_task_delete(self)

@ft.control
class TodoApp(ft.Column):
 # application's root control is a Column containing all other controls
 def init(self):
 self.new_task = ft.TextField(hint_text="What needs to be done?", expand=True)
 self.tasks = ft.Column()
 self.width = 600
 self.controls = [
 ft.Row(
 controls=[
 self.new_task,
 ft.FloatingActionButton(
 icon=ft.Icons.ADD, on_click=self.add_clicked
),
],
),
 self.tasks,
]

 def add_clicked(self, e):
 task = Task(task_name=self.new_task.value, on_task_delete=self.task_delete)
 self.tasks.controls.append(task)
 self.new_task.value = ""
 self.update()

 def task_delete(self, task):
 self.tasks.controls.remove(task)
 self.update()

def main(page: ft.Page):
 page.title = "To-Do App"
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.update()

 # create application instance
 app = TodoApp()

 # add application's root control to the page
 page.add(app)

ft.run(main)

```

Run the app and try to edit and delete tasks:



## Filtering list items

We already have a functional To-Do app where we can create, edit, and delete tasks. To be even more productive, we want to be able to filter tasks by their status.

Copy the entire code for this step from here. Below we will explain the changes we've done to implement filtering.

Tabs control is used to display filter:

todo.py

```
...

class TodoApp(ft.Column):
 # application's root control is a Column containing all other controls
 def init(self):
 self.new_task = ft.TextField(hint_text="Whats needs to be done?", expand=True)
 self.tasks = ft.Column()

 self.filter = ft.TabBar(
 scrollable=False,
 tabs=[
 ft.Tab(label="all"),
 ft.Tab(label="active"),
 ft.Tab(label="completed"),
],
)

 self.filter_tabs = ft.Tabs(
 length=3,
 selected_index=0,
 on_change=lambda e: self.update(),
 content=self.filter,
)

...
```

To display different lists of tasks depending on their statuses, we could maintain three lists with "All", "Active" and "Completed" tasks. We, however, chose an easier approach where we maintain the same list and only change a task's visibility depending on its status.

In TodoApp class we overrode `before_update()` method called every time when the control is being updated. It iterates through all the tasks and updates their visible property depending on the status of the task:

todo.py

```
class TodoApp(ft.Column):

 # ...

 def before_update(self):
 status = self.filter.tabs[self.filter.selected_index].text
 for task in self.tasks.controls:
 task.visible = (
 status == "all"
 or (status == "active" and task.completed == False)
 or (status == "completed" and task.completed)
)
```

Filtering should occur when we click on a tab or change a task status. `TodoApp.before_update()` method is called when Tabs selected value is changed or Task item checkbox is clicked:

todo.py

```
class TodoApp(ft.Column):

 # ...

 def tabs_changed(self, e):
 self.update()

 def task_status_change(self, e):
 self.update()

 def add_clicked(self, e):
 task = Task(
 task_name=self.new_task.value,
 on_status_change=self.task_status_change,
 on_delete=self.task_delete,
)
 self.tasks.controls.append(task)
 self.new_task.value = ""
```

```

 self.update()
 # ...

class Task(ft.Column):
 task_name: str = ""
 on_status_change: Callable[[], None] = field(default=lambda: None)
 on_delete: Callable[["Task"], None] = field(default=lambda task: None)

 # ...

 def status_changed(self, e):
 self.completed = self.display_task.value
 self.task_status_change()

```

Run the app and try filtering tasks by clicking on the tabs:

Image: Image - <https://flet.dev/docs/examples/tutorials/todo/media/filtering.gif>

## Final touches

Our Todo app is almost complete now. As a final touch, we will add a footer (Column control) displaying the number of incomplete tasks (Text control) and a "Clear completed" button.

Full code

```

from dataclasses import field
from typing import Callable

import flet as ft

@ft.control
class Task(ft.Column):
 task_name: str = ""
 on_status_change: Callable[[], None] = field(default=lambda: None)
 on_delete: Callable[["Task"], None] = field(default=lambda task: None)

 def init(self):
 self.completed = False
 self.display_task = ft.Checkbox(
 value=False, label=self.task_name, on_change=self.status_changed
)
 self.edit_name = ft.TextField(expand=1)

 self.display_view = ft.Row(
 alignment=ft.MainAxisAlignment.SPACE_BETWEEN,
 vertical_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 self.display_task,
 ft.Row(
 spacing=0,
 controls=[
 ft.IconButton(
 icon=ft.Icons.CREATE_OUTLINED,
 tooltip="Edit To-Do",
 on_click=self.edit_clicked,
),
 ft.IconButton(
 ft.Icons.DELETE_OUTLINE,
 tooltip="Delete To-Do",
 on_click=self.delete_clicked,
),
],
),
],
)

 self.edit_view = ft.Row(
 visible=False,
 alignment=ft.MainAxisAlignment.SPACE_BETWEEN,
 vertical_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 self.edit_name,
 ft.IconButton(
 icon=ft.Icons.DONE_OUTLINE_OUTLINED,
 icon_color=ft.Colors.GREEN,
 tooltip="Update To-Do",
 on_click=self.save_clicked,
),
],
)

 self.controls = [self.display_view, self.edit_view]

 def edit_clicked(self, e):
 self.edit_name.value = self.display_task.label
 self.display_view.visible = False

```

```

 self.edit_view.visible = True
 self.update()

 def save_clicked(self, e):
 self.display_task.label = self.edit_name.value
 self.display_view.visible = True
 self.edit_view.visible = False
 self.update()

 def status_changed(self, e):
 self.completed = self.display_task.value
 self.on_status_change()

 def delete_clicked(self, e):
 self.on_delete(self)

@ft.control
class TodoApp(ft.Column):
 # application's root control is a Column containing all other controls
 def init(self):
 self.new_task = ft.TextField(hint_text="Whats needs to be done?", expand=True)
 self.tasks = ft.Column()

 self.filter = ft.TabBar(
 scrollable=False,
 tabs=[
 ft.Tab(label="all"),
 ft.Tab(label="active"),
 ft.Tab(label="completed"),
],
)

 self.filter_tabs = ft.Tabs(
 length=3,
 selected_index=0,
 on_change=lambda e: self.update(),
 content=self.filter,
)

 self.width = 600
 self.controls = [
 ft.Row(
 controls=[
 self.new_task,
 ft.FloatingActionButton(
 icon=ft.Icons.ADD, on_click=self.add_clicked
),
],
),
 ft.Column(
 spacing=25,
 controls=[
 self.filter_tabs,
 self.tasks,
],
),
]

 def add_clicked(self, e):
 task = Task(
 task_name=self.new_task.value,
 on_status_change=self.task_status_change,
 on_delete=self.task_delete,
)
 self.tasks.controls.append(task)
 self.new_task.value = ""
 self.update()

 def task_status_change(self):
 self.update()

 def task_delete(self, task):
 self.tasks.controls.remove(task)
 self.update()

 def before_update(self):
 status = self.filter_tabs[self.filter_tabs.selected_index].label
 for task in self.tasks.controls:
 task.visible = (
 status == "all"
 or (status == "active" and not task.completed)
 or (status == "completed" and task.completed)
)

 def tabs_changed(self, e):
 self.update()

```

```
def main(page: ft.Page):
 page.title = "To-Do App"
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.update()

 # create application instance
 app = TodoApp()

 # add application's root control to the page
 page.add(app)

ft.run(main)
```

Image: app-4 - <https://flet.dev/docs/examples/tutorials/todo/media/app-4.png>

## Publishing the app

Congratulations! You have created your first Python app with Flet, and it looks awesome!

Now it's time to share your app with the world!

Follow these instructions to publish your Flet app as a mobile, desktop or web app.

## Summary

In this tutorial, you have learnt how to:

- Create a simple Flet app;
- Work with Reusable UI components;
- Design UI layout using Column and Row controls;
- Work with lists: view, edit and delete items, filtering;
- Publish your Flet app to multiple platforms;

For further reading you can explore controls and examples.

# AccelerometerReadingEvent

Discrete reading from an accelerometer. Accelerometers measure the velocity of the device. Note that these readings include the effects of gravity. Put simply, you can use accelerometer readings to tell if the device is moving in a particular direction.

Inherits: Event

## Properties

- timestamp - Event timestamp.
- x - Acceleration along the X axis, in m/s<sup>2</sup>.
- y - Acceleration along the Y axis, in m/s<sup>2</sup>.
- z - Acceleration along the Z axis, in m/s<sup>2</sup>.

## Properties

### buildtimestampinstance-attribute

timestamp: datetime

Event timestamp.

### buildxinstance-attribute

x: float

Acceleration along the X axis, in m/s<sup>2</sup>.

### buildyinstance-attribute

y: float

Acceleration along the Y axis, in m/s<sup>2</sup>.

### buildzinstance-attribute

z: float

Acceleration along the Z axis, in m/s<sup>2</sup>.

# Aliases

## AppCallable

Type alias for Flet app lifecycle callbacks.

Represents a callable (synchronous or asynchronous) that accepts a single argument of type Page. The return value is ignored.

Used for both main and before\_main handlers.

`Callable[[Page], Any | Awaitable[Any]]`

## AnimationValue

Type alias for animation configuration values.

Represents animation input as either:

- True or False to enable or disable animation,
- an int duration in milliseconds,
- or an explicit Animation configuration.

`bool | int | Animation`

## BadgeValue

Type alias for badge content values.

Represents a badge as either:

- a str value, rendered as a text label badge,
- or a Badge.

`str | Badge`

## BlurValue

Type alias for blur configuration values.

Represents blur as either:

- a single sigma value applied to both axes,
- a (sigma\_x, sigma\_y) tuple,
- or an explicit Blur object.

`Number | tuple[Number, Number] | Blur`

## BorderRadiusValue

Type alias for border radius values.

Represents radius as either:

- a single numeric radius applied to all corners,
- or an explicit BorderRadius configuration.

`Number | BorderRadius`

## BorderSideStrokeAlignValue

Type alias for border stroke alignment values.

Represents stroke alignment as either:

- a BorderSideStrokeAlign enum value,
- or a numeric alignment value.

`BorderSideStrokeAlign | Number`

## BoxShadowValue

Type alias for box shadow values.

Represents shadows as either:

- a single BoxShadow object,
- or a list of BoxShadow objects.

`BoxShadow` | `list[BoxShadow]`

## ColorValue

Type alias for color values.

Represents a color and can be:

- a string (representing a color name or hex value),
- a material color from the Colors enum,
- or a Cupertino color from the CupertinoColors enum.

More information here.

`str` | `Colors` | `CupertinoColors`

## context

Global context object for the running Flet app.

Use context to access the current page and control auto-update behavior inside callbacks.

`Context()`

## ControlEvent

Type alias for an event emitted by any base control.

`Event[_BaseControlType]`

## ControlEventHandler

Type alias for typed control event callback handlers.

Represents a callback that accepts either:

- no arguments,
- or a typed Event for a specific control type.

`Callable[[], Any]` | `Callable[[Event[EventControlType]], Any]`

## ControlStateValue

Type alias for state-dependent control values.

Represents either:

- a single value applied to all (supported) states,
- or a mapping from ControlState to per-state values.

`T` | `dict[ControlState, T]`

## DateTimeValue

Type alias for date/time values.

Represents either a datetime or a date.

`datetime` | `date`

## DurationValue

Type alias for duration values.

Represents duration as either:

- a Duration object,
- or an integer number of milliseconds (or seconds, depending on context).

`Duration | int`

## EventControlType

Type variable bound to a control type for typed event payloads.

`TypeVar('EventControlType', bound=_BaseControlType)`

## EventHandler

Type alias for generic event callback handlers.

Represents a callback that accepts either:

- no arguments,
- or an Event-derived payload.

`Callable[[], Any] | Callable[[EventType], Any]`

## IconData

Represents an icon used in the UI.

An icon can come from:

- the Material icon set via the Icons icon collection,
- the Cupertino icon set via the CupertinoIcons icon collection,
- or a custom icon set defined by the developer.

Internally, an icon is stored as an integer that encodes icon's index in its originating code set.

Encoding structure:

- Lower 16 bits (bits 0-15): the icon's index.
- Third byte (bits 16-24): the icon set identifier (set ID), which distinguishes between icon sets like Material, Cupertino, etc.

This encoding scheme allows a single integer to uniquely represent any icon across multiple icon sets.

Inherits: `enum.IntEnum`

## Methods

- `random` - Selects a random icon from the subclass enum, with optional exclusions and weights.

## IconDataOrControl

Type alias for icon-like values.

Represents either:

- an IconData value (for example, a member of Icons or CupertinoIcons),
- or a custom icon Control.

`IconData | 'Control'`

## KeyValue

Type alias for control key values.

Represents keys as either:

- a ValueKey or ScrollKey object,
- or a primitive str, int, float, or bool value.

`ValueKey | ScrollKey | str | int | float | bool`



## MarginValue

Type alias for margin values.

Represents margin as either:

- a single numeric value applied to all sides,
- or an explicit Margin configuration.

Number | Margin

## Number

Type alias for numeric values (int or float).

int | float

## OffsetValue

Type alias for offset values.

Represents offset as either:

- an Offset object,
- or an (x, y) tuple.

Offset | tuple[Number, Number]

## PaddingValue

Type alias for padding values.

Represents padding as either:

- a single numeric value applied to all sides,
- or an explicit Padding configuration.

Number | Padding

## ResponsiveNumber

Type alias for responsive numeric values.

Represents either:

- a single numeric value used for all breakpoints,
- or a breakpoint-to-value mapping keyed by string or ResponsiveRowBreakpoint.

dict[str | ResponsiveRowBreakpoint, Number] | Number

## RotateValue

Type alias for rotation values.

Represents rotation as either:

- a numeric angle in radians,
- or an explicit Rotate transform.

Number | Rotate

## ScaleValue

Type alias for scale values.

Represents scale as either:

- a numeric uniform scale factor,
- or an explicit Scale transform.

Number | Scale

## StrOrControl

Type alias for string or control values.

Represents a string or a control and can be:

- a string, which will be converted internally into a Text control,
- or a control.

`str | 'Control'`

## TooltipValue

Type alias for tooltip values.

Represents a tooltip as either:

- a str message,
- or a Tooltip.

`str | Tooltip`

# Alignment

Defines an alignment relative to the center.

Image: Alignment - <https://flet.dev/docs/examples/controls/core/types/alignment/media/overview.png>

Alignment explained

## Properties

- **BOTTOM\_CENTER** - Represents the bottom center and is equivalent to `Alignment(0.0, 1.0)`.
- **BOTTOM\_LEFT** - Represents the bottom left corner and is equivalent to `Alignment(-1.0, 1.0)`.
- **BOTTOM\_RIGHT** - Represents the bottom right corner and is equivalent to `Alignment(1.0, 1.0)`.
- **CENTER** - Represents the center and is equivalent to `Alignment(0.0, 0.0)`.
- **CENTER\_LEFT** - Represents the center left and is equivalent to `Alignment(-1.0, 0.0)`.
- **CENTER\_RIGHT** - Represents the center right and is equivalent to `Alignment(1.0, 0.0)`.
- **TOP\_CENTER** - Represents the top center and is equivalent to `Alignment(0.0, -1.0)`.
- **TOP\_LEFT** - Represents the top left corner and is equivalent to `Alignment(-1.0, -1.0)`.
- **TOP\_RIGHT** - Represents the top right corner and is equivalent to `Alignment(1.0, -1.0)`.
- **x** - Represents the horizontal distance from the center.
- **y** - Represents the vertical distance from the center.

## Methods

- **copy** - Returns a copy of this object with the specified properties overridden.

## Properties

### **buildBOTTOM\_CENTER**class-attribute

`BOTTOM_CENTER: AlignmentProperty`

Represents the bottom center and is equivalent to `Alignment(0.0, 1.0)`.

### **buildBOTTOM\_LEFT**class-attribute

`BOTTOM_LEFT: AlignmentProperty`

Represents the bottom left corner and is equivalent to `Alignment(-1.0, 1.0)`.

### **buildBOTTOM\_RIGHT**class-attribute

`BOTTOM_RIGHT: AlignmentProperty`

Represents the bottom right corner and is equivalent to `Alignment(1.0, 1.0)`.

### **buildCENTER**class-attribute

`CENTER: AlignmentProperty`

Represents the center and is equivalent to `Alignment(0.0, 0.0)`.

### **buildCENTER\_LEFT**class-attribute

`CENTER_LEFT: AlignmentProperty`

Represents the center left and is equivalent to `Alignment(-1.0, 0.0)`.

### **buildCENTER\_RIGHT**class-attribute

`CENTER_RIGHT: AlignmentProperty`

Represents the center right and is equivalent to `Alignment(1.0, 0.0)`.

### **buildTOP\_CENTER**class-attribute

`TOP_CENTER: AlignmentProperty`

Represents the top center and is equivalent to `Alignment(0.0, -1.0)`.

## buildTOP\_LEFTclass-attribute

TOP\_LEFT: AlignmentProperty

Represents the top left corner and is equivalent to Alignment(-1.0, -1.0).

## buildTOP\_RIGHTclass-attribute

TOP\_RIGHT: AlignmentProperty

Represents the top right corner and is equivalent to Alignment(1.0, -1.0).

## buildxinstance-attribute

x: Number

Represents the horizontal distance from the center.

It's value ranges between -1.0 and 1.0 inclusive.

## buildyinstance-attribute

y: Number

Represents the vertical distance from the center.

It's value ranges between -1.0 and 1.0 inclusive.

# Methods

## deployed\_codecopy

```
copy(
 x: Number | None = None, y: Number | None = None
) -> Alignment
```

Returns a copy of this object with the specified properties overridden.

# Examples

## Container alignment

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Containers with different alignments"

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Row(
 controls=[
 ft.Container(
 bgcolor=ft.Colors.AMBER,
 padding=15,
 alignment=ft.Alignment.CENTER,
 width=150,
 height=150,
 content=ft.Button("Center"),
),
 ft.Container(
 bgcolor=ft.Colors.AMBER,
 padding=15,
 alignment=ft.Alignment.TOP_LEFT,
 width=150,
 height=150,
 content=ft.Button("Top left"),
),
 ft.Container(
 bgcolor=ft.Colors.AMBER,
 padding=15,
 alignment=ft.Alignment(-0.5, -0.5),
 width=150,
 height=150,
 content=ft.Button("-0.5, -0.5"),
),
],
),
],
 wrap=True,
)
)
)
```

```
)
)
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

Image: Image - <https://flet.dev/docs/examples/controls/core/types/alignment/media/container.png>

# AndroidBuildVersion

Android OS version details derived from `android.os.Build.VERSION`.

This object is exposed as `AndroidDeviceInfo.version`.

## Properties

- `base_os` - The base OS build the product is based on.
- `code_name` - The current development codename, or the string "REL" if this is a release build.
- `incremental` - The internal value used by the underlying source control to represent this build.
- `preview_sdk` - The developer preview revision of a pre-release SDK.
- `release` - The user-visible version string.
- `sdk` - The user-visible SDK version of the framework.
- `security_patch` - The user-visible security patch level.

## Properties

### `buildbase_osclass-attributeinstance-attribute`

`base_os: str | None = None`

The base OS build the product is based on.

Note

Available only on Android M (API 23) and newer.

### `buildcode_nameinstance-attribute`

`code_name: str`

The current development codename, or the string "REL" if this is a release build.

### `buildincrementalinstance-attribute`

`incremental: str`

The internal value used by the underlying source control to represent this build.

Note

Available only on Android M (API 23) and newer.

### `buildpreview_sdkclass-attributeinstance-attribute`

`preview_sdk: int | None = None`

The developer preview revision of a pre-release SDK.

### `buildreleaseinstance-attribute`

`release: str`

The user-visible version string.

### `buildsdkinstance-attribute`

`sdk: int`

The user-visible SDK version of the framework.

Possible values are defined in: [https://developer.android.com/reference/android/os/Build.VERSION\\_CODES.html](https://developer.android.com/reference/android/os/Build.VERSION_CODES.html)

### `buildsecurity_patchclass-attributeinstance-attribute`

`security_patch: str | None = None`

The user-visible security patch level.

Note

Available only on Android M (API 23) and newer.

# AndroidDeviceInfo

Device information snapshot for Android devices and emulators.

Returned by `Page.get_device_info` on Android.

Inherits: `DeviceInfo`

## Properties

- `available_ram_size` - Total available RAM size in bytes.
- `board` - The name of the underlying board, like "goldfish".
- `bootloader` - The system bootloader version number.
- `brand` - The consumer-visible brand with which the product/hardware will be associated, if any.
- `device` - The name of the industrial design.
- `display` - A build ID string meant for displaying to the user.
- `fingerprint` - A string that uniquely identifies this build.
- `free_disk_size` - Free disk size in bytes.
- `hardware` - The name of the hardware (from the kernel command line or `/proc`).
- `host` - Hostname.
- `id` - Either a changelist number, or a label like "M4-rc20".
- `is_low_ram_device` - True if the application is running on a low-RAM device, False otherwise.
- `is_physical_device` - False if the application is running in an emulator, True otherwise.
- `manufacturer` - The manufacturer of the product/hardware.
- `model` - The end-user-visible name for the end product.
- `name` - The name of the device.
- `physical_ram_size` - Total physical RAM size in bytes.
- `product` - The name of the overall product.
- `supported_32_bit_abis` - An ordered list of 32 bit ABIs supported by this device.
- `supported_64_bit_abis` - An ordered list of 64 bit ABIs supported by this device.
- `supported_abis` - An ordered list of ABIs supported by this device.
- `system_features` - Describes what features are available on the current device.
- `tags` - Comma-separated tags describing the build, like "unsigned,debug".
- `total_disk_size` - Total disk size in bytes.
- `type` - The type of build, like "user" or "eng".
- `version` - Android operating system version values derived from `android.os.Build.VERSION`.

## Properties

### `buildavailable_ram_sizeinstance-attribute`

`available_ram_size: int`

Total available RAM size in bytes.

### `buildboardinstance-attribute`

`board: str`

The name of the underlying board, like "goldfish".

More info: <https://developer.android.com/reference/android/os/Build#BOARD>

### buildbootloaderinstance-attribute

bootloader: str

The system bootloader version number.

More info: <https://developer.android.com/reference/android/os/Build#BOOTLOADER>

### buildbrandinstance-attribute

brand: str

The consumer-visible brand with which the product/hardware will be associated, if any.

More info: <https://developer.android.com/reference/android/os/Build#BRAND>

### builddeviceinstance-attribute

device: str

The name of the industrial design.

More info: <https://developer.android.com/reference/android/os/Build#DEVICE>

### bulddisplayinstance-attribute

display: str

A build ID string meant for displaying to the user.

More info: <https://developer.android.com/reference/android/os/Build#DISPLAY>

### buildfingerprintinstance-attribute ☐ ☐

fingerprint: str

A string that uniquely identifies this build.

More info: <https://developer.android.com/reference/android/os/Build#FINGERPRINT>

### buildfree\_disk\_sizeinstance-attribute

free\_disk\_size: int

Free disk size in bytes.

### buildhardwareinstance-attribute

hardware: str

The name of the hardware (from the kernel command line or /proc).

More info: <https://developer.android.com/reference/android/os/Build#HARDWARE>

### buildhostinstance-attribute

host: str

Hostname.

More info: <https://developer.android.com/reference/android/os/Build#HOST>

### buildidinstance-attribute

id: str

Either a changelist number, or a label like "M4-rc20".

More info: <https://developer.android.com/reference/android/os/Build#ID>

### buildis\_low\_ram\_deviceinstance-attribute

is\_low\_ram\_device: bool

True if the application is running on a low-RAM device, False otherwise.

### buildis\_physical\_deviceinstance-attribute

is\_physical\_device: bool

False if the application is running in an emulator, True otherwise.

### buildmanufacturerinstance-attribute

manufacturer: str



The manufacturer of the product/hardware.

More info: <https://developer.android.com/reference/android/os/Build#MANUFACTURER>

### **buildmodelinstance-attribute**

`model: str`

The end-user-visible name for the end product.

More info: <https://developer.android.com/reference/android/os/Build#MODEL>

### **buildnameinstance-attribute**

`name: str`

The name of the device.

### **buildphysical\_ram\_sizeinstance-attribute**

`physical_ram_size: int`

Total physical RAM size in bytes.

### **buildproductinstance-attribute**

`product: str`

The name of the overall product.

More info: <https://developer.android.com/reference/android/os/Build#PRODUCT>

### **buildsupported\_32\_bit\_abisinstance-attribute**

`supported_32_bit_abis: list[str]`

An ordered list of 32 bit ABIs supported by this device. Available only on Android L (API 21) and newer.

More info: [https://developer.android.com/reference/android/os/Build#SUPPORTED\\_32\\_BIT\\_ABIS](https://developer.android.com/reference/android/os/Build#SUPPORTED_32_BIT_ABIS)

### **buildsupported\_64\_bit\_abisinstance-attribute**

`supported_64_bit_abis: list[str]`

An ordered list of 64 bit ABIs supported by this device. Available only on Android L (API 21) and newer.

More info: [https://developer.android.com/reference/android/os/Build#SUPPORTED\\_64\\_BIT\\_ABIS](https://developer.android.com/reference/android/os/Build#SUPPORTED_64_BIT_ABIS)

### **buildsupported\_abisinstance-attribute**

`supported_abis: list[str]`

An ordered list of ABIs supported by this device. Available only on Android L (API 21) and newer.

More info: [https://developer.android.com/reference/android/os/Build#SUPPORTED\\_ABIS](https://developer.android.com/reference/android/os/Build#SUPPORTED_ABIS)

### **buildsystem\_featuresinstance-attribute**

`system_features: list[str]`

Describes what features are available on the current device.

This can be used to check if the device has, for example, a front-facing camera, or a touchscreen. However, in many cases this is not the best API to use. For example, if you are interested in bluetooth, this API can tell you if the device has a bluetooth radio, but it cannot tell you if bluetooth is currently enabled, or if you have been granted the necessary permissions to use it. Please only use this if there is no other way to determine if a feature is supported.

This data comes from Android's `PackageManager.getSystemAvailableFeatures`, and many of the common feature strings to look for are available in `PackageManager`'s public documentation:

<https://developer.android.com/reference/android/content/pm/PackageManager>

### **buildtagsinstance-attribute**

`tags: str`

Comma-separated tags describing the build, like "unsigned,debug".

More info: <https://developer.android.com/reference/android/os/Build#TAGS>

### **buildtotal\_disk\_sizeinstance-attribute**

`total_disk_size: int`

Total disk size in bytes.

### **buildtypeinstance-attribute**

type: str

The type of build, like "user" or "eng".

More info: <https://developer.android.com/reference/android/os/Build#TYPE>

### **buildversioninstance-attribute**

version: AndroidBuildVersion

Android operating system version values derived from android.os.Build.VERSION.

# AnimatedSwitcherTransition

Visual transition strategy used by AnimatedSwitcher.transition.

Controls how old and new content are animated while the switch occurs.

Inherits: enum.Enum

## Properties

- FADE - Cross-fades between outgoing and incoming content.
- ROTATION - Rotates content during the switch.
- SCALE - Scales content during the switch.

## Examples

### Animated Switcher Transition showcase

play\_arrowTry Online

```
import flet as ft

def showcase_card(transition: ft.AnimatedSwitcherTransition) -> ft.Container:
 state = 0
 values = ["A", "B"]

 switcher = ft.AnimatedSwitcher(
 duration=500,
 reverse_duration=300,
 transition=transition,
 content=ft.Text(
 values[0],
 size=40,
 weight=ft.FontWeight.BOLD,
),
)

 def swap(_):
 nonlocal state
 state = 1 - state
 switcher.content = ft.Text(
 values[state],
 size=40,
 weight=ft.FontWeight.BOLD,
)
 switcher.update()

 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(transition.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=240,
 height=90,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 alignment=ft.Alignment.CENTER,
 content=switcher,
),
 ft.Button("Swap", on_click=swap),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="AnimatedSwitcherTransition Showcase")
 page.add(
 ft.SafeArea(
```

```

 content=ft.Column(
 controls=[
 ft.Text("Swap content to compare switcher transition effects."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(transition)
 for transition in ft.AnimatedSwitcherTransition
],
),
],
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

**buildFADE**class-attributeinstance-attribute

Cross-fades between outgoing and incoming content.

**buildROTATION**class-attributeinstance-attribute

Rotates content during the switch.

**buildSCALE**class-attributeinstance-attribute

Scales content during the switch.

# Animation

Explicit animation configuration for animatable control properties.

Properties that accept animation usually also support shorthand values via `AnimationValue`:

- `True`: enables a default 1000ms LINEAR animation.
- `int`: interpreted as animation duration in milliseconds with a linear curve.

## Properties

- `curve` - Easing curve that shapes interpolation over time.
- `duration` - The duration of the animation.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildcurveclass-attributeinstance-attribute`

```
curve: AnimationCurve = AnimationCurve.LINEAR
```

Easing curve that shapes interpolation over time.

### `bulldurationclass-attributeinstance-attribute`

```
duration: DurationValue = field(
 default_factory=lambda: Duration()
)
```

The duration of the animation.

If provided as an integer, it is considered/assumed to be in milliseconds. For more control and flexibility, use `Duration` instead.

## Methods

### `deployed_codecopy`

```
copy(
 duration: DurationValue | None = None,
 curve: AnimationCurve | None = None,
) -> Animation
```

Returns a copy of this object with the specified properties overridden.

# AnimationCurve

Animation curves.

Inherits: `enum.Enum`

## Properties

- `BOUNCE_IN` - Easing that enters with a bounce effect.
- `BOUNCE_IN_OUT` - Easing that bounces both at the beginning and the end.
- `BOUNCE_OUT` - Easing that exits with a bounce effect.
- `DECELERATE` - Starts quickly, then slows down toward the end.
- `EASE` - Standard symmetric ease-in/ease-out curve.
- `EASE_IN` - Starts slowly and accelerates.
- `EASE_IN_BACK` - Ease-in curve with a slight initial backward overshoot.
- `EASE_IN_CIRC` - Circular ease-in curve.
- `EASE_IN_CUBIC` - Cubic ease-in curve.
- `EASE_IN_EXPO` - Exponential ease-in curve.
- `EASE_IN_OUT` - Symmetric ease-in/ease-out curve.
- `EASE_IN_OUT_BACK` - Ease-in/ease-out curve with overshoot near both ends.
- `EASE_IN_OUT_CIRC` - Circular ease-in/ease-out curve.
- `EASE_IN_OUT_CUBIC` - Cubic ease-in/ease-out curve.
- `EASE_IN_OUT_CUBIC_EMPHASIZED` - Material 3 emphasized cubic ease-in/ease-out curve.
- `EASE_IN_OUT_EXPO` - Exponential ease-in/ease-out curve.
- `EASE_IN_OUT_QUAD` - Quadratic ease-in/ease-out curve.
- `EASE_IN_OUT_QUART` - Quartic ease-in/ease-out curve.
- `EASE_IN_OUT_QUINT` - Quintic ease-in/ease-out curve.
- `EASE_IN_OUT_SINE` - Sinusoidal ease-in/ease-out curve.
- `EASE_IN_QUAD` - Quadratic ease-in curve.
- `EASE_IN_QUART` - Quartic ease-in curve.
- `EASE_IN_QUINT` - Quintic ease-in curve.
- `EASE_IN_SINE` - Sinusoidal ease-in curve.
- `EASE_IN_TO_LINEAR` - Transitions from ease-in motion into linear motion.
- `EASE_OUT` - Starts quickly and decelerates to the end.
- `EASE_OUT_BACK` - Ease-out curve with trailing overshoot.
- `EASE_OUT_CIRC` - Circular ease-out curve.
- `EASE_OUT_CUBIC` - Cubic ease-out curve.
- `EASE_OUT_EXPO` - Exponential ease-out curve.
- `EASE_OUT_QUAD` - Quadratic ease-out curve.
- `EASE_OUT_QUART` - Quartic ease-out curve.
- `EASE_OUT_QUINT` - Quintic ease-out curve.
- `EASE_OUT_SINE` - Sinusoidal ease-out curve.
- `ELASTIC_IN` - Elastic spring-like ease-in curve.

- ELASTIC\_IN\_OUT - Elastic spring-like ease-in/ease-out curve.
- ELASTIC\_OUT - Elastic spring-like ease-out curve.
- FAST\_LINEAR\_TO\_SLOW\_EASE\_IN - Starts linear and then eases into a slower end.
- FAST\_OUT\_SLOWIN - Material motion curve: quick start, gentle finish.
- LINEAR - Constant speed with no easing.
- LINEAR\_TO\_EASE\_OUT - Starts linear and transitions to an ease-out tail.
- SLOW\_MIDDLE - Slows in the middle section of the animation.

## Examples

### Animation Curve showcase

play\_arrowTry Online

```
import asyncio

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 duration_ms = 1200
 track_width = 248
 racer_size = 26
 lane_padding = 10
 lane_inner_width = track_width
 lane_width = lane_inner_width + lane_padding * 2
 travel_units = (lane_inner_width - racer_size) / racer_size
 card_animations: list = []

 def showcase_card(curve: ft.AnimationCurve) -> ft.Container:
 state = {"forward": False}
 progress = ft.Container(
 width=0,
 height=6,
 border_radius=3,
 bgcolor=ft.Colors.PRIMARY_CONTAINER,
 animate=ft.Animation(duration_ms, curve=curve),
)
 racer = ft.Container(
 width=racer_size,
 height=racer_size,
 border_radius=13,
 bgcolor=ft.Colors.PRIMARY,
 shadow=ft.BoxShadow(
 blur_radius=12, spread_radius=1, color=ft.Colors.PRIMARY
),
 alignment=ft.Alignment.CENTER,
 content=ft.Icon(ft.Icons.BOLT, size=14, color=ft.Colors.ON_PRIMARY),
 offset=ft.Offset(0, 0),
 rotate=0,
 scale=1,
 animate_offset=ft.Animation(duration_ms, curve=curve),
 animate_rotation=ft.Animation(duration_ms, curve=curve),
 animate_scale=ft.Animation(duration_ms, curve=curve),
)
 status = ft.Text("idle", size=11, color=ft.Colors.ON_SURFACE_VARIANT)

 def animate(forward: bool):
 state["forward"] = forward
 racer.offset = ft.Offset(travel_units if forward else 0, 0)
 racer.rotate = 1 if forward else 0
 racer.scale = 1.25 if forward else 1
 progress.width = track_width if forward else 0
 status.value = "forward" if forward else "reverse"
 racer.update()
 progress.update()
 status.update()

 def replay(e):
 animate(not state["forward"])

 card_animations.append(animate)

 return ft.Container(
 width=340,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
```

```

 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=9,
 controls=[
 ft.Text(curve.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=track_width,
 height=6,
 border_radius=3,
 bgcolor=ft.Colors.SURFACE_CONTAINER_HIGH,
 content=progress,
),
 ft.Container(
 width=lane_width,
 height=48,
 padding=lane_padding,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 clip_behavior=ft.ClipBehavior.HARD_EDGE,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Stack(
 controls=[
 ft.Row(
 spacing=0,
 alignment=ft.MainAxisAlignment.SPACE_BETWEEN,
 controls=[
 ft.Container(
 width=4,
 height=22,
 bgcolor=ft.Colors.OUTLINE,
),
 ft.Container(
 width=4,
 height=22,
 bgcolor=ft.Colors.OUTLINE,
),
],
),
 racer,
],
),
),
],
),
),
 ft.Row(
 alignment=ft.MainAxisAlignment.SPACE_BETWEEN,
 controls=[
 status,
 ft.Button("Replay", icon=ft.Icons.REPLAY, on_click=replay),
],
),
],
),
),
)

def play_all(e):
 for animate in card_animations:
 animate(True)

def reverse_all(e):
 for animate in card_animations:
 animate(False)

async def wave_all():
 for animate in card_animations:
 animate(True)
 await asyncio.sleep(0.04)

page.appbar = ft.AppBar(title="AnimationCurve Showcase")
page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Curve Lab: compare timing profiles across motion, "
 "progress, and spin."
),
 ft.Row(
 wrap=True,
 spacing=8,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 ft.Button(
 "Play all", icon=ft.Icons.PLAY_ARROW, on_click=play_all
),
 ft.Button(
 "Reverse all",
 icon=ft.Icons.REPLAY,
 on_click=reverse_all,

```



```

),
 ft.Button(
 "Wave", on_click=lambda e: page.run_task(wave_all)
),
],
),
ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(curve) for curve in ft.AnimationCurve],
),
]
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

**buildBOUNCE\_IN**class-attributeinstance-attribute

Easing that enters with a bounce effect.

**buildBOUNCE\_IN\_OUT**class-attributeinstance-attribute

Easing that bounces both at the beginning and the end.

**buildBOUNCE\_OUT**class-attributeinstance-attribute

Easing that exits with a bounce effect.

**buildDECELERATE**class-attributeinstance-attribute

Starts quickly, then slows down toward the end.

**buildEASE**class-attributeinstance-attribute

Standard symmetric ease-in/ease-out curve.

**buildEASE\_IN**class-attributeinstance-attribute

Starts slowly and accelerates.

**buildEASE\_IN\_BACK**class-attributeinstance-attribute

Ease-in curve with a slight initial backward overshoot.

**buildEASE\_IN\_CIRC**class-attributeinstance-attribute

Circular ease-in curve.

**buildEASE\_IN\_CUBIC**class-attributeinstance-attribute

Cubic ease-in curve.

**buildEASE\_IN\_EXPO**class-attributeinstance-attribute

Exponential ease-in curve.

**buildEASE\_IN\_OUT**class-attributeinstance-attribute

Symmetric ease-in/ease-out curve.

**buildEASE\_IN\_OUT\_BACK**class-attributeinstance-attribute

Ease-in/ease-out curve with overshoot near both ends.

**buildEASE\_IN\_OUT\_CIRC**class-attributeinstance-attribute

Circular ease-in/ease-out curve.

**buildEASE\_IN\_OUT\_CUBIC**class-attributeinstance-attribute

Cubic ease-in/ease-out curve.

`buildEASE_IN_OUT_CUBIC_EMPHASIZED`class-attributeinstance-attribute

Material 3 emphasized cubic ease-in/ease-out curve.

`buildEASE_IN_OUT_EXPO`class-attributeinstance-attribute

Exponential ease-in/ease-out curve.

`buildEASE_IN_OUT_QUAD`class-attributeinstance-attribute

Quadratic ease-in/ease-out curve.

`buildEASE_IN_OUT_QUART`class-attributeinstance-attribute

Quartic ease-in/ease-out curve.

`buildEASE_IN_OUT_QUINT`class-attributeinstance-attribute

Quintic ease-in/ease-out curve.

`buildEASE_IN_OUT_SINE`class-attributeinstance-attribute

Sinusoidal ease-in/ease-out curve.

`buildEASE_IN_QUAD`class-attributeinstance-attribute

Quadratic ease-in curve.

`buildEASE_IN_QUART`class-attributeinstance-attribute

Quartic ease-in curve.

`buildEASE_IN_QUINT`class-attributeinstance-attribute

Quintic ease-in curve.

`buildEASE_IN_SINE`class-attributeinstance-attribute

Sinusoidal ease-in curve.

`buildEASE_IN_TO_LINEAR`class-attributeinstance-attribute

Transitions from ease-in motion into linear motion.

`buildEASE_OUT`class-attributeinstance-attribute

Starts quickly and decelerates to the end.

`buildEASE_OUT_BACK`class-attributeinstance-attribute

Ease-out curve with trailing overshoot.

`buildEASE_OUT_CIRC`class-attributeinstance-attribute

Circular ease-out curve.

`buildEASE_OUT_CUBIC`class-attributeinstance-attribute

Cubic ease-out curve.

`buildEASE_OUT_EXPO`class-attributeinstance-attribute

Exponential ease-out curve.

`buildEASE_OUT_QUAD`class-attributeinstance-attribute

Quadratic ease-out curve.

`buildEASE_OUT_QUART`class-attributeinstance-attribute

Quartic ease-out curve.

`buildEASE_OUT_QUINT`class-attributeinstance-attribute

Quintic ease-out curve.

**buildEASE\_OUT\_SINE**class-attributeinstance-attribute

Sinusoidal ease-out curve.

**buildELASTIC\_IN**class-attributeinstance-attribute

Elastic spring-like ease-in curve.

**buildELASTIC\_IN\_OUT**class-attributeinstance-attribute

Elastic spring-like ease-in/ease-out curve.

**buildELASTIC\_OUT**class-attributeinstance-attribute

Elastic spring-like ease-out curve.

**buildFAST\_LINEAR\_TO\_SLOW\_EASE\_IN**class-attributeinstance-attribute

Starts linear and then eases into a slower end.

**buildFAST\_OUT\_SLOWIN**class-attributeinstance-attribute

Material motion curve: quick start, gentle finish.

**buildLINEAR**class-attributeinstance-attribute

Constant speed with no easing.

**buildLINEAR\_TO\_EASE\_OUT**class-attributeinstance-attribute

Starts linear and transitions to an ease-out tail.

**buildSLOW\_MIDDLE**class-attributeinstance-attribute

Slows in the middle section of the animation.

# AnimationStyle

Used to override the default parameters of an animation.

## Note

If duration and reverse\_duration are set to Duration, the corresponding animation will be disabled. See no\_animation method for a convenient way to create such an instance.

## Properties

- curve - The curve to use for the animation.
- duration - The duration of the animation.
- reverse\_curve - The curve to use for the reverse animation.
- reverse\_duration - The duration of the reverse animation.

## Methods

- copy - Returns a copy of this object with the specified properties overridden.
- no\_animation - Creates an instance of AnimationStyle with no animation.

## Properties

### buildcurveclass-attributeinstance-attribute

curve: AnimationCurve | None = None

The curve to use for the animation.

### bulldurationclass-attributeinstance-attribute

duration: DurationValue | None = None

The duration of the animation.

### buildreverse\_curveclass-attributeinstance-attribute

reverse\_curve: AnimationCurve | None = None

The curve to use for the reverse animation.

### buildreverse\_durationclass-attributeinstance-attribute

reverse\_duration: DurationValue | None = None

The duration of the reverse animation.

## Methods

### deployed\_codecopy

```
copy(
 duration: DurationValue | None = None,
 reverse_duration: DurationValue | None = None,
 curve: AnimationCurve | None = None,
 reverse_curve: AnimationCurve | None = None,
) -> AnimationStyle
```

Returns a copy of this object with the specified properties overridden.

### deployed\_codeno\_animationstaticmethod

```
no_animation() -> AnimationStyle
```

Creates an instance of AnimationStyle with no animation.

# AppBarTheme

Customizes the appearance of AppBar controls across the app.

## Properties

- `actions_padding` - Overrides the default value of `AppBar.actions_padding` in all descendant AppBar controls.
- `bgcolor` - Overrides the default value of `AppBar.bgcolor` in all descendant AppBar controls.
- `center_title` - Overrides the default value of `AppBar.center_title` in all descendant AppBar controls.
- `color` - Overrides the default value of `AppBar.color` in all descendant AppBar controls.
- `elevation` - Overrides the default value of `AppBar.elevation` in all descendant AppBar controls.
- `elevation_on_scroll` - Overrides the default value of `AppBar.elevation_on_scroll` in all descendant AppBar controls.
- `shadow_color` - Overrides the default value of `AppBar.shadow_color` in all descendant AppBar controls.
- `shape` - Overrides the default value of `AppBar.shape` in all descendant AppBar controls.
- `title_spacing` - Overrides the default value of `AppBar.title_spacing` in all descendant AppBar controls.
- `title_text_style` - Overrides the default value of `AppBar.title_text_style` in all descendant AppBar controls.
- `toolbar_height` - Overrides the default value of `AppBar.toolbar_height` in all descendant AppBar controls.
- `toolbar_text_style` - Overrides the default value of `AppBar.toolbar_text_style` in all descendant AppBar controls.

## Properties

### `buildactions_paddingclass-attributeinstance-attribute`

`actions_padding: PaddingValue | None = None`

Overrides the default value of `AppBar.actions_padding` in all descendant AppBar controls.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `AppBar.bgcolor` in all descendant AppBar controls.

### `buildcenter_titleclass-attributeinstance-attribute`

`center_title: bool | None = None`

Overrides the default value of `AppBar.center_title` in all descendant AppBar controls.

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

Overrides the default value of `AppBar.color` in all descendant AppBar controls.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value of `AppBar.elevation` in all descendant AppBar controls.

### `buildelevation_on_scrollclass-attributeinstance-attribute`

`elevation_on_scroll: Number | None = None`

Overrides the default value of `AppBar.elevation_on_scroll` in all descendant AppBar controls.

### `buildshadow_colorclass-attributeinstance-attribute`

`shadow_color: ColorValue | None = None`

Overrides the default value of `AppBar.shadow_color` in all descendant AppBar controls.

### `buildshapeclass-attributeinstance-attribute`

`shape: OutlinedBorder | None = None`

Overrides the default value of `AppBar.shape` in all descendant AppBar controls.

### **buildtitle\_spacingclass-attributeinstance-attribute**

`title_spacing: Number | None = None`

Overrides the default value of `AppBar.title_spacing` in all descendant `AppBar` controls.

### **buildtitle\_text\_styleclass-attributeinstance-attribute**

`title_text_style: TextStyle | None = None`

Overrides the default value of `AppBar.title_text_style` in all descendant `AppBar` controls.

### **buildtoolbar\_heightclass-attributeinstance-attribute**

`toolbar_height: Number | None = None`

Overrides the default value of `AppBar.toolbar_height` in all descendant `AppBar` controls.

### **buildtoolbar\_text\_styleclass-attributeinstance-attribute**

`toolbar_text_style: TextStyle | None = None`

Overrides the default value of `AppBar.toolbar_text_style` in all descendant `AppBar` controls.

# AppLifecycleState

States that an application can be in once it is running.

Inherits: `enum.Enum`

## Properties

- DETACH - The application has exited, and detached all host views from the engine.
- HIDE - The application is hidden.
- INACTIVE - The application loses input focus.
- PAUSE - The application is paused.
- RESTART - The application is resumed after being paused.
- RESUME - The application gains input focus.
- SHOW - The application is shown.

## Examples

### App Lifecycle State showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 log = ft.Column(
 spacing=4,
 scroll=ft.ScrollMode.AUTO,
 height=220,
)

 def add_log(message: str):
 log.controls.insert(0, ft.Text(message, size=12))
 if len(log.controls) > 20:
 log.controls.pop()
 log.update()

 def on_lifecycle(e: ft.AppLifecycleStateChangeEvent):
 add_log(f"Received: {e.state.name}")

 page.on_app_lifecycle_state_change = on_lifecycle

 page.appbar = ft.AppBar(title="AppLifecycleState Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Switch app focus/visibility to see lifecycle state changes."
),
 ft.Text(
 "Ex: minimize/restore app, switch tabs, or "
 "background/foreground app.",
 size=11,
),
 ft.Row(
 wrap=True,
 spacing=8,
 controls=[
 ft.Container(
 padding=ft.Padding.symmetric(horizontal=8, vertical=4),
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=12,
 content=ft.Text(state.name, size=11),
)
 for state in ft.AppLifecycleState
],
),
 ft.Container(
 width=720,
 padding=12,
```

```

 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=log,
),
],
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildDETACHclass-attributeinstance-attribute

The application has exited, and detached all host views from the engine.

This callback is only called on iOS and Android.

### buildHIDEclass-attributeinstance-attribute

The application is hidden.

On mobile platforms, this is usually just before the application is replaced by another application in the foreground.

On desktop platforms, this is just before the application is hidden by being minimized or otherwise hiding all views of the application.

On the web, this is just before a window (or tab) is hidden.

### buildINACTIVEclass-attributeinstance-attribute

The application loses input focus.

On mobile platforms, this can be during a phone call or when a system dialog is visible.

On desktop platforms, this is when all views in an application have lost input focus but at least one view of the application is still visible.

On the web, this is when the window (or tab) has lost input focus.

### buildPAUSEclass-attributeinstance-attribute

The application is paused.

On mobile platforms, this happens right before the application is replaced by another application.

On desktop platforms and the web, this function is not called.

### buildRESTARTclass-attributeinstance-attribute

The application is resumed after being paused.

On mobile platforms, this happens just before this application takes over as the active application.

On desktop platforms and the web, this function is not called.

### buildRESUMEclass-attributeinstance-attribute

The application gains input focus. Indicates that the application is entering a state where it is visible, active, and accepting user input.

### buildSHOWclass-attributeinstance-attribute

The application is shown.

On mobile platforms, this is usually just before the application replaces another application in the foreground.

On desktop platforms, this is just before the application is shown after being minimized or otherwise made to show at least one view of the application.

On the web, this is just before a window (or tab) is shown.



# AppLifecycleStateChangeEvent

Event payload for app lifecycle transitions.

Delivered to `Page.on_app_lifecycle_state_change`.

Inherits: `Event`

## Properties

- `state` - New application lifecycle state.

## Properties

### `buildstateinstance-attribute`

`state`: `AppLifecycleState`

New application lifecycle state.

# AppView

TBD

Inherits: `enum.Enum`

## Properties

- `FLET_APP`
- `FLET_APP_HIDDEN`
- `FLET_APP_WEB`
- `WEB_BROWSER`

## Properties

`buildFLET_APP`class-attributeinstance-attribute

`buildFLET_APP_HIDDEN`class-attributeinstance-attribute

`buildFLET_APP_WEB`class-attributeinstance-attribute

`buildWEB_BROWSER`class-attributeinstance-attribute

# Assertiveness

Determines the assertiveness level of the accessibility announcement.

Inherits: `enum.Enum`

## Properties

- **ASSERTIVE** - The assistive technology will interrupt any announcement that it is currently making to notify the user about the change.
- **POLITE** - The assistive technology will speak changes whenever the user is idle.

## Examples

### Assertiveness showcase

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 semantics = ft.SemanticsService()

 def showcase_card(assertiveness: ft.Assertiveness) -> ft.Container:
 status = ft.Text("Send an accessibility announcement.", size=11)

 async def announce():
 try:
 await semantics.announce_message(
 f"Announcement with {assertiveness.name} assertiveness.",
 assertiveness=assertiveness,
)
 status.value = f"Sent with {assertiveness.value}."
 except Exception as ex:
 status.value = f"Error: {ex}"
 status.update()

 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(assertiveness.name, weight=ft.FontWeight.BOLD),
 ft.Text(
 assertiveness.value,
 size=11,
 color=ft.Colors.ON_SURFACE_VARIANT,
),
 ft.Button(
 "Announce message",
 icon=ft.Icons.RECORD_VOICE_OVER,
 on_click=lambda: page.run_task(announce),
),
 status,
],
),
)

 page.appbar = ft.AppBar(title="Assertiveness Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare announcement assertiveness levels. "
 "Enable a screen reader to hear the difference."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
```

```

 controls=[
 showcase_card(assertiveness)
 for assertiveness in ft.Assertiveness
],
),
]

)

)

)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildASSERTIVEclass-attributeinstance-attribute

The assistive technology will interrupt any announcement that it is currently making to notify the user about the change.

It should only be used for time-sensitive/critical notifications.

### buildPOLITEclass-attributeinstance-attribute

The assistive technology will speak changes whenever the user is idle.

# AutoCompleteSelectEvent

Event representing the selection of a suggestion in the AutoComplete control.

Inherits: Event

## Properties

- `index` - The index of the selected suggestion from the corresponding `AutoComplete.suggestions` list.
- `selection` - The selected suggestion.

## Properties

### `buildindexinstance-attribute`

`index: int`

The index of the selected suggestion from the corresponding `AutoComplete.suggestions` list.

### `buildselectioninstance-attribute`

`selection: AutoCompleteSuggestion`

The selected suggestion.

# AutoCompleteSuggestion

Represents a suggestion item for the AutoComplete control.

## Properties

- **key** - A unique identifier or value used for filtering and selection.
- **value** - The display text shown to the user.

## Properties

### buildkeyinstance-attribute

`key: str`

A unique identifier or value used for filtering and selection.

### buildvalueinstance-attribute

`value: str`

The display text shown to the user.

# AutoFillGroupDisposeAction

Predefined autofill context clean-up actions.

Inherits: `enum.Enum`

## Properties

- `CANCEL` - Destroys the current autofill context without saving the user input.
- `COMMIT` - Destroys the current autofill context after informing the platform to save the user input from it.

## Properties

`buildCANCEL`class-attributeinstance-attribute

Destroys the current autofill context without saving the user input.

`buildCOMMIT`class-attributeinstance-attribute

Destroys the current autofill context after informing the platform to save the user input from it.

# AutoFillHint

Predefined autofill hint identifiers for text fields.

Each enum value represents semantic input meaning (for example city, email, or password) and is translated to platform-specific autofill constants when available. On unsupported platforms, the raw hint string value is used as-is.

Inherits: `enum.Enum`

## Properties

- `ADDRESS_CITY` - The input field expects an address locality (city/town).
- `ADDRESS_CITY_AND_STATE` - The input field expects a city name combined with a state name.
- `ADDRESS_STATE` - The input field expects a region/state.
- `BIRTHDAY` - The input field expects a person's full birth date.
- `BIRTHDAY_DAY` - The input field expects a person's birth day(of the month).
- `BIRTHDAY_MONTH` - The input field expects a person's birth month.
- `BIRTHDAY_YEAR` - The input field expects a person's birth year.
- `COUNTRY_CODE` - The input field expects an ISO 3166-1-alpha-2 country code.
- `COUNTRY_NAME` - The input field expects a country name.
- `CREDIT_CARD_EXPIRATION_DATE` - The input field expects a credit card expiration date.
- `CREDIT_CARD_EXPIRATION_DAY` - The input field expects a credit card expiration day.
- `CREDIT_CARD_EXPIRATION_MONTH` - The input field expects a credit card expiration month.
- `CREDIT_CARD_EXPIRATION_YEAR` - The input field expects a credit card expiration year.
- `CREDIT_CARD_FAMILY_NAME` - The input field expects the holder's last/family name as given on a credit card.
- `CREDIT_CARD_GIVEN_NAME` - The input field expects the holder's first/given name as given on a credit card.
- `CREDIT_CARD_MIDDLE_NAME` - The input field expects the holder's middle name as given on a credit card.
- `CREDIT_CARD_NAME` - The input field expects the holder's full name as given on a credit card.
- `CREDIT_CARD_NUMBER` - The input field expects a credit card number.
- `CREDIT_CARD_SECURITY_CODE` - The input field expects a credit card security code.
- `CREDIT_CARD_TYPE` - The input field expects the type of a credit card, for example "Visa".
- `EMAIL` - The input field expects an email address.
- `FAMILY_NAME` - The input field expects a person's last/family name.
- `FULL_STREET_ADDRESS` - The input field expects a street address that fully identifies a location.
- `GENDER` - The input field expects a gender.
- `GIVEN_NAME` - The input field expects a person's first/given name.
- `IMPP` - The input field expects a URL representing an instant messaging protocol endpoint.
- `JOB_TITLE` - The input field expects a job title.
- `LANGUAGE` - The input field expects the preferred language of the user.
- `LOCATION` - The input field expects a location, such as a point of interest, an address,or another way to identify a location.
- `MIDDLE_INITIAL` - The input field expects a person's middle initial.
- `MIDDLE_NAME` - The input field expects a person's middle name.
- `NAME` - The input field expects a person's full name.
- `NAME_PREFIX` - The input field expects a person's name prefix or title, such as "Dr.".



- NAME\_SUFFIX - The input field expects a person's name suffix, such as "Jr."
- NEW\_PASSWORD - The input field expects a newly created password for save/update.
- NEW\_USERNAME - The input field expects a newly created username for save/update.
- NICKNAME - The input field expects a nickname.
- ONE\_TIME\_CODE - The input field expects a SMS one-time code.
- ORGANIZATION\_NAME - The input field expects an organization name corresponding to the person, address, or contact information in the other fields associated with this field.
- PASSWORD - The input field expects a password.
- PHOTO - The input field expects a photograph, icon, or other image corresponding to the company, person, address, or contact information in the other fields associated with this field.
- POSTAL\_ADDRESS - The input field expects a postal address.
- POSTAL\_ADDRESS\_EXTENDED - The input field expects an auxiliary address details.
- POSTAL\_ADDRESS\_EXTENDED\_POSTAL\_CODE - The input field expects an extended ZIP/POSTAL code.
- POSTAL\_CODE - The input field expects a postal code.
- STREET\_ADDRESS\_LEVEL1 - The first administrative level in the address.
- STREET\_ADDRESS\_LEVEL2 - The second administrative level, in addresses with at least two of them.
- STREET\_ADDRESS\_LEVEL3 - The third administrative level, in addresses with at least three administrative levels.
- STREET\_ADDRESS\_LEVEL4 - The finest-grained administrative level, in addresses which have four levels.
- STREET\_ADDRESS\_LINE1 - The input field expects the first line of a street address.
- STREET\_ADDRESS\_LINE2 - The input field expects the second line of a street address.
- STREET\_ADDRESS\_LINE3 - The input field expects the third line of a street address.
- SUB\_LOCALITY - The input field expects a sublocality.
- TELEPHONE\_NUMBER - The input field expects a telephone number.
- TELEPHONE\_NUMBER\_AREA\_CODE - The input field expects a phone number's area code, with a country-internal prefix applied if applicable.
- TELEPHONE\_NUMBER\_COUNTRY\_CODE - The input field expects a phone number's country code.
- TELEPHONE\_NUMBER\_DEVICE - The input field expects the current device's phone number, usually for Sign Up / OTP flows.
- TELEPHONE\_NUMBER\_EXTENSION - The input field expects a phone number's internal extension code.
- TELEPHONE\_NUMBER\_LOCAL - The input field expects a phone number without the country code and area code components.
- TELEPHONE\_NUMBER\_LOCAL\_PREFIX - The input field expects the first part of the component of the telephone number that follows the area code, when that component is split into two components.
- TELEPHONE\_NUMBER\_LOCAL\_SUFFIX - The input field expects the second part of the component of the telephone number that follows the area code, when that component is split into two components.
- TELEPHONE\_NUMBER\_NATIONAL - The input field expects a phone number without country code.
- TRANSACTION\_AMOUNT - The amount that the user would like for the transaction (e.g.
- TRANSACTION\_CURRENCY - The currency that the user would prefer the transaction to use, in ISO 4217 currency code.
- URL - The input field expects a URL.
- USERNAME - The input field expects a username or an account name.

## Properties

buildADDRESS\_CITYclass-attributeinstance-attribute

The input field expects an address locality (city/town).

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_POSTAL_ADDRESS_LOCALITY`.
- iOS: `addressCity`.
- Otherwise, the hint string will be used as-is.

### **buildADDRESS\_CITY\_AND\_STATEclass-attributeinstance-attribute**

The input field expects a city name combined with a state name.

This hint will be translated to the below values on different platforms:

- iOS: `addressCityAndState`.
- Otherwise, the hint string will be used as-is.

### **buildADDRESS\_STATEclass-attributeinstance-attribute**

The input field expects a region/state.

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_POSTAL_ADDRESS_REGION`.
- iOS: `addressState`.
- Otherwise, the hint string will be used as-is.

### **buildBIRTHDAYclass-attributeinstance-attribute**

The input field expects a person's full birth date.

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_BIRTH_DATE_FULL`.
- web: `"bday"`.
- Otherwise, the hint string will be used as-is.

### **buildBIRTHDAY\_DAYclass-attributeinstance-attribute**

The input field expects a person's birth day(of the month).

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_BIRTH_DATE_DAY`.
- web: `"bday-day"`.
- Otherwise, the hint string will be used as-is.

### **buildBIRTHDAY\_MONTHclass-attributeinstance-attribute**

The input field expects a person's birth month.

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_BIRTH_DATE_MONTH`.
- web: `"bday-month"`.
- Otherwise, the hint string will be used as-is.

### **buildBIRTHDAY\_YEARclass-attributeinstance-attribute**

The input field expects a person's birth year.

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_BIRTH_DATE_YEAR`.
- web: `"bday-year"`.
- Otherwise, the hint string will be used as-is.

### **buildCOUNTRY\_CODEclass-attributeinstance-attribute**

The input field expects an ISO 3166-1-alpha-2 country code.

This hint will be translated to the below values on different platforms:

- web: "country".
- Otherwise, the hint string will be used as-is.

### **buildCOUNTRY\_NAMEclass-attributeinstance-attribute**

The input field expects a country name.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_POSTAL\_ADDRESS\_COUNTRY.
- iOS: countryName.
- web: "country-name".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_EXPIRATION\_DATEclass-attributeinstance-attribute**

The input field expects a credit card expiration date.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_CREDIT\_CARD\_NUMBER.
- web: "cc-exp".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_EXPIRATION\_DAYclass-attributeinstance-attribute**

The input field expects a credit card expiration day.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_CREDIT\_CARD\_EXPIRATION\_DAY.
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_EXPIRATION\_MONTHclass-attributeinstance-attribute**

The input field expects a credit card expiration month.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_CREDIT\_CARD\_EXPIRATION\_MONTH.
- web: "cc-exp-month".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_EXPIRATION\_YEARclass-attributeinstance-attribute**

The input field expects a credit card expiration year.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_CREDIT\_CARD\_EXPIRATION\_YEAR.
- web: "cc-exp-year".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_FAMILY\_NAMEclass-attributeinstance-attribute**

The input field expects the holder's last/family name as given on a credit card.

This hint will be translated to the below values on different platforms:

- web: "cc-family-name".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_GIVEN\_NAMEclass-attributeinstance-attribute**

The input field expects the holder's first/given name as given on a credit card.

This hint will be translated to the below values on different platforms:

- web: "cc-given-name".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_MIDDLE\_NAMEclass-attributeinstance-attribute**

The input field expects the holder's middle name as given on a credit card.

This hint will be translated to the below values on different platforms:

- web: "cc-additional-name".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_NAMEclass-attributeinstance-attribute**

The input field expects the holder's full name as given on a credit card.

This hint will be translated to the below values on different platforms:

- web: "cc-name".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_NUMBERclass-attributeinstance-attribute**

The input field expects a credit card number.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_CREDIT\_CARD\_NUMBER.
- iOS: creditCardNumber.
- web: "cc-number".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_SECURITY\_CODEclass-attributeinstance-attribute**

The input field expects a credit card security code.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_CREDIT\_CARD\_SECURITY\_CODE.
- web: "cc-csc".
- Otherwise, the hint string will be used as-is.

### **buildCREDIT\_CARD\_TYPEclass-attributeinstance-attribute**

The input field expects the type of a credit card, for example "Visa".

This hint will be translated to the below values on different platforms:

- web: "cc-type".
- Otherwise, the hint string will be used as-is.

### **buildEMAILclass-attributeinstance-attribute**

The input field expects an email address.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_EMAIL\_ADDRESS.
- iOS: emailAddress.
- web: "email".
- Otherwise, the hint string will be used as-is.

### **buildFAMILY\_NAMEclass-attributeinstance-attribute**

The input field expects a person's last/family name.

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_PERSON_NAME_FAMILY`.
- iOS: `familyName`.
- web: `"family-name"`.
- Otherwise, the hint string will be used as-is.

### **buildFULL\_STREET\_ADDRESSclass-attributeinstance-attribute**

The input field expects a street address that fully identifies a location.

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_POSTAL_ADDRESS_STREET_ADDRESS`.
- iOS: `fullStreetAddress`.
- web: `"street-address"`.
- Otherwise, the hint string will be used as-is.

### **buildGENDERclass-attributeinstance-attribute**

The input field expects a gender.

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_GENDER`.
- web: `"sex"`.
- Otherwise, the hint string will be used as-is.

### **buildGIVEN\_NAMEclass-attributeinstance-attribute**

The input field expects a person's first/given name.

This hint will be translated to the below values on different platforms:

- Android: `AUTOFILL_HINT_PERSON_NAME_GIVEN`.
- iOS: `givenName`.
- web: `"given-name"`.
- Otherwise, the hint string will be used as-is.

### **buildIMPPclass-attributeinstance-attribute**

The input field expects a URL representing an instant messaging protocol endpoint.

This hint will be translated to the below values on different platforms:

- web: `"impp"`.
- Otherwise, the hint string will be used as-is.

### **buildJOB\_TITLEclass-attributeinstance-attribute**

The input field expects a job title.

This hint will be translated to the below values on different platforms:

- iOS: `jobTitle`.
- web: `"organization-title"`.
- Otherwise, the hint string will be used as-is.

### **buildLANGUAGEclass-attributeinstance-attribute**

The input field expects the preferred language of the user.

This hint will be translated to the below values on different platforms:

- web: "language".
- Otherwise, the hint string will be used as-is.

### **buildLOCATIONclass-attributeinstance-attribute**

The input field expects a location, such as a point of interest, an address, or another way to identify a location.

This hint will be translated to the below values on different platforms:

- iOS: location.
- Otherwise, the hint string will be used as-is.

### **buildMIDDLE\_INITIALclass-attributeinstance-attribute**

The input field expects a person's middle initial.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PERSON\_NAME\_MIDDLE\_INITIAL.
- Otherwise, the hint string will be used as-is.

### **buildMIDDLE\_NAMEclass-attributeinstance-attribute**

The input field expects a person's middle name.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PERSON\_NAME\_MIDDLE.
- iOS: middleName.
- web: "additional-name".
- Otherwise, the hint string will be used as-is.

### **buildNAMEclass-attributeinstance-attribute**

The input field expects a person's full name.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PERSON\_NAME.
- iOS: name.
- web: "name".
- Otherwise, the hint string will be used as-is.

### **buildNAME\_PREFIXclass-attributeinstance-attribute**

The input field expects a person's name prefix or title, such as "Dr.".

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PERSON\_NAME\_PREFIX.
- iOS: namePrefix.
- web: "honorific-prefix".
- Otherwise, the hint string will be used as-is.

### **buildNAME\_SUFFIXclass-attributeinstance-attribute**

The input field expects a person's name suffix, such as "Jr.".

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PERSON\_NAME\_SUFFIX.
- iOS: nameSuffix.
- web: "honorific-suffix".

- Otherwise, the hint string will be used as-is.

### **buildNEW\_PASSWORDclass-attributeinstance-attribute**

The input field expects a newly created password for save/update.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_NEW\_PASSWORD.
- iOS: newPassword.
- web: "new-password".
- Otherwise, the hint string will be used as-is.

### **buildNEW\_USERNAMEclass-attributeinstance-attribute**

The input field expects a newly created username for save/update.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_NEW\_USERNAME.
- Otherwise, the hint string will be used as-is.

### **buildNICKNAMEclass-attributeinstance-attribute**

The input field expects a nickname.

This hint will be translated to the below values on different platforms:

- iOS: nickname.
- web: "nickname".
- Otherwise, the hint string will be used as-is.

### **buildONE\_TIME\_CODEclass-attributeinstance-attribute**

The input field expects a SMS one-time code.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_SMS\_OTP.
- iOS: oneTimeCode.
- web: "one-time-code".
- Otherwise, the hint string will be used as-is.

### **buildORGANIZATION\_NAMEclass-attributeinstance-attribute**

The input field expects an organization name corresponding to the person, address, or contact information in the other fields associated with this field.

This hint will be translated to the below values on different platforms:

- iOS: organizationName.
- web: "organization".
- Otherwise, the hint string will be used as-is.

### **buildPASSWORDclass-attributeinstance-attribute**

The input field expects a password.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PASSWORD.
- iOS: password.
- web: "current-password".
- Otherwise, the hint string will be used as-is.

## buildPHOTOclass-attributeinstance-attribute

The input field expects a photograph, icon, or other image corresponding to the company, person, address, or contact information in the other fields associated with this field.

This hint will be translated to the below values on different platforms:

- web: "photo".
- Otherwise, the hint string will be used as-is.

## buildPOSTAL\_ADDRESSclass-attributeinstance-attribute

The input field expects a postal address.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_POSTAL\_ADDRESS.
- Otherwise, the hint string will be used as-is.

## buildPOSTAL\_ADDRESS\_EXTENDEDclass-attributeinstance-attribute

The input field expects an auxiliary address details.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_POSTAL\_ADDRESS\_EXTENDED\_ADDRESS.
- Otherwise, the hint string will be used as-is.

## buildPOSTAL\_ADDRESS\_EXTENDED\_POSTAL\_CODEclass-attributeinstance-attribute

The input field expects an extended ZIP/POSTAL code.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_POSTAL\_ADDRESS\_EXTENDED\_POSTAL\_CODE.
- Otherwise, the hint string will be used as-is.

## buildPOSTAL\_CODEclass-attributeinstance-attribute

The input field expects a postal code.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_POSTAL\_CODE.
- iOS: postalCode.
- web: "postal-code".
- Otherwise, the hint string will be used as-is.

## buildSTREET\_ADDRESS\_LEVEL1class-attributeinstance-attribute

The first administrative level in the address.

This is typically the province in which the address is located. In the United States, this would be the state. In Switzerland, the canton. In the United Kingdom, the post town.

This hint will be translated to the below values on different platforms:

- web: "address-level1".
- Otherwise, the hint string will be used as-is.

## buildSTREET\_ADDRESS\_LEVEL2class-attributeinstance-attribute

The second administrative level, in addresses with at least two of them. In countries with two administrative levels, this would typically be the city, town, village, or other locality in which the address is located.

This hint will be translated to the below values on different platforms:

- web: "address-level2".
- Otherwise, the hint string will be used as-is.



### **buildSTREET\_ADDRESS\_LEVEL3class-attributeinstance-attribute**

The third administrative level, in addresses with at least three administrative levels.

This hint will be translated to the below values on different platforms:

- web: "address-level3".
- Otherwise, the hint string will be used as-is.

### **buildSTREET\_ADDRESS\_LEVEL4class-attributeinstance-attribute**

The finest-grained administrative level, in addresses which have four levels.

This hint will be translated to the below values on different platforms:

- web: "address-level4".
- Otherwise, the hint string will be used as-is.

### **buildSTREET\_ADDRESS\_LINE1class-attributeinstance-attribute**

The input field expects the first line of a street address.

This hint will be translated to the below values on different platforms:

- iOS: streetAddressLine1.
- web: "address-line1".
- Otherwise, the hint string will be used as-is.

### **buildSTREET\_ADDRESS\_LINE2class-attributeinstance-attribute**

The input field expects the second line of a street address.

This hint will be translated to the below values on different platforms:

- iOS: streetAddressLine2. As of iOS 14.2 this hint does not trigger autofill.
- web: "address-line2".
- Otherwise, the hint string will be used as-is.

### **buildSTREET\_ADDRESS\_LINE3class-attributeinstance-attribute**

The input field expects the third line of a street address.

This hint will be translated to the below values on different platforms:

- web: "address-line3".
- Otherwise, the hint string will be used as-is.

### **buildSUB\_LOCALITYclass-attributeinstance-attribute**

The input field expects a sublocality.

This hint will be translated to the below values on different platforms:

- iOS: sublocality.
- Otherwise, the hint string will be used as-is.

### **buildTELEPHONE\_NUMBERclass-attributeinstance-attribute**

The input field expects a telephone number.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PHONE\_NUMBER.
- iOS: telephoneNumber.
- web: "tel".
- Otherwise, the hint string will be used as-is.

### **buildTELEPHONE\_NUMBER\_AREA\_CODEclass-attributeinstance-attribute**

The input field expects a phone number's area code, with a country-internal prefix applied if applicable.

This hint will be translated to the below values on different platforms:

- web: "tel-area-code".
- Otherwise, the hint string will be used as-is.

#### **buildTELEPHONE\_NUMBER\_COUNTRY\_CODEclass-attributeinstance-attribute**

The input field expects a phone number's country code.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PHONE\_COUNTRY\_CODE.
- web: "tel-country-code".
- Otherwise, the hint string will be used as-is.

#### **buildTELEPHONE\_NUMBER\_DEVICEclass-attributeinstance-attribute**

The input field expects the current device's phone number, usually for Sign Up / OTP flows.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PHONE\_NUMBER\_DEVICE.
- Otherwise, the hint string will be used as-is.

#### **buildTELEPHONE\_NUMBER\_EXTENSIONclass-attributeinstance-attribute**

The input field expects a phone number's internal extension code.

This hint will be translated to the below values on different platforms:

- web: "tel-extension".
- Otherwise, the hint string will be used as-is.

#### **buildTELEPHONE\_NUMBER\_LOCALclass-attributeinstance-attribute**

The input field expects a phone number without the country code and area code components.

This hint will be translated to the below values on different platforms:

- web: "tel-local".
- Otherwise, the hint string will be used as-is.

#### **buildTELEPHONE\_NUMBER\_LOCAL\_PREFIXclass-attributeinstance-attribute**

The input field expects the first part of the component of the telephone number that follows the area code, when that component is split into two components.

This hint will be translated to the below values on different platforms:

- web: "tel-local-prefix".
- Otherwise, the hint string will be used as-is.

#### **buildTELEPHONE\_NUMBER\_LOCAL\_SUFFIXclass-attributeinstance-attribute**

The input field expects the second part of the component of the telephone number that follows the area code, when that component is split into two components.

This hint will be translated to the below values on different platforms:

- web: "tel-local-suffix".
- Otherwise, the hint string will be used as-is.

#### **buildTELEPHONE\_NUMBER\_NATIONALclass-attributeinstance-attribute**

The input field expects a phone number without country code.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_PHONE\_NATIONAL.

- web: "tel-national".
- Otherwise, the hint string will be used as-is.

### **buildTRANSACTION\_AMOUNTclass-attributeinstance-attribute**

The amount that the user would like for the transaction (e.g. when entering a bid or sale price).

This hint will be translated to the below values on different platforms:

- web: "transaction-amount".
- Otherwise, the hint string will be used as-is.

### **buildTRANSACTION\_CURRENCYclass-attributeinstance-attribute**

The currency that the user would prefer the transaction to use, in ISO 4217 currency code.

This hint will be translated to the below values on different platforms:

- web: "transaction-currency".
- Otherwise, the hint string will be used as-is.

### **buildURLclass-attributeinstance-attribute**

The input field expects a URL.

This hint will be translated to the below values on different platforms:

- iOS: URL.
- web: "url".
- Otherwise, the hint string will be used as-is.

### **buildUSERNAMEclass-attributeinstance-attribute**

The input field expects a username or an account name.

This hint will be translated to the below values on different platforms:

- Android: AUTOFILL\_HINT\_USERNAME.
- iOS: username.
- web: "username".
- Otherwise, the hint string will be used as-is.

# AutomaticNotchShape

A notch shape created from ShapeBorders.

Inherits: NotchShape

## Properties

- guest
- host

## Properties

**buildguestclass-attributeinstance-attribute**

guest: ShapeBorder | None = None

**buildhostinstance-attribute**

host: ShapeBorder

# Axis

The two cardinal directions in two dimensions.

The axis is always relative to the current coordinate space. This means, for example, that a [horizontal] axis might actually be diagonally from top right to bottom left, due to some local transform applied to the scene.

Inherits: enum.Enum

## Properties

- HORIZONTAL - Left and right.
- VERTICAL - Up and down.

## Examples

### Axis showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(axis: ft.Axis) -> ft.Container:
 return ft.Container(
 width=350 if axis == ft.Axis.HORIZONTAL else 220,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Text(axis.name, weight=ft.FontWeight.BOLD),
 ft.SegmentedButton(
 direction=axis,
 selected=["medium"],
 segments=[
 ft.Segment(value="small", label="Small"),
 ft.Segment(value="medium", label="Medium"),
 ft.Segment(value="large", label="Large"),
],
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="Axis Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare horizontal vs vertical segment layout."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(axis) for axis in ft.Axis],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

buildHORIZONTALclass-attributeinstance-attribute

Left and right.

buildVERTICALclass-attributeinstance-attribute

Up and down.

# Badge

Badges are used to show notifications, counts, or status information on navigation items such as `NavigationBar` or `NavigationRail` destinations or a button's icon.

```
ft.FilledIconButton(
 icon=ft.Icons.PHONE,
 badge=ft.Badge(label="3"),
)
```

Image: Badge - [https://flet.dev/docs/test-images/examples/controls/material/golden/macros/badge/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/controls/material/golden/macros/badge/image_for_docs.png)

## Badge

Inherits: `BaseControl`

## Properties

- `alignment` - Aligns the label relative to the content of the badge.
- `bgcolor` - The background color of the label.
- `label` - The label of this badge.
- `label_visible` - Whether the label should be visible.
- `large_size` - The badge's label height if label is provided.
- `offset` - Combined with `alignment` to determine the location of the label relative to the content.
- `padding` - The padding added to the label.
- `small_size` - The badge's label diameter if label is not provided.
- `text_color` - The color of the text shown in the label.
- `text_style` - The text style to use for text in the label.

## Examples

### Badge

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Badge example"
 page.navigation_bar = ft.NavigationBar(
 destinations=[
 ft.NavigationBarDestination(
 icon=ft.Icon(
 ft.Icons.EXPLORE,
 badge=ft.Badge(small_size=10),
),
 label="Explore",
),
 ft.NavigationBarDestination(
 icon=ft.Icons.COMMUTE,
 label="Commute",
),
 ft.NavigationBarDestination(
 icon=ft.Icon(
 ft.Icons.PHONE,
 badge="10",
),
),
],
)
 page.add(ft.SafeArea(content=ft.Column(controls=[ft.Text("Body!")])))

if __name__ == "__main__":
 ft.run(main)
```

Image: Image - <https://flet.dev/docs/test-images/examples/controls/material/golden/macros/badge/basic.png>

## Properties

## buildalignmentclass-attributeinstance-attribute

`alignment: Alignment | None = None`

Aligns the label relative to the content of the badge.

The alignment positions the label in similar way `Container.content` is positioned using `Container.alignment`, except that the badge alignment is resolved as if the label was a `large_size` square and offset is added to the result.

Note

Has effect only used if label is also provided.

## buildbgcolorclass-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

The background color of the label.

## buildlabelclass-attributeinstance-attribute

`label: StrOrControl | None = None`

The label of this badge.

Typically a 1 to 4 characters text.

If the label is not provided, the badge is shown as a filled circle of `small_size` diameter.

If label is provided, it is a `StadiumBorder` shaped badge with height equal to `large_size`.

## buildlabel\_visibleclass-attributeinstance-attribute

`label_visible: bool = True`

Whether the label should be visible.

It can be used to create a badge only shown under certain conditions.

## buildlarge\_sizeclass-attributeinstance-attribute

`large_size: Number | None = None`

The badge's label height if label is provided.

If the default value is overridden then it may be useful to also override padding and alignment.

Defaults to `BadgeTheme.large_size`, or if that is `None`, falls back to 16.

## buildoffsetclass-attributeinstance-attribute

`offset: OffsetValue | None = None`

Combined with alignment to determine the location of the label relative to the content.

Note

Has effect only used if label is also provided.

## buildpaddingclass-attributeinstance-attribute

`padding: PaddingValue | None = None`

The padding added to the label.

Defaults to `BadgeTheme.padding`, or if that is `None`, falls back to 4 pixels on the left and right.

Note

Has effect only if label is not `None`.

## buildsmall\_sizeclass-attributeinstance-attribute

`small_size: Number | None = None`

The badge's label diameter if label is not provided.

Defaults to `BadgeTheme.small_size`, or if that is `None`, falls back to 6.

## buildtext\_colorclass-attributeinstance-attribute

`text_color: ColorValue | None = None`

The color of the text shown in the label.



It overrides the color of the label's `text_style`.

**buildtext\_style**class-attributeinstance-attribute

`text_style: TextStyle | None = None`

The text style to use for text in the label.

# BadgeTheme

Defines default property values for descendant Badge controls.

## Properties

- `alignment` - Overrides the default value of `Badge.alignment` in all descendant Badge controls.
- `bgcolor` - Overrides the default value of `Badge.bgcolor` in all descendant Badge controls.
- `large_size` - Overrides the default value of `Badge.large_size` in all descendant Badge controls.
- `offset` - Overrides the default value of `Badge.offset` in all descendant Badge controls.
- `padding` - Overrides the default value of `Badge.padding` in all descendant Badge controls.
- `small_size` - Overrides the default value of `Badge.small_size` in all descendant Badge controls.
- `text_color` - Overrides the default value of `Badge.text_color` in all descendant Badge controls.
- `text_style` - Overrides the default value of `Badge.text_style` in all descendant Badge controls.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

`alignment: Alignment | None = None`

Overrides the default value of `Badge.alignment` in all descendant Badge controls.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `Badge.bgcolor` in all descendant Badge controls.

### `buildlarge_sizeclass-attributeinstance-attribute`

`large_size: Number | None = None`

Overrides the default value of `Badge.large_size` in all descendant Badge controls.

### `buildoffsetclass-attributeinstance-attribute`

`offset: OffsetValue | None = None`

Overrides the default value of `Badge.offset` in all descendant Badge controls.

### `buildpaddingclass-attributeinstance-attribute`

`padding: PaddingValue | None = None`

Overrides the default value of `Badge.padding` in all descendant Badge controls.

### `buildsmall_sizeclass-attributeinstance-attribute`

`small_size: Number | None = None`

Overrides the default value of `Badge.small_size` in all descendant Badge controls.

### `buildtext_colorclass-attributeinstance-attribute`

`text_color: ColorValue | None = None`

Overrides the default value of `Badge.text_color` in all descendant Badge controls.

### `buildtext_styleclass-attributeinstance-attribute`

`text_style: TextStyle | None = None`

Overrides the default value of `Badge.text_style` in all descendant Badge controls.

# BannerTheme

Defines default property values for descendant Banner controls.

## Properties

- `bgcolor` - Overrides the default value of `Banner.bgcolor` in all descendant Banner controls.
- `content_text_style` - Overrides the default value of `Banner.content_text_style` in all descendant Banner controls.
- `divider_color` - Overrides the default value of `Banner.divider_color` in all descendant Banner controls.
- `elevation` - Overrides the default value of `Banner.elevation` in all descendant Banner controls.
- `leading_padding` - Overrides the default value of `Banner.leading_padding` in all descendant Banner controls.
- `padding` - Overrides the default value of `Banner.content_padding` in all descendant Banner controls.
- `shadow_color` - Overrides the default value of `Banner.shadow_color` in all descendant Banner controls.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `Banner.bgcolor` in all descendant Banner controls.

### `buildcontent_text_styleclass-attributeinstance-attribute`

`content_text_style: TextStyle | None = None`

Overrides the default value of `Banner.content_text_style` in all descendant Banner controls.

### `bullddivider_colorclass-attributeinstance-attribute`

`divider_color: ColorValue | None = None`

Overrides the default value of `Banner.divider_color` in all descendant Banner controls.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value of `Banner.elevation` in all descendant Banner controls.

### `buildleading_paddingclass-attributeinstance-attribute`

`leading_padding: PaddingValue | None = None`

Overrides the default value of `Banner.leading_padding` in all descendant Banner controls.

### `buildpaddingclass-attributeinstance-attribute`

`padding: PaddingValue | None = None`

Overrides the default value of `Banner.content_padding` in all descendant Banner controls.

### `buildshadow_colorclass-attributeinstance-attribute`

`shadow_color: ColorValue | None = None`

Overrides the default value of `Banner.shadow_color` in all descendant Banner controls.

# BarometerReadingEvent

A sensor sample from a barometer.

Barometers measure the atmospheric pressure surrounding the sensor, returning values in hectopascals hPa.

Consider that these samples may be affected by altitude and weather conditions, and can be used to predict short-term weather changes or determine altitude.

Note that water-resistant phones or similar sealed devices may experience pressure fluctuations as the device is held or used, due to changes in pressure caused by handling the device.

An altimeter is an example of a general utility for barometer data.

Inherits: Event

## Properties

- pressure - Atmospheric pressure reading, in hectopascals (hPa).
- timestamp - Event timestamp.

## Properties

### buildpressureinstance-attribute

pressure: float

Atmospheric pressure reading, in hectopascals (hPa).

### buildtimestampinstance-attribute

timestamp: datetime

Event timestamp.

# BatteryState

Battery state.

Inherits: `enum.Enum`

## Properties

- `CHARGING` - The battery is currently charging.
- `CONNECTED_NOT_CHARGING` - The battery is connected to a power source but not charging.
- `DISCHARGING` - The battery is discharging.
- `FULL` - The battery is fully charged.
- `UNKNOWN` - The battery state is unknown.

## Properties

`buildCHARGINGclass-attributeinstance-attribute`

The battery is currently charging.

`buildCONNECTED_NOT_CHARGINGclass-attributeinstance-attribute`

The battery is connected to a power source but not charging.

`buildDISCHARGINGclass-attributeinstance-attribute`

The battery is discharging.

`buildFULLclass-attributeinstance-attribute`

The battery is fully charged.

`buildUNKNOWNclass-attributeinstance-attribute`

The battery state is unknown.

# BatteryStateChangeEvent

Event fired when battery state changes.

Inherits: Event

## Properties

- state - Current battery state.

## Properties

### buildstateinstance-attribute

state: BatteryState

Current battery state.

# BeveledRectangleBorder

A border with beveled rectangle corners.

Inherits: `RoundedRectangleBorder`

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Methods

### `deployed_code``copy`

```
copy(
 side: BorderSide | None = None,
 radius: BorderRadiusValue | None = None,
) -> BeveledRectangleBorder
```

Returns a copy of this object with the specified properties overridden.

# BlendMode

See BlendMode from Flutter documentation for blend mode examples.

Inherits: `enum.Enum`

## Properties

- CLEAR
- COLOR
- COLOR\_BURN
- COLOR\_DODGE
- DARKEN
- DIFFERENCE
- DST
- DST\_A\_TOP
- DST\_IN
- DST\_OUT
- DST\_OVER
- EXCLUSION
- HARD\_LIGHT
- HUE
- LIGHTEN
- LUMINOSITY
- MODULATE
- MULTIPLY
- OVERLAY
- PLUS
- SATURATION
- SCREEN
- SOFT\_LIGHT
- SRC
- SRC\_A\_TOP
- SRC\_IN
- SRC\_OUT
- SRC\_OVER
- VALUES
- XOR

## Examples

### Blend Mode showcase

`play_arrow` Try Online

```
import flet as ft
```

```
def showcase_card(blend_mode: ft.BlendMode) -> ft.Container:
```



```

return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(blend_mode.name, weight=ft.FontWeight.BOLD),
 ft.Image(
 src="https://picsum.photos/id/237/200/300",
 width=240,
 height=130,
 fit=ft.BoxFit.COVER,
 color=ft.Colors.LIGHT_GREEN_200,
 color_blend_mode=blend_mode,
 border_radius=8,
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="BlendMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare color blending results for each BlendMode value."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(blend_mode) for blend_mode in ft.BlendMode
],
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

buildCLEARclass-attributeinstance-attribute

buildCOLORclass-attributeinstance-attribute

buildCOLOR\_BURNclass-attributeinstance-attribute

buildCOLOR\_DODGEclass-attributeinstance-attribute

buildDARKENclass-attributeinstance-attribute

buildDIFFERENCEclass-attributeinstance-attribute

buildDSTclass-attributeinstance-attribute

buildDST\_A\_TOPclass-attributeinstance-attribute

buildDST\_INclass-attributeinstance-attribute

buildDST\_OUTclass-attributeinstance-attribute

buildDST\_OVERclass-attributeinstance-attribute

buildEXCLUSIONclass-attributeinstance-attribute

buildHARD\_LIGHTclass-attributeinstance-attribute

buildHUEclass-attributeinstance-attribute  
buildLIGHTENclass-attributeinstance-attribute  
buildLUMINOSITYclass-attributeinstance-attribute  
buildMODULATEclass-attributeinstance-attribute  
buildMULTIPLYclass-attributeinstance-attribute  
buildOVERLAYclass-attributeinstance-attribute  
buildPLUSclass-attributeinstance-attribute  
buildSATURATIONclass-attributeinstance-attribute  
buildSCREENclass-attributeinstance-attribute  
buildSOFT\_LIGHTclass-attributeinstance-attribute  
buildSRCclass-attributeinstance-attribute  
buildSRC\_A\_TOPclass-attributeinstance-attribute  
buildSRC\_INclass-attributeinstance-attribute  
buildSRC\_OUTclass-attributeinstance-attribute  
buildSRC\_OVERclass-attributeinstance-attribute  
buildVALUESclass-attributeinstance-attribute  
buildXORclass-attributeinstance-attribute

# Blur

Gaussian blur configuration.

## Properties

- `sigma_x` - Horizontal Gaussian sigma.
- `sigma_y` - Vertical Gaussian sigma.
- `tile_mode` - How sampling outside source bounds is handled during blur.

## Properties

### `buildsigma_xinstance-attribute`

`sigma_x`: Number

Horizontal Gaussian sigma.

Larger values produce stronger blur along the X axis.

### `buildsigma_yinstance-attribute`

`sigma_y`: Number

Vertical Gaussian sigma.

Larger values produce stronger blur along the Y axis.

### `buildtile_modeclass-attributeinstance-attribute`

`tile_mode`: `BlurTileMode` | `None` = `None`

How sampling outside source bounds is handled during blur.

## Examples

### Example 1

`play_arrow` Try Online

```
import flet as ft

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.LIGHT
 i = 1

 img_container = ft.Container(
 image=ft.DecorationImage(src="https://picsum.photos/300/300"),
 width=300,
 height=300,
)

 def handle_button_click(e: ft.Event[ft.Button]):
 nonlocal i
 img_container.image = ft.DecorationImage(
 src=f"https://picsum.photos/300/300?random={i}"
)
 i += 1
 page.update()

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Stack(
 controls=[
 img_container,
 ft.Container(
 width=100,
 height=100,
 blur=10,
 bgcolor="#22CCCC00",
),
 ft.Container(
 width=100,
 height=100,
```



# BlurStyle

Styles to use for blurs

Inherits: enum.Enum

## Properties

- INNER - Fuzzy inside, nothing outside.
- NORMAL - Fuzzy inside and outside.
- OUTER - Nothing inside, fuzzy outside.
- SOLID - Solid inside, fuzzy outside.

## Examples

### Blur Style showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(style: ft.BlurStyle) -> ft.Container:
 sample = ft.Container(
 width=130,
 height=90,
 border_radius=12,
 bgcolor=ft.Colors.BLUE_400,
 shadow=ft.BoxShadow(
 blur_radius=28,
 spread_radius=3,
 color=ft.Colors.RED_400,
 offset=ft.Offset(10, 10),
 blur_style=style,
),
)

 return ft.Container(
 width=300,
 height=180,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=12,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Text(style.name, weight=ft.FontWeight.BOLD),
 sample,
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="BlurStyle Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare shadow blur rendering styles. "
 "The blue box uses red shadow with selected blur style."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(style) for style in ft.BlurStyle],
),
],
)
)
)
```

)

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildINNERclass-attributeinstance-attribute**

Fuzzy inside, nothing outside. This can make shapes appear to be lit from within.

### **buildNORMALclass-attributeinstance-attribute**

Fuzzy inside and outside. This is useful for painting shadows that are offset from the shape that ostensibly is casting the shadow.

### **buildOUTERclass-attributeinstance-attribute**

Nothing inside, fuzzy outside. This is useful for painting shadows for partially transparent shapes, when they are painted separately but without an offset, so that the shadow doesn't paint below the shape.

### **buildSOLIDclass-attributeinstance-attribute**

Solid inside, fuzzy outside. This corresponds to drawing the shape, and additionally drawing the blur. This can make objects appear brighter, maybe even as if they were fluorescent.

# BlurTileMode

Edge sampling mode used when applying blur beyond source bounds.

Inherits: enum.Enum

## Properties

- CLAMP - Extends edge pixels outward.
- DECAL - Treats samples outside bounds as transparent.
- MIRROR - Repeats the image by mirroring at each edge.
- REPEATED - Repeats the image pattern without mirroring.

## Examples

### Blur Tile Mode showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def checkerboard() -> ft.Column:
 colors = [ft.Colors.TEAL_300, ft.Colors.AMBER_300]
 return ft.Column(
 spacing=0,
 controls=[
 ft.Row(
 spacing=0,
 controls=[
 ft.Container(
 width=24,
 height=24,
 bgcolor=colors[(row + col) % 2],
)
 for col in range(9)
],
)
 for row in range(6)
],
)

def showcase_card(mode: ft.BlurTileMode) -> ft.Container:
 preview = ft.Container(
 width=216,
 height=144,
 border_radius=8,
 clip_behavior=ft.ClipBehavior.ANTI_ALIAS,
 content=ft.Stack(
 controls=[
 checkerboard(),
 ft.Container(
 left=8,
 top=28,
 width=120,
 height=80,
 blur=ft.Blur(8, 8, mode),
 bgcolor="#55FFFFFF",
 border_radius=8,
 border=ft.Border.all(1, ft.Colors.WHITE),
),
],
),
)

 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=10,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Text(mode.name, weight=ft.FontWeight.BOLD),
 preview,
],
),
)
```

```

],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="BlurTileMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare blur edge sampling outside source bounds."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(mode) for mode in ft.BlurTileMode],
),
]
)
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildCLAMPclass-attributeinstance-attribute

Extends edge pixels outward.

### buildDECALclass-attributeinstance-attribute

Treats samples outside bounds as transparent.

### buildMIRRORclass-attributeinstance-attribute

Repeats the image by mirroring at each edge.

### buildREPEATEDclass-attributeinstance-attribute

Repeats the image pattern without mirroring.



# Border

A border comprised of four sides: top, right, bottom, left.

Each side of the border is an instance of BorderSide.

## Properties

- bottom - Bottom side of the border.
- left - Left side of the border.
- right - Right side of the border.
- top - Top side of the border.

## Methods

- all - Creates a border whose sides are all the same.
- copy - Returns a copy of this object with the specified properties overridden.
- only - Creates a border with explicit sides.
- symmetric - Creates a border with symmetrical vertical and horizontal sides.

## Properties

### buildbottomclass-attributeinstance-attribute

```
bottom: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

Bottom side of the border.

### buildleftclass-attributeinstance-attribute

```
left: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

Left side of the border.

### buildrightclass-attributeinstance-attribute

```
right: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

Right side of the border.

### buildtopclass-attributeinstance-attribute

```
top: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

Top side of the border.

## Methods

### deployed\_codeallclassmethod

```
all(
 width: Number | None = None,
 color: ColorValue | None = None,
 side: BorderSide | None = None,
) -> Border
```

Creates a border whose sides are all the same.

If side is not None, it gets used and both width and color are ignored.

Parameters:

- width (Number | None, default: None) - The side width. Used only when side is None.

- color (ColorValue | None, default: None) - The side color. Used only when side is None.
- side (BorderSide | None, default: None) - The BorderSide to apply to all sides. If set, it has precedence over width and color.

Returns:

- A (Border) - class:~flet.Border with identical sides.

## deployed\_codecopy

```
copy(
 left: BorderSide | None = None,
 top: BorderSide | None = None,
 right: BorderSide | None = None,
 bottom: BorderSide | None = None,
) -> Border
```

Returns a copy of this object with the specified properties overridden.

## deployed\_codeonlyclassmethod

```
only(
 left: BorderSide | None = None,
 top: BorderSide | None = None,
 right: BorderSide | None = None,
 bottom: BorderSide | None = None,
) -> Border
```

Creates a border with explicit sides.

Parameters:

- left (BorderSide | None, default: None) - The left side. Defaults to no border.
- top (BorderSide | None, default: None) - The top side. Defaults to no border.
- right (BorderSide | None, default: None) - The right side. Defaults to no border.
- bottom (BorderSide | None, default: None) - The bottom side. Defaults to no border.

Returns:

- A (Border) - class:~flet.Border built from the provided sides.

## deployed\_codesymmetricclassmethod

```
symmetric(
 vertical: BorderSide | None = None,
 horizontal: BorderSide | None = None,
) -> Border
```

Creates a border with symmetrical vertical and horizontal sides.

The vertical argument applies to the left and right sides, and the horizontal argument applies to the top and bottom sides.

Parameters:

- vertical (BorderSide | None, default: None) - The side applied to the left and right.
- horizontal (BorderSide | None, default: None) - The side applied to the top and bottom.

Returns:

- A (Border) - class:~flet.Border with mirrored side pairs.

# Examples

## Example 1

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Containers with different borders"

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Row(
```

```

controls=[
 ft.Container(
 bgcolor=ft.Colors.AMBER,
 padding=15,
 border=ft.Border.all(10, ft.Colors.PINK_600),
 border_radius=ft.BorderRadius.all(30),
 width=150,
 height=150,
),
 ft.Container(
 bgcolor=ft.Colors.DEEP_PURPLE,
 padding=15,
 border=ft.Border.all(3, ft.Colors.LIGHT_GREEN_ACCENT),
 border_radius=ft.BorderRadius.only(
 top_left=10, bottom_right=10
),
 width=150,
 height=150,
),
 ft.Container(
 bgcolor=ft.Colors.BLUE_GREY_900,
 padding=15,
 border=ft.Border.symmetric(
 vertical=ft.BorderSide(8, ft.Colors.YELLOW_800)
),
 width=150,
 height=150,
),
]

)
)
)
]
)

if __name__ == "__main__":
 ft.run(main)

```

# BorderRadius

Corner radii for a rectangle, defined for all four corners.

## Properties

- `bottom_left` - Radius of the bottom left border corner.
- `bottom_right` - Radius of the bottom right border corner.
- `top_left` - Radius of the top left border corner.
- `top_right` - Radius of the top right border corner.

## Methods

- `all` - Creates a `BorderRadius` where all radii are radius.
- `copy` - Returns a copy of this object with the specified properties overridden.
- `horizontal` - Creates a horizontally symmetrical `BorderRadius` where the left and right sides of the rectangle have the same radii.
- `only` - Creates a border radius with only the given values.
- `vertical` - Creates a vertically symmetric `BorderRadius` where the top and bottom sides of the rectangle have the same radii.

## Properties

### `buildbottom_leftinstance-attribute`

`bottom_left: Number`

Radius of the bottom left border corner.

### `buildbottom_rightinstance-attribute`

`bottom_right: Number`

Radius of the bottom right border corner.

### `buildtop_leftinstance-attribute`

`top_left: Number`

Radius of the top left border corner.

### `buildtop_rightinstance-attribute`

`top_right: Number`

Radius of the top right border corner.

## Methods

### `deployed_codeallclassmethod`

`all(value: Number) -> BorderRadius`

Creates a `BorderRadius` where all radii are radius.

### `deployed_codecopy`

```
copy(
 top_left: Number | None = None,
 top_right: Number | None = None,
 bottom_left: Number | None = None,
 bottom_right: Number | None = None,
) -> BorderRadius
```

Returns a copy of this object with the specified properties overridden.

### `deployed_codehorizontalclassmethod`

```
horizontal(
 left: Number = 0, right: Number = 0
```

```
) -> BorderRadius
```

Creates a horizontally symmetrical BorderRadius where the left and right sides of the rectangle have the same radii.

### deployed\_codeonlyclassmethod

```
only(
 top_left: Number = 0,
 top_right: Number = 0,
 bottom_left: Number = 0,
 bottom_right: Number = 0,
) -> BorderRadius
```

Creates a border radius with only the given values. The other corners will be right angles.

### deployed\_codeverticalclassmethod

```
vertical(
 top: Number = 0, bottom: Number = 0
) -> BorderRadius
```

Creates a vertically symmetric BorderRadius where the top and bottom sides of the rectangle have the same radii.

## Usage example

```
container_1.border_radius= ft.BorderRadius.all(30)
```

# BorderSide

Creates the side of a border.

By default, the border is 1.0 logical pixels wide and solid black color.

To omit (not show) a side, set style to `BorderStyle.NONE`. It skips painting the border, but the border still has a width.

## Properties

- `color` - The color of this side of the border.
- `stroke_align` - The relative position of the stroke on a `BorderSide` in an `OutlinedBorder` or `Border`.
- `stroke_inset` - The amount of the stroke width that lies inside this `BorderSide`.
- `stroke_offset` - The offset of the stroke, taking into account the stroke alignment.
- `stroke_outset` - The amount of the stroke width that lies outside this `BorderSide`.
- `style` - The style of this side of the border.
- `width` - The width of this side of the border, in logical pixels.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.
- `none` - Creates a border side that is not rendered.

## Properties

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue = Colors.BLACK
```

The color of this side of the border.

### `buildstroke_alignclass-attributeinstance-attribute`

```
stroke_align: BorderSideStrokeAlignValue = (
 BorderSideStrokeAlign.INSIDE
)
```

The relative position of the stroke on a `BorderSide` in an `OutlinedBorder` or `Border`.

### `buildstroke_insetproperty`

The amount of the stroke width that lies inside this `BorderSide`.

For example, this will return the width for a `stroke_align` of -1, half the width for a `stroke_align` of 0, and 0 for a `stroke_align` of 1.

### `buildstroke_offsetproperty`

The offset of the stroke, taking into account the stroke alignment.

For example, this will return the negative width of the stroke for a `stroke_align` of -1, 0 for a `stroke_align` of 0, and the width for a `stroke_align` of 1.

### `buildstroke_outsetproperty`

The amount of the stroke width that lies outside this `BorderSide`.

For example, this will return 0 for a `stroke_align` of -1, half the width for a `stroke_align` of 0, and the width for a `stroke_align` of 1.

### `buildstyleclass-attributeinstance-attribute`

```
style: BorderStyle = BorderStyle.SOLID
```

The style of this side of the border.

### `buildwidthclass-attributeinstance-attribute`

```
width: Number = 1.0
```

The width of this side of the border, in logical pixels.

Setting width to 0.0 will result in a hairline border. This means that the border will have the width of one physical pixel. Hairline rendering takes shortcuts when the path overlaps a pixel more than once. This means that it will render faster than otherwise, but it might double-hit pixels, giving it a slightly darker/lighter result.

Tip

To omit the border entirely, set the style to `BorderStyle.NONE`.

Raises:

- `ValueError` - If it is not greater than or equal to 0.

## Methods

### `deployed_code``copy`

```
copy(
 width: Number | None = None,
 color: ColorValue | None = None,
 stroke_align: BorderSideStrokeAlignValue | None = None,
) -> BorderSide
```

Returns a copy of this object with the specified properties overridden.

### `deployed_code``none``staticmethod`

```
none() -> BorderSide
```

Creates a border side that is not rendered.

Returns:

- `BorderSide` - A hairline black `BorderSide` with style set to `BorderStyle.NONE`.

# BorderSideStrokeAlign

Defines where a border stroke is painted relative to the border path.

Inherits: float, enum.Enum

## Properties

- CENTER - The border is drawn on the center of the border path, with half of the BorderSide.width on the inside, and the other half on the outside of the path.
- INSIDE - The border is drawn fully inside of the border path.
- OUTSIDE - The border is drawn on the outside of the border path.

## Examples

### Border Side Stroke Align showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(align: ft.BorderSideStrokeAlign) -> ft.Container:
 preview = ft.Container(
 width=150,
 height=150,
 alignment=ft.Alignment.CENTER,
 content=ft.Stack(
 controls=[
 ft.Container(
 width=90,
 height=90,
 border_radius=45,
 bgcolor=ft.Colors.ON_SURFACE,
),
 ft.Container(
 width=90,
 height=90,
 border_radius=45,
 border=ft.Border.all(
 side=ft.BorderSide(
 width=18,
 color=ft.Colors.RED,
 stroke_align=align,
)
),
),
],
),
)

 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Text(align.name, weight=ft.FontWeight.BOLD),
 preview,
 ft.Text(f"value={align.value}", size=11),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="BorderSideStrokeAlign Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
```



```

 "Compare how thick borders are painted relative to "
 "the shape path."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(aligned) for aligned in ft.BorderSideStrokeAlign
],
),
]

)
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildCENTERclass-attributeinstance-attribute

The border is drawn on the center of the border path, with half of the BorderSide.width on the inside, and the other half on the outside of the path.

### buildINSIDEclass-attributeinstance-attribute

The border is drawn fully inside of the border path.

### buildOUTSIDEclass-attributeinstance-attribute

The border is drawn on the outside of the border path.

# BorderStyle

Defines how a border side is rendered.

Inherits: `enum.Enum`

## Properties

- NONE - Skip (doesn't paint) the border.
- SOLID - Draw the border as a solid line.

## Examples

### Border Style showcase

[play\\_arrow](#)Try Online

```
import flet as ft

def showcase_card(style: ft.BorderStyle) -> ft.Container:
 side = ft.BorderSide(width=4, color=ft.Colors.PRIMARY, style=style)
 border = ft.Border(left=side, top=side, right=side, bottom=side)

 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(style.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=240,
 height=100,
 border=border,
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 alignment=ft.Alignment.CENTER,
 content=ft.Text("Border preview"),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="BorderStyle Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare rendered border sides for each BorderStyle value."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(style) for style in ft.BorderStyle],
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildNONE**class-attributeinstance-attribute

Skip (doesn't paint) the border.

**buildSOLID**class-attributeinstance-attribute

Draw the border as a solid line.

# BottomAppBarTheme

Customizes the appearance of BottomAppBar controls across the app.

## Properties

- `bgcolor` - Overrides the default value of BottomAppBar.bgcolor in all descendant BottomAppBar controls.
- `elevation` - Overrides the default value of BottomAppBar.elevation in all descendant BottomAppBar controls.
- `height` - Overrides the default value of BottomAppBar.height in all descendant BottomAppBar controls.
- `padding` - Overrides the default value of BottomAppBar.padding in all descendant BottomAppBar controls.
- `shadow_color` - Overrides the default value of BottomAppBar.shadow\_color in all descendant BottomAppBar controls.
- `shape` - Overrides the default value of BottomAppBar.shape in all descendant BottomAppBar controls.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of BottomAppBar.bgcolor in all descendant BottomAppBar controls.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value of BottomAppBar.elevation in all descendant BottomAppBar controls.

### `buildheightclass-attributeinstance-attribute`

`height: Number | None = None`

Overrides the default value of BottomAppBar.height in all descendant BottomAppBar controls.

### `buildpaddingclass-attributeinstance-attribute`

`padding: PaddingValue | None = None`

Overrides the default value of BottomAppBar.padding in all descendant BottomAppBar controls.

### `buildshadow_colorclass-attributeinstance-attribute`

`shadow_color: ColorValue | None = None`

Overrides the default value of BottomAppBar.shadow\_color in all descendant BottomAppBar controls.

### `buildshapeclass-attributeinstance-attribute`

`shape: NotchShape | None = None`

Overrides the default value of BottomAppBar.shape in all descendant BottomAppBar controls.

# BottomSheetTheme

Customizes the appearance of BottomSheet across the app.

## Properties

- `barrier_color` - Overrides the default value of `BottomSheet.barrier_color` in all descendant `BottomSheet` controls.
- `bgcolor` - Overrides the default value of `BottomSheet.bgcolor` in all descendant `BottomSheet` controls.
- `clip_behavior` - Overrides the default value of `BottomSheet.clip_behavior` in all descendant `BottomSheet` controls.
- `drag_handle_color` - Overrides the default value of drag handle color in all descendant `BottomSheet` controls.
- `elevation` - Overrides the default value of `BottomSheet.elevation` in all descendant `BottomSheet` controls.
- `shadow_color` - Overrides the default value of shadow color in all descendant `BottomSheet` controls.
- `shape` - Overrides the default value of `BottomSheet.shape` in all descendant `BottomSheet` controls.
- `show_drag_handle` - Overrides the default value of `BottomSheet.show_drag_handle` in all descendant `BottomSheet` controls.
- `size_constraints` - Overrides the default value of `BottomSheet.size_constraints` in all descendant `BottomSheet` controls.

## Properties

### `buildbarrier_colorclass-attributeinstance-attribute`

`barrier_color: ColorValue | None = None`

Overrides the default value of `BottomSheet.barrier_color` in all descendant `BottomSheet` controls.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `BottomSheet.bgcolor` in all descendant `BottomSheet` controls.

### `buildclip_behaviorclass-attributeinstance-attribute`

`clip_behavior: ClipBehavior | None = None`

Overrides the default value of `BottomSheet.clip_behavior` in all descendant `BottomSheet` controls.

### `bulldrag_handle_colorclass-attributeinstance-attribute`

`drag_handle_color: ColorValue | None = None`

Overrides the default value of drag handle color in all descendant `BottomSheet` controls.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value of `BottomSheet.elevation` in all descendant `BottomSheet` controls.

### `buildshadow_colorclass-attributeinstance-attribute`

`shadow_color: ColorValue | None = None`

Overrides the default value of shadow color in all descendant `BottomSheet` controls.

### `buildshapeclass-attributeinstance-attribute`

`shape: OutlinedBorder | None = None`

Overrides the default value of `BottomSheet.shape` in all descendant `BottomSheet` controls.

### `buildshow_drag_handleclass-attributeinstance-attribute`

`show_drag_handle: bool | None = None`

Overrides the default value of `BottomSheet.show_drag_handle` in all descendant `BottomSheet` controls.

### `buildsize_constraintsclass-attributeinstance-attribute`

`size_constraints: BoxConstraints | None = None`

Overrides the default value of `BottomSheet.size_constraints` in all descendant `BottomSheet` controls.

# BoxConstraints

Constraints that must be respected by a size of a box.

A Size respects a BoxConstraints if, and only if, all of the following relations hold:

$\text{min\_width} \leq \text{Size.width} \leq \text{max\_width}$   $\text{min\_height} \leq \text{Size.height} \leq \text{max\_height}$

Read more about BoxConstraints here.

## Properties

- `max_height` - The maximum height that satisfies the constraints, such that  $\text{min\_height} \leq \text{max\_height} \leq \text{float}(\text{"inf"})$ .
- `max_width` - The maximum width that satisfies the constraints, such that  $\text{min\_width} \leq \text{max\_width} \leq \text{float}(\text{"inf"})$ .
- `min_height` - The minimum height that satisfies the constraints, such that  $0.0 \leq \text{min\_height} \leq \text{max\_height}$ .
- `min_width` - The minimum width that satisfies the constraints, such that  $0.0 \leq \text{min\_width} \leq \text{max\_width}$ .

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

`buildmax_height`class-attributeinstance-attribute

```
max_height: Number = float('inf')
```

The maximum height that satisfies the constraints, such that  $\text{min\_height} \leq \text{max\_height} \leq \text{float}(\text{"inf"})$ .

`buildmax_width`class-attributeinstance-attribute

```
max_width: Number = float('inf')
```

The maximum width that satisfies the constraints, such that  $\text{min\_width} \leq \text{max\_width} \leq \text{float}(\text{"inf"})$ .

`buildmin_height`class-attributeinstance-attribute

```
min_height: Number = 0
```

The minimum height that satisfies the constraints, such that  $0.0 \leq \text{min\_height} \leq \text{max\_height}$ .

`buildmin_width`class-attributeinstance-attribute

```
min_width: Number = 0
```

The minimum width that satisfies the constraints, such that  $0.0 \leq \text{min\_width} \leq \text{max\_width}$ .

## Methods

`deployed_code`copy

```
copy(
 min_width: Number | None = None,
 min_height: Number | None = None,
 max_width: Number | None = None,
 max_height: Number | None = None,
) -> BoxConstraints
```

Returns a copy of this object with the specified properties overridden.

# BoxDecoration

BoxDecoration provides a description of how to paint a box. The box has a border, a body, and may cast a shadow.

## Properties

- `bgcolor` - The color to fill in the background of the box.
- `blend_mode` - The blend mode to apply to the background `bgcolor` or gradient.
- `border` - A border to draw above the background `bgcolor`, gradient, and image.
- `border_radius` - The border radius of the box.
- `gradient` - A gradient to use when filling the box.
- `image` - An image to paint above the background `bgcolor` or gradient.
- `shadows` - A list of shadows cast by the box.
- `shape` - The shape to fill the `bgcolor`, gradient, and image into and to cast as the shadows.

## Methods

- `copy` - Returns a new `BoxDecoration` with selected fields overridden.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

The color to fill in the background of the box.

### `buildblend_modeclass-attributeinstance-attribute`

`blend_mode: BlendMode | None = None`

The blend mode to apply to the background `bgcolor` or gradient.

### `buildborderclass-attributeinstance-attribute`

`border: Border | None = None`

A border to draw above the background `bgcolor`, gradient, and image.

### `buildborder_radiusclass-attributeinstance-attribute`

`border_radius: BorderRadiusValue | None = None`

The border radius of the box.

### `buildgradientclass-attributeinstance-attribute`

`gradient: Gradient | None = None`

A gradient to use when filling the box.

### `buildimageclass-attributeinstance-attribute`

`image: DecorationImage | None = None`

An image to paint above the background `bgcolor` or gradient.

### `buildshadowsclass-attributeinstance-attribute`

`shadows: BoxShadowValue | None = None`

A list of shadows cast by the box.

### `buildshapeclass-attributeinstance-attribute`

`shape: BoxShape = BoxShape.RECTANGLE`

The shape to fill the `bgcolor`, gradient, and image into and to cast as the shadows.

## Methods



## deployed\_codecopy

```
copy(
 bgcolor: ColorValue | None = None,
 image: DecorationImage | None = None,
 border: Border | None = None,
 border_radius: BorderRadiusValue | None = None,
 shadows: BoxShadowValue | None = None,
 gradient: Gradient | None = None,
 shape: BoxShape | None = None,
 blend_mode: BlendMode | None = None,
)
```

Returns a new BoxDecoration with selected fields overridden.

# BoxFit

How a box should be inscribed into another box.

Inherits: `enum.Enum`

## Properties

- **CONTAIN** - As large as possible while still containing the source entirely within the target box.
- **COVER** - As small as possible while still covering the entire target box.
- **FILL** - Fill the target box by distorting the source's aspect ratio.
- **FIT\_HEIGHT** - Make sure the full height of the source is shown, regardless of whether this means the source overflows the target box horizontally.
- **FIT\_WIDTH** - Make sure the full width of the source is shown, regardless of whether this means the source overflows the target box vertically.
- **NONE** - Align the source within the target box (by default, centering) and discard any portions of the source that lie outside the box.
- **SCALE\_DOWN** - Align the source within the target box (by default, centering) and, if necessary, scale the source down to ensure that the source fits within the box.

## Examples

### Box Fit showcase

[play\\_arrow](#) [Try Online](#)

```
import flet as ft

def showcase_card(fit: ft.BoxFit) -> ft.Container:
 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(fit.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=240,
 height=120,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 clip_behavior=ft.ClipBehavior.HARD_EDGE,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Image(
 src="https://picsum.photos/id/1025/420/220",
 width=240,
 height=120,
 fit=fit,
),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="BoxFit Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare how the same image is inscribed into a fixed frame."
),
 ft.Row(
 wrap=True,
 spacing=12,
```

```

 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(fit) for fit in ft.BoxFit],
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildCONTAINclass-attributeinstance-attribute

As large as possible while still containing the source entirely within the target box.

Image:  - [https://flutter.github.io/assets-for-api-docs/assets/painting/box\\_fit\\_contain.png](https://flutter.github.io/assets-for-api-docs/assets/painting/box_fit_contain.png)

### buildCOVERclass-attributeinstance-attribute

As small as possible while still covering the entire target box.

Image:  - [https://flutter.github.io/assets-for-api-docs/assets/painting/box\\_fit\\_cover.png](https://flutter.github.io/assets-for-api-docs/assets/painting/box_fit_cover.png)

### buildFILLclass-attributeinstance-attribute

Fill the target box by distorting the source's aspect ratio.

Image:  - [https://flutter.github.io/assets-for-api-docs/assets/painting/box\\_fit\\_fill.png](https://flutter.github.io/assets-for-api-docs/assets/painting/box_fit_fill.png)

### buildFIT\_HEIGHTclass-attributeinstance-attribute

Make sure the full height of the source is shown, regardless of whether this means the source overflows the target box horizontally.

Image:  - [https://flutter.github.io/assets-for-api-docs/assets/painting/box\\_fit\\_fitHeight.png](https://flutter.github.io/assets-for-api-docs/assets/painting/box_fit_fitHeight.png)

### buildFIT\_WIDTHclass-attributeinstance-attribute

Make sure the full width of the source is shown, regardless of whether this means the source overflows the target box vertically.

Image:  - [https://flutter.github.io/assets-for-api-docs/assets/painting/box\\_fit\\_fitWidth.png](https://flutter.github.io/assets-for-api-docs/assets/painting/box_fit_fitWidth.png)

### buildNONEclass-attributeinstance-attribute

Align the source within the target box (by default, centering) and discard any portions of the source that lie outside the box.

The source image is not resized.

Image:  - [https://flutter.github.io/assets-for-api-docs/assets/painting/box\\_fit\\_none.png](https://flutter.github.io/assets-for-api-docs/assets/painting/box_fit_none.png)

### buildSCALE\_DOWNclass-attributeinstance-attribute

Align the source within the target box (by default, centering) and, if necessary, scale the source down to ensure that the source fits within the box.

This is the same as CONTAIN if that would shrink the image, otherwise it is the same as NONE.

Image:  - [https://flutter.github.io/assets-for-api-docs/assets/painting/box\\_fit\\_scaleDown.png](https://flutter.github.io/assets-for-api-docs/assets/painting/box_fit_scaleDown.png)

# BoxShadow

Configuration for a box's shadow.

## Properties

- `blur_radius` - The standard deviation of the Gaussian to convolve with the shadow's shape.
- `blur_style` - The blur style to apply to this shadow.
- `color` - Color used to draw the shadow.
- `offset` - The displacement of the shadow from the casting element.
- `spread_radius` - The amount the box should be inflated prior to applying the blur.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildblur_radiusclass-attributeinstance-attribute`

```
blur_radius: Number = 0.0
```

The standard deviation of the Gaussian to convolve with the shadow's shape.

### `buildblur_styleclass-attributeinstance-attribute`

```
blur_style: BlurStyle = BlurStyle.NORMAL
```

The blur style to apply to this shadow.

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue = Colors.BLACK
```

Color used to draw the shadow.

### `buildoffsetclass-attributeinstance-attribute`

```
offset: OffsetValue = field(
 default_factory=lambda: Offset()
)
```

The displacement of the shadow from the casting element. Positive x/y offsets will shift the shadow to the right and down, while negative offsets shift the shadow to the left and up. The offsets are relative to the position of the element that is casting it.

### `buildspread_radiusclass-attributeinstance-attribute`

```
spread_radius: Number = 0.0
```

The amount the box should be inflated prior to applying the blur.

## Methods

### `deployed_code` `copy`

```
copy(
 spread_radius: Number | None = None,
 blur_radius: Number | None = None,
 color: ColorValue | None = None,
 offset: OffsetValue | None = None,
 blur_style: BlurStyle | None = None,
) -> BoxShadow
```

Returns a copy of this object with the specified properties overridden.

## Examples

### Example 1

`play_arrow` Try Online

```
import flet as ft
```

```

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Container(
 bgcolor=ft.Colors.YELLOW,
 width=100,
 height=100,
 border_radius=50,
 shadow=[
 ft.BoxShadow(
 spread_radius=1,
 blur_radius=15,
 color=ft.Colors.WHITE,
 offset=ft.Offset(-5, -5),
),
 ft.BoxShadow(
 spread_radius=1,
 blur_radius=15,
 color=ft.Colors.GREY_600,
 offset=ft.Offset(5, 5),
),
],
),
 ft.Container(
 # bgcolor=ft.Colors.WHITE,
 border_radius=10,
 width=100,
 height=100,
 shadow=ft.BoxShadow(
 spread_radius=1,
 blur_radius=15,
 color=ft.Colors.BLUE_GREY_300,
 offset=ft.Offset(0, 0),
 blur_style=ft.BlurStyle.OUTER,
),
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

# BoxShape

The shape to use when rendering a Border or BoxDecoration.

Inherits: enum.Enum

## Properties

- CIRCLE - A circle centered in the middle of the box into which the Border or BoxDecoration is painted.
- RECTANGLE - An axis-aligned rectangle, optionally with rounded corners.

## Examples

### Box Shape showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(shape: ft.BoxShape) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(shape.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=200,
 height=120,
 alignment=ft.Alignment.CENTER,
 bgcolor=ft.Colors.SURFACE,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 content=ft.Container(
 width=120,
 height=90,
 alignment=ft.Alignment.CENTER,
 shape=shape,
 bgcolor=ft.Colors.PRIMARY_CONTAINER,
 border=ft.Border.all(2, ft.Colors.PRIMARY),
 border_radius=(
 ft.BorderRadius.all(16)
 if shape == ft.BoxShape.RECTANGLE
 else None
),
 content=ft.Text(
 "Shape",
 color=ft.Colors.ON_PRIMARY_CONTAINER,
 weight=ft.FontWeight.BOLD,
),
),
],
),
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="BoxShape Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare rectangular and circular box decoration shapes."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(shape) for shape in ft.BoxShape],
),
],
),
),
)
```

```
)
)
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

### buildCIRCLEclass-attributeinstance-attribute

A circle centered in the middle of the box into which the Border or BoxDecoration is painted. The diameter of the circle is the shortest dimension of the box, either the width or the height, such that the circle touches the edges of the box.

### buildRECTANGLEclass-attributeinstance-attribute

An axis-aligned rectangle, optionally with rounded corners.

The amount of corner rounding, if any, is determined by the border radius specified by classes such as BoxDecoration or Border. The rectangle's edges match those of the box in which it is painted.

# Brightness

Describes the contrast of a theme or color palette.

Inherits: `enum.Enum`

## Properties

- **DARK** - The color is dark and will require a light text color to achieve readable contrast.
- **LIGHT** - The color is light and will require a dark text color to achieve readable contrast.

## Properties

### **buildDARK**`class-attributeinstance-attribute`

The color is dark and will require a light text color to achieve readable contrast.

For example, the color might be dark grey, requiring white text.

### **buildLIGHT**`class-attributeinstance-attribute`

The color is light and will require a dark text color to achieve readable contrast.

For example, the color might be bright white, requiring black text.



# BrowserConfiguration

Configuration options for in-app browser views.

## Properties

- `show_title` - Whether the browser view should display the page title.

## Properties

`buildshow_titleclass-attributeinstance-attribute`

`show_title: bool = False`

Whether the browser view should display the page title.

# ButtonStyle

Allows controlling all visual aspects of a button, such as shape, foreground, background and shadow colors, content padding, border width and radius.

Most of these style attributes could be configured for all or particular `ControlState` of a button, such as `HOVERED`, `FOCUSED`, `DISABLED` and others.

## Properties

- `alignment` - The alignment of the button's content.
- `animation_duration` - Defines the duration in milliseconds of animated changes for shape and elevation.
- `bgcolor` - The button's background fill color.
- `color` - The color for the button's Text and Icon control descendants.
- `elevation` - The elevation of the button's Material.
- `enable_feedback` - Whether detected gestures should provide acoustic and/or haptic feedback.
- `icon_color` - The icon's color inside the button.
- `icon_size` - The icon's size inside of the button.
- `mouse_cursor` - The cursor to be displayed when the mouse pointer enters or is hovering over the button.
- `overlay_color` - The highlight color that's typically used to indicate that the button is focused, hovered, or pressed.
- `padding` - The padding between the button's boundary and its content.
- `shadow_color` - The shadow color of the button's Material.
- `shape` - The shape of the button's underlying Material.
- `side` - Defines the button's border outline.
- `text_style` - The text style of the button's Text control descendants.
- `visual_density` - Defines how compact the button's layout will be.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

`alignment: Alignment | None = None`

The alignment of the button's content.

### `buildanimation_durationclass-attributeinstance-attribute`

`animation_duration: DurationValue | None = None`

Defines the duration in milliseconds of animated changes for shape and elevation.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ControlStateValue[ColorValue] | None = None`

The button's background fill color.

### `buildcolorclass-attributeinstance-attribute`

`color: ControlStateValue[ColorValue] | None = None`

The color for the button's Text and Icon control descendants.

### `buildelevationclass-attributeinstance-attribute`

`elevation: ControlStateValue[Number | None] | None = None`

The elevation of the button's Material.

### **buildenable\_feedbackclass-attributeinstance-attribute**

`enable_feedback: bool | None = None`

Whether detected gestures should provide acoustic and/or haptic feedback.

### **buildicon\_colorclass-attributeinstance-attribute**

`icon_color: ControlStateValue[ColorValue] | None = None`

The icon's color inside the button.

If not set or None, then the color will be used.

### **buildicon\_sizeclass-attributeinstance-attribute**

`icon_size: ControlStateValue[Number | None] | None = None`

The icon's size inside of the button.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

The cursor to be displayed when the mouse pointer enters or is hovering over the button.

### **buildoverlay\_colorclass-attributeinstance-attribute**

`overlay_color: ControlStateValue[ColorValue] | None = None`

The highlight color that's typically used to indicate that the button is focused, hovered, or pressed.

### **buildpaddingclass-attributeinstance-attribute**

`padding: ControlStateValue[PaddingValue] | None = None`

The padding between the button's boundary and its content.

### **buildshadow\_colorclass-attributeinstance-attribute**

`shadow_color: ControlStateValue[ColorValue] | None = None`

The shadow color of the button's Material.

### **buildshapeclass-attributeinstance-attribute**

`shape: ControlStateValue[OutlinedBorder] | None = None`

The shape of the button's underlying Material.

### **buildsideclass-attributeinstance-attribute**

`side: ControlStateValue[BorderSide] | None = None`

Defines the button's border outline.

### **buildtext\_styleclass-attributeinstance-attribute**

`text_style: ControlStateValue[TextStyle] | None = None`

The text style of the button's Text control descendants.

### **buildvisual\_densityclass-attributeinstance-attribute**

`visual_density: VisualDensity | None = None`

Defines how compact the button's layout will be.

## **Methods**

### **deployed\_codecopy**

```
copy(
 color: ControlStateValue[ColorValue] | None = None,
 bgcolor: ControlStateValue[ColorValue] | None = None,
 overlay_color: ControlStateValue[ColorValue]
 | None = None,
 shadow_color: ControlStateValue[ColorValue]
 | None = None,
 elevation: ControlStateValue[Number | None]
 | None = None,
 animation_duration: DurationValue | None = None,
 padding: ControlStateValue[PaddingValue] | None = None,
 side: ControlStateValue[BorderSide] | None = None,
 shape: ControlStateValue[OutlinedBorder] | None = None,
)
```

```

alignment: Alignment | None = None,
enable_feedback: bool | None = None,
text_style: ControlStateValue[TextStyle] | None = None,
icon_size: ControlStateValue[Number | None]
| None = None,
icon_color: ControlStateValue[ColorValue] | None = None,
visual_density: VisualDensity | None = None,
mouse_cursor: ControlStateValue[MouseCursor]
| None = None,
) -> ButtonStyle

```

Returns a copy of this object with the specified properties overridden.

## Usage example

You can configure a different shape, background color for a hovered state and configure fallback values for all other states.

To configure style attribute for all Material states set its value to a literal (or class instance). For example, if you set color property to a literal the value will be applied to all button states:

```

ButtonStyle(
 color=ft.Colors.WHITE
)

```

To configure style attribute for specific Material states set its value to a dictionary where the key is state name. For example, to configure different background colors for HOVERED and FOCUSED states and another colors for all other states:

```

ButtonStyle(
 color={
 ft.ControlState.HOVERED: ft.Colors.WHITE,
 ft.ControlState.FOCUSED: ft.Colors.BLUE,
 ft.ControlState.DEFAULT: ft.Colors.BLACK,
 }
)

```

## Various button shapes example

Image:  - <https://flet.dev/img/blog/gradients/button-shapes.png>

```

import flet as ft

def main(page: ft.Page):
 page.padding = 30
 page.spacing = 30
 page.add(
 ft.FilledButton(
 "Stadium",
 style=ft.ButtonStyle(
 shape=ft.StadiumBorder(),
),
),
 ft.FilledButton(
 "Rounded rectangle",
 style=ft.ButtonStyle(
 shape=ft.RoundedRectangleBorder(radius=10),
),
),
 ft.FilledButton(
 "Continuous rectangle",
 style=ft.ButtonStyle(
 shape=ft.ContinuousRectangleBorder(radius=30),
),
),
 ft.FilledButton(
 "Beveled rectangle",
 style=ft.ButtonStyle(
 shape=ft.BeveledRectangleBorder(radius=10),
),
),
 ft.FilledButton(
 "Circle",
 style=ft.ButtonStyle(shape=ft.CircleBorder(), padding=30),
),
)

ft.run(main)

```

## Styled button example

Check the following example:

Image:  - <https://flet.dev/img/blog/gradients/styled-button.gif>

```

import flet as ft

```

```

def main(page: ft.Page):
 page.add(
 ft.Button(
 "Styled button 1",
 style=ft.ButtonStyle(
 color={
 ft.ControlState.HOVERED: ft.Colors.WHITE,
 ft.ControlState.FOCUSED: ft.Colors.BLUE,
 ft.ControlState.DEFAULT: ft.Colors.BLACK,
 },
 bgcolor={ft.ControlState.FOCUSED: ft.Colors.PINK_200, "": ft.Colors.YELLOW},
 padding={ft.ControlState.HOVERED: 20},
 overlay_color=ft.Colors.TRANSPARENT,
 elevation={"pressed": 0, "": 1},
 animation_duration=500,
 side={
 ft.ControlState.DEFAULT: ft.BorderSide(1, ft.Colors.BLUE),
 ft.ControlState.HOVERED: ft.BorderSide(2, ft.Colors.BLUE),
 },
 shape={
 ft.ControlState.HOVERED: ft.RoundedRectangleBorder(radius=20),
 ft.ControlState.DEFAULT: ft.RoundedRectangleBorder(radius=2),
 },
),
),
)
ft.run(main)

```

# ButtonTheme

Customizes the appearance of Button across the app.

## Properties

- `style` - Overrides the default value of `Button.style` in all descendant Button controls.

## Properties

**buildstyleclass-attributeinstance-attribute**

`style: ButtonStyle | None = None`

Overrides the default value of `Button.style` in all descendant Button controls.

# CanvasResizeEvent

Event emitted when a Canvas reports a new rendered size.

This event is delivered through `on_resize` and carries the latest measured dimensions in logical pixels.

Inherits: `Event`

## Properties

- `height` - New height of the canvas.
- `width` - New width of the canvas.

## Properties

### `buildheightclass-attributeinstance-attribute`

```
height: float = field(metadata={'data_field': 'h'})
```

New height of the canvas.

### `buildwidthclass-attributeinstance-attribute`

```
width: float = field(metadata={'data_field': 'w'})
```

New width of the canvas.

# CardTheme

Customizes the appearance of Card across the app.

## Properties

- `clip_behavior` - Overrides the default value of `Card.clip_behavior` in all descendant Card controls.
- `color` - Overrides the default value of `Card.clip_behavior` in all descendant Card controls.
- `elevation` - Overrides the default value of `Card.elevation` in all descendant Card controls.
- `margin` - Overrides the default value of `Card.margin` in all descendant Card controls.
- `shadow_color` - Overrides the default value of `Card.shadow_color` in all descendant Card controls.
- `shape` - Overrides the default value of `Card.shape` in all descendant Card controls.

## Properties

### `buildclip_behaviorclass-attributeinstance-attribute`

`clip_behavior: ClipBehavior | None = None`

Overrides the default value of `Card.clip_behavior` in all descendant Card controls.

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

Overrides the default value of `Card.clip_behavior` in all descendant Card controls.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value of `Card.elevation` in all descendant Card controls.

### `buildmarginclass-attributeinstance-attribute`

`margin: MarginValue | None = None`

Overrides the default value of `Card.margin` in all descendant Card controls.

### `buildshadow_colorclass-attributeinstance-attribute`

`shadow_color: ColorValue | None = None`

Overrides the default value of `Card.shadow_color` in all descendant Card controls.

### `buildshapeclass-attributeinstance-attribute`

`shape: OutlinedBorder | None = None`

Overrides the default value of `Card.shape` in all descendant Card controls.



# CardVariant

Material card visual style preset.

Used to define the Card.variant.

Inherits: enum.Enum

## Properties

- ELEVATED - Standard elevated card style.
- FILLED - Filled card style emphasizing container color.
- OUTLINED - Outlined card style with a visible border.

## Examples

### Card Variant showcase

[play\\_arrow](#)Try Online

```
import flet as ft

def showcase_card(variant: ft.CardVariant) -> ft.Container:
 card = ft.Card(
 variant=variant,
 content=ft.Container(
 width=260,
 padding=12,
 content=ft.Column(
 spacing=6,
 controls=[
 ft.Text("Quarterly Report", weight=ft.FontWeight.BOLD),
 ft.Text(
 "Revenue increased by 18% compared to last quarter.",
 size=12,
),
],
),
),
)

 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(variant.name, weight=ft.FontWeight.BOLD),
 card,
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="CardVariant Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare Material card visual variants."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(variant) for variant in ft.CardVariant],
),
],
),
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildELEVATED**class-attributeinstance-attribute

Standard elevated card style.

**buildFILLED**class-attributeinstance-attribute

Filled card style emphasizing container color.

**buildOUTLINED**class-attributeinstance-attribute

Outlined card style with a visible border.

# CheckboxTheme

Defines default property values for descendant Checkbox controls.

## Properties

- `border_side` - Overrides the default value of `Checkbox.border_side` in all descendant Checkbox controls.
- `check_color` - Overrides the default value of `Checkbox.check_color` in all descendant Checkbox controls.
- `fill_color` - Overrides the default value of `Checkbox.fill_color` in all descendant Checkbox controls.
- `mouse_cursor` - Overrides the default value of `Checkbox.mouse_cursor` in all descendant Checkbox controls.
- `overlay_color` - Overrides the default value of `Checkbox.overlay_color` in all descendant Checkbox controls.
- `shape` - Overrides the default value of `Checkbox.shape` in all descendant Checkbox controls.
- `splash_radius` - Overrides the default value of `Checkbox.splash_radius` in all descendant Checkbox controls.
- `visual_density` - Overrides the default value of `Checkbox.visual_density` in all descendant Checkbox controls.

## Properties

### `buildborder_sideclass-attributeinstance-attribute`

`border_side: BorderSide | None = None`

Overrides the default value of `Checkbox.border_side` in all descendant Checkbox controls.

### `buildcheck_colorclass-attributeinstance-attribute`

`check_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Checkbox.check_color` in all descendant Checkbox controls.

### `buildfill_colorclass-attributeinstance-attribute`

`fill_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Checkbox.fill_color` in all descendant Checkbox controls.

### `buildmouse_cursorclass-attributeinstance-attribute`

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

Overrides the default value of `Checkbox.mouse_cursor` in all descendant Checkbox controls.

### `buildoverlay_colorclass-attributeinstance-attribute`

`overlay_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Checkbox.overlay_color` in all descendant Checkbox controls.

### `buildshapeclass-attributeinstance-attribute`

`shape: OutlinedBorder | None = None`

Overrides the default value of `Checkbox.shape` in all descendant Checkbox controls.

### `buildsplash_radiusclass-attributeinstance-attribute`

`splash_radius: Number | None = None`

Overrides the default value of `Checkbox.splash_radius` in all descendant Checkbox controls.

### `buildvisual_densityclass-attributeinstance-attribute`

`visual_density: VisualDensity | None = None`

Overrides the default value of `Checkbox.visual_density` in all descendant Checkbox controls.

# ChipTheme

Customizes the appearance of Chip across the app.

## Properties

- `bgcolor` - Overrides the default value of `Chip.bgcolor` in all descendant Chip controls.
- `border_side` - Overrides the default value of `Chip.border_side` in all descendant Chip controls.
- `brightness` - Overrides the default value for all chips which affects various base material color choices in the chip rendering.
- `check_color` - Overrides the default value of `Chip.check_color` in all descendant Chip controls.
- `color` - Overrides the default value of `Chip.color` in all descendant Chip controls.
- `delete_icon_color` - Overrides the default value of `Chip.delete_icon_color` in all descendant Chip controls.
- `delete_icon_size_constraints` - Overrides the default value of `Chip.delete_icon_size_constraints` in all descendant Chip controls.
- `disabled_color` - Overrides the default value of `Chip.disabled_color` in all descendant Chip controls.
- `elevation` - Overrides the default value of `Chip.elevation` in all descendant Chip controls.
- `elevation_on_click` - Overrides the default value of `Chip.elevation_on_click` in all descendant Chip controls.
- `label_padding` - Overrides the default value of `Chip.label_padding` in all descendant Chip controls.
- `label_text_style` - Overrides the default value of `Chip.label_text_style` in all descendant Chip controls.
- `leading_size_constraints` - Overrides the default value of `Chip.leading_size_constraints` in all descendant Chip controls.
- `padding` - Overrides the default value of `Chip.padding` in all descendant Chip controls.
- `selected_color` - Overrides the default value of `Chip.selected_color` in all descendant Chip controls.
- `selected_shadow_color` - Overrides the default value of `Chip.selected_shadow_color` in all descendant Chip controls.
- `shadow_color` - Overrides the default value of `Chip.shadow_color` in all descendant Chip controls.
- `shape` - Overrides the default value of `Chip.shape` in all descendant Chip controls.
- `show_checkmark` - Overrides the default value of `Chip.show_checkmark` in all descendant Chip controls.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `Chip.bgcolor` in all descendant Chip controls.

### `buildborder_sideclass-attributeinstance-attribute`

`border_side: BorderSide | None = None`

Overrides the default value of `Chip.border_side` in all descendant Chip controls.

### `buildbrightnessclass-attributeinstance-attribute`

`brightness: Brightness | None = None`

Overrides the default value for all chips which affects various base material color choices in the chip rendering.

### `buildcheck_colorclass-attributeinstance-attribute`

`check_color: ColorValue | None = None`

Overrides the default value of `Chip.check_color` in all descendant Chip controls.

### `buildcolorclass-attributeinstance-attribute`

`color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Chip.color` in all descendant Chip controls.

### **builddelete\_icon\_colorclass-attributeinstance-attribute**

`delete_icon_color: ColorValue | None = None`

Overrides the default value of `Chip.delete_icon_color` in all descendant `Chip` controls.

### **builddelete\_icon\_size\_constraintsclass-attributeinstance-attribute**

`delete_icon_size_constraints: BoxConstraints | None = None`

Overrides the default value of `Chip.delete_icon_size_constraints` in all descendant `Chip` controls.

### **bulddisabled\_colorclass-attributeinstance-attribute**

`disabled_color: ColorValue | None = None`

Overrides the default value of `Chip.disabled_color` in all descendant `Chip` controls.

### **buildelevationclass-attributeinstance-attribute**

`elevation: Number | None = None`

Overrides the default value of `Chip.elevation` in all descendant `Chip` controls.

### **buildelevation\_on\_clickclass-attributeinstance-attribute**

`elevation_on_click: Number | None = None`

Overrides the default value of `Chip.elevation_on_click` in all descendant `Chip` controls.

### **buildlabel\_paddingclass-attributeinstance-attribute**

`label_padding: PaddingValue | None = None`

Overrides the default value of `Chip.label_padding` in all descendant `Chip` controls.

### **buildlabel\_text\_styleclass-attributeinstance-attribute**

`label_text_style: TextStyle | None = None`

Overrides the default value of `Chip.label_text_style` in all descendant `Chip` controls.

### **buildleading\_size\_constraintsclass-attributeinstance-attribute**

`leading_size_constraints: BoxConstraints | None = None`

Overrides the default value of `Chip.leading_size_constraints` in all descendant `Chip` controls.

### **buildpaddingclass-attributeinstance-attribute**

`padding: PaddingValue | None = None`

Overrides the default value of `Chip.padding` in all descendant `Chip` controls.

### **buildselected\_colorclass-attributeinstance-attribute**

`selected_color: ColorValue | None = None`

Overrides the default value of `Chip.selected_color` in all descendant `Chip` controls.

### **buildselected\_shadow\_colorclass-attributeinstance-attribute**

`selected_shadow_color: ColorValue | None = None`

Overrides the default value of `Chip.selected_shadow_color` in all descendant `Chip` controls.

### **buildshadow\_colorclass-attributeinstance-attribute**

`shadow_color: ColorValue | None = None`

Overrides the default value of `Chip.shadow_color` in all descendant `Chip` controls.

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

Overrides the default value of `Chip.shape` in all descendant `Chip` controls.

### **buildshow\_checkmarkclass-attributeinstance-attribute**

`show_checkmark: bool | None = None`

Overrides the default value of `Chip.show_checkmark` in all descendant `Chip` controls.

# CircleBorder

A border with a circle shape.

Inherits: OutlinedBorder

## Properties

- eccentricity

## Methods

- copy - Returns a copy of this object with the specified properties overridden.

## Properties

buildeccentricityclass-attributeinstance-attribute

eccentricity: Number = 0.0

## Methods

deployed\_codecopy

```
copy(
 side: BorderSide | None = None,
 eccentricity: Number | None = None,
) -> CircleBorder
```

Returns a copy of this object with the specified properties overridden.

# CircularRectangleNotchShape

A rectangle with a smooth circular notch.

Inherits: NotchShape

## Properties

- `inverted` - Whether the notch should be placed at the bottom of the rectangle.

## Properties

`buildinvertedclass-attributeinstance-attribute`

`inverted: bool = False`

Whether the notch should be placed at the bottom of the rectangle.

# ClipBehavior

Different ways to clip content.

See Clip from Flutter documentation for ClipBehavior examples.

Inherits: enum.Enum

## Properties

- ANTI\_ALIAS - Clip with anti-aliasing.
- ANTI\_ALIAS\_WITH\_SAVE\_LAYER - Clip with anti-aliasing and saveLayer immediately following the clip.
- HARD\_EDGE - Clip, but do not apply anti-aliasing.
- NONE - No clip at all.

## Examples

### Clip Behavior showcase

play\_arrowTry Online

```
import flet as ft

def clip_preview(clip_behavior: ft.ClipBehavior) -> ft.Container:
 return ft.Container(
 width=240,
 height=130,
 border=ft.Border.all(2, ft.Colors.OUTLINE),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Stack(
 clip_behavior=clip_behavior,
 controls=[
 ft.Container(
 width=160,
 height=70,
 left=-25,
 top=30,
 bgcolor=ft.Colors.PRIMARY_CONTAINER,
 border_radius=16,
),
 ft.Container(
 width=90,
 height=90,
 left=150,
 top=-20,
 bgcolor=ft.Colors.TERTIARY_CONTAINER,
 border_radius=45,
),
 ft.Container(
 width=90,
 height=90,
 left=70,
 top=22,
 bgcolor=ft.Colors.SECONDARY,
 border_radius=45,
 alignment=ft.Alignment.CENTER,
 content=ft.Icon(ft.Icons.CROP, color=ft.Colors.WHITE),
),
],
),
)

def showcase_card(clip_behavior: ft.ClipBehavior) -> ft.Container:
 return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(clip_behavior.name, weight=ft.FontWeight.BOLD),
 clip_preview(clip_behavior),
],
),
)
```



```

),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="ClipBehavior Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare how overflow is clipped for each ClipBehavior value."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(cb) for cb in ft.ClipBehavior],
),
]
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildANTI\_ALIASclass-attributeinstance-attribute

Clip with anti-aliasing.

This mode has anti-aliased clipping edges, which reduces jagged edges when the clip shape itself has edges that are diagonal, curved, or otherwise not axis-aligned.

### buildANTI\_ALIAS\_WITH\_SAVE\_LAYERclass-attributeinstance-attribute

Clip with anti-aliasing and saveLayer immediately following the clip.

### buildHARD\_EDGEclass-attributeinstance-attribute

Clip, but do not apply anti-aliasing.

This mode enables clipping, but curves and non-axis-aligned straight lines will be jagged as no effort is made to anti-alias.

### buildNONEclass-attributeinstance-attribute

No clip at all.

This is the default option for most widgets: if the content does not overflow the widget boundary, don't pay any performance cost for clipping.

# ColorFilter

Defines a color filter.

## Properties

- `blend_mode` - The blend mode to apply to the color filter.
- `color` - The color to use when applying the filter.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildblend_modeclass-attributeinstance-attribute`

`blend_mode: BlendMode | None = None`

The blend mode to apply to the color filter.

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color to use when applying the filter.

## Methods

### `deployed_code``copy`

```
copy(
 color: ColorValue | None = None,
 blend_mode: BlendMode | None = None,
) -> ColorFilter
```

Returns a copy of this object with the specified properties overridden.

# Colors

Named Material colors.

Inherits: str, enum.Enum

## Properties

- AMBER
- AMBER\_100
- AMBER\_200
- AMBER\_300
- AMBER\_400
- AMBER\_50
- AMBER\_500
- AMBER\_600
- AMBER\_700
- AMBER\_800
- AMBER\_900
- AMBER\_ACCENT
- AMBER\_ACCENT\_100
- AMBER\_ACCENT\_200
- AMBER\_ACCENT\_400
- AMBER\_ACCENT\_700
- BLACK
- BLACK\_12
- BLACK\_26
- BLACK\_38
- BLACK\_45
- BLACK\_54
- BLACK\_87
- BLUE
- BLUE\_100
- BLUE\_200
- BLUE\_300
- BLUE\_400
- BLUE\_50
- BLUE\_500
- BLUE\_600
- BLUE\_700
- BLUE\_800
- BLUE\_900
- BLUE\_ACCENT

- BLUE\_ACCENT\_100
- BLUE\_ACCENT\_200
- BLUE\_ACCENT\_400
- BLUE\_ACCENT\_700
- BLUE\_GREY
- BLUE\_GREY\_100
- BLUE\_GREY\_200
- BLUE\_GREY\_300
- BLUE\_GREY\_400
- BLUE\_GREY\_50
- BLUE\_GREY\_500
- BLUE\_GREY\_600
- BLUE\_GREY\_700
- BLUE\_GREY\_800
- BLUE\_GREY\_900
- BROWN
- BROWN\_100
- BROWN\_200
- BROWN\_300
- BROWN\_400
- BROWN\_50
- BROWN\_500
- BROWN\_600
- BROWN\_700
- BROWN\_800
- BROWN\_900
- CYAN
- CYAN\_100
- CYAN\_200
- CYAN\_300
- CYAN\_400
- CYAN\_50
- CYAN\_500
- CYAN\_600
- CYAN\_700
- CYAN\_800
- CYAN\_900
- CYAN\_ACCENT
- CYAN\_ACCENT\_100
- CYAN\_ACCENT\_200
- CYAN\_ACCENT\_400

- CYAN\_ACCENT\_700
- DEEP\_ORANGE
- DEEP\_ORANGE\_100
- DEEP\_ORANGE\_200
- DEEP\_ORANGE\_300
- DEEP\_ORANGE\_400
- DEEP\_ORANGE\_50
- DEEP\_ORANGE\_500
- DEEP\_ORANGE\_600
- DEEP\_ORANGE\_700
- DEEP\_ORANGE\_800
- DEEP\_ORANGE\_900
- DEEP\_ORANGE\_ACCENT
- DEEP\_ORANGE\_ACCENT\_100
- DEEP\_ORANGE\_ACCENT\_200
- DEEP\_ORANGE\_ACCENT\_400
- DEEP\_ORANGE\_ACCENT\_700
- DEEP\_PURPLE
- DEEP\_PURPLE\_100
- DEEP\_PURPLE\_200
- DEEP\_PURPLE\_300
- DEEP\_PURPLE\_400
- DEEP\_PURPLE\_50
- DEEP\_PURPLE\_500
- DEEP\_PURPLE\_600
- DEEP\_PURPLE\_700
- DEEP\_PURPLE\_800
- DEEP\_PURPLE\_900
- DEEP\_PURPLE\_ACCENT
- DEEP\_PURPLE\_ACCENT\_100
- DEEP\_PURPLE\_ACCENT\_200
- DEEP\_PURPLE\_ACCENT\_400
- DEEP\_PURPLE\_ACCENT\_700
- ERROR
- ERROR\_CONTAINER
- GREEN
- GREEN\_100
- GREEN\_200
- GREEN\_300
- GREEN\_400
- GREEN\_50

- GREEN\_500
- GREEN\_600
- GREEN\_700
- GREEN\_800
- GREEN\_900
- GREEN\_ACCENT
- GREEN\_ACCENT\_100
- GREEN\_ACCENT\_200
- GREEN\_ACCENT\_400
- GREEN\_ACCENT\_700
- GREY
- GREY\_100
- GREY\_200
- GREY\_300
- GREY\_400
- GREY\_50
- GREY\_500
- GREY\_600
- GREY\_700
- GREY\_800
- GREY\_900
- INDIGO
- INDIGO\_100
- INDIGO\_200
- INDIGO\_300
- INDIGO\_400
- INDIGO\_50
- INDIGO\_500
- INDIGO\_600
- INDIGO\_700
- INDIGO\_800
- INDIGO\_900
- INDIGO\_ACCENT
- INDIGO\_ACCENT\_100
- INDIGO\_ACCENT\_200
- INDIGO\_ACCENT\_400
- INDIGO\_ACCENT\_700
- INVERSE\_PRIMARY
- INVERSE\_SURFACE
- LIGHT\_BLUE
- LIGHT\_BLUE\_100

- LIGHT\_BLUE\_200
- LIGHT\_BLUE\_300
- LIGHT\_BLUE\_400
- LIGHT\_BLUE\_50
- LIGHT\_BLUE\_500
- LIGHT\_BLUE\_600
- LIGHT\_BLUE\_700
- LIGHT\_BLUE\_800
- LIGHT\_BLUE\_900
- LIGHT\_BLUE\_ACCENT
- LIGHT\_BLUE\_ACCENT\_100
- LIGHT\_BLUE\_ACCENT\_200
- LIGHT\_BLUE\_ACCENT\_400
- LIGHT\_BLUE\_ACCENT\_700
- LIGHT\_GREEN
- LIGHT\_GREEN\_100
- LIGHT\_GREEN\_200
- LIGHT\_GREEN\_300
- LIGHT\_GREEN\_400
- LIGHT\_GREEN\_50
- LIGHT\_GREEN\_500
- LIGHT\_GREEN\_600
- LIGHT\_GREEN\_700
- LIGHT\_GREEN\_800
- LIGHT\_GREEN\_900
- LIGHT\_GREEN\_ACCENT
- LIGHT\_GREEN\_ACCENT\_100
- LIGHT\_GREEN\_ACCENT\_200
- LIGHT\_GREEN\_ACCENT\_400
- LIGHT\_GREEN\_ACCENT\_700
- LIME
- LIME\_100
- LIME\_200
- LIME\_300
- LIME\_400
- LIME\_50
- LIME\_500
- LIME\_600
- LIME\_700
- LIME\_800
- LIME\_900

- LIME\_ACCENT
- LIME\_ACCENT\_100
- LIME\_ACCENT\_200
- LIME\_ACCENT\_400
- LIME\_ACCENT\_700
- ON\_ERROR
- ON\_ERROR\_CONTAINER
- ON\_INVERSE\_SURFACE
- ON\_PRIMARY
- ON\_PRIMARY\_CONTAINER
- ON\_PRIMARY\_FIXED
- ON\_PRIMARY\_FIXED\_VARIANT
- ON\_SECONDARY
- ON\_SECONDARY\_CONTAINER
- ON\_SECONDARY\_FIXED
- ON\_SECONDARY\_FIXED\_VARIANT
- ON\_SURFACE
- ON\_SURFACE\_VARIANT
- ON\_TERTIARY
- ON\_TERTIARY\_CONTAINER
- ON\_TERTIARY\_FIXED
- ON\_TERTIARY\_FIXED\_VARIANT
- ORANGE
- ORANGE\_100
- ORANGE\_200
- ORANGE\_300
- ORANGE\_400
- ORANGE\_50
- ORANGE\_500
- ORANGE\_600
- ORANGE\_700
- ORANGE\_800
- ORANGE\_900
- ORANGE\_ACCENT
- ORANGE\_ACCENT\_100
- ORANGE\_ACCENT\_200
- ORANGE\_ACCENT\_400
- ORANGE\_ACCENT\_700
- OUTLINE
- OUTLINE\_VARIANT
- PINK



- PINK\_100
- PINK\_200
- PINK\_300
- PINK\_400
- PINK\_50
- PINK\_500
- PINK\_600
- PINK\_700
- PINK\_800
- PINK\_900
- PINK\_ACCENT
- PINK\_ACCENT\_100
- PINK\_ACCENT\_200
- PINK\_ACCENT\_400
- PINK\_ACCENT\_700
- PRIMARY
- PRIMARY\_CONTAINER
- PRIMARY\_FIXED
- PRIMARY\_FIXED\_DIM
- PURPLE
- PURPLE\_100
- PURPLE\_200
- PURPLE\_300
- PURPLE\_400
- PURPLE\_50
- PURPLE\_500
- PURPLE\_600
- PURPLE\_700
- PURPLE\_800
- PURPLE\_900
- PURPLE\_ACCENT
- PURPLE\_ACCENT\_100
- PURPLE\_ACCENT\_200
- PURPLE\_ACCENT\_400
- PURPLE\_ACCENT\_700
- RED
- RED\_100
- RED\_200
- RED\_300
- RED\_400
- RED\_50

- RED\_500
- RED\_600
- RED\_700
- RED\_800
- RED\_900
- RED\_ACCENT
- RED\_ACCENT\_100
- RED\_ACCENT\_200
- RED\_ACCENT\_400
- RED\_ACCENT\_700
- SCRIM
- SECONDARY
- SECONDARY\_CONTAINER
- SECONDARY\_FIXED
- SECONDARY\_FIXED\_DIM
- SHADOW
- SURFACE
- SURFACE\_BRIGHT
- SURFACE\_CONTAINER
- SURFACE\_CONTAINER\_HIGH
- SURFACE\_CONTAINER\_HIGHEST
- SURFACE\_CONTAINER\_LOW
- SURFACE\_CONTAINER\_LOWEST
- SURFACE\_DIM
- SURFACE\_TINT
- TEAL
- TEAL\_100
- TEAL\_200
- TEAL\_300
- TEAL\_400
- TEAL\_50
- TEAL\_500
- TEAL\_600
- TEAL\_700
- TEAL\_800
- TEAL\_900
- TEAL\_ACCENT
- TEAL\_ACCENT\_100
- TEAL\_ACCENT\_200
- TEAL\_ACCENT\_400
- TEAL\_ACCENT\_700

- TERTIARY
- TERTIARY\_CONTAINER
- TERTIARY\_FIXED
- TERTIARY\_FIXED\_DIM
- TRANSPARENT
- WHITE
- WHITE\_10
- WHITE\_12
- WHITE\_24
- WHITE\_30
- WHITE\_38
- WHITE\_54
- WHITE\_60
- WHITE\_70
- YELLOW
- YELLOW\_100
- YELLOW\_200
- YELLOW\_300
- YELLOW\_400
- YELLOW\_50
- YELLOW\_500
- YELLOW\_600
- YELLOW\_700
- YELLOW\_800
- YELLOW\_900
- YELLOW\_ACCENT
- YELLOW\_ACCENT\_100
- YELLOW\_ACCENT\_200
- YELLOW\_ACCENT\_400
- YELLOW\_ACCENT\_700

## Methods

- random - Selects a random color, with optional exclusions and weights.
- with\_opacity - Returns a color with the given opacity.

## Properties

buildAMBERclass-attributeinstance-attribute

buildAMBER\_100class-attributeinstance-attribute

buildAMBER\_200class-attributeinstance-attribute

buildAMBER\_300class-attributeinstance-attribute

buildAMBER\_400class-attributeinstance-attribute

buildAMBER\_50class-attributeinstance-attribute

buildAMBER\_500class-attributeinstance-attribute  
buildAMBER\_600class-attributeinstance-attribute  
buildAMBER\_700class-attributeinstance-attribute  
buildAMBER\_800class-attributeinstance-attribute  
buildAMBER\_900class-attributeinstance-attribute  
buildAMBER\_ACCENTclass-attributeinstance-attribute  
buildAMBER\_ACCENT\_100class-attributeinstance-attribute  
buildAMBER\_ACCENT\_200class-attributeinstance-attribute  
buildAMBER\_ACCENT\_400class-attributeinstance-attribute  
buildAMBER\_ACCENT\_700class-attributeinstance-attribute  
buildBLACKclass-attributeinstance-attribute  
buildBLACK\_12class-attributeinstance-attribute  
buildBLACK\_26class-attributeinstance-attribute  
buildBLACK\_38class-attributeinstance-attribute  
buildBLACK\_45class-attributeinstance-attribute  
buildBLACK\_54class-attributeinstance-attribute  
buildBLACK\_87class-attributeinstance-attribute  
buildBLUEclass-attributeinstance-attribute  
buildBLUE\_100class-attributeinstance-attribute  
buildBLUE\_200class-attributeinstance-attribute  
buildBLUE\_300class-attributeinstance-attribute  
buildBLUE\_400class-attributeinstance-attribute  
buildBLUE\_50class-attributeinstance-attribute  
buildBLUE\_500class-attributeinstance-attribute  
buildBLUE\_600class-attributeinstance-attribute  
buildBLUE\_700class-attributeinstance-attribute  
buildBLUE\_800class-attributeinstance-attribute  
buildBLUE\_900class-attributeinstance-attribute  
buildBLUE\_ACCENTclass-attributeinstance-attribute  
buildBLUE\_ACCENT\_100class-attributeinstance-attribute  
buildBLUE\_ACCENT\_200class-attributeinstance-attribute  
buildBLUE\_ACCENT\_400class-attributeinstance-attribute  
buildBLUE\_ACCENT\_700class-attributeinstance-attribute  
buildBLUE\_GREYclass-attributeinstance-attribute  
buildBLUE\_GREY\_100class-attributeinstance-attribute

buildBLUE\_GREY\_200class-attributeinstance-attribute  
buildBLUE\_GREY\_300class-attributeinstance-attribute  
buildBLUE\_GREY\_400class-attributeinstance-attribute  
buildBLUE\_GREY\_50class-attributeinstance-attribute  
buildBLUE\_GREY\_500class-attributeinstance-attribute  
buildBLUE\_GREY\_600class-attributeinstance-attribute  
buildBLUE\_GREY\_700class-attributeinstance-attribute  
buildBLUE\_GREY\_800class-attributeinstance-attribute  
buildBLUE\_GREY\_900class-attributeinstance-attribute  
buildBROWNclass-attributeinstance-attribute  
buildBROWN\_100class-attributeinstance-attribute  
buildBROWN\_200class-attributeinstance-attribute  
buildBROWN\_300class-attributeinstance-attribute  
buildBROWN\_400class-attributeinstance-attribute  
buildBROWN\_50class-attributeinstance-attribute  
buildBROWN\_500class-attributeinstance-attribute  
buildBROWN\_600class-attributeinstance-attribute  
buildBROWN\_700class-attributeinstance-attribute  
buildBROWN\_800class-attributeinstance-attribute  
buildBROWN\_900class-attributeinstance-attribute  
buildCYANclass-attributeinstance-attribute  
buildCYAN\_100class-attributeinstance-attribute  
buildCYAN\_200class-attributeinstance-attribute  
buildCYAN\_300class-attributeinstance-attribute  
buildCYAN\_400class-attributeinstance-attribute  
buildCYAN\_50class-attributeinstance-attribute  
buildCYAN\_500class-attributeinstance-attribute  
buildCYAN\_600class-attributeinstance-attribute  
buildCYAN\_700class-attributeinstance-attribute  
buildCYAN\_800class-attributeinstance-attribute  
buildCYAN\_900class-attributeinstance-attribute  
buildCYAN\_ACCENTclass-attributeinstance-attribute  
buildCYAN\_ACCENT\_100class-attributeinstance-attribute  
buildCYAN\_ACCENT\_200class-attributeinstance-attribute  
buildCYAN\_ACCENT\_400class-attributeinstance-attribute

buildCYAN\_ACCENT\_700class-attributeinstance-attribute  
buildDEEP\_ORANGEclass-attributeinstance-attribute  
buildDEEP\_ORANGE\_100class-attributeinstance-attribute  
buildDEEP\_ORANGE\_200class-attributeinstance-attribute  
buildDEEP\_ORANGE\_300class-attributeinstance-attribute  
buildDEEP\_ORANGE\_400class-attributeinstance-attribute  
buildDEEP\_ORANGE\_50class-attributeinstance-attribute  
buildDEEP\_ORANGE\_500class-attributeinstance-attribute  
buildDEEP\_ORANGE\_600class-attributeinstance-attribute  
buildDEEP\_ORANGE\_700class-attributeinstance-attribute  
buildDEEP\_ORANGE\_800class-attributeinstance-attribute  
buildDEEP\_ORANGE\_900class-attributeinstance-attribute  
buildDEEP\_ORANGE\_ACCENTclass-attributeinstance-attribute  
buildDEEP\_ORANGE\_ACCENT\_100class-attributeinstance-attribute  
buildDEEP\_ORANGE\_ACCENT\_200class-attributeinstance-attribute  
buildDEEP\_ORANGE\_ACCENT\_400class-attributeinstance-attribute  
buildDEEP\_ORANGE\_ACCENT\_700class-attributeinstance-attribute  
buildDEEP\_PURPLEclass-attributeinstance-attribute  
buildDEEP\_PURPLE\_100class-attributeinstance-attribute  
buildDEEP\_PURPLE\_200class-attributeinstance-attribute  
buildDEEP\_PURPLE\_300class-attributeinstance-attribute  
buildDEEP\_PURPLE\_400class-attributeinstance-attribute  
buildDEEP\_PURPLE\_50class-attributeinstance-attribute  
buildDEEP\_PURPLE\_500class-attributeinstance-attribute  
buildDEEP\_PURPLE\_600class-attributeinstance-attribute  
buildDEEP\_PURPLE\_700class-attributeinstance-attribute  
buildDEEP\_PURPLE\_800class-attributeinstance-attribute  
buildDEEP\_PURPLE\_900class-attributeinstance-attribute  
buildDEEP\_PURPLE\_ACCENTclass-attributeinstance-attribute  
buildDEEP\_PURPLE\_ACCENT\_100class-attributeinstance-attribute  
buildDEEP\_PURPLE\_ACCENT\_200class-attributeinstance-attribute  
buildDEEP\_PURPLE\_ACCENT\_400class-attributeinstance-attribute  
buildDEEP\_PURPLE\_ACCENT\_700class-attributeinstance-attribute  
buildERRORclass-attributeinstance-attribute  
buildERROR\_CONTAINERclass-attributeinstance-attribute

buildGREENclass-attributeinstance-attribute  
buildGREEN\_100class-attributeinstance-attribute  
buildGREEN\_200class-attributeinstance-attribute  
buildGREEN\_300class-attributeinstance-attribute  
buildGREEN\_400class-attributeinstance-attribute  
buildGREEN\_50class-attributeinstance-attribute  
buildGREEN\_500class-attributeinstance-attribute  
buildGREEN\_600class-attributeinstance-attribute  
buildGREEN\_700class-attributeinstance-attribute  
buildGREEN\_800class-attributeinstance-attribute  
buildGREEN\_900class-attributeinstance-attribute  
buildGREEN\_ACCENTclass-attributeinstance-attribute  
buildGREEN\_ACCENT\_100class-attributeinstance-attribute  
buildGREEN\_ACCENT\_200class-attributeinstance-attribute  
buildGREEN\_ACCENT\_400class-attributeinstance-attribute  
buildGREEN\_ACCENT\_700class-attributeinstance-attribute  
buildGREYclass-attributeinstance-attribute  
buildGREY\_100class-attributeinstance-attribute  
buildGREY\_200class-attributeinstance-attribute  
buildGREY\_300class-attributeinstance-attribute  
buildGREY\_400class-attributeinstance-attribute  
buildGREY\_50class-attributeinstance-attribute  
buildGREY\_500class-attributeinstance-attribute  
buildGREY\_600class-attributeinstance-attribute  
buildGREY\_700class-attributeinstance-attribute  
buildGREY\_800class-attributeinstance-attribute  
buildGREY\_900class-attributeinstance-attribute  
buildINDIGOclass-attributeinstance-attribute  
buildINDIGO\_100class-attributeinstance-attribute  
buildINDIGO\_200class-attributeinstance-attribute  
buildINDIGO\_300class-attributeinstance-attribute  
buildINDIGO\_400class-attributeinstance-attribute  
buildINDIGO\_50class-attributeinstance-attribute  
buildINDIGO\_500class-attributeinstance-attribute  
buildINDIGO\_600class-attributeinstance-attribute

buildINDIGO\_700class-attributeinstance-attribute  
buildINDIGO\_800class-attributeinstance-attribute  
buildINDIGO\_900class-attributeinstance-attribute  
buildINDIGO\_ACCENTclass-attributeinstance-attribute  
buildINDIGO\_ACCENT\_100class-attributeinstance-attribute  
buildINDIGO\_ACCENT\_200class-attributeinstance-attribute  
buildINDIGO\_ACCENT\_400class-attributeinstance-attribute  
buildINDIGO\_ACCENT\_700class-attributeinstance-attribute  
buildINVERSE\_PRIMARYclass-attributeinstance-attribute  
buildINVERSE\_SURFACEclass-attributeinstance-attribute  
buildLIGHT\_BLUEclass-attributeinstance-attribute  
buildLIGHT\_BLUE\_100class-attributeinstance-attribute  
buildLIGHT\_BLUE\_200class-attributeinstance-attribute  
buildLIGHT\_BLUE\_300class-attributeinstance-attribute  
buildLIGHT\_BLUE\_400class-attributeinstance-attribute  
buildLIGHT\_BLUE\_50class-attributeinstance-attribute  
buildLIGHT\_BLUE\_500class-attributeinstance-attribute  
buildLIGHT\_BLUE\_600class-attributeinstance-attribute  
buildLIGHT\_BLUE\_700class-attributeinstance-attribute  
buildLIGHT\_BLUE\_800class-attributeinstance-attribute  
buildLIGHT\_BLUE\_900class-attributeinstance-attribute  
buildLIGHT\_BLUE\_ACCENTclass-attributeinstance-attribute  
buildLIGHT\_BLUE\_ACCENT\_100class-attributeinstance-attribute  
buildLIGHT\_BLUE\_ACCENT\_200class-attributeinstance-attribute  
buildLIGHT\_BLUE\_ACCENT\_400class-attributeinstance-attribute  
buildLIGHT\_BLUE\_ACCENT\_700class-attributeinstance-attribute  
buildLIGHT\_GREENclass-attributeinstance-attribute  
buildLIGHT\_GREEN\_100class-attributeinstance-attribute  
buildLIGHT\_GREEN\_200class-attributeinstance-attribute  
buildLIGHT\_GREEN\_300class-attributeinstance-attribute  
buildLIGHT\_GREEN\_400class-attributeinstance-attribute  
buildLIGHT\_GREEN\_50class-attributeinstance-attribute  
buildLIGHT\_GREEN\_500class-attributeinstance-attribute  
buildLIGHT\_GREEN\_600class-attributeinstance-attribute  
buildLIGHT\_GREEN\_700class-attributeinstance-attribute



buildLIGHT\_GREEN\_800class-attributeinstance-attribute  
buildLIGHT\_GREEN\_900class-attributeinstance-attribute  
buildLIGHT\_GREEN\_ACCENTclass-attributeinstance-attribute  
buildLIGHT\_GREEN\_ACCENT\_100class-attributeinstance-attribute  
buildLIGHT\_GREEN\_ACCENT\_200class-attributeinstance-attribute  
buildLIGHT\_GREEN\_ACCENT\_400class-attributeinstance-attribute  
buildLIGHT\_GREEN\_ACCENT\_700class-attributeinstance-attribute  
buildLIMEclass-attributeinstance-attribute  
buildLIME\_100class-attributeinstance-attribute  
buildLIME\_200class-attributeinstance-attribute  
buildLIME\_300class-attributeinstance-attribute  
buildLIME\_400class-attributeinstance-attribute  
buildLIME\_50class-attributeinstance-attribute  
buildLIME\_500class-attributeinstance-attribute  
buildLIME\_600class-attributeinstance-attribute  
buildLIME\_700class-attributeinstance-attribute  
buildLIME\_800class-attributeinstance-attribute  
buildLIME\_900class-attributeinstance-attribute  
buildLIME\_ACCENTclass-attributeinstance-attribute  
buildLIME\_ACCENT\_100class-attributeinstance-attribute  
buildLIME\_ACCENT\_200class-attributeinstance-attribute  
buildLIME\_ACCENT\_400class-attributeinstance-attribute  
buildLIME\_ACCENT\_700class-attributeinstance-attribute  
buildON\_ERRORclass-attributeinstance-attribute  
buildON\_ERROR\_CONTAINERclass-attributeinstance-attribute  
buildON\_INVERSE\_SURFACEclass-attributeinstance-attribute  
buildON\_PRIMARYclass-attributeinstance-attribute  
buildON\_PRIMARY\_CONTAINERclass-attributeinstance-attribute  
buildON\_PRIMARY\_FIXEDclass-attributeinstance-attribute  
buildON\_PRIMARY\_FIXED\_VARIANTclass-attributeinstance-attribute  
buildON\_SECONDARYclass-attributeinstance-attribute  
buildON\_SECONDARY\_CONTAINERclass-attributeinstance-attribute  
buildON\_SECONDARY\_FIXEDclass-attributeinstance-attribute  
buildON\_SECONDARY\_FIXED\_VARIANTclass-attributeinstance-attribute  
buildON\_SURFACEclass-attributeinstance-attribute

buildON\_SURFACE\_VARIANTclass-attributeinstance-attribute  
buildON\_TERTIARYclass-attributeinstance-attribute  
buildON\_TERTIARY\_CONTAINERclass-attributeinstance-attribute  
buildON\_TERTIARY\_FIXEDclass-attributeinstance-attribute  
buildON\_TERTIARY\_FIXED\_VARIANTclass-attributeinstance-attribute  
buildORANGEclass-attributeinstance-attribute  
buildORANGE\_100class-attributeinstance-attribute  
buildORANGE\_200class-attributeinstance-attribute  
buildORANGE\_300class-attributeinstance-attribute  
buildORANGE\_400class-attributeinstance-attribute  
buildORANGE\_50class-attributeinstance-attribute  
buildORANGE\_500class-attributeinstance-attribute  
buildORANGE\_600class-attributeinstance-attribute  
buildORANGE\_700class-attributeinstance-attribute  
buildORANGE\_800class-attributeinstance-attribute  
buildORANGE\_900class-attributeinstance-attribute  
buildORANGE\_ACCENTclass-attributeinstance-attribute  
buildORANGE\_ACCENT\_100class-attributeinstance-attribute  
buildORANGE\_ACCENT\_200class-attributeinstance-attribute  
buildORANGE\_ACCENT\_400class-attributeinstance-attribute  
buildORANGE\_ACCENT\_700class-attributeinstance-attribute  
buildOUTLINEclass-attributeinstance-attribute  
buildOUTLINE\_VARIANTclass-attributeinstance-attribute  
buildPINKclass-attributeinstance-attribute  
buildPINK\_100class-attributeinstance-attribute  
buildPINK\_200class-attributeinstance-attribute  
buildPINK\_300class-attributeinstance-attribute  
buildPINK\_400class-attributeinstance-attribute  
buildPINK\_50class-attributeinstance-attribute  
buildPINK\_500class-attributeinstance-attribute  
buildPINK\_600class-attributeinstance-attribute  
buildPINK\_700class-attributeinstance-attribute  
buildPINK\_800class-attributeinstance-attribute  
buildPINK\_900class-attributeinstance-attribute  
buildPINK\_ACCENTclass-attributeinstance-attribute

buildPINK\_ACCENT\_100class-attributeinstance-attribute  
buildPINK\_ACCENT\_200class-attributeinstance-attribute  
buildPINK\_ACCENT\_400class-attributeinstance-attribute  
buildPINK\_ACCENT\_700class-attributeinstance-attribute  
buildPRIMARYclass-attributeinstance-attribute  
buildPRIMARY\_CONTAINERclass-attributeinstance-attribute  
buildPRIMARY\_FIXEDclass-attributeinstance-attribute  
buildPRIMARY\_FIXED\_DIMclass-attributeinstance-attribute  
buildPURPLEclass-attributeinstance-attribute  
buildPURPLE\_100class-attributeinstance-attribute  
buildPURPLE\_200class-attributeinstance-attribute  
buildPURPLE\_300class-attributeinstance-attribute  
buildPURPLE\_400class-attributeinstance-attribute  
buildPURPLE\_50class-attributeinstance-attribute  
buildPURPLE\_500class-attributeinstance-attribute  
buildPURPLE\_600class-attributeinstance-attribute  
buildPURPLE\_700class-attributeinstance-attribute  
buildPURPLE\_800class-attributeinstance-attribute  
buildPURPLE\_900class-attributeinstance-attribute  
buildPURPLE\_ACCENTclass-attributeinstance-attribute  
buildPURPLE\_ACCENT\_100class-attributeinstance-attribute  
buildPURPLE\_ACCENT\_200class-attributeinstance-attribute  
buildPURPLE\_ACCENT\_400class-attributeinstance-attribute  
buildPURPLE\_ACCENT\_700class-attributeinstance-attribute  
buildREDclass-attributeinstance-attribute  
buildRED\_100class-attributeinstance-attribute  
buildRED\_200class-attributeinstance-attribute  
buildRED\_300class-attributeinstance-attribute  
buildRED\_400class-attributeinstance-attribute  
buildRED\_50class-attributeinstance-attribute  
buildRED\_500class-attributeinstance-attribute  
buildRED\_600class-attributeinstance-attribute  
buildRED\_700class-attributeinstance-attribute  
buildRED\_800class-attributeinstance-attribute  
buildRED\_900class-attributeinstance-attribute

buildRED\_ACCENTclass-attributeinstance-attribute  
buildRED\_ACCENT\_100class-attributeinstance-attribute  
buildRED\_ACCENT\_200class-attributeinstance-attribute  
buildRED\_ACCENT\_400class-attributeinstance-attribute  
buildRED\_ACCENT\_700class-attributeinstance-attribute  
buildSCRIMclass-attributeinstance-attribute  
buildSECONDARYclass-attributeinstance-attribute  
buildSECONDARY\_CONTAINERclass-attributeinstance-attribute  
buildSECONDARY\_FIXEDclass-attributeinstance-attribute  
buildSECONDARY\_FIXED\_DIMclass-attributeinstance-attribute  
buildSHADOWclass-attributeinstance-attribute  
buildSURFACEclass-attributeinstance-attribute  
buildSURFACE\_BRIGHTclass-attributeinstance-attribute  
buildSURFACE\_CONTAINERclass-attributeinstance-attribute  
buildSURFACE\_CONTAINER\_HIGHclass-attributeinstance-attribute  
buildSURFACE\_CONTAINER\_HIGHESTclass-attributeinstance-attribute  
buildSURFACE\_CONTAINER\_LOWclass-attributeinstance-attribute  
buildSURFACE\_CONTAINER\_LOWESTclass-attributeinstance-attribute  
buildSURFACE\_DIMclass-attributeinstance-attribute  
buildSURFACE\_TINTclass-attributeinstance-attribute  
buildTEALclass-attributeinstance-attribute  
buildTEAL\_100class-attributeinstance-attribute  
buildTEAL\_200class-attributeinstance-attribute  
buildTEAL\_300class-attributeinstance-attribute  
buildTEAL\_400class-attributeinstance-attribute  
buildTEAL\_50class-attributeinstance-attribute  
buildTEAL\_500class-attributeinstance-attribute  
buildTEAL\_600class-attributeinstance-attribute  
buildTEAL\_700class-attributeinstance-attribute  
buildTEAL\_800class-attributeinstance-attribute  
buildTEAL\_900class-attributeinstance-attribute  
buildTEAL\_ACCENTclass-attributeinstance-attribute  
buildTEAL\_ACCENT\_100class-attributeinstance-attribute  
buildTEAL\_ACCENT\_200class-attributeinstance-attribute  
buildTEAL\_ACCENT\_400class-attributeinstance-attribute

buildTEAL\_ACCENT\_700class-attributeinstance-attribute  
buildTERTIARYclass-attributeinstance-attribute  
buildTERTIARY\_CONTAINERclass-attributeinstance-attribute  
buildTERTIARY\_FIXEDclass-attributeinstance-attribute  
buildTERTIARY\_FIXED\_DIMclass-attributeinstance-attribute  
buildTRANSPARENTclass-attributeinstance-attribute  
buildWHITEclass-attributeinstance-attribute  
buildWHITE\_10class-attributeinstance-attribute  
buildWHITE\_12class-attributeinstance-attribute  
buildWHITE\_24class-attributeinstance-attribute  
buildWHITE\_30class-attributeinstance-attribute  
buildWHITE\_38class-attributeinstance-attribute  
buildWHITE\_54class-attributeinstance-attribute  
buildWHITE\_60class-attributeinstance-attribute  
buildWHITE\_70class-attributeinstance-attribute  
buildYELLOWclass-attributeinstance-attribute  
buildYELLOW\_100class-attributeinstance-attribute  
buildYELLOW\_200class-attributeinstance-attribute  
buildYELLOW\_300class-attributeinstance-attribute  
buildYELLOW\_400class-attributeinstance-attribute  
buildYELLOW\_50class-attributeinstance-attribute  
buildYELLOW\_500class-attributeinstance-attribute  
buildYELLOW\_600class-attributeinstance-attribute  
buildYELLOW\_700class-attributeinstance-attribute  
buildYELLOW\_800class-attributeinstance-attribute  
buildYELLOW\_900class-attributeinstance-attribute  
buildYELLOW\_ACCENTclass-attributeinstance-attribute  
buildYELLOW\_ACCENT\_100class-attributeinstance-attribute  
buildYELLOW\_ACCENT\_200class-attributeinstance-attribute  
buildYELLOW\_ACCENT\_400class-attributeinstance-attribute  
buildYELLOW\_ACCENT\_700class-attributeinstance-attribute

## Methods

deployed\_coderandomstaticmethod

```
random(
 exclude: list[Colors] | None = None,
 weights: dict[Colors, int] | None = None,
) -> Colors | None
```

Selects a random color, with optional exclusions and weights.

Parameters:

- `exclude` (list[Colors] | None, default: None) - A list of Colors to exclude from the selection.
- `weights` (dict[Colors, int] | None, default: None) - A dictionary mapping color members to their respective weights for weighted random selection.

Returns:

- Colors | None - A randomly selected color, or None if all members are excluded.

### `deployed_codewith_opacitystaticmethod`

```
with_opacity(
 opacity: int | float, color: ColorValue
) -> str
```

Returns a color with the given opacity.

Parameters:

- `opacity` (int | float) - The opacity value between 0.0 and 1.0.
- `color` (ColorValue) - The color to apply opacity to.

Returns:

- str - A string representing the color with opacity, in the format
- str - "color,opacity".

Raises:

- ValueError - If the opacity is not between 0 and 1 (inclusive).

# ColorScheme

A set of more than 40 colors based on the Material spec that can be used to configure the color properties of most components. Read more about color schemes in [here](#).

## Properties

- `error` - The color to use for input validation errors, e.g.
- `error_container` - A color used for error elements needing less emphasis than `error`.
- `inverse_primary` - An accent color used for displaying a highlight color on `inverse_surface` backgrounds, like button text in a `SnackBar`.
- `inverse_surface` - A surface color used for displaying the reverse of what's seen in the surrounding UI, for example in a `SnackBar` to bring attention to an alert.
- `outline` - A utility color that creates boundaries and emphasis to improve usability.
- `outline_variant` - A utility color that creates boundaries for decorative elements when a 3:1 contrast isn't required, such as for dividers or decorative elements.
- `primary` - The color displayed most frequently across your app's screens and components.
- `primary_container` - A color used for elements needing less emphasis than `primary`.
- `primary_fixed` - A substitute for `primary_container` that's the same color for the dark and light themes.
- `primary_fixed_dim` - A color used for elements needing more emphasis than `primary_fixed`.
- `scrim` - A color use to paint the scrim around of modal components.
- `secondary` - An accent color used for less prominent components in the UI, such as filter Chips, while expanding the opportunity for color expression.
- `secondary_container` - A color used for elements needing less emphasis than `secondary`.
- `secondary_fixed` - A substitute for `secondary_container` that's the same color for the dark and light themes.
- `secondary_fixed_dim` - A color used for elements needing more emphasis than `secondary_fixed`.
- `shadow` - A color use to paint the drop shadows of elevated components.
- `surface` - The background color for widgets like `Card`.
- `surface_bright` - A color that's always the lightest in the dark or light theme.
- `surface_container` - A recommended color role for a distinct area within the surface.
- `surface_container_high` - A surface container color with a darker tone.
- `surface_container_highest` - A surface container color with the darkest tone.
- `surface_container_low` - A surface container color with a lighter tone that creates less emphasis than `surface_container` but more emphasis than `surface_container_lowest`.
- `surface_container_lowest` - A surface container color with the lightest tone and the least emphasis relative to the surface.
- `surface_dim` - A color that's always darkest in the dark or light theme.
- `surface_tint` - A color used as an overlay on a surface color to indicate a component's elevation.
- `tertiary` - A color used as a contrasting accent that can balance primary and secondary colors or bring heightened attention to an element, such as an input field.
- `tertiary_container` - A color used for elements needing less emphasis than `tertiary`.
- `tertiary_fixed` - A substitute for `tertiary_container` that's the same color for dark and light themes.
- `tertiary_fixed_dim` - A color used for elements needing more emphasis than `tertiary_fixed`.

## Events

- `on_error` - A color that's clearly legible when drawn on `error`.

- `on_error_container` - A color that's clearly legible when drawn on `error_container`.
- `on_inverse_surface` - A color that's clearly legible when drawn on `inverse_surface`.
- `on_primary` - A color that's clearly legible when drawn on `primary`.
- `on_primary_container` - A color that's clearly legible when drawn on `primary_container`.
- `on_primary_fixed` - A color that is used for text and icons that exist on top of elements having `primary_fixed` color.
- `on_primary_fixed_variant` - A color that provides a lower-emphasis option for text and icons than `on_primary_fixed`.
- `on_secondary` - A color that's clearly legible when drawn on `secondary`.
- `on_secondary_container` - A color that's clearly legible when drawn on `secondary_container`.
- `on_secondary_fixed` - A color that is used for text and icons that exist on top of elements having `secondary_fixed` color.
- `on_secondary_fixed_variant` - A color that provides a lower-emphasis option for text and icons than `on_secondary_fixed`.
- `on_surface` - A color that's clearly legible when drawn on `surface`.
- `on_surface_variant` - A color that's clearly legible when drawn on `surface_container_highest`.
- `on_tertiary` - A color that's clearly legible when drawn on `tertiary`.
- `on_tertiary_container` - A color that's clearly legible when drawn on `tertiary_container`.
- `on_tertiary_fixed` - A color that is used for text and icons that exist on top of elements having `tertiary_fixed` color.
- `on_tertiary_fixed_variant` - A color that provides a lower-emphasis option for text and icons than `on_tertiary_fixed`.

## Properties

### `builderrorclass-attributeinstance-attribute`

`error: ColorValue | None = None`

The color to use for input validation errors, e.g. for `FormFieldControl.error`.

### `builderror_containerclass-attributeinstance-attribute`

`error_container: ColorValue | None = None`

A color used for error elements needing less emphasis than `error`.

### `buildinverse_primaryclass-attributeinstance-attribute`

`inverse_primary: ColorValue | None = None`

An accent color used for displaying a highlight color on `inverse_surface` backgrounds, like button text in a `SnackBar`.

### `buildinverse_surfaceclass-attributeinstance-attribute`

`inverse_surface: ColorValue | None = None`

A surface color used for displaying the reverse of what's seen in the surrounding UI, for example in a `SnackBar` to bring attention to an alert.

### `buildoutlineclass-attributeinstance-attribute`

`outline: ColorValue | None = None`

A utility color that creates boundaries and emphasis to improve usability.

### `buildoutline_variantclass-attributeinstance-attribute`

`outline_variant: ColorValue | None = None`

A utility color that creates boundaries for decorative elements when a 3:1 contrast isn't required, such as for dividers or decorative elements.

### `buildprimaryclass-attributeinstance-attribute`

`primary: ColorValue | None = None`

The color displayed most frequently across your app's screens and components.

### `buildprimary_containerclass-attributeinstance-attribute`

`primary_container: ColorValue | None = None`



A color used for elements needing less emphasis than primary.

### **buildprimary\_fixedclass-attributeinstance-attribute**

`primary_fixed: ColorValue | None = None`

A substitute for `primary_container` that's the same color for the dark and light themes.

### **buildprimary\_fixed\_dimclass-attributeinstance-attribute**

`primary_fixed_dim: ColorValue | None = None`

A color used for elements needing more emphasis than `primary_fixed`.

### **buildscrimclass-attributeinstance-attribute**

`scrim: ColorValue | None = None`

A color use to paint the scrim around of modal components.

### **buildsecondaryclass-attributeinstance-attribute**

`secondary: ColorValue | None = None`

An accent color used for less prominent components in the UI, such as filter Chips, while expanding the opportunity for color expression.

### **buildsecondary\_containerclass-attributeinstance-attribute**

`secondary_container: ColorValue | None = None`

A color used for elements needing less emphasis than secondary.

### **buildsecondary\_fixedclass-attributeinstance-attribute**

`secondary_fixed: ColorValue | None = None`

A substitute for `secondary_container` that's the same color for the dark and light themes.

### **buildsecondary\_fixed\_dimclass-attributeinstance-attribute**

`secondary_fixed_dim: ColorValue | None = None`

A color used for elements needing more emphasis than `secondary_fixed`.

### **buildshadowclass-attributeinstance-attribute**

`shadow: ColorValue | None = None`

A color use to paint the drop shadows of elevated components.

### **buildsurfaceclass-attributeinstance-attribute**

`surface: ColorValue | None = None`

The background color for widgets like Card.

### **buildsurface\_brightclass-attributeinstance-attribute**

`surface_bright: ColorValue | None = None`

A color that's always the lightest in the dark or light theme.

### **buildsurface\_containerclass-attributeinstance-attribute**

`surface_container: ColorValue | None = None`

A recommended color role for a distinct area within the surface.

### **buildsurface\_container\_highclass-attributeinstance-attribute**

`surface_container_high: ColorValue | None = None`

A surface container color with a darker tone.

### **buildsurface\_container\_highestclass-attributeinstance-attribute**

`surface_container_highest: ColorValue | None = None`

A surface container color with the darkest tone. It is used to create the most emphasis against the surface.

### **buildsurface\_container\_lowclass-attributeinstance-attribute**

`surface_container_low: ColorValue | None = None`

A surface container color with a lighter tone that creates less emphasis than `surface_container` but more emphasis than `surface_container_lowest`.

#### **buildsurface\_container\_lowestclass-attributeinstance-attribute**

```
surface_container_lowest: ColorValue | None = None
```

A surface container color with the lightest tone and the least emphasis relative to the surface.

#### **buildsurface\_dimclass-attributeinstance-attribute**

```
surface_dim: ColorValue | None = None
```

A color that's always darkest in the dark or light theme.

#### **buildsurface\_tintclass-attributeinstance-attribute**

```
surface_tint: ColorValue | None = None
```

A color used as an overlay on a surface color to indicate a component's elevation.

#### **buildtertiaryclass-attributeinstance-attribute**

```
tertiary: ColorValue | None = None
```

A color used as a contrasting accent that can balance primary and secondary colors or bring heightened attention to an element, such as an input field.

#### **buildtertiary\_containerclass-attributeinstance-attribute**

```
tertiary_container: ColorValue | None = None
```

A color used for elements needing less emphasis than tertiary.

#### **buildtertiary\_fixedclass-attributeinstance-attribute**

```
tertiary_fixed: ColorValue | None = None
```

A substitute for `tertiary_container` that's the same color for dark and light themes.

#### **buildtertiary\_fixed\_dimclass-attributeinstance-attribute**

```
tertiary_fixed_dim: ColorValue | None = None
```

A color used for elements needing more emphasis than `tertiary_fixed`.

## **Events**

#### **bolton\_errorclass-attributeinstance-attribute**

```
on_error: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on error.

#### **bolton\_error\_containerclass-attributeinstance-attribute**

```
on_error_container: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on `error_container`.

#### **bolton\_inverse\_surfaceclass-attributeinstance-attribute**

```
on_inverse_surface: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on `inverse_surface`.

#### **bolton\_primaryclass-attributeinstance-attribute**

```
on_primary: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on primary.

#### **bolton\_primary\_containerclass-attributeinstance-attribute**

```
on_primary_container: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

```
)
```

A color that's clearly legible when drawn on `primary_container`.

### **`bolton_primary_fixed`class-attributeinstance-attribute**

```
on_primary_fixed: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that is used for text and icons that exist on top of elements having `primary_fixed` color.

### **`bolton_primary_fixed_variant`class-attributeinstance-attribute**

```
on_primary_fixed_variant: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that provides a lower-emphasis option for text and icons than `on_primary_fixed`.

### **`bolton_secondary`class-attributeinstance-attribute**

```
on_secondary: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on `secondary`.

### **`bolton_secondary_container`class-attributeinstance-attribute**

```
on_secondary_container: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on `secondary_container`.

### **`bolton_secondary_fixed`class-attributeinstance-attribute**

```
on_secondary_fixed: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that is used for text and icons that exist on top of elements having `secondary_fixed` color.

### **`bolton_secondary_fixed_variant`class-attributeinstance-attribute**

```
on_secondary_fixed_variant: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that provides a lower-emphasis option for text and icons than `on_secondary_fixed`.

### **`bolton_surface`class-attributeinstance-attribute**

```
on_surface: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on `surface`.

### **`bolton_surface_variant`class-attributeinstance-attribute**

```
on_surface_variant: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on `surface_container_highest`.

### **`bolton_tertiary`class-attributeinstance-attribute**

```
on_tertiary: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on `tertiary`.

### **`bolton_tertiary_container`class-attributeinstance-attribute**

```
on_tertiary_container: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that's clearly legible when drawn on `tertiary_container`.

### **`bolton_tertiary_fixed`class-attributeinstance-attribute**

```
on_tertiary_fixed: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that is used for text and icons that exist on top of elements having tertiary\_fixed color.

### **bolton\_tertiary\_fixed\_variant**class-attributeinstance-attribute

```
on_tertiary_fixed_variant: ColorValue | None = field(
 default=None, metadata={"event": False}
)
```

A color that provides a lower-emphasis option for text and icons than on\_tertiary\_fixed.

## @component

Marks a function as a component.

# ConnectivityChangeEvent

Event fired when connectivity changes.

Inherits: Event

## Properties

- `connectivity` - Current connectivity type(s).

## Properties

### `buildconnectivityinstance-attribute`

`connectivity: list[ConnectivityType]`

Current connectivity type(s).

# ConnectivityType

Connectivity states.

Inherits: `enum.Enum`

## Properties

- `BLUETOOTH` - Bluetooth connectivity.
- `ETHERNET` - Ethernet connectivity.
- `MOBILE` - Mobile data connectivity.
- `NONE` - No connectivity.
- `OTHER` - Other connectivity.
- `VPN` - VPN connectivity.
- `WIFI` - Wi-Fi connectivity.

## Properties

`buildBLUETOOTHclass-attributeinstance-attribute`

Bluetooth connectivity.

`buildETHERNETclass-attributeinstance-attribute`

Ethernet connectivity.

`buildMOBILEclass-attributeinstance-attribute`

Mobile data connectivity.

`buildNONEclass-attributeinstance-attribute`

No connectivity.

`buildOTHERclass-attributeinstance-attribute`

Other connectivity.

`buildVPNclass-attributeinstance-attribute`

VPN connectivity.

`buildWIFIclass-attributeinstance-attribute`

Wi-Fi connectivity.

# Context

Manages the context for Flet controls, including page reference and auto-update behavior.

Context instance is accessed via `context`.

## Properties

- `page` - Returns the current Page associated with the context.

## Methods

- `auto_update_enabled` - Returns whether auto-update is enabled in the current context.
- `disable_auto_update` - Disables auto-update behavior for the current context.
- `disable_components_mode` - Disables components mode in the current context.
- `enable_auto_update` - Enables auto-update behavior for the current context.
- `enable_components_mode` - Enables components mode in the current context.
- `is_components_mode` - Returns whether the current context is in components mode.
- `mark_update_called` - Marks that `.update()` was explicitly called during the current handler.
- `reset_auto_update` - Copies the parent auto-update state into the current context.
- `reset_update_called` - Resets the update-called flag for the current context.
- `was_update_called` - Returns whether `.update()` was explicitly called during the current handler.

## Properties

### `buildpageproperty`

`page: Page`

Returns the current Page associated with the context.

#### Example

```
take page width anywhere in the app
width = ft.context.page.width
```

Returns:

- Page - The current page.

Raises:

- `RuntimeError` - If the property is accessed outside a running Flet app.

## Methods

### `deployed_codeauto_update_enabled`

`auto_update_enabled() -> bool`

Returns whether auto-update is enabled in the current context.

Returns:

- bool - True if auto-update is enabled, False otherwise.

### `deployed_codedisable_auto_update`

`disable_auto_update()`

Disables auto-update behavior for the current context.

#### Example

```
import flet as ft

def main(page: ft.Page):
 def button_click():
```



```

 ft.context.disable_auto_update()
 b.content = "Button clicked!"
 # update just the button
 b.update()

 page.controls.append(ft.Text("This won't appear"))
 # no page.update() will be called here

page.controls.append(b := ft.Button("Action!", on_click=button_click))
page.update() - auto-update is enabled by default

```

```
ft.run(main)
```

## deployed\_codedisable\_components\_mode

```
disable_components_mode()
```

Disables components mode in the current context.

## deployed\_codeenable\_auto\_update

```
enable_auto_update()
```

Enables auto-update behavior for the current context.

### Example

```

import flet as ft

disable auto-update globally for the app
ft.context.disable_auto_update()

def main(page: ft.Page):
 # enable auto-update just inside main
 ft.context.enable_auto_update()

 page.controls.append(ft.Text("Hello, world!"))
 # page.update() - we don't need to call it explicitly

ft.run(main)

```

## deployed\_codeenable\_components\_mode

```
enable_components_mode()
```

Enables components mode in the current context.

## deployed\_codeis\_components\_mode

```
is_components_mode() -> bool
```

Returns whether the current context is in components mode.

Returns:

- bool - True if in components mode, False otherwise.

## deployed\_codemark\_update\_called

```
mark_update_called()
```

Marks that `.update()` was explicitly called during the current handler.

## deployed\_codereset\_auto\_update

```
reset_auto_update()
```

Copies the parent auto-update state into the current context.

## deployed\_codereset\_update\_called

```
reset_update_called()
```

Resets the update-called flag for the current context.

## deployed\_codewas\_update\_called

```
was_update_called() -> bool
```

Returns whether `.update()` was explicitly called during the current handler.

Returns:

- bool - True if `.update()` was called, False otherwise.

# ContextMenuDismissEvent

Event fired when a ContextMenu is dismissed.

Inherits: Event

## Properties

- `button` - Mouse button that triggered the menu.
- `global_position` - Global pointer position in logical pixels.
- `item_count` - Total number of entries displayed in the corresponding context menu.
- `local_position` - Local pointer position relative to the wrapped control.
- `trigger` - The trigger mode that opened the menu.

## Properties

### buildbuttonclass-attributeinstance-attribute

```
button: str | None = field(
 default=None, metadata={"data_field": "b"}
)
```

Mouse button that triggered the menu.

If a string, can be one of: "primary" (linked to ContextMenu.primary\_items), "secondary" (linked to ContextMenu.secondary\_items), or "tertiary" (linked to ContextMenu.tertiary\_items).

### buildglobal\_positionclass-attributeinstance-attribute

```
global_position: Offset = field(
 metadata={"data_field": "g"}
)
```

Global pointer position in logical pixels.

### builditem\_countclass-attributeinstance-attribute

```
item_count: int | None = field(
 default=None, metadata={"data_field": "ic"}
)
```

Total number of entries displayed in the corresponding context menu.

### buildlocal\_positionclass-attributeinstance-attribute

```
local_position: Offset | None = field(
 default=None, metadata={"data_field": "l"}
)
```

Local pointer position relative to the wrapped control.

### buildtriggerclass-attributeinstance-attribute

```
trigger: ContextMenuTrigger | None = field(
 default=None, metadata={"data_field": "tr"}
)
```

The trigger mode that opened the menu.

# ContextMenuSelectEvent

Event fired when a ContextMenu item is selected.

Inherits: ContextMenuDismissEvent

## Properties

- `item` - The selected menu item.
- `item_id` - Internal numeric identifier of the selected menu item.
- `item_index` - Index of the selected menu entry within the rendered list.

## Properties

### `builditemproperty`

`item: PopupMenuItem | None`

The selected menu item.

### `builditem_idclass-attributeinstance-attribute`

```
item_id: int | None = field(
 default=None, metadata={"data_field": "id"}
)
```

Internal numeric identifier of the selected menu item.

### `builditem_indexclass-attributeinstance-attribute`

```
item_index: int | None = field(
 default=None, metadata={"data_field": "idx"}
)
```

Index of the selected menu entry within the rendered list.

# ContextMenuTrigger

Defines how a menu is shown for a specific mouse button.

Inherits: `enum.Enum`

## Properties

- **DOWN** - Represents a trigger mode where the menu is shown when the mouse button is pressed down.
- **LONG\_PRESS** - Represents a trigger mode where the menu is shown after a long press of the mouse button.

## Examples

### Context Menu Trigger showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(trigger: ft.ContextMenuTrigger) -> ft.Container:
 def on_select(e: ft.ContextMenuSelectEvent):
 item = e.item.content if e.item else f"item #{e.item_index}"
 status.value = f"Selected: {item}"
 status.update()

 def on_dismiss(_):
 if status.value == "Open the menu in the area below.":
 status.value = "Menu dismissed."
 status.update()

 menu = ft.ContextMenu(
 primary_trigger=trigger,
 primary_items=[
 ft.PopupMenuItem(icon=ft.Icons.EDIT, content="Edit"),
 ft.PopupMenuItem(icon=ft.Icons.CONTENT_COPY, content="Duplicate"),
 ft.PopupMenuItem(icon=ft.Icons.DELETE, content="Delete"),
],
 on_select=on_select,
 on_dismiss=on_dismiss,
 content=ft.Container(
 width=280,
 height=120,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 alignment=ft.Alignment.CENTER,
 content=ft.Text(
 text_align=ft.TextAlign.CENTER,
 value=(
 "Press and hold inside the area."
 if trigger == ft.ContextMenuTrigger.LONG_PRESS
 else "Press mouse button down inside the area."
),
),
),
)

 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(trigger.name, weight=ft.FontWeight.BOLD),
 menu,
 status := ft.Text("Open the menu in the area below.", size=11),
],
),
)

async def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 if page.web:
 await ft.BrowserContextMenu().disable()
```

```

page.appbar = ft.AppBar(title="ContextMenuTrigger Showcase")
page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare context-menu open behavior for primary trigger modes."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(trigger) for trigger in ft.ContextMenuTrigger
],
),
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

## Properties

**buildDOWN**class-attributeinstance-attribute

Represents a trigger mode where the menu is shown when the mouse button is pressed down.

**buildLONG\_PRESS**class-attributeinstance-attribute

Represents a trigger mode where the menu is shown after a long press of the mouse button.

# ContinuousRectangleBorder

A border with continuous rectangle corners.

Inherits: RoundedRectangleBorder

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Methods

### `deployed_codecopy`

```
copy(
 side: BorderSide | None = None,
 radius: BorderRadiusValue | None = None,
) -> ContinuousRectangleBorder
```

Returns a copy of this object with the specified properties overridden.

## @control

Decorator to optionally set widget name and 'isolated' while behaving like `dataclasses.dataclass`.

Parameters:

- `dart_widget_name` (`type[T] | str | None`, default: `None`) - The name of widget on Dart side.
- `isolated` (`bool | None`, default: `None`) - If `True`, marks the control as isolated. An isolated control is excluded from page updates when its parent control is updated.
- `post_init_args` (`int`, default: `1`) - Number of `InitVar` arguments to pass to `post_init`.
- `dataclass_kwargs` (`Any`, default: `{}`) - Additional keyword arguments passed to `@dataclass`.

Usage:

- Supports `@control` (without parentheses)
- Supports `@control("WidgetName")` (with optional arguments)
- Supports `@control("WidgetName", post_init_args=1, isolated=True)` to specify the number of `InitVar` arguments and isolation



# ControlState

Interactive states that some controls can take on when receiving input from the user.

States are defined by the Material 3 design specification, but are not limited to it.

Not all states are always applicable to all controls.

Inherits: `enum.Enum`

## Properties

- **DEFAULT** - The default state.
- **DISABLED** - The state when this control is disabled and cannot be interacted with.
- **DRAGGED** - The state when this control is being dragged from one place to another by the user.
- **ERROR** - The state when the control has entered some form of invalid state.
- **FOCUSED** - The state when the user navigates with the keyboard to a given control.
- **HOVERED** - The state when the user drags their mouse cursor over the given control.
- **PRESSED** - The state when the user is actively pressing down on the given control.
- **SCROLLED\_UNDER** - The state when this control overlaps the content of a scrollable below.
- **SELECTED** - The state when this item has been selected.

## Properties

### **buildDEFAULT**`class-attributeinstance-attribute`

The default state. Will be used for undeclared states.

### **buildDISABLED**`class-attributeinstance-attribute`

The state when this control is disabled and cannot be interacted with.

Disabled controls should not respond to hover, focus, press, or drag interactions.

See material docs.

### **buildDRAGGED**`class-attributeinstance-attribute`

The state when this control is being dragged from one place to another by the user.

See material docs.

### **buildERROR**`class-attributeinstance-attribute`

The state when the control has entered some form of invalid state.

See material docs.

### **buildFOCUSED**`class-attributeinstance-attribute`

The state when the user navigates with the keyboard to a given control.

This can also sometimes be triggered when a control is tapped. For example, when a `[TextField]` is tapped, it becomes **FOCUSED**.

See material docs.

### **buildHOVERED**`class-attributeinstance-attribute`

The state when the user drags their mouse cursor over the given control.

See material docs.

### **buildPRESSED**`class-attributeinstance-attribute`

The state when the user is actively pressing down on the given control.

See material docs.

## buildSCROLLED\_UNDERclass-attributeinstance-attribute

The state when this control overlaps the content of a scrollable below.

Used by AppBar to indicate that the primary scrollable's content has scrolled up and behind the app bar.

## buildSELECTEDclass-attributeinstance-attribute

The state when this item has been selected.

This applies to things that can be toggled (such as chips and checkboxes) and things that are selected from a set of options (such as tabs and radio buttons).

See material docs.

## Usage Example

Configuring fill\_color can be done in two ways:

### 1. Single Value for All States

If you want to use the same color across all Material states, simply assign fill\_color to a single color:

```
ft.Radio(fill_color=ft.Colors.GREEN)
```

### 2. Specific Values for Each State

For more control, you can provide a dictionary where the key is the state name and the value is the corresponding color.

## Ordering Matters

- The order in which states appear in the dictionary will determine their priority, allowing for flexibility and customization.
- The position of DEFAULT does not affect behavior—it always has the least priority and can be placed anywhere in the dictionary.

## Example

To configure different fill colors of a Radio button for ControlState.HOVERED and ControlState.FOCUSED, with a fallback color for all other states:

```
ft.Radio(
 fill_color={
 ft.ControlState.HOVERED: ft.Colors.GREEN,
 ft.ControlState.FOCUSED: ft.Colors.RED,
 ft.ControlState.DEFAULT: ft.Colors.BLACK,
 }
)
```

This setup ensures that HOVERED and FOCUSED states take precedence, while all unspecified states default to BLACK.

## create\_context

Create a context provider.

Parameters:

- `default_value (T)` - Default context value when no provider is found.

Returns:

- `ContextProvider[T]` - A context provider that can be used with `use_context` and to wrap parts of the component tree that should share the same context value.

# CrossAxisAlignment

How the children should be placed along the cross axis

Inherits: `enum.Enum`

## Properties

- **BASELINE** - Place the children along the cross axis such that their baselines match.
- **CENTER** - Place the children so that their centers align with the middle of the cross axis.
- **END** - Place the children as close to the end of the cross axis as possible.
- **START** - Place the children with their start edge aligned with the start side of the cross axis.
- **STRETCH** - Require the children to fill the cross axis.

## Examples

### Cross Axis Alignment showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(alignment: ft.CrossAxisAlignment) -> ft.Container:
 if alignment == ft.CrossAxisAlignment.STRETCH:
 preview = ft.Row(
 alignment=ft.MainAxisAlignment.SPACE_AROUND,
 vertical_alignment=alignment,
 controls=[
 ft.Container(width=40, bgcolor=ft.Colors.PRIMARY_CONTAINER),
 ft.Container(width=40, bgcolor=ft.Colors.TERTIARY_CONTAINER),
 ft.Container(width=40, bgcolor=ft.Colors.SECONDARY_CONTAINER),
],
)
 else:
 effective_alignment = (
 ft.CrossAxisAlignment.CENTER
 if alignment == ft.CrossAxisAlignment.BASELINE
 else alignment
)
 preview = ft.Row(
 alignment=ft.MainAxisAlignment.SPACE_AROUND,
 vertical_alignment=effective_alignment,
 controls=[
 ft.Container(
 width=40,
 height=24,
 bgcolor=ft.Colors.PRIMARY_CONTAINER,
),
 ft.Container(
 width=40,
 height=46,
 bgcolor=ft.Colors.TERTIARY_CONTAINER,
),
 ft.Container(
 width=40,
 height=34,
 bgcolor=ft.Colors.SECONDARY_CONTAINER,
),
],
)
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(alignment.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=250,
 height=120,
 padding=8,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
)
]
)
)
```

```

 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=preview,
),
 ft.Text(
 "Baseline uses center fallback in this showcase.",
 size=11,
 visible=alignment == ft.CrossAxisAlignment.BASELINE,
 color=ft.Colors.ON_SURFACE_VARIANT,
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="CrossAxisAlignment Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare vertical alignment behavior inside the same "
 "row height."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(alignment)
 for alignment in ft.CrossAxisAlignment
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildBASELINEclass-attributeinstance-attribute

Place the children along the cross axis such that their baselines match.

### buildCENTERclass-attributeinstance-attribute

Place the children so that their centers align with the middle of the cross axis.

### buildENDclass-attributeinstance-attribute

Place the children as close to the end of the cross axis as possible.

### buildSTARTclass-attributeinstance-attribute

Place the children with their start edge aligned with the start side of the cross axis.

### buildSTRETCHclass-attributeinstance-attribute

Require the children to fill the cross axis.

# CupertinoButtonSize

Preset size style for CupertinoButton.

Influences defaults such as minimum size, padding, border radius, and text style.

Inherits: enum.Enum

## Properties

- LARGE - Classic large Cupertino button style.
- MEDIUM - Medium button style with regular text and balanced sizing.
- SMALL - Compact button style with smaller text and tighter sizing.

## Examples

### Cupertino Button Size showcase

play\_arrowTry Online

```
import flet as ft

def showcase_card(size: ft.CupertinoButtonSize) -> ft.Container:
 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=10,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Text(size.name, weight=ft.FontWeight.BOLD),
 ft.CupertinoButton(
 content="Continue",
 icon=ft.CupertinoIcons.RIGHT_CHEVRON,
 size=size,
 bgcolor=ft.Colors.BLUE_600,
 color=ft.Colors.WHITE,
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="CupertinoButtonSize Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare iOS button size presets."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(size) for size in ft.CupertinoButtonSize
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildLARGEclass-attributeinstance-attribute**

Classic large Cupertino button style.

**buildMEDIUMclass-attributeinstance-attribute**

Medium button style with regular text and balanced sizing.

**buildSMALLclass-attributeinstance-attribute**

Compact button style with smaller text and tighter sizing.

# CupertinoColors

Named cupertino colors.

Inherits: str, enum.Enum

## Properties

- ACTIVE\_BLUE
- ACTIVE\_GREEN
- ACTIVE\_ORANGE
- BLACK
- DARK\_BACKGROUND\_GRAY
- DESTRUCTIVE\_RED
- EXTRA\_LIGHT\_BACKGROUND\_GRAY
- INACTIVE\_GRAY
- LABEL
- LIGHT\_BACKGROUND\_GRAY
- LINK
- ON\_PRIMARY
- OPAQUE\_SEPARATOR
- PLACEHOLDER\_TEXT
- PRIMARY
- QUATERNARY\_LABEL
- QUATERNARY\_SYSTEM\_FILL
- SECONDARY\_LABEL
- SECONDARY\_SYSTEM\_BACKGROUND
- SECONDARY\_SYSTEM\_FILL
- SECONDARY\_SYSTEM\_GROUPED\_BACKGROUND
- SEPARATOR
- SYSTEM\_BACKGROUND
- SYSTEM\_BLUE
- SYSTEM\_BROWN
- SYSTEM\_CYAN
- SYSTEM\_FILL
- SYSTEM\_GREEN
- SYSTEM\_GREY
- SYSTEM\_GREY2
- SYSTEM\_GREY3
- SYSTEM\_GREY4
- SYSTEM\_GREY5
- SYSTEM\_GREY6
- SYSTEM\_GROUPED\_BACKGROUND



- SYSTEM\_INDIGO
- SYSTEM\_MINT
- SYSTEM\_ORANGE
- SYSTEM\_PINK
- SYSTEM\_PURPLE
- SYSTEM\_RED
- SYSTEM\_TEAL
- SYSTEM\_YELLOW
- TERTIARY\_LABEL
- TERTIARY\_SYSTEM\_BACKGROUND
- TERTIARY\_SYSTEM\_FILL
- TERTIARY\_SYSTEM\_GROUPED\_BACKGROUND
- WHITE

## Methods

- random - Selects a random color, with optional exclusions and weights.
- with\_opacity - Returns the color with the specified opacity.

## Properties

buildACTIVE\_BLUEclass-attributeinstance-attribute

buildACTIVE\_GREENclass-attributeinstance-attribute

buildACTIVE\_ORANGEcass-attributeinstance-attribute

buildBLACKclass-attributeinstance-attribute

buildDARK\_BACKGROUND\_GRAYclass-attributeinstance-attribute

buildDESTRUCTIVE\_REDClass-attributeinstance-attribute

buildEXTRA\_LIGHT\_BACKGROUND\_GRAYclass-attributeinstance-attribute

buildINACTIVE\_GRAYclass-attributeinstance-attribute

buildLABELclass-attributeinstance-attribute

buildLIGHT\_BACKGROUND\_GRAYclass-attributeinstance-attribute

buildLINKclass-attributeinstance-attribute

buildON\_PRIMARYclass-attributeinstance-attribute

buildOPAQUE\_SEPARATORclass-attributeinstance-attribute

buildPLACEHOLDER\_TEXTclass-attributeinstance-attribute

buildPRIMARYclass-attributeinstance-attribute

buildQUATERNARY\_LABELclass-attributeinstance-attribute

buildQUATERNARY\_SYSTEM\_FILLclass-attributeinstance-attribute

buildSECONDARY\_LABELclass-attributeinstance-attribute

buildSECONDARY\_SYSTEM\_BACKGROUNDclass-attributeinstance-attribute

buildSECONDARY\_SYSTEM\_FILLclass-attributeinstance-attribute

buildSECONDARY\_SYSTEM\_GROUPED\_BACKGROUNDclass-attributeinstance-attribute  
 buildSEPARATORclass-attributeinstance-attribute  
 buildSYSTEM\_BACKGROUNDclass-attributeinstance-attribute  
 buildSYSTEM\_BLUEclass-attributeinstance-attribute  
 buildSYSTEM\_BROWNclass-attributeinstance-attribute  
 buildSYSTEM\_CYANclass-attributeinstance-attribute  
 buildSYSTEM\_FILLclass-attributeinstance-attribute  
 buildSYSTEM\_GREENclass-attributeinstance-attribute  
 buildSYSTEM\_GREYclass-attributeinstance-attribute  
 buildSYSTEM\_GREY2class-attributeinstance-attribute  
 buildSYSTEM\_GREY3class-attributeinstance-attribute  
 buildSYSTEM\_GREY4class-attributeinstance-attribute  
 buildSYSTEM\_GREY5class-attributeinstance-attribute  
 buildSYSTEM\_GREY6class-attributeinstance-attribute  
 buildSYSTEM\_GROUPED\_BACKGROUNDclass-attributeinstance-attribute  
 buildSYSTEM\_INDIGOclass-attributeinstance-attribute  
 buildSYSTEM\_MINTclass-attributeinstance-attribute  
 buildSYSTEM\_ORANGEclass-attributeinstance-attribute  
 buildSYSTEM\_PINKclass-attributeinstance-attribute  
 buildSYSTEM\_PURPLEclass-attributeinstance-attribute  
 buildSYSTEM\_REDclass-attributeinstance-attribute  
 buildSYSTEM\_TEALclass-attributeinstance-attribute  
 buildSYSTEM\_YELLOWclass-attributeinstance-attribute  
 buildTERTIARY\_LABELclass-attributeinstance-attribute  
 buildTERTIARY\_SYSTEM\_BACKGROUNDclass-attributeinstance-attribute  
 buildTERTIARY\_SYSTEM\_FILLclass-attributeinstance-attribute  
 buildTERTIARY\_SYSTEM\_GROUPED\_BACKGROUNDclass-attributeinstance-attribute  
 buildWHITEclass-attributeinstance-attribute

## Methods

deployed\_coderandomstaticmethod

```

random(
 exclude: list[CupertinoColors] | None = None,
 weights: dict[CupertinoColors, int] | None = None,
) -> CupertinoColors | None

```

Selects a random color, with optional exclusions and weights.

Parameters:

- exclude (list[CupertinoColors] | None, default: None) - A list of colors members to exclude from the selection.

- `weights` (`dict[CupertinoColors, int] | None`, default: `None`) - A dictionary mapping color members to their respective weights for weighted random selection.

Returns:

- `CupertinoColors | None` - A randomly selected color, or `None` if all members are excluded.

### `deployed_codewith_opacitystaticmethod`

```
with_opacity(
 opacity: int | float, color: ColorValue
) -> str
```

Returns the color with the specified opacity.

Parameters:

- `opacity` (`int | float`) - The opacity value between 0.0 and 1.0.
- `color` (`ColorValue`) - The color to apply opacity to.

Returns:

- `str` - A string representing the color with opacity, in the format "color,opacity".

Raises:

- `ValueError` - If the opacity is not between 0 and 1 (inclusive).

# CupertinoDatePickerDateOrder

Determines the order of the columns inside CupertinoDatePicker in date mode.

Inherits: `enum.Enum`

## Properties

- `DAY_MONTH_YEAR` - Order of the columns, from left to right: day, month, year.
- `MONTH_DAY_YEAR` - Order of the columns, from left to right: month, day, year.
- `YEAR_DAY_MONTH` - Order of the columns, from left to right: year, day, month.
- `YEAR_MONTH_DAY` - Order of the columns, from left to right: year, month, day.

## Examples

### Cupertino Date Picker Date Order showcase

`play_arrow` Try Online

```
from datetime import datetime

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 def showcase_card(date_order: ft.CupertinoDatePickerDateOrder) -> ft.Container:
 def open_picker(_):
 picker = ft.CupertinoDatePicker(
 date_picker_mode=ft.CupertinoDatePickerMode.DATE,
 date_order=date_order,
 value=datetime(2024, 3, 12, 10, 0),
)
 page.show_dialog(
 ft.CupertinoBottomSheet(
 content=picker,
 height=216,
 padding=ft.Padding.only(top=6),
)
)

 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(date_order.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open picker",
 icon=ft.CupertinoIcons.CALENDAR,
 on_click=open_picker,
),
],
),
)

 page.appbar = ft.AppBar(title="CupertinoDatePickerDateOrder Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Open each variant in CupertinoBottomSheet."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(date_order)
 for date_order in ft.CupertinoDatePickerDateOrder
],
),
],
),
),
)
```

```
), 1,
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildDAY\_MONTH\_YEAR**class-attributeinstance-attribute

Order of the columns, from left to right: day, month, year.

Example: 12 | March | 1996

**buildMONTH\_DAY\_YEAR**class-attributeinstance-attribute

Order of the columns, from left to right: month, day, year.

Example: March | 12 | 1996

**buildYEAR\_DAY\_MONTH**class-attributeinstance-attribute

Order of the columns, from left to right: year, day, month.

Example: 1996 | 12 | March

**buildYEAR\_MONTH\_DAY**class-attributeinstance-attribute

Order of the columns, from left to right: year, month, day.

Example: 1996 | March | 12

# CupertinoDatePickerMode

Different display modes of CupertinoDatePicker.

Inherits: `enum.Enum`

## Properties

- **DATE** - Mode that shows the date in month, day of month, and year.
- **DATE\_AND\_TIME** - Mode that shows the date as day of the week, month, day of month and the time in hour, minute, and (optional) an AM/PM designation.
- **MONTH\_YEAR** - Mode that shows the date in month and year.
- **TIME** - Mode that shows the date in hour, minute, and (optional) an AM/PM designation.

## Examples

### Cupertino Date Picker Mode showcase

[play\\_arrow](#) Try Online

```
from datetime import datetime

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 def showcase_card(mode: ft.CupertinoDatePickerMode) -> ft.Container:
 def open_picker(_):
 picker = ft.CupertinoDatePicker(
 date_picker_mode=mode,
 value=datetime(2024, 7, 13, 16, 14),
 use_24h_format=True,
)
 page.show_dialog(
 ft.CupertinoBottomSheet(
 content=picker,
 height=216,
 padding=ft.Padding.only(top=6),
)
)

 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(mode.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open picker",
 icon=ft.CupertinoIcons.CALENDAR,
 on_click=open_picker,
),
],
),
)

 page.appbar = ft.AppBar(title="CupertinoDatePickerMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Open each mode in CupertinoBottomSheet to compare behavior."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
```

```

 showcase_card(mode) for mode in ft.CupertinoDatePickerMode
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildDATEclass-attributeinstance-attribute

Mode that shows the date in month, day of month, and year. Name of month is spelled in full. Column order is subject to internationalization.

Example: July | 13 | 2012

### buildDATE\_AND\_TIMEclass-attributeinstance-attribute

Mode that shows the date as day of the week, month, day of month and the time in hour, minute, and (optional) an AM/PM designation. The AM/PM designation is shown only if CupertinoDatePicker does not use 24h format, i.e. if `use_24h_format` is False. Column order is subject to internationalization.

Example: Fri Jul 13 | 4 | 14 | PM

### buildMONTH\_YEARclass-attributeinstance-attribute

Mode that shows the date in month and year. Name of month is spelled in full. Column order is subject to internationalization.

Example: July | 2012

### buildTIMEclass-attributeinstance-attribute

Mode that shows the date in hour, minute, and (optional) an AM/PM designation. The AM/PM designation is shown only if CupertinoDatePicker does not use 24h format, i.e. if `use_24h_format` is False. Column order is subject to internationalization.

Example: 4 | 14 | PM

# CupertinoIcons

## Properties

- ADD
- ADD\_CIRCLED
- ADD\_CIRCLED\_SOLID
- AIRPLANE
- ALARM
- ALARM\_FILL
- ALT
- ANT
- ANT\_CIRCLE
- ANT\_CIRCLE\_FILL
- ANT\_FILL
- ANTENNA\_RADIOWAVES\_LEFT\_RIGHT
- APP
- APP\_BADGE
- APP\_BADGE\_FILL
- APP\_FILL
- ARCHIVEBOX
- ARCHIVEBOX\_FILL
- ARROW\_2\_CIRCLEPATH
- ARROW\_2\_CIRCLEPATH\_CIRCLE
- ARROW\_2\_CIRCLEPATH\_CIRCLE\_FILL
- ARROW\_2\_SQUAREPATH
- ARROW\_3\_TRIANGLEPATH
- ARROW\_BRANCH
- ARROW\_CLOCKWISE
- ARROW\_CLOCKWISE\_CIRCLE
- ARROW\_CLOCKWISE\_CIRCLE\_FILL
- ARROW\_COUNTERCLOCKWISE
- ARROW\_COUNTERCLOCKWISE\_CIRCLE
- ARROW\_COUNTERCLOCKWISE\_CIRCLE\_FILL
- ARROW\_DOWN
- ARROW\_DOWN\_CIRCLE
- ARROW\_DOWN\_CIRCLE\_FILL
- ARROW\_DOWN\_DOC
- ARROW\_DOWN\_DOC\_FILL
- ARROW\_DOWN\_LEFT
- ARROW\_DOWN\_LEFT\_CIRCLE



- ARROW\_DOWN\_LEFT\_CIRCLE\_FILL
- ARROW\_DOWN\_LEFT\_SQUARE
- ARROW\_DOWN\_LEFT\_SQUARE\_FILL
- ARROW\_DOWN\_RIGHT
- ARROW\_DOWN\_RIGHT\_ARROW\_UP\_LEFT
- ARROW\_DOWN\_RIGHT\_CIRCLE
- ARROW\_DOWN\_RIGHT\_CIRCLE\_FILL
- ARROW\_DOWN\_RIGHT\_SQUARE
- ARROW\_DOWN\_RIGHT\_SQUARE\_FILL
- ARROW\_DOWN\_SQUARE
- ARROW\_DOWN\_SQUARE\_FILL
- ARROW\_DOWN\_TO\_LINE
- ARROW\_DOWN\_TO\_LINE\_ALT
- ARROW\_LEFT
- ARROW\_LEFT\_CIRCLE
- ARROW\_LEFT\_CIRCLE\_FILL
- ARROW\_LEFT\_RIGHT
- ARROW\_LEFT\_RIGHT\_CIRCLE
- ARROW\_LEFT\_RIGHT\_CIRCLE\_FILL
- ARROW\_LEFT\_RIGHT\_SQUARE
- ARROW\_LEFT\_RIGHT\_SQUARE\_FILL
- ARROW\_LEFT\_SQUARE
- ARROW\_LEFT\_SQUARE\_FILL
- ARROW\_LEFT\_TO\_LINE
- ARROW\_LEFT\_TO\_LINE\_ALT
- ARROW\_MERGE
- ARROW\_RIGHT
- ARROW\_RIGHT\_ARROW\_LEFT
- ARROW\_RIGHT\_ARROW\_LEFT\_CIRCLE
- ARROW\_RIGHT\_ARROW\_LEFT\_CIRCLE\_FILL
- ARROW\_RIGHT\_ARROW\_LEFT\_SQUARE
- ARROW\_RIGHT\_ARROW\_LEFT\_SQUARE\_FILL
- ARROW\_RIGHT\_CIRCLE
- ARROW\_RIGHT\_CIRCLE\_FILL
- ARROW\_RIGHT\_SQUARE
- ARROW\_RIGHT\_SQUARE\_FILL
- ARROW\_RIGHT\_TO\_LINE
- ARROW\_RIGHT\_TO\_LINE\_ALT
- ARROW\_SWAP
- ARROW\_TURN\_DOWN\_LEFT
- ARROW\_TURN\_DOWN\_RIGHT

- ARROW\_TURN\_LEFT\_DOWN
- ARROW\_TURN\_LEFT\_UP
- ARROW\_TURN\_RIGHT\_DOWN
- ARROW\_TURN\_RIGHT\_UP
- ARROW\_TURN\_UP\_LEFT
- ARROW\_TURN\_UP\_RIGHT
- ARROW\_UP
- ARROW\_UP\_ARROW\_DOWN
- ARROW\_UP\_ARROW\_DOWN\_CIRCLE
- ARROW\_UP\_ARROW\_DOWN\_CIRCLE\_FILL
- ARROW\_UP\_ARROW\_DOWN\_SQUARE
- ARROW\_UP\_ARROW\_DOWN\_SQUARE\_FILL
- ARROW\_UP\_BIN
- ARROW\_UP\_BIN\_FILL
- ARROW\_UP\_CIRCLE
- ARROW\_UP\_CIRCLE\_FILL
- ARROW\_UP\_DOC
- ARROW\_UP\_DOC\_FILL
- ARROW\_UP\_DOWN
- ARROW\_UP\_DOWN\_CIRCLE
- ARROW\_UP\_DOWN\_CIRCLE\_FILL
- ARROW\_UP\_DOWN\_SQUARE
- ARROW\_UP\_DOWN\_SQUARE\_FILL
- ARROW\_UP\_LEFT
- ARROW\_UP\_LEFT\_ARROW\_DOWN\_RIGHT
- ARROW\_UP\_LEFT\_CIRCLE
- ARROW\_UP\_LEFT\_CIRCLE\_FILL
- ARROW\_UP\_LEFT\_SQUARE
- ARROW\_UP\_LEFT\_SQUARE\_FILL
- ARROW\_UP\_RIGHT
- ARROW\_UP\_RIGHT\_CIRCLE
- ARROW\_UP\_RIGHT\_CIRCLE\_FILL
- ARROW\_UP\_RIGHT\_DIAMOND
- ARROW\_UP\_RIGHT\_DIAMOND\_FILL
- ARROW\_UP\_RIGHT\_SQUARE
- ARROW\_UP\_RIGHT\_SQUARE\_FILL
- ARROW\_UP\_SQUARE
- ARROW\_UP\_SQUARE\_FILL
- ARROW\_UP\_TO\_LINE
- ARROW\_UP\_TO\_LINE\_ALT
- ARROW\_UTURN\_DOWN

- ARROW\_UTURN\_DOWN\_CIRCLE
- ARROW\_UTURN\_DOWN\_CIRCLE\_FILL
- ARROW\_UTURN\_DOWN\_SQUARE
- ARROW\_UTURN\_DOWN\_SQUARE\_FILL
- ARROW\_UTURN\_LEFT
- ARROW\_UTURN\_LEFT\_CIRCLE
- ARROW\_UTURN\_LEFT\_CIRCLE\_FILL
- ARROW\_UTURN\_LEFT\_SQUARE
- ARROW\_UTURN\_LEFT\_SQUARE\_FILL
- ARROW\_UTURN\_RIGHT
- ARROW\_UTURN\_RIGHT\_CIRCLE
- ARROW\_UTURN\_RIGHT\_CIRCLE\_FILL
- ARROW\_UTURN\_RIGHT\_SQUARE
- ARROW\_UTURN\_RIGHT\_SQUARE\_FILL
- ARROW\_UTURN\_UP
- ARROW\_UTURN\_UP\_CIRCLE
- ARROW\_UTURN\_UP\_CIRCLE\_FILL
- ARROW\_UTURN\_UP\_SQUARE
- ARROW\_UTURN\_UP\_SQUARE\_FILL
- ARROWSHAPE\_TURN\_UP\_LEFT
- ARROWSHAPE\_TURN\_UP\_LEFT\_2
- ARROWSHAPE\_TURN\_UP\_LEFT\_2\_FILL
- ARROWSHAPE\_TURN\_UP\_LEFT\_CIRCLE
- ARROWSHAPE\_TURN\_UP\_LEFT\_CIRCLE\_FILL
- ARROWSHAPE\_TURN\_UP\_LEFT\_FILL
- ARROWSHAPE\_TURN\_UP\_RIGHT
- ARROWSHAPE\_TURN\_UP\_RIGHT\_CIRCLE
- ARROWSHAPE\_TURN\_UP\_RIGHT\_CIRCLE\_FILL
- ARROWSHAPE\_TURN\_UP\_RIGHT\_FILL
- ARROWTRIANGLE\_DOWN
- ARROWTRIANGLE\_DOWN\_CIRCLE
- ARROWTRIANGLE\_DOWN\_CIRCLE\_FILL
- ARROWTRIANGLE\_DOWN\_FILL
- ARROWTRIANGLE\_DOWN\_SQUARE
- ARROWTRIANGLE\_DOWN\_SQUARE\_FILL
- ARROWTRIANGLE\_LEFT
- ARROWTRIANGLE\_LEFT\_CIRCLE
- ARROWTRIANGLE\_LEFT\_CIRCLE\_FILL
- ARROWTRIANGLE\_LEFT\_FILL
- ARROWTRIANGLE\_LEFT\_SQUARE
- ARROWTRIANGLE\_LEFT\_SQUARE\_FILL

- ARROWTRIANGLE\_RIGHT
- ARROWTRIANGLE\_RIGHT\_CIRCLE
- ARROWTRIANGLE\_RIGHT\_CIRCLE\_FILL
- ARROWTRIANGLE\_RIGHT\_FILL
- ARROWTRIANGLE\_RIGHT\_SQUARE
- ARROWTRIANGLE\_RIGHT\_SQUARE\_FILL
- ARROWTRIANGLE\_UP
- ARROWTRIANGLE\_UP\_CIRCLE
- ARROWTRIANGLE\_UP\_CIRCLE\_FILL
- ARROWTRIANGLE\_UP\_FILL
- ARROWTRIANGLE\_UP\_SQUARE
- ARROWTRIANGLE\_UP\_SQUARE\_FILL
- ASTERISK\_CIRCLE
- ASTERISK\_CIRCLE\_FILL
- AT
- AT\_BADGE\_MINUS
- AT\_BADGE\_PLUS
- AT\_CIRCLE
- AT\_CIRCLE\_FILL
- BACK
- BACKWARD
- BACKWARD\_END
- BACKWARD\_END\_ALT
- BACKWARD\_END\_ALT\_FILL
- BACKWARD\_END\_FILL
- BACKWARD\_FILL
- BADGE\_PLUS\_RADIO WAVES\_RIGHT
- BAG
- BAG\_BADGE\_MINUS
- BAG\_BADGE\_PLUS
- BAG\_FILL
- BAG\_FILL\_BADGE\_MINUS
- BAG\_FILL\_BADGE\_PLUS
- BANDAGE
- BANDAGE\_FILL
- BARCODE
- BARCODE\_VIEWFINDER
- BARS
- BATTERY\_0
- BATTERY\_100
- BATTERY\_25

- BATTERY\_25\_PERCENT
- BATTERY\_75\_PERCENT
- BATTERY\_CHARGING
- BATTERY\_EMPTY
- BATTERY\_FULL
- BED\_DOUBLE
- BED\_DOUBLE\_FILL
- BELL
- BELL\_CIRCLE
- BELL\_CIRCLE\_FILL
- BELL\_FILL
- BELL\_SLASH
- BELL\_SLASH\_FILL
- BELL\_SOLID
- BIN\_XMARK
- BIN\_XMARK\_FILL
- BITCOIN
- BITCOIN\_CIRCLE
- BITCOIN\_CIRCLE\_FILL
- BLUETOOTH
- BOLD
- BOLD\_ITALIC\_UNDERLINE
- BOLD\_UNDERLINE
- BOLT
- BOLT\_BADGE\_A
- BOLT\_BADGE\_A\_FILL
- BOLT\_CIRCLE
- BOLT\_CIRCLE\_FILL
- BOLT\_FILL
- BOLT\_HORIZONTAL
- BOLT\_HORIZONTAL\_CIRCLE
- BOLT\_HORIZONTAL\_CIRCLE\_FILL
- BOLT\_HORIZONTAL\_FILL
- BOLT\_SLASH
- BOLT\_SLASH\_FILL
- BOOK
- BOOK\_CIRCLE
- BOOK\_CIRCLE\_FILL
- BOOK\_FILL
- BOOK\_SOLID
- BOOKMARK

- BOOKMARK\_FILL
- BOOKMARK\_SOLID
- BRIEFCASE
- BRIEFCASE\_FILL
- BRIGHTNESS
- BRIGHTNESS\_SOLID
- BUBBLE\_LEFT
- BUBBLE\_LEFT\_BUBBLE\_RIGHT
- BUBBLE\_LEFT\_BUBBLE\_RIGHT\_FILL
- BUBBLE\_LEFT\_FILL
- BUBBLE\_MIDDLE\_BOTTOM
- BUBBLE\_MIDDLE\_BOTTOM\_FILL
- BUBBLE\_MIDDLE\_TOP
- BUBBLE\_MIDDLE\_TOP\_FILL
- BUBBLE\_RIGHT
- BUBBLE\_RIGHT\_FILL
- BUILDING\_2\_FILL
- BURN
- BURST
- BURST\_FILL
- BUS
- CALENDAR
- CALENDAR\_BADGE\_MINUS
- CALENDAR\_BADGE\_PLUS
- CALENDAR\_CIRCLE
- CALENDAR\_CIRCLE\_FILL
- CALENDAR\_TODAY
- CAMERA
- CAMERA\_CIRCLE
- CAMERA\_CIRCLE\_FILL
- CAMERA\_FILL
- CAMERA\_ON\_RECTANGLE
- CAMERA\_ON\_RECTANGLE\_FILL
- CAMERA\_ROTATE
- CAMERA\_ROTATE\_FILL
- CAMERA\_VIEWFINDER
- CAPSLOCK
- CAPSLOCK\_FILL
- CAPSULE
- CAPSULE\_FILL
- CAPTIONS\_BUBBLE

- CAPTIONS\_BUBBLE\_FILL
- CAR
- CAR\_DETAILED
- CAR\_FILL
- CART
- CART\_BADGE\_MINUS
- CART\_BADGE\_PLUS
- CART\_FILL
- CART\_FILL\_BADGE\_MINUS
- CART\_FILL\_BADGE\_PLUS
- CHART\_BAR
- CHART\_BAR\_ALT\_FILL
- CHART\_BAR\_CIRCLE
- CHART\_BAR\_CIRCLE\_FILL
- CHART\_BAR\_FILL
- CHART\_BAR\_SQUARE
- CHART\_BAR\_SQUARE\_FILL
- CHART\_PIE
- CHART\_PIE\_FILL
- CHAT\_BUBBLE
- CHAT\_BUBBLE\_2
- CHAT\_BUBBLE\_2\_FILL
- CHAT\_BUBBLE\_FILL
- CHAT\_BUBBLE\_TEXT
- CHAT\_BUBBLE\_TEXT\_FILL
- CHECK\_MARK
- CHECK\_MARK\_CIRCLED
- CHECK\_MARK\_CIRCLED\_SOLID
- CHECKMARK
- CHECKMARK\_ALT
- CHECKMARK\_ALT\_CIRCLE
- CHECKMARK\_ALT\_CIRCLE\_FILL
- CHECKMARK\_CIRCLE
- CHECKMARK\_CIRCLE\_FILL
- CHECKMARK\_RECTANGLE
- CHECKMARK\_RECTANGLE\_FILL
- CHECKMARK\_SEAL
- CHECKMARK\_SEAL\_FILL
- CHECKMARK\_SHIELD
- CHECKMARK\_SHIELD\_FILL
- CHECKMARK\_SQUARE

- CHECKMARK\_SQUARE\_FILL
- CHEVRON\_BACK
- CHEVRON\_COMPACT\_DOWN
- CHEVRON\_COMPACT\_LEFT
- CHEVRON\_COMPACT\_RIGHT
- CHEVRON\_COMPACT\_UP
- CHEVRON\_DOWN
- CHEVRON\_DOWN\_CIRCLE
- CHEVRON\_DOWN\_CIRCLE\_FILL
- CHEVRON\_DOWN\_SQUARE
- CHEVRON\_DOWN\_SQUARE\_FILL
- CHEVRON\_FORWARD
- CHEVRON\_LEFT
- CHEVRON\_LEFT\_2
- CHEVRON\_LEFT\_CIRCLE
- CHEVRON\_LEFT\_CIRCLE\_FILL
- CHEVRON\_LEFT\_SLASH\_CHEVRON\_RIGHT
- CHEVRON\_LEFT\_SQUARE
- CHEVRON\_LEFT\_SQUARE\_FILL
- CHEVRON\_RIGHT
- CHEVRON\_RIGHT\_2
- CHEVRON\_RIGHT\_CIRCLE
- CHEVRON\_RIGHT\_CIRCLE\_FILL
- CHEVRON\_RIGHT\_SQUARE
- CHEVRON\_RIGHT\_SQUARE\_FILL
- CHEVRON\_UP
- CHEVRON\_UP\_CHEVRON\_DOWN
- CHEVRON\_UP\_CIRCLE
- CHEVRON\_UP\_CIRCLE\_FILL
- CHEVRON\_UP\_SQUARE
- CHEVRON\_UP\_SQUARE\_FILL
- CIRCLE
- CIRCLE\_BOTTOMTHIRD\_SPLIT
- CIRCLE\_FILL
- CIRCLE\_FILLED
- CIRCLE\_GRID\_3X3
- CIRCLE\_GRID\_3X3\_FILL
- CIRCLE\_GRID\_HEX
- CIRCLE\_GRID\_HEX\_FILL
- CIRCLE\_LEFTHALF\_FILL
- CIRCLE\_RIGHTHALF\_FILL



- CLEAR
- CLEAR\_CIRCLED
- CLEAR\_CIRCLED\_SOLID
- CLEAR\_FILL
- CLEAR\_THICK
- CLEAR\_THICK\_CIRCLED
- CLOCK
- CLOCK\_FILL
- CLOCK\_SOLID
- CLOUD
- CLOUD\_BOLT
- CLOUD\_BOLT\_FILL
- CLOUD\_BOLT\_RAIN
- CLOUD\_BOLT\_RAIN\_FILL
- CLOUD\_DOWNLOAD
- CLOUD\_DOWNLOAD\_FILL
- CLOUD\_DRIZZLE
- CLOUD\_DRIZZLE\_FILL
- CLOUD\_FILL
- CLOUD\_FOG
- CLOUD\_FOG\_FILL
- CLOUD\_HAIL
- CLOUD\_HAIL\_FILL
- CLOUD\_HEAVYRAIN
- CLOUD\_HEAVYRAIN\_FILL
- CLOUD\_MOON
- CLOUD\_MOON\_BOLT
- CLOUD\_MOON\_BOLT\_FILL
- CLOUD\_MOON\_FILL
- CLOUD\_MOON\_RAIN
- CLOUD\_MOON\_RAIN\_FILL
- CLOUD\_RAIN
- CLOUD\_RAIN\_FILL
- CLOUD\_SLEET
- CLOUD\_SLEET\_FILL
- CLOUD\_SNOW
- CLOUD\_SNOW\_FILL
- CLOUD\_SUN
- CLOUD\_SUN\_BOLT
- CLOUD\_SUN\_BOLT\_FILL
- CLOUD\_SUN\_FILL

- CLOUD\_SUN\_RAIN
- CLOUD\_SUN\_RAIN\_FILL
- CLOUD\_UPLOAD
- CLOUD\_UPLOAD\_FILL
- COLLECTIONS
- COLLECTIONS\_SOLID
- COLOR\_FILTER
- COLOR\_FILTER\_FILL
- COMMAND
- COMPASS
- COMPASS\_FILL
- CONTROL
- CONVERSATION\_BUBBLE
- CREATE
- CREATE\_SOLID
- CREDITCARD
- CREDITCARD\_FILL
- CROP
- CROP\_ROTATE
- CUBE
- CUBE\_BOX
- CUBE\_BOX\_FILL
- CUBE\_FILL
- CURSOR\_RAYS
- DECREASE\_INDENT
- DECREASE\_QUOTELEVEL
- DELETE
- DELETE\_LEFT
- DELETE\_LEFT\_FILL
- DELETE\_RIGHT
- DELETE\_RIGHT\_FILL
- DELETE\_SIMPLE
- DELETE\_SOLID
- DESKTOPCOMPUTER
- DEVICE\_DESKTOP
- DEVICE\_LAPTOP
- DEVICE\_PHONE\_LANDSCAPE
- DEVICE\_PHONE\_PORTRAIT
- DIAL
- DIAL\_FILL
- DIVIDE

- DIVIDE\_CIRCLE
- DIVIDE\_CIRCLE\_FILL
- DIVIDE\_SQUARE
- DIVIDE\_SQUARE\_FILL
- DOC
- DOC\_APPEND
- DOC\_CHART
- DOC\_CHART\_FILL
- DOC\_CHECKMARK
- DOC\_CHECKMARK\_FILL
- DOC\_CIRCLE
- DOC\_CIRCLE\_FILL
- DOC\_FILL
- DOC\_ON\_CLIPBOARD
- DOC\_ON\_CLIPBOARD\_FILL
- DOC\_ON\_DOC
- DOC\_ON\_DOC\_FILL
- DOC\_PERSON
- DOC\_PERSON\_FILL
- DOC\_PLAINTEXT
- DOC\_RICHTEXT
- DOC\_TEXT
- DOC\_TEXT\_FILL
- DOC\_TEXT\_SEARCH
- DOC\_TEXT\_VIEWFINDER
- DOT\_RADIO WAVES\_LEFT\_RIGHT
- DOT\_RADIO WAVES\_RIGHT
- DOT\_SQUARE
- DOT\_SQUARE\_FILL
- DOUBLE\_MUSIC\_NOTE
- DOWN\_ARROW
- DOWNLOAD\_CIRCLE
- DOWNLOAD\_CIRCLE\_FILL
- DROP
- DROP\_FILL
- DROP\_TRIANGLE
- DROP\_TRIANGLE\_FILL
- EAR
- EJECT
- EJECT\_FILL
- ELLIPSES\_BUBBLE

- ELLIPSES\_BUBBLE\_FILL
- ELLIPSIS
- ELLIPSIS\_CIRCLE
- ELLIPSIS\_CIRCLE\_FILL
- ELLIPSIS\_VERTICAL
- ELLIPSIS\_VERTICAL\_CIRCLE
- ELLIPSIS\_VERTICAL\_CIRCLE\_FILL
- ENVELOPE
- ENVELOPE\_BADGE
- ENVELOPE\_BADGE\_FILL
- ENVELOPE\_CIRCLE
- ENVELOPE\_CIRCLE\_FILL
- ENVELOPE\_FILL
- ENVELOPE\_OPEN
- ENVELOPE\_OPEN\_FILL
- EQUAL
- EQUAL\_CIRCLE
- EQUAL\_CIRCLE\_FILL
- EQUAL\_SQUARE
- EQUAL\_SQUARE\_FILL
- ESCAPE
- EXCLAMATIONMARK
- EXCLAMATIONMARK\_BUBBLE
- EXCLAMATIONMARK\_BUBBLE\_FILL
- EXCLAMATIONMARK\_CIRCLE
- EXCLAMATIONMARK\_CIRCLE\_FILL
- EXCLAMATIONMARK\_OCTAGON
- EXCLAMATIONMARK\_OCTAGON\_FILL
- EXCLAMATIONMARK\_SHIELD
- EXCLAMATIONMARK\_SHIELD\_FILL
- EXCLAMATIONMARK\_SQUARE
- EXCLAMATIONMARK\_SQUARE\_FILL
- EXCLAMATIONMARK\_TRIANGLE
- EXCLAMATIONMARK\_TRIANGLE\_FILL
- EYE
- EYE\_FILL
- EYE\_SLASH
- EYE\_SLASH\_FILL
- EYE\_SOLID
- EYEDROPPER
- EYEDROPPER\_FULL

- EYEDROPPER\_HALFFULL
- EYEGLASSES
- F\_CURSIVE
- F\_CURSIVE\_CIRCLE
- F\_CURSIVE\_CIRCLE\_FILL
- FILM
- FILM\_FILL
- FLAG
- FLAG\_CIRCLE
- FLAG\_CIRCLE\_FILL
- FLAG\_FILL
- FLAG\_SLASH
- FLAG\_SLASH\_FILL
- FLAME
- FLAME\_FILL
- FLOPPY\_DISK
- FLOWCHART
- FLOWCHART\_FILL
- FOLDER
- FOLDER\_BADGE\_MINUS
- FOLDER\_BADGE\_PERSON\_CROP
- FOLDER\_BADGE\_PLUS
- FOLDER\_CIRCLE
- FOLDER\_CIRCLE\_FILL
- FOLDER\_FILL
- FOLDER\_FILL\_BADGE\_MINUS
- FOLDER\_FILL\_BADGE\_PERSON\_CROP
- FOLDER\_FILL\_BADGE\_PLUS
- FOLDER\_OPEN
- FOLDER\_SOLID
- FORWARD
- FORWARD\_END
- FORWARD\_END\_ALT
- FORWARD\_END\_ALT\_FILL
- FORWARD\_END\_FILL
- FORWARD\_FILL
- FULLSCREEN
- FULLSCREEN\_EXIT
- FUNCTION
- FX
- GAME\_CONTROLLER

- GAME\_CONTROLLER\_SOLID
- GAMECONTROLLER
- GAMECONTROLLER\_ALT\_FILL
- GAMECONTROLLER\_FILL
- GAUGE
- GAUGE\_BADGE\_MINUS
- GAUGE\_BADGE\_PLUS
- GEAR
- GEAR\_ALT
- GEAR\_ALT\_FILL
- GEAR\_BIG
- GEAR\_SOLID
- GIFT
- GIFT\_ALT
- GIFT\_ALT\_FILL
- GIFT\_FILL
- GLOBE
- GOBACKWARD
- GOBACKWARD\_10
- GOBACKWARD\_15
- GOBACKWARD\_30
- GOBACKWARD\_45
- GOBACKWARD\_60
- GOBACKWARD\_75
- GOBACKWARD\_90
- GOBACKWARD\_MINUS
- GOFORWARD
- GOFORWARD\_10
- GOFORWARD\_15
- GOFORWARD\_30
- GOFORWARD\_45
- GOFORWARD\_60
- GOFORWARD\_75
- GOFORWARD\_90
- GOFORWARD\_PLUS
- GRAPH\_CIRCLE
- GRAPH\_CIRCLE\_FILL
- GRAPH\_SQUARE
- GRAPH\_SQUARE\_FILL
- GREATERTHAN
- GREATERTHAN\_CIRCLE

- GREATERTHAN\_CIRCLE\_FILL
- GREATERTHAN\_SQUARE
- GREATERTHAN\_SQUARE\_FILL
- GRID
- GRID\_CIRCLE
- GRID\_CIRCLE\_FILL
- GROUP
- GROUP\_SOLID
- GUITARS
- HAMMER
- HAMMER\_FILL
- HAND\_DRAW
- HAND\_DRAW\_FILL
- HAND\_POINT\_LEFT
- HAND\_POINT\_LEFT\_FILL
- HAND\_POINT\_RIGHT
- HAND\_POINT\_RIGHT\_FILL
- HAND\_RAISED
- HAND\_RAISED\_FILL
- HAND\_RAISED\_SLASH
- HAND\_RAISED\_SLASH\_FILL
- HAND\_THUMBSDOWN
- HAND\_THUMBSDOWN\_FILL
- HAND\_THUMBSUP
- HAND\_THUMBSUP\_FILL
- HARE
- HARE\_FILL
- HEADPHONES
- HEART
- HEART\_CIRCLE
- HEART\_CIRCLE\_FILL
- HEART\_FILL
- HEART\_SLASH
- HEART\_SLASH\_CIRCLE
- HEART\_SLASH\_CIRCLE\_FILL
- HEART\_SLASH\_FILL
- HEART\_SOLID
- HELM
- HEXAGON
- HEXAGON\_FILL
- HIFISPEAKER

- HIFISPEAKER\_FILL
- HOME
- HOURGLASS
- HOURGLASS\_BOTTOMHALF\_FILL
- HOURGLASS\_TOPHALF\_FILL
- HOUSE
- HOUSE\_ALT
- HOUSE\_ALT\_FILL
- HOUSE\_FILL
- HURRICANE
- INCREASE\_INDENT
- INCREASE\_QUOTELEVEL
- INFINITE
- INFO
- INFO\_CIRCLE
- INFO\_CIRCLE\_FILL
- ITALIC
- KEYBOARD
- KEYBOARD\_CHEVRON\_COMPACT\_DOWN
- LAB\_FLASK
- LAB\_FLASK\_SOLID
- LARGE\_CIRCLE\_FILL\_CIRCLE
- LASSO
- LAYERS
- LAYERS\_ALT
- LAYERS\_ALT\_FILL
- LAYERS\_FILL
- LEAF\_ARROW\_CIRCLEPATH
- LEFT\_CHEVRON
- LESSTHAN
- LESSTHAN\_CIRCLE
- LESSTHAN\_CIRCLE\_FILL
- LESSTHAN\_SQUARE
- LESSTHAN\_SQUARE\_FILL
- LIGHT\_MAX
- LIGHT\_MIN
- LIGHTBULB
- LIGHTBULB\_FILL
- LIGHTBULB\_SLASH
- LIGHTBULB\_SLASH\_FILL
- LINE\_HORIZONTAL\_3



- LINE\_HORIZONTAL\_3\_DECREASE
- LINE\_HORIZONTAL\_3\_DECREASE\_CIRCLE
- LINE\_HORIZONTAL\_3\_DECREASE\_CIRCLE\_FILL
- LINK
- LINK\_CIRCLE
- LINK\_CIRCLE\_FILL
- LIST\_BULLET
- LIST\_BULLET\_BELOW\_RECTANGLE
- LIST\_BULLET\_INDENT
- LIST\_DASH
- LIST\_NUMBER
- LIST\_NUMBER\_RTL
- LOCATION
- LOCATION\_CIRCLE
- LOCATION\_CIRCLE\_FILL
- LOCATION\_FILL
- LOCATION\_NORTH
- LOCATION\_NORTH\_FILL
- LOCATION\_NORTH\_LINE
- LOCATION\_NORTH\_LINE\_FILL
- LOCATION\_SLASH
- LOCATION\_SLASH\_FILL
- LOCATION\_SOLID
- LOCK
- LOCK\_CIRCLE
- LOCK\_CIRCLE\_FILL
- LOCK\_FILL
- LOCK\_OPEN
- LOCK\_OPEN\_FILL
- LOCK\_ROTATION
- LOCK\_ROTATION\_OPEN
- LOCK\_SHIELD
- LOCK\_SHIELD\_FILL
- LOCK\_SLASH
- LOCK\_SLASH\_FILL
- LOOP
- LOOP\_THICK
- MACWINDOW
- MAIL
- MAIL\_SOLID
- MAP

- MAP\_FILL
- MAP\_PIN
- MAP\_PIN\_ELLIPSE
- MAP\_PIN\_SLASH
- MEMORIES
- MEMORIES\_BADGE\_MINUS
- MEMORIES\_BADGE\_PLUS
- METRONOME
- MIC
- MIC\_CIRCLE
- MIC\_CIRCLE\_FILL
- MIC\_FILL
- MIC\_OFF
- MIC\_SLASH
- MIC\_SLASH\_FILL
- MIC\_SOLID
- MINUS
- MINUS\_CIRCLE
- MINUS\_CIRCLE\_FILL
- MINUS\_CIRCLED
- MINUS\_RECTANGLE
- MINUS\_RECTANGLE\_FILL
- MINUS\_SLASH\_PLUS
- MINUS\_SQUARE
- MINUS\_SQUARE\_FILL
- MONEY\_DOLLAR
- MONEY\_DOLLAR\_CIRCLE
- MONEY\_DOLLAR\_CIRCLE\_FILL
- MONEY\_EURO
- MONEY\_EURO\_CIRCLE
- MONEY\_EURO\_CIRCLE\_FILL
- MONEY\_POUND
- MONEY\_POUND\_CIRCLE
- MONEY\_POUND\_CIRCLE\_FILL
- MONEY\_RUBL
- MONEY\_RUBL\_CIRCLE
- MONEY\_RUBL\_CIRCLE\_FILL
- MONEY\_YEN
- MONEY\_YEN\_CIRCLE
- MONEY\_YEN\_CIRCLE\_FILL
- MOON

- MOON\_CIRCLE
- MOON\_CIRCLE\_FILL
- MOON\_FILL
- MOON\_STARS
- MOON\_STARS\_FILL
- MOON\_ZZZ
- MOON\_ZZZ\_FILL
- MOVE
- MULTIPLY
- MULTIPLY\_CIRCLE
- MULTIPLY\_CIRCLE\_FILL
- MULTIPLY\_SQUARE
- MULTIPLY\_SQUARE\_FILL
- MUSIC\_ALBUMS
- MUSIC\_ALBUMS\_FILL
- MUSIC\_HOUSE
- MUSIC\_HOUSE\_FILL
- MUSIC\_MIC
- MUSIC\_NOTE
- MUSIC\_NOTE\_2
- MUSIC\_NOTE\_LIST
- NEWS
- NEWS\_SOLID
- NOSIGN
- NUMBER
- NUMBER\_CIRCLE
- NUMBER\_CIRCLE\_FILL
- NUMBER\_SQUARE
- NUMBER\_SQUARE\_FILL
- OPTION
- PADLOCK
- PADLOCK\_SOLID
- PAINTBRUSH
- PAINTBRUSH\_FILL
- PANO
- PANO\_FILL
- PAPERCLIP
- PAPERPLANE
- PAPERPLANE\_FILL
- PARAGRAPH
- PAUSE

- PAUSE\_CIRCLE
- PAUSE\_CIRCLE\_FILL
- PAUSE\_FILL
- PAUSE\_RECTANGLE
- PAUSE\_RECTANGLE\_FILL
- PAUSE\_SOLID
- PAW
- PAW\_SOLID
- PEN
- PENCIL
- PENCIL\_CIRCLE
- PENCIL\_CIRCLE\_FILL
- PENCIL\_ELLIPSIS\_RECTANGLE
- PENCIL\_OUTLINE
- PENCIL\_SLASH
- PERCENT
- PERSON
- PERSON\_2
- PERSON\_2\_ALT
- PERSON\_2\_FILL
- PERSON\_2\_SQUARE\_STACK
- PERSON\_2\_SQUARE\_STACK\_FILL
- PERSON\_3
- PERSON\_3\_FILL
- PERSON\_ADD
- PERSON\_ADD\_SOLID
- PERSON\_ALT
- PERSON\_ALT\_CIRCLE
- PERSON\_ALT\_CIRCLE\_FILL
- PERSON\_BADGE\_MINUS
- PERSON\_BADGE\_MINUS\_FILL
- PERSON\_BADGE\_PLUS
- PERSON\_BADGE\_PLUS\_FILL
- PERSON\_CIRCLE
- PERSON\_CIRCLE\_FILL
- PERSON\_CROP\_CIRCLE
- PERSON\_CROP\_CIRCLE\_BADGE\_CHECKMARK
- PERSON\_CROP\_CIRCLE\_BADGE\_EXCLAM
- PERSON\_CROP\_CIRCLE\_BADGE\_MINUS
- PERSON\_CROP\_CIRCLE\_BADGE\_PLUS
- PERSON\_CROP\_CIRCLE\_BADGE\_XMARK

- PERSON\_CROP\_CIRCLE\_FILL
- PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_CHECKMARK
- PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_EXCLAM
- PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_MINUS
- PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_PLUS
- PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_XMARK
- PERSON\_CROP\_RECTANGLE
- PERSON\_CROP\_RECTANGLE\_FILL
- PERSON\_CROP\_SQUARE
- PERSON\_CROP\_SQUARE\_FILL
- PERSON\_FILL
- PERSON\_SOLID
- PERSONALHOTSPOT
- PERSPECTIVE
- PHONE
- PHONE\_ARROW\_DOWN\_LEFT
- PHONE\_ARROW\_RIGHT
- PHONE\_ARROW\_UP\_RIGHT
- PHONE\_BADGE\_PLUS
- PHONE\_CIRCLE
- PHONE\_CIRCLE\_FILL
- PHONE\_DOWN
- PHONE\_DOWN\_CIRCLE
- PHONE\_DOWN\_CIRCLE\_FILL
- PHONE\_DOWN\_FILL
- PHONE\_FILL
- PHONE\_FILL\_ARROW\_DOWN\_LEFT
- PHONE\_FILL\_ARROW\_RIGHT
- PHONE\_FILL\_ARROW\_UP\_RIGHT
- PHONE\_FILL\_BADGE\_PLUS
- PHONE\_SOLID
- PHOTO
- PHOTO\_CAMERA
- PHOTO\_CAMERA\_SOLID
- PHOTO\_FILL
- PHOTO\_FILL\_ON\_RECTANGLE\_FILL
- PHOTO\_ON\_RECTANGLE
- PIANO
- PIN
- PIN\_FILL
- PIN\_SLASH

- PIN\_SLASH\_FILL
- PLACEMARK
- PLACEMARK\_FILL
- PLAY
- PLAY\_ARROW
- PLAY\_ARROW\_SOLID
- PLAY\_CIRCLE
- PLAY\_CIRCLE\_FILL
- PLAY\_FILL
- PLAY\_RECTANGLE
- PLAY\_RECTANGLE\_FILL
- PLAYPAUSE
- PLAYPAUSE\_FILL
- PLUS
- PLUS\_APP
- PLUS\_APP\_FILL
- PLUS\_BUBBLE
- PLUS\_BUBBLE\_FILL
- PLUS\_CIRCLE
- PLUS\_CIRCLE\_FILL
- PLUS\_CIRCLED
- PLUS\_RECTANGLE
- PLUS\_RECTANGLE\_FILL
- PLUS\_RECTANGLE\_FILL\_ON\_RECTANGLE\_FILL
- PLUS\_RECTANGLE\_ON\_RECTANGLE
- PLUS\_SLASH\_MINUS
- PLUS\_SQUARE
- PLUS\_SQUARE\_FILL
- PLUS\_SQUARE\_FILL\_ON\_SQUARE\_FILL
- PLUS\_SQUARE\_ON\_SQUARE
- PLUSMINUS
- PLUSMINUS\_CIRCLE
- PLUSMINUS\_CIRCLE\_FILL
- POWER
- PRINTER
- PRINTER\_FILL
- PROFILE\_CIRCLED
- PROJECTIVE
- PURCHASED
- PURCHASED\_CIRCLE
- PURCHASED\_CIRCLE\_FILL

- QRCODE
- QRCODE\_VIEWFINDER
- QUESTION
- QUESTION\_CIRCLE
- QUESTION\_CIRCLE\_FILL
- QUESTION\_DIAMOND
- QUESTION\_DIAMOND\_FILL
- QUESTION\_SQUARE
- QUESTION\_SQUARE\_FILL
- QUOTE\_BUBBLE
- QUOTE\_BUBBLE\_FILL
- RADIOWAVES\_LEFT
- RADIOWAVES\_RIGHT
- RAYS
- RECORDINGTAPE
- RECTANGLE
- RECTANGLE\_3\_OFFGRID
- RECTANGLE\_3\_OFFGRID\_FILL
- RECTANGLE\_ARROW\_UP\_RIGHT\_ARROW\_DOWN\_LEFT
- RECTANGLE\_ARROW\_UP\_RIGHT\_ARROW\_DOWN\_LEFT\_SLASH
- RECTANGLE\_BADGE\_CHECKMARK
- RECTANGLE\_BADGE\_XMARK
- RECTANGLE\_COMPRESS\_VERTICAL
- RECTANGLE\_DOCK
- RECTANGLE\_EXPAND\_VERTICAL
- RECTANGLE\_FILL
- RECTANGLE\_FILL\_BADGE\_CHECKMARK
- RECTANGLE\_FILL\_BADGE\_XMARK
- RECTANGLE\_FILL\_ON\_RECTANGLE\_ANGLED\_FILL
- RECTANGLE\_FILL\_ON\_RECTANGLE\_FILL
- RECTANGLE\_GRID\_1X2
- RECTANGLE\_GRID\_1X2\_FILL
- RECTANGLE\_GRID\_2X2
- RECTANGLE\_GRID\_2X2\_FILL
- RECTANGLE\_GRID\_3X2
- RECTANGLE\_GRID\_3X2\_FILL
- RECTANGLE\_ON\_RECTANGLE
- RECTANGLE\_ON\_RECTANGLE\_ANGLED
- RECTANGLE\_PAPERCLIP
- RECTANGLE\_SPLIT\_3X1
- RECTANGLE\_SPLIT\_3X1\_FILL

- RECTANGLE\_SPLIT\_3X3
- RECTANGLE\_SPLIT\_3X3\_FILL
- RECTANGLE\_STACK
- RECTANGLE\_STACK\_BADGE\_MINUS
- RECTANGLE\_STACK\_BADGE\_PERSON\_CROP
- RECTANGLE\_STACK\_BADGE\_PLUS
- RECTANGLE\_STACK\_FILL
- RECTANGLE\_STACK\_FILL\_BADGE\_MINUS
- RECTANGLE\_STACK\_FILL\_BADGE\_PERSON\_CROP
- RECTANGLE\_STACK\_FILL\_BADGE\_PLUS
- RECTANGLE\_STACK\_PERSON\_CROP
- RECTANGLE\_STACK\_PERSON\_CROP\_FILL
- REFRESH
- REFRESH\_BOLD
- REFRESH\_CIRCLED
- REFRESH\_CIRCLED\_SOLID
- REFRESH\_THICK
- REFRESH\_THIN
- REPEAT
- REPEAT\_1
- REPLY
- REPLY\_ALL
- REPLY\_THICK\_SOLID
- RESIZE
- RESIZE\_H
- RESIZE\_V
- RESTART
- RETURN\_ICON
- RHOMBUS
- RHOMBUS\_FILL
- RIGHT\_CHEVRON
- ROCKET
- ROCKET\_FILL
- ROSETTE
- ROTATE\_LEFT
- ROTATE\_LEFT\_FILL
- ROTATE\_RIGHT
- ROTATE\_RIGHT\_FILL
- SCISSORS
- SCISSORS\_ALT
- SCOPE



- SCRIBBLE
- SEARCH
- SEARCH\_CIRCLE
- SEARCH\_CIRCLE\_FILL
- SELECTION\_PIN\_IN\_OUT
- SETTINGS
- SETTINGS\_SOLID
- SHARE
- SHARE\_SOLID
- SHARE\_UP
- SHIELD
- SHIELD\_FILL
- SHIELD\_LEFTHALF\_FILL
- SHIELD\_SLASH
- SHIELD\_SLASH\_FILL
- SHIFT
- SHIFT\_FILL
- SHOPPING\_CART
- SHUFFLE
- SHUFFLE\_MEDIUM
- SHUFFLE\_THICK
- SIDEBAR\_LEFT
- SIDEBAR\_RIGHT
- SIGNATURE
- SKEW
- SLASH\_CIRCLE
- SLASH\_CIRCLE\_FILL
- SLIDER\_HORIZONTAL\_3
- SLIDER\_HORIZONTAL\_BELOW\_RECTANGLE
- SLOWMO
- SMALLCIRCLE\_CIRCLE
- SMALLCIRCLE\_CIRCLE\_FILL
- SMALLCIRCLE\_FILL\_CIRCLE
- SMALLCIRCLE\_FILL\_CIRCLE\_FILL
- SMILEY
- SMILEY\_FILL
- SMOKE
- SMOKE\_FILL
- SNOW
- SORT\_DOWN
- SORT\_DOWN\_CIRCLE

- SORT\_DOWN\_CIRCLE\_FILL
- SORT\_UP
- SORT\_UP\_CIRCLE
- SORT\_UP\_CIRCLE\_FILL
- SPARKLES
- SPEAKER
- SPEAKER\_1
- SPEAKER\_1\_FILL
- SPEAKER\_2
- SPEAKER\_2\_FILL
- SPEAKER\_3
- SPEAKER\_3\_FILL
- SPEAKER\_FILL
- SPEAKER\_SLASH
- SPEAKER\_SLASH\_FILL
- SPEAKER\_SLASH\_FILL\_RTL
- SPEAKER\_SLASH\_RTL
- SPEAKER\_ZZZ
- SPEAKER\_ZZZ\_FILL
- SPEAKER\_ZZZ\_FILL\_RTL
- SPEAKER\_ZZZ\_RTL
- SPEEDOMETER
- SPORTSCOURT
- SPORTSCOURT\_FILL
- SQUARE
- SQUARE\_ARROW\_DOWN
- SQUARE\_ARROW\_DOWN\_FILL
- SQUARE\_ARROW\_DOWN\_ON\_SQUARE
- SQUARE\_ARROW\_DOWN\_ON\_SQUARE\_FILL
- SQUARE\_ARROW\_LEFT
- SQUARE\_ARROW\_LEFT\_FILL
- SQUARE\_ARROW\_RIGHT
- SQUARE\_ARROW\_RIGHT\_FILL
- SQUARE\_ARROW\_UP
- SQUARE\_ARROW\_UP\_FILL
- SQUARE\_ARROW\_UP\_ON\_SQUARE
- SQUARE\_ARROW\_UP\_ON\_SQUARE\_FILL
- SQUARE\_FAVORITES
- SQUARE\_FAVORITES\_ALT
- SQUARE\_FAVORITES\_ALT\_FILL
- SQUARE\_FAVORITES\_FILL

- SQUARE\_FILL
- SQUARE\_FILL\_LINE\_VERTICAL\_SQUARE
- SQUARE\_FILL\_LINE\_VERTICAL\_SQUARE\_FILL
- SQUARE\_FILL\_ON\_CIRCLE\_FILL
- SQUARE\_FILL\_ON\_SQUARE\_FILL
- SQUARE\_GRID\_2X2
- SQUARE\_GRID\_2X2\_FILL
- SQUARE\_GRID\_3X2
- SQUARE\_GRID\_3X2\_FILL
- SQUARE\_GRID\_4X3\_FILL
- SQUARE\_LEFTHALF\_FILL
- SQUARE\_LINE\_VERTICAL\_SQUARE
- SQUARE\_LINE\_VERTICAL\_SQUARE\_FILL
- SQUARE\_LIST
- SQUARE\_LIST\_FILL
- SQUARE\_ON\_CIRCLE
- SQUARE\_ON\_SQUARE
- SQUARE\_PENCIL
- SQUARE\_PENCIL\_FILL
- SQUARE\_RIGHTHALF\_FILL
- SQUARE\_SPLIT\_1X2
- SQUARE\_SPLIT\_1X2\_FILL
- SQUARE\_SPLIT\_2X1
- SQUARE\_SPLIT\_2X1\_FILL
- SQUARE\_SPLIT\_2X2
- SQUARE\_SPLIT\_2X2\_FILL
- SQUARE\_STACK
- SQUARE\_STACK\_3D\_DOWN\_DOTTEDLINE
- SQUARE\_STACK\_3D\_DOWN\_RIGHT
- SQUARE\_STACK\_3D\_DOWN\_RIGHT\_FILL
- SQUARE\_STACK\_3D\_UP
- SQUARE\_STACK\_3D\_UP\_FILL
- SQUARE\_STACK\_3D\_UP\_SLASH
- SQUARE\_STACK\_3D\_UP\_SLASH\_FILL
- SQUARE\_STACK\_FILL
- SQUARES\_BELOW\_RECTANGLE
- STAR
- STAR\_CIRCLE
- STAR\_CIRCLE\_FILL
- STAR\_FILL
- STAR\_LEFTHALF\_FILL

- STAR\_SLASH
- STAR\_SLASH\_FILL
- STAROFLIFE
- STAROFLIFE\_FILL
- STOP
- STOP\_CIRCLE
- STOP\_CIRCLE\_FILL
- STOP\_FILL
- STOPWATCH
- STOPWATCH\_FILL
- STRIKETHROUGH
- SUIT\_CLUB
- SUIT\_CLUB\_FILL
- SUIT\_DIAMOND
- SUIT\_DIAMOND\_FILL
- SUIT\_HEART
- SUIT\_HEART\_FILL
- SUIT\_SPADE
- SUIT\_SPADE\_FILL
- SUM
- SUN\_DUST
- SUN\_DUST\_FILL
- SUN\_HAZE
- SUN\_HAZE\_FILL
- SUN\_MAX
- SUN\_MAX\_FILL
- SUN\_MIN
- SUN\_MIN\_FILL
- SUNRISE
- SUNRISE\_FILL
- SUNSET
- SUNSET\_FILL
- SWITCH\_CAMERA
- SWITCH\_CAMERA\_SOLID
- T\_BUBBLE
- T\_BUBBLE\_FILL
- TABLE
- TABLE\_BADGE\_MORE
- TABLE\_BADGE\_MORE\_FILL
- TABLE\_FILL
- TAG

- TAG\_CIRCLE
- TAG\_CIRCLE\_FILL
- TAG\_FILL
- TAG\_SOLID
- TAGS
- TAGS\_SOLID
- TEXT\_ALIGNCENTER
- TEXT\_ALIGNLEFT
- TEXT\_ALIGNRIGHT
- TEXT\_APPEND
- TEXT\_BADGE\_CHECKMARK
- TEXT\_BADGE\_MINUS
- TEXT\_BADGE\_PLUS
- TEXT\_BADGE\_STAR
- TEXT\_BADGE\_XMARK
- TEXT\_BUBBLE
- TEXT\_BUBBLE\_FILL
- TEXT\_CURSOR
- TEXT\_INSERT
- TEXT\_JUSTIFY
- TEXT\_JUSTIFYLEFT
- TEXT\_JUSTIFYRIGHT
- TEXT\_QUOTE
- TEXTBOX
- TEXTFORMAT
- TEXTFORMAT\_123
- TEXTFORMAT\_ABC
- TEXTFORMAT\_ABC\_DOTTEDUNDERLINE
- TEXTFORMAT\_ALT
- TEXTFORMAT\_SIZE
- TEXTFORMAT\_SUBSCRIPT
- TEXTFORMAT\_SUPERScript
- THERMOMETER
- THERMOMETER\_SNOWFLAKE
- THERMOMETER\_SUN
- TICKET
- TICKET\_FILL
- TICKETS
- TICKETS\_FILL
- TIME
- TIME\_SOLID

- TIMELAPSE
- TIMER
- TIMER\_FILL
- TODAY
- TODAY\_FILL
- TORNADO
- TORTOISE
- TORTOISE\_FILL
- TRAIN\_STYLE\_ONE
- TRAIN\_STYLE\_TWO
- TRAM\_FILL
- TRASH
- TRASH\_CIRCLE
- TRASH\_CIRCLE\_FILL
- TRASH\_FILL
- TRASH\_SLASH
- TRASH\_SLASH\_FILL
- TRAY
- TRAY\_2
- TRAY\_2\_FILL
- TRAY\_ARROW\_DOWN
- TRAY\_ARROW\_DOWN\_FILL
- TRAY\_ARROW\_UP
- TRAY\_ARROW\_UP\_FILL
- TRAY\_FILL
- TRAY\_FULL
- TRAY\_FULL\_FILL
- TREE
- TRIANGLE
- TRIANGLE\_FILL
- TRIANGLE\_LEFTHALF\_FILL
- TRIANGLE\_RIGHTHALF\_FILL
- TROPICALSTORM
- TUNINGFORK
- TV
- TV\_CIRCLE
- TV\_CIRCLE\_FILL
- TV\_FILL
- TV\_MUSIC\_NOTE
- TV\_MUSIC\_NOTE\_FILL
- UIWINDOW\_SPLIT\_2X1

- UMBRELLA
- UMBRELLA\_FILL
- UNDERLINE
- UP\_ARROW
- UPLOAD\_CIRCLE
- UPLOAD\_CIRCLE\_FILL
- VIDEO\_CAMERA
- VIDEO\_CAMERA\_SOLID
- VIDEOCAM
- VIDEOCAM\_CIRCLE
- VIDEOCAM\_CIRCLE\_FILL
- VIDEOCAM\_FILL
- VIEW\_2D
- VIEW\_3D
- VIEWFINDER
- VIEWFINDER\_CIRCLE
- VIEWFINDER\_CIRCLE\_FILL
- VOLUME\_DOWN
- VOLUME\_MUTE
- VOLUME\_OFF
- VOLUME\_UP
- WAND\_RAYS
- WAND\_RAYS\_INVERSE
- WAND\_STARS
- WAND\_STARS\_INVERSE
- WAVEFORM
- WAVEFORM\_CIRCLE
- WAVEFORM\_CIRCLE\_FILL
- WAVEFORM\_PATH
- WAVEFORM\_PATH\_BADGE\_MINUS
- WAVEFORM\_PATH\_BADGE\_PLUS
- WAVEFORM\_PATH\_ECG
- WIFI
- WIFI\_EXCLAMATIONMARK
- WIFI\_SLASH
- WIND
- WIND\_SNOW
- WRENCH
- WRENCH\_FILL
- XMARK
- XMARK\_CIRCLE

- XMARK\_CIRCLE\_FILL
- XMARK\_OCTAGON
- XMARK\_OCTAGON\_FILL
- XMARK\_RECTANGLE
- XMARK\_RECTANGLE\_FILL
- XMARK\_SEAL
- XMARK\_SEAL\_FILL
- XMARK\_SHIELD
- XMARK\_SHIELD\_FILL
- XMARK\_SQUARE
- XMARK\_SQUARE\_FILL
- ZOOM\_IN
- ZOOM\_OUT
- ZZZ

## Properties

### buildADDInstance-attribute

ADD: IconData

### buildADD\_CIRCLEDInstance-attribute

ADD\_CIRCLED: IconData

### buildADD\_CIRCLED\_SOLIDInstance-attribute

ADD\_CIRCLED\_SOLID: IconData

### buildAIRPLANEInstance-attribute

AIRPLANE: IconData

### buildALARMinstance-attribute

ALARM: IconData

### buildALARM\_FILLInstance-attribute

ALARM\_FILL: IconData

### buildALTInstance-attribute

ALT: IconData

### buildANTInstance-attribute

ANT: IconData

### buildANT\_CIRCLEInstance-attribute

ANT\_CIRCLE: IconData

### buildANT\_CIRCLE\_FILLInstance-attribute

ANT\_CIRCLE\_FILL: IconData

### buildANT\_FILLInstance-attribute

ANT\_FILL: IconData

### buildANTENNA\_RADIOWAVES\_LEFT\_RIGHTInstance-attribute

ANTENNA\_RADIOWAVES\_LEFT\_RIGHT: IconData

### buildAPPInstance-attribute

APP: IconData

### buildAPP\_BADGEInstance-attribute

APP\_BADGE: IconData



**buildAPP\_BADGE\_FILLinstance-attribute**

APP\_BADGE\_FILL: IconData

**buildAPP\_FILLinstance-attribute**

APP\_FILL: IconData

**buildARCHIVEBOXinstance-attribute**

ARCHIVEBOX: IconData

**buildARCHIVEBOX\_FILLinstance-attribute**

ARCHIVEBOX\_FILL: IconData

**buildARROW\_2\_CIRCLEPATHinstance-attribute**

ARROW\_2\_CIRCLEPATH: IconData

**buildARROW\_2\_CIRCLEPATH\_CIRCLEinstance-attribute**

ARROW\_2\_CIRCLEPATH\_CIRCLE: IconData

**buildARROW\_2\_CIRCLEPATH\_CIRCLE\_FILLinstance-attribute**

ARROW\_2\_CIRCLEPATH\_CIRCLE\_FILL: IconData

**buildARROW\_2\_SQUAREPATHinstance-attribute**

ARROW\_2\_SQUAREPATH: IconData

**buildARROW\_3\_TRIANGLEPATHinstance-attribute**

ARROW\_3\_TRIANGLEPATH: IconData

**buildARROW\_BRANCHinstance-attribute**

ARROW\_BRANCH: IconData

**buildARROW\_CLOCKWISEinstance-attribute**

ARROW\_CLOCKWISE: IconData

**buildARROW\_CLOCKWISE\_CIRCLEinstance-attribute**

ARROW\_CLOCKWISE\_CIRCLE: IconData

**buildARROW\_CLOCKWISE\_CIRCLE\_FILLinstance-attribute**

ARROW\_CLOCKWISE\_CIRCLE\_FILL: IconData

**buildARROW\_COUNTERCLOCKWISEinstance-attribute**

ARROW\_COUNTERCLOCKWISE: IconData

**buildARROW\_COUNTERCLOCKWISE\_CIRCLEinstance-attribute**

ARROW\_COUNTERCLOCKWISE\_CIRCLE: IconData

**buildARROW\_COUNTERCLOCKWISE\_CIRCLE\_FILLinstance-attribute**

ARROW\_COUNTERCLOCKWISE\_CIRCLE\_FILL: IconData

**buildARROW\_DOWNinstance-attribute**

ARROW\_DOWN: IconData

**buildARROW\_DOWN\_CIRCLEinstance-attribute**

ARROW\_DOWN\_CIRCLE: IconData

**buildARROW\_DOWN\_CIRCLE\_FILLinstance-attribute**

ARROW\_DOWN\_CIRCLE\_FILL: IconData

**buildARROW\_DOWN\_DOCinstance-attribute**

ARROW\_DOWN\_DOC: IconData

**buildARROW\_DOWN\_DOC\_FILLinstance-attribute**

ARROW\_DOWN\_DOC\_FILL: IconData

**buildARROW\_DOWN\_LEFTinstance-attribute**

ARROW\_DOWN\_LEFT: IconData

buildARROW\_DOWN\_LEFT\_CIRCLEinstance-attribute

ARROW\_DOWN\_LEFT\_CIRCLE: IconData

buildARROW\_DOWN\_LEFT\_CIRCLE\_FILLinstance-attribute

ARROW\_DOWN\_LEFT\_CIRCLE\_FILL: IconData

buildARROW\_DOWN\_LEFT\_SQUAREinstance-attribute

ARROW\_DOWN\_LEFT\_SQUARE: IconData

buildARROW\_DOWN\_LEFT\_SQUARE\_FILLinstance-attribute

ARROW\_DOWN\_LEFT\_SQUARE\_FILL: IconData

buildARROW\_DOWN\_RIGHTinstance-attribute

ARROW\_DOWN\_RIGHT: IconData

buildARROW\_DOWN\_RIGHT\_ARROW\_UP\_LEFTinstance-attribute

ARROW\_DOWN\_RIGHT\_ARROW\_UP\_LEFT: IconData

buildARROW\_DOWN\_RIGHT\_CIRCLEinstance-attribute

ARROW\_DOWN\_RIGHT\_CIRCLE: IconData

buildARROW\_DOWN\_RIGHT\_CIRCLE\_FILLinstance-attribute

ARROW\_DOWN\_RIGHT\_CIRCLE\_FILL: IconData

buildARROW\_DOWN\_RIGHT\_SQUAREinstance-attribute

ARROW\_DOWN\_RIGHT\_SQUARE: IconData

buildARROW\_DOWN\_RIGHT\_SQUARE\_FILLinstance-attribute

ARROW\_DOWN\_RIGHT\_SQUARE\_FILL: IconData

buildARROW\_DOWN\_SQUAREinstance-attribute

ARROW\_DOWN\_SQUARE: IconData

buildARROW\_DOWN\_SQUARE\_FILLinstance-attribute

ARROW\_DOWN\_SQUARE\_FILL: IconData

buildARROW\_DOWN\_TO\_LINEinstance-attribute

ARROW\_DOWN\_TO\_LINE: IconData

buildARROW\_DOWN\_TO\_LINE\_ALTinstance-attribute

ARROW\_DOWN\_TO\_LINE\_ALT: IconData

buildARROW\_LEFTinstance-attribute

ARROW\_LEFT: IconData

buildARROW\_LEFT\_CIRCLEinstance-attribute

ARROW\_LEFT\_CIRCLE: IconData

buildARROW\_LEFT\_CIRCLE\_FILLinstance-attribute

ARROW\_LEFT\_CIRCLE\_FILL: IconData

buildARROW\_LEFT\_RIGHTinstance-attribute

ARROW\_LEFT\_RIGHT: IconData

buildARROW\_LEFT\_RIGHT\_CIRCLEinstance-attribute

ARROW\_LEFT\_RIGHT\_CIRCLE: IconData

buildARROW\_LEFT\_RIGHT\_CIRCLE\_FILLinstance-attribute

ARROW\_LEFT\_RIGHT\_CIRCLE\_FILL: IconData

buildARROW\_LEFT\_RIGHT\_SQUAREinstance-attribute

ARROW\_LEFT\_RIGHT\_SQUARE: IconData

buildARROW\_LEFT\_RIGHT\_SQUARE\_FILLinstance-attribute

ARROW\_LEFT\_RIGHT\_SQUARE\_FILL: IconData

buildARROW\_LEFT\_SQUAREinstance-attribute

ARROW\_LEFT\_SQUARE: IconData

buildARROW\_LEFT\_SQUARE\_FILLinstance-attribute

ARROW\_LEFT\_SQUARE\_FILL: IconData

buildARROW\_LEFT\_TO\_LINEinstance-attribute

ARROW\_LEFT\_TO\_LINE: IconData

buildARROW\_LEFT\_TO\_LINE\_ALTinstance-attribute

ARROW\_LEFT\_TO\_LINE\_ALT: IconData

buildARROW\_MERGEinstance-attribute

ARROW\_MERGE: IconData

buildARROW\_RIGHTinstance-attribute

ARROW\_RIGHT: IconData

buildARROW\_RIGHT\_ARROW\_LEFTinstance-attribute

ARROW\_RIGHT\_ARROW\_LEFT: IconData

buildARROW\_RIGHT\_ARROW\_LEFT\_CIRCLEinstance-attribute

ARROW\_RIGHT\_ARROW\_LEFT\_CIRCLE: IconData

buildARROW\_RIGHT\_ARROW\_LEFT\_CIRCLE\_FILLinstance-attribute

ARROW\_RIGHT\_ARROW\_LEFT\_CIRCLE\_FILL: IconData

buildARROW\_RIGHT\_ARROW\_LEFT\_SQUAREinstance-attribute

ARROW\_RIGHT\_ARROW\_LEFT\_SQUARE: IconData

buildARROW\_RIGHT\_ARROW\_LEFT\_SQUARE\_FILLinstance-attribute

ARROW\_RIGHT\_ARROW\_LEFT\_SQUARE\_FILL: IconData

buildARROW\_RIGHT\_CIRCLEinstance-attribute

ARROW\_RIGHT\_CIRCLE: IconData

buildARROW\_RIGHT\_CIRCLE\_FILLinstance-attribute

ARROW\_RIGHT\_CIRCLE\_FILL: IconData

buildARROW\_RIGHT\_SQUAREinstance-attribute

ARROW\_RIGHT\_SQUARE: IconData

buildARROW\_RIGHT\_SQUARE\_FILLinstance-attribute

ARROW\_RIGHT\_SQUARE\_FILL: IconData

buildARROW\_RIGHT\_TO\_LINEinstance-attribute

ARROW\_RIGHT\_TO\_LINE: IconData

buildARROW\_RIGHT\_TO\_LINE\_ALTinstance-attribute

ARROW\_RIGHT\_TO\_LINE\_ALT: IconData

buildARROW\_SWAPinstance-attribute

ARROW\_SWAP: IconData

buildARROW\_TURN\_DOWN\_LEFTinstance-attribute

ARROW\_TURN\_DOWN\_LEFT: IconData

buildARROW\_TURN\_DOWN\_RIGHTinstance-attribute

ARROW\_TURN\_DOWN\_RIGHT: IconData

buildARROW\_TURN\_LEFT\_DOWNinstance-attribute

ARROW\_TURN\_LEFT\_DOWN: IconData

buildARROW\_TURN\_LEFT\_UPinstance-attribute

ARROW\_TURN\_LEFT\_UP: IconData

buildARROW\_TURN\_RIGHT\_DOWNinstance-attribute

ARROW\_TURN\_RIGHT\_DOWN: IconData

buildARROW\_TURN\_RIGHT\_UPinstance-attribute

ARROW\_TURN\_RIGHT\_UP: IconData

buildARROW\_TURN\_UP\_LEFTinstance-attribute

ARROW\_TURN\_UP\_LEFT: IconData

buildARROW\_TURN\_UP\_RIGHTinstance-attribute

ARROW\_TURN\_UP\_RIGHT: IconData

buildARROW\_UPinstance-attribute

ARROW\_UP: IconData

buildARROW\_UP\_ARROW\_DOWNinstance-attribute

ARROW\_UP\_ARROW\_DOWN: IconData

buildARROW\_UP\_ARROW\_DOWN\_CIRCLEinstance-attribute

ARROW\_UP\_ARROW\_DOWN\_CIRCLE: IconData

buildARROW\_UP\_ARROW\_DOWN\_CIRCLE\_FILLinstance-attribute

ARROW\_UP\_ARROW\_DOWN\_CIRCLE\_FILL: IconData

buildARROW\_UP\_ARROW\_DOWN\_SQUAREinstance-attribute

ARROW\_UP\_ARROW\_DOWN\_SQUARE: IconData

buildARROW\_UP\_ARROW\_DOWN\_SQUARE\_FILLinstance-attribute

ARROW\_UP\_ARROW\_DOWN\_SQUARE\_FILL: IconData

buildARROW\_UP\_BINinstance-attribute

ARROW\_UP\_BIN: IconData

buildARROW\_UP\_BIN\_FILLinstance-attribute

ARROW\_UP\_BIN\_FILL: IconData

buildARROW\_UP\_CIRCLEinstance-attribute

ARROW\_UP\_CIRCLE: IconData

buildARROW\_UP\_CIRCLE\_FILLinstance-attribute

ARROW\_UP\_CIRCLE\_FILL: IconData

buildARROW\_UP\_DOCinstance-attribute

ARROW\_UP\_DOC: IconData

buildARROW\_UP\_DOC\_FILLinstance-attribute

ARROW\_UP\_DOC\_FILL: IconData

buildARROW\_UP\_DOWNinstance-attribute

ARROW\_UP\_DOWN: IconData

buildARROW\_UP\_DOWN\_CIRCLEinstance-attribute

ARROW\_UP\_DOWN\_CIRCLE: IconData

buildARROW\_UP\_DOWN\_CIRCLE\_FILLinstance-attribute

ARROW\_UP\_DOWN\_CIRCLE\_FILL: IconData

buildARROW\_UP\_DOWN\_SQUAREinstance-attribute

ARROW\_UP\_DOWN\_SQUARE: IconData

buildARROW\_UP\_DOWN\_SQUARE\_FILLinstance-attribute

ARROW\_UP\_DOWN\_SQUARE\_FILL: IconData

buildARROW\_UP\_LEFTinstance-attribute

ARROW\_UP\_LEFT: IconData

buildARROW\_UP\_LEFT\_ARROW\_DOWN\_RIGHTinstance-attribute

ARROW\_UP\_LEFT\_ARROW\_DOWN\_RIGHT: IconData

buildARROW\_UP\_LEFT\_CIRCLEinstance-attribute

ARROW\_UP\_LEFT\_CIRCLE: IconData

buildARROW\_UP\_LEFT\_CIRCLE\_FILLinstance-attribute

ARROW\_UP\_LEFT\_CIRCLE\_FILL: IconData

buildARROW\_UP\_LEFT\_SQUAREinstance-attribute

ARROW\_UP\_LEFT\_SQUARE: IconData

buildARROW\_UP\_LEFT\_SQUARE\_FILLinstance-attribute

ARROW\_UP\_LEFT\_SQUARE\_FILL: IconData

buildARROW\_UP\_RIGHTinstance-attribute

ARROW\_UP\_RIGHT: IconData

buildARROW\_UP\_RIGHT\_CIRCLEinstance-attribute

ARROW\_UP\_RIGHT\_CIRCLE: IconData

buildARROW\_UP\_RIGHT\_CIRCLE\_FILLinstance-attribute

ARROW\_UP\_RIGHT\_CIRCLE\_FILL: IconData

buildARROW\_UP\_RIGHT\_DIAMONDinstance-attribute

ARROW\_UP\_RIGHT\_DIAMOND: IconData

buildARROW\_UP\_RIGHT\_DIAMOND\_FILLinstance-attribute

ARROW\_UP\_RIGHT\_DIAMOND\_FILL: IconData

buildARROW\_UP\_RIGHT\_SQUAREinstance-attribute

ARROW\_UP\_RIGHT\_SQUARE: IconData

buildARROW\_UP\_RIGHT\_SQUARE\_FILLinstance-attribute

ARROW\_UP\_RIGHT\_SQUARE\_FILL: IconData

buildARROW\_UP\_SQUAREinstance-attribute

ARROW\_UP\_SQUARE: IconData

buildARROW\_UP\_SQUARE\_FILLinstance-attribute

ARROW\_UP\_SQUARE\_FILL: IconData

buildARROW\_UP\_TO\_LINEinstance-attribute

ARROW\_UP\_TO\_LINE: IconData

buildARROW\_UP\_TO\_LINE\_ALTinstance-attribute

ARROW\_UP\_TO\_LINE\_ALT: IconData

buildARROW\_UTURN\_DOWNinstance-attribute

ARROW\_UTURN\_DOWN: IconData

buildARROW\_UTURN\_DOWN\_CIRCLEinstance-attribute

ARROW\_UTURN\_DOWN\_CIRCLE: IconData

buildARROW\_UTURN\_DOWN\_CIRCLE\_FILLinstance-attribute

ARROW\_UTURN\_DOWN\_CIRCLE\_FILL: IconData

buildARROW\_UTURN\_DOWN\_SQUAREinstance-attribute

ARROW\_UTURN\_DOWN\_SQUARE: IconData

buildARROW\_UTURN\_DOWN\_SQUARE\_FILLinstance-attribute

ARROW\_UTURN\_DOWN\_SQUARE\_FILL: IconData

buildARROW\_UTURN\_LEFTinstance-attribute

ARROW\_UTURN\_LEFT: IconData

buildARROW\_UTURN\_LEFT\_CIRCLEinstance-attribute

ARROW\_UTURN\_LEFT\_CIRCLE: IconData

buildARROW\_UTURN\_LEFT\_CIRCLE\_FILLinstance-attribute

ARROW\_UTURN\_LEFT\_CIRCLE\_FILL: IconData

buildARROW\_UTURN\_LEFT\_SQUAREinstance-attribute

ARROW\_UTURN\_LEFT\_SQUARE: IconData

buildARROW\_UTURN\_LEFT\_SQUARE\_FILLinstance-attribute

ARROW\_UTURN\_LEFT\_SQUARE\_FILL: IconData

buildARROW\_UTURN\_RIGHTinstance-attribute

ARROW\_UTURN\_RIGHT: IconData

buildARROW\_UTURN\_RIGHT\_CIRCLEinstance-attribute

ARROW\_UTURN\_RIGHT\_CIRCLE: IconData

buildARROW\_UTURN\_RIGHT\_CIRCLE\_FILLinstance-attribute

ARROW\_UTURN\_RIGHT\_CIRCLE\_FILL: IconData

buildARROW\_UTURN\_RIGHT\_SQUAREinstance-attribute

ARROW\_UTURN\_RIGHT\_SQUARE: IconData

buildARROW\_UTURN\_RIGHT\_SQUARE\_FILLinstance-attribute

ARROW\_UTURN\_RIGHT\_SQUARE\_FILL: IconData

buildARROW\_UTURN\_UPinstance-attribute

ARROW\_UTURN\_UP: IconData

buildARROW\_UTURN\_UP\_CIRCLEinstance-attribute

ARROW\_UTURN\_UP\_CIRCLE: IconData

buildARROW\_UTURN\_UP\_CIRCLE\_FILLinstance-attribute

ARROW\_UTURN\_UP\_CIRCLE\_FILL: IconData

buildARROW\_UTURN\_UP\_SQUAREinstance-attribute

ARROW\_UTURN\_UP\_SQUARE: IconData

buildARROW\_UTURN\_UP\_SQUARE\_FILLinstance-attribute

ARROW\_UTURN\_UP\_SQUARE\_FILL: IconData

buildARROWSHAPE\_TURN\_UP\_LEFTinstance-attribute

ARROWSHAPE\_TURN\_UP\_LEFT: IconData

buildARROWSHAPE\_TURN\_UP\_LEFT\_2instance-attribute

ARROWSHAPE\_TURN\_UP\_LEFT\_2: IconData

buildARROWSHAPE\_TURN\_UP\_LEFT\_2\_FILLinstance-attribute

ARROWSHAPE\_TURN\_UP\_LEFT\_2\_FILL: IconData

buildARROWSHAPE\_TURN\_UP\_LEFT\_CIRCLEinstance-attribute

ARROWSHAPE\_TURN\_UP\_LEFT\_CIRCLE: IconData

buildARROWSHAPE\_TURN\_UP\_LEFT\_CIRCLE\_FILLinstance-attribute

ARROWSHAPE\_TURN\_UP\_LEFT\_CIRCLE\_FILL: IconData

buildARROWSHAPE\_TURN\_UP\_LEFT\_FILLinstance-attribute

ARROWSHAPE\_TURN\_UP\_LEFT\_FILL: IconData

buildARROWSHAPE\_TURN\_UP\_RIGHTinstance-attribute

ARROWSHAPE\_TURN\_UP\_RIGHT: IconData

buildARROWSHAPE\_TURN\_UP\_RIGHT\_CIRCLEinstance-attribute

ARROWSHAPE\_TURN\_UP\_RIGHT\_CIRCLE: IconData

**buildARROWSHAPE\_TURN\_UP\_RIGHT\_CIRCLE\_FILLinstance-attribute**

ARROWSHAPE\_TURN\_UP\_RIGHT\_CIRCLE\_FILL: IconData

**buildARROWSHAPE\_TURN\_UP\_RIGHT\_FILLinstance-attribute**

ARROWSHAPE\_TURN\_UP\_RIGHT\_FILL: IconData

**buildARROWTRIANGLE\_DOWNinstance-attribute**

ARROWTRIANGLE\_DOWN: IconData

**buildARROWTRIANGLE\_DOWN\_CIRCLEinstance-attribute**

ARROWTRIANGLE\_DOWN\_CIRCLE: IconData

**buildARROWTRIANGLE\_DOWN\_CIRCLE\_FILLinstance-attribute**

ARROWTRIANGLE\_DOWN\_CIRCLE\_FILL: IconData

**buildARROWTRIANGLE\_DOWN\_FILLinstance-attribute**

ARROWTRIANGLE\_DOWN\_FILL: IconData

**buildARROWTRIANGLE\_DOWN\_SQUAREinstance-attribute**

ARROWTRIANGLE\_DOWN\_SQUARE: IconData

**buildARROWTRIANGLE\_DOWN\_SQUARE\_FILLinstance-attribute**

ARROWTRIANGLE\_DOWN\_SQUARE\_FILL: IconData

**buildARROWTRIANGLE\_LEFTinstance-attribute**

ARROWTRIANGLE\_LEFT: IconData

**buildARROWTRIANGLE\_LEFT\_CIRCLEinstance-attribute**

ARROWTRIANGLE\_LEFT\_CIRCLE: IconData

**buildARROWTRIANGLE\_LEFT\_CIRCLE\_FILLinstance-attribute**

ARROWTRIANGLE\_LEFT\_CIRCLE\_FILL: IconData

**buildARROWTRIANGLE\_LEFT\_FILLinstance-attribute**

ARROWTRIANGLE\_LEFT\_FILL: IconData

**buildARROWTRIANGLE\_LEFT\_SQUAREinstance-attribute**

ARROWTRIANGLE\_LEFT\_SQUARE: IconData

**buildARROWTRIANGLE\_LEFT\_SQUARE\_FILLinstance-attribute**

ARROWTRIANGLE\_LEFT\_SQUARE\_FILL: IconData

**buildARROWTRIANGLE\_RIGHTinstance-attribute**

ARROWTRIANGLE\_RIGHT: IconData

**buildARROWTRIANGLE\_RIGHT\_CIRCLEinstance-attribute**

ARROWTRIANGLE\_RIGHT\_CIRCLE: IconData

**buildARROWTRIANGLE\_RIGHT\_CIRCLE\_FILLinstance-attribute**

ARROWTRIANGLE\_RIGHT\_CIRCLE\_FILL: IconData

**buildARROWTRIANGLE\_RIGHT\_FILLinstance-attribute**

ARROWTRIANGLE\_RIGHT\_FILL: IconData

**buildARROWTRIANGLE\_RIGHT\_SQUAREinstance-attribute**

ARROWTRIANGLE\_RIGHT\_SQUARE: IconData

**buildARROWTRIANGLE\_RIGHT\_SQUARE\_FILLinstance-attribute**

ARROWTRIANGLE\_RIGHT\_SQUARE\_FILL: IconData

**buildARROWTRIANGLE\_UPinstance-attribute**

ARROWTRIANGLE\_UP: IconData

**buildARROWTRIANGLE\_UP\_CIRCLEinstance-attribute**

ARROWTRIANGLE\_UP\_CIRCLE: IconData

**buildARROWTRIANGLE\_UP\_CIRCLE\_FILLinstance-attribute**

ARROWTRIANGLE\_UP\_CIRCLE\_FILL: IconData

**buildARROWTRIANGLE\_UP\_FILLinstance-attribute**

ARROWTRIANGLE\_UP\_FILL: IconData

**buildARROWTRIANGLE\_UP\_SQUAREinstance-attribute**

ARROWTRIANGLE\_UP\_SQUARE: IconData

**buildARROWTRIANGLE\_UP\_SQUARE\_FILLinstance-attribute**

ARROWTRIANGLE\_UP\_SQUARE\_FILL: IconData

**buildASTERISK\_CIRCLEinstance-attribute**

ASTERISK\_CIRCLE: IconData

**buildASTERISK\_CIRCLE\_FILLinstance-attribute**

ASTERISK\_CIRCLE\_FILL: IconData

**buildATinstance-attribute**

AT: IconData

**buildAT\_BADGE\_MINUSinstance-attribute**

AT\_BADGE\_MINUS: IconData

**buildAT\_BADGE\_PLUSinstance-attribute**

AT\_BADGE\_PLUS: IconData

**buildAT\_CIRCLEinstance-attribute**

AT\_CIRCLE: IconData

**buildAT\_CIRCLE\_FILLinstance-attribute**

AT\_CIRCLE\_FILL: IconData

**buildBACKinstance-attribute**

BACK: IconData

**buildBACKWARDinstance-attribute**

BACKWARD: IconData

**buildBACKWARD\_ENDinstance-attribute**

BACKWARD\_END: IconData

**buildBACKWARD\_END\_ALTinstance-attribute**

BACKWARD\_END\_ALT: IconData

**buildBACKWARD\_END\_ALT\_FILLinstance-attribute**

BACKWARD\_END\_ALT\_FILL: IconData

**buildBACKWARD\_END\_FILLinstance-attribute**

BACKWARD\_END\_FILL: IconData

**buildBACKWARD\_FILLinstance-attribute**

BACKWARD\_FILL: IconData

**buildBADGE\_PLUS\_RADIO WAVES\_RIGHTinstance-attribute**

BADGE\_PLUS\_RADIO WAVES\_RIGHT: IconData

**buildBAGinstance-attribute**

BAG: IconData

**buildBAG\_BADGE\_MINUSinstance-attribute**

BAG\_BADGE\_MINUS: IconData

**buildBAG\_BADGE\_PLUSinstance-attribute**

BAG\_BADGE\_PLUS: IconData



**buildBAG\_FILLinstance-attribute**

BAG\_FILL: IconData

**buildBAG\_FILL\_BADGE\_MINUSinstance-attribute**

BAG\_FILL\_BADGE\_MINUS: IconData

**buildBAG\_FILL\_BADGE\_PLUSinstance-attribute**

BAG\_FILL\_BADGE\_PLUS: IconData

**buildBANDAGEinstance-attribute**

BANDAGE: IconData

**buildBANDAGE\_FILLinstance-attribute**

BANDAGE\_FILL: IconData

**buildBARCODEinstance-attribute**

BARCODE: IconData

**buildBARCODE\_VIEWFINDERinstance-attribute**

BARCODE\_VIEWFINDER: IconData

**buildBARSinstance-attribute**

BARS: IconData

**buildBATTERY\_0instance-attribute**

BATTERY\_0: IconData

**buildBATTERY\_100instance-attribute**

BATTERY\_100: IconData

**buildBATTERY\_25instance-attribute**

BATTERY\_25: IconData

**buildBATTERY\_25\_PERCENTinstance-attribute**

BATTERY\_25\_PERCENT: IconData

**buildBATTERY\_75\_PERCENTinstance-attribute**

BATTERY\_75\_PERCENT: IconData

**buildBATTERY\_CHARGINGinstance-attribute**

BATTERY\_CHARGING: IconData

**buildBATTERY\_EMPTYinstance-attribute**

BATTERY\_EMPTY: IconData

**buildBATTERY\_FULLinstance-attribute**

BATTERY\_FULL: IconData

**buildBED\_DOUBLEinstance-attribute**

BED\_DOUBLE: IconData

**buildBED\_DOUBLE\_FILLinstance-attribute**

BED\_DOUBLE\_FILL: IconData

**buildBELLinstance-attribute**

BELL: IconData

**buildBELL\_CIRCLEinstance-attribute**

BELL\_CIRCLE: IconData

**buildBELL\_CIRCLE\_FILLinstance-attribute**

BELL\_CIRCLE\_FILL: IconData

**buildBELL\_FILLinstance-attribute**

BELL\_FILL: IconData

**buildBELL\_SLASHinstance-attribute**

BELL\_SLASH: IconData

**buildBELL\_SLASH\_FILLinstance-attribute**

BELL\_SLASH\_FILL: IconData

**buildBELL\_SOLIDinstance-attribute**

BELL\_SOLID: IconData

**buildBIN\_XMARKinstance-attribute**

BIN\_XMARK: IconData

**buildBIN\_XMARK\_FILLinstance-attribute**

BIN\_XMARK\_FILL: IconData

**buildBITCOINinstance-attribute**

BITCOIN: IconData

**buildBITCOIN\_CIRCLEinstance-attribute**

BITCOIN\_CIRCLE: IconData

**buildBITCOIN\_CIRCLE\_FILLinstance-attribute**

BITCOIN\_CIRCLE\_FILL: IconData

**buildBLUETOOTHinstance-attribute**

BLUETOOTH: IconData

**buildBOLDinstance-attribute**

BOLD: IconData

**buildBOLD\_ITALIC\_UNDERLINEinstance-attribute**

BOLD\_ITALIC\_UNDERLINE: IconData

**buildBOLD\_UNDERLINEinstance-attribute**

BOLD\_UNDERLINE: IconData

**buildBOLTinstance-attribute**

BOLT: IconData

**buildBOLT\_BADGE\_Ainstance-attribute**

BOLT\_BADGE\_A: IconData

**buildBOLT\_BADGE\_A\_FILLinstance-attribute**

BOLT\_BADGE\_A\_FILL: IconData

**buildBOLT\_CIRCLEinstance-attribute**

BOLT\_CIRCLE: IconData

**buildBOLT\_CIRCLE\_FILLinstance-attribute**

BOLT\_CIRCLE\_FILL: IconData

**buildBOLT\_FILLinstance-attribute**

BOLT\_FILL: IconData

**buildBOLT\_HORIZONTALinstance-attribute**

BOLT\_HORIZONTAL: IconData

**buildBOLT\_HORIZONTAL\_CIRCLEinstance-attribute**

BOLT\_HORIZONTAL\_CIRCLE: IconData

**buildBOLT\_HORIZONTAL\_CIRCLE\_FILLinstance-attribute**

BOLT\_HORIZONTAL\_CIRCLE\_FILL: IconData

**buildBOLT\_HORIZONTAL\_FILLinstance-attribute**

BOLT\_HORIZONTAL\_FILL: IconData

**buildBOLT\_SLASHinstance-attribute**

BOLT\_SLASH: IconData

**buildBOLT\_SLASH\_FILLinstance-attribute**

BOLT\_SLASH\_FILL: IconData

**buildBOOKinstance-attribute**

BOOK: IconData

**buildBOOK\_CIRCLEinstance-attribute**

BOOK\_CIRCLE: IconData

**buildBOOK\_CIRCLE\_FILLinstance-attribute**

BOOK\_CIRCLE\_FILL: IconData

**buildBOOK\_FILLinstance-attribute**

BOOK\_FILL: IconData

**buildBOOK\_SOLIDinstance-attribute**

BOOK\_SOLID: IconData

**buildBOOKMARKinstance-attribute**

BOOKMARK: IconData

**buildBOOKMARK\_FILLinstance-attribute**

BOOKMARK\_FILL: IconData

**buildBOOKMARK\_SOLIDinstance-attribute**

BOOKMARK\_SOLID: IconData

**buildBRIEFCASEinstance-attribute**

BRIEFCASE: IconData

**buildBRIEFCASE\_FILLinstance-attribute**

BRIEFCASE\_FILL: IconData

**buildBRIGHTNESSinstance-attribute**

BRIGHTNESS: IconData

**buildBRIGHTNESS\_SOLIDinstance-attribute**

BRIGHTNESS\_SOLID: IconData

**buildBUBBLE\_LEFTinstance-attribute**

BUBBLE\_LEFT: IconData

**buildBUBBLE\_LEFT\_BUBBLE\_RIGHTinstance-attribute**

BUBBLE\_LEFT\_BUBBLE\_RIGHT: IconData

**buildBUBBLE\_LEFT\_BUBBLE\_RIGHT\_FILLinstance-attribute**

BUBBLE\_LEFT\_BUBBLE\_RIGHT\_FILL: IconData

**buildBUBBLE\_LEFT\_FILLinstance-attribute**

BUBBLE\_LEFT\_FILL: IconData

**buildBUBBLE\_MIDDLE\_BOTTOMinstance-attribute**

BUBBLE\_MIDDLE\_BOTTOM: IconData

**buildBUBBLE\_MIDDLE\_BOTTOM\_FILLinstance-attribute**

BUBBLE\_MIDDLE\_BOTTOM\_FILL: IconData

**buildBUBBLE\_MIDDLE\_TOPinstance-attribute**

BUBBLE\_MIDDLE\_TOP: IconData

**buildBUBBLE\_MIDDLE\_TOP\_FILLinstance-attribute**

BUBBLE\_MIDDLE\_TOP\_FILL: IconData

**buildBUBBLE\_RIGHT**instance-attribute

BUBBLE\_RIGHT: IconData

**buildBUBBLE\_RIGHT\_FILL**instance-attribute

BUBBLE\_RIGHT\_FILL: IconData

**buildBUILDING\_2\_FILL**instance-attribute

BUILDING\_2\_FILL: IconData

**buildBURN**instance-attribute

BURN: IconData

**buildBURST**instance-attribute

BURST: IconData

**buildBURST\_FILL**instance-attribute

BURST\_FILL: IconData

**buildBUS**instance-attribute

BUS: IconData

**buildCALENDAR**instance-attribute

CALENDAR: IconData

**buildCALENDAR\_BADGE\_MINUS**instance-attribute

CALENDAR\_BADGE\_MINUS: IconData

**buildCALENDAR\_BADGE\_PLUS**instance-attribute

CALENDAR\_BADGE\_PLUS: IconData

**buildCALENDAR\_CIRCLE**instance-attribute

CALENDAR\_CIRCLE: IconData

**buildCALENDAR\_CIRCLE\_FILL**instance-attribute

CALENDAR\_CIRCLE\_FILL: IconData

**buildCALENDAR\_TODAY**instance-attribute

CALENDAR\_TODAY: IconData

**buildCAMERA**instance-attribute

CAMERA: IconData

**buildCAMERA\_CIRCLE**instance-attribute

CAMERA\_CIRCLE: IconData

**buildCAMERA\_CIRCLE\_FILL**instance-attribute

CAMERA\_CIRCLE\_FILL: IconData

**buildCAMERA\_FILL**instance-attribute

CAMERA\_FILL: IconData

**buildCAMERA\_ON\_RECTANGLE**instance-attribute

CAMERA\_ON\_RECTANGLE: IconData

**buildCAMERA\_ON\_RECTANGLE\_FILL**instance-attribute

CAMERA\_ON\_RECTANGLE\_FILL: IconData

**buildCAMERA\_ROTATE**instance-attribute

CAMERA\_ROTATE: IconData

**buildCAMERA\_ROTATE\_FILL**instance-attribute

CAMERA\_ROTATE\_FILL: IconData

**buildCAMERA\_VIEWFINDER**instance-attribute

CAMERA\_VIEWFINDER: IconData

**buildCAPSLOCKinstance-attribute**

CAPSLOCK: IconData

**buildCAPSLOCK\_FILLinstance-attribute**

CAPSLOCK\_FILL: IconData

**buildCAPSULEinstance-attribute**

CAPSULE: IconData

**buildCAPSULE\_FILLinstance-attribute**

CAPSULE\_FILL: IconData

**buildCAPTIONS\_BUBBLEinstance-attribute**

CAPTIONS\_BUBBLE: IconData

**buildCAPTIONS\_BUBBLE\_FILLinstance-attribute**

CAPTIONS\_BUBBLE\_FILL: IconData

**buildCARinstance-attribute**

CAR: IconData

**buildCAR\_DETAILEDinstance-attribute**

CAR\_DETAILED: IconData

**buildCAR\_FILLinstance-attribute**

CAR\_FILL: IconData

**buildCARTinstance-attribute**

CART: IconData

**buildCART\_BADGE\_MINUSinstance-attribute**

CART\_BADGE\_MINUS: IconData

**buildCART\_BADGE\_PLUSinstance-attribute**

CART\_BADGE\_PLUS: IconData

**buildCART\_FILLinstance-attribute**

CART\_FILL: IconData

**buildCART\_FILL\_BADGE\_MINUSinstance-attribute**

CART\_FILL\_BADGE\_MINUS: IconData

**buildCART\_FILL\_BADGE\_PLUSinstance-attribute**

CART\_FILL\_BADGE\_PLUS: IconData

**buildCHART\_BARinstance-attribute**

CHART\_BAR: IconData

**buildCHART\_BAR\_ALT\_FILLinstance-attribute**

CHART\_BAR\_ALT\_FILL: IconData

**buildCHART\_BAR\_CIRCLEinstance-attribute**

CHART\_BAR\_CIRCLE: IconData

**buildCHART\_BAR\_CIRCLE\_FILLinstance-attribute**

CHART\_BAR\_CIRCLE\_FILL: IconData

**buildCHART\_BAR\_FILLinstance-attribute**

CHART\_BAR\_FILL: IconData

**buildCHART\_BAR\_SQUAREinstance-attribute**

CHART\_BAR\_SQUARE: IconData

**buildCHART\_BAR\_SQUARE\_FILLinstance-attribute**

CHART\_BAR\_SQUARE\_FILL: IconData

**buildCHART\_PIEinstance-attribute**

CHART\_PIE: IconData

**buildCHART\_PIE\_FILLinstance-attribute**

CHART\_PIE\_FILL: IconData

**buildCHAT\_BUBBLEinstance-attribute**

CHAT\_BUBBLE: IconData

**buildCHAT\_BUBBLE\_2instance-attribute**

CHAT\_BUBBLE\_2: IconData

**buildCHAT\_BUBBLE\_2\_FILLinstance-attribute**

CHAT\_BUBBLE\_2\_FILL: IconData

**buildCHAT\_BUBBLE\_FILLinstance-attribute**

CHAT\_BUBBLE\_FILL: IconData

**buildCHAT\_BUBBLE\_TEXTinstance-attribute**

CHAT\_BUBBLE\_TEXT: IconData

**buildCHAT\_BUBBLE\_TEXT\_FILLinstance-attribute**

CHAT\_BUBBLE\_TEXT\_FILL: IconData

**buildCHECK\_MARKinstance-attribute**

CHECK\_MARK: IconData

**buildCHECK\_MARK\_CIRCLEDinstance-attribute**

CHECK\_MARK\_CIRCLED: IconData

**buildCHECK\_MARK\_CIRCLED\_SOLIDinstance-attribute**

CHECK\_MARK\_CIRCLED\_SOLID: IconData

**buildCHECKMARKinstance-attribute**

CHECKMARK: IconData

**buildCHECKMARK\_ALTinstance-attribute**

CHECKMARK\_ALT: IconData

**buildCHECKMARK\_ALT\_CIRCLEinstance-attribute**

CHECKMARK\_ALT\_CIRCLE: IconData

**buildCHECKMARK\_ALT\_CIRCLE\_FILLinstance-attribute**

CHECKMARK\_ALT\_CIRCLE\_FILL: IconData

**buildCHECKMARK\_CIRCLEinstance-attribute**

CHECKMARK\_CIRCLE: IconData

**buildCHECKMARK\_CIRCLE\_FILLinstance-attribute**

CHECKMARK\_CIRCLE\_FILL: IconData

**buildCHECKMARK\_RECTANGLEinstance-attribute**

CHECKMARK\_RECTANGLE: IconData

**buildCHECKMARK\_RECTANGLE\_FILLinstance-attribute**

CHECKMARK\_RECTANGLE\_FILL: IconData

**buildCHECKMARK\_SEALinstance-attribute**

CHECKMARK\_SEAL: IconData

**buildCHECKMARK\_SEAL\_FILLinstance-attribute**

CHECKMARK\_SEAL\_FILL: IconData

**buildCHECKMARK\_SHIELDinstance-attribute**

CHECKMARK\_SHIELD: IconData

**buildCHECKMARK\_SHIELD\_FILLinstance-attribute**

CHECKMARK\_SHIELD\_FILL: IconData

**buildCHECKMARK\_SQUAREinstance-attribute**

CHECKMARK\_SQUARE: IconData

**buildCHECKMARK\_SQUARE\_FILLinstance-attribute**

CHECKMARK\_SQUARE\_FILL: IconData

**buildCHEVRON\_BACKinstance-attribute**

CHEVRON\_BACK: IconData

**buildCHEVRON\_COMPACT\_DOWNinstance-attribute**

CHEVRON\_COMPACT\_DOWN: IconData

**buildCHEVRON\_COMPACT\_LEFTinstance-attribute**

CHEVRON\_COMPACT\_LEFT: IconData

**buildCHEVRON\_COMPACT\_RIGHTinstance-attribute**

CHEVRON\_COMPACT\_RIGHT: IconData

**buildCHEVRON\_COMPACT\_UPinstance-attribute**

CHEVRON\_COMPACT\_UP: IconData

**buildCHEVRON\_DOWNinstance-attribute**

CHEVRON\_DOWN: IconData

**buildCHEVRON\_DOWN\_CIRCLEinstance-attribute**

CHEVRON\_DOWN\_CIRCLE: IconData

**buildCHEVRON\_DOWN\_CIRCLE\_FILLinstance-attribute**

CHEVRON\_DOWN\_CIRCLE\_FILL: IconData

**buildCHEVRON\_DOWN\_SQUAREinstance-attribute**

CHEVRON\_DOWN\_SQUARE: IconData

**buildCHEVRON\_DOWN\_SQUARE\_FILLinstance-attribute**

CHEVRON\_DOWN\_SQUARE\_FILL: IconData

**buildCHEVRON\_FORWARDinstance-attribute**

CHEVRON\_FORWARD: IconData

**buildCHEVRON\_LEFTinstance-attribute**

CHEVRON\_LEFT: IconData

**buildCHEVRON\_LEFT\_2instance-attribute**

CHEVRON\_LEFT\_2: IconData

**buildCHEVRON\_LEFT\_CIRCLEinstance-attribute**

CHEVRON\_LEFT\_CIRCLE: IconData

**buildCHEVRON\_LEFT\_CIRCLE\_FILLinstance-attribute**

CHEVRON\_LEFT\_CIRCLE\_FILL: IconData

**buildCHEVRON\_LEFT\_SLASH\_CHEVRON\_RIGHTinstance-attribute**

CHEVRON\_LEFT\_SLASH\_CHEVRON\_RIGHT: IconData

**buildCHEVRON\_LEFT\_SQUAREinstance-attribute**

CHEVRON\_LEFT\_SQUARE: IconData

**buildCHEVRON\_LEFT\_SQUARE\_FILLinstance-attribute**

CHEVRON\_LEFT\_SQUARE\_FILL: IconData

**buildCHEVRON\_RIGHTinstance-attribute**

CHEVRON\_RIGHT: IconData

**buildCHEVRON\_RIGHT\_2instance-attribute**

CHEVRON\_RIGHT\_2: IconData

**buildCHEVRON\_RIGHT\_CIRCLEinstance-attribute**

CHEVRON\_RIGHT\_CIRCLE: IconData

**buildCHEVRON\_RIGHT\_CIRCLE\_FILLinstance-attribute**

CHEVRON\_RIGHT\_CIRCLE\_FILL: IconData

**buildCHEVRON\_RIGHT\_SQUAREinstance-attribute**

CHEVRON\_RIGHT\_SQUARE: IconData

**buildCHEVRON\_RIGHT\_SQUARE\_FILLinstance-attribute**

CHEVRON\_RIGHT\_SQUARE\_FILL: IconData

**buildCHEVRON\_UPinstance-attribute**

CHEVRON\_UP: IconData

**buildCHEVRON\_UP\_CHEVRON\_DOWNinstance-attribute**

CHEVRON\_UP\_CHEVRON\_DOWN: IconData

**buildCHEVRON\_UP\_CIRCLEinstance-attribute**

CHEVRON\_UP\_CIRCLE: IconData

**buildCHEVRON\_UP\_CIRCLE\_FILLinstance-attribute**

CHEVRON\_UP\_CIRCLE\_FILL: IconData

**buildCHEVRON\_UP\_SQUAREinstance-attribute**

CHEVRON\_UP\_SQUARE: IconData

**buildCHEVRON\_UP\_SQUARE\_FILLinstance-attribute**

CHEVRON\_UP\_SQUARE\_FILL: IconData

**buildCIRCLEinstance-attribute**

CIRCLE: IconData

**buildCIRCLE\_BOTTOMTHIRD\_SPLITinstance-attribute**

CIRCLE\_BOTTOMTHIRD\_SPLIT: IconData

**buildCIRCLE\_FILLinstance-attribute**

CIRCLE\_FILL: IconData

**buildCIRCLE\_FILLEDinstance-attribute**

CIRCLE\_FILLED: IconData

**buildCIRCLE\_GRID\_3X3instance-attribute**

CIRCLE\_GRID\_3X3: IconData

**buildCIRCLE\_GRID\_3X3\_FILLinstance-attribute**

CIRCLE\_GRID\_3X3\_FILL: IconData

**buildCIRCLE\_GRID\_HEXinstance-attribute**

CIRCLE\_GRID\_HEX: IconData

**buildCIRCLE\_GRID\_HEX\_FILLinstance-attribute**

CIRCLE\_GRID\_HEX\_FILL: IconData

**buildCIRCLE\_LEFTHALF\_FILLinstance-attribute**

CIRCLE\_LEFTHALF\_FILL: IconData

**buildCIRCLE\_RIGHTHALF\_FILLinstance-attribute**

CIRCLE\_RIGHTHALF\_FILL: IconData

**buildCLEARinstance-attribute**

CLEAR: IconData



**buildCLEAR\_CIRCLEDinstance-attribute**

CLEAR\_CIRCLED: IconData

**buildCLEAR\_CIRCLED\_SOLIDinstance-attribute**

CLEAR\_CIRCLED\_SOLID: IconData

**buildCLEAR\_FILLinstance-attribute**

CLEAR\_FILL: IconData

**buildCLEAR\_THICKinstance-attribute**

CLEAR\_THICK: IconData

**buildCLEAR\_THICK\_CIRCLEDinstance-attribute**

CLEAR\_THICK\_CIRCLED: IconData

**buildCLOCKinstance-attribute**

CLOCK: IconData

**buildCLOCK\_FILLinstance-attribute**

CLOCK\_FILL: IconData

**buildCLOCK\_SOLIDinstance-attribute**

CLOCK\_SOLID: IconData

**buildCLOUDinstance-attribute**

CLOUD: IconData

**buildCLOUD\_BOLTinstance-attribute**

CLOUD\_BOLT: IconData

**buildCLOUD\_BOLT\_FILLinstance-attribute**

CLOUD\_BOLT\_FILL: IconData

**buildCLOUD\_BOLT\_RAINinstance-attribute**

CLOUD\_BOLT\_RAIN: IconData

**buildCLOUD\_BOLT\_RAIN\_FILLinstance-attribute**

CLOUD\_BOLT\_RAIN\_FILL: IconData

**buildCLOUD\_DOWNLOADinstance-attribute**

CLOUD\_DOWNLOAD: IconData

**buildCLOUD\_DOWNLOAD\_FILLinstance-attribute**

CLOUD\_DOWNLOAD\_FILL: IconData

**buildCLOUD\_DRIZZLEinstance-attribute**

CLOUD\_DRIZZLE: IconData

**buildCLOUD\_DRIZZLE\_FILLinstance-attribute**

CLOUD\_DRIZZLE\_FILL: IconData

**buildCLOUD\_FILLinstance-attribute**

CLOUD\_FILL: IconData

**buildCLOUD\_FOGinstance-attribute**

CLOUD\_FOG: IconData

**buildCLOUD\_FOG\_FILLinstance-attribute**

CLOUD\_FOG\_FILL: IconData

**buildCLOUD\_HAILinstance-attribute**

CLOUD\_HAIL: IconData

**buildCLOUD\_HAIL\_FILLinstance-attribute**

CLOUD\_HAIL\_FILL: IconData

**buildCLOUD\_HEAVYRAINinstance-attribute**

CLOUD\_HEAVYRAIN: IconData

**buildCLOUD\_HEAVYRAIN\_FILLinstance-attribute**

CLOUD\_HEAVYRAIN\_FILL: IconData

**buildCLOUD\_MOONinstance-attribute**

CLOUD\_MOON: IconData

**buildCLOUD\_MOON\_BOLTinstance-attribute**

CLOUD\_MOON\_BOLT: IconData

**buildCLOUD\_MOON\_BOLT\_FILLinstance-attribute**

CLOUD\_MOON\_BOLT\_FILL: IconData

**buildCLOUD\_MOON\_FILLinstance-attribute**

CLOUD\_MOON\_FILL: IconData

**buildCLOUD\_MOON\_RAINinstance-attribute**

CLOUD\_MOON\_RAIN: IconData

**buildCLOUD\_MOON\_RAIN\_FILLinstance-attribute**

CLOUD\_MOON\_RAIN\_FILL: IconData

**buildCLOUD\_RAINinstance-attribute**

CLOUD\_RAIN: IconData

**buildCLOUD\_RAIN\_FILLinstance-attribute**

CLOUD\_RAIN\_FILL: IconData

**buildCLOUD\_SLEETinstance-attribute**

CLOUD\_SLEET: IconData

**buildCLOUD\_SLEET\_FILLinstance-attribute**

CLOUD\_SLEET\_FILL: IconData

**buildCLOUD\_SNOWinstance-attribute**

CLOUD\_SNOW: IconData

**buildCLOUD\_SNOW\_FILLinstance-attribute**

CLOUD\_SNOW\_FILL: IconData

**buildCLOUD\_SUNinstance-attribute**

CLOUD\_SUN: IconData

**buildCLOUD\_SUN\_BOLTinstance-attribute**

CLOUD\_SUN\_BOLT: IconData

**buildCLOUD\_SUN\_BOLT\_FILLinstance-attribute**

CLOUD\_SUN\_BOLT\_FILL: IconData

**buildCLOUD\_SUN\_FILLinstance-attribute**

CLOUD\_SUN\_FILL: IconData

**buildCLOUD\_SUN\_RAINinstance-attribute**

CLOUD\_SUN\_RAIN: IconData

**buildCLOUD\_SUN\_RAIN\_FILLinstance-attribute**

CLOUD\_SUN\_RAIN\_FILL: IconData

**buildCLOUD\_UPLOADinstance-attribute**

CLOUD\_UPLOAD: IconData

**buildCLOUD\_UPLOAD\_FILLinstance-attribute**

CLOUD\_UPLOAD\_FILL: IconData

**buildCOLLECTIONSinstance-attribute**

COLLECTIONS: IconData

**buildCOLLECTIONS\_SOLIDinstance-attribute**

COLLECTIONS\_SOLID: IconData

**buildCOLOR\_FILTERinstance-attribute**

COLOR\_FILTER: IconData

**buildCOLOR\_FILTER\_FILLinstance-attribute**

COLOR\_FILTER\_FILL: IconData

**buildCOMMANDinstance-attribute**

COMMAND: IconData

**buildCOMPASSinstance-attribute**

COMPASS: IconData

**buildCOMPASS\_FILLinstance-attribute**

COMPASS\_FILL: IconData

**buildCONTROLinstance-attribute**

CONTROL: IconData

**buildCONVERSATION\_BUBBLEinstance-attribute**

CONVERSATION\_BUBBLE: IconData

**buildCREATEinstance-attribute**

CREATE: IconData

**buildCREATE\_SOLIDinstance-attribute**

CREATE\_SOLID: IconData

**buildCREDITCARDinstance-attribute**

CREDITCARD: IconData

**buildCREDITCARD\_FILLinstance-attribute**

CREDITCARD\_FILL: IconData

**buildCROPinstance-attribute**

CROP: IconData

**buildCROP\_ROTATEinstance-attribute**

CROP\_ROTATE: IconData

**buildCUBEinstance-attribute**

CUBE: IconData

**buildCUBE\_BOXinstance-attribute**

CUBE\_BOX: IconData

**buildCUBE\_BOX\_FILLinstance-attribute**

CUBE\_BOX\_FILL: IconData

**buildCUBE\_FILLinstance-attribute**

CUBE\_FILL: IconData

**buildCURSOR\_RAYSinstance-attribute**

CURSOR\_RAYS: IconData

**buildDECREASE\_INDENTinstance-attribute**

DECREASE\_INDENT: IconData

**buildDECREASE\_QUOTELEVELinstance-attribute**

DECREASE\_QUOTELEVEL: IconData

**buildDELETEinstance-attribute**

DELETE: IconData

**buildDELETE\_LEFTinstance-attribute**

DELETE\_LEFT: IconData

**buildDELETE\_LEFT\_FILLinstance-attribute**

DELETE\_LEFT\_FILL: IconData

**buildDELETE\_RIGHTinstance-attribute**

DELETE\_RIGHT: IconData

**buildDELETE\_RIGHT\_FILLinstance-attribute**

DELETE\_RIGHT\_FILL: IconData

**buildDELETE\_SIMPLEinstance-attribute**

DELETE\_SIMPLE: IconData

**buildDELETE\_SOLIDinstance-attribute**

DELETE\_SOLID: IconData

**buildDESKTOPCOMPUTERinstance-attribute**

DESKTOPCOMPUTER: IconData

**buildDEVICE\_DESKTOPinstance-attribute**

DEVICE\_DESKTOP: IconData

**buildDEVICE\_LAPTOPinstance-attribute**

DEVICE\_LAPTOP: IconData

**buildDEVICE\_PHONE\_LANDSCAPEinstance-attribute**

DEVICE\_PHONE\_LANDSCAPE: IconData

**buildDEVICE\_PHONE\_PORTRAITinstance-attribute**

DEVICE\_PHONE\_PORTRAIT: IconData

**buildDIALinstance-attribute**

DIAL: IconData

**buildDIAL\_FILLinstance-attribute**

DIAL\_FILL: IconData

**buildDIVIDEinstance-attribute**

DIVIDE: IconData

**buildDIVIDE\_CIRCLEinstance-attribute**

DIVIDE\_CIRCLE: IconData

**buildDIVIDE\_CIRCLE\_FILLinstance-attribute**

DIVIDE\_CIRCLE\_FILL: IconData

**buildDIVIDE\_SQUAREinstance-attribute**

DIVIDE\_SQUARE: IconData

**buildDIVIDE\_SQUARE\_FILLinstance-attribute**

DIVIDE\_SQUARE\_FILL: IconData

**buildDOCinstance-attribute**

DOC: IconData

**buildDOC\_APPENDinstance-attribute**

DOC\_APPEND: IconData

**buildDOC\_CHARTinstance-attribute**

DOC\_CHART: IconData

**buildDOC\_CHART\_FILLinstance-attribute**

DOC\_CHART\_FILL: IconData

**buildDOC\_CHECKMARKinstance-attribute**

DOC\_CHECKMARK: IconData

**buildDOC\_CHECKMARK\_FILLinstance-attribute**

DOC\_CHECKMARK\_FILL: IconData

**buildDOC\_CIRCLEinstance-attribute**

DOC\_CIRCLE: IconData

**buildDOC\_CIRCLE\_FILLinstance-attribute**

DOC\_CIRCLE\_FILL: IconData

**buildDOC\_FILLinstance-attribute**

DOC\_FILL: IconData

**buildDOC\_ON\_CLIPBOARDinstance-attribute**

DOC\_ON\_CLIPBOARD: IconData

**buildDOC\_ON\_CLIPBOARD\_FILLinstance-attribute**

DOC\_ON\_CLIPBOARD\_FILL: IconData

**buildDOC\_ON\_DOCinstance-attribute**

DOC\_ON\_DOC: IconData

**buildDOC\_ON\_DOC\_FILLinstance-attribute**

DOC\_ON\_DOC\_FILL: IconData

**buildDOC\_PERSONinstance-attribute**

DOC\_PERSON: IconData

**buildDOC\_PERSON\_FILLinstance-attribute**

DOC\_PERSON\_FILL: IconData

**buildDOC\_PLAINTEXTinstance-attribute**

DOC\_PLAINTEXT: IconData

**buildDOC\_RICHTEXTinstance-attribute**

DOC\_RICHTEXT: IconData

**buildDOC\_TEXTinstance-attribute**

DOC\_TEXT: IconData

**buildDOC\_TEXT\_FILLinstance-attribute**

DOC\_TEXT\_FILL: IconData

**buildDOC\_TEXT\_SEARCHinstance-attribute**

DOC\_TEXT\_SEARCH: IconData

**buildDOC\_TEXT\_VIEWFINDERinstance-attribute**

DOC\_TEXT\_VIEWFINDER: IconData

**buildDOT\_RADIOWAVES\_LEFT\_RIGHTinstance-attribute**

DOT\_RADIOWAVES\_LEFT\_RIGHT: IconData

**buildDOT\_RADIOWAVES\_RIGHTinstance-attribute**

DOT\_RADIOWAVES\_RIGHT: IconData

**buildDOT\_SQUAREinstance-attribute**

DOT\_SQUARE: IconData

**buildDOT\_SQUARE\_FILLinstance-attribute**

DOT\_SQUARE\_FILL: IconData

**buildDOUBLE\_MUSIC\_NOTEinstance-attribute**

DOUBLE\_MUSIC\_NOTE: IconData

**buildDOWN\_ARROWinstance-attribute**

DOWN\_ARROW: IconData

**buildDOWNLOAD\_CIRCLEinstance-attribute**

DOWNLOAD\_CIRCLE: IconData

**buildDOWNLOAD\_CIRCLE\_FILLinstance-attribute**

DOWNLOAD\_CIRCLE\_FILL: IconData

**buildDROPinstance-attribute**

DROP: IconData

**buildDROP\_FILLinstance-attribute**

DROP\_FILL: IconData

**buildDROP\_TRIANGLEinstance-attribute**

DROP\_TRIANGLE: IconData

**buildDROP\_TRIANGLE\_FILLinstance-attribute**

DROP\_TRIANGLE\_FILL: IconData

**buildEARinstance-attribute**

EAR: IconData

**buildEJECTinstance-attribute**

EJECT: IconData

**buildEJECT\_FILLinstance-attribute**

EJECT\_FILL: IconData

**buildELLIPSES\_BUBBLEinstance-attribute**

ELLIPSES\_BUBBLE: IconData

**buildELLIPSES\_BUBBLE\_FILLinstance-attribute**

ELLIPSES\_BUBBLE\_FILL: IconData

**buildELLIPSISinstance-attribute**

ELLIPSIS: IconData

**buildELLIPSIS\_CIRCLEinstance-attribute**

ELLIPSIS\_CIRCLE: IconData

**buildELLIPSIS\_CIRCLE\_FILLinstance-attribute**

ELLIPSIS\_CIRCLE\_FILL: IconData

**buildELLIPSIS\_VERTICALinstance-attribute**

ELLIPSIS\_VERTICAL: IconData

**buildELLIPSIS\_VERTICAL\_CIRCLEinstance-attribute**

ELLIPSIS\_VERTICAL\_CIRCLE: IconData

**buildELLIPSIS\_VERTICAL\_CIRCLE\_FILLinstance-attribute**

ELLIPSIS\_VERTICAL\_CIRCLE\_FILL: IconData

**buildENVELOPEinstance-attribute**

ENVELOPE: IconData

**buildENVELOPE\_BADGEinstance-attribute**

ENVELOPE\_BADGE: IconData

**buildENVELOPE\_BADGE\_FILLinstance-attribute**

ENVELOPE\_BADGE\_FILL: IconData

**buildENVELOPE\_CIRCLEinstance-attribute**

ENVELOPE\_CIRCLE: IconData

**buildENVELOPE\_CIRCLE\_FILLinstance-attribute**

ENVELOPE\_CIRCLE\_FILL: IconData

**buildENVELOPE\_FILLinstance-attribute**

ENVELOPE\_FILL: IconData

**buildENVELOPE\_OPENinstance-attribute**

ENVELOPE\_OPEN: IconData

**buildENVELOPE\_OPEN\_FILLinstance-attribute**

ENVELOPE\_OPEN\_FILL: IconData

**buildEQUALinstance-attribute**

EQUAL: IconData

**buildEQUAL\_CIRCLEinstance-attribute**

EQUAL\_CIRCLE: IconData

**buildEQUAL\_CIRCLE\_FILLinstance-attribute**

EQUAL\_CIRCLE\_FILL: IconData

**buildEQUAL\_SQUAREinstance-attribute**

EQUAL\_SQUARE: IconData

**buildEQUAL\_SQUARE\_FILLinstance-attribute**

EQUAL\_SQUARE\_FILL: IconData

**buildESCAPEinstance-attribute**

ESCAPE: IconData

**buildEXCLAMATIONMARKinstance-attribute**

EXCLAMATIONMARK: IconData

**buildEXCLAMATIONMARK\_BUBBLEinstance-attribute**

EXCLAMATIONMARK\_BUBBLE: IconData

**buildEXCLAMATIONMARK\_BUBBLE\_FILLinstance-attribute**

EXCLAMATIONMARK\_BUBBLE\_FILL: IconData

**buildEXCLAMATIONMARK\_CIRCLEinstance-attribute**

EXCLAMATIONMARK\_CIRCLE: IconData

**buildEXCLAMATIONMARK\_CIRCLE\_FILLinstance-attribute**

EXCLAMATIONMARK\_CIRCLE\_FILL: IconData

**buildEXCLAMATIONMARK\_OCTAGONinstance-attribute**

EXCLAMATIONMARK\_OCTAGON: IconData

**buildEXCLAMATIONMARK\_OCTAGON\_FILLinstance-attribute**

EXCLAMATIONMARK\_OCTAGON\_FILL: IconData

**buildEXCLAMATIONMARK\_SHIELDinstance-attribute**

EXCLAMATIONMARK\_SHIELD: IconData

**buildEXCLAMATIONMARK\_SHIELD\_FILLinstance-attribute**

EXCLAMATIONMARK\_SHIELD\_FILL: IconData

**buildEXCLAMATIONMARK\_SQUAREinstance-attribute**

EXCLAMATIONMARK\_SQUARE: IconData

**buildEXCLAMATIONMARK\_SQUARE\_FILLinstance-attribute**

EXCLAMATIONMARK\_SQUARE\_FILL: IconData

**buildEXCLAMATIONMARK\_TRIANGLEinstance-attribute**

EXCLAMATIONMARK\_TRIANGLE: IconData

**buildEXCLAMATIONMARK\_TRIANGLE\_FILLinstance-attribute**

EXCLAMATIONMARK\_TRIANGLE\_FILL: IconData

**buildEYEinstance-attribute**

EYE: IconData

**buildEYE\_FILLinstance-attribute**

EYE\_FILL: IconData

**buildEYE\_SLASHinstance-attribute**

EYE\_SLASH: IconData

**buildEYE\_SLASH\_FILLinstance-attribute**

EYE\_SLASH\_FILL: IconData

**buildEYE\_SOLIDinstance-attribute**

EYE\_SOLID: IconData

**buildEYEDROPPERinstance-attribute**

EYEDROPPER: IconData

**buildEYEDROPPER\_FULLinstance-attribute**

EYEDROPPER\_FULL: IconData

**buildEYEDROPPER\_HALFFULLinstance-attribute**

EYEDROPPER\_HALFFULL: IconData

**buildEYEGLASSESinstance-attribute**

EYEGLASSES: IconData

**buildF\_CURSIVEinstance-attribute**

F\_CURSIVE: IconData

**buildF\_CURSIVE\_CIRCLEinstance-attribute**

F\_CURSIVE\_CIRCLE: IconData

**buildF\_CURSIVE\_CIRCLE\_FILLinstance-attribute**

F\_CURSIVE\_CIRCLE\_FILL: IconData

**buildFILMinstance-attribute**

FILM: IconData

**buildFILM\_FILLinstance-attribute**

FILM\_FILL: IconData

**buildFLAGinstance-attribute**

FLAG: IconData

**buildFLAG\_CIRCLEinstance-attribute**

FLAG\_CIRCLE: IconData

**buildFLAG\_CIRCLE\_FILLinstance-attribute**

FLAG\_CIRCLE\_FILL: IconData

**buildFLAG\_FILLinstance-attribute**

FLAG\_FILL: IconData

**buildFLAG\_SLASHinstance-attribute**

FLAG\_SLASH: IconData

**buildFLAG\_SLASH\_FILLinstance-attribute**

FLAG\_SLASH\_FILL: IconData



**buildFLAMEinstance-attribute**

FLAME: IconData

**buildFLAME\_FILLinstance-attribute**

FLAME\_FILL: IconData

**buildFLOPPY\_DISKinstance-attribute**

FLOPPY\_DISK: IconData

**buildFLOWCHARTinstance-attribute**

FLOWCHART: IconData

**buildFLOWCHART\_FILLinstance-attribute**

FLOWCHART\_FILL: IconData

**buildFOLDERinstance-attribute**

FOLDER: IconData

**buildFOLDER\_BADGE\_MINUSinstance-attribute**

FOLDER\_BADGE\_MINUS: IconData

**buildFOLDER\_BADGE\_PERSON\_CROPinstance-attribute**

FOLDER\_BADGE\_PERSON\_CROP: IconData

**buildFOLDER\_BADGE\_PLUSinstance-attribute**

FOLDER\_BADGE\_PLUS: IconData

**buildFOLDER\_CIRCLEinstance-attribute**

FOLDER\_CIRCLE: IconData

**buildFOLDER\_CIRCLE\_FILLinstance-attribute**

FOLDER\_CIRCLE\_FILL: IconData

**buildFOLDER\_FILLinstance-attribute**

FOLDER\_FILL: IconData

**buildFOLDER\_FILL\_BADGE\_MINUSinstance-attribute**

FOLDER\_FILL\_BADGE\_MINUS: IconData

**buildFOLDER\_FILL\_BADGE\_PERSON\_CROPinstance-attribute**

FOLDER\_FILL\_BADGE\_PERSON\_CROP: IconData

**buildFOLDER\_FILL\_BADGE\_PLUSinstance-attribute**

FOLDER\_FILL\_BADGE\_PLUS: IconData

**buildFOLDER\_OPENinstance-attribute**

FOLDER\_OPEN: IconData

**buildFOLDER\_SOLIDinstance-attribute**

FOLDER\_SOLID: IconData

**buildFORWARDinstance-attribute**

FORWARD: IconData

**buildFORWARD\_ENDinstance-attribute**

FORWARD\_END: IconData

**buildFORWARD\_END\_ALTinstance-attribute**

FORWARD\_END\_ALT: IconData

**buildFORWARD\_END\_ALT\_FILLinstance-attribute**

FORWARD\_END\_ALT\_FILL: IconData

**buildFORWARD\_END\_FILLinstance-attribute**

FORWARD\_END\_FILL: IconData

**buildFORWARD\_FILLinstance-attribute**

FORWARD\_FILL: IconData

**buildFULLSCREENinstance-attribute**

FULLSCREEN: IconData

**buildFULLSCREEN\_EXITinstance-attribute**

FULLSCREEN\_EXIT: IconData

**buildFUNCTIONinstance-attribute**

FUNCTION: IconData

**buildFXinstance-attribute**

FX: IconData

**buildGAME\_CONTROLLERinstance-attribute**

GAME\_CONTROLLER: IconData

**buildGAME\_CONTROLLER\_SOLIDinstance-attribute**

GAME\_CONTROLLER\_SOLID: IconData

**buildGAMECONTROLLERinstance-attribute**

GAMECONTROLLER: IconData

**buildGAMECONTROLLER\_ALT\_FILLinstance-attribute**

GAMECONTROLLER\_ALT\_FILL: IconData

**buildGAMECONTROLLER\_FILLinstance-attribute**

GAMECONTROLLER\_FILL: IconData

**buildGAUGEinstance-attribute**

GAUGE: IconData

**buildGAUGE\_BADGE\_MINUSinstance-attribute**

GAUGE\_BADGE\_MINUS: IconData

**buildGAUGE\_BADGE\_PLUSinstance-attribute**

GAUGE\_BADGE\_PLUS: IconData

**buildGEARinstance-attribute**

GEAR: IconData

**buildGEAR\_ALTinstance-attribute**

GEAR\_ALT: IconData

**buildGEAR\_ALT\_FILLinstance-attribute**

GEAR\_ALT\_FILL: IconData

**buildGEAR\_BIGinstance-attribute**

GEAR\_BIG: IconData

**buildGEAR\_SOLIDinstance-attribute**

GEAR\_SOLID: IconData

**buildGIFTinstance-attribute**

GIFT: IconData

**buildGIFT\_ALTinstance-attribute**

GIFT\_ALT: IconData

**buildGIFT\_ALT\_FILLinstance-attribute**

GIFT\_ALT\_FILL: IconData

**buildGIFT\_FILLinstance-attribute**

GIFT\_FILL: IconData

**buildGLOBEinstance-attribute**

GLOBE: IconData

**buildGOBACKWARDinstance-attribute**

GOBACKWARD: IconData

**buildGOBACKWARD\_10instance-attribute**

GOBACKWARD\_10: IconData

**buildGOBACKWARD\_15instance-attribute**

GOBACKWARD\_15: IconData

**buildGOBACKWARD\_30instance-attribute**

GOBACKWARD\_30: IconData

**buildGOBACKWARD\_45instance-attribute**

GOBACKWARD\_45: IconData

**buildGOBACKWARD\_60instance-attribute**

GOBACKWARD\_60: IconData

**buildGOBACKWARD\_75instance-attribute**

GOBACKWARD\_75: IconData

**buildGOBACKWARD\_90instance-attribute**

GOBACKWARD\_90: IconData

**buildGOBACKWARD\_MINUSinstance-attribute**

GOBACKWARD\_MINUS: IconData

**buildGOFORWARDinstance-attribute**

GOFORWARD: IconData

**buildGOFORWARD\_10instance-attribute**

GOFORWARD\_10: IconData

**buildGOFORWARD\_15instance-attribute**

GOFORWARD\_15: IconData

**buildGOFORWARD\_30instance-attribute**

GOFORWARD\_30: IconData

**buildGOFORWARD\_45instance-attribute**

GOFORWARD\_45: IconData

**buildGOFORWARD\_60instance-attribute**

GOFORWARD\_60: IconData

**buildGOFORWARD\_75instance-attribute**

GOFORWARD\_75: IconData

**buildGOFORWARD\_90instance-attribute**

GOFORWARD\_90: IconData

**buildGOFORWARD\_PLUSinstance-attribute**

GOFORWARD\_PLUS: IconData

**buildGRAPH\_CIRCLEinstance-attribute**

GRAPH\_CIRCLE: IconData

**buildGRAPH\_CIRCLE\_FILLinstance-attribute**

GRAPH\_CIRCLE\_FILL: IconData

**buildGRAPH\_SQUAREinstance-attribute**

GRAPH\_SQUARE: IconData

**buildGRAPH\_SQUARE\_FILLinstance-attribute**  
GRAPH\_SQUARE\_FILL: IconData

**buildGREATERTHANinstance-attribute**  
GREATERTHAN: IconData

**buildGREATERTHAN\_CIRCLEinstance-attribute**  
GREATERTHAN\_CIRCLE: IconData

**buildGREATERTHAN\_CIRCLE\_FILLinstance-attribute**  
GREATERTHAN\_CIRCLE\_FILL: IconData

**buildGREATERTHAN\_SQUAREinstance-attribute**  
GREATERTHAN\_SQUARE: IconData

**buildGREATERTHAN\_SQUARE\_FILLinstance-attribute**  
GREATERTHAN\_SQUARE\_FILL: IconData

**buildGRIDinstance-attribute**  
GRID: IconData

**buildGRID\_CIRCLEinstance-attribute**  
GRID\_CIRCLE: IconData

**buildGRID\_CIRCLE\_FILLinstance-attribute**  
GRID\_CIRCLE\_FILL: IconData

**buildGROUPinstance-attribute**  
GROUP: IconData

**buildGROUP\_SOLIDinstance-attribute**  
GROUP\_SOLID: IconData

**buildGUITARSinstance-attribute**  
GUITARS: IconData

**buildHAMMERinstance-attribute**  
HAMMER: IconData

**buildHAMMER\_FILLinstance-attribute**  
HAMMER\_FILL: IconData

**buildHAND\_DRAWinstance-attribute**  
HAND\_DRAW: IconData

**buildHAND\_DRAW\_FILLinstance-attribute**  
HAND\_DRAW\_FILL: IconData

**buildHAND\_POINT\_LEFTinstance-attribute**  
HAND\_POINT\_LEFT: IconData

**buildHAND\_POINT\_LEFT\_FILLinstance-attribute**  
HAND\_POINT\_LEFT\_FILL: IconData

**buildHAND\_POINT\_RIGHTinstance-attribute**  
HAND\_POINT\_RIGHT: IconData

**buildHAND\_POINT\_RIGHT\_FILLinstance-attribute**  
HAND\_POINT\_RIGHT\_FILL: IconData

**buildHAND\_RAISEDinstance-attribute**  
HAND\_RAISED: IconData

**buildHAND\_RAISED\_FILLinstance-attribute**  
HAND\_RAISED\_FILL: IconData

**buildHAND\_RAISED\_SLASHinstance-attribute**

HAND\_RAISED\_SLASH: IconData

**buildHAND\_RAISED\_SLASH\_FILLinstance-attribute**

HAND\_RAISED\_SLASH\_FILL: IconData

**buildHAND\_THUMBSDOWNinstance-attribute**

HAND\_THUMBSDOWN: IconData

**buildHAND\_THUMBSDOWN\_FILLinstance-attribute**

HAND\_THUMBSDOWN\_FILL: IconData

**buildHAND\_THUMBSUPinstance-attribute**

HAND\_THUMBSUP: IconData

**buildHAND\_THUMBSUP\_FILLinstance-attribute**

HAND\_THUMBSUP\_FILL: IconData

**buildHAREinstance-attribute**

HARE: IconData

**buildHARE\_FILLinstance-attribute**

HARE\_FILL: IconData

**buildHEADPHONESinstance-attribute**

HEADPHONES: IconData

**buildHEARTinstance-attribute**

HEART: IconData

**buildHEART\_CIRCLEinstance-attribute**

HEART\_CIRCLE: IconData

**buildHEART\_CIRCLE\_FILLinstance-attribute**

HEART\_CIRCLE\_FILL: IconData

**buildHEART\_FILLinstance-attribute**

HEART\_FILL: IconData

**buildHEART\_SLASHinstance-attribute**

HEART\_SLASH: IconData

**buildHEART\_SLASH\_CIRCLEinstance-attribute**

HEART\_SLASH\_CIRCLE: IconData

**buildHEART\_SLASH\_CIRCLE\_FILLinstance-attribute**

HEART\_SLASH\_CIRCLE\_FILL: IconData

**buildHEART\_SLASH\_FILLinstance-attribute**

HEART\_SLASH\_FILL: IconData

**buildHEART\_SOLIDinstance-attribute**

HEART\_SOLID: IconData

**buildHELMinstance-attribute**

HELM: IconData

**buildHEXAGONinstance-attribute**

HEXAGON: IconData

**buildHEXAGON\_FILLinstance-attribute**

HEXAGON\_FILL: IconData

**buildHIFISPEAKERinstance-attribute**

HIFISPEAKER: IconData

**buildHIFISPEAKER\_FILLinstance-attribute**

HIFISPEAKER\_FILL: IconData

**buildHOMEinstance-attribute**

HOME: IconData

**buildHOURGLASSinstance-attribute**

HOURGLASS: IconData

**buildHOURGLASS\_BOTTOMHALF\_FILLinstance-attribute**

HOURGLASS\_BOTTOMHALF\_FILL: IconData

**buildHOURGLASS\_TOPHALF\_FILLinstance-attribute**

HOURGLASS\_TOPHALF\_FILL: IconData

**buildHOUSEinstance-attribute**

HOUSE: IconData

**buildHOUSE\_ALTinstance-attribute**

HOUSE\_ALT: IconData

**buildHOUSE\_ALT\_FILLinstance-attribute**

HOUSE\_ALT\_FILL: IconData

**buildHOUSE\_FILLinstance-attribute**

HOUSE\_FILL: IconData

**buildHURRICANEinstance-attribute**

HURRICANE: IconData

**buildINCREASE\_INDENTinstance-attribute**

INCREASE\_INDENT: IconData

**buildINCREASE\_QUOTELEVELinstance-attribute**

INCREASE\_QUOTELEVEL: IconData

**buildINFINITEinstance-attribute**

INFINITE: IconData

**buildINFOinstance-attribute**

INFO: IconData

**buildINFO\_CIRCLEinstance-attribute**

INFO\_CIRCLE: IconData

**buildINFO\_CIRCLE\_FILLinstance-attribute**

INFO\_CIRCLE\_FILL: IconData

**buildITALICinstance-attribute**

ITALIC: IconData

**buildKEYBOARDinstance-attribute**

KEYBOARD: IconData

**buildKEYBOARD\_CHEVRON\_COMPACT\_DOWNinstance-attribute**

KEYBOARD\_CHEVRON\_COMPACT\_DOWN: IconData

**buildLAB\_FLASKinstance-attribute**

LAB\_FLASK: IconData

**buildLAB\_FLASK\_SOLIDinstance-attribute**

LAB\_FLASK\_SOLID: IconData

**buildLARGECIRCLE\_FILL\_CIRCLEinstance-attribute**

LARGECIRCLE\_FILL\_CIRCLE: IconData

**buildLASSOinstance-attribute**

LASSO: IconData

**buildLAYERSinstance-attribute**

LAYERS: IconData

**buildLAYERS\_ALTinstance-attribute**

LAYERS\_ALT: IconData

**buildLAYERS\_ALT\_FILLinstance-attribute**

LAYERS\_ALT\_FILL: IconData

**buildLAYERS\_FILLinstance-attribute**

LAYERS\_FILL: IconData

**buildLEAF\_ARROW\_CIRCLEPATHinstance-attribute**

LEAF\_ARROW\_CIRCLEPATH: IconData

**buildLEFT\_CHEVRONinstance-attribute**

LEFT\_CHEVRON: IconData

**buildLESSTHANinstance-attribute**

LESSTHAN: IconData

**buildLESSTHAN\_CIRCLEinstance-attribute**

LESSTHAN\_CIRCLE: IconData

**buildLESSTHAN\_CIRCLE\_FILLinstance-attribute**

LESSTHAN\_CIRCLE\_FILL: IconData

**buildLESSTHAN\_SQUAREinstance-attribute**

LESSTHAN\_SQUARE: IconData

**buildLESSTHAN\_SQUARE\_FILLinstance-attribute**

LESSTHAN\_SQUARE\_FILL: IconData

**buildLIGHT\_MAXinstance-attribute**

LIGHT\_MAX: IconData

**buildLIGHT\_MINinstance-attribute**

LIGHT\_MIN: IconData

**buildLIGHTBULBinstance-attribute**

LIGHTBULB: IconData

**buildLIGHTBULB\_FILLinstance-attribute**

LIGHTBULB\_FILL: IconData

**buildLIGHTBULB\_SLASHinstance-attribute**

LIGHTBULB\_SLASH: IconData

**buildLIGHTBULB\_SLASH\_FILLinstance-attribute**

LIGHTBULB\_SLASH\_FILL: IconData

**buildLINE\_HORIZONTAL\_3instance-attribute**

LINE\_HORIZONTAL\_3: IconData

**buildLINE\_HORIZONTAL\_3\_DECREASEinstance-attribute**

LINE\_HORIZONTAL\_3\_DECREASE: IconData

**buildLINE\_HORIZONTAL\_3\_DECREASE\_CIRCLEinstance-attribute**

LINE\_HORIZONTAL\_3\_DECREASE\_CIRCLE: IconData

**buildLINE\_HORIZONTAL\_3\_DECREASE\_CIRCLE\_FILLinstance-attribute**

LINE\_HORIZONTAL\_3\_DECREASE\_CIRCLE\_FILL: IconData

**buildLINKinstance-attribute**

LINK: IconData

**buildLINK\_CIRCLEinstance-attribute**

LINK\_CIRCLE: IconData

**buildLINK\_CIRCLE\_FILLinstance-attribute**

LINK\_CIRCLE\_FILL: IconData

**buildLIST\_BULLETinstance-attribute**

LIST\_BULLET: IconData

**buildLIST\_BULLET\_BELOW\_RECTANGLEinstance-attribute**

LIST\_BULLET\_BELOW\_RECTANGLE: IconData

**buildLIST\_BULLET\_INDENTinstance-attribute**

LIST\_BULLET\_INDENT: IconData

**buildLIST\_DASHinstance-attribute**

LIST\_DASH: IconData

**buildLIST\_NUMBERinstance-attribute**

LIST\_NUMBER: IconData

**buildLIST\_NUMBER\_RTListance-attribute**

LIST\_NUMBER\_RTL: IconData

**buildLOCATIONinstance-attribute**

LOCATION: IconData

**buildLOCATION\_CIRCLEinstance-attribute**

LOCATION\_CIRCLE: IconData

**buildLOCATION\_CIRCLE\_FILLinstance-attribute**

LOCATION\_CIRCLE\_FILL: IconData

**buildLOCATION\_FILLinstance-attribute**

LOCATION\_FILL: IconData

**buildLOCATION\_NORTHinstance-attribute**

LOCATION\_NORTH: IconData

**buildLOCATION\_NORTH\_FILLinstance-attribute**

LOCATION\_NORTH\_FILL: IconData

**buildLOCATION\_NORTH\_LINEinstance-attribute**

LOCATION\_NORTH\_LINE: IconData

**buildLOCATION\_NORTH\_LINE\_FILLinstance-attribute**

LOCATION\_NORTH\_LINE\_FILL: IconData

**buildLOCATION\_SLASHinstance-attribute**

LOCATION\_SLASH: IconData

**buildLOCATION\_SLASH\_FILLinstance-attribute**

LOCATION\_SLASH\_FILL: IconData

**buildLOCATION\_SOLIDinstance-attribute**

LOCATION\_SOLID: IconData

**buildLOCKinstance-attribute**

LOCK: IconData

**buildLOCK\_CIRCLEinstance-attribute**

LOCK\_CIRCLE: IconData



**buildLOCK\_CIRCLE\_FILLinstance-attribute**

LOCK\_CIRCLE\_FILL: IconData

**buildLOCK\_FILLinstance-attribute**

LOCK\_FILL: IconData

**buildLOCK\_OPENinstance-attribute**

LOCK\_OPEN: IconData

**buildLOCK\_OPEN\_FILLinstance-attribute**

LOCK\_OPEN\_FILL: IconData

**buildLOCK\_ROTATIONinstance-attribute**

LOCK\_ROTATION: IconData

**buildLOCK\_ROTATION\_OPENinstance-attribute**

LOCK\_ROTATION\_OPEN: IconData

**buildLOCK\_SHIELDinstance-attribute**

LOCK\_SHIELD: IconData

**buildLOCK\_SHIELD\_FILLinstance-attribute**

LOCK\_SHIELD\_FILL: IconData

**buildLOCK\_SLASHinstance-attribute**

LOCK\_SLASH: IconData

**buildLOCK\_SLASH\_FILLinstance-attribute**

LOCK\_SLASH\_FILL: IconData

**buildLOOPinstance-attribute**

LOOP: IconData

**buildLOOP\_THICKinstance-attribute**

LOOP\_THICK: IconData

**buildMACWINDOWinstance-attribute**

MACWINDOW: IconData

**buildMAILinstance-attribute**

MAIL: IconData

**buildMAIL\_SOLIDinstance-attribute**

MAIL\_SOLID: IconData

**buildMAPinstance-attribute**

MAP: IconData

**buildMAP\_FILLinstance-attribute**

MAP\_FILL: IconData

**buildMAP\_PINinstance-attribute**

MAP\_PIN: IconData

**buildMAP\_PIN\_ELLIPSEinstance-attribute**

MAP\_PIN\_ELLIPSE: IconData

**buildMAP\_PIN\_SLASHinstance-attribute**

MAP\_PIN\_SLASH: IconData

**buildMEMORIESinstance-attribute**

MEMORIES: IconData

**buildMEMORIES\_BADGE\_MINUSinstance-attribute**

MEMORIES\_BADGE\_MINUS: IconData

**buildMEMORIES\_BADGE\_PLUSinstance-attribute**

MEMORIES\_BADGE\_PLUS: IconData

**buildMETRONOMEinstance-attribute**

METRONOME: IconData

**buildMICinstance-attribute**

MIC: IconData

**buildMIC\_CIRCLEinstance-attribute**

MIC\_CIRCLE: IconData

**buildMIC\_CIRCLE\_FILLinstance-attribute**

MIC\_CIRCLE\_FILL: IconData

**buildMIC\_FILLinstance-attribute**

MIC\_FILL: IconData

**buildMIC\_OFFinstance-attribute**

MIC\_OFF: IconData

**buildMIC\_SLASHinstance-attribute**

MIC\_SLASH: IconData

**buildMIC\_SLASH\_FILLinstance-attribute**

MIC\_SLASH\_FILL: IconData

**buildMIC\_SOLIDinstance-attribute**

MIC\_SOLID: IconData

**buildMINUSinstance-attribute**

MINUS: IconData

**buildMINUS\_CIRCLEinstance-attribute**

MINUS\_CIRCLE: IconData

**buildMINUS\_CIRCLE\_FILLinstance-attribute**

MINUS\_CIRCLE\_FILL: IconData

**buildMINUS\_CIRCLEDinstance-attribute**

MINUS\_CIRCLED: IconData

**buildMINUS\_RECTANGLEinstance-attribute**

MINUS\_RECTANGLE: IconData

**buildMINUS\_RECTANGLE\_FILLinstance-attribute**

MINUS\_RECTANGLE\_FILL: IconData

**buildMINUS\_SLASH\_PLUSinstance-attribute**

MINUS\_SLASH\_PLUS: IconData

**buildMINUS\_SQUAREinstance-attribute**

MINUS\_SQUARE: IconData

**buildMINUS\_SQUARE\_FILLinstance-attribute**

MINUS\_SQUARE\_FILL: IconData

**buildMONEY\_DOLLARinstance-attribute**

MONEY\_DOLLAR: IconData

**buildMONEY\_DOLLAR\_CIRCLEinstance-attribute**

MONEY\_DOLLAR\_CIRCLE: IconData

**buildMONEY\_DOLLAR\_CIRCLE\_FILLinstance-attribute**

MONEY\_DOLLAR\_CIRCLE\_FILL: IconData

**buildMONEY\_EUROinstance-attribute**

MONEY\_EURO: IconData

**buildMONEY\_EURO\_CIRCLEinstance-attribute**

MONEY\_EURO\_CIRCLE: IconData

**buildMONEY\_EURO\_CIRCLE\_FILLinstance-attribute**

MONEY\_EURO\_CIRCLE\_FILL: IconData

**buildMONEY\_POUNDinstance-attribute**

MONEY\_POUND: IconData

**buildMONEY\_POUND\_CIRCLEinstance-attribute**

MONEY\_POUND\_CIRCLE: IconData

**buildMONEY\_POUND\_CIRCLE\_FILLinstance-attribute**

MONEY\_POUND\_CIRCLE\_FILL: IconData

**buildMONEY\_RUBLinstance-attribute**

MONEY\_RUBL: IconData

**buildMONEY\_RUBL\_CIRCLEinstance-attribute**

MONEY\_RUBL\_CIRCLE: IconData

**buildMONEY\_RUBL\_CIRCLE\_FILLinstance-attribute**

MONEY\_RUBL\_CIRCLE\_FILL: IconData

**buildMONEY\_YENinstance-attribute**

MONEY\_YEN: IconData

**buildMONEY\_YEN\_CIRCLEinstance-attribute**

MONEY\_YEN\_CIRCLE: IconData

**buildMONEY\_YEN\_CIRCLE\_FILLinstance-attribute**

MONEY\_YEN\_CIRCLE\_FILL: IconData

**buildMOONinstance-attribute**

MOON: IconData

**buildMOON\_CIRCLEinstance-attribute**

MOON\_CIRCLE: IconData

**buildMOON\_CIRCLE\_FILLinstance-attribute**

MOON\_CIRCLE\_FILL: IconData

**buildMOON\_FILLinstance-attribute**

MOON\_FILL: IconData

**buildMOON\_STARSinstance-attribute**

MOON\_STARS: IconData

**buildMOON\_STARS\_FILLinstance-attribute**

MOON\_STARS\_FILL: IconData

**buildMOON\_ZZZinstance-attribute**

MOON\_ZZZ: IconData

**buildMOON\_ZZZ\_FILLinstance-attribute**

MOON\_ZZZ\_FILL: IconData

**buildMOVEinstance-attribute**

MOVE: IconData

**buildMULTIPLYinstance-attribute**

MULTIPLY: IconData

**buildMULTIPLY\_CIRCLEinstance-attribute**

MULTIPLY\_CIRCLE: IconData

**buildMULTIPLY\_CIRCLE\_FILLinstance-attribute**

MULTIPLY\_CIRCLE\_FILL: IconData

**buildMULTIPLY\_SQUAREinstance-attribute**

MULTIPLY\_SQUARE: IconData

**buildMULTIPLY\_SQUARE\_FILLinstance-attribute**

MULTIPLY\_SQUARE\_FILL: IconData

**buildMUSIC\_ALBUMSinstance-attribute**

MUSIC\_ALBUMS: IconData

**buildMUSIC\_ALBUMS\_FILLinstance-attribute**

MUSIC\_ALBUMS\_FILL: IconData

**buildMUSIC\_HOUSEinstance-attribute**

MUSIC\_HOUSE: IconData

**buildMUSIC\_HOUSE\_FILLinstance-attribute**

MUSIC\_HOUSE\_FILL: IconData

**buildMUSIC\_MICinstance-attribute**

MUSIC\_MIC: IconData

**buildMUSIC\_NOTEinstance-attribute**

MUSIC\_NOTE: IconData

**buildMUSIC\_NOTE\_2instance-attribute**

MUSIC\_NOTE\_2: IconData

**buildMUSIC\_NOTE\_LISTinstance-attribute**

MUSIC\_NOTE\_LIST: IconData

**buildNEWSinstance-attribute**

NEWS: IconData

**buildNEWS\_SOLIDinstance-attribute**

NEWS\_SOLID: IconData

**buildNOSIGNinstance-attribute**

NOSIGN: IconData

**buildNUMBERinstance-attribute**

NUMBER: IconData

**buildNUMBER\_CIRCLEinstance-attribute**

NUMBER\_CIRCLE: IconData

**buildNUMBER\_CIRCLE\_FILLinstance-attribute**

NUMBER\_CIRCLE\_FILL: IconData

**buildNUMBER\_SQUAREinstance-attribute**

NUMBER\_SQUARE: IconData

**buildNUMBER\_SQUARE\_FILLinstance-attribute**

NUMBER\_SQUARE\_FILL: IconData

**buildOPTIONinstance-attribute**

OPTION: IconData

**buildPADLOCKinstance-attribute**

PADLOCK: IconData

**buildPADLOCK\_SOLIDinstance-attribute**

PADLOCK\_SOLID: IconData

**buildPAINTBRUSHinstance-attribute**

PAINTBRUSH: IconData

**buildPAINTBRUSH\_FILLinstance-attribute**

PAINTBRUSH\_FILL: IconData

**buildPANOinstance-attribute**

PANO: IconData

**buildPANO\_FILLinstance-attribute**

PANO\_FILL: IconData

**buildPAPERCLIPinstance-attribute**

PAPERCLIP: IconData

**buildPAPERPLANEinstance-attribute**

PAPERPLANE: IconData

**buildPAPERPLANE\_FILLinstance-attribute**

PAPERPLANE\_FILL: IconData

**buildPARAGRAPHinstance-attribute**

PARAGRAPH: IconData

**buildPAUSEinstance-attribute**

PAUSE: IconData

**buildPAUSE\_CIRCLEinstance-attribute**

PAUSE\_CIRCLE: IconData

**buildPAUSE\_CIRCLE\_FILLinstance-attribute**

PAUSE\_CIRCLE\_FILL: IconData

**buildPAUSE\_FILLinstance-attribute**

PAUSE\_FILL: IconData

**buildPAUSE\_RECTANGLEinstance-attribute**

PAUSE\_RECTANGLE: IconData

**buildPAUSE\_RECTANGLE\_FILLinstance-attribute**

PAUSE\_RECTANGLE\_FILL: IconData

**buildPAUSE\_SOLIDinstance-attribute**

PAUSE\_SOLID: IconData

**buildPAWinstance-attribute**

PAW: IconData

**buildPAW\_SOLIDinstance-attribute**

PAW\_SOLID: IconData

**buildPENinstance-attribute**

PEN: IconData

**buildPENCILinstance-attribute**

PENCIL: IconData

**buildPENCIL\_CIRCLEinstance-attribute**

PENCIL\_CIRCLE: IconData

**buildPENCIL\_CIRCLE\_FILLinstance-attribute**

PENCIL\_CIRCLE\_FILL: IconData

buildPENCIL\_ELLIPSIS\_RECTANGLEinstance-attribute

PENCIL\_ELLIPSIS\_RECTANGLE: IconData

buildPENCIL\_OUTLINEinstance-attribute

PENCIL\_OUTLINE: IconData

buildPENCIL\_SLASHinstance-attribute

PENCIL\_SLASH: IconData

buildPERCENTinstance-attribute

PERCENT: IconData

buildPERSONinstance-attribute

PERSON: IconData

buildPERSON\_2instance-attribute

PERSON\_2: IconData

buildPERSON\_2\_ALTinstance-attribute

PERSON\_2\_ALT: IconData

buildPERSON\_2\_FILLinstance-attribute

PERSON\_2\_FILL: IconData

buildPERSON\_2\_SQUARE\_STACKinstance-attribute

PERSON\_2\_SQUARE\_STACK: IconData

buildPERSON\_2\_SQUARE\_STACK\_FILLinstance-attribute

PERSON\_2\_SQUARE\_STACK\_FILL: IconData

buildPERSON\_3instance-attribute

PERSON\_3: IconData

buildPERSON\_3\_FILLinstance-attribute

PERSON\_3\_FILL: IconData

buildPERSON\_ADDinstance-attribute

PERSON\_ADD: IconData

buildPERSON\_ADD\_SOLIDinstance-attribute

PERSON\_ADD\_SOLID: IconData

buildPERSON\_ALTinstance-attribute

PERSON\_ALT: IconData

buildPERSON\_ALT\_CIRCLEinstance-attribute

PERSON\_ALT\_CIRCLE: IconData

buildPERSON\_ALT\_CIRCLE\_FILLinstance-attribute

PERSON\_ALT\_CIRCLE\_FILL: IconData

buildPERSON\_BADGE\_MINUSinstance-attribute

PERSON\_BADGE\_MINUS: IconData

buildPERSON\_BADGE\_MINUS\_FILLinstance-attribute

PERSON\_BADGE\_MINUS\_FILL: IconData

buildPERSON\_BADGE\_PLUSinstance-attribute

PERSON\_BADGE\_PLUS: IconData

buildPERSON\_BADGE\_PLUS\_FILLinstance-attribute

PERSON\_BADGE\_PLUS\_FILL: IconData

buildPERSON\_CIRCLEinstance-attribute

PERSON\_CIRCLE: IconData

**buildPERSON\_CIRCLE\_FILLinstance-attribute**

PERSON\_CIRCLE\_FILL: IconData

**buildPERSON\_CROP\_CIRCLEinstance-attribute**

PERSON\_CROP\_CIRCLE: IconData

**buildPERSON\_CROP\_CIRCLE\_BADGE\_CHECKMARKinstance-attribute**

PERSON\_CROP\_CIRCLE\_BADGE\_CHECKMARK: IconData

**buildPERSON\_CROP\_CIRCLE\_BADGE\_EXCLAMinstance-attribute**

PERSON\_CROP\_CIRCLE\_BADGE\_EXCLAM: IconData

**buildPERSON\_CROP\_CIRCLE\_BADGE\_MINUSinstance-attribute**

PERSON\_CROP\_CIRCLE\_BADGE\_MINUS: IconData

**buildPERSON\_CROP\_CIRCLE\_BADGE\_PLUSinstance-attribute**

PERSON\_CROP\_CIRCLE\_BADGE\_PLUS: IconData

**buildPERSON\_CROP\_CIRCLE\_BADGE\_XMARKinstance-attribute**

PERSON\_CROP\_CIRCLE\_BADGE\_XMARK: IconData

**buildPERSON\_CROP\_CIRCLE\_FILLinstance-attribute**

PERSON\_CROP\_CIRCLE\_FILL: IconData

**buildPERSON\_CROP\_CIRCLE\_FILL\_BADGE\_CHECKMARKinstance-attribute**

PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_CHECKMARK: IconData

**buildPERSON\_CROP\_CIRCLE\_FILL\_BADGE\_EXCLAMinstance-attribute**

PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_EXCLAM: IconData

**buildPERSON\_CROP\_CIRCLE\_FILL\_BADGE\_MINUSinstance-attribute**

PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_MINUS: IconData

**buildPERSON\_CROP\_CIRCLE\_FILL\_BADGE\_PLUSinstance-attribute**

PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_PLUS: IconData

**buildPERSON\_CROP\_CIRCLE\_FILL\_BADGE\_XMARKinstance-attribute**

PERSON\_CROP\_CIRCLE\_FILL\_BADGE\_XMARK: IconData

**buildPERSON\_CROP\_RECTANGLEinstance-attribute**

PERSON\_CROP\_RECTANGLE: IconData

**buildPERSON\_CROP\_RECTANGLE\_FILLinstance-attribute**

PERSON\_CROP\_RECTANGLE\_FILL: IconData

**buildPERSON\_CROP\_SQUAREinstance-attribute**

PERSON\_CROP\_SQUARE: IconData

**buildPERSON\_CROP\_SQUARE\_FILLinstance-attribute**

PERSON\_CROP\_SQUARE\_FILL: IconData

**buildPERSON\_FILLinstance-attribute**

PERSON\_FILL: IconData

**buildPERSON\_SOLIDinstance-attribute**

PERSON\_SOLID: IconData

**buildPERSONALHOTSPOTinstance-attribute**

PERSONALHOTSPOT: IconData

**buildPERSPECTIVEinstance-attribute**

PERSPECTIVE: IconData

**buildPHONEinstance-attribute**

PHONE: IconData

**buildPHONE\_ARROW\_DOWN\_LEFTinstance-attribute**

PHONE\_ARROW\_DOWN\_LEFT: IconData

**buildPHONE\_ARROW\_RIGHTinstance-attribute**

PHONE\_ARROW\_RIGHT: IconData

**buildPHONE\_ARROW\_UP\_RIGHTinstance-attribute**

PHONE\_ARROW\_UP\_RIGHT: IconData

**buildPHONE\_BADGE\_PLUSinstance-attribute**

PHONE\_BADGE\_PLUS: IconData

**buildPHONE\_CIRCLEinstance-attribute**

PHONE\_CIRCLE: IconData

**buildPHONE\_CIRCLE\_FILLinstance-attribute**

PHONE\_CIRCLE\_FILL: IconData

**buildPHONE\_DOWNinstance-attribute**

PHONE\_DOWN: IconData

**buildPHONE\_DOWN\_CIRCLEinstance-attribute**

PHONE\_DOWN\_CIRCLE: IconData

**buildPHONE\_DOWN\_CIRCLE\_FILLinstance-attribute**

PHONE\_DOWN\_CIRCLE\_FILL: IconData

**buildPHONE\_DOWN\_FILLinstance-attribute**

PHONE\_DOWN\_FILL: IconData

**buildPHONE\_FILLinstance-attribute**

PHONE\_FILL: IconData

**buildPHONE\_FILL\_ARROW\_DOWN\_LEFTinstance-attribute**

PHONE\_FILL\_ARROW\_DOWN\_LEFT: IconData

**buildPHONE\_FILL\_ARROW\_RIGHTinstance-attribute**

PHONE\_FILL\_ARROW\_RIGHT: IconData

**buildPHONE\_FILL\_ARROW\_UP\_RIGHTinstance-attribute**

PHONE\_FILL\_ARROW\_UP\_RIGHT: IconData

**buildPHONE\_FILL\_BADGE\_PLUSinstance-attribute**

PHONE\_FILL\_BADGE\_PLUS: IconData

**buildPHONE\_SOLIDinstance-attribute**

PHONE\_SOLID: IconData

**buildPHOTOinstance-attribute**

PHOTO: IconData

**buildPHOTO\_CAMERAinstance-attribute**

PHOTO\_CAMERA: IconData

**buildPHOTO\_CAMERA\_SOLIDinstance-attribute**

PHOTO\_CAMERA\_SOLID: IconData

**buildPHOTO\_FILLinstance-attribute**

PHOTO\_FILL: IconData

**buildPHOTO\_FILL\_ON\_RECTANGLE\_FILLinstance-attribute**

PHOTO\_FILL\_ON\_RECTANGLE\_FILL: IconData

**buildPHOTO\_ON\_RECTANGLEinstance-attribute**

PHOTO\_ON\_RECTANGLE: IconData



**buildPIANOinstance-attribute**

PIANO: IconData

**buildPINinstance-attribute**

PIN: IconData

**buildPIN\_FILLinstance-attribute**

PIN\_FILL: IconData

**buildPIN\_SLASHinstance-attribute**

PIN\_SLASH: IconData

**buildPIN\_SLASH\_FILLinstance-attribute**

PIN\_SLASH\_FILL: IconData

**buildPLACEMARKinstance-attribute**

PLACEMARK: IconData

**buildPLACEMARK\_FILLinstance-attribute**

PLACEMARK\_FILL: IconData

**buildPLAYinstance-attribute**

PLAY: IconData

**buildPLAY\_ARROWinstance-attribute**

PLAY\_ARROW: IconData

**buildPLAY\_ARROW\_SOLIDinstance-attribute**

PLAY\_ARROW\_SOLID: IconData

**buildPLAY\_CIRCLEinstance-attribute**

PLAY\_CIRCLE: IconData

**buildPLAY\_CIRCLE\_FILLinstance-attribute**

PLAY\_CIRCLE\_FILL: IconData

**buildPLAY\_FILLinstance-attribute**

PLAY\_FILL: IconData

**buildPLAY\_RECTANGLEinstance-attribute**

PLAY\_RECTANGLE: IconData

**buildPLAY\_RECTANGLE\_FILLinstance-attribute**

PLAY\_RECTANGLE\_FILL: IconData

**buildPLAYPAUSEinstance-attribute**

PLAYPAUSE: IconData

**buildPLAYPAUSE\_FILLinstance-attribute**

PLAYPAUSE\_FILL: IconData

**buildPLUSinstance-attribute**

PLUS: IconData

**buildPLUS\_APPinstance-attribute**

PLUS\_APP: IconData

**buildPLUS\_APP\_FILLinstance-attribute**

PLUS\_APP\_FILL: IconData

**buildPLUS\_BUBBLEinstance-attribute**

PLUS\_BUBBLE: IconData

**buildPLUS\_BUBBLE\_FILLinstance-attribute**

PLUS\_BUBBLE\_FILL: IconData

**buildPLUS\_CIRCLEinstance-attribute**

PLUS\_CIRCLE: IconData

**buildPLUS\_CIRCLE\_FILLinstance-attribute**

PLUS\_CIRCLE\_FILL: IconData

**buildPLUS\_CIRCLEDinstance-attribute**

PLUS\_CIRCLED: IconData

**buildPLUS\_RECTANGLEinstance-attribute**

PLUS\_RECTANGLE: IconData

**buildPLUS\_RECTANGLE\_FILLinstance-attribute**

PLUS\_RECTANGLE\_FILL: IconData

**buildPLUS\_RECTANGLE\_FILL\_ON\_RECTANGLE\_FILLinstance-attribute**

PLUS\_RECTANGLE\_FILL\_ON\_RECTANGLE\_FILL: IconData

**buildPLUS\_RECTANGLE\_ON\_RECTANGLEinstance-attribute**

PLUS\_RECTANGLE\_ON\_RECTANGLE: IconData

**buildPLUS\_SLASH\_MINUSinstance-attribute**

PLUS\_SLASH\_MINUS: IconData

**buildPLUS\_SQUAREinstance-attribute**

PLUS\_SQUARE: IconData

**buildPLUS\_SQUARE\_FILLinstance-attribute**

PLUS\_SQUARE\_FILL: IconData

**buildPLUS\_SQUARE\_FILL\_ON\_SQUARE\_FILLinstance-attribute**

PLUS\_SQUARE\_FILL\_ON\_SQUARE\_FILL: IconData

**buildPLUS\_SQUARE\_ON\_SQUAREinstance-attribute**

PLUS\_SQUARE\_ON\_SQUARE: IconData

**buildPLUSMINUSinstance-attribute**

PLUSMINUS: IconData

**buildPLUSMINUS\_CIRCLEinstance-attribute**

PLUSMINUS\_CIRCLE: IconData

**buildPLUSMINUS\_CIRCLE\_FILLinstance-attribute**

PLUSMINUS\_CIRCLE\_FILL: IconData

**buildPOWERinstance-attribute**

POWER: IconData

**buildPRINTERinstance-attribute**

PRINTER: IconData

**buildPRINTER\_FILLinstance-attribute**

PRINTER\_FILL: IconData

**buildPROFILE\_CIRCLEDinstance-attribute**

PROFILE\_CIRCLED: IconData

**buildPROJECTIVEinstance-attribute**

PROJECTIVE: IconData

**buildPURCHASEDinstance-attribute**

PURCHASED: IconData

**buildPURCHASED\_CIRCLEinstance-attribute**

PURCHASED\_CIRCLE: IconData

**buildPURCHASED\_CIRCLE\_FILLinstance-attribute**

PURCHASED\_CIRCLE\_FILL: IconData

**buildQRCODEinstance-attribute**

QRCODE: IconData

**buildQRCODE\_VIEWFINDERinstance-attribute**

QRCODE\_VIEWFINDER: IconData

**buildQUESTIONinstance-attribute**

QUESTION: IconData

**buildQUESTION\_CIRCLEinstance-attribute**

QUESTION\_CIRCLE: IconData

**buildQUESTION\_CIRCLE\_FILLinstance-attribute**

QUESTION\_CIRCLE\_FILL: IconData

**buildQUESTION\_DIAMONDinstance-attribute**

QUESTION\_DIAMOND: IconData

**buildQUESTION\_DIAMOND\_FILLinstance-attribute**

QUESTION\_DIAMOND\_FILL: IconData

**buildQUESTION\_SQUAREinstance-attribute**

QUESTION\_SQUARE: IconData

**buildQUESTION\_SQUARE\_FILLinstance-attribute**

QUESTION\_SQUARE\_FILL: IconData

**buildQUOTE\_BUBBLEinstance-attribute**

QUOTE\_BUBBLE: IconData

**buildQUOTE\_BUBBLE\_FILLinstance-attribute**

QUOTE\_BUBBLE\_FILL: IconData

**buildRADIOWAVES\_LEFTinstance-attribute**

RADIOWAVES\_LEFT: IconData

**buildRADIOWAVES\_RIGHTinstance-attribute**

RADIOWAVES\_RIGHT: IconData

**buildRAYSinstance-attribute**

RAYS: IconData

**buildRECORDINGTAPEinstance-attribute**

RECORDINGTAPE: IconData

**buildRECTANGLEinstance-attribute**

RECTANGLE: IconData

**buildRECTANGLE\_3\_OFFGRIDinstance-attribute**

RECTANGLE\_3\_OFFGRID: IconData

**buildRECTANGLE\_3\_OFFGRID\_FILLinstance-attribute**

RECTANGLE\_3\_OFFGRID\_FILL: IconData

**buildRECTANGLE\_ARROW\_UP\_RIGHT\_ARROW\_DOWN\_LEFTinstance-attribute**

RECTANGLE\_ARROW\_UP\_RIGHT\_ARROW\_DOWN\_LEFT: IconData

**buildRECTANGLE\_ARROW\_UP\_RIGHT\_ARROW\_DOWN\_LEFT\_SLASHinstance-attribute**

RECTANGLE\_ARROW\_UP\_RIGHT\_ARROW\_DOWN\_LEFT\_SLASH: IconData

**buildRECTANGLE\_BADGE\_CHECKMARKinstance-attribute**

RECTANGLE\_BADGE\_CHECKMARK: IconData

**buildRECTANGLE\_BADGE\_XMARKinstance-attribute**

RECTANGLE\_BADGE\_XMARK: IconData

**buildRECTANGLE\_COMPRESS\_VERTICALinstance-attribute**

RECTANGLE\_COMPRESS\_VERTICAL: IconData

**buildRECTANGLE\_DOCKinstance-attribute**

RECTANGLE\_DOCK: IconData

**buildRECTANGLE\_EXPAND\_VERTICALinstance-attribute**

RECTANGLE\_EXPAND\_VERTICAL: IconData

**buildRECTANGLE\_FILLinstance-attribute**

RECTANGLE\_FILL: IconData

**buildRECTANGLE\_FILL\_BADGE\_CHECKMARKinstance-attribute**

RECTANGLE\_FILL\_BADGE\_CHECKMARK: IconData

**buildRECTANGLE\_FILL\_BADGE\_XMARKinstance-attribute**

RECTANGLE\_FILL\_BADGE\_XMARK: IconData

**buildRECTANGLE\_FILL\_ON\_RECTANGLE\_ANGLED\_FILLinstance-attribute**

RECTANGLE\_FILL\_ON\_RECTANGLE\_ANGLED\_FILL: IconData

**buildRECTANGLE\_FILL\_ON\_RECTANGLE\_FILLinstance-attribute**

RECTANGLE\_FILL\_ON\_RECTANGLE\_FILL: IconData

**buildRECTANGLE\_GRID\_1X2instance-attribute**

RECTANGLE\_GRID\_1X2: IconData

**buildRECTANGLE\_GRID\_1X2\_FILLinstance-attribute**

RECTANGLE\_GRID\_1X2\_FILL: IconData

**buildRECTANGLE\_GRID\_2X2instance-attribute**

RECTANGLE\_GRID\_2X2: IconData

**buildRECTANGLE\_GRID\_2X2\_FILLinstance-attribute**

RECTANGLE\_GRID\_2X2\_FILL: IconData

**buildRECTANGLE\_GRID\_3X2instance-attribute**

RECTANGLE\_GRID\_3X2: IconData

**buildRECTANGLE\_GRID\_3X2\_FILLinstance-attribute**

RECTANGLE\_GRID\_3X2\_FILL: IconData

**buildRECTANGLE\_ON\_RECTANGLEinstance-attribute**

RECTANGLE\_ON\_RECTANGLE: IconData

**buildRECTANGLE\_ON\_RECTANGLE\_ANGLEDinstance-attribute**

RECTANGLE\_ON\_RECTANGLE\_ANGLED: IconData

**buildRECTANGLE\_PAPERCLIPinstance-attribute**

RECTANGLE\_PAPERCLIP: IconData

**buildRECTANGLE\_SPLIT\_3X1instance-attribute**

RECTANGLE\_SPLIT\_3X1: IconData

**buildRECTANGLE\_SPLIT\_3X1\_FILLinstance-attribute**

RECTANGLE\_SPLIT\_3X1\_FILL: IconData

**buildRECTANGLE\_SPLIT\_3X3instance-attribute**

RECTANGLE\_SPLIT\_3X3: IconData

**buildRECTANGLE\_SPLIT\_3X3\_FILLinstance-attribute**

RECTANGLE\_SPLIT\_3X3\_FILL: IconData

**buildRECTANGLE\_STACK**instance-attribute

RECTANGLE\_STACK: IconData

**buildRECTANGLE\_STACK\_BADGE\_MINUS**instance-attribute

RECTANGLE\_STACK\_BADGE\_MINUS: IconData

**buildRECTANGLE\_STACK\_BADGE\_PERSON\_CROP**instance-attribute

RECTANGLE\_STACK\_BADGE\_PERSON\_CROP: IconData

**buildRECTANGLE\_STACK\_BADGE\_PLUS**instance-attribute

RECTANGLE\_STACK\_BADGE\_PLUS: IconData

**buildRECTANGLE\_STACK\_FILL**instance-attribute

RECTANGLE\_STACK\_FILL: IconData

**buildRECTANGLE\_STACK\_FILL\_BADGE\_MINUS**instance-attribute

RECTANGLE\_STACK\_FILL\_BADGE\_MINUS: IconData

**buildRECTANGLE\_STACK\_FILL\_BADGE\_PERSON\_CROP**instance-attribute

RECTANGLE\_STACK\_FILL\_BADGE\_PERSON\_CROP: IconData

**buildRECTANGLE\_STACK\_FILL\_BADGE\_PLUS**instance-attribute

RECTANGLE\_STACK\_FILL\_BADGE\_PLUS: IconData

**buildRECTANGLE\_STACK\_PERSON\_CROP**instance-attribute

RECTANGLE\_STACK\_PERSON\_CROP: IconData

**buildRECTANGLE\_STACK\_PERSON\_CROP\_FILL**instance-attribute

RECTANGLE\_STACK\_PERSON\_CROP\_FILL: IconData

**buildREFRESH**instance-attribute

REFRESH: IconData

**buildREFRESH\_BOLD**instance-attribute

REFRESH\_BOLD: IconData

**buildREFRESH\_CIRCLED**instance-attribute

REFRESH\_CIRCLED: IconData

**buildREFRESH\_CIRCLED\_SOLID**instance-attribute

REFRESH\_CIRCLED\_SOLID: IconData

**buildREFRESH\_THICK**instance-attribute

REFRESH\_THICK: IconData

**buildREFRESH\_THIN**instance-attribute

REFRESH\_THIN: IconData

**buildREPEAT**instance-attribute

REPEAT: IconData

**buildREPEAT\_1**instance-attribute

REPEAT\_1: IconData

**buildREPLY**instance-attribute

REPLY: IconData

**buildREPLY\_ALL**instance-attribute

REPLY\_ALL: IconData

**buildREPLY\_THICK\_SOLID**instance-attribute

REPLY\_THICK\_SOLID: IconData

**buildRESIZE**instance-attribute

RESIZE: IconData

**buildRESIZE\_Hinstance-attribute**

RESIZE\_H: IconData

**buildRESIZE\_Vinstance-attribute**

RESIZE\_V: IconData

**buildRESTARTinstance-attribute**

RESTART: IconData

**buildRETURN\_ICONinstance-attribute**

RETURN\_ICON: IconData

**buildRHOMBUSinstance-attribute**

RHOMBUS: IconData

**buildRHOMBUS\_FILLinstance-attribute**

RHOMBUS\_FILL: IconData

**buildRIGHT\_CHEVRONinstance-attribute**

RIGHT\_CHEVRON: IconData

**buildROCKETinstance-attribute**

ROCKET: IconData

**buildROCKET\_FILLinstance-attribute**

ROCKET\_FILL: IconData

**buildROSETTEinstance-attribute**

ROSETTE: IconData

**buildROTATE\_LEFTinstance-attribute**

ROTATE\_LEFT: IconData

**buildROTATE\_LEFT\_FILLinstance-attribute**

ROTATE\_LEFT\_FILL: IconData

**buildROTATE\_RIGHTinstance-attribute**

ROTATE\_RIGHT: IconData

**buildROTATE\_RIGHT\_FILLinstance-attribute**

ROTATE\_RIGHT\_FILL: IconData

**buildSCISSORSinstance-attribute**

SCISSORS: IconData

**buildSCISSORS\_ALTinstance-attribute**

SCISSORS\_ALT: IconData

**buildSCOPEinstance-attribute**

SCOPE: IconData

**buildSCRIBBLEinstance-attribute**

SCRIBBLE: IconData

**buildSEARCHinstance-attribute**

SEARCH: IconData

**buildSEARCH\_CIRCLEinstance-attribute**

SEARCH\_CIRCLE: IconData

**buildSEARCH\_CIRCLE\_FILLinstance-attribute**

SEARCH\_CIRCLE\_FILL: IconData

**buildSELECTION\_PIN\_IN\_OUTinstance-attribute**

SELECTION\_PIN\_IN\_OUT: IconData

**buildSETTINGSinstance-attribute**

SETTINGS: IconData

**buildSETTINGS\_SOLIDinstance-attribute**

SETTINGS\_SOLID: IconData

**buildSHAREinstance-attribute**

SHARE: IconData

**buildSHARE\_SOLIDinstance-attribute**

SHARE\_SOLID: IconData

**buildSHARE\_UPinstance-attribute**

SHARE\_UP: IconData

**buildSHIELDinstance-attribute**

SHIELD: IconData

**buildSHIELD\_FILLinstance-attribute**

SHIELD\_FILL: IconData

**buildSHIELD\_LEFTHALF\_FILLinstance-attribute**

SHIELD\_LEFTHALF\_FILL: IconData

**buildSHIELD\_SLASHinstance-attribute**

SHIELD\_SLASH: IconData

**buildSHIELD\_SLASH\_FILLinstance-attribute**

SHIELD\_SLASH\_FILL: IconData

**buildSHIFTinstance-attribute**

SHIFT: IconData

**buildSHIFT\_FILLinstance-attribute**

SHIFT\_FILL: IconData

**buildSHOPPING\_CARTinstance-attribute**

SHOPPING\_CART: IconData

**buildSHUFFLEinstance-attribute**

SHUFFLE: IconData

**buildSHUFFLE\_MEDIUMinstance-attribute**

SHUFFLE\_MEDIUM: IconData

**buildSHUFFLE\_THICKinstance-attribute**

SHUFFLE\_THICK: IconData

**buildSIDEBAR\_LEFTinstance-attribute**

SIDEBAR\_LEFT: IconData

**buildSIDEBAR\_RIGHTinstance-attribute**

SIDEBAR\_RIGHT: IconData

**buildSIGNATUREinstance-attribute**

SIGNATURE: IconData

**buildSKEWinstance-attribute**

SKEW: IconData

**buildSLASH\_CIRCLEinstance-attribute**

SLASH\_CIRCLE: IconData

**buildSLASH\_CIRCLE\_FILLinstance-attribute**

SLASH\_CIRCLE\_FILL: IconData

**buildSLIDER\_HORIZONTAL\_3instance-attribute**

SLIDER\_HORIZONTAL\_3: IconData

**buildSLIDER\_HORIZONTAL\_BELOW\_RECTANGLEinstance-attribute**

SLIDER\_HORIZONTAL\_BELOW\_RECTANGLE: IconData

**buildSLOWMOinstance-attribute**

SLOWMO: IconData

**buildSMALLCIRCLE\_CIRCLEinstance-attribute**

SMALLCIRCLE\_CIRCLE: IconData

**buildSMALLCIRCLE\_CIRCLE\_FILLinstance-attribute**

SMALLCIRCLE\_CIRCLE\_FILL: IconData

**buildSMALLCIRCLE\_FILL\_CIRCLEinstance-attribute**

SMALLCIRCLE\_FILL\_CIRCLE: IconData

**buildSMALLCIRCLE\_FILL\_CIRCLE\_FILLinstance-attribute**

SMALLCIRCLE\_FILL\_CIRCLE\_FILL: IconData

**buildSMILEYinstance-attribute**

SMILEY: IconData

**buildSMILEY\_FILLinstance-attribute**

SMILEY\_FILL: IconData

**buildSMOKEinstance-attribute**

SMOKE: IconData

**buildSMOKE\_FILLinstance-attribute**

SMOKE\_FILL: IconData

**buildSNOWinstance-attribute**

SNOW: IconData

**buildSORT\_DOWNinstance-attribute**

SORT\_DOWN: IconData

**buildSORT\_DOWN\_CIRCLEinstance-attribute**

SORT\_DOWN\_CIRCLE: IconData

**buildSORT\_DOWN\_CIRCLE\_FILLinstance-attribute**

SORT\_DOWN\_CIRCLE\_FILL: IconData

**buildSORT\_UPinstance-attribute**

SORT\_UP: IconData

**buildSORT\_UP\_CIRCLEinstance-attribute**

SORT\_UP\_CIRCLE: IconData

**buildSORT\_UP\_CIRCLE\_FILLinstance-attribute**

SORT\_UP\_CIRCLE\_FILL: IconData

**buildSPARKLESinstance-attribute**

SPARKLES: IconData

**buildSPEAKERinstance-attribute**

SPEAKER: IconData

**buildSPEAKER\_1instance-attribute**

SPEAKER\_1: IconData

**buildSPEAKER\_1\_FILLinstance-attribute**

SPEAKER\_1\_FILL: IconData



**buildSPEAKER\_2instance-attribute**

SPEAKER\_2: IconData

**buildSPEAKER\_2\_FILLinstance-attribute**

SPEAKER\_2\_FILL: IconData

**buildSPEAKER\_3instance-attribute**

SPEAKER\_3: IconData

**buildSPEAKER\_3\_FILLinstance-attribute**

SPEAKER\_3\_FILL: IconData

**buildSPEAKER\_FILLinstance-attribute**

SPEAKER\_FILL: IconData

**buildSPEAKER\_SLASHinstance-attribute**

SPEAKER\_SLASH: IconData

**buildSPEAKER\_SLASH\_FILLinstance-attribute**

SPEAKER\_SLASH\_FILL: IconData

**buildSPEAKER\_SLASH\_FILL\_RTInstance-attribute**

SPEAKER\_SLASH\_FILL\_RTL: IconData

**buildSPEAKER\_SLASH\_RTInstance-attribute**

SPEAKER\_SLASH\_RTL: IconData

**buildSPEAKER\_ZZZinstance-attribute**

SPEAKER\_ZZZ: IconData

**buildSPEAKER\_ZZZ\_FILLinstance-attribute**

SPEAKER\_ZZZ\_FILL: IconData

**buildSPEAKER\_ZZZ\_FILL\_RTInstance-attribute**

SPEAKER\_ZZZ\_FILL\_RTL: IconData

**buildSPEAKER\_ZZZ\_RTInstance-attribute**

SPEAKER\_ZZZ\_RTL: IconData

**buildSPEEDOMETERinstance-attribute**

SPEEDOMETER: IconData

**buildSPORTSCOURTinstance-attribute**

SPORTSCOURT: IconData

**buildSPORTSCOURT\_FILLinstance-attribute**

SPORTSCOURT\_FILL: IconData

**buildSQUAREinstance-attribute**

SQUARE: IconData

**buildSQUARE\_ARROW\_DOWNinstance-attribute**

SQUARE\_ARROW\_DOWN: IconData

**buildSQUARE\_ARROW\_DOWN\_FILLinstance-attribute**

SQUARE\_ARROW\_DOWN\_FILL: IconData

**buildSQUARE\_ARROW\_DOWN\_ON\_SQUAREinstance-attribute**

SQUARE\_ARROW\_DOWN\_ON\_SQUARE: IconData

**buildSQUARE\_ARROW\_DOWN\_ON\_SQUARE\_FILLinstance-attribute**

SQUARE\_ARROW\_DOWN\_ON\_SQUARE\_FILL: IconData

**buildSQUARE\_ARROW\_LEFTinstance-attribute**

SQUARE\_ARROW\_LEFT: IconData

**buildSQUARE\_ARROW\_LEFT\_FILLinstance-attribute**

SQUARE\_ARROW\_LEFT\_FILL: IconData

**buildSQUARE\_ARROW\_RIGHTinstance-attribute**

SQUARE\_ARROW\_RIGHT: IconData

**buildSQUARE\_ARROW\_RIGHT\_FILLinstance-attribute**

SQUARE\_ARROW\_RIGHT\_FILL: IconData

**buildSQUARE\_ARROW\_UPinstance-attribute**

SQUARE\_ARROW\_UP: IconData

**buildSQUARE\_ARROW\_UP\_FILLinstance-attribute**

SQUARE\_ARROW\_UP\_FILL: IconData

**buildSQUARE\_ARROW\_UP\_ON\_SQUAREinstance-attribute**

SQUARE\_ARROW\_UP\_ON\_SQUARE: IconData

**buildSQUARE\_ARROW\_UP\_ON\_SQUARE\_FILLinstance-attribute**

SQUARE\_ARROW\_UP\_ON\_SQUARE\_FILL: IconData

**buildSQUARE\_FAVORITESinstance-attribute**

SQUARE\_FAVORITES: IconData

**buildSQUARE\_FAVORITES\_ALTinstance-attribute**

SQUARE\_FAVORITES\_ALT: IconData

**buildSQUARE\_FAVORITES\_ALT\_FILLinstance-attribute**

SQUARE\_FAVORITES\_ALT\_FILL: IconData

**buildSQUARE\_FAVORITES\_FILLinstance-attribute**

SQUARE\_FAVORITES\_FILL: IconData

**buildSQUARE\_FILLinstance-attribute**

SQUARE\_FILL: IconData

**buildSQUARE\_FILL\_LINE\_VERTICAL\_SQUAREinstance-attribute**

SQUARE\_FILL\_LINE\_VERTICAL\_SQUARE: IconData

**buildSQUARE\_FILL\_LINE\_VERTICAL\_SQUARE\_FILLinstance-attribute**

SQUARE\_FILL\_LINE\_VERTICAL\_SQUARE\_FILL: IconData

**buildSQUARE\_FILL\_ON\_CIRCLE\_FILLinstance-attribute**

SQUARE\_FILL\_ON\_CIRCLE\_FILL: IconData

**buildSQUARE\_FILL\_ON\_SQUARE\_FILLinstance-attribute**

SQUARE\_FILL\_ON\_SQUARE\_FILL: IconData

**buildSQUARE\_GRID\_2X2instance-attribute**

SQUARE\_GRID\_2X2: IconData

**buildSQUARE\_GRID\_2X2\_FILLinstance-attribute**

SQUARE\_GRID\_2X2\_FILL: IconData

**buildSQUARE\_GRID\_3X2instance-attribute**

SQUARE\_GRID\_3X2: IconData

**buildSQUARE\_GRID\_3X2\_FILLinstance-attribute**

SQUARE\_GRID\_3X2\_FILL: IconData

**buildSQUARE\_GRID\_4X3\_FILLinstance-attribute**

SQUARE\_GRID\_4X3\_FILL: IconData

**buildSQUARE\_LEFTHALF\_FILLinstance-attribute**

SQUARE\_LEFTHALF\_FILL: IconData

**buildSQUARE\_LINE\_VERTICAL\_SQUAREinstance-attribute**

SQUARE\_LINE\_VERTICAL\_SQUARE: IconData

**buildSQUARE\_LINE\_VERTICAL\_SQUARE\_FILLinstance-attribute**

SQUARE\_LINE\_VERTICAL\_SQUARE\_FILL: IconData

**buildSQUARE\_LISTinstance-attribute**

SQUARE\_LIST: IconData

**buildSQUARE\_LIST\_FILLinstance-attribute**

SQUARE\_LIST\_FILL: IconData

**buildSQUARE\_ON\_CIRCLEinstance-attribute**

SQUARE\_ON\_CIRCLE: IconData

**buildSQUARE\_ON\_SQUAREinstance-attribute**

SQUARE\_ON\_SQUARE: IconData

**buildSQUARE\_PENCILinstance-attribute**

SQUARE\_PENCIL: IconData

**buildSQUARE\_PENCIL\_FILLinstance-attribute**

SQUARE\_PENCIL\_FILL: IconData

**buildSQUARE\_RIGHTHALF\_FILLinstance-attribute**

SQUARE\_RIGHTHALF\_FILL: IconData

**buildSQUARE\_SPLIT\_1X2instance-attribute**

SQUARE\_SPLIT\_1X2: IconData

**buildSQUARE\_SPLIT\_1X2\_FILLinstance-attribute**

SQUARE\_SPLIT\_1X2\_FILL: IconData

**buildSQUARE\_SPLIT\_2X1instance-attribute**

SQUARE\_SPLIT\_2X1: IconData

**buildSQUARE\_SPLIT\_2X1\_FILLinstance-attribute**

SQUARE\_SPLIT\_2X1\_FILL: IconData

**buildSQUARE\_SPLIT\_2X2instance-attribute**

SQUARE\_SPLIT\_2X2: IconData

**buildSQUARE\_SPLIT\_2X2\_FILLinstance-attribute**

SQUARE\_SPLIT\_2X2\_FILL: IconData

**buildSQUARE\_STACKinstance-attribute**

SQUARE\_STACK: IconData

**buildSQUARE\_STACK\_3D\_DOWN\_DOTTEDLINEinstance-attribute**

SQUARE\_STACK\_3D\_DOWN\_DOTTEDLINE: IconData

**buildSQUARE\_STACK\_3D\_DOWN\_RIGHTinstance-attribute**

SQUARE\_STACK\_3D\_DOWN\_RIGHT: IconData

**buildSQUARE\_STACK\_3D\_DOWN\_RIGHT\_FILLinstance-attribute**

SQUARE\_STACK\_3D\_DOWN\_RIGHT\_FILL: IconData

**buildSQUARE\_STACK\_3D\_UPinstance-attribute**

SQUARE\_STACK\_3D\_UP: IconData

**buildSQUARE\_STACK\_3D\_UP\_FILLinstance-attribute**

SQUARE\_STACK\_3D\_UP\_FILL: IconData

**buildSQUARE\_STACK\_3D\_UP\_SLASHinstance-attribute**

SQUARE\_STACK\_3D\_UP\_SLASH: IconData

**buildSQUARE\_STACK\_3D\_UP\_SLASH\_FILLinstance-attribute**

SQUARE\_STACK\_3D\_UP\_SLASH\_FILL: IconData

**buildSQUARE\_STACK\_FILLinstance-attribute**

SQUARE\_STACK\_FILL: IconData

**buildSQUARES\_BELOW\_RECTANGLEinstance-attribute**

SQUARES\_BELOW\_RECTANGLE: IconData

**buildSTARinstance-attribute**

STAR: IconData

**buildSTAR\_CIRCLEinstance-attribute**

STAR\_CIRCLE: IconData

**buildSTAR\_CIRCLE\_FILLinstance-attribute**

STAR\_CIRCLE\_FILL: IconData

**buildSTAR\_FILLinstance-attribute**

STAR\_FILL: IconData

**buildSTAR\_LEFTHALF\_FILLinstance-attribute**

STAR\_LEFTHALF\_FILL: IconData

**buildSTAR\_SLASHinstance-attribute**

STAR\_SLASH: IconData

**buildSTAR\_SLASH\_FILLinstance-attribute**

STAR\_SLASH\_FILL: IconData

**buildSTAROFLIFEinstance-attribute**

STAROFLIFE: IconData

**buildSTAROFLIFE\_FILLinstance-attribute**

STAROFLIFE\_FILL: IconData

**buildSTOPinstance-attribute**

STOP: IconData

**buildSTOP\_CIRCLEinstance-attribute**

STOP\_CIRCLE: IconData

**buildSTOP\_CIRCLE\_FILLinstance-attribute**

STOP\_CIRCLE\_FILL: IconData

**buildSTOP\_FILLinstance-attribute**

STOP\_FILL: IconData

**buildSTOPWATCHinstance-attribute**

STOPWATCH: IconData

**buildSTOPWATCH\_FILLinstance-attribute**

STOPWATCH\_FILL: IconData

**buildSTRIKETHROUGHinstance-attribute**

STRIKETHROUGH: IconData

**buildSUIT\_CLUBinstance-attribute**

SUIT\_CLUB: IconData

**buildSUIT\_CLUB\_FILLinstance-attribute**

SUIT\_CLUB\_FILL: IconData

**buildSUIT\_DIAMONDinstance-attribute**

SUIT\_DIAMOND: IconData

**buildSUIT\_DIAMOND\_FILLinstance-attribute**

SUIT\_DIAMOND\_FILL: IconData

**buildSUIT\_HEARTinstance-attribute**

SUIT\_HEART: IconData

**buildSUIT\_HEART\_FILLinstance-attribute**

SUIT\_HEART\_FILL: IconData

**buildSUIT\_SPADEinstance-attribute**

SUIT\_SPADE: IconData

**buildSUIT\_SPADE\_FILLinstance-attribute**

SUIT\_SPADE\_FILL: IconData

**buildSUMinstance-attribute**

SUM: IconData

**buildSUN\_DUSTinstance-attribute**

SUN\_DUST: IconData

**buildSUN\_DUST\_FILLinstance-attribute**

SUN\_DUST\_FILL: IconData

**buildSUN\_HAZEinstance-attribute**

SUN\_HAZE: IconData

**buildSUN\_HAZE\_FILLinstance-attribute**

SUN\_HAZE\_FILL: IconData

**buildSUN\_MAXinstance-attribute**

SUN\_MAX: IconData

**buildSUN\_MAX\_FILLinstance-attribute**

SUN\_MAX\_FILL: IconData

**buildSUN\_MINinstance-attribute**

SUN\_MIN: IconData

**buildSUN\_MIN\_FILLinstance-attribute**

SUN\_MIN\_FILL: IconData

**buildSUNRISEinstance-attribute**

SUNRISE: IconData

**buildSUNRISE\_FILLinstance-attribute**

SUNRISE\_FILL: IconData

**buildSUNSETinstance-attribute**

SUNSET: IconData

**buildSUNSET\_FILLinstance-attribute**

SUNSET\_FILL: IconData

**buildSWITCH\_CAMERAinstance-attribute**

SWITCH\_CAMERA: IconData

**buildSWITCH\_CAMERA\_SOLIDinstance-attribute**

SWITCH\_CAMERA\_SOLID: IconData

**buildT\_BUBBLEinstance-attribute**

T\_BUBBLE: IconData

**buildT\_BUBBLE\_FILLinstance-attribute**

T\_BUBBLE\_FILL: IconData

**buildTABLEinstance-attribute**

TABLE: IconData

**buildTABLE\_BADGE\_MOREinstance-attribute**

TABLE\_BADGE\_MORE: IconData

**buildTABLE\_BADGE\_MORE\_FILLinstance-attribute**

TABLE\_BADGE\_MORE\_FILL: IconData

**buildTABLE\_FILLinstance-attribute**

TABLE\_FILL: IconData

**buildTAGinstance-attribute**

TAG: IconData

**buildTAG\_CIRCLEinstance-attribute**

TAG\_CIRCLE: IconData

**buildTAG\_CIRCLE\_FILLinstance-attribute**

TAG\_CIRCLE\_FILL: IconData

**buildTAG\_FILLinstance-attribute**

TAG\_FILL: IconData

**buildTAG\_SOLIDinstance-attribute**

TAG\_SOLID: IconData

**buildTAGSinstance-attribute**

TAGS: IconData

**buildTAGS\_SOLIDinstance-attribute**

TAGS\_SOLID: IconData

**buildTEXT\_ALIGNCENTERinstance-attribute**

TEXT\_ALIGNCENTER: IconData

**buildTEXT\_ALIGNLEFTinstance-attribute**

TEXT\_ALIGNLEFT: IconData

**buildTEXT\_ALIGNRIGHTinstance-attribute**

TEXT\_ALIGNRIGHT: IconData

**buildTEXT\_APPENDinstance-attribute**

TEXT\_APPEND: IconData

**buildTEXT\_BADGE\_CHECKMARKinstance-attribute**

TEXT\_BADGE\_CHECKMARK: IconData

**buildTEXT\_BADGE\_MINUSinstance-attribute**

TEXT\_BADGE\_MINUS: IconData

**buildTEXT\_BADGE\_PLUSinstance-attribute**

TEXT\_BADGE\_PLUS: IconData

**buildTEXT\_BADGE\_STARinstance-attribute**

TEXT\_BADGE\_STAR: IconData

**buildTEXT\_BADGE\_XMARKinstance-attribute**

TEXT\_BADGE\_XMARK: IconData

**buildTEXT\_BUBBLEinstance-attribute**

TEXT\_BUBBLE: IconData

**buildTEXT\_BUBBLE\_FILLinstance-attribute**

TEXT\_BUBBLE\_FILL: IconData

**buildTEXT\_CURSORinstance-attribute**

TEXT\_CURSOR: IconData

**buildTEXT\_INSERTinstance-attribute**

TEXT\_INSERT: IconData

**buildTEXT\_JUSTIFYinstance-attribute**

TEXT\_JUSTIFY: IconData

**buildTEXT\_JUSTIFYLEFTinstance-attribute**

TEXT\_JUSTIFYLEFT: IconData

**buildTEXT\_JUSTIFYRIGHTinstance-attribute**

TEXT\_JUSTIFYRIGHT: IconData

**buildTEXT\_QUOTEinstance-attribute**

TEXT\_QUOTE: IconData

**buildTEXTBOXinstance-attribute**

TEXTBOX: IconData

**buildTEXTFORMATinstance-attribute**

TEXTFORMAT: IconData

**buildTEXTFORMAT\_123instance-attribute**

TEXTFORMAT\_123: IconData

**buildTEXTFORMAT\_ABCinstance-attribute**

TEXTFORMAT\_ABC: IconData

**buildTEXTFORMAT\_ABC\_DOTTEDUNDERLINEinstance-attribute**

TEXTFORMAT\_ABC\_DOTTEDUNDERLINE: IconData

**buildTEXTFORMAT\_ALTinstance-attribute**

TEXTFORMAT\_ALT: IconData

**buildTEXTFORMAT\_SIZEinstance-attribute**

TEXTFORMAT\_SIZE: IconData

**buildTEXTFORMAT\_SUBSCRIPTinstance-attribute**

TEXTFORMAT\_SUBSCRIPT: IconData

**buildTEXTFORMAT\_SUPERSCRIPTinstance-attribute**

TEXTFORMAT\_SUPERSCRIPT: IconData

**buildTHERMOMETERinstance-attribute**

THERMOMETER: IconData

**buildTHERMOMETER\_SNOWFLAKEinstance-attribute**

THERMOMETER\_SNOWFLAKE: IconData

**buildTHERMOMETER\_SUNinstance-attribute**

THERMOMETER\_SUN: IconData

**buildTICKETinstance-attribute**

TICKET: IconData

**buildTICKET\_FILLinstance-attribute**

TICKET\_FILL: IconData

**buildTICKETSinstance-attribute**

TICKETS: IconData

**buildTICKETS\_FILLinstance-attribute**

TICKETS\_FILL: IconData

**buildTIMEinstance-attribute**

TIME: IconData

**buildTIME\_SOLIDinstance-attribute**

TIME\_SOLID: IconData

**buildTIMELAPSEinstance-attribute**

TIMELAPSE: IconData

**buildTIMERinstance-attribute**

TIMER: IconData

**buildTIMER\_FILLinstance-attribute**

TIMER\_FILL: IconData

**buildTODAYinstance-attribute**

TODAY: IconData

**buildTODAY\_FILLinstance-attribute**

TODAY\_FILL: IconData

**buildTORNADOinstance-attribute**

TORNADO: IconData

**buildTORTOISEinstance-attribute**

TORTOISE: IconData

**buildTORTOISE\_FILLinstance-attribute**

TORTOISE\_FILL: IconData

**buildTRAIN\_STYLE\_ONEinstance-attribute**

TRAIN\_STYLE\_ONE: IconData

**buildTRAIN\_STYLE\_TWOinstance-attribute**

TRAIN\_STYLE\_TWO: IconData

**buildTRAM\_FILLinstance-attribute**

TRAM\_FILL: IconData

**buildTRASHinstance-attribute**

TRASH: IconData

**buildTRASH\_CIRCLEinstance-attribute**

TRASH\_CIRCLE: IconData

**buildTRASH\_CIRCLE\_FILLinstance-attribute**

TRASH\_CIRCLE\_FILL: IconData

**buildTRASH\_FILLinstance-attribute**

TRASH\_FILL: IconData

**buildTRASH\_SLASHinstance-attribute**

TRASH\_SLASH: IconData

**buildTRASH\_SLASH\_FILLinstance-attribute**

TRASH\_SLASH\_FILL: IconData

**buildTRAYinstance-attribute**

TRAY: IconData

**buildTRAY\_2instance-attribute**

TRAY\_2: IconData

**buildTRAY\_2\_FILLinstance-attribute**

TRAY\_2\_FILL: IconData



**buildTRAY\_ARROW\_DOWN**instance-attribute  
TRAY\_ARROW\_DOWN: IconData

**buildTRAY\_ARROW\_DOWN\_FILL**instance-attribute  
TRAY\_ARROW\_DOWN\_FILL: IconData

**buildTRAY\_ARROW\_UP**instance-attribute  
TRAY\_ARROW\_UP: IconData

**buildTRAY\_ARROW\_UP\_FILL**instance-attribute  
TRAY\_ARROW\_UP\_FILL: IconData

**buildTRAY\_FILL**instance-attribute  
TRAY\_FILL: IconData

**buildTRAY\_FULL**instance-attribute  
TRAY\_FULL: IconData

**buildTRAY\_FULL\_FILL**instance-attribute  
TRAY\_FULL\_FILL: IconData

**buildTREE**instance-attribute  
TREE: IconData

**buildTRIANGLE**instance-attribute  
TRIANGLE: IconData

**buildTRIANGLE\_FILL**instance-attribute  
TRIANGLE\_FILL: IconData

**buildTRIANGLE\_LEFTHALF\_FILL**instance-attribute  
TRIANGLE\_LEFTHALF\_FILL: IconData

**buildTRIANGLE\_RIGHTHALF\_FILL**instance-attribute  
TRIANGLE\_RIGHTHALF\_FILL: IconData

**buildTROPICALSTORM**instance-attribute  
TROPICALSTORM: IconData

**buildTUNINGFORK**instance-attribute  
TUNINGFORK: IconData

**buildTV**instance-attribute  
TV: IconData

**buildTV\_CIRCLE**instance-attribute  
TV\_CIRCLE: IconData

**buildTV\_CIRCLE\_FILL**instance-attribute  
TV\_CIRCLE\_FILL: IconData

**buildTV\_FILL**instance-attribute  
TV\_FILL: IconData

**buildTV\_MUSIC\_NOTE**instance-attribute  
TV\_MUSIC\_NOTE: IconData

**buildTV\_MUSIC\_NOTE\_FILL**instance-attribute  
TV\_MUSIC\_NOTE\_FILL: IconData

**buildUIWINDOW\_SPLIT\_2X1**instance-attribute  
UIWINDOW\_SPLIT\_2X1: IconData

**buildUMBRELLA**instance-attribute  
UMBRELLA: IconData

**buildUMBRELLA\_FILLinstance-attribute**

UMBRELLA\_FILL: IconData

**buildUNDERLINEinstance-attribute**

UNDERLINE: IconData

**buildUP\_ARROWinstance-attribute**

UP\_ARROW: IconData

**buildUPLOAD\_CIRCLEinstance-attribute**

UPLOAD\_CIRCLE: IconData

**buildUPLOAD\_CIRCLE\_FILLinstance-attribute**

UPLOAD\_CIRCLE\_FILL: IconData

**buildVIDEO\_CAMERAinstance-attribute**

VIDEO\_CAMERA: IconData

**buildVIDEO\_CAMERA\_SOLIDinstance-attribute**

VIDEO\_CAMERA\_SOLID: IconData

**buildVIDEOCAMinstance-attribute**

VIDEOCAM: IconData

**buildVIDEOCAM\_CIRCLEinstance-attribute**

VIDEOCAM\_CIRCLE: IconData

**buildVIDEOCAM\_CIRCLE\_FILLinstance-attribute**

VIDEOCAM\_CIRCLE\_FILL: IconData

**buildVIDEOCAM\_FILLinstance-attribute**

VIDEOCAM\_FILL: IconData

**buildVIEW\_2Dinstance-attribute**

VIEW\_2D: IconData

**buildVIEW\_3Dinstance-attribute**

VIEW\_3D: IconData

**buildVIEWFINDERinstance-attribute**

VIEWFINDER: IconData

**buildVIEWFINDER\_CIRCLEinstance-attribute**

VIEWFINDER\_CIRCLE: IconData

**buildVIEWFINDER\_CIRCLE\_FILLinstance-attribute**

VIEWFINDER\_CIRCLE\_FILL: IconData

**buildVOLUME\_DOWNinstance-attribute**

VOLUME\_DOWN: IconData

**buildVOLUME\_MUTEinstance-attribute**

VOLUME\_MUTE: IconData

**buildVOLUME\_OFFinstance-attribute**

VOLUME\_OFF: IconData

**buildVOLUME\_UPinstance-attribute**

VOLUME\_UP: IconData

**buildWAND\_RAYSinstance-attribute**

WAND\_RAYS: IconData

**buildWAND\_RAYS\_INVERSEinstance-attribute**

WAND\_RAYS\_INVERSE: IconData

**buildWAND\_STARSinstance-attribute**

WAND\_STARS: IconData

**buildWAND\_STARS\_INVERSEinstance-attribute**

WAND\_STARS\_INVERSE: IconData

**buildWAVEFORMinstance-attribute**

WAVEFORM: IconData

**buildWAVEFORM\_CIRCLEinstance-attribute**

WAVEFORM\_CIRCLE: IconData

**buildWAVEFORM\_CIRCLE\_FILLinstance-attribute**

WAVEFORM\_CIRCLE\_FILL: IconData

**buildWAVEFORM\_PATHinstance-attribute**

WAVEFORM\_PATH: IconData

**buildWAVEFORM\_PATH\_BADGE\_MINUSinstance-attribute**

WAVEFORM\_PATH\_BADGE\_MINUS: IconData

**buildWAVEFORM\_PATH\_BADGE\_PLUSinstance-attribute**

WAVEFORM\_PATH\_BADGE\_PLUS: IconData

**buildWAVEFORM\_PATH\_ECGinstance-attribute**

WAVEFORM\_PATH\_ECG: IconData

**buildWIFIinstance-attribute**

WIFI: IconData

**buildWIFI\_EXCLAMATIONMARKinstance-attribute**

WIFI\_EXCLAMATIONMARK: IconData

**buildWIFI\_SLASHinstance-attribute**

WIFI\_SLASH: IconData

**buildWINDinstance-attribute**

WIND: IconData

**buildWIND\_SNOWinstance-attribute**

WIND\_SNOW: IconData

**buildWRENCHinstance-attribute**

WRENCH: IconData

**buildWRENCH\_FILLinstance-attribute**

WRENCH\_FILL: IconData

**buildXMARKinstance-attribute**

XMARK: IconData

**buildXMARK\_CIRCLEinstance-attribute**

XMARK\_CIRCLE: IconData

**buildXMARK\_CIRCLE\_FILLinstance-attribute**

XMARK\_CIRCLE\_FILL: IconData

**buildXMARK\_OCTAGONinstance-attribute**

XMARK\_OCTAGON: IconData

**buildXMARK\_OCTAGON\_FILLinstance-attribute**

XMARK\_OCTAGON\_FILL: IconData

**buildXMARK\_RECTANGLEinstance-attribute**

XMARK\_RECTANGLE: IconData

**buildXMARK\_RECTANGLE\_FILLinstance-attribute**

XMARK\_RECTANGLE\_FILL: IconData

**buildXMARK\_SEALinstance-attribute**

XMARK\_SEAL: IconData

**buildXMARK\_SEAL\_FILLinstance-attribute**

XMARK\_SEAL\_FILL: IconData

**buildXMARK\_SHIELDinstance-attribute**

XMARK\_SHIELD: IconData

**buildXMARK\_SHIELD\_FILLinstance-attribute**

XMARK\_SHIELD\_FILL: IconData

**buildXMARK\_SQUAREinstance-attribute**

XMARK\_SQUARE: IconData

**buildXMARK\_SQUARE\_FILLinstance-attribute**

XMARK\_SQUARE\_FILL: IconData

**buildZOOM\_INinstance-attribute**

ZOOM\_IN: IconData

**buildZOOM\_OUTinstance-attribute**

ZOOM\_OUT: IconData

**buildZZZinstance-attribute**

ZZZ: IconData

# CupertinoTimerPickerMode

Different modes of CupertinoTimerPicker.

Inherits: `enum.Enum`

## Properties

- `HOURL_MINUTE` - Mode that shows the timer duration in hour and minute.
- `HOURL_MINUTE_SECONDS` - Mode that shows the timer duration in hour, minute, and second.
- `MINUTE_SECONDS` - Mode that shows the timer duration in minute and second.

## Examples

### Cupertino Timer Picker Mode showcase

`play_arrow`Try Online

```
import flet as ft

def showcase_card(mode: ft.CupertinoTimerPickerMode) -> ft.Container:
 return ft.Container(
 width=340,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(mode.name, weight=ft.FontWeight.BOLD),
 ft.CupertinoTimerPicker(
 mode=mode,
 value=ft.Duration(seconds=754),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="CupertinoTimerPickerMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare timer picker layouts."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(mode) for mode in ft.CupertinoTimerPickerMode
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### `buildHOURL_MINUTE`class-attributeinstance-attribute

Mode that shows the timer duration in hour and minute.

Examples: 16 hours | 14 min

**buildHOUR\_MINUTE\_SECONDS**class-attributeinstance-attribute

Mode that shows the timer duration in hour, minute, and second.

Examples: 16 hours | 14 min | 43 sec

**buildMINUTE\_SECONDS**class-attributeinstance-attribute

Mode that shows the timer duration in minute and second.

Examples: 14 min | 43 sec

# DataColumnSortEvent

Event emitted when a DataColumn requests sorting.

This event is delivered to DataColumn.on\_sort when the user triggers sorting from a column header. Use column\_index and ascending to update table state (for example, sort\_column\_index and sort\_ascending on DataTable).

Inherits: Event

## Properties

- ascending - Requested sort direction: True for ascending, False for descending.
- column\_index - Index of the column that requested sorting.

## Properties

### buildascendingclass-attributeinstance-attribute

```
ascending: bool = field(metadata={'data_field': 'asc'})
```

Requested sort direction: True for ascending, False for descending.

### buildcolumn\_indexclass-attributeinstance-attribute

```
column_index: int = field(metadata={'data_field': 'ci'})
```

Index of the column that requested sorting.

# DataTableTheme

Customizes the appearance of DataTable controls across the app.

## Properties

- `checkbox_horizontal_margin` - Overrides the default value for `DataTable.checkbox_horizontal_margin`.
- `column_spacing` - Overrides the default value for `DataTable.column_spacing`.
- `data_row_color` - Overrides the default value for `DataTable.data_row_color`.
- `data_row_cursor` - Overrides the default value for `DataRow` mouse cursor.
- `data_row_max_height` - Overrides the default value for `DataTable.data_row_max_height`.
- `data_row_min_height` - Overrides the default value for `DataTable.data_row_min_height`.
- `data_text_style` - Overrides the default value for `DataTable.data_text_style`.
- `decoration` - Overrides the default value for `DataTable` decoration.
- `divider_thickness` - Overrides the default value for `DataTable.divider_thickness`.
- `heading_cell_cursor` - Overrides the default value for `DataColumn` mouse cursor.
- `heading_row_alignment` - Overrides the default value for `DataColumn.heading_row_alignment`.
- `heading_row_color` - Overrides the default value for `DataTable.heading_row_color`.
- `heading_row_height` - Overrides the default value for `DataTable.heading_row_height`.
- `heading_text_style` - Overrides the default value for `DataTable.heading_text_style`.
- `horizontal_margin` - Overrides the default value for `DataTable.horizontal_margin`.

## Properties

### `buildcheckbox_horizontal_marginclass-attributeinstance-attribute`

```
checkbox_horizontal_margin: Number | None = None
```

Overrides the default value for `DataTable.checkbox_horizontal_margin`.

### `buildcolumn_spacingclass-attributeinstance-attribute`

```
column_spacing: Number | None = None
```

Overrides the default value for `DataTable.column_spacing`.

### `bulldata_row_colorclass-attributeinstance-attribute`

```
data_row_color: ControlStateValue[ColorValue] | None = None
```

Overrides the default value for `DataTable.data_row_color`.

### `bulldata_row_cursorclass-attributeinstance-attribute`

```
data_row_cursor: ControlStateValue[MouseCursor] | None = (
 None
)
```

Overrides the default value for `DataRow` mouse cursor.

### `bulldata_row_max_heightclass-attributeinstance-attribute`

```
data_row_max_height: Number | None = None
```

Overrides the default value for `DataTable.data_row_max_height`.

### `bulldata_row_min_heightclass-attributeinstance-attribute`

```
data_row_min_height: Number | None = None
```

Overrides the default value for `DataTable.data_row_min_height`.

### `bulldata_text_styleclass-attributeinstance-attribute`

```
data_text_style: TextStyle | None = None
```



Overrides the default value for DataTable.data\_text\_style.

### **builddecorationclass-attributeinstance-attribute**

```
decoration: BoxDecoration | None = None
```

Overrides the default value for DataTable.decoration.

### **builddivider\_thicknessclass-attributeinstance-attribute**

```
divider_thickness: Number | None = None
```

Overrides the default value for DataTable.divider\_thickness.

### **buildheading\_cell\_cursorclass-attributeinstance-attribute**

```
heading_cell_cursor: (
 ControlStateValue[MouseCursor] | None
) = None
```

Overrides the default value for DataColumn.mouse cursor.

### **buildheading\_row\_alignmentclass-attributeinstance-attribute**

```
heading_row_alignment: MainAxisAlignment | None = None
```

Overrides the default value for DataColumn.heading\_row\_alignment.

### **buildheading\_row\_colorclass-attributeinstance-attribute**

```
heading_row_color: ControlStateValue[ColorValue] | None = (
 None
)
```

Overrides the default value for DataTable.heading\_row\_color.

### **buildheading\_row\_heightclass-attributeinstance-attribute**

```
heading_row_height: Number | None = None
```

Overrides the default value for DataTable.heading\_row\_height.

### **buildheading\_text\_styleclass-attributeinstance-attribute**

```
heading_text_style: TextStyle | None = None
```

Overrides the default value for DataTable.heading\_text\_style.

### **buildhorizontal\_marginclass-attributeinstance-attribute**

```
horizontal_margin: Number | None = None
```

Overrides the default value for DataTable.horizontal\_margin.

# DatePickerEntryMode

Mode of date entry method for the date picker dialog.

Inherits: enum.Enum

## Properties

- CALENDAR - User picks a date from calendar grid.
- CALENDAR\_ONLY - User can only pick a date from calendar grid.
- INPUT - User can input the date by typing it into a text field.
- INPUT\_ONLY - User can only input the date by typing it into a text field.

## Examples

### Date Picker Entry Mode showcase

[play\\_arrowTry Online](#)

```
import datetime

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 today = datetime.datetime.now()
 picker = ft.DatePicker(
 first_date=datetime.datetime(year=today.year - 1, month=1, day=1),
 last_date=datetime.datetime(year=today.year + 1, month=12, day=31),
)

 def open_picker(entry_mode: ft.DatePickerEntryMode):
 picker.entry_mode = entry_mode
 picker.date_picker_mode = ft.DatePickerMode.DAY
 page.show_dialog(picker)

 def showcase_card(entry_mode: ft.DatePickerEntryMode) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(entry_mode.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open DatePicker",
 icon=ft.Icons.CALENDAR_MONTH,
 on_click=lambda _, m=entry_mode: open_picker(m),
),
],
),
)

 page.appbar = ft.AppBar(title="DatePickerEntryMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Open the picker to compare calendar and input entry modes."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(m) for m in ft.DatePickerEntryMode],
),
],
),
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildCALENDARclass-attributeinstance-attribute**

User picks a date from calendar grid.

Can switch to INPUT by activating a mode button in the dialog.

### **buildCALENDAR\_ONLYclass-attributeinstance-attribute**

User can only pick a date from calendar grid.

There is no user interface to switch to another mode.

### **buildINPUTclass-attributeinstance-attribute**

User can input the date by typing it into a text field.

Can switch to CALENDAR by activating a mode button in the dialog.

### **buildINPUT\_ONLYclass-attributeinstance-attribute**

User can only input the date by typing it into a text field.

There is no user interface to switch to another mode.

# DatePickerEntryModeChangeEvent

Event fired when the DatePicker entry mode is changed.

Inherits: Event

## Properties

- `entry_mode` - The new date picker entry mode.

## Properties

`buildentry_modeinstance-attribute`

`entry_mode: DatePickerEntryMode`

The new date picker entry mode.

# DatePickerMode

Initial display of a calendar date picker.

Inherits: enum.Enum

## Properties

- DAY - Choosing a month and day.
- YEAR - Choosing a year.

## Examples

### Date Picker Mode showcase

play\_arrowTry Online

```
import datetime

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 today = datetime.datetime.now()
 picker = ft.DatePicker(
 first_date=datetime.datetime(year=today.year - 1, month=1, day=1),
 last_date=datetime.datetime(year=today.year + 1, month=12, day=31),
 entry_mode=ft.DatePickerEntryMode.CALENDAR,
)

 def open_picker(date_picker_mode: ft.DatePickerMode):
 picker.date_picker_mode = date_picker_mode
 page.show_dialog(picker)

 def showcase_card(date_picker_mode: ft.DatePickerMode) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(date_picker_mode.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open DatePicker",
 icon=ft.Icons.CALENDAR_MONTH,
 on_click=lambda _, m=date_picker_mode: open_picker(m),
),
],
),
)

 page.appbar = ft.AppBar(title="DatePickerMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Open the picker to compare initial day vs year display mode."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(m) for m in ft.DatePickerMode],
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildDAY**class-attributeinstance-attribute

Choosing a month and day.

**buildYEAR**class-attributeinstance-attribute

Choosing a year.

# DatePickerTheme

Customizes the appearance of DatePicker controls across the app.

## Properties

- `bgcolor` - Overrides the default background color of the surface in all descendant DatePicker controls.
- `cancel_button_style` - Overrides the default style of the cancel button of a DatePicker.
- `confirm_button_style` - Overrides the default style of the confirm (OK) button of a DatePicker.
- `day_bgcolor` - Overrides the default color used to paint the background of the day labels in the grid of the DatePicker.
- `day_foreground_color` - Overrides the default color used to paint the day labels in the grid of the DatePicker.
- `day_overlay_color` - Overrides the default highlight color that's typically used to indicate that a day in the grid is focused, hovered, or pressed.
- `day_shape` - Overrides the default shape used to paint the shape decoration of the day labels in the grid of the DatePicker.
- `day_text_style` - Overrides the default text style used for each individual day label in the grid of the DatePicker.
- `divider_color` - Overrides the default color used to paint the divider in all descendant DatePicker controls.
- `elevation` - Overrides the default value of DatePicker elevation.
- `header_bgcolor` - Overrides the header's default background fill color.
- `header_foreground_color` - Overrides the header's default color used for text labels and icons.
- `header_headline_text_style` - Overrides the header's default headline text style.
- `header_help_text_style` - Overrides the header's default help text style.
- `range_picker_bgcolor` - Overrides the default background color for DateRangePicker.
- `range_picker_elevation` - Overrides the default elevation of the full screen DateRangePicker (TBD).
- `range_picker_header_bgcolor` - Overrides the default background fill color for DateRangePicker.
- `range_picker_header_foreground_color` - Overrides the default color used for text labels and icons in the header of a full screen DateRangePicker.
- `range_picker_header_headline_text_style` - Overrides the default text style used for the headline text in the header of a full screen DateRangePicker.
- `range_picker_header_help_text_style` - Overrides the default text style used for the help text of the header of a full screen DateRangePicker (TBD).
- `range_picker_shape` - Overrides the default overall shape of a full screen DateRangePicker (TBD).
- `range_selection_bgcolor` - Overrides the default background color used to paint days selected between the start and end dates in a DateRangePicker.
- `range_selection_overlay_color` - Overrides the default highlight color that's typically used to indicate that a date in the selected range of a DateRangePicker is focused, hovered, or pressed.
- `shadow_color` - Overrides the default shadow color in all descendant DatePicker controls.
- `shape` - Overrides the default value of DatePicker shape.
- `today_bgcolor` - Overrides the default color used to paint the background of the `[DatePicker.current_date].[flet.DatePicker.current_date]` label in the grid of the DatePicker.
- `today_border_side` - Overrides the border used to paint the `DatePicker.current_date` label in the grid of the DatePicker.
- `today_foreground_color` - Overrides the default color used to paint the `DatePicker.current_date` label in the grid of the dialog's `CalendarDatePicker` and the corresponding year in the dialog's `YearPicker`.
- `weekday_text_style` - Overrides the default text style used for the row of weekday labels at the top of the DatePicker grid.

- `year_bgcolor` - Overrides the default color used to paint the background of the year labels in the year selector of the of the DatePicker.
- `year_foreground_color` - Overrides the default color used to paint the year labels in the year selector of the date picker.
- `year_overlay_color` - Overrides the default highlight color that's typically used to indicate that a year in the year selector is focused, hovered, or pressed.
- `year_text_style` - Overrides the default text style used to paint each of the year entries in the year selector of the DatePicker.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default background color of the surface in all descendant DatePicker controls.

### `buildcancel_button_styleclass-attributeinstance-attribute`

`cancel_button_style: ButtonStyle | None = None`

Overrides the default style of the cancel button of a DatePicker.

### `buildconfirm_button_styleclass-attributeinstance-attribute`

`confirm_button_style: ButtonStyle | None = None`

Overrides the default style of the confirm (OK) button of a DatePicker.

### `bullday_bgcolorclass-attributeinstance-attribute`

`day_bgcolor: ControlStateValue[ColorValue] | None = None`

Overrides the default color used to paint the background of the day labels in the grid of the DatePicker.

### `bullday_foreground_colorclass-attributeinstance-attribute`

`day_foreground_color: (ControlStateValue[ColorValue] | None) = None`

Overrides the default color used to paint the day labels in the grid of the DatePicker.

This will be used instead of the color provided in `DatePickerTheme.day_text_style`.

### `bullday_overlay_colorclass-attributeinstance-attribute`

`day_overlay_color: ControlStateValue[ColorValue] | None = (None)`

Overrides the default highlight color that's typically used to indicate that a day in the grid is focused, hovered, or pressed.

### `bullday_shapeclass-attributeinstance-attribute`

`day_shape: ControlStateValue[OutlinedBorder] | None = None`

Overrides the default shape used to paint the shape decoration of the day labels in the grid of the DatePicker.

If the selected day is the current day, the provided shape with the value of `DatePickerTheme.today_bgcolor` is used to paint the shape decoration of the day label and the value of `DatePickerTheme.today_border_side` and `DatePickerTheme.today_foreground_color` is used to paint the border.

If the selected day is not the current day, the provided shape with the value of `DatePickerTheme.day_bgcolor` is used to paint the shape decoration of the day label.

### `bullday_text_styleclass-attributeinstance-attribute`

`day_text_style: TextStyle | None = None`

Overrides the default text style used for each individual day label in the grid of the DatePicker.

The color in `DatePickerTheme.day_text_style` is not used, `DatePickerTheme.day_foreground_color` is used instead.

### `bulldivider_colorclass-attributeinstance-attribute`

`divider_color: ColorValue | None = None`

Overrides the default color used to paint the divider in all descendant DatePicker controls.



## **buildelevationclass-attributeinstance-attribute**

`elevation: Number | None = None`

Overrides the default value of DatePicker elevation.

## **buildheader\_bgcolorclass-attributeinstance-attribute**

`header_bgcolor: ColorValue | None = None`

Overrides the header's default background fill color.

The DatePicker's header displays the currently selected date.

## **buildheader\_foreground\_colorclass-attributeinstance-attribute**

`header_foreground_color: ColorValue | None = None`

Overrides the header's default color used for text labels and icons.

The dialog's header displays the currently selected date.

This is used instead of the color property of DatePickerTheme.header\_headline\_text\_style and DatePickerTheme.header\_help\_text\_style.

## **buildheader\_headline\_text\_styleclass-attributeinstance-attribute**

`header_headline_text_style: TextStyle | None = None`

Overrides the header's default headline text style.

The dialog's header displays the currently selected date.

The color of the DatePickerTheme.header\_headline\_text\_style is not used, DatePickerTheme.header\_foreground\_color is used instead.

## **buildheader\_help\_text\_styleclass-attributeinstance-attribute**

`header_help_text_style: TextStyle | None = None`

Overrides the header's default help text style.

The help text (also referred to as "supporting text" in the Material spec) is usually a prompt to the user at the top of the header (i.e. 'Select date').

The color of the header\_help\_style is not used, DatePickerTheme.header\_foreground\_color is used instead.

## **buildrange\_picker\_bgcolorclass-attributeinstance-attribute**

`range_picker_bgcolor: ColorValue | None = None`

Overrides the default background color for DateRangePicker.

## **buildrange\_picker\_elevationclass-attributeinstance-attribute**

`range_picker_elevation: Number | None = None`

Overrides the default elevation of the full screen DateRangePicker (TBD).

## **buildrange\_picker\_header\_bgcolorclass-attributeinstance-attribute**

`range_picker_header_bgcolor: ColorValue | None = None`

Overrides the default background fill color for DateRangePicker.

The dialog's header displays the currently selected date range.

## **buildrange\_picker\_header\_foreground\_colorclass-attributeinstance-attribute**

`range_picker_header_foreground_color: ColorValue | None = (  
 None  
)`

Overrides the default color used for text labels and icons in the header of a full screen DateRangePicker.

The dialog's header displays the currently selected date range.

This is used instead of any colors provided by range\_picker\_header\_headline\_text\_style or range\_picker\_header\_help\_text\_style.

## **buildrange\_picker\_header\_headline\_text\_styleclass-attributeinstance-attribute**

`range_picker_header_headline_text_style: (  
 TextStyle | None`

```
) = None
```

Overrides the default text style used for the headline text in the header of a full screen `DateRangePicker`.

The dialog's header displays the currently selected date range.

The color of `range_picker_header_headline_text_style` is not used, `range_picker_header_foreground_color` is used instead.

### **buildrange\_picker\_header\_help\_text\_styleclass-attributeinstance-attribute**

```
range_picker_header_help_text_style: TextStyle | None = None
```

Overrides the default text style used for the help text of the header of a full screen `DateRangePicker` (TBD).

The help text (also referred to as "supporting text" in the Material spec) is usually a prompt to the user at the top of the header (i.e. 'Select date').

The color of the `range_picker_header_help_text_style` is not used, `range_picker_header_foreground_color` is used instead.

### **buildrange\_picker\_shapeclass-attributeinstance-attribute**

```
range_picker_shape: OutlinedBorder | None = None
```

Overrides the default overall shape of a full screen `DateRangePicker` (TBD).

If elevation is greater than zero then a shadow is shown and the shadow's shape mirrors the shape of the dialog.

### **buildrange\_selection\_bgcolorclass-attributeinstance-attribute**

```
range_selection_bgcolor: ColorValue | None = None
```

Overrides the default background color used to paint days selected between the start and end dates in a `DateRangePicker`.

### **buildrange\_selection\_overlay\_colorclass-attributeinstance-attribute**

```
range_selection_overlay_color: (
 ControlStateValue[ColorValue] | None
) = None
```

Overrides the default highlight color that's typically used to indicate that a date in the selected range of a `DateRangePicker` is focused, hovered, or pressed.

### **buildshadow\_colorclass-attributeinstance-attribute**

```
shadow_color: ColorValue | None = None
```

Overrides the default shadow color in all descendant `DatePicker` controls.

### **buildshapeclass-attributeinstance-attribute**

```
shape: OutlinedBorder | None = None
```

Overrides the default value of `DatePicker` shape.

If elevation is greater than zero then a shadow is shown and the shadow's shape mirrors the shape of the dialog.

### **buildtoday\_bgcolorclass-attributeinstance-attribute**

```
today_bgcolor: ControlStateValue[ColorValue] | None = None
```

Overrides the default color used to paint the background of the `[DatePicker.current_date].[flet.DatePicker.current_date]` label in the grid of the `DatePicker`.

### **buildtoday\_border\_sideclass-attributeinstance-attribute**

```
today_border_side: BorderSide | None = None
```

Overrides the border used to paint the `DatePicker.current_date` label in the grid of the `DatePicker`.

The border side's `[BorderSide.color]` is not used, `DatePickerTheme.today_foreground_color` is used instead.

### **buildtoday\_foreground\_colorclass-attributeinstance-attribute**

```
today_foreground_color: (
 ControlStateValue[ColorValue] | None
) = None
```

Overrides the default color used to paint the `DatePicker.current_date` label in the grid of the dialog's `CalendarDatePicker` and the corresponding year in the dialog's `YearPicker`.

This will be used instead of the color provided in `DatePickerTheme.day_text_style`.

### **buildweekday\_text\_styleclass-attributeinstance-attribute**

`weekday_text_style: TextStyle | None = None`

Overrides the default text style used for the row of weekday labels at the top of the DatePicker grid.

### **buildyear\_bgcolorclass-attributeinstance-attribute**

`year_bgcolor: ControlStateValue[ColorValue] | None = None`

Overrides the default color used to paint the background of the year labels in the year selector of the of the DatePicker.

### **buildyear\_foreground\_colorclass-attributeinstance-attribute**

`year_foreground_color: (ControlStateValue[ColorValue] | None) = None`

Overrides the default color used to paint the year labels in the year selector of the date picker.

This will be used instead of the color provided in `DatePickerTheme.year_text_style`.

### **buildyear\_overlay\_colorclass-attributeinstance-attribute**

`year_overlay_color: ControlStateValue[ColorValue] | None = (None)`

Overrides the default highlight color that's typically used to indicate that a year in the year selector is focused, hovered, or pressed.

### **buildyear\_text\_styleclass-attributeinstance-attribute**

`year_text_style: TextStyle | None = None`

Overrides the default text style used to paint each of the year entries in the year selector of the DatePicker.

The color of the `DatePickerTheme.year_text_style` is not used, `DatePickerTheme.year_foreground_color` is used instead.

# DecorationImage

An image for a box decoration.

## Properties

- `alignment` - The alignment of the image within its bounds.
- `anti_alias` - Whether to paint the image in anti-aliased quality.
- `color_filter` - A color filter to apply to the image before painting it.
- `filter_quality` - The quality of the image filter.
- `fit` - How the image should be inscribed into the box.
- `invert_colors` - Whether to invert the colors of the image while drawing.
- `match_text_direction` - Whether to paint the image in the direction of the `TextDirection`.
- `opacity` - The opacity of the image.
- `repeat` - How the image should be repeated to fill the box.
- `scale` - The `scale(image pixels to be shown per logical pixels)` to apply to the image.
- `src` - The image source to paint.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

```
alignment: Alignment = field(
 default_factory=lambda: Alignment.CENTER
)
```

The alignment of the image within its bounds.

### `buildanti_aliasclass-attributeinstance-attribute`

```
anti_alias: bool = False
```

Whether to paint the image in anti-aliased quality.

### `buildcolor_filterclass-attributeinstance-attribute`

```
color_filter: ColorFilter | None = None
```

A color filter to apply to the image before painting it.

### `buildfilter_qualityclass-attributeinstance-attribute`

```
filter_quality: FilterQuality = FilterQuality.MEDIUM
```

The quality of the image filter.

### `buildfitclass-attributeinstance-attribute`

```
fit: BoxFit | None = None
```

How the image should be inscribed into the box.

### `buildinvert_colorsclass-attributeinstance-attribute`

```
invert_colors: bool = False
```

Whether to invert the colors of the image while drawing.

### `buildmatch_text_directionclass-attributeinstance-attribute`

```
match_text_direction: bool = False
```

Whether to paint the image in the direction of the `TextDirection`.

### `buildopacityclass-attributeinstance-attribute`

```
opacity: Number = 1.0
```

The opacity of the image.

### buildrepeatclass-attributeinstance-attribute

```
repeat: ImageRepeat = ImageRepeat.NO_REPEAT
```

How the image should be repeated to fill the box.

### buildscaleclass-attributeinstance-attribute

```
scale: Number = 1.0
```

The scale(image pixels to be shown per logical pixels) to apply to the image.

### buildsrcclass-attributeinstance-attribute

```
src: str | bytes | None = None
```

The image source to paint.

Accepts URLs, asset paths, base64 strings (with or without data: prefixes), or raw bytes.

## Methods

### deployed\_codecopy

```
copy(
 src: str | bytes | None = None,
 color_filter: ColorFilter | None = None,
 fit: BoxFit | None = None,
 alignment: Alignment | None = None,
 repeat: ImageRepeat | None = None,
 match_text_direction: bool | None = None,
 scale: Number | None = None,
 opacity: Number | None = None,
 filter_quality: FilterQuality | None = None,
 invert_colors: bool | None = None,
 anti_alias: bool | None = None,
) -> DecorationImage
```

Returns a copy of this object with the specified properties overridden.

# DeviceInfo

Base class for device information.

Platform-specific classes include:

- `AndroidDeviceInfo`
- `IosDeviceInfo`
- `LinuxDeviceInfo`
- `MacOsDeviceInfo`
- `WebDeviceInfo`
- `WindowsDeviceInfo`

## Properties

- `locales` - The full system-reported supported locales of the device.

## Properties

### `buildlocalesinstance-attribute`

```
locales: list[Locale]
```

The full system-reported supported locales of the device.

This establishes the language and formatting conventions that application should, if possible, use to render their user interface.

The list is ordered in order of priority, with lower-indexed locales being preferred over higher-indexed ones. The first element is the primary locale.

The `Page.on_locale_change` event is called whenever this value changes.

# DeviceOrientation

Supported physical orientations for mobile devices.

Inherits: `enum.Enum`

## Properties

- `LANDSCAPE_LEFT` - Device rotated 90° counter-clockwise (home button or primary edge on the right).
- `LANDSCAPE_RIGHT` - Device rotated 90° clockwise (home button or primary edge on the left).
- `PORTRAIT_DOWN` - Device held upside-down in portrait mode.
- `PORTRAIT_UP` - Device held upright in portrait mode.

## Examples

### Mobile device orientation configuration

See the [Page mobile device orientation example](#), which demonstrates how to use `DeviceOrientation` with `Page.set_allowed_device_orientations()`.

## Properties

`buildLANDSCAPE_LEFT`class-attributeinstance-attribute

Device rotated 90° counter-clockwise (home button or primary edge on the right).

`buildLANDSCAPE_RIGHT`class-attributeinstance-attribute

Device rotated 90° clockwise (home button or primary edge on the left).

`buildPORTRAIT_DOWN`class-attributeinstance-attribute

Device held upside-down in portrait mode.

`buildPORTRAIT_UP`class-attributeinstance-attribute

Device held upright in portrait mode.

# DialogTheme

Customizes the appearance of AlertDialog across the app.

## Properties

- `actions_padding` - Overrides the default value of `AlertDialog.actions_padding` in all descendant `AlertDialog` controls.
- `alignment` - Overrides the default value of `AlertDialog.alignment` in all descendant `AlertDialog` controls.
- `barrier_color` - Overrides the default value of `AlertDialog.barrier_color` in all descendant `AlertDialog` controls.
- `bgcolor` - Overrides the default value of `AlertDialog.bgcolor` in all descendant `AlertDialog` controls.
- `clip_behavior` - Overrides the default value of `AlertDialog.clip_behavior` in all descendant `AlertDialog` controls.
- `content_text_style` - Overrides the default value of `AlertDialog.content_text_style` in all descendant `AlertDialog` controls.
- `elevation` - Overrides the default value of `AlertDialog.elevation` in all descendant dialog controls.
- `icon_color` - Used to configure the `IconTheme` for the `AlertDialog.icon` control.
- `inset_padding` - Overrides the default value of `AlertDialog.inset_padding` in all descendant `AlertDialog` controls.
- `shadow_color` - Overrides the default value of `AlertDialog.shadow_color` in all descendant `AlertDialog` controls.
- `shape` - Overrides the default value of `AlertDialog.shape` in all descendant `AlertDialog` controls.
- `title_text_style` - Overrides the default value of `AlertDialog.title_text_style` in all descendant `AlertDialog` controls.

## Properties

### `buildactions_paddingclass-attributeinstance-attribute`

`actions_padding: PaddingValue | None = None`

Overrides the default value of `AlertDialog.actions_padding` in all descendant `AlertDialog` controls.

### `buildalignmentclass-attributeinstance-attribute`

`alignment: Alignment | None = None`

Overrides the default value of `AlertDialog.alignment` in all descendant `AlertDialog` controls.

### `buildbarrier_colorclass-attributeinstance-attribute`

`barrier_color: ColorValue | None = None`

Overrides the default value of `AlertDialog.barrier_color` in all descendant `AlertDialog` controls.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `AlertDialog.bgcolor` in all descendant `AlertDialog` controls.

### `buildclip_behaviorclass-attributeinstance-attribute`

`clip_behavior: ClipBehavior | None = None`

Overrides the default value of `AlertDialog.clip_behavior` in all descendant `AlertDialog` controls.

### `buildcontent_text_styleclass-attributeinstance-attribute`

`content_text_style: TextStyle | None = None`

Overrides the default value of `AlertDialog.content_text_style` in all descendant `AlertDialog` controls.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value of `AlertDialog.elevation` in all descendant dialog controls.

### `buildicon_colorclass-attributeinstance-attribute`

`icon_color: ColorValue | None = None`

Used to configure the `IconTheme` for the `AlertDialog.icon` control.



### **buildinset\_paddingclass-attributeinstance-attribute**

`inset_padding: PaddingValue | None = None`

Overrides the default value of `AlertDialog.inset_padding` in all descendant `AlertDialog` controls.

### **buildshadow\_colorclass-attributeinstance-attribute**

`shadow_color: ColorValue | None = None`

Overrides the default value of `AlertDialog.shadow_color` in all descendant `AlertDialog` controls.

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

Overrides the default value of `AlertDialog.shape` in all descendant `AlertDialog` controls.

### **buildtitle\_text\_styleclass-attributeinstance-attribute**

`title_text_style: TextStyle | None = None`

Overrides the default value of `AlertDialog.title_text_style` in all descendant `AlertDialog` controls.

# DismissDirection

Defines swipe directions allowed for dismissing a `SnackBar`.

The direction controls which drag gestures can close the snack bar when dismissal is enabled.

Inherits: `enum.Enum`

## Properties

- `DOWN` - Allows dismissing only by swiping downward.
- `END_TO_START` - Allows dismissing toward the end-to-start reading direction.
- `HORIZONTAL` - Allows dismissing by swiping horizontally.
- `NONE` - Disables swipe-to-dismiss gestures.
- `START_TO_END` - Allows dismissing toward the start-to-end reading direction.
- `UP` - Allows dismissing only by swiping upward.
- `VERTICAL` - Allows dismissing by swiping vertically.

## Examples

### Dismiss Direction showcase

[play\\_arrow](#) Try Online

```
import flet as ft

def showcase_card(direction: ft.DismissDirection) -> ft.Container:
 status = ft.Text("Swipe the tile", size=12, color=ft.Colors.ON_SURFACE_VARIANT)

 def create_item() -> ft.Dismissible:
 return ft.Dismissible(
 dismiss_direction=direction,
 on_dismiss=lambda _: on_dismiss(),
 background=ft.Container(
 bgcolor=ft.Colors.GREEN_200,
 alignment=ft.Alignment.CENTER_LEFT,
 padding=10,
 content=ft.Icon(ft.Icons.CHECK, color=ft.Colors.GREEN_900),
),
 secondary_background=ft.Container(
 bgcolor=ft.Colors.RED_200,
 alignment=ft.Alignment.CENTER_RIGHT,
 padding=10,
 content=ft.Icon(ft.Icons.DELETE, color=ft.Colors.RED_900),
),
 content=ft.Container(
 height=52,
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 alignment=ft.Alignment.CENTER,
 content=ft.Text("Dismiss me"),
),
)

 def on_dismiss():
 slot.content = ft.Container(
 height=52,
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 alignment=ft.Alignment.CENTER,
 content=ft.Text("Dismissed"),
)
 status.value = f"Dismissed via {direction.name}"
 slot.update()
 status.update()

 def reset_item(_):
 slot.content = create_item()
 status.value = "Swipe the tile"
 slot.update()
 status.update()
```

```

slot = ft.Container(content=create_item())

return ft.Container(
 width=360,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(direction.name, weight=ft.FontWeight.BOLD),
 status,
 slot,
 ft.Button("Reset", on_click=reset_item),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="DismissDirection Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Try swipe directions to see which ones are allowed."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(direction)
 for direction in ft.DismissDirection
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### **buildDOWN**class-attributeinstance-attribute

Allows dismissing only by swiping downward.

### **buildEND\_TO\_START**class-attributeinstance-attribute

Allows dismissing toward the end-to-start reading direction.

For left-to-right locales this is right-to-left; for right-to-left locales this is left-to-right.

### **buildHORIZONTAL**class-attributeinstance-attribute

Allows dismissing by swiping horizontally.

Users can drag either left or right to dismiss.

### **buildNONE**class-attributeinstance-attribute

Disables swipe-to-dismiss gestures.

### **buildSTART\_TO\_END**class-attributeinstance-attribute

Allows dismissing toward the start-to-end reading direction.

For left-to-right locales this is left-to-right; for right-to-left locales this is right-to-left.

### **buildUP**class-attributeinstance-attribute

Allows dismissing only by swiping upward.

### **buildVERTICAL**class-attributeinstance-attribute

Allows dismissing by swiping vertically.

Users can drag either up or down to dismiss.

# DismissibleDismissEvent

Event payload for dismissal confirmation and completion callbacks.

Used by `on_confirm_dismiss` and `on_dismiss`.

Inherits: `Event`

## Properties

- `direction` - Direction in which the control is being (or was) dismissed.

## Properties

### `builddirectioninstance-attribute`

`direction: DismissDirection`

Direction in which the control is being (or was) dismissed.

# DismissibleUpdateEvent

Event payload emitted while a dismiss gesture is in progress.

Inherits: Event

## Properties

- `direction` - Direction of the current drag gesture.
- `previous_reached` - Whether threshold was reached on the previous update event.
- `progress` - Drag progress from 0.0 to 1.0 relative to dismissal threshold.
- `reached` - Whether the dismiss threshold is currently reached.

## Properties

### `builddirectioninstance-attribute`

`direction: DismissDirection`

Direction of the current drag gesture.

### `buildprevious_reachedinstance-attribute`

`previous_reached: bool`

Whether threshold was reached on the previous update event.

### `buildprogressinstance-attribute`

`progress: float`

Drag progress from 0.0 to 1.0 relative to dismissal threshold.

### `buildreachedinstance-attribute`

`reached: bool`

Whether the dismiss threshold is currently reached.

# DividerTheme

Defines the visual properties of Divider, VerticalDivider, dividers between ListTiles, and dividers between rows in DataTable.

## Properties

- **color** - The color of Dividers and VerticalDividers, also used between ListTiles, between rows in DataTables, and so forth.
- **leading\_indent** - The amount of empty space at the leading edge of Divider or top edge of VerticalDivider.
- **space** - The Divider's height or the VerticalDivider's width.
- **thickness** - The thickness of the line drawn within the divider.
- **trailing\_indent** - The amount of empty space at the trailing edge of Divider or bottom edge of VerticalDivider.

## Properties

### **buildcolorclass-attributeinstance-attribute**

`color: ColorValue | None = None`

The color of Dividers and VerticalDividers, also used between ListTiles, between rows in DataTables, and so forth.

### **buildleading\_indentclass-attributeinstance-attribute**

`leading_indent: Number | None = None`

The amount of empty space at the leading edge of Divider or top edge of VerticalDivider.

### **buildspaceclass-attributeinstance-attribute**

`space: Number | None = None`

The Divider's height or the VerticalDivider's width.

This represents the amount of horizontal or vertical space the divider takes up.

### **buildthicknessclass-attributeinstance-attribute**

`thickness: Number | None = None`

The thickness of the line drawn within the divider.

### **buildtrailing\_indentclass-attributeinstance-attribute**

`trailing_indent: Number | None = None`

The amount of empty space at the trailing edge of Divider or bottom edge of VerticalDivider.

# DragEndEvent

Completion payload for drag gestures.

Inherits: Event

## Properties

- `global_position` - The global position the pointer is located at when the drag gesture has been completed.
- `local_position` - The local position in the coordinate system of the event receiver when the drag gesture has been completed.
- `primary_velocity` - The velocity the pointer was moving along the primary axis when it stopped contacting the screen, in logical pixels per second.
- `velocity` - The velocity vector the pointer was moving when it stopped contacting the screen, in logical pixels per second.

## Properties

### `buildglobal_positionclass-attributeinstance-attribute`

```
global_position: Offset = field(
 metadata={"data_field": "g"}
)
```

The global position the pointer is located at when the drag gesture has been completed.

### `buildlocal_positionclass-attributeinstance-attribute`

```
local_position: Offset = field(metadata={"data_field": "l"})
```

The local position in the coordinate system of the event receiver when the drag gesture has been completed.

### `buildprimary_velocityclass-attributeinstance-attribute`

```
primary_velocity: float | None = field(
 default=None, metadata={"data_field": "pv"}
)
```

The velocity the pointer was moving along the primary axis when it stopped contacting the screen, in logical pixels per second.

### `buildvelocityclass-attributeinstance-attribute`

```
velocity: Offset = field(metadata={'data_field': 'v'})
```

The velocity vector the pointer was moving when it stopped contacting the screen, in logical pixels per second.



# DragStartEvent

Start payload for recognized drag gestures.

Inherits: Event

## Properties

- `global_position` - The global position at which the pointer contacted the screen.
- `kind` - The kind of the device that initiated the event.
- `local_position` - The local position in the coordinate system of the event receiver at which the pointer contacted the screen.
- `timestamp` - Recorded timestamp of the source pointer event that triggered the drag event.

## Properties

### `buildglobal_positionclass-attributeinstance-attribute`

```
global_position: Offset = field(
 metadata={"data_field": "g"}
)
```

The global position at which the pointer contacted the screen.

Defaults to the origin if not specified in the constructor.

### `buildkindclass-attributeinstance-attribute`

```
kind: PointerDeviceType = field(
 metadata={"data_field": "k"}
)
```

The kind of the device that initiated the event.

### `buildlocal_positionclass-attributeinstance-attribute`

```
local_position: Offset = field(metadata={"data_field": "l"})
```

The local position in the coordinate system of the event receiver at which the pointer contacted the screen.

### `buildtimestampclass-attributeinstance-attribute`

```
timestamp: Duration | None = field(
 default=None, metadata={"data_field": "ts"}
)
```

Recorded timestamp of the source pointer event that triggered the drag event.

Could be None if triggered from proxied events such as accessibility.

# DragTargetEvent

Event payload for drag move and accepted-drop callbacks.

Inherits: Event

## Properties

- `global_position` - Pointer position in the global coordinate space.
- `local_position` - Pointer position relative to the target bounds.
- `offset` deprecated - Pointer position in global coordinates.
- `src` - Source Draggable control resolved from `src_id`.
- `src_id` - ID of the Draggable source control, if available.
- `x` deprecated - Horizontal pointer position in the global coordinate space.
- `y` deprecated - Vertical pointer position in the global coordinate space.

## Properties

### `buildglobal_positionclass-attributeinstance-attribute`

```
global_position: Offset = field(
 metadata={"data_field": "g"}
)
```

Pointer position in the global coordinate space.

### `buildlocal_positionclass-attributeinstance-attribute`

```
local_position: Offset = field(metadata={"data_field": "l"})
```

Pointer position relative to the target bounds.

### `buildoffsetdeprecatedproperty`

```
offset: Offset
```

Deprecated

Deprecated since version 0.85.0 and scheduled for removal in 0.88.0. Use `local_position` for target-relative coordinates or `global_position` for global coordinates instead.

Pointer position in global coordinates.

### `buildsrcclass-attributeinstance-attribute`

```
src: Draggable = field(init=False)
```

Source Draggable control resolved from `src_id`.

### `buildsrc_idinstance-attribute`

```
src_id: int | None
```

ID of the Draggable source control, if available.

### `buildxdeprecatedproperty`

```
x: float
```

Deprecated

Deprecated since version 0.85.0 and scheduled for removal in 0.88.0. Use `local_position.x` for target-relative coordinates or `global_position.x` for global coordinates instead.

Horizontal pointer position in the global coordinate space.

### `buildydeprecatedproperty`

```
y: float
```

Deprecated

Deprecated since version 0.85.0 and scheduled for removal in 0.88.0. Use `local_position.y` for target-relative coordinates or `global_position.y` for global coordinates instead.

Vertical pointer position in the global coordinate space.

# DragTargetLeaveEvent

Event payload for DragTarget.on\_leave.

Inherits: Event

## Properties

- src - Source Draggable control resolved from src\_id.
- src\_id - ID of the Draggable source control, if available.

## Properties

### buildsrcclass-attributeinstance-attribute

```
src: Draggable = field(init=False)
```

Source Draggable control resolved from src\_id.

### buildsrc\_idinstance-attribute

```
src_id: int | None
```

ID of the Draggable source control, if available.

# DragUpdateEvent

Incremental payload for drag progress updates.

Inherits: Event

## Properties

- `global_delta` - The amount the pointer has moved in the global coordinate space since the start of the drag.
- `global_position` - The pointer's global position when it triggered this update.
- `local_delta` - The amount the pointer has moved in the local coordinate space of the event receiver since the start of the drag.
- `local_position` - The local position in the coordinate system of the event receiver at which the pointer contacted the screen.
- `primary_delta` - The amount the pointer has moved along the primary axis in the coordinate space of the event receiver since the previous update.
- `timestamp` - Recorded timestamp of the source pointer event that triggered the drag event.

## Properties

### `buildglobal_delta`class-attributeinstance-attribute

```
global_delta: Offset | None = field(
 default=None, metadata={"data_field": "gd"}
)
```

The amount the pointer has moved in the global coordinate space since the start of the drag.

### `buildglobal_position`class-attributeinstance-attribute

```
global_position: Offset = field(
 metadata={"data_field": "g"}
)
```

The pointer's global position when it triggered this update.

### `buildlocal_delta`class-attributeinstance-attribute

```
local_delta: Offset | None = field(
 default=None, metadata={"data_field": "ld"}
)
```

The amount the pointer has moved in the local coordinate space of the event receiver since the start of the drag.

### `buildlocal_position`class-attributeinstance-attribute

```
local_position: Offset = field(metadata={"data_field": "l"})
```

The local position in the coordinate system of the event receiver at which the pointer contacted the screen.

### `buildprimary_delta`class-attributeinstance-attribute

```
primary_delta: float | None = field(
 default=None, metadata={"data_field": "pd"}
)
```

The amount the pointer has moved along the primary axis in the coordinate space of the event receiver since the previous update.

### `buildtimestamp`class-attributeinstance-attribute

```
timestamp: Duration | None = field(
 default=None, metadata={"data_field": "ts"}
)
```

Recorded timestamp of the source pointer event that triggered the drag event.

Could be None if triggered from proxied events such as accessibility.

# DragWillAcceptEvent

Event payload for DragTarget.on\_will\_accept.

Inherits: Event

## Properties

- `accept` - Whether this target will accept the dragged source.
- `src` - Source Draggable control resolved from `src_id`.
- `src_id` - ID of the Draggable source control, if available.

## Properties

### `buildacceptinstance-attribute`

`accept: bool`

Whether this target will accept the dragged source.

### `buildsrcclass-attributeinstance-attribute`

`src: Draggable = field(init=False)`

Source Draggable control resolved from `src_id`.

### `buildsrc_idinstance-attribute`

`src_id: int | None`

ID of the Draggable source control, if available.

# DropdownTheme

Customizes the appearance of Dropdown across the app.

## Properties

- `menu_style` - Overrides the default value for `Dropdown.menu_style`.
- `text_style` - Overrides the default value for `Dropdown.text_style`.

## Properties

### `buildmenu_styleclass-attributeinstance-attribute`

`menu_style: MenuStyle | None = None`

Overrides the default value for `Dropdown.menu_style`.

### `buildtext_styleclass-attributeinstance-attribute`

`text_style: TextStyle | None = None`

Overrides the default value for `Dropdown.text_style`.

# Duration

A span of time, such as 27 days, 4 hours, 12 minutes, and 3 seconds.

A Duration represents a difference from one point in time to another. The duration may be "negative" if the difference is from a later time to an earlier.

## Properties

- `days` - The number of days in the duration.
- `hours` - The number of hours in the duration.
- `in_days` - Returns total duration in days.
- `in_hours` - Returns total duration in hours.
- `in_microseconds` - Returns total duration in microseconds.
- `in_milliseconds` - Returns total duration in milliseconds.
- `in_minutes` - Returns total duration in minutes.
- `in_seconds` - Returns total duration in seconds.
- `is_negative` - Returns True if duration is negative.
- `microseconds` - The number of microseconds in the duration.
- `milliseconds` - The number of milliseconds in the duration.
- `minutes` - The number of minutes in the duration.
- `seconds` - The number of seconds in the duration.

## Methods

- `abs` - Returns absolute (non-negative) Duration.
- `copy` - Returns a copy of this Duration instance with the given fields replaced with the new values.
- `from_unit` - Creates a Duration from individual time units.

## Properties

### `builddaysclass-attributeinstance-attribute`

`days: int = 0`

The number of days in the duration.

### `buildhoursclass-attributeinstance-attribute`

`hours: int = 0`

The number of hours in the duration.

### `buildin_daysproperty`

`in_days: int`

Returns total duration in days.

### `buildin_hoursproperty`

`in_hours: int`

Returns total duration in hours.

### `buildin_microsecondsproperty`

`in_microseconds: int`

Returns total duration in microseconds.

### `buildin_millisecondsproperty`

`in_milliseconds: int`



Returns total duration in milliseconds.

### **buildin\_minutesproperty**

```
in_minutes: int
```

Returns total duration in minutes.

### **buildin\_secondsproperty**

```
in_seconds: int
```

Returns total duration in seconds.

### **buildis\_negativeproperty**

```
is_negative: bool
```

Returns True if duration is negative.

### **buildmicrosecondsclass-attributeinstance-attribute**

```
microseconds: int = 0
```

The number of microseconds in the duration.

### **buildmillisecondsclass-attributeinstance-attribute**

```
milliseconds: int = 0
```

The number of milliseconds in the duration.

### **buildminutesclass-attributeinstance-attribute**

```
minutes: int = 0
```

The number of minutes in the duration.

### **buildsecondsclass-attributeinstance-attribute**

```
seconds: int = 0
```

The number of seconds in the duration.

## **Methods**

### **deployed\_codeabs**

```
abs() -> Duration
```

Returns absolute (non-negative) Duration.

### **deployed\_codecopy**

```
copy(
 microseconds: int | None = None,
 milliseconds: int | None = None,
 seconds: int | None = None,
 minutes: int | None = None,
 hours: int | None = None,
 days: int | None = None,
) -> Duration
```

Returns a copy of this Duration instance with the given fields replaced with the new values.

### **deployed\_codefrom\_unitclassmethod**

```
from_unit(
 days: int = 0,
 hours: int = 0,
 minutes: int = 0,
 seconds: int = 0,
 milliseconds: int = 0,
 microseconds: int = 0,
) -> Duration
```

Creates a Duration from individual time units.

# Event

Base event payload passed to control event handlers.

Inherits: typing.Generic

## Properties

- control
- data
- name
- page - Page that owns the event source control.
- target - Internal id of the control that emitted this event.

## Properties

**buildcontrolclass-attributeinstance-attribute**

control: EventControlType = field(repr=False)

**bulddataclass-attributeinstance-attribute**

data: Any | None = field(default=None, kw\_only=True)

**buildnameinstance-attribute**

name: str

**buildpageproperty**

page: Page | BasePage

Page that owns the event source control.

**buildtargetproperty**

target: int

Internal id of the control that emitted this event.

# ExpansionPanellistChangeEvent

Payload for ExpansionPanellist.on\_change event.

Inherits: Event

## Properties

- expanded - Whether the toggled panel is expanded (True) or collapsed (False) after the event.
- index - The index of the panel in ExpansionPanellist.controls that was toggled.

## Properties

### buildexpandedinstance-attribute

expanded: bool

Whether the toggled panel is expanded (True) or collapsed (False) after the event.

### buildindexinstance-attribute

index: int

The index of the panel in ExpansionPanellist.controls that was toggled.

Panels with visible set to False are not counted/indexed. This means the value may differ from the panel's position in the original ExpansionPanellist.controls list when some panels are invisible. To map it back, filter ExpansionPanellist.controls to only visible panels and use this index on that filtered list.

# ExpansionTileTheme

Customizes the appearance of descendant ExpansionTile controls.

## Properties

- `animation_style` - Overrides the default value for ExpansionTile.animation\_style.
- `bgcolor` - Overrides the default value for ExpansionTile.bgcolor.
- `clip_behavior` - Overrides the default value for ExpansionTile.clip\_behavior.
- `collapsed_bgcolor` - Overrides the default value for ExpansionTile.collapsed\_bgcolor.
- `collapsed_icon_color` - Overrides the default value for ExpansionTile.collapsed\_icon\_color.
- `collapsed_shape` - Overrides the default value for ExpansionTile.collapsed\_shape.
- `collapsed_text_color` - Overrides the default value for ExpansionTile.collapsed\_text\_color.
- `controls_padding` - Overrides the default value for ExpansionTile.controls\_padding.
- `expanded_alignment` - Overrides the default value for ExpansionTile.expanded\_alignment.
- `icon_color` - Overrides the default value for ExpansionTile.icon\_color.
- `shape` - Overrides the default value for ExpansionTile.shape.
- `text_color` - Overrides the default value for ExpansionTile.text\_color.
- `tile_padding` - Overrides the default value for ExpansionTile.tile\_padding.

## Properties

### **buildanimation\_styleclass-attributeinstance-attribute**

`animation_style: AnimationStyle | None = None`

Overrides the default value for ExpansionTile.animation\_style.

### **buildbgcolorclass-attributeinstance-attribute**

`bgcolor: ColorValue | None = None`

Overrides the default value for ExpansionTile.bgcolor.

### **buildclip\_behaviorclass-attributeinstance-attribute**

`clip_behavior: ClipBehavior | None = None`

Overrides the default value for ExpansionTile.clip\_behavior.

### **buildcollapsed\_bgcolorclass-attributeinstance-attribute**

`collapsed_bgcolor: ColorValue | None = None`

Overrides the default value for ExpansionTile.collapsed\_bgcolor.

### **buildcollapsed\_icon\_colorclass-attributeinstance-attribute**

`collapsed_icon_color: ColorValue | None = None`

Overrides the default value for ExpansionTile.collapsed\_icon\_color.

### **buildcollapsed\_shapeclass-attributeinstance-attribute**

`collapsed_shape: OutlinedBorder | None = None`

Overrides the default value for ExpansionTile.collapsed\_shape.

### **buildcollapsed\_text\_colorclass-attributeinstance-attribute**

`collapsed_text_color: ColorValue | None = None`

Overrides the default value for ExpansionTile.collapsed\_text\_color.

### **buildcontrols\_paddingclass-attributeinstance-attribute**

`controls_padding: PaddingValue | None = None`

Overrides the default value for ExpansionTile.controls\_padding.

### **buildexpanded\_alignmentclass-attributeinstance-attribute**

expanded\_alignment: Alignment | None = None

Overrides the default value for ExpansionTile.expanded\_alignment.

### **buildicon\_colorclass-attributeinstance-attribute**

icon\_color: ColorValue | None = None

Overrides the default value for ExpansionTile.icon\_color.

### **buildshapeclass-attributeinstance-attribute**

shape: OutlinedBorder | None = None

Overrides the default value for ExpansionTile.shape.

### **buildtext\_colorclass-attributeinstance-attribute**

text\_color: ColorValue | None = None

Overrides the default value for ExpansionTile.text\_color.

### **buildtile\_paddingclass-attributeinstance-attribute**

tile\_padding: PaddingValue | None = None

Overrides the default value for ExpansionTile.tile\_padding.

# FilePickerFile

Metadata for a file selected via `FilePicker.pick_files`.

Returned by `FilePicker.pick_files` and used as input context for `FilePickerUploadFile` when uploading selected files.

## Properties

- `bytes` - File contents.
- `id` - Selection-scoped file identifier assigned by Flet.
- `name` - File name (basename), without directory path.
- `path` - Absolute path to the selected file, when available.
- `size` - File size in bytes.

## Properties

### `buildbytesclass-attributeinstance-attribute`

`bytes: bytes | None = None`

File contents.

Returned only when `pick_files` is called with `with_data=True`. Otherwise this value is `None`.

### `buildidinstance-attribute`

`id: int`

Selection-scoped file identifier assigned by Flet.

This value is stable for the current picker selection and is preferred for upload matching in `FilePickerUploadFile`.

### `buildnameinstance-attribute`

`name: str`

File name (basename), without directory path.

### `buildpathclass-attributeinstance-attribute`

`path: str | None = None`

Absolute path to the selected file, when available.

Note

- Web mode always returns `None`.
- On native platforms, this can still be `None` if the platform picker does not expose a filesystem path.

### `buildsizeinstance-attribute`

`size: int`

File size in bytes.

# FilePickerFileType

Defines the file types that can be selected using the FilePicker.

Inherits: `enum.Enum`

## Properties

- ANY - Allows any file type.
- AUDIO - Allows only audio files.
- CUSTOM - Allows only the custom file types specified in the `allowed_extensions` list.
- IMAGE - Allows only image files.
- MEDIA - A combination of VIDEO and IMAGE.
- VIDEO - Allows only video files.

## Properties

**buildANYclass-attributeinstance-attribute**

Allows any file type.

**buildAUDIOclass-attributeinstance-attribute**

Allows only audio files.

**buildCUSTOMclass-attributeinstance-attribute**

Allows only the custom file types specified in the `allowed_extensions` list.

**buildIMAGEclass-attributeinstance-attribute**

Allows only image files.

**buildMEDIAClass-attributeinstance-attribute**

A combination of VIDEO and IMAGE.

**buildVIDEOclass-attributeinstance-attribute**

Allows only video files.

# FilePickerUploadEvent

Event emitted when a file is uploaded via FilePicker.upload method.

Inherits: Event

## Properties

- `error` - An error message if the upload failed.
- `file_name` - The name of the uploaded file.
- `progress` - A value from 0.0 to 1.0 representing the progress of the upload.

## Properties

### builderrorclass-attributeinstance-attribute

`error: str | None = None`

An error message if the upload failed.

### buildfile\_nameinstance-attribute

`file_name: str`

The name of the uploaded file.

### buildprogressclass-attributeinstance-attribute

`progress: float | None = None`

A value from 0.0 to 1.0 representing the progress of the upload.



# FilePickerUploadFile

Upload descriptor for one file selected by FilePicker.

Instances are passed to FilePicker.upload. During upload, Flet resolves the selected file by id first and, when id is absent or not found, falls back to name.

At least one of id or name should be provided.

## Properties

- `id` - Selected file identifier returned by FilePicker.pick\_files.
- `method` - HTTP method used for the upload request, usually PUT or POST.
- `name` - Selected file name used as fallback lookup when id is missing or does not match any currently selected file.
- `upload_url` - Upload destination URL.

## Properties

### buildidclass-attributeinstance-attribute

`id: int | None = None`

Selected file identifier returned by FilePicker.pick\_files.

This is the preferred lookup key when both id and name are specified.

### buildmethodclass-attributeinstance-attribute

`method: str = 'PUT'`

HTTP method used for the upload request, usually PUT or POST.

### buildnameclass-attributeinstance-attribute

`name: str | None = None`

Selected file name used as fallback lookup when id is missing or does not match any currently selected file.

### buildupload\_urlinstance-attribute

`upload_url: str`

Upload destination URL.

Can be an absolute URL or a page-relative URL returned by Page.get\_upload\_url.

# FilledButtonTheme

Customizes the appearance of FilledButton across the app.

## Properties

- `style` - Overrides the default value of `Button.style` in all descendant `FilledButton` controls.

## Properties

**buildstyleclass-attributeinstance-attribute**

`style: ButtonStyle | None = None`

Overrides the default value of `Button.style` in all descendant `FilledButton` controls.

# FilterQuality

Quality levels for image sampling in Image and DecorationImage objects.

Inherits: enum.Enum

## Properties

- HIGH - Best possible quality when scaling up images by scale factors larger than 5-10x.
- LOW - Better quality than none, faster than medium.
- MEDIUM - The best all around filtering method that is only worse than high at extremely large scale factors.
- NONE - The fastest filtering method, albeit also the lowest quality.

## Examples

### Filter Quality showcase

[play\\_arrow](#)Try Online

```
import flet as ft

IMAGE_URL = "https://picsum.photos/id/1025/64/64"

def showcase_card(quality: ft.FilterQuality) -> ft.Container:
 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(quality.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=240,
 height=120,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 clip_behavior=ft.ClipBehavior.HARD_EDGE,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Image(
 src=IMAGE_URL,
 width=240,
 height=120,
 fit=ft.BoxFit.FILL,
 filter_quality=quality,
),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="FilterQuality Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare image sampling quality while scaling the "
 "same source image."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(quality) for quality in ft.FilterQuality
],
),
],
),
),
)
```

```
),
)
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildHIGHclass-attributeinstance-attribute**

Best possible quality when scaling up images by scale factors larger than 5-10x. When images are scaled down, this can be worse than medium for scales smaller than 0.5x, or when animating the scale factor. This option is also the slowest.

### **buildLOWclass-attributeinstance-attribute**

Better quality than none, faster than medium.

### **buildMEDIUMclass-attributeinstance-attribute**

The best all around filtering method that is only worse than high at extremely large scale factors.

### **buildNONEclass-attributeinstance-attribute**

The fastest filtering method, albeit also the lowest quality.

# Finder

::: flet.testing.finder.Finder

# FletException

Base class for all Flet exceptions.

See these subclasses/implementations:

- `FletUnsupportedPlatformException`
- `FletUnimplementedPlatformException`
- `FletPageDisconnectedException`

Inherits: `Exception`

# FletPageDisconnectedException

Thrown when the page is disconnected.

Inherits: `FletException`

<https://flet.dev/docs/types/flettestapp>

# FletTestApp

::: flet.testing.flet\_test\_app.FletTestApp



# FletUnimplementedPlatformException

Thrown by operations that have not been implemented yet.

Inherits: `FletUnsupportedPlatformException`

# FletUnsupportedPlatformException

Thrown by operations that are not supported on the current platform.

Inherits: `FletException`

# Flip

Configuration for `LayoutControl.flip`.

Mirrors a control across its x and/or y axis.

## Properties

- `filter_quality` - The filter quality with which to apply this transform as a bitmap operation.
- `flip_x` - Whether to flip the x axis.
- `flip_y` - Whether to flip the y axis.
- `origin` - The origin of the coordinate system (relative to the upper left corner).
- `transform_hit_tests` - Whether to apply the transformation when performing hit tests.

## Properties

### `buildfilter_qualityclass-attributeinstance-attribute`

`filter_quality: FilterQuality | None = None`

The filter quality with which to apply this transform as a bitmap operation.

### `buildflip_xclass-attributeinstance-attribute`

`flip_x: bool = False`

Whether to flip the x axis.

### `buildflip_yclass-attributeinstance-attribute`

`flip_y: bool = False`

Whether to flip the y axis.

### `buildoriginclass-attributeinstance-attribute`

`origin: Offset | None = None`

The origin of the coordinate system (relative to the upper left corner).

### `buildtransform_hit_testsclass-attributeinstance-attribute`

`transform_hit_tests: bool = True`

Whether to apply the transformation when performing hit tests.

# FloatingActionButtonLocation

Defines a position for the FloatingActionButton.

See FloatingActionButtonLocation from Flutter documentation for placement location examples.

Inherits: enum.Enum

## Properties

- CENTER\_DOCKED
- CENTER\_FLOAT
- CENTER\_TOP
- END\_CONTAINED
- END\_DOCKED
- END\_FLOAT
- END\_TOP
- MINI\_CENTER\_DOCKED
- MINI\_CENTER\_FLOAT
- MINI\_CENTER\_TOP
- MINI\_END\_DOCKED
- MINI\_END\_FLOAT
- MINI\_END\_TOP
- MINI\_START\_DOCKED
- MINI\_START\_FLOAT
- MINI\_START\_TOP
- START\_DOCKED
- START\_FLOAT
- START\_TOP

## Examples

### Floating Action Button Location showcase

play\_arrowTry Online

```
import flet as ft

def showcase_card(location: ft.FloatingActionButtonLocation) -> ft.Container:
 mini = location.name.startswith("MINI_")
 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(location.name, weight=ft.FontWeight.BOLD),
 ft.Pagelet(
 width=260,
 height=200,
 bgcolor=ft.Colors.SURFACE,
 appbar=ft.AppBar(
 title=ft.Text("AppBar", size=12),
 center_title=True,
 toolbar_height=38,
 bgcolor=ft.Colors.PRIMARY_CONTAINER,
),
```

```

 content=ft.Container(
 alignment=ft.Alignment.CENTER,
 content=ft.Text("Body", size=12),
),
 bottom_appbar=ft.BottomAppBar(
 height=42,
 bgcolor=ft.Colors.SECONDARY_CONTAINER,
),
 floating_action_button=ft.FloatingActionButton(
 icon=ft.Icons.ADD,
 mini=mini,
 bgcolor=ft.Colors.LIGHT_BLUE_400,
),
 floating_action_button_location=location,
],
),
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="FloatingActionButtonLocation Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare FloatingActionButton placement presets."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(location)
 for location in ft.FloatingActionButtonLocation
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

buildCENTER\_DOCKEDclass-attributeinstance-attribute

buildCENTER\_FLOATclass-attributeinstance-attribute

buildCENTER\_TOPclass-attributeinstance-attribute

buildEND\_CONTAINEDclass-attributeinstance-attribute

buildEND\_DOCKEDclass-attributeinstance-attribute

buildEND\_FLOATclass-attributeinstance-attribute

buildEND\_TOPclass-attributeinstance-attribute

buildMINI\_CENTER\_DOCKEDclass-attributeinstance-attribute

buildMINI\_CENTER\_FLOATclass-attributeinstance-attribute

buildMINI\_CENTER\_TOPclass-attributeinstance-attribute

buildMINI\_END\_DOCKEDclass-attributeinstance-attribute

buildMINI\_END\_FLOATclass-attributeinstance-attribute

buildMINI\_END\_TOPclass-attributeinstance-attribute

buildMINI\_START\_DOCKEDclass-attributeinstance-attribute

buildMINI\_START\_FLOATclass-attributeinstance-attribute  
buildMINI\_START\_TOPclass-attributeinstance-attribute  
buildSTART\_DOCKEDclass-attributeinstance-attribute  
buildSTART\_FLOATclass-attributeinstance-attribute  
buildSTART\_TOPclass-attributeinstance-attribute

# FloatingActionButtonTheme

Customizes the appearance of FloatingActionButton across the app.

## Properties

- `bgcolor` - Color to be used for the unselected, enabled FloatingActionButton's background.
- `disabled_elevation` - The z-coordinate to be used for the disabled FloatingActionButton's elevation foreground.
- `elevation` - The z-coordinate to be used for the unselected, enabled FloatingActionButton's elevation foreground.
- `enable_feedback` - If specified, defines the feedback property for FloatingActionButton.
- `extended_padding` - The padding for a FloatingActionButton's that has both icon and content.
- `extended_size_constraints` - Overrides the default size constraints of FloatingActionButton that has both icon and content.
- `focus_color` - The color to use for filling the button when the button has input focus.
- `focus_elevation` - The z-coordinate at which to place this button relative to its parent when the button has the input focus.
- `foreground_color` - Color to be used for the unselected, enabled FloatingActionButton's foreground.
- `highlight_elevation` - The z-coordinate to be used for the selected, enabled FloatingActionButton's elevation foreground.
- `hover_color` - The color to use for filling the button when the button has a pointer hovering over it.
- `hover_elevation` - The z-coordinate at which to place this button relative to its parent when the button is enabled and has a pointer hovering over it.
- `icon_label_spacing` - The spacing between the icon and the label for FloatingActionButton.
- `shape` - Overrides the default value of FloatingActionButton.shape in all descendant FloatingActionButton controls.
- `size_constraints` - Overrides the default size constraints of FloatingActionButton that has either icon or content and is not a mini button.
- `splash_color` - Overrides the default value of FloatingActionButton.splash\_color in all descendant FloatingActionButton controls.
- `text_style` - Text style merged into default text style of FloatingActionButton.content.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Color to be used for the unselected, enabled FloatingActionButton's background.

### `builddisabled_elevationclass-attributeinstance-attribute`

`disabled_elevation: Number | None = None`

The z-coordinate to be used for the disabled FloatingActionButton's elevation foreground.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

The z-coordinate to be used for the unselected, enabled FloatingActionButton's elevation foreground.

### `buildenable_feedbackclass-attributeinstance-attribute`

`enable_feedback: bool | None = None`

If specified, defines the feedback property for FloatingActionButton.

### `buildextended_paddingclass-attributeinstance-attribute`

`extended_padding: PaddingValue | None = None`

The padding for a FloatingActionButton's that has both icon and content.

### **buildextended\_size\_constraintsclass-attributeinstance-attribute**

`extended_size_constraints: BoxConstraints | None = None`

Overrides the default size constraints of `FloatingActionButton` that has both icon and content.

### **buildfocus\_colorclass-attributeinstance-attribute**

`focus_color: ColorValue | None = None`

The color to use for filling the button when the button has input focus.

### **buildfocus\_elevationclass-attributeinstance-attribute**

`focus_elevation: Number | None = None`

The z-coordinate at which to place this button relative to its parent when the button has the input focus.

### **buildforeground\_colorclass-attributeinstance-attribute**

`foreground_color: ColorValue | None = None`

Color to be used for the unselected, enabled `FloatingActionButton`'s foreground.

### **buildhighlight\_elevationclass-attributeinstance-attribute**

`highlight_elevation: Number | None = None`

The z-coordinate to be used for the selected, enabled `FloatingActionButton`'s elevation foreground.

### **buildhover\_colorclass-attributeinstance-attribute**

`hover_color: ColorValue | None = None`

The color to use for filling the button when the button has a pointer hovering over it.

### **buildhover\_elevationclass-attributeinstance-attribute**

`hover_elevation: Number | None = None`

The z-coordinate at which to place this button relative to its parent when the button is enabled and has a pointer hovering over it.

### **buildicon\_label\_spacingclass-attributeinstance-attribute**

`icon_label_spacing: Number | None = None`

The spacing between the icon and the label for `FloatingActionButton`.

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

Overrides the default value of `FloatingActionButton.shape` in all descendant `FloatingActionButton` controls.

### **buildsize\_constraintsclass-attributeinstance-attribute**

`size_constraints: BoxConstraints | None = None`

Overrides the default size constraints of `FloatingActionButton` that has either icon or content and is not a mini button.

### **buildsplash\_colorclass-attributeinstance-attribute**

`splash_color: ColorValue | None = None`

Overrides the default value of `FloatingActionButton.splash_color` in all descendant `FloatingActionButton` controls.

### **buildtext\_styleclass-attributeinstance-attribute**

`text_style: TextStyle | None = None`

Text style merged into default text style of `FloatingActionButton.content`.



# FontWeight

The thickness of the glyphs used to draw the text.

Inherits: `enum.Enum`

## Properties

- **BOLD** - A commonly used font weight that is heavier than normal, equal to w700.
- **NORMAL** - The default font weight, equal to w400.
- **W\_100** - Thin, the least thick.
- **W\_200** - Extra-light.
- **W\_300** - Light.
- **W\_400** - Normal / regular / plain.
- **W\_500** - Medium.
- **W\_600** - Semi-bold.
- **W\_700** - Bold.
- **W\_800** - Extra-bold.
- **W\_900** - Black, the most thick.

## Examples

### Font Weight showcase

`play_arrow` Try Online

```
import flet as ft

SAMPLE_TEXT = "Sphinx of black quartz"

def showcase_card(weight: ft.FontWeight) -> ft.Container:
 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(weight.name, size=12, color=ft.Colors.ON_SURFACE_VARIANT),
 ft.Text(SAMPLE_TEXT, weight=weight, size=24),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="FontWeight Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare text thickness across all FontWeight values."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(weight) for weight in ft.FontWeight],
),
],
),
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildBOLD**class-attributeinstance-attribute

A commonly used font weight that is heavier than normal, equal to w700.

**buildNORMAL**class-attributeinstance-attribute

The default font weight, equal to w400.

**buildW\_100**class-attributeinstance-attribute

Thin, the least thick.

**buildW\_200**class-attributeinstance-attribute

Extra-light.

**buildW\_300**class-attributeinstance-attribute

Light.

**buildW\_400**class-attributeinstance-attribute

Normal / regular / plain.

**buildW\_500**class-attributeinstance-attribute

Medium.

**buildW\_600**class-attributeinstance-attribute

Semi-bold.

**buildW\_700**class-attributeinstance-attribute

Bold.

**buildW\_800**class-attributeinstance-attribute

Extra-bold.

**buildW\_900**class-attributeinstance-attribute

Black, the most thick.

# Gradient

A shader that renders a color gradient.

There are several types of gradients:

- LinearGradient
- RadialGradient
- SweepGradient

## Properties

- colors - The colors the gradient should obtain at each of the stops.
- rotation - The rotation of the gradient in radians, around the center-point of its bounding box.
- stops - A list of values from 0.0 to 1.0 that denote fractions along the gradient.
- tile\_mode - How this gradient should tile the plane beyond in the region before begin and after end.

## Properties

### buildcolorsinstance-attribute

colors: list[str]

The colors the gradient should obtain at each of the stops. This list must contain at least two colors.

If stops is provided, this list must have the same length as it.

### buildrotationclass-attributeinstance-attribute

rotation: Number | None = None

The rotation of the gradient in radians, around the center-point of its bounding box.

### buildstopsclass-attributeinstance-attribute

stops: list[Number] | None = None

A list of values from 0.0 to 1.0 that denote fractions along the gradient.

If provided, this list must have the same length as colors. If the first value is not 0.0, then a stop with position 0.0 and a color equal to the first color in colors is implied. If the last value is not 1.0, then a stop with position 1.0 and a color equal to the last color in colors is implied.

### buildtile\_modeclass-attributeinstance-attribute

tile\_mode: GradientTileMode = GradientTileMode.CLAMP

How this gradient should tile the plane beyond in the region before begin and after end.

## Examples

### Containers with gradients

play\_arrowTry Online

```
import math

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.vertical_alignment = ft.MainAxisAlignment.CENTER

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Row(
 alignment=ft.MainAxisAlignment.CENTER,
 scroll=ft.ScrollMode.AUTO,
 controls=[
 ft.Container(
 content=ft.Text("Linear gradient"),
```



# GradientTileMode

Defines what happens at the edge of a gradient. More information about GradientTileMode here.

Inherits: enum.Enum

## Properties

- CLAMP - Samples beyond the edge are clamped to the nearest color in the defined inner area.
- DECAL - Samples beyond the edge are treated as transparent black.
- MIRROR - Samples beyond the edge are mirrored back and forth across the defined area.
- REPEATED - Samples beyond the edge are repeated from the far end of the defined area.

## Examples

### Gradient Tile Mode showcase

[play\\_arrowTry Online](#)

```
import flet as ft
```

```
def showcase_card(mode: ft.GradientTileMode) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(mode.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=240,
 height=120,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 gradient=ft.RadialGradient(
 radius=0.22,
 colors=[ft.Colors.PRIMARY, ft.Colors.AMBER_400],
 tile_mode=mode,
),
],
),
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="GradientTileMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare how gradients behave outside their defined "
 "paint region."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(mode) for mode in ft.GradientTileMode],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildCLAMPclass-attributeinstance-attribute**

Samples beyond the edge are clamped to the nearest color in the defined inner area. [More information here.](#)

### **buildDECALclass-attributeinstance-attribute**

Samples beyond the edge are treated as transparent black. [More information here.](#)

### **buildMIRRORclass-attributeinstance-attribute**

Samples beyond the edge are mirrored back and forth across the defined area. [More information here.](#)

### **buildREPEATEDclass-attributeinstance-attribute**

Samples beyond the edge are repeated from the far end of the defined area. [More information here.](#)

# GyroscopeReadingEvent

Discrete reading from a gyroscope.

Gyroscope sample containing device rotation rate (rad/s) around each axis plus the microsecond timestamp.

Inherits: Event

## Properties

- timestamp - Event timestamp.
- x - Rotation rate around the X axis, in rad/s.
- y - Rotation rate around the Y axis, in rad/s.
- z - Rotation rate around the Z axis, in rad/s.

## Properties

### buildtimestampinstance-attribute

timestamp: datetime

Event timestamp.

### buildxinstance-attribute

x: float

Rotation rate around the X axis, in rad/s.

### buildyinstance-attribute

y: float

Rotation rate around the Y axis, in rad/s.

### buildzinstance-attribute

z: float

Rotation rate around the Z axis, in rad/s.

# HoverEvent

PointerEvent



# IconButtonTheme

Customizes the appearance of IconButton across the app.

## Properties

- `style` - Overrides the default value of `IconButton.style` in all descendant `IconButton` controls.

## Properties

**buildstyleclass-attributeinstance-attribute**

`style: ButtonStyle | None = None`

Overrides the default value of `IconButton.style` in all descendant `IconButton` controls.

# Icons

## Properties

- ABC
- ABC\_OUTLINED
- ABC\_ROUNDED
- ABC\_SHARP
- AC\_UNIT
- AC\_UNIT\_OUTLINED
- AC\_UNIT\_ROUNDED
- AC\_UNIT\_SHARP
- ACCESS\_ALARM
- ACCESS\_ALARM\_OUTLINED
- ACCESS\_ALARM\_ROUNDED
- ACCESS\_ALARM\_SHARP
- ACCESS\_ALARMS
- ACCESS\_ALARMS\_OUTLINED
- ACCESS\_ALARMS\_ROUNDED
- ACCESS\_ALARMS\_SHARP
- ACCESS\_TIME
- ACCESS\_TIME\_FILLED
- ACCESS\_TIME\_FILLED\_OUTLINED
- ACCESS\_TIME\_FILLED\_ROUNDED
- ACCESS\_TIME\_FILLED\_SHARP
- ACCESS\_TIME\_OUTLINED
- ACCESS\_TIME\_ROUNDED
- ACCESS\_TIME\_SHARP
- ACCESSIBILITY
- ACCESSIBILITY\_NEW
- ACCESSIBILITY\_NEW\_OUTLINED
- ACCESSIBILITY\_NEW\_ROUNDED
- ACCESSIBILITY\_NEW\_SHARP
- ACCESSIBILITY\_OUTLINED
- ACCESSIBILITY\_ROUNDED
- ACCESSIBILITY\_SHARP
- ACCESSIBLE
- ACCESSIBLE\_FORWARD
- ACCESSIBLE\_FORWARD\_OUTLINED
- ACCESSIBLE\_FORWARD\_ROUNDED
- ACCESSIBLE\_FORWARD\_SHARP

- ACCESSIBLE\_OUTLINED
- ACCESSIBLE\_ROUNDED
- ACCESSIBLE\_SHARP
- ACCOUNT\_BALANCE
- ACCOUNT\_BALANCE\_OUTLINED
- ACCOUNT\_BALANCE\_ROUNDED
- ACCOUNT\_BALANCE\_SHARP
- ACCOUNT\_BALANCE\_WALLET
- ACCOUNT\_BALANCE\_WALLET\_OUTLINED
- ACCOUNT\_BALANCE\_WALLET\_ROUNDED
- ACCOUNT\_BALANCE\_WALLET\_SHARP
- ACCOUNT\_BOX
- ACCOUNT\_BOX\_OUTLINED
- ACCOUNT\_BOX\_ROUNDED
- ACCOUNT\_BOX\_SHARP
- ACCOUNT\_CIRCLE
- ACCOUNT\_CIRCLE\_OUTLINED
- ACCOUNT\_CIRCLE\_ROUNDED
- ACCOUNT\_CIRCLE\_SHARP
- ACCOUNT\_TREE
- ACCOUNT\_TREE\_OUTLINED
- ACCOUNT\_TREE\_ROUNDED
- ACCOUNT\_TREE\_SHARP
- AD\_UNITS
- AD\_UNITS\_OUTLINED
- AD\_UNITS\_ROUNDED
- AD\_UNITS\_SHARP
- ADB
- ADB\_OUTLINED
- ADB\_ROUNDED
- ADB\_SHARP
- ADD
- ADD\_A\_PHOTO
- ADD\_A\_PHOTO\_OUTLINED
- ADD\_A\_PHOTO\_ROUNDED
- ADD\_A\_PHOTO\_SHARP
- ADD\_ALARM
- ADD\_ALARM\_OUTLINED
- ADD\_ALARM\_ROUNDED
- ADD\_ALARM\_SHARP
- ADD\_ALERT

- ADD\_ALERT\_OUTLINED
- ADD\_ALERT\_ROUNDED
- ADD\_ALERT\_SHARP
- ADD\_BOX
- ADD\_BOX\_OUTLINED
- ADD\_BOX\_ROUNDED
- ADD\_BOX\_SHARP
- ADD\_BUSINESS
- ADD\_BUSINESS\_OUTLINED
- ADD\_BUSINESS\_ROUNDED
- ADD\_BUSINESS\_SHARP
- ADD\_CALL
- ADD\_CARD
- ADD\_CARD\_OUTLINED
- ADD\_CARD\_ROUNDED
- ADD\_CARD\_SHARP
- ADD\_CHART
- ADD\_CHART\_OUTLINED
- ADD\_CHART\_ROUNDED
- ADD\_CHART\_SHARP
- ADD\_CIRCLE
- ADD\_CIRCLE\_OUTLINE
- ADD\_CIRCLE\_OUTLINE\_OUTLINED
- ADD\_CIRCLE\_OUTLINE\_ROUNDED
- ADD\_CIRCLE\_OUTLINE\_SHARP
- ADD\_CIRCLE\_OUTLINED
- ADD\_CIRCLE\_ROUNDED
- ADD\_CIRCLE\_SHARP
- ADD\_COMMENT
- ADD\_COMMENT\_OUTLINED
- ADD\_COMMENT\_ROUNDED
- ADD\_COMMENT\_SHARP
- ADD\_HOME
- ADD\_HOME\_OUTLINED
- ADD\_HOME\_ROUNDED
- ADD\_HOME\_SHARP
- ADD\_HOME\_WORK
- ADD\_HOME\_WORK\_OUTLINED
- ADD\_HOME\_WORK\_ROUNDED
- ADD\_HOME\_WORK\_SHARP
- ADD\_IC\_CALL

- ADD\_IC\_CALL\_OUTLINED
- ADD\_IC\_CALL\_ROUNDED
- ADD\_IC\_CALL\_SHARP
- ADD\_LINK
- ADD\_LINK\_OUTLINED
- ADD\_LINK\_ROUNDED
- ADD\_LINK\_SHARP
- ADD\_LOCATION
- ADD\_LOCATION\_ALT
- ADD\_LOCATION\_ALT\_OUTLINED
- ADD\_LOCATION\_ALT\_ROUNDED
- ADD\_LOCATION\_ALT\_SHARP
- ADD\_LOCATION\_OUTLINED
- ADD\_LOCATION\_ROUNDED
- ADD\_LOCATION\_SHARP
- ADD\_MODERATOR
- ADD\_MODERATOR\_OUTLINED
- ADD\_MODERATOR\_ROUNDED
- ADD\_MODERATOR\_SHARP
- ADD\_OUTLINED
- ADD\_PHOTO\_ALTERNATE
- ADD\_PHOTO\_ALTERNATE\_OUTLINED
- ADD\_PHOTO\_ALTERNATE\_ROUNDED
- ADD\_PHOTO\_ALTERNATE\_SHARP
- ADD\_REACTION
- ADD\_REACTION\_OUTLINED
- ADD\_REACTION\_ROUNDED
- ADD\_REACTION\_SHARP
- ADD\_ROAD
- ADD\_ROAD\_OUTLINED
- ADD\_ROAD\_ROUNDED
- ADD\_ROAD\_SHARP
- ADD\_ROUNDED
- ADD\_SHARP
- ADD\_SHOPPING\_CART
- ADD\_SHOPPING\_CART\_OUTLINED
- ADD\_SHOPPING\_CART\_ROUNDED
- ADD\_SHOPPING\_CART\_SHARP
- ADD\_TASK
- ADD\_TASK\_OUTLINED
- ADD\_TASK\_ROUNDED

- ADD\_TASK\_SHARP
- ADD\_TO\_DRIVE
- ADD\_TO\_DRIVE\_OUTLINED
- ADD\_TO\_DRIVE\_ROUNDED
- ADD\_TO\_DRIVE\_SHARP
- ADD\_TO\_HOME\_SCREEN
- ADD\_TO\_HOME\_SCREEN\_OUTLINED
- ADD\_TO\_HOME\_SCREEN\_ROUNDED
- ADD\_TO\_HOME\_SCREEN\_SHARP
- ADD\_TO\_PHOTOS
- ADD\_TO\_PHOTOS\_OUTLINED
- ADD\_TO\_PHOTOS\_ROUNDED
- ADD\_TO\_PHOTOS\_SHARP
- ADD\_TO\_QUEUE
- ADD\_TO\_QUEUE\_OUTLINED
- ADD\_TO\_QUEUE\_ROUNDED
- ADD\_TO\_QUEUE\_SHARP
- ADDCHART
- ADDCHART\_OUTLINED
- ADDCHART\_ROUNDED
- ADDCHART\_SHARP
- ADF\_SCANNER
- ADF\_SCANNER\_OUTLINED
- ADF\_SCANNER\_ROUNDED
- ADF\_SCANNER\_SHARP
- ADJUST
- ADJUST\_OUTLINED
- ADJUST\_ROUNDED
- ADJUST\_SHARP
- ADMIN\_PANEL\_SETTINGS
- ADMIN\_PANEL\_SETTINGS\_OUTLINED
- ADMIN\_PANEL\_SETTINGS\_ROUNDED
- ADMIN\_PANEL\_SETTINGS\_SHARP
- ADOBE
- ADOBE\_OUTLINED
- ADOBE\_ROUNDED
- ADOBE\_SHARP
- ADS\_CLICK
- ADS\_CLICK\_OUTLINED
- ADS\_CLICK\_ROUNDED
- ADS\_CLICK\_SHARP

- AGRICULTURE
- AGRICULTURE\_OUTLINED
- AGRICULTURE\_ROUNDED
- AGRICULTURE\_SHARP
- AIR
- AIR\_OUTLINED
- AIR\_ROUNDED
- AIR\_SHARP
- AIRLINE\_SEAT\_FLAT
- AIRLINE\_SEAT\_FLAT\_ANGLED
- AIRLINE\_SEAT\_FLAT\_ANGLED\_OUTLINED
- AIRLINE\_SEAT\_FLAT\_ANGLED\_ROUNDED
- AIRLINE\_SEAT\_FLAT\_ANGLED\_SHARP
- AIRLINE\_SEAT\_FLAT\_OUTLINED
- AIRLINE\_SEAT\_FLAT\_ROUNDED
- AIRLINE\_SEAT\_FLAT\_SHARP
- AIRLINE\_SEAT\_INDIVIDUAL\_SUITE
- AIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_OUTLINED
- AIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_ROUNDED
- AIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_SHARP
- AIRLINE\_SEAT\_LEGROOM\_EXTRA
- AIRLINE\_SEAT\_LEGROOM\_EXTRA\_OUTLINED
- AIRLINE\_SEAT\_LEGROOM\_EXTRA\_ROUNDED
- AIRLINE\_SEAT\_LEGROOM\_EXTRA\_SHARP
- AIRLINE\_SEAT\_LEGROOM\_NORMAL
- AIRLINE\_SEAT\_LEGROOM\_NORMAL\_OUTLINED
- AIRLINE\_SEAT\_LEGROOM\_NORMAL\_ROUNDED
- AIRLINE\_SEAT\_LEGROOM\_NORMAL\_SHARP
- AIRLINE\_SEAT\_LEGROOM\_REDUCED
- AIRLINE\_SEAT\_LEGROOM\_REDUCED\_OUTLINED
- AIRLINE\_SEAT\_LEGROOM\_REDUCED\_ROUNDED
- AIRLINE\_SEAT\_LEGROOM\_REDUCED\_SHARP
- AIRLINE\_SEAT\_RECLINE\_EXTRA
- AIRLINE\_SEAT\_RECLINE\_EXTRA\_OUTLINED
- AIRLINE\_SEAT\_RECLINE\_EXTRA\_ROUNDED
- AIRLINE\_SEAT\_RECLINE\_EXTRA\_SHARP
- AIRLINE\_SEAT\_RECLINE\_NORMAL
- AIRLINE\_SEAT\_RECLINE\_NORMAL\_OUTLINED
- AIRLINE\_SEAT\_RECLINE\_NORMAL\_ROUNDED
- AIRLINE\_SEAT\_RECLINE\_NORMAL\_SHARP
- AIRLINE\_STOPS

- AIRLINE\_STOPS\_OUTLINED
- AIRLINE\_STOPS\_ROUNDED
- AIRLINE\_STOPS\_SHARP
- AIRLINES
- AIRLINES\_OUTLINED
- AIRLINES\_ROUNDED
- AIRLINES\_SHARP
- AIRPLANE\_TICKET
- AIRPLANE\_TICKET\_OUTLINED
- AIRPLANE\_TICKET\_ROUNDED
- AIRPLANE\_TICKET\_SHARP
- AIRPLANEMODE\_ACTIVE
- AIRPLANEMODE\_ACTIVE\_OUTLINED
- AIRPLANEMODE\_ACTIVE\_ROUNDED
- AIRPLANEMODE\_ACTIVE\_SHARP
- AIRPLANEMODE\_INACTIVE
- AIRPLANEMODE\_INACTIVE\_OUTLINED
- AIRPLANEMODE\_INACTIVE\_ROUNDED
- AIRPLANEMODE\_INACTIVE\_SHARP
- AIRPLANEMODE\_OFF
- AIRPLANEMODE\_OFF\_OUTLINED
- AIRPLANEMODE\_OFF\_ROUNDED
- AIRPLANEMODE\_OFF\_SHARP
- AIRPLANEMODE\_ON
- AIRPLANEMODE\_ON\_OUTLINED
- AIRPLANEMODE\_ON\_ROUNDED
- AIRPLANEMODE\_ON\_SHARP
- AIRPLAY
- AIRPLAY\_OUTLINED
- AIRPLAY\_ROUNDED
- AIRPLAY\_SHARP
- AIRPORT\_SHUTTLE
- AIRPORT\_SHUTTLE\_OUTLINED
- AIRPORT\_SHUTTLE\_ROUNDED
- AIRPORT\_SHUTTLE\_SHARP
- ALARM
- ALARM\_ADD
- ALARM\_ADD\_OUTLINED
- ALARM\_ADD\_ROUNDED
- ALARM\_ADD\_SHARP
- ALARM\_OFF



- ALARM\_OFF\_OUTLINED
- ALARM\_OFF\_ROUNDED
- ALARM\_OFF\_SHARP
- ALARM\_ON
- ALARM\_ON\_OUTLINED
- ALARM\_ON\_ROUNDED
- ALARM\_ON\_SHARP
- ALARM\_OUTLINED
- ALARM\_ROUNDED
- ALARM\_SHARP
- ALBUM
- ALBUM\_OUTLINED
- ALBUM\_ROUNDED
- ALBUM\_SHARP
- ALIGN\_HORIZONTAL\_CENTER
- ALIGN\_HORIZONTAL\_CENTER\_OUTLINED
- ALIGN\_HORIZONTAL\_CENTER\_ROUNDED
- ALIGN\_HORIZONTAL\_CENTER\_SHARP
- ALIGN\_HORIZONTAL\_LEFT
- ALIGN\_HORIZONTAL\_LEFT\_OUTLINED
- ALIGN\_HORIZONTAL\_LEFT\_ROUNDED
- ALIGN\_HORIZONTAL\_LEFT\_SHARP
- ALIGN\_HORIZONTAL\_RIGHT
- ALIGN\_HORIZONTAL\_RIGHT\_OUTLINED
- ALIGN\_HORIZONTAL\_RIGHT\_ROUNDED
- ALIGN\_HORIZONTAL\_RIGHT\_SHARP
- ALIGN\_VERTICAL\_BOTTOM
- ALIGN\_VERTICAL\_BOTTOM\_OUTLINED
- ALIGN\_VERTICAL\_BOTTOM\_ROUNDED
- ALIGN\_VERTICAL\_BOTTOM\_SHARP
- ALIGN\_VERTICAL\_CENTER
- ALIGN\_VERTICAL\_CENTER\_OUTLINED
- ALIGN\_VERTICAL\_CENTER\_ROUNDED
- ALIGN\_VERTICAL\_CENTER\_SHARP
- ALIGN\_VERTICAL\_TOP
- ALIGN\_VERTICAL\_TOP\_OUTLINED
- ALIGN\_VERTICAL\_TOP\_ROUNDED
- ALIGN\_VERTICAL\_TOP\_SHARP
- ALL\_INBOX
- ALL\_INBOX\_OUTLINED
- ALL\_INBOX\_ROUNDED

- ALL\_INBOX\_SHARP
- ALL\_INCLUSIVE
- ALL\_INCLUSIVE\_OUTLINED
- ALL\_INCLUSIVE\_ROUNDED
- ALL\_INCLUSIVE\_SHARP
- ALL\_OUT
- ALL\_OUT\_OUTLINED
- ALL\_OUT\_ROUNDED
- ALL\_OUT\_SHARP
- ALT\_ROUTE
- ALT\_ROUTE\_OUTLINED
- ALT\_ROUTE\_ROUNDED
- ALT\_ROUTE\_SHARP
- ALTERNATE\_EMAIL
- ALTERNATE\_EMAIL\_OUTLINED
- ALTERNATE\_EMAIL\_ROUNDED
- ALTERNATE\_EMAIL\_SHARP
- AMP\_STORIES
- AMP\_STORIES\_OUTLINED
- AMP\_STORIES\_ROUNDED
- AMP\_STORIES\_SHARP
- ANALYTICS
- ANALYTICS\_OUTLINED
- ANALYTICS\_ROUNDED
- ANALYTICS\_SHARP
- ANCHOR
- ANCHOR\_OUTLINED
- ANCHOR\_ROUNDED
- ANCHOR\_SHARP
- ANDROID
- ANDROID\_OUTLINED
- ANDROID\_ROUNDED
- ANDROID\_SHARP
- ANIMATION
- ANIMATION\_OUTLINED
- ANIMATION\_ROUNDED
- ANIMATION\_SHARP
- ANNOUNCEMENT
- ANNOUNCEMENT\_OUTLINED
- ANNOUNCEMENT\_ROUNDED
- ANNOUNCEMENT\_SHARP

- AOD
- AOD\_OUTLINED
- AOD\_ROUNDED
- AOD\_SHARP
- APARTMENT
- APARTMENT\_OUTLINED
- APARTMENT\_ROUNDED
- APARTMENT\_SHARP
- API
- API\_OUTLINED
- API\_ROUNDED
- API\_SHARP
- APP\_BLOCKING
- APP\_BLOCKING\_OUTLINED
- APP\_BLOCKING\_ROUNDED
- APP\_BLOCKING\_SHARP
- APP\_REGISTRATION
- APP\_REGISTRATION\_OUTLINED
- APP\_REGISTRATION\_ROUNDED
- APP\_REGISTRATION\_SHARP
- APP\_SETTINGS\_ALT
- APP\_SETTINGS\_ALT\_OUTLINED
- APP\_SETTINGS\_ALT\_ROUNDED
- APP\_SETTINGS\_ALT\_SHARP
- APP\_SHORTCUT
- APP\_SHORTCUT\_OUTLINED
- APP\_SHORTCUT\_ROUNDED
- APP\_SHORTCUT\_SHARP
- APPLE
- APPLE\_OUTLINED
- APPLE\_ROUNDED
- APPLE\_SHARP
- APPROVAL
- APPROVAL\_OUTLINED
- APPROVAL\_ROUNDED
- APPROVAL\_SHARP
- APPS
- APPS\_OUTAGE
- APPS\_OUTAGE\_OUTLINED
- APPS\_OUTAGE\_ROUNDED
- APPS\_OUTAGE\_SHARP

- APPS\_OUTLINED
- APPS\_ROUNDED
- APPS\_SHARP
- ARCHITECTURE
- ARCHITECTURE\_OUTLINED
- ARCHITECTURE\_ROUNDED
- ARCHITECTURE\_SHARP
- ARCHIVE
- ARCHIVE\_OUTLINED
- ARCHIVE\_ROUNDED
- ARCHIVE\_SHARP
- AREA\_CHART
- AREA\_CHART\_OUTLINED
- AREA\_CHART\_ROUNDED
- AREA\_CHART\_SHARP
- ARROW\_BACK
- ARROW\_BACK\_IOS
- ARROW\_BACK\_IOS\_NEW
- ARROW\_BACK\_IOS\_NEW\_OUTLINED
- ARROW\_BACK\_IOS\_NEW\_ROUNDED
- ARROW\_BACK\_IOS\_NEW\_SHARP
- ARROW\_BACK\_IOS\_OUTLINED
- ARROW\_BACK\_IOS\_ROUNDED
- ARROW\_BACK\_IOS\_SHARP
- ARROW\_BACK\_OUTLINED
- ARROW\_BACK\_ROUNDED
- ARROW\_BACK\_SHARP
- ARROW\_CIRCLE\_DOWN
- ARROW\_CIRCLE\_DOWN\_OUTLINED
- ARROW\_CIRCLE\_DOWN\_ROUNDED
- ARROW\_CIRCLE\_DOWN\_SHARP
- ARROW\_CIRCLE\_LEFT
- ARROW\_CIRCLE\_LEFT\_OUTLINED
- ARROW\_CIRCLE\_LEFT\_ROUNDED
- ARROW\_CIRCLE\_LEFT\_SHARP
- ARROW\_CIRCLE\_RIGHT
- ARROW\_CIRCLE\_RIGHT\_OUTLINED
- ARROW\_CIRCLE\_RIGHT\_ROUNDED
- ARROW\_CIRCLE\_RIGHT\_SHARP
- ARROW\_CIRCLE\_UP
- ARROW\_CIRCLE\_UP\_OUTLINED

- ARROW\_CIRCLE\_UP\_ROUNDED
- ARROW\_CIRCLE\_UP\_SHARP
- ARROW\_DOWNWARD
- ARROW\_DOWNWARD\_OUTLINED
- ARROW\_DOWNWARD\_ROUNDED
- ARROW\_DOWNWARD\_SHARP
- ARROW\_DROP\_DOWN
- ARROW\_DROP\_DOWN\_CIRCLE
- ARROW\_DROP\_DOWN\_CIRCLE\_OUTLINED
- ARROW\_DROP\_DOWN\_CIRCLE\_ROUNDED
- ARROW\_DROP\_DOWN\_CIRCLE\_SHARP
- ARROW\_DROP\_DOWN\_OUTLINED
- ARROW\_DROP\_DOWN\_ROUNDED
- ARROW\_DROP\_DOWN\_SHARP
- ARROW\_DROP\_UP
- ARROW\_DROP\_UP\_OUTLINED
- ARROW\_DROP\_UP\_ROUNDED
- ARROW\_DROP\_UP\_SHARP
- ARROW\_FORWARD
- ARROW\_FORWARD\_IOS
- ARROW\_FORWARD\_IOS\_OUTLINED
- ARROW\_FORWARD\_IOS\_ROUNDED
- ARROW\_FORWARD\_IOS\_SHARP
- ARROW\_FORWARD\_OUTLINED
- ARROW\_FORWARD\_ROUNDED
- ARROW\_FORWARD\_SHARP
- ARROW\_LEFT
- ARROW\_LEFT\_OUTLINED
- ARROW\_LEFT\_ROUNDED
- ARROW\_LEFT\_SHARP
- ARROW\_OUTWARD
- ARROW\_OUTWARD\_OUTLINED
- ARROW\_OUTWARD\_ROUNDED
- ARROW\_OUTWARD\_SHARP
- ARROW\_RIGHT
- ARROW\_RIGHT\_ALT
- ARROW\_RIGHT\_ALT\_OUTLINED
- ARROW\_RIGHT\_ALT\_ROUNDED
- ARROW\_RIGHT\_ALT\_SHARP
- ARROW\_RIGHT\_OUTLINED
- ARROW\_RIGHT\_ROUNDED

- ARROW\_RIGHT\_SHARP
- ARROW\_UPWARD
- ARROW\_UPWARD\_OUTLINED
- ARROW\_UPWARD\_ROUNDED
- ARROW\_UPWARD\_SHARP
- ART\_TRACK
- ART\_TRACK\_OUTLINED
- ART\_TRACK\_ROUNDED
- ART\_TRACK\_SHARP
- ARTICLE
- ARTICLE\_OUTLINED
- ARTICLE\_ROUNDED
- ARTICLE\_SHARP
- ASPECT\_RATIO
- ASPECT\_RATIO\_OUTLINED
- ASPECT\_RATIO\_ROUNDED
- ASPECT\_RATIO\_SHARP
- ASSESSMENT
- ASSESSMENT\_OUTLINED
- ASSESSMENT\_ROUNDED
- ASSESSMENT\_SHARP
- ASSIGNMENT
- ASSIGNMENT\_ADD
- ASSIGNMENT\_IND
- ASSIGNMENT\_IND\_OUTLINED
- ASSIGNMENT\_IND\_ROUNDED
- ASSIGNMENT\_IND\_SHARP
- ASSIGNMENT\_LATE
- ASSIGNMENT\_LATE\_OUTLINED
- ASSIGNMENT\_LATE\_ROUNDED
- ASSIGNMENT\_LATE\_SHARP
- ASSIGNMENT\_OUTLINED
- ASSIGNMENT\_RETURN
- ASSIGNMENT\_RETURN\_OUTLINED
- ASSIGNMENT\_RETURN\_ROUNDED
- ASSIGNMENT\_RETURN\_SHARP
- ASSIGNMENT\_RETURNED
- ASSIGNMENT\_RETURNED\_OUTLINED
- ASSIGNMENT\_RETURNED\_ROUNDED
- ASSIGNMENT\_RETURNED\_SHARP
- ASSIGNMENT\_ROUNDED

- ASSIGNMENT\_SHARP
- ASSIGNMENT\_TURNED\_IN
- ASSIGNMENT\_TURNED\_IN\_OUTLINED
- ASSIGNMENT\_TURNED\_IN\_ROUNDED
- ASSIGNMENT\_TURNED\_IN\_SHARP
- ASSIST\_WALKER
- ASSIST\_WALKER\_OUTLINED
- ASSIST\_WALKER\_ROUNDED
- ASSIST\_WALKER\_SHARP
- ASSISTANT
- ASSISTANT\_DIRECTION
- ASSISTANT\_DIRECTION\_OUTLINED
- ASSISTANT\_DIRECTION\_ROUNDED
- ASSISTANT\_DIRECTION\_SHARP
- ASSISTANT\_NAVIGATION
- ASSISTANT\_OUTLINED
- ASSISTANT\_PHOTO
- ASSISTANT\_PHOTO\_OUTLINED
- ASSISTANT\_PHOTO\_ROUNDED
- ASSISTANT\_PHOTO\_SHARP
- ASSISTANT\_ROUNDED
- ASSISTANT\_SHARP
- ASSURED\_WORKLOAD
- ASSURED\_WORKLOAD\_OUTLINED
- ASSURED\_WORKLOAD\_ROUNDED
- ASSURED\_WORKLOAD\_SHARP
- ATM
- ATM\_OUTLINED
- ATM\_ROUNDED
- ATM\_SHARP
- ATTACH\_EMAIL
- ATTACH\_EMAIL\_OUTLINED
- ATTACH\_EMAIL\_ROUNDED
- ATTACH\_EMAIL\_SHARP
- ATTACH\_FILE
- ATTACH\_FILE\_OUTLINED
- ATTACH\_FILE\_ROUNDED
- ATTACH\_FILE\_SHARP
- ATTACH\_MONEY
- ATTACH\_MONEY\_OUTLINED
- ATTACH\_MONEY\_ROUNDED

- ATTACH\_MONEY\_SHARP
- ATTACHMENT
- ATTACHMENT\_OUTLINED
- ATTACHMENT\_ROUNDED
- ATTACHMENT\_SHARP
- ATTRACTIONS
- ATTRACTIONS\_OUTLINED
- ATTRACTIONS\_ROUNDED
- ATTRACTIONS\_SHARP
- ATTRIBUTION
- ATTRIBUTION\_OUTLINED
- ATTRIBUTION\_ROUNDED
- ATTRIBUTION\_SHARP
- AUDIO\_FILE
- AUDIO\_FILE\_OUTLINED
- AUDIO\_FILE\_ROUNDED
- AUDIO\_FILE\_SHARP
- AUDIOTRACK
- AUDIOTRACK\_OUTLINED
- AUDIOTRACK\_ROUNDED
- AUDIOTRACK\_SHARP
- AUTO\_AWESOME
- AUTO\_AWESOME\_MOSAIC
- AUTO\_AWESOME\_MOSAIC\_OUTLINED
- AUTO\_AWESOME\_MOSAIC\_ROUNDED
- AUTO\_AWESOME\_MOSAIC\_SHARP
- AUTO\_AWESOME\_MOTION
- AUTO\_AWESOME\_MOTION\_OUTLINED
- AUTO\_AWESOME\_MOTION\_ROUNDED
- AUTO\_AWESOME\_MOTION\_SHARP
- AUTO\_AWESOME\_OUTLINED
- AUTO\_AWESOME\_ROUNDED
- AUTO\_AWESOME\_SHARP
- AUTO\_DELETE
- AUTO\_DELETE\_OUTLINED
- AUTO\_DELETE\_ROUNDED
- AUTO\_DELETE\_SHARP
- AUTO\_FIX\_HIGH
- AUTO\_FIX\_HIGH\_OUTLINED
- AUTO\_FIX\_HIGH\_ROUNDED
- AUTO\_FIX\_HIGH\_SHARP



- AUTO\_FIX\_NORMAL
- AUTO\_FIX\_NORMAL\_OUTLINED
- AUTO\_FIX\_NORMAL\_ROUNDED
- AUTO\_FIX\_NORMAL\_SHARP
- AUTO\_FIX\_OFF
- AUTO\_FIX\_OFF\_OUTLINED
- AUTO\_FIX\_OFF\_ROUNDED
- AUTO\_FIX\_OFF\_SHARP
- AUTO\_GRAPH
- AUTO\_GRAPH\_OUTLINED
- AUTO\_GRAPH\_ROUNDED
- AUTO\_GRAPH\_SHARP
- AUTO\_MODE
- AUTO\_MODE\_OUTLINED
- AUTO\_MODE\_ROUNDED
- AUTO\_MODE\_SHARP
- AUTO\_STORIES
- AUTO\_STORIES\_OUTLINED
- AUTO\_STORIES\_ROUNDED
- AUTO\_STORIES\_SHARP
- AUTOFPS\_SELECT
- AUTOFPS\_SELECT\_OUTLINED
- AUTOFPS\_SELECT\_ROUNDED
- AUTOFPS\_SELECT\_SHARP
- AUTORENEW
- AUTORENEW\_OUTLINED
- AUTORENEW\_ROUNDED
- AUTORENEW\_SHARP
- AV\_TIMER
- AV\_TIMER\_OUTLINED
- AV\_TIMER\_ROUNDED
- AV\_TIMER\_SHARP
- BABY\_CHANGING\_STATION
- BABY\_CHANGING\_STATION\_OUTLINED
- BABY\_CHANGING\_STATION\_ROUNDED
- BABY\_CHANGING\_STATION\_SHARP
- BACK\_HAND
- BACK\_HAND\_OUTLINED
- BACK\_HAND\_ROUNDED
- BACK\_HAND\_SHARP
- BACKPACK

- BACKPACK\_OUTLINED
- BACKPACK\_ROUNDED
- BACKPACK\_SHARP
- BACKSPACE
- BACKSPACE\_OUTLINED
- BACKSPACE\_ROUNDED
- BACKSPACE\_SHARP
- BACKUP
- BACKUP\_OUTLINED
- BACKUP\_ROUNDED
- BACKUP\_SHARP
- BACKUP\_TABLE
- BACKUP\_TABLE\_OUTLINED
- BACKUP\_TABLE\_ROUNDED
- BACKUP\_TABLE\_SHARP
- BADGE
- BADGE\_OUTLINED
- BADGE\_ROUNDED
- BADGE\_SHARP
- BAKERY\_DINING
- BAKERY\_DINING\_OUTLINED
- BAKERY\_DINING\_ROUNDED
- BAKERY\_DINING\_SHARP
- BALANCE
- BALANCE\_OUTLINED
- BALANCE\_ROUNDED
- BALANCE\_SHARP
- BALCONY
- BALCONY\_OUTLINED
- BALCONY\_ROUNDED
- BALCONY\_SHARP
- BALLOT
- BALLOT\_OUTLINED
- BALLOT\_ROUNDED
- BALLOT\_SHARP
- BAR\_CHART
- BAR\_CHART\_OUTLINED
- BAR\_CHART\_ROUNDED
- BAR\_CHART\_SHARP
- BARCODE\_READER
- BATCH\_PREDICTION

- BATCH\_PREDICTION\_OUTLINED
- BATCH\_PREDICTION\_ROUNDED
- BATCH\_PREDICTION\_SHARP
- BATHROOM
- BATHROOM\_OUTLINED
- BATHROOM\_ROUNDED
- BATHROOM\_SHARP
- BATHTUB
- BATHTUB\_OUTLINED
- BATHTUB\_ROUNDED
- BATHTUB\_SHARP
- BATTERY\_0\_BAR
- BATTERY\_0\_BAR\_OUTLINED
- BATTERY\_0\_BAR\_ROUNDED
- BATTERY\_0\_BAR\_SHARP
- BATTERY\_1\_BAR
- BATTERY\_1\_BAR\_OUTLINED
- BATTERY\_1\_BAR\_ROUNDED
- BATTERY\_1\_BAR\_SHARP
- BATTERY\_2\_BAR
- BATTERY\_2\_BAR\_OUTLINED
- BATTERY\_2\_BAR\_ROUNDED
- BATTERY\_2\_BAR\_SHARP
- BATTERY\_3\_BAR
- BATTERY\_3\_BAR\_OUTLINED
- BATTERY\_3\_BAR\_ROUNDED
- BATTERY\_3\_BAR\_SHARP
- BATTERY\_4\_BAR
- BATTERY\_4\_BAR\_OUTLINED
- BATTERY\_4\_BAR\_ROUNDED
- BATTERY\_4\_BAR\_SHARP
- BATTERY\_5\_BAR
- BATTERY\_5\_BAR\_OUTLINED
- BATTERY\_5\_BAR\_ROUNDED
- BATTERY\_5\_BAR\_SHARP
- BATTERY\_6\_BAR
- BATTERY\_6\_BAR\_OUTLINED
- BATTERY\_6\_BAR\_ROUNDED
- BATTERY\_6\_BAR\_SHARP
- BATTERY\_ALERT
- BATTERY\_ALERT\_OUTLINED

- BATTERY\_ALERT\_ROUNDED
- BATTERY\_ALERT\_SHARP
- BATTERY\_CHARGING\_FULL
- BATTERY\_CHARGING\_FULL\_OUTLINED
- BATTERY\_CHARGING\_FULL\_ROUNDED
- BATTERY\_CHARGING\_FULL\_SHARP
- BATTERY\_FULL
- BATTERY\_FULL\_OUTLINED
- BATTERY\_FULL\_ROUNDED
- BATTERY\_FULL\_SHARP
- BATTERY\_SAVER
- BATTERY\_SAVER\_OUTLINED
- BATTERY\_SAVER\_ROUNDED
- BATTERY\_SAVER\_SHARP
- BATTERY\_STD
- BATTERY\_STD\_OUTLINED
- BATTERY\_STD\_ROUNDED
- BATTERY\_STD\_SHARP
- BATTERY\_UNKNOWN
- BATTERY\_UNKNOWN\_OUTLINED
- BATTERY\_UNKNOWN\_ROUNDED
- BATTERY\_UNKNOWN\_SHARP
- BEACH\_ACCESS
- BEACH\_ACCESS\_OUTLINED
- BEACH\_ACCESS\_ROUNDED
- BEACH\_ACCESS\_SHARP
- BED
- BED\_OUTLINED
- BED\_ROUNDED
- BED\_SHARP
- BEDROOM\_BABY
- BEDROOM\_BABY\_OUTLINED
- BEDROOM\_BABY\_ROUNDED
- BEDROOM\_BABY\_SHARP
- BEDROOM\_CHILD
- BEDROOM\_CHILD\_OUTLINED
- BEDROOM\_CHILD\_ROUNDED
- BEDROOM\_CHILD\_SHARP
- BEDROOM\_PARENT
- BEDROOM\_PARENT\_OUTLINED
- BEDROOM\_PARENT\_ROUNDED

- BEDROOM\_PARENT\_SHARP
- BEDTIME
- BEDTIME\_OFF
- BEDTIME\_OFF\_OUTLINED
- BEDTIME\_OFF\_ROUNDED
- BEDTIME\_OFF\_SHARP
- BEDTIME\_OUTLINED
- BEDTIME\_ROUNDED
- BEDTIME\_SHARP
- BEENHERE
- BEENHERE\_OUTLINED
- BEENHERE\_ROUNDED
- BEENHERE\_SHARP
- BENTO
- BENTO\_OUTLINED
- BENTO\_ROUNDED
- BENTO\_SHARP
- BIKE\_SCOOTER
- BIKE\_SCOOTER\_OUTLINED
- BIKE\_SCOOTER\_ROUNDED
- BIKE\_SCOOTER\_SHARP
- BIOTECH
- BIOTECH\_OUTLINED
- BIOTECH\_ROUNDED
- BIOTECH\_SHARP
- BLENDER
- BLENDER\_OUTLINED
- BLENDER\_ROUNDED
- BLENDER\_SHARP
- BLIND
- BLIND\_OUTLINED
- BLIND\_ROUNDED
- BLIND\_SHARP
- BLINDS
- BLINDS\_CLOSED
- BLINDS\_CLOSED\_OUTLINED
- BLINDS\_CLOSED\_ROUNDED
- BLINDS\_CLOSED\_SHARP
- BLINDS\_OUTLINED
- BLINDS\_ROUNDED
- BLINDS\_SHARP

- BLOCK
- BLOCK\_FLIPPED
- BLOCK\_OUTLINED
- BLOCK\_ROUNDED
- BLOCK\_SHARP
- BLOODTYPE
- BLOODTYPE\_OUTLINED
- BLOODTYPE\_ROUNDED
- BLOODTYPE\_SHARP
- BLUETOOTH
- BLUETOOTH\_AUDIO
- BLUETOOTH\_AUDIO\_OUTLINED
- BLUETOOTH\_AUDIO\_ROUNDED
- BLUETOOTH\_AUDIO\_SHARP
- BLUETOOTH\_CONNECTED
- BLUETOOTH\_CONNECTED\_OUTLINED
- BLUETOOTH\_CONNECTED\_ROUNDED
- BLUETOOTH\_CONNECTED\_SHARP
- BLUETOOTH\_DISABLED
- BLUETOOTH\_DISABLED\_OUTLINED
- BLUETOOTH\_DISABLED\_ROUNDED
- BLUETOOTH\_DISABLED\_SHARP
- BLUETOOTH\_DRIVE
- BLUETOOTH\_DRIVE\_OUTLINED
- BLUETOOTH\_DRIVE\_ROUNDED
- BLUETOOTH\_DRIVE\_SHARP
- BLUETOOTH\_OUTLINED
- BLUETOOTH\_ROUNDED
- BLUETOOTH\_SEARCHING
- BLUETOOTH\_SEARCHING\_OUTLINED
- BLUETOOTH\_SEARCHING\_ROUNDED
- BLUETOOTH\_SEARCHING\_SHARP
- BLUETOOTH\_SHARP
- BLUR\_CIRCULAR
- BLUR\_CIRCULAR\_OUTLINED
- BLUR\_CIRCULAR\_ROUNDED
- BLUR\_CIRCULAR\_SHARP
- BLUR\_LINEAR
- BLUR\_LINEAR\_OUTLINED
- BLUR\_LINEAR\_ROUNDED
- BLUR\_LINEAR\_SHARP

- BLUR\_OFF
- BLUR\_OFF\_OUTLINED
- BLUR\_OFF\_ROUNDED
- BLUR\_OFF\_SHARP
- BLUR\_ON
- BLUR\_ON\_OUTLINED
- BLUR\_ON\_ROUNDED
- BLUR\_ON\_SHARP
- BOLT
- BOLT\_OUTLINED
- BOLT\_ROUNDED
- BOLT\_SHARP
- BOOK
- BOOK\_ONLINE
- BOOK\_ONLINE\_OUTLINED
- BOOK\_ONLINE\_ROUNDED
- BOOK\_ONLINE\_SHARP
- BOOK\_OUTLINED
- BOOK\_ROUNDED
- BOOK\_SHARP
- BOOKMARK
- BOOKMARK\_ADD
- BOOKMARK\_ADD\_OUTLINED
- BOOKMARK\_ADD\_ROUNDED
- BOOKMARK\_ADD\_SHARP
- BOOKMARK\_ADDED
- BOOKMARK\_ADDED\_OUTLINED
- BOOKMARK\_ADDED\_ROUNDED
- BOOKMARK\_ADDED\_SHARP
- BOOKMARK\_BORDER
- BOOKMARK\_BORDER\_OUTLINED
- BOOKMARK\_BORDER\_ROUNDED
- BOOKMARK\_BORDER\_SHARP
- BOOKMARK\_OUTLINE
- BOOKMARK\_OUTLINE\_OUTLINED
- BOOKMARK\_OUTLINE\_ROUNDED
- BOOKMARK\_OUTLINE\_SHARP
- BOOKMARK\_OUTLINED
- BOOKMARK\_REMOVE
- BOOKMARK\_REMOVE\_OUTLINED
- BOOKMARK\_REMOVE\_ROUNDED

- BOOKMARK\_REMOVE\_SHARP
- BOOKMARK\_ROUNDED
- BOOKMARK\_SHARP
- BOOKMARKS
- BOOKMARKS\_OUTLINED
- BOOKMARKS\_ROUNDED
- BOOKMARKS\_SHARP
- BORDER\_ALL
- BORDER\_ALL\_OUTLINED
- BORDER\_ALL\_ROUNDED
- BORDER\_ALL\_SHARP
- BORDER\_BOTTOM
- BORDER\_BOTTOM\_OUTLINED
- BORDER\_BOTTOM\_ROUNDED
- BORDER\_BOTTOM\_SHARP
- BORDER\_CLEAR
- BORDER\_CLEAR\_OUTLINED
- BORDER\_CLEAR\_ROUNDED
- BORDER\_CLEAR\_SHARP
- BORDER\_COLOR
- BORDER\_COLOR\_OUTLINED
- BORDER\_COLOR\_ROUNDED
- BORDER\_COLOR\_SHARP
- BORDER\_HORIZONTAL
- BORDER\_HORIZONTAL\_OUTLINED
- BORDER\_HORIZONTAL\_ROUNDED
- BORDER\_HORIZONTAL\_SHARP
- BORDER\_INNER
- BORDER\_INNER\_OUTLINED
- BORDER\_INNER\_ROUNDED
- BORDER\_INNER\_SHARP
- BORDER\_LEFT
- BORDER\_LEFT\_OUTLINED
- BORDER\_LEFT\_ROUNDED
- BORDER\_LEFT\_SHARP
- BORDER\_OUTER
- BORDER\_OUTER\_OUTLINED
- BORDER\_OUTER\_ROUNDED
- BORDER\_OUTER\_SHARP
- BORDER\_RIGHT
- BORDER\_RIGHT\_OUTLINED



- BORDER\_RIGHT\_ROUNDED
- BORDER\_RIGHT\_SHARP
- BORDER\_STYLE
- BORDER\_STYLE\_OUTLINED
- BORDER\_STYLE\_ROUNDED
- BORDER\_STYLE\_SHARP
- BORDER\_TOP
- BORDER\_TOP\_OUTLINED
- BORDER\_TOP\_ROUNDED
- BORDER\_TOP\_SHARP
- BORDER\_VERTICAL
- BORDER\_VERTICAL\_OUTLINED
- BORDER\_VERTICAL\_ROUNDED
- BORDER\_VERTICAL\_SHARP
- BOY
- BOY\_OUTLINED
- BOY\_ROUNDED
- BOY\_SHARP
- BRANDING\_WATERMARK
- BRANDING\_WATERMARK\_OUTLINED
- BRANDING\_WATERMARK\_ROUNDED
- BRANDING\_WATERMARK\_SHARP
- BREAKFAST\_DINING
- BREAKFAST\_DINING\_OUTLINED
- BREAKFAST\_DINING\_ROUNDED
- BREAKFAST\_DINING\_SHARP
- BRIGHTNESS\_1
- BRIGHTNESS\_1\_OUTLINED
- BRIGHTNESS\_1\_ROUNDED
- BRIGHTNESS\_1\_SHARP
- BRIGHTNESS\_2
- BRIGHTNESS\_2\_OUTLINED
- BRIGHTNESS\_2\_ROUNDED
- BRIGHTNESS\_2\_SHARP
- BRIGHTNESS\_3
- BRIGHTNESS\_3\_OUTLINED
- BRIGHTNESS\_3\_ROUNDED
- BRIGHTNESS\_3\_SHARP
- BRIGHTNESS\_4
- BRIGHTNESS\_4\_OUTLINED
- BRIGHTNESS\_4\_ROUNDED

- BRIGHTNESS\_4\_SHARP
- BRIGHTNESS\_5
- BRIGHTNESS\_5\_OUTLINED
- BRIGHTNESS\_5\_ROUNDED
- BRIGHTNESS\_5\_SHARP
- BRIGHTNESS\_6
- BRIGHTNESS\_6\_OUTLINED
- BRIGHTNESS\_6\_ROUNDED
- BRIGHTNESS\_6\_SHARP
- BRIGHTNESS\_7
- BRIGHTNESS\_7\_OUTLINED
- BRIGHTNESS\_7\_ROUNDED
- BRIGHTNESS\_7\_SHARP
- BRIGHTNESS\_AUTO
- BRIGHTNESS\_AUTO\_OUTLINED
- BRIGHTNESS\_AUTO\_ROUNDED
- BRIGHTNESS\_AUTO\_SHARP
- BRIGHTNESS\_HIGH
- BRIGHTNESS\_HIGH\_OUTLINED
- BRIGHTNESS\_HIGH\_ROUNDED
- BRIGHTNESS\_HIGH\_SHARP
- BRIGHTNESS\_LOW
- BRIGHTNESS\_LOW\_OUTLINED
- BRIGHTNESS\_LOW\_ROUNDED
- BRIGHTNESS\_LOW\_SHARP
- BRIGHTNESS\_MEDIUM
- BRIGHTNESS\_MEDIUM\_OUTLINED
- BRIGHTNESS\_MEDIUM\_ROUNDED
- BRIGHTNESS\_MEDIUM\_SHARP
- BROADCAST\_ON\_HOME
- BROADCAST\_ON\_HOME\_OUTLINED
- BROADCAST\_ON\_HOME\_ROUNDED
- BROADCAST\_ON\_HOME\_SHARP
- BROADCAST\_ON\_PERSONAL
- BROADCAST\_ON\_PERSONAL\_OUTLINED
- BROADCAST\_ON\_PERSONAL\_ROUNDED
- BROADCAST\_ON\_PERSONAL\_SHARP
- BROKEN\_IMAGE
- BROKEN\_IMAGE\_OUTLINED
- BROKEN\_IMAGE\_ROUNDED
- BROKEN\_IMAGE\_SHARP

- BROWSE\_GALLERY
- BROWSE\_GALLERY\_OUTLINED
- BROWSE\_GALLERY\_ROUNDED
- BROWSE\_GALLERY\_SHARP
- BROWSER\_NOT\_SUPPORTED
- BROWSER\_NOT\_SUPPORTED\_OUTLINED
- BROWSER\_NOT\_SUPPORTED\_ROUNDED
- BROWSER\_NOT\_SUPPORTED\_SHARP
- BROWSER\_UPDATED
- BROWSER\_UPDATED\_OUTLINED
- BROWSER\_UPDATED\_ROUNDED
- BROWSER\_UPDATED\_SHARP
- BRUNCH\_DINING
- BRUNCH\_DINING\_OUTLINED
- BRUNCH\_DINING\_ROUNDED
- BRUNCH\_DINING\_SHARP
- BRUSH
- BRUSH\_OUTLINED
- BRUSH\_ROUNDED
- BRUSH\_SHARP
- BUBBLE\_CHART
- BUBBLE\_CHART\_OUTLINED
- BUBBLE\_CHART\_ROUNDED
- BUBBLE\_CHART\_SHARP
- BUG\_REPORT
- BUG\_REPORT\_OUTLINED
- BUG\_REPORT\_ROUNDED
- BUG\_REPORT\_SHARP
- BUILD
- BUILD\_CIRCLE
- BUILD\_CIRCLE\_OUTLINED
- BUILD\_CIRCLE\_ROUNDED
- BUILD\_CIRCLE\_SHARP
- BUILD\_OUTLINED
- BUILD\_ROUNDED
- BUILD\_SHARP
- BUNGALOW
- BUNGALOW\_OUTLINED
- BUNGALOW\_ROUNDED
- BUNGALOW\_SHARP
- BURST\_MODE

- BURST\_MODE\_OUTLINED
- BURST\_MODE\_ROUNDED
- BURST\_MODE\_SHARP
- BUS\_ALERT
- BUS\_ALERT\_OUTLINED
- BUS\_ALERT\_ROUNDED
- BUS\_ALERT\_SHARP
- BUSINESS
- BUSINESS\_CENTER
- BUSINESS\_CENTER\_OUTLINED
- BUSINESS\_CENTER\_ROUNDED
- BUSINESS\_CENTER\_SHARP
- BUSINESS\_OUTLINED
- BUSINESS\_ROUNDED
- BUSINESS\_SHARP
- CABIN
- CABIN\_OUTLINED
- CABIN\_ROUNDED
- CABIN\_SHARP
- CABLE
- CABLE\_OUTLINED
- CABLE\_ROUNDED
- CABLE\_SHARP
- CACHED
- CACHED\_OUTLINED
- CACHED\_ROUNDED
- CACHED\_SHARP
- CAKE
- CAKE\_OUTLINED
- CAKE\_ROUNDED
- CAKE\_SHARP
- CALCULATE
- CALCULATE\_OUTLINED
- CALCULATE\_ROUNDED
- CALCULATE\_SHARP
- CALENDAR\_MONTH
- CALENDAR\_MONTH\_OUTLINED
- CALENDAR\_MONTH\_ROUNDED
- CALENDAR\_MONTH\_SHARP
- CALENDAR\_TODAY
- CALENDAR\_TODAY\_OUTLINED

- CALENDAR\_TODAY\_ROUNDED
- CALENDAR\_TODAY\_SHARP
- CALENDAR\_VIEW\_DAY
- CALENDAR\_VIEW\_DAY\_OUTLINED
- CALENDAR\_VIEW\_DAY\_ROUNDED
- CALENDAR\_VIEW\_DAY\_SHARP
- CALENDAR\_VIEW\_MONTH
- CALENDAR\_VIEW\_MONTH\_OUTLINED
- CALENDAR\_VIEW\_MONTH\_ROUNDED
- CALENDAR\_VIEW\_MONTH\_SHARP
- CALENDAR\_VIEW\_WEEK
- CALENDAR\_VIEW\_WEEK\_OUTLINED
- CALENDAR\_VIEW\_WEEK\_ROUNDED
- CALENDAR\_VIEW\_WEEK\_SHARP
- CALL
- CALL\_END
- CALL\_END\_OUTLINED
- CALL\_END\_ROUNDED
- CALL\_END\_SHARP
- CALL\_MADE
- CALL\_MADE\_OUTLINED
- CALL\_MADE\_ROUNDED
- CALL\_MADE\_SHARP
- CALL\_MERGE
- CALL\_MERGE\_OUTLINED
- CALL\_MERGE\_ROUNDED
- CALL\_MERGE\_SHARP
- CALL\_MISSED
- CALL\_MISSED\_OUTGOING
- CALL\_MISSED\_OUTGOING\_OUTLINED
- CALL\_MISSED\_OUTGOING\_ROUNDED
- CALL\_MISSED\_OUTGOING\_SHARP
- CALL\_MISSED\_OUTLINED
- CALL\_MISSED\_ROUNDED
- CALL\_MISSED\_SHARP
- CALL\_OUTLINED
- CALL\_RECEIVED
- CALL\_RECEIVED\_OUTLINED
- CALL\_RECEIVED\_ROUNDED
- CALL\_RECEIVED\_SHARP
- CALL\_ROUNDED

- CALL\_SHARP
- CALL\_SPLIT
- CALL\_SPLIT\_OUTLINED
- CALL\_SPLIT\_ROUNDED
- CALL\_SPLIT\_SHARP
- CALL\_TO\_ACTION
- CALL\_TO\_ACTION\_OUTLINED
- CALL\_TO\_ACTION\_ROUNDED
- CALL\_TO\_ACTION\_SHARP
- CAMERA
- CAMERA\_ALT
- CAMERA\_ALT\_OUTLINED
- CAMERA\_ALT\_ROUNDED
- CAMERA\_ALT\_SHARP
- CAMERA\_ENHANCE
- CAMERA\_ENHANCE\_OUTLINED
- CAMERA\_ENHANCE\_ROUNDED
- CAMERA\_ENHANCE\_SHARP
- CAMERA\_FRONT
- CAMERA\_FRONT\_OUTLINED
- CAMERA\_FRONT\_ROUNDED
- CAMERA\_FRONT\_SHARP
- CAMERA\_INDOOR
- CAMERA\_INDOOR\_OUTLINED
- CAMERA\_INDOOR\_ROUNDED
- CAMERA\_INDOOR\_SHARP
- CAMERA\_OUTDOOR
- CAMERA\_OUTDOOR\_OUTLINED
- CAMERA\_OUTDOOR\_ROUNDED
- CAMERA\_OUTDOOR\_SHARP
- CAMERA\_OUTLINED
- CAMERA\_REAR
- CAMERA\_REAR\_OUTLINED
- CAMERA\_REAR\_ROUNDED
- CAMERA\_REAR\_SHARP
- CAMERA\_ROLL
- CAMERA\_ROLL\_OUTLINED
- CAMERA\_ROLL\_ROUNDED
- CAMERA\_ROLL\_SHARP
- CAMERA\_ROUNDED
- CAMERA\_SHARP

- CAMERASWITCH
- CAMERASWITCH\_OUTLINED
- CAMERASWITCH\_ROUNDED
- CAMERASWITCH\_SHARP
- CAMPAIGN
- CAMPAIGN\_OUTLINED
- CAMPAIGN\_ROUNDED
- CAMPAIGN\_SHARP
- CANCEL
- CANCEL\_OUTLINED
- CANCEL\_PRESENTATION
- CANCEL\_PRESENTATION\_OUTLINED
- CANCEL\_PRESENTATION\_ROUNDED
- CANCEL\_PRESENTATION\_SHARP
- CANCEL\_ROUNDED
- CANCEL\_SCHEDULE\_SEND
- CANCEL\_SCHEDULE\_SEND\_OUTLINED
- CANCEL\_SCHEDULE\_SEND\_ROUNDED
- CANCEL\_SCHEDULE\_SEND\_SHARP
- CANCEL\_SHARP
- CANDLESTICK\_CHART
- CANDLESTICK\_CHART\_OUTLINED
- CANDLESTICK\_CHART\_ROUNDED
- CANDLESTICK\_CHART\_SHARP
- CAR\_CRASH
- CAR\_CRASH\_OUTLINED
- CAR\_CRASH\_ROUNDED
- CAR\_CRASH\_SHARP
- CAR\_RENTAL
- CAR\_RENTAL\_OUTLINED
- CAR\_RENTAL\_ROUNDED
- CAR\_RENTAL\_SHARP
- CAR\_REPAIR
- CAR\_REPAIR\_OUTLINED
- CAR\_REPAIR\_ROUNDED
- CAR\_REPAIR\_SHARP
- CARD\_GIFTCARD
- CARD\_GIFTCARD\_OUTLINED
- CARD\_GIFTCARD\_ROUNDED
- CARD\_GIFTCARD\_SHARP
- CARD\_MEMBERSHIP

- CARD\_MEMBERSHIP\_OUTLINED
- CARD\_MEMBERSHIP\_ROUNDED
- CARD\_MEMBERSHIP\_SHARP
- CARD\_TRAVEL
- CARD\_TRAVEL\_OUTLINED
- CARD\_TRAVEL\_ROUNDED
- CARD\_TRAVEL\_SHARP
- CARPENTER
- CARPENTER\_OUTLINED
- CARPENTER\_ROUNDED
- CARPENTER\_SHARP
- CASES
- CASES\_OUTLINED
- CASES\_ROUNDED
- CASES\_SHARP
- CASINO
- CASINO\_OUTLINED
- CASINO\_ROUNDED
- CASINO\_SHARP
- CAST
- CAST\_CONNECTED
- CAST\_CONNECTED\_OUTLINED
- CAST\_CONNECTED\_ROUNDED
- CAST\_CONNECTED\_SHARP
- CAST\_FOR\_EDUCATION
- CAST\_FOR\_EDUCATION\_OUTLINED
- CAST\_FOR\_EDUCATION\_ROUNDED
- CAST\_FOR\_EDUCATION\_SHARP
- CAST\_OUTLINED
- CAST\_ROUNDED
- CAST\_SHARP
- CASTLE
- CASTLE\_OUTLINED
- CASTLE\_ROUNDED
- CASTLE\_SHARP
- CATCHING\_POKEMON
- CATCHING\_POKEMON\_OUTLINED
- CATCHING\_POKEMON\_ROUNDED
- CATCHING\_POKEMON\_SHARP
- CATEGORY
- CATEGORY\_OUTLINED



- CATEGORY\_ROUNDED
- CATEGORY\_SHARP
- CELEBRATION
- CELEBRATION\_OUTLINED
- CELEBRATION\_ROUNDED
- CELEBRATION\_SHARP
- CELL\_TOWER
- CELL\_TOWER\_OUTLINED
- CELL\_TOWER\_ROUNDED
- CELL\_TOWER\_SHARP
- CELL\_WIFI
- CELL\_WIFI\_OUTLINED
- CELL\_WIFI\_ROUNDED
- CELL\_WIFI\_SHARP
- CENTER\_FOCUS\_STRONG
- CENTER\_FOCUS\_STRONG\_OUTLINED
- CENTER\_FOCUS\_STRONG\_ROUNDED
- CENTER\_FOCUS\_STRONG\_SHARP
- CENTER\_FOCUS\_WEAK
- CENTER\_FOCUS\_WEAK\_OUTLINED
- CENTER\_FOCUS\_WEAK\_ROUNDED
- CENTER\_FOCUS\_WEAK\_SHARP
- CHAIR
- CHAIR\_ALT
- CHAIR\_ALT\_OUTLINED
- CHAIR\_ALT\_ROUNDED
- CHAIR\_ALT\_SHARP
- CHAIR\_OUTLINED
- CHAIR\_ROUNDED
- CHAIR\_SHARP
- CHALET
- CHALET\_OUTLINED
- CHALET\_ROUNDED
- CHALET\_SHARP
- CHANGE\_CIRCLE
- CHANGE\_CIRCLE\_OUTLINED
- CHANGE\_CIRCLE\_ROUNDED
- CHANGE\_CIRCLE\_SHARP
- CHANGE\_HISTORY
- CHANGE\_HISTORY\_OUTLINED
- CHANGE\_HISTORY\_ROUNDED

- CHANGE\_HISTORY\_SHARP
- CHARGING\_STATION
- CHARGING\_STATION\_OUTLINED
- CHARGING\_STATION\_ROUNDED
- CHARGING\_STATION\_SHARP
- CHAT
- CHAT\_BUBBLE
- CHAT\_BUBBLE\_OUTLINE
- CHAT\_BUBBLE\_OUTLINE\_OUTLINED
- CHAT\_BUBBLE\_OUTLINE\_ROUNDED
- CHAT\_BUBBLE\_OUTLINE\_SHARP
- CHAT\_BUBBLE\_OUTLINED
- CHAT\_BUBBLE\_ROUNDED
- CHAT\_BUBBLE\_SHARP
- CHAT\_OUTLINED
- CHAT\_ROUNDED
- CHAT\_SHARP
- CHECK
- CHECK\_BOX
- CHECK\_BOX\_OUTLINE\_BLANK
- CHECK\_BOX\_OUTLINE\_BLANK\_OUTLINED
- CHECK\_BOX\_OUTLINE\_BLANK\_ROUNDED
- CHECK\_BOX\_OUTLINE\_BLANK\_SHARP
- CHECK\_BOX\_OUTLINED
- CHECK\_BOX\_ROUNDED
- CHECK\_BOX\_SHARP
- CHECK\_CIRCLE
- CHECK\_CIRCLE\_OUTLINE
- CHECK\_CIRCLE\_OUTLINE\_OUTLINED
- CHECK\_CIRCLE\_OUTLINE\_ROUNDED
- CHECK\_CIRCLE\_OUTLINE\_SHARP
- CHECK\_CIRCLE\_OUTLINED
- CHECK\_CIRCLE\_ROUNDED
- CHECK\_CIRCLE\_SHARP
- CHECK\_OUTLINED
- CHECK\_ROUNDED
- CHECK\_SHARP
- CHECKLIST
- CHECKLIST\_OUTLINED
- CHECKLIST\_ROUNDED
- CHECKLIST\_RTL

- CHECKLIST\_RTL\_OUTLINED
- CHECKLIST\_RTL\_ROUNDED
- CHECKLIST\_RTL\_SHARP
- CHECKLIST\_SHARP
- CHECKROOM
- CHECKROOM\_OUTLINED
- CHECKROOM\_ROUNDED
- CHECKROOM\_SHARP
- CHEVRON\_LEFT
- CHEVRON\_LEFT\_OUTLINED
- CHEVRON\_LEFT\_ROUNDED
- CHEVRON\_LEFT\_SHARP
- CHEVRON\_RIGHT
- CHEVRON\_RIGHT\_OUTLINED
- CHEVRON\_RIGHT\_ROUNDED
- CHEVRON\_RIGHT\_SHARP
- CHILD\_CARE
- CHILD\_CARE\_OUTLINED
- CHILD\_CARE\_ROUNDED
- CHILD\_CARE\_SHARP
- CHILD\_FRIENDLY
- CHILD\_FRIENDLY\_OUTLINED
- CHILD\_FRIENDLY\_ROUNDED
- CHILD\_FRIENDLY\_SHARP
- CHROME\_READER\_MODE
- CHROME\_READER\_MODE\_OUTLINED
- CHROME\_READER\_MODE\_ROUNDED
- CHROME\_READER\_MODE\_SHARP
- CHURCH
- CHURCH\_OUTLINED
- CHURCH\_ROUNDED
- CHURCH\_SHARP
- CIRCLE
- CIRCLE\_NOTIFICATIONS
- CIRCLE\_NOTIFICATIONS\_OUTLINED
- CIRCLE\_NOTIFICATIONS\_ROUNDED
- CIRCLE\_NOTIFICATIONS\_SHARP
- CIRCLE\_OUTLINED
- CIRCLE\_ROUNDED
- CIRCLE\_SHARP
- CLASS\_

- CLASS\_OUTLINED
- CLASS\_ROUNDED
- CLASS\_SHARP
- CLEAN\_HANDS
- CLEAN\_HANDS\_OUTLINED
- CLEAN\_HANDS\_ROUNDED
- CLEAN\_HANDS\_SHARP
- CLEANING\_SERVICES
- CLEANING\_SERVICES\_OUTLINED
- CLEANING\_SERVICES\_ROUNDED
- CLEANING\_SERVICES\_SHARP
- CLEAR
- CLEAR\_ALL
- CLEAR\_ALL\_OUTLINED
- CLEAR\_ALL\_ROUNDED
- CLEAR\_ALL\_SHARP
- CLEAR\_OUTLINED
- CLEAR\_ROUNDED
- CLEAR\_SHARP
- CLOSE
- CLOSE\_FULLSCREEN
- CLOSE\_FULLSCREEN\_OUTLINED
- CLOSE\_FULLSCREEN\_ROUNDED
- CLOSE\_FULLSCREEN\_SHARP
- CLOSE\_OUTLINED
- CLOSE\_ROUNDED
- CLOSE\_SHARP
- CLOSED\_CAPTION
- CLOSED\_CAPTION\_DISABLED
- CLOSED\_CAPTION\_DISABLED\_OUTLINED
- CLOSED\_CAPTION\_DISABLED\_ROUNDED
- CLOSED\_CAPTION\_DISABLED\_SHARP
- CLOSED\_CAPTION\_OFF
- CLOSED\_CAPTION\_OFF\_OUTLINED
- CLOSED\_CAPTION\_OFF\_ROUNDED
- CLOSED\_CAPTION\_OFF\_SHARP
- CLOSED\_CAPTION\_OUTLINED
- CLOSED\_CAPTION\_ROUNDED
- CLOSED\_CAPTION\_SHARP
- CLOUD
- CLOUD\_CIRCLE

- CLOUD\_CIRCLE\_OUTLINED
- CLOUD\_CIRCLE\_ROUNDED
- CLOUD\_CIRCLE\_SHARP
- CLOUD\_DONE
- CLOUD\_DONE\_OUTLINED
- CLOUD\_DONE\_ROUNDED
- CLOUD\_DONE\_SHARP
- CLOUD\_DOWNLOAD
- CLOUD\_DOWNLOAD\_OUTLINED
- CLOUD\_DOWNLOAD\_ROUNDED
- CLOUD\_DOWNLOAD\_SHARP
- CLOUD\_OFF
- CLOUD\_OFF\_OUTLINED
- CLOUD\_OFF\_ROUNDED
- CLOUD\_OFF\_SHARP
- CLOUD\_OUTLINED
- CLOUD\_QUEUE
- CLOUD\_QUEUE\_OUTLINED
- CLOUD\_QUEUE\_ROUNDED
- CLOUD\_QUEUE\_SHARP
- CLOUD\_ROUNDED
- CLOUD\_SHARP
- CLOUD\_SYNC
- CLOUD\_SYNC\_OUTLINED
- CLOUD\_SYNC\_ROUNDED
- CLOUD\_SYNC\_SHARP
- CLOUD\_UPLOAD
- CLOUD\_UPLOAD\_OUTLINED
- CLOUD\_UPLOAD\_ROUNDED
- CLOUD\_UPLOAD\_SHARP
- CLOUDY\_SNOWING
- CO2
- CO2\_OUTLINED
- CO2\_ROUNDED
- CO2\_SHARP
- CO\_PRESENT
- CO\_PRESENT\_OUTLINED
- CO\_PRESENT\_ROUNDED
- CO\_PRESENT\_SHARP
- CODE
- CODE\_OFF

- CODE\_OFF\_OUTLINED
- CODE\_OFF\_ROUNDED
- CODE\_OFF\_SHARP
- CODE\_OUTLINED
- CODE\_ROUNDED
- CODE\_SHARP
- COFFEE
- COFFEE\_MAKER
- COFFEE\_MAKER\_OUTLINED
- COFFEE\_MAKER\_ROUNDED
- COFFEE\_MAKER\_SHARP
- COFFEE\_OUTLINED
- COFFEE\_ROUNDED
- COFFEE\_SHARP
- COLLECTIONS
- COLLECTIONS\_BOOKMARK
- COLLECTIONS\_BOOKMARK\_OUTLINED
- COLLECTIONS\_BOOKMARK\_ROUNDED
- COLLECTIONS\_BOOKMARK\_SHARP
- COLLECTIONS\_OUTLINED
- COLLECTIONS\_ROUNDED
- COLLECTIONS\_SHARP
- COLOR\_LENS
- COLOR\_LENS\_OUTLINED
- COLOR\_LENS\_ROUNDED
- COLOR\_LENS\_SHARP
- COLORIZE
- COLORIZE\_OUTLINED
- COLORIZE\_ROUNDED
- COLORIZE\_SHARP
- COMMENT
- COMMENT\_BANK
- COMMENT\_BANK\_OUTLINED
- COMMENT\_BANK\_ROUNDED
- COMMENT\_BANK\_SHARP
- COMMENT\_OUTLINED
- COMMENT\_ROUNDED
- COMMENT\_SHARP
- COMMENTS\_DISABLED
- COMMENTS\_DISABLED\_OUTLINED
- COMMENTS\_DISABLED\_ROUNDED

- COMMENTS\_DISABLED\_SHARP
- COMMIT
- COMMIT\_OUTLINED
- COMMIT\_ROUNDED
- COMMIT\_SHARP
- COMMUTE
- COMMUTE\_OUTLINED
- COMMUTE\_ROUNDED
- COMMUTE\_SHARP
- COMPARE
- COMPARE\_ARROWS
- COMPARE\_ARROWS\_OUTLINED
- COMPARE\_ARROWS\_ROUNDED
- COMPARE\_ARROWS\_SHARP
- COMPARE\_OUTLINED
- COMPARE\_ROUNDED
- COMPARE\_SHARP
- COMPASS\_CALIBRATION
- COMPASS\_CALIBRATION\_OUTLINED
- COMPASS\_CALIBRATION\_ROUNDED
- COMPASS\_CALIBRATION\_SHARP
- COMPOST
- COMPOST\_OUTLINED
- COMPOST\_ROUNDED
- COMPOST\_SHARP
- COMPRESS
- COMPRESS\_OUTLINED
- COMPRESS\_ROUNDED
- COMPRESS\_SHARP
- COMPUTER
- COMPUTER\_OUTLINED
- COMPUTER\_ROUNDED
- COMPUTER\_SHARP
- CONFIRMATION\_NUM
- CONFIRMATION\_NUM\_OUTLINED
- CONFIRMATION\_NUM\_ROUNDED
- CONFIRMATION\_NUM\_SHARP
- CONFIRMATION\_NUMBER
- CONFIRMATION\_NUMBER\_OUTLINED
- CONFIRMATION\_NUMBER\_ROUNDED
- CONFIRMATION\_NUMBER\_SHARP

- CONNECT\_WITHOUT\_CONTACT
- CONNECT\_WITHOUT\_CONTACT\_OUTLINED
- CONNECT\_WITHOUT\_CONTACT\_ROUNDED
- CONNECT\_WITHOUT\_CONTACT\_SHARP
- CONNECTED\_TV
- CONNECTED\_TV\_OUTLINED
- CONNECTED\_TV\_ROUNDED
- CONNECTED\_TV\_SHARP
- CONNECTING\_AIRPORTS
- CONNECTING\_AIRPORTS\_OUTLINED
- CONNECTING\_AIRPORTS\_ROUNDED
- CONNECTING\_AIRPORTS\_SHARP
- CONSTRUCTION
- CONSTRUCTION\_OUTLINED
- CONSTRUCTION\_ROUNDED
- CONSTRUCTION\_SHARP
- CONTACT\_EMERGENCY
- CONTACT\_EMERGENCY\_OUTLINED
- CONTACT\_EMERGENCY\_ROUNDED
- CONTACT\_EMERGENCY\_SHARP
- CONTACT\_MAIL
- CONTACT\_MAIL\_OUTLINED
- CONTACT\_MAIL\_ROUNDED
- CONTACT\_MAIL\_SHARP
- CONTACT\_PAGE
- CONTACT\_PAGE\_OUTLINED
- CONTACT\_PAGE\_ROUNDED
- CONTACT\_PAGE\_SHARP
- CONTACT\_PHONE
- CONTACT\_PHONE\_OUTLINED
- CONTACT\_PHONE\_ROUNDED
- CONTACT\_PHONE\_SHARP
- CONTACT\_SUPPORT
- CONTACT\_SUPPORT\_OUTLINED
- CONTACT\_SUPPORT\_ROUNDED
- CONTACT\_SUPPORT\_SHARP
- CONTACTLESS
- CONTACTLESS\_OUTLINED
- CONTACTLESS\_ROUNDED
- CONTACTLESS\_SHARP
- CONTACTS



- CONTACTS\_OUTLINED
- CONTACTS\_ROUNDED
- CONTACTS\_SHARP
- CONTENT\_COPY
- CONTENT\_COPY\_OUTLINED
- CONTENT\_COPY\_ROUNDED
- CONTENT\_COPY\_SHARP
- CONTENT\_CUT
- CONTENT\_CUT\_OUTLINED
- CONTENT\_CUT\_ROUNDED
- CONTENT\_CUT\_SHARP
- CONTENT\_PASTE
- CONTENT\_PASTE\_GO
- CONTENT\_PASTE\_GO\_OUTLINED
- CONTENT\_PASTE\_GO\_ROUNDED
- CONTENT\_PASTE\_GO\_SHARP
- CONTENT\_PASTE\_OFF
- CONTENT\_PASTE\_OFF\_OUTLINED
- CONTENT\_PASTE\_OFF\_ROUNDED
- CONTENT\_PASTE\_OFF\_SHARP
- CONTENT\_PASTE\_OUTLINED
- CONTENT\_PASTE\_ROUNDED
- CONTENT\_PASTE\_SEARCH
- CONTENT\_PASTE\_SEARCH\_OUTLINED
- CONTENT\_PASTE\_SEARCH\_ROUNDED
- CONTENT\_PASTE\_SEARCH\_SHARP
- CONTENT\_PASTE\_SHARP
- CONTRAST
- CONTRAST\_OUTLINED
- CONTRAST\_ROUNDED
- CONTRAST\_SHARP
- CONTROL\_CAMERA
- CONTROL\_CAMERA\_OUTLINED
- CONTROL\_CAMERA\_ROUNDED
- CONTROL\_CAMERA\_SHARP
- CONTROL\_POINT
- CONTROL\_POINT\_DUPLICATE
- CONTROL\_POINT\_DUPLICATE\_OUTLINED
- CONTROL\_POINT\_DUPLICATE\_ROUNDED
- CONTROL\_POINT\_DUPLICATE\_SHARP
- CONTROL\_POINT\_OUTLINED

- CONTROL\_POINT\_ROUNDED
- CONTROL\_POINT\_SHARP
- CONVEYOR\_BELT
- COOKIE
- COOKIE\_OUTLINED
- COOKIE\_ROUNDED
- COOKIE\_SHARP
- COPY
- COPY\_ALL
- COPY\_ALL\_OUTLINED
- COPY\_ALL\_ROUNDED
- COPY\_ALL\_SHARP
- COPY\_OUTLINED
- COPY\_ROUNDED
- COPY\_SHARP
- COPYRIGHT
- COPYRIGHT\_OUTLINED
- COPYRIGHT\_ROUNDED
- COPYRIGHT\_SHARP
- CORONAVIRUS
- CORONAVIRUS\_OUTLINED
- CORONAVIRUS\_ROUNDED
- CORONAVIRUS\_SHARP
- CORPORATE\_FARE
- CORPORATE\_FARE\_OUTLINED
- CORPORATE\_FARE\_ROUNDED
- CORPORATE\_FARE\_SHARP
- COTTAGE
- COTTAGE\_OUTLINED
- COTTAGE\_ROUNDED
- COTTAGE\_SHARP
- COUNTERTOPS
- COUNTERTOPS\_OUTLINED
- COUNTERTOPS\_ROUNDED
- COUNTERTOPS\_SHARP
- CREATE
- CREATE\_NEW\_FOLDER
- CREATE\_NEW\_FOLDER\_OUTLINED
- CREATE\_NEW\_FOLDER\_ROUNDED
- CREATE\_NEW\_FOLDER\_SHARP
- CREATE\_OUTLINED

- CREATE\_ROUNDED
- CREATE\_SHARP
- CREDIT\_CARD
- CREDIT\_CARD\_OFF
- CREDIT\_CARD\_OFF\_OUTLINED
- CREDIT\_CARD\_OFF\_ROUNDED
- CREDIT\_CARD\_OFF\_SHARP
- CREDIT\_CARD\_OUTLINED
- CREDIT\_CARD\_ROUNDED
- CREDIT\_CARD\_SHARP
- CREDIT\_SCORE
- CREDIT\_SCORE\_OUTLINED
- CREDIT\_SCORE\_ROUNDED
- CREDIT\_SCORE\_SHARP
- CRIB
- CRIB\_OUTLINED
- CRIB\_ROUNDED
- CRIB\_SHARP
- CRISIS\_ALERT
- CRISIS\_ALERT\_OUTLINED
- CRISIS\_ALERT\_ROUNDED
- CRISIS\_ALERT\_SHARP
- CROP
- CROP\_16\_9
- CROP\_16\_9\_OUTLINED
- CROP\_16\_9\_ROUNDED
- CROP\_16\_9\_SHARP
- CROP\_3\_2
- CROP\_3\_2\_OUTLINED
- CROP\_3\_2\_ROUNDED
- CROP\_3\_2\_SHARP
- CROP\_5\_4
- CROP\_5\_4\_OUTLINED
- CROP\_5\_4\_ROUNDED
- CROP\_5\_4\_SHARP
- CROP\_7\_5
- CROP\_7\_5\_OUTLINED
- CROP\_7\_5\_ROUNDED
- CROP\_7\_5\_SHARP
- CROP\_DIN
- CROP\_DIN\_OUTLINED

- CROP\_DIN\_ROUNDED
- CROP\_DIN\_SHARP
- CROP\_FREE
- CROP\_FREE\_OUTLINED
- CROP\_FREE\_ROUNDED
- CROP\_FREE\_SHARP
- CROP\_LANDSCAPE
- CROP\_LANDSCAPE\_OUTLINED
- CROP\_LANDSCAPE\_ROUNDED
- CROP\_LANDSCAPE\_SHARP
- CROP\_ORIGINAL
- CROP\_ORIGINAL\_OUTLINED
- CROP\_ORIGINAL\_ROUNDED
- CROP\_ORIGINAL\_SHARP
- CROP\_OUTLINED
- CROP\_PORTRAIT
- CROP\_PORTRAIT\_OUTLINED
- CROP\_PORTRAIT\_ROUNDED
- CROP\_PORTRAIT\_SHARP
- CROP\_ROTATE
- CROP\_ROTATE\_OUTLINED
- CROP\_ROTATE\_ROUNDED
- CROP\_ROTATE\_SHARP
- CROP\_ROUNDED
- CROP\_SHARP
- CROP\_SQUARE
- CROP\_SQUARE\_OUTLINED
- CROP\_SQUARE\_ROUNDED
- CROP\_SQUARE\_SHARP
- CRUELTY\_FREE
- CRUELTY\_FREE\_OUTLINED
- CRUELTY\_FREE\_ROUNDED
- CRUELTY\_FREE\_SHARP
- CSS
- CSS\_OUTLINED
- CSS\_ROUNDED
- CSS\_SHARP
- CURRENCY\_BITCOIN
- CURRENCY\_BITCOIN\_OUTLINED
- CURRENCY\_BITCOIN\_ROUNDED
- CURRENCY\_BITCOIN\_SHARP

- CURRENCY\_EXCHANGE
- CURRENCY\_EXCHANGE\_OUTLINED
- CURRENCY\_EXCHANGE\_ROUNDED
- CURRENCY\_EXCHANGE\_SHARP
- CURRENCY\_FRANC
- CURRENCY\_FRANC\_OUTLINED
- CURRENCY\_FRANC\_ROUNDED
- CURRENCY\_FRANC\_SHARP
- CURRENCY\_LIRA
- CURRENCY\_LIRA\_OUTLINED
- CURRENCY\_LIRA\_ROUNDED
- CURRENCY\_LIRA\_SHARP
- CURRENCY\_POUND
- CURRENCY\_POUND\_OUTLINED
- CURRENCY\_POUND\_ROUNDED
- CURRENCY\_POUND\_SHARP
- CURRENCY\_RUBLE
- CURRENCY\_RUBLE\_OUTLINED
- CURRENCY\_RUBLE\_ROUNDED
- CURRENCY\_RUBLE\_SHARP
- CURRENCY\_RUPEE
- CURRENCY\_RUPEE\_OUTLINED
- CURRENCY\_RUPEE\_ROUNDED
- CURRENCY\_RUPEE\_SHARP
- CURRENCY\_YEN
- CURRENCY\_YEN\_OUTLINED
- CURRENCY\_YEN\_ROUNDED
- CURRENCY\_YEN\_SHARP
- CURRENCY\_YUAN
- CURRENCY\_YUAN\_OUTLINED
- CURRENCY\_YUAN\_ROUNDED
- CURRENCY\_YUAN\_SHARP
- CURTAINS
- CURTAINS\_CLOSED
- CURTAINS\_CLOSED\_OUTLINED
- CURTAINS\_CLOSED\_ROUNDED
- CURTAINS\_CLOSED\_SHARP
- CURTAINS\_OUTLINED
- CURTAINS\_ROUNDED
- CURTAINS\_SHARP
- CUT

- CUT\_OUTLINED
- CUT\_ROUNDED
- CUT\_SHARP
- CYCLONE
- CYCLONE\_OUTLINED
- CYCLONE\_ROUNDED
- CYCLONE\_SHARP
- DANGEROUS
- DANGEROUS\_OUTLINED
- DANGEROUS\_ROUNDED
- DANGEROUS\_SHARP
- DARK\_MODE
- DARK\_MODE\_OUTLINED
- DARK\_MODE\_ROUNDED
- DARK\_MODE\_SHARP
- DASHBOARD
- DASHBOARD\_CUSTOMIZE
- DASHBOARD\_CUSTOMIZE\_OUTLINED
- DASHBOARD\_CUSTOMIZE\_ROUNDED
- DASHBOARD\_CUSTOMIZE\_SHARP
- DASHBOARD\_OUTLINED
- DASHBOARD\_ROUNDED
- DASHBOARD\_SHARP
- DATA\_ARRAY
- DATA\_ARRAY\_OUTLINED
- DATA\_ARRAY\_ROUNDED
- DATA\_ARRAY\_SHARP
- DATA\_EXPLORATION
- DATA\_EXPLORATION\_OUTLINED
- DATA\_EXPLORATION\_ROUNDED
- DATA\_EXPLORATION\_SHARP
- DATA\_OBJECT
- DATA\_OBJECT\_OUTLINED
- DATA\_OBJECT\_ROUNDED
- DATA\_OBJECT\_SHARP
- DATA\_SAVER\_OFF
- DATA\_SAVER\_OFF\_OUTLINED
- DATA\_SAVER\_OFF\_ROUNDED
- DATA\_SAVER\_OFF\_SHARP
- DATA\_SAVER\_ON
- DATA\_SAVER\_ON\_OUTLINED

- DATA\_SAVER\_ON\_ROUNDED
- DATA\_SAVER\_ON\_SHARP
- DATA\_THRESHOLDING
- DATA\_THRESHOLDING\_OUTLINED
- DATA\_THRESHOLDING\_ROUNDED
- DATA\_THRESHOLDING\_SHARP
- DATA\_USAGE
- DATA\_USAGE\_OUTLINED
- DATA\_USAGE\_ROUNDED
- DATA\_USAGE\_SHARP
- DATASET
- DATASET\_LINKED
- DATASET\_LINKED\_OUTLINED
- DATASET\_LINKED\_ROUNDED
- DATASET\_LINKED\_SHARP
- DATASET\_OUTLINED
- DATASET\_ROUNDED
- DATASET\_SHARP
- DATE\_RANGE
- DATE\_RANGE\_OUTLINED
- DATE\_RANGE\_ROUNDED
- DATE\_RANGE\_SHARP
- DEBLUR
- DEBLUR\_OUTLINED
- DEBLUR\_ROUNDED
- DEBLUR\_SHARP
- DECK
- DECK\_OUTLINED
- DECK\_ROUNDED
- DECK\_SHARP
- DEHAZE
- DEHAZE\_OUTLINED
- DEHAZE\_ROUNDED
- DEHAZE\_SHARP
- DELETE
- DELETE\_FOREVER
- DELETE\_FOREVER\_OUTLINED
- DELETE\_FOREVER\_ROUNDED
- DELETE\_FOREVER\_SHARP
- DELETE\_OUTLINE
- DELETE\_OUTLINE\_OUTLINED

- DELETE\_OUTLINE\_ROUNDED
- DELETE\_OUTLINE\_SHARP
- DELETE\_OUTLINED
- DELETE\_ROUNDED
- DELETE\_SHARP
- DELETE\_SWEEP
- DELETE\_SWEEP\_OUTLINED
- DELETE\_SWEEP\_ROUNDED
- DELETE\_SWEEP\_SHARP
- DELIVERY\_DINING
- DELIVERY\_DINING\_OUTLINED
- DELIVERY\_DINING\_ROUNDED
- DELIVERY\_DINING\_SHARP
- DENSITY\_LARGE
- DENSITY\_LARGE\_OUTLINED
- DENSITY\_LARGE\_ROUNDED
- DENSITY\_LARGE\_SHARP
- DENSITY\_MEDIUM
- DENSITY\_MEDIUM\_OUTLINED
- DENSITY\_MEDIUM\_ROUNDED
- DENSITY\_MEDIUM\_SHARP
- DENSITY\_SMALL
- DENSITY\_SMALL\_OUTLINED
- DENSITY\_SMALL\_ROUNDED
- DENSITY\_SMALL\_SHARP
- DEPARTURE\_BOARD
- DEPARTURE\_BOARD\_OUTLINED
- DEPARTURE\_BOARD\_ROUNDED
- DEPARTURE\_BOARD\_SHARP
- DESCRIPTION
- DESCRIPTION\_OUTLINED
- DESCRIPTION\_ROUNDED
- DESCRIPTION\_SHARP
- DESELECT
- DESELECT\_OUTLINED
- DESELECT\_ROUNDED
- DESELECT\_SHARP
- DESIGN\_SERVICES
- DESIGN\_SERVICES\_OUTLINED
- DESIGN\_SERVICES\_ROUNDED
- DESIGN\_SERVICES\_SHARP



- DESK
- DESK\_OUTLINED
- DESK\_ROUNDED
- DESK\_SHARP
- DESKTOP\_ACCESS\_DISABLED
- DESKTOP\_ACCESS\_DISABLED\_OUTLINED
- DESKTOP\_ACCESS\_DISABLED\_ROUNDED
- DESKTOP\_ACCESS\_DISABLED\_SHARP
- DESKTOP\_MAC
- DESKTOP\_MAC\_OUTLINED
- DESKTOP\_MAC\_ROUNDED
- DESKTOP\_MAC\_SHARP
- DESKTOP\_WINDOWS
- DESKTOP\_WINDOWS\_OUTLINED
- DESKTOP\_WINDOWS\_ROUNDED
- DESKTOP\_WINDOWS\_SHARP
- DETAILS
- DETAILS\_OUTLINED
- DETAILS\_ROUNDED
- DETAILS\_SHARP
- DEVELOPER\_BOARD
- DEVELOPER\_BOARD\_OFF
- DEVELOPER\_BOARD\_OFF\_OUTLINED
- DEVELOPER\_BOARD\_OFF\_ROUNDED
- DEVELOPER\_BOARD\_OFF\_SHARP
- DEVELOPER\_BOARD\_OUTLINED
- DEVELOPER\_BOARD\_ROUNDED
- DEVELOPER\_BOARD\_SHARP
- DEVELOPER\_MODE
- DEVELOPER\_MODE\_OUTLINED
- DEVELOPER\_MODE\_ROUNDED
- DEVELOPER\_MODE\_SHARP
- DEVICE\_HUB
- DEVICE\_HUB\_OUTLINED
- DEVICE\_HUB\_ROUNDED
- DEVICE\_HUB\_SHARP
- DEVICE\_THERMOSTAT
- DEVICE\_THERMOSTAT\_OUTLINED
- DEVICE\_THERMOSTAT\_ROUNDED
- DEVICE\_THERMOSTAT\_SHARP
- DEVICE\_UNKNOWN

- DEVICE\_UNKNOWN\_OUTLINED
- DEVICE\_UNKNOWN\_ROUNDED
- DEVICE\_UNKNOWN\_SHARP
- DEVICES
- DEVICES\_FOLD
- DEVICES\_FOLD\_OUTLINED
- DEVICES\_FOLD\_ROUNDED
- DEVICES\_FOLD\_SHARP
- DEVICES\_OTHER
- DEVICES\_OTHER\_OUTLINED
- DEVICES\_OTHER\_ROUNDED
- DEVICES\_OTHER\_SHARP
- DEVICES\_OUTLINED
- DEVICES\_ROUNDED
- DEVICES\_SHARP
- DEW\_POINT
- DIALER\_SIP
- DIALER\_SIP\_OUTLINED
- DIALER\_SIP\_ROUNDED
- DIALER\_SIP\_SHARP
- DIALPAD
- DIALPAD\_OUTLINED
- DIALPAD\_ROUNDED
- DIALPAD\_SHARP
- DIAMOND
- DIAMOND\_OUTLINED
- DIAMOND\_ROUNDED
- DIAMOND\_SHARP
- DIFFERENCE
- DIFFERENCE\_OUTLINED
- DIFFERENCE\_ROUNDED
- DIFFERENCE\_SHARP
- DINING
- DINING\_OUTLINED
- DINING\_ROUNDED
- DINING\_SHARP
- DINNER\_DINING
- DINNER\_DINING\_OUTLINED
- DINNER\_DINING\_ROUNDED
- DINNER\_DINING\_SHARP
- DIRECTIONS

- DIRECTIONS\_BIKE
- DIRECTIONS\_BIKE\_OUTLINED
- DIRECTIONS\_BIKE\_ROUNDED
- DIRECTIONS\_BIKE\_SHARP
- DIRECTIONS\_BOAT
- DIRECTIONS\_BOAT\_FILLED
- DIRECTIONS\_BOAT\_FILLED\_OUTLINED
- DIRECTIONS\_BOAT\_FILLED\_ROUNDED
- DIRECTIONS\_BOAT\_FILLED\_SHARP
- DIRECTIONS\_BOAT\_OUTLINED
- DIRECTIONS\_BOAT\_ROUNDED
- DIRECTIONS\_BOAT\_SHARP
- DIRECTIONS\_BUS
- DIRECTIONS\_BUS\_FILLED
- DIRECTIONS\_BUS\_FILLED\_OUTLINED
- DIRECTIONS\_BUS\_FILLED\_ROUNDED
- DIRECTIONS\_BUS\_FILLED\_SHARP
- DIRECTIONS\_BUS\_OUTLINED
- DIRECTIONS\_BUS\_ROUNDED
- DIRECTIONS\_BUS\_SHARP
- DIRECTIONS\_CAR
- DIRECTIONS\_CAR\_FILLED
- DIRECTIONS\_CAR\_FILLED\_OUTLINED
- DIRECTIONS\_CAR\_FILLED\_ROUNDED
- DIRECTIONS\_CAR\_FILLED\_SHARP
- DIRECTIONS\_CAR\_OUTLINED
- DIRECTIONS\_CAR\_ROUNDED
- DIRECTIONS\_CAR\_SHARP
- DIRECTIONS\_FERRY
- DIRECTIONS\_FERRY\_OUTLINED
- DIRECTIONS\_FERRY\_ROUNDED
- DIRECTIONS\_FERRY\_SHARP
- DIRECTIONS\_OFF
- DIRECTIONS\_OFF\_OUTLINED
- DIRECTIONS\_OFF\_ROUNDED
- DIRECTIONS\_OFF\_SHARP
- DIRECTIONS\_OUTLINED
- DIRECTIONS\_RAILWAY
- DIRECTIONS\_RAILWAY\_FILLED
- DIRECTIONS\_RAILWAY\_FILLED\_OUTLINED
- DIRECTIONS\_RAILWAY\_FILLED\_ROUNDED

- DIRECTIONS\_RAILWAY\_FILLED\_SHARP
- DIRECTIONS\_RAILWAY\_OUTLINED
- DIRECTIONS\_RAILWAY\_ROUNDED
- DIRECTIONS\_RAILWAY\_SHARP
- DIRECTIONS\_ROUNDED
- DIRECTIONS\_RUN
- DIRECTIONS\_RUN\_OUTLINED
- DIRECTIONS\_RUN\_ROUNDED
- DIRECTIONS\_RUN\_SHARP
- DIRECTIONS\_SHARP
- DIRECTIONS\_SUBWAY
- DIRECTIONS\_SUBWAY\_FILLED
- DIRECTIONS\_SUBWAY\_FILLED\_OUTLINED
- DIRECTIONS\_SUBWAY\_FILLED\_ROUNDED
- DIRECTIONS\_SUBWAY\_FILLED\_SHARP
- DIRECTIONS\_SUBWAY\_OUTLINED
- DIRECTIONS\_SUBWAY\_ROUNDED
- DIRECTIONS\_SUBWAY\_SHARP
- DIRECTIONS\_TRAIN
- DIRECTIONS\_TRAIN\_OUTLINED
- DIRECTIONS\_TRAIN\_ROUNDED
- DIRECTIONS\_TRAIN\_SHARP
- DIRECTIONS\_TRANSIT
- DIRECTIONS\_TRANSIT\_FILLED
- DIRECTIONS\_TRANSIT\_FILLED\_OUTLINED
- DIRECTIONS\_TRANSIT\_FILLED\_ROUNDED
- DIRECTIONS\_TRANSIT\_FILLED\_SHARP
- DIRECTIONS\_TRANSIT\_OUTLINED
- DIRECTIONS\_TRANSIT\_ROUNDED
- DIRECTIONS\_TRANSIT\_SHARP
- DIRECTIONS\_WALK
- DIRECTIONS\_WALK\_OUTLINED
- DIRECTIONS\_WALK\_ROUNDED
- DIRECTIONS\_WALK\_SHARP
- DIRTY\_LENS
- DIRTY\_LENS\_OUTLINED
- DIRTY\_LENS\_ROUNDED
- DIRTY\_LENS\_SHARP
- DISABLED\_BY\_DEFAULT
- DISABLED\_BY\_DEFAULT\_OUTLINED
- DISABLED\_BY\_DEFAULT\_ROUNDED

- DISABLED\_BY\_DEFAULT\_SHARP
- DISABLED\_VISIBLE
- DISABLED\_VISIBLE\_OUTLINED
- DISABLED\_VISIBLE\_ROUNDED
- DISABLED\_VISIBLE\_SHARP
- DISC\_FULL
- DISC\_FULL\_OUTLINED
- DISC\_FULL\_ROUNDED
- DISC\_FULL\_SHARP
- DISCORD
- DISCORD\_OUTLINED
- DISCORD\_ROUNDED
- DISCORD\_SHARP
- DISCOUNT
- DISCOUNT\_OUTLINED
- DISCOUNT\_ROUNDED
- DISCOUNT\_SHARP
- DISPLAY\_SETTINGS
- DISPLAY\_SETTINGS\_OUTLINED
- DISPLAY\_SETTINGS\_ROUNDED
- DISPLAY\_SETTINGS\_SHARP
- DIVERSITY\_1
- DIVERSITY\_1\_OUTLINED
- DIVERSITY\_1\_ROUNDED
- DIVERSITY\_1\_SHARP
- DIVERSITY\_2
- DIVERSITY\_2\_OUTLINED
- DIVERSITY\_2\_ROUNDED
- DIVERSITY\_2\_SHARP
- DIVERSITY\_3
- DIVERSITY\_3\_OUTLINED
- DIVERSITY\_3\_ROUNDED
- DIVERSITY\_3\_SHARP
- DND\_FORWARDSLASH
- DND\_FORWARDSLASH\_OUTLINED
- DND\_FORWARDSLASH\_ROUNDED
- DND\_FORWARDSLASH\_SHARP
- DNS
- DNS\_OUTLINED
- DNS\_ROUNDED
- DNS\_SHARP

- DO\_DISTURB
- DO\_DISTURB\_ALT
- DO\_DISTURB\_ALT\_OUTLINED
- DO\_DISTURB\_ALT\_ROUNDED
- DO\_DISTURB\_ALT\_SHARP
- DO\_DISTURB\_OFF
- DO\_DISTURB\_OFF\_OUTLINED
- DO\_DISTURB\_OFF\_ROUNDED
- DO\_DISTURB\_OFF\_SHARP
- DO\_DISTURB\_ON
- DO\_DISTURB\_ON\_OUTLINED
- DO\_DISTURB\_ON\_ROUNDED
- DO\_DISTURB\_ON\_SHARP
- DO\_DISTURB\_OUTLINED
- DO\_DISTURB\_ROUNDED
- DO\_DISTURB\_SHARP
- DO\_NOT\_DISTURB
- DO\_NOT\_DISTURB\_ALT
- DO\_NOT\_DISTURB\_ALT\_OUTLINED
- DO\_NOT\_DISTURB\_ALT\_ROUNDED
- DO\_NOT\_DISTURB\_ALT\_SHARP
- DO\_NOT\_DISTURB\_OFF
- DO\_NOT\_DISTURB\_OFF\_OUTLINED
- DO\_NOT\_DISTURB\_OFF\_ROUNDED
- DO\_NOT\_DISTURB\_OFF\_SHARP
- DO\_NOT\_DISTURB\_ON
- DO\_NOT\_DISTURB\_ON\_OUTLINED
- DO\_NOT\_DISTURB\_ON\_ROUNDED
- DO\_NOT\_DISTURB\_ON\_SHARP
- DO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE
- DO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_OUTLINED
- DO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_ROUNDED
- DO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_SHARP
- DO\_NOT\_DISTURB\_OUTLINED
- DO\_NOT\_DISTURB\_ROUNDED
- DO\_NOT\_DISTURB\_SHARP
- DO\_NOT\_STEP
- DO\_NOT\_STEP\_OUTLINED
- DO\_NOT\_STEP\_ROUNDED
- DO\_NOT\_STEP\_SHARP
- DO\_NOT\_TOUCH

- DO\_NOT\_TOUCH\_OUTLINED
- DO\_NOT\_TOUCH\_ROUNDED
- DO\_NOT\_TOUCH\_SHARP
- DOCK
- DOCK\_OUTLINED
- DOCK\_ROUNDED
- DOCK\_SHARP
- DOCUMENT\_SCANNER
- DOCUMENT\_SCANNER\_OUTLINED
- DOCUMENT\_SCANNER\_ROUNDED
- DOCUMENT\_SCANNER\_SHARP
- DOMAIN
- DOMAIN\_ADD
- DOMAIN\_ADD\_OUTLINED
- DOMAIN\_ADD\_ROUNDED
- DOMAIN\_ADD\_SHARP
- DOMAIN\_DISABLED
- DOMAIN\_DISABLED\_OUTLINED
- DOMAIN\_DISABLED\_ROUNDED
- DOMAIN\_DISABLED\_SHARP
- DOMAIN\_OUTLINED
- DOMAIN\_ROUNDED
- DOMAIN\_SHARP
- DOMAIN\_VERIFICATION
- DOMAIN\_VERIFICATION\_OUTLINED
- DOMAIN\_VERIFICATION\_ROUNDED
- DOMAIN\_VERIFICATION\_SHARP
- DONE
- DONE\_ALL
- DONE\_ALL\_OUTLINED
- DONE\_ALL\_ROUNDED
- DONE\_ALL\_SHARP
- DONE\_OUTLINE
- DONE\_OUTLINE\_OUTLINED
- DONE\_OUTLINE\_ROUNDED
- DONE\_OUTLINE\_SHARP
- DONE\_OUTLINED
- DONE\_ROUNDED
- DONE\_SHARP
- DONUT\_LARGE
- DONUT\_LARGE\_OUTLINED

- DONUT\_LARGE\_ROUNDED
- DONUT\_LARGE\_SHARP
- DONUT\_SMALL
- DONUT\_SMALL\_OUTLINED
- DONUT\_SMALL\_ROUNDED
- DONUT\_SMALL\_SHARP
- DOOR\_BACK\_DOOR
- DOOR\_BACK\_DOOR\_OUTLINED
- DOOR\_BACK\_DOOR\_ROUNDED
- DOOR\_BACK\_DOOR\_SHARP
- DOOR\_FRONT\_DOOR
- DOOR\_FRONT\_DOOR\_OUTLINED
- DOOR\_FRONT\_DOOR\_ROUNDED
- DOOR\_FRONT\_DOOR\_SHARP
- DOOR\_SLIDING
- DOOR\_SLIDING\_OUTLINED
- DOOR\_SLIDING\_ROUNDED
- DOOR\_SLIDING\_SHARP
- DOORBELL
- DOORBELL\_OUTLINED
- DOORBELL\_ROUNDED
- DOORBELL\_SHARP
- DOUBLE\_ARROW
- DOUBLE\_ARROW\_OUTLINED
- DOUBLE\_ARROW\_ROUNDED
- DOUBLE\_ARROW\_SHARP
- DOWNHILL\_SKIING
- DOWNHILL\_SKIING\_OUTLINED
- DOWNHILL\_SKIING\_ROUNDED
- DOWNHILL\_SKIING\_SHARP
- DOWNLOAD
- DOWNLOAD\_DONE
- DOWNLOAD\_DONE\_OUTLINED
- DOWNLOAD\_DONE\_ROUNDED
- DOWNLOAD\_DONE\_SHARP
- DOWNLOAD\_FOR\_OFFLINE
- DOWNLOAD\_FOR\_OFFLINE\_OUTLINED
- DOWNLOAD\_FOR\_OFFLINE\_ROUNDED
- DOWNLOAD\_FOR\_OFFLINE\_SHARP
- DOWNLOAD\_OUTLINED
- DOWNLOAD\_ROUNDED



- DOWNLOAD\_SHARP
- DOWNLOADING
- DOWNLOADING\_OUTLINED
- DOWNLOADING\_ROUNDED
- DOWNLOADING\_SHARP
- DRAFTS
- DRAFTS\_OUTLINED
- DRAFTS\_ROUNDED
- DRAFTS\_SHARP
- DRAG\_HANDLE
- DRAG\_HANDLE\_OUTLINED
- DRAG\_HANDLE\_ROUNDED
- DRAG\_HANDLE\_SHARP
- DRAG\_INDICATOR
- DRAG\_INDICATOR\_OUTLINED
- DRAG\_INDICATOR\_ROUNDED
- DRAG\_INDICATOR\_SHARP
- DRAW
- DRAW\_OUTLINED
- DRAW\_ROUNDED
- DRAW\_SHARP
- DRIVE\_ETA
- DRIVE\_ETA\_OUTLINED
- DRIVE\_ETA\_ROUNDED
- DRIVE\_ETA\_SHARP
- DRIVE\_FILE\_MOVE
- DRIVE\_FILE\_MOVE\_OUTLINE
- DRIVE\_FILE\_MOVE\_OUTLINED
- DRIVE\_FILE\_MOVE\_ROUNDED
- DRIVE\_FILE\_MOVE\_RTL
- DRIVE\_FILE\_MOVE\_RTL\_OUTLINED
- DRIVE\_FILE\_MOVE\_RTL\_ROUNDED
- DRIVE\_FILE\_MOVE\_RTL\_SHARP
- DRIVE\_FILE\_MOVE\_SHARP
- DRIVE\_FILE\_RENAME\_OUTLINE
- DRIVE\_FILE\_RENAME\_OUTLINE\_OUTLINED
- DRIVE\_FILE\_RENAME\_OUTLINE\_ROUNDED
- DRIVE\_FILE\_RENAME\_OUTLINE\_SHARP
- DRIVE\_FOLDER\_UPLOAD
- DRIVE\_FOLDER\_UPLOAD\_OUTLINED
- DRIVE\_FOLDER\_UPLOAD\_ROUNDED

- DRIVE\_FOLDER\_UPLOAD\_SHARP
- DRY
- DRY\_CLEANING
- DRY\_CLEANING\_OUTLINED
- DRY\_CLEANING\_ROUNDED
- DRY\_CLEANING\_SHARP
- DRY\_OUTLINED
- DRY\_ROUNDED
- DRY\_SHARP
- DUO
- DUO\_OUTLINED
- DUO\_ROUNDED
- DUO\_SHARP
- DVR
- DVR\_OUTLINED
- DVR\_ROUNDED
- DVR\_SHARP
- DYNAMIC\_FEED
- DYNAMIC\_FEED\_OUTLINED
- DYNAMIC\_FEED\_ROUNDED
- DYNAMIC\_FEED\_SHARP
- DYNAMIC\_FORM
- DYNAMIC\_FORM\_OUTLINED
- DYNAMIC\_FORM\_ROUNDED
- DYNAMIC\_FORM\_SHARP
- E\_MOBILEDATA
- E\_MOBILEDATA\_OUTLINED
- E\_MOBILEDATA\_ROUNDED
- E\_MOBILEDATA\_SHARP
- EARBUDS
- EARBUDS\_BATTERY
- EARBUDS\_BATTERY\_OUTLINED
- EARBUDS\_BATTERY\_ROUNDED
- EARBUDS\_BATTERY\_SHARP
- EARBUDS\_OUTLINED
- EARBUDS\_ROUNDED
- EARBUDS\_SHARP
- EAST
- EAST\_OUTLINED
- EAST\_ROUNDED
- EAST\_SHARP

- ECO
- ECO\_OUTLINED
- ECO\_ROUNDED
- ECO\_SHARP
- EDGESENSOR\_HIGH
- EDGESENSOR\_HIGH\_OUTLINED
- EDGESENSOR\_HIGH\_ROUNDED
- EDGESENSOR\_HIGH\_SHARP
- EDGESENSOR\_LOW
- EDGESENSOR\_LOW\_OUTLINED
- EDGESENSOR\_LOW\_ROUNDED
- EDGESENSOR\_LOW\_SHARP
- EDIT
- EDIT\_ATTRIBUTES
- EDIT\_ATTRIBUTES\_OUTLINED
- EDIT\_ATTRIBUTES\_ROUNDED
- EDIT\_ATTRIBUTES\_SHARP
- EDIT\_CALENDAR
- EDIT\_CALENDAR\_OUTLINED
- EDIT\_CALENDAR\_ROUNDED
- EDIT\_CALENDAR\_SHARP
- EDIT\_DOCUMENT
- EDIT\_LOCATION
- EDIT\_LOCATION\_ALT
- EDIT\_LOCATION\_ALT\_OUTLINED
- EDIT\_LOCATION\_ALT\_ROUNDED
- EDIT\_LOCATION\_ALT\_SHARP
- EDIT\_LOCATION\_OUTLINED
- EDIT\_LOCATION\_ROUNDED
- EDIT\_LOCATION\_SHARP
- EDIT\_NOTE
- EDIT\_NOTE\_OUTLINED
- EDIT\_NOTE\_ROUNDED
- EDIT\_NOTE\_SHARP
- EDIT\_NOTIFICATIONS
- EDIT\_NOTIFICATIONS\_OUTLINED
- EDIT\_NOTIFICATIONS\_ROUNDED
- EDIT\_NOTIFICATIONS\_SHARP
- EDIT\_OFF
- EDIT\_OFF\_OUTLINED
- EDIT\_OFF\_ROUNDED

- EDIT\_OFF\_SHARP
- EDIT\_OUTLINED
- EDIT\_ROAD
- EDIT\_ROAD\_OUTLINED
- EDIT\_ROAD\_ROUNDED
- EDIT\_ROAD\_SHARP
- EDIT\_ROUNDED
- EDIT\_SHARP
- EDIT\_SQUARE
- EGG
- EGG\_ALT
- EGG\_ALT\_OUTLINED
- EGG\_ALT\_ROUNDED
- EGG\_ALT\_SHARP
- EGG\_OUTLINED
- EGG\_ROUNDED
- EGG\_SHARP
- EIGHT\_K
- EIGHT\_K\_OUTLINED
- EIGHT\_K\_PLUS
- EIGHT\_K\_PLUS\_OUTLINED
- EIGHT\_K\_PLUS\_ROUNDED
- EIGHT\_K\_PLUS\_SHARP
- EIGHT\_K\_ROUNDED
- EIGHT\_K\_SHARP
- EIGHT\_MP
- EIGHT\_MP\_OUTLINED
- EIGHT\_MP\_ROUNDED
- EIGHT\_MP\_SHARP
- EIGHTEEN\_MP
- EIGHTEEN\_MP\_OUTLINED
- EIGHTEEN\_MP\_ROUNDED
- EIGHTEEN\_MP\_SHARP
- EIGHTEEN\_UP\_RATING
- EIGHTEEN\_UP\_RATING\_OUTLINED
- EIGHTEEN\_UP\_RATING\_ROUNDED
- EIGHTEEN\_UP\_RATING\_SHARP
- EJECT
- EJECT\_OUTLINED
- EJECT\_ROUNDED
- EJECT\_SHARP

- ELDERLY
- ELDERLY\_OUTLINED
- ELDERLY\_ROUNDED
- ELDERLY\_SHARP
- ELDERLY\_WOMAN
- ELDERLY\_WOMAN\_OUTLINED
- ELDERLY\_WOMAN\_ROUNDED
- ELDERLY\_WOMAN\_SHARP
- ELECTRIC\_BIKE
- ELECTRIC\_BIKE\_OUTLINED
- ELECTRIC\_BIKE\_ROUNDED
- ELECTRIC\_BIKE\_SHARP
- ELECTRIC\_BOLT
- ELECTRIC\_BOLT\_OUTLINED
- ELECTRIC\_BOLT\_ROUNDED
- ELECTRIC\_BOLT\_SHARP
- ELECTRIC\_CAR
- ELECTRIC\_CAR\_OUTLINED
- ELECTRIC\_CAR\_ROUNDED
- ELECTRIC\_CAR\_SHARP
- ELECTRIC\_METER
- ELECTRIC\_METER\_OUTLINED
- ELECTRIC\_METER\_ROUNDED
- ELECTRIC\_METER\_SHARP
- ELECTRIC\_MOPED
- ELECTRIC\_MOPED\_OUTLINED
- ELECTRIC\_MOPED\_ROUNDED
- ELECTRIC\_MOPED\_SHARP
- ELECTRIC\_RICKSHAW
- ELECTRIC\_RICKSHAW\_OUTLINED
- ELECTRIC\_RICKSHAW\_ROUNDED
- ELECTRIC\_RICKSHAW\_SHARP
- ELECTRIC\_SCOOTER
- ELECTRIC\_SCOOTER\_OUTLINED
- ELECTRIC\_SCOOTER\_ROUNDED
- ELECTRIC\_SCOOTER\_SHARP
- ELECTRICAL\_SERVICES
- ELECTRICAL\_SERVICES\_OUTLINED
- ELECTRICAL\_SERVICES\_ROUNDED
- ELECTRICAL\_SERVICES\_SHARP
- ELEVATOR

- ELEVATOR\_OUTLINED
- ELEVATOR\_ROUNDED
- ELEVATOR\_SHARP
- ELEVEN\_MP
- ELEVEN\_MP\_OUTLINED
- ELEVEN\_MP\_ROUNDED
- ELEVEN\_MP\_SHARP
- EMAIL
- EMAIL\_OUTLINED
- EMAIL\_ROUNDED
- EMAIL\_SHARP
- EMERGENCY
- EMERGENCY\_OUTLINED
- EMERGENCY\_RECORDING
- EMERGENCY\_RECORDING\_OUTLINED
- EMERGENCY\_RECORDING\_ROUNDED
- EMERGENCY\_RECORDING\_SHARP
- EMERGENCY\_ROUNDED
- EMERGENCY\_SHARE
- EMERGENCY\_SHARE\_OUTLINED
- EMERGENCY\_SHARE\_ROUNDED
- EMERGENCY\_SHARE\_SHARP
- EMERGENCY\_SHARP
- EMOJI\_EMOTIONS
- EMOJI\_EMOTIONS\_OUTLINED
- EMOJI\_EMOTIONS\_ROUNDED
- EMOJI\_EMOTIONS\_SHARP
- EMOJI\_EVENTS
- EMOJI\_EVENTS\_OUTLINED
- EMOJI\_EVENTS\_ROUNDED
- EMOJI\_EVENTS\_SHARP
- EMOJI\_FLAGS
- EMOJI\_FLAGS\_OUTLINED
- EMOJI\_FLAGS\_ROUNDED
- EMOJI\_FLAGS\_SHARP
- EMOJI\_FOOD\_BEVERAGE
- EMOJI\_FOOD\_BEVERAGE\_OUTLINED
- EMOJI\_FOOD\_BEVERAGE\_ROUNDED
- EMOJI\_FOOD\_BEVERAGE\_SHARP
- EMOJI\_NATURE
- EMOJI\_NATURE\_OUTLINED

- EMOJI\_NATURE\_ROUNDED
- EMOJI\_NATURE\_SHARP
- EMOJI\_OBJECTS
- EMOJI\_OBJECTS\_OUTLINED
- EMOJI\_OBJECTS\_ROUNDED
- EMOJI\_OBJECTS\_SHARP
- EMOJI\_PEOPLE
- EMOJI\_PEOPLE\_OUTLINED
- EMOJI\_PEOPLE\_ROUNDED
- EMOJI\_PEOPLE\_SHARP
- EMOJI\_SYMBOLS
- EMOJI\_SYMBOLS\_OUTLINED
- EMOJI\_SYMBOLS\_ROUNDED
- EMOJI\_SYMBOLS\_SHARP
- EMOJI\_TRANSPORTATION
- EMOJI\_TRANSPORTATION\_OUTLINED
- EMOJI\_TRANSPORTATION\_ROUNDED
- EMOJI\_TRANSPORTATION\_SHARP
- ENERGY\_SAVINGS\_LEAF
- ENERGY\_SAVINGS\_LEAF\_OUTLINED
- ENERGY\_SAVINGS\_LEAF\_ROUNDED
- ENERGY\_SAVINGS\_LEAF\_SHARP
- ENGINEERING
- ENGINEERING\_OUTLINED
- ENGINEERING\_ROUNDED
- ENGINEERING\_SHARP
- ENHANCE\_PHOTO\_TRANSLATE
- ENHANCE\_PHOTO\_TRANSLATE\_OUTLINED
- ENHANCE\_PHOTO\_TRANSLATE\_ROUNDED
- ENHANCE\_PHOTO\_TRANSLATE\_SHARP
- ENHANCED\_ENCRYPTION
- ENHANCED\_ENCRYPTION\_OUTLINED
- ENHANCED\_ENCRYPTION\_ROUNDED
- ENHANCED\_ENCRYPTION\_SHARP
- EQUALIZER
- EQUALIZER\_OUTLINED
- EQUALIZER\_ROUNDED
- EQUALIZER\_SHARP
- ERROR
- ERROR\_OUTLINE
- ERROR\_OUTLINE\_OUTLINED

- ERROR\_OUTLINE\_ROUNDED
- ERROR\_OUTLINE\_SHARP
- ERROR\_OUTLINED
- ERROR\_ROUNDED
- ERROR\_SHARP
- ESCALATOR
- ESCALATOR\_OUTLINED
- ESCALATOR\_ROUNDED
- ESCALATOR\_SHARP
- ESCALATOR\_WARNING
- ESCALATOR\_WARNING\_OUTLINED
- ESCALATOR\_WARNING\_ROUNDED
- ESCALATOR\_WARNING\_SHARP
- EURO
- EURO\_OUTLINED
- EURO\_ROUNDED
- EURO\_SHARP
- EURO\_SYMBOL
- EURO\_SYMBOL\_OUTLINED
- EURO\_SYMBOL\_ROUNDED
- EURO\_SYMBOL\_SHARP
- EV\_STATION
- EV\_STATION\_OUTLINED
- EV\_STATION\_ROUNDED
- EV\_STATION\_SHARP
- EVENT
- EVENT\_AVAILABLE
- EVENT\_AVAILABLE\_OUTLINED
- EVENT\_AVAILABLE\_ROUNDED
- EVENT\_AVAILABLE\_SHARP
- EVENT\_BUSY
- EVENT\_BUSY\_OUTLINED
- EVENT\_BUSY\_ROUNDED
- EVENT\_BUSY\_SHARP
- EVENT\_NOTE
- EVENT\_NOTE\_OUTLINED
- EVENT\_NOTE\_ROUNDED
- EVENT\_NOTE\_SHARP
- EVENT\_OUTLINED
- EVENT\_REPEAT
- EVENT\_REPEAT\_OUTLINED



- EVENT\_REPEAT\_ROUNDED
- EVENT\_REPEAT\_SHARP
- EVENT\_ROUNDED
- EVENT\_SEAT
- EVENT\_SEAT\_OUTLINED
- EVENT\_SEAT\_ROUNDED
- EVENT\_SEAT\_SHARP
- EVENT\_SHARP
- EXIT\_TO\_APP
- EXIT\_TO\_APP\_OUTLINED
- EXIT\_TO\_APP\_ROUNDED
- EXIT\_TO\_APP\_SHARP
- EXPAND
- EXPAND\_CIRCLE\_DOWN
- EXPAND\_CIRCLE\_DOWN\_OUTLINED
- EXPAND\_CIRCLE\_DOWN\_ROUNDED
- EXPAND\_CIRCLE\_DOWN\_SHARP
- EXPAND\_LESS
- EXPAND\_LESS\_OUTLINED
- EXPAND\_LESS\_ROUNDED
- EXPAND\_LESS\_SHARP
- EXPAND\_MORE
- EXPAND\_MORE\_OUTLINED
- EXPAND\_MORE\_ROUNDED
- EXPAND\_MORE\_SHARP
- EXPAND\_OUTLINED
- EXPAND\_ROUNDED
- EXPAND\_SHARP
- EXPLICIT
- EXPLICIT\_OUTLINED
- EXPLICIT\_ROUNDED
- EXPLICIT\_SHARP
- EXPLORE
- EXPLORE\_OFF
- EXPLORE\_OFF\_OUTLINED
- EXPLORE\_OFF\_ROUNDED
- EXPLORE\_OFF\_SHARP
- EXPLORE\_OUTLINED
- EXPLORE\_ROUNDED
- EXPLORE\_SHARP
- EXPOSURE

- EXPOSURE\_MINUS\_1
- EXPOSURE\_MINUS\_1\_OUTLINED
- EXPOSURE\_MINUS\_1\_ROUNDED
- EXPOSURE\_MINUS\_1\_SHARP
- EXPOSURE\_MINUS\_2
- EXPOSURE\_MINUS\_2\_OUTLINED
- EXPOSURE\_MINUS\_2\_ROUNDED
- EXPOSURE\_MINUS\_2\_SHARP
- EXPOSURE\_NEG\_1
- EXPOSURE\_NEG\_1\_OUTLINED
- EXPOSURE\_NEG\_1\_ROUNDED
- EXPOSURE\_NEG\_1\_SHARP
- EXPOSURE\_NEG\_2
- EXPOSURE\_NEG\_2\_OUTLINED
- EXPOSURE\_NEG\_2\_ROUNDED
- EXPOSURE\_NEG\_2\_SHARP
- EXPOSURE\_OUTLINED
- EXPOSURE\_PLUS\_1
- EXPOSURE\_PLUS\_1\_OUTLINED
- EXPOSURE\_PLUS\_1\_ROUNDED
- EXPOSURE\_PLUS\_1\_SHARP
- EXPOSURE\_PLUS\_2
- EXPOSURE\_PLUS\_2\_OUTLINED
- EXPOSURE\_PLUS\_2\_ROUNDED
- EXPOSURE\_PLUS\_2\_SHARP
- EXPOSURE\_ROUNDED
- EXPOSURE\_SHARP
- EXPOSURE\_ZERO
- EXPOSURE\_ZERO\_OUTLINED
- EXPOSURE\_ZERO\_ROUNDED
- EXPOSURE\_ZERO\_SHARP
- EXTENSION
- EXTENSION\_OFF
- EXTENSION\_OFF\_OUTLINED
- EXTENSION\_OFF\_ROUNDED
- EXTENSION\_OFF\_SHARP
- EXTENSION\_OUTLINED
- EXTENSION\_ROUNDED
- EXTENSION\_SHARP
- FACE
- FACE\_2

- FACE\_2\_OUTLINED
- FACE\_2\_ROUNDED
- FACE\_2\_SHARP
- FACE\_3
- FACE\_3\_OUTLINED
- FACE\_3\_ROUNDED
- FACE\_3\_SHARP
- FACE\_4
- FACE\_4\_OUTLINED
- FACE\_4\_ROUNDED
- FACE\_4\_SHARP
- FACE\_5
- FACE\_5\_OUTLINED
- FACE\_5\_ROUNDED
- FACE\_5\_SHARP
- FACE\_6
- FACE\_6\_OUTLINED
- FACE\_6\_ROUNDED
- FACE\_6\_SHARP
- FACE\_OUTLINED
- FACE\_RETOUCHING\_NATURAL
- FACE\_RETOUCHING\_NATURAL\_OUTLINED
- FACE\_RETOUCHING\_NATURAL\_ROUNDED
- FACE\_RETOUCHING\_NATURAL\_SHARP
- FACE\_RETOUCHING\_OFF
- FACE\_RETOUCHING\_OFF\_OUTLINED
- FACE\_RETOUCHING\_OFF\_ROUNDED
- FACE\_RETOUCHING\_OFF\_SHARP
- FACE\_ROUNDED
- FACE\_SHARP
- FACE\_UNLOCK\_OUTLINED
- FACE\_UNLOCK\_ROUNDED
- FACE\_UNLOCK\_SHARP
- FACEBOOK
- FACEBOOK\_OUTLINED
- FACEBOOK\_ROUNDED
- FACEBOOK\_SHARP
- FACT\_CHECK
- FACT\_CHECK\_OUTLINED
- FACT\_CHECK\_ROUNDED
- FACT\_CHECK\_SHARP

- FACTORY
- FACTORY\_OUTLINED
- FACTORY\_ROUNDED
- FACTORY\_SHARP
- FAMILY\_RESTROOM
- FAMILY\_RESTROOM\_OUTLINED
- FAMILY\_RESTROOM\_ROUNDED
- FAMILY\_RESTROOM\_SHARP
- FAST\_FORWARD
- FAST\_FORWARD\_OUTLINED
- FAST\_FORWARD\_ROUNDED
- FAST\_FORWARD\_SHARP
- FAST\_REWIND
- FAST\_REWIND\_OUTLINED
- FAST\_REWIND\_ROUNDED
- FAST\_REWIND\_SHARP
- FASTFOOD
- FASTFOOD\_OUTLINED
- FASTFOOD\_ROUNDED
- FASTFOOD\_SHARP
- FAVORITE
- FAVORITE\_BORDER
- FAVORITE\_BORDER\_OUTLINED
- FAVORITE\_BORDER\_ROUNDED
- FAVORITE\_BORDER\_SHARP
- FAVORITE\_OUTLINE
- FAVORITE\_OUTLINE\_OUTLINED
- FAVORITE\_OUTLINE\_ROUNDED
- FAVORITE\_OUTLINE\_SHARP
- FAVORITE\_OUTLINED
- FAVORITE\_ROUNDED
- FAVORITE\_SHARP
- FAX
- FAX\_OUTLINED
- FAX\_ROUNDED
- FAX\_SHARP
- FEATURED\_PLAY\_LIST
- FEATURED\_PLAY\_LIST\_OUTLINED
- FEATURED\_PLAY\_LIST\_ROUNDED
- FEATURED\_PLAY\_LIST\_SHARP
- FEATURED\_VIDEO

- FEATURED\_VIDEO\_OUTLINED
- FEATURED\_VIDEO\_ROUNDED
- FEATURED\_VIDEO\_SHARP
- FEED
- FEED\_OUTLINED
- FEED\_ROUNDED
- FEED\_SHARP
- FEEDBACK
- FEEDBACK\_OUTLINED
- FEEDBACK\_ROUNDED
- FEEDBACK\_SHARP
- FEMALE
- FEMALE\_OUTLINED
- FEMALE\_ROUNDED
- FEMALE\_SHARP
- FENCE
- FENCE\_OUTLINED
- FENCE\_ROUNDED
- FENCE\_SHARP
- FESTIVAL
- FESTIVAL\_OUTLINED
- FESTIVAL\_ROUNDED
- FESTIVAL\_SHARP
- FIBER\_DVR
- FIBER\_DVR\_OUTLINED
- FIBER\_DVR\_ROUNDED
- FIBER\_DVR\_SHARP
- FIBER\_MANUAL\_RECORD
- FIBER\_MANUAL\_RECORD\_OUTLINED
- FIBER\_MANUAL\_RECORD\_ROUNDED
- FIBER\_MANUAL\_RECORD\_SHARP
- FIBER\_NEW
- FIBER\_NEW\_OUTLINED
- FIBER\_NEW\_ROUNDED
- FIBER\_NEW\_SHARP
- FIBER\_PIN
- FIBER\_PIN\_OUTLINED
- FIBER\_PIN\_ROUNDED
- FIBER\_PIN\_SHARP
- FIBER\_SMART\_RECORD
- FIBER\_SMART\_RECORD\_OUTLINED

- FIBER\_SMART\_RECORD\_ROUNDED
- FIBER\_SMART\_RECORD\_SHARP
- FIFTEEN\_MP
- FIFTEEN\_MP\_OUTLINED
- FIFTEEN\_MP\_ROUNDED
- FIFTEEN\_MP\_SHARP
- FILE\_COPY
- FILE\_COPY\_OUTLINED
- FILE\_COPY\_ROUNDED
- FILE\_COPY\_SHARP
- FILE\_DOWNLOAD
- FILE\_DOWNLOAD\_DONE
- FILE\_DOWNLOAD\_DONE\_OUTLINED
- FILE\_DOWNLOAD\_DONE\_ROUNDED
- FILE\_DOWNLOAD\_DONE\_SHARP
- FILE\_DOWNLOAD\_OFF
- FILE\_DOWNLOAD\_OFF\_OUTLINED
- FILE\_DOWNLOAD\_OFF\_ROUNDED
- FILE\_DOWNLOAD\_OFF\_SHARP
- FILE\_DOWNLOAD\_OUTLINED
- FILE\_DOWNLOAD\_ROUNDED
- FILE\_DOWNLOAD\_SHARP
- FILE\_OPEN
- FILE\_OPEN\_OUTLINED
- FILE\_OPEN\_ROUNDED
- FILE\_OPEN\_SHARP
- FILE\_PRESENT
- FILE\_PRESENT\_OUTLINED
- FILE\_PRESENT\_ROUNDED
- FILE\_PRESENT\_SHARP
- FILE\_UPLOAD
- FILE\_UPLOAD\_OFF
- FILE\_UPLOAD\_OUTLINED
- FILE\_UPLOAD\_ROUNDED
- FILE\_UPLOAD\_SHARP
- FILTER
- FILTER\_1
- FILTER\_1\_OUTLINED
- FILTER\_1\_ROUNDED
- FILTER\_1\_SHARP
- FILTER\_2

- FILTER\_2\_OUTLINED
- FILTER\_2\_ROUNDED
- FILTER\_2\_SHARP
- FILTER\_3
- FILTER\_3\_OUTLINED
- FILTER\_3\_ROUNDED
- FILTER\_3\_SHARP
- FILTER\_4
- FILTER\_4\_OUTLINED
- FILTER\_4\_ROUNDED
- FILTER\_4\_SHARP
- FILTER\_5
- FILTER\_5\_OUTLINED
- FILTER\_5\_ROUNDED
- FILTER\_5\_SHARP
- FILTER\_6
- FILTER\_6\_OUTLINED
- FILTER\_6\_ROUNDED
- FILTER\_6\_SHARP
- FILTER\_7
- FILTER\_7\_OUTLINED
- FILTER\_7\_ROUNDED
- FILTER\_7\_SHARP
- FILTER\_8
- FILTER\_8\_OUTLINED
- FILTER\_8\_ROUNDED
- FILTER\_8\_SHARP
- FILTER\_9
- FILTER\_9\_OUTLINED
- FILTER\_9\_PLUS
- FILTER\_9\_PLUS\_OUTLINED
- FILTER\_9\_PLUS\_ROUNDED
- FILTER\_9\_PLUS\_SHARP
- FILTER\_9\_ROUNDED
- FILTER\_9\_SHARP
- FILTER\_ALT
- FILTER\_ALT\_OFF
- FILTER\_ALT\_OFF\_OUTLINED
- FILTER\_ALT\_OFF\_ROUNDED
- FILTER\_ALT\_OFF\_SHARP
- FILTER\_ALT\_OUTLINED

- FILTER\_ALT\_ROUNDED
- FILTER\_ALT\_SHARP
- FILTER\_B\_AND\_W
- FILTER\_B\_AND\_W\_OUTLINED
- FILTER\_B\_AND\_W\_ROUNDED
- FILTER\_B\_AND\_W\_SHARP
- FILTER\_CENTER\_FOCUS
- FILTER\_CENTER\_FOCUS\_OUTLINED
- FILTER\_CENTER\_FOCUS\_ROUNDED
- FILTER\_CENTER\_FOCUS\_SHARP
- FILTER\_DRAMA
- FILTER\_DRAMA\_OUTLINED
- FILTER\_DRAMA\_ROUNDED
- FILTER\_DRAMA\_SHARP
- FILTER\_FRAMES
- FILTER\_FRAMES\_OUTLINED
- FILTER\_FRAMES\_ROUNDED
- FILTER\_FRAMES\_SHARP
- FILTER\_HDR
- FILTER\_HDR\_OUTLINED
- FILTER\_HDR\_ROUNDED
- FILTER\_HDR\_SHARP
- FILTER\_LIST
- FILTER\_LIST\_ALT
- FILTER\_LIST\_OFF
- FILTER\_LIST\_OFF\_OUTLINED
- FILTER\_LIST\_OFF\_ROUNDED
- FILTER\_LIST\_OFF\_SHARP
- FILTER\_LIST\_OUTLINED
- FILTER\_LIST\_ROUNDED
- FILTER\_LIST\_SHARP
- FILTER\_NONE
- FILTER\_NONE\_OUTLINED
- FILTER\_NONE\_ROUNDED
- FILTER\_NONE\_SHARP
- FILTER\_OUTLINED
- FILTER\_ROUNDED
- FILTER\_SHARP
- FILTER\_TILT\_SHIFT
- FILTER\_TILT\_SHIFT\_OUTLINED
- FILTER\_TILT\_SHIFT\_ROUNDED



- FILTER\_TILT\_SHIFT\_SHARP
- FILTER\_VINTAGE
- FILTER\_VINTAGE\_OUTLINED
- FILTER\_VINTAGE\_ROUNDED
- FILTER\_VINTAGE\_SHARP
- FIND\_IN\_PAGE
- FIND\_IN\_PAGE\_OUTLINED
- FIND\_IN\_PAGE\_ROUNDED
- FIND\_IN\_PAGE\_SHARP
- FIND\_REPLACE
- FIND\_REPLACE\_OUTLINED
- FIND\_REPLACE\_ROUNDED
- FIND\_REPLACE\_SHARP
- FINGERPRINT
- FINGERPRINT\_OUTLINED
- FINGERPRINT\_ROUNDED
- FINGERPRINT\_SHARP
- FIRE\_EXTINGUISHER
- FIRE\_EXTINGUISHER\_OUTLINED
- FIRE\_EXTINGUISHER\_ROUNDED
- FIRE\_EXTINGUISHER\_SHARP
- FIRE\_HYDRANT
- FIRE\_HYDRANT\_ALT
- FIRE\_HYDRANT\_ALT\_OUTLINED
- FIRE\_HYDRANT\_ALT\_ROUNDED
- FIRE\_HYDRANT\_ALT\_SHARP
- FIRE\_TRUCK
- FIRE\_TRUCK\_OUTLINED
- FIRE\_TRUCK\_ROUNDED
- FIRE\_TRUCK\_SHARP
- FIREPLACE
- FIREPLACE\_OUTLINED
- FIREPLACE\_ROUNDED
- FIREPLACE\_SHARP
- FIRST\_PAGE
- FIRST\_PAGE\_OUTLINED
- FIRST\_PAGE\_ROUNDED
- FIRST\_PAGE\_SHARP
- FIT\_SCREEN
- FIT\_SCREEN\_OUTLINED
- FIT\_SCREEN\_ROUNDED

- FIT\_SCREEN\_SHARP
- FITBIT
- FITBIT\_OUTLINED
- FITBIT\_ROUNDED
- FITBIT\_SHARP
- FITNESS\_CENTER
- FITNESS\_CENTER\_OUTLINED
- FITNESS\_CENTER\_ROUNDED
- FITNESS\_CENTER\_SHARP
- FIVE\_G
- FIVE\_G\_OUTLINED
- FIVE\_G\_ROUNDED
- FIVE\_G\_SHARP
- FIVE\_K
- FIVE\_K\_OUTLINED
- FIVE\_K\_PLUS
- FIVE\_K\_PLUS\_OUTLINED
- FIVE\_K\_PLUS\_ROUNDED
- FIVE\_K\_PLUS\_SHARP
- FIVE\_K\_ROUNDED
- FIVE\_K\_SHARP
- FIVE\_MP
- FIVE\_MP\_OUTLINED
- FIVE\_MP\_ROUNDED
- FIVE\_MP\_SHARP
- FLAG
- FLAG\_CIRCLE
- FLAG\_CIRCLE\_OUTLINED
- FLAG\_CIRCLE\_ROUNDED
- FLAG\_CIRCLE\_SHARP
- FLAG\_OUTLINED
- FLAG\_ROUNDED
- FLAG\_SHARP
- FLAKY
- FLAKY\_OUTLINED
- FLAKY\_ROUNDED
- FLAKY\_SHARP
- FLARE
- FLARE\_OUTLINED
- FLARE\_ROUNDED
- FLARE\_SHARP

- FLASH\_AUTO
- FLASH\_AUTO\_OUTLINED
- FLASH\_AUTO\_ROUNDED
- FLASH\_AUTO\_SHARP
- FLASH\_OFF
- FLASH\_OFF\_OUTLINED
- FLASH\_OFF\_ROUNDED
- FLASH\_OFF\_SHARP
- FLASH\_ON
- FLASH\_ON\_OUTLINED
- FLASH\_ON\_ROUNDED
- FLASH\_ON\_SHARP
- FLASHLIGHT\_OFF
- FLASHLIGHT\_OFF\_OUTLINED
- FLASHLIGHT\_OFF\_ROUNDED
- FLASHLIGHT\_OFF\_SHARP
- FLASHLIGHT\_ON
- FLASHLIGHT\_ON\_OUTLINED
- FLASHLIGHT\_ON\_ROUNDED
- FLASHLIGHT\_ON\_SHARP
- FLATWARE
- FLATWARE\_OUTLINED
- FLATWARE\_ROUNDED
- FLATWARE\_SHARP
- FLIGHT
- FLIGHT\_CLASS
- FLIGHT\_CLASS\_OUTLINED
- FLIGHT\_CLASS\_ROUNDED
- FLIGHT\_CLASS\_SHARP
- FLIGHT\_LAND
- FLIGHT\_LAND\_OUTLINED
- FLIGHT\_LAND\_ROUNDED
- FLIGHT\_LAND\_SHARP
- FLIGHT\_OUTLINED
- FLIGHT\_ROUNDED
- FLIGHT\_SHARP
- FLIGHT\_TAKEOFF
- FLIGHT\_TAKEOFF\_OUTLINED
- FLIGHT\_TAKEOFF\_ROUNDED
- FLIGHT\_TAKEOFF\_SHARP
- FLIP

- FLIP\_CAMERA\_ANDROID
- FLIP\_CAMERA\_ANDROID\_OUTLINED
- FLIP\_CAMERA\_ANDROID\_ROUNDED
- FLIP\_CAMERA\_ANDROID\_SHARP
- FLIP\_CAMERA\_IOS
- FLIP\_CAMERA\_IOS\_OUTLINED
- FLIP\_CAMERA\_IOS\_ROUNDED
- FLIP\_CAMERA\_IOS\_SHARP
- FLIP\_OUTLINED
- FLIP\_ROUNDED
- FLIP\_SHARP
- FLIP\_TO\_BACK
- FLIP\_TO\_BACK\_OUTLINED
- FLIP\_TO\_BACK\_ROUNDED
- FLIP\_TO\_BACK\_SHARP
- FLIP\_TO\_FRONT
- FLIP\_TO\_FRONT\_OUTLINED
- FLIP\_TO\_FRONT\_ROUNDED
- FLIP\_TO\_FRONT\_SHARP
- FLOOD
- FLOOD\_OUTLINED
- FLOOD\_ROUNDED
- FLOOD\_SHARP
- FLOURESCENT
- FLOURESCENT\_OUTLINED
- FLOURESCENT\_ROUNDED
- FLOURESCENT\_SHARP
- FLUORESCENT
- FLUORESCENT\_OUTLINED
- FLUORESCENT\_ROUNDED
- FLUORESCENT\_SHARP
- FLUTTER\_DASH
- FLUTTER\_DASH\_OUTLINED
- FLUTTER\_DASH\_ROUNDED
- FLUTTER\_DASH\_SHARP
- FMD\_BAD
- FMD\_BAD\_OUTLINED
- FMD\_BAD\_ROUNDED
- FMD\_BAD\_SHARP
- FMD\_GOOD
- FMD\_GOOD\_OUTLINED

- FMD\_GOOD\_ROUNDED
- FMD\_GOOD\_SHARP
- FOGGY
- FOLDER
- FOLDER\_COPY
- FOLDER\_COPY\_OUTLINED
- FOLDER\_COPY\_ROUNDED
- FOLDER\_COPY\_SHARP
- FOLDER\_DELETE
- FOLDER\_DELETE\_OUTLINED
- FOLDER\_DELETE\_ROUNDED
- FOLDER\_DELETE\_SHARP
- FOLDER\_OFF
- FOLDER\_OFF\_OUTLINED
- FOLDER\_OFF\_ROUNDED
- FOLDER\_OFF\_SHARP
- FOLDER\_OPEN
- FOLDER\_OPEN\_OUTLINED
- FOLDER\_OPEN\_ROUNDED
- FOLDER\_OPEN\_SHARP
- FOLDER\_OUTLINED
- FOLDER\_ROUNDED
- FOLDER\_SHARED
- FOLDER\_SHARED\_OUTLINED
- FOLDER\_SHARED\_ROUNDED
- FOLDER\_SHARED\_SHARP
- FOLDER\_SHARP
- FOLDER\_SPECIAL
- FOLDER\_SPECIAL\_OUTLINED
- FOLDER\_SPECIAL\_ROUNDED
- FOLDER\_SPECIAL\_SHARP
- FOLDER\_ZIP
- FOLDER\_ZIP\_OUTLINED
- FOLDER\_ZIP\_ROUNDED
- FOLDER\_ZIP\_SHARP
- FOLLOW\_THE\_SIGNS
- FOLLOW\_THE\_SIGNS\_OUTLINED
- FOLLOW\_THE\_SIGNS\_ROUNDED
- FOLLOW\_THE\_SIGNS\_SHARP
- FONT\_DOWNLOAD
- FONT\_DOWNLOAD\_OFF

- FONT\_DOWNLOAD\_OFF\_OUTLINED
- FONT\_DOWNLOAD\_OFF\_ROUNDED
- FONT\_DOWNLOAD\_OFF\_SHARP
- FONT\_DOWNLOAD\_OUTLINED
- FONT\_DOWNLOAD\_ROUNDED
- FONT\_DOWNLOAD\_SHARP
- FOOD\_BANK
- FOOD\_BANK\_OUTLINED
- FOOD\_BANK\_ROUNDED
- FOOD\_BANK\_SHARP
- FOREST
- FOREST\_OUTLINED
- FOREST\_ROUNDED
- FOREST\_SHARP
- FORK\_LEFT
- FORK\_LEFT\_OUTLINED
- FORK\_LEFT\_ROUNDED
- FORK\_LEFT\_SHARP
- FORK\_RIGHT
- FORK\_RIGHT\_OUTLINED
- FORK\_RIGHT\_ROUNDED
- FORK\_RIGHT\_SHARP
- FORKLIFT
- FORMAT\_ALIGN\_CENTER
- FORMAT\_ALIGN\_CENTER\_OUTLINED
- FORMAT\_ALIGN\_CENTER\_ROUNDED
- FORMAT\_ALIGN\_CENTER\_SHARP
- FORMAT\_ALIGN\_JUSTIFY
- FORMAT\_ALIGN\_JUSTIFY\_OUTLINED
- FORMAT\_ALIGN\_JUSTIFY\_ROUNDED
- FORMAT\_ALIGN\_JUSTIFY\_SHARP
- FORMAT\_ALIGN\_LEFT
- FORMAT\_ALIGN\_LEFT\_OUTLINED
- FORMAT\_ALIGN\_LEFT\_ROUNDED
- FORMAT\_ALIGN\_LEFT\_SHARP
- FORMAT\_ALIGN\_RIGHT
- FORMAT\_ALIGN\_RIGHT\_OUTLINED
- FORMAT\_ALIGN\_RIGHT\_ROUNDED
- FORMAT\_ALIGN\_RIGHT\_SHARP
- FORMAT\_BOLD
- FORMAT\_BOLD\_OUTLINED

- FORMAT\_BOLD\_ROUNDED
- FORMAT\_BOLD\_SHARP
- FORMAT\_CLEAR
- FORMAT\_CLEAR\_OUTLINED
- FORMAT\_CLEAR\_ROUNDED
- FORMAT\_CLEAR\_SHARP
- FORMAT\_COLOR\_FILL
- FORMAT\_COLOR\_FILL\_OUTLINED
- FORMAT\_COLOR\_FILL\_ROUNDED
- FORMAT\_COLOR\_FILL\_SHARP
- FORMAT\_COLOR\_RESET
- FORMAT\_COLOR\_RESET\_OUTLINED
- FORMAT\_COLOR\_RESET\_ROUNDED
- FORMAT\_COLOR\_RESET\_SHARP
- FORMAT\_COLOR\_TEXT
- FORMAT\_COLOR\_TEXT\_OUTLINED
- FORMAT\_COLOR\_TEXT\_ROUNDED
- FORMAT\_COLOR\_TEXT\_SHARP
- FORMAT\_INDENT\_DECREASE
- FORMAT\_INDENT\_DECREASE\_OUTLINED
- FORMAT\_INDENT\_DECREASE\_ROUNDED
- FORMAT\_INDENT\_DECREASE\_SHARP
- FORMAT\_INDENT\_INCREASE
- FORMAT\_INDENT\_INCREASE\_OUTLINED
- FORMAT\_INDENT\_INCREASE\_ROUNDED
- FORMAT\_INDENT\_INCREASE\_SHARP
- FORMAT\_ITALIC
- FORMAT\_ITALIC\_OUTLINED
- FORMAT\_ITALIC\_ROUNDED
- FORMAT\_ITALIC\_SHARP
- FORMAT\_LINE\_SPACING
- FORMAT\_LINE\_SPACING\_OUTLINED
- FORMAT\_LINE\_SPACING\_ROUNDED
- FORMAT\_LINE\_SPACING\_SHARP
- FORMAT\_LIST\_BULLETED
- FORMAT\_LIST\_BULLETED\_ADD
- FORMAT\_LIST\_BULLETED\_OUTLINED
- FORMAT\_LIST\_BULLETED\_ROUNDED
- FORMAT\_LIST\_BULLETED\_SHARP
- FORMAT\_LIST\_NUMBERED
- FORMAT\_LIST\_NUMBERED\_OUTLINED

- FORMAT\_LIST\_NUMBERED\_ROUNDED
- FORMAT\_LIST\_NUMBERED\_RTL
- FORMAT\_LIST\_NUMBERED\_RTL\_OUTLINED
- FORMAT\_LIST\_NUMBERED\_RTL\_ROUNDED
- FORMAT\_LIST\_NUMBERED\_RTL\_SHARP
- FORMAT\_LIST\_NUMBERED\_SHARP
- FORMAT\_OVERLINE
- FORMAT\_OVERLINE\_OUTLINED
- FORMAT\_OVERLINE\_ROUNDED
- FORMAT\_OVERLINE\_SHARP
- FORMAT\_PAINT
- FORMAT\_PAINT\_OUTLINED
- FORMAT\_PAINT\_ROUNDED
- FORMAT\_PAINT\_SHARP
- FORMAT\_QUOTE
- FORMAT\_QUOTE\_OUTLINED
- FORMAT\_QUOTE\_ROUNDED
- FORMAT\_QUOTE\_SHARP
- FORMAT\_SHAPES
- FORMAT\_SHAPES\_OUTLINED
- FORMAT\_SHAPES\_ROUNDED
- FORMAT\_SHAPES\_SHARP
- FORMAT\_SIZE
- FORMAT\_SIZE\_OUTLINED
- FORMAT\_SIZE\_ROUNDED
- FORMAT\_SIZE\_SHARP
- FORMAT\_STRIKETHROUGH
- FORMAT\_STRIKETHROUGH\_OUTLINED
- FORMAT\_STRIKETHROUGH\_ROUNDED
- FORMAT\_STRIKETHROUGH\_SHARP
- FORMAT\_TEXTDIRECTION\_L\_TO\_R
- FORMAT\_TEXTDIRECTION\_L\_TO\_R\_OUTLINED
- FORMAT\_TEXTDIRECTION\_L\_TO\_R\_ROUNDED
- FORMAT\_TEXTDIRECTION\_L\_TO\_R\_SHARP
- FORMAT\_TEXTDIRECTION\_R\_TO\_L
- FORMAT\_TEXTDIRECTION\_R\_TO\_L\_OUTLINED
- FORMAT\_TEXTDIRECTION\_R\_TO\_L\_ROUNDED
- FORMAT\_TEXTDIRECTION\_R\_TO\_L\_SHARP
- FORMAT\_UNDERLINE
- FORMAT\_UNDERLINE\_OUTLINED
- FORMAT\_UNDERLINE\_ROUNDED



- FORMAT\_UNDERLINE\_SHARP
- FORMAT\_UNDERLINED
- FORMAT\_UNDERLINED\_OUTLINED
- FORMAT\_UNDERLINED\_ROUNDED
- FORMAT\_UNDERLINED\_SHARP
- FORT
- FORT\_OUTLINED
- FORT\_ROUNDED
- FORT\_SHARP
- FORUM
- FORUM\_OUTLINED
- FORUM\_ROUNDED
- FORUM\_SHARP
- FORWARD
- FORWARD\_10
- FORWARD\_10\_OUTLINED
- FORWARD\_10\_ROUNDED
- FORWARD\_10\_SHARP
- FORWARD\_30
- FORWARD\_30\_OUTLINED
- FORWARD\_30\_ROUNDED
- FORWARD\_30\_SHARP
- FORWARD\_5
- FORWARD\_5\_OUTLINED
- FORWARD\_5\_ROUNDED
- FORWARD\_5\_SHARP
- FORWARD\_OUTLINED
- FORWARD\_ROUNDED
- FORWARD\_SHARP
- FORWARD\_TO\_INBOX
- FORWARD\_TO\_INBOX\_OUTLINED
- FORWARD\_TO\_INBOX\_ROUNDED
- FORWARD\_TO\_INBOX\_SHARP
- FOUNDATION
- FOUNDATION\_OUTLINED
- FOUNDATION\_ROUNDED
- FOUNDATION\_SHARP
- FOUR\_G\_MOBILEDATA
- FOUR\_G\_MOBILEDATA\_OUTLINED
- FOUR\_G\_MOBILEDATA\_ROUNDED
- FOUR\_G\_MOBILEDATA\_SHARP

- FOUR\_G\_PLUS\_MOBILEDATA
- FOUR\_G\_PLUS\_MOBILEDATA\_OUTLINED
- FOUR\_G\_PLUS\_MOBILEDATA\_ROUNDED
- FOUR\_G\_PLUS\_MOBILEDATA\_SHARP
- FOUR\_K
- FOUR\_K\_OUTLINED
- FOUR\_K\_PLUS
- FOUR\_K\_PLUS\_OUTLINED
- FOUR\_K\_PLUS\_ROUNDED
- FOUR\_K\_PLUS\_SHARP
- FOUR\_K\_ROUNDED
- FOUR\_K\_SHARP
- FOUR\_MP
- FOUR\_MP\_OUTLINED
- FOUR\_MP\_ROUNDED
- FOUR\_MP\_SHARP
- FOURTEEN\_MP
- FOURTEEN\_MP\_OUTLINED
- FOURTEEN\_MP\_ROUNDED
- FOURTEEN\_MP\_SHARP
- FREE\_BREAKFAST
- FREE\_BREAKFAST\_OUTLINED
- FREE\_BREAKFAST\_ROUNDED
- FREE\_BREAKFAST\_SHARP
- FREE\_CANCELLATION
- FREE\_CANCELLATION\_OUTLINED
- FREE\_CANCELLATION\_ROUNDED
- FREE\_CANCELLATION\_SHARP
- FRONT\_HAND
- FRONT\_HAND\_OUTLINED
- FRONT\_HAND\_ROUNDED
- FRONT\_HAND\_SHARP
- FRONT\_LOADER
- FULLSCREEN
- FULLSCREEN\_EXIT
- FULLSCREEN\_EXIT\_OUTLINED
- FULLSCREEN\_EXIT\_ROUNDED
- FULLSCREEN\_EXIT\_SHARP
- FULLSCREEN\_OUTLINED
- FULLSCREEN\_ROUNDED
- FULLSCREEN\_SHARP

- FUNCTIONS
- FUNCTIONS\_OUTLINED
- FUNCTIONS\_ROUNDED
- FUNCTIONS\_SHARP
- G\_MOBILEDATA
- G\_MOBILEDATA\_OUTLINED
- G\_MOBILEDATA\_ROUNDED
- G\_MOBILEDATA\_SHARP
- G\_TRANSLATE
- G\_TRANSLATE\_OUTLINED
- G\_TRANSLATE\_ROUNDED
- G\_TRANSLATE\_SHARP
- GAMEPAD
- GAMEPAD\_OUTLINED
- GAMEPAD\_ROUNDED
- GAMEPAD\_SHARP
- GAMES
- GAMES\_OUTLINED
- GAMES\_ROUNDED
- GAMES\_SHARP
- GARAGE
- GARAGE\_OUTLINED
- GARAGE\_ROUNDED
- GARAGE\_SHARP
- GAS\_METER
- GAS\_METER\_OUTLINED
- GAS\_METER\_ROUNDED
- GAS\_METER\_SHARP
- GAVEL
- GAVEL\_OUTLINED
- GAVEL\_ROUNDED
- GAVEL\_SHARP
- GENERATING\_TOKENS
- GENERATING\_TOKENS\_OUTLINED
- GENERATING\_TOKENS\_ROUNDED
- GENERATING\_TOKENS\_SHARP
- GESTURE
- GESTURE\_OUTLINED
- GESTURE\_ROUNDED
- GESTURE\_SHARP
- GET\_APP

- GET\_APP\_OUTLINED
- GET\_APP\_ROUNDED
- GET\_APP\_SHARP
- GIF
- GIF\_BOX
- GIF\_BOX\_OUTLINED
- GIF\_BOX\_ROUNDED
- GIF\_BOX\_SHARP
- GIF\_OUTLINED
- GIF\_ROUNDED
- GIF\_SHARP
- GIRL
- GIRL\_OUTLINED
- GIRL\_ROUNDED
- GIRL\_SHARP
- GITE
- GITE\_OUTLINED
- GITE\_ROUNDED
- GITE\_SHARP
- GOLF\_COURSE
- GOLF\_COURSE\_OUTLINED
- GOLF\_COURSE\_ROUNDED
- GOLF\_COURSE\_SHARP
- GPP\_BAD
- GPP\_BAD\_OUTLINED
- GPP\_BAD\_ROUNDED
- GPP\_BAD\_SHARP
- GPP\_GOOD
- GPP\_GOOD\_OUTLINED
- GPP\_GOOD\_ROUNDED
- GPP\_GOOD\_SHARP
- GPP\_MAYBE
- GPP\_MAYBE\_OUTLINED
- GPP\_MAYBE\_ROUNDED
- GPP\_MAYBE\_SHARP
- GPS\_FIXED
- GPS\_FIXED\_OUTLINED
- GPS\_FIXED\_ROUNDED
- GPS\_FIXED\_SHARP
- GPS\_NOT\_FIXED
- GPS\_NOT\_FIXED\_OUTLINED

- GPS\_NOT\_FIXED\_ROUNDED
- GPS\_NOT\_FIXED\_SHARP
- GPS\_OFF
- GPS\_OFF\_OUTLINED
- GPS\_OFF\_ROUNDED
- GPS\_OFF\_SHARP
- GRADE
- GRADE\_OUTLINED
- GRADE\_ROUNDED
- GRADE\_SHARP
- GRADIENT
- GRADIENT\_OUTLINED
- GRADIENT\_ROUNDED
- GRADIENT\_SHARP
- GRADING
- GRADING\_OUTLINED
- GRADING\_ROUNDED
- GRADING\_SHARP
- GRAIN
- GRAIN\_OUTLINED
- GRAIN\_ROUNDED
- GRAIN\_SHARP
- GRAPHIC\_EQ
- GRAPHIC\_EQ\_OUTLINED
- GRAPHIC\_EQ\_ROUNDED
- GRAPHIC\_EQ\_SHARP
- GRASS
- GRASS\_OUTLINED
- GRASS\_ROUNDED
- GRASS\_SHARP
- GRID\_3X3
- GRID\_3X3\_OUTLINED
- GRID\_3X3\_ROUNDED
- GRID\_3X3\_SHARP
- GRID\_4X4
- GRID\_4X4\_OUTLINED
- GRID\_4X4\_ROUNDED
- GRID\_4X4\_SHARP
- GRID\_GOLDENRATIO
- GRID\_GOLDENRATIO\_OUTLINED
- GRID\_GOLDENRATIO\_ROUNDED

- GRID\_GOLDENRATIO\_SHARP
- GRID\_OFF
- GRID\_OFF\_OUTLINED
- GRID\_OFF\_ROUNDED
- GRID\_OFF\_SHARP
- GRID\_ON
- GRID\_ON\_OUTLINED
- GRID\_ON\_ROUNDED
- GRID\_ON\_SHARP
- GRID\_VIEW
- GRID\_VIEW\_OUTLINED
- GRID\_VIEW\_ROUNDED
- GRID\_VIEW\_SHARP
- GROUP
- GROUP\_ADD
- GROUP\_ADD\_OUTLINED
- GROUP\_ADD\_ROUNDED
- GROUP\_ADD\_SHARP
- GROUP\_OFF
- GROUP\_OFF\_OUTLINED
- GROUP\_OFF\_ROUNDED
- GROUP\_OFF\_SHARP
- GROUP\_OUTLINED
- GROUP\_REMOVE
- GROUP\_REMOVE\_OUTLINED
- GROUP\_REMOVE\_ROUNDED
- GROUP\_REMOVE\_SHARP
- GROUP\_ROUNDED
- GROUP\_SHARP
- GROUP\_WORK
- GROUP\_WORK\_OUTLINED
- GROUP\_WORK\_ROUNDED
- GROUP\_WORK\_SHARP
- GROUPS
- GROUPS\_2
- GROUPS\_2\_OUTLINED
- GROUPS\_2\_ROUNDED
- GROUPS\_2\_SHARP
- GROUPS\_3
- GROUPS\_3\_OUTLINED
- GROUPS\_3\_ROUNDED

- GROUPS\_3\_SHARP
- GROUPS\_OUTLINED
- GROUPS\_ROUNDED
- GROUPS\_SHARP
- H\_MOBILEDATA
- H\_MOBILEDATA\_OUTLINED
- H\_MOBILEDATA\_ROUNDED
- H\_MOBILEDATA\_SHARP
- H\_PLUS\_MOBILEDATA
- H\_PLUS\_MOBILEDATA\_OUTLINED
- H\_PLUS\_MOBILEDATA\_ROUNDED
- H\_PLUS\_MOBILEDATA\_SHARP
- HAIL
- HAIL\_OUTLINED
- HAIL\_ROUNDED
- HAIL\_SHARP
- HANDSHAKE
- HANDSHAKE\_OUTLINED
- HANDSHAKE\_ROUNDED
- HANDSHAKE\_SHARP
- HANDYMAN
- HANDYMAN\_OUTLINED
- HANDYMAN\_ROUNDED
- HANDYMAN\_SHARP
- HARDWARE
- HARDWARE\_OUTLINED
- HARDWARE\_ROUNDED
- HARDWARE\_SHARP
- HD
- HD\_OUTLINED
- HD\_ROUNDED
- HD\_SHARP
- HDR\_AUTO
- HDR\_AUTO\_OUTLINED
- HDR\_AUTO\_ROUNDED
- HDR\_AUTO\_SELECT
- HDR\_AUTO\_SELECT\_OUTLINED
- HDR\_AUTO\_SELECT\_ROUNDED
- HDR\_AUTO\_SELECT\_SHARP
- HDR\_AUTO\_SHARP
- HDR\_ENHANCED\_SELECT

- HDR\_ENHANCED\_SELECT\_OUTLINED
- HDR\_ENHANCED\_SELECT\_ROUNDED
- HDR\_ENHANCED\_SELECT\_SHARP
- HDR\_OFF
- HDR\_OFF\_OUTLINED
- HDR\_OFF\_ROUNDED
- HDR\_OFF\_SELECT
- HDR\_OFF\_SELECT\_OUTLINED
- HDR\_OFF\_SELECT\_ROUNDED
- HDR\_OFF\_SELECT\_SHARP
- HDR\_OFF\_SHARP
- HDR\_ON
- HDR\_ON\_OUTLINED
- HDR\_ON\_ROUNDED
- HDR\_ON\_SELECT
- HDR\_ON\_SELECT\_OUTLINED
- HDR\_ON\_SELECT\_ROUNDED
- HDR\_ON\_SELECT\_SHARP
- HDR\_ON\_SHARP
- HDR\_PLUS
- HDR\_PLUS\_OUTLINED
- HDR\_PLUS\_ROUNDED
- HDR\_PLUS\_SHARP
- HDR\_STRONG
- HDR\_STRONG\_OUTLINED
- HDR\_STRONG\_ROUNDED
- HDR\_STRONG\_SHARP
- HDR\_WEAK
- HDR\_WEAK\_OUTLINED
- HDR\_WEAK\_ROUNDED
- HDR\_WEAK\_SHARP
- HEADPHONES
- HEADPHONES\_BATTERY
- HEADPHONES\_BATTERY\_OUTLINED
- HEADPHONES\_BATTERY\_ROUNDED
- HEADPHONES\_BATTERY\_SHARP
- HEADPHONES\_OUTLINED
- HEADPHONES\_ROUNDED
- HEADPHONES\_SHARP
- HEADSET
- HEADSET\_MIC



- HEADSET\_MIC\_OUTLINED
- HEADSET\_MIC\_ROUNDED
- HEADSET\_MIC\_SHARP
- HEADSET\_OFF
- HEADSET\_OFF\_OUTLINED
- HEADSET\_OFF\_ROUNDED
- HEADSET\_OFF\_SHARP
- HEADSET\_OUTLINED
- HEADSET\_ROUNDED
- HEADSET\_SHARP
- HEALING
- HEALING\_OUTLINED
- HEALING\_ROUNDED
- HEALING\_SHARP
- HEALTH\_AND\_SAFETY
- HEALTH\_AND\_SAFETY\_OUTLINED
- HEALTH\_AND\_SAFETY\_ROUNDED
- HEALTH\_AND\_SAFETY\_SHARP
- HEARING
- HEARING\_DISABLED
- HEARING\_DISABLED\_OUTLINED
- HEARING\_DISABLED\_ROUNDED
- HEARING\_DISABLED\_SHARP
- HEARING\_OUTLINED
- HEARING\_ROUNDED
- HEARING\_SHARP
- HEART\_BROKEN
- HEART\_BROKEN\_OUTLINED
- HEART\_BROKEN\_ROUNDED
- HEART\_BROKEN\_SHARP
- HEAT\_PUMP
- HEAT\_PUMP\_OUTLINED
- HEAT\_PUMP\_ROUNDED
- HEAT\_PUMP\_SHARP
- HEIGHT
- HEIGHT\_OUTLINED
- HEIGHT\_ROUNDED
- HEIGHT\_SHARP
- HELP
- HELP\_CENTER
- HELP\_CENTER\_OUTLINED

- HELP\_CENTER\_ROUNDED
- HELP\_CENTER\_SHARP
- HELP\_OUTLINE
- HELP\_OUTLINE\_OUTLINED
- HELP\_OUTLINE\_ROUNDED
- HELP\_OUTLINE\_SHARP
- HELP\_OUTLINED
- HELP\_ROUNDED
- HELP\_SHARP
- HEVC
- HEVC\_OUTLINED
- HEVC\_ROUNDED
- HEVC\_SHARP
- HEXAGON
- HEXAGON\_OUTLINED
- HEXAGON\_ROUNDED
- HEXAGON\_SHARP
- HIDE\_IMAGE
- HIDE\_IMAGE\_OUTLINED
- HIDE\_IMAGE\_ROUNDED
- HIDE\_IMAGE\_SHARP
- HIDE\_SOURCE
- HIDE\_SOURCE\_OUTLINED
- HIDE\_SOURCE\_ROUNDED
- HIDE\_SOURCE\_SHARP
- HIGH\_QUALITY
- HIGH\_QUALITY\_OUTLINED
- HIGH\_QUALITY\_ROUNDED
- HIGH\_QUALITY\_SHARP
- HIGHLIGHT
- HIGHLIGHT\_ALT
- HIGHLIGHT\_ALT\_OUTLINED
- HIGHLIGHT\_ALT\_ROUNDED
- HIGHLIGHT\_ALT\_SHARP
- HIGHLIGHT\_OFF
- HIGHLIGHT\_OFF\_OUTLINED
- HIGHLIGHT\_OFF\_ROUNDED
- HIGHLIGHT\_OFF\_SHARP
- HIGHLIGHT\_OUTLINED
- HIGHLIGHT\_REMOVE
- HIGHLIGHT\_REMOVE\_OUTLINED

- HIGHLIGHT\_REMOVE\_ROUNDED
- HIGHLIGHT\_REMOVE\_SHARP
- HIGHLIGHT\_ROUNDED
- HIGHLIGHT\_SHARP
- HIKING
- HIKING\_OUTLINED
- HIKING\_ROUNDED
- HIKING\_SHARP
- HISTORY
- HISTORY\_EDU
- HISTORY\_EDU\_OUTLINED
- HISTORY\_EDU\_ROUNDED
- HISTORY\_EDU\_SHARP
- HISTORY\_OUTLINED
- HISTORY\_ROUNDED
- HISTORY\_SHARP
- HISTORY\_TOGGLE\_OFF
- HISTORY\_TOGGLE\_OFF\_OUTLINED
- HISTORY\_TOGGLE\_OFF\_ROUNDED
- HISTORY\_TOGGLE\_OFF\_SHARP
- HIVE
- HIVE\_OUTLINED
- HIVE\_ROUNDED
- HIVE\_SHARP
- HLS
- HLS\_OFF
- HLS\_OFF\_OUTLINED
- HLS\_OFF\_ROUNDED
- HLS\_OFF\_SHARP
- HLS\_OUTLINED
- HLS\_ROUNDED
- HLS\_SHARP
- HOLIDAY\_VILLAGE
- HOLIDAY\_VILLAGE\_OUTLINED
- HOLIDAY\_VILLAGE\_ROUNDED
- HOLIDAY\_VILLAGE\_SHARP
- HOME
- HOME\_FILLED
- HOME\_MAX
- HOME\_MAX\_OUTLINED
- HOME\_MAX\_ROUNDED

- HOME\_MAX\_SHARP
- HOME\_MINI
- HOME\_MINI\_OUTLINED
- HOME\_MINI\_ROUNDED
- HOME\_MINI\_SHARP
- HOME\_OUTLINED
- HOME\_REPAIR\_SERVICE
- HOME\_REPAIR\_SERVICE\_OUTLINED
- HOME\_REPAIR\_SERVICE\_ROUNDED
- HOME\_REPAIR\_SERVICE\_SHARP
- HOME\_ROUNDED
- HOME\_SHARP
- HOME\_WORK
- HOME\_WORK\_OUTLINED
- HOME\_WORK\_ROUNDED
- HOME\_WORK\_SHARP
- HORIZONTAL\_DISTRIBUTE
- HORIZONTAL\_DISTRIBUTE\_OUTLINED
- HORIZONTAL\_DISTRIBUTE\_ROUNDED
- HORIZONTAL\_DISTRIBUTE\_SHARP
- HORIZONTAL\_RULE
- HORIZONTAL\_RULE\_OUTLINED
- HORIZONTAL\_RULE\_ROUNDED
- HORIZONTAL\_RULE\_SHARP
- HORIZONTAL\_SPLIT
- HORIZONTAL\_SPLIT\_OUTLINED
- HORIZONTAL\_SPLIT\_ROUNDED
- HORIZONTAL\_SPLIT\_SHARP
- HOT\_TUB
- HOT\_TUB\_OUTLINED
- HOT\_TUB\_ROUNDED
- HOT\_TUB\_SHARP
- HOTEL
- HOTEL\_CLASS
- HOTEL\_CLASS\_OUTLINED
- HOTEL\_CLASS\_ROUNDED
- HOTEL\_CLASS\_SHARP
- HOTEL\_OUTLINED
- HOTEL\_ROUNDED
- HOTEL\_SHARP
- HOURGLASS\_BOTTOM

- HOURGLASS\_BOTTOM\_OUTLINED
- HOURGLASS\_BOTTOM\_ROUNDED
- HOURGLASS\_BOTTOM\_SHARP
- HOURGLASS\_DISABLED
- HOURGLASS\_DISABLED\_OUTLINED
- HOURGLASS\_DISABLED\_ROUNDED
- HOURGLASS\_DISABLED\_SHARP
- HOURGLASS\_EMPTY
- HOURGLASS\_EMPTY\_OUTLINED
- HOURGLASS\_EMPTY\_ROUNDED
- HOURGLASS\_EMPTY\_SHARP
- HOURGLASS\_FULL
- HOURGLASS\_FULL\_OUTLINED
- HOURGLASS\_FULL\_ROUNDED
- HOURGLASS\_FULL\_SHARP
- HOURGLASS\_TOP
- HOURGLASS\_TOP\_OUTLINED
- HOURGLASS\_TOP\_ROUNDED
- HOURGLASS\_TOP\_SHARP
- HOUSE
- HOUSE\_OUTLINED
- HOUSE\_ROUNDED
- HOUSE\_SHARP
- HOUSE\_SIDING
- HOUSE\_SIDING\_OUTLINED
- HOUSE\_SIDING\_ROUNDED
- HOUSE\_SIDING\_SHARP
- HOUSEBOAT
- HOUSEBOAT\_OUTLINED
- HOUSEBOAT\_ROUNDED
- HOUSEBOAT\_SHARP
- HOW\_TO\_REG
- HOW\_TO\_REG\_OUTLINED
- HOW\_TO\_REG\_ROUNDED
- HOW\_TO\_REG\_SHARP
- HOW\_TO\_VOTE
- HOW\_TO\_VOTE\_OUTLINED
- HOW\_TO\_VOTE\_ROUNDED
- HOW\_TO\_VOTE\_SHARP
- HTML
- HTML\_OUTLINED

- HTML\_ROUNDED
- HTML\_SHARP
- HTTP
- HTTP\_OUTLINED
- HTTP\_ROUNDED
- HTTP\_SHARP
- HTTPS
- HTTPS\_OUTLINED
- HTTPS\_ROUNDED
- HTTPS\_SHARP
- HUB
- HUB\_OUTLINED
- HUB\_ROUNDED
- HUB\_SHARP
- HVAC
- HVAC\_OUTLINED
- HVAC\_ROUNDED
- HVAC\_SHARP
- ICE\_SKATING
- ICE\_SKATING\_OUTLINED
- ICE\_SKATING\_ROUNDED
- ICE\_SKATING\_SHARP
- ICECREAM
- ICECREAM\_OUTLINED
- ICECREAM\_ROUNDED
- ICECREAM\_SHARP
- IMAGE
- IMAGE\_ASPECT\_RATIO
- IMAGE\_ASPECT\_RATIO\_OUTLINED
- IMAGE\_ASPECT\_RATIO\_ROUNDED
- IMAGE\_ASPECT\_RATIO\_SHARP
- IMAGE\_NOT\_SUPPORTED
- IMAGE\_NOT\_SUPPORTED\_OUTLINED
- IMAGE\_NOT\_SUPPORTED\_ROUNDED
- IMAGE\_NOT\_SUPPORTED\_SHARP
- IMAGE\_OUTLINED
- IMAGE\_ROUNDED
- IMAGE\_SEARCH
- IMAGE\_SEARCH\_OUTLINED
- IMAGE\_SEARCH\_ROUNDED
- IMAGE\_SEARCH\_SHARP

- IMAGE\_SHARP
- IMAGESEARCH\_ROLLER
- IMAGESEARCH\_ROLLER\_OUTLINED
- IMAGESEARCH\_ROLLER\_ROUNDED
- IMAGESEARCH\_ROLLER\_SHARP
- IMPORT\_CONTACTS
- IMPORT\_CONTACTS\_OUTLINED
- IMPORT\_CONTACTS\_ROUNDED
- IMPORT\_CONTACTS\_SHARP
- IMPORT\_EXPORT
- IMPORT\_EXPORT\_OUTLINED
- IMPORT\_EXPORT\_ROUNDED
- IMPORT\_EXPORT\_SHARP
- IMPORTANT\_DEVICES
- IMPORTANT\_DEVICES\_OUTLINED
- IMPORTANT\_DEVICES\_ROUNDED
- IMPORTANT\_DEVICES\_SHARP
- INBOX
- INBOX\_OUTLINED
- INBOX\_ROUNDED
- INBOX\_SHARP
- INCOMPLETE\_CIRCLE
- INCOMPLETE\_CIRCLE\_OUTLINED
- INCOMPLETE\_CIRCLE\_ROUNDED
- INCOMPLETE\_CIRCLE\_SHARP
- INDETERMINATE\_CHECK\_BOX
- INDETERMINATE\_CHECK\_BOX\_OUTLINED
- INDETERMINATE\_CHECK\_BOX\_ROUNDED
- INDETERMINATE\_CHECK\_BOX\_SHARP
- INFO
- INFO\_OUTLINE
- INFO\_OUTLINE\_ROUNDED
- INFO\_OUTLINE\_SHARP
- INFO\_OUTLINED
- INFO\_ROUNDED
- INFO\_SHARP
- INPUT
- INPUT\_OUTLINED
- INPUT\_ROUNDED
- INPUT\_SHARP
- INSERT\_CHART

- INSERT\_CHART\_OUTLINED
- INSERT\_CHART\_OUTLINED\_OUTLINED
- INSERT\_CHART\_OUTLINED\_ROUNDED
- INSERT\_CHART\_OUTLINED\_SHARP
- INSERT\_CHART\_ROUNDED
- INSERT\_CHART\_SHARP
- INSERT\_COMMENT
- INSERT\_COMMENT\_OUTLINED
- INSERT\_COMMENT\_ROUNDED
- INSERT\_COMMENT\_SHARP
- INSERT\_DRIVE\_FILE
- INSERT\_DRIVE\_FILE\_OUTLINED
- INSERT\_DRIVE\_FILE\_ROUNDED
- INSERT\_DRIVE\_FILE\_SHARP
- INSERT\_EMOTICON
- INSERT\_EMOTICON\_OUTLINED
- INSERT\_EMOTICON\_ROUNDED
- INSERT\_EMOTICON\_SHARP
- INSERT\_INVITATION
- INSERT\_INVITATION\_OUTLINED
- INSERT\_INVITATION\_ROUNDED
- INSERT\_INVITATION\_SHARP
- INSERT\_LINK
- INSERT\_LINK\_OUTLINED
- INSERT\_LINK\_ROUNDED
- INSERT\_LINK\_SHARP
- INSERT\_PAGE\_BREAK
- INSERT\_PAGE\_BREAK\_OUTLINED
- INSERT\_PAGE\_BREAK\_ROUNDED
- INSERT\_PAGE\_BREAK\_SHARP
- INSERT\_PHOTO
- INSERT\_PHOTO\_OUTLINED
- INSERT\_PHOTO\_ROUNDED
- INSERT\_PHOTO\_SHARP
- INSIGHTS
- INSIGHTS\_OUTLINED
- INSIGHTS\_ROUNDED
- INSIGHTS\_SHARP
- INSTALL\_DESKTOP
- INSTALL\_DESKTOP\_OUTLINED
- INSTALL\_DESKTOP\_ROUNDED



- INSTALL\_DESKTOP\_SHARP
- INSTALL\_MOBILE
- INSTALL\_MOBILE\_OUTLINED
- INSTALL\_MOBILE\_ROUNDED
- INSTALL\_MOBILE\_SHARP
- INTEGRATION\_INSTRUCTIONS
- INTEGRATION\_INSTRUCTIONS\_OUTLINED
- INTEGRATION\_INSTRUCTIONS\_ROUNDED
- INTEGRATION\_INSTRUCTIONS\_SHARP
- INTERESTS
- INTERESTS\_OUTLINED
- INTERESTS\_ROUNDED
- INTERESTS\_SHARP
- INTERPRETER\_MODE
- INTERPRETER\_MODE\_OUTLINED
- INTERPRETER\_MODE\_ROUNDED
- INTERPRETER\_MODE\_SHARP
- INVENTORY
- INVENTORY\_2
- INVENTORY\_2\_OUTLINED
- INVENTORY\_2\_ROUNDED
- INVENTORY\_2\_SHARP
- INVENTORY\_OUTLINED
- INVENTORY\_ROUNDED
- INVENTORY\_SHARP
- INVERT\_COLORS
- INVERT\_COLORS\_OFF
- INVERT\_COLORS\_OFF\_OUTLINED
- INVERT\_COLORS\_OFF\_ROUNDED
- INVERT\_COLORS\_OFF\_SHARP
- INVERT\_COLORS\_ON
- INVERT\_COLORS\_ON\_OUTLINED
- INVERT\_COLORS\_ON\_ROUNDED
- INVERT\_COLORS\_ON\_SHARP
- INVERT\_COLORS\_OUTLINED
- INVERT\_COLORS\_ROUNDED
- INVERT\_COLORS\_SHARP
- IOS\_SHARE
- IOS\_SHARE\_OUTLINED
- IOS\_SHARE\_ROUNDED
- IOS\_SHARE\_SHARP

- IRON
- IRON\_OUTLINED
- IRON\_ROUNDED
- IRON\_SHARP
- ISO
- ISO\_OUTLINED
- ISO\_ROUNDED
- ISO\_SHARP
- JAVASCRIPT
- JAVASCRIPT\_OUTLINED
- JAVASCRIPT\_ROUNDED
- JAVASCRIPT\_SHARP
- JOIN\_FULL
- JOIN\_FULL\_OUTLINED
- JOIN\_FULL\_ROUNDED
- JOIN\_FULL\_SHARP
- JOIN\_INNER
- JOIN\_INNER\_OUTLINED
- JOIN\_INNER\_ROUNDED
- JOIN\_INNER\_SHARP
- JOIN\_LEFT
- JOIN\_LEFT\_OUTLINED
- JOIN\_LEFT\_ROUNDED
- JOIN\_LEFT\_SHARP
- JOIN\_RIGHT
- JOIN\_RIGHT\_OUTLINED
- JOIN\_RIGHT\_ROUNDED
- JOIN\_RIGHT\_SHARP
- KAYAKING
- KAYAKING\_OUTLINED
- KAYAKING\_ROUNDED
- KAYAKING\_SHARP
- KEBAB\_DINING
- KEBAB\_DINING\_OUTLINED
- KEBAB\_DINING\_ROUNDED
- KEBAB\_DINING\_SHARP
- KEY
- KEY\_OFF
- KEY\_OFF\_OUTLINED
- KEY\_OFF\_ROUNDED
- KEY\_OFF\_SHARP

- KEY\_OUTLINED
- KEY\_ROUNDED
- KEY\_SHARP
- KEYBOARD
- KEYBOARD\_ALT
- KEYBOARD\_ALT\_OUTLINED
- KEYBOARD\_ALT\_ROUNDED
- KEYBOARD\_ALT\_SHARP
- KEYBOARD\_ARROW\_DOWN
- KEYBOARD\_ARROW\_DOWN\_OUTLINED
- KEYBOARD\_ARROW\_DOWN\_ROUNDED
- KEYBOARD\_ARROW\_DOWN\_SHARP
- KEYBOARD\_ARROW\_LEFT
- KEYBOARD\_ARROW\_LEFT\_OUTLINED
- KEYBOARD\_ARROW\_LEFT\_ROUNDED
- KEYBOARD\_ARROW\_LEFT\_SHARP
- KEYBOARD\_ARROW\_RIGHT
- KEYBOARD\_ARROW\_RIGHT\_OUTLINED
- KEYBOARD\_ARROW\_RIGHT\_ROUNDED
- KEYBOARD\_ARROW\_RIGHT\_SHARP
- KEYBOARD\_ARROW\_UP
- KEYBOARD\_ARROW\_UP\_OUTLINED
- KEYBOARD\_ARROW\_UP\_ROUNDED
- KEYBOARD\_ARROW\_UP\_SHARP
- KEYBOARD\_BACKSPACE
- KEYBOARD\_BACKSPACE\_OUTLINED
- KEYBOARD\_BACKSPACE\_ROUNDED
- KEYBOARD\_BACKSPACE\_SHARP
- KEYBOARD\_CAPSLOCK
- KEYBOARD\_CAPSLOCK\_OUTLINED
- KEYBOARD\_CAPSLOCK\_ROUNDED
- KEYBOARD\_CAPSLOCK\_SHARP
- KEYBOARD\_COMMAND\_KEY
- KEYBOARD\_COMMAND\_KEY\_OUTLINED
- KEYBOARD\_COMMAND\_KEY\_ROUNDED
- KEYBOARD\_COMMAND\_KEY\_SHARP
- KEYBOARD\_CONTROL
- KEYBOARD\_CONTROL\_KEY
- KEYBOARD\_CONTROL\_KEY\_OUTLINED
- KEYBOARD\_CONTROL\_KEY\_ROUNDED
- KEYBOARD\_CONTROL\_KEY\_SHARP

- KEYBOARD\_CONTROL\_OUTLINED
- KEYBOARD\_CONTROL\_ROUNDED
- KEYBOARD\_CONTROL\_SHARP
- KEYBOARD\_DOUBLE\_ARROW\_DOWN
- KEYBOARD\_DOUBLE\_ARROW\_DOWN\_OUTLINED
- KEYBOARD\_DOUBLE\_ARROW\_DOWN\_ROUNDED
- KEYBOARD\_DOUBLE\_ARROW\_DOWN\_SHARP
- KEYBOARD\_DOUBLE\_ARROW\_LEFT
- KEYBOARD\_DOUBLE\_ARROW\_LEFT\_OUTLINED
- KEYBOARD\_DOUBLE\_ARROW\_LEFT\_ROUNDED
- KEYBOARD\_DOUBLE\_ARROW\_LEFT\_SHARP
- KEYBOARD\_DOUBLE\_ARROW\_RIGHT
- KEYBOARD\_DOUBLE\_ARROW\_RIGHT\_OUTLINED
- KEYBOARD\_DOUBLE\_ARROW\_RIGHT\_ROUNDED
- KEYBOARD\_DOUBLE\_ARROW\_RIGHT\_SHARP
- KEYBOARD\_DOUBLE\_ARROW\_UP
- KEYBOARD\_DOUBLE\_ARROW\_UP\_OUTLINED
- KEYBOARD\_DOUBLE\_ARROW\_UP\_ROUNDED
- KEYBOARD\_DOUBLE\_ARROW\_UP\_SHARP
- KEYBOARD\_HIDE
- KEYBOARD\_HIDE\_OUTLINED
- KEYBOARD\_HIDE\_ROUNDED
- KEYBOARD\_HIDE\_SHARP
- KEYBOARD\_OPTION\_KEY
- KEYBOARD\_OPTION\_KEY\_OUTLINED
- KEYBOARD\_OPTION\_KEY\_ROUNDED
- KEYBOARD\_OPTION\_KEY\_SHARP
- KEYBOARD\_OUTLINED
- KEYBOARD\_RETURN
- KEYBOARD\_RETURN\_OUTLINED
- KEYBOARD\_RETURN\_ROUNDED
- KEYBOARD\_RETURN\_SHARP
- KEYBOARD\_ROUNDED
- KEYBOARD\_SHARP
- KEYBOARD\_TAB
- KEYBOARD\_TAB\_OUTLINED
- KEYBOARD\_TAB\_ROUNDED
- KEYBOARD\_TAB\_SHARP
- KEYBOARD\_VOICE
- KEYBOARD\_VOICE\_OUTLINED
- KEYBOARD\_VOICE\_ROUNDED

- KEYBOARD\_VOICE\_SHARP
- KING\_BED
- KING\_BED\_OUTLINED
- KING\_BED\_ROUNDED
- KING\_BED\_SHARP
- KITCHEN
- KITCHEN\_OUTLINED
- KITCHEN\_ROUNDED
- KITCHEN\_SHARP
- KITESURFING
- KITESURFING\_OUTLINED
- KITESURFING\_ROUNDED
- KITESURFING\_SHARP
- LABEL
- LABEL\_IMPORTANT
- LABEL\_IMPORTANT\_OUTLINE
- LABEL\_IMPORTANT\_OUTLINE\_ROUNDED
- LABEL\_IMPORTANT\_OUTLINE\_SHARP
- LABEL\_IMPORTANT\_OUTLINED
- LABEL\_IMPORTANT\_ROUNDED
- LABEL\_IMPORTANT\_SHARP
- LABEL\_OFF
- LABEL\_OFF\_OUTLINED
- LABEL\_OFF\_ROUNDED
- LABEL\_OFF\_SHARP
- LABEL\_OUTLINE
- LABEL\_OUTLINE\_ROUNDED
- LABEL\_OUTLINE\_SHARP
- LABEL\_OUTLINED
- LABEL\_ROUNDED
- LABEL\_SHARP
- LAN
- LAN\_OUTLINED
- LAN\_ROUNDED
- LAN\_SHARP
- LANDSCAPE
- LANDSCAPE\_OUTLINED
- LANDSCAPE\_ROUNDED
- LANDSCAPE\_SHARP
- LANDSLIDE
- LANDSLIDE\_OUTLINED

- LANDSLIDE\_ROUNDED
- LANDSLIDE\_SHARP
- LANGUAGE
- LANGUAGE\_OUTLINED
- LANGUAGE\_ROUNDED
- LANGUAGE\_SHARP
- LAPTOP
- LAPTOP\_CHROMEBOOK
- LAPTOP\_CHROMEBOOK\_OUTLINED
- LAPTOP\_CHROMEBOOK\_ROUNDED
- LAPTOP\_CHROMEBOOK\_SHARP
- LAPTOP\_MAC
- LAPTOP\_MAC\_OUTLINED
- LAPTOP\_MAC\_ROUNDED
- LAPTOP\_MAC\_SHARP
- LAPTOP\_OUTLINED
- LAPTOP\_ROUNDED
- LAPTOP\_SHARP
- LAPTOP\_WINDOWS
- LAPTOP\_WINDOWS\_OUTLINED
- LAPTOP\_WINDOWS\_ROUNDED
- LAPTOP\_WINDOWS\_SHARP
- LAST\_PAGE
- LAST\_PAGE\_OUTLINED
- LAST\_PAGE\_ROUNDED
- LAST\_PAGE\_SHARP
- LAUNCH
- LAUNCH\_OUTLINED
- LAUNCH\_ROUNDED
- LAUNCH\_SHARP
- LAYERS
- LAYERS\_CLEAR
- LAYERS\_CLEAR\_OUTLINED
- LAYERS\_CLEAR\_ROUNDED
- LAYERS\_CLEAR\_SHARP
- LAYERS\_OUTLINED
- LAYERS\_ROUNDED
- LAYERS\_SHARP
- LEADERBOARD
- LEADERBOARD\_OUTLINED
- LEADERBOARD\_ROUNDED

- LEADERBOARD\_SHARP
- LEAK\_ADD
- LEAK\_ADD\_OUTLINED
- LEAK\_ADD\_ROUNDED
- LEAK\_ADD\_SHARP
- LEAK\_REMOVE
- LEAK\_REMOVE\_OUTLINED
- LEAK\_REMOVE\_ROUNDED
- LEAK\_REMOVE\_SHARP
- LEAVE\_BAGS\_AT\_HOME
- LEAVE\_BAGS\_AT\_HOME\_OUTLINED
- LEAVE\_BAGS\_AT\_HOME\_ROUNDED
- LEAVE\_BAGS\_AT\_HOME\_SHARP
- LEGEND\_TOGGLE
- LEGEND\_TOGGLE\_OUTLINED
- LEGEND\_TOGGLE\_ROUNDED
- LEGEND\_TOGGLE\_SHARP
- LENS
- LENS\_BLUR
- LENS\_BLUR\_OUTLINED
- LENS\_BLUR\_ROUNDED
- LENS\_BLUR\_SHARP
- LENS\_OUTLINED
- LENS\_ROUNDED
- LENS\_SHARP
- LIBRARY\_ADD
- LIBRARY\_ADD\_CHECK
- LIBRARY\_ADD\_CHECK\_OUTLINED
- LIBRARY\_ADD\_CHECK\_ROUNDED
- LIBRARY\_ADD\_CHECK\_SHARP
- LIBRARY\_ADD\_OUTLINED
- LIBRARY\_ADD\_ROUNDED
- LIBRARY\_ADD\_SHARP
- LIBRARY\_BOOKS
- LIBRARY\_BOOKS\_OUTLINED
- LIBRARY\_BOOKS\_ROUNDED
- LIBRARY\_BOOKS\_SHARP
- LIBRARY\_MUSIC
- LIBRARY\_MUSIC\_OUTLINED
- LIBRARY\_MUSIC\_ROUNDED
- LIBRARY\_MUSIC\_SHARP

- LIGHT
- LIGHT\_MODE
- LIGHT\_MODE\_OUTLINED
- LIGHT\_MODE\_ROUNDED
- LIGHT\_MODE\_SHARP
- LIGHT\_OUTLINED
- LIGHT\_ROUNDED
- LIGHT\_SHARP
- LIGHTBULB
- LIGHTBULB\_CIRCLE
- LIGHTBULB\_CIRCLE\_OUTLINED
- LIGHTBULB\_CIRCLE\_ROUNDED
- LIGHTBULB\_CIRCLE\_SHARP
- LIGHTBULB\_OUTLINE
- LIGHTBULB\_OUTLINE\_ROUNDED
- LIGHTBULB\_OUTLINE\_SHARP
- LIGHTBULB\_OUTLINED
- LIGHTBULB\_ROUNDED
- LIGHTBULB\_SHARP
- LINE\_AXIS
- LINE\_AXIS\_OUTLINED
- LINE\_AXIS\_ROUNDED
- LINE\_AXIS\_SHARP
- LINE\_STYLE
- LINE\_STYLE\_OUTLINED
- LINE\_STYLE\_ROUNDED
- LINE\_STYLE\_SHARP
- LINE\_WEIGHT
- LINE\_WEIGHT\_OUTLINED
- LINE\_WEIGHT\_ROUNDED
- LINE\_WEIGHT\_SHARP
- LINEAR\_SCALE
- LINEAR\_SCALE\_OUTLINED
- LINEAR\_SCALE\_ROUNDED
- LINEAR\_SCALE\_SHARP
- LINK
- LINK\_OFF
- LINK\_OFF\_OUTLINED
- LINK\_OFF\_ROUNDED
- LINK\_OFF\_SHARP
- LINK\_OUTLINED



- LINK\_ROUNDED
- LINK\_SHARP
- LINKED\_CAMERA
- LINKED\_CAMERA\_OUTLINED
- LINKED\_CAMERA\_ROUNDED
- LINKED\_CAMERA\_SHARP
- LIQUOR
- LIQUOR\_OUTLINED
- LIQUOR\_ROUNDED
- LIQUOR\_SHARP
- LIST
- LIST\_ALT
- LIST\_ALT\_OUTLINED
- LIST\_ALT\_ROUNDED
- LIST\_ALT\_SHARP
- LIST\_OUTLINED
- LIST\_ROUNDED
- LIST\_SHARP
- LIVE\_HELP
- LIVE\_HELP\_OUTLINED
- LIVE\_HELP\_ROUNDED
- LIVE\_HELP\_SHARP
- LIVE\_TV
- LIVE\_TV\_OUTLINED
- LIVE\_TV\_ROUNDED
- LIVE\_TV\_SHARP
- LIVING
- LIVING\_OUTLINED
- LIVING\_ROUNDED
- LIVING\_SHARP
- LOCAL\_ACTIVITY
- LOCAL\_ACTIVITY\_OUTLINED
- LOCAL\_ACTIVITY\_ROUNDED
- LOCAL\_ACTIVITY\_SHARP
- LOCAL\_AIRPORT
- LOCAL\_AIRPORT\_OUTLINED
- LOCAL\_AIRPORT\_ROUNDED
- LOCAL\_AIRPORT\_SHARP
- LOCAL\_ATM
- LOCAL\_ATM\_OUTLINED
- LOCAL\_ATM\_ROUNDED

- LOCAL\_ATM\_SHARP
- LOCAL\_ATTRACTION
- LOCAL\_ATTRACTION\_OUTLINED
- LOCAL\_ATTRACTION\_ROUNDED
- LOCAL\_ATTRACTION\_SHARP
- LOCAL\_BAR
- LOCAL\_BAR\_OUTLINED
- LOCAL\_BAR\_ROUNDED
- LOCAL\_BAR\_SHARP
- LOCAL\_CAFE
- LOCAL\_CAFE\_OUTLINED
- LOCAL\_CAFE\_ROUNDED
- LOCAL\_CAFE\_SHARP
- LOCAL\_CAR\_WASH
- LOCAL\_CAR\_WASH\_OUTLINED
- LOCAL\_CAR\_WASH\_ROUNDED
- LOCAL\_CAR\_WASH\_SHARP
- LOCAL\_CONVENIENCE\_STORE
- LOCAL\_CONVENIENCE\_STORE\_OUTLINED
- LOCAL\_CONVENIENCE\_STORE\_ROUNDED
- LOCAL\_CONVENIENCE\_STORE\_SHARP
- LOCAL\_DINING
- LOCAL\_DINING\_OUTLINED
- LOCAL\_DINING\_ROUNDED
- LOCAL\_DINING\_SHARP
- LOCAL\_DRINK
- LOCAL\_DRINK\_OUTLINED
- LOCAL\_DRINK\_ROUNDED
- LOCAL\_DRINK\_SHARP
- LOCAL\_FIRE\_DEPARTMENT
- LOCAL\_FIRE\_DEPARTMENT\_OUTLINED
- LOCAL\_FIRE\_DEPARTMENT\_ROUNDED
- LOCAL\_FIRE\_DEPARTMENT\_SHARP
- LOCAL\_FLORIST
- LOCAL\_FLORIST\_OUTLINED
- LOCAL\_FLORIST\_ROUNDED
- LOCAL\_FLORIST\_SHARP
- LOCAL\_GAS\_STATION
- LOCAL\_GAS\_STATION\_OUTLINED
- LOCAL\_GAS\_STATION\_ROUNDED
- LOCAL\_GAS\_STATION\_SHARP

- LOCAL\_GROCERY\_STORE
- LOCAL\_GROCERY\_STORE\_OUTLINED
- LOCAL\_GROCERY\_STORE\_ROUNDED
- LOCAL\_GROCERY\_STORE\_SHARP
- LOCAL\_HOSPITAL
- LOCAL\_HOSPITAL\_OUTLINED
- LOCAL\_HOSPITAL\_ROUNDED
- LOCAL\_HOSPITAL\_SHARP
- LOCAL\_HOTEL
- LOCAL\_HOTEL\_OUTLINED
- LOCAL\_HOTEL\_ROUNDED
- LOCAL\_HOTEL\_SHARP
- LOCAL\_LAUNDRY\_SERVICE
- LOCAL\_LAUNDRY\_SERVICE\_OUTLINED
- LOCAL\_LAUNDRY\_SERVICE\_ROUNDED
- LOCAL\_LAUNDRY\_SERVICE\_SHARP
- LOCAL\_LIBRARY
- LOCAL\_LIBRARY\_OUTLINED
- LOCAL\_LIBRARY\_ROUNDED
- LOCAL\_LIBRARY\_SHARP
- LOCAL\_MALL
- LOCAL\_MALL\_OUTLINED
- LOCAL\_MALL\_ROUNDED
- LOCAL\_MALL\_SHARP
- LOCAL\_MOVIES
- LOCAL\_MOVIES\_OUTLINED
- LOCAL\_MOVIES\_ROUNDED
- LOCAL\_MOVIES\_SHARP
- LOCAL\_OFFER
- LOCAL\_OFFER\_OUTLINED
- LOCAL\_OFFER\_ROUNDED
- LOCAL\_OFFER\_SHARP
- LOCAL\_PARKING
- LOCAL\_PARKING\_OUTLINED
- LOCAL\_PARKING\_ROUNDED
- LOCAL\_PARKING\_SHARP
- LOCAL\_PHARMACY
- LOCAL\_PHARMACY\_OUTLINED
- LOCAL\_PHARMACY\_ROUNDED
- LOCAL\_PHARMACY\_SHARP
- LOCAL\_PHONE

- LOCAL\_PHONE\_OUTLINED
- LOCAL\_PHONE\_ROUNDED
- LOCAL\_PHONE\_SHARP
- LOCAL\_PIZZA
- LOCAL\_PIZZA\_OUTLINED
- LOCAL\_PIZZA\_ROUNDED
- LOCAL\_PIZZA\_SHARP
- LOCAL\_PLAY
- LOCAL\_PLAY\_OUTLINED
- LOCAL\_PLAY\_ROUNDED
- LOCAL\_PLAY\_SHARP
- LOCAL\_POLICE
- LOCAL\_POLICE\_OUTLINED
- LOCAL\_POLICE\_ROUNDED
- LOCAL\_POLICE\_SHARP
- LOCAL\_POST\_OFFICE
- LOCAL\_POST\_OFFICE\_OUTLINED
- LOCAL\_POST\_OFFICE\_ROUNDED
- LOCAL\_POST\_OFFICE\_SHARP
- LOCAL\_PRINT\_SHOP
- LOCAL\_PRINT\_SHOP\_OUTLINED
- LOCAL\_PRINT\_SHOP\_ROUNDED
- LOCAL\_PRINT\_SHOP\_SHARP
- LOCAL\_PRINTSHOP
- LOCAL\_PRINTSHOP\_OUTLINED
- LOCAL\_PRINTSHOP\_ROUNDED
- LOCAL\_PRINTSHOP\_SHARP
- LOCAL\_RESTAURANT
- LOCAL\_RESTAURANT\_OUTLINED
- LOCAL\_RESTAURANT\_ROUNDED
- LOCAL\_RESTAURANT\_SHARP
- LOCAL\_SEE
- LOCAL\_SEE\_OUTLINED
- LOCAL\_SEE\_ROUNDED
- LOCAL\_SEE\_SHARP
- LOCAL\_SHIPPING
- LOCAL\_SHIPPING\_OUTLINED
- LOCAL\_SHIPPING\_ROUNDED
- LOCAL\_SHIPPING\_SHARP
- LOCAL\_TAXI
- LOCAL\_TAXI\_OUTLINED

- LOCAL\_TAXI\_ROUNDED
- LOCAL\_TAXI\_SHARP
- LOCATION\_CITY
- LOCATION\_CITY\_OUTLINED
- LOCATION\_CITY\_ROUNDED
- LOCATION\_CITY\_SHARP
- LOCATION\_DISABLED
- LOCATION\_DISABLED\_OUTLINED
- LOCATION\_DISABLED\_ROUNDED
- LOCATION\_DISABLED\_SHARP
- LOCATION\_HISTORY
- LOCATION\_HISTORY\_OUTLINED
- LOCATION\_HISTORY\_ROUNDED
- LOCATION\_HISTORY\_SHARP
- LOCATION\_OFF
- LOCATION\_OFF\_OUTLINED
- LOCATION\_OFF\_ROUNDED
- LOCATION\_OFF\_SHARP
- LOCATION\_ON
- LOCATION\_ON\_OUTLINED
- LOCATION\_ON\_ROUNDED
- LOCATION\_ON\_SHARP
- LOCATION\_PIN
- LOCATION\_SEARCHING
- LOCATION\_SEARCHING\_OUTLINED
- LOCATION\_SEARCHING\_ROUNDED
- LOCATION\_SEARCHING\_SHARP
- LOCK
- LOCK\_CLOCK
- LOCK\_CLOCK\_OUTLINED
- LOCK\_CLOCK\_ROUNDED
- LOCK\_CLOCK\_SHARP
- LOCK\_OPEN
- LOCK\_OPEN\_OUTLINED
- LOCK\_OPEN\_ROUNDED
- LOCK\_OPEN\_SHARP
- LOCK\_OUTLINE
- LOCK\_OUTLINE\_ROUNDED
- LOCK\_OUTLINE\_SHARP
- LOCK\_OUTLINED
- LOCK\_PERSON

- LOCK\_PERSON\_OUTLINED
- LOCK\_PERSON\_ROUNDED
- LOCK\_PERSON\_SHARP
- LOCK\_RESET
- LOCK\_RESET\_OUTLINED
- LOCK\_RESET\_ROUNDED
- LOCK\_RESET\_SHARP
- LOCK\_ROUNDED
- LOCK\_SHARP
- LOGIN
- LOGIN\_OUTLINED
- LOGIN\_ROUNDED
- LOGIN\_SHARP
- LOGO\_DEV
- LOGO\_DEV\_OUTLINED
- LOGO\_DEV\_ROUNDED
- LOGO\_DEV\_SHARP
- LOGOUT
- LOGOUT\_OUTLINED
- LOGOUT\_ROUNDED
- LOGOUT\_SHARP
- LOOKS
- LOOKS\_3
- LOOKS\_3\_OUTLINED
- LOOKS\_3\_ROUNDED
- LOOKS\_3\_SHARP
- LOOKS\_4
- LOOKS\_4\_OUTLINED
- LOOKS\_4\_ROUNDED
- LOOKS\_4\_SHARP
- LOOKS\_5
- LOOKS\_5\_OUTLINED
- LOOKS\_5\_ROUNDED
- LOOKS\_5\_SHARP
- LOOKS\_6
- LOOKS\_6\_OUTLINED
- LOOKS\_6\_ROUNDED
- LOOKS\_6\_SHARP
- LOOKS\_ONE
- LOOKS\_ONE\_OUTLINED
- LOOKS\_ONE\_ROUNDED

- LOOKS\_ONE\_SHARP
- LOOKS\_OUTLINED
- LOOKS\_ROUNDED
- LOOKS\_SHARP
- LOOKS\_TWO
- LOOKS\_TWO\_OUTLINED
- LOOKS\_TWO\_ROUNDED
- LOOKS\_TWO\_SHARP
- LOOP
- LOOP\_OUTLINED
- LOOP\_ROUNDED
- LOOP\_SHARP
- LOUPE
- LOUPE\_OUTLINED
- LOUPE\_ROUNDED
- LOUPE\_SHARP
- LOW\_PRIORITY
- LOW\_PRIORITY\_OUTLINED
- LOW\_PRIORITY\_ROUNDED
- LOW\_PRIORITY\_SHARP
- LOYALTY
- LOYALTY\_OUTLINED
- LOYALTY\_ROUNDED
- LOYALTY\_SHARP
- LTE\_MOBILEDATA
- LTE\_MOBILEDATA\_OUTLINED
- LTE\_MOBILEDATA\_ROUNDED
- LTE\_MOBILEDATA\_SHARP
- LTE\_PLUS\_MOBILEDATA
- LTE\_PLUS\_MOBILEDATA\_OUTLINED
- LTE\_PLUS\_MOBILEDATA\_ROUNDED
- LTE\_PLUS\_MOBILEDATA\_SHARP
- LUGGAGE
- LUGGAGE\_OUTLINED
- LUGGAGE\_ROUNDED
- LUGGAGE\_SHARP
- LUNCH\_DINING
- LUNCH\_DINING\_OUTLINED
- LUNCH\_DINING\_ROUNDED
- LUNCH\_DINING\_SHARP
- LYRICS

- LYRICS\_OUTLINED
- LYRICS\_ROUNDED
- LYRICS\_SHARP
- MACRO\_OFF
- MACRO\_OFF\_OUTLINED
- MACRO\_OFF\_ROUNDED
- MACRO\_OFF\_SHARP
- MAIL
- MAIL\_LOCK
- MAIL\_LOCK\_OUTLINED
- MAIL\_LOCK\_ROUNDED
- MAIL\_LOCK\_SHARP
- MAIL\_OUTLINE
- MAIL\_OUTLINE\_OUTLINED
- MAIL\_OUTLINE\_ROUNDED
- MAIL\_OUTLINE\_SHARP
- MAIL\_OUTLINED
- MAIL\_ROUNDED
- MAIL\_SHARP
- MALE
- MALE\_OUTLINED
- MALE\_ROUNDED
- MALE\_SHARP
- MAN
- MAN\_2
- MAN\_2\_OUTLINED
- MAN\_2\_ROUNDED
- MAN\_2\_SHARP
- MAN\_3
- MAN\_3\_OUTLINED
- MAN\_3\_ROUNDED
- MAN\_3\_SHARP
- MAN\_4
- MAN\_4\_OUTLINED
- MAN\_4\_ROUNDED
- MAN\_4\_SHARP
- MAN\_OUTLINED
- MAN\_ROUNDED
- MAN\_SHARP
- MANAGE\_ACCOUNTS
- MANAGE\_ACCOUNTS\_OUTLINED



- MANAGE\_ACCOUNTS\_ROUNDED
- MANAGE\_ACCOUNTS\_SHARP
- MANAGE\_HISTORY
- MANAGE\_HISTORY\_OUTLINED
- MANAGE\_HISTORY\_ROUNDED
- MANAGE\_HISTORY\_SHARP
- MANAGE\_SEARCH
- MANAGE\_SEARCH\_OUTLINED
- MANAGE\_SEARCH\_ROUNDED
- MANAGE\_SEARCH\_SHARP
- MAP
- MAP\_OUTLINED
- MAP\_ROUNDED
- MAP\_SHARP
- MAPS\_HOME\_WORK
- MAPS\_HOME\_WORK\_OUTLINED
- MAPS\_HOME\_WORK\_ROUNDED
- MAPS\_HOME\_WORK\_SHARP
- MAPS\_UGC
- MAPS\_UGC\_OUTLINED
- MAPS\_UGC\_ROUNDED
- MAPS\_UGC\_SHARP
- MARGIN
- MARGIN\_OUTLINED
- MARGIN\_ROUNDED
- MARGIN\_SHARP
- MARK\_AS\_UNREAD
- MARK\_AS\_UNREAD\_OUTLINED
- MARK\_AS\_UNREAD\_ROUNDED
- MARK\_AS\_UNREAD\_SHARP
- MARK\_CHAT\_READ
- MARK\_CHAT\_READ\_OUTLINED
- MARK\_CHAT\_READ\_ROUNDED
- MARK\_CHAT\_READ\_SHARP
- MARK\_CHAT\_UNREAD
- MARK\_CHAT\_UNREAD\_OUTLINED
- MARK\_CHAT\_UNREAD\_ROUNDED
- MARK\_CHAT\_UNREAD\_SHARP
- MARK\_EMAIL\_READ
- MARK\_EMAIL\_READ\_OUTLINED
- MARK\_EMAIL\_READ\_ROUNDED

- MARK\_EMAIL\_READ\_SHARP
- MARK\_EMAIL\_UNREAD
- MARK\_EMAIL\_UNREAD\_OUTLINED
- MARK\_EMAIL\_UNREAD\_ROUNDED
- MARK\_EMAIL\_UNREAD\_SHARP
- MARK\_UNREAD\_CHAT\_ALT
- MARK\_UNREAD\_CHAT\_ALT\_OUTLINED
- MARK\_UNREAD\_CHAT\_ALT\_ROUNDED
- MARK\_UNREAD\_CHAT\_ALT\_SHARP
- MARKUNREAD
- MARKUNREAD\_MAILBOX
- MARKUNREAD\_MAILBOX\_OUTLINED
- MARKUNREAD\_MAILBOX\_ROUNDED
- MARKUNREAD\_MAILBOX\_SHARP
- MARKUNREAD\_OUTLINED
- MARKUNREAD\_ROUNDED
- MARKUNREAD\_SHARP
- MASKS
- MASKS\_OUTLINED
- MASKS\_ROUNDED
- MASKS\_SHARP
- MAXIMIZE
- MAXIMIZE\_OUTLINED
- MAXIMIZE\_ROUNDED
- MAXIMIZE\_SHARP
- MEDIA\_BLUETOOTH\_OFF
- MEDIA\_BLUETOOTH\_OFF\_OUTLINED
- MEDIA\_BLUETOOTH\_OFF\_ROUNDED
- MEDIA\_BLUETOOTH\_OFF\_SHARP
- MEDIA\_BLUETOOTH\_ON
- MEDIA\_BLUETOOTH\_ON\_OUTLINED
- MEDIA\_BLUETOOTH\_ON\_ROUNDED
- MEDIA\_BLUETOOTH\_ON\_SHARP
- MEDIATION
- MEDIATION\_OUTLINED
- MEDIATION\_ROUNDED
- MEDIATION\_SHARP
- MEDICAL\_INFORMATION
- MEDICAL\_INFORMATION\_OUTLINED
- MEDICAL\_INFORMATION\_ROUNDED
- MEDICAL\_INFORMATION\_SHARP

- MEDICAL\_SERVICES
- MEDICAL\_SERVICES\_OUTLINED
- MEDICAL\_SERVICES\_ROUNDED
- MEDICAL\_SERVICES\_SHARP
- MEDICATION
- MEDICATION\_LIQUID
- MEDICATION\_LIQUID\_OUTLINED
- MEDICATION\_LIQUID\_ROUNDED
- MEDICATION\_LIQUID\_SHARP
- MEDICATION\_OUTLINED
- MEDICATION\_ROUNDED
- MEDICATION\_SHARP
- MEETING\_ROOM
- MEETING\_ROOM\_OUTLINED
- MEETING\_ROOM\_ROUNDED
- MEETING\_ROOM\_SHARP
- MEMORY
- MEMORY\_OUTLINED
- MEMORY\_ROUNDED
- MEMORY\_SHARP
- MENU
- MENU\_BOOK
- MENU\_BOOK\_OUTLINED
- MENU\_BOOK\_ROUNDED
- MENU\_BOOK\_SHARP
- MENU\_OPEN
- MENU\_OPEN\_OUTLINED
- MENU\_OPEN\_ROUNDED
- MENU\_OPEN\_SHARP
- MENU\_OUTLINED
- MENU\_ROUNDED
- MENU\_SHARP
- MERGE
- MERGE\_OUTLINED
- MERGE\_ROUNDED
- MERGE\_SHARP
- MERGE\_TYPE
- MERGE\_TYPE\_OUTLINED
- MERGE\_TYPE\_ROUNDED
- MERGE\_TYPE\_SHARP
- MESSAGE

- MESSAGE\_OUTLINED
- MESSAGE\_ROUNDED
- MESSAGE\_SHARP
- MESSENGER
- MESSENGER\_OUTLINE
- MESSENGER\_OUTLINE\_OUTLINED
- MESSENGER\_OUTLINE\_ROUNDED
- MESSENGER\_OUTLINE\_SHARP
- MESSENGER\_OUTLINED
- MESSENGER\_ROUNDED
- MESSENGER\_SHARP
- MIC
- MIC\_EXTERNAL\_OFF
- MIC\_EXTERNAL\_OFF\_OUTLINED
- MIC\_EXTERNAL\_OFF\_ROUNDED
- MIC\_EXTERNAL\_OFF\_SHARP
- MIC\_EXTERNAL\_ON
- MIC\_EXTERNAL\_ON\_OUTLINED
- MIC\_EXTERNAL\_ON\_ROUNDED
- MIC\_EXTERNAL\_ON\_SHARP
- MIC\_NONE
- MIC\_NONE\_OUTLINED
- MIC\_NONE\_ROUNDED
- MIC\_NONE\_SHARP
- MIC\_OFF
- MIC\_OFF\_OUTLINED
- MIC\_OFF\_ROUNDED
- MIC\_OFF\_SHARP
- MIC\_OUTLINED
- MIC\_ROUNDED
- MIC\_SHARP
- MICROWAVE
- MICROWAVE\_OUTLINED
- MICROWAVE\_ROUNDED
- MICROWAVE\_SHARP
- MILITARY\_TECH
- MILITARY\_TECH\_OUTLINED
- MILITARY\_TECH\_ROUNDED
- MILITARY\_TECH\_SHARP
- MINIMIZE
- MINIMIZE\_OUTLINED

- MINIMIZE\_ROUNDED
- MINIMIZE\_SHARP
- MINOR\_CRASH
- MINOR\_CRASH\_OUTLINED
- MINOR\_CRASH\_ROUNDED
- MINOR\_CRASH\_SHARP
- MISCELLANEOUS\_SERVICES
- MISCELLANEOUS\_SERVICES\_OUTLINED
- MISCELLANEOUS\_SERVICES\_ROUNDED
- MISCELLANEOUS\_SERVICES\_SHARP
- MISSED\_VIDEO\_CALL
- MISSED\_VIDEO\_CALL\_OUTLINED
- MISSED\_VIDEO\_CALL\_ROUNDED
- MISSED\_VIDEO\_CALL\_SHARP
- MMS
- MMS\_OUTLINED
- MMS\_ROUNDED
- MMS\_SHARP
- MOBILE\_FRIENDLY
- MOBILE\_FRIENDLY\_OUTLINED
- MOBILE\_FRIENDLY\_ROUNDED
- MOBILE\_FRIENDLY\_SHARP
- MOBILE\_OFF
- MOBILE\_OFF\_OUTLINED
- MOBILE\_OFF\_ROUNDED
- MOBILE\_OFF\_SHARP
- MOBILE\_SCREEN\_SHARE
- MOBILE\_SCREEN\_SHARE\_OUTLINED
- MOBILE\_SCREEN\_SHARE\_ROUNDED
- MOBILE\_SCREEN\_SHARE\_SHARP
- MOBILEDATA\_OFF
- MOBILEDATA\_OFF\_OUTLINED
- MOBILEDATA\_OFF\_ROUNDED
- MOBILEDATA\_OFF\_SHARP
- MODE
- MODE\_COMMENT
- MODE\_COMMENT\_OUTLINED
- MODE\_COMMENT\_ROUNDED
- MODE\_COMMENT\_SHARP
- MODE\_EDIT
- MODE\_EDIT\_OUTLINE

- MODE\_EDIT\_OUTLINE\_OUTLINED
- MODE\_EDIT\_OUTLINE\_ROUNDED
- MODE\_EDIT\_OUTLINE\_SHARP
- MODE\_EDIT\_OUTLINED
- MODE\_EDIT\_ROUNDED
- MODE\_EDIT\_SHARP
- MODE\_FAN\_OFF
- MODE\_FAN\_OFF\_OUTLINED
- MODE\_FAN\_OFF\_ROUNDED
- MODE\_FAN\_OFF\_SHARP
- MODE\_NIGHT
- MODE\_NIGHT\_OUTLINED
- MODE\_NIGHT\_ROUNDED
- MODE\_NIGHT\_SHARP
- MODE\_OF\_TRAVEL
- MODE\_OF\_TRAVEL\_OUTLINED
- MODE\_OF\_TRAVEL\_ROUNDED
- MODE\_OF\_TRAVEL\_SHARP
- MODE\_OUTLINED
- MODE\_ROUNDED
- MODE\_SHARP
- MODE\_STANDBY
- MODE\_STANDBY\_OUTLINED
- MODE\_STANDBY\_ROUNDED
- MODE\_STANDBY\_SHARP
- MODEL\_TRAINING
- MODEL\_TRAINING\_OUTLINED
- MODEL\_TRAINING\_ROUNDED
- MODEL\_TRAINING\_SHARP
- MONETIZATION\_ON
- MONETIZATION\_ON\_OUTLINED
- MONETIZATION\_ON\_ROUNDED
- MONETIZATION\_ON\_SHARP
- MONEY
- MONEY\_OFF
- MONEY\_OFF\_CSRED
- MONEY\_OFF\_CSRED\_OUTLINED
- MONEY\_OFF\_CSRED\_ROUNDED
- MONEY\_OFF\_CSRED\_SHARP
- MONEY\_OFF\_OUTLINED
- MONEY\_OFF\_ROUNDED

- MONEY\_OFF\_SHARP
- MONEY\_OUTLINED
- MONEY\_ROUNDED
- MONEY\_SHARP
- MONITOR
- MONITOR\_HEART
- MONITOR\_HEART\_OUTLINED
- MONITOR\_HEART\_ROUNDED
- MONITOR\_HEART\_SHARP
- MONITOR\_OUTLINED
- MONITOR\_ROUNDED
- MONITOR\_SHARP
- MONITOR\_WEIGHT
- MONITOR\_WEIGHT\_OUTLINED
- MONITOR\_WEIGHT\_ROUNDED
- MONITOR\_WEIGHT\_SHARP
- MONOCHROME\_PHOTOS
- MONOCHROME\_PHOTOS\_OUTLINED
- MONOCHROME\_PHOTOS\_ROUNDED
- MONOCHROME\_PHOTOS\_SHARP
- MOOD
- MOOD\_BAD
- MOOD\_BAD\_OUTLINED
- MOOD\_BAD\_ROUNDED
- MOOD\_BAD\_SHARP
- MOOD\_OUTLINED
- MOOD\_ROUNDED
- MOOD\_SHARP
- MOPED
- MOPED\_OUTLINED
- MOPED\_ROUNDED
- MOPED\_SHARP
- MORE
- MORE\_HORIZ
- MORE\_HORIZ\_OUTLINED
- MORE\_HORIZ\_ROUNDED
- MORE\_HORIZ\_SHARP
- MORE\_OUTLINED
- MORE\_ROUNDED
- MORE\_SHARP
- MORE\_TIME

- MORE\_TIME\_OUTLINED
- MORE\_TIME\_ROUNDED
- MORE\_TIME\_SHARP
- MORE\_VERT
- MORE\_VERT\_OUTLINED
- MORE\_VERT\_ROUNDED
- MORE\_VERT\_SHARP
- MOSQUE
- MOSQUE\_OUTLINED
- MOSQUE\_ROUNDED
- MOSQUE\_SHARP
- MOTION\_PHOTOS\_AUTO
- MOTION\_PHOTOS\_AUTO\_OUTLINED
- MOTION\_PHOTOS\_AUTO\_ROUNDED
- MOTION\_PHOTOS\_AUTO\_SHARP
- MOTION\_PHOTOS\_OFF
- MOTION\_PHOTOS\_OFF\_OUTLINED
- MOTION\_PHOTOS\_OFF\_ROUNDED
- MOTION\_PHOTOS\_OFF\_SHARP
- MOTION\_PHOTOS\_ON
- MOTION\_PHOTOS\_ON\_OUTLINED
- MOTION\_PHOTOS\_ON\_ROUNDED
- MOTION\_PHOTOS\_ON\_SHARP
- MOTION\_PHOTOS\_PAUSE
- MOTION\_PHOTOS\_PAUSE\_OUTLINED
- MOTION\_PHOTOS\_PAUSE\_ROUNDED
- MOTION\_PHOTOS\_PAUSE\_SHARP
- MOTION\_PHOTOS\_PAUSED
- MOTION\_PHOTOS\_PAUSED\_OUTLINED
- MOTION\_PHOTOS\_PAUSED\_ROUNDED
- MOTION\_PHOTOS\_PAUSED\_SHARP
- MOTORCYCLE
- MOTORCYCLE\_OUTLINED
- MOTORCYCLE\_ROUNDED
- MOTORCYCLE\_SHARP
- MOUSE
- MOUSE\_OUTLINED
- MOUSE\_ROUNDED
- MOUSE\_SHARP
- MOVE\_DOWN
- MOVE\_DOWN\_OUTLINED



- MOVE\_DOWN\_ROUNDED
- MOVE\_DOWN\_SHARP
- MOVE\_TO\_INBOX
- MOVE\_TO\_INBOX\_OUTLINED
- MOVE\_TO\_INBOX\_ROUNDED
- MOVE\_TO\_INBOX\_SHARP
- MOVE\_UP
- MOVE\_UP\_OUTLINED
- MOVE\_UP\_ROUNDED
- MOVE\_UP\_SHARP
- MOVIE
- MOVIE\_CREATION
- MOVIE\_CREATION\_OUTLINED
- MOVIE\_CREATION\_ROUNDED
- MOVIE\_CREATION\_SHARP
- MOVIE\_EDIT
- MOVIE\_FILTER
- MOVIE\_FILTER\_OUTLINED
- MOVIE\_FILTER\_ROUNDED
- MOVIE\_FILTER\_SHARP
- MOVIE\_OUTLINED
- MOVIE\_ROUNDED
- MOVIE\_SHARP
- MOVING
- MOVING\_OUTLINED
- MOVING\_ROUNDED
- MOVING\_SHARP
- MP
- MP\_OUTLINED
- MP\_ROUNDED
- MP\_SHARP
- MULTILINE\_CHART
- MULTILINE\_CHART\_OUTLINED
- MULTILINE\_CHART\_ROUNDED
- MULTILINE\_CHART\_SHARP
- MULTIPLE\_STOP
- MULTIPLE\_STOP\_OUTLINED
- MULTIPLE\_STOP\_ROUNDED
- MULTIPLE\_STOP\_SHARP
- MULTITRACK\_AUDIO
- MULTITRACK\_AUDIO\_OUTLINED

- MULTITRACK\_AUDIO\_ROUNDED
- MULTITRACK\_AUDIO\_SHARP
- MUSEUM
- MUSEUM\_OUTLINED
- MUSEUM\_ROUNDED
- MUSEUM\_SHARP
- MUSIC\_NOTE
- MUSIC\_NOTE\_OUTLINED
- MUSIC\_NOTE\_ROUNDED
- MUSIC\_NOTE\_SHARP
- MUSIC\_OFF
- MUSIC\_OFF\_OUTLINED
- MUSIC\_OFF\_ROUNDED
- MUSIC\_OFF\_SHARP
- MUSIC\_VIDEO
- MUSIC\_VIDEO\_OUTLINED
- MUSIC\_VIDEO\_ROUNDED
- MUSIC\_VIDEO\_SHARP
- MY\_LIBRARY\_ADD
- MY\_LIBRARY\_ADD\_OUTLINED
- MY\_LIBRARY\_ADD\_ROUNDED
- MY\_LIBRARY\_ADD\_SHARP
- MY\_LIBRARY\_BOOKS
- MY\_LIBRARY\_BOOKS\_OUTLINED
- MY\_LIBRARY\_BOOKS\_ROUNDED
- MY\_LIBRARY\_BOOKS\_SHARP
- MY\_LIBRARY\_MUSIC
- MY\_LIBRARY\_MUSIC\_OUTLINED
- MY\_LIBRARY\_MUSIC\_ROUNDED
- MY\_LIBRARY\_MUSIC\_SHARP
- MY\_LOCATION
- MY\_LOCATION\_OUTLINED
- MY\_LOCATION\_ROUNDED
- MY\_LOCATION\_SHARP
- NAT
- NAT\_OUTLINED
- NAT\_ROUNDED
- NAT\_SHARP
- NATURE
- NATURE\_OUTLINED
- NATURE\_PEOPLE

- NATURE\_PEOPLE\_OUTLINED
- NATURE\_PEOPLE\_ROUNDED
- NATURE\_PEOPLE\_SHARP
- NATURE\_ROUNDED
- NATURE\_SHARP
- NAVIGATE\_BEFORE
- NAVIGATE\_BEFORE\_OUTLINED
- NAVIGATE\_BEFORE\_ROUNDED
- NAVIGATE\_BEFORE\_SHARP
- NAVIGATE\_NEXT
- NAVIGATE\_NEXT\_OUTLINED
- NAVIGATE\_NEXT\_ROUNDED
- NAVIGATE\_NEXT\_SHARP
- NAVIGATION
- NAVIGATION\_OUTLINED
- NAVIGATION\_ROUNDED
- NAVIGATION\_SHARP
- NEAR\_ME
- NEAR\_ME\_DISABLED
- NEAR\_ME\_DISABLED\_OUTLINED
- NEAR\_ME\_DISABLED\_ROUNDED
- NEAR\_ME\_DISABLED\_SHARP
- NEAR\_ME\_OUTLINED
- NEAR\_ME\_ROUNDED
- NEAR\_ME\_SHARP
- NEARBY\_ERROR
- NEARBY\_ERROR\_OUTLINED
- NEARBY\_ERROR\_ROUNDED
- NEARBY\_ERROR\_SHARP
- NEARBY\_OFF
- NEARBY\_OFF\_OUTLINED
- NEARBY\_OFF\_ROUNDED
- NEARBY\_OFF\_SHARP
- NEST\_CAM\_WIRED\_STAND
- NEST\_CAM\_WIRED\_STAND\_OUTLINED
- NEST\_CAM\_WIRED\_STAND\_ROUNDED
- NEST\_CAM\_WIRED\_STAND\_SHARP
- NETWORK\_CELL
- NETWORK\_CELL\_OUTLINED
- NETWORK\_CELL\_ROUNDED
- NETWORK\_CELL\_SHARP

- NETWORK\_CHECK
- NETWORK\_CHECK\_OUTLINED
- NETWORK\_CHECK\_ROUNDED
- NETWORK\_CHECK\_SHARP
- NETWORK\_LOCKED
- NETWORK\_LOCKED\_OUTLINED
- NETWORK\_LOCKED\_ROUNDED
- NETWORK\_LOCKED\_SHARP
- NETWORK\_PING
- NETWORK\_PING\_OUTLINED
- NETWORK\_PING\_ROUNDED
- NETWORK\_PING\_SHARP
- NETWORK\_WIFI
- NETWORK\_WIFI\_1\_BAR
- NETWORK\_WIFI\_1\_BAR\_OUTLINED
- NETWORK\_WIFI\_1\_BAR\_ROUNDED
- NETWORK\_WIFI\_1\_BAR\_SHARP
- NETWORK\_WIFI\_2\_BAR
- NETWORK\_WIFI\_2\_BAR\_OUTLINED
- NETWORK\_WIFI\_2\_BAR\_ROUNDED
- NETWORK\_WIFI\_2\_BAR\_SHARP
- NETWORK\_WIFI\_3\_BAR
- NETWORK\_WIFI\_3\_BAR\_OUTLINED
- NETWORK\_WIFI\_3\_BAR\_ROUNDED
- NETWORK\_WIFI\_3\_BAR\_SHARP
- NETWORK\_WIFI\_OUTLINED
- NETWORK\_WIFI\_ROUNDED
- NETWORK\_WIFI\_SHARP
- NEW\_LABEL
- NEW\_LABEL\_OUTLINED
- NEW\_LABEL\_ROUNDED
- NEW\_LABEL\_SHARP
- NEW\_RELEASES
- NEW\_RELEASES\_OUTLINED
- NEW\_RELEASES\_ROUNDED
- NEW\_RELEASES\_SHARP
- NEWSPAPER
- NEWSPAPER\_OUTLINED
- NEWSPAPER\_ROUNDED
- NEWSPAPER\_SHARP
- NEXT\_PLAN

- NEXT\_PLAN\_OUTLINED
- NEXT\_PLAN\_ROUNDED
- NEXT\_PLAN\_SHARP
- NEXT\_WEEK
- NEXT\_WEEK\_OUTLINED
- NEXT\_WEEK\_ROUNDED
- NEXT\_WEEK\_SHARP
- NFC
- NFC\_OUTLINED
- NFC\_ROUNDED
- NFC\_SHARP
- NIGHT\_SHELTER
- NIGHT\_SHELTER\_OUTLINED
- NIGHT\_SHELTER\_ROUNDED
- NIGHT\_SHELTER\_SHARP
- NIGHTLIFE
- NIGHTLIFE\_OUTLINED
- NIGHTLIFE\_ROUNDED
- NIGHTLIFE\_SHARP
- NIGHTLIGHT
- NIGHTLIGHT\_OUTLINED
- NIGHTLIGHT\_ROUND
- NIGHTLIGHT\_ROUND\_OUTLINED
- NIGHTLIGHT\_ROUND\_ROUNDED
- NIGHTLIGHT\_ROUND\_SHARP
- NIGHTLIGHT\_ROUNDED
- NIGHTLIGHT\_SHARP
- NIGHTS\_STAY
- NIGHTS\_STAY\_OUTLINED
- NIGHTS\_STAY\_ROUNDED
- NIGHTS\_STAY\_SHARP
- NINE\_K
- NINE\_K\_OUTLINED
- NINE\_K\_PLUS
- NINE\_K\_PLUS\_OUTLINED
- NINE\_K\_PLUS\_ROUNDED
- NINE\_K\_PLUS\_SHARP
- NINE\_K\_ROUNDED
- NINE\_K\_SHARP
- NINE\_MP
- NINE\_MP\_OUTLINED

- NINE\_MP\_ROUNDED
- NINE\_MP\_SHARP
- NINETEEN\_MP
- NINETEEN\_MP\_OUTLINED
- NINETEEN\_MP\_ROUNDED
- NINETEEN\_MP\_SHARP
- NO\_ACCOUNTS
- NO\_ACCOUNTS\_OUTLINED
- NO\_ACCOUNTS\_ROUNDED
- NO\_ACCOUNTS\_SHARP
- NO\_ADULT\_CONTENT
- NO\_ADULT\_CONTENT\_OUTLINED
- NO\_ADULT\_CONTENT\_ROUNDED
- NO\_ADULT\_CONTENT\_SHARP
- NO\_BACKPACK
- NO\_BACKPACK\_OUTLINED
- NO\_BACKPACK\_ROUNDED
- NO\_BACKPACK\_SHARP
- NO\_CELL
- NO\_CELL\_OUTLINED
- NO\_CELL\_ROUNDED
- NO\_CELL\_SHARP
- NO\_CRASH
- NO\_CRASH\_OUTLINED
- NO\_CRASH\_ROUNDED
- NO\_CRASH\_SHARP
- NO\_DRINKS
- NO\_DRINKS\_OUTLINED
- NO\_DRINKS\_ROUNDED
- NO\_DRINKS\_SHARP
- NO\_ENCRYPTION
- NO\_ENCRYPTION\_GMAILERRORRED
- NO\_ENCRYPTION\_GMAILERRORRED\_OUTLINED
- NO\_ENCRYPTION\_GMAILERRORRED\_ROUNDED
- NO\_ENCRYPTION\_GMAILERRORRED\_SHARP
- NO\_ENCRYPTION\_OUTLINED
- NO\_ENCRYPTION\_ROUNDED
- NO\_ENCRYPTION\_SHARP
- NO\_FLASH
- NO\_FLASH\_OUTLINED
- NO\_FLASH\_ROUNDED

- NO\_FLASH\_SHARP
- NO\_FOOD
- NO\_FOOD\_OUTLINED
- NO\_FOOD\_ROUNDED
- NO\_FOOD\_SHARP
- NO\_LUGGAGE
- NO\_LUGGAGE\_OUTLINED
- NO\_LUGGAGE\_ROUNDED
- NO\_LUGGAGE\_SHARP
- NO\_MEALS
- NO\_MEALS\_OULINE
- NO\_MEALS\_OUTLINED
- NO\_MEALS\_ROUNDED
- NO\_MEALS\_SHARP
- NO\_MEETING\_ROOM
- NO\_MEETING\_ROOM\_OUTLINED
- NO\_MEETING\_ROOM\_ROUNDED
- NO\_MEETING\_ROOM\_SHARP
- NO\_PHOTOGRAPHY
- NO\_PHOTOGRAPHY\_OUTLINED
- NO\_PHOTOGRAPHY\_ROUNDED
- NO\_PHOTOGRAPHY\_SHARP
- NO\_SIM
- NO\_SIM\_OUTLINED
- NO\_SIM\_ROUNDED
- NO\_SIM\_SHARP
- NO\_STROLLER
- NO\_STROLLER\_OUTLINED
- NO\_STROLLER\_ROUNDED
- NO\_STROLLER\_SHARP
- NO\_TRANSFER
- NO\_TRANSFER\_OUTLINED
- NO\_TRANSFER\_ROUNDED
- NO\_TRANSFER\_SHARP
- NOISE\_AWARE
- NOISE\_AWARE\_OUTLINED
- NOISE\_AWARE\_ROUNDED
- NOISE\_AWARE\_SHARP
- NOISE\_CONTROL\_OFF
- NOISE\_CONTROL\_OFF\_OUTLINED
- NOISE\_CONTROL\_OFF\_ROUNDED

- NOISE\_CONTROL\_OFF\_SHARP
- NORDIC\_WALKING
- NORDIC\_WALKING\_OUTLINED
- NORDIC\_WALKING\_ROUNDED
- NORDIC\_WALKING\_SHARP
- NORTH
- NORTH\_EAST
- NORTH\_EAST\_OUTLINED
- NORTH\_EAST\_ROUNDED
- NORTH\_EAST\_SHARP
- NORTH\_OUTLINED
- NORTH\_ROUNDED
- NORTH\_SHARP
- NORTH\_WEST
- NORTH\_WEST\_OUTLINED
- NORTH\_WEST\_ROUNDED
- NORTH\_WEST\_SHARP
- NOT\_ACCESSIBLE
- NOT\_ACCESSIBLE\_OUTLINED
- NOT\_ACCESSIBLE\_ROUNDED
- NOT\_ACCESSIBLE\_SHARP
- NOT\_INTERESTED
- NOT\_INTERESTED\_OUTLINED
- NOT\_INTERESTED\_ROUNDED
- NOT\_INTERESTED\_SHARP
- NOT\_LISTED\_LOCATION
- NOT\_LISTED\_LOCATION\_OUTLINED
- NOT\_LISTED\_LOCATION\_ROUNDED
- NOT\_LISTED\_LOCATION\_SHARP
- NOT\_STARTED
- NOT\_STARTED\_OUTLINED
- NOT\_STARTED\_ROUNDED
- NOT\_STARTED\_SHARP
- NOTE
- NOTE\_ADD
- NOTE\_ADD\_OUTLINED
- NOTE\_ADD\_ROUNDED
- NOTE\_ADD\_SHARP
- NOTE\_ALT
- NOTE\_ALT\_OUTLINED
- NOTE\_ALT\_ROUNDED



- NOTE\_ALT\_SHARP
- NOTE\_OUTLINED
- NOTE\_ROUNDED
- NOTE\_SHARP
- NOTES
- NOTES\_OUTLINED
- NOTES\_ROUNDED
- NOTES\_SHARP
- NOTIFICATION\_ADD
- NOTIFICATION\_ADD\_OUTLINED
- NOTIFICATION\_ADD\_ROUNDED
- NOTIFICATION\_ADD\_SHARP
- NOTIFICATION\_IMPORTANT
- NOTIFICATION\_IMPORTANT\_OUTLINED
- NOTIFICATION\_IMPORTANT\_ROUNDED
- NOTIFICATION\_IMPORTANT\_SHARP
- NOTIFICATIONS
- NOTIFICATIONS\_ACTIVE
- NOTIFICATIONS\_ACTIVE\_OUTLINED
- NOTIFICATIONS\_ACTIVE\_ROUNDED
- NOTIFICATIONS\_ACTIVE\_SHARP
- NOTIFICATIONS\_NONE
- NOTIFICATIONS\_NONE\_OUTLINED
- NOTIFICATIONS\_NONE\_ROUNDED
- NOTIFICATIONS\_NONE\_SHARP
- NOTIFICATIONS\_OFF
- NOTIFICATIONS\_OFF\_OUTLINED
- NOTIFICATIONS\_OFF\_ROUNDED
- NOTIFICATIONS\_OFF\_SHARP
- NOTIFICATIONS\_ON
- NOTIFICATIONS\_ON\_OUTLINED
- NOTIFICATIONS\_ON\_ROUNDED
- NOTIFICATIONS\_ON\_SHARP
- NOTIFICATIONS\_OUTLINED
- NOTIFICATIONS\_PAUSED
- NOTIFICATIONS\_PAUSED\_OUTLINED
- NOTIFICATIONS\_PAUSED\_ROUNDED
- NOTIFICATIONS\_PAUSED\_SHARP
- NOTIFICATIONS\_ROUNDED
- NOTIFICATIONS\_SHARP
- NOW\_WALLPAPER

- NOW\_WALLPAPER\_OUTLINED
- NOW\_WALLPAPER\_ROUNDED
- NOW\_WALLPAPER\_SHARP
- NOW\_WIDGETS
- NOW\_WIDGETS\_OUTLINED
- NOW\_WIDGETS\_ROUNDED
- NOW\_WIDGETS\_SHARP
- NUMBERS
- NUMBERS\_OUTLINED
- NUMBERS\_ROUNDED
- NUMBERS\_SHARP
- OFFLINE\_BOLT
- OFFLINE\_BOLT\_OUTLINED
- OFFLINE\_BOLT\_ROUNDED
- OFFLINE\_BOLT\_SHARP
- OFFLINE\_PIN
- OFFLINE\_PIN\_OUTLINED
- OFFLINE\_PIN\_ROUNDED
- OFFLINE\_PIN\_SHARP
- OFFLINE\_SHARE
- OFFLINE\_SHARE\_OUTLINED
- OFFLINE\_SHARE\_ROUNDED
- OFFLINE\_SHARE\_SHARP
- OIL\_BARREL
- OIL\_BARREL\_OUTLINED
- OIL\_BARREL\_ROUNDED
- OIL\_BARREL\_SHARP
- ON\_DEVICE\_TRAINING
- ON\_DEVICE\_TRAINING\_OUTLINED
- ON\_DEVICE\_TRAINING\_ROUNDED
- ON\_DEVICE\_TRAINING\_SHARP
- ONDEMAND\_VIDEO
- ONDEMAND\_VIDEO\_OUTLINED
- ONDEMAND\_VIDEO\_ROUNDED
- ONDEMAND\_VIDEO\_SHARP
- ONE\_K
- ONE\_K\_OUTLINED
- ONE\_K\_PLUS
- ONE\_K\_PLUS\_OUTLINED
- ONE\_K\_PLUS\_ROUNDED
- ONE\_K\_PLUS\_SHARP

- ONE\_K\_ROUNDED
- ONE\_K\_SHARP
- ONE\_X\_MOBILEDATA
- ONE\_X\_MOBILEDATA\_OUTLINED
- ONE\_X\_MOBILEDATA\_ROUNDED
- ONE\_X\_MOBILEDATA\_SHARP
- ONETWOTHREE
- ONETWOTHREE\_OUTLINED
- ONETWOTHREE\_ROUNDED
- ONETWOTHREE\_SHARP
- ONLINE\_PREDICTION
- ONLINE\_PREDICTION\_OUTLINED
- ONLINE\_PREDICTION\_ROUNDED
- ONLINE\_PREDICTION\_SHARP
- OPACITY
- OPACITY\_OUTLINED
- OPACITY\_ROUNDED
- OPACITY\_SHARP
- OPEN\_IN\_BROWSER
- OPEN\_IN\_BROWSER\_OUTLINED
- OPEN\_IN\_BROWSER\_ROUNDED
- OPEN\_IN\_BROWSER\_SHARP
- OPEN\_IN\_FULL
- OPEN\_IN\_FULL\_OUTLINED
- OPEN\_IN\_FULL\_ROUNDED
- OPEN\_IN\_FULL\_SHARP
- OPEN\_IN\_NEW
- OPEN\_IN\_NEW\_OFF
- OPEN\_IN\_NEW\_OFF\_OUTLINED
- OPEN\_IN\_NEW\_OFF\_ROUNDED
- OPEN\_IN\_NEW\_OFF\_SHARP
- OPEN\_IN\_NEW\_OUTLINED
- OPEN\_IN\_NEW\_ROUNDED
- OPEN\_IN\_NEW\_SHARP
- OPEN\_WITH
- OPEN\_WITH\_OUTLINED
- OPEN\_WITH\_ROUNDED
- OPEN\_WITH\_SHARP
- OTHER\_HOUSES
- OTHER\_HOUSES\_OUTLINED
- OTHER\_HOUSES\_ROUNDED

- OTHER\_HOUSES\_SHARP
- OUTBOND
- OUTBOND\_OUTLINED
- OUTBOND\_ROUNDED
- OUTBOND\_SHARP
- OUTBOUND
- OUTBOUND\_OUTLINED
- OUTBOUND\_ROUNDED
- OUTBOUND\_SHARP
- OUTBOX
- OUTBOX\_OUTLINED
- OUTBOX\_ROUNDED
- OUTBOX\_SHARP
- OUTDOOR\_GRILL
- OUTDOOR\_GRILL\_OUTLINED
- OUTDOOR\_GRILL\_ROUNDED
- OUTDOOR\_GRILL\_SHARP
- OUTGOING\_MAIL
- OUTLET
- OUTLET\_OUTLINED
- OUTLET\_ROUNDED
- OUTLET\_SHARP
- OUTLINED\_FLAG
- OUTLINED\_FLAG\_OUTLINED
- OUTLINED\_FLAG\_ROUNDED
- OUTLINED\_FLAG\_SHARP
- OUTPUT
- OUTPUT\_OUTLINED
- OUTPUT\_ROUNDED
- OUTPUT\_SHARP
- PADDING
- PADDING\_OUTLINED
- PADDING\_ROUNDED
- PADDING\_SHARP
- PAGES
- PAGES\_OUTLINED
- PAGES\_ROUNDED
- PAGES\_SHARP
- PAGEVIEW
- PAGEVIEW\_OUTLINED
- PAGEVIEW\_ROUNDED

- PAGEVIEW\_SHARP
- PAID
- PAID\_OUTLINED
- PAID\_ROUNDED
- PAID\_SHARP
- PALETTE
- PALETTE\_OUTLINED
- PALETTE\_ROUNDED
- PALETTE\_SHARP
- PALLET
- PAN\_TOOL
- PAN\_TOOL\_ALT
- PAN\_TOOL\_ALT\_OUTLINED
- PAN\_TOOL\_ALT\_ROUNDED
- PAN\_TOOL\_ALT\_SHARP
- PAN\_TOOL\_OUTLINED
- PAN\_TOOL\_ROUNDED
- PAN\_TOOL\_SHARP
- PANORAMA
- PANORAMA\_FISH\_EYE
- PANORAMA\_FISH\_EYE\_OUTLINED
- PANORAMA\_FISH\_EYE\_ROUNDED
- PANORAMA\_FISH\_EYE\_SHARP
- PANORAMA\_FISHEYE
- PANORAMA\_FISHEYE\_OUTLINED
- PANORAMA\_FISHEYE\_ROUNDED
- PANORAMA\_FISHEYE\_SHARP
- PANORAMA\_HORIZONTAL
- PANORAMA\_HORIZONTAL\_OUTLINED
- PANORAMA\_HORIZONTAL\_ROUNDED
- PANORAMA\_HORIZONTAL\_SELECT
- PANORAMA\_HORIZONTAL\_SELECT\_OUTLINED
- PANORAMA\_HORIZONTAL\_SELECT\_ROUNDED
- PANORAMA\_HORIZONTAL\_SELECT\_SHARP
- PANORAMA\_HORIZONTAL\_SHARP
- PANORAMA\_OUTLINED
- PANORAMA\_PHOTOSPHERE
- PANORAMA\_PHOTOSPHERE\_OUTLINED
- PANORAMA\_PHOTOSPHERE\_ROUNDED
- PANORAMA\_PHOTOSPHERE\_SELECT
- PANORAMA\_PHOTOSPHERE\_SELECT\_OUTLINED

- PANORAMA\_PHOTOSPHERE\_SELECT\_ROUNDED
- PANORAMA\_PHOTOSPHERE\_SELECT\_SHARP
- PANORAMA\_PHOTOSPHERE\_SHARP
- PANORAMA\_ROUNDED
- PANORAMA\_SHARP
- PANORAMA\_VERTICAL
- PANORAMA\_VERTICAL\_OUTLINED
- PANORAMA\_VERTICAL\_ROUNDED
- PANORAMA\_VERTICAL\_SELECT
- PANORAMA\_VERTICAL\_SELECT\_OUTLINED
- PANORAMA\_VERTICAL\_SELECT\_ROUNDED
- PANORAMA\_VERTICAL\_SELECT\_SHARP
- PANORAMA\_VERTICAL\_SHARP
- PANORAMA\_WIDE\_ANGLE
- PANORAMA\_WIDE\_ANGLE\_OUTLINED
- PANORAMA\_WIDE\_ANGLE\_ROUNDED
- PANORAMA\_WIDE\_ANGLE\_SELECT
- PANORAMA\_WIDE\_ANGLE\_SELECT\_OUTLINED
- PANORAMA\_WIDE\_ANGLE\_SELECT\_ROUNDED
- PANORAMA\_WIDE\_ANGLE\_SELECT\_SHARP
- PANORAMA\_WIDE\_ANGLE\_SHARP
- PARAGLIDING
- PARAGLIDING\_OUTLINED
- PARAGLIDING\_ROUNDED
- PARAGLIDING\_SHARP
- PARK
- PARK\_OUTLINED
- PARK\_ROUNDED
- PARK\_SHARP
- PARTY\_MODE
- PARTY\_MODE\_OUTLINED
- PARTY\_MODE\_ROUNDED
- PARTY\_MODE\_SHARP
- PASSWORD
- PASSWORD\_OUTLINED
- PASSWORD\_ROUNDED
- PASSWORD\_SHARP
- PASTE
- PASTE\_OUTLINED
- PASTE\_ROUNDED
- PASTE\_SHARP

- PATTERN
- PATTERN\_OUTLINED
- PATTERN\_ROUNDED
- PATTERN\_SHARP
- PAUSE
- PAUSE\_CIRCLE
- PAUSE\_CIRCLE\_FILLED
- PAUSE\_CIRCLE\_FILLED\_OUTLINED
- PAUSE\_CIRCLE\_FILLED\_ROUNDED
- PAUSE\_CIRCLE\_FILLED\_SHARP
- PAUSE\_CIRCLE\_OUTLINE
- PAUSE\_CIRCLE\_OUTLINE\_OUTLINED
- PAUSE\_CIRCLE\_OUTLINE\_ROUNDED
- PAUSE\_CIRCLE\_OUTLINE\_SHARP
- PAUSE\_CIRCLE\_OUTLINED
- PAUSE\_CIRCLE\_ROUNDED
- PAUSE\_CIRCLE\_SHARP
- PAUSE\_OUTLINED
- PAUSE\_PRESENTATION
- PAUSE\_PRESENTATION\_OUTLINED
- PAUSE\_PRESENTATION\_ROUNDED
- PAUSE\_PRESENTATION\_SHARP
- PAUSE\_ROUNDED
- PAUSE\_SHARP
- PAYMENT
- PAYMENT\_OUTLINED
- PAYMENT\_ROUNDED
- PAYMENT\_SHARP
- PAYMENTS
- PAYMENTS\_OUTLINED
- PAYMENTS\_ROUNDED
- PAYMENTS\_SHARP
- PAYPAL
- PAYPAL\_OUTLINED
- PAYPAL\_ROUNDED
- PAYPAL\_SHARP
- PEDAL\_BIKE
- PEDAL\_BIKE\_OUTLINED
- PEDAL\_BIKE\_ROUNDED
- PEDAL\_BIKE\_SHARP
- PENDING

- PENDING\_ACTIONS
- PENDING\_ACTIONS\_OUTLINED
- PENDING\_ACTIONS\_ROUNDED
- PENDING\_ACTIONS\_SHARP
- PENDING\_OUTLINED
- PENDING\_ROUNDED
- PENDING\_SHARP
- PENTAGON
- PENTAGON\_OUTLINED
- PENTAGON\_ROUNDED
- PENTAGON\_SHARP
- PEOPLE
- PEOPLE\_ALT
- PEOPLE\_ALT\_OUTLINED
- PEOPLE\_ALT\_ROUNDED
- PEOPLE\_ALT\_SHARP
- PEOPLE\_OUTLINE
- PEOPLE\_OUTLINE\_OUTLINED
- PEOPLE\_OUTLINE\_ROUNDED
- PEOPLE\_OUTLINE\_SHARP
- PEOPLE\_OUTLINED
- PEOPLE\_ROUNDED
- PEOPLE\_SHARP
- PERCENT
- PERCENT\_OUTLINED
- PERCENT\_ROUNDED
- PERCENT\_SHARP
- PERM\_CAMERA\_MIC
- PERM\_CAMERA\_MIC\_OUTLINED
- PERM\_CAMERA\_MIC\_ROUNDED
- PERM\_CAMERA\_MIC\_SHARP
- PERM\_CONTACT\_CAL
- PERM\_CONTACT\_CAL\_OUTLINED
- PERM\_CONTACT\_CAL\_ROUNDED
- PERM\_CONTACT\_CAL\_SHARP
- PERM\_CONTACT\_CALENDAR
- PERM\_CONTACT\_CALENDAR\_OUTLINED
- PERM\_CONTACT\_CALENDAR\_ROUNDED
- PERM\_CONTACT\_CALENDAR\_SHARP
- PERM\_DATA\_SETTING
- PERM\_DATA\_SETTING\_OUTLINED



- PERM\_DATA\_SETTING\_ROUNDED
- PERM\_DATA\_SETTING\_SHARP
- PERM\_DEVICE\_INFO
- PERM\_DEVICE\_INFO\_OUTLINED
- PERM\_DEVICE\_INFO\_ROUNDED
- PERM\_DEVICE\_INFO\_SHARP
- PERM\_DEVICE\_INFORMATION
- PERM\_DEVICE\_INFORMATION\_OUTLINED
- PERM\_DEVICE\_INFORMATION\_ROUNDED
- PERM\_DEVICE\_INFORMATION\_SHARP
- PERM\_IDENTITY
- PERM\_IDENTITY\_OUTLINED
- PERM\_IDENTITY\_ROUNDED
- PERM\_IDENTITY\_SHARP
- PERM\_MEDIA
- PERM\_MEDIA\_OUTLINED
- PERM\_MEDIA\_ROUNDED
- PERM\_MEDIA\_SHARP
- PERM\_PHONE\_MSG
- PERM\_PHONE\_MSG\_OUTLINED
- PERM\_PHONE\_MSG\_ROUNDED
- PERM\_PHONE\_MSG\_SHARP
- PERM\_SCAN\_WIFI
- PERM\_SCAN\_WIFI\_OUTLINED
- PERM\_SCAN\_WIFI\_ROUNDED
- PERM\_SCAN\_WIFI\_SHARP
- PERSON
- PERSON\_2
- PERSON\_2\_OUTLINED
- PERSON\_2\_ROUNDED
- PERSON\_2\_SHARP
- PERSON\_3
- PERSON\_3\_OUTLINED
- PERSON\_3\_ROUNDED
- PERSON\_3\_SHARP
- PERSON\_4
- PERSON\_4\_OUTLINED
- PERSON\_4\_ROUNDED
- PERSON\_4\_SHARP
- PERSON\_ADD
- PERSON\_ADD\_ALT

- PERSON\_ADD\_ALT\_1
- PERSON\_ADD\_ALT\_1\_OUTLINED
- PERSON\_ADD\_ALT\_1\_ROUNDED
- PERSON\_ADD\_ALT\_1\_SHARP
- PERSON\_ADD\_ALT\_OUTLINED
- PERSON\_ADD\_ALT\_ROUNDED
- PERSON\_ADD\_ALT\_SHARP
- PERSON\_ADD\_DISABLED
- PERSON\_ADD\_DISABLED\_OUTLINED
- PERSON\_ADD\_DISABLED\_ROUNDED
- PERSON\_ADD\_DISABLED\_SHARP
- PERSON\_ADD\_OUTLINED
- PERSON\_ADD\_ROUNDED
- PERSON\_ADD\_SHARP
- PERSON\_OFF
- PERSON\_OFF\_OUTLINED
- PERSON\_OFF\_ROUNDED
- PERSON\_OFF\_SHARP
- PERSON\_OUTLINE
- PERSON\_OUTLINE\_OUTLINED
- PERSON\_OUTLINE\_ROUNDED
- PERSON\_OUTLINE\_SHARP
- PERSON\_OUTLINED
- PERSON\_PIN
- PERSON\_PIN\_CIRCLE
- PERSON\_PIN\_CIRCLE\_OUTLINED
- PERSON\_PIN\_CIRCLE\_ROUNDED
- PERSON\_PIN\_CIRCLE\_SHARP
- PERSON\_PIN\_OUTLINED
- PERSON\_PIN\_ROUNDED
- PERSON\_PIN\_SHARP
- PERSON\_REMOVE
- PERSON\_REMOVE\_ALT\_1
- PERSON\_REMOVE\_ALT\_1\_OUTLINED
- PERSON\_REMOVE\_ALT\_1\_ROUNDED
- PERSON\_REMOVE\_ALT\_1\_SHARP
- PERSON\_REMOVE\_OUTLINED
- PERSON\_REMOVE\_ROUNDED
- PERSON\_REMOVE\_SHARP
- PERSON\_ROUNDED
- PERSON\_SEARCH

- PERSON\_SEARCH\_OUTLINED
- PERSON\_SEARCH\_ROUNDED
- PERSON\_SEARCH\_SHARP
- PERSON\_SHARP
- PERSONAL\_INJURY
- PERSONAL\_INJURY\_OUTLINED
- PERSONAL\_INJURY\_ROUNDED
- PERSONAL\_INJURY\_SHARP
- PERSONAL\_VIDEO
- PERSONAL\_VIDEO\_OUTLINED
- PERSONAL\_VIDEO\_ROUNDED
- PERSONAL\_VIDEO\_SHARP
- PEST\_CONTROL
- PEST\_CONTROL\_OUTLINED
- PEST\_CONTROL\_RODENT
- PEST\_CONTROL\_RODENT\_OUTLINED
- PEST\_CONTROL\_RODENT\_ROUNDED
- PEST\_CONTROL\_RODENT\_SHARP
- PEST\_CONTROL\_ROUNDED
- PEST\_CONTROL\_SHARP
- PETS
- PETS\_OUTLINED
- PETS\_ROUNDED
- PETS\_SHARP
- PHISHING
- PHISHING\_OUTLINED
- PHISHING\_ROUNDED
- PHISHING\_SHARP
- PHONE
- PHONE\_ANDROID
- PHONE\_ANDROID\_OUTLINED
- PHONE\_ANDROID\_ROUNDED
- PHONE\_ANDROID\_SHARP
- PHONE\_BLUETOOTH\_SPEAKER
- PHONE\_BLUETOOTH\_SPEAKER\_OUTLINED
- PHONE\_BLUETOOTH\_SPEAKER\_ROUNDED
- PHONE\_BLUETOOTH\_SPEAKER\_SHARP
- PHONE\_CALLBACK
- PHONE\_CALLBACK\_OUTLINED
- PHONE\_CALLBACK\_ROUNDED
- PHONE\_CALLBACK\_SHARP

- PHONE\_DISABLED
- PHONE\_DISABLED\_OUTLINED
- PHONE\_DISABLED\_ROUNDED
- PHONE\_DISABLED\_SHARP
- PHONE\_ENABLED
- PHONE\_ENABLED\_OUTLINED
- PHONE\_ENABLED\_ROUNDED
- PHONE\_ENABLED\_SHARP
- PHONE\_FORWARDED
- PHONE\_FORWARDED\_OUTLINED
- PHONE\_FORWARDED\_ROUNDED
- PHONE\_FORWARDED\_SHARP
- PHONE\_IN\_TALK
- PHONE\_IN\_TALK\_OUTLINED
- PHONE\_IN\_TALK\_ROUNDED
- PHONE\_IN\_TALK\_SHARP
- PHONE\_IPHONE
- PHONE\_IPHONE\_OUTLINED
- PHONE\_IPHONE\_ROUNDED
- PHONE\_IPHONE\_SHARP
- PHONE\_LOCKED
- PHONE\_LOCKED\_OUTLINED
- PHONE\_LOCKED\_ROUNDED
- PHONE\_LOCKED\_SHARP
- PHONE\_MISSED
- PHONE\_MISSED\_OUTLINED
- PHONE\_MISSED\_ROUNDED
- PHONE\_MISSED\_SHARP
- PHONE\_OUTLINED
- PHONE\_PAUSED
- PHONE\_PAUSED\_OUTLINED
- PHONE\_PAUSED\_ROUNDED
- PHONE\_PAUSED\_SHARP
- PHONE\_ROUNDED
- PHONE\_SHARP
- PHONELINK
- PHONELINK\_ERASE
- PHONELINK\_ERASE\_OUTLINED
- PHONELINK\_ERASE\_ROUNDED
- PHONELINK\_ERASE\_SHARP
- PHONELINK\_LOCK

- PHONELINK\_LOCK\_OUTLINED
- PHONELINK\_LOCK\_ROUNDED
- PHONELINK\_LOCK\_SHARP
- PHONELINK\_OFF
- PHONELINK\_OFF\_OUTLINED
- PHONELINK\_OFF\_ROUNDED
- PHONELINK\_OFF\_SHARP
- PHONELINK\_OUTLINED
- PHONELINK\_RING
- PHONELINK\_RING\_OUTLINED
- PHONELINK\_RING\_ROUNDED
- PHONELINK\_RING\_SHARP
- PHONELINK\_ROUNDED
- PHONELINK\_SETUP
- PHONELINK\_SETUP\_OUTLINED
- PHONELINK\_SETUP\_ROUNDED
- PHONELINK\_SETUP\_SHARP
- PHONELINK\_SHARP
- PHOTO
- PHOTO\_ALBUM
- PHOTO\_ALBUM\_OUTLINED
- PHOTO\_ALBUM\_ROUNDED
- PHOTO\_ALBUM\_SHARP
- PHOTO\_CAMERA
- PHOTO\_CAMERA\_BACK
- PHOTO\_CAMERA\_BACK\_OUTLINED
- PHOTO\_CAMERA\_BACK\_ROUNDED
- PHOTO\_CAMERA\_BACK\_SHARP
- PHOTO\_CAMERA\_FRONT
- PHOTO\_CAMERA\_FRONT\_OUTLINED
- PHOTO\_CAMERA\_FRONT\_ROUNDED
- PHOTO\_CAMERA\_FRONT\_SHARP
- PHOTO\_CAMERA\_OUTLINED
- PHOTO\_CAMERA\_ROUNDED
- PHOTO\_CAMERA\_SHARP
- PHOTO\_FILTER
- PHOTO\_FILTER\_OUTLINED
- PHOTO\_FILTER\_ROUNDED
- PHOTO\_FILTER\_SHARP
- PHOTO\_LIBRARY
- PHOTO\_LIBRARY\_OUTLINED

- PHOTO\_LIBRARY\_ROUNDED
- PHOTO\_LIBRARY\_SHARP
- PHOTO\_OUTLINED
- PHOTO\_ROUNDED
- PHOTO\_SHARP
- PHOTO\_SIZE\_SELECT\_ACTUAL
- PHOTO\_SIZE\_SELECT\_ACTUAL\_OUTLINED
- PHOTO\_SIZE\_SELECT\_ACTUAL\_ROUNDED
- PHOTO\_SIZE\_SELECT\_ACTUAL\_SHARP
- PHOTO\_SIZE\_SELECT\_LARGE
- PHOTO\_SIZE\_SELECT\_LARGE\_OUTLINED
- PHOTO\_SIZE\_SELECT\_LARGE\_ROUNDED
- PHOTO\_SIZE\_SELECT\_LARGE\_SHARP
- PHOTO\_SIZE\_SELECT\_SMALL
- PHOTO\_SIZE\_SELECT\_SMALL\_OUTLINED
- PHOTO\_SIZE\_SELECT\_SMALL\_ROUNDED
- PHOTO\_SIZE\_SELECT\_SMALL\_SHARP
- PHP
- PHP\_OUTLINED
- PHP\_ROUNDED
- PHP\_SHARP
- PIANO
- PIANO\_OFF
- PIANO\_OFF\_OUTLINED
- PIANO\_OFF\_ROUNDED
- PIANO\_OFF\_SHARP
- PIANO\_OUTLINED
- PIANO\_ROUNDED
- PIANO\_SHARP
- PICTURE\_AS\_PDF
- PICTURE\_AS\_PDF\_OUTLINED
- PICTURE\_AS\_PDF\_ROUNDED
- PICTURE\_AS\_PDF\_SHARP
- PICTURE\_IN\_PICTURE
- PICTURE\_IN\_PICTURE\_ALT
- PICTURE\_IN\_PICTURE\_ALT\_OUTLINED
- PICTURE\_IN\_PICTURE\_ALT\_ROUNDED
- PICTURE\_IN\_PICTURE\_ALT\_SHARP
- PICTURE\_IN\_PICTURE\_OUTLINED
- PICTURE\_IN\_PICTURE\_ROUNDED
- PICTURE\_IN\_PICTURE\_SHARP

- PIE\_CHART
- PIE\_CHART\_OUTLINE
- PIE\_CHART\_OUTLINE\_OUTLINED
- PIE\_CHART\_OUTLINE\_ROUNDED
- PIE\_CHART\_OUTLINE\_SHARP
- PIE\_CHART\_ROUNDED
- PIE\_CHART\_SHARP
- PIN
- PIN\_DROP
- PIN\_DROP\_OUTLINED
- PIN\_DROP\_ROUNDED
- PIN\_DROP\_SHARP
- PIN\_END
- PIN\_END\_OUTLINED
- PIN\_END\_ROUNDED
- PIN\_END\_SHARP
- PIN\_INVOKE
- PIN\_INVOKE\_OUTLINED
- PIN\_INVOKE\_ROUNDED
- PIN\_INVOKE\_SHARP
- PIN\_OUTLINED
- PIN\_ROUNDED
- PIN\_SHARP
- PINCH
- PINCH\_OUTLINED
- PINCH\_ROUNDED
- PINCH\_SHARP
- PIVOT\_TABLE\_CHART
- PIVOT\_TABLE\_CHART\_OUTLINED
- PIVOT\_TABLE\_CHART\_ROUNDED
- PIVOT\_TABLE\_CHART\_SHARP
- PIX
- PIX\_OUTLINED
- PIX\_ROUNDED
- PIX\_SHARP
- PLACE
- PLACE\_OUTLINED
- PLACE\_ROUNDED
- PLACE\_SHARP
- PLAGIARISM
- PLAGIARISM\_OUTLINED

- PLAGIARISM\_ROUNDED
- PLAGIARISM\_SHARP
- PLAY\_ARROW
- PLAY\_ARROW\_OUTLINED
- PLAY\_ARROW\_ROUNDED
- PLAY\_ARROW\_SHARP
- PLAY\_CIRCLE
- PLAY\_CIRCLE\_FILL
- PLAY\_CIRCLE\_FILL\_OUTLINED
- PLAY\_CIRCLE\_FILL\_ROUNDED
- PLAY\_CIRCLE\_FILL\_SHARP
- PLAY\_CIRCLE\_FILLED
- PLAY\_CIRCLE\_FILLED\_OUTLINED
- PLAY\_CIRCLE\_FILLED\_ROUNDED
- PLAY\_CIRCLE\_FILLED\_SHARP
- PLAY\_CIRCLE\_OUTLINE
- PLAY\_CIRCLE\_OUTLINE\_OUTLINED
- PLAY\_CIRCLE\_OUTLINE\_ROUNDED
- PLAY\_CIRCLE\_OUTLINE\_SHARP
- PLAY\_CIRCLE\_OUTLINED
- PLAY\_CIRCLE\_ROUNDED
- PLAY\_CIRCLE\_SHARP
- PLAY\_DISABLED
- PLAY\_DISABLED\_OUTLINED
- PLAY\_DISABLED\_ROUNDED
- PLAY\_DISABLED\_SHARP
- PLAY\_FOR\_WORK
- PLAY\_FOR\_WORK\_OUTLINED
- PLAY\_FOR\_WORK\_ROUNDED
- PLAY\_FOR\_WORK\_SHARP
- PLAY\_LESSON
- PLAY\_LESSON\_OUTLINED
- PLAY\_LESSON\_ROUNDED
- PLAY\_LESSON\_SHARP
- PLAYLIST\_ADD
- PLAYLIST\_ADD\_CHECK
- PLAYLIST\_ADD\_CHECK\_CIRCLE
- PLAYLIST\_ADD\_CHECK\_CIRCLE\_OUTLINED
- PLAYLIST\_ADD\_CHECK\_CIRCLE\_ROUNDED
- PLAYLIST\_ADD\_CHECK\_CIRCLE\_SHARP
- PLAYLIST\_ADD\_CHECK\_OUTLINED



- PLAYLIST\_ADD\_CHECK\_ROUNDED
- PLAYLIST\_ADD\_CHECK\_SHARP
- PLAYLIST\_ADD\_CIRCLE
- PLAYLIST\_ADD\_CIRCLE\_OUTLINED
- PLAYLIST\_ADD\_CIRCLE\_ROUNDED
- PLAYLIST\_ADD\_CIRCLE\_SHARP
- PLAYLIST\_ADD\_OUTLINED
- PLAYLIST\_ADD\_ROUNDED
- PLAYLIST\_ADD\_SHARP
- PLAYLIST\_PLAY
- PLAYLIST\_PLAY\_OUTLINED
- PLAYLIST\_PLAY\_ROUNDED
- PLAYLIST\_PLAY\_SHARP
- PLAYLIST\_REMOVE
- PLAYLIST\_REMOVE\_OUTLINED
- PLAYLIST\_REMOVE\_ROUNDED
- PLAYLIST\_REMOVE\_SHARP
- PLUMBING
- PLUMBING\_OUTLINED
- PLUMBING\_ROUNDED
- PLUMBING\_SHARP
- PLUS\_ONE
- PLUS\_ONE\_OUTLINED
- PLUS\_ONE\_ROUNDED
- PLUS\_ONE\_SHARP
- PODCASTS
- PODCASTS\_OUTLINED
- PODCASTS\_ROUNDED
- PODCASTS\_SHARP
- POINT\_OF\_SALE
- POINT\_OF\_SALE\_OUTLINED
- POINT\_OF\_SALE\_ROUNDED
- POINT\_OF\_SALE\_SHARP
- POLICY
- POLICY\_OUTLINED
- POLICY\_ROUNDED
- POLICY\_SHARP
- POLL
- POLL\_OUTLINED
- POLL\_ROUNDED
- POLL\_SHARP

- POLYLINE
- POLYLINE\_OUTLINED
- POLYLINE\_ROUNDED
- POLYLINE\_SHARP
- POLYMER
- POLYMER\_OUTLINED
- POLYMER\_ROUNDED
- POLYMER\_SHARP
- POOL
- POOL\_OUTLINED
- POOL\_ROUNDED
- POOL\_SHARP
- PORTABLE\_WIFI\_OFF
- PORTABLE\_WIFI\_OFF\_OUTLINED
- PORTABLE\_WIFI\_OFF\_ROUNDED
- PORTABLE\_WIFI\_OFF\_SHARP
- PORTRAIT
- PORTRAIT\_OUTLINED
- PORTRAIT\_ROUNDED
- PORTRAIT\_SHARP
- POST\_ADD
- POST\_ADD\_OUTLINED
- POST\_ADD\_ROUNDED
- POST\_ADD\_SHARP
- POWER
- POWER\_INPUT
- POWER\_INPUT\_OUTLINED
- POWER\_INPUT\_ROUNDED
- POWER\_INPUT\_SHARP
- POWER\_OFF
- POWER\_OFF\_OUTLINED
- POWER\_OFF\_ROUNDED
- POWER\_OFF\_SHARP
- POWER\_OUTLINED
- POWER\_ROUNDED
- POWER\_SETTINGS\_NEW
- POWER\_SETTINGS\_NEW\_OUTLINED
- POWER\_SETTINGS\_NEW\_ROUNDED
- POWER\_SETTINGS\_NEW\_SHARP
- POWER\_SHARP
- PRECISION\_MANUFACTURING

- PRECISION\_MANUFACTURING\_OUTLINED
- PRECISION\_MANUFACTURING\_ROUNDED
- PRECISION\_MANUFACTURING\_SHARP
- PREGNANT\_WOMAN
- PREGNANT\_WOMAN\_OUTLINED
- PREGNANT\_WOMAN\_ROUNDED
- PREGNANT\_WOMAN\_SHARP
- PRESENT\_TO\_ALL
- PRESENT\_TO\_ALL\_OUTLINED
- PRESENT\_TO\_ALL\_ROUNDED
- PRESENT\_TO\_ALL\_SHARP
- PREVIEW
- PREVIEW\_OUTLINED
- PREVIEW\_ROUNDED
- PREVIEW\_SHARP
- PRICE\_CHANGE
- PRICE\_CHANGE\_OUTLINED
- PRICE\_CHANGE\_ROUNDED
- PRICE\_CHANGE\_SHARP
- PRICE\_CHECK
- PRICE\_CHECK\_OUTLINED
- PRICE\_CHECK\_ROUNDED
- PRICE\_CHECK\_SHARP
- PRINT
- PRINT\_DISABLED
- PRINT\_DISABLED\_OUTLINED
- PRINT\_DISABLED\_ROUNDED
- PRINT\_DISABLED\_SHARP
- PRINT\_OUTLINED
- PRINT\_ROUNDED
- PRINT\_SHARP
- PRIORITY\_HIGH
- PRIORITY\_HIGH\_OUTLINED
- PRIORITY\_HIGH\_ROUNDED
- PRIORITY\_HIGH\_SHARP
- PRIVACY\_TIP
- PRIVACY\_TIP\_OUTLINED
- PRIVACY\_TIP\_ROUNDED
- PRIVACY\_TIP\_SHARP
- PRIVATE\_CONNECTIVITY
- PRIVATE\_CONNECTIVITY\_OUTLINED

- PRIVATE\_CONNECTIVITY\_ROUNDED
- PRIVATE\_CONNECTIVITY\_SHARP
- PRODUCTION\_QUANTITY\_LIMITS
- PRODUCTION\_QUANTITY\_LIMITS\_OUTLINED
- PRODUCTION\_QUANTITY\_LIMITS\_ROUNDED
- PRODUCTION\_QUANTITY\_LIMITS\_SHARP
- PROPANE
- PROPANE\_OUTLINED
- PROPANE\_ROUNDED
- PROPANE\_SHARP
- PROPANE\_TANK
- PROPANE\_TANK\_OUTLINED
- PROPANE\_TANK\_ROUNDED
- PROPANE\_TANK\_SHARP
- PSYCHOLOGY
- PSYCHOLOGY\_ALT
- PSYCHOLOGY\_ALT\_OUTLINED
- PSYCHOLOGY\_ALT\_ROUNDED
- PSYCHOLOGY\_ALT\_SHARP
- PSYCHOLOGY\_OUTLINED
- PSYCHOLOGY\_ROUNDED
- PSYCHOLOGY\_SHARP
- PUBLIC
- PUBLIC\_OFF
- PUBLIC\_OFF\_OUTLINED
- PUBLIC\_OFF\_ROUNDED
- PUBLIC\_OFF\_SHARP
- PUBLIC\_OUTLINED
- PUBLIC\_ROUNDED
- PUBLIC\_SHARP
- PUBLISH
- PUBLISH\_OUTLINED
- PUBLISH\_ROUNDED
- PUBLISH\_SHARP
- PUBLISHED\_WITH\_CHANGES
- PUBLISHED\_WITH\_CHANGES\_OUTLINED
- PUBLISHED\_WITH\_CHANGES\_ROUNDED
- PUBLISHED\_WITH\_CHANGES\_SHARP
- PUNCH\_CLOCK
- PUNCH\_CLOCK\_OUTLINED
- PUNCH\_CLOCK\_ROUNDED

- PUNCH\_CLOCK\_SHARP
- PUSH\_PIN
- PUSH\_PIN\_OUTLINED
- PUSH\_PIN\_ROUNDED
- PUSH\_PIN\_SHARP
- QR\_CODE
- QR\_CODE\_2
- QR\_CODE\_2\_OUTLINED
- QR\_CODE\_2\_ROUNDED
- QR\_CODE\_2\_SHARP
- QR\_CODE\_OUTLINED
- QR\_CODE\_ROUNDED
- QR\_CODE\_SCANNER
- QR\_CODE\_SCANNER\_OUTLINED
- QR\_CODE\_SCANNER\_ROUNDED
- QR\_CODE\_SCANNER\_SHARP
- QR\_CODE\_SHARP
- QUERY\_BUILDER
- QUERY\_BUILDER\_OUTLINED
- QUERY\_BUILDER\_ROUNDED
- QUERY\_BUILDER\_SHARP
- QUERY\_STATS
- QUERY\_STATS\_OUTLINED
- QUERY\_STATS\_ROUNDED
- QUERY\_STATS\_SHARP
- QUESTION\_ANSWER
- QUESTION\_ANSWER\_OUTLINED
- QUESTION\_ANSWER\_ROUNDED
- QUESTION\_ANSWER\_SHARP
- QUESTION\_MARK
- QUESTION\_MARK\_OUTLINED
- QUESTION\_MARK\_ROUNDED
- QUESTION\_MARK\_SHARP
- QUEUE
- QUEUE\_MUSIC
- QUEUE\_MUSIC\_OUTLINED
- QUEUE\_MUSIC\_ROUNDED
- QUEUE\_MUSIC\_SHARP
- QUEUE\_OUTLINED
- QUEUE\_PLAY\_NEXT
- QUEUE\_PLAY\_NEXT\_OUTLINED

- QUEUE\_PLAY\_NEXT\_ROUNDED
- QUEUE\_PLAY\_NEXT\_SHARP
- QUEUE\_ROUNDED
- QUEUE\_SHARP
- QUICK\_CONTACTS\_DIALER
- QUICK\_CONTACTS\_DIALER\_OUTLINED
- QUICK\_CONTACTS\_DIALER\_ROUNDED
- QUICK\_CONTACTS\_DIALER\_SHARP
- QUICK\_CONTACTS\_MAIL
- QUICK\_CONTACTS\_MAIL\_OUTLINED
- QUICK\_CONTACTS\_MAIL\_ROUNDED
- QUICK\_CONTACTS\_MAIL\_SHARP
- QUICKREPLY
- QUICKREPLY\_OUTLINED
- QUICKREPLY\_ROUNDED
- QUICKREPLY\_SHARP
- QUIZ
- QUIZ\_OUTLINED
- QUIZ\_ROUNDED
- QUIZ\_SHARP
- QUORA
- QUORA\_OUTLINED
- QUORA\_ROUNDED
- QUORA\_SHARP
- R\_MOBILEDATA
- R\_MOBILEDATA\_OUTLINED
- R\_MOBILEDATA\_ROUNDED
- R\_MOBILEDATA\_SHARP
- RADAR
- RADAR\_OUTLINED
- RADAR\_ROUNDED
- RADAR\_SHARP
- RADIO
- RADIO\_BUTTON\_CHECKED
- RADIO\_BUTTON\_CHECKED\_OUTLINED
- RADIO\_BUTTON\_CHECKED\_ROUNDED
- RADIO\_BUTTON\_CHECKED\_SHARP
- RADIO\_BUTTON\_OFF
- RADIO\_BUTTON\_OFF\_OUTLINED
- RADIO\_BUTTON\_OFF\_ROUNDED
- RADIO\_BUTTON\_OFF\_SHARP

- RADIO\_BUTTON\_ON
- RADIO\_BUTTON\_ON\_OUTLINED
- RADIO\_BUTTON\_ON\_ROUNDED
- RADIO\_BUTTON\_ON\_SHARP
- RADIO\_BUTTON\_UNCHECKED
- RADIO\_BUTTON\_UNCHECKED\_OUTLINED
- RADIO\_BUTTON\_UNCHECKED\_ROUNDED
- RADIO\_BUTTON\_UNCHECKED\_SHARP
- RADIO\_OUTLINED
- RADIO\_ROUNDED
- RADIO\_SHARP
- RAILWAY\_ALERT
- RAILWAY\_ALERT\_OUTLINED
- RAILWAY\_ALERT\_ROUNDED
- RAILWAY\_ALERT\_SHARP
- RAMEN\_DINING
- RAMEN\_DINING\_OUTLINED
- RAMEN\_DINING\_ROUNDED
- RAMEN\_DINING\_SHARP
- RAMP\_LEFT
- RAMP\_LEFT\_OUTLINED
- RAMP\_LEFT\_ROUNDED
- RAMP\_LEFT\_SHARP
- RAMP\_RIGHT
- RAMP\_RIGHT\_OUTLINED
- RAMP\_RIGHT\_ROUNDED
- RAMP\_RIGHT\_SHARP
- RATE\_REVIEW
- RATE\_REVIEW\_OUTLINED
- RATE\_REVIEW\_ROUNDED
- RATE\_REVIEW\_SHARP
- RAW\_OFF
- RAW\_OFF\_OUTLINED
- RAW\_OFF\_ROUNDED
- RAW\_OFF\_SHARP
- RAW\_ON
- RAW\_ON\_OUTLINED
- RAW\_ON\_ROUNDED
- RAW\_ON\_SHARP
- READ\_MORE
- READ\_MORE\_OUTLINED

- READ\_MORE\_ROUNDED
- READ\_MORE\_SHARP
- REAL\_ESTATE\_AGENT
- REAL\_ESTATE\_AGENT\_OUTLINED
- REAL\_ESTATE\_AGENT\_ROUNDED
- REAL\_ESTATE\_AGENT\_SHARP
- REBASE\_EDIT
- RECEIPT
- RECEIPT\_LONG
- RECEIPT\_LONG\_OUTLINED
- RECEIPT\_LONG\_ROUNDED
- RECEIPT\_LONG\_SHARP
- RECEIPT\_OUTLINED
- RECEIPT\_ROUNDED
- RECEIPT\_SHARP
- RECENT\_ACTORS
- RECENT\_ACTORS\_OUTLINED
- RECENT\_ACTORS\_ROUNDED
- RECENT\_ACTORS\_SHARP
- RECOMMEND
- RECOMMEND\_OUTLINED
- RECOMMEND\_ROUNDED
- RECOMMEND\_SHARP
- RECORD\_VOICE\_OVER
- RECORD\_VOICE\_OVER\_OUTLINED
- RECORD\_VOICE\_OVER\_ROUNDED
- RECORD\_VOICE\_OVER\_SHARP
- RECTANGLE
- RECTANGLE\_OUTLINED
- RECTANGLE\_ROUNDED
- RECTANGLE\_SHARP
- RECYCLING
- RECYCLING\_OUTLINED
- RECYCLING\_ROUNDED
- RECYCLING\_SHARP
- REDDIT
- REDDIT\_OUTLINED
- REDDIT\_ROUNDED
- REDDIT\_SHARP
- REDEEM
- REDEEM\_OUTLINED



- REDEEM\_ROUNDED
- REDEEM\_SHARP
- REDO
- REDO\_OUTLINED
- REDO\_ROUNDED
- REDO\_SHARP
- REDUCE\_CAPACITY
- REDUCE\_CAPACITY\_OUTLINED
- REDUCE\_CAPACITY\_ROUNDED
- REDUCE\_CAPACITY\_SHARP
- REFRESH
- REFRESH\_OUTLINED
- REFRESH\_ROUNDED
- REFRESH\_SHARP
- REMEMBER\_ME
- REMEMBER\_ME\_OUTLINED
- REMEMBER\_ME\_ROUNDED
- REMEMBER\_ME\_SHARP
- REMOVE
- REMOVE\_CIRCLE
- REMOVE\_CIRCLE\_OUTLINE
- REMOVE\_CIRCLE\_OUTLINE\_OUTLINED
- REMOVE\_CIRCLE\_OUTLINE\_ROUNDED
- REMOVE\_CIRCLE\_OUTLINE\_SHARP
- REMOVE\_CIRCLE\_OUTLINED
- REMOVE\_CIRCLE\_ROUNDED
- REMOVE\_CIRCLE\_SHARP
- REMOVE\_DONE
- REMOVE\_DONE\_OUTLINED
- REMOVE\_DONE\_ROUNDED
- REMOVE\_DONE\_SHARP
- REMOVE\_FROM\_QUEUE
- REMOVE\_FROM\_QUEUE\_OUTLINED
- REMOVE\_FROM\_QUEUE\_ROUNDED
- REMOVE\_FROM\_QUEUE\_SHARP
- REMOVE\_MODERATOR
- REMOVE\_MODERATOR\_OUTLINED
- REMOVE\_MODERATOR\_ROUNDED
- REMOVE\_MODERATOR\_SHARP
- REMOVE\_OUTLINED
- REMOVE\_RED\_EYE

- REMOVE\_RED\_EYE\_OUTLINED
- REMOVE\_RED\_EYE\_ROUNDED
- REMOVE\_RED\_EYE\_SHARP
- REMOVE\_ROAD
- REMOVE\_ROAD\_OUTLINED
- REMOVE\_ROAD\_ROUNDED
- REMOVE\_ROAD\_SHARP
- REMOVE\_ROUNDED
- REMOVE\_SHARP
- REMOVE\_SHOPPING\_CART
- REMOVE\_SHOPPING\_CART\_OUTLINED
- REMOVE\_SHOPPING\_CART\_ROUNDED
- REMOVE\_SHOPPING\_CART\_SHARP
- REORDER
- REORDER\_OUTLINED
- REORDER\_ROUNDED
- REORDER\_SHARP
- REPARTITION
- REPARTITION\_OUTLINED
- REPARTITION\_ROUNDED
- REPARTITION\_SHARP
- REPEAT
- REPEAT\_ON
- REPEAT\_ON\_OUTLINED
- REPEAT\_ON\_ROUNDED
- REPEAT\_ON\_SHARP
- REPEAT\_ONE
- REPEAT\_ONE\_ON
- REPEAT\_ONE\_ON\_OUTLINED
- REPEAT\_ONE\_ON\_ROUNDED
- REPEAT\_ONE\_ON\_SHARP
- REPEAT\_ONE\_OUTLINED
- REPEAT\_ONE\_ROUNDED
- REPEAT\_ONE\_SHARP
- REPEAT\_OUTLINED
- REPEAT\_ROUNDED
- REPEAT\_SHARP
- REPLAY
- REPLAY\_10
- REPLAY\_10\_OUTLINED
- REPLAY\_10\_ROUNDED

- REPLAY\_10\_SHARP
- REPLAY\_30
- REPLAY\_30\_OUTLINED
- REPLAY\_30\_ROUNDED
- REPLAY\_30\_SHARP
- REPLAY\_5
- REPLAY\_5\_OUTLINED
- REPLAY\_5\_ROUNDED
- REPLAY\_5\_SHARP
- REPLAY\_CIRCLE\_FILLED
- REPLAY\_CIRCLE\_FILLED\_OUTLINED
- REPLAY\_CIRCLE\_FILLED\_ROUNDED
- REPLAY\_CIRCLE\_FILLED\_SHARP
- REPLAY\_OUTLINED
- REPLAY\_ROUNDED
- REPLAY\_SHARP
- REPLY
- REPLY\_ALL
- REPLY\_ALL\_OUTLINED
- REPLY\_ALL\_ROUNDED
- REPLY\_ALL\_SHARP
- REPLY\_OUTLINED
- REPLY\_ROUNDED
- REPLY\_SHARP
- REPORT
- REPORT\_GMAILERRORRED
- REPORT\_GMAILERRORRED\_OUTLINED
- REPORT\_GMAILERRORRED\_ROUNDED
- REPORT\_GMAILERRORRED\_SHARP
- REPORT\_OFF
- REPORT\_OFF\_OUTLINED
- REPORT\_OFF\_ROUNDED
- REPORT\_OFF\_SHARP
- REPORT\_OUTLINED
- REPORT\_PROBLEM
- REPORT\_PROBLEM\_OUTLINED
- REPORT\_PROBLEM\_ROUNDED
- REPORT\_PROBLEM\_SHARP
- REPORT\_ROUNDED
- REPORT\_SHARP
- REQUEST\_PAGE

- REQUEST\_PAGE\_OUTLINED
- REQUEST\_PAGE\_ROUNDED
- REQUEST\_PAGE\_SHARP
- REQUEST\_QUOTE
- REQUEST\_QUOTE\_OUTLINED
- REQUEST\_QUOTE\_ROUNDED
- REQUEST\_QUOTE\_SHARP
- RESET\_TV
- RESET\_TV\_OUTLINED
- RESET\_TV\_ROUNDED
- RESET\_TV\_SHARP
- RESTART\_ALT
- RESTART\_ALT\_OUTLINED
- RESTART\_ALT\_ROUNDED
- RESTART\_ALT\_SHARP
- RESTAURANT
- RESTAURANT\_MENU
- RESTAURANT\_MENU\_OUTLINED
- RESTAURANT\_MENU\_ROUNDED
- RESTAURANT\_MENU\_SHARP
- RESTAURANT\_OUTLINED
- RESTAURANT\_ROUNDED
- RESTAURANT\_SHARP
- RESTORE
- RESTORE\_FROM\_TRASH
- RESTORE\_FROM\_TRASH\_OUTLINED
- RESTORE\_FROM\_TRASH\_ROUNDED
- RESTORE\_FROM\_TRASH\_SHARP
- RESTORE\_OUTLINED
- RESTORE\_PAGE
- RESTORE\_PAGE\_OUTLINED
- RESTORE\_PAGE\_ROUNDED
- RESTORE\_PAGE\_SHARP
- RESTORE\_ROUNDED
- RESTORE\_SHARP
- REVIEWS
- REVIEWS\_OUTLINED
- REVIEWS\_ROUNDED
- REVIEWS\_SHARP
- RICE\_BOWL
- RICE\_BOWL\_OUTLINED

- RICE\_BOWL\_ROUNDED
- RICE\_BOWL\_SHARP
- RING\_VOLUME
- RING\_VOLUME\_OUTLINED
- RING\_VOLUME\_ROUNDED
- RING\_VOLUME\_SHARP
- ROCKET
- ROCKET\_LAUNCH
- ROCKET\_LAUNCH\_OUTLINED
- ROCKET\_LAUNCH\_ROUNDED
- ROCKET\_LAUNCH\_SHARP
- ROCKET\_OUTLINED
- ROCKET\_ROUNDED
- ROCKET\_SHARP
- ROLLER\_SHADES
- ROLLER\_SHADES\_CLOSED
- ROLLER\_SHADES\_CLOSED\_OUTLINED
- ROLLER\_SHADES\_CLOSED\_ROUNDED
- ROLLER\_SHADES\_CLOSED\_SHARP
- ROLLER\_SHADES\_OUTLINED
- ROLLER\_SHADES\_ROUNDED
- ROLLER\_SHADES\_SHARP
- ROLLER\_SKATING
- ROLLER\_SKATING\_OUTLINED
- ROLLER\_SKATING\_ROUNDED
- ROLLER\_SKATING\_SHARP
- ROOFING
- ROOFING\_OUTLINED
- ROOFING\_ROUNDED
- ROOFING\_SHARP
- ROOM
- ROOM\_OUTLINED
- ROOM\_PREFERENCES
- ROOM\_PREFERENCES\_OUTLINED
- ROOM\_PREFERENCES\_ROUNDED
- ROOM\_PREFERENCES\_SHARP
- ROOM\_ROUNDED
- ROOM\_SERVICE
- ROOM\_SERVICE\_OUTLINED
- ROOM\_SERVICE\_ROUNDED
- ROOM\_SERVICE\_SHARP

- ROOM\_SHARP
- ROTATE\_90\_DEGREES\_CCW
- ROTATE\_90\_DEGREES\_CCW\_OUTLINED
- ROTATE\_90\_DEGREES\_CCW\_ROUNDED
- ROTATE\_90\_DEGREES\_CCW\_SHARP
- ROTATE\_90\_DEGREES\_CW
- ROTATE\_90\_DEGREES\_CW\_OUTLINED
- ROTATE\_90\_DEGREES\_CW\_ROUNDED
- ROTATE\_90\_DEGREES\_CW\_SHARP
- ROTATE\_LEFT
- ROTATE\_LEFT\_OUTLINED
- ROTATE\_LEFT\_ROUNDED
- ROTATE\_LEFT\_SHARP
- ROTATE\_RIGHT
- ROTATE\_RIGHT\_OUTLINED
- ROTATE\_RIGHT\_ROUNDED
- ROTATE\_RIGHT\_SHARP
- ROUNDABOUT\_LEFT
- ROUNDABOUT\_LEFT\_OUTLINED
- ROUNDABOUT\_LEFT\_ROUNDED
- ROUNDABOUT\_LEFT\_SHARP
- ROUNDABOUT\_RIGHT
- ROUNDABOUT\_RIGHT\_OUTLINED
- ROUNDABOUT\_RIGHT\_ROUNDED
- ROUNDABOUT\_RIGHT\_SHARP
- ROUNDED\_CORNER
- ROUNDED\_CORNER\_OUTLINED
- ROUNDED\_CORNER\_ROUNDED
- ROUNDED\_CORNER\_SHARP
- ROUTE
- ROUTE\_OUTLINED
- ROUTE\_ROUNDED
- ROUTE\_SHARP
- ROUTER
- ROUTER\_OUTLINED
- ROUTER\_ROUNDED
- ROUTER\_SHARP
- ROWING
- ROWING\_OUTLINED
- ROWING\_ROUNDED
- ROWING\_SHARP

- RSS\_FEED
- RSS\_FEED\_OUTLINED
- RSS\_FEED\_ROUNDED
- RSS\_FEED\_SHARP
- RSVP
- RSVP\_OUTLINED
- RSVP\_ROUNDED
- RSVP\_SHARP
- RTT
- RTT\_OUTLINED
- RTT\_ROUNDED
- RTT\_SHARP
- RULE
- RULE\_FOLDER
- RULE\_FOLDER\_OUTLINED
- RULE\_FOLDER\_ROUNDED
- RULE\_FOLDER\_SHARP
- RULE\_OUTLINED
- RULE\_ROUNDED
- RULE\_SHARP
- RUN\_CIRCLE
- RUN\_CIRCLE\_OUTLINED
- RUN\_CIRCLE\_ROUNDED
- RUN\_CIRCLE\_SHARP
- RUNNING\_WITH\_ERRORS
- RUNNING\_WITH\_ERRORS\_OUTLINED
- RUNNING\_WITH\_ERRORS\_ROUNDED
- RUNNING\_WITH\_ERRORS\_SHARP
- RV\_HOOKUP
- RV\_HOOKUP\_OUTLINED
- RV\_HOOKUP\_ROUNDED
- RV\_HOOKUP\_SHARP
- SAFETY\_CHECK
- SAFETY\_CHECK\_OUTLINED
- SAFETY\_CHECK\_ROUNDED
- SAFETY\_CHECK\_SHARP
- SAFETY\_DIVIDER
- SAFETY\_DIVIDER\_OUTLINED
- SAFETY\_DIVIDER\_ROUNDED
- SAFETY\_DIVIDER\_SHARP
- SAILING

- SAILING\_OUTLINED
- SAILING\_ROUNDED
- SAILING\_SHARP
- SANITIZER
- SANITIZER\_OUTLINED
- SANITIZER\_ROUNDED
- SANITIZER\_SHARP
- SATELLITE
- SATELLITE\_ALT
- SATELLITE\_ALT\_OUTLINED
- SATELLITE\_ALT\_ROUNDED
- SATELLITE\_ALT\_SHARP
- SATELLITE\_OUTLINED
- SATELLITE\_ROUNDED
- SATELLITE\_SHARP
- SAVE
- SAVE\_ALT
- SAVE\_ALT\_OUTLINED
- SAVE\_ALT\_ROUNDED
- SAVE\_ALT\_SHARP
- SAVE\_AS
- SAVE\_AS\_OUTLINED
- SAVE\_AS\_ROUNDED
- SAVE\_AS\_SHARP
- SAVE\_OUTLINED
- SAVE\_ROUNDED
- SAVE\_SHARP
- SAVED\_SEARCH
- SAVED\_SEARCH\_OUTLINED
- SAVED\_SEARCH\_ROUNDED
- SAVED\_SEARCH\_SHARP
- SAVINGS
- SAVINGS\_OUTLINED
- SAVINGS\_ROUNDED
- SAVINGS\_SHARP
- SCALE
- SCALE\_OUTLINED
- SCALE\_ROUNDED
- SCALE\_SHARP
- SCANNER
- SCANNER\_OUTLINED



- SCANNER\_ROUNDED
- SCANNER\_SHARP
- SCATTER\_PLOT
- SCATTER\_PLOT\_OUTLINED
- SCATTER\_PLOT\_ROUNDED
- SCATTER\_PLOT\_SHARP
- SCHEDULE
- SCHEDULE\_OUTLINED
- SCHEDULE\_ROUNDED
- SCHEDULE\_SEND
- SCHEDULE\_SEND\_OUTLINED
- SCHEDULE\_SEND\_ROUNDED
- SCHEDULE\_SEND\_SHARP
- SCHEDULE\_SHARP
- SCHEMA
- SCHEMA\_OUTLINED
- SCHEMA\_ROUNDED
- SCHEMA\_SHARP
- SCHOOL
- SCHOOL\_OUTLINED
- SCHOOL\_ROUNDED
- SCHOOL\_SHARP
- SCIENCE
- SCIENCE\_OUTLINED
- SCIENCE\_ROUNDED
- SCIENCE\_SHARP
- SCORE
- SCORE\_OUTLINED
- SCORE\_ROUNDED
- SCORE\_SHARP
- SCOREBOARD
- SCOREBOARD\_OUTLINED
- SCOREBOARD\_ROUNDED
- SCOREBOARD\_SHARP
- SCREEN\_LOCK\_LANDSCAPE
- SCREEN\_LOCK\_LANDSCAPE\_OUTLINED
- SCREEN\_LOCK\_LANDSCAPE\_ROUNDED
- SCREEN\_LOCK\_LANDSCAPE\_SHARP
- SCREEN\_LOCK\_PORTRAIT
- SCREEN\_LOCK\_PORTRAIT\_OUTLINED
- SCREEN\_LOCK\_PORTRAIT\_ROUNDED

- SCREEN\_LOCK\_PORTRAIT\_SHARP
- SCREEN\_LOCK\_ROTATION
- SCREEN\_LOCK\_ROTATION\_OUTLINED
- SCREEN\_LOCK\_ROTATION\_ROUNDED
- SCREEN\_LOCK\_ROTATION\_SHARP
- SCREEN\_ROTATION
- SCREEN\_ROTATION\_ALT
- SCREEN\_ROTATION\_ALT\_OUTLINED
- SCREEN\_ROTATION\_ALT\_ROUNDED
- SCREEN\_ROTATION\_ALT\_SHARP
- SCREEN\_ROTATION\_OUTLINED
- SCREEN\_ROTATION\_ROUNDED
- SCREEN\_ROTATION\_SHARP
- SCREEN\_SEARCH\_DESKTOP
- SCREEN\_SEARCH\_DESKTOP\_OUTLINED
- SCREEN\_SEARCH\_DESKTOP\_ROUNDED
- SCREEN\_SEARCH\_DESKTOP\_SHARP
- SCREEN\_SHARE
- SCREEN\_SHARE\_OUTLINED
- SCREEN\_SHARE\_ROUNDED
- SCREEN\_SHARE\_SHARP
- SCREENSHOT
- SCREENSHOT\_MONITOR
- SCREENSHOT\_MONITOR\_OUTLINED
- SCREENSHOT\_MONITOR\_ROUNDED
- SCREENSHOT\_MONITOR\_SHARP
- SCREENSHOT\_OUTLINED
- SCREENSHOT\_ROUNDED
- SCREENSHOT\_SHARP
- SCUBA\_DIVING
- SCUBA\_DIVING\_OUTLINED
- SCUBA\_DIVING\_ROUNDED
- SCUBA\_DIVING\_SHARP
- SD
- SD\_CARD
- SD\_CARD\_ALERT
- SD\_CARD\_ALERT\_OUTLINED
- SD\_CARD\_ALERT\_ROUNDED
- SD\_CARD\_ALERT\_SHARP
- SD\_CARD\_OUTLINED
- SD\_CARD\_ROUNDED

- SD\_CARD\_SHARP
- SD\_OUTLINED
- SD\_ROUNDED
- SD\_SHARP
- SD\_STORAGE
- SD\_STORAGE\_OUTLINED
- SD\_STORAGE\_ROUNDED
- SD\_STORAGE\_SHARP
- SEARCH
- SEARCH\_OFF
- SEARCH\_OFF\_OUTLINED
- SEARCH\_OFF\_ROUNDED
- SEARCH\_OFF\_SHARP
- SEARCH\_OUTLINED
- SEARCH\_ROUNDED
- SEARCH\_SHARP
- SECURITY
- SECURITY\_OUTLINED
- SECURITY\_ROUNDED
- SECURITY\_SHARP
- SECURITY\_UPDATE
- SECURITY\_UPDATE\_GOOD
- SECURITY\_UPDATE\_GOOD\_OUTLINED
- SECURITY\_UPDATE\_GOOD\_ROUNDED
- SECURITY\_UPDATE\_GOOD\_SHARP
- SECURITY\_UPDATE\_OUTLINED
- SECURITY\_UPDATE\_ROUNDED
- SECURITY\_UPDATE\_SHARP
- SECURITY\_UPDATE\_WARNING
- SECURITY\_UPDATE\_WARNING\_OUTLINED
- SECURITY\_UPDATE\_WARNING\_ROUNDED
- SECURITY\_UPDATE\_WARNING\_SHARP
- SEGMENT
- SEGMENT\_OUTLINED
- SEGMENT\_ROUNDED
- SEGMENT\_SHARP
- SELECT\_ALL
- SELECT\_ALL\_OUTLINED
- SELECT\_ALL\_ROUNDED
- SELECT\_ALL\_SHARP
- SELF\_IMPROVEMENT

- SELF\_IMPROVEMENT\_OUTLINED
- SELF\_IMPROVEMENT\_ROUNDED
- SELF\_IMPROVEMENT\_SHARP
- SELL
- SELL\_OUTLINED
- SELL\_ROUNDED
- SELL\_SHARP
- SEND
- SEND\_AND\_ARCHIVE
- SEND\_AND\_ARCHIVE\_OUTLINED
- SEND\_AND\_ARCHIVE\_ROUNDED
- SEND\_AND\_ARCHIVE\_SHARP
- SEND\_OUTLINED
- SEND\_ROUNDED
- SEND\_SHARP
- SEND\_TIME\_EXTENSION
- SEND\_TIME\_EXTENSION\_OUTLINED
- SEND\_TIME\_EXTENSION\_ROUNDED
- SEND\_TIME\_EXTENSION\_SHARP
- SEND\_TO\_MOBILE
- SEND\_TO\_MOBILE\_OUTLINED
- SEND\_TO\_MOBILE\_ROUNDED
- SEND\_TO\_MOBILE\_SHARP
- SENSOR\_DOOR
- SENSOR\_DOOR\_OUTLINED
- SENSOR\_DOOR\_ROUNDED
- SENSOR\_DOOR\_SHARP
- SENSOR\_OCCUPIED
- SENSOR\_OCCUPIED\_OUTLINED
- SENSOR\_OCCUPIED\_ROUNDED
- SENSOR\_OCCUPIED\_SHARP
- SENSOR\_WINDOW
- SENSOR\_WINDOW\_OUTLINED
- SENSOR\_WINDOW\_ROUNDED
- SENSOR\_WINDOW\_SHARP
- SENSORS
- SENSORS\_OFF
- SENSORS\_OFF\_OUTLINED
- SENSORS\_OFF\_ROUNDED
- SENSORS\_OFF\_SHARP
- SENSORS\_OUTLINED

- SENSORS\_ROUNDED
- SENSORS\_SHARP
- SENTIMENT\_DISSATISFIED
- SENTIMENT\_DISSATISFIED\_OUTLINED
- SENTIMENT\_DISSATISFIED\_ROUNDED
- SENTIMENT\_DISSATISFIED\_SHARP
- SENTIMENT\_NEUTRAL
- SENTIMENT\_NEUTRAL\_OUTLINED
- SENTIMENT\_NEUTRAL\_ROUNDED
- SENTIMENT\_NEUTRAL\_SHARP
- SENTIMENT\_SATISFIED
- SENTIMENT\_SATISFIED\_ALT
- SENTIMENT\_SATISFIED\_ALT\_OUTLINED
- SENTIMENT\_SATISFIED\_ALT\_ROUNDED
- SENTIMENT\_SATISFIED\_ALT\_SHARP
- SENTIMENT\_SATISFIED\_OUTLINED
- SENTIMENT\_SATISFIED\_ROUNDED
- SENTIMENT\_SATISFIED\_SHARP
- SENTIMENT\_VERY\_DISSATISFIED
- SENTIMENT\_VERY\_DISSATISFIED\_OUTLINED
- SENTIMENT\_VERY\_DISSATISFIED\_ROUNDED
- SENTIMENT\_VERY\_DISSATISFIED\_SHARP
- SENTIMENT\_VERY\_SATISFIED
- SENTIMENT\_VERY\_SATISFIED\_OUTLINED
- SENTIMENT\_VERY\_SATISFIED\_ROUNDED
- SENTIMENT\_VERY\_SATISFIED\_SHARP
- SET\_MEAL
- SET\_MEAL\_OUTLINED
- SET\_MEAL\_ROUNDED
- SET\_MEAL\_SHARP
- SETTINGS
- SETTINGS\_ACCESSIBILITY
- SETTINGS\_ACCESSIBILITY\_OUTLINED
- SETTINGS\_ACCESSIBILITY\_ROUNDED
- SETTINGS\_ACCESSIBILITY\_SHARP
- SETTINGS\_APPLICATIONS
- SETTINGS\_APPLICATIONS\_OUTLINED
- SETTINGS\_APPLICATIONS\_ROUNDED
- SETTINGS\_APPLICATIONS\_SHARP
- SETTINGS\_BACKUP\_RESTORE
- SETTINGS\_BACKUP\_RESTORE\_OUTLINED

- SETTINGS\_BACKUP\_RESTORE\_ROUNDED
- SETTINGS\_BACKUP\_RESTORE\_SHARP
- SETTINGS\_BLUETOOTH
- SETTINGS\_BLUETOOTH\_OUTLINED
- SETTINGS\_BLUETOOTH\_ROUNDED
- SETTINGS\_BLUETOOTH\_SHARP
- SETTINGS\_BRIGHTNESS
- SETTINGS\_BRIGHTNESS\_OUTLINED
- SETTINGS\_BRIGHTNESS\_ROUNDED
- SETTINGS\_BRIGHTNESS\_SHARP
- SETTINGS\_CELL
- SETTINGS\_CELL\_OUTLINED
- SETTINGS\_CELL\_ROUNDED
- SETTINGS\_CELL\_SHARP
- SETTINGS\_DISPLAY
- SETTINGS\_DISPLAY\_OUTLINED
- SETTINGS\_DISPLAY\_ROUNDED
- SETTINGS\_DISPLAY\_SHARP
- SETTINGS\_ETHERNET
- SETTINGS\_ETHERNET\_OUTLINED
- SETTINGS\_ETHERNET\_ROUNDED
- SETTINGS\_ETHERNET\_SHARP
- SETTINGS\_INPUT\_ANTENNA
- SETTINGS\_INPUT\_ANTENNA\_OUTLINED
- SETTINGS\_INPUT\_ANTENNA\_ROUNDED
- SETTINGS\_INPUT\_ANTENNA\_SHARP
- SETTINGS\_INPUT\_COMPONENT
- SETTINGS\_INPUT\_COMPONENT\_OUTLINED
- SETTINGS\_INPUT\_COMPONENT\_ROUNDED
- SETTINGS\_INPUT\_COMPONENT\_SHARP
- SETTINGS\_INPUT\_COMPOSITE
- SETTINGS\_INPUT\_COMPOSITE\_OUTLINED
- SETTINGS\_INPUT\_COMPOSITE\_ROUNDED
- SETTINGS\_INPUT\_COMPOSITE\_SHARP
- SETTINGS\_INPUT\_HDMI
- SETTINGS\_INPUT\_HDMI\_OUTLINED
- SETTINGS\_INPUT\_HDMI\_ROUNDED
- SETTINGS\_INPUT\_HDMI\_SHARP
- SETTINGS\_INPUT\_SVIDEO
- SETTINGS\_INPUT\_SVIDEO\_OUTLINED
- SETTINGS\_INPUT\_SVIDEO\_ROUNDED

- SETTINGS\_INPUT\_SVIDEO\_SHARP
- SETTINGS\_OUTLINED
- SETTINGS\_OVERSCAN
- SETTINGS\_OVERSCAN\_OUTLINED
- SETTINGS\_OVERSCAN\_ROUNDED
- SETTINGS\_OVERSCAN\_SHARP
- SETTINGS\_PHONE
- SETTINGS\_PHONE\_OUTLINED
- SETTINGS\_PHONE\_ROUNDED
- SETTINGS\_PHONE\_SHARP
- SETTINGS\_POWER
- SETTINGS\_POWER\_OUTLINED
- SETTINGS\_POWER\_ROUNDED
- SETTINGS\_POWER\_SHARP
- SETTINGS\_REMOTE
- SETTINGS\_REMOTE\_OUTLINED
- SETTINGS\_REMOTE\_ROUNDED
- SETTINGS\_REMOTE\_SHARP
- SETTINGS\_ROUNDED
- SETTINGS\_SHARP
- SETTINGS\_SUGGEST
- SETTINGS\_SUGGEST\_OUTLINED
- SETTINGS\_SUGGEST\_ROUNDED
- SETTINGS\_SUGGEST\_SHARP
- SETTINGS\_SYSTEM\_DAYDREAM
- SETTINGS\_SYSTEM\_DAYDREAM\_OUTLINED
- SETTINGS\_SYSTEM\_DAYDREAM\_ROUNDED
- SETTINGS\_SYSTEM\_DAYDREAM\_SHARP
- SETTINGS\_VOICE
- SETTINGS\_VOICE\_OUTLINED
- SETTINGS\_VOICE\_ROUNDED
- SETTINGS\_VOICE\_SHARP
- SEVEN\_K
- SEVEN\_K\_OUTLINED
- SEVEN\_K\_PLUS
- SEVEN\_K\_PLUS\_OUTLINED
- SEVEN\_K\_PLUS\_ROUNDED
- SEVEN\_K\_PLUS\_SHARP
- SEVEN\_K\_ROUNDED
- SEVEN\_K\_SHARP
- SEVEN\_MP

- SEVEN\_MP\_OUTLINED
- SEVEN\_MP\_ROUNDED
- SEVEN\_MP\_SHARP
- SEVENTEEN\_MP
- SEVENTEEN\_MP\_OUTLINED
- SEVENTEEN\_MP\_ROUNDED
- SEVENTEEN\_MP\_SHARP
- SEVERE\_COLD
- SEVERE\_COLD\_OUTLINED
- SEVERE\_COLD\_ROUNDED
- SEVERE\_COLD\_SHARP
- SHAPE\_LINE
- SHAPE\_LINE\_OUTLINED
- SHAPE\_LINE\_ROUNDED
- SHAPE\_LINE\_SHARP
- SHARE
- SHARE\_ARRIVAL\_TIME
- SHARE\_ARRIVAL\_TIME\_OUTLINED
- SHARE\_ARRIVAL\_TIME\_ROUNDED
- SHARE\_ARRIVAL\_TIME\_SHARP
- SHARE\_LOCATION
- SHARE\_LOCATION\_OUTLINED
- SHARE\_LOCATION\_ROUNDED
- SHARE\_LOCATION\_SHARP
- SHARE\_OUTLINED
- SHARE\_ROUNDED
- SHARE\_SHARP
- SHELVES
- SHIELD
- SHIELD\_MOON
- SHIELD\_MOON\_OUTLINED
- SHIELD\_MOON\_ROUNDED
- SHIELD\_MOON\_SHARP
- SHIELD\_OUTLINED
- SHIELD\_ROUNDED
- SHIELD\_SHARP
- SHOP
- SHOP\_2
- SHOP\_2\_OUTLINED
- SHOP\_2\_ROUNDED
- SHOP\_2\_SHARP



- SHOP\_OUTLINED
- SHOP\_ROUNDED
- SHOP\_SHARP
- SHOP\_TWO
- SHOP\_TWO\_OUTLINED
- SHOP\_TWO\_ROUNDED
- SHOP\_TWO\_SHARP
- SHOIFY
- SHOIFY\_OUTLINED
- SHOIFY\_ROUNDED
- SHOIFY\_SHARP
- SHOPPING\_BAG
- SHOPPING\_BAG\_OUTLINED
- SHOPPING\_BAG\_ROUNDED
- SHOPPING\_BAG\_SHARP
- SHOPPING\_BASKET
- SHOPPING\_BASKET\_OUTLINED
- SHOPPING\_BASKET\_ROUNDED
- SHOPPING\_BASKET\_SHARP
- SHOPPING\_CART
- SHOPPING\_CART\_CHECKOUT
- SHOPPING\_CART\_CHECKOUT\_OUTLINED
- SHOPPING\_CART\_CHECKOUT\_ROUNDED
- SHOPPING\_CART\_CHECKOUT\_SHARP
- SHOPPING\_CART\_OUTLINED
- SHOPPING\_CART\_ROUNDED
- SHOPPING\_CART\_SHARP
- SHORT\_TEXT
- SHORT\_TEXT\_OUTLINED
- SHORT\_TEXT\_ROUNDED
- SHORT\_TEXT\_SHARP
- SHORTCUT
- SHORTCUT\_OUTLINED
- SHORTCUT\_ROUNDED
- SHORTCUT\_SHARP
- SHOW\_CHART
- SHOW\_CHART\_OUTLINED
- SHOW\_CHART\_ROUNDED
- SHOW\_CHART\_SHARP
- SHOWER
- SHOWER\_OUTLINED

- SHOWER\_ROUNDED
- SHOWER\_SHARP
- SHUFFLE
- SHUFFLE\_ON
- SHUFFLE\_ON\_OUTLINED
- SHUFFLE\_ON\_ROUNDED
- SHUFFLE\_ON\_SHARP
- SHUFFLE\_OUTLINED
- SHUFFLE\_ROUNDED
- SHUFFLE\_SHARP
- SHUTTER\_SPEED
- SHUTTER\_SPEED\_OUTLINED
- SHUTTER\_SPEED\_ROUNDED
- SHUTTER\_SPEED\_SHARP
- SICK
- SICK\_OUTLINED
- SICK\_ROUNDED
- SICK\_SHARP
- SIGN\_LANGUAGE
- SIGN\_LANGUAGE\_OUTLINED
- SIGN\_LANGUAGE\_ROUNDED
- SIGN\_LANGUAGE\_SHARP
- SIGNAL\_CELLULAR\_0\_BAR
- SIGNAL\_CELLULAR\_0\_BAR\_OUTLINED
- SIGNAL\_CELLULAR\_0\_BAR\_ROUNDED
- SIGNAL\_CELLULAR\_0\_BAR\_SHARP
- SIGNAL\_CELLULAR\_4\_BAR
- SIGNAL\_CELLULAR\_4\_BAR\_OUTLINED
- SIGNAL\_CELLULAR\_4\_BAR\_ROUNDED
- SIGNAL\_CELLULAR\_4\_BAR\_SHARP
- SIGNAL\_CELLULAR\_ALT
- SIGNAL\_CELLULAR\_ALT\_1\_BAR
- SIGNAL\_CELLULAR\_ALT\_1\_BAR\_OUTLINED
- SIGNAL\_CELLULAR\_ALT\_1\_BAR\_ROUNDED
- SIGNAL\_CELLULAR\_ALT\_1\_BAR\_SHARP
- SIGNAL\_CELLULAR\_ALT\_2\_BAR
- SIGNAL\_CELLULAR\_ALT\_2\_BAR\_OUTLINED
- SIGNAL\_CELLULAR\_ALT\_2\_BAR\_ROUNDED
- SIGNAL\_CELLULAR\_ALT\_2\_BAR\_SHARP
- SIGNAL\_CELLULAR\_ALT\_OUTLINED
- SIGNAL\_CELLULAR\_ALT\_ROUNDED

- SIGNAL\_CELLULAR\_ALT\_SHARP
- SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR
- SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_OUTLINED
- SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_ROUNDED
- SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_SHARP
- SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR
- SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_OUTLINED
- SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_ROUNDED
- SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_SHARP
- SIGNAL\_CELLULAR\_NO\_SIM
- SIGNAL\_CELLULAR\_NO\_SIM\_OUTLINED
- SIGNAL\_CELLULAR\_NO\_SIM\_ROUNDED
- SIGNAL\_CELLULAR\_NO\_SIM\_SHARP
- SIGNAL\_CELLULAR\_NODATA
- SIGNAL\_CELLULAR\_NODATA\_OUTLINED
- SIGNAL\_CELLULAR\_NODATA\_ROUNDED
- SIGNAL\_CELLULAR\_NODATA\_SHARP
- SIGNAL\_CELLULAR\_NULL
- SIGNAL\_CELLULAR\_NULL\_OUTLINED
- SIGNAL\_CELLULAR\_NULL\_ROUNDED
- SIGNAL\_CELLULAR\_NULL\_SHARP
- SIGNAL\_CELLULAR\_OFF
- SIGNAL\_CELLULAR\_OFF\_OUTLINED
- SIGNAL\_CELLULAR\_OFF\_ROUNDED
- SIGNAL\_CELLULAR\_OFF\_SHARP
- SIGNAL\_WIFI\_0\_BAR
- SIGNAL\_WIFI\_0\_BAR\_OUTLINED
- SIGNAL\_WIFI\_0\_BAR\_ROUNDED
- SIGNAL\_WIFI\_0\_BAR\_SHARP
- SIGNAL\_WIFI\_4\_BAR
- SIGNAL\_WIFI\_4\_BAR\_LOCK
- SIGNAL\_WIFI\_4\_BAR\_LOCK\_OUTLINED
- SIGNAL\_WIFI\_4\_BAR\_LOCK\_ROUNDED
- SIGNAL\_WIFI\_4\_BAR\_LOCK\_SHARP
- SIGNAL\_WIFI\_4\_BAR\_OUTLINED
- SIGNAL\_WIFI\_4\_BAR\_ROUNDED
- SIGNAL\_WIFI\_4\_BAR\_SHARP
- SIGNAL\_WIFI\_BAD
- SIGNAL\_WIFI\_BAD\_OUTLINED
- SIGNAL\_WIFI\_BAD\_ROUNDED
- SIGNAL\_WIFI\_BAD\_SHARP

- SIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4
- SIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_OUTLINED
- SIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_ROUNDED
- SIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_SHARP
- SIGNAL\_WIFI\_OFF
- SIGNAL\_WIFI\_OFF\_OUTLINED
- SIGNAL\_WIFI\_OFF\_ROUNDED
- SIGNAL\_WIFI\_OFF\_SHARP
- SIGNAL\_WIFI\_STATUSBAR\_4\_BAR
- SIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_OUTLINED
- SIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_ROUNDED
- SIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_SHARP
- SIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4
- SIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_OUTLINED
- SIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_ROUNDED
- SIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_SHARP
- SIGNAL\_WIFI\_STATUSBAR\_NULL
- SIGNAL\_WIFI\_STATUSBAR\_NULL\_OUTLINED
- SIGNAL\_WIFI\_STATUSBAR\_NULL\_ROUNDED
- SIGNAL\_WIFI\_STATUSBAR\_NULL\_SHARP
- SIGNPOST
- SIGNPOST\_OUTLINED
- SIGNPOST\_ROUNDED
- SIGNPOST\_SHARP
- SIM\_CARD
- SIM\_CARD\_ALERT
- SIM\_CARD\_ALERT\_OUTLINED
- SIM\_CARD\_ALERT\_ROUNDED
- SIM\_CARD\_ALERT\_SHARP
- SIM\_CARD\_DOWNLOAD
- SIM\_CARD\_DOWNLOAD\_OUTLINED
- SIM\_CARD\_DOWNLOAD\_ROUNDED
- SIM\_CARD\_DOWNLOAD\_SHARP
- SIM\_CARD\_OUTLINED
- SIM\_CARD\_ROUNDED
- SIM\_CARD\_SHARP
- SINGLE\_BED
- SINGLE\_BED\_OUTLINED
- SINGLE\_BED\_ROUNDED
- SINGLE\_BED\_SHARP
- SIP

- SIP\_OUTLINED
- SIP\_ROUNDED
- SIP\_SHARP
- SIX\_FT\_APART
- SIX\_FT\_APART\_OUTLINED
- SIX\_FT\_APART\_ROUNDED
- SIX\_FT\_APART\_SHARP
- SIX\_K
- SIX\_K\_OUTLINED
- SIX\_K\_PLUS
- SIX\_K\_PLUS\_OUTLINED
- SIX\_K\_PLUS\_ROUNDED
- SIX\_K\_PLUS\_SHARP
- SIX\_K\_ROUNDED
- SIX\_K\_SHARP
- SIX\_MP
- SIX\_MP\_OUTLINED
- SIX\_MP\_ROUNDED
- SIX\_MP\_SHARP
- SIXTEEN\_MP
- SIXTEEN\_MP\_OUTLINED
- SIXTEEN\_MP\_ROUNDED
- SIXTEEN\_MP\_SHARP
- SIXTY\_FPS
- SIXTY\_FPS\_OUTLINED
- SIXTY\_FPS\_ROUNDED
- SIXTY\_FPS\_SELECT
- SIXTY\_FPS\_SELECT\_OUTLINED
- SIXTY\_FPS\_SELECT\_ROUNDED
- SIXTY\_FPS\_SELECT\_SHARP
- SIXTY\_FPS\_SHARP
- SKATEBOARDING
- SKATEBOARDING\_OUTLINED
- SKATEBOARDING\_ROUNDED
- SKATEBOARDING\_SHARP
- SKIP\_NEXT
- SKIP\_NEXT\_OUTLINED
- SKIP\_NEXT\_ROUNDED
- SKIP\_NEXT\_SHARP
- SKIP\_PREVIOUS
- SKIP\_PREVIOUS\_OUTLINED

- SKIP\_PREVIOUS\_ROUNDED
- SKIP\_PREVIOUS\_SHARP
- SLEDDING
- SLEDDING\_OUTLINED
- SLEDDING\_ROUNDED
- SLEDDING\_SHARP
- SLIDESHOW
- SLIDESHOW\_OUTLINED
- SLIDESHOW\_ROUNDED
- SLIDESHOW\_SHARP
- SLOW\_MOTION\_VIDEO
- SLOW\_MOTION\_VIDEO\_OUTLINED
- SLOW\_MOTION\_VIDEO\_ROUNDED
- SLOW\_MOTION\_VIDEO\_SHARP
- SMART\_BUTTON
- SMART\_BUTTON\_OUTLINED
- SMART\_BUTTON\_ROUNDED
- SMART\_BUTTON\_SHARP
- SMART\_DISPLAY
- SMART\_DISPLAY\_OUTLINED
- SMART\_DISPLAY\_ROUNDED
- SMART\_DISPLAY\_SHARP
- SMART\_SCREEN
- SMART\_SCREEN\_OUTLINED
- SMART\_SCREEN\_ROUNDED
- SMART\_SCREEN\_SHARP
- SMART\_TOY
- SMART\_TOY\_OUTLINED
- SMART\_TOY\_ROUNDED
- SMART\_TOY\_SHARP
- SMARTPHONE
- SMARTPHONE\_OUTLINED
- SMARTPHONE\_ROUNDED
- SMARTPHONE\_SHARP
- SMOKE\_FREE
- SMOKE\_FREE\_OUTLINED
- SMOKE\_FREE\_ROUNDED
- SMOKE\_FREE\_SHARP
- SMOKING\_ROOMS
- SMOKING\_ROOMS\_OUTLINED
- SMOKING\_ROOMS\_ROUNDED

- SMOKING\_ROOMS\_SHARP
- SMS
- SMS\_FAILED
- SMS\_FAILED\_OUTLINED
- SMS\_FAILED\_ROUNDED
- SMS\_FAILED\_SHARP
- SMS\_OUTLINED
- SMS\_ROUNDED
- SMS\_SHARP
- SNAPCHAT
- SNAPCHAT\_OUTLINED
- SNAPCHAT\_ROUNDED
- SNAPCHAT\_SHARP
- SNIPPET\_FOLDER
- SNIPPET\_FOLDER\_OUTLINED
- SNIPPET\_FOLDER\_ROUNDED
- SNIPPET\_FOLDER\_SHARP
- SNOOZE
- SNOOZE\_OUTLINED
- SNOOZE\_ROUNDED
- SNOOZE\_SHARP
- SNOWBOARDING
- SNOWBOARDING\_OUTLINED
- SNOWBOARDING\_ROUNDED
- SNOWBOARDING\_SHARP
- SNOWING
- SNOWMOBILE
- SNOWMOBILE\_OUTLINED
- SNOWMOBILE\_ROUNDED
- SNOWMOBILE\_SHARP
- SNOWSHOEING
- SNOWSHOEING\_OUTLINED
- SNOWSHOEING\_ROUNDED
- SNOWSHOEING\_SHARP
- SOAP
- SOAP\_OUTLINED
- SOAP\_ROUNDED
- SOAP\_SHARP
- SOCIAL\_DISTANCE
- SOCIAL\_DISTANCE\_OUTLINED
- SOCIAL\_DISTANCE\_ROUNDED

- SOCIAL\_DISTANCE\_SHARP
- SOLAR\_POWER
- SOLAR\_POWER\_OUTLINED
- SOLAR\_POWER\_ROUNDED
- SOLAR\_POWER\_SHARP
- SORT
- SORT\_BY\_ALPHA
- SORT\_BY\_ALPHA\_OUTLINED
- SORT\_BY\_ALPHA\_ROUNDED
- SORT\_BY\_ALPHA\_SHARP
- SORT\_OUTLINED
- SORT\_ROUNDED
- SORT\_SHARP
- SOS
- SOS\_OUTLINED
- SOS\_ROUNDED
- SOS\_SHARP
- SOUP\_KITCHEN
- SOUP\_KITCHEN\_OUTLINED
- SOUP\_KITCHEN\_ROUNDED
- SOUP\_KITCHEN\_SHARP
- SOURCE
- SOURCE\_OUTLINED
- SOURCE\_ROUNDED
- SOURCE\_SHARP
- SOUTH
- SOUTH\_AMERICA
- SOUTH\_AMERICA\_OUTLINED
- SOUTH\_AMERICA\_ROUNDED
- SOUTH\_AMERICA\_SHARP
- SOUTH\_EAST
- SOUTH\_EAST\_OUTLINED
- SOUTH\_EAST\_ROUNDED
- SOUTH\_EAST\_SHARP
- SOUTH\_OUTLINED
- SOUTH\_ROUNDED
- SOUTH\_SHARP
- SOUTH\_WEST
- SOUTH\_WEST\_OUTLINED
- SOUTH\_WEST\_ROUNDED
- SOUTH\_WEST\_SHARP



- SPA
- SPA\_OUTLINED
- SPA\_ROUNDED
- SPA\_SHARP
- SPACE\_BAR
- SPACE\_BAR\_OUTLINED
- SPACE\_BAR\_ROUNDED
- SPACE\_BAR\_SHARP
- SPACE\_DASHBOARD
- SPACE\_DASHBOARD\_OUTLINED
- SPACE\_DASHBOARD\_ROUNDED
- SPACE\_DASHBOARD\_SHARP
- SPATIAL\_AUDIO
- SPATIAL\_AUDIO\_OFF
- SPATIAL\_AUDIO\_OFF\_OUTLINED
- SPATIAL\_AUDIO\_OFF\_ROUNDED
- SPATIAL\_AUDIO\_OFF\_SHARP
- SPATIAL\_AUDIO\_OUTLINED
- SPATIAL\_AUDIO\_ROUNDED
- SPATIAL\_AUDIO\_SHARP
- SPATIAL\_TRACKING
- SPATIAL\_TRACKING\_OUTLINED
- SPATIAL\_TRACKING\_ROUNDED
- SPATIAL\_TRACKING\_SHARP
- SPEAKER
- SPEAKER\_GROUP
- SPEAKER\_GROUP\_OUTLINED
- SPEAKER\_GROUP\_ROUNDED
- SPEAKER\_GROUP\_SHARP
- SPEAKER\_NOTES
- SPEAKER\_NOTES\_OFF
- SPEAKER\_NOTES\_OFF\_OUTLINED
- SPEAKER\_NOTES\_OFF\_ROUNDED
- SPEAKER\_NOTES\_OFF\_SHARP
- SPEAKER\_NOTES\_OUTLINED
- SPEAKER\_NOTES\_ROUNDED
- SPEAKER\_NOTES\_SHARP
- SPEAKER\_OUTLINED
- SPEAKER\_PHONE
- SPEAKER\_PHONE\_OUTLINED
- SPEAKER\_PHONE\_ROUNDED

- SPEAKER\_PHONE\_SHARP
- SPEAKER\_ROUNDED
- SPEAKER\_SHARP
- SPEED
- SPEED\_OUTLINED
- SPEED\_ROUNDED
- SPEED\_SHARP
- SPELLCHECK
- SPELLCHECK\_OUTLINED
- SPELLCHECK\_ROUNDED
- SPELLCHECK\_SHARP
- SPLITSCREEN
- SPLITSCREEN\_OUTLINED
- SPLITSCREEN\_ROUNDED
- SPLITSCREEN\_SHARP
- SPOKE
- SPOKE\_OUTLINED
- SPOKE\_ROUNDED
- SPOKE\_SHARP
- SPORTS
- SPORTS\_BAR
- SPORTS\_BAR\_OUTLINED
- SPORTS\_BAR\_ROUNDED
- SPORTS\_BAR\_SHARP
- SPORTS\_BASEBALL
- SPORTS\_BASEBALL\_OUTLINED
- SPORTS\_BASEBALL\_ROUNDED
- SPORTS\_BASEBALL\_SHARP
- SPORTS\_BASKETBALL
- SPORTS\_BASKETBALL\_OUTLINED
- SPORTS\_BASKETBALL\_ROUNDED
- SPORTS\_BASKETBALL\_SHARP
- SPORTS\_CRICKET
- SPORTS\_CRICKET\_OUTLINED
- SPORTS\_CRICKET\_ROUNDED
- SPORTS\_CRICKET\_SHARP
- SPORTS\_ESPORTS
- SPORTS\_ESPORTS\_OUTLINED
- SPORTS\_ESPORTS\_ROUNDED
- SPORTS\_ESPORTS\_SHARP
- SPORTS\_FOOTBALL

- SPORTS\_FOOTBALL\_OUTLINED
- SPORTS\_FOOTBALL\_ROUNDED
- SPORTS\_FOOTBALL\_SHARP
- SPORTS\_GOLF
- SPORTS\_GOLF\_OUTLINED
- SPORTS\_GOLF\_ROUNDED
- SPORTS\_GOLF\_SHARP
- SPORTS\_GYMNASTICS
- SPORTS\_GYMNASTICS\_OUTLINED
- SPORTS\_GYMNASTICS\_ROUNDED
- SPORTS\_GYMNASTICS\_SHARP
- SPORTS\_HANDBALL
- SPORTS\_HANDBALL\_OUTLINED
- SPORTS\_HANDBALL\_ROUNDED
- SPORTS\_HANDBALL\_SHARP
- SPORTS\_HOCKEY
- SPORTS\_HOCKEY\_OUTLINED
- SPORTS\_HOCKEY\_ROUNDED
- SPORTS\_HOCKEY\_SHARP
- SPORTS\_KABADDI
- SPORTS\_KABADDI\_OUTLINED
- SPORTS\_KABADDI\_ROUNDED
- SPORTS\_KABADDI\_SHARP
- SPORTS\_MARTIAL\_ARTS
- SPORTS\_MARTIAL\_ARTS\_OUTLINED
- SPORTS\_MARTIAL\_ARTS\_ROUNDED
- SPORTS\_MARTIAL\_ARTS\_SHARP
- SPORTS\_MMA
- SPORTS\_MMA\_OUTLINED
- SPORTS\_MMA\_ROUNDED
- SPORTS\_MMA\_SHARP
- SPORTS\_MOTORSPORTS
- SPORTS\_MOTORSPORTS\_OUTLINED
- SPORTS\_MOTORSPORTS\_ROUNDED
- SPORTS\_MOTORSPORTS\_SHARP
- SPORTS\_OUTLINED
- SPORTS\_ROUNDED
- SPORTS\_RUGBY
- SPORTS\_RUGBY\_OUTLINED
- SPORTS\_RUGBY\_ROUNDED
- SPORTS\_RUGBY\_SHARP

- SPORTS\_SCORE
- SPORTS\_SCORE\_OUTLINED
- SPORTS\_SCORE\_ROUNDED
- SPORTS\_SCORE\_SHARP
- SPORTS\_SHARP
- SPORTS\_SOCCER
- SPORTS\_SOCCER\_OUTLINED
- SPORTS\_SOCCER\_ROUNDED
- SPORTS\_SOCCER\_SHARP
- SPORTS\_TENNIS
- SPORTS\_TENNIS\_OUTLINED
- SPORTS\_TENNIS\_ROUNDED
- SPORTS\_TENNIS\_SHARP
- SPORTS\_VOLLEYBALL
- SPORTS\_VOLLEYBALL\_OUTLINED
- SPORTS\_VOLLEYBALL\_ROUNDED
- SPORTS\_VOLLEYBALL\_SHARP
- SQUARE
- SQUARE\_FOOT
- SQUARE\_FOOT\_OUTLINED
- SQUARE\_FOOT\_ROUNDED
- SQUARE\_FOOT\_SHARP
- SQUARE\_OUTLINED
- SQUARE\_ROUNDED
- SQUARE\_SHARP
- SSID\_CHART
- SSID\_CHART\_OUTLINED
- SSID\_CHART\_ROUNDED
- SSID\_CHART\_SHARP
- STACKED\_BAR\_CHART
- STACKED\_BAR\_CHART\_OUTLINED
- STACKED\_BAR\_CHART\_ROUNDED
- STACKED\_BAR\_CHART\_SHARP
- STACKED\_LINE\_CHART
- STACKED\_LINE\_CHART\_OUTLINED
- STACKED\_LINE\_CHART\_ROUNDED
- STACKED\_LINE\_CHART\_SHARP
- STADIUM
- STADIUM\_OUTLINED
- STADIUM\_ROUNDED
- STADIUM\_SHARP

- STAIRS
- STAIRS\_OUTLINED
- STAIRS\_ROUNDED
- STAIRS\_SHARP
- STAR
- STAR\_BORDER
- STAR\_BORDER\_OUTLINED
- STAR\_BORDER\_PURPLE500
- STAR\_BORDER\_PURPLE500\_OUTLINED
- STAR\_BORDER\_PURPLE500\_ROUNDED
- STAR\_BORDER\_PURPLE500\_SHARP
- STAR\_BORDER\_ROUNDED
- STAR\_BORDER\_SHARP
- STAR\_HALF
- STAR\_HALF\_OUTLINED
- STAR\_HALF\_ROUNDED
- STAR\_HALF\_SHARP
- STAR\_OUTLINE
- STAR\_OUTLINE\_OUTLINED
- STAR\_OUTLINE\_ROUNDED
- STAR\_OUTLINE\_SHARP
- STAR\_OUTLINED
- STAR\_PURPLE500
- STAR\_PURPLE500\_OUTLINED
- STAR\_PURPLE500\_ROUNDED
- STAR\_PURPLE500\_SHARP
- STAR\_RATE
- STAR\_RATE\_OUTLINED
- STAR\_RATE\_ROUNDED
- STAR\_RATE\_SHARP
- STAR\_ROUNDED
- STAR\_SHARP
- STARS
- STARS\_OUTLINED
- STARS\_ROUNDED
- STARS\_SHARP
- START
- START\_OUTLINED
- START\_ROUNDED
- START\_SHARP
- STAY\_CURRENT\_LANDSCAPE

- STAY\_CURRENT\_LANDSCAPE\_OUTLINED
- STAY\_CURRENT\_LANDSCAPE\_ROUNDED
- STAY\_CURRENT\_LANDSCAPE\_SHARP
- STAY\_CURRENT\_PORTRAIT
- STAY\_CURRENT\_PORTRAIT\_OUTLINED
- STAY\_CURRENT\_PORTRAIT\_ROUNDED
- STAY\_CURRENT\_PORTRAIT\_SHARP
- STAY\_PRIMARY\_LANDSCAPE
- STAY\_PRIMARY\_LANDSCAPE\_OUTLINED
- STAY\_PRIMARY\_LANDSCAPE\_ROUNDED
- STAY\_PRIMARY\_LANDSCAPE\_SHARP
- STAY\_PRIMARY\_PORTRAIT
- STAY\_PRIMARY\_PORTRAIT\_OUTLINED
- STAY\_PRIMARY\_PORTRAIT\_ROUNDED
- STAY\_PRIMARY\_PORTRAIT\_SHARP
- STICKY\_NOTE\_2
- STICKY\_NOTE\_2\_OUTLINED
- STICKY\_NOTE\_2\_ROUNDED
- STICKY\_NOTE\_2\_SHARP
- STOP
- STOP\_CIRCLE
- STOP\_CIRCLE\_OUTLINED
- STOP\_CIRCLE\_ROUNDED
- STOP\_CIRCLE\_SHARP
- STOP\_OUTLINED
- STOP\_ROUNDED
- STOP\_SCREEN\_SHARE
- STOP\_SCREEN\_SHARE\_OUTLINED
- STOP\_SCREEN\_SHARE\_ROUNDED
- STOP\_SCREEN\_SHARE\_SHARP
- STOP\_SHARP
- STORAGE
- STORAGE\_OUTLINED
- STORAGE\_ROUNDED
- STORAGE\_SHARP
- STORE
- STORE\_MALL\_DIRECTORY
- STORE\_MALL\_DIRECTORY\_OUTLINED
- STORE\_MALL\_DIRECTORY\_ROUNDED
- STORE\_MALL\_DIRECTORY\_SHARP
- STORE\_OUTLINED

- STORE\_ROUNDED
- STORE\_SHARP
- STOREFRONT
- STOREFRONT\_OUTLINED
- STOREFRONT\_ROUNDED
- STOREFRONT\_SHARP
- STORM
- STORM\_OUTLINED
- STORM\_ROUNDED
- STORM\_SHARP
- STRAIGHT
- STRAIGHT\_OUTLINED
- STRAIGHT\_ROUNDED
- STRAIGHT\_SHARP
- STRAIGHTEN
- STRAIGHTEN\_OUTLINED
- STRAIGHTEN\_ROUNDED
- STRAIGHTEN\_SHARP
- STREAM
- STREAM\_OUTLINED
- STREAM\_ROUNDED
- STREAM\_SHARP
- STREETVIEW
- STREETVIEW\_OUTLINED
- STREETVIEW\_ROUNDED
- STREETVIEW\_SHARP
- STRIKETHROUGH\_S
- STRIKETHROUGH\_S\_OUTLINED
- STRIKETHROUGH\_S\_ROUNDED
- STRIKETHROUGH\_S\_SHARP
- STROLLER
- STROLLER\_OUTLINED
- STROLLER\_ROUNDED
- STROLLER\_SHARP
- STYLE
- STYLE\_OUTLINED
- STYLE\_ROUNDED
- STYLE\_SHARP
- SUBDIRECTORY\_ARROW\_LEFT
- SUBDIRECTORY\_ARROW\_LEFT\_OUTLINED
- SUBDIRECTORY\_ARROW\_LEFT\_ROUNDED

- SUBDIRECTORY\_ARROW\_LEFT\_SHARP
- SUBDIRECTORY\_ARROW\_RIGHT
- SUBDIRECTORY\_ARROW\_RIGHT\_OUTLINED
- SUBDIRECTORY\_ARROW\_RIGHT\_ROUNDED
- SUBDIRECTORY\_ARROW\_RIGHT\_SHARP
- SUBJECT
- SUBJECT\_OUTLINED
- SUBJECT\_ROUNDED
- SUBJECT\_SHARP
- SUBSCRIPT
- SUBSCRIPT\_OUTLINED
- SUBSCRIPT\_ROUNDED
- SUBSCRIPT\_SHARP
- SUBSCRIPTIONS
- SUBSCRIPTIONS\_OUTLINED
- SUBSCRIPTIONS\_ROUNDED
- SUBSCRIPTIONS\_SHARP
- SUBTITLES
- SUBTITLES\_OFF
- SUBTITLES\_OFF\_OUTLINED
- SUBTITLES\_OFF\_ROUNDED
- SUBTITLES\_OFF\_SHARP
- SUBTITLES\_OUTLINED
- SUBTITLES\_ROUNDED
- SUBTITLES\_SHARP
- SUBWAY
- SUBWAY\_OUTLINED
- SUBWAY\_ROUNDED
- SUBWAY\_SHARP
- SUMMARIZE
- SUMMARIZE\_OUTLINED
- SUMMARIZE\_ROUNDED
- SUMMARIZE\_SHARP
- SUNNY
- SUNNY\_SNOWING
- SUPERSCRIPIT
- SUPERSCRIPIT\_OUTLINED
- SUPERSCRIPIT\_ROUNDED
- SUPERSCRIPIT\_SHARP
- SUPERVISED\_USER\_CIRCLE
- SUPERVISED\_USER\_CIRCLE\_OUTLINED



- SUPERVISED\_USER\_CIRCLE\_ROUNDED
- SUPERVISED\_USER\_CIRCLE\_SHARP
- SUPERVISOR\_ACCOUNT
- SUPERVISOR\_ACCOUNT\_OUTLINED
- SUPERVISOR\_ACCOUNT\_ROUNDED
- SUPERVISOR\_ACCOUNT\_SHARP
- SUPPORT
- SUPPORT\_AGENT
- SUPPORT\_AGENT\_OUTLINED
- SUPPORT\_AGENT\_ROUNDED
- SUPPORT\_AGENT\_SHARP
- SUPPORT\_OUTLINED
- SUPPORT\_ROUNDED
- SUPPORT\_SHARP
- SURFING
- SURFING\_OUTLINED
- SURFING\_ROUNDED
- SURFING\_SHARP
- SURROUND\_SOUND
- SURROUND\_SOUND\_OUTLINED
- SURROUND\_SOUND\_ROUNDED
- SURROUND\_SOUND\_SHARP
- SWAP\_CALLS
- SWAP\_CALLS\_OUTLINED
- SWAP\_CALLS\_ROUNDED
- SWAP\_CALLS\_SHARP
- SWAP\_HORIZ
- SWAP\_HORIZ\_OUTLINED
- SWAP\_HORIZ\_ROUNDED
- SWAP\_HORIZ\_SHARP
- SWAP\_HORIZONTAL\_CIRCLE
- SWAP\_HORIZONTAL\_CIRCLE\_OUTLINED
- SWAP\_HORIZONTAL\_CIRCLE\_ROUNDED
- SWAP\_HORIZONTAL\_CIRCLE\_SHARP
- SWAP\_VERT
- SWAP\_VERT\_CIRCLE
- SWAP\_VERT\_CIRCLE\_OUTLINED
- SWAP\_VERT\_CIRCLE\_ROUNDED
- SWAP\_VERT\_CIRCLE\_SHARP
- SWAP\_VERT\_OUTLINED
- SWAP\_VERT\_ROUNDED

- SWAP\_VERT\_SHARP
- SWAP\_VERTICAL\_CIRCLE
- SWAP\_VERTICAL\_CIRCLE\_OUTLINED
- SWAP\_VERTICAL\_CIRCLE\_ROUNDED
- SWAP\_VERTICAL\_CIRCLE\_SHARP
- SWIPE
- SWIPE\_DOWN
- SWIPE\_DOWN\_ALT
- SWIPE\_DOWN\_ALT\_OUTLINED
- SWIPE\_DOWN\_ALT\_ROUNDED
- SWIPE\_DOWN\_ALT\_SHARP
- SWIPE\_DOWN\_OUTLINED
- SWIPE\_DOWN\_ROUNDED
- SWIPE\_DOWN\_SHARP
- SWIPE\_LEFT
- SWIPE\_LEFT\_ALT
- SWIPE\_LEFT\_ALT\_OUTLINED
- SWIPE\_LEFT\_ALT\_ROUNDED
- SWIPE\_LEFT\_ALT\_SHARP
- SWIPE\_LEFT\_OUTLINED
- SWIPE\_LEFT\_ROUNDED
- SWIPE\_LEFT\_SHARP
- SWIPE\_OUTLINED
- SWIPE\_RIGHT
- SWIPE\_RIGHT\_ALT
- SWIPE\_RIGHT\_ALT\_OUTLINED
- SWIPE\_RIGHT\_ALT\_ROUNDED
- SWIPE\_RIGHT\_ALT\_SHARP
- SWIPE\_RIGHT\_OUTLINED
- SWIPE\_RIGHT\_ROUNDED
- SWIPE\_RIGHT\_SHARP
- SWIPE\_ROUNDED
- SWIPE\_SHARP
- SWIPE\_UP
- SWIPE\_UP\_ALT
- SWIPE\_UP\_ALT\_OUTLINED
- SWIPE\_UP\_ALT\_ROUNDED
- SWIPE\_UP\_ALT\_SHARP
- SWIPE\_UP\_OUTLINED
- SWIPE\_UP\_ROUNDED
- SWIPE\_UP\_SHARP

- SWIPE\_VERTICAL
- SWIPE\_VERTICAL\_OUTLINED
- SWIPE\_VERTICAL\_ROUNDED
- SWIPE\_VERTICAL\_SHARP
- SWITCH\_ACCESS\_SHORTCUT
- SWITCH\_ACCESS\_SHORTCUT\_ADD
- SWITCH\_ACCESS\_SHORTCUT\_ADD\_OUTLINED
- SWITCH\_ACCESS\_SHORTCUT\_ADD\_ROUNDED
- SWITCH\_ACCESS\_SHORTCUT\_ADD\_SHARP
- SWITCH\_ACCESS\_SHORTCUT\_OUTLINED
- SWITCH\_ACCESS\_SHORTCUT\_ROUNDED
- SWITCH\_ACCESS\_SHORTCUT\_SHARP
- SWITCH\_ACCOUNT
- SWITCH\_ACCOUNT\_OUTLINED
- SWITCH\_ACCOUNT\_ROUNDED
- SWITCH\_ACCOUNT\_SHARP
- SWITCH\_CAMERA
- SWITCH\_CAMERA\_OUTLINED
- SWITCH\_CAMERA\_ROUNDED
- SWITCH\_CAMERA\_SHARP
- SWITCH\_LEFT
- SWITCH\_LEFT\_OUTLINED
- SWITCH\_LEFT\_ROUNDED
- SWITCH\_LEFT\_SHARP
- SWITCH\_RIGHT
- SWITCH\_RIGHT\_OUTLINED
- SWITCH\_RIGHT\_ROUNDED
- SWITCH\_RIGHT\_SHARP
- SWITCH\_VIDEO
- SWITCH\_VIDEO\_OUTLINED
- SWITCH\_VIDEO\_ROUNDED
- SWITCH\_VIDEO\_SHARP
- SYNAGOGUE
- SYNAGOGUE\_OUTLINED
- SYNAGOGUE\_ROUNDED
- SYNAGOGUE\_SHARP
- SYNC
- SYNC\_ALT
- SYNC\_ALT\_OUTLINED
- SYNC\_ALT\_ROUNDED
- SYNC\_ALT\_SHARP

- SYNC\_DISABLED
- SYNC\_DISABLED\_OUTLINED
- SYNC\_DISABLED\_ROUNDED
- SYNC\_DISABLED\_SHARP
- SYNC\_LOCK
- SYNC\_LOCK\_OUTLINED
- SYNC\_LOCK\_ROUNDED
- SYNC\_LOCK\_SHARP
- SYNC\_OUTLINED
- SYNC\_PROBLEM
- SYNC\_PROBLEM\_OUTLINED
- SYNC\_PROBLEM\_ROUNDED
- SYNC\_PROBLEM\_SHARP
- SYNC\_ROUNDED
- SYNC\_SHARP
- SYSTEM\_SECURITY\_UPDATE
- SYSTEM\_SECURITY\_UPDATE\_GOOD
- SYSTEM\_SECURITY\_UPDATE\_GOOD\_OUTLINED
- SYSTEM\_SECURITY\_UPDATE\_GOOD\_ROUNDED
- SYSTEM\_SECURITY\_UPDATE\_GOOD\_SHARP
- SYSTEM\_SECURITY\_UPDATE\_OUTLINED
- SYSTEM\_SECURITY\_UPDATE\_ROUNDED
- SYSTEM\_SECURITY\_UPDATE\_SHARP
- SYSTEM\_SECURITY\_UPDATE\_WARNING
- SYSTEM\_SECURITY\_UPDATE\_WARNING\_OUTLINED
- SYSTEM\_SECURITY\_UPDATE\_WARNING\_ROUNDED
- SYSTEM\_SECURITY\_UPDATE\_WARNING\_SHARP
- SYSTEM\_UPDATE
- SYSTEM\_UPDATE\_ALT
- SYSTEM\_UPDATE\_ALT\_OUTLINED
- SYSTEM\_UPDATE\_ALT\_ROUNDED
- SYSTEM\_UPDATE\_ALT\_SHARP
- SYSTEM\_UPDATE\_OUTLINED
- SYSTEM\_UPDATE\_ROUNDED
- SYSTEM\_UPDATE\_SHARP
- SYSTEM\_UPDATE\_TV
- SYSTEM\_UPDATE\_TV\_OUTLINED
- SYSTEM\_UPDATE\_TV\_ROUNDED
- SYSTEM\_UPDATE\_TV\_SHARP
- TAB
- TAB\_OUTLINED

- TAB\_ROUNDED
- TAB\_SHARP
- TAB\_UNSELECTED
- TAB\_UNSELECTED\_OUTLINED
- TAB\_UNSELECTED\_ROUNDED
- TAB\_UNSELECTED\_SHARP
- TABLE\_BAR
- TABLE\_BAR\_OUTLINED
- TABLE\_BAR\_ROUNDED
- TABLE\_BAR\_SHARP
- TABLE\_CHART
- TABLE\_CHART\_OUTLINED
- TABLE\_CHART\_ROUNDED
- TABLE\_CHART\_SHARP
- TABLE\_RESTAURANT
- TABLE\_RESTAURANT\_OUTLINED
- TABLE\_RESTAURANT\_ROUNDED
- TABLE\_RESTAURANT\_SHARP
- TABLE\_ROWS
- TABLE\_ROWS\_OUTLINED
- TABLE\_ROWS\_ROUNDED
- TABLE\_ROWS\_SHARP
- TABLE\_VIEW
- TABLE\_VIEW\_OUTLINED
- TABLE\_VIEW\_ROUNDED
- TABLE\_VIEW\_SHARP
- TABLET
- TABLET\_ANDROID
- TABLET\_ANDROID\_OUTLINED
- TABLET\_ANDROID\_ROUNDED
- TABLET\_ANDROID\_SHARP
- TABLET\_MAC
- TABLET\_MAC\_OUTLINED
- TABLET\_MAC\_ROUNDED
- TABLET\_MAC\_SHARP
- TABLET\_OUTLINED
- TABLET\_ROUNDED
- TABLET\_SHARP
- TAG
- TAG\_FACES
- TAG\_FACES\_OUTLINED

- TAG\_FACES\_ROUNDED
- TAG\_FACES\_SHARP
- TAG\_OUTLINED
- TAG\_ROUNDED
- TAG\_SHARP
- TAKEOUT\_DINING
- TAKEOUT\_DINING\_OUTLINED
- TAKEOUT\_DINING\_ROUNDED
- TAKEOUT\_DINING\_SHARP
- TAP\_AND\_PLAY
- TAP\_AND\_PLAY\_OUTLINED
- TAP\_AND\_PLAY\_ROUNDED
- TAP\_AND\_PLAY\_SHARP
- TAPAS
- TAPAS\_OUTLINED
- TAPAS\_ROUNDED
- TAPAS\_SHARP
- TASK
- TASK\_ALT
- TASK\_ALT\_OUTLINED
- TASK\_ALT\_ROUNDED
- TASK\_ALT\_SHARP
- TASK\_OUTLINED
- TASK\_ROUNDED
- TASK\_SHARP
- TAXI\_ALERT
- TAXI\_ALERT\_OUTLINED
- TAXI\_ALERT\_ROUNDED
- TAXI\_ALERT\_SHARP
- TELEGRAM
- TELEGRAM\_OUTLINED
- TELEGRAM\_ROUNDED
- TELEGRAM\_SHARP
- TEMPLE\_BUDDHIST
- TEMPLE\_BUDDHIST\_OUTLINED
- TEMPLE\_BUDDHIST\_ROUNDED
- TEMPLE\_BUDDHIST\_SHARP
- TEMPLE\_HINDU
- TEMPLE\_HINDU\_OUTLINED
- TEMPLE\_HINDU\_ROUNDED
- TEMPLE\_HINDU\_SHARP

- TEN\_K
- TEN\_K\_OUTLINED
- TEN\_K\_ROUNDED
- TEN\_K\_SHARP
- TEN\_MP
- TEN\_MP\_OUTLINED
- TEN\_MP\_ROUNDED
- TEN\_MP\_SHARP
- TERMINAL
- TERMINAL\_OUTLINED
- TERMINAL\_ROUNDED
- TERMINAL\_SHARP
- TERRAIN
- TERRAIN\_OUTLINED
- TERRAIN\_ROUNDED
- TERRAIN\_SHARP
- TEXT\_DECREASE
- TEXT\_DECREASE\_OUTLINED
- TEXT\_DECREASE\_ROUNDED
- TEXT\_DECREASE\_SHARP
- TEXT\_FIELDS
- TEXT\_FIELDS\_OUTLINED
- TEXT\_FIELDS\_ROUNDED
- TEXT\_FIELDS\_SHARP
- TEXT\_FORMAT
- TEXT\_FORMAT\_OUTLINED
- TEXT\_FORMAT\_ROUNDED
- TEXT\_FORMAT\_SHARP
- TEXT\_INCREASE
- TEXT\_INCREASE\_OUTLINED
- TEXT\_INCREASE\_ROUNDED
- TEXT\_INCREASE\_SHARP
- TEXT\_ROTATE\_UP
- TEXT\_ROTATE\_UP\_OUTLINED
- TEXT\_ROTATE\_UP\_ROUNDED
- TEXT\_ROTATE\_UP\_SHARP
- TEXT\_ROTATE\_VERTICAL
- TEXT\_ROTATE\_VERTICAL\_OUTLINED
- TEXT\_ROTATE\_VERTICAL\_ROUNDED
- TEXT\_ROTATE\_VERTICAL\_SHARP
- TEXT\_ROTATION\_ANGLEDOWN

- TEXT\_ROTATION\_ANGLEDOWN\_OUTLINED
- TEXT\_ROTATION\_ANGLEDOWN\_ROUNDED
- TEXT\_ROTATION\_ANGLEDOWN\_SHARP
- TEXT\_ROTATION\_ANGLEUP
- TEXT\_ROTATION\_ANGLEUP\_OUTLINED
- TEXT\_ROTATION\_ANGLEUP\_ROUNDED
- TEXT\_ROTATION\_ANGLEUP\_SHARP
- TEXT\_ROTATION\_DOWN
- TEXT\_ROTATION\_DOWN\_OUTLINED
- TEXT\_ROTATION\_DOWN\_ROUNDED
- TEXT\_ROTATION\_DOWN\_SHARP
- TEXT\_ROTATION\_NONE
- TEXT\_ROTATION\_NONE\_OUTLINED
- TEXT\_ROTATION\_NONE\_ROUNDED
- TEXT\_ROTATION\_NONE\_SHARP
- TEXT\_SNIPPET
- TEXT\_SNIPPET\_OUTLINED
- TEXT\_SNIPPET\_ROUNDED
- TEXT\_SNIPPET\_SHARP
- TEXTSMS
- TEXTSMS\_OUTLINED
- TEXTSMS\_ROUNDED
- TEXTSMS\_SHARP
- TEXTURE
- TEXTURE\_OUTLINED
- TEXTURE\_ROUNDED
- TEXTURE\_SHARP
- THEATER\_COMEDY
- THEATER\_COMEDY\_OUTLINED
- THEATER\_COMEDY\_ROUNDED
- THEATER\_COMEDY\_SHARP
- THEATERS
- THEATERS\_OUTLINED
- THEATERS\_ROUNDED
- THEATERS\_SHARP
- THERMOSTAT
- THERMOSTAT\_AUTO
- THERMOSTAT\_AUTO\_OUTLINED
- THERMOSTAT\_AUTO\_ROUNDED
- THERMOSTAT\_AUTO\_SHARP
- THERMOSTAT\_OUTLINED



- THERMOSTAT\_ROUNDED
- THERMOSTAT\_SHARP
- THIRTEEN\_MP
- THIRTEEN\_MP\_OUTLINED
- THIRTEEN\_MP\_ROUNDED
- THIRTEEN\_MP\_SHARP
- THIRTY\_FPS
- THIRTY\_FPS\_OUTLINED
- THIRTY\_FPS\_ROUNDED
- THIRTY\_FPS\_SELECT
- THIRTY\_FPS\_SELECT\_OUTLINED
- THIRTY\_FPS\_SELECT\_ROUNDED
- THIRTY\_FPS\_SELECT\_SHARP
- THIRTY\_FPS\_SHARP
- THREE\_G\_MOBILEDATA
- THREE\_G\_MOBILEDATA\_OUTLINED
- THREE\_G\_MOBILEDATA\_ROUNDED
- THREE\_G\_MOBILEDATA\_SHARP
- THREE\_K
- THREE\_K\_OUTLINED
- THREE\_K\_PLUS
- THREE\_K\_PLUS\_OUTLINED
- THREE\_K\_PLUS\_ROUNDED
- THREE\_K\_PLUS\_SHARP
- THREE\_K\_ROUNDED
- THREE\_K\_SHARP
- THREE\_MP
- THREE\_MP\_OUTLINED
- THREE\_MP\_ROUNDED
- THREE\_MP\_SHARP
- THREE\_P
- THREE\_P\_OUTLINED
- THREE\_P\_ROUNDED
- THREE\_P\_SHARP
- THREED\_ROTATION
- THREED\_ROTATION\_OUTLINED
- THREED\_ROTATION\_ROUNDED
- THREED\_ROTATION\_SHARP
- THREESIXTY
- THREESIXTY\_OUTLINED
- THREESIXTY\_ROUNDED

- THREESIXTY\_SHARP
- THUMB\_DOWN
- THUMB\_DOWN\_ALT
- THUMB\_DOWN\_ALT\_OUTLINED
- THUMB\_DOWN\_ALT\_ROUNDED
- THUMB\_DOWN\_ALT\_SHARP
- THUMB\_DOWN\_OFF\_ALT
- THUMB\_DOWN\_OFF\_ALT\_OUTLINED
- THUMB\_DOWN\_OFF\_ALT\_ROUNDED
- THUMB\_DOWN\_OFF\_ALT\_SHARP
- THUMB\_DOWN\_OUTLINED
- THUMB\_DOWN\_ROUNDED
- THUMB\_DOWN\_SHARP
- THUMB\_UP
- THUMB\_UP\_ALT
- THUMB\_UP\_ALT\_OUTLINED
- THUMB\_UP\_ALT\_ROUNDED
- THUMB\_UP\_ALT\_SHARP
- THUMB\_UP\_OFF\_ALT
- THUMB\_UP\_OFF\_ALT\_OUTLINED
- THUMB\_UP\_OFF\_ALT\_ROUNDED
- THUMB\_UP\_OFF\_ALT\_SHARP
- THUMB\_UP\_OUTLINED
- THUMB\_UP\_ROUNDED
- THUMB\_UP\_SHARP
- THUMBS\_UP\_DOWN
- THUMBS\_UP\_DOWN\_OUTLINED
- THUMBS\_UP\_DOWN\_ROUNDED
- THUMBS\_UP\_DOWN\_SHARP
- THUNDERSTORM
- THUNDERSTORM\_OUTLINED
- THUNDERSTORM\_ROUNDED
- THUNDERSTORM\_SHARP
- TIKTOK
- TIKTOK\_OUTLINED
- TIKTOK\_ROUNDED
- TIKTOK\_SHARP
- TIME\_TO\_LEAVE
- TIME\_TO\_LEAVE\_OUTLINED
- TIME\_TO\_LEAVE\_ROUNDED
- TIME\_TO\_LEAVE\_SHARP

- TIMELAPSE
- TIMELAPSE\_OUTLINED
- TIMELAPSE\_ROUNDED
- TIMELAPSE\_SHARP
- TIMELINE
- TIMELINE\_OUTLINED
- TIMELINE\_ROUNDED
- TIMELINE\_SHARP
- TIMER
- TIMER\_10
- TIMER\_10\_OUTLINED
- TIMER\_10\_ROUNDED
- TIMER\_10\_SELECT
- TIMER\_10\_SELECT\_OUTLINED
- TIMER\_10\_SELECT\_ROUNDED
- TIMER\_10\_SELECT\_SHARP
- TIMER\_10\_SHARP
- TIMER\_3
- TIMER\_3\_OUTLINED
- TIMER\_3\_ROUNDED
- TIMER\_3\_SELECT
- TIMER\_3\_SELECT\_OUTLINED
- TIMER\_3\_SELECT\_ROUNDED
- TIMER\_3\_SELECT\_SHARP
- TIMER\_3\_SHARP
- TIMER\_OFF
- TIMER\_OFF\_OUTLINED
- TIMER\_OFF\_ROUNDED
- TIMER\_OFF\_SHARP
- TIMER\_OUTLINED
- TIMER\_ROUNDED
- TIMER\_SHARP
- TIPS\_AND\_UPDATES
- TIPS\_AND\_UPDATES\_OUTLINED
- TIPS\_AND\_UPDATES\_ROUNDED
- TIPS\_AND\_UPDATES\_SHARP
- TIRE\_REPAIR
- TIRE\_REPAIR\_OUTLINED
- TIRE\_REPAIR\_ROUNDED
- TIRE\_REPAIR\_SHARP
- TITLE

- TITLE\_OUTLINED
- TITLE\_ROUNDED
- TITLE\_SHARP
- TOC
- TOC\_OUTLINED
- TOC\_ROUNDED
- TOC\_SHARP
- TODAY
- TODAY\_OUTLINED
- TODAY\_ROUNDED
- TODAY\_SHARP
- TOGGLE\_OFF
- TOGGLE\_OFF\_OUTLINED
- TOGGLE\_OFF\_ROUNDED
- TOGGLE\_OFF\_SHARP
- TOGGLE\_ON
- TOGGLE\_ON\_OUTLINED
- TOGGLE\_ON\_ROUNDED
- TOGGLE\_ON\_SHARP
- TOKEN
- TOKEN\_OUTLINED
- TOKEN\_ROUNDED
- TOKEN\_SHARP
- TOLL
- TOLL\_OUTLINED
- TOLL\_ROUNDED
- TOLL\_SHARP
- TONALITY
- TONALITY\_OUTLINED
- TONALITY\_ROUNDED
- TONALITY\_SHARP
- TOPIC
- TOPIC\_OUTLINED
- TOPIC\_ROUNDED
- TOPIC\_SHARP
- TORNADO
- TORNADO\_OUTLINED
- TORNADO\_ROUNDED
- TORNADO\_SHARP
- TOUCH\_APP
- TOUCH\_APP\_OUTLINED

- TOUCH\_APP\_ROUNDED
- TOUCH\_APP\_SHARP
- TOUR
- TOUR\_OUTLINED
- TOUR\_ROUNDED
- TOUR\_SHARP
- TOYS
- TOYS\_OUTLINED
- TOYS\_ROUNDED
- TOYS\_SHARP
- TRACK\_CHANGES
- TRACK\_CHANGES\_OUTLINED
- TRACK\_CHANGES\_ROUNDED
- TRACK\_CHANGES\_SHARP
- TRAFFIC
- TRAFFIC\_OUTLINED
- TRAFFIC\_ROUNDED
- TRAFFIC\_SHARP
- TRAIN
- TRAIN\_OUTLINED
- TRAIN\_ROUNDED
- TRAIN\_SHARP
- TRAM
- TRAM\_OUTLINED
- TRAM\_ROUNDED
- TRAM\_SHARP
- TRANSCRIBE
- TRANSCRIBE\_OUTLINED
- TRANSCRIBE\_ROUNDED
- TRANSCRIBE\_SHARP
- TRANSFER\_WITHIN\_A\_STATION
- TRANSFER\_WITHIN\_A\_STATION\_OUTLINED
- TRANSFER\_WITHIN\_A\_STATION\_ROUNDED
- TRANSFER\_WITHIN\_A\_STATION\_SHARP
- TRANSFORM
- TRANSFORM\_OUTLINED
- TRANSFORM\_ROUNDED
- TRANSFORM\_SHARP
- TRANSGENDER
- TRANSGENDER\_OUTLINED
- TRANSGENDER\_ROUNDED

- TRANSGENDER\_SHARP
- TRANSIT\_ENTEREXIT
- TRANSIT\_ENTEREXIT\_OUTLINED
- TRANSIT\_ENTEREXIT\_ROUNDED
- TRANSIT\_ENTEREXIT\_SHARP
- TRANSLATE
- TRANSLATE\_OUTLINED
- TRANSLATE\_ROUNDED
- TRANSLATE\_SHARP
- TRAVEL\_EXPLORE
- TRAVEL\_EXPLORE\_OUTLINED
- TRAVEL\_EXPLORE\_ROUNDED
- TRAVEL\_EXPLORE\_SHARP
- TRENDING\_DOWN
- TRENDING\_DOWN\_OUTLINED
- TRENDING\_DOWN\_ROUNDED
- TRENDING\_DOWN\_SHARP
- TRENDING\_FLAT
- TRENDING\_FLAT\_OUTLINED
- TRENDING\_FLAT\_ROUNDED
- TRENDING\_FLAT\_SHARP
- TRENDING\_NEUTRAL
- TRENDING\_NEUTRAL\_OUTLINED
- TRENDING\_NEUTRAL\_ROUNDED
- TRENDING\_NEUTRAL\_SHARP
- TRENDING\_UP
- TRENDING\_UP\_OUTLINED
- TRENDING\_UP\_ROUNDED
- TRENDING\_UP\_SHARP
- TRIP\_ORIGIN
- TRIP\_ORIGIN\_OUTLINED
- TRIP\_ORIGIN\_ROUNDED
- TRIP\_ORIGIN\_SHARP
- TROLLEY
- TROUBLESHOOT
- TROUBLESHOOT\_OUTLINED
- TROUBLESHOOT\_ROUNDED
- TROUBLESHOOT\_SHARP
- TRY\_SMS\_STAR
- TRY\_SMS\_STAR\_OUTLINED
- TRY\_SMS\_STAR\_ROUNDED

- TRY\_SMS\_STAR\_SHARP
- TSUNAMI
- TSUNAMI\_OUTLINED
- TSUNAMI\_ROUNDED
- TSUNAMI\_SHARP
- TTY
- TTY\_OUTLINED
- TTY\_ROUNDED
- TTY\_SHARP
- TUNE
- TUNE\_OUTLINED
- TUNE\_ROUNDED
- TUNE\_SHARP
- TUNGSTEN
- TUNGSTEN\_OUTLINED
- TUNGSTEN\_ROUNDED
- TUNGSTEN\_SHARP
- TURN\_LEFT
- TURN\_LEFT\_OUTLINED
- TURN\_LEFT\_ROUNDED
- TURN\_LEFT\_SHARP
- TURN\_RIGHT
- TURN\_RIGHT\_OUTLINED
- TURN\_RIGHT\_ROUNDED
- TURN\_RIGHT\_SHARP
- TURN\_SHARP\_LEFT
- TURN\_SHARP\_LEFT\_OUTLINED
- TURN\_SHARP\_LEFT\_ROUNDED
- TURN\_SHARP\_LEFT\_SHARP
- TURN\_SHARP\_RIGHT
- TURN\_SHARP\_RIGHT\_OUTLINED
- TURN\_SHARP\_RIGHT\_ROUNDED
- TURN\_SHARP\_RIGHT\_SHARP
- TURN\_SLIGHT\_LEFT
- TURN\_SLIGHT\_LEFT\_OUTLINED
- TURN\_SLIGHT\_LEFT\_ROUNDED
- TURN\_SLIGHT\_LEFT\_SHARP
- TURN\_SLIGHT\_RIGHT
- TURN\_SLIGHT\_RIGHT\_OUTLINED
- TURN\_SLIGHT\_RIGHT\_ROUNDED
- TURN\_SLIGHT\_RIGHT\_SHARP

- TURNED\_IN
- TURNED\_IN\_NOT
- TURNED\_IN\_NOT\_OUTLINED
- TURNED\_IN\_NOT\_ROUNDED
- TURNED\_IN\_NOT\_SHARP
- TURNED\_IN\_OUTLINED
- TURNED\_IN\_ROUNDED
- TURNED\_IN\_SHARP
- TV
- TV\_OFF
- TV\_OFF\_OUTLINED
- TV\_OFF\_ROUNDED
- TV\_OFF\_SHARP
- TV\_OUTLINED
- TV\_ROUNDED
- TV\_SHARP
- TWELVE\_MP
- TWELVE\_MP\_OUTLINED
- TWELVE\_MP\_ROUNDED
- TWELVE\_MP\_SHARP
- TWENTY\_FOUR\_MP
- TWENTY\_FOUR\_MP\_OUTLINED
- TWENTY\_FOUR\_MP\_ROUNDED
- TWENTY\_FOUR\_MP\_SHARP
- TWENTY\_MP
- TWENTY\_MP\_OUTLINED
- TWENTY\_MP\_ROUNDED
- TWENTY\_MP\_SHARP
- TWENTY\_ONE\_MP
- TWENTY\_ONE\_MP\_OUTLINED
- TWENTY\_ONE\_MP\_ROUNDED
- TWENTY\_ONE\_MP\_SHARP
- TWENTY\_THREE\_MP
- TWENTY\_THREE\_MP\_OUTLINED
- TWENTY\_THREE\_MP\_ROUNDED
- TWENTY\_THREE\_MP\_SHARP
- TWENTY\_TWO\_MP
- TWENTY\_TWO\_MP\_OUTLINED
- TWENTY\_TWO\_MP\_ROUNDED
- TWENTY\_TWO\_MP\_SHARP
- TWO\_K



- TWO\_K\_OUTLINED
- TWO\_K\_PLUS
- TWO\_K\_PLUS\_OUTLINED
- TWO\_K\_PLUS\_ROUNDED
- TWO\_K\_PLUS\_SHARP
- TWO\_K\_ROUNDED
- TWO\_K\_SHARP
- TWO\_MP
- TWO\_MP\_OUTLINED
- TWO\_MP\_ROUNDED
- TWO\_MP\_SHARP
- TWO\_WHEELER
- TWO\_WHEELER\_OUTLINED
- TWO\_WHEELER\_ROUNDED
- TWO\_WHEELER\_SHARP
- TYPE\_SPECIMEN
- TYPE\_SPECIMEN\_OUTLINED
- TYPE\_SPECIMEN\_ROUNDED
- TYPE\_SPECIMEN\_SHARP
- U\_TURN\_LEFT
- U\_TURN\_LEFT\_OUTLINED
- U\_TURN\_LEFT\_ROUNDED
- U\_TURN\_LEFT\_SHARP
- U\_TURN\_RIGHT
- U\_TURN\_RIGHT\_OUTLINED
- U\_TURN\_RIGHT\_ROUNDED
- U\_TURN\_RIGHT\_SHARP
- UMBRELLA
- UMBRELLA\_OUTLINED
- UMBRELLA\_ROUNDED
- UMBRELLA\_SHARP
- UNARCHIVE
- UNARCHIVE\_OUTLINED
- UNARCHIVE\_ROUNDED
- UNARCHIVE\_SHARP
- UNDO
- UNDO\_OUTLINED
- UNDO\_ROUNDED
- UNDO\_SHARP
- UNFOLD\_LESS
- UNFOLD\_LESS\_DOUBLE

- UNFOLD\_LESS\_DOUBLE\_OUTLINED
- UNFOLD\_LESS\_DOUBLE\_ROUNDED
- UNFOLD\_LESS\_DOUBLE\_SHARP
- UNFOLD\_LESS\_OUTLINED
- UNFOLD\_LESS\_ROUNDED
- UNFOLD\_LESS\_SHARP
- UNFOLD\_MORE
- UNFOLD\_MORE\_DOUBLE
- UNFOLD\_MORE\_DOUBLE\_OUTLINED
- UNFOLD\_MORE\_DOUBLE\_ROUNDED
- UNFOLD\_MORE\_DOUBLE\_SHARP
- UNFOLD\_MORE\_OUTLINED
- UNFOLD\_MORE\_ROUNDED
- UNFOLD\_MORE\_SHARP
- UNPUBLISHED
- UNPUBLISHED\_OUTLINED
- UNPUBLISHED\_ROUNDED
- UNPUBLISHED\_SHARP
- UNSUBSCRIBE
- UNSUBSCRIBE\_OUTLINED
- UNSUBSCRIBE\_ROUNDED
- UNSUBSCRIBE\_SHARP
- UPCOMING
- UPCOMING\_OUTLINED
- UPCOMING\_ROUNDED
- UPCOMING\_SHARP
- UPDATE
- UPDATE\_DISABLED
- UPDATE\_DISABLED\_OUTLINED
- UPDATE\_DISABLED\_ROUNDED
- UPDATE\_DISABLED\_SHARP
- UPDATE\_OUTLINED
- UPDATE\_ROUNDED
- UPDATE\_SHARP
- UPGRADE
- UPGRADE\_OUTLINED
- UPGRADE\_ROUNDED
- UPGRADE\_SHARP
- UPLOAD
- UPLOAD\_FILE
- UPLOAD\_FILE\_OUTLINED

- UPLOAD\_FILE\_ROUNDED
- UPLOAD\_FILE\_SHARP
- UPLOAD\_OUTLINED
- UPLOAD\_ROUNDED
- UPLOAD\_SHARP
- USB
- USB\_OFF
- USB\_OFF\_OUTLINED
- USB\_OFF\_ROUNDED
- USB\_OFF\_SHARP
- USB\_OUTLINED
- USB\_ROUNDED
- USB\_SHARP
- VACCINES
- VACCINES\_OUTLINED
- VACCINES\_ROUNDED
- VACCINES\_SHARP
- VAPE\_FREE
- VAPE\_FREE\_OUTLINED
- VAPE\_FREE\_ROUNDED
- VAPE\_FREE\_SHARP
- VAPING\_ROOMS
- VAPING\_ROOMS\_OUTLINED
- VAPING\_ROOMS\_ROUNDED
- VAPING\_ROOMS\_SHARP
- VERIFIED
- VERIFIED\_OUTLINED
- VERIFIED\_ROUNDED
- VERIFIED\_SHARP
- VERIFIED\_USER
- VERIFIED\_USER\_OUTLINED
- VERIFIED\_USER\_ROUNDED
- VERIFIED\_USER\_SHARP
- VERTICAL\_ALIGN\_BOTTOM
- VERTICAL\_ALIGN\_BOTTOM\_OUTLINED
- VERTICAL\_ALIGN\_BOTTOM\_ROUNDED
- VERTICAL\_ALIGN\_BOTTOM\_SHARP
- VERTICAL\_ALIGN\_CENTER
- VERTICAL\_ALIGN\_CENTER\_OUTLINED
- VERTICAL\_ALIGN\_CENTER\_ROUNDED
- VERTICAL\_ALIGN\_CENTER\_SHARP

- VERTICAL\_ALIGN\_TOP
- VERTICAL\_ALIGN\_TOP\_OUTLINED
- VERTICAL\_ALIGN\_TOP\_ROUNDED
- VERTICAL\_ALIGN\_TOP\_SHARP
- VERTICAL\_DISTRIBUTE
- VERTICAL\_DISTRIBUTE\_OUTLINED
- VERTICAL\_DISTRIBUTE\_ROUNDED
- VERTICAL\_DISTRIBUTE\_SHARP
- VERTICAL\_SHADES
- VERTICAL\_SHADES\_CLOSED
- VERTICAL\_SHADES\_CLOSED\_OUTLINED
- VERTICAL\_SHADES\_CLOSED\_ROUNDED
- VERTICAL\_SHADES\_CLOSED\_SHARP
- VERTICAL\_SHADES\_OUTLINED
- VERTICAL\_SHADES\_ROUNDED
- VERTICAL\_SHADES\_SHARP
- VERTICAL\_SPLIT
- VERTICAL\_SPLIT\_OUTLINED
- VERTICAL\_SPLIT\_ROUNDED
- VERTICAL\_SPLIT\_SHARP
- VIBRATION
- VIBRATION\_OUTLINED
- VIBRATION\_ROUNDED
- VIBRATION\_SHARP
- VIDEO\_CALL
- VIDEO\_CALL\_OUTLINED
- VIDEO\_CALL\_ROUNDED
- VIDEO\_CALL\_SHARP
- VIDEO\_CAMERA\_BACK
- VIDEO\_CAMERA\_BACK\_OUTLINED
- VIDEO\_CAMERA\_BACK\_ROUNDED
- VIDEO\_CAMERA\_BACK\_SHARP
- VIDEO\_CAMERA\_FRONT
- VIDEO\_CAMERA\_FRONT\_OUTLINED
- VIDEO\_CAMERA\_FRONT\_ROUNDED
- VIDEO\_CAMERA\_FRONT\_SHARP
- VIDEO\_CHAT
- VIDEO\_CHAT\_OUTLINED
- VIDEO\_CHAT\_ROUNDED
- VIDEO\_CHAT\_SHARP
- VIDEO\_COLLECTION

- VIDEO\_COLLECTION\_OUTLINED
- VIDEO\_COLLECTION\_ROUNDED
- VIDEO\_COLLECTION\_SHARP
- VIDEO\_FILE
- VIDEO\_FILE\_OUTLINED
- VIDEO\_FILE\_ROUNDED
- VIDEO\_FILE\_SHARP
- VIDEO\_LABEL
- VIDEO\_LABEL\_OUTLINED
- VIDEO\_LABEL\_ROUNDED
- VIDEO\_LABEL\_SHARP
- VIDEO\_LIBRARY
- VIDEO\_LIBRARY\_OUTLINED
- VIDEO\_LIBRARY\_ROUNDED
- VIDEO\_LIBRARY\_SHARP
- VIDEO\_SETTINGS
- VIDEO\_SETTINGS\_OUTLINED
- VIDEO\_SETTINGS\_ROUNDED
- VIDEO\_SETTINGS\_SHARP
- VIDEO\_STABLE
- VIDEO\_STABLE\_OUTLINED
- VIDEO\_STABLE\_ROUNDED
- VIDEO\_STABLE\_SHARP
- VIDEOCAM
- VIDEOCAM\_OFF
- VIDEOCAM\_OFF\_OUTLINED
- VIDEOCAM\_OFF\_ROUNDED
- VIDEOCAM\_OFF\_SHARP
- VIDEOCAM\_OUTLINED
- VIDEOCAM\_ROUNDED
- VIDEOCAM\_SHARP
- VIDEOGAME\_ASSET
- VIDEOGAME\_ASSET\_OFF
- VIDEOGAME\_ASSET\_OFF\_OUTLINED
- VIDEOGAME\_ASSET\_OFF\_ROUNDED
- VIDEOGAME\_ASSET\_OFF\_SHARP
- VIDEOGAME\_ASSET\_OUTLINED
- VIDEOGAME\_ASSET\_ROUNDED
- VIDEOGAME\_ASSET\_SHARP
- VIEW\_AGENDA
- VIEW\_AGENDA\_OUTLINED

- VIEW\_AGENDA\_ROUNDED
- VIEW\_AGENDA\_SHARP
- VIEW\_ARRAY
- VIEW\_ARRAY\_OUTLINED
- VIEW\_ARRAY\_ROUNDED
- VIEW\_ARRAY\_SHARP
- VIEW\_CAROUSEL
- VIEW\_CAROUSEL\_OUTLINED
- VIEW\_CAROUSEL\_ROUNDED
- VIEW\_CAROUSEL\_SHARP
- VIEW\_COLUMN
- VIEW\_COLUMN\_OUTLINED
- VIEW\_COLUMN\_ROUNDED
- VIEW\_COLUMN\_SHARP
- VIEW\_COMFORTABLE
- VIEW\_COMFORTABLE\_OUTLINED
- VIEW\_COMFORTABLE\_ROUNDED
- VIEW\_COMFORTABLE\_SHARP
- VIEW\_COMFY
- VIEW\_COMFY\_ALT
- VIEW\_COMFY\_ALT\_OUTLINED
- VIEW\_COMFY\_ALT\_ROUNDED
- VIEW\_COMFY\_ALT\_SHARP
- VIEW\_COMFY\_OUTLINED
- VIEW\_COMFY\_ROUNDED
- VIEW\_COMFY\_SHARP
- VIEW\_COMPACT
- VIEW\_COMPACT\_ALT
- VIEW\_COMPACT\_ALT\_OUTLINED
- VIEW\_COMPACT\_ALT\_ROUNDED
- VIEW\_COMPACT\_ALT\_SHARP
- VIEW\_COMPACT\_OUTLINED
- VIEW\_COMPACT\_ROUNDED
- VIEW\_COMPACT\_SHARP
- VIEW\_COZY
- VIEW\_COZY\_OUTLINED
- VIEW\_COZY\_ROUNDED
- VIEW\_COZY\_SHARP
- VIEW\_DAY
- VIEW\_DAY\_OUTLINED
- VIEW\_DAY\_ROUNDED

- VIEW\_DAY\_SHARP
- VIEW\_HEADLINE
- VIEW\_HEADLINE\_OUTLINED
- VIEW\_HEADLINE\_ROUNDED
- VIEW\_HEADLINE\_SHARP
- VIEW\_IN\_AR
- VIEW\_IN\_AR\_OUTLINED
- VIEW\_IN\_AR\_ROUNDED
- VIEW\_IN\_AR\_SHARP
- VIEW\_KANBAN
- VIEW\_KANBAN\_OUTLINED
- VIEW\_KANBAN\_ROUNDED
- VIEW\_KANBAN\_SHARP
- VIEW\_LIST
- VIEW\_LIST\_OUTLINED
- VIEW\_LIST\_ROUNDED
- VIEW\_LIST\_SHARP
- VIEW\_MODULE
- VIEW\_MODULE\_OUTLINED
- VIEW\_MODULE\_ROUNDED
- VIEW\_MODULE\_SHARP
- VIEW\_QUILT
- VIEW\_QUILT\_OUTLINED
- VIEW\_QUILT\_ROUNDED
- VIEW\_QUILT\_SHARP
- VIEW\_SIDEBAR
- VIEW\_SIDEBAR\_OUTLINED
- VIEW\_SIDEBAR\_ROUNDED
- VIEW\_SIDEBAR\_SHARP
- VIEW\_STREAM
- VIEW\_STREAM\_OUTLINED
- VIEW\_STREAM\_ROUNDED
- VIEW\_STREAM\_SHARP
- VIEW\_TIMELINE
- VIEW\_TIMELINE\_OUTLINED
- VIEW\_TIMELINE\_ROUNDED
- VIEW\_TIMELINE\_SHARP
- VIEW\_WEEK
- VIEW\_WEEK\_OUTLINED
- VIEW\_WEEK\_ROUNDED
- VIEW\_WEEK\_SHARP

- VIGNETTE
- VIGNETTE\_OUTLINED
- VIGNETTE\_ROUNDED
- VIGNETTE\_SHARP
- VILLA
- VILLA\_OUTLINED
- VILLA\_ROUNDED
- VILLA\_SHARP
- VISIBILITY
- VISIBILITY\_OFF
- VISIBILITY\_OFF\_OUTLINED
- VISIBILITY\_OFF\_ROUNDED
- VISIBILITY\_OFF\_SHARP
- VISIBILITY\_OUTLINED
- VISIBILITY\_ROUNDED
- VISIBILITY\_SHARP
- VOICE\_CHAT
- VOICE\_CHAT\_OUTLINED
- VOICE\_CHAT\_ROUNDED
- VOICE\_CHAT\_SHARP
- VOICE\_OVER\_OFF
- VOICE\_OVER\_OFF\_OUTLINED
- VOICE\_OVER\_OFF\_ROUNDED
- VOICE\_OVER\_OFF\_SHARP
- VOICEMAIL
- VOICEMAIL\_OUTLINED
- VOICEMAIL\_ROUNDED
- VOICEMAIL\_SHARP
- VOLCANO
- VOLCANO\_OUTLINED
- VOLCANO\_ROUNDED
- VOLCANO\_SHARP
- VOLUME\_DOWN
- VOLUME\_DOWN\_ALT
- VOLUME\_DOWN\_OUTLINED
- VOLUME\_DOWN\_ROUNDED
- VOLUME\_DOWN\_SHARP
- VOLUME\_MUTE
- VOLUME\_MUTE\_OUTLINED
- VOLUME\_MUTE\_ROUNDED
- VOLUME\_MUTE\_SHARP



- VOLUME\_OFF
- VOLUME\_OFF\_OUTLINED
- VOLUME\_OFF\_ROUNDED
- VOLUME\_OFF\_SHARP
- VOLUME\_UP
- VOLUME\_UP\_OUTLINED
- VOLUME\_UP\_ROUNDED
- VOLUME\_UP\_SHARP
- VOLUNTEER\_ACTIVISM
- VOLUNTEER\_ACTIVISM\_OUTLINED
- VOLUNTEER\_ACTIVISM\_ROUNDED
- VOLUNTEER\_ACTIVISM\_SHARP
- VPN\_KEY
- VPN\_KEY\_OFF
- VPN\_KEY\_OFF\_OUTLINED
- VPN\_KEY\_OFF\_ROUNDED
- VPN\_KEY\_OFF\_SHARP
- VPN\_KEY\_OUTLINED
- VPN\_KEY\_ROUNDED
- VPN\_KEY\_SHARP
- VPN\_LOCK
- VPN\_LOCK\_OUTLINED
- VPN\_LOCK\_ROUNDED
- VPN\_LOCK\_SHARP
- VRPANO
- VRPANO\_OUTLINED
- VRPANO\_ROUNDED
- VRPANO\_SHARP
- WALLET
- WALLET\_GIFTCARD
- WALLET\_GIFTCARD\_OUTLINED
- WALLET\_GIFTCARD\_ROUNDED
- WALLET\_GIFTCARD\_SHARP
- WALLET\_MEMBERSHIP
- WALLET\_MEMBERSHIP\_OUTLINED
- WALLET\_MEMBERSHIP\_ROUNDED
- WALLET\_MEMBERSHIP\_SHARP
- WALLET\_OUTLINED
- WALLET\_ROUNDED
- WALLET\_SHARP
- WALLET\_TRAVEL

- WALLET\_TRAVEL\_OUTLINED
- WALLET\_TRAVEL\_ROUNDED
- WALLET\_TRAVEL\_SHARP
- WALLPAPER
- WALLPAPER\_OUTLINED
- WALLPAPER\_ROUNDED
- WALLPAPER\_SHARP
- WAREHOUSE
- WAREHOUSE\_OUTLINED
- WAREHOUSE\_ROUNDED
- WAREHOUSE\_SHARP
- WARNING
- WARNING\_AMBER
- WARNING\_AMBER\_OUTLINED
- WARNING\_AMBER\_ROUNDED
- WARNING\_AMBER\_SHARP
- WARNING\_OUTLINED
- WARNING\_ROUNDED
- WARNING\_SHARP
- WASH
- WASH\_OUTLINED
- WASH\_ROUNDED
- WASH\_SHARP
- WATCH
- WATCH\_LATER
- WATCH\_LATER\_OUTLINED
- WATCH\_LATER\_ROUNDED
- WATCH\_LATER\_SHARP
- WATCH\_OFF
- WATCH\_OFF\_OUTLINED
- WATCH\_OFF\_ROUNDED
- WATCH\_OFF\_SHARP
- WATCH\_OUTLINED
- WATCH\_ROUNDED
- WATCH\_SHARP
- WATER
- WATER\_DAMAGE
- WATER\_DAMAGE\_OUTLINED
- WATER\_DAMAGE\_ROUNDED
- WATER\_DAMAGE\_SHARP
- WATER\_DROP

- WATER\_DROP\_OUTLINED
- WATER\_DROP\_ROUNDED
- WATER\_DROP\_SHARP
- WATER\_OUTLINED
- WATER\_ROUNDED
- WATER\_SHARP
- WATERFALL\_CHART
- WATERFALL\_CHART\_OUTLINED
- WATERFALL\_CHART\_ROUNDED
- WATERFALL\_CHART\_SHARP
- WAVES
- WAVES\_OUTLINED
- WAVES\_ROUNDED
- WAVES\_SHARP
- WAVING\_HAND
- WAVING\_HAND\_OUTLINED
- WAVING\_HAND\_ROUNDED
- WAVING\_HAND\_SHARP
- WB\_AUTO
- WB\_AUTO\_OUTLINED
- WB\_AUTO\_ROUNDED
- WB\_AUTO\_SHARP
- WB\_CLOUDY
- WB\_CLOUDY\_OUTLINED
- WB\_CLOUDY\_ROUNDED
- WB\_CLOUDY\_SHARP
- WB\_INCANDESCENT
- WB\_INCANDESCENT\_OUTLINED
- WB\_INCANDESCENT\_ROUNDED
- WB\_INCANDESCENT\_SHARP
- WB\_IRIDESCENT
- WB\_IRIDESCENT\_OUTLINED
- WB\_IRIDESCENT\_ROUNDED
- WB\_IRIDESCENT\_SHARP
- WB\_SHADE
- WB\_SHADE\_OUTLINED
- WB\_SHADE\_ROUNDED
- WB\_SHADE\_SHARP
- WB\_SUNNY
- WB\_SUNNY\_OUTLINED
- WB\_SUNNY\_ROUNDED

- WB\_SUNNY\_SHARP
- WB\_TWIGLIGHT
- WB\_TWILIGHT
- WB\_TWILIGHT\_OUTLINED
- WB\_TWILIGHT\_ROUNDED
- WB\_TWILIGHT\_SHARP
- WC
- WC\_OUTLINED
- WC\_ROUNDED
- WC\_SHARP
- WEB
- WEB\_ASSET
- WEB\_ASSET\_OFF
- WEB\_ASSET\_OFF\_OUTLINED
- WEB\_ASSET\_OFF\_ROUNDED
- WEB\_ASSET\_OFF\_SHARP
- WEB\_ASSET\_OUTLINED
- WEB\_ASSET\_ROUNDED
- WEB\_ASSET\_SHARP
- WEB\_OUTLINED
- WEB\_ROUNDED
- WEB\_SHARP
- WEB\_STORIES
- WEB\_STORIES\_OUTLINED
- WEB\_STORIES\_ROUNDED
- WEB\_STORIES\_SHARP
- WEBHOOK
- WEBHOOK\_OUTLINED
- WEBHOOK\_ROUNDED
- WEBHOOK\_SHARP
- WECHAT
- WECHAT\_OUTLINED
- WECHAT\_ROUNDED
- WECHAT\_SHARP
- WEEKEND
- WEEKEND\_OUTLINED
- WEEKEND\_ROUNDED
- WEEKEND\_SHARP
- WEST
- WEST\_OUTLINED
- WEST\_ROUNDED

- WEST\_SHARP
- WHATSHOT
- WHATSHOT\_OUTLINED
- WHATSHOT\_ROUNDED
- WHATSHOT\_SHARP
- WHEELCHAIR\_PICKUP
- WHEELCHAIR\_PICKUP\_OUTLINED
- WHEELCHAIR\_PICKUP\_ROUNDED
- WHEELCHAIR\_PICKUP\_SHARP
- WHERE\_TO\_VOTE
- WHERE\_TO\_VOTE\_OUTLINED
- WHERE\_TO\_VOTE\_ROUNDED
- WHERE\_TO\_VOTE\_SHARP
- WIDGETS
- WIDGETS\_OUTLINED
- WIDGETS\_ROUNDED
- WIDGETS\_SHARP
- WIDTH\_FULL
- WIDTH\_FULL\_OUTLINED
- WIDTH\_FULL\_ROUNDED
- WIDTH\_FULL\_SHARP
- WIDTH\_NORMAL
- WIDTH\_NORMAL\_OUTLINED
- WIDTH\_NORMAL\_ROUNDED
- WIDTH\_NORMAL\_SHARP
- WIDTH\_WIDE
- WIDTH\_WIDE\_OUTLINED
- WIDTH\_WIDE\_ROUNDED
- WIDTH\_WIDE\_SHARP
- WIFI
- WIFI\_1\_BAR
- WIFI\_1\_BAR\_OUTLINED
- WIFI\_1\_BAR\_ROUNDED
- WIFI\_1\_BAR\_SHARP
- WIFI\_2\_BAR
- WIFI\_2\_BAR\_OUTLINED
- WIFI\_2\_BAR\_ROUNDED
- WIFI\_2\_BAR\_SHARP
- WIFI\_CALLING
- WIFI\_CALLING\_3
- WIFI\_CALLING\_3\_OUTLINED

- WIFI\_CALLING\_3\_ROUNDED
- WIFI\_CALLING\_3\_SHARP
- WIFI\_CALLING\_OUTLINED
- WIFI\_CALLING\_ROUNDED
- WIFI\_CALLING\_SHARP
- WIFI\_CHANNEL
- WIFI\_CHANNEL\_OUTLINED
- WIFI\_CHANNEL\_ROUNDED
- WIFI\_CHANNEL\_SHARP
- WIFI\_FIND
- WIFI\_FIND\_OUTLINED
- WIFI\_FIND\_ROUNDED
- WIFI\_FIND\_SHARP
- WIFI\_LOCK
- WIFI\_LOCK\_OUTLINED
- WIFI\_LOCK\_ROUNDED
- WIFI\_LOCK\_SHARP
- WIFI\_OFF
- WIFI\_OFF\_OUTLINED
- WIFI\_OFF\_ROUNDED
- WIFI\_OFF\_SHARP
- WIFI\_OUTLINED
- WIFI\_PASSWORD
- WIFI\_PASSWORD\_OUTLINED
- WIFI\_PASSWORD\_ROUNDED
- WIFI\_PASSWORD\_SHARP
- WIFI\_PROTECTED\_SETUP
- WIFI\_PROTECTED\_SETUP\_OUTLINED
- WIFI\_PROTECTED\_SETUP\_ROUNDED
- WIFI\_PROTECTED\_SETUP\_SHARP
- WIFI\_ROUNDED
- WIFI\_SHARP
- WIFI\_TETHERING
- WIFI\_TETHERING\_ERROR
- WIFI\_TETHERING\_ERROR\_OUTLINED
- WIFI\_TETHERING\_ERROR\_ROUNDED
- WIFI\_TETHERING\_ERROR\_ROUNDED\_OUTLINED
- WIFI\_TETHERING\_ERROR\_ROUNDED\_ROUNDED
- WIFI\_TETHERING\_ERROR\_ROUNDED\_SHARP
- WIFI\_TETHERING\_ERROR\_SHARP
- WIFI\_TETHERING\_OFF

- WIFI\_TETHERING\_OFF\_OUTLINED
- WIFI\_TETHERING\_OFF\_ROUNDED
- WIFI\_TETHERING\_OFF\_SHARP
- WIFI\_TETHERING\_OUTLINED
- WIFI\_TETHERING\_ROUNDED
- WIFI\_TETHERING\_SHARP
- WIND\_POWER
- WIND\_POWER\_OUTLINED
- WIND\_POWER\_ROUNDED
- WIND\_POWER\_SHARP
- WINDOW
- WINDOW\_OUTLINED
- WINDOW\_ROUNDED
- WINDOW\_SHARP
- WINE\_BAR
- WINE\_BAR\_OUTLINED
- WINE\_BAR\_ROUNDED
- WINE\_BAR\_SHARP
- WOMAN
- WOMAN\_2
- WOMAN\_2\_OUTLINED
- WOMAN\_2\_ROUNDED
- WOMAN\_2\_SHARP
- WOMAN\_OUTLINED
- WOMAN\_ROUNDED
- WOMAN\_SHARP
- WOO\_COMMERCE
- WOO\_COMMERCE\_OUTLINED
- WOO\_COMMERCE\_ROUNDED
- WOO\_COMMERCE\_SHARP
- WORDPRESS
- WORDPRESS\_OUTLINED
- WORDPRESS\_ROUNDED
- WORDPRESS\_SHARP
- WORK
- WORK\_HISTORY
- WORK\_HISTORY\_OUTLINED
- WORK\_HISTORY\_ROUNDED
- WORK\_HISTORY\_SHARP
- WORK\_OFF
- WORK\_OFF\_OUTLINED

- WORK\_OFF\_ROUNDED
- WORK\_OFF\_SHARP
- WORK\_OUTLINE
- WORK\_OUTLINE\_OUTLINED
- WORK\_OUTLINE\_ROUNDED
- WORK\_OUTLINE\_SHARP
- WORK\_OUTLINED
- WORK\_ROUNDED
- WORK\_SHARP
- WORKSPACE\_PREMIUM
- WORKSPACE\_PREMIUM\_OUTLINED
- WORKSPACE\_PREMIUM\_ROUNDED
- WORKSPACE\_PREMIUM\_SHARP
- WORKSPACES
- WORKSPACES\_FILLED
- WORKSPACES\_OUTLINE
- WORKSPACES\_OUTLINED
- WORKSPACES\_ROUNDED
- WORKSPACES\_SHARP
- WRAP\_TEXT
- WRAP\_TEXT\_OUTLINED
- WRAP\_TEXT\_ROUNDED
- WRAP\_TEXT\_SHARP
- WRONG\_LOCATION
- WRONG\_LOCATION\_OUTLINED
- WRONG\_LOCATION\_ROUNDED
- WRONG\_LOCATION\_SHARP
- WYSIWYG
- WYSIWYG\_OUTLINED
- WYSIWYG\_ROUNDED
- WYSIWYG\_SHARP
- YARD
- YARD\_OUTLINED
- YARD\_ROUNDED
- YARD\_SHARP
- YOUTUBE\_SEARCHED\_FOR
- YOUTUBE\_SEARCHED\_FOR\_OUTLINED
- YOUTUBE\_SEARCHED\_FOR\_ROUNDED
- YOUTUBE\_SEARCHED\_FOR\_SHARP
- ZOOM\_IN
- ZOOM\_IN\_MAP



- ZOOM\_IN\_MAP\_OUTLINED
- ZOOM\_IN\_MAP\_ROUNDED
- ZOOM\_IN\_MAP\_SHARP
- ZOOM\_IN\_OUTLINED
- ZOOM\_IN\_ROUNDED
- ZOOM\_IN\_SHARP
- ZOOM\_OUT
- ZOOM\_OUT\_MAP
- ZOOM\_OUT\_MAP\_OUTLINED
- ZOOM\_OUT\_MAP\_ROUNDED
- ZOOM\_OUT\_MAP\_SHARP
- ZOOM\_OUT\_OUTLINED
- ZOOM\_OUT\_ROUNDED
- ZOOM\_OUT\_SHARP

## Properties

**buildABCinstance-attribute**

ABC: IconData

**buildABC\_OUTLINEDinstance-attribute**

ABC\_OUTLINED: IconData

**buildABC\_ROUNDEDinstance-attribute**

ABC\_ROUNDED: IconData

**buildABC\_SHARPinstance-attribute**

ABC\_SHARP: IconData

**buildAC\_UNITinstance-attribute**

AC\_UNIT: IconData

**buildAC\_UNIT\_OUTLINEDinstance-attribute**

AC\_UNIT\_OUTLINED: IconData

**buildAC\_UNIT\_ROUNDEDinstance-attribute**

AC\_UNIT\_ROUNDED: IconData

**buildAC\_UNIT\_SHARPinstance-attribute**

AC\_UNIT\_SHARP: IconData

**buildACCESS\_ALARMinstance-attribute**□□

ACCESS\_ALARM: IconData

**buildACCESS\_ALARM\_OUTLINEDinstance-attribute**

ACCESS\_ALARM\_OUTLINED: IconData

**buildACCESS\_ALARM\_ROUNDEDinstance-attribute**

ACCESS\_ALARM\_ROUNDED: IconData

**buildACCESS\_ALARM\_SHARPinstance-attribute**

ACCESS\_ALARM\_SHARP: IconData

**buildACCESS\_ALARMSinstance-attribute**

ACCESS\_ALARMS: IconData

**buildACCESS\_ALARMS\_OUTLINEDinstance-attribute**

ACCESS\_ALARMS\_OUTLINED: IconData

**buildACCESS\_ALARMS\_ROUNDEDinstance-attribute**

ACCESS\_ALARMS\_ROUNDED: IconData

**buildACCESS\_ALARMS\_SHARPinstance-attribute**

ACCESS\_ALARMS\_SHARP: IconData

**buildACCESS\_TIMEinstance-attribute**

ACCESS\_TIME: IconData

**buildACCESS\_TIME\_FILLEDinstance-attribute**

ACCESS\_TIME\_FILLED: IconData

**buildACCESS\_TIME\_FILLED\_OUTLINEDinstance-attribute**

ACCESS\_TIME\_FILLED\_OUTLINED: IconData

**buildACCESS\_TIME\_FILLED\_ROUNDEDinstance-attribute**

ACCESS\_TIME\_FILLED\_ROUNDED: IconData

**buildACCESS\_TIME\_FILLED\_SHARPinstance-attribute**

ACCESS\_TIME\_FILLED\_SHARP: IconData

**buildACCESS\_TIME\_OUTLINEDinstance-attribute**

ACCESS\_TIME\_OUTLINED: IconData

**buildACCESS\_TIME\_ROUNDEDinstance-attribute**

ACCESS\_TIME\_ROUNDED: IconData

**buildACCESS\_TIME\_SHARPinstance-attribute**

ACCESS\_TIME\_SHARP: IconData

**buildACCESSIBILITYinstance-attribute**

ACCESSIBILITY: IconData

**buildACCESSIBILITY\_NEWinstance-attribute**

ACCESSIBILITY\_NEW: IconData

**buildACCESSIBILITY\_NEW\_OUTLINEDinstance-attribute**

ACCESSIBILITY\_NEW\_OUTLINED: IconData

**buildACCESSIBILITY\_NEW\_ROUNDEDinstance-attribute**

ACCESSIBILITY\_NEW\_ROUNDED: IconData

**buildACCESSIBILITY\_NEW\_SHARPinstance-attribute**

ACCESSIBILITY\_NEW\_SHARP: IconData

**buildACCESSIBILITY\_OUTLINEDinstance-attribute**

ACCESSIBILITY\_OUTLINED: IconData

**buildACCESSIBILITY\_ROUNDEDinstance-attribute**

ACCESSIBILITY\_ROUNDED: IconData

**buildACCESSIBILITY\_SHARPinstance-attribute**

ACCESSIBILITY\_SHARP: IconData

**buildACCESSIBLEinstance-attribute**

ACCESSIBLE: IconData

**buildACCESSIBLE\_FORWARDinstance-attribute**

ACCESSIBLE\_FORWARD: IconData

**buildACCESSIBLE\_FORWARD\_OUTLINEDinstance-attribute**

ACCESSIBLE\_FORWARD\_OUTLINED: IconData

**buildACCESSIBLE\_FORWARD\_ROUNDEDinstance-attribute**

ACCESSIBLE\_FORWARD\_ROUNDED: IconData

**buildACCESSIBLE\_FORWARD\_SHARPinstance-attribute**

ACCESSIBLE\_FORWARD\_SHARP: IconData

**buildACCESSIBLE\_OUTLINEDinstance-attribute**

ACCESSIBLE\_OUTLINED: IconData

**buildACCESSIBLE\_ROUNDEDinstance-attribute**

ACCESSIBLE\_ROUNDED: IconData

**buildACCESSIBLE\_SHARPinstance-attribute**

ACCESSIBLE\_SHARP: IconData

**buildACCOUNT\_BALANCEinstance-attribute**

ACCOUNT\_BALANCE: IconData

**buildACCOUNT\_BALANCE\_OUTLINEDinstance-attribute**

ACCOUNT\_BALANCE\_OUTLINED: IconData

**buildACCOUNT\_BALANCE\_ROUNDEDinstance-attribute**

ACCOUNT\_BALANCE\_ROUNDED: IconData

**buildACCOUNT\_BALANCE\_SHARPinstance-attribute**

ACCOUNT\_BALANCE\_SHARP: IconData

**buildACCOUNT\_BALANCE\_WALLETinstance-attribute**

ACCOUNT\_BALANCE\_WALLET: IconData

**buildACCOUNT\_BALANCE\_WALLET\_OUTLINEDinstance-attribute**

ACCOUNT\_BALANCE\_WALLET\_OUTLINED: IconData

**buildACCOUNT\_BALANCE\_WALLET\_ROUNDEDinstance-attribute**

ACCOUNT\_BALANCE\_WALLET\_ROUNDED: IconData

**buildACCOUNT\_BALANCE\_WALLET\_SHARPinstance-attribute**

ACCOUNT\_BALANCE\_WALLET\_SHARP: IconData

**buildACCOUNT\_BOXinstance-attribute**

ACCOUNT\_BOX: IconData

**buildACCOUNT\_BOX\_OUTLINEDinstance-attribute**

ACCOUNT\_BOX\_OUTLINED: IconData

**buildACCOUNT\_BOX\_ROUNDEDinstance-attribute**

ACCOUNT\_BOX\_ROUNDED: IconData

**buildACCOUNT\_BOX\_SHARPinstance-attribute**

ACCOUNT\_BOX\_SHARP: IconData

**buildACCOUNT\_CIRCLEinstance-attribute**

ACCOUNT\_CIRCLE: IconData

**buildACCOUNT\_CIRCLE\_OUTLINEDinstance-attribute**

ACCOUNT\_CIRCLE\_OUTLINED: IconData

**buildACCOUNT\_CIRCLE\_ROUNDEDinstance-attribute**

ACCOUNT\_CIRCLE\_ROUNDED: IconData

**buildACCOUNT\_CIRCLE\_SHARPinstance-attribute**

ACCOUNT\_CIRCLE\_SHARP: IconData

**buildACCOUNT\_TREEinstance-attribute**

ACCOUNT\_TREE: IconData

**buildACCOUNT\_TREE\_OUTLINEDinstance-attribute**

ACCOUNT\_TREE\_OUTLINED: IconData

**buildACCOUNT\_TREE\_ROUNDEDinstance-attribute**

ACCOUNT\_TREE\_ROUNDED: IconData

**buildACCOUNT\_TREE\_SHARPinstance-attribute**

ACCOUNT\_TREE\_SHARP: IconData

**buildAD\_UNITSinstance-attribute**

AD\_UNITS: IconData

**buildAD\_UNITS\_OUTLINEDinstance-attribute**

AD\_UNITS\_OUTLINED: IconData

**buildAD\_UNITS\_ROUNDEDinstance-attribute**

AD\_UNITS\_ROUNDED: IconData

**buildAD\_UNITS\_SHARPinstance-attribute**

AD\_UNITS\_SHARP: IconData

**buildADBinstance-attribute**

ADB: IconData

**buildADB\_OUTLINEDinstance-attribute**

ADB\_OUTLINED: IconData

**buildADB\_ROUNDEDinstance-attribute**

ADB\_ROUNDED: IconData

**buildADB\_SHARPinstance-attribute**

ADB\_SHARP: IconData

**buildADDinstance-attribute**

ADD: IconData

**buildADD\_A\_PHOTOinstance-attribute**

ADD\_A\_PHOTO: IconData

**buildADD\_A\_PHOTO\_OUTLINEDinstance-attribute**

ADD\_A\_PHOTO\_OUTLINED: IconData

**buildADD\_A\_PHOTO\_ROUNDEDinstance-attribute**

ADD\_A\_PHOTO\_ROUNDED: IconData

**buildADD\_A\_PHOTO\_SHARPinstance-attribute**

ADD\_A\_PHOTO\_SHARP: IconData

**buildADD\_ALARMinstance-attribute**

ADD\_ALARM: IconData

**buildADD\_ALARM\_OUTLINEDinstance-attribute**

ADD\_ALARM\_OUTLINED: IconData

**buildADD\_ALARM\_ROUNDEDinstance-attribute**

ADD\_ALARM\_ROUNDED: IconData

**buildADD\_ALARM\_SHARPinstance-attribute**

ADD\_ALARM\_SHARP: IconData

**buildADD\_ALERTinstance-attribute**

ADD\_ALERT: IconData

**buildADD\_ALERT\_OUTLINEDinstance-attribute**

ADD\_ALERT\_OUTLINED: IconData

**buildADD\_ALERT\_ROUNDEDinstance-attribute**

ADD\_ALERT\_ROUNDED: IconData

**buildADD\_ALERT\_SHARPinstance-attribute**

ADD\_ALERT\_SHARP: IconData

**buildADD\_BOXinstance-attribute**

ADD\_BOX: IconData

**buildADD\_BOX\_OUTLINEDinstance-attribute**

ADD\_BOX\_OUTLINED: IconData

**buildADD\_BOX\_ROUNDEDinstance-attribute**

ADD\_BOX\_ROUNDED: IconData

**buildADD\_BOX\_SHARPinstance-attribute**

ADD\_BOX\_SHARP: IconData

**buildADD\_BUSINESSinstance-attribute**

ADD\_BUSINESS: IconData

**buildADD\_BUSINESS\_OUTLINEDinstance-attribute**

ADD\_BUSINESS\_OUTLINED: IconData

**buildADD\_BUSINESS\_ROUNDEDinstance-attribute**

ADD\_BUSINESS\_ROUNDED: IconData

**buildADD\_BUSINESS\_SHARPinstance-attribute**

ADD\_BUSINESS\_SHARP: IconData

**buildADD\_CALLinstance-attribute**

ADD\_CALL: IconData

**buildADD\_CARDinstance-attribute**

ADD\_CARD: IconData

**buildADD\_CARD\_OUTLINEDinstance-attribute**

ADD\_CARD\_OUTLINED: IconData

**buildADD\_CARD\_ROUNDEDinstance-attribute**

ADD\_CARD\_ROUNDED: IconData

**buildADD\_CARD\_SHARPinstance-attribute**

ADD\_CARD\_SHARP: IconData

**buildADD\_CHARTinstance-attribute**

ADD\_CHART: IconData

**buildADD\_CHART\_OUTLINEDinstance-attribute**

ADD\_CHART\_OUTLINED: IconData

**buildADD\_CHART\_ROUNDEDinstance-attribute**

ADD\_CHART\_ROUNDED: IconData

**buildADD\_CHART\_SHARPinstance-attribute**

ADD\_CHART\_SHARP: IconData

**buildADD\_CIRCLEinstance-attribute**

ADD\_CIRCLE: IconData

**buildADD\_CIRCLE\_OUTLINEinstance-attribute**

ADD\_CIRCLE\_OUTLINE: IconData

**buildADD\_CIRCLE\_OUTLINE\_OUTLINEDinstance-attribute**

ADD\_CIRCLE\_OUTLINE\_OUTLINED: IconData

**buildADD\_CIRCLE\_OUTLINE\_ROUNDEDinstance-attribute**

ADD\_CIRCLE\_OUTLINE\_ROUNDED: IconData

**buildADD\_CIRCLE\_OUTLINE\_SHARPinstance-attribute**

ADD\_CIRCLE\_OUTLINE\_SHARP: IconData

**buildADD\_CIRCLE\_OUTLINEDinstance-attribute**

ADD\_CIRCLE\_OUTLINED: IconData

**buildADD\_CIRCLE\_ROUNDEDinstance-attribute**

ADD\_CIRCLE\_ROUNDED: IconData

**buildADD\_CIRCLE\_SHARPinstance-attribute**

ADD\_CIRCLE\_SHARP: IconData

**buildADD\_COMMENTinstance-attribute**

ADD\_COMMENT: IconData

**buildADD\_COMMENT\_OUTLINEDinstance-attribute**

ADD\_COMMENT\_OUTLINED: IconData

**buildADD\_COMMENT\_ROUNDEDinstance-attribute**

ADD\_COMMENT\_ROUNDED: IconData

**buildADD\_COMMENT\_SHARPinstance-attribute**

ADD\_COMMENT\_SHARP: IconData

**buildADD\_HOMEinstance-attribute**

ADD\_HOME: IconData

**buildADD\_HOME\_OUTLINEDinstance-attribute**

ADD\_HOME\_OUTLINED: IconData

**buildADD\_HOME\_ROUNDEDinstance-attribute**

ADD\_HOME\_ROUNDED: IconData

**buildADD\_HOME\_SHARPinstance-attribute**

ADD\_HOME\_SHARP: IconData

**buildADD\_HOME\_WORKinstance-attribute**

ADD\_HOME\_WORK: IconData

**buildADD\_HOME\_WORK\_OUTLINEDinstance-attribute**

ADD\_HOME\_WORK\_OUTLINED: IconData

**buildADD\_HOME\_WORK\_ROUNDEDinstance-attribute**

ADD\_HOME\_WORK\_ROUNDED: IconData

**buildADD\_HOME\_WORK\_SHARPinstance-attribute**

ADD\_HOME\_WORK\_SHARP: IconData

**buildADD\_IC\_CALLinstance-attribute**

ADD\_IC\_CALL: IconData

**buildADD\_IC\_CALL\_OUTLINEDinstance-attribute**

ADD\_IC\_CALL\_OUTLINED: IconData

**buildADD\_IC\_CALL\_ROUNDEDinstance-attribute**

ADD\_IC\_CALL\_ROUNDED: IconData

**buildADD\_IC\_CALL\_SHARPinstance-attribute**

ADD\_IC\_CALL\_SHARP: IconData

**buildADD\_LINKinstance-attribute**

ADD\_LINK: IconData

**buildADD\_LINK\_OUTLINEDinstance-attribute**

ADD\_LINK\_OUTLINED: IconData

**buildADD\_LINK\_ROUNDEDinstance-attribute**

ADD\_LINK\_ROUNDED: IconData

**buildADD\_LINK\_SHARPinstance-attribute**

ADD\_LINK\_SHARP: IconData

**buildADD\_LOCATIONinstance-attribute**

ADD\_LOCATION: IconData

**buildADD\_LOCATION\_ALTinstance-attribute**

ADD\_LOCATION\_ALT: IconData

**buildADD\_LOCATION\_ALT\_OUTLINEDinstance-attribute**

ADD\_LOCATION\_ALT\_OUTLINED: IconData

**buildADD\_LOCATION\_ALT\_ROUNDEDinstance-attribute**

ADD\_LOCATION\_ALT\_ROUNDED: IconData

**buildADD\_LOCATION\_ALT\_SHARPinstance-attribute**

ADD\_LOCATION\_ALT\_SHARP: IconData

**buildADD\_LOCATION\_OUTLINEDinstance-attribute**

ADD\_LOCATION\_OUTLINED: IconData

**buildADD\_LOCATION\_ROUNDEDinstance-attribute**

ADD\_LOCATION\_ROUNDED: IconData

**buildADD\_LOCATION\_SHARPinstance-attribute**

ADD\_LOCATION\_SHARP: IconData

**buildADD\_MODERATORinstance-attribute**

ADD\_MODERATOR: IconData

**buildADD\_MODERATOR\_OUTLINEDinstance-attribute**

ADD\_MODERATOR\_OUTLINED: IconData

**buildADD\_MODERATOR\_ROUNDEDinstance-attribute**

ADD\_MODERATOR\_ROUNDED: IconData

**buildADD\_MODERATOR\_SHARPinstance-attribute**

ADD\_MODERATOR\_SHARP: IconData

**buildADD\_OUTLINEDinstance-attribute**

ADD\_OUTLINED: IconData

**buildADD\_PHOTO\_ALTERNATEinstance-attribute**

ADD\_PHOTO\_ALTERNATE: IconData

**buildADD\_PHOTO\_ALTERNATE\_OUTLINEDinstance-attribute**

ADD\_PHOTO\_ALTERNATE\_OUTLINED: IconData

**buildADD\_PHOTO\_ALTERNATE\_ROUNDEDinstance-attribute**

ADD\_PHOTO\_ALTERNATE\_ROUNDED: IconData

**buildADD\_PHOTO\_ALTERNATE\_SHARPinstance-attribute**

ADD\_PHOTO\_ALTERNATE\_SHARP: IconData

**buildADD\_REACTIONinstance-attribute**

ADD\_REACTION: IconData

**buildADD\_REACTION\_OUTLINEDinstance-attribute**

ADD\_REACTION\_OUTLINED: IconData

**buildADD\_REACTION\_ROUNDEDinstance-attribute**

ADD\_REACTION\_ROUNDED: IconData

**buildADD\_REACTION\_SHARPinstance-attribute**

ADD\_REACTION\_SHARP: IconData

**buildADD\_ROADinstance-attribute**

ADD\_ROAD: IconData

**buildADD\_ROAD\_OUTLINEDinstance-attribute**

ADD\_ROAD\_OUTLINED: IconData

**buildADD\_ROAD\_ROUNDEDinstance-attribute**

ADD\_ROAD\_ROUNDED: IconData

**buildADD\_ROAD\_SHARPinstance-attribute**

ADD\_ROAD\_SHARP: IconData

**buildADD\_ROUNDEDinstance-attribute**

ADD\_ROUNDED: IconData

**buildADD\_SHARPinstance-attribute**

ADD\_SHARP: IconData

**buildADD\_SHOPPING\_CARTinstance-attribute**

ADD\_SHOPPING\_CART: IconData

**buildADD\_SHOPPING\_CART\_OUTLINEDinstance-attribute**

ADD\_SHOPPING\_CART\_OUTLINED: IconData

**buildADD\_SHOPPING\_CART\_ROUNDEDinstance-attribute**

ADD\_SHOPPING\_CART\_ROUNDED: IconData

**buildADD\_SHOPPING\_CART\_SHARPinstance-attribute**

ADD\_SHOPPING\_CART\_SHARP: IconData

**buildADD\_TASKinstance-attribute**

ADD\_TASK: IconData

**buildADD\_TASK\_OUTLINEDinstance-attribute**

ADD\_TASK\_OUTLINED: IconData

**buildADD\_TASK\_ROUNDEDinstance-attribute**

ADD\_TASK\_ROUNDED: IconData

**buildADD\_TASK\_SHARPinstance-attribute**

ADD\_TASK\_SHARP: IconData

**buildADD\_TO\_DRIVEinstance-attribute**

ADD\_TO\_DRIVE: IconData

**buildADD\_TO\_DRIVE\_OUTLINEDinstance-attribute**

ADD\_TO\_DRIVE\_OUTLINED: IconData

**buildADD\_TO\_DRIVE\_ROUNDEDinstance-attribute**

ADD\_TO\_DRIVE\_ROUNDED: IconData

**buildADD\_TO\_DRIVE\_SHARPinstance-attribute**

ADD\_TO\_DRIVE\_SHARP: IconData

**buildADD\_TO\_HOME\_SCREENinstance-attribute**

ADD\_TO\_HOME\_SCREEN: IconData

**buildADD\_TO\_HOME\_SCREEN\_OUTLINEDinstance-attribute**

ADD\_TO\_HOME\_SCREEN\_OUTLINED: IconData

**buildADD\_TO\_HOME\_SCREEN\_ROUNDEDinstance-attribute**

ADD\_TO\_HOME\_SCREEN\_ROUNDED: IconData



**buildADD\_TO\_HOME\_SCREEN\_SHARPinstance-attribute**

ADD\_TO\_HOME\_SCREEN\_SHARP: IconData

**buildADD\_TO\_PHOTOSinstance-attribute**

ADD\_TO\_PHOTOS: IconData

**buildADD\_TO\_PHOTOS\_OUTLINEDinstance-attribute**

ADD\_TO\_PHOTOS\_OUTLINED: IconData

**buildADD\_TO\_PHOTOS\_ROUNDEDinstance-attribute**

ADD\_TO\_PHOTOS\_ROUNDED: IconData

**buildADD\_TO\_PHOTOS\_SHARPinstance-attribute**

ADD\_TO\_PHOTOS\_SHARP: IconData

**buildADD\_TO\_QUEUEinstance-attribute**

ADD\_TO\_QUEUE: IconData

**buildADD\_TO\_QUEUE\_OUTLINEDinstance-attribute**

ADD\_TO\_QUEUE\_OUTLINED: IconData

**buildADD\_TO\_QUEUE\_ROUNDEDinstance-attribute**

ADD\_TO\_QUEUE\_ROUNDED: IconData

**buildADD\_TO\_QUEUE\_SHARPinstance-attribute**

ADD\_TO\_QUEUE\_SHARP: IconData

**buildADDCHARTinstance-attribute**

ADDCHART: IconData

**buildADDCHART\_OUTLINEDinstance-attribute**

ADDCHART\_OUTLINED: IconData

**buildADDCHART\_ROUNDEDinstance-attribute**

ADDCHART\_ROUNDED: IconData

**buildADDCHART\_SHARPinstance-attribute**

ADDCHART\_SHARP: IconData

**buildADF\_SCANNERinstance-attribute**

ADF\_SCANNER: IconData

**buildADF\_SCANNER\_OUTLINEDinstance-attribute**

ADF\_SCANNER\_OUTLINED: IconData

**buildADF\_SCANNER\_ROUNDEDinstance-attribute**

ADF\_SCANNER\_ROUNDED: IconData

**buildADF\_SCANNER\_SHARPinstance-attribute**

ADF\_SCANNER\_SHARP: IconData

**buildADJUSTinstance-attribute**

ADJUST: IconData

**buildADJUST\_OUTLINEDinstance-attribute**

ADJUST\_OUTLINED: IconData

**buildADJUST\_ROUNDEDinstance-attribute**

ADJUST\_ROUNDED: IconData

**buildADJUST\_SHARPinstance-attribute**

ADJUST\_SHARP: IconData

**buildADMIN\_PANEL\_SETTINGSinstance-attribute**

ADMIN\_PANEL\_SETTINGS: IconData

**buildADMIN\_PANEL\_SETTINGS\_OUTLINED**instance-attribute

ADMIN\_PANEL\_SETTINGS\_OUTLINED: IconData

**buildADMIN\_PANEL\_SETTINGS\_ROUNDED**instance-attribute

ADMIN\_PANEL\_SETTINGS\_ROUNDED: IconData

**buildADMIN\_PANEL\_SETTINGS\_SHARP**instance-attribute

ADMIN\_PANEL\_SETTINGS\_SHARP: IconData

**buildADOBE**instance-attribute

ADOBE: IconData

**buildADOBE\_OUTLINED**instance-attribute

ADOBE\_OUTLINED: IconData

**buildADOBE\_ROUNDED**instance-attribute

ADOBE\_ROUNDED: IconData

**buildADOBE\_SHARP**instance-attribute

ADOBE\_SHARP: IconData

**buildADS\_CLICK**instance-attribute

ADS\_CLICK: IconData

**buildADS\_CLICK\_OUTLINED**instance-attribute

ADS\_CLICK\_OUTLINED: IconData

**buildADS\_CLICK\_ROUNDED**instance-attribute

ADS\_CLICK\_ROUNDED: IconData

**buildADS\_CLICK\_SHARP**instance-attribute

ADS\_CLICK\_SHARP: IconData

**buildAGRICULTURE**instance-attribute

AGRICULTURE: IconData

**buildAGRICULTURE\_OUTLINED**instance-attribute

AGRICULTURE\_OUTLINED: IconData

**buildAGRICULTURE\_ROUNDED**instance-attribute

AGRICULTURE\_ROUNDED: IconData

**buildAGRICULTURE\_SHARP**instance-attribute

AGRICULTURE\_SHARP: IconData

**buildAIR**instance-attribute

AIR: IconData

**buildAIR\_OUTLINED**instance-attribute

AIR\_OUTLINED: IconData

**buildAIR\_ROUNDED**instance-attribute

AIR\_ROUNDED: IconData

**buildAIR\_SHARP**instance-attribute

AIR\_SHARP: IconData

**buildAIRLINE\_SEAT\_FLAT**instance-attribute

AIRLINE\_SEAT\_FLAT: IconData

**buildAIRLINE\_SEAT\_FLAT\_ANGLED**instance-attribute

AIRLINE\_SEAT\_FLAT\_ANGLED: IconData

**buildAIRLINE\_SEAT\_FLAT\_ANGLED\_OUTLINED**instance-attribute

AIRLINE\_SEAT\_FLAT\_ANGLED\_OUTLINED: IconData

buildAIRLINE\_SEAT\_FLAT\_ANGLED\_ROUNDEDinstance-attribute

AIRLINE\_SEAT\_FLAT\_ANGLED\_ROUNDED: IconData

buildAIRLINE\_SEAT\_FLAT\_ANGLED\_SHARPinstance-attribute

AIRLINE\_SEAT\_FLAT\_ANGLED\_SHARP: IconData

buildAIRLINE\_SEAT\_FLAT\_OUTLINEDinstance-attribute

AIRLINE\_SEAT\_FLAT\_OUTLINED: IconData

buildAIRLINE\_SEAT\_FLAT\_ROUNDEDinstance-attribute

AIRLINE\_SEAT\_FLAT\_ROUNDED: IconData

buildAIRLINE\_SEAT\_FLAT\_SHARPinstance-attribute

AIRLINE\_SEAT\_FLAT\_SHARP: IconData

buildAIRLINE\_SEAT\_INDIVIDUAL\_SUITEinstance-attribute

AIRLINE\_SEAT\_INDIVIDUAL\_SUITE: IconData

buildAIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_OUTLINEDinstance-attribute

AIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_OUTLINED: IconData

buildAIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_ROUNDEDinstance-attribute

AIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_ROUNDED: IconData

buildAIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_SHARPinstance-attribute

AIRLINE\_SEAT\_INDIVIDUAL\_SUITE\_SHARP: IconData

buildAIRLINE\_SEAT\_LEGROOM\_EXTRAinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_EXTRA: IconData

buildAIRLINE\_SEAT\_LEGROOM\_EXTRA\_OUTLINEDinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_EXTRA\_OUTLINED: IconData

buildAIRLINE\_SEAT\_LEGROOM\_EXTRA\_ROUNDEDinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_EXTRA\_ROUNDED: IconData

buildAIRLINE\_SEAT\_LEGROOM\_EXTRA\_SHARPinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_EXTRA\_SHARP: IconData

buildAIRLINE\_SEAT\_LEGROOM\_NORMALinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_NORMAL: IconData

buildAIRLINE\_SEAT\_LEGROOM\_NORMAL\_OUTLINEDinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_NORMAL\_OUTLINED: IconData

buildAIRLINE\_SEAT\_LEGROOM\_NORMAL\_ROUNDEDinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_NORMAL\_ROUNDED: IconData

buildAIRLINE\_SEAT\_LEGROOM\_NORMAL\_SHARPinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_NORMAL\_SHARP: IconData

buildAIRLINE\_SEAT\_LEGROOM\_REDUCEDinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_REDUCED: IconData

buildAIRLINE\_SEAT\_LEGROOM\_REDUCED\_OUTLINEDinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_REDUCED\_OUTLINED: IconData

buildAIRLINE\_SEAT\_LEGROOM\_REDUCED\_ROUNDEDinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_REDUCED\_ROUNDED: IconData

buildAIRLINE\_SEAT\_LEGROOM\_REDUCED\_SHARPinstance-attribute

AIRLINE\_SEAT\_LEGROOM\_REDUCED\_SHARP: IconData

buildAIRLINE\_SEAT\_RECLINE\_EXTRAinstance-attribute

AIRLINE\_SEAT\_RECLINE\_EXTRA: IconData

**buildAIRLINE\_SEAT\_RECLINE\_EXTRA\_OUTLINED**instance-attribute

AIRLINE\_SEAT\_RECLINE\_EXTRA\_OUTLINED: IconData

**buildAIRLINE\_SEAT\_RECLINE\_EXTRA\_ROUNDED**instance-attribute

AIRLINE\_SEAT\_RECLINE\_EXTRA\_ROUNDED: IconData

**buildAIRLINE\_SEAT\_RECLINE\_EXTRA\_SHARP**instance-attribute

AIRLINE\_SEAT\_RECLINE\_EXTRA\_SHARP: IconData

**buildAIRLINE\_SEAT\_RECLINE\_NORMAL**instance-attribute

AIRLINE\_SEAT\_RECLINE\_NORMAL: IconData

**buildAIRLINE\_SEAT\_RECLINE\_NORMAL\_OUTLINED**instance-attribute

AIRLINE\_SEAT\_RECLINE\_NORMAL\_OUTLINED: IconData

**buildAIRLINE\_SEAT\_RECLINE\_NORMAL\_ROUNDED**instance-attribute

AIRLINE\_SEAT\_RECLINE\_NORMAL\_ROUNDED: IconData

**buildAIRLINE\_SEAT\_RECLINE\_NORMAL\_SHARP**instance-attribute

AIRLINE\_SEAT\_RECLINE\_NORMAL\_SHARP: IconData

**buildAIRLINE\_STOPS**instance-attribute

AIRLINE\_STOPS: IconData

**buildAIRLINE\_STOPS\_OUTLINED**instance-attribute

AIRLINE\_STOPS\_OUTLINED: IconData

**buildAIRLINE\_STOPS\_ROUNDED**instance-attribute

AIRLINE\_STOPS\_ROUNDED: IconData

**buildAIRLINE\_STOPS\_SHARP**instance-attribute

AIRLINE\_STOPS\_SHARP: IconData

**buildAIRLINES**instance-attribute

AIRLINES: IconData

**buildAIRLINES\_OUTLINED**instance-attribute

AIRLINES\_OUTLINED: IconData

**buildAIRLINES\_ROUNDED**instance-attribute

AIRLINES\_ROUNDED: IconData

**buildAIRLINES\_SHARP**instance-attribute

AIRLINES\_SHARP: IconData

**buildAIRPLANE\_TICKET**instance-attribute

AIRPLANE\_TICKET: IconData

**buildAIRPLANE\_TICKET\_OUTLINED**instance-attribute

AIRPLANE\_TICKET\_OUTLINED: IconData

**buildAIRPLANE\_TICKET\_ROUNDED**instance-attribute

AIRPLANE\_TICKET\_ROUNDED: IconData

**buildAIRPLANE\_TICKET\_SHARP**instance-attribute

AIRPLANE\_TICKET\_SHARP: IconData

**buildAIRPLANEMODE\_ACTIVE**instance-attribute

AIRPLANEMODE\_ACTIVE: IconData

**buildAIRPLANEMODE\_ACTIVE\_OUTLINED**instance-attribute

AIRPLANEMODE\_ACTIVE\_OUTLINED: IconData

**buildAIRPLANEMODE\_ACTIVE\_ROUNDED**instance-attribute

AIRPLANEMODE\_ACTIVE\_ROUNDED: IconData

**buildAIRPLANEMODE\_ACTIVE\_SHARPinstance-attribute**

AIRPLANEMODE\_ACTIVE\_SHARP: IconData

**buildAIRPLANEMODE\_INACTIVEinstance-attribute**

AIRPLANEMODE\_INACTIVE: IconData

**buildAIRPLANEMODE\_INACTIVE\_OUTLINEDinstance-attribute**

AIRPLANEMODE\_INACTIVE\_OUTLINED: IconData

**buildAIRPLANEMODE\_INACTIVE\_ROUNDEDinstance-attribute**

AIRPLANEMODE\_INACTIVE\_ROUNDED: IconData

**buildAIRPLANEMODE\_INACTIVE\_SHARPinstance-attribute**

AIRPLANEMODE\_INACTIVE\_SHARP: IconData

**buildAIRPLANEMODE\_OFFinstance-attribute**

AIRPLANEMODE\_OFF: IconData

**buildAIRPLANEMODE\_OFF\_OUTLINEDinstance-attribute**

AIRPLANEMODE\_OFF\_OUTLINED: IconData

**buildAIRPLANEMODE\_OFF\_ROUNDEDinstance-attribute**

AIRPLANEMODE\_OFF\_ROUNDED: IconData

**buildAIRPLANEMODE\_OFF\_SHARPinstance-attribute**

AIRPLANEMODE\_OFF\_SHARP: IconData

**buildAIRPLANEMODE\_ONinstance-attribute**

AIRPLANEMODE\_ON: IconData

**buildAIRPLANEMODE\_ON\_OUTLINEDinstance-attribute**

AIRPLANEMODE\_ON\_OUTLINED: IconData

**buildAIRPLANEMODE\_ON\_ROUNDEDinstance-attribute**

AIRPLANEMODE\_ON\_ROUNDED: IconData

**buildAIRPLANEMODE\_ON\_SHARPinstance-attribute**

AIRPLANEMODE\_ON\_SHARP: IconData

**buildAIRPLAYinstance-attribute**

AIRPLAY: IconData

**buildAIRPLAY\_OUTLINEDinstance-attribute**

AIRPLAY\_OUTLINED: IconData

**buildAIRPLAY\_ROUNDEDinstance-attribute**

AIRPLAY\_ROUNDED: IconData

**buildAIRPLAY\_SHARPinstance-attribute**

AIRPLAY\_SHARP: IconData

**buildAIRPORT\_SHUTTLEinstance-attribute**

AIRPORT\_SHUTTLE: IconData

**buildAIRPORT\_SHUTTLE\_OUTLINEDinstance-attribute**

AIRPORT\_SHUTTLE\_OUTLINED: IconData

**buildAIRPORT\_SHUTTLE\_ROUNDEDinstance-attribute**

AIRPORT\_SHUTTLE\_ROUNDED: IconData

**buildAIRPORT\_SHUTTLE\_SHARPinstance-attribute**

AIRPORT\_SHUTTLE\_SHARP: IconData

**buildALARMinstance-attribute**

ALARM: IconData

**buildALARM\_ADDInstance-attribute**

ALARM\_ADD: IconData

**buildALARM\_ADD\_OUTLINEDInstance-attribute**

ALARM\_ADD\_OUTLINED: IconData

**buildALARM\_ADD\_ROUNDEDInstance-attribute**

ALARM\_ADD\_ROUNDED: IconData

**buildALARM\_ADD\_SHARPInstance-attribute**

ALARM\_ADD\_SHARP: IconData

**buildALARM\_OFFInstance-attribute**

ALARM\_OFF: IconData

**buildALARM\_OFF\_OUTLINEDInstance-attribute**

ALARM\_OFF\_OUTLINED: IconData

**buildALARM\_OFF\_ROUNDEDInstance-attribute**

ALARM\_OFF\_ROUNDED: IconData

**buildALARM\_OFF\_SHARPInstance-attribute**

ALARM\_OFF\_SHARP: IconData

**buildALARM\_ONInstance-attribute**

ALARM\_ON: IconData

**buildALARM\_ON\_OUTLINEDInstance-attribute**

ALARM\_ON\_OUTLINED: IconData

**buildALARM\_ON\_ROUNDEDInstance-attribute**

ALARM\_ON\_ROUNDED: IconData

**buildALARM\_ON\_SHARPInstance-attribute**

ALARM\_ON\_SHARP: IconData

**buildALARM\_OUTLINEDInstance-attribute**

ALARM\_OUTLINED: IconData

**buildALARM\_ROUNDEDInstance-attribute**

ALARM\_ROUNDED: IconData

**buildALARM\_SHARPInstance-attribute**

ALARM\_SHARP: IconData

**buildALBUMInstance-attribute**

ALBUM: IconData

**buildALBUM\_OUTLINEDInstance-attribute**

ALBUM\_OUTLINED: IconData

**buildALBUM\_ROUNDEDInstance-attribute**

ALBUM\_ROUNDED: IconData

**buildALBUM\_SHARPInstance-attribute**

ALBUM\_SHARP: IconData

**buildALIGN\_HORIZONTAL\_CENTERInstance-attribute**

ALIGN\_HORIZONTAL\_CENTER: IconData

**buildALIGN\_HORIZONTAL\_CENTER\_OUTLINEDInstance-attribute**

ALIGN\_HORIZONTAL\_CENTER\_OUTLINED: IconData

**buildALIGN\_HORIZONTAL\_CENTER\_ROUNDEDInstance-attribute**

ALIGN\_HORIZONTAL\_CENTER\_ROUNDED: IconData

**buildALIGN\_HORIZONTAL\_CENTER\_SHARPinstance-attribute**

ALIGN\_HORIZONTAL\_CENTER\_SHARP: IconData

**buildALIGN\_HORIZONTAL\_LEFTinstance-attribute**

ALIGN\_HORIZONTAL\_LEFT: IconData

**buildALIGN\_HORIZONTAL\_LEFT\_OUTLINEDinstance-attribute**

ALIGN\_HORIZONTAL\_LEFT\_OUTLINED: IconData

**buildALIGN\_HORIZONTAL\_LEFT\_ROUNDEDinstance-attribute**

ALIGN\_HORIZONTAL\_LEFT\_ROUNDED: IconData

**buildALIGN\_HORIZONTAL\_LEFT\_SHARPinstance-attribute**

ALIGN\_HORIZONTAL\_LEFT\_SHARP: IconData

**buildALIGN\_HORIZONTAL\_RIGHTinstance-attribute**

ALIGN\_HORIZONTAL\_RIGHT: IconData

**buildALIGN\_HORIZONTAL\_RIGHT\_OUTLINEDinstance-attribute**

ALIGN\_HORIZONTAL\_RIGHT\_OUTLINED: IconData

**buildALIGN\_HORIZONTAL\_RIGHT\_ROUNDEDinstance-attribute**

ALIGN\_HORIZONTAL\_RIGHT\_ROUNDED: IconData

**buildALIGN\_HORIZONTAL\_RIGHT\_SHARPinstance-attribute**

ALIGN\_HORIZONTAL\_RIGHT\_SHARP: IconData

**buildALIGN\_VERTICAL\_BOTTOMinstance-attribute**

ALIGN\_VERTICAL\_BOTTOM: IconData

**buildALIGN\_VERTICAL\_BOTTOM\_OUTLINEDinstance-attribute**

ALIGN\_VERTICAL\_BOTTOM\_OUTLINED: IconData

**buildALIGN\_VERTICAL\_BOTTOM\_ROUNDEDinstance-attribute**

ALIGN\_VERTICAL\_BOTTOM\_ROUNDED: IconData

**buildALIGN\_VERTICAL\_BOTTOM\_SHARPinstance-attribute**

ALIGN\_VERTICAL\_BOTTOM\_SHARP: IconData

**buildALIGN\_VERTICAL\_CENTERinstance-attribute**

ALIGN\_VERTICAL\_CENTER: IconData

**buildALIGN\_VERTICAL\_CENTER\_OUTLINEDinstance-attribute**

ALIGN\_VERTICAL\_CENTER\_OUTLINED: IconData

**buildALIGN\_VERTICAL\_CENTER\_ROUNDEDinstance-attribute**

ALIGN\_VERTICAL\_CENTER\_ROUNDED: IconData

**buildALIGN\_VERTICAL\_CENTER\_SHARPinstance-attribute**

ALIGN\_VERTICAL\_CENTER\_SHARP: IconData

**buildALIGN\_VERTICAL\_TOPinstance-attribute**

ALIGN\_VERTICAL\_TOP: IconData

**buildALIGN\_VERTICAL\_TOP\_OUTLINEDinstance-attribute**

ALIGN\_VERTICAL\_TOP\_OUTLINED: IconData

**buildALIGN\_VERTICAL\_TOP\_ROUNDEDinstance-attribute**

ALIGN\_VERTICAL\_TOP\_ROUNDED: IconData

**buildALIGN\_VERTICAL\_TOP\_SHARPinstance-attribute**

ALIGN\_VERTICAL\_TOP\_SHARP: IconData

**buildALL\_INBOXinstance-attribute**

ALL\_INBOX: IconData

**buildALL\_INBOX\_OUTLINEDinstance-attribute**

ALL\_INBOX\_OUTLINED: IconData

**buildALL\_INBOX\_ROUNDEDinstance-attribute**

ALL\_INBOX\_ROUNDED: IconData

**buildALL\_INBOX\_SHARPinstance-attribute**

ALL\_INBOX\_SHARP: IconData

**buildALL\_INCLUSIVEinstance-attribute**

ALL\_INCLUSIVE: IconData

**buildALL\_INCLUSIVE\_OUTLINEDinstance-attribute**

ALL\_INCLUSIVE\_OUTLINED: IconData

**buildALL\_INCLUSIVE\_ROUNDEDinstance-attribute**

ALL\_INCLUSIVE\_ROUNDED: IconData

**buildALL\_INCLUSIVE\_SHARPinstance-attribute**

ALL\_INCLUSIVE\_SHARP: IconData

**buildALL\_OUTinstance-attribute**

ALL\_OUT: IconData

**buildALL\_OUT\_OUTLINEDinstance-attribute**

ALL\_OUT\_OUTLINED: IconData

**buildALL\_OUT\_ROUNDEDinstance-attribute**

ALL\_OUT\_ROUNDED: IconData

**buildALL\_OUT\_SHARPinstance-attribute**

ALL\_OUT\_SHARP: IconData

**buildALT\_ROUTEinstance-attribute**

ALT\_ROUTE: IconData

**buildALT\_ROUTE\_OUTLINEDinstance-attribute**

ALT\_ROUTE\_OUTLINED: IconData

**buildALT\_ROUTE\_ROUNDEDinstance-attribute**

ALT\_ROUTE\_ROUNDED: IconData

**buildALT\_ROUTE\_SHARPinstance-attribute**

ALT\_ROUTE\_SHARP: IconData

**buildALTERNATE\_EMAILinstance-attribute**

ALTERNATE\_EMAIL: IconData

**buildALTERNATE\_EMAIL\_OUTLINEDinstance-attribute**

ALTERNATE\_EMAIL\_OUTLINED: IconData

**buildALTERNATE\_EMAIL\_ROUNDEDinstance-attribute**

ALTERNATE\_EMAIL\_ROUNDED: IconData

**buildALTERNATE\_EMAIL\_SHARPinstance-attribute**

ALTERNATE\_EMAIL\_SHARP: IconData

**buildAMP\_STORIESinstance-attribute**

AMP\_STORIES: IconData

**buildAMP\_STORIES\_OUTLINEDinstance-attribute**

AMP\_STORIES\_OUTLINED: IconData

**buildAMP\_STORIES\_ROUNDEDinstance-attribute**

AMP\_STORIES\_ROUNDED: IconData



**buildAMP\_STORIES\_SHARPinstance-attribute**

AMP\_STORIES\_SHARP: IconData

**buildANALYTICSinstance-attribute**

ANALYTICS: IconData

**buildANALYTICS\_OUTLINEDinstance-attribute**

ANALYTICS\_OUTLINED: IconData

**buildANALYTICS\_ROUNDEDinstance-attribute**

ANALYTICS\_ROUNDED: IconData

**buildANALYTICS\_SHARPinstance-attribute**

ANALYTICS\_SHARP: IconData

**buildANCHORinstance-attribute**

ANCHOR: IconData

**buildANCHOR\_OUTLINEDinstance-attribute**

ANCHOR\_OUTLINED: IconData

**buildANCHOR\_ROUNDEDinstance-attribute**

ANCHOR\_ROUNDED: IconData

**buildANCHOR\_SHARPinstance-attribute**

ANCHOR\_SHARP: IconData

**buildANDROIDinstance-attribute**

ANDROID: IconData

**buildANDROID\_OUTLINEDinstance-attribute**

ANDROID\_OUTLINED: IconData

**buildANDROID\_ROUNDEDinstance-attribute**

ANDROID\_ROUNDED: IconData

**buildANDROID\_SHARPinstance-attribute**

ANDROID\_SHARP: IconData

**buildANIMATIONinstance-attribute**

ANIMATION: IconData

**buildANIMATION\_OUTLINEDinstance-attribute**

ANIMATION\_OUTLINED: IconData

**buildANIMATION\_ROUNDEDinstance-attribute**

ANIMATION\_ROUNDED: IconData

**buildANIMATION\_SHARPinstance-attribute**

ANIMATION\_SHARP: IconData

**buildANNOUNCEMENTinstance-attribute**

ANNOUNCEMENT: IconData

**buildANNOUNCEMENT\_OUTLINEDinstance-attribute**

ANNOUNCEMENT\_OUTLINED: IconData

**buildANNOUNCEMENT\_ROUNDEDinstance-attribute**

ANNOUNCEMENT\_ROUNDED: IconData

**buildANNOUNCEMENT\_SHARPinstance-attribute**

ANNOUNCEMENT\_SHARP: IconData

**buildAODinstance-attribute**

AOD: IconData

**buildAOD\_OUTLINEDinstance-attribute**

AOD\_OUTLINED: IconData

**buildAOD\_ROUNDEDinstance-attribute**

AOD\_ROUNDED: IconData

**buildAOD\_SHARPinstance-attribute**

AOD\_SHARP: IconData

**buildAPARTMENTinstance-attribute**

APARTMENT: IconData

**buildAPARTMENT\_OUTLINEDinstance-attribute**

APARTMENT\_OUTLINED: IconData

**buildAPARTMENT\_ROUNDEDinstance-attribute**

APARTMENT\_ROUNDED: IconData

**buildAPARTMENT\_SHARPinstance-attribute**

APARTMENT\_SHARP: IconData

**buildAPIinstance-attribute**

API: IconData

**buildAPI\_OUTLINEDinstance-attribute**

API\_OUTLINED: IconData

**buildAPI\_ROUNDEDinstance-attribute**

API\_ROUNDED: IconData

**buildAPI\_SHARPinstance-attribute**

API\_SHARP: IconData

**buildAPP\_BLOCKINGinstance-attribute**

APP\_BLOCKING: IconData

**buildAPP\_BLOCKING\_OUTLINEDinstance-attribute**

APP\_BLOCKING\_OUTLINED: IconData

**buildAPP\_BLOCKING\_ROUNDEDinstance-attribute**

APP\_BLOCKING\_ROUNDED: IconData

**buildAPP\_BLOCKING\_SHARPinstance-attribute**

APP\_BLOCKING\_SHARP: IconData

**buildAPP\_REGISTRATIONinstance-attribute**

APP\_REGISTRATION: IconData

**buildAPP\_REGISTRATION\_OUTLINEDinstance-attribute**

APP\_REGISTRATION\_OUTLINED: IconData

**buildAPP\_REGISTRATION\_ROUNDEDinstance-attribute**

APP\_REGISTRATION\_ROUNDED: IconData

**buildAPP\_REGISTRATION\_SHARPinstance-attribute**

APP\_REGISTRATION\_SHARP: IconData

**buildAPP\_SETTINGS\_ALTinstance-attribute**

APP\_SETTINGS\_ALT: IconData

**buildAPP\_SETTINGS\_ALT\_OUTLINEDinstance-attribute**

APP\_SETTINGS\_ALT\_OUTLINED: IconData

**buildAPP\_SETTINGS\_ALT\_ROUNDEDinstance-attribute**

APP\_SETTINGS\_ALT\_ROUNDED: IconData

**buildAPP\_SETTINGS\_ALT\_SHARPinstance-attribute**

APP\_SETTINGS\_ALT\_SHARP: IconData

**buildAPP\_SHORTCUTinstance-attribute**

APP\_SHORTCUT: IconData

**buildAPP\_SHORTCUT\_OUTLINEDinstance-attribute**

APP\_SHORTCUT\_OUTLINED: IconData

**buildAPP\_SHORTCUT\_ROUNDEDinstance-attribute**

APP\_SHORTCUT\_ROUNDED: IconData

**buildAPP\_SHORTCUT\_SHARPinstance-attribute**

APP\_SHORTCUT\_SHARP: IconData

**buildAPPLEinstance-attribute**

APPLE: IconData

**buildAPPLE\_OUTLINEDinstance-attribute**

APPLE\_OUTLINED: IconData

**buildAPPLE\_ROUNDEDinstance-attribute**

APPLE\_ROUNDED: IconData

**buildAPPLE\_SHARPinstance-attribute**

APPLE\_SHARP: IconData

**buildAPPROVALinstance-attribute**

APPROVAL: IconData

**buildAPPROVAL\_OUTLINEDinstance-attribute**

APPROVAL\_OUTLINED: IconData

**buildAPPROVAL\_ROUNDEDinstance-attribute**

APPROVAL\_ROUNDED: IconData

**buildAPPROVAL\_SHARPinstance-attribute**

APPROVAL\_SHARP: IconData

**buildAPPSinstance-attribute**

APPS: IconData

**buildAPPS\_OUTAGEinstance-attribute**

APPS\_OUTAGE: IconData

**buildAPPS\_OUTAGE\_OUTLINEDinstance-attribute**

APPS\_OUTAGE\_OUTLINED: IconData

**buildAPPS\_OUTAGE\_ROUNDEDinstance-attribute**

APPS\_OUTAGE\_ROUNDED: IconData

**buildAPPS\_OUTAGE\_SHARPinstance-attribute**

APPS\_OUTAGE\_SHARP: IconData

**buildAPPS\_OUTLINEDinstance-attribute**

APPS\_OUTLINED: IconData

**buildAPPS\_ROUNDEDinstance-attribute**

APPS\_ROUNDED: IconData

**buildAPPS\_SHARPinstance-attribute**

APPS\_SHARP: IconData

**buildARCHITECTUREinstance-attribute**

ARCHITECTURE: IconData

**buildARCHITECTURE\_OUTLINED**instance-attribute

ARCHITECTURE\_OUTLINED: IconData

**buildARCHITECTURE\_ROUNDED**instance-attribute

ARCHITECTURE\_ROUNDED: IconData

**buildARCHITECTURE\_SHARP**instance-attribute

ARCHITECTURE\_SHARP: IconData

**buildARCHIVE**instance-attribute

ARCHIVE: IconData

**buildARCHIVE\_OUTLINED**instance-attribute

ARCHIVE\_OUTLINED: IconData

**buildARCHIVE\_ROUNDED**instance-attribute

ARCHIVE\_ROUNDED: IconData

**buildARCHIVE\_SHARP**instance-attribute

ARCHIVE\_SHARP: IconData

**buildAREA\_CHART**instance-attribute

AREA\_CHART: IconData

**buildAREA\_CHART\_OUTLINED**instance-attribute

AREA\_CHART\_OUTLINED: IconData

**buildAREA\_CHART\_ROUNDED**instance-attribute

AREA\_CHART\_ROUNDED: IconData

**buildAREA\_CHART\_SHARP**instance-attribute

AREA\_CHART\_SHARP: IconData

**buildARROW\_BACK**instance-attribute

ARROW\_BACK: IconData

**buildARROW\_BACK\_IOS**instance-attribute

ARROW\_BACK\_IOS: IconData

**buildARROW\_BACK\_IOS\_NEW**instance-attribute

ARROW\_BACK\_IOS\_NEW: IconData

**buildARROW\_BACK\_IOS\_NEW\_OUTLINED**instance-attribute

ARROW\_BACK\_IOS\_NEW\_OUTLINED: IconData

**buildARROW\_BACK\_IOS\_NEW\_ROUNDED**instance-attribute

ARROW\_BACK\_IOS\_NEW\_ROUNDED: IconData

**buildARROW\_BACK\_IOS\_NEW\_SHARP**instance-attribute

ARROW\_BACK\_IOS\_NEW\_SHARP: IconData

**buildARROW\_BACK\_IOS\_OUTLINED**instance-attribute

ARROW\_BACK\_IOS\_OUTLINED: IconData

**buildARROW\_BACK\_IOS\_ROUNDED**instance-attribute

ARROW\_BACK\_IOS\_ROUNDED: IconData

**buildARROW\_BACK\_IOS\_SHARP**instance-attribute

ARROW\_BACK\_IOS\_SHARP: IconData

**buildARROW\_BACK\_OUTLINED**instance-attribute

ARROW\_BACK\_OUTLINED: IconData

**buildARROW\_BACK\_ROUNDED**instance-attribute

ARROW\_BACK\_ROUNDED: IconData

**buildARROW\_BACK\_SHARPinstance-attribute**

ARROW\_BACK\_SHARP: IconData

**buildARROW\_CIRCLE\_DOWNinstance-attribute**

ARROW\_CIRCLE\_DOWN: IconData

**buildARROW\_CIRCLE\_DOWN\_OUTLINEDinstance-attribute**

ARROW\_CIRCLE\_DOWN\_OUTLINED: IconData

**buildARROW\_CIRCLE\_DOWN\_ROUNDEDinstance-attribute**

ARROW\_CIRCLE\_DOWN\_ROUNDED: IconData

**buildARROW\_CIRCLE\_DOWN\_SHARPinstance-attribute**

ARROW\_CIRCLE\_DOWN\_SHARP: IconData

**buildARROW\_CIRCLE\_LEFTinstance-attribute**

ARROW\_CIRCLE\_LEFT: IconData

**buildARROW\_CIRCLE\_LEFT\_OUTLINEDinstance-attribute**

ARROW\_CIRCLE\_LEFT\_OUTLINED: IconData

**buildARROW\_CIRCLE\_LEFT\_ROUNDEDinstance-attribute**

ARROW\_CIRCLE\_LEFT\_ROUNDED: IconData

**buildARROW\_CIRCLE\_LEFT\_SHARPinstance-attribute**

ARROW\_CIRCLE\_LEFT\_SHARP: IconData

**buildARROW\_CIRCLE\_RIGHTinstance-attribute**

ARROW\_CIRCLE\_RIGHT: IconData

**buildARROW\_CIRCLE\_RIGHT\_OUTLINEDinstance-attribute**

ARROW\_CIRCLE\_RIGHT\_OUTLINED: IconData

**buildARROW\_CIRCLE\_RIGHT\_ROUNDEDinstance-attribute**

ARROW\_CIRCLE\_RIGHT\_ROUNDED: IconData

**buildARROW\_CIRCLE\_RIGHT\_SHARPinstance-attribute**

ARROW\_CIRCLE\_RIGHT\_SHARP: IconData

**buildARROW\_CIRCLE\_UPinstance-attribute**

ARROW\_CIRCLE\_UP: IconData

**buildARROW\_CIRCLE\_UP\_OUTLINEDinstance-attribute**

ARROW\_CIRCLE\_UP\_OUTLINED: IconData

**buildARROW\_CIRCLE\_UP\_ROUNDEDinstance-attribute**

ARROW\_CIRCLE\_UP\_ROUNDED: IconData

**buildARROW\_CIRCLE\_UP\_SHARPinstance-attribute**

ARROW\_CIRCLE\_UP\_SHARP: IconData

**buildARROW\_DOWNWARDinstance-attribute**

ARROW\_DOWNWARD: IconData

**buildARROW\_DOWNWARD\_OUTLINEDinstance-attribute**

ARROW\_DOWNWARD\_OUTLINED: IconData

**buildARROW\_DOWNWARD\_ROUNDEDinstance-attribute**

ARROW\_DOWNWARD\_ROUNDED: IconData

**buildARROW\_DOWNWARD\_SHARPinstance-attribute**

ARROW\_DOWNWARD\_SHARP: IconData

**buildARROW\_DROP\_DOWNinstance-attribute**

ARROW\_DROP\_DOWN: IconData

**buildARROW\_DROP\_DOWN\_CIRCLE**instance-attribute

ARROW\_DROP\_DOWN\_CIRCLE: IconData

**buildARROW\_DROP\_DOWN\_CIRCLE\_OUTLINED**instance-attribute

ARROW\_DROP\_DOWN\_CIRCLE\_OUTLINED: IconData

**buildARROW\_DROP\_DOWN\_CIRCLE\_ROUNDED**instance-attribute

ARROW\_DROP\_DOWN\_CIRCLE\_ROUNDED: IconData

**buildARROW\_DROP\_DOWN\_CIRCLE\_SHARP**instance-attribute

ARROW\_DROP\_DOWN\_CIRCLE\_SHARP: IconData

**buildARROW\_DROP\_DOWN\_OUTLINED**instance-attribute

ARROW\_DROP\_DOWN\_OUTLINED: IconData

**buildARROW\_DROP\_DOWN\_ROUNDED**instance-attribute

ARROW\_DROP\_DOWN\_ROUNDED: IconData

**buildARROW\_DROP\_DOWN\_SHARP**instance-attribute

ARROW\_DROP\_DOWN\_SHARP: IconData

**buildARROW\_DROP\_UP**instance-attribute

ARROW\_DROP\_UP: IconData

**buildARROW\_DROP\_UP\_OUTLINED**instance-attribute

ARROW\_DROP\_UP\_OUTLINED: IconData

**buildARROW\_DROP\_UP\_ROUNDED**instance-attribute

ARROW\_DROP\_UP\_ROUNDED: IconData

**buildARROW\_DROP\_UP\_SHARP**instance-attribute

ARROW\_DROP\_UP\_SHARP: IconData

**buildARROW\_FORWARD**instance-attribute

ARROW\_FORWARD: IconData

**buildARROW\_FORWARD\_IOS**instance-attribute

ARROW\_FORWARD\_IOS: IconData

**buildARROW\_FORWARD\_IOS\_OUTLINED**instance-attribute

ARROW\_FORWARD\_IOS\_OUTLINED: IconData

**buildARROW\_FORWARD\_IOS\_ROUNDED**instance-attribute

ARROW\_FORWARD\_IOS\_ROUNDED: IconData

**buildARROW\_FORWARD\_IOS\_SHARP**instance-attribute

ARROW\_FORWARD\_IOS\_SHARP: IconData

**buildARROW\_FORWARD\_OUTLINED**instance-attribute

ARROW\_FORWARD\_OUTLINED: IconData

**buildARROW\_FORWARD\_ROUNDED**instance-attribute

ARROW\_FORWARD\_ROUNDED: IconData

**buildARROW\_FORWARD\_SHARP**instance-attribute

ARROW\_FORWARD\_SHARP: IconData

**buildARROW\_LEFT**instance-attribute

ARROW\_LEFT: IconData

**buildARROW\_LEFT\_OUTLINED**instance-attribute

ARROW\_LEFT\_OUTLINED: IconData

**buildARROW\_LEFT\_ROUNDED**instance-attribute

ARROW\_LEFT\_ROUNDED: IconData

**buildARROW\_LEFT\_SHARP**instance-attribute

ARROW\_LEFT\_SHARP: IconData

**buildARROW\_OUTWARD**instance-attribute

ARROW\_OUTWARD: IconData

**buildARROW\_OUTWARD\_OUTLINED**instance-attribute

ARROW\_OUTWARD\_OUTLINED: IconData

**buildARROW\_OUTWARD\_ROUNDED**instance-attribute

ARROW\_OUTWARD\_ROUNDED: IconData

**buildARROW\_OUTWARD\_SHARP**instance-attribute

ARROW\_OUTWARD\_SHARP: IconData

**buildARROW\_RIGHT**instance-attribute

ARROW\_RIGHT: IconData

**buildARROW\_RIGHT\_ALT**instance-attribute

ARROW\_RIGHT\_ALT: IconData

**buildARROW\_RIGHT\_ALT\_OUTLINED**instance-attribute

ARROW\_RIGHT\_ALT\_OUTLINED: IconData

**buildARROW\_RIGHT\_ALT\_ROUNDED**instance-attribute

ARROW\_RIGHT\_ALT\_ROUNDED: IconData

**buildARROW\_RIGHT\_ALT\_SHARP**instance-attribute

ARROW\_RIGHT\_ALT\_SHARP: IconData

**buildARROW\_RIGHT\_OUTLINED**instance-attribute

ARROW\_RIGHT\_OUTLINED: IconData

**buildARROW\_RIGHT\_ROUNDED**instance-attribute

ARROW\_RIGHT\_ROUNDED: IconData

**buildARROW\_RIGHT\_SHARP**instance-attribute

ARROW\_RIGHT\_SHARP: IconData

**buildARROW\_UPWARD**instance-attribute

ARROW\_UPWARD: IconData

**buildARROW\_UPWARD\_OUTLINED**instance-attribute

ARROW\_UPWARD\_OUTLINED: IconData

**buildARROW\_UPWARD\_ROUNDED**instance-attribute

ARROW\_UPWARD\_ROUNDED: IconData

**buildARROW\_UPWARD\_SHARP**instance-attribute

ARROW\_UPWARD\_SHARP: IconData

**buildART\_TRACK**instance-attribute

ART\_TRACK: IconData

**buildART\_TRACK\_OUTLINED**instance-attribute

ART\_TRACK\_OUTLINED: IconData

**buildART\_TRACK\_ROUNDED**instance-attribute

ART\_TRACK\_ROUNDED: IconData

**buildART\_TRACK\_SHARP**instance-attribute

ART\_TRACK\_SHARP: IconData

**buildARTICLE**instance-attribute

ARTICLE: IconData

**buildARTICLE\_OUTLINEDinstance-attribute**

ARTICLE\_OUTLINED: IconData

**buildARTICLE\_ROUNDEDinstance-attribute**

ARTICLE\_ROUNDED: IconData

**buildARTICLE\_SHARPinstance-attribute**

ARTICLE\_SHARP: IconData

**buildASPECT\_RATIOinstance-attribute**

ASPECT\_RATIO: IconData

**buildASPECT\_RATIO\_OUTLINEDinstance-attribute**

ASPECT\_RATIO\_OUTLINED: IconData

**buildASPECT\_RATIO\_ROUNDEDinstance-attribute**

ASPECT\_RATIO\_ROUNDED: IconData

**buildASPECT\_RATIO\_SHARPinstance-attribute**

ASPECT\_RATIO\_SHARP: IconData

**buildASSESSMENTinstance-attribute**

ASSESSMENT: IconData

**buildASSESSMENT\_OUTLINEDinstance-attribute**

ASSESSMENT\_OUTLINED: IconData

**buildASSESSMENT\_ROUNDEDinstance-attribute**

ASSESSMENT\_ROUNDED: IconData

**buildASSESSMENT\_SHARPinstance-attribute**

ASSESSMENT\_SHARP: IconData

**buildASSIGNMENTinstance-attribute**

ASSIGNMENT: IconData

**buildASSIGNMENT\_ADDinstance-attribute**

ASSIGNMENT\_ADD: IconData

**buildASSIGNMENT\_INDinstance-attribute**

ASSIGNMENT\_IND: IconData

**buildASSIGNMENT\_IND\_OUTLINEDinstance-attribute**

ASSIGNMENT\_IND\_OUTLINED: IconData

**buildASSIGNMENT\_IND\_ROUNDEDinstance-attribute**

ASSIGNMENT\_IND\_ROUNDED: IconData

**buildASSIGNMENT\_IND\_SHARPinstance-attribute**

ASSIGNMENT\_IND\_SHARP: IconData

**buildASSIGNMENT\_LATEinstance-attribute**

ASSIGNMENT\_LATE: IconData

**buildASSIGNMENT\_LATE\_OUTLINEDinstance-attribute**

ASSIGNMENT\_LATE\_OUTLINED: IconData

**buildASSIGNMENT\_LATE\_ROUNDEDinstance-attribute**□

ASSIGNMENT\_LATE\_ROUNDED: IconData

**buildASSIGNMENT\_LATE\_SHARPinstance-attribute**

ASSIGNMENT\_LATE\_SHARP: IconData

**buildASSIGNMENT\_OUTLINEDinstance-attribute**

ASSIGNMENT\_OUTLINED: IconData



**buildASSIGNMENT\_RETURNinstance-attribute**

ASSIGNMENT\_RETURN: IconData

**buildASSIGNMENT\_RETURN\_OUTLINEDinstance-attribute**

ASSIGNMENT\_RETURN\_OUTLINED: IconData

**buildASSIGNMENT\_RETURN\_ROUNDEDinstance-attribute**

ASSIGNMENT\_RETURN\_ROUNDED: IconData

**buildASSIGNMENT\_RETURN\_SHARPinstance-attribute**

ASSIGNMENT\_RETURN\_SHARP: IconData

**buildASSIGNMENT\_RETURNEDinstance-attribute**

ASSIGNMENT\_RETURNED: IconData

**buildASSIGNMENT\_RETURNED\_OUTLINEDinstance-attribute**

ASSIGNMENT\_RETURNED\_OUTLINED: IconData

**buildASSIGNMENT\_RETURNED\_ROUNDEDinstance-attribute**

ASSIGNMENT\_RETURNED\_ROUNDED: IconData

**buildASSIGNMENT\_RETURNED\_SHARPinstance-attribute**

ASSIGNMENT\_RETURNED\_SHARP: IconData

**buildASSIGNMENT\_ROUNDEDinstance-attribute**

ASSIGNMENT\_ROUNDED: IconData

**buildASSIGNMENT\_SHARPinstance-attribute**

ASSIGNMENT\_SHARP: IconData

**buildASSIGNMENT\_TURNED\_INinstance-attribute**

ASSIGNMENT\_TURNED\_IN: IconData

**buildASSIGNMENT\_TURNED\_IN\_OUTLINEDinstance-attribute**

ASSIGNMENT\_TURNED\_IN\_OUTLINED: IconData

**buildASSIGNMENT\_TURNED\_IN\_ROUNDEDinstance-attribute**

ASSIGNMENT\_TURNED\_IN\_ROUNDED: IconData

**buildASSIGNMENT\_TURNED\_IN\_SHARPinstance-attribute**

ASSIGNMENT\_TURNED\_IN\_SHARP: IconData

**buildASSIST\_WALKERinstance-attribute**

ASSIST\_WALKER: IconData

**buildASSIST\_WALKER\_OUTLINEDinstance-attribute**

ASSIST\_WALKER\_OUTLINED: IconData

**buildASSIST\_WALKER\_ROUNDEDinstance-attribute**

ASSIST\_WALKER\_ROUNDED: IconData

**buildASSIST\_WALKER\_SHARPinstance-attribute**

ASSIST\_WALKER\_SHARP: IconData

**buildASSISTANTinstance-attribute**

ASSISTANT: IconData

**buildASSISTANT\_DIRECTIONinstance-attribute**

ASSISTANT\_DIRECTION: IconData

**buildASSISTANT\_DIRECTION\_OUTLINEDinstance-attribute**

ASSISTANT\_DIRECTION\_OUTLINED: IconData

**buildASSISTANT\_DIRECTION\_ROUNDEDinstance-attribute**

ASSISTANT\_DIRECTION\_ROUNDED: IconData

**buildASSISTANT\_DIRECTION\_SHARP**instance-attribute

ASSISTANT\_DIRECTION\_SHARP: IconData

**buildASSISTANT\_NAVIGATION**instance-attribute

ASSISTANT\_NAVIGATION: IconData

**buildASSISTANT\_OUTLINED**instance-attribute

ASSISTANT\_OUTLINED: IconData

**buildASSISTANT\_PHOTO**instance-attribute

ASSISTANT\_PHOTO: IconData

**buildASSISTANT\_PHOTO\_OUTLINED**instance-attribute

ASSISTANT\_PHOTO\_OUTLINED: IconData

**buildASSISTANT\_PHOTO\_ROUNDED**instance-attribute

ASSISTANT\_PHOTO\_ROUNDED: IconData

**buildASSISTANT\_PHOTO\_SHARP**instance-attribute

ASSISTANT\_PHOTO\_SHARP: IconData

**buildASSISTANT\_ROUNDED**instance-attribute

ASSISTANT\_ROUNDED: IconData

**buildASSISTANT\_SHARP**instance-attribute

ASSISTANT\_SHARP: IconData

**buildASSURED\_WORKLOAD**instance-attribute

ASSURED\_WORKLOAD: IconData

**buildASSURED\_WORKLOAD\_OUTLINED**instance-attribute

ASSURED\_WORKLOAD\_OUTLINED: IconData

**buildASSURED\_WORKLOAD\_ROUNDED**instance-attribute

ASSURED\_WORKLOAD\_ROUNDED: IconData

**buildASSURED\_WORKLOAD\_SHARP**instance-attribute

ASSURED\_WORKLOAD\_SHARP: IconData

**buildATM**instance-attribute

ATM: IconData

**buildATM\_OUTLINED**instance-attribute

ATM\_OUTLINED: IconData

**buildATM\_ROUNDED**instance-attribute

ATM\_ROUNDED: IconData

**buildATM\_SHARP**instance-attribute

ATM\_SHARP: IconData

**buildATTACH\_EMAIL**instance-attribute

ATTACH\_EMAIL: IconData

**buildATTACH\_EMAIL\_OUTLINED**instance-attribute

ATTACH\_EMAIL\_OUTLINED: IconData

**buildATTACH\_EMAIL\_ROUNDED**instance-attribute

ATTACH\_EMAIL\_ROUNDED: IconData

**buildATTACH\_EMAIL\_SHARP**instance-attribute

ATTACH\_EMAIL\_SHARP: IconData

**buildATTACH\_FILE**instance-attribute

ATTACH\_FILE: IconData

**buildATTACH\_FILE\_OUTLINEDinstance-attribute**

ATTACH\_FILE\_OUTLINED: IconData

**buildATTACH\_FILE\_ROUNDEDinstance-attribute**

ATTACH\_FILE\_ROUNDED: IconData

**buildATTACH\_FILE\_SHARPinstance-attribute**

ATTACH\_FILE\_SHARP: IconData

**buildATTACH\_MONEYinstance-attribute**

ATTACH\_MONEY: IconData

**buildATTACH\_MONEY\_OUTLINEDinstance-attribute**

ATTACH\_MONEY\_OUTLINED: IconData

**buildATTACH\_MONEY\_ROUNDEDinstance-attribute**

ATTACH\_MONEY\_ROUNDED: IconData

**buildATTACH\_MONEY\_SHARPinstance-attribute**

ATTACH\_MONEY\_SHARP: IconData

**buildATTACHMENTinstance-attribute**

ATTACHMENT: IconData

**buildATTACHMENT\_OUTLINEDinstance-attribute**

ATTACHMENT\_OUTLINED: IconData

**buildATTACHMENT\_ROUNDEDinstance-attribute**

ATTACHMENT\_ROUNDED: IconData

**buildATTACHMENT\_SHARPinstance-attribute**

ATTACHMENT\_SHARP: IconData

**buildATTRACTIONSinstance-attribute**

ATTRACTIONS: IconData

**buildATTRACTIONS\_OUTLINEDinstance-attribute**

ATTRACTIONS\_OUTLINED: IconData

**buildATTRACTIONS\_ROUNDEDinstance-attribute**

ATTRACTIONS\_ROUNDED: IconData

**buildATTRACTIONS\_SHARPinstance-attribute**

ATTRACTIONS\_SHARP: IconData

**buildATTRIBUTIONinstance-attribute**

ATTRIBUTION: IconData

**buildATTRIBUTION\_OUTLINEDinstance-attribute**

ATTRIBUTION\_OUTLINED: IconData

**buildATTRIBUTION\_ROUNDEDinstance-attribute**

ATTRIBUTION\_ROUNDED: IconData

**buildATTRIBUTION\_SHARPinstance-attribute**

ATTRIBUTION\_SHARP: IconData

**buildAUDIO\_FILEinstance-attribute**

AUDIO\_FILE: IconData

**buildAUDIO\_FILE\_OUTLINEDinstance-attribute**

AUDIO\_FILE\_OUTLINED: IconData

**buildAUDIO\_FILE\_ROUNDEDinstance-attribute**

AUDIO\_FILE\_ROUNDED: IconData

**buildAUDIO\_FILE\_SHARPinstance-attribute**

AUDIO\_FILE\_SHARP: IconData

**buildAUDIOTRACKinstance-attribute**

AUDIOTRACK: IconData

**buildAUDIOTRACK\_OUTLINEDinstance-attribute**

AUDIOTRACK\_OUTLINED: IconData

**buildAUDIOTRACK\_ROUNDEDinstance-attribute**

AUDIOTRACK\_ROUNDED: IconData

**buildAUDIOTRACK\_SHARPinstance-attribute**

AUDIOTRACK\_SHARP: IconData

**buildAUTO\_AWESOMEinstance-attribute**

AUTO\_AWESOME: IconData

**buildAUTO\_AWESOME\_MOSAICinstance-attribute**

AUTO\_AWESOME\_MOSAIC: IconData

**buildAUTO\_AWESOME\_MOSAIC\_OUTLINEDinstance-attribute**

AUTO\_AWESOME\_MOSAIC\_OUTLINED: IconData

**buildAUTO\_AWESOME\_MOSAIC\_ROUNDEDinstance-attribute**

AUTO\_AWESOME\_MOSAIC\_ROUNDED: IconData

**buildAUTO\_AWESOME\_MOSAIC\_SHARPinstance-attribute**

AUTO\_AWESOME\_MOSAIC\_SHARP: IconData

**buildAUTO\_AWESOME\_MOTIONinstance-attribute**

AUTO\_AWESOME\_MOTION: IconData

**buildAUTO\_AWESOME\_MOTION\_OUTLINEDinstance-attribute**

AUTO\_AWESOME\_MOTION\_OUTLINED: IconData

**buildAUTO\_AWESOME\_MOTION\_ROUNDEDinstance-attribute**

AUTO\_AWESOME\_MOTION\_ROUNDED: IconData

**buildAUTO\_AWESOME\_MOTION\_SHARPinstance-attribute**

AUTO\_AWESOME\_MOTION\_SHARP: IconData

**buildAUTO\_AWESOME\_OUTLINEDinstance-attribute**

AUTO\_AWESOME\_OUTLINED: IconData

**buildAUTO\_AWESOME\_ROUNDEDinstance-attribute**

AUTO\_AWESOME\_ROUNDED: IconData

**buildAUTO\_AWESOME\_SHARPinstance-attribute**

AUTO\_AWESOME\_SHARP: IconData

**buildAUTO\_DELETEinstance-attribute**

AUTO\_DELETE: IconData

**buildAUTO\_DELETE\_OUTLINEDinstance-attribute**

AUTO\_DELETE\_OUTLINED: IconData

**buildAUTO\_DELETE\_ROUNDEDinstance-attribute**

AUTO\_DELETE\_ROUNDED: IconData

**buildAUTO\_DELETE\_SHARPinstance-attribute**

AUTO\_DELETE\_SHARP: IconData

**buildAUTO\_FIX\_HIGHinstance-attribute**

AUTO\_FIX\_HIGH: IconData

**buildAUTO\_FIX\_HIGH\_OUTLINEDinstance-attribute**

AUTO\_FIX\_HIGH\_OUTLINED: IconData

**buildAUTO\_FIX\_HIGH\_ROUNDEDinstance-attribute**

AUTO\_FIX\_HIGH\_ROUNDED: IconData

**buildAUTO\_FIX\_HIGH\_SHARPinstance-attribute**

AUTO\_FIX\_HIGH\_SHARP: IconData

**buildAUTO\_FIX\_NORMALinstance-attribute**

AUTO\_FIX\_NORMAL: IconData

**buildAUTO\_FIX\_NORMAL\_OUTLINEDinstance-attribute**

AUTO\_FIX\_NORMAL\_OUTLINED: IconData

**buildAUTO\_FIX\_NORMAL\_ROUNDEDinstance-attribute**

AUTO\_FIX\_NORMAL\_ROUNDED: IconData

**buildAUTO\_FIX\_NORMAL\_SHARPinstance-attribute**

AUTO\_FIX\_NORMAL\_SHARP: IconData

**buildAUTO\_FIX\_OFFinstance-attribute**

AUTO\_FIX\_OFF: IconData

**buildAUTO\_FIX\_OFF\_OUTLINEDinstance-attribute**

AUTO\_FIX\_OFF\_OUTLINED: IconData

**buildAUTO\_FIX\_OFF\_ROUNDEDinstance-attribute**

AUTO\_FIX\_OFF\_ROUNDED: IconData

**buildAUTO\_FIX\_OFF\_SHARPinstance-attribute**

AUTO\_FIX\_OFF\_SHARP: IconData

**buildAUTO\_GRAPHinstance-attribute**

AUTO\_GRAPH: IconData

**buildAUTO\_GRAPH\_OUTLINEDinstance-attribute**

AUTO\_GRAPH\_OUTLINED: IconData

**buildAUTO\_GRAPH\_ROUNDEDinstance-attribute**

AUTO\_GRAPH\_ROUNDED: IconData

**buildAUTO\_GRAPH\_SHARPinstance-attribute**

AUTO\_GRAPH\_SHARP: IconData

**buildAUTO\_MODEinstance-attribute**

AUTO\_MODE: IconData

**buildAUTO\_MODE\_OUTLINEDinstance-attribute**

AUTO\_MODE\_OUTLINED: IconData

**buildAUTO\_MODE\_ROUNDEDinstance-attribute**

AUTO\_MODE\_ROUNDED: IconData

**buildAUTO\_MODE\_SHARPinstance-attribute**

AUTO\_MODE\_SHARP: IconData

**buildAUTO\_STORIESinstance-attribute**

AUTO\_STORIES: IconData

**buildAUTO\_STORIES\_OUTLINEDinstance-attribute**

AUTO\_STORIES\_OUTLINED: IconData

**buildAUTO\_STORIES\_ROUNDEDinstance-attribute**

AUTO\_STORIES\_ROUNDED: IconData

**buildAUTO\_STORIES\_SHARPinstance-attribute**

AUTO\_STORIES\_SHARP: IconData

**buildAUTOFPS\_SELECTinstance-attribute**

AUTOFPS\_SELECT: IconData

**buildAUTOFPS\_SELECT\_OUTLINEDinstance-attribute**

AUTOFPS\_SELECT\_OUTLINED: IconData

**buildAUTOFPS\_SELECT\_ROUNDEDinstance-attribute**

AUTOFPS\_SELECT\_ROUNDED: IconData

**buildAUTOFPS\_SELECT\_SHARPinstance-attribute**

AUTOFPS\_SELECT\_SHARP: IconData

**buildAUTORENEWinstance-attribute**

AUTORENEW: IconData

**buildAUTORENEW\_OUTLINEDinstance-attribute**

AUTORENEW\_OUTLINED: IconData

**buildAUTORENEW\_ROUNDEDinstance-attribute**

AUTORENEW\_ROUNDED: IconData

**buildAUTORENEW\_SHARPinstance-attribute**

AUTORENEW\_SHARP: IconData

**buildAV\_TIMERinstance-attribute**

AV\_TIMER: IconData

**buildAV\_TIMER\_OUTLINEDinstance-attribute**

AV\_TIMER\_OUTLINED: IconData

**buildAV\_TIMER\_ROUNDEDinstance-attribute**

AV\_TIMER\_ROUNDED: IconData

**buildAV\_TIMER\_SHARPinstance-attribute**

AV\_TIMER\_SHARP: IconData

**buildBABY\_CHANGING\_STATIONinstance-attribute**

BABY\_CHANGING\_STATION: IconData

**buildBABY\_CHANGING\_STATION\_OUTLINEDinstance-attribute**

BABY\_CHANGING\_STATION\_OUTLINED: IconData

**buildBABY\_CHANGING\_STATION\_ROUNDEDinstance-attribute**

BABY\_CHANGING\_STATION\_ROUNDED: IconData

**buildBABY\_CHANGING\_STATION\_SHARPinstance-attribute**

BABY\_CHANGING\_STATION\_SHARP: IconData

**buildBACK\_HANDinstance-attribute**

BACK\_HAND: IconData

**buildBACK\_HAND\_OUTLINEDinstance-attribute**

BACK\_HAND\_OUTLINED: IconData

**buildBACK\_HAND\_ROUNDEDinstance-attribute**

BACK\_HAND\_ROUNDED: IconData

**buildBACK\_HAND\_SHARPinstance-attribute**

BACK\_HAND\_SHARP: IconData

**buildBACKPACKinstance-attribute**

BACKPACK: IconData

**buildBACKPACK\_OUTLINEDinstance-attribute**

BACKPACK\_OUTLINED: IconData

**buildBACKPACK\_ROUNDEDinstance-attribute**

BACKPACK\_ROUNDED: IconData

**buildBACKPACK\_SHARPinstance-attribute**

BACKPACK\_SHARP: IconData

**buildBACKSPACEinstance-attribute**

BACKSPACE: IconData

**buildBACKSPACE\_OUTLINEDinstance-attribute**

BACKSPACE\_OUTLINED: IconData

**buildBACKSPACE\_ROUNDEDinstance-attribute**

BACKSPACE\_ROUNDED: IconData

**buildBACKSPACE\_SHARPinstance-attribute**

BACKSPACE\_SHARP: IconData

**buildBACKUPinstance-attribute**

BACKUP: IconData

**buildBACKUP\_OUTLINEDinstance-attribute**

BACKUP\_OUTLINED: IconData

**buildBACKUP\_ROUNDEDinstance-attribute**

BACKUP\_ROUNDED: IconData

**buildBACKUP\_SHARPinstance-attribute**

BACKUP\_SHARP: IconData

**buildBACKUP\_TABLEinstance-attribute**

BACKUP\_TABLE: IconData

**buildBACKUP\_TABLE\_OUTLINEDinstance-attribute**

BACKUP\_TABLE\_OUTLINED: IconData

**buildBACKUP\_TABLE\_ROUNDEDinstance-attribute**

BACKUP\_TABLE\_ROUNDED: IconData

**buildBACKUP\_TABLE\_SHARPinstance-attribute**

BACKUP\_TABLE\_SHARP: IconData

**buildBADGEinstance-attribute**

BADGE: IconData

**buildBADGE\_OUTLINEDinstance-attribute**

BADGE\_OUTLINED: IconData

**buildBADGE\_ROUNDEDinstance-attribute**

BADGE\_ROUNDED: IconData

**buildBADGE\_SHARPinstance-attribute**

BADGE\_SHARP: IconData

**buildBAKERY\_DININGinstance-attribute**

BAKERY\_DINING: IconData

**buildBAKERY\_DINING\_OUTLINEDinstance-attribute**

BAKERY\_DINING\_OUTLINED: IconData

**buildBAKERY\_DINING\_ROUNDEDinstance-attribute**

BAKERY\_DINING\_ROUNDED: IconData

**buildBAKERY\_DINING\_SHARPinstance-attribute**

BAKERY\_DINING\_SHARP: IconData

**buildBALANCEinstance-attribute**

BALANCE: IconData

**buildBALANCE\_OUTLINEDinstance-attribute**

BALANCE\_OUTLINED: IconData

**buildBALANCE\_ROUNDEDinstance-attribute**

BALANCE\_ROUNDED: IconData

**buildBALANCE\_SHARPinstance-attribute**

BALANCE\_SHARP: IconData

**buildBALCONYinstance-attribute**

BALCONY: IconData

**buildBALCONY\_OUTLINEDinstance-attribute**

BALCONY\_OUTLINED: IconData

**buildBALCONY\_ROUNDEDinstance-attribute**

BALCONY\_ROUNDED: IconData

**buildBALCONY\_SHARPinstance-attribute**

BALCONY\_SHARP: IconData

**buildBALLOTinstance-attribute**

BALLOT: IconData

**buildBALLOT\_OUTLINEDinstance-attribute**

BALLOT\_OUTLINED: IconData

**buildBALLOT\_ROUNDEDinstance-attribute**

BALLOT\_ROUNDED: IconData

**buildBALLOT\_SHARPinstance-attribute**

BALLOT\_SHARP: IconData

**buildBAR\_CHARTinstance-attribute**

BAR\_CHART: IconData

**buildBAR\_CHART\_OUTLINEDinstance-attribute**

BAR\_CHART\_OUTLINED: IconData

**buildBAR\_CHART\_ROUNDEDinstance-attribute**

BAR\_CHART\_ROUNDED: IconData

**buildBAR\_CHART\_SHARPinstance-attribute**

BAR\_CHART\_SHARP: IconData

**buildBARCODE\_READERinstance-attribute**

BARCODE\_READER: IconData

**buildBATCH\_PREDICTIONinstance-attribute**

BATCH\_PREDICTION: IconData

**buildBATCH\_PREDICTION\_OUTLINEDinstance-attribute**

BATCH\_PREDICTION\_OUTLINED: IconData

**buildBATCH\_PREDICTION\_ROUNDEDinstance-attribute**

BATCH\_PREDICTION\_ROUNDED: IconData

**buildBATCH\_PREDICTION\_SHARPinstance-attribute**

BATCH\_PREDICTION\_SHARP: IconData



**buildBATHROOMinstance-attribute**

BATHROOM: IconData

**buildBATHROOM\_OUTLINEDinstance-attribute**

BATHROOM\_OUTLINED: IconData

**buildBATHROOM\_ROUNDEDinstance-attribute**

BATHROOM\_ROUNDED: IconData

**buildBATHROOM\_SHARPinstance-attribute**

BATHROOM\_SHARP: IconData

**buildBATHTUBinstance-attribute**

BATHTUB: IconData

**buildBATHTUB\_OUTLINEDinstance-attribute**

BATHTUB\_OUTLINED: IconData

**buildBATHTUB\_ROUNDEDinstance-attribute**

BATHTUB\_ROUNDED: IconData

**buildBATHTUB\_SHARPinstance-attribute**

BATHTUB\_SHARP: IconData

**buildBATTERY\_0\_BARinstance-attribute**

BATTERY\_0\_BAR: IconData

**buildBATTERY\_0\_BAR\_OUTLINEDinstance-attribute**

BATTERY\_0\_BAR\_OUTLINED: IconData

**buildBATTERY\_0\_BAR\_ROUNDEDinstance-attribute**

BATTERY\_0\_BAR\_ROUNDED: IconData

**buildBATTERY\_0\_BAR\_SHARPinstance-attribute**

BATTERY\_0\_BAR\_SHARP: IconData

**buildBATTERY\_1\_BARinstance-attribute**

BATTERY\_1\_BAR: IconData

**buildBATTERY\_1\_BAR\_OUTLINEDinstance-attribute**

BATTERY\_1\_BAR\_OUTLINED: IconData

**buildBATTERY\_1\_BAR\_ROUNDEDinstance-attribute**

BATTERY\_1\_BAR\_ROUNDED: IconData

**buildBATTERY\_1\_BAR\_SHARPinstance-attribute**

BATTERY\_1\_BAR\_SHARP: IconData

**buildBATTERY\_2\_BARinstance-attribute**

BATTERY\_2\_BAR: IconData

**buildBATTERY\_2\_BAR\_OUTLINEDinstance-attribute**

BATTERY\_2\_BAR\_OUTLINED: IconData

**buildBATTERY\_2\_BAR\_ROUNDEDinstance-attribute**

BATTERY\_2\_BAR\_ROUNDED: IconData

**buildBATTERY\_2\_BAR\_SHARPinstance-attribute**

BATTERY\_2\_BAR\_SHARP: IconData

**buildBATTERY\_3\_BARinstance-attribute**

BATTERY\_3\_BAR: IconData

**buildBATTERY\_3\_BAR\_OUTLINEDinstance-attribute**

BATTERY\_3\_BAR\_OUTLINED: IconData

buildBATTERY\_3\_BAR\_ROUNDEDinstance-attribute

BATTERY\_3\_BAR\_ROUNDED: IconData

buildBATTERY\_3\_BAR\_SHARPinstance-attribute

BATTERY\_3\_BAR\_SHARP: IconData

buildBATTERY\_4\_BARinstance-attribute

BATTERY\_4\_BAR: IconData

buildBATTERY\_4\_BAR\_OUTLINEDinstance-attribute

BATTERY\_4\_BAR\_OUTLINED: IconData

buildBATTERY\_4\_BAR\_ROUNDEDinstance-attribute

BATTERY\_4\_BAR\_ROUNDED: IconData

buildBATTERY\_4\_BAR\_SHARPinstance-attribute

BATTERY\_4\_BAR\_SHARP: IconData

buildBATTERY\_5\_BARinstance-attribute

BATTERY\_5\_BAR: IconData

buildBATTERY\_5\_BAR\_OUTLINEDinstance-attribute

BATTERY\_5\_BAR\_OUTLINED: IconData

buildBATTERY\_5\_BAR\_ROUNDEDinstance-attribute

BATTERY\_5\_BAR\_ROUNDED: IconData

buildBATTERY\_5\_BAR\_SHARPinstance-attribute

BATTERY\_5\_BAR\_SHARP: IconData

buildBATTERY\_6\_BARinstance-attribute

BATTERY\_6\_BAR: IconData

buildBATTERY\_6\_BAR\_OUTLINEDinstance-attribute

BATTERY\_6\_BAR\_OUTLINED: IconData

buildBATTERY\_6\_BAR\_ROUNDEDinstance-attribute

BATTERY\_6\_BAR\_ROUNDED: IconData

buildBATTERY\_6\_BAR\_SHARPinstance-attribute

BATTERY\_6\_BAR\_SHARP: IconData

buildBATTERY\_ALERTinstance-attribute

BATTERY\_ALERT: IconData

buildBATTERY\_ALERT\_OUTLINEDinstance-attribute

BATTERY\_ALERT\_OUTLINED: IconData

buildBATTERY\_ALERT\_ROUNDEDinstance-attribute

BATTERY\_ALERT\_ROUNDED: IconData

buildBATTERY\_ALERT\_SHARPinstance-attribute

BATTERY\_ALERT\_SHARP: IconData

buildBATTERY\_CHARGING\_FULLinstance-attribute

BATTERY\_CHARGING\_FULL: IconData

buildBATTERY\_CHARGING\_FULL\_OUTLINEDinstance-attribute

BATTERY\_CHARGING\_FULL\_OUTLINED: IconData

buildBATTERY\_CHARGING\_FULL\_ROUNDEDinstance-attribute

BATTERY\_CHARGING\_FULL\_ROUNDED: IconData

buildBATTERY\_CHARGING\_FULL\_SHARPinstance-attribute

BATTERY\_CHARGING\_FULL\_SHARP: IconData

**buildBATTERY\_FULLInstance-attribute**

BATTERY\_FULL: IconData

**buildBATTERY\_FULL\_OUTLINEDInstance-attribute**

BATTERY\_FULL\_OUTLINED: IconData

**buildBATTERY\_FULL\_ROUNDEDInstance-attribute**

BATTERY\_FULL\_ROUNDED: IconData

**buildBATTERY\_FULL\_SHARPInstance-attribute**

BATTERY\_FULL\_SHARP: IconData

**buildBATTERY\_SAVERInstance-attribute**

BATTERY\_SAVER: IconData

**buildBATTERY\_SAVER\_OUTLINEDInstance-attribute**

BATTERY\_SAVER\_OUTLINED: IconData

**buildBATTERY\_SAVER\_ROUNDEDInstance-attribute**

BATTERY\_SAVER\_ROUNDED: IconData

**buildBATTERY\_SAVER\_SHARPInstance-attribute**

BATTERY\_SAVER\_SHARP: IconData

**buildBATTERY\_STDInstance-attribute**

BATTERY\_STD: IconData

**buildBATTERY\_STD\_OUTLINEDInstance-attribute**

BATTERY\_STD\_OUTLINED: IconData

**buildBATTERY\_STD\_ROUNDEDInstance-attribute**

BATTERY\_STD\_ROUNDED: IconData

**buildBATTERY\_STD\_SHARPInstance-attribute**

BATTERY\_STD\_SHARP: IconData

**buildBATTERY\_UNKNOWNInstance-attribute**

BATTERY\_UNKNOWN: IconData

**buildBATTERY\_UNKNOWN\_OUTLINEDInstance-attribute**

BATTERY\_UNKNOWN\_OUTLINED: IconData

**buildBATTERY\_UNKNOWN\_ROUNDEDInstance-attribute**

BATTERY\_UNKNOWN\_ROUNDED: IconData

**buildBATTERY\_UNKNOWN\_SHARPInstance-attribute**

BATTERY\_UNKNOWN\_SHARP: IconData

**buildBEACH\_ACCESSInstance-attribute**

BEACH\_ACCESS: IconData

**buildBEACH\_ACCESS\_OUTLINEDInstance-attribute**

BEACH\_ACCESS\_OUTLINED: IconData

**buildBEACH\_ACCESS\_ROUNDEDInstance-attribute**

BEACH\_ACCESS\_ROUNDED: IconData

**buildBEACH\_ACCESS\_SHARPInstance-attribute**

BEACH\_ACCESS\_SHARP: IconData

**buildBEDInstance-attribute**

BED: IconData

**buildBED\_OUTLINEDInstance-attribute**

BED\_OUTLINED: IconData

buildBED\_ROUNDEDinstance-attribute

BED\_ROUNDED: IconData

buildBED\_SHARPinstance-attribute

BED\_SHARP: IconData

buildBEDROOM\_BABYinstance-attribute

BEDROOM\_BABY: IconData

buildBEDROOM\_BABY\_OUTLINEDinstance-attribute

BEDROOM\_BABY\_OUTLINED: IconData

buildBEDROOM\_BABY\_ROUNDEDinstance-attribute

BEDROOM\_BABY\_ROUNDED: IconData

buildBEDROOM\_BABY\_SHARPinstance-attribute

BEDROOM\_BABY\_SHARP: IconData

buildBEDROOM\_CHILDinstance-attribute

BEDROOM\_CHILD: IconData

buildBEDROOM\_CHILD\_OUTLINEDinstance-attribute

BEDROOM\_CHILD\_OUTLINED: IconData

buildBEDROOM\_CHILD\_ROUNDEDinstance-attribute

BEDROOM\_CHILD\_ROUNDED: IconData

buildBEDROOM\_CHILD\_SHARPinstance-attribute

BEDROOM\_CHILD\_SHARP: IconData

buildBEDROOM\_PARENTinstance-attribute

BEDROOM\_PARENT: IconData

buildBEDROOM\_PARENT\_OUTLINEDinstance-attribute

BEDROOM\_PARENT\_OUTLINED: IconData

buildBEDROOM\_PARENT\_ROUNDEDinstance-attribute

BEDROOM\_PARENT\_ROUNDED: IconData

buildBEDROOM\_PARENT\_SHARPinstance-attribute

BEDROOM\_PARENT\_SHARP: IconData

buildBEDTIMEinstance-attribute

BEDTIME: IconData

buildBEDTIME\_OFFinstance-attribute

BEDTIME\_OFF: IconData

buildBEDTIME\_OFF\_OUTLINEDinstance-attribute

BEDTIME\_OFF\_OUTLINED: IconData

buildBEDTIME\_OFF\_ROUNDEDinstance-attribute

BEDTIME\_OFF\_ROUNDED: IconData

buildBEDTIME\_OFF\_SHARPinstance-attribute

BEDTIME\_OFF\_SHARP: IconData

buildBEDTIME\_OUTLINEDinstance-attribute

BEDTIME\_OUTLINED: IconData

buildBEDTIME\_ROUNDEDinstance-attribute

BEDTIME\_ROUNDED: IconData

buildBEDTIME\_SHARPinstance-attribute

BEDTIME\_SHARP: IconData

**buildBEENHEREinstance-attribute**

BEENHERE: IconData

**buildBEENHERE\_OUTLINEDinstance-attribute**

BEENHERE\_OUTLINED: IconData

**buildBEENHERE\_ROUNDEDinstance-attribute**

BEENHERE\_ROUNDED: IconData

**buildBEENHERE\_SHARPinstance-attribute**

BEENHERE\_SHARP: IconData

**buildBENTOinstance-attribute**

BENTO: IconData

**buildBENTO\_OUTLINEDinstance-attribute**

BENTO\_OUTLINED: IconData

**buildBENTO\_ROUNDEDinstance-attribute**

BENTO\_ROUNDED: IconData

**buildBENTO\_SHARPinstance-attribute**

BENTO\_SHARP: IconData

**buildBIKE\_SCOOTERinstance-attribute**

BIKE\_SCOOTER: IconData

**buildBIKE\_SCOOTER\_OUTLINEDinstance-attribute**

BIKE\_SCOOTER\_OUTLINED: IconData

**buildBIKE\_SCOOTER\_ROUNDEDinstance-attribute**

BIKE\_SCOOTER\_ROUNDED: IconData

**buildBIKE\_SCOOTER\_SHARPinstance-attribute**

BIKE\_SCOOTER\_SHARP: IconData

**buildBIOTECHinstance-attribute**

BIOTECH: IconData

**buildBIOTECH\_OUTLINEDinstance-attribute**

BIOTECH\_OUTLINED: IconData

**buildBIOTECH\_ROUNDEDinstance-attribute**

BIOTECH\_ROUNDED: IconData

**buildBIOTECH\_SHARPinstance-attribute**

BIOTECH\_SHARP: IconData

**buildBLENDERinstance-attribute**

BLENDER: IconData

**buildBLENDER\_OUTLINEDinstance-attribute**

BLENDER\_OUTLINED: IconData

**buildBLENDER\_ROUNDEDinstance-attribute**

BLENDER\_ROUNDED: IconData

**buildBLENDER\_SHARPinstance-attribute**

BLENDER\_SHARP: IconData

**buildBLINDinstance-attribute**

BLIND: IconData

**buildBLIND\_OUTLINEDinstance-attribute**

BLIND\_OUTLINED: IconData

**buildBLIND\_ROUNDEDinstance-attribute**

BLIND\_ROUNDED: IconData

**buildBLIND\_SHARPinstance-attribute**

BLIND\_SHARP: IconData

**buildBLINDSinstance-attribute**

BLINDS: IconData

**buildBLINDS\_CLOSEDinstance-attribute**

BLINDS\_CLOSED: IconData

**buildBLINDS\_CLOSED\_OUTLINEDinstance-attribute**

BLINDS\_CLOSED\_OUTLINED: IconData

**buildBLINDS\_CLOSED\_ROUNDEDinstance-attribute**

BLINDS\_CLOSED\_ROUNDED: IconData

**buildBLINDS\_CLOSED\_SHARPinstance-attribute**

BLINDS\_CLOSED\_SHARP: IconData

**buildBLINDS\_OUTLINEDinstance-attribute**

BLINDS\_OUTLINED: IconData

**buildBLINDS\_ROUNDEDinstance-attribute**

BLINDS\_ROUNDED: IconData

**buildBLINDS\_SHARPinstance-attribute**

BLINDS\_SHARP: IconData

**buildBLOCKinstance-attribute**

BLOCK: IconData

**buildBLOCK\_FLIPPEDinstance-attribute**

BLOCK\_FLIPPED: IconData

**buildBLOCK\_OUTLINEDinstance-attribute**

BLOCK\_OUTLINED: IconData

**buildBLOCK\_ROUNDEDinstance-attribute**

BLOCK\_ROUNDED: IconData

**buildBLOCK\_SHARPinstance-attribute**

BLOCK\_SHARP: IconData

**buildBLOODTYPEinstance-attribute**

BLOODTYPE: IconData

**buildBLOODTYPE\_OUTLINEDinstance-attribute**

BLOODTYPE\_OUTLINED: IconData

**buildBLOODTYPE\_ROUNDEDinstance-attribute**

BLOODTYPE\_ROUNDED: IconData

**buildBLOODTYPE\_SHARPinstance-attribute**

BLOODTYPE\_SHARP: IconData

**buildBLUETOOTHinstance-attribute**

BLUETOOTH: IconData

**buildBLUETOOTH\_AUDIOinstance-attribute**

BLUETOOTH\_AUDIO: IconData

**buildBLUETOOTH\_AUDIO\_OUTLINEDinstance-attribute**

BLUETOOTH\_AUDIO\_OUTLINED: IconData

**buildBLUETOOTH\_AUDIO\_ROUNDEDinstance-attribute**

BLUETOOTH\_AUDIO\_ROUNDED: IconData

**buildBLUETOOTH\_AUDIO\_SHARPinstance-attribute**

BLUETOOTH\_AUDIO\_SHARP: IconData

**buildBLUETOOTH\_CONNECTEDinstance-attribute**

BLUETOOTH\_CONNECTED: IconData

**buildBLUETOOTH\_CONNECTED\_OUTLINEDinstance-attribute**

BLUETOOTH\_CONNECTED\_OUTLINED: IconData

**buildBLUETOOTH\_CONNECTED\_ROUNDEDinstance-attribute**

BLUETOOTH\_CONNECTED\_ROUNDED: IconData

**buildBLUETOOTH\_CONNECTED\_SHARPinstance-attribute**

BLUETOOTH\_CONNECTED\_SHARP: IconData

**buildBLUETOOTH\_DISABLEDinstance-attribute**

BLUETOOTH\_DISABLED: IconData

**buildBLUETOOTH\_DISABLED\_OUTLINEDinstance-attribute**

BLUETOOTH\_DISABLED\_OUTLINED: IconData

**buildBLUETOOTH\_DISABLED\_ROUNDEDinstance-attribute**

BLUETOOTH\_DISABLED\_ROUNDED: IconData

**buildBLUETOOTH\_DISABLED\_SHARPinstance-attribute**

BLUETOOTH\_DISABLED\_SHARP: IconData

**buildBLUETOOTH\_DRIVEinstance-attribute**

BLUETOOTH\_DRIVE: IconData

**buildBLUETOOTH\_DRIVE\_OUTLINEDinstance-attribute**

BLUETOOTH\_DRIVE\_OUTLINED: IconData

**buildBLUETOOTH\_DRIVE\_ROUNDEDinstance-attribute**

BLUETOOTH\_DRIVE\_ROUNDED: IconData

**buildBLUETOOTH\_DRIVE\_SHARPinstance-attribute**

BLUETOOTH\_DRIVE\_SHARP: IconData

**buildBLUETOOTH\_OUTLINEDinstance-attribute**

BLUETOOTH\_OUTLINED: IconData

**buildBLUETOOTH\_ROUNDEDinstance-attribute**

BLUETOOTH\_ROUNDED: IconData

**buildBLUETOOTH\_SEARCHINGinstance-attribute**

BLUETOOTH\_SEARCHING: IconData

**buildBLUETOOTH\_SEARCHING\_OUTLINEDinstance-attribute**

BLUETOOTH\_SEARCHING\_OUTLINED: IconData

**buildBLUETOOTH\_SEARCHING\_ROUNDEDinstance-attribute**

BLUETOOTH\_SEARCHING\_ROUNDED: IconData

**buildBLUETOOTH\_SEARCHING\_SHARPinstance-attribute**

BLUETOOTH\_SEARCHING\_SHARP: IconData

**buildBLUETOOTH\_SHARPinstance-attribute**

BLUETOOTH\_SHARP: IconData

**buildBLUR\_CIRCULARinstance-attribute**

BLUR\_CIRCULAR: IconData

**buildBLUR\_CIRCULAR\_OUTLINED**instance-attribute

BLUR\_CIRCULAR\_OUTLINED: IconData

**buildBLUR\_CIRCULAR\_ROUNDED**instance-attribute

BLUR\_CIRCULAR\_ROUNDED: IconData

**buildBLUR\_CIRCULAR\_SHARP**instance-attribute

BLUR\_CIRCULAR\_SHARP: IconData

**buildBLUR\_LINEAR**instance-attribute

BLUR\_LINEAR: IconData

**buildBLUR\_LINEAR\_OUTLINED**instance-attribute

BLUR\_LINEAR\_OUTLINED: IconData

**buildBLUR\_LINEAR\_ROUNDED**instance-attribute

BLUR\_LINEAR\_ROUNDED: IconData

**buildBLUR\_LINEAR\_SHARP**instance-attribute

BLUR\_LINEAR\_SHARP: IconData

**buildBLUR\_OFF**instance-attribute

BLUR\_OFF: IconData

**buildBLUR\_OFF\_OUTLINED**instance-attribute

BLUR\_OFF\_OUTLINED: IconData

**buildBLUR\_OFF\_ROUNDED**instance-attribute

BLUR\_OFF\_ROUNDED: IconData

**buildBLUR\_OFF\_SHARP**instance-attribute

BLUR\_OFF\_SHARP: IconData

**buildBLUR\_ON**instance-attribute

BLUR\_ON: IconData

**buildBLUR\_ON\_OUTLINED**instance-attribute

BLUR\_ON\_OUTLINED: IconData

**buildBLUR\_ON\_ROUNDED**instance-attribute

BLUR\_ON\_ROUNDED: IconData

**buildBLUR\_ON\_SHARP**instance-attribute

BLUR\_ON\_SHARP: IconData

**buildBOLT**instance-attribute

BOLT: IconData

**buildBOLT\_OUTLINED**instance-attribute

BOLT\_OUTLINED: IconData

**buildBOLT\_ROUNDED**instance-attribute

BOLT\_ROUNDED: IconData

**buildBOLT\_SHARP**instance-attribute

BOLT\_SHARP: IconData

**buildBOOK**instance-attribute

BOOK: IconData

**buildBOOK\_ONLINE**instance-attribute

BOOK\_ONLINE: IconData

**buildBOOK\_ONLINE\_OUTLINED**instance-attribute

BOOK\_ONLINE\_OUTLINED: IconData



buildBOOK\_ONLINE\_ROUNDEDinstance-attribute

BOOK\_ONLINE\_ROUNDED: IconData

buildBOOK\_ONLINE\_SHARPinstance-attribute

BOOK\_ONLINE\_SHARP: IconData

buildBOOK\_OUTLINEDinstance-attribute

BOOK\_OUTLINED: IconData

buildBOOK\_ROUNDEDinstance-attribute

BOOK\_ROUNDED: IconData

buildBOOK\_SHARPinstance-attribute

BOOK\_SHARP: IconData

buildBOOKMARKinstance-attribute

BOOKMARK: IconData

buildBOOKMARK\_ADDinstance-attribute

BOOKMARK\_ADD: IconData

buildBOOKMARK\_ADD\_OUTLINEDinstance-attribute

BOOKMARK\_ADD\_OUTLINED: IconData

buildBOOKMARK\_ADD\_ROUNDEDinstance-attribute

BOOKMARK\_ADD\_ROUNDED: IconData

buildBOOKMARK\_ADD\_SHARPinstance-attribute

BOOKMARK\_ADD\_SHARP: IconData

buildBOOKMARK\_ADDEDinstance-attribute

BOOKMARK\_ADDED: IconData

buildBOOKMARK\_ADDED\_OUTLINEDinstance-attribute

BOOKMARK\_ADDED\_OUTLINED: IconData

buildBOOKMARK\_ADDED\_ROUNDEDinstance-attribute

BOOKMARK\_ADDED\_ROUNDED: IconData

buildBOOKMARK\_ADDED\_SHARPinstance-attribute

BOOKMARK\_ADDED\_SHARP: IconData

buildBOOKMARK\_BORDERinstance-attribute

BOOKMARK\_BORDER: IconData

buildBOOKMARK\_BORDER\_OUTLINEDinstance-attribute

BOOKMARK\_BORDER\_OUTLINED: IconData

buildBOOKMARK\_BORDER\_ROUNDEDinstance-attribute

BOOKMARK\_BORDER\_ROUNDED: IconData

buildBOOKMARK\_BORDER\_SHARPinstance-attribute

BOOKMARK\_BORDER\_SHARP: IconData

buildBOOKMARK\_OUTLINEinstance-attribute

BOOKMARK\_OUTLINE: IconData

buildBOOKMARK\_OUTLINE\_OUTLINEDinstance-attribute

BOOKMARK\_OUTLINE\_OUTLINED: IconData

buildBOOKMARK\_OUTLINE\_ROUNDEDinstance-attribute

BOOKMARK\_OUTLINE\_ROUNDED: IconData

buildBOOKMARK\_OUTLINE\_SHARPinstance-attribute

BOOKMARK\_OUTLINE\_SHARP: IconData

buildBOOKMARK\_OUTLINEDinstance-attribute

BOOKMARK\_OUTLINED: IconData

buildBOOKMARK\_REMOVEinstance-attribute

BOOKMARK\_REMOVE: IconData

buildBOOKMARK\_REMOVE\_OUTLINEDinstance-attribute

BOOKMARK\_REMOVE\_OUTLINED: IconData

buildBOOKMARK\_REMOVE\_ROUNDEDinstance-attribute

BOOKMARK\_REMOVE\_ROUNDED: IconData

buildBOOKMARK\_REMOVE\_SHARPinstance-attribute

BOOKMARK\_REMOVE\_SHARP: IconData

buildBOOKMARK\_ROUNDEDinstance-attribute

BOOKMARK\_ROUNDED: IconData

buildBOOKMARK\_SHARPinstance-attribute

BOOKMARK\_SHARP: IconData

buildBOOKMARKSinstance-attribute

BOOKMARKS: IconData

buildBOOKMARKS\_OUTLINEDinstance-attribute

BOOKMARKS\_OUTLINED: IconData

buildBOOKMARKS\_ROUNDEDinstance-attribute

BOOKMARKS\_ROUNDED: IconData

buildBOOKMARKS\_SHARPinstance-attribute

BOOKMARKS\_SHARP: IconData

buildBORDER\_ALLinstance-attribute

BORDER\_ALL: IconData

buildBORDER\_ALL\_OUTLINEDinstance-attribute

BORDER\_ALL\_OUTLINED: IconData

buildBORDER\_ALL\_ROUNDEDinstance-attribute

BORDER\_ALL\_ROUNDED: IconData

buildBORDER\_ALL\_SHARPinstance-attribute

BORDER\_ALL\_SHARP: IconData

buildBORDER\_BOTTOMinstance-attribute

BORDER\_BOTTOM: IconData

buildBORDER\_BOTTOM\_OUTLINEDinstance-attribute

BORDER\_BOTTOM\_OUTLINED: IconData

buildBORDER\_BOTTOM\_ROUNDEDinstance-attribute

BORDER\_BOTTOM\_ROUNDED: IconData

buildBORDER\_BOTTOM\_SHARPinstance-attribute

BORDER\_BOTTOM\_SHARP: IconData

buildBORDER\_CLEARinstance-attribute

BORDER\_CLEAR: IconData

buildBORDER\_CLEAR\_OUTLINEDinstance-attribute

BORDER\_CLEAR\_OUTLINED: IconData

buildBORDER\_CLEAR\_ROUNDEDinstance-attribute

BORDER\_CLEAR\_ROUNDED: IconData

**buildBORDER\_CLEAR\_SHARPinstance-attribute**

BORDER\_CLEAR\_SHARP: IconData

**buildBORDER\_COLORinstance-attribute**

BORDER\_COLOR: IconData

**buildBORDER\_COLOR\_OUTLINEDinstance-attribute**

BORDER\_COLOR\_OUTLINED: IconData

**buildBORDER\_COLOR\_ROUNDEDinstance-attribute**

BORDER\_COLOR\_ROUNDED: IconData

**buildBORDER\_COLOR\_SHARPinstance-attribute**

BORDER\_COLOR\_SHARP: IconData

**buildBORDER\_HORIZONTALinstance-attribute**

BORDER\_HORIZONTAL: IconData

**buildBORDER\_HORIZONTAL\_OUTLINEDinstance-attribute**

BORDER\_HORIZONTAL\_OUTLINED: IconData

**buildBORDER\_HORIZONTAL\_ROUNDEDinstance-attribute**

BORDER\_HORIZONTAL\_ROUNDED: IconData

**buildBORDER\_HORIZONTAL\_SHARPinstance-attribute**

BORDER\_HORIZONTAL\_SHARP: IconData

**buildBORDER\_INNERinstance-attribute**

BORDER\_INNER: IconData

**buildBORDER\_INNER\_OUTLINEDinstance-attribute**

BORDER\_INNER\_OUTLINED: IconData

**buildBORDER\_INNER\_ROUNDEDinstance-attribute**

BORDER\_INNER\_ROUNDED: IconData

**buildBORDER\_INNER\_SHARPinstance-attribute**

BORDER\_INNER\_SHARP: IconData

**buildBORDER\_LEFTinstance-attribute**

BORDER\_LEFT: IconData

**buildBORDER\_LEFT\_OUTLINEDinstance-attribute**

BORDER\_LEFT\_OUTLINED: IconData

**buildBORDER\_LEFT\_ROUNDEDinstance-attribute**

BORDER\_LEFT\_ROUNDED: IconData

**buildBORDER\_LEFT\_SHARPinstance-attribute**

BORDER\_LEFT\_SHARP: IconData

**buildBORDER\_OUTERinstance-attribute**

BORDER\_OUTER: IconData

**buildBORDER\_OUTER\_OUTLINEDinstance-attribute**

BORDER\_OUTER\_OUTLINED: IconData

**buildBORDER\_OUTER\_ROUNDEDinstance-attribute**

BORDER\_OUTER\_ROUNDED: IconData

**buildBORDER\_OUTER\_SHARPinstance-attribute**

BORDER\_OUTER\_SHARP: IconData

**buildBORDER\_RIGHTinstance-attribute**

BORDER\_RIGHT: IconData

**buildBORDER\_RIGHT\_OUTLINED**instance-attribute

BORDER\_RIGHT\_OUTLINED: IconData

**buildBORDER\_RIGHT\_ROUNDED**instance-attribute

BORDER\_RIGHT\_ROUNDED: IconData

**buildBORDER\_RIGHT\_SHARP**instance-attribute

BORDER\_RIGHT\_SHARP: IconData

**buildBORDER\_STYLE**instance-attribute

BORDER\_STYLE: IconData

**buildBORDER\_STYLE\_OUTLINED**instance-attribute

BORDER\_STYLE\_OUTLINED: IconData

**buildBORDER\_STYLE\_ROUNDED**instance-attribute

BORDER\_STYLE\_ROUNDED: IconData

**buildBORDER\_STYLE\_SHARP**instance-attribute

BORDER\_STYLE\_SHARP: IconData

**buildBORDER\_TOP**instance-attribute

BORDER\_TOP: IconData

**buildBORDER\_TOP\_OUTLINED**instance-attribute

BORDER\_TOP\_OUTLINED: IconData

**buildBORDER\_TOP\_ROUNDED**instance-attribute

BORDER\_TOP\_ROUNDED: IconData

**buildBORDER\_TOP\_SHARP**instance-attribute

BORDER\_TOP\_SHARP: IconData

**buildBORDER\_VERTICAL**instance-attribute

BORDER\_VERTICAL: IconData

**buildBORDER\_VERTICAL\_OUTLINED**instance-attribute

BORDER\_VERTICAL\_OUTLINED: IconData

**buildBORDER\_VERTICAL\_ROUNDED**instance-attribute

BORDER\_VERTICAL\_ROUNDED: IconData

**buildBORDER\_VERTICAL\_SHARP**instance-attribute

BORDER\_VERTICAL\_SHARP: IconData

**buildBOY**instance-attribute

BOY: IconData

**buildBOY\_OUTLINED**instance-attribute

BOY\_OUTLINED: IconData

**buildBOY\_ROUNDED**instance-attribute

BOY\_ROUNDED: IconData

**buildBOY\_SHARP**instance-attribute

BOY\_SHARP: IconData

**buildBRANDING\_WATERMARK**instance-attribute

BRANDING\_WATERMARK: IconData

**buildBRANDING\_WATERMARK\_OUTLINED**instance-attribute

BRANDING\_WATERMARK\_OUTLINED: IconData

**buildBRANDING\_WATERMARK\_ROUNDED**instance-attribute

BRANDING\_WATERMARK\_ROUNDED: IconData

**buildBRANDING\_WATERMARK\_SHARPinstance-attribute**

BRANDING\_WATERMARK\_SHARP: IconData

**buildBREAKFAST\_DININGinstance-attribute**

BREAKFAST\_DINING: IconData

**buildBREAKFAST\_DINING\_OUTLINEDinstance-attribute**

BREAKFAST\_DINING\_OUTLINED: IconData

**buildBREAKFAST\_DINING\_ROUNDEDinstance-attribute**

BREAKFAST\_DINING\_ROUNDED: IconData

**buildBREAKFAST\_DINING\_SHARPinstance-attribute**

BREAKFAST\_DINING\_SHARP: IconData

**buildBRIGHTNESS\_1instance-attribute**

BRIGHTNESS\_1: IconData

**buildBRIGHTNESS\_1\_OUTLINEDinstance-attribute**

BRIGHTNESS\_1\_OUTLINED: IconData

**buildBRIGHTNESS\_1\_ROUNDEDinstance-attribute**

BRIGHTNESS\_1\_ROUNDED: IconData

**buildBRIGHTNESS\_1\_SHARPinstance-attribute**

BRIGHTNESS\_1\_SHARP: IconData

**buildBRIGHTNESS\_2instance-attribute**

BRIGHTNESS\_2: IconData

**buildBRIGHTNESS\_2\_OUTLINEDinstance-attribute**

BRIGHTNESS\_2\_OUTLINED: IconData

**buildBRIGHTNESS\_2\_ROUNDEDinstance-attribute**

BRIGHTNESS\_2\_ROUNDED: IconData

**buildBRIGHTNESS\_2\_SHARPinstance-attribute**

BRIGHTNESS\_2\_SHARP: IconData

**buildBRIGHTNESS\_3instance-attribute**

BRIGHTNESS\_3: IconData

**buildBRIGHTNESS\_3\_OUTLINEDinstance-attribute**

BRIGHTNESS\_3\_OUTLINED: IconData

**buildBRIGHTNESS\_3\_ROUNDEDinstance-attribute**

BRIGHTNESS\_3\_ROUNDED: IconData

**buildBRIGHTNESS\_3\_SHARPinstance-attribute**

BRIGHTNESS\_3\_SHARP: IconData

**buildBRIGHTNESS\_4instance-attribute**

BRIGHTNESS\_4: IconData

**buildBRIGHTNESS\_4\_OUTLINEDinstance-attribute**

BRIGHTNESS\_4\_OUTLINED: IconData

**buildBRIGHTNESS\_4\_ROUNDEDinstance-attribute**

BRIGHTNESS\_4\_ROUNDED: IconData

**buildBRIGHTNESS\_4\_SHARPinstance-attribute**

BRIGHTNESS\_4\_SHARP: IconData

**buildBRIGHTNESS\_5instance-attribute**

BRIGHTNESS\_5: IconData

**buildBRIGHTNESS\_5\_OUTLINED**instance-attribute

BRIGHTNESS\_5\_OUTLINED: IconData

**buildBRIGHTNESS\_5\_ROUNDED**instance-attribute

BRIGHTNESS\_5\_ROUNDED: IconData

**buildBRIGHTNESS\_5\_SHARP**instance-attribute

BRIGHTNESS\_5\_SHARP: IconData

**buildBRIGHTNESS\_6**instance-attribute

BRIGHTNESS\_6: IconData

**buildBRIGHTNESS\_6\_OUTLINED**instance-attribute

BRIGHTNESS\_6\_OUTLINED: IconData

**buildBRIGHTNESS\_6\_ROUNDED**instance-attribute

BRIGHTNESS\_6\_ROUNDED: IconData

**buildBRIGHTNESS\_6\_SHARP**instance-attribute

BRIGHTNESS\_6\_SHARP: IconData

**buildBRIGHTNESS\_7**instance-attribute

BRIGHTNESS\_7: IconData

**buildBRIGHTNESS\_7\_OUTLINED**instance-attribute

BRIGHTNESS\_7\_OUTLINED: IconData

**buildBRIGHTNESS\_7\_ROUNDED**instance-attribute

BRIGHTNESS\_7\_ROUNDED: IconData

**buildBRIGHTNESS\_7\_SHARP**instance-attribute

BRIGHTNESS\_7\_SHARP: IconData

**buildBRIGHTNESS\_AUTO**instance-attribute

BRIGHTNESS\_AUTO: IconData

**buildBRIGHTNESS\_AUTO\_OUTLINED**instance-attribute

BRIGHTNESS\_AUTO\_OUTLINED: IconData

**buildBRIGHTNESS\_AUTO\_ROUNDED**instance-attribute

BRIGHTNESS\_AUTO\_ROUNDED: IconData

**buildBRIGHTNESS\_AUTO\_SHARP**instance-attribute

BRIGHTNESS\_AUTO\_SHARP: IconData

**buildBRIGHTNESS\_HIGH**instance-attribute

BRIGHTNESS\_HIGH: IconData

**buildBRIGHTNESS\_HIGH\_OUTLINED**instance-attribute

BRIGHTNESS\_HIGH\_OUTLINED: IconData

**buildBRIGHTNESS\_HIGH\_ROUNDED**instance-attribute

BRIGHTNESS\_HIGH\_ROUNDED: IconData

**buildBRIGHTNESS\_HIGH\_SHARP**instance-attribute

BRIGHTNESS\_HIGH\_SHARP: IconData

**buildBRIGHTNESS\_LOW**instance-attribute

BRIGHTNESS\_LOW: IconData

**buildBRIGHTNESS\_LOW\_OUTLINED**instance-attribute

BRIGHTNESS\_LOW\_OUTLINED: IconData

**buildBRIGHTNESS\_LOW\_ROUNDED**instance-attribute

BRIGHTNESS\_LOW\_ROUNDED: IconData

**buildBRIGHTNESS\_LOW\_SHARPinstance-attribute**

BRIGHTNESS\_LOW\_SHARP: IconData

**buildBRIGHTNESS\_MEDIUMinstance-attribute**

BRIGHTNESS\_MEDIUM: IconData

**buildBRIGHTNESS\_MEDIUM\_OUTLINEDinstance-attribute**

BRIGHTNESS\_MEDIUM\_OUTLINED: IconData

**buildBRIGHTNESS\_MEDIUM\_ROUNDEDinstance-attribute**

BRIGHTNESS\_MEDIUM\_ROUNDED: IconData

**buildBRIGHTNESS\_MEDIUM\_SHARPinstance-attribute**

BRIGHTNESS\_MEDIUM\_SHARP: IconData

**buildBROADCAST\_ON\_HOMEinstance-attribute**

BROADCAST\_ON\_HOME: IconData

**buildBROADCAST\_ON\_HOME\_OUTLINEDinstance-attribute**

BROADCAST\_ON\_HOME\_OUTLINED: IconData

**buildBROADCAST\_ON\_HOME\_ROUNDEDinstance-attribute**

BROADCAST\_ON\_HOME\_ROUNDED: IconData

**buildBROADCAST\_ON\_HOME\_SHARPinstance-attribute**

BROADCAST\_ON\_HOME\_SHARP: IconData

**buildBROADCAST\_ON\_PERSONALinstance-attribute**

BROADCAST\_ON\_PERSONAL: IconData

**buildBROADCAST\_ON\_PERSONAL\_OUTLINEDinstance-attribute**

BROADCAST\_ON\_PERSONAL\_OUTLINED: IconData

**buildBROADCAST\_ON\_PERSONAL\_ROUNDEDinstance-attribute**

BROADCAST\_ON\_PERSONAL\_ROUNDED: IconData

**buildBROADCAST\_ON\_PERSONAL\_SHARPinstance-attribute**

BROADCAST\_ON\_PERSONAL\_SHARP: IconData

**buildBROKEN\_IMAGEinstance-attribute**

BROKEN\_IMAGE: IconData

**buildBROKEN\_IMAGE\_OUTLINEDinstance-attribute**

BROKEN\_IMAGE\_OUTLINED: IconData

**buildBROKEN\_IMAGE\_ROUNDEDinstance-attribute**

BROKEN\_IMAGE\_ROUNDED: IconData

**buildBROKEN\_IMAGE\_SHARPinstance-attribute**

BROKEN\_IMAGE\_SHARP: IconData

**buildBROWSE\_GALLERYinstance-attribute**

BROWSE\_GALLERY: IconData

**buildBROWSE\_GALLERY\_OUTLINEDinstance-attribute**

BROWSE\_GALLERY\_OUTLINED: IconData

**buildBROWSE\_GALLERY\_ROUNDEDinstance-attribute**

BROWSE\_GALLERY\_ROUNDED: IconData

**buildBROWSE\_GALLERY\_SHARPinstance-attribute**

BROWSE\_GALLERY\_SHARP: IconData

**buildBROWSER\_NOT\_SUPPORTEDinstance-attribute**

BROWSER\_NOT\_SUPPORTED: IconData

**buildBROWSER\_NOT\_SUPPORTED\_OUTLINED**instance-attribute

BROWSER\_NOT\_SUPPORTED\_OUTLINED: IconData

**buildBROWSER\_NOT\_SUPPORTED\_ROUNDED**instance-attribute

BROWSER\_NOT\_SUPPORTED\_ROUNDED: IconData

**buildBROWSER\_NOT\_SUPPORTED\_SHARP**instance-attribute

BROWSER\_NOT\_SUPPORTED\_SHARP: IconData

**buildBROWSER\_UPDATED**instance-attribute

BROWSER\_UPDATED: IconData

**buildBROWSER\_UPDATED\_OUTLINED**instance-attribute

BROWSER\_UPDATED\_OUTLINED: IconData

**buildBROWSER\_UPDATED\_ROUNDED**instance-attribute

BROWSER\_UPDATED\_ROUNDED: IconData

**buildBROWSER\_UPDATED\_SHARP**instance-attribute

BROWSER\_UPDATED\_SHARP: IconData

**buildBRUNCH\_DINING**instance-attribute

BRUNCH\_DINING: IconData

**buildBRUNCH\_DINING\_OUTLINED**instance-attribute

BRUNCH\_DINING\_OUTLINED: IconData

**buildBRUNCH\_DINING\_ROUNDED**instance-attribute

BRUNCH\_DINING\_ROUNDED: IconData

**buildBRUNCH\_DINING\_SHARP**instance-attribute

BRUNCH\_DINING\_SHARP: IconData

**buildBRUSH**instance-attribute

BRUSH: IconData

**buildBRUSH\_OUTLINED**instance-attribute

BRUSH\_OUTLINED: IconData

**buildBRUSH\_ROUNDED**instance-attribute

BRUSH\_ROUNDED: IconData

**buildBRUSH\_SHARP**instance-attribute

BRUSH\_SHARP: IconData

**buildBUBBLE\_CHART**instance-attribute

BUBBLE\_CHART: IconData

**buildBUBBLE\_CHART\_OUTLINED**instance-attribute

BUBBLE\_CHART\_OUTLINED: IconData

**buildBUBBLE\_CHART\_ROUNDED**instance-attribute

BUBBLE\_CHART\_ROUNDED: IconData

**buildBUBBLE\_CHART\_SHARP**instance-attribute

BUBBLE\_CHART\_SHARP: IconData

**buildBUG\_REPORT**instance-attribute

BUG\_REPORT: IconData

**buildBUG\_REPORT\_OUTLINED**instance-attribute

BUG\_REPORT\_OUTLINED: IconData

**buildBUG\_REPORT\_ROUNDED**instance-attribute

BUG\_REPORT\_ROUNDED: IconData



**buildBUG\_REPORT\_SHARPinstance-attribute**  
BUG\_REPORT\_SHARP: IconData

**buildBUILDinstance-attribute**  
BUILD: IconData

**buildBUILD\_CIRCLEinstance-attribute**  
BUILD\_CIRCLE: IconData

**buildBUILD\_CIRCLE\_OUTLINEDinstance-attribute**  
BUILD\_CIRCLE\_OUTLINED: IconData

**buildBUILD\_CIRCLE\_ROUNDEDinstance-attribute**  
BUILD\_CIRCLE\_ROUNDED: IconData

**buildBUILD\_CIRCLE\_SHARPinstance-attribute**  
BUILD\_CIRCLE\_SHARP: IconData

**buildBUILD\_OUTLINEDinstance-attribute**  
BUILD\_OUTLINED: IconData

**buildBUILD\_ROUNDEDinstance-attribute**  
BUILD\_ROUNDED: IconData

**buildBUILD\_SHARPinstance-attribute**  
BUILD\_SHARP: IconData

**buildBUNGALOWinstance-attribute**  
BUNGALOW: IconData

**buildBUNGALOW\_OUTLINEDinstance-attribute**  
BUNGALOW\_OUTLINED: IconData

**buildBUNGALOW\_ROUNDEDinstance-attribute**  
BUNGALOW\_ROUNDED: IconData

**buildBUNGALOW\_SHARPinstance-attribute**  
BUNGALOW\_SHARP: IconData

**buildBURST\_MODEinstance-attribute**  
BURST\_MODE: IconData

**buildBURST\_MODE\_OUTLINEDinstance-attribute**  
BURST\_MODE\_OUTLINED: IconData

**buildBURST\_MODE\_ROUNDEDinstance-attribute**  
BURST\_MODE\_ROUNDED: IconData

**buildBURST\_MODE\_SHARPinstance-attribute**  
BURST\_MODE\_SHARP: IconData

**buildBUS\_ALERTinstance-attribute**  
BUS\_ALERT: IconData

**buildBUS\_ALERT\_OUTLINEDinstance-attribute**  
BUS\_ALERT\_OUTLINED: IconData

**buildBUS\_ALERT\_ROUNDEDinstance-attribute**  
BUS\_ALERT\_ROUNDED: IconData

**buildBUS\_ALERT\_SHARPinstance-attribute**  
BUS\_ALERT\_SHARP: IconData

**buildBUSINESSinstance-attribute**  
BUSINESS: IconData

**buildBUSINESS\_CENTERinstance-attribute**

BUSINESS\_CENTER: IconData

**buildBUSINESS\_CENTER\_OUTLINEDinstance-attribute**

BUSINESS\_CENTER\_OUTLINED: IconData

**buildBUSINESS\_CENTER\_ROUNDEDinstance-attribute**

BUSINESS\_CENTER\_ROUNDED: IconData

**buildBUSINESS\_CENTER\_SHARPinstance-attribute**

BUSINESS\_CENTER\_SHARP: IconData

**buildBUSINESS\_OUTLINEDinstance-attribute**

BUSINESS\_OUTLINED: IconData

**buildBUSINESS\_ROUNDEDinstance-attribute**

BUSINESS\_ROUNDED: IconData

**buildBUSINESS\_SHARPinstance-attribute**

BUSINESS\_SHARP: IconData

**buildCABINinstance-attribute**

CABIN: IconData

**buildCABIN\_OUTLINEDinstance-attribute**

CABIN\_OUTLINED: IconData

**buildCABIN\_ROUNDEDinstance-attribute**

CABIN\_ROUNDED: IconData

**buildCABIN\_SHARPinstance-attribute**

CABIN\_SHARP: IconData

**buildCABLEinstance-attribute**

CABLE: IconData

**buildCABLE\_OUTLINEDinstance-attribute**

CABLE\_OUTLINED: IconData

**buildCABLE\_ROUNDEDinstance-attribute**

CABLE\_ROUNDED: IconData

**buildCABLE\_SHARPinstance-attribute**

CABLE\_SHARP: IconData

**buildCACHEDinstance-attribute**

CACHED: IconData

**buildCACHED\_OUTLINEDinstance-attribute**

CACHED\_OUTLINED: IconData

**buildCACHED\_ROUNDEDinstance-attribute**

CACHED\_ROUNDED: IconData

**buildCACHED\_SHARPinstance-attribute**

CACHED\_SHARP: IconData

**buildCAKEinstance-attribute**

CAKE: IconData

**buildCAKE\_OUTLINEDinstance-attribute**

CAKE\_OUTLINED: IconData

**buildCAKE\_ROUNDEDinstance-attribute**

CAKE\_ROUNDED: IconData

**buildCAKE\_SHARPinstance-attribute**

CAKE\_SHARP: IconData

**buildCALCULATEinstance-attribute**

CALCULATE: IconData

**buildCALCULATE\_OUTLINEDinstance-attribute**

CALCULATE\_OUTLINED: IconData

**buildCALCULATE\_ROUNDEDinstance-attribute**

CALCULATE\_ROUNDED: IconData

**buildCALCULATE\_SHARPinstance-attribute**

CALCULATE\_SHARP: IconData

**buildCALENDAR\_MONTHinstance-attribute**

CALENDAR\_MONTH: IconData

**buildCALENDAR\_MONTH\_OUTLINEDinstance-attribute**

CALENDAR\_MONTH\_OUTLINED: IconData

**buildCALENDAR\_MONTH\_ROUNDEDinstance-attribute**

CALENDAR\_MONTH\_ROUNDED: IconData

**buildCALENDAR\_MONTH\_SHARPinstance-attribute**

CALENDAR\_MONTH\_SHARP: IconData

**buildCALENDAR\_TODAYinstance-attribute**

CALENDAR\_TODAY: IconData

**buildCALENDAR\_TODAY\_OUTLINEDinstance-attribute**

CALENDAR\_TODAY\_OUTLINED: IconData

**buildCALENDAR\_TODAY\_ROUNDEDinstance-attribute**

CALENDAR\_TODAY\_ROUNDED: IconData

**buildCALENDAR\_TODAY\_SHARPinstance-attribute**

CALENDAR\_TODAY\_SHARP: IconData

**buildCALENDAR\_VIEW\_DAYinstance-attribute**

CALENDAR\_VIEW\_DAY: IconData

**buildCALENDAR\_VIEW\_DAY\_OUTLINEDinstance-attribute**

CALENDAR\_VIEW\_DAY\_OUTLINED: IconData

**buildCALENDAR\_VIEW\_DAY\_ROUNDEDinstance-attribute**

CALENDAR\_VIEW\_DAY\_ROUNDED: IconData

**buildCALENDAR\_VIEW\_DAY\_SHARPinstance-attribute**

CALENDAR\_VIEW\_DAY\_SHARP: IconData

**buildCALENDAR\_VIEW\_MONTHinstance-attribute**

CALENDAR\_VIEW\_MONTH: IconData

**buildCALENDAR\_VIEW\_MONTH\_OUTLINEDinstance-attribute**

CALENDAR\_VIEW\_MONTH\_OUTLINED: IconData

**buildCALENDAR\_VIEW\_MONTH\_ROUNDEDinstance-attribute**

CALENDAR\_VIEW\_MONTH\_ROUNDED: IconData

**buildCALENDAR\_VIEW\_MONTH\_SHARPinstance-attribute**

CALENDAR\_VIEW\_MONTH\_SHARP: IconData

**buildCALENDAR\_VIEW\_WEEKinstance-attribute**

CALENDAR\_VIEW\_WEEK: IconData

**buildCALENDAR\_VIEW\_WEEK\_OUTLINEDinstance-attribute**

CALENDAR\_VIEW\_WEEK\_OUTLINED: IconData

**buildCALENDAR\_VIEW\_WEEK\_ROUNDEDinstance-attribute**

CALENDAR\_VIEW\_WEEK\_ROUNDED: IconData

**buildCALENDAR\_VIEW\_WEEK\_SHARPinstance-attribute**

CALENDAR\_VIEW\_WEEK\_SHARP: IconData

**buildCALLinstance-attribute**

CALL: IconData

**buildCALL\_ENDinstance-attribute**

CALL\_END: IconData

**buildCALL\_END\_OUTLINEDinstance-attribute**

CALL\_END\_OUTLINED: IconData

**buildCALL\_END\_ROUNDEDinstance-attribute**

CALL\_END\_ROUNDED: IconData

**buildCALL\_END\_SHARPinstance-attribute**

CALL\_END\_SHARP: IconData

**buildCALL\_MADEinstance-attribute**

CALL\_MADE: IconData

**buildCALL\_MADE\_OUTLINEDinstance-attribute**

CALL\_MADE\_OUTLINED: IconData

**buildCALL\_MADE\_ROUNDEDinstance-attribute**

CALL\_MADE\_ROUNDED: IconData

**buildCALL\_MADE\_SHARPinstance-attribute**

CALL\_MADE\_SHARP: IconData

**buildCALL\_MERGEinstance-attribute**

CALL\_MERGE: IconData

**buildCALL\_MERGE\_OUTLINEDinstance-attribute**

CALL\_MERGE\_OUTLINED: IconData

**buildCALL\_MERGE\_ROUNDEDinstance-attribute**

CALL\_MERGE\_ROUNDED: IconData

**buildCALL\_MERGE\_SHARPinstance-attribute**

CALL\_MERGE\_SHARP: IconData

**buildCALL\_MISSEDinstance-attribute**

CALL\_MISSED: IconData

**buildCALL\_MISSED\_OUTGOINGinstance-attribute**

CALL\_MISSED\_OUTGOING: IconData

**buildCALL\_MISSED\_OUTGOING\_OUTLINEDinstance-attribute**

CALL\_MISSED\_OUTGOING\_OUTLINED: IconData

**buildCALL\_MISSED\_OUTGOING\_ROUNDEDinstance-attribute**

CALL\_MISSED\_OUTGOING\_ROUNDED: IconData

**buildCALL\_MISSED\_OUTGOING\_SHARPinstance-attribute**

CALL\_MISSED\_OUTGOING\_SHARP: IconData

**buildCALL\_MISSED\_OUTLINEDinstance-attribute**

CALL\_MISSED\_OUTLINED: IconData

**buildCALL\_MISSED\_ROUNDEDinstance-attribute**

CALL\_MISSED\_ROUNDED: IconData

**buildCALL\_MISSED\_SHARPinstance-attribute**

CALL\_MISSED\_SHARP: IconData

**buildCALL\_OUTLINEDinstance-attribute**

CALL\_OUTLINED: IconData

**buildCALL\_RECEIVEDinstance-attribute**

CALL\_RECEIVED: IconData

**buildCALL\_RECEIVED\_OUTLINEDinstance-attribute**

CALL\_RECEIVED\_OUTLINED: IconData

**buildCALL\_RECEIVED\_ROUNDEDinstance-attribute**

CALL\_RECEIVED\_ROUNDED: IconData

**buildCALL\_RECEIVED\_SHARPinstance-attribute**

CALL\_RECEIVED\_SHARP: IconData

**buildCALL\_ROUNDEDinstance-attribute**

CALL\_ROUNDED: IconData

**buildCALL\_SHARPinstance-attribute**

CALL\_SHARP: IconData

**buildCALL\_SPLITinstance-attribute**

CALL\_SPLIT: IconData

**buildCALL\_SPLIT\_OUTLINEDinstance-attribute**

CALL\_SPLIT\_OUTLINED: IconData

**buildCALL\_SPLIT\_ROUNDEDinstance-attribute**

CALL\_SPLIT\_ROUNDED: IconData

**buildCALL\_SPLIT\_SHARPinstance-attribute**

CALL\_SPLIT\_SHARP: IconData

**buildCALL\_TO\_ACTIONinstance-attribute**

CALL\_TO\_ACTION: IconData

**buildCALL\_TO\_ACTION\_OUTLINEDinstance-attribute**

CALL\_TO\_ACTION\_OUTLINED: IconData

**buildCALL\_TO\_ACTION\_ROUNDEDinstance-attribute**

CALL\_TO\_ACTION\_ROUNDED: IconData

**buildCALL\_TO\_ACTION\_SHARPinstance-attribute**

CALL\_TO\_ACTION\_SHARP: IconData

**buildCAMERAinstance-attribute**

CAMERA: IconData

**buildCAMERA\_ALTinstance-attribute**

CAMERA\_ALT: IconData

**buildCAMERA\_ALT\_OUTLINEDinstance-attribute**

CAMERA\_ALT\_OUTLINED: IconData

**buildCAMERA\_ALT\_ROUNDEDinstance-attribute**

CAMERA\_ALT\_ROUNDED: IconData

**buildCAMERA\_ALT\_SHARPinstance-attribute**

CAMERA\_ALT\_SHARP: IconData

**buildCAMERA\_ENHANCEinstance-attribute**

CAMERA\_ENHANCE: IconData

**buildCAMERA\_ENHANCE\_OUTLINEDinstance-attribute**

CAMERA\_ENHANCE\_OUTLINED: IconData

**buildCAMERA\_ENHANCE\_ROUNDEDinstance-attribute**

CAMERA\_ENHANCE\_ROUNDED: IconData

**buildCAMERA\_ENHANCE\_SHARPinstance-attribute**

CAMERA\_ENHANCE\_SHARP: IconData

**buildCAMERA\_FRONTinstance-attribute**

CAMERA\_FRONT: IconData

**buildCAMERA\_FRONT\_OUTLINEDinstance-attribute**

CAMERA\_FRONT\_OUTLINED: IconData

**buildCAMERA\_FRONT\_ROUNDEDinstance-attribute**

CAMERA\_FRONT\_ROUNDED: IconData

**buildCAMERA\_FRONT\_SHARPinstance-attribute**

CAMERA\_FRONT\_SHARP: IconData

**buildCAMERA\_INDOORinstance-attribute**

CAMERA\_INDOOR: IconData

**buildCAMERA\_INDOOR\_OUTLINEDinstance-attribute**

CAMERA\_INDOOR\_OUTLINED: IconData

**buildCAMERA\_INDOOR\_ROUNDEDinstance-attribute**

CAMERA\_INDOOR\_ROUNDED: IconData

**buildCAMERA\_INDOOR\_SHARPinstance-attribute**

CAMERA\_INDOOR\_SHARP: IconData

**buildCAMERA\_OUTDOORinstance-attribute**

CAMERA\_OUTDOOR: IconData

**buildCAMERA\_OUTDOOR\_OUTLINEDinstance-attribute**

CAMERA\_OUTDOOR\_OUTLINED: IconData

**buildCAMERA\_OUTDOOR\_ROUNDEDinstance-attribute**

CAMERA\_OUTDOOR\_ROUNDED: IconData

**buildCAMERA\_OUTDOOR\_SHARPinstance-attribute**

CAMERA\_OUTDOOR\_SHARP: IconData

**buildCAMERA\_OUTLINEDinstance-attribute**

CAMERA\_OUTLINED: IconData

**buildCAMERA\_REARinstance-attribute**

CAMERA\_REAR: IconData

**buildCAMERA\_REAR\_OUTLINEDinstance-attribute**

CAMERA\_REAR\_OUTLINED: IconData

**buildCAMERA\_REAR\_ROUNDEDinstance-attribute**

CAMERA\_REAR\_ROUNDED: IconData

**buildCAMERA\_REAR\_SHARPinstance-attribute**

CAMERA\_REAR\_SHARP: IconData

**buildCAMERA\_ROLLinstance-attribute**

CAMERA\_ROLL: IconData

**buildCAMERA\_ROLL\_OUTLINEDinstance-attribute**

CAMERA\_ROLL\_OUTLINED: IconData

**buildCAMERA\_ROLL\_ROUNDEDinstance-attribute**

CAMERA\_ROLL\_ROUNDED: IconData

**buildCAMERA\_ROLL\_SHARPinstance-attribute**

CAMERA\_ROLL\_SHARP: IconData

**buildCAMERA\_ROUNDEDinstance-attribute**

CAMERA\_ROUNDED: IconData

**buildCAMERA\_SHARPinstance-attribute**

CAMERA\_SHARP: IconData

**buildCAMERASWITCHinstance-attribute**

CAMERASWITCH: IconData

**buildCAMERASWITCH\_OUTLINEDinstance-attribute**

CAMERASWITCH\_OUTLINED: IconData

**buildCAMERASWITCH\_ROUNDEDinstance-attribute**

CAMERASWITCH\_ROUNDED: IconData

**buildCAMERASWITCH\_SHARPinstance-attribute**

CAMERASWITCH\_SHARP: IconData

**buildCAMPAIGNinstance-attribute**

CAMPAIGN: IconData

**buildCAMPAIGN\_OUTLINEDinstance-attribute**

CAMPAIGN\_OUTLINED: IconData

**buildCAMPAIGN\_ROUNDEDinstance-attribute**

CAMPAIGN\_ROUNDED: IconData

**buildCAMPAIGN\_SHARPinstance-attribute**

CAMPAIGN\_SHARP: IconData

**buildCANCELinstance-attribute**

CANCEL: IconData

**buildCANCEL\_OUTLINEDinstance-attribute**

CANCEL\_OUTLINED: IconData

**buildCANCEL\_PRESENTATIONinstance-attribute**

CANCEL\_PRESENTATION: IconData

**buildCANCEL\_PRESENTATION\_OUTLINEDinstance-attribute**

CANCEL\_PRESENTATION\_OUTLINED: IconData

**buildCANCEL\_PRESENTATION\_ROUNDEDinstance-attribute**

CANCEL\_PRESENTATION\_ROUNDED: IconData

**buildCANCEL\_PRESENTATION\_SHARPinstance-attribute**

CANCEL\_PRESENTATION\_SHARP: IconData

**buildCANCEL\_ROUNDEDinstance-attribute**

CANCEL\_ROUNDED: IconData

**buildCANCEL\_SCHEDULE\_SENDinstance-attribute**

CANCEL\_SCHEDULE\_SEND: IconData

**buildCANCEL\_SCHEDULE\_SEND\_OUTLINEDinstance-attribute**

CANCEL\_SCHEDULE\_SEND\_OUTLINED: IconData

**buildCANCEL\_SCHEDULE\_SEND\_ROUNDEDinstance-attribute**

CANCEL\_SCHEDULE\_SEND\_ROUNDED: IconData

**buildCANCEL\_SCHEDULE\_SEND\_SHARPinstance-attribute**

CANCEL\_SCHEDULE\_SEND\_SHARP: IconData

**buildCANCEL\_SHARPinstance-attribute**

CANCEL\_SHARP: IconData

**buildCANDLESTICK\_CHARTinstance-attribute**

CANDLESTICK\_CHART: IconData

**buildCANDLESTICK\_CHART\_OUTLINEDinstance-attribute**

CANDLESTICK\_CHART\_OUTLINED: IconData

**buildCANDLESTICK\_CHART\_ROUNDEDinstance-attribute**

CANDLESTICK\_CHART\_ROUNDED: IconData

**buildCANDLESTICK\_CHART\_SHARPinstance-attribute**

CANDLESTICK\_CHART\_SHARP: IconData

**buildCAR\_CRASHinstance-attribute**

CAR\_CRASH: IconData

**buildCAR\_CRASH\_OUTLINEDinstance-attribute**

CAR\_CRASH\_OUTLINED: IconData

**buildCAR\_CRASH\_ROUNDEDinstance-attribute**

CAR\_CRASH\_ROUNDED: IconData

**buildCAR\_CRASH\_SHARPinstance-attribute**

CAR\_CRASH\_SHARP: IconData

**buildCAR\_RENTALinstance-attribute**

CAR\_RENTAL: IconData

**buildCAR\_RENTAL\_OUTLINEDinstance-attribute**

CAR\_RENTAL\_OUTLINED: IconData

**buildCAR\_RENTAL\_ROUNDEDinstance-attribute**

CAR\_RENTAL\_ROUNDED: IconData

**buildCAR\_RENTAL\_SHARPinstance-attribute**

CAR\_RENTAL\_SHARP: IconData

**buildCAR\_REPAIRinstance-attribute**

CAR\_REPAIR: IconData

**buildCAR\_REPAIR\_OUTLINEDinstance-attribute**

CAR\_REPAIR\_OUTLINED: IconData

**buildCAR\_REPAIR\_ROUNDEDinstance-attribute**

CAR\_REPAIR\_ROUNDED: IconData

**buildCAR\_REPAIR\_SHARPinstance-attribute**

CAR\_REPAIR\_SHARP: IconData

**buildCARD\_GIFTCARDinstance-attribute**

CARD\_GIFTCARD: IconData

**buildCARD\_GIFTCARD\_OUTLINEDinstance-attribute**

CARD\_GIFTCARD\_OUTLINED: IconData

**buildCARD\_GIFTCARD\_ROUNDEDinstance-attribute**

CARD\_GIFTCARD\_ROUNDED: IconData



**buildCARD\_GIFTCARD\_SHARPinstance-attribute**

CARD\_GIFTCARD\_SHARP: IconData

**buildCARD\_MEMBERSHIPinstance-attribute**

CARD\_MEMBERSHIP: IconData

**buildCARD\_MEMBERSHIP\_OUTLINEDinstance-attribute**

CARD\_MEMBERSHIP\_OUTLINED: IconData

**buildCARD\_MEMBERSHIP\_ROUNDEDinstance-attribute**

CARD\_MEMBERSHIP\_ROUNDED: IconData

**buildCARD\_MEMBERSHIP\_SHARPinstance-attribute**

CARD\_MEMBERSHIP\_SHARP: IconData

**buildCARD\_TRAVELinstance-attribute**

CARD\_TRAVEL: IconData

**buildCARD\_TRAVEL\_OUTLINEDinstance-attribute**

CARD\_TRAVEL\_OUTLINED: IconData

**buildCARD\_TRAVEL\_ROUNDEDinstance-attribute**

CARD\_TRAVEL\_ROUNDED: IconData

**buildCARD\_TRAVEL\_SHARPinstance-attribute**

CARD\_TRAVEL\_SHARP: IconData

**buildCARPENTERinstance-attribute**

CARPENTER: IconData

**buildCARPENTER\_OUTLINEDinstance-attribute**

CARPENTER\_OUTLINED: IconData

**buildCARPENTER\_ROUNDEDinstance-attribute**

CARPENTER\_ROUNDED: IconData

**buildCARPENTER\_SHARPinstance-attribute**

CARPENTER\_SHARP: IconData

**buildCASESinstance-attribute**

CASES: IconData

**buildCASES\_OUTLINEDinstance-attribute**

CASES\_OUTLINED: IconData

**buildCASES\_ROUNDEDinstance-attribute**

CASES\_ROUNDED: IconData

**buildCASES\_SHARPinstance-attribute**

CASES\_SHARP: IconData

**buildCASINOinstance-attribute**

CASINO: IconData

**buildCASINO\_OUTLINEDinstance-attribute**

CASINO\_OUTLINED: IconData

**buildCASINO\_ROUNDEDinstance-attribute**

CASINO\_ROUNDED: IconData

**buildCASINO\_SHARPinstance-attribute**

CASINO\_SHARP: IconData

**buildCASTinstance-attribute**

CAST: IconData

**buildCAST\_CONNECTED**Instance-attribute

CAST\_CONNECTED: IconData

**buildCAST\_CONNECTED\_OUTLINED**Instance-attribute

CAST\_CONNECTED\_OUTLINED: IconData

**buildCAST\_CONNECTED\_ROUNDED**Instance-attribute

CAST\_CONNECTED\_ROUNDED: IconData

**buildCAST\_CONNECTED\_SHARP**Instance-attribute

CAST\_CONNECTED\_SHARP: IconData

**buildCAST\_FOR\_EDUCATION**Instance-attribute

CAST\_FOR\_EDUCATION: IconData

**buildCAST\_FOR\_EDUCATION\_OUTLINED**Instance-attribute

CAST\_FOR\_EDUCATION\_OUTLINED: IconData

**buildCAST\_FOR\_EDUCATION\_ROUNDED**Instance-attribute

CAST\_FOR\_EDUCATION\_ROUNDED: IconData

**buildCAST\_FOR\_EDUCATION\_SHARP**Instance-attribute

CAST\_FOR\_EDUCATION\_SHARP: IconData

**buildCAST\_OUTLINED**Instance-attribute

CAST\_OUTLINED: IconData

**buildCAST\_ROUNDED**Instance-attribute

CAST\_ROUNDED: IconData

**buildCAST\_SHARP**Instance-attribute

CAST\_SHARP: IconData

**buildCASTLE**Instance-attribute

CASTLE: IconData

**buildCASTLE\_OUTLINED**Instance-attribute

CASTLE\_OUTLINED: IconData

**buildCASTLE\_ROUNDED**Instance-attribute

CASTLE\_ROUNDED: IconData

**buildCASTLE\_SHARP**Instance-attribute

CASTLE\_SHARP: IconData

**buildCATCHING\_POKEMON**Instance-attribute

CATCHING\_POKEMON: IconData

**buildCATCHING\_POKEMON\_OUTLINED**Instance-attribute

CATCHING\_POKEMON\_OUTLINED: IconData

**buildCATCHING\_POKEMON\_ROUNDED**Instance-attribute

CATCHING\_POKEMON\_ROUNDED: IconData

**buildCATCHING\_POKEMON\_SHARP**Instance-attribute

CATCHING\_POKEMON\_SHARP: IconData

**buildCATEGORY**Instance-attribute

CATEGORY: IconData

**buildCATEGORY\_OUTLINED**Instance-attribute

CATEGORY\_OUTLINED: IconData

**buildCATEGORY\_ROUNDED**Instance-attribute

CATEGORY\_ROUNDED: IconData

**buildCATEGORY\_SHARPinstance-attribute**

CATEGORY\_SHARP: IconData

**buildCELEBRATIONinstance-attribute**

CELEBRATION: IconData

**buildCELEBRATION\_OUTLINEDinstance-attribute**

CELEBRATION\_OUTLINED: IconData

**buildCELEBRATION\_ROUNDEDinstance-attribute**

CELEBRATION\_ROUNDED: IconData

**buildCELEBRATION\_SHARPinstance-attribute**

CELEBRATION\_SHARP: IconData

**buildCELL\_TOWERinstance-attribute**

CELL\_TOWER: IconData

**buildCELL\_TOWER\_OUTLINEDinstance-attribute**

CELL\_TOWER\_OUTLINED: IconData

**buildCELL\_TOWER\_ROUNDEDinstance-attribute**

CELL\_TOWER\_ROUNDED: IconData

**buildCELL\_TOWER\_SHARPinstance-attribute**

CELL\_TOWER\_SHARP: IconData

**buildCELL\_WIFIinstance-attribute**

CELL\_WIFI: IconData

**buildCELL\_WIFI\_OUTLINEDinstance-attribute**

CELL\_WIFI\_OUTLINED: IconData

**buildCELL\_WIFI\_ROUNDEDinstance-attribute**

CELL\_WIFI\_ROUNDED: IconData

**buildCELL\_WIFI\_SHARPinstance-attribute**

CELL\_WIFI\_SHARP: IconData

**buildCENTER\_FOCUS\_STRONGinstance-attribute**

CENTER\_FOCUS\_STRONG: IconData

**buildCENTER\_FOCUS\_STRONG\_OUTLINEDinstance-attribute**

CENTER\_FOCUS\_STRONG\_OUTLINED: IconData

**buildCENTER\_FOCUS\_STRONG\_ROUNDEDinstance-attribute**

CENTER\_FOCUS\_STRONG\_ROUNDED: IconData

**buildCENTER\_FOCUS\_STRONG\_SHARPinstance-attribute**

CENTER\_FOCUS\_STRONG\_SHARP: IconData

**buildCENTER\_FOCUS\_WEAKinstance-attribute**

CENTER\_FOCUS\_WEAK: IconData

**buildCENTER\_FOCUS\_WEAK\_OUTLINEDinstance-attribute**

CENTER\_FOCUS\_WEAK\_OUTLINED: IconData

**buildCENTER\_FOCUS\_WEAK\_ROUNDEDinstance-attribute**

CENTER\_FOCUS\_WEAK\_ROUNDED: IconData

**buildCENTER\_FOCUS\_WEAK\_SHARPinstance-attribute**

CENTER\_FOCUS\_WEAK\_SHARP: IconData

**buildCHAIRinstance-attribute**

CHAIR: IconData

**buildCHAIR\_ALTInstance-attribute**

CHAIR\_ALT: IconData

**buildCHAIR\_ALT\_OUTLINEDInstance-attribute**

CHAIR\_ALT\_OUTLINED: IconData

**buildCHAIR\_ALT\_ROUNDEDInstance-attribute**

CHAIR\_ALT\_ROUNDED: IconData

**buildCHAIR\_ALT\_SHARPInstance-attribute**

CHAIR\_ALT\_SHARP: IconData

**buildCHAIR\_OUTLINEDInstance-attribute**

CHAIR\_OUTLINED: IconData

**buildCHAIR\_ROUNDEDInstance-attribute**

CHAIR\_ROUNDED: IconData

**buildCHAIR\_SHARPInstance-attribute**

CHAIR\_SHARP: IconData

**buildCHALETInstance-attribute**

CHALET: IconData

**buildCHALET\_OUTLINEDInstance-attribute**

CHALET\_OUTLINED: IconData

**buildCHALET\_ROUNDEDInstance-attribute**

CHALET\_ROUNDED: IconData

**buildCHALET\_SHARPInstance-attribute**

CHALET\_SHARP: IconData

**buildCHANGE\_CIRCLEInstance-attribute**

CHANGE\_CIRCLE: IconData

**buildCHANGE\_CIRCLE\_OUTLINEDInstance-attribute**

CHANGE\_CIRCLE\_OUTLINED: IconData

**buildCHANGE\_CIRCLE\_ROUNDEDInstance-attribute**

CHANGE\_CIRCLE\_ROUNDED: IconData

**buildCHANGE\_CIRCLE\_SHARPInstance-attribute**

CHANGE\_CIRCLE\_SHARP: IconData

**buildCHANGE\_HISTORYInstance-attribute**

CHANGE\_HISTORY: IconData

**buildCHANGE\_HISTORY\_OUTLINEDInstance-attribute**

CHANGE\_HISTORY\_OUTLINED: IconData

**buildCHANGE\_HISTORY\_ROUNDEDInstance-attribute**

CHANGE\_HISTORY\_ROUNDED: IconData

**buildCHANGE\_HISTORY\_SHARPInstance-attribute**

CHANGE\_HISTORY\_SHARP: IconData

**buildCHARGING\_STATIONInstance-attribute**

CHARGING\_STATION: IconData

**buildCHARGING\_STATION\_OUTLINEDInstance-attribute**

CHARGING\_STATION\_OUTLINED: IconData

**buildCHARGING\_STATION\_ROUNDEDInstance-attribute**

CHARGING\_STATION\_ROUNDED: IconData

**buildCHARGING\_STATION\_SHARPinstance-attribute**

CHARGING\_STATION\_SHARP: IconData

**buildCHATinstance-attribute**

CHAT: IconData

**buildCHAT\_BUBBLEinstance-attribute**

CHAT\_BUBBLE: IconData

**buildCHAT\_BUBBLE\_OUTLINEinstance-attribute**

CHAT\_BUBBLE\_OUTLINE: IconData

**buildCHAT\_BUBBLE\_OUTLINE\_OUTLINEDinstance-attribute**

CHAT\_BUBBLE\_OUTLINE\_OUTLINED: IconData

**buildCHAT\_BUBBLE\_OUTLINE\_ROUNDEDinstance-attribute**

CHAT\_BUBBLE\_OUTLINE\_ROUNDED: IconData

**buildCHAT\_BUBBLE\_OUTLINE\_SHARPinstance-attribute**

CHAT\_BUBBLE\_OUTLINE\_SHARP: IconData

**buildCHAT\_BUBBLE\_OUTLINEDinstance-attribute**

CHAT\_BUBBLE\_OUTLINED: IconData

**buildCHAT\_BUBBLE\_ROUNDEDinstance-attribute**

CHAT\_BUBBLE\_ROUNDED: IconData

**buildCHAT\_BUBBLE\_SHARPinstance-attribute**

CHAT\_BUBBLE\_SHARP: IconData

**buildCHAT\_OUTLINEDinstance-attribute**

CHAT\_OUTLINED: IconData

**buildCHAT\_ROUNDEDinstance-attribute**

CHAT\_ROUNDED: IconData

**buildCHAT\_SHARPinstance-attribute**

CHAT\_SHARP: IconData

**buildCHECKinstance-attribute**

CHECK: IconData

**buildCHECK\_BOXinstance-attribute**

CHECK\_BOX: IconData

**buildCHECK\_BOX\_OUTLINE\_BLANKinstance-attribute**

CHECK\_BOX\_OUTLINE\_BLANK: IconData

**buildCHECK\_BOX\_OUTLINE\_BLANK\_OUTLINEDinstance-attribute**

CHECK\_BOX\_OUTLINE\_BLANK\_OUTLINED: IconData

**buildCHECK\_BOX\_OUTLINE\_BLANK\_ROUNDEDinstance-attribute**

CHECK\_BOX\_OUTLINE\_BLANK\_ROUNDED: IconData

**buildCHECK\_BOX\_OUTLINE\_BLANK\_SHARPinstance-attribute**

CHECK\_BOX\_OUTLINE\_BLANK\_SHARP: IconData

**buildCHECK\_BOX\_OUTLINEDinstance-attribute**

CHECK\_BOX\_OUTLINED: IconData

**buildCHECK\_BOX\_ROUNDEDinstance-attribute**

CHECK\_BOX\_ROUNDED: IconData

**buildCHECK\_BOX\_SHARPinstance-attribute**

CHECK\_BOX\_SHARP: IconData

**buildCHECK\_CIRCLEinstance-attribute**

CHECK\_CIRCLE: IconData

**buildCHECK\_CIRCLE\_OUTLINEinstance-attribute**

CHECK\_CIRCLE\_OUTLINE: IconData

**buildCHECK\_CIRCLE\_OUTLINE\_OUTLINEDinstance-attribute**

CHECK\_CIRCLE\_OUTLINE\_OUTLINED: IconData

**buildCHECK\_CIRCLE\_OUTLINE\_ROUNDEDinstance-attribute**

CHECK\_CIRCLE\_OUTLINE\_ROUNDED: IconData

**buildCHECK\_CIRCLE\_OUTLINE\_SHARPinstance-attribute**

CHECK\_CIRCLE\_OUTLINE\_SHARP: IconData

**buildCHECK\_CIRCLE\_OUTLINEDinstance-attribute**

CHECK\_CIRCLE\_OUTLINED: IconData

**buildCHECK\_CIRCLE\_ROUNDEDinstance-attribute**

CHECK\_CIRCLE\_ROUNDED: IconData

**buildCHECK\_CIRCLE\_SHARPinstance-attribute**

CHECK\_CIRCLE\_SHARP: IconData

**buildCHECK\_OUTLINEDinstance-attribute**

CHECK\_OUTLINED: IconData

**buildCHECK\_ROUNDEDinstance-attribute**

CHECK\_ROUNDED: IconData

**buildCHECK\_SHARPinstance-attribute**

CHECK\_SHARP: IconData

**buildCHECKLISTinstance-attribute**

CHECKLIST: IconData

**buildCHECKLIST\_OUTLINEDinstance-attribute**

CHECKLIST\_OUTLINED: IconData

**buildCHECKLIST\_ROUNDEDinstance-attribute**

CHECKLIST\_ROUNDED: IconData

**buildCHECKLIST\_RTListance-attribute**

CHECKLIST\_RTL: IconData

**buildCHECKLIST\_RTL\_OUTLINEDinstance-attribute**

CHECKLIST\_RTL\_OUTLINED: IconData

**buildCHECKLIST\_RTL\_ROUNDEDinstance-attribute**

CHECKLIST\_RTL\_ROUNDED: IconData

**buildCHECKLIST\_RTL\_SHARPinstance-attribute**

CHECKLIST\_RTL\_SHARP: IconData

**buildCHECKLIST\_SHARPinstance-attribute**

CHECKLIST\_SHARP: IconData

**buildCHECKROOMinstance-attribute**

CHECKROOM: IconData

**buildCHECKROOM\_OUTLINEDinstance-attribute**

CHECKROOM\_OUTLINED: IconData

**buildCHECKROOM\_ROUNDEDinstance-attribute**

CHECKROOM\_ROUNDED: IconData

**buildCHECKROOM\_SHARPinstance-attribute**

CHECKROOM\_SHARP: IconData

**buildCHEVRON\_LEFTinstance-attribute**

CHEVRON\_LEFT: IconData

**buildCHEVRON\_LEFT\_OUTLINEDinstance-attribute**

CHEVRON\_LEFT\_OUTLINED: IconData

**buildCHEVRON\_LEFT\_ROUNDEDinstance-attribute**

CHEVRON\_LEFT\_ROUNDED: IconData

**buildCHEVRON\_LEFT\_SHARPinstance-attribute**

CHEVRON\_LEFT\_SHARP: IconData

**buildCHEVRON\_RIGHTinstance-attribute**

CHEVRON\_RIGHT: IconData

**buildCHEVRON\_RIGHT\_OUTLINEDinstance-attribute**

CHEVRON\_RIGHT\_OUTLINED: IconData

**buildCHEVRON\_RIGHT\_ROUNDEDinstance-attribute**

CHEVRON\_RIGHT\_ROUNDED: IconData

**buildCHEVRON\_RIGHT\_SHARPinstance-attribute**

CHEVRON\_RIGHT\_SHARP: IconData

**buildCHILD\_CAREinstance-attribute**

CHILD\_CARE: IconData

**buildCHILD\_CARE\_OUTLINEDinstance-attribute**

CHILD\_CARE\_OUTLINED: IconData

**buildCHILD\_CARE\_ROUNDEDinstance-attribute**

CHILD\_CARE\_ROUNDED: IconData

**buildCHILD\_CARE\_SHARPinstance-attribute**

CHILD\_CARE\_SHARP: IconData

**buildCHILD\_FRIENDLYinstance-attribute**

CHILD\_FRIENDLY: IconData

**buildCHILD\_FRIENDLY\_OUTLINEDinstance-attribute**

CHILD\_FRIENDLY\_OUTLINED: IconData

**buildCHILD\_FRIENDLY\_ROUNDEDinstance-attribute**

CHILD\_FRIENDLY\_ROUNDED: IconData

**buildCHILD\_FRIENDLY\_SHARPinstance-attribute**

CHILD\_FRIENDLY\_SHARP: IconData

**buildCHROME\_READER\_MODEinstance-attribute**

CHROME\_READER\_MODE: IconData

**buildCHROME\_READER\_MODE\_OUTLINEDinstance-attribute**

CHROME\_READER\_MODE\_OUTLINED: IconData

**buildCHROME\_READER\_MODE\_ROUNDEDinstance-attribute**

CHROME\_READER\_MODE\_ROUNDED: IconData

**buildCHROME\_READER\_MODE\_SHARPinstance-attribute**

CHROME\_READER\_MODE\_SHARP: IconData

**buildCHURCHinstance-attribute**

CHURCH: IconData

**buildCHURCH\_OUTLINEDinstance-attribute**

CHURCH\_OUTLINED: IconData

**buildCHURCH\_ROUNDEDinstance-attribute**

CHURCH\_ROUNDED: IconData

**buildCHURCH\_SHARPinstance-attribute**

CHURCH\_SHARP: IconData

**buildCIRCLEinstance-attribute**

CIRCLE: IconData

**buildCIRCLE\_NOTIFICATIONSinstance-attribute**

CIRCLE\_NOTIFICATIONS: IconData

**buildCIRCLE\_NOTIFICATIONS\_OUTLINEDinstance-attribute**

CIRCLE\_NOTIFICATIONS\_OUTLINED: IconData

**buildCIRCLE\_NOTIFICATIONS\_ROUNDEDinstance-attribute**

CIRCLE\_NOTIFICATIONS\_ROUNDED: IconData

**buildCIRCLE\_NOTIFICATIONS\_SHARPinstance-attribute**

CIRCLE\_NOTIFICATIONS\_SHARP: IconData

**buildCIRCLE\_OUTLINEDinstance-attribute**

CIRCLE\_OUTLINED: IconData

**buildCIRCLE\_ROUNDEDinstance-attribute**

CIRCLE\_ROUNDED: IconData

**buildCIRCLE\_SHARPinstance-attribute**

CIRCLE\_SHARP: IconData

**buildCLASS\_instance-attribute**

CLASS\_: IconData

**buildCLASS\_OUTLINEDinstance-attribute**

CLASS\_OUTLINED: IconData

**buildCLASS\_ROUNDEDinstance-attribute**

CLASS\_ROUNDED: IconData

**buildCLASS\_SHARPinstance-attribute**

CLASS\_SHARP: IconData

**buildCLEAN\_HANDSinstance-attribute**

CLEAN\_HANDS: IconData

**buildCLEAN\_HANDS\_OUTLINEDinstance-attribute**

CLEAN\_HANDS\_OUTLINED: IconData

**buildCLEAN\_HANDS\_ROUNDEDinstance-attribute**

CLEAN\_HANDS\_ROUNDED: IconData

**buildCLEAN\_HANDS\_SHARPinstance-attribute**

CLEAN\_HANDS\_SHARP: IconData

**buildCLEANING\_SERVICESinstance-attribute**

CLEANING\_SERVICES: IconData

**buildCLEANING\_SERVICES\_OUTLINEDinstance-attribute**

CLEANING\_SERVICES\_OUTLINED: IconData

**buildCLEANING\_SERVICES\_ROUNDEDinstance-attribute**

CLEANING\_SERVICES\_ROUNDED: IconData



**buildCLEANING\_SERVICES\_SHARPinstance-attribute**

CLEANING\_SERVICES\_SHARP: IconData

**buildCLEARinstance-attribute**

CLEAR: IconData

**buildCLEAR\_ALLinstance-attribute**

CLEAR\_ALL: IconData

**buildCLEAR\_ALL\_OUTLINEDinstance-attribute**

CLEAR\_ALL\_OUTLINED: IconData

**buildCLEAR\_ALL\_ROUNDEDinstance-attribute**

CLEAR\_ALL\_ROUNDED: IconData

**buildCLEAR\_ALL\_SHARPinstance-attribute**

CLEAR\_ALL\_SHARP: IconData

**buildCLEAR\_OUTLINEDinstance-attribute**

CLEAR\_OUTLINED: IconData

**buildCLEAR\_ROUNDEDinstance-attribute**

CLEAR\_ROUNDED: IconData

**buildCLEAR\_SHARPinstance-attribute**

CLEAR\_SHARP: IconData

**buildCLOSEinstance-attribute**

CLOSE: IconData

**buildCLOSE\_FULLSCREENinstance-attribute**

CLOSE\_FULLSCREEN: IconData

**buildCLOSE\_FULLSCREEN\_OUTLINEDinstance-attribute**

CLOSE\_FULLSCREEN\_OUTLINED: IconData

**buildCLOSE\_FULLSCREEN\_ROUNDEDinstance-attribute**

CLOSE\_FULLSCREEN\_ROUNDED: IconData

**buildCLOSE\_FULLSCREEN\_SHARPinstance-attribute**

CLOSE\_FULLSCREEN\_SHARP: IconData

**buildCLOSE\_OUTLINEDinstance-attribute**

CLOSE\_OUTLINED: IconData

**buildCLOSE\_ROUNDEDinstance-attribute**

CLOSE\_ROUNDED: IconData

**buildCLOSE\_SHARPinstance-attribute**

CLOSE\_SHARP: IconData

**buildCLOSED\_CAPTIONinstance-attribute**

CLOSED\_CAPTION: IconData

**buildCLOSED\_CAPTION\_DISABLEDinstance-attribute**

CLOSED\_CAPTION\_DISABLED: IconData

**buildCLOSED\_CAPTION\_DISABLED\_OUTLINEDinstance-attribute**

CLOSED\_CAPTION\_DISABLED\_OUTLINED: IconData

**buildCLOSED\_CAPTION\_DISABLED\_ROUNDEDinstance-attribute**

CLOSED\_CAPTION\_DISABLED\_ROUNDED: IconData

**buildCLOSED\_CAPTION\_DISABLED\_SHARPinstance-attribute**

CLOSED\_CAPTION\_DISABLED\_SHARP: IconData

**buildCLOSED\_CAPTION\_OFFInstance-attribute**

CLOSED\_CAPTION\_OFF: IconData

**buildCLOSED\_CAPTION\_OFF\_OUTLINEDInstance-attribute**

CLOSED\_CAPTION\_OFF\_OUTLINED: IconData

**buildCLOSED\_CAPTION\_OFF\_ROUNDEDInstance-attribute**

CLOSED\_CAPTION\_OFF\_ROUNDED: IconData

**buildCLOSED\_CAPTION\_OFF\_SHARPInstance-attribute**

CLOSED\_CAPTION\_OFF\_SHARP: IconData

**buildCLOSED\_CAPTION\_OUTLINEDInstance-attribute**

CLOSED\_CAPTION\_OUTLINED: IconData

**buildCLOSED\_CAPTION\_ROUNDEDInstance-attribute**

CLOSED\_CAPTION\_ROUNDED: IconData

**buildCLOSED\_CAPTION\_SHARPInstance-attribute**

CLOSED\_CAPTION\_SHARP: IconData

**buildCLOUDInstance-attribute**

CLOUD: IconData

**buildCLOUD\_CIRCLEInstance-attribute**

CLOUD\_CIRCLE: IconData

**buildCLOUD\_CIRCLE\_OUTLINEDInstance-attribute**

CLOUD\_CIRCLE\_OUTLINED: IconData

**buildCLOUD\_CIRCLE\_ROUNDEDInstance-attribute**

CLOUD\_CIRCLE\_ROUNDED: IconData

**buildCLOUD\_CIRCLE\_SHARPInstance-attribute**

CLOUD\_CIRCLE\_SHARP: IconData

**buildCLOUD\_DONEInstance-attribute**

CLOUD\_DONE: IconData

**buildCLOUD\_DONE\_OUTLINEDInstance-attribute**

CLOUD\_DONE\_OUTLINED: IconData

**buildCLOUD\_DONE\_ROUNDEDInstance-attribute**

CLOUD\_DONE\_ROUNDED: IconData

**buildCLOUD\_DONE\_SHARPInstance-attribute**

CLOUD\_DONE\_SHARP: IconData

**buildCLOUD\_DOWNLOADInstance-attribute**

CLOUD\_DOWNLOAD: IconData

**buildCLOUD\_DOWNLOAD\_OUTLINEDInstance-attribute**

CLOUD\_DOWNLOAD\_OUTLINED: IconData

**buildCLOUD\_DOWNLOAD\_ROUNDEDInstance-attribute**

CLOUD\_DOWNLOAD\_ROUNDED: IconData

**buildCLOUD\_DOWNLOAD\_SHARPInstance-attribute**

CLOUD\_DOWNLOAD\_SHARP: IconData

**buildCLOUD\_OFFInstance-attribute**

CLOUD\_OFF: IconData

**buildCLOUD\_OFF\_OUTLINEDInstance-attribute**

CLOUD\_OFF\_OUTLINED: IconData

**buildCLOUD\_OFF\_ROUNDEDinstance-attribute**

CLOUD\_OFF\_ROUNDED: IconData

**buildCLOUD\_OFF\_SHARPinstance-attribute**

CLOUD\_OFF\_SHARP: IconData

**buildCLOUD\_OUTLINEDinstance-attribute**

CLOUD\_OUTLINED: IconData

**buildCLOUD\_QUEUEinstance-attribute**

CLOUD\_QUEUE: IconData

**buildCLOUD\_QUEUE\_OUTLINEDinstance-attribute**

CLOUD\_QUEUE\_OUTLINED: IconData

**buildCLOUD\_QUEUE\_ROUNDEDinstance-attribute**

CLOUD\_QUEUE\_ROUNDED: IconData

**buildCLOUD\_QUEUE\_SHARPinstance-attribute**

CLOUD\_QUEUE\_SHARP: IconData

**buildCLOUD\_ROUNDEDinstance-attribute**

CLOUD\_ROUNDED: IconData

**buildCLOUD\_SHARPinstance-attribute**

CLOUD\_SHARP: IconData

**buildCLOUD\_SYNCinstance-attribute**

CLOUD\_SYNC: IconData

**buildCLOUD\_SYNC\_OUTLINEDinstance-attribute**

CLOUD\_SYNC\_OUTLINED: IconData

**buildCLOUD\_SYNC\_ROUNDEDinstance-attribute**

CLOUD\_SYNC\_ROUNDED: IconData

**buildCLOUD\_SYNC\_SHARPinstance-attribute**

CLOUD\_SYNC\_SHARP: IconData

**buildCLOUD\_UPLOADinstance-attribute**

CLOUD\_UPLOAD: IconData

**buildCLOUD\_UPLOAD\_OUTLINEDinstance-attribute**

CLOUD\_UPLOAD\_OUTLINED: IconData

**buildCLOUD\_UPLOAD\_ROUNDEDinstance-attribute**

CLOUD\_UPLOAD\_ROUNDED: IconData

**buildCLOUD\_UPLOAD\_SHARPinstance-attribute**

CLOUD\_UPLOAD\_SHARP: IconData

**buildCLOUDY\_SNOWINGinstance-attribute**

CLOUDY\_SNOWING: IconData

**buildCO2instance-attribute**

CO2: IconData

**buildCO2\_OUTLINEDinstance-attribute**

CO2\_OUTLINED: IconData

**buildCO2\_ROUNDEDinstance-attribute**

CO2\_ROUNDED: IconData

**buildCO2\_SHARPinstance-attribute**

CO2\_SHARP: IconData

**buildCO\_PRESENTinstance-attribute**

CO\_PRESENT: IconData

**buildCO\_PRESENT\_OUTLINEDinstance-attribute**

CO\_PRESENT\_OUTLINED: IconData

**buildCO\_PRESENT\_ROUNDEDinstance-attribute**

CO\_PRESENT\_ROUNDED: IconData

**buildCO\_PRESENT\_SHARPinstance-attribute**

CO\_PRESENT\_SHARP: IconData

**buildCODEinstance-attribute**

CODE: IconData

**buildCODE\_OFFinstance-attribute**

CODE\_OFF: IconData

**buildCODE\_OFF\_OUTLINEDinstance-attribute**

CODE\_OFF\_OUTLINED: IconData

**buildCODE\_OFF\_ROUNDEDinstance-attribute**

CODE\_OFF\_ROUNDED: IconData

**buildCODE\_OFF\_SHARPinstance-attribute**

CODE\_OFF\_SHARP: IconData

**buildCODE\_OUTLINEDinstance-attribute**

CODE\_OUTLINED: IconData

**buildCODE\_ROUNDEDinstance-attribute**

CODE\_ROUNDED: IconData

**buildCODE\_SHARPinstance-attribute**

CODE\_SHARP: IconData

**buildCOFFEEinstance-attribute**

COFFEE: IconData

**buildCOFFEE\_MAKERinstance-attribute**

COFFEE\_MAKER: IconData

**buildCOFFEE\_MAKER\_OUTLINEDinstance-attribute**

COFFEE\_MAKER\_OUTLINED: IconData

**buildCOFFEE\_MAKER\_ROUNDEDinstance-attribute**

COFFEE\_MAKER\_ROUNDED: IconData

**buildCOFFEE\_MAKER\_SHARPinstance-attribute**

COFFEE\_MAKER\_SHARP: IconData

**buildCOFFEE\_OUTLINEDinstance-attribute**

COFFEE\_OUTLINED: IconData

**buildCOFFEE\_ROUNDEDinstance-attribute**

COFFEE\_ROUNDED: IconData

**buildCOFFEE\_SHARPinstance-attribute**

COFFEE\_SHARP: IconData

**buildCOLLECTIONSinstance-attribute**

COLLECTIONS: IconData

**buildCOLLECTIONS\_BOOKMARKinstance-attribute**

COLLECTIONS\_BOOKMARK: IconData

**buildCOLLECTIONS\_BOOKMARK\_OUTLINED**instance-attribute

COLLECTIONS\_BOOKMARK\_OUTLINED: IconData

**buildCOLLECTIONS\_BOOKMARK\_ROUNDED**instance-attribute

COLLECTIONS\_BOOKMARK\_ROUNDED: IconData

**buildCOLLECTIONS\_BOOKMARK\_SHARP**instance-attribute

COLLECTIONS\_BOOKMARK\_SHARP: IconData

**buildCOLLECTIONS\_OUTLINED**instance-attribute

COLLECTIONS\_OUTLINED: IconData

**buildCOLLECTIONS\_ROUNDED**instance-attribute

COLLECTIONS\_ROUNDED: IconData

**buildCOLLECTIONS\_SHARP**instance-attribute

COLLECTIONS\_SHARP: IconData

**buildCOLOR\_LENS**instance-attribute

COLOR\_LENS: IconData

**buildCOLOR\_LENS\_OUTLINED**instance-attribute

COLOR\_LENS\_OUTLINED: IconData

**buildCOLOR\_LENS\_ROUNDED**instance-attribute

COLOR\_LENS\_ROUNDED: IconData

**buildCOLOR\_LENS\_SHARP**instance-attribute

COLOR\_LENS\_SHARP: IconData

**buildCOLORIZE**instance-attribute

COLORIZE: IconData

**buildCOLORIZE\_OUTLINED**instance-attribute

COLORIZE\_OUTLINED: IconData

**buildCOLORIZE\_ROUNDED**instance-attribute

COLORIZE\_ROUNDED: IconData

**buildCOLORIZE\_SHARP**instance-attribute

COLORIZE\_SHARP: IconData

**buildCOMMENT**instance-attribute

COMMENT: IconData

**buildCOMMENT\_BANK**instance-attribute

COMMENT\_BANK: IconData

**buildCOMMENT\_BANK\_OUTLINED**instance-attribute

COMMENT\_BANK\_OUTLINED: IconData

**buildCOMMENT\_BANK\_ROUNDED**instance-attribute

COMMENT\_BANK\_ROUNDED: IconData

**buildCOMMENT\_BANK\_SHARP**instance-attribute

COMMENT\_BANK\_SHARP: IconData

**buildCOMMENT\_OUTLINED**instance-attribute

COMMENT\_OUTLINED: IconData

**buildCOMMENT\_ROUNDED**instance-attribute

COMMENT\_ROUNDED: IconData

**buildCOMMENT\_SHARP**instance-attribute

COMMENT\_SHARP: IconData

**buildCOMMENTS\_DISABLEDinstance-attribute**

COMMENTS\_DISABLED: IconData

**buildCOMMENTS\_DISABLED\_OUTLINEDinstance-attribute**

COMMENTS\_DISABLED\_OUTLINED: IconData

**buildCOMMENTS\_DISABLED\_ROUNDEDinstance-attribute**

COMMENTS\_DISABLED\_ROUNDED: IconData

**buildCOMMENTS\_DISABLED\_SHARPinstance-attribute**

COMMENTS\_DISABLED\_SHARP: IconData

**buildCOMMITinstance-attribute**

COMMIT: IconData

**buildCOMMIT\_OUTLINEDinstance-attribute**

COMMIT\_OUTLINED: IconData

**buildCOMMIT\_ROUNDEDinstance-attribute**

COMMIT\_ROUNDED: IconData

**buildCOMMIT\_SHARPinstance-attribute**

COMMIT\_SHARP: IconData

**buildCOMMUTEinstance-attribute**

COMMUTE: IconData

**buildCOMMUTE\_OUTLINEDinstance-attribute**

COMMUTE\_OUTLINED: IconData

**buildCOMMUTE\_ROUNDEDinstance-attribute**

COMMUTE\_ROUNDED: IconData

**buildCOMMUTE\_SHARPinstance-attribute**

COMMUTE\_SHARP: IconData

**buildCOMPAREinstance-attribute**  

COMPARE: IconData

**buildCOMPARE\_ARROWSinstance-attribute**

COMPARE\_ARROWS: IconData

**buildCOMPARE\_ARROWS\_OUTLINEDinstance-attribute**

COMPARE\_ARROWS\_OUTLINED: IconData

**buildCOMPARE\_ARROWS\_ROUNDEDinstance-attribute**

COMPARE\_ARROWS\_ROUNDED: IconData

**buildCOMPARE\_ARROWS\_SHARPinstance-attribute**

COMPARE\_ARROWS\_SHARP: IconData

**buildCOMPARE\_OUTLINEDinstance-attribute**

COMPARE\_OUTLINED: IconData

**buildCOMPARE\_ROUNDEDinstance-attribute**

COMPARE\_ROUNDED: IconData

**buildCOMPARE\_SHARPinstance-attribute**

COMPARE\_SHARP: IconData

**buildCOMPASS\_CALIBRATIONinstance-attribute**

COMPASS\_CALIBRATION: IconData

**buildCOMPASS\_CALIBRATION\_OUTLINEDinstance-attribute**

COMPASS\_CALIBRATION\_OUTLINED: IconData

**buildCOMPASS\_CALIBRATION\_ROUNDEDinstance-attribute**

COMPASS\_CALIBRATION\_ROUNDED: IconData

**buildCOMPASS\_CALIBRATION\_SHARPinstance-attribute**

COMPASS\_CALIBRATION\_SHARP: IconData

**buildCOMPOSTinstance-attribute**

COMPOST: IconData

**buildCOMPOST\_OUTLINEDinstance-attribute**

COMPOST\_OUTLINED: IconData

**buildCOMPOST\_ROUNDEDinstance-attribute**

COMPOST\_ROUNDED: IconData

**buildCOMPOST\_SHARPinstance-attribute**

COMPOST\_SHARP: IconData

**buildCOMPRESSinstance-attribute**

COMPRESS: IconData

**buildCOMPRESS\_OUTLINEDinstance-attribute**

COMPRESS\_OUTLINED: IconData

**buildCOMPRESS\_ROUNDEDinstance-attribute**

COMPRESS\_ROUNDED: IconData

**buildCOMPRESS\_SHARPinstance-attribute**

COMPRESS\_SHARP: IconData

**buildCOMPUTERinstance-attribute**

COMPUTER: IconData

**buildCOMPUTER\_OUTLINEDinstance-attribute**

COMPUTER\_OUTLINED: IconData

**buildCOMPUTER\_ROUNDEDinstance-attribute**

COMPUTER\_ROUNDED: IconData

**buildCOMPUTER\_SHARPinstance-attribute**

COMPUTER\_SHARP: IconData

**buildCONFIRMATION\_NUMinstance-attribute**

CONFIRMATION\_NUM: IconData

**buildCONFIRMATION\_NUM\_OUTLINEDinstance-attribute**

CONFIRMATION\_NUM\_OUTLINED: IconData

**buildCONFIRMATION\_NUM\_ROUNDEDinstance-attribute**

CONFIRMATION\_NUM\_ROUNDED: IconData

**buildCONFIRMATION\_NUM\_SHARPinstance-attribute**

CONFIRMATION\_NUM\_SHARP: IconData

**buildCONFIRMATION\_NUMBERinstance-attribute**

CONFIRMATION\_NUMBER: IconData

**buildCONFIRMATION\_NUMBER\_OUTLINEDinstance-attribute**

CONFIRMATION\_NUMBER\_OUTLINED: IconData

**buildCONFIRMATION\_NUMBER\_ROUNDEDinstance-attribute**

CONFIRMATION\_NUMBER\_ROUNDED: IconData

**buildCONFIRMATION\_NUMBER\_SHARPinstance-attribute**

CONFIRMATION\_NUMBER\_SHARP: IconData

**buildCONNECT\_WITHOUT\_CONTACT**instance-attribute

CONNECT\_WITHOUT\_CONTACT: IconData

**buildCONNECT\_WITHOUT\_CONTACT\_OUTLINED**instance-attribute

CONNECT\_WITHOUT\_CONTACT\_OUTLINED: IconData

**buildCONNECT\_WITHOUT\_CONTACT\_ROUNDED**instance-attribute

CONNECT\_WITHOUT\_CONTACT\_ROUNDED: IconData

**buildCONNECT\_WITHOUT\_CONTACT\_SHARP**instance-attribute

CONNECT\_WITHOUT\_CONTACT\_SHARP: IconData

**buildCONNECTED\_TV**instance-attribute

CONNECTED\_TV: IconData

**buildCONNECTED\_TV\_OUTLINED**instance-attribute

CONNECTED\_TV\_OUTLINED: IconData

**buildCONNECTED\_TV\_ROUNDED**instance-attribute

CONNECTED\_TV\_ROUNDED: IconData

**buildCONNECTED\_TV\_SHARP**instance-attribute

CONNECTED\_TV\_SHARP: IconData

**buildCONNECTING\_AIRPORTS**instance-attribute

CONNECTING\_AIRPORTS: IconData

**buildCONNECTING\_AIRPORTS\_OUTLINED**instance-attribute

CONNECTING\_AIRPORTS\_OUTLINED: IconData

**buildCONNECTING\_AIRPORTS\_ROUNDED**instance-attribute

CONNECTING\_AIRPORTS\_ROUNDED: IconData

**buildCONNECTING\_AIRPORTS\_SHARP**instance-attribute

CONNECTING\_AIRPORTS\_SHARP: IconData

**buildCONSTRUCTION**instance-attribute

CONSTRUCTION: IconData

**buildCONSTRUCTION\_OUTLINED**instance-attribute

CONSTRUCTION\_OUTLINED: IconData

**buildCONSTRUCTION\_ROUNDED**instance-attribute

CONSTRUCTION\_ROUNDED: IconData

**buildCONSTRUCTION\_SHARP**instance-attribute

CONSTRUCTION\_SHARP: IconData

**buildCONTACT\_EMERGENCY**instance-attribute

CONTACT\_EMERGENCY: IconData

**buildCONTACT\_EMERGENCY\_OUTLINED**instance-attribute

CONTACT\_EMERGENCY\_OUTLINED: IconData

**buildCONTACT\_EMERGENCY\_ROUNDED**instance-attribute

CONTACT\_EMERGENCY\_ROUNDED: IconData

**buildCONTACT\_EMERGENCY\_SHARP**instance-attribute

CONTACT\_EMERGENCY\_SHARP: IconData

**buildCONTACT\_MAIL**instance-attribute

CONTACT\_MAIL: IconData

**buildCONTACT\_MAIL\_OUTLINED**instance-attribute

CONTACT\_MAIL\_OUTLINED: IconData



**buildCONTACT\_MAIL\_ROUNDEDinstance-attribute**

CONTACT\_MAIL\_ROUNDED: IconData

**buildCONTACT\_MAIL\_SHARPinstance-attribute**

CONTACT\_MAIL\_SHARP: IconData

**buildCONTACT\_PAGEinstance-attribute**

CONTACT\_PAGE: IconData

**buildCONTACT\_PAGE\_OUTLINEDinstance-attribute**

CONTACT\_PAGE\_OUTLINED: IconData

**buildCONTACT\_PAGE\_ROUNDEDinstance-attribute**

CONTACT\_PAGE\_ROUNDED: IconData

**buildCONTACT\_PAGE\_SHARPinstance-attribute**

CONTACT\_PAGE\_SHARP: IconData

**buildCONTACT\_PHONEinstance-attribute**

CONTACT\_PHONE: IconData

**buildCONTACT\_PHONE\_OUTLINEDinstance-attribute**

CONTACT\_PHONE\_OUTLINED: IconData

**buildCONTACT\_PHONE\_ROUNDEDinstance-attribute**

CONTACT\_PHONE\_ROUNDED: IconData

**buildCONTACT\_PHONE\_SHARPinstance-attribute**

CONTACT\_PHONE\_SHARP: IconData

**buildCONTACT\_SUPPORTinstance-attribute**

CONTACT\_SUPPORT: IconData

**buildCONTACT\_SUPPORT\_OUTLINEDinstance-attribute**

CONTACT\_SUPPORT\_OUTLINED: IconData

**buildCONTACT\_SUPPORT\_ROUNDEDinstance-attribute**

CONTACT\_SUPPORT\_ROUNDED: IconData

**buildCONTACT\_SUPPORT\_SHARPinstance-attribute**

CONTACT\_SUPPORT\_SHARP: IconData

**buildCONTACTLESSinstance-attribute**

CONTACTLESS: IconData

**buildCONTACTLESS\_OUTLINEDinstance-attribute**

CONTACTLESS\_OUTLINED: IconData

**buildCONTACTLESS\_ROUNDEDinstance-attribute**

CONTACTLESS\_ROUNDED: IconData

**buildCONTACTLESS\_SHARPinstance-attribute**

CONTACTLESS\_SHARP: IconData

**buildCONTACTSinstance-attribute**

CONTACTS: IconData

**buildCONTACTS\_OUTLINEDinstance-attribute**

CONTACTS\_OUTLINED: IconData

**buildCONTACTS\_ROUNDEDinstance-attribute**

CONTACTS\_ROUNDED: IconData

**buildCONTACTS\_SHARPinstance-attribute**

CONTACTS\_SHARP: IconData

**buildCONTENT\_COPYinstance-attribute**

CONTENT\_COPY: IconData

**buildCONTENT\_COPY\_OUTLINEDinstance-attribute**

CONTENT\_COPY\_OUTLINED: IconData

**buildCONTENT\_COPY\_ROUNDEDinstance-attribute**

CONTENT\_COPY\_ROUNDED: IconData

**buildCONTENT\_COPY\_SHARPinstance-attribute**

CONTENT\_COPY\_SHARP: IconData

**buildCONTENT\_CUTinstance-attribute**

CONTENT\_CUT: IconData

**buildCONTENT\_CUT\_OUTLINEDinstance-attribute**

CONTENT\_CUT\_OUTLINED: IconData

**buildCONTENT\_CUT\_ROUNDEDinstance-attribute**

CONTENT\_CUT\_ROUNDED: IconData

**buildCONTENT\_CUT\_SHARPinstance-attribute**

CONTENT\_CUT\_SHARP: IconData

**buildCONTENT\_PASTEinstance-attribute**

CONTENT\_PASTE: IconData

**buildCONTENT\_PASTE\_GOinstance-attribute**

CONTENT\_PASTE\_GO: IconData

**buildCONTENT\_PASTE\_GO\_OUTLINEDinstance-attribute**

CONTENT\_PASTE\_GO\_OUTLINED: IconData

**buildCONTENT\_PASTE\_GO\_ROUNDEDinstance-attribute**

CONTENT\_PASTE\_GO\_ROUNDED: IconData

**buildCONTENT\_PASTE\_GO\_SHARPinstance-attribute**

CONTENT\_PASTE\_GO\_SHARP: IconData

**buildCONTENT\_PASTE\_OFFinstance-attribute**

CONTENT\_PASTE\_OFF: IconData

**buildCONTENT\_PASTE\_OFF\_OUTLINEDinstance-attribute**

CONTENT\_PASTE\_OFF\_OUTLINED: IconData

**buildCONTENT\_PASTE\_OFF\_ROUNDEDinstance-attribute**

CONTENT\_PASTE\_OFF\_ROUNDED: IconData

**buildCONTENT\_PASTE\_OFF\_SHARPinstance-attribute**

CONTENT\_PASTE\_OFF\_SHARP: IconData

**buildCONTENT\_PASTE\_OUTLINEDinstance-attribute**

CONTENT\_PASTE\_OUTLINED: IconData

**buildCONTENT\_PASTE\_ROUNDEDinstance-attribute**

CONTENT\_PASTE\_ROUNDED: IconData

**buildCONTENT\_PASTE\_SEARCHinstance-attribute**

CONTENT\_PASTE\_SEARCH: IconData

**buildCONTENT\_PASTE\_SEARCH\_OUTLINEDinstance-attribute**

CONTENT\_PASTE\_SEARCH\_OUTLINED: IconData

**buildCONTENT\_PASTE\_SEARCH\_ROUNDEDinstance-attribute**

CONTENT\_PASTE\_SEARCH\_ROUNDED: IconData

**buildCONTENT\_PASTE\_SEARCH\_SHARPinstance-attribute**

CONTENT\_PASTE\_SEARCH\_SHARP: IconData

**buildCONTENT\_PASTE\_SHARPinstance-attribute**

CONTENT\_PASTE\_SHARP: IconData

**buildCONTRASTinstance-attribute**

CONTRAST: IconData

**buildCONTRAST\_OUTLINEDinstance-attribute**

CONTRAST\_OUTLINED: IconData

**buildCONTRAST\_ROUNDEDinstance-attribute**

CONTRAST\_ROUNDED: IconData

**buildCONTRAST\_SHARPinstance-attribute**

CONTRAST\_SHARP: IconData

**buildCONTROL\_CAMERAinstance-attribute**

CONTROL\_CAMERA: IconData

**buildCONTROL\_CAMERA\_OUTLINEDinstance-attribute**

CONTROL\_CAMERA\_OUTLINED: IconData

**buildCONTROL\_CAMERA\_ROUNDEDinstance-attribute**

CONTROL\_CAMERA\_ROUNDED: IconData

**buildCONTROL\_CAMERA\_SHARPinstance-attribute**

CONTROL\_CAMERA\_SHARP: IconData

**buildCONTROL\_POINTinstance-attribute**

CONTROL\_POINT: IconData

**buildCONTROL\_POINT\_DUPLICATEinstance-attribute**

CONTROL\_POINT\_DUPLICATE: IconData

**buildCONTROL\_POINT\_DUPLICATE\_OUTLINEDinstance-attribute**

CONTROL\_POINT\_DUPLICATE\_OUTLINED: IconData

**buildCONTROL\_POINT\_DUPLICATE\_ROUNDEDinstance-attribute**

CONTROL\_POINT\_DUPLICATE\_ROUNDED: IconData

**buildCONTROL\_POINT\_DUPLICATE\_SHARPinstance-attribute**

CONTROL\_POINT\_DUPLICATE\_SHARP: IconData

**buildCONTROL\_POINT\_OUTLINEDinstance-attribute**

CONTROL\_POINT\_OUTLINED: IconData

**buildCONTROL\_POINT\_ROUNDEDinstance-attribute**

CONTROL\_POINT\_ROUNDED: IconData

**buildCONTROL\_POINT\_SHARPinstance-attribute**

CONTROL\_POINT\_SHARP: IconData

**buildCONVEYOR\_BELTinstance-attribute**

CONVEYOR\_BELT: IconData

**buildCOOKIEinstance-attribute**

COOKIE: IconData

**buildCOOKIE\_OUTLINEDinstance-attribute**

COOKIE\_OUTLINED: IconData

**buildCOOKIE\_ROUNDEDinstance-attribute**

COOKIE\_ROUNDED: IconData

**buildCOOKIE\_SHARPinstance-attribute**

COOKIE\_SHARP: IconData

**buildCOPYinstance-attribute**

COPY: IconData

**buildCOPY\_ALLinstance-attribute**

COPY\_ALL: IconData

**buildCOPY\_ALL\_OUTLINEDinstance-attribute**

COPY\_ALL\_OUTLINED: IconData

**buildCOPY\_ALL\_ROUNDEDinstance-attribute**

COPY\_ALL\_ROUNDED: IconData

**buildCOPY\_ALL\_SHARPinstance-attribute**

COPY\_ALL\_SHARP: IconData

**buildCOPY\_OUTLINEDinstance-attribute**

COPY\_OUTLINED: IconData

**buildCOPY\_ROUNDEDinstance-attribute**

COPY\_ROUNDED: IconData

**buildCOPY\_SHARPinstance-attribute**

COPY\_SHARP: IconData

**buildCOPYRIGHTinstance-attribute**

COPYRIGHT: IconData

**buildCOPYRIGHT\_OUTLINEDinstance-attribute**

COPYRIGHT\_OUTLINED: IconData

**buildCOPYRIGHT\_ROUNDEDinstance-attribute**

COPYRIGHT\_ROUNDED: IconData

**buildCOPYRIGHT\_SHARPinstance-attribute**

COPYRIGHT\_SHARP: IconData

**buildCORONAVIRUSinstance-attribute**

CORONAVIRUS: IconData

**buildCORONAVIRUS\_OUTLINEDinstance-attribute**

CORONAVIRUS\_OUTLINED: IconData

**buildCORONAVIRUS\_ROUNDEDinstance-attribute**

CORONAVIRUS\_ROUNDED: IconData

**buildCORONAVIRUS\_SHARPinstance-attribute**

CORONAVIRUS\_SHARP: IconData

**buildCORPORATE\_FAREinstance-attribute**

CORPORATE\_FARE: IconData

**buildCORPORATE\_FARE\_OUTLINEDinstance-attribute**

CORPORATE\_FARE\_OUTLINED: IconData

**buildCORPORATE\_FARE\_ROUNDEDinstance-attribute**

CORPORATE\_FARE\_ROUNDED: IconData

**buildCORPORATE\_FARE\_SHARPinstance-attribute**

CORPORATE\_FARE\_SHARP: IconData

**buildCOTTAGEinstance-attribute**

COTTAGE: IconData

**buildCOTTAGE\_OUTLINED**instance-attribute

COTTAGE\_OUTLINED: IconData

**buildCOTTAGE\_ROUNDED**instance-attribute

COTTAGE\_ROUNDED: IconData

**buildCOTTAGE\_SHARP**instance-attribute

COTTAGE\_SHARP: IconData

**buildCOUNTERTOPS**instance-attribute

COUNTERTOPS: IconData

**buildCOUNTERTOPS\_OUTLINED**instance-attribute

COUNTERTOPS\_OUTLINED: IconData

**buildCOUNTERTOPS\_ROUNDED**instance-attribute

COUNTERTOPS\_ROUNDED: IconData

**buildCOUNTERTOPS\_SHARP**instance-attribute

COUNTERTOPS\_SHARP: IconData

**buildCREATE**instance-attribute

CREATE: IconData

**buildCREATE\_NEW\_FOLDER**instance-attribute

CREATE\_NEW\_FOLDER: IconData

**buildCREATE\_NEW\_FOLDER\_OUTLINED**instance-attribute

CREATE\_NEW\_FOLDER\_OUTLINED: IconData

**buildCREATE\_NEW\_FOLDER\_ROUNDED**instance-attribute

CREATE\_NEW\_FOLDER\_ROUNDED: IconData

**buildCREATE\_NEW\_FOLDER\_SHARP**instance-attribute

CREATE\_NEW\_FOLDER\_SHARP: IconData

**buildCREATE\_OUTLINED**instance-attribute

CREATE\_OUTLINED: IconData

**buildCREATE\_ROUNDED**instance-attribute

CREATE\_ROUNDED: IconData

**buildCREATE\_SHARP**instance-attribute

CREATE\_SHARP: IconData

**buildCREDIT\_CARD**instance-attribute

CREDIT\_CARD: IconData

**buildCREDIT\_CARD\_OFF**instance-attribute

CREDIT\_CARD\_OFF: IconData

**buildCREDIT\_CARD\_OFF\_OUTLINED**instance-attribute

CREDIT\_CARD\_OFF\_OUTLINED: IconData

**buildCREDIT\_CARD\_OFF\_ROUNDED**instance-attribute

CREDIT\_CARD\_OFF\_ROUNDED: IconData

**buildCREDIT\_CARD\_OFF\_SHARP**instance-attribute

CREDIT\_CARD\_OFF\_SHARP: IconData

**buildCREDIT\_CARD\_OUTLINED**instance-attribute

CREDIT\_CARD\_OUTLINED: IconData

**buildCREDIT\_CARD\_ROUNDED**instance-attribute

CREDIT\_CARD\_ROUNDED: IconData

**buildCREDIT\_CARD\_SHARPinstance-attribute**

CREDIT\_CARD\_SHARP: IconData

**buildCREDIT\_SCOREinstance-attribute**

CREDIT\_SCORE: IconData

**buildCREDIT\_SCORE\_OUTLINEDinstance-attribute**

CREDIT\_SCORE\_OUTLINED: IconData

**buildCREDIT\_SCORE\_ROUNDEDinstance-attribute**

CREDIT\_SCORE\_ROUNDED: IconData

**buildCREDIT\_SCORE\_SHARPinstance-attribute**

CREDIT\_SCORE\_SHARP: IconData

**buildCRIBinstance-attribute**

CRIB: IconData

**buildCRIB\_OUTLINEDinstance-attribute**

CRIB\_OUTLINED: IconData

**buildCRIB\_ROUNDEDinstance-attribute**

CRIB\_ROUNDED: IconData

**buildCRIB\_SHARPinstance-attribute**

CRIB\_SHARP: IconData

**buildCRISIS\_ALERTinstance-attribute**

CRISIS\_ALERT: IconData

**buildCRISIS\_ALERT\_OUTLINEDinstance-attribute**

CRISIS\_ALERT\_OUTLINED: IconData

**buildCRISIS\_ALERT\_ROUNDEDinstance-attribute**

CRISIS\_ALERT\_ROUNDED: IconData

**buildCRISIS\_ALERT\_SHARPinstance-attribute**

CRISIS\_ALERT\_SHARP: IconData

**buildCROPinstance-attribute**

CROP: IconData

**buildCROP\_16\_9instance-attribute**

CROP\_16\_9: IconData

**buildCROP\_16\_9\_OUTLINEDinstance-attribute**

CROP\_16\_9\_OUTLINED: IconData

**buildCROP\_16\_9\_ROUNDEDinstance-attribute**

CROP\_16\_9\_ROUNDED: IconData

**buildCROP\_16\_9\_SHARPinstance-attribute**

CROP\_16\_9\_SHARP: IconData

**buildCROP\_3\_2instance-attribute**

CROP\_3\_2: IconData

**buildCROP\_3\_2\_OUTLINEDinstance-attribute**

CROP\_3\_2\_OUTLINED: IconData

**buildCROP\_3\_2\_ROUNDEDinstance-attribute**

CROP\_3\_2\_ROUNDED: IconData

**buildCROP\_3\_2\_SHARPinstance-attribute**

CROP\_3\_2\_SHARP: IconData

**buildCROP\_5\_4instance-attribute**

CROP\_5\_4: IconData

**buildCROP\_5\_4\_OUTLINEDinstance-attribute**

CROP\_5\_4\_OUTLINED: IconData

**buildCROP\_5\_4\_ROUNDEDinstance-attribute**

CROP\_5\_4\_ROUNDED: IconData

**buildCROP\_5\_4\_SHARPinstance-attribute**

CROP\_5\_4\_SHARP: IconData

**buildCROP\_7\_5instance-attribute**

CROP\_7\_5: IconData

**buildCROP\_7\_5\_OUTLINEDinstance-attribute**

CROP\_7\_5\_OUTLINED: IconData

**buildCROP\_7\_5\_ROUNDEDinstance-attribute**

CROP\_7\_5\_ROUNDED: IconData

**buildCROP\_7\_5\_SHARPinstance-attribute**

CROP\_7\_5\_SHARP: IconData

**buildCROP\_DINinstance-attribute**

CROP\_DIN: IconData

**buildCROP\_DIN\_OUTLINEDinstance-attribute**

CROP\_DIN\_OUTLINED: IconData

**buildCROP\_DIN\_ROUNDEDinstance-attribute**

CROP\_DIN\_ROUNDED: IconData

**buildCROP\_DIN\_SHARPinstance-attribute**

CROP\_DIN\_SHARP: IconData

**buildCROP\_FREEinstance-attribute**

CROP\_FREE: IconData

**buildCROP\_FREE\_OUTLINEDinstance-attribute**

CROP\_FREE\_OUTLINED: IconData

**buildCROP\_FREE\_ROUNDEDinstance-attribute**

CROP\_FREE\_ROUNDED: IconData

**buildCROP\_FREE\_SHARPinstance-attribute**

CROP\_FREE\_SHARP: IconData

**buildCROP\_LANDSCAPEinstance-attribute**

CROP\_LANDSCAPE: IconData

**buildCROP\_LANDSCAPE\_OUTLINEDinstance-attribute**

CROP\_LANDSCAPE\_OUTLINED: IconData

**buildCROP\_LANDSCAPE\_ROUNDEDinstance-attribute**

CROP\_LANDSCAPE\_ROUNDED: IconData

**buildCROP\_LANDSCAPE\_SHARPinstance-attribute**

CROP\_LANDSCAPE\_SHARP: IconData

**buildCROP\_ORIGINALinstance-attribute**

CROP\_ORIGINAL: IconData

**buildCROP\_ORIGINAL\_OUTLINEDinstance-attribute**

CROP\_ORIGINAL\_OUTLINED: IconData

**buildCROP\_ORIGINAL\_ROUNDEDinstance-attribute**

CROP\_ORIGINAL\_ROUNDED: IconData

**buildCROP\_ORIGINAL\_SHARPinstance-attribute**

CROP\_ORIGINAL\_SHARP: IconData

**buildCROP\_OUTLINEDinstance-attribute**

CROP\_OUTLINED: IconData

**buildCROP\_PORTRAITinstance-attribute**

CROP\_PORTRAIT: IconData

**buildCROP\_PORTRAIT\_OUTLINEDinstance-attribute**

CROP\_PORTRAIT\_OUTLINED: IconData

**buildCROP\_PORTRAIT\_ROUNDEDinstance-attribute**

CROP\_PORTRAIT\_ROUNDED: IconData

**buildCROP\_PORTRAIT\_SHARPinstance-attribute**

CROP\_PORTRAIT\_SHARP: IconData

**buildCROP\_ROTATEinstance-attribute**

CROP\_ROTATE: IconData

**buildCROP\_ROTATE\_OUTLINEDinstance-attribute**

CROP\_ROTATE\_OUTLINED: IconData

**buildCROP\_ROTATE\_ROUNDEDinstance-attribute**

CROP\_ROTATE\_ROUNDED: IconData

**buildCROP\_ROTATE\_SHARPinstance-attribute**

CROP\_ROTATE\_SHARP: IconData

**buildCROP\_ROUNDEDinstance-attribute**

CROP\_ROUNDED: IconData

**buildCROP\_SHARPinstance-attribute**

CROP\_SHARP: IconData

**buildCROP\_SQUAREinstance-attribute**

CROP\_SQUARE: IconData

**buildCROP\_SQUARE\_OUTLINEDinstance-attribute**

CROP\_SQUARE\_OUTLINED: IconData

**buildCROP\_SQUARE\_ROUNDEDinstance-attribute**

CROP\_SQUARE\_ROUNDED: IconData

**buildCROP\_SQUARE\_SHARPinstance-attribute**

CROP\_SQUARE\_SHARP: IconData

**buildCRUELTY\_FREEinstance-attribute**

CRUELTY\_FREE: IconData

**buildCRUELTY\_FREE\_OUTLINEDinstance-attribute**

CRUELTY\_FREE\_OUTLINED: IconData

**buildCRUELTY\_FREE\_ROUNDEDinstance-attribute**

CRUELTY\_FREE\_ROUNDED: IconData

**buildCRUELTY\_FREE\_SHARPinstance-attribute**

CRUELTY\_FREE\_SHARP: IconData

**buildCSSinstance-attribute**

CSS: IconData



**buildCSS\_OUTLINED**instance-attribute

CSS\_OUTLINED: IconData

**buildCSS\_ROUNDED**instance-attribute

CSS\_ROUNDED: IconData

**buildCSS\_SHARP**instance-attribute

CSS\_SHARP: IconData

**buildCURRENCY\_BITCOIN**instance-attribute

CURRENCY\_BITCOIN: IconData

**buildCURRENCY\_BITCOIN\_OUTLINED**instance-attribute

CURRENCY\_BITCOIN\_OUTLINED: IconData

**buildCURRENCY\_BITCOIN\_ROUNDED**instance-attribute

CURRENCY\_BITCOIN\_ROUNDED: IconData

**buildCURRENCY\_BITCOIN\_SHARP**instance-attribute

CURRENCY\_BITCOIN\_SHARP: IconData

**buildCURRENCY\_EXCHANGE**instance-attribute

CURRENCY\_EXCHANGE: IconData

**buildCURRENCY\_EXCHANGE\_OUTLINED**instance-attribute

CURRENCY\_EXCHANGE\_OUTLINED: IconData

**buildCURRENCY\_EXCHANGE\_ROUNDED**instance-attribute

CURRENCY\_EXCHANGE\_ROUNDED: IconData

**buildCURRENCY\_EXCHANGE\_SHARP**instance-attribute

CURRENCY\_EXCHANGE\_SHARP: IconData

**buildCURRENCY\_FRANC**instance-attribute

CURRENCY\_FRANC: IconData

**buildCURRENCY\_FRANC\_OUTLINED**instance-attribute

CURRENCY\_FRANC\_OUTLINED: IconData

**buildCURRENCY\_FRANC\_ROUNDED**instance-attribute

CURRENCY\_FRANC\_ROUNDED: IconData

**buildCURRENCY\_FRANC\_SHARP**instance-attribute

CURRENCY\_FRANC\_SHARP: IconData

**buildCURRENCY\_LIRA**instance-attribute

CURRENCY\_LIRA: IconData

**buildCURRENCY\_LIRA\_OUTLINED**instance-attribute

CURRENCY\_LIRA\_OUTLINED: IconData

**buildCURRENCY\_LIRA\_ROUNDED**instance-attribute

CURRENCY\_LIRA\_ROUNDED: IconData

**buildCURRENCY\_LIRA\_SHARP**instance-attribute

CURRENCY\_LIRA\_SHARP: IconData

**buildCURRENCY\_POUND**instance-attribute

CURRENCY\_POUND: IconData

**buildCURRENCY\_POUND\_OUTLINED**instance-attribute

CURRENCY\_POUND\_OUTLINED: IconData

**buildCURRENCY\_POUND\_ROUNDED**instance-attribute

CURRENCY\_POUND\_ROUNDED: IconData

**buildCURRENCY\_POUND\_SHARPinstance-attribute**

CURRENCY\_POUND\_SHARP: IconData

**buildCURRENCY\_RUBLEinstance-attribute**

CURRENCY\_RUBLE: IconData

**buildCURRENCY\_RUBLE\_OUTLINEDinstance-attribute**

CURRENCY\_RUBLE\_OUTLINED: IconData

**buildCURRENCY\_RUBLE\_ROUNDEDinstance-attribute**

CURRENCY\_RUBLE\_ROUNDED: IconData

**buildCURRENCY\_RUBLE\_SHARPinstance-attribute**

CURRENCY\_RUBLE\_SHARP: IconData

**buildCURRENCY\_RUPEEinstance-attribute**

CURRENCY\_RUPEE: IconData

**buildCURRENCY\_RUPEE\_OUTLINEDinstance-attribute**

CURRENCY\_RUPEE\_OUTLINED: IconData

**buildCURRENCY\_RUPEE\_ROUNDEDinstance-attribute**

CURRENCY\_RUPEE\_ROUNDED: IconData

**buildCURRENCY\_RUPEE\_SHARPinstance-attribute**

CURRENCY\_RUPEE\_SHARP: IconData

**buildCURRENCY\_YENinstance-attribute**

CURRENCY\_YEN: IconData

**buildCURRENCY\_YEN\_OUTLINEDinstance-attribute**

CURRENCY\_YEN\_OUTLINED: IconData

**buildCURRENCY\_YEN\_ROUNDEDinstance-attribute**

CURRENCY\_YEN\_ROUNDED: IconData

**buildCURRENCY\_YEN\_SHARPinstance-attribute**

CURRENCY\_YEN\_SHARP: IconData

**buildCURRENCY\_YUANinstance-attribute**

CURRENCY\_YUAN: IconData

**buildCURRENCY\_YUAN\_OUTLINEDinstance-attribute**

CURRENCY\_YUAN\_OUTLINED: IconData

**buildCURRENCY\_YUAN\_ROUNDEDinstance-attribute**

CURRENCY\_YUAN\_ROUNDED: IconData

**buildCURRENCY\_YUAN\_SHARPinstance-attribute**

CURRENCY\_YUAN\_SHARP: IconData

**buildCURTAINSinstance-attribute**

CURTAINS: IconData

**buildCURTAINS\_CLOSEDinstance-attribute**

CURTAINS\_CLOSED: IconData

**buildCURTAINS\_CLOSED\_OUTLINEDinstance-attribute**

CURTAINS\_CLOSED\_OUTLINED: IconData

**buildCURTAINS\_CLOSED\_ROUNDEDinstance-attribute**

CURTAINS\_CLOSED\_ROUNDED: IconData

**buildCURTAINS\_CLOSED\_SHARPinstance-attribute**

CURTAINS\_CLOSED\_SHARP: IconData

**buildCURTAINS\_OUTLINEDinstance-attribute**

CURTAINS\_OUTLINED: IconData

**buildCURTAINS\_ROUNDEDinstance-attribute**

CURTAINS\_ROUNDED: IconData

**buildCURTAINS\_SHARPinstance-attribute**

CURTAINS\_SHARP: IconData

**buildCUTinstance-attribute**

CUT: IconData

**buildCUT\_OUTLINEDinstance-attribute**

CUT\_OUTLINED: IconData

**buildCUT\_ROUNDEDinstance-attribute**

CUT\_ROUNDED: IconData

**buildCUT\_SHARPinstance-attribute**

CUT\_SHARP: IconData

**buildCYCLONEinstance-attribute**

CYCLONE: IconData

**buildCYCLONE\_OUTLINEDinstance-attribute**

CYCLONE\_OUTLINED: IconData

**buildCYCLONE\_ROUNDEDinstance-attribute**

CYCLONE\_ROUNDED: IconData

**buildCYCLONE\_SHARPinstance-attribute**

CYCLONE\_SHARP: IconData

**buildDANGEROUSinstance-attribute**

DANGEROUS: IconData

**buildDANGEROUS\_OUTLINEDinstance-attribute**

DANGEROUS\_OUTLINED: IconData

**buildDANGEROUS\_ROUNDEDinstance-attribute**

DANGEROUS\_ROUNDED: IconData

**buildDANGEROUS\_SHARPinstance-attribute**

DANGEROUS\_SHARP: IconData

**buildDARK\_MODEinstance-attribute**

DARK\_MODE: IconData

**buildDARK\_MODE\_OUTLINEDinstance-attribute**

DARK\_MODE\_OUTLINED: IconData

**buildDARK\_MODE\_ROUNDEDinstance-attribute**

DARK\_MODE\_ROUNDED: IconData

**buildDARK\_MODE\_SHARPinstance-attribute**

DARK\_MODE\_SHARP: IconData

**buildDASHBOARDinstance-attribute**

DASHBOARD: IconData

**buildDASHBOARD\_CUSTOMIZEinstance-attribute**

DASHBOARD\_CUSTOMIZE: IconData

**buildDASHBOARD\_CUSTOMIZE\_OUTLINEDinstance-attribute**

DASHBOARD\_CUSTOMIZE\_OUTLINED: IconData

**buildDASHBOARD\_CUSTOMIZE\_ROUNDED**instance-attribute

DASHBOARD\_CUSTOMIZE\_ROUNDED: IconData

**buildDASHBOARD\_CUSTOMIZE\_SHARP**instance-attribute

DASHBOARD\_CUSTOMIZE\_SHARP: IconData

**buildDASHBOARD\_OUTLINED**instance-attribute

DASHBOARD\_OUTLINED: IconData

**buildDASHBOARD\_ROUNDED**instance-attribute

DASHBOARD\_ROUNDED: IconData

**buildDASHBOARD\_SHARP**instance-attribute

DASHBOARD\_SHARP: IconData

**buildDATA\_ARRAY**instance-attribute

DATA\_ARRAY: IconData

**buildDATA\_ARRAY\_OUTLINED**instance-attribute

DATA\_ARRAY\_OUTLINED: IconData

**buildDATA\_ARRAY\_ROUNDED**instance-attribute

DATA\_ARRAY\_ROUNDED: IconData

**buildDATA\_ARRAY\_SHARP**instance-attribute

DATA\_ARRAY\_SHARP: IconData

**buildDATA\_EXPLORATION**instance-attribute

DATA\_EXPLORATION: IconData

**buildDATA\_EXPLORATION\_OUTLINED**instance-attribute

DATA\_EXPLORATION\_OUTLINED: IconData

**buildDATA\_EXPLORATION\_ROUNDED**instance-attribute

DATA\_EXPLORATION\_ROUNDED: IconData

**buildDATA\_EXPLORATION\_SHARP**instance-attribute

DATA\_EXPLORATION\_SHARP: IconData

**buildDATA\_OBJECT**instance-attribute

DATA\_OBJECT: IconData

**buildDATA\_OBJECT\_OUTLINED**instance-attribute

DATA\_OBJECT\_OUTLINED: IconData

**buildDATA\_OBJECT\_ROUNDED**instance-attribute

DATA\_OBJECT\_ROUNDED: IconData

**buildDATA\_OBJECT\_SHARP**instance-attribute

DATA\_OBJECT\_SHARP: IconData

**buildDATA\_SAVER\_OFF**instance-attribute ☐

DATA\_SAVER\_OFF: IconData

**buildDATA\_SAVER\_OFF\_OUTLINED**instance-attribute

DATA\_SAVER\_OFF\_OUTLINED: IconData

**buildDATA\_SAVER\_OFF\_ROUNDED**instance-attribute

DATA\_SAVER\_OFF\_ROUNDED: IconData

**buildDATA\_SAVER\_OFF\_SHARP**instance-attribute

DATA\_SAVER\_OFF\_SHARP: IconData

**buildDATA\_SAVER\_ON**instance-attribute

DATA\_SAVER\_ON: IconData

**buildDATA\_SAVER\_ON\_OUTLINEDinstance-attribute**

DATA\_SAVER\_ON\_OUTLINED: IconData

**buildDATA\_SAVER\_ON\_ROUNDEDinstance-attribute**

DATA\_SAVER\_ON\_ROUNDED: IconData

**buildDATA\_SAVER\_ON\_SHARPinstance-attribute**

DATA\_SAVER\_ON\_SHARP: IconData

**buildDATA\_THRESHOLDINGinstance-attribute**

DATA\_THRESHOLDING: IconData

**buildDATA\_THRESHOLDING\_OUTLINEDinstance-attribute**

DATA\_THRESHOLDING\_OUTLINED: IconData

**buildDATA\_THRESHOLDING\_ROUNDEDinstance-attribute**

DATA\_THRESHOLDING\_ROUNDED: IconData

**buildDATA\_THRESHOLDING\_SHARPinstance-attribute**

DATA\_THRESHOLDING\_SHARP: IconData

**buildDATA\_USAGEinstance-attribute**

DATA\_USAGE: IconData

**buildDATA\_USAGE\_OUTLINEDinstance-attribute**

DATA\_USAGE\_OUTLINED: IconData

**buildDATA\_USAGE\_ROUNDEDinstance-attribute**

DATA\_USAGE\_ROUNDED: IconData

**buildDATA\_USAGE\_SHARPinstance-attribute**

DATA\_USAGE\_SHARP: IconData

**buildDATASETinstance-attribute**

DATASET: IconData

**buildDATASET\_LINKEDinstance-attribute**

DATASET\_LINKED: IconData

**buildDATASET\_LINKED\_OUTLINEDinstance-attribute**

DATASET\_LINKED\_OUTLINED: IconData

**buildDATASET\_LINKED\_ROUNDEDinstance-attribute**

DATASET\_LINKED\_ROUNDED: IconData

**buildDATASET\_LINKED\_SHARPinstance-attribute**

DATASET\_LINKED\_SHARP: IconData

**buildDATASET\_OUTLINEDinstance-attribute**

DATASET\_OUTLINED: IconData

**buildDATASET\_ROUNDEDinstance-attribute**

DATASET\_ROUNDED: IconData

**buildDATASET\_SHARPinstance-attribute**

DATASET\_SHARP: IconData

**buildDATE\_RANGEinstance-attribute**

DATE\_RANGE: IconData

**buildDATE\_RANGE\_OUTLINEDinstance-attribute**

DATE\_RANGE\_OUTLINED: IconData

**buildDATE\_RANGE\_ROUNDEDinstance-attribute**

DATE\_RANGE\_ROUNDED: IconData

**buildDATE\_RANGE\_SHARPinstance-attribute**

DATE\_RANGE\_SHARP: IconData

**buildDEBLURinstance-attribute**

DEBLUR: IconData

**buildDEBLUR\_OUTLINEDinstance-attribute**

DEBLUR\_OUTLINED: IconData

**buildDEBLUR\_ROUNDEDinstance-attribute**

DEBLUR\_ROUNDED: IconData

**buildDEBLUR\_SHARPinstance-attribute**

DEBLUR\_SHARP: IconData

**buildDECKinstance-attribute**

DECK: IconData

**buildDECK\_OUTLINEDinstance-attribute**

DECK\_OUTLINED: IconData

**buildDECK\_ROUNDEDinstance-attribute**

DECK\_ROUNDED: IconData

**buildDECK\_SHARPinstance-attribute**

DECK\_SHARP: IconData

**buildDEHAZEinstance-attribute**

DEHAZE: IconData

**buildDEHAZE\_OUTLINEDinstance-attribute**

DEHAZE\_OUTLINED: IconData

**buildDEHAZE\_ROUNDEDinstance-attribute**

DEHAZE\_ROUNDED: IconData

**buildDEHAZE\_SHARPinstance-attribute**

DEHAZE\_SHARP: IconData

**buildDELETEinstance-attribute**

DELETE: IconData

**buildDELETE\_FOREVERinstance-attribute**

DELETE\_FOREVER: IconData

**buildDELETE\_FOREVER\_OUTLINEDinstance-attribute**

DELETE\_FOREVER\_OUTLINED: IconData

**buildDELETE\_FOREVER\_ROUNDEDinstance-attribute**

DELETE\_FOREVER\_ROUNDED: IconData

**buildDELETE\_FOREVER\_SHARPinstance-attribute**

DELETE\_FOREVER\_SHARP: IconData

**buildDELETE\_OUTLINEinstance-attribute**

DELETE\_OUTLINE: IconData

**buildDELETE\_OUTLINE\_OUTLINEDinstance-attribute**

DELETE\_OUTLINE\_OUTLINED: IconData

**buildDELETE\_OUTLINE\_ROUNDEDinstance-attribute**

DELETE\_OUTLINE\_ROUNDED: IconData

**buildDELETE\_OUTLINE\_SHARPinstance-attribute**

DELETE\_OUTLINE\_SHARP: IconData

**buildDELETE\_OUTLINEDinstance-attribute**

DELETE\_OUTLINED: IconData

**buildDELETE\_ROUNDEDinstance-attribute**

DELETE\_ROUNDED: IconData

**buildDELETE\_SHARPinstance-attribute**

DELETE\_SHARP: IconData

**buildDELETE\_SWEEPinstance-attribute**

DELETE\_SWEEP: IconData

**buildDELETE\_SWEEP\_OUTLINEDinstance-attribute**

DELETE\_SWEEP\_OUTLINED: IconData

**buildDELETE\_SWEEP\_ROUNDEDinstance-attribute**

DELETE\_SWEEP\_ROUNDED: IconData

**buildDELETE\_SWEEP\_SHARPinstance-attribute**

DELETE\_SWEEP\_SHARP: IconData

**buildDELIVERY\_DININGinstance-attribute**

DELIVERY\_DINING: IconData

**buildDELIVERY\_DINING\_OUTLINEDinstance-attribute**

DELIVERY\_DINING\_OUTLINED: IconData

**buildDELIVERY\_DINING\_ROUNDEDinstance-attribute**

DELIVERY\_DINING\_ROUNDED: IconData

**buildDELIVERY\_DINING\_SHARPinstance-attribute**

DELIVERY\_DINING\_SHARP: IconData

**buildDENSITY\_LARGEinstance-attribute**

DENSITY\_LARGE: IconData

**buildDENSITY\_LARGE\_OUTLINEDinstance-attribute**

DENSITY\_LARGE\_OUTLINED: IconData

**buildDENSITY\_LARGE\_ROUNDEDinstance-attribute**

DENSITY\_LARGE\_ROUNDED: IconData

**buildDENSITY\_LARGE\_SHARPinstance-attribute**

DENSITY\_LARGE\_SHARP: IconData

**buildDENSITY\_MEDIUMinstance-attribute**

DENSITY\_MEDIUM: IconData

**buildDENSITY\_MEDIUM\_OUTLINEDinstance-attribute**

DENSITY\_MEDIUM\_OUTLINED: IconData

**buildDENSITY\_MEDIUM\_ROUNDEDinstance-attribute**

DENSITY\_MEDIUM\_ROUNDED: IconData

**buildDENSITY\_MEDIUM\_SHARPinstance-attribute**

DENSITY\_MEDIUM\_SHARP: IconData

**buildDENSITY\_SMALLinstance-attribute**

DENSITY\_SMALL: IconData

**buildDENSITY\_SMALL\_OUTLINEDinstance-attribute**

DENSITY\_SMALL\_OUTLINED: IconData

**buildDENSITY\_SMALL\_ROUNDEDinstance-attribute**

DENSITY\_SMALL\_ROUNDED: IconData

**buildDENSITY\_SMALL\_SHARPinstance-attribute**

DENSITY\_SMALL\_SHARP: IconData

**buildDEPARTURE\_BOARDinstance-attribute**

DEPARTURE\_BOARD: IconData

**buildDEPARTURE\_BOARD\_OUTLINEDinstance-attribute**

DEPARTURE\_BOARD\_OUTLINED: IconData

**buildDEPARTURE\_BOARD\_ROUNDEDinstance-attribute**

DEPARTURE\_BOARD\_ROUNDED: IconData

**buildDEPARTURE\_BOARD\_SHARPinstance-attribute**

DEPARTURE\_BOARD\_SHARP: IconData

**buildDESCRIPTIONinstance-attribute**

DESCRIPTION: IconData

**buildDESCRIPTION\_OUTLINEDinstance-attribute**

DESCRIPTION\_OUTLINED: IconData

**buildDESCRIPTION\_ROUNDEDinstance-attribute**

DESCRIPTION\_ROUNDED: IconData

**buildDESCRIPTION\_SHARPinstance-attribute**

DESCRIPTION\_SHARP: IconData

**buildDESELECTinstance-attribute**

DESELECT: IconData

**buildDESELECT\_OUTLINEDinstance-attribute**

DESELECT\_OUTLINED: IconData

**buildDESELECT\_ROUNDEDinstance-attribute**

DESELECT\_ROUNDED: IconData

**buildDESELECT\_SHARPinstance-attribute**

DESELECT\_SHARP: IconData

**buildDESIGN\_SERVICESinstance-attribute**

DESIGN\_SERVICES: IconData

**buildDESIGN\_SERVICES\_OUTLINEDinstance-attribute**

DESIGN\_SERVICES\_OUTLINED: IconData

**buildDESIGN\_SERVICES\_ROUNDEDinstance-attribute**

DESIGN\_SERVICES\_ROUNDED: IconData

**buildDESIGN\_SERVICES\_SHARPinstance-attribute**

DESIGN\_SERVICES\_SHARP: IconData

**buildDESKinstance-attribute**

DESK: IconData

**buildDESK\_OUTLINEDinstance-attribute**

DESK\_OUTLINED: IconData

**buildDESK\_ROUNDEDinstance-attribute**

DESK\_ROUNDED: IconData

**buildDESK\_SHARPinstance-attribute**

DESK\_SHARP: IconData

**buildDESKTOP\_ACCESS\_DISABLEDinstance-attribute**

DESKTOP\_ACCESS\_DISABLED: IconData



**buildDESKTOP\_ACCESS\_DISABLED\_OUTLINED**instance-attribute

DESKTOP\_ACCESS\_DISABLED\_OUTLINED: IconData

**buildDESKTOP\_ACCESS\_DISABLED\_ROUNDED**instance-attribute

DESKTOP\_ACCESS\_DISABLED\_ROUNDED: IconData

**buildDESKTOP\_ACCESS\_DISABLED\_SHARP**instance-attribute

DESKTOP\_ACCESS\_DISABLED\_SHARP: IconData

**buildDESKTOP\_MAC**instance-attribute

DESKTOP\_MAC: IconData

**buildDESKTOP\_MAC\_OUTLINED**instance-attribute

DESKTOP\_MAC\_OUTLINED: IconData

**buildDESKTOP\_MAC\_ROUNDED**instance-attribute

DESKTOP\_MAC\_ROUNDED: IconData

**buildDESKTOP\_MAC\_SHARP**instance-attribute

DESKTOP\_MAC\_SHARP: IconData

**buildDESKTOP\_WINDOWS**instance-attribute

DESKTOP\_WINDOWS: IconData

**buildDESKTOP\_WINDOWS\_OUTLINED**instance-attribute

DESKTOP\_WINDOWS\_OUTLINED: IconData

**buildDESKTOP\_WINDOWS\_ROUNDED**instance-attribute

DESKTOP\_WINDOWS\_ROUNDED: IconData

**buildDESKTOP\_WINDOWS\_SHARP**instance-attribute

DESKTOP\_WINDOWS\_SHARP: IconData

**buildDETAILS**instance-attribute

DETAILS: IconData

**buildDETAILS\_OUTLINED**instance-attribute

DETAILS\_OUTLINED: IconData

**buildDETAILS\_ROUNDED**instance-attribute

DETAILS\_ROUNDED: IconData

**buildDETAILS\_SHARP**instance-attribute

DETAILS\_SHARP: IconData

**buildDEVELOPER\_BOARD**instance-attribute

DEVELOPER\_BOARD: IconData

**buildDEVELOPER\_BOARD\_OFF**instance-attribute

DEVELOPER\_BOARD\_OFF: IconData

**buildDEVELOPER\_BOARD\_OFF\_OUTLINED**instance-attribute

DEVELOPER\_BOARD\_OFF\_OUTLINED: IconData

**buildDEVELOPER\_BOARD\_OFF\_ROUNDED**instance-attribute

DEVELOPER\_BOARD\_OFF\_ROUNDED: IconData

**buildDEVELOPER\_BOARD\_OFF\_SHARP**instance-attribute

DEVELOPER\_BOARD\_OFF\_SHARP: IconData

**buildDEVELOPER\_BOARD\_OUTLINED**instance-attribute

DEVELOPER\_BOARD\_OUTLINED: IconData

**buildDEVELOPER\_BOARD\_ROUNDED**instance-attribute

DEVELOPER\_BOARD\_ROUNDED: IconData

**buildDEVELOPER\_BOARD\_SHARPinstance-attribute**

DEVELOPER\_BOARD\_SHARP: IconData

**buildDEVELOPER\_MODEinstance-attribute**

DEVELOPER\_MODE: IconData

**buildDEVELOPER\_MODE\_OUTLINEDinstance-attribute**

DEVELOPER\_MODE\_OUTLINED: IconData

**buildDEVELOPER\_MODE\_ROUNDEDinstance-attribute**

DEVELOPER\_MODE\_ROUNDED: IconData

**buildDEVELOPER\_MODE\_SHARPinstance-attribute**

DEVELOPER\_MODE\_SHARP: IconData

**buildDEVICE\_HUBinstance-attribute**

DEVICE\_HUB: IconData

**buildDEVICE\_HUB\_OUTLINEDinstance-attribute**

DEVICE\_HUB\_OUTLINED: IconData

**buildDEVICE\_HUB\_ROUNDEDinstance-attribute**

DEVICE\_HUB\_ROUNDED: IconData

**buildDEVICE\_HUB\_SHARPinstance-attribute**

DEVICE\_HUB\_SHARP: IconData

**buildDEVICE\_THERMOSTATinstance-attribute**

DEVICE\_THERMOSTAT: IconData

**buildDEVICE\_THERMOSTAT\_OUTLINEDinstance-attribute**

DEVICE\_THERMOSTAT\_OUTLINED: IconData

**buildDEVICE\_THERMOSTAT\_ROUNDEDinstance-attribute**

DEVICE\_THERMOSTAT\_ROUNDED: IconData

**buildDEVICE\_THERMOSTAT\_SHARPinstance-attribute**

DEVICE\_THERMOSTAT\_SHARP: IconData

**buildDEVICE\_UNKNOWNinstance-attribute**

DEVICE\_UNKNOWN: IconData

**buildDEVICE\_UNKNOWN\_OUTLINEDinstance-attribute**

DEVICE\_UNKNOWN\_OUTLINED: IconData

**buildDEVICE\_UNKNOWN\_ROUNDEDinstance-attribute**

DEVICE\_UNKNOWN\_ROUNDED: IconData

**buildDEVICE\_UNKNOWN\_SHARPinstance-attribute**

DEVICE\_UNKNOWN\_SHARP: IconData

**buildDEVICESinstance-attribute**

DEVICES: IconData

**buildDEVICES\_FOLDinstance-attribute**

DEVICES\_FOLD: IconData

**buildDEVICES\_FOLD\_OUTLINEDinstance-attribute**

DEVICES\_FOLD\_OUTLINED: IconData

**buildDEVICES\_FOLD\_ROUNDEDinstance-attribute**

DEVICES\_FOLD\_ROUNDED: IconData

**buildDEVICES\_FOLD\_SHARPinstance-attribute**

DEVICES\_FOLD\_SHARP: IconData

**buildDEVICES\_OTHERinstance-attribute**

DEVICES\_OTHER: IconData

**buildDEVICES\_OTHER\_OUTLINEDinstance-attribute**

DEVICES\_OTHER\_OUTLINED: IconData

**buildDEVICES\_OTHER\_ROUNDEDinstance-attribute**

DEVICES\_OTHER\_ROUNDED: IconData

**buildDEVICES\_OTHER\_SHARPinstance-attribute**

DEVICES\_OTHER\_SHARP: IconData

**buildDEVICES\_OUTLINEDinstance-attribute**

DEVICES\_OUTLINED: IconData

**buildDEVICES\_ROUNDEDinstance-attribute**

DEVICES\_ROUNDED: IconData

**buildDEVICES\_SHARPinstance-attribute**

DEVICES\_SHARP: IconData

**buildDEW\_POINTinstance-attribute**

DEW\_POINT: IconData

**buildDIALER\_SIPinstance-attribute**

DIALER\_SIP: IconData

**buildDIALER\_SIP\_OUTLINEDinstance-attribute**

DIALER\_SIP\_OUTLINED: IconData

**buildDIALER\_SIP\_ROUNDEDinstance-attribute**

DIALER\_SIP\_ROUNDED: IconData

**buildDIALER\_SIP\_SHARPinstance-attribute**

DIALER\_SIP\_SHARP: IconData

**buildDIALPADinstance-attribute**

DIALPAD: IconData

**buildDIALPAD\_OUTLINEDinstance-attribute**

DIALPAD\_OUTLINED: IconData

**buildDIALPAD\_ROUNDEDinstance-attribute**

DIALPAD\_ROUNDED: IconData

**buildDIALPAD\_SHARPinstance-attribute**

DIALPAD\_SHARP: IconData

**buildDIAMONDinstance-attribute**

DIAMOND: IconData

**buildDIAMOND\_OUTLINEDinstance-attribute**

DIAMOND\_OUTLINED: IconData

**buildDIAMOND\_ROUNDEDinstance-attribute**

DIAMOND\_ROUNDED: IconData

**buildDIAMOND\_SHARPinstance-attribute**

DIAMOND\_SHARP: IconData

**buildDIFFERENCEinstance-attribute**

DIFFERENCE: IconData

**buildDIFFERENCE\_OUTLINEDinstance-attribute**

DIFFERENCE\_OUTLINED: IconData

**buildDIFFERENCE\_ROUNDEDinstance-attribute**

DIFFERENCE\_ROUNDED: IconData

**buildDIFFERENCE\_SHARPinstance-attribute**

DIFFERENCE\_SHARP: IconData

**buildDININGinstance-attribute**

DINING: IconData

**buildDINING\_OUTLINEDinstance-attribute**

DINING\_OUTLINED: IconData

**buildDINING\_ROUNDEDinstance-attribute**

DINING\_ROUNDED: IconData

**buildDINING\_SHARPinstance-attribute**

DINING\_SHARP: IconData

**buildDINNER\_DININGinstance-attribute**

DINNER\_DINING: IconData

**buildDINNER\_DINING\_OUTLINEDinstance-attribute**

DINNER\_DINING\_OUTLINED: IconData

**buildDINNER\_DINING\_ROUNDEDinstance-attribute**

DINNER\_DINING\_ROUNDED: IconData

**buildDINNER\_DINING\_SHARPinstance-attribute**

DINNER\_DINING\_SHARP: IconData

**buildDIRECTIONSinstance-attribute**

DIRECTIONS: IconData

**buildDIRECTIONS\_BIKEinstance-attribute**

DIRECTIONS\_BIKE: IconData

**buildDIRECTIONS\_BIKE\_OUTLINEDinstance-attribute**

DIRECTIONS\_BIKE\_OUTLINED: IconData

**buildDIRECTIONS\_BIKE\_ROUNDEDinstance-attribute**

DIRECTIONS\_BIKE\_ROUNDED: IconData

**buildDIRECTIONS\_BIKE\_SHARPinstance-attribute**

DIRECTIONS\_BIKE\_SHARP: IconData

**buildDIRECTIONS\_BOATinstance-attribute**

DIRECTIONS\_BOAT: IconData

**buildDIRECTIONS\_BOAT\_FILLEDinstance-attribute**

DIRECTIONS\_BOAT\_FILLED: IconData

**buildDIRECTIONS\_BOAT\_FILLED\_OUTLINEDinstance-attribute**

DIRECTIONS\_BOAT\_FILLED\_OUTLINED: IconData

**buildDIRECTIONS\_BOAT\_FILLED\_ROUNDEDinstance-attribute**

DIRECTIONS\_BOAT\_FILLED\_ROUNDED: IconData

**buildDIRECTIONS\_BOAT\_FILLED\_SHARPinstance-attribute**

DIRECTIONS\_BOAT\_FILLED\_SHARP: IconData

**buildDIRECTIONS\_BOAT\_OUTLINEDinstance-attribute**

DIRECTIONS\_BOAT\_OUTLINED: IconData

**buildDIRECTIONS\_BOAT\_ROUNDEDinstance-attribute**

DIRECTIONS\_BOAT\_ROUNDED: IconData

**buildDIRECTIONS\_BOAT\_SHARPinstance-attribute**

DIRECTIONS\_BOAT\_SHARP: IconData

**buildDIRECTIONS\_BUSinstance-attribute**

DIRECTIONS\_BUS: IconData

**buildDIRECTIONS\_BUS\_FILLEDinstance-attribute**

DIRECTIONS\_BUS\_FILLED: IconData

**buildDIRECTIONS\_BUS\_FILLED\_OUTLINEDinstance-attribute**

DIRECTIONS\_BUS\_FILLED\_OUTLINED: IconData

**buildDIRECTIONS\_BUS\_FILLED\_ROUNDEDinstance-attribute**

DIRECTIONS\_BUS\_FILLED\_ROUNDED: IconData

**buildDIRECTIONS\_BUS\_FILLED\_SHARPinstance-attribute**

DIRECTIONS\_BUS\_FILLED\_SHARP: IconData

**buildDIRECTIONS\_BUS\_OUTLINEDinstance-attribute**

DIRECTIONS\_BUS\_OUTLINED: IconData

**buildDIRECTIONS\_BUS\_ROUNDEDinstance-attribute**

DIRECTIONS\_BUS\_ROUNDED: IconData

**buildDIRECTIONS\_BUS\_SHARPinstance-attribute**

DIRECTIONS\_BUS\_SHARP: IconData

**buildDIRECTIONS\_CARinstance-attribute**

DIRECTIONS\_CAR: IconData

**buildDIRECTIONS\_CAR\_FILLEDinstance-attribute**

DIRECTIONS\_CAR\_FILLED: IconData

**buildDIRECTIONS\_CAR\_FILLED\_OUTLINEDinstance-attribute**

DIRECTIONS\_CAR\_FILLED\_OUTLINED: IconData

**buildDIRECTIONS\_CAR\_FILLED\_ROUNDEDinstance-attribute**

DIRECTIONS\_CAR\_FILLED\_ROUNDED: IconData

**buildDIRECTIONS\_CAR\_FILLED\_SHARPinstance-attribute**

DIRECTIONS\_CAR\_FILLED\_SHARP: IconData

**buildDIRECTIONS\_CAR\_OUTLINEDinstance-attribute**

DIRECTIONS\_CAR\_OUTLINED: IconData

**buildDIRECTIONS\_CAR\_ROUNDEDinstance-attribute**

DIRECTIONS\_CAR\_ROUNDED: IconData

**buildDIRECTIONS\_CAR\_SHARPinstance-attribute**

DIRECTIONS\_CAR\_SHARP: IconData

**buildDIRECTIONS\_FERRYinstance-attribute**

DIRECTIONS\_FERRY: IconData

**buildDIRECTIONS\_FERRY\_OUTLINEDinstance-attribute**

DIRECTIONS\_FERRY\_OUTLINED: IconData

**buildDIRECTIONS\_FERRY\_ROUNDEDinstance-attribute**

DIRECTIONS\_FERRY\_ROUNDED: IconData

**buildDIRECTIONS\_FERRY\_SHARPinstance-attribute**

DIRECTIONS\_FERRY\_SHARP: IconData

**buildDIRECTIONS\_OFFinstance-attribute**

DIRECTIONS\_OFF: IconData

**buildDIRECTIONS\_OFF\_OUTLINEDinstance-attribute**

DIRECTIONS\_OFF\_OUTLINED: IconData

**buildDIRECTIONS\_OFF\_ROUNDEDinstance-attribute**

DIRECTIONS\_OFF\_ROUNDED: IconData

**buildDIRECTIONS\_OFF\_SHARPinstance-attribute**

DIRECTIONS\_OFF\_SHARP: IconData

**buildDIRECTIONS\_OUTLINEDinstance-attribute**

DIRECTIONS\_OUTLINED: IconData

**buildDIRECTIONS\_RAILWAYinstance-attribute**

DIRECTIONS\_RAILWAY: IconData

**buildDIRECTIONS\_RAILWAY\_FILLEDinstance-attribute**

DIRECTIONS\_RAILWAY\_FILLED: IconData

**buildDIRECTIONS\_RAILWAY\_FILLED\_OUTLINEDinstance-attribute**

DIRECTIONS\_RAILWAY\_FILLED\_OUTLINED: IconData

**buildDIRECTIONS\_RAILWAY\_FILLED\_ROUNDEDinstance-attribute**

DIRECTIONS\_RAILWAY\_FILLED\_ROUNDED: IconData

**buildDIRECTIONS\_RAILWAY\_FILLED\_SHARPinstance-attribute**

DIRECTIONS\_RAILWAY\_FILLED\_SHARP: IconData

**buildDIRECTIONS\_RAILWAY\_OUTLINEDinstance-attribute**

DIRECTIONS\_RAILWAY\_OUTLINED: IconData

**buildDIRECTIONS\_RAILWAY\_ROUNDEDinstance-attribute**

DIRECTIONS\_RAILWAY\_ROUNDED: IconData

**buildDIRECTIONS\_RAILWAY\_SHARPinstance-attribute**

DIRECTIONS\_RAILWAY\_SHARP: IconData

**buildDIRECTIONS\_ROUNDEDinstance-attribute**

DIRECTIONS\_ROUNDED: IconData

**buildDIRECTIONS\_RUNinstance-attribute**

DIRECTIONS\_RUN: IconData

**buildDIRECTIONS\_RUN\_OUTLINEDinstance-attribute**

DIRECTIONS\_RUN\_OUTLINED: IconData

**buildDIRECTIONS\_RUN\_ROUNDEDinstance-attribute**

DIRECTIONS\_RUN\_ROUNDED: IconData

**buildDIRECTIONS\_RUN\_SHARPinstance-attribute**

DIRECTIONS\_RUN\_SHARP: IconData

**buildDIRECTIONS\_SHARPinstance-attribute**

DIRECTIONS\_SHARP: IconData

**buildDIRECTIONS\_SUBWAYinstance-attribute**

DIRECTIONS\_SUBWAY: IconData

**buildDIRECTIONS\_SUBWAY\_FILLEDinstance-attribute**

DIRECTIONS\_SUBWAY\_FILLED: IconData

**buildDIRECTIONS\_SUBWAY\_FILLED\_OUTLINEDinstance-attribute**

DIRECTIONS\_SUBWAY\_FILLED\_OUTLINED: IconData

**buildDIRECTIONS\_SUBWAY\_FILLED\_ROUNDEDinstance-attribute**

DIRECTIONS\_SUBWAY\_FILLED\_ROUNDED: IconData

**buildDIRECTIONS\_SUBWAY\_FILLED\_SHARPinstance-attribute**

DIRECTIONS\_SUBWAY\_FILLED\_SHARP: IconData

**buildDIRECTIONS\_SUBWAY\_OUTLINEDinstance-attribute**

DIRECTIONS\_SUBWAY\_OUTLINED: IconData

**buildDIRECTIONS\_SUBWAY\_ROUNDEDinstance-attribute**

DIRECTIONS\_SUBWAY\_ROUNDED: IconData

**buildDIRECTIONS\_SUBWAY\_SHARPinstance-attribute**

DIRECTIONS\_SUBWAY\_SHARP: IconData

**buildDIRECTIONS\_TRAINinstance-attribute**

DIRECTIONS\_TRAIN: IconData

**buildDIRECTIONS\_TRAIN\_OUTLINEDinstance-attribute**

DIRECTIONS\_TRAIN\_OUTLINED: IconData

**buildDIRECTIONS\_TRAIN\_ROUNDEDinstance-attribute**

DIRECTIONS\_TRAIN\_ROUNDED: IconData

**buildDIRECTIONS\_TRAIN\_SHARPinstance-attribute**

DIRECTIONS\_TRAIN\_SHARP: IconData

**buildDIRECTIONS\_TRANSITinstance-attribute**

DIRECTIONS\_TRANSIT: IconData

**buildDIRECTIONS\_TRANSIT\_FILLEDinstance-attribute**

DIRECTIONS\_TRANSIT\_FILLED: IconData

**buildDIRECTIONS\_TRANSIT\_FILLED\_OUTLINEDinstance-attribute**

DIRECTIONS\_TRANSIT\_FILLED\_OUTLINED: IconData

**buildDIRECTIONS\_TRANSIT\_FILLED\_ROUNDEDinstance-attribute**

DIRECTIONS\_TRANSIT\_FILLED\_ROUNDED: IconData

**buildDIRECTIONS\_TRANSIT\_FILLED\_SHARPinstance-attribute**

DIRECTIONS\_TRANSIT\_FILLED\_SHARP: IconData

**buildDIRECTIONS\_TRANSIT\_OUTLINEDinstance-attribute**

DIRECTIONS\_TRANSIT\_OUTLINED: IconData

**buildDIRECTIONS\_TRANSIT\_ROUNDEDinstance-attribute**

DIRECTIONS\_TRANSIT\_ROUNDED: IconData

**buildDIRECTIONS\_TRANSIT\_SHARPinstance-attribute**

DIRECTIONS\_TRANSIT\_SHARP: IconData

**buildDIRECTIONS\_WALKinstance-attribute**

DIRECTIONS\_WALK: IconData

**buildDIRECTIONS\_WALK\_OUTLINEDinstance-attribute** □ □

DIRECTIONS\_WALK\_OUTLINED: IconData

**buildDIRECTIONS\_WALK\_ROUNDEDinstance-attribute**

DIRECTIONS\_WALK\_ROUNDED: IconData

**buildDIRECTIONS\_WALK\_SHARPinstance-attribute**

DIRECTIONS\_WALK\_SHARP: IconData

**buildDIRTY\_LENSinstance-attribute**

DIRTY\_LENS: IconData

**buildDIRTY\_LENS\_OUTLINEDinstance-attribute**

DIRTY\_LENS\_OUTLINED: IconData

**buildDIRTY\_LENS\_ROUNDEDinstance-attribute**

DIRTY\_LENS\_ROUNDED: IconData

**buildDIRTY\_LENS\_SHARPinstance-attribute**

DIRTY\_LENS\_SHARP: IconData

**buildDISABLED\_BY\_DEFAULTinstance-attribute**

DISABLED\_BY\_DEFAULT: IconData

**buildDISABLED\_BY\_DEFAULT\_OUTLINEDinstance-attribute**

DISABLED\_BY\_DEFAULT\_OUTLINED: IconData

**buildDISABLED\_BY\_DEFAULT\_ROUNDEDinstance-attribute**

DISABLED\_BY\_DEFAULT\_ROUNDED: IconData

**buildDISABLED\_BY\_DEFAULT\_SHARPinstance-attribute**

DISABLED\_BY\_DEFAULT\_SHARP: IconData

**buildDISABLED\_VISIBLEinstance-attribute**

DISABLED\_VISIBLE: IconData

**buildDISABLED\_VISIBLE\_OUTLINEDinstance-attribute**

DISABLED\_VISIBLE\_OUTLINED: IconData

**buildDISABLED\_VISIBLE\_ROUNDEDinstance-attribute**

DISABLED\_VISIBLE\_ROUNDED: IconData

**buildDISABLED\_VISIBLE\_SHARPinstance-attribute**

DISABLED\_VISIBLE\_SHARP: IconData

**buildDISC\_FULLinstance-attribute**

DISC\_FULL: IconData

**buildDISC\_FULL\_OUTLINEDinstance-attribute**

DISC\_FULL\_OUTLINED: IconData

**buildDISC\_FULL\_ROUNDEDinstance-attribute**

DISC\_FULL\_ROUNDED: IconData

**buildDISC\_FULL\_SHARPinstance-attribute**

DISC\_FULL\_SHARP: IconData

**buildDISCORDinstance-attribute**

DISCORD: IconData

**buildDISCORD\_OUTLINEDinstance-attribute**

DISCORD\_OUTLINED: IconData

**buildDISCORD\_ROUNDEDinstance-attribute**

DISCORD\_ROUNDED: IconData

**buildDISCORD\_SHARPinstance-attribute**

DISCORD\_SHARP: IconData

**buildDISCOUNTinstance-attribute**

DISCOUNT: IconData

**buildDISCOUNT\_OUTLINEDinstance-attribute**

DISCOUNT\_OUTLINED: IconData

**buildDISCOUNT\_ROUNDEDinstance-attribute**

DISCOUNT\_ROUNDED: IconData

**buildDISCOUNT\_SHARPinstance-attribute**

DISCOUNT\_SHARP: IconData



**buildDISPLAY\_SETTINGSinstance-attribute**

DISPLAY\_SETTINGS: IconData

**buildDISPLAY\_SETTINGS\_OUTLINEDinstance-attribute**

DISPLAY\_SETTINGS\_OUTLINED: IconData

**buildDISPLAY\_SETTINGS\_ROUNDEDinstance-attribute**

DISPLAY\_SETTINGS\_ROUNDED: IconData

**buildDISPLAY\_SETTINGS\_SHARPinstance-attribute**

DISPLAY\_SETTINGS\_SHARP: IconData

**buildDIVERSITY\_1instance-attribute**

DIVERSITY\_1: IconData

**buildDIVERSITY\_1\_OUTLINEDinstance-attribute**

DIVERSITY\_1\_OUTLINED: IconData

**buildDIVERSITY\_1\_ROUNDEDinstance-attribute**

DIVERSITY\_1\_ROUNDED: IconData

**buildDIVERSITY\_1\_SHARPinstance-attribute**

DIVERSITY\_1\_SHARP: IconData

**buildDIVERSITY\_2instance-attribute**

DIVERSITY\_2: IconData

**buildDIVERSITY\_2\_OUTLINEDinstance-attribute**

DIVERSITY\_2\_OUTLINED: IconData

**buildDIVERSITY\_2\_ROUNDEDinstance-attribute**

DIVERSITY\_2\_ROUNDED: IconData

**buildDIVERSITY\_2\_SHARPinstance-attribute**

DIVERSITY\_2\_SHARP: IconData

**buildDIVERSITY\_3instance-attribute**

DIVERSITY\_3: IconData

**buildDIVERSITY\_3\_OUTLINEDinstance-attribute**

DIVERSITY\_3\_OUTLINED: IconData

**buildDIVERSITY\_3\_ROUNDEDinstance-attribute**

DIVERSITY\_3\_ROUNDED: IconData

**buildDIVERSITY\_3\_SHARPinstance-attribute**

DIVERSITY\_3\_SHARP: IconData

**buildDND\_FORWARDSLASHinstance-attribute**

DND\_FORWARDSLASH: IconData

**buildDND\_FORWARDSLASH\_OUTLINEDinstance-attribute**

DND\_FORWARDSLASH\_OUTLINED: IconData

**buildDND\_FORWARDSLASH\_ROUNDEDinstance-attribute**

DND\_FORWARDSLASH\_ROUNDED: IconData

**buildDND\_FORWARDSLASH\_SHARPinstance-attribute**

DND\_FORWARDSLASH\_SHARP: IconData

**buildDNSinstance-attribute**

DNS: IconData

**buildDNS\_OUTLINEDinstance-attribute**

DNS\_OUTLINED: IconData

**buildDNS\_ROUNDEDinstance-attribute**

DNS\_ROUNDED: IconData

**buildDNS\_SHARPinstance-attribute**

DNS\_SHARP: IconData

**buildDO\_DISTURBinstance-attribute**

DO\_DISTURB: IconData

**buildDO\_DISTURB\_ALTinstance-attribute**

DO\_DISTURB\_ALT: IconData

**buildDO\_DISTURB\_ALT\_OUTLINEDinstance-attribute**

DO\_DISTURB\_ALT\_OUTLINED: IconData

**buildDO\_DISTURB\_ALT\_ROUNDEDinstance-attribute**

DO\_DISTURB\_ALT\_ROUNDED: IconData

**buildDO\_DISTURB\_ALT\_SHARPinstance-attribute**

DO\_DISTURB\_ALT\_SHARP: IconData

**buildDO\_DISTURB\_OFFinstance-attribute**

DO\_DISTURB\_OFF: IconData

**buildDO\_DISTURB\_OFF\_OUTLINEDinstance-attribute**

DO\_DISTURB\_OFF\_OUTLINED: IconData

**buildDO\_DISTURB\_OFF\_ROUNDEDinstance-attribute**

DO\_DISTURB\_OFF\_ROUNDED: IconData

**buildDO\_DISTURB\_OFF\_SHARPinstance-attribute**

DO\_DISTURB\_OFF\_SHARP: IconData

**buildDO\_DISTURB\_ONinstance-attribute**

DO\_DISTURB\_ON: IconData

**buildDO\_DISTURB\_ON\_OUTLINEDinstance-attribute**

DO\_DISTURB\_ON\_OUTLINED: IconData

**buildDO\_DISTURB\_ON\_ROUNDEDinstance-attribute**

DO\_DISTURB\_ON\_ROUNDED: IconData

**buildDO\_DISTURB\_ON\_SHARPinstance-attribute**

DO\_DISTURB\_ON\_SHARP: IconData

**buildDO\_DISTURB\_OUTLINEDinstance-attribute**

DO\_DISTURB\_OUTLINED: IconData

**buildDO\_DISTURB\_ROUNDEDinstance-attribute**

DO\_DISTURB\_ROUNDED: IconData

**buildDO\_DISTURB\_SHARPinstance-attribute**

DO\_DISTURB\_SHARP: IconData

**buildDO\_NOT\_DISTURBinstance-attribute**

DO\_NOT\_DISTURB: IconData

**buildDO\_NOT\_DISTURB\_ALTinstance-attribute**

DO\_NOT\_DISTURB\_ALT: IconData

**buildDO\_NOT\_DISTURB\_ALT\_OUTLINEDinstance-attribute**

DO\_NOT\_DISTURB\_ALT\_OUTLINED: IconData

**buildDO\_NOT\_DISTURB\_ALT\_ROUNDEDinstance-attribute**

DO\_NOT\_DISTURB\_ALT\_ROUNDED: IconData

**buildDO\_NOT\_DISTURB\_ALT\_SHARPinstance-attribute**

DO\_NOT\_DISTURB\_ALT\_SHARP: IconData

**buildDO\_NOT\_DISTURB\_OFFinstance-attribute**

DO\_NOT\_DISTURB\_OFF: IconData

**buildDO\_NOT\_DISTURB\_OFF\_OUTLINEDinstance-attribute**

DO\_NOT\_DISTURB\_OFF\_OUTLINED: IconData

**buildDO\_NOT\_DISTURB\_OFF\_ROUNDEDinstance-attribute**

DO\_NOT\_DISTURB\_OFF\_ROUNDED: IconData

**buildDO\_NOT\_DISTURB\_OFF\_SHARPinstance-attribute**

DO\_NOT\_DISTURB\_OFF\_SHARP: IconData

**buildDO\_NOT\_DISTURB\_ONinstance-attribute**

DO\_NOT\_DISTURB\_ON: IconData

**buildDO\_NOT\_DISTURB\_ON\_OUTLINEDinstance-attribute**

DO\_NOT\_DISTURB\_ON\_OUTLINED: IconData

**buildDO\_NOT\_DISTURB\_ON\_ROUNDEDinstance-attribute**

DO\_NOT\_DISTURB\_ON\_ROUNDED: IconData

**buildDO\_NOT\_DISTURB\_ON\_SHARPinstance-attribute**

DO\_NOT\_DISTURB\_ON\_SHARP: IconData

**buildDO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCEinstance-attribute**

DO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE: IconData

**buildDO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_OUTLINEDinstance-attribute**

DO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_OUTLINED: IconData

**buildDO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_ROUNDEDinstance-attribute**

DO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_ROUNDED: IconData

**buildDO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_SHARPinstance-attribute**

DO\_NOT\_DISTURB\_ON\_TOTAL\_SILENCE\_SHARP: IconData

**buildDO\_NOT\_DISTURB\_OUTLINEDinstance-attribute**

DO\_NOT\_DISTURB\_OUTLINED: IconData

**buildDO\_NOT\_DISTURB\_ROUNDEDinstance-attribute**

DO\_NOT\_DISTURB\_ROUNDED: IconData

**buildDO\_NOT\_DISTURB\_SHARPinstance-attribute**

DO\_NOT\_DISTURB\_SHARP: IconData

**buildDO\_NOT\_STEPinstance-attribute**

DO\_NOT\_STEP: IconData

**buildDO\_NOT\_STEP\_OUTLINEDinstance-attribute**

DO\_NOT\_STEP\_OUTLINED: IconData

**buildDO\_NOT\_STEP\_ROUNDEDinstance-attribute**

DO\_NOT\_STEP\_ROUNDED: IconData

**buildDO\_NOT\_STEP\_SHARPinstance-attribute**

DO\_NOT\_STEP\_SHARP: IconData

**buildDO\_NOT\_TOUCHinstance-attribute**

DO\_NOT\_TOUCH: IconData

**buildDO\_NOT\_TOUCH\_OUTLINEDinstance-attribute**

DO\_NOT\_TOUCH\_OUTLINED: IconData

**buildDO\_NOT\_TOUCH\_ROUNDEDinstance-attribute**

DO\_NOT\_TOUCH\_ROUNDED: IconData

**buildDO\_NOT\_TOUCH\_SHARPinstance-attribute**

DO\_NOT\_TOUCH\_SHARP: IconData

**buildDOCKinstance-attribute**

DOCK: IconData

**buildDOCK\_OUTLINEDinstance-attribute**

DOCK\_OUTLINED: IconData

**buildDOCK\_ROUNDEDinstance-attribute**

DOCK\_ROUNDED: IconData

**buildDOCK\_SHARPinstance-attribute**

DOCK\_SHARP: IconData

**buildDOCUMENT\_SCANNERinstance-attribute**

DOCUMENT\_SCANNER: IconData

**buildDOCUMENT\_SCANNER\_OUTLINEDinstance-attribute**

DOCUMENT\_SCANNER\_OUTLINED: IconData

**buildDOCUMENT\_SCANNER\_ROUNDEDinstance-attribute**

DOCUMENT\_SCANNER\_ROUNDED: IconData

**buildDOCUMENT\_SCANNER\_SHARPinstance-attribute**

DOCUMENT\_SCANNER\_SHARP: IconData

**buildDOMAINinstance-attribute**

DOMAIN: IconData

**buildDOMAIN\_ADDinstance-attribute**

DOMAIN\_ADD: IconData

**buildDOMAIN\_ADD\_OUTLINEDinstance-attribute**

DOMAIN\_ADD\_OUTLINED: IconData

**buildDOMAIN\_ADD\_ROUNDEDinstance-attribute**

DOMAIN\_ADD\_ROUNDED: IconData

**buildDOMAIN\_ADD\_SHARPinstance-attribute**

DOMAIN\_ADD\_SHARP: IconData

**buildDOMAIN\_DISABLEDinstance-attribute**

DOMAIN\_DISABLED: IconData

**buildDOMAIN\_DISABLED\_OUTLINEDinstance-attribute**

DOMAIN\_DISABLED\_OUTLINED: IconData

**buildDOMAIN\_DISABLED\_ROUNDEDinstance-attribute**

DOMAIN\_DISABLED\_ROUNDED: IconData

**buildDOMAIN\_DISABLED\_SHARPinstance-attribute**

DOMAIN\_DISABLED\_SHARP: IconData

**buildDOMAIN\_OUTLINEDinstance-attribute**

DOMAIN\_OUTLINED: IconData

**buildDOMAIN\_ROUNDEDinstance-attribute**

DOMAIN\_ROUNDED: IconData

**buildDOMAIN\_SHARPinstance-attribute**

DOMAIN\_SHARP: IconData

**buildDOMAIN\_VERIFICATIONinstance-attribute**

DOMAIN\_VERIFICATION: IconData

**buildDOMAIN\_VERIFICATION\_OUTLINEDinstance-attribute**

DOMAIN\_VERIFICATION\_OUTLINED: IconData

**buildDOMAIN\_VERIFICATION\_ROUNDEDinstance-attribute**

DOMAIN\_VERIFICATION\_ROUNDED: IconData

**buildDOMAIN\_VERIFICATION\_SHARPinstance-attribute**

DOMAIN\_VERIFICATION\_SHARP: IconData

**buildDONEinstance-attribute**

DONE: IconData

**buildDONE\_ALLinstance-attribute**

DONE\_ALL: IconData

**buildDONE\_ALL\_OUTLINEDinstance-attribute**

DONE\_ALL\_OUTLINED: IconData

**buildDONE\_ALL\_ROUNDEDinstance-attribute**

DONE\_ALL\_ROUNDED: IconData

**buildDONE\_ALL\_SHARPinstance-attribute**

DONE\_ALL\_SHARP: IconData

**buildDONE\_OUTLINEinstance-attribute**

DONE\_OUTLINE: IconData

**buildDONE\_OUTLINE\_OUTLINEDinstance-attribute**

DONE\_OUTLINE\_OUTLINED: IconData

**buildDONE\_OUTLINE\_ROUNDEDinstance-attribute**

DONE\_OUTLINE\_ROUNDED: IconData

**buildDONE\_OUTLINE\_SHARPinstance-attribute**

DONE\_OUTLINE\_SHARP: IconData

**buildDONE\_OUTLINEDinstance-attribute**

DONE\_OUTLINED: IconData

**buildDONE\_ROUNDEDinstance-attribute**

DONE\_ROUNDED: IconData

**buildDONE\_SHARPinstance-attribute**

DONE\_SHARP: IconData

**buildDONUT\_LARGEinstance-attribute**

DONUT\_LARGE: IconData

**buildDONUT\_LARGE\_OUTLINEDinstance-attribute**

DONUT\_LARGE\_OUTLINED: IconData

**buildDONUT\_LARGE\_ROUNDEDinstance-attribute**

DONUT\_LARGE\_ROUNDED: IconData

**buildDONUT\_LARGE\_SHARPinstance-attribute**

DONUT\_LARGE\_SHARP: IconData

**buildDONUT\_SMALLinstance-attribute**

DONUT\_SMALL: IconData

**buildDONUT\_SMALL\_OUTLINEDinstance-attribute**

DONUT\_SMALL\_OUTLINED: IconData

**buildDONUT\_SMALL\_ROUNDEDinstance-attribute**

DONUT\_SMALL\_ROUNDED: IconData

**buildDONUT\_SMALL\_SHARPinstance-attribute**

DONUT\_SMALL\_SHARP: IconData

**buildDOOR\_BACK\_DOORinstance-attribute**

DOOR\_BACK\_DOOR: IconData

**buildDOOR\_BACK\_DOOR\_OUTLINEDinstance-attribute**

DOOR\_BACK\_DOOR\_OUTLINED: IconData

**buildDOOR\_BACK\_DOOR\_ROUNDEDinstance-attribute**

DOOR\_BACK\_DOOR\_ROUNDED: IconData

**buildDOOR\_BACK\_DOOR\_SHARPinstance-attribute**

DOOR\_BACK\_DOOR\_SHARP: IconData

**buildDOOR\_FRONT\_DOORinstance-attribute**

DOOR\_FRONT\_DOOR: IconData

**buildDOOR\_FRONT\_DOOR\_OUTLINEDinstance-attribute**

DOOR\_FRONT\_DOOR\_OUTLINED: IconData

**buildDOOR\_FRONT\_DOOR\_ROUNDEDinstance-attribute**

DOOR\_FRONT\_DOOR\_ROUNDED: IconData

**buildDOOR\_FRONT\_DOOR\_SHARPinstance-attribute**

DOOR\_FRONT\_DOOR\_SHARP: IconData

**buildDOOR\_SLIDINGinstance-attribute**

DOOR\_SLIDING: IconData

**buildDOOR\_SLIDING\_OUTLINEDinstance-attribute**

DOOR\_SLIDING\_OUTLINED: IconData

**buildDOOR\_SLIDING\_ROUNDEDinstance-attribute**

DOOR\_SLIDING\_ROUNDED: IconData

**buildDOOR\_SLIDING\_SHARPinstance-attribute**

DOOR\_SLIDING\_SHARP: IconData

**buildDOORBELLinstance-attribute**

DOORBELL: IconData

**buildDOORBELL\_OUTLINEDinstance-attribute**

DOORBELL\_OUTLINED: IconData

**buildDOORBELL\_ROUNDEDinstance-attribute**

DOORBELL\_ROUNDED: IconData

**buildDOORBELL\_SHARPinstance-attribute**

DOORBELL\_SHARP: IconData

**buildDOUBLE\_ARROWinstance-attribute**

DOUBLE\_ARROW: IconData

**buildDOUBLE\_ARROW\_OUTLINEDinstance-attribute**

DOUBLE\_ARROW\_OUTLINED: IconData

**buildDOUBLE\_ARROW\_ROUNDEDinstance-attribute**

DOUBLE\_ARROW\_ROUNDED: IconData

**buildDOUBLE\_ARROW\_SHARPinstance-attribute**

DOUBLE\_ARROW\_SHARP: IconData

**buildDOWNHILL\_SKIINGinstance-attribute**

DOWNHILL\_SKIING: IconData

**buildDOWNHILL\_SKIING\_OUTLINEDinstance-attribute**

DOWNHILL\_SKIING\_OUTLINED: IconData

**buildDOWNHILL\_SKIING\_ROUNDEDinstance-attribute**

DOWNHILL\_SKIING\_ROUNDED: IconData

**buildDOWNHILL\_SKIING\_SHARPinstance-attribute**

DOWNHILL\_SKIING\_SHARP: IconData

**buildDOWNLOADinstance-attribute**

DOWNLOAD: IconData

**buildDOWNLOAD\_DONEinstance-attribute**

DOWNLOAD\_DONE: IconData

**buildDOWNLOAD\_DONE\_OUTLINEDinstance-attribute**

DOWNLOAD\_DONE\_OUTLINED: IconData

**buildDOWNLOAD\_DONE\_ROUNDEDinstance-attribute**

DOWNLOAD\_DONE\_ROUNDED: IconData

**buildDOWNLOAD\_DONE\_SHARPinstance-attribute**

DOWNLOAD\_DONE\_SHARP: IconData

**buildDOWNLOAD\_FOR\_OFFLINEinstance-attribute**

DOWNLOAD\_FOR\_OFFLINE: IconData

**buildDOWNLOAD\_FOR\_OFFLINE\_OUTLINEDinstance-attribute**

DOWNLOAD\_FOR\_OFFLINE\_OUTLINED: IconData

**buildDOWNLOAD\_FOR\_OFFLINE\_ROUNDEDinstance-attribute**

DOWNLOAD\_FOR\_OFFLINE\_ROUNDED: IconData

**buildDOWNLOAD\_FOR\_OFFLINE\_SHARPinstance-attribute**

DOWNLOAD\_FOR\_OFFLINE\_SHARP: IconData

**buildDOWNLOAD\_OUTLINEDinstance-attribute**

DOWNLOAD\_OUTLINED: IconData

**buildDOWNLOAD\_ROUNDEDinstance-attribute**

DOWNLOAD\_ROUNDED: IconData

**buildDOWNLOAD\_SHARPinstance-attribute**

DOWNLOAD\_SHARP: IconData

**buildDOWNLOADINGinstance-attribute**

DOWNLOADING: IconData

**buildDOWNLOADING\_OUTLINEDinstance-attribute**

DOWNLOADING\_OUTLINED: IconData

**buildDOWNLOADING\_ROUNDEDinstance-attribute**

DOWNLOADING\_ROUNDED: IconData

**buildDOWNLOADING\_SHARPinstance-attribute**

DOWNLOADING\_SHARP: IconData

**buildDRAFTSinstance-attribute**

DRAFTS: IconData

**buildDRAFTS\_OUTLINEDinstance-attribute**

DRAFTS\_OUTLINED: IconData

**buildDRAFTS\_ROUNDED**instance-attribute

DRAFTS\_ROUNDED: IconData

**buildDRAFTS\_SHARP**instance-attribute

DRAFTS\_SHARP: IconData

**buildDRAG\_HANDLE**instance-attribute

DRAG\_HANDLE: IconData

**buildDRAG\_HANDLE\_OUTLINED**instance-attribute

DRAG\_HANDLE\_OUTLINED: IconData

**buildDRAG\_HANDLE\_ROUNDED**instance-attribute

DRAG\_HANDLE\_ROUNDED: IconData

**buildDRAG\_HANDLE\_SHARP**instance-attribute

DRAG\_HANDLE\_SHARP: IconData

**buildDRAG\_INDICATOR**instance-attribute

DRAG\_INDICATOR: IconData

**buildDRAG\_INDICATOR\_OUTLINED**instance-attribute

DRAG\_INDICATOR\_OUTLINED: IconData

**buildDRAG\_INDICATOR\_ROUNDED**instance-attribute

DRAG\_INDICATOR\_ROUNDED: IconData

**buildDRAG\_INDICATOR\_SHARP**instance-attribute□□

DRAG\_INDICATOR\_SHARP: IconData

**buildDRAW**instance-attribute

DRAW: IconData

**buildDRAW\_OUTLINED**instance-attribute

DRAW\_OUTLINED: IconData

**buildDRAW\_ROUNDED**instance-attribute

DRAW\_ROUNDED: IconData

**buildDRAW\_SHARP**instance-attribute

DRAW\_SHARP: IconData

**buildDRIVE\_ETA**instance-attribute

DRIVE\_ETA: IconData

**buildDRIVE\_ETA\_OUTLINED**instance-attribute

DRIVE\_ETA\_OUTLINED: IconData

**buildDRIVE\_ETA\_ROUNDED**instance-attribute

DRIVE\_ETA\_ROUNDED: IconData

**buildDRIVE\_ETA\_SHARP**instance-attribute

DRIVE\_ETA\_SHARP: IconData

**buildDRIVE\_FILE\_MOVE**instance-attribute

DRIVE\_FILE\_MOVE: IconData

**buildDRIVE\_FILE\_MOVE\_OUTLINE**instance-attribute

DRIVE\_FILE\_MOVE\_OUTLINE: IconData

**buildDRIVE\_FILE\_MOVE\_OUTLINED**instance-attribute

DRIVE\_FILE\_MOVE\_OUTLINED: IconData

**buildDRIVE\_FILE\_MOVE\_ROUNDED**instance-attribute

DRIVE\_FILE\_MOVE\_ROUNDED: IconData



**buildDRIVE\_FILE\_MOVE\_RTLOnstance-attribute**

DRIVE\_FILE\_MOVE\_RTLOnstance: IconData

**buildDRIVE\_FILE\_MOVE\_RTLOUTLINEDInstance-attribute**

DRIVE\_FILE\_MOVE\_RTLOUTLINEDInstance: IconData

**buildDRIVE\_FILE\_MOVE\_RTLOUNDEDInstance-attribute**

DRIVE\_FILE\_MOVE\_RTLOUNDEDInstance: IconData

**buildDRIVE\_FILE\_MOVE\_RTLSHARPInstance-attribute**

DRIVE\_FILE\_MOVE\_RTLSHARPInstance: IconData

**buildDRIVE\_FILE\_MOVE\_SHARPInstance-attribute**

DRIVE\_FILE\_MOVE\_SHARPInstance: IconData

**buildDRIVE\_FILE\_RENAME\_OUTLINEInstance-attribute**

DRIVE\_FILE\_RENAME\_OUTLINEInstance: IconData

**buildDRIVE\_FILE\_RENAME\_OUTLINE\_OUTLINEDInstance-attribute**

DRIVE\_FILE\_RENAME\_OUTLINE\_OUTLINEDInstance: IconData

**buildDRIVE\_FILE\_RENAME\_OUTLINE\_ROUNDEDInstance-attribute**

DRIVE\_FILE\_RENAME\_OUTLINE\_ROUNDEDInstance: IconData

**buildDRIVE\_FILE\_RENAME\_OUTLINE\_SHARPInstance-attribute**

DRIVE\_FILE\_RENAME\_OUTLINE\_SHARPInstance: IconData

**buildDRIVE\_FOLDER\_UPLOADInstance-attribute**

DRIVE\_FOLDER\_UPLOADInstance: IconData

**buildDRIVE\_FOLDER\_UPLOAD\_OUTLINEDInstance-attribute**

DRIVE\_FOLDER\_UPLOAD\_OUTLINEDInstance: IconData

**buildDRIVE\_FOLDER\_UPLOAD\_ROUNDEDInstance-attribute**

DRIVE\_FOLDER\_UPLOAD\_ROUNDEDInstance: IconData

**buildDRIVE\_FOLDER\_UPLOAD\_SHARPInstance-attribute**

DRIVE\_FOLDER\_UPLOAD\_SHARPInstance: IconData

**buildDRYInstance-attribute**

DRYInstance: IconData

**buildDRY\_CLEANINGInstance-attribute**

DRY\_CLEANINGInstance: IconData

**buildDRY\_CLEANING\_OUTLINEDInstance-attribute**

DRY\_CLEANING\_OUTLINEDInstance: IconData

**buildDRY\_CLEANING\_ROUNDEDInstance-attribute**

DRY\_CLEANING\_ROUNDEDInstance: IconData

**buildDRY\_CLEANING\_SHARPInstance-attribute**

DRY\_CLEANING\_SHARPInstance: IconData

**buildDRY\_OUTLINEDInstance-attribute**

DRY\_OUTLINEDInstance: IconData

**buildDRY\_ROUNDEDInstance-attribute**

DRY\_ROUNDEDInstance: IconData

**buildDRY\_SHARPInstance-attribute**

DRY\_SHARPInstance: IconData

**buildDUOInstance-attribute**

DUOInstance: IconData

**buildDUO\_OUTLINEDinstance-attribute**

DUO\_OUTLINED: IconData

**buildDUO\_ROUNDEDinstance-attribute**

DUO\_ROUNDED: IconData

**buildDUO\_SHARPinstance-attribute**

DUO\_SHARP: IconData

**buildDVRinstance-attribute**

DVR: IconData

**buildDVR\_OUTLINEDinstance-attribute**

DVR\_OUTLINED: IconData

**buildDVR\_ROUNDEDinstance-attribute**

DVR\_ROUNDED: IconData

**buildDVR\_SHARPinstance-attribute**

DVR\_SHARP: IconData

**buildDYNAMIC\_FEEDinstance-attribute**

DYNAMIC\_FEED: IconData

**buildDYNAMIC\_FEED\_OUTLINEDinstance-attribute**

DYNAMIC\_FEED\_OUTLINED: IconData

**buildDYNAMIC\_FEED\_ROUNDEDinstance-attribute**

DYNAMIC\_FEED\_ROUNDED: IconData

**buildDYNAMIC\_FEED\_SHARPinstance-attribute**

DYNAMIC\_FEED\_SHARP: IconData

**buildDYNAMIC\_FORMinstance-attribute**

DYNAMIC\_FORM: IconData

**buildDYNAMIC\_FORM\_OUTLINEDinstance-attribute**

DYNAMIC\_FORM\_OUTLINED: IconData

**buildDYNAMIC\_FORM\_ROUNDEDinstance-attribute**

DYNAMIC\_FORM\_ROUNDED: IconData

**buildDYNAMIC\_FORM\_SHARPinstance-attribute**

DYNAMIC\_FORM\_SHARP: IconData

**buildE\_MOBILEDATAinstance-attribute**

E\_MOBILEDATA: IconData

**buildE\_MOBILEDATA\_OUTLINEDinstance-attribute**

E\_MOBILEDATA\_OUTLINED: IconData

**buildE\_MOBILEDATA\_ROUNDEDinstance-attribute**

E\_MOBILEDATA\_ROUNDED: IconData

**buildE\_MOBILEDATA\_SHARPinstance-attribute**

E\_MOBILEDATA\_SHARP: IconData

**buildEARBUDSinstance-attribute**

EARBUDS: IconData

**buildEARBUDS\_BATTERYinstance-attribute**

EARBUDS\_BATTERY: IconData

**buildEARBUDS\_BATTERY\_OUTLINEDinstance-attribute**

EARBUDS\_BATTERY\_OUTLINED: IconData

**buildEARBUDS\_BATTERY\_ROUNDEDinstance-attribute**

EARBUDS\_BATTERY\_ROUNDED: IconData

**buildEARBUDS\_BATTERY\_SHARPinstance-attribute**

EARBUDS\_BATTERY\_SHARP: IconData

**buildEARBUDS\_OUTLINEDinstance-attribute**

EARBUDS\_OUTLINED: IconData

**buildEARBUDS\_ROUNDEDinstance-attribute**

EARBUDS\_ROUNDED: IconData

**buildEARBUDS\_SHARPinstance-attribute**

EARBUDS\_SHARP: IconData

**buildEASTinstance-attribute**

EAST: IconData

**buildEAST\_OUTLINEDinstance-attribute**

EAST\_OUTLINED: IconData

**buildEAST\_ROUNDEDinstance-attribute**

EAST\_ROUNDED: IconData

**buildEAST\_SHARPinstance-attribute**

EAST\_SHARP: IconData

**buildECOinstance-attribute**

ECO: IconData

**buildECO\_OUTLINEDinstance-attribute**

ECO\_OUTLINED: IconData

**buildECO\_ROUNDEDinstance-attribute**

ECO\_ROUNDED: IconData

**buildECO\_SHARPinstance-attribute**

ECO\_SHARP: IconData

**buildEDGESENSOR\_HIGHinstance-attribute**

EDGESENSOR\_HIGH: IconData

**buildEDGESENSOR\_HIGH\_OUTLINEDinstance-attribute**

EDGESENSOR\_HIGH\_OUTLINED: IconData

**buildEDGESENSOR\_HIGH\_ROUNDEDinstance-attribute**

EDGESENSOR\_HIGH\_ROUNDED: IconData

**buildEDGESENSOR\_HIGH\_SHARPinstance-attribute**

EDGESENSOR\_HIGH\_SHARP: IconData

**buildEDGESENSOR\_LOWinstance-attribute**

EDGESENSOR\_LOW: IconData

**buildEDGESENSOR\_LOW\_OUTLINEDinstance-attribute**

EDGESENSOR\_LOW\_OUTLINED: IconData

**buildEDGESENSOR\_LOW\_ROUNDEDinstance-attribute**

EDGESENSOR\_LOW\_ROUNDED: IconData

**buildEDGESENSOR\_LOW\_SHARPinstance-attribute**

EDGESENSOR\_LOW\_SHARP: IconData

**buildEDITinstance-attribute**

EDIT: IconData

**buildEDIT\_ATTRIBUTESinstance-attribute**

EDIT\_ATTRIBUTES: IconData

**buildEDIT\_ATTRIBUTES\_OUTLINEDinstance-attribute**

EDIT\_ATTRIBUTES\_OUTLINED: IconData

**buildEDIT\_ATTRIBUTES\_ROUNDEDinstance-attribute**

EDIT\_ATTRIBUTES\_ROUNDED: IconData

**buildEDIT\_ATTRIBUTES\_SHARPinstance-attribute**

EDIT\_ATTRIBUTES\_SHARP: IconData

**buildEDIT\_CALENDARinstance-attribute**

EDIT\_CALENDAR: IconData

**buildEDIT\_CALENDAR\_OUTLINEDinstance-attribute**

EDIT\_CALENDAR\_OUTLINED: IconData

**buildEDIT\_CALENDAR\_ROUNDEDinstance-attribute**

EDIT\_CALENDAR\_ROUNDED: IconData

**buildEDIT\_CALENDAR\_SHARPinstance-attribute**

EDIT\_CALENDAR\_SHARP: IconData

**buildEDIT\_DOCUMENTinstance-attribute**

EDIT\_DOCUMENT: IconData

**buildEDIT\_LOCATIONinstance-attribute**

EDIT\_LOCATION: IconData

**buildEDIT\_LOCATION\_ALTinstance-attribute**

EDIT\_LOCATION\_ALT: IconData

**buildEDIT\_LOCATION\_ALT\_OUTLINEDinstance-attribute**

EDIT\_LOCATION\_ALT\_OUTLINED: IconData

**buildEDIT\_LOCATION\_ALT\_ROUNDEDinstance-attribute**

EDIT\_LOCATION\_ALT\_ROUNDED: IconData

**buildEDIT\_LOCATION\_ALT\_SHARPinstance-attribute**

EDIT\_LOCATION\_ALT\_SHARP: IconData

**buildEDIT\_LOCATION\_OUTLINEDinstance-attribute**

EDIT\_LOCATION\_OUTLINED: IconData

**buildEDIT\_LOCATION\_ROUNDEDinstance-attribute**

EDIT\_LOCATION\_ROUNDED: IconData

**buildEDIT\_LOCATION\_SHARPinstance-attribute**

EDIT\_LOCATION\_SHARP: IconData

**buildEDIT\_NOTEinstance-attribute**

EDIT\_NOTE: IconData

**buildEDIT\_NOTE\_OUTLINEDinstance-attribute**

EDIT\_NOTE\_OUTLINED: IconData

**buildEDIT\_NOTE\_ROUNDEDinstance-attribute**

EDIT\_NOTE\_ROUNDED: IconData

**buildEDIT\_NOTE\_SHARPinstance-attribute**

EDIT\_NOTE\_SHARP: IconData

**buildEDIT\_NOTIFICATIONSinstance-attribute**

EDIT\_NOTIFICATIONS: IconData

**buildEDIT\_NOTIFICATIONS\_OUTLINED**Instance-attribute

EDIT\_NOTIFICATIONS\_OUTLINED: IconData

**buildEDIT\_NOTIFICATIONS\_ROUNDED**Instance-attribute

EDIT\_NOTIFICATIONS\_ROUNDED: IconData

**buildEDIT\_NOTIFICATIONS\_SHARP**Instance-attribute

EDIT\_NOTIFICATIONS\_SHARP: IconData

**buildEDIT\_OFF**Instance-attribute

EDIT\_OFF: IconData

**buildEDIT\_OFF\_OUTLINED**Instance-attribute

EDIT\_OFF\_OUTLINED: IconData

**buildEDIT\_OFF\_ROUNDED**Instance-attribute

EDIT\_OFF\_ROUNDED: IconData

**buildEDIT\_OFF\_SHARP**Instance-attribute

EDIT\_OFF\_SHARP: IconData

**buildEDIT\_OUTLINED**Instance-attribute

EDIT\_OUTLINED: IconData

**buildEDIT\_ROAD**Instance-attribute

EDIT\_ROAD: IconData

**buildEDIT\_ROAD\_OUTLINED**Instance-attribute

EDIT\_ROAD\_OUTLINED: IconData

**buildEDIT\_ROAD\_ROUNDED**Instance-attribute

EDIT\_ROAD\_ROUNDED: IconData

**buildEDIT\_ROAD\_SHARP**Instance-attribute

EDIT\_ROAD\_SHARP: IconData

**buildEDIT\_ROUNDED**Instance-attribute

EDIT\_ROUNDED: IconData

**buildEDIT\_SHARP**Instance-attribute

EDIT\_SHARP: IconData

**buildEDIT\_SQUARE**Instance-attribute

EDIT\_SQUARE: IconData

**buildEGG**Instance-attribute

EGG: IconData

**buildEGG\_ALT**Instance-attribute

EGG\_ALT: IconData

**buildEGG\_ALT\_OUTLINED**Instance-attribute

EGG\_ALT\_OUTLINED: IconData

**buildEGG\_ALT\_ROUNDED**Instance-attribute

EGG\_ALT\_ROUNDED: IconData

**buildEGG\_ALT\_SHARP**Instance-attribute

EGG\_ALT\_SHARP: IconData

**buildEGG\_OUTLINED**Instance-attribute

EGG\_OUTLINED: IconData

**buildEGG\_ROUNDED**Instance-attribute

EGG\_ROUNDED: IconData

**buildEGG\_SHARPinstance-attribute**

EGG\_SHARP: IconData

**buildEIGHT\_Kinstance-attribute**

EIGHT\_K: IconData

**buildEIGHT\_K\_OUTLINEDinstance-attribute**

EIGHT\_K\_OUTLINED: IconData

**buildEIGHT\_K\_PLUSinstance-attribute**

EIGHT\_K\_PLUS: IconData

**buildEIGHT\_K\_PLUS\_OUTLINEDinstance-attribute**

EIGHT\_K\_PLUS\_OUTLINED: IconData

**buildEIGHT\_K\_PLUS\_ROUNDEDinstance-attribute**

EIGHT\_K\_PLUS\_ROUNDED: IconData

**buildEIGHT\_K\_PLUS\_SHARPinstance-attribute**

EIGHT\_K\_PLUS\_SHARP: IconData

**buildEIGHT\_K\_ROUNDEDinstance-attribute**

EIGHT\_K\_ROUNDED: IconData

**buildEIGHT\_K\_SHARPinstance-attribute**

EIGHT\_K\_SHARP: IconData

**buildEIGHT\_MPinstance-attribute**

EIGHT\_MP: IconData

**buildEIGHT\_MP\_OUTLINEDinstance-attribute**

EIGHT\_MP\_OUTLINED: IconData

**buildEIGHT\_MP\_ROUNDEDinstance-attribute**

EIGHT\_MP\_ROUNDED: IconData

**buildEIGHT\_MP\_SHARPinstance-attribute**

EIGHT\_MP\_SHARP: IconData

**buildEIGHTEEN\_MPinstance-attribute**

EIGHTEEN\_MP: IconData

**buildEIGHTEEN\_MP\_OUTLINEDinstance-attribute**

EIGHTEEN\_MP\_OUTLINED: IconData

**buildEIGHTEEN\_MP\_ROUNDEDinstance-attribute**

EIGHTEEN\_MP\_ROUNDED: IconData

**buildEIGHTEEN\_MP\_SHARPinstance-attribute**

EIGHTEEN\_MP\_SHARP: IconData

**buildEIGHTEEN\_UP\_RATINGinstance-attribute**

EIGHTEEN\_UP\_RATING: IconData

**buildEIGHTEEN\_UP\_RATING\_OUTLINEDinstance-attribute**

EIGHTEEN\_UP\_RATING\_OUTLINED: IconData

**buildEIGHTEEN\_UP\_RATING\_ROUNDEDinstance-attribute**

EIGHTEEN\_UP\_RATING\_ROUNDED: IconData

**buildEIGHTEEN\_UP\_RATING\_SHARPinstance-attribute**

EIGHTEEN\_UP\_RATING\_SHARP: IconData

**buildEJECTinstance-attribute**

EJECT: IconData

**buildEJECT\_OUTLINEDinstance-attribute**

EJECT\_OUTLINED: IconData

**buildEJECT\_ROUNDEDinstance-attribute**

EJECT\_ROUNDED: IconData

**buildEJECT\_SHARPinstance-attribute**

EJECT\_SHARP: IconData

**buildELDERLYinstance-attribute**

ELDERLY: IconData

**buildELDERLY\_OUTLINEDinstance-attribute**

ELDERLY\_OUTLINED: IconData

**buildELDERLY\_ROUNDEDinstance-attribute**

ELDERLY\_ROUNDED: IconData

**buildELDERLY\_SHARPinstance-attribute**

ELDERLY\_SHARP: IconData

**buildELDERLY\_WOMANinstance-attribute**

ELDERLY\_WOMAN: IconData

**buildELDERLY\_WOMAN\_OUTLINEDinstance-attribute**

ELDERLY\_WOMAN\_OUTLINED: IconData

**buildELDERLY\_WOMAN\_ROUNDEDinstance-attribute**

ELDERLY\_WOMAN\_ROUNDED: IconData

**buildELDERLY\_WOMAN\_SHARPinstance-attribute**

ELDERLY\_WOMAN\_SHARP: IconData

**buildELECTRIC\_BIKEinstance-attribute**

ELECTRIC\_BIKE: IconData

**buildELECTRIC\_BIKE\_OUTLINEDinstance-attribute**

ELECTRIC\_BIKE\_OUTLINED: IconData

**buildELECTRIC\_BIKE\_ROUNDEDinstance-attribute**

ELECTRIC\_BIKE\_ROUNDED: IconData

**buildELECTRIC\_BIKE\_SHARPinstance-attribute**

ELECTRIC\_BIKE\_SHARP: IconData

**buildELECTRIC\_BOLTinstance-attribute**

ELECTRIC\_BOLT: IconData

**buildELECTRIC\_BOLT\_OUTLINEDinstance-attribute**

ELECTRIC\_BOLT\_OUTLINED: IconData

**buildELECTRIC\_BOLT\_ROUNDEDinstance-attribute**

ELECTRIC\_BOLT\_ROUNDED: IconData

**buildELECTRIC\_BOLT\_SHARPinstance-attribute**

ELECTRIC\_BOLT\_SHARP: IconData

**buildELECTRIC\_CARinstance-attribute**

ELECTRIC\_CAR: IconData

**buildELECTRIC\_CAR\_OUTLINEDinstance-attribute**

ELECTRIC\_CAR\_OUTLINED: IconData

**buildELECTRIC\_CAR\_ROUNDEDinstance-attribute**

ELECTRIC\_CAR\_ROUNDED: IconData

**buildELECTRIC\_CAR\_SHARPinstance-attribute**

ELECTRIC\_CAR\_SHARP: IconData

**buildELECTRIC\_METERinstance-attribute**

ELECTRIC\_METER: IconData

**buildELECTRIC\_METER\_OUTLINEDinstance-attribute**

ELECTRIC\_METER\_OUTLINED: IconData

**buildELECTRIC\_METER\_ROUNDEDinstance-attribute**

ELECTRIC\_METER\_ROUNDED: IconData

**buildELECTRIC\_METER\_SHARPinstance-attribute**

ELECTRIC\_METER\_SHARP: IconData

**buildELECTRIC\_MOPEDinstance-attribute**

ELECTRIC\_MOPED: IconData

**buildELECTRIC\_MOPED\_OUTLINEDinstance-attribute**

ELECTRIC\_MOPED\_OUTLINED: IconData

**buildELECTRIC\_MOPED\_ROUNDEDinstance-attribute**

ELECTRIC\_MOPED\_ROUNDED: IconData

**buildELECTRIC\_MOPED\_SHARPinstance-attribute**

ELECTRIC\_MOPED\_SHARP: IconData

**buildELECTRIC\_RICKSHAWinstance-attribute**

ELECTRIC\_RICKSHAW: IconData

**buildELECTRIC\_RICKSHAW\_OUTLINEDinstance-attribute**

ELECTRIC\_RICKSHAW\_OUTLINED: IconData

**buildELECTRIC\_RICKSHAW\_ROUNDEDinstance-attribute**

ELECTRIC\_RICKSHAW\_ROUNDED: IconData

**buildELECTRIC\_RICKSHAW\_SHARPinstance-attribute**

ELECTRIC\_RICKSHAW\_SHARP: IconData

**buildELECTRIC\_SCOOTERinstance-attribute**

ELECTRIC\_SCOOTER: IconData

**buildELECTRIC\_SCOOTER\_OUTLINEDinstance-attribute**

ELECTRIC\_SCOOTER\_OUTLINED: IconData

**buildELECTRIC\_SCOOTER\_ROUNDEDinstance-attribute**

ELECTRIC\_SCOOTER\_ROUNDED: IconData

**buildELECTRIC\_SCOOTER\_SHARPinstance-attribute**

ELECTRIC\_SCOOTER\_SHARP: IconData

**buildELECTRICAL\_SERVICESinstance-attribute**

ELECTRICAL\_SERVICES: IconData

**buildELECTRICAL\_SERVICES\_OUTLINEDinstance-attribute**

ELECTRICAL\_SERVICES\_OUTLINED: IconData

**buildELECTRICAL\_SERVICES\_ROUNDEDinstance-attribute**

ELECTRICAL\_SERVICES\_ROUNDED: IconData

**buildELECTRICAL\_SERVICES\_SHARPinstance-attribute**

ELECTRICAL\_SERVICES\_SHARP: IconData

**buildELEVATORinstance-attribute**

ELEVATOR: IconData



**buildELEVATOR\_OUTLINED**instance-attribute  
ELEVATOR\_OUTLINED: IconData

**buildELEVATOR\_ROUNDED**instance-attribute  
ELEVATOR\_ROUNDED: IconData

**buildELEVATOR\_SHARP**instance-attribute  
ELEVATOR\_SHARP: IconData

**buildELEVEN\_MP**instance-attribute  
ELEVEN\_MP: IconData

**buildELEVEN\_MP\_OUTLINED**instance-attribute  
ELEVEN\_MP\_OUTLINED: IconData

**buildELEVEN\_MP\_ROUNDED**instance-attribute  
ELEVEN\_MP\_ROUNDED: IconData

**buildELEVEN\_MP\_SHARP**instance-attribute  
ELEVEN\_MP\_SHARP: IconData

**buildEMAIL**instance-attribute  
EMAIL: IconData

**buildEMAIL\_OUTLINED**instance-attribute  
EMAIL\_OUTLINED: IconData

**buildEMAIL\_ROUNDED**instance-attribute  
EMAIL\_ROUNDED: IconData

**buildEMAIL\_SHARP**instance-attribute  
EMAIL\_SHARP: IconData

**buildEMERGENCY**instance-attribute  
EMERGENCY: IconData

**buildEMERGENCY\_OUTLINED**instance-attribute  
EMERGENCY\_OUTLINED: IconData

**buildEMERGENCY\_RECORDING**instance-attribute  
EMERGENCY\_RECORDING: IconData

**buildEMERGENCY\_RECORDING\_OUTLINED**instance-attribute  
EMERGENCY\_RECORDING\_OUTLINED: IconData

**buildEMERGENCY\_RECORDING\_ROUNDED**instance-attribute  
EMERGENCY\_RECORDING\_ROUNDED: IconData

**buildEMERGENCY\_RECORDING\_SHARP**instance-attribute  
EMERGENCY\_RECORDING\_SHARP: IconData

**buildEMERGENCY\_ROUNDED**instance-attribute  
EMERGENCY\_ROUNDED: IconData

**buildEMERGENCY\_SHARE**instance-attribute  
EMERGENCY\_SHARE: IconData

**buildEMERGENCY\_SHARE\_OUTLINED**instance-attribute  
EMERGENCY\_SHARE\_OUTLINED: IconData

**buildEMERGENCY\_SHARE\_ROUNDED**instance-attribute  
EMERGENCY\_SHARE\_ROUNDED: IconData

**buildEMERGENCY\_SHARE\_SHARP**instance-attribute  
EMERGENCY\_SHARE\_SHARP: IconData

buildEMERGENCY\_SHARPinstance-attribute

EMERGENCY\_SHARP: IconData

buildEMOJI\_EMOTIONSinstance-attribute

EMOJI\_EMOTIONS: IconData

buildEMOJI\_EMOTIONS\_OUTLINEDinstance-attribute

EMOJI\_EMOTIONS\_OUTLINED: IconData

buildEMOJI\_EMOTIONS\_ROUNDEDinstance-attribute

EMOJI\_EMOTIONS\_ROUNDED: IconData

buildEMOJI\_EMOTIONS\_SHARPinstance-attribute

EMOJI\_EMOTIONS\_SHARP: IconData

buildEMOJI\_EVENTSinstance-attribute

EMOJI\_EVENTS: IconData

buildEMOJI\_EVENTS\_OUTLINEDinstance-attribute

EMOJI\_EVENTS\_OUTLINED: IconData

buildEMOJI\_EVENTS\_ROUNDEDinstance-attribute

EMOJI\_EVENTS\_ROUNDED: IconData

buildEMOJI\_EVENTS\_SHARPinstance-attribute

EMOJI\_EVENTS\_SHARP: IconData

buildEMOJI\_FLAGSinstance-attribute

EMOJI\_FLAGS: IconData

buildEMOJI\_FLAGS\_OUTLINEDinstance-attribute

EMOJI\_FLAGS\_OUTLINED: IconData

buildEMOJI\_FLAGS\_ROUNDEDinstance-attribute

EMOJI\_FLAGS\_ROUNDED: IconData

buildEMOJI\_FLAGS\_SHARPinstance-attribute

EMOJI\_FLAGS\_SHARP: IconData

buildEMOJI\_FOOD\_BEVERAGEinstance-attribute

EMOJI\_FOOD\_BEVERAGE: IconData

buildEMOJI\_FOOD\_BEVERAGE\_OUTLINEDinstance-attribute

EMOJI\_FOOD\_BEVERAGE\_OUTLINED: IconData

buildEMOJI\_FOOD\_BEVERAGE\_ROUNDEDinstance-attribute

EMOJI\_FOOD\_BEVERAGE\_ROUNDED: IconData

buildEMOJI\_FOOD\_BEVERAGE\_SHARPinstance-attribute

EMOJI\_FOOD\_BEVERAGE\_SHARP: IconData

buildEMOJI\_NATUREinstance-attribute

EMOJI\_NATURE: IconData

buildEMOJI\_NATURE\_OUTLINEDinstance-attribute

EMOJI\_NATURE\_OUTLINED: IconData

buildEMOJI\_NATURE\_ROUNDEDinstance-attribute

EMOJI\_NATURE\_ROUNDED: IconData

buildEMOJI\_NATURE\_SHARPinstance-attribute

EMOJI\_NATURE\_SHARP: IconData

buildEMOJI\_OBJECTSinstance-attribute

EMOJI\_OBJECTS: IconData

**buildEMOJI\_OBJECTS\_OUTLINEDinstance-attribute**

EMOJI\_OBJECTS\_OUTLINED: IconData

**buildEMOJI\_OBJECTS\_ROUNDEDinstance-attribute**

EMOJI\_OBJECTS\_ROUNDED: IconData

**buildEMOJI\_OBJECTS\_SHARPinstance-attribute**

EMOJI\_OBJECTS\_SHARP: IconData

**buildEMOJI\_PEOPLEinstance-attribute**

EMOJI\_PEOPLE: IconData

**buildEMOJI\_PEOPLE\_OUTLINEDinstance-attribute**

EMOJI\_PEOPLE\_OUTLINED: IconData

**buildEMOJI\_PEOPLE\_ROUNDEDinstance-attribute**

EMOJI\_PEOPLE\_ROUNDED: IconData

**buildEMOJI\_PEOPLE\_SHARPinstance-attribute**

EMOJI\_PEOPLE\_SHARP: IconData

**buildEMOJI\_SYMBOLSinstance-attribute**

EMOJI\_SYMBOLS: IconData

**buildEMOJI\_SYMBOLS\_OUTLINEDinstance-attribute**

EMOJI\_SYMBOLS\_OUTLINED: IconData

**buildEMOJI\_SYMBOLS\_ROUNDEDinstance-attribute**

EMOJI\_SYMBOLS\_ROUNDED: IconData

**buildEMOJI\_SYMBOLS\_SHARPinstance-attribute**

EMOJI\_SYMBOLS\_SHARP: IconData

**buildEMOJI\_TRANSPORTATIONinstance-attribute**

EMOJI\_TRANSPORTATION: IconData

**buildEMOJI\_TRANSPORTATION\_OUTLINEDinstance-attribute**

EMOJI\_TRANSPORTATION\_OUTLINED: IconData

**buildEMOJI\_TRANSPORTATION\_ROUNDEDinstance-attribute**

EMOJI\_TRANSPORTATION\_ROUNDED: IconData

**buildEMOJI\_TRANSPORTATION\_SHARPinstance-attribute**

EMOJI\_TRANSPORTATION\_SHARP: IconData

**buildENERGY\_SAVINGS\_LEAFinstance-attribute**

ENERGY\_SAVINGS\_LEAF: IconData

**buildENERGY\_SAVINGS\_LEAF\_OUTLINEDinstance-attribute**

ENERGY\_SAVINGS\_LEAF\_OUTLINED: IconData

**buildENERGY\_SAVINGS\_LEAF\_ROUNDEDinstance-attribute**

ENERGY\_SAVINGS\_LEAF\_ROUNDED: IconData

**buildENERGY\_SAVINGS\_LEAF\_SHARPinstance-attribute**

ENERGY\_SAVINGS\_LEAF\_SHARP: IconData

**buildENGINEERINGinstance-attribute**

ENGINEERING: IconData

**buildENGINEERING\_OUTLINEDinstance-attribute**

ENGINEERING\_OUTLINED: IconData

**buildENGINEERING\_ROUNDEDinstance-attribute**

ENGINEERING\_ROUNDED: IconData

**buildENGINEERING\_SHARPinstance-attribute**

ENGINEERING\_SHARP: IconData

**buildENHANCE\_PHOTO\_TRANSLATEinstance-attribute**

ENHANCE\_PHOTO\_TRANSLATE: IconData

**buildENHANCE\_PHOTO\_TRANSLATE\_OUTLINEDinstance-attribute**

ENHANCE\_PHOTO\_TRANSLATE\_OUTLINED: IconData

**buildENHANCE\_PHOTO\_TRANSLATE\_ROUNDEDinstance-attribute**

ENHANCE\_PHOTO\_TRANSLATE\_ROUNDED: IconData

**buildENHANCE\_PHOTO\_TRANSLATE\_SHARPinstance-attribute**

ENHANCE\_PHOTO\_TRANSLATE\_SHARP: IconData

**buildENHANCED\_ENCRYPTIONinstance-attribute**

ENHANCED\_ENCRYPTION: IconData

**buildENHANCED\_ENCRYPTION\_OUTLINEDinstance-attribute**

ENHANCED\_ENCRYPTION\_OUTLINED: IconData

**buildENHANCED\_ENCRYPTION\_ROUNDEDinstance-attribute**

ENHANCED\_ENCRYPTION\_ROUNDED: IconData

**buildENHANCED\_ENCRYPTION\_SHARPinstance-attribute**

ENHANCED\_ENCRYPTION\_SHARP: IconData

**buildEQUALIZERinstance-attribute**

EQUALIZER: IconData

**buildEQUALIZER\_OUTLINEDinstance-attribute**

EQUALIZER\_OUTLINED: IconData

**buildEQUALIZER\_ROUNDEDinstance-attribute**

EQUALIZER\_ROUNDED: IconData

**buildEQUALIZER\_SHARPinstance-attribute**

EQUALIZER\_SHARP: IconData

**buildERRORinstance-attribute**

ERROR: IconData

**buildERROR\_OUTLINEinstance-attribute**

ERROR\_OUTLINE: IconData

**buildERROR\_OUTLINE\_OUTLINEDinstance-attribute**

ERROR\_OUTLINE\_OUTLINED: IconData

**buildERROR\_OUTLINE\_ROUNDEDinstance-attribute**

ERROR\_OUTLINE\_ROUNDED: IconData

**buildERROR\_OUTLINE\_SHARPinstance-attribute**

ERROR\_OUTLINE\_SHARP: IconData

**buildERROR\_OUTLINEDinstance-attribute**

ERROR\_OUTLINED: IconData

**buildERROR\_ROUNDEDinstance-attribute**

ERROR\_ROUNDED: IconData

**buildERROR\_SHARPinstance-attribute**

ERROR\_SHARP: IconData

**buildESCALATORinstance-attribute**

ESCALATOR: IconData

**buildESCALATOR\_OUTLINEDinstance-attribute**

ESCALATOR\_OUTLINED: IconData

**buildESCALATOR\_ROUNDEDinstance-attribute**

ESCALATOR\_ROUNDED: IconData

**buildESCALATOR\_SHARPinstance-attribute**

ESCALATOR\_SHARP: IconData

**buildESCALATOR\_WARNINGinstance-attribute**

ESCALATOR\_WARNING: IconData

**buildESCALATOR\_WARNING\_OUTLINEDinstance-attribute**

ESCALATOR\_WARNING\_OUTLINED: IconData

**buildESCALATOR\_WARNING\_ROUNDEDinstance-attribute**

ESCALATOR\_WARNING\_ROUNDED: IconData

**buildESCALATOR\_WARNING\_SHARPinstance-attribute**

ESCALATOR\_WARNING\_SHARP: IconData

**buildEUROinstance-attribute**

EURO: IconData

**buildEURO\_OUTLINEDinstance-attribute**

EURO\_OUTLINED: IconData

**buildEURO\_ROUNDEDinstance-attribute**

EURO\_ROUNDED: IconData

**buildEURO\_SHARPinstance-attribute**

EURO\_SHARP: IconData

**buildEURO\_SYMBOLinstance-attribute**

EURO\_SYMBOL: IconData

**buildEURO\_SYMBOL\_OUTLINEDinstance-attribute**

EURO\_SYMBOL\_OUTLINED: IconData

**buildEURO\_SYMBOL\_ROUNDEDinstance-attribute**

EURO\_SYMBOL\_ROUNDED: IconData

**buildEURO\_SYMBOL\_SHARPinstance-attribute**

EURO\_SYMBOL\_SHARP: IconData

**buildEV\_STATIONinstance-attribute**

EV\_STATION: IconData

**buildEV\_STATION\_OUTLINEDinstance-attribute**

EV\_STATION\_OUTLINED: IconData

**buildEV\_STATION\_ROUNDEDinstance-attribute**

EV\_STATION\_ROUNDED: IconData

**buildEV\_STATION\_SHARPinstance-attribute**

EV\_STATION\_SHARP: IconData

**buildEVENTinstance-attribute**

EVENT: IconData

**buildEVENT\_AVAILABLEinstance-attribute**

EVENT\_AVAILABLE: IconData

**buildEVENT\_AVAILABLE\_OUTLINEDinstance-attribute**

EVENT\_AVAILABLE\_OUTLINED: IconData

**buildEVENT\_AVAILABLE\_ROUNDEDinstance-attribute**

EVENT\_AVAILABLE\_ROUNDED: IconData

**buildEVENT\_AVAILABLE\_SHARPinstance-attribute**

EVENT\_AVAILABLE\_SHARP: IconData

**buildEVENT\_BUSYinstance-attribute**

EVENT\_BUSY: IconData

**buildEVENT\_BUSY\_OUTLINEDinstance-attribute**

EVENT\_BUSY\_OUTLINED: IconData

**buildEVENT\_BUSY\_ROUNDEDinstance-attribute**

EVENT\_BUSY\_ROUNDED: IconData

**buildEVENT\_BUSY\_SHARPinstance-attribute**

EVENT\_BUSY\_SHARP: IconData

**buildEVENT\_NOTEinstance-attribute**

EVENT\_NOTE: IconData

**buildEVENT\_NOTE\_OUTLINEDinstance-attribute**

EVENT\_NOTE\_OUTLINED: IconData

**buildEVENT\_NOTE\_ROUNDEDinstance-attribute**

EVENT\_NOTE\_ROUNDED: IconData

**buildEVENT\_NOTE\_SHARPinstance-attribute**

EVENT\_NOTE\_SHARP: IconData

**buildEVENT\_OUTLINEDinstance-attribute**

EVENT\_OUTLINED: IconData

**buildEVENT\_REPEATinstance-attribute**

EVENT\_REPEAT: IconData

**buildEVENT\_REPEAT\_OUTLINEDinstance-attribute**□

EVENT\_REPEAT\_OUTLINED: IconData

**buildEVENT\_REPEAT\_ROUNDEDinstance-attribute**

EVENT\_REPEAT\_ROUNDED: IconData

**buildEVENT\_REPEAT\_SHARPinstance-attribute**

EVENT\_REPEAT\_SHARP: IconData

**buildEVENT\_ROUNDEDinstance-attribute**

EVENT\_ROUNDED: IconData

**buildEVENT\_SEATinstance-attribute**

EVENT\_SEAT: IconData

**buildEVENT\_SEAT\_OUTLINEDinstance-attribute**

EVENT\_SEAT\_OUTLINED: IconData

**buildEVENT\_SEAT\_ROUNDEDinstance-attribute**

EVENT\_SEAT\_ROUNDED: IconData

**buildEVENT\_SEAT\_SHARPinstance-attribute**

EVENT\_SEAT\_SHARP: IconData

**buildEVENT\_SHARPinstance-attribute**

EVENT\_SHARP: IconData

**buildEXIT\_TO\_APPinstance-attribute**

EXIT\_TO\_APP: IconData

**buildEXIT\_TO\_APP\_OUTLINEDinstance-attribute**

EXIT\_TO\_APP\_OUTLINED: IconData

**buildEXIT\_TO\_APP\_ROUNDEDinstance-attribute**

EXIT\_TO\_APP\_ROUNDED: IconData

**buildEXIT\_TO\_APP\_SHARPinstance-attribute**

EXIT\_TO\_APP\_SHARP: IconData

**buildEXPANDinstance-attribute**

EXPAND: IconData

**buildEXPAND\_CIRCLE\_DOWNinstance-attribute**

EXPAND\_CIRCLE\_DOWN: IconData

**buildEXPAND\_CIRCLE\_DOWN\_OUTLINEDinstance-attribute**

EXPAND\_CIRCLE\_DOWN\_OUTLINED: IconData

**buildEXPAND\_CIRCLE\_DOWN\_ROUNDEDinstance-attribute**

EXPAND\_CIRCLE\_DOWN\_ROUNDED: IconData

**buildEXPAND\_CIRCLE\_DOWN\_SHARPinstance-attribute**

EXPAND\_CIRCLE\_DOWN\_SHARP: IconData

**buildEXPAND\_LESSinstance-attribute**

EXPAND\_LESS: IconData

**buildEXPAND\_LESS\_OUTLINEDinstance-attribute**

EXPAND\_LESS\_OUTLINED: IconData

**buildEXPAND\_LESS\_ROUNDEDinstance-attribute**

EXPAND\_LESS\_ROUNDED: IconData

**buildEXPAND\_LESS\_SHARPinstance-attribute**

EXPAND\_LESS\_SHARP: IconData

**buildEXPAND\_MOREinstance-attribute**

EXPAND\_MORE: IconData

**buildEXPAND\_MORE\_OUTLINEDinstance-attribute**

EXPAND\_MORE\_OUTLINED: IconData

**buildEXPAND\_MORE\_ROUNDEDinstance-attribute**

EXPAND\_MORE\_ROUNDED: IconData

**buildEXPAND\_MORE\_SHARPinstance-attribute**

EXPAND\_MORE\_SHARP: IconData

**buildEXPAND\_OUTLINEDinstance-attribute**

EXPAND\_OUTLINED: IconData

**buildEXPAND\_ROUNDEDinstance-attribute**

EXPAND\_ROUNDED: IconData

**buildEXPAND\_SHARPinstance-attribute**

EXPAND\_SHARP: IconData

**buildEXPLICITinstance-attribute**

EXPLICIT: IconData

**buildEXPLICIT\_OUTLINEDinstance-attribute**

EXPLICIT\_OUTLINED: IconData

**buildEXPLICIT\_ROUNDEDinstance-attribute**

EXPLICIT\_ROUNDED: IconData

**buildEXPLICIT\_SHARPinstance-attribute**

EXPLICIT\_SHARP: IconData

**buildEXPLOREinstance-attribute**

EXPLORE: IconData

**buildEXPLORE\_OFFinstance-attribute**

EXPLORE\_OFF: IconData

**buildEXPLORE\_OFF\_OUTLINEDinstance-attribute**

EXPLORE\_OFF\_OUTLINED: IconData

**buildEXPLORE\_OFF\_ROUNDEDinstance-attribute**

EXPLORE\_OFF\_ROUNDED: IconData

**buildEXPLORE\_OFF\_SHARPinstance-attribute**

EXPLORE\_OFF\_SHARP: IconData

**buildEXPLORE\_OUTLINEDinstance-attribute**

EXPLORE\_OUTLINED: IconData

**buildEXPLORE\_ROUNDEDinstance-attribute**

EXPLORE\_ROUNDED: IconData

**buildEXPLORE\_SHARPinstance-attribute**

EXPLORE\_SHARP: IconData

**buildEXPOSUREinstance-attribute**

EXPOSURE: IconData

**buildEXPOSURE\_MINUS\_1instance-attribute**

EXPOSURE\_MINUS\_1: IconData

**buildEXPOSURE\_MINUS\_1\_OUTLINEDinstance-attribute**

EXPOSURE\_MINUS\_1\_OUTLINED: IconData

**buildEXPOSURE\_MINUS\_1\_ROUNDEDinstance-attribute**

EXPOSURE\_MINUS\_1\_ROUNDED: IconData

**buildEXPOSURE\_MINUS\_1\_SHARPinstance-attribute**

EXPOSURE\_MINUS\_1\_SHARP: IconData

**buildEXPOSURE\_MINUS\_2instance-attribute**

EXPOSURE\_MINUS\_2: IconData

**buildEXPOSURE\_MINUS\_2\_OUTLINEDinstance-attribute**

EXPOSURE\_MINUS\_2\_OUTLINED: IconData

**buildEXPOSURE\_MINUS\_2\_ROUNDEDinstance-attribute**

EXPOSURE\_MINUS\_2\_ROUNDED: IconData

**buildEXPOSURE\_MINUS\_2\_SHARPinstance-attribute**

EXPOSURE\_MINUS\_2\_SHARP: IconData

**buildEXPOSURE\_NEG\_1instance-attribute**

EXPOSURE\_NEG\_1: IconData

**buildEXPOSURE\_NEG\_1\_OUTLINEDinstance-attribute**

EXPOSURE\_NEG\_1\_OUTLINED: IconData

**buildEXPOSURE\_NEG\_1\_ROUNDEDinstance-attribute**

EXPOSURE\_NEG\_1\_ROUNDED: IconData

**buildEXPOSURE\_NEG\_1\_SHARPinstance-attribute**

EXPOSURE\_NEG\_1\_SHARP: IconData



**buildEXPOSURE\_NEG\_2instance-attribute**

EXPOSURE\_NEG\_2: IconData

**buildEXPOSURE\_NEG\_2\_OUTLINEDinstance-attribute**

EXPOSURE\_NEG\_2\_OUTLINED: IconData

**buildEXPOSURE\_NEG\_2\_ROUNDEDinstance-attribute**

EXPOSURE\_NEG\_2\_ROUNDED: IconData

**buildEXPOSURE\_NEG\_2\_SHARPinstance-attribute**

EXPOSURE\_NEG\_2\_SHARP: IconData

**buildEXPOSURE\_OUTLINEDinstance-attribute**

EXPOSURE\_OUTLINED: IconData

**buildEXPOSURE\_PLUS\_1instance-attribute**

EXPOSURE\_PLUS\_1: IconData

**buildEXPOSURE\_PLUS\_1\_OUTLINEDinstance-attribute**

EXPOSURE\_PLUS\_1\_OUTLINED: IconData

**buildEXPOSURE\_PLUS\_1\_ROUNDEDinstance-attribute**

EXPOSURE\_PLUS\_1\_ROUNDED: IconData

**buildEXPOSURE\_PLUS\_1\_SHARPinstance-attribute**

EXPOSURE\_PLUS\_1\_SHARP: IconData

**buildEXPOSURE\_PLUS\_2instance-attribute**

EXPOSURE\_PLUS\_2: IconData

**buildEXPOSURE\_PLUS\_2\_OUTLINEDinstance-attribute**

EXPOSURE\_PLUS\_2\_OUTLINED: IconData

**buildEXPOSURE\_PLUS\_2\_ROUNDEDinstance-attribute**

EXPOSURE\_PLUS\_2\_ROUNDED: IconData

**buildEXPOSURE\_PLUS\_2\_SHARPinstance-attribute**

EXPOSURE\_PLUS\_2\_SHARP: IconData

**buildEXPOSURE\_ROUNDEDinstance-attribute**

EXPOSURE\_ROUNDED: IconData

**buildEXPOSURE\_SHARPinstance-attribute**

EXPOSURE\_SHARP: IconData

**buildEXPOSURE\_ZEROinstance-attribute**

EXPOSURE\_ZERO: IconData

**buildEXPOSURE\_ZERO\_OUTLINEDinstance-attribute**

EXPOSURE\_ZERO\_OUTLINED: IconData

**buildEXPOSURE\_ZERO\_ROUNDEDinstance-attribute**

EXPOSURE\_ZERO\_ROUNDED: IconData

**buildEXPOSURE\_ZERO\_SHARPinstance-attribute**

EXPOSURE\_ZERO\_SHARP: IconData

**buildEXTENSIONinstance-attribute**

EXTENSION: IconData

**buildEXTENSION\_OFFinstance-attribute**

EXTENSION\_OFF: IconData

**buildEXTENSION\_OFF\_OUTLINEDinstance-attribute**

EXTENSION\_OFF\_OUTLINED: IconData

**buildEXTENSION\_OFF\_ROUNDEDinstance-attribute**

EXTENSION\_OFF\_ROUNDED: IconData

**buildEXTENSION\_OFF\_SHARPinstance-attribute**

EXTENSION\_OFF\_SHARP: IconData

**buildEXTENSION\_OUTLINEDinstance-attribute**

EXTENSION\_OUTLINED: IconData

**buildEXTENSION\_ROUNDEDinstance-attribute**

EXTENSION\_ROUNDED: IconData

**buildEXTENSION\_SHARPinstance-attribute**

EXTENSION\_SHARP: IconData

**buildFACEinstance-attribute**

FACE: IconData

**buildFACE\_2instance-attribute**

FACE\_2: IconData

**buildFACE\_2\_OUTLINEDinstance-attribute**

FACE\_2\_OUTLINED: IconData

**buildFACE\_2\_ROUNDEDinstance-attribute**

FACE\_2\_ROUNDED: IconData

**buildFACE\_2\_SHARPinstance-attribute**

FACE\_2\_SHARP: IconData

**buildFACE\_3instance-attribute**

FACE\_3: IconData

**buildFACE\_3\_OUTLINEDinstance-attribute**

FACE\_3\_OUTLINED: IconData

**buildFACE\_3\_ROUNDEDinstance-attribute**

FACE\_3\_ROUNDED: IconData

**buildFACE\_3\_SHARPinstance-attribute**

FACE\_3\_SHARP: IconData

**buildFACE\_4instance-attribute**

FACE\_4: IconData

**buildFACE\_4\_OUTLINEDinstance-attribute**

FACE\_4\_OUTLINED: IconData

**buildFACE\_4\_ROUNDEDinstance-attribute**

FACE\_4\_ROUNDED: IconData

**buildFACE\_4\_SHARPinstance-attribute**

FACE\_4\_SHARP: IconData

**buildFACE\_5instance-attribute**

FACE\_5: IconData

**buildFACE\_5\_OUTLINEDinstance-attribute**

FACE\_5\_OUTLINED: IconData

**buildFACE\_5\_ROUNDEDinstance-attribute**

FACE\_5\_ROUNDED: IconData

**buildFACE\_5\_SHARPinstance-attribute**

FACE\_5\_SHARP: IconData

**buildFACE\_6instance-attribute**

FACE\_6: IconData

**buildFACE\_6\_OUTLINEDinstance-attribute**

FACE\_6\_OUTLINED: IconData

**buildFACE\_6\_ROUNDEDinstance-attribute**

FACE\_6\_ROUNDED: IconData

**buildFACE\_6\_SHARPinstance-attribute**

FACE\_6\_SHARP: IconData

**buildFACE\_OUTLINEDinstance-attribute**

FACE\_OUTLINED: IconData

**buildFACE\_RETOUCHING\_NATURALinstance-attribute**

FACE\_RETOUCHING\_NATURAL: IconData

**buildFACE\_RETOUCHING\_NATURAL\_OUTLINEDinstance-attribute**

FACE\_RETOUCHING\_NATURAL\_OUTLINED: IconData

**buildFACE\_RETOUCHING\_NATURAL\_ROUNDEDinstance-attribute**

FACE\_RETOUCHING\_NATURAL\_ROUNDED: IconData

**buildFACE\_RETOUCHING\_NATURAL\_SHARPinstance-attribute**

FACE\_RETOUCHING\_NATURAL\_SHARP: IconData

**buildFACE\_RETOUCHING\_OFFinstance-attribute**

FACE\_RETOUCHING\_OFF: IconData

**buildFACE\_RETOUCHING\_OFF\_OUTLINEDinstance-attribute**

FACE\_RETOUCHING\_OFF\_OUTLINED: IconData

**buildFACE\_RETOUCHING\_OFF\_ROUNDEDinstance-attribute**

FACE\_RETOUCHING\_OFF\_ROUNDED: IconData

**buildFACE\_RETOUCHING\_OFF\_SHARPinstance-attribute**

FACE\_RETOUCHING\_OFF\_SHARP: IconData

**buildFACE\_ROUNDEDinstance-attribute**

FACE\_ROUNDED: IconData

**buildFACE\_SHARPinstance-attribute**

FACE\_SHARP: IconData

**buildFACE\_UNLOCK\_OUTLINEDinstance-attribute**

FACE\_UNLOCK\_OUTLINED: IconData

**buildFACE\_UNLOCK\_ROUNDEDinstance-attribute**

FACE\_UNLOCK\_ROUNDED: IconData

**buildFACE\_UNLOCK\_SHARPinstance-attribute**

FACE\_UNLOCK\_SHARP: IconData

**buildFACEBOOKinstance-attribute**

FACEBOOK: IconData

**buildFACEBOOK\_OUTLINEDinstance-attribute**

FACEBOOK\_OUTLINED: IconData

**buildFACEBOOK\_ROUNDEDinstance-attribute**

FACEBOOK\_ROUNDED: IconData

**buildFACEBOOK\_SHARPinstance-attribute**

FACEBOOK\_SHARP: IconData

**buildFACT\_CHECKinstance-attribute**

FACT\_CHECK: IconData

**buildFACT\_CHECK\_OUTLINEDinstance-attribute**

FACT\_CHECK\_OUTLINED: IconData

**buildFACT\_CHECK\_ROUNDEDinstance-attribute**

FACT\_CHECK\_ROUNDED: IconData

**buildFACT\_CHECK\_SHARPinstance-attribute**

FACT\_CHECK\_SHARP: IconData

**buildFACTORYinstance-attribute**

FACTORY: IconData

**buildFACTORY\_OUTLINEDinstance-attribute** 

FACTORY\_OUTLINED: IconData

**buildFACTORY\_ROUNDEDinstance-attribute**

FACTORY\_ROUNDED: IconData

**buildFACTORY\_SHARPinstance-attribute**

FACTORY\_SHARP: IconData

**buildFAMILY\_RESTROOMinstance-attribute**

FAMILY\_RESTROOM: IconData

**buildFAMILY\_RESTROOM\_OUTLINEDinstance-attribute**

FAMILY\_RESTROOM\_OUTLINED: IconData

**buildFAMILY\_RESTROOM\_ROUNDEDinstance-attribute**

FAMILY\_RESTROOM\_ROUNDED: IconData

**buildFAMILY\_RESTROOM\_SHARPinstance-attribute**

FAMILY\_RESTROOM\_SHARP: IconData

**buildFAST\_FORWARDinstance-attribute**

FAST\_FORWARD: IconData

**buildFAST\_FORWARD\_OUTLINEDinstance-attribute**

FAST\_FORWARD\_OUTLINED: IconData

**buildFAST\_FORWARD\_ROUNDEDinstance-attribute**

FAST\_FORWARD\_ROUNDED: IconData

**buildFAST\_FORWARD\_SHARPinstance-attribute**

FAST\_FORWARD\_SHARP: IconData

**buildFAST\_REWINDinstance-attribute**

FAST\_REWIND: IconData

**buildFAST\_REWIND\_OUTLINEDinstance-attribute**

FAST\_REWIND\_OUTLINED: IconData

**buildFAST\_REWIND\_ROUNDEDinstance-attribute**

FAST\_REWIND\_ROUNDED: IconData

**buildFAST\_REWIND\_SHARPinstance-attribute**

FAST\_REWIND\_SHARP: IconData

**buildFASTFOODinstance-attribute**

FASTFOOD: IconData

**buildFASTFOOD\_OUTLINEDinstance-attribute**

FASTFOOD\_OUTLINED: IconData

**buildFASTFOOD\_ROUNDEDinstance-attribute**

FASTFOOD\_ROUNDED: IconData

**buildFASTFOOD\_SHARPinstance-attribute**

FASTFOOD\_SHARP: IconData

**buildFAVORITEinstance-attribute**

FAVORITE: IconData

**buildFAVORITE\_BORDERinstance-attribute**

FAVORITE\_BORDER: IconData

**buildFAVORITE\_BORDER\_OUTLINEDinstance-attribute**

FAVORITE\_BORDER\_OUTLINED: IconData

**buildFAVORITE\_BORDER\_ROUNDEDinstance-attribute**

FAVORITE\_BORDER\_ROUNDED: IconData

**buildFAVORITE\_BORDER\_SHARPinstance-attribute**

FAVORITE\_BORDER\_SHARP: IconData

**buildFAVORITE\_OUTLINEinstance-attribute**

FAVORITE\_OUTLINE: IconData

**buildFAVORITE\_OUTLINE\_OUTLINEDinstance-attribute**

FAVORITE\_OUTLINE\_OUTLINED: IconData

**buildFAVORITE\_OUTLINE\_ROUNDEDinstance-attribute**

FAVORITE\_OUTLINE\_ROUNDED: IconData

**buildFAVORITE\_OUTLINE\_SHARPinstance-attribute**

FAVORITE\_OUTLINE\_SHARP: IconData

**buildFAVORITE\_OUTLINEDinstance-attribute**

FAVORITE\_OUTLINED: IconData

**buildFAVORITE\_ROUNDEDinstance-attribute**

FAVORITE\_ROUNDED: IconData

**buildFAVORITE\_SHARPinstance-attribute**

FAVORITE\_SHARP: IconData

**buildFAXinstance-attribute**

FAX: IconData

**buildFAX\_OUTLINEDinstance-attribute**

FAX\_OUTLINED: IconData

**buildFAX\_ROUNDEDinstance-attribute**

FAX\_ROUNDED: IconData

**buildFAX\_SHARPinstance-attribute**

FAX\_SHARP: IconData

**buildFEATURED\_PLAY\_LISTinstance-attribute**

FEATURED\_PLAY\_LIST: IconData

**buildFEATURED\_PLAY\_LIST\_OUTLINEDinstance-attribute**

FEATURED\_PLAY\_LIST\_OUTLINED: IconData

**buildFEATURED\_PLAY\_LIST\_ROUNDEDinstance-attribute**

FEATURED\_PLAY\_LIST\_ROUNDED: IconData

**buildFEATURED\_PLAY\_LIST\_SHARPinstance-attribute**

FEATURED\_PLAY\_LIST\_SHARP: IconData

**buildFEATURED\_VIDEO**instance-attribute

FEATURED\_VIDEO: IconData

**buildFEATURED\_VIDEO\_OUTLINED**instance-attribute

FEATURED\_VIDEO\_OUTLINED: IconData

**buildFEATURED\_VIDEO\_ROUNDED**instance-attribute

FEATURED\_VIDEO\_ROUNDED: IconData

**buildFEATURED\_VIDEO\_SHARP**instance-attribute

FEATURED\_VIDEO\_SHARP: IconData

**buildFEED**instance-attribute

FEED: IconData

**buildFEED\_OUTLINED**instance-attribute

FEED\_OUTLINED: IconData

**buildFEED\_ROUNDED**instance-attribute

FEED\_ROUNDED: IconData

**buildFEED\_SHARP**instance-attribute

FEED\_SHARP: IconData

**buildFEEDBACK**instance-attribute

FEEDBACK: IconData

**buildFEEDBACK\_OUTLINED**instance-attribute

FEEDBACK\_OUTLINED: IconData

**buildFEEDBACK\_ROUNDED**instance-attribute

FEEDBACK\_ROUNDED: IconData

**buildFEEDBACK\_SHARP**instance-attribute

FEEDBACK\_SHARP: IconData

**buildFEMALE**instance-attribute

FEMALE: IconData

**buildFEMALE\_OUTLINED**instance-attribute

FEMALE\_OUTLINED: IconData

**buildFEMALE\_ROUNDED**instance-attribute

FEMALE\_ROUNDED: IconData

**buildFEMALE\_SHARP**instance-attribute

FEMALE\_SHARP: IconData

**buildFENCE**instance-attribute

FENCE: IconData

**buildFENCE\_OUTLINED**instance-attribute

FENCE\_OUTLINED: IconData

**buildFENCE\_ROUNDED**instance-attribute

FENCE\_ROUNDED: IconData

**buildFENCE\_SHARP**instance-attribute

FENCE\_SHARP: IconData

**buildFESTIVAL**instance-attribute

FESTIVAL: IconData

**buildFESTIVAL\_OUTLINED**instance-attribute

FESTIVAL\_OUTLINED: IconData

buildFESTIVAL\_ROUNDEDinstance-attribute

FESTIVAL\_ROUNDED: IconData

buildFESTIVAL\_SHARPinstance-attribute

FESTIVAL\_SHARP: IconData

buildFIBER\_DVRinstance-attribute

FIBER\_DVR: IconData

buildFIBER\_DVR\_OUTLINEDinstance-attribute

FIBER\_DVR\_OUTLINED: IconData

buildFIBER\_DVR\_ROUNDEDinstance-attribute

FIBER\_DVR\_ROUNDED: IconData

buildFIBER\_DVR\_SHARPinstance-attribute

FIBER\_DVR\_SHARP: IconData

buildFIBER\_MANUAL\_RECORDinstance-attribute

FIBER\_MANUAL\_RECORD: IconData

buildFIBER\_MANUAL\_RECORD\_OUTLINEDinstance-attribute

FIBER\_MANUAL\_RECORD\_OUTLINED: IconData

buildFIBER\_MANUAL\_RECORD\_ROUNDEDinstance-attribute

FIBER\_MANUAL\_RECORD\_ROUNDED: IconData

buildFIBER\_MANUAL\_RECORD\_SHARPinstance-attribute

FIBER\_MANUAL\_RECORD\_SHARP: IconData

buildFIBER\_NEWinstance-attribute

FIBER\_NEW: IconData

buildFIBER\_NEW\_OUTLINEDinstance-attribute

FIBER\_NEW\_OUTLINED: IconData

buildFIBER\_NEW\_ROUNDEDinstance-attribute

FIBER\_NEW\_ROUNDED: IconData

buildFIBER\_NEW\_SHARPinstance-attribute

FIBER\_NEW\_SHARP: IconData

buildFIBER\_PINinstance-attribute

FIBER\_PIN: IconData

buildFIBER\_PIN\_OUTLINEDinstance-attribute

FIBER\_PIN\_OUTLINED: IconData

buildFIBER\_PIN\_ROUNDEDinstance-attribute

FIBER\_PIN\_ROUNDED: IconData

buildFIBER\_PIN\_SHARPinstance-attribute

FIBER\_PIN\_SHARP: IconData

buildFIBER\_SMART\_RECORDinstance-attribute

FIBER\_SMART\_RECORD: IconData

buildFIBER\_SMART\_RECORD\_OUTLINEDinstance-attribute

FIBER\_SMART\_RECORD\_OUTLINED: IconData

buildFIBER\_SMART\_RECORD\_ROUNDEDinstance-attribute

FIBER\_SMART\_RECORD\_ROUNDED: IconData

buildFIBER\_SMART\_RECORD\_SHARPinstance-attribute

FIBER\_SMART\_RECORD\_SHARP: IconData

**buildFIFTEEN\_MPinstance-attribute**

FIFTEEN\_MP: IconData

**buildFIFTEEN\_MP\_OUTLINEDinstance-attribute**

FIFTEEN\_MP\_OUTLINED: IconData

**buildFIFTEEN\_MP\_ROUNDEDinstance-attribute**

FIFTEEN\_MP\_ROUNDED: IconData

**buildFIFTEEN\_MP\_SHARPinstance-attribute**

FIFTEEN\_MP\_SHARP: IconData

**buildFILE\_COPYinstance-attribute**

FILE\_COPY: IconData

**buildFILE\_COPY\_OUTLINEDinstance-attribute**

FILE\_COPY\_OUTLINED: IconData

**buildFILE\_COPY\_ROUNDEDinstance-attribute**

FILE\_COPY\_ROUNDED: IconData

**buildFILE\_COPY\_SHARPinstance-attribute**

FILE\_COPY\_SHARP: IconData

**buildFILE\_DOWNLOADinstance-attribute**

FILE\_DOWNLOAD: IconData

**buildFILE\_DOWNLOAD\_DONEinstance-attribute**

FILE\_DOWNLOAD\_DONE: IconData

**buildFILE\_DOWNLOAD\_DONE\_OUTLINEDinstance-attribute**

FILE\_DOWNLOAD\_DONE\_OUTLINED: IconData

**buildFILE\_DOWNLOAD\_DONE\_ROUNDEDinstance-attribute**

FILE\_DOWNLOAD\_DONE\_ROUNDED: IconData

**buildFILE\_DOWNLOAD\_DONE\_SHARPinstance-attribute**

FILE\_DOWNLOAD\_DONE\_SHARP: IconData

**buildFILE\_DOWNLOAD\_OFFinstance-attribute**

FILE\_DOWNLOAD\_OFF: IconData

**buildFILE\_DOWNLOAD\_OFF\_OUTLINEDinstance-attribute**

FILE\_DOWNLOAD\_OFF\_OUTLINED: IconData

**buildFILE\_DOWNLOAD\_OFF\_ROUNDEDinstance-attribute**

FILE\_DOWNLOAD\_OFF\_ROUNDED: IconData

**buildFILE\_DOWNLOAD\_OFF\_SHARPinstance-attribute**

FILE\_DOWNLOAD\_OFF\_SHARP: IconData

**buildFILE\_DOWNLOAD\_OUTLINEDinstance-attribute**

FILE\_DOWNLOAD\_OUTLINED: IconData

**buildFILE\_DOWNLOAD\_ROUNDEDinstance-attribute**

FILE\_DOWNLOAD\_ROUNDED: IconData

**buildFILE\_DOWNLOAD\_SHARPinstance-attribute**

FILE\_DOWNLOAD\_SHARP: IconData

**buildFILE\_OPENinstance-attribute**

FILE\_OPEN: IconData

**buildFILE\_OPEN\_OUTLINEDinstance-attribute**

FILE\_OPEN\_OUTLINED: IconData



**buildFILE\_OPEN\_ROUNDEDinstance-attribute**

FILE\_OPEN\_ROUNDED: IconData

**buildFILE\_OPEN\_SHARPinstance-attribute**

FILE\_OPEN\_SHARP: IconData

**buildFILE\_PRESENTinstance-attribute**

FILE\_PRESENT: IconData

**buildFILE\_PRESENT\_OUTLINEDinstance-attribute**

FILE\_PRESENT\_OUTLINED: IconData

**buildFILE\_PRESENT\_ROUNDEDinstance-attribute**

FILE\_PRESENT\_ROUNDED: IconData

**buildFILE\_PRESENT\_SHARPinstance-attribute**

FILE\_PRESENT\_SHARP: IconData

**buildFILE\_UPLOADinstance-attribute**

FILE\_UPLOAD: IconData

**buildFILE\_UPLOAD\_OFFinstance-attribute**

FILE\_UPLOAD\_OFF: IconData

**buildFILE\_UPLOAD\_OUTLINEDinstance-attribute**

FILE\_UPLOAD\_OUTLINED: IconData

**buildFILE\_UPLOAD\_ROUNDEDinstance-attribute**

FILE\_UPLOAD\_ROUNDED: IconData

**buildFILE\_UPLOAD\_SHARPinstance-attribute**

FILE\_UPLOAD\_SHARP: IconData

**buildFILTERinstance-attribute**

FILTER: IconData

**buildFILTER\_1instance-attribute**

FILTER\_1: IconData

**buildFILTER\_1\_OUTLINEDinstance-attribute**

FILTER\_1\_OUTLINED: IconData

**buildFILTER\_1\_ROUNDEDinstance-attribute**

FILTER\_1\_ROUNDED: IconData

**buildFILTER\_1\_SHARPinstance-attribute**

FILTER\_1\_SHARP: IconData

**buildFILTER\_2instance-attribute**

FILTER\_2: IconData

**buildFILTER\_2\_OUTLINEDinstance-attribute**

FILTER\_2\_OUTLINED: IconData

**buildFILTER\_2\_ROUNDEDinstance-attribute**

FILTER\_2\_ROUNDED: IconData

**buildFILTER\_2\_SHARPinstance-attribute**

FILTER\_2\_SHARP: IconData

**buildFILTER\_3instance-attribute**

FILTER\_3: IconData

**buildFILTER\_3\_OUTLINEDinstance-attribute**

FILTER\_3\_OUTLINED: IconData

**buildFILTER\_3\_ROUNDEDinstance-attribute**

**FILTER\_3\_ROUNDED:** IconData

**buildFILTER\_3\_SHARPinstance-attribute**

**FILTER\_3\_SHARP:** IconData

**buildFILTER\_4instance-attribute**

**FILTER\_4:** IconData

**buildFILTER\_4\_OUTLINEDinstance-attribute**

**FILTER\_4\_OUTLINED:** IconData

**buildFILTER\_4\_ROUNDEDinstance-attribute**

**FILTER\_4\_ROUNDED:** IconData

**buildFILTER\_4\_SHARPinstance-attribute**

**FILTER\_4\_SHARP:** IconData

**buildFILTER\_5instance-attribute**

**FILTER\_5:** IconData

**buildFILTER\_5\_OUTLINEDinstance-attribute**

**FILTER\_5\_OUTLINED:** IconData

**buildFILTER\_5\_ROUNDEDinstance-attribute**

**FILTER\_5\_ROUNDED:** IconData

**buildFILTER\_5\_SHARPinstance-attribute**

**FILTER\_5\_SHARP:** IconData

**buildFILTER\_6instance-attribute**

**FILTER\_6:** IconData

**buildFILTER\_6\_OUTLINEDinstance-attribute**

**FILTER\_6\_OUTLINED:** IconData

**buildFILTER\_6\_ROUNDEDinstance-attribute**

**FILTER\_6\_ROUNDED:** IconData

**buildFILTER\_6\_SHARPinstance-attribute**

**FILTER\_6\_SHARP:** IconData

**buildFILTER\_7instance-attribute**

**FILTER\_7:** IconData

**buildFILTER\_7\_OUTLINEDinstance-attribute**

**FILTER\_7\_OUTLINED:** IconData

**buildFILTER\_7\_ROUNDEDinstance-attribute**

**FILTER\_7\_ROUNDED:** IconData

**buildFILTER\_7\_SHARPinstance-attribute**

**FILTER\_7\_SHARP:** IconData

**buildFILTER\_8instance-attribute**

**FILTER\_8:** IconData

**buildFILTER\_8\_OUTLINEDinstance-attribute**

**FILTER\_8\_OUTLINED:** IconData

**buildFILTER\_8\_ROUNDEDinstance-attribute**

**FILTER\_8\_ROUNDED:** IconData

**buildFILTER\_8\_SHARPinstance-attribute**

**FILTER\_8\_SHARP:** IconData

**buildFILTER\_9instance-attribute**

**FILTER\_9:** IconData

**buildFILTER\_9\_OUTLINEDinstance-attribute**

**FILTER\_9\_OUTLINED:** IconData

**buildFILTER\_9\_PLUSinstance-attribute**

**FILTER\_9\_PLUS:** IconData

**buildFILTER\_9\_PLUS\_OUTLINEDinstance-attribute**

**FILTER\_9\_PLUS\_OUTLINED:** IconData

**buildFILTER\_9\_PLUS\_ROUNDEDinstance-attribute**

**FILTER\_9\_PLUS\_ROUNDED:** IconData

**buildFILTER\_9\_PLUS\_SHARPinstance-attribute**

**FILTER\_9\_PLUS\_SHARP:** IconData

**buildFILTER\_9\_ROUNDEDinstance-attribute**

**FILTER\_9\_ROUNDED:** IconData

**buildFILTER\_9\_SHARPinstance-attribute**

**FILTER\_9\_SHARP:** IconData

**buildFILTER\_ALTinstance-attribute**

**FILTER\_ALT:** IconData

**buildFILTER\_ALT\_OFFinstance-attribute**

**FILTER\_ALT\_OFF:** IconData

**buildFILTER\_ALT\_OFF\_OUTLINEDinstance-attribute**

**FILTER\_ALT\_OFF\_OUTLINED:** IconData

**buildFILTER\_ALT\_OFF\_ROUNDEDinstance-attribute**

**FILTER\_ALT\_OFF\_ROUNDED:** IconData

**buildFILTER\_ALT\_OFF\_SHARPinstance-attribute**

**FILTER\_ALT\_OFF\_SHARP:** IconData

**buildFILTER\_ALT\_OUTLINEDinstance-attribute**

**FILTER\_ALT\_OUTLINED:** IconData

**buildFILTER\_ALT\_ROUNDEDinstance-attribute**

**FILTER\_ALT\_ROUNDED:** IconData

**buildFILTER\_ALT\_SHARPinstance-attribute**

**FILTER\_ALT\_SHARP:** IconData

**buildFILTER\_B\_AND\_Winstance-attribute**

**FILTER\_B\_AND\_W:** IconData

**buildFILTER\_B\_AND\_W\_OUTLINEDinstance-attribute**

**FILTER\_B\_AND\_W\_OUTLINED:** IconData

**buildFILTER\_B\_AND\_W\_ROUNDEDinstance-attribute**

**FILTER\_B\_AND\_W\_ROUNDED:** IconData

**buildFILTER\_B\_AND\_W\_SHARPinstance-attribute**

**FILTER\_B\_AND\_W\_SHARP:** IconData

**buildFILTER\_CENTER\_FOCUSinstance-attribute**

**FILTER\_CENTER\_FOCUS:** IconData

**buildFILTER\_CENTER\_FOCUS\_OUTLINEDinstance-attribute**

**FILTER\_CENTER\_FOCUS\_OUTLINED:** IconData

**buildFILTER\_CENTER\_FOCUS\_ROUNDEDinstance-attribute**

**FILTER\_CENTER\_FOCUS\_ROUNDED:** IconData

**buildFILTER\_CENTER\_FOCUS\_SHARPinstance-attribute**

**FILTER\_CENTER\_FOCUS\_SHARP:** IconData

**buildFILTER\_DRAMAinstance-attribute**

**FILTER\_DRAMA:** IconData

**buildFILTER\_DRAMA\_OUTLINEDinstance-attribute**

**FILTER\_DRAMA\_OUTLINED:** IconData

**buildFILTER\_DRAMA\_ROUNDEDinstance-attribute**

**FILTER\_DRAMA\_ROUNDED:** IconData

**buildFILTER\_DRAMA\_SHARPinstance-attribute**

**FILTER\_DRAMA\_SHARP:** IconData

**buildFILTER\_FRAMESinstance-attribute**

**FILTER\_FRAMES:** IconData

**buildFILTER\_FRAMES\_OUTLINEDinstance-attribute**

**FILTER\_FRAMES\_OUTLINED:** IconData

**buildFILTER\_FRAMES\_ROUNDEDinstance-attribute**

**FILTER\_FRAMES\_ROUNDED:** IconData

**buildFILTER\_FRAMES\_SHARPinstance-attribute**

**FILTER\_FRAMES\_SHARP:** IconData

**buildFILTER\_HDRinstance-attribute**

**FILTER\_HDR:** IconData

**buildFILTER\_HDR\_OUTLINEDinstance-attribute**

**FILTER\_HDR\_OUTLINED:** IconData

**buildFILTER\_HDR\_ROUNDEDinstance-attribute**

**FILTER\_HDR\_ROUNDED:** IconData

**buildFILTER\_HDR\_SHARPinstance-attribute**

**FILTER\_HDR\_SHARP:** IconData

**buildFILTER\_LISTinstance-attribute**

**FILTER\_LIST:** IconData

**buildFILTER\_LIST\_ALTinstance-attribute**

**FILTER\_LIST\_ALT:** IconData

**buildFILTER\_LIST\_OFFinstance-attribute**

**FILTER\_LIST\_OFF:** IconData

**buildFILTER\_LIST\_OFF\_OUTLINEDinstance-attribute**

**FILTER\_LIST\_OFF\_OUTLINED:** IconData

**buildFILTER\_LIST\_OFF\_ROUNDEDinstance-attribute**

**FILTER\_LIST\_OFF\_ROUNDED:** IconData

**buildFILTER\_LIST\_OFF\_SHARPinstance-attribute**

**FILTER\_LIST\_OFF\_SHARP:** IconData

**buildFILTER\_LIST\_OUTLINEDinstance-attribute**

**FILTER\_LIST\_OUTLINED:** IconData

**buildFILTER\_LIST\_ROUNDEDinstance-attribute**

**FILTER\_LIST\_ROUNDED:** IconData

**buildFILTER\_LIST\_SHARPinstance-attribute**

**FILTER\_LIST\_SHARP:** IconData

**buildFILTER\_NONEinstance-attribute**

**FILTER\_NONE:** IconData

**buildFILTER\_NONE\_OUTLINEDinstance-attribute**

**FILTER\_NONE\_OUTLINED:** IconData

**buildFILTER\_NONE\_ROUNDEDinstance-attribute**

**FILTER\_NONE\_ROUNDED:** IconData

**buildFILTER\_NONE\_SHARPinstance-attribute**

**FILTER\_NONE\_SHARP:** IconData

**buildFILTER\_OUTLINEDinstance-attribute**

**FILTER\_OUTLINED:** IconData

**buildFILTER\_ROUNDEDinstance-attribute**

**FILTER\_ROUNDED:** IconData

**buildFILTER\_SHARPinstance-attribute**

**FILTER\_SHARP:** IconData

**buildFILTER\_TILT\_SHIFTinstance-attribute**

**FILTER\_TILT\_SHIFT:** IconData

**buildFILTER\_TILT\_SHIFT\_OUTLINEDinstance-attribute**

**FILTER\_TILT\_SHIFT\_OUTLINED:** IconData

**buildFILTER\_TILT\_SHIFT\_ROUNDEDinstance-attribute**

**FILTER\_TILT\_SHIFT\_ROUNDED:** IconData

**buildFILTER\_TILT\_SHIFT\_SHARPinstance-attribute**

**FILTER\_TILT\_SHIFT\_SHARP:** IconData

**buildFILTER\_VINTAGEinstance-attribute**

**FILTER\_VINTAGE:** IconData

**buildFILTER\_VINTAGE\_OUTLINEDinstance-attribute**

**FILTER\_VINTAGE\_OUTLINED:** IconData

**buildFILTER\_VINTAGE\_ROUNDEDinstance-attribute**

**FILTER\_VINTAGE\_ROUNDED:** IconData

**buildFILTER\_VINTAGE\_SHARPinstance-attribute**

**FILTER\_VINTAGE\_SHARP:** IconData

**buildFIND\_IN\_PAGEinstance-attribute**

**FIND\_IN\_PAGE:** IconData

**buildFIND\_IN\_PAGE\_OUTLINEDinstance-attribute**

**FIND\_IN\_PAGE\_OUTLINED:** IconData

**buildFIND\_IN\_PAGE\_ROUNDEDinstance-attribute**

**FIND\_IN\_PAGE\_ROUNDED:** IconData

**buildFIND\_IN\_PAGE\_SHARPinstance-attribute**

**FIND\_IN\_PAGE\_SHARP:** IconData

**buildFIND\_REPLACEinstance-attribute**

**FIND\_REPLACE:** IconData

**buildFIND\_REPLACE\_OUTLINEDinstance-attribute**

**FIND\_REPLACE\_OUTLINED:** IconData

**buildFIND\_REPLACE\_ROUNDED**instance-attribute

FIND\_REPLACE\_ROUNDED: IconData

**buildFIND\_REPLACE\_SHARP**instance-attribute

FIND\_REPLACE\_SHARP: IconData

**buildFINGERPRINT**instance-attribute

FINGERPRINT: IconData

**buildFINGERPRINT\_OUTLINED**instance-attribute

FINGERPRINT\_OUTLINED: IconData

**buildFINGERPRINT\_ROUNDED**instance-attribute

FINGERPRINT\_ROUNDED: IconData

**buildFINGERPRINT\_SHARP**instance-attribute

FINGERPRINT\_SHARP: IconData

**buildFIRE\_EXTINGUISHER**instance-attribute

FIRE\_EXTINGUISHER: IconData

**buildFIRE\_EXTINGUISHER\_OUTLINED**instance-attribute

FIRE\_EXTINGUISHER\_OUTLINED: IconData

**buildFIRE\_EXTINGUISHER\_ROUNDED**instance-attribute

FIRE\_EXTINGUISHER\_ROUNDED: IconData

**buildFIRE\_EXTINGUISHER\_SHARP**instance-attribute

FIRE\_EXTINGUISHER\_SHARP: IconData

**buildFIRE\_HYDRANT**instance-attribute

FIRE\_HYDRANT: IconData

**buildFIRE\_HYDRANT\_ALT**instance-attribute

FIRE\_HYDRANT\_ALT: IconData

**buildFIRE\_HYDRANT\_ALT\_OUTLINED**instance-attribute

FIRE\_HYDRANT\_ALT\_OUTLINED: IconData

**buildFIRE\_HYDRANT\_ALT\_ROUNDED**instance-attribute

FIRE\_HYDRANT\_ALT\_ROUNDED: IconData

**buildFIRE\_HYDRANT\_ALT\_SHARP**instance-attribute

FIRE\_HYDRANT\_ALT\_SHARP: IconData

**buildFIRE\_TRUCK**instance-attribute

FIRE\_TRUCK: IconData

**buildFIRE\_TRUCK\_OUTLINED**instance-attribute

FIRE\_TRUCK\_OUTLINED: IconData

**buildFIRE\_TRUCK\_ROUNDED**instance-attribute

FIRE\_TRUCK\_ROUNDED: IconData

**buildFIRE\_TRUCK\_SHARP**instance-attribute

FIRE\_TRUCK\_SHARP: IconData

**buildFIREPLACE**instance-attribute

FIREPLACE: IconData

**buildFIREPLACE\_OUTLINED**instance-attribute

FIREPLACE\_OUTLINED: IconData

**buildFIREPLACE\_ROUNDED**instance-attribute

FIREPLACE\_ROUNDED: IconData

**buildFIREPLACE\_SHARPinstance-attribute**

FIREPLACE\_SHARP: IconData

**buildFIRST\_PAGEinstance-attribute**

FIRST\_PAGE: IconData

**buildFIRST\_PAGE\_OUTLINEDinstance-attribute**

FIRST\_PAGE\_OUTLINED: IconData

**buildFIRST\_PAGE\_ROUNDEDinstance-attribute**

FIRST\_PAGE\_ROUNDED: IconData

**buildFIRST\_PAGE\_SHARPinstance-attribute**

FIRST\_PAGE\_SHARP: IconData

**buildFIT\_SCREENinstance-attribute**

FIT\_SCREEN: IconData

**buildFIT\_SCREEN\_OUTLINEDinstance-attribute**

FIT\_SCREEN\_OUTLINED: IconData

**buildFIT\_SCREEN\_ROUNDEDinstance-attribute**

FIT\_SCREEN\_ROUNDED: IconData

**buildFIT\_SCREEN\_SHARPinstance-attribute**

FIT\_SCREEN\_SHARP: IconData

**buildFITBITinstance-attribute**

FITBIT: IconData

**buildFITBIT\_OUTLINEDinstance-attribute**

FITBIT\_OUTLINED: IconData

**buildFITBIT\_ROUNDEDinstance-attribute**

FITBIT\_ROUNDED: IconData

**buildFITBIT\_SHARPinstance-attribute**

FITBIT\_SHARP: IconData

**buildFITNESS\_CENTERinstance-attribute**

FITNESS\_CENTER: IconData

**buildFITNESS\_CENTER\_OUTLINEDinstance-attribute**

FITNESS\_CENTER\_OUTLINED: IconData

**buildFITNESS\_CENTER\_ROUNDEDinstance-attribute**

FITNESS\_CENTER\_ROUNDED: IconData

**buildFITNESS\_CENTER\_SHARPinstance-attribute**

FITNESS\_CENTER\_SHARP: IconData

**buildFIVE\_Ginstance-attribute**

FIVE\_G: IconData

**buildFIVE\_G\_OUTLINEDinstance-attribute**

FIVE\_G\_OUTLINED: IconData

**buildFIVE\_G\_ROUNDEDinstance-attribute**

FIVE\_G\_ROUNDED: IconData

**buildFIVE\_G\_SHARPinstance-attribute**

FIVE\_G\_SHARP: IconData

**buildFIVE\_Kinstance-attribute**

FIVE\_K: IconData

**buildFIVE\_K\_OUTLINEDinstance-attribute**

FIVE\_K\_OUTLINED: IconData

**buildFIVE\_K\_PLUSinstance-attribute**

FIVE\_K\_PLUS: IconData

**buildFIVE\_K\_PLUS\_OUTLINEDinstance-attribute**

FIVE\_K\_PLUS\_OUTLINED: IconData

**buildFIVE\_K\_PLUS\_ROUNDEDinstance-attribute**

FIVE\_K\_PLUS\_ROUNDED: IconData

**buildFIVE\_K\_PLUS\_SHARPinstance-attribute**

FIVE\_K\_PLUS\_SHARP: IconData

**buildFIVE\_K\_ROUNDEDinstance-attribute**

FIVE\_K\_ROUNDED: IconData

**buildFIVE\_K\_SHARPinstance-attribute**

FIVE\_K\_SHARP: IconData

**buildFIVE\_MPinstance-attribute**

FIVE\_MP: IconData

**buildFIVE\_MP\_OUTLINEDinstance-attribute**

FIVE\_MP\_OUTLINED: IconData

**buildFIVE\_MP\_ROUNDEDinstance-attribute**

FIVE\_MP\_ROUNDED: IconData

**buildFIVE\_MP\_SHARPinstance-attribute**

FIVE\_MP\_SHARP: IconData

**buildFLAGinstance-attribute**

FLAG: IconData

**buildFLAG\_CIRCLEinstance-attribute**

FLAG\_CIRCLE: IconData

**buildFLAG\_CIRCLE\_OUTLINEDinstance-attribute**

FLAG\_CIRCLE\_OUTLINED: IconData

**buildFLAG\_CIRCLE\_ROUNDEDinstance-attribute**

FLAG\_CIRCLE\_ROUNDED: IconData

**buildFLAG\_CIRCLE\_SHARPinstance-attribute**

FLAG\_CIRCLE\_SHARP: IconData

**buildFLAG\_OUTLINEDinstance-attribute**

FLAG\_OUTLINED: IconData

**buildFLAG\_ROUNDEDinstance-attribute**

FLAG\_ROUNDED: IconData

**buildFLAG\_SHARPinstance-attribute**

FLAG\_SHARP: IconData

**buildFLAKYinstance-attribute**

FLAKY: IconData

**buildFLAKY\_OUTLINEDinstance-attribute**

FLAKY\_OUTLINED: IconData

**buildFLAKY\_ROUNDEDinstance-attribute**

FLAKY\_ROUNDED: IconData



**buildFLAKY\_SHARPinstance-attribute**

FLAKY\_SHARP: IconData

**buildFLAREinstance-attribute**

FLARE: IconData

**buildFLARE\_OUTLINEDinstance-attribute**

FLARE\_OUTLINED: IconData

**buildFLARE\_ROUNDEDinstance-attribute**

FLARE\_ROUNDED: IconData

**buildFLARE\_SHARPinstance-attribute**

FLARE\_SHARP: IconData

**buildFLASH\_AUTOinstance-attribute**

FLASH\_AUTO: IconData

**buildFLASH\_AUTO\_OUTLINEDinstance-attribute**

FLASH\_AUTO\_OUTLINED: IconData

**buildFLASH\_AUTO\_ROUNDEDinstance-attribute**

FLASH\_AUTO\_ROUNDED: IconData

**buildFLASH\_AUTO\_SHARPinstance-attribute**

FLASH\_AUTO\_SHARP: IconData

**buildFLASH\_OFFinstance-attribute**

FLASH\_OFF: IconData

**buildFLASH\_OFF\_OUTLINEDinstance-attribute**

FLASH\_OFF\_OUTLINED: IconData

**buildFLASH\_OFF\_ROUNDEDinstance-attribute**

FLASH\_OFF\_ROUNDED: IconData

**buildFLASH\_OFF\_SHARPinstance-attribute**

FLASH\_OFF\_SHARP: IconData

**buildFLASH\_ONinstance-attribute**

FLASH\_ON: IconData

**buildFLASH\_ON\_OUTLINEDinstance-attribute**

FLASH\_ON\_OUTLINED: IconData

**buildFLASH\_ON\_ROUNDEDinstance-attribute**

FLASH\_ON\_ROUNDED: IconData

**buildFLASH\_ON\_SHARPinstance-attribute**

FLASH\_ON\_SHARP: IconData

**buildFLASHLIGHT\_OFFinstance-attribute**

FLASHLIGHT\_OFF: IconData

**buildFLASHLIGHT\_OFF\_OUTLINEDinstance-attribute**

FLASHLIGHT\_OFF\_OUTLINED: IconData

**buildFLASHLIGHT\_OFF\_ROUNDEDinstance-attribute**

FLASHLIGHT\_OFF\_ROUNDED: IconData

**buildFLASHLIGHT\_OFF\_SHARPinstance-attribute**

FLASHLIGHT\_OFF\_SHARP: IconData

**buildFLASHLIGHT\_ONinstance-attribute**

FLASHLIGHT\_ON: IconData

**buildFLASHLIGHT\_ON\_OUTLINEDinstance-attribute**

FLASHLIGHT\_ON\_OUTLINED: IconData

**buildFLASHLIGHT\_ON\_ROUNDEDinstance-attribute**

FLASHLIGHT\_ON\_ROUNDED: IconData

**buildFLASHLIGHT\_ON\_SHARPinstance-attribute**

FLASHLIGHT\_ON\_SHARP: IconData

**buildFLATWAREinstance-attribute**

FLATWARE: IconData

**buildFLATWARE\_OUTLINEDinstance-attribute**

FLATWARE\_OUTLINED: IconData

**buildFLATWARE\_ROUNDEDinstance-attribute**

FLATWARE\_ROUNDED: IconData

**buildFLATWARE\_SHARPinstance-attribute**

FLATWARE\_SHARP: IconData

**buildFLIGHTinstance-attribute**

FLIGHT: IconData

**buildFLIGHT\_CLASSinstance-attribute**

FLIGHT\_CLASS: IconData

**buildFLIGHT\_CLASS\_OUTLINEDinstance-attribute**

FLIGHT\_CLASS\_OUTLINED: IconData

**buildFLIGHT\_CLASS\_ROUNDEDinstance-attribute**

FLIGHT\_CLASS\_ROUNDED: IconData

**buildFLIGHT\_CLASS\_SHARPinstance-attribute**

FLIGHT\_CLASS\_SHARP: IconData

**buildFLIGHT\_LANDinstance-attribute**

FLIGHT\_LAND: IconData

**buildFLIGHT\_LAND\_OUTLINEDinstance-attribute**

FLIGHT\_LAND\_OUTLINED: IconData

**buildFLIGHT\_LAND\_ROUNDEDinstance-attribute**

FLIGHT\_LAND\_ROUNDED: IconData

**buildFLIGHT\_LAND\_SHARPinstance-attribute**

FLIGHT\_LAND\_SHARP: IconData

**buildFLIGHT\_OUTLINEDinstance-attribute**

FLIGHT\_OUTLINED: IconData

**buildFLIGHT\_ROUNDEDinstance-attribute**

FLIGHT\_ROUNDED: IconData

**buildFLIGHT\_SHARPinstance-attribute**

FLIGHT\_SHARP: IconData

**buildFLIGHT\_TAKEOFFinstance-attribute**

FLIGHT\_TAKEOFF: IconData

**buildFLIGHT\_TAKEOFF\_OUTLINEDinstance-attribute**

FLIGHT\_TAKEOFF\_OUTLINED: IconData

**buildFLIGHT\_TAKEOFF\_ROUNDEDinstance-attribute**

FLIGHT\_TAKEOFF\_ROUNDED: IconData

**buildFLIGHT\_TAKEOFF\_SHARPinstance-attribute**

FLIGHT\_TAKEOFF\_SHARP: IconData

**buildFLIPinstance-attribute**

FLIP: IconData

**buildFLIP\_CAMERA\_ANDROIDinstance-attribute**

FLIP\_CAMERA\_ANDROID: IconData

**buildFLIP\_CAMERA\_ANDROID\_OUTLINEDinstance-attribute**

FLIP\_CAMERA\_ANDROID\_OUTLINED: IconData

**buildFLIP\_CAMERA\_ANDROID\_ROUNDEDinstance-attribute**

FLIP\_CAMERA\_ANDROID\_ROUNDED: IconData

**buildFLIP\_CAMERA\_ANDROID\_SHARPinstance-attribute**

FLIP\_CAMERA\_ANDROID\_SHARP: IconData

**buildFLIP\_CAMERA\_IOSinstance-attribute**

FLIP\_CAMERA\_IOS: IconData

**buildFLIP\_CAMERA\_IOS\_OUTLINEDinstance-attribute**

FLIP\_CAMERA\_IOS\_OUTLINED: IconData

**buildFLIP\_CAMERA\_IOS\_ROUNDEDinstance-attribute**

FLIP\_CAMERA\_IOS\_ROUNDED: IconData

**buildFLIP\_CAMERA\_IOS\_SHARPinstance-attribute**

FLIP\_CAMERA\_IOS\_SHARP: IconData

**buildFLIP\_OUTLINEDinstance-attribute**

FLIP\_OUTLINED: IconData

**buildFLIP\_ROUNDEDinstance-attribute**

FLIP\_ROUNDED: IconData

**buildFLIP\_SHARPinstance-attribute**

FLIP\_SHARP: IconData

**buildFLIP\_TO\_BACKinstance-attribute**

FLIP\_TO\_BACK: IconData

**buildFLIP\_TO\_BACK\_OUTLINEDinstance-attribute**

FLIP\_TO\_BACK\_OUTLINED: IconData

**buildFLIP\_TO\_BACK\_ROUNDEDinstance-attribute**

FLIP\_TO\_BACK\_ROUNDED: IconData

**buildFLIP\_TO\_BACK\_SHARPinstance-attribute**

FLIP\_TO\_BACK\_SHARP: IconData

**buildFLIP\_TO\_FRONTinstance-attribute**

FLIP\_TO\_FRONT: IconData

**buildFLIP\_TO\_FRONT\_OUTLINEDinstance-attribute**

FLIP\_TO\_FRONT\_OUTLINED: IconData

**buildFLIP\_TO\_FRONT\_ROUNDEDinstance-attribute**

FLIP\_TO\_FRONT\_ROUNDED: IconData

**buildFLIP\_TO\_FRONT\_SHARPinstance-attribute**

FLIP\_TO\_FRONT\_SHARP: IconData

**buildFLOODinstance-attribute**

FLOOD: IconData

**buildFLOOD\_OUTLINEDinstance-attribute**

FLOOD\_OUTLINED: IconData

**buildFLOOD\_ROUNDEDinstance-attribute**

FLOOD\_ROUNDED: IconData

**buildFLOOD\_SHARPinstance-attribute**

FLOOD\_SHARP: IconData

**buildFLOUORESCENTinstance-attribute**

FLOUORESCENT: IconData

**buildFLOUORESCENT\_OUTLINEDinstance-attribute**

FLOUORESCENT\_OUTLINED: IconData

**buildFLOUORESCENT\_ROUNDEDinstance-attribute**

FLOUORESCENT\_ROUNDED: IconData

**buildFLOUORESCENT\_SHARPinstance-attribute**

FLOUORESCENT\_SHARP: IconData

**buildFLUORESCENTinstance-attribute**

FLUORESCENT: IconData

**buildFLUORESCENT\_OUTLINEDinstance-attribute**

FLUORESCENT\_OUTLINED: IconData

**buildFLUORESCENT\_ROUNDEDinstance-attribute**

FLUORESCENT\_ROUNDED: IconData

**buildFLUORESCENT\_SHARPinstance-attribute**

FLUORESCENT\_SHARP: IconData

**buildFLUTTER\_DASHinstance-attribute**

FLUTTER\_DASH: IconData

**buildFLUTTER\_DASH\_OUTLINEDinstance-attribute**

FLUTTER\_DASH\_OUTLINED: IconData

**buildFLUTTER\_DASH\_ROUNDEDinstance-attribute**

FLUTTER\_DASH\_ROUNDED: IconData

**buildFLUTTER\_DASH\_SHARPinstance-attribute**

FLUTTER\_DASH\_SHARP: IconData

**buildFMD\_BADinstance-attribute**

FMD\_BAD: IconData

**buildFMD\_BAD\_OUTLINEDinstance-attribute**

FMD\_BAD\_OUTLINED: IconData

**buildFMD\_BAD\_ROUNDEDinstance-attribute**

FMD\_BAD\_ROUNDED: IconData

**buildFMD\_BAD\_SHARPinstance-attribute**

FMD\_BAD\_SHARP: IconData

**buildFMD\_GOODinstance-attribute**

FMD\_GOOD: IconData

**buildFMD\_GOOD\_OUTLINEDinstance-attribute**

FMD\_GOOD\_OUTLINED: IconData

**buildFMD\_GOOD\_ROUNDEDinstance-attribute**

FMD\_GOOD\_ROUNDED: IconData

**buildFMD\_GOOD\_SHARPinstance-attribute**

FMD\_GOOD\_SHARP: IconData

**buildFOGGYinstance-attribute**

FOGGY: IconData

**buildFOLDERinstance-attribute**

FOLDER: IconData

**buildFOLDER\_COPYinstance-attribute**

FOLDER\_COPY: IconData

**buildFOLDER\_COPY\_OUTLINEDinstance-attribute**

FOLDER\_COPY\_OUTLINED: IconData

**buildFOLDER\_COPY\_ROUNDEDinstance-attribute**

FOLDER\_COPY\_ROUNDED: IconData

**buildFOLDER\_COPY\_SHARPinstance-attribute**

FOLDER\_COPY\_SHARP: IconData

**buildFOLDER\_DELETEinstance-attribute**

FOLDER\_DELETE: IconData

**buildFOLDER\_DELETE\_OUTLINEDinstance-attribute**

FOLDER\_DELETE\_OUTLINED: IconData

**buildFOLDER\_DELETE\_ROUNDEDinstance-attribute**

FOLDER\_DELETE\_ROUNDED: IconData

**buildFOLDER\_DELETE\_SHARPinstance-attribute**

FOLDER\_DELETE\_SHARP: IconData

**buildFOLDER\_OFFinstance-attribute**

FOLDER\_OFF: IconData

**buildFOLDER\_OFF\_OUTLINEDinstance-attribute**

FOLDER\_OFF\_OUTLINED: IconData

**buildFOLDER\_OFF\_ROUNDEDinstance-attribute**

FOLDER\_OFF\_ROUNDED: IconData

**buildFOLDER\_OFF\_SHARPinstance-attribute**

FOLDER\_OFF\_SHARP: IconData

**buildFOLDER\_OPENinstance-attribute**

FOLDER\_OPEN: IconData

**buildFOLDER\_OPEN\_OUTLINEDinstance-attribute**

FOLDER\_OPEN\_OUTLINED: IconData

**buildFOLDER\_OPEN\_ROUNDEDinstance-attribute**

FOLDER\_OPEN\_ROUNDED: IconData

**buildFOLDER\_OPEN\_SHARPinstance-attribute**

FOLDER\_OPEN\_SHARP: IconData

**buildFOLDER\_OUTLINEDinstance-attribute**

FOLDER\_OUTLINED: IconData

**buildFOLDER\_ROUNDEDinstance-attribute**

FOLDER\_ROUNDED: IconData

**buildFOLDER\_SHAREDinstance-attribute**

FOLDER\_SHARED: IconData

**buildFOLDER\_SHARED\_OUTLINED**instance-attribute

FOLDER\_SHARED\_OUTLINED: IconData

**buildFOLDER\_SHARED\_ROUNDED**instance-attribute

FOLDER\_SHARED\_ROUNDED: IconData

**buildFOLDER\_SHARED\_SHARP**instance-attribute

FOLDER\_SHARED\_SHARP: IconData

**buildFOLDER\_SHARP**instance-attribute

FOLDER\_SHARP: IconData

**buildFOLDER\_SPECIAL**instance-attribute

FOLDER\_SPECIAL: IconData

**buildFOLDER\_SPECIAL\_OUTLINED**instance-attribute

FOLDER\_SPECIAL\_OUTLINED: IconData

**buildFOLDER\_SPECIAL\_ROUNDED**instance-attribute

FOLDER\_SPECIAL\_ROUNDED: IconData

**buildFOLDER\_SPECIAL\_SHARP**instance-attribute

FOLDER\_SPECIAL\_SHARP: IconData

**buildFOLDER\_ZIP**instance-attribute

FOLDER\_ZIP: IconData

**buildFOLDER\_ZIP\_OUTLINED**instance-attribute

FOLDER\_ZIP\_OUTLINED: IconData

**buildFOLDER\_ZIP\_ROUNDED**instance-attribute

FOLDER\_ZIP\_ROUNDED: IconData

**buildFOLDER\_ZIP\_SHARP**instance-attribute

FOLDER\_ZIP\_SHARP: IconData

**buildFOLLOW\_THE\_SIGNS**instance-attribute

FOLLOW\_THE\_SIGNS: IconData

**buildFOLLOW\_THE\_SIGNS\_OUTLINED**instance-attribute

FOLLOW\_THE\_SIGNS\_OUTLINED: IconData

**buildFOLLOW\_THE\_SIGNS\_ROUNDED**instance-attribute

FOLLOW\_THE\_SIGNS\_ROUNDED: IconData

**buildFOLLOW\_THE\_SIGNS\_SHARP**instance-attribute

FOLLOW\_THE\_SIGNS\_SHARP: IconData

**buildFONT\_DOWNLOAD**instance-attribute

FONT\_DOWNLOAD: IconData

**buildFONT\_DOWNLOAD\_OFF**instance-attribute

FONT\_DOWNLOAD\_OFF: IconData

**buildFONT\_DOWNLOAD\_OFF\_OUTLINED**instance-attribute

FONT\_DOWNLOAD\_OFF\_OUTLINED: IconData

**buildFONT\_DOWNLOAD\_OFF\_ROUNDED**instance-attribute

FONT\_DOWNLOAD\_OFF\_ROUNDED: IconData

**buildFONT\_DOWNLOAD\_OFF\_SHARP**instance-attribute

FONT\_DOWNLOAD\_OFF\_SHARP: IconData

**buildFONT\_DOWNLOAD\_OUTLINED**instance-attribute

FONT\_DOWNLOAD\_OUTLINED: IconData

**buildFONT\_DOWNLOAD\_ROUNDEDinstance-attribute**

FONT\_DOWNLOAD\_ROUNDED: IconData

**buildFONT\_DOWNLOAD\_SHARPinstance-attribute**

FONT\_DOWNLOAD\_SHARP: IconData

**buildFOOD\_BANKinstance-attribute**

FOOD\_BANK: IconData

**buildFOOD\_BANK\_OUTLINEDinstance-attribute**

FOOD\_BANK\_OUTLINED: IconData

**buildFOOD\_BANK\_ROUNDEDinstance-attribute**

FOOD\_BANK\_ROUNDED: IconData

**buildFOOD\_BANK\_SHARPinstance-attribute**

FOOD\_BANK\_SHARP: IconData

**buildFORESTinstance-attribute**

FOREST: IconData

**buildFOREST\_OUTLINEDinstance-attribute**

FOREST\_OUTLINED: IconData

**buildFOREST\_ROUNDEDinstance-attribute**

FOREST\_ROUNDED: IconData

**buildFOREST\_SHARPinstance-attribute**

FOREST\_SHARP: IconData

**buildFORK\_LEFTinstance-attribute**

FORK\_LEFT: IconData

**buildFORK\_LEFT\_OUTLINEDinstance-attribute**

FORK\_LEFT\_OUTLINED: IconData

**buildFORK\_LEFT\_ROUNDEDinstance-attribute**

FORK\_LEFT\_ROUNDED: IconData

**buildFORK\_LEFT\_SHARPinstance-attribute**

FORK\_LEFT\_SHARP: IconData

**buildFORK\_RIGHTinstance-attribute**

FORK\_RIGHT: IconData

**buildFORK\_RIGHT\_OUTLINEDinstance-attribute**

FORK\_RIGHT\_OUTLINED: IconData

**buildFORK\_RIGHT\_ROUNDEDinstance-attribute**

FORK\_RIGHT\_ROUNDED: IconData

**buildFORK\_RIGHT\_SHARPinstance-attribute**

FORK\_RIGHT\_SHARP: IconData

**buildFORKLIFTinstance-attribute**

FORKLIFT: IconData

**buildFORMAT\_ALIGN\_CENTERinstance-attribute**

FORMAT\_ALIGN\_CENTER: IconData

**buildFORMAT\_ALIGN\_CENTER\_OUTLINEDinstance-attribute**

FORMAT\_ALIGN\_CENTER\_OUTLINED: IconData

**buildFORMAT\_ALIGN\_CENTER\_ROUNDEDinstance-attribute**

FORMAT\_ALIGN\_CENTER\_ROUNDED: IconData

**buildFORMAT\_ALIGN\_CENTER\_SHARPinstance-attribute**

FORMAT\_ALIGN\_CENTER\_SHARP: IconData

**buildFORMAT\_ALIGN\_JUSTIFYinstance-attribute**

FORMAT\_ALIGN\_JUSTIFY: IconData

**buildFORMAT\_ALIGN\_JUSTIFY\_OUTLINEDinstance-attribute**

FORMAT\_ALIGN\_JUSTIFY\_OUTLINED: IconData

**buildFORMAT\_ALIGN\_JUSTIFY\_ROUNDEDinstance-attribute**

FORMAT\_ALIGN\_JUSTIFY\_ROUNDED: IconData

**buildFORMAT\_ALIGN\_JUSTIFY\_SHARPinstance-attribute**

FORMAT\_ALIGN\_JUSTIFY\_SHARP: IconData

**buildFORMAT\_ALIGN\_LEFTinstance-attribute**

FORMAT\_ALIGN\_LEFT: IconData

**buildFORMAT\_ALIGN\_LEFT\_OUTLINEDinstance-attribute**

FORMAT\_ALIGN\_LEFT\_OUTLINED: IconData

**buildFORMAT\_ALIGN\_LEFT\_ROUNDEDinstance-attribute**

FORMAT\_ALIGN\_LEFT\_ROUNDED: IconData

**buildFORMAT\_ALIGN\_LEFT\_SHARPinstance-attribute**

FORMAT\_ALIGN\_LEFT\_SHARP: IconData

**buildFORMAT\_ALIGN\_RIGHTinstance-attribute**

FORMAT\_ALIGN\_RIGHT: IconData

**buildFORMAT\_ALIGN\_RIGHT\_OUTLINEDinstance-attribute**

FORMAT\_ALIGN\_RIGHT\_OUTLINED: IconData

**buildFORMAT\_ALIGN\_RIGHT\_ROUNDEDinstance-attribute**

FORMAT\_ALIGN\_RIGHT\_ROUNDED: IconData

**buildFORMAT\_ALIGN\_RIGHT\_SHARPinstance-attribute**

FORMAT\_ALIGN\_RIGHT\_SHARP: IconData

**buildFORMAT\_BOLDinstance-attribute**

FORMAT\_BOLD: IconData

**buildFORMAT\_BOLD\_OUTLINEDinstance-attribute**

FORMAT\_BOLD\_OUTLINED: IconData

**buildFORMAT\_BOLD\_ROUNDEDinstance-attribute**

FORMAT\_BOLD\_ROUNDED: IconData

**buildFORMAT\_BOLD\_SHARPinstance-attribute**

FORMAT\_BOLD\_SHARP: IconData

**buildFORMAT\_CLEARinstance-attribute**

FORMAT\_CLEAR: IconData

**buildFORMAT\_CLEAR\_OUTLINEDinstance-attribute**

FORMAT\_CLEAR\_OUTLINED: IconData

**buildFORMAT\_CLEAR\_ROUNDEDinstance-attribute**

FORMAT\_CLEAR\_ROUNDED: IconData

**buildFORMAT\_CLEAR\_SHARPinstance-attribute**

FORMAT\_CLEAR\_SHARP: IconData

**buildFORMAT\_COLOR\_FILLinstance-attribute**

FORMAT\_COLOR\_FILL: IconData



**buildFORMAT\_COLOR\_FILL\_OUTLINEDinstance-attribute**

FORMAT\_COLOR\_FILL\_OUTLINED: IconData

**buildFORMAT\_COLOR\_FILL\_ROUNDEDinstance-attribute**

FORMAT\_COLOR\_FILL\_ROUNDED: IconData

**buildFORMAT\_COLOR\_FILL\_SHARPinstance-attribute**

FORMAT\_COLOR\_FILL\_SHARP: IconData

**buildFORMAT\_COLOR\_RESETinstance-attribute**

FORMAT\_COLOR\_RESET: IconData

**buildFORMAT\_COLOR\_RESET\_OUTLINEDinstance-attribute**

FORMAT\_COLOR\_RESET\_OUTLINED: IconData

**buildFORMAT\_COLOR\_RESET\_ROUNDEDinstance-attribute**

FORMAT\_COLOR\_RESET\_ROUNDED: IconData

**buildFORMAT\_COLOR\_RESET\_SHARPinstance-attribute**

FORMAT\_COLOR\_RESET\_SHARP: IconData

**buildFORMAT\_COLOR\_TEXTinstance-attribute**

FORMAT\_COLOR\_TEXT: IconData

**buildFORMAT\_COLOR\_TEXT\_OUTLINEDinstance-attribute**

FORMAT\_COLOR\_TEXT\_OUTLINED: IconData

**buildFORMAT\_COLOR\_TEXT\_ROUNDEDinstance-attribute**

FORMAT\_COLOR\_TEXT\_ROUNDED: IconData

**buildFORMAT\_COLOR\_TEXT\_SHARPinstance-attribute**

FORMAT\_COLOR\_TEXT\_SHARP: IconData

**buildFORMAT\_INDENT\_DECREASEinstance-attribute**

FORMAT\_INDENT\_DECREASE: IconData

**buildFORMAT\_INDENT\_DECREASE\_OUTLINEDinstance-attribute**

FORMAT\_INDENT\_DECREASE\_OUTLINED: IconData

**buildFORMAT\_INDENT\_DECREASE\_ROUNDEDinstance-attribute**

FORMAT\_INDENT\_DECREASE\_ROUNDED: IconData

**buildFORMAT\_INDENT\_DECREASE\_SHARPinstance-attribute**

FORMAT\_INDENT\_DECREASE\_SHARP: IconData

**buildFORMAT\_INDENT\_INCREASEinstance-attribute**

FORMAT\_INDENT\_INCREASE: IconData

**buildFORMAT\_INDENT\_INCREASE\_OUTLINEDinstance-attribute**

FORMAT\_INDENT\_INCREASE\_OUTLINED: IconData

**buildFORMAT\_INDENT\_INCREASE\_ROUNDEDinstance-attribute**

FORMAT\_INDENT\_INCREASE\_ROUNDED: IconData

**buildFORMAT\_INDENT\_INCREASE\_SHARPinstance-attribute**

FORMAT\_INDENT\_INCREASE\_SHARP: IconData

**buildFORMAT\_ITALICinstance-attribute**

FORMAT\_ITALIC: IconData

**buildFORMAT\_ITALIC\_OUTLINEDinstance-attribute**

FORMAT\_ITALIC\_OUTLINED: IconData

**buildFORMAT\_ITALIC\_ROUNDEDinstance-attribute**

FORMAT\_ITALIC\_ROUNDED: IconData

**buildFORMAT\_ITALIC\_SHARPinstance-attribute**

FORMAT\_ITALIC\_SHARP: IconData

**buildFORMAT\_LINE\_SPACINGinstance-attribute**

FORMAT\_LINE\_SPACING: IconData

**buildFORMAT\_LINE\_SPACING\_OUTLINEDinstance-attribute**

FORMAT\_LINE\_SPACING\_OUTLINED: IconData

**buildFORMAT\_LINE\_SPACING\_ROUNDEDinstance-attribute**

FORMAT\_LINE\_SPACING\_ROUNDED: IconData

**buildFORMAT\_LINE\_SPACING\_SHARPinstance-attribute**

FORMAT\_LINE\_SPACING\_SHARP: IconData

**buildFORMAT\_LIST\_BULLETEDinstance-attribute**

FORMAT\_LIST\_BULLETED: IconData

**buildFORMAT\_LIST\_BULLETED\_ADDinstance-attribute**

FORMAT\_LIST\_BULLETED\_ADD: IconData

**buildFORMAT\_LIST\_BULLETED\_OUTLINEDinstance-attribute**

FORMAT\_LIST\_BULLETED\_OUTLINED: IconData

**buildFORMAT\_LIST\_BULLETED\_ROUNDEDinstance-attribute**

FORMAT\_LIST\_BULLETED\_ROUNDED: IconData

**buildFORMAT\_LIST\_BULLETED\_SHARPinstance-attribute**

FORMAT\_LIST\_BULLETED\_SHARP: IconData

**buildFORMAT\_LIST\_NUMBEREDinstance-attribute**

FORMAT\_LIST\_NUMBERED: IconData

**buildFORMAT\_LIST\_NUMBERED\_OUTLINEDinstance-attribute**

FORMAT\_LIST\_NUMBERED\_OUTLINED: IconData

**buildFORMAT\_LIST\_NUMBERED\_ROUNDEDinstance-attribute**

FORMAT\_LIST\_NUMBERED\_ROUNDED: IconData

**buildFORMAT\_LIST\_NUMBERED\_RTListance-attribute**

FORMAT\_LIST\_NUMBERED\_RTL: IconData

**buildFORMAT\_LIST\_NUMBERED\_RTL\_OUTLINEDinstance-attribute**

FORMAT\_LIST\_NUMBERED\_RTL\_OUTLINED: IconData

**buildFORMAT\_LIST\_NUMBERED\_RTL\_ROUNDEDinstance-attribute**

FORMAT\_LIST\_NUMBERED\_RTL\_ROUNDED: IconData

**buildFORMAT\_LIST\_NUMBERED\_RTL\_SHARPinstance-attribute**

FORMAT\_LIST\_NUMBERED\_RTL\_SHARP: IconData

**buildFORMAT\_LIST\_NUMBERED\_SHARPinstance-attribute**

FORMAT\_LIST\_NUMBERED\_SHARP: IconData

**buildFORMAT\_OVERLINEinstance-attribute**

FORMAT\_OVERLINE: IconData

**buildFORMAT\_OVERLINE\_OUTLINEDinstance-attribute**

FORMAT\_OVERLINE\_OUTLINED: IconData

**buildFORMAT\_OVERLINE\_ROUNDEDinstance-attribute**

FORMAT\_OVERLINE\_ROUNDED: IconData

**buildFORMAT\_OVERLINE\_SHARPinstance-attribute**

FORMAT\_OVERLINE\_SHARP: IconData

**buildFORMAT\_PAINTinstance-attribute**

FORMAT\_PAINT: IconData

**buildFORMAT\_PAINT\_OUTLINEDinstance-attribute**

FORMAT\_PAINT\_OUTLINED: IconData

**buildFORMAT\_PAINT\_ROUNDEDinstance-attribute**

FORMAT\_PAINT\_ROUNDED: IconData

**buildFORMAT\_PAINT\_SHARPinstance-attribute**

FORMAT\_PAINT\_SHARP: IconData

**buildFORMAT\_QUOTEinstance-attribute**

FORMAT\_QUOTE: IconData

**buildFORMAT\_QUOTE\_OUTLINEDinstance-attribute**

FORMAT\_QUOTE\_OUTLINED: IconData

**buildFORMAT\_QUOTE\_ROUNDEDinstance-attribute**

FORMAT\_QUOTE\_ROUNDED: IconData

**buildFORMAT\_QUOTE\_SHARPinstance-attribute**

FORMAT\_QUOTE\_SHARP: IconData

**buildFORMAT\_SHAPESinstance-attribute**

FORMAT\_SHAPES: IconData

**buildFORMAT\_SHAPES\_OUTLINEDinstance-attribute**

FORMAT\_SHAPES\_OUTLINED: IconData

**buildFORMAT\_SHAPES\_ROUNDEDinstance-attribute**

FORMAT\_SHAPES\_ROUNDED: IconData

**buildFORMAT\_SHAPES\_SHARPinstance-attribute**

FORMAT\_SHAPES\_SHARP: IconData

**buildFORMAT\_SIZEinstance-attribute**

FORMAT\_SIZE: IconData

**buildFORMAT\_SIZE\_OUTLINEDinstance-attribute**

FORMAT\_SIZE\_OUTLINED: IconData

**buildFORMAT\_SIZE\_ROUNDEDinstance-attribute**

FORMAT\_SIZE\_ROUNDED: IconData

**buildFORMAT\_SIZE\_SHARPinstance-attribute**

FORMAT\_SIZE\_SHARP: IconData

**buildFORMAT\_STRIKETHROUGHinstance-attribute**

FORMAT\_STRIKETHROUGH: IconData

**buildFORMAT\_STRIKETHROUGH\_OUTLINEDinstance-attribute**

FORMAT\_STRIKETHROUGH\_OUTLINED: IconData

**buildFORMAT\_STRIKETHROUGH\_ROUNDEDinstance-attribute**

FORMAT\_STRIKETHROUGH\_ROUNDED: IconData

**buildFORMAT\_STRIKETHROUGH\_SHARPinstance-attribute**

FORMAT\_STRIKETHROUGH\_SHARP: IconData

**buildFORMAT\_TEXTDIRECTION\_L\_TO\_Rinstance-attribute**

FORMAT\_TEXTDIRECTION\_L\_TO\_R: IconData

**buildFORMAT\_TEXTDIRECTION\_L\_TO\_R\_OUTLINEDinstance-attribute**

FORMAT\_TEXTDIRECTION\_L\_TO\_R\_OUTLINED: IconData

**buildFORMAT\_TEXTDIRECTION\_L\_TO\_R\_ROUNDED**instance-attribute

FORMAT\_TEXTDIRECTION\_L\_TO\_R\_ROUNDED: IconData

**buildFORMAT\_TEXTDIRECTION\_L\_TO\_R\_SHARP**instance-attribute

FORMAT\_TEXTDIRECTION\_L\_TO\_R\_SHARP: IconData

**buildFORMAT\_TEXTDIRECTION\_R\_TO\_L**instance-attribute

FORMAT\_TEXTDIRECTION\_R\_TO\_L: IconData

**buildFORMAT\_TEXTDIRECTION\_R\_TO\_L\_OUTLINED**instance-attribute

FORMAT\_TEXTDIRECTION\_R\_TO\_L\_OUTLINED: IconData

**buildFORMAT\_TEXTDIRECTION\_R\_TO\_L\_ROUNDED**instance-attribute

FORMAT\_TEXTDIRECTION\_R\_TO\_L\_ROUNDED: IconData

**buildFORMAT\_TEXTDIRECTION\_R\_TO\_L\_SHARP**instance-attribute

FORMAT\_TEXTDIRECTION\_R\_TO\_L\_SHARP: IconData

**buildFORMAT\_UNDERLINE**instance-attribute

FORMAT\_UNDERLINE: IconData

**buildFORMAT\_UNDERLINE\_OUTLINED**instance-attribute

FORMAT\_UNDERLINE\_OUTLINED: IconData

**buildFORMAT\_UNDERLINE\_ROUNDED**instance-attribute

FORMAT\_UNDERLINE\_ROUNDED: IconData

**buildFORMAT\_UNDERLINE\_SHARP**instance-attribute

FORMAT\_UNDERLINE\_SHARP: IconData

**buildFORMAT\_UNDERLINED**instance-attribute

FORMAT\_UNDERLINED: IconData

**buildFORMAT\_UNDERLINED\_OUTLINED**instance-attribute

FORMAT\_UNDERLINED\_OUTLINED: IconData

**buildFORMAT\_UNDERLINED\_ROUNDED**instance-attribute

FORMAT\_UNDERLINED\_ROUNDED: IconData

**buildFORMAT\_UNDERLINED\_SHARP**instance-attribute

FORMAT\_UNDERLINED\_SHARP: IconData

**buildFORT**instance-attribute

FORT: IconData

**buildFORT\_OUTLINED**instance-attribute

FORT\_OUTLINED: IconData

**buildFORT\_ROUNDED**instance-attribute

FORT\_ROUNDED: IconData

**buildFORT\_SHARP**instance-attribute

FORT\_SHARP: IconData

**buildFORUM**instance-attribute

FORUM: IconData

**buildFORUM\_OUTLINED**instance-attribute

FORUM\_OUTLINED: IconData

**buildFORUM\_ROUNDED**instance-attribute

FORUM\_ROUNDED: IconData

**buildFORUM\_SHARP**instance-attribute

FORUM\_SHARP: IconData

**buildFORWARDinstance-attribute**

FORWARD: IconData

**buildFORWARD\_10instance-attribute**

FORWARD\_10: IconData

**buildFORWARD\_10\_OUTLINEDinstance-attribute**

FORWARD\_10\_OUTLINED: IconData

**buildFORWARD\_10\_ROUNDEDinstance-attribute**

FORWARD\_10\_ROUNDED: IconData

**buildFORWARD\_10\_SHARPinstance-attribute**

FORWARD\_10\_SHARP: IconData

**buildFORWARD\_30instance-attribute**

FORWARD\_30: IconData

**buildFORWARD\_30\_OUTLINEDinstance-attribute**

FORWARD\_30\_OUTLINED: IconData

**buildFORWARD\_30\_ROUNDEDinstance-attribute**

FORWARD\_30\_ROUNDED: IconData

**buildFORWARD\_30\_SHARPinstance-attribute**

FORWARD\_30\_SHARP: IconData

**buildFORWARD\_5instance-attribute**

FORWARD\_5: IconData

**buildFORWARD\_5\_OUTLINEDinstance-attribute**

FORWARD\_5\_OUTLINED: IconData

**buildFORWARD\_5\_ROUNDEDinstance-attribute**

FORWARD\_5\_ROUNDED: IconData

**buildFORWARD\_5\_SHARPinstance-attribute**

FORWARD\_5\_SHARP: IconData

**buildFORWARD\_OUTLINEDinstance-attribute**

FORWARD\_OUTLINED: IconData

**buildFORWARD\_ROUNDEDinstance-attribute**

FORWARD\_ROUNDED: IconData

**buildFORWARD\_SHARPinstance-attribute**

FORWARD\_SHARP: IconData

**buildFORWARD\_TO\_INBOXinstance-attribute**

FORWARD\_TO\_INBOX: IconData

**buildFORWARD\_TO\_INBOX\_OUTLINEDinstance-attribute**

FORWARD\_TO\_INBOX\_OUTLINED: IconData

**buildFORWARD\_TO\_INBOX\_ROUNDEDinstance-attribute**

FORWARD\_TO\_INBOX\_ROUNDED: IconData

**buildFORWARD\_TO\_INBOX\_SHARPinstance-attribute**

FORWARD\_TO\_INBOX\_SHARP: IconData

**buildFOUNDATIONinstance-attribute**

FOUNDATION: IconData

**buildFOUNDATION\_OUTLINEDinstance-attribute**

FOUNDATION\_OUTLINED: IconData

**buildFOUNDATION\_ROUNDEDinstance-attribute**

FOUNDATION\_ROUNDED: IconData

**buildFOUNDATION\_SHARPinstance-attribute**

FOUNDATION\_SHARP: IconData

**buildFOUR\_G\_MOBILEDATAinstance-attribute**

FOUR\_G\_MOBILEDATA: IconData

**buildFOUR\_G\_MOBILEDATA\_OUTLINEDinstance-attribute**

FOUR\_G\_MOBILEDATA\_OUTLINED: IconData

**buildFOUR\_G\_MOBILEDATA\_ROUNDEDinstance-attribute**

FOUR\_G\_MOBILEDATA\_ROUNDED: IconData

**buildFOUR\_G\_MOBILEDATA\_SHARPinstance-attribute**

FOUR\_G\_MOBILEDATA\_SHARP: IconData

**buildFOUR\_G\_PLUS\_MOBILEDATAinstance-attribute**

FOUR\_G\_PLUS\_MOBILEDATA: IconData

**buildFOUR\_G\_PLUS\_MOBILEDATA\_OUTLINEDinstance-attribute**

FOUR\_G\_PLUS\_MOBILEDATA\_OUTLINED: IconData

**buildFOUR\_G\_PLUS\_MOBILEDATA\_ROUNDEDinstance-attribute**

FOUR\_G\_PLUS\_MOBILEDATA\_ROUNDED: IconData

**buildFOUR\_G\_PLUS\_MOBILEDATA\_SHARPinstance-attribute**

FOUR\_G\_PLUS\_MOBILEDATA\_SHARP: IconData

**buildFOUR\_Kinstance-attribute**

FOUR\_K: IconData

**buildFOUR\_K\_OUTLINEDinstance-attribute**

FOUR\_K\_OUTLINED: IconData

**buildFOUR\_K\_PLUSinstance-attribute**

FOUR\_K\_PLUS: IconData

**buildFOUR\_K\_PLUS\_OUTLINEDinstance-attribute**

FOUR\_K\_PLUS\_OUTLINED: IconData

**buildFOUR\_K\_PLUS\_ROUNDEDinstance-attribute**

FOUR\_K\_PLUS\_ROUNDED: IconData

**buildFOUR\_K\_PLUS\_SHARPinstance-attribute**

FOUR\_K\_PLUS\_SHARP: IconData

**buildFOUR\_K\_ROUNDEDinstance-attribute**

FOUR\_K\_ROUNDED: IconData

**buildFOUR\_K\_SHARPinstance-attribute**

FOUR\_K\_SHARP: IconData

**buildFOUR\_MPinstance-attribute**

FOUR\_MP: IconData

**buildFOUR\_MP\_OUTLINEDinstance-attribute**

FOUR\_MP\_OUTLINED: IconData

**buildFOUR\_MP\_ROUNDEDinstance-attribute**

FOUR\_MP\_ROUNDED: IconData

**buildFOUR\_MP\_SHARPinstance-attribute**

FOUR\_MP\_SHARP: IconData

**buildFOURTEEN\_MPinstance-attribute**

FOURTEEN\_MP: IconData

**buildFOURTEEN\_MP\_OUTLINEDinstance-attribute**

FOURTEEN\_MP\_OUTLINED: IconData

**buildFOURTEEN\_MP\_ROUNDEDinstance-attribute**

FOURTEEN\_MP\_ROUNDED: IconData

**buildFOURTEEN\_MP\_SHARPinstance-attribute**

FOURTEEN\_MP\_SHARP: IconData

**buildFREE\_BREAKFASTinstance-attribute**

FREE\_BREAKFAST: IconData

**buildFREE\_BREAKFAST\_OUTLINEDinstance-attribute**

FREE\_BREAKFAST\_OUTLINED: IconData

**buildFREE\_BREAKFAST\_ROUNDEDinstance-attribute**

FREE\_BREAKFAST\_ROUNDED: IconData

**buildFREE\_BREAKFAST\_SHARPinstance-attribute**

FREE\_BREAKFAST\_SHARP: IconData

**buildFREE\_CANCELLATIONinstance-attribute**

FREE\_CANCELLATION: IconData

**buildFREE\_CANCELLATION\_OUTLINEDinstance-attribute**

FREE\_CANCELLATION\_OUTLINED: IconData

**buildFREE\_CANCELLATION\_ROUNDEDinstance-attribute**

FREE\_CANCELLATION\_ROUNDED: IconData

**buildFREE\_CANCELLATION\_SHARPinstance-attribute**

FREE\_CANCELLATION\_SHARP: IconData

**buildFRONT\_HANDinstance-attribute**

FRONT\_HAND: IconData

**buildFRONT\_HAND\_OUTLINEDinstance-attribute**

FRONT\_HAND\_OUTLINED: IconData

**buildFRONT\_HAND\_ROUNDEDinstance-attribute**

FRONT\_HAND\_ROUNDED: IconData

**buildFRONT\_HAND\_SHARPinstance-attribute**

FRONT\_HAND\_SHARP: IconData

**buildFRONT\_LOADERinstance-attribute**

FRONT\_LOADER: IconData

**buildFULLSCREENinstance-attribute**

FULLSCREEN: IconData

**buildFULLSCREEN\_EXITinstance-attribute**

FULLSCREEN\_EXIT: IconData

**buildFULLSCREEN\_EXIT\_OUTLINEDinstance-attribute**

FULLSCREEN\_EXIT\_OUTLINED: IconData

**buildFULLSCREEN\_EXIT\_ROUNDEDinstance-attribute**

FULLSCREEN\_EXIT\_ROUNDED: IconData

**buildFULLSCREEN\_EXIT\_SHARPinstance-attribute**

FULLSCREEN\_EXIT\_SHARP: IconData

**buildFULLSCREEN\_OUTLINED**instance-attribute

FULLSCREEN\_OUTLINED: IconData

**buildFULLSCREEN\_ROUNDED**instance-attribute

FULLSCREEN\_ROUNDED: IconData

**buildFULLSCREEN\_SHARP**instance-attribute

FULLSCREEN\_SHARP: IconData

**buildFUNCTIONS**instance-attribute

FUNCTIONS: IconData

**buildFUNCTIONS\_OUTLINED**instance-attribute

FUNCTIONS\_OUTLINED: IconData

**buildFUNCTIONS\_ROUNDED**instance-attribute

FUNCTIONS\_ROUNDED: IconData

**buildFUNCTIONS\_SHARP**instance-attribute

FUNCTIONS\_SHARP: IconData

**buildG\_MOBILEDATA**instance-attribute

G\_MOBILEDATA: IconData

**buildG\_MOBILEDATA\_OUTLINED**instance-attribute

G\_MOBILEDATA\_OUTLINED: IconData

**buildG\_MOBILEDATA\_ROUNDED**instance-attribute

G\_MOBILEDATA\_ROUNDED: IconData

**buildG\_MOBILEDATA\_SHARP**instance-attribute

G\_MOBILEDATA\_SHARP: IconData

**buildG\_TRANSLATE**instance-attribute

G\_TRANSLATE: IconData

**buildG\_TRANSLATE\_OUTLINED**instance-attribute

G\_TRANSLATE\_OUTLINED: IconData

**buildG\_TRANSLATE\_ROUNDED**instance-attribute

G\_TRANSLATE\_ROUNDED: IconData

**buildG\_TRANSLATE\_SHARP**instance-attribute

G\_TRANSLATE\_SHARP: IconData

**buildGAMEPAD**instance-attribute

GAMEPAD: IconData

**buildGAMEPAD\_OUTLINED**instance-attribute

GAMEPAD\_OUTLINED: IconData

**buildGAMEPAD\_ROUNDED**instance-attribute

GAMEPAD\_ROUNDED: IconData

**buildGAMEPAD\_SHARP**instance-attribute

GAMEPAD\_SHARP: IconData

**buildGAMES**instance-attribute

GAMES: IconData

**buildGAMES\_OUTLINED**instance-attribute

GAMES\_OUTLINED: IconData

**buildGAMES\_ROUNDED**instance-attribute

GAMES\_ROUNDED: IconData



**buildGAMES\_SHARPinstance-attribute**

GAMES\_SHARP: IconData

**buildGARAGEinstance-attribute**

GARAGE: IconData

**buildGARAGE\_OUTLINEDinstance-attribute**

GARAGE\_OUTLINED: IconData

**buildGARAGE\_ROUNDEDinstance-attribute**

GARAGE\_ROUNDED: IconData

**buildGARAGE\_SHARPinstance-attribute**

GARAGE\_SHARP: IconData

**buildGAS\_METERinstance-attribute**

GAS\_METER: IconData

**buildGAS\_METER\_OUTLINEDinstance-attribute**

GAS\_METER\_OUTLINED: IconData

**buildGAS\_METER\_ROUNDEDinstance-attribute**

GAS\_METER\_ROUNDED: IconData

**buildGAS\_METER\_SHARPinstance-attribute**

GAS\_METER\_SHARP: IconData

**buildGAVELinstance-attribute**

GAVEL: IconData

**buildGAVEL\_OUTLINEDinstance-attribute**

GAVEL\_OUTLINED: IconData

**buildGAVEL\_ROUNDEDinstance-attribute**

GAVEL\_ROUNDED: IconData

**buildGAVEL\_SHARPinstance-attribute**

GAVEL\_SHARP: IconData

**buildGENERATING\_TOKENSinstance-attribute**

GENERATING\_TOKENS: IconData

**buildGENERATING\_TOKENS\_OUTLINEDinstance-attribute**

GENERATING\_TOKENS\_OUTLINED: IconData

**buildGENERATING\_TOKENS\_ROUNDEDinstance-attribute**

GENERATING\_TOKENS\_ROUNDED: IconData

**buildGENERATING\_TOKENS\_SHARPinstance-attribute**

GENERATING\_TOKENS\_SHARP: IconData

**buildGESTUREinstance-attribute**

GESTURE: IconData

**buildGESTURE\_OUTLINEDinstance-attribute**

GESTURE\_OUTLINED: IconData

**buildGESTURE\_ROUNDEDinstance-attribute**

GESTURE\_ROUNDED: IconData

**buildGESTURE\_SHARPinstance-attribute**

GESTURE\_SHARP: IconData

**buildGET\_APPinstance-attribute**

GET\_APP: IconData

**buildGET\_APP\_OUTLINEDinstance-attribute**

GET\_APP\_OUTLINED: IconData

**buildGET\_APP\_ROUNDEDinstance-attribute**

GET\_APP\_ROUNDED: IconData

**buildGET\_APP\_SHARPinstance-attribute**

GET\_APP\_SHARP: IconData

**buildGIFinstance-attribute**

GIF: IconData

**buildGIF\_BOXinstance-attribute**

GIF\_BOX: IconData

**buildGIF\_BOX\_OUTLINEDinstance-attribute**

GIF\_BOX\_OUTLINED: IconData

**buildGIF\_BOX\_ROUNDEDinstance-attribute**

GIF\_BOX\_ROUNDED: IconData

**buildGIF\_BOX\_SHARPinstance-attribute**

GIF\_BOX\_SHARP: IconData

**buildGIF\_OUTLINEDinstance-attribute**

GIF\_OUTLINED: IconData

**buildGIF\_ROUNDEDinstance-attribute**

GIF\_ROUNDED: IconData

**buildGIF\_SHARPinstance-attribute**

GIF\_SHARP: IconData

**buildGIRLinstance-attribute**

GIRL: IconData

**buildGIRL\_OUTLINEDinstance-attribute**

GIRL\_OUTLINED: IconData

**buildGIRL\_ROUNDEDinstance-attribute**

GIRL\_ROUNDED: IconData

**buildGIRL\_SHARPinstance-attribute**

GIRL\_SHARP: IconData

**buildGITEinstance-attribute**

GITE: IconData

**buildGITE\_OUTLINEDinstance-attribute**

GITE\_OUTLINED: IconData

**buildGITE\_ROUNDEDinstance-attribute**

GITE\_ROUNDED: IconData

**buildGITE\_SHARPinstance-attribute**

GITE\_SHARP: IconData

**buildGOLF\_COURSEinstance-attribute**

GOLF\_COURSE: IconData

**buildGOLF\_COURSE\_OUTLINEDinstance-attribute**

GOLF\_COURSE\_OUTLINED: IconData

**buildGOLF\_COURSE\_ROUNDEDinstance-attribute**

GOLF\_COURSE\_ROUNDED: IconData

**buildGOLF\_COURSE\_SHARPinstance-attribute**

GOLF\_COURSE\_SHARP: IconData

**buildGPP\_BADinstance-attribute**

GPP\_BAD: IconData

**buildGPP\_BAD\_OUTLINEDinstance-attribute**

GPP\_BAD\_OUTLINED: IconData

**buildGPP\_BAD\_ROUNDEDinstance-attribute**

GPP\_BAD\_ROUNDED: IconData

**buildGPP\_BAD\_SHARPinstance-attribute**

GPP\_BAD\_SHARP: IconData

**buildGPP\_GOODinstance-attribute**

GPP\_GOOD: IconData

**buildGPP\_GOOD\_OUTLINEDinstance-attribute**

GPP\_GOOD\_OUTLINED: IconData

**buildGPP\_GOOD\_ROUNDEDinstance-attribute**

GPP\_GOOD\_ROUNDED: IconData

**buildGPP\_GOOD\_SHARPinstance-attribute**

GPP\_GOOD\_SHARP: IconData

**buildGPP\_MAYBEinstance-attribute**

GPP\_MAYBE: IconData

**buildGPP\_MAYBE\_OUTLINEDinstance-attribute**

GPP\_MAYBE\_OUTLINED: IconData

**buildGPP\_MAYBE\_ROUNDEDinstance-attribute**

GPP\_MAYBE\_ROUNDED: IconData

**buildGPP\_MAYBE\_SHARPinstance-attribute**

GPP\_MAYBE\_SHARP: IconData

**buildGPS\_FIXEDinstance-attribute**

GPS\_FIXED: IconData

**buildGPS\_FIXED\_OUTLINEDinstance-attribute**

GPS\_FIXED\_OUTLINED: IconData

**buildGPS\_FIXED\_ROUNDEDinstance-attribute**

GPS\_FIXED\_ROUNDED: IconData

**buildGPS\_FIXED\_SHARPinstance-attribute**

GPS\_FIXED\_SHARP: IconData

**buildGPS\_NOT\_FIXEDinstance-attribute**

GPS\_NOT\_FIXED: IconData

**buildGPS\_NOT\_FIXED\_OUTLINEDinstance-attribute**

GPS\_NOT\_FIXED\_OUTLINED: IconData

**buildGPS\_NOT\_FIXED\_ROUNDEDinstance-attribute**

GPS\_NOT\_FIXED\_ROUNDED: IconData

**buildGPS\_NOT\_FIXED\_SHARPinstance-attribute**

GPS\_NOT\_FIXED\_SHARP: IconData

**buildGPS\_OFFinstance-attribute**

GPS\_OFF: IconData

**buildGPS\_OFF\_OUTLINEDinstance-attribute**

GPS\_OFF\_OUTLINED: IconData

**buildGPS\_OFF\_ROUNDEDinstance-attribute**

GPS\_OFF\_ROUNDED: IconData

**buildGPS\_OFF\_SHARPinstance-attribute**

GPS\_OFF\_SHARP: IconData

**buildGRADEinstance-attribute**

GRADE: IconData

**buildGRADE\_OUTLINEDinstance-attribute**

GRADE\_OUTLINED: IconData

**buildGRADE\_ROUNDEDinstance-attribute**

GRADE\_ROUNDED: IconData

**buildGRADE\_SHARPinstance-attribute**

GRADE\_SHARP: IconData

**buildGRADIENTinstance-attribute**

GRADIENT: IconData

**buildGRADIENT\_OUTLINEDinstance-attribute**

GRADIENT\_OUTLINED: IconData

**buildGRADIENT\_ROUNDEDinstance-attribute**

GRADIENT\_ROUNDED: IconData

**buildGRADIENT\_SHARPinstance-attribute**

GRADIENT\_SHARP: IconData

**buildGRADINGinstance-attribute**

GRADING: IconData

**buildGRADING\_OUTLINEDinstance-attribute**

GRADING\_OUTLINED: IconData

**buildGRADING\_ROUNDEDinstance-attribute**

GRADING\_ROUNDED: IconData

**buildGRADING\_SHARPinstance-attribute**

GRADING\_SHARP: IconData

**buildGRAINinstance-attribute**

GRAIN: IconData

**buildGRAIN\_OUTLINEDinstance-attribute**

GRAIN\_OUTLINED: IconData

**buildGRAIN\_ROUNDEDinstance-attribute**

GRAIN\_ROUNDED: IconData

**buildGRAIN\_SHARPinstance-attribute**

GRAIN\_SHARP: IconData

**buildGRAPHIC\_EQinstance-attribute**

GRAPHIC\_EQ: IconData

**buildGRAPHIC\_EQ\_OUTLINEDinstance-attribute**

GRAPHIC\_EQ\_OUTLINED: IconData

**buildGRAPHIC\_EQ\_ROUNDEDinstance-attribute**

GRAPHIC\_EQ\_ROUNDED: IconData

**buildGRAPHIC\_EQ\_SHARPinstance-attribute**

GRAPHIC\_EQ\_SHARP: IconData

**buildGRASSinstance-attribute**

GRASS: IconData

**buildGRASS\_OUTLINEDinstance-attribute**

GRASS\_OUTLINED: IconData

**buildGRASS\_ROUNDEDinstance-attribute**

GRASS\_ROUNDED: IconData

**buildGRASS\_SHARPinstance-attribute**

GRASS\_SHARP: IconData

**buildGRID\_3X3instance-attribute**

GRID\_3X3: IconData

**buildGRID\_3X3\_OUTLINEDinstance-attribute**

GRID\_3X3\_OUTLINED: IconData

**buildGRID\_3X3\_ROUNDEDinstance-attribute**

GRID\_3X3\_ROUNDED: IconData

**buildGRID\_3X3\_SHARPinstance-attribute**

GRID\_3X3\_SHARP: IconData

**buildGRID\_4X4instance-attribute**

GRID\_4X4: IconData

**buildGRID\_4X4\_OUTLINEDinstance-attribute**

GRID\_4X4\_OUTLINED: IconData

**buildGRID\_4X4\_ROUNDEDinstance-attribute**

GRID\_4X4\_ROUNDED: IconData

**buildGRID\_4X4\_SHARPinstance-attribute**

GRID\_4X4\_SHARP: IconData

**buildGRID\_GOLDENRATIOinstance-attribute**

GRID\_GOLDENRATIO: IconData

**buildGRID\_GOLDENRATIO\_OUTLINEDinstance-attribute**

GRID\_GOLDENRATIO\_OUTLINED: IconData

**buildGRID\_GOLDENRATIO\_ROUNDEDinstance-attribute**

GRID\_GOLDENRATIO\_ROUNDED: IconData

**buildGRID\_GOLDENRATIO\_SHARPinstance-attribute**

GRID\_GOLDENRATIO\_SHARP: IconData

**buildGRID\_OFFinstance-attribute**

GRID\_OFF: IconData

**buildGRID\_OFF\_OUTLINEDinstance-attribute**

GRID\_OFF\_OUTLINED: IconData

**buildGRID\_OFF\_ROUNDEDinstance-attribute**

GRID\_OFF\_ROUNDED: IconData

**buildGRID\_OFF\_SHARPinstance-attribute**

GRID\_OFF\_SHARP: IconData

**buildGRID\_ONinstance-attribute**

GRID\_ON: IconData

**buildGRID\_ON\_OUTLINEDinstance-attribute**

GRID\_ON\_OUTLINED: IconData

**buildGRID\_ON\_ROUNDEDinstance-attribute**

GRID\_ON\_ROUNDED: IconData

**buildGRID\_ON\_SHARPinstance-attribute**

GRID\_ON\_SHARP: IconData

**buildGRID\_VIEWinstance-attribute**

GRID\_VIEW: IconData

**buildGRID\_VIEW\_OUTLINEDinstance-attribute**

GRID\_VIEW\_OUTLINED: IconData

**buildGRID\_VIEW\_ROUNDEDinstance-attribute**

GRID\_VIEW\_ROUNDED: IconData

**buildGRID\_VIEW\_SHARPinstance-attribute**

GRID\_VIEW\_SHARP: IconData

**buildGROUPinstance-attribute**

GROUP: IconData

**buildGROUP\_ADDinstance-attribute**

GROUP\_ADD: IconData

**buildGROUP\_ADD\_OUTLINEDinstance-attribute**

GROUP\_ADD\_OUTLINED: IconData

**buildGROUP\_ADD\_ROUNDEDinstance-attribute**

GROUP\_ADD\_ROUNDED: IconData

**buildGROUP\_ADD\_SHARPinstance-attribute**

GROUP\_ADD\_SHARP: IconData

**buildGROUP\_OFFinstance-attribute**

GROUP\_OFF: IconData

**buildGROUP\_OFF\_OUTLINEDinstance-attribute**

GROUP\_OFF\_OUTLINED: IconData

**buildGROUP\_OFF\_ROUNDEDinstance-attribute**

GROUP\_OFF\_ROUNDED: IconData

**buildGROUP\_OFF\_SHARPinstance-attribute**

GROUP\_OFF\_SHARP: IconData

**buildGROUP\_OUTLINEDinstance-attribute**

GROUP\_OUTLINED: IconData

**buildGROUP\_REMOVEinstance-attribute**

GROUP\_REMOVE: IconData

**buildGROUP\_REMOVE\_OUTLINEDinstance-attribute**

GROUP\_REMOVE\_OUTLINED: IconData

**buildGROUP\_REMOVE\_ROUNDEDinstance-attribute**

GROUP\_REMOVE\_ROUNDED: IconData

**buildGROUP\_REMOVE\_SHARPinstance-attribute**

GROUP\_REMOVE\_SHARP: IconData

**buildGROUP\_ROUNDEDinstance-attribute**

GROUP\_ROUNDED: IconData

**buildGROUP\_SHARPinstance-attribute**

GROUP\_SHARP: IconData

**buildGROUP\_WORKinstance-attribute**

GROUP\_WORK: IconData

**buildGROUP\_WORK\_OUTLINEDinstance-attribute**

GROUP\_WORK\_OUTLINED: IconData

**buildGROUP\_WORK\_ROUNDEDinstance-attribute**

GROUP\_WORK\_ROUNDED: IconData

**buildGROUP\_WORK\_SHARPinstance-attribute**

GROUP\_WORK\_SHARP: IconData

**buildGROUPSinstance-attribute**

GROUPS: IconData

**buildGROUPS\_2instance-attribute**

GROUPS\_2: IconData

**buildGROUPS\_2\_OUTLINEDinstance-attribute**

GROUPS\_2\_OUTLINED: IconData

**buildGROUPS\_2\_ROUNDEDinstance-attribute**

GROUPS\_2\_ROUNDED: IconData

**buildGROUPS\_2\_SHARPinstance-attribute**

GROUPS\_2\_SHARP: IconData

**buildGROUPS\_3instance-attribute**

GROUPS\_3: IconData

**buildGROUPS\_3\_OUTLINEDinstance-attribute**

GROUPS\_3\_OUTLINED: IconData

**buildGROUPS\_3\_ROUNDEDinstance-attribute**

GROUPS\_3\_ROUNDED: IconData

**buildGROUPS\_3\_SHARPinstance-attribute**

GROUPS\_3\_SHARP: IconData

**buildGROUPS\_OUTLINEDinstance-attribute**

GROUPS\_OUTLINED: IconData

**buildGROUPS\_ROUNDEDinstance-attribute**

GROUPS\_ROUNDED: IconData

**buildGROUPS\_SHARPinstance-attribute**

GROUPS\_SHARP: IconData

**buildH\_MOBILEDATAinstance-attribute**

H\_MOBILEDATA: IconData

**buildH\_MOBILEDATA\_OUTLINEDinstance-attribute**

H\_MOBILEDATA\_OUTLINED: IconData

**buildH\_MOBILEDATA\_ROUNDEDinstance-attribute**

H\_MOBILEDATA\_ROUNDED: IconData

**buildH\_MOBILEDATA\_SHARPinstance-attribute**

H\_MOBILEDATA\_SHARP: IconData

**buildH\_PLUS\_MOBILEDATAinstance-attribute**

H\_PLUS\_MOBILEDATA: IconData

**buildH\_PLUS\_MOBILEDATA\_OUTLINEDinstance-attribute**

H\_PLUS\_MOBILEDATA\_OUTLINED: IconData

**buildH\_PLUS\_MOBILEDATA\_ROUNDEDinstance-attribute**

H\_PLUS\_MOBILEDATA\_ROUNDED: IconData

**buildH\_PLUS\_MOBILEDATA\_SHARPinstance-attribute**

H\_PLUS\_MOBILEDATA\_SHARP: IconData

**buildHAILinstance-attribute**

HAIL: IconData

**buildHAIL\_OUTLINEDinstance-attribute**

HAIL\_OUTLINED: IconData

**buildHAIL\_ROUNDEDinstance-attribute**

HAIL\_ROUNDED: IconData

**buildHAIL\_SHARPinstance-attribute**

HAIL\_SHARP: IconData

**buildHANDSHAKEinstance-attribute**

HANDSHAKE: IconData

**buildHANDSHAKE\_OUTLINEDinstance-attribute**

HANDSHAKE\_OUTLINED: IconData

**buildHANDSHAKE\_ROUNDEDinstance-attribute**

HANDSHAKE\_ROUNDED: IconData

**buildHANDSHAKE\_SHARPinstance-attribute**

HANDSHAKE\_SHARP: IconData

**buildHANDYMANinstance-attribute**

HANDYMAN: IconData

**buildHANDYMAN\_OUTLINEDinstance-attribute**

HANDYMAN\_OUTLINED: IconData

**buildHANDYMAN\_ROUNDEDinstance-attribute**

HANDYMAN\_ROUNDED: IconData

**buildHANDYMAN\_SHARPinstance-attribute**

HANDYMAN\_SHARP: IconData

**buildHARDWAREinstance-attribute**

HARDWARE: IconData

**buildHARDWARE\_OUTLINEDinstance-attribute**

HARDWARE\_OUTLINED: IconData

**buildHARDWARE\_ROUNDEDinstance-attribute**

HARDWARE\_ROUNDED: IconData

**buildHARDWARE\_SHARPinstance-attribute**

HARDWARE\_SHARP: IconData

**buildHDinstance-attribute**

HD: IconData

**buildHD\_OUTLINEDinstance-attribute**

HD\_OUTLINED: IconData

**buildHD\_ROUNDEDinstance-attribute**

HD\_ROUNDED: IconData



**buildHD\_SHARPinstance-attribute**

HD\_SHARP: IconData

**buildHDR\_AUTOinstance-attribute**

HDR\_AUTO: IconData

**buildHDR\_AUTO\_OUTLINEDinstance-attribute**

HDR\_AUTO\_OUTLINED: IconData

**buildHDR\_AUTO\_ROUNDEDinstance-attribute**

HDR\_AUTO\_ROUNDED: IconData

**buildHDR\_AUTO\_SELECTinstance-attribute**

HDR\_AUTO\_SELECT: IconData

**buildHDR\_AUTO\_SELECT\_OUTLINEDinstance-attribute**

HDR\_AUTO\_SELECT\_OUTLINED: IconData

**buildHDR\_AUTO\_SELECT\_ROUNDEDinstance-attribute**

HDR\_AUTO\_SELECT\_ROUNDED: IconData

**buildHDR\_AUTO\_SELECT\_SHARPinstance-attribute**

HDR\_AUTO\_SELECT\_SHARP: IconData

**buildHDR\_AUTO\_SHARPinstance-attribute**

HDR\_AUTO\_SHARP: IconData

**buildHDR\_ENHANCED\_SELECTinstance-attribute**

HDR\_ENHANCED\_SELECT: IconData

**buildHDR\_ENHANCED\_SELECT\_OUTLINEDinstance-attribute**

HDR\_ENHANCED\_SELECT\_OUTLINED: IconData

**buildHDR\_ENHANCED\_SELECT\_ROUNDEDinstance-attribute**

HDR\_ENHANCED\_SELECT\_ROUNDED: IconData

**buildHDR\_ENHANCED\_SELECT\_SHARPinstance-attribute**

HDR\_ENHANCED\_SELECT\_SHARP: IconData

**buildHDR\_OFFinstance-attribute**

HDR\_OFF: IconData

**buildHDR\_OFF\_OUTLINEDinstance-attribute**

HDR\_OFF\_OUTLINED: IconData

**buildHDR\_OFF\_ROUNDEDinstance-attribute**

HDR\_OFF\_ROUNDED: IconData

**buildHDR\_OFF\_SELECTinstance-attribute**

HDR\_OFF\_SELECT: IconData

**buildHDR\_OFF\_SELECT\_OUTLINEDinstance-attribute**

HDR\_OFF\_SELECT\_OUTLINED: IconData

**buildHDR\_OFF\_SELECT\_ROUNDEDinstance-attribute**

HDR\_OFF\_SELECT\_ROUNDED: IconData

**buildHDR\_OFF\_SELECT\_SHARPinstance-attribute**

HDR\_OFF\_SELECT\_SHARP: IconData

**buildHDR\_OFF\_SHARPinstance-attribute**

HDR\_OFF\_SHARP: IconData

**buildHDR\_ONinstance-attribute**

HDR\_ON: IconData

**buildHDR\_ON\_OUTLINEDinstance-attribute**

HDR\_ON\_OUTLINED: IconData

**buildHDR\_ON\_ROUNDEDinstance-attribute**

HDR\_ON\_ROUNDED: IconData

**buildHDR\_ON\_SELECTinstance-attribute**

HDR\_ON\_SELECT: IconData

**buildHDR\_ON\_SELECT\_OUTLINEDinstance-attribute**

HDR\_ON\_SELECT\_OUTLINED: IconData

**buildHDR\_ON\_SELECT\_ROUNDEDinstance-attribute**

HDR\_ON\_SELECT\_ROUNDED: IconData

**buildHDR\_ON\_SELECT\_SHARPinstance-attribute**

HDR\_ON\_SELECT\_SHARP: IconData

**buildHDR\_ON\_SHARPinstance-attribute**

HDR\_ON\_SHARP: IconData

**buildHDR\_PLUSinstance-attribute**

HDR\_PLUS: IconData

**buildHDR\_PLUS\_OUTLINEDinstance-attribute**

HDR\_PLUS\_OUTLINED: IconData

**buildHDR\_PLUS\_ROUNDEDinstance-attribute**

HDR\_PLUS\_ROUNDED: IconData

**buildHDR\_PLUS\_SHARPinstance-attribute**

HDR\_PLUS\_SHARP: IconData

**buildHDR\_STRONGinstance-attribute**

HDR\_STRONG: IconData

**buildHDR\_STRONG\_OUTLINEDinstance-attribute**

HDR\_STRONG\_OUTLINED: IconData

**buildHDR\_STRONG\_ROUNDEDinstance-attribute**

HDR\_STRONG\_ROUNDED: IconData

**buildHDR\_STRONG\_SHARPinstance-attribute**

HDR\_STRONG\_SHARP: IconData

**buildHDR\_WEAKinstance-attribute**

HDR\_WEAK: IconData

**buildHDR\_WEAK\_OUTLINEDinstance-attribute**

HDR\_WEAK\_OUTLINED: IconData

**buildHDR\_WEAK\_ROUNDEDinstance-attribute**

HDR\_WEAK\_ROUNDED: IconData

**buildHDR\_WEAK\_SHARPinstance-attribute**

HDR\_WEAK\_SHARP: IconData

**buildHEADPHONESinstance-attribute**

HEADPHONES: IconData

**buildHEADPHONES\_BATTERYinstance-attribute**

HEADPHONES\_BATTERY: IconData

**buildHEADPHONES\_BATTERY\_OUTLINEDinstance-attribute**

HEADPHONES\_BATTERY\_OUTLINED: IconData

**buildHEADPHONES\_BATTERY\_ROUNDEDinstance-attribute**

HEADPHONES\_BATTERY\_ROUNDED: IconData

**buildHEADPHONES\_BATTERY\_SHARPinstance-attribute**

HEADPHONES\_BATTERY\_SHARP: IconData

**buildHEADPHONES\_OUTLINEDinstance-attribute**

HEADPHONES\_OUTLINED: IconData

**buildHEADPHONES\_ROUNDEDinstance-attribute**

HEADPHONES\_ROUNDED: IconData

**buildHEADPHONES\_SHARPinstance-attribute**

HEADPHONES\_SHARP: IconData

**buildHEADSETinstance-attribute**

HEADSET: IconData

**buildHEADSET\_MICinstance-attribute**

HEADSET\_MIC: IconData

**buildHEADSET\_MIC\_OUTLINEDinstance-attribute**

HEADSET\_MIC\_OUTLINED: IconData

**buildHEADSET\_MIC\_ROUNDEDinstance-attribute**

HEADSET\_MIC\_ROUNDED: IconData

**buildHEADSET\_MIC\_SHARPinstance-attribute**

HEADSET\_MIC\_SHARP: IconData

**buildHEADSET\_OFFinstance-attribute**

HEADSET\_OFF: IconData

**buildHEADSET\_OFF\_OUTLINEDinstance-attribute**

HEADSET\_OFF\_OUTLINED: IconData

**buildHEADSET\_OFF\_ROUNDEDinstance-attribute**

HEADSET\_OFF\_ROUNDED: IconData

**buildHEADSET\_OFF\_SHARPinstance-attribute**

HEADSET\_OFF\_SHARP: IconData

**buildHEADSET\_OUTLINEDinstance-attribute**

HEADSET\_OUTLINED: IconData

**buildHEADSET\_ROUNDEDinstance-attribute**

HEADSET\_ROUNDED: IconData

**buildHEADSET\_SHARPinstance-attribute**

HEADSET\_SHARP: IconData

**buildHEALINGinstance-attribute**

HEALING: IconData

**buildHEALING\_OUTLINEDinstance-attribute**

HEALING\_OUTLINED: IconData

**buildHEALING\_ROUNDEDinstance-attribute**

HEALING\_ROUNDED: IconData

**buildHEALING\_SHARPinstance-attribute**

HEALING\_SHARP: IconData

**buildHEALTH\_AND\_SAFETYinstance-attribute**

HEALTH\_AND\_SAFETY: IconData

**buildHEALTH\_AND\_SAFETY\_OUTLINEDinstance-attribute**

HEALTH\_AND\_SAFETY\_OUTLINED: IconData

**buildHEALTH\_AND\_SAFETY\_ROUNDEDinstance-attribute**

HEALTH\_AND\_SAFETY\_ROUNDED: IconData

**buildHEALTH\_AND\_SAFETY\_SHARPinstance-attribute**

HEALTH\_AND\_SAFETY\_SHARP: IconData

**buildHEARINGinstance-attribute**

HEARING: IconData

**buildHEARING\_DISABLEDinstance-attribute**

HEARING\_DISABLED: IconData

**buildHEARING\_DISABLED\_OUTLINEDinstance-attribute**

HEARING\_DISABLED\_OUTLINED: IconData

**buildHEARING\_DISABLED\_ROUNDEDinstance-attribute**

HEARING\_DISABLED\_ROUNDED: IconData

**buildHEARING\_DISABLED\_SHARPinstance-attribute**

HEARING\_DISABLED\_SHARP: IconData

**buildHEARING\_OUTLINEDinstance-attribute**

HEARING\_OUTLINED: IconData

**buildHEARING\_ROUNDEDinstance-attribute**

HEARING\_ROUNDED: IconData

**buildHEARING\_SHARPinstance-attribute**

HEARING\_SHARP: IconData

**buildHEART\_BROKENinstance-attribute**

HEART\_BROKEN: IconData

**buildHEART\_BROKEN\_OUTLINEDinstance-attribute**

HEART\_BROKEN\_OUTLINED: IconData

**buildHEART\_BROKEN\_ROUNDEDinstance-attribute**

HEART\_BROKEN\_ROUNDED: IconData

**buildHEART\_BROKEN\_SHARPinstance-attribute**

HEART\_BROKEN\_SHARP: IconData

**buildHEAT\_PUMPinstance-attribute**

HEAT\_PUMP: IconData

**buildHEAT\_PUMP\_OUTLINEDinstance-attribute**

HEAT\_PUMP\_OUTLINED: IconData

**buildHEAT\_PUMP\_ROUNDEDinstance-attribute**

HEAT\_PUMP\_ROUNDED: IconData

**buildHEAT\_PUMP\_SHARPinstance-attribute**

HEAT\_PUMP\_SHARP: IconData

**buildHEIGHTinstance-attribute**

HEIGHT: IconData

**buildHEIGHT\_OUTLINEDinstance-attribute**

HEIGHT\_OUTLINED: IconData

**buildHEIGHT\_ROUNDEDinstance-attribute**

HEIGHT\_ROUNDED: IconData

**buildHEIGHT\_SHARPinstance-attribute**

HEIGHT\_SHARP: IconData

**buildHELPinstance-attribute**

HELP: IconData

**buildHELP\_CENTERinstance-attribute**

HELP\_CENTER: IconData

**buildHELP\_CENTER\_OUTLINEDinstance-attribute**

HELP\_CENTER\_OUTLINED: IconData

**buildHELP\_CENTER\_ROUNDEDinstance-attribute**

HELP\_CENTER\_ROUNDED: IconData

**buildHELP\_CENTER\_SHARPinstance-attribute**

HELP\_CENTER\_SHARP: IconData

**buildHELP\_OUTLINEinstance-attribute**

HELP\_OUTLINE: IconData

**buildHELP\_OUTLINE\_OUTLINEDinstance-attribute**

HELP\_OUTLINE\_OUTLINED: IconData

**buildHELP\_OUTLINE\_ROUNDEDinstance-attribute**

HELP\_OUTLINE\_ROUNDED: IconData

**buildHELP\_OUTLINE\_SHARPinstance-attribute**

HELP\_OUTLINE\_SHARP: IconData

**buildHELP\_OUTLINEDinstance-attribute**

HELP\_OUTLINED: IconData

**buildHELP\_ROUNDEDinstance-attribute**

HELP\_ROUNDED: IconData

**buildHELP\_SHARPinstance-attribute**

HELP\_SHARP: IconData

**buildHEVCinstance-attribute**

HEVC: IconData

**buildHEVC\_OUTLINEDinstance-attribute**

HEVC\_OUTLINED: IconData

**buildHEVC\_ROUNDEDinstance-attribute**

HEVC\_ROUNDED: IconData

**buildHEVC\_SHARPinstance-attribute**

HEVC\_SHARP: IconData

**buildHEXAGONinstance-attribute**

HEXAGON: IconData

**buildHEXAGON\_OUTLINEDinstance-attribute**

HEXAGON\_OUTLINED: IconData

**buildHEXAGON\_ROUNDEDinstance-attribute**

HEXAGON\_ROUNDED: IconData

**buildHEXAGON\_SHARPinstance-attribute**

HEXAGON\_SHARP: IconData

**buildHIDE\_IMAGEinstance-attribute**

HIDE\_IMAGE: IconData

**buildHIDE\_IMAGE\_OUTLINEDinstance-attribute**

HIDE\_IMAGE\_OUTLINED: IconData

**buildHIDE\_IMAGE\_ROUNDEDinstance-attribute**

HIDE\_IMAGE\_ROUNDED: IconData

**buildHIDE\_IMAGE\_SHARPinstance-attribute**

HIDE\_IMAGE\_SHARP: IconData

**buildHIDE\_SOURCEinstance-attribute**

HIDE\_SOURCE: IconData

**buildHIDE\_SOURCE\_OUTLINEDinstance-attribute**

HIDE\_SOURCE\_OUTLINED: IconData

**buildHIDE\_SOURCE\_ROUNDEDinstance-attribute**

HIDE\_SOURCE\_ROUNDED: IconData

**buildHIDE\_SOURCE\_SHARPinstance-attribute**

HIDE\_SOURCE\_SHARP: IconData

**buildHIGH\_QUALITYinstance-attribute**

HIGH\_QUALITY: IconData

**buildHIGH\_QUALITY\_OUTLINEDinstance-attribute**

HIGH\_QUALITY\_OUTLINED: IconData

**buildHIGH\_QUALITY\_ROUNDEDinstance-attribute**

HIGH\_QUALITY\_ROUNDED: IconData

**buildHIGH\_QUALITY\_SHARPinstance-attribute**

HIGH\_QUALITY\_SHARP: IconData

**buildHIGHLIGHTinstance-attribute**

HIGHLIGHT: IconData

**buildHIGHLIGHT\_ALTinstance-attribute**

HIGHLIGHT\_ALT: IconData

**buildHIGHLIGHT\_ALT\_OUTLINEDinstance-attribute**

HIGHLIGHT\_ALT\_OUTLINED: IconData

**buildHIGHLIGHT\_ALT\_ROUNDEDinstance-attribute**

HIGHLIGHT\_ALT\_ROUNDED: IconData

**buildHIGHLIGHT\_ALT\_SHARPinstance-attribute**

HIGHLIGHT\_ALT\_SHARP: IconData

**buildHIGHLIGHT\_OFFinstance-attribute**

HIGHLIGHT\_OFF: IconData

**buildHIGHLIGHT\_OFF\_OUTLINEDinstance-attribute**

HIGHLIGHT\_OFF\_OUTLINED: IconData

**buildHIGHLIGHT\_OFF\_ROUNDEDinstance-attribute**

HIGHLIGHT\_OFF\_ROUNDED: IconData

**buildHIGHLIGHT\_OFF\_SHARPinstance-attribute**

HIGHLIGHT\_OFF\_SHARP: IconData

**buildHIGHLIGHT\_OUTLINEDinstance-attribute**

HIGHLIGHT\_OUTLINED: IconData

**buildHIGHLIGHT\_REMOVEinstance-attribute**

HIGHLIGHT\_REMOVE: IconData

**buildHIGHLIGHT\_REMOVE\_OUTLINED**instance-attribute  
HIGHLIGHT\_REMOVE\_OUTLINED: IconData

**buildHIGHLIGHT\_REMOVE\_ROUNDED**instance-attribute  
HIGHLIGHT\_REMOVE\_ROUNDED: IconData

**buildHIGHLIGHT\_REMOVE\_SHARP**instance-attribute  
HIGHLIGHT\_REMOVE\_SHARP: IconData

**buildHIGHLIGHT\_ROUNDED**instance-attribute  
HIGHLIGHT\_ROUNDED: IconData

**buildHIGHLIGHT\_SHARP**instance-attribute  
HIGHLIGHT\_SHARP: IconData

**buildHIKING**instance-attribute  
HIKING: IconData

**buildHIKING\_OUTLINED**instance-attribute  
HIKING\_OUTLINED: IconData

**buildHIKING\_ROUNDED**instance-attribute  
HIKING\_ROUNDED: IconData

**buildHIKING\_SHARP**instance-attribute  
HIKING\_SHARP: IconData

**buildHISTORY**instance-attribute  
HISTORY: IconData

**buildHISTORY\_EDU**instance-attribute  
HISTORY\_EDU: IconData

**buildHISTORY\_EDU\_OUTLINED**instance-attribute  
HISTORY\_EDU\_OUTLINED: IconData

**buildHISTORY\_EDU\_ROUNDED**instance-attribute  
HISTORY\_EDU\_ROUNDED: IconData

**buildHISTORY\_EDU\_SHARP**instance-attribute  
HISTORY\_EDU\_SHARP: IconData

**buildHISTORY\_OUTLINED**instance-attribute  
HISTORY\_OUTLINED: IconData

**buildHISTORY\_ROUNDED**instance-attribute  
HISTORY\_ROUNDED: IconData

**buildHISTORY\_SHARP**instance-attribute  
HISTORY\_SHARP: IconData

**buildHISTORY\_TOGGLE\_OFF**instance-attribute  
HISTORY\_TOGGLE\_OFF: IconData

**buildHISTORY\_TOGGLE\_OFF\_OUTLINED**instance-attribute  
HISTORY\_TOGGLE\_OFF\_OUTLINED: IconData

**buildHISTORY\_TOGGLE\_OFF\_ROUNDED**instance-attribute  
HISTORY\_TOGGLE\_OFF\_ROUNDED: IconData

**buildHISTORY\_TOGGLE\_OFF\_SHARP**instance-attribute  
HISTORY\_TOGGLE\_OFF\_SHARP: IconData

**buildHIVE**instance-attribute  
HIVE: IconData

**buildHIVE\_OUTLINED**instance-attribute

HIVE\_OUTLINED: IconData

**buildHIVE\_ROUNDED**instance-attribute

HIVE\_ROUNDED: IconData

**buildHIVE\_SHARP**instance-attribute

HIVE\_SHARP: IconData

**buildHLS**instance-attribute

HLS: IconData

**buildHLS\_OFF**instance-attribute

HLS\_OFF: IconData

**buildHLS\_OFF\_OUTLINED**instance-attribute

HLS\_OFF\_OUTLINED: IconData

**buildHLS\_OFF\_ROUNDED**instance-attribute

HLS\_OFF\_ROUNDED: IconData

**buildHLS\_OFF\_SHARP**instance-attribute

HLS\_OFF\_SHARP: IconData

**buildHLS\_OUTLINED**instance-attribute

HLS\_OUTLINED: IconData

**buildHLS\_ROUNDED**instance-attribute

HLS\_ROUNDED: IconData

**buildHLS\_SHARP**instance-attribute

HLS\_SHARP: IconData

**buildHOLIDAY\_VILLAGE**instance-attribute

HOLIDAY\_VILLAGE: IconData

**buildHOLIDAY\_VILLAGE\_OUTLINED**instance-attribute

HOLIDAY\_VILLAGE\_OUTLINED: IconData

**buildHOLIDAY\_VILLAGE\_ROUNDED**instance-attribute

HOLIDAY\_VILLAGE\_ROUNDED: IconData

**buildHOLIDAY\_VILLAGE\_SHARP**instance-attribute

HOLIDAY\_VILLAGE\_SHARP: IconData

**buildHOME**instance-attribute

HOME: IconData

**buildHOME\_FILLED**instance-attribute

HOME\_FILLED: IconData

**buildHOME\_MAX**instance-attribute

HOME\_MAX: IconData

**buildHOME\_MAX\_OUTLINED**instance-attribute

HOME\_MAX\_OUTLINED: IconData

**buildHOME\_MAX\_ROUNDED**instance-attribute

HOME\_MAX\_ROUNDED: IconData

**buildHOME\_MAX\_SHARP**instance-attribute

HOME\_MAX\_SHARP: IconData

**buildHOME\_MINI**instance-attribute

HOME\_MINI: IconData



buildHOME\_MINI\_OUTLINEDinstance-attribute

HOME\_MINI\_OUTLINED: IconData

buildHOME\_MINI\_ROUNDEDinstance-attribute

HOME\_MINI\_ROUNDED: IconData

buildHOME\_MINI\_SHARPinstance-attribute

HOME\_MINI\_SHARP: IconData

buildHOME\_OUTLINEDinstance-attribute

HOME\_OUTLINED: IconData

buildHOME\_REPAIR\_SERVICEinstance-attribute

HOME\_REPAIR\_SERVICE: IconData

buildHOME\_REPAIR\_SERVICE\_OUTLINEDinstance-attribute

HOME\_REPAIR\_SERVICE\_OUTLINED: IconData

buildHOME\_REPAIR\_SERVICE\_ROUNDEDinstance-attribute

HOME\_REPAIR\_SERVICE\_ROUNDED: IconData

buildHOME\_REPAIR\_SERVICE\_SHARPinstance-attribute

HOME\_REPAIR\_SERVICE\_SHARP: IconData

buildHOME\_ROUNDEDinstance-attribute

HOME\_ROUNDED: IconData

buildHOME\_SHARPinstance-attribute

HOME\_SHARP: IconData

buildHOME\_WORKinstance-attribute

HOME\_WORK: IconData

buildHOME\_WORK\_OUTLINEDinstance-attribute

HOME\_WORK\_OUTLINED: IconData

buildHOME\_WORK\_ROUNDEDinstance-attribute

HOME\_WORK\_ROUNDED: IconData

buildHOME\_WORK\_SHARPinstance-attribute

HOME\_WORK\_SHARP: IconData

buildHORIZONTAL\_DISTRIBUTEinstance-attribute

HORIZONTAL\_DISTRIBUTE: IconData

buildHORIZONTAL\_DISTRIBUTE\_OUTLINEDinstance-attribute

HORIZONTAL\_DISTRIBUTE\_OUTLINED: IconData

buildHORIZONTAL\_DISTRIBUTE\_ROUNDEDinstance-attribute

HORIZONTAL\_DISTRIBUTE\_ROUNDED: IconData

buildHORIZONTAL\_DISTRIBUTE\_SHARPinstance-attribute

HORIZONTAL\_DISTRIBUTE\_SHARP: IconData

buildHORIZONTAL\_RULEinstance-attribute

HORIZONTAL\_RULE: IconData

buildHORIZONTAL\_RULE\_OUTLINEDinstance-attribute

HORIZONTAL\_RULE\_OUTLINED: IconData

buildHORIZONTAL\_RULE\_ROUNDEDinstance-attribute

HORIZONTAL\_RULE\_ROUNDED: IconData

buildHORIZONTAL\_RULE\_SHARPinstance-attribute

HORIZONTAL\_RULE\_SHARP: IconData

**buildHORIZONTAL\_SPLITinstance-attribute**

HORIZONTAL\_SPLIT: IconData

**buildHORIZONTAL\_SPLIT\_OUTLINEDinstance-attribute**

HORIZONTAL\_SPLIT\_OUTLINED: IconData

**buildHORIZONTAL\_SPLIT\_ROUNDEDinstance-attribute**

HORIZONTAL\_SPLIT\_ROUNDED: IconData

**buildHORIZONTAL\_SPLIT\_SHARPinstance-attribute**

HORIZONTAL\_SPLIT\_SHARP: IconData

**buildHOT\_TUBinstance-attribute**

HOT\_TUB: IconData

**buildHOT\_TUB\_OUTLINEDinstance-attribute**

HOT\_TUB\_OUTLINED: IconData

**buildHOT\_TUB\_ROUNDEDinstance-attribute**

HOT\_TUB\_ROUNDED: IconData

**buildHOT\_TUB\_SHARPinstance-attribute**

HOT\_TUB\_SHARP: IconData

**buildHOTELinstance-attribute**

HOTEL: IconData

**buildHOTEL\_CLASSinstance-attribute**

HOTEL\_CLASS: IconData

**buildHOTEL\_CLASS\_OUTLINEDinstance-attribute**

HOTEL\_CLASS\_OUTLINED: IconData

**buildHOTEL\_CLASS\_ROUNDEDinstance-attribute**

HOTEL\_CLASS\_ROUNDED: IconData

**buildHOTEL\_CLASS\_SHARPinstance-attribute**

HOTEL\_CLASS\_SHARP: IconData

**buildHOTEL\_OUTLINEDinstance-attribute**

HOTEL\_OUTLINED: IconData

**buildHOTEL\_ROUNDEDinstance-attribute**

HOTEL\_ROUNDED: IconData

**buildHOTEL\_SHARPinstance-attribute**

HOTEL\_SHARP: IconData

**buildHOURGLASS\_BOTTOMinstance-attribute**

HOURGLASS\_BOTTOM: IconData

**buildHOURGLASS\_BOTTOM\_OUTLINEDinstance-attribute**

HOURGLASS\_BOTTOM\_OUTLINED: IconData

**buildHOURGLASS\_BOTTOM\_ROUNDEDinstance-attribute**

HOURGLASS\_BOTTOM\_ROUNDED: IconData

**buildHOURGLASS\_BOTTOM\_SHARPinstance-attribute**

HOURGLASS\_BOTTOM\_SHARP: IconData

**buildHOURGLASS\_DISABLEDinstance-attribute**

HOURGLASS\_DISABLED: IconData

**buildHOURGLASS\_DISABLED\_OUTLINEDinstance-attribute**

HOURGLASS\_DISABLED\_OUTLINED: IconData

**buildHOURGLASS\_DISABLED\_ROUNDEDinstance-attribute**

HOURGLASS\_DISABLED\_ROUNDED: IconData

**buildHOURGLASS\_DISABLED\_SHARPinstance-attribute**

HOURGLASS\_DISABLED\_SHARP: IconData

**buildHOURGLASS\_EMPTYinstance-attribute**

HOURGLASS\_EMPTY: IconData

**buildHOURGLASS\_EMPTY\_OUTLINEDinstance-attribute**

HOURGLASS\_EMPTY\_OUTLINED: IconData

**buildHOURGLASS\_EMPTY\_ROUNDEDinstance-attribute**

HOURGLASS\_EMPTY\_ROUNDED: IconData

**buildHOURGLASS\_EMPTY\_SHARPinstance-attribute**

HOURGLASS\_EMPTY\_SHARP: IconData

**buildHOURGLASS\_FULLinstance-attribute**

HOURGLASS\_FULL: IconData

**buildHOURGLASS\_FULL\_OUTLINEDinstance-attribute**

HOURGLASS\_FULL\_OUTLINED: IconData

**buildHOURGLASS\_FULL\_ROUNDEDinstance-attribute**

HOURGLASS\_FULL\_ROUNDED: IconData

**buildHOURGLASS\_FULL\_SHARPinstance-attribute**

HOURGLASS\_FULL\_SHARP: IconData

**buildHOURGLASS\_TOPinstance-attribute**

HOURGLASS\_TOP: IconData

**buildHOURGLASS\_TOP\_OUTLINEDinstance-attribute**

HOURGLASS\_TOP\_OUTLINED: IconData

**buildHOURGLASS\_TOP\_ROUNDEDinstance-attribute**

HOURGLASS\_TOP\_ROUNDED: IconData

**buildHOURGLASS\_TOP\_SHARPinstance-attribute**

HOURGLASS\_TOP\_SHARP: IconData

**buildHOUSEinstance-attribute**

HOUSE: IconData

**buildHOUSE\_OUTLINEDinstance-attribute**

HOUSE\_OUTLINED: IconData

**buildHOUSE\_ROUNDEDinstance-attribute**

HOUSE\_ROUNDED: IconData

**buildHOUSE\_SHARPinstance-attribute**

HOUSE\_SHARP: IconData

**buildHOUSE\_SIDINGinstance-attribute**

HOUSE\_SIDING: IconData

**buildHOUSE\_SIDING\_OUTLINEDinstance-attribute**

HOUSE\_SIDING\_OUTLINED: IconData

**buildHOUSE\_SIDING\_ROUNDEDinstance-attribute**

HOUSE\_SIDING\_ROUNDED: IconData

**buildHOUSE\_SIDING\_SHARPinstance-attribute**

HOUSE\_SIDING\_SHARP: IconData

**buildHOUSEBOATinstance-attribute**

HOUSEBOAT: IconData

**buildHOUSEBOAT\_OUTLINEDinstance-attribute**

HOUSEBOAT\_OUTLINED: IconData

**buildHOUSEBOAT\_ROUNDEDinstance-attribute**

HOUSEBOAT\_ROUNDED: IconData

**buildHOUSEBOAT\_SHARPinstance-attribute**

HOUSEBOAT\_SHARP: IconData

**buildHOW\_TO\_REGinstance-attribute**

HOW\_TO\_REG: IconData

**buildHOW\_TO\_REG\_OUTLINEDinstance-attribute**

HOW\_TO\_REG\_OUTLINED: IconData

**buildHOW\_TO\_REG\_ROUNDEDinstance-attribute**

HOW\_TO\_REG\_ROUNDED: IconData

**buildHOW\_TO\_REG\_SHARPinstance-attribute**

HOW\_TO\_REG\_SHARP: IconData

**buildHOW\_TO\_VOTEinstance-attribute**

HOW\_TO\_VOTE: IconData

**buildHOW\_TO\_VOTE\_OUTLINEDinstance-attribute**

HOW\_TO\_VOTE\_OUTLINED: IconData

**buildHOW\_TO\_VOTE\_ROUNDEDinstance-attribute**

HOW\_TO\_VOTE\_ROUNDED: IconData

**buildHOW\_TO\_VOTE\_SHARPinstance-attribute**

HOW\_TO\_VOTE\_SHARP: IconData

**buildHTMLinstance-attribute**

HTML: IconData

**buildHTML\_OUTLINEDinstance-attribute**

HTML\_OUTLINED: IconData

**buildHTML\_ROUNDEDinstance-attribute**

HTML\_ROUNDED: IconData

**buildHTML\_SHARPinstance-attribute**

HTML\_SHARP: IconData

**buildHTTPinstance-attribute**

HTTP: IconData

**buildHTTP\_OUTLINEDinstance-attribute**

HTTP\_OUTLINED: IconData

**buildHTTP\_ROUNDEDinstance-attribute**

HTTP\_ROUNDED: IconData

**buildHTTP\_SHARPinstance-attribute**

HTTP\_SHARP: IconData

**buildHTTPSinstance-attribute**

HTTPS: IconData

**buildHTTPS\_OUTLINEDinstance-attribute**

HTTPS\_OUTLINED: IconData

**buildHTTPS\_ROUNDEDinstance-attribute**

HTTPS\_ROUNDED: IconData

**buildHTTPS\_SHARPinstance-attribute**

HTTPS\_SHARP: IconData

**buildHUBinstance-attribute**

HUB: IconData

**buildHUB\_OUTLINEDinstance-attribute**

HUB\_OUTLINED: IconData

**buildHUB\_ROUNDEDinstance-attribute**

HUB\_ROUNDED: IconData

**buildHUB\_SHARPinstance-attribute**

HUB\_SHARP: IconData

**buildHVACinstance-attribute**

HVAC: IconData

**buildHVAC\_OUTLINEDinstance-attribute**

HVAC\_OUTLINED: IconData

**buildHVAC\_ROUNDEDinstance-attribute**

HVAC\_ROUNDED: IconData

**buildHVAC\_SHARPinstance-attribute**

HVAC\_SHARP: IconData

**buildICE\_SKATINGinstance-attribute**

ICE\_SKATING: IconData

**buildICE\_SKATING\_OUTLINEDinstance-attribute**

ICE\_SKATING\_OUTLINED: IconData

**buildICE\_SKATING\_ROUNDEDinstance-attribute**

ICE\_SKATING\_ROUNDED: IconData

**buildICE\_SKATING\_SHARPinstance-attribute**

ICE\_SKATING\_SHARP: IconData

**buildICECREAMinstance-attribute**

ICECREAM: IconData

**buildICECREAM\_OUTLINEDinstance-attribute**

ICECREAM\_OUTLINED: IconData

**buildICECREAM\_ROUNDEDinstance-attribute**

ICECREAM\_ROUNDED: IconData

**buildICECREAM\_SHARPinstance-attribute**

ICECREAM\_SHARP: IconData

**buildIMAGEinstance-attribute**

IMAGE: IconData

**buildIMAGE\_ASPECT\_RATIOinstance-attribute**

IMAGE\_ASPECT\_RATIO: IconData

**buildIMAGE\_ASPECT\_RATIO\_OUTLINEDinstance-attribute**

IMAGE\_ASPECT\_RATIO\_OUTLINED: IconData

**buildIMAGE\_ASPECT\_RATIO\_ROUNDEDinstance-attribute**

IMAGE\_ASPECT\_RATIO\_ROUNDED: IconData

**buildIMAGE\_ASPECT\_RATIO\_SHARPinstance-attribute**

IMAGE\_ASPECT\_RATIO\_SHARP: IconData

**buildIMAGE\_NOT\_SUPPORTEDinstance-attribute**

IMAGE\_NOT\_SUPPORTED: IconData

**buildIMAGE\_NOT\_SUPPORTED\_OUTLINEDinstance-attribute**

IMAGE\_NOT\_SUPPORTED\_OUTLINED: IconData

**buildIMAGE\_NOT\_SUPPORTED\_ROUNDEDinstance-attribute**

IMAGE\_NOT\_SUPPORTED\_ROUNDED: IconData

**buildIMAGE\_NOT\_SUPPORTED\_SHARPinstance-attribute**

IMAGE\_NOT\_SUPPORTED\_SHARP: IconData

**buildIMAGE\_OUTLINEDinstance-attribute**

IMAGE\_OUTLINED: IconData

**buildIMAGE\_ROUNDEDinstance-attribute**

IMAGE\_ROUNDED: IconData

**buildIMAGE\_SEARCHinstance-attribute**

IMAGE\_SEARCH: IconData

**buildIMAGE\_SEARCH\_OUTLINEDinstance-attribute**

IMAGE\_SEARCH\_OUTLINED: IconData

**buildIMAGE\_SEARCH\_ROUNDEDinstance-attribute**

IMAGE\_SEARCH\_ROUNDED: IconData

**buildIMAGE\_SEARCH\_SHARPinstance-attribute**

IMAGE\_SEARCH\_SHARP: IconData

**buildIMAGE\_SHARPinstance-attribute**

IMAGE\_SHARP: IconData

**buildIMAGESEARCH\_ROLLERinstance-attribute**

IMAGESEARCH\_ROLLER: IconData

**buildIMAGESEARCH\_ROLLER\_OUTLINEDinstance-attribute**

IMAGESEARCH\_ROLLER\_OUTLINED: IconData

**buildIMAGESEARCH\_ROLLER\_ROUNDEDinstance-attribute**

IMAGESEARCH\_ROLLER\_ROUNDED: IconData

**buildIMAGESEARCH\_ROLLER\_SHARPinstance-attribute**

IMAGESEARCH\_ROLLER\_SHARP: IconData

**buildIMPORT\_CONTACTSinstance-attribute**

IMPORT\_CONTACTS: IconData

**buildIMPORT\_CONTACTS\_OUTLINEDinstance-attribute**

IMPORT\_CONTACTS\_OUTLINED: IconData

**buildIMPORT\_CONTACTS\_ROUNDEDinstance-attribute**

IMPORT\_CONTACTS\_ROUNDED: IconData

**buildIMPORT\_CONTACTS\_SHARPinstance-attribute**

IMPORT\_CONTACTS\_SHARP: IconData

**buildIMPORT\_EXPORTinstance-attribute**

IMPORT\_EXPORT: IconData

**buildIMPORT\_EXPORT\_OUTLINEDinstance-attribute**

IMPORT\_EXPORT\_OUTLINED: IconData

**buildIMPORT\_EXPORT\_ROUNDED**instance-attribute

IMPORT\_EXPORT\_ROUNDED: IconData

**buildIMPORT\_EXPORT\_SHARP**instance-attribute

IMPORT\_EXPORT\_SHARP: IconData

**buildIMPORTANT\_DEVICES**instance-attribute

IMPORTANT\_DEVICES: IconData

**buildIMPORTANT\_DEVICES\_OUTLINED**instance-attribute

IMPORTANT\_DEVICES\_OUTLINED: IconData

**buildIMPORTANT\_DEVICES\_ROUNDED**instance-attribute

IMPORTANT\_DEVICES\_ROUNDED: IconData

**buildIMPORTANT\_DEVICES\_SHARP**instance-attribute

IMPORTANT\_DEVICES\_SHARP: IconData

**buildINBOX**instance-attribute

INBOX: IconData

**buildINBOX\_OUTLINED**instance-attribute

INBOX\_OUTLINED: IconData

**buildINBOX\_ROUNDED**instance-attribute

INBOX\_ROUNDED: IconData

**buildINBOX\_SHARP**instance-attribute

INBOX\_SHARP: IconData

**buildINCOMPLETE\_CIRCLE**instance-attribute

INCOMPLETE\_CIRCLE: IconData

**buildINCOMPLETE\_CIRCLE\_OUTLINED**instance-attribute

INCOMPLETE\_CIRCLE\_OUTLINED: IconData

**buildINCOMPLETE\_CIRCLE\_ROUNDED**instance-attribute

INCOMPLETE\_CIRCLE\_ROUNDED: IconData

**buildINCOMPLETE\_CIRCLE\_SHARP**instance-attribute

INCOMPLETE\_CIRCLE\_SHARP: IconData

**buildINDETERMINATE\_CHECK\_BOX**instance-attribute

INDETERMINATE\_CHECK\_BOX: IconData

**buildINDETERMINATE\_CHECK\_BOX\_OUTLINED**instance-attribute

INDETERMINATE\_CHECK\_BOX\_OUTLINED: IconData

**buildINDETERMINATE\_CHECK\_BOX\_ROUNDED**instance-attribute

INDETERMINATE\_CHECK\_BOX\_ROUNDED: IconData

**buildINDETERMINATE\_CHECK\_BOX\_SHARP**instance-attribute

INDETERMINATE\_CHECK\_BOX\_SHARP: IconData

**buildINFO**instance-attribute

INFO: IconData

**buildINFO\_OUTLINE**instance-attribute

INFO\_OUTLINE: IconData

**buildINFO\_OUTLINE\_ROUNDED**instance-attribute

INFO\_OUTLINE\_ROUNDED: IconData

**buildINFO\_OUTLINE\_SHARP**instance-attribute

INFO\_OUTLINE\_SHARP: IconData

**buildINFO\_OUTLINED**Instance-attribute

INFO\_OUTLINED: IconData

**buildINFO\_ROUNDED**Instance-attribute

INFO\_ROUNDED: IconData

**buildINFO\_SHARP**Instance-attribute

INFO\_SHARP: IconData

**buildINPUT**Instance-attribute

INPUT: IconData

**buildINPUT\_OUTLINED**Instance-attribute

INPUT\_OUTLINED: IconData

**buildINPUT\_ROUNDED**Instance-attribute

INPUT\_ROUNDED: IconData

**buildINPUT\_SHARP**Instance-attribute

INPUT\_SHARP: IconData

**buildINSERT\_CHART**Instance-attribute

INSERT\_CHART: IconData

**buildINSERT\_CHART\_OUTLINED**Instance-attribute

INSERT\_CHART\_OUTLINED: IconData

**buildINSERT\_CHART\_OUTLINED\_OUTLINED**Instance-attribute

INSERT\_CHART\_OUTLINED\_OUTLINED: IconData

**buildINSERT\_CHART\_OUTLINED\_ROUNDED**Instance-attribute

INSERT\_CHART\_OUTLINED\_ROUNDED: IconData

**buildINSERT\_CHART\_OUTLINED\_SHARP**Instance-attribute

INSERT\_CHART\_OUTLINED\_SHARP: IconData

**buildINSERT\_CHART\_ROUNDED**Instance-attribute

INSERT\_CHART\_ROUNDED: IconData

**buildINSERT\_CHART\_SHARP**Instance-attribute

INSERT\_CHART\_SHARP: IconData

**buildINSERT\_COMMENT**Instance-attribute

INSERT\_COMMENT: IconData

**buildINSERT\_COMMENT\_OUTLINED**Instance-attribute

INSERT\_COMMENT\_OUTLINED: IconData

**buildINSERT\_COMMENT\_ROUNDED**Instance-attribute

INSERT\_COMMENT\_ROUNDED: IconData

**buildINSERT\_COMMENT\_SHARP**Instance-attribute

INSERT\_COMMENT\_SHARP: IconData

**buildINSERT\_DRIVE\_FILE**Instance-attribute

INSERT\_DRIVE\_FILE: IconData

**buildINSERT\_DRIVE\_FILE\_OUTLINED**Instance-attribute

INSERT\_DRIVE\_FILE\_OUTLINED: IconData

**buildINSERT\_DRIVE\_FILE\_ROUNDED**Instance-attribute

INSERT\_DRIVE\_FILE\_ROUNDED: IconData

**buildINSERT\_DRIVE\_FILE\_SHARP**Instance-attribute

INSERT\_DRIVE\_FILE\_SHARP: IconData



buildINSERT\_EMOTICONinstance-attribute

INSERT\_EMOTICON: IconData

buildINSERT\_EMOTICON\_OUTLINEDinstance-attribute

INSERT\_EMOTICON\_OUTLINED: IconData

buildINSERT\_EMOTICON\_ROUNDEDinstance-attribute

INSERT\_EMOTICON\_ROUNDED: IconData

buildINSERT\_EMOTICON\_SHARPinstance-attribute

INSERT\_EMOTICON\_SHARP: IconData

buildINSERT\_INVITATIONinstance-attribute

INSERT\_INVITATION: IconData

buildINSERT\_INVITATION\_OUTLINEDinstance-attribute

INSERT\_INVITATION\_OUTLINED: IconData

buildINSERT\_INVITATION\_ROUNDEDinstance-attribute

INSERT\_INVITATION\_ROUNDED: IconData

buildINSERT\_INVITATION\_SHARPinstance-attribute

INSERT\_INVITATION\_SHARP: IconData

buildINSERT\_LINKinstance-attribute

INSERT\_LINK: IconData

buildINSERT\_LINK\_OUTLINEDinstance-attribute

INSERT\_LINK\_OUTLINED: IconData

buildINSERT\_LINK\_ROUNDEDinstance-attribute

INSERT\_LINK\_ROUNDED: IconData

buildINSERT\_LINK\_SHARPinstance-attribute

INSERT\_LINK\_SHARP: IconData

buildINSERT\_PAGE\_BREAKinstance-attribute

INSERT\_PAGE\_BREAK: IconData

buildINSERT\_PAGE\_BREAK\_OUTLINEDinstance-attribute

INSERT\_PAGE\_BREAK\_OUTLINED: IconData

buildINSERT\_PAGE\_BREAK\_ROUNDEDinstance-attribute

INSERT\_PAGE\_BREAK\_ROUNDED: IconData

buildINSERT\_PAGE\_BREAK\_SHARPinstance-attribute

INSERT\_PAGE\_BREAK\_SHARP: IconData

buildINSERT\_PHOTOinstance-attribute

INSERT\_PHOTO: IconData

buildINSERT\_PHOTO\_OUTLINEDinstance-attribute

INSERT\_PHOTO\_OUTLINED: IconData

buildINSERT\_PHOTO\_ROUNDEDinstance-attribute

INSERT\_PHOTO\_ROUNDED: IconData

buildINSERT\_PHOTO\_SHARPinstance-attribute

INSERT\_PHOTO\_SHARP: IconData

buildINSIGHTSinstance-attribute

INSIGHTS: IconData

buildINSIGHTS\_OUTLINEDinstance-attribute

INSIGHTS\_OUTLINED: IconData

**buildINSIGHTS\_ROUNDEDinstance-attribute**

INSIGHTS\_ROUNDED: IconData

**buildINSIGHTS\_SHARPinstance-attribute**

INSIGHTS\_SHARP: IconData

**buildINSTALL\_DESKTOPinstance-attribute**

INSTALL\_DESKTOP: IconData

**buildINSTALL\_DESKTOP\_OUTLINEDinstance-attribute**

INSTALL\_DESKTOP\_OUTLINED: IconData

**buildINSTALL\_DESKTOP\_ROUNDEDinstance-attribute**

INSTALL\_DESKTOP\_ROUNDED: IconData

**buildINSTALL\_DESKTOP\_SHARPinstance-attribute**

INSTALL\_DESKTOP\_SHARP: IconData

**buildINSTALL\_MOBILEinstance-attribute**

INSTALL\_MOBILE: IconData

**buildINSTALL\_MOBILE\_OUTLINEDinstance-attribute**

INSTALL\_MOBILE\_OUTLINED: IconData

**buildINSTALL\_MOBILE\_ROUNDEDinstance-attribute**

INSTALL\_MOBILE\_ROUNDED: IconData

**buildINSTALL\_MOBILE\_SHARPinstance-attribute**

INSTALL\_MOBILE\_SHARP: IconData

**buildINTEGRATION\_INSTRUCTIONSinstance-attribute**

INTEGRATION\_INSTRUCTIONS: IconData

**buildINTEGRATION\_INSTRUCTIONS\_OUTLINEDinstance-attribute**

INTEGRATION\_INSTRUCTIONS\_OUTLINED: IconData

**buildINTEGRATION\_INSTRUCTIONS\_ROUNDEDinstance-attribute**

INTEGRATION\_INSTRUCTIONS\_ROUNDED: IconData

**buildINTEGRATION\_INSTRUCTIONS\_SHARPinstance-attribute**

INTEGRATION\_INSTRUCTIONS\_SHARP: IconData

**buildINTERESTSinstance-attribute**

INTERESTS: IconData

**buildINTERESTS\_OUTLINEDinstance-attribute**

INTERESTS\_OUTLINED: IconData

**buildINTERESTS\_ROUNDEDinstance-attribute**

INTERESTS\_ROUNDED: IconData

**buildINTERESTS\_SHARPinstance-attribute**

INTERESTS\_SHARP: IconData

**buildINTERPRETER\_MODEinstance-attribute**

INTERPRETER\_MODE: IconData

**buildINTERPRETER\_MODE\_OUTLINEDinstance-attribute**

INTERPRETER\_MODE\_OUTLINED: IconData

**buildINTERPRETER\_MODE\_ROUNDEDinstance-attribute**

INTERPRETER\_MODE\_ROUNDED: IconData

**buildINTERPRETER\_MODE\_SHARPinstance-attribute**

INTERPRETER\_MODE\_SHARP: IconData

**buildINVENTORYinstance-attribute**

INVENTORY: IconData

**buildINVENTORY\_2instance-attribute**

INVENTORY\_2: IconData

**buildINVENTORY\_2\_OUTLINEDinstance-attribute**

INVENTORY\_2\_OUTLINED: IconData

**buildINVENTORY\_2\_ROUNDEDinstance-attribute**

INVENTORY\_2\_ROUNDED: IconData

**buildINVENTORY\_2\_SHARPinstance-attribute**

INVENTORY\_2\_SHARP: IconData

**buildINVENTORY\_OUTLINEDinstance-attribute**

INVENTORY\_OUTLINED: IconData

**buildINVENTORY\_ROUNDEDinstance-attribute**

INVENTORY\_ROUNDED: IconData

**buildINVENTORY\_SHARPinstance-attribute**

INVENTORY\_SHARP: IconData

**buildINVERT\_COLORSinstance-attribute**

INVERT\_COLORS: IconData

**buildINVERT\_COLORS\_OFFinstance-attribute**

INVERT\_COLORS\_OFF: IconData

**buildINVERT\_COLORS\_OFF\_OUTLINEDinstance-attribute**

INVERT\_COLORS\_OFF\_OUTLINED: IconData

**buildINVERT\_COLORS\_OFF\_ROUNDEDinstance-attribute**

INVERT\_COLORS\_OFF\_ROUNDED: IconData

**buildINVERT\_COLORS\_OFF\_SHARPinstance-attribute**

INVERT\_COLORS\_OFF\_SHARP: IconData

**buildINVERT\_COLORS\_ONinstance-attribute**

INVERT\_COLORS\_ON: IconData

**buildINVERT\_COLORS\_ON\_OUTLINEDinstance-attribute**

INVERT\_COLORS\_ON\_OUTLINED: IconData

**buildINVERT\_COLORS\_ON\_ROUNDEDinstance-attribute**

INVERT\_COLORS\_ON\_ROUNDED: IconData

**buildINVERT\_COLORS\_ON\_SHARPinstance-attribute**

INVERT\_COLORS\_ON\_SHARP: IconData

**buildINVERT\_COLORS\_OUTLINEDinstance-attribute**

INVERT\_COLORS\_OUTLINED: IconData

**buildINVERT\_COLORS\_ROUNDEDinstance-attribute**

INVERT\_COLORS\_ROUNDED: IconData

**buildINVERT\_COLORS\_SHARPinstance-attribute**

INVERT\_COLORS\_SHARP: IconData

**buildIOS\_SHAREinstance-attribute**

IOS\_SHARE: IconData

**buildIOS\_SHARE\_OUTLINEDinstance-attribute**

IOS\_SHARE\_OUTLINED: IconData

**buildIOS\_SHARE\_ROUNDEDinstance-attribute**

IOS\_SHARE\_ROUNDED: IconData

**buildIOS\_SHARE\_SHARPinstance-attribute**

IOS\_SHARE\_SHARP: IconData

**buildIRONinstance-attribute**

IRON: IconData

**buildIRON\_OUTLINEDinstance-attribute**

IRON\_OUTLINED: IconData

**buildIRON\_ROUNDEDinstance-attribute**

IRON\_ROUNDED: IconData

**buildIRON\_SHARPinstance-attribute**

IRON\_SHARP: IconData

**buildISOinstance-attribute**

ISO: IconData

**buildISO\_OUTLINEDinstance-attribute**

ISO\_OUTLINED: IconData

**buildISO\_ROUNDEDinstance-attribute**

ISO\_ROUNDED: IconData

**buildISO\_SHARPinstance-attribute**

ISO\_SHARP: IconData

**buildJAVASCRIPTinstance-attribute**

JAVASCRIPT: IconData

**buildJAVASCRIPT\_OUTLINEDinstance-attribute**

JAVASCRIPT\_OUTLINED: IconData

**buildJAVASCRIPT\_ROUNDEDinstance-attribute**

JAVASCRIPT\_ROUNDED: IconData

**buildJAVASCRIPT\_SHARPinstance-attribute**

JAVASCRIPT\_SHARP: IconData

**buildJOIN\_FULLinstance-attribute**

JOIN\_FULL: IconData

**buildJOIN\_FULL\_OUTLINEDinstance-attribute**

JOIN\_FULL\_OUTLINED: IconData

**buildJOIN\_FULL\_ROUNDEDinstance-attribute**

JOIN\_FULL\_ROUNDED: IconData

**buildJOIN\_FULL\_SHARPinstance-attribute**

JOIN\_FULL\_SHARP: IconData

**buildJOIN\_INNERinstance-attribute**

JOIN\_INNER: IconData

**buildJOIN\_INNER\_OUTLINEDinstance-attribute**

JOIN\_INNER\_OUTLINED: IconData

**buildJOIN\_INNER\_ROUNDEDinstance-attribute**

JOIN\_INNER\_ROUNDED: IconData

**buildJOIN\_INNER\_SHARPinstance-attribute**

JOIN\_INNER\_SHARP: IconData

**buildJOIN\_LEFTinstance-attribute**

JOIN\_LEFT: IconData

**buildJOIN\_LEFT\_OUTLINEDinstance-attribute**

JOIN\_LEFT\_OUTLINED: IconData

**buildJOIN\_LEFT\_ROUNDEDinstance-attribute**

JOIN\_LEFT\_ROUNDED: IconData

**buildJOIN\_LEFT\_SHARPinstance-attribute**

JOIN\_LEFT\_SHARP: IconData

**buildJOIN\_RIGHTinstance-attribute**

JOIN\_RIGHT: IconData

**buildJOIN\_RIGHT\_OUTLINEDinstance-attribute**

JOIN\_RIGHT\_OUTLINED: IconData

**buildJOIN\_RIGHT\_ROUNDEDinstance-attribute**

JOIN\_RIGHT\_ROUNDED: IconData

**buildJOIN\_RIGHT\_SHARPinstance-attribute**

JOIN\_RIGHT\_SHARP: IconData

**buildKAYAKINGinstance-attribute**

KAYAKING: IconData

**buildKAYAKING\_OUTLINEDinstance-attribute**

KAYAKING\_OUTLINED: IconData

**buildKAYAKING\_ROUNDEDinstance-attribute**

KAYAKING\_ROUNDED: IconData

**buildKAYAKING\_SHARPinstance-attribute**

KAYAKING\_SHARP: IconData

**buildKEBAB\_DININGinstance-attribute**

KEBAB\_DINING: IconData

**buildKEBAB\_DINING\_OUTLINEDinstance-attribute**

KEBAB\_DINING\_OUTLINED: IconData

**buildKEBAB\_DINING\_ROUNDEDinstance-attribute**

KEBAB\_DINING\_ROUNDED: IconData

**buildKEBAB\_DINING\_SHARPinstance-attribute**

KEBAB\_DINING\_SHARP: IconData

**buildKEYinstance-attribute**

KEY: IconData

**buildKEY\_OFFinstance-attribute**

KEY\_OFF: IconData

**buildKEY\_OFF\_OUTLINEDinstance-attribute**

KEY\_OFF\_OUTLINED: IconData

**buildKEY\_OFF\_ROUNDEDinstance-attribute**

KEY\_OFF\_ROUNDED: IconData

**buildKEY\_OFF\_SHARPinstance-attribute**

KEY\_OFF\_SHARP: IconData

**buildKEY\_OUTLINEDinstance-attribute**

KEY\_OUTLINED: IconData

**buildKEY\_ROUNDEDinstance-attribute**

KEY\_ROUNDED: IconData

**buildKEY\_SHARPinstance-attribute**

KEY\_SHARP: IconData

**buildKEYBOARDinstance-attribute**

KEYBOARD: IconData

**buildKEYBOARD\_ALTinstance-attribute**

KEYBOARD\_ALT: IconData

**buildKEYBOARD\_ALT\_OUTLINEDinstance-attribute**

KEYBOARD\_ALT\_OUTLINED: IconData

**buildKEYBOARD\_ALT\_ROUNDEDinstance-attribute**

KEYBOARD\_ALT\_ROUNDED: IconData

**buildKEYBOARD\_ALT\_SHARPinstance-attribute**

KEYBOARD\_ALT\_SHARP: IconData

**buildKEYBOARD\_ARROW\_DOWNinstance-attribute**

KEYBOARD\_ARROW\_DOWN: IconData

**buildKEYBOARD\_ARROW\_DOWN\_OUTLINEDinstance-attribute**

KEYBOARD\_ARROW\_DOWN\_OUTLINED: IconData

**buildKEYBOARD\_ARROW\_DOWN\_ROUNDEDinstance-attribute**

KEYBOARD\_ARROW\_DOWN\_ROUNDED: IconData

**buildKEYBOARD\_ARROW\_DOWN\_SHARPinstance-attribute**

KEYBOARD\_ARROW\_DOWN\_SHARP: IconData

**buildKEYBOARD\_ARROW\_LEFTinstance-attribute**

KEYBOARD\_ARROW\_LEFT: IconData

**buildKEYBOARD\_ARROW\_LEFT\_OUTLINEDinstance-attribute**

KEYBOARD\_ARROW\_LEFT\_OUTLINED: IconData

**buildKEYBOARD\_ARROW\_LEFT\_ROUNDEDinstance-attribute**

KEYBOARD\_ARROW\_LEFT\_ROUNDED: IconData

**buildKEYBOARD\_ARROW\_LEFT\_SHARPinstance-attribute**

KEYBOARD\_ARROW\_LEFT\_SHARP: IconData

**buildKEYBOARD\_ARROW\_RIGHTinstance-attribute**

KEYBOARD\_ARROW\_RIGHT: IconData

**buildKEYBOARD\_ARROW\_RIGHT\_OUTLINEDinstance-attribute**

KEYBOARD\_ARROW\_RIGHT\_OUTLINED: IconData

**buildKEYBOARD\_ARROW\_RIGHT\_ROUNDEDinstance-attribute**

KEYBOARD\_ARROW\_RIGHT\_ROUNDED: IconData

**buildKEYBOARD\_ARROW\_RIGHT\_SHARPinstance-attribute**

KEYBOARD\_ARROW\_RIGHT\_SHARP: IconData

**buildKEYBOARD\_ARROW\_UPinstance-attribute**

KEYBOARD\_ARROW\_UP: IconData

**buildKEYBOARD\_ARROW\_UP\_OUTLINEDinstance-attribute**

KEYBOARD\_ARROW\_UP\_OUTLINED: IconData

**buildKEYBOARD\_ARROW\_UP\_ROUNDEDinstance-attribute**

KEYBOARD\_ARROW\_UP\_ROUNDED: IconData

**buildKEYBOARD\_ARROW\_UP\_SHARPinstance-attribute**

KEYBOARD\_ARROW\_UP\_SHARP: IconData

**buildKEYBOARD\_BACKSPACEinstance-attribute**

KEYBOARD\_BACKSPACE: IconData

**buildKEYBOARD\_BACKSPACE\_OUTLINEDinstance-attribute**

KEYBOARD\_BACKSPACE\_OUTLINED: IconData

**buildKEYBOARD\_BACKSPACE\_ROUNDEDinstance-attribute**

KEYBOARD\_BACKSPACE\_ROUNDED: IconData

**buildKEYBOARD\_BACKSPACE\_SHARPinstance-attribute**

KEYBOARD\_BACKSPACE\_SHARP: IconData

**buildKEYBOARD\_CAPSLOCKinstance-attribute**

KEYBOARD\_CAPSLOCK: IconData

**buildKEYBOARD\_CAPSLOCK\_OUTLINEDinstance-attribute**

KEYBOARD\_CAPSLOCK\_OUTLINED: IconData

**buildKEYBOARD\_CAPSLOCK\_ROUNDEDinstance-attribute**

KEYBOARD\_CAPSLOCK\_ROUNDED: IconData

**buildKEYBOARD\_CAPSLOCK\_SHARPinstance-attribute**

KEYBOARD\_CAPSLOCK\_SHARP: IconData

**buildKEYBOARD\_COMMAND\_KEYinstance-attribute**

KEYBOARD\_COMMAND\_KEY: IconData

**buildKEYBOARD\_COMMAND\_KEY\_OUTLINEDinstance-attribute**

KEYBOARD\_COMMAND\_KEY\_OUTLINED: IconData

**buildKEYBOARD\_COMMAND\_KEY\_ROUNDEDinstance-attribute**

KEYBOARD\_COMMAND\_KEY\_ROUNDED: IconData

**buildKEYBOARD\_COMMAND\_KEY\_SHARPinstance-attribute**

KEYBOARD\_COMMAND\_KEY\_SHARP: IconData

**buildKEYBOARD\_CONTROLinstance-attribute**

KEYBOARD\_CONTROL: IconData

**buildKEYBOARD\_CONTROL\_KEYinstance-attribute**

KEYBOARD\_CONTROL\_KEY: IconData

**buildKEYBOARD\_CONTROL\_KEY\_OUTLINEDinstance-attribute**

KEYBOARD\_CONTROL\_KEY\_OUTLINED: IconData

**buildKEYBOARD\_CONTROL\_KEY\_ROUNDEDinstance-attribute**

KEYBOARD\_CONTROL\_KEY\_ROUNDED: IconData

**buildKEYBOARD\_CONTROL\_KEY\_SHARPinstance-attribute**

KEYBOARD\_CONTROL\_KEY\_SHARP: IconData

**buildKEYBOARD\_CONTROL\_OUTLINEDinstance-attribute**

KEYBOARD\_CONTROL\_OUTLINED: IconData

**buildKEYBOARD\_CONTROL\_ROUNDEDinstance-attribute**

KEYBOARD\_CONTROL\_ROUNDED: IconData

**buildKEYBOARD\_CONTROL\_SHARPinstance-attribute**

KEYBOARD\_CONTROL\_SHARP: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_DOWNinstance-attribute**

KEYBOARD\_DOUBLE\_ARROW\_DOWN: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_DOWN\_OUTLINED**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_DOWN\_OUTLINED: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_DOWN\_ROUNDED**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_DOWN\_ROUNDED: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_DOWN\_SHARP**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_DOWN\_SHARP: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_LEFT**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_LEFT: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_LEFT\_OUTLINED**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_LEFT\_OUTLINED: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_LEFT\_ROUNDED**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_LEFT\_ROUNDED: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_LEFT\_SHARP**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_LEFT\_SHARP: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_RIGHT**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_RIGHT: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_RIGHT\_OUTLINED**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_RIGHT\_OUTLINED: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_RIGHT\_ROUNDED**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_RIGHT\_ROUNDED: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_RIGHT\_SHARP**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_RIGHT\_SHARP: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_UP**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_UP: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_UP\_OUTLINED**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_UP\_OUTLINED: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_UP\_ROUNDED**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_UP\_ROUNDED: IconData

**buildKEYBOARD\_DOUBLE\_ARROW\_UP\_SHARP**instance-attribute

KEYBOARD\_DOUBLE\_ARROW\_UP\_SHARP: IconData

**buildKEYBOARD\_HIDE**instance-attribute

KEYBOARD\_HIDE: IconData

**buildKEYBOARD\_HIDE\_OUTLINED**instance-attribute

KEYBOARD\_HIDE\_OUTLINED: IconData

**buildKEYBOARD\_HIDE\_ROUNDED**instance-attribute

KEYBOARD\_HIDE\_ROUNDED: IconData

**buildKEYBOARD\_HIDE\_SHARP**instance-attribute

KEYBOARD\_HIDE\_SHARP: IconData

**buildKEYBOARD\_OPTION\_KEY**instance-attribute

KEYBOARD\_OPTION\_KEY: IconData

**buildKEYBOARD\_OPTION\_KEY\_OUTLINED**instance-attribute

KEYBOARD\_OPTION\_KEY\_OUTLINED: IconData

**buildKEYBOARD\_OPTION\_KEY\_ROUNDED**instance-attribute

KEYBOARD\_OPTION\_KEY\_ROUNDED: IconData



**buildKEYBOARD\_OPTION\_KEY\_SHARPinstance-attribute**

KEYBOARD\_OPTION\_KEY\_SHARP: IconData

**buildKEYBOARD\_OUTLINEDinstance-attribute**

KEYBOARD\_OUTLINED: IconData

**buildKEYBOARD\_RETURNinstance-attribute**

KEYBOARD\_RETURN: IconData

**buildKEYBOARD\_RETURN\_OUTLINEDinstance-attribute**

KEYBOARD\_RETURN\_OUTLINED: IconData

**buildKEYBOARD\_RETURN\_ROUNDEDinstance-attribute**

KEYBOARD\_RETURN\_ROUNDED: IconData

**buildKEYBOARD\_RETURN\_SHARPinstance-attribute**

KEYBOARD\_RETURN\_SHARP: IconData

**buildKEYBOARD\_ROUNDEDinstance-attribute**

KEYBOARD\_ROUNDED: IconData

**buildKEYBOARD\_SHARPinstance-attribute**

KEYBOARD\_SHARP: IconData

**buildKEYBOARD\_TABinstance-attribute**

KEYBOARD\_TAB: IconData

**buildKEYBOARD\_TAB\_OUTLINEDinstance-attribute**

KEYBOARD\_TAB\_OUTLINED: IconData

**buildKEYBOARD\_TAB\_ROUNDEDinstance-attribute**

KEYBOARD\_TAB\_ROUNDED: IconData

**buildKEYBOARD\_TAB\_SHARPinstance-attribute**

KEYBOARD\_TAB\_SHARP: IconData

**buildKEYBOARD\_VOICEinstance-attribute**

KEYBOARD\_VOICE: IconData

**buildKEYBOARD\_VOICE\_OUTLINEDinstance-attribute**

KEYBOARD\_VOICE\_OUTLINED: IconData

**buildKEYBOARD\_VOICE\_ROUNDEDinstance-attribute**

KEYBOARD\_VOICE\_ROUNDED: IconData

**buildKEYBOARD\_VOICE\_SHARPinstance-attribute**

KEYBOARD\_VOICE\_SHARP: IconData

**buildKING\_BEDinstance-attribute**

KING\_BED: IconData

**buildKING\_BED\_OUTLINEDinstance-attribute**

KING\_BED\_OUTLINED: IconData

**buildKING\_BED\_ROUNDEDinstance-attribute**

KING\_BED\_ROUNDED: IconData

**buildKING\_BED\_SHARPinstance-attribute**

KING\_BED\_SHARP: IconData

**buildKITCHENinstance-attribute**

KITCHEN: IconData

**buildKITCHEN\_OUTLINEDinstance-attribute**

KITCHEN\_OUTLINED: IconData

**buildKITCHEN\_ROUNDEDinstance-attribute**

KITCHEN\_ROUNDED: IconData

**buildKITCHEN\_SHARPinstance-attribute**

KITCHEN\_SHARP: IconData

**buildKITESURFINGinstance-attribute**

KITESURFING: IconData

**buildKITESURFING\_OUTLINEDinstance-attribute**

KITESURFING\_OUTLINED: IconData

**buildKITESURFING\_ROUNDEDinstance-attribute**

KITESURFING\_ROUNDED: IconData

**buildKITESURFING\_SHARPinstance-attribute**

KITESURFING\_SHARP: IconData

**buildLABELinstance-attribute**

LABEL: IconData

**buildLABEL\_IMPORTANTinstance-attribute**

LABEL\_IMPORTANT: IconData

**buildLABEL\_IMPORTANT\_OUTLINEinstance-attribute**

LABEL\_IMPORTANT\_OUTLINE: IconData

**buildLABEL\_IMPORTANT\_OUTLINE\_ROUNDEDinstance-attribute**

LABEL\_IMPORTANT\_OUTLINE\_ROUNDED: IconData

**buildLABEL\_IMPORTANT\_OUTLINE\_SHARPinstance-attribute**

LABEL\_IMPORTANT\_OUTLINE\_SHARP: IconData

**buildLABEL\_IMPORTANT\_OUTLINEDinstance-attribute**

LABEL\_IMPORTANT\_OUTLINED: IconData

**buildLABEL\_IMPORTANT\_ROUNDEDinstance-attribute**

LABEL\_IMPORTANT\_ROUNDED: IconData

**buildLABEL\_IMPORTANT\_SHARPinstance-attribute**

LABEL\_IMPORTANT\_SHARP: IconData

**buildLABEL\_OFFinstance-attribute**

LABEL\_OFF: IconData

**buildLABEL\_OFF\_OUTLINEDinstance-attribute**

LABEL\_OFF\_OUTLINED: IconData

**buildLABEL\_OFF\_ROUNDEDinstance-attribute**

LABEL\_OFF\_ROUNDED: IconData

**buildLABEL\_OFF\_SHARPinstance-attribute**

LABEL\_OFF\_SHARP: IconData

**buildLABEL\_OUTLINEinstance-attribute**

LABEL\_OUTLINE: IconData

**buildLABEL\_OUTLINE\_ROUNDEDinstance-attribute**

LABEL\_OUTLINE\_ROUNDED: IconData

**buildLABEL\_OUTLINE\_SHARPinstance-attribute**

LABEL\_OUTLINE\_SHARP: IconData

**buildLABEL\_OUTLINEDinstance-attribute**

LABEL\_OUTLINED: IconData

**buildLABEL\_ROUNDEDinstance-attribute**

LABEL\_ROUNDED: IconData

**buildLABEL\_SHARPinstance-attribute**

LABEL\_SHARP: IconData

**buildLANinstance-attribute**

LAN: IconData

**buildLAN\_OUTLINEDinstance-attribute**

LAN\_OUTLINED: IconData

**buildLAN\_ROUNDEDinstance-attribute**

LAN\_ROUNDED: IconData

**buildLAN\_SHARPinstance-attribute**

LAN\_SHARP: IconData

**buildLANDSCAPEinstance-attribute**

LANDSCAPE: IconData

**buildLANDSCAPE\_OUTLINEDinstance-attribute**

LANDSCAPE\_OUTLINED: IconData

**buildLANDSCAPE\_ROUNDEDinstance-attribute**

LANDSCAPE\_ROUNDED: IconData

**buildLANDSCAPE\_SHARPinstance-attribute**

LANDSCAPE\_SHARP: IconData

**buildLANDSLIDEinstance-attribute**

LANDSLIDE: IconData

**buildLANDSLIDE\_OUTLINEDinstance-attribute**

LANDSLIDE\_OUTLINED: IconData

**buildLANDSLIDE\_ROUNDEDinstance-attribute**

LANDSLIDE\_ROUNDED: IconData

**buildLANDSLIDE\_SHARPinstance-attribute**

LANDSLIDE\_SHARP: IconData

**buildLANGUAGEinstance-attribute**

LANGUAGE: IconData

**buildLANGUAGE\_OUTLINEDinstance-attribute**

LANGUAGE\_OUTLINED: IconData

**buildLANGUAGE\_ROUNDEDinstance-attribute**

LANGUAGE\_ROUNDED: IconData

**buildLANGUAGE\_SHARPinstance-attribute**

LANGUAGE\_SHARP: IconData

**buildLAPTOPinstance-attribute**

LAPTOP: IconData

**buildLAPTOP\_CHROMEBOOKinstance-attribute**

LAPTOP\_CHROMEBOOK: IconData

**buildLAPTOP\_CHROMEBOOK\_OUTLINEDinstance-attribute**

LAPTOP\_CHROMEBOOK\_OUTLINED: IconData

**buildLAPTOP\_CHROMEBOOK\_ROUNDEDinstance-attribute**

LAPTOP\_CHROMEBOOK\_ROUNDED: IconData

**buildLAPTOP\_CHROMEBOOK\_SHARPinstance-attribute**

LAPTOP\_CHROMEBOOK\_SHARP: IconData

**buildLAPTOP\_MACinstance-attribute**

LAPTOP\_MAC: IconData

**buildLAPTOP\_MAC\_OUTLINEDinstance-attribute**

LAPTOP\_MAC\_OUTLINED: IconData

**buildLAPTOP\_MAC\_ROUNDEDinstance-attribute**

LAPTOP\_MAC\_ROUNDED: IconData

**buildLAPTOP\_MAC\_SHARPinstance-attribute**

LAPTOP\_MAC\_SHARP: IconData

**buildLAPTOP\_OUTLINEDinstance-attribute**

LAPTOP\_OUTLINED: IconData

**buildLAPTOP\_ROUNDEDinstance-attribute**

LAPTOP\_ROUNDED: IconData

**buildLAPTOP\_SHARPinstance-attribute**

LAPTOP\_SHARP: IconData

**buildLAPTOP\_WINDOWSinstance-attribute**

LAPTOP\_WINDOWS: IconData

**buildLAPTOP\_WINDOWS\_OUTLINEDinstance-attribute**

LAPTOP\_WINDOWS\_OUTLINED: IconData

**buildLAPTOP\_WINDOWS\_ROUNDEDinstance-attribute**

LAPTOP\_WINDOWS\_ROUNDED: IconData

**buildLAPTOP\_WINDOWS\_SHARPinstance-attribute**

LAPTOP\_WINDOWS\_SHARP: IconData

**buildLAST\_PAGEinstance-attribute**

LAST\_PAGE: IconData

**buildLAST\_PAGE\_OUTLINEDinstance-attribute**

LAST\_PAGE\_OUTLINED: IconData

**buildLAST\_PAGE\_ROUNDEDinstance-attribute**

LAST\_PAGE\_ROUNDED: IconData

**buildLAST\_PAGE\_SHARPinstance-attribute**

LAST\_PAGE\_SHARP: IconData

**buildLAUNCHinstance-attribute**

LAUNCH: IconData

**buildLAUNCH\_OUTLINEDinstance-attribute**

LAUNCH\_OUTLINED: IconData

**buildLAUNCH\_ROUNDEDinstance-attribute**

LAUNCH\_ROUNDED: IconData

**buildLAUNCH\_SHARPinstance-attribute**

LAUNCH\_SHARP: IconData

**buildLAYERSinstance-attribute**

LAYERS: IconData

**buildLAYERS\_CLEARinstance-attribute**

LAYERS\_CLEAR: IconData

**buildLAYERS\_CLEAR\_OUTLINED**instance-attribute

LAYERS\_CLEAR\_OUTLINED: IconData

**buildLAYERS\_CLEAR\_ROUNDED**instance-attribute

LAYERS\_CLEAR\_ROUNDED: IconData

**buildLAYERS\_CLEAR\_SHARP**instance-attribute

LAYERS\_CLEAR\_SHARP: IconData

**buildLAYERS\_OUTLINED**instance-attribute

LAYERS\_OUTLINED: IconData

**buildLAYERS\_ROUNDED**instance-attribute

LAYERS\_ROUNDED: IconData

**buildLAYERS\_SHARP**instance-attribute

LAYERS\_SHARP: IconData

**buildLEADERBOARD**instance-attribute

LEADERBOARD: IconData

**buildLEADERBOARD\_OUTLINED**instance-attribute

LEADERBOARD\_OUTLINED: IconData

**buildLEADERBOARD\_ROUNDED**instance-attribute

LEADERBOARD\_ROUNDED: IconData

**buildLEADERBOARD\_SHARP**instance-attribute

LEADERBOARD\_SHARP: IconData

**buildLEAK\_ADD**instance-attribute

LEAK\_ADD: IconData

**buildLEAK\_ADD\_OUTLINED**instance-attribute

LEAK\_ADD\_OUTLINED: IconData

**buildLEAK\_ADD\_ROUNDED**instance-attribute

LEAK\_ADD\_ROUNDED: IconData

**buildLEAK\_ADD\_SHARP**instance-attribute

LEAK\_ADD\_SHARP: IconData

**buildLEAK\_REMOVE**instance-attribute

LEAK\_REMOVE: IconData

**buildLEAK\_REMOVE\_OUTLINED**instance-attribute

LEAK\_REMOVE\_OUTLINED: IconData

**buildLEAK\_REMOVE\_ROUNDED**instance-attribute

LEAK\_REMOVE\_ROUNDED: IconData

**buildLEAK\_REMOVE\_SHARP**instance-attribute

LEAK\_REMOVE\_SHARP: IconData

**buildLEAVE\_BAGS\_AT\_HOME**instance-attribute

LEAVE\_BAGS\_AT\_HOME: IconData

**buildLEAVE\_BAGS\_AT\_HOME\_OUTLINED**instance-attribute

LEAVE\_BAGS\_AT\_HOME\_OUTLINED: IconData

**buildLEAVE\_BAGS\_AT\_HOME\_ROUNDED**instance-attribute

LEAVE\_BAGS\_AT\_HOME\_ROUNDED: IconData

**buildLEAVE\_BAGS\_AT\_HOME\_SHARP**instance-attribute

LEAVE\_BAGS\_AT\_HOME\_SHARP: IconData

**buildLEGEND\_TOGGLEinstance-attribute**

LEGEND\_TOGGLE: IconData

**buildLEGEND\_TOGGLE\_OUTLINEDinstance-attribute**

LEGEND\_TOGGLE\_OUTLINED: IconData

**buildLEGEND\_TOGGLE\_ROUNDEDinstance-attribute**

LEGEND\_TOGGLE\_ROUNDED: IconData

**buildLEGEND\_TOGGLE\_SHARPinstance-attribute**

LEGEND\_TOGGLE\_SHARP: IconData

**buildLENSinstance-attribute**

LENS: IconData

**buildLENS\_BLURinstance-attribute**

LENS\_BLUR: IconData

**buildLENS\_BLUR\_OUTLINEDinstance-attribute**

LENS\_BLUR\_OUTLINED: IconData

**buildLENS\_BLUR\_ROUNDEDinstance-attribute**

LENS\_BLUR\_ROUNDED: IconData

**buildLENS\_BLUR\_SHARPinstance-attribute**

LENS\_BLUR\_SHARP: IconData

**buildLENS\_OUTLINEDinstance-attribute**

LENS\_OUTLINED: IconData

**buildLENS\_ROUNDEDinstance-attribute**

LENS\_ROUNDED: IconData

**buildLENS\_SHARPinstance-attribute**

LENS\_SHARP: IconData

**buildLIBRARY\_ADDinstance-attribute**

LIBRARY\_ADD: IconData

**buildLIBRARY\_ADD\_CHECKinstance-attribute**

LIBRARY\_ADD\_CHECK: IconData

**buildLIBRARY\_ADD\_CHECK\_OUTLINEDinstance-attribute**

LIBRARY\_ADD\_CHECK\_OUTLINED: IconData

**buildLIBRARY\_ADD\_CHECK\_ROUNDEDinstance-attribute**

LIBRARY\_ADD\_CHECK\_ROUNDED: IconData

**buildLIBRARY\_ADD\_CHECK\_SHARPinstance-attribute**

LIBRARY\_ADD\_CHECK\_SHARP: IconData

**buildLIBRARY\_ADD\_OUTLINEDinstance-attribute**

LIBRARY\_ADD\_OUTLINED: IconData

**buildLIBRARY\_ADD\_ROUNDEDinstance-attribute**

LIBRARY\_ADD\_ROUNDED: IconData

**buildLIBRARY\_ADD\_SHARPinstance-attribute**

LIBRARY\_ADD\_SHARP: IconData

**buildLIBRARY\_BOOKSinstance-attribute**

LIBRARY\_BOOKS: IconData

**buildLIBRARY\_BOOKS\_OUTLINEDinstance-attribute**

LIBRARY\_BOOKS\_OUTLINED: IconData

**buildLIBRARY\_BOOKS\_ROUNDEDinstance-attribute**

LIBRARY\_BOOKS\_ROUNDED: IconData

**buildLIBRARY\_BOOKS\_SHARPinstance-attribute**

LIBRARY\_BOOKS\_SHARP: IconData

**buildLIBRARY\_MUSICinstance-attribute**

LIBRARY\_MUSIC: IconData

**buildLIBRARY\_MUSIC\_OUTLINEDinstance-attribute**

LIBRARY\_MUSIC\_OUTLINED: IconData

**buildLIBRARY\_MUSIC\_ROUNDEDinstance-attribute**

LIBRARY\_MUSIC\_ROUNDED: IconData

**buildLIBRARY\_MUSIC\_SHARPinstance-attribute**

LIBRARY\_MUSIC\_SHARP: IconData

**buildLIGHTinstance-attribute**

LIGHT: IconData

**buildLIGHT\_MODEinstance-attribute**

LIGHT\_MODE: IconData

**buildLIGHT\_MODE\_OUTLINEDinstance-attribute**

LIGHT\_MODE\_OUTLINED: IconData

**buildLIGHT\_MODE\_ROUNDEDinstance-attribute**

LIGHT\_MODE\_ROUNDED: IconData

**buildLIGHT\_MODE\_SHARPinstance-attribute**

LIGHT\_MODE\_SHARP: IconData

**buildLIGHT\_OUTLINEDinstance-attribute**

LIGHT\_OUTLINED: IconData

**buildLIGHT\_ROUNDEDinstance-attribute**

LIGHT\_ROUNDED: IconData

**buildLIGHT\_SHARPinstance-attribute**

LIGHT\_SHARP: IconData

**buildLIGHTBULBinstance-attribute**

LIGHTBULB: IconData

**buildLIGHTBULB\_CIRCLEinstance-attribute**

LIGHTBULB\_CIRCLE: IconData

**buildLIGHTBULB\_CIRCLE\_OUTLINEDinstance-attribute**

LIGHTBULB\_CIRCLE\_OUTLINED: IconData

**buildLIGHTBULB\_CIRCLE\_ROUNDEDinstance-attribute**

LIGHTBULB\_CIRCLE\_ROUNDED: IconData

**buildLIGHTBULB\_CIRCLE\_SHARPinstance-attribute**

LIGHTBULB\_CIRCLE\_SHARP: IconData

**buildLIGHTBULB\_OUTLINEinstance-attribute**

LIGHTBULB\_OUTLINE: IconData

**buildLIGHTBULB\_OUTLINE\_ROUNDEDinstance-attribute**

LIGHTBULB\_OUTLINE\_ROUNDED: IconData

**buildLIGHTBULB\_OUTLINE\_SHARPinstance-attribute**

LIGHTBULB\_OUTLINE\_SHARP: IconData

**buildLIGHTBULB\_OUTLINEDinstance-attribute**

LIGHTBULB\_OUTLINED: IconData

**buildLIGHTBULB\_ROUNDEDinstance-attribute**

LIGHTBULB\_ROUNDED: IconData

**buildLIGHTBULB\_SHARPinstance-attribute**

LIGHTBULB\_SHARP: IconData

**buildLINE\_AXISinstance-attribute**

LINE\_AXIS: IconData

**buildLINE\_AXIS\_OUTLINEDinstance-attribute**

LINE\_AXIS\_OUTLINED: IconData

**buildLINE\_AXIS\_ROUNDEDinstance-attribute**

LINE\_AXIS\_ROUNDED: IconData

**buildLINE\_AXIS\_SHARPinstance-attribute**

LINE\_AXIS\_SHARP: IconData

**buildLINE\_STYLEinstance-attribute**

LINE\_STYLE: IconData

**buildLINE\_STYLE\_OUTLINEDinstance-attribute**

LINE\_STYLE\_OUTLINED: IconData

**buildLINE\_STYLE\_ROUNDEDinstance-attribute**

LINE\_STYLE\_ROUNDED: IconData

**buildLINE\_STYLE\_SHARPinstance-attribute**

LINE\_STYLE\_SHARP: IconData

**buildLINE\_WEIGHTinstance-attribute**

LINE\_WEIGHT: IconData

**buildLINE\_WEIGHT\_OUTLINEDinstance-attribute**

LINE\_WEIGHT\_OUTLINED: IconData

**buildLINE\_WEIGHT\_ROUNDEDinstance-attribute**

LINE\_WEIGHT\_ROUNDED: IconData

**buildLINE\_WEIGHT\_SHARPinstance-attribute**

LINE\_WEIGHT\_SHARP: IconData

**buildLINEAR\_SCALEinstance-attribute**

LINEAR\_SCALE: IconData

**buildLINEAR\_SCALE\_OUTLINEDinstance-attribute**

LINEAR\_SCALE\_OUTLINED: IconData

**buildLINEAR\_SCALE\_ROUNDEDinstance-attribute**

LINEAR\_SCALE\_ROUNDED: IconData

**buildLINEAR\_SCALE\_SHARPinstance-attribute**

LINEAR\_SCALE\_SHARP: IconData

**buildLINKinstance-attribute**

LINK: IconData

**buildLINK\_OFFinstance-attribute**

LINK\_OFF: IconData

**buildLINK\_OFF\_OUTLINEDinstance-attribute**

LINK\_OFF\_OUTLINED: IconData



**buildLINK\_OFF\_ROUNDEDinstance-attribute**

LINK\_OFF\_ROUNDED: IconData

**buildLINK\_OFF\_SHARPinstance-attribute**

LINK\_OFF\_SHARP: IconData

**buildLINK\_OUTLINEDinstance-attribute**

LINK\_OUTLINED: IconData

**buildLINK\_ROUNDEDinstance-attribute**

LINK\_ROUNDED: IconData

**buildLINK\_SHARPinstance-attribute**

LINK\_SHARP: IconData

**buildLINKED\_CAMERAinstance-attribute**

LINKED\_CAMERA: IconData

**buildLINKED\_CAMERA\_OUTLINEDinstance-attribute**

LINKED\_CAMERA\_OUTLINED: IconData

**buildLINKED\_CAMERA\_ROUNDEDinstance-attribute**

LINKED\_CAMERA\_ROUNDED: IconData

**buildLINKED\_CAMERA\_SHARPinstance-attribute**

LINKED\_CAMERA\_SHARP: IconData

**buildLIQUORinstance-attribute**

LIQUOR: IconData

**buildLIQUOR\_OUTLINEDinstance-attribute**

LIQUOR\_OUTLINED: IconData

**buildLIQUOR\_ROUNDEDinstance-attribute**

LIQUOR\_ROUNDED: IconData

**buildLIQUOR\_SHARPinstance-attribute**

LIQUOR\_SHARP: IconData

**buildLISTinstance-attribute**

LIST: IconData

**buildLIST\_ALTinstance-attribute**

LIST\_ALT: IconData

**buildLIST\_ALT\_OUTLINEDinstance-attribute**

LIST\_ALT\_OUTLINED: IconData

**buildLIST\_ALT\_ROUNDEDinstance-attribute**

LIST\_ALT\_ROUNDED: IconData

**buildLIST\_ALT\_SHARPinstance-attribute**

LIST\_ALT\_SHARP: IconData

**buildLIST\_OUTLINEDinstance-attribute**

LIST\_OUTLINED: IconData

**buildLIST\_ROUNDEDinstance-attribute**

LIST\_ROUNDED: IconData

**buildLIST\_SHARPinstance-attribute**

LIST\_SHARP: IconData

**buildLIVE\_HELPinstance-attribute**

LIVE\_HELP: IconData

**buildLIVE\_HELP\_OUTLINEDinstance-attribute**

LIVE\_HELP\_OUTLINED: IconData

**buildLIVE\_HELP\_ROUNDEDinstance-attribute**

LIVE\_HELP\_ROUNDED: IconData

**buildLIVE\_HELP\_SHARPinstance-attribute**

LIVE\_HELP\_SHARP: IconData

**buildLIVE\_TVinstance-attribute**

LIVE\_TV: IconData

**buildLIVE\_TV\_OUTLINEDinstance-attribute**

LIVE\_TV\_OUTLINED: IconData

**buildLIVE\_TV\_ROUNDEDinstance-attribute**

LIVE\_TV\_ROUNDED: IconData

**buildLIVE\_TV\_SHARPinstance-attribute**

LIVE\_TV\_SHARP: IconData

**buildLIVINGinstance-attribute**

LIVING: IconData

**buildLIVING\_OUTLINEDinstance-attribute**

LIVING\_OUTLINED: IconData

**buildLIVING\_ROUNDEDinstance-attribute**

LIVING\_ROUNDED: IconData

**buildLIVING\_SHARPinstance-attribute**

LIVING\_SHARP: IconData

**buildLOCAL\_ACTIVITYinstance-attribute**

LOCAL\_ACTIVITY: IconData

**buildLOCAL\_ACTIVITY\_OUTLINEDinstance-attribute**

LOCAL\_ACTIVITY\_OUTLINED: IconData

**buildLOCAL\_ACTIVITY\_ROUNDEDinstance-attribute**

LOCAL\_ACTIVITY\_ROUNDED: IconData

**buildLOCAL\_ACTIVITY\_SHARPinstance-attribute**

LOCAL\_ACTIVITY\_SHARP: IconData

**buildLOCAL\_AIRPORTinstance-attribute**

LOCAL\_AIRPORT: IconData

**buildLOCAL\_AIRPORT\_OUTLINEDinstance-attribute**

LOCAL\_AIRPORT\_OUTLINED: IconData

**buildLOCAL\_AIRPORT\_ROUNDEDinstance-attribute**

LOCAL\_AIRPORT\_ROUNDED: IconData

**buildLOCAL\_AIRPORT\_SHARPinstance-attribute**

LOCAL\_AIRPORT\_SHARP: IconData

**buildLOCAL\_ATMinstance-attribute**

LOCAL\_ATM: IconData

**buildLOCAL\_ATM\_OUTLINEDinstance-attribute**

LOCAL\_ATM\_OUTLINED: IconData

**buildLOCAL\_ATM\_ROUNDEDinstance-attribute**

LOCAL\_ATM\_ROUNDED: IconData

**buildLOCAL\_ATM\_SHARPinstance-attribute**

LOCAL\_ATM\_SHARP: IconData

**buildLOCAL\_ATTRACTIONinstance-attribute**

LOCAL\_ATTRACTION: IconData

**buildLOCAL\_ATTRACTION\_OUTLINEDinstance-attribute**

LOCAL\_ATTRACTION\_OUTLINED: IconData

**buildLOCAL\_ATTRACTION\_ROUNDEDinstance-attribute**

LOCAL\_ATTRACTION\_ROUNDED: IconData

**buildLOCAL\_ATTRACTION\_SHARPinstance-attribute**

LOCAL\_ATTRACTION\_SHARP: IconData

**buildLOCAL\_BARinstance-attribute**

LOCAL\_BAR: IconData

**buildLOCAL\_BAR\_OUTLINEDinstance-attribute**

LOCAL\_BAR\_OUTLINED: IconData

**buildLOCAL\_BAR\_ROUNDEDinstance-attribute**

LOCAL\_BAR\_ROUNDED: IconData

**buildLOCAL\_BAR\_SHARPinstance-attribute**

LOCAL\_BAR\_SHARP: IconData

**buildLOCAL\_CAFEinstance-attribute**

LOCAL\_CAFE: IconData

**buildLOCAL\_CAFE\_OUTLINEDinstance-attribute**

LOCAL\_CAFE\_OUTLINED: IconData

**buildLOCAL\_CAFE\_ROUNDEDinstance-attribute**

LOCAL\_CAFE\_ROUNDED: IconData

**buildLOCAL\_CAFE\_SHARPinstance-attribute**

LOCAL\_CAFE\_SHARP: IconData

**buildLOCAL\_CAR\_WASHinstance-attribute**

LOCAL\_CAR\_WASH: IconData

**buildLOCAL\_CAR\_WASH\_OUTLINEDinstance-attribute**

LOCAL\_CAR\_WASH\_OUTLINED: IconData

**buildLOCAL\_CAR\_WASH\_ROUNDEDinstance-attribute**

LOCAL\_CAR\_WASH\_ROUNDED: IconData

**buildLOCAL\_CAR\_WASH\_SHARPinstance-attribute**

LOCAL\_CAR\_WASH\_SHARP: IconData

**buildLOCAL\_CONVENIENCE\_STOREinstance-attribute**

LOCAL\_CONVENIENCE\_STORE: IconData

**buildLOCAL\_CONVENIENCE\_STORE\_OUTLINEDinstance-attribute**

LOCAL\_CONVENIENCE\_STORE\_OUTLINED: IconData

**buildLOCAL\_CONVENIENCE\_STORE\_ROUNDEDinstance-attribute**

LOCAL\_CONVENIENCE\_STORE\_ROUNDED: IconData

**buildLOCAL\_CONVENIENCE\_STORE\_SHARPinstance-attribute**

LOCAL\_CONVENIENCE\_STORE\_SHARP: IconData

**buildLOCAL\_DININGinstance-attribute**

LOCAL\_DINING: IconData

**buildLOCAL\_DINING\_OUTLINEDinstance-attribute**

LOCAL\_DINING\_OUTLINED: IconData

**buildLOCAL\_DINING\_ROUNDEDinstance-attribute**

LOCAL\_DINING\_ROUNDED: IconData

**buildLOCAL\_DINING\_SHARPinstance-attribute**

LOCAL\_DINING\_SHARP: IconData

**buildLOCAL\_DRINKinstance-attribute**

LOCAL\_DRINK: IconData

**buildLOCAL\_DRINK\_OUTLINEDinstance-attribute**

LOCAL\_DRINK\_OUTLINED: IconData

**buildLOCAL\_DRINK\_ROUNDEDinstance-attribute**

LOCAL\_DRINK\_ROUNDED: IconData

**buildLOCAL\_DRINK\_SHARPinstance-attribute**

LOCAL\_DRINK\_SHARP: IconData

**buildLOCAL\_FIRE\_DEPARTMENTinstance-attribute**

LOCAL\_FIRE\_DEPARTMENT: IconData

**buildLOCAL\_FIRE\_DEPARTMENT\_OUTLINEDinstance-attribute**

LOCAL\_FIRE\_DEPARTMENT\_OUTLINED: IconData

**buildLOCAL\_FIRE\_DEPARTMENT\_ROUNDEDinstance-attribute**

LOCAL\_FIRE\_DEPARTMENT\_ROUNDED: IconData

**buildLOCAL\_FIRE\_DEPARTMENT\_SHARPinstance-attribute**

LOCAL\_FIRE\_DEPARTMENT\_SHARP: IconData

**buildLOCAL\_FLORISTinstance-attribute**

LOCAL\_FLORIST: IconData

**buildLOCAL\_FLORIST\_OUTLINEDinstance-attribute**

LOCAL\_FLORIST\_OUTLINED: IconData

**buildLOCAL\_FLORIST\_ROUNDEDinstance-attribute**

LOCAL\_FLORIST\_ROUNDED: IconData

**buildLOCAL\_FLORIST\_SHARPinstance-attribute**

LOCAL\_FLORIST\_SHARP: IconData

**buildLOCAL\_GAS\_STATIONinstance-attribute**

LOCAL\_GAS\_STATION: IconData

**buildLOCAL\_GAS\_STATION\_OUTLINEDinstance-attribute**

LOCAL\_GAS\_STATION\_OUTLINED: IconData

**buildLOCAL\_GAS\_STATION\_ROUNDEDinstance-attribute**

LOCAL\_GAS\_STATION\_ROUNDED: IconData

**buildLOCAL\_GAS\_STATION\_SHARPinstance-attribute**

LOCAL\_GAS\_STATION\_SHARP: IconData

**buildLOCAL\_GROCERY\_STOREinstance-attribute**

LOCAL\_GROCERY\_STORE: IconData

**buildLOCAL\_GROCERY\_STORE\_OUTLINEDinstance-attribute**

LOCAL\_GROCERY\_STORE\_OUTLINED: IconData

**buildLOCAL\_GROCERY\_STORE\_ROUNDEDinstance-attribute**

LOCAL\_GROCERY\_STORE\_ROUNDED: IconData

**buildLOCAL\_GROCERY\_STORE\_SHARPinstance-attribute**

LOCAL\_GROCERY\_STORE\_SHARP: IconData

**buildLOCAL\_HOSPITALinstance-attribute**

LOCAL\_HOSPITAL: IconData

**buildLOCAL\_HOSPITAL\_OUTLINEDinstance-attribute**

LOCAL\_HOSPITAL\_OUTLINED: IconData

**buildLOCAL\_HOSPITAL\_ROUNDEDinstance-attribute**

LOCAL\_HOSPITAL\_ROUNDED: IconData

**buildLOCAL\_HOSPITAL\_SHARPinstance-attribute**

LOCAL\_HOSPITAL\_SHARP: IconData

**buildLOCAL\_HOTELinstance-attribute**

LOCAL\_HOTEL: IconData

**buildLOCAL\_HOTEL\_OUTLINEDinstance-attribute**

LOCAL\_HOTEL\_OUTLINED: IconData

**buildLOCAL\_HOTEL\_ROUNDEDinstance-attribute**

LOCAL\_HOTEL\_ROUNDED: IconData

**buildLOCAL\_HOTEL\_SHARPinstance-attribute**

LOCAL\_HOTEL\_SHARP: IconData

**buildLOCAL\_LAUNDRY\_SERVICEinstance-attribute**

LOCAL\_LAUNDRY\_SERVICE: IconData

**buildLOCAL\_LAUNDRY\_SERVICE\_OUTLINEDinstance-attribute**

LOCAL\_LAUNDRY\_SERVICE\_OUTLINED: IconData

**buildLOCAL\_LAUNDRY\_SERVICE\_ROUNDEDinstance-attribute**

LOCAL\_LAUNDRY\_SERVICE\_ROUNDED: IconData

**buildLOCAL\_LAUNDRY\_SERVICE\_SHARPinstance-attribute**

LOCAL\_LAUNDRY\_SERVICE\_SHARP: IconData

**buildLOCAL\_LIBRARYinstance-attribute**

LOCAL\_LIBRARY: IconData

**buildLOCAL\_LIBRARY\_OUTLINEDinstance-attribute**

LOCAL\_LIBRARY\_OUTLINED: IconData

**buildLOCAL\_LIBRARY\_ROUNDEDinstance-attribute**

LOCAL\_LIBRARY\_ROUNDED: IconData

**buildLOCAL\_LIBRARY\_SHARPinstance-attribute**

LOCAL\_LIBRARY\_SHARP: IconData

**buildLOCAL\_MALLinstance-attribute**

LOCAL\_MALL: IconData

**buildLOCAL\_MALL\_OUTLINEDinstance-attribute**

LOCAL\_MALL\_OUTLINED: IconData

**buildLOCAL\_MALL\_ROUNDEDinstance-attribute**

LOCAL\_MALL\_ROUNDED: IconData

**buildLOCAL\_MALL\_SHARPinstance-attribute**

LOCAL\_MALL\_SHARP: IconData

**buildLOCAL\_MOVIESinstance-attribute**

LOCAL\_MOVIES: IconData

**buildLOCAL\_MOVIES\_OUTLINEDinstance-attribute**

LOCAL\_MOVIES\_OUTLINED: IconData

**buildLOCAL\_MOVIES\_ROUNDEDinstance-attribute**

LOCAL\_MOVIES\_ROUNDED: IconData

**buildLOCAL\_MOVIES\_SHARPinstance-attribute**

LOCAL\_MOVIES\_SHARP: IconData

**buildLOCAL\_OFFERinstance-attribute**

LOCAL\_OFFER: IconData

**buildLOCAL\_OFFER\_OUTLINEDinstance-attribute**

LOCAL\_OFFER\_OUTLINED: IconData

**buildLOCAL\_OFFER\_ROUNDEDinstance-attribute**

LOCAL\_OFFER\_ROUNDED: IconData

**buildLOCAL\_OFFER\_SHARPinstance-attribute**

LOCAL\_OFFER\_SHARP: IconData

**buildLOCAL\_PARKINGinstance-attribute**

LOCAL\_PARKING: IconData

**buildLOCAL\_PARKING\_OUTLINEDinstance-attribute**

LOCAL\_PARKING\_OUTLINED: IconData

**buildLOCAL\_PARKING\_ROUNDEDinstance-attribute**

LOCAL\_PARKING\_ROUNDED: IconData

**buildLOCAL\_PARKING\_SHARPinstance-attribute**

LOCAL\_PARKING\_SHARP: IconData

**buildLOCAL\_PHARMACYinstance-attribute**

LOCAL\_PHARMACY: IconData

**buildLOCAL\_PHARMACY\_OUTLINEDinstance-attribute**

LOCAL\_PHARMACY\_OUTLINED: IconData

**buildLOCAL\_PHARMACY\_ROUNDEDinstance-attribute**

LOCAL\_PHARMACY\_ROUNDED: IconData

**buildLOCAL\_PHARMACY\_SHARPinstance-attribute**

LOCAL\_PHARMACY\_SHARP: IconData

**buildLOCAL\_PHONEinstance-attribute**

LOCAL\_PHONE: IconData

**buildLOCAL\_PHONE\_OUTLINEDinstance-attribute**

LOCAL\_PHONE\_OUTLINED: IconData

**buildLOCAL\_PHONE\_ROUNDEDinstance-attribute**

LOCAL\_PHONE\_ROUNDED: IconData

**buildLOCAL\_PHONE\_SHARPinstance-attribute**

LOCAL\_PHONE\_SHARP: IconData

**buildLOCAL\_PIZZAinstance-attribute**

LOCAL\_PIZZA: IconData

**buildLOCAL\_PIZZA\_OUTLINEDinstance-attribute**

LOCAL\_PIZZA\_OUTLINED: IconData

**buildLOCAL\_PIZZA\_ROUNDEDinstance-attribute**

LOCAL\_PIZZA\_ROUNDED: IconData

**buildLOCAL\_PIZZA\_SHARPinstance-attribute**

LOCAL\_PIZZA\_SHARP: IconData

**buildLOCAL\_PLAYinstance-attribute**

LOCAL\_PLAY: IconData

**buildLOCAL\_PLAY\_OUTLINEDinstance-attribute**

LOCAL\_PLAY\_OUTLINED: IconData

**buildLOCAL\_PLAY\_ROUNDEDinstance-attribute**

LOCAL\_PLAY\_ROUNDED: IconData

**buildLOCAL\_PLAY\_SHARPinstance-attribute**

LOCAL\_PLAY\_SHARP: IconData

**buildLOCAL\_POLICEinstance-attribute**

LOCAL\_POLICE: IconData

**buildLOCAL\_POLICE\_OUTLINEDinstance-attribute**

LOCAL\_POLICE\_OUTLINED: IconData

**buildLOCAL\_POLICE\_ROUNDEDinstance-attribute**

LOCAL\_POLICE\_ROUNDED: IconData

**buildLOCAL\_POLICE\_SHARPinstance-attribute**

LOCAL\_POLICE\_SHARP: IconData

**buildLOCAL\_POST\_OFFICEinstance-attribute**

LOCAL\_POST\_OFFICE: IconData

**buildLOCAL\_POST\_OFFICE\_OUTLINEDinstance-attribute**

LOCAL\_POST\_OFFICE\_OUTLINED: IconData

**buildLOCAL\_POST\_OFFICE\_ROUNDEDinstance-attribute**

LOCAL\_POST\_OFFICE\_ROUNDED: IconData

**buildLOCAL\_POST\_OFFICE\_SHARPinstance-attribute**

LOCAL\_POST\_OFFICE\_SHARP: IconData

**buildLOCAL\_PRINT\_SHOPinstance-attribute**

LOCAL\_PRINT\_SHOP: IconData

**buildLOCAL\_PRINT\_SHOP\_OUTLINEDinstance-attribute**

LOCAL\_PRINT\_SHOP\_OUTLINED: IconData

**buildLOCAL\_PRINT\_SHOP\_ROUNDEDinstance-attribute**

LOCAL\_PRINT\_SHOP\_ROUNDED: IconData

**buildLOCAL\_PRINT\_SHOP\_SHARPinstance-attribute**

LOCAL\_PRINT\_SHOP\_SHARP: IconData

**buildLOCAL\_PRINTSHOPinstance-attribute**

LOCAL\_PRINTSHOP: IconData

**buildLOCAL\_PRINTSHOP\_OUTLINEDinstance-attribute**

LOCAL\_PRINTSHOP\_OUTLINED: IconData

**buildLOCAL\_PRINTSHOP\_ROUNDEDinstance-attribute**

LOCAL\_PRINTSHOP\_ROUNDED: IconData

**buildLOCAL\_PRINTSHOP\_SHARPinstance-attribute**

LOCAL\_PRINTSHOP\_SHARP: IconData

**buildLOCAL\_RESTAURANTinstance-attribute**

LOCAL\_RESTAURANT: IconData

**buildLOCAL\_RESTAURANT\_OUTLINEDinstance-attribute**

LOCAL\_RESTAURANT\_OUTLINED: IconData

**buildLOCAL\_RESTAURANT\_ROUNDEDinstance-attribute**

LOCAL\_RESTAURANT\_ROUNDED: IconData

**buildLOCAL\_RESTAURANT\_SHARPinstance-attribute**

LOCAL\_RESTAURANT\_SHARP: IconData

**buildLOCAL\_SEEinstance-attribute**

LOCAL\_SEE: IconData

**buildLOCAL\_SEE\_OUTLINEDinstance-attribute**

LOCAL\_SEE\_OUTLINED: IconData

**buildLOCAL\_SEE\_ROUNDEDinstance-attribute**

LOCAL\_SEE\_ROUNDED: IconData

**buildLOCAL\_SEE\_SHARPinstance-attribute**

LOCAL\_SEE\_SHARP: IconData

**buildLOCAL\_SHIPPINGinstance-attribute**

LOCAL\_SHIPPING: IconData

**buildLOCAL\_SHIPPING\_OUTLINEDinstance-attribute**

LOCAL\_SHIPPING\_OUTLINED: IconData

**buildLOCAL\_SHIPPING\_ROUNDEDinstance-attribute**

LOCAL\_SHIPPING\_ROUNDED: IconData

**buildLOCAL\_SHIPPING\_SHARPinstance-attribute**

LOCAL\_SHIPPING\_SHARP: IconData

**buildLOCAL\_TAXIinstance-attribute**

LOCAL\_TAXI: IconData

**buildLOCAL\_TAXI\_OUTLINEDinstance-attribute**

LOCAL\_TAXI\_OUTLINED: IconData

**buildLOCAL\_TAXI\_ROUNDEDinstance-attribute**

LOCAL\_TAXI\_ROUNDED: IconData

**buildLOCAL\_TAXI\_SHARPinstance-attribute**

LOCAL\_TAXI\_SHARP: IconData

**buildLOCATION\_CITYinstance-attribute**

LOCATION\_CITY: IconData

**buildLOCATION\_CITY\_OUTLINEDinstance-attribute**

LOCATION\_CITY\_OUTLINED: IconData

**buildLOCATION\_CITY\_ROUNDEDinstance-attribute**

LOCATION\_CITY\_ROUNDED: IconData

**buildLOCATION\_CITY\_SHARPinstance-attribute**

LOCATION\_CITY\_SHARP: IconData

**buildLOCATION\_DISABLEDinstance-attribute**

LOCATION\_DISABLED: IconData

**buildLOCATION\_DISABLED\_OUTLINEDinstance-attribute**

LOCATION\_DISABLED\_OUTLINED: IconData

**buildLOCATION\_DISABLED\_ROUNDEDinstance-attribute**

LOCATION\_DISABLED\_ROUNDED: IconData



**buildLOCATION\_DISABLED\_SHARPinstance-attribute**

LOCATION\_DISABLED\_SHARP: IconData

**buildLOCATION\_HISTORYinstance-attribute**

LOCATION\_HISTORY: IconData

**buildLOCATION\_HISTORY\_OUTLINEDinstance-attribute**

LOCATION\_HISTORY\_OUTLINED: IconData

**buildLOCATION\_HISTORY\_ROUNDEDinstance-attribute**

LOCATION\_HISTORY\_ROUNDED: IconData

**buildLOCATION\_HISTORY\_SHARPinstance-attribute**

LOCATION\_HISTORY\_SHARP: IconData

**buildLOCATION\_OFFinstance-attribute**

LOCATION\_OFF: IconData

**buildLOCATION\_OFF\_OUTLINEDinstance-attribute**

LOCATION\_OFF\_OUTLINED: IconData

**buildLOCATION\_OFF\_ROUNDEDinstance-attribute**

LOCATION\_OFF\_ROUNDED: IconData

**buildLOCATION\_OFF\_SHARPinstance-attribute**

LOCATION\_OFF\_SHARP: IconData

**buildLOCATION\_ONinstance-attribute**

LOCATION\_ON: IconData

**buildLOCATION\_ON\_OUTLINEDinstance-attribute**

LOCATION\_ON\_OUTLINED: IconData

**buildLOCATION\_ON\_ROUNDEDinstance-attribute**

LOCATION\_ON\_ROUNDED: IconData

**buildLOCATION\_ON\_SHARPinstance-attribute**

LOCATION\_ON\_SHARP: IconData

**buildLOCATION\_PINinstance-attribute**

LOCATION\_PIN: IconData

**buildLOCATION\_SEARCHINGinstance-attribute**

LOCATION\_SEARCHING: IconData

**buildLOCATION\_SEARCHING\_OUTLINEDinstance-attribute**

LOCATION\_SEARCHING\_OUTLINED: IconData

**buildLOCATION\_SEARCHING\_ROUNDEDinstance-attribute**

LOCATION\_SEARCHING\_ROUNDED: IconData

**buildLOCATION\_SEARCHING\_SHARPinstance-attribute**

LOCATION\_SEARCHING\_SHARP: IconData

**buildLOCKinstance-attribute**

LOCK: IconData

**buildLOCK\_CLOCKinstance-attribute**

LOCK\_CLOCK: IconData

**buildLOCK\_CLOCK\_OUTLINEDinstance-attribute**

LOCK\_CLOCK\_OUTLINED: IconData

**buildLOCK\_CLOCK\_ROUNDEDinstance-attribute**

LOCK\_CLOCK\_ROUNDED: IconData

**buildLOCK\_CLOCK\_SHARPinstance-attribute**

LOCK\_CLOCK\_SHARP: IconData

**buildLOCK\_OPENinstance-attribute**

LOCK\_OPEN: IconData

**buildLOCK\_OPEN\_OUTLINEDinstance-attribute**

LOCK\_OPEN\_OUTLINED: IconData

**buildLOCK\_OPEN\_ROUNDEDinstance-attribute**

LOCK\_OPEN\_ROUNDED: IconData

**buildLOCK\_OPEN\_SHARPinstance-attribute**

LOCK\_OPEN\_SHARP: IconData

**buildLOCK\_OUTLINEinstance-attribute**

LOCK\_OUTLINE: IconData

**buildLOCK\_OUTLINE\_ROUNDEDinstance-attribute**

LOCK\_OUTLINE\_ROUNDED: IconData

**buildLOCK\_OUTLINE\_SHARPinstance-attribute**

LOCK\_OUTLINE\_SHARP: IconData

**buildLOCK\_OUTLINEDinstance-attribute**

LOCK\_OUTLINED: IconData

**buildLOCK\_PERSONinstance-attribute**

LOCK\_PERSON: IconData

**buildLOCK\_PERSON\_OUTLINEDinstance-attribute**

LOCK\_PERSON\_OUTLINED: IconData

**buildLOCK\_PERSON\_ROUNDEDinstance-attribute**

LOCK\_PERSON\_ROUNDED: IconData

**buildLOCK\_PERSON\_SHARPinstance-attribute**

LOCK\_PERSON\_SHARP: IconData

**buildLOCK\_RESETinstance-attribute**

LOCK\_RESET: IconData

**buildLOCK\_RESET\_OUTLINEDinstance-attribute**

LOCK\_RESET\_OUTLINED: IconData

**buildLOCK\_RESET\_ROUNDEDinstance-attribute**

LOCK\_RESET\_ROUNDED: IconData

**buildLOCK\_RESET\_SHARPinstance-attribute**

LOCK\_RESET\_SHARP: IconData

**buildLOCK\_ROUNDEDinstance-attribute**

LOCK\_ROUNDED: IconData

**buildLOCK\_SHARPinstance-attribute**

LOCK\_SHARP: IconData

**buildLOGINinstance-attribute**

LOGIN: IconData

**buildLOGIN\_OUTLINEDinstance-attribute**

LOGIN\_OUTLINED: IconData

**buildLOGIN\_ROUNDEDinstance-attribute**

LOGIN\_ROUNDED: IconData

**buildLOGIN\_SHARPinstance-attribute**

LOGIN\_SHARP: IconData

**buildLOGO\_DEVinstance-attribute**

LOGO\_DEV: IconData

**buildLOGO\_DEV\_OUTLINEDinstance-attribute**

LOGO\_DEV\_OUTLINED: IconData

**buildLOGO\_DEV\_ROUNDEDinstance-attribute**

LOGO\_DEV\_ROUNDED: IconData

**buildLOGO\_DEV\_SHARPinstance-attribute**

LOGO\_DEV\_SHARP: IconData

**buildLOGOUTinstance-attribute**

LOGOUT: IconData

**buildLOGOUT\_OUTLINEDinstance-attribute**

LOGOUT\_OUTLINED: IconData

**buildLOGOUT\_ROUNDEDinstance-attribute**

LOGOUT\_ROUNDED: IconData

**buildLOGOUT\_SHARPinstance-attribute**

LOGOUT\_SHARP: IconData

**buildLOOKSinstance-attribute**

LOOKS: IconData

**buildLOOKS\_3instance-attribute**

LOOKS\_3: IconData

**buildLOOKS\_3\_OUTLINEDinstance-attribute**

LOOKS\_3\_OUTLINED: IconData

**buildLOOKS\_3\_ROUNDEDinstance-attribute**

LOOKS\_3\_ROUNDED: IconData

**buildLOOKS\_3\_SHARPinstance-attribute**

LOOKS\_3\_SHARP: IconData

**buildLOOKS\_4instance-attribute**

LOOKS\_4: IconData

**buildLOOKS\_4\_OUTLINEDinstance-attribute**

LOOKS\_4\_OUTLINED: IconData

**buildLOOKS\_4\_ROUNDEDinstance-attribute**

LOOKS\_4\_ROUNDED: IconData

**buildLOOKS\_4\_SHARPinstance-attribute**

LOOKS\_4\_SHARP: IconData

**buildLOOKS\_5instance-attribute**

LOOKS\_5: IconData

**buildLOOKS\_5\_OUTLINEDinstance-attribute**

LOOKS\_5\_OUTLINED: IconData

**buildLOOKS\_5\_ROUNDEDinstance-attribute**

LOOKS\_5\_ROUNDED: IconData

**buildLOOKS\_5\_SHARPinstance-attribute**

LOOKS\_5\_SHARP: IconData

**buildLOOKS\_6instance-attribute**

LOOKS\_6: IconData

**buildLOOKS\_6\_OUTLINEDinstance-attribute**

LOOKS\_6\_OUTLINED: IconData

**buildLOOKS\_6\_ROUNDEDinstance-attribute**

LOOKS\_6\_ROUNDED: IconData

**buildLOOKS\_6\_SHARPinstance-attribute**

LOOKS\_6\_SHARP: IconData

**buildLOOKS\_ONEinstance-attribute**

LOOKS\_ONE: IconData

**buildLOOKS\_ONE\_OUTLINEDinstance-attribute**

LOOKS\_ONE\_OUTLINED: IconData

**buildLOOKS\_ONE\_ROUNDEDinstance-attribute**

LOOKS\_ONE\_ROUNDED: IconData

**buildLOOKS\_ONE\_SHARPinstance-attribute**

LOOKS\_ONE\_SHARP: IconData

**buildLOOKS\_OUTLINEDinstance-attribute**

LOOKS\_OUTLINED: IconData

**buildLOOKS\_ROUNDEDinstance-attribute**

LOOKS\_ROUNDED: IconData

**buildLOOKS\_SHARPinstance-attribute**

LOOKS\_SHARP: IconData

**buildLOOKS\_TWOinstance-attribute**

LOOKS\_TWO: IconData

**buildLOOKS\_TWO\_OUTLINEDinstance-attribute**

LOOKS\_TWO\_OUTLINED: IconData

**buildLOOKS\_TWO\_ROUNDEDinstance-attribute**

LOOKS\_TWO\_ROUNDED: IconData

**buildLOOKS\_TWO\_SHARPinstance-attribute**

LOOKS\_TWO\_SHARP: IconData

**buildLOOPinstance-attribute**

LOOP: IconData

**buildLOOP\_OUTLINEDinstance-attribute**

LOOP\_OUTLINED: IconData

**buildLOOP\_ROUNDEDinstance-attribute**

LOOP\_ROUNDED: IconData

**buildLOOP\_SHARPinstance-attribute**

LOOP\_SHARP: IconData

**buildLOUPEinstance-attribute**

LOUPE: IconData

**buildLOUPE\_OUTLINEDinstance-attribute**

LOUPE\_OUTLINED: IconData

**buildLOUPE\_ROUNDEDinstance-attribute**

LOUPE\_ROUNDED: IconData

**buildLOUPE\_SHARPinstance-attribute**

LOUPE\_SHARP: IconData

**buildLOW\_PRIORITYinstance-attribute**

LOW\_PRIORITY: IconData

**buildLOW\_PRIORITY\_OUTLINEDinstance-attribute**

LOW\_PRIORITY\_OUTLINED: IconData

**buildLOW\_PRIORITY\_ROUNDEDinstance-attribute**

LOW\_PRIORITY\_ROUNDED: IconData

**buildLOW\_PRIORITY\_SHARPinstance-attribute**

LOW\_PRIORITY\_SHARP: IconData

**buildLOYALTYinstance-attribute**

LOYALTY: IconData

**buildLOYALTY\_OUTLINEDinstance-attribute**

LOYALTY\_OUTLINED: IconData

**buildLOYALTY\_ROUNDEDinstance-attribute**

LOYALTY\_ROUNDED: IconData

**buildLOYALTY\_SHARPinstance-attribute**

LOYALTY\_SHARP: IconData

**buildLTE\_MOBILEDATAinstance-attribute**

LTE\_MOBILEDATA: IconData

**buildLTE\_MOBILEDATA\_OUTLINEDinstance-attribute**

LTE\_MOBILEDATA\_OUTLINED: IconData

**buildLTE\_MOBILEDATA\_ROUNDEDinstance-attribute**

LTE\_MOBILEDATA\_ROUNDED: IconData

**buildLTE\_MOBILEDATA\_SHARPinstance-attribute**

LTE\_MOBILEDATA\_SHARP: IconData

**buildLTE\_PLUS\_MOBILEDATAinstance-attribute**

LTE\_PLUS\_MOBILEDATA: IconData

**buildLTE\_PLUS\_MOBILEDATA\_OUTLINEDinstance-attribute**

LTE\_PLUS\_MOBILEDATA\_OUTLINED: IconData

**buildLTE\_PLUS\_MOBILEDATA\_ROUNDEDinstance-attribute**

LTE\_PLUS\_MOBILEDATA\_ROUNDED: IconData

**buildLTE\_PLUS\_MOBILEDATA\_SHARPinstance-attribute**

LTE\_PLUS\_MOBILEDATA\_SHARP: IconData

**buildLUGGAGEinstance-attribute**

LUGGAGE: IconData

**buildLUGGAGE\_OUTLINEDinstance-attribute**

LUGGAGE\_OUTLINED: IconData

**buildLUGGAGE\_ROUNDEDinstance-attribute**

LUGGAGE\_ROUNDED: IconData

**buildLUGGAGE\_SHARPinstance-attribute**

LUGGAGE\_SHARP: IconData

**buildLUNCH\_DININGinstance-attribute**

LUNCH\_DINING: IconData

**buildLUNCH\_DINING\_OUTLINEDinstance-attribute**

LUNCH\_DINING\_OUTLINED: IconData

**buildLUNCH\_DINING\_ROUNDEDinstance-attribute**

LUNCH\_DINING\_ROUNDED: IconData

**buildLUNCH\_DINING\_SHARPinstance-attribute**

LUNCH\_DINING\_SHARP: IconData

**buildLYRICSinstance-attribute**

LYRICS: IconData

**buildLYRICS\_OUTLINEDinstance-attribute**

LYRICS\_OUTLINED: IconData

**buildLYRICS\_ROUNDEDinstance-attribute**

LYRICS\_ROUNDED: IconData

**buildLYRICS\_SHARPinstance-attribute**

LYRICS\_SHARP: IconData

**buildMACRO\_OFFinstance-attribute**

MACRO\_OFF: IconData

**buildMACRO\_OFF\_OUTLINEDinstance-attribute**

MACRO\_OFF\_OUTLINED: IconData

**buildMACRO\_OFF\_ROUNDEDinstance-attribute**

MACRO\_OFF\_ROUNDED: IconData

**buildMACRO\_OFF\_SHARPinstance-attribute**

MACRO\_OFF\_SHARP: IconData

**buildMAILinstance-attribute**

MAIL: IconData

**buildMAIL\_LOCKinstance-attribute**

MAIL\_LOCK: IconData

**buildMAIL\_LOCK\_OUTLINEDinstance-attribute**

MAIL\_LOCK\_OUTLINED: IconData

**buildMAIL\_LOCK\_ROUNDEDinstance-attribute**

MAIL\_LOCK\_ROUNDED: IconData

**buildMAIL\_LOCK\_SHARPinstance-attribute**

MAIL\_LOCK\_SHARP: IconData

**buildMAIL\_OUTLINEinstance-attribute**

MAIL\_OUTLINE: IconData

**buildMAIL\_OUTLINE\_OUTLINEDinstance-attribute**

MAIL\_OUTLINE\_OUTLINED: IconData

**buildMAIL\_OUTLINE\_ROUNDEDinstance-attribute**

MAIL\_OUTLINE\_ROUNDED: IconData

**buildMAIL\_OUTLINE\_SHARPinstance-attribute**

MAIL\_OUTLINE\_SHARP: IconData

**buildMAIL\_OUTLINEDinstance-attribute**

MAIL\_OUTLINED: IconData

**buildMAIL\_ROUNDEDinstance-attribute**

MAIL\_ROUNDED: IconData

**buildMAIL\_SHARPinstance-attribute**

MAIL\_SHARP: IconData

**buildMALEinstance-attribute**

MALE: IconData

**buildMALE\_OUTLINEDinstance-attribute**

MALE\_OUTLINED: IconData

**buildMALE\_ROUNDEDinstance-attribute**

MALE\_ROUNDED: IconData

**buildMALE\_SHARPinstance-attribute**

MALE\_SHARP: IconData

**buildMANinstance-attribute**

MAN: IconData

**buildMAN\_2instance-attribute**

MAN\_2: IconData

**buildMAN\_2\_OUTLINEDinstance-attribute**

MAN\_2\_OUTLINED: IconData

**buildMAN\_2\_ROUNDEDinstance-attribute**

MAN\_2\_ROUNDED: IconData

**buildMAN\_2\_SHARPinstance-attribute**

MAN\_2\_SHARP: IconData

**buildMAN\_3instance-attribute**

MAN\_3: IconData

**buildMAN\_3\_OUTLINEDinstance-attribute**

MAN\_3\_OUTLINED: IconData

**buildMAN\_3\_ROUNDEDinstance-attribute**

MAN\_3\_ROUNDED: IconData

**buildMAN\_3\_SHARPinstance-attribute**

MAN\_3\_SHARP: IconData

**buildMAN\_4instance-attribute**

MAN\_4: IconData

**buildMAN\_4\_OUTLINEDinstance-attribute**

MAN\_4\_OUTLINED: IconData

**buildMAN\_4\_ROUNDEDinstance-attribute**

MAN\_4\_ROUNDED: IconData

**buildMAN\_4\_SHARPinstance-attribute**

MAN\_4\_SHARP: IconData

**buildMAN\_OUTLINEDinstance-attribute**

MAN\_OUTLINED: IconData

**buildMAN\_ROUNDEDinstance-attribute**

MAN\_ROUNDED: IconData

**buildMAN\_SHARPinstance-attribute**

MAN\_SHARP: IconData

**buildMANAGE\_ACCOUNTSinstance-attribute**

MANAGE\_ACCOUNTS: IconData

**buildMANAGE\_ACCOUNTS\_OUTLINED**instance-attribute

MANAGE\_ACCOUNTS\_OUTLINED: IconData

**buildMANAGE\_ACCOUNTS\_ROUNDED**instance-attribute

MANAGE\_ACCOUNTS\_ROUNDED: IconData

**buildMANAGE\_ACCOUNTS\_SHARP**instance-attribute

MANAGE\_ACCOUNTS\_SHARP: IconData

**buildMANAGE\_HISTORY**instance-attribute

MANAGE\_HISTORY: IconData

**buildMANAGE\_HISTORY\_OUTLINED**instance-attribute

MANAGE\_HISTORY\_OUTLINED: IconData

**buildMANAGE\_HISTORY\_ROUNDED**instance-attribute

MANAGE\_HISTORY\_ROUNDED: IconData

**buildMANAGE\_HISTORY\_SHARP**instance-attribute

MANAGE\_HISTORY\_SHARP: IconData

**buildMANAGE\_SEARCH**instance-attribute

MANAGE\_SEARCH: IconData

**buildMANAGE\_SEARCH\_OUTLINED**instance-attribute

MANAGE\_SEARCH\_OUTLINED: IconData

**buildMANAGE\_SEARCH\_ROUNDED**instance-attribute

MANAGE\_SEARCH\_ROUNDED: IconData

**buildMANAGE\_SEARCH\_SHARP**instance-attribute

MANAGE\_SEARCH\_SHARP: IconData

**buildMAP**instance-attribute

MAP: IconData

**buildMAP\_OUTLINED**instance-attribute

MAP\_OUTLINED: IconData

**buildMAP\_ROUNDED**instance-attribute

MAP\_ROUNDED: IconData

**buildMAP\_SHARP**instance-attribute

MAP\_SHARP: IconData

**buildMAPS\_HOME\_WORK**instance-attribute

MAPS\_HOME\_WORK: IconData

**buildMAPS\_HOME\_WORK\_OUTLINED**instance-attribute

MAPS\_HOME\_WORK\_OUTLINED: IconData

**buildMAPS\_HOME\_WORK\_ROUNDED**instance-attribute

MAPS\_HOME\_WORK\_ROUNDED: IconData

**buildMAPS\_HOME\_WORK\_SHARP**instance-attribute

MAPS\_HOME\_WORK\_SHARP: IconData

**buildMAPS\_UGC**instance-attribute

MAPS\_UGC: IconData

**buildMAPS\_UGC\_OUTLINED**instance-attribute

MAPS\_UGC\_OUTLINED: IconData

**buildMAPS\_UGC\_ROUNDED**instance-attribute

MAPS\_UGC\_ROUNDED: IconData



**buildMAPS\_UGC\_SHARPinstance-attribute**

MAPS\_UGC\_SHARP: IconData

**buildMARGINinstance-attribute**

MARGIN: IconData

**buildMARGIN\_OUTLINEDinstance-attribute**

MARGIN\_OUTLINED: IconData

**buildMARGIN\_ROUNDEDinstance-attribute**

MARGIN\_ROUNDED: IconData

**buildMARGIN\_SHARPinstance-attribute**

MARGIN\_SHARP: IconData

**buildMARK\_AS\_UNREADinstance-attribute**

MARK\_AS\_UNREAD: IconData

**buildMARK\_AS\_UNREAD\_OUTLINEDinstance-attribute**

MARK\_AS\_UNREAD\_OUTLINED: IconData

**buildMARK\_AS\_UNREAD\_ROUNDEDinstance-attribute**

MARK\_AS\_UNREAD\_ROUNDED: IconData

**buildMARK\_AS\_UNREAD\_SHARPinstance-attribute**

MARK\_AS\_UNREAD\_SHARP: IconData

**buildMARK\_CHAT\_READinstance-attribute**

MARK\_CHAT\_READ: IconData

**buildMARK\_CHAT\_READ\_OUTLINEDinstance-attribute**

MARK\_CHAT\_READ\_OUTLINED: IconData

**buildMARK\_CHAT\_READ\_ROUNDEDinstance-attribute**

MARK\_CHAT\_READ\_ROUNDED: IconData

**buildMARK\_CHAT\_READ\_SHARPinstance-attribute**

MARK\_CHAT\_READ\_SHARP: IconData

**buildMARK\_CHAT\_UNREADinstance-attribute**

MARK\_CHAT\_UNREAD: IconData

**buildMARK\_CHAT\_UNREAD\_OUTLINEDinstance-attribute**

MARK\_CHAT\_UNREAD\_OUTLINED: IconData

**buildMARK\_CHAT\_UNREAD\_ROUNDEDinstance-attribute**

MARK\_CHAT\_UNREAD\_ROUNDED: IconData

**buildMARK\_CHAT\_UNREAD\_SHARPinstance-attribute**

MARK\_CHAT\_UNREAD\_SHARP: IconData

**buildMARK\_EMAIL\_READinstance-attribute**

MARK\_EMAIL\_READ: IconData

**buildMARK\_EMAIL\_READ\_OUTLINEDinstance-attribute**

MARK\_EMAIL\_READ\_OUTLINED: IconData

**buildMARK\_EMAIL\_READ\_ROUNDEDinstance-attribute**

MARK\_EMAIL\_READ\_ROUNDED: IconData

**buildMARK\_EMAIL\_READ\_SHARPinstance-attribute**

MARK\_EMAIL\_READ\_SHARP: IconData

**buildMARK\_EMAIL\_UNREADinstance-attribute**

MARK\_EMAIL\_UNREAD: IconData

**buildMARK\_EMAIL\_UNREAD\_OUTLINEDinstance-attribute**

MARK\_EMAIL\_UNREAD\_OUTLINED: IconData

**buildMARK\_EMAIL\_UNREAD\_ROUNDEDinstance-attribute**

MARK\_EMAIL\_UNREAD\_ROUNDED: IconData

**buildMARK\_EMAIL\_UNREAD\_SHARPinstance-attribute**

MARK\_EMAIL\_UNREAD\_SHARP: IconData

**buildMARK\_UNREAD\_CHAT\_ALTinstance-attribute**

MARK\_UNREAD\_CHAT\_ALT: IconData

**buildMARK\_UNREAD\_CHAT\_ALT\_OUTLINEDinstance-attribute**

MARK\_UNREAD\_CHAT\_ALT\_OUTLINED: IconData

**buildMARK\_UNREAD\_CHAT\_ALT\_ROUNDEDinstance-attribute**

MARK\_UNREAD\_CHAT\_ALT\_ROUNDED: IconData

**buildMARK\_UNREAD\_CHAT\_ALT\_SHARPinstance-attribute**

MARK\_UNREAD\_CHAT\_ALT\_SHARP: IconData

**buildMARKUNREADinstance-attribute**

MARKUNREAD: IconData

**buildMARKUNREAD\_MAILBOXinstance-attribute**

MARKUNREAD\_MAILBOX: IconData

**buildMARKUNREAD\_MAILBOX\_OUTLINEDinstance-attribute**

MARKUNREAD\_MAILBOX\_OUTLINED: IconData

**buildMARKUNREAD\_MAILBOX\_ROUNDEDinstance-attribute**

MARKUNREAD\_MAILBOX\_ROUNDED: IconData

**buildMARKUNREAD\_MAILBOX\_SHARPinstance-attribute**

MARKUNREAD\_MAILBOX\_SHARP: IconData

**buildMARKUNREAD\_OUTLINEDinstance-attribute**

MARKUNREAD\_OUTLINED: IconData

**buildMARKUNREAD\_ROUNDEDinstance-attribute**

MARKUNREAD\_ROUNDED: IconData

**buildMARKUNREAD\_SHARPinstance-attribute**

MARKUNREAD\_SHARP: IconData

**buildMASKSinstance-attribute**

MASKS: IconData

**buildMASKS\_OUTLINEDinstance-attribute**

MASKS\_OUTLINED: IconData

**buildMASKS\_ROUNDEDinstance-attribute**

MASKS\_ROUNDED: IconData

**buildMASKS\_SHARPinstance-attribute**

MASKS\_SHARP: IconData

**buildMAXIMIZEinstance-attribute**

MAXIMIZE: IconData

**buildMAXIMIZE\_OUTLINEDinstance-attribute**

MAXIMIZE\_OUTLINED: IconData

**buildMAXIMIZE\_ROUNDEDinstance-attribute**

MAXIMIZE\_ROUNDED: IconData

**buildMAXIMIZE\_SHARPinstance-attribute**

MAXIMIZE\_SHARP: IconData

**buildMEDIA\_BLUETOOTH\_OFFinstance-attribute**

MEDIA\_BLUETOOTH\_OFF: IconData

**buildMEDIA\_BLUETOOTH\_OFF\_OUTLINEDinstance-attribute**

MEDIA\_BLUETOOTH\_OFF\_OUTLINED: IconData

**buildMEDIA\_BLUETOOTH\_OFF\_ROUNDEDinstance-attribute**

MEDIA\_BLUETOOTH\_OFF\_ROUNDED: IconData

**buildMEDIA\_BLUETOOTH\_OFF\_SHARPinstance-attribute**

MEDIA\_BLUETOOTH\_OFF\_SHARP: IconData

**buildMEDIA\_BLUETOOTH\_ONinstance-attribute**

MEDIA\_BLUETOOTH\_ON: IconData

**buildMEDIA\_BLUETOOTH\_ON\_OUTLINEDinstance-attribute**

MEDIA\_BLUETOOTH\_ON\_OUTLINED: IconData

**buildMEDIA\_BLUETOOTH\_ON\_ROUNDEDinstance-attribute**

MEDIA\_BLUETOOTH\_ON\_ROUNDED: IconData

**buildMEDIA\_BLUETOOTH\_ON\_SHARPinstance-attribute**

MEDIA\_BLUETOOTH\_ON\_SHARP: IconData

**buildMEDIATIONinstance-attribute**

MEDIATION: IconData

**buildMEDIATION\_OUTLINEDinstance-attribute**

MEDIATION\_OUTLINED: IconData

**buildMEDIATION\_ROUNDEDinstance-attribute**

MEDIATION\_ROUNDED: IconData

**buildMEDIATION\_SHARPinstance-attribute**

MEDIATION\_SHARP: IconData

**buildMEDICAL\_INFORMATIONinstance-attribute**

MEDICAL\_INFORMATION: IconData

**buildMEDICAL\_INFORMATION\_OUTLINEDinstance-attribute**

MEDICAL\_INFORMATION\_OUTLINED: IconData

**buildMEDICAL\_INFORMATION\_ROUNDEDinstance-attribute**

MEDICAL\_INFORMATION\_ROUNDED: IconData

**buildMEDICAL\_INFORMATION\_SHARPinstance-attribute**

MEDICAL\_INFORMATION\_SHARP: IconData

**buildMEDICAL\_SERVICESinstance-attribute**

MEDICAL\_SERVICES: IconData

**buildMEDICAL\_SERVICES\_OUTLINEDinstance-attribute**

MEDICAL\_SERVICES\_OUTLINED: IconData

**buildMEDICAL\_SERVICES\_ROUNDEDinstance-attribute**

MEDICAL\_SERVICES\_ROUNDED: IconData

**buildMEDICAL\_SERVICES\_SHARPinstance-attribute**

MEDICAL\_SERVICES\_SHARP: IconData

**buildMEDICATIONinstance-attribute**

MEDICATION: IconData

**buildMEDICATION\_LIQUIDinstance-attribute**

MEDICATION\_LIQUID: IconData

**buildMEDICATION\_LIQUID\_OUTLINEDinstance-attribute**

MEDICATION\_LIQUID\_OUTLINED: IconData

**buildMEDICATION\_LIQUID\_ROUNDEDinstance-attribute**

MEDICATION\_LIQUID\_ROUNDED: IconData

**buildMEDICATION\_LIQUID\_SHARPinstance-attribute**

MEDICATION\_LIQUID\_SHARP: IconData

**buildMEDICATION\_OUTLINEDinstance-attribute**

MEDICATION\_OUTLINED: IconData

**buildMEDICATION\_ROUNDEDinstance-attribute**

MEDICATION\_ROUNDED: IconData

**buildMEDICATION\_SHARPinstance-attribute**

MEDICATION\_SHARP: IconData

**buildMEETING\_ROOMinstance-attribute**

MEETING\_ROOM: IconData

**buildMEETING\_ROOM\_OUTLINEDinstance-attribute**

MEETING\_ROOM\_OUTLINED: IconData

**buildMEETING\_ROOM\_ROUNDEDinstance-attribute**

MEETING\_ROOM\_ROUNDED: IconData

**buildMEETING\_ROOM\_SHARPinstance-attribute**

MEETING\_ROOM\_SHARP: IconData

**buildMEMORYinstance-attribute**

MEMORY: IconData

**buildMEMORY\_OUTLINEDinstance-attribute**

MEMORY\_OUTLINED: IconData

**buildMEMORY\_ROUNDEDinstance-attribute**

MEMORY\_ROUNDED: IconData

**buildMEMORY\_SHARPinstance-attribute**

MEMORY\_SHARP: IconData

**buildMENUinstance-attribute**

MENU: IconData

**buildMENU\_BOOKinstance-attribute**

MENU\_BOOK: IconData

**buildMENU\_BOOK\_OUTLINEDinstance-attribute**

MENU\_BOOK\_OUTLINED: IconData

**buildMENU\_BOOK\_ROUNDEDinstance-attribute**

MENU\_BOOK\_ROUNDED: IconData

**buildMENU\_BOOK\_SHARPinstance-attribute**

MENU\_BOOK\_SHARP: IconData

**buildMENU\_OPENinstance-attribute**

MENU\_OPEN: IconData

**buildMENU\_OPEN\_OUTLINEDinstance-attribute**

MENU\_OPEN\_OUTLINED: IconData

**buildMENU\_OPEN\_ROUNDEDinstance-attribute**

MENU\_OPEN\_ROUNDED: IconData

**buildMENU\_OPEN\_SHARPinstance-attribute**

MENU\_OPEN\_SHARP: IconData

**buildMENU\_OUTLINEDinstance-attribute**

MENU\_OUTLINED: IconData

**buildMENU\_ROUNDEDinstance-attribute**

MENU\_ROUNDED: IconData

**buildMENU\_SHARPinstance-attribute**

MENU\_SHARP: IconData

**buildMERGEinstance-attribute**

MERGE: IconData

**buildMERGE\_OUTLINEDinstance-attribute**

MERGE\_OUTLINED: IconData

**buildMERGE\_ROUNDEDinstance-attribute**

MERGE\_ROUNDED: IconData

**buildMERGE\_SHARPinstance-attribute**

MERGE\_SHARP: IconData

**buildMERGE\_TYPEinstance-attribute**

MERGE\_TYPE: IconData

**buildMERGE\_TYPE\_OUTLINEDinstance-attribute**

MERGE\_TYPE\_OUTLINED: IconData

**buildMERGE\_TYPE\_ROUNDEDinstance-attribute**

MERGE\_TYPE\_ROUNDED: IconData

**buildMERGE\_TYPE\_SHARPinstance-attribute**

MERGE\_TYPE\_SHARP: IconData

**buildMESSAGEinstance-attribute**

MESSAGE: IconData

**buildMESSAGE\_OUTLINEDinstance-attribute**

MESSAGE\_OUTLINED: IconData

**buildMESSAGE\_ROUNDEDinstance-attribute**

MESSAGE\_ROUNDED: IconData

**buildMESSAGE\_SHARPinstance-attribute**

MESSAGE\_SHARP: IconData

**buildMESSENGERinstance-attribute**

MESSENGER: IconData

**buildMESSENGER\_OUTLINEinstance-attribute**

MESSENGER\_OUTLINE: IconData

**buildMESSENGER\_OUTLINE\_OUTLINEDinstance-attribute**

MESSENGER\_OUTLINE\_OUTLINED: IconData

**buildMESSENGER\_OUTLINE\_ROUNDEDinstance-attribute**

MESSENGER\_OUTLINE\_ROUNDED: IconData

**buildMESSENGER\_OUTLINE\_SHARPinstance-attribute**

MESSENGER\_OUTLINE\_SHARP: IconData

**buildMESSENGER\_OUTLINEDinstance-attribute**

MESSENGER\_OUTLINED: IconData

**buildMESSENGER\_ROUNDEDinstance-attribute**

MESSENGER\_ROUNDED: IconData

**buildMESSENGER\_SHARPinstance-attribute**

MESSENGER\_SHARP: IconData

**buildMICinstance-attribute**

MIC: IconData

**buildMIC\_EXTERNAL\_OFFinstance-attribute**

MIC\_EXTERNAL\_OFF: IconData

**buildMIC\_EXTERNAL\_OFF\_OUTLINEDinstance-attribute**

MIC\_EXTERNAL\_OFF\_OUTLINED: IconData

**buildMIC\_EXTERNAL\_OFF\_ROUNDEDinstance-attribute**

MIC\_EXTERNAL\_OFF\_ROUNDED: IconData

**buildMIC\_EXTERNAL\_OFF\_SHARPinstance-attribute**

MIC\_EXTERNAL\_OFF\_SHARP: IconData

**buildMIC\_EXTERNAL\_ONinstance-attribute**

MIC\_EXTERNAL\_ON: IconData

**buildMIC\_EXTERNAL\_ON\_OUTLINEDinstance-attribute**

MIC\_EXTERNAL\_ON\_OUTLINED: IconData

**buildMIC\_EXTERNAL\_ON\_ROUNDEDinstance-attribute**

MIC\_EXTERNAL\_ON\_ROUNDED: IconData

**buildMIC\_EXTERNAL\_ON\_SHARPinstance-attribute**

MIC\_EXTERNAL\_ON\_SHARP: IconData

**buildMIC\_NONEinstance-attribute**

MIC\_NONE: IconData

**buildMIC\_NONE\_OUTLINEDinstance-attribute**

MIC\_NONE\_OUTLINED: IconData

**buildMIC\_NONE\_ROUNDEDinstance-attribute**

MIC\_NONE\_ROUNDED: IconData

**buildMIC\_NONE\_SHARPinstance-attribute**

MIC\_NONE\_SHARP: IconData

**buildMIC\_OFFinstance-attribute**

MIC\_OFF: IconData

**buildMIC\_OFF\_OUTLINEDinstance-attribute**

MIC\_OFF\_OUTLINED: IconData

**buildMIC\_OFF\_ROUNDEDinstance-attribute**

MIC\_OFF\_ROUNDED: IconData

**buildMIC\_OFF\_SHARPinstance-attribute**

MIC\_OFF\_SHARP: IconData

**buildMIC\_OUTLINEDinstance-attribute**

MIC\_OUTLINED: IconData

**buildMIC\_ROUNDEDinstance-attribute**

MIC\_ROUNDED: IconData

**buildMIC\_SHARPinstance-attribute**

MIC\_SHARP: IconData

**buildMICROWAVEinstance-attribute**

MICROWAVE: IconData

**buildMICROWAVE\_OUTLINEDinstance-attribute**

MICROWAVE\_OUTLINED: IconData

**buildMICROWAVE\_ROUNDEDinstance-attribute**

MICROWAVE\_ROUNDED: IconData

**buildMICROWAVE\_SHARPinstance-attribute**

MICROWAVE\_SHARP: IconData

**buildMILITARY\_TECHinstance-attribute**

MILITARY\_TECH: IconData

**buildMILITARY\_TECH\_OUTLINEDinstance-attribute**

MILITARY\_TECH\_OUTLINED: IconData

**buildMILITARY\_TECH\_ROUNDEDinstance-attribute**

MILITARY\_TECH\_ROUNDED: IconData

**buildMILITARY\_TECH\_SHARPinstance-attribute**

MILITARY\_TECH\_SHARP: IconData

**buildMINIMIZEinstance-attribute**

MINIMIZE: IconData

**buildMINIMIZE\_OUTLINEDinstance-attribute**

MINIMIZE\_OUTLINED: IconData

**buildMINIMIZE\_ROUNDEDinstance-attribute**

MINIMIZE\_ROUNDED: IconData

**buildMINIMIZE\_SHARPinstance-attribute**

MINIMIZE\_SHARP: IconData

**buildMINOR\_CRASHinstance-attribute**

MINOR\_CRASH: IconData

**buildMINOR\_CRASH\_OUTLINEDinstance-attribute**

MINOR\_CRASH\_OUTLINED: IconData

**buildMINOR\_CRASH\_ROUNDEDinstance-attribute**

MINOR\_CRASH\_ROUNDED: IconData

**buildMINOR\_CRASH\_SHARPinstance-attribute**

MINOR\_CRASH\_SHARP: IconData

**buildMISCELLANEOUS\_SERVICESinstance-attribute**

MISCELLANEOUS\_SERVICES: IconData

**buildMISCELLANEOUS\_SERVICES\_OUTLINEDinstance-attribute**

MISCELLANEOUS\_SERVICES\_OUTLINED: IconData

**buildMISCELLANEOUS\_SERVICES\_ROUNDEDinstance-attribute**

MISCELLANEOUS\_SERVICES\_ROUNDED: IconData

**buildMISCELLANEOUS\_SERVICES\_SHARPinstance-attribute**

MISCELLANEOUS\_SERVICES\_SHARP: IconData

**buildMISSED\_VIDEO\_CALLinstance-attribute**

MISSED\_VIDEO\_CALL: IconData

**buildMISSED\_VIDEO\_CALL\_OUTLINED**Instance-attribute

MISSED\_VIDEO\_CALL\_OUTLINED: IconData

**buildMISSED\_VIDEO\_CALL\_ROUNDED**Instance-attribute

MISSED\_VIDEO\_CALL\_ROUNDED: IconData

**buildMISSED\_VIDEO\_CALL\_SHARP**Instance-attribute

MISSED\_VIDEO\_CALL\_SHARP: IconData

**buildMMS**Instance-attribute

MMS: IconData

**buildMMS\_OUTLINED**Instance-attribute

MMS\_OUTLINED: IconData

**buildMMS\_ROUNDED**Instance-attribute

MMS\_ROUNDED: IconData

**buildMMS\_SHARP**Instance-attribute

MMS\_SHARP: IconData

**buildMOBILE\_FRIENDLY**Instance-attribute

MOBILE\_FRIENDLY: IconData

**buildMOBILE\_FRIENDLY\_OUTLINED**Instance-attribute

MOBILE\_FRIENDLY\_OUTLINED: IconData

**buildMOBILE\_FRIENDLY\_ROUNDED**Instance-attribute

MOBILE\_FRIENDLY\_ROUNDED: IconData

**buildMOBILE\_FRIENDLY\_SHARP**Instance-attribute

MOBILE\_FRIENDLY\_SHARP: IconData

**buildMOBILE\_OFF**Instance-attribute

MOBILE\_OFF: IconData

**buildMOBILE\_OFF\_OUTLINED**Instance-attribute

MOBILE\_OFF\_OUTLINED: IconData

**buildMOBILE\_OFF\_ROUNDED**Instance-attribute

MOBILE\_OFF\_ROUNDED: IconData

**buildMOBILE\_OFF\_SHARP**Instance-attribute

MOBILE\_OFF\_SHARP: IconData

**buildMOBILE\_SCREEN\_SHARE**Instance-attribute

MOBILE\_SCREEN\_SHARE: IconData

**buildMOBILE\_SCREEN\_SHARE\_OUTLINED**Instance-attribute

MOBILE\_SCREEN\_SHARE\_OUTLINED: IconData

**buildMOBILE\_SCREEN\_SHARE\_ROUNDED**Instance-attribute

MOBILE\_SCREEN\_SHARE\_ROUNDED: IconData

**buildMOBILE\_SCREEN\_SHARE\_SHARP**Instance-attribute

MOBILE\_SCREEN\_SHARE\_SHARP: IconData

**buildMOBILEDATA\_OFF**Instance-attribute

MOBILEDATA\_OFF: IconData

**buildMOBILEDATA\_OFF\_OUTLINED**Instance-attribute

MOBILEDATA\_OFF\_OUTLINED: IconData

**buildMOBILEDATA\_OFF\_ROUNDED**Instance-attribute

MOBILEDATA\_OFF\_ROUNDED: IconData



**buildMOBILEDATA\_OFF\_SHARPinstance-attribute**

MOBILEDATA\_OFF\_SHARP: IconData

**buildMODEinstance-attribute**

MODE: IconData

**buildMODE\_COMMENTinstance-attribute**

MODE\_COMMENT: IconData

**buildMODE\_COMMENT\_OUTLINEDinstance-attribute**

MODE\_COMMENT\_OUTLINED: IconData

**buildMODE\_COMMENT\_ROUNDEDinstance-attribute**

MODE\_COMMENT\_ROUNDED: IconData

**buildMODE\_COMMENT\_SHARPinstance-attribute**

MODE\_COMMENT\_SHARP: IconData

**buildMODE\_EDITinstance-attribute**

MODE\_EDIT: IconData

**buildMODE\_EDIT\_OUTLINEinstance-attribute**

MODE\_EDIT\_OUTLINE: IconData

**buildMODE\_EDIT\_OUTLINE\_OUTLINEDinstance-attribute**

MODE\_EDIT\_OUTLINE\_OUTLINED: IconData

**buildMODE\_EDIT\_OUTLINE\_ROUNDEDinstance-attribute**

MODE\_EDIT\_OUTLINE\_ROUNDED: IconData

**buildMODE\_EDIT\_OUTLINE\_SHARPinstance-attribute**

MODE\_EDIT\_OUTLINE\_SHARP: IconData

**buildMODE\_EDIT\_OUTLINEDinstance-attribute**

MODE\_EDIT\_OUTLINED: IconData

**buildMODE\_EDIT\_ROUNDEDinstance-attribute**

MODE\_EDIT\_ROUNDED: IconData

**buildMODE\_EDIT\_SHARPinstance-attribute**

MODE\_EDIT\_SHARP: IconData

**buildMODE\_FAN\_OFFinstance-attribute**

MODE\_FAN\_OFF: IconData

**buildMODE\_FAN\_OFF\_OUTLINEDinstance-attribute**

MODE\_FAN\_OFF\_OUTLINED: IconData

**buildMODE\_FAN\_OFF\_ROUNDEDinstance-attribute**

MODE\_FAN\_OFF\_ROUNDED: IconData

**buildMODE\_FAN\_OFF\_SHARPinstance-attribute**

MODE\_FAN\_OFF\_SHARP: IconData

**buildMODE\_NIGHTinstance-attribute**

MODE\_NIGHT: IconData

**buildMODE\_NIGHT\_OUTLINEDinstance-attribute**

MODE\_NIGHT\_OUTLINED: IconData

**buildMODE\_NIGHT\_ROUNDEDinstance-attribute**

MODE\_NIGHT\_ROUNDED: IconData

**buildMODE\_NIGHT\_SHARPinstance-attribute**

MODE\_NIGHT\_SHARP: IconData

**buildMODE\_OF\_TRAVELinstance-attribute**

MODE\_OF\_TRAVEL: IconData

**buildMODE\_OF\_TRAVEL\_OUTLINEDinstance-attribute**

MODE\_OF\_TRAVEL\_OUTLINED: IconData

**buildMODE\_OF\_TRAVEL\_ROUNDEDinstance-attribute**

MODE\_OF\_TRAVEL\_ROUNDED: IconData

**buildMODE\_OF\_TRAVEL\_SHARPinstance-attribute**

MODE\_OF\_TRAVEL\_SHARP: IconData

**buildMODE\_OUTLINEDinstance-attribute**

MODE\_OUTLINED: IconData

**buildMODE\_ROUNDEDinstance-attribute**

MODE\_ROUNDED: IconData

**buildMODE\_SHARPinstance-attribute**

MODE\_SHARP: IconData

**buildMODE\_STANDBYinstance-attribute**

MODE\_STANDBY: IconData

**buildMODE\_STANDBY\_OUTLINEDinstance-attribute**

MODE\_STANDBY\_OUTLINED: IconData

**buildMODE\_STANDBY\_ROUNDEDinstance-attribute**

MODE\_STANDBY\_ROUNDED: IconData

**buildMODE\_STANDBY\_SHARPinstance-attribute**

MODE\_STANDBY\_SHARP: IconData

**buildMODEL\_TRAININGinstance-attribute**

MODEL\_TRAINING: IconData

**buildMODEL\_TRAINING\_OUTLINEDinstance-attribute**

MODEL\_TRAINING\_OUTLINED: IconData

**buildMODEL\_TRAINING\_ROUNDEDinstance-attribute**

MODEL\_TRAINING\_ROUNDED: IconData

**buildMODEL\_TRAINING\_SHARPinstance-attribute**

MODEL\_TRAINING\_SHARP: IconData

**buildMONETIZATION\_ONinstance-attribute**

MONETIZATION\_ON: IconData

**buildMONETIZATION\_ON\_OUTLINEDinstance-attribute**

MONETIZATION\_ON\_OUTLINED: IconData

**buildMONETIZATION\_ON\_ROUNDEDinstance-attribute**

MONETIZATION\_ON\_ROUNDED: IconData

**buildMONETIZATION\_ON\_SHARPinstance-attribute**

MONETIZATION\_ON\_SHARP: IconData

**buildMONEYinstance-attribute**

MONEY: IconData

**buildMONEY\_OFFinstance-attribute**

MONEY\_OFF: IconData

**buildMONEY\_OFF\_CSREDinstance-attribute**

MONEY\_OFF\_CSRED: IconData

**buildMONEY\_OFF\_CSRED\_OUTLINEDinstance-attribute**

MONEY\_OFF\_CSRED\_OUTLINED: IconData

**buildMONEY\_OFF\_CSRED\_ROUNDEDinstance-attribute**

MONEY\_OFF\_CSRED\_ROUNDED: IconData

**buildMONEY\_OFF\_CSRED\_SHARPinstance-attribute**

MONEY\_OFF\_CSRED\_SHARP: IconData

**buildMONEY\_OFF\_OUTLINEDinstance-attribute**

MONEY\_OFF\_OUTLINED: IconData

**buildMONEY\_OFF\_ROUNDEDinstance-attribute**

MONEY\_OFF\_ROUNDED: IconData

**buildMONEY\_OFF\_SHARPinstance-attribute**

MONEY\_OFF\_SHARP: IconData

**buildMONEY\_OUTLINEDinstance-attribute**

MONEY\_OUTLINED: IconData

**buildMONEY\_ROUNDEDinstance-attribute**

MONEY\_ROUNDED: IconData

**buildMONEY\_SHARPinstance-attribute**

MONEY\_SHARP: IconData

**buildMONITORinstance-attribute**

MONITOR: IconData

**buildMONITOR\_HEARTinstance-attribute**

MONITOR\_HEART: IconData

**buildMONITOR\_HEART\_OUTLINEDinstance-attribute**

MONITOR\_HEART\_OUTLINED: IconData

**buildMONITOR\_HEART\_ROUNDEDinstance-attribute**

MONITOR\_HEART\_ROUNDED: IconData

**buildMONITOR\_HEART\_SHARPinstance-attribute**

MONITOR\_HEART\_SHARP: IconData

**buildMONITOR\_OUTLINEDinstance-attribute**

MONITOR\_OUTLINED: IconData

**buildMONITOR\_ROUNDEDinstance-attribute**

MONITOR\_ROUNDED: IconData

**buildMONITOR\_SHARPinstance-attribute**

MONITOR\_SHARP: IconData

**buildMONITOR\_WEIGHTinstance-attribute**

MONITOR\_WEIGHT: IconData

**buildMONITOR\_WEIGHT\_OUTLINEDinstance-attribute**

MONITOR\_WEIGHT\_OUTLINED: IconData

**buildMONITOR\_WEIGHT\_ROUNDEDinstance-attribute**

MONITOR\_WEIGHT\_ROUNDED: IconData

**buildMONITOR\_WEIGHT\_SHARPinstance-attribute**

MONITOR\_WEIGHT\_SHARP: IconData

**buildMONOCHROME\_PHOTOSinstance-attribute**

MONOCHROME\_PHOTOS: IconData

**buildMONOCHROME\_PHOTOS\_OUTLINEDinstance-attribute**

MONOCHROME\_PHOTOS\_OUTLINED: IconData

**buildMONOCHROME\_PHOTOS\_ROUNDEDinstance-attribute**

MONOCHROME\_PHOTOS\_ROUNDED: IconData

**buildMONOCHROME\_PHOTOS\_SHARPinstance-attribute**

MONOCHROME\_PHOTOS\_SHARP: IconData

**buildMOODinstance-attribute**

MOOD: IconData

**buildMOOD\_BADinstance-attribute**

MOOD\_BAD: IconData

**buildMOOD\_BAD\_OUTLINEDinstance-attribute**

MOOD\_BAD\_OUTLINED: IconData

**buildMOOD\_BAD\_ROUNDEDinstance-attribute**

MOOD\_BAD\_ROUNDED: IconData

**buildMOOD\_BAD\_SHARPinstance-attribute**

MOOD\_BAD\_SHARP: IconData

**buildMOOD\_OUTLINEDinstance-attribute**

MOOD\_OUTLINED: IconData

**buildMOOD\_ROUNDEDinstance-attribute**

MOOD\_ROUNDED: IconData

**buildMOOD\_SHARPinstance-attribute**

MOOD\_SHARP: IconData

**buildMOPEDinstance-attribute**

MOPED: IconData

**buildMOPED\_OUTLINEDinstance-attribute**

MOPED\_OUTLINED: IconData

**buildMOPED\_ROUNDEDinstance-attribute**

MOPED\_ROUNDED: IconData

**buildMOPED\_SHARPinstance-attribute**

MOPED\_SHARP: IconData

**buildMOREinstance-attribute**

MORE: IconData

**buildMORE\_HORIZinstance-attribute**

MORE\_HORIZ: IconData

**buildMORE\_HORIZ\_OUTLINEDinstance-attribute**

MORE\_HORIZ\_OUTLINED: IconData

**buildMORE\_HORIZ\_ROUNDEDinstance-attribute**

MORE\_HORIZ\_ROUNDED: IconData

**buildMORE\_HORIZ\_SHARPinstance-attribute**

MORE\_HORIZ\_SHARP: IconData

**buildMORE\_OUTLINEDinstance-attribute**

MORE\_OUTLINED: IconData

**buildMORE\_ROUNDEDinstance-attribute**

MORE\_ROUNDED: IconData

**buildMORE\_SHARPinstance-attribute**

MORE\_SHARP: IconData

**buildMORE\_TIMEinstance-attribute**

MORE\_TIME: IconData

**buildMORE\_TIME\_OUTLINEDinstance-attribute**

MORE\_TIME\_OUTLINED: IconData

**buildMORE\_TIME\_ROUNDEDinstance-attribute**

MORE\_TIME\_ROUNDED: IconData

**buildMORE\_TIME\_SHARPinstance-attribute**

MORE\_TIME\_SHARP: IconData

**buildMORE\_VERTinstance-attribute**

MORE\_VERT: IconData

**buildMORE\_VERT\_OUTLINEDinstance-attribute**

MORE\_VERT\_OUTLINED: IconData

**buildMORE\_VERT\_ROUNDEDinstance-attribute**

MORE\_VERT\_ROUNDED: IconData

**buildMORE\_VERT\_SHARPinstance-attribute**

MORE\_VERT\_SHARP: IconData

**buildMOSQUEinstance-attribute**

MOSQUE: IconData

**buildMOSQUE\_OUTLINEDinstance-attribute**

MOSQUE\_OUTLINED: IconData

**buildMOSQUE\_ROUNDEDinstance-attribute**

MOSQUE\_ROUNDED: IconData

**buildMOSQUE\_SHARPinstance-attribute**

MOSQUE\_SHARP: IconData

**buildMOTION\_PHOTOS\_AUTOinstance-attribute**

MOTION\_PHOTOS\_AUTO: IconData

**buildMOTION\_PHOTOS\_AUTO\_OUTLINEDinstance-attribute**

MOTION\_PHOTOS\_AUTO\_OUTLINED: IconData

**buildMOTION\_PHOTOS\_AUTO\_ROUNDEDinstance-attribute**

MOTION\_PHOTOS\_AUTO\_ROUNDED: IconData

**buildMOTION\_PHOTOS\_AUTO\_SHARPinstance-attribute**

MOTION\_PHOTOS\_AUTO\_SHARP: IconData

**buildMOTION\_PHOTOS\_OFFinstance-attribute**

MOTION\_PHOTOS\_OFF: IconData

**buildMOTION\_PHOTOS\_OFF\_OUTLINEDinstance-attribute**

MOTION\_PHOTOS\_OFF\_OUTLINED: IconData

**buildMOTION\_PHOTOS\_OFF\_ROUNDEDinstance-attribute**

MOTION\_PHOTOS\_OFF\_ROUNDED: IconData

**buildMOTION\_PHOTOS\_OFF\_SHARPinstance-attribute**

MOTION\_PHOTOS\_OFF\_SHARP: IconData

**buildMOTION\_PHOTOS\_ONinstance-attribute**

MOTION\_PHOTOS\_ON: IconData

**buildMOTION\_PHOTOS\_ON\_OUTLINED**instance-attribute

MOTION\_PHOTOS\_ON\_OUTLINED: IconData

**buildMOTION\_PHOTOS\_ON\_ROUNDED**instance-attribute

MOTION\_PHOTOS\_ON\_ROUNDED: IconData

**buildMOTION\_PHOTOS\_ON\_SHARP**instance-attribute

MOTION\_PHOTOS\_ON\_SHARP: IconData

**buildMOTION\_PHOTOS\_PAUSE**instance-attribute

MOTION\_PHOTOS\_PAUSE: IconData

**buildMOTION\_PHOTOS\_PAUSE\_OUTLINED**instance-attribute

MOTION\_PHOTOS\_PAUSE\_OUTLINED: IconData

**buildMOTION\_PHOTOS\_PAUSE\_ROUNDED**instance-attribute

MOTION\_PHOTOS\_PAUSE\_ROUNDED: IconData

**buildMOTION\_PHOTOS\_PAUSE\_SHARP**instance-attribute

MOTION\_PHOTOS\_PAUSE\_SHARP: IconData

**buildMOTION\_PHOTOS\_PAUSED**instance-attribute

MOTION\_PHOTOS\_PAUSED: IconData

**buildMOTION\_PHOTOS\_PAUSED\_OUTLINED**instance-attribute

MOTION\_PHOTOS\_PAUSED\_OUTLINED: IconData

**buildMOTION\_PHOTOS\_PAUSED\_ROUNDED**instance-attribute

MOTION\_PHOTOS\_PAUSED\_ROUNDED: IconData

**buildMOTION\_PHOTOS\_PAUSED\_SHARP**instance-attribute

MOTION\_PHOTOS\_PAUSED\_SHARP: IconData

**buildMOTORCYCLE**instance-attribute

MOTORCYCLE: IconData

**buildMOTORCYCLE\_OUTLINED**instance-attribute

MOTORCYCLE\_OUTLINED: IconData

**buildMOTORCYCLE\_ROUNDED**instance-attribute

MOTORCYCLE\_ROUNDED: IconData

**buildMOTORCYCLE\_SHARP**instance-attribute

MOTORCYCLE\_SHARP: IconData

**buildMOUSE**instance-attribute

MOUSE: IconData

**buildMOUSE\_OUTLINED**instance-attribute

MOUSE\_OUTLINED: IconData

**buildMOUSE\_ROUNDED**instance-attribute

MOUSE\_ROUNDED: IconData

**buildMOUSE\_SHARP**instance-attribute

MOUSE\_SHARP: IconData

**buildMOVE\_DOWN**instance-attribute

MOVE\_DOWN: IconData

**buildMOVE\_DOWN\_OUTLINED**instance-attribute

MOVE\_DOWN\_OUTLINED: IconData

**buildMOVE\_DOWN\_ROUNDED**instance-attribute

MOVE\_DOWN\_ROUNDED: IconData

**buildMOVE\_DOWN\_SHARPinstance-attribute**

MOVE\_DOWN\_SHARP: IconData

**buildMOVE\_TO\_INBOXinstance-attribute**

MOVE\_TO\_INBOX: IconData

**buildMOVE\_TO\_INBOX\_OUTLINEDinstance-attribute**

MOVE\_TO\_INBOX\_OUTLINED: IconData

**buildMOVE\_TO\_INBOX\_ROUNDEDinstance-attribute**

MOVE\_TO\_INBOX\_ROUNDED: IconData

**buildMOVE\_TO\_INBOX\_SHARPinstance-attribute**

MOVE\_TO\_INBOX\_SHARP: IconData

**buildMOVE\_UPinstance-attribute**

MOVE\_UP: IconData

**buildMOVE\_UP\_OUTLINEDinstance-attribute**

MOVE\_UP\_OUTLINED: IconData

**buildMOVE\_UP\_ROUNDEDinstance-attribute**

MOVE\_UP\_ROUNDED: IconData

**buildMOVE\_UP\_SHARPinstance-attribute**

MOVE\_UP\_SHARP: IconData

**buildMOVIEinstance-attribute**

MOVIE: IconData

**buildMOVIE\_CREATIONinstance-attribute**

MOVIE\_CREATION: IconData

**buildMOVIE\_CREATION\_OUTLINEDinstance-attribute**

MOVIE\_CREATION\_OUTLINED: IconData

**buildMOVIE\_CREATION\_ROUNDEDinstance-attribute**

MOVIE\_CREATION\_ROUNDED: IconData

**buildMOVIE\_CREATION\_SHARPinstance-attribute**

MOVIE\_CREATION\_SHARP: IconData

**buildMOVIE\_EDITinstance-attribute**

MOVIE\_EDIT: IconData

**buildMOVIE\_FILTERinstance-attribute**

MOVIE\_FILTER: IconData

**buildMOVIE\_FILTER\_OUTLINEDinstance-attribute**

MOVIE\_FILTER\_OUTLINED: IconData

**buildMOVIE\_FILTER\_ROUNDEDinstance-attribute**

MOVIE\_FILTER\_ROUNDED: IconData

**buildMOVIE\_FILTER\_SHARPinstance-attribute**

MOVIE\_FILTER\_SHARP: IconData

**buildMOVIE\_OUTLINEDinstance-attribute**

MOVIE\_OUTLINED: IconData

**buildMOVIE\_ROUNDEDinstance-attribute**

MOVIE\_ROUNDED: IconData

**buildMOVIE\_SHARPinstance-attribute**

MOVIE\_SHARP: IconData

**buildMOVINGinstance-attribute**

MOVING: IconData

**buildMOVING\_OUTLINEDinstance-attribute**

MOVING\_OUTLINED: IconData

**buildMOVING\_ROUNDEDinstance-attribute**

MOVING\_ROUNDED: IconData

**buildMOVING\_SHARPinstance-attribute**

MOVING\_SHARP: IconData

**buildMPinstance-attribute**

MP: IconData

**buildMP\_OUTLINEDinstance-attribute**

MP\_OUTLINED: IconData

**buildMP\_ROUNDEDinstance-attribute**

MP\_ROUNDED: IconData

**buildMP\_SHARPinstance-attribute**

MP\_SHARP: IconData

**buildMULTILINE\_CHARTinstance-attribute**

MULTILINE\_CHART: IconData

**buildMULTILINE\_CHART\_OUTLINEDinstance-attribute**

MULTILINE\_CHART\_OUTLINED: IconData

**buildMULTILINE\_CHART\_ROUNDEDinstance-attribute**

MULTILINE\_CHART\_ROUNDED: IconData

**buildMULTILINE\_CHART\_SHARPinstance-attribute**

MULTILINE\_CHART\_SHARP: IconData

**buildMULTIPLE\_STOPinstance-attribute**

MULTIPLE\_STOP: IconData

**buildMULTIPLE\_STOP\_OUTLINEDinstance-attribute**

MULTIPLE\_STOP\_OUTLINED: IconData

**buildMULTIPLE\_STOP\_ROUNDEDinstance-attribute**

MULTIPLE\_STOP\_ROUNDED: IconData

**buildMULTIPLE\_STOP\_SHARPinstance-attribute**

MULTIPLE\_STOP\_SHARP: IconData

**buildMULTITRACK\_AUDIOinstance-attribute**

MULTITRACK\_AUDIO: IconData

**buildMULTITRACK\_AUDIO\_OUTLINEDinstance-attribute**

MULTITRACK\_AUDIO\_OUTLINED: IconData

**buildMULTITRACK\_AUDIO\_ROUNDEDinstance-attribute**

MULTITRACK\_AUDIO\_ROUNDED: IconData

**buildMULTITRACK\_AUDIO\_SHARPinstance-attribute**

MULTITRACK\_AUDIO\_SHARP: IconData

**buildMUSEUMinstance-attribute**

MUSEUM: IconData

**buildMUSEUM\_OUTLINEDinstance-attribute**

MUSEUM\_OUTLINED: IconData



**buildMUSEUM\_ROUNDEDinstance-attribute**

MUSEUM\_ROUNDED: IconData

**buildMUSEUM\_SHARPinstance-attribute**

MUSEUM\_SHARP: IconData

**buildMUSIC\_NOTEinstance-attribute**

MUSIC\_NOTE: IconData

**buildMUSIC\_NOTE\_OUTLINEDinstance-attribute**

MUSIC\_NOTE\_OUTLINED: IconData

**buildMUSIC\_NOTE\_ROUNDEDinstance-attribute**

MUSIC\_NOTE\_ROUNDED: IconData

**buildMUSIC\_NOTE\_SHARPinstance-attribute**

MUSIC\_NOTE\_SHARP: IconData

**buildMUSIC\_OFFinstance-attribute**

MUSIC\_OFF: IconData

**buildMUSIC\_OFF\_OUTLINEDinstance-attribute**

MUSIC\_OFF\_OUTLINED: IconData

**buildMUSIC\_OFF\_ROUNDEDinstance-attribute**

MUSIC\_OFF\_ROUNDED: IconData

**buildMUSIC\_OFF\_SHARPinstance-attribute**

MUSIC\_OFF\_SHARP: IconData

**buildMUSIC\_VIDEOinstance-attribute**

MUSIC\_VIDEO: IconData

**buildMUSIC\_VIDEO\_OUTLINEDinstance-attribute**

MUSIC\_VIDEO\_OUTLINED: IconData

**buildMUSIC\_VIDEO\_ROUNDEDinstance-attribute**

MUSIC\_VIDEO\_ROUNDED: IconData

**buildMUSIC\_VIDEO\_SHARPinstance-attribute**

MUSIC\_VIDEO\_SHARP: IconData

**buildMY\_LIBRARY\_ADDinstance-attribute**

MY\_LIBRARY\_ADD: IconData

**buildMY\_LIBRARY\_ADD\_OUTLINEDinstance-attribute**

MY\_LIBRARY\_ADD\_OUTLINED: IconData

**buildMY\_LIBRARY\_ADD\_ROUNDEDinstance-attribute**

MY\_LIBRARY\_ADD\_ROUNDED: IconData

**buildMY\_LIBRARY\_ADD\_SHARPinstance-attribute**

MY\_LIBRARY\_ADD\_SHARP: IconData

**buildMY\_LIBRARY\_BOOKSinstance-attribute**

MY\_LIBRARY\_BOOKS: IconData

**buildMY\_LIBRARY\_BOOKS\_OUTLINEDinstance-attribute**

MY\_LIBRARY\_BOOKS\_OUTLINED: IconData

**buildMY\_LIBRARY\_BOOKS\_ROUNDEDinstance-attribute**

MY\_LIBRARY\_BOOKS\_ROUNDED: IconData

**buildMY\_LIBRARY\_BOOKS\_SHARPinstance-attribute**

MY\_LIBRARY\_BOOKS\_SHARP: IconData

**buildMY\_LIBRARY\_MUSICinstance-attribute**

MY\_LIBRARY\_MUSIC: IconData

**buildMY\_LIBRARY\_MUSIC\_OUTLINEDinstance-attribute**

MY\_LIBRARY\_MUSIC\_OUTLINED: IconData

**buildMY\_LIBRARY\_MUSIC\_ROUNDEDinstance-attribute**

MY\_LIBRARY\_MUSIC\_ROUNDED: IconData

**buildMY\_LIBRARY\_MUSIC\_SHARPinstance-attribute**

MY\_LIBRARY\_MUSIC\_SHARP: IconData

**buildMY\_LOCATIONinstance-attribute**

MY\_LOCATION: IconData

**buildMY\_LOCATION\_OUTLINEDinstance-attribute**

MY\_LOCATION\_OUTLINED: IconData

**buildMY\_LOCATION\_ROUNDEDinstance-attribute**

MY\_LOCATION\_ROUNDED: IconData

**buildMY\_LOCATION\_SHARPinstance-attribute**

MY\_LOCATION\_SHARP: IconData

**buildNATinstance-attribute**

NAT: IconData

**buildNAT\_OUTLINEDinstance-attribute**

NAT\_OUTLINED: IconData

**buildNAT\_ROUNDEDinstance-attribute**

NAT\_ROUNDED: IconData

**buildNAT\_SHARPinstance-attribute**

NAT\_SHARP: IconData

**buildNATUREinstance-attribute**

NATURE: IconData

**buildNATURE\_OUTLINEDinstance-attribute**

NATURE\_OUTLINED: IconData

**buildNATURE\_PEOPLEinstance-attribute**

NATURE\_PEOPLE: IconData

**buildNATURE\_PEOPLE\_OUTLINEDinstance-attribute**

NATURE\_PEOPLE\_OUTLINED: IconData

**buildNATURE\_PEOPLE\_ROUNDEDinstance-attribute**

NATURE\_PEOPLE\_ROUNDED: IconData

**buildNATURE\_PEOPLE\_SHARPinstance-attribute**

NATURE\_PEOPLE\_SHARP: IconData

**buildNATURE\_ROUNDEDinstance-attribute**

NATURE\_ROUNDED: IconData

**buildNATURE\_SHARPinstance-attribute**

NATURE\_SHARP: IconData

**buildNAVIGATE\_BEFOREinstance-attribute**

NAVIGATE\_BEFORE: IconData

**buildNAVIGATE\_BEFORE\_OUTLINEDinstance-attribute**

NAVIGATE\_BEFORE\_OUTLINED: IconData

**buildNAVIGATE\_BEFORE\_ROUNDEDinstance-attribute**

NAVIGATE\_BEFORE\_ROUNDED: IconData

**buildNAVIGATE\_BEFORE\_SHARPinstance-attribute**

NAVIGATE\_BEFORE\_SHARP: IconData

**buildNAVIGATE\_NEXTinstance-attribute**

NAVIGATE\_NEXT: IconData

**buildNAVIGATE\_NEXT\_OUTLINEDinstance-attribute**

NAVIGATE\_NEXT\_OUTLINED: IconData

**buildNAVIGATE\_NEXT\_ROUNDEDinstance-attribute**

NAVIGATE\_NEXT\_ROUNDED: IconData

**buildNAVIGATE\_NEXT\_SHARPinstance-attribute**

NAVIGATE\_NEXT\_SHARP: IconData

**buildNAVIGATIONinstance-attribute**

NAVIGATION: IconData

**buildNAVIGATION\_OUTLINEDinstance-attribute**

NAVIGATION\_OUTLINED: IconData

**buildNAVIGATION\_ROUNDEDinstance-attribute**

NAVIGATION\_ROUNDED: IconData

**buildNAVIGATION\_SHARPinstance-attribute**

NAVIGATION\_SHARP: IconData

**buildNEAR\_MEinstance-attribute**

NEAR\_ME: IconData

**buildNEAR\_ME\_DISABLEDinstance-attribute**

NEAR\_ME\_DISABLED: IconData

**buildNEAR\_ME\_DISABLED\_OUTLINEDinstance-attribute**

NEAR\_ME\_DISABLED\_OUTLINED: IconData

**buildNEAR\_ME\_DISABLED\_ROUNDEDinstance-attribute**

NEAR\_ME\_DISABLED\_ROUNDED: IconData

**buildNEAR\_ME\_DISABLED\_SHARPinstance-attribute**

NEAR\_ME\_DISABLED\_SHARP: IconData

**buildNEAR\_ME\_OUTLINEDinstance-attribute**

NEAR\_ME\_OUTLINED: IconData

**buildNEAR\_ME\_ROUNDEDinstance-attribute**

NEAR\_ME\_ROUNDED: IconData

**buildNEAR\_ME\_SHARPinstance-attribute**

NEAR\_ME\_SHARP: IconData

**buildNEARBY\_ERRORinstance-attribute**

NEARBY\_ERROR: IconData

**buildNEARBY\_ERROR\_OUTLINEDinstance-attribute**

NEARBY\_ERROR\_OUTLINED: IconData

**buildNEARBY\_ERROR\_ROUNDEDinstance-attribute**

NEARBY\_ERROR\_ROUNDED: IconData

**buildNEARBY\_ERROR\_SHARPinstance-attribute**

NEARBY\_ERROR\_SHARP: IconData

**buildNEARBY\_OFF**Instance-attribute

NEARBY\_OFF: IconData

**buildNEARBY\_OFF\_OUTLINED**Instance-attribute

NEARBY\_OFF\_OUTLINED: IconData

**buildNEARBY\_OFF\_ROUNDED**Instance-attribute

NEARBY\_OFF\_ROUNDED: IconData

**buildNEARBY\_OFF\_SHARP**Instance-attribute

NEARBY\_OFF\_SHARP: IconData

**buildNEST\_CAM\_WIRED\_STAND**Instance-attribute

NEST\_CAM\_WIRED\_STAND: IconData

**buildNEST\_CAM\_WIRED\_STAND\_OUTLINED**Instance-attribute

NEST\_CAM\_WIRED\_STAND\_OUTLINED: IconData

**buildNEST\_CAM\_WIRED\_STAND\_ROUNDED**Instance-attribute

NEST\_CAM\_WIRED\_STAND\_ROUNDED: IconData

**buildNEST\_CAM\_WIRED\_STAND\_SHARP**Instance-attribute

NEST\_CAM\_WIRED\_STAND\_SHARP: IconData

**buildNETWORK\_CELL**Instance-attribute

NETWORK\_CELL: IconData

**buildNETWORK\_CELL\_OUTLINED**Instance-attribute

NETWORK\_CELL\_OUTLINED: IconData

**buildNETWORK\_CELL\_ROUNDED**Instance-attribute

NETWORK\_CELL\_ROUNDED: IconData

**buildNETWORK\_CELL\_SHARP**Instance-attribute

NETWORK\_CELL\_SHARP: IconData

**buildNETWORK\_CHECK**Instance-attribute

NETWORK\_CHECK: IconData

**buildNETWORK\_CHECK\_OUTLINED**Instance-attribute

NETWORK\_CHECK\_OUTLINED: IconData

**buildNETWORK\_CHECK\_ROUNDED**Instance-attribute

NETWORK\_CHECK\_ROUNDED: IconData

**buildNETWORK\_CHECK\_SHARP**Instance-attribute

NETWORK\_CHECK\_SHARP: IconData

**buildNETWORK\_LOCKED**Instance-attribute

NETWORK\_LOCKED: IconData

**buildNETWORK\_LOCKED\_OUTLINED**Instance-attribute

NETWORK\_LOCKED\_OUTLINED: IconData

**buildNETWORK\_LOCKED\_ROUNDED**Instance-attribute

NETWORK\_LOCKED\_ROUNDED: IconData

**buildNETWORK\_LOCKED\_SHARP**Instance-attribute

NETWORK\_LOCKED\_SHARP: IconData

**buildNETWORK\_PING**Instance-attribute

NETWORK\_PING: IconData

**buildNETWORK\_PING\_OUTLINED**Instance-attribute

NETWORK\_PING\_OUTLINED: IconData

buildNETWORK\_PING\_ROUNDEDinstance-attribute

NETWORK\_PING\_ROUNDED: IconData

buildNETWORK\_PING\_SHARPinstance-attribute

NETWORK\_PING\_SHARP: IconData

buildNETWORK\_WIFIinstance-attribute

NETWORK\_WIFI: IconData

buildNETWORK\_WIFI\_1\_BARinstance-attribute

NETWORK\_WIFI\_1\_BAR: IconData

buildNETWORK\_WIFI\_1\_BAR\_OUTLINEDinstance-attribute

NETWORK\_WIFI\_1\_BAR\_OUTLINED: IconData

buildNETWORK\_WIFI\_1\_BAR\_ROUNDEDinstance-attribute

NETWORK\_WIFI\_1\_BAR\_ROUNDED: IconData

buildNETWORK\_WIFI\_1\_BAR\_SHARPinstance-attribute

NETWORK\_WIFI\_1\_BAR\_SHARP: IconData

buildNETWORK\_WIFI\_2\_BARinstance-attribute

NETWORK\_WIFI\_2\_BAR: IconData

buildNETWORK\_WIFI\_2\_BAR\_OUTLINEDinstance-attribute

NETWORK\_WIFI\_2\_BAR\_OUTLINED: IconData

buildNETWORK\_WIFI\_2\_BAR\_ROUNDEDinstance-attribute

NETWORK\_WIFI\_2\_BAR\_ROUNDED: IconData

buildNETWORK\_WIFI\_2\_BAR\_SHARPinstance-attribute

NETWORK\_WIFI\_2\_BAR\_SHARP: IconData

buildNETWORK\_WIFI\_3\_BARinstance-attribute

NETWORK\_WIFI\_3\_BAR: IconData

buildNETWORK\_WIFI\_3\_BAR\_OUTLINEDinstance-attribute

NETWORK\_WIFI\_3\_BAR\_OUTLINED: IconData

buildNETWORK\_WIFI\_3\_BAR\_ROUNDEDinstance-attribute

NETWORK\_WIFI\_3\_BAR\_ROUNDED: IconData

buildNETWORK\_WIFI\_3\_BAR\_SHARPinstance-attribute

NETWORK\_WIFI\_3\_BAR\_SHARP: IconData

buildNETWORK\_WIFI\_OUTLINEDinstance-attribute

NETWORK\_WIFI\_OUTLINED: IconData

buildNETWORK\_WIFI\_ROUNDEDinstance-attribute

NETWORK\_WIFI\_ROUNDED: IconData

buildNETWORK\_WIFI\_SHARPinstance-attribute

NETWORK\_WIFI\_SHARP: IconData

buildNEW\_LABELinstance-attribute

NEW\_LABEL: IconData

buildNEW\_LABEL\_OUTLINEDinstance-attribute

NEW\_LABEL\_OUTLINED: IconData

buildNEW\_LABEL\_ROUNDEDinstance-attribute

NEW\_LABEL\_ROUNDED: IconData

buildNEW\_LABEL\_SHARPinstance-attribute

NEW\_LABEL\_SHARP: IconData

**buildNEW\_RELEASES**Instance-attribute

NEW\_RELEASES: IconData

**buildNEW\_RELEASES\_OUTLINED**Instance-attribute

NEW\_RELEASES\_OUTLINED: IconData

**buildNEW\_RELEASES\_ROUNDED**Instance-attribute

NEW\_RELEASES\_ROUNDED: IconData

**buildNEW\_RELEASES\_SHARP**Instance-attribute

NEW\_RELEASES\_SHARP: IconData

**buildNEWSPAPER**Instance-attribute

NEWSPAPER: IconData

**buildNEWSPAPER\_OUTLINED**Instance-attribute

NEWSPAPER\_OUTLINED: IconData

**buildNEWSPAPER\_ROUNDED**Instance-attribute

NEWSPAPER\_ROUNDED: IconData

**buildNEWSPAPER\_SHARP**Instance-attribute

NEWSPAPER\_SHARP: IconData

**buildNEXT\_PLAN**Instance-attribute

NEXT\_PLAN: IconData

**buildNEXT\_PLAN\_OUTLINED**Instance-attribute

NEXT\_PLAN\_OUTLINED: IconData

**buildNEXT\_PLAN\_ROUNDED**Instance-attribute

NEXT\_PLAN\_ROUNDED: IconData

**buildNEXT\_PLAN\_SHARP**Instance-attribute

NEXT\_PLAN\_SHARP: IconData

**buildNEXT\_WEEK**Instance-attribute

NEXT\_WEEK: IconData

**buildNEXT\_WEEK\_OUTLINED**Instance-attribute

NEXT\_WEEK\_OUTLINED: IconData

**buildNEXT\_WEEK\_ROUNDED**Instance-attribute

NEXT\_WEEK\_ROUNDED: IconData

**buildNEXT\_WEEK\_SHARP**Instance-attribute

NEXT\_WEEK\_SHARP: IconData

**buildNFC**Instance-attribute

NFC: IconData

**buildNFC\_OUTLINED**Instance-attribute

NFC\_OUTLINED: IconData

**buildNFC\_ROUNDED**Instance-attribute

NFC\_ROUNDED: IconData

**buildNFC\_SHARP**Instance-attribute

NFC\_SHARP: IconData

**buildNIGHT\_SHELTER**Instance-attribute

NIGHT\_SHELTER: IconData

**buildNIGHT\_SHELTER\_OUTLINED**Instance-attribute

NIGHT\_SHELTER\_OUTLINED: IconData

buildNIGHT\_SHELTER\_ROUNDEDinstance-attribute

NIGHT\_SHELTER\_ROUNDED: IconData

buildNIGHT\_SHELTER\_SHARPinstance-attribute

NIGHT\_SHELTER\_SHARP: IconData

buildNIGHTLIFEinstance-attribute

NIGHTLIFE: IconData

buildNIGHTLIFE\_OUTLINEDinstance-attribute

NIGHTLIFE\_OUTLINED: IconData

buildNIGHTLIFE\_ROUNDEDinstance-attribute

NIGHTLIFE\_ROUNDED: IconData

buildNIGHTLIFE\_SHARPinstance-attribute

NIGHTLIFE\_SHARP: IconData

buildNIGHTLIGHTinstance-attribute

NIGHTLIGHT: IconData

buildNIGHTLIGHT\_OUTLINEDinstance-attribute

NIGHTLIGHT\_OUTLINED: IconData

buildNIGHTLIGHT\_ROUNDinstance-attribute

NIGHTLIGHT\_ROUND: IconData

buildNIGHTLIGHT\_ROUND\_OUTLINEDinstance-attribute

NIGHTLIGHT\_ROUND\_OUTLINED: IconData

buildNIGHTLIGHT\_ROUND\_ROUNDEDinstance-attribute

NIGHTLIGHT\_ROUND\_ROUNDED: IconData

buildNIGHTLIGHT\_ROUND\_SHARPinstance-attribute

NIGHTLIGHT\_ROUND\_SHARP: IconData

buildNIGHTLIGHT\_ROUNDEDinstance-attribute

NIGHTLIGHT\_ROUNDED: IconData

buildNIGHTLIGHT\_SHARPinstance-attribute

NIGHTLIGHT\_SHARP: IconData

buildNIGHTS\_STAYinstance-attribute

NIGHTS\_STAY: IconData

buildNIGHTS\_STAY\_OUTLINEDinstance-attribute

NIGHTS\_STAY\_OUTLINED: IconData

buildNIGHTS\_STAY\_ROUNDEDinstance-attribute

NIGHTS\_STAY\_ROUNDED: IconData

buildNIGHTS\_STAY\_SHARPinstance-attribute

NIGHTS\_STAY\_SHARP: IconData

buildNINE\_Kinstance-attribute

NINE\_K: IconData

buildNINE\_K\_OUTLINEDinstance-attribute

NINE\_K\_OUTLINED: IconData

buildNINE\_K\_PLUSinstance-attribute

NINE\_K\_PLUS: IconData

buildNINE\_K\_PLUS\_OUTLINEDinstance-attribute

NINE\_K\_PLUS\_OUTLINED: IconData

buildNINE\_K\_PLUS\_ROUNDEDinstance-attribute

NINE\_K\_PLUS\_ROUNDED: IconData

buildNINE\_K\_PLUS\_SHARPinstance-attribute

NINE\_K\_PLUS\_SHARP: IconData

buildNINE\_K\_ROUNDEDinstance-attribute

NINE\_K\_ROUNDED: IconData

buildNINE\_K\_SHARPinstance-attribute

NINE\_K\_SHARP: IconData

buildNINE\_MPinstance-attribute

NINE\_MP: IconData

buildNINE\_MP\_OUTLINEDinstance-attribute

NINE\_MP\_OUTLINED: IconData

buildNINE\_MP\_ROUNDEDinstance-attribute

NINE\_MP\_ROUNDED: IconData

buildNINE\_MP\_SHARPinstance-attribute

NINE\_MP\_SHARP: IconData

buildNINETEEN\_MPinstance-attribute

NINETEEN\_MP: IconData

buildNINETEEN\_MP\_OUTLINEDinstance-attribute

NINETEEN\_MP\_OUTLINED: IconData

buildNINETEEN\_MP\_ROUNDEDinstance-attribute

NINETEEN\_MP\_ROUNDED: IconData

buildNINETEEN\_MP\_SHARPinstance-attribute

NINETEEN\_MP\_SHARP: IconData

buildNO\_ACCOUNTSinstance-attribute

NO\_ACCOUNTS: IconData

buildNO\_ACCOUNTS\_OUTLINEDinstance-attribute

NO\_ACCOUNTS\_OUTLINED: IconData

buildNO\_ACCOUNTS\_ROUNDEDinstance-attribute

NO\_ACCOUNTS\_ROUNDED: IconData

buildNO\_ACCOUNTS\_SHARPinstance-attribute

NO\_ACCOUNTS\_SHARP: IconData

buildNO\_ADULT\_CONTENTinstance-attribute

NO\_ADULT\_CONTENT: IconData

buildNO\_ADULT\_CONTENT\_OUTLINEDinstance-attribute

NO\_ADULT\_CONTENT\_OUTLINED: IconData

buildNO\_ADULT\_CONTENT\_ROUNDEDinstance-attribute

NO\_ADULT\_CONTENT\_ROUNDED: IconData

buildNO\_ADULT\_CONTENT\_SHARPinstance-attribute

NO\_ADULT\_CONTENT\_SHARP: IconData

buildNO\_BACKPACKinstance-attribute

NO\_BACKPACK: IconData

buildNO\_BACKPACK\_OUTLINEDinstance-attribute

NO\_BACKPACK\_OUTLINED: IconData



**buildNO\_BACKPACK\_ROUNDEDinstance-attribute**

NO\_BACKPACK\_ROUNDED: IconData

**buildNO\_BACKPACK\_SHARPinstance-attribute**

NO\_BACKPACK\_SHARP: IconData

**buildNO\_CELLinstance-attribute**

NO\_CELL: IconData

**buildNO\_CELL\_OUTLINEDinstance-attribute**

NO\_CELL\_OUTLINED: IconData

**buildNO\_CELL\_ROUNDEDinstance-attribute**

NO\_CELL\_ROUNDED: IconData

**buildNO\_CELL\_SHARPinstance-attribute**

NO\_CELL\_SHARP: IconData

**buildNO\_CRASHinstance-attribute**

NO\_CRASH: IconData

**buildNO\_CRASH\_OUTLINEDinstance-attribute**

NO\_CRASH\_OUTLINED: IconData

**buildNO\_CRASH\_ROUNDEDinstance-attribute**

NO\_CRASH\_ROUNDED: IconData

**buildNO\_CRASH\_SHARPinstance-attribute**

NO\_CRASH\_SHARP: IconData

**buildNO\_DRINKSinstance-attribute**

NO\_DRINKS: IconData

**buildNO\_DRINKS\_OUTLINEDinstance-attribute**

NO\_DRINKS\_OUTLINED: IconData

**buildNO\_DRINKS\_ROUNDEDinstance-attribute**

NO\_DRINKS\_ROUNDED: IconData

**buildNO\_DRINKS\_SHARPinstance-attribute**

NO\_DRINKS\_SHARP: IconData

**buildNO\_ENCRYPTIONinstance-attribute**

NO\_ENCRYPTION: IconData

**buildNO\_ENCRYPTION\_GMAILERRORREDinstance-attribute**

NO\_ENCRYPTION\_GMAILERRORRED: IconData

**buildNO\_ENCRYPTION\_GMAILERRORRED\_OUTLINEDinstance-attribute**

NO\_ENCRYPTION\_GMAILERRORRED\_OUTLINED: IconData

**buildNO\_ENCRYPTION\_GMAILERRORRED\_ROUNDEDinstance-attribute**

NO\_ENCRYPTION\_GMAILERRORRED\_ROUNDED: IconData

**buildNO\_ENCRYPTION\_GMAILERRORRED\_SHARPinstance-attribute**

NO\_ENCRYPTION\_GMAILERRORRED\_SHARP: IconData

**buildNO\_ENCRYPTION\_OUTLINEDinstance-attribute**

NO\_ENCRYPTION\_OUTLINED: IconData

**buildNO\_ENCRYPTION\_ROUNDEDinstance-attribute**

NO\_ENCRYPTION\_ROUNDED: IconData

**buildNO\_ENCRYPTION\_SHARPinstance-attribute**

NO\_ENCRYPTION\_SHARP: IconData

**buildNO\_FLASHinstance-attribute**

NO\_FLASH: IconData

**buildNO\_FLASH\_OUTLINEDinstance-attribute**

NO\_FLASH\_OUTLINED: IconData

**buildNO\_FLASH\_ROUNDEDinstance-attribute**

NO\_FLASH\_ROUNDED: IconData

**buildNO\_FLASH\_SHARPinstance-attribute**

NO\_FLASH\_SHARP: IconData

**buildNO\_FOODinstance-attribute**

NO\_FOOD: IconData

**buildNO\_FOOD\_OUTLINEDinstance-attribute**

NO\_FOOD\_OUTLINED: IconData

**buildNO\_FOOD\_ROUNDEDinstance-attribute**

NO\_FOOD\_ROUNDED: IconData

**buildNO\_FOOD\_SHARPinstance-attribute**

NO\_FOOD\_SHARP: IconData

**buildNO\_LUGGAGEinstance-attribute**

NO\_LUGGAGE: IconData

**buildNO\_LUGGAGE\_OUTLINEDinstance-attribute**

NO\_LUGGAGE\_OUTLINED: IconData

**buildNO\_LUGGAGE\_ROUNDEDinstance-attribute**

NO\_LUGGAGE\_ROUNDED: IconData

**buildNO\_LUGGAGE\_SHARPinstance-attribute**

NO\_LUGGAGE\_SHARP: IconData

**buildNO\_MEALSinstance-attribute**

NO\_MEALS: IconData

**buildNO\_MEALS\_OULINEinstance-attribute**

NO\_MEALS\_OULINE: IconData

**buildNO\_MEALS\_OUTLINEDinstance-attribute**

NO\_MEALS\_OUTLINED: IconData

**buildNO\_MEALS\_ROUNDEDinstance-attribute**

NO\_MEALS\_ROUNDED: IconData

**buildNO\_MEALS\_SHARPinstance-attribute**

NO\_MEALS\_SHARP: IconData

**buildNO\_MEETING\_ROOMinstance-attribute**

NO\_MEETING\_ROOM: IconData

**buildNO\_MEETING\_ROOM\_OUTLINEDinstance-attribute**

NO\_MEETING\_ROOM\_OUTLINED: IconData

**buildNO\_MEETING\_ROOM\_ROUNDEDinstance-attribute**

NO\_MEETING\_ROOM\_ROUNDED: IconData

**buildNO\_MEETING\_ROOM\_SHARPinstance-attribute**

NO\_MEETING\_ROOM\_SHARP: IconData

**buildNO\_PHOTOGRAPHYinstance-attribute**

NO\_PHOTOGRAPHY: IconData

**buildNO\_PHOTOGRAPHY\_OUTLINEDinstance-attribute**

NO\_PHOTOGRAPHY\_OUTLINED: IconData

**buildNO\_PHOTOGRAPHY\_ROUNDEDinstance-attribute**

NO\_PHOTOGRAPHY\_ROUNDED: IconData

**buildNO\_PHOTOGRAPHY\_SHARPinstance-attribute**

NO\_PHOTOGRAPHY\_SHARP: IconData

**buildNO\_SIMinstance-attribute**

NO\_SIM: IconData

**buildNO\_SIM\_OUTLINEDinstance-attribute**

NO\_SIM\_OUTLINED: IconData

**buildNO\_SIM\_ROUNDEDinstance-attribute**

NO\_SIM\_ROUNDED: IconData

**buildNO\_SIM\_SHARPinstance-attribute**

NO\_SIM\_SHARP: IconData

**buildNO\_STROLLERinstance-attribute**

NO\_STROLLER: IconData

**buildNO\_STROLLER\_OUTLINEDinstance-attribute**

NO\_STROLLER\_OUTLINED: IconData

**buildNO\_STROLLER\_ROUNDEDinstance-attribute**

NO\_STROLLER\_ROUNDED: IconData

**buildNO\_STROLLER\_SHARPinstance-attribute**

NO\_STROLLER\_SHARP: IconData

**buildNO\_TRANSFERinstance-attribute**

NO\_TRANSFER: IconData

**buildNO\_TRANSFER\_OUTLINEDinstance-attribute**

NO\_TRANSFER\_OUTLINED: IconData

**buildNO\_TRANSFER\_ROUNDEDinstance-attribute**

NO\_TRANSFER\_ROUNDED: IconData

**buildNO\_TRANSFER\_SHARPinstance-attribute**

NO\_TRANSFER\_SHARP: IconData

**buildNOISE\_AWAREinstance-attribute**

NOISE\_AWARE: IconData

**buildNOISE\_AWARE\_OUTLINEDinstance-attribute**

NOISE\_AWARE\_OUTLINED: IconData

**buildNOISE\_AWARE\_ROUNDEDinstance-attribute**

NOISE\_AWARE\_ROUNDED: IconData

**buildNOISE\_AWARE\_SHARPinstance-attribute**

NOISE\_AWARE\_SHARP: IconData

**buildNOISE\_CONTROL\_OFFinstance-attribute**

NOISE\_CONTROL\_OFF: IconData

**buildNOISE\_CONTROL\_OFF\_OUTLINEDinstance-attribute**

NOISE\_CONTROL\_OFF\_OUTLINED: IconData

**buildNOISE\_CONTROL\_OFF\_ROUNDEDinstance-attribute**

NOISE\_CONTROL\_OFF\_ROUNDED: IconData

**buildNOISE\_CONTROL\_OFF\_SHARPinstance-attribute**

NOISE\_CONTROL\_OFF\_SHARP: IconData

**buildNORDIC\_WALKINGinstance-attribute**

NORDIC\_WALKING: IconData

**buildNORDIC\_WALKING\_OUTLINEDinstance-attribute**

NORDIC\_WALKING\_OUTLINED: IconData

**buildNORDIC\_WALKING\_ROUNDEDinstance-attribute**

NORDIC\_WALKING\_ROUNDED: IconData

**buildNORDIC\_WALKING\_SHARPinstance-attribute**

NORDIC\_WALKING\_SHARP: IconData

**buildNORTHinstance-attribute**

NORTH: IconData

**buildNORTH\_EASTinstance-attribute**

NORTH\_EAST: IconData

**buildNORTH\_EAST\_OUTLINEDinstance-attribute**

NORTH\_EAST\_OUTLINED: IconData

**buildNORTH\_EAST\_ROUNDEDinstance-attribute**

NORTH\_EAST\_ROUNDED: IconData

**buildNORTH\_EAST\_SHARPinstance-attribute**

NORTH\_EAST\_SHARP: IconData

**buildNORTH\_OUTLINEDinstance-attribute**

NORTH\_OUTLINED: IconData

**buildNORTH\_ROUNDEDinstance-attribute**

NORTH\_ROUNDED: IconData

**buildNORTH\_SHARPinstance-attribute**

NORTH\_SHARP: IconData

**buildNORTH\_WESTinstance-attribute**

NORTH\_WEST: IconData

**buildNORTH\_WEST\_OUTLINEDinstance-attribute**

NORTH\_WEST\_OUTLINED: IconData

**buildNORTH\_WEST\_ROUNDEDinstance-attribute**

NORTH\_WEST\_ROUNDED: IconData

**buildNORTH\_WEST\_SHARPinstance-attribute**

NORTH\_WEST\_SHARP: IconData

**buildNOT\_ACCESSIBLEinstance-attribute**

NOT\_ACCESSIBLE: IconData

**buildNOT\_ACCESSIBLE\_OUTLINEDinstance-attribute**

NOT\_ACCESSIBLE\_OUTLINED: IconData

**buildNOT\_ACCESSIBLE\_ROUNDEDinstance-attribute**

NOT\_ACCESSIBLE\_ROUNDED: IconData

**buildNOT\_ACCESSIBLE\_SHARPinstance-attribute**

NOT\_ACCESSIBLE\_SHARP: IconData

**buildNOT\_INTERESTEDinstance-attribute**

NOT\_INTERESTED: IconData

**buildNOT\_INTERESTED\_OUTLINEDinstance-attribute**

NOT\_INTERESTED\_OUTLINED: IconData

**buildNOT\_INTERESTED\_ROUNDEDinstance-attribute**

NOT\_INTERESTED\_ROUNDED: IconData

**buildNOT\_INTERESTED\_SHARPinstance-attribute**

NOT\_INTERESTED\_SHARP: IconData

**buildNOT\_LISTED\_LOCATIONinstance-attribute**

NOT\_LISTED\_LOCATION: IconData

**buildNOT\_LISTED\_LOCATION\_OUTLINEDinstance-attribute**

NOT\_LISTED\_LOCATION\_OUTLINED: IconData

**buildNOT\_LISTED\_LOCATION\_ROUNDEDinstance-attribute**

NOT\_LISTED\_LOCATION\_ROUNDED: IconData

**buildNOT\_LISTED\_LOCATION\_SHARPinstance-attribute**

NOT\_LISTED\_LOCATION\_SHARP: IconData

**buildNOT\_STARTEDinstance-attribute**

NOT\_STARTED: IconData

**buildNOT\_STARTED\_OUTLINEDinstance-attribute**

NOT\_STARTED\_OUTLINED: IconData

**buildNOT\_STARTED\_ROUNDEDinstance-attribute**

NOT\_STARTED\_ROUNDED: IconData

**buildNOT\_STARTED\_SHARPinstance-attribute**

NOT\_STARTED\_SHARP: IconData

**buildNOTEinstance-attribute**

NOTE: IconData

**buildNOTE\_ADDinstance-attribute**

NOTE\_ADD: IconData

**buildNOTE\_ADD\_OUTLINEDinstance-attribute**

NOTE\_ADD\_OUTLINED: IconData

**buildNOTE\_ADD\_ROUNDEDinstance-attribute**

NOTE\_ADD\_ROUNDED: IconData

**buildNOTE\_ADD\_SHARPinstance-attribute**

NOTE\_ADD\_SHARP: IconData

**buildNOTE\_ALTinstance-attribute**

NOTE\_ALT: IconData

**buildNOTE\_ALT\_OUTLINEDinstance-attribute**

NOTE\_ALT\_OUTLINED: IconData

**buildNOTE\_ALT\_ROUNDEDinstance-attribute**

NOTE\_ALT\_ROUNDED: IconData

**buildNOTE\_ALT\_SHARPinstance-attribute**

NOTE\_ALT\_SHARP: IconData

**buildNOTE\_OUTLINEDinstance-attribute**

NOTE\_OUTLINED: IconData

**buildNOTE\_ROUNDEDinstance-attribute**

NOTE\_ROUNDED: IconData

**buildNOTE\_SHARPinstance-attribute**

NOTE\_SHARP: IconData

**buildNOTESinstance-attribute**

NOTES: IconData

**buildNOTES\_OUTLINEDinstance-attribute**

NOTES\_OUTLINED: IconData

**buildNOTES\_ROUNDEDinstance-attribute**

NOTES\_ROUNDED: IconData

**buildNOTES\_SHARPinstance-attribute**

NOTES\_SHARP: IconData

**buildNOTIFICATION\_ADDinstance-attribute**

NOTIFICATION\_ADD: IconData

**buildNOTIFICATION\_ADD\_OUTLINEDinstance-attribute**

NOTIFICATION\_ADD\_OUTLINED: IconData

**buildNOTIFICATION\_ADD\_ROUNDEDinstance-attribute**

NOTIFICATION\_ADD\_ROUNDED: IconData

**buildNOTIFICATION\_ADD\_SHARPinstance-attribute**

NOTIFICATION\_ADD\_SHARP: IconData

**buildNOTIFICATION\_IMPORTANTinstance-attribute**

NOTIFICATION\_IMPORTANT: IconData

**buildNOTIFICATION\_IMPORTANT\_OUTLINEDinstance-attribute**

NOTIFICATION\_IMPORTANT\_OUTLINED: IconData

**buildNOTIFICATION\_IMPORTANT\_ROUNDEDinstance-attribute**

NOTIFICATION\_IMPORTANT\_ROUNDED: IconData

**buildNOTIFICATION\_IMPORTANT\_SHARPinstance-attribute**

NOTIFICATION\_IMPORTANT\_SHARP: IconData

**buildNOTIFICATIONSinstance-attribute**

NOTIFICATIONS: IconData

**buildNOTIFICATIONS\_ACTIVEinstance-attribute**

NOTIFICATIONS\_ACTIVE: IconData

**buildNOTIFICATIONS\_ACTIVE\_OUTLINEDinstance-attribute**

NOTIFICATIONS\_ACTIVE\_OUTLINED: IconData

**buildNOTIFICATIONS\_ACTIVE\_ROUNDEDinstance-attribute**

NOTIFICATIONS\_ACTIVE\_ROUNDED: IconData

**buildNOTIFICATIONS\_ACTIVE\_SHARPinstance-attribute**

NOTIFICATIONS\_ACTIVE\_SHARP: IconData

**buildNOTIFICATIONS\_NONEinstance-attribute**

NOTIFICATIONS\_NONE: IconData

**buildNOTIFICATIONS\_NONE\_OUTLINEDinstance-attribute**

NOTIFICATIONS\_NONE\_OUTLINED: IconData

**buildNOTIFICATIONS\_NONE\_ROUNDEDinstance-attribute**

NOTIFICATIONS\_NONE\_ROUNDED: IconData

**buildNOTIFICATIONS\_NONE\_SHARPinstance-attribute**

NOTIFICATIONS\_NONE\_SHARP: IconData

**buildNOTIFICATIONS\_OFFinstance-attribute**

NOTIFICATIONS\_OFF: IconData

**buildNOTIFICATIONS\_OFF\_OUTLINEDinstance-attribute**

NOTIFICATIONS\_OFF\_OUTLINED: IconData

**buildNOTIFICATIONS\_OFF\_ROUNDEDinstance-attribute**

NOTIFICATIONS\_OFF\_ROUNDED: IconData

**buildNOTIFICATIONS\_OFF\_SHARPinstance-attribute**

NOTIFICATIONS\_OFF\_SHARP: IconData

**buildNOTIFICATIONS\_ONinstance-attribute**

NOTIFICATIONS\_ON: IconData

**buildNOTIFICATIONS\_ON\_OUTLINEDinstance-attribute**

NOTIFICATIONS\_ON\_OUTLINED: IconData

**buildNOTIFICATIONS\_ON\_ROUNDEDinstance-attribute**

NOTIFICATIONS\_ON\_ROUNDED: IconData

**buildNOTIFICATIONS\_ON\_SHARPinstance-attribute**

NOTIFICATIONS\_ON\_SHARP: IconData

**buildNOTIFICATIONS\_OUTLINEDinstance-attribute**

NOTIFICATIONS\_OUTLINED: IconData

**buildNOTIFICATIONS\_PAUSEDinstance-attribute**

NOTIFICATIONS\_PAUSED: IconData

**buildNOTIFICATIONS\_PAUSED\_OUTLINEDinstance-attribute**

NOTIFICATIONS\_PAUSED\_OUTLINED: IconData

**buildNOTIFICATIONS\_PAUSED\_ROUNDEDinstance-attribute**

NOTIFICATIONS\_PAUSED\_ROUNDED: IconData

**buildNOTIFICATIONS\_PAUSED\_SHARPinstance-attribute**

NOTIFICATIONS\_PAUSED\_SHARP: IconData

**buildNOTIFICATIONS\_ROUNDEDinstance-attribute**

NOTIFICATIONS\_ROUNDED: IconData

**buildNOTIFICATIONS\_SHARPinstance-attribute**

NOTIFICATIONS\_SHARP: IconData

**buildNOW\_WALLPAPERinstance-attribute**

NOW\_WALLPAPER: IconData

**buildNOW\_WALLPAPER\_OUTLINEDinstance-attribute**

NOW\_WALLPAPER\_OUTLINED: IconData

**buildNOW\_WALLPAPER\_ROUNDEDinstance-attribute**

NOW\_WALLPAPER\_ROUNDED: IconData

**buildNOW\_WALLPAPER\_SHARPinstance-attribute**

NOW\_WALLPAPER\_SHARP: IconData

**buildNOW\_WIDGETSinstance-attribute**

NOW\_WIDGETS: IconData

**buildNOW\_WIDGETS\_OUTLINEDinstance-attribute**

NOW\_WIDGETS\_OUTLINED: IconData

**buildNOW\_WIDGETS\_ROUNDEDinstance-attribute**

NOW\_WIDGETS\_ROUNDED: IconData

**buildNOW\_WIDGETS\_SHARPinstance-attribute**

NOW\_WIDGETS\_SHARP: IconData

**buildNUMBERSinstance-attribute**

NUMBERS: IconData

**buildNUMBERS\_OUTLINEDinstance-attribute**

NUMBERS\_OUTLINED: IconData

**buildNUMBERS\_ROUNDEDinstance-attribute**

NUMBERS\_ROUNDED: IconData

**buildNUMBERS\_SHARPinstance-attribute**

NUMBERS\_SHARP: IconData

**buildOFFLINE\_BOLTinstance-attribute**

OFFLINE\_BOLT: IconData

**buildOFFLINE\_BOLT\_OUTLINEDinstance-attribute**

OFFLINE\_BOLT\_OUTLINED: IconData

**buildOFFLINE\_BOLT\_ROUNDEDinstance-attribute**

OFFLINE\_BOLT\_ROUNDED: IconData

**buildOFFLINE\_BOLT\_SHARPinstance-attribute**

OFFLINE\_BOLT\_SHARP: IconData

**buildOFFLINE\_PINinstance-attribute**

OFFLINE\_PIN: IconData

**buildOFFLINE\_PIN\_OUTLINEDinstance-attribute**

OFFLINE\_PIN\_OUTLINED: IconData

**buildOFFLINE\_PIN\_ROUNDEDinstance-attribute**

OFFLINE\_PIN\_ROUNDED: IconData

**buildOFFLINE\_PIN\_SHARPinstance-attribute**

OFFLINE\_PIN\_SHARP: IconData

**buildOFFLINE\_SHAREinstance-attribute**

OFFLINE\_SHARE: IconData

**buildOFFLINE\_SHARE\_OUTLINEDinstance-attribute**

OFFLINE\_SHARE\_OUTLINED: IconData

**buildOFFLINE\_SHARE\_ROUNDEDinstance-attribute**

OFFLINE\_SHARE\_ROUNDED: IconData

**buildOFFLINE\_SHARE\_SHARPinstance-attribute**

OFFLINE\_SHARE\_SHARP: IconData

**buildOIL\_BARRELinstance-attribute**

OIL\_BARREL: IconData

**buildOIL\_BARREL\_OUTLINEDinstance-attribute**

OIL\_BARREL\_OUTLINED: IconData

**buildOIL\_BARREL\_ROUNDEDinstance-attribute**

OIL\_BARREL\_ROUNDED: IconData

**buildOIL\_BARREL\_SHARPinstance-attribute**

OIL\_BARREL\_SHARP: IconData

**buildON\_DEVICE\_TRAININGinstance-attribute**

ON\_DEVICE\_TRAINING: IconData



buildON\_DEVICE\_TRAINING\_OUTLINEDinstance-attribute

ON\_DEVICE\_TRAINING\_OUTLINED: IconData

buildON\_DEVICE\_TRAINING\_ROUNDEDinstance-attribute

ON\_DEVICE\_TRAINING\_ROUNDED: IconData

buildON\_DEVICE\_TRAINING\_SHARPinstance-attribute

ON\_DEVICE\_TRAINING\_SHARP: IconData

buildONDEMAND\_VIDEOinstance-attribute

ONDEMAND\_VIDEO: IconData

buildONDEMAND\_VIDEO\_OUTLINEDinstance-attribute

ONDEMAND\_VIDEO\_OUTLINED: IconData

buildONDEMAND\_VIDEO\_ROUNDEDinstance-attribute

ONDEMAND\_VIDEO\_ROUNDED: IconData

buildONDEMAND\_VIDEO\_SHARPinstance-attribute

ONDEMAND\_VIDEO\_SHARP: IconData

buildONE\_Kinstance-attribute

ONE\_K: IconData

buildONE\_K\_OUTLINEDinstance-attribute

ONE\_K\_OUTLINED: IconData

buildONE\_K\_PLUSinstance-attribute

ONE\_K\_PLUS: IconData

buildONE\_K\_PLUS\_OUTLINEDinstance-attribute

ONE\_K\_PLUS\_OUTLINED: IconData

buildONE\_K\_PLUS\_ROUNDEDinstance-attribute

ONE\_K\_PLUS\_ROUNDED: IconData

buildONE\_K\_PLUS\_SHARPinstance-attribute

ONE\_K\_PLUS\_SHARP: IconData

buildONE\_K\_ROUNDEDinstance-attribute

ONE\_K\_ROUNDED: IconData

buildONE\_K\_SHARPinstance-attribute

ONE\_K\_SHARP: IconData

buildONE\_X\_MOBILEDATAinstance-attribute

ONE\_X\_MOBILEDATA: IconData

buildONE\_X\_MOBILEDATA\_OUTLINEDinstance-attribute

ONE\_X\_MOBILEDATA\_OUTLINED: IconData

buildONE\_X\_MOBILEDATA\_ROUNDEDinstance-attribute

ONE\_X\_MOBILEDATA\_ROUNDED: IconData

buildONE\_X\_MOBILEDATA\_SHARPinstance-attribute

ONE\_X\_MOBILEDATA\_SHARP: IconData

buildONETWOTHREEinstance-attribute

ONETWOTHREE: IconData

buildONETWOTHREE\_OUTLINEDinstance-attribute

ONETWOTHREE\_OUTLINED: IconData

buildONETWOTHREE\_ROUNDEDinstance-attribute

ONETWOTHREE\_ROUNDED: IconData

**buildONETWOTHREE\_SHARPinstance-attribute**

ONETWOTHREE\_SHARP: IconData

**buildONLINE\_PREDICTIONinstance-attribute**

ONLINE\_PREDICTION: IconData

**buildONLINE\_PREDICTION\_OUTLINEDinstance-attribute**

ONLINE\_PREDICTION\_OUTLINED: IconData

**buildONLINE\_PREDICTION\_ROUNDEDinstance-attribute**

ONLINE\_PREDICTION\_ROUNDED: IconData

**buildONLINE\_PREDICTION\_SHARPinstance-attribute**

ONLINE\_PREDICTION\_SHARP: IconData

**buildOPACITYinstance-attribute**

OPACITY: IconData

**buildOPACITY\_OUTLINEDinstance-attribute**

OPACITY\_OUTLINED: IconData

**buildOPACITY\_ROUNDEDinstance-attribute**

OPACITY\_ROUNDED: IconData

**buildOPACITY\_SHARPinstance-attribute**

OPACITY\_SHARP: IconData

**buildOPEN\_IN\_BROWSERinstance-attribute**

OPEN\_IN\_BROWSER: IconData

**buildOPEN\_IN\_BROWSER\_OUTLINEDinstance-attribute**

OPEN\_IN\_BROWSER\_OUTLINED: IconData

**buildOPEN\_IN\_BROWSER\_ROUNDEDinstance-attribute**

OPEN\_IN\_BROWSER\_ROUNDED: IconData

**buildOPEN\_IN\_BROWSER\_SHARPinstance-attribute**

OPEN\_IN\_BROWSER\_SHARP: IconData

**buildOPEN\_IN\_FULLinstance-attribute**

OPEN\_IN\_FULL: IconData

**buildOPEN\_IN\_FULL\_OUTLINEDinstance-attribute**

OPEN\_IN\_FULL\_OUTLINED: IconData

**buildOPEN\_IN\_FULL\_ROUNDEDinstance-attribute**

OPEN\_IN\_FULL\_ROUNDED: IconData

**buildOPEN\_IN\_FULL\_SHARPinstance-attribute**

OPEN\_IN\_FULL\_SHARP: IconData

**buildOPEN\_IN\_NEWinstance-attribute**

OPEN\_IN\_NEW: IconData

**buildOPEN\_IN\_NEW\_OFFinstance-attribute**

OPEN\_IN\_NEW\_OFF: IconData

**buildOPEN\_IN\_NEW\_OFF\_OUTLINEDinstance-attribute**

OPEN\_IN\_NEW\_OFF\_OUTLINED: IconData

**buildOPEN\_IN\_NEW\_OFF\_ROUNDEDinstance-attribute**

OPEN\_IN\_NEW\_OFF\_ROUNDED: IconData

**buildOPEN\_IN\_NEW\_OFF\_SHARPinstance-attribute**

OPEN\_IN\_NEW\_OFF\_SHARP: IconData

**buildOPEN\_IN\_NEW\_OUTLINEDinstance-attribute**

OPEN\_IN\_NEW\_OUTLINED: IconData

**buildOPEN\_IN\_NEW\_ROUNDEDinstance-attribute**

OPEN\_IN\_NEW\_ROUNDED: IconData

**buildOPEN\_IN\_NEW\_SHARPinstance-attribute**

OPEN\_IN\_NEW\_SHARP: IconData

**buildOPEN\_WITHinstance-attribute**

OPEN\_WITH: IconData

**buildOPEN\_WITH\_OUTLINEDinstance-attribute**

OPEN\_WITH\_OUTLINED: IconData

**buildOPEN\_WITH\_ROUNDEDinstance-attribute**

OPEN\_WITH\_ROUNDED: IconData

**buildOPEN\_WITH\_SHARPinstance-attribute**

OPEN\_WITH\_SHARP: IconData

**buildOTHER\_HOUSESinstance-attribute**

OTHER\_HOUSES: IconData

**buildOTHER\_HOUSES\_OUTLINEDinstance-attribute**

OTHER\_HOUSES\_OUTLINED: IconData

**buildOTHER\_HOUSES\_ROUNDEDinstance-attribute**

OTHER\_HOUSES\_ROUNDED: IconData

**buildOTHER\_HOUSES\_SHARPinstance-attribute**

OTHER\_HOUSES\_SHARP: IconData

**buildOUTBONDinstance-attribute**

OUTBOND: IconData

**buildOUTBOND\_OUTLINEDinstance-attribute**

OUTBOND\_OUTLINED: IconData

**buildOUTBOND\_ROUNDEDinstance-attribute**

OUTBOND\_ROUNDED: IconData

**buildOUTBOND\_SHARPinstance-attribute**

OUTBOND\_SHARP: IconData

**buildOUTBOUNDinstance-attribute**

OUTBOUND: IconData

**buildOUTBOUND\_OUTLINEDinstance-attribute**

OUTBOUND\_OUTLINED: IconData

**buildOUTBOUND\_ROUNDEDinstance-attribute**

OUTBOUND\_ROUNDED: IconData

**buildOUTBOUND\_SHARPinstance-attribute**

OUTBOUND\_SHARP: IconData

**buildOUTBOXinstance-attribute**

OUTBOX: IconData

**buildOUTBOX\_OUTLINEDinstance-attribute**

OUTBOX\_OUTLINED: IconData

**buildOUTBOX\_ROUNDEDinstance-attribute**

OUTBOX\_ROUNDED: IconData

**buildOUTBOX\_SHARPinstance-attribute**

OUTBOX\_SHARP: IconData

**buildOUTDOOR\_GRILLinstance-attribute**

OUTDOOR\_GRILL: IconData

**buildOUTDOOR\_GRILL\_OUTLINEDinstance-attribute**

OUTDOOR\_GRILL\_OUTLINED: IconData

**buildOUTDOOR\_GRILL\_ROUNDEDinstance-attribute**

OUTDOOR\_GRILL\_ROUNDED: IconData

**buildOUTDOOR\_GRILL\_SHARPinstance-attribute**

OUTDOOR\_GRILL\_SHARP: IconData

**buildOUTGOING\_MAILinstance-attribute**

OUTGOING\_MAIL: IconData

**buildOUTLETinstance-attribute**

OUTLET: IconData

**buildOUTLET\_OUTLINEDinstance-attribute**

OUTLET\_OUTLINED: IconData

**buildOUTLET\_ROUNDEDinstance-attribute**

OUTLET\_ROUNDED: IconData

**buildOUTLET\_SHARPinstance-attribute**

OUTLET\_SHARP: IconData

**buildOUTLINED\_FLAGinstance-attribute**

OUTLINED\_FLAG: IconData

**buildOUTLINED\_FLAG\_OUTLINEDinstance-attribute**

OUTLINED\_FLAG\_OUTLINED: IconData

**buildOUTLINED\_FLAG\_ROUNDEDinstance-attribute**

OUTLINED\_FLAG\_ROUNDED: IconData

**buildOUTLINED\_FLAG\_SHARPinstance-attribute**

OUTLINED\_FLAG\_SHARP: IconData

**buildOUTPUTinstance-attribute**

OUTPUT: IconData

**buildOUTPUT\_OUTLINEDinstance-attribute**

OUTPUT\_OUTLINED: IconData

**buildOUTPUT\_ROUNDEDinstance-attribute**

OUTPUT\_ROUNDED: IconData

**buildOUTPUT\_SHARPinstance-attribute**

OUTPUT\_SHARP: IconData

**buildPADDINGinstance-attribute**

PADDING: IconData

**buildPADDING\_OUTLINEDinstance-attribute**

PADDING\_OUTLINED: IconData

**buildPADDING\_ROUNDEDinstance-attribute**

PADDING\_ROUNDED: IconData

**buildPADDING\_SHARPinstance-attribute**

PADDING\_SHARP: IconData

**buildPAGESinstance-attribute**

PAGES: IconData

**buildPAGES\_OUTLINEDinstance-attribute**

PAGES\_OUTLINED: IconData

**buildPAGES\_ROUNDEDinstance-attribute**

PAGES\_ROUNDED: IconData

**buildPAGES\_SHARPinstance-attribute**

PAGES\_SHARP: IconData

**buildPAGEVIEWinstance-attribute**

PAGEVIEW: IconData

**buildPAGEVIEW\_OUTLINEDinstance-attribute**

PAGEVIEW\_OUTLINED: IconData

**buildPAGEVIEW\_ROUNDEDinstance-attribute**

PAGEVIEW\_ROUNDED: IconData

**buildPAGEVIEW\_SHARPinstance-attribute**

PAGEVIEW\_SHARP: IconData

**buildPAIDinstance-attribute**

PAID: IconData

**buildPAID\_OUTLINEDinstance-attribute**

PAID\_OUTLINED: IconData

**buildPAID\_ROUNDEDinstance-attribute**

PAID\_ROUNDED: IconData

**buildPAID\_SHARPinstance-attribute**

PAID\_SHARP: IconData

**buildPALETTEinstance-attribute**

PALETTE: IconData

**buildPALETTE\_OUTLINEDinstance-attribute**

PALETTE\_OUTLINED: IconData

**buildPALETTE\_ROUNDEDinstance-attribute**

PALETTE\_ROUNDED: IconData

**buildPALETTE\_SHARPinstance-attribute**

PALETTE\_SHARP: IconData

**buildPALLETinstance-attribute**

PALLET: IconData

**buildPAN\_TOOLinstance-attribute**

PAN\_TOOL: IconData

**buildPAN\_TOOL\_ALTinstance-attribute**

PAN\_TOOL\_ALT: IconData

**buildPAN\_TOOL\_ALT\_OUTLINEDinstance-attribute**

PAN\_TOOL\_ALT\_OUTLINED: IconData

**buildPAN\_TOOL\_ALT\_ROUNDEDinstance-attribute**

PAN\_TOOL\_ALT\_ROUNDED: IconData

**buildPAN\_TOOL\_ALT\_SHARPinstance-attribute**

PAN\_TOOL\_ALT\_SHARP: IconData

buildPAN\_TOOL\_OUTLINEDinstance-attribute

PAN\_TOOL\_OUTLINED: IconData

buildPAN\_TOOL\_ROUNDEDinstance-attribute

PAN\_TOOL\_ROUNDED: IconData

buildPAN\_TOOL\_SHARPinstance-attribute

PAN\_TOOL\_SHARP: IconData

buildPANORAMAinstance-attribute

PANORAMA: IconData

buildPANORAMA\_FISH\_EYEinstance-attribute

PANORAMA\_FISH\_EYE: IconData

buildPANORAMA\_FISH\_EYE\_OUTLINEDinstance-attribute

PANORAMA\_FISH\_EYE\_OUTLINED: IconData

buildPANORAMA\_FISH\_EYE\_ROUNDEDinstance-attribute

PANORAMA\_FISH\_EYE\_ROUNDED: IconData

buildPANORAMA\_FISH\_EYE\_SHARPinstance-attribute

PANORAMA\_FISH\_EYE\_SHARP: IconData

buildPANORAMA\_FISHEYEinstance-attribute

PANORAMA\_FISHEYE: IconData

buildPANORAMA\_FISHEYE\_OUTLINEDinstance-attribute

PANORAMA\_FISHEYE\_OUTLINED: IconData

buildPANORAMA\_FISHEYE\_ROUNDEDinstance-attribute

PANORAMA\_FISHEYE\_ROUNDED: IconData

buildPANORAMA\_FISHEYE\_SHARPinstance-attribute

PANORAMA\_FISHEYE\_SHARP: IconData

buildPANORAMA\_HORIZONTALinstance-attribute

PANORAMA\_HORIZONTAL: IconData

buildPANORAMA\_HORIZONTAL\_OUTLINEDinstance-attribute

PANORAMA\_HORIZONTAL\_OUTLINED: IconData

buildPANORAMA\_HORIZONTAL\_ROUNDEDinstance-attribute

PANORAMA\_HORIZONTAL\_ROUNDED: IconData

buildPANORAMA\_HORIZONTAL\_SELECTinstance-attribute

PANORAMA\_HORIZONTAL\_SELECT: IconData

buildPANORAMA\_HORIZONTAL\_SELECT\_OUTLINEDinstance-attribute

PANORAMA\_HORIZONTAL\_SELECT\_OUTLINED: IconData

buildPANORAMA\_HORIZONTAL\_SELECT\_ROUNDEDinstance-attribute

PANORAMA\_HORIZONTAL\_SELECT\_ROUNDED: IconData

buildPANORAMA\_HORIZONTAL\_SELECT\_SHARPinstance-attribute

PANORAMA\_HORIZONTAL\_SELECT\_SHARP: IconData

buildPANORAMA\_HORIZONTAL\_SHARPinstance-attribute

PANORAMA\_HORIZONTAL\_SHARP: IconData

buildPANORAMA\_OUTLINEDinstance-attribute

PANORAMA\_OUTLINED: IconData

buildPANORAMA\_PHOTOSPHEREinstance-attribute

PANORAMA\_PHOTOSPHERE: IconData

buildPANORAMA\_PHOTOSPHERE\_OUTLINEDinstance-attribute

PANORAMA\_PHOTOSPHERE\_OUTLINED: IconData

buildPANORAMA\_PHOTOSPHERE\_ROUNDEDinstance-attribute

PANORAMA\_PHOTOSPHERE\_ROUNDED: IconData

buildPANORAMA\_PHOTOSPHERE\_SELECTinstance-attribute

PANORAMA\_PHOTOSPHERE\_SELECT: IconData

buildPANORAMA\_PHOTOSPHERE\_SELECT\_OUTLINEDinstance-attribute

PANORAMA\_PHOTOSPHERE\_SELECT\_OUTLINED: IconData

buildPANORAMA\_PHOTOSPHERE\_SELECT\_ROUNDEDinstance-attribute

PANORAMA\_PHOTOSPHERE\_SELECT\_ROUNDED: IconData

buildPANORAMA\_PHOTOSPHERE\_SELECT\_SHARPinstance-attribute

PANORAMA\_PHOTOSPHERE\_SELECT\_SHARP: IconData

buildPANORAMA\_PHOTOSPHERE\_SHARPinstance-attribute

PANORAMA\_PHOTOSPHERE\_SHARP: IconData

buildPANORAMA\_ROUNDEDinstance-attribute

PANORAMA\_ROUNDED: IconData

buildPANORAMA\_SHARPinstance-attribute

PANORAMA\_SHARP: IconData

buildPANORAMA\_VERTICALinstance-attribute

PANORAMA\_VERTICAL: IconData

buildPANORAMA\_VERTICAL\_OUTLINEDinstance-attribute

PANORAMA\_VERTICAL\_OUTLINED: IconData

buildPANORAMA\_VERTICAL\_ROUNDEDinstance-attribute

PANORAMA\_VERTICAL\_ROUNDED: IconData

buildPANORAMA\_VERTICAL\_SELECTinstance-attribute

PANORAMA\_VERTICAL\_SELECT: IconData

buildPANORAMA\_VERTICAL\_SELECT\_OUTLINEDinstance-attribute

PANORAMA\_VERTICAL\_SELECT\_OUTLINED: IconData

buildPANORAMA\_VERTICAL\_SELECT\_ROUNDEDinstance-attribute

PANORAMA\_VERTICAL\_SELECT\_ROUNDED: IconData

buildPANORAMA\_VERTICAL\_SELECT\_SHARPinstance-attribute

PANORAMA\_VERTICAL\_SELECT\_SHARP: IconData

buildPANORAMA\_VERTICAL\_SHARPinstance-attribute

PANORAMA\_VERTICAL\_SHARP: IconData

buildPANORAMA\_WIDE\_ANGLEinstance-attribute

PANORAMA\_WIDE\_ANGLE: IconData

buildPANORAMA\_WIDE\_ANGLE\_OUTLINEDinstance-attribute

PANORAMA\_WIDE\_ANGLE\_OUTLINED: IconData

buildPANORAMA\_WIDE\_ANGLE\_ROUNDEDinstance-attribute

PANORAMA\_WIDE\_ANGLE\_ROUNDED: IconData

buildPANORAMA\_WIDE\_ANGLE\_SELECTinstance-attribute

PANORAMA\_WIDE\_ANGLE\_SELECT: IconData

buildPANORAMA\_WIDE\_ANGLE\_SELECT\_OUTLINEDinstance-attribute

PANORAMA\_WIDE\_ANGLE\_SELECT\_OUTLINED: IconData

**buildPANORAMA\_WIDE\_ANGLE\_SELECT\_ROUNDEDinstance-attribute**

PANORAMA\_WIDE\_ANGLE\_SELECT\_ROUNDED: IconData

**buildPANORAMA\_WIDE\_ANGLE\_SELECT\_SHARPinstance-attribute**

PANORAMA\_WIDE\_ANGLE\_SELECT\_SHARP: IconData

**buildPANORAMA\_WIDE\_ANGLE\_SHARPinstance-attribute**

PANORAMA\_WIDE\_ANGLE\_SHARP: IconData

**buildPARAGLIDINGinstance-attribute**

PARAGLIDING: IconData

**buildPARAGLIDING\_OUTLINEDinstance-attribute**

PARAGLIDING\_OUTLINED: IconData

**buildPARAGLIDING\_ROUNDEDinstance-attribute**

PARAGLIDING\_ROUNDED: IconData

**buildPARAGLIDING\_SHARPinstance-attribute**

PARAGLIDING\_SHARP: IconData

**buildPARKinstance-attribute**

PARK: IconData

**buildPARK\_OUTLINEDinstance-attribute**

PARK\_OUTLINED: IconData

**buildPARK\_ROUNDEDinstance-attribute**

PARK\_ROUNDED: IconData

**buildPARK\_SHARPinstance-attribute**

PARK\_SHARP: IconData

**buildPARTY\_MODEinstance-attribute**

PARTY\_MODE: IconData

**buildPARTY\_MODE\_OUTLINEDinstance-attribute**

PARTY\_MODE\_OUTLINED: IconData

**buildPARTY\_MODE\_ROUNDEDinstance-attribute**

PARTY\_MODE\_ROUNDED: IconData

**buildPARTY\_MODE\_SHARPinstance-attribute**

PARTY\_MODE\_SHARP: IconData

**buildPASSWORDinstance-attribute**

PASSWORD: IconData

**buildPASSWORD\_OUTLINEDinstance-attribute**

PASSWORD\_OUTLINED: IconData

**buildPASSWORD\_ROUNDEDinstance-attribute**

PASSWORD\_ROUNDED: IconData

**buildPASSWORD\_SHARPinstance-attribute**

PASSWORD\_SHARP: IconData

**buildPASTEinstance-attribute**

PASTE: IconData

**buildPASTE\_OUTLINEDinstance-attribute**

PASTE\_OUTLINED: IconData

**buildPASTE\_ROUNDEDinstance-attribute**

PASTE\_ROUNDED: IconData



**buildPASTE\_SHARPinstance-attribute**

PASTE\_SHARP: IconData

**buildPATTERNinstance-attribute**

PATTERN: IconData

**buildPATTERN\_OUTLINEDinstance-attribute**

PATTERN\_OUTLINED: IconData

**buildPATTERN\_ROUNDEDinstance-attribute**

PATTERN\_ROUNDED: IconData

**buildPATTERN\_SHARPinstance-attribute**

PATTERN\_SHARP: IconData

**buildPAUSEinstance-attribute**

PAUSE: IconData

**buildPAUSE\_CIRCLEinstance-attribute**

PAUSE\_CIRCLE: IconData

**buildPAUSE\_CIRCLE\_FILLEDinstance-attribute**

PAUSE\_CIRCLE\_FILLED: IconData

**buildPAUSE\_CIRCLE\_FILLED\_OUTLINEDinstance-attribute**

PAUSE\_CIRCLE\_FILLED\_OUTLINED: IconData

**buildPAUSE\_CIRCLE\_FILLED\_ROUNDEDinstance-attribute**

PAUSE\_CIRCLE\_FILLED\_ROUNDED: IconData

**buildPAUSE\_CIRCLE\_FILLED\_SHARPinstance-attribute**

PAUSE\_CIRCLE\_FILLED\_SHARP: IconData

**buildPAUSE\_CIRCLE\_OUTLINEinstance-attribute**

PAUSE\_CIRCLE\_OUTLINE: IconData

**buildPAUSE\_CIRCLE\_OUTLINE\_OUTLINEDinstance-attribute**

PAUSE\_CIRCLE\_OUTLINE\_OUTLINED: IconData

**buildPAUSE\_CIRCLE\_OUTLINE\_ROUNDEDinstance-attribute**

PAUSE\_CIRCLE\_OUTLINE\_ROUNDED: IconData

**buildPAUSE\_CIRCLE\_OUTLINE\_SHARPinstance-attribute**

PAUSE\_CIRCLE\_OUTLINE\_SHARP: IconData

**buildPAUSE\_CIRCLE\_OUTLINEDinstance-attribute**

PAUSE\_CIRCLE\_OUTLINED: IconData

**buildPAUSE\_CIRCLE\_ROUNDEDinstance-attribute**

PAUSE\_CIRCLE\_ROUNDED: IconData

**buildPAUSE\_CIRCLE\_SHARPinstance-attribute**

PAUSE\_CIRCLE\_SHARP: IconData

**buildPAUSE\_OUTLINEDinstance-attribute**

PAUSE\_OUTLINED: IconData

**buildPAUSE\_PRESENTATIONinstance-attribute**

PAUSE\_PRESENTATION: IconData

**buildPAUSE\_PRESENTATION\_OUTLINEDinstance-attribute**

PAUSE\_PRESENTATION\_OUTLINED: IconData

**buildPAUSE\_PRESENTATION\_ROUNDEDinstance-attribute**

PAUSE\_PRESENTATION\_ROUNDED: IconData

**buildPAUSE\_PRESENTATION\_SHARPinstance-attribute**

PAUSE\_PRESENTATION\_SHARP: IconData

**buildPAUSE\_ROUNDEDinstance-attribute**

PAUSE\_ROUNDED: IconData

**buildPAUSE\_SHARPinstance-attribute**

PAUSE\_SHARP: IconData

**buildPAYMENTinstance-attribute**

PAYMENT: IconData

**buildPAYMENT\_OUTLINEDinstance-attribute**

PAYMENT\_OUTLINED: IconData

**buildPAYMENT\_ROUNDEDinstance-attribute**

PAYMENT\_ROUNDED: IconData

**buildPAYMENT\_SHARPinstance-attribute**

PAYMENT\_SHARP: IconData

**buildPAYMENTSinstance-attribute**

PAYMENTS: IconData

**buildPAYMENTS\_OUTLINEDinstance-attribute**

PAYMENTS\_OUTLINED: IconData

**buildPAYMENTS\_ROUNDEDinstance-attribute**

PAYMENTS\_ROUNDED: IconData

**buildPAYMENTS\_SHARPinstance-attribute**

PAYMENTS\_SHARP: IconData

**buildPAYPALinstance-attribute**

PAYPAL: IconData

**buildPAYPAL\_OUTLINEDinstance-attribute**

PAYPAL\_OUTLINED: IconData

**buildPAYPAL\_ROUNDEDinstance-attribute**

PAYPAL\_ROUNDED: IconData

**buildPAYPAL\_SHARPinstance-attribute**

PAYPAL\_SHARP: IconData

**buildPEDAL\_BIKEinstance-attribute**

PEDAL\_BIKE: IconData

**buildPEDAL\_BIKE\_OUTLINEDinstance-attribute**

PEDAL\_BIKE\_OUTLINED: IconData

**buildPEDAL\_BIKE\_ROUNDEDinstance-attribute**

PEDAL\_BIKE\_ROUNDED: IconData

**buildPEDAL\_BIKE\_SHARPinstance-attribute**

PEDAL\_BIKE\_SHARP: IconData

**buildPENDINGinstance-attribute**

PENDING: IconData

**buildPENDING\_ACTIONSinstance-attribute**

PENDING\_ACTIONS: IconData

**buildPENDING\_ACTIONS\_OUTLINEDinstance-attribute**

PENDING\_ACTIONS\_OUTLINED: IconData

**buildPENDING\_ACTIONS\_ROUNDED**instance-attribute

PENDING\_ACTIONS\_ROUNDED: IconData

**buildPENDING\_ACTIONS\_SHARP**instance-attribute

PENDING\_ACTIONS\_SHARP: IconData

**buildPENDING\_OUTLINED**instance-attribute

PENDING\_OUTLINED: IconData

**buildPENDING\_ROUNDED**instance-attribute

PENDING\_ROUNDED: IconData

**buildPENDING\_SHARP**instance-attribute

PENDING\_SHARP: IconData

**buildPENTAGON**instance-attribute

PENTAGON: IconData

**buildPENTAGON\_OUTLINED**instance-attribute

PENTAGON\_OUTLINED: IconData

**buildPENTAGON\_ROUNDED**instance-attribute

PENTAGON\_ROUNDED: IconData

**buildPENTAGON\_SHARP**instance-attribute

PENTAGON\_SHARP: IconData

**buildPEOPLE**instance-attribute

PEOPLE: IconData

**buildPEOPLE\_ALT**instance-attribute

PEOPLE\_ALT: IconData

**buildPEOPLE\_ALT\_OUTLINED**instance-attribute

PEOPLE\_ALT\_OUTLINED: IconData

**buildPEOPLE\_ALT\_ROUNDED**instance-attribute

PEOPLE\_ALT\_ROUNDED: IconData

**buildPEOPLE\_ALT\_SHARP**instance-attribute

PEOPLE\_ALT\_SHARP: IconData

**buildPEOPLE\_OUTLINE**instance-attribute

PEOPLE\_OUTLINE: IconData

**buildPEOPLE\_OUTLINE\_OUTLINED**instance-attribute

PEOPLE\_OUTLINE\_OUTLINED: IconData

**buildPEOPLE\_OUTLINE\_ROUNDED**instance-attribute

PEOPLE\_OUTLINE\_ROUNDED: IconData

**buildPEOPLE\_OUTLINE\_SHARP**instance-attribute

PEOPLE\_OUTLINE\_SHARP: IconData

**buildPEOPLE\_OUTLINED**instance-attribute

PEOPLE\_OUTLINED: IconData

**buildPEOPLE\_ROUNDED**instance-attribute

PEOPLE\_ROUNDED: IconData

**buildPEOPLE\_SHARP**instance-attribute

PEOPLE\_SHARP: IconData

**buildPERCENT**instance-attribute

PERCENT: IconData

buildPERCENT\_OUTLINEDinstance-attribute

PERCENT\_OUTLINED: IconData

buildPERCENT\_ROUNDEDinstance-attribute□

PERCENT\_ROUNDED: IconData

buildPERCENT\_SHARPinstance-attribute

PERCENT\_SHARP: IconData

buildPERM\_CAMERA\_MICinstance-attribute

PERM\_CAMERA\_MIC: IconData

buildPERM\_CAMERA\_MIC\_OUTLINEDinstance-attribute

PERM\_CAMERA\_MIC\_OUTLINED: IconData

buildPERM\_CAMERA\_MIC\_ROUNDEDinstance-attribute

PERM\_CAMERA\_MIC\_ROUNDED: IconData

buildPERM\_CAMERA\_MIC\_SHARPinstance-attribute

PERM\_CAMERA\_MIC\_SHARP: IconData

buildPERM\_CONTACT\_CALinstance-attribute

PERM\_CONTACT\_CAL: IconData

buildPERM\_CONTACT\_CAL\_OUTLINEDinstance-attribute

PERM\_CONTACT\_CAL\_OUTLINED: IconData

buildPERM\_CONTACT\_CAL\_ROUNDEDinstance-attribute

PERM\_CONTACT\_CAL\_ROUNDED: IconData

buildPERM\_CONTACT\_CAL\_SHARPinstance-attribute

PERM\_CONTACT\_CAL\_SHARP: IconData

buildPERM\_CONTACT\_CALENDARinstance-attribute

PERM\_CONTACT\_CALENDAR: IconData

buildPERM\_CONTACT\_CALENDAR\_OUTLINEDinstance-attribute

PERM\_CONTACT\_CALENDAR\_OUTLINED: IconData

buildPERM\_CONTACT\_CALENDAR\_ROUNDEDinstance-attribute

PERM\_CONTACT\_CALENDAR\_ROUNDED: IconData

buildPERM\_CONTACT\_CALENDAR\_SHARPinstance-attribute

PERM\_CONTACT\_CALENDAR\_SHARP: IconData

buildPERM\_DATA\_SETTINGinstance-attribute

PERM\_DATA\_SETTING: IconData

buildPERM\_DATA\_SETTING\_OUTLINEDinstance-attribute

PERM\_DATA\_SETTING\_OUTLINED: IconData

buildPERM\_DATA\_SETTING\_ROUNDEDinstance-attribute

PERM\_DATA\_SETTING\_ROUNDED: IconData

buildPERM\_DATA\_SETTING\_SHARPinstance-attribute

PERM\_DATA\_SETTING\_SHARP: IconData

buildPERM\_DEVICE\_INFOinstance-attribute

PERM\_DEVICE\_INFO: IconData

buildPERM\_DEVICE\_INFO\_OUTLINEDinstance-attribute

PERM\_DEVICE\_INFO\_OUTLINED: IconData

buildPERM\_DEVICE\_INFO\_ROUNDEDinstance-attribute

PERM\_DEVICE\_INFO\_ROUNDED: IconData

**buildPERM\_DEVICE\_INFO\_SHARPinstance-attribute**

PERM\_DEVICE\_INFO\_SHARP: IconData

**buildPERM\_DEVICE\_INFORMATIONinstance-attribute**

PERM\_DEVICE\_INFORMATION: IconData

**buildPERM\_DEVICE\_INFORMATION\_OUTLINEDinstance-attribute**

PERM\_DEVICE\_INFORMATION\_OUTLINED: IconData

**buildPERM\_DEVICE\_INFORMATION\_ROUNDEDinstance-attribute**

PERM\_DEVICE\_INFORMATION\_ROUNDED: IconData

**buildPERM\_DEVICE\_INFORMATION\_SHARPinstance-attribute**

PERM\_DEVICE\_INFORMATION\_SHARP: IconData

**buildPERM\_IDENTITYinstance-attribute**

PERM\_IDENTITY: IconData

**buildPERM\_IDENTITY\_OUTLINEDinstance-attribute**

PERM\_IDENTITY\_OUTLINED: IconData

**buildPERM\_IDENTITY\_ROUNDEDinstance-attribute**

PERM\_IDENTITY\_ROUNDED: IconData

**buildPERM\_IDENTITY\_SHARPinstance-attribute**

PERM\_IDENTITY\_SHARP: IconData

**buildPERM\_MEDIAinstance-attribute**

PERM\_MEDIA: IconData

**buildPERM\_MEDIA\_OUTLINEDinstance-attribute**

PERM\_MEDIA\_OUTLINED: IconData

**buildPERM\_MEDIA\_ROUNDEDinstance-attribute**

PERM\_MEDIA\_ROUNDED: IconData

**buildPERM\_MEDIA\_SHARPinstance-attribute**

PERM\_MEDIA\_SHARP: IconData

**buildPERM\_PHONE\_MSGinstance-attribute**

PERM\_PHONE\_MSG: IconData

**buildPERM\_PHONE\_MSG\_OUTLINEDinstance-attribute**

PERM\_PHONE\_MSG\_OUTLINED: IconData

**buildPERM\_PHONE\_MSG\_ROUNDEDinstance-attribute**

PERM\_PHONE\_MSG\_ROUNDED: IconData

**buildPERM\_PHONE\_MSG\_SHARPinstance-attribute**

PERM\_PHONE\_MSG\_SHARP: IconData

**buildPERM\_SCAN\_WIFIinstance-attribute**

PERM\_SCAN\_WIFI: IconData

**buildPERM\_SCAN\_WIFI\_OUTLINEDinstance-attribute**

PERM\_SCAN\_WIFI\_OUTLINED: IconData

**buildPERM\_SCAN\_WIFI\_ROUNDEDinstance-attribute**

PERM\_SCAN\_WIFI\_ROUNDED: IconData

**buildPERM\_SCAN\_WIFI\_SHARPinstance-attribute**

PERM\_SCAN\_WIFI\_SHARP: IconData

**buildPERSONinstance-attribute**

PERSON: IconData

**buildPERSON\_2instance-attribute**

PERSON\_2: IconData

**buildPERSON\_2\_OUTLINEDinstance-attribute**

PERSON\_2\_OUTLINED: IconData

**buildPERSON\_2\_ROUNDEDinstance-attribute**

PERSON\_2\_ROUNDED: IconData

**buildPERSON\_2\_SHARPinstance-attribute**

PERSON\_2\_SHARP: IconData

**buildPERSON\_3instance-attribute**

PERSON\_3: IconData

**buildPERSON\_3\_OUTLINEDinstance-attribute**

PERSON\_3\_OUTLINED: IconData

**buildPERSON\_3\_ROUNDEDinstance-attribute**

PERSON\_3\_ROUNDED: IconData

**buildPERSON\_3\_SHARPinstance-attribute**

PERSON\_3\_SHARP: IconData

**buildPERSON\_4instance-attribute**

PERSON\_4: IconData

**buildPERSON\_4\_OUTLINEDinstance-attribute**

PERSON\_4\_OUTLINED: IconData

**buildPERSON\_4\_ROUNDEDinstance-attribute**

PERSON\_4\_ROUNDED: IconData

**buildPERSON\_4\_SHARPinstance-attribute**

PERSON\_4\_SHARP: IconData

**buildPERSON\_ADDinstance-attribute**

PERSON\_ADD: IconData

**buildPERSON\_ADD\_ALTinstance-attribute**

PERSON\_ADD\_ALT: IconData

**buildPERSON\_ADD\_ALT\_1instance-attribute**

PERSON\_ADD\_ALT\_1: IconData

**buildPERSON\_ADD\_ALT\_1\_OUTLINEDinstance-attribute**

PERSON\_ADD\_ALT\_1\_OUTLINED: IconData

**buildPERSON\_ADD\_ALT\_1\_ROUNDEDinstance-attribute**

PERSON\_ADD\_ALT\_1\_ROUNDED: IconData

**buildPERSON\_ADD\_ALT\_1\_SHARPinstance-attribute**

PERSON\_ADD\_ALT\_1\_SHARP: IconData

**buildPERSON\_ADD\_ALT\_OUTLINEDinstance-attribute**

PERSON\_ADD\_ALT\_OUTLINED: IconData

**buildPERSON\_ADD\_ALT\_ROUNDEDinstance-attribute**

PERSON\_ADD\_ALT\_ROUNDED: IconData

**buildPERSON\_ADD\_ALT\_SHARPinstance-attribute**

PERSON\_ADD\_ALT\_SHARP: IconData

**buildPERSON\_ADD\_DISABLEDinstance-attribute**

PERSON\_ADD\_DISABLED: IconData

buildPERSON\_ADD\_DISABLED\_OUTLINEDinstance-attribute

PERSON\_ADD\_DISABLED\_OUTLINED: IconData

buildPERSON\_ADD\_DISABLED\_ROUNDEDinstance-attribute

PERSON\_ADD\_DISABLED\_ROUNDED: IconData

buildPERSON\_ADD\_DISABLED\_SHARPinstance-attribute

PERSON\_ADD\_DISABLED\_SHARP: IconData

buildPERSON\_ADD\_OUTLINEDinstance-attribute

PERSON\_ADD\_OUTLINED: IconData

buildPERSON\_ADD\_ROUNDEDinstance-attribute

PERSON\_ADD\_ROUNDED: IconData

buildPERSON\_ADD\_SHARPinstance-attribute

PERSON\_ADD\_SHARP: IconData

buildPERSON\_OFFinstance-attribute

PERSON\_OFF: IconData

buildPERSON\_OFF\_OUTLINEDinstance-attribute

PERSON\_OFF\_OUTLINED: IconData

buildPERSON\_OFF\_ROUNDEDinstance-attribute

PERSON\_OFF\_ROUNDED: IconData

buildPERSON\_OFF\_SHARPinstance-attribute

PERSON\_OFF\_SHARP: IconData

buildPERSON\_OUTLINEinstance-attribute

PERSON\_OUTLINE: IconData

buildPERSON\_OUTLINE\_OUTLINEDinstance-attribute

PERSON\_OUTLINE\_OUTLINED: IconData

buildPERSON\_OUTLINE\_ROUNDEDinstance-attribute

PERSON\_OUTLINE\_ROUNDED: IconData

buildPERSON\_OUTLINE\_SHARPinstance-attribute

PERSON\_OUTLINE\_SHARP: IconData

buildPERSON\_OUTLINEDinstance-attribute

PERSON\_OUTLINED: IconData

buildPERSON\_PINinstance-attribute

PERSON\_PIN: IconData

buildPERSON\_PIN\_CIRCLEinstance-attribute

PERSON\_PIN\_CIRCLE: IconData

buildPERSON\_PIN\_CIRCLE\_OUTLINEDinstance-attribute

PERSON\_PIN\_CIRCLE\_OUTLINED: IconData

buildPERSON\_PIN\_CIRCLE\_ROUNDEDinstance-attribute

PERSON\_PIN\_CIRCLE\_ROUNDED: IconData

buildPERSON\_PIN\_CIRCLE\_SHARPinstance-attribute

PERSON\_PIN\_CIRCLE\_SHARP: IconData

buildPERSON\_PIN\_OUTLINEDinstance-attribute

PERSON\_PIN\_OUTLINED: IconData

buildPERSON\_PIN\_ROUNDEDinstance-attribute

PERSON\_PIN\_ROUNDED: IconData

**buildPERSON\_PIN\_SHARPinstance-attribute**

PERSON\_PIN\_SHARP: IconData

**buildPERSON\_REMOVEinstance-attribute**

PERSON\_REMOVE: IconData

**buildPERSON\_REMOVE\_ALT\_1instance-attribute**

PERSON\_REMOVE\_ALT\_1: IconData

**buildPERSON\_REMOVE\_ALT\_1\_OUTLINEDinstance-attribute**

PERSON\_REMOVE\_ALT\_1\_OUTLINED: IconData

**buildPERSON\_REMOVE\_ALT\_1\_ROUNDEDinstance-attribute**

PERSON\_REMOVE\_ALT\_1\_ROUNDED: IconData

**buildPERSON\_REMOVE\_ALT\_1\_SHARPinstance-attribute**

PERSON\_REMOVE\_ALT\_1\_SHARP: IconData

**buildPERSON\_REMOVE\_OUTLINEDinstance-attribute**

PERSON\_REMOVE\_OUTLINED: IconData

**buildPERSON\_REMOVE\_ROUNDEDinstance-attribute**

PERSON\_REMOVE\_ROUNDED: IconData

**buildPERSON\_REMOVE\_SHARPinstance-attribute**

PERSON\_REMOVE\_SHARP: IconData

**buildPERSON\_ROUNDEDinstance-attribute**

PERSON\_ROUNDED: IconData

**buildPERSON\_SEARCHinstance-attribute**

PERSON\_SEARCH: IconData

**buildPERSON\_SEARCH\_OUTLINEDinstance-attribute**

PERSON\_SEARCH\_OUTLINED: IconData

**buildPERSON\_SEARCH\_ROUNDEDinstance-attribute**

PERSON\_SEARCH\_ROUNDED: IconData

**buildPERSON\_SEARCH\_SHARPinstance-attribute**

PERSON\_SEARCH\_SHARP: IconData

**buildPERSON\_SHARPinstance-attribute**

PERSON\_SHARP: IconData

**buildPERSONAL\_INJURYinstance-attribute**

PERSONAL\_INJURY: IconData

**buildPERSONAL\_INJURY\_OUTLINEDinstance-attribute**

PERSONAL\_INJURY\_OUTLINED: IconData

**buildPERSONAL\_INJURY\_ROUNDEDinstance-attribute**

PERSONAL\_INJURY\_ROUNDED: IconData

**buildPERSONAL\_INJURY\_SHARPinstance-attribute**

PERSONAL\_INJURY\_SHARP: IconData

**buildPERSONAL\_VIDEOinstance-attribute**

PERSONAL\_VIDEO: IconData

**buildPERSONAL\_VIDEO\_OUTLINEDinstance-attribute**

PERSONAL\_VIDEO\_OUTLINED: IconData

**buildPERSONAL\_VIDEO\_ROUNDEDinstance-attribute**

PERSONAL\_VIDEO\_ROUNDED: IconData



**buildPERSONAL\_VIDEO\_SHARPinstance-attribute**

PERSONAL\_VIDEO\_SHARP: IconData

**buildPEST\_CONTROLinstance-attribute**

PEST\_CONTROL: IconData

**buildPEST\_CONTROL\_OUTLINEDinstance-attribute**

PEST\_CONTROL\_OUTLINED: IconData

**buildPEST\_CONTROL\_RODENTinstance-attribute**

PEST\_CONTROL\_RODENT: IconData

**buildPEST\_CONTROL\_RODENT\_OUTLINEDinstance-attribute**

PEST\_CONTROL\_RODENT\_OUTLINED: IconData

**buildPEST\_CONTROL\_RODENT\_ROUNDEDinstance-attribute**

PEST\_CONTROL\_RODENT\_ROUNDED: IconData

**buildPEST\_CONTROL\_RODENT\_SHARPinstance-attribute**

PEST\_CONTROL\_RODENT\_SHARP: IconData

**buildPEST\_CONTROL\_ROUNDEDinstance-attribute**

PEST\_CONTROL\_ROUNDED: IconData

**buildPEST\_CONTROL\_SHARPinstance-attribute**

PEST\_CONTROL\_SHARP: IconData

**buildPETSinstance-attribute**

PETS: IconData

**buildPETS\_OUTLINEDinstance-attribute**

PETS\_OUTLINED: IconData

**buildPETS\_ROUNDEDinstance-attribute**

PETS\_ROUNDED: IconData

**buildPETS\_SHARPinstance-attribute**

PETS\_SHARP: IconData

**buildPHISHINGinstance-attribute**

PHISHING: IconData

**buildPHISHING\_OUTLINEDinstance-attribute**

PHISHING\_OUTLINED: IconData

**buildPHISHING\_ROUNDEDinstance-attribute**

PHISHING\_ROUNDED: IconData

**buildPHISHING\_SHARPinstance-attribute**

PHISHING\_SHARP: IconData

**buildPHONEinstance-attribute**

PHONE: IconData

**buildPHONE\_ANDROIDinstance-attribute**

PHONE\_ANDROID: IconData

**buildPHONE\_ANDROID\_OUTLINEDinstance-attribute**

PHONE\_ANDROID\_OUTLINED: IconData

**buildPHONE\_ANDROID\_ROUNDEDinstance-attribute**

PHONE\_ANDROID\_ROUNDED: IconData

**buildPHONE\_ANDROID\_SHARPinstance-attribute**

PHONE\_ANDROID\_SHARP: IconData

buildPHONE\_BLUETOOTH\_SPEAKERinstance-attribute

PHONE\_BLUETOOTH\_SPEAKER: IconData

buildPHONE\_BLUETOOTH\_SPEAKER\_OUTLINEDinstance-attribute

PHONE\_BLUETOOTH\_SPEAKER\_OUTLINED: IconData

buildPHONE\_BLUETOOTH\_SPEAKER\_ROUNDEDinstance-attribute

PHONE\_BLUETOOTH\_SPEAKER\_ROUNDED: IconData

buildPHONE\_BLUETOOTH\_SPEAKER\_SHARPinstance-attribute

PHONE\_BLUETOOTH\_SPEAKER\_SHARP: IconData

buildPHONE\_CALLBACKinstance-attribute

PHONE\_CALLBACK: IconData

buildPHONE\_CALLBACK\_OUTLINEDinstance-attribute

PHONE\_CALLBACK\_OUTLINED: IconData

buildPHONE\_CALLBACK\_ROUNDEDinstance-attribute

PHONE\_CALLBACK\_ROUNDED: IconData

buildPHONE\_CALLBACK\_SHARPinstance-attribute

PHONE\_CALLBACK\_SHARP: IconData

buildPHONE\_DISABLEDinstance-attribute

PHONE\_DISABLED: IconData

buildPHONE\_DISABLED\_OUTLINEDinstance-attribute

PHONE\_DISABLED\_OUTLINED: IconData

buildPHONE\_DISABLED\_ROUNDEDinstance-attribute

PHONE\_DISABLED\_ROUNDED: IconData

buildPHONE\_DISABLED\_SHARPinstance-attribute

PHONE\_DISABLED\_SHARP: IconData

buildPHONE\_ENABLEDinstance-attribute

PHONE\_ENABLED: IconData

buildPHONE\_ENABLED\_OUTLINEDinstance-attribute

PHONE\_ENABLED\_OUTLINED: IconData

buildPHONE\_ENABLED\_ROUNDEDinstance-attribute

PHONE\_ENABLED\_ROUNDED: IconData

buildPHONE\_ENABLED\_SHARPinstance-attribute

PHONE\_ENABLED\_SHARP: IconData

buildPHONE\_FORWARDEDinstance-attribute

PHONE\_FORWARDED: IconData

buildPHONE\_FORWARDED\_OUTLINEDinstance-attribute

PHONE\_FORWARDED\_OUTLINED: IconData

buildPHONE\_FORWARDED\_ROUNDEDinstance-attribute

PHONE\_FORWARDED\_ROUNDED: IconData

buildPHONE\_FORWARDED\_SHARPinstance-attribute

PHONE\_FORWARDED\_SHARP: IconData

buildPHONE\_IN\_TALKinstance-attribute

PHONE\_IN\_TALK: IconData

buildPHONE\_IN\_TALK\_OUTLINEDinstance-attribute

PHONE\_IN\_TALK\_OUTLINED: IconData

**buildPHONE\_IN\_TALK\_ROUNDEDinstance-attribute**

PHONE\_IN\_TALK\_ROUNDED: IconData

**buildPHONE\_IN\_TALK\_SHARPinstance-attribute**

PHONE\_IN\_TALK\_SHARP: IconData

**buildPHONE\_IPHONEinstance-attribute**

PHONE\_IPHONE: IconData

**buildPHONE\_IPHONE\_OUTLINEDinstance-attribute**

PHONE\_IPHONE\_OUTLINED: IconData

**buildPHONE\_IPHONE\_ROUNDEDinstance-attribute**

PHONE\_IPHONE\_ROUNDED: IconData

**buildPHONE\_IPHONE\_SHARPinstance-attribute**

PHONE\_IPHONE\_SHARP: IconData

**buildPHONE\_LOCKEDinstance-attribute**

PHONE\_LOCKED: IconData

**buildPHONE\_LOCKED\_OUTLINEDinstance-attribute**

PHONE\_LOCKED\_OUTLINED: IconData

**buildPHONE\_LOCKED\_ROUNDEDinstance-attribute**

PHONE\_LOCKED\_ROUNDED: IconData

**buildPHONE\_LOCKED\_SHARPinstance-attribute**

PHONE\_LOCKED\_SHARP: IconData

**buildPHONE\_MISSEDinstance-attribute**

PHONE\_MISSED: IconData

**buildPHONE\_MISSED\_OUTLINEDinstance-attribute**

PHONE\_MISSED\_OUTLINED: IconData

**buildPHONE\_MISSED\_ROUNDEDinstance-attribute**

PHONE\_MISSED\_ROUNDED: IconData

**buildPHONE\_MISSED\_SHARPinstance-attribute**

PHONE\_MISSED\_SHARP: IconData

**buildPHONE\_OUTLINEDinstance-attribute**

PHONE\_OUTLINED: IconData

**buildPHONE\_PAUSEDinstance-attribute**

PHONE\_PAUSED: IconData

**buildPHONE\_PAUSED\_OUTLINEDinstance-attribute**

PHONE\_PAUSED\_OUTLINED: IconData

**buildPHONE\_PAUSED\_ROUNDEDinstance-attribute**

PHONE\_PAUSED\_ROUNDED: IconData

**buildPHONE\_PAUSED\_SHARPinstance-attribute**

PHONE\_PAUSED\_SHARP: IconData

**buildPHONE\_ROUNDEDinstance-attribute**

PHONE\_ROUNDED: IconData

**buildPHONE\_SHARPinstance-attribute**

PHONE\_SHARP: IconData

**buildPHONELINKinstance-attribute**

PHONELINK: IconData

buildPHONELINK\_ERASEinstance-attribute

PHONELINK\_ERASE: IconData

buildPHONELINK\_ERASE\_OUTLINEDinstance-attribute

PHONELINK\_ERASE\_OUTLINED: IconData

buildPHONELINK\_ERASE\_ROUNDEDinstance-attribute

PHONELINK\_ERASE\_ROUNDED: IconData

buildPHONELINK\_ERASE\_SHARPinstance-attribute

PHONELINK\_ERASE\_SHARP: IconData

buildPHONELINK\_LOCKinstance-attribute

PHONELINK\_LOCK: IconData

buildPHONELINK\_LOCK\_OUTLINEDinstance-attribute

PHONELINK\_LOCK\_OUTLINED: IconData

buildPHONELINK\_LOCK\_ROUNDEDinstance-attribute

PHONELINK\_LOCK\_ROUNDED: IconData

buildPHONELINK\_LOCK\_SHARPinstance-attribute

PHONELINK\_LOCK\_SHARP: IconData

buildPHONELINK\_OFFinstance-attribute

PHONELINK\_OFF: IconData

buildPHONELINK\_OFF\_OUTLINEDinstance-attribute

PHONELINK\_OFF\_OUTLINED: IconData

buildPHONELINK\_OFF\_ROUNDEDinstance-attribute

PHONELINK\_OFF\_ROUNDED: IconData

buildPHONELINK\_OFF\_SHARPinstance-attribute

PHONELINK\_OFF\_SHARP: IconData

buildPHONELINK\_OUTLINEDinstance-attribute

PHONELINK\_OUTLINED: IconData

buildPHONELINK\_RINGinstance-attribute

PHONELINK\_RING: IconData

buildPHONELINK\_RING\_OUTLINEDinstance-attribute

PHONELINK\_RING\_OUTLINED: IconData

buildPHONELINK\_RING\_ROUNDEDinstance-attribute

PHONELINK\_RING\_ROUNDED: IconData

buildPHONELINK\_RING\_SHARPinstance-attribute

PHONELINK\_RING\_SHARP: IconData

buildPHONELINK\_ROUNDEDinstance-attribute

PHONELINK\_ROUNDED: IconData

buildPHONELINK\_SETUPinstance-attribute

PHONELINK\_SETUP: IconData

buildPHONELINK\_SETUP\_OUTLINEDinstance-attribute

PHONELINK\_SETUP\_OUTLINED: IconData

buildPHONELINK\_SETUP\_ROUNDEDinstance-attribute

PHONELINK\_SETUP\_ROUNDED: IconData

buildPHONELINK\_SETUP\_SHARPinstance-attribute

PHONELINK\_SETUP\_SHARP: IconData

**buildPHONELINK\_SHARPinstance-attribute**

PHONELINK\_SHARP: IconData

**buildPHOTOinstance-attribute**

PHOTO: IconData

**buildPHOTO\_ALBUMinstance-attribute**

PHOTO\_ALBUM: IconData

**buildPHOTO\_ALBUM\_OUTLINEDinstance-attribute**

PHOTO\_ALBUM\_OUTLINED: IconData

**buildPHOTO\_ALBUM\_ROUNDEDinstance-attribute**

PHOTO\_ALBUM\_ROUNDED: IconData

**buildPHOTO\_ALBUM\_SHARPinstance-attribute**

PHOTO\_ALBUM\_SHARP: IconData

**buildPHOTO\_CAMERAinstance-attribute**

PHOTO\_CAMERA: IconData

**buildPHOTO\_CAMERA\_BACKinstance-attribute**

PHOTO\_CAMERA\_BACK: IconData

**buildPHOTO\_CAMERA\_BACK\_OUTLINEDinstance-attribute**

PHOTO\_CAMERA\_BACK\_OUTLINED: IconData

**buildPHOTO\_CAMERA\_BACK\_ROUNDEDinstance-attribute**

PHOTO\_CAMERA\_BACK\_ROUNDED: IconData

**buildPHOTO\_CAMERA\_BACK\_SHARPinstance-attribute**

PHOTO\_CAMERA\_BACK\_SHARP: IconData

**buildPHOTO\_CAMERA\_FRONTinstance-attribute**

PHOTO\_CAMERA\_FRONT: IconData

**buildPHOTO\_CAMERA\_FRONT\_OUTLINEDinstance-attribute**

PHOTO\_CAMERA\_FRONT\_OUTLINED: IconData

**buildPHOTO\_CAMERA\_FRONT\_ROUNDEDinstance-attribute**

PHOTO\_CAMERA\_FRONT\_ROUNDED: IconData

**buildPHOTO\_CAMERA\_FRONT\_SHARPinstance-attribute**

PHOTO\_CAMERA\_FRONT\_SHARP: IconData

**buildPHOTO\_CAMERA\_OUTLINEDinstance-attribute**

PHOTO\_CAMERA\_OUTLINED: IconData

**buildPHOTO\_CAMERA\_ROUNDEDinstance-attribute**

PHOTO\_CAMERA\_ROUNDED: IconData

**buildPHOTO\_CAMERA\_SHARPinstance-attribute**

PHOTO\_CAMERA\_SHARP: IconData

**buildPHOTO\_FILTERinstance-attribute**

PHOTO\_FILTER: IconData

**buildPHOTO\_FILTER\_OUTLINEDinstance-attribute**

PHOTO\_FILTER\_OUTLINED: IconData

**buildPHOTO\_FILTER\_ROUNDEDinstance-attribute**

PHOTO\_FILTER\_ROUNDED: IconData

**buildPHOTO\_FILTER\_SHARPinstance-attribute**

PHOTO\_FILTER\_SHARP: IconData

**buildPHOTO\_LIBRARYinstance-attribute**

PHOTO\_LIBRARY: IconData

**buildPHOTO\_LIBRARY\_OUTLINEDinstance-attribute**

PHOTO\_LIBRARY\_OUTLINED: IconData

**buildPHOTO\_LIBRARY\_ROUNDEDinstance-attribute**

PHOTO\_LIBRARY\_ROUNDED: IconData

**buildPHOTO\_LIBRARY\_SHARPinstance-attribute**

PHOTO\_LIBRARY\_SHARP: IconData

**buildPHOTO\_OUTLINEDinstance-attribute**

PHOTO\_OUTLINED: IconData

**buildPHOTO\_ROUNDEDinstance-attribute**

PHOTO\_ROUNDED: IconData

**buildPHOTO\_SHARPinstance-attribute**

PHOTO\_SHARP: IconData

**buildPHOTO\_SIZE\_SELECT\_ACTUALinstance-attribute**

PHOTO\_SIZE\_SELECT\_ACTUAL: IconData

**buildPHOTO\_SIZE\_SELECT\_ACTUAL\_OUTLINEDinstance-attribute**

PHOTO\_SIZE\_SELECT\_ACTUAL\_OUTLINED: IconData

**buildPHOTO\_SIZE\_SELECT\_ACTUAL\_ROUNDEDinstance-attribute**

PHOTO\_SIZE\_SELECT\_ACTUAL\_ROUNDED: IconData

**buildPHOTO\_SIZE\_SELECT\_ACTUAL\_SHARPinstance-attribute**

PHOTO\_SIZE\_SELECT\_ACTUAL\_SHARP: IconData

**buildPHOTO\_SIZE\_SELECT\_LARGEinstance-attribute**

PHOTO\_SIZE\_SELECT\_LARGE: IconData

**buildPHOTO\_SIZE\_SELECT\_LARGE\_OUTLINEDinstance-attribute**

PHOTO\_SIZE\_SELECT\_LARGE\_OUTLINED: IconData

**buildPHOTO\_SIZE\_SELECT\_LARGE\_ROUNDEDinstance-attribute**

PHOTO\_SIZE\_SELECT\_LARGE\_ROUNDED: IconData

**buildPHOTO\_SIZE\_SELECT\_LARGE\_SHARPinstance-attribute**

PHOTO\_SIZE\_SELECT\_LARGE\_SHARP: IconData

**buildPHOTO\_SIZE\_SELECT\_SMALLinstance-attribute**

PHOTO\_SIZE\_SELECT\_SMALL: IconData

**buildPHOTO\_SIZE\_SELECT\_SMALL\_OUTLINEDinstance-attribute**

PHOTO\_SIZE\_SELECT\_SMALL\_OUTLINED: IconData

**buildPHOTO\_SIZE\_SELECT\_SMALL\_ROUNDEDinstance-attribute**

PHOTO\_SIZE\_SELECT\_SMALL\_ROUNDED: IconData

**buildPHOTO\_SIZE\_SELECT\_SMALL\_SHARPinstance-attribute**

PHOTO\_SIZE\_SELECT\_SMALL\_SHARP: IconData

**buildPHPinstance-attribute**

PHP: IconData

**buildPHP\_OUTLINEDinstance-attribute**

PHP\_OUTLINED: IconData

**buildPHP\_ROUNDEDinstance-attribute**

PHP\_ROUNDED: IconData

**buildPHP\_SHARPinstance-attribute**

PHP\_SHARP: IconData

**buildPIANOinstance-attribute**

PIANO: IconData

**buildPIANO\_OFFinstance-attribute**

PIANO\_OFF: IconData

**buildPIANO\_OFF\_OUTLINEDinstance-attribute**

PIANO\_OFF\_OUTLINED: IconData

**buildPIANO\_OFF\_ROUNDEDinstance-attribute**

PIANO\_OFF\_ROUNDED: IconData

**buildPIANO\_OFF\_SHARPinstance-attribute**

PIANO\_OFF\_SHARP: IconData

**buildPIANO\_OUTLINEDinstance-attribute**

PIANO\_OUTLINED: IconData

**buildPIANO\_ROUNDEDinstance-attribute**

PIANO\_ROUNDED: IconData

**buildPIANO\_SHARPinstance-attribute**

PIANO\_SHARP: IconData

**buildPICTURE\_AS\_PDFinstance-attribute**

PICTURE\_AS\_PDF: IconData

**buildPICTURE\_AS\_PDF\_OUTLINEDinstance-attribute**

PICTURE\_AS\_PDF\_OUTLINED: IconData

**buildPICTURE\_AS\_PDF\_ROUNDEDinstance-attribute**

PICTURE\_AS\_PDF\_ROUNDED: IconData

**buildPICTURE\_AS\_PDF\_SHARPinstance-attribute**

PICTURE\_AS\_PDF\_SHARP: IconData

**buildPICTURE\_IN\_PICTUREinstance-attribute**

PICTURE\_IN\_PICTURE: IconData

**buildPICTURE\_IN\_PICTURE\_ALTinstance-attribute**

PICTURE\_IN\_PICTURE\_ALT: IconData

**buildPICTURE\_IN\_PICTURE\_ALT\_OUTLINEDinstance-attribute**

PICTURE\_IN\_PICTURE\_ALT\_OUTLINED: IconData

**buildPICTURE\_IN\_PICTURE\_ALT\_ROUNDEDinstance-attribute**

PICTURE\_IN\_PICTURE\_ALT\_ROUNDED: IconData

**buildPICTURE\_IN\_PICTURE\_ALT\_SHARPinstance-attribute**

PICTURE\_IN\_PICTURE\_ALT\_SHARP: IconData

**buildPICTURE\_IN\_PICTURE\_OUTLINEDinstance-attribute**

PICTURE\_IN\_PICTURE\_OUTLINED: IconData

**buildPICTURE\_IN\_PICTURE\_ROUNDEDinstance-attribute**

PICTURE\_IN\_PICTURE\_ROUNDED: IconData

**buildPICTURE\_IN\_PICTURE\_SHARPinstance-attribute**

PICTURE\_IN\_PICTURE\_SHARP: IconData

**buildPIE\_CHARTinstance-attribute**

PIE\_CHART: IconData

buildPIE\_CHART\_OUTLINEinstance-attribute

PIE\_CHART\_OUTLINE: IconData

buildPIE\_CHART\_OUTLINE\_OUTLINEDinstance-attribute

PIE\_CHART\_OUTLINE\_OUTLINED: IconData

buildPIE\_CHART\_OUTLINE\_ROUNDEDinstance-attribute

PIE\_CHART\_OUTLINE\_ROUNDED: IconData

buildPIE\_CHART\_OUTLINE\_SHARPinstance-attribute

PIE\_CHART\_OUTLINE\_SHARP: IconData

buildPIE\_CHART\_ROUNDEDinstance-attribute

PIE\_CHART\_ROUNDED: IconData

buildPIE\_CHART\_SHARPinstance-attribute

PIE\_CHART\_SHARP: IconData

buildPINinstance-attribute

PIN: IconData

buildPIN\_DROPinstance-attribute

PIN\_DROP: IconData

buildPIN\_DROP\_OUTLINEDinstance-attribute

PIN\_DROP\_OUTLINED: IconData

buildPIN\_DROP\_ROUNDEDinstance-attribute

PIN\_DROP\_ROUNDED: IconData

buildPIN\_DROP\_SHARPinstance-attribute

PIN\_DROP\_SHARP: IconData

buildPIN\_ENDinstance-attribute

PIN\_END: IconData

buildPIN\_END\_OUTLINEDinstance-attribute

PIN\_END\_OUTLINED: IconData

buildPIN\_END\_ROUNDEDinstance-attribute

PIN\_END\_ROUNDED: IconData

buildPIN\_END\_SHARPinstance-attribute

PIN\_END\_SHARP: IconData

buildPIN\_INVOKEinstance-attribute

PIN\_INVOKE: IconData

buildPIN\_INVOKE\_OUTLINEDinstance-attribute

PIN\_INVOKE\_OUTLINED: IconData

buildPIN\_INVOKE\_ROUNDEDinstance-attribute

PIN\_INVOKE\_ROUNDED: IconData

buildPIN\_INVOKE\_SHARPinstance-attribute

PIN\_INVOKE\_SHARP: IconData

buildPIN\_OUTLINEDinstance-attribute

PIN\_OUTLINED: IconData

buildPIN\_ROUNDEDinstance-attribute

PIN\_ROUNDED: IconData

buildPIN\_SHARPinstance-attribute

PIN\_SHARP: IconData



**buildPINCHinstance-attribute**

PINCH: IconData

**buildPINCH\_OUTLINEDinstance-attribute**

PINCH\_OUTLINED: IconData

**buildPINCH\_ROUNDEDinstance-attribute**

PINCH\_ROUNDED: IconData

**buildPINCH\_SHARPinstance-attribute**

PINCH\_SHARP: IconData

**buildPIVOT\_TABLE\_CHARTinstance-attribute**

PIVOT\_TABLE\_CHART: IconData

**buildPIVOT\_TABLE\_CHART\_OUTLINEDinstance-attribute**

PIVOT\_TABLE\_CHART\_OUTLINED: IconData

**buildPIVOT\_TABLE\_CHART\_ROUNDEDinstance-attribute**

PIVOT\_TABLE\_CHART\_ROUNDED: IconData

**buildPIVOT\_TABLE\_CHART\_SHARPinstance-attribute**

PIVOT\_TABLE\_CHART\_SHARP: IconData

**buildPIXinstance-attribute**

PIX: IconData

**buildPIX\_OUTLINEDinstance-attribute**

PIX\_OUTLINED: IconData

**buildPIX\_ROUNDEDinstance-attribute**

PIX\_ROUNDED: IconData

**buildPIX\_SHARPinstance-attribute**

PIX\_SHARP: IconData

**buildPLACEinstance-attribute**

PLACE: IconData

**buildPLACE\_OUTLINEDinstance-attribute**

PLACE\_OUTLINED: IconData

**buildPLACE\_ROUNDEDinstance-attribute**

PLACE\_ROUNDED: IconData

**buildPLACE\_SHARPinstance-attribute**

PLACE\_SHARP: IconData

**buildPLAGIARISMinstance-attribute**

PLAGIARISM: IconData

**buildPLAGIARISM\_OUTLINEDinstance-attribute**

PLAGIARISM\_OUTLINED: IconData

**buildPLAGIARISM\_ROUNDEDinstance-attribute**

PLAGIARISM\_ROUNDED: IconData

**buildPLAGIARISM\_SHARPinstance-attribute**

PLAGIARISM\_SHARP: IconData

**buildPLAY\_ARROWinstance-attribute**

PLAY\_ARROW: IconData

**buildPLAY\_ARROW\_OUTLINEDinstance-attribute**

PLAY\_ARROW\_OUTLINED: IconData

**buildPLAY\_ARROW\_ROUNDEDinstance-attribute**

PLAY\_ARROW\_ROUNDED: IconData

**buildPLAY\_ARROW\_SHARPinstance-attribute**

PLAY\_ARROW\_SHARP: IconData

**buildPLAY\_CIRCLEinstance-attribute**

PLAY\_CIRCLE: IconData

**buildPLAY\_CIRCLE\_FILLinstance-attribute**

PLAY\_CIRCLE\_FILL: IconData

**buildPLAY\_CIRCLE\_FILL\_OUTLINEDinstance-attribute**

PLAY\_CIRCLE\_FILL\_OUTLINED: IconData

**buildPLAY\_CIRCLE\_FILL\_ROUNDEDinstance-attribute**

PLAY\_CIRCLE\_FILL\_ROUNDED: IconData

**buildPLAY\_CIRCLE\_FILL\_SHARPinstance-attribute**

PLAY\_CIRCLE\_FILL\_SHARP: IconData

**buildPLAY\_CIRCLE\_FILLEDinstance-attribute**

PLAY\_CIRCLE\_FILLED: IconData

**buildPLAY\_CIRCLE\_FILLED\_OUTLINEDinstance-attribute**

PLAY\_CIRCLE\_FILLED\_OUTLINED: IconData

**buildPLAY\_CIRCLE\_FILLED\_ROUNDEDinstance-attribute**

PLAY\_CIRCLE\_FILLED\_ROUNDED: IconData

**buildPLAY\_CIRCLE\_FILLED\_SHARPinstance-attribute**

PLAY\_CIRCLE\_FILLED\_SHARP: IconData

**buildPLAY\_CIRCLE\_OUTLINEinstance-attribute**

PLAY\_CIRCLE\_OUTLINE: IconData

**buildPLAY\_CIRCLE\_OUTLINE\_OUTLINEDinstance-attribute**

PLAY\_CIRCLE\_OUTLINE\_OUTLINED: IconData

**buildPLAY\_CIRCLE\_OUTLINE\_ROUNDEDinstance-attribute**

PLAY\_CIRCLE\_OUTLINE\_ROUNDED: IconData

**buildPLAY\_CIRCLE\_OUTLINE\_SHARPinstance-attribute**

PLAY\_CIRCLE\_OUTLINE\_SHARP: IconData

**buildPLAY\_CIRCLE\_OUTLINEDinstance-attribute**

PLAY\_CIRCLE\_OUTLINED: IconData

**buildPLAY\_CIRCLE\_ROUNDEDinstance-attribute**

PLAY\_CIRCLE\_ROUNDED: IconData

**buildPLAY\_CIRCLE\_SHARPinstance-attribute**

PLAY\_CIRCLE\_SHARP: IconData

**buildPLAY\_DISABLEDinstance-attribute**

PLAY\_DISABLED: IconData

**buildPLAY\_DISABLED\_OUTLINEDinstance-attribute**

PLAY\_DISABLED\_OUTLINED: IconData

**buildPLAY\_DISABLED\_ROUNDEDinstance-attribute**

PLAY\_DISABLED\_ROUNDED: IconData

**buildPLAY\_DISABLED\_SHARPinstance-attribute**

PLAY\_DISABLED\_SHARP: IconData

**buildPLAY\_FOR\_WORKinstance-attribute**

PLAY\_FOR\_WORK: IconData

**buildPLAY\_FOR\_WORK\_OUTLINEDinstance-attribute**

PLAY\_FOR\_WORK\_OUTLINED: IconData

**buildPLAY\_FOR\_WORK\_ROUNDEDinstance-attribute**

PLAY\_FOR\_WORK\_ROUNDED: IconData

**buildPLAY\_FOR\_WORK\_SHARPinstance-attribute**

PLAY\_FOR\_WORK\_SHARP: IconData

**buildPLAY\_LESSONinstance-attribute**

PLAY\_LESSON: IconData

**buildPLAY\_LESSON\_OUTLINEDinstance-attribute**

PLAY\_LESSON\_OUTLINED: IconData

**buildPLAY\_LESSON\_ROUNDEDinstance-attribute**

PLAY\_LESSON\_ROUNDED: IconData

**buildPLAY\_LESSON\_SHARPinstance-attribute**

PLAY\_LESSON\_SHARP: IconData

**buildPLAYLIST\_ADDinstance-attribute**

PLAYLIST\_ADD: IconData

**buildPLAYLIST\_ADD\_CHECKinstance-attribute**

PLAYLIST\_ADD\_CHECK: IconData

**buildPLAYLIST\_ADD\_CHECK\_CIRCLEinstance-attribute**

PLAYLIST\_ADD\_CHECK\_CIRCLE: IconData

**buildPLAYLIST\_ADD\_CHECK\_CIRCLE\_OUTLINEDinstance-attribute**

PLAYLIST\_ADD\_CHECK\_CIRCLE\_OUTLINED: IconData

**buildPLAYLIST\_ADD\_CHECK\_CIRCLE\_ROUNDEDinstance-attribute**

PLAYLIST\_ADD\_CHECK\_CIRCLE\_ROUNDED: IconData

**buildPLAYLIST\_ADD\_CHECK\_CIRCLE\_SHARPinstance-attribute**

PLAYLIST\_ADD\_CHECK\_CIRCLE\_SHARP: IconData

**buildPLAYLIST\_ADD\_CHECK\_OUTLINEDinstance-attribute**

PLAYLIST\_ADD\_CHECK\_OUTLINED: IconData

**buildPLAYLIST\_ADD\_CHECK\_ROUNDEDinstance-attribute**

PLAYLIST\_ADD\_CHECK\_ROUNDED: IconData

**buildPLAYLIST\_ADD\_CHECK\_SHARPinstance-attribute**

PLAYLIST\_ADD\_CHECK\_SHARP: IconData

**buildPLAYLIST\_ADD\_CIRCLEinstance-attribute**

PLAYLIST\_ADD\_CIRCLE: IconData

**buildPLAYLIST\_ADD\_CIRCLE\_OUTLINEDinstance-attribute**

PLAYLIST\_ADD\_CIRCLE\_OUTLINED: IconData

**buildPLAYLIST\_ADD\_CIRCLE\_ROUNDEDinstance-attribute**

PLAYLIST\_ADD\_CIRCLE\_ROUNDED: IconData

**buildPLAYLIST\_ADD\_CIRCLE\_SHARPinstance-attribute**

PLAYLIST\_ADD\_CIRCLE\_SHARP: IconData

**buildPLAYLIST\_ADD\_OUTLINEDinstance-attribute**

PLAYLIST\_ADD\_OUTLINED: IconData

**buildPLAYLIST\_ADD\_ROUNDEDinstance-attribute**

PLAYLIST\_ADD\_ROUNDED: IconData

**buildPLAYLIST\_ADD\_SHARPinstance-attribute**

PLAYLIST\_ADD\_SHARP: IconData

**buildPLAYLIST\_PLAYinstance-attribute**

PLAYLIST\_PLAY: IconData

**buildPLAYLIST\_PLAY\_OUTLINEDinstance-attribute**

PLAYLIST\_PLAY\_OUTLINED: IconData

**buildPLAYLIST\_PLAY\_ROUNDEDinstance-attribute**

PLAYLIST\_PLAY\_ROUNDED: IconData

**buildPLAYLIST\_PLAY\_SHARPinstance-attribute**

PLAYLIST\_PLAY\_SHARP: IconData

**buildPLAYLIST\_REMOVEinstance-attribute**

PLAYLIST\_REMOVE: IconData

**buildPLAYLIST\_REMOVE\_OUTLINEDinstance-attribute**

PLAYLIST\_REMOVE\_OUTLINED: IconData

**buildPLAYLIST\_REMOVE\_ROUNDEDinstance-attribute**

PLAYLIST\_REMOVE\_ROUNDED: IconData

**buildPLAYLIST\_REMOVE\_SHARPinstance-attribute**

PLAYLIST\_REMOVE\_SHARP: IconData

**buildPLUMBINGinstance-attribute**

PLUMBING: IconData

**buildPLUMBING\_OUTLINEDinstance-attribute**

PLUMBING\_OUTLINED: IconData

**buildPLUMBING\_ROUNDEDinstance-attribute**

PLUMBING\_ROUNDED: IconData

**buildPLUMBING\_SHARPinstance-attribute**

PLUMBING\_SHARP: IconData

**buildPLUS\_ONEinstance-attribute**

PLUS\_ONE: IconData

**buildPLUS\_ONE\_OUTLINEDinstance-attribute**

PLUS\_ONE\_OUTLINED: IconData

**buildPLUS\_ONE\_ROUNDEDinstance-attribute**

PLUS\_ONE\_ROUNDED: IconData

**buildPLUS\_ONE\_SHARPinstance-attribute**

PLUS\_ONE\_SHARP: IconData

**buildPODCASTSinstance-attribute**

PODCASTS: IconData

**buildPODCASTS\_OUTLINEDinstance-attribute**

PODCASTS\_OUTLINED: IconData

**buildPODCASTS\_ROUNDEDinstance-attribute**

PODCASTS\_ROUNDED: IconData

**buildPODCASTS\_SHARPinstance-attribute**

PODCASTS\_SHARP: IconData

**buildPOINT\_OF\_SALEinstance-attribute**

POINT\_OF\_SALE: IconData

**buildPOINT\_OF\_SALE\_OUTLINEDinstance-attribute**

POINT\_OF\_SALE\_OUTLINED: IconData

**buildPOINT\_OF\_SALE\_ROUNDEDinstance-attribute**

POINT\_OF\_SALE\_ROUNDED: IconData

**buildPOINT\_OF\_SALE\_SHARPinstance-attribute**

POINT\_OF\_SALE\_SHARP: IconData

**buildPOLICYinstance-attribute**

POLICY: IconData

**buildPOLICY\_OUTLINEDinstance-attribute**

POLICY\_OUTLINED: IconData

**buildPOLICY\_ROUNDEDinstance-attribute**

POLICY\_ROUNDED: IconData

**buildPOLICY\_SHARPinstance-attribute**

POLICY\_SHARP: IconData

**buildPOLLinstance-attribute**

POLL: IconData

**buildPOLL\_OUTLINEDinstance-attribute**

POLL\_OUTLINED: IconData

**buildPOLL\_ROUNDEDinstance-attribute**

POLL\_ROUNDED: IconData

**buildPOLL\_SHARPinstance-attribute**

POLL\_SHARP: IconData

**buildPOLYLINEinstance-attribute**

POLYLINE: IconData

**buildPOLYLINE\_OUTLINEDinstance-attribute**

POLYLINE\_OUTLINED: IconData

**buildPOLYLINE\_ROUNDEDinstance-attribute**

POLYLINE\_ROUNDED: IconData

**buildPOLYLINE\_SHARPinstance-attribute**

POLYLINE\_SHARP: IconData

**buildPOLYMERinstance-attribute**

POLYMER: IconData

**buildPOLYMER\_OUTLINEDinstance-attribute**

POLYMER\_OUTLINED: IconData

**buildPOLYMER\_ROUNDEDinstance-attribute**

POLYMER\_ROUNDED: IconData

**buildPOLYMER\_SHARPinstance-attribute**

POLYMER\_SHARP: IconData

**buildPOOLinstance-attribute**

POOL: IconData

**buildPOOL\_OUTLINEDinstance-attribute**

POOL\_OUTLINED: IconData

**buildPOOL\_ROUNDEDinstance-attribute**

POOL\_ROUNDED: IconData

**buildPOOL\_SHARPinstance-attribute**

POOL\_SHARP: IconData

**buildPORTABLE\_WIFI\_OFFinstance-attribute**

PORTABLE\_WIFI\_OFF: IconData

**buildPORTABLE\_WIFI\_OFF\_OUTLINEDinstance-attribute**

PORTABLE\_WIFI\_OFF\_OUTLINED: IconData

**buildPORTABLE\_WIFI\_OFF\_ROUNDEDinstance-attribute**

PORTABLE\_WIFI\_OFF\_ROUNDED: IconData

**buildPORTABLE\_WIFI\_OFF\_SHARPinstance-attribute**

PORTABLE\_WIFI\_OFF\_SHARP: IconData

**buildPORTRAITinstance-attribute**

PORTRAIT: IconData

**buildPORTRAIT\_OUTLINEDinstance-attribute**

PORTRAIT\_OUTLINED: IconData

**buildPORTRAIT\_ROUNDEDinstance-attribute**

PORTRAIT\_ROUNDED: IconData

**buildPORTRAIT\_SHARPinstance-attribute**

PORTRAIT\_SHARP: IconData

**buildPOST\_ADDinstance-attribute**

POST\_ADD: IconData

**buildPOST\_ADD\_OUTLINEDinstance-attribute**

POST\_ADD\_OUTLINED: IconData

**buildPOST\_ADD\_ROUNDEDinstance-attribute**

POST\_ADD\_ROUNDED: IconData

**buildPOST\_ADD\_SHARPinstance-attribute**

POST\_ADD\_SHARP: IconData

**buildPOWERinstance-attribute**

POWER: IconData

**buildPOWER\_INPUTinstance-attribute**

POWER\_INPUT: IconData

**buildPOWER\_INPUT\_OUTLINEDinstance-attribute**

POWER\_INPUT\_OUTLINED: IconData

**buildPOWER\_INPUT\_ROUNDEDinstance-attribute**

POWER\_INPUT\_ROUNDED: IconData

**buildPOWER\_INPUT\_SHARPinstance-attribute**

POWER\_INPUT\_SHARP: IconData

**buildPOWER\_OFFinstance-attribute**

POWER\_OFF: IconData

**buildPOWER\_OFF\_OUTLINEDinstance-attribute**

POWER\_OFF\_OUTLINED: IconData

**buildPOWER\_OFF\_ROUNDEDinstance-attribute**

POWER\_OFF\_ROUNDED: IconData

**buildPOWER\_OFF\_SHARPinstance-attribute**

POWER\_OFF\_SHARP: IconData

**buildPOWER\_OUTLINEDinstance-attribute**

POWER\_OUTLINED: IconData

**buildPOWER\_ROUNDEDinstance-attribute**

POWER\_ROUNDED: IconData

**buildPOWER\_SETTINGS\_NEWinstance-attribute**

POWER\_SETTINGS\_NEW: IconData

**buildPOWER\_SETTINGS\_NEW\_OUTLINEDinstance-attribute**

POWER\_SETTINGS\_NEW\_OUTLINED: IconData

**buildPOWER\_SETTINGS\_NEW\_ROUNDEDinstance-attribute**

POWER\_SETTINGS\_NEW\_ROUNDED: IconData

**buildPOWER\_SETTINGS\_NEW\_SHARPinstance-attribute**

POWER\_SETTINGS\_NEW\_SHARP: IconData

**buildPOWER\_SHARPinstance-attribute**

POWER\_SHARP: IconData

**buildPRECISION\_MANUFACTURINGinstance-attribute**

PRECISION\_MANUFACTURING: IconData

**buildPRECISION\_MANUFACTURING\_OUTLINEDinstance-attribute**

PRECISION\_MANUFACTURING\_OUTLINED: IconData

**buildPRECISION\_MANUFACTURING\_ROUNDEDinstance-attribute**

PRECISION\_MANUFACTURING\_ROUNDED: IconData

**buildPRECISION\_MANUFACTURING\_SHARPinstance-attribute**

PRECISION\_MANUFACTURING\_SHARP: IconData

**buildPREGNANT\_WOMANinstance-attribute**

PREGNANT\_WOMAN: IconData

**buildPREGNANT\_WOMAN\_OUTLINEDinstance-attribute**

PREGNANT\_WOMAN\_OUTLINED: IconData

**buildPREGNANT\_WOMAN\_ROUNDEDinstance-attribute**

PREGNANT\_WOMAN\_ROUNDED: IconData

**buildPREGNANT\_WOMAN\_SHARPinstance-attribute**

PREGNANT\_WOMAN\_SHARP: IconData

**buildPRESENT\_TO\_ALLinstance-attribute**

PRESENT\_TO\_ALL: IconData

**buildPRESENT\_TO\_ALL\_OUTLINEDinstance-attribute**

PRESENT\_TO\_ALL\_OUTLINED: IconData

**buildPRESENT\_TO\_ALL\_ROUNDEDinstance-attribute**

PRESENT\_TO\_ALL\_ROUNDED: IconData

**buildPRESENT\_TO\_ALL\_SHARPinstance-attribute**

PRESENT\_TO\_ALL\_SHARP: IconData

**buildPREVIEWinstance-attribute**

PREVIEW: IconData

**buildPREVIEW\_OUTLINEDinstance-attribute**

PREVIEW\_OUTLINED: IconData

**buildPREVIEW\_ROUNDEDinstance-attribute**

PREVIEW\_ROUNDED: IconData

**buildPREVIEW\_SHARPinstance-attribute**

PREVIEW\_SHARP: IconData

**buildPRICE\_CHANGEinstance-attribute**

PRICE\_CHANGE: IconData

**buildPRICE\_CHANGE\_OUTLINEDinstance-attribute**

PRICE\_CHANGE\_OUTLINED: IconData

**buildPRICE\_CHANGE\_ROUNDEDinstance-attribute**

PRICE\_CHANGE\_ROUNDED: IconData

**buildPRICE\_CHANGE\_SHARPinstance-attribute**

PRICE\_CHANGE\_SHARP: IconData

**buildPRICE\_CHECKinstance-attribute**

PRICE\_CHECK: IconData

**buildPRICE\_CHECK\_OUTLINEDinstance-attribute**

PRICE\_CHECK\_OUTLINED: IconData

**buildPRICE\_CHECK\_ROUNDEDinstance-attribute**

PRICE\_CHECK\_ROUNDED: IconData

**buildPRICE\_CHECK\_SHARPinstance-attribute**

PRICE\_CHECK\_SHARP: IconData

**buildPRINTinstance-attribute**

PRINT: IconData

**buildPRINT\_DISABLEDinstance-attribute**

PRINT\_DISABLED: IconData

**buildPRINT\_DISABLED\_OUTLINEDinstance-attribute**

PRINT\_DISABLED\_OUTLINED: IconData

**buildPRINT\_DISABLED\_ROUNDEDinstance-attribute**

PRINT\_DISABLED\_ROUNDED: IconData

**buildPRINT\_DISABLED\_SHARPinstance-attribute**

PRINT\_DISABLED\_SHARP: IconData

**buildPRINT\_OUTLINEDinstance-attribute**

PRINT\_OUTLINED: IconData

**buildPRINT\_ROUNDEDinstance-attribute**

PRINT\_ROUNDED: IconData

**buildPRINT\_SHARPinstance-attribute**

PRINT\_SHARP: IconData

**buildPRIORITY\_HIGHinstance-attribute**

PRIORITY\_HIGH: IconData

**buildPRIORITY\_HIGH\_OUTLINEDinstance-attribute**

PRIORITY\_HIGH\_OUTLINED: IconData

**buildPRIORITY\_HIGH\_ROUNDEDinstance-attribute**

PRIORITY\_HIGH\_ROUNDED: IconData

**buildPRIORITY\_HIGH\_SHARPinstance-attribute**

PRIORITY\_HIGH\_SHARP: IconData



**buildPRIVACY\_TIPinstance-attribute**

PRIVACY\_TIP: IconData

**buildPRIVACY\_TIP\_OUTLINEDinstance-attribute**

PRIVACY\_TIP\_OUTLINED: IconData

**buildPRIVACY\_TIP\_ROUNDEDinstance-attribute**

PRIVACY\_TIP\_ROUNDED: IconData

**buildPRIVACY\_TIP\_SHARPinstance-attribute**

PRIVACY\_TIP\_SHARP: IconData

**buildPRIVATE\_CONNECTIVITYinstance-attribute**

PRIVATE\_CONNECTIVITY: IconData

**buildPRIVATE\_CONNECTIVITY\_OUTLINEDinstance-attribute**

PRIVATE\_CONNECTIVITY\_OUTLINED: IconData

**buildPRIVATE\_CONNECTIVITY\_ROUNDEDinstance-attribute**

PRIVATE\_CONNECTIVITY\_ROUNDED: IconData

**buildPRIVATE\_CONNECTIVITY\_SHARPinstance-attribute**

PRIVATE\_CONNECTIVITY\_SHARP: IconData

**buildPRODUCTION\_QUANTITY\_LIMITSinstance-attribute**

PRODUCTION\_QUANTITY\_LIMITS: IconData

**buildPRODUCTION\_QUANTITY\_LIMITS\_OUTLINEDinstance-attribute**

PRODUCTION\_QUANTITY\_LIMITS\_OUTLINED: IconData

**buildPRODUCTION\_QUANTITY\_LIMITS\_ROUNDEDinstance-attribute**

PRODUCTION\_QUANTITY\_LIMITS\_ROUNDED: IconData

**buildPRODUCTION\_QUANTITY\_LIMITS\_SHARPinstance-attribute**

PRODUCTION\_QUANTITY\_LIMITS\_SHARP: IconData

**buildPROPANEinstance-attribute**

PROPANE: IconData

**buildPROPANE\_OUTLINEDinstance-attribute**

PROPANE\_OUTLINED: IconData

**buildPROPANE\_ROUNDEDinstance-attribute**

PROPANE\_ROUNDED: IconData

**buildPROPANE\_SHARPinstance-attribute**

PROPANE\_SHARP: IconData

**buildPROPANE\_TANKinstance-attribute**

PROPANE\_TANK: IconData

**buildPROPANE\_TANK\_OUTLINEDinstance-attribute**

PROPANE\_TANK\_OUTLINED: IconData

**buildPROPANE\_TANK\_ROUNDEDinstance-attribute**

PROPANE\_TANK\_ROUNDED: IconData

**buildPROPANE\_TANK\_SHARPinstance-attribute**

PROPANE\_TANK\_SHARP: IconData

**buildPSYCHOLOGYinstance-attribute**

PSYCHOLOGY: IconData

**buildPSYCHOLOGY\_ALTinstance-attribute**

PSYCHOLOGY\_ALT: IconData

**buildPSYCHOLOGY\_ALT\_OUTLINED**instance-attribute

PSYCHOLOGY\_ALT\_OUTLINED: IconData

**buildPSYCHOLOGY\_ALT\_ROUNDED**instance-attribute

PSYCHOLOGY\_ALT\_ROUNDED: IconData

**buildPSYCHOLOGY\_ALT\_SHARP**instance-attribute

PSYCHOLOGY\_ALT\_SHARP: IconData

**buildPSYCHOLOGY\_OUTLINED**instance-attribute

PSYCHOLOGY\_OUTLINED: IconData

**buildPSYCHOLOGY\_ROUNDED**instance-attribute

PSYCHOLOGY\_ROUNDED: IconData

**buildPSYCHOLOGY\_SHARP**instance-attribute

PSYCHOLOGY\_SHARP: IconData

**buildPUBLIC**instance-attribute

PUBLIC: IconData

**buildPUBLIC\_OFF**instance-attribute

PUBLIC\_OFF: IconData

**buildPUBLIC\_OFF\_OUTLINED**instance-attribute

PUBLIC\_OFF\_OUTLINED: IconData

**buildPUBLIC\_OFF\_ROUNDED**instance-attribute

PUBLIC\_OFF\_ROUNDED: IconData

**buildPUBLIC\_OFF\_SHARP**instance-attribute

PUBLIC\_OFF\_SHARP: IconData

**buildPUBLIC\_OUTLINED**instance-attribute

PUBLIC\_OUTLINED: IconData

**buildPUBLIC\_ROUNDED**instance-attribute

PUBLIC\_ROUNDED: IconData

**buildPUBLIC\_SHARP**instance-attribute

PUBLIC\_SHARP: IconData

**buildPUBLISH**instance-attribute

PUBLISH: IconData

**buildPUBLISH\_OUTLINED**instance-attribute

PUBLISH\_OUTLINED: IconData

**buildPUBLISH\_ROUNDED**instance-attribute

PUBLISH\_ROUNDED: IconData

**buildPUBLISH\_SHARP**instance-attribute

PUBLISH\_SHARP: IconData

**buildPUBLISHED\_WITH\_CHANGES**instance-attribute

PUBLISHED\_WITH\_CHANGES: IconData

**buildPUBLISHED\_WITH\_CHANGES\_OUTLINED**instance-attribute

PUBLISHED\_WITH\_CHANGES\_OUTLINED: IconData

**buildPUBLISHED\_WITH\_CHANGES\_ROUNDED**instance-attribute

PUBLISHED\_WITH\_CHANGES\_ROUNDED: IconData

**buildPUBLISHED\_WITH\_CHANGES\_SHARP**instance-attribute

PUBLISHED\_WITH\_CHANGES\_SHARP: IconData

**buildPUNCH\_CLOCKinstance-attribute**

PUNCH\_CLOCK: IconData

**buildPUNCH\_CLOCK\_OUTLINEDinstance-attribute**

PUNCH\_CLOCK\_OUTLINED: IconData

**buildPUNCH\_CLOCK\_ROUNDEDinstance-attribute**

PUNCH\_CLOCK\_ROUNDED: IconData

**buildPUNCH\_CLOCK\_SHARPinstance-attribute**

PUNCH\_CLOCK\_SHARP: IconData

**buildPUSH\_PINinstance-attribute**

PUSH\_PIN: IconData

**buildPUSH\_PIN\_OUTLINEDinstance-attribute**

PUSH\_PIN\_OUTLINED: IconData

**buildPUSH\_PIN\_ROUNDEDinstance-attribute**

PUSH\_PIN\_ROUNDED: IconData

**buildPUSH\_PIN\_SHARPinstance-attribute**

PUSH\_PIN\_SHARP: IconData

**buildQR\_CODEinstance-attribute**

QR\_CODE: IconData

**buildQR\_CODE\_2instance-attribute**

QR\_CODE\_2: IconData

**buildQR\_CODE\_2\_OUTLINEDinstance-attribute**

QR\_CODE\_2\_OUTLINED: IconData

**buildQR\_CODE\_2\_ROUNDEDinstance-attribute**

QR\_CODE\_2\_ROUNDED: IconData

**buildQR\_CODE\_2\_SHARPinstance-attribute**

QR\_CODE\_2\_SHARP: IconData

**buildQR\_CODE\_OUTLINEDinstance-attribute**

QR\_CODE\_OUTLINED: IconData

**buildQR\_CODE\_ROUNDEDinstance-attribute**

QR\_CODE\_ROUNDED: IconData

**buildQR\_CODE\_SCANNERinstance-attribute**

QR\_CODE\_SCANNER: IconData

**buildQR\_CODE\_SCANNER\_OUTLINEDinstance-attribute**

QR\_CODE\_SCANNER\_OUTLINED: IconData

**buildQR\_CODE\_SCANNER\_ROUNDEDinstance-attribute**

QR\_CODE\_SCANNER\_ROUNDED: IconData

**buildQR\_CODE\_SCANNER\_SHARPinstance-attribute**

QR\_CODE\_SCANNER\_SHARP: IconData

**buildQR\_CODE\_SHARPinstance-attribute**

QR\_CODE\_SHARP: IconData

**buildQUERY\_BUILDERinstance-attribute**

QUERY\_BUILDER: IconData

**buildQUERY\_BUILDER\_OUTLINEDinstance-attribute**

QUERY\_BUILDER\_OUTLINED: IconData

**buildQUERY\_BUILDER\_ROUNDED**instance-attribute

QUERY\_BUILDER\_ROUNDED: IconData

**buildQUERY\_BUILDER\_SHARP**instance-attribute

QUERY\_BUILDER\_SHARP: IconData

**buildQUERY\_STATS**instance-attribute

QUERY\_STATS: IconData

**buildQUERY\_STATS\_OUTLINED**instance-attribute

QUERY\_STATS\_OUTLINED: IconData

**buildQUERY\_STATS\_ROUNDED**instance-attribute

QUERY\_STATS\_ROUNDED: IconData

**buildQUERY\_STATS\_SHARP**instance-attribute

QUERY\_STATS\_SHARP: IconData

**buildQUESTION\_ANSWER**instance-attribute

QUESTION\_ANSWER: IconData

**buildQUESTION\_ANSWER\_OUTLINED**instance-attribute

QUESTION\_ANSWER\_OUTLINED: IconData

**buildQUESTION\_ANSWER\_ROUNDED**instance-attribute

QUESTION\_ANSWER\_ROUNDED: IconData

**buildQUESTION\_ANSWER\_SHARP**instance-attribute

QUESTION\_ANSWER\_SHARP: IconData

**buildQUESTION\_MARK**instance-attribute

QUESTION\_MARK: IconData

**buildQUESTION\_MARK\_OUTLINED**instance-attribute

QUESTION\_MARK\_OUTLINED: IconData

**buildQUESTION\_MARK\_ROUNDED**instance-attribute

QUESTION\_MARK\_ROUNDED: IconData

**buildQUESTION\_MARK\_SHARP**instance-attribute

QUESTION\_MARK\_SHARP: IconData

**buildQUEUE**instance-attribute

QUEUE: IconData

**buildQUEUE\_MUSIC**instance-attribute

QUEUE\_MUSIC: IconData

**buildQUEUE\_MUSIC\_OUTLINED**instance-attribute

QUEUE\_MUSIC\_OUTLINED: IconData

**buildQUEUE\_MUSIC\_ROUNDED**instance-attribute

QUEUE\_MUSIC\_ROUNDED: IconData

**buildQUEUE\_MUSIC\_SHARP**instance-attribute

QUEUE\_MUSIC\_SHARP: IconData

**buildQUEUE\_OUTLINED**instance-attribute

QUEUE\_OUTLINED: IconData

**buildQUEUE\_PLAY\_NEXT**instance-attribute

QUEUE\_PLAY\_NEXT: IconData

**buildQUEUE\_PLAY\_NEXT\_OUTLINED**instance-attribute

QUEUE\_PLAY\_NEXT\_OUTLINED: IconData

**buildQUEUE\_PLAY\_NEXT\_ROUNDEDinstance-attribute**

QUEUE\_PLAY\_NEXT\_ROUNDED: IconData

**buildQUEUE\_PLAY\_NEXT\_SHARPinstance-attribute**

QUEUE\_PLAY\_NEXT\_SHARP: IconData

**buildQUEUE\_ROUNDEDinstance-attribute**

QUEUE\_ROUNDED: IconData

**buildQUEUE\_SHARPinstance-attribute**

QUEUE\_SHARP: IconData

**buildQUICK\_CONTACTS\_DIALERinstance-attribute**

QUICK\_CONTACTS\_DIALER: IconData

**buildQUICK\_CONTACTS\_DIALER\_OUTLINEDinstance-attribute**

QUICK\_CONTACTS\_DIALER\_OUTLINED: IconData

**buildQUICK\_CONTACTS\_DIALER\_ROUNDEDinstance-attribute**

QUICK\_CONTACTS\_DIALER\_ROUNDED: IconData

**buildQUICK\_CONTACTS\_DIALER\_SHARPinstance-attribute**

QUICK\_CONTACTS\_DIALER\_SHARP: IconData

**buildQUICK\_CONTACTS\_MAILinstance-attribute**

QUICK\_CONTACTS\_MAIL: IconData

**buildQUICK\_CONTACTS\_MAIL\_OUTLINEDinstance-attribute**

QUICK\_CONTACTS\_MAIL\_OUTLINED: IconData

**buildQUICK\_CONTACTS\_MAIL\_ROUNDEDinstance-attribute**

QUICK\_CONTACTS\_MAIL\_ROUNDED: IconData

**buildQUICK\_CONTACTS\_MAIL\_SHARPinstance-attribute**

QUICK\_CONTACTS\_MAIL\_SHARP: IconData

**buildQUICKREPLYinstance-attribute**

QUICKREPLY: IconData

**buildQUICKREPLY\_OUTLINEDinstance-attribute**

QUICKREPLY\_OUTLINED: IconData

**buildQUICKREPLY\_ROUNDEDinstance-attribute**

QUICKREPLY\_ROUNDED: IconData

**buildQUICKREPLY\_SHARPinstance-attribute**

QUICKREPLY\_SHARP: IconData

**buildQUIZinstance-attribute**

QUIZ: IconData

**buildQUIZ\_OUTLINEDinstance-attribute**

QUIZ\_OUTLINED: IconData

**buildQUIZ\_ROUNDEDinstance-attribute**

QUIZ\_ROUNDED: IconData

**buildQUIZ\_SHARPinstance-attribute**

QUIZ\_SHARP: IconData

**buildQUORAinstance-attribute**

QUORA: IconData

**buildQUORA\_OUTLINEDinstance-attribute**

QUORA\_OUTLINED: IconData

**buildQUORA\_ROUNDEDinstance-attribute**

QUORA\_ROUNDED: IconData

**buildQUORA\_SHARPinstance-attribute**

QUORA\_SHARP: IconData

**buildR\_MOBILEDATAinstance-attribute**

R\_MOBILEDATA: IconData

**buildR\_MOBILEDATA\_OUTLINEDinstance-attribute**

R\_MOBILEDATA\_OUTLINED: IconData

**buildR\_MOBILEDATA\_ROUNDEDinstance-attribute**

R\_MOBILEDATA\_ROUNDED: IconData

**buildR\_MOBILEDATA\_SHARPinstance-attribute**

R\_MOBILEDATA\_SHARP: IconData

**buildRADARinstance-attribute**

RADAR: IconData

**buildRADAR\_OUTLINEDinstance-attribute**

RADAR\_OUTLINED: IconData

**buildRADAR\_ROUNDEDinstance-attribute**

RADAR\_ROUNDED: IconData

**buildRADAR\_SHARPinstance-attribute**

RADAR\_SHARP: IconData

**buildRADIOinstance-attribute**

RADIO: IconData

**buildRADIO\_BUTTON\_CHECKEDinstance-attribute**

RADIO\_BUTTON\_CHECKED: IconData

**buildRADIO\_BUTTON\_CHECKED\_OUTLINEDinstance-attribute**

RADIO\_BUTTON\_CHECKED\_OUTLINED: IconData

**buildRADIO\_BUTTON\_CHECKED\_ROUNDEDinstance-attribute**

RADIO\_BUTTON\_CHECKED\_ROUNDED: IconData

**buildRADIO\_BUTTON\_CHECKED\_SHARPinstance-attribute**

RADIO\_BUTTON\_CHECKED\_SHARP: IconData

**buildRADIO\_BUTTON\_OFFinstance-attribute**

RADIO\_BUTTON\_OFF: IconData

**buildRADIO\_BUTTON\_OFF\_OUTLINEDinstance-attribute**

RADIO\_BUTTON\_OFF\_OUTLINED: IconData

**buildRADIO\_BUTTON\_OFF\_ROUNDEDinstance-attribute**

RADIO\_BUTTON\_OFF\_ROUNDED: IconData

**buildRADIO\_BUTTON\_OFF\_SHARPinstance-attribute**

RADIO\_BUTTON\_OFF\_SHARP: IconData

**buildRADIO\_BUTTON\_ONinstance-attribute**

RADIO\_BUTTON\_ON: IconData

**buildRADIO\_BUTTON\_ON\_OUTLINEDinstance-attribute**

RADIO\_BUTTON\_ON\_OUTLINED: IconData

**buildRADIO\_BUTTON\_ON\_ROUNDEDinstance-attribute**

RADIO\_BUTTON\_ON\_ROUNDED: IconData

buildRADIO\_BUTTON\_ON\_SHARPinstance-attribute

RADIO\_BUTTON\_ON\_SHARP: IconData

buildRADIO\_BUTTON\_UNCHECKEDinstance-attribute

RADIO\_BUTTON\_UNCHECKED: IconData

buildRADIO\_BUTTON\_UNCHECKED\_OUTLINEDinstance-attribute

RADIO\_BUTTON\_UNCHECKED\_OUTLINED: IconData

buildRADIO\_BUTTON\_UNCHECKED\_ROUNDEDinstance-attribute

RADIO\_BUTTON\_UNCHECKED\_ROUNDED: IconData

buildRADIO\_BUTTON\_UNCHECKED\_SHARPinstance-attribute

RADIO\_BUTTON\_UNCHECKED\_SHARP: IconData

buildRADIO\_OUTLINEDinstance-attribute

RADIO\_OUTLINED: IconData

buildRADIO\_ROUNDEDinstance-attribute

RADIO\_ROUNDED: IconData

buildRADIO\_SHARPinstance-attribute

RADIO\_SHARP: IconData

buildRAILWAY\_ALERTinstance-attribute

RAILWAY\_ALERT: IconData

buildRAILWAY\_ALERT\_OUTLINEDinstance-attribute

RAILWAY\_ALERT\_OUTLINED: IconData

buildRAILWAY\_ALERT\_ROUNDEDinstance-attribute

RAILWAY\_ALERT\_ROUNDED: IconData

buildRAILWAY\_ALERT\_SHARPinstance-attribute

RAILWAY\_ALERT\_SHARP: IconData

buildRAMEN\_DININGinstance-attribute

RAMEN\_DINING: IconData

buildRAMEN\_DINING\_OUTLINEDinstance-attribute

RAMEN\_DINING\_OUTLINED: IconData

buildRAMEN\_DINING\_ROUNDEDinstance-attribute

RAMEN\_DINING\_ROUNDED: IconData

buildRAMEN\_DINING\_SHARPinstance-attribute

RAMEN\_DINING\_SHARP: IconData

buildRAMP\_LEFTinstance-attribute

RAMP\_LEFT: IconData

buildRAMP\_LEFT\_OUTLINEDinstance-attribute

RAMP\_LEFT\_OUTLINED: IconData

buildRAMP\_LEFT\_ROUNDEDinstance-attribute

RAMP\_LEFT\_ROUNDED: IconData

buildRAMP\_LEFT\_SHARPinstance-attribute

RAMP\_LEFT\_SHARP: IconData

buildRAMP\_RIGHTinstance-attribute

RAMP\_RIGHT: IconData

buildRAMP\_RIGHT\_OUTLINEDinstance-attribute

RAMP\_RIGHT\_OUTLINED: IconData

**buildRAMP\_RIGHT\_ROUNDED**instance-attribute

RAMP\_RIGHT\_ROUNDED: IconData

**buildRAMP\_RIGHT\_SHARP**instance-attribute

RAMP\_RIGHT\_SHARP: IconData

**buildRATE\_REVIEW**instance-attribute

RATE\_REVIEW: IconData

**buildRATE\_REVIEW\_OUTLINED**instance-attribute

RATE\_REVIEW\_OUTLINED: IconData

**buildRATE\_REVIEW\_ROUNDED**instance-attribute

RATE\_REVIEW\_ROUNDED: IconData

**buildRATE\_REVIEW\_SHARP**instance-attribute

RATE\_REVIEW\_SHARP: IconData

**buildRAW\_OFF**instance-attribute

RAW\_OFF: IconData

**buildRAW\_OFF\_OUTLINED**instance-attribute

RAW\_OFF\_OUTLINED: IconData

**buildRAW\_OFF\_ROUNDED**instance-attribute

RAW\_OFF\_ROUNDED: IconData

**buildRAW\_OFF\_SHARP**instance-attribute

RAW\_OFF\_SHARP: IconData

**buildRAW\_ON**instance-attribute

RAW\_ON: IconData

**buildRAW\_ON\_OUTLINED**instance-attribute

RAW\_ON\_OUTLINED: IconData

**buildRAW\_ON\_ROUNDED**instance-attribute

RAW\_ON\_ROUNDED: IconData

**buildRAW\_ON\_SHARP**instance-attribute

RAW\_ON\_SHARP: IconData

**buildREAD\_MORE**instance-attribute

READ\_MORE: IconData

**buildREAD\_MORE\_OUTLINED**instance-attribute

READ\_MORE\_OUTLINED: IconData

**buildREAD\_MORE\_ROUNDED**instance-attribute

READ\_MORE\_ROUNDED: IconData

**buildREAD\_MORE\_SHARP**instance-attribute

READ\_MORE\_SHARP: IconData

**buildREAL\_ESTATE\_AGENT**instance-attribute

REAL\_ESTATE\_AGENT: IconData

**buildREAL\_ESTATE\_AGENT\_OUTLINED**instance-attribute

REAL\_ESTATE\_AGENT\_OUTLINED: IconData

**buildREAL\_ESTATE\_AGENT\_ROUNDED**instance-attribute

REAL\_ESTATE\_AGENT\_ROUNDED: IconData

**buildREAL\_ESTATE\_AGENT\_SHARP**instance-attribute

REAL\_ESTATE\_AGENT\_SHARP: IconData



**buildREBASE\_EDITinstance-attribute**

REBASE\_EDIT: IconData

**buildRECEIPTinstance-attribute**

RECEIPT: IconData

**buildRECEIPT\_LONGinstance-attribute**

RECEIPT\_LONG: IconData

**buildRECEIPT\_LONG\_OUTLINEDinstance-attribute**

RECEIPT\_LONG\_OUTLINED: IconData

**buildRECEIPT\_LONG\_ROUNDEDinstance-attribute**

RECEIPT\_LONG\_ROUNDED: IconData

**buildRECEIPT\_LONG\_SHARPinstance-attribute**

RECEIPT\_LONG\_SHARP: IconData

**buildRECEIPT\_OUTLINEDinstance-attribute**

RECEIPT\_OUTLINED: IconData

**buildRECEIPT\_ROUNDEDinstance-attribute**

RECEIPT\_ROUNDED: IconData

**buildRECEIPT\_SHARPinstance-attribute**

RECEIPT\_SHARP: IconData

**buildRECENT\_ACTORSinstance-attribute**

RECENT\_ACTORS: IconData

**buildRECENT\_ACTORS\_OUTLINEDinstance-attribute**

RECENT\_ACTORS\_OUTLINED: IconData

**buildRECENT\_ACTORS\_ROUNDEDinstance-attribute**

RECENT\_ACTORS\_ROUNDED: IconData

**buildRECENT\_ACTORS\_SHARPinstance-attribute**

RECENT\_ACTORS\_SHARP: IconData

**buildRECOMMENDinstance-attribute**

RECOMMEND: IconData

**buildRECOMMEND\_OUTLINEDinstance-attribute**

RECOMMEND\_OUTLINED: IconData

**buildRECOMMEND\_ROUNDEDinstance-attribute**

RECOMMEND\_ROUNDED: IconData

**buildRECOMMEND\_SHARPinstance-attribute**

RECOMMEND\_SHARP: IconData

**buildRECORD\_VOICE\_OVERinstance-attribute**

RECORD\_VOICE\_OVER: IconData

**buildRECORD\_VOICE\_OVER\_OUTLINEDinstance-attribute**

RECORD\_VOICE\_OVER\_OUTLINED: IconData

**buildRECORD\_VOICE\_OVER\_ROUNDEDinstance-attribute**

RECORD\_VOICE\_OVER\_ROUNDED: IconData

**buildRECORD\_VOICE\_OVER\_SHARPinstance-attribute**

RECORD\_VOICE\_OVER\_SHARP: IconData

**buildRECTANGLEinstance-attribute**

RECTANGLE: IconData

**buildRECTANGLE\_OUTLINEDinstance-attribute**

RECTANGLE\_OUTLINED: IconData

**buildRECTANGLE\_ROUNDEDinstance-attribute**

RECTANGLE\_ROUNDED: IconData

**buildRECTANGLE\_SHARPinstance-attribute**

RECTANGLE\_SHARP: IconData

**buildRECYCLINGinstance-attribute**

RECYCLING: IconData

**buildRECYCLING\_OUTLINEDinstance-attribute**

RECYCLING\_OUTLINED: IconData

**buildRECYCLING\_ROUNDEDinstance-attribute**

RECYCLING\_ROUNDED: IconData

**buildRECYCLING\_SHARPinstance-attribute**

RECYCLING\_SHARP: IconData

**buildREDDITinstance-attribute**

REDDIT: IconData

**buildREDDIT\_OUTLINEDinstance-attribute**

REDDIT\_OUTLINED: IconData

**buildREDDIT\_ROUNDEDinstance-attribute**

REDDIT\_ROUNDED: IconData

**buildREDDIT\_SHARPinstance-attribute**

REDDIT\_SHARP: IconData

**buildREDEEMinstance-attribute**

REDEEM: IconData

**buildREDEEM\_OUTLINEDinstance-attribute**

REDEEM\_OUTLINED: IconData

**buildREDEEM\_ROUNDEDinstance-attribute**

REDEEM\_ROUNDED: IconData

**buildREDEEM\_SHARPinstance-attribute**

REDEEM\_SHARP: IconData

**buildREDOinstance-attribute**

REDO: IconData

**buildREDO\_OUTLINEDinstance-attribute**

REDO\_OUTLINED: IconData

**buildREDO\_ROUNDEDinstance-attribute**

REDO\_ROUNDED: IconData

**buildREDO\_SHARPinstance-attribute**

REDO\_SHARP: IconData

**buildREDUCE\_CAPACITYinstance-attribute**

REDUCE\_CAPACITY: IconData

**buildREDUCE\_CAPACITY\_OUTLINEDinstance-attribute**

REDUCE\_CAPACITY\_OUTLINED: IconData

**buildREDUCE\_CAPACITY\_ROUNDEDinstance-attribute**

REDUCE\_CAPACITY\_ROUNDED: IconData

**buildREDUCE\_CAPACITY\_SHARPinstance-attribute**

REDUCE\_CAPACITY\_SHARP: IconData

**buildREFRESHinstance-attribute**

REFRESH: IconData

**buildREFRESH\_OUTLINEDinstance-attribute**

REFRESH\_OUTLINED: IconData

**buildREFRESH\_ROUNDEDinstance-attribute**

REFRESH\_ROUNDED: IconData

**buildREFRESH\_SHARPinstance-attribute**

REFRESH\_SHARP: IconData

**buildREMEMBER\_MEinstance-attribute**

REMEMBER\_ME: IconData

**buildREMEMBER\_ME\_OUTLINEDinstance-attribute**

REMEMBER\_ME\_OUTLINED: IconData

**buildREMEMBER\_ME\_ROUNDEDinstance-attribute**

REMEMBER\_ME\_ROUNDED: IconData

**buildREMEMBER\_ME\_SHARPinstance-attribute**

REMEMBER\_ME\_SHARP: IconData

**buildREMOVEinstance-attribute**

REMOVE: IconData

**buildREMOVE\_CIRCLEinstance-attribute**

REMOVE\_CIRCLE: IconData

**buildREMOVE\_CIRCLE\_OUTLINEinstance-attribute**

REMOVE\_CIRCLE\_OUTLINE: IconData

**buildREMOVE\_CIRCLE\_OUTLINE\_OUTLINEDinstance-attribute**

REMOVE\_CIRCLE\_OUTLINE\_OUTLINED: IconData

**buildREMOVE\_CIRCLE\_OUTLINE\_ROUNDEDinstance-attribute**

REMOVE\_CIRCLE\_OUTLINE\_ROUNDED: IconData

**buildREMOVE\_CIRCLE\_OUTLINE\_SHARPinstance-attribute**

REMOVE\_CIRCLE\_OUTLINE\_SHARP: IconData

**buildREMOVE\_CIRCLE\_OUTLINEDinstance-attribute**

REMOVE\_CIRCLE\_OUTLINED: IconData

**buildREMOVE\_CIRCLE\_ROUNDEDinstance-attribute**

REMOVE\_CIRCLE\_ROUNDED: IconData

**buildREMOVE\_CIRCLE\_SHARPinstance-attribute**

REMOVE\_CIRCLE\_SHARP: IconData

**buildREMOVE\_DONEinstance-attribute**

REMOVE\_DONE: IconData

**buildREMOVE\_DONE\_OUTLINEDinstance-attribute**

REMOVE\_DONE\_OUTLINED: IconData

**buildREMOVE\_DONE\_ROUNDEDinstance-attribute**

REMOVE\_DONE\_ROUNDED: IconData

**buildREMOVE\_DONE\_SHARPinstance-attribute**

REMOVE\_DONE\_SHARP: IconData

**buildREMOVE\_FROM\_QUEUEinstance-attribute**  
REMOVE\_FROM\_QUEUE: IconData

**buildREMOVE\_FROM\_QUEUE\_OUTLINEDinstance-attribute**  
REMOVE\_FROM\_QUEUE\_OUTLINED: IconData

**buildREMOVE\_FROM\_QUEUE\_ROUNDEDinstance-attribute**  
REMOVE\_FROM\_QUEUE\_ROUNDED: IconData

**buildREMOVE\_FROM\_QUEUE\_SHARPinstance-attribute**  
REMOVE\_FROM\_QUEUE\_SHARP: IconData

**buildREMOVE\_MODERATORinstance-attribute**  
REMOVE\_MODERATOR: IconData

**buildREMOVE\_MODERATOR\_OUTLINEDinstance-attribute**  
REMOVE\_MODERATOR\_OUTLINED: IconData

**buildREMOVE\_MODERATOR\_ROUNDEDinstance-attribute**  
REMOVE\_MODERATOR\_ROUNDED: IconData

**buildREMOVE\_MODERATOR\_SHARPinstance-attribute**  
REMOVE\_MODERATOR\_SHARP: IconData

**buildREMOVE\_OUTLINEDinstance-attribute**  
REMOVE\_OUTLINED: IconData

**buildREMOVE\_RED\_EYEinstance-attribute**  
REMOVE\_RED\_EYE: IconData

**buildREMOVE\_RED\_EYE\_OUTLINEDinstance-attribute**  
REMOVE\_RED\_EYE\_OUTLINED: IconData

**buildREMOVE\_RED\_EYE\_ROUNDEDinstance-attribute**  
REMOVE\_RED\_EYE\_ROUNDED: IconData

**buildREMOVE\_RED\_EYE\_SHARPinstance-attribute**  
REMOVE\_RED\_EYE\_SHARP: IconData

**buildREMOVE\_ROADinstance-attribute**  
REMOVE\_ROAD: IconData

**buildREMOVE\_ROAD\_OUTLINEDinstance-attribute**  
REMOVE\_ROAD\_OUTLINED: IconData

**buildREMOVE\_ROAD\_ROUNDEDinstance-attribute**  
REMOVE\_ROAD\_ROUNDED: IconData

**buildREMOVE\_ROAD\_SHARPinstance-attribute**  
REMOVE\_ROAD\_SHARP: IconData

**buildREMOVE\_ROUNDEDinstance-attribute**  
REMOVE\_ROUNDED: IconData

**buildREMOVE\_SHARPinstance-attribute**  
REMOVE\_SHARP: IconData

**buildREMOVE\_SHOPPING\_CARTinstance-attribute**  
REMOVE\_SHOPPING\_CART: IconData

**buildREMOVE\_SHOPPING\_CART\_OUTLINEDinstance-attribute**  
REMOVE\_SHOPPING\_CART\_OUTLINED: IconData

**buildREMOVE\_SHOPPING\_CART\_ROUNDEDinstance-attribute**  
REMOVE\_SHOPPING\_CART\_ROUNDED: IconData

**buildREMOVE\_SHOPPING\_CART\_SHARPinstance-attribute**

REMOVE\_SHOPPING\_CART\_SHARP: IconData

**buildREORDERinstance-attribute**

REORDER: IconData

**buildREORDER\_OUTLINEDinstance-attribute**

REORDER\_OUTLINED: IconData

**buildREORDER\_ROUNDEDinstance-attribute**

REORDER\_ROUNDED: IconData

**buildREORDER\_SHARPinstance-attribute**

REORDER\_SHARP: IconData

**buildREPARTITIONinstance-attribute**

REPARTITION: IconData

**buildREPARTITION\_OUTLINEDinstance-attribute**

REPARTITION\_OUTLINED: IconData

**buildREPARTITION\_ROUNDEDinstance-attribute**

REPARTITION\_ROUNDED: IconData

**buildREPARTITION\_SHARPinstance-attribute**

REPARTITION\_SHARP: IconData

**buildREPEATinstance-attribute**

REPEAT: IconData

**buildREPEAT\_ONinstance-attribute**

REPEAT\_ON: IconData

**buildREPEAT\_ON\_OUTLINEDinstance-attribute**

REPEAT\_ON\_OUTLINED: IconData

**buildREPEAT\_ON\_ROUNDEDinstance-attribute**

REPEAT\_ON\_ROUNDED: IconData

**buildREPEAT\_ON\_SHARPinstance-attribute**

REPEAT\_ON\_SHARP: IconData

**buildREPEAT\_ONEinstance-attribute**

REPEAT\_ONE: IconData

**buildREPEAT\_ONE\_ONinstance-attribute**

REPEAT\_ONE\_ON: IconData

**buildREPEAT\_ONE\_ON\_OUTLINEDinstance-attribute**

REPEAT\_ONE\_ON\_OUTLINED: IconData

**buildREPEAT\_ONE\_ON\_ROUNDEDinstance-attribute**

REPEAT\_ONE\_ON\_ROUNDED: IconData

**buildREPEAT\_ONE\_ON\_SHARPinstance-attribute**

REPEAT\_ONE\_ON\_SHARP: IconData

**buildREPEAT\_ONE\_OUTLINEDinstance-attribute**

REPEAT\_ONE\_OUTLINED: IconData

**buildREPEAT\_ONE\_ROUNDEDinstance-attribute**

REPEAT\_ONE\_ROUNDED: IconData

**buildREPEAT\_ONE\_SHARPinstance-attribute**

REPEAT\_ONE\_SHARP: IconData

**buildREPEAT\_OUTLINED**instance-attribute

REPEAT\_OUTLINED: IconData

**buildREPEAT\_ROUNDED**instance-attribute

REPEAT\_ROUNDED: IconData

**buildREPEAT\_SHARP**instance-attribute

REPEAT\_SHARP: IconData

**buildREPLAY**instance-attribute

REPLAY: IconData

**buildREPLAY\_10**instance-attribute

REPLAY\_10: IconData

**buildREPLAY\_10\_OUTLINED**instance-attribute

REPLAY\_10\_OUTLINED: IconData

**buildREPLAY\_10\_ROUNDED**instance-attribute

REPLAY\_10\_ROUNDED: IconData

**buildREPLAY\_10\_SHARP**instance-attribute

REPLAY\_10\_SHARP: IconData

**buildREPLAY\_30**instance-attribute

REPLAY\_30: IconData

**buildREPLAY\_30\_OUTLINED**instance-attribute

REPLAY\_30\_OUTLINED: IconData

**buildREPLAY\_30\_ROUNDED**instance-attribute

REPLAY\_30\_ROUNDED: IconData

**buildREPLAY\_30\_SHARP**instance-attribute

REPLAY\_30\_SHARP: IconData

**buildREPLAY\_5**instance-attribute

REPLAY\_5: IconData

**buildREPLAY\_5\_OUTLINED**instance-attribute

REPLAY\_5\_OUTLINED: IconData

**buildREPLAY\_5\_ROUNDED**instance-attribute

REPLAY\_5\_ROUNDED: IconData

**buildREPLAY\_5\_SHARP**instance-attribute

REPLAY\_5\_SHARP: IconData

**buildREPLAY\_CIRCLE\_FILLED**instance-attribute

REPLAY\_CIRCLE\_FILLED: IconData

**buildREPLAY\_CIRCLE\_FILLED\_OUTLINED**instance-attribute

REPLAY\_CIRCLE\_FILLED\_OUTLINED: IconData

**buildREPLAY\_CIRCLE\_FILLED\_ROUNDED**instance-attribute

REPLAY\_CIRCLE\_FILLED\_ROUNDED: IconData

**buildREPLAY\_CIRCLE\_FILLED\_SHARP**instance-attribute

REPLAY\_CIRCLE\_FILLED\_SHARP: IconData

**buildREPLAY\_OUTLINED**instance-attribute

REPLAY\_OUTLINED: IconData

**buildREPLAY\_ROUNDED**instance-attribute

REPLAY\_ROUNDED: IconData

**buildREPLAY\_SHARPinstance-attribute**

REPLAY\_SHARP: IconData

**buildREPLYinstance-attribute**

REPLY: IconData

**buildREPLY\_ALLinstance-attribute**

REPLY\_ALL: IconData

**buildREPLY\_ALL\_OUTLINEDinstance-attribute**

REPLY\_ALL\_OUTLINED: IconData

**buildREPLY\_ALL\_ROUNDEDinstance-attribute**

REPLY\_ALL\_ROUNDED: IconData

**buildREPLY\_ALL\_SHARPinstance-attribute**

REPLY\_ALL\_SHARP: IconData

**buildREPLY\_OUTLINEDinstance-attribute**

REPLY\_OUTLINED: IconData

**buildREPLY\_ROUNDEDinstance-attribute**

REPLY\_ROUNDED: IconData

**buildREPLY\_SHARPinstance-attribute**

REPLY\_SHARP: IconData

**buildREPORTinstance-attribute**

REPORT: IconData

**buildREPORT\_GMAILERRORREDinstance-attribute**

REPORT\_GMAILERRORRED: IconData

**buildREPORT\_GMAILERRORRED\_OUTLINEDinstance-attribute**

REPORT\_GMAILERRORRED\_OUTLINED: IconData

**buildREPORT\_GMAILERRORRED\_ROUNDEDinstance-attribute**

REPORT\_GMAILERRORRED\_ROUNDED: IconData

**buildREPORT\_GMAILERRORRED\_SHARPinstance-attribute**

REPORT\_GMAILERRORRED\_SHARP: IconData

**buildREPORT\_OFFinstance-attribute**

REPORT\_OFF: IconData

**buildREPORT\_OFF\_OUTLINEDinstance-attribute**

REPORT\_OFF\_OUTLINED: IconData

**buildREPORT\_OFF\_ROUNDEDinstance-attribute**

REPORT\_OFF\_ROUNDED: IconData

**buildREPORT\_OFF\_SHARPinstance-attribute**

REPORT\_OFF\_SHARP: IconData

**buildREPORT\_OUTLINEDinstance-attribute**

REPORT\_OUTLINED: IconData

**buildREPORT\_PROBLEMinstance-attribute**

REPORT\_PROBLEM: IconData

**buildREPORT\_PROBLEM\_OUTLINEDinstance-attribute**

REPORT\_PROBLEM\_OUTLINED: IconData

**buildREPORT\_PROBLEM\_ROUNDEDinstance-attribute**

REPORT\_PROBLEM\_ROUNDED: IconData

**buildREPORT\_PROBLEM\_SHARPinstance-attribute**

REPORT\_PROBLEM\_SHARP: IconData

**buildREPORT\_ROUNDEDinstance-attribute**

REPORT\_ROUNDED: IconData

**buildREPORT\_SHARPinstance-attribute**

REPORT\_SHARP: IconData

**buildREQUEST\_PAGEinstance-attribute**

REQUEST\_PAGE: IconData

**buildREQUEST\_PAGE\_OUTLINEDinstance-attribute**

REQUEST\_PAGE\_OUTLINED: IconData

**buildREQUEST\_PAGE\_ROUNDEDinstance-attribute**

REQUEST\_PAGE\_ROUNDED: IconData

**buildREQUEST\_PAGE\_SHARPinstance-attribute**

REQUEST\_PAGE\_SHARP: IconData

**buildREQUEST\_QUOTEinstance-attribute**

REQUEST\_QUOTE: IconData

**buildREQUEST\_QUOTE\_OUTLINEDinstance-attribute**

REQUEST\_QUOTE\_OUTLINED: IconData

**buildREQUEST\_QUOTE\_ROUNDEDinstance-attribute**

REQUEST\_QUOTE\_ROUNDED: IconData

**buildREQUEST\_QUOTE\_SHARPinstance-attribute**

REQUEST\_QUOTE\_SHARP: IconData

**buildRESET\_TVinstance-attribute**

RESET\_TV: IconData

**buildRESET\_TV\_OUTLINEDinstance-attribute**

RESET\_TV\_OUTLINED: IconData

**buildRESET\_TV\_ROUNDEDinstance-attribute**

RESET\_TV\_ROUNDED: IconData

**buildRESET\_TV\_SHARPinstance-attribute**

RESET\_TV\_SHARP: IconData

**buildRESTART\_ALTinstance-attribute**

RESTART\_ALT: IconData

**buildRESTART\_ALT\_OUTLINEDinstance-attribute**

RESTART\_ALT\_OUTLINED: IconData

**buildRESTART\_ALT\_ROUNDEDinstance-attribute**

RESTART\_ALT\_ROUNDED: IconData

**buildRESTART\_ALT\_SHARPinstance-attribute**

RESTART\_ALT\_SHARP: IconData

**buildRESTAURANTinstance-attribute**

RESTAURANT: IconData

**buildRESTAURANT\_MENUinstance-attribute**

RESTAURANT\_MENU: IconData

**buildRESTAURANT\_MENU\_OUTLINEDinstance-attribute**

RESTAURANT\_MENU\_OUTLINED: IconData



**buildRESTAURANT\_MENU\_ROUNDEDinstance-attribute**

RESTAURANT\_MENU\_ROUNDED: IconData

**buildRESTAURANT\_MENU\_SHARPinstance-attribute**

RESTAURANT\_MENU\_SHARP: IconData

**buildRESTAURANT\_OUTLINEDinstance-attribute**

RESTAURANT\_OUTLINED: IconData

**buildRESTAURANT\_ROUNDEDinstance-attribute**

RESTAURANT\_ROUNDED: IconData

**buildRESTAURANT\_SHARPinstance-attribute**

RESTAURANT\_SHARP: IconData

**buildRESTOREinstance-attribute**

RESTORE: IconData

**buildRESTORE\_FROM\_TRASHinstance-attribute**

RESTORE\_FROM\_TRASH: IconData

**buildRESTORE\_FROM\_TRASH\_OUTLINEDinstance-attribute**

RESTORE\_FROM\_TRASH\_OUTLINED: IconData

**buildRESTORE\_FROM\_TRASH\_ROUNDEDinstance-attribute**

RESTORE\_FROM\_TRASH\_ROUNDED: IconData

**buildRESTORE\_FROM\_TRASH\_SHARPinstance-attribute**

RESTORE\_FROM\_TRASH\_SHARP: IconData

**buildRESTORE\_OUTLINEDinstance-attribute**

RESTORE\_OUTLINED: IconData

**buildRESTORE\_PAGEinstance-attribute**

RESTORE\_PAGE: IconData

**buildRESTORE\_PAGE\_OUTLINEDinstance-attribute**

RESTORE\_PAGE\_OUTLINED: IconData

**buildRESTORE\_PAGE\_ROUNDEDinstance-attribute**

RESTORE\_PAGE\_ROUNDED: IconData

**buildRESTORE\_PAGE\_SHARPinstance-attribute**

RESTORE\_PAGE\_SHARP: IconData

**buildRESTORE\_ROUNDEDinstance-attribute**

RESTORE\_ROUNDED: IconData

**buildRESTORE\_SHARPinstance-attribute**

RESTORE\_SHARP: IconData

**buildREVIEWSinstance-attribute**

REVIEWS: IconData

**buildREVIEWS\_OUTLINEDinstance-attribute**

REVIEWS\_OUTLINED: IconData

**buildREVIEWS\_ROUNDEDinstance-attribute**

REVIEWS\_ROUNDED: IconData

**buildREVIEWS\_SHARPinstance-attribute**

REVIEWS\_SHARP: IconData

**buildRICE\_BOWLinstance-attribute**

RICE\_BOWL: IconData

**buildRICE\_BOWL\_OUTLINED**instance-attribute

RICE\_BOWL\_OUTLINED: IconData

**buildRICE\_BOWL\_ROUNDED**instance-attribute

RICE\_BOWL\_ROUNDED: IconData

**buildRICE\_BOWL\_SHARP**instance-attribute

RICE\_BOWL\_SHARP: IconData

**buildRING\_VOLUME**instance-attribute

RING\_VOLUME: IconData

**buildRING\_VOLUME\_OUTLINED**instance-attribute

RING\_VOLUME\_OUTLINED: IconData

**buildRING\_VOLUME\_ROUNDED**instance-attribute

RING\_VOLUME\_ROUNDED: IconData

**buildRING\_VOLUME\_SHARP**instance-attribute

RING\_VOLUME\_SHARP: IconData

**buildROCKET**instance-attribute

ROCKET: IconData

**buildROCKET\_LAUNCH**instance-attribute

ROCKET\_LAUNCH: IconData

**buildROCKET\_LAUNCH\_OUTLINED**instance-attribute

ROCKET\_LAUNCH\_OUTLINED: IconData

**buildROCKET\_LAUNCH\_ROUNDED**instance-attribute

ROCKET\_LAUNCH\_ROUNDED: IconData

**buildROCKET\_LAUNCH\_SHARP**instance-attribute

ROCKET\_LAUNCH\_SHARP: IconData

**buildROCKET\_OUTLINED**instance-attribute

ROCKET\_OUTLINED: IconData

**buildROCKET\_ROUNDED**instance-attribute

ROCKET\_ROUNDED: IconData

**buildROCKET\_SHARP**instance-attribute

ROCKET\_SHARP: IconData

**buildROLLER\_SHADES**instance-attribute

ROLLER\_SHADES: IconData

**buildROLLER\_SHADES\_CLOSED**instance-attribute

ROLLER\_SHADES\_CLOSED: IconData

**buildROLLER\_SHADES\_CLOSED\_OUTLINED**instance-attribute

ROLLER\_SHADES\_CLOSED\_OUTLINED: IconData

**buildROLLER\_SHADES\_CLOSED\_ROUNDED**instance-attribute

ROLLER\_SHADES\_CLOSED\_ROUNDED: IconData

**buildROLLER\_SHADES\_CLOSED\_SHARP**instance-attribute

ROLLER\_SHADES\_CLOSED\_SHARP: IconData

**buildROLLER\_SHADES\_OUTLINED**instance-attribute

ROLLER\_SHADES\_OUTLINED: IconData

**buildROLLER\_SHADES\_ROUNDED**instance-attribute

ROLLER\_SHADES\_ROUNDED: IconData

**buildROLLER\_SHADES\_SHARPinstance-attribute**

ROLLER\_SHADES\_SHARP: IconData

**buildROLLER\_SKATINGinstance-attribute**

ROLLER\_SKATING: IconData

**buildROLLER\_SKATING\_OUTLINEDinstance-attribute**

ROLLER\_SKATING\_OUTLINED: IconData

**buildROLLER\_SKATING\_ROUNDEDinstance-attribute**

ROLLER\_SKATING\_ROUNDED: IconData

**buildROLLER\_SKATING\_SHARPinstance-attribute**

ROLLER\_SKATING\_SHARP: IconData

**buildROOFINGinstance-attribute**

ROOFING: IconData

**buildROOFING\_OUTLINEDinstance-attribute**

ROOFING\_OUTLINED: IconData

**buildROOFING\_ROUNDEDinstance-attribute**

ROOFING\_ROUNDED: IconData

**buildROOFING\_SHARPinstance-attribute**

ROOFING\_SHARP: IconData

**buildROOMinstance-attribute**

ROOM: IconData

**buildROOM\_OUTLINEDinstance-attribute**

ROOM\_OUTLINED: IconData

**buildROOM\_PREFERENCESinstance-attribute**

ROOM\_PREFERENCES: IconData

**buildROOM\_PREFERENCES\_OUTLINEDinstance-attribute**

ROOM\_PREFERENCES\_OUTLINED: IconData

**buildROOM\_PREFERENCES\_ROUNDEDinstance-attribute**

ROOM\_PREFERENCES\_ROUNDED: IconData

**buildROOM\_PREFERENCES\_SHARPinstance-attribute**

ROOM\_PREFERENCES\_SHARP: IconData

**buildROOM\_ROUNDEDinstance-attribute**

ROOM\_ROUNDED: IconData

**buildROOM\_SERVICEinstance-attribute**

ROOM\_SERVICE: IconData

**buildROOM\_SERVICE\_OUTLINEDinstance-attribute**

ROOM\_SERVICE\_OUTLINED: IconData

**buildROOM\_SERVICE\_ROUNDEDinstance-attribute**

ROOM\_SERVICE\_ROUNDED: IconData

**buildROOM\_SERVICE\_SHARPinstance-attribute**

ROOM\_SERVICE\_SHARP: IconData

**buildROOM\_SHARPinstance-attribute**

ROOM\_SHARP: IconData

**buildROTATE\_90\_DEGREES\_CCWinstance-attribute**

ROTATE\_90\_DEGREES\_CCW: IconData

buildROTATE\_90\_DEGREES\_CCW\_OUTLINEDinstance-attribute

ROTATE\_90\_DEGREES\_CCW\_OUTLINED: IconData

buildROTATE\_90\_DEGREES\_CCW\_ROUNDEDinstance-attribute

ROTATE\_90\_DEGREES\_CCW\_ROUNDED: IconData

buildROTATE\_90\_DEGREES\_CCW\_SHARPinstance-attribute

ROTATE\_90\_DEGREES\_CCW\_SHARP: IconData

buildROTATE\_90\_DEGREES\_CWinstance-attribute

ROTATE\_90\_DEGREES\_CW: IconData

buildROTATE\_90\_DEGREES\_CW\_OUTLINEDinstance-attribute

ROTATE\_90\_DEGREES\_CW\_OUTLINED: IconData

buildROTATE\_90\_DEGREES\_CW\_ROUNDEDinstance-attribute

ROTATE\_90\_DEGREES\_CW\_ROUNDED: IconData

buildROTATE\_90\_DEGREES\_CW\_SHARPinstance-attribute

ROTATE\_90\_DEGREES\_CW\_SHARP: IconData

buildROTATE\_LEFTinstance-attribute

ROTATE\_LEFT: IconData

buildROTATE\_LEFT\_OUTLINEDinstance-attribute

ROTATE\_LEFT\_OUTLINED: IconData

buildROTATE\_LEFT\_ROUNDEDinstance-attribute

ROTATE\_LEFT\_ROUNDED: IconData

buildROTATE\_LEFT\_SHARPinstance-attribute

ROTATE\_LEFT\_SHARP: IconData

buildROTATE\_RIGHTinstance-attribute

ROTATE\_RIGHT: IconData

buildROTATE\_RIGHT\_OUTLINEDinstance-attribute

ROTATE\_RIGHT\_OUTLINED: IconData

buildROTATE\_RIGHT\_ROUNDEDinstance-attribute

ROTATE\_RIGHT\_ROUNDED: IconData

buildROTATE\_RIGHT\_SHARPinstance-attribute

ROTATE\_RIGHT\_SHARP: IconData

buildROUNDBOUT\_LEFTinstance-attribute

ROUNDBOUT\_LEFT: IconData

buildROUNDBOUT\_LEFT\_OUTLINEDinstance-attribute

ROUNDBOUT\_LEFT\_OUTLINED: IconData

buildROUNDBOUT\_LEFT\_ROUNDEDinstance-attribute

ROUNDBOUT\_LEFT\_ROUNDED: IconData

buildROUNDBOUT\_LEFT\_SHARPinstance-attribute

ROUNDBOUT\_LEFT\_SHARP: IconData

buildROUNDBOUT\_RIGHTinstance-attribute

ROUNDBOUT\_RIGHT: IconData

buildROUNDBOUT\_RIGHT\_OUTLINEDinstance-attribute

ROUNDBOUT\_RIGHT\_OUTLINED: IconData

buildROUNDBOUT\_RIGHT\_ROUNDEDinstance-attribute

ROUNDBOUT\_RIGHT\_ROUNDED: IconData

**buildROUNDAABOUT\_RIGHT\_SHARPinstance-attribute**

ROUNDAABOUT\_RIGHT\_SHARP: IconData

**buildROUNDED\_CORNERinstance-attribute**

ROUNDED\_CORNER: IconData

**buildROUNDED\_CORNER\_OUTLINEDinstance-attribute**

ROUNDED\_CORNER\_OUTLINED: IconData

**buildROUNDED\_CORNER\_ROUNDEDinstance-attribute**

ROUNDED\_CORNER\_ROUNDED: IconData

**buildROUNDED\_CORNER\_SHARPinstance-attribute**

ROUNDED\_CORNER\_SHARP: IconData

**buildROUTEinstance-attribute**

ROUTE: IconData

**buildROUTE\_OUTLINEDinstance-attribute**

ROUTE\_OUTLINED: IconData

**buildROUTE\_ROUNDEDinstance-attribute**

ROUTE\_ROUNDED: IconData

**buildROUTE\_SHARPinstance-attribute**

ROUTE\_SHARP: IconData

**buildROUTERinstance-attribute**

ROUTER: IconData

**buildROUTER\_OUTLINEDinstance-attribute**

ROUTER\_OUTLINED: IconData

**buildROUTER\_ROUNDEDinstance-attribute**

ROUTER\_ROUNDED: IconData

**buildROUTER\_SHARPinstance-attribute**

ROUTER\_SHARP: IconData

**buildROWINGinstance-attribute**

ROWING: IconData

**buildROWING\_OUTLINEDinstance-attribute**

ROWING\_OUTLINED: IconData

**buildROWING\_ROUNDEDinstance-attribute**

ROWING\_ROUNDED: IconData

**buildROWING\_SHARPinstance-attribute**

ROWING\_SHARP: IconData

**buildRSS\_FEEDinstance-attribute**

RSS\_FEED: IconData

**buildRSS\_FEED\_OUTLINEDinstance-attribute**

RSS\_FEED\_OUTLINED: IconData

**buildRSS\_FEED\_ROUNDEDinstance-attribute**

RSS\_FEED\_ROUNDED: IconData

**buildRSS\_FEED\_SHARPinstance-attribute**

RSS\_FEED\_SHARP: IconData

**buildRSVPinstance-attribute**

RSVP: IconData

**buildRSVP\_OUTLINEDinstance-attribute**

RSVP\_OUTLINED: IconData

**buildRSVP\_ROUNDEDinstance-attribute**

RSVP\_ROUNDED: IconData

**buildRSVP\_SHARPinstance-attribute**

RSVP\_SHARP: IconData

**buildRTTinstance-attribute**

RTT: IconData

**buildRTT\_OUTLINEDinstance-attribute**

RTT\_OUTLINED: IconData

**buildRTT\_ROUNDEDinstance-attribute**

RTT\_ROUNDED: IconData

**buildRTT\_SHARPinstance-attribute**

RTT\_SHARP: IconData

**buildRULEinstance-attribute**

RULE: IconData

**buildRULE\_FOLDERinstance-attribute**

RULE\_FOLDER: IconData

**buildRULE\_FOLDER\_OUTLINEDinstance-attribute**

RULE\_FOLDER\_OUTLINED: IconData

**buildRULE\_FOLDER\_ROUNDEDinstance-attribute**

RULE\_FOLDER\_ROUNDED: IconData

**buildRULE\_FOLDER\_SHARPinstance-attribute**

RULE\_FOLDER\_SHARP: IconData

**buildRULE\_OUTLINEDinstance-attribute**

RULE\_OUTLINED: IconData

**buildRULE\_ROUNDEDinstance-attribute**

RULE\_ROUNDED: IconData

**buildRULE\_SHARPinstance-attribute**

RULE\_SHARP: IconData

**buildRUN\_CIRCLEinstance-attribute**

RUN\_CIRCLE: IconData

**buildRUN\_CIRCLE\_OUTLINEDinstance-attribute**

RUN\_CIRCLE\_OUTLINED: IconData

**buildRUN\_CIRCLE\_ROUNDEDinstance-attribute**

RUN\_CIRCLE\_ROUNDED: IconData

**buildRUN\_CIRCLE\_SHARPinstance-attribute**

RUN\_CIRCLE\_SHARP: IconData

**buildRUNNING\_WITH\_ERRORSinstance-attribute**

RUNNING\_WITH\_ERRORS: IconData

**buildRUNNING\_WITH\_ERRORS\_OUTLINEDinstance-attribute**

RUNNING\_WITH\_ERRORS\_OUTLINED: IconData

**buildRUNNING\_WITH\_ERRORS\_ROUNDEDinstance-attribute**

RUNNING\_WITH\_ERRORS\_ROUNDED: IconData

**buildRUNNING\_WITH\_ERRORS\_SHARPinstance-attribute**

RUNNING\_WITH\_ERRORS\_SHARP: IconData

**buildRV\_HOOKUPinstance-attribute**

RV\_HOOKUP: IconData

**buildRV\_HOOKUP\_OUTLINEDinstance-attribute**

RV\_HOOKUP\_OUTLINED: IconData

**buildRV\_HOOKUP\_ROUNDEDinstance-attribute**

RV\_HOOKUP\_ROUNDED: IconData

**buildRV\_HOOKUP\_SHARPinstance-attribute**

RV\_HOOKUP\_SHARP: IconData

**buildSAFETY\_CHECKinstance-attribute**

SAFETY\_CHECK: IconData

**buildSAFETY\_CHECK\_OUTLINEDinstance-attribute**

SAFETY\_CHECK\_OUTLINED: IconData

**buildSAFETY\_CHECK\_ROUNDEDinstance-attribute**

SAFETY\_CHECK\_ROUNDED: IconData

**buildSAFETY\_CHECK\_SHARPinstance-attribute**

SAFETY\_CHECK\_SHARP: IconData

**buildSAFETY\_DIVIDERinstance-attribute**

SAFETY\_DIVIDER: IconData

**buildSAFETY\_DIVIDER\_OUTLINEDinstance-attribute**

SAFETY\_DIVIDER\_OUTLINED: IconData

**buildSAFETY\_DIVIDER\_ROUNDEDinstance-attribute**

SAFETY\_DIVIDER\_ROUNDED: IconData

**buildSAFETY\_DIVIDER\_SHARPinstance-attribute**

SAFETY\_DIVIDER\_SHARP: IconData

**buildSAILINGinstance-attribute**

SAILING: IconData

**buildSAILING\_OUTLINEDinstance-attribute**

SAILING\_OUTLINED: IconData

**buildSAILING\_ROUNDEDinstance-attribute**

SAILING\_ROUNDED: IconData

**buildSAILING\_SHARPinstance-attribute**

SAILING\_SHARP: IconData

**buildSANITIZERinstance-attribute**

SANITIZER: IconData

**buildSANITIZER\_OUTLINEDinstance-attribute**

SANITIZER\_OUTLINED: IconData

**buildSANITIZER\_ROUNDEDinstance-attribute**

SANITIZER\_ROUNDED: IconData

**buildSANITIZER\_SHARPinstance-attribute**

SANITIZER\_SHARP: IconData

**buildSATELLITEinstance-attribute**

SATELLITE: IconData

**buildSATELLITE\_ALTinstance-attribute**

SATELLITE\_ALT: IconData

**buildSATELLITE\_ALT\_OUTLINEDinstance-attribute**

SATELLITE\_ALT\_OUTLINED: IconData

**buildSATELLITE\_ALT\_ROUNDEDinstance-attribute**

SATELLITE\_ALT\_ROUNDED: IconData

**buildSATELLITE\_ALT\_SHARPinstance-attribute**

SATELLITE\_ALT\_SHARP: IconData

**buildSATELLITE\_OUTLINEDinstance-attribute**

SATELLITE\_OUTLINED: IconData

**buildSATELLITE\_ROUNDEDinstance-attribute**

SATELLITE\_ROUNDED: IconData

**buildSATELLITE\_SHARPinstance-attribute**

SATELLITE\_SHARP: IconData

**buildSAVEinstance-attribute**

SAVE: IconData

**buildSAVE\_ALTinstance-attribute**

SAVE\_ALT: IconData

**buildSAVE\_ALT\_OUTLINEDinstance-attribute**

SAVE\_ALT\_OUTLINED: IconData

**buildSAVE\_ALT\_ROUNDEDinstance-attribute**

SAVE\_ALT\_ROUNDED: IconData

**buildSAVE\_ALT\_SHARPinstance-attribute**

SAVE\_ALT\_SHARP: IconData

**buildSAVE\_ASinstance-attribute**

SAVE\_AS: IconData

**buildSAVE\_AS\_OUTLINEDinstance-attribute**

SAVE\_AS\_OUTLINED: IconData

**buildSAVE\_AS\_ROUNDEDinstance-attribute**

SAVE\_AS\_ROUNDED: IconData

**buildSAVE\_AS\_SHARPinstance-attribute**

SAVE\_AS\_SHARP: IconData

**buildSAVE\_OUTLINEDinstance-attribute**

SAVE\_OUTLINED: IconData

**buildSAVE\_ROUNDEDinstance-attribute**

SAVE\_ROUNDED: IconData

**buildSAVE\_SHARPinstance-attribute**

SAVE\_SHARP: IconData

**buildSAVED\_SEARCHinstance-attribute**

SAVED\_SEARCH: IconData

**buildSAVED\_SEARCH\_OUTLINEDinstance-attribute**

SAVED\_SEARCH\_OUTLINED: IconData

**buildSAVED\_SEARCH\_ROUNDEDinstance-attribute**

SAVED\_SEARCH\_ROUNDED: IconData



**buildSAVED\_SEARCH\_SHARPinstance-attribute**

SAVED\_SEARCH\_SHARP: IconData

**buildSAVINGSinstance-attribute**

SAVINGS: IconData

**buildSAVINGS\_OUTLINEDinstance-attribute**

SAVINGS\_OUTLINED: IconData

**buildSAVINGS\_ROUNDEDinstance-attribute**

SAVINGS\_ROUNDED: IconData

**buildSAVINGS\_SHARPinstance-attribute**

SAVINGS\_SHARP: IconData

**buildSCALEinstance-attribute**

SCALE: IconData

**buildSCALE\_OUTLINEDinstance-attribute**

SCALE\_OUTLINED: IconData

**buildSCALE\_ROUNDEDinstance-attribute**

SCALE\_ROUNDED: IconData

**buildSCALE\_SHARPinstance-attribute**

SCALE\_SHARP: IconData

**buildSCANNERinstance-attribute**

SCANNER: IconData

**buildSCANNER\_OUTLINEDinstance-attribute**

SCANNER\_OUTLINED: IconData

**buildSCANNER\_ROUNDEDinstance-attribute**

SCANNER\_ROUNDED: IconData

**buildSCANNER\_SHARPinstance-attribute**

SCANNER\_SHARP: IconData

**buildSCATTER\_PLOTinstance-attribute**

SCATTER\_PLOT: IconData

**buildSCATTER\_PLOT\_OUTLINEDinstance-attribute**

SCATTER\_PLOT\_OUTLINED: IconData

**buildSCATTER\_PLOT\_ROUNDEDinstance-attribute**

SCATTER\_PLOT\_ROUNDED: IconData

**buildSCATTER\_PLOT\_SHARPinstance-attribute**

SCATTER\_PLOT\_SHARP: IconData

**buildSCHEDULEinstance-attribute**

SCHEDULE: IconData

**buildSCHEDULE\_OUTLINEDinstance-attribute**

SCHEDULE\_OUTLINED: IconData

**buildSCHEDULE\_ROUNDEDinstance-attribute**

SCHEDULE\_ROUNDED: IconData

**buildSCHEDULE\_SENDinstance-attribute**

SCHEDULE\_SEND: IconData

**buildSCHEDULE\_SEND\_OUTLINEDinstance-attribute**

SCHEDULE\_SEND\_OUTLINED: IconData

**buildSCHEDULE\_SEND\_ROUNDEDinstance-attribute**

SCHEDULE\_SEND\_ROUNDED: IconData

**buildSCHEDULE\_SEND\_SHARPinstance-attribute**

SCHEDULE\_SEND\_SHARP: IconData

**buildSCHEDULE\_SHARPinstance-attribute**

SCHEDULE\_SHARP: IconData

**buildSCHEMAinstance-attribute**

SCHEMA: IconData

**buildSCHEMA\_OUTLINEDinstance-attribute**

SCHEMA\_OUTLINED: IconData

**buildSCHEMA\_ROUNDEDinstance-attribute**

SCHEMA\_ROUNDED: IconData

**buildSCHEMA\_SHARPinstance-attribute**

SCHEMA\_SHARP: IconData

**buildSCHOOLinstance-attribute**

SCHOOL: IconData

**buildSCHOOL\_OUTLINEDinstance-attribute**

SCHOOL\_OUTLINED: IconData

**buildSCHOOL\_ROUNDEDinstance-attribute**

SCHOOL\_ROUNDED: IconData

**buildSCHOOL\_SHARPinstance-attribute**

SCHOOL\_SHARP: IconData

**buildSCIENCEinstance-attribute**

SCIENCE: IconData

**buildSCIENCE\_OUTLINEDinstance-attribute**

SCIENCE\_OUTLINED: IconData

**buildSCIENCE\_ROUNDEDinstance-attribute**

SCIENCE\_ROUNDED: IconData

**buildSCIENCE\_SHARPinstance-attribute**

SCIENCE\_SHARP: IconData

**buildSCOREinstance-attribute**

SCORE: IconData

**buildSCORE\_OUTLINEDinstance-attribute**

SCORE\_OUTLINED: IconData

**buildSCORE\_ROUNDEDinstance-attribute**

SCORE\_ROUNDED: IconData

**buildSCORE\_SHARPinstance-attribute**

SCORE\_SHARP: IconData

**buildSCOREBOARDinstance-attribute**

SCOREBOARD: IconData

**buildSCOREBOARD\_OUTLINEDinstance-attribute**

SCOREBOARD\_OUTLINED: IconData

**buildSCOREBOARD\_ROUNDEDinstance-attribute**

SCOREBOARD\_ROUNDED: IconData

**buildSCOREBOARD\_SHARPinstance-attribute**

SCOREBOARD\_SHARP: IconData

**buildSCREEN\_LOCK\_LANDSCAPEinstance-attribute**

SCREEN\_LOCK\_LANDSCAPE: IconData

**buildSCREEN\_LOCK\_LANDSCAPE\_OUTLINEDinstance-attribute**

SCREEN\_LOCK\_LANDSCAPE\_OUTLINED: IconData

**buildSCREEN\_LOCK\_LANDSCAPE\_ROUNDEDinstance-attribute**

SCREEN\_LOCK\_LANDSCAPE\_ROUNDED: IconData

**buildSCREEN\_LOCK\_LANDSCAPE\_SHARPinstance-attribute**

SCREEN\_LOCK\_LANDSCAPE\_SHARP: IconData

**buildSCREEN\_LOCK\_PORTRAITinstance-attribute**

SCREEN\_LOCK\_PORTRAIT: IconData

**buildSCREEN\_LOCK\_PORTRAIT\_OUTLINEDinstance-attribute**

SCREEN\_LOCK\_PORTRAIT\_OUTLINED: IconData

**buildSCREEN\_LOCK\_PORTRAIT\_ROUNDEDinstance-attribute**

SCREEN\_LOCK\_PORTRAIT\_ROUNDED: IconData

**buildSCREEN\_LOCK\_PORTRAIT\_SHARPinstance-attribute**

SCREEN\_LOCK\_PORTRAIT\_SHARP: IconData

**buildSCREEN\_LOCK\_ROTATIONinstance-attribute**

SCREEN\_LOCK\_ROTATION: IconData

**buildSCREEN\_LOCK\_ROTATION\_OUTLINEDinstance-attribute**

SCREEN\_LOCK\_ROTATION\_OUTLINED: IconData

**buildSCREEN\_LOCK\_ROTATION\_ROUNDEDinstance-attribute**

SCREEN\_LOCK\_ROTATION\_ROUNDED: IconData

**buildSCREEN\_LOCK\_ROTATION\_SHARPinstance-attribute**

SCREEN\_LOCK\_ROTATION\_SHARP: IconData

**buildSCREEN\_ROTATIONinstance-attribute**

SCREEN\_ROTATION: IconData

**buildSCREEN\_ROTATION\_ALTinstance-attribute**

SCREEN\_ROTATION\_ALT: IconData

**buildSCREEN\_ROTATION\_ALT\_OUTLINEDinstance-attribute**

SCREEN\_ROTATION\_ALT\_OUTLINED: IconData

**buildSCREEN\_ROTATION\_ALT\_ROUNDEDinstance-attribute**

SCREEN\_ROTATION\_ALT\_ROUNDED: IconData

**buildSCREEN\_ROTATION\_ALT\_SHARPinstance-attribute**

SCREEN\_ROTATION\_ALT\_SHARP: IconData

**buildSCREEN\_ROTATION\_OUTLINEDinstance-attribute**

SCREEN\_ROTATION\_OUTLINED: IconData

**buildSCREEN\_ROTATION\_ROUNDEDinstance-attribute**

SCREEN\_ROTATION\_ROUNDED: IconData

**buildSCREEN\_ROTATION\_SHARPinstance-attribute**

SCREEN\_ROTATION\_SHARP: IconData

**buildSCREEN\_SEARCH\_DESKTOPinstance-attribute**

SCREEN\_SEARCH\_DESKTOP: IconData

**buildSCREEN\_SEARCH\_DESKTOP\_OUTLINEDinstance-attribute**

SCREEN\_SEARCH\_DESKTOP\_OUTLINED: IconData

**buildSCREEN\_SEARCH\_DESKTOP\_ROUNDEDinstance-attribute**

SCREEN\_SEARCH\_DESKTOP\_ROUNDED: IconData

**buildSCREEN\_SEARCH\_DESKTOP\_SHARPinstance-attribute**

SCREEN\_SEARCH\_DESKTOP\_SHARP: IconData

**buildSCREEN\_SHAREinstance-attribute**

SCREEN\_SHARE: IconData

**buildSCREEN\_SHARE\_OUTLINEDinstance-attribute**

SCREEN\_SHARE\_OUTLINED: IconData

**buildSCREEN\_SHARE\_ROUNDEDinstance-attribute**

SCREEN\_SHARE\_ROUNDED: IconData

**buildSCREEN\_SHARE\_SHARPinstance-attribute**

SCREEN\_SHARE\_SHARP: IconData

**buildSCREENSHOTinstance-attribute**

SCREENSHOT: IconData

**buildSCREENSHOT\_MONITORinstance-attribute**

SCREENSHOT\_MONITOR: IconData

**buildSCREENSHOT\_MONITOR\_OUTLINEDinstance-attribute**

SCREENSHOT\_MONITOR\_OUTLINED: IconData

**buildSCREENSHOT\_MONITOR\_ROUNDEDinstance-attribute**

SCREENSHOT\_MONITOR\_ROUNDED: IconData

**buildSCREENSHOT\_MONITOR\_SHARPinstance-attribute**

SCREENSHOT\_MONITOR\_SHARP: IconData

**buildSCREENSHOT\_OUTLINEDinstance-attribute**

SCREENSHOT\_OUTLINED: IconData

**buildSCREENSHOT\_ROUNDEDinstance-attribute**

SCREENSHOT\_ROUNDED: IconData

**buildSCREENSHOT\_SHARPinstance-attribute**

SCREENSHOT\_SHARP: IconData

**buildSCUBA\_DIVINGinstance-attribute**

SCUBA\_DIVING: IconData

**buildSCUBA\_DIVING\_OUTLINEDinstance-attribute**

SCUBA\_DIVING\_OUTLINED: IconData

**buildSCUBA\_DIVING\_ROUNDEDinstance-attribute**

SCUBA\_DIVING\_ROUNDED: IconData

**buildSCUBA\_DIVING\_SHARPinstance-attribute**

SCUBA\_DIVING\_SHARP: IconData

**buildSDinstance-attribute**

SD: IconData

**buildSD\_CARDinstance-attribute**

SD\_CARD: IconData

**buildSD\_CARD\_ALERTinstance-attribute**

SD\_CARD\_ALERT: IconData

**buildSD\_CARD\_ALERT\_OUTLINED**instance-attribute

SD\_CARD\_ALERT\_OUTLINED: IconData

**buildSD\_CARD\_ALERT\_ROUNDED**instance-attribute

SD\_CARD\_ALERT\_ROUNDED: IconData

**buildSD\_CARD\_ALERT\_SHARP**instance-attribute

SD\_CARD\_ALERT\_SHARP: IconData

**buildSD\_CARD\_OUTLINED**instance-attribute

SD\_CARD\_OUTLINED: IconData

**buildSD\_CARD\_ROUNDED**instance-attribute

SD\_CARD\_ROUNDED: IconData

**buildSD\_CARD\_SHARP**instance-attribute

SD\_CARD\_SHARP: IconData

**buildSD\_OUTLINED**instance-attribute

SD\_OUTLINED: IconData

**buildSD\_ROUNDED**instance-attribute

SD\_ROUNDED: IconData

**buildSD\_SHARP**instance-attribute

SD\_SHARP: IconData

**buildSD\_STORAGE**instance-attribute

SD\_STORAGE: IconData

**buildSD\_STORAGE\_OUTLINED**instance-attribute

SD\_STORAGE\_OUTLINED: IconData

**buildSD\_STORAGE\_ROUNDED**instance-attribute

SD\_STORAGE\_ROUNDED: IconData

**buildSD\_STORAGE\_SHARP**instance-attribute

SD\_STORAGE\_SHARP: IconData

**buildSEARCH**instance-attribute

SEARCH: IconData

**buildSEARCH\_OFF**instance-attribute

SEARCH\_OFF: IconData

**buildSEARCH\_OFF\_OUTLINED**instance-attribute

SEARCH\_OFF\_OUTLINED: IconData

**buildSEARCH\_OFF\_ROUNDED**instance-attribute

SEARCH\_OFF\_ROUNDED: IconData

**buildSEARCH\_OFF\_SHARP**instance-attribute

SEARCH\_OFF\_SHARP: IconData

**buildSEARCH\_OUTLINED**instance-attribute

SEARCH\_OUTLINED: IconData

**buildSEARCH\_ROUNDED**instance-attribute

SEARCH\_ROUNDED: IconData

**buildSEARCH\_SHARP**instance-attribute

SEARCH\_SHARP: IconData

**buildSECURITY**instance-attribute

SECURITY: IconData

**buildSECURITY\_OUTLINEDinstance-attribute**

SECURITY\_OUTLINED: IconData

**buildSECURITY\_ROUNDEDinstance-attribute**

SECURITY\_ROUNDED: IconData

**buildSECURITY\_SHARPinstance-attribute**

SECURITY\_SHARP: IconData

**buildSECURITY\_UPDATEinstance-attribute**

SECURITY\_UPDATE: IconData

**buildSECURITY\_UPDATE\_GOODinstance-attribute**

SECURITY\_UPDATE\_GOOD: IconData

**buildSECURITY\_UPDATE\_GOOD\_OUTLINEDinstance-attribute**

SECURITY\_UPDATE\_GOOD\_OUTLINED: IconData

**buildSECURITY\_UPDATE\_GOOD\_ROUNDEDinstance-attribute**

SECURITY\_UPDATE\_GOOD\_ROUNDED: IconData

**buildSECURITY\_UPDATE\_GOOD\_SHARPinstance-attribute**

SECURITY\_UPDATE\_GOOD\_SHARP: IconData

**buildSECURITY\_UPDATE\_OUTLINEDinstance-attribute**

SECURITY\_UPDATE\_OUTLINED: IconData

**buildSECURITY\_UPDATE\_ROUNDEDinstance-attribute**

SECURITY\_UPDATE\_ROUNDED: IconData

**buildSECURITY\_UPDATE\_SHARPinstance-attribute**

SECURITY\_UPDATE\_SHARP: IconData

**buildSECURITY\_UPDATE\_WARNINGinstance-attribute**

SECURITY\_UPDATE\_WARNING: IconData

**buildSECURITY\_UPDATE\_WARNING\_OUTLINEDinstance-attribute**

SECURITY\_UPDATE\_WARNING\_OUTLINED: IconData

**buildSECURITY\_UPDATE\_WARNING\_ROUNDEDinstance-attribute**

SECURITY\_UPDATE\_WARNING\_ROUNDED: IconData

**buildSECURITY\_UPDATE\_WARNING\_SHARPinstance-attribute**

SECURITY\_UPDATE\_WARNING\_SHARP: IconData

**buildSEGMENTinstance-attribute**

SEGMENT: IconData

**buildSEGMENT\_OUTLINEDinstance-attribute**

SEGMENT\_OUTLINED: IconData

**buildSEGMENT\_ROUNDEDinstance-attribute**

SEGMENT\_ROUNDED: IconData

**buildSEGMENT\_SHARPinstance-attribute**

SEGMENT\_SHARP: IconData

**buildSELECT\_ALLinstance-attribute**

SELECT\_ALL: IconData

**buildSELECT\_ALL\_OUTLINEDinstance-attribute**

SELECT\_ALL\_OUTLINED: IconData

**buildSELECT\_ALL\_ROUNDEDinstance-attribute**

SELECT\_ALL\_ROUNDED: IconData

**buildSELECT\_ALL\_SHARPinstance-attribute**

SELECT\_ALL\_SHARP: IconData

**buildSELF\_IMPROVEMENTinstance-attribute**

SELF\_IMPROVEMENT: IconData

**buildSELF\_IMPROVEMENT\_OUTLINEDinstance-attribute**

SELF\_IMPROVEMENT\_OUTLINED: IconData

**buildSELF\_IMPROVEMENT\_ROUNDEDinstance-attribute**

SELF\_IMPROVEMENT\_ROUNDED: IconData

**buildSELF\_IMPROVEMENT\_SHARPinstance-attribute**

SELF\_IMPROVEMENT\_SHARP: IconData

**buildSELLinstance-attribute**

SELL: IconData

**buildSELL\_OUTLINEDinstance-attribute**

SELL\_OUTLINED: IconData

**buildSELL\_ROUNDEDinstance-attribute**

SELL\_ROUNDED: IconData

**buildSELL\_SHARPinstance-attribute**

SELL\_SHARP: IconData

**buildSENDinstance-attribute**

SEND: IconData

**buildSEND\_AND\_ARCHIVEinstance-attribute**

SEND\_AND\_ARCHIVE: IconData

**buildSEND\_AND\_ARCHIVE\_OUTLINEDinstance-attribute**

SEND\_AND\_ARCHIVE\_OUTLINED: IconData

**buildSEND\_AND\_ARCHIVE\_ROUNDEDinstance-attribute**

SEND\_AND\_ARCHIVE\_ROUNDED: IconData

**buildSEND\_AND\_ARCHIVE\_SHARPinstance-attribute**

SEND\_AND\_ARCHIVE\_SHARP: IconData

**buildSEND\_OUTLINEDinstance-attribute**

SEND\_OUTLINED: IconData

**buildSEND\_ROUNDEDinstance-attribute**

SEND\_ROUNDED: IconData

**buildSEND\_SHARPinstance-attribute**

SEND\_SHARP: IconData

**buildSEND\_TIME\_EXTENSIONinstance-attribute**

SEND\_TIME\_EXTENSION: IconData

**buildSEND\_TIME\_EXTENSION\_OUTLINEDinstance-attribute**

SEND\_TIME\_EXTENSION\_OUTLINED: IconData

**buildSEND\_TIME\_EXTENSION\_ROUNDEDinstance-attribute**

SEND\_TIME\_EXTENSION\_ROUNDED: IconData

**buildSEND\_TIME\_EXTENSION\_SHARPinstance-attribute**

SEND\_TIME\_EXTENSION\_SHARP: IconData

**buildSEND\_TO\_MOBILEinstance-attribute**

SEND\_TO\_MOBILE: IconData

**buildSEND\_TO\_MOBILE\_OUTLINEDinstance-attribute**

SEND\_TO\_MOBILE\_OUTLINED: IconData

**buildSEND\_TO\_MOBILE\_ROUNDEDinstance-attribute**

SEND\_TO\_MOBILE\_ROUNDED: IconData

**buildSEND\_TO\_MOBILE\_SHARPinstance-attribute**

SEND\_TO\_MOBILE\_SHARP: IconData

**buildSENSOR\_DOORinstance-attribute**

SENSOR\_DOOR: IconData

**buildSENSOR\_DOOR\_OUTLINEDinstance-attribute**

SENSOR\_DOOR\_OUTLINED: IconData

**buildSENSOR\_DOOR\_ROUNDEDinstance-attribute**

SENSOR\_DOOR\_ROUNDED: IconData

**buildSENSOR\_DOOR\_SHARPinstance-attribute**

SENSOR\_DOOR\_SHARP: IconData

**buildSENSOR\_OCCUPIEDinstance-attribute**

SENSOR\_OCCUPIED: IconData

**buildSENSOR\_OCCUPIED\_OUTLINEDinstance-attribute**

SENSOR\_OCCUPIED\_OUTLINED: IconData

**buildSENSOR\_OCCUPIED\_ROUNDEDinstance-attribute**

SENSOR\_OCCUPIED\_ROUNDED: IconData

**buildSENSOR\_OCCUPIED\_SHARPinstance-attribute**

SENSOR\_OCCUPIED\_SHARP: IconData

**buildSENSOR\_WINDOWinstance-attribute**

SENSOR\_WINDOW: IconData

**buildSENSOR\_WINDOW\_OUTLINEDinstance-attribute**

SENSOR\_WINDOW\_OUTLINED: IconData

**buildSENSOR\_WINDOW\_ROUNDEDinstance-attribute**

SENSOR\_WINDOW\_ROUNDED: IconData

**buildSENSOR\_WINDOW\_SHARPinstance-attribute**

SENSOR\_WINDOW\_SHARP: IconData

**buildSENSORSinstance-attribute**

SENSORS: IconData

**buildSENSORS\_OFFinstance-attribute**

SENSORS\_OFF: IconData

**buildSENSORS\_OFF\_OUTLINEDinstance-attribute**

SENSORS\_OFF\_OUTLINED: IconData

**buildSENSORS\_OFF\_ROUNDEDinstance-attribute**

SENSORS\_OFF\_ROUNDED: IconData

**buildSENSORS\_OFF\_SHARPinstance-attribute**

SENSORS\_OFF\_SHARP: IconData

**buildSENSORS\_OUTLINEDinstance-attribute**

SENSORS\_OUTLINED: IconData

**buildSENSORS\_ROUNDEDinstance-attribute**

SENSORS\_ROUNDED: IconData



**buildSENSORS\_SHARPinstance-attribute**

SENSORS\_SHARP: IconData

**buildSENTIMENT\_DISSATISFIEDinstance-attribute**

SENTIMENT\_DISSATISFIED: IconData

**buildSENTIMENT\_DISSATISFIED\_OUTLINEDinstance-attribute**

SENTIMENT\_DISSATISFIED\_OUTLINED: IconData

**buildSENTIMENT\_DISSATISFIED\_ROUNDEDinstance-attribute**

SENTIMENT\_DISSATISFIED\_ROUNDED: IconData

**buildSENTIMENT\_DISSATISFIED\_SHARPinstance-attribute**

SENTIMENT\_DISSATISFIED\_SHARP: IconData

**buildSENTIMENT\_NEUTRALinstance-attribute**

SENTIMENT\_NEUTRAL: IconData

**buildSENTIMENT\_NEUTRAL\_OUTLINEDinstance-attribute**

SENTIMENT\_NEUTRAL\_OUTLINED: IconData

**buildSENTIMENT\_NEUTRAL\_ROUNDEDinstance-attribute**

SENTIMENT\_NEUTRAL\_ROUNDED: IconData

**buildSENTIMENT\_NEUTRAL\_SHARPinstance-attribute**

SENTIMENT\_NEUTRAL\_SHARP: IconData

**buildSENTIMENT\_SATISFIEDinstance-attribute**

SENTIMENT\_SATISFIED: IconData

**buildSENTIMENT\_SATISFIED\_ALTinstance-attribute**

SENTIMENT\_SATISFIED\_ALT: IconData

**buildSENTIMENT\_SATISFIED\_ALT\_OUTLINEDinstance-attribute**

SENTIMENT\_SATISFIED\_ALT\_OUTLINED: IconData

**buildSENTIMENT\_SATISFIED\_ALT\_ROUNDEDinstance-attribute**

SENTIMENT\_SATISFIED\_ALT\_ROUNDED: IconData

**buildSENTIMENT\_SATISFIED\_ALT\_SHARPinstance-attribute**

SENTIMENT\_SATISFIED\_ALT\_SHARP: IconData

**buildSENTIMENT\_SATISFIED\_OUTLINEDinstance-attribute**

SENTIMENT\_SATISFIED\_OUTLINED: IconData

**buildSENTIMENT\_SATISFIED\_ROUNDEDinstance-attribute**

SENTIMENT\_SATISFIED\_ROUNDED: IconData

**buildSENTIMENT\_SATISFIED\_SHARPinstance-attribute**

SENTIMENT\_SATISFIED\_SHARP: IconData

**buildSENTIMENT\_VERY\_DISSATISFIEDinstance-attribute**

SENTIMENT\_VERY\_DISSATISFIED: IconData

**buildSENTIMENT\_VERY\_DISSATISFIED\_OUTLINEDinstance-attribute**

SENTIMENT\_VERY\_DISSATISFIED\_OUTLINED: IconData

**buildSENTIMENT\_VERY\_DISSATISFIED\_ROUNDEDinstance-attribute**

SENTIMENT\_VERY\_DISSATISFIED\_ROUNDED: IconData

**buildSENTIMENT\_VERY\_DISSATISFIED\_SHARPinstance-attribute**

SENTIMENT\_VERY\_DISSATISFIED\_SHARP: IconData

**buildSENTIMENT\_VERY\_SATISFIEDinstance-attribute**

SENTIMENT\_VERY\_SATISFIED: IconData

**buildSENTIMENT\_VERY\_SATISFIED\_OUTLINEDinstance-attribute**

SENTIMENT\_VERY\_SATISFIED\_OUTLINED: IconData

**buildSENTIMENT\_VERY\_SATISFIED\_ROUNDEDinstance-attribute**

SENTIMENT\_VERY\_SATISFIED\_ROUNDED: IconData

**buildSENTIMENT\_VERY\_SATISFIED\_SHARPinstance-attribute**

SENTIMENT\_VERY\_SATISFIED\_SHARP: IconData

**buildSET\_MEALinstance-attribute**

SET\_MEAL: IconData

**buildSET\_MEAL\_OUTLINEDinstance-attribute**

SET\_MEAL\_OUTLINED: IconData

**buildSET\_MEAL\_ROUNDEDinstance-attribute**

SET\_MEAL\_ROUNDED: IconData

**buildSET\_MEAL\_SHARPinstance-attribute**

SET\_MEAL\_SHARP: IconData

**buildSETTINGSinstance-attribute**

SETTINGS: IconData

**buildSETTINGS\_ACCESSIBILITYinstance-attribute**

SETTINGS\_ACCESSIBILITY: IconData

**buildSETTINGS\_ACCESSIBILITY\_OUTLINEDinstance-attribute**

SETTINGS\_ACCESSIBILITY\_OUTLINED: IconData

**buildSETTINGS\_ACCESSIBILITY\_ROUNDEDinstance-attribute**

SETTINGS\_ACCESSIBILITY\_ROUNDED: IconData

**buildSETTINGS\_ACCESSIBILITY\_SHARPinstance-attribute**

SETTINGS\_ACCESSIBILITY\_SHARP: IconData

**buildSETTINGS\_APPLICATIONSinstance-attribute**

SETTINGS\_APPLICATIONS: IconData

**buildSETTINGS\_APPLICATIONS\_OUTLINEDinstance-attribute**

SETTINGS\_APPLICATIONS\_OUTLINED: IconData

**buildSETTINGS\_APPLICATIONS\_ROUNDEDinstance-attribute**

SETTINGS\_APPLICATIONS\_ROUNDED: IconData

**buildSETTINGS\_APPLICATIONS\_SHARPinstance-attribute**

SETTINGS\_APPLICATIONS\_SHARP: IconData

**buildSETTINGS\_BACKUP\_RESTOREinstance-attribute**

SETTINGS\_BACKUP\_RESTORE: IconData

**buildSETTINGS\_BACKUP\_RESTORE\_OUTLINEDinstance-attribute**

SETTINGS\_BACKUP\_RESTORE\_OUTLINED: IconData

**buildSETTINGS\_BACKUP\_RESTORE\_ROUNDEDinstance-attribute**

SETTINGS\_BACKUP\_RESTORE\_ROUNDED: IconData

**buildSETTINGS\_BACKUP\_RESTORE\_SHARPinstance-attribute**

SETTINGS\_BACKUP\_RESTORE\_SHARP: IconData

**buildSETTINGS\_BLUETOOTHinstance-attribute**

SETTINGS\_BLUETOOTH: IconData

**buildSETTINGS\_BLUETOOTH\_OUTLINEDinstance-attribute**

SETTINGS\_BLUETOOTH\_OUTLINED: IconData

**buildSETTINGS\_BLUETOOTH\_ROUNDEDinstance-attribute**

SETTINGS\_BLUETOOTH\_ROUNDED: IconData

**buildSETTINGS\_BLUETOOTH\_SHARPinstance-attribute**

SETTINGS\_BLUETOOTH\_SHARP: IconData

**buildSETTINGS\_BRIGHTNESSinstance-attribute**

SETTINGS\_BRIGHTNESS: IconData

**buildSETTINGS\_BRIGHTNESS\_OUTLINEDinstance-attribute**

SETTINGS\_BRIGHTNESS\_OUTLINED: IconData

**buildSETTINGS\_BRIGHTNESS\_ROUNDEDinstance-attribute**

SETTINGS\_BRIGHTNESS\_ROUNDED: IconData

**buildSETTINGS\_BRIGHTNESS\_SHARPinstance-attribute**

SETTINGS\_BRIGHTNESS\_SHARP: IconData

**buildSETTINGS\_CELLinstance-attribute**

SETTINGS\_CELL: IconData

**buildSETTINGS\_CELL\_OUTLINEDinstance-attribute**

SETTINGS\_CELL\_OUTLINED: IconData

**buildSETTINGS\_CELL\_ROUNDEDinstance-attribute**

SETTINGS\_CELL\_ROUNDED: IconData

**buildSETTINGS\_CELL\_SHARPinstance-attribute**

SETTINGS\_CELL\_SHARP: IconData

**buildSETTINGS\_DISPLAYinstance-attribute**

SETTINGS\_DISPLAY: IconData

**buildSETTINGS\_DISPLAY\_OUTLINEDinstance-attribute**

SETTINGS\_DISPLAY\_OUTLINED: IconData

**buildSETTINGS\_DISPLAY\_ROUNDEDinstance-attribute**

SETTINGS\_DISPLAY\_ROUNDED: IconData

**buildSETTINGS\_DISPLAY\_SHARPinstance-attribute**

SETTINGS\_DISPLAY\_SHARP: IconData

**buildSETTINGS\_ETHERNETinstance-attribute**

SETTINGS\_ETHERNET: IconData

**buildSETTINGS\_ETHERNET\_OUTLINEDinstance-attribute**

SETTINGS\_ETHERNET\_OUTLINED: IconData

**buildSETTINGS\_ETHERNET\_ROUNDEDinstance-attribute**

SETTINGS\_ETHERNET\_ROUNDED: IconData

**buildSETTINGS\_ETHERNET\_SHARPinstance-attribute**

SETTINGS\_ETHERNET\_SHARP: IconData

**buildSETTINGS\_INPUT\_ANTENNAinstance-attribute**

SETTINGS\_INPUT\_ANTENNA: IconData

**buildSETTINGS\_INPUT\_ANTENNA\_OUTLINEDinstance-attribute**

SETTINGS\_INPUT\_ANTENNA\_OUTLINED: IconData

**buildSETTINGS\_INPUT\_ANTENNA\_ROUNDEDinstance-attribute**

SETTINGS\_INPUT\_ANTENNA\_ROUNDED: IconData

**buildSETTINGS\_INPUT\_ANTENNA\_SHARPinstance-attribute**

SETTINGS\_INPUT\_ANTENNA\_SHARP: IconData

**buildSETTINGS\_INPUT\_COMPONENTinstance-attribute**

SETTINGS\_INPUT\_COMPONENT: IconData

**buildSETTINGS\_INPUT\_COMPONENT\_OUTLINEDinstance-attribute**

SETTINGS\_INPUT\_COMPONENT\_OUTLINED: IconData

**buildSETTINGS\_INPUT\_COMPONENT\_ROUNDEDinstance-attribute**

SETTINGS\_INPUT\_COMPONENT\_ROUNDED: IconData

**buildSETTINGS\_INPUT\_COMPONENT\_SHARPinstance-attribute**

SETTINGS\_INPUT\_COMPONENT\_SHARP: IconData

**buildSETTINGS\_INPUT\_COMPOSITEinstance-attribute**

SETTINGS\_INPUT\_COMPOSITE: IconData

**buildSETTINGS\_INPUT\_COMPOSITE\_OUTLINEDinstance-attribute**

SETTINGS\_INPUT\_COMPOSITE\_OUTLINED: IconData

**buildSETTINGS\_INPUT\_COMPOSITE\_ROUNDEDinstance-attribute**

SETTINGS\_INPUT\_COMPOSITE\_ROUNDED: IconData

**buildSETTINGS\_INPUT\_COMPOSITE\_SHARPinstance-attribute**

SETTINGS\_INPUT\_COMPOSITE\_SHARP: IconData

**buildSETTINGS\_INPUT\_HDMIinstance-attribute**

SETTINGS\_INPUT\_HDMI: IconData

**buildSETTINGS\_INPUT\_HDMI\_OUTLINEDinstance-attribute**

SETTINGS\_INPUT\_HDMI\_OUTLINED: IconData

**buildSETTINGS\_INPUT\_HDMI\_ROUNDEDinstance-attribute**

SETTINGS\_INPUT\_HDMI\_ROUNDED: IconData

**buildSETTINGS\_INPUT\_HDMI\_SHARPinstance-attribute**

SETTINGS\_INPUT\_HDMI\_SHARP: IconData

**buildSETTINGS\_INPUT\_SVIDEOinstance-attribute**

SETTINGS\_INPUT\_SVIDEO: IconData

**buildSETTINGS\_INPUT\_SVIDEO\_OUTLINEDinstance-attribute**

SETTINGS\_INPUT\_SVIDEO\_OUTLINED: IconData

**buildSETTINGS\_INPUT\_SVIDEO\_ROUNDEDinstance-attribute**

SETTINGS\_INPUT\_SVIDEO\_ROUNDED: IconData

**buildSETTINGS\_INPUT\_SVIDEO\_SHARPinstance-attribute**

SETTINGS\_INPUT\_SVIDEO\_SHARP: IconData

**buildSETTINGS\_OUTLINEDinstance-attribute**

SETTINGS\_OUTLINED: IconData

**buildSETTINGS\_OVERSCANinstance-attribute**

SETTINGS\_OVERSCAN: IconData

**buildSETTINGS\_OVERSCAN\_OUTLINEDinstance-attribute**

SETTINGS\_OVERSCAN\_OUTLINED: IconData

**buildSETTINGS\_OVERSCAN\_ROUNDEDinstance-attribute**

SETTINGS\_OVERSCAN\_ROUNDED: IconData

**buildSETTINGS\_OVERSCAN\_SHARPinstance-attribute**

SETTINGS\_OVERSCAN\_SHARP: IconData

**buildSETTINGS\_PHONEinstance-attribute**

SETTINGS\_PHONE: IconData

**buildSETTINGS\_PHONE\_OUTLINEDinstance-attribute**

SETTINGS\_PHONE\_OUTLINED: IconData

**buildSETTINGS\_PHONE\_ROUNDEDinstance-attribute**

SETTINGS\_PHONE\_ROUNDED: IconData

**buildSETTINGS\_PHONE\_SHARPinstance-attribute**

SETTINGS\_PHONE\_SHARP: IconData

**buildSETTINGS\_POWERinstance-attribute**

SETTINGS\_POWER: IconData

**buildSETTINGS\_POWER\_OUTLINEDinstance-attribute**

SETTINGS\_POWER\_OUTLINED: IconData

**buildSETTINGS\_POWER\_ROUNDEDinstance-attribute**

SETTINGS\_POWER\_ROUNDED: IconData

**buildSETTINGS\_POWER\_SHARPinstance-attribute**

SETTINGS\_POWER\_SHARP: IconData

**buildSETTINGS\_REMOTEinstance-attribute**

SETTINGS\_REMOTE: IconData

**buildSETTINGS\_REMOTE\_OUTLINEDinstance-attribute**

SETTINGS\_REMOTE\_OUTLINED: IconData

**buildSETTINGS\_REMOTE\_ROUNDEDinstance-attribute**

SETTINGS\_REMOTE\_ROUNDED: IconData

**buildSETTINGS\_REMOTE\_SHARPinstance-attribute**

SETTINGS\_REMOTE\_SHARP: IconData

**buildSETTINGS\_ROUNDEDinstance-attribute**

SETTINGS\_ROUNDED: IconData

**buildSETTINGS\_SHARPinstance-attribute**

SETTINGS\_SHARP: IconData

**buildSETTINGS\_SUGGESTinstance-attribute**

SETTINGS\_SUGGEST: IconData

**buildSETTINGS\_SUGGEST\_OUTLINEDinstance-attribute**

SETTINGS\_SUGGEST\_OUTLINED: IconData

**buildSETTINGS\_SUGGEST\_ROUNDEDinstance-attribute**

SETTINGS\_SUGGEST\_ROUNDED: IconData

**buildSETTINGS\_SUGGEST\_SHARPinstance-attribute**

SETTINGS\_SUGGEST\_SHARP: IconData

**buildSETTINGS\_SYSTEM\_DAYDREAMinstance-attribute**

SETTINGS\_SYSTEM\_DAYDREAM: IconData

**buildSETTINGS\_SYSTEM\_DAYDREAM\_OUTLINEDinstance-attribute**

SETTINGS\_SYSTEM\_DAYDREAM\_OUTLINED: IconData

**buildSETTINGS\_SYSTEM\_DAYDREAM\_ROUNDEDinstance-attribute**

SETTINGS\_SYSTEM\_DAYDREAM\_ROUNDED: IconData

**buildSETTINGS\_SYSTEM\_DAYDREAM\_SHARPinstance-attribute**

SETTINGS\_SYSTEM\_DAYDREAM\_SHARP: IconData

**buildSETTINGS\_VOICEinstance-attribute**

SETTINGS\_VOICE: IconData

buildSETTINGS\_VOICE\_OUTLINEDinstance-attribute

SETTINGS\_VOICE\_OUTLINED: IconData

buildSETTINGS\_VOICE\_ROUNDEDinstance-attribute

SETTINGS\_VOICE\_ROUNDED: IconData

buildSETTINGS\_VOICE\_SHARPinstance-attribute

SETTINGS\_VOICE\_SHARP: IconData

buildSEVEN\_Kinstance-attribute

SEVEN\_K: IconData

buildSEVEN\_K\_OUTLINEDinstance-attribute

SEVEN\_K\_OUTLINED: IconData

buildSEVEN\_K\_PLUSinstance-attribute

SEVEN\_K\_PLUS: IconData

buildSEVEN\_K\_PLUS\_OUTLINEDinstance-attribute

SEVEN\_K\_PLUS\_OUTLINED: IconData

buildSEVEN\_K\_PLUS\_ROUNDEDinstance-attribute

SEVEN\_K\_PLUS\_ROUNDED: IconData

buildSEVEN\_K\_PLUS\_SHARPinstance-attribute

SEVEN\_K\_PLUS\_SHARP: IconData

buildSEVEN\_K\_ROUNDEDinstance-attribute

SEVEN\_K\_ROUNDED: IconData

buildSEVEN\_K\_SHARPinstance-attribute

SEVEN\_K\_SHARP: IconData

buildSEVEN\_MPinstance-attribute

SEVEN\_MP: IconData

buildSEVEN\_MP\_OUTLINEDinstance-attribute

SEVEN\_MP\_OUTLINED: IconData

buildSEVEN\_MP\_ROUNDEDinstance-attribute

SEVEN\_MP\_ROUNDED: IconData

buildSEVEN\_MP\_SHARPinstance-attribute

SEVEN\_MP\_SHARP: IconData

buildSEVENTEEN\_MPinstance-attribute

SEVENTEEN\_MP: IconData

buildSEVENTEEN\_MP\_OUTLINEDinstance-attribute

SEVENTEEN\_MP\_OUTLINED: IconData

buildSEVENTEEN\_MP\_ROUNDEDinstance-attribute

SEVENTEEN\_MP\_ROUNDED: IconData

buildSEVENTEEN\_MP\_SHARPinstance-attribute

SEVENTEEN\_MP\_SHARP: IconData

buildSEVERE\_COLDinstance-attribute

SEVERE\_COLD: IconData

buildSEVERE\_COLD\_OUTLINEDinstance-attribute

SEVERE\_COLD\_OUTLINED: IconData

buildSEVERE\_COLD\_ROUNDEDinstance-attribute

SEVERE\_COLD\_ROUNDED: IconData

**buildSEVERE\_COLD\_SHARPinstance-attribute**

SEVERE\_COLD\_SHARP: IconData

**buildSHAPE\_LINEinstance-attribute**

SHAPE\_LINE: IconData

**buildSHAPE\_LINE\_OUTLINEDinstance-attribute**

SHAPE\_LINE\_OUTLINED: IconData

**buildSHAPE\_LINE\_ROUNDEDinstance-attribute**

SHAPE\_LINE\_ROUNDED: IconData

**buildSHAPE\_LINE\_SHARPinstance-attribute**

SHAPE\_LINE\_SHARP: IconData

**buildSHAREinstance-attribute**

SHARE: IconData

**buildSHARE\_ARRIVAL\_TIMEinstance-attribute**

SHARE\_ARRIVAL\_TIME: IconData

**buildSHARE\_ARRIVAL\_TIME\_OUTLINEDinstance-attribute**

SHARE\_ARRIVAL\_TIME\_OUTLINED: IconData

**buildSHARE\_ARRIVAL\_TIME\_ROUNDEDinstance-attribute**

SHARE\_ARRIVAL\_TIME\_ROUNDED: IconData

**buildSHARE\_ARRIVAL\_TIME\_SHARPinstance-attribute**

SHARE\_ARRIVAL\_TIME\_SHARP: IconData

**buildSHARE\_LOCATIONinstance-attribute**

SHARE\_LOCATION: IconData

**buildSHARE\_LOCATION\_OUTLINEDinstance-attribute**

SHARE\_LOCATION\_OUTLINED: IconData

**buildSHARE\_LOCATION\_ROUNDEDinstance-attribute**

SHARE\_LOCATION\_ROUNDED: IconData

**buildSHARE\_LOCATION\_SHARPinstance-attribute**

SHARE\_LOCATION\_SHARP: IconData

**buildSHARE\_OUTLINEDinstance-attribute**

SHARE\_OUTLINED: IconData

**buildSHARE\_ROUNDEDinstance-attribute**

SHARE\_ROUNDED: IconData

**buildSHARE\_SHARPinstance-attribute**

SHARE\_SHARP: IconData

**buildSHELVESinstance-attribute**

SHELVES: IconData

**buildSHIELDinstance-attribute**

SHIELD: IconData

**buildSHIELD\_MOONinstance-attribute**

SHIELD\_MOON: IconData

**buildSHIELD\_MOON\_OUTLINEDinstance-attribute**

SHIELD\_MOON\_OUTLINED: IconData

**buildSHIELD\_MOON\_ROUNDEDinstance-attribute**

SHIELD\_MOON\_ROUNDED: IconData

**buildSHIELD\_MOON\_SHARPinstance-attribute**

SHIELD\_MOON\_SHARP: IconData

**buildSHIELD\_OUTLINEDinstance-attribute**

SHIELD\_OUTLINED: IconData

**buildSHIELD\_ROUNDEDinstance-attribute**

SHIELD\_ROUNDED: IconData

**buildSHIELD\_SHARPinstance-attribute**

SHIELD\_SHARP: IconData

**buildSHOPinstance-attribute**

SHOP: IconData

**buildSHOP\_2instance-attribute**

SHOP\_2: IconData

**buildSHOP\_2\_OUTLINEDinstance-attribute**

SHOP\_2\_OUTLINED: IconData

**buildSHOP\_2\_ROUNDEDinstance-attribute**

SHOP\_2\_ROUNDED: IconData

**buildSHOP\_2\_SHARPinstance-attribute**

SHOP\_2\_SHARP: IconData

**buildSHOP\_OUTLINEDinstance-attribute**

SHOP\_OUTLINED: IconData

**buildSHOP\_ROUNDEDinstance-attribute**

SHOP\_ROUNDED: IconData

**buildSHOP\_SHARPinstance-attribute**

SHOP\_SHARP: IconData

**buildSHOP\_TWOinstance-attribute**

SHOP\_TWO: IconData

**buildSHOP\_TWO\_OUTLINEDinstance-attribute**

SHOP\_TWO\_OUTLINED: IconData

**buildSHOP\_TWO\_ROUNDEDinstance-attribute**

SHOP\_TWO\_ROUNDED: IconData

**buildSHOP\_TWO\_SHARPinstance-attribute**

SHOP\_TWO\_SHARP: IconData

**buildSHOPIFYinstance-attribute**

SHOPIFY: IconData

**buildSHOPIFY\_OUTLINEDinstance-attribute**

SHOPIFY\_OUTLINED: IconData

**buildSHOPIFY\_ROUNDEDinstance-attribute**

SHOPIFY\_ROUNDED: IconData

**buildSHOPIFY\_SHARPinstance-attribute**

SHOPIFY\_SHARP: IconData

**buildSHOPPING\_BAGinstance-attribute**

SHOPPING\_BAG: IconData

**buildSHOPPING\_BAG\_OUTLINEDinstance-attribute**

SHOPPING\_BAG\_OUTLINED: IconData



**buildSHOPPING\_BAG\_ROUNDEDinstance-attribute**

SHOPPING\_BAG\_ROUNDED: IconData

**buildSHOPPING\_BAG\_SHARPinstance-attribute**

SHOPPING\_BAG\_SHARP: IconData

**buildSHOPPING\_BASKETinstance-attribute**

SHOPPING\_BASKET: IconData

**buildSHOPPING\_BASKET\_OUTLINEDinstance-attribute**

SHOPPING\_BASKET\_OUTLINED: IconData

**buildSHOPPING\_BASKET\_ROUNDEDinstance-attribute**

SHOPPING\_BASKET\_ROUNDED: IconData

**buildSHOPPING\_BASKET\_SHARPinstance-attribute**

SHOPPING\_BASKET\_SHARP: IconData

**buildSHOPPING\_CARTinstance-attribute**

SHOPPING\_CART: IconData

**buildSHOPPING\_CART\_CHECKOUTinstance-attribute**

SHOPPING\_CART\_CHECKOUT: IconData

**buildSHOPPING\_CART\_CHECKOUT\_OUTLINEDinstance-attribute**

SHOPPING\_CART\_CHECKOUT\_OUTLINED: IconData

**buildSHOPPING\_CART\_CHECKOUT\_ROUNDEDinstance-attribute**

SHOPPING\_CART\_CHECKOUT\_ROUNDED: IconData

**buildSHOPPING\_CART\_CHECKOUT\_SHARPinstance-attribute**

SHOPPING\_CART\_CHECKOUT\_SHARP: IconData

**buildSHOPPING\_CART\_OUTLINEDinstance-attribute**

SHOPPING\_CART\_OUTLINED: IconData

**buildSHOPPING\_CART\_ROUNDEDinstance-attribute**

SHOPPING\_CART\_ROUNDED: IconData

**buildSHOPPING\_CART\_SHARPinstance-attribute**

SHOPPING\_CART\_SHARP: IconData

**buildSHORT\_TEXTinstance-attribute**

SHORT\_TEXT: IconData

**buildSHORT\_TEXT\_OUTLINEDinstance-attribute**

SHORT\_TEXT\_OUTLINED: IconData

**buildSHORT\_TEXT\_ROUNDEDinstance-attribute**

SHORT\_TEXT\_ROUNDED: IconData

**buildSHORT\_TEXT\_SHARPinstance-attribute**

SHORT\_TEXT\_SHARP: IconData

**buildSHORTCUTinstance-attribute**

SHORTCUT: IconData

**buildSHORTCUT\_OUTLINEDinstance-attribute**

SHORTCUT\_OUTLINED: IconData

**buildSHORTCUT\_ROUNDEDinstance-attribute**

SHORTCUT\_ROUNDED: IconData

**buildSHORTCUT\_SHARPinstance-attribute**

SHORTCUT\_SHARP: IconData

**buildSHOW\_CHARTinstance-attribute**

SHOW\_CHART: IconData

**buildSHOW\_CHART\_OUTLINEDinstance-attribute**

SHOW\_CHART\_OUTLINED: IconData

**buildSHOW\_CHART\_ROUNDEDinstance-attribute**

SHOW\_CHART\_ROUNDED: IconData

**buildSHOW\_CHART\_SHARPinstance-attribute**

SHOW\_CHART\_SHARP: IconData

**buildSHOWERinstance-attribute**

SHOWER: IconData

**buildSHOWER\_OUTLINEDinstance-attribute**

SHOWER\_OUTLINED: IconData

**buildSHOWER\_ROUNDEDinstance-attribute**

SHOWER\_ROUNDED: IconData

**buildSHOWER\_SHARPinstance-attribute**

SHOWER\_SHARP: IconData

**buildSHUFFLEinstance-attribute**

SHUFFLE: IconData

**buildSHUFFLE\_ONinstance-attribute**

SHUFFLE\_ON: IconData

**buildSHUFFLE\_ON\_OUTLINEDinstance-attribute**

SHUFFLE\_ON\_OUTLINED: IconData

**buildSHUFFLE\_ON\_ROUNDEDinstance-attribute**

SHUFFLE\_ON\_ROUNDED: IconData

**buildSHUFFLE\_ON\_SHARPinstance-attribute**

SHUFFLE\_ON\_SHARP: IconData

**buildSHUFFLE\_OUTLINEDinstance-attribute**

SHUFFLE\_OUTLINED: IconData

**buildSHUFFLE\_ROUNDEDinstance-attribute**

SHUFFLE\_ROUNDED: IconData

**buildSHUFFLE\_SHARPinstance-attribute**

SHUFFLE\_SHARP: IconData

**buildSHUTTER\_SPEEDinstance-attribute**

SHUTTER\_SPEED: IconData

**buildSHUTTER\_SPEED\_OUTLINEDinstance-attribute**

SHUTTER\_SPEED\_OUTLINED: IconData

**buildSHUTTER\_SPEED\_ROUNDEDinstance-attribute**

SHUTTER\_SPEED\_ROUNDED: IconData

**buildSHUTTER\_SPEED\_SHARPinstance-attribute**

SHUTTER\_SPEED\_SHARP: IconData

**buildSICKinstance-attribute**

SICK: IconData

**buildSICK\_OUTLINEDinstance-attribute**

SICK\_OUTLINED: IconData

buildSICK\_ROUNDEDinstance-attribute

SICK\_ROUNDED: IconData

buildSICK\_SHARPinstance-attribute

SICK\_SHARP: IconData

buildSIGN\_LANGUAGEinstance-attribute

SIGN\_LANGUAGE: IconData

buildSIGN\_LANGUAGE\_OUTLINEDinstance-attribute

SIGN\_LANGUAGE\_OUTLINED: IconData

buildSIGN\_LANGUAGE\_ROUNDEDinstance-attribute

SIGN\_LANGUAGE\_ROUNDED: IconData

buildSIGN\_LANGUAGE\_SHARPinstance-attribute

SIGN\_LANGUAGE\_SHARP: IconData

buildSIGNAL\_CELLULAR\_0\_BARinstance-attribute

SIGNAL\_CELLULAR\_0\_BAR: IconData

buildSIGNAL\_CELLULAR\_0\_BAR\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_0\_BAR\_OUTLINED: IconData

buildSIGNAL\_CELLULAR\_0\_BAR\_ROUNDEDinstance-attribute

SIGNAL\_CELLULAR\_0\_BAR\_ROUNDED: IconData

buildSIGNAL\_CELLULAR\_0\_BAR\_SHARPinstance-attribute

SIGNAL\_CELLULAR\_0\_BAR\_SHARP: IconData

buildSIGNAL\_CELLULAR\_4\_BARinstance-attribute

SIGNAL\_CELLULAR\_4\_BAR: IconData

buildSIGNAL\_CELLULAR\_4\_BAR\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_4\_BAR\_OUTLINED: IconData

buildSIGNAL\_CELLULAR\_4\_BAR\_ROUNDEDinstance-attribute

SIGNAL\_CELLULAR\_4\_BAR\_ROUNDED: IconData

buildSIGNAL\_CELLULAR\_4\_BAR\_SHARPinstance-attribute

SIGNAL\_CELLULAR\_4\_BAR\_SHARP: IconData

buildSIGNAL\_CELLULAR\_ALTinstance-attribute

SIGNAL\_CELLULAR\_ALT: IconData

buildSIGNAL\_CELLULAR\_ALT\_1\_BARinstance-attribute

SIGNAL\_CELLULAR\_ALT\_1\_BAR: IconData

buildSIGNAL\_CELLULAR\_ALT\_1\_BAR\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_ALT\_1\_BAR\_OUTLINED: IconData

buildSIGNAL\_CELLULAR\_ALT\_1\_BAR\_ROUNDEDinstance-attribute

SIGNAL\_CELLULAR\_ALT\_1\_BAR\_ROUNDED: IconData

buildSIGNAL\_CELLULAR\_ALT\_1\_BAR\_SHARPinstance-attribute

SIGNAL\_CELLULAR\_ALT\_1\_BAR\_SHARP: IconData

buildSIGNAL\_CELLULAR\_ALT\_2\_BARinstance-attribute

SIGNAL\_CELLULAR\_ALT\_2\_BAR: IconData

buildSIGNAL\_CELLULAR\_ALT\_2\_BAR\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_ALT\_2\_BAR\_OUTLINED: IconData

buildSIGNAL\_CELLULAR\_ALT\_2\_BAR\_ROUNDEDinstance-attribute

SIGNAL\_CELLULAR\_ALT\_2\_BAR\_ROUNDED: IconData

buildSIGNAL\_CELLULAR\_ALT\_2\_BAR\_SHARPinstance-attribute

SIGNAL\_CELLULAR\_ALT\_2\_BAR\_SHARP: IconData

buildSIGNAL\_CELLULAR\_ALT\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_ALT\_OUTLINED: IconData

buildSIGNAL\_CELLULAR\_ALT\_ROUNDEDinstance-attribute

SIGNAL\_CELLULAR\_ALT\_ROUNDED: IconData

buildSIGNAL\_CELLULAR\_ALT\_SHARPinstance-attribute

SIGNAL\_CELLULAR\_ALT\_SHARP: IconData

buildSIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BARinstance-attribute

SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR: IconData

buildSIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_OUTLINED: (  
IconData  
)

buildSIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_ROUNDEDinstance-attribute

SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_ROUNDED: (  
IconData  
)

buildSIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_SHARPinstance-attribute

SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_0\_BAR\_SHARP: IconData

buildSIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BARinstance-attribute

SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR: IconData

buildSIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_OUTLINED: (  
IconData  
)

buildSIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_ROUNDEDinstance-attribute

SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_ROUNDED: (  
IconData  
)

buildSIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_SHARPinstance-attribute

SIGNAL\_CELLULAR\_CONNECTED\_NO\_INTERNET\_4\_BAR\_SHARP: IconData

buildSIGNAL\_CELLULAR\_NO\_SIMinstance-attribute

SIGNAL\_CELLULAR\_NO\_SIM: IconData

buildSIGNAL\_CELLULAR\_NO\_SIM\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_NO\_SIM\_OUTLINED: IconData

buildSIGNAL\_CELLULAR\_NO\_SIM\_ROUNDEDinstance-attribute

SIGNAL\_CELLULAR\_NO\_SIM\_ROUNDED: IconData

buildSIGNAL\_CELLULAR\_NO\_SIM\_SHARPinstance-attribute

SIGNAL\_CELLULAR\_NO\_SIM\_SHARP: IconData

buildSIGNAL\_CELLULAR\_NODATAinstance-attribute

SIGNAL\_CELLULAR\_NODATA: IconData

buildSIGNAL\_CELLULAR\_NODATA\_OUTLINEDinstance-attribute

SIGNAL\_CELLULAR\_NODATA\_OUTLINED: IconData

**buildSIGNAL\_CELLULAR\_NODATA\_ROUNDEDinstance-attribute**

SIGNAL\_CELLULAR\_NODATA\_ROUNDED: IconData

**buildSIGNAL\_CELLULAR\_NODATA\_SHARPinstance-attribute**

SIGNAL\_CELLULAR\_NODATA\_SHARP: IconData

**buildSIGNAL\_CELLULAR\_NULLinstance-attribute**

SIGNAL\_CELLULAR\_NULL: IconData

**buildSIGNAL\_CELLULAR\_NULL\_OUTLINEDinstance-attribute**

SIGNAL\_CELLULAR\_NULL\_OUTLINED: IconData

**buildSIGNAL\_CELLULAR\_NULL\_ROUNDEDinstance-attribute**

SIGNAL\_CELLULAR\_NULL\_ROUNDED: IconData

**buildSIGNAL\_CELLULAR\_NULL\_SHARPinstance-attribute**

SIGNAL\_CELLULAR\_NULL\_SHARP: IconData

**buildSIGNAL\_CELLULAR\_OFFinstance-attribute**

SIGNAL\_CELLULAR\_OFF: IconData

**buildSIGNAL\_CELLULAR\_OFF\_OUTLINEDinstance-attribute**

SIGNAL\_CELLULAR\_OFF\_OUTLINED: IconData

**buildSIGNAL\_CELLULAR\_OFF\_ROUNDEDinstance-attribute**

SIGNAL\_CELLULAR\_OFF\_ROUNDED: IconData

**buildSIGNAL\_CELLULAR\_OFF\_SHARPinstance-attribute**

SIGNAL\_CELLULAR\_OFF\_SHARP: IconData

**buildSIGNAL\_WIFI\_0\_BARinstance-attribute**

SIGNAL\_WIFI\_0\_BAR: IconData

**buildSIGNAL\_WIFI\_0\_BAR\_OUTLINEDinstance-attribute**

SIGNAL\_WIFI\_0\_BAR\_OUTLINED: IconData

**buildSIGNAL\_WIFI\_0\_BAR\_ROUNDEDinstance-attribute**

SIGNAL\_WIFI\_0\_BAR\_ROUNDED: IconData

**buildSIGNAL\_WIFI\_0\_BAR\_SHARPinstance-attribute**

SIGNAL\_WIFI\_0\_BAR\_SHARP: IconData

**buildSIGNAL\_WIFI\_4\_BARinstance-attribute**

SIGNAL\_WIFI\_4\_BAR: IconData

**buildSIGNAL\_WIFI\_4\_BAR\_LOCKinstance-attribute**

SIGNAL\_WIFI\_4\_BAR\_LOCK: IconData

**buildSIGNAL\_WIFI\_4\_BAR\_LOCK\_OUTLINEDinstance-attribute**

SIGNAL\_WIFI\_4\_BAR\_LOCK\_OUTLINED: IconData

**buildSIGNAL\_WIFI\_4\_BAR\_LOCK\_ROUNDEDinstance-attribute**

SIGNAL\_WIFI\_4\_BAR\_LOCK\_ROUNDED: IconData

**buildSIGNAL\_WIFI\_4\_BAR\_LOCK\_SHARPinstance-attribute**

SIGNAL\_WIFI\_4\_BAR\_LOCK\_SHARP: IconData

**buildSIGNAL\_WIFI\_4\_BAR\_OUTLINEDinstance-attribute**

SIGNAL\_WIFI\_4\_BAR\_OUTLINED: IconData

**buildSIGNAL\_WIFI\_4\_BAR\_ROUNDEDinstance-attribute**

SIGNAL\_WIFI\_4\_BAR\_ROUNDED: IconData

**buildSIGNAL\_WIFI\_4\_BAR\_SHARPinstance-attribute**

SIGNAL\_WIFI\_4\_BAR\_SHARP: IconData

buildSIGNAL\_WIFI\_BADInstance-attribute

SIGNAL\_WIFI\_BAD: IconData

buildSIGNAL\_WIFI\_BAD\_OUTLINEDInstance-attribute

SIGNAL\_WIFI\_BAD\_OUTLINED: IconData

buildSIGNAL\_WIFI\_BAD\_ROUNDEDInstance-attribute

SIGNAL\_WIFI\_BAD\_ROUNDED: IconData

buildSIGNAL\_WIFI\_BAD\_SHARPInstance-attribute

SIGNAL\_WIFI\_BAD\_SHARP: IconData

buildSIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4Instance-attribute

SIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4: IconData

buildSIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_OUTLINEDInstance-attribute

SIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_OUTLINED: IconData

buildSIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_ROUNDEDInstance-attribute

SIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_ROUNDED: IconData

buildSIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_SHARPInstance-attribute

SIGNAL\_WIFI\_CONNECTED\_NO\_INTERNET\_4\_SHARP: IconData

buildSIGNAL\_WIFI\_OFFInstance-attribute

SIGNAL\_WIFI\_OFF: IconData

buildSIGNAL\_WIFI\_OFF\_OUTLINEDInstance-attribute

SIGNAL\_WIFI\_OFF\_OUTLINED: IconData

buildSIGNAL\_WIFI\_OFF\_ROUNDEDInstance-attribute

SIGNAL\_WIFI\_OFF\_ROUNDED: IconData

buildSIGNAL\_WIFI\_OFF\_SHARPInstance-attribute

SIGNAL\_WIFI\_OFF\_SHARP: IconData

buildSIGNAL\_WIFI\_STATUSBAR\_4\_BARInstance-attribute

SIGNAL\_WIFI\_STATUSBAR\_4\_BAR: IconData

buildSIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_OUTLINEDInstance-attribute

SIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_OUTLINED: IconData

buildSIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_ROUNDEDInstance-attribute

SIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_ROUNDED: IconData

buildSIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_SHARPInstance-attribute

SIGNAL\_WIFI\_STATUSBAR\_4\_BAR\_SHARP: IconData

buildSIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4Instance-attribute

SIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4: IconData

buildSIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_OUTLINEDInstance-attribute

SIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_OUTLINED: (  
IconData  
)

buildSIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_ROUNDEDInstance-attribute

SIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_ROUNDED: (  
IconData  
)

buildSIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_SHARPInstance-attribute

SIGNAL\_WIFI\_STATUSBAR\_CONNECTED\_NO\_INTERNET\_4\_SHARP: (  
IconData  
)

```

 IconData
)

buildSIGNAL_WIFI_STATUSBAR_NULLInstance-attribute
 SIGNAL_WIFI_STATUSBAR_NULL: IconData

buildSIGNAL_WIFI_STATUSBAR_NULL_OUTLINEDInstance-attribute
 SIGNAL_WIFI_STATUSBAR_NULL_OUTLINED: IconData

buildSIGNAL_WIFI_STATUSBAR_NULL_ROUNDEDInstance-attribute
 SIGNAL_WIFI_STATUSBAR_NULL_ROUNDED: IconData

buildSIGNAL_WIFI_STATUSBAR_NULL_SHARPInstance-attribute
 SIGNAL_WIFI_STATUSBAR_NULL_SHARP: IconData

buildSIGNPOSTInstance-attribute
 SIGNPOST: IconData

buildSIGNPOST_OUTLINEDInstance-attribute
 SIGNPOST_OUTLINED: IconData

buildSIGNPOST_ROUNDEDInstance-attribute
 SIGNPOST_ROUNDED: IconData

buildSIGNPOST_SHARPInstance-attribute
 SIGNPOST_SHARP: IconData

buildSIM_CARDInstance-attribute
 SIM_CARD: IconData

buildSIM_CARD_ALERTInstance-attribute
 SIM_CARD_ALERT: IconData

buildSIM_CARD_ALERT_OUTLINEDInstance-attribute
 SIM_CARD_ALERT_OUTLINED: IconData

buildSIM_CARD_ALERT_ROUNDEDInstance-attribute
 SIM_CARD_ALERT_ROUNDED: IconData

buildSIM_CARD_ALERT_SHARPInstance-attribute
 SIM_CARD_ALERT_SHARP: IconData

buildSIM_CARD_DOWNLOADInstance-attribute
 SIM_CARD_DOWNLOAD: IconData

buildSIM_CARD_DOWNLOAD_OUTLINEDInstance-attribute
 SIM_CARD_DOWNLOAD_OUTLINED: IconData

buildSIM_CARD_DOWNLOAD_ROUNDEDInstance-attribute
 SIM_CARD_DOWNLOAD_ROUNDED: IconData

buildSIM_CARD_DOWNLOAD_SHARPInstance-attribute
 SIM_CARD_DOWNLOAD_SHARP: IconData

buildSIM_CARD_OUTLINEDInstance-attribute
 SIM_CARD_OUTLINED: IconData

buildSIM_CARD_ROUNDEDInstance-attribute
 SIM_CARD_ROUNDED: IconData

buildSIM_CARD_SHARPInstance-attribute
 SIM_CARD_SHARP: IconData

buildSINGLE_BEDInstance-attribute
 SINGLE_BED: IconData

buildSINGLE_BED_OUTLINEDInstance-attribute

```

SINGLE\_BED\_OUTLINED: IconData

buildSINGLE\_BED\_ROUNDEDinstance-attribute

SINGLE\_BED\_ROUNDED: IconData

buildSINGLE\_BED\_SHARPinstance-attribute

SINGLE\_BED\_SHARP: IconData

buildSIPinstance-attribute

SIP: IconData

buildSIP\_OUTLINEDinstance-attribute

SIP\_OUTLINED: IconData

buildSIP\_ROUNDEDinstance-attribute

SIP\_ROUNDED: IconData

buildSIP\_SHARPinstance-attribute

SIP\_SHARP: IconData

buildSIX\_FT\_APARTinstance-attribute

SIX\_FT\_APART: IconData

buildSIX\_FT\_APART\_OUTLINEDinstance-attribute

SIX\_FT\_APART\_OUTLINED: IconData

buildSIX\_FT\_APART\_ROUNDEDinstance-attribute

SIX\_FT\_APART\_ROUNDED: IconData

buildSIX\_FT\_APART\_SHARPinstance-attribute

SIX\_FT\_APART\_SHARP: IconData

buildSIX\_Kinstance-attribute

SIX\_K: IconData

buildSIX\_K\_OUTLINEDinstance-attribute

SIX\_K\_OUTLINED: IconData

buildSIX\_K\_PLUSinstance-attribute

SIX\_K\_PLUS: IconData

buildSIX\_K\_PLUS\_OUTLINEDinstance-attribute

SIX\_K\_PLUS\_OUTLINED: IconData

buildSIX\_K\_PLUS\_ROUNDEDinstance-attribute

SIX\_K\_PLUS\_ROUNDED: IconData

buildSIX\_K\_PLUS\_SHARPinstance-attribute

SIX\_K\_PLUS\_SHARP: IconData

buildSIX\_K\_ROUNDEDinstance-attribute

SIX\_K\_ROUNDED: IconData

buildSIX\_K\_SHARPinstance-attribute

SIX\_K\_SHARP: IconData

buildSIX\_MPinstance-attribute

SIX\_MP: IconData

buildSIX\_MP\_OUTLINEDinstance-attribute

SIX\_MP\_OUTLINED: IconData

buildSIX\_MP\_ROUNDEDinstance-attribute

SIX\_MP\_ROUNDED: IconData

buildSIX\_MP\_SHARPinstance-attribute



SIX\_MP\_SHARP: IconData

**buildSIXTEEN\_MPinstance-attribute**

SIXTEEN\_MP: IconData

**buildSIXTEEN\_MP\_OUTLINEDinstance-attribute**

SIXTEEN\_MP\_OUTLINED: IconData

**buildSIXTEEN\_MP\_ROUNDEDinstance-attribute**

SIXTEEN\_MP\_ROUNDED: IconData

**buildSIXTEEN\_MP\_SHARPinstance-attribute**

SIXTEEN\_MP\_SHARP: IconData

**buildSIXTY\_FPSinstance-attribute**

SIXTY\_FPS: IconData

**buildSIXTY\_FPS\_OUTLINEDinstance-attribute**

SIXTY\_FPS\_OUTLINED: IconData

**buildSIXTY\_FPS\_ROUNDEDinstance-attribute**

SIXTY\_FPS\_ROUNDED: IconData

**buildSIXTY\_FPS\_SELECTinstance-attribute**

SIXTY\_FPS\_SELECT: IconData

**buildSIXTY\_FPS\_SELECT\_OUTLINEDinstance-attribute**

SIXTY\_FPS\_SELECT\_OUTLINED: IconData

**buildSIXTY\_FPS\_SELECT\_ROUNDEDinstance-attribute**

SIXTY\_FPS\_SELECT\_ROUNDED: IconData

**buildSIXTY\_FPS\_SELECT\_SHARPinstance-attribute**

SIXTY\_FPS\_SELECT\_SHARP: IconData

**buildSIXTY\_FPS\_SHARPinstance-attribute**

SIXTY\_FPS\_SHARP: IconData

**buildSKATEBOARDINGinstance-attribute**

SKATEBOARDING: IconData

**buildSKATEBOARDING\_OUTLINEDinstance-attribute**

SKATEBOARDING\_OUTLINED: IconData

**buildSKATEBOARDING\_ROUNDEDinstance-attribute**

SKATEBOARDING\_ROUNDED: IconData

**buildSKATEBOARDING\_SHARPinstance-attribute**

SKATEBOARDING\_SHARP: IconData

**buildSKIP\_NEXTinstance-attribute**

SKIP\_NEXT: IconData

**buildSKIP\_NEXT\_OUTLINEDinstance-attribute**

SKIP\_NEXT\_OUTLINED: IconData

**buildSKIP\_NEXT\_ROUNDEDinstance-attribute**

SKIP\_NEXT\_ROUNDED: IconData

**buildSKIP\_NEXT\_SHARPinstance-attribute**

SKIP\_NEXT\_SHARP: IconData

**buildSKIP\_PREVIOUSinstance-attribute**

SKIP\_PREVIOUS: IconData

**buildSKIP\_PREVIOUS\_OUTLINEDinstance-attribute**

SKIP\_PREVIOUS\_OUTLINED: IconData

**buildSKIP\_PREVIOUS\_ROUNDEDinstance-attribute**

SKIP\_PREVIOUS\_ROUNDED: IconData

**buildSKIP\_PREVIOUS\_SHARPinstance-attribute**

SKIP\_PREVIOUS\_SHARP: IconData

**buildSLEDDINGinstance-attribute**

SLEDDING: IconData

**buildSLEDDING\_OUTLINEDinstance-attribute**

SLEDDING\_OUTLINED: IconData

**buildSLEDDING\_ROUNDEDinstance-attribute**

SLEDDING\_ROUNDED: IconData

**buildSLEDDING\_SHARPinstance-attribute**

SLEDDING\_SHARP: IconData

**buildSLIDESHOWinstance-attribute**

SLIDESHOW: IconData

**buildSLIDESHOW\_OUTLINEDinstance-attribute**

SLIDESHOW\_OUTLINED: IconData

**buildSLIDESHOW\_ROUNDEDinstance-attribute**

SLIDESHOW\_ROUNDED: IconData

**buildSLIDESHOW\_SHARPinstance-attribute**

SLIDESHOW\_SHARP: IconData

**buildSLOW\_MOTION\_VIDEOinstance-attribute**

SLOW\_MOTION\_VIDEO: IconData

**buildSLOW\_MOTION\_VIDEO\_OUTLINEDinstance-attribute**

SLOW\_MOTION\_VIDEO\_OUTLINED: IconData

**buildSLOW\_MOTION\_VIDEO\_ROUNDEDinstance-attribute**

SLOW\_MOTION\_VIDEO\_ROUNDED: IconData

**buildSLOW\_MOTION\_VIDEO\_SHARPinstance-attribute**

SLOW\_MOTION\_VIDEO\_SHARP: IconData

**buildSMART\_BUTTONinstance-attribute**

SMART\_BUTTON: IconData

**buildSMART\_BUTTON\_OUTLINEDinstance-attribute**

SMART\_BUTTON\_OUTLINED: IconData

**buildSMART\_BUTTON\_ROUNDEDinstance-attribute**

SMART\_BUTTON\_ROUNDED: IconData

**buildSMART\_BUTTON\_SHARPinstance-attribute**

SMART\_BUTTON\_SHARP: IconData

**buildSMART\_DISPLAYinstance-attribute**

SMART\_DISPLAY: IconData

**buildSMART\_DISPLAY\_OUTLINEDinstance-attribute**

SMART\_DISPLAY\_OUTLINED: IconData

**buildSMART\_DISPLAY\_ROUNDEDinstance-attribute**

SMART\_DISPLAY\_ROUNDED: IconData

**buildSMART\_DISPLAY\_SHARPinstance-attribute**

SMART\_DISPLAY\_SHARP: IconData

buildSMART\_SCREENinstance-attribute

SMART\_SCREEN: IconData

buildSMART\_SCREEN\_OUTLINEDinstance-attribute

SMART\_SCREEN\_OUTLINED: IconData

buildSMART\_SCREEN\_ROUNDEDinstance-attribute

SMART\_SCREEN\_ROUNDED: IconData

buildSMART\_SCREEN\_SHARPinstance-attribute

SMART\_SCREEN\_SHARP: IconData

buildSMART\_TOYinstance-attribute

SMART\_TOY: IconData

buildSMART\_TOY\_OUTLINEDinstance-attribute

SMART\_TOY\_OUTLINED: IconData

buildSMART\_TOY\_ROUNDEDinstance-attribute

SMART\_TOY\_ROUNDED: IconData

buildSMART\_TOY\_SHARPinstance-attribute

SMART\_TOY\_SHARP: IconData

buildSMARTPHONEinstance-attribute

SMARTPHONE: IconData

buildSMARTPHONE\_OUTLINEDinstance-attribute

SMARTPHONE\_OUTLINED: IconData

buildSMARTPHONE\_ROUNDEDinstance-attribute

SMARTPHONE\_ROUNDED: IconData

buildSMARTPHONE\_SHARPinstance-attribute

SMARTPHONE\_SHARP: IconData

buildSMOKE\_FREEinstance-attribute

SMOKE\_FREE: IconData

buildSMOKE\_FREE\_OUTLINEDinstance-attribute

SMOKE\_FREE\_OUTLINED: IconData

buildSMOKE\_FREE\_ROUNDEDinstance-attribute

SMOKE\_FREE\_ROUNDED: IconData

buildSMOKE\_FREE\_SHARPinstance-attribute

SMOKE\_FREE\_SHARP: IconData

buildSMOKING\_ROOMSinstance-attribute

SMOKING\_ROOMS: IconData

buildSMOKING\_ROOMS\_OUTLINEDinstance-attribute

SMOKING\_ROOMS\_OUTLINED: IconData

buildSMOKING\_ROOMS\_ROUNDEDinstance-attribute

SMOKING\_ROOMS\_ROUNDED: IconData

buildSMOKING\_ROOMS\_SHARPinstance-attribute

SMOKING\_ROOMS\_SHARP: IconData

buildSMSinstance-attribute

SMS: IconData

buildSMS\_FAILEDinstance-attribute

SMS\_FAILED: IconData

buildSMS\_FAILED\_OUTLINEDinstance-attribute

SMS\_FAILED\_OUTLINED: IconData

buildSMS\_FAILED\_ROUNDEDinstance-attribute

SMS\_FAILED\_ROUNDED: IconData

buildSMS\_FAILED\_SHARPinstance-attribute

SMS\_FAILED\_SHARP: IconData

buildSMS\_OUTLINEDinstance-attribute

SMS\_OUTLINED: IconData

buildSMS\_ROUNDEDinstance-attribute

SMS\_ROUNDED: IconData

buildSMS\_SHARPinstance-attribute

SMS\_SHARP: IconData

buildSNAPCHATinstance-attribute

SNAPCHAT: IconData

buildSNAPCHAT\_OUTLINEDinstance-attribute

SNAPCHAT\_OUTLINED: IconData

buildSNAPCHAT\_ROUNDEDinstance-attribute

SNAPCHAT\_ROUNDED: IconData

buildSNAPCHAT\_SHARPinstance-attribute

SNAPCHAT\_SHARP: IconData

buildSNIPPET\_FOLDERinstance-attribute

SNIPPET\_FOLDER: IconData

buildSNIPPET\_FOLDER\_OUTLINEDinstance-attribute

SNIPPET\_FOLDER\_OUTLINED: IconData

buildSNIPPET\_FOLDER\_ROUNDEDinstance-attribute

SNIPPET\_FOLDER\_ROUNDED: IconData

buildSNIPPET\_FOLDER\_SHARPinstance-attribute

SNIPPET\_FOLDER\_SHARP: IconData

buildSNOOZEinstance-attribute

SNOOZE: IconData

buildSNOOZE\_OUTLINEDinstance-attribute

SNOOZE\_OUTLINED: IconData

buildSNOOZE\_ROUNDEDinstance-attribute

SNOOZE\_ROUNDED: IconData

buildSNOOZE\_SHARPinstance-attribute

SNOOZE\_SHARP: IconData

buildSNOWBOARDINGinstance-attribute

SNOWBOARDING: IconData

buildSNOWBOARDING\_OUTLINEDinstance-attribute

SNOWBOARDING\_OUTLINED: IconData

buildSNOWBOARDING\_ROUNDEDinstance-attribute

SNOWBOARDING\_ROUNDED: IconData

buildSNOWBOARDING\_SHARPinstance-attribute

SNOWBOARDING\_SHARP: IconData

**buildSNOWINGinstance-attribute**

SNOWING: IconData

**buildSNOWMOBILEinstance-attribute**

SNOWMOBILE: IconData

**buildSNOWMOBILE\_OUTLINEDinstance-attribute**

SNOWMOBILE\_OUTLINED: IconData

**buildSNOWMOBILE\_ROUNDEDinstance-attribute**

SNOWMOBILE\_ROUNDED: IconData

**buildSNOWMOBILE\_SHARPinstance-attribute**

SNOWMOBILE\_SHARP: IconData

**buildSNOWSHOEINGinstance-attribute**

SNOWSHOEING: IconData

**buildSNOWSHOEING\_OUTLINEDinstance-attribute**

SNOWSHOEING\_OUTLINED: IconData

**buildSNOWSHOEING\_ROUNDEDinstance-attribute**

SNOWSHOEING\_ROUNDED: IconData

**buildSNOWSHOEING\_SHARPinstance-attribute**

SNOWSHOEING\_SHARP: IconData

**buildSOAPinstance-attribute**

SOAP: IconData

**buildSOAP\_OUTLINEDinstance-attribute**

SOAP\_OUTLINED: IconData

**buildSOAP\_ROUNDEDinstance-attribute**

SOAP\_ROUNDED: IconData

**buildSOAP\_SHARPinstance-attribute**

SOAP\_SHARP: IconData

**buildSOCIAL\_DISTANCEinstance-attribute**

SOCIAL\_DISTANCE: IconData

**buildSOCIAL\_DISTANCE\_OUTLINEDinstance-attribute**

SOCIAL\_DISTANCE\_OUTLINED: IconData

**buildSOCIAL\_DISTANCE\_ROUNDEDinstance-attribute**

SOCIAL\_DISTANCE\_ROUNDED: IconData

**buildSOCIAL\_DISTANCE\_SHARPinstance-attribute**

SOCIAL\_DISTANCE\_SHARP: IconData

**buildSOLAR\_POWERinstance-attribute**

SOLAR\_POWER: IconData

**buildSOLAR\_POWER\_OUTLINEDinstance-attribute**

SOLAR\_POWER\_OUTLINED: IconData

**buildSOLAR\_POWER\_ROUNDEDinstance-attribute**

SOLAR\_POWER\_ROUNDED: IconData

**buildSOLAR\_POWER\_SHARPinstance-attribute**

SOLAR\_POWER\_SHARP: IconData

**buildSORTinstance-attribute**

**SORT:** IconData

**buildSORT\_BY\_ALPHA**instance-attribute

**SORT\_BY\_ALPHA:** IconData

**buildSORT\_BY\_ALPHA\_OUTLINED**instance-attribute

**SORT\_BY\_ALPHA\_OUTLINED:** IconData

**buildSORT\_BY\_ALPHA\_ROUNDED**instance-attribute

**SORT\_BY\_ALPHA\_ROUNDED:** IconData

**buildSORT\_BY\_ALPHA\_SHARP**instance-attribute

**SORT\_BY\_ALPHA\_SHARP:** IconData

**buildSORT\_OUTLINED**instance-attribute

**SORT\_OUTLINED:** IconData

**buildSORT\_ROUNDED**instance-attribute

**SORT\_ROUNDED:** IconData

**buildSORT\_SHARP**instance-attribute

**SORT\_SHARP:** IconData

**buildSOS**instance-attribute

**SOS:** IconData

**buildSOS\_OUTLINED**instance-attribute

**SOS\_OUTLINED:** IconData

**buildSOS\_ROUNDED**instance-attribute

**SOS\_ROUNDED:** IconData

**buildSOS\_SHARP**instance-attribute

**SOS\_SHARP:** IconData

**buildSOUP\_KITCHEN**instance-attribute

**SOUP\_KITCHEN:** IconData

**buildSOUP\_KITCHEN\_OUTLINED**instance-attribute

**SOUP\_KITCHEN\_OUTLINED:** IconData

**buildSOUP\_KITCHEN\_ROUNDED**instance-attribute

**SOUP\_KITCHEN\_ROUNDED:** IconData

**buildSOUP\_KITCHEN\_SHARP**instance-attribute

**SOUP\_KITCHEN\_SHARP:** IconData

**buildSOURCE**instance-attribute

**SOURCE:** IconData

**buildSOURCE\_OUTLINED**instance-attribute

**SOURCE\_OUTLINED:** IconData

**buildSOURCE\_ROUNDED**instance-attribute

**SOURCE\_ROUNDED:** IconData

**buildSOURCE\_SHARP**instance-attribute

**SOURCE\_SHARP:** IconData

**buildSOUTH**instance-attribute

**SOUTH:** IconData

**buildSOUTH\_AMERICA**instance-attribute

**SOUTH\_AMERICA:** IconData

**buildSOUTH\_AMERICA\_OUTLINED**instance-attribute

SOUTH\_AMERICA\_OUTLINED: IconData

buildSOUTH\_AMERICA\_ROUNDEDinstance-attribute

SOUTH\_AMERICA\_ROUNDED: IconData

buildSOUTH\_AMERICA\_SHARPinstance-attribute

SOUTH\_AMERICA\_SHARP: IconData

buildSOUTH\_EASTinstance-attribute

SOUTH\_EAST: IconData

buildSOUTH\_EAST\_OUTLINEDinstance-attribute

SOUTH\_EAST\_OUTLINED: IconData

buildSOUTH\_EAST\_ROUNDEDinstance-attribute

SOUTH\_EAST\_ROUNDED: IconData

buildSOUTH\_EAST\_SHARPinstance-attribute

SOUTH\_EAST\_SHARP: IconData

buildSOUTH\_OUTLINEDinstance-attribute

SOUTH\_OUTLINED: IconData

buildSOUTH\_ROUNDEDinstance-attribute

SOUTH\_ROUNDED: IconData

buildSOUTH\_SHARPinstance-attribute

SOUTH\_SHARP: IconData

buildSOUTH\_WESTinstance-attribute

SOUTH\_WEST: IconData

buildSOUTH\_WEST\_OUTLINEDinstance-attribute

SOUTH\_WEST\_OUTLINED: IconData

buildSOUTH\_WEST\_ROUNDEDinstance-attribute

SOUTH\_WEST\_ROUNDED: IconData

buildSOUTH\_WEST\_SHARPinstance-attribute

SOUTH\_WEST\_SHARP: IconData

buildSPAinstance-attribute

SPA: IconData

buildSPA\_OUTLINEDinstance-attribute

SPA\_OUTLINED: IconData

buildSPA\_ROUNDEDinstance-attribute

SPA\_ROUNDED: IconData

buildSPA\_SHARPinstance-attribute

SPA\_SHARP: IconData

buildSPACE\_BARinstance-attribute

SPACE\_BAR: IconData

buildSPACE\_BAR\_OUTLINEDinstance-attribute

SPACE\_BAR\_OUTLINED: IconData

buildSPACE\_BAR\_ROUNDEDinstance-attribute

SPACE\_BAR\_ROUNDED: IconData

buildSPACE\_BAR\_SHARPinstance-attribute

SPACE\_BAR\_SHARP: IconData

buildSPACE\_DASHBOARDinstance-attribute

SPACE\_DASHBOARD: IconData

buildSPACE\_DASHBOARD\_OUTLINEDinstance-attribute

SPACE\_DASHBOARD\_OUTLINED: IconData

buildSPACE\_DASHBOARD\_ROUNDEDinstance-attribute

SPACE\_DASHBOARD\_ROUNDED: IconData

buildSPACE\_DASHBOARD\_SHARPinstance-attribute

SPACE\_DASHBOARD\_SHARP: IconData

buildSPATIAL\_AUDIOinstance-attribute

SPATIAL\_AUDIO: IconData

buildSPATIAL\_AUDIO\_OFFinstance-attribute

SPATIAL\_AUDIO\_OFF: IconData

buildSPATIAL\_AUDIO\_OFF\_OUTLINEDinstance-attribute

SPATIAL\_AUDIO\_OFF\_OUTLINED: IconData

buildSPATIAL\_AUDIO\_OFF\_ROUNDEDinstance-attribute

SPATIAL\_AUDIO\_OFF\_ROUNDED: IconData

buildSPATIAL\_AUDIO\_OFF\_SHARPinstance-attribute

SPATIAL\_AUDIO\_OFF\_SHARP: IconData

buildSPATIAL\_AUDIO\_OUTLINEDinstance-attribute

SPATIAL\_AUDIO\_OUTLINED: IconData

buildSPATIAL\_AUDIO\_ROUNDEDinstance-attribute

SPATIAL\_AUDIO\_ROUNDED: IconData

buildSPATIAL\_AUDIO\_SHARPinstance-attribute

SPATIAL\_AUDIO\_SHARP: IconData

buildSPATIAL\_TRACKINGinstance-attribute

SPATIAL\_TRACKING: IconData

buildSPATIAL\_TRACKING\_OUTLINEDinstance-attribute

SPATIAL\_TRACKING\_OUTLINED: IconData

buildSPATIAL\_TRACKING\_ROUNDEDinstance-attribute

SPATIAL\_TRACKING\_ROUNDED: IconData

buildSPATIAL\_TRACKING\_SHARPinstance-attribute

SPATIAL\_TRACKING\_SHARP: IconData

buildSPEAKERinstance-attribute

SPEAKER: IconData

buildSPEAKER\_GROUPinstance-attribute

SPEAKER\_GROUP: IconData

buildSPEAKER\_GROUP\_OUTLINEDinstance-attribute

SPEAKER\_GROUP\_OUTLINED: IconData

buildSPEAKER\_GROUP\_ROUNDEDinstance-attribute

SPEAKER\_GROUP\_ROUNDED: IconData

buildSPEAKER\_GROUP\_SHARPinstance-attribute

SPEAKER\_GROUP\_SHARP: IconData

buildSPEAKER\_NOTESinstance-attribute

SPEAKER\_NOTES: IconData

buildSPEAKER\_NOTES\_OFFinstance-attribute



SPEAKER\_NOTES\_OFF: IconData

buildSPEAKER\_NOTES\_OFF\_OUTLINEDinstance-attribute

SPEAKER\_NOTES\_OFF\_OUTLINED: IconData

buildSPEAKER\_NOTES\_OFF\_ROUNDEDinstance-attribute

SPEAKER\_NOTES\_OFF\_ROUNDED: IconData

buildSPEAKER\_NOTES\_OFF\_SHARPinstance-attribute

SPEAKER\_NOTES\_OFF\_SHARP: IconData

buildSPEAKER\_NOTES\_OUTLINEDinstance-attribute

SPEAKER\_NOTES\_OUTLINED: IconData

buildSPEAKER\_NOTES\_ROUNDEDinstance-attribute

SPEAKER\_NOTES\_ROUNDED: IconData

buildSPEAKER\_NOTES\_SHARPinstance-attribute

SPEAKER\_NOTES\_SHARP: IconData

buildSPEAKER\_OUTLINEDinstance-attribute

SPEAKER\_OUTLINED: IconData

buildSPEAKER\_PHONEinstance-attribute

SPEAKER\_PHONE: IconData

buildSPEAKER\_PHONE\_OUTLINEDinstance-attribute

SPEAKER\_PHONE\_OUTLINED: IconData

buildSPEAKER\_PHONE\_ROUNDEDinstance-attribute

SPEAKER\_PHONE\_ROUNDED: IconData

buildSPEAKER\_PHONE\_SHARPinstance-attribute

SPEAKER\_PHONE\_SHARP: IconData

buildSPEAKER\_ROUNDEDinstance-attribute

SPEAKER\_ROUNDED: IconData

buildSPEAKER\_SHARPinstance-attribute

SPEAKER\_SHARP: IconData

buildSPEEDinstance-attribute

SPEED: IconData

buildSPEED\_OUTLINEDinstance-attribute

SPEED\_OUTLINED: IconData

buildSPEED\_ROUNDEDinstance-attribute

SPEED\_ROUNDED: IconData

buildSPEED\_SHARPinstance-attribute

SPEED\_SHARP: IconData

buildSPELLCHECKinstance-attribute

SPELLCHECK: IconData

buildSPELLCHECK\_OUTLINEDinstance-attribute

SPELLCHECK\_OUTLINED: IconData

buildSPELLCHECK\_ROUNDEDinstance-attribute

SPELLCHECK\_ROUNDED: IconData

buildSPELLCHECK\_SHARPinstance-attribute

SPELLCHECK\_SHARP: IconData

buildSPLITSCREENinstance-attribute

SPLITSCREEN: IconData

buildSPLITSCREEN\_OUTLINEDinstance-attribute

SPLITSCREEN\_OUTLINED: IconData

buildSPLITSCREEN\_ROUNDEDinstance-attribute

SPLITSCREEN\_ROUNDED: IconData

buildSPLITSCREEN\_SHARPinstance-attribute

SPLITSCREEN\_SHARP: IconData

buildSPOKEinstance-attribute

SPOKE: IconData

buildSPOKE\_OUTLINEDinstance-attribute

SPOKE\_OUTLINED: IconData

buildSPOKE\_ROUNDEDinstance-attribute

SPOKE\_ROUNDED: IconData

buildSPOKE\_SHARPinstance-attribute

SPOKE\_SHARP: IconData

buildSPORTSinstance-attribute

SPORTS: IconData

buildSPORTS\_BARinstance-attribute

SPORTS\_BAR: IconData

buildSPORTS\_BAR\_OUTLINEDinstance-attribute

SPORTS\_BAR\_OUTLINED: IconData

buildSPORTS\_BAR\_ROUNDEDinstance-attribute

SPORTS\_BAR\_ROUNDED: IconData

buildSPORTS\_BAR\_SHARPinstance-attribute

SPORTS\_BAR\_SHARP: IconData

buildSPORTS\_BASEBALLinstance-attribute

SPORTS\_BASEBALL: IconData

buildSPORTS\_BASEBALL\_OUTLINEDinstance-attribute

SPORTS\_BASEBALL\_OUTLINED: IconData

buildSPORTS\_BASEBALL\_ROUNDEDinstance-attribute

SPORTS\_BASEBALL\_ROUNDED: IconData

buildSPORTS\_BASEBALL\_SHARPinstance-attribute

SPORTS\_BASEBALL\_SHARP: IconData

buildSPORTS\_BASKETBALLinstance-attribute

SPORTS\_BASKETBALL: IconData

buildSPORTS\_BASKETBALL\_OUTLINEDinstance-attribute

SPORTS\_BASKETBALL\_OUTLINED: IconData

buildSPORTS\_BASKETBALL\_ROUNDEDinstance-attribute

SPORTS\_BASKETBALL\_ROUNDED: IconData

buildSPORTS\_BASKETBALL\_SHARPinstance-attribute

SPORTS\_BASKETBALL\_SHARP: IconData

buildSPORTS\_CRICKETinstance-attribute

SPORTS\_CRICKET: IconData

buildSPORTS\_CRICKET\_OUTLINEDinstance-attribute

SPORTS\_CRICKET\_OUTLINED: IconData

buildSPORTS\_CRICKET\_ROUNDEDinstance-attribute

SPORTS\_CRICKET\_ROUNDED: IconData

buildSPORTS\_CRICKET\_SHARPinstance-attribute

SPORTS\_CRICKET\_SHARP: IconData

buildSPORTS\_ESPORTSinstance-attribute

SPORTS\_ESPORTS: IconData

buildSPORTS\_ESPORTS\_OUTLINEDinstance-attribute

SPORTS\_ESPORTS\_OUTLINED: IconData

buildSPORTS\_ESPORTS\_ROUNDEDinstance-attribute

SPORTS\_ESPORTS\_ROUNDED: IconData

buildSPORTS\_ESPORTS\_SHARPinstance-attribute

SPORTS\_ESPORTS\_SHARP: IconData

buildSPORTS\_FOOTBALLinstance-attribute

SPORTS\_FOOTBALL: IconData

buildSPORTS\_FOOTBALL\_OUTLINEDinstance-attribute

SPORTS\_FOOTBALL\_OUTLINED: IconData

buildSPORTS\_FOOTBALL\_ROUNDEDinstance-attribute

SPORTS\_FOOTBALL\_ROUNDED: IconData

buildSPORTS\_FOOTBALL\_SHARPinstance-attribute

SPORTS\_FOOTBALL\_SHARP: IconData

buildSPORTS\_GOLFinstance-attribute

SPORTS\_GOLF: IconData

buildSPORTS\_GOLF\_OUTLINEDinstance-attribute

SPORTS\_GOLF\_OUTLINED: IconData

buildSPORTS\_GOLF\_ROUNDEDinstance-attribute

SPORTS\_GOLF\_ROUNDED: IconData

buildSPORTS\_GOLF\_SHARPinstance-attribute

SPORTS\_GOLF\_SHARP: IconData

buildSPORTS\_GYMNASTICSinstance-attribute

SPORTS\_GYMNASTICS: IconData

buildSPORTS\_GYMNASTICS\_OUTLINEDinstance-attribute

SPORTS\_GYMNASTICS\_OUTLINED: IconData

buildSPORTS\_GYMNASTICS\_ROUNDEDinstance-attribute

SPORTS\_GYMNASTICS\_ROUNDED: IconData

buildSPORTS\_GYMNASTICS\_SHARPinstance-attribute

SPORTS\_GYMNASTICS\_SHARP: IconData

buildSPORTS\_HANDBALLinstance-attribute

SPORTS\_HANDBALL: IconData

buildSPORTS\_HANDBALL\_OUTLINEDinstance-attribute

SPORTS\_HANDBALL\_OUTLINED: IconData

buildSPORTS\_HANDBALL\_ROUNDEDinstance-attribute

SPORTS\_HANDBALL\_ROUNDED: IconData

buildSPORTS\_HANDBALL\_SHARPinstance-attribute

SPORTS\_HANDBALL\_SHARP: IconData

buildSPORTS\_HOCKEYinstance-attribute

SPORTS\_HOCKEY: IconData

buildSPORTS\_HOCKEY\_OUTLINEDinstance-attribute

SPORTS\_HOCKEY\_OUTLINED: IconData

buildSPORTS\_HOCKEY\_ROUNDEDinstance-attribute

SPORTS\_HOCKEY\_ROUNDED: IconData

buildSPORTS\_HOCKEY\_SHARPinstance-attribute

SPORTS\_HOCKEY\_SHARP: IconData

buildSPORTS\_KABADDIinstance-attribute

SPORTS\_KABADDI: IconData

buildSPORTS\_KABADDI\_OUTLINEDinstance-attribute

SPORTS\_KABADDI\_OUTLINED: IconData

buildSPORTS\_KABADDI\_ROUNDEDinstance-attribute

SPORTS\_KABADDI\_ROUNDED: IconData

buildSPORTS\_KABADDI\_SHARPinstance-attribute

SPORTS\_KABADDI\_SHARP: IconData

buildSPORTS\_MARTIAL\_ARTSinstance-attribute

SPORTS\_MARTIAL\_ARTS: IconData

buildSPORTS\_MARTIAL\_ARTS\_OUTLINEDinstance-attribute

SPORTS\_MARTIAL\_ARTS\_OUTLINED: IconData

buildSPORTS\_MARTIAL\_ARTS\_ROUNDEDinstance-attribute

SPORTS\_MARTIAL\_ARTS\_ROUNDED: IconData

buildSPORTS\_MARTIAL\_ARTS\_SHARPinstance-attribute

SPORTS\_MARTIAL\_ARTS\_SHARP: IconData

buildSPORTS\_MMAinstance-attribute

SPORTS\_MMA: IconData

buildSPORTS\_MMA\_OUTLINEDinstance-attribute

SPORTS\_MMA\_OUTLINED: IconData

buildSPORTS\_MMA\_ROUNDEDinstance-attribute

SPORTS\_MMA\_ROUNDED: IconData

buildSPORTS\_MMA\_SHARPinstance-attribute

SPORTS\_MMA\_SHARP: IconData

buildSPORTS\_MOTORSPORTSinstance-attribute

SPORTS\_MOTORSPORTS: IconData

buildSPORTS\_MOTORSPORTS\_OUTLINEDinstance-attribute

SPORTS\_MOTORSPORTS\_OUTLINED: IconData

buildSPORTS\_MOTORSPORTS\_ROUNDEDinstance-attribute

SPORTS\_MOTORSPORTS\_ROUNDED: IconData

buildSPORTS\_MOTORSPORTS\_SHARPinstance-attribute

SPORTS\_MOTORSPORTS\_SHARP: IconData

buildSPORTS\_OUTLINEDinstance-attribute

SPORTS\_OUTLINED: IconData

buildSPORTS\_ROUNDEDinstance-attribute

SPORTS\_ROUNDED: IconData

buildSPORTS\_RUGBYinstance-attribute

SPORTS\_RUGBY: IconData

buildSPORTS\_RUGBY\_OUTLINEDinstance-attribute

SPORTS\_RUGBY\_OUTLINED: IconData

buildSPORTS\_RUGBY\_ROUNDEDinstance-attribute

SPORTS\_RUGBY\_ROUNDED: IconData

buildSPORTS\_RUGBY\_SHARPinstance-attribute

SPORTS\_RUGBY\_SHARP: IconData

buildSPORTS\_SCOREinstance-attribute

SPORTS\_SCORE: IconData

buildSPORTS\_SCORE\_OUTLINEDinstance-attribute

SPORTS\_SCORE\_OUTLINED: IconData

buildSPORTS\_SCORE\_ROUNDEDinstance-attribute

SPORTS\_SCORE\_ROUNDED: IconData

buildSPORTS\_SCORE\_SHARPinstance-attribute

SPORTS\_SCORE\_SHARP: IconData

buildSPORTS\_SHARPinstance-attribute

SPORTS\_SHARP: IconData

buildSPORTS\_SOCCERinstance-attribute

SPORTS\_SOCCER: IconData

buildSPORTS\_SOCCER\_OUTLINEDinstance-attribute

SPORTS\_SOCCER\_OUTLINED: IconData

buildSPORTS\_SOCCER\_ROUNDEDinstance-attribute

SPORTS\_SOCCER\_ROUNDED: IconData

buildSPORTS\_SOCCER\_SHARPinstance-attribute

SPORTS\_SOCCER\_SHARP: IconData

buildSPORTS\_TENNISinstance-attribute

SPORTS\_TENNIS: IconData

buildSPORTS\_TENNIS\_OUTLINEDinstance-attribute

SPORTS\_TENNIS\_OUTLINED: IconData

buildSPORTS\_TENNIS\_ROUNDEDinstance-attribute

SPORTS\_TENNIS\_ROUNDED: IconData

buildSPORTS\_TENNIS\_SHARPinstance-attribute

SPORTS\_TENNIS\_SHARP: IconData

buildSPORTS\_VOLLEYBALLinstance-attribute

SPORTS\_VOLLEYBALL: IconData

buildSPORTS\_VOLLEYBALL\_OUTLINEDinstance-attribute

SPORTS\_VOLLEYBALL\_OUTLINED: IconData

buildSPORTS\_VOLLEYBALL\_ROUNDEDinstance-attribute

SPORTS\_VOLLEYBALL\_ROUNDED: IconData

buildSPORTS\_VOLLEYBALL\_SHARPinstance-attribute

SPORTS\_VOLLEYBALL\_SHARP: IconData

buildSQUAREinstance-attribute

SQUARE: IconData

buildSQUARE\_FOOTInstance-attribute

SQUARE\_FOOT: IconData

buildSQUARE\_FOOT\_OUTLINEDInstance-attribute

SQUARE\_FOOT\_OUTLINED: IconData

buildSQUARE\_FOOT\_ROUNDEDInstance-attribute

SQUARE\_FOOT\_ROUNDED: IconData

buildSQUARE\_FOOT\_SHARPInstance-attribute

SQUARE\_FOOT\_SHARP: IconData

buildSQUARE\_OUTLINEDInstance-attribute

SQUARE\_OUTLINED: IconData

buildSQUARE\_ROUNDEDInstance-attribute

SQUARE\_ROUNDED: IconData

buildSQUARE\_SHARPInstance-attribute

SQUARE\_SHARP: IconData

buildSSID\_CHARTInstance-attribute

SSID\_CHART: IconData

buildSSID\_CHART\_OUTLINEDInstance-attribute

SSID\_CHART\_OUTLINED: IconData

buildSSID\_CHART\_ROUNDEDInstance-attribute

SSID\_CHART\_ROUNDED: IconData

buildSSID\_CHART\_SHARPInstance-attribute

SSID\_CHART\_SHARP: IconData

buildSTACKED\_BAR\_CHARTInstance-attribute

STACKED\_BAR\_CHART: IconData

buildSTACKED\_BAR\_CHART\_OUTLINEDInstance-attribute

STACKED\_BAR\_CHART\_OUTLINED: IconData

buildSTACKED\_BAR\_CHART\_ROUNDEDInstance-attribute

STACKED\_BAR\_CHART\_ROUNDED: IconData

buildSTACKED\_BAR\_CHART\_SHARPInstance-attribute

STACKED\_BAR\_CHART\_SHARP: IconData

buildSTACKED\_LINE\_CHARTInstance-attribute

STACKED\_LINE\_CHART: IconData

buildSTACKED\_LINE\_CHART\_OUTLINEDInstance-attribute

STACKED\_LINE\_CHART\_OUTLINED: IconData

buildSTACKED\_LINE\_CHART\_ROUNDEDInstance-attribute

STACKED\_LINE\_CHART\_ROUNDED: IconData

buildSTACKED\_LINE\_CHART\_SHARPInstance-attribute

STACKED\_LINE\_CHART\_SHARP: IconData

buildSTADIUMInstance-attribute

STADIUM: IconData

buildSTADIUM\_OUTLINEDInstance-attribute

STADIUM\_OUTLINED: IconData

buildSTADIUM\_ROUNDEDInstance-attribute

STADIUM\_ROUNDED: IconData

buildSTADIUM\_SHARPinstance-attribute

STADIUM\_SHARP: IconData

buildSTAIRSinstance-attribute

STAIRS: IconData

buildSTAIRS\_OUTLINEDinstance-attribute

STAIRS\_OUTLINED: IconData

buildSTAIRS\_ROUNDEDinstance-attribute

STAIRS\_ROUNDED: IconData

buildSTAIRS\_SHARPinstance-attribute

STAIRS\_SHARP: IconData

buildSTARinstance-attribute

STAR: IconData

buildSTAR\_BORDERinstance-attribute

STAR\_BORDER: IconData

buildSTAR\_BORDER\_OUTLINEDinstance-attribute

STAR\_BORDER\_OUTLINED: IconData

buildSTAR\_BORDER\_PURPLE500instance-attribute

STAR\_BORDER\_PURPLE500: IconData

buildSTAR\_BORDER\_PURPLE500\_OUTLINEDinstance-attribute

STAR\_BORDER\_PURPLE500\_OUTLINED: IconData

buildSTAR\_BORDER\_PURPLE500\_ROUNDEDinstance-attribute

STAR\_BORDER\_PURPLE500\_ROUNDED: IconData

buildSTAR\_BORDER\_PURPLE500\_SHARPinstance-attribute

STAR\_BORDER\_PURPLE500\_SHARP: IconData

buildSTAR\_BORDER\_ROUNDEDinstance-attribute

STAR\_BORDER\_ROUNDED: IconData

buildSTAR\_BORDER\_SHARPinstance-attribute

STAR\_BORDER\_SHARP: IconData

buildSTAR\_HALFinstance-attribute

STAR\_HALF: IconData

buildSTAR\_HALF\_OUTLINEDinstance-attribute

STAR\_HALF\_OUTLINED: IconData

buildSTAR\_HALF\_ROUNDEDinstance-attribute

STAR\_HALF\_ROUNDED: IconData

buildSTAR\_HALF\_SHARPinstance-attribute

STAR\_HALF\_SHARP: IconData

buildSTAR\_OUTLINEinstance-attribute

STAR\_OUTLINE: IconData

buildSTAR\_OUTLINE\_OUTLINEDinstance-attribute

STAR\_OUTLINE\_OUTLINED: IconData

buildSTAR\_OUTLINE\_ROUNDEDinstance-attribute

STAR\_OUTLINE\_ROUNDED: IconData

buildSTAR\_OUTLINE\_SHARPinstance-attribute

STAR\_OUTLINE\_SHARP: IconData

buildSTAR\_OUTLINEDinstance-attribute

STAR\_OUTLINED: IconData

buildSTAR\_PURPLE500instance-attribute

STAR\_PURPLE500: IconData

buildSTAR\_PURPLE500\_OUTLINEDinstance-attribute

STAR\_PURPLE500\_OUTLINED: IconData

buildSTAR\_PURPLE500\_ROUNDEDinstance-attribute

STAR\_PURPLE500\_ROUNDED: IconData

buildSTAR\_PURPLE500\_SHARPinstance-attribute

STAR\_PURPLE500\_SHARP: IconData

buildSTAR\_RATEinstance-attribute

STAR\_RATE: IconData

buildSTAR\_RATE\_OUTLINEDinstance-attribute

STAR\_RATE\_OUTLINED: IconData

buildSTAR\_RATE\_ROUNDEDinstance-attribute

STAR\_RATE\_ROUNDED: IconData

buildSTAR\_RATE\_SHARPinstance-attribute

STAR\_RATE\_SHARP: IconData

buildSTAR\_ROUNDEDinstance-attribute

STAR\_ROUNDED: IconData

buildSTAR\_SHARPinstance-attribute

STAR\_SHARP: IconData

buildSTARSinstance-attribute

STARS: IconData

buildSTARS\_OUTLINEDinstance-attribute

STARS\_OUTLINED: IconData

buildSTARS\_ROUNDEDinstance-attribute

STARS\_ROUNDED: IconData

buildSTARS\_SHARPinstance-attribute

STARS\_SHARP: IconData

buildSTARTinstance-attribute

START: IconData

buildSTART\_OUTLINEDinstance-attribute

START\_OUTLINED: IconData

buildSTART\_ROUNDEDinstance-attribute

START\_ROUNDED: IconData

buildSTART\_SHARPinstance-attribute

START\_SHARP: IconData

buildSTAY\_CURRENT\_LANDSCAPEinstance-attribute

STAY\_CURRENT\_LANDSCAPE: IconData

buildSTAY\_CURRENT\_LANDSCAPE\_OUTLINEDinstance-attribute

STAY\_CURRENT\_LANDSCAPE\_OUTLINED: IconData

buildSTAY\_CURRENT\_LANDSCAPE\_ROUNDEDinstance-attribute



STAY\_CURRENT\_LANDSCAPE\_ROUNDED: IconData

**buildSTAY\_CURRENT\_LANDSCAPE\_SHARP**instance-attribute

STAY\_CURRENT\_LANDSCAPE\_SHARP: IconData

**buildSTAY\_CURRENT\_PORTRAIT**instance-attribute

STAY\_CURRENT\_PORTRAIT: IconData

**buildSTAY\_CURRENT\_PORTRAIT\_OUTLINED**instance-attribute

STAY\_CURRENT\_PORTRAIT\_OUTLINED: IconData

**buildSTAY\_CURRENT\_PORTRAIT\_ROUNDED**instance-attribute

STAY\_CURRENT\_PORTRAIT\_ROUNDED: IconData

**buildSTAY\_CURRENT\_PORTRAIT\_SHARP**instance-attribute

STAY\_CURRENT\_PORTRAIT\_SHARP: IconData

**buildSTAY\_PRIMARY\_LANDSCAPE**instance-attribute

STAY\_PRIMARY\_LANDSCAPE: IconData

**buildSTAY\_PRIMARY\_LANDSCAPE\_OUTLINED**instance-attribute

STAY\_PRIMARY\_LANDSCAPE\_OUTLINED: IconData

**buildSTAY\_PRIMARY\_LANDSCAPE\_ROUNDED**instance-attribute

STAY\_PRIMARY\_LANDSCAPE\_ROUNDED: IconData

**buildSTAY\_PRIMARY\_LANDSCAPE\_SHARP**instance-attribute

STAY\_PRIMARY\_LANDSCAPE\_SHARP: IconData

**buildSTAY\_PRIMARY\_PORTRAIT**instance-attribute

STAY\_PRIMARY\_PORTRAIT: IconData

**buildSTAY\_PRIMARY\_PORTRAIT\_OUTLINED**instance-attribute

STAY\_PRIMARY\_PORTRAIT\_OUTLINED: IconData

**buildSTAY\_PRIMARY\_PORTRAIT\_ROUNDED**instance-attribute

STAY\_PRIMARY\_PORTRAIT\_ROUNDED: IconData

**buildSTAY\_PRIMARY\_PORTRAIT\_SHARP**instance-attribute

STAY\_PRIMARY\_PORTRAIT\_SHARP: IconData

**buildSTICKY\_NOTE\_2**instance-attribute

STICKY\_NOTE\_2: IconData

**buildSTICKY\_NOTE\_2\_OUTLINED**instance-attribute

STICKY\_NOTE\_2\_OUTLINED: IconData

**buildSTICKY\_NOTE\_2\_ROUNDED**instance-attribute

STICKY\_NOTE\_2\_ROUNDED: IconData

**buildSTICKY\_NOTE\_2\_SHARP**instance-attribute

STICKY\_NOTE\_2\_SHARP: IconData

**buildSTOP**instance-attribute

STOP: IconData

**buildSTOP\_CIRCLE**instance-attribute

STOP\_CIRCLE: IconData

**buildSTOP\_CIRCLE\_OUTLINED**instance-attribute

STOP\_CIRCLE\_OUTLINED: IconData

**buildSTOP\_CIRCLE\_ROUNDED**instance-attribute

STOP\_CIRCLE\_ROUNDED: IconData

**buildSTOP\_CIRCLE\_SHARP**instance-attribute

STOP\_CIRCLE\_SHARP: IconData

buildSTOP\_OUTLINEDinstance-attribute

STOP\_OUTLINED: IconData

buildSTOP\_ROUNDEDinstance-attribute

STOP\_ROUNDED: IconData

buildSTOP\_SCREEN\_SHAREinstance-attribute

STOP\_SCREEN\_SHARE: IconData

buildSTOP\_SCREEN\_SHARE\_OUTLINEDinstance-attribute

STOP\_SCREEN\_SHARE\_OUTLINED: IconData

buildSTOP\_SCREEN\_SHARE\_ROUNDEDinstance-attribute

STOP\_SCREEN\_SHARE\_ROUNDED: IconData

buildSTOP\_SCREEN\_SHARE\_SHARPinstance-attribute

STOP\_SCREEN\_SHARE\_SHARP: IconData

buildSTOP\_SHARPinstance-attribute

STOP\_SHARP: IconData

buildSTORAGEinstance-attribute

STORAGE: IconData

buildSTORAGE\_OUTLINEDinstance-attribute

STORAGE\_OUTLINED: IconData

buildSTORAGE\_ROUNDEDinstance-attribute

STORAGE\_ROUNDED: IconData

buildSTORAGE\_SHARPinstance-attribute

STORAGE\_SHARP: IconData

buildSTOREinstance-attribute

STORE: IconData

buildSTORE\_MALL\_DIRECTORYinstance-attribute

STORE\_MALL\_DIRECTORY: IconData

buildSTORE\_MALL\_DIRECTORY\_OUTLINEDinstance-attribute

STORE\_MALL\_DIRECTORY\_OUTLINED: IconData

buildSTORE\_MALL\_DIRECTORY\_ROUNDEDinstance-attribute

STORE\_MALL\_DIRECTORY\_ROUNDED: IconData

buildSTORE\_MALL\_DIRECTORY\_SHARPinstance-attribute

STORE\_MALL\_DIRECTORY\_SHARP: IconData

buildSTORE\_OUTLINEDinstance-attribute

STORE\_OUTLINED: IconData

buildSTORE\_ROUNDEDinstance-attribute

STORE\_ROUNDED: IconData

buildSTORE\_SHARPinstance-attribute

STORE\_SHARP: IconData

buildSTOREFRONTinstance-attribute

STOREFRONT: IconData

buildSTOREFRONT\_OUTLINEDinstance-attribute

STOREFRONT\_OUTLINED: IconData

buildSTOREFRONT\_ROUNDEDinstance-attribute

STOREFRONT\_ROUNDED: IconData

buildSTOREFRONT\_SHARPinstance-attribute

STOREFRONT\_SHARP: IconData

buildSTORMinstance-attribute

STORM: IconData

buildSTORM\_OUTLINEDinstance-attribute

STORM\_OUTLINED: IconData

buildSTORM\_ROUNDEDinstance-attribute

STORM\_ROUNDED: IconData

buildSTORM\_SHARPinstance-attribute

STORM\_SHARP: IconData

buildSTRAIGHTinstance-attribute

STRAIGHT: IconData

buildSTRAIGHT\_OUTLINEDinstance-attribute

STRAIGHT\_OUTLINED: IconData

buildSTRAIGHT\_ROUNDEDinstance-attribute

STRAIGHT\_ROUNDED: IconData

buildSTRAIGHT\_SHARPinstance-attribute

STRAIGHT\_SHARP: IconData

buildSTRAIGHTENinstance-attribute

STRAIGHTEN: IconData

buildSTRAIGHTEN\_OUTLINEDinstance-attribute

STRAIGHTEN\_OUTLINED: IconData

buildSTRAIGHTEN\_ROUNDEDinstance-attribute

STRAIGHTEN\_ROUNDED: IconData

buildSTRAIGHTEN\_SHARPinstance-attribute

STRAIGHTEN\_SHARP: IconData

buildSTREAMinstance-attribute

STREAM: IconData

buildSTREAM\_OUTLINEDinstance-attribute

STREAM\_OUTLINED: IconData

buildSTREAM\_ROUNDEDinstance-attribute

STREAM\_ROUNDED: IconData

buildSTREAM\_SHARPinstance-attribute

STREAM\_SHARP: IconData

buildSTREETVIEWinstance-attribute

STREETVIEW: IconData

buildSTREETVIEW\_OUTLINEDinstance-attribute

STREETVIEW\_OUTLINED: IconData

buildSTREETVIEW\_ROUNDEDinstance-attribute

STREETVIEW\_ROUNDED: IconData

buildSTREETVIEW\_SHARPinstance-attribute

STREETVIEW\_SHARP: IconData

buildSTRIKETHROUGH\_Sinstance-attribute

STRIKETHROUGH\_S: IconData

**buildSTRIKETHROUGH\_S\_OUTLINED**instance-attribute

STRIKETHROUGH\_S\_OUTLINED: IconData

**buildSTRIKETHROUGH\_S\_ROUNDED**instance-attribute

STRIKETHROUGH\_S\_ROUNDED: IconData

**buildSTRIKETHROUGH\_S\_SHARP**instance-attribute

STRIKETHROUGH\_S\_SHARP: IconData

**buildSTROLLER**instance-attribute

STROLLER: IconData

**buildSTROLLER\_OUTLINED**instance-attribute

STROLLER\_OUTLINED: IconData

**buildSTROLLER\_ROUNDED**instance-attribute

STROLLER\_ROUNDED: IconData

**buildSTROLLER\_SHARP**instance-attribute

STROLLER\_SHARP: IconData

**buildSTYLE**instance-attribute

STYLE: IconData

**buildSTYLE\_OUTLINED**instance-attribute

STYLE\_OUTLINED: IconData

**buildSTYLE\_ROUNDED**instance-attribute

STYLE\_ROUNDED: IconData

**buildSTYLE\_SHARP**instance-attribute

STYLE\_SHARP: IconData

**buildSUBDIRECTORY\_ARROW\_LEFT**instance-attribute

SUBDIRECTORY\_ARROW\_LEFT: IconData

**buildSUBDIRECTORY\_ARROW\_LEFT\_OUTLINED**instance-attribute

SUBDIRECTORY\_ARROW\_LEFT\_OUTLINED: IconData

**buildSUBDIRECTORY\_ARROW\_LEFT\_ROUNDED**instance-attribute

SUBDIRECTORY\_ARROW\_LEFT\_ROUNDED: IconData

**buildSUBDIRECTORY\_ARROW\_LEFT\_SHARP**instance-attribute

SUBDIRECTORY\_ARROW\_LEFT\_SHARP: IconData

**buildSUBDIRECTORY\_ARROW\_RIGHT**instance-attribute

SUBDIRECTORY\_ARROW\_RIGHT: IconData

**buildSUBDIRECTORY\_ARROW\_RIGHT\_OUTLINED**instance-attribute

SUBDIRECTORY\_ARROW\_RIGHT\_OUTLINED: IconData

**buildSUBDIRECTORY\_ARROW\_RIGHT\_ROUNDED**instance-attribute

SUBDIRECTORY\_ARROW\_RIGHT\_ROUNDED: IconData

**buildSUBDIRECTORY\_ARROW\_RIGHT\_SHARP**instance-attribute

SUBDIRECTORY\_ARROW\_RIGHT\_SHARP: IconData

**buildSUBJECT**instance-attribute

SUBJECT: IconData

**buildSUBJECT\_OUTLINED**instance-attribute

SUBJECT\_OUTLINED: IconData

**buildSUBJECT\_ROUNDED**instance-attribute

SUBJECT\_ROUNDED: IconData

buildSUBJECT\_SHARPinstance-attribute

SUBJECT\_SHARP: IconData

buildSUBSCRIPTinstance-attribute

SUBSCRIPT: IconData

buildSUBSCRIPT\_OUTLINEDinstance-attribute

SUBSCRIPT\_OUTLINED: IconData

buildSUBSCRIPT\_ROUNDEDinstance-attribute

SUBSCRIPT\_ROUNDED: IconData

buildSUBSCRIPT\_SHARPinstance-attribute

SUBSCRIPT\_SHARP: IconData

buildSUBSCRIPTIONSinstance-attribute

SUBSCRIPTIONS: IconData

buildSUBSCRIPTIONS\_OUTLINEDinstance-attribute

SUBSCRIPTIONS\_OUTLINED: IconData

buildSUBSCRIPTIONS\_ROUNDEDinstance-attribute

SUBSCRIPTIONS\_ROUNDED: IconData

buildSUBSCRIPTIONS\_SHARPinstance-attribute

SUBSCRIPTIONS\_SHARP: IconData

buildSUBTITLESinstance-attribute

SUBTITLES: IconData

buildSUBTITLES\_OFFinstance-attribute

SUBTITLES\_OFF: IconData

buildSUBTITLES\_OFF\_OUTLINEDinstance-attribute

SUBTITLES\_OFF\_OUTLINED: IconData

buildSUBTITLES\_OFF\_ROUNDEDinstance-attribute

SUBTITLES\_OFF\_ROUNDED: IconData

buildSUBTITLES\_OFF\_SHARPinstance-attribute

SUBTITLES\_OFF\_SHARP: IconData

buildSUBTITLES\_OUTLINEDinstance-attribute

SUBTITLES\_OUTLINED: IconData

buildSUBTITLES\_ROUNDEDinstance-attribute

SUBTITLES\_ROUNDED: IconData

buildSUBTITLES\_SHARPinstance-attribute

SUBTITLES\_SHARP: IconData

buildSUBWAYinstance-attribute

SUBWAY: IconData

buildSUBWAY\_OUTLINEDinstance-attribute

SUBWAY\_OUTLINED: IconData

buildSUBWAY\_ROUNDEDinstance-attribute

SUBWAY\_ROUNDED: IconData

buildSUBWAY\_SHARPinstance-attribute

SUBWAY\_SHARP: IconData

buildSUMMARIZEinstance-attribute

SUMMARIZE: IconData

**buildSUMMARIZE\_OUTLINED**instance-attribute

SUMMARIZE\_OUTLINED: IconData

**buildSUMMARIZE\_ROUNDED**instance-attribute

SUMMARIZE\_ROUNDED: IconData

**buildSUMMARIZE\_SHARP**instance-attribute

SUMMARIZE\_SHARP: IconData

**buildSUNNY**instance-attribute

SUNNY: IconData

**buildSUNNY\_SNOWING**instance-attribute

SUNNY\_SNOWING: IconData

**buildSUPERSCRIPT**instance-attribute

SUPERSCRIPT: IconData

**buildSUPERSCRIPT\_OUTLINED**instance-attribute

SUPERSCRIPT\_OUTLINED: IconData

**buildSUPERSCRIPT\_ROUNDED**instance-attribute

SUPERSCRIPT\_ROUNDED: IconData

**buildSUPERSCRIPT\_SHARP**instance-attribute

SUPERSCRIPT\_SHARP: IconData

**buildSUPERVISED\_USER\_CIRCLE**instance-attribute

SUPERVISED\_USER\_CIRCLE: IconData

**buildSUPERVISED\_USER\_CIRCLE\_OUTLINED**instance-attribute

SUPERVISED\_USER\_CIRCLE\_OUTLINED: IconData

**buildSUPERVISED\_USER\_CIRCLE\_ROUNDED**instance-attribute

SUPERVISED\_USER\_CIRCLE\_ROUNDED: IconData

**buildSUPERVISED\_USER\_CIRCLE\_SHARP**instance-attribute

SUPERVISED\_USER\_CIRCLE\_SHARP: IconData

**buildSUPERVISOR\_ACCOUNT**instance-attribute

SUPERVISOR\_ACCOUNT: IconData

**buildSUPERVISOR\_ACCOUNT\_OUTLINED**instance-attribute

SUPERVISOR\_ACCOUNT\_OUTLINED: IconData

**buildSUPERVISOR\_ACCOUNT\_ROUNDED**instance-attribute

SUPERVISOR\_ACCOUNT\_ROUNDED: IconData

**buildSUPERVISOR\_ACCOUNT\_SHARP**instance-attribute

SUPERVISOR\_ACCOUNT\_SHARP: IconData

**buildSUPPORT**instance-attribute

SUPPORT: IconData

**buildSUPPORT\_AGENT**instance-attribute

SUPPORT\_AGENT: IconData

**buildSUPPORT\_AGENT\_OUTLINED**instance-attribute

SUPPORT\_AGENT\_OUTLINED: IconData

**buildSUPPORT\_AGENT\_ROUNDED**instance-attribute

SUPPORT\_AGENT\_ROUNDED: IconData

**buildSUPPORT\_AGENT\_SHARP**instance-attribute

SUPPORT\_AGENT\_SHARP: IconData

buildSUPPORT\_OUTLINEDinstance-attribute

SUPPORT\_OUTLINED: IconData

buildSUPPORT\_ROUNDEDinstance-attribute

SUPPORT\_ROUNDED: IconData

buildSUPPORT\_SHARPinstance-attribute

SUPPORT\_SHARP: IconData

buildSURFINGinstance-attribute

SURFING: IconData

buildSURFING\_OUTLINEDinstance-attribute

SURFING\_OUTLINED: IconData

buildSURFING\_ROUNDEDinstance-attribute

SURFING\_ROUNDED: IconData

buildSURFING\_SHARPinstance-attribute

SURFING\_SHARP: IconData

buildSURROUND\_SOUNDinstance-attribute

SURROUND\_SOUND: IconData

buildSURROUND\_SOUND\_OUTLINEDinstance-attribute

SURROUND\_SOUND\_OUTLINED: IconData

buildSURROUND\_SOUND\_ROUNDEDinstance-attribute

SURROUND\_SOUND\_ROUNDED: IconData

buildSURROUND\_SOUND\_SHARPinstance-attribute

SURROUND\_SOUND\_SHARP: IconData

buildSWAP\_CALLSinstance-attribute

SWAP\_CALLS: IconData

buildSWAP\_CALLS\_OUTLINEDinstance-attribute

SWAP\_CALLS\_OUTLINED: IconData

buildSWAP\_CALLS\_ROUNDEDinstance-attribute

SWAP\_CALLS\_ROUNDED: IconData

buildSWAP\_CALLS\_SHARPinstance-attribute

SWAP\_CALLS\_SHARP: IconData

buildSWAP\_HORIZinstance-attribute

SWAP\_HORIZ: IconData

buildSWAP\_HORIZ\_OUTLINEDinstance-attribute

SWAP\_HORIZ\_OUTLINED: IconData

buildSWAP\_HORIZ\_ROUNDEDinstance-attribute

SWAP\_HORIZ\_ROUNDED: IconData

buildSWAP\_HORIZ\_SHARPinstance-attribute

SWAP\_HORIZ\_SHARP: IconData

buildSWAP\_HORIZONTAL\_CIRCLEinstance-attribute

SWAP\_HORIZONTAL\_CIRCLE: IconData

buildSWAP\_HORIZONTAL\_CIRCLE\_OUTLINEDinstance-attribute

SWAP\_HORIZONTAL\_CIRCLE\_OUTLINED: IconData

buildSWAP\_HORIZONTAL\_CIRCLE\_ROUNDEDinstance-attribute

SWAP\_HORIZONTAL\_CIRCLE\_ROUNDED: IconData

buildSWAP\_HORIZONTAL\_CIRCLE\_SHARPinstance-attribute

SWAP\_HORIZONTAL\_CIRCLE\_SHARP: IconData

buildSWAP\_VERTinstance-attribute

SWAP\_VERT: IconData

buildSWAP\_VERT\_CIRCLEinstance-attribute

SWAP\_VERT\_CIRCLE: IconData

buildSWAP\_VERT\_CIRCLE\_OUTLINEDinstance-attribute

SWAP\_VERT\_CIRCLE\_OUTLINED: IconData

buildSWAP\_VERT\_CIRCLE\_ROUNDEDinstance-attribute

SWAP\_VERT\_CIRCLE\_ROUNDED: IconData

buildSWAP\_VERT\_CIRCLE\_SHARPinstance-attribute

SWAP\_VERT\_CIRCLE\_SHARP: IconData

buildSWAP\_VERT\_OUTLINEDinstance-attribute

SWAP\_VERT\_OUTLINED: IconData

buildSWAP\_VERT\_ROUNDEDinstance-attribute

SWAP\_VERT\_ROUNDED: IconData

buildSWAP\_VERT\_SHARPinstance-attribute

SWAP\_VERT\_SHARP: IconData

buildSWAP\_VERTICAL\_CIRCLEinstance-attribute

SWAP\_VERTICAL\_CIRCLE: IconData

buildSWAP\_VERTICAL\_CIRCLE\_OUTLINEDinstance-attribute

SWAP\_VERTICAL\_CIRCLE\_OUTLINED: IconData

buildSWAP\_VERTICAL\_CIRCLE\_ROUNDEDinstance-attribute

SWAP\_VERTICAL\_CIRCLE\_ROUNDED: IconData

buildSWAP\_VERTICAL\_CIRCLE\_SHARPinstance-attribute

SWAP\_VERTICAL\_CIRCLE\_SHARP: IconData

buildSWIPEinstance-attribute

SWIPE: IconData

buildSWIPE\_DOWNinstance-attribute

SWIPE\_DOWN: IconData

buildSWIPE\_DOWN\_ALTinstance-attribute

SWIPE\_DOWN\_ALT: IconData

buildSWIPE\_DOWN\_ALT\_OUTLINEDinstance-attribute

SWIPE\_DOWN\_ALT\_OUTLINED: IconData

buildSWIPE\_DOWN\_ALT\_ROUNDEDinstance-attribute

SWIPE\_DOWN\_ALT\_ROUNDED: IconData

buildSWIPE\_DOWN\_ALT\_SHARPinstance-attribute

SWIPE\_DOWN\_ALT\_SHARP: IconData

buildSWIPE\_DOWN\_OUTLINEDinstance-attribute

SWIPE\_DOWN\_OUTLINED: IconData

buildSWIPE\_DOWN\_ROUNDEDinstance-attribute

SWIPE\_DOWN\_ROUNDED: IconData

buildSWIPE\_DOWN\_SHARPinstance-attribute



SWIPE\_DOWN\_SHARP: IconData

buildSWIPE\_LEFTInstance-attribute

SWIPE\_LEFT: IconData

buildSWIPE\_LEFT\_ALTInstance-attribute

SWIPE\_LEFT\_ALT: IconData

buildSWIPE\_LEFT\_ALT\_OUTLINEDInstance-attribute

SWIPE\_LEFT\_ALT\_OUTLINED: IconData

buildSWIPE\_LEFT\_ALT\_ROUNDEDInstance-attribute

SWIPE\_LEFT\_ALT\_ROUNDED: IconData

buildSWIPE\_LEFT\_ALT\_SHARPInstance-attribute

SWIPE\_LEFT\_ALT\_SHARP: IconData

buildSWIPE\_LEFT\_OUTLINEDInstance-attribute

SWIPE\_LEFT\_OUTLINED: IconData

buildSWIPE\_LEFT\_ROUNDEDInstance-attribute

SWIPE\_LEFT\_ROUNDED: IconData

buildSWIPE\_LEFT\_SHARPInstance-attribute

SWIPE\_LEFT\_SHARP: IconData

buildSWIPE\_OUTLINEDInstance-attribute

SWIPE\_OUTLINED: IconData

buildSWIPE\_RIGHTInstance-attribute

SWIPE\_RIGHT: IconData

buildSWIPE\_RIGHT\_ALTInstance-attribute

SWIPE\_RIGHT\_ALT: IconData

buildSWIPE\_RIGHT\_ALT\_OUTLINEDInstance-attribute

SWIPE\_RIGHT\_ALT\_OUTLINED: IconData

buildSWIPE\_RIGHT\_ALT\_ROUNDEDInstance-attribute

SWIPE\_RIGHT\_ALT\_ROUNDED: IconData

buildSWIPE\_RIGHT\_ALT\_SHARPInstance-attribute

SWIPE\_RIGHT\_ALT\_SHARP: IconData

buildSWIPE\_RIGHT\_OUTLINEDInstance-attribute

SWIPE\_RIGHT\_OUTLINED: IconData

buildSWIPE\_RIGHT\_ROUNDEDInstance-attribute

SWIPE\_RIGHT\_ROUNDED: IconData

buildSWIPE\_RIGHT\_SHARPInstance-attribute

SWIPE\_RIGHT\_SHARP: IconData

buildSWIPE\_ROUNDEDInstance-attribute

SWIPE\_ROUNDED: IconData

buildSWIPE\_SHARPInstance-attribute

SWIPE\_SHARP: IconData

buildSWIPE\_UPInstance-attribute

SWIPE\_UP: IconData

buildSWIPE\_UP\_ALTInstance-attribute

SWIPE\_UP\_ALT: IconData

buildSWIPE\_UP\_ALT\_OUTLINEDInstance-attribute

SWIPE\_UP\_ALT\_OUTLINED: IconData

buildSWIPE\_UP\_ALT\_ROUNDEDinstance-attribute

SWIPE\_UP\_ALT\_ROUNDED: IconData

buildSWIPE\_UP\_ALT\_SHARPinstance-attribute

SWIPE\_UP\_ALT\_SHARP: IconData

buildSWIPE\_UP\_OUTLINEDinstance-attribute

SWIPE\_UP\_OUTLINED: IconData

buildSWIPE\_UP\_ROUNDEDinstance-attribute

SWIPE\_UP\_ROUNDED: IconData

buildSWIPE\_UP\_SHARPinstance-attribute

SWIPE\_UP\_SHARP: IconData

buildSWIPE\_VERTICALinstance-attribute

SWIPE\_VERTICAL: IconData

buildSWIPE\_VERTICAL\_OUTLINEDinstance-attribute

SWIPE\_VERTICAL\_OUTLINED: IconData

buildSWIPE\_VERTICAL\_ROUNDEDinstance-attribute

SWIPE\_VERTICAL\_ROUNDED: IconData

buildSWIPE\_VERTICAL\_SHARPinstance-attribute

SWIPE\_VERTICAL\_SHARP: IconData

buildSWITCH\_ACCESS\_SHORTCUTinstance-attribute

SWITCH\_ACCESS\_SHORTCUT: IconData

buildSWITCH\_ACCESS\_SHORTCUT\_ADDinstance-attribute

SWITCH\_ACCESS\_SHORTCUT\_ADD: IconData

buildSWITCH\_ACCESS\_SHORTCUT\_ADD\_OUTLINEDinstance-attribute

SWITCH\_ACCESS\_SHORTCUT\_ADD\_OUTLINED: IconData

buildSWITCH\_ACCESS\_SHORTCUT\_ADD\_ROUNDEDinstance-attribute

SWITCH\_ACCESS\_SHORTCUT\_ADD\_ROUNDED: IconData

buildSWITCH\_ACCESS\_SHORTCUT\_ADD\_SHARPinstance-attribute

SWITCH\_ACCESS\_SHORTCUT\_ADD\_SHARP: IconData

buildSWITCH\_ACCESS\_SHORTCUT\_OUTLINEDinstance-attribute

SWITCH\_ACCESS\_SHORTCUT\_OUTLINED: IconData

buildSWITCH\_ACCESS\_SHORTCUT\_ROUNDEDinstance-attribute

SWITCH\_ACCESS\_SHORTCUT\_ROUNDED: IconData

buildSWITCH\_ACCESS\_SHORTCUT\_SHARPinstance-attribute

SWITCH\_ACCESS\_SHORTCUT\_SHARP: IconData

buildSWITCH\_ACCOUNTinstance-attribute

SWITCH\_ACCOUNT: IconData

buildSWITCH\_ACCOUNT\_OUTLINEDinstance-attribute

SWITCH\_ACCOUNT\_OUTLINED: IconData

buildSWITCH\_ACCOUNT\_ROUNDEDinstance-attribute

SWITCH\_ACCOUNT\_ROUNDED: IconData

buildSWITCH\_ACCOUNT\_SHARPinstance-attribute

SWITCH\_ACCOUNT\_SHARP: IconData

buildSWITCH\_CAMERAinstance-attribute

SWITCH\_CAMERA: IconData

buildSWITCH\_CAMERA\_OUTLINEDinstance-attribute

SWITCH\_CAMERA\_OUTLINED: IconData

buildSWITCH\_CAMERA\_ROUNDEDinstance-attribute

SWITCH\_CAMERA\_ROUNDED: IconData

buildSWITCH\_CAMERA\_SHARPinstance-attribute

SWITCH\_CAMERA\_SHARP: IconData

buildSWITCH\_LEFTinstance-attribute

SWITCH\_LEFT: IconData

buildSWITCH\_LEFT\_OUTLINEDinstance-attribute

SWITCH\_LEFT\_OUTLINED: IconData

buildSWITCH\_LEFT\_ROUNDEDinstance-attribute

SWITCH\_LEFT\_ROUNDED: IconData

buildSWITCH\_LEFT\_SHARPinstance-attribute

SWITCH\_LEFT\_SHARP: IconData

buildSWITCH\_RIGHTinstance-attribute

SWITCH\_RIGHT: IconData

buildSWITCH\_RIGHT\_OUTLINEDinstance-attribute

SWITCH\_RIGHT\_OUTLINED: IconData

buildSWITCH\_RIGHT\_ROUNDEDinstance-attribute

SWITCH\_RIGHT\_ROUNDED: IconData

buildSWITCH\_RIGHT\_SHARPinstance-attribute

SWITCH\_RIGHT\_SHARP: IconData

buildSWITCH\_VIDEOinstance-attribute

SWITCH\_VIDEO: IconData

buildSWITCH\_VIDEO\_OUTLINEDinstance-attribute

SWITCH\_VIDEO\_OUTLINED: IconData

buildSWITCH\_VIDEO\_ROUNDEDinstance-attribute

SWITCH\_VIDEO\_ROUNDED: IconData

buildSWITCH\_VIDEO\_SHARPinstance-attribute

SWITCH\_VIDEO\_SHARP: IconData

buildSYNAGOGUEinstance-attribute

SYNAGOGUE: IconData

buildSYNAGOGUE\_OUTLINEDinstance-attribute

SYNAGOGUE\_OUTLINED: IconData

buildSYNAGOGUE\_ROUNDEDinstance-attribute

SYNAGOGUE\_ROUNDED: IconData

buildSYNAGOGUE\_SHARPinstance-attribute

SYNAGOGUE\_SHARP: IconData

buildSYNCinstance-attribute

SYNC: IconData

buildSYNC\_ALTinstance-attribute

SYNC\_ALT: IconData

buildSYNC\_ALT\_OUTLINEDinstance-attribute

SYNC\_ALT\_OUTLINED: IconData

**buildSYNC\_ALT\_ROUNDEDinstance-attribute**

SYNC\_ALT\_ROUNDED: IconData

**buildSYNC\_ALT\_SHARPinstance-attribute**

SYNC\_ALT\_SHARP: IconData

**buildSYNC\_DISABLEDinstance-attribute**

SYNC\_DISABLED: IconData

**buildSYNC\_DISABLED\_OUTLINEDinstance-attribute**

SYNC\_DISABLED\_OUTLINED: IconData

**buildSYNC\_DISABLED\_ROUNDEDinstance-attribute**

SYNC\_DISABLED\_ROUNDED: IconData

**buildSYNC\_DISABLED\_SHARPinstance-attribute**

SYNC\_DISABLED\_SHARP: IconData

**buildSYNC\_LOCKinstance-attribute**

SYNC\_LOCK: IconData

**buildSYNC\_LOCK\_OUTLINEDinstance-attribute**

SYNC\_LOCK\_OUTLINED: IconData

**buildSYNC\_LOCK\_ROUNDEDinstance-attribute**

SYNC\_LOCK\_ROUNDED: IconData

**buildSYNC\_LOCK\_SHARPinstance-attribute**

SYNC\_LOCK\_SHARP: IconData

**buildSYNC\_OUTLINEDinstance-attribute**

SYNC\_OUTLINED: IconData

**buildSYNC\_PROBLEMinstance-attribute**

SYNC\_PROBLEM: IconData

**buildSYNC\_PROBLEM\_OUTLINEDinstance-attribute**

SYNC\_PROBLEM\_OUTLINED: IconData

**buildSYNC\_PROBLEM\_ROUNDEDinstance-attribute**

SYNC\_PROBLEM\_ROUNDED: IconData

**buildSYNC\_PROBLEM\_SHARPinstance-attribute**

SYNC\_PROBLEM\_SHARP: IconData

**buildSYNC\_ROUNDEDinstance-attribute**

SYNC\_ROUNDED: IconData

**buildSYNC\_SHARPinstance-attribute**

SYNC\_SHARP: IconData

**buildSYSTEM\_SECURITY\_UPDATEinstance-attribute**

SYSTEM\_SECURITY\_UPDATE: IconData

**buildSYSTEM\_SECURITY\_UPDATE\_GOODinstance-attribute**

SYSTEM\_SECURITY\_UPDATE\_GOOD: IconData

**buildSYSTEM\_SECURITY\_UPDATE\_GOOD\_OUTLINEDinstance-attribute**

SYSTEM\_SECURITY\_UPDATE\_GOOD\_OUTLINED: IconData

**buildSYSTEM\_SECURITY\_UPDATE\_GOOD\_ROUNDEDinstance-attribute**

SYSTEM\_SECURITY\_UPDATE\_GOOD\_ROUNDED: IconData

**buildSYSTEM\_SECURITY\_UPDATE\_GOOD\_SHARPinstance-attribute**

SYSTEM\_SECURITY\_UPDATE\_GOOD\_SHARP: IconData

buildSYSTEM\_SECURITY\_UPDATE\_OUTLINEDinstance-attribute

SYSTEM\_SECURITY\_UPDATE\_OUTLINED: IconData

buildSYSTEM\_SECURITY\_UPDATE\_ROUNDEDinstance-attribute

SYSTEM\_SECURITY\_UPDATE\_ROUNDED: IconData

buildSYSTEM\_SECURITY\_UPDATE\_SHARPinstance-attribute

SYSTEM\_SECURITY\_UPDATE\_SHARP: IconData

buildSYSTEM\_SECURITY\_UPDATE\_WARNINGinstance-attribute

SYSTEM\_SECURITY\_UPDATE\_WARNING: IconData

buildSYSTEM\_SECURITY\_UPDATE\_WARNING\_OUTLINEDinstance-attribute

SYSTEM\_SECURITY\_UPDATE\_WARNING\_OUTLINED: IconData

buildSYSTEM\_SECURITY\_UPDATE\_WARNING\_ROUNDEDinstance-attribute

SYSTEM\_SECURITY\_UPDATE\_WARNING\_ROUNDED: IconData

buildSYSTEM\_SECURITY\_UPDATE\_WARNING\_SHARPinstance-attribute

SYSTEM\_SECURITY\_UPDATE\_WARNING\_SHARP: IconData

buildSYSTEM\_UPDATEinstance-attribute

SYSTEM\_UPDATE: IconData

buildSYSTEM\_UPDATE\_ALTinstance-attribute

SYSTEM\_UPDATE\_ALT: IconData

buildSYSTEM\_UPDATE\_ALT\_OUTLINEDinstance-attribute

SYSTEM\_UPDATE\_ALT\_OUTLINED: IconData

buildSYSTEM\_UPDATE\_ALT\_ROUNDEDinstance-attribute

SYSTEM\_UPDATE\_ALT\_ROUNDED: IconData

buildSYSTEM\_UPDATE\_ALT\_SHARPinstance-attribute

SYSTEM\_UPDATE\_ALT\_SHARP: IconData

buildSYSTEM\_UPDATE\_OUTLINEDinstance-attribute

SYSTEM\_UPDATE\_OUTLINED: IconData

buildSYSTEM\_UPDATE\_ROUNDEDinstance-attribute

SYSTEM\_UPDATE\_ROUNDED: IconData

buildSYSTEM\_UPDATE\_SHARPinstance-attribute

SYSTEM\_UPDATE\_SHARP: IconData

buildSYSTEM\_UPDATE\_TVinstance-attribute

SYSTEM\_UPDATE\_TV: IconData

buildSYSTEM\_UPDATE\_TV\_OUTLINEDinstance-attribute

SYSTEM\_UPDATE\_TV\_OUTLINED: IconData

buildSYSTEM\_UPDATE\_TV\_ROUNDEDinstance-attribute

SYSTEM\_UPDATE\_TV\_ROUNDED: IconData

buildSYSTEM\_UPDATE\_TV\_SHARPinstance-attribute

SYSTEM\_UPDATE\_TV\_SHARP: IconData

buildTABinstance-attribute

TAB: IconData

buildTAB\_OUTLINEDinstance-attribute

TAB\_OUTLINED: IconData

buildTAB\_ROUNDEDinstance-attribute

TAB\_ROUNDED: IconData

buildTAB\_SHARPinstance-attribute

TAB\_SHARP: IconData

buildTAB\_UNSELECTEDinstance-attribute

TAB\_UNSELECTED: IconData

buildTAB\_UNSELECTED\_OUTLINEDinstance-attribute

TAB\_UNSELECTED\_OUTLINED: IconData

buildTAB\_UNSELECTED\_ROUNDEDinstance-attribute

TAB\_UNSELECTED\_ROUNDED: IconData

buildTAB\_UNSELECTED\_SHARPinstance-attribute

TAB\_UNSELECTED\_SHARP: IconData

buildTABLE\_BARinstance-attribute

TABLE\_BAR: IconData

buildTABLE\_BAR\_OUTLINEDinstance-attribute

TABLE\_BAR\_OUTLINED: IconData

buildTABLE\_BAR\_ROUNDEDinstance-attribute

TABLE\_BAR\_ROUNDED: IconData

buildTABLE\_BAR\_SHARPinstance-attribute

TABLE\_BAR\_SHARP: IconData

buildTABLE\_CHARTinstance-attribute

TABLE\_CHART: IconData

buildTABLE\_CHART\_OUTLINEDinstance-attribute

TABLE\_CHART\_OUTLINED: IconData

buildTABLE\_CHART\_ROUNDEDinstance-attribute

TABLE\_CHART\_ROUNDED: IconData

buildTABLE\_CHART\_SHARPinstance-attribute

TABLE\_CHART\_SHARP: IconData

buildTABLE\_RESTAURANTinstance-attribute

TABLE\_RESTAURANT: IconData

buildTABLE\_RESTAURANT\_OUTLINEDinstance-attribute

TABLE\_RESTAURANT\_OUTLINED: IconData

buildTABLE\_RESTAURANT\_ROUNDEDinstance-attribute

TABLE\_RESTAURANT\_ROUNDED: IconData

buildTABLE\_RESTAURANT\_SHARPinstance-attribute

TABLE\_RESTAURANT\_SHARP: IconData

buildTABLE\_ROWSinstance-attribute

TABLE\_ROWS: IconData

buildTABLE\_ROWS\_OUTLINEDinstance-attribute

TABLE\_ROWS\_OUTLINED: IconData

buildTABLE\_ROWS\_ROUNDEDinstance-attribute

TABLE\_ROWS\_ROUNDED: IconData

buildTABLE\_ROWS\_SHARPinstance-attribute

TABLE\_ROWS\_SHARP: IconData

buildTABLE\_VIEWinstance-attribute

TABLE\_VIEW: IconData

buildTABLE\_VIEW\_OUTLINEDinstance-attribute

TABLE\_VIEW\_OUTLINED: IconData

buildTABLE\_VIEW\_ROUNDEDinstance-attribute

TABLE\_VIEW\_ROUNDED: IconData

buildTABLE\_VIEW\_SHARPinstance-attribute

TABLE\_VIEW\_SHARP: IconData

buildTABLETinstance-attribute

TABLET: IconData

buildTABLET\_ANDROIDinstance-attribute

TABLET\_ANDROID: IconData

buildTABLET\_ANDROID\_OUTLINEDinstance-attribute

TABLET\_ANDROID\_OUTLINED: IconData

buildTABLET\_ANDROID\_ROUNDEDinstance-attribute

TABLET\_ANDROID\_ROUNDED: IconData

buildTABLET\_ANDROID\_SHARPinstance-attribute

TABLET\_ANDROID\_SHARP: IconData

buildTABLET\_MACinstance-attribute

TABLET\_MAC: IconData

buildTABLET\_MAC\_OUTLINEDinstance-attribute

TABLET\_MAC\_OUTLINED: IconData

buildTABLET\_MAC\_ROUNDEDinstance-attribute

TABLET\_MAC\_ROUNDED: IconData

buildTABLET\_MAC\_SHARPinstance-attribute

TABLET\_MAC\_SHARP: IconData

buildTABLET\_OUTLINEDinstance-attribute

TABLET\_OUTLINED: IconData

buildTABLET\_ROUNDEDinstance-attribute

TABLET\_ROUNDED: IconData

buildTABLET\_SHARPinstance-attribute

TABLET\_SHARP: IconData

buildTAGinstance-attribute

TAG: IconData

buildTAG\_FACESinstance-attribute

TAG\_FACES: IconData

buildTAG\_FACES\_OUTLINEDinstance-attribute

TAG\_FACES\_OUTLINED: IconData

buildTAG\_FACES\_ROUNDEDinstance-attribute

TAG\_FACES\_ROUNDED: IconData

buildTAG\_FACES\_SHARPinstance-attribute

TAG\_FACES\_SHARP: IconData

buildTAG\_OUTLINEDinstance-attribute

TAG\_OUTLINED: IconData

buildTAG\_ROUNDEDinstance-attribute

**TAG\_ROUNDED:** IconData

**buildTAG\_SHARP**instance-attribute

**TAG\_SHARP:** IconData

**buildTAKEOUT\_DINING**instance-attribute

**TAKEOUT\_DINING:** IconData

**buildTAKEOUT\_DINING\_OUTLINED**instance-attribute

**TAKEOUT\_DINING\_OUTLINED:** IconData

**buildTAKEOUT\_DINING\_ROUNDED**instance-attribute

**TAKEOUT\_DINING\_ROUNDED:** IconData

**buildTAKEOUT\_DINING\_SHARP**instance-attribute

**TAKEOUT\_DINING\_SHARP:** IconData

**buildTAP\_AND\_PLAY**instance-attribute

**TAP\_AND\_PLAY:** IconData

**buildTAP\_AND\_PLAY\_OUTLINED**instance-attribute

**TAP\_AND\_PLAY\_OUTLINED:** IconData

**buildTAP\_AND\_PLAY\_ROUNDED**instance-attribute

**TAP\_AND\_PLAY\_ROUNDED:** IconData

**buildTAP\_AND\_PLAY\_SHARP**instance-attribute

**TAP\_AND\_PLAY\_SHARP:** IconData

**buildTAPAS**instance-attribute

**TAPAS:** IconData

**buildTAPAS\_OUTLINED**instance-attribute

**TAPAS\_OUTLINED:** IconData

**buildTAPAS\_ROUNDED**instance-attribute

**TAPAS\_ROUNDED:** IconData

**buildTAPAS\_SHARP**instance-attribute

**TAPAS\_SHARP:** IconData

**buildTASK**instance-attribute

**TASK:** IconData

**buildTASK\_ALT**instance-attribute

**TASK\_ALT:** IconData

**buildTASK\_ALT\_OUTLINED**instance-attribute

**TASK\_ALT\_OUTLINED:** IconData

**buildTASK\_ALT\_ROUNDED**instance-attribute

**TASK\_ALT\_ROUNDED:** IconData

**buildTASK\_ALT\_SHARP**instance-attribute

**TASK\_ALT\_SHARP:** IconData

**buildTASK\_OUTLINED**instance-attribute

**TASK\_OUTLINED:** IconData

**buildTASK\_ROUNDED**instance-attribute

**TASK\_ROUNDED:** IconData

**buildTASK\_SHARP**instance-attribute

**TASK\_SHARP:** IconData

**buildTAXI\_ALERT**instance-attribute



TAXI\_ALERT: IconData

buildTAXI\_ALERT\_OUTLINEDinstance-attribute

TAXI\_ALERT\_OUTLINED: IconData

buildTAXI\_ALERT\_ROUNDEDinstance-attribute

TAXI\_ALERT\_ROUNDED: IconData

buildTAXI\_ALERT\_SHARPinstance-attribute

TAXI\_ALERT\_SHARP: IconData

buildTELEGRAMinstance-attribute

TELEGRAM: IconData

buildTELEGRAM\_OUTLINEDinstance-attribute

TELEGRAM\_OUTLINED: IconData

buildTELEGRAM\_ROUNDEDinstance-attribute

TELEGRAM\_ROUNDED: IconData

buildTELEGRAM\_SHARPinstance-attribute

TELEGRAM\_SHARP: IconData

buildTEMPLE\_BUDDHISTinstance-attribute

TEMPLE\_BUDDHIST: IconData

buildTEMPLE\_BUDDHIST\_OUTLINEDinstance-attribute

TEMPLE\_BUDDHIST\_OUTLINED: IconData

buildTEMPLE\_BUDDHIST\_ROUNDEDinstance-attribute

TEMPLE\_BUDDHIST\_ROUNDED: IconData

buildTEMPLE\_BUDDHIST\_SHARPinstance-attribute

TEMPLE\_BUDDHIST\_SHARP: IconData

buildTEMPLE\_HINDUinstance-attribute

TEMPLE\_HINDU: IconData

buildTEMPLE\_HINDU\_OUTLINEDinstance-attribute

TEMPLE\_HINDU\_OUTLINED: IconData

buildTEMPLE\_HINDU\_ROUNDEDinstance-attribute

TEMPLE\_HINDU\_ROUNDED: IconData

buildTEMPLE\_HINDU\_SHARPinstance-attribute

TEMPLE\_HINDU\_SHARP: IconData

buildTEN\_Kinstance-attribute

TEN\_K: IconData

buildTEN\_K\_OUTLINEDinstance-attribute

TEN\_K\_OUTLINED: IconData

buildTEN\_K\_ROUNDEDinstance-attribute

TEN\_K\_ROUNDED: IconData

buildTEN\_K\_SHARPinstance-attribute

TEN\_K\_SHARP: IconData

buildTEN\_MPinstance-attribute

TEN\_MP: IconData

buildTEN\_MP\_OUTLINEDinstance-attribute

TEN\_MP\_OUTLINED: IconData

buildTEN\_MP\_ROUNDEDinstance-attribute

TEN\_MP\_ROUNDED: IconData

**buildTEN\_MP\_SHARPinstance-attribute**

TEN\_MP\_SHARP: IconData

**buildTERMINALinstance-attribute**

TERMINAL: IconData

**buildTERMINAL\_OUTLINEDinstance-attribute**

TERMINAL\_OUTLINED: IconData

**buildTERMINAL\_ROUNDEDinstance-attribute**

TERMINAL\_ROUNDED: IconData

**buildTERMINAL\_SHARPinstance-attribute**

TERMINAL\_SHARP: IconData

**buildTERRAINinstance-attribute**

TERRAIN: IconData

**buildTERRAIN\_OUTLINEDinstance-attribute**

TERRAIN\_OUTLINED: IconData

**buildTERRAIN\_ROUNDEDinstance-attribute**

TERRAIN\_ROUNDED: IconData

**buildTERRAIN\_SHARPinstance-attribute**

TERRAIN\_SHARP: IconData

**buildTEXT\_DECREASEinstance-attribute**

TEXT\_DECREASE: IconData

**buildTEXT\_DECREASE\_OUTLINEDinstance-attribute**

TEXT\_DECREASE\_OUTLINED: IconData

**buildTEXT\_DECREASE\_ROUNDEDinstance-attribute**

TEXT\_DECREASE\_ROUNDED: IconData

**buildTEXT\_DECREASE\_SHARPinstance-attribute**

TEXT\_DECREASE\_SHARP: IconData

**buildTEXT\_FIELDSinstance-attribute**

TEXT\_FIELDS: IconData

**buildTEXT\_FIELDS\_OUTLINEDinstance-attribute**

TEXT\_FIELDS\_OUTLINED: IconData

**buildTEXT\_FIELDS\_ROUNDEDinstance-attribute**

TEXT\_FIELDS\_ROUNDED: IconData

**buildTEXT\_FIELDS\_SHARPinstance-attribute**

TEXT\_FIELDS\_SHARP: IconData

**buildTEXT\_FORMATinstance-attribute**

TEXT\_FORMAT: IconData

**buildTEXT\_FORMAT\_OUTLINEDinstance-attribute**

TEXT\_FORMAT\_OUTLINED: IconData

**buildTEXT\_FORMAT\_ROUNDEDinstance-attribute**

TEXT\_FORMAT\_ROUNDED: IconData

**buildTEXT\_FORMAT\_SHARPinstance-attribute**

TEXT\_FORMAT\_SHARP: IconData

**buildTEXT\_INCREASEinstance-attribute**

TEXT\_INCREASE: IconData

buildTEXT\_INCREASE\_OUTLINEDinstance-attribute

TEXT\_INCREASE\_OUTLINED: IconData

buildTEXT\_INCREASE\_ROUNDEDinstance-attribute

TEXT\_INCREASE\_ROUNDED: IconData

buildTEXT\_INCREASE\_SHARPinstance-attribute

TEXT\_INCREASE\_SHARP: IconData

buildTEXT\_ROTATE\_UPinstance-attribute

TEXT\_ROTATE\_UP: IconData

buildTEXT\_ROTATE\_UP\_OUTLINEDinstance-attribute

TEXT\_ROTATE\_UP\_OUTLINED: IconData

buildTEXT\_ROTATE\_UP\_ROUNDEDinstance-attribute

TEXT\_ROTATE\_UP\_ROUNDED: IconData

buildTEXT\_ROTATE\_UP\_SHARPinstance-attribute

TEXT\_ROTATE\_UP\_SHARP: IconData

buildTEXT\_ROTATE\_VERTICALinstance-attribute

TEXT\_ROTATE\_VERTICAL: IconData

buildTEXT\_ROTATE\_VERTICAL\_OUTLINEDinstance-attribute

TEXT\_ROTATE\_VERTICAL\_OUTLINED: IconData

buildTEXT\_ROTATE\_VERTICAL\_ROUNDEDinstance-attribute

TEXT\_ROTATE\_VERTICAL\_ROUNDED: IconData

buildTEXT\_ROTATE\_VERTICAL\_SHARPinstance-attribute

TEXT\_ROTATE\_VERTICAL\_SHARP: IconData

buildTEXT\_ROTATION\_ANGLEDOWNinstance-attribute

TEXT\_ROTATION\_ANGLEDOWN: IconData

buildTEXT\_ROTATION\_ANGLEDOWN\_OUTLINEDinstance-attribute

TEXT\_ROTATION\_ANGLEDOWN\_OUTLINED: IconData

buildTEXT\_ROTATION\_ANGLEDOWN\_ROUNDEDinstance-attribute

TEXT\_ROTATION\_ANGLEDOWN\_ROUNDED: IconData

buildTEXT\_ROTATION\_ANGLEDOWN\_SHARPinstance-attribute

TEXT\_ROTATION\_ANGLEDOWN\_SHARP: IconData

buildTEXT\_ROTATION\_ANGLEUPinstance-attribute

TEXT\_ROTATION\_ANGLEUP: IconData

buildTEXT\_ROTATION\_ANGLEUP\_OUTLINEDinstance-attribute

TEXT\_ROTATION\_ANGLEUP\_OUTLINED: IconData

buildTEXT\_ROTATION\_ANGLEUP\_ROUNDEDinstance-attribute

TEXT\_ROTATION\_ANGLEUP\_ROUNDED: IconData

buildTEXT\_ROTATION\_ANGLEUP\_SHARPinstance-attribute

TEXT\_ROTATION\_ANGLEUP\_SHARP: IconData

buildTEXT\_ROTATION\_DOWNinstance-attribute

TEXT\_ROTATION\_DOWN: IconData

buildTEXT\_ROTATION\_DOWN\_OUTLINEDinstance-attribute

TEXT\_ROTATION\_DOWN\_OUTLINED: IconData

buildTEXT\_ROTATION\_DOWN\_ROUNDEDinstance-attribute

TEXT\_ROTATION\_DOWN\_ROUNDED: IconData

buildTEXT\_ROTATION\_DOWN\_SHARPinstance-attribute

TEXT\_ROTATION\_DOWN\_SHARP: IconData

buildTEXT\_ROTATION\_NONEinstance-attribute

TEXT\_ROTATION\_NONE: IconData

buildTEXT\_ROTATION\_NONE\_OUTLINEDinstance-attribute

TEXT\_ROTATION\_NONE\_OUTLINED: IconData

buildTEXT\_ROTATION\_NONE\_ROUNDEDinstance-attribute

TEXT\_ROTATION\_NONE\_ROUNDED: IconData

buildTEXT\_ROTATION\_NONE\_SHARPinstance-attribute

TEXT\_ROTATION\_NONE\_SHARP: IconData

buildTEXT\_SNIPPETinstance-attribute

TEXT\_SNIPPET: IconData

buildTEXT\_SNIPPET\_OUTLINEDinstance-attribute

TEXT\_SNIPPET\_OUTLINED: IconData

buildTEXT\_SNIPPET\_ROUNDEDinstance-attribute

TEXT\_SNIPPET\_ROUNDED: IconData

buildTEXT\_SNIPPET\_SHARPinstance-attribute

TEXT\_SNIPPET\_SHARP: IconData

buildTEXTSMSinstance-attribute

TEXTSMS: IconData

buildTEXTSMS\_OUTLINEDinstance-attribute

TEXTSMS\_OUTLINED: IconData

buildTEXTSMS\_ROUNDEDinstance-attribute

TEXTSMS\_ROUNDED: IconData

buildTEXTSMS\_SHARPinstance-attribute

TEXTSMS\_SHARP: IconData

buildTEXTUREinstance-attribute

TEXTURE: IconData

buildTEXTURE\_OUTLINEDinstance-attribute

TEXTURE\_OUTLINED: IconData

buildTEXTURE\_ROUNDEDinstance-attribute

TEXTURE\_ROUNDED: IconData

buildTEXTURE\_SHARPinstance-attribute

TEXTURE\_SHARP: IconData

buildTHEATER\_COMEDYinstance-attribute

THEATER\_COMEDY: IconData

buildTHEATER\_COMEDY\_OUTLINEDinstance-attribute

THEATER\_COMEDY\_OUTLINED: IconData

buildTHEATER\_COMEDY\_ROUNDEDinstance-attribute

THEATER\_COMEDY\_ROUNDED: IconData

buildTHEATER\_COMEDY\_SHARPinstance-attribute

THEATER\_COMEDY\_SHARP: IconData

buildTHEATERSinstance-attribute

THEATERS: IconData

buildTHEATERS\_OUTLINEDinstance-attribute

THEATERS\_OUTLINED: IconData

buildTHEATERS\_ROUNDEDinstance-attribute

THEATERS\_ROUNDED: IconData

buildTHEATERS\_SHARPinstance-attribute

THEATERS\_SHARP: IconData

buildTHERMOSTATinstance-attribute

THERMOSTAT: IconData

buildTHERMOSTAT\_AUTOinstance-attribute

THERMOSTAT\_AUTO: IconData

buildTHERMOSTAT\_AUTO\_OUTLINEDinstance-attribute

THERMOSTAT\_AUTO\_OUTLINED: IconData

buildTHERMOSTAT\_AUTO\_ROUNDEDinstance-attribute

THERMOSTAT\_AUTO\_ROUNDED: IconData

buildTHERMOSTAT\_AUTO\_SHARPinstance-attribute

THERMOSTAT\_AUTO\_SHARP: IconData

buildTHERMOSTAT\_OUTLINEDinstance-attribute

THERMOSTAT\_OUTLINED: IconData

buildTHERMOSTAT\_ROUNDEDinstance-attribute

THERMOSTAT\_ROUNDED: IconData

buildTHERMOSTAT\_SHARPinstance-attribute

THERMOSTAT\_SHARP: IconData

buildTHIRTEEN\_MPinstance-attribute

THIRTEEN\_MP: IconData

buildTHIRTEEN\_MP\_OUTLINEDinstance-attribute

THIRTEEN\_MP\_OUTLINED: IconData

buildTHIRTEEN\_MP\_ROUNDEDinstance-attribute

THIRTEEN\_MP\_ROUNDED: IconData

buildTHIRTEEN\_MP\_SHARPinstance-attribute

THIRTEEN\_MP\_SHARP: IconData

buildTHIRTY\_FPSinstance-attribute

THIRTY\_FPS: IconData

buildTHIRTY\_FPS\_OUTLINEDinstance-attribute

THIRTY\_FPS\_OUTLINED: IconData

buildTHIRTY\_FPS\_ROUNDEDinstance-attribute

THIRTY\_FPS\_ROUNDED: IconData

buildTHIRTY\_FPS\_SELECTinstance-attribute

THIRTY\_FPS\_SELECT: IconData

buildTHIRTY\_FPS\_SELECT\_OUTLINEDinstance-attribute

THIRTY\_FPS\_SELECT\_OUTLINED: IconData

buildTHIRTY\_FPS\_SELECT\_ROUNDEDinstance-attribute

THIRTY\_FPS\_SELECT\_ROUNDED: IconData

buildTHIRTY\_FPS\_SELECT\_SHARPinstance-attribute

THIRTY\_FPS\_SELECT\_SHARP: IconData

buildTHIRTY\_FPS\_SHARPinstance-attribute

THIRTY\_FPS\_SHARP: IconData

buildTHREE\_G\_MOBILEDATAinstance-attribute

THREE\_G\_MOBILEDATA: IconData

buildTHREE\_G\_MOBILEDATA\_OUTLINEDinstance-attribute

THREE\_G\_MOBILEDATA\_OUTLINED: IconData

buildTHREE\_G\_MOBILEDATA\_ROUNDEDinstance-attribute

THREE\_G\_MOBILEDATA\_ROUNDED: IconData

buildTHREE\_G\_MOBILEDATA\_SHARPinstance-attribute

THREE\_G\_MOBILEDATA\_SHARP: IconData

buildTHREE\_Kinstance-attribute

THREE\_K: IconData

buildTHREE\_K\_OUTLINEDinstance-attribute

THREE\_K\_OUTLINED: IconData

buildTHREE\_K\_PLUSinstance-attribute

THREE\_K\_PLUS: IconData

buildTHREE\_K\_PLUS\_OUTLINEDinstance-attribute

THREE\_K\_PLUS\_OUTLINED: IconData

buildTHREE\_K\_PLUS\_ROUNDEDinstance-attribute

THREE\_K\_PLUS\_ROUNDED: IconData

buildTHREE\_K\_PLUS\_SHARPinstance-attribute

THREE\_K\_PLUS\_SHARP: IconData

buildTHREE\_K\_ROUNDEDinstance-attribute

THREE\_K\_ROUNDED: IconData

buildTHREE\_K\_SHARPinstance-attribute

THREE\_K\_SHARP: IconData

buildTHREE\_MPinstance-attribute

THREE\_MP: IconData

buildTHREE\_MP\_OUTLINEDinstance-attribute

THREE\_MP\_OUTLINED: IconData

buildTHREE\_MP\_ROUNDEDinstance-attribute

THREE\_MP\_ROUNDED: IconData

buildTHREE\_MP\_SHARPinstance-attribute

THREE\_MP\_SHARP: IconData

buildTHREE\_Pinstance-attribute

THREE\_P: IconData

buildTHREE\_P\_OUTLINEDinstance-attribute

THREE\_P\_OUTLINED: IconData

buildTHREE\_P\_ROUNDEDinstance-attribute

THREE\_P\_ROUNDED: IconData

buildTHREE\_P\_SHARPinstance-attribute

THREE\_P\_SHARP: IconData

buildTHREED\_ROTATIONinstance-attribute

THREED\_ROTATION: IconData

buildTHREED\_ROTATION\_OUTLINEDinstance-attribute

THREED\_ROTATION\_OUTLINED: IconData

buildTHREED\_ROTATION\_ROUNDEDinstance-attribute

THREED\_ROTATION\_ROUNDED: IconData

buildTHREED\_ROTATION\_SHARPinstance-attribute

THREED\_ROTATION\_SHARP: IconData

buildTHREESIXTYinstance-attribute

THREESIXTY: IconData

buildTHREESIXTY\_OUTLINEDinstance-attribute

THREESIXTY\_OUTLINED: IconData

buildTHREESIXTY\_ROUNDEDinstance-attribute

THREESIXTY\_ROUNDED: IconData

buildTHREESIXTY\_SHARPinstance-attribute

THREESIXTY\_SHARP: IconData

buildTHUMB\_DOWNinstance-attribute

THUMB\_DOWN: IconData

buildTHUMB\_DOWN\_ALTinstance-attribute

THUMB\_DOWN\_ALT: IconData

buildTHUMB\_DOWN\_ALT\_OUTLINEDinstance-attribute

THUMB\_DOWN\_ALT\_OUTLINED: IconData

buildTHUMB\_DOWN\_ALT\_ROUNDEDinstance-attribute

THUMB\_DOWN\_ALT\_ROUNDED: IconData

buildTHUMB\_DOWN\_ALT\_SHARPinstance-attribute

THUMB\_DOWN\_ALT\_SHARP: IconData

buildTHUMB\_DOWN\_OFF\_ALTinstance-attribute

THUMB\_DOWN\_OFF\_ALT: IconData

buildTHUMB\_DOWN\_OFF\_ALT\_OUTLINEDinstance-attribute

THUMB\_DOWN\_OFF\_ALT\_OUTLINED: IconData

buildTHUMB\_DOWN\_OFF\_ALT\_ROUNDEDinstance-attribute

THUMB\_DOWN\_OFF\_ALT\_ROUNDED: IconData

buildTHUMB\_DOWN\_OFF\_ALT\_SHARPinstance-attribute

THUMB\_DOWN\_OFF\_ALT\_SHARP: IconData

buildTHUMB\_DOWN\_OUTLINEDinstance-attribute

THUMB\_DOWN\_OUTLINED: IconData

buildTHUMB\_DOWN\_ROUNDEDinstance-attribute

THUMB\_DOWN\_ROUNDED: IconData

buildTHUMB\_DOWN\_SHARPinstance-attribute

THUMB\_DOWN\_SHARP: IconData

buildTHUMB\_UPinstance-attribute

THUMB\_UP: IconData

buildTHUMB\_UP\_ALTinstance-attribute

THUMB\_UP\_ALT: IconData

buildTHUMB\_UP\_ALT\_OUTLINEDinstance-attribute

THUMB\_UP\_ALT\_OUTLINED: IconData

buildTHUMB\_UP\_ALT\_ROUNDEDinstance-attribute

THUMB\_UP\_ALT\_ROUNDED: IconData

buildTHUMB\_UP\_ALT\_SHARPinstance-attribute

THUMB\_UP\_ALT\_SHARP: IconData

buildTHUMB\_UP\_OFF\_ALTinstance-attribute

THUMB\_UP\_OFF\_ALT: IconData

buildTHUMB\_UP\_OFF\_ALT\_OUTLINEDinstance-attribute

THUMB\_UP\_OFF\_ALT\_OUTLINED: IconData

buildTHUMB\_UP\_OFF\_ALT\_ROUNDEDinstance-attribute

THUMB\_UP\_OFF\_ALT\_ROUNDED: IconData

buildTHUMB\_UP\_OFF\_ALT\_SHARPinstance-attribute

THUMB\_UP\_OFF\_ALT\_SHARP: IconData

buildTHUMB\_UP\_OUTLINEDinstance-attribute

THUMB\_UP\_OUTLINED: IconData

buildTHUMB\_UP\_ROUNDEDinstance-attribute

THUMB\_UP\_ROUNDED: IconData

buildTHUMB\_UP\_SHARPinstance-attribute

THUMB\_UP\_SHARP: IconData

buildTHUMBS\_UP\_DOWNinstance-attribute

THUMBS\_UP\_DOWN: IconData

buildTHUMBS\_UP\_DOWN\_OUTLINEDinstance-attribute

THUMBS\_UP\_DOWN\_OUTLINED: IconData

buildTHUMBS\_UP\_DOWN\_ROUNDEDinstance-attribute

THUMBS\_UP\_DOWN\_ROUNDED: IconData

buildTHUMBS\_UP\_DOWN\_SHARPinstance-attribute

THUMBS\_UP\_DOWN\_SHARP: IconData

buildTHUNDERSTORMinstance-attribute

THUNDERSTORM: IconData

buildTHUNDERSTORM\_OUTLINEDinstance-attribute

THUNDERSTORM\_OUTLINED: IconData

buildTHUNDERSTORM\_ROUNDEDinstance-attribute

THUNDERSTORM\_ROUNDED: IconData

buildTHUNDERSTORM\_SHARPinstance-attribute

THUNDERSTORM\_SHARP: IconData

buildTIKTOKinstance-attribute

TIKTOK: IconData

buildTIKTOK\_OUTLINEDinstance-attribute

TIKTOK\_OUTLINED: IconData

buildTIKTOK\_ROUNDEDinstance-attribute

TIKTOK\_ROUNDED: IconData

buildTIKTOK\_SHARPinstance-attribute

TIKTOK\_SHARP: IconData

buildTIME\_TO\_LEAVEinstance-attribute



TIME\_TO\_LEAVE: IconData

buildTIME\_TO\_LEAVE\_OUTLINEDinstance-attribute

TIME\_TO\_LEAVE\_OUTLINED: IconData

buildTIME\_TO\_LEAVE\_ROUNDEDinstance-attribute

TIME\_TO\_LEAVE\_ROUNDED: IconData

buildTIME\_TO\_LEAVE\_SHARPinstance-attribute

TIME\_TO\_LEAVE\_SHARP: IconData

buildTIMELAPSEinstance-attribute

TIMELAPSE: IconData

buildTIMELAPSE\_OUTLINEDinstance-attribute

TIMELAPSE\_OUTLINED: IconData

buildTIMELAPSE\_ROUNDEDinstance-attribute

TIMELAPSE\_ROUNDED: IconData

buildTIMELAPSE\_SHARPinstance-attribute

TIMELAPSE\_SHARP: IconData

buildTIMELINEinstance-attribute

TIMELINE: IconData

buildTIMELINE\_OUTLINEDinstance-attribute

TIMELINE\_OUTLINED: IconData

buildTIMELINE\_ROUNDEDinstance-attribute

TIMELINE\_ROUNDED: IconData

buildTIMELINE\_SHARPinstance-attribute

TIMELINE\_SHARP: IconData

buildTIMERinstance-attribute

TIMER: IconData

buildTIMER\_10instance-attribute

TIMER\_10: IconData

buildTIMER\_10\_OUTLINEDinstance-attribute

TIMER\_10\_OUTLINED: IconData

buildTIMER\_10\_ROUNDEDinstance-attribute

TIMER\_10\_ROUNDED: IconData

buildTIMER\_10\_SELECTinstance-attribute

TIMER\_10\_SELECT: IconData

buildTIMER\_10\_SELECT\_OUTLINEDinstance-attribute

TIMER\_10\_SELECT\_OUTLINED: IconData

buildTIMER\_10\_SELECT\_ROUNDEDinstance-attribute

TIMER\_10\_SELECT\_ROUNDED: IconData

buildTIMER\_10\_SELECT\_SHARPinstance-attribute

TIMER\_10\_SELECT\_SHARP: IconData

buildTIMER\_10\_SHARPinstance-attribute

TIMER\_10\_SHARP: IconData

buildTIMER\_3instance-attribute

TIMER\_3: IconData

buildTIMER\_3\_OUTLINEDinstance-attribute

TIMER\_3\_OUTLINED: IconData

buildTIMER\_3\_ROUNDEDinstance-attribute

TIMER\_3\_ROUNDED: IconData

buildTIMER\_3\_SELECTinstance-attribute

TIMER\_3\_SELECT: IconData

buildTIMER\_3\_SELECT\_OUTLINEDinstance-attribute

TIMER\_3\_SELECT\_OUTLINED: IconData

buildTIMER\_3\_SELECT\_ROUNDEDinstance-attribute

TIMER\_3\_SELECT\_ROUNDED: IconData

buildTIMER\_3\_SELECT\_SHARPinstance-attribute

TIMER\_3\_SELECT\_SHARP: IconData

buildTIMER\_3\_SHARPinstance-attribute

TIMER\_3\_SHARP: IconData

buildTIMER\_OFFinstance-attribute

TIMER\_OFF: IconData

buildTIMER\_OFF\_OUTLINEDinstance-attribute

TIMER\_OFF\_OUTLINED: IconData

buildTIMER\_OFF\_ROUNDEDinstance-attribute

TIMER\_OFF\_ROUNDED: IconData

buildTIMER\_OFF\_SHARPinstance-attribute

TIMER\_OFF\_SHARP: IconData

buildTIMER\_OUTLINEDinstance-attribute

TIMER\_OUTLINED: IconData

buildTIMER\_ROUNDEDinstance-attribute

TIMER\_ROUNDED: IconData

buildTIMER\_SHARPinstance-attribute

TIMER\_SHARP: IconData

buildTIPS\_AND\_UPDATESinstance-attribute

TIPS\_AND\_UPDATES: IconData

buildTIPS\_AND\_UPDATES\_OUTLINEDinstance-attribute

TIPS\_AND\_UPDATES\_OUTLINED: IconData

buildTIPS\_AND\_UPDATES\_ROUNDEDinstance-attribute

TIPS\_AND\_UPDATES\_ROUNDED: IconData

buildTIPS\_AND\_UPDATES\_SHARPinstance-attribute

TIPS\_AND\_UPDATES\_SHARP: IconData

buildTIRE\_REPAIRinstance-attribute

TIRE\_REPAIR: IconData

buildTIRE\_REPAIR\_OUTLINEDinstance-attribute

TIRE\_REPAIR\_OUTLINED: IconData

buildTIRE\_REPAIR\_ROUNDEDinstance-attribute

TIRE\_REPAIR\_ROUNDED: IconData

buildTIRE\_REPAIR\_SHARPinstance-attribute

TIRE\_REPAIR\_SHARP: IconData

buildTITLEinstance-attribute

TITLE: IconData

buildTITLE\_OUTLINEDinstance-attribute

TITLE\_OUTLINED: IconData

buildTITLE\_ROUNDEDinstance-attribute

TITLE\_ROUNDED: IconData

buildTITLE\_SHARPinstance-attribute

TITLE\_SHARP: IconData

buildTOCinstance-attribute

TOC: IconData

buildTOC\_OUTLINEDinstance-attribute

TOC\_OUTLINED: IconData

buildTOC\_ROUNDEDinstance-attribute

TOC\_ROUNDED: IconData

buildTOC\_SHARPinstance-attribute

TOC\_SHARP: IconData

buildTODAYinstance-attribute

TODAY: IconData

buildTODAY\_OUTLINEDinstance-attribute

TODAY\_OUTLINED: IconData

buildTODAY\_ROUNDEDinstance-attribute

TODAY\_ROUNDED: IconData

buildTODAY\_SHARPinstance-attribute

TODAY\_SHARP: IconData

buildTOGGLE\_OFFinstance-attribute

TOGGLE\_OFF: IconData

buildTOGGLE\_OFF\_OUTLINEDinstance-attribute

TOGGLE\_OFF\_OUTLINED: IconData

buildTOGGLE\_OFF\_ROUNDEDinstance-attribute

TOGGLE\_OFF\_ROUNDED: IconData

buildTOGGLE\_OFF\_SHARPinstance-attribute

TOGGLE\_OFF\_SHARP: IconData

buildTOGGLE\_ONinstance-attribute

TOGGLE\_ON: IconData

buildTOGGLE\_ON\_OUTLINEDinstance-attribute

TOGGLE\_ON\_OUTLINED: IconData

buildTOGGLE\_ON\_ROUNDEDinstance-attribute

TOGGLE\_ON\_ROUNDED: IconData

buildTOGGLE\_ON\_SHARPinstance-attribute

TOGGLE\_ON\_SHARP: IconData

buildTOKENinstance-attribute

TOKEN: IconData

buildTOKEN\_OUTLINEDinstance-attribute

TOKEN\_OUTLINED: IconData

buildTOKEN\_ROUNDEDinstance-attribute

TOKEN\_ROUNDED: IconData

**buildTOKEN\_SHARPinstance-attribute**

TOKEN\_SHARP: IconData

**buildTOLLinstance-attribute**

TOLL: IconData

**buildTOLL\_OUTLINEDinstance-attribute**

TOLL\_OUTLINED: IconData

**buildTOLL\_ROUNDEDinstance-attribute**

TOLL\_ROUNDED: IconData

**buildTOLL\_SHARPinstance-attribute**

TOLL\_SHARP: IconData

**buildTONALITYinstance-attribute**

TONALITY: IconData

**buildTONALITY\_OUTLINEDinstance-attribute**

TONALITY\_OUTLINED: IconData

**buildTONALITY\_ROUNDEDinstance-attribute**

TONALITY\_ROUNDED: IconData

**buildTONALITY\_SHARPinstance-attribute**

TONALITY\_SHARP: IconData

**buildTOPICinstance-attribute**

TOPIC: IconData

**buildTOPIC\_OUTLINEDinstance-attribute**

TOPIC\_OUTLINED: IconData

**buildTOPIC\_ROUNDEDinstance-attribute**

TOPIC\_ROUNDED: IconData

**buildTOPIC\_SHARPinstance-attribute**

TOPIC\_SHARP: IconData

**buildTORNADOinstance-attribute**

TORNADO: IconData

**buildTORNADO\_OUTLINEDinstance-attribute**

TORNADO\_OUTLINED: IconData

**buildTORNADO\_ROUNDEDinstance-attribute**

TORNADO\_ROUNDED: IconData

**buildTORNADO\_SHARPinstance-attribute**

TORNADO\_SHARP: IconData

**buildTOUCH\_APPinstance-attribute**

TOUCH\_APP: IconData

**buildTOUCH\_APP\_OUTLINEDinstance-attribute**

TOUCH\_APP\_OUTLINED: IconData

**buildTOUCH\_APP\_ROUNDEDinstance-attribute**

TOUCH\_APP\_ROUNDED: IconData

**buildTOUCH\_APP\_SHARPinstance-attribute**

TOUCH\_APP\_SHARP: IconData

**buildTOURinstance-attribute**

TOUR: IconData

buildTOUR\_OUTLINEDinstance-attribute

TOUR\_OUTLINED: IconData

buildTOUR\_ROUNDEDinstance-attribute

TOUR\_ROUNDED: IconData

buildTOUR\_SHARPinstance-attribute

TOUR\_SHARP: IconData

buildTOYSinstance-attribute

TOYS: IconData

buildTOYS\_OUTLINEDinstance-attribute

TOYS\_OUTLINED: IconData

buildTOYS\_ROUNDEDinstance-attribute

TOYS\_ROUNDED: IconData

buildTOYS\_SHARPinstance-attribute

TOYS\_SHARP: IconData

buildTRACK\_CHANGESinstance-attribute

TRACK\_CHANGES: IconData

buildTRACK\_CHANGES\_OUTLINEDinstance-attribute

TRACK\_CHANGES\_OUTLINED: IconData

buildTRACK\_CHANGES\_ROUNDEDinstance-attribute

TRACK\_CHANGES\_ROUNDED: IconData

buildTRACK\_CHANGES\_SHARPinstance-attribute

TRACK\_CHANGES\_SHARP: IconData

buildTRAFFICinstance-attribute

TRAFFIC: IconData

buildTRAFFIC\_OUTLINEDinstance-attribute

TRAFFIC\_OUTLINED: IconData

buildTRAFFIC\_ROUNDEDinstance-attribute

TRAFFIC\_ROUNDED: IconData

buildTRAFFIC\_SHARPinstance-attribute

TRAFFIC\_SHARP: IconData

buildTRAINinstance-attribute

TRAIN: IconData

buildTRAIN\_OUTLINEDinstance-attribute

TRAIN\_OUTLINED: IconData

buildTRAIN\_ROUNDEDinstance-attribute

TRAIN\_ROUNDED: IconData

buildTRAIN\_SHARPinstance-attribute

TRAIN\_SHARP: IconData

buildTRAMinstance-attribute

TRAM: IconData

buildTRAM\_OUTLINEDinstance-attribute

TRAM\_OUTLINED: IconData

buildTRAM\_ROUNDEDinstance-attribute

TRAM\_ROUNDED: IconData

buildTRAM\_SHARPinstance-attribute

TRAM\_SHARP: IconData

buildTRANSCRIBEinstance-attribute

TRANSCRIBE: IconData

buildTRANSCRIBE\_OUTLINEDinstance-attribute

TRANSCRIBE\_OUTLINED: IconData

buildTRANSCRIBE\_ROUNDEDinstance-attribute

TRANSCRIBE\_ROUNDED: IconData

buildTRANSCRIBE\_SHARPinstance-attribute

TRANSCRIBE\_SHARP: IconData

buildTRANSFER\_WITHIN\_A\_STATIONinstance-attribute

TRANSFER\_WITHIN\_A\_STATION: IconData

buildTRANSFER\_WITHIN\_A\_STATION\_OUTLINEDinstance-attribute

TRANSFER\_WITHIN\_A\_STATION\_OUTLINED: IconData

buildTRANSFER\_WITHIN\_A\_STATION\_ROUNDEDinstance-attribute

TRANSFER\_WITHIN\_A\_STATION\_ROUNDED: IconData

buildTRANSFER\_WITHIN\_A\_STATION\_SHARPinstance-attribute

TRANSFER\_WITHIN\_A\_STATION\_SHARP: IconData

buildTRANSFORMinstance-attribute

TRANSFORM: IconData

buildTRANSFORM\_OUTLINEDinstance-attribute

TRANSFORM\_OUTLINED: IconData

buildTRANSFORM\_ROUNDEDinstance-attribute

TRANSFORM\_ROUNDED: IconData

buildTRANSFORM\_SHARPinstance-attribute

TRANSFORM\_SHARP: IconData

buildTRANSGENDERinstance-attribute

TRANSGENDER: IconData

buildTRANSGENDER\_OUTLINEDinstance-attribute

TRANSGENDER\_OUTLINED: IconData

buildTRANSGENDER\_ROUNDEDinstance-attribute

TRANSGENDER\_ROUNDED: IconData

buildTRANSGENDER\_SHARPinstance-attribute

TRANSGENDER\_SHARP: IconData

buildTRANSIT\_ENTEREXITinstance-attribute

TRANSIT\_ENTEREXIT: IconData

buildTRANSIT\_ENTEREXIT\_OUTLINEDinstance-attribute

TRANSIT\_ENTEREXIT\_OUTLINED: IconData

buildTRANSIT\_ENTEREXIT\_ROUNDEDinstance-attribute

TRANSIT\_ENTEREXIT\_ROUNDED: IconData

buildTRANSIT\_ENTEREXIT\_SHARPinstance-attribute

TRANSIT\_ENTEREXIT\_SHARP: IconData

buildTRANSLATEinstance-attribute

TRANSLATE: IconData

buildTRANSLATE\_OUTLINEDinstance-attribute

TRANSLATE\_OUTLINED: IconData

buildTRANSLATE\_ROUNDEDinstance-attribute

TRANSLATE\_ROUNDED: IconData

buildTRANSLATE\_SHARPinstance-attribute

TRANSLATE\_SHARP: IconData

buildTRAVEL\_EXPLOREinstance-attribute

TRAVEL\_EXPLORE: IconData

buildTRAVEL\_EXPLORE\_OUTLINEDinstance-attribute

TRAVEL\_EXPLORE\_OUTLINED: IconData

buildTRAVEL\_EXPLORE\_ROUNDEDinstance-attribute

TRAVEL\_EXPLORE\_ROUNDED: IconData

buildTRAVEL\_EXPLORE\_SHARPinstance-attribute

TRAVEL\_EXPLORE\_SHARP: IconData

buildTRENDING\_DOWNinstance-attribute

TRENDING\_DOWN: IconData

buildTRENDING\_DOWN\_OUTLINEDinstance-attribute

TRENDING\_DOWN\_OUTLINED: IconData

buildTRENDING\_DOWN\_ROUNDEDinstance-attribute

TRENDING\_DOWN\_ROUNDED: IconData

buildTRENDING\_DOWN\_SHARPinstance-attribute

TRENDING\_DOWN\_SHARP: IconData

buildTRENDING\_FLATinstance-attribute

TRENDING\_FLAT: IconData

buildTRENDING\_FLAT\_OUTLINEDinstance-attribute

TRENDING\_FLAT\_OUTLINED: IconData

buildTRENDING\_FLAT\_ROUNDEDinstance-attribute

TRENDING\_FLAT\_ROUNDED: IconData

buildTRENDING\_FLAT\_SHARPinstance-attribute

TRENDING\_FLAT\_SHARP: IconData

buildTRENDING\_NEUTRALinstance-attribute

TRENDING\_NEUTRAL: IconData

buildTRENDING\_NEUTRAL\_OUTLINEDinstance-attribute

TRENDING\_NEUTRAL\_OUTLINED: IconData

buildTRENDING\_NEUTRAL\_ROUNDEDinstance-attribute

TRENDING\_NEUTRAL\_ROUNDED: IconData

buildTRENDING\_NEUTRAL\_SHARPinstance-attribute

TRENDING\_NEUTRAL\_SHARP: IconData

buildTRENDING\_UPinstance-attribute

TRENDING\_UP: IconData

buildTRENDING\_UP\_OUTLINEDinstance-attribute

TRENDING\_UP\_OUTLINED: IconData

buildTRENDING\_UP\_ROUNDEDinstance-attribute

TRENDING\_UP\_ROUNDED: IconData

buildTRENDING\_UP\_SHARPinstance-attribute

TRENDING\_UP\_SHARP: IconData

buildTRIP\_ORIGINinstance-attribute

TRIP\_ORIGIN: IconData

buildTRIP\_ORIGIN\_OUTLINEDinstance-attribute

TRIP\_ORIGIN\_OUTLINED: IconData

buildTRIP\_ORIGIN\_ROUNDEDinstance-attribute

TRIP\_ORIGIN\_ROUNDED: IconData

buildTRIP\_ORIGIN\_SHARPinstance-attribute

TRIP\_ORIGIN\_SHARP: IconData

buildTROLLEYinstance-attribute

TROLLEY: IconData

buildTROUBLESHOOTinstance-attribute

TROUBLESHOOT: IconData

buildTROUBLESHOOT\_OUTLINEDinstance-attribute

TROUBLESHOOT\_OUTLINED: IconData

buildTROUBLESHOOT\_ROUNDEDinstance-attribute

TROUBLESHOOT\_ROUNDED: IconData

buildTROUBLESHOOT\_SHARPinstance-attribute

TROUBLESHOOT\_SHARP: IconData

buildTRY\_SMS\_STARinstance-attribute

TRY\_SMS\_STAR: IconData

buildTRY\_SMS\_STAR\_OUTLINEDinstance-attribute

TRY\_SMS\_STAR\_OUTLINED: IconData

buildTRY\_SMS\_STAR\_ROUNDEDinstance-attribute

TRY\_SMS\_STAR\_ROUNDED: IconData

buildTRY\_SMS\_STAR\_SHARPinstance-attribute

TRY\_SMS\_STAR\_SHARP: IconData

buildTSUNAMIinstance-attribute

TSUNAMI: IconData

buildTSUNAMI\_OUTLINEDinstance-attribute

TSUNAMI\_OUTLINED: IconData

buildTSUNAMI\_ROUNDEDinstance-attribute

TSUNAMI\_ROUNDED: IconData

buildTSUNAMI\_SHARPinstance-attribute

TSUNAMI\_SHARP: IconData

buildTTYinstance-attribute

TTY: IconData

buildTTY\_OUTLINEDinstance-attribute

TTY\_OUTLINED: IconData

buildTTY\_ROUNDEDinstance-attribute

TTY\_ROUNDED: IconData

buildTTY\_SHARPinstance-attribute



TTY\_SHARP: IconData

**buildTUNEinstance-attribute**

TUNE: IconData

**buildTUNE\_OUTLINEDinstance-attribute**

TUNE\_OUTLINED: IconData

**buildTUNE\_ROUNDEDinstance-attribute**

TUNE\_ROUNDED: IconData

**buildTUNE\_SHARPinstance-attribute**

TUNE\_SHARP: IconData

**buildTUNGSTENinstance-attribute**

TUNGSTEN: IconData

**buildTUNGSTEN\_OUTLINEDinstance-attribute**

TUNGSTEN\_OUTLINED: IconData

**buildTUNGSTEN\_ROUNDEDinstance-attribute**

TUNGSTEN\_ROUNDED: IconData

**buildTUNGSTEN\_SHARPinstance-attribute**

TUNGSTEN\_SHARP: IconData

**buildTURN\_LEFTinstance-attribute**

TURN\_LEFT: IconData

**buildTURN\_LEFT\_OUTLINEDinstance-attribute**

TURN\_LEFT\_OUTLINED: IconData

**buildTURN\_LEFT\_ROUNDEDinstance-attribute**

TURN\_LEFT\_ROUNDED: IconData

**buildTURN\_LEFT\_SHARPinstance-attribute**

TURN\_LEFT\_SHARP: IconData

**buildTURN\_RIGHTinstance-attribute**

TURN\_RIGHT: IconData

**buildTURN\_RIGHT\_OUTLINEDinstance-attribute**

TURN\_RIGHT\_OUTLINED: IconData

**buildTURN\_RIGHT\_ROUNDEDinstance-attribute**

TURN\_RIGHT\_ROUNDED: IconData

**buildTURN\_RIGHT\_SHARPinstance-attribute**

TURN\_RIGHT\_SHARP: IconData

**buildTURN\_SHARP\_LEFTinstance-attribute**

TURN\_SHARP\_LEFT: IconData

**buildTURN\_SHARP\_LEFT\_OUTLINEDinstance-attribute**

TURN\_SHARP\_LEFT\_OUTLINED: IconData

**buildTURN\_SHARP\_LEFT\_ROUNDEDinstance-attribute**

TURN\_SHARP\_LEFT\_ROUNDED: IconData

**buildTURN\_SHARP\_LEFT\_SHARPinstance-attribute**

TURN\_SHARP\_LEFT\_SHARP: IconData

**buildTURN\_SHARP\_RIGHTinstance-attribute**

TURN\_SHARP\_RIGHT: IconData

**buildTURN\_SHARP\_RIGHT\_OUTLINEDinstance-attribute**

TURN\_SHARP\_RIGHT\_OUTLINED: IconData

buildTURN\_SHARP\_RIGHT\_ROUNDEDinstance-attribute

TURN\_SHARP\_RIGHT\_ROUNDED: IconData

buildTURN\_SHARP\_RIGHT\_SHARPinstance-attribute

TURN\_SHARP\_RIGHT\_SHARP: IconData

buildTURN\_SLIGHT\_LEFTinstance-attribute

TURN\_SLIGHT\_LEFT: IconData

buildTURN\_SLIGHT\_LEFT\_OUTLINEDinstance-attribute

TURN\_SLIGHT\_LEFT\_OUTLINED: IconData

buildTURN\_SLIGHT\_LEFT\_ROUNDEDinstance-attribute

TURN\_SLIGHT\_LEFT\_ROUNDED: IconData

buildTURN\_SLIGHT\_LEFT\_SHARPinstance-attribute

TURN\_SLIGHT\_LEFT\_SHARP: IconData

buildTURN\_SLIGHT\_RIGHTinstance-attribute

TURN\_SLIGHT\_RIGHT: IconData

buildTURN\_SLIGHT\_RIGHT\_OUTLINEDinstance-attribute

TURN\_SLIGHT\_RIGHT\_OUTLINED: IconData

buildTURN\_SLIGHT\_RIGHT\_ROUNDEDinstance-attribute

TURN\_SLIGHT\_RIGHT\_ROUNDED: IconData

buildTURN\_SLIGHT\_RIGHT\_SHARPinstance-attribute

TURN\_SLIGHT\_RIGHT\_SHARP: IconData

buildTURNED\_INinstance-attribute

TURNED\_IN: IconData

buildTURNED\_IN\_NOTinstance-attribute

TURNED\_IN\_NOT: IconData

buildTURNED\_IN\_NOT\_OUTLINEDinstance-attribute

TURNED\_IN\_NOT\_OUTLINED: IconData

buildTURNED\_IN\_NOT\_ROUNDEDinstance-attribute

TURNED\_IN\_NOT\_ROUNDED: IconData

buildTURNED\_IN\_NOT\_SHARPinstance-attribute

TURNED\_IN\_NOT\_SHARP: IconData

buildTURNED\_IN\_OUTLINEDinstance-attribute

TURNED\_IN\_OUTLINED: IconData

buildTURNED\_IN\_ROUNDEDinstance-attribute

TURNED\_IN\_ROUNDED: IconData

buildTURNED\_IN\_SHARPinstance-attribute

TURNED\_IN\_SHARP: IconData

buildTVinstance-attribute

TV: IconData

buildTV\_OFFinstance-attribute

TV\_OFF: IconData

buildTV\_OFF\_OUTLINEDinstance-attribute

TV\_OFF\_OUTLINED: IconData

buildTV\_OFF\_ROUNDEDinstance-attribute

TV\_OFF\_ROUNDED: IconData

buildTV\_OFF\_SHARPinstance-attribute

TV\_OFF\_SHARP: IconData

buildTV\_OUTLINEDinstance-attribute

TV\_OUTLINED: IconData

buildTV\_ROUNDEDinstance-attribute

TV\_ROUNDED: IconData

buildTV\_SHARPinstance-attribute

TV\_SHARP: IconData

buildTWELVE\_MPinstance-attribute

TWELVE\_MP: IconData

buildTWELVE\_MP\_OUTLINEDinstance-attribute

TWELVE\_MP\_OUTLINED: IconData

buildTWELVE\_MP\_ROUNDEDinstance-attribute

TWELVE\_MP\_ROUNDED: IconData

buildTWELVE\_MP\_SHARPinstance-attribute

TWELVE\_MP\_SHARP: IconData

buildTWENTY\_FOUR\_MPinstance-attribute

TWENTY\_FOUR\_MP: IconData

buildTWENTY\_FOUR\_MP\_OUTLINEDinstance-attribute

TWENTY\_FOUR\_MP\_OUTLINED: IconData

buildTWENTY\_FOUR\_MP\_ROUNDEDinstance-attribute

TWENTY\_FOUR\_MP\_ROUNDED: IconData

buildTWENTY\_FOUR\_MP\_SHARPinstance-attribute

TWENTY\_FOUR\_MP\_SHARP: IconData

buildTWENTY\_MPinstance-attribute

TWENTY\_MP: IconData

buildTWENTY\_MP\_OUTLINEDinstance-attribute

TWENTY\_MP\_OUTLINED: IconData

buildTWENTY\_MP\_ROUNDEDinstance-attribute

TWENTY\_MP\_ROUNDED: IconData

buildTWENTY\_MP\_SHARPinstance-attribute

TWENTY\_MP\_SHARP: IconData

buildTWENTY\_ONE\_MPinstance-attribute

TWENTY\_ONE\_MP: IconData

buildTWENTY\_ONE\_MP\_OUTLINEDinstance-attribute

TWENTY\_ONE\_MP\_OUTLINED: IconData

buildTWENTY\_ONE\_MP\_ROUNDEDinstance-attribute

TWENTY\_ONE\_MP\_ROUNDED: IconData

buildTWENTY\_ONE\_MP\_SHARPinstance-attribute

TWENTY\_ONE\_MP\_SHARP: IconData

buildTWENTY\_THREE\_MPinstance-attribute

TWENTY\_THREE\_MP: IconData

buildTWENTY\_THREE\_MP\_OUTLINEDinstance-attribute

TWENTY\_THREE\_MP\_OUTLINED: IconData

buildTWENTY\_THREE\_MP\_ROUNDEDinstance-attribute

TWENTY\_THREE\_MP\_ROUNDED: IconData

buildTWENTY\_THREE\_MP\_SHARPinstance-attribute

TWENTY\_THREE\_MP\_SHARP: IconData

buildTWENTY\_TWO\_MPinstance-attribute

TWENTY\_TWO\_MP: IconData

buildTWENTY\_TWO\_MP\_OUTLINEDinstance-attribute

TWENTY\_TWO\_MP\_OUTLINED: IconData

buildTWENTY\_TWO\_MP\_ROUNDEDinstance-attribute

TWENTY\_TWO\_MP\_ROUNDED: IconData

buildTWENTY\_TWO\_MP\_SHARPinstance-attribute

TWENTY\_TWO\_MP\_SHARP: IconData

buildTWO\_Kinstance-attribute

TWO\_K: IconData

buildTWO\_K\_OUTLINEDinstance-attribute

TWO\_K\_OUTLINED: IconData

buildTWO\_K\_PLUSinstance-attribute

TWO\_K\_PLUS: IconData

buildTWO\_K\_PLUS\_OUTLINEDinstance-attribute

TWO\_K\_PLUS\_OUTLINED: IconData

buildTWO\_K\_PLUS\_ROUNDEDinstance-attribute

TWO\_K\_PLUS\_ROUNDED: IconData

buildTWO\_K\_PLUS\_SHARPinstance-attribute

TWO\_K\_PLUS\_SHARP: IconData

buildTWO\_K\_ROUNDEDinstance-attribute

TWO\_K\_ROUNDED: IconData

buildTWO\_K\_SHARPinstance-attribute

TWO\_K\_SHARP: IconData

buildTWO\_MPinstance-attribute

TWO\_MP: IconData

buildTWO\_MP\_OUTLINEDinstance-attribute

TWO\_MP\_OUTLINED: IconData

buildTWO\_MP\_ROUNDEDinstance-attribute

TWO\_MP\_ROUNDED: IconData

buildTWO\_MP\_SHARPinstance-attribute

TWO\_MP\_SHARP: IconData

buildTWO\_WHEELERinstance-attribute

TWO\_WHEELER: IconData

buildTWO\_WHEELER\_OUTLINEDinstance-attribute

TWO\_WHEELER\_OUTLINED: IconData

buildTWO\_WHEELER\_ROUNDEDinstance-attribute

TWO\_WHEELER\_ROUNDED: IconData

buildTWO\_WHEELER\_SHARPinstance-attribute

TWO\_WHEELER\_SHARP: IconData

**buildTYPE\_SPECIMENinstance-attribute**

TYPE\_SPECIMEN: IconData

**buildTYPE\_SPECIMEN\_OUTLINEDinstance-attribute**

TYPE\_SPECIMEN\_OUTLINED: IconData

**buildTYPE\_SPECIMEN\_ROUNDEDinstance-attribute**

TYPE\_SPECIMEN\_ROUNDED: IconData

**buildTYPE\_SPECIMEN\_SHARPinstance-attribute**

TYPE\_SPECIMEN\_SHARP: IconData

**buildU\_TURN\_LEFTinstance-attribute**

U\_TURN\_LEFT: IconData

**buildU\_TURN\_LEFT\_OUTLINEDinstance-attribute**

U\_TURN\_LEFT\_OUTLINED: IconData

**buildU\_TURN\_LEFT\_ROUNDEDinstance-attribute**

U\_TURN\_LEFT\_ROUNDED: IconData

**buildU\_TURN\_LEFT\_SHARPinstance-attribute**

U\_TURN\_LEFT\_SHARP: IconData

**buildU\_TURN\_RIGHTinstance-attribute**

U\_TURN\_RIGHT: IconData

**buildU\_TURN\_RIGHT\_OUTLINEDinstance-attribute**

U\_TURN\_RIGHT\_OUTLINED: IconData

**buildU\_TURN\_RIGHT\_ROUNDEDinstance-attribute**

U\_TURN\_RIGHT\_ROUNDED: IconData

**buildU\_TURN\_RIGHT\_SHARPinstance-attribute**

U\_TURN\_RIGHT\_SHARP: IconData

**buildUMBRELLAinstance-attribute**

UMBRELLA: IconData

**buildUMBRELLA\_OUTLINEDinstance-attribute**

UMBRELLA\_OUTLINED: IconData

**buildUMBRELLA\_ROUNDEDinstance-attribute**

UMBRELLA\_ROUNDED: IconData

**buildUMBRELLA\_SHARPinstance-attribute**

UMBRELLA\_SHARP: IconData

**buildUNARCHIVEinstance-attribute**

UNARCHIVE: IconData

**buildUNARCHIVE\_OUTLINEDinstance-attribute**

UNARCHIVE\_OUTLINED: IconData

**buildUNARCHIVE\_ROUNDEDinstance-attribute**

UNARCHIVE\_ROUNDED: IconData

**buildUNARCHIVE\_SHARPinstance-attribute**

UNARCHIVE\_SHARP: IconData

**buildUNDOinstance-attribute**

UNDO: IconData

**buildUNDO\_OUTLINEDinstance-attribute**

UNDO\_OUTLINED: IconData

buildUNDO\_ROUNDEDinstance-attribute

UNDO\_ROUNDED: IconData

buildUNDO\_SHARPinstance-attribute

UNDO\_SHARP: IconData

buildUNFOLD\_LESSinstance-attribute

UNFOLD\_LESS: IconData

buildUNFOLD\_LESS\_DOUBLEinstance-attribute

UNFOLD\_LESS\_DOUBLE: IconData

buildUNFOLD\_LESS\_DOUBLE\_OUTLINEDinstance-attribute

UNFOLD\_LESS\_DOUBLE\_OUTLINED: IconData

buildUNFOLD\_LESS\_DOUBLE\_ROUNDEDinstance-attribute

UNFOLD\_LESS\_DOUBLE\_ROUNDED: IconData

buildUNFOLD\_LESS\_DOUBLE\_SHARPinstance-attribute

UNFOLD\_LESS\_DOUBLE\_SHARP: IconData

buildUNFOLD\_LESS\_OUTLINEDinstance-attribute

UNFOLD\_LESS\_OUTLINED: IconData

buildUNFOLD\_LESS\_ROUNDEDinstance-attribute

UNFOLD\_LESS\_ROUNDED: IconData

buildUNFOLD\_LESS\_SHARPinstance-attribute

UNFOLD\_LESS\_SHARP: IconData

buildUNFOLD\_MOREinstance-attribute

UNFOLD\_MORE: IconData

buildUNFOLD\_MORE\_DOUBLEinstance-attribute

UNFOLD\_MORE\_DOUBLE: IconData

buildUNFOLD\_MORE\_DOUBLE\_OUTLINEDinstance-attribute

UNFOLD\_MORE\_DOUBLE\_OUTLINED: IconData

buildUNFOLD\_MORE\_DOUBLE\_ROUNDEDinstance-attribute

UNFOLD\_MORE\_DOUBLE\_ROUNDED: IconData

buildUNFOLD\_MORE\_DOUBLE\_SHARPinstance-attribute

UNFOLD\_MORE\_DOUBLE\_SHARP: IconData

buildUNFOLD\_MORE\_OUTLINEDinstance-attribute

UNFOLD\_MORE\_OUTLINED: IconData

buildUNFOLD\_MORE\_ROUNDEDinstance-attribute

UNFOLD\_MORE\_ROUNDED: IconData

buildUNFOLD\_MORE\_SHARPinstance-attribute

UNFOLD\_MORE\_SHARP: IconData

buildUNPUBLISHEDinstance-attribute

UNPUBLISHED: IconData

buildUNPUBLISHED\_OUTLINEDinstance-attribute

UNPUBLISHED\_OUTLINED: IconData

buildUNPUBLISHED\_ROUNDEDinstance-attribute

UNPUBLISHED\_ROUNDED: IconData

buildUNPUBLISHED\_SHARPinstance-attribute

UNPUBLISHED\_SHARP: IconData

**buildUNSUBSCRIBEinstance-attribute**

UNSUBSCRIBE: IconData

**buildUNSUBSCRIBE\_OUTLINEDinstance-attribute**

UNSUBSCRIBE\_OUTLINED: IconData

**buildUNSUBSCRIBE\_ROUNDEDinstance-attribute**

UNSUBSCRIBE\_ROUNDED: IconData

**buildUNSUBSCRIBE\_SHARPinstance-attribute**

UNSUBSCRIBE\_SHARP: IconData

**buildUPCOMINGinstance-attribute**

UPCOMING: IconData

**buildUPCOMING\_OUTLINEDinstance-attribute**

UPCOMING\_OUTLINED: IconData

**buildUPCOMING\_ROUNDEDinstance-attribute**

UPCOMING\_ROUNDED: IconData

**buildUPCOMING\_SHARPinstance-attribute**

UPCOMING\_SHARP: IconData

**buildUPDATEinstance-attribute**

UPDATE: IconData

**buildUPDATE\_DISABLEDinstance-attribute**

UPDATE\_DISABLED: IconData

**buildUPDATE\_DISABLED\_OUTLINEDinstance-attribute**

UPDATE\_DISABLED\_OUTLINED: IconData

**buildUPDATE\_DISABLED\_ROUNDEDinstance-attribute**

UPDATE\_DISABLED\_ROUNDED: IconData

**buildUPDATE\_DISABLED\_SHARPinstance-attribute**

UPDATE\_DISABLED\_SHARP: IconData

**buildUPDATE\_OUTLINEDinstance-attribute**

UPDATE\_OUTLINED: IconData

**buildUPDATE\_ROUNDEDinstance-attribute**

UPDATE\_ROUNDED: IconData

**buildUPDATE\_SHARPinstance-attribute**

UPDATE\_SHARP: IconData

**buildUPGRADEinstance-attribute**

UPGRADE: IconData

**buildUPGRADE\_OUTLINEDinstance-attribute**

UPGRADE\_OUTLINED: IconData

**buildUPGRADE\_ROUNDEDinstance-attribute**

UPGRADE\_ROUNDED: IconData

**buildUPGRADE\_SHARPinstance-attribute**

UPGRADE\_SHARP: IconData

**buildUPLOADinstance-attribute**

UPLOAD: IconData

**buildUPLOAD\_FILEinstance-attribute**

UPLOAD\_FILE: IconData

buildUPLOAD\_FILE\_OUTLINEDinstance-attribute

UPLOAD\_FILE\_OUTLINED: IconData

buildUPLOAD\_FILE\_ROUNDEDinstance-attribute

UPLOAD\_FILE\_ROUNDED: IconData

buildUPLOAD\_FILE\_SHARPinstance-attribute

UPLOAD\_FILE\_SHARP: IconData

buildUPLOAD\_OUTLINEDinstance-attribute

UPLOAD\_OUTLINED: IconData

buildUPLOAD\_ROUNDEDinstance-attribute

UPLOAD\_ROUNDED: IconData

buildUPLOAD\_SHARPinstance-attribute

UPLOAD\_SHARP: IconData

buildUSBinstance-attribute

USB: IconData

buildUSB\_OFFinstance-attribute

USB\_OFF: IconData

buildUSB\_OFF\_OUTLINEDinstance-attribute

USB\_OFF\_OUTLINED: IconData

buildUSB\_OFF\_ROUNDEDinstance-attribute

USB\_OFF\_ROUNDED: IconData

buildUSB\_OFF\_SHARPinstance-attribute

USB\_OFF\_SHARP: IconData

buildUSB\_OUTLINEDinstance-attribute

USB\_OUTLINED: IconData

buildUSB\_ROUNDEDinstance-attribute

USB\_ROUNDED: IconData

buildUSB\_SHARPinstance-attribute

USB\_SHARP: IconData

buildVACCINESinstance-attribute

VACCINES: IconData

buildVACCINES\_OUTLINEDinstance-attribute

VACCINES\_OUTLINED: IconData

buildVACCINES\_ROUNDEDinstance-attribute

VACCINES\_ROUNDED: IconData

buildVACCINES\_SHARPinstance-attribute

VACCINES\_SHARP: IconData

buildVAPE\_FREEinstance-attribute

VAPE\_FREE: IconData

buildVAPE\_FREE\_OUTLINEDinstance-attribute

VAPE\_FREE\_OUTLINED: IconData

buildVAPE\_FREE\_ROUNDEDinstance-attribute

VAPE\_FREE\_ROUNDED: IconData

buildVAPE\_FREE\_SHARPinstance-attribute



VAPE\_FREE\_SHARP: IconData

**buildVAPING\_ROOMSinstance-attribute**

VAPING\_ROOMS: IconData

**buildVAPING\_ROOMS\_OUTLINEDinstance-attribute**

VAPING\_ROOMS\_OUTLINED: IconData

**buildVAPING\_ROOMS\_ROUNDEDinstance-attribute**

VAPING\_ROOMS\_ROUNDED: IconData

**buildVAPING\_ROOMS\_SHARPinstance-attribute**

VAPING\_ROOMS\_SHARP: IconData

**buildVERIFIEDinstance-attribute**

VERIFIED: IconData

**buildVERIFIED\_OUTLINEDinstance-attribute**

VERIFIED\_OUTLINED: IconData

**buildVERIFIED\_ROUNDEDinstance-attribute**

VERIFIED\_ROUNDED: IconData

**buildVERIFIED\_SHARPinstance-attribute**

VERIFIED\_SHARP: IconData

**buildVERIFIED\_USERinstance-attribute**

VERIFIED\_USER: IconData

**buildVERIFIED\_USER\_OUTLINEDinstance-attribute**

VERIFIED\_USER\_OUTLINED: IconData

**buildVERIFIED\_USER\_ROUNDEDinstance-attribute**

VERIFIED\_USER\_ROUNDED: IconData

**buildVERIFIED\_USER\_SHARPinstance-attribute**

VERIFIED\_USER\_SHARP: IconData

**buildVERTICAL\_ALIGN\_BOTTOMinstance-attribute**

VERTICAL\_ALIGN\_BOTTOM: IconData

**buildVERTICAL\_ALIGN\_BOTTOM\_OUTLINEDinstance-attribute**

VERTICAL\_ALIGN\_BOTTOM\_OUTLINED: IconData

**buildVERTICAL\_ALIGN\_BOTTOM\_ROUNDEDinstance-attribute**

VERTICAL\_ALIGN\_BOTTOM\_ROUNDED: IconData

**buildVERTICAL\_ALIGN\_BOTTOM\_SHARPinstance-attribute**

VERTICAL\_ALIGN\_BOTTOM\_SHARP: IconData

**buildVERTICAL\_ALIGN\_CENTERinstance-attribute**

VERTICAL\_ALIGN\_CENTER: IconData

**buildVERTICAL\_ALIGN\_CENTER\_OUTLINEDinstance-attribute**

VERTICAL\_ALIGN\_CENTER\_OUTLINED: IconData

**buildVERTICAL\_ALIGN\_CENTER\_ROUNDEDinstance-attribute**

VERTICAL\_ALIGN\_CENTER\_ROUNDED: IconData

**buildVERTICAL\_ALIGN\_CENTER\_SHARPinstance-attribute**

VERTICAL\_ALIGN\_CENTER\_SHARP: IconData

**buildVERTICAL\_ALIGN\_TOPinstance-attribute**

VERTICAL\_ALIGN\_TOP: IconData

**buildVERTICAL\_ALIGN\_TOP\_OUTLINEDinstance-attribute**

VERTICAL\_ALIGN\_TOP\_OUTLINED: IconData

buildVERTICAL\_ALIGN\_TOP\_ROUNDEDinstance-attribute

VERTICAL\_ALIGN\_TOP\_ROUNDED: IconData

buildVERTICAL\_ALIGN\_TOP\_SHARPinstance-attribute

VERTICAL\_ALIGN\_TOP\_SHARP: IconData

buildVERTICAL\_DISTRIBUTEinstance-attribute

VERTICAL\_DISTRIBUTE: IconData

buildVERTICAL\_DISTRIBUTE\_OUTLINEDinstance-attribute

VERTICAL\_DISTRIBUTE\_OUTLINED: IconData

buildVERTICAL\_DISTRIBUTE\_ROUNDEDinstance-attribute

VERTICAL\_DISTRIBUTE\_ROUNDED: IconData

buildVERTICAL\_DISTRIBUTE\_SHARPinstance-attribute

VERTICAL\_DISTRIBUTE\_SHARP: IconData

buildVERTICAL\_SHADESinstance-attribute

VERTICAL\_SHADES: IconData

buildVERTICAL\_SHADES\_CLOSEDinstance-attribute

VERTICAL\_SHADES\_CLOSED: IconData

buildVERTICAL\_SHADES\_CLOSED\_OUTLINEDinstance-attribute

VERTICAL\_SHADES\_CLOSED\_OUTLINED: IconData

buildVERTICAL\_SHADES\_CLOSED\_ROUNDEDinstance-attribute

VERTICAL\_SHADES\_CLOSED\_ROUNDED: IconData

buildVERTICAL\_SHADES\_CLOSED\_SHARPinstance-attribute

VERTICAL\_SHADES\_CLOSED\_SHARP: IconData

buildVERTICAL\_SHADES\_OUTLINEDinstance-attribute

VERTICAL\_SHADES\_OUTLINED: IconData

buildVERTICAL\_SHADES\_ROUNDEDinstance-attribute

VERTICAL\_SHADES\_ROUNDED: IconData

buildVERTICAL\_SHADES\_SHARPinstance-attribute

VERTICAL\_SHADES\_SHARP: IconData

buildVERTICAL\_SPLITinstance-attribute

VERTICAL\_SPLIT: IconData

buildVERTICAL\_SPLIT\_OUTLINEDinstance-attribute

VERTICAL\_SPLIT\_OUTLINED: IconData

buildVERTICAL\_SPLIT\_ROUNDEDinstance-attribute

VERTICAL\_SPLIT\_ROUNDED: IconData

buildVERTICAL\_SPLIT\_SHARPinstance-attribute

VERTICAL\_SPLIT\_SHARP: IconData

buildVIBRATIONinstance-attribute

VIBRATION: IconData

buildVIBRATION\_OUTLINEDinstance-attribute

VIBRATION\_OUTLINED: IconData

buildVIBRATION\_ROUNDEDinstance-attribute

VIBRATION\_ROUNDED: IconData

buildVIBRATION\_SHARPinstance-attribute

VIBRATION\_SHARP: IconData

**buildVIDEO\_CALLInstance-attribute**

VIDEO\_CALL: IconData

**buildVIDEO\_CALL\_OUTLINEDInstance-attribute**

VIDEO\_CALL\_OUTLINED: IconData

**buildVIDEO\_CALL\_ROUNDEDInstance-attribute**

VIDEO\_CALL\_ROUNDED: IconData

**buildVIDEO\_CALL\_SHARPInstance-attribute**

VIDEO\_CALL\_SHARP: IconData

**buildVIDEO\_CAMERA\_BACKInstance-attribute**

VIDEO\_CAMERA\_BACK: IconData

**buildVIDEO\_CAMERA\_BACK\_OUTLINEDInstance-attribute**

VIDEO\_CAMERA\_BACK\_OUTLINED: IconData

**buildVIDEO\_CAMERA\_BACK\_ROUNDEDInstance-attribute**

VIDEO\_CAMERA\_BACK\_ROUNDED: IconData

**buildVIDEO\_CAMERA\_BACK\_SHARPInstance-attribute**

VIDEO\_CAMERA\_BACK\_SHARP: IconData

**buildVIDEO\_CAMERA\_FRONTInstance-attribute**

VIDEO\_CAMERA\_FRONT: IconData

**buildVIDEO\_CAMERA\_FRONT\_OUTLINEDInstance-attribute**

VIDEO\_CAMERA\_FRONT\_OUTLINED: IconData

**buildVIDEO\_CAMERA\_FRONT\_ROUNDEDInstance-attribute**

VIDEO\_CAMERA\_FRONT\_ROUNDED: IconData

**buildVIDEO\_CAMERA\_FRONT\_SHARPInstance-attribute**

VIDEO\_CAMERA\_FRONT\_SHARP: IconData

**buildVIDEO\_CHATInstance-attribute**

VIDEO\_CHAT: IconData

**buildVIDEO\_CHAT\_OUTLINEDInstance-attribute**

VIDEO\_CHAT\_OUTLINED: IconData

**buildVIDEO\_CHAT\_ROUNDEDInstance-attribute**

VIDEO\_CHAT\_ROUNDED: IconData

**buildVIDEO\_CHAT\_SHARPInstance-attribute**

VIDEO\_CHAT\_SHARP: IconData

**buildVIDEO\_COLLECTIONInstance-attribute**

VIDEO\_COLLECTION: IconData

**buildVIDEO\_COLLECTION\_OUTLINEDInstance-attribute**

VIDEO\_COLLECTION\_OUTLINED: IconData

**buildVIDEO\_COLLECTION\_ROUNDEDInstance-attribute**

VIDEO\_COLLECTION\_ROUNDED: IconData

**buildVIDEO\_COLLECTION\_SHARPInstance-attribute**

VIDEO\_COLLECTION\_SHARP: IconData

**buildVIDEO\_FILEInstance-attribute**

VIDEO\_FILE: IconData

**buildVIDEO\_FILE\_OUTLINEDInstance-attribute**

VIDEO\_FILE\_OUTLINED: IconData

buildVIDEO\_FILE\_ROUNDEDinstance-attribute

VIDEO\_FILE\_ROUNDED: IconData

buildVIDEO\_FILE\_SHARPinstance-attribute

VIDEO\_FILE\_SHARP: IconData

buildVIDEO\_LABELinstance-attribute

VIDEO\_LABEL: IconData

buildVIDEO\_LABEL\_OUTLINEDinstance-attribute

VIDEO\_LABEL\_OUTLINED: IconData

buildVIDEO\_LABEL\_ROUNDEDinstance-attribute

VIDEO\_LABEL\_ROUNDED: IconData

buildVIDEO\_LABEL\_SHARPinstance-attribute

VIDEO\_LABEL\_SHARP: IconData

buildVIDEO\_LIBRARYinstance-attribute

VIDEO\_LIBRARY: IconData

buildVIDEO\_LIBRARY\_OUTLINEDinstance-attribute

VIDEO\_LIBRARY\_OUTLINED: IconData

buildVIDEO\_LIBRARY\_ROUNDEDinstance-attribute

VIDEO\_LIBRARY\_ROUNDED: IconData

buildVIDEO\_LIBRARY\_SHARPinstance-attribute

VIDEO\_LIBRARY\_SHARP: IconData

buildVIDEO\_SETTINGSinstance-attribute

VIDEO\_SETTINGS: IconData

buildVIDEO\_SETTINGS\_OUTLINEDinstance-attribute

VIDEO\_SETTINGS\_OUTLINED: IconData

buildVIDEO\_SETTINGS\_ROUNDEDinstance-attribute

VIDEO\_SETTINGS\_ROUNDED: IconData

buildVIDEO\_SETTINGS\_SHARPinstance-attribute

VIDEO\_SETTINGS\_SHARP: IconData

buildVIDEO\_STABLEinstance-attribute

VIDEO\_STABLE: IconData

buildVIDEO\_STABLE\_OUTLINEDinstance-attribute

VIDEO\_STABLE\_OUTLINED: IconData

buildVIDEO\_STABLE\_ROUNDEDinstance-attribute

VIDEO\_STABLE\_ROUNDED: IconData

buildVIDEO\_STABLE\_SHARPinstance-attribute

VIDEO\_STABLE\_SHARP: IconData

buildVIDEOCAMinstance-attribute

VIDEOCAM: IconData

buildVIDEOCAM\_OFFinstance-attribute

VIDEOCAM\_OFF: IconData

buildVIDEOCAM\_OFF\_OUTLINEDinstance-attribute

VIDEOCAM\_OFF\_OUTLINED: IconData

buildVIDEOCAM\_OFF\_ROUNDEDinstance-attribute

VIDEOCAM\_OFF\_ROUNDED: IconData

buildVIDEOCAM\_OFF\_SHARPinstance-attribute

VIDEOCAM\_OFF\_SHARP: IconData

buildVIDEOCAM\_OUTLINEDinstance-attribute

VIDEOCAM\_OUTLINED: IconData

buildVIDEOCAM\_ROUNDEDinstance-attribute

VIDEOCAM\_ROUNDED: IconData

buildVIDEOCAM\_SHARPinstance-attribute

VIDEOCAM\_SHARP: IconData

buildVIDEOGAME\_ASSETinstance-attribute

VIDEOGAME\_ASSET: IconData

buildVIDEOGAME\_ASSET\_OFFinstance-attribute

VIDEOGAME\_ASSET\_OFF: IconData

buildVIDEOGAME\_ASSET\_OFF\_OUTLINEDinstance-attribute

VIDEOGAME\_ASSET\_OFF\_OUTLINED: IconData

buildVIDEOGAME\_ASSET\_OFF\_ROUNDEDinstance-attribute

VIDEOGAME\_ASSET\_OFF\_ROUNDED: IconData

buildVIDEOGAME\_ASSET\_OFF\_SHARPinstance-attribute

VIDEOGAME\_ASSET\_OFF\_SHARP: IconData

buildVIDEOGAME\_ASSET\_OUTLINEDinstance-attribute

VIDEOGAME\_ASSET\_OUTLINED: IconData

buildVIDEOGAME\_ASSET\_ROUNDEDinstance-attribute

VIDEOGAME\_ASSET\_ROUNDED: IconData

buildVIDEOGAME\_ASSET\_SHARPinstance-attribute

VIDEOGAME\_ASSET\_SHARP: IconData

buildVIEW\_AGENDAinstance-attribute

VIEW\_AGENDA: IconData

buildVIEW\_AGENDA\_OUTLINEDinstance-attribute

VIEW\_AGENDA\_OUTLINED: IconData

buildVIEW\_AGENDA\_ROUNDEDinstance-attribute

VIEW\_AGENDA\_ROUNDED: IconData

buildVIEW\_AGENDA\_SHARPinstance-attribute

VIEW\_AGENDA\_SHARP: IconData

buildVIEW\_ARRAYinstance-attribute

VIEW\_ARRAY: IconData

buildVIEW\_ARRAY\_OUTLINEDinstance-attribute

VIEW\_ARRAY\_OUTLINED: IconData

buildVIEW\_ARRAY\_ROUNDEDinstance-attribute

VIEW\_ARRAY\_ROUNDED: IconData

buildVIEW\_ARRAY\_SHARPinstance-attribute

VIEW\_ARRAY\_SHARP: IconData

buildVIEW\_CAROUSELinstance-attribute

VIEW\_CAROUSEL: IconData

buildVIEW\_CAROUSEL\_OUTLINEDinstance-attribute

VIEW\_CAROUSEL\_OUTLINED: IconData

buildVIEW\_CAROUSEL\_ROUNDEDinstance-attribute

VIEW\_CAROUSEL\_ROUNDED: IconData

buildVIEW\_CAROUSEL\_SHARPinstance-attribute

VIEW\_CAROUSEL\_SHARP: IconData

buildVIEW\_COLUMNinstance-attribute

VIEW\_COLUMN: IconData

buildVIEW\_COLUMN\_OUTLINEDinstance-attribute

VIEW\_COLUMN\_OUTLINED: IconData

buildVIEW\_COLUMN\_ROUNDEDinstance-attribute

VIEW\_COLUMN\_ROUNDED: IconData

buildVIEW\_COLUMN\_SHARPinstance-attribute

VIEW\_COLUMN\_SHARP: IconData

buildVIEW\_COMFORTABLEinstance-attribute

VIEW\_COMFORTABLE: IconData

buildVIEW\_COMFORTABLE\_OUTLINEDinstance-attribute

VIEW\_COMFORTABLE\_OUTLINED: IconData

buildVIEW\_COMFORTABLE\_ROUNDEDinstance-attribute

VIEW\_COMFORTABLE\_ROUNDED: IconData

buildVIEW\_COMFORTABLE\_SHARPinstance-attribute

VIEW\_COMFORTABLE\_SHARP: IconData

buildVIEW\_COMFYinstance-attribute

VIEW\_COMFY: IconData

buildVIEW\_COMFY\_ALTinstance-attribute

VIEW\_COMFY\_ALT: IconData

buildVIEW\_COMFY\_ALT\_OUTLINEDinstance-attribute

VIEW\_COMFY\_ALT\_OUTLINED: IconData

buildVIEW\_COMFY\_ALT\_ROUNDEDinstance-attribute

VIEW\_COMFY\_ALT\_ROUNDED: IconData

buildVIEW\_COMFY\_ALT\_SHARPinstance-attribute

VIEW\_COMFY\_ALT\_SHARP: IconData

buildVIEW\_COMFY\_OUTLINEDinstance-attribute

VIEW\_COMFY\_OUTLINED: IconData

buildVIEW\_COMFY\_ROUNDEDinstance-attribute

VIEW\_COMFY\_ROUNDED: IconData

buildVIEW\_COMFY\_SHARPinstance-attribute

VIEW\_COMFY\_SHARP: IconData

buildVIEW\_COMPACTinstance-attribute

VIEW\_COMPACT: IconData

buildVIEW\_COMPACT\_ALTinstance-attribute

VIEW\_COMPACT\_ALT: IconData

buildVIEW\_COMPACT\_ALT\_OUTLINEDinstance-attribute

VIEW\_COMPACT\_ALT\_OUTLINED: IconData

buildVIEW\_COMPACT\_ALT\_ROUNDEDinstance-attribute

VIEW\_COMPACT\_ALT\_ROUNDED: IconData

buildVIEW\_COMPACT\_ALT\_SHARPinstance-attribute

VIEW\_COMPACT\_ALT\_SHARP: IconData

buildVIEW\_COMPACT\_OUTLINEDinstance-attribute

VIEW\_COMPACT\_OUTLINED: IconData

buildVIEW\_COMPACT\_ROUNDEDinstance-attribute

VIEW\_COMPACT\_ROUNDED: IconData

buildVIEW\_COMPACT\_SHARPinstance-attribute

VIEW\_COMPACT\_SHARP: IconData

buildVIEW\_COZYinstance-attribute

VIEW\_COZY: IconData

buildVIEW\_COZY\_OUTLINEDinstance-attribute

VIEW\_COZY\_OUTLINED: IconData

buildVIEW\_COZY\_ROUNDEDinstance-attribute

VIEW\_COZY\_ROUNDED: IconData

buildVIEW\_COZY\_SHARPinstance-attribute

VIEW\_COZY\_SHARP: IconData

buildVIEW\_DAYinstance-attribute

VIEW\_DAY: IconData

buildVIEW\_DAY\_OUTLINEDinstance-attribute

VIEW\_DAY\_OUTLINED: IconData

buildVIEW\_DAY\_ROUNDEDinstance-attribute

VIEW\_DAY\_ROUNDED: IconData

buildVIEW\_DAY\_SHARPinstance-attribute

VIEW\_DAY\_SHARP: IconData

buildVIEW\_HEADLINEinstance-attribute

VIEW\_HEADLINE: IconData

buildVIEW\_HEADLINE\_OUTLINEDinstance-attribute

VIEW\_HEADLINE\_OUTLINED: IconData

buildVIEW\_HEADLINE\_ROUNDEDinstance-attribute

VIEW\_HEADLINE\_ROUNDED: IconData

buildVIEW\_HEADLINE\_SHARPinstance-attribute

VIEW\_HEADLINE\_SHARP: IconData

buildVIEW\_IN\_ARinstance-attribute

VIEW\_IN\_AR: IconData

buildVIEW\_IN\_AR\_OUTLINEDinstance-attribute

VIEW\_IN\_AR\_OUTLINED: IconData

buildVIEW\_IN\_AR\_ROUNDEDinstance-attribute

VIEW\_IN\_AR\_ROUNDED: IconData

buildVIEW\_IN\_AR\_SHARPinstance-attribute

VIEW\_IN\_AR\_SHARP: IconData

buildVIEW\_KANBANinstance-attribute

VIEW\_KANBAN: IconData

buildVIEW\_KANBAN\_OUTLINEDinstance-attribute

VIEW\_KANBAN\_OUTLINED: IconData

buildVIEW\_KANBAN\_ROUNDEDinstance-attribute

VIEW\_KANBAN\_ROUNDED: IconData

buildVIEW\_KANBAN\_SHARPinstance-attribute

VIEW\_KANBAN\_SHARP: IconData

buildVIEW\_LISTinstance-attribute

VIEW\_LIST: IconData

buildVIEW\_LIST\_OUTLINEDinstance-attribute

VIEW\_LIST\_OUTLINED: IconData

buildVIEW\_LIST\_ROUNDEDinstance-attribute

VIEW\_LIST\_ROUNDED: IconData

buildVIEW\_LIST\_SHARPinstance-attribute

VIEW\_LIST\_SHARP: IconData

buildVIEW\_MODULEinstance-attribute

VIEW\_MODULE: IconData

buildVIEW\_MODULE\_OUTLINEDinstance-attribute

VIEW\_MODULE\_OUTLINED: IconData

buildVIEW\_MODULE\_ROUNDEDinstance-attribute

VIEW\_MODULE\_ROUNDED: IconData

buildVIEW\_MODULE\_SHARPinstance-attribute

VIEW\_MODULE\_SHARP: IconData

buildVIEW\_QUILTinstance-attribute

VIEW\_QUILT: IconData

buildVIEW\_QUILT\_OUTLINEDinstance-attribute

VIEW\_QUILT\_OUTLINED: IconData

buildVIEW\_QUILT\_ROUNDEDinstance-attribute

VIEW\_QUILT\_ROUNDED: IconData

buildVIEW\_QUILT\_SHARPinstance-attribute

VIEW\_QUILT\_SHARP: IconData

buildVIEW\_SIDEBARinstance-attribute

VIEW\_SIDEBAR: IconData

buildVIEW\_SIDEBAR\_OUTLINEDinstance-attribute

VIEW\_SIDEBAR\_OUTLINED: IconData

buildVIEW\_SIDEBAR\_ROUNDEDinstance-attribute

VIEW\_SIDEBAR\_ROUNDED: IconData

buildVIEW\_SIDEBAR\_SHARPinstance-attribute

VIEW\_SIDEBAR\_SHARP: IconData

buildVIEW\_STREAMinstance-attribute

VIEW\_STREAM: IconData

buildVIEW\_STREAM\_OUTLINEDinstance-attribute

VIEW\_STREAM\_OUTLINED: IconData

buildVIEW\_STREAM\_ROUNDEDinstance-attribute

VIEW\_STREAM\_ROUNDED: IconData

buildVIEW\_STREAM\_SHARPinstance-attribute



VIEW\_STREAM\_SHARP: IconData

**buildVIEW\_TIMELINE**Instance-attribute

VIEW\_TIMELINE: IconData

**buildVIEW\_TIMELINE\_OUTLINED**Instance-attribute

VIEW\_TIMELINE\_OUTLINED: IconData

**buildVIEW\_TIMELINE\_ROUNDED**Instance-attribute

VIEW\_TIMELINE\_ROUNDED: IconData

**buildVIEW\_TIMELINE\_SHARP**Instance-attribute

VIEW\_TIMELINE\_SHARP: IconData

**buildVIEW\_WEEK**Instance-attribute

VIEW\_WEEK: IconData

**buildVIEW\_WEEK\_OUTLINED**Instance-attribute

VIEW\_WEEK\_OUTLINED: IconData

**buildVIEW\_WEEK\_ROUNDED**Instance-attribute

VIEW\_WEEK\_ROUNDED: IconData

**buildVIEW\_WEEK\_SHARP**Instance-attribute

VIEW\_WEEK\_SHARP: IconData

**buildVIGNETTE**Instance-attribute

VIGNETTE: IconData

**buildVIGNETTE\_OUTLINED**Instance-attribute

VIGNETTE\_OUTLINED: IconData

**buildVIGNETTE\_ROUNDED**Instance-attribute

VIGNETTE\_ROUNDED: IconData

**buildVIGNETTE\_SHARP**Instance-attribute

VIGNETTE\_SHARP: IconData

**buildVILLA**Instance-attribute

VILLA: IconData

**buildVILLA\_OUTLINED**Instance-attribute

VILLA\_OUTLINED: IconData

**buildVILLA\_ROUNDED**Instance-attribute

VILLA\_ROUNDED: IconData

**buildVILLA\_SHARP**Instance-attribute

VILLA\_SHARP: IconData

**buildVISIBILITY**Instance-attribute

VISIBILITY: IconData

**buildVISIBILITY\_OFF**Instance-attribute

VISIBILITY\_OFF: IconData

**buildVISIBILITY\_OFF\_OUTLINED**Instance-attribute

VISIBILITY\_OFF\_OUTLINED: IconData

**buildVISIBILITY\_OFF\_ROUNDED**Instance-attribute

VISIBILITY\_OFF\_ROUNDED: IconData

**buildVISIBILITY\_OFF\_SHARP**Instance-attribute

VISIBILITY\_OFF\_SHARP: IconData

**buildVISIBILITY\_OUTLINED**Instance-attribute

VISIBILITY\_OUTLINED: IconData

**buildVISIBILITY\_ROUNDEDinstance-attribute**

VISIBILITY\_ROUNDED: IconData

**buildVISIBILITY\_SHARPinstance-attribute**

VISIBILITY\_SHARP: IconData

**buildVOICE\_CHATinstance-attribute**

VOICE\_CHAT: IconData

**buildVOICE\_CHAT\_OUTLINEDinstance-attribute**

VOICE\_CHAT\_OUTLINED: IconData

**buildVOICE\_CHAT\_ROUNDEDinstance-attribute**

VOICE\_CHAT\_ROUNDED: IconData

**buildVOICE\_CHAT\_SHARPinstance-attribute**

VOICE\_CHAT\_SHARP: IconData

**buildVOICE\_OVER\_OFFinstance-attribute**

VOICE\_OVER\_OFF: IconData

**buildVOICE\_OVER\_OFF\_OUTLINEDinstance-attribute**

VOICE\_OVER\_OFF\_OUTLINED: IconData

**buildVOICE\_OVER\_OFF\_ROUNDEDinstance-attribute**

VOICE\_OVER\_OFF\_ROUNDED: IconData

**buildVOICE\_OVER\_OFF\_SHARPinstance-attribute**

VOICE\_OVER\_OFF\_SHARP: IconData

**buildVOICEMAILinstance-attribute**

VOICEMAIL: IconData

**buildVOICEMAIL\_OUTLINEDinstance-attribute**

VOICEMAIL\_OUTLINED: IconData

**buildVOICEMAIL\_ROUNDEDinstance-attribute**

VOICEMAIL\_ROUNDED: IconData

**buildVOICEMAIL\_SHARPinstance-attribute**

VOICEMAIL\_SHARP: IconData

**buildVOLCANOinstance-attribute**

VOLCANO: IconData

**buildVOLCANO\_OUTLINEDinstance-attribute**

VOLCANO\_OUTLINED: IconData

**buildVOLCANO\_ROUNDEDinstance-attribute**

VOLCANO\_ROUNDED: IconData

**buildVOLCANO\_SHARPinstance-attribute**

VOLCANO\_SHARP: IconData

**buildVOLUME\_DOWNinstance-attribute**

VOLUME\_DOWN: IconData

**buildVOLUME\_DOWN\_ALTinstance-attribute**

VOLUME\_DOWN\_ALT: IconData

**buildVOLUME\_DOWN\_OUTLINEDinstance-attribute**

VOLUME\_DOWN\_OUTLINED: IconData

**buildVOLUME\_DOWN\_ROUNDEDinstance-attribute**

VOLUME\_DOWN\_ROUNDED: IconData

buildVOLUME\_DOWN\_SHARPinstance-attribute

VOLUME\_DOWN\_SHARP: IconData

buildVOLUME\_MUTEinstance-attribute

VOLUME\_MUTE: IconData

buildVOLUME\_MUTE\_OUTLINEDinstance-attribute

VOLUME\_MUTE\_OUTLINED: IconData

buildVOLUME\_MUTE\_ROUNDEDinstance-attribute

VOLUME\_MUTE\_ROUNDED: IconData

buildVOLUME\_MUTE\_SHARPinstance-attribute

VOLUME\_MUTE\_SHARP: IconData

buildVOLUME\_OFFinstance-attribute

VOLUME\_OFF: IconData

buildVOLUME\_OFF\_OUTLINEDinstance-attribute

VOLUME\_OFF\_OUTLINED: IconData

buildVOLUME\_OFF\_ROUNDEDinstance-attribute

VOLUME\_OFF\_ROUNDED: IconData

buildVOLUME\_OFF\_SHARPinstance-attribute

VOLUME\_OFF\_SHARP: IconData

buildVOLUME\_UPinstance-attribute

VOLUME\_UP: IconData

buildVOLUME\_UP\_OUTLINEDinstance-attribute

VOLUME\_UP\_OUTLINED: IconData

buildVOLUME\_UP\_ROUNDEDinstance-attribute

VOLUME\_UP\_ROUNDED: IconData

buildVOLUME\_UP\_SHARPinstance-attribute

VOLUME\_UP\_SHARP: IconData

buildVOLUNTEER\_ACTIVISMinstance-attribute

VOLUNTEER\_ACTIVISM: IconData

buildVOLUNTEER\_ACTIVISM\_OUTLINEDinstance-attribute

VOLUNTEER\_ACTIVISM\_OUTLINED: IconData

buildVOLUNTEER\_ACTIVISM\_ROUNDEDinstance-attribute

VOLUNTEER\_ACTIVISM\_ROUNDED: IconData

buildVOLUNTEER\_ACTIVISM\_SHARPinstance-attribute

VOLUNTEER\_ACTIVISM\_SHARP: IconData

buildVPN\_KEYinstance-attribute

VPN\_KEY: IconData

buildVPN\_KEY\_OFFinstance-attribute

VPN\_KEY\_OFF: IconData

buildVPN\_KEY\_OFF\_OUTLINEDinstance-attribute

VPN\_KEY\_OFF\_OUTLINED: IconData

buildVPN\_KEY\_OFF\_ROUNDEDinstance-attribute

VPN\_KEY\_OFF\_ROUNDED: IconData

buildVPN\_KEY\_OFF\_SHARPinstance-attribute

VPN\_KEY\_OFF\_SHARP: IconData

buildVPN\_KEY\_OUTLINEDinstance-attribute

VPN\_KEY\_OUTLINED: IconData

buildVPN\_KEY\_ROUNDEDinstance-attribute

VPN\_KEY\_ROUNDED: IconData

buildVPN\_KEY\_SHARPinstance-attribute

VPN\_KEY\_SHARP: IconData

buildVPN\_LOCKinstance-attribute

VPN\_LOCK: IconData

buildVPN\_LOCK\_OUTLINEDinstance-attribute

VPN\_LOCK\_OUTLINED: IconData

buildVPN\_LOCK\_ROUNDEDinstance-attribute

VPN\_LOCK\_ROUNDED: IconData

buildVPN\_LOCK\_SHARPinstance-attribute

VPN\_LOCK\_SHARP: IconData

buildVRPANOinstance-attribute

VRPANO: IconData

buildVRPANO\_OUTLINEDinstance-attribute

VRPANO\_OUTLINED: IconData

buildVRPANO\_ROUNDEDinstance-attribute

VRPANO\_ROUNDED: IconData

buildVRPANO\_SHARPinstance-attribute

VRPANO\_SHARP: IconData

buildWALLETinstance-attribute

WALLET: IconData

buildWALLET\_GIFTCARDinstance-attribute

WALLET\_GIFTCARD: IconData

buildWALLET\_GIFTCARD\_OUTLINEDinstance-attribute

WALLET\_GIFTCARD\_OUTLINED: IconData

buildWALLET\_GIFTCARD\_ROUNDEDinstance-attribute

WALLET\_GIFTCARD\_ROUNDED: IconData

buildWALLET\_GIFTCARD\_SHARPinstance-attribute

WALLET\_GIFTCARD\_SHARP: IconData

buildWALLET\_MEMBERSHIPinstance-attribute

WALLET\_MEMBERSHIP: IconData

buildWALLET\_MEMBERSHIP\_OUTLINEDinstance-attribute

WALLET\_MEMBERSHIP\_OUTLINED: IconData

buildWALLET\_MEMBERSHIP\_ROUNDEDinstance-attribute

WALLET\_MEMBERSHIP\_ROUNDED: IconData

buildWALLET\_MEMBERSHIP\_SHARPinstance-attribute

WALLET\_MEMBERSHIP\_SHARP: IconData

buildWALLET\_OUTLINEDinstance-attribute

WALLET\_OUTLINED: IconData

buildWALLET\_ROUNDEDinstance-attribute

WALLET\_ROUNDED: IconData

buildWALLET\_SHARPinstance-attribute

WALLET\_SHARP: IconData

buildWALLET\_TRAVELinstance-attribute

WALLET\_TRAVEL: IconData

buildWALLET\_TRAVEL\_OUTLINEDinstance-attribute

WALLET\_TRAVEL\_OUTLINED: IconData

buildWALLET\_TRAVEL\_ROUNDEDinstance-attribute

WALLET\_TRAVEL\_ROUNDED: IconData

buildWALLET\_TRAVEL\_SHARPinstance-attribute

WALLET\_TRAVEL\_SHARP: IconData

buildWALLPAPERinstance-attribute

WALLPAPER: IconData

buildWALLPAPER\_OUTLINEDinstance-attribute

WALLPAPER\_OUTLINED: IconData

buildWALLPAPER\_ROUNDEDinstance-attribute

WALLPAPER\_ROUNDED: IconData

buildWALLPAPER\_SHARPinstance-attribute

WALLPAPER\_SHARP: IconData

buildWAREHOUSEinstance-attribute

WAREHOUSE: IconData

buildWAREHOUSE\_OUTLINEDinstance-attribute

WAREHOUSE\_OUTLINED: IconData

buildWAREHOUSE\_ROUNDEDinstance-attribute

WAREHOUSE\_ROUNDED: IconData

buildWAREHOUSE\_SHARPinstance-attribute

WAREHOUSE\_SHARP: IconData

buildWARNINGinstance-attribute

WARNING: IconData

buildWARNING\_AMBERinstance-attribute

WARNING\_AMBER: IconData

buildWARNING\_AMBER\_OUTLINEDinstance-attribute

WARNING\_AMBER\_OUTLINED: IconData

buildWARNING\_AMBER\_ROUNDEDinstance-attribute

WARNING\_AMBER\_ROUNDED: IconData

buildWARNING\_AMBER\_SHARPinstance-attribute

WARNING\_AMBER\_SHARP: IconData

buildWARNING\_OUTLINEDinstance-attribute

WARNING\_OUTLINED: IconData

buildWARNING\_ROUNDEDinstance-attribute

WARNING\_ROUNDED: IconData

buildWARNING\_SHARPinstance-attribute

WARNING\_SHARP: IconData

buildWASHinstance-attribute

WASH: IconData

buildWASH\_OUTLINEDinstance-attribute

WASH\_OUTLINED: IconData

buildWASH\_ROUNDEDinstance-attribute

WASH\_ROUNDED: IconData

buildWASH\_SHARPinstance-attribute

WASH\_SHARP: IconData

buildWATCHinstance-attribute

WATCH: IconData

buildWATCH\_LATERinstance-attribute

WATCH\_LATER: IconData

buildWATCH\_LATER\_OUTLINEDinstance-attribute

WATCH\_LATER\_OUTLINED: IconData

buildWATCH\_LATER\_ROUNDEDinstance-attribute

WATCH\_LATER\_ROUNDED: IconData

buildWATCH\_LATER\_SHARPinstance-attribute

WATCH\_LATER\_SHARP: IconData

buildWATCH\_OFFinstance-attribute

WATCH\_OFF: IconData

buildWATCH\_OFF\_OUTLINEDinstance-attribute

WATCH\_OFF\_OUTLINED: IconData

buildWATCH\_OFF\_ROUNDEDinstance-attribute

WATCH\_OFF\_ROUNDED: IconData

buildWATCH\_OFF\_SHARPinstance-attribute

WATCH\_OFF\_SHARP: IconData

buildWATCH\_OUTLINEDinstance-attribute

WATCH\_OUTLINED: IconData

buildWATCH\_ROUNDEDinstance-attribute

WATCH\_ROUNDED: IconData

buildWATCH\_SHARPinstance-attribute

WATCH\_SHARP: IconData

buildWATERinstance-attribute

WATER: IconData

buildWATER\_DAMAGEinstance-attribute

WATER\_DAMAGE: IconData

buildWATER\_DAMAGE\_OUTLINEDinstance-attribute

WATER\_DAMAGE\_OUTLINED: IconData

buildWATER\_DAMAGE\_ROUNDEDinstance-attribute

WATER\_DAMAGE\_ROUNDED: IconData

buildWATER\_DAMAGE\_SHARPinstance-attribute

WATER\_DAMAGE\_SHARP: IconData

buildWATER\_DROPinstance-attribute

WATER\_DROP: IconData

buildWATER\_DROP\_OUTLINEDinstance-attribute

WATER\_DROP\_OUTLINED: IconData

buildWATER\_DROP\_ROUNDEDinstance-attribute

WATER\_DROP\_ROUNDED: IconData

buildWATER\_DROP\_SHARPinstance-attribute

WATER\_DROP\_SHARP: IconData

buildWATER\_OUTLINEDinstance-attribute

WATER\_OUTLINED: IconData

buildWATER\_ROUNDEDinstance-attribute

WATER\_ROUNDED: IconData

buildWATER\_SHARPinstance-attribute

WATER\_SHARP: IconData

buildWATERFALL\_CHARTinstance-attribute

WATERFALL\_CHART: IconData

buildWATERFALL\_CHART\_OUTLINEDinstance-attribute

WATERFALL\_CHART\_OUTLINED: IconData

buildWATERFALL\_CHART\_ROUNDEDinstance-attribute

WATERFALL\_CHART\_ROUNDED: IconData

buildWATERFALL\_CHART\_SHARPinstance-attribute

WATERFALL\_CHART\_SHARP: IconData

buildWAVESinstance-attribute

WAVES: IconData

buildWAVES\_OUTLINEDinstance-attribute

WAVES\_OUTLINED: IconData

buildWAVES\_ROUNDEDinstance-attribute

WAVES\_ROUNDED: IconData

buildWAVES\_SHARPinstance-attribute

WAVES\_SHARP: IconData

buildWAVING\_HANDinstance-attribute

WAVING\_HAND: IconData

buildWAVING\_HAND\_OUTLINEDinstance-attribute

WAVING\_HAND\_OUTLINED: IconData

buildWAVING\_HAND\_ROUNDEDinstance-attribute

WAVING\_HAND\_ROUNDED: IconData

buildWAVING\_HAND\_SHARPinstance-attribute

WAVING\_HAND\_SHARP: IconData

buildWB\_AUTOinstance-attribute

WB\_AUTO: IconData

buildWB\_AUTO\_OUTLINEDinstance-attribute

WB\_AUTO\_OUTLINED: IconData

buildWB\_AUTO\_ROUNDEDinstance-attribute

WB\_AUTO\_ROUNDED: IconData

buildWB\_AUTO\_SHARPinstance-attribute

WB\_AUTO\_SHARP: IconData

buildWB\_CLOUDYinstance-attribute

WB\_CLOUDY: IconData

buildWB\_CLOUDY\_OUTLINEDinstance-attribute

WB\_CLOUDY\_OUTLINED: IconData

buildWB\_CLOUDY\_ROUNDEDinstance-attribute

WB\_CLOUDY\_ROUNDED: IconData

buildWB\_CLOUDY\_SHARPinstance-attribute

WB\_CLOUDY\_SHARP: IconData

buildWB\_INCANDESCENTinstance-attribute

WB\_INCANDESCENT: IconData

buildWB\_INCANDESCENT\_OUTLINEDinstance-attribute

WB\_INCANDESCENT\_OUTLINED: IconData

buildWB\_INCANDESCENT\_ROUNDEDinstance-attribute

WB\_INCANDESCENT\_ROUNDED: IconData

buildWB\_INCANDESCENT\_SHARPinstance-attribute

WB\_INCANDESCENT\_SHARP: IconData

buildWB\_IRIDESCENTinstance-attribute

WB\_IRIDESCENT: IconData

buildWB\_IRIDESCENT\_OUTLINEDinstance-attribute

WB\_IRIDESCENT\_OUTLINED: IconData

buildWB\_IRIDESCENT\_ROUNDEDinstance-attribute

WB\_IRIDESCENT\_ROUNDED: IconData

buildWB\_IRIDESCENT\_SHARPinstance-attribute

WB\_IRIDESCENT\_SHARP: IconData

buildWB\_SHADEinstance-attribute

WB\_SHADE: IconData

buildWB\_SHADE\_OUTLINEDinstance-attribute

WB\_SHADE\_OUTLINED: IconData

buildWB\_SHADE\_ROUNDEDinstance-attribute

WB\_SHADE\_ROUNDED: IconData

buildWB\_SHADE\_SHARPinstance-attribute

WB\_SHADE\_SHARP: IconData

buildWB\_SUNNYinstance-attribute

WB\_SUNNY: IconData

buildWB\_SUNNY\_OUTLINEDinstance-attribute

WB\_SUNNY\_OUTLINED: IconData

buildWB\_SUNNY\_ROUNDEDinstance-attribute

WB\_SUNNY\_ROUNDED: IconData

buildWB\_SUNNY\_SHARPinstance-attribute

WB\_SUNNY\_SHARP: IconData

buildWB\_TWIGHLIGHTinstance-attribute

WB\_TWIGHLIGHT: IconData

buildWB\_TWILIGHTinstance-attribute

WB\_TWILIGHT: IconData

buildWB\_TWILIGHT\_OUTLINEDinstance-attribute



WB\_TWILIGHT\_OUTLINED: IconData

buildWB\_TWILIGHT\_ROUNDEDinstance-attribute

WB\_TWILIGHT\_ROUNDED: IconData

buildWB\_TWILIGHT\_SHARPinstance-attribute

WB\_TWILIGHT\_SHARP: IconData

buildWCinstance-attribute

WC: IconData

buildWC\_OUTLINEDinstance-attribute

WC\_OUTLINED: IconData

buildWC\_ROUNDEDinstance-attribute

WC\_ROUNDED: IconData

buildWC\_SHARPinstance-attribute

WC\_SHARP: IconData

buildWEBinstance-attribute

WEB: IconData

buildWEB\_ASSETinstance-attribute

WEB\_ASSET: IconData

buildWEB\_ASSET\_OFFinstance-attribute

WEB\_ASSET\_OFF: IconData

buildWEB\_ASSET\_OFF\_OUTLINEDinstance-attribute

WEB\_ASSET\_OFF\_OUTLINED: IconData

buildWEB\_ASSET\_OFF\_ROUNDEDinstance-attribute

WEB\_ASSET\_OFF\_ROUNDED: IconData

buildWEB\_ASSET\_OFF\_SHARPinstance-attribute

WEB\_ASSET\_OFF\_SHARP: IconData

buildWEB\_ASSET\_OUTLINEDinstance-attribute

WEB\_ASSET\_OUTLINED: IconData

buildWEB\_ASSET\_ROUNDEDinstance-attribute

WEB\_ASSET\_ROUNDED: IconData

buildWEB\_ASSET\_SHARPinstance-attribute

WEB\_ASSET\_SHARP: IconData

buildWEB\_OUTLINEDinstance-attribute

WEB\_OUTLINED: IconData

buildWEB\_ROUNDEDinstance-attribute

WEB\_ROUNDED: IconData

buildWEB\_SHARPinstance-attribute

WEB\_SHARP: IconData

buildWEB\_STORIESinstance-attribute

WEB\_STORIES: IconData

buildWEB\_STORIES\_OUTLINEDinstance-attribute

WEB\_STORIES\_OUTLINED: IconData

buildWEB\_STORIES\_ROUNDEDinstance-attribute

WEB\_STORIES\_ROUNDED: IconData

buildWEB\_STORIES\_SHARPinstance-attribute

WEB\_STORIES\_SHARP: IconData

**buildWEBHOOKinstance-attribute**

WEBHOOK: IconData

**buildWEBHOOK\_OUTLINEDinstance-attribute**

WEBHOOK\_OUTLINED: IconData

**buildWEBHOOK\_ROUNDEDinstance-attribute**

WEBHOOK\_ROUNDED: IconData

**buildWEBHOOK\_SHARPinstance-attribute**

WEBHOOK\_SHARP: IconData

**buildWECHATinstance-attribute**

WECHAT: IconData

**buildWECHAT\_OUTLINEDinstance-attribute**

WECHAT\_OUTLINED: IconData

**buildWECHAT\_ROUNDEDinstance-attribute**

WECHAT\_ROUNDED: IconData

**buildWECHAT\_SHARPinstance-attribute**

WECHAT\_SHARP: IconData

**buildWEEKENDinstance-attribute**

WEEKEND: IconData

**buildWEEKEND\_OUTLINEDinstance-attribute**

WEEKEND\_OUTLINED: IconData

**buildWEEKEND\_ROUNDEDinstance-attribute**

WEEKEND\_ROUNDED: IconData

**buildWEEKEND\_SHARPinstance-attribute**

WEEKEND\_SHARP: IconData

**buildWESTinstance-attribute**

WEST: IconData

**buildWEST\_OUTLINEDinstance-attribute**

WEST\_OUTLINED: IconData

**buildWEST\_ROUNDEDinstance-attribute**

WEST\_ROUNDED: IconData

**buildWEST\_SHARPinstance-attribute**

WEST\_SHARP: IconData

**buildWHATSHOTinstance-attribute**

WHATSHOT: IconData

**buildWHATSHOT\_OUTLINEDinstance-attribute**

WHATSHOT\_OUTLINED: IconData

**buildWHATSHOT\_ROUNDEDinstance-attribute**

WHATSHOT\_ROUNDED: IconData

**buildWHATSHOT\_SHARPinstance-attribute**

WHATSHOT\_SHARP: IconData

**buildWHEELCHAIR\_PICKUPinstance-attribute**

WHEELCHAIR\_PICKUP: IconData

**buildWHEELCHAIR\_PICKUP\_OUTLINEDinstance-attribute**

WHEELCHAIR\_PICKUP\_OUTLINED: IconData

buildWHEELCHAIR\_PICKUP\_ROUNDEDinstance-attribute

WHEELCHAIR\_PICKUP\_ROUNDED: IconData

buildWHEELCHAIR\_PICKUP\_SHARPinstance-attribute

WHEELCHAIR\_PICKUP\_SHARP: IconData

buildWHERE\_TO\_VOTEinstance-attribute

WHERE\_TO\_VOTE: IconData

buildWHERE\_TO\_VOTE\_OUTLINEDinstance-attribute

WHERE\_TO\_VOTE\_OUTLINED: IconData

buildWHERE\_TO\_VOTE\_ROUNDEDinstance-attribute

WHERE\_TO\_VOTE\_ROUNDED: IconData

buildWHERE\_TO\_VOTE\_SHARPinstance-attribute

WHERE\_TO\_VOTE\_SHARP: IconData

buildWIDGETSinstance-attribute

WIDGETS: IconData

buildWIDGETS\_OUTLINEDinstance-attribute

WIDGETS\_OUTLINED: IconData

buildWIDGETS\_ROUNDEDinstance-attribute

WIDGETS\_ROUNDED: IconData

buildWIDGETS\_SHARPinstance-attribute

WIDGETS\_SHARP: IconData

buildWIDTH\_FULLinstance-attribute

WIDTH\_FULL: IconData

buildWIDTH\_FULL\_OUTLINEDinstance-attribute

WIDTH\_FULL\_OUTLINED: IconData

buildWIDTH\_FULL\_ROUNDEDinstance-attribute

WIDTH\_FULL\_ROUNDED: IconData

buildWIDTH\_FULL\_SHARPinstance-attribute

WIDTH\_FULL\_SHARP: IconData

buildWIDTH\_NORMALinstance-attribute

WIDTH\_NORMAL: IconData

buildWIDTH\_NORMAL\_OUTLINEDinstance-attribute

WIDTH\_NORMAL\_OUTLINED: IconData

buildWIDTH\_NORMAL\_ROUNDEDinstance-attribute

WIDTH\_NORMAL\_ROUNDED: IconData

buildWIDTH\_NORMAL\_SHARPinstance-attribute

WIDTH\_NORMAL\_SHARP: IconData

buildWIDTH\_WIDEinstance-attribute

WIDTH\_WIDE: IconData

buildWIDTH\_WIDE\_OUTLINEDinstance-attribute

WIDTH\_WIDE\_OUTLINED: IconData

buildWIDTH\_WIDE\_ROUNDEDinstance-attribute

WIDTH\_WIDE\_ROUNDED: IconData

buildWIDTH\_WIDE\_SHARPinstance-attribute

WIDTH\_WIDE\_SHARP: IconData

## buildWIFIinstance-attribute

WIFI: IconData

## buildWIFI\_1\_BARinstance-attribute

WIFI\_1\_BAR: IconData

## buildWIFI\_1\_BAR\_OUTLINEDinstance-attribute

WIFI\_1\_BAR\_OUTLINED: IconData

## buildWIFI\_1\_BAR\_ROUNDEDinstance-attribute

WIFI\_1\_BAR\_ROUNDED: IconData

## buildWIFI\_1\_BAR\_SHARPinstance-attribute

WIFI\_1\_BAR\_SHARP: IconData

## buildWIFI\_2\_BARinstance-attribute

WIFI\_2\_BAR: IconData

## buildWIFI\_2\_BAR\_OUTLINEDinstance-attribute

WIFI\_2\_BAR\_OUTLINED: IconData

## buildWIFI\_2\_BAR\_ROUNDEDinstance-attribute

WIFI\_2\_BAR\_ROUNDED: IconData

## buildWIFI\_2\_BAR\_SHARPinstance-attribute

WIFI\_2\_BAR\_SHARP: IconData

## buildWIFI\_CALLINGinstance-attribute

WIFI\_CALLING: IconData

## buildWIFI\_CALLING\_3instance-attribute

WIFI\_CALLING\_3: IconData

## buildWIFI\_CALLING\_3\_OUTLINEDinstance-attribute

WIFI\_CALLING\_3\_OUTLINED: IconData

## buildWIFI\_CALLING\_3\_ROUNDEDinstance-attribute

WIFI\_CALLING\_3\_ROUNDED: IconData

## buildWIFI\_CALLING\_3\_SHARPinstance-attribute

WIFI\_CALLING\_3\_SHARP: IconData

## buildWIFI\_CALLING\_OUTLINEDinstance-attribute

WIFI\_CALLING\_OUTLINED: IconData

## buildWIFI\_CALLING\_ROUNDEDinstance-attribute

WIFI\_CALLING\_ROUNDED: IconData

## buildWIFI\_CALLING\_SHARPinstance-attribute

WIFI\_CALLING\_SHARP: IconData

## buildWIFI\_CHANNELinstance-attribute

WIFI\_CHANNEL: IconData

## buildWIFI\_CHANNEL\_OUTLINEDinstance-attribute

WIFI\_CHANNEL\_OUTLINED: IconData

## buildWIFI\_CHANNEL\_ROUNDEDinstance-attribute

WIFI\_CHANNEL\_ROUNDED: IconData

## buildWIFI\_CHANNEL\_SHARPinstance-attribute

WIFI\_CHANNEL\_SHARP: IconData

## buildWIFI\_FINDinstance-attribute

WIFI\_FIND: IconData

buildWIFI\_FIND\_OUTLINEDinstance-attribute

WIFI\_FIND\_OUTLINED: IconData

buildWIFI\_FIND\_ROUNDEDinstance-attribute

WIFI\_FIND\_ROUNDED: IconData

buildWIFI\_FIND\_SHARPinstance-attribute

WIFI\_FIND\_SHARP: IconData

buildWIFI\_LOCKinstance-attribute

WIFI\_LOCK: IconData

buildWIFI\_LOCK\_OUTLINEDinstance-attribute

WIFI\_LOCK\_OUTLINED: IconData

buildWIFI\_LOCK\_ROUNDEDinstance-attribute

WIFI\_LOCK\_ROUNDED: IconData

buildWIFI\_LOCK\_SHARPinstance-attribute

WIFI\_LOCK\_SHARP: IconData

buildWIFI\_OFFinstance-attribute

WIFI\_OFF: IconData

buildWIFI\_OFF\_OUTLINEDinstance-attribute

WIFI\_OFF\_OUTLINED: IconData

buildWIFI\_OFF\_ROUNDEDinstance-attribute

WIFI\_OFF\_ROUNDED: IconData

buildWIFI\_OFF\_SHARPinstance-attribute

WIFI\_OFF\_SHARP: IconData

buildWIFI\_OUTLINEDinstance-attribute

WIFI\_OUTLINED: IconData

buildWIFI\_PASSWORDinstance-attribute

WIFI\_PASSWORD: IconData

buildWIFI\_PASSWORD\_OUTLINEDinstance-attribute

WIFI\_PASSWORD\_OUTLINED: IconData

buildWIFI\_PASSWORD\_ROUNDEDinstance-attribute

WIFI\_PASSWORD\_ROUNDED: IconData

buildWIFI\_PASSWORD\_SHARPinstance-attribute

WIFI\_PASSWORD\_SHARP: IconData

buildWIFI\_PROTECTED\_SETUPinstance-attribute

WIFI\_PROTECTED\_SETUP: IconData

buildWIFI\_PROTECTED\_SETUP\_OUTLINEDinstance-attribute

WIFI\_PROTECTED\_SETUP\_OUTLINED: IconData

buildWIFI\_PROTECTED\_SETUP\_ROUNDEDinstance-attribute

WIFI\_PROTECTED\_SETUP\_ROUNDED: IconData

buildWIFI\_PROTECTED\_SETUP\_SHARPinstance-attribute

WIFI\_PROTECTED\_SETUP\_SHARP: IconData

buildWIFI\_ROUNDEDinstance-attribute

WIFI\_ROUNDED: IconData

buildWIFI\_SHARPinstance-attribute

WIFI\_SHARP: IconData

**buildWIFI\_TETHERINGInstance-attribute**

WIFI\_TETHERING: IconData

**buildWIFI\_TETHERING\_ERRORInstance-attribute**

WIFI\_TETHERING\_ERROR: IconData

**buildWIFI\_TETHERING\_ERROR\_OUTLINEDInstance-attribute**

WIFI\_TETHERING\_ERROR\_OUTLINED: IconData

**buildWIFI\_TETHERING\_ERROR\_ROUNDEDInstance-attribute**

WIFI\_TETHERING\_ERROR\_ROUNDED: IconData

**buildWIFI\_TETHERING\_ERROR\_ROUNDED\_OUTLINEDInstance-attribute**

WIFI\_TETHERING\_ERROR\_ROUNDED\_OUTLINED: IconData

**buildWIFI\_TETHERING\_ERROR\_ROUNDED\_ROUNDEDInstance-attribute**

WIFI\_TETHERING\_ERROR\_ROUNDED\_ROUNDED: IconData

**buildWIFI\_TETHERING\_ERROR\_ROUNDED\_SHARPInstance-attribute**

WIFI\_TETHERING\_ERROR\_ROUNDED\_SHARP: IconData

**buildWIFI\_TETHERING\_ERROR\_SHARPInstance-attribute**

WIFI\_TETHERING\_ERROR\_SHARP: IconData

**buildWIFI\_TETHERING\_OFFInstance-attribute**

WIFI\_TETHERING\_OFF: IconData

**buildWIFI\_TETHERING\_OFF\_OUTLINEDInstance-attribute**

WIFI\_TETHERING\_OFF\_OUTLINED: IconData

**buildWIFI\_TETHERING\_OFF\_ROUNDEDInstance-attribute**

WIFI\_TETHERING\_OFF\_ROUNDED: IconData

**buildWIFI\_TETHERING\_OFF\_SHARPInstance-attribute**

WIFI\_TETHERING\_OFF\_SHARP: IconData

**buildWIFI\_TETHERING\_OUTLINEDInstance-attribute**

WIFI\_TETHERING\_OUTLINED: IconData

**buildWIFI\_TETHERING\_ROUNDEDInstance-attribute**

WIFI\_TETHERING\_ROUNDED: IconData

**buildWIFI\_TETHERING\_SHARPInstance-attribute**

WIFI\_TETHERING\_SHARP: IconData

**buildWIND\_POWERInstance-attribute**

WIND\_POWER: IconData

**buildWIND\_POWER\_OUTLINEDInstance-attribute**

WIND\_POWER\_OUTLINED: IconData

**buildWIND\_POWER\_ROUNDEDInstance-attribute**

WIND\_POWER\_ROUNDED: IconData

**buildWIND\_POWER\_SHARPInstance-attribute**

WIND\_POWER\_SHARP: IconData

**buildWINDOWInstance-attribute**

WINDOW: IconData

**buildWINDOW\_OUTLINEDInstance-attribute**

WINDOW\_OUTLINED: IconData

**buildWINDOW\_ROUNDEDInstance-attribute**

WINDOW\_ROUNDED: IconData

buildWINDOW\_SHARPinstance-attribute

WINDOW\_SHARP: IconData

buildWINE\_BARinstance-attribute

WINE\_BAR: IconData

buildWINE\_BAR\_OUTLINEDinstance-attribute

WINE\_BAR\_OUTLINED: IconData

buildWINE\_BAR\_ROUNDEDinstance-attribute

WINE\_BAR\_ROUNDED: IconData

buildWINE\_BAR\_SHARPinstance-attribute

WINE\_BAR\_SHARP: IconData

buildWOMANinstance-attribute

WOMAN: IconData

buildWOMAN\_2instance-attribute

WOMAN\_2: IconData

buildWOMAN\_2\_OUTLINEDinstance-attribute

WOMAN\_2\_OUTLINED: IconData

buildWOMAN\_2\_ROUNDEDinstance-attribute

WOMAN\_2\_ROUNDED: IconData

buildWOMAN\_2\_SHARPinstance-attribute

WOMAN\_2\_SHARP: IconData

buildWOMAN\_OUTLINEDinstance-attribute

WOMAN\_OUTLINED: IconData

buildWOMAN\_ROUNDEDinstance-attribute

WOMAN\_ROUNDED: IconData

buildWOMAN\_SHARPinstance-attribute

WOMAN\_SHARP: IconData

buildWOO\_COMMERCEinstance-attribute

WOO\_COMMERCE: IconData

buildWOO\_COMMERCE\_OUTLINEDinstance-attribute

WOO\_COMMERCE\_OUTLINED: IconData

buildWOO\_COMMERCE\_ROUNDEDinstance-attribute

WOO\_COMMERCE\_ROUNDED: IconData

buildWOO\_COMMERCE\_SHARPinstance-attribute

WOO\_COMMERCE\_SHARP: IconData

buildWORDPRESSinstance-attribute

WORDPRESS: IconData

buildWORDPRESS\_OUTLINEDinstance-attribute

WORDPRESS\_OUTLINED: IconData

buildWORDPRESS\_ROUNDEDinstance-attribute

WORDPRESS\_ROUNDED: IconData

buildWORDPRESS\_SHARPinstance-attribute

WORDPRESS\_SHARP: IconData

buildWORKinstance-attribute

WORK: IconData

buildWORK\_HISTORYinstance-attribute

WORK\_HISTORY: IconData

buildWORK\_HISTORY\_OUTLINEDinstance-attribute

WORK\_HISTORY\_OUTLINED: IconData

buildWORK\_HISTORY\_ROUNDEDinstance-attribute

WORK\_HISTORY\_ROUNDED: IconData

buildWORK\_HISTORY\_SHARPinstance-attribute

WORK\_HISTORY\_SHARP: IconData

buildWORK\_OFFinstance-attribute

WORK\_OFF: IconData

buildWORK\_OFF\_OUTLINEDinstance-attribute

WORK\_OFF\_OUTLINED: IconData

buildWORK\_OFF\_ROUNDEDinstance-attribute

WORK\_OFF\_ROUNDED: IconData

buildWORK\_OFF\_SHARPinstance-attribute

WORK\_OFF\_SHARP: IconData

buildWORK\_OUTLINEinstance-attribute

WORK\_OUTLINE: IconData

buildWORK\_OUTLINE\_OUTLINEDinstance-attribute

WORK\_OUTLINE\_OUTLINED: IconData

buildWORK\_OUTLINE\_ROUNDEDinstance-attribute

WORK\_OUTLINE\_ROUNDED: IconData

buildWORK\_OUTLINE\_SHARPinstance-attribute

WORK\_OUTLINE\_SHARP: IconData

buildWORK\_OUTLINEDinstance-attribute

WORK\_OUTLINED: IconData

buildWORK\_ROUNDEDinstance-attribute

WORK\_ROUNDED: IconData

buildWORK\_SHARPinstance-attribute

WORK\_SHARP: IconData

buildWORKSPACE\_PREMIUMinstance-attribute

WORKSPACE\_PREMIUM: IconData

buildWORKSPACE\_PREMIUM\_OUTLINEDinstance-attribute

WORKSPACE\_PREMIUM\_OUTLINED: IconData

buildWORKSPACE\_PREMIUM\_ROUNDEDinstance-attribute

WORKSPACE\_PREMIUM\_ROUNDED: IconData

buildWORKSPACE\_PREMIUM\_SHARPinstance-attribute

WORKSPACE\_PREMIUM\_SHARP: IconData

buildWORKSPACESinstance-attribute

WORKSPACES: IconData

buildWORKSPACES\_FILLEDinstance-attribute

WORKSPACES\_FILLED: IconData

buildWORKSPACES\_OUTLINEinstance-attribute



WORKSPACES\_OUTLINE: IconData

buildWORKSPACES\_OUTLINEDinstance-attribute

WORKSPACES\_OUTLINED: IconData

buildWORKSPACES\_ROUNDEDinstance-attribute

WORKSPACES\_ROUNDED: IconData

buildWORKSPACES\_SHARPinstance-attribute

WORKSPACES\_SHARP: IconData

buildWRAP\_TEXTinstance-attribute

WRAP\_TEXT: IconData

buildWRAP\_TEXT\_OUTLINEDinstance-attribute

WRAP\_TEXT\_OUTLINED: IconData

buildWRAP\_TEXT\_ROUNDEDinstance-attribute

WRAP\_TEXT\_ROUNDED: IconData

buildWRAP\_TEXT\_SHARPinstance-attribute

WRAP\_TEXT\_SHARP: IconData

buildWRONG\_LOCATIONinstance-attribute

WRONG\_LOCATION: IconData

buildWRONG\_LOCATION\_OUTLINEDinstance-attribute

WRONG\_LOCATION\_OUTLINED: IconData

buildWRONG\_LOCATION\_ROUNDEDinstance-attribute

WRONG\_LOCATION\_ROUNDED: IconData

buildWRONG\_LOCATION\_SHARPinstance-attribute

WRONG\_LOCATION\_SHARP: IconData

buildWYSIWYGinstance-attribute

WYSIWYG: IconData

buildWYSIWYG\_OUTLINEDinstance-attribute

WYSIWYG\_OUTLINED: IconData

buildWYSIWYG\_ROUNDEDinstance-attribute

WYSIWYG\_ROUNDED: IconData

buildWYSIWYG\_SHARPinstance-attribute

WYSIWYG\_SHARP: IconData

buildYARDinstance-attribute

YARD: IconData

buildYARD\_OUTLINEDinstance-attribute

YARD\_OUTLINED: IconData

buildYARD\_ROUNDEDinstance-attribute

YARD\_ROUNDED: IconData

buildYARD\_SHARPinstance-attribute

YARD\_SHARP: IconData

buildYOUTUBE\_SEARCHED\_FORinstance-attribute

YOUTUBE\_SEARCHED\_FOR: IconData

buildYOUTUBE\_SEARCHED\_FOR\_OUTLINEDinstance-attribute

YOUTUBE\_SEARCHED\_FOR\_OUTLINED: IconData

buildYOUTUBE\_SEARCHED\_FOR\_ROUNDEDinstance-attribute

YOUTUBE\_SEARCHED\_FOR\_ROUNDED: IconData

buildYOUTUBE\_SEARCHED\_FOR\_SHARPinstance-attribute

YOUTUBE\_SEARCHED\_FOR\_SHARP: IconData

buildZOOM\_INinstance-attribute

ZOOM\_IN: IconData

buildZOOM\_IN\_MAPinstance-attribute

ZOOM\_IN\_MAP: IconData

buildZOOM\_IN\_MAP\_OUTLINEDinstance-attribute

ZOOM\_IN\_MAP\_OUTLINED: IconData

buildZOOM\_IN\_MAP\_ROUNDEDinstance-attribute

ZOOM\_IN\_MAP\_ROUNDED: IconData

buildZOOM\_IN\_MAP\_SHARPinstance-attribute

ZOOM\_IN\_MAP\_SHARP: IconData

buildZOOM\_IN\_OUTLINEDinstance-attribute

ZOOM\_IN\_OUTLINED: IconData

buildZOOM\_IN\_ROUNDEDinstance-attribute

ZOOM\_IN\_ROUNDED: IconData

buildZOOM\_IN\_SHARPinstance-attribute

ZOOM\_IN\_SHARP: IconData

buildZOOM\_OUTinstance-attribute

ZOOM\_OUT: IconData

buildZOOM\_OUT\_MAPinstance-attribute

ZOOM\_OUT\_MAP: IconData

buildZOOM\_OUT\_MAP\_OUTLINEDinstance-attribute

ZOOM\_OUT\_MAP\_OUTLINED: IconData

buildZOOM\_OUT\_MAP\_ROUNDEDinstance-attribute

ZOOM\_OUT\_MAP\_ROUNDED: IconData

buildZOOM\_OUT\_MAP\_SHARPinstance-attribute

ZOOM\_OUT\_MAP\_SHARP: IconData

buildZOOM\_OUT\_OUTLINEDinstance-attribute

ZOOM\_OUT\_OUTLINED: IconData

buildZOOM\_OUT\_ROUNDEDinstance-attribute

ZOOM\_OUT\_ROUNDED: IconData

buildZOOM\_OUT\_SHARPinstance-attribute

ZOOM\_OUT\_SHARP: IconData

# IconTheme

Customizes the appearance of Icon controls across the app.

## Properties

- `apply_text_scaling` - Overrides the default value for `Icon.apply_text_scaling`.
- `color` - Overrides the default value for `Icon.color`.
- `fill` - Overrides the default value for `Icon.fill`.
- `grade` - Overrides the default value for `Icon.grade`.
- `opacity` - An opacity to apply to both explicit and default icon colors.
- `optical_size` - Overrides the default value for `Icon.optical_size`.
- `shadows` - Overrides the default value for `Icon.shadows`.
- `size` - Overrides the default value for `Icon.size`.
- `weight` - Overrides the default value for `Icon.weight`.

## Properties

### `buildapply_text_scalingclass-attributeinstance-attribute`

`apply_text_scaling: bool | None = None`

Overrides the default value for `Icon.apply_text_scaling`.

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

Overrides the default value for `Icon.color`.

### `buildfillclass-attributeinstance-attribute`

`fill: Number | None = None`

Overrides the default value for `Icon.fill`.

### `buildgradeclass-attributeinstance-attribute`

`grade: Number | None = None`

Overrides the default value for `Icon.grade`.

### `buildopacityclass-attributeinstance-attribute`

`opacity: Number | None = None`

An opacity to apply to both explicit and default icon colors.

### `buildoptical_sizeclass-attributeinstance-attribute`

`optical_size: Number | None = None`

Overrides the default value for `Icon.optical_size`.

### `buildshadowsclass-attributeinstance-attribute`

`shadows: BoxShadowValue | None = None`

Overrides the default value for `Icon.shadows`.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

Overrides the default value for `Icon.size`.

### `buildweightclass-attributeinstance-attribute`

`weight: Number | None = None`

Overrides the default value for `Icon.weight`.

# ImageRepeat

How to paint any portions of a box not covered by an image.

Inherits: `enum.Enum`

## Properties

- `NO_REPEAT` - Leave uncovered portions of the box transparent.
- `REPEAT` - Repeat the image in both the x and y directions until the box is filled.
- `REPEAT_X` - Repeat the image in the x direction until the box is filled horizontally.
- `REPEAT_Y` - Repeat the image in the y direction until the box is filled vertically.

## Examples

### Image Repeat showcase

`play_arrow` Try Online

```
import flet as ft
```

```
def showcase_card(repeat: ft.ImageRepeat) -> ft.Container:
 return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(repeat.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=240,
 height=130,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Image(
 src="https://picsum.photos/id/237/200/300",
 width=240,
 height=130,
 repeat=repeat,
),
],
),
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="ImageRepeat Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare how an image fills uncovered space using repeat modes."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(repeat) for repeat in ft.ImageRepeat],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildNO\_REPEAT**class-attributeinstance-attribute

Leave uncovered portions of the box transparent.

**buildREPEAT**class-attributeinstance-attribute

Repeat the image in both the x and y directions until the box is filled.

**buildREPEAT\_X**class-attributeinstance-attribute

Repeat the image in the x direction until the box is filled horizontally.

**buildREPEAT\_Y**class-attributeinstance-attribute

Repeat the image in the y direction until the box is filled vertically.

# InputBorder

Inherits: `enum.Enum`

## Properties

- `NONE`
- `OUTLINE`
- `UNDERLINE`

## Properties

`buildNONE`class-attributeinstance-attribute

`buildOUTLINE`class-attributeinstance-attribute

`buildUNDERLINE`class-attributeinstance-attribute

# InputFilter

An input filter that uses a regular expression to allow or deny/block certain patterns in the input.

## Properties

- `allow` - A boolean value indicating whether to allow or deny/block the matched patterns.
- `case_sensitive` - Whether this regular expression is case sensitive.
- `dot_all` - Whether "." in this regular expression matches line terminators.
- `multiline` - Whether this regular expression matches multiple lines.
- `regex_string` - A regular expression pattern for the filter.
- `replacement_string` - A string used to replace banned/denied patterns.
- `unicode` - Whether this regular expression uses Unicode mode.

## Properties

### `buildallowclass-attributeinstance-attribute`

```
allow: bool = True
```

A boolean value indicating whether to allow or deny/block the matched patterns.

### `buildcase_sensitiveclass-attributeinstance-attribute`

```
case_sensitive: bool = True
```

Whether this regular expression is case sensitive.

If the regular expression is not case sensitive, it will match an input letter with a pattern letter even if the two letters are different case versions of the same letter.

### `bulldot_allclass-attributeinstance-attribute`

```
dot_all: bool = False
```

Whether "." in this regular expression matches line terminators.

When false, the "." character matches a single character, unless that character terminates a line. When true, then the "." character will match any single character including line terminators.

This feature is distinct from multiline. They affect the behavior of different pattern characters, so they can be used together or separately.

### `buildmultilineclass-attributeinstance-attribute`

```
multiline: bool = False
```

Whether this regular expression matches multiple lines.

If the regexp does match multiple lines, the "^" and "\$" characters match the beginning and end of lines. If not, the characters match the beginning and end of the input.

### `buildregex_stringinstance-attribute`

```
regex_string: str
```

A regular expression pattern for the filter.

It is recommended to use raw strings (prefix your string with r) for the pattern, ex: r"pattern".

### `buildreplacement_stringclass-attributeinstance-attribute`

```
replacement_string: str = ''
```

A string used to replace banned/denied patterns.

Defaults to an empty string.

### `buildunicodeclass-attributeinstance-attribute`

```
unicode: bool = False
```

Whether this regular expression uses Unicode mode.

## Predefined filters

### NumbersOnlyInputFilter

Allows only numbers.

Inherits: InputFilter

### TextOnlyInputFilter

Allows only text.

Inherits: InputFilter

## Usage example

```
ft.CupertinoTextField(
 placeholder_text="Only numbers are allowed",
 input_filter=ft.InputFilter(allow=True, regex_string=r"^[0-9]*$", replacement_string="")
)

ft.TextField(
 label="Only letters are allowed",
 input_filter=ft.TextOnlyInputFilter()
)
```



# IosDeviceInfo

Device information snapshot for iOS/iPadOS runtimes.

Returned by `Page.get_device_info` on iOS.

Inherits: `DeviceInfo`

## Properties

- `available_ram_size` - Current unallocated RAM size of the device in megabytes.
- `free_disk_size` - Free disk size in bytes.
- `identifier_for_vendor` - Unique UUID value identifying the current device.
- `is_ios_app_on_mac` - Indicates whether the process is an iPhone or iPad app running on a Mac.
- `is_physical_device` - False if the application is running in a simulator, True otherwise.
- `localized_model` - Localized name of the device model.
- `model` - Device model according to OS.
- `model_name` - Commercial or user-known model name.
- `name` - The device name.
- `physical_ram_size` - Total physical RAM size of the device in megabytes.
- `system_name` - The name of the current operating system.
- `system_version` - The current operating system version.
- `total_disk_size` - Total disk size in bytes.
- `utsname` - Operating system information derived from `sys/utsname.h`.

## Properties

### `buildavailable_ram_sizeinstance-attribute`

`available_ram_size: int`

Current unallocated RAM size of the device in megabytes.

### `buildfree_disk_sizeinstance-attribute`

`free_disk_size: int`

Free disk size in bytes.

### `buildidentifier_for_vendorclass-attributeinstance-attribute`

`identifier_for_vendor: str | None = None`

Unique UUID value identifying the current device.

More info: <https://developer.apple.com/documentation/uikit/uidevice/1620059-identifierforvendor>

### `buildis_ios_app_on_macinstance-attribute`

`is_ios_app_on_mac: bool`

Indicates whether the process is an iPhone or iPad app running on a Mac.

More info: <https://developer.apple.com/documentation/foundation/nsprocessinfo/3608556-iosapponmac>

### `buildis_physical_deviceinstance-attribute`

`is_physical_device: bool`

False if the application is running in a simulator, True otherwise.

### `buildlocalized_modelinstance-attribute`

`localized_model: str`

Localized name of the device model.

More info: <https://developer.apple.com/documentation/uikit/uidevice/1620029-localizedmodel>

### **buildmodelinstance-attribute**

`model: str`

Device model according to OS.

More info: <https://developer.apple.com/documentation/uikit/uidevice/1620044-model>

### **buildmodel\_nameinstance-attribute**

`model_name: str`

Commercial or user-known model name.

For example: "iPhone 16 Pro", "iPad Pro 11-Inch 3"

### **buildnameinstance-attribute**

`name: str`

The device name.

Note

- On iOS < 16 returns user-assigned device name.
- On iOS >= 16 returns a generic device name if project has no entitlement to get user-assigned device name.

More info: <https://developer.apple.com/documentation/uikit/uidevice/1620015-name>

### **buildphysical\_ram\_sizeinstance-attribute**

`physical_ram_size: int`

Total physical RAM size of the device in megabytes.

### **buildsystem\_nameinstance-attribute**

`system_name: str`

The name of the current operating system.

More info: <https://developer.apple.com/documentation/uikit/uidevice/1620054-systemname>

### **buildsystem\_versioninstance-attribute**

`system_version: str`

The current operating system version.

More info: <https://developer.apple.com/documentation/uikit/uidevice/1620043-systemversion>

### **buildtotal\_disk\_sizeinstance-attribute**

`total_disk_size: int`

Total disk size in bytes.

### **buildutsnameinstance-attribute**

`utsname: IosUtsname`

Operating system information derived from sys/utsname.h.

# IosUtsname

Information derived from utsname.

More info: <http://pubs.opengroup.org/onlinepubs/7908799/xsh/sysutsname.h.html>

## Properties

- machine - Hardware type (e.g.
- node\_name - Network node name.
- release - Release level.
- sys\_name - Operating system name.
- version - Version level.

## Properties

### buildmachineinstance-attribute

machine: str

Hardware type (e.g. "iPhone7,1" for iPhone 6 Plus).

### buildnode\_nameinstance-attribute

node\_name: str

Network node name.

### buildreleaseinstance-attribute

release: str

Release level.

### buildsys\_nameinstance-attribute

sys\_name: str

Operating system name.

### buildversioninstance-attribute

version: str

Version level.

## is\_route\_active

Check whether the given path matches the current location.

Useful for highlighting active navigation links or tabs.

Parameters:

- path (str) - Path to check against the current location.
- exact (bool, default: False) - If True, requires an exact match. If False (default), a prefix match is used (e.g. "/products" matches "/products/42").

Returns:

- bool - True if the path matches the current location.

# Key

Base class for control keys.

Concrete subclasses define `_type` and therefore behavior on the Flutter side. Use `ValueKey` for general control identity and `ScrollKey` for scroll-target lookups.

## Properties

- `value` - Stable primitive identifier used to match a control.

## Properties

### `buildvalueinstance-attribute`

`value: str | int | float | bool`

Stable primitive identifier used to match a control.

# KeyboardEvent

Event payload for keyboard key-down notifications.

Delivered to `Page.on_keyboard_event`.

Inherits: `Event`

## Properties

- `alt` - Whether Alt was pressed when the key event was emitted.
- `ctrl` - Whether Control was pressed when the key event was emitted.
- `key` - Human-readable key label for the pressed key.
- `meta` - Whether Meta (Command/Windows) was pressed when the key event was emitted.
- `shift` - Whether Shift was pressed when the key event was emitted.

## Properties

### `buildaltinstance-attribute`

`alt: bool`

Whether Alt was pressed when the key event was emitted.

### `buildctrlinstance-attribute`

`ctrl: bool`

Whether Control was pressed when the key event was emitted.

### `buildkeyinstance-attribute`

`key: str`

Human-readable key label for the pressed key.

### `buildmetainstance-attribute`

`meta: bool`

Whether Meta (Command/Windows) was pressed when the key event was emitted.

### `buildshiftinstance-attribute`

`shift: bool`

Whether Shift was pressed when the key event was emitted.

# KeyboardType

The type of information for which to optimize the text input control.

On Android, behavior may vary across device and keyboard provider.

Inherits: `enum.Enum`

## Properties

- `DATETIME` - Optimized for date and time information.
- `EMAIL` - Optimized for email addresses.
- `MULTILINE` - Optimized for multiline textual information.
- `NAME` - Optimized for a person's name.
- `NONE` - Prevents the OS from showing the on-screen virtual keyboard.
- `NUMBER` - Optimized for unsigned numerical information without a decimal point.
- `PHONE` - Optimized for telephone numbers.
- `STREET_ADDRESS` - Optimized for postal mailing addresses.
- `TEXT` - Optimized for textual information.
- `TWITTER` - Optimized for social media.
- `URL` - Optimized for URLs.
- `VISIBLE_PASSWORD` - Optimized for passwords that are visible to the user.
- `WEB_SEARCH` - Optimized for web searches.

## Properties

### `buildDATETIMEclass-attributeinstance-attribute`

Optimized for date and time information.

- On iOS, requests the default keyboard.
- On Android, requests a keyboard with ready access to the number keys, ":", and "-".

### `buildEMAILclass-attributeinstance-attribute`

Optimized for email addresses.

Requests a keyboard with ready access to the "@" and "." keys.

### `buildMULTILINEclass-attributeinstance-attribute`

Optimized for multiline textual information.

Requests the default platform keyboard, but accepts newlines when the enter key is pressed. This is the input type used for all multiline text fields.

### `buildNAMEclass-attributeinstance-attribute`

Optimized for a person's name.

- On iOS, requests the `UIKeyboardType.namePhonePad` keyboard, a keyboard optimized for entering a person's name or phone number. Does not support auto-capitalization.
- On Android, requests a keyboard optimized for `TYPE_TEXT_VARIATION_PERSON_NAME`.

### `buildNONEclass-attributeinstance-attribute`

Prevents the OS from showing the on-screen virtual keyboard.

### `buildNUMBERclass-attributeinstance-attribute`

Optimized for unsigned numerical information without a decimal point.

Requests a default keyboard with ready access to the number keys.

### **buildPHONEclass-attributeinstance-attribute**

Optimized for telephone numbers.

Requests a keyboard with ready access to the number keys, "\*", and "#".

### **buildSTREET\_ADDRESSclass-attributeinstance-attribute**

Optimized for postal mailing addresses.

- On iOS, requests the default keyboard.
- On Android, requests a keyboard optimized for TYPE\_TEXT\_VARIATION\_POSTAL\_ADDRESS.

### **buildTEXTclass-attributeinstance-attribute**

Optimized for textual information.

Requests the default platform keyboard.

### **buildTWITTERclass-attributeinstance-attribute**

Optimized for social media.

Requests a keyboard that includes keys useful for handles and tags.

- On iOS, requests a default keyboard with ready access to the "@" and "#" keys.
- On Android this is remapped to the EMAIL keyboard type as it always shows the "@" key.

### **buildURLclass-attributeinstance-attribute**

Optimized for URLs.

Requests a keyboard with ready access to the "/" and "." keys.

### **buildVISIBLE\_PASSWORDclass-attributeinstance-attribute**

Optimized for passwords that are visible to the user.

Requests a keyboard with ready access to both letters and numbers.

### **buildWEB\_SEARCHclass-attributeinstance-attribute**

Optimized for web searches.

Requests a keyboard that includes keys useful for web searches as well as URLs.

- On iOS, requests a default keyboard with ready access to the "." key. In contrast to URL, a space bar is available.
- On Android this is remapped to the URL keyboard type as it always shows a space bar.



# KeyDownEvent

Event triggered when a key is pressed down.

Typically used to detect the initial press of a key before it is released or repeated.

Inherits: Event

## Properties

- `key` - The key that was pressed down.

## Properties

### buildkeyinstance-attribute

`key: str`

The key that was pressed down.

Represents the physical key (e.g., A, Enter, Shift) that triggered the key down event.

# KeyRepeatEvent

Event triggered when a key is held down and repeating.

This event fires continuously while the key remains pressed, depending on the system's key repeat rate.

Inherits: Event

## Properties

- `key` - The key that is being held down and repeating.

## Properties

### buildkeyinstance-attribute

`key: str`

The key that is being held down and repeating.

Represents the physical key that is generating repeat events (e.g., ArrowDown, Backspace).

# KeyUpEvent

Event triggered when a key is released.

Useful for tracking when a key is no longer being pressed after a key down or repeat event.

Inherits: Event

## Properties

- `key` - The key that was released.

## Properties

### buildkeyinstance-attribute

`key: str`

The key that was released.

Indicates which key was previously pressed and has now been lifted.

# LabelPosition

Position of label in a Checkbox, Radio or Switch

Inherits: enum.Enum

## Properties

- LEFT - The label is positioned to the left of the control.
- RIGHT - The label is positioned to the right of the control.

## Examples

### Label Position showcase

[play\\_arrow](#)Try Online

```
import flet as ft

def showcase_card(label_position: ft.LabelPosition) -> ft.Container:
 return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=10,
 controls=[
 ft.Text(label_position.name, weight=ft.FontWeight.BOLD),
 ft.Checkbox(
 label="Checkbox label",
 value=True,
 label_position=label_position,
),
 ft.RadioGroup(
 content=ft.Row(
 controls=[
 ft.Radio(
 label=f"{i}",
 value=f"{i}",
 label_position=label_position,
)
 for i in range(1, 4)
],
),
),
 ft.Switch(
 label="Switch label",
 value=True,
 label_position=label_position,
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="LabelPosition Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare left/right label placement for form controls."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(label_position)
 for label_position in ft.LabelPosition
],
),
],
),
),
)
```

```
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildLEFT**class-attributeinstance-attribute

The label is positioned to the left of the control.

**buildRIGHT**class-attributeinstance-attribute

The label is positioned to the right of the control.

# LaunchMode

Preferred launch mode for opening a URL.

Inherits: `enum.Enum`

## Properties

- `EXTERNAL_APPLICATION` - Pass the URL to another application to handle.
- `EXTERNAL_NON_BROWSER_APPLICATION` - Pass the URL to a non-browser application to handle.
- `IN_APP_BROWSER_VIEW` - Load the URL inside an in-app browser view (e.g., custom tabs).
- `IN_APP_WEB_VIEW` - Load the URL inside an in-app web view.
- `PLATFORM_DEFAULT` - Platform decides how to open the URL.

## Examples

### Launch Mode showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 launcher = ft.UrlLauncher()

 def showcase_card(mode: ft.LaunchMode) -> ft.Container:
 status = ft.Text("Click to check support on this platform.", size=11)

 async def check_support():
 try:
 supported = await launcher.supports_launch_mode(mode)
 closable = await launcher.supports_close_for_launch_mode(mode)
 status.value = (
 f"supports_launch_mode={supported} / supports_close={closable}"
)
 except Exception as ex:
 status.value = f"Error: {ex}"
 status.update()

 return ft.Container(
 width=360,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(mode.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Check support",
 icon=ft.Icons.CHECK_CIRCLE_OUTLINE,
 on_click=lambda: page.run_task(check_support),
),
 status,
],
),
)

 page.appbar = ft.AppBar(title="LaunchMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Check launch-mode support reported by the current platform."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(mode) for mode in ft.LaunchMode],
),
],
),
),
)
```

```

),
],
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

**buildEXTERNAL\_APPLICATION**class-attributeinstance-attribute

Pass the URL to another application to handle.

**buildEXTERNAL\_NON\_BROWSER\_APPLICATION**class-attributeinstance-attribute

Pass the URL to a non-browser application to handle.

**buildIN\_APP\_BROWSER\_VIEW**class-attributeinstance-attribute

Load the URL inside an in-app browser view (e.g., custom tabs).

**buildIN\_APP\_WEB\_VIEW**class-attributeinstance-attribute

Load the URL inside an in-app web view.

**buildPLATFORM\_DEFAULT**class-attributeinstance-attribute

Platform decides how to open the URL.

# LinearGradient

Creates a linear gradient from begin to end.

More information on [here](#).

Inherits: Gradient

## Properties

- `begin` - The offset at which stop 0.0 of the gradient is placed.
- `end` - The offset at which stop 1.0 of the gradient is placed.

## Properties

### `buildbeginclass-attributeinstance-attribute`

```
begin: Alignment = field(
 default_factory=lambda: Alignment.CENTER_LEFT
)
```

The offset at which stop 0.0 of the gradient is placed.

### `buildendclass-attributeinstance-attribute`

```
end: Alignment = field(
 default_factory=lambda: Alignment.CENTER_RIGHT
)
```

The offset at which stop 1.0 of the gradient is placed.

## Examples

### Container with linear gradient

[play\\_arrow](#)Try Online

```
import math

import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Container(
 alignment=ft.Alignment.CENTER,
 width=150,
 height=150,
 border_radius=ft.BorderRadius.all(5),
 gradient=ft.LinearGradient(
 begin=ft.Alignment.TOP_LEFT,
 end=ft.Alignment(0.8, 1),
 tile_mode=ft.GradientTileMode.MIRROR,
 rotation=math.pi / 3,
 colors=[
 "0xff1f005c",
 "0xff5b0060",
 "0xff870160",
 "0xffac255e",
 "0xffca485c",
 "0xffe16b5c",
 "0xffff39060",
 "0xfffffb56b",
],
),
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: container - [https://flet.dev/docs/examples/controls/core/types/gradient/linear\\_gradient/media/container.png](https://flet.dev/docs/examples/controls/core/types/gradient/linear_gradient/media/container.png)



# LinuxDeviceInfo

Device information for a Linux system.

More info:

- <https://www.freedesktop.org/software/systemd/man/os-release.html>
- <https://www.freedesktop.org/software/systemd/man/machine-id.html>

Inherits: DeviceInfo

## Properties

- **build\_id** - A string uniquely identifying the system image used as the origin for a distribution (it is not updated with system updates).
- **id** - A lower-case string identifying the operating system, excluding any version information and suitable for processing by scripts or usage in generated filenames.
- **id\_like** - A space-separated list of operating system identifiers in the same syntax as the id value.
- **machine\_id** - A unique machine ID of the local system that is set during installation or boot.
- **name** - A string identifying the operating system, without a version component, and suitable for presentation to the user.
- **pretty\_name** - A pretty operating system name in a format suitable for presentation to the user.
- **variant** - A string identifying a specific variant or edition of the operating system suitable for presentation to the user.
- **variant\_id** - A lower-case string identifying a specific variant or edition of the operating system.
- **version** - A string identifying the operating system version, excluding any OS name information, possibly including a release code name, and suitable for presentation to the user.
- **version\_code\_name** - A lower-case string identifying the operating system release code name, excluding any OS name information or release version, and suitable for processing by scripts or usage in generated filenames.
- **version\_id** - A lower-case string identifying the operating system version, excluding any OS name information or release code name, and suitable for processing by scripts or usage in generated filenames.

## Properties

### build\_idclass-attributeinstance-attribute

`build_id: str | None = None`

A string uniquely identifying the system image used as the origin for a distribution (it is not updated with system updates). The field can be identical between different version\_id values as build\_id is only a unique identifier to a specific version.

Examples: "2013-03-20.3", "201303203".

May be None on some systems.

### build\_idinstance-attribute

`id: str`

A lower-case string identifying the operating system, excluding any version information and suitable for processing by scripts or usage in generated filenames.

The ID contains no spaces or other characters outside of 0–9, a–z, '.', '\_' and '-'.

Examples: "fedora", "debian".

If not set, defaults to "linux".

### build\_id\_likeclass-attributeinstance-attribute

`id_like: list[str] | None = None`

A space-separated list of operating system identifiers in the same syntax as the id value. It lists identifiers of operating systems that are closely related to the local operating system in regards to packaging and programming interfaces, for example listing one or more OS identifiers the local OS is a derivative from.

Examples: an operating system with id "centos", would list "rhel" and "fedora", and an operating system with id "ubuntu" would list "debian".

May be None on some systems.

### **buildmachine\_idclass-attributeinstance-attribute**

machine\_id: str | None = None

A unique machine ID of the local system that is set during installation or boot. The machine ID is hexadecimal, 32-character, lowercase ID. When decoded from hexadecimal, this corresponds to a 16-byte/128-bit value.

### **buildnameinstance-attribute**

name: str

A string identifying the operating system, without a version component, and suitable for presentation to the user.

Examples: "Fedora", "Debian GNU/Linux".

If not set, defaults to "Linux".

### **buildpretty\_nameinstance-attribute**

pretty\_name: str

A pretty operating system name in a format suitable for presentation to the user. May or may not contain a release code name or OS version of some kind, as suitable.

Examples: "Fedora 17 (Beefy Miracle)".

If not set, defaults to "Linux".

### **buildvariantclass-attributeinstance-attribute**

variant: str | None = None

A string identifying a specific variant or edition of the operating system suitable for presentation to the user. This field may be used to inform the user that the configuration of this system is subject to a specific divergent set of rules or default configuration settings.

Examples: "Server Edition", "Smart Refrigerator Edition".

Note: this field is for display purposes only. The variant\_id field should be used for making programmatic decisions.

May be None on some systems.

### **buildvariant\_idclass-attributeinstance-attribute**

variant\_id: str | None = None

A lower-case string identifying a specific variant or edition of the operating system. This may be interpreted in order to determine a divergent default configuration.

The variant ID contains no spaces or other characters outside of 0–9, a–z, '.', '\_' and '-'.

Examples: "server", "embedded".

May be None on some systems.

### **buildversionclass-attributeinstance-attribute**

version: str | None = None

A string identifying the operating system version, excluding any OS name information, possibly including a release code name, and suitable for presentation to the user.

Examples: "17", "17 (Beefy Miracle)".

May be None on some systems.

### **buildversion\_code\_nameclass-attributeinstance-attribute**

version\_code\_name: str | None = None

A lower-case string identifying the operating system release code name, excluding any OS name information or release version, and suitable for processing by scripts or usage in generated filenames.

The codename contains no spaces or other characters outside of 0–9, a–z, '.', '\_' and '-'.

Examples: "buster", "xenial".

May be None on some systems.

## buildversion\_idclass-attributeinstance-attribute

version\_id: str | None = None

A lower-case string identifying the operating system version, excluding any OS name information or release code name, and suitable for processing by scripts or usage in generated filenames.

The version is mostly numeric, and contains no spaces or other characters outside of 0–9, a–z, '.', '\_' and '-'.

Examples: "17", "11.04".

May be None on some systems.

# ListTileStyle

Defines the title font used for ListTile descendants of a ListTileTheme.

List tiles that appear in a Drawer use the theme's TextTheme.body\_large text style, which is a little smaller than the theme's TextTheme.title\_medium text style, which is used by default.

Inherits: enum.Enum

## Properties

- DRAWER
- LIST

## Examples

### List Tile Style showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(style: ft.ListTileStyle) -> ft.Container:
 return ft.Container(
 width=360,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(style.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.ListTile(
 style=style,
 leading=ft.Icon(ft.Icons.PERSON),
 title=ft.Text("Jane Doe"),
 subtitle=ft.Text("Product Manager"),
 trailing=ft.Icon(ft.Icons.CHEVRON_RIGHT),
),
],
),
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="ListTileStyle Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare list tile typography presets."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(style) for style in ft.ListTileStyle],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

buildDRAWERclass-attributeinstance-attribute

buildLISTclass-attributeinstance-attribute

# ListTileTheme

Customizes the appearance of descendant ListTile controls.

## Properties

- `affinity` - Overrides the default value for `ExpansionTile.affinity`.
- `bgcolor` - Overrides the default value for `ListTile.bgcolor`.
- `content_padding` - Overrides the default value for `ListTile.content_padding`.
- `dense` - Overrides the default value for `ListTile.dense`.
- `enable_feedback` - Overrides the default value for `ListTile.enable_feedback`.
- `horizontal_spacing` - Overrides the default value for `ListTile.horizontal_spacing`.
- `icon_color` - Overrides the default value for `ListTile.icon_color`.
- `is_three_line` - Overrides the default value for `ListTile.is_three_line`.
- `leading_and_trailing_text_style` - Overrides the default value for `ListTile.leading_and_trailing_text_style`.
- `min_height` - Overrides the default value for `ListTile.min_height`.
- `min_leading_width` - Overrides the default value for `ListTile.min_leading_width`.
- `min_vertical_padding` - Overrides the default value for `ListTile.min_vertical_padding`.
- `mouse_cursor` - Overrides the default value for `ListTile.mouse_cursor`.
- `selected_color` - Overrides the default value for `ListTile.selected_color`.
- `selected_tile_color` - Overrides the default value for `ListTile.selected_tile_color`.
- `shape` - Overrides the default value for `ListTile.shape`.
- `style` - Overrides the default value for `ListTile.style`.
- `subtitle_text_style` - Overrides the default value for `ListTile.subtitle_text_style`.
- `text_color` - Overrides the default value for `ListTile.text_color`.
- `title_alignment` - Overrides the default value for `ListTile.title_alignment`.
- `title_text_style` - Overrides the default value for `ListTile.title_text_style`.
- `visual_density` - Overrides the default value for `ListTile.visual_density`.

## Properties

### `buildaffinityclass-attributeinstance-attribute`

`affinity: TileAffinity | None = None`

Overrides the default value for `ExpansionTile.affinity`.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value for `ListTile.bgcolor`.

### `buildcontent_paddingclass-attributeinstance-attribute`

`content_padding: PaddingValue | None = None`

Overrides the default value for `ListTile.content_padding`.

### `builddenseclass-attributeinstance-attribute`

`dense: bool | None = None`

Overrides the default value for `ListTile.dense`.

### `buildenable_feedbackclass-attributeinstance-attribute`

`enable_feedback: bool | None = None`

Overrides the default value for `ListTile.enable_feedback`.

### **buildhorizontal\_spacingclass-attributeinstance-attribute**

`horizontal_spacing: Number | None = None`

Overrides the default value for `ListTile.horizontal_spacing`.

### **buildicon\_colorclass-attributeinstance-attribute**

`icon_color: ColorValue | None = None`

Overrides the default value for `ListTile.icon_color`.

### **buildis\_three\_lineclass-attributeinstance-attribute**

`is_three_line: bool | None = None`

Overrides the default value for `ListTile.is_three_line`.

### **buildleading\_and\_trailing\_text\_styleclass-attributeinstance-attribute**

`leading_and_trailing_text_style: TextStyle | None = None`

Overrides the default value for `ListTile.leading_and_trailing_text_style`.

### **buildmin\_heightclass-attributeinstance-attribute**

`min_height: Number | None = None`

Overrides the default value for `ListTile.min_height`.

### **buildmin\_leading\_widthclass-attributeinstance-attribute**

`min_leading_width: Number | None = None`

Overrides the default value for `ListTile.min_leading_width`.

### **buildmin\_vertical\_paddingclass-attributeinstance-attribute**

`min_vertical_padding: PaddingValue | None = None`

Overrides the default value for `ListTile.min_vertical_padding`.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

Overrides the default value for `ListTile.mouse_cursor`.

### **buildselected\_colorclass-attributeinstance-attribute**

`selected_color: ColorValue | None = None`

Overrides the default value for `ListTile.selected_color`.

### **buildselected\_tile\_colorclass-attributeinstance-attribute**

`selected_tile_color: ColorValue | None = None`

Overrides the default value for `ListTile.selected_tile_color`.

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

Overrides the default value for `ListTile.shape`.

### **buildstyleclass-attributeinstance-attribute**

`style: ListTileStyle | None = None`

Overrides the default value for `ListTile.style`.

### **buildsubtitle\_text\_styleclass-attributeinstance-attribute**

`subtitle_text_style: TextStyle | None = None`

Overrides the default value for `ListTile.subtitle_text_style`.

### **buildtext\_colorclass-attributeinstance-attribute**

`text_color: ColorValue | None = None`

Overrides the default value for `ListTile.text_color`.

### **buildtitle\_alignmentclass-attributeinstance-attribute**

title\_alignment: ListTileTitleAlignment | None = None

Overrides the default value for ListTile.title\_alignment.

### **buildtitle\_text\_styleclass-attributeinstance-attribute**

title\_text\_style: TextStyle | None = None

Overrides the default value for ListTile.title\_text\_style.

### **buildvisual\_densityclass-attributeinstance-attribute**

visual\_density: VisualDensity | None = None

Overrides the default value for ListTile.visual\_density.



# ListTileTitleAlignment

Defines how ListTile aligns leading and trailing relative to the tile's title area.

The alignment is computed against the text block formed by title and subtitle. Use this to tune the visual balance between icon/avatar controls and text, especially when tiles switch between one-line, two-line, and three-line layouts.

Inherits: `enum.Enum`

## Properties

- **BOTTOM** - Aligns leading and trailing toward the bottom of the title area.
- **CENTER** - Centers leading and trailing relative to the title/subtitle block.
- **THREE\_LINE** - Uses alignment behavior optimized for three-line list tile layouts.
- **TITLE\_HEIGHT** - Uses alignment behavior based on title-height rules from legacy list tile layouts.
- **TOP** - Aligns leading and trailing toward the top of the title area.

## Examples

### List Tile Title Alignment showcase

[play\\_arrow](#) Try Online

```
import flet as ft

def showcase_card(alignment: ft.ListTileTitleAlignment) -> ft.Container:
 return ft.Container(
 width=380,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(alignment.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.ListTile(
 title_alignment=alignment,
 leading=ft.CircleAvatar(content=ft.Text("JD")),
 title=ft.Text("Jane Doe"),
 subtitle=ft.Text(
 "This subtitle helps visualize vertical alignment."
),
 is_three_line=True,
 trailing=ft.Icon(ft.Icons.MORE_VERT),
),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="ListTileTitleAlignment Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare leading/trailing alignment against title area."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(alignment)
 for alignment in ft.ListTileTitleAlignment
],
),
],
),
),
)
```

```

),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### **buildBOTTOM**class-attributeinstance-attribute

Aligns leading and trailing toward the bottom of the title area.

Bottom placement respects ListTile.min\_vertical\_padding.

### **buildCENTER**class-attributeinstance-attribute

Centers leading and trailing relative to the title/subtitle block.

### **buildTHREE\_LINE**class-attributeinstance-attribute

Uses alignment behavior optimized for three-line list tile layouts.

This is the default alignment style in Material 3.

### **buildTITLE\_HEIGHT**class-attributeinstance-attribute

Uses alignment behavior based on title-height rules from legacy list tile layouts.

This is the default alignment style in Material 2.

### **buildTOP**class-attributeinstance-attribute

Aligns leading and trailing toward the top of the title area.

Top placement respects ListTile.min\_vertical\_padding.

# Locale

An identifier used to select a user's language and formatting preferences.

## Properties

- `country_code` - The country code or region subtag of the locale, e.g US for the United States.
- `language_code` - The primary language subtag/code of the locale, e.g en for English.
- `language_tag` - Returns a syntactically valid Unicode BCP47 Locale Identifier.
- `script_code` - The script code/subtag of the locale, e.g.

## Properties

### `buildcountry_code`class-attributeinstance-attribute

`country_code: str | None = None`

The country code or region subtag of the locale, e.g US for the United States.

Its value is case-sensitive and must be a registered subtag in the IANA Language Subtag Registry with the type "region". If you use a deprecated subtag, it will be internally modified to its "Preferred-Value", if it has one. For example: `Locale("de", "DE")` and `Locale("de", "DD")` are equal, and both have the `country_code="DE"`, because DD is a deprecated region subtag which has been replaced by DE.

Locales may also be defined without this, to specify a generic fallback for a particular script.

### `buildlanguage_code`class-attributeinstance-attribute

`language_code: str = 'und'`

The primary language subtag/code of the locale, e.g en for English.

Its value is case-sensitive and must be a registered subtag in the IANA Language Subtag Registry with the type "language". If you use a deprecated subtag, it will be internally modified to its "Preferred-Value", if it has one. For example: `Locale("he")` and `Locale("iw")` are equal, and both have the `language_code="he"`, because iw is a deprecated language subtag which has been replaced by he.

When there is no language subtag, this parameter should be set to "und" (the default), which represents an undefined language code.

### `buildlanguage_tag`property

`language_tag: str`

Returns a syntactically valid Unicode BCP47 Locale Identifier.

See this for technical details.

Examples: en, es-419, hi-Deva-IN, zh-Hans-CN

### `buildscript_code`class-attributeinstance-attribute

`script_code: str | None = None`

The script code/subtag of the locale, e.g. Hant for Traditional Chinese or Hans for Simplified Chinese. It is especially recommended to set this property explicitly for languages with more than one script.

Its value must be a valid Unicode Language Identifier script subtag as listed in Unicode CLDR supplemental data.

# LocaleChangeEvent

Event payload describing a change in the host platform's locale preferences.

See also:

- `Page.on_locale_change`: event called when locale preferences/settings of the host platform have changed.

Inherits: `Event`

## Properties

- `locales` - The full, ordered list of locales reported by the host platform.

## Properties

### `buildlocalesinstance-attribute`

`locales: list[Locale]`

The full, ordered list of locales reported by the host platform.

The first item represents the highest-priority locale.

# LocaleConfiguration

Represents the configuration for supported locales and the current locale.

## Properties

- `current_locale` - The current locale.
- `supported_locales` - A list of locales that the application supports for localization.

## Properties

### `buildcurrent_localeclass-attributeinstance-attribute`

```
current_locale: Locale | None = None
```

The current locale.

Note

- Must be an item of `supported_locales` to take effect.
- If `None` or invalid/unsupported, the first supported locale in `supported_locales` is used.

### `buildsupported_localesclass-attributeinstance-attribute`

```
supported_locales: list[Locale] = field(
 default_factory=lambda: [Locale("en", "US")]
)
```

A list of locales that the application supports for localization.

Note

- Locales unsupported/invalid are ignored.

Order matters: Locale resolution is performed by progressively relaxing specificity until a match is found. Matching is attempted in the following priority order:

- language + script + country
- language + script
- language + country
- language only
- country only (only if no language-based match is found)

When multiple supported locales match at the same priority level, the first matching locale in `supported_locales` is selected. If no supported locale matches at any level, the first entry in `supported_locales` is used as the final fallback.

- For languages with multiple scripts or regional variants, include locales in increasing order of specificity: language-only → language+script → language+script+country. Defining all of these variants is not strictly required, but doing so greatly improves locale resolution and reduces ambiguous matches. Omitting intermediate fallbacks can lead to incorrect script or regional selection in edge cases—for example, a Simplified Chinese user in Taiwan (`zh_Hans_TW`) may incorrectly resolve to Traditional Chinese if `zh_Hans` and `zh_Hans_CN` are not present.

Example: Full Chinese support for CN (Mainland China), TW (Taiwan), and HK (Hong Kong)

```
supported_locales = [
 # Generic Chinese (fallback for all Chinese variants)
 Locale(language_code="zh"),

 # Script-level variants
 Locale(language_code="zh", script_code="Hans"), # Simplified Chinese zh_Hans
 Locale(language_code="zh", script_code="Hant"), # Traditional Chinese zh_Hant

 # Region-specific variants
 Locale(language_code="zh", script_code="Hans", country_code="CN"), # zh_Hans_CN
 Locale(language_code="zh", script_code="Hant", country_code="TW"), # zh_Hant_TW
 Locale(language_code="zh", script_code="Hant", country_code="HK"), # zh_Hant_HK
]
```

# LocationInfo

Describes the current location parsed from the page route.

Parameters:

- pathname (str) - URL path portion (e.g. /products/42).
- search (str, default: "") - Query string portion without leading ?.
- hash (str, default: "") - Fragment portion without leading #.

## Properties

- hash
- pathname
- search

## Properties

buildhashclass-attributeinstance-attribute

```
hash: str = ''
```

buildpathnameinstance-attribute

```
pathname: str
```

buildsearchclass-attributeinstance-attribute

```
search: str = ''
```

# LoginEvent

Event payload for OAuth login completion.

Emitted to `Page.on_login` for both successful and failed authorization attempts.

Inherits: `Event`

## Properties

- `error` - Error code or message when login failed; empty/None on success.
- `error_description` - Provider-specific error details when login failed.

## Properties

### `builderrorinstance-attribute`

`error: str | None`

Error code or message when login failed; empty/None on success.

### `builderror_descriptioninstance-attribute`

`error_description: str | None`

Provider-specific error details when login failed.

# LongPressEndEvent

End payload for a completed long-press gesture.

Inherits: Event

## Properties

- `global_position` - The global position at which the pointer lifted from the screen.
- `local_position` - The local position at which the pointer contacted the screen.
- `velocity` - The pointer's velocity when it stopped contacting the screen, in pixels per second.

## Properties

### `buildglobal_positionclass-attributeinstance-attribute`

```
global_position: Offset | None = field(
 default=None, metadata={"data_field": "g"}
)
```

The global position at which the pointer lifted from the screen.

### `buildlocal_positionclass-attributeinstance-attribute`

```
local_position: Offset | None = field(
 default=None, metadata={"data_field": "l"}
)
```

The local position at which the pointer contacted the screen.

### `buildvelocityclass-attributeinstance-attribute`

```
velocity: Offset = field(metadata={'data_field': 'v'})
```

The pointer's velocity when it stopped contacting the screen, in pixels per second.

Defaults to zero if not specified in the constructor.



# LongPressStartEvent

Start payload for an accepted long-press gesture.

Inherits: Event

## Properties

- `global_position` - The global position at which the pointer initially contacted the screen.
- `local_position` - The local position at which the pointer initially contacted the screen.

## Properties

### `buildglobal_position`class-attributeinstance-attribute

```
global_position: Offset | None = field(
 default=None, metadata={"data_field": "g"}
)
```

The global position at which the pointer initially contacted the screen.

### `buildlocal_position`class-attributeinstance-attribute

```
local_position: Offset | None = field(
 default=None, metadata={"data_field": "l"}
)
```

The local position at which the pointer initially contacted the screen.

# MacOsDeviceInfo

Device information snapshot for macOS hosts.

Returned by `Page.get_device_info` when the current platform is macOS. Includes CPU, memory, model, and OS version fields collected from native system APIs.

Inherits: `DeviceInfo`

## Properties

- `active_cpus` - Number of active CPUs.
- `arch` - Machine CPU architecture.
- `computer_name` - Name given to the local machine.
- `cpu_frequency` - Device CPU frequency.
- `host_name` - Operating system type.
- `kernel_version` - Machine kernel version.
- `major_version` - The major release number, such as 10 in version 10.9.3.
- `memory_size` - Machine's memory size.
- `minor_version` - The minor release number, such as 9 in version 10.9.3.
- `model` - Device model identifier.
- `model_name` - Device model name.
- `os_release` - Operating system release number.
- `patch_version` - The update release number, such as 3 in version 10.9.3.
- `system_guid` - Device GUID.

## Properties

### `buildactive_cpusinstance-attribute`

`active_cpus: int`

Number of active CPUs.

### `buildarchinstance-attribute`

`arch: str`

Machine CPU architecture.

Note

Apple Silicon Macs can return "x86\_64" if app runs via Rosetta.

### `buildcomputer_nameinstance-attribute`

`computer_name: str`

Name given to the local machine.

### `buildcpu_frequencyinstance-attribute`

`cpu_frequency: int`

Device CPU frequency.

### `buildhost_nameinstance-attribute`

`host_name: str`

Operating system type.

### `buildkernel_versioninstance-attribute`

`kernel_version: str`

Machine kernel version.

### **buildmajor\_versioninstance-attribute**

major\_version: int

The major release number, such as 10 in version 10.9.3.

### **buildmemory\_sizeinstance-attribute**

memory\_size: int

Machine's memory size.

### **buildminor\_versioninstance-attribute**

minor\_version: int

The minor release number, such as 9 in version 10.9.3.

### **buildmodelinstance-attribute**

model: str

Device model identifier.

For example: "MacBookPro18,3", "Mac16,2".

### **buildmodel\_nameinstance-attribute**

model\_name: str

Device model name.

For example: "MacBook Pro (16-inch, 2021)", "iMac (24-inch, 2024)".

### **buildos\_releaseinstance-attribute**

os\_release: str

Operating system release number.

### **buildpatch\_versioninstance-attribute**

patch\_version: int

The update release number, such as 3 in version 10.9.3.

### **buildsystem\_guidclass-attributeinstance-attribute**

system\_guid: str | None = None

Device GUID.

# MagnetometerReadingEvent

A sensor sample from a magnetometer.

Magnetometers measure the ambient magnetic field surrounding the sensor, returning values in microteslas  $\mu\text{T}$  for each three-dimensional axis.

Consider that these samples may bear effects of Earth's magnetic field as well as local factors such as the metal of the device itself or nearby magnets, though most devices compensate for these factors.

A compass is an example of a general utility for magnetometer data.

Inherits: Event

## Properties

- timestamp - Event timestamp.
- x - Ambient magnetic field on the X axis, in microteslas (uT).
- y - Ambient magnetic field on the Y axis, in uT.
- z - Ambient magnetic field on the Z axis, in uT.

## Properties

### buildtimestampinstance-attribute

timestamp: datetime

Event timestamp.

### buildxinstance-attribute

x: float

Ambient magnetic field on the X axis, in microteslas (uT).

### buildyinstance-attribute

y: float

Ambient magnetic field on the Y axis, in uT.

### buildzinstance-attribute

z: float

Ambient magnetic field on the Z axis, in uT.

# MainAxisAlignment

How the children should be placed along the main axis.

Inherits: `enum.Enum`

## Properties

- **CENTER** - Place the children as close to the middle of the main axis as possible.
- **END** - Place the children as close to the end of the main axis as possible.
- **SPACE\_AROUND** - Place the free space evenly between the children as well as half of that space before and after the first and last child.
- **SPACE\_BETWEEN** - Place the free space evenly between the children.
- **SPACE\_EVENLY** - Place the free space evenly between the children as well as before and after the first and last child.
- **START** - Place the children as close to the start of the main axis as possible.

## Examples

### Main Axis Alignment showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def dot(label: str) -> ft.Container:
 return ft.Container(
 width=36,
 height=36,
 border_radius=18,
 bgcolor=ft.Colors.PRIMARY_CONTAINER,
 alignment=ft.Alignment.CENTER,
 content=ft.Text(label, size=12, color=ft.Colors.ON_PRIMARY_CONTAINER),
)

def showcase_card(alignment: ft.MainAxisAlignment) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(alignment.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=250,
 height=70,
 padding=8,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Row(
 alignment=alignment,
 vertical_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[dot("1"), dot("2"), dot("3")],
),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="MainAxisAlignment Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare horizontal distribution of children in a "
```

```

 "fixed-width row."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(alignment)
 for alignment in ft.MainAxisAlignment
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### **buildCENTER**class-attributeinstance-attribute

Place the children as close to the middle of the main axis as possible.

### **buildEND**class-attributeinstance-attribute

Place the children as close to the end of the main axis as possible.

### **buildSPACE\_AROUND**class-attributeinstance-attribute

Place the free space evenly between the children as well as half of that space before and after the first and last child.

### **buildSPACE\_BETWEEN**class-attributeinstance-attribute

Place the free space evenly between the children.

### **buildSPACE\_EVENLY**class-attributeinstance-attribute

Place the free space evenly between the children as well as before and after the first and last child.

### **buildSTART**class-attributeinstance-attribute

Place the children as close to the start of the main axis as possible.

# Margin

Margin class has the properties to set margins for all sides of the rectangle.

## Properties

- bottom - The margin applied to the bottom.
- left - The margin applied to the left.
- right - The margin applied to the right.
- top - The margin applied to the top.

## Methods

- all - Applies the same margin to all sides.
- only - Applies margin to the specified sides.
- symmetric - Applies vertical margin to top and bottom sides and horizontal margin to left and right sides.

## Properties

### buildbottomclass-attributeinstance-attribute

```
bottom: Number = 0
```

The margin applied to the bottom.

### buildleftclass-attributeinstance-attribute

```
left: Number = 0
```

The margin applied to the left.

### buildrightclass-attributeinstance-attribute

```
right: Number = 0
```

The margin applied to the right.

### buildtopclass-attributeinstance-attribute

```
top: Number = 0
```

The margin applied to the top.

## Methods

### deployed\_codeallclassmethod

```
all(value: Number) -> Margin
```

Applies the same margin to all sides.

### deployed\_codeonlyclassmethod

```
only(
 left: Number = 0,
 top: Number = 0,
 right: Number = 0,
 bottom: Number = 0,
) -> Margin
```

Applies margin to the specified sides.

### deployed\_codesymmetricclassmethod

```
symmetric(
 vertical: Number = 0, horizontal: Number = 0
) -> Margin
```

Applies vertical margin to top and bottom sides and horizontal margin to left and right sides.

## Examples

## Example 1

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Margin Example"

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Row(
 spacing=0,
 controls=[
 ft.Container(
 content=ft.Button("container_1"),
 bgcolor=ft.Colors.AMBER,
 # padding=ft.Padding.all(10),
 margin=ft.Margin.all(10),
 width=200,
 height=200,
),
 ft.Container(
 content=ft.Button("container_2"),
 bgcolor=ft.Colors.AMBER,
 # padding=ft.Padding.all(20),
 margin=ft.Margin.all(20),
 width=200,
 height=200,
),
 ft.Container(
 content=ft.Button("container_3"),
 bgcolor=ft.Colors.AMBER,
 # padding=ft.Padding.symmetric(horizontal=10),
 margin=ft.Margin.symmetric(vertical=10),
 width=200,
 height=200,
),
 ft.Container(
 content=ft.Button("container_4"),
 bgcolor=ft.Colors.AMBER,
 # padding=ft.Padding.only(left=10),
 margin=ft.Margin.only(left=10),
 width=200,
 height=200,
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```



# MarkdownCodeTheme

Built-in code highlighting themes for markdown code blocks.

Use with `Markdown.code_theme` to choose a predefined syntax highlighting palette by name.

Inherits: `enum.Enum`

## Properties

- `A11Y_DARK` - Uses the a11y-dark syntax highlighting theme.
- `A11Y_LIGHT` - Uses the a11y-light syntax highlighting theme.
- `AGATE` - Uses the agate syntax highlighting theme.
- `AN_OLD_HOPE` - Uses the an-old-hope syntax highlighting theme.
- `ANDROID_STUDIO` - Uses the androidstudio syntax highlighting theme.
- `ARDUINO_LIGHT` - Uses the arduino-light syntax highlighting theme.
- `ARTA` - Uses the arta syntax highlighting theme.
- `ASCETIC` - Uses the ascetic syntax highlighting theme.
- `ATELIER_CAVE_DARK` - Uses the atelier-cave-dark syntax highlighting theme.
- `ATELIER_CAVE_LIGHT` - Uses the atelier-cave-light syntax highlighting theme.
- `ATELIER_DUNE_DARK` - Uses the atelier-dune-dark syntax highlighting theme.
- `ATELIER_DUNE_LIGHT` - Uses the atelier-dune-light syntax highlighting theme.
- `ATELIER_ESTUARY_DARK` - Uses the atelier-estuary-dark syntax highlighting theme.
- `ATELIER_ESTUARY_LIGHT` - Uses the atelier-estuary-light syntax highlighting theme.
- `ATELIER_FOREST_DARK` - Uses the atelier-forest-dark syntax highlighting theme.
- `ATELIER_FOREST_LIGHT` - Uses the atelier-forest-light syntax highlighting theme.
- `ATELIER_HEATH_DARK` - Uses the atelier-heath-dark syntax highlighting theme.
- `ATELIER_HEATH_LIGHT` - Uses the atelier-heath-light syntax highlighting theme.
- `ATELIER_LAKESIDE_DARK` - Uses the atelier-lakeside-dark syntax highlighting theme.
- `ATELIER_LAKESIDE_LIGHT` - Uses the atelier-lakeside-light syntax highlighting theme.
- `ATELIER_PLATEAU_DARK` - Uses the atelier-plateau-dark syntax highlighting theme.
- `ATELIER_PLATEAU_LIGHT` - Uses the atelier-plateau-light syntax highlighting theme.
- `ATELIER_SAVANNA_DARK` - Uses the atelier-savanna-dark syntax highlighting theme.
- `ATELIER_SAVANNA_LIGHT` - Uses the atelier-savanna-light syntax highlighting theme.
- `ATELIER_SEASIDE_DARK` - Uses the atelier-seaside-dark syntax highlighting theme.
- `ATELIER_SEASIDE_LIGHT` - Uses the atelier-seaside-light syntax highlighting theme.
- `ATELIER_SULPHURPOOL_DARK` - Uses the atelier-sulphurpool-dark syntax highlighting theme.
- `ATELIER_SULPHURPOOL_LIGHT` - Uses the atelier-sulphurpool-light syntax highlighting theme.
- `ATOM_ONE_DARK` - Uses the atom-one-dark syntax highlighting theme.
- `ATOM_ONE_DARK_REASONABLE` - Uses the atom-one-dark-reasonable syntax highlighting theme.
- `ATOM_ONE_LIGHT` - Uses the atom-one-light syntax highlighting theme.
- `BROWN_PAPER` - Uses the brown-paper syntax highlighting theme.
- `CODEPEN_EMBED` - Uses the codepen-embed syntax highlighting theme.
- `COLOR_BREWER` - Uses the color-brewer syntax highlighting theme.

- DARK - Uses the dark syntax highlighting theme.
- DEFAULT - Uses the default syntax highlighting theme.
- DOCCO - Uses the docco syntax highlighting theme.
- DRACULA - Uses the dracula syntax highlighting theme.
- DRAGULA - Uses the dracula syntax highlighting theme.
- FAR - Uses the far syntax highlighting theme.
- FOUNDATION - Uses the foundation syntax highlighting theme.
- GITHUB - Uses the github syntax highlighting theme.
- GITHUB\_GIST - Uses the github-gist syntax highlighting theme.
- GML - Uses the gml syntax highlighting theme.
- GOOGLE\_CODE - Uses the googlecode syntax highlighting theme.
- GRADIENT\_DARK - Uses the gradient-dark syntax highlighting theme.
- GRAYSCALE - Uses the grayscale syntax highlighting theme.
- GRUVBOX\_DARK - Uses the gruvbox-dark syntax highlighting theme.
- GRUVBOX\_LIGHT - Uses the gruvbox-light syntax highlighting theme.
- HOPSCOTCH - Uses the hopscotch syntax highlighting theme.
- HYBRID - Uses the hybrid syntax highlighting theme.
- IDEA - Uses the idea syntax highlighting theme.
- IR\_BLACK - Uses the ir-black syntax highlighting theme.
- ISBL\_EDITOR\_DARK - Uses the isbl-editor-dark syntax highlighting theme.
- ISBL\_EDITOR\_LIGHT - Uses the isbl-editor-light syntax highlighting theme.
- KIMBIE\_DARK - Uses the kimbie.dark syntax highlighting theme.
- KIMBIE\_LIGHT - Uses the kimbie.light syntax highlighting theme.
- LIGHTFAIR - Uses the lightfair syntax highlighting theme.
- MAGULA - Uses the magula syntax highlighting theme.
- MONO\_BLUE - Uses the mono-blue syntax highlighting theme.
- MONOKAI - Uses the monokai syntax highlighting theme.
- MONOKAI\_SUBLIME - Uses the monokai-sublime syntax highlighting theme.
- NIGHT\_OWL - Uses the night-owl syntax highlighting theme.
- NORD - Uses the nord syntax highlighting theme.
- OBSIDIAN - Uses the obsidian syntax highlighting theme.
- OCEAN - Uses the ocean syntax highlighting theme.
- PARASEO\_DARK - Uses the paraiso-dark syntax highlighting theme.
- PARASEO\_LIGHT - Uses the paraiso-light syntax highlighting theme.
- POJOAQUE - Uses the pojoaque syntax highlighting theme.
- PURE\_BASIC - Uses the purebasic syntax highlighting theme.
- QT\_CREATOR\_DARK - Uses the qtcreator\_dark syntax highlighting theme.
- QT\_CREATOR\_LIGHT - Uses the qtcreator\_light syntax highlighting theme.
- RAILSCASTS - Uses the railscasts syntax highlighting theme.
- RAINBOW - Uses the rainbow syntax highlighting theme.
- ROUTERS - Uses the routers syntax highlighting theme.

- SCHOOL\_BOOK - Uses the school-book syntax highlighting theme.
- SHADES\_OF\_PURPLE - Uses the shades-of-purple syntax highlighting theme.
- SOLARIZED\_DARK - Uses the solarized-dark syntax highlighting theme.
- SOLARIZED\_LIGHT - Uses the solarized-light syntax highlighting theme.
- SUNBURST - Uses the sunburst syntax highlighting theme.
- TOMORROW - Uses the tomorrow syntax highlighting theme.
- TOMORROW\_NIGHT - Uses the tomorrow-night syntax highlighting theme.
- TOMORROW\_NIGHT\_BLUE - Uses the tomorrow-night-blue syntax highlighting theme.
- TOMORROW\_NIGHT\_BRIGHT - Uses the tomorrow-night-bright syntax highlighting theme.
- TOMORROW\_NIGHT\_EIGHTIES - Uses the tomorrow-night-eighties syntax highlighting theme.
- VS - Uses the vs syntax highlighting theme.
- VS2015 - Uses the vs2015 syntax highlighting theme.
- XCODE - Uses the xcode syntax highlighting theme.
- XT256 - Uses the xt256 syntax highlighting theme.
- ZENBURN - Uses the zenburn syntax highlighting theme.

## Properties

**buildA11Y\_DARK**class-attributeinstance-attribute

Uses the a11y-dark syntax highlighting theme.

**buildA11Y\_LIGHT**class-attributeinstance-attribute

Uses the a11y-light syntax highlighting theme.

**buildAGATE**class-attributeinstance-attribute

Uses the agate syntax highlighting theme.

**buildAN\_OLD\_HOPE**class-attributeinstance-attribute

Uses the an-old-hope syntax highlighting theme.

**buildANDROID\_STUDIO**class-attributeinstance-attribute

Uses the androidstudio syntax highlighting theme.

**buildARDUINO\_LIGHT**class-attributeinstance-attribute

Uses the arduino-light syntax highlighting theme.

**buildART**class-attributeinstance-attribute

Uses the arta syntax highlighting theme.

**buildASCETIC**class-attributeinstance-attribute

Uses the ascetic syntax highlighting theme.

**buildATELIER\_CAVE\_DARK**class-attributeinstance-attribute

Uses the atelier-cave-dark syntax highlighting theme.

**buildATELIER\_CAVE\_LIGHT**class-attributeinstance-attribute

Uses the atelier-cave-light syntax highlighting theme.

**buildATELIER\_DUNE\_DARK**class-attributeinstance-attribute

Uses the atelier-dune-dark syntax highlighting theme.

**buildATELIER\_DUNE\_LIGHT**class-attributeinstance-attribute

Uses the atelier-dune-light syntax highlighting theme.

**buildATELIER\_ESTUARY\_DARK**class-attributeinstance-attribute

Uses the atelier-estuary-dark syntax highlighting theme.

**buildATELIER\_ESTUARY\_LIGHT**class-attributeinstance-attribute

Uses the atelier-estuary-light syntax highlighting theme.

**buildATELIER\_FOREST\_DARK**class-attributeinstance-attribute

Uses the atelier-forest-dark syntax highlighting theme.

**buildATELIER\_FOREST\_LIGHT**class-attributeinstance-attribute

Uses the atelier-forest-light syntax highlighting theme.

**buildATELIER\_HEATH\_DARK**class-attributeinstance-attribute

Uses the atelier-heath-dark syntax highlighting theme.

**buildATELIER\_HEATH\_LIGHT**class-attributeinstance-attribute

Uses the atelier-heath-light syntax highlighting theme.

**buildATELIER\_LAKESIDE\_DARK**class-attributeinstance-attribute

Uses the atelier-lakeside-dark syntax highlighting theme.

**buildATELIER\_LAKESIDE\_LIGHT**class-attributeinstance-attribute

Uses the atelier-lakeside-light syntax highlighting theme.

**buildATELIER\_PLATEAU\_DARK**class-attributeinstance-attribute

Uses the atelier-plateau-dark syntax highlighting theme.

**buildATELIER\_PLATEAU\_LIGHT**class-attributeinstance-attribute

Uses the atelier-plateau-light syntax highlighting theme.

**buildATELIER\_SAVANNA\_DARK**class-attributeinstance-attribute

Uses the atelier-savanna-dark syntax highlighting theme.

**buildATELIER\_SAVANNA\_LIGHT**class-attributeinstance-attribute

Uses the atelier-savanna-light syntax highlighting theme.

**buildATELIER\_SEASIDE\_DARK**class-attributeinstance-attribute

Uses the atelier-seaside-dark syntax highlighting theme.

**buildATELIER\_SEASIDE\_LIGHT**class-attributeinstance-attribute

Uses the atelier-seaside-light syntax highlighting theme.

**buildATELIER\_SULPHURPOOL\_DARK**class-attributeinstance-attribute

Uses the atelier-sulphurpool-dark syntax highlighting theme.

**buildATELIER\_SULPHURPOOL\_LIGHT**class-attributeinstance-attribute

Uses the atelier-sulphurpool-light syntax highlighting theme.

**buildATOM\_ONE\_DARK**class-attributeinstance-attribute

Uses the atom-one-dark syntax highlighting theme.

**buildATOM\_ONE\_DARK\_REASONABLE**class-attributeinstance-attribute

Uses the atom-one-dark-reasonable syntax highlighting theme.

**buildATOM\_ONE\_LIGHT**class-attributeinstance-attribute

Uses the atom-one-light syntax highlighting theme.

**buildBROWN\_PAPERclass-attributeinstance-attribute**

Uses the brown-paper syntax highlighting theme.

**buildCODEPEN\_EMBEDclass-attributeinstance-attribute**

Uses the codepen-embed syntax highlighting theme.

**buildCOLOR\_BREWERclass-attributeinstance-attribute**

Uses the color-brewer syntax highlighting theme.

**buildDARKclass-attributeinstance-attribute**

Uses the dark syntax highlighting theme.

**buildDEFAULTclass-attributeinstance-attribute**

Uses the default syntax highlighting theme.

**buildDOCCOclass-attributeinstance-attribute**

Uses the docco syntax highlighting theme.

**buildDRACULAcass-attributeinstance-attribute**

Uses the dracula syntax highlighting theme.

**buildDRAGULAcass-attributeinstance-attribute**

Uses the dracula syntax highlighting theme.

**buildFARclass-attributeinstance-attribute**

Uses the far syntax highlighting theme.

**buildFOUNDATIONclass-attributeinstance-attribute**

Uses the foundation syntax highlighting theme.

**buildGITHUBclass-attributeinstance-attribute**

Uses the github syntax highlighting theme.

**buildGITHUB\_GISTclass-attributeinstance-attribute**

Uses the github-gist syntax highlighting theme.

**buildGMLclass-attributeinstance-attribute**

Uses the gml syntax highlighting theme.

**buildGOOGLE\_CODEclass-attributeinstance-attribute**

Uses the googlecode syntax highlighting theme.

**buildGRADIENT\_DARKclass-attributeinstance-attribute**

Uses the gradient-dark syntax highlighting theme.

**buildGRAYSCALEclass-attributeinstance-attribute**

Uses the grayscale syntax highlighting theme.

**buildGRUVBOX\_DARKclass-attributeinstance-attribute**

Uses the gruvbox-dark syntax highlighting theme.

**buildGRUVBOX\_LIGHTclass-attributeinstance-attribute**

Uses the gruvbox-light syntax highlighting theme.

**buildHOPSCOTCHclass-attributeinstance-attribute**

Uses the hopscotch syntax highlighting theme.

**buildHYBRIDclass-attributeinstance-attribute**

Uses the hybrid syntax highlighting theme.

**buildIDEAclass-attributeinstance-attribute**

Uses the idea syntax highlighting theme.

**buildIR\_BLACKclass-attributeinstance-attribute**

Uses the ir-black syntax highlighting theme.

**buildISBL\_EDITOR\_DARKclass-attributeinstance-attribute**

Uses the isbl-editor-dark syntax highlighting theme.

**buildISBL\_EDITOR\_LIGHTclass-attributeinstance-attribute**

Uses the isbl-editor-light syntax highlighting theme.

**buildKIMBIE\_DARKclass-attributeinstance-attribute**

Uses the kimbie.dark syntax highlighting theme.

**buildKIMBIE\_LIGHTclass-attributeinstance-attribute**

Uses the kimbie.light syntax highlighting theme.

**buildLIGHTFAIRclass-attributeinstance-attribute**

Uses the lightfair syntax highlighting theme.

**buildMAGULAcass-attributeinstance-attribute**

Uses the magula syntax highlighting theme.

**buildMONO\_BLUEclass-attributeinstance-attribute**

Uses the mono-blue syntax highlighting theme.

**buildMONOKAIclass-attributeinstance-attribute**

Uses the monokai syntax highlighting theme.

**buildMONOKAI\_SUBLIMEclass-attributeinstance-attribute**

Uses the monokai-sublime syntax highlighting theme.

**buildNIGHT\_OWLclass-attributeinstance-attribute**

Uses the night-owl syntax highlighting theme.

**buildNORDclass-attributeinstance-attribute**

Uses the nord syntax highlighting theme.

**buildOBSIDIANclass-attributeinstance-attribute**

Uses the obsidian syntax highlighting theme.

**buildOCEANclass-attributeinstance-attribute**

Uses the ocean syntax highlighting theme.

**buildPARAISEO\_DARKclass-attributeinstance-attribute**

Uses the paraiso-dark syntax highlighting theme.

**buildPARAISEO\_LIGHTclass-attributeinstance-attribute**

Uses the paraiso-light syntax highlighting theme.

**buildPOJOAQUEclass-attributeinstance-attribute**

Uses the pojopaque syntax highlighting theme.

**buildPURE\_BASICclass-attributeinstance-attribute**

Uses the purebasic syntax highlighting theme.

**buildQT\_CREATOR\_DARKclass-attributeinstance-attribute**

Uses the qtcreator\_dark syntax highlighting theme.

**buildQT\_CREATOR\_LIGHTclass-attributeinstance-attribute**

Uses the qtcreator\_light syntax highlighting theme.

**buildRAILSCASTSclass-attributeinstance-attribute**

Uses the railscasts syntax highlighting theme.

**buildRAINBOWclass-attributeinstance-attribute**

Uses the rainbow syntax highlighting theme.

**buildROUTERSclass-attributeinstance-attribute**

Uses the routers syntax highlighting theme.

**buildSCHOOL\_BOOKclass-attributeinstance-attribute**

Uses the school-book syntax highlighting theme.

**buildSHADES\_OF\_PURPLEclass-attributeinstance-attribute**

Uses the shades-of-purple syntax highlighting theme.

**buildSOLARIZED\_DARKclass-attributeinstance-attribute**

Uses the solarized-dark syntax highlighting theme.

**buildSOLARIZED\_LIGHTclass-attributeinstance-attribute**

Uses the solarized-light syntax highlighting theme.

**buildSUNBURSTclass-attributeinstance-attribute**

Uses the sunburst syntax highlighting theme.

**buildTOMORROWclass-attributeinstance-attribute**

Uses the tomorrow syntax highlighting theme.

**buildTOMORROW\_NIGHTclass-attributeinstance-attribute**

Uses the tomorrow-night syntax highlighting theme.

**buildTOMORROW\_NIGHT\_BLUEclass-attributeinstance-attribute**

Uses the tomorrow-night-blue syntax highlighting theme.

**buildTOMORROW\_NIGHT\_BRIGHTclass-attributeinstance-attribute**

Uses the tomorrow-night-bright syntax highlighting theme.

**buildTOMORROW\_NIGHT\_EIGHTIESclass-attributeinstance-attribute**

Uses the tomorrow-night-eighties syntax highlighting theme.

**buildVSclass-attributeinstance-attribute**

Uses the vs syntax highlighting theme.

**buildVS2015class-attributeinstance-attribute**

Uses the vs2015 syntax highlighting theme.

**buildXCODEclass-attributeinstance-attribute**

Uses the xcode syntax highlighting theme.

**buildXT256class-attributeinstance-attribute**

Uses the xt256 syntax highlighting theme.

**buildZENBURNclass-attributeinstance-attribute**

Uses the zenburn syntax highlighting theme.



# MarkdownCustomCodeTheme

Custom text-style mapping for code token highlighting.

Use this as `Markdown.code_theme` when you need per-token styling instead of a built-in `MarkdownCodeTheme`.

Field names correspond to code token kinds (for example keyword, string, comment). Unspecified fields keep default token styling.

## Properties

- `addition` - Style for inserted/added diff tokens.
- `attr` - Style for attribute tokens.
- `attribute` - Style for alternative attribute token name.
- `built_in` - Style for built-in symbol tokens.
- `builtin_name` - Style for built-in name tokens.
- `bullet` - Style for bullet/list marker tokens.
- `class_name` - Style for class name tokens.
- `code` - Base style for code text.
- `comment` - Style for comment tokens.
- `deletion` - Style for removed/deleted diff tokens.
- `doctag` - Style for documentation tag tokens.
- `emphasis` - Style for emphasized tokens.
- `formula` - Style for formula/math tokens.
- `function` - Style for function identifier tokens.
- `keyword` - Style for keyword tokens.
- `link` - Style for link tokens.
- `link_label` - Style for link label tokens.
- `literal` - Style for literal constant tokens.
- `meta` - Style for metadata tokens.
- `meta_keyword` - Style for metadata keyword tokens.
- `meta_string` - Style for metadata string tokens.
- `name` - Style for generic name/identifier tokens.
- `number` - Style for numeric literal tokens.
- `operator` - Style for operator tokens.
- `params` - Style for parameter list tokens.
- `pattern_match` - Style for pattern matching tokens.
- `quote` - Style for quote/blockquote tokens.
- `regexp` - Style for regular expression tokens.
- `root` - Base style for root code container.
- `section` - Style for section heading tokens.
- `selector_attr` - Style for CSS selector attribute tokens.
- `selector_class` - Style for CSS selector class tokens.
- `selector_id` - Style for CSS selector id tokens.
- `selector_pseudo` - Style for CSS selector pseudo-class tokens.

- `selector_tag` - Style for CSS selector tag tokens.
- `string` - Style for string literal tokens.
- `strong` - Style for strong/bold tokens.
- `stronge` - Legacy/alternate token key style for strong text.
- `subst` - Style for substitution/interpolation tokens.
- `subtr` - Legacy/alternate token key style for substitution tokens.
- `symbol` - Style for symbol tokens.
- `tag` - Style for markup tag tokens.
- `template_tag` - Style for template tag tokens.
- `template_variable` - Style for template variable tokens.
- `title` - Style for title/name tokens.
- `type` - Style for type annotation tokens.
- `variable` - Style for variable identifier tokens.

## Properties

### `buildadditionclass-attributeinstance-attribute`

`addition: TextStyle | None = None`

Style for inserted/added diff tokens.

### `buildattrclass-attributeinstance-attribute`

`attr: TextStyle | None = None`

Style for attribute tokens.

### `buildattributeclass-attributeinstance-attribute`

`attribute: TextStyle | None = None`

Style for alternative attribute token name.

### `buildbuiltin_inclass-attributeinstance-attribute`

`builtin_in: TextStyle | None = None`

Style for built-in symbol tokens.

### `buildbuiltin_nameclass-attributeinstance-attribute`

`builtin_name: TextStyle | None = None`

Style for built-in name tokens.

### `buildbulletclass-attributeinstance-attribute`

`bullet: TextStyle | None = None`

Style for bullet/list marker tokens.

### `buildclass_nameclass-attributeinstance-attribute`

`class_name: TextStyle | None = None`

Style for class name tokens.

### `buildcodeclass-attributeinstance-attribute`

`code: TextStyle | None = None`

Base style for code text.

### `buildcommentclass-attributeinstance-attribute`

`comment: TextStyle | None = None`

Style for comment tokens.

### `builddeletionclass-attributeinstance-attribute`

deletion: TextStyle | None = None

Style for removed/deleted diff tokens.

### **builddoctagclass-attributeinstance-attribute**

doctag: TextStyle | None = None

Style for documentation tag tokens.

### **buildemphasisclass-attributeinstance-attribute**

emphasis: TextStyle | None = None

Style for emphasized tokens.

### **buildformulaclass-attributeinstance-attribute**

formula: TextStyle | None = None

Style for formula/math tokens.

### **buildfunctionclass-attributeinstance-attribute**

function: TextStyle | None = None

Style for function identifier tokens.

### **buildkeywordclass-attributeinstance-attribute**

keyword: TextStyle | None = None

Style for keyword tokens.

### **buildlinkclass-attributeinstance-attribute**

link: TextStyle | None = None

Style for link tokens.

### **buildlink\_labelclass-attributeinstance-attribute**

link\_label: TextStyle | None = None

Style for link label tokens.

### **builliteralclass-attributeinstance-attribute**

literal: TextStyle | None = None

Style for literal constant tokens.

### **buildmetaclass-attributeinstance-attribute**

meta: TextStyle | None = None

Style for metadata tokens.

### **buildmeta\_keywordclass-attributeinstance-attribute**

meta\_keyword: TextStyle | None = None

Style for metadata keyword tokens.

### **buildmeta\_stringclass-attributeinstance-attribute**

meta\_string: TextStyle | None = None

Style for metadata string tokens.

### **buildnameclass-attributeinstance-attribute**

name: TextStyle | None = None

Style for generic name/identifier tokens.

### **buildnumberclass-attributeinstance-attribute**

number: TextStyle | None = None

Style for numeric literal tokens.

### **buildoperatorclass-attributeinstance-attribute**

operator: TextStyle | None = None

Style for operator tokens.

### **buildparamsclass-attributeinstance-attribute**

params: TextStyle | None = None

Style for parameter list tokens.

### **buildpattern\_matchclass-attributeinstance-attribute**

pattern\_match: TextStyle | None = None

Style for pattern matching tokens.

### **buildquoteclass-attributeinstance-attribute**

quote: TextStyle | None = None

Style for quote/blockquote tokens.

### **buildregexprclass-attributeinstance-attribute**

regex: TextStyle | None = None

Style for regular expression tokens.

### **buildrootclass-attributeinstance-attribute**

root: TextStyle | None = None

Base style for root code container.

### **buildsectionclass-attributeinstance-attribute**

section: TextStyle | None = None

Style for section heading tokens.

### **buildselector\_attrclass-attributeinstance-attribute**

selector\_attr: TextStyle | None = None

Style for CSS selector attribute tokens.

### **buildselector\_classclass-attributeinstance-attribute**

selector\_class: TextStyle | None = None

Style for CSS selector class tokens.

### **buildselector\_idclass-attributeinstance-attribute**

selector\_id: TextStyle | None = None

Style for CSS selector id tokens.

### **buildselector\_pseudoclass-attributeinstance-attribute**

selector\_pseudo: TextStyle | None = None

Style for CSS selector pseudo-class tokens.

### **buildselector\_tagclass-attributeinstance-attribute**

selector\_tag: TextStyle | None = None

Style for CSS selector tag tokens.

### **buildstringclass-attributeinstance-attribute**

string: TextStyle | None = None

Style for string literal tokens.

### **buildstrongclass-attributeinstance-attribute**

strong: TextStyle | None = None

Style for strong/bold tokens.

### **buildstrongeclass-attributeinstance-attribute**

stronge: TextStyle | None = None

Legacy/alternate token key style for strong text.

### **buildsubstclass-attributeinstance-attribute**

subst: TextStyle | None = None

Style for substitution/interpolation tokens.

### **buildsubtrclass-attributeinstance-attribute**

subtr: TextStyle | None = None

Legacy/alternate token key style for substitution tokens.

### **buildsymbolclass-attributeinstance-attribute**

symbol: TextStyle | None = None

Style for symbol tokens.

### **buildtagclass-attributeinstance-attribute**

tag: TextStyle | None = None

Style for markup tag tokens.

### **buildtemplate\_tagclass-attributeinstance-attribute**

template\_tag: TextStyle | None = None

Style for template tag tokens.

### **buildtemplate\_variableclass-attributeinstance-attribute**

template\_variable: TextStyle | None = None

Style for template variable tokens.

### **buildtitleclass-attributeinstance-attribute**

title: TextStyle | None = None

Style for title/name tokens.

### **buildtypeclass-attributeinstance-attribute**

type: TextStyle | None = None

Style for type annotation tokens.

### **buildvariableclass-attributeinstance-attribute**

variable: TextStyle | None = None

Style for variable identifier tokens.

# MarkdownExtensionSet

Preset markdown syntax extension bundles for Markdown.

The selected set controls which block/inline syntaxes are enabled while parsing `Markdown.value`.

Inherits: `enum.Enum`

## Properties

- `COMMON_MARK` - CommonMark-compatible extension set.
- `GITHUB_FLAVORED` - GitHub Flavored Markdown (GFM) extension set.
- `GITHUB_WEB` - GitHub web renderer style extension set.
- `NONE` - Basic markdown parsing without additional extension bundle.

## Properties

`buildCOMMON_MARKclass-attributeinstance-attribute`

CommonMark-compatible extension set.

`buildGITHUB_FLAVOREDclass-attributeinstance-attribute`

GitHub Flavored Markdown (GFM) extension set.

`buildGITHUB_WEBclass-attributeinstance-attribute`

GitHub web renderer style extension set.

`buildNONEclass-attributeinstance-attribute`

Basic markdown parsing without additional extension bundle.

# MarkdownStyleSheet

Style overrides for markdown element rendering.

Each field customizes one element group (headings, paragraphs, tables, lists, block quotes, code blocks, etc.). Any None field keeps the control's default styling for that property.

## Properties

- `a_text_style` - Text style for links (<a>).
- `block_spacing` - Vertical spacing between markdown block elements.
- `blockquote_alignment` - Alignment for blockquote block content.
- `blockquote_decoration` - Decoration applied to blockquote container.
- `blockquote_padding` - Inner padding for blockquote container.
- `blockquote_text_style` - Text style for blockquote content.
- `checkbox_text_style` - Text style for task-list checkbox labels.
- `code_text_style` - Text style for inline code spans.
- `codeblock_alignment` - Alignment for code block content.
- `codeblock_decoration` - Decoration applied to fenced code blocks.
- `codeblock_padding` - Inner padding for fenced code blocks.
- `del_text_style` - Text style for deleted/strikethrough text.
- `em_text_style` - Text style for emphasized text.
- `h1_alignment` - Alignment for level-1 heading blocks.
- `h1_padding` - Padding around level-1 headings.
- `h1_text_style` - Text style for level-1 headings.
- `h2_alignment` - Alignment for level-2 heading blocks.
- `h2_padding` - Padding around level-2 headings.
- `h2_text_style` - Text style for level-2 headings.
- `h3_alignment` - Alignment for level-3 heading blocks.
- `h3_padding` - Padding around level-3 headings.
- `h3_text_style` - Text style for level-3 headings.
- `h4_alignment` - Alignment for level-4 heading blocks.
- `h4_padding` - Padding around level-4 headings.
- `h4_text_style` - Text style for level-4 headings.
- `h5_alignment` - Alignment for level-5 heading blocks.
- `h5_padding` - Padding around level-5 headings.
- `h5_text_style` - Text style for level-5 headings.
- `h6_alignment` - Alignment for level-6 heading blocks.
- `h6_padding` - Padding around level-6 headings.
- `h6_text_style` - Text style for level-6 headings.
- `horizontal_rule_decoration` - Decoration used to render horizontal rule separators.
- `img_text_style` - Text style for image alt text / fallback labels.
- `list_bullet_padding` - Padding around list bullet markers.
- `list_bullet_text_style` - Text style for list bullets/markers.

- `list_indent` - Indentation width for list items.
- `ordered_list_alignment` - Alignment for ordered list blocks.
- `p_padding` - Padding around paragraph blocks.
- `p_text_style` - Text style for paragraph blocks.
- `strong_text_style` - Text style for strong (bold) text.
- `table_body_text_style` - Text style for table body cells.
- `table_cells_decoration` - Decoration applied to table cells.
- `table_cells_padding` - Inner padding for each table cell.
- `table_head_text_align` - Text alignment for table header cells.
- `table_head_text_style` - Text style for table header cells.
- `table_padding` - Outer padding around rendered table blocks.
- `text_alignment` - Default alignment for regular text blocks.
- `unordered_list_alignment` - Alignment for unordered list blocks.

## Properties

### `builda_text_style`class-attributeinstance-attribute

`a_text_style`: TextStyle | None = None

Text style for links (<a>).

### `buildblock_spacing`class-attributeinstance-attribute

`block_spacing`: Number | None = None

Vertical spacing between markdown block elements.

### `buildblockquote_alignment`class-attributeinstance-attribute

`blockquote_alignment`: MainAxisAlignment | None = None

Alignment for blockquote block content.

### `buildblockquote_decoration`class-attributeinstance-attribute

`blockquote_decoration`: BoxDecoration | None = None

Decoration applied to blockquote container.

### `buildblockquote_padding`class-attributeinstance-attribute

`blockquote_padding`: PaddingValue | None = None

Inner padding for blockquote container.

### `buildblockquote_text_style`class-attributeinstance-attribute

`blockquote_text_style`: TextStyle | None = None

Text style for blockquote content.

### `buildcheckbox_text_style`class-attributeinstance-attribute

`checkbox_text_style`: TextStyle | None = None

Text style for task-list checkbox labels.

### `buildcode_text_style`class-attributeinstance-attribute

`code_text_style`: TextStyle | None = None

Text style for inline code spans.

### `buildcodeblock_alignment`class-attributeinstance-attribute

`codeblock_alignment`: MainAxisAlignment | None = None

Alignment for code block content.

### `buildcodeblock_decoration`class-attributeinstance-attribute



`codeblock_decoration: BoxDecoration | None = None`

Decoration applied to fenced code blocks.

## **buildcodeblock\_paddingclass-attributeinstance-attribute**

`codeblock_padding: PaddingValue | None = None`

Inner padding for fenced code blocks.

## **builddel\_text\_styleclass-attributeinstance-attribute**

`del_text_style: TextStyle | None = None`

Text style for deleted/strikethrough text.

## **builдем\_text\_styleclass-attributeinstance-attribute**

`em_text_style: TextStyle | None = None`

Text style for emphasized text.

## **buildh1\_alignmentclass-attributeinstance-attribute**

`h1_alignment: MainAxisAlignment | None = None`

Alignment for level-1 heading blocks.

## **buildh1\_paddingclass-attributeinstance-attribute**

`h1_padding: PaddingValue | None = None`

Padding around level-1 headings.

## **buildh1\_text\_styleclass-attributeinstance-attribute**

`h1_text_style: TextStyle | None = None`

Text style for level-1 headings.

## **buildh2\_alignmentclass-attributeinstance-attribute**

`h2_alignment: MainAxisAlignment | None = None`

Alignment for level-2 heading blocks.

## **buildh2\_paddingclass-attributeinstance-attribute**

`h2_padding: PaddingValue | None = None`

Padding around level-2 headings.

## **buildh2\_text\_styleclass-attributeinstance-attribute**

`h2_text_style: TextStyle | None = None`

Text style for level-2 headings.

## **buildh3\_alignmentclass-attributeinstance-attribute**

`h3_alignment: MainAxisAlignment | None = None`

Alignment for level-3 heading blocks.

## **buildh3\_paddingclass-attributeinstance-attribute**

`h3_padding: PaddingValue | None = None`

Padding around level-3 headings.

## **buildh3\_text\_styleclass-attributeinstance-attribute**

`h3_text_style: TextStyle | None = None`

Text style for level-3 headings.

## **buildh4\_alignmentclass-attributeinstance-attribute**

`h4_alignment: MainAxisAlignment | None = None`

Alignment for level-4 heading blocks.

## **buildh4\_paddingclass-attributeinstance-attribute**

`h4_padding: PaddingValue | None = None`

Padding around level-4 headings.

**buildh4\_text\_style**class-attributeinstance-attribute

h4\_text\_style: TextStyle | None = None

Text style for level-4 headings.

**buildh5\_alignment**class-attributeinstance-attribute

h5\_alignment: MainAxisAlignment | None = None

Alignment for level-5 heading blocks.

**buildh5\_padding**class-attributeinstance-attribute

h5\_padding: PaddingValue | None = None

Padding around level-5 headings.

**buildh5\_text\_style**class-attributeinstance-attribute

h5\_text\_style: TextStyle | None = None

Text style for level-5 headings.

**buildh6\_alignment**class-attributeinstance-attribute

h6\_alignment: MainAxisAlignment | None = None

Alignment for level-6 heading blocks.

**buildh6\_padding**class-attributeinstance-attribute

h6\_padding: PaddingValue | None = None

Padding around level-6 headings.

**buildh6\_text\_style**class-attributeinstance-attribute

h6\_text\_style: TextStyle | None = None

Text style for level-6 headings.

**buildhorizontal\_rule\_decoration**class-attributeinstance-attribute

horizontal\_rule\_decoration: BoxDecoration | None = None

Decoration used to render horizontal rule separators.

**buildimg\_text\_style**class-attributeinstance-attribute

img\_text\_style: TextStyle | None = None

Text style for image alt text / fallback labels.

**buildlist\_bullet\_padding**class-attributeinstance-attribute

list\_bullet\_padding: PaddingValue | None = None

Padding around list bullet markers.

**buildlist\_bullet\_text\_style**class-attributeinstance-attribute

list\_bullet\_text\_style: TextStyle | None = None

Text style for list bullets/markers.

**buildlist\_indent**class-attributeinstance-attribute

list\_indent: Number | None = None

Indentation width for list items.

**buildordered\_list\_alignment**class-attributeinstance-attribute

ordered\_list\_alignment: MainAxisAlignment | None = None

Alignment for ordered list blocks.

**buildp\_padding**class-attributeinstance-attribute

p\_padding: PaddingValue | None = None

Padding around paragraph blocks.

### **buildp\_text\_styleclass-attributeinstance-attribute**

`p_text_style: TextStyle | None = None`

Text style for paragraph blocks.

### **buildstrong\_text\_styleclass-attributeinstance-attribute**

`strong_text_style: TextStyle | None = None`

Text style for strong (bold) text.

### **buildtable\_body\_text\_styleclass-attributeinstance-attribute**

`table_body_text_style: TextStyle | None = None`

Text style for table body cells.

### **buildtable\_cells\_decorationclass-attributeinstance-attribute**

`table_cells_decoration: BoxDecoration | None = None`

Decoration applied to table cells.

### **buildtable\_cells\_paddingclass-attributeinstance-attribute**

`table_cells_padding: PaddingValue | None = None`

Inner padding for each table cell.

### **buildtable\_head\_text\_alignclass-attributeinstance-attribute**

`table_head_text_align: TextAlign | None = None`

Text alignment for table header cells.

### **buildtable\_head\_text\_styleclass-attributeinstance-attribute**

`table_head_text_style: TextStyle | None = None`

Text style for table header cells.

### **buildtable\_paddingclass-attributeinstance-attribute**

`table_padding: PaddingValue | None = None`

Outer padding around rendered table blocks.

### **buildtext\_alignmentclass-attributeinstance-attribute**

`text_alignment: MainAxisAlignment | None = None`

Default alignment for regular text blocks.

### **buildunordered\_list\_alignmentclass-attributeinstance-attribute**

`unordered_list_alignment: MainAxisAlignment | None = None`

Alignment for unordered list blocks.

# Matrix4

A recorded, replayable Matrix4 transform intent.

This class records constructors and mutating method calls, then Flet replays them in Flutter to build a real Matrix4 for Transform.matrix.

## Properties

- `ctor` - Recorded constructor call.
- `ops` - Ordered list of recorded mutating operations.

## Methods

- `diagonal3_values` - Creates a matrix initialized with diagonal scale values.
- `identity` - Creates a matrix initialized with identity transform.
- `multiply` - Appends multiplication by another recorded matrix.
- `rotate_x` - Appends x-axis rotation operation in radians.
- `rotate_y` - Appends y-axis rotation operation in radians.
- `rotate_z` - Appends z-axis rotation operation in radians.
- `rotation_z` - Creates a matrix initialized with a z-axis rotation in radians.
- `scale` - Appends scale operation.
- `set_entry` - Appends raw matrix entry mutation.
- `skew_x` - Creates a matrix initialized with x-axis skew in radians.
- `skew_y` - Creates a matrix initialized with y-axis skew in radians.
- `translate` - Appends translation operation.
- `translation_values` - Creates a matrix initialized with translation components.

## Properties

### buildctorclass-attributeinstance-attribute

```
ctor: _Matrix4Call = field(
 default_factory=lambda: _Matrix4Call(name="identity")
)
```

Recorded constructor call.

### buildopsclass-attributeinstance-attribute

```
ops: list[_Matrix4Call] = field(default_factory=list)
```

Ordered list of recorded mutating operations.

## Methods

### deployed\_codediagonal3\_valuesclassmethod

```
diagonal3_values(
 x: Number, y: Number, z: Number
) -> Matrix4
```

Creates a matrix initialized with diagonal scale values.

### deployed\_codeidentityclassmethod

```
identity() -> Matrix4
```

Creates a matrix initialized with identity transform.

### deployed\_codemultiply

```
multiply(other: Matrix4) -> Matrix4
```

Appends multiplication by another recorded matrix.

### deployed\_coderotate\_x

```
rotate_x(angle: Number) -> Matrix4
```

Appends x-axis rotation operation in radians.

### deployed\_coderotate\_y

```
rotate_y(angle: Number) -> Matrix4
```

Appends y-axis rotation operation in radians.

### deployed\_coderotate\_z

```
rotate_z(angle: Number) -> Matrix4
```

Appends z-axis rotation operation in radians.

### deployed\_coderotation\_zclassmethod

```
rotation_z(angle: Number) -> Matrix4
```

Creates a matrix initialized with a z-axis rotation in radians.

### deployed\_codescale

```
scale(
 x: Number,
 y: Number | None = None,
 z: Number | None = None,
) -> Matrix4
```

Appends scale operation.

If only x is provided then uniform scale is used. If x and y are provided then 2D scale is used. If all three are provided then 3D scale is used.

### deployed\_codeset\_entry

```
set_entry(row: int, col: int, value: Number) -> Matrix4
```

Appends raw matrix entry mutation.

### deployed\_codeskew\_xclassmethod

```
skew_x(angle: Number) -> Matrix4
```

Creates a matrix initialized with x-axis skew in radians.

### deployed\_codeskew\_yclassmethod

```
skew_y(angle: Number) -> Matrix4
```

Creates a matrix initialized with y-axis skew in radians.

### deployed\_codetranslate

```
translate(
 x: Number, y: Number, z: Number = 0
) -> Matrix4
```

Appends translation operation.

### deployed\_codetranslation\_valuesclassmethod

```
translation_values(
 x: Number, y: Number, z: Number
) -> Matrix4
```

Creates a matrix initialized with translation components.

## memo

Lets you skip re-rendering a component when its props are unchanged.

### Example

```
import flet as ft

@ft.component
def MyComponent(x, y):
 return ft.Text(f"x={x}, y={y}")

MemoizedMyComponent = ft.memo(MyComponent)

flet.run(
 lambda page: page.render(
 lambda: MemoizedMyComponent(x=1, y=2),
),
)
```

# MenuStyle

Defines the visual style/appearance of a menu.

## Properties

- `alignment` - Determines the desired alignment of the submenu when opened relative to the button that opens it.
- `bgcolor` - The menu's background fill color.
- `elevation` - The menu's elevation, i.e.
- `fixed_size` - The menu's size.
- `max_size` - The maximum size of the menu itself.
- `min_size` - The minimum size of the menu itself.
- `mouse_cursor` - The cursor for a mouse pointer when it enters or is hovering over the menu.
- `padding` - The padding between the menu's boundary and its child.
- `shadow_color` - The shadow color of the menu
- `shape` - The menu's shape.
- `side` - The color and weight of the menu's outline.
- `visual_density` - Defines how compact the menu's layout will be.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

`alignment: Alignment | None = None`

Determines the desired alignment of the submenu when opened relative to the button that opens it.

If there isn't sufficient space to open the menu with the given alignment, and there's space on the other side of the button, then the alignment is swapped to it's opposite (1 becomes -1, etc.), and the menu will try to appear on the other side of the button. If there isn't enough space there either, then the menu will be pushed as far over as necessary to display as much of itself as possible, possibly overlapping the parent button.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ControlStateValue[ColorValue] | None = None`

The menu's background fill color.

### `buildelevationclass-attributeinstance-attribute`

`elevation: ControlStateValue[Number | None] | None = None`

The menu's elevation, i.e. the size of the shadow below the menu.

### `buildfixed_sizeclass-attributeinstance-attribute`

`fixed_size: ControlStateValue[Size] | None = None`

The menu's size.

This size is still constrained by the style's `min_size` and `max_size`. Fixed size dimensions whose value is `float('inf')` are ignored.

To specify menus with a fixed width and the default height use `Size.from_width(320)`. Similarly, to specify a fixed height and the default width use `Size.from_height(100)`.

### `buildmax_sizeclass-attributeinstance-attribute`

`max_size: ControlStateValue[Size] | None = None`

The maximum size of the menu itself.

A `Size.infinite` or `None` value for this property means that the menu's maximum size is not constrained.

This value must be greater than or equal to `min_size`.

### **buildmin\_sizeclass-attributeinstance-attribute**

`min_size: ControlStateValue[Size] | None = None`

The minimum size of the menu itself.

This value must be less than or equal to `max_size`.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

The cursor for a mouse pointer when it enters or is hovering over the menu.

### **buildpaddingclass-attributeinstance-attribute**

`padding: ControlStateValue[PaddingValue] | None = None`

The padding between the menu's boundary and its child.

### **buildshadow\_colorclass-attributeinstance-attribute**

`shadow_color: ControlStateValue[ColorValue] | None = None`

The shadow color of the menu

The material's elevation shadow can be difficult to see for dark themes, so by default the menu classes add a semi-transparent overlay to indicate elevation.

### **buildshapeclass-attributeinstance-attribute**

`shape: ControlStateValue[OutlinedBorder] | None = None`

The menu's shape.

This shape is combined with `side` to create a shape decorated with an outline.

### **buildsideclass-attributeinstance-attribute**

`side: ControlStateValue[BorderSide] | None = None`

The color and weight of the menu's outline.

This value is combined with `shape` to create a shape decorated with an outline.

### **buildvisual\_densityclass-attributeinstance-attribute**

`visual_density: VisualDensity | None = None`

Defines how compact the menu's layout will be.



# MouseCursor

Various mouse cursor types that represent different operations or states.

Inherits: `enum.Enum`

## Properties

- ALIAS - A cursor indicating that the current operation will create an alias of, or a shortcut of the item.
- ALL\_SCROLL - A cursor indicating scrolling in any direction.
- BASIC - The platform-dependent basic cursor.
- CELL - A cursor indicating selectable table cells.
- CLICK - A cursor that emphasizes an element being clickable, such as a hyperlink.
- CONTEXT\_MENU - A cursor indicating somewhere the user can trigger a context menu.
- COPY - A cursor indicating that the current operation will copy the item.
- DISAPPEARING - A cursor indicating that the current operation will result in the disappearance of the item.
- FORBIDDEN - A cursor indicating an operation that will not be carried out.
- GRAB - A cursor indicating something that can be dragged.
- GRABBING - A cursor indicating something that is being dragged.
- HELP - A cursor indicating help information.
- MOVE - A cursor indicating moving something.
- NO\_DROP - A cursor indicating somewhere that the current item may not be dropped.
- NONE - Hide the cursor.
- PRECISE - A cursor indicating precise selection, such as selecting a pixel in a bitmap.
- PROGRESS - A cursor indicating the status that the program is busy but can still be interacted with.
- RESIZE\_COLUMN - A cursor indicating resizing a column, or an item horizontally.
- RESIZE\_DOWN - A cursor indicating resizing an object from its bottom edge.
- RESIZE\_DOWN\_LEFT - A cursor indicating resizing an object from its bottom-left corner.
- RESIZE\_DOWN\_RIGHT - A cursor indicating resizing an object from its bottom-right corner.
- RESIZE\_LEFT - A cursor indicating resizing an object from its left edge.
- RESIZE\_LEFT\_RIGHT - A cursor indicating resizing an object bidirectionally from its left or right edge.
- RESIZE\_RIGHT - A cursor indicating resizing an object from its right edge.
- RESIZE\_ROW - A cursor indicating resizing a row, or an item vertically.
- RESIZE\_UP - A cursor indicating resizing an object from its top edge.
- RESIZE\_UP\_DOWN - A cursor indicating resizing an object bidirectionally from its top or bottom edge.
- RESIZE\_UP\_LEFT - A cursor indicating resizing an object from its top-left corner.
- RESIZE\_UP\_LEFT\_DOWN\_RIGHT - A cursor indicating resizing an object bidirectionally from its top left or bottom right corner.
- RESIZE\_UP\_RIGHT - A cursor indicating resizing an object from its top-right corner.
- RESIZE\_UP\_RIGHT\_DOWN\_LEFT - A cursor indicating resizing an object bidirectionally from its top right or bottom left corner.
- TEXT - A cursor indicating selectable text.
- VERTICAL\_TEXT - A cursor indicating selectable vertical text.
- WAIT - A cursor indicating the status that the program is busy and therefore can not be interacted with.

- ZOOM\_IN - A cursor indicating zooming in.
- ZOOM\_OUT - A cursor indicating zooming out.

## Examples

### Mouse Cursor showcase

play\_arrowTry Online

```
import flet as ft

def showcase_card(cursor: ft.MouseCursor) -> ft.GestureDetector:
 return ft.GestureDetector(
 mouse_cursor=cursor,
 content=ft.Container(
 width=250,
 height=100,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=4,
 alignment=ft.MainAxisAlignment.CENTER,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Text(cursor.name, weight=ft.FontWeight.BOLD),
],
),
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="MouseCursor Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Hover each card to compare cursor styles. "
 "Cursor rendering can vary by OS and browser.",
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(cursor) for cursor in ft.MouseCursor],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### buildALIASclass-attributeinstance-attribute

A cursor indicating that the current operation will create an alias of, or a shortcut of the item. Typically the shape of an arrow with a shortcut icon at the corner.

### buildALL\_SCROLLclass-attributeinstance-attribute

A cursor indicating scrolling in any direction. Typically the shape of a dot surrounded by 4 arrows.

### buildBASICclass-attributeinstance-attribute

The platform-dependent basic cursor. Typically the shape of an arrow.

### buildCELLclass-attributeinstance-attribute

A cursor indicating selectable table cells. Typically the shape of a hollow plus sign.

#### **buildCLICKclass-attributeinstance-attribute**

A cursor that emphasizes an element being clickable, such as a hyperlink. Typically the shape of a pointing hand.

#### **buildCONTEXT\_MENUclass-attributeinstance-attribute**

A cursor indicating somewhere the user can trigger a context menu. Typically the shape of an arrow with a small menu at the corner.

#### **buildCOPYclass-attributeinstance-attribute**

A cursor indicating that the current operation will copy the item. Typically the shape of an arrow with a boxed plus sign at the corner.

#### **buildDISAPPEARINGclass-attributeinstance-attribute**

A cursor indicating that the current operation will result in the disappearance of the item. Typically the shape of an arrow with a cloud of smoke at the corner.

#### **buildFORBIDDENclass-attributeinstance-attribute**

A cursor indicating an operation that will not be carried out. Typically the shape of a circle with a diagonal line.

#### **buildGRABclass-attributeinstance-attribute**

A cursor indicating something that can be dragged. Typically the shape of an open hand.

#### **buildGRABBINGclass-attributeinstance-attribute**

A cursor indicating something that is being dragged. Typically the shape of a closed hand.

#### **buildHELPclass-attributeinstance-attribute**

A cursor indicating help information. Typically the shape of a question mark, or an arrow therewith.

#### **buildMOVEclass-attributeinstance-attribute**

A cursor indicating moving something. Typically the shape of four-way arrow.

#### **buildNO\_DROPclass-attributeinstance-attribute**

A cursor indicating somewhere that the current item may not be dropped. Typically the shape of a hand with a forbidden sign at the corner.

#### **buildNONEclass-attributeinstance-attribute**

Hide the cursor.

#### **buildPRECISEclass-attributeinstance-attribute**

A cursor indicating precise selection, such as selecting a pixel in a bitmap. Typically the shape of a crosshair.

#### **buildPROGRESSclass-attributeinstance-attribute**

A cursor indicating the status that the program is busy but can still be interacted with. Typically the shape of an arrow with an hourglass or a watch at the corner.

#### **buildRESIZE\_COLUMNclass-attributeinstance-attribute**

A cursor indicating resizing a column, or an item horizontally. Typically the shape of arrows pointing left and right with a vertical bar separating them.

#### **buildRESIZE\_DOWNclass-attributeinstance-attribute**

A cursor indicating resizing an object from its bottom edge. Typically the shape of an arrow pointing down.

#### **buildRESIZE\_DOWN\_LEFTclass-attributeinstance-attribute**

A cursor indicating resizing an object from its bottom-left corner. Typically the shape of an arrow pointing lower left.

#### **buildRESIZE\_DOWN\_RIGHTclass-attributeinstance-attribute**

A cursor indicating resizing an object from its bottom-right corner. Typically the shape of an arrow pointing lower right.

**buildRESIZE\_LEFT**class-attributeinstance-attribute

A cursor indicating resizing an object from its left edge. Typically the shape of an arrow pointing left.

**buildRESIZE\_LEFT\_RIGHT**class-attributeinstance-attribute

A cursor indicating resizing an object bidirectionally from its left or right edge. Typically the shape of a bidirectional arrow pointing left and right.

**buildRESIZE\_RIGHT**class-attributeinstance-attribute

A cursor indicating resizing an object from its right edge. Typically the shape of an arrow pointing right.

**buildRESIZE\_ROW**class-attributeinstance-attribute

A cursor indicating resizing a row, or an item vertically. Typically the shape of arrows pointing up and down with a horizontal bar separating them.

**buildRESIZE\_UP**class-attributeinstance-attribute

A cursor indicating resizing an object from its top edge. Typically the shape of an arrow pointing up.

**buildRESIZE\_UP\_DOWN**class-attributeinstance-attribute

A cursor indicating resizing an object bidirectionally from its top or bottom edge. Typically the shape of a bidirectional arrow pointing up and down.

**buildRESIZE\_UP\_LEFT**class-attributeinstance-attribute

A cursor indicating resizing an object from its top-left corner. Typically the shape of an arrow pointing upper left.

**buildRESIZE\_UP\_LEFT\_DOWN\_RIGHT**class-attributeinstance-attribute

A cursor indicating resizing an object bidirectionally from its top left or bottom right corner. Typically the shape of a bidirectional arrow pointing upper left and lower right.

**buildRESIZE\_UP\_RIGHT**class-attributeinstance-attribute

A cursor indicating resizing an object from its top-right corner. Typically the shape of an arrow pointing upper right.

**buildRESIZE\_UP\_RIGHT\_DOWN\_LEFT**class-attributeinstance-attribute

A cursor indicating resizing an object bidirectionally from its top right or bottom left corner. Typically the shape of a bidirectional arrow pointing upper right and lower left.

**buildTEXT**class-attributeinstance-attribute

A cursor indicating selectable text. Typically the shape of a capital I.

**buildVERTICAL\_TEXT**class-attributeinstance-attribute

A cursor indicating selectable vertical text. Typically the shape of a capital I rotated to be horizontal.

**buildWAIT**class-attributeinstance-attribute

A cursor indicating the status that the program is busy and therefore can not be interacted with. Typically the shape of an hourglass or a watch.

This cursor is not available as a system cursor on macOS. Although macOS displays a "spinning ball" cursor when busy, it's handled by the OS and not exposed for applications to choose.

**buildZOOM\_IN**class-attributeinstance-attribute

A cursor indicating zooming in. Typically a magnifying glass with a plus sign.

**buildZOOM\_OUT**class-attributeinstance-attribute

A cursor indicating zooming out. Typically a magnifying glass with a minus sign.

# MultiTapEvent

Result payload for multi-touch tap detection.

Inherits: `flet.controls.control_event.ControlEvent`

## Properties

- `correct_touches` - True when the active touch count reaches `GestureDetector.multi_tap_touches`; otherwise False is emitted when the sequence is interrupted.

## Properties

`buildcorrect_touchesclass-attributeinstance-attribute`

```
correct_touches: bool = field(metadata={"data_field": "ct"})
```

True when the active touch count reaches `GestureDetector.multi_tap_touches`; otherwise False is emitted when the sequence is interrupted.

# MultiViewAddEvent

Event payload emitted when a new multi-view is created.

Delivered to `Page.on_multi_view_add`.

Inherits: `Event`

## Properties

- `initial_data` - Optional initial payload provided when the view was opened.
- `view_id` - Unique identifier of the newly created view.

## Properties

### `buildinitial_datainstance-attribute`

`initial_data`: Any

Optional initial payload provided when the view was opened.

### `buildview_idinstance-attribute`

`view_id`: int

Unique identifier of the newly created view.

# MultiViewRemoveEvent

Event payload emitted when a multi-view is removed.

Delivered to `Page.on_multi_view_remove`.

Inherits: `Event`

## Properties

- `view_id` - Unique identifier of the removed view.

## Properties

`buildview_idinstance-attribute`

`view_id: int`

Unique identifier of the removed view.

# NavigationBarLabelBehavior

Defines how the destinations' labels will be laid out and when they'll be displayed.

Inherits: `enum.Enum`

## Properties

- ALWAYS\_HIDE
- ALWAYS\_SHOW
- ONLY\_SHOW\_SELECTED

## Examples

### Navigation Bar Label Behavior showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(behavior: ft.NavigationBarLabelBehavior) -> ft.Container:
 bar = ft.NavigationBar(
 label_behavior=behavior,
 selected_index=1,
 destinations=[
 ft.NavigationBarDestination(icon=ft.Icons.HOME, label="Home"),
 ft.NavigationBarDestination(icon=ft.Icons.SEARCH, label="Search"),
 ft.NavigationBarDestination(icon=ft.Icons.PERSON, label="Profile"),
],
)

 return ft.Container(
 width=380,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(behavior.name, weight=ft.FontWeight.BOLD),
 bar,
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="NavigationBarLabelBehavior Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare destination label visibility strategies."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(behavior)
 for behavior in ft.NavigationBarLabelBehavior
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties



buildALWAYS\_HIDEclass-attributeinstance-attribute

buildALWAYS\_SHOWclass-attributeinstance-attribute

buildONLY\_SHOW\_SELECTEDclass-attributeinstance-attribute

# NavigationBarTheme

Customizes the appearance of NavigationBar controls across the app.

## Properties

- `bgcolor` - Overrides the default value for `NavigationBar.bgcolor`.
- `elevation` - Overrides the default value for `NavigationBar.elevation`.
- `height` - Overrides the default value for `NavigationBar` height.
- `indicator_color` - Overrides the default value for `NavigationBar.indicator_color`.
- `indicator_shape` - Overrides the default value for `NavigationBar.indicator_shape`.
- `label_behavior` - Overrides the default value for `NavigationBar.label_behavior`.
- `label_padding` - Overrides the default value for `NavigationBar.label_padding`.
- `label_text_style` - The style to merge with the default text style for `NavigationBarDestination` labels.
- `overlay_color` - Overrides the default value for `NavigationBar.overlay_color`.
- `shadow_color` - Overrides the default value for `NavigationBar.shadow_color`.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value for `NavigationBar.bgcolor`.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value for `NavigationBar.elevation`.

### `buildheightclass-attributeinstance-attribute`

`height: Number | None = None`

Overrides the default value for `NavigationBar` height.

### `buildindicator_colorclass-attributeinstance-attribute`

`indicator_color: ColorValue | None = None`

Overrides the default value for `NavigationBar.indicator_color`.

### `buildindicator_shapeclass-attributeinstance-attribute`

`indicator_shape: OutlinedBorder | None = None`

Overrides the default value for `NavigationBar.indicator_shape`.

### `buildlabel_behaviorclass-attributeinstance-attribute`

`label_behavior: NavigationBarLabelBehavior | None = None`

Overrides the default value for `NavigationBar.label_behavior`.

### `buildlabel_paddingclass-attributeinstance-attribute`

`label_padding: PaddingValue | None = None`

Overrides the default value for `NavigationBar.label_padding`.

### `buildlabel_text_styleclass-attributeinstance-attribute`

`label_text_style: ControlStateValue[TextStyle] | None = None`

The style to merge with the default text style for `NavigationBarDestination` labels.

### `buildoverlay_colorclass-attributeinstance-attribute`

`overlay_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value for `NavigationBar.overlay_color`.

**buildshadow\_color**class-attributeinstance-attribute

`shadow_color: ColorValue | None = None`

Overrides the default value for `NavigationBar.shadow_color`.

# NavigationDrawerTheme

Customizes the appearance of descendant NavigationDrawer controls.

## Properties

- `bgcolor` - Overrides the default value for `NavigationDrawer.bgcolor`.
- `elevation` - Overrides the default value for `NavigationDrawer.elevation`.
- `icon_theme` - The theme to merge with the default icon theme for `NavigationDrawerDestination` icons.
- `indicator_color` - Overrides the default value for `NavigationDrawer.indicator_color`.
- `indicator_shape` - Overrides the default value for `NavigationDrawer.indicator_shape`.
- `indicator_size` - Overrides the default size of the `NavigationDrawer`'s selection indicator.
- `label_text_style` - The style to merge with the default text style for `NavigationDrawerDestination` labels.
- `shadow_color` - Overrides the default value for `NavigationDrawer.shadow_color`.
- `tile_height` - Overrides the default height of `NavigationDrawerDestination`.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value for `NavigationDrawer.bgcolor`.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value for `NavigationDrawer.elevation`.

### `buildicon_themeclass-attributeinstance-attribute`

`icon_theme: ControlStateValue[IconTheme] | None = None`

The theme to merge with the default icon theme for `NavigationDrawerDestination` icons.

Can be used to specify a different icon theme when the icon is selected.

### `buildindicator_colorclass-attributeinstance-attribute`

`indicator_color: ColorValue | None = None`

Overrides the default value for `NavigationDrawer.indicator_color`.

### `buildindicator_shapeclass-attributeinstance-attribute`

`indicator_shape: OutlinedBorder | None = None`

Overrides the default value for `NavigationDrawer.indicator_shape`.

### `buildindicator_sizeclass-attributeinstance-attribute`

`indicator_size: Size | None = None`

Overrides the default size of the `NavigationDrawer`'s selection indicator.

### `buildlabel_text_styleclass-attributeinstance-attribute`

`label_text_style: ControlStateValue[TextStyle] | None = None`

The style to merge with the default text style for `NavigationDrawerDestination` labels.

Can be used to specify a different style when the label is selected.

### `buildshadow_colorclass-attributeinstance-attribute`

`shadow_color: ColorValue | None = None`

Overrides the default value for `NavigationDrawer.shadow_color`.

### `buildtile_heightclass-attributeinstance-attribute`

`tile_height: Number | None = None`

Overrides the default height of `NavigationDrawerDestination`.

# NavigationRailLabelType

Defines how destination labels are shown in a NavigationRail.

This setting affects the non-extended rail layout (extended). When the rail is extended, labels are shown next to icons regardless of this value.

Inherits: enum.Enum

## Properties

- ALL - Shows labels for all destinations.
- NONE - Hides labels for all destinations.
- SELECTED - Shows the label only for the selected destination.

## Examples

### Navigation Rail Label Type showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(label_type: ft.NavigationRailLabelType) -> ft.Container:
 rail = ft.NavigationRail(
 width=100,
 height=220,
 selected_index=1,
 label_type=label_type,
 destinations=[
 ft.NavigationRailDestination(icon=ft.Icons.HOME, label="Home"),
 ft.NavigationRailDestination(icon=ft.Icons.SEARCH, label="Search"),
 ft.NavigationRailDestination(icon=ft.Icons.PERSON, label="Profile"),
],
)

 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 controls=[
 ft.Text(label_type.name, weight=ft.FontWeight.BOLD),
 rail,
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="NavigationRailLabelType Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare label visibility in compact navigation rail."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(t) for t in ft.NavigationRailLabelType],
),
],
),
),
)

if __name__ == "__main__":
```

```
ft.run(main)
```

## Properties

**buildALLclass-attributeinstance-attribute**

Shows labels for all destinations.

**buildNONEclass-attributeinstance-attribute**

Hides labels for all destinations.

Only destination icons are shown.

**buildSELECTEDclass-attributeinstance-attribute**

Shows the label only for the selected destination.

As selection changes, the visible label transitions to the newly selected destination.

# NavigationRailTheme

Customizes the appearance of NavigationRail across the app.

## Properties

- `bgcolor` - Color to be used for the NavigationRail's background.
- `elevation` - The z-coordinate to be used for the NavigationRail's elevation.
- `group_alignment` - The alignment for the NavigationRail.destinations as they are positioned within the NavigationRail.
- `indicator_color` - Overrides the default value of NavigationRail.indicator\_color in all descendant NavigationRail controls.
- `indicator_shape` - Overrides the default value of NavigationRail.indicator\_shape in all descendant NavigationRail controls.
- `label_type` - The type that defines the layout and behavior of the labels in the NavigationRail.
- `min_extended_width` - Overrides the default value of NavigationRail.min\_extended\_width in all descendant NavigationRail controls when they are extended.
- `min_width` - Overrides the default value of NavigationRail.min\_width in all descendant NavigationRail controls when they are not extended.
- `selected_label_text_style` - Overrides the default value of NavigationRail.selected\_label\_text\_style in all descendant NavigationRail controls.
- `unselected_label_text_style` - Overrides the default value of NavigationRail.unselected\_label\_text\_style in all descendant NavigationRail controls.
- `use_indicator` - Overrides the default value of NavigationRail.use\_indicator in all descendant NavigationRail controls.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Color to be used for the NavigationRail's background.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

The z-coordinate to be used for the NavigationRail's elevation.

### `buildgroup_alignmentclass-attributeinstance-attribute`

`group_alignment: Number | None = None`

The alignment for the NavigationRail.destinations as they are positioned within the NavigationRail.

### `buildindicator_colorclass-attributeinstance-attribute`

`indicator_color: ColorValue | None = None`

Overrides the default value of NavigationRail.indicator\_color in all descendant NavigationRail controls. when NavigationRailTheme.use\_indicator is true.

### `buildindicator_shapeclass-attributeinstance-attribute`

`indicator_shape: OutlinedBorder | None = None`

Overrides the default value of NavigationRail.indicator\_shape in all descendant NavigationRail controls.

### `buildlabel_typeclass-attributeinstance-attribute`

`label_type: NavigationRailLabelType | None = None`

The type that defines the layout and behavior of the labels in the NavigationRail.

### `buildmin_extended_widthclass-attributeinstance-attribute`

`min_extended_width: Number | None = None`



Overrides the default value of `NavigationRail.min_extended_width` in all descendant `NavigationRail` controls when they are extended.

### **buildmin\_widthclass-attributeinstance-attribute**

`min_width: Number | None = None`

Overrides the default value of `NavigationRail.min_width` in all descendant `NavigationRail` controls when they are not extended.

### **buildselected\_label\_text\_styleclass-attributeinstance-attribute**

`selected_label_text_style: TextStyle | None = None`

Overrides the default value of `NavigationRail.selected_label_text_style` in all descendant `NavigationRail` controls.

### **buildunselected\_label\_text\_styleclass-attributeinstance-attribute**

`unselected_label_text_style: TextStyle | None = None`

Overrides the default value of `NavigationRail.unselected_label_text_style` in all descendant `NavigationRail` controls.

### **builduse\_indicatorclass-attributeinstance-attribute**

`use_indicator: bool | None = None`

Overrides the default value of `NavigationRail.use_indicator` in all descendant `NavigationRail` controls.

# NotchShape

A shape with a notch in its outline.

Typically used as the outline of a 'host' control to make a notch that accommodates a 'guest' control. e.g the BottomAppBar may have a notch to accommodate the FloatingActionButton.

This class is not intended to be used directly. See usable derivatives:

- AutomaticNotchShape
- CircularRectangleNotchShape

# Observable

Mixin: notifies when fields change; auto-wraps lists/dicts to be observable.

## Example

```
import flet as ft
from dataclasses import dataclass

@ft.observable
@dataclass
class MyDataClass:
 x: int
 y: int

obj = MyDataClass(1, 2)

def listener(sender, field):
 print(f"Changed: {field} in {sender}")

obj.subscribe(listener)
obj.x = 3
obj.y = 4
```

## Methods

- `notify` - Manually notify listeners that something changed.
- `subscribe` - Register a change listener.

## Methods

### `deployed_codenotify`

```
notify()
```

Manually notify listeners that something changed.

### `deployed_codesubscribe`

```
subscribe(fn: Listener) -> Callable[[], None]
```

Register a change listener.

Parameters:

- `fn (Listener)` - Listener callback receiving (sender, field).

Returns:

- `Callable[[], None]` - A disposer function that unsubscribes the listener.

## @observable

Makes a class observable by mixing in Observable. Can be applied to any class including `dataclasses.dataclass`. For dataclasses, decorator can be placed either above or below the `@dataclass` decorator.

### Example

```
from dataclasses import dataclass
import flet as ft
```

```
@ft.observable
@dataclass
class MyDataClass:
 x: int
 y: int
```

# Offset

A 2D floating-point offset.

## Properties

- `distance` - The magnitude of the offset.
- `filter_quality` - The filter quality with which to apply this transform as a bitmap operation.
- `transform_hit_tests` - Whether to apply the transformation when performing hit tests.
- `x` - The horizontal offset.
- `y` - The vertical offset.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `builddistanceproperty`

`distance: float`

The magnitude of the offset.

### `buildfilter_qualityclass-attributeinstance-attribute`

`filter_quality: FilterQuality | None = None`

The filter quality with which to apply this transform as a bitmap operation.

### `buildtransform_hit_testsclass-attributeinstance-attribute`

`transform_hit_tests: bool = True`

Whether to apply the transformation when performing hit tests.

### `buildxclass-attributeinstance-attribute`

`x: Number = 0`

The horizontal offset.

### `buildyclass-attributeinstance-attribute`

`y: Number = 0`

The vertical offset.

## Methods

### `deployed_codecopy`

```
copy(
 x: Number | None = None,
 y: Number | None = None,
 transform_hit_tests: bool | None = None,
 filter_quality: FilterQuality | None = None,
) -> Offset
```

Returns a copy of this object with the specified properties overridden.

## on\_mounted

Run exactly once after the component mounts.

Parameters:

- `fn` - A function to run after the component mounts.

`on_mounted`

# OnReorderEvent

Payload for ReorderableView events related to item reordering.

Inherits: Event

## Properties

- `new_index` - The new position of the item after the reordering, if available.
- `old_index` - The previous position of the item before the reordering, if available.

## Properties

**buildnew\_index**class-attributeinstance-attribute

`new_index: int | None = None`

The new position of the item after the reordering, if available.

Will be non-None only for the following events: `on_reorder`, `on_reorder_end`.

**buildold\_index**class-attributeinstance-attribute

`old_index: int | None = None`

The previous position of the item before the reordering, if available.

Will be non-None only for the following events: `on_reorder`, `on_reorder_start`.

# OnScrollEvent

Payload for ScrollableControl.on\_scroll handlers.

Inherits: Event

## Properties

- `at_edge` - Whether pixels is exactly at either scroll edge.
- `direction` - User scroll direction reported by Flutter.
- `event_type` - Logical type of the scroll notification.
- `extent_after` - Quantity of content conceptually "after" the viewport.
- `extent_before` - Quantity of content conceptually "before" the viewport.
- `extent_total` - Total conceptual content extent available to the scrollable.
- `max_scroll_extent` - Maximum in-range value for pixels.
- `min_scroll_extent` - Minimum in-range value for pixels.
- `out_of_range` - Whether pixels is outside scroll extents.
- `overscroll` - Logical pixels that were prevented from being applied to pixels.
- `pixels` - Current scroll offset in logical pixels.
- `scroll_delta` - Delta in logical pixels since the previous update.
- `velocity` - Scroll velocity when overscroll occurred, in logical pixels per second.
- `viewport_dimension` - Visible viewport extent along the scroll axis, in logical pixels.

## Properties

### `buildat_edgeproperty`

`at_edge: bool`

Whether pixels is exactly at either scroll edge.

Returns:

- `bool` - True when pixels equals `min_scroll_extent` or `max_scroll_extent`; otherwise False.

### `builddirectionclass-attributeinstance-attribute`

`direction: ScrollDirection | None = None`

User scroll direction reported by Flutter.

Populated for ScrollType.USER notifications.

### `buildevent_typeinstance-attribute`

`event_type: ScrollType`

Logical type of the scroll notification.

Determines which optional fields are populated:

- `ScrollType.UPDATE`: `scroll_delta`
- `ScrollType.USER`: `direction`
- `ScrollType.OVERSCROLL`: `overscroll` and `velocity`

### `buildextent_afterproperty`

`extent_after: float`

Quantity of content conceptually "after" the viewport.

This corresponds to the content following the visible region in the current scroll direction.



## buildextent\_beforeproperty

extent\_before: float

Quantity of content conceptually "before" the viewport.

This corresponds to the content preceding the visible region in the current scroll direction.

## buildextent\_totalproperty

extent\_total: float

Total conceptual content extent available to the scrollable.

Equivalent to:  $\text{max\_scroll\_extent} - \text{min\_scroll\_extent}$

- viewport\_dimension.

## buildmax\_scroll\_extentinstance-attribute

max\_scroll\_extent: float

Maximum in-range value for pixels.

pixels may still be out\_of\_range during overscroll. For unbounded scrollables this value can be positive infinity.

## buildmin\_scroll\_extentinstance-attribute

min\_scroll\_extent: float

Minimum in-range value for pixels.

pixels may still be out\_of\_range during overscroll. For unbounded scrollables this value can be negative infinity.

## buildout\_of\_rangeproperty

out\_of\_range: bool

Whether pixels is outside scroll extents.

Returns:

- bool - True if  $\text{pixels} < \text{min\_scroll\_extent}$  or  $\text{pixels} > \text{max\_scroll\_extent}$ ; otherwise False.

## buildoverscrollclass-attributeinstance-attribute

overscroll: float | None = None

Logical pixels that were prevented from being applied to pixels.

Negative values indicate overscroll on the start side; positive values indicate overscroll on the end side. Populated for ScrollType.OVERSCROLL.

## buildpixelsinstance-attribute

pixels: float

Current scroll offset in logical pixels.

## buildscroll\_deltainstance-attribute

scroll\_delta: float | None = None

Delta in logical pixels since the previous update.

Populated for ScrollType.UPDATE notifications.

## buildvelocityclass-attributeinstance-attribute

velocity: float | None = None

Scroll velocity when overscroll occurred, in logical pixels per second.

Populated for ScrollType.OVERSCROLL.

## buildviewport\_dimensioninstance-attribute

viewport\_dimension: float

Visible viewport extent along the scroll axis, in logical pixels.

## on\_unmounted

Run exactly once when the component unmounts.

Parameters:

- fn - A function to run when the component unmounts.

`on_unmounted`

## on\_updated

Run after each post-mount render (or when dependencies change). With dependencies=None this fires every update; with dependencies=[...] only on changes.

Parameters:

- fn - A function to run after each post-mount render (or when dependencies change).
- dependencies - If present, fn is only run when one of the dependencies has changed. If absent, fn is run after every render.

on\_updated

# Orientation

Represents the layout orientation.

Inherits: `enum.Enum`

## Properties

- LANDSCAPE - Orientation with greater width than height.
- PORTRAIT - Orientation with greater height than width.

## Examples

### Orientation showcase

[play\\_arrowTry Online](#)

```
from datetime import time

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 def showcase_card(orientation: ft.Orientation) -> ft.Container:
 picker = ft.TimePicker(
 value=time(hour=10, minute=30),
 help_text=f"{orientation.name} time picker",
 orientation=orientation,
)

 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(orientation.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open TimePicker",
 icon=ft.Icons.ACCESS_TIME,
 on_click=lambda _, p=picker: page.show_dialog(p),
),
],
),
)

 page.appbar = ft.AppBar(title="Orientation Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("TimePicker supports PORTRAIT and LANDSCAPE layouts."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(orientation) for orientation in ft.Orientation
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildLANDSCAPEclass-attributeinstance-attribute**

Orientation with greater width than height.

**buildPORTRAITclass-attributeinstance-attribute**

Orientation with greater height than width.

# OutlinedBorder

An abstract class that can be used to create custom borders.

See subclasses/implementations:

- BeveledRectangleBorder
- ContinuousRectangleBorder
- CircleBorder
- RoundedRectangleBorder
- StadiumBorder

Inherits: ShapeBorder

## Properties

- side - The border outline's color and weight.

## Properties

buildsideclass-attributeinstance-attribute

side: BorderSide | None = None

The border outline's color and weight.

# OutlinedButtonTheme

Customizes the appearance of OutlinedButton across the app.

## Properties

- `style` - Overrides the default value of `OutlinedButton.style` in all descendant `OutlinedButton` controls.

## Properties

`buildstyleclass-attributeinstance-attribute`

`style: ButtonStyle | None = None`

Overrides the default value of `OutlinedButton.style` in all descendant `OutlinedButton` controls.

# OverlayVisibilityMode

Visibility of text field overlays based on the state of the current text entry.

Inherits: `enum.Enum`

## Properties

- ALWAYS - Always show the overlay regardless of the text entry state.
- EDITING - Overlay will only appear when the current text entry is not empty.
- NEVER - Overlay will never appear regardless of the text entry state.
- NOT\_EDITING - Overlay will only appear when the current text entry is empty.

## Examples

### Overlay Visibility Mode showcase

[play\\_arrow](#) Try Online

```
import flet as ft

def showcase_card(mode: ft.OverlayVisibilityMode) -> ft.Container:
 return ft.Container(
 width=380,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(mode.name, weight=ft.FontWeight.BOLD),
 ft.Text("Prefix/suffix visibility (prefilled):", size=11),
 ft.CupertinoTextField(
 value="Flet",
 placeholder_text="Search",
 prefix=ft.Icon(ft.CupertinoIcons.SEARCH),
 suffix=ft.Icon(ft.CupertinoIcons.MIC),
 prefix_visibility_mode=mode,
 suffix_visibility_mode=mode,
),
 ft.Text("Clear button visibility:", size=11),
 ft.CupertinoTextField(
 value="Clear me",
 placeholder_text="Type text",
 clear_button_visibility_mode=mode,
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="OverlayVisibilityMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare when Cupertino text field overlays appear."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(mode) for mode in ft.OverlayVisibilityMode
],
),
],
),
),
)

if __name__ == "__main__":
```



```
ft.run(main)
```

## Properties

**buildALWAYS**class-attributeinstance-attribute

Always show the overlay regardless of the text entry state.

**buildEDITING**class-attributeinstance-attribute

Overlay will only appear when the current text entry is not empty.

This includes prefilled text that the user did not type in manually. But does not include text in placeholders.

**buildNEVER**class-attributeinstance-attribute

Overlay will never appear regardless of the text entry state.

**buildNOT\_EDITING**class-attributeinstance-attribute

Overlay will only appear when the current text entry is empty.

This also includes not having prefilled text that the user did not type in manually. Texts in placeholders are ignored.

# Padding

Defines padding for all sides of a rectangle.

## Properties

- bottom - The padding value for the bottom side of the rectangle.
- left - The padding value for the left side of the rectangle.
- right - The padding value for the right side of the rectangle.
- top - The padding value for the top side of the rectangle.

## Methods

- all - Applies the same padding to all sides.
- only - Applies padding to the specified sides.
- symmetric - Applies vertical padding to top and bottom sides and horizontal padding to left and right sides.
- zero

## Properties

### buildbottomclass-attributeinstance-attribute

```
bottom: Number = 0
```

The padding value for the bottom side of the rectangle.

### buildleftclass-attributeinstance-attribute

```
left: Number = 0
```

The padding value for the left side of the rectangle.

### buildrightclass-attributeinstance-attribute

```
right: Number = 0
```

The padding value for the right side of the rectangle.

### buildtopclass-attributeinstance-attribute

```
top: Number = 0
```

The padding value for the top side of the rectangle.

## Methods

### deployed\_codeallclassmethod

```
all(value: Number) -> Padding
```

Applies the same padding to all sides.

### deployed\_codeonlyclassmethod

```
only(
 left: Number = 0,
 top: Number = 0,
 right: Number = 0,
 bottom: Number = 0,
) -> Padding
```

Applies padding to the specified sides.

### deployed\_codesymmetricclassmethod

```
symmetric(
 vertical: Number = 0, horizontal: Number = 0
) -> Padding
```

Applies vertical padding to top and bottom sides and horizontal padding to left and right sides.

### deployed\_codezeroclassmethod

zero() -> Padding

## Examples

### Example 1

play\_arrowTry Online

```
import flet as ft

def main(page: ft.Page):
 page.title = "Containers with different padding"

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Row(
 controls=[
 ft.Container(
 content=ft.Button("container_1"),
 bgcolor=ft.Colors.AMBER,
 padding=ft.Padding.all(10),
 width=150,
 height=150,
),
 ft.Container(
 content=ft.Button("container_2"),
 bgcolor=ft.Colors.AMBER,
 padding=ft.Padding.all(20),
 width=150,
 height=150,
),
 ft.Container(
 content=ft.Button("container_3"),
 bgcolor=ft.Colors.AMBER,
 padding=ft.Padding.symmetric(horizontal=10),
 width=150,
 height=150,
),
 ft.Container(
 content=ft.Button("container_4"),
 bgcolor=ft.Colors.AMBER,
 padding=ft.Padding.only(left=10),
 width=150,
 height=150,
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: container - <https://flet.dev/docs/examples/controls/core/types/padding/media/container.png>

# PageMediaData

Represents the environmental metrics of a page or window.

This data is updated whenever the platform window or layout changes, such as when rotating a device, resizing a browser window, or adjusting system UI elements like the keyboard or safe areas.

## Properties

- `always_use_24_hour_format` - Whether to use 24-hour format when formatting time.
- `device_pixel_ratio` - The number of device pixels for each logical pixel.
- `orientation` - The orientation of the page.
- `padding` - The space surrounding the entire display, accounting for system UI like notches and status bars.
- `view_insets` - Areas obscured by system UI overlays, such as the on-screen keyboard or system gesture areas.
- `view_padding` - Similar to padding, but includes padding that is always reserved (even when the system UI is hidden).

## Properties

### `buildalways_use_24_hour_format`class-attributeinstance-attribute

`always_use_24_hour_format: bool = False`

Whether to use 24-hour format when formatting time.

#### Note

The behavior of this flag is different across platforms:

- On Android this flag is reported directly from the user settings called "Use 24-hour format". It applies to any locale used by the application, whether it is the system-wide locale, or the custom locale set by the application.
- On iOS this flag is set to true when the user setting called "24-Hour Time" is set or the system-wide locale's default uses 24-hour formatting.

### `builddevice_pixel_ratio`instance-attribute

`device_pixel_ratio: float`

The number of device pixels for each logical pixel.

### `buildorientation`instance-attribute

`orientation: Orientation`

The orientation of the page.

### `buildpadding`instance-attribute

`padding: Padding`

The space surrounding the entire display, accounting for system UI like notches and status bars.

### `buildview_insets`instance-attribute

`view_insets: Padding`

Areas obscured by system UI overlays, such as the on-screen keyboard or system gesture areas.

### `buildview_padding`instance-attribute

`view_padding: Padding`

Similar to padding, but includes padding that is always reserved (even when the system UI is hidden).

# PagePlatform

Supported platforms for a page, including mobile and desktop systems. Each platform corresponds to a specific operating system or environment.

Inherits: `enum.Enum`

## Properties

- `ANDROID`
- `ANDROID_TV`
- `IOS`
- `LINUX`
- `MACOS`
- `WINDOWS`

## Methods

- `is_apple` - Whether this PagePlatform instance is an Apple (iOS or macOS) platform.
- `is_desktop` - Whether this PagePlatform instance is a desktop (macOS, Windows, Linux) platform.
- `is_mobile` - Whether this PagePlatform instance is a mobile (iOS or Android) platform.

## Properties

`buildANDROIDclass-attributeinstance-attribute`

`buildANDROID_TVclass-attributeinstance-attribute`

`buildIOSclass-attributeinstance-attribute`

`buildLINUXclass-attributeinstance-attribute`

`buildMACOSclass-attributeinstance-attribute`

`buildWINDOWSclass-attributeinstance-attribute`

## Methods

`deployed_codeis_apple`

`is_apple() -> bool`

Whether this PagePlatform instance is an Apple (iOS or macOS) platform.

`deployed_codeis_desktop`

`is_desktop() -> bool`

Whether this PagePlatform instance is a desktop (macOS, Windows, Linux) platform.

`deployed_codeis_mobile`

`is_mobile() -> bool`

Whether this PagePlatform instance is a mobile (iOS or Android) platform.

# PageResizeEvent

Event fired when the size of the containing window or browser is changed.

Typically used to adapt layout dynamically in response to resizes, such as switching between compact and expanded views in a responsive design.

Inherits: Event

## Properties

- height - The new height of the page in logical pixels.
- width - The new width of the page in logical pixels.

## Properties

### buildheightinstance-attribute

height: float

The new height of the page in logical pixels.

### buildwidthinstance-attribute

width: float

The new width of the page in logical pixels.

# PageTransitionsTheme

Per-platform mapping of route transition presets.

Assigned to `Theme.page_transitions` to override how Material routes animate on each target platform.

## Properties

- `android` - Transition preset for Android routes.
- `ios` - Transition preset for iOS routes.
- `linux` - Transition preset for Linux desktop routes.
- `macos` - Transition preset for macOS desktop routes.
- `windows` - Transition preset for Windows desktop routes.

## Properties

### `buildandroidclass-attributeinstance-attribute`

`android: PageTransitionTheme | None = None`

Transition preset for Android routes.

If `None`, defaults to `PageTransitionTheme.FADE_UPWARDS`.

### `buildiosclass-attributeinstance-attribute`

`ios: PageTransitionTheme | None = None`

Transition preset for iOS routes.

If `None`, defaults to `PageTransitionTheme.CUPERTINO`.

### `buildlinuxclass-attributeinstance-attribute`

`linux: PageTransitionTheme | None = None`

Transition preset for Linux desktop routes.

If `None`, defaults to `PageTransitionTheme.ZOOM`.

### `buildmacosclass-attributeinstance-attribute`

`macos: PageTransitionTheme | None = None`

Transition preset for macOS desktop routes.

If `None`, defaults to `PageTransitionTheme.ZOOM`.

### `buildwindowsclass-attributeinstance-attribute`

`windows: PageTransitionTheme | None = None`

Transition preset for Windows desktop routes.

If `None`, defaults to `PageTransitionTheme.ZOOM`.

# PageTransitionTheme

Page transition preset for route changes.

These could, for example, be used together with PageTransitionsTheme to define per-platform navigation animations.

Inherits: enum.Enum

## Properties

- CUPERTINO - Cupertino-style horizontal page transition, which matches native iOS page transitions.
- FADE\_FORWARDS - Fade-forward Material route transition, similar to the one provided by Android U.
- FADE\_UPWARDS - Fade-and-slide transition where the incoming page moves upward, similar to the one provided by Android O.
- NONE - Disables route transition animation.
- OPEN\_UPWARDS - Upward reveal transition with clipping/scrim, matching the transition used on Android P.
- PREDICTIVE - Predictive-back transition that allows peeking behind the current route on Android.
- ZOOM - Zoom/fade transition used by modern Material route animations, similar to the one provided in Android Q.

## Properties

### buildCUPERTINOclass-attributeinstance-attribute

Cupertino-style horizontal page transition, which matches native iOS page transitions.

### buildFADE\_FORWARDSclass-attributeinstance-attribute

Fade-forward Material route transition, similar to the one provided by Android U.

### buildFADE\_UPWARDSclass-attributeinstance-attribute

Fade-and-slide transition where the incoming page moves upward, similar to the one provided by Android O.

### buildNONEclass-attributeinstance-attribute

Disables route transition animation.

### buildOPEN\_UPWARDSclass-attributeinstance-attribute

Upward reveal transition with clipping/scrim, matching the transition used on Android P.

### buildPREDICTIVEclass-attributeinstance-attribute

Predictive-back transition that allows peeking behind the current route on Android.

### buildZOOMclass-attributeinstance-attribute

Zoom/fade transition used by modern Material route animations, similar to the one provided in Android Q.



# Paint

A description of the style to use when drawing a shape on the canvas.

## Properties

- `anti_alias` - Whether to apply anti-aliasing to lines and images drawn on the canvas.
- `blend_mode` - A blend mode to apply when a shape is drawn or a layer is composited.
- `blur_image` - Blur image when drawing it on a canvas.
- `color` - The color to use when stroking or filling a shape.
- `gradient` - Configures gradient paint.
- `stroke_cap` - TBD
- `stroke_dash_pattern` - TBD
- `stroke_join` - TBD
- `stroke_miter_limit` - TBD
- `stroke_width` - TBD
- `style` - TBD

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildanti_aliasclass-attributeinstance-attribute`

`anti_alias: bool | None = None`

Whether to apply anti-aliasing to lines and images drawn on the canvas.

Defaults to `True`.

### `buildblend_modeclass-attributeinstance-attribute`

`blend_mode: BlendMode | None = None`

A blend mode to apply when a shape is drawn or a layer is composited.

Defaults to `BlendMode.SRC_OVER`.

### `buildblur_imageclass-attributeinstance-attribute`

`blur_image: BlurValue | None = None`

Blur image when drawing it on a canvas.

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color to use when stroking or filling a shape.

Defaults to opaque black.

### `buildgradientclass-attributeinstance-attribute`

`gradient: PaintGradient | None = None`

Configures gradient paint.

### `buildstroke_capclass-attributeinstance-attribute`

`stroke_cap: StrokeCap | None = None`

TBD

### `buildstroke_dash_patternclass-attributeinstance-attribute`

`stroke_dash_pattern: list[Number] | None = None`

TBD

### buildstroke\_joinclass-attributeinstance-attribute

stroke\_join: StrokeJoin | None = None

TBD

### buildstroke\_miter\_limitclass-attributeinstance-attribute

stroke\_miter\_limit: Number | None = None

TBD

### buildstroke\_widthclass-attributeinstance-attribute

stroke\_width: Number | None = None

TBD

### buildstyleclass-attributeinstance-attribute

style: PaintingStyle | None = None

TBD

## Methods

### deployed\_codecopy

```
copy(
 color: ColorValue | None = None,
 blend_mode: BlendMode | None = None,
 blur_image: BlurValue | None = None,
 anti_alias: bool | None = None,
 gradient: PaintGradient | None = None,
 stroke_cap: StrokeCap | None = None,
 stroke_join: StrokeJoin | None = None,
 stroke_miter_limit: Number | None = None,
 stroke_width: Number | None = None,
 stroke_dash_pattern: list[Number] | None = None,
 style: PaintingStyle | None = None,
) -> Paint
```

Returns a copy of this object with the specified properties overridden.

# PaintGradient

Base class for gradients applied via `Paint.gradient`.

Use one of its concrete variants:

- `PaintLinearGradient`
- `PaintRadialGradient`
- `PaintSweepGradient`

## Examples

### Canvas paint

[play\\_arrowTry Online](#)

```
import math

import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Rect(
 x=10,
 y=10,
 width=100,
 height=100,
 border_radius=5,
 paint=ft.Paint(
 style=ft.PaintingStyle.FILL,
 gradient=ft.PaintLinearGradient(
 begin=(0, 10),
 end=(100, 50),
 colors=[ft.Colors.BLUE, ft.Colors.YELLOW],
),
),
),
],
),
 cv.Circle(
 x=60,
 y=170,
 radius=50,
 paint=ft.Paint(
 style=ft.PaintingStyle.FILL,
 gradient=ft.PaintRadialGradient(
 radius=50,
 center=(60, 170),
 colors=[ft.Colors.YELLOW, ft.Colors.BLUE],
),
),
),
 cv.Path(
 elements=[
 cv.Path.Arc(
 x=10,
 y=230,
 width=100,
 height=100,
 start_angle=3 * math.pi / 4,
 sweep_angle=3 * math.pi / 2,
),
],
 paint=ft.Paint(
 stroke_width=15,
 stroke_join=ft.StrokeJoin.ROUND,
 style=ft.PaintingStyle.STROKE,
 gradient=ft.PaintSweepGradient(
 start_angle=0,
 end_angle=math.pi * 2,
 rotation=3 * math.pi / 4,
 center=(60, 280),
```

```

 colors=[ft.Colors.YELLOW, ft.Colors.PURPLE],
 color_stops=[0.0, 1.0],
),
),
),
),
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

# PaintingStyle

Strategy used by Paint.style when drawing geometry.

Determines whether a shape is rendered as a filled interior or as an outlined contour.

Inherits: enum.Enum

## Properties

- FILL - Paint the interior of the shape.
- STROKE - Paint only the shape outline.

## Examples

### Painting Style showcase

play\_arrowTry Online

```
import flet as ft
import flet.canvas as cv

def showcase_card(style: ft.PaintingStyle) -> ft.Container:
 return ft.Container(
 width=250,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(style.name, weight=ft.FontWeight.BOLD),
 cv.Canvas(
 width=240,
 height=110,
 shapes=[
 cv.Circle(
 x=120,
 y=55,
 radius=34,
 paint=ft.Paint(
 style=style,
 stroke_width=10,
 color=ft.Colors.PRIMARY,
),
),
],
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="PaintingStyle Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare filled vs outlined rendering for the same shape."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(style) for style in ft.PaintingStyle],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildFILLclass-attributeinstance-attribute**

Paint the interior of the shape.

For example, circles and polygons are rendered as solid filled regions.

### **buildSTROKEclass-attributeinstance-attribute**

Paint only the shape outline.

Stroke thickness and joins/caps are controlled by properties such as `Paint.stroke_width`, `Paint.stroke_join`, and `Paint.stroke_cap`.

# PaintLinearGradient

More information on Linear gradient here.

Inherits: PaintGradient

## Properties

- **begin** - The offset at which stop 0.0 of the gradient is placed.
- **color\_stops** - A list of values from 0.0 to 1.0 that denote fractions along the gradient.
- **colors** - The colors the gradient should obtain at each of the stops.
- **end** - The offset at which stop 1.0 of the gradient is placed.
- **tile\_mode** - How this gradient should tile the plane beyond in the region before begin and after end.

## Methods

- **copy** - Returns a copy of this object with the specified properties overridden.

## Properties

### buildbegininstance-attribute

`begin: OffsetValue | None`

The offset at which stop 0.0 of the gradient is placed.

### buildcolor\_stopsclass-attributeinstance-attribute

`color_stops: list[Number] | None = None`

A list of values from 0.0 to 1.0 that denote fractions along the gradient.

#### Note

If non-none, this list must have the same length as colors. If the first value is not 0.0, then a stop with position 0.0 and a color equal to the first color in colors is implied. If the last value is not 1.0, then a stop with position 1.0 and a color equal to the last color in colors is implied.

### buildcolorsinstance-attribute

`colors: list[ColorValue]`

The colors the gradient should obtain at each of the stops. This list must contain at least two colors.

#### Note

If color\_stops is not None, this list must have the same length as color\_stops.

### buildendinstance-attribute

`end: OffsetValue | None`

The offset at which stop 1.0 of the gradient is placed.

### buildtile\_modeclass-attributeinstance-attribute

`tile_mode: GradientTileMode = GradientTileMode.CLAMP`

How this gradient should tile the plane beyond in the region before begin and after end.

## Methods

### deployed\_codecopy

```
copy(
 begin: OffsetValue | None = None,
 end: OffsetValue | None = None,
 colors: list[str] | None = None,
 color_stops: list[Number] | None = None,
 tile_mode: GradientTileMode | None = None,
) -> PaintLinearGradient
```

Returns a copy of this object with the specified properties overridden.

# Examples

## Canvas paint

play\_arrowTry Online

```
import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Rect(
 x=10,
 y=10,
 width=100,
 height=100,
 border_radius=5,
 paint=ft.Paint(
 style=ft.PaintingStyle.FILL,
 gradient=ft.PaintLinearGradient(
 begin=(0, 10),
 end=(100, 50),
 colors=[ft.Colors.BLUE, ft.Colors.YELLOW],
),
),
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: canvas-paint - [https://flet.dev/docs/examples/controls/core/types/paint\\_gradient/paint\\_linear\\_gradient/media/canvas\\_paint.png](https://flet.dev/docs/examples/controls/core/types/paint_gradient/paint_linear_gradient/media/canvas_paint.png)



# PaintRadialGradient

More information on Radial gradient <https://api.flutter.dev/flutter/dart-ui/Gradient/Gradient.radial.html>

Inherits: PaintGradient

## Properties

- center - The center of the gradient.
- color\_stops - A list of values from 0.0 to 1.0 that denote fractions along the gradient.
- colors - The colors the gradient should obtain at each of the stops.
- focal - The focal point of the gradient.
- focal\_radius - The radius of the focal point of gradient, as a fraction of the shortest side of the paint box.
- radius - The radius of the gradient.
- tile\_mode - How this gradient should tile the plane beyond in the region before begin and after end.

## Methods

- copy - Returns a copy of this object with the specified properties overridden.

## Properties

### buildcenterinstance-attribute

center: OffsetValue | None

The center of the gradient.

### buildcolor\_stopsclass-attributeinstance-attribute

color\_stops: list[Number] | None = None

A list of values from 0.0 to 1.0 that denote fractions along the gradient.

If provided, this list must have the same length as colors. If the first value is not 0.0, then a stop with position 0.0 and a color equal to the first color in colors is implied. If the last value is not 1.0, then a stop with position 1.0 and a color equal to the last color in colors is implied.

### buildcolorsinstance-attribute

colors: list[ColorValue]

The colors the gradient should obtain at each of the stops. This list must contain at least two colors.

If stops is provided, this list must have the same length as stops.

### buildfocalclass-attributeinstance-attribute

focal: OffsetValue | None = None

The focal point of the gradient. If specified, the gradient will appear to be focused along the vector from center to focal.

### buildfocal\_radiusclass-attributeinstance-attribute

focal\_radius: Number = 0.0

The radius of the focal point of gradient, as a fraction of the shortest side of the paint box.

For example, if a radial gradient is painted on a box that is 100.0 pixels wide and 200.0 pixels tall, then a radius of 1.0 will place the 1.0 stop at 100.0 pixels from the focal point.

### buildradiusinstance-attribute

radius: Number

The radius of the gradient.

### buildtile\_modeclass-attributeinstance-attribute

tile\_mode: GradientTileMode = GradientTileMode.CLAMP

How this gradient should tile the plane beyond in the region before begin and after end.

# Methods

## deployed\_codecopy

```
copy(
 center: OffsetValue | None = None,
 radius: Number | None = None,
 colors: list[str] | None = None,
 color_stops: list[Number] | None = None,
 tile_mode: GradientTileMode | None = None,
 focal: OffsetValue | None = None,
 focal_radius: Number | None = None,
) -> PaintRadialGradient
```

Returns a copy of this object with the specified properties overridden.

## Examples

### Canvas paint

play\_arrowTry Online

```
import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Circle(
 x=60,
 y=170,
 radius=50,
 paint=ft.Paint(
 style=ft.PaintingStyle.FILL,
 gradient=ft.PaintRadialGradient(
 radius=50,
 center=(60, 170),
 colors=[ft.Colors.YELLOW, ft.Colors.BLUE],
),
),
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: canvas-paint - [https://flet.dev/docs/examples/controls/core/types/paint\\_gradient/paint\\_radial\\_gradient/media/canvas\\_paint.png](https://flet.dev/docs/examples/controls/core/types/paint_gradient/paint_radial_gradient/media/canvas_paint.png)

# PaintSweepGradient

More information on Sweep gradient <https://api.flutter.dev/flutter/dart-ui/Gradient/Gradient.sweep.html>

Inherits: PaintGradient

## Properties

- `center` - The center of the gradient.
- `color_stops` - A list of values from 0.0 to 1.0 that denote fractions along the gradient.
- `colors` - The colors the gradient should obtain at each of the stops.
- `end_angle` - The angle in radians at which stop 1.0 of the gradient is placed.
- `rotation` - The rotation of the gradient in <https://en.wikipedia.org/wiki/Radian>, around the center-point of its bounding box.
- `start_angle` - The angle in <https://en.wikipedia.org/wiki/Radian> at which stop 0.0 of the gradient is placed.
- `tile_mode` - How this gradient should tile the plane beyond in the region before begin and after end.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### buildcenterinstance-attribute

`center: OffsetValue | None`

The center of the gradient.

### buildcolor\_stopsclass-attributeinstance-attribute

`color_stops: list[Number] | None = None`

A list of values from 0.0 to 1.0 that denote fractions along the gradient.

If provided, this list must have the same length as colors. If the first value is not 0.0, then a stop with position 0.0 and a color equal to the first color in colors is implied. If the last value is not 1.0, then a stop with position 1.0 and a color equal to the last color in colors is implied.

### buildcolorsinstance-attribute

`colors: list[str]`

The colors the gradient should obtain at each of the stops. This list must contain at least two colors.

If `color_stops` is provided, this list must have the same length as `color_stops`.

### buildend\_angleclass-attributeinstance-attribute

`end_angle: Number = math.pi * 2`

The angle in radians at which stop 1.0 of the gradient is placed. Defaults to  $\text{math.pi} * 2$ .

### buildrotationclass-attributeinstance-attribute

`rotation: Number | None = None`

The rotation of the gradient in <https://en.wikipedia.org/wiki/Radian>, around the center-point of its bounding box.

### buildstart\_angleclass-attributeinstance-attribute

`start_angle: Number = 0.0`

The angle in <https://en.wikipedia.org/wiki/Radian> at which stop 0.0 of the gradient is placed. Defaults to 0.0.

### buildtile\_modeclass-attributeinstance-attribute

`tile_mode: GradientTileMode = GradientTileMode.CLAMP`

How this gradient should tile the plane beyond in the region before begin and after end.

## Methods

## deployed\_codecopy

```
copy(
 center: OffsetValue | None = None,
 colors: list[str] | None = None,
 color_stops: list[Number] | None = None,
 tile_mode: GradientTileMode | None = None,
 start_angle: Number | None = None,
 end_angle: Number | None = None,
 rotation: Number | None = None,
) -> PaintSweepGradient
```

Returns a copy of this object with the specified properties overridden.

## Examples

### Canvas paint

play\_arrowTry Online

```
import math

import flet as ft
import flet.canvas as cv

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 cv.Canvas(
 width=float("inf"),
 expand=True,
 shapes=[
 cv.Path(
 elements=[
 cv.Path.Arc(
 x=10,
 y=230,
 width=100,
 height=100,
 start_angle=3 * math.pi / 4,
 sweep_angle=3 * math.pi / 2,
),
],
 paint=ft.Paint(
 stroke_width=15,
 stroke_join=ft.StrokeJoin.ROUND,
 style=ft.PaintingStyle.STROKE,
 gradient=ft.PaintSweepGradient(
 start_angle=0,
 end_angle=math.pi * 2,
 rotation=3 * math.pi / 4,
 center=(60, 280),
 colors=[ft.Colors.YELLOW, ft.Colors.PURPLE],
 color_stops=[0.0, 1.0],
),
),
),
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

Image: canvas-paint - [https://flet.dev/docs/examples/controls/core/types/paint\\_gradient/paint\\_sweep\\_gradient/media/canvas\\_paint.png](https://flet.dev/docs/examples/controls/core/types/paint_gradient/paint_sweep_gradient/media/canvas_paint.png)

# PlatformBrightnessChangeEvent

Event payload for platform brightness changes.

Delivered to `Page.on_platform_brightness_change`.

Inherits: `Event`

## Properties

- `brightness` - Current platform brightness mode.

## Properties

### `buildbrightnessinstance-attribute`

`brightness`: `Brightness`

Current platform brightness mode.

# PointerDeviceType

The kind of pointer device.

Inherits: `enum.Enum`

## Properties

- `INVERTED_STYLUS` - A pointer device with a stylus that has been inverted.
- `MOUSE` - A mouse-based pointer device.
- `STYLUS` - A pointer device with a stylus.
- `TOUCH` - A touch-based pointer device.
- `TRACKPAD` - Gestures from a trackpad.
- `UNKNOWN` - An unknown pointer device.

## Properties

`buildINVERTED_STYLUS`class-attributeinstance-attribute

A pointer device with a stylus that has been inverted.

`buildMOUSE`class-attributeinstance-attribute

A mouse-based pointer device.

`buildSTYLUS`class-attributeinstance-attribute

A pointer device with a stylus.

`buildTOUCH`class-attributeinstance-attribute

A touch-based pointer device.

`buildTRACKPAD`class-attributeinstance-attribute

Gestures from a trackpad.

`buildUNKNOWN`class-attributeinstance-attribute

An unknown pointer device.

# PointerEvent

Low-level pointer payload with detailed device metrics.

Inherits: Event

## Properties

- **device** - Unique identifier for the pointing device, reused across interactions.
- **distance** - The distance of the detected object from the input surface.
- **distance\_max** - The maximum value that distance can return for this pointer.
- **global\_delta** - The delta of the pointer's position since the event start, in logical pixels, within the global coordinate space.
- **global\_position** - Coordinate of the position of the pointer, in logical pixels in the global coordinate space.
- **kind** - The kind of input device for which the event was generated.
- **local\_delta** - The delta of the pointer's position since the event start, in logical pixels, within the local coordinate space.
- **local\_position** - The position transformed into the event receiver's local coordinate system according to transform.
- **orientation** - The orientation angle of the detected object, in radians.
- **pressure** - The pressure of the touch.
- **pressure\_max** - The maximum value that pressure can return for this pointer.
- **pressure\_min** - The minimum value that pressure can return for this pointer.
- **radius\_major** - The radius of the contact ellipse along the major axis, in logical pixels.
- **radius\_max** - The maximum value that could be reported for radius\_major and radius\_minor for this pointer, in logical pixels.
- **radius\_min** - The minimum value that could be reported for radius\_major and radius\_minor for this pointer, in logical pixels.
- **radius\_minor** - The radius of the contact ellipse along the minor axis, in logical pixels.
- **size** - The area of the screen being pressed.
- **tilt** - The tilt angle of the detected object, in radians.
- **timestamp** - Time of event dispatch, relative to an arbitrary timeline.

## Properties

### builddeviceclass-attributeinstance-attribute

```
device: float = field(metadata={'data_field': 'dev'})
```

Unique identifier for the pointing device, reused across interactions.

### builddistanceclass-attributeinstance-attribute

```
distance: float = field(metadata={'data_field': 'dist'})
```

The distance of the detected object from the input surface. For instance, this value could be the distance of a stylus or finger from a touch screen, in arbitrary units on an arbitrary (not necessarily linear) scale. If the pointer is down, this is 0.0 by definition.

### builddistance\_maxclass-attributeinstance-attribute

```
distance_max: float = field(
 metadata={"data_field": "distMax"}
)
```

The maximum value that distance can return for this pointer.

If this input device cannot detect "hover touch" input events, then this will be 0.0.

### buildglobal\_deltaclass-attributeinstance-attribute

```
global_delta: Offset | None = field(
 default=None, metadata={"data_field": "gd"}
)
```

The delta of the pointer's position since the event start, in logical pixels, within the global coordinate space.

### buildglobal\_positionclass-attributeinstance-attribute

```
global_position: Offset = field(
 metadata={"data_field": "g"}
)
```

Coordinate of the position of the pointer, in logical pixels in the global coordinate space.

### buildkindclass-attributeinstance-attribute

```
kind: PointerDeviceType = field(
 metadata={"data_field": "k"}
)
```

The kind of input device for which the event was generated.

### buildlocal\_deltaclass-attributeinstance-attribute

```
local_delta: Offset | None = field(
 default=None, metadata={"data_field": "ld"}
)
```

The delta of the pointer's position since the event start, in logical pixels, within the local coordinate space.

### buildlocal\_positionclass-attributeinstance-attribute

```
local_position: Offset = field(metadata={"data_field": "l"})
```

The position transformed into the event receiver's local coordinate system according to transform.

### buildorientationclass-attributeinstance-attribute

```
orientation: float = field(metadata={"data_field": "or"})
```

The orientation angle of the detected object, in radians.

### buildpressureclass-attributeinstance-attribute

```
pressure: float = field(metadata={"data_field": "ps"})
```

The pressure of the touch.

This value is a number ranging from 0.0, indicating a touch with no discernible pressure, to 1.0, indicating a touch with "normal" pressure, and possibly beyond, indicating a stronger touch. For devices that do not detect pressure (e.g. mice), returns 1.0.

### buildpressure\_maxclass-attributeinstance-attribute

```
pressure_max: float = field(metadata={"data_field": "pMax"})
```

The maximum value that pressure can return for this pointer. For devices that do not detect pressure (e.g. mice), returns 1.0. This will always be a greater than or equal to 1.0.

### buildpressure\_minclass-attributeinstance-attribute

```
pressure_min: float = field(metadata={"data_field": "pMin"})
```

The minimum value that pressure can return for this pointer.

For devices that do not detect pressure (e.g. mice), returns 1.0. This will always be a number less than or equal to 1.0.

### buildradius\_majorclass-attributeinstance-attribute

```
radius_major: float = field(metadata={"data_field": "rMj"})
```

The radius of the contact ellipse along the major axis, in logical pixels.

### buildradius\_maxclass-attributeinstance-attribute

```
radius_max: float = field(metadata={"data_field": "rMax"})
```

The maximum value that could be reported for radius\_major and radius\_minor for this pointer, in logical pixels.

### buildradius\_minclass-attributeinstance-attribute

```
radius_min: float = field(metadata={"data_field": "rMin"})
```

The minimum value that could be reported for radius\_major and radius\_minor for this pointer, in logical pixels.



### **buildradius\_minorclass-attributeinstance-attribute**

```
radius_minor: float = field(metadata={'data_field': 'rMn'})
```

The radius of the contact ellipse along the minor axis, in logical pixels.

### **buildsizeclass-attributeinstance-attribute**

```
size: float = field(metadata={'data_field': 'size'})
```

The area of the screen being pressed.

This value is scaled to a range between 0 and 1. It can be used to determine fat touch events. This value is only set on Android and is a device specific approximation within the range of detectable values. So, for example, the value of 0.1 could mean a touch with the tip of the finger, 0.2 a touch with full finger, and 0.3 the full palm.

### **buildtiltclass-attributeinstance-attribute**

```
tilt: float = field(metadata={'data_field': 'tilt'})
```

The tilt angle of the detected object, in radians.

### **buildtimestampclass-attributeinstance-attribute**

```
timestamp: Duration = field(metadata={'data_field': 'ts'})
```

Time of event dispatch, relative to an arbitrary timeline.

# PointMode

Defines how a list of points is interpreted when drawing a set of points.

Inherits: `enum.Enum`

## Properties

- LINES - Draw each sequence of two points as a line segment.
- POINTS - Draw each point separately.
- POLYGON - Draw the entire sequence of point as one line.

## Examples

### Point Mode showcase

[play\\_arrow](#) [Try Online](#)

```
import flet as ft
import flet.canvas as cv

POINTS = [
 (25, 75),
 (70, 30),
 (115, 75),
 (160, 30),
 (205, 75),
]

def showcase_card(mode: cv.PointMode) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(mode.name, weight=ft.FontWeight.BOLD),
 cv.Canvas(
 width=240,
 height=110,
 shapes=[
 cv.Points(
 points=POINTS,
 point_mode=mode,
 paint=ft.Paint(
 color=ft.Colors.PRIMARY,
 stroke_width=12,
 stroke_cap=ft.StrokeCap.ROUND,
),
),
],
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="PointMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare how the same coordinate list is interpreted "
 "by each point mode."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
```

```

 controls=[showcase_card(mode) for mode in cv.PointMode],
),
 1,
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildLINEsclass-attributeinstance-attribute

Draw each sequence of two points as a line segment. If the number of points is odd, then the last point is ignored. The lines are stroked as described by the Paint (ignoring Paint.style).

### buildPOINTSclass-attributeinstance-attribute

Draw each point separately. If the Paint.stroke\_cap is StrokeCap.ROUND, then each point is drawn as a circle with the diameter of the Paint.stroke\_width, filled as described by the Paint (ignoring Paint.style). Otherwise, each point is drawn as an axis-aligned square with sides of length Paint.stroke\_width, filled as described by the Paint (ignoring Paint.style).

### buildPOLYGONclass-attributeinstance-attribute

Draw the entire sequence of point as one line. The lines are stroked as described by the Paint (ignoring Paint.style).

# PopupMenuPosition

Used to configure how the PopupMenuButton positions its popup menu.

Inherits: enum.Enum

## Properties

- OVER - Menu is positioned over the anchor.
- UNDER - Menu is positioned under the anchor.

## Examples

### Popup Menu Position showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(position: ft.PopupMenuPosition) -> ft.Container:
 menu = ft.PopupMenuButton(
 menu_position=position,
 icon=ft.Text("Click me to open menu", color=ft.Colors.BLUE),
 items=[
 ft.PopupMenuItem(icon=ft.Icons.EDIT, content="Rename"),
 ft.PopupMenuItem(icon=ft.Icons.CONTENT_COPY, content="Duplicate"),
 ft.PopupMenuItem(icon=ft.Icons.DELETE, content="Delete"),
],
)

 return ft.Container(
 width=300,
 height=220,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 horizontal_alignment=ft.CrossAxisAlignment.CENTER,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 ft.Text(position.name, weight=ft.FontWeight.BOLD),
 menu,
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="PopupMenuPosition Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Open each popup menu to compare their positioning."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(position) for position in ft.PopupMenuPosition
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildOVERclass-attributeinstance-attribute**

Menu is positioned over the anchor.

**buildUNDERclass-attributeinstance-attribute**

Menu is positioned under the anchor.

# PopupMenuTheme

Customizes the appearance of PopupMenuButton across the app.

## Properties

- **color** - Overrides the default value of PopupMenuButton.bgcolor in all descendant PopupMenuButton controls.
- **elevation** - Overrides the default value of PopupMenuButton.elevation in all descendant PopupMenuButton controls.
- **enable\_feedback** - Overrides the default value of PopupMenuButton.enable\_feedback in all descendant PopupMenuButton controls
- **icon\_color** - Overrides the default value of PopupMenuButton.icon\_color in all descendant PopupMenuButton controls.
- **icon\_size** - Overrides the default value of PopupMenuButton.icon\_size in all descendant PopupMenuButton controls.
- **label\_text\_style** - Overrides the default value of PopupMenuitem.label\_text\_style in all descendant PopupMenuitem controls.
- **menu\_padding** - Overrides the default value of PopupMenuButton.menu\_padding in all descendant PopupMenuButton controls.
- **menu\_position** - Overrides the default value of PopupMenuButton.menu\_position in all descendant PopupMenuButton controls.
- **mouse\_cursor** - Overrides the default value of PopupMenuitem.mouse\_cursor in all descendant PopupMenuitem controls.
- **shadow\_color** - Overrides the default value of PopupMenuButton.shadow\_color in all descendant PopupMenuButton controls.
- **shape** - Overrides the default value of PopupMenuButton.shape in all descendant PopupMenuButton controls.

## Properties

### buildcolorclass-attributeinstance-attribute

`color: ColorValue | None = None`

Overrides the default value of PopupMenuButton.bgcolor in all descendant PopupMenuButton controls.

### buildelevationclass-attributeinstance-attribute

`elevation: Number | None = None`

Overrides the default value of PopupMenuButton.elevation in all descendant PopupMenuButton controls.

### buildenable\_feedbackclass-attributeinstance-attribute

`enable_feedback: bool | None = None`

Overrides the default value of PopupMenuButton.enable\_feedback in all descendant PopupMenuButton controls

### buildicon\_colorclass-attributeinstance-attribute

`icon_color: ColorValue | None = None`

Overrides the default value of PopupMenuButton.icon\_color in all descendant PopupMenuButton controls.

### buildicon\_sizeclass-attributeinstance-attribute

`icon_size: Number | None = None`

Overrides the default value of PopupMenuButton.icon\_size in all descendant PopupMenuButton controls.

### buildlabel\_text\_styleclass-attributeinstance-attribute

`label_text_style: TextStyle | None = None`

Overrides the default value of PopupMenuitem.label\_text\_style in all descendant PopupMenuitem controls.

### buildmenu\_paddingclass-attributeinstance-attribute

`menu_padding: PaddingValue | None = None`

Overrides the default value of PopupMenuButton.menu\_padding in all descendant PopupMenuButton controls.

### **buildmenu\_positionclass-attributeinstance-attribute**

`menu_position: PopupMenuPosition | None = None`

Overrides the default value of `PopupMenuButton.menu_position` in all descendant `PopupMenuButton` controls.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

Overrides the default value of `PopupMenuItem.mouse_cursor` in all descendant `PopupMenuItem` controls.

### **buildshadow\_colorclass-attributeinstance-attribute**

`shadow_color: ColorValue | None = None`

Overrides the default value of `PopupMenuButton.shadow_color` in all descendant `PopupMenuButton` controls.

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

Overrides the default value of `PopupMenuButton.shape` in all descendant `PopupMenuButton` controls.

# ProgressIndicatorTheme

Customizes the appearance of progress indicators (ProgressBar, ProgressRing) across the app.

## Properties

- `border_radius` - Overrides the default value for `ProgressBar.border_radius`.
- `circular_track_color` - Overrides the default value for `ProgressRing.bgcolor`.
- `circular_track_padding` - Overrides the default value for `ProgressRing.padding`.
- `color` - Overrides the default values for `ProgressBar.color` and `ProgressRing.color`.
- `linear_min_height` - Overrides the default value for `ProgressBar.bar_height`.
- `linear_track_color` - Overrides the default value for `ProgressBar.bgcolor`.
- `refresh_bgcolor` - Background color of that fills the circle under the RefreshIndicator (TBD).
- `size_constraints` - Overrides the default value for `ProgressRing.size_constraints`.
- `stop_indicator_color` - Overrides the default value for `ProgressBar.stop_indicator_color`.
- `stop_indicator_radius` - Overrides the default value for `ProgressBar.stop_indicator_radius`.
- `stroke_align` - Overrides the default value for `ProgressRing.stroke_align`.
- `stroke_cap` - Overrides the default value for `ProgressRing.stroke_cap`.
- `stroke_width` - Overrides the default value for `ProgressRing.stroke_width`.
- `track_gap` - Overrides the default values for `ProgressBar.track_gap` and `ProgressRing.track_gap`.
- `year_2023` - Overrides the default values for `ProgressBar.year_2023` and `ProgressRing.year_2023`.

## Properties

### `buildborder_radiusclass-attributeinstance-attribute`

`border_radius: BorderRadiusValue | None = None`

Overrides the default value for `ProgressBar.border_radius`.

### `buildcircular_track_colorclass-attributeinstance-attribute`

`circular_track_color: ColorValue | None = None`

Overrides the default value for `ProgressRing.bgcolor`.

### `buildcircular_track_paddingclass-attributeinstance-attribute`

`circular_track_padding: PaddingValue | None = None`

Overrides the default value for `ProgressRing.padding`.

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

Overrides the default values for `ProgressBar.color` and `ProgressRing.color`.

### `buildlinear_min_heightclass-attributeinstance-attribute`

`linear_min_height: Number | None = None`

Overrides the default value for `ProgressBar.bar_height`.

### `buildlinear_track_colorclass-attributeinstance-attribute`

`linear_track_color: ColorValue | None = None`

Overrides the default value for `ProgressBar.bgcolor`.

### `buildrefresh_bgcolorclass-attributeinstance-attribute`

`refresh_bgcolor: ColorValue | None = None`

Background color of that fills the circle under the RefreshIndicator (TBD).



### **buildsize\_constraintsclass-attributeinstance-attribute**

size\_constraints: BoxConstraints | None = None

Overrides the default value for ProgressRing.size\_constraints.

### **buildstop\_indicator\_colorclass-attributeinstance-attribute**

stop\_indicator\_color: ColorValue | None = None

Overrides the default value for ProgressBar.stop\_indicator\_color.

### **buildstop\_indicator\_radiusclass-attributeinstance-attribute**

stop\_indicator\_radius: Number | None = None

Overrides the default value for ProgressBar.stop\_indicator\_radius.

### **buildstroke\_alignclass-attributeinstance-attribute**

stroke\_align: Number | None = None

Overrides the default value for ProgressRing.stroke\_align.

### **buildstroke\_capclass-attributeinstance-attribute**

stroke\_cap: StrokeCap | None = None

Overrides the default value for ProgressRing.stroke\_cap.

### **buildstroke\_widthclass-attributeinstance-attribute**

stroke\_width: Number | None = None

Overrides the default value for ProgressRing.stroke\_width.

### **buildtrack\_gapclass-attributeinstance-attribute**

track\_gap: Number | None = None

Overrides the default values for ProgressBar.track\_gap and ProgressRing.track\_gap.

### **buildyear\_2023class-attributeinstance-attribute**

year\_2023: bool = False

Overrides the default values for ProgressBar.year\_2023 and ProgressRing.year\_2023.

# RadialGradient

Creates a radial gradient centered at center that ends at radius distance from the center.

More information [here](#).

Inherits: Gradient

## Properties

- **center** - The center of the gradient, as an offset into the (-1.0, -1.0) x (1.0, 1.0) square describing the gradient which will be mapped onto the paint box.
- **focal** - The focal point of the gradient.
- **focal\_radius** - The radius of the focal point of gradient, as a fraction of the shortest side of the paint box.
- **radius** - The radius of the gradient, as a fraction of the shortest side of the paint box.

## Properties

### buildcenterclass-attributeinstance-attribute

```
center: Alignment = field(
 default_factory=lambda: Alignment.CENTER
)
```

The center of the gradient, as an offset into the (-1.0, -1.0) x (1.0, 1.0) square describing the gradient which will be mapped onto the paint box. For example, an alignment of (0.0, 0.0) will place the radial gradient in the center of the box.

### buildfocalclass-attributeinstance-attribute

```
focal: Alignment | None = None
```

The focal point of the gradient. If specified, the gradient will appear to be focused along the vector from center to focal.

### buildfocal\_radiusclass-attributeinstance-attribute

```
focal_radius: Number = 0.0
```

The radius of the focal point of gradient, as a fraction of the shortest side of the paint box. For example, if a radial gradient is painted on a box that is 100.0 pixels wide and 200.0 pixels tall, then a radius of 1.0 will place the 1.0 stop at 100.0 pixels from the focal point.

### buildradiusclass-attributeinstance-attribute

```
radius: Number = 0.5
```

The radius of the gradient, as a fraction of the shortest side of the paint box. For example, if a radial gradient is painted on a box that is 100.0 pixels wide and 200.0 pixels tall, then a radius of 1.0 will place the 1.0 stop at 100.0 pixels from the center.

## Examples

### Container with radial gradient

[play\\_arrow](#)Try Online

```
import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Container(
 alignment=ft.Alignment.CENTER,
 width=150,
 height=150,
 border_radius=ft.BorderRadius.all(5),
 gradient=ft.RadialGradient(
 center=ft.Alignment(0.7, -0.6),
 radius=0.2,
 colors=["0xFFFFFFFF00", "0xFF0099FF"],
 stops=[0.4, 1.0],
),
),
],
),
),
)
```

```
)
),
 1,
)
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

Image: container - [https://flet.dev/docs/examples/controls/core/types/gradient/radial\\_gradient/media/container.png](https://flet.dev/docs/examples/controls/core/types/gradient/radial_gradient/media/container.png)

# RadioTheme

Defines default property values for descendant Radio controls.

## Properties

- `fill_color` - Overrides the default value of `Radio.fill_color` in all descendant Radio controls.
- `mouse_cursor` - Overrides the default value of `Radio.mouse_cursor` in all descendant Radio controls.
- `overlay_color` - Overrides the default value of `Radio.overlay_color` in all descendant Radio controls.
- `splash_radius` - Overrides the default value of `Radio.splash_radius` in all descendant Radio controls.
- `visual_density` - Overrides the default value of `Radio.visual_density` in all descendant Radio controls.

## Properties

### `buildfill_colorclass=attributeinstance=attribute`

`fill_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Radio.fill_color` in all descendant Radio controls.

### `buildmouse_cursorclass=attributeinstance=attribute`

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

Overrides the default value of `Radio.mouse_cursor` in all descendant Radio controls.

### `buildoverlay_colorclass=attributeinstance=attribute`

`overlay_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Radio.overlay_color` in all descendant Radio controls.

### `buildsplash_radiusclass=attributeinstance=attribute`

`splash_radius: Number | None = None`

Overrides the default value of `Radio.splash_radius` in all descendant Radio controls.

### `buildvisual_densityclass=attributeinstance=attribute`

`visual_density: VisualDensity | None = None`

Overrides the default value of `Radio.visual_density` in all descendant Radio controls.

# Rect

A 2D, axis-aligned, floating-point rectangle whose coordinates are relative to a given origin.

## Properties

- **bottom** - The offset of the bottom edge of this rectangle from the y-axis.
- **height** - The distance between the top and bottom edges of this rectangle.
- **left** - The offset of the left edge of this rectangle from the x-axis.
- **right** - The offset of the right edge of this rectangle from the x-axis.
- **size** - The distance between the upper-left corner and the lower-right corner of this rectangle.
- **top** - The offset of the top edge of this rectangle from the y-axis.
- **width** - The distance between the left and right edges of this rectangle.

## Methods

- **copy** - Returns a copy of this object with the specified properties overridden.
- **from\_center** - Constructs a rectangle from its center point, width, and height.
- **from\_lwth** - Construct a rectangle from its left and top edges, its width, and its height.
- **from\_points** - Construct the smallest rectangle that encloses the given offsets, treating them as vectors from the origin.

## Properties

### buildbottominstance-attribute

**bottom:** Number

The offset of the bottom edge of this rectangle from the y-axis.

### buildheightproperty

**height:** Number

The distance between the top and bottom edges of this rectangle.

### buildleftinstance-attribute

**left:** Number

The offset of the left edge of this rectangle from the x-axis.

### buildrightinstance-attribute

**right:** Number

The offset of the right edge of this rectangle from the x-axis.

### buildsizeproperty

**size:** Size

The distance between the upper-left corner and the lower-right corner of this rectangle.

### buildtopinstance-attribute

**top:** Number

The offset of the top edge of this rectangle from the y-axis.

### buildwidthproperty

**width:** Number

The distance between the left and right edges of this rectangle.

## Methods

### deployed\_codecopy

```
copy(
 left: Number | None = None,
 top: Number | None = None,
 right: Number | None = None,
 bottom: Number | None = None,
) -> Rect
```

Returns a copy of this object with the specified properties overridden.

### deployed\_codefrom\_centerclassmethod

```
from_center(
 center: Offset, width: Number, height: Number
)
```

Constructs a rectangle from its center point, width, and height. The center argument is assumed to be an offset from the origin.

### deployed\_codefrom\_lwthclassmethod

```
from_lwth(
 left: Number, top: Number, width: Number, height: Number
) -> Rect
```

Construct a rectangle from its left and top edges, its width, and its height.

### deployed\_codefrom\_pointsclassmethod

```
from_points(a: Offset, b: Offset)
```

Construct the smallest rectangle that encloses the given offsets, treating them as vectors from the origin.

# Ref

Utility class which allows defining a reference to a control.

Inherits: `typing.Generic`

## Properties

- `current` - Referenced object, or `None` if no object is set or it was garbage-collected.

## Properties

`buildcurrentpropertywritable`

`current: T | None`

Referenced object, or `None` if no object is set or it was garbage-collected.

# ResponsiveRowBreakpoint

Breakpoint names used by ResponsiveRow and responsive properties such as Control.col.

To define custom breakpoints, see ResponsiveRow.breakpoints.

Inherits: enum.Enum

## Properties

- LG - Large screens.
- MD - Medium screens.
- SM - Small screens.
- XL - Extra-large screens.
- XS - Extra small screens.
- XXL - Double extra-large screens.

## Properties

**buildLGclass-attributeinstance-attribute**

Large screens. Default min width: 992 px.

**buildMDclass-attributeinstance-attribute**

Medium screens. Default min width: 768 px.

**buildSMclass-attributeinstance-attribute**

Small screens. Default min width: 576 px.

**buildXLclass-attributeinstance-attribute**

Extra-large screens. Default min width: 1200 px.

**buildXSclass-attributeinstance-attribute**

Extra small screens. Default min width: 0 px.

**buildXXLclass-attributeinstance-attribute**

Double extra-large screens. Default min width: 1400 px.



# Rotate

Rotation configuration of an object.

## Properties

- `alignment` - The alignment of the rotation.
- `angle` - The rotation in clockwise radians.
- `filter_quality` - The filter quality with which to apply this transform as a bitmap operation.
- `origin` - The origin of the coordinate system (relative to the upper left corner), in logical pixels.
- `transform_hit_tests` - Whether to apply the transformation when performing hit tests.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

`alignment: Alignment | None = None`

The alignment of the rotation.

### `buildangleinstance-attribute`

`angle: Number`

The rotation in clockwise radians.

### `buildfilter_qualityclass-attributeinstance-attribute`

`filter_quality: FilterQuality | None = None`

The filter quality with which to apply this transform as a bitmap operation.

### `buildoriginclass-attributeinstance-attribute`

`origin: Offset | None = None`

The origin of the coordinate system (relative to the upper left corner), in logical pixels.

### `buildtransform_hit_testsclass-attributeinstance-attribute`

`transform_hit_tests: bool = True`

Whether to apply the transformation when performing hit tests.

## Methods

### `deployed_codecopy`

```
copy(
 angle: Number | None = None,
 alignment: Alignment | None = None,
 origin: Offset | None = None,
 transform_hit_tests: bool | None = None,
 filter_quality: FilterQuality | None = None,
) -> Rotate
```

Returns a copy of this object with the specified properties overridden.

# RoundedRectangleBorder

A border with rounded rectangle corners.

Inherits: OutlinedBorder

## Properties

- radius - The radius for each corner.

## Methods

- copy - Returns a copy of this object with the specified properties overridden.

## Properties

### buildradiusclass-attributeinstance-attribute

radius: BorderRadiusValue | None = None

The radius for each corner.

## Methods

### deployed\_codecopy

```
copy(
 side: BorderSide | None = None,
 radius: BorderRadiusValue | None = None,
) -> RoundedRectangleBorder
```

Returns a copy of this object with the specified properties overridden.

# Route

Defines a single route in the route tree.

Routes can be nested via children to create layout hierarchies. A route with component and children acts as a layout route — its component should call `use_route_outlet` to render the matched child.

Parameters:

- `path` (str | None, default: None) - Relative path segment. Supports dynamic segments (`:name`), optional segments (`:name?`), splats (`:name*`), and custom regex constraints (`:name(\\d+)`). None for pathless layout routes.
- `index` (bool, default: False) - When True, this route matches when the parent path matches exactly (no further segments). Index routes must not have path or children.
- `component` (Callable | None, default: None) - A `@component` function to render when this route matches.
- `children` (list[Route] | None, default: None) - Nested child routes.
- `loader` (Callable[..., Any] | None, default: None) - Optional data loader function. Called with the matched params dict when the route matches. Result is available via `use_route_loader_data`.
- `outlet` (bool, default: False) - When True and `manage_views=True`, this route acts as a layout that wraps its matched child via `use_route_outlet` within a single View, instead of each child becoming a separate View.
- `modal` (bool, default: False) - When True and `manage_views=True`, the route's View is rendered as a modal overlay on top of the existing stack instead of replacing it. The route's component should set `fullscreen_dialog=True` on its returned View so Flutter renders the slide-up presentation and a close (X) icon. Placement controls the base stack: \* Declared at the top level: a global modal. The base stack is rebuilt from the last non-modal location the Router saw (defaults to "/" on first render). \* Declared as a child of a non-modal route: a local modal. The base stack is the chain above the modal in the route tree, so deep-link works from the URL alone. On pop, the Router navigates to the resolved URL of the non-modal parent — never to `views[-2].route` — so a modal close is always a real navigation back to the base location.
- `recursive` (bool, default: False) - When True, the route can match itself as its own descendant — one matched `_RouteMatch` per consumed segment. Useful for tree-shaped URLs with unbounded depth (e.g. `/folder/a/b/c`) where each segment should become its own stack entry. Non-recursive children are tried before self-recursion at every depth so a more specific sibling (e.g. `example/:gp*`) wins over the recursive `:slug`.

## Properties

- `children`
- `component`
- `index`
- `loader`
- `modal`
- `outlet`
- `path`
- `recursive`

## Properties

### buildchildrenclass-attributeinstance-attribute

`children: list[Route] | None = field(default=None)`

### buildcomponentclass-attributeinstance-attribute

`component: Callable | None = None`

### buildindexclass-attributeinstance-attribute

`index: bool = False`

### buildloaderclass-attributeinstance-attribute

`loader: Callable[..., Any] | None = None`

**buildmodalclass-attributeinstance-attribute**

modal: bool = False

**buildoutletclass-attributeinstance-attribute**

outlet: bool = False

**buildpathclass-attributeinstance-attribute**

path: str | None = None

**buildrecursiveclass-attributeinstance-attribute**

recursive: bool = False

# RouteChangeEvent

Event payload for Page.on\_route\_change.

Inherits: Event

## Properties

- route - New route value after navigation state changed.

## Properties

buildrouteinstance-attribute

route: str

New route value after navigation state changed.

# RouteUrlStrategy

TBD

Inherits: `enum.Enum`

## Properties

- `HASH`
- `PATH`

## Properties

`buildHASHclass-attributeinstance-attribute`

`buildPATHclass-attributeinstance-attribute`

# Scale

Scaling configuration for an object.

## Properties

- `alignment` - Gives the origin of scale.
- `filter_quality` - The filter quality with which to apply this transform as a bitmap operation.
- `origin` - The origin of the coordinate system (relative to the upper left corner), in logical pixels.
- `scale` - `scale_x` and `scale_y` get the value of scale if scale is provided.
- `scale_x` - The scalar by which to multiply the x-axis.
- `scale_y` - The scalar by which to multiply the y-axis.
- `transform_hit_tests` - Whether to apply the transformation when performing hit tests.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

`alignment: Alignment | None = None`

Gives the origin of scale.

### `buildfilter_qualityclass-attributeinstance-attribute`

`filter_quality: FilterQuality | None = None`

The filter quality with which to apply this transform as a bitmap operation.

### `buildoriginclass-attributeinstance-attribute`

`origin: Offset | None = None`

The origin of the coordinate system (relative to the upper left corner), in logical pixels.

### `buildscaleclass-attributeinstance-attribute`

`scale: Number | None = None`

`scale_x` and `scale_y` get the value of scale if scale is provided.

### `buildscale_xclass-attributeinstance-attribute`

`scale_x: Number | None = None`

The scalar by which to multiply the x-axis.

### `buildscale_yclass-attributeinstance-attribute`

`scale_y: Number | None = None`

The scalar by which to multiply the y-axis.

### `buildtransform_hit_testsclass-attributeinstance-attribute`

`transform_hit_tests: bool = True`

Whether to apply the transformation when performing hit tests.

## Methods

### `deployed_code` `copy`

```
copy(
 scale: Number | None = None,
 scale_x: Number | None = None,
 scale_y: Number | None = None,
 alignment: Alignment | None = None,
 origin: Offset | None = None,
)
```

```
 transform_hit_tests: bool | None = None,
 filter_quality: FilterQuality | None = None,
) -> Scale
```

Returns a copy of this object with the specified properties overridden.



# ScaleEndEvent

End payload for scale/transform interaction sequences.

Inherits: Event

## Properties

- `pointer_count` - The number of pointers being tracked by the gesture recognizer.
- `velocity` - The velocity of the last pointer to be lifted off of the screen, in pixels per second.

## Properties

### `buildpointer_count`class-attributeinstance-attribute

```
pointer_count: int = field(metadata={'data_field': 'pc'})
```

The number of pointers being tracked by the gesture recognizer.

Typically this is the number of fingers being used to pan the control using the gesture recognizer.

### `buildvelocity`class-attributeinstance-attribute

```
velocity: Offset = field(metadata={'data_field': 'v'})
```

The velocity of the last pointer to be lifted off of the screen, in pixels per second.

# ScaleStartEvent

Start payload for scale/transform interaction sequences.

Inherits: Event

## Properties

- `global_focal_point` - The initial focal point of the pointers in contact with the screen, in global coordinates.
- `local_focal_point` - The initial focal point of the pointers in contact with the screen, in local coordinates.
- `pointer_count` - The number of pointers being tracked by the gesture recognizer.
- `timestamp` - Recorded timestamp of the source pointer event that triggered the scale event.

## Properties

### `buildglobal_focal_pointclass-attributeinstance-attribute`

```
global_focal_point: Offset = field(
 metadata={"data_field": "gfp"}
)
```

The initial focal point of the pointers in contact with the screen, in global coordinates.

### `buildlocal_focal_pointclass-attributeinstance-attribute`

```
local_focal_point: Offset = field(
 metadata={"data_field": "lfp"}
)
```

The initial focal point of the pointers in contact with the screen, in local coordinates.

### `buildpointer_countclass-attributeinstance-attribute`

```
pointer_count: int = field(metadata={"data_field": "pc"})
```

The number of pointers being tracked by the gesture recognizer.

Typically this is the number of fingers being used to pan the control using the gesture recognizer.

### `buildtimestampclass-attributeinstance-attribute`

```
timestamp: Duration | None = field(
 default=None, metadata={"data_field": "ts"}
)
```

Recorded timestamp of the source pointer event that triggered the scale event.

Could be None if triggered from proxied events such as accessibility.

# ScaleUpdateEvent

Continuous payload for scale, pan, and rotation updates.

Inherits: Event

## Properties

- `focal_point_delta` - The amount the gesture's focal point has moved in the coordinate space of the event receiver since the previous update.
- `global_focal_point` - The focal point of the pointers in contact with the screen, in global coordinates.
- `horizontal_scale` - The scale implied by the average distance along the horizontal axis between the pointers in contact with the screen.
- `local_focal_point` - The focal point of the pointers in contact with the screen, in local coordinates.
- `pointer_count` - The number of pointers being tracked by the gesture recognizer.
- `rotation` - The angle (in radians) implied by the first two pointers to enter in contact with the screen.
- `scale` - The scale implied by the average distance between the pointers in contact with the screen.
- `timestamp` - Recorded timestamp of the source pointer event that triggered the scale event.
- `vertical_scale` - The scale implied by the average distance along the vertical axis between the pointers in contact with the screen.

## Properties

### `buildfocal_point_delta`class-attributeinstance-attribute

```
focal_point_delta: Offset = field(
 metadata={"data_field": "fpd"}
)
```

The amount the gesture's focal point has moved in the coordinate space of the event receiver since the previous update.

### `buildglobal_focal_point`class-attributeinstance-attribute

```
global_focal_point: Offset = field(
 metadata={"data_field": "gfp"}
)
```

The focal point of the pointers in contact with the screen, in global coordinates.

### `buildhorizontal_scale`class-attributeinstance-attribute

```
horizontal_scale: float = field(
 metadata={"data_field": "hs"}
)
```

The scale implied by the average distance along the horizontal axis between the pointers in contact with the screen.

This value must be greater than or equal to zero.

### `buildlocal_focal_point`class-attributeinstance-attribute

```
local_focal_point: Offset = field(
 metadata={"data_field": "lfp"}
)
```

The focal point of the pointers in contact with the screen, in local coordinates.

### `buildpointer_count`class-attributeinstance-attribute

```
pointer_count: int = field(metadata={'data_field': 'pc'})
```

The number of pointers being tracked by the gesture recognizer.

Typically this is the number of fingers being used to pan the widget using the gesture recognizer. Due to platform limitations, trackpad gestures count as two fingers even if more than two fingers are used.

### `buildrotation`class-attributeinstance-attribute

```
rotation: float = field(metadata={'data_field': 'rot'})
```

The angle (in radians) implied by the first two pointers to enter in contact with the screen.

### **buildscaleclass-attributeinstance-attribute**

```
scale: float = field(metadata={'data_field': 's'})
```

The scale implied by the average distance between the pointers in contact with the screen.

This value must be greater than or equal to zero.

### **buildtimestampclass-attributeinstance-attribute**

```
timestamp: Duration | None = field(
 default=None, metadata={"data_field": "ts"}
)
```

Recorded timestamp of the source pointer event that triggered the scale event.

Could be None if triggered from proxied events such as accessibility.

### **buildvertical\_scaleclass-attributeinstance-attribute**

```
vertical_scale: float = field(metadata={"data_field": "vs"})
```

The scale implied by the average distance along the vertical axis between the pointers in contact with the screen.

This value must be greater than or equal to zero.

# ScreenBrightnessChangeEvent

Event fired when screen brightness changes.

Inherits: Event

## Properties

- brightness - The new screen brightness, in range 0.0..1.0.

## Properties

**buildbrightnessinstance-attribute**

brightness: float

The new screen brightness, in range 0.0..1.0.

# Scrollbar

Configures the scrollbar that scrollable controls render for their content.

## Properties

- **interactive** - Whether this scroll bar should be interactive and respond to dragging on the thumb, or tapping in the track area.
- **orientation** - Specifies where the scrollbar should appear relative to the scrollable.
- **radius** - Circular radius of the scrollbar thumb's rounded rectangle corners in logical pixels.
- **thickness** - Controls the cross-axis size of the scrollbar in logical pixels.
- **thumb\_visibility** - Whether this scrollbar's thumb should be always be visible, even when not being scrolled.
- **track\_visibility** - Indicates whether the scrollbar track should be visible, so long as the thumb\_visibility is visible.

## Examples

### Scroll Bar showcase

[play\\_arrowTry Online](#)

```
from typing import Optional

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.appbar = ft.AppBar(title="Scrollbar Dataclass Showcase")

 def get_scrollbar() -> ft.Scrollbar:
 def str_as_bool(value: Optional[str]) -> Optional[bool]:
 return True if value == "true" else False if value == "false" else None

 return ft.Scrollbar(
 thumb_visibility=str_as_bool(thumb_visibility.value),
 track_visibility=str_as_bool(track_visibility.value),
 interactive=str_as_bool(interactive.value),
 thickness=thickness_value.value if use_thickness.value else None,
 radius=radius_value.value if use_radius.value else None,
 orientation=None
 if orientation.value == "none"
 else ft.ScrollbarOrientation(orientation.value),
)

 def get_preview_content(scrollbar: ft.Scrollbar) -> tuple[str, ft.Control]:
 selected_orientation = scrollbar.orientation
 is_horizontal = selected_orientation in (
 ft.ScrollbarOrientation.TOP,
 ft.ScrollbarOrientation.BOTTOM,
)
 if is_horizontal:
 return (
 "Horizontal preview (TOP/BOTTOM orientation)",
 ft.Row(
 spacing=8,
 scroll=scrollbar,
 controls=[
 ft.Container(
 width=110,
 height=80,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
 alignment=ft.Alignment.CENTER,
 content=ft.Text(f"Tile {i + 1}"),
)
 for i in range(18)
],
),
)
 return (
 "Vertical preview (None/LEFT/RIGHT orientation)",
 ft.Column(
 spacing=4,
```

```

 scrollbar=scrollbar,
 controls=[ft.Text(f"Item {i + 1}") for i in range(40)],
),
)

def update_preview():
 thickness_value.disabled = not use_thickness.value
 radius_value.disabled = not use_radius.value

 scrollbar = get_scrollbar()
 title, content = get_preview_content(scrollbar)
 preview_title.value = title
 preview_viewport.content = content
 page.update()

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Interactive playground for the Scrollbar dataclass. "
 "Change each property and inspect the live result."
),
 ft.Row(
 wrap=True,
 spacing=12,
 run_spacing=12,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 ft.Container(
 width=360,
 padding=12,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=10,
 controls=[
 ft.Text(
 "Configuration", weight=ft.FontWeight.BOLD
),
 thumb_visibility := ft.Dropdown(
 label="thumb_visibility",
 value="none",
 on_select=update_preview,
 options=[
 ft.DropdownOption(
 "none", "None (theme default)"
),
 ft.DropdownOption("true", "True"),
 ft.DropdownOption("false", "False"),
],
),
 track_visibility := ft.Dropdown(
 label="track_visibility",
 value="none",
 on_select=update_preview,
 options=[
 ft.DropdownOption(
 "none", "None (theme default)"
),
 ft.DropdownOption("true", "True"),
 ft.DropdownOption("false", "False"),
],
),
 interactive := ft.Dropdown(
 label="interactive",
 value="none",
 on_select=update_preview,
 options=[
 ft.DropdownOption(
 "none", "None (platform default)"
),
 ft.DropdownOption("true", "True"),
 ft.DropdownOption("false", "False"),
],
),
 orientation := ft.Dropdown(
 label="orientation",
 value="none",
 on_select=update_preview,
 options=[
 ft.DropdownOption(
 "none", "None (auto side)"
)
]
)
]
)
)
]
)
]
)
)
)
+ [

```

```

 ft.DropdownOption(o.value, o.name)
 for o in ft.ScrollbarOrientation
],
),
use_thickness := ft.Checkbox(
 label="Set thickness",
 value=False,
 on_change=update_preview,
),
thickness_value := ft.Slider(
 min=0,
 max=20,
 divisions=20,
 value=8,
 label="{value}",
 on_change=update_preview,
),
use_radius := ft.Checkbox(
 label="Set radius",
 value=False,
 on_change=update_preview,

```

## Properties

### buildinteractiveclass-attributeinstance-attribute

`interactive: bool | None = None`

Whether this scroll bar should be interactive and respond to dragging on the thumb, or tapping in the track area.

When False, the scrollbar will not respond to gesture or hover events, and will allow to click through it.

If None, defaults to True, unless on Android, where it defaults to False.

### buildorientationclass-attributeinstance-attribute

`orientation: ScrollbarOrientation | None = None`

Specifies where the scrollbar should appear relative to the scrollable.

If None, for a vertical scroll, defaults to ScrollbarOrientation.RIGHT for left-to-right text direction and ScrollbarOrientation.LEFT for right-to-left text direction, while for a horizontal scroll, it defaults to ScrollbarOrientation.BOTTOM.

#### Note

ScrollbarOrientation.TOP and ScrollbarOrientation.BOTTOM can only be used with a horizontal scroll; ScrollbarOrientation.LEFT and ScrollbarOrientation.RIGHT can only be used with a vertical scroll.

### buildradiusclass-attributeinstance-attribute

`radius: Number | None = None`

Circular radius of the scrollbar thumb's rounded rectangle corners in logical pixels. If None, platform defaults are used.

The radius of the scrollbar thumb's rounded rectangle corners.

If None, the default value is platform dependent: no radius is applied on Android (Page.platform == PagePlatform.ANDROID); 1.5 pixels on iOS (Page.platform == PagePlatform.IOS); 8.0 pixels on the remaining platforms.

### buildthicknessclass-attributeinstance-attribute

`thickness: Number | None = None`

Controls the cross-axis size of the scrollbar in logical pixels. The thickness of the scrollbar in the cross axis of the scrollable.

If None, the default value is platform dependent: 4.0 pixels on Android (Page.platform == PagePlatform.ANDROID) and iOS (Page.platform == PagePlatform.IOS); ScrollbarTheme.thickness on the remaining platforms.

### buildthumb\_visibilityclass-attributeinstance-attribute

`thumb_visibility: bool | None = None`

Whether this scrollbar's thumb should be always be visible, even when not being scrolled. When False, the scrollbar will be shown during scrolling and will fade out otherwise.

If None, then ScrollbarTheme.thumb\_visibility is used. If that is also None, defaults to False.

### buildtrack\_visibilityclass-attributeinstance-attribute

`track_visibility: bool | None = None`



Indicates whether the scrollbar track should be visible, so long as the thumb\_visibility is visible.

If None, then ScrollbarTheme.track\_visibility is used. If that is also None, defaults to False.

# ScrollbarOrientation

Defines the edge/side of the viewport where the Scrollbar is shown.

Inherits: enum.Enum

## Properties

- BOTTOM - Places the scrollbar below a horizontal scrollable.
- LEFT - Places the scrollbar on the left/leading edge of a vertical scrollable.
- RIGHT - Places the scrollbar on the right/trailing edge of a vertical scrollable.
- TOP - Places the scrollbar above a horizontal scrollable.

## Examples

### Scroll Bar Orientation showcase

play\_arrowTry Online

```
import flet as ft

def showcase_card(orientation: ft.ScrollbarOrientation) -> ft.Container:
 is_vertical = orientation in (
 ft.ScrollbarOrientation.LEFT,
 ft.ScrollbarOrientation.RIGHT,
)
 scrollbar = ft.Scrollbar(
 orientation=orientation,
 thumb_visibility=True,
 track_visibility=True,
 thickness=10,
 radius=8,
)

 if is_vertical:
 viewport = ft.Container(
 height=220,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 padding=8,
 content=ft.Column(
 spacing=4,
 scroll=scrollbar,
 controls=[ft.Text(f"Item {i + 1}") for i in range(35)],
),
)
 else:
 viewport = ft.Container(
 height=130,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 padding=8,
 content=ft.Row(
 spacing=8,
 scroll=scrollbar,
 controls=[
 ft.Container(
 width=84,
 height=72,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE_CONTAINER_HIGHEST,
 alignment=ft.Alignment.CENTER,
 content=ft.Text(f"{i + 1}"),
)
 for i in range(20)
],
),
)

 return ft.Container(
 width=330,
 padding=12,
 border=ft.Border.all(1, ft.Colors.OUTLINE_VARIANT),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
```

```

 spacing=8,
 controls=[
 ft.Text(orientation.name, weight=ft.FontWeight.BOLD),
 viewport,
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="ScrollbarOrientation Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "LEFT/RIGHT apply to vertical scrollables, TOP/BOTTOM apply to "
 "horizontal scrollables.",
),
 ft.Row(
 wrap=True,
 spacing=12,
 run_spacing=12,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(orientation)
 for orientation in ft.ScrollbarOrientation
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### **buildBOTTOM**class-attributeinstance-attribute

Places the scrollbar below a horizontal scrollable.

### **buildLEFT**class-attributeinstance-attribute

Places the scrollbar on the left/leading edge of a vertical scrollable.

### **buildRIGHT**class-attributeinstance-attribute

Places the scrollbar on the right/trailing edge of a vertical scrollable.

### **buildTOP**class-attributeinstance-attribute

Places the scrollbar above a horizontal scrollable.

# ScrollBarTheme

Customizes the colors, thickness, and shape of scrollbars across the app.

## Properties

- `cross_axis_margin` - Distance from the scrollbar thumb to the nearest cross axis edge in logical pixels.
- `interactive` - Whether the Scrollbar should be interactive and respond to dragging on the thumb, or tapping in the track area.
- `main_axis_margin` - Distance from the scrollbar thumb's start and end to the edge of the viewport in logical pixels.
- `min_thumb_length` - The preferred smallest size the scrollbar thumb can shrink to when the total scrollable extent is large, the current visible viewport is small, and the viewport is not overscrolled.
- `radius` - The Radius of the scrollbar thumb's rounded rectangle corners.
- `thickness` - The thickness of the scrollbar in the cross axis of the scrollable.
- `thumb_color` - Overrides the default Color of the Scrollbar thumb.
- `thumb_visibility` - Indicates that the scrollbar thumb should be visible, even when a scroll is not underway.
- `track_border_color` - Overrides the default Color of the Scrollbar track border.
- `track_color` - Overrides the default Color of the Scrollbar track.
- `track_visibility` - Indicates that the scrollbar track should be visible.

## Properties

### `buildcross_axis_marginclass-attributeinstance-attribute`

`cross_axis_margin: Number | None = None`

Distance from the scrollbar thumb to the nearest cross axis edge in logical pixels. The scrollbar track consumes this space. Must not be null and defaults to 0.

### `buildinteractiveclass-attributeinstance-attribute`

`interactive: bool | None = None`

Whether the Scrollbar should be interactive and respond to dragging on the thumb, or tapping in the track area. When False, the scrollbar will not respond to gesture or hover events, and will allow to click through it. Defaults to True when None, unless on Android, which will default to False when None.

### `buildmain_axis_marginclass-attributeinstance-attribute`

`main_axis_margin: Number | None = None`

Distance from the scrollbar thumb's start and end to the edge of the viewport in logical pixels. It affects the amount of available paint area. The scrollbar track consumes this space. Mustn't be null and defaults to 0.

### `buildmin_thumb_lengthclass-attributeinstance-attribute`

`min_thumb_length: Number | None = None`

The preferred smallest size the scrollbar thumb can shrink to when the total scrollable extent is large, the current visible viewport is small, and the viewport is not overscrolled.

### `buildradiusclass-attributeinstance-attribute`

`radius: Number | None = None`

The Radius of the scrollbar thumb's rounded rectangle corners.

### `buildthicknessclass-attributeinstance-attribute`

`thickness: ControlStateValue[Number | None] | None = None`

The thickness of the scrollbar in the cross axis of the scrollable. Property value could be either a single float value or a dictionary with `ft.ControlState` as keys and float as values.

### `buildthumb_colorclass-attributeinstance-attribute`

```
thumb_color: ControlStateValue[ColorValue] | None = None
```

Overrides the default Color of the Scrollbar thumb. The value is either a single color string or ft.ControlState dictionary.

### **buildthumb\_visibilityclass-attributeinstance-attribute**

```
thumb_visibility: ControlStateValue[bool] | None = None
```

Indicates that the scrollbar thumb should be visible, even when a scroll is not underway. When False, the scrollbar will be shown during scrolling and will fade out otherwise. When True, the scrollbar will always be visible and never fade out.

Property value could be either a single boolean value or a dictionary with ft.ControlState as keys and boolean as values.

### **buildtrack\_border\_colorclass-attributeinstance-attribute**

```
track_border_color: ControlStateValue[ColorValue] | None = (
 None
)
```

Overrides the default Color of the Scrollbar track border. The value is either a single color string or ft.ControlState dictionary.

### **buildtrack\_colorclass-attributeinstance-attribute**

```
track_color: ControlStateValue[ColorValue] | None = None
```

Overrides the default Color of the Scrollbar track. The value is either a single color string or ft.ControlState dictionary.

### **buildtrack\_visibilityclass-attributeinstance-attribute**

```
track_visibility: ControlStateValue[bool] | None = None
```

Indicates that the scrollbar track should be visible. When True, the scrollbar track will always be visible so long as the thumb is visible. If the scrollbar thumb is not visible, the track will not be visible either. Defaults to False when None. If this property is None, then ScrollbarTheme.track\_visibility of Theme.scrollbar\_theme is used. If that is also None, the default value is False. Property value could be either a single boolean value or a dictionary with ft.ControlState as keys and boolean as values.

# ScrollDirection

User scroll direction reported by Flutter user-scroll notifications.

Used by `OnScrollEvent.direction` when `OnScrollEvent.event_type` is `ScrollType.USER`.

Inherits: `enum.Enum`

## Properties

- FORWARD - Scrolling in the forward axis direction.
- IDLE - No active user-driven scroll direction.
- REVERSE - Scrolling in the reverse axis direction.

## Examples

### Scroll Direction showcase

`play_arrow` Try Online

```
import flet as ft
```

```
def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 counts = {direction: 0 for direction in ft.ScrollDirection}
 count_labels = {
 direction: ft.Text(f"{direction.name}: 0") for direction in ft.ScrollDirection
 }

 def on_scroll(e: ft.OnScrollEvent):
 if e.event_type == ft.ScrollType.USER and e.direction is not None:
 counts[e.direction] += 1
 count_labels[
 e.direction
].value = f"{e.direction.name}: {counts[e.direction]}"
 last.value = f"Last USER direction: {e.direction.name}"
 for label in count_labels.values():
 label.update()
 last.update()

 page.appbar = ft.AppBar(title="ScrollDirection Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Scroll the list and watch USER direction notifications."),
 last := ft.Text("Last USER direction: none"),
 ft.Row(
 wrap=True,
 spacing=10,
 controls=[count_labels[d] for d in ft.ScrollDirection],
 alignment=ft.MainAxisAlignment.CENTER,
),
 ft.Container(
 width=360,
 height=240,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 padding=8,
 content=ft.Column(
 scroll=ft.ScrollMode.ALWAYS,
 on_scroll=on_scroll,
 controls=[
 ft.Text(f"Scrollable item {i + 1}") for i in range(60)
],
),
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildFORWARD**class-attributeinstance-attribute

Scrolling in the forward axis direction.

**buildIDLE**class-attributeinstance-attribute

No active user-driven scroll direction.

**buildREVERSE**class-attributeinstance-attribute

Scrolling in the reverse axis direction.

# ScrollEvent

The pointer issued a scroll event.

Inherits: Event

## Properties

- `global_position` - The coordinate of the position of the pointer, in logical pixels in the global coordinate space.
- `local_position` - The coordinate of the position of the pointer, in logical pixels in the local coordinate space.
- `scroll_delta` - The amount to scroll, in logical pixels.

## Properties

### `buildglobal_positionclass-attributeinstance-attribute`

```
global_position: Offset = field(
 metadata={"data_field": "g"}
)
```

The coordinate of the position of the pointer, in logical pixels in the global coordinate space.

### `buildlocal_positionclass-attributeinstance-attribute`

```
local_position: Offset = field(metadata={"data_field": "l"})
```

The coordinate of the position of the pointer, in logical pixels in the local coordinate space.

### `buildscroll_deltaclass-attributeinstance-attribute`

```
scroll_delta: Offset = field(metadata={"data_field": "sd"})
```

The amount to scroll, in logical pixels.



# ScrollKey

Key type used for imperative scroll targeting.

Use this key to identify an item when calling `ScrollableControl.scroll_to` with `scroll_key`.

Inherits: `Key`

# ScrollMode

Defines scrolling behavior and scroll bar visibility for scrollable controls.

When assigned to ScrollableControl.scroll, for example, each value internally maps to a specific Scrollbar configuration.

Inherits: enum.Enum

## Properties

- ADAPTIVE - Scrolling is enabled and scroll bar is always shown when running app as web or desktop.
- ALWAYS - Scrolling is enabled and scroll bar is always shown.
- AUTO - Scrolling is enabled and scroll bar is only shown when scrolling occurs.
- HIDDEN - Scrolling is enabled, but scroll bar is always hidden.

## Examples

### Scroll Mode showcase

play\_arrowTry Online

```
import flet as ft

def showcase_card(scroll_mode: ft.ScrollMode) -> ft.Container:
 return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(scroll_mode.name, weight=ft.FontWeight.BOLD),
 ft.Text("Scroll inside this panel", size=12),
 ft.Container(
 height=170,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 padding=8,
 content=ft.Column(
 spacing=4,
 scroll=scroll_mode,
 controls=[ft.Text(f"Item {i + 1}") for i in range(24)],
),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="ScrollMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare scrollbar visibility and scrolling behavior modes."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(scroll_mode) for scroll_mode in ft.ScrollMode
],
),
],
),
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

### buildADAPTIVEclass-attributeinstance-attribute

Scrolling is enabled and scroll bar is always shown when running app as web or desktop.

Scrollbar equivalent:

```
ft.Scrollbar(
 thumb_visibility=page.web or not page.platform.is_mobile(),
 thickness=4.0 if page.platform.is_mobile() and not page.web else None,
)
```

### buildALWAYSclass-attributeinstance-attribute

Scrolling is enabled and scroll bar is always shown.

Scrollbar equivalent:

```
ft.Scrollbar(
 thumb_visibility=True,
 thickness=4.0 if page.platform.is_mobile() and not page.web else None,
)
```

### buildAUTOCclass-attributeinstance-attribute

Scrolling is enabled and scroll bar is only shown when scrolling occurs.

Scrollbar equivalent:

```
ft.Scrollbar(
 thickness=4.0 if page.platform.is_mobile() and not page.web else None,
)
```

### buildHIDDENclass-attributeinstance-attribute

Scrolling is enabled, but scroll bar is always hidden.

Scrollbar equivalent:

```
ft.Scrollbar(
 thickness=0,
)
```

# ScrollType

The kind of scroll notification emitted by ScrollableControl.

Inherits: enum.Enum

## Properties

- END - Scrolling has ended.
- OVERSCROLL - Viewport was overscrolled.
- START - Scrolling has started.
- UPDATE - Scroll position changed.
- USER - User scroll direction changed.

## Examples

### Scroll Type showcase

[play\\_arrow](#)Try Online

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 counts = {event_type: 0 for event_type in ft.ScrollType}
 count_labels = {
 event_type: ft.Text(f"{event_type.name}: 0") for event_type in ft.ScrollType
 }

 def on_scroll(e: ft.OnScrollEvent):
 counts[e.event_type] += 1
 count_labels[
 e.event_type
].value = f"{e.event_type.name}: {counts[e.event_type]}"
 last.value = (
 f"Last event: {e.event_type.name}, pixels={round(e.pixels, 1)}"
 f"{'', direction=' ' if e.direction else ''}"
)
 for label in count_labels.values():
 label.update()
 last.update()

 page.appbar = ft.AppBar(title="ScrollType Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Scroll the list to generate start/update/user/end "
 "notifications."
),
 last := ft.Text(),
 ft.Row(
 wrap=True,
 spacing=10,
 controls=[count_labels[t] for t in ft.ScrollType],
 alignment=ft.MainAxisAlignment.CENTER,
),
 ft.Container(
 width=360,
 height=240,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 padding=8,
 content=ft.Column(
 scroll=ft.ScrollMode.ALWAYS,
 on_scroll=on_scroll,
 controls=[
 ft.Text(f"Scrollable item {i + 1}") for i in range(60)
],
),
),
],
),
),
)
```

)

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildEND**class-attributeinstance-attribute

Scrolling has ended.

**buildOVERSCROLL**class-attributeinstance-attribute

Viewport was overscrolled.

See `overscroll` and `velocity` are available.

**buildSTART**class-attributeinstance-attribute

Scrolling has started.

**buildUPDATE**class-attributeinstance-attribute

Scroll position changed.

See `OnScrollEvent.scroll_delta`.

**buildUSER**class-attributeinstance-attribute

User scroll direction changed.

See `OnScrollEvent.direction`.

# SearchBarTheme

Customizes the appearance of SearchBar controls across the app.

## Properties

- `bgcolor` - Overrides the default value of `SearchBar.bar_bgcolor` in all descendant `SearchBar` controls.
- `border_side` - Overrides the default value of `SearchBar.bar_border_side` in all descendant `SearchBar` controls.
- `elevation` - Overrides the default value of `SearchBar.bar_elevation` in all descendant `SearchBar` controls.
- `hint_style` - Overrides the default value of `SearchBar.bar_hint_text_style` in all descendant `SearchBar` controls.
- `overlay_color` - Overrides the default value of `SearchBar.bar_overlay_color` in all descendant `SearchBar` controls.
- `padding` - Overrides the default value of `SearchBar.bar_padding` in all descendant `SearchBar` controls.
- `shadow_color` - Overrides the default value of `SearchBar.bar_shadow_color` in all descendant `SearchBar` controls.
- `shape` - Overrides the default value of `SearchBar.bar_shape` in all descendant `SearchBar` controls.
- `size_constraints` - Overrides the default value of `SearchBar.bar_size_constraints` in all descendant `SearchBar` controls.
- `text_capitalization` - Overrides the default value of `SearchBar.capitalization` in all descendant `SearchBar` controls.
- `text_style` - Overrides the default value of `SearchBar.bar_text_style` in all descendant `SearchBar` controls.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `SearchBar.bar_bgcolor` in all descendant `SearchBar` controls.

### `buildborder_sideclass-attributeinstance-attribute`

`border_side: ControlStateValue[BorderSide] | None = None`

Overrides the default value of `SearchBar.bar_border_side` in all descendant `SearchBar` controls.

### `buildelevationclass-attributeinstance-attribute`

`elevation: ControlStateValue[Number | None] | None = None`

Overrides the default value of `SearchBar.bar_elevation` in all descendant `SearchBar` controls.

### `buildhint_styleclass-attributeinstance-attribute`

`hint_style: ControlStateValue[TextStyle] | None = None`

Overrides the default value of `SearchBar.bar_hint_text_style` in all descendant `SearchBar` controls.

### `buildoverlay_colorclass-attributeinstance-attribute`

`overlay_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `SearchBar.bar_overlay_color` in all descendant `SearchBar` controls.

### `buildpaddingclass-attributeinstance-attribute`

`padding: ControlStateValue[PaddingValue] | None = None`

Overrides the default value of `SearchBar.bar_padding` in all descendant `SearchBar` controls.

### `buildshadow_colorclass-attributeinstance-attribute`

`shadow_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `SearchBar.bar_shadow_color` in all descendant `SearchBar` controls.

### `buildshapeclass-attributeinstance-attribute`

`shape: ControlStateValue[OutlinedBorder] | None = None`

Overrides the default value of `SearchBar.bar_shape` in all descendant `SearchBar` controls.

### `buildsize_constraintsclass-attributeinstance-attribute`

`size_constraints: BoxConstraints | None = None`

Overrides the default value of `SearchBar.bar_size_constraints` in all descendant `SearchBar` controls.

### **buildtext\_capitalizationclass-attributeinstance-attribute**

`text_capitalization: TextCapitalization | None = None`

Overrides the default value of `SearchBar.capitalization` in all descendant `SearchBar` controls.

### **buildtext\_styleclass-attributeinstance-attribute**

`text_style: ControlStateValue[TextStyle] | None = None`

Overrides the default value of `SearchBar.bar_text_style` in all descendant `SearchBar` controls.

# SearchViewTheme

Customizes the appearance of SearchBar controls across the app.

## Properties

- `bar_padding` - Overrides the default value of `SearchBar.view_bar_padding` in all descendant `SearchBar` controls.
- `bgcolor` - Overrides the default value of `SearchBar.view_bgcolor` in all descendant `SearchBar` controls.
- `border_side` - Overrides the default value of `SearchBar.view_side` in all descendant `SearchBar` controls.
- `divider_color` - Overrides the default value of `SearchBar.divider_color` in all descendant `SearchBar` controls.
- `elevation` - Overrides the default value of `SearchBar.view_elevation` in all descendant `SearchBar` controls.
- `header_height` - Overrides the default value of `SearchBar.view_header_height` in all descendant `SearchBar` controls.
- `header_hint_text_style` - Overrides the default value of `SearchBar.view_hint_text_style` in all descendant `SearchBar` controls.
- `header_text_style` - Overrides the default value of `SearchBar.view_header_text_style` in all descendant `SearchBar` controls.
- `padding` - Overrides the default value of `SearchBar.view_padding` in all descendant `SearchBar` controls.
- `shape` - Overrides the default value of `SearchBar.view_shape` in all descendant `SearchBar` controls.
- `shrink_wrap` - Overrides the default value of `SearchBar.shrink_wrap` in all descendant `SearchBar` controls.
- `size_constraints` - Overrides the default value of `SearchBar.view_size_constraints` in all descendant `SearchBar` controls.

## Properties

### `buildbar_paddingclass-attributeinstance-attribute`

`bar_padding: PaddingValue | None = None`

Overrides the default value of `SearchBar.view_bar_padding` in all descendant `SearchBar` controls.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `SearchBar.view_bgcolor` in all descendant `SearchBar` controls.

### `buildborder_sideclass-attributeinstance-attribute`

`border_side: BorderSide | None = None`

Overrides the default value of `SearchBar.view_side` in all descendant `SearchBar` controls.

### `bullddivider_colorclass-attributeinstance-attribute`

`divider_color: ColorValue | None = None`

Overrides the default value of `SearchBar.divider_color` in all descendant `SearchBar` controls.

### `buildelevationclass-attributeinstance-attribute`

`elevation: Number | None = None`

Overrides the default value of `SearchBar.view_elevation` in all descendant `SearchBar` controls.

### `buildheader_heightclass-attributeinstance-attribute`

`header_height: Number | None = None`

Overrides the default value of `SearchBar.view_header_height` in all descendant `SearchBar` controls.

### `buildheader_hint_text_styleclass-attributeinstance-attribute`

`header_hint_text_style: TextStyle | None = None`

Overrides the default value of `SearchBar.view_hint_text_style` in all descendant `SearchBar` controls.

### `buildheader_text_styleclass-attributeinstance-attribute`

`header_text_style: TextStyle | None = None`



Overrides the default value of `SearchBar.view_header_text_style` in all descendant `SearchBar` controls.

### **buildpaddingclass-attributeinstance-attribute**

`padding: PaddingValue | None = None`

Overrides the default value of `SearchBar.view_padding` in all descendant `SearchBar` controls.

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

Overrides the default value of `SearchBar.view_shape` in all descendant `SearchBar` controls.

### **buildshrink\_wrapclass-attributeinstance-attribute**

`shrink_wrap: bool | None = None`

Overrides the default value of `SearchBar.shrink_wrap` in all descendant `SearchBar` controls.

### **buildsize\_constraintsclass-attributeinstance-attribute**

`size_constraints: BoxConstraints | None = None`

Overrides the default value of `SearchBar.view_size_constraints` in all descendant `SearchBar` controls.

# SegmentedButtonTheme

Customizes the appearance of SegmentedButton controls across the app.

## Properties

- `selected_icon` - Overrides the default value for `SegmentedButton.selected_icon`.
- `style` - Overrides the default value for `SegmentedButton.style`.

## Properties

### `buildselected_iconclass-attributeinstance-attribute`

`selected_icon: IconData | None = None`

Overrides the default value for `SegmentedButton.selected_icon`.

### `buildstyleclass-attributeinstance-attribute`

`style: ButtonStyle | None = None`

Overrides the default value for `SegmentedButton.style`.

# SensorErrorEvent

Generic sensor error event. message contains the platform error text.

Inherits: Event

## Properties

- message - Human-readable description of the sensor error.

## Properties

buildmessageinstance-attribute

message: str

Human-readable description of the sensor error.

# ShapeBorder

Base class for shape outlines. Not intended to be used directly.

See subclasses/implementations:

- OutlinedBorder

# ShareCupertinoActivityType

iOS/macOS activity types that can be excluded from the share sheet.

Inherits: `enum.Enum`

## Properties

- `ADD_TO_HOME_SCREEN` - Add to home screen.
- `ADD_TO_READING_LIST` - Add to reading list.
- `AIR_DROP` - AirDrop.
- `ASSIGN_TO_CONTACT` - Assign to contact.
- `COLLABORATION_COPY_LINK` - Collaboration copy link.
- `COLLABORATION_INVITE_WITH_LINK` - Collaboration invite with link.
- `COPY_TO_PASTEBOARD` - Copy to clipboard.
- `MAIL` - Share via Mail.
- `MARKUP_AS_PDF` - Markup as PDF.
- `MESSAGE` - Share via Messages.
- `OPEN_IN_IBOOKS` - Open in iBooks.
- `POST_TO_FACEBOOK` - Share to Facebook.
- `POST_TO_FLICKR` - Post to Flickr.
- `POST_TO_TENCENT_WEIBO` - Post to Tencent Weibo.
- `POST_TO_TWITTER` - Share to Twitter.
- `POST_TO_VIMEO` - Post to Vimeo.
- `POST_TO_WEIBO` - Share to Weibo.
- `PRINT` - Send to printer.
- `SAVE_TO_CAMERA_ROLL` - Save to camera roll.
- `SHARE_PLAY` - SharePlay.

## Properties

`buildADD_TO_HOME_SCREENclass-attributeinstance-attribute`

Add to home screen.

`buildADD_TO_READING_LISTclass-attributeinstance-attribute`

Add to reading list.

`buildAIR_DROPclass-attributeinstance-attribute`

AirDrop.

`buildASSIGN_TO_CONTACTclass-attributeinstance-attribute`

Assign to contact.

`buildCOLLABORATION_COPY_LINKclass-attributeinstance-attribute`

Collaboration copy link.

`buildCOLLABORATION_INVITE_WITH_LINKclass-attributeinstance-attribute`

Collaboration invite with link.

`buildCOPY_TO_PASTEBOARDclass-attributeinstance-attribute`

Copy to clipboard.

**buildMAILclass-attributeinstance-attribute**

Share via Mail.

**buildMARKUP\_AS\_PDFclass-attributeinstance-attribute**

Markup as PDF.

**buildMESSAGEclass-attributeinstance-attribute**

Share via Messages.

**buildOPEN\_IN\_IBOOKSclass-attributeinstance-attribute**

Open in iBooks.

**buildPOST\_TO\_FACEBOOKclass-attributeinstance-attribute**

Share to Facebook.

**buildPOST\_TO\_FLICKRclass-attributeinstance-attribute**

Post to Flickr.

**buildPOST\_TO\_TENCENT\_WEIBOclass-attributeinstance-attribute**

Post to Tencent Weibo.

**buildPOST\_TO\_TWITTERclass-attributeinstance-attribute**

Share to Twitter.

**buildPOST\_TO\_VIMEOclass-attributeinstance-attribute**

Post to Vimeo.

**buildPOST\_TO\_WEIBOclass-attributeinstance-attribute**

Share to Weibo.

**buildPRINTclass-attributeinstance-attribute**

Send to printer.

**buildSAVE\_TO\_CAMERA\_ROLLclass-attributeinstance-attribute**

Save to camera roll.

**buildSHARE\_PLAYclass-attributeinstance-attribute**

SharePlay.

# ShareFile

Represents a file to share, either from disk or in-memory bytes.

## Properties

- data - Raw bytes of the file to share.
- mime\_type - MIME type of the file.
- name - Optional name of the file.
- path - Filesystem path to the file to share.

## Methods

- from\_bytes - Create ShareFile from raw bytes.
- from\_path - Create ShareFile from a filesystem path.

## Properties

### builddataclass-attributeinstance-attribute

data: bytes | None = None

Raw bytes of the file to share.

### buildmime\_typeclass-attributeinstance-attribute

mime\_type: str | None = None

MIME type of the file.

### buildnameclass-attributeinstance-attribute

name: str | None = None

Optional name of the file.

### buildpathclass-attributeinstance-attribute

path: str | None = None

Filesystem path to the file to share.

## Methods

### deployed\_codefrom\_bytesstaticmethod

```
from_bytes(
 data: bytes,
 mime_type: str | None = None,
 name: str | None = None,
) -> ShareFile
```

Create ShareFile from raw bytes.

Parameters:

- data (bytes) - Raw bytes of the file.
- mime\_type (str | None, default: None) - Optional MIME type of the file.
- name (str | None, default: None) - Optional name of the file.

### deployed\_codefrom\_pathstaticmethod

```
from_path(
 path: str, name: str | None = None
) -> ShareFile
```

Create ShareFile from a filesystem path.

Parameters:

- path (str) - Filesystem path to the file.

- name (str | None, default: None) - Optional name of the file.



# ShareResult

Result returned from the native share sheet.

## Properties

- `raw` - The raw string result from the platform.
- `status` - The outcome of the share operation.

## Properties

### `buildrawinstance-attribute`

`raw: str`

The raw string result from the platform.

### `buildstatusinstance-attribute`

`status: ShareResultStatus`

The outcome of the share operation.

# ShareResultStatus

Outcome of a share operation.

Inherits: `enum.Enum`

## Properties

- `DISMISSED` - The user dismissed the share sheet.
- `SUCCESS` - The user selected an action.
- `UNAVAILABLE` - Platform cannot report a definite result.

## Properties

`buildDISMISSED`class-attributeinstance-attribute

The user dismissed the share sheet.

`buildSUCCESS`class-attributeinstance-attribute

The user selected an action.

`buildUNAVAILABLE`class-attributeinstance-attribute

Platform cannot report a definite result.

# Size

A 2D size with width and height.

## Properties

- `aspect_ratio` - Returns the aspect ratio (width / height).
- `height`
- `width`

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.
- `from_height` - Creates a Size with the given height and an infinite width.
- `from_radius` - Creates a square Size whose width and height are twice the given radius.
- `from_width` - Creates a Size with the given width and an infinite height.
- `infinite` - Creates a Size whose width and height are both positive infinity.
- `is_finite` - Checks if both dimensions are finite.
- `is_infinite` - Checks if either dimension is infinite.
- `square` - Creates a square Size where width and height are the same.
- `zero` - Creates a Size whose width and height are both 0.0.

## Properties

### `buildaspect_ratio`property

`aspect_ratio: float`

Returns the aspect ratio (width / height).

### `buildheight`instance-attribute

`height: Number`

### `buildwidth`instance-attribute

`width: Number`

## Methods

### `deployed_code`copy

```
copy(
 width: Number | None = None,
 height: Number | None = None,
) -> Size
```

Returns a copy of this object with the specified properties overridden.

### `deployed_code`from\_heightclassmethod

`from_height(height: Number) -> Size`

Creates a Size with the given height and an infinite width.

### `deployed_code`from\_radiusclassmethod

`from_radius(radius: Number) -> Size`

Creates a square Size whose width and height are twice the given radius.

### `deployed_code`from\_widthclassmethod

`from_width(width: Number) -> Size`

Creates a Size with the given width and an infinite height.

### `deployed_code`infiniteclassmethod

```
infinite()
```

Creates a Size whose width and height are both positive infinity.

### **deployed\_codeis\_finite**

```
is_finite() -> bool
```

Checks if both dimensions are finite.

### **deployed\_codeis\_infinite**

```
is_infinite() -> bool
```

Checks if either dimension is infinite.

### **deployed\_codesquareclassmethod**

```
square(dimension: Number) -> Size
```

Creates a square Size where width and height are the same.

### **deployed\_codezeroclassmethod**

```
zero()
```

Creates a Size whose width and height are both 0.0.

# SliderInteraction

Defines how users can interact with a Slider to change its value.

Each mode controls whether taps and drag gestures on the slider track are accepted, and whether interaction is restricted to dragging the thumb.

Inherits: `enum.Enum`

## Properties

- `SLIDE_ONLY` - Allows only sliding interaction to change the value.
- `SLIDE_THUMB` - Allows interaction only by dragging the slider thumb.
- `TAP_AND_SLIDE` - Allows the user to interact with a slider by tapping or sliding anywhere on the track.
- `TAP_ONLY` - Allows only tap interaction on the slider track.

## Examples

### Slider Interaction showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(interaction: ft.SliderInteraction) -> ft.Container:
 value_text = ft.Text("Value: 50")

 def on_change(e: ft.Event[ft.Slider]):
 value_text.value = f"Value: {round(e.control.value)}"
 value_text.update()

 return ft.Container(
 width=340,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(interaction.name, weight=ft.FontWeight.BOLD),
 value_text,
 ft.Container(
 width=260,
 padding=ft.Padding.symmetric(horizontal=8),
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Slider(
 min=0,
 max=100,
 value=50,
 divisions=20,
 interaction=interaction,
 on_change=on_change,
),
),
 ft.Text("Try tapping track and dragging thumb.", size=11),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="SliderInteraction Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare which gestures are accepted by each slider mode."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
```

```

 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(i) for i in ft.SliderInteraction],
),
 1,
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildSLIDE\_ONLYclass-attributeinstance-attribute

Allows only sliding interaction to change the value.

Track taps are ignored.

### buildSLIDE\_THUMBclass-attributeinstance-attribute

Allows interaction only by dragging the slider thumb.

Taps and drag/slide gestures that begin on the track are ignored.

### buildTAP\_AND\_SLIDEclass-attributeinstance-attribute

Allows the user to interact with a slider by tapping or sliding anywhere on the track.

Essentially all possible interactions are allowed.

It is different from SLIDE\_ONLY, in that when you try to slide anywhere other than the thumb, the thumb will move to the first point of contact.

### buildTAP\_ONLYclass-attributeinstance-attribute

Allows only tap interaction on the slider track.

Drag/sliding gestures are ignored.

# SliderTheme

Customizes the appearance of descendant Slider controls.

## Properties

- `active_tick_mark_color` - The color of the track's tick marks that are drawn between the `Slider.min` position and the current thumb position.
- `active_track_color` - Overrides the default value for `Slider.active_color`.
- `disabled_active_tick_mark_color` - The color of the track's tick marks that are drawn between the current thumb position and the `Slider.max` position when the Slider is disabled.
- `disabled_active_track_color` - The color of the Slider track between the `Slider.min` position and the current thumb position when the Slider is disabled.
- `disabled_inactive_tick_mark_color` - The color of the track's tick marks that are drawn between the current thumb position and the `Slider.max` position when the Slider is disabled.
- `disabled_inactive_track_color` - The color of the Slider track between the current thumb position and the `Slider.max` position when the Slider is disabled.
- `disabled_secondary_active_track_color` - The color of the Slider track between the current thumb position and the `Slider.secondary_track_value` position when the Slider is disabled.
- `disabled_thumb_color` - The color given to the thumb to draw itself with when the Slider is disabled.
- `inactive_tick_mark_color` - The color of the track's tick marks that are drawn between the current thumb position and the `Slider.max` position.
- `inactive_track_color` - Overrides the default value for `Slider.inactive_color`.
- `interaction` - Overrides the default value for `Slider.interaction`.
- `min_thumb_separation` - Limits the thumb's separation distance.
- `mouse_cursor` - Overrides the default value for `Slider.mouse_cursor`.
- `overlapping_shape_stroke_color` - The color given to the perimeter of the top range thumbs of a `RangeSlider` when the thumbs are overlapping and the top range value indicator when the value indicators are overlapping.
- `overlay_color` - Overrides the default value for `Slider.overlay_color`.
- `padding` - Overrides the default value for `Slider.padding`.
- `secondary_active_track_color` - Overrides the default value for `Slider.secondary_active_color`.
- `thumb_color` - Overrides the default value for `Slider.thumb_color`.
- `thumb_size` - The size of the handle thumb shape thumb.
- `track_gap` - The size of the gap between the active and inactive tracks of the gapped slider track shape.
- `track_height` - The height of the Slider track.
- `value_indicator_color` - The color given to the Slider's value indicator to draw itself with.
- `value_indicator_stroke_color` - The color given to the value indicator shape stroke.
- `value_indicator_text_style` - The `TextStyle` for the text on the value indicator.
- `year_2023` - Overrides the default value for `Slider.year_2023`.

## Properties

`buildactive_tick_mark_color`class-attributeinstance-attribute

`active_tick_mark_color`: `ColorValue` | `None` = `None`

The color of the track's tick marks that are drawn between the `Slider.min` position and the current thumb position.

`buildactive_track_color`class-attributeinstance-attribute

`active_track_color`: `ColorValue` | `None` = `None`

Overrides the default value for `Slider.active_color`.

### **builddisabled\_active\_tick\_mark\_color**class-attributeinstance-attribute

`disabled_active_tick_mark_color: ColorValue | None = None`

The color of the track's tick marks that are drawn between the current thumb position and the `Slider.max` position when the Slider is disabled.

### **builddisabled\_active\_track\_color**class-attributeinstance-attribute

`disabled_active_track_color: ColorValue | None = None`

The color of the Slider track between the `Slider.min` position and the current thumb position when the Slider is disabled.

### **builddisabled\_inactive\_tick\_mark\_color**class-attributeinstance-attribute

`disabled_inactive_tick_mark_color: ColorValue | None = None`

The color of the track's tick marks that are drawn between the current thumb position and the `Slider.max` position when the Slider is disabled.

### **builddisabled\_inactive\_track\_color**class-attributeinstance-attribute

`disabled_inactive_track_color: ColorValue | None = None`

The color of the Slider track between the current thumb position and the `Slider.max` position when the Slider is disabled.

### **builddisabled\_secondary\_active\_track\_color**class-attributeinstance-attribute

`disabled_secondary_active_track_color: ColorValue | None = (None)`

The color of the Slider track between the current thumb position and the `Slider.secondary_track_value` position when the Slider is disabled.

### **builddisabled\_thumb\_color**class-attributeinstance-attribute

`disabled_thumb_color: ColorValue | None = None`

The color given to the thumb to draw itself with when the Slider is disabled.

### **buildinactive\_tick\_mark\_color**class-attributeinstance-attribute

`inactive_tick_mark_color: ColorValue | None = None`

The color of the track's tick marks that are drawn between the current thumb position and the `Slider.max` position.

### **buildinactive\_track\_color**class-attributeinstance-attribute

`inactive_track_color: ColorValue | None = None`

Overrides the default value for `Slider.inactive_color`.

### **buildinteraction**class-attributeinstance-attribute

`interaction: SliderInteraction | None = None`

Overrides the default value for `Slider.interaction`.

### **buildmin\_thumb\_separation**class-attributeinstance-attribute

`min_thumb_separation: Number | None = None`

Limits the thumb's separation distance.

Use this only if you want to control the visual appearance of the thumbs in terms of a logical pixel value. This can be done when you want a specific look for thumbs when they are close together. To limit with the real values, rather than logical pixels, the values can be restricted by the parent.

### **buildmouse\_cursor**class-attributeinstance-attribute

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

Overrides the default value for `Slider.mouse_cursor`.

### **buildoverlapping\_shape\_stroke\_color**class-attributeinstance-attribute

`overlapping_shape_stroke_color: ColorValue | None = None`

The color given to the perimeter of the top range thumbs of a `RangeSlider` when the thumbs are overlapping and the top range value indicator when the value indicators are overlapping.



### **buildoverlay\_colorclass-attributeinstance-attribute**

overlay\_color: ColorValue | None = None

Overrides the default value for Slider.overlay\_color.

### **buildpaddingclass-attributeinstance-attribute**

padding: PaddingValue | None = None

Overrides the default value for Slider.padding.

### **buildsecondary\_active\_track\_colorclass-attributeinstance-attribute**

secondary\_active\_track\_color: ColorValue | None = None

Overrides the default value for Slider.secondary\_active\_color.

### **buildthumb\_colorclass-attributeinstance-attribute**

thumb\_color: ColorValue | None = None

Overrides the default value for Slider.thumb\_color.

### **buildthumb\_sizeclass-attributeinstance-attribute**

thumb\_size: ControlStateValue[Size] | None = None

The size of the handle thumb shape thumb.

### **buildtrack\_gapclass-attributeinstance-attribute**

track\_gap: Number | None = None

The size of the gap between the active and inactive tracks of the gapped slider track shape.

### **buildtrack\_heightclass-attributeinstance-attribute**

track\_height: Number | None = None

The height of the Slider track.

### **buildvalue\_indicator\_colorclass-attributeinstance-attribute**

value\_indicator\_color: ColorValue | None = None

The color given to the Slider's value indicator to draw itself with.

### **buildvalue\_indicator\_stroke\_colorclass-attributeinstance-attribute**

value\_indicator\_stroke\_color: ColorValue | None = None

The color given to the value indicator shape stroke.

### **buildvalue\_indicator\_text\_styleclass-attributeinstance-attribute**

value\_indicator\_text\_style: TextStyle | None = None

The TextStyle for the text on the value indicator.

### **buildyear\_2023class-attributeinstance-attribute**

year\_2023: bool = False

Overrides the default value for Slider.year\_2023.

# SnackBarBehavior

Defines where a SnackBar appears within a page and how it is positioned relative to bottom UI elements.

Inherits: `enum.Enum`

## Properties

- **FIXED** - Anchors the snack bar to the bottom of the page.
- **FLOATING** - Displays the snack bar as a floating surface above page content.

## Examples

### Snack Bar Behavior showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 def open_snack(behavior: ft.SnackBarBehavior):
 page.show_dialog(
 ft.SnackBar(
 content=ft.Text(f"Behavior: {behavior.name}"),
 behavior=behavior,
 margin=ft.Margin.only(bottom=100),
 action=ft.SnackBarAction(label="Close"),
)
)

 def showcase_card(behavior: ft.SnackBarBehavior) -> ft.Container:
 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(behavior.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open SnackBar",
 icon=ft.Icons.MESSAGE,
 on_click=lambda _, b=behavior: open_snack(b),
),
],
),
)

 page.appbar = ft.AppBar(title="SnackBarBehavior Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare snack bar placement: fixed vs floating."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(behavior) for behavior in ft.SnackBarBehavior
],
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

## **buildFIXEDclass-attributeinstance-attribute**

Anchors the snack bar to the bottom of the page.

If a `NavigationBar` is present, the snack bar is shown above it. Other non-fixed content can be pushed upward while the snack bar is visible.

## **buildFLOATINGclass-attributeinstance-attribute**

Displays the snack bar as a floating surface above page content.

This mode can overlay bottom widgets, such as a `NavigationBar` and a bottom-positioned `FloatingActionButton`.

# SnackBarTheme

Defines default property values for descendant SnackBar controls.

## Properties

- `action_bgcolor` - Overrides the default value of `bgcolor` of `SnackBar.action` in all descendant `SnackBar` controls.
- `action_overflow_threshold` - Overrides the default value of `SnackBar.action_overflow_threshold` in all descendant `SnackBar` controls.
- `action_text_color` - Overrides the default value of `text_color` of `SnackBar.action` in all descendant `SnackBar` controls.
- `behavior` - Overrides the default value of `SnackBar.behavior` in all descendant `SnackBar` controls.
- `bgcolor` - Overrides the default value of `SnackBar.bgcolor` in all descendant `SnackBar` controls.
- `close_icon_color` - Overrides the default value of `SnackBar.close_icon_color` in all descendant `SnackBar` controls.
- `content_text_style` - Used to configure the `text_style` property for the `[SnackBar.content]` control.
- `disabled_action_bgcolor` - Overrides the default value of `disabled_color` of `SnackBar.action` in all descendant `SnackBar` controls.
- `disabled_action_text_color` - Overrides the default value of `disabled_text_color` of `SnackBar.action` in all descendant `SnackBar` controls.
- `dismiss_direction` - Overrides the default value of `SnackBar.dismiss_direction` in all descendant `SnackBar` controls.
- `elevation` - Overrides the default value of `SnackBar.elevation` in all descendant `SnackBar` controls.
- `inset_padding` - Overrides the default value for `SnackBar.margin`.
- `shape` - Overrides the default value of `SnackBar.shape` in all descendant `SnackBar` controls.
- `show_close_icon` - Overrides the default value of `SnackBar.show_close_icon` in all descendant `SnackBar` controls.
- `width` - Overrides the default value of `SnackBar.width` in all descendant `SnackBar` controls.

## Properties

### `buildaction_bgcolorclass-attributeinstance-attribute`

`action_bgcolor: ColorValue | None = None`

Overrides the default value of `bgcolor` of `SnackBar.action` in all descendant `SnackBar` controls.

### `buildaction_overflow_thresholdclass-attributeinstance-attribute`

`action_overflow_threshold: Number | None = None`

Overrides the default value of `SnackBar.action_overflow_threshold` in all descendant `SnackBar` controls.

### `buildaction_text_colorclass-attributeinstance-attribute`

`action_text_color: ColorValue | None = None`

Overrides the default value of `text_color` of `SnackBar.action` in all descendant `SnackBar` controls.

### `buildbehaviorclass-attributeinstance-attribute`

`behavior: SnackBarBehavior | None = None`

Overrides the default value of `SnackBar.behavior` in all descendant `SnackBar` controls.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Overrides the default value of `SnackBar.bgcolor` in all descendant `SnackBar` controls.

### `buildclose_icon_colorclass-attributeinstance-attribute`

`close_icon_color: ColorValue | None = None`

Overrides the default value of `SnackBar.close_icon_color` in all descendant `SnackBar` controls.

### **buildcontent\_text\_styleclass-attributeinstance-attribute**

`content_text_style: TextStyle | None = None`

Used to configure the `text_style` property for the `[SnackBar.content]` control.

### **builddisabled\_action\_bgcolorclass-attributeinstance-attribute**

`disabled_action_bgcolor: ColorValue | None = None`

Overrides the default value of `disabled_color` of `SnackBar.action` in all descendant `SnackBar` controls.

### **builddisabled\_action\_text\_colorclass-attributeinstance-attribute**

`disabled_action_text_color: ColorValue | None = None`

Overrides the default value of `disabled_text_color` of `SnackBar.action` in all descendant `SnackBar` controls.

### **builddismiss\_directionclass-attributeinstance-attribute**

`dismiss_direction: DismissDirection | None = None`

Overrides the default value of `SnackBar.dismiss_direction` in all descendant `SnackBar` controls.

### **buildelevationclass-attributeinstance-attribute**

`elevation: Number | None = None`

Overrides the default value of `SnackBar.elevation` in all descendant `SnackBar` controls.

### **buildinset\_paddingclass-attributeinstance-attribute**

`inset_padding: PaddingValue | None = None`

Overrides the default value for `SnackBar.margin`.

This value is only used when behavior is `SnackBarBehavior.floating`.

### **buildshapeclass-attributeinstance-attribute**

`shape: OutlinedBorder | None = None`

Overrides the default value of `SnackBar.shape` in all descendant `SnackBar` controls.

### **buildshow\_close\_iconclass-attributeinstance-attribute**

`show_close_icon: bool | None = None`

Overrides the default value of `SnackBar.show_close_icon` in all descendant `SnackBar` controls.

### **buildwidthclass-attributeinstance-attribute**

`width: Number | None = None`

Overrides the default value of `SnackBar.width` in all descendant `SnackBar` controls.

# StackFit

How to size the non-positioned children of a Stack.

Inherits: `enum.Enum`

## Properties

- **EXPAND** - The constraints passed to the stack from its parent are tightened to the biggest size allowed.
- **LOOSE** - The constraints passed to the stack from its parent are loosened.
- **PASS\_THROUGH** - The constraints passed to the stack from its parent are passed unmodified to the non-positioned children.

## Properties

### **buildEXPAND**`class-attributeinstance-attribute`

The constraints passed to the stack from its parent are tightened to the biggest size allowed.

For example, if the Stack has loose constraints with a width in the range 10 to 100 and a height in the range 0 to 600, then the non-positioned children of the stack would all be sized as 100 pixels wide and 600 high.

### **buildLOOSE**`class-attributeinstance-attribute`

The constraints passed to the stack from its parent are loosened.

For example, if the stack has constraints that force it to 350x600, then this would allow the non-positioned children of the stack to have any width from zero to 350 and any height from zero to 600.

### **buildPASS\_THROUGH**`class-attributeinstance-attribute`

The constraints passed to the stack from its parent are passed unmodified to the non-positioned children.

For example, if a Stack is an expanded child of a Row, the horizontal constraints will be tight and the vertical constraints will be loose.

# StadiumBorder

A border that looks like a stadium.

Inherits: OutlinedBorder

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Methods

### `deployed_codecopy`

`copy(side: BorderSide | None = None) -> StadiumBorder`

Returns a copy of this object with the specified properties overridden.

# StrokeCap

Styles to use for line endings.

Inherits: `enum.Enum`

## Properties

- BUTT - Begin and end contours with a flat edge and no extension.
- ROUND - Begin and end contours with a semi-circle extension.
- SQUARE - Begin and end contours with a half square extension.

## Examples

### Stroke Cap showcase

[play\\_arrow](#) [Try Online](#)

```
import flet as ft
import flet.canvas as cv

def showcase_card(stroke_cap: ft.StrokeCap) -> ft.Container:
 return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(stroke_cap.name, weight=ft.FontWeight.BOLD),
 cv.Canvas(
 width=240,
 height=90,
 shapes=[
 cv.Line(
 x1=40,
 y1=45,
 x2=200,
 y2=45,
 paint=ft.Paint(
 stroke_width=24,
 color=ft.Colors.PRIMARY,
 stroke_cap=stroke_cap,
),
),
 cv.Line(
 x1=40,
 y1=16,
 x2=40,
 y2=74,
 paint=ft.Paint(stroke_width=2, color=ft.Colors.RED),
),
 cv.Line(
 x1=200,
 y1=16,
 x2=200,
 y2=74,
 paint=ft.Paint(stroke_width=2, color=ft.Colors.RED),
),
],
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="StrokeCap Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare line endings for each StrokeCap value."),
 ft.Row(
```



```

 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(stroke_cap) for stroke_cap in ft.StrokeCap
],
),
],
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

**buildBUTTclass-attributeinstance-attribute**

Begin and end contours with a flat edge and no extension.

**buildROUNDclass-attributeinstance-attribute**

Begin and end contours with a semi-circle extension.

**buildSQUAREclass-attributeinstance-attribute**

Begin and end contours with a half square extension.

# StrokeJoin

Styles to use for line segment joins.

Inherits: `enum.Enum`

## Properties

- BEVEL - Joins between line segments connect the corners of the butt ends of the line segments to give a beveled appearance.
- MITER - Joins between line segments form sharp corners.
- ROUND - Joins between line segments are semi-circular.

## Examples

### Stroke Join showcase

[play\\_arrowTry Online](#)

```
import flet as ft
import flet.canvas as cv

def showcase_card(stroke_join: ft.StrokeJoin) -> ft.Container:
 return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(stroke_join.name, weight=ft.FontWeight.BOLD),
 cv.Canvas(
 width=240,
 height=120,
 shapes=[
 cv.Path(
 elements=[
 cv.Path.MoveTo(40, 95),
 cv.Path.LineTo(120, 25),
 cv.Path.LineTo(200, 95),
],
 paint=ft.Paint(
 style=ft.PaintingStyle.STROKE,
 stroke_width=24,
 color=ft.Colors.PRIMARY,
 stroke_cap=ft.StrokeCap.BUTT,
 stroke_join=stroke_join,
),
),
],
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="StrokeJoin Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare corner rendering for each StrokeJoin value."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(stroke_join) for stroke_join in ft.StrokeJoin
],
),
],
),
),
)
```

```

),
 1,
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildBEVELclass-attributeinstance-attribute

Joins between line segments connect the corners of the butt ends of the line segments to give a beveled appearance.

### buildMITERclass-attributeinstance-attribute

Joins between line segments form sharp corners.

### buildROUNDclass-attributeinstance-attribute

Joins between line segments are semi-circular.

# StrutStyle

TBD

## Properties

- `font_family` - See `Text.font_family`.
- `force_strut_height` - Whether the strut height should be forced.
- `height` - The minimum height of the strut, as a multiple of size.
- `italic` - Whether to use italic typeface.
- `leading` - The amount of additional space to place between lines when rendering text.
- `size` - The size of text (in logical pixels) to use when getting metrics from the font.
- `weight` - The typeface thickness to use when calculating the strut.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

**buildfont\_family**class-attributeinstance-attribute

`font_family: str | None = None`

See `Text.font_family`.

**buildforce\_strut\_height**class-attributeinstance-attribute

`force_strut_height: bool | None = None`

Whether the strut height should be forced.

Defaults to `False`.

**buildheight**class-attributeinstance-attribute

`height: Number | None = None`

The minimum height of the strut, as a multiple of size.

See detailed explanation here: <https://api.flutter.dev/flutter/painting/StrutStyle/height.html>

**builditalic**class-attributeinstance-attribute

`italic: bool = False`

Whether to use italic typeface.

**buildleading**class-attributeinstance-attribute

`leading: Number | None = None`

The amount of additional space to place between lines when rendering text.

Defaults to using the font-specified leading value.

**buildsize**class-attributeinstance-attribute

`size: Number | None = None`

The size of text (in logical pixels) to use when getting metrics from the font.

Defaults to 14.

**buildweight**class-attributeinstance-attribute

`weight: FontWeight | None = None`

The typeface thickness to use when calculating the strut.

Defaults to `FontWeight.W_400`.

## Methods

### deployed\_codecopy

```
copy(
 size: Number | None = None,
 height: Number | None = None,
 weight: FontWeight | None = None,
 italic: bool | None = None,
 font_family: str | None = None,
 leading: Number | None = None,
 force_strut_height: bool | None = None,
) -> StrutStyle
```

Returns a copy of this object with the specified properties overridden.

# SweepGradient

Creates a sweep gradient centered at center that starts at start\_angle and ends at end\_angle.

More information here.

Inherits: Gradient

## Properties

- center - The center of the gradient, as an offset into the (-1.0, -1.0) x (1.0, 1.0) square describing the gradient which will be mapped onto the paint box.
- end\_angle - The angle in radians at which stop 1.0 of the gradient is placed.
- start\_angle - The angle in radians at which stop 0.0 of the gradient is placed.

## Properties

### buildcenterclass-attributeinstance-attribute

```
center: Alignment = field(
 default_factory=lambda: Alignment.CENTER
)
```

The center of the gradient, as an offset into the (-1.0, -1.0) x (1.0, 1.0) square describing the gradient which will be mapped onto the paint box.

For example, an Alignment.CENTER will place the sweep gradient in the center of the box.

### buildend\_angleclass-attributeinstance-attribute

```
end_angle: Number = math.pi * 2
```

The angle in radians at which stop 1.0 of the gradient is placed.

### buildstart\_angleclass-attributeinstance-attribute

```
start_angle: Number = 0.0
```

The angle in radians at which stop 0.0 of the gradient is placed.

## Examples

### Container with sweep gradient

play\_arrowTry Online

```
import math

import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Container(
 alignment=ft.Alignment.CENTER,
 width=150,
 height=150,
 border_radius=ft.BorderRadius.all(5),
 gradient=ft.SweepGradient(
 center=ft.Alignment.CENTER,
 start_angle=0.0,
 end_angle=math.pi * 2,
 stops=[0.0, 0.25, 0.5, 0.75, 1.0],
 colors=[
 "0xFF4285F4",
 "0xFF34A853",
 "0xFFFFB800",
 "0xFFE44335",
 "0xFF4285F4",
],
),
),
],
),
),
)
```

```
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

Image: container - [https://flet.dev/docs/examples/controls/core/types/gradient/sweep\\_gradient/media/container.png](https://flet.dev/docs/examples/controls/core/types/gradient/sweep_gradient/media/container.png)

# SwitchTheme

Defines default property values for descendant Switch controls.

## Properties

- `mouse_cursor` - Overrides the default value of `Switch.mouse_cursor` in all descendant Switch controls.
- `overlay_color` - Overrides the default value of `Switch.overlay_color` in all descendant Switch controls.
- `padding` - Overrides the default value of `Switch.padding` in all descendant Switch controls.
- `splash_radius` - Overrides the default value of `Switch.splash_radius` in all descendant Switch controls.
- `thumb_color` - Overrides the default value of `Switch.thumb_color` in all descendant Switch controls.
- `thumb_icon` - Overrides the default value of `Switch.thumb_icon` in all descendant Switch controls.
- `track_color` - Overrides the default value of `Switch.track_color` in all descendant Switch controls.
- `track_outline_color` - Overrides the default value of `Switch.track_outline_color` in all descendant Switch controls.
- `track_outline_width` - Overrides the default value of `Switch.track_outline_width` in all descendant Switch controls.

## Properties

### `buildmouse_cursor`class-attributeinstance-attribute

`mouse_cursor: ControlStateValue[MouseCursor] | None = None`

Overrides the default value of `Switch.mouse_cursor` in all descendant Switch controls.

### `buildoverlay_color`class-attributeinstance-attribute

`overlay_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Switch.overlay_color` in all descendant Switch controls.

### `buildpadding`class-attributeinstance-attribute

`padding: PaddingValue | None = None`

Overrides the default value of `Switch.padding` in all descendant Switch controls.

### `buildsplash_radius`class-attributeinstance-attribute

`splash_radius: Number | None = None`

Overrides the default value of `Switch.splash_radius` in all descendant Switch controls.

### `buildthumb_color`class-attributeinstance-attribute

`thumb_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Switch.thumb_color` in all descendant Switch controls.

### `buildthumb_icon`class-attributeinstance-attribute

`thumb_icon: ControlStateValue[IconData] | None = None`

Overrides the default value of `Switch.thumb_icon` in all descendant Switch controls.

### `buildtrack_color`class-attributeinstance-attribute

`track_color: ControlStateValue[ColorValue] | None = None`

Overrides the default value of `Switch.track_color` in all descendant Switch controls.

### `buildtrack_outline_color`class-attributeinstance-attribute

`track_outline_color: (ControlStateValue[ColorValue] | None) = None`

Overrides the default value of `Switch.track_outline_color` in all descendant Switch controls.

### `buildtrack_outline_width`class-attributeinstance-attribute

`track_outline_width: (ControlStateValue[Number | None] | None)`



) = None

Overrides the default value of `Switch.track_outline_width` in all descendant Switch controls.

# SystemOverlayStyle

Allows the customization of the mobile's system overlay (which consists of the system status and navigation bars) appearance.

## Properties

- `enforce_system_navigation_bar_contrast` - Indicates whether the system should enforce contrast for the status bar when setting a transparent status bar.
- `enforce_system_status_bar_contrast` - Indicates whether the system should enforce contrast for the navigation bar when setting a transparent navigation bar.
- `status_bar_brightness` - The brightness of the status bar.
- `status_bar_color` - The color of the status bar.
- `status_bar_icon_brightness` - The brightness of the status bar icons.
- `system_navigation_bar_color` - The color of the system navigation bar.
- `system_navigation_bar_divider_color` - The color of the divider between the system navigation bar and the app content.
- `system_navigation_bar_icon_brightness` - The brightness of the system navigation bar icons.

## Properties

### `buildenforce_system_navigation_bar_contrastclass-attributeinstance-attribute`

`enforce_system_navigation_bar_contrast: bool | None = None`

Indicates whether the system should enforce contrast for the status bar when setting a transparent status bar.

### `buildenforce_system_status_bar_contrastclass-attributeinstance-attribute`

`enforce_system_status_bar_contrast: bool | None = None`

Indicates whether the system should enforce contrast for the navigation bar when setting a transparent navigation bar.

### `buildstatus_bar_brightnessclass-attributeinstance-attribute`

`status_bar_brightness: Brightness | None = None`

The brightness of the status bar.

### `buildstatus_bar_colorclass-attributeinstance-attribute`

`status_bar_color: ColorValue | None = None`

The color of the status bar.

### `buildstatus_bar_icon_brightnessclass-attributeinstance-attribute`

`status_bar_icon_brightness: Brightness | None = None`

The brightness of the status bar icons.

### `buildsystem_navigation_bar_colorclass-attributeinstance-attribute`

`system_navigation_bar_color: ColorValue | None = None`

The color of the system navigation bar.

### `buildsystem_navigation_bar_divider_colorclass-attributeinstance-attribute`

`system_navigation_bar_divider_color: ColorValue | None = (None)`

The color of the divider between the system navigation bar and the app content.

### `buildsystem_navigation_bar_icon_brightnessclass-attributeinstance-attribute`

`system_navigation_bar_icon_brightness: Brightness | None = (None)`

The brightness of the system navigation bar icons.

# TabAlignment

Defines how tabs are aligned horizontally in a Tabs.

Inherits: enum.Enum

## Properties

- CENTER - Tabs are aligned to the center of the TabBar.
- FILL - If TabBar.scrollable is False, tabs are stretched to fill the TabBar.
- START - If TabBar.scrollable is True, tabs are aligned to the start of the TabBar.
- START\_OFFSET - If TabBar.scrollable is True, tabs are aligned to the start of the TabBar with an offset of 52.0 pixels.

## Examples

### Tab Alignment showcase

[play\\_arrow](#)Try Online

```
import flet as ft

def showcase_card(alignment: ft.TabAlignment) -> ft.Container:
 scrollable = alignment != ft.TabAlignment.FILL
 tabs_count = 4

 tab_bar = ft.TabBar(
 scrollable=scrollable,
 tab_alignment=alignment,
 tabs=[ft.Tab(label=f"Tab {i + 1}") for i in range(tabs_count)],
)

 tab_view = ft.Container(
 height=70,
 alignment=ft.Alignment.CENTER,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.TabBarView(
 controls=[
 ft.Container(
 alignment=ft.Alignment.CENTER,
 content=ft.Text(f"View {i + 1}"),
)
 for i in range(tabs_count)
]
),
)

 return ft.Container(
 width=380,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(alignment.name, weight=ft.FontWeight.BOLD),
 ft.Tabs(
 length=tabs_count,
 selected_index=1,
 content=ft.Column(spacing=8, controls=[tab_bar, tab_view]),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="TabAlignment Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare how tabs are positioned within the tab bar."),
]
)
)
)
```

```

 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(alignment) for alignment in ft.TabAlignment
],
),
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

**buildCENTER**class-attributeinstance-attribute

Tabs are aligned to the center of the TabBar.

**buildFILL**class-attributeinstance-attribute

If TabBar.scrollable is False, tabs are stretched to fill the TabBar.

**buildSTART**class-attributeinstance-attribute

If TabBar.scrollable is True, tabs are aligned to the start of the TabBar.

**buildSTART\_OFFSET**class-attributeinstance-attribute

If TabBar.scrollable is True, tabs are aligned to the start of the TabBar with an offset of 52.0 pixels.

# TabBarHoverEvent

Represents a hover-state change for a tab in a TabBar.

This event is emitted by TabBar.on\_hover when a pointer enters or exits a tab item. When moving directly from one tab to another, the handler is typically called twice: first with hovering set to False for the previous tab, then with hovering set to True for the next tab.

Inherits: Event

## Properties

- hovering - Whether a pointer has entered (True) or exited (False) the tab bar at index.
- index - The index of the tab that is being hovered over.

## Properties

### buildhoveringinstance-attribute

hovering: bool

Whether a pointer has entered (True) or exited (False) the tab bar at index.

### buildindexinstance-attribute

index: int

The index of the tab that is being hovered over.

# TabBarIndicatorSize

Defines how the bounds of the selected tab indicator are computed.

Inherits: `enum.Enum`

## Properties

- LABEL - The tab's bounds are only as wide as the (centered) tab widget itself.
- TAB - The tab indicator's bounds are as wide as the space occupied by the tab in the tab bar: from the right edge of the previous tab to the left edge of the next tab.

## Examples

### Tab Bar Indicator Size showcase

[play\\_arrow](#)Try Online

```
import flet as ft

def showcase_card(indicator_size: ft.TabBarIndicatorSize) -> ft.Container:
 tabs_count = 3

 tab_bar = ft.TabBar(
 indicator_size=indicator_size,
 indicator=ft.UnderlineTabIndicator(
 border_side=ft.BorderSide(width=4, color=ft.Colors.PRIMARY),
 insets=ft.Padding.only(bottom=3),
),
 tabs=[
 ft.Tab(label="A"),
 ft.Tab(label="Long Label"),
 ft.Tab(label="Mid"),
],
)

 tab_view = ft.Container(
 height=70,
 alignment=ft.Alignment.CENTER,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.TabBarView(
 controls=[
 ft.Container(
 alignment=ft.Alignment.CENTER,
 content=ft.Text(f"View {i + 1}"),
)
 for i in range(tabs_count)
]
),
)

 return ft.Container(
 width=380,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(indicator_size.name, weight=ft.FontWeight.BOLD),
 ft.Tabs(
 length=tabs_count,
 selected_index=1,
 content=ft.Column(spacing=8, controls=[tab_bar, tab_view]),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="TabBarIndicatorSize Showcase")
 page.add()
```

```

ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare indicator width when matching tab bounds vs "
 "label bounds."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(indicator_size)
 for indicator_size in ft.TabBarIndicatorSize
],
),
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildLABELclass-attributeinstance-attribute

The tab's bounds are only as wide as the (centered) tab widget itself.

This value is used to align the tab's label, typically a [Tab] widget's text or icon, with the selected tab indicator.

### buildTABclass-attributeinstance-attribute

The tab indicator's bounds are as wide as the space occupied by the tab in the tab bar: from the right edge of the previous tab to the left edge of the next tab.

# TabBarTheme

Customizes the appearance of TabBar control across the app.

## Properties

- `divider_color` - Overrides the default value for `TabBar.divider_color`.
- `divider_height` - Overrides the default value for `TabBar.divider_height`.
- `indicator` - Overrides the default value for `TabBar.indicator`.
- `indicator_animation` - Overrides the default value for `TabBar.indicator_animation`.
- `indicator_color` - Overrides the default value for `TabBar.indicator_color`.
- `indicator_size` - Overrides the default value for `TabBar.indicator_size`.
- `label_color` - Overrides the default value for `TabBar.label_color`.
- `label_padding` - Overrides the default value for `TabBar.label_padding`.
- `label_text_style` - Overrides the default value for `TabBar.label_text_style`.
- `mouse_cursor` - Overrides the default value for `TabBar.mouse_cursor`.
- `overlay_color` - Overrides the default value for `TabBar.overlay_color`.
- `splash_border_radius` - Overrides the default value for `TabBar.splash_border_radius`.
- `tab_alignment` - Overrides the default value for `TabBar.tab_alignment`.
- `unselected_label_color` - Overrides the default value for `TabBar.unselected_label_color`.
- `unselected_label_text_style` - Overrides the default value for `TabBar.unselected_label_text_style`.

## Properties

### `builddivider_color`class-attributeinstance-attribute

`divider_color: ColorValue | None = None`

Overrides the default value for `TabBar.divider_color`.

### `builddivider_height`class-attributeinstance-attribute

`divider_height: Number | None = None`

Overrides the default value for `TabBar.divider_height`.

### `buildindicator`class-attributeinstance-attribute

`indicator: UnderlineTabIndicator | None = None`

Overrides the default value for `TabBar.indicator`.

### `buildindicator_animation`class-attributeinstance-attribute

`indicator_animation: TabIndicatorAnimation | None = None`

Overrides the default value for `TabBar.indicator_animation`.

### `buildindicator_color`class-attributeinstance-attribute

`indicator_color: ColorValue | None = None`

Overrides the default value for `TabBar.indicator_color`.

### `buildindicator_size`class-attributeinstance-attribute

`indicator_size: TabBarIndicatorSize | None = None`

Overrides the default value for `TabBar.indicator_size`.

### `buildlabel_color`class-attributeinstance-attribute

`label_color: ColorValue | None = None`

Overrides the default value for `TabBar.label_color`.



### buildlabel\_paddingclass-attributeinstance-attribute

label\_padding: PaddingValue | None = None

Overrides the default value for TabBar.label\_padding.

### buildlabel\_text\_styleclass-attributeinstance-attribute

label\_text\_style: TextStyle | None = None

Overrides the default value for TabBar.label\_text\_style.

### buildmouse\_cursorclass-attributeinstance-attribute

```
mouse_cursor: (
 ControlStateValue[MouseCursor | None] | None
) = None
```

Overrides the default value for TabBar.mouse\_cursor.

### buildoverlay\_colorclass-attributeinstance-attribute

overlay\_color: ControlStateValue[ColorValue] | None = None

Overrides the default value for TabBar.overlay\_color.

### buildsplash\_border\_radiusclass-attributeinstance-attribute

splash\_border\_radius: BorderRadiusValue | None = None

Overrides the default value for TabBar.splash\_border\_radius.

### buildtab\_alignmentclass-attributeinstance-attribute

tab\_alignment: TabAlignment | None = None

Overrides the default value for TabBar.tab\_alignment.

### buildunselected\_label\_colorclass-attributeinstance-attribute

unselected\_label\_color: ColorValue | None = None

Overrides the default value for TabBar.unselected\_label\_color.

### buildunselected\_label\_text\_styleclass-attributeinstance-attribute

unselected\_label\_text\_style: TextStyle | None = None

Overrides the default value for TabBar.unselected\_label\_text\_style.

## Examples

### Example 1

```
page.theme = ft.Theme(
 tabs_theme=ft.TabBarTheme(
 divider_color=ft.Colors.BLUE,
 indicator_color=ft.Colors.RED,
 indicator_tab_size=True,
 label_color=ft.Colors.GREEN,
 unselected_label_color=ft.Colors.AMBER,
 overlay_color={
 ft.MaterialState.FOCUSED: ft.Colors.with_opacity(0.2, ft.Colors.GREEN),
 ft.MaterialState.DEFAULT: ft.Colors.with_opacity(0.2, ft.Colors.PINK),
 },
),
)
```

# TabIndicatorAnimation

Defines how the tab indicator animates when the selected tab changes.

Inherits: `enum.Enum`

## Properties

- **ELASTIC** - The tab indicator animates with an elastic effect.
- **LINEAR** - The tab indicator animates linearly.

## Examples

### Tab Indicator Animation showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(animation: ft.TabIndicatorAnimation) -> ft.Container:
 tabs_count = 3

 tab_bar = ft.TabBar(
 indicator_animation=animation,
 indicator=ft.UnderlineTabIndicator(
 border_side=ft.BorderSide(width=4, color=ft.Colors.PRIMARY),
 insets=ft.Padding.only(bottom=3),
),
 tabs=[
 ft.Tab(label="Home"),
 ft.Tab(label="Search"),
 ft.Tab(label="Profile"),
],
)

 tab_view = ft.Container(
 height=70,
 alignment=ft.Alignment.CENTER,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.TabBarView(
 controls=[
 ft.Container(
 alignment=ft.Alignment.CENTER,
 content=ft.Text(f"View {i + 1}"),
)
 for i in range(tabs_count)
]
),
)

 return ft.Container(
 width=360,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(animation.name, weight=ft.FontWeight.BOLD),
 ft.Tabs(
 length=tabs_count,
 selected_index=1,
 content=ft.Column(spacing=8, controls=[tab_bar, tab_view]),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="TabIndicatorAnimation Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
```

```

 controls=[
 ft.Text("Click tabs to compare indicator movement animation."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(animation)
 for animation in ft.TabIndicatorAnimation
],
),
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

**buildELASTIC**class-attributeinstance-attribute

The tab indicator animates with an elastic effect.

**buildLINEAR**class-attributeinstance-attribute

The tab indicator animates linearly.

# TapEvent

Payload for tap press/release events.

Inherits: Event

## Properties

- `global_position` - The global position at which the pointer contacted the screen.
- `kind` - The kind of the device that initiated the event.
- `local_position` - The local position at which the pointer contacted the screen.

## Properties

### `buildglobal_positionclass-attributeinstance-attribute`

```
global_position: Offset | None = field(
 default=None, metadata={"data_field": "g"}
)
```

The global position at which the pointer contacted the screen.

### `buildkindclass-attributeinstance-attribute`

```
kind: PointerDeviceType | None = field(
 default=None, metadata={"data_field": "k"}
)
```

The kind of the device that initiated the event.

### `buildlocal_positionclass-attributeinstance-attribute`

```
local_position: Offset | None = field(
 default=None, metadata={"data_field": "l"}
)
```

The local position at which the pointer contacted the screen.

# TemplateRoute

Matches a concrete route string against parameterized route templates.

Instances keep the original route in `route` and expose matched template parameters as dynamic attributes after a successful call to `match()`. Parameter attributes captured by a previous match are reset to `None` before each new matching attempt, preventing stale values from leaking between checks.

## Properties

- `route`

## Methods

- `match` - Tries to match this instance route against a route template.

## Properties

`buildrouteinstance-attribute`

## Methods

### `deployed_codematch`

```
match(route_template: str) -> bool
```

Tries to match this instance route against a route template.

The template is compiled with `repath.pattern()`. If matching succeeds, named parameters are stored and also assigned as attributes on this object (for example, `self.user_id`). If matching fails, previously captured attributes remain cleared.

Parameters:

- `route_template (str)` - Route template in a format supported by `repath.pattern()`.

Returns:

- `bool` - True if the route matches the template; otherwise False.

# Tester

::: flet.testing.testers.Tester

# TextAffinity

Defines the permissions which can be checked and requested.

Inherits: `enum.Enum`

## Properties

- `DOWNSTREAM` - The position has affinity for the upstream side of the text position, i.e.
- `UPSTREAM` - The position has affinity for the downstream side of the text position, i.e.

## Properties

`buildDOWNSTREAMclass-attributeinstance-attribute`

The position has affinity for the upstream side of the text position, i.e. in the direction of the beginning of the string.

`buildUPSTREAMclass-attributeinstance-attribute`

The position has affinity for the downstream side of the text position, i.e. in the direction of the end of the string.

# TextAlign

The horizontal alignment of text within an input box.

Inherits: `enum.Enum`

## Properties

- **CENTER** - Align the text in the center of the container.
- **END** - Align the text on the trailing edge of the container.
- **JUSTIFY** - Stretch lines of text that end with a soft line break to fill the width of the container.
- **LEFT** - Align the text on the left edge of the container.
- **RIGHT** - Align the text on the right edge of the container.
- **START** - Align the text on the leading edge of the container.

## Examples

### Text Align showcase

[play\\_arrow](#) Try Online

```
import flet as ft

def showcase_card(text_align: ft.TextAlign) -> ft.Container:
 return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(text_align.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=240,
 height=130,
 padding=10,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Text(
 text_align=text_align,
 value="Flet helps you build cross-platform Python apps from one codebase.", # noqa: E501
),
],
),
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="TextAlign Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare horizontal text alignment modes in the same "
 "text block."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(text_align) for text_align in ft.TextAlign
],
),
],
),
),
)
```



```
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildCENTER**class-attributeinstance-attribute

Align the text in the center of the container.

**buildEND**class-attributeinstance-attribute

Align the text on the trailing edge of the container.

**buildJUSTIFY**class-attributeinstance-attribute

Stretch lines of text that end with a soft line break to fill the width of the container.

**buildLEFT**class-attributeinstance-attribute

Align the text on the left edge of the container.

**buildRIGHT**class-attributeinstance-attribute

Align the text on the right edge of the container.

**buildSTART**class-attributeinstance-attribute

Align the text on the leading edge of the container.

# TextBaseline

A horizontal line used for aligning text.

Inherits: `enum.Enum`

## Properties

- `ALPHABETIC` - The horizontal line used to align the bottom of glyphs for alphabetic characters.
- `IDEOGRAPHIC` - The horizontal line used to align ideographic characters.

## Properties

`buildALPHABETIC`class-attributeinstance-attribute

The horizontal line used to align the bottom of glyphs for alphabetic characters.

`buildIDEOGRAPHIC`class-attributeinstance-attribute

The horizontal line used to align ideographic characters.

# TextButtonTheme

Customizes the appearance of TextButton across the app.

## Properties

- `style` - Overrides the default value of `TextButton.style` in all descendant `TextButton` controls.

## Properties

**buildstyleclass-attributeinstance-attribute**

`style: ButtonStyle | None = None`

Overrides the default value of `TextButton.style` in all descendant `TextButton` controls.

# TextCapitalization

Configures how the platform keyboard will select an uppercase or lowercase keyboard.

Only supports text keyboards, other keyboard types will ignore this configuration. Capitalization is locale-aware.

Inherits: `enum.Enum`

## Properties

- **CHARACTERS** - Uppercase keyboard for each character.
- **NONE** - Lowercase keyboard.
- **SENTENCES** - Uppercase keyboard for the first letter of each sentence.
- **WORDS** - Uppercase keyboard for the first letter of each word.

## Examples

### Text Capitalization showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(cap: ft.TextCapitalization) -> ft.Container:
 field = ft.TextField(
 width=260,
 label="Type here",
 capitalization=cap,
 border=ft.InputBorder.OUTLINE,
)

 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(cap.name, weight=ft.FontWeight.BOLD),
 field,
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="TextCapitalization Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare keyboard capitalization preferences for text fields."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(c) for c in ft.TextCapitalization],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildCHARACTERSclass-attributeinstance-attribute**

Uppercase keyboard for each character.

Info

Corresponds to `InputType.TYPE_TEXT_FLAG_CAP_CHARACTERS` on Android, and `UITextAutocapitalizationTypeAllCharacters` on iOS.

### **buildNONEclass-attributeinstance-attribute**

Lowercase keyboard.

### **buildSENTENCESclass-attributeinstance-attribute**

Uppercase keyboard for the first letter of each sentence.

Info

Corresponds to `InputType.TYPE_TEXT_FLAG_CAP_SENTENCES` on Android, and `UITextAutocapitalizationTypeSentences` on iOS.

### **buildWORDSclass-attributeinstance-attribute**

Uppercase keyboard for the first letter of each word.

Info

Corresponds to `InputType.TYPE_TEXT_FLAG_CAP_WORDS` on Android, and `UITextAutocapitalizationTypeWords` on iOS.

# TextDecoration

A linear decoration to draw near the text.

Inherits: `enum.IntFlag`

## Properties

- `LINE_THROUGH` - Draw a line through each line of text.
- `NONE` - Do not draw a decoration.
- `OVERLINE` - Draw a line above each line of text.
- `UNDERLINE` - Draw a line underneath each line of text.

## Methods

- `combine` - Creates a decoration that paints the union of all the given decorations.

## Properties

`buildLINE_THROUGH`class-attributeinstance-attribute

Draw a line through each line of text.

`buildNONE`class-attributeinstance-attribute

Do not draw a decoration.

`buildOVERLINE`class-attributeinstance-attribute

Draw a line above each line of text.

`buildUNDERLINE`class-attributeinstance-attribute

Draw a line underneath each line of text.

## Methods

`deployed_codecombine`classmethod

```
combine(
 decorations: list[TextDecoration],
) -> TextDecoration
```

Creates a decoration that paints the union of all the given decorations.

## Usage Example

The enum is a flag, so multiple decorations can be combined together as follows:

```
style = ft.TextStyle(decoration=ft.TextDecoration.UNDERLINE | ft.TextDecoration.OVERLINE)
```

# TextDecorationStyle

The style in which to draw a text decoration.

Inherits: `enum.Enum`

## Properties

- DASHED - Draw a dashed line.
- DOTTED - Draw a dotted line.
- DOUBLE - Draw two lines.
- SOLID - Draw a solid line.
- WAVY - Draw a sinusoidal line.

## Examples

### Text Decoration Style showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(style: ft.TextDecorationStyle) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(style.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=240,
 height=80,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 alignment=ft.Alignment.CENTER,
 content=ft.Text(
 "Decorated text sample",
 style=ft.TextStyle(
 size=22,
 decoration=ft.TextDecoration.UNDERLINE,
 decoration_style=style,
 decoration_thickness=2.4,
 decoration_color=ft.Colors.PRIMARY,
),
),
],
),
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="TextDecorationStyle Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare underline rendering for each decoration style."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(style) for style in ft.TextDecorationStyle
],
),
],
),
),
)
```

```
)
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildDASHED**class-attributeinstance-attribute

Draw a dashed line.

**buildDOTTED**class-attributeinstance-attribute

Draw a dotted line.

**buildDOUBLE**class-attributeinstance-attribute

Draw two lines.

**buildSOLID**class-attributeinstance-attribute

Draw a solid line.

**buildWAVY**class-attributeinstance-attribute

Draw a sinusoidal line.



# TextOverflow

How overflowing text should be handled.

Inherits: `enum.Enum`

## Properties

- CLIP - Clip the overflowing text to fit its container.
- ELLIPSIS - Use an ellipsis to indicate that the text has overflowed.
- FADE - Fade the overflowing text to transparent.
- VISIBLE - Render overflowing text outside of its container.

## Examples

### Text Overflow showcase

[play\\_arrow](#) Try Online

```
import flet as ft
```

```
def showcase_card(overflow: ft.TextOverflow) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(overflow.name, weight=ft.FontWeight.BOLD),
 ft.Container(
 width=250,
 height=46,
 padding=8,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=8,
 bgcolor=ft.Colors.SURFACE,
 content=ft.Text(
 "The quick brown fox jumps over the lazy dog near the Flet bridge.", # noqa: E501
 max_lines=1,
 overflow=overflow,
),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="TextOverflow Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare how one-line text behaves when content exceeds width."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(overflow) for overflow in ft.TextOverflow
],
),
],
),
),
)

if __name__ == "__main__":
```

```
ft.run(main)
```

## Properties

**buildCLIPclass-attributeinstance-attribute**

Clip the overflowing text to fix its container.

**buildELLIPSISclass-attributeinstance-attribute**

Use an ellipsis to indicate that the text has overflowed.

**buildFADEclass-attributeinstance-attribute**

Fade the overflowing text to transparent.

**buildVISIBLEclass-attributeinstance-attribute**

Render overflowing text outside of its container.

# TextSelection

A range of text that represents a selection.

## Properties

- **affinity** - If the text range is collapsed and has more than one visual location (e.g., occurs at a line break), which of the two locations to use when painting the caret.
- **base\_offset** - The offset at which the selection originates.
- **directional** - Whether this selection has disambiguated its base and extent.
- **end** - The next index after the characters in this range.
- **extent\_offset** - The offset at which the selection terminates.
- **is\_collapsed** - Whether this range is empty (but still potentially placed inside the text).
- **is\_normalized** - Whether the start of this range precedes the end.
- **is\_valid** - Whether this range represents a valid position in the text.
- **start** - The index of the first character in the range.

## Methods

- **get\_selected\_text** - Returns the selected text from the given full text.

## Properties

### buildaffinityclass-attributeinstance-attribute

`affinity: TextAffinity = TextAffinity.DOWNSTREAM`

If the text range is collapsed and has more than one visual location (e.g., occurs at a line break), which of the two locations to use when painting the caret.

### buildbase\_offsetinstance-attribute

`base_offset: int`

The offset at which the selection originates.

### bulldirectionalclass-attributeinstance-attribute

`directional: bool = False`

Whether this selection has disambiguated its base and extent.

### buildendproperty

`end: int`

The next index after the characters in this range.

#### Note

This property is read-only.

### buildextent\_offsetinstance-attribute

`extent_offset: int`

The offset at which the selection terminates.

### buildis\_collapsedproperty

`is_collapsed: bool`

Whether this range is empty (but still potentially placed inside the text).

#### Note

This property is read-only.

### buildis\_normalizedproperty

`is_normalized: bool`

Whether the start of this range precedes the end.

Note

This property is read-only.

### **buildis\_validproperty**

`is_valid: bool`

Whether this range represents a valid position in the text.

Note

This property is read-only.

### **buildstartproperty**

`start: int`

The index of the first character in the range.

Note

This property is read-only.

## **Methods**

### **deployed\_codeget\_selected\_text**

`get_selected_text(source_text: str) -> str`

Returns the selected text from the given full text.

Parameters:

- `source_text` (str) - The full text to get the selection from.

Raises:

- `AssertionError` - If the selection is not valid, i.e. `is_valid` is `False`.

# TextSelectionChangeCause

Indicates what triggered the change in selected text.

Inherits: `enum.Enum`

## Properties

- `DOUBLE_TAP` - The user tapped twice in quick succession on the text and that caused the selection (or the location of the cursor) to change.
- `DRAG` - The user used the mouse to change the selection by dragging over a piece of text.
- `FORCE_PRESS` - The user force-pressed the text and that caused the selection (or the location of the cursor) to change.
- `KEYBOARD` - The user used the keyboard to change the selection or the location of the cursor.
- `LONG_PRESS` - The user long-pressed the text and that caused the selection (or the location of the cursor) to change.
- `SCRIBBLE` - The user used iPadOS 14+ Scribble to change the selection.
- `TAP` - The user tapped on the text and that caused the selection (or the location of the cursor) to change.
- `TOOLBAR` - The user used the selection toolbar to change the selection or the location of the cursor.
- `UNKNOWN` - The cause of the selection change is unknown or could not be determined.

## Properties

### `buildDOUBLE_TAPclass-attributeinstance-attribute`

The user tapped twice in quick succession on the text and that caused the selection (or the location of the cursor) to change.

### `buildDRAGclass-attributeinstance-attribute`

The user used the mouse to change the selection by dragging over a piece of text.

### `buildFORCE_PRESSclass-attributeinstance-attribute`

The user force-pressed the text and that caused the selection (or the location of the cursor) to change.

### `buildKEYBOARDclass-attributeinstance-attribute`

The user used the keyboard to change the selection or the location of the cursor.

Keyboard-triggered selection changes may be caused by the IME as well as by accessibility tools (e.g. TalkBack on Android).

### `buildLONG_PRESSclass-attributeinstance-attribute`

The user long-pressed the text and that caused the selection (or the location of the cursor) to change.

### `buildSCRIBBLEclass-attributeinstance-attribute`

The user used iPadOS 14+ Scribble to change the selection.

### `buildTAPclass-attributeinstance-attribute`

The user tapped on the text and that caused the selection (or the location of the cursor) to change.

### `buildTOOLBARclass-attributeinstance-attribute`

The user used the selection toolbar to change the selection or the location of the cursor.

An example is when the user taps on select all in the tool bar.

### `buildUNKNOWNclass-attributeinstance-attribute`

The cause of the selection change is unknown or could not be determined.

# TextSelectionChangeEvent

An event emitted when the text selection changes.

Inherits: Event

## Properties

- `cause` - The cause of the selection change.
- `selected_text` - The selected text.
- `selection` - The new text selection.

## Properties

### `buildcauseclass-attributeinstance-attribute`

`cause: TextSelectionChangeCause | None = None`

The cause of the selection change.

### `buildselected_textinstance-attribute`

`selected_text: str`

The selected text.

### `buildselectioninstance-attribute`

`selection: TextSelection`

The new text selection.

# TextSpan

A text span.

Usage Example: As a child of Text.spans.

For the object to be useful, at least one of text or spans should be set.

Inherits: Control

## Properties

- `semantics_label` - An alternative semantics label for this text.
- `spans` - Additional spans to include as children.
- `spell_out` - Whether the assistive technologies should spell out this text character by character.
- `style` - Defines the style of this text span.
- `text` - The text contained in this span.
- `url` - The URL to open when this button is clicked.

## Events

- `on_click` - Called when this span is clicked.
- `on_enter` - Called when a mouse pointer has entered this span.
- `on_exit` - Called when a mouse pointer has exited this span.

## Properties

### `buildsemantics_labelclass-attributeinstance-attribute`

`semantics_label: str | None = None`

An alternative semantics label for this text.

If present, the semantics of this control will contain this value instead of the actual text.

Raises:

- `ValueError` - If it is set when text is None.

### `buildspansclass-attributeinstance-attribute`

`spans: list[TextSpan] | None = None`

Additional spans to include as children.

Note

If both spans and text are defined, the text takes precedence.

### `buildspell_outclass-attributeinstance-attribute`

`spell_out: bool | None = None`

Whether the assistive technologies should spell out this text character by character.

If the text is 'hello world', setting this to true causes the assistive technologies, such as VoiceOver or TalkBack, to pronounce 'h-e-l-l-o-space-w-o-r-l-d' instead of complete words. This is useful for texts, such as passwords or verification codes.

If this span contains other text span children, they also inherit the property from this span unless explicitly set.

If the property is not set, this text span inherits the spell out setting from its parent. If this text span does not have a parent or the parent does not have a spell out setting, this text span does not spell out the text by default.

### `buildstyleclass-attributeinstance-attribute`

`style: TextStyle | None = None`

Defines the style of this text span.

### **buildtextclass-attributeinstance-attribute**

`text: str | None = None`

The text contained in this span.

Note

If both text and spans are defined, the text takes precedence.

### **buildurlclass-attributeinstance-attribute**

`url: str | Url | None = None`

The URL to open when this button is clicked.

Additionally, if `on_click` event callback is provided, it is fired after that.

## **Events**

### **bolton\_clickclass-attributeinstance-attribute**

`on_click: ControlEventHandler[TextSpan] | None = None`

Called when this span is clicked.

### **bolton\_enterclass-attributeinstance-attribute**

`on_enter: ControlEventHandler[TextSpan] | None = None`

Called when a mouse pointer has entered this span.

### **bolton\_exitclass-attributeinstance-attribute**

`on_exit: ControlEventHandler[TextSpan] | None = None`

Called when a mouse pointer has exited this span.



# TextStyle

A style describing how to format and paint text.

## Properties

- **baseline** - The common baseline that should be aligned between this text span and its parent text span, or, for the root text spans, with the line box.
- **bgcolor** - Text background <https://flet.dev/docs/types/colors>.
- **color** - Text foreground <https://flet.dev/docs/types/colors>.
- **decoration** - The decorations to paint near the text (e.g., an underline).
- **decoration\_color** - The color in which to paint the text decorations.
- **decoration\_style** - The style in which to paint the text decorations (e.g., dashed).
- **decoration\_thickness** - The thickness of the decoration stroke as a multiplier of the thickness defined by the font.
- **font\_family** - See [https://flet.dev/docs/controls/text#font\\_family](https://flet.dev/docs/controls/text#font_family).
- **font\_family\_fallback** - Ordered fallback font families to use when glyphs are not available in **font\_family**.
- **foreground** - The paint drawn as a foreground for the text.
- **height** - The height of this text span, as a multiple of the font size.
- **italic** - Whether to use italic typeface.
- **letter\_spacing** - The amount of space (in logical pixels) to add between each letter.
- **overflow** - How visual text overflow should be handled.
- **shadow** - TBD
- **size** - The size of glyphs (in logical pixels) to use when painting the text.
- **weight** - The typeface thickness to use when painting the text (e.g., bold).
- **word\_spacing** - The amount of space (in logical pixels) to add at each sequence of white-space (i.e.

## Methods

- **copy** - Returns a copy of this object with the specified properties overridden.

## Properties

### buildbaselineclass-attributeinstance-attribute

**baseline:** `TextBaseline` | `None` = `None`

The common baseline that should be aligned between this text span and its parent text span, or, for the root text spans, with the line box.

### buildbgcolorclass-attributeinstance-attribute

**bgcolor:** `ColorValue` | `None` = `None`

Text background <https://flet.dev/docs/types/colors>.

### buildcolorclass-attributeinstance-attribute

**color:** `ColorValue` | `None` = `None`

Text foreground <https://flet.dev/docs/types/colors>.

### builddecorationclass-attributeinstance-attribute

**decoration:** `TextDecoration` | `None` = `None`

The decorations to paint near the text (e.g., an underline).

### builddecoration\_colorclass-attributeinstance-attribute

**decoration\_color:** `ColorValue` | `None` = `None`

The color in which to paint the text decorations.

### **builddecoration\_styleclass-attributeinstance-attribute**

`decoration_style: TextStyle | None = None`

The style in which to paint the text decorations (e.g., dashed).

Defaults to `TextStyle.SOLID`.

### **builddecoration\_thicknessclass-attributeinstance-attribute**

`decoration_thickness: Number | None = None`

The thickness of the decoration stroke as a multiplier of the thickness defined by the font.

### **buildfont\_familyclass-attributeinstance-attribute**

`font_family: str | None = None`

See [https://flet.dev/docs/controls/text#font\\_family](https://flet.dev/docs/controls/text#font_family).

### **buildfont\_family\_fallbackclass-attributeinstance-attribute**

`font_family_fallback: list[str] | None = None`

Ordered fallback font families to use when glyphs are not available in `font_family`.

### **buildforegroundclass-attributeinstance-attribute**

`foreground: Paint | None = None`

The paint drawn as a foreground for the text.

### **buildheightclass-attributeinstance-attribute**

`height: Number | None = None`

The height of this text span, as a multiple of the font size.

See detailed explanation [here](#).

### **builditalicclass-attributeinstance-attribute**

`italic: bool = False`

Whether to use italic typeface.

### **buildletter\_spacingclass-attributeinstance-attribute**

`letter_spacing: Number | None = None`

The amount of space (in logical pixels) to add between each letter. A negative value can be used to bring the letters closer.

### **buildoverflowclass-attributeinstance-attribute**

`overflow: TextOverflow | None = None`

How visual text overflow should be handled.

### **buildshadowclass-attributeinstance-attribute**

`shadow: BoxShadowValue | None = None`

TBD

### **buildsizeclass-attributeinstance-attribute**

`size: Number | None = None`

The size of glyphs (in logical pixels) to use when painting the text.

Defaults to 14.

### **buildweightclass-attributeinstance-attribute**

`weight: FontWeight | None = None`

The typeface thickness to use when painting the text (e.g., bold).

Defaults to `FontWeight.NORMAL`.

### **buildword\_spacingclass-attributeinstance-attribute**

`word_spacing: Number | None = None`

The amount of space (in logical pixels) to add at each sequence of white-space (i.e. between each word). A negative value can be used to bring the words closer.

## Methods

### deployed\_codecopy

```
copy(
 size: Number | None = None,
 height: Number | None = None,
 weight: FontWeight | None = None,
 italic: bool | None = None,
 decoration: TextDecoration | None = None,
 decoration_color: ColorValue | None = None,
 decoration_thickness: Number | None = None,
 decoration_style: TextDecorationStyle | None = None,
 font_family: str | None = None,
 font_family_fallback: list[str] | None = None,
 color: ColorValue | None = None,
 bgcolor: ColorValue | None = None,
 shadow: BoxShadowValue | None = None,
 foreground: Paint | None = None,
 letter_spacing: Number | None = None,
 word_spacing: Number | None = None,
 overflow: TextOverflow | None = None,
 baseline: TextBaseline | None = None,
)
```

Returns a copy of this object with the specified properties overridden.

# TextTheme

Customizes Text styles.

Material 3 design defines 5 groups of text styles with 3 sizes in each group: "Display", "Headline", "Title", "Label" and "Body" which are used across Flet controls.

## Properties

- `body_large` - Largest of the body styles.
- `body_medium` - Middle size of the body styles.
- `body_small` - Smallest of the body styles.
- `display_large` - Largest of the display styles.
- `display_medium` - Middle size of the display styles.
- `display_small` - Smallest of the display styles.
- `headline_large` - Largest of the headline styles.
- `headline_medium` - Middle size of the headline styles.
- `headline_small` - Smallest of the headline styles.
- `label_large` - Largest of the label styles.
- `label_medium` - Middle size of the label styles.
- `label_small` - Smallest of the label styles.
- `title_large` - Largest of the title styles.
- `title_medium` - Middle size of the title styles.
- `title_small` - Smallest of the title styles.

## Properties

### `body_large`class-attributeinstance-attribute

`body_large: TextStyle | None = None`

Largest of the body styles. Body styles are used for longer passages of text.

### `body_medium`class-attributeinstance-attribute

`body_medium: TextStyle | None = None`

Middle size of the body styles. Body styles are used for longer passages of text. The default text style for Material.

### `body_small`class-attributeinstance-attribute

`body_small: TextStyle | None = None`

Smallest of the body styles.

### `display_large`class-attributeinstance-attribute

`display_large: TextStyle | None = None`

Largest of the display styles. As the largest text on the screen, display styles are reserved for short, important text or numerals. They work best on large screens.

### `display_medium`class-attributeinstance-attribute

`display_medium: TextStyle | None = None`

Middle size of the display styles.

### `display_small`class-attributeinstance-attribute

`display_small: TextStyle | None = None`

Smallest of the display styles.

## **buildheadline\_largeclass-attributeinstance-attribute**

`headline_large: TextStyle | None = None`

Largest of the headline styles. Headline styles are smaller than display styles. They're best-suited for short, high-emphasis text on smaller screens.

## **buildheadline\_mediumclass-attributeinstance-attribute**

`headline_medium: TextStyle | None = None`

Middle size of the headline styles.

## **buildheadline\_smallclass-attributeinstance-attribute**

`headline_small: TextStyle | None = None`

Smallest of the headline styles.

## **buildlabel\_largeclass-attributeinstance-attribute**

`label_large: TextStyle | None = None`

Largest of the label styles. Label styles are smaller, utilitarian styles, used for areas of the UI such as text inside of components or very small supporting text in the content body, like captions. Used for text on Button, TextButton and OutlinedButton.

## **buildlabel\_mediumclass-attributeinstance-attribute**

`label_medium: TextStyle | None = None`

Middle size of the label styles.

## **buildlabel\_smallclass-attributeinstance-attribute**

`label_small: TextStyle | None = None`

Smallest of the label styles.

## **buildtitle\_largeclass-attributeinstance-attribute**

`title_large: TextStyle | None = None`

Largest of the title styles. Titles are smaller than headline styles and should be used for shorter, medium-emphasis text.

## **buildtitle\_mediumclass-attributeinstance-attribute**

`title_medium: TextStyle | None = None`

Middle size of the title styles.

## **buildtitle\_smallclass-attributeinstance-attribute**

`title_small: TextStyle | None = None`

Smallest of the title styles.

# TextThemeStyle

Predefined Material text style roles from the active theme.

Use these values with properties such as `Text.theme_style` to reference semantic typography slots (display, headline, title, body, label) instead of hard-coding font metrics in each control.

Inherits: `enum.Enum`

## Properties

- `BODY_LARGE` - Largest body style.
- `BODY_MEDIUM` - Medium body style.
- `BODY_SMALL` - Smallest body style.
- `DISPLAY_LARGE` - Largest display style.
- `DISPLAY_MEDIUM` - Medium display style.
- `DISPLAY_SMALL` - Smallest display style.
- `HEADLINE_LARGE` - Largest headline style.
- `HEADLINE_MEDIUM` - Medium headline style.
- `HEADLINE_SMALL` - Smallest headline style.
- `LABEL_LARGE` - Largest label style.
- `LABEL_MEDIUM` - Medium label style.
- `LABEL_SMALL` - Smallest label style.
- `TITLE_LARGE` - Largest title style.
- `TITLE_MEDIUM` - Medium title style.
- `TITLE_SMALL` - Smallest title style.

## Properties

`buildBODY_LARGE`class-attributeinstance-attribute

Largest body style.

Body styles are intended for longer passages and primary reading content.

`buildBODY_MEDIUM`class-attributeinstance-attribute

Medium body style.

Common default style for standard paragraph text in Material-themed UIs.

`buildBODY_SMALL`class-attributeinstance-attribute

Smallest body style.

Suitable for secondary body text, footnotes, and compact long-form content.

`buildDISPLAY_LARGE`class-attributeinstance-attribute

Largest display style.

Intended for very prominent, short text on large surfaces, such as hero headlines.

`buildDISPLAY_MEDIUM`class-attributeinstance-attribute

Medium display style.

Intended for high-impact short text when `DISPLAY_LARGE` is too dominant.

`buildDISPLAY_SMALL`class-attributeinstance-attribute

Smallest display style.

Intended for concise, emphasized text that still needs display-level prominence.

**buildHEADLINE\_LARGEclass-attributeinstance-attribute**

Largest headline style.

Headline styles are below display styles and suit short, high-emphasis headings.

**buildHEADLINE\_MEDIUMclass-attributeinstance-attribute**

Medium headline style.

Suitable for section headers and prominent in-content headings.

**buildHEADLINE\_SMALLclass-attributeinstance-attribute**

Smallest headline style.

Suitable for compact headline usage on dense layouts.

**buildLABEL\_LARGEclass-attributeinstance-attribute**

Largest label style.

Commonly used for component text, such as button labels and prominent captions.

**buildLABEL\_MEDIUMclass-attributeinstance-attribute**

Medium label style.

Suitable for compact UI labels in controls and supporting interface text.

**buildLABEL\_SMALLclass-attributeinstance-attribute**

Smallest label style.

Suitable for dense, low-footprint labels and small supporting annotations.

**buildTITLE\_LARGEclass-attributeinstance-attribute**

Largest title style.

Titles are typically used for medium-emphasis, short text such as card titles.

**buildTITLE\_MEDIUMclass-attributeinstance-attribute**

Medium title style.

Suitable for secondary titles and emphasized labels in structured content.

**buildTITLE\_SMALLclass-attributeinstance-attribute**

Smallest title style.

Suitable for compact title text where space is limited.

# Theme

Customizes the overall appearance of the application.

## Properties

- `appbar_theme`
- `badge_theme`
- `banner_theme`
- `bottom_appbar_theme`
- `bottom_sheet_theme`
- `button_theme` - Customizes the appearance of `Button` across the app.
- `canvas_color`
- `card_bgcolor`
- `card_theme`
- `checkbox_theme`
- `chip_theme` - Customizes the appearance of `Chip` across the app.
- `color_scheme` - Overrides the default `ColorScheme` used for the application.
- `color_scheme_seed` - Overrides the default color scheme seed used to generate `ColorScheme`.
- `data_table_theme` - Customizes the appearance of `DataTable` across the app.
- `date_picker_theme` - Customizes the appearance of `DatePicker` across the app.
- `dialog_theme` - Customizes the appearance of `AlertDialog` across the app.
- `disabled_color`
- `divider_color` - Overrides the default color of dividers used in `Divider`, `VerticalDivider`, dividers between `ListTiles`, and dividers between rows in `DataTable`.
- `divider_theme` - Defines the visual properties of `Divider`, `VerticalDivider`, dividers between `ListTiles`, and dividers between rows in `DataTable`.
- `dropdown_theme` - Customizes the appearance of `Dropdown` across the app.
- `expansion_tile_theme` - Customizes the appearance of `ExpansionTile` across the app.
- `filled_button_theme` - Customizes the appearance of `FilledButton` across the app.
- `floating_action_button_theme` - Customizes the appearance of `FloatingActionButton` across the app.
- `focus_color`
- `font_family` - Overrides a default font for the app.
- `highlight_color`
- `hint_color`
- `hover_color`
- `icon_button_theme` - Customizes the appearance of `IconButton` across the app.
- `icon_theme` - Customizes the appearance of `Icon` across the app.
- `list_tile_theme` - Customizes the appearance of `ListTile` across the app.
- `navigation_bar_theme`
- `navigation_drawer_theme`
- `navigation_rail_theme`
- `outlined_button_theme` - Customizes the appearance of `OutlinedButton` across the app.



- `page_transitions`
- `popup_menu_theme`
- `primary_text_theme`
- `progress_indicator_theme`
- `radio_theme`
- `scaffold_bgcolor` - Customizes the page background color.
- `scrollbar_theme`
- `search_bar_theme`
- `search_view_theme`
- `secondary_header_color`
- `segmented_button_theme`
- `slider_theme`
- `snackbar_theme`
- `splash_color`
- `switch_theme`
- `system_overlay_style`
- `tab_bar_theme`
- `text_button_theme` - Customizes the appearance of `TextButton` across the app.
- `text_theme`
- `time_picker_theme`
- `tooltip_theme`
- `unselected_control_color`
- `use_material3` - A temporary flag that can be used to opt-out of Material 3 features.
- `visual_density`

## Properties

### `buildappbar_theme`class-attributeinstance-attribute

`appbar_theme`: `AppBarTheme` | `None` = `None`

### `buildbadge_theme`class-attributeinstance-attribute

`badge_theme`: `BadgeTheme` | `None` = `None`

### `buildbanner_theme`class-attributeinstance-attribute

`banner_theme`: `BannerTheme` | `None` = `None`

### `buildbottom_appbar_theme`class-attributeinstance-attribute

`bottom_appbar_theme`: `BottomAppBarTheme` | `None` = `None`

### `buildbottom_sheet_theme`class-attributeinstance-attribute

`bottom_sheet_theme`: `BottomSheetTheme` | `None` = `None`

### `buildbutton_theme`class-attributeinstance-attribute

`button_theme`: `ButtonTheme` | `None` = `None`

Customizes the appearance of `Button` across the app.

### `buildcanvas_color`class-attributeinstance-attribute

`canvas_color`: `ColorValue` | `None` = `None`

### `buildcard_bgcolor`class-attributeinstance-attribute

`card_bgcolor`: `ColorValue` | `None` = `None`

### **buildcard\_themeclass-attributeinstance-attribute**

card\_theme: CardTheme | None = None

### **buildcheckbox\_themeclass-attributeinstance-attribute**

checkbox\_theme: CheckboxTheme | None = None

### **buildchip\_themeclass-attributeinstance-attribute**

chip\_theme: ChipTheme | None = None

Customizes the appearance of Chip across the app.

### **buildcolor\_schemeclass-attributeinstance-attribute**

color\_scheme: ColorScheme | None = None

Overrides the default ColorScheme used for the application.

### **buildcolor\_scheme\_seedclass-attributeinstance-attribute**

color\_scheme\_seed: ColorValue | None = None

Overrides the default color scheme seed used to generate ColorScheme. The default color is blue.

### **bulddata\_table\_themeclass-attributeinstance-attribute**

data\_table\_theme: DataTableTheme | None = None

Customizes the appearance of DataTable across the app.

### **bulddate\_picker\_themeclass-attributeinstance-attribute**

date\_picker\_theme: DatePickerTheme | None = None

Customizes the appearance of DatePicker across the app.

### **bulddialog\_themeclass-attributeinstance-attribute**

dialog\_theme: DialogTheme | None = None

Customizes the appearance of AlertDialog across the app.

### **bulddisabled\_colorclass-attributeinstance-attribute**

disabled\_color: ColorValue | None = None

### **bulddivider\_colorclass-attributeinstance-attribute**

divider\_color: ColorValue | None = None

Overrides the default color of dividers used in Divider, VerticalDivider, dividers between ListTiles, and dividers between rows in DataTable.

### **bulddivider\_themeclass-attributeinstance-attribute**

divider\_theme: DividerTheme | None = None

Defines the visual properties of Divider, VerticalDivider, dividers between ListTiles, and dividers between rows in DataTable.

### **bulddropdown\_themeclass-attributeinstance-attribute**

dropdown\_theme: DropdownTheme | None = None

Customizes the appearance of Dropdown across the app.

### **buildexpansion\_tile\_themeclass-attributeinstance-attribute**

expansion\_tile\_theme: ExpansionTileTheme | None = None

Customizes the appearance of ExpansionTile across the app.

### **buildfilled\_button\_themeclass-attributeinstance-attribute**

filled\_button\_theme: FilledButtonTheme | None = None

Customizes the appearance of FilledButton across the app.

### **buildfloating\_action\_button\_themeclass-attributeinstance-attribute**

```
floating_action_button_theme: (
 FloatingActionButtonTheme | None
) = None
```

Customizes the appearance of FloatingActionButton across the app.

### **buildfocus\_colorclass-attributeinstance-attribute**

focus\_color: ColorValue | None = None

### **buildfont\_familyclass-attributeinstance-attribute**

font\_family: str | None = None

Overrides a default font for the app.

### **buildhighlight\_colorclass-attributeinstance-attribute**

highlight\_color: ColorValue | None = None

### **buildhint\_colorclass-attributeinstance-attribute**

hint\_color: ColorValue | None = None

### **buildhover\_colorclass-attributeinstance-attribute**

hover\_color: ColorValue | None = None

### **buildicon\_button\_themeclass-attributeinstance-attribute**

icon\_button\_theme: IconButtonTheme | None = None

Customizes the appearance of IconButton across the app.

### **buildicon\_themeclass-attributeinstance-attribute**

icon\_theme: IconTheme | None = None

Customizes the appearance of Icon across the app.

### **buildlist\_tile\_themeclass-attributeinstance-attribute**

list\_tile\_theme: ListTileTheme | None = None

Customizes the appearance of ListTile across the app.

### **buildnavigation\_bar\_themeclass-attributeinstance-attribute**

navigation\_bar\_theme: NavigationBarTheme | None = None

### **buildnavigation\_drawer\_themeclass-attributeinstance-attribute**

navigation\_drawer\_theme: NavigationDrawerTheme | None = None

### **buildnavigation\_rail\_themeclass-attributeinstance-attribute**

navigation\_rail\_theme: NavigationRailTheme | None = None

### **buildoutlined\_button\_themeclass-attributeinstance-attribute**

outlined\_button\_theme: OutlinedButtonTheme | None = None

Customizes the appearance of OutlinedButton across the app.

### **buildpage\_transitionsclass-attributeinstance-attribute**

```
page_transitions: PageTransitionsTheme = field(
 default_factory=PageTransitionsTheme
)
```

### **buildpopup\_menu\_themeclass-attributeinstance-attribute**

popup\_menu\_theme: PopupMenuTheme | None = None

### **buildprimary\_text\_themeclass-attributeinstance-attribute**

primary\_text\_theme: TextTheme | None = None

### **buildprogress\_indicator\_themeclass-attributeinstance-attribute**

```
progress_indicator_theme: ProgressIndicatorTheme | None = (
 None
)
```

### **buildradio\_themeclass-attributeinstance-attribute**

radio\_theme: RadioTheme | None = None

### **buildscaffold\_bgcolorclass-attributeinstance-attribute**

scaffold\_bgcolor: ColorValue | None = None

Customizes the page background color.

### **buildscrollbar\_themeclass-attributeinstance-attribute**

scrollbar\_theme: ScrollbarTheme | None = None

### **buildsearch\_bar\_themeclass-attributeinstance-attribute**

search\_bar\_theme: SearchBarTheme | None = None

### **buildsearch\_view\_themeclass-attributeinstance-attribute**

search\_view\_theme: SearchViewTheme | None = None

### **buildsecondary\_header\_colorclass-attributeinstance-attribute**

secondary\_header\_color: ColorValue | None = None

### **buildsegmented\_button\_themeclass-attributeinstance-attribute**

segmented\_button\_theme: SegmentedButtonTheme | None = None

### **buildslider\_themeclass-attributeinstance-attribute**

slider\_theme: SliderTheme | None = None

### **buildsnackbar\_themeclass-attributeinstance-attribute**

snackbar\_theme: SnackbarTheme | None = None

### **buildsplash\_colorclass-attributeinstance-attribute**

splash\_color: ColorValue | None = None

### **buildswitch\_themeclass-attributeinstance-attribute**

switch\_theme: SwitchTheme | None = None

### **buildsystem\_overlay\_styleclass-attributeinstance-attribute**

system\_overlay\_style: SystemOverlayStyle | None = None

### **buildtab\_bar\_themeclass-attributeinstance-attribute**

tab\_bar\_theme: TabBarTheme | None = None

### **buildtext\_button\_themeclass-attributeinstance-attribute**

text\_button\_theme: TextButtonTheme | None = None

Customizes the appearance of TextButton across the app.

### **buildtext\_themeclass-attributeinstance-attribute**

text\_theme: TextTheme | None = None

### **buildtime\_picker\_themeclass-attributeinstance-attribute**

time\_picker\_theme: TimePickerTheme | None = None

### **buildtooltip\_themeclass-attributeinstance-attribute**

tooltip\_theme: TooltipTheme | None = None

### **buildunselected\_control\_colorclass-attributeinstance-attribute**

unselected\_control\_color: ColorValue | None = None

### **builduse\_material3class-attributeinstance-attribute**

use\_material3: bool | None = None

A temporary flag that can be used to opt-out of Material 3 features.

### **buildvisual\_densityclass-attributeinstance-attribute**

visual\_density: VisualDensity | None = None

# ThemeMode

Describes which theme will be used by Flet app.

Inherits: `enum.Enum`

## Properties

- DARK - Always use the dark mode (if available) regardless of system preference.
- LIGHT - Always use the light mode regardless of system preference.
- SYSTEM - Use either the light or dark theme based on what the user has selected in the system settings.

## Examples

### Theme Mode showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 page.theme_mode = ft.ThemeMode.SYSTEM
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 def apply_theme_mode(theme_mode: ft.ThemeMode):
 page.theme_mode = theme_mode
 status.value = f"Active mode: {theme_mode.name}"
 page.update()

 def showcase_card(theme_mode: ft.ThemeMode) -> ft.Container:
 return ft.Container(
 width=250,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(theme_mode.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Apply",
 on_click=lambda _, m=theme_mode: apply_theme_mode(m),
),
],
),
)

 page.appbar = ft.AppBar(title="ThemeMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Switch the app theme mode and inspect the preview below."),
 status := ft.Text(f"Active mode: {page.theme_mode.name}"),
 ft.Container(
 width=500,
 padding=12,
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=10,
 content=ft.Row(
 alignment=ft.MainAxisAlignment.SPACE_AROUND,
 controls=[
 ft.OutlinedButton("Outlined"),
 ft.FilledButton("Filled"),
 ft.Switch(label="Switch", value=True),
],
),
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(theme_mode) for theme_mode in ft.ThemeMode
],
),
],
),
),
)
```

```

),
),
)

apply_theme_mode(ft.ThemeMode.SYSTEM)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildDARKclass-attributeinstance-attribute

Always use the dark mode (if available) regardless of system preference.

### buildLIGHTclass-attributeinstance-attribute

Always use the light mode regardless of system preference.

### buildSYSTEMclass-attributeinstance-attribute

Use either the light or dark theme based on what the user has selected in the system settings.

# TileAffinity

Where to place a control in controls that use ListTile to position a control next to a label.

Inherits: enum.Enum

## Properties

- LEADING - Positions the control on the leading edge, and the secondary control, if any, on the trailing edge.
- PLATFORM - Positions the control relative to the text in the fashion that is typical for the current platform, and place the secondary control on the opposite side.
- TRAILING - Positions the control on the trailing edge, and the secondary control, if any, on the leading edge.

## Examples

### Tile Affinity showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def showcase_card(affinity: ft.TileAffinity) -> ft.Container:
 tile = ft.ExpansionTile(
 title="Project settings",
 subtitle="Compare where the expand arrow appears.",
 affinity=affinity,
 expanded=True,
 controls=[
 ft.ListTile(title="General"),
 ft.ListTile(title="Notifications"),
],
)

 return ft.Container(
 width=360,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(affinity.name, weight=ft.FontWeight.BOLD),
 tile,
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="TileAffinity Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Compare expand-arrow placement in ExpansionTile."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(affinity) for affinity in ft.TileAffinity
],
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildLEADINGclass-attributeinstance-attribute**

Positions the control on the leading edge, and the secondary control, if any, on the trailing edge.

**buildPLATFORMclass-attributeinstance-attribute**

Positions the control relative to the text in the fashion that is typical for the current platform, and place the secondary control on the opposite side.

**buildTRAILINGclass-attributeinstance-attribute**

Positions the control on the trailing edge, and the secondary control, if any, on the leading edge.



# TimePickerEntryMode

Interactive input mode of the TimePicker dialog.

In DIAL mode, a clock dial is displayed, and the user taps or drags the time they wish to select. In INPUT mode, TextFields are displayed and the user types in the time they wish to select.

Inherits: enum.Enum

## Properties

- DIAL - User picks time from a clock dial.
- DIAL\_ONLY - User can only pick time from a clock dial.
- INPUT - User can input the time by typing it into text fields.
- INPUT\_ONLY - User can only input the time by typing it into text fields.

## Examples

### Time Picker Entry Mode showcase

[play\\_arrowTry Online](#)

```
from datetime import time

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 picker = ft.TimePicker(value=time(hour=19, minute=30))

 def open_picker(entry_mode: ft.TimePickerEntryMode):
 picker.entry_mode = entry_mode
 page.show_dialog(picker)

 def showcase_card(entry_mode: ft.TimePickerEntryMode) -> ft.Container:
 return ft.Container(
 width=320,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(entry_mode.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open TimePicker",
 icon=ft.Icons.SCHEDULE,
 on_click=lambda _, m=entry_mode: open_picker(m),
),
],
),
)

 page.appbar = ft.AppBar(title="TimePickerEntryMode Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Open the picker to compare dial/input entry modes."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(m) for m in ft.TimePickerEntryMode],
),
],
),
),
)

if __name__ == "__main__":
```

```
ft.run(main)
```

## Properties

### **buildDIALclass-attributeinstance-attribute**

User picks time from a clock dial.

Can switch to INPUT by activating a mode button in the time picker dialog.

### **buildDIAL\_ONLYclass-attributeinstance-attribute**

User can only pick time from a clock dial.

There is no user interface to switch to another mode.

### **buildINPUTclass-attributeinstance-attribute**

User can input the time by typing it into text fields.

Can switch to DIAL by activating a mode button in the time picker dialog.

### **buildINPUT\_ONLYclass-attributeinstance-attribute**

User can only input the time by typing it into text fields.

There is no user interface to switch to another mode.

# TimePickerEntryModeChangeEvent

Represents the event triggered when the entry mode of a TimePicker changes.

Inherits: Event

## Properties

- `entry_mode` - The new entry mode.

## Properties

**buildentry\_modeinstance-attribute**

`entry_mode: TimePickerEntryMode`

The new entry mode.

# TimePickerHourFormat

Defines the hour format for the TimePicker control.

Inherits: `enum.Enum`

## Properties

- H12 - A 12-hour clock with an AM/PM selector.
- H24 - A 24-hour clock without an AM/PM selector.
- SYSTEM - Respect the host platform setting.

## Examples

### Time Picker Hour Format showcase

[play\\_arrow](#) [Try Online](#)

```
from datetime import time

import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 time_picker = ft.TimePicker(
 value=time(hour=19, minute=30),
 help_text="Pick meeting time",
)

 def open_picker(hour_format: ft.TimePickerHourFormat):
 time_picker.hour_format = hour_format
 page.show_dialog(time_picker)

 def showcase_card(hour_format: ft.TimePickerHourFormat) -> ft.Container:
 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(hour_format.name, weight=ft.FontWeight.BOLD),
 ft.Button(
 "Open TimePicker",
 icon=ft.Icons.SCHEDULE,
 on_click=lambda _, f=hour_format: open_picker(f),
),
],
),
)

 page.appbar = ft.AppBar(title="TimePickerHourFormat Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text("Open the picker to compare 12h, 24h, and system modes."),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[
 showcase_card(hour_format)
 for hour_format in ft.TimePickerHourFormat
],
),
],
),
),
)

if __name__ == "__main__":
```

```
ft.run(main)
```

## Properties

**buildH12class-attributeinstance-attribute**

A 12-hour clock with an AM/PM selector.

**buildH24class-attributeinstance-attribute**

A 24-hour clock without an AM/PM selector.

**buildSYSTEMclass-attributeinstance-attribute**

Respect the host platform setting.

# TimePickerTheme

Customizes the appearance of TimePicker controls across the app.

## Properties

- `bgcolor` - The background color of a TimePicker.
- `cancel_button_style` - The style of the cancel button of a TimePicker.
- `confirm_button_style` - The style of the confirm (OK) button of a TimePicker.
- `day_period_border_side` - The color and weight of the day period's outline.
- `day_period_button_style` - The style of the AM/PM toggle control of a TimePicker.
- `day_period_color` - The background color of the AM/PM toggle.
- `day_period_shape` - The shape of the day period that the TimePicker uses.
- `day_period_text_color` - The color of the day period text that represents AM/PM.
- `day_period_text_style` - Used to configure the TextStyle for the AM/PM toggle control.
- `dial_bgcolor` - The background color of the time picker dial when the entry mode is `TimePickerEntryMode.DIAL` or `TimePickerEntryMode.DIAL_ONLY`.
- `dial_hand_color` - The color of the time picker dial's hand when the entry mode is `TimePickerEntryMode.DIAL` or `TimePickerEntryMode.DIAL_ONLY`.
- `dial_text_color` - The color of the dial text that represents specific hours and minutes.
- `dial_text_style` - The TextStyle for the numbers on the time selection dial.
- `elevation` - The Material elevation for the time picker dialog.
- `entry_mode_icon_color` - The color of the entry mode IconButton.
- `help_text_style` - Used to configure the TextStyle for the helper text in the header.
- `hour_minute_color` - The background color of the hour and minute header segments.
- `hour_minute_shape` - The shape of the hour and minute controls that the TimePicker uses.
- `hour_minute_text_color` - The color of the header text that represents hours and minutes.
- `hour_minute_text_style` - Used to configure the TextStyle for the hour/minute controls.
- `padding` - The padding around the time picker dialog when the entry mode is `TimePickerEntryMode.DIAL` or `TimePickerEntryMode.DIAL_ONLY`.
- `shape` - The shape of the Dialog that the time picker is presented in.
- `time_selector_separator_color` - The color of the time selector separator between the hour and minute controls.
- `time_selector_separator_text_style` - Used to configure the text style for the time selector separator between the hour and minute controls.

## Properties

### `buildbgcolor`class-attributeinstance-attribute

`bgcolor: ColorValue | None = None`

The background color of a TimePicker.

If this is null, the time picker defaults to the overall theme's `ColorScheme.surface_container_high`.

### `buildcancel_button_style`class-attributeinstance-attribute

`cancel_button_style: ButtonStyle | None = None`

The style of the cancel button of a TimePicker.

### `buildconfirm_button_style`class-attributeinstance-attribute

`confirm_button_style: ButtonStyle | None = None`

The style of the confirm (OK) button of a TimePicker.

### **buildday\_period\_border\_sideclass-attributeinstance-attribute**

`day_period_border_side: BorderSide | None = None`

The color and weight of the day period's outline.

### **buildday\_period\_button\_styleclass-attributeinstance-attribute**

`day_period_button_style: ButtonStyle | None = None`

The style of the AM/PM toggle control of a TimePicker.

### **buildday\_period\_colorclass-attributeinstance-attribute**

`day_period_color: ColorValue | None = None`

The background color of the AM/PM toggle.

### **buildday\_period\_shapeclass-attributeinstance-attribute**

`day_period_shape: OutlinedBorder | None = None`

The shape of the day period that the TimePicker uses.

### **buildday\_period\_text\_colorclass-attributeinstance-attribute**

`day_period_text_color: ColorValue | None = None`

The color of the day period text that represents AM/PM.

### **buildday\_period\_text\_styleclass-attributeinstance-attribute**

`day_period_text_style: TextStyle | None = None`

Used to configure the TextStyle for the AM/PM toggle control.

If this is null, the time picker defaults to the overall theme's TextTheme.title\_medium.

### **builddial\_bgcolorclass-attributeinstance-attribute**

`dial_bgcolor: ColorValue | None = None`

The background color of the time picker dial when the entry mode is TimePickerEntryMode.DIAL or TimePickerEntryMode.DIAL\_ONLY.

### **builddial\_hand\_colorclass-attributeinstance-attribute**

`dial_hand_color: ColorValue | None = None`

The color of the time picker dial's hand when the entry mode is TimePickerEntryMode.DIAL or TimePickerEntryMode.DIAL\_ONLY.

### **builddial\_text\_colorclass-attributeinstance-attribute**

`dial_text_color: ColorValue | None = None`

The color of the dial text that represents specific hours and minutes.

### **builddial\_text\_styleclass-attributeinstance-attribute**

`dial_text_style: TextStyle | None = None`

The TextStyle for the numbers on the time selection dial.

### **buildelevationclass-attributeinstance-attribute**

`elevation: Number | None = None`

The Material elevation for the time picker dialog.

### **builentry\_mode\_icon\_colorclass-attributeinstance-attribute**

`entry_mode_icon_color: ColorValue | None = None`

The color of the entry mode IconButton.

### **buildhelp\_text\_styleclass-attributeinstance-attribute**

`help_text_style: TextStyle | None = None`

Used to configure the TextStyle for the helper text in the header.

### **buildhour\_minute\_colorclass-attributeinstance-attribute**

```
hour_minute_color: ColorValue | None = None
```

The background color of the hour and minute header segments.

### **buildhour\_minute\_shapeclass-attributeinstance-attribute**

```
hour_minute_shape: OutlinedBorder | None = None
```

The shape of the hour and minute controls that the TimePicker uses.

### **buildhour\_minute\_text\_colorclass-attributeinstance-attribute**

```
hour_minute_text_color: ColorValue | None = None
```

The color of the header text that represents hours and minutes.

### **buildhour\_minute\_text\_styleclass-attributeinstance-attribute**

```
hour_minute_text_style: TextStyle | None = None
```

Used to configure the TextStyle for the hour/minute controls.

### **buildpaddingclass-attributeinstance-attribute**

```
padding: PaddingValue | None = None
```

The padding around the time picker dialog when the entry mode is TimePickerEntryMode.DIAL or TimePickerEntryMode.DIAL\_ONLY.

### **buildshapeclass-attributeinstance-attribute**

```
shape: OutlinedBorder | None = None
```

The shape of the Dialog that the time picker is presented in.

### **buildtime\_selector\_separator\_colorclass-attributeinstance-attribute**

```
time_selector_separator_color: (
 ControlStateValue[ColorValue] | None
) = None
```

The color of the time selector separator between the hour and minute controls.

### **buildtime\_selector\_separator\_text\_styleclass-attributeinstance-attribute**

```
time_selector_separator_text_style: (
 ControlStateValue[TextStyle] | None
) = None
```

Used to configure the text style for the time selector separator between the hour and minute controls.



# Tooltip

Provide text labels which help explain the function of a button or other user interface action.

## Properties

- `bgcolor` - Background color of the tooltip.
- `decoration` - The background decoration of this tooltip.
- `enable_feedback` - When True (default) the tooltip should provide acoustic and/or haptic feedback.
- `exclude_from_semantics` - Whether the tooltip's message should be excluded from the semantics tree.
- `exit_duration` - The length of time that the tooltip will be shown after a long press is released or a tap is released or mouse pointer exits the control.
- `margin` - The empty space that surrounds the tooltip.
- `message` - The text to display in this tooltip.
- `mouse_cursor` - The cursor for a mouse pointer when it enters or is hovering over the content.
- `padding` - The amount of space by which to inset the tooltip's content.
- `prefer_below` - Whether the tooltip defaults to being displayed below the control.
- `show_duration` - The length of time that the tooltip will be shown after a long press is released (if `triggerMode` is `TooltipTriggerMode.LONG_PRESS`) or a tap is released (if `triggerMode` is `TooltipTriggerMode.TAP`).
- `size_constraints` - Defines the constraints on the size of this tooltip.
- `tap_to_dismiss` - Whether the tooltip can be dismissed by tapping on it.
- `text_align` - How the message of the tooltip is aligned horizontally.
- `text_style` - The `TextStyle` to use for the message of the tooltip.
- `trigger_mode` - The mode of the tooltip's trigger.
- `vertical_offset` - The vertical gap between the control and the displayed tooltip.
- `wait_duration` - The length of time, in milliseconds, that a pointer must hover over a tooltip's control before the tooltip will be shown.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

Background color of the tooltip.

### `builddecorationclass-attributeinstance-attribute`

`decoration: BoxDecoration | None = None`

The background decoration of this tooltip.

If None, `TooltipTheme.decoration` is used. If that is also None, a default decoration will be picked based on the current theme mode:

In light theme mode:

```
ft.BoxDecoration(
 border_radius=ft.BorderRadius.all(4.0),
 bgcolor=ft.Colors.with_opacity(0.9, ft.Colors.GREY_700),
)
```

In dark theme mode:

```
ft.BoxDecoration(
 border_radius=ft.BorderRadius.all(4.0),
 bgcolor=ft.Colors.with_opacity(0.9, ft.Colors.WHITE),
)
```

### `buildenable_feedbackclass-attributeinstance-attribute`

`enable_feedback: bool | None = None`

When True (default) the tooltip should provide acoustic and/or haptic feedback.

For example, on Android a tap will produce a clicking sound and a long-press will produce a short vibration, when feedback is enabled.

### **buildexclude\_from\_semanticsclass-attributeinstance-attribute**

`exclude_from_semantics: bool | None = False`

Whether the tooltip's message should be excluded from the semantics tree.

### **buildexit\_durationclass-attributeinstance-attribute**

`exit_duration: DurationValue | None = None`

The length of time that the tooltip will be shown after a long press is released or a tap is released or mouse pointer exits the control.

If None, `TooltipTheme.exit_duration` is used. If that's is also None, defaults to 0 milliseconds - no delay.

### **buildmarginclass-attributeinstance-attribute**

`margin: MarginValue | None = None`

The empty space that surrounds the tooltip.

If None, `TooltipTheme.margin` is used. If that's is also None, defaults to `Margin.all(0.0)`.

### **buildmessageinstance-attribute**

`message: str`

The text to display in this tooltip.

### **buildmouse\_cursorclass-attributeinstance-attribute**

`mouse_cursor: MouseCursor | None = None`

The cursor for a mouse pointer when it enters or is hovering over the content.

### **buildpaddingclass-attributeinstance-attribute**

`padding: PaddingValue | None = None`

The amount of space by which to inset the tooltip's content.

It has the following default values based on the current platform:

- On mobile platforms: `Padding.symmetric(horizontal=16.0, vertical=4.0)`
- On desktop platforms: `Padding.symmetric(horizontal=8.0, vertical=4.0)`

### **buildprefer\_belowclass-attributeinstance-attribute**

`prefer_below: bool | None = None`

Whether the tooltip defaults to being displayed below the control.

If there is insufficient space to display the tooltip in the preferred direction, the tooltip will be displayed in the opposite direction.

If None, `TooltipTheme.prefer_below` is used. If that's is also None, defaults to True.

### **buildshow\_durationclass-attributeinstance-attribute**

`show_duration: DurationValue | None = None`

The length of time that the tooltip will be shown after a long press is released (if `triggerMode` is `TooltipTriggerMode.LONG_PRESS`) or a tap is released (if `triggerMode` is `TooltipTriggerMode.TAP`). This property does not affect mouse pointer devices.

If None, `TooltipTheme.show_duration` is used. If that's is also None, defaults to 1.5 seconds for long press and tap released

### **buildsize\_constraintsclass-attributeinstance-attribute**

`size_constraints: BoxConstraints | None = None`

Defines the constraints on the size of this tooltip.

If None, `TooltipTheme.size_constraints` is used. If that's is also None, then a default value will be picked based on the current platform:

- on desktop platforms: `BoxConstraints(min_height=24.0)`
- on mobile platforms: `BoxConstraints(min_height=32.0)`

### `buildtap_to_dismiss`class-attributeinstance-attribute

`tap_to_dismiss: bool = True`

Whether the tooltip can be dismissed by tapping on it.

### `buildtext_align`class-attributeinstance-attribute

`text_align: TextAlign | None = None`

How the message of the tooltip is aligned horizontally.

If None, `TooltipTheme.text_align` is used. If that's is also None, defaults to `TextAlign.START`.

### `buildtext_style`class-attributeinstance-attribute

`text_style: TextStyle | None = None`

The `TextStyle` to use for the message of the tooltip.

### `buildtrigger_mode`class-attributeinstance-attribute

`trigger_mode: TooltipTriggerMode | None = None`

The mode of the tooltip's trigger.

If None, `TooltipTheme.trigger_mode` is used. If that's is also None, defaults to `TooltipTriggerMode.LONG_PRESS`.

### `buildvertical_offset`class-attributeinstance-attribute

`vertical_offset: Number | None = None`

The vertical gap between the control and the displayed tooltip.

When `prefer_below` is set to `True` and tooltips have sufficient space to display themselves, this property defines how much vertical space tooltips will position themselves under their corresponding controls. Otherwise, tooltips will position themselves above their corresponding controls with the given offset.

### `buildwait_duration`class-attributeinstance-attribute

`wait_duration: DurationValue | None = None`

The length of time, in milliseconds, that a pointer must hover over a tooltip's control before the tooltip will be shown.

If None, `TooltipTheme.wait_duration` is used. If that's is also None, defaults to 100 milliseconds.

## Examples

### Tooltip with decoration

`play_arrow`[Try Online](#)

```
import math

import flet as ft

def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Hover to see the simple tooltip",
 tooltip="This is a simple tooltip",
),
 ft.Text(
 value="Hover to see the complex tooltip",
 tooltip=ft.Tooltip(
 message="This is a complex tooltip",
 padding=20,
 text_style=ft.TextStyle(size=20, color=ft.Colors.WHITE),
 decoration=ft.BoxDecoration(
 border_radius=10,
 gradient=ft.LinearGradient(
 begin=ft.Alignment.TOP_LEFT,
```

) ) , 1, ), , , , 1,

2108

# TooltipTheme

Customizes the appearance of descendant Tooltip controls.

## Properties

- `decoration` - Overrides the default value for `Tooltip.decoration`.
- `enable_feedback` - Overrides the default value for `Tooltip.enable_feedback`.
- `exclude_from_semantics` - Overrides the default value for `Tooltip.exclude_from_semantics`.
- `exit_duration` - Overrides the default value for `Tooltip.exit_duration`.
- `margin` - Overrides the default value for `Tooltip.margin`.
- `padding` - Overrides the default value for `Tooltip.padding`.
- `prefer_below` - Overrides the default value for `Tooltip.prefer_below`.
- `show_duration` - Overrides the default value for `Tooltip.show_duration`.
- `size_constraints` - Overrides the default value for `Tooltip.size_constraints`.
- `text_align` - Overrides the default value for `Tooltip.text_align`.
- `text_style` - Overrides the default value for `Tooltip.text_style`.
- `trigger_mode` - Overrides the default value for `Tooltip.trigger_mode`.
- `vertical_offset` - Overrides the default value for `Tooltip.vertical_offset`.
- `wait_duration` - Overrides the default value for `Tooltip.wait_duration`.

## Properties

### `builddecorationclass-attributeinstance-attribute`

`decoration: BoxDecoration | None = None`

Overrides the default value for `Tooltip.decoration`.

### `buildenable_feedbackclass-attributeinstance-attribute`

`enable_feedback: bool | None = None`

Overrides the default value for `Tooltip.enable_feedback`.

### `buildexclude_from_semanticsclass-attributeinstance-attribute`

`exclude_from_semantics: bool | None = None`

Overrides the default value for `Tooltip.exclude_from_semantics`.

### `buildexit_durationclass-attributeinstance-attribute`

`exit_duration: DurationValue | None = None`

Overrides the default value for `Tooltip.exit_duration`.

### `buildmarginclass-attributeinstance-attribute`

`margin: MarginValue | None = None`

Overrides the default value for `Tooltip.margin`.

### `buildpaddingclass-attributeinstance-attribute`

`padding: PaddingValue | None = None`

Overrides the default value for `Tooltip.padding`.

### `buildprefer_belowclass-attributeinstance-attribute`

`prefer_below: bool | None = None`

Overrides the default value for `Tooltip.prefer_below`.

### `buildshow_durationclass-attributeinstance-attribute`

```
show_duration: DurationValue | None = None
```

Overrides the default value for `Tooltip.show_duration`.

### **buildsize\_constraintsclass-attributeinstance-attribute**

```
size_constraints: BoxConstraints | None = None
```

Overrides the default value for `Tooltip.size_constraints`.

### **buildtext\_alignclass-attributeinstance-attribute**

```
text_align: TextAlign | None = None
```

Overrides the default value for `Tooltip.text_align`.

### **buildtext\_styleclass-attributeinstance-attribute**

```
text_style: TextStyle | None = None
```

Overrides the default value for `Tooltip.text_style`.

### **buildtrigger\_modeclass-attributeinstance-attribute**

```
trigger_mode: TooltipTriggerMode | None = None
```

Overrides the default value for `Tooltip.trigger_mode`.

### **buildvertical\_offsetclass-attributeinstance-attribute**

```
vertical_offset: Number | None = None
```

Overrides the default value for `Tooltip.vertical_offset`.

### **buildwait\_durationclass-attributeinstance-attribute**

```
wait_duration: DurationValue | None = None
```

Overrides the default value for `Tooltip.wait_duration`.

# TooltipTriggerMode

The method of interaction that will trigger a tooltip. Used in `Tooltip.trigger_mode` and `TooltipTheme.trigger_mode`.

On desktop, a tooltip will be shown as soon as a pointer hovers over the widget, regardless of the value of `Tooltip.trigger_mode`.

## Properties

- `LONG_PRESS` - Tooltip will be shown after a long press.
- `MANUAL`
- `TAP` - Tooltip will be shown after a single tap.

## Properties

`buildLONG_PRESS`class-attributeinstance-attribute

Tooltip will be shown after a long press.

`buildMANUAL`class-attributeinstance-attribute

`buildTAP`class-attributeinstance-attribute

Tooltip will be shown after a single tap.

# Transform

Configuration for `LayoutControl.transform`.

Applies a generic matrix transform backed by recorded `Matrix4`.

## Properties

- `alignment` - The alignment of the origin, relative to the size of the box.
- `filter_quality` - The filter quality with which to apply this transform as a bitmap operation.
- `matrix` - Matrix transform intent recorded in Python and replayed in Flutter.
- `origin` - The origin of the coordinate system (relative to the upper left corner).
- `transform_hit_tests` - Whether to apply the transformation when performing hit tests.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

`alignment: Alignment | None = None`

The alignment of the origin, relative to the size of the box.

### `buildfilter_qualityclass-attributeinstance-attribute`

`filter_quality: FilterQuality | None = None`

The filter quality with which to apply this transform as a bitmap operation.

### `buildmatrixinstance-attribute`

`matrix: Matrix4`

Matrix transform intent recorded in Python and replayed in Flutter.

### `buildoriginclass-attributeinstance-attribute`

`origin: Offset | None = None`

The origin of the coordinate system (relative to the upper left corner).

### `buildtransform_hit_testsclass-attributeinstance-attribute`

`transform_hit_tests: bool = True`

Whether to apply the transformation when performing hit tests.



# UnderlineTabIndicator

## Properties

- `border_radius` - The radius of the indicator's corners.
- `border_side` - The color and weight of the horizontal line drawn below the selected tab.
- `insets` - Locates the selected tab's underline relative to the tab's boundary.

## Properties

### `buildborder_radiusclass-attributeinstance-attribute`

```
border_radius: BorderRadiusValue | None = None
```

The radius of the indicator's corners.

If this value is not None, a rounded rectangular tab indicator is drawn, otherwise rectangular tab indicator is drawn.

### `buildborder_sideclass-attributeinstance-attribute`

```
border_side: BorderSide = field(
 default_factory=lambda: BorderSide(
 width=2.0, color=Colors.WHITE
)
)
```

The color and weight of the horizontal line drawn below the selected tab.

### `buildinsetsclass-attributeinstance-attribute`

```
insets: PaddingValue = field(
 default_factory=lambda: Padding.zero()
)
```

Locates the selected tab's underline relative to the tab's boundary.

The `TabBar.indicator_size` property can be used to define the tab indicator's bounds in terms of its (centered) tab control with `TabBarIndicatorSize.LABEL`, or the entire tab with `TabBarIndicatorSize.TAB`.

# Url

URL descriptor used by APIs that open links in a browser context.

## Properties

- `target` - Where to open URL in the web mode.
- `url` - The url to open.

## Properties

### `buildtargetclass-attributeinstance-attribute`

`target: UrlTarget | str | None = None`

Where to open URL in the web mode.

### `buildurlinstance-attribute`

`url: str`

The url to open.

# UrlTarget

Specifies where to open a URL.

Inherits: `enum.Enum`

## Properties

- BLANK - Opens the URL in a new browser tab or window.
- PARENT - Opens in the parent frame, useful with nested iframes.
- SELF - Opens in the same browsing context (i.e., same tab).
- TOP - Opens in the topmost frame, breaking out of any iframe.

## Examples

### URL Target showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 async def open_url(target: ft.UrlTarget):
 url = "https://flet.dev"
 await ft.UrlLauncher().launch_url(ft.Url(url=url, target=target))
 status.value = f"Opened {url} with target {target.name}"
 status.update()

 def showcase_card(target: ft.UrlTarget) -> ft.Container:
 return ft.Container(
 width=280,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(target.name, weight=ft.FontWeight.BOLD),
 ft.Text(target.value, size=12, color=ft.Colors.ON_SURFACE_VARIANT),
 ft.Button(
 "Open flet.dev",
 icon=ft.Icons.OPEN_IN_NEW,
 on_click=lambda: page.run_task(open_url, target),
),
],
),
)

 page.appbar = ft.AppBar(title="UrlTarget Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Click a card to launch URL with the selected browser target."
),
 status := ft.Text(),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[showcase_card(target) for target in ft.UrlTarget],
),
],
),
)
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

**buildBLANK**class-attributeinstance-attribute

Opens the URL in a new browser tab or window.

**buildPARENT**class-attributeinstance-attribute

Opens in the parent frame, useful with nested iframes.

**buildSELF**class-attributeinstance-attribute

Opens in the same browsing context (i.e., same tab).

**buildTOP**class-attributeinstance-attribute

Opens in the topmost frame, breaking out of any iframe.

## use\_route\_loader\_data

Returns the data from the current route's loader function.

Must be called inside a component whose route has a loader defined.

Returns:

- The return value of the route's loader function, or None.

## use\_route\_location

Returns the current location pathname.

Must be called inside a component rendered by a Router. Returns an empty string if called outside a Router tree.

Returns:

- str - The current URL pathname (e.g. "/products/42").

## use\_route\_outlet

Returns the matched child route's rendered component.

Used inside layout route components to render the active child route. Must be called inside a component rendered by a Router. Returns None if called outside a Router tree.

Returns:

- Control - The rendered child component, or None if no child matches.

## use\_route\_params

Returns all dynamic segment parameters from the current matched route chain.

Must be called inside a component rendered by a Router. Returns an empty dict if called outside a Router tree (e.g. during a stale observable re-render).

Returns:

- dict[str, str] - Dictionary mapping parameter names to their matched string values.



## use\_view\_path

Returns the resolved URL for the current view level.

This differs from `use_route_location` which returns the full current URL. `use_view_path` returns the URL up to the current view level (the leaf route of the view in `manage_views=True` mode).

Useful as the route value for `View` to produce a unique `Navigator` key per view in the stack.

Must be called inside a component rendered by a `Router`. Returns an empty string if called outside a `Router` tree.

Returns:

- `str` - The resolved URL for this view level (e.g. `"/products/42"`).

## use\_callback

Memoize a function identity between renders.

Parameters:

- `fn (Callable[P, R])` - A function to memoize.
- `dependencies (Sequence[Any] | None, default: None)` - If present, `fn` is only re-memoized when one of the dependencies has changed. If absent, `fn` is only memoized on initial render.

Returns:

- `Callable[P, R]` - A memoized version of the function whose identity is stable between renders.

## use\_context

Access the current context value.

Parameters:

- context (ContextProvider[T]) - A context provider created by create\_context.

Returns:

- T - The current context value, or the default value if no provider is found.

## use\_dialog

Portal a DialogControl to the page's dialog overlay.

Call this hook inside a component function on every render. Pass a DialogControl instance to show the dialog, or None to hide/remove it.

The hook automatically sets open=True on the dialog when it is added to the overlay and removes it when None is passed or the component unmounts.

Parameters:

- dialog (DialogControl | None, default: None) - A DialogControl to display, or None to hide it.

## use\_effect

Perform side effects in function components.

Parameters:

- `setup` (`Callable[[], Any | Awaitable[Any]]`) - A function that performs the side effect. It may optionally return a cleanup function.
- `dependencies` (`Sequence[Any] | None`, default: `None`) - If present, the effect is only re-run when one of the dependencies has changed. If absent, the effect is only run on initial render.
- `cleanup` (`Callable[[], Any | Awaitable[Any]] | None`, default: `None`) - An optional function that cleans up after the effect. It is run before the effect is re-run, and when the component unmounts.

## use\_memo

Memoize a computed value between renders.

Parameters:

- `calculate_value` (`Callable[[], MemoValueT]`) - A function that computes the value to be memoized.
- `dependencies` (`Sequence[Any] | None`, default: `None`) - If present, the value is only recomputed when one of the dependencies has changed. If absent, the value is only computed on the initial render.

Returns:

- `MemoValueT` - A memoized value whose identity is stable between renders.

# UserAccelerometerReadingEvent

Like AccelerometerReadingEvent, this is a discrete reading from an accelerometer and measures the velocity of the device. However, unlike AccelerometerReadingEvent, this event does not include the effects of gravity.

Inherits: Event

## Properties

- timestamp - Event timestamp.
- x - Linear acceleration along the X axis, gravity removed, in m/s<sup>2</sup>.
- y - Linear acceleration along the Y axis, gravity removed, in m/s<sup>2</sup>.
- z - Linear acceleration along the Z axis, gravity removed, in m/s<sup>2</sup>.

## Properties

### buildtimestampinstance-attribute

timestamp: datetime

Event timestamp.

### buildxinstance-attribute

x: float

Linear acceleration along the X axis, gravity removed, in m/s<sup>2</sup>.

### buildyinstance-attribute

y: float

Linear acceleration along the Y axis, gravity removed, in m/s<sup>2</sup>.

### buildzinstance-attribute

z: float

Linear acceleration along the Z axis, gravity removed, in m/s<sup>2</sup>.

## use\_ref

Preserve a mutable value for the lifetime of the component without causing re-renders.

Parameters:

- `initial_value` (`RefValueT` | `Callable[[], RefValueT]` | `None`, default: `None`) - Optional value or callable returning the value assigned to `ref.current`.

Returns:

- `MutableRef[RefValueT]` - A `MutableRef` whose `.current` property can be read and written freely.



## use\_state

Adds state to a function component, similar to React's `useState()`.

The returned setter accepts either:

- a new value, or
- a function receiving the previous state and returning the next one.

Parameters:

- `initial (StateT | Callable[[], StateT])` - Initial state value or a function returning it.

Returns:

- `tuple[StateT, Callable[[StateT | Updater], None]]` - (value, set\_value) tuple.

## @value

Decorator for non-control value types to enable sparse `_values` tracking.

Applies `@dataclass` (passing `dataclass_kwargs`) and installs Prop descriptors via `_install_props`. No base class is required — the decorator wraps `__init__` to inject `_values` and `_dirty` before the first `Prop.__set__` call, and registers the class as a `Value` subclass for `isinstance` checks.

Usage::

```
@value class TextStyle: color: Optional[str] = None size: Optional[float] = None
```

## or with explicit dataclass kwargs:

```
@value(eq=False) class TextStyle: ...
```

# ValueKey

General-purpose key for control identity.

Prefer it when you need a stable identifier across rebuilds but do not need scroll-specific behavior.

Inherits: Key

# VerticalAlignment

The vertical alignment of text within an input box.

Inherits: `enum.Enum`

## Properties

- `CENTER` - Aligns the text vertically in the center of the `TextField`.
- `END` - Aligns the text vertically at the bottommost location of the `TextField`.
- `NONE`
- `START` - Aligns the text vertically at the topmost location of the `TextField`.

## Properties

`buildCENTER`class-attributeinstance-attribute

Aligns the text vertically in the center of the `TextField`.

`buildEND`class-attributeinstance-attribute

Aligns the text vertically at the bottommost location of the `TextField`.

`buildNONE`class-attributeinstance-attribute

`buildSTART`class-attributeinstance-attribute

Aligns the text vertically at the topmost location of the `TextField`.

# ViewPopEvent

Event payload for view-pop navigation actions.

Delivered to `Page.on_view_pop` when the top view is being popped by system or app-bar back behavior.

Inherits: `Event`

## Properties

- `route` - Route of the view being popped.
- `view` - Matched View instance for route, if found on the page.

## Properties

### `buildrouteinstance-attribute`

`route: str`

Route of the view being popped.

### `buildviewclass-attributeinstance-attribute`

`view: View | None = None`

Matched View instance for route, if found on the page.

# VisualDensity

Defines the visual density of user interface components.

Inherits: `enum.Enum`

## Properties

- `ADAPTIVE_PLATFORM_DENSITY` - Visual density that is adaptive based on the given platform.
- `COMFORTABLE` - The profile for a "comfortable" interpretation of visual density.
- `COMPACT` - The profile for a "compact" interpretation of visual density.
- `STANDARD` - The default/standard profile for visual density.

## Examples

### Visual Density showcase

[play\\_arrowTry Online](#)

```
import flet as ft

def density_card(visual_density: ft.VisualDensity) -> ft.Container:
 return ft.Container(
 width=300,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 theme=ft.Theme(visual_density=visual_density),
 dark_theme=ft.Theme(visual_density=visual_density),
 content=ft.Column(
 spacing=8,
 controls=[
 ft.Text(visual_density.name, weight=ft.FontWeight.BOLD),
 ft.IconButton(icon=ft.Icons.ADD),
 ft.Checkbox(label="Checkbox", value=True),
 ft.Chip(
 label="Explore topics",
 leading=ft.Icon(ft.Icons.EXPLORE_OUTLINED),
),
 ft.RadioGroup(
 value="1",
 content=ft.Row(
 controls=[
 ft.Radio(label=f"{i}", value=f"{i}") for i in range(1, 4)
],
),
),
],
),
)

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 page.appbar = ft.AppBar(title="VisualDensity Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Compare component density presets across Material controls."
),
 ft.Row(
 wrap=True,
 spacing=12,
 expand=True,
 scroll=ft.ScrollMode.AUTO,
 alignment=ft.MainAxisAlignment.CENTER,
 controls=[density_card(vd) for vd in ft.VisualDensity],
),
],
),
),
)
```

```
if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildADAPTIVE\_PLATFORM\_DENSITY**class-attributeinstance-attribute

Visual density that is adaptive based on the given platform.

For desktop platforms, this returns COMPACT, and for other platforms, it returns a default-constructed visual density.

### **buildCOMFORTABLE**class-attributeinstance-attribute

The profile for a "comfortable" interpretation of visual density.

Individual components will interpret the density value independently, making themselves more visually dense than STANDARD and less dense than COMPACT to different degrees based on the Material Design specification of the comfortable setting for their particular use case.

It corresponds to a density value of -1 in both axes.

### **buildCOMPACT**class-attributeinstance-attribute

The profile for a "compact" interpretation of visual density.

Individual components will interpret the density value independently, making themselves more visually dense than STANDARD and COMFORTABLE to different degrees based on the Material Design specification of the COMFORTABLE setting for their particular use case.

It corresponds to a density value of -2 in both axes.

### **buildSTANDARD**class-attributeinstance-attribute

The default/standard profile for visual density.

This default value represents a visual density that is less dense than either COMFORTABLE or COMPACT, and corresponds to density values of zero in both axes.

# WebBrowserName

Represents commonly used browsers.

Inherits: `enum.Enum`

## Properties

- CHROME - Google Chrome
- EDGE - Microsoft Edge
- FIREFOX - Mozilla Firefox
- MSIE - Microsoft Internet Explorer
- OPERA - Opera Web Browser
- SAFARI - Apple Safari
- SAMSUNG\_INTERNET - Samsung Internet Browser
- UNKNOWN - Unknown web browser

## Properties

`buildCHROMEclass-attributeinstance-attribute`

Google Chrome

`buildEDGEclass-attributeinstance-attribute`

Microsoft Edge

`buildFIREFOXclass-attributeinstance-attribute`

Mozilla Firefox

`buildMSIEclass-attributeinstance-attribute`

Microsoft Internet Explorer

`buildOPERAclass-attributeinstance-attribute`

Opera Web Browser

`buildSAFARIclass-attributeinstance-attribute`

Apple Safari

`buildSAMSUNG_INTERNETclass-attributeinstance-attribute`

Samsung Internet Browser

`buildUNKNOWNclass-attributeinstance-attribute`

Unknown web browser



# WebDeviceInfo

Information derived from navigator.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Window/navigator>

Inherits: DeviceInfo

## Properties

- `app_code_name` - The internal 'code' name of the current browser.
- `app_name` - A string with the official name of the browser.
- `app_version` - The version of the browser as a string.
- `browser_name` - The name of the browser in use.
- `device_memory` - The amount of device memory in gigabytes.
- `hardware_concurrency` - The number of logical processor cores available.
- `language` - A string representing the preferred language of the user, usually the language of the browser UI.
- `languages` - A list of strings representing the languages known to the user, by order of preference.
- `max_touch_points` - The maximum number of simultaneous touch contact points supported by the current device.
- `platform` - A string representing the platform of the browser.
- `product` - Always returns "Gecko", on any browser.
- `product_sub` - The build number of the current browser.
- `user_agent` - The user agent string for the current browser (e.g., 'Mozilla/5.0 ...').
- `vendor` - The vendor name of the current browser.
- `vendor_sub` - Returns the vendor version number (e.g., '6.1').

## Properties

### `buildapp_code_name`class-attributeinstance-attribute

`app_code_name: str | None = None`

The internal 'code' name of the current browser.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/appCodeName>

Note

Do not rely on this property to return the correct value.

### `buildapp_name`class-attributeinstance-attribute

`app_name: str | None = None`

A string with the official name of the browser.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/appName>

Note

Do not rely on this property to return the correct value.

### `buildapp_version`class-attributeinstance-attribute

`app_version: str | None = None`

The version of the browser as a string.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/appVersion>

Note

Do not rely on this property to return the correct value.

## **buildbrowser\_nameinstance-attribute**

`browser_name: WebBrowserName`

The name of the browser in use.

## **builddevice\_memoryclass-attributeinstance-attribute**

`device_memory: float | None = None`

The amount of device memory in gigabytes.

This value is an approximation given by rounding to the nearest power of 2 and dividing that number by 1024.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/deviceMemory>

## **buildhardware\_concurrencyclass-attributeinstance-attribute**

`hardware_concurrency: int | None = None`

The number of logical processor cores available.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/hardwareConcurrency>

## **buildlanguageclass-attributeinstance-attribute**

`language: str | None = None`

A string representing the preferred language of the user, usually the language of the browser UI.

Will be None if unknown.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/language>

## **buildlanguagesclass-attributeinstance-attribute**

`languages: list[str] | None = None`

A list of strings representing the languages known to the user, by order of preference.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/languages>

## **buildmax\_touch\_pointsclass-attributeinstance-attribute**

`max_touch_points: int | None = None`

The maximum number of simultaneous touch contact points supported by the current device.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/maxTouchPoints>

## **buildplatformclass-attributeinstance-attribute**

`platform: str | None = None`

A string representing the platform of the browser.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/platform>

### **Note**

Do not rely on this property to return the correct value.

## **buildproductclass-attributeinstance-attribute**

`product: str | None = None`

Always returns "Gecko", on any browser.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/product>

### **Note**

Do not rely on this property to return the correct value. This property is kept only for compatibility purposes.

## **buildproduct\_subclass-attributeinstance-attribute**

`product_sub: str | None = None`

The build number of the current browser.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/productSub>

### **Note**

Do not rely on this property to return the correct value.

### **builduser\_agentclass-attributeinstance-attribute**

`user_agent: str | None = None`

The user agent string for the current browser (e.g., 'Mozilla/5.0 ...').

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/userAgent>

### **buildvendorclass-attributeinstance-attribute**

`vendor: str | None = None`

The vendor name of the current browser.

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/vendor>

### **buildvendor\_subclass-attributeinstance-attribute**

`vendor_sub: str | None = None`

Returns the vendor version number (e.g., '6.1').

More info: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator/vendorSub>

#### **Note**

Do not rely on this property to return the correct value.

# WebRenderer

TBD

Inherits: `enum.Enum`

## Properties

- `AUTO`
- `CANVAS_KIT`
- `SKWASM`

## Properties

`buildAUTO`class-attributeinstance-attribute

`buildCANVAS_KIT`class-attributeinstance-attribute

`buildSKWASM`class-attributeinstance-attribute

## Usage Example

You can explicitly set what renderer to use when running a Flet program in web mode using the `web_renderer` parameter of the `ft.run`.

```
import flet as ft

def main(page: ft.Page):
 ...

ft.run(main, view=ft.AppView.WEB_BROWSER, web_renderer=ft.WebRenderer.CANVAS_KIT)
```

# WebViewConfiguration

Configuration options for in-app web views.

## Properties

- `enable_dom_storage` - Whether DOM storage is enabled.
- `enable_javascript` - Whether JavaScript execution is allowed.
- `headers` - Additional HTTP headers to include with the request.

## Properties

### `buildenable_dom_storageclass-attributeinstance-attribute`

```
enable_dom_storage: bool = True
```

Whether DOM storage is enabled.

### `buildenable_javascriptclass-attributeinstance-attribute`

```
enable_javascript: bool = True
```

Whether JavaScript execution is allowed.

### `buildheadersclass-attributeinstance-attribute`

```
headers: dict[str, str] = field(default_factory=dict)
```

Additional HTTP headers to include with the request.

# Window

Controls the app window.

Limitation:

This control is for Desktop platforms (macOS, Windows, Linux) only.

Inherits: `BaseControl`

## Properties

- `alignment` - Defines the alignment of the app window.
- `always_on_bottom` - Whether the app window should always be displayed below other windows.
- `always_on_top` - Whether the app window should always be displayed on top of other windows/apps.
- `aspect_ratio` - Defines the aspect ratio of the app window.
- `badge_label` - Sets a badge label on the app window.
- `bgcolor` - Sets background color of an application window.
- `brightness` - The brightness of a app window.
- `focused` - Whether the app window should be focused.
- `frameless` - Whether the app window should be frameless.
- `full_screen` - Whether to switch the app's window to fullscreen mode.
- `height` - Defines the height of the app window.
- `icon` - The icon of the app window.
- `ignore_mouse_events` - Whether the app window should ignore mouse events, passing them to the window below it.
- `left` - Defines the horizontal position of the app window - a distance in virtual pixels from the left edge of the screen.
- `max_height` - Defines the maximum height of the app window.
- `max_width` - Defines the maximum width of the app window.
- `maximizable` - Whether to hide/disable app window's "Maximize" button.
- `maximized` - Whether the app window is maximized.
- `min_height` - Defines the minimum height of the app window.
- `min_width` - Defines the minimum width of the app window.
- `minimizable` - Whether the app window can be minimized through the window's "Minimize" button.
- `minimized` - Whether the app window is minimized.
- `movable` - Whether the app window can be moved.
- `opacity` - Defines the opacity of the app window.
- `prevent_close` - Set to True to intercept the native close signal.
- `progress_bar` - The value from 0.0 to 1.0 to display a progress bar on Task Bar or Dock.
- `resizable` - Whether the app window can be resized.
- `shadow` - Whether to display a shadow around the app window.
- `skip_task_bar` - Whether the app window should be hidden from the Task Bar (on Windows) or Dock (on macOS).
- `title_bar_buttons_hidden` - Whether to hide the app window's title bar buttons.
- `title_bar_hidden` - Whether to hide the app window's title bar.
- `top` - Defines the vertical position of a native OS window - a distance in virtual pixels from the top edge of the screen.
- `visible` - Whether to make the app window visible.

- width - Defines the width of the app window.

## Events

- on\_event - Called when app window changes its state.

## Methods

- center - Centers the app window.
- close - Requests graceful closing of the app window.
- destroy - Destroys the app window.
- start\_dragging - Starts dragging the app window.
- start\_resizing - Starts resizing the app window.
- to\_front - Brings the app window to the front.
- wait\_until\_ready\_to\_show - Waits until the app window is ready to show.

## Properties

### buildalignmentclass-attributeinstance-attribute

alignment: Alignment | None = None

Defines the alignment of the app window.

### buildalways\_on\_bottomclass-attributeinstance-attribute

always\_on\_bottom: bool = False

Whether the app window should always be displayed below other windows.

Limitation:

Has effect on Linux and Windows only.

### buildalways\_on\_topclass-attributeinstance-attribute

always\_on\_top: bool = False

Whether the app window should always be displayed on top of other windows/apps.

### buildaspect\_ratiorclass-attributeinstance-attribute

aspect\_ratio: Number | None = None

Defines the aspect ratio of the app window.

### buildbadge\_labelclass-attributeinstance-attribute

badge\_label: str | None = None

Sets a badge label on the app window.

Limitation:

Has effect on macOS only.

### buildbgcolorclass-attributeinstance-attribute

bgcolor: ColorValue | None = None

Sets background color of an application window.

Tip

Can be used together with Page.bgcolor to make a window transparent.

### buildbrightnessclass-attributeinstance-attribute

brightness: Brightness | None = None

The brightness of a app window.

### buildfocusedclass-attributeinstance-attribute

focused: bool = True

Whether the app window should be focused.

Set to True to focus programmatically.

### **buildframelessclass-attributeinstance-attribute**

```
frameless: bool = False
```

Whether the app window should be frameless.

### **buildfull\_screenclass-attributeinstance-attribute**

```
full_screen: bool = False
```

Whether to switch the app's window to fullscreen mode.

### **buildheightclass-attributeinstance-attribute**

```
height: Number | None = None
```

Defines the height of the app window.

### **buildiconclass-attributeinstance-attribute**

```
icon: str | None = None
```

The icon of the app window.

The file should have the .ico extension.

Limitation:

Has effect on Windows only.

### **buildignore\_mouse\_eventsclass-attributeinstance-attribute**

```
ignore_mouse_events: bool = False
```

Whether the app window should ignore mouse events, passing them to the window below it. If this window has focus, it will still receive keyboard events.

### **buildleftclass-attributeinstance-attribute**

```
left: Number | None = None
```

Defines the horizontal position of the app window - a distance in virtual pixels from the left edge of the screen.

### **buildmax\_heightclass-attributeinstance-attribute**

```
max_height: Number | None = None
```

Defines the maximum height of the app window.

### **buildmax\_widthclass-attributeinstance-attribute**

```
max_width: Number | None = None
```

Defines the maximum width of the app window.

### **buildmaximizableclass-attributeinstance-attribute**

```
maximizable: bool = True
```

Whether to hide/disable app window's "Maximize" button.

### **buildmaximizedclass-attributeinstance-attribute**

```
maximized: bool = False
```

Whether the app window is maximized.

Set to True to maximize programmatically.

### **buildmin\_heightclass-attributeinstance-attribute**

```
min_height: Number | None = None
```

Defines the minimum height of the app window.

### **buildmin\_widthclass-attributeinstance-attribute**

```
min_width: Number | None = None
```

Defines the minimum width of the app window.



## **buildminimizableclass-attributeinstance-attribute**

`minimizable: bool = True`

Whether the app window can be minimized through the window's "Minimize" button.

## **buildminimizedclass-attributeinstance-attribute**

`minimized: bool = False`

Whether the app window is minimized.

Set to True to minimize programmatically.

## **buildmovableclass-attributeinstance-attribute**

`movable: bool = True`

Whether the app window can be moved.

Limitation:

Has effect on macOS only.

## **buildopacityclass-attributeinstance-attribute**

`opacity: Number = 1.0`

Defines the opacity of the app window.

Raises:

- `ValueError` - If it is not between 0.0 and 1.0 inclusive.

## **buildprevent\_closeclass-attributeinstance-attribute**

`prevent_close: bool = False`

Set to True to intercept the native close signal.

Could be used to implement app exit confirmation logic.

## **buildprogress\_barclass-attributeinstance-attribute**

`progress_bar: Number | None = None`

The value from 0.0 to 1.0 to display a progress bar on Task Bar or Dock.

## **buildresizableclass-attributeinstance-attribute**

`resizable: bool = True`

Whether the app window can be resized.

## **buildshadowclass-attributeinstance-attribute**

`shadow: bool = True`

Whether to display a shadow around the app window.

## **buildskip\_task\_barclass-attributeinstance-attribute**

`skip_task_bar: bool = False`

Whether the app window should be hidden from the Task Bar (on Windows) or Dock (on macOS).

## **buildtitle\_bar\_buttons\_hiddenclass-attributeinstance-attribute**

`title_bar_buttons_hidden: bool = False`

Whether to hide the app window's title bar buttons.

Limitation:

Has effect on macOS only.

## **buildtitle\_bar\_hiddenclass-attributeinstance-attribute**

`title_bar_hidden: bool = False`

Whether to hide the app window's title bar.

## **buildtopclass-attributeinstance-attribute**

`top: Number | None = None`

Defines the vertical position of a native OS window - a distance in virtual pixels from the top edge of the screen.

### **buildvisibleclass-attributeinstance-attribute**

`visible: bool = True`

Whether to make the app window visible.

Can be of use when the app starts as hidden.

### **buildwidthclass-attributeinstance-attribute**

`width: Number | None = None`

Defines the width of the app window.

## **Events**

### **bolton\_eventclass-attributeinstance-attribute**

`on_event: EventHandler[WindowEvent] | None = None`

Called when app window changes its state. For example, when the window is maximized or minimized.

## **Methods**

### **deployed\_codecenterasync**

`center()`

Centers the app window.

### **deployed\_codecloseasync**

`close()`

Requests graceful closing of the app window.

This sends a native close request equivalent to pressing the window close button. If `prevent_close` is enabled, the close may be intercepted and reported via `on_event` with `WindowEventType.CLOSE`.

### **deployed\_codedestroyasync**

`destroy()`

Destroys the app window.

### **deployed\_codestart\_draggingasync**

`start_dragging()`

Starts dragging the app window.

### **deployed\_codestart\_resizingasync**

`start_resizing(edge: WindowResizeEdge)`

Starts resizing the app window.

### **deployed\_codeto\_frontasync**

`to_front()`

Brings the app window to the front.

### **deployed\_codewait\_until\_ready\_to\_showasync**

`wait_until_ready_to_show()`

Waits until the app window is ready to show.

# WindowEvent

Payload for Window.on\_event callbacks.

Inherits: Event

## Properties

- type - Native event kind emitted by the desktop window backend.

## Properties

buildtypeinstance-attribute

type: WindowEventType

Native event kind emitted by the desktop window backend.

# WindowEventType

Type of native desktop window event.

Values are reported in `WindowEvent.type` and delivered by `Window.on_event`. Event availability depends on the operating system and underlying window manager.

Inherits: `enum.Enum`

## Properties

- BLUR - The window lost focus.
- CLOSE - The OS requested the window to close.
- ENTER\_FULL\_SCREEN - The window entered full-screen mode.
- FOCUS - The window became focused.
- HIDE - The window became hidden.
- LEAVE\_FULL\_SCREEN - The window exited full-screen mode.
- MAXIMIZE - The window was maximized.
- MINIMIZE - The window was minimized.
- MOVE - The window position is changing.
- MOVED - The window position changed.
- RESIZE - The window size is changing.
- RESIZED - The window size changed.
- RESTORE - The window was restored from a minimized state.
- SHOW - The window became visible.
- UNMAXIMIZE - The window exited maximized state.

## Examples

### Window Event Type showcase

[play\\_arrow](#)Try Online

```
import flet as ft

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 log = ft.Column(
 spacing=4,
 scroll=ft.ScrollMode.AUTO,
 height=220,
)

 def add_log(message: str):
 log.controls.insert(0, ft.Text(message, size=12))
 if len(log.controls) > 20:
 log.controls.pop()
 log.update()

 def on_window_event(e: ft.WindowEvent):
 add_log(f"Received: {e.type.name}")

 page.window.on_event = on_window_event

 page.appbar = ft.AppBar(title="WindowEventType Showcase")
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Interact with the app window and watch incoming event types."
),
 ft.Text(
```

```

 "Desktop only. Try focus, blur, resize, move, "
 "minimize, maximize.",
 size=11,
),
 ft.Row(
 wrap=True,
 spacing=8,
 controls=[
 ft.Container(
 padding=ft.Padding.symmetric(horizontal=8, vertical=4),
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=12,
 content=ft.Text(event_type.name, size=11),
)
 for event_type in ft.WindowEventType
],
),
 ft.Container(
 width=720,
 padding=12,
 border=ft.Border.all(1, ft.Colors.RED),
 border_radius=10,
 bgcolor=ft.Colors.SURFACE_CONTAINER_LOW,
 content=log,
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildBLURclass-attributeinstance-attribute

The window lost focus.

### buildCLOSEclass-attributeinstance-attribute

The OS requested the window to close.

### buildENTER\_FULL\_SCREENclass-attributeinstance-attribute

The window entered full-screen mode.

### buildFOCUSclass-attributeinstance-attribute

The window became focused.

### buildHIDEclass-attributeinstance-attribute

The window became hidden.

### buildLEAVE\_FULL\_SCREENclass-attributeinstance-attribute

The window exited full-screen mode.

### buildMAXIMIZEclass-attributeinstance-attribute

The window was maximized.

### buildMINIMIZEclass-attributeinstance-attribute

The window was minimized.

### buildMOVEclass-attributeinstance-attribute

The window position is changing.

### buildMOVEDclass-attributeinstance-attribute

The window position changed.

### buildRESIZEclass-attributeinstance-attribute

The window size is changing.

**buildRESIZEDclass-attributeinstance-attribute**

The window size changed.

**buildRESTOREclass-attributeinstance-attribute**

The window was restored from a minimized state.

**buildSHOWclass-attributeinstance-attribute**

The window became visible.

**buildUNMAXIMIZEclass-attributeinstance-attribute**

The window exited maximized state.

# WindowResizeEdge

Edge or corner used by `Window.start_resizing`.

The selected value defines which resize handle is used when initiating native window resizing programmatically.

Inherits: `enum.Enum`

## Properties

- `BOTTOM` - Bottom edge.
- `BOTTOM_LEFT` - Bottom-left corner.
- `BOTTOM_RIGHT` - Bottom-right corner.
- `LEFT` - Left edge.
- `RIGHT` - Right edge.
- `TOP` - Top edge.
- `TOP_LEFT` - Top-left corner.
- `TOP_RIGHT` - Top-right corner.

## Properties

`buildBOTTOM`class-attributeinstance-attribute

Bottom edge.

`buildBOTTOM_LEFT`class-attributeinstance-attribute

Bottom-left corner.

`buildBOTTOM_RIGHT`class-attributeinstance-attribute

Bottom-right corner.

`buildLEFT`class-attributeinstance-attribute

Left edge.

`buildRIGHT`class-attributeinstance-attribute

Right edge.

`buildTOP`class-attributeinstance-attribute

Top edge.

`buildTOP_LEFT`class-attributeinstance-attribute

Top-left corner.

`buildTOP_RIGHT`class-attributeinstance-attribute

Top-right corner.

# WindowsDeviceInfo

Device information snapshot for Windows systems.

Returned by `Page.get_device_info` on Windows.

Inherits: `DeviceInfo`

## Properties

- `build_lab` - Value of `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\BuildLab` registry key.
- `build_lab_ex` - Value of `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\BuildLabEx` registry key.
- `build_number` - The build number of the operating system.
- `computer_name` - The computer's fully-qualified DNS name, where available.
- `csd_version` - The service-pack version string.
- `device_id` - Displayed as "Device ID" in Windows Settings.
- `display_version` - Value of `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\DisplayVersion` registry key.
- `edition_id` - Value of `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\EditionID` registry key.
- `install_date` - Value of `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\InstallDate` registry key.
- `major_version` - The major version number of the operating system.
- `minor_version` - The minor version number of the operating system.
- `number_of_cores` - Number of CPU cores on the local machine.
- `platform_id` - The operating system platform.
- `product_id` - Displayed as "Product ID" in Windows Settings.
- `product_name` - Value of `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProductName` registry key.
- `product_type` - The product type.
- `registered_owner` - Value of the `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\RegisteredOwner` registry key.
- `release_id` - Value of the `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ReleaseId` registry key.
- `reserved` - Reserved for future use.
- `service_pack_major` - The major version number of the latest service pack installed on the system.
- `service_pack_minor` - The minor version number of the latest service pack installed on the system.
- `suit_mask` - The product suites available on the system.
- `system_memory` - The physically installed memory in the computer, in megabytes.
- `user_name`

## Properties

### `buildbuild_labinstance-attribute`

`build_lab`: `str`

Value of `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\BuildLab` registry key.

For example: "22000.co\_release.210604-1628".



### **buildbuild\_lab\_exinstance-attribute**

build\_lab\_ex: str

Value of HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\BuildLabEx registry key.

For example: "22000.1.amd64free.co\_release.210604-1628".

### **buildbuild\_numberinstance-attribute**

build\_number: int

The build number of the operating system.

### **buildcomputer\_nameinstance-attribute**

computer\_name: str

The computer's fully-qualified DNS name, where available.

### **buildcsd\_versioninstance-attribute**

csd\_version: str

The service-pack version string.

This member contains a string, such as "Service Pack 3", which indicates the latest service pack installed on the system.

### **builddevice\_idinstance-attribute**

device\_id: str

Displayed as "Device ID" in Windows Settings.

Value of HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\SQMClient\MachineId registry key.

### **bulddisplay\_versioninstance-attribute**

display\_version: str

Value of HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\DisplayVersion registry key.

For example: "21H2".

### **buldedition\_idinstance-attribute**

edition\_id: str

Value of HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\EditionID registry key.

### **buildinstall\_dateinstance-attribute**

install\_date: datetime

Value of HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\InstallDate registry key.

### **buildmajor\_versioninstance-attribute**

major\_version: int

The major version number of the operating system.

For example, for Windows 2000, the major version number is 5.

For more info, see the table in Remarks:

[https://docs.microsoft.com/en-us/windows-hardware/drivers/ddi/wdm/ns-wdm-\\_osversioninfoexw#remarks](https://docs.microsoft.com/en-us/windows-hardware/drivers/ddi/wdm/ns-wdm-_osversioninfoexw#remarks)

### **buildminor\_versioninstance-attribute**

minor\_version: int

The minor version number of the operating system.

For example, for Windows 2000, the minor version number is 0.

For more info, see the table in Remarks:

[https://docs.microsoft.com/en-us/windows-hardware/drivers/ddi/wdm/ns-wdm-\\_osversioninfoexw#remarks](https://docs.microsoft.com/en-us/windows-hardware/drivers/ddi/wdm/ns-wdm-_osversioninfoexw#remarks)

### **buildnumber\_of\_coresinstance-attribute**

number\_of\_cores: int

Number of CPU cores on the local machine.

### **buildplatform\_idinstance-attribute**

platform\_id: int

The operating system platform.

For Win32 on NT-based operating systems, RtlGetVersion returns the value VER\_PLATFORM\_WIN32\_NT.

### **buildproduct\_idinstance-attribute**

product\_id: str

Displayed as "Product ID" in Windows Settings.

Value of the HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProductId registry key.

For example: "00000-00000-0000-AAAAA".

### **buildproduct\_nameinstance-attribute**

product\_name: str

Value of HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProductName registry key.

For example: "Windows 10 Home Single Language".

### **buildproduct\_typeinstance-attribute**

product\_type: int

The product type.

This member contains additional information about the system.

### **buildregistered\_ownerinstance-attribute**

registered\_owner: str

Value of the HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\RegisteredOwner registry key.

For example: "Microsoft Corporation".

### **buildrelease\_idinstance-attribute**

release\_id: str

Value of the HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ReleaseId registry key.

For example: "1903".

### **buildreservedinstance-attribute**

reserved: int

Reserved for future use.

### **buildservice\_pack\_majorinstance-attribute**

service\_pack\_major: int

The major version number of the latest service pack installed on the system.

For example, for Service Pack 3, the major version number is three. If no service pack has been installed, the value is zero.

### **buildservice\_pack\_minorinstance-attribute**

service\_pack\_minor: int

The minor version number of the latest service pack installed on the system.

For example, for Service Pack 3, the minor version number is zero.

### **buildsuit\_maskinstance-attribute**

suit\_mask: int

The product suites available on the system.

### **buildsystem\_memoryinstance-attribute**

system\_memory: int

The physically installed memory in the computer, in megabytes.

This may not be the same as available memory.

**builduser\_nameinstance-attribute**

user\_name: str

# Breaking changes and deprecations

Breaking changes and deprecations are release changes that can require code, dependency, or configuration updates when you upgrade. Removed APIs are listed here as breaking changes. Deprecated APIs still work for now, but they are scheduled for removal.

note

These breaking change and deprecation guides are accurate as of the release that introduced the change. Later releases might add new APIs or additional migration paths.

This page lists the guides created for each release.

## By release

The following guides are available. They're sorted by release, with the most recent release first. Each guide explains the change, the reason for it, and how to migrate your code.

### Released in Flet 0.85.0

#### Breaking changes

- Deprecated spacing and border helper functions removed

#### Deprecations

- DragTargetEvent coordinate fields deprecated
- Video control APIs deprecated

# Compatibility policy

Flet tries to balance API stability with the need to keep improving the framework, fixing bugs, and simplifying APIs before and after Flet 1.0.

## Breaking changes

A breaking change is a change that can require users to update app code, configuration, dependencies, or build settings when upgrading Flet.

Examples include:

- removing a public API;
- changing a default behavior in a way that affects existing apps;
- changing event payloads or method return values;
- changing minimum supported Python, Flutter, platform, or dependency versions;
- changing build, packaging, or publishing requirements.

When a breaking change needs migration guidance, it should be listed in Breaking changes and deprecations and linked from Release notes.

## Deprecation policy

When possible, Flet deprecates APIs before removing them. Deprecated APIs remain available during the deprecation period, emit runtime warnings where supported, and appear with deprecation labels in API docs.

By default, deprecated APIs are removed after three minor releases. For example, an API deprecated in 0.85.0 is normally scheduled for removal in 0.88.0.

Exceptions can happen when a deprecation needs more migration time or when compatibility would block an important fix. Any exception should be called out in the related release notes or migration guide.

## Where changes are documented

- Release notes list each release and link to announcements, changelogs, and migration guides.
- Breaking changes and deprecations explain changes that require migration, grouped by release.
- API reference pages show deprecation details next to affected APIs.

# Release notes

This page links release announcements, changelogs, and migration notes for Flet releases.

## Stable releases

### 0.85.x

- 0.85.1: Changelog
- 0.85.0: Announcement, Changelog, Breaking changes and deprecations

### 0.84.x

- 0.84.0: Announcement, Changelog

### 0.83.x

- 0.83.1: Changelog
- 0.83.0: Announcement, Changelog

### 0.82.x

- 0.82.2: Changelog
- 0.82.0: Changelog

### 0.81.x

- 0.81.0: Announcement, Changelog

### 0.80.x

- 0.80.5: Changelog
- 0.80.4: Changelog
- 0.80.3: Changelog
- 0.80.2: Changelog
- 0.80.1: Changelog
- 0.80.0: Changelog

### 0.70.x

- 0.70.0: Changelog

### 0.28.x

- 0.28.3: Changelog
- 0.28.2: Changelog
- 0.28.0: Changelog

### 0.27.x

- 0.27.6: Changelog
- 0.27.5: Changelog
- 0.27.4: Changelog
- 0.27.3: Changelog
- 0.27.2: Changelog
- 0.27.1: Changelog
- 0.27.0: Announcement, Changelog

## 0.26.x

- 0.26.0: Announcement, Changelog

## 0.25.x

- 0.25.2: Changelog
- 0.25.1: Changelog
- 0.25.0: Announcement, Changelog

## 0.24.x

- 0.24.1: Changelog
- 0.24.0: Announcement, Changelog

## 0.23.x

- 0.23.2: Changelog
- 0.23.1: Changelog
- 0.23.0: Announcement, Changelog

## 0.22.x

- 0.22.1: Changelog
- 0.22.0: Changelog

## 0.21.x

- 0.21.2: Changelog
- 0.21.1: Changelog
- 0.21.0: Changelog

## 0.20.x

- 0.20.2: Changelog
- 0.20.1: Changelog
- 0.20.0: Changelog

## 0.19.x

- 0.19.0: Changelog

## 0.18.x

- 0.18.0: Changelog

## 0.17.x

- 0.17.0: Changelog

## 0.16.x

- 0.16.0: Changelog

## 0.15.x

- 0.15.0: Changelog

## 0.14.x

- 0.14.0: Changelog

## 0.13.x

- 0.13.0: Changelog

## 0.12.x

- 0.12.2: Changelog
- 0.12.1: Changelog
- 0.12.0: Changelog

#### 0.11.x

- 0.11.0: Changelog

#### 0.10.x

- 0.10.3: Changelog
- 0.10.2: Changelog
- 0.10.1: Changelog
- 0.10.0: Changelog

#### 0.9.x

- 0.9.0: Changelog

#### 0.8.x

- 0.8.4: Changelog
- 0.8.3: Changelog
- 0.8.2: Changelog
- 0.8.1: Changelog
- 0.8.0: Changelog

#### 0.7.x

- 0.7.4: Changelog
- 0.7.3: Changelog
- 0.7.2: Changelog
- 0.7.1: Changelog
- 0.7.0: Changelog

#### 0.6.x

- 0.6.2: Changelog
- 0.6.1: Changelog
- 0.6.0: Changelog

#### 0.5.x

- 0.5.2: Changelog
- 0.5.1: Changelog
- 0.5.0: Changelog

#### 0.4.x

- 0.4.2: Changelog
- 0.4.1: Changelog
- 0.4.0: Changelog

#### 0.3.x

- 0.3.2: Changelog
- 0.3.1: Changelog
- 0.3.0: Changelog



## 0.2.x

- 0.2.4: Changelog
- 0.2.2: Changelog

## 0.1.x

- 0.1.65: Changelog
- 0.1.64: Changelog
- 0.1.63: Changelog
- 0.1.62: Changelog
- 0.1.61: Changelog
- 0.1.60: Changelog
- 0.1.59: Changelog
- 0.1.58: Changelog
- 0.1.57: Changelog
- 0.1.56: Changelog
- 0.1.55: Changelog
- 0.1.54: Changelog
- 0.1.52: Changelog
- 0.1.51: Changelog
- 0.1.50: Changelog
- 0.1.49: Changelog
- 0.1.48: Changelog
- 0.1.47: Changelog
- 0.1.43: Changelog
- 0.1.42: Changelog
- 0.1.41: Changelog
- 0.1.40: Changelog
- 0.1.39: Changelog
- 0.1.38: Changelog
- 0.1.37: Changelog
- 0.1.36: Changelog
- 0.1.35: Changelog
- 0.1.34: Changelog
- 0.1.33: Changelog
- 0.1.32: Changelog
- 0.1.31: Changelog
- 0.1.30: Changelog
- 0.1.29: Changelog
- 0.1.28: Changelog
- 0.1.27: Changelog
- 0.1.26: Changelog
- 0.1.25: Changelog

- 0.1.24: Changelog
- 0.1.23: Changelog
- 0.1.22: Changelog
- 0.1.20: Changelog
- 0.1.19: Changelog
- 0.1.18: Changelog
- 0.1.17: Changelog
- 0.1.16: Changelog
- 0.1.14: Changelog
- 0.1.13: Changelog
- 0.1.12: Changelog
- 0.1.11: Changelog
- 0.1.10: Changelog
- 0.1.7: Changelog
- 0.1.5: Changelog
- 0.1.4: Changelog

# BannerAd

Displays a banner ad.

Test IDs

AdMob provides unit IDs for testing purposes. Set `unit_id` to the appropriate value based on the platform you're testing on:

- Android: "ca-app-pub-3940256099942544/9214589741"
- iOS: "ca-app-pub-3940256099942544/2435281174"

Remember to replace them in production.

Raises:

- `FletUnsupportedPlatformException` - When this control is used on a web and/or non-mobile platform.

Inherits: `LayoutControl`, `BaseAd`

## Events

- `on_paid` - Called when this ad is estimated to have earned money.
- `on_will_dismiss` - Called before dismissing a full screen view.

## Examples

### BannerAd and InterstitialAd

```
import flet as ft
import flet_ads as fta

Test ad unit IDs
ids = {
 ft.PagePlatform.ANDROID: {
 "banner": "ca-app-pub-3940256099942544/6300978111",
 "interstitial": "ca-app-pub-3940256099942544/1033173712",
 },
 ft.PagePlatform.IOS: {
 "banner": "ca-app-pub-3940256099942544/2934735716",
 "interstitial": "ca-app-pub-3940256099942544/4411468910",
 },
}

def main(page: ft.Page):
 page.appbar = ft.AppBar(
 adaptive=True,
 title="Mobile Ads Playground",
 bgcolor=ft.Colors.LIGHT_BLUE_300,
)
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.scroll = ft.ScrollMode.AUTO

 def show_new_interstitial_ad():
 async def show_iad(e: ft.Event[fta.InterstitialAd]):
 await iad.show()

 iad = fta.InterstitialAd(
 unit_id=ids[page.platform]["interstitial"],
 on_load=show_iad,
 on_error=lambda e: print("InterstitialAd error", e.data),
 on_open=lambda e: print("InterstitialAd opened"),
 on_close=lambda e: print("InterstitialAd closed"),
 on_impression=lambda e: print("InterstitialAd impression"),
 on_click=lambda e: print("InterstitialAd clicked"),
)

 def get_new_banner_ad() -> fta.BannerAd:
 return fta.BannerAd(
 unit_id=ids[page.platform]["banner"],
 width=320,
 height=50,
 on_click=lambda e: print("BannerAd clicked"),
 on_load=lambda e: print("BannerAd loaded"),
 on_error=lambda e: print("BannerAd error", e.data),
 on_open=lambda e: print("BannerAd opened"),
 on_close=lambda e: print("BannerAd closed"),
 on_impression=lambda e: print("BannerAd impression"),
```

```

 on_will_dismiss=lambda e: print("BannerAd will dismiss"),
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.OutlinedButton(
 content="Show InterstitialAd",
 on_click=show_new_interstitial_ad,
 disabled=page.web
 or not page.platform.is_mobile(), # mobile only
),
 ft.OutlinedButton(
 content="Show BannerAd",
 on_click=lambda e: page.add(get_new_banner_ad()),
 disabled=page.web
 or not page.platform.is_mobile(), # mobile only
),
]
),
),
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/examples/extensions/ads/media/example\\_1.gif](https://flet.dev/docs/examples/extensions/ads/media/example_1.gif)

## Events

### `bolton_paid`class-attributeinstance-attribute

```

on_paid: (
 ControlEventHandler[PaidAdEvent[BannerAd]] | None
) = None

```

Called when this ad is estimated to have earned money.

Available for allowlisted accounts only.

### `bolton_will_dismiss`class-attributeinstance-attribute

```

on_will_dismiss: ControlEventHandler[BannerAd] | None = (
 None
)

```

Called before dismissing a full screen view.

#### Note

Only available on iOS.

# BaseAd

Base class for all Ad controls.

This class defines shared ad request and lifecycle event properties for concrete ad controls/services.

Raises:

- `FletUnsupportedPlatformException` - When using this control on a web and/or non-mobile platform.

Inherits: `BaseControl`

## Properties

- `request` - Targeting information used to fetch an Ad.
- `unit_id` - Ad unit ID for this ad.

## Events

- `on_click` - Called when this ad is clicked.
- `on_close` - Called when the full screen view has been closed.
- `on_error` - Called when an ad request failed.
- `on_impression` - Called when an impression occurs on this ad.
- `on_load` - Called when this ad is loaded successfully.
- `on_open` - Called when this ad opens up.

## Properties

### `buildrequestclass-attributeinstance-attribute`

```
request: AdRequest = field(
 default_factory=lambda: AdRequest()
)
```

Targeting information used to fetch an Ad.

### `buildunit_idinstance-attribute`

```
unit_id: str
```

Ad unit ID for this ad.

## Events

### `bolton_clickclass-attributeinstance-attribute`

```
on_click: ControlEventHandler[BaseAd] | None = None
```

Called when this ad is clicked.

### `bolton_closeclass-attributeinstance-attribute`

```
on_close: ControlEventHandler[BaseAd] | None = None
```

Called when the full screen view has been closed. You should restart anything paused while handling `on_open`.

### `bolton_errorclass-attributeinstance-attribute`

```
on_error: ControlEventHandler[BaseAd] | None = None
```

Called when an ad request failed.

Event handler argument data property contains information about the error.

### `bolton_impressionclass-attributeinstance-attribute`

```
on_impression: ControlEventHandler[BaseAd] | None = None
```

Called when an impression occurs on this ad.

### `bolton_loadclass-attributeinstance-attribute`

`on_load: ControlEventHandler[BaseAd] | None = None`

Called when this ad is loaded successfully.

## **bolton\_openclass-attributeinstance-attribute**

`on_open: ControlEventHandler[BaseAd] | None = None`

Called when this ad opens up.

A full screen view/overlay is presented in response to the user clicking on an ad. You may want to pause animations and time sensitive interactions.

# InterstitialAd

Displays a full-screen interstitial ad.

## Note

Each instance is allowed to be shown (using show) only once. To show another ad, create a new instance. Reusing an already-shown instance will result in errors or unexpected behavior.

## Test IDs

AdMob provides unit IDs for testing purposes. Set unit\_id to the appropriate value based on the platform you're testing on:

- Android: "ca-app-pub-3940256099942544/1033173712"
- iOS: "ca-app-pub-3940256099942544/4411468910"

Remember to replace them in production.

## Raises:

- `FletUnsupportedPlatformException` - When using this control on a web and/or non-mobile platform.

Inherits: Service, BaseAd

## Methods

- `show` - Present the loaded interstitial ad as a full-screen overlay.

## Examples

### BannerAd and InterstitialAd

```
import flet as ft
import flet_ads as fta

Test ad unit IDs
ids = {
 ft.PagePlatform.ANDROID: {
 "banner": "ca-app-pub-3940256099942544/6300978111",
 "interstitial": "ca-app-pub-3940256099942544/1033173712",
 },
 ft.PagePlatform.IOS: {
 "banner": "ca-app-pub-3940256099942544/2934735716",
 "interstitial": "ca-app-pub-3940256099942544/4411468910",
 },
}

def main(page: ft.Page):
 page.appbar = ft.AppBar(
 adaptive=True,
 title="Mobile Ads Playground",
 bgcolor=ft.Colors.LIGHT_BLUE_300,
)
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
 page.scroll = ft.ScrollMode.AUTO

 def show_new_interstitial_ad():
 async def show_iad(e: ft.Event[fta.InterstitialAd]):
 await iad.show()

 iad = fta.InterstitialAd(
 unit_id=ids[page.platform]["interstitial"],
 on_load=show_iad,
 on_error=lambda e: print("InterstitialAd error", e.data),
 on_open=lambda e: print("InterstitialAd opened"),
 on_close=lambda e: print("InterstitialAd closed"),
 on_impression=lambda e: print("InterstitialAd impression"),
 on_click=lambda e: print("InterstitialAd clicked"),
)

 def get_new_banner_ad() -> fta.BannerAd:
 return fta.BannerAd(
 unit_id=ids[page.platform]["banner"],
 width=320,
 height=50,
 on_click=lambda e: print("BannerAd clicked"),
 on_load=lambda e: print("BannerAd loaded"),
 on_error=lambda e: print("BannerAd error", e.data),
)
```

```

 on_open=lambda e: print("BannerAd opened"),
 on_close=lambda e: print("BannerAd closed"),
 on_impression=lambda e: print("BannerAd impression"),
 on_will_dismiss=lambda e: print("BannerAd will dismiss"),
)
 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.OutlinedButton(
 content="Show InterstitialAd",
 on_click=show_new_interstitial_ad,
 disabled=page.web
 or not page.platform.is_mobile(), # mobile only
),
 ft.OutlinedButton(
 content="Show BannerAd",
 on_click=lambda e: page.add(get_new_banner_ad()),
 disabled=page.web
 or not page.platform.is_mobile(), # mobile only
),
]
),
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/examples/extensions/ads/media/example\\_1.gif](https://flet.dev/docs/examples/extensions/ads/media/example_1.gif)

## Methods

### deployed\_codeshowasync

`show()`

Present the loaded interstitial ad as a full-screen overlay.

The ad must be loaded before this method is called.



# Arc

Draws an arc scaled to fit inside the given rectangle.

It starts from `start_angle` radians around the oval up to `start_angle + sweep_angle` radians around the oval, with zero radians being the point on the right hand side of the oval that crosses the horizontal line that intersects the center of the rectangle and with positive angles going clockwise around the oval. If `use_center` is `True`, the arc is closed back to the center, forming a circle sector. Otherwise, the arc is not closed, forming a circle segment.

<https://api.flutter.dev/flutter/dart-ui/Canvas/drawArc.html>

Inherits: `canvas.Shape`

## Properties

- `height` - The height of the rectangle containing the arc's oval.
- `paint` - A style to draw an arc with.
- `start_angle` - The starting angle in radians to draw arc from.
- `sweep_angle` - The length of the arc in radians.
- `use_center` - Whether this arc is closed back to the center, forming a circle sector.
- `width` - The width of the rectangle containing the arc's oval.
- `x` - The x-axis coordinate of the arc's top left point.
- `y` - The y-axis coordinate of the arc's top left point.

## Properties

### `buildheightclass-attributeinstance-attribute`

`height: Number = 0`

The height of the rectangle containing the arc's oval.

### `buildpaintclass-attributeinstance-attribute`

`paint: Paint = field(default_factory=lambda: Paint())`

A style to draw an arc with.

### `buildstart_angleclass-attributeinstance-attribute`

`start_angle: Number = 0`

The starting angle in radians to draw arc from.

### `buildsweep_angleclass-attributeinstance-attribute`

`sweep_angle: Number = 0`

The length of the arc in radians.

### `builduse_centerclass-attributeinstance-attribute`

`use_center: bool = False`

Whether this arc is closed back to the center, forming a circle sector. If not closed (`False`), this arc forms a circle segment.

### `buildwidthclass-attributeinstance-attribute`

`width: Number = 0`

The width of the rectangle containing the arc's oval.

### `buildxinstance-attribute`

`x: Number`

The x-axis coordinate of the arc's top left point.

### `buildyinstance-attribute`

`y: Number`

The y-axis coordinate of the arc's top left point.

# Circle

Draws a circle.

Inherits: `canvas.Shape`

## Properties

- `paint` - A style to draw a circle with.
- `radius` - Circle's radius.
- `x` - The x-axis coordinate of the circle's center point.
- `y` - The y-axis coordinate of the circle's center point.

## Properties

### **buildpaintclass-attributeinstance-attribute**

```
paint: Paint = field(default_factory=lambda: Paint())
```

A style to draw a circle with.

### **buildradiusclass-attributeinstance-attribute**

```
radius: Number = 0
```

Circle's radius.

### **buildxinstance-attribute**

```
x: Number
```

The x-axis coordinate of the circle's center point.

### **buildyinstance-attribute**

```
y: Number
```

The y-axis coordinate of the circle's center point.

# Color

Paints the given color onto the canvas, applying the given `blend_mode`, with the given color being the source and the background being the destination.

Inherits: `canvas.Shape`

## Properties

- `blend_mode` - Blend mode to apply.
- `color` - Color to paint onto the canvas.

## Properties

**buildblend\_mode**class-attributeinstance-attribute

```
blend_mode: BlendMode = BlendMode.SRC_OVER
```

Blend mode to apply.

**buildcolor**class-attributeinstance-attribute

```
color: ColorValue = Colors.BLACK
```

Color to paint onto the canvas.

# Fill

Fills the canvas with the given paint.

To fill the canvas with a solid color and blend mode, consider Color shape instead.

Inherits: canvas.Shape

## Properties

- paint - A style to fill the canvas with.

## Properties

buildpaintclass-attributeinstance-attribute

```
paint: Paint = field(default_factory=lambda: Paint())
```

A style to fill the canvas with.

# Image

Draws an image.

Inherits: `canvas.Shape`

## Properties

- `height` - The height of the rectangle to draw the image into.
- `paint` - A paint to composite the image into canvas.
- `src` - Draws an image from a source.
- `width` - The width of the rectangle to draw the image into.
- `x` - The x-axis coordinate of the image's top-left corner.
- `y` - The y-axis coordinate of the image's top-left corner.

## Properties

### `buildheightclass-attributeinstance-attribute`

`height: Number | None = None`

The height of the rectangle to draw the image into. Use image height if None.

### `buildpaintclass-attributeinstance-attribute`

`paint: Paint | None = None`

A paint to composite the image into canvas.

### `buildsrcclass-attributeinstance-attribute`

`src: str | bytes | None = None`

Draws an image from a source.

Accepts URLs/paths, base64 strings, or raw bytes.

### `buildwidthclass-attributeinstance-attribute`

`width: Number | None = None`

The width of the rectangle to draw the image into. Use image width if None.

### `buildxclass-attributeinstance-attribute`

`x: Number | None = None`

The x-axis coordinate of the image's top-left corner.

### `buildyclass-attributeinstance-attribute`

`y: Number | None = None`

The y-axis coordinate of the image's top-left corner.

# Line

Draws a line between the given points using the given paint.

Note

The line is always rendered as a stroke, regardless of the value of paint's style property.

Inherits: `canvas.Shape`

## Properties

- `paint` - A style to draw a line with.
- `x1` - The x-axis coordinate of the line's starting point.
- `x2` - The x-axis coordinate of the line's end point.
- `y1` - The y-axis coordinate of the line's starting point.
- `y2` - The y-axis coordinate of the line's end point.

## Properties

### `buildpaintclass-attributeinstance-attribute`

```
paint: Paint = field(default_factory=lambda: Paint())
```

A style to draw a line with.

### `buildx1instance-attribute`

```
x1: Number
```

The x-axis coordinate of the line's starting point.

### `buildx2instance-attribute`

```
x2: Number
```

The x-axis coordinate of the line's end point.

### `buildy1instance-attribute`

```
y1: Number
```

The y-axis coordinate of the line's starting point.

### `buildy2instance-attribute`

```
y2: Number
```

The y-axis coordinate of the line's end point.

# Oval

Draws an axis-aligned oval that fills the given axis-aligned rectangle with the given paint.

The style property of paint indicates whether this oval is filled, stroked, or both.

Inherits: canvas.Shape

## Properties

- height - The height of the rectangle containing the oval.
- paint - A style to draw an oval with.
- width - The width of the rectangle containing the oval.
- x - The x-axis coordinate of the oval's top left point.
- y - The y-axis coordinate of the oval's top left point.

## Properties

### buildheightclass-attributeinstance-attribute

```
height: Number = 0
```

The height of the rectangle containing the oval.

### buildpaintclass-attributeinstance-attribute

```
paint: Paint = field(default_factory=lambda: Paint())
```

A style to draw an oval with.

### buildwidthclass-attributeinstance-attribute

```
width: Number = 0
```

The width of the rectangle containing the oval.

### buildxinstance-attribute

```
x: Number
```

The x-axis coordinate of the oval's top left point.

### buildyinstance-attribute

```
y: Number
```

The y-axis coordinate of the oval's top left point.



# Path

Draws a path with given elements with the given paint.

Whether this shape is filled, stroked, or both, is controlled by `paint.style`. If the path is filled, then sub-paths within it are implicitly closed (see `Path.Close`).

Inherits: `canvas.Shape`

## Properties

- `elements` - The list of path elements.
- `paint` - A style to draw a path with.

## Properties

### `buildelementsclass-attributeinstance-attribute`

```
elements: list[PathElement] = field(default_factory=list)
```

The list of path elements.

### `buildpaintclass-attributeinstance-attribute`

```
paint: Paint = field(default_factory=lambda: Paint())
```

A style to draw a path with.

# Points

Draws a sequence of points according to the given `point_mode`.

Inherits: `canvas.Shape`

## Properties

- `paint` - A style to draw points with.
- `point_mode` - Defines how a list of points is interpreted when drawing a set of points.
- `points` - The list of offsets describing points.

## Properties

### `buildpaintclass-attributeinstance-attribute`

```
paint: Paint = field(default_factory=lambda: Paint())
```

A style to draw points with.

### `buildpoint_modeclass-attributeinstance-attribute`

```
point_mode: PointMode = PointMode.POINTS
```

Defines how a list of points is interpreted when drawing a set of points.

### `buildpointsclass-attributeinstance-attribute`

```
points: list[OffsetValue] | None = None
```

The list of offsets describing points.

# Rect

Draws a rectangle.

Inherits: `canvas.Shape`

## Properties

- `border_radius` - The border radius of this rectangle.
- `height` - The height of this rectangle.
- `paint` - A style to draw this rectangle with.
- `width` - The width of this rectangle.
- `x` - The x-axis coordinate of this rectangle's top left point.
- `y` - The y-axis coordinate of this rectangle's top left point.

## Properties

### `buildborder_radiusclass-attributeinstance-attribute`

```
border_radius: BorderRadiusValue = field(
 default_factory=lambda: BorderRadius.all(0)
)
```

The border radius of this rectangle.

### `buildheightclass-attributeinstance-attribute`

```
height: Number = 0
```

The height of this rectangle.

### `buildpaintclass-attributeinstance-attribute`

```
paint: Paint = field(default_factory=lambda: Paint())
```

A style to draw this rectangle with.

### `buildwidthclass-attributeinstance-attribute`

```
width: Number = 0
```

The width of this rectangle.

### `buildxinstance-attribute`

```
x: Number
```

The x-axis coordinate of this rectangle's top left point.

### `buildyinstance-attribute`

```
y: Number
```

The y-axis coordinate of this rectangle's top left point.

# Shadow

Draws a shadow for a path representing the given material elevation.

Note

The `transparent_occluder` argument should be `True` if the occluding object is not opaque.

Inherits: `canvas.Shape`

## Properties

- `color` - The shadow's color.
- `elevation` - The shadow's elevation.
- `path` - The list of elements describing the path of this shape.
- `transparent_occluder` - Whether the occluding object is transparent (not opaque).

## Properties

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue = Colors.BLACK
```

The shadow's color.

### `buildelevationclass-attributeinstance-attribute`

```
elevation: Number = 0
```

The shadow's elevation.

### `buildpathclass-attributeinstance-attribute`

```
path: list[Path.PathElement] = field(default_factory=list)
```

The list of elements describing the path of this shape.

### `buildtransparent_occluderclass-attributeinstance-attribute`

```
transparent_occluder: bool = False
```

Whether the occluding object is transparent (not opaque).

# Shape

A base class for canvas shapes.

See subclasses/implementations:

- Arc
- Circle
- Color
- Fill
- Line
- Oval
- Path
- Points
- Rect
- Text

Inherits: BaseControl

# Text

Draws value with style at the given (x, y) point.

Inherits: `canvas.Shape`

## Properties

- `alignment` - A point within a text rectangle to determine its position and rotation center.
- `ellipsis` - String used to ellipsize overflowing text.
- `max_lines` - The maximum number of lines painted.
- `max_width` - The maximum width of the painted text.
- `rotate` - The rotation of this text in radians.
- `spans` - The list of `TextSpan` objects to build a rich text paragraph.
- `style` - A text style to draw text and spans with.
- `text_align` - Text horizontal align.
- `value` - The text to draw.
- `x` - The x-axis coordinate of the text's alignment point.
- `y` - The y-axis coordinate of the text's alignment point.

## Properties

### `buildalignmentclass-attributeinstance-attribute`

```
alignment: Alignment = field(
 default_factory=lambda: Alignment.TOP_LEFT
)
```

A point within a text rectangle to determine its position and rotation center.

### `buildellipsisclass-attributeinstance-attribute`

```
ellipsis: str | None = None
```

String used to ellipsize overflowing text.

### `buildmax_linesclass-attributeinstance-attribute`

```
max_lines: int | None = None
```

The maximum number of lines painted. Lines beyond this number are silently dropped. For example, if `max_lines = 1`, then only one line is rendered. If `max_lines = None`, but `ellipsis != None`, then lines after the first one that overflows the width constraints are dropped.

### `buildmax_widthclass-attributeinstance-attribute`

```
max_width: Number | None = None
```

The maximum width of the painted text.

Defaults to `None` - infinity.

### `buildrotateclass-attributeinstance-attribute`

```
rotate: Number = 0
```

The rotation of this text in radians. Text is rotated around the point determined by alignment.

### `buildspansclass-attributeinstance-attribute`

```
spans: list[TextSpan] | None = None
```

The list of `TextSpan` objects to build a rich text paragraph.

### `buildstyleclass-attributeinstance-attribute`

```
style: TextStyle | None = None
```

A text style to draw text and spans with.

### **buildtext\_alignclass-attributeinstance-attribute**

`text_align: TextAlign = TextAlign.START`

Text horizontal align.

### **buildvalueclass-attributeinstance-attribute**

`value: str | None = None`

The text to draw.

### **buildxinstance-attribute**

`x: Number`

The x-axis coordinate of the text's alignment point.

### **buildyinstance-attribute**

`y: Number`

The y-axis coordinate of the text's alignment point.

# BarChart

Draws a bar chart.

```
fch.BarChart(
 border=ft.Border.all(1, ft.Colors.GREY_400),
 groups=[
 fch.BarChartGroup(
 x=0,
 rods=[
 fch.BarChartRod(
 from_y=0,
 to_y=40,
 color=ft.Colors.BLUE_GREY_200,
),
],
),
 ...,
],
)
```

Image: BarChart - <https://flet.dev/docs/assets/controls/charts/bar-chart-diagram.svg>

Bar chart

Inherits: `LayoutControl`

## Properties

- `animation` - Controls chart implicit animation.
- `baseline_y` - Baseline value for Y axis.
- `bgcolor` - Background color of the chart.
- `border` - The border around the chart.
- `bottom_axis` - The appearance of the bottom axis, its title and labels.
- `group_alignment` - The alignment of the bar groups within this chart.
- `group_spacing` - An amount of space between bar groups.
- `groups` - The list of `BarChartGroups` to draw.
- `horizontal_grid_lines` - Controls drawing of chart's horizontal lines.
- `interactive` - Enables automatic tooltips when hovering chart bars.
- `left_axis` - The appearance of the left axis, its title and labels.
- `max_y` - The maximum displayed value for Y axis.
- `min_y` - The minimum displayed value for Y axis.
- `right_axis` - The appearance of the right axis, its title and labels.
- `tooltip` - The tooltip configuration for this chart.
- `top_axis` - The appearance of the top axis, its title and labels.
- `vertical_grid_lines` - Controls drawing of chart's vertical lines.

## Events

- `on_event` - Called when an event occurs on this chart, such as a click or hover.

## Examples

### Interactive bar chart

`play_arrow` Try Online

```
import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 page.add(
```



```

ft.SafeArea(
 content=fch.BarChart(
 expand=True,
 interactive=True,
 max_y=110,
 border=ft.Border.all(1, ft.Colors.GREY_400),
 horizontal_grid_lines=fch.ChartGridLines(
 color=ft.Colors.GREY_300, width=1, dash_pattern=[3, 3]
),
 tooltip=fch.BarChartTooltip(
 bgcolor=ft.Colors.with_opacity(0.5, ft.Colors.GREY_300),
 border_radius=ft.BorderRadius.all(20),
),
 left_axis=fch.ChartAxis(
 label_size=40, title=ft.Text("Fruit supply"), title_size=40
),
 right_axis=fch.ChartAxis(show_labels=False),
 bottom_axis=fch.ChartAxis(
 label_size=40,
 labels=[
 fch.ChartAxisLabel(
 value=0, label=ft.Container(ft.Text("Apple"), padding=10)
),
 fch.ChartAxisLabel(
 value=1,
 label=ft.Container(ft.Text("Blueberry"), padding=10),
),
 fch.ChartAxisLabel(
 value=2, label=ft.Container(ft.Text("Cherry"), padding=10)
),
 fch.ChartAxisLabel(
 value=3, label=ft.Container(ft.Text("Orange"), padding=10)
),
],
),
),
 groups=[
 fch.BarChartGroup(
 x=0,
 rods=[
 fch.BarChartRod(
 from_y=0,
 to_y=40,
 width=40,
 color=ft.Colors.GREEN,
 border_radius=0,
),
],
),
 fch.BarChartGroup(
 x=1,
 rods=[
 fch.BarChartRod(
 from_y=0,
 to_y=100,
 width=40,
 color=ft.Colors.BLUE,
 tooltip=fch.BarChartRodTooltip("Blueberry"),
 border_radius=0,
),
],
),
 fch.BarChartGroup(
 x=2,
 rods=[
 fch.BarChartRod(
 from_y=0,
 to_y=30,
 width=40,
 color=ft.Colors.RED,
 border_radius=0,
),
],
),
 fch.BarChartGroup(
 x=3,
 rods=[
 fch.BarChartRod(
 from_y=0,
 to_y=60,
 width=40,
 color=ft.Colors.ORANGE,
 tooltip=fch.BarChartRodTooltip("Orange"),
 border_radius=0,
),
],
),
],
),
],

```

```

),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/bar\\_chart/example\\_1.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/bar_chart/example_1.png)

## Bar chart with hover

play\_arrowTry Online

```

import flet as ft
import flet_charts as fch

class CustomRod(fch.BarChartRod):
 def __init__(self, y: float, hovered: bool = False):
 super().__init__()
 self.hovered = hovered
 self.y = y
 self.width = 22
 self.color = ft.Colors.WHITE
 self.bg_to_y = 20
 self.bg_color = ft.Colors.GREEN_300

 def before_update(self):
 super().before_update()
 self.to_y = self.y + 0.5 if self.hovered else self.y
 self.color = ft.Colors.YELLOW if self.hovered else ft.Colors.WHITE
 self.border_side = (
 ft.BorderSide(width=1, color=ft.Colors.RED)
 if self.hovered
 else ft.BorderSide(width=1, color=ft.Colors.BLUE)
)

def main(page: ft.Page):
 def on_chart_event(e: fch.BarChartEvent):
 if e.type == fch.ChartEventType.POINTER_HOVER:
 for group_index, group in enumerate(chart.groups):
 for rod_index, rod in enumerate(group.rods):
 rod.hovered = (
 e.group_index == group_index and e.rod_index == rod_index
)
 chart.update()

 chart = fch.BarChart(
 on_event=on_chart_event,
 interactive=True,
 groups=[
 fch.BarChartGroup(x=0, rods=[CustomRod(5)]),
 fch.BarChartGroup(x=1, rods=[CustomRod(6.5)]),
 fch.BarChartGroup(x=2, rods=[CustomRod(15)]),
 fch.BarChartGroup(x=3, rods=[CustomRod(7.5)]),
 fch.BarChartGroup(x=4, rods=[CustomRod(9)]),
 fch.BarChartGroup(x=5, rods=[CustomRod(11.5)]),
 fch.BarChartGroup(x=6, rods=[CustomRod(6)]),
],
 bottom_axis=fch.ChartAxis(
 labels=[
 fch.ChartAxisLabel(value=0, label=ft.Text("M", color=ft.Colors.BLUE)),
 fch.ChartAxisLabel(value=1, label=ft.Text("T", color=ft.Colors.YELLOW)),
 fch.ChartAxisLabel(value=2, label=ft.Text("W", color=ft.Colors.BLUE)),
 fch.ChartAxisLabel(value=3, label=ft.Text("T", color=ft.Colors.YELLOW)),
 fch.ChartAxisLabel(value=4, label=ft.Text("F", color=ft.Colors.BLUE)),
 fch.ChartAxisLabel(value=5, label=ft.Text("S", color=ft.Colors.YELLOW)),
 fch.ChartAxisLabel(value=6, label=ft.Text("S", color=ft.Colors.BLUE)),
],
),
),
)

page.add(
 ft.SafeArea(
 content=ft.Container(
 bgcolor=ft.Colors.GREEN_200,
 padding=10,
 border_radius=5,
 expand=True,
 content=chart,
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildanimationclass-attributeinstance-attribute

```
animation: AnimationValue = field(
 default_factory=lambda: Animation(
 duration=Duration(milliseconds=150),
 curve=AnimationCurve.LINEAR,
)
)
```

Controls chart implicit animation.

### buildbaseline\_yclass-attributeinstance-attribute

```
baseline_y: Number | None = None
```

Baseline value for Y axis.

### buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue | None = None
```

Background color of the chart.

### buildborderclass-attributeinstance-attribute

```
border: Border | None = None
```

The border around the chart.

### buildbottom\_axisclass-attributeinstance-attribute

```
bottom_axis: ChartAxis | None = None
```

The appearance of the bottom axis, its title and labels.

### buildgroup\_alignmentclass-attributeinstance-attribute

```
group_alignment: MainAxisAlignment = (
 MainAxisAlignment.SPACE_EVENLY
)
```

The alignment of the bar groups within this chart.

If set to MainAxisAlignment.CENTER, the space between the groups can be specified using group\_spacing.

### buildgroup\_spacingclass-attributeinstance-attribute

```
group_spacing: Number = 16.0
```

An amount of space between bar groups.

### buildgroupsclass-attributeinstance-attribute

```
groups: list[BarChartGroup] = field(default_factory=list)
```

The list of BarChartGroups to draw.

### buildhorizontal\_grid\_linesclass-attributeinstance-attribute

```
horizontal_grid_lines: ChartGridLines | None = None
```

Controls drawing of chart's horizontal lines.

### buildinteractiveclass-attributeinstance-attribute

```
interactive: bool = True
```

Enables automatic tooltips when hovering chart bars.

### buildleft\_axisclass-attributeinstance-attribute

```
left_axis: ChartAxis | None = None
```

The appearance of the left axis, its title and labels.

### buildmax\_yclass-attributeinstance-attribute

```
max_y: Number | None = None
```

The maximum displayed value for Y axis.

### **buildmin\_yclass-attributeinstance-attribute**

`min_y: Number | None = None`

The minimum displayed value for Y axis.

### **buildright\_axisclass-attributeinstance-attribute**

`right_axis: ChartAxis | None = None`

The appearance of the right axis, its title and labels.

### **buildtooltipclass-attributeinstance-attribute**

```
tooltip: BarChartTooltip | None = field(
 default_factory=lambda: BarChartTooltip()
)
```

The tooltip configuration for this chart.

If set to None, tooltips will not shown throughout this chart.

### **buildtop\_axisclass-attributeinstance-attribute**

`top_axis: ChartAxis | None = None`

The appearance of the top axis, its title and labels.

### **buildvertical\_grid\_linesclass-attributeinstance-attribute**

`vertical_grid_lines: ChartGridLines | None = None`

Controls drawing of chart's vertical lines.

## **Events**

### **bolton\_eventclass-attributeinstance-attribute**

`on_event: EventHandler[BarChartEvent] | None = None`

Called when an event occurs on this chart, such as a click or hover.

# CandlestickChart

Draws a candlestick chart representing OHLC values.

```
fch.CandlestickChart(
 min_x=-1,
 max_x=2,
 min_y=20,
 max_y=30,
 bgcolor=ft.Colors.AMBER_200,
 spots=[
 fch.CandlestickChartSpot(
 x=0,
 open=22.6,
 high=28.3,
 low=21.4,
 close=24.1,
),
 ...
],
)
```

Image: CandlestickChart - [https://flet.dev/docs/test-images-charts/examples/golden/macros/candlestick\\_chart/image\\_for\\_docs.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/candlestick_chart/image_for_docs.png)

Candlestick chart

Inherits: `LayoutControl`

## Properties

- `animation` - Controls chart implicit animations.
- `baseline_x` - Baseline value on the X axis.
- `baseline_y` - Baseline value on the Y axis.
- `bgcolor` - Background color of the chart.
- `border` - Border drawn around the chart.
- `bottom_axis` - Appearance of the bottom axis, its title and labels.
- `horizontal_grid_lines` - Horizontal grid lines configuration.
- `interactive` - Enables automatic tooltips and highlighting when hovering the chart.
- `left_axis` - Appearance of the left axis, its title and labels.
- `long_press_duration` - The duration of a long press on the chart.
- `max_x` - Maximum value displayed on the X axis.
- `max_y` - Maximum value displayed on the Y axis.
- `min_x` - Minimum value displayed on the X axis.
- `min_y` - Minimum value displayed on the Y axis.
- `right_axis` - Appearance of the right axis, its title and labels.
- `rotation_quarter_turns` - Number of quarter turns (90-degree increments) to rotate the chart.
- `show_tooltips_for_selected_spots_only` - Whether to permanently and only show the tooltips of spots with their selected property set to True.
- `spots` - Candlesticks to display on the chart.
- `tooltip` - Tooltip configuration for the chart.
- `top_axis` - Appearance of the top axis, its title and labels.
- `touch_spot_threshold` - The distance threshold to consider a touch near a candlestick.
- `vertical_grid_lines` - Vertical grid lines configuration.

## Events

- `on_event` - Called when an event occurs on this chart.

# Examples

## Candlestick chart

play\_arrowTry Online

```
import flet as ft
import flet_charts as fch

CANDLE_DATA = [
 ("Mon", 24.8, 28.6, 23.9, 27.2),
 ("Tue", 27.2, 30.1, 25.8, 28.4),
 ("Wed", 28.4, 31.2, 26.5, 29.1),
 ("Thu", 29.1, 32.4, 27.9, 31.8),
 ("Fri", 31.8, 34.0, 29.7, 30.2),
 ("Sat", 30.2, 33.6, 28.3, 32.7),
 ("Sun", 32.7, 35.5, 30.1, 34.6),
]

def build_spots() -> list[fch.CandlestickChartSpot]:
 """Create candlestick spots from the static data."""
 spots: list[fch.CandlestickChartSpot] = []
 for index, (label, open_, high, low, close) in enumerate(CANDLE_DATA):
 spots.append(
 fch.CandlestickChartSpot(
 x=float(index),
 open=open_,
 high=high,
 low=low,
 close=close,
 selected=index == len(CANDLE_DATA) - 1,
 tooltip=fch.CandlestickChartSpotTooltip(
 text=(
 f"{label}\n"
 f"Open: {open_:0.1f}\n"
 f"High: {high:0.1f}\n"
 f"Low : {low:0.1f}\n"
 f"Close: {close:0.1f}"
),
 bottom_margin=12,
),
)
)
 return spots

def main(page: ft.Page):
 page.title = "Candlestick chart"
 page.padding = 24
 page.theme_mode = ft.ThemeMode.DARK

 info = ft.Text("Interact with the chart to see event details.")

 spots = build_spots()
 min_x = -0.5
 max_x = len(spots) - 0.5
 min_y = min(low for _, _, _, low, _ in CANDLE_DATA) - 1
 max_y = max(high for _, _, _, _, high in CANDLE_DATA) + 1

 def handle_event(e: fch.CandlestickChartEvent):
 if e.spot_index is not None and e.spot_index >= 0:
 label, open_, high, low, close = CANDLE_DATA[e.spot_index]
 info.value = (
 f"{e.type.value} • {label}: "
 f"O {open_:0.1f} H {high:0.1f} L {low:0.1f} C {close:0.1f}"
)
 else:
 info.value = f"{e.type.value} • outside candlesticks"
 info.update()

 chart = fch.CandlestickChart(
 expand=True,
 min_x=min_x,
 max_x=max_x,
 min_y=min_y,
 max_y=max_y,
 baseline_x=0,
 baseline_y=min_y,
 bgcolor=ft.Colors.with_opacity(0.2, ft.Colors.BLUE_GREY_900),
 horizontal_grid_lines=fch.ChartGridLines(interval=2, dash_pattern=[2, 2]),
 vertical_grid_lines=fch.ChartGridLines(interval=1, dash_pattern=[2, 2]),
 left_axis=fch.ChartAxis(
 label_spacing=2,
 label_size=60,
 title=ft.Text("Price (k USD)", color=ft.Colors.GREY_300),
),
)
```

```

 show_min=False,
),
 bottom_axis=fch.ChartAxis(
 labels=[
 fch.ChartAxisLabel(
 value=index,
 label=ft.Text(name, color=ft.Colors.GREY_300),
)
 for index, (name, *_rest) in enumerate(CANDLE_DATA)
],
 label_spacing=1,
 label_size=40,
 show_min=False,
 show_max=False,
),
 spots=spots,
 tooltip=fch.CandlestickChartTooltip(
 bgcolor=ft.Colors.BLUE_GREY_800,
 horizontal_alignment=fch.HorizontalAlignment.CENTER,
 fit_inside_horizontally=True,
),
 on_event=handle_event,
)

page.add(
 ft.SafeArea(
 content=ft.Container(
 expand=True,
 border_radius=16,
 padding=20,
 content=ft.Column(
 expand=True,
 spacing=20,
 controls=[
 ft.Text(
 "Weekly OHLC snapshot (demo data)",
 size=20,
 weight=ft.FontWeight.BOLD,
),
 chart,
 info,
],
),
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: [Image - https://flet.dev/docs/test-images-charts/examples/golden/macros/candlestick\\_chart/example\\_1.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/candlestick_chart/example_1.png)

## Properties

### buildanimationclass-attributeinstance-attribute

```

animation: AnimationValue = field(
 default_factory=lambda: Animation(
 duration=Duration(milliseconds=150),
 curve=AnimationCurve.LINEAR,
)
)

```

Controls chart implicit animations.

### buildbaseline\_xclass-attributeinstance-attribute

```
baseline_x: Number | None = None
```

Baseline value on the X axis.

### buildbaseline\_yclass-attributeinstance-attribute

```
baseline_y: Number | None = None
```

Baseline value on the Y axis.

### buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue | None = None
```

Background color of the chart.

### buildborderclass-attributeinstance-attribute

```
border: Border | None = None
```

Border drawn around the chart.

### **buildbottom\_axisclass-attributeinstance-attribute**

```
bottom_axis: ChartAxis | None = None
```

Appearance of the bottom axis, its title and labels.

### **buildhorizontal\_grid\_linesclass-attributeinstance-attribute**

```
horizontal_grid_lines: ChartGridLines | None = None
```

Horizontal grid lines configuration.

### **buildinteractiveclass-attributeinstance-attribute**

```
interactive: bool = True
```

Enables automatic tooltips and highlighting when hovering the chart.

### **buildleft\_axisclass-attributeinstance-attribute**

```
left_axis: ChartAxis | None = None
```

Appearance of the left axis, its title and labels.

### **buildlong\_press\_durationclass-attributeinstance-attribute**

```
long_press_duration: DurationValue | None = None
```

The duration of a long press on the chart.

### **buildmax\_xclass-attributeinstance-attribute**

```
max_x: Number | None = None
```

Maximum value displayed on the X axis.

### **buildmax\_yclass-attributeinstance-attribute**

```
max_y: Number | None = None
```

Maximum value displayed on the Y axis.

### **buildmin\_xclass-attributeinstance-attribute**

```
min_x: Number | None = None
```

Minimum value displayed on the X axis.

### **buildmin\_yclass-attributeinstance-attribute**

```
min_y: Number | None = None
```

Minimum value displayed on the Y axis.

### **buildright\_axisclass-attributeinstance-attribute**

```
right_axis: ChartAxis | None = None
```

Appearance of the right axis, its title and labels.

### **buildrotation\_quarter\_turnsclass-attributeinstance-attribute**

```
rotation_quarter_turns: Number = 0
```

Number of quarter turns (90-degree increments) to rotate the chart.

### **buildshow\_tooltips\_for\_selected\_spots\_onlyclass-attributeinstance-attribute**

```
show_tooltips_for_selected_spots_only: bool = False
```

Whether to permanently and only show the tooltips of spots with their selected property set to True.

### **buildspotsclass-attributeinstance-attribute**

```
spots: list[CandlestickChartSpot] = field(
 default_factory=list
)
```

Candlesticks to display on the chart.

### **buildtooltipclass-attributeinstance-attribute**



```
tooltip: CandlestickChartTooltip | None = field(
 default_factory=lambda: CandlestickChartTooltip()
)
```

Tooltip configuration for the chart.

### buildtop\_axisclass-attributeinstance-attribute

```
top_axis: ChartAxis | None = None
```

Appearance of the top axis, its title and labels.

### buildtouch\_spot\_thresholdclass-attributeinstance-attribute

```
touch_spot_threshold: Number = 4
```

The distance threshold to consider a touch near a candlestick.

### buildvertical\_grid\_linesclass-attributeinstance-attribute

```
vertical_grid_lines: ChartGridLines | None = None
```

Vertical grid lines configuration.

## Events

### bolton\_eventclass-attributeinstance-attribute

```
on_event: EventHandler[CandlestickChartEvent] | None = (
 None
)
```

Called when an event occurs on this chart.

# LineChart

Draws a line chart.

Example:

```
fch.LineChart(
 min_y=0,
 max_y=3,
 min_x=0,
 max_x=5,
 data_series=[
 fch.LineChartData(
 color=ft.Colors.BLUE_GREY_500,
 curved=True,
 points=[
 fch.LineChartDataPoint(1, 0.5),
 fch.LineChartDataPoint(2, 1.5),
 fch.LineChartDataPoint(3, 1),
],
),
 ...
],
)
```

Image: LineChart - <https://flet.dev/docs/assets/controls/charts/line-chart-diagram.svg>

Line chart

Inherits: `LayoutControl`

## Properties

- `animation` - Controls chart implicit animation.
- `baseline_x` - Baseline value for X axis.
- `baseline_y` - Baseline value for Y axis.
- `bgcolor` - Background color of the chart.
- `border` - The border around the chart.
- `bottom_axis` - Defines the appearance of the bottom axis, its title and labels.
- `data_series` - A list of `LineChartData` controls drawn as separate lines on a chart.
- `horizontal_grid_lines` - Controls drawing of chart's horizontal lines.
- `interactive` - Enables automatic tooltips and points highlighting when hovering over the chart.
- `left_axis` - Defines the appearance of the left axis, its title and labels.
- `max_x` - Defines the maximum displayed value for X axis.
- `max_y` - Defines the maximum displayed value for Y axis.
- `min_x` - Defines the minimum displayed value for X axis.
- `min_y` - Defines the minimum displayed value for Y axis.
- `point_line_end` - The end of the vertical line drawn at selected point position.
- `point_line_start` - The start of the vertical line drawn under the selected point.
- `right_axis` - Defines the appearance of the right axis, its title and labels.
- `tooltip` - The tooltip configuration for this chart.
- `top_axis` - Defines the appearance of the top axis, its title and labels.
- `vertical_grid_lines` - Controls drawing of chart's vertical lines.

## Events

- `on_event` - Fires when a chart line is hovered or clicked.

## Examples

## Multi-series line chart

play\_arrow Try Online

```
import flet as ft
import flet_charts as fch

class State:
 toggled = True

state = State()

def main(page: ft.Page):
 data_1 = [
 fch.LineChartData(
 stroke_width=8,
 color=ft.Colors.LIGHT_GREEN,
 curved=True,
 rounded_stroke_cap=True,
 points=[
 fch.LineChartDataPoint(1, 1),
 fch.LineChartDataPoint(3, 1.5),
 fch.LineChartDataPoint(5, 1.4),
 fch.LineChartDataPoint(7, 3.4),
 fch.LineChartDataPoint(10, 2),
 fch.LineChartDataPoint(12, 2.2),
 fch.LineChartDataPoint(13, 1.8),
],
),
 fch.LineChartData(
 color=ft.Colors.PINK,
 below_line_bgcolor=ft.Colors.with_opacity(0, ft.Colors.PINK),
 stroke_width=8,
 curved=True,
 rounded_stroke_cap=True,
 points=[
 fch.LineChartDataPoint(1, 1),
 fch.LineChartDataPoint(3, 2.8),
 fch.LineChartDataPoint(7, 1.2),
 fch.LineChartDataPoint(10, 2.8),
 fch.LineChartDataPoint(12, 2.6),
 fch.LineChartDataPoint(13, 3.9),
],
),
 fch.LineChartData(
 color=ft.Colors.CYAN,
 stroke_width=8,
 curved=True,
 rounded_stroke_cap=True,
 points=[
 fch.LineChartDataPoint(1, 2.8),
 fch.LineChartDataPoint(3, 1.9),
 fch.LineChartDataPoint(6, 3),
 fch.LineChartDataPoint(10, 1.3),
 fch.LineChartDataPoint(13, 2.5),
],
),
]

 data_2 = [
 fch.LineChartData(
 stroke_width=4,
 color=ft.Colors.with_opacity(0.5, ft.Colors.LIGHT_GREEN),
 rounded_stroke_cap=True,
 points=[
 fch.LineChartDataPoint(1, 1),
 fch.LineChartDataPoint(3, 4),
 fch.LineChartDataPoint(5, 1.8),
 fch.LineChartDataPoint(7, 5),
 fch.LineChartDataPoint(10, 2),
 fch.LineChartDataPoint(12, 2.2),
 fch.LineChartDataPoint(13, 1.8),
],
),
 fch.LineChartData(
 color=ft.Colors.with_opacity(0.5, ft.Colors.PINK),
 below_line_bgcolor=ft.Colors.with_opacity(0.2, ft.Colors.PINK),
 stroke_width=4,
 curved=True,
 rounded_stroke_cap=True,
 points=[
 fch.LineChartDataPoint(1, 1),
 fch.LineChartDataPoint(3, 2.8),
 fch.LineChartDataPoint(7, 1.2),
],
),
]
```

```

 fch.LineChartDataPoint(10, 2.8),
 fch.LineChartDataPoint(12, 2.6),
 fch.LineChartDataPoint(13, 3.9),
],
),
fch.LineChartData(
 color=ft.Colors.with_opacity(0.5, ft.Colors.CYAN),
 stroke_width=4,
 rounded_stroke_cap=True,
 points=[
 fch.LineChartDataPoint(1, 3.8),
 fch.LineChartDataPoint(3, 1.9),
 fch.LineChartDataPoint(6, 5),
 fch.LineChartDataPoint(10, 3.3),
 fch.LineChartDataPoint(13, 4.5),
],
),
]

chart = fch.LineChart(
 data_series=data_1,
 border=ft.Border(
 bottom=ft.BorderSide(4, ft.Colors.with_opacity(0.5, ft.Colors.ON_SURFACE))
),
 tooltip=fch.LineChartTooltip(
 bgcolor=ft.Colors.with_opacity(0.8, ft.Colors.BLUE_GREY)
),
 min_y=0,
 max_y=4,
 min_x=0,
 max_x=14,
 expand=True,
 right_axis=fch.ChartAxis(show_labels=False),
 left_axis=fch.ChartAxis(
 label_size=40,
 labels=[
 fch.ChartAxisLabel(
 value=1,
 label=ft.Text("1m", size=14, weight=ft.FontWeight.BOLD),
),
 fch.ChartAxisLabel(
 value=2,
 label=ft.Text("2m", size=14, weight=ft.FontWeight.BOLD),
),
 fch.ChartAxisLabel(
 value=3,
 label=ft.Text("3m", size=14, weight=ft.FontWeight.BOLD),
),
 fch.ChartAxisLabel(
 value=4,
 label=ft.Text("4m", size=14, weight=ft.FontWeight.BOLD),
),
 fch.ChartAxisLabel(
 value=5,
 label=ft.Text("5m", size=14, weight=ft.FontWeight.BOLD),
),
 fch.ChartAxisLabel(
 value=6,
 label=ft.Text("6m", size=14, weight=ft.FontWeight.BOLD),
),
],
),
 bottom_axis=fch.ChartAxis(
 label_size=32,
 labels=[
 fch.ChartAxisLabel(
 value=2,
 label=ft.Container(
 margin=ft.Margin(top=10),
 content=ft.Text(
 value="SEP",
 size=16,
 weight=ft.FontWeight.BOLD,
 color=ft.Colors.with_opacity(0.5, ft.Colors.ON_SURFACE),
),
),
),
 fch.ChartAxisLabel(
 value=7,
 label=ft.Container(
 margin=ft.Margin(top=10),
 content=ft.Text(
 value="OCT",
 size=16,
 weight=ft.FontWeight.BOLD,
 color=ft.Colors.with_opacity(0.5, ft.Colors.ON_SURFACE),
),
),
),
],
),
)

```

```

),
 fch.ChartAxisLabel(
 value=12,
 label=ft.Container(
 margin=ft.Margin(top=10),
 content=ft.Text(
 value="DEC",
 size=16,
 weight=ft.FontWeight.BOLD,
 color=ft.Colors.with_opacity(0.5, ft.Colors.ON_SURFACE),
),
),
),
],
),
)

def toggle_data(e: ft.Event[ft.IconButton]):
 if state.toggled:
 chart.data_series = data_2
 chart.data_series[2].point = True
 chart.max_y = 6
 chart.interactive = False
 else:
 chart.data_series = data_1
 chart.max_y = 4
 chart.interactive = True
 state.toggled = not state.toggled
 chart.update()

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.IconButton(ft.Icons.REFRESH, on_click=toggle_data),
 chart,
]
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macOS/line\\_chart/example\\_1.png](https://flet.dev/docs/test-images-charts/examples/golden/macOS/line_chart/example_1.png)

## Line chart with custom axes

play\_arrowTry Online

```

import flet as ft
import flet_charts as fch

class State:
 toggled = True

state = State()

def main(page: ft.Page):
 data_1 = [
 fch.LineChartData(
 stroke_width=5,
 color=ft.Colors.CYAN,
 curved=True,
 rounded_stroke_cap=True,
 points=[
 fch.LineChartDataPoint(0, 3),
 fch.LineChartDataPoint(2.6, 2),
 fch.LineChartDataPoint(4.9, 5),
 fch.LineChartDataPoint(6.8, 3.1),
 fch.LineChartDataPoint(8, 4),
 fch.LineChartDataPoint(9.5, 3),
 fch.LineChartDataPoint(11, 4),
],
)
]

 data_2 = [
 fch.LineChartData(
 stroke_width=5,
 color=ft.Colors.CYAN,
 curved=True,

```

```

 rounded_stroke_cap=True,
 points=[
 fch.LineChartDataPoint(0, 3.44),
 fch.LineChartDataPoint(2.6, 3.44),
 fch.LineChartDataPoint(4.9, 3.44),
 fch.LineChartDataPoint(6.8, 3.44),
 fch.LineChartDataPoint(8, 3.44),
 fch.LineChartDataPoint(9.5, 3.44),
 fch.LineChartDataPoint(11, 3.44),
],
)
]

chart = fch.LineChart(
 expand=True,
 data_series=data_1,
 min_y=0,
 max_y=6,
 min_x=0,
 max_x=11,
 border=ft.Border.all(3, ft.Colors.with_opacity(0.2, ft.Colors.ON_SURFACE)),
 horizontal_grid_lines=fch.ChartGridLines(
 interval=1, color=ft.Colors.with_opacity(0.2, ft.Colors.ON_SURFACE), width=1
),
 vertical_grid_lines=fch.ChartGridLines(
 interval=1, color=ft.Colors.with_opacity(0.2, ft.Colors.ON_SURFACE), width=1
),
 tooltip=fch.LineChartTooltip(
 bgcolor=ft.Colors.with_opacity(0.8, ft.Colors.BLUE_GREY)
),
 left_axis=fch.ChartAxis(
 label_size=40,
 labels=[
 fch.ChartAxisLabel(
 value=1,
 label=ft.Text("10K", size=14, weight=ft.FontWeight.BOLD),
),
 fch.ChartAxisLabel(
 value=3,
 label=ft.Text("30K", size=14, weight=ft.FontWeight.BOLD),
),
 fch.ChartAxisLabel(
 value=5,
 label=ft.Text("50K", size=14, weight=ft.FontWeight.BOLD),
),
],
),
 bottom_axis=fch.ChartAxis(
 label_size=32,
 labels=[
 fch.ChartAxisLabel(
 value=2,
 label=ft.Container(
 margin=ft.Margin(top=10),
 content=ft.Text(
 value="MAR",
 size=16,
 weight=ft.FontWeight.BOLD,
 color=ft.Colors.with_opacity(0.5, ft.Colors.ON_SURFACE),
),
),
),
 fch.ChartAxisLabel(
 value=5,
 label=ft.Container(
 margin=ft.Margin(top=10),
 content=ft.Text(
 value="JUN",
 size=16,
 weight=ft.FontWeight.BOLD,
 color=ft.Colors.with_opacity(0.5, ft.Colors.ON_SURFACE),
),
),
),
 fch.ChartAxisLabel(
 value=8,
 label=ft.Container(
 margin=ft.Margin(top=10),
 content=ft.Text(
 value="SEP",
 size=16,
 weight=ft.FontWeight.BOLD,
 color=ft.Colors.with_opacity(0.5, ft.Colors.ON_SURFACE),
),
),
),
],
),
)

```

```

),
)

def toggle_data(e: ft.Event[ft.ElevatedButton]):
 if state.toggled:
 chart.data_series = data_2
 chart.interactive = False
 else:
 chart.data_series = data_1
 chart.interactive = True
 state.toggled = not state.toggled
 chart.update()

page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Button("avg", on_click=toggle_data),
 chart,
]
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/line\\_chart/example\\_2.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/line_chart/example_2.png)

## Properties

### buildanimationclass-attributeinstance-attribute

```

animation: AnimationValue = field(
 default_factory=lambda: Animation(
 duration=Duration(milliseconds=150),
 curve=AnimationCurve.LINEAR,
)
)

```

Controls chart implicit animation.

### buildbaseline\_xclass-attributeinstance-attribute

```
baseline_x: Number | None = None
```

Baseline value for X axis.

### buildbaseline\_yclass-attributeinstance-attribute

```
baseline_y: Number | None = None
```

Baseline value for Y axis.

### buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue | None = None
```

Background color of the chart.

### buildborderclass-attributeinstance-attribute

```
border: Border | None = None
```

The border around the chart.

### buildbottom\_axisclass-attributeinstance-attribute

```
bottom_axis: ChartAxis | None = None
```

Defines the appearance of the bottom axis, its title and labels.

### bulddata\_seriesclass-attributeinstance-attribute

```

data_series: list[LineChartData] = field(
 default_factory=list
)

```

A list of LineChartData controls drawn as separate lines on a chart.

### buildhorizontal\_grid\_linesclass-attributeinstance-attribute

```
horizontal_grid_lines: ChartGridLines | None = None
```

Controls drawing of chart's horizontal lines.

## **buildinteractiveclass-attributeinstance-attribute**

`interactive: bool = True`

Enables automatic tooltips and points highlighting when hovering over the chart.

## **buildleft\_axisclass-attributeinstance-attribute**

`left_axis: ChartAxis | None = None`

Defines the appearance of the left axis, its title and labels.

## **buildmax\_xclass-attributeinstance-attribute**

`max_x: Number | None = None`

Defines the maximum displayed value for X axis.

## **buildmax\_yclass-attributeinstance-attribute**

`max_y: Number | None = None`

Defines the maximum displayed value for Y axis.

## **buildmin\_xclass-attributeinstance-attribute**

`min_x: Number | None = None`

Defines the minimum displayed value for X axis.

## **buildmin\_yclass-attributeinstance-attribute**

`min_y: Number | None = None`

Defines the minimum displayed value for Y axis.

## **buildpoint\_line\_endclass-attributeinstance-attribute**

`point_line_end: Number | None = None`

The end of the vertical line drawn at selected point position.

Defaults to data point's y value.

## **buildpoint\_line\_startclass-attributeinstance-attribute**

`point_line_start: Number | None = None`

The start of the vertical line drawn under the selected point.

Defaults to chart's bottom edge.

## **buildright\_axisclass-attributeinstance-attribute**

`right_axis: ChartAxis | None = None`

Defines the appearance of the right axis, its title and labels.

## **buildtooltipclass-attributeinstance-attribute**

```
tooltip: LineChartTooltip | None = field(
 default_factory=lambda: LineChartTooltip()
)
```

The tooltip configuration for this chart.

If set to None, no tooltips will be shown throughout this chart.

## **buildtop\_axisclass-attributeinstance-attribute**

`top_axis: ChartAxis | None = None`

Defines the appearance of the top axis, its title and labels.

## **buildvertical\_grid\_linesclass-attributeinstance-attribute**

`vertical_grid_lines: ChartGridLines | None = None`

Controls drawing of chart's vertical lines.

# **Events**

## **bolton\_eventclass-attributeinstance-attribute**

`on_event: EventHandler[LineChartEvent] | None = None`



Fires when a chart line is hovered or clicked.

# MatplotlibChart

Displays a Matplotlib chart.

To display a Matplotlib figure with a built-in toolbar UI, use `MatplotlibChartWithToolbar`.

## Warning

This control requires the matplotlib Python package to be installed.

See this [installation guide](#) for more information.

Inherits: `GestureDetector`

## Properties

- `figure` - Matplotlib figure to draw - an instance of `matplotlib.figure.Figure`.

## Events

- `on_message` - The event is triggered on figure message update.
- `on_toolbar_buttons_update` - Triggers when toolbar buttons status is updated.

## Methods

- `back` - Goes back to the previous view.
- `build`
- `download` - Downloads the current figure in the specified format.
- `forward` - Goes forward to the next view.
- `home` - Resets the view to the original state.
- `pan` - Activates the pan tool.
- `send_binary` - Sends a binary message to the front end.
- `send_json` - Sends a JSON message to the front end.
- `send_message` - Sends a message to the figure's canvas manager.
- `will_unmount` - Called when the control is about to be removed from the page.
- `zoom` - Activates the zoom tool.

## Examples

### Bar Chart

Based on an official Matplotlib example.

`play_arrow` Try Online

```
import matplotlib.pyplot as plt

import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 fig, ax = plt.subplots()

 fruits = ["apple", "blueberry", "cherry", "orange"]
 counts = [40, 100, 30, 55]
 bar_labels = ["red", "blue", "red", "orange"]
 bar_colors = ["tab:red", "tab:blue", "tab:red", "tab:orange"]

 ax.bar(fruits, counts, label=bar_labels, color=bar_colors)

 ax.set_ylabel("fruit supply")
 ax.set_title("Fruit supply by kind and color")
 ax.legend(title="Fruit color")

 page.add(
 ft.SafeArea(
```

```

 expand=True,
 content=fch.MatplotlibChart(figure=fig, expand=True),
)
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macos/matplotlib\\_chart/bar\\_chart.png](https://flet.dev/docs/test-images-charts/examples/golden/macos/matplotlib_chart/bar_chart.png)

## Properties

### buildfigureclass-attributeinstance-attribute

```
figure: Figure = field(metadata={'skip': True})
```

Matplotlib figure to draw - an instance of matplotlib.figure.Figure.

## Events

### bolton\_messageclass-attributeinstance-attribute

```

on_message: (
 EventHandler[MatplotlibChartMessageEvent] | None
) = None

```

The event is triggered on figure message update.

### bolton\_toolbar\_buttons\_updateclass-attributeinstance-attribute

```

on_toolbar_buttons_update: (
 EventHandler[
 MatplotlibChartToolbarButtonsUpdateEvent
]
 | None
) = None

```

Triggers when toolbar buttons status is updated.

## Methods

### deployed\_codeback

```
back()
```

Goes back to the previous view.

### deployed\_codebuild

```
build()
```

### deployed\_codedownload

```
download(format) -> bytes
```

Downloads the current figure in the specified format. Args: format (str): The format to download the figure in (e.g., 'png', 'jpg', 'svg', etc.). Returns: bytes: The figure image in the specified format as a byte array.

### deployed\_codeforward

```
forward()
```

Goes forward to the next view.

### deployed\_codehome

```
home()
```

Resets the view to the original state.

### deployed\_codepan

```
pan()
```

Activates the pan tool.

### deployed\_codesend\_binary

```
send_binary(blob)
```

Sends a binary message to the front end.

### **deployed\_codesend\_json**

`send_json(content)`

Sends a JSON message to the front end.

### **deployed\_codesend\_message**

`send_message(message)`

Sends a message to the figure's canvas manager.

### **deployed\_codewill\_unmount**

`will_unmount()`

Called when the control is about to be removed from the page.

### **deployed\_codezoom**

`zoom()`

Activates the zoom tool.

# MatplotlibChartWithToolbar

Composite control that combines a MatplotlibChart with a ready-made toolbar UI.

## Warning

This control requires the matplotlib Python package to be installed.

See this installation guide for more information.

Image: MatplotlibChartWithToolbar - [https://flet.dev/docs/test-images-charts/examples/golden/macros/matplotlib\\_chart/toolbar.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/matplotlib_chart/toolbar.png)

Inherits: Column

## Properties

- figure - Matplotlib figure to draw - an instance of matplotlib.figure.Figure.

## Methods

- build
- download\_click - Export the figure in the selected format and prompt user to save it.
- on\_message - Show status text produced by the underlying chart toolbar backend.
- on\_toolbar\_update - Synchronize back/forward button enabled state with chart history.
- pan\_click - Toggle pan mode and ensure zoom mode is turned off.
- zoom\_click - Toggle zoom mode and ensure pan mode is turned off.

## Examples

### Toolbar

Based on an official Matplotlib example.

play\_arrowTry Online

```
import matplotlib.pyplot as plt
import numpy as np

import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 # Fixing random state for reproducibility
 np.random.seed(19680801)

 dt = 0.01
 t = np.arange(0, 30, dt)
 nse1 = np.random.randn(len(t)) # white noise 1
 nse2 = np.random.randn(len(t)) # white noise 2

 # Two signals with a coherent part at 10Hz and a random part
 s1 = np.sin(2 * np.pi * 10 * t) + nse1
 s2 = np.sin(2 * np.pi * 10 * t) + nse2

 fig, axs = plt.subplots(2, 1)
 axs[0].plot(t, s1, t, s2)
 axs[0].set_xlim(0, 2)
 axs[0].set_xlabel("time")
 axs[0].set_ylabel("s1 and s2")
 axs[0].grid(True)

 cxy, f = axs[1].cohere(s1, s2, 256, 1.0 / dt)
 axs[1].set_ylabel("coherence")

 fig.tight_layout()

 page.add(
 ft.SafeArea(
 expand=True,
 content=fch.MatplotlibChartWithToolbar(figure=fig, expand=True),
)
)

if __name__ == "__main__":
```

```
ft.run(main)
```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/matplotlib\\_chart/toolbar.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/matplotlib_chart/toolbar.png)

## Three D

play\_arrowTry Online

```
import logging

import matplotlib.pyplot as plt

import flet as ft
import flet_charts

logging.basicConfig(level=logging.INFO)

def main(page: ft.Page):
 from mpl_toolkits.mplot3d import axes3d

 fig = plt.figure()
 ax = fig.add_subplot(projection="3d")
 X, Y, Z = axes3d.get_test_data(0.1)

 # Plot the 3D surface
 ax.plot_surface(
 X, Y, Z, edgecolor="royalblue", lw=0.5, rstride=8, cstride=8, alpha=0.3
)

 # Plot projections of the contours for each dimension. By choosing offsets
 # that match the appropriate axes limits, the projected contours will sit on
 # the 'walls' of the graph
 ax.contourf(X, Y, Z, zdir="z", offset=-100, cmap="coolwarm")
 ax.contourf(X, Y, Z, zdir="x", offset=-40, cmap="coolwarm")
 ax.contourf(X, Y, Z, zdir="y", offset=40, cmap="coolwarm")

 ax.set(
 xlim=(-40, 40),
 ylim=(-40, 40),
 zlim=(-100, 100),
 xlabel="X",
 ylabel="Y",
 zlabel="Z",
)

 page.add(
 ft.SafeArea(
 content=flet_charts.MatplotlibChartWithToolbar(figure=fig, expand=True),
 expand=True,
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/matplotlib\\_chart/three\\_d.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/matplotlib_chart/three_d.png)

## Handle Events

play\_arrowTry Online

```
import matplotlib.pyplot as plt
import numpy as np

import flet as ft
import flet_charts as fch

state = {}

def main(page: ft.Page):
 # Fixing random state for reproducibility
 np.random.seed(19680801)

 X = np.random.rand(100, 200)
 xs = np.mean(X, axis=1)
 ys = np.std(X, axis=1)

 fig, (ax, ax2) = plt.subplots(2, 1)
 ax.set_title("click on point to plot time series")
 (line,) = ax.plot(xs, ys, "o", picker=True, pickradius=5)

 class PointBrowser:
 """
 Click on a point to select and highlight it -- the data that
```

generated the point will be shown in the lower Axes. Use the 'n' and 'p' keys to browse through the next and previous points

```

"""
def __init__(self):
 self.lastind = 0

 self.text = ax.text(
 0.05, 0.95, "selected: none", transform=ax.transAxes, va="top"
)
 (self.selected,) = ax.plot(
 [xs[0]], [ys[0]], "o", ms=12, alpha=0.4, color="yellow", visible=False
)

def on_press(self, event):
 if self.lastind is None:
 return
 if event.key not in ("n", "p"):
 return
 inc = 1 if event.key == "n" else -1

 self.lastind += inc
 self.lastind = np.clip(self.lastind, 0, len(xs) - 1)
 self.update()

def on_pick(self, event):
 if event.artist != line:
 return True

 N = len(event.ind)
 if not N:
 return True

 # the click locations
 x = event.mouseevent.xdata
 y = event.mouseevent.ydata

 distances = np.hypot(x - xs[event.ind], y - ys[event.ind])
 indmin = distances.argmin()
 dataind = event.ind[indmin]

 self.lastind = dataind
 self.update()

def update(self):
 if self.lastind is None:
 return

 dataind = self.lastind

 ax2.clear()
 ax2.plot(X[dataind])

 ax2.text(
 0.05,
 0.9,
 f"mu={xs[dataind]:1.3f}\nsigma={ys[dataind]:1.3f}",
 transform=ax2.transAxes,
 va="top",
)
 ax2.set_ylim(-0.5, 1.5)
 self.selected.set_visible(True)
 self.selected.set_data([xs[dataind]], [ys[dataind]])

 self.text.set_text(f"selected: {dataind:d}")
 fig.canvas.draw()

browser = PointBrowser()
state["browser"] = browser

fig.canvas.mpl_connect("pick_event", browser.on_pick)
fig.canvas.mpl_connect("key_press_event", browser.on_press)

plt.show()

page.add(
 ft.SafeArea(
 expand=True,
 content=fch.MatplotlibChartWithToolbar(figure=fig, expand=True),
)
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/matplotlib\\_chart/handle\\_events.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/matplotlib_chart/handle_events.png)

## Animate

### play\_arrowTry Online

```
import logging

import matplotlib.pyplot as plt
import numpy as np

import flet as ft
import flet_charts

logging.basicConfig(level=logging.INFO)

state = {}

def main(page: ft.Page):
 import matplotlib.animation as animation

 # Fixing random state for reproducibility
 np.random.seed(19680801)

 def random_walk(num_steps, max_step=0.05):
 """Return a 3D random walk as (num_steps, 3) array."""
 start_pos = np.random.random(3)
 steps = np.random.uniform(-max_step, max_step, size=(num_steps, 3))
 walk = start_pos + np.cumsum(steps, axis=0)
 return walk

 def update_lines(num, walks, lines):
 for line, walk in zip(lines, walks):
 line.set_data_3d(walk[:num, :].T)
 return lines

 # Data: 40 random walks as (num_steps, 3) arrays
 num_steps = 30
 walks = [random_walk(num_steps) for index in range(40)]

 # Attaching 3D axis to the figure
 fig = plt.figure()
 ax = fig.add_subplot(projection="3d")

 # Create lines initially without data
 lines = [ax.plot([], [], [], [0])[0] for _ in walks]

 # Setting the Axes properties
 ax.set(xlim3d=(0, 1), xlabel="X")
 ax.set(ylim3d=(0, 1), ylabel="Y")
 ax.set(zlim3d=(0, 1), zlabel="Z")

 # Creating the Animation object
 state["anim"] = animation.FuncAnimation(
 fig, update_lines, num_steps, fargs=(walks, lines), interval=100
)

 page.add(
 ft.SafeArea(
 expand=True,
 content=flet_charts.MatplotlibChartWithToolbar(figure=fig, expand=True),
)
)

if __name__ == "__main__":
 ft.run(main)
```

Image: Image - [https://flet.dev/docs/examples/extensions/charts/matplotlib\\_chart/media/animate.png](https://flet.dev/docs/examples/extensions/charts/matplotlib_chart/media/animate.png)

## Properties

### buildfigureclass-attributeinstance-attribute

figure: Figure = field(metadata={'skip': True})

Matplotlib figure to draw - an instance of matplotlib.figure.Figure.

## Methods

### deployed\_codebuild

build()

### deployed\_codedownload\_clickasync



```
download_click()
```

Export the figure in the selected format and prompt user to save it.

### **deployed\_codeon\_message**

```
on_message(e: flet_charts.MatplotlibChartMessageEvent)
```

Show status text produced by the underlying chart toolbar backend.

Parameters:

- e (flet\_charts.MatplotlibChartMessageEvent) - Message event emitted by MatplotlibChart.

### **deployed\_codeon\_toolbar\_update**

```
on_toolbar_update(
 e: flet_charts.MatplotlibChartToolbarButtonsUpdateEvent,
)
```

Synchronize back/forward button enabled state with chart history.

Parameters:

- e (flet\_charts.MatplotlibChartToolbarButtonsUpdateEvent) - Toolbar state update event from MatplotlibChart.

### **deployed\_codepan\_click**

```
pan_click()
```

Toggle pan mode and ensure zoom mode is turned off.

### **deployed\_codezoom\_click**

```
zoom_click()
```

Toggle zoom mode and ensure pan mode is turned off.

# PieChart

A pie chart control displaying multiple sections as slices of a circle.

Image: PieChart - <https://flet.dev/docs/assets/controls/charts/pie-chart-diagram.svg>

Inherits: `LayoutControl`

## Properties

- `animation` - Controls chart implicit animation.
- `center_space_color` - Free space color in the middle of a chart.
- `center_space_radius` - Free space radius in the middle of a chart.
- `sections` - A list of `PieChartSection` controls drawn in a circle.
- `sections_space` - A gap between sections.
- `start_degree_offset` - By default, sections are drawn from zero degree (right side of the circle) clockwise.

## Events

- `on_event` - Fires when a chart section is hovered or clicked.

## Examples

### Pie chart with hover borders

`play_arrow` Try Online

```
import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 normal_border = ft.BorderSide(0, ft.Colors.with_opacity(0, ft.Colors.WHITE))
 hovered_border = ft.BorderSide(6, ft.Colors.SECONDARY)

 def on_chart_event(e: fch.PieChartEvent):
 for idx, section in enumerate(chart.sections):
 section.border_side = (
 hovered_border if idx == e.section_index else normal_border
)
 chart.update()

 chart = fch.PieChart(
 sections_space=1,
 center_space_radius=0,
 on_event=on_chart_event,
 expand=True,
 sections=[
 fch.PieChartSection(
 value=25,
 color=ft.Colors.BLUE,
 radius=80,
 border_side=normal_border,
),
 fch.PieChartSection(
 value=25,
 color=ft.Colors.YELLOW,
 radius=65,
 border_side=normal_border,
),
 fch.PieChartSection(
 value=25,
 color=ft.Colors.PINK,
 radius=60,
 border_side=normal_border,
),
 fch.PieChartSection(
 value=25,
 color=ft.Colors.GREEN,
 radius=70,
 border_side=normal_border,
),
],
)
```

```
page.add(ft.SafeArea(content=chart))
```

```
if __name__ == "__main__":
 ft.run(main)
```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/pie\\_chart/example\\_1.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/pie_chart/example_1.png)

## Pie chart with hover sections

play\_arrowTry Online

```
import flet as ft
import flet_charts as fch

NORMAL_RADIUS = 50
HOVER_RADIUS = 60
NORMAL_TITLE_STYLE = ft.TextStyle(
 size=16, color=ft.Colors.WHITE, weight=ft.FontWeight.BOLD
)
HOVER_TITLE_STYLE = ft.TextStyle(
 size=22,
 color=ft.Colors.WHITE,
 weight=ft.FontWeight.BOLD,
 shadow=ft.BoxShadow(blur_radius=2, color=ft.Colors.BLACK_54),
)

def main(page: ft.Page):
 def on_chart_event(e: fch.PieChartEvent):
 for idx, section in enumerate(chart.sections):
 if idx == e.section_index:
 section.radius = HOVER_RADIUS
 section.title_style = HOVER_TITLE_STYLE
 else:
 section.radius = NORMAL_RADIUS
 section.title_style = NORMAL_TITLE_STYLE
 chart.update()

 chart = fch.PieChart(
 expand=True,
 sections_space=0,
 center_space_radius=40,
 on_event=on_chart_event,
 sections=[
 fch.PieChartSection(
 value=40,
 title="40%",
 title_style=NORMAL_TITLE_STYLE,
 color=ft.Colors.BLUE,
 radius=NORMAL_RADIUS,
),
 fch.PieChartSection(
 value=30,
 title="30%",
 title_style=NORMAL_TITLE_STYLE,
 color=ft.Colors.YELLOW,
 radius=NORMAL_RADIUS,
),
 fch.PieChartSection(
 value=15,
 title="15%",
 title_style=NORMAL_TITLE_STYLE,
 color=ft.Colors.PURPLE,
 radius=NORMAL_RADIUS,
),
 fch.PieChartSection(
 value=15,
 title="15%",
 title_style=NORMAL_TITLE_STYLE,
 color=ft.Colors.GREEN,
 radius=NORMAL_RADIUS,
),
],
)

 page.add(ft.SafeArea(content=chart))

if __name__ == "__main__":
 ft.run(main)
```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/pie\\_chart/example\\_2.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/pie_chart/example_2.png)

## Pie chart with icon badges

play\_arrowTry Online

```

import flet as ft
import flet_charts as fch

NORMAL_RADIUS = 100
HOVER_RADIUS = 110
NORMAL_TITLE_STYLE = ft.TextStyle(
 size=12, color=ft.Colors.WHITE, weight=ft.FontWeight.BOLD
)
HOVER_TITLE_STYLE = ft.TextStyle(
 size=16,
 color=ft.Colors.WHITE,
 weight=ft.FontWeight.BOLD,
 shadow=ft.BoxShadow(blur_radius=2, color=ft.Colors.BLACK_54),
)
NORMAL_BADGE_SIZE = 40
HOVER_BADGE_SIZE = 50

class SectionBadge(ft.Container):
 def __init__(self, icon: ft.IconData, size: int = NORMAL_BADGE_SIZE):
 super().__init__(
 content=ft.Icon(icon),
 width=size,
 height=size,
 border=ft.Border.all(1, ft.Colors.BROWN),
 border_radius=size / 2,
 bgcolor=ft.Colors.WHITE,
)

def main(page: ft.Page):
 def on_chart_event(e: fch.PieChartEvent):
 for idx, section in enumerate(chart.sections):
 if idx == e.section_index:
 section.radius = HOVER_RADIUS
 section.title_style = HOVER_TITLE_STYLE
 else:
 section.radius = NORMAL_RADIUS
 section.title_style = NORMAL_TITLE_STYLE
 chart.update()

 chart = fch.PieChart(
 sections_space=0,
 center_space_radius=0,
 on_event=on_chart_event,
 expand=True,
 sections=[
 fch.PieChartSection(
 value=40,
 title="40%",
 title_style=NORMAL_TITLE_STYLE,
 color=ft.Colors.BLUE,
 radius=NORMAL_RADIUS,
 badge=SectionBadge(ft.Icons.AC_UNIT),
 badge_position=0.98,
),
 fch.PieChartSection(
 value=30,
 title="30%",
 title_style=NORMAL_TITLE_STYLE,
 color=ft.Colors.YELLOW,
 radius=NORMAL_RADIUS,
 badge=SectionBadge(ft.Icons.ACCESS_ALARM),
 badge_position=0.98,
),
 fch.PieChartSection(
 value=15,
 title="15%",
 title_style=NORMAL_TITLE_STYLE,
 color=ft.Colors.PURPLE,
 radius=NORMAL_RADIUS,
 badge=SectionBadge(ft.Icons.APPLE),
 badge_position=0.98,
),
 fch.PieChartSection(
 value=15,
 title="15%",
 title_style=NORMAL_TITLE_STYLE,
 color=ft.Colors.GREEN,
 radius=NORMAL_RADIUS,
 badge=SectionBadge(ft.Icons.PEDAL_BIKE),
 badge_position=0.98,
),
],
)

 page.add(ft.SafeArea(content=chart))

```

```
if __name__ == "__main__":
 ft.run(main)
```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/pie\\_chart/example\\_3.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/pie_chart/example_3.png)

## Properties

### buildanimationclass-attributeinstance-attribute

```
animation: AnimationValue = field(
 default_factory=lambda: Animation(
 duration=Duration(milliseconds=150),
 curve=AnimationCurve.LINEAR,
)
)
```

Controls chart implicit animation.

### buildcenter\_space\_colorclass-attributeinstance-attribute

```
center_space_color: ColorValue | None = None
```

Free space color in the middle of a chart.

### buildcenter\_space\_radiusclass-attributeinstance-attribute

```
center_space_radius: Number | None = None
```

Free space radius in the middle of a chart.

### buildsectionsclass-attributeinstance-attribute

```
sections: list[PieChartSection] = field(
 default_factory=list
)
```

A list of PieChartSection controls drawn in a circle.

### buildsections\_spaceclass-attributeinstance-attribute

```
sections_space: Number | None = None
```

A gap between sections.

### buildstart\_degree\_offsetclass-attributeinstance-attribute

```
start_degree_offset: Number | None = None
```

By default, sections are drawn from zero degree (right side of the circle) clockwise. You can change the starting point by setting start\_degree\_offset (in degrees).

## Events

### bolton\_eventclass-attributeinstance-attribute

```
on_event: EventHandler[PieChartEvent] | None = None
```

Fires when a chart section is hovered or clicked.

# PlotlyChart

Displays a Plotly chart.

## Warning

This control requires the plotly and kaleido Python packages to be installed.

See this [installation guide](#) for more information.

```
import plotly.express as px

data_canada = px.data.gapminder().query("country == 'Canada'")
fig = px.bar(data_canada, x="year", y="pop")
fch.PlotlyChart(figure=fig, expand=True)
```

Image: PlotlyChart - [https://flet.dev/docs/test-images-charts/examples/golden/macros/plotly\\_chart/image\\_for\\_docs.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/plotly_chart/image_for_docs.png)

Bar chart based on official [Plotly example](<https://plotly.com/python/bar-charts>)

Inherits: Container

## Properties

- figure - Plotly figure to draw.
- original\_size - Whether to display this chart in original size.

## Examples

### Plotly line chart

Based on an official Plotly example.

```
import plotly.express as px

import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 df = px.data.gapminder().query("continent=='Oceania'")
 fig = px.line(df, x="year", y="lifeExp", color="country")

 page.add(ft.SafeArea(content=fch.PlotlyChart(figure=fig, expand=True)))

if __name__ == "__main__":
 ft.run(main)
```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/plotly\\_chart/example\\_1.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/plotly_chart/example_1.png)

### Plotly bar chart

Based on an official Plotly example.

```
import plotly.express as px

import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 df = px.data.gapminder().query("continent == 'Oceania'")
 fig = px.bar(
 df,
 x="year",
 y="pop",
 hover_data=["lifeExp", "gdpPercap"],
 color="country",
 labels={"pop": "population of Canada"},
 height=400,
)

 page.add(ft.SafeArea(content=fch.PlotlyChart(figure=fig, expand=True)))

if __name__ == "__main__":
 ft.run(main)
```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/plotly\\_chart/example\\_2.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/plotly_chart/example_2.png)

## Plotly pie chart

Based on an official Plotly example.

```
import plotly.graph_objects as go

import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 labels = ["Oxygen", "Hydrogen", "Carbon_Dioxide", "Nitrogen"]
 values = [4500, 2500, 1053, 500]

 fig = go.Figure(data=[go.Pie(labels=labels, values=values)])

 page.add(ft.SafeArea(content=fch.PlotlyChart(figure=fig, expand=True)))

if __name__ == "__main__":
 ft.run(main)
```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/plotly\\_chart/example\\_3.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/plotly_chart/example_3.png)

## Plotly box plot

Based on an official Plotly example.

```
import plotly.graph_objects as go

import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 x = [
 "day 1",
 "day 1",
 "day 1",
 "day 1",
 "day 1",
 "day 1",
 "day 1",
 "day 2",
 "day 2",
 "day 2",
 "day 2",
 "day 2",
 "day 2",
]

 fig = go.Figure()

 fig.add_trace(
 go.Box(
 y=[0.2, 0.2, 0.6, 1.0, 0.5, 0.4, 0.2, 0.7, 0.9, 0.1, 0.5, 0.3],
 x=x,
 name="kale",
 marker_color="#3D9970",
)
)
 fig.add_trace(
 go.Box(
 y=[0.6, 0.7, 0.3, 0.6, 0.0, 0.5, 0.7, 0.9, 0.5, 0.8, 0.7, 0.2],
 x=x,
 name="radishes",
 marker_color="#FF4136",
)
)
 fig.add_trace(
 go.Box(
 y=[0.1, 0.3, 0.1, 0.9, 0.6, 0.6, 0.9, 1.0, 0.3, 0.6, 0.8, 0.5],
 x=x,
 name="carrots",
 marker_color="#FF851B",
)
)

 fig.update_layout(
 yaxis_title="normalized moisture",
 boxmode="group", # group together boxes of the different traces
)

 page.add(ft.SafeArea(content=fch.PlotlyChart(figure=fig, expand=True)))

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### buildfigureclass-attributeinstance-attribute

`figure: Figure = field(metadata={'skip': True})`

Plotly figure to draw.

The value is an instance of `plotly.graph_objects.Figure`.

### buildoriginal\_sizeclass-attributeinstance-attribute

`original_size: bool = False`

Whether to display this chart in original size.

Set to `False` for it to fit it's configured bounds.



# RadarChart

A radar chart made of multiple datasets.

```
fch.RadarChart(
 titles=[fch.RadarChartTitle(text="winter"), ...],
 radar_shape=fch.RadarShape.CIRCLE,
 data_sets=[
 fch.RadarDataSet(
 fill_color=ft.Colors.with_opacity(0.2, ft.Colors.BLUE_GREY_700),
 entries=[fch.RadarDataSetEntry(130), ...],
),
 ...,
],
)
```

Image: RadarChart - [https://flet.dev/docs/test-images-charts/examples/golden/macos/radar\\_chart/image\\_for\\_docs.png](https://flet.dev/docs/test-images-charts/examples/golden/macos/radar_chart/image_for_docs.png)

Radar chart

Inherits: LayoutControl

## Properties

- **animation** - Controls the implicit animation applied when updating this chart.
- **border** - The border drawn around this chart.
- **center\_min\_value** - Whether minimum entry values should be positioned at the center of this chart.
- **data\_sets** - A list of RadarDataSet controls rendered on the chart.
- **grid\_border\_side** - The style of the radar grid lines.
- **interactive** - Enables touch interactions and event notifications.
- **long\_press\_duration** - The duration before a long-press event fires.
- **radar\_bgcolor** - The background color of the radar area.
- **radar\_border\_side** - The outline drawn around the radar area.
- **radar\_shape** - The shape of the radar area.
- **tick\_border\_side** - The style of the tick rings.
- **tick\_count** - Number of tick rings drawn from the centre to the edge.
- **ticks\_text\_style** - The text style used to draw tick labels.
- **title\_position\_percentage\_offset** - Defines the relative distance of titles from the chart center.
- **title\_text\_style** - The text style applied to titles around this chart.
- **titles** - The titles shown around this chart, matching the number of entries per set.
- **touch\_spot\_threshold** - The radius (in logical pixels) used to detect nearby entries for touches.

## Events

- **on\_event** - Called when the chart is interacted with.

## Examples

### Radar chart

play\_arrowTry Online

```
import flet as ft
import flet_charts as fch

def main(page: ft.Page):
 page.title = "Radar chart"
 page.padding = 20
 # page.vertical_alignment = page.horizontal_alignment = "center"
 page.theme_mode = ft.ThemeMode.LIGHT

 categories = ["macOS", "Linux", "Windows"]
```

```

page.add(
 ft.SafeArea(
 content=fch.RadarChart(
 expand=True,
 titles=[fch.RadarChartTitle(text=label) for label in categories],
 center_min_value=True,
 tick_count=4,
 ticks_text_style=ft.TextStyle(size=20, color=ft.Colors.ON_SURFACE),
 title_text_style=ft.TextStyle(
 size=24, weight=ft.FontWeight.BOLD, color=ft.Colors.ON_SURFACE
),
 on_event=lambda e: print(e.type),
 data_sets=[
 fch.RadarDataSet(
 fill_color=ft.Colors.with_opacity(0.2, ft.Colors.DEEP_PURPLE),
 border_color=ft.Colors.DEEP_PURPLE,
 entry_radius=4,
 entries=[
 fch.RadarDataSetEntry(300),
 fch.RadarDataSetEntry(50),
 fch.RadarDataSetEntry(250),
],
),
 fch.RadarDataSet(
 fill_color=ft.Colors.with_opacity(0.15, ft.Colors.PINK),
 border_color=ft.Colors.PINK,
 entry_radius=4,
 entries=[
 fch.RadarDataSetEntry(250),
 fch.RadarDataSetEntry(100),
 fch.RadarDataSetEntry(200),
],
),
 fch.RadarDataSet(
 fill_color=ft.Colors.with_opacity(0.12, ft.Colors.CYAN),
 border_color=ft.Colors.CYAN,
 entry_radius=4,
 entries=[
 fch.RadarDataSetEntry(200),
 fch.RadarDataSetEntry(150),
 fch.RadarDataSetEntry(50),
],
),
],
),
),
)

```

```

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macros/radar\\_chart/example\\_1.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/radar_chart/example_1.png)

## Properties

### buildanimationclass-attributeinstance-attribute

```

animation: AnimationValue = field(
 default_factory=lambda: Animation(
 duration=Duration(milliseconds=150),
 curve=AnimationCurve.LINEAR,
)
)

```

Controls the implicit animation applied when updating this chart.

### buildborderclass-attributeinstance-attribute

```
border: Border | None = None
```

The border drawn around this chart.

### buildcenter\_min\_valueclass-attributeinstance-attribute

```
center_min_value: bool = False
```

Whether minimum entry values should be positioned at the center of this chart.

### bulldata\_setsclass-attributeinstance-attribute

```
data_sets: list[RadarDataSet] = field(default_factory=list)
```

A list of RadarDataSet controls rendered on the chart.

### buildgrid\_border\_sideclass-attributeinstance-attribute

```
grid_border_side: BorderSide = field(
 default_factory=lambda: BorderSide(width=2.0)
)
```

The style of the radar grid lines.

### buildinteractiveclass-attributeinstance-attribute

```
interactive: bool = True
```

Enables touch interactions and event notifications.

### buildlong\_press\_durationclass-attributeinstance-attribute

```
long_press_duration: DurationValue | None = None
```

The duration before a long-press event fires.

### buildradar\_bgcolorclass-attributeinstance-attribute

```
radar_bgcolor: ColorValue = Colors.TRANSPARENT
```

The background color of the radar area.

### buildradar\_border\_sideclass-attributeinstance-attribute

```
radar_border_side: BorderSide = field(
 default_factory=lambda: BorderSide(width=2.0)
)
```

The outline drawn around the radar area.

### buildradar\_shapeclass-attributeinstance-attribute

```
radar_shape: RadarShape = RadarShape.POLYGON
```

The shape of the radar area.

### buildtick\_border\_sideclass-attributeinstance-attribute

```
tick_border_side: BorderSide = field(
 default_factory=lambda: BorderSide(width=2.0)
)
```

The style of the tick rings.

### buildtick\_countclass-attributeinstance-attribute

```
tick_count: Number = 1
```

Number of tick rings drawn from the centre to the edge.

Must be greater than or equal to 1.

Raises:

- ValueError - If set to a value less than 1.

### buildticks\_text\_styleclass-attributeinstance-attribute

```
ticks_text_style: TextStyle | None = None
```

The text style used to draw tick labels.

### buildtitle\_position\_percentage\_offsetclass-attributeinstance-attribute

```
title_position_percentage_offset: Number = 0.2
```

Defines the relative distance of titles from the chart center.

- 0 draws titles near the inside edge of each section.
- 1 draws titles near the outside edge of each section.

Must be between 0 and 1 (inclusive).

Raises:

- ValueError - If set to a value less than 0 or greater than 1.

### buildtitle\_text\_styleclass-attributeinstance-attribute

```
title_text_style: TextStyle | None = None
```

The text style applied to titles around this chart.

### **buildtitlesclass-attributeinstance-attribute**

```
titles: list[RadarChartTitle] = field(default_factory=list)
```

The titles shown around this chart, matching the number of entries per set.

### **buildtouch\_spot\_thresholdclass-attributeinstance-attribute**

```
touch_spot_threshold: Number = 10
```

The radius (in logical pixels) used to detect nearby entries for touches.

## **Events**

### **bolton\_eventclass-attributeinstance-attribute**

```
on_event: EventHandler[RadarChartEvent] | None = None
```

Called when the chart is interacted with.

# ScatterChart

A scatter chart control.

ScatterChart draws some points in a square space, points are defined by ScatterChartSpots.

```
fch.ScatterChart(
 aspect_ratio=1.0,
 spots=[
 fch.ScatterChartSpot(
 x=random.uniform(4, 50),
 y=random.uniform(4, 50),
)
 for _ in range(30)
],
)
```

Image: ScatterChart - [https://flet.dev/docs/test-images-charts/examples/golden/macros/scatter\\_chart/image\\_for\\_docs.png](https://flet.dev/docs/test-images-charts/examples/golden/macros/scatter_chart/image_for_docs.png)

Scatter chart

Inherits: LayoutControl

## Properties

- animation - Controls chart implicit animation.
- baseline\_x - The baseline value for X axis.
- baseline\_y - Baseline value for Y axis.
- bgcolor - The chart's background color.
- border - The border around the chart.
- bottom\_axis - Configures the appearance of the bottom axis, its title and labels.
- horizontal\_grid\_lines - Controls drawing of chart's horizontal lines.
- interactive - Enables automatic tooltips when hovering chart bars.
- left\_axis - Configures the appearance of the left axis, its title and labels.
- long\_press\_duration - The duration of a long press on the chart.
- max\_x - The maximum displayed value for X axis.
- max\_y - The maximum displayed value for Y axis.
- min\_x - The minimum displayed value for X axis.
- min\_y - The minimum displayed value for Y axis.
- right\_axis - Configures the appearance of the right axis, its title and labels.
- rotation\_quarter\_turns - Number of quarter turns (90-degree increments) to rotate the chart.
- show\_tooltips\_for\_selected\_spots\_only - Whether to permanently and only show the tooltips of spots with their selected property set to True.
- spots - List of ScatterChartSpots to show on the chart.
- tooltip - The tooltip configuration for the chart.
- top\_axis - Configures the appearance of the top axis, its title and labels.
- vertical\_grid\_lines - Controls drawing of chart's vertical lines.

## Events

- on\_event - Called when an event occurs on this chart.

## Examples

### Scatter chart

[play\\_arrow](#)Try Online

```

import random

import flet as ft
import flet_charts as fch

class MySpot(fch.ScatterChartSpot):
 def __init__(
 self,
 x: float,
 y: float,
 radius: float = 8.0,
 color: ft.Colors = None,
 show_tooltip: bool = False,
):
 super().__init__(
 x=x,
 y=y,
 radius=radius,
 color=color,
 show_tooltip=show_tooltip,
 selected=y == 43,
)

flutter_logo_spots = [
 MySpot(20, 14.5),
 MySpot(20, 14.5),
 MySpot(22, 16.5),
 MySpot(24, 18.5),
 MySpot(22, 12.5),
 MySpot(24, 14.5),
 MySpot(26, 16.5),
 MySpot(24, 10.5),
 MySpot(26, 12.5),
 MySpot(28, 14.5),
 MySpot(26, 8.5),
 MySpot(28, 10.5),
 MySpot(30, 12.5),
 MySpot(28, 6.5),
 MySpot(30, 8.5),
 MySpot(32, 10.5),
 MySpot(30, 4.5),
 MySpot(32, 6.5),
 MySpot(34, 8.5),
 MySpot(34, 4.5),
 MySpot(36, 6.5),
 MySpot(38, 4.5),
 # section 2
 MySpot(20, 14.5),
 MySpot(22, 12.5),
 MySpot(24, 10.5),
 MySpot(22, 16.5),
 MySpot(24, 14.5),
 MySpot(26, 12.5),
 MySpot(24, 18.5),
 MySpot(26, 16.5),
 MySpot(28, 14.5),
 MySpot(26, 20.5),
 MySpot(28, 18.5),
 MySpot(30, 16.5),
 MySpot(28, 22.5),
 MySpot(30, 20.5),
 MySpot(32, 18.5),
 MySpot(30, 24.5),
 MySpot(32, 22.5),
 MySpot(34, 20.5),
 MySpot(34, 24.5),
 MySpot(36, 22.5),
 MySpot(38, 24.5),
 # section 3
 MySpot(10, 25),
 MySpot(12, 23),
 MySpot(14, 21),
 MySpot(12, 27),
 MySpot(14, 25),
 MySpot(16, 23),
 MySpot(14, 29),
 MySpot(16, 27),
 MySpot(18, 25),
 MySpot(16, 31),
 MySpot(18, 29),
 MySpot(20, 27),
 MySpot(18, 33),
 MySpot(20, 31),
 MySpot(22, 29),
 MySpot(20, 35),

```

```

MySpot(22, 33),
MySpot(24, 31),
MySpot(22, 37),
MySpot(24, 35),
MySpot(26, 33),
MySpot(24, 39),
MySpot(26, 37),
MySpot(28, 35),
MySpot(26, 41),
MySpot(28, 39),
MySpot(30, 37),
MySpot(28, 43),
MySpot(30, 41),
MySpot(32, 39),
MySpot(30, 45),
MySpot(32, 43),
MySpot(34, 41),
MySpot(34, 45),
MySpot(36, 43),
MySpot(38, 45),
]

def get_random_spots():
 """Generates random spots for the scatter chart."""
 return [
 MySpot(
 x=random.uniform(4, 50),
 y=random.uniform(4, 50),
 radius=random.uniform(4, 20),
)
 for _ in range(len(flutter_logo_spots))
]

def main(page: ft.Page):
 page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

 def handle_event(e: fch.ScatterChartEvent):
 if e.type == fch.ChartEventType.TAP_DOWN:
 e.control.spots = (
 flutter_logo_spots
 if (e.control.spots != flutter_logo_spots)
 else get_random_spots()
)

 page.add(
 ft.SafeArea(
 content=ft.Column(
 controls=[
 ft.Text(
 "Tap on the chart to toggle between random spots and Flutter "
 "logo spots."
),
 fch.ScatterChart(
 expand=True,
 aspect_ratio=1.0,
 min_x=0.0,
 max_x=50.0,
 min_y=0.0,
 max_y=50.0,
 left_axis=fch.ChartAxis(show_labels=False),
 right_axis=fch.ChartAxis(show_labels=False),
 top_axis=fch.ChartAxis(show_labels=False),
 bottom_axis=fch.ChartAxis(show_labels=False),
 show_tooltips_for_selected_spots_only=False,
 on_event=handle_event,
 animation=ft.Animation(
 duration=ft.Duration(milliseconds=600),
 curve=ft.AnimationCurve.FAST_OUT_SLOWIN,
),
 spots=flutter_logo_spots,
),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)

```

Image: Image - [https://flet.dev/docs/test-images-charts/examples/golden/macOS/scatter\\_chart/example\\_1.png](https://flet.dev/docs/test-images-charts/examples/golden/macOS/scatter_chart/example_1.png)

## Properties

## buildanimationclass-attributeinstance-attribute

```
animation: AnimationValue = field(
 default_factory=lambda: Animation(
 duration=Duration(milliseconds=150),
 curve=AnimationCurve.LINEAR,
)
)
```

Controls chart implicit animation.

## buildbaseline\_xclass-attributeinstance-attribute

```
baseline_x: Number | None = None
```

The baseline value for X axis.

## buildbaseline\_yclass-attributeinstance-attribute

```
baseline_y: Number | None = None
```

Baseline value for Y axis.

## buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue | None = None
```

The chart's background color.

## buildborderclass-attributeinstance-attribute

```
border: Border | None = None
```

The border around the chart.

## buildbottom\_axisclass-attributeinstance-attribute

```
bottom_axis: ChartAxis | None = None
```

Configures the appearance of the bottom axis, its title and labels.

## buildhorizontal\_grid\_linesclass-attributeinstance-attribute

```
horizontal_grid_lines: ChartGridLines | None = None
```

Controls drawing of chart's horizontal lines.

## buildinteractiveclass-attributeinstance-attribute

```
interactive: bool = True
```

Enables automatic tooltips when hovering chart bars.

## buildleft\_axisclass-attributeinstance-attribute

```
left_axis: ChartAxis | None = None
```

Configures the appearance of the left axis, its title and labels.

## buildlong\_press\_durationclass-attributeinstance-attribute

```
long_press_duration: DurationValue | None = None
```

The duration of a long press on the chart.

## buildmax\_xclass-attributeinstance-attribute

```
max_x: Number | None = None
```

The maximum displayed value for X axis.

## buildmax\_yclass-attributeinstance-attribute

```
max_y: Number | None = None
```

The maximum displayed value for Y axis.

## buildmin\_xclass-attributeinstance-attribute

```
min_x: Number | None = None
```

The minimum displayed value for X axis.

## buildmin\_yclass-attributeinstance-attribute

```
min_y: Number | None = None
```



The minimum displayed value for Y axis.

### **buildright\_axisclass-attributeinstance-attribute**

```
right_axis: ChartAxis | None = None
```

Configures the appearance of the right axis, its title and labels.

### **buildrotation\_quarter\_turnsclass-attributeinstance-attribute**

```
rotation_quarter_turns: Number = 0
```

Number of quarter turns (90-degree increments) to rotate the chart. Ex: 1 rotates the chart 90 degrees clockwise, 2 rotates 180 degrees and 0 for no rotation.

### **buildshow\_tooltips\_for\_selected\_spots\_onlyclass-attributeinstance-attribute**

```
show_tooltips_for_selected_spots_only: bool = False
```

Whether to permanently and only show the tooltips of spots with their selected property set to True.

### **buildspotsclass-attributeinstance-attribute**

```
spots: list[ScatterChartSpot] = field(default_factory=list)
```

List of ScatterChartSpots to show on the chart.

### **buildtooltipclass-attributeinstance-attribute**

```
tooltip: ScatterChartTooltip = field(
 default_factory=lambda: ScatterChartTooltip()
)
```

The tooltip configuration for the chart.

### **buildtop\_axisclass-attributeinstance-attribute**

```
top_axis: ChartAxis | None = None
```

Configures the appearance of the top axis, its title and labels.

### **buildvertical\_grid\_linesclass-attributeinstance-attribute**

```
vertical_grid_lines: ChartGridLines | None = None
```

Controls drawing of chart's vertical lines.

## **Events**

### **bolton\_eventclass-attributeinstance-attribute**

```
on_event: EventHandler[ScatterChartEvent] | None = None
```

Called when an event occurs on this chart.

# BlockPicker

A color picker that lets users choose a color from a grid of available colors.

Image: BlockPicker - [https://flet.dev/docs/test-images/examples/extensions/color\\_pickers/golden/macros/color\\_pickers/block\\_picker.png](https://flet.dev/docs/test-images/examples/extensions/color_pickers/golden/macros/color_pickers/block_picker.png)

BlockPicker

Inherits: LayoutControl

## Properties

- `available_colors` - A list of available colors to pick from.
- `color` - The currently selected color.

## Events

- `on_color_change` - Called when the picker color is changed.

## Example

### BlockPicker

`play_arrow`Try Online

```
import flet as ft
from flet_color_pickers import BlockPicker

def main(page: ft.Page):
 page.title = "BlockPicker"
 page.padding = 20

 def on_color_change(e: ft.ControlEvent):
 print(f"color: {e.data}")

 dialog_picker = BlockPicker(
 color="#9c27b0",
 available_colors=[
 "#f44336",
 "#e91e63",
 "#9c27b0",
 "#3f51b5",
 "#2196f3",
 "#009688",
 "#4caf50",
 "#795548",
],
 on_color_change=on_color_change,
)

 dialog = ft.AlertDialog(
 modal=True,
 title=ft.Text("Pick a color"),
 content=dialog_picker,
 actions=[
 ft.TextButton("Close", on_click=lambda e: page.pop_dialog()),
],
)

 page.add(
 ft.SafeArea(
 content=ft.IconButton(
 icon=ft.Icons.BRUSH,
 on_click=lambda e: page.show_dialog(dialog),
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

`buildavailable_colors`class-attributeinstance-attribute

`available_colors`: list[ColorValue] | None = None

A list of available colors to pick from.

## buildcolorclass-attributeinstance-attribute

```
color: ColorValue | None = None
```

The currently selected color.

## Events

### bolton\_color\_changeclass-attributeinstance-attribute

```
on_color_change: (
 ControlEventHandler[BlockPicker] | None
) = None
```

Called when the picker color is changed.

The data property of the event handler argument contains the color value as a hex string.

# ColorPicker

A full-featured color picker with palette, sliders, labels, and optional history.

Image: ColorPicker - [https://flet.dev/docs/test-images/examples/extensions/color\\_pickers/golden/macos/color\\_pickers/image\\_for\\_docs.png](https://flet.dev/docs/test-images/examples/extensions/color_pickers/golden/macos/color_pickers/image_for_docs.png)

ColorPicker

Inherits: LayoutControl

## Properties

- `color` - The currently selected color.
- `color_history` - A list of colors to display as a history palette.
- `color_picker_width` - Width of the color picker in virtual pixels.
- `display_thumb_color` - Whether to display the thumb color in slider.
- `enable_alpha` - Whether to enable alpha (opacity) slider.
- `hex_input_bar` - Whether to show the hex input bar.
- `hsv_color` - The currently selected HSV color.
- `label_text_style` - Text style for labels.
- `label_types` - Color label types to display.
- `palette_type` - Palette type for the picker area.
- `picker_area_border_radius` - Border radius for the picker area.
- `picker_area_height_percent` - Height of the picker area as a percentage of the picker width.

## Events

- `on_color_change` - Called when the picker color is changed.
- `on_history_change` - Called when the history palette is changed.
- `on_hsv_color_change` - Called when the picker HSV color is changed.

## Example

### ColorPicker

play\_arrowTry Online

```
import flet as ft
from flet_color_pickers import ColorLabelType, ColorPicker, PaletteType

def main(page: ft.Page):
 page.title = "ColorPicker"
 page.padding = 20

 def on_color_change(e: ft.ControlEvent):
 print(f"color: {e.data}")

 def on_history_change(e: ft.ControlEvent):
 # e.data is a list of hex strings
 print(f"history: {e.data}")

 def on_hsv_color_change(e: ft.ControlEvent):
 print("hsv: ", e.control.hsv_color)

 picker = ColorPicker(
 color="#ff0000",
 # hsv_color=HsvColor(alpha=1, hue=0, saturation=1, value=1),
 color_history=[
 "#ff0000",
 "#00ff00",
 "#0000ff",
 "#ffff00",
 "#00ffff",
 "#ff00ff",
],
),
```

```

 on_color_change=on_color_change,
 palette_type=PaletteType.RGB_WITH_GREEN,
 on_history_change=on_history_change,
 on_hsv_color_change=on_hsv_color_change,
 label_types=[
 ColorLabelType.HEX,
 ColorLabelType.RGB,
],
 picker_area_border_radius=ft.BorderRadius.all(20),
)

 page.add(ft.SafeArea(content=picker))

```

```

if __name__ == "__main__":
 ft.run(main)

```

## Properties

### buildcolorclass-attributeinstance-attribute

color: ColorValue | None = None

The currently selected color.

### buildcolor\_historyclass-attributeinstance-attribute

color\_history: list[ColorValue] | None = None

A list of colors to display as a history palette.

To be notified when the palette changes, set on\_history\_change.

### buildcolor\_picker\_widthclass-attributeinstance-attribute

color\_picker\_width: Number | None = None

Width of the color picker in virtual pixels.

### bulddisplay\_thumb\_colorclass-attributeinstance-attribute

display\_thumb\_color: bool = True

Whether to display the thumb color in slider.

### buildenable\_alphaclass-attributeinstance-attribute

enable\_alpha: bool = True

Whether to enable alpha (opacity) slider.

### buildhex\_input\_barclass-attributeinstance-attribute

hex\_input\_bar: bool = True

Whether to show the hex input bar.

### buildhsv\_colorclass-attributeinstance-attribute

hsv\_color: HsvColor | None = None

The currently selected HSV color.

Provide an HsvColor instance with fields: alpha, hue, saturation, value.

### buildlabel\_text\_styleclass-attributeinstance-attribute

label\_text\_style: TextStyle | None = None

Text style for labels.

### buildlabel\_typesclass-attributeinstance-attribute

label\_types: list[ColorLabelType] | None = None

Color label types to display.

### buildpalette\_typeclass-attributeinstance-attribute

palette\_type: PaletteType | None = None

Palette type for the picker area.

### buildpicker\_area\_border\_radiusclass-attributeinstance-attribute

```
picker_area_border_radius: BorderRadiusValue | None = (
 None
)
```

Border radius for the picker area.

### **buildpicker\_area\_height\_percentclass-attributeinstance-attribute**

```
picker_area_height_percent: Number | None = None
```

Height of the picker area as a percentage of the picker width.

## **Events**

### **bolton\_color\_change**class-attributeinstance-attribute

```
on_color_change: (
 ControlEventHandler[ColorPicker] | None
) = None
```

Called when the picker color is changed.

The data property of the event handler argument contains the color value as a hex string.

### **bolton\_history\_change**class-attributeinstance-attribute

```
on_history_change: (
 ControlEventHandler[ColorPicker] | None
) = None
```

Called when the history palette is changed.

The data property of the event handler argument contains the list of color values as hex strings.

### **bolton\_hsv\_color\_change**class-attributeinstance-attribute

```
on_hsv_color_change: (
 ControlEventHandler[ColorPicker] | None
) = None
```

Called when the picker HSV color is changed.

The data property of the event handler argument contains the HSV values as a dict with keys: alpha, hue, saturation, value.

# HueRingPicker

A hue ring color picker that lets users select a hue on a circular ring, with an optional alpha slider.

Image: HueRingPicker - [https://flet.dev/docs/test-images/examples/extensions/color\\_pickers/golden/macros/color\\_pickers/hue\\_ring\\_picker.png](https://flet.dev/docs/test-images/examples/extensions/color_pickers/golden/macros/color_pickers/hue_ring_picker.png)

HueRingPicker

Inherits: `LayoutControl`

## Properties

- `color` - The currently selected color.
- `color_picker_height` - Height of the color picker in virtual pixels.
- `enable_alpha` - Whether to enable alpha (opacity) slider.
- `hue_ring_stroke_width` - Stroke width for the hue ring.
- `picker_area_border_radius` - Border radius for the picker area.
- `portrait_only` - Whether to force portrait layout.

## Events

- `on_color_change` - Called when the picker color is changed.

## Example

### HueRingPicker

`play_arrow`Try Online

```
import flet as ft
from flet_color_pickers import HueRingPicker

def main(page: ft.Page):
 page.title = "HueRingPicker"
 page.padding = 20

 def on_color_change(e: ft.ControlEvent):
 print(f"color: {e.data}")

 picker = HueRingPicker(
 color="#00ff00",
 hue_ring_stroke_width=20,
 picker_area_border_radius=ft.BorderRadius.all(5),
 on_color_change=on_color_change,
)

 page.add(ft.SafeArea(content=picker))

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color`: `ColorValue` | `None` = `None`

The currently selected color.

### `buildcolor_picker_heightclass-attributeinstance-attribute`

`color_picker_height`: `Number` | `None` = `None`

Height of the color picker in virtual pixels.

### `buildenable_alphaclass-attributeinstance-attribute`

`enable_alpha`: `bool` = `False`

Whether to enable alpha (opacity) slider.

### buildhue\_ring\_stroke\_widthclass-attributeinstance-attribute

```
hue_ring_stroke_width: Number | None = None
```

Stroke width for the hue ring.

### buildpicker\_area\_border\_radiusclass-attributeinstance-attribute

```
picker_area_border_radius: BorderRadiusValue | None = (
 None
)
```

Border radius for the picker area.

### buildportrait\_onlyclass-attributeinstance-attribute

```
portrait_only: bool = False
```

Whether to force portrait layout.

## Events

### bolton\_color\_changeclass-attributeinstance-attribute

```
on_color_change: (
 ControlEventHandler[HueRingPicker] | None
) = None
```

Called when the picker color is changed.

The data property of the event handler argument contains the color value as a hex string.



# MaterialPicker

A material palette picker for selecting primary and shade colors, with optional shade labels.

Image: MaterialPicker - [https://flet.dev/docs/test-images/examples/extensions/color\\_pickers/golden/macros/color\\_pickers/material\\_picker.png](https://flet.dev/docs/test-images/examples/extensions/color_pickers/golden/macros/color_pickers/material_picker.png)

MaterialPicker

Inherits: LayoutControl

## Properties

- `color` - The currently selected color.
- `enable_label` - Whether to show color shade labels.
- `portrait_only` - Whether to force portrait layout.

## Events

- `on_color_change` - Called when the picker color is changed.
- `on_primary_change` - Called when the primary color is changed.

## Example

### MaterialPicker

play\_arrowTry Online

```
import flet as ft
from flet_color_pickers import MaterialPicker

def main(page: ft.Page):
 page.title = "MaterialPicker"
 page.padding = 20

 def on_color_change(e: ft.ControlEvent):
 print(f"color: {e.data}")

 def on_primary_change(e: ft.ControlEvent):
 print(f"primary: {e.data}")

 picker = MaterialPicker(
 color="#ff9800",
 on_color_change=on_color_change,
 on_primary_change=on_primary_change,
)

 page.add(ft.SafeArea(content=picker))

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### build\_colorclass=attributeinstance=attribute

`color: ColorValue | None = None`

The currently selected color.

### build\_enable\_labelclass=attributeinstance=attribute

`enable_label: bool = False`

Whether to show color shade labels.

### build\_portrait\_onlyclass=attributeinstance=attribute

`portrait_only: bool = False`

Whether to force portrait layout.

## Events

## **bolton\_color\_change**class-attributeinstance-attribute

```
on_color_change: (
 ControlEventHandler[MaterialPicker] | None
) = None
```

Called when the picker color is changed.

The data property of the event handler argument contains the color value as a hex string.

## **bolton\_primary\_change**class-attributeinstance-attribute

```
on_primary_change: (
 ControlEventHandler[MaterialPicker] | None
) = None
```

Called when the primary color is changed.

The data property of the event handler argument contains the color value as a hex string.

# MultipleChoiceBlockPicker

A color picker that lets users choose multiple colors from a grid of available colors.

Image: MultipleChoiceBlockPicker -

[https://flet.dev/docs/test-images/examples/extensions/color\\_pickers/golden/macos/color\\_pickers/multiple\\_choice\\_block\\_picker.png](https://flet.dev/docs/test-images/examples/extensions/color_pickers/golden/macos/color_pickers/multiple_choice_block_picker.png)

MultipleChoiceBlockPicker

Inherits: LayoutControl

## Properties

- `available_colors` - A list of available colors to pick from.
- `colors` - The currently selected colors.

## Events

- `on_colors_change` - Called when the picker colors are changed.

## Example

### MultipleChoiceBlockPicker

[play\\_arrow](#)Try Online

```
import flet as ft
from flet_color_pickers import MultipleChoiceBlockPicker

def main(page: ft.Page):
 page.title = "MultipleChoiceBlockPicker"
 page.padding = 20

 def on_colors_change(e: ft.ControlEvent):
 print(f"colors: {e.data}")

 dialog_picker = MultipleChoiceBlockPicker(
 colors=["#03a9f4", "#4caf50"],
 available_colors=[
 "#f44336",
 "#e91e63",
 "#9c27b0",
 "#3f51b5",
 "#2196f3",
 "#03a9f4",
 "#009688",
 "#4caf50",
 "#ff9800",
 "#795548",
],
 on_colors_change=on_colors_change,
)

 dialog = ft.AlertDialog(
 modal=True,
 title=ft.Text("Pick colors"),
 content=dialog_picker,
 actions=[
 ft.TextButton("Close", on_click=lambda e: page.pop_dialog()),
],
)

 page.add(
 ft.SafeArea(
 content=ft.IconButton(
 icon=ft.Icons.BRUSH,
 on_click=lambda e: page.show_dialog(dialog),
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildavailable\_colorsclass-attributeinstance-attribute**

```
available_colors: list[ColorValue] | None = None
```

A list of available colors to pick from.

### **buildcolorsclass-attributeinstance-attribute**

```
colors: list[ColorValue] | None = None
```

The currently selected colors.

## **Events**

### **bolton\_colors\_changeclass-attributeinstance-attribute**

```
on_colors_change: (
 ControlEventHandler[MultipleChoiceBlockPicker] | None
) = None
```

Called when the picker colors are changed.

The data property of the event handler argument contains the list of color values as hex strings.

# SlidePicker

A slider-based color picker that exposes RGB/HSV/HSL channels with optional labels and indicators.

Image: SlidePicker - [https://flet.dev/docs/test-images/examples/extensions/color\\_pickers/golden/macos/color\\_pickers/slide\\_picker.png](https://flet.dev/docs/test-images/examples/extensions/color_pickers/golden/macos/color_pickers/slide_picker.png)

SlidePicker

Inherits: `LayoutControl`

## Properties

- `color` - The currently selected color.
- `color_model` - The color model used by the sliders.
- `display_thumb_color` - Whether to display the thumb color in sliders.
- `enable_alpha` - Whether to enable alpha (opacity) slider.
- `indicator_alignment_begin` - Alignment for the indicator split begin.
- `indicator_alignment_end` - Alignment for the indicator split end.
- `indicator_border_radius` - Border radius for the indicator.
- `indicator_size` - Size of the indicator.
- `label_text_style` - Text style for labels.
- `label_types` - Color label types to display.
- `show_indicator` - Whether to show the color indicator.
- `show_label` - Whether to show labels.
- `show_params` - Whether to show parameter values.
- `show_slider_text` - Whether to show slider text.
- `slider_size` - Size of the sliders.
- `slider_text_style` - Text style for slider text.

## Events

- `on_color_change` - Called when the picker color is changed.

# Example

## SlidePicker

`play_arrow` Try Online

```
import flet as ft
from flet_color_pickers import ColorModel, SlidePicker

def main(page: ft.Page):
 page.title = "SlidePicker"
 page.padding = 20

 def on_color_change(e: ft.ControlEvent):
 print(f"color: {e.data}")

 picker = SlidePicker(
 color="#0000ff",
 color_model=ColorModel.RGB,
 indicator_border_radius=ft.BorderRadius.all(5),
 on_color_change=on_color_change,
)

 page.add(ft.SafeArea(content=picker))

if __name__ == "__main__":
 ft.run(main)
```

## Properties

### **buildcolorclass-attributeinstance-attribute**

`color: ColorValue | None = None`

The currently selected color.

### **buildcolor\_modelclass-attributeinstance-attribute**

`color_model: ColorModel | None = None`

The color model used by the sliders.

### **bulddisplay\_thumb\_colorclass-attributeinstance-attribute**

`display_thumb_color: bool = True`

Whether to display the thumb color in sliders.

### **buildenable\_alphaclass-attributeinstance-attribute**

`enable_alpha: bool = True`

Whether to enable alpha (opacity) slider.

### **buildindicator\_alignment\_beginclass-attributeinstance-attribute**

`indicator_alignment_begin: Alignment | None = None`

Alignment for the indicator split begin.

### **buildindicator\_alignment\_endclass-attributeinstance-attribute**

`indicator_alignment_end: Alignment | None = None`

Alignment for the indicator split end.

### **buildindicator\_border\_radiusclass-attributeinstance-attribute**

`indicator_border_radius: BorderRadiusValue | None = None`

Border radius for the indicator.

### **buildindicator\_sizeclass-attributeinstance-attribute**

`indicator_size: Size | None = None`

Size of the indicator.

### **buildlabel\_text\_styleclass-attributeinstance-attribute**

`label_text_style: TextStyle | None = None`

Text style for labels.

### **buildlabel\_typesclass-attributeinstance-attribute**

`label_types: list[ColorLabelType] | None = None`

Color label types to display.

### **buildshow\_indicatorclass-attributeinstance-attribute**

`show_indicator: bool = True`

Whether to show the color indicator.

### **buildshow\_labelclass-attributeinstance-attribute**

`show_label: bool = True`

Whether to show labels.

### **buildshow\_paramsclass-attributeinstance-attribute**

`show_params: bool = True`

Whether to show parameter values.

### **buildshow\_slider\_textclass-attributeinstance-attribute**

`show_slider_text: bool = True`

Whether to show slider text.

### **buildslider\_sizeclass-attributeinstance-attribute**

`slider_size: Size | None = None`

Size of the sliders.

### **buildslider\_text\_styleclass-attributeinstance-attribute**

`slider_text_style: TextStyle | None = None`

Text style for slider text.

## **Events**

### **bolton\_color\_changeclass-attributeinstance-attribute**

`on_color_change: (ControlEventHandler[SlidePicker] | None) = None`

Called when the picker color is changed.

The data property of the event handler argument contains the color value as a hex string.

# DataColumn2

Extends DataColumn, adding the ability to set relative column size and fixed column width.

Meant to be used as an item of DataTable2.columns.

Inherits: DataColumn

## Properties

- `fixed_width` - Defines absolute width of the column in pixels (as opposed to relative size used by default).
- `size` - Column sizes are determined based on available width by distributing it to individual columns accounting for their relative sizes.

## Properties

### `fixed_width`class-attributeinstance-attribute

`fixed_width: Number | None = None`

Defines absolute width of the column in pixels (as opposed to relative size used by default).

### `size`class-attributeinstance-attribute

`size: DataColumnSize | None = DataColumnSize.S`

Column sizes are determined based on available width by distributing it to individual columns accounting for their relative sizes.



# DataRow2

Extends DataRow, adding row-level tap events.

There are also `on_secondary_tap` and `on_secondary_tap_down`, which are not available in `DataCells` and can be useful in desktop settings to handle right-click actions.

Inherits: DataRow

## Properties

- `decoration` - Decoration to be applied to this row.
- `specific_row_height` - Specific row height.

## Events

- `on_double_tap` - Fires when the row is double-tapped.
- `on_secondary_tap` - Fires when the row is right-clicked (secondary tap).
- `on_secondary_tap_down` - Fires when the row is right-clicked (secondary tap down).
- `on_tap` - Fires when the row is tapped.

## Properties

### `builddecorationclass-attributeinstance-attribute`

```
decoration: BoxDecoration | None = None
```

Decoration to be applied to this row.

Note

If provided, `DataTable.divider_thickness` has no effect.

### `buildspecific_row_heightclass-attributeinstance-attribute`

```
specific_row_height: Number | None = None
```

Specific row height.

Falls back to `DataTable2.data_row_height` if not set.

## Events

### `bolton_double_tapclass-attributeinstance-attribute`

```
on_double_tap: ControlEventHandler[DataRow2] | None = (
 None
)
```

Fires when the row is double-tapped.

Note

Won't be called if tapped cell has any tap event handlers (`on_tap`, `on_double_tap`, `on_long_press`, `on_tap_cancel`, `on_tap_down`) set.

### `bolton_secondary_tapclass-attributeinstance-attribute`

```
on_secondary_tap: (
 ControlEventHandler[DataRow2] | None
) = None
```

Fires when the row is right-clicked (secondary tap).

Note

Won't be called if tapped cell has any tap event handlers (`on_tap`, `on_double_tap`, `on_long_press`, `on_tap_cancel`, `on_tap_down`) set.

### `bolton_secondary_tap_downclass-attributeinstance-attribute`

```
on_secondary_tap_down: (
```

```
ControlEventHandler[DataRow2] | None
) = None
```

Fires when the row is right-clicked (secondary tap down).

Note

Won't be called if tapped cell has any tap event handlers (on\_tap, on\_double\_tap, on\_long\_press, on\_tap\_cancel, on\_tap\_down) set.

### **bolton\_tapclass-attributeinstance-attribute**

```
on_tap: EventHandler[TapEvent[DataRow2]] | None = None
```

Fires when the row is tapped.

Note

Won't be called if tapped cell has any tap event handlers (on\_tap, on\_double\_tap, on\_long\_press, on\_tap\_cancel, on\_tap\_down) set.

# BaseOverlayImage

Abstract class for image overlays displayed through OverlayImageLayer.

The following overlay image types are available:

- OverlayImage
- RotatedOverlayImage

Inherits: BaseControl

## Properties

- `filter_quality` - The rendering quality of the image.
- `gapless_playback` - Whether to continue showing the old image (True), or briefly show nothing (False), when the image provider changes.
- `opacity` - The opacity in which the image should get rendered on the map.
- `src` - The image source.
- `visible` - Whether this overlay image is rendered on the map.

## Properties

### `buildfilter_qualityclass-attributeinstance-attribute`

`filter_quality: FilterQuality = FilterQuality.MEDIUM`

The rendering quality of the image.

### `buildgapless_playbackclass-attributeinstance-attribute`

`gapless_playback: bool = False`

Whether to continue showing the old image (True), or briefly show nothing (False), when the image provider changes.

### `buildopacityclass-attributeinstance-attribute`

`opacity: Number = 1.0`

The opacity in which the image should get rendered on the map.

Raises:

- `ValueError` - If it is not between 0.0 and 1.0, inclusive.

### `buildsrcinstance-attribute`

`src: str | bytes`

The image source.

It can be one of the following:

- A URL or local asset file path;
- A base64 string;
- Raw bytes.

### `buildvisibleclass-attributeinstance-attribute`

`visible: bool = True`

Whether this overlay image is rendered on the map.

# CircleLayer

A layer to display CircleMarker.

Inherits: MapLayer

## Properties

- circles - A list of CircleMarker to display.

## Properties

**buildcirclesinstance-attribute**

circles: list[CircleMarker]

A list of CircleMarker to display.

# CircleMarker

A circular marker displayed on the Map at the specified location through the CircleLayer.

Inherits: BaseControl

## Properties

- `border_color` - The color of the circle border line.
- `border_stroke_width` - The stroke width for the circle border.
- `color` - The color of the circle area.
- `coordinates` - The center coordinates of the circle
- `radius` - The radius of the circle
- `use_radius_in_meter` - Whether the radius should use the unit meters.
- `visible` - Whether this marker is rendered on the map.

## Properties

### `buildborder_colorclass-attributeinstance-attribute`

`border_color: ColorValue | None = None`

The color of the circle border line.

Tip

`border_stroke_width` must be greater than 0.0 in order for this color to be visible.

### `buildborder_stroke_widthclass-attributeinstance-attribute`

`border_stroke_width: Number = 0.0`

The stroke width for the circle border.

Raises:

- `ValueError` - If it is less than 0.0.

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the circle area.

### `buildcoordinatesinstance-attribute`

`coordinates: MapLatitudeLongitude`

The center coordinates of the circle

### `buildradiusinstance-attribute`

`radius: Number`

The radius of the circle

### `builduse_radius_in_meterclass-attributeinstance-attribute`

`use_radius_in_meter: bool = False`

Whether the radius should use the unit meters.

### `buildvisibleclass-attributeinstance-attribute`

`visible: bool = True`

Whether this marker is rendered on the map.

# ImageSourceAttribution

An image attribution permanently displayed adjacent to the open/close icon of a RichAttribution control. For it to be displayed, it should be part of a RichAttribution.attributions list.

Inherits: SourceAttribution

## Properties

- height - The height of the image.
- image - The Image to be displayed.
- tooltip - Tooltip text to be displayed when the image is hovered over.

## Events

- on\_click - Fired when this attribution is clicked/pressed.

## Properties

### buildheightclass-attributeinstance-attribute

height: Number = 24.0

The height of the image. Should be the same as RichAttribution.permanent\_height, otherwise layout issues may occur.

### buildimageinstance-attribute

image: Image

The Image to be displayed.

Raises:

- ValueError - If the image is not visible.

### buildtooltipclass-attributeinstance-attribute

tooltip: str | None = None

Tooltip text to be displayed when the image is hovered over.

## Events

### bolton\_clickclass-attributeinstance-attribute

```
on_click: (
 ControlEventHandler[ImageSourceAttribution] | None
) = None
```

Fired when this attribution is clicked/pressed.

# Map

An interactive map that displays various layers.

Inherits: `LayoutControl`

## Properties

- `animation_curve` - The default animation curve to be used for map-animations when calling instance methods like `zoom_in`, `rotate_from`, `move_to` etc.
- `animation_duration` - The default animation duration to be used for map-animations when calling instance methods like `zoom_in`, `rotate_from`, `move_to` etc.
- `bgcolor` - The background color of this control.
- `initial_camera_fit` - Defines the visible bounds when the map is first loaded.
- `initial_center` - The initial center of the map.
- `initial_rotation` - The rotation (in degrees) when the map is first loaded.
- `initial_zoom` - The zoom when the map is first loaded.
- `interaction_configuration` - The interaction configuration.
- `keep_alive` - Whether to enable the built in keep-alive functionality.
- `layers` - A list of layers to be displayed (stack-like) on the map.
- `max_zoom` - The maximum (highest) zoom level of every layer.
- `min_zoom` - The minimum (smallest) zoom level of every layer.

## Events

- `on_event` - Fires when any map events occurs.
- `on_hover` - Fires when a hover event occurs.
- `on_init` - Fires when the map is initialized.
- `on_long_press` - Fires when a long press event occurs.
- `on_pointer_cancel` - Fires when a pointer cancel event occurs.
- `on_pointer_down` - Fires when a pointer down event occurs.
- `on_pointer_up` - Fires when a pointer up event occurs.
- `on_position_change` - Fires when the map position changes, e.g.
- `on_secondary_tap` - Fires when a secondary tap event occurs.
- `on_tap` - Fires when a tap event occurs.

## Methods

- `center_on` - Centers the map on the given point.
- `get_camera` - Gets the current camera snapshot of the map.
- `move_to` - Moves to a specific location.
- `reset_rotation` - Resets the map's rotation to 0 degrees.
- `rotate_from` - Applies a rotation of degree to the current rotation.
- `zoom_in` - Zooms in by one zoom-level from the current one.
- `zoom_out` - Zooms out by one zoom-level from the current one.
- `zoom_to` - Zoom the map to a specific zoom level.

## Properties

### buildanimation\_curveclass-attributeinstance-attribute

```
animation_curve: AnimationCurve = (
 AnimationCurve.FAST_OUT_SLOWIN
)
```

The default animation curve to be used for map-animations when calling instance methods like `zoom_in`, `rotate_from`, `move_to` etc.

### buildanimation\_durationclass-attributeinstance-attribute

```
animation_duration: DurationValue = field(
 default_factory=lambda: Duration(milliseconds=500)
)
```

The default animation duration to be used for map-animations when calling instance methods like `zoom_in`, `rotate_from`, `move_to` etc.

### buildbgcolorclass-attributeinstance-attribute

```
bgcolor: ColorValue = Colors.GREY_300
```

The background color of this control.

### buildinitial\_camera\_fitclass-attributeinstance-attribute

```
initial_camera_fit: CameraFit | None = None
```

Defines the visible bounds when the map is first loaded. Takes precedence over `initial_center`/`initial_zoom`.

### buildinitial\_centerclass-attributeinstance-attribute

```
initial_center: MapLatitudeLongitude = field(
 default_factory=lambda: MapLatitudeLongitude(
 latitude=50.5, longitude=30.51
)
)
```

The initial center of the map.

### buildinitial\_rotationclass-attributeinstance-attribute

```
initial_rotation: Number = 0.0
```

The rotation (in degrees) when the map is first loaded.

### buildinitial\_zoomclass-attributeinstance-attribute

```
initial_zoom: Number = 13.0
```

The zoom when the map is first loaded.

If `initial_camera_fit` is non-None this has no effect.

### buildinteraction\_configurationclass-attributeinstance-attribute

```
interaction_configuration: InteractionConfiguration = field(
 default_factory=lambda: InteractionConfiguration()
)
```

The interaction configuration.

### buildkeep\_aliveclass-attributeinstance-attribute

```
keep_alive: bool = False
```

Whether to enable the built in keep-alive functionality.

If the map is within a complex layout, such as a `ListView`, the map will reset to its initial position after it appears back into view. To ensure this doesn't happen, enable this flag to prevent it from rebuilding.

### buildlayersinstance-attribute

```
layers: list[MapLayer]
```

A list of layers to be displayed (stack-like) on the map.

### buildmax\_zoomclass-attributeinstance-attribute

```
max_zoom: Number | None = None
```

The maximum (highest) zoom level of every layer. Each layer can specify additional zoom level restrictions.

### buildmin\_zoomclass-attributeinstance-attribute



`min_zoom: Number | None = None`

The minimum (smallest) zoom level of every layer. Each layer can specify additional zoom level restrictions.

## Events

### `bolton_eventclass-attributeinstance-attribute`

`on_event: EventHandler[MapEvent] | None = None`

Fires when any map events occurs.

### `bolton_hoverclass-attributeinstance-attribute`

`on_hover: EventHandler[MapHoverEvent] | None = None`

Fires when a hover event occurs.

### `bolton_initclass-attributeinstance-attribute`

`on_init: ControlEventHandler[Map] | None = None`

Fires when the map is initialized.

### `bolton_long_pressclass-attributeinstance-attribute`

`on_long_press: EventHandler[MapTapEvent] | None = None`

Fires when a long press event occurs.

### `bolton_pointer_cancelclass-attributeinstance-attribute`

```
on_pointer_cancel: (
 EventHandler[MapPointerEvent] | None
) = None
```

Fires when a pointer cancel event occurs.

### `bolton_pointer_downclass-attributeinstance-attribute`

```
on_pointer_down: EventHandler[MapPointerEvent] | None = (
 None
)
```

Fires when a pointer down event occurs.

### `bolton_pointer_upclass-attributeinstance-attribute`

```
on_pointer_up: EventHandler[MapPointerEvent] | None = (
 None
)
```

Fires when a pointer up event occurs.

### `bolton_position_changeclass-attributeinstance-attribute`

```
on_position_change: (
 EventHandler[MapPositionChangeEvent] | None
) = None
```

Fires when the map position changes, e.g. when the user pans or zooms the map.

### `bolton_secondary_tapclass-attributeinstance-attribute`

`on_secondary_tap: EventHandler[MapTapEvent] | None = None`

Fires when a secondary tap event occurs.

### `bolton_tapclass-attributeinstance-attribute`

`on_tap: EventHandler[MapTapEvent] | None = None`

Fires when a tap event occurs.

## Methods

### `deployed_codecenter_onasync`

```
center_on(
 point: MapLatitudeLongitude,
 zoom: Number | None,
 animation_curve: AnimationCurve | None = None,
 animation_duration: DurationValue | None = None,
 cancel_ongoing_animations: bool = False,
```

```
) -> None
```

Centers the map on the given point.

Parameters:

- point (MapLatitudeLongitude) - The point on which to center the map.
- zoom (Number | None) - The zoom level to be applied.
- animation\_curve (AnimationCurve | None, default: None) - The curve of the animation. If None (the default), Map.animation\_curve will be used.
- animation\_duration (DurationValue | None, default: None) - The duration of the animation. If None (the default), Map.animation\_duration will be used.
- cancel\_ongoing\_animations (bool, default: False) - Whether to cancel/stop all ongoing map-animations before starting this new one.

## deployed\_codeget\_cameraasync

```
get_camera() -> Camera
```

Gets the current camera snapshot of the map.

Returns:

- Camera - The current camera state.

## deployed\_codemove\_toasync

```
move_to(
 destination: MapLatitudeLongitude | None = None,
 zoom: Number | None = None,
 rotation: Number | None = None,
 animation_curve: AnimationCurve | None = None,
 animation_duration: DurationValue | None = None,
 offset: OffsetValue = (0, 0),
 cancel_ongoing_animations: bool = False,
) -> None
```

Moves to a specific location.

Parameters:

- destination (MapLatitudeLongitude | None, default: None) - The destination point to move to.
- zoom (Number | None, default: None) - The zoom level to be applied. If provided, must be greater than or equal to 0.0.
- rotation (Number | None, default: None) - Rotation (in degrees) to be applied.
- offset (OffsetValue, default: (0, 0)) - The offset to be used. Only works when rotation is None.
- animation\_curve (AnimationCurve | None, default: None) - The curve of the animation. If None (the default), Map.animation\_curve will be used.
- animation\_duration (DurationValue | None, default: None) - The duration of the animation. If None (the default), Map.animation\_duration will be used.
- cancel\_ongoing\_animations (bool, default: False) - Whether to cancel/stop all ongoing map-animations before starting this new one.

Raises:

- ValueError - If zoom is not None and is negative.

## deployed\_codereset\_rotationasync

```
reset_rotation(
 animation_curve: AnimationCurve | None = None,
 animation_duration: DurationValue | None = None,
 cancel_ongoing_animations: bool = False,
) -> None
```

Resets the map's rotation to 0 degrees.

Parameters:

- animation\_curve (AnimationCurve | None, default: None) - The curve of the animation. If None (the default), Map.animation\_curve will be used.

- `animation_duration` (`DurationValue` | `None`, default: `None`) - The duration of the animation. If `None` (the default), `Map.animation_duration` will be used.
- `cancel_ongoing_animations` (`bool`, default: `False`) - Whether to cancel/stop all ongoing map-animations before starting this new one.

### `deployed_coderotate_fromasync`

```
rotate_from(
 degree: Number,
 animation_curve: AnimationCurve | None = None,
 animation_duration: DurationValue | None = None,
 cancel_ongoing_animations: bool = False,
) -> None
```

Applies a rotation of degree to the current rotation.

Parameters:

- `degree` (`Number`) - The number of degrees to increment to the current rotation.
- `animation_curve` (`AnimationCurve` | `None`, default: `None`) - The curve of the animation. If `None` (the default), `Map.animation_curve` will be used.
- `animation_duration` (`DurationValue` | `None`, default: `None`) - The duration of the animation. If `None` (the default), `Map.animation_duration` will be used.
- `cancel_ongoing_animations` (`bool`, default: `False`) - Whether to cancel/stop all ongoing map-animations before starting this new one.

### `deployed_codezoom_inasync`

```
zoom_in(
 animation_curve: AnimationCurve | None = None,
 animation_duration: DurationValue | None = None,
 cancel_ongoing_animations: bool = False,
) -> None
```

Zooms in by one zoom-level from the current one.

Parameters:

- `animation_curve` (`AnimationCurve` | `None`, default: `None`) - The curve of the animation. If `None` (the default), `Map.animation_curve` will be used.
- `animation_duration` (`DurationValue` | `None`, default: `None`) - The duration of the animation. If `None` (the default), `Map.animation_duration` will be used.
- `cancel_ongoing_animations` (`bool`, default: `False`) - Whether to cancel/stop all ongoing map-animations before starting this new one.

### `deployed_codezoom_outasync`

```
zoom_out(
 animation_curve: AnimationCurve | None = None,
 animation_duration: DurationValue | None = None,
 cancel_ongoing_animations: bool = False,
) -> None
```

Zooms out by one zoom-level from the current one.

Parameters:

- `animation_curve` (`AnimationCurve` | `None`, default: `None`) - The curve of the animation. If `None` (the default), `Map.animation_curve` will be used.
- `animation_duration` (`DurationValue` | `None`, default: `None`) - The duration of the animation. If `None` (the default), `Map.animation_duration` will be used.
- `cancel_ongoing_animations` (`bool`, default: `False`) - Whether to cancel/stop all ongoing map-animations before starting this new one.

### `deployed_codezoom_toasync`

```
zoom_to(
 zoom: Number,
 animation_curve: AnimationCurve | None = None,
 animation_duration: DurationValue | None = None,
 cancel_ongoing_animations: bool = False,
) -> None
```

Zoom the map to a specific zoom level.

Parameters:

- `zoom` (Number) - The zoom level to zoom to.
- `animation_curve` (AnimationCurve | None, default: None) - The curve of the animation. If None (the default), `Map.animation_curve` will be used.
- `animation_duration` (DurationValue | None, default: None) - The duration of the animation. If None (the default), `Map.animation_duration` will be used.
- `cancel_ongoing_animations` (bool, default: False) - Whether to cancel/stop all ongoing map-animations before starting this new one.

# MapLayer

Abstract class for all map layers.

The following layers are available:

- CircleLayer
- MarkerLayer
- OverlayImageLayer
- PolygonLayer
- PolylineLayer
- RichAttribution
- SimpleAttribution
- TileLayer

Inherits: Control

# Marker

A marker displayed on the Map at the specified location through the MarkerLayer.

Inherits: BaseControl

## Properties

- alignment - Alignment of the marker relative to the normal center at coordinates.
- content - The content to be displayed at coordinates.
- coordinates - The coordinates of the marker.
- height - The height of the content Control.
- rotate - Whether to counter rotate this marker to the map's rotation, to keep a fixed orientation.
- visible - Whether this marker is rendered on the map.
- width - The width of the content Control.

## Properties

### buildalignmentclass-attributeinstance-attribute

alignment: Alignment | None = None

Alignment of the marker relative to the normal center at coordinates.

Defaults to the value of the parent MarkerLayer.alignment.

### buildcontentinstance-attribute

content: Control

The content to be displayed at coordinates.

Raises:

- ValueError - If it is not visible.

### buildcoordinatesinstance-attribute

coordinates: MapLatitudeLongitude

The coordinates of the marker.

This will be the center of the marker, if alignment is Alignment.CENTER.

### buildheightclass-attributeinstance-attribute

height: Number = 30.0

The height of the content Control.

Raises:

- ValueError - If it is less than 0.0.

### buildrotateclass-attributeinstance-attribute

rotate: bool | None = None

Whether to counter rotate this marker to the map's rotation, to keep a fixed orientation. So, when True, this marker will always appear upright and vertical from the user's perspective.

If None, defaults to the value of the parent MarkerLayer.rotate.

Note

This is not used to apply a custom rotation in degrees to this marker.

### buildvisibleclass-attributeinstance-attribute

visible: bool = True

Whether this marker is rendered on the map.

## buildwidthclass-attributeinstance-attribute

width: Number = 30.0

The width of the content Control.

Raises:

- ValueError - If it is less than 0.0.

# MarkerLayer

A layer to display Markers.

Inherits: MapLayer

## Properties

- alignment - The alignment of each marker relative to its normal center at Marker.coordinates.
- markers - A list of Markers to display.
- rotate - Whether to counter-rotate markers to the map's rotation, to keep a fixed orientation.

## Properties

### buildalignmentclass-attributeinstance-attribute

```
alignment: Alignment | None = field(
 default_factory=lambda: Alignment.CENTER
)
```

The alignment of each marker relative to its normal center at Marker.coordinates.

### buildmarkersinstance-attribute

```
markers: list[Marker]
```

A list of Markers to display.

### buildrotateclass-attributeinstance-attribute

```
rotate: bool = False
```

Whether to counter-rotate markers to the map's rotation, to keep a fixed orientation.



# OverlayImage

An unrotated image overlay that spans between a given bounding box.

Inherits: BaseOverlayImage

## Properties

- bounds - The latitude and longitude bounds where this image will be displayed.

## Properties

### buildboundsinstance-attribute

bounds: MapLatitudeLongitudeBounds

The latitude and longitude bounds where this image will be displayed.

## Examples

### Overlay Images

play\_arrow Try Online

```
import flet as ft
import flet_map as ftm
```

```
base64_image = "iVBORw0KGgoAAAANSUhEUgAAABkAAAAgCAYAAADnnNMGAAACXBjIwXMAAA0RAAADkQFnq8zdAAAAGXRFWHRTb2Z0d2FyZQB3d3cuaw5rc2NhcGUubG9vc3QAAACAAQAA"

```

```
def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 expand=True,
 content=ftm.Map(
 expand=True,
 initial_center=ftm.MapLatitudeLongitude(51.5, -0.09),
 initial_zoom=6,
 layers=[
 ftm.TileLayer(
 url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
 user_agent_package_name="flet-map-examples/1.0",
),
 ftm.OverlayImageLayer(
 overlay_images=[
 ftm.OverlayImage(
 src=base64_image,
 bounds=ftm.MapLatitudeLongitudeBounds(
 corner_1=ftm.MapLatitudeLongitude(51.5, -0.09),
 corner_2=ftm.MapLatitudeLongitude(48.8566, 2.3522),
),
 opacity=0.8,
),
 ftm.RotatedOverlayImage(
 src=base64_image,
 top_left_corner=ftm.MapLatitudeLongitude(
 53.377, -2.999
),
 bottom_left_corner=ftm.MapLatitudeLongitude(
 52.503, -1.868
),
 bottom_right_corner=ftm.MapLatitudeLongitude(
 53.475, 0.275
),
 opacity=0.8,
),
],
),
 ftm.SimpleAttribution(text="OpenStreetMap contributors"),
],
),
),
)

if __name__ == "__main__":
 ft.run(main)
```

# OverlayImageLayer

A layer to display image overlays.

Tip

Place this layer after every non-translucent layer that should appear below it. Layers rendered after this one may cover its overlay images.

Inherits: MapLayer

## Properties

- `overlay_images` - A list of image overlays to display.

## Properties

## buildoverlay\_imagesinstance-attribute

overlay\_images: list[BaseOverlayImage]

A list of image overlays to display.

## Examples

## Overlay Images

play\_arrow Try Online

```
import flet as ft
import flet_map as ftm
```

```
base64_image = "iVBORw0KGgoAAAANSUhEUgAAABkAAAAGCAYAAADnnNMGAAAACXBIXMAAAORAAADkQFnq8zdAAAAGXRFWHRTb2Z0d2FyZQB3d3cuaw5rc2NhcGUub"

```

```
def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 expand=True,
 content=ft.Map(
 expand=True,
 initial_center=ft.MapLatitudeLongitude(51.5, -0.09),
 initial_zoom=6,
 layers=[
 ftm.TileLayer(
 url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
 user_agent_package_name="flet-map-examples/1.0",
),
 ftm.OverlayImageLayer(
 overlay_images=[
 ftm.OverlayImage(
 src=base64_image,
 bounds=ftm.MapLatitudeLongitudeBounds(
 corner_1=ftm.MapLatitudeLongitude(51.5, -0.09),
 corner_2=ftm.MapLatitudeLongitude(48.8566, 2.3522),
),
 opacity=0.8,
),
 ftm.RotatedOverlayImage(
 src=base64_image,
 top_left_corner=ftm.MapLatitudeLongitude(
 53.377, -2.999
),
 bottom_left_corner=ftm.MapLatitudeLongitude(
 52.503, -1.868
),
 bottom_right_corner=ftm.MapLatitudeLongitude(
 53.475, 0.275
),
 opacity=0.8,
),
],
),
 ftm.SimpleAttribution(text="OpenStreetMap contributors"),
],
),
),
)

if __name__ == "__main__":
```

```
ft.run(main)
```

# PolygonLayer

A layer to display PolygonMarkers.

Inherits: MapLayer

## Properties

- `draw_labels_last` - Whether to draw labels last and thus over all the polygons.
- `polygon_culling` - Whether to cull polygons and polygon sections that are outside of the viewport.
- `polygon_labels` - Whether to draw per-polygon labels.
- `polygons` - A list of PolygonMarkers to display.
- `simplification_tolerance` - The tolerance value used to simplify polygon outlines before rendering.
- `use_alternative_rendering` - Whether to use an alternative rendering pathway to draw polygons onto the underlying Canvas, which can be more performant in 'some' circumstances.

## Properties

### `builddraw_labels_lastclass-attributeinstance-attribute`

```
draw_labels_last: bool = False
```

Whether to draw labels last and thus over all the polygons.

### `buildpolygon_cullingclass-attributeinstance-attribute`

```
polygon_culling: bool = True
```

Whether to cull polygons and polygon sections that are outside of the viewport.

### `buildpolygon_labelsclass-attributeinstance-attribute`

```
polygon_labels: bool = True
```

Whether to draw per-polygon labels.

### `buildpolygonsinstance-attribute`

```
polygons: list[PolygonMarker]
```

A list of PolygonMarkers to display.

### `buildsimplification_toleranceclass-attributeinstance-attribute`

```
simplification_tolerance: Number = 0.3
```

The tolerance value used to simplify polygon outlines before rendering.

Higher values will result in polygons with fewer points, which can improve rendering performance at the cost of reduced geometric accuracy. Lower values preserve more detail but may decrease performance, especially with complex polygons.

Set to 0 to disable simplification.

### `builduse_alternative_renderingclass-attributeinstance-attribute`

```
use_alternative_rendering: bool = False
```

Whether to use an alternative rendering pathway to draw polygons onto the underlying Canvas, which can be more performant in 'some' circumstances.

This will not always improve performance, and there are other important considerations before enabling it. It is intended for use when prior profiling indicates more performance is required after other methods are already in use. For example, it may worsen performance when there are a huge number of polygons to triangulate - and so this is best used in conjunction with simplification, not as a replacement.

# PolygonMarker

A marker for the PolygonLayer.

Inherits: BaseControl

## Properties

- `border_color` - The color of the border outline.
- `border_stroke_width` - The width of the border outline.
- `color` - The color of the polygon.
- `coordinates` - The points for the outline of this polygon.
- `disable_holes_border` - Whether holes should have borders.
- `label` - An optional label for this polygon.
- `label_text_style` - The text style for the label.
- `rotate_label` - Whether to rotate the label counter to the camera's rotation, to ensure it remains upright.
- `stroke_cap` - Style to use for line endings.
- `stroke_join` - Style to use for line segment joins.
- `visible` - Whether this marker is rendered on the map.

## Properties

### `buildborder_colorclass-attributeinstance-attribute`

```
border_color: ColorValue = Colors.GREEN
```

The color of the border outline.

### `buildborder_stroke_widthclass-attributeinstance-attribute`

```
border_stroke_width: Number = 0.0
```

The width of the border outline.

Raises:

- `ValueError` - If it is less than 0.0.

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue = Colors.GREEN
```

The color of the polygon.

### `buildcoordinatesinstance-attribute`

```
coordinates: list[MapLatitudeLongitude]
```

The points for the outline of this polygon.

### `bulddisable_holes_borderclass-attributeinstance-attribute`

```
disable_holes_border: bool = False
```

Whether holes should have borders.

### `buildlabelclass-attributeinstance-attribute`

```
label: str | None = None
```

An optional label for this polygon.

Note

Specifying a label will reduce performance, as the internal canvas must be drawn to and 'saved' more frequently to ensure the proper stacking order is maintained. This can be avoided, potentially at the expense of appearance, by setting `PolygonLayer.draw_labels_last`.

### **buildlabel\_text\_styleclass-attributeinstance-attribute**

label\_text\_style: TextStyle | None = None

The text style for the label.

### **buildrotate\_labelclass-attributeinstance-attribute**

rotate\_label: bool = False

Whether to rotate the label counter to the camera's rotation, to ensure it remains upright.

### **buildstroke\_capclass-attributeinstance-attribute**

stroke\_cap: StrokeCap = StrokeCap.ROUND

Style to use for line endings.

### **buildstroke\_joinclass-attributeinstance-attribute**

stroke\_join: StrokeJoin = StrokeJoin.ROUND

Style to use for line segment joins.

### **buildvisibleclass-attributeinstance-attribute**

visible: bool = True

Whether this marker is rendered on the map.

# PolylineLayer

A layer to display PolylineMarkers.

Inherits: MapLayer

## Properties

- `culling_margin` - Acceptable extent outside of viewport before culling polyline segments.
- `min_hittable_radius` - The minimum radius of the hittable area around each polyline in logical pixels.
- `polylines` - List of PolylineMarkers to be drawn.
- `simplification_tolerance` - The tolerance (in map units) used to simplify polylines for rendering.

## Properties

### `buildculling_marginclass-attributeinstance-attribute`

`culling_margin: Number = 10.0`

Acceptable extent outside of viewport before culling polyline segments.

### `buildmin_hittable_radiusclass-attributeinstance-attribute`

`min_hittable_radius: Number = 10.0`

The minimum radius of the hittable area around each polyline in logical pixels.

The entire visible area is always hittable, but if the visible area is smaller than this, then this will be the hittable area.

### `buildpolylinesinstance-attribute`

`polylines: list[PolylineMarker]`

List of PolylineMarkers to be drawn.

### `buildsimplification_tolerancelass-attributeinstance-attribute`

`simplification_tolerance: Number = 0.3`

The tolerance (in map units) used to simplify polylines for rendering.

Higher values result in more aggressive simplification, which can improve performance but may reduce the accuracy of the displayed polyline.

# PolylineMarker

A marker for the PolylineLayer.

Inherits: BaseControl

## Properties

- `border_color` - The border's color.
- `border_stroke_width` - The width of the stroke with of the line border.
- `color` - The color of the line stroke.
- `colors_stop` - The stops for the `gradient_colors`.
- `coordinates` - The list of coordinates for the polyline.
- `gradient_colors` - The List of colors in case a gradient should get used.
- `stroke_cap` - Style to use for line endings.
- `stroke_join` - Style to use for line segment joins.
- `stroke_pattern` - Determines whether the line should be solid, dotted, or dashed, and the exact characteristics of each.
- `stroke_width` - The width of the stroke.
- `use_stroke_width_in_meter` - Whether the stroke's width should have meters as unit.
- `visible` - Whether this marker is rendered on the map.

## Properties

### `buildborder_colorclass=attributeinstance=attribute`

```
border_color: ColorValue = Colors.YELLOW
```

The border's color.

### `buildborder_stroke_widthclass=attributeinstance=attribute`

```
border_stroke_width: Number = 0.0
```

The width of the stroke with of the line border.

Raises:

- `ValueError` - If it is less than 0.0.

### `buildcolorclass=attributeinstance=attribute`

```
color: ColorValue = Colors.YELLOW
```

The color of the line stroke.

### `buildcolors_stopclass=attributeinstance=attribute`

```
colors_stop: list[Number] | None = None
```

The stops for the `gradient_colors`.

### `buildcoordinatesinstance=attribute`

```
coordinates: list[MapLatitudeLongitude]
```

The list of coordinates for the polyline.

### `buildgradient_colorsclass=attributeinstance=attribute`

```
gradient_colors: list[ColorValue] | None = None
```

The List of colors in case a gradient should get used.

### `buildstroke_capclass=attributeinstance=attribute`

```
stroke_cap: StrokeCap = StrokeCap.ROUND
```

Style to use for line endings.



### **buildstroke\_joinclass-attributeinstance-attribute**

```
stroke_join: StrokeJoin = StrokeJoin.ROUND
```

Style to use for line segment joins.

### **buildstroke\_patternclass-attributeinstance-attribute**

```
stroke_pattern: StrokePattern = field(
 default_factory=lambda: SolidStrokePattern()
)
```

Determines whether the line should be solid, dotted, or dashed, and the exact characteristics of each.

### **buildstroke\_widthclass-attributeinstance-attribute**

```
stroke_width: Number = 1.0
```

The width of the stroke.

Raises:

- `ValueError` - If it is less than 0.0.

### **builduse\_stroke\_width\_in\_meterclass-attributeinstance-attribute**

```
use_stroke_width_in_meter: bool = False
```

Whether the stroke's width should have meters as unit.

### **buildvisibleclass-attributeinstance-attribute**

```
visible: bool = True
```

Whether this marker is rendered on the map.

# RichAttribution

An animated and interactive attribution layer that supports both images and text (displayed in a popup controlled by an icon button adjacent to the images).

Inherits: MapLayer

## Properties

- **alignment** - The position in which to anchor this attribution control.
- **attributions** - List of attributions to display.
- **permanent\_height** - The height of the permanent row in which is found the popup menu toggle button.
- **popup\_bgcolor** - The color to use as the popup box's background color.
- **popup\_border\_radius** - The radius of the edges of the popup box.
- **popup\_initial\_display\_duration** - The popup box will be open by default and be hidden this long after the map is initialised.
- **show\_flutter\_map\_attribution** - Whether to add an additional attribution logo and text for flutter-map, on which 'flet-map' package is based for map-renderings.

## Properties

### buildalignmentclass-attributeinstance-attribute

`alignment: AttributionAlignment | None = None`

The position in which to anchor this attribution control.

### buildattributionsinstance-attribute

`attributions: list[SourceAttribution]`

List of attributions to display.

TextSourceAttributions are shown in a popup box, unlike ImageSourceAttribution, which are visible permanently.

### buildpermanent\_heightclass-attributeinstance-attribute

`permanent_height: Number = 24.0`

The height of the permanent row in which is found the popup menu toggle button. Also determines spacing between the items within the row.

### buildpopup\_bgcolorclass-attributeinstance-attribute

`popup_bgcolor: ColorValue | None = Colors.SURFACE`

The color to use as the popup box's background color.

### buildpopup\_border\_radiusclass-attributeinstance-attribute

`popup_border_radius: BorderRadiusValue | None = None`

The radius of the edges of the popup box.

### buildpopup\_initial\_display\_durationclass-attributeinstance-attribute

`popup_initial_display_duration: DurationValue = field(  
 default_factory=lambda: Duration()  
)`

The popup box will be open by default and be hidden this long after the map is initialised.

This is useful with certain sources/tile servers that make immediate attribution mandatory and are not attributed with a permanently visible ImageSourceAttribution.

### buildshow\_flutter\_map\_attributionclass-attributeinstance-attribute

`show_flutter_map_attribution: bool = True`

Whether to add an additional attribution logo and text for flutter-map, on which 'flet-map' package is based for map-renderings.

# RotatedOverlayImage

An image overlay transformed across three corner points.

The top-right corner is derived from `top_left_corner`, `bottom_left_corner`, and `bottom_right_corner`.

Inherits: `BaseOverlayImage`

## Properties

- `bottom_left_corner` - The coordinates of the bottom-left corner of the image.
- `bottom_right_corner` - The coordinates of the bottom-right corner of the image.
- `top_left_corner` - The coordinates of the top-left corner of the image.

## Properties

### `buildbottom_left_cornerinstance-attribute`

`bottom_left_corner: MapLatitudeLongitude`

The coordinates of the bottom-left corner of the image.

### `buildbottom_right_cornerinstance-attribute`

`bottom_right_corner: MapLatitudeLongitude`

The coordinates of the bottom-right corner of the image.

### `buildtop_left_cornerinstance-attribute`

`top_left_corner: MapLatitudeLongitude`

The coordinates of the top-left corner of the image.

## Examples

### Overlay Images

`play_arrow` Try Online

```
import flet as ft
import flet_map as ftm
```

```
base64_image = "iVBORw0KGgoAAAANSUhEUgAAABkAAAgCAYAAADnnNMGAAACXBIIWMAAAOAAADkQFnq8zdAAAAGXRFWHRTb2Z0d2FyZQB3d3cuaw5rc2NhcGUubG9vc2Rlcj0AAAC0AAAA"

```

```
def main(page: ft.Page):
 page.add(
 ft.SafeArea(
 expand=True,
 content=ftm.Map(
 expand=True,
 initial_center=ftm.MapLatitudeLongitude(51.5, -0.09),
 initial_zoom=6,
 layers=[
 ftm.TileLayer(
 url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
 user_agent_package_name="flet-map-examples/1.0",
),
 ftm.OverlayImageLayer(
 overlay_images=[
 ftm.OverlayImage(
 src=base64_image,
 bounds=ftm.MapLatitudeLongitudeBounds(
 corner_1=ftm.MapLatitudeLongitude(51.5, -0.09),
 corner_2=ftm.MapLatitudeLongitude(48.8566, 2.3522),
),
 opacity=0.8,
),
 ftm.RotatedOverlayImage(
 src=base64_image,
 top_left_corner=ftm.MapLatitudeLongitude(
 53.377, -2.999
),
 bottom_left_corner=ftm.MapLatitudeLongitude(
 52.503, -1.868
),

```

```

 bottom_right_corner=ftm.MapLatitudeLongitude(
 53.475, 0.275
),
 opacity=0.8,
],
),
ftm.SimpleAttribution(text="OpenStreetMap contributors"),
],
),
)

if __name__ == "__main__":
 ft.run(main)

```

# SimpleAttribution

A simple attribution layer displayed on the Map.

Inherits: MapLayer

## Properties

- `alignment` - The alignment of this attribution on the map.
- `bgcolor` - The color of the box containing the text.
- `text` - The attribution message to be displayed.
- `text_style` - The style to use for text.

## Events

- `on_click` - Fired when this attribution is clicked/pressed.

## Properties

### `buildalignmentclass-attributeinstance-attribution`

```
alignment: Alignment = field(
 default_factory=lambda: Alignment.BOTTOM_RIGHT
)
```

The alignment of this attribution on the map.

### `buildbgcolorclass-attributeinstance-attribution`

```
bgcolor: ColorValue = Colors.SURFACE
```

The color of the box containing the text.

### `buildtextinstance-attribution`

```
text: str
```

The attribution message to be displayed.

### `buildtext_styleclass-attributeinstance-attribution`

```
text_style: TextStyle | None = None
```

The style to use for text.

## Events

### `bolton_clickclass-attributeinstance-attribution`

```
on_click: (
 ControlEventHandler[SimpleAttribution] | None
) = None
```

Fired when this attribution is clicked/pressed.

# SourceAttribution

Abstract class for source attribution controls:

- ImageSourceAttribution
- TextSourceAttribution

Inherits: BaseControl

# TextSourceAttribution

A text source attribution displayed on the Map. For it to be displayed, it should be part of a RichAttribution.attributions list.

Inherits: SourceAttribution

## Properties

- `prepend_copyright` - Whether to add the '©' character to the start of text automatically.
- `text` - The text to display as attribution, styled with `text_style`.
- `text_style` - Style used to display the text.

## Events

- `on_click` - Fired when this attribution is clicked/pressed.

## Properties

### `buildprepend_copyrightclass-attributeinstance-attribute`

`prepend_copyright: bool = True`

Whether to add the '©' character to the start of text automatically.

### `buildtextinstance-attribute`

`text: str`

The text to display as attribution, styled with `text_style`.

### `buildtext_styleclass-attributeinstance-attribute`

`text_style: TextStyle | None = None`

Style used to display the text.

## Events

### `bolton_clickclass-attributeinstance-attribute`

`on_click: (ControlEventHandler[TextSourceAttribution] | None) = None`

Fired when this attribution is clicked/pressed.

# TileLayer

Displays square raster images in a continuous grid, sourced from the provided `url_template` and `fallback_url`.

Typically, the first layer to be added to a Map, as it provides the tiles on which other layers are displayed.

## Caching

This control supports basic map tile caching (for compatible tile providers). On non-web platforms, built-in caching is automatically enabled with a default soft limit of 1 GB. On web platforms, caching is typically handled by the browser.

No guarantees are provided regarding the persistence or reliability of cached tiles. Cached data may become unavailable or be cleared at any time. For example, do not rely on this caching mechanism in scenarios where missing tiles could create risk or unsafe conditions (for example, offline or safety-critical mapping applications).

It aims to:

Improve developer experience by:

- Reducing the costs of using tile servers by reducing duplicate tile requests
- Keep your app lightweight - the built-in cache doesn't ship any binaries or databases, just a couple extra libraries you probably already use

Improve user experience by:

- Reducing tile loading durations, as fetching from the cache is very quick
- Reducing network/Internet usage, which may be limited or metered/expensive (eg. mobile broadband)
- Improve compliance with tile server requirements, by reducing the strain on them
- Be extensible, customizable, and integrate with multiple tile providers

But it comes at the expense of usage of on-device storage capacity.

## Supported sources

flet-map doesn't provide tiles, so you'll need to bring your own raster tiles. There are multiple different supported sources.

- Slippy Map/CARTO (XYZ): This is the most common format for raster tiles, although many satellite tiles will instead use WMS. Typically, a URL with placeholders for X, Y, and Z values. Set the `url_template` to the template provided by the tile server - usually it can be copied directly from an account portal or documentation. Additional information, like API/access keys, can be passed in using the `additional_options` property. It's also possible to specify a `fallback_url` template, used if fetching a tile from the primary `url_template` fails.
- Tile Map Service (TMS): This is also supported. Follow the instructions for the XYZ source above, then set the `enable_tms` property to True. Read more on WMS [here](#).
- Web Map Services (WMS): This is also supported. Use `wms_configuration` to specify the necessary configuration for WMS tile servers. Read more on WMS [here](#).

Example:

```
ftm.TileLayer(
 url_template="https://tile.openstreetmap.org/{z}/{x}/{y}.png",
 user_agent_package_name="MyTownMaps/1.4 (+https://example.org; contact: )",
)
```

Inherits: MapLayer

## Properties

- `additional_options` - Static information that should replace placeholders in the `url_template`.
- `display_mode` - Defines how tiles are displayed on the map.
- `enable_retina_mode` - Whether to enable retina mode.
- `enable_tms` - Whether to inverse Y-axis numbering for tiles.
- `error_image_src` - The source of the tile image to show in place of the tile that failed to load.
- `evict_error_tile_strategy` - If a tile was loaded with error, the tile provider will be asked to evict the image based on this strategy.



- `fallback_url` - Fallback URL template used if fetching tiles from `url_template` fails.
- `keep_buffer` - When panning the map, keep this many rows and columns of tiles before unloading them.
- `max_native_zoom` - Maximum zoom number supported by the tile source has available.
- `max_zoom` - The maximum zoom level up to which this layer will be displayed (inclusive).
- `min_native_zoom` - Minimum zoom level supported by the tile source.
- `min_zoom` - The minimum zoom level at which this layer is displayed (inclusive).
- `pan_buffer` - When loading tiles only visible tiles are loaded by default.
- `subdomains` - List of subdomains used in the URL template.
- `tile_bounds` - Defines the bounds of the map.
- `tile_size` - The size in pixels of each tile image.
- `url_template` - The URL template is a string that contains placeholders, which, when filled in, create a URL/URI to a specific tile.
- `user_agent_package_name` - The package name of the user agent to use when fetching tiles from the tile server.
- `wms_configuration` - The configuration for WMS tile servers.
- `zoom_offset` - The zoom number used in tile URLs will be offset with this value.
- `zoom_reverse` - Whether the zoom number used in tile URLs will be reversed (`max_zoom - zoom` instead of `zoom`).

## Events

- `on_image_error` - Fires if an error occurs when fetching the tiles.

## Properties

### `buildadditional_optionsclass-attributeinstance-attribute`

```
additional_options: dict[str, str] = field(
 default_factory=dict
)
```

Static information that should replace placeholders in the `url_template`. Applying API keys, for example, is a good usecase of this parameter.

#### Example

```
TileLayer(
 url_template="https://api.tiles.mapbox.com/v4/{id}/{z}/{x}/{y}/{r}.png?access_token={accessToken}",
 additional_options={
 'accessToken': '<ACCESS_TOKEN_HERE>',
 'id': 'mapbox.streets',
 },
)
```

### `bulddisplay_modeclass-attributeinstance-attribute`

```
display_mode: TileDisplay = field(
 default_factory=lambda: FadeInTileDisplay()
)
```

Defines how tiles are displayed on the map.

### `buildenable_retina_modeclass-attributeinstance-attribute`

```
enable_retina_mode: bool = False
```

Whether to enable retina mode. Retina mode improves the resolution of map tiles, particularly on high density displays.

### `buildenable_tmsclass-attributeinstance-attribute`

```
enable_tms: bool = False
```

Whether to inverse Y-axis numbering for tiles. Turn this on for TMS services.

### `builderror_image_srcclass-attributeinstance-attribute`

```
error_image_src: str | None = None
```

The source of the tile image to show in place of the tile that failed to load.

See `on_image_error` property for details on the error.

## **builderror\_tile\_strategyclass-attributeinstance-attribute**

```
evict_error_tile_strategy: (
 TileLayerEvictErrorTileStrategy | None
) = TileLayerEvictErrorTileStrategy.NONE
```

If a tile was loaded with error, the tile provider will be asked to evict the image based on this strategy.

## **buildfallback\_urlclass-attributeinstance-attribute**

```
fallback_url: str | None = None
```

Fallback URL template used if fetching tiles from `url_template` fails.

The template must follow the same format and support the same placeholders as `url_template`.

### **Note**

When this is specified, tiles will not be cached in memory, to prevent inconsistencies when `url_template` is unreliable, avoiding situations where tiles from different sources are displayed simultaneously. Disabling caching may negatively impact performance and efficiency, hence the recommendation to only specify a fallback URL when really necessary.

## **buildkeep\_bufferclass-attributeinstance-attribute**

```
keep_buffer: int = 2
```

When panning the map, keep this many rows and columns of tiles before unloading them.

## **buildmax\_native\_zoomclass-attributeinstance-attribute**

```
max_native_zoom: int = 19
```

Maximum zoom number supported by the tile source has available.

Tiles from above this zoom level will not be displayed, instead tiles at this zoom level will be displayed and scaled.

Most tile servers support up to zoom level 19, which is the default. Otherwise, this should be specified.

You can also set `max_zoom`, which is an absolute zoom limit for users. It is recommended to set it to a few levels greater than the maximum zoom level covered by any of your tile layers.

Raises:

- `ValueError` - If it is less than 0.0.

## **buildmax\_zoomclass-attributeinstance-attribute**

```
max_zoom: Number = float('inf')
```

The maximum zoom level up to which this layer will be displayed (inclusive). The main usage for this property is to display a different `TileLayer` when zoomed far in.

Prefer `max_native_zoom` for setting the maximum zoom level supported by the tile source.

Typically set to infinity so that there are tiles always displayed.

Raises:

- `ValueError` - If it is less than 0.0.

## **buildmin\_native\_zoomclass-attributeinstance-attribute**

```
min_native_zoom: int = 0
```

Minimum zoom level supported by the tile source.

Tiles from below this zoom level will not be displayed, instead tiles at this zoom level will be displayed and scaled.

This should usually be 0 (as default), as most tile sources will support zoom levels onwards from this.

Raises:

- `ValueError` - If it is less than 0.0.

## **buildmin\_zoomclass-attributeinstance-attribute**

```
min_zoom: Number = 0.0
```

The minimum zoom level at which this layer is displayed (inclusive).

Typically 0.0.

Raises:

- `ValueError` - If it is less than 0.0.

### `buildpan_bufferclass-attributeinstance-attribute`

```
pan_buffer: int = 1
```

When loading tiles only visible tiles are loaded by default.

This option increases the loaded tiles by the given number on both axis which can help prevent the user from seeing loading tiles whilst panning. Setting the pan buffer too high can impact performance, typically this is set to 0 or 1.

### `buildsubdomainsclass-attributeinstance-attribute`

```
subdomains: list[str] = field(
 default_factory=lambda: ["a", "b", "c"]
)
```

List of subdomains used in the URL template.

To use subdomains, add the `{s}` placeholder to the URL template (`url_template` and `fallback_url`)

Note

Subdomains are now usually considered redundant due to the usage of HTTP/2 & HTTP/3 which don't have the same restrictions. Usage of subdomains will also hinder the ability to cache tiles, potentially leading to increased tile requests and costs. Hence, if the server supports HTTP/2 or HTTP/3 (how to check), avoid using subdomains.

Example

If subdomains is set to `["a", "b", "c"]` and the `url_template` is `"https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png"`, the resulting tile URLs will be:

- `"https://a.tile.openstreetmap.org/{z}/{x}/{y}.png"`
- `"https://b.tile.openstreetmap.org/{z}/{x}/{y}.png"`
- `"https://c.tile.openstreetmap.org/{z}/{x}/{y}.png"`

### `buildtile_boundsclass-attributeinstance-attribute`

```
tile_bounds: MapLatitudeLongitudeBounds | None = None
```

Defines the bounds of the map. Only tiles that fall within these bounds will be loaded.

### `buildtile_sizeclass-attributeinstance-attribute`

```
tile_size: int = 256
```

The size in pixels of each tile image. Should be a positive power of 2.

Note

Some tile servers will use 512x512px tiles instead of 256x256px, such as Mapbox. Using these larger tiles can help reduce tile requests, and when combined with Retina Mode, it can give the same resolution.

To use these tiles, set `tile_size` to the actual dimensions of the tiles (otherwise they will appear too small), such as 512. Also set `zoom_offset` to the result of  $-(d/256) - 1$  - ie. -1 for 512px tiles (otherwise they will appear at the wrong geographical locations).

The `{d}` placeholder may also be used in the URL template (`url_template` and `fallback_url`) to pass through the value of `tile_size`.

Raises:

- `ValueError` - If it is less than 0.0.

### `buildurl_templateinstance-attribute`

```
url_template: str
```

The URL template is a string that contains placeholders, which, when filled in, create a URL/URI to a specific tile.

Provider Examples: [https://wiki.openstreetmap.org/wiki/Raster\\_tile\\_providers](https://wiki.openstreetmap.org/wiki/Raster_tile_providers)

Placeholders:

As well as the standard XYZ placeholders in the template, the following placeholders may also be used:

- `{s}`: see subdomains property

- {r}: retina scaling factor (2 or 1)
- {d}: reflects the tile\_size property

Additional placeholders can also be added freely to the template, and are filled in with the specified values in `additional_options`. This can be used to easier add switchable styles or access tokens.

Compliance with tile server requirements:

It is your own responsibility to comply with any appropriate restrictions and requirements set by your chosen tile server/provider. Always read their terms of service. Failure to do so may lead to any punishment, at the tile server's discretion.

Production apps should be extremely cautious about using this tile server; other projects, libraries, and packages suggesting that OpenStreetMap provides free-to-use map tiles are incorrect.

Case Example - OpenStreetMap (direct): OpenStreetMap (OSM) is one of the most popular sources for map tiles and data. Their data is free for everyone to use (under ODbL), but their public tile server is not free for everyone to use. It is without cost (for users), but, "without cost" ≠ "without restriction" ≠ "open". Due to excessive usage, the OSM Foundation (running OSM as a not-for-profit) has implemented some measures to prevent abuse and ensure the sustainability of their service.

For example: on non-web platforms (ex: desktop), they require a proper User-Agent header to be set. See the `user_agent_package_name` property for details and recommended best practices. This does not apply to the web platform, because you cannot set a User-Agent header different to what is provided by the browser.

Read more on their tile usage policy [here](#).

## builduser\_agent\_package\_nameclass-attributeinstance-attribute

```
user_agent_package_name: str = 'unknown'
```

The package name of the user agent to use when fetching tiles from the tile server.

This is used to identify your app to the tile server, and is important for compliance with tile server usage policies.

OSM best practice recommendations

OpenStreetMap (OSM) recommends the following:

Use a clear, unique User-Agent string that names your app and optionally includes a contact URL or email.

- Good Example: MyTownMaps/1.4 (+<https://example.org>; contact: [email protected])
- Bad Example: com.example.app
- For web apps, the browsers will use the browser's default User-Agent header.
- Do not use a library default User-Agent, and never impersonate another app or a browser.
- If your platform automatically sets an X-Requested-With header with an app ID, that is acceptable, but a proper User-Agent is still recommended.
- Referer (web only): Browsers are expected to send a valid Referer header. Native apps usually do not have a referer, this is ok.

## buildwms\_configurationclass-attributeinstance-attribute

```
wms_configuration: WMSTileLayerConfiguration | None = None
```

The configuration for WMS tile servers.

## buildzoom\_offsetclass-attributeinstance-attribute

```
zoom_offset: Number = 0.0
```

The zoom number used in tile URLs will be offset with this value.

Raises:

- ValueError - If it is less than 0.0.

## buildzoom\_reverseclass-attributeinstance-attribute

```
zoom_reverse: bool = False
```

Whether the zoom number used in tile URLs will be reversed (`max_zoom - zoom` instead of `zoom`).

## Events

### `bolton_image_error`class-attributeinstance-attribute

```
on_image_error: ControlEventHandler[TileLayer] | None = (
 None
)
```

Fires if an error occurs when fetching the tiles.

Event handler argument data property contains information about the error.

# ChasingDots

Two dots that chase each other in a circular path.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# Circle

A ring of dots that fade sequentially around a circle.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# CubeGrid

A 3x3 grid of cubes that scale and fade in a wave.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.



# DancingSquare

A square that dances by rotating and scaling.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# DoubleBounce

Two overlapping circles that bounce in and out.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# DualRing

Two concentric rings that spin in opposite directions.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `line_width` - The stroke width of the rings.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildline_widthclass-attributeinstance-attribute`

`line_width: Number | None = None`

The stroke width of the rings. Defaults to 7.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# FadingCircle

A ring of dots that fade sequentially.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# FadingCube

A cube made of four smaller cubes that fade and rotate.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# FadingFour

Four dots arranged in a square that fade in sequence.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# FadingGrid

A 3x3 grid of dots that fade in a spiral sequence.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# FoldingCube

Four cubes that fold and unfold in sequence.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.



# HourGlass

An hourglass shape that rotates.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# PianoWave

A row of bars that animate like piano keys.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `item_count` - Number of bars.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `builditem_countclass-attributeinstance-attribute`

`item_count: int | None = None`

Number of bars. Defaults to 5.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# PouringHourGlass

An hourglass that pours dots from top to bottom.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# PouringHourGlassRefined

A refined version of the pouring hourglass with smoother animation.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# Pulse

A circle that pulses in and out.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# PulsingGrid

A 3x3 grid of dots that pulse in sequence.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# PumpingHeart

A heart shape that pumps in and out.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# Ring

A ring that spins with a gap.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `line_width` - The stroke width of the ring.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildline_widthclass-attributeinstance-attribute`

`line_width: Number | None = None`

The stroke width of the ring. Defaults to 7.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.



# Ripple

Two concentric circles that expand outward like a ripple.

Inherits: `LayoutControl`

## Properties

- `border_width` - The stroke width of the ripple rings.
- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

**buildborder\_widthclass-attributeinstance-attribute**

`border_width: Number | None = None`

The stroke width of the ripple rings. Defaults to 6.

**buildcolorclass-attributeinstance-attribute**

`color: ColorValue | None = None`

The color of the spinner.

**bulddurationclass-attributeinstance-attribute**

`duration: DurationValue | None = None`

The duration of one animation cycle.

**buildsizeclass-attributeinstance-attribute**

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# RotatingCircle

A circle that rotates with a color gradient effect.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# RotatingPlain

A plain rotating square spinner.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# SpinningCircle

A circle that spins continuously.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# SpinningLines

Multiple lines radiating from the center that spin.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `line_width` - The stroke width of the lines.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildline_widthclass-attributeinstance-attribute`

`line_width: Number | None = None`

The stroke width of the lines. Defaults to 2.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# SquareCircle

A square and a circle that morph between shapes.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# ThreeBounce

Three dots that bounce sequentially.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# ThreeInOut

Three dots that move in and out of view.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.



# WanderingCubes

Two cubes that rotate and wander around each other.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# Wave

A row of bars that animate in a wave pattern.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `item_count` - Number of bars in the wave.
- `size` - The size of the spinner in pixels.
- `wave_type` - Controls the wave propagation direction.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `builddurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `builditem_countclass-attributeinstance-attribute`

`item_count: int | None = None`

Number of bars in the wave. Defaults to 5.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

### `buildwave_typeclass-attributeinstance-attribute`

`wave_type: WaveType | None = None`

Controls the wave propagation direction.

Defaults to `WaveType.START`.

# WaveSpinner

A circular wave spinner.

Inherits: `LayoutControl`

## Properties

- `color` - The color of the spinner.
- `duration` - The duration of one animation cycle.
- `size` - The size of the spinner in pixels.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of the spinner.

### `bulldurationclass-attributeinstance-attribute`

`duration: DurationValue | None = None`

The duration of one animation cycle.

### `buildsizeclass-attributeinstance-attribute`

`size: Number | None = None`

The size of the spinner in pixels. Defaults to 50.

# WaveType

Controls how the wave animation is aligned.

Inherits: `enum.Enum`

## Properties

- **CENTER** - Wave animation radiates from the center.
- **END** - Wave animation starts from the last item.
- **START** - Wave animation starts from the first item.

## Properties

**buildCENTER**`class-attributeinstance-attribute`

Wave animation radiates from the center.

**buildEND**`class-attributeinstance-attribute`

Wave animation starts from the last item.

**buildSTART**`class-attributeinstance-attribute`

Wave animation starts from the first item.

# Publish web app as dynamic website

Flet runs your app as a FastAPI (ASGI) app to serve a dynamic website.

It uses Uvicorn by default, but any ASGI-compatible server can be used instead.

## Sync and async handlers

In a Flet web app you can mix sync and async handlers.

For example, you can write an app like this:

```
import flet as ft
import time
import asyncio

def main(page: ft.Page):

 def handler(e):
 time.sleep(3)
 page.add(ft.Text("Handler clicked"))

 async def handler_async(e):
 await asyncio.sleep(3)
 page.add(ft.Text("Async handler clicked"))

 page.add(
 ft.Button("Call handler", on_click=handler),
 ft.Button("Call async handler", on_click=handler_async)
)

ft.run(main)
```

In the example, a click on one button is handled by a blocking handler while a click on the second button calls an async handler. The first handler runs in a `threading.Thread`, while the second runs as an `asyncio.Task`.

In web apps, threads are a finite resource. A thread pool is usually used and can become a bottleneck as the number of users grows.

If your app is mostly doing I/O (database, web API) and you can use async-ready libraries, prefer async handlers.

Check FastAPI's article about `async/await` to better understand the differences between concurrency and parallelism.

## Running the app locally

Use `--web (-w)` option to start a Flet app as a web app:

```
flet run --web app.py
```

A new browser window/tab will be opened and the app will use a random TCP port.

To run on a fixed port use `--port (-p)`:

```
flet run --web --port 8000 app.py
```

## Running the app in production

You can run your program directly with python:

```
python app.py
```

Uvicorn web server is used by default to host the web app.

If Flet detects a headless Linux environment (such as a Docker container or EC2 VM):

- Port 8000 will be used to run the app.
- A browser window will not be opened.

If Flet cannot detect a headless environment, you can force that behavior by defining the following environment variable:

```
FLET_FORCE_WEB_SERVER=true
```

## ASGI web server

You can host your Flet web app with any ASGI-compatible server such as Uvicorn (used by default), Hypercorn or Daphne.

Just tell Flet to export ASGI app:

main.py

```
import flet as ft

def main(page: ft.Page):
 page.add(ft.Text("Hello ASGI!"))

app = ft.run(main, export_asgi_app=True)
```

## Hypercorn

Hypercorn is another ASGI web server inspired by Gunicorn.

To run the app with Hypercorn:

```
hypercorn main:app --bind 0.0.0.0:8000
```

## Daphne

Daphne is a HTTP, HTTP2 and WebSocket protocol server for ASGI and ASGI-HTTP, developed to power Django Channels.

To run the app with Daphne:

```
daphne -b 0.0.0.0 -p 8000 main:app
```

## Gunicorn

Gunicorn is popular web server to run Python web applications. While it implements WSGI specification it's possible to run ASGI FastAPI apps with a worker process class provided by Uvicorn:

```
gunicorn --bind 0.0.0.0:8000 -k uvicorn.workers.UvicornWorker counter:app
```

## Assets

When you open a Flet app in the browser, its index.html, Flutter engine, favicon, images and other web app resources are served by a web server. These resources are bundled with flet Python package. However, there are situations when you need to modify the contents of certain files to customize the appearance of your app or its behavior, for example:

- Favicon.
- App loading animation.
- manifest.json with PWA details.

You can specify `assets_dir` in `flet.run()` to set the location of assets that should be available to the application. `assets_dir` should be relative to your `main.py` directory or an absolute path. Default value for `assets_dir` argument is `assets`.

For example, consider the following program structure:

```
/assets
 /images/my-image.png
main.py
```

You can access your images in the application as follows:

```
import flet as ft

def main(page: ft.Page):
 page.add(ft.Image(src=f"/images/my-image.png"))

ft.run(main, assets_dir="assets")
```

## Customizing web app

### Favicon

To override favicon with your own put `favicon.png` file into the root of `assets` directory. It should be a PNG image with the size of at least 32x32 pixels.

### Loading animation

To override the Flet animation image, put `icons/loading-animation.png` with your own app logo in the root of the `assets` directory.

### PWA

Progressive Web Apps, or PWAs, offer a way to turn app-like websites into website-like apps.

Check PWAs Turn Websites Into Apps: Here's How for the PWA introduction.

Browsers that support PWA (installation instructions):

- Chrome on all platforms
- Edge on all platforms
- Firefox on Android
- Safari on iOS and iPadOS

Info

The information in this section is based on the following sources (check them out for more details):

- General information about PWAs
- PWA manifests

## Manifest

You can change PWA's name, description, colors and other information in manifest.json that must be put in the root of assets directory.

Here are the links to the most common manifest items that you'd like to customize:

- name - the name of the web application as it is usually displayed to the user.
- short\_name - the name of the web application displayed to the user if there is not enough space to display name.
- description - explains what the application does.
- theme\_color - defines the default theme color for the application.
- background\_color - defines a placeholder background color for the application page to display before its stylesheet is loaded.

## Icons

Custom icons should be placed in assets/icons directory:

- icon-192.png, icon-512.png - app icons displayed in Windows taskbar.
- icon-maskable-192.png, icon-maskable-512.png - app icons displayed in Android.
- apple-touch-icon-192.png - app icon displayed in iOS.

## Environment variables

Every aspect of web app hosting can be additionally controlled with environment variables.

## Advanced FastAPI scenarios

### Flet FastAPI app

flet.fastapi.app() creates a FastAPI application to handle Flet sessions and mounts the following endpoints at the app root:

- /ws (WS) - WebSocket handler for the Flet app. It calls main() when a new WebSocket connection is established and a new app session is created.
- /upload (PUT) - file uploads handler.
- /oauth\_callback (GET) - OAuth flow callback handler.
- / (GET) - Flet app static files with SPA catch-all handler.

### Hosting multiple Flet apps under the same domain

```
import flet as ft
import flet.fastapi as flet_fastapi

async def root_main(page: ft.Page):
 page.add(ft.Text("This is root app!"))

async def sub_main(page: ft.Page):
 page.add(ft.Text("This is sub app!"))

app = flet_fastapi.FastAPI()
app.mount("/sub-app", flet_fastapi.app(sub_main))
```

```
app.mount("/", flet_fastapi.app(root_main))
```

Sub-apps must be mapped before the root Flet app as it configures catch-all index.html for SPA.

Run the app with uvicorn and visit <http://127.0.0.1:8000> and then <http://127.0.0.1:8000/sub-app/> to see both Flet apps running.

Notice the trailing slash in /sub-app/ - without it the request will be routed to the root app.

## Adding Flet to the existing FastAPI app

```
from contextlib import asynccontextmanager

import flet as ft
import flet.fastapi as flet_fastapi
from fastapi import FastAPI

@asynccontextmanager
async def lifespan(app: FastAPI):
 await flet_fastapi.app_manager.start()
 yield
 await flet_fastapi.app_manager.shutdown()

app = FastAPI(lifespan=lifespan)

async def main(page: ft.Page):
 page.add(ft.Text("Hello, Flet!"))

app.mount("/flet-app", flet_fastapi.app(main))
```

When adding a Flet app to an existing FastAPI app you need to call `flet_fastapi.app_manager.start()` on app start and `flet_fastapi.app_manager.shutdown()` on shutdown. Use the way that best suits you: `lifespan` (in the example above) or `app events`.

`app_manager.start()` method starts background tasks cleaning up expired sessions and OAuth flow states.

`app_manager.shutdown()` method removes any temporary files created by a Flet app.

## Configuring individual Flet endpoints

### Static files

A FastAPI app to serve static Flet app files (index.html, manifest.json, Flutter JS app, etc.) and user assets.

```
from flet.fastapi import FastAPI, FletStaticFiles

app = FastAPI()

mount to the root of web app
app.mount(path="/", app=FletStaticFiles())
```

### WebSocket handler

Handles WebSocket connections from the Flet client running in the browser. The WebSocket channel is used to send events from the browser to the backend and receive page updates.

```
import asyncio
import flet as ft
from flet.fastapi import FletApp, app_manager

async def main(page: ft.Page):
 page.add(ft.Text("Hello, Flet!"))

@app.websocket("/app1/ws")
async def flet_app(websocket: WebSocket):
 await FletApp(
 loop=asyncio.get_running_loop(),
 executor=app_manager.executor,
 main=main,
 before_main=None,
).handle(websocket)
```

### Uploads handler

Handles file uploads by `FilePicker` control. This endpoint is optional - if your app doesn't use `FilePicker`, then it's unnecessary.

```
from flet.fastapi import FletUpload

@app.put("/upload")
async def flet_uploads(request: Request):
 await FletUpload("/Users/feodor/Downloads/123").handle(request)
```



## OAuth callback handler

Handles OAuth flow callback requests. If your app doesn't use authentication, then this endpoint is unnecessary.

```
from flet.fastapi import FletOAuth

@app.get("/oauth_callback")
async def flet_oauth(request: Request):
 return await FletOAuth().handle(request)
```

# Publish web app as static website

Instructions for publishing a Flet app as a standalone static website (SPA) that runs entirely in the browser with Pyodide. No Python code runs on the server.

Pyodide is a port of CPython to WebAssembly (WASM) and has some limitations.

### Native Python packages

Native Python packages (vs "pure" Python packages written in Python only) are packages partially written in C, Rust, or other languages producing native code. Example packages are numpy, cryptography, lxml, pydantic-core.

Pyodide comes with a list of built-in packages. To use a package from PyPI, it must be pure Python or provide a wheel built for Emscripten.

### Async and threading

Static websites run in a single browser thread. You can use sync and async handlers, but long-running CPU work or blocking calls will freeze the UI. Prefer async I/O, or move heavy work to a server and call it via a web API.

### Micropip

Pyodide installs packages with micropip. You can use the Micropip API directly in your Flet app:

```
import sys

if sys.platform == "emscripten":
 import micropip
 await micropip.install("regex")
```

## Differences

There are two ways to publish a static website: flet build web and flet publish. Both produce a static site that runs in the browser via Pyodide. They differ mainly in how and when Python dependencies are installed:

|                              | No                                                        | Yes                                                            |
|------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|
| Flutter required             | No                                                        | Yes                                                            |
| Dependency install           | At runtime, in the browser ()                             | At build time, on your machine ()                              |
| Build time                   | Faster — no Flutter compilation, no local pip install     | Slower — Flutter build + local dependency install              |
| Initial load time            | Slower — wheels are fetched from PyPI on page load        | Faster — dependencies are already bundled                      |
| Pure-Python wheels           | <input type="checkbox"/>                                  | <input type="checkbox"/>                                       |
| Pyodide-built binary wheels  | <input type="checkbox"/> (from Pyodide CDN)               | <input type="checkbox"/> (auto-detected from Pyodide registry) |
| Source distributions (sdist) | <input type="checkbox"/> can't build sdist in the browser | <input type="checkbox"/> pure-Python sdist, opt in via         |

flet publishflet build webmicropippipmicropipssource\_packages

### Sdist-only dependencies

A common failure mode with flet publish is a (transitive) dependency that ships only a source distribution (.tar.gz, no wheel) — for example, docopt. micropip cannot build sdist, so load fails with:

```
ValueError: Can't find a pure Python 3 wheel for '<package>'
```

In such cases, we suggest switching to flet build web and adding the package to source\_packages. Note that source\_packages only works for pure-Python sdist. Sdist with C/Rust extensions (e.g. numpy, cryptography) cannot be built for Pyodide — use Pyodide's built-in packages instead.

## flet publish

Does not require Flutter. It packages your app and installs dependencies in the browser at runtime via micropip.

To publish an app, run:

```
flet publish <path-to-app.py>
```

The website is published to --distpath (default: ./dist).

### Testing the site

You can try published Flet app using flet serve command:

```
flet serve dist
```

Then, open <http://localhost:8000> in your browser to check the published app.

## Assets

If the assets directory exists (default: `./assets`), its contents are copied to the published site root. Use `--assets` to point to a different folder. Assets are not packaged inside the `app.tar.gz`.

## flet build web

Uses Flutter and Pyodide. Dependencies are resolved and installed locally at build time (via pip), then bundled into the output archive.

Note

Complementary and more general information is available [here](#).

## Testing the site

flet serve serves the default output directory (`./build/web`):

```
flet serve
```

## Configuration options

These settings apply to flet build web and flet publish, unless noted.

### Base URL

Use a base URL when hosting your app in a subdirectory. Flet normalizes it to `<value>/` and uses `/` when unset.

### Resolution order

Its value is determined in the following order of precedence:

- `--base-url`
- `[tool.flet.web].base_url`
- `"/`

### Example

- flet build
- pyproject.toml

```
flet build web --base-url /myapp/

[tool.flet.web]
base_url = "/myapp/"
```

### Route URL strategy

Controls how routes are represented in the URL:

- `path` - clean paths; requires SPA-capable hosting.
- `hash` - uses the URL hash; works on static hosts without SPA support.

### Resolution order

Its value is determined in the following order of precedence:

- `--route-url-strategy`
- `[tool.flet.web].route_url_strategy`
- `"path"`

### Example

- flet build
- pyproject.toml

```
flet build web --route-url-strategy hash
```

```
[tool.flet.web]
route_url_strategy = "hash"
```

## Web renderer

Selects the Flutter web renderer:

- auto (default) - let Flutter choose the best renderer
- canvaskit
- skwasm

## Resolution order

Its value is determined in the following order of precedence:

- --web-renderer
- [tool.flet.web].renderer
- "auto"

## Example

- flet build
- pyproject.toml

```
flet build web --web-renderer canvaskit

[tool.flet.web]
renderer = "canvaskit"
```

## CDN assets

By default, Pyodide, CanvasKit, and fonts are loaded from CDNs to keep the output small. Disable CDN loading for offline or air-gapped deployments.

## Resolution order

CDN loading is disabled in the following order of precedence:

- --no-cdn
- [tool.flet.web].cdn = false
- default: CDN enabled

## Example

- flet build
- pyproject.toml

```
flet build web --no-cdn

[tool.flet.web]
cdn = false
```

## PWA colors

Configure PWA colors used in manifest.json and browser UI.

## Resolution order

For each setting:

- --pwa-background-color / --pwa-theme-color
- [tool.flet.web].pwa\_background\_color / [tool.flet.web].pwa\_theme\_color
- #FFFFFF / #0175C2

## Example

- flet build
- pyproject.toml

```
flet build web --pwa-background-color "#000000" --pwa-theme-color "#FF0000"
```

```
[tool.flet.web]
pwa_background_color = "#000000"
pwa_theme_color = "#FF0000"
```

## WASM output

By default, flet build web enables Flutter's WASM output.

Note

flet build web only.

## Resolution order

The WASM output is disabled in the following order of precedence:

- --no-wasm
- [tool.flet.web].wasm = false
- default: WASM enabled

## Example

- flet build
- pyproject.toml

```
flet build web --no-wasm

[tool.flet.web]
wasm = false
```

# Auth0OAuthProvider

OAuth provider preset for Auth0.

Configures Auth0 authorization/token endpoints and user-info retrieval via /userinfo, using the sub claim as the user id.

Inherits: `auth.OAuthProvider`

## Properties

- `audience`
- `domain`

## Properties

`buildaudienceinstance-attribute`

`bulddomaininstance-attribute`

# Authorization

Abstract authorization contract used by Flet authentication workflows.

Implementations provide OAuth-style authorization URL/state generation, token exchange, token hydration from persisted storage, and token retrieval.

## Methods

- `dehydrate_token` - Restore token state from previously persisted token data.
- `get_authorization_data` - Build authorization URL and state for starting auth flow.
- `get_token` - Return the current token, refreshing it when needed.
- `request_token` - Exchange authorization code for access/refresh token data.

## Methods

### `deployed_code``dehydrate_token``as_async`

```
dehydrate_token(saved_token: str) -> None
```

Restore token state from previously persisted token data.

Parameters:

- `saved_token (str)` - Serialized token payload.

### `deployed_code``get_authorization_data`

```
get_authorization_data() -> tuple[str, str]
```

Build authorization URL and state for starting auth flow.

Returns:

- `tuple[str, str]` - A tuple containing authorization URL and generated state value.

### `deployed_code``get_token``as_async`

```
get_token() -> Any
```

Return the current token, refreshing it when needed.

Returns:

- `Any` - Current authorization token object.

### `deployed_code``request_token``as_async`

```
request_token(code: str) -> None
```

Exchange authorization code for access/refresh token data.

Parameters:

- `code (str)` - Authorization code returned by provider redirect.

# AuthorizationService

OAuth authorization implementation used by Page.login.

The service coordinates authorization URL generation, token exchange, token refresh, and optional user/group resolution using the configured OAuthProvider.

Parameters:

- provider (OAuthProvider) - Configured OAuthProvider describing OAuth endpoints, credentials, and optional user/group APIs.
- fetch\_user (bool) - Whether to request provider user profile information.
- fetch\_groups (bool) - Whether to request user groups/roles.
- scope (list[str] | None, default: None) - Initial OAuth scopes. The service augments this list with provider defaults (provider.scopes) and, when enabled, provider user/group scopes.

Inherits: auth.Authorization

## Properties

- fetch\_groups
- fetch\_user
- provider
- scope
- user

## Methods

- dehydrate\_token - Restore and validate previously persisted token state.
- get\_authorization\_data - Generate authorization URL and CSRF state for OAuth redirect flow.
- get\_token - Return current token after applying refresh logic when required.
- request\_token - Exchange authorization code for access token and optional profile data.

## Properties

buildfetch\_groupsinstance-attribute

buildfetch\_userinstance-attribute

buildproviderinstance-attribute

buildscopeinstance-attribute

builduserinstance-attribute

user: User | None = None

## Methods

deployed\_codehydrate\_tokenasync

hydrate\_token(saved\_token: str)

Restore and validate previously persisted token state.

The token is deserialized, refreshed when expired, and optionally used to load user and group metadata.

Parameters:

- saved\_token (str) - JSON-serialized token data produced by auth.oauth\_token.OAuthToken.to\_json.

deployed\_codeget\_authorization\_data

get\_authorization\_data() -> tuple[str, str]



Generate authorization URL and CSRF state for OAuth redirect flow.

Returns:

- tuple[str, str] - A tuple of (authorization\_url, state).

### deployed\_codeget\_tokenasync

get\_token() -> OAuthToken | None

Return current token after applying refresh logic when required.

Returns:

- Current (OAuthToken | None) - class:~flet.auth.OAuthToken, or None if no token is available yet.

### deployed\_coderequest\_tokenasync

request\_token(code: str)

Exchange authorization code for access token and optional profile data.

Parameters:

- code (str) - Provider-issued authorization code returned to redirect URL.

Raises:

- httpx.HTTPStatusError - If token endpoint returns a non-success status.

# AzureOAuthProvider

OAuth provider preset for Microsoft Entra ID (Azure AD).

Uses v2 endpoints for the selected tenant and Microsoft Graph /me for user profile retrieval, with Graph id used as the user id.

Inherits: `auth.OAuthProvider`

## Properties

- `tenant`

## Properties

`buildtenantinstance-attribute`

# GitHubOAuthProvider

OAuth provider preset for GitHub.

Uses GitHub OAuth endpoints and fetches user profile/groups via GitHub REST APIs (/user, /user/emails, /user/teams).

Inherits: `auth.OAuthProvider`

# GoogleOAuthProvider

OAuth provider preset for Google accounts.

Configures Google OAuth endpoints and user-info retrieval via the OpenID Connect userinfo endpoint, using sub as the user id.

Inherits: `auth.OAuthProvider`

# Group

Group or role entry associated with an authenticated User.

The instance behaves like a mutable mapping with provider-specific metadata, while exposing a normalized name attribute commonly used by app logic.

Parameters:

- `kwargs` (`dict[str, Any]`) - Provider-specific group fields to store in the mapping.
- `name` (`str`) - Group name used for display and matching.

Inherits: `dict`

## Properties

- `name` - Human-readable group name.

## Properties

### `buildnameinstance-attribute`

```
name: str = name
```

Human-readable group name.

# OAuthProvider

Base configuration and extension point for OAuth providers in Flet.

Instances describe OAuth endpoints, client credentials, optional PKCE parameters, and optional user/group retrieval behavior used by AuthorizationService.

Parameters:

- `client_id` (str) - OAuth client/application ID issued by the provider.
- `client_secret` (str) - OAuth client secret issued by the provider.
- `authorization_endpoint` (str) - Authorization endpoint URL used to build the login redirect URL.
- `token_endpoint` (str) - Token endpoint URL used for authorization-code and refresh-token exchange.
- `redirect_url` (str) - Redirect/callback URL registered with the provider.
- `scopes` (list[str] | None, default: None) - Base OAuth scopes always requested during login.
- `user_scopes` (list[str] | None, default: None) - Additional scopes requested when user profile loading is enabled.
- `user_endpoint` (str | None, default: None) - Endpoint used to fetch raw user profile data after login.
- `user_id_fn` (Callable | None, default: None) - Function extracting a stable user id from user\_endpoint response data.
- `group_scopes` (list[str] | None, default: None) - Additional scopes requested when group loading is enabled.
- `code_challenge` (str | None, default: None) - PKCE code challenge.
- `code_challenge_method` (str | None, default: None) - PKCE challenge method. For example, S256.
- `code_verifier` (str | None, default: None) - PKCE code verifier used during token exchange.
- `authorization_params` (dict[str, str] | None, default: None) - Extra query parameters appended to authorization URL generation by the OAuth authorization service.

## Properties

- `authorization_endpoint`
- `authorization_params`
- `client_id`
- `client_secret`
- `code_challenge`
- `code_challenge_method`
- `code_verifier`
- `group_scopes`
- `redirect_url`
- `scopes`
- `token_endpoint`
- `user_endpoint`
- `user_id_fn`
- `user_scopes`

## Properties

`buildauthorization_endpointinstance-attribute`

`buildauthorization_paramsinstance-attribute`

`buildclient_idinstance-attribute`

buildclient\_secretinstance-attribute  
buildcode\_challengeinstance-attribute  
buildcode\_challenge\_methodinstance-attribute  
buildcode\_verifierinstance-attribute  
buildgroup\_scopesinstance-attribute  
buildredirect\_urlinstance-attribute  
buildscopesinstance-attribute  
buildtoken\_endpointinstance-attribute  
builduser\_endpointinstance-attribute  
builduser\_id\_fninstance-attribute  
builduser\_scopesinstance-attribute

# OAuthToken

OAuth token payload used by Flet authentication flows.

Stores access/refresh token fields and expiry metadata returned by OAuth token endpoints.

## Properties

- `access_token`
- `expires_at`
- `expires_in`
- `refresh_token`
- `scope`
- `token_type`

## Methods

- `from_json` - Deserializes a token from JSON.
- `to_json` - Serializes this token to a compact JSON string.

## Properties

`buildaccess_tokeninstance-attribute`

`buildexpires_atinstance-attribute`

`buildexpires_ininstance-attribute`

`buildrefresh_tokeninstance-attribute`

`buildscopeinstance-attribute`

`buildtoken_typeinstance-attribute`

## Methods

`deployed_codefrom_jsonstaticmethod`

`from_json(data: str) -> OAuthToken`

Deserializes a token from JSON.

Parameters:

- `data (str)` - JSON produced by `to_json`.

Returns:

- `OAuthToken` - A new `OAuthToken` instance.

`deployed_codeto_json`

`to_json() -> str`

Serializes this token to a compact JSON string.

Returns:

- `str` - JSON representation suitable for persistence and later hydration via `from_json`.



# User

Authenticated user profile used by Flet authorization flows.

The instance is a mutable mapping with provider-specific user fields and normalized attributes such as id and groups.

Parameters:

- `kwargs` (`dict[str, Any]`) - Provider-specific user fields to store in the mapping.
- `id` (`str`) - Stable user identifier.

Inherits: `dict`

## Properties

- `groups` - Groups loaded for this user when group retrieval is enabled.
- `id` - Provider user identifier represented as a string.

## Properties

### `buildgroupsinstance-attribute`

```
groups: list[Group] = []
```

Groups loaded for this user when group retrieval is enabled.

### `buildidinstance-attribute`

```
id: str = id
```

Provider user identifier represented as a string.

# PubSubClient

Session-scoped facade over PubSubHub.

This client binds all pub/sub operations to one session ID so callers can publish and subscribe without passing their session identity explicitly on each call.

## Methods

- `send_all` - Broadcasts a global message to all sessions.
- `send_all_on_topic` - Broadcasts a topic message to all subscribers of topic.
- `send_others` - Broadcasts a global message to all sessions except this client session.
- `send_others_on_topic` - Broadcasts a topic message excluding this client session.
- `subscribe` - Subscribes this session to global messages.
- `subscribe_topic` - Subscribes this session to a topic.
- `unsubscribe` - Removes global subscriptions for this session.
- `unsubscribe_all` - Removes all global and topic subscriptions for this session.
- `unsubscribe_topic` - Removes this session's subscriptions for a specific topic.

## Methods

### `deployed_codesend_all`

```
send_all(message: Any)
```

Broadcasts a global message to all sessions.

Parameters:

- `message (Any)` - Payload to publish.

### `deployed_codesend_all_on_topic`

```
send_all_on_topic(topic: str, message: Any)
```

Broadcasts a topic message to all subscribers of topic.

Parameters:

- `topic (str)` - Topic name to publish on.
- `message (Any)` - Payload to publish.

### `deployed_codesend_others`

```
send_others(message: Any)
```

Broadcasts a global message to all sessions except this client session.

Parameters:

- `message (Any)` - Payload to publish.

### `deployed_codesend_others_on_topic`

```
send_others_on_topic(topic: str, message: Any)
```

Broadcasts a topic message excluding this client session.

Parameters:

- `topic (str)` - Topic name to publish on.
- `message (Any)` - Payload to publish.

### `deployed_codesubscribe`

```
subscribe(handler: Callable)
```

Subscribes this session to global messages.

The handler is invoked with one positional argument: message.

Parameters:

- handler (Callable) - Sync or async callback for global messages.

### **deployed\_codesubscribe\_topic**

```
subscribe_topic(topic: str, handler: Callable)
```

Subscribes this session to a topic.

The handler is invoked with two positional arguments: (topic, message).

Parameters:

- topic (str) - Topic name to subscribe to.
- handler (Callable) - Sync or async callback for topic messages.

### **deployed\_codeunsubscribe**

```
unsubscribe()
```

Removes global subscriptions for this session.

### **deployed\_codeunsubscribe\_all**

```
unsubscribe_all()
```

Removes all global and topic subscriptions for this session.

### **deployed\_codeunsubscribe\_topic**

```
unsubscribe_topic(topic: str)
```

Removes this session's subscriptions for a specific topic.

Parameters:

- topic (str) - Topic to unsubscribe from.

# PubSubHub

Thread-safe in-memory pub/sub router scoped to a Flet server process.

Subscribers are grouped by session ID and optionally by topic. Handlers can be synchronous callables or async coroutine functions:

- global subscribers receive (message);
- topic subscribers receive (topic, message).

This hub is used by session-scoped PubSubClient instances to fan out messages between connected sessions.

## Methods

- `send_all` - Sends a message to all global subscribers across all sessions.
- `send_all_on_topic` - Sends a topic message to all subscribers of topic.
- `send_others` - Sends a global message to all sessions except one.
- `send_others_on_topic` - Sends a topic message to all subscribers except one session.
- `subscribe` - Registers a global subscriber for a session.
- `subscribe_topic` - Registers a topic subscriber for a session.
- `unsubscribe` - Removes all global subscribers for a session.
- `unsubscribe_all` - Removes both global and topic subscriptions for a session.
- `unsubscribe_topic` - Removes all handlers for a specific session/topic pair.

## Methods

### `deployed_codesend_all`

```
send_all(message: Any)
```

Sends a message to all global subscribers across all sessions.

Parameters:

- `message (Any)` - Payload to deliver.

### `deployed_codesend_all_on_topic`

```
send_all_on_topic(topic: str, message: Any)
```

Sends a topic message to all subscribers of topic.

Parameters:

- `topic (str)` - Topic name to broadcast on.
- `message (Any)` - Payload to deliver.

### `deployed_codesend_others`

```
send_others(except_session_id: str, message: Any)
```

Sends a global message to all sessions except one.

Parameters:

- `except_session_id (str)` - Session ID to exclude from delivery.
- `message (Any)` - Payload to deliver.

### `deployed_codesend_others_on_topic`

```
send_others_on_topic(
 except_session_id: str, topic: str, message: Any
)
```

Sends a topic message to all subscribers except one session.

Parameters:

- `except_session_id` (str) - Session ID to exclude from delivery.
- `topic` (str) - Topic name to publish on.
- `message` (Any) - Payload to deliver.

### `deployed_codesubscribe`

```
subscribe(session_id: str, handler: Callable)
```

Registers a global subscriber for a session.

The handler will receive one positional argument: `message`. Duplicate registrations of the same handler are ignored because handlers are stored in a set.

Parameters:

- `session_id` (str) - Session identifier that owns this subscription.
- `handler` (Callable) - Sync or async callback invoked for global messages.

### `deployed_codesubscribe_topic`

```
subscribe_topic(
 session_id: str,
 topic: str,
 handler: Callable | Callable[..., Awaitable[Any]],
)
```

Registers a topic subscriber for a session.

The handler will receive two positional arguments: (`topic`, `message`).

Parameters:

- `session_id` (str) - Session identifier that owns this subscription.
- `topic` (str) - Topic name to subscribe to.
- `handler` (Callable | Callable[..., Awaitable[Any]]) - Sync or async callback invoked for topic messages.

### `deployed_codeunsubscribe`

```
unsubscribe(session_id: str)
```

Removes all global subscribers for a session.

This does not remove topic subscriptions; use `unsubscribe_all` to remove both.

Parameters:

- `session_id` (str) - Session identifier to remove.

### `deployed_codeunsubscribe_all`

```
unsubscribe_all(session_id: str)
```

Removes both global and topic subscriptions for a session.

Parameters:

- `session_id` (str) - Session identifier to fully unsubscribe.

### `deployed_codeunsubscribe_topic`

```
unsubscribe_topic(session_id: str, topic: str)
```

Removes all handlers for a specific session/topic pair.

Parameters:

- `session_id` (str) - Session identifier to remove from the topic.
- `topic` (str) - Topic to unsubscribe from.

# DragTargetEvent coordinate fields deprecated

note

This guide is accurate as of Flet 0.85.0. Later releases might add new APIs or additional migration paths.

The breaking changes and deprecations index lists the guides created for each release.

## Summary

Flet 0.85.0 deprecated `DragTargetEvent.x`, `DragTargetEvent.y`, and `DragTargetEvent.offset`.

Use `DragTargetEvent.local_position` for target-relative coordinates, or `DragTargetEvent.global_position` for page-level coordinates.

## Context

The old `x`, `y`, and `offset` fields did not make their coordinate space explicit. The new position fields separate target-relative coordinates from global coordinates, which makes drag and drop code easier to read and less ambiguous.

## Migration guide

Code before migration:

```
def on_accept(event: ft.DragTargetEvent):
 target_x = event.x
 target_y = event.y
 target_position = event.offset
```

Code after migration:

```
def on_accept(event: ft.DragTargetEvent):
 target_x = event.local_position.x
 target_y = event.local_position.y
 target_position = event.local_position
```

If your code needs page-level coordinates instead, use `event.global_position.x`, `event.global_position.y`, or `event.global_position`.

## Timeline

- Deprecated in: 0.85.0
- Removal in: 0.88.0

## References

- API documentation: `DragTargetEvent`
- Issues and PRs: #6387, #6401
- Release notes: Flet 0.85.0

# Video control APIs deprecated

note

This guide is accurate as of Flet 0.85.0. Later releases might add new APIs or additional migration paths.

The breaking changes and deprecations index lists the guides created for each release.

## Summary

Flet 0.85.0 deprecated `Video.show_controls`, `Video.playlist_add()`, and `Video.playlist_remove()`.

Use `Video.controls` to configure or hide video controls. Mutate `Video.playlist` directly with standard list methods.

## Context

`Video.controls` provides a single configuration point for built-in and custom video controls. It also supports the old `hide_controls` behavior by setting the value to `None`.

The playlist is now the source of truth for video items. Mutating the list directly keeps the API aligned with other list-backed Flet controls.

## Migration guide

### Hiding controls

Code before migration:

```
video = ft.Video(
 playlist=[media],
 show_controls=False,
)
```

Code after migration:

```
video = ft.Video(
 playlist=[media],
 controls=None,
)
```

### Updating the playlist

Code before migration:

```
video.playlist_add(media)
video.playlist_remove(media_index)
```

Code after migration:

```
video.playlist.append(media)
video.playlist.pop(media_index)
video.update()
```

Call `video.update()` when the mutation happens after the control has already been added to the page.

## Timeline

- Deprecated in: 0.85.0
- Removal in: 0.88.0

## References

- API documentation: [Video](#)
- Issues and PRs: [PR #6463](#)
- Release notes: [Flet 0.85.0](#)

# Deprecated spacing and border helper functions removed

note

This guide is accurate as of Flet 0.85.0. Later releases might add new APIs or additional migration paths.

The breaking changes and deprecations index lists the guides created for each release.

## Summary

Flet 0.85.0 removed deprecated module-level helper functions from `ft.margin`, `ft.padding`, `ft.border`, and `ft.border_radius`. Replace module-level helper calls with class methods:

| Removed helper | Replacement |
|----------------|-------------|
|                |             |
|                |             |
|                |             |
|                |             |
|                |             |
|                |             |
|                |             |
|                |             |
|                |             |
|                |             |

`margin.all(...)``Margin.all(...)``margin.symmetric(...)``Margin.symmetric(...)``margin.only(...)``Margin.only(...)``padding.all(...)``Padding.all(...)``padding.symmetric(...)``Padding.symmetric(...)``padding.only(...)``Padding.only(...)``border.all(...)``Border.all(...)``border.symmetric(...)``Border.symmetric(...)``border.only(...)``Border.only(...)``border_radius.all(...)``BorderRadius.all(...)``border_radius.horizontal(...)``BorderRadius.horizontal(...)``border_radius.vertical(...)``BorderRadius.vertical(...)``border_radius.only(...)``BorderRadius.only(...)`

## Background

The helper functions were replaced by class methods to keep the public API consistent with the concrete value types they create. For example, `Padding.all(...)` now clearly returns a `Padding` value, and `BorderRadius.only(...)` returns a `BorderRadius` value.

## Migration guide

Code before migration:

```
import flet as ft

card = ft.Container(
 padding=ft.padding.all(12),
 border=ft.border.all(1, ft.Colors.OUTLINE),
 border_radius=ft.border_radius.all(8),
)
```

Code after migration:

```
import flet as ft

card = ft.Container(
 padding=ft.Padding.all(12),
 border=ft.Border.all(1, ft.Colors.OUTLINE),
 border_radius=ft.BorderRadius.all(8),
)
```

## Timeline

- Removed in: 0.85.0

## References

- API documentation: `Border`, `BorderRadius`, `Padding`
- Issues and PRs: #6425
- Release notes: Flet 0.85.0



# AdRequest

Targeting info per the AdMob API.

This class's properties mirror the native AdRequest API. See for example: `AdRequest.Builder` for Android.

## Properties

- `content_url` - URL string for a webpage whose content matches the app's primary content.
- `extras` - Extras to pass to the AdMob adapter.
- `http_timeout` - A custom timeout (in milliseconds) for HTTPS calls during an ad request.
- `keywords` - Words or phrases describing the current user activity.
- `neighboring_content_urls` - URLs representing web content near an ad.
- `non_personalized_ads` - Non-personalized ads are ads that are not based on a user's past behavior.

## Properties

### `buildcontent_url`class-attributeinstance-attribute

`content_url: str | None = None`

URL string for a webpage whose content matches the app's primary content.

This webpage content is used for targeting and brand safety purposes.

### `buildextras`class-attributeinstance-attribute

`extras: dict[str, str] | None = None`

Extras to pass to the AdMob adapter.

### `buildhttp_timeout`class-attributeinstance-attribute

`http_timeout: int | None = None`

A custom timeout (in milliseconds) for HTTPS calls during an ad request.

Note

This is only supported in Android. (ignored on iOS)

### `buildkeywords`class-attributeinstance-attribute

`keywords: list[str] | None = None`

Words or phrases describing the current user activity.

### `buildneighboring_content_urls`class-attributeinstance-attribute

`neighboring_content_urls: list[str] | None = None`

URLs representing web content near an ad.

### `buildnon_personalized_ads`class-attributeinstance-attribute

`non_personalized_ads: bool | None = None`

Non-personalized ads are ads that are not based on a user's past behavior.

For more information: <https://support.google.com/admob/answer/7676680?hl=en>

# PaidAdRequest

Event data for paid ad events.

Inherits: Event

## Properties

- `currency_code` - The currency code of the ad value.
- `precision` - The precision of the ad value.
- `value` - The monetary value of the ad.

## Properties

### `buildcurrency_codeinstance-attribute`

`currency_code: str`

The currency code of the ad value.

### `buildprecisioninstance-attribute`

`precision: PrecisionType`

The precision of the ad value.

### `buildvalueinstance-attribute`

`value: float`

The monetary value of the ad.

# PrecisionType

Describes how accurately a paid ad value is reported.

Inherits: `enum.Enum`

## Properties

- `ESTIMATED` - An ad value estimated from aggregated data.
- `PRECISE` - The precise value paid for this ad.
- `PUBLISHER_PROVIDED` - A publisher-provided ad value, such as manual CPMs in a mediation group.
- `UNKNOWN` - An ad value with unknown precision.

## Properties

`buildESTIMATEDclass-attributeinstance-attribute`

An ad value estimated from aggregated data.

`buildPRECISEclass-attributeinstance-attribute`

The precise value paid for this ad.

`buildPUBLISHER_PROVIDEDclass-attributeinstance-attribute`

A publisher-provided ad value, such as manual CPMs in a mediation group.

`buildUNKNOWNclass-attributeinstance-attribute`

An ad value with unknown precision.

# CameraDescription

Properties of a camera device.

## Properties

- `lens_direction` - Physical lens direction (front, back, external).
- `lens_type` - Lens hardware type (wide, telephoto, ultra-wide, or unknown).
- `name` - Human-readable identifier of the camera device.
- `sensor_orientation` - Clockwise angle (0, 90, 180, 270) needed to display upright output.

## Properties

### `buildlens_directioninstance-attribute`

`lens_direction: CameraLensDirection`

Physical lens direction (front, back, external).

### `buildlens_typeclass-attributeinstance-attribute`

`lens_type: CameraLensType = CameraLensType.UNKNOWN`

Lens hardware type (wide, telephoto, ultra-wide, or unknown).

### `buildnameinstance-attribute`

`name: str`

Human-readable identifier of the camera device.

### `buildsensor_orientationinstance-attribute`

`sensor_orientation: int`

Clockwise angle (0, 90, 180, 270) needed to display upright output.

# CameraImageEvent

Image data produced by the camera stream.

Inherits: Event

## Properties

- `bytes` - Encoded image bytes.
- `encoded_format` - Encoding used for the output bytes (e.g., jpeg).
- `format` - Raw image format group of the source frame.
- `height` - Pixel height of the frame.
- `lens_aperture` - Lens aperture value for the frame.
- `sensor_exposure_time` - Exposure time in nanoseconds.
- `sensor_sensitivity` - Sensor sensitivity (ISO).
- `width` - Pixel width of the frame.

## Properties

### `buildbytesinstance-attribute`

`bytes: bytes`

Encoded image bytes.

### `buildencoded_formatinstance-attribute`

`encoded_format: str`

Encoding used for the output bytes (e.g., jpeg).

### `buildformatinstance-attribute`

`format: ImageFormatGroup | None`

Raw image format group of the source frame.

### `buildheightinstance-attribute`

`height: int`

Pixel height of the frame.

### `buildlens_apertureclass-attributeinstance-attribute`

`lens_aperture: Number | None = None`

Lens aperture value for the frame.

### `buildsensor_exposure_timeclass-attributeinstance-attribute`

`sensor_exposure_time: int | None = None`

Exposure time in nanoseconds.

### `buildsensor_sensitivityclass-attributeinstance-attribute`

`sensor_sensitivity: Number | None = None`

Sensor sensitivity (ISO).

### `buildwidthinstance-attribute`

`width: int`

Pixel width of the frame.

# CameraLensDirection

Direction the camera is facing.

Inherits: `enum.Enum`

## Properties

- `BACK` - Back facing camera.
- `EXTERNAL` - External camera device.
- `FRONT` - Front facing camera.

## Properties

`buildBACK`class-attributeinstance-attribute

Back facing camera.

`buildEXTERNAL`class-attributeinstance-attribute

External camera device.

`buildFRONT`class-attributeinstance-attribute

Front facing camera.

# CameraLensType

Type of the camera lens.

Inherits: `enum.Enum`

## Properties

- `TELEPHOTO` - Telephoto lens.
- `ULTRA_WIDE` - Ultra-wide lens.
- `UNKNOWN` - Unknown lens type.
- `WIDE` - Wide-angle lens.

## Properties

`buildTELEPHOTOclass-attributeinstance-attribute`

Telephoto lens.

`buildULTRA_WIDEclass-attributeinstance-attribute`

Ultra-wide lens.

`buildUNKNOWNclass-attributeinstance-attribute`

Unknown lens type.

`buildWIDEclass-attributeinstance-attribute`

Wide-angle lens.

# CameraPreviewSize

Dimensions of a camera preview.

## Properties

- `height` - Preview height in logical pixels.
- `width` - Preview width in logical pixels.

## Properties

### `buildheightinstance-attribute`

`height`: `Number`

Preview height in logical pixels.

### `buildwidthinstance-attribute`

`width`: `Number`

Preview width in logical pixels.



# CameraStateEvent

Snapshot of the camera controller state.

Inherits: Event

## Properties

- `aspect_ratio` - Preview aspect ratio, when available.
- `description` - The underlying camera description.
- `device_orientation` - Current device UI orientation.
- `error_description` - Error message when the controller has an error.
- `exposure_mode` - Active exposure mode.
- `exposure_point_supported` - Whether custom exposure points are supported.
- `flash_mode` - Active flash mode.
- `focus_mode` - Active focus mode.
- `focus_point_supported` - Whether custom focus points are supported.
- `has_error` - Indicates if the controller is in an error state.
- `is_capture_orientation_locked` - True if capture orientation is locked.
- `is_initialized` - Whether the controller has been initialized.
- `is_preview_paused` - True when the preview has been manually paused.
- `is_recording_paused` - Whether an active recording is currently paused.
- `is_recording_video` - Whether a video recording is in progress.
- `is_streaming_images` - True when image streaming is running.
- `is_taking_picture` - True while a still capture is underway.
- `locked_capture_orientation` - Orientation used when capture orientation is locked.
- `preview_pause_orientation` - Orientation used when the preview was paused.
- `preview_size` - Preview dimensions, when available.
- `recording_orientation` - Orientation used for the current recording.

## Properties

### `buildaspect_ratio`class-attributeinstance-attribute

`aspect_ratio`: Number | None = None

Preview aspect ratio, when available.

### `builddescription`class-attributeinstance-attribute

`description`: CameraDescription | None = None

The underlying camera description.

### `builddevice_orientation`class-attributeinstance-attribute

`device_orientation`: DeviceOrientation | None = None

Current device UI orientation.

### `builderror_description`class-attributeinstance-attribute

`error_description`: str | None = None

Error message when the controller has an error.

### `buildexposure_mode`class-attributeinstance-attribute

`exposure_mode`: ExposureMode | None = None

Active exposure mode.

### **buildexposure\_point\_supportedclass-attributeinstance-attribute**

exposure\_point\_supported: bool | None = None

Whether custom exposure points are supported.

### **buildflash\_modeclass-attributeinstance-attribute**

flash\_mode: FlashMode | None = None

Active flash mode.

### **buildfocus\_modeclass-attributeinstance-attribute**

focus\_mode: FocusMode | None = None

Active focus mode.

### **buildfocus\_point\_supportedclass-attributeinstance-attribute**

focus\_point\_supported: bool | None = None

Whether custom focus points are supported.

### **buildhas\_errorclass-attributeinstance-attribute**

has\_error: bool | None = None

Indicates if the controller is in an error state.

### **buildis\_capture\_orientation\_lockedinstance-attribute**

is\_capture\_orientation\_locked: bool

True if capture orientation is locked.

### **buildis\_initializedinstance-attribute**

is\_initialized: bool

Whether the controller has been initialized.

### **buildis\_preview\_pausedinstance-attribute**

is\_preview\_paused: bool

True when the preview has been manually paused.

### **buildis\_recording\_pausedinstance-attribute**

is\_recording\_paused: bool

Whether an active recording is currently paused.

### **buildis\_recording\_videoinstance-attribute**

is\_recording\_video: bool

Whether a video recording is in progress.

### **buildis\_streaming\_imagesinstance-attribute**

is\_streaming\_images: bool

True when image streaming is running.

### **buildis\_taking\_pictureinstance-attribute**

is\_taking\_picture: bool

True while a still capture is underway.

### **buildlocked\_capture\_orientationclass-attributeinstance-attribute**

locked\_capture\_orientation: DeviceOrientation | None = (  
 None  
)

Orientation used when capture orientation is locked.

### **buildpreview\_pause\_orientationclass-attributeinstance-attribute**

preview\_pause\_orientation: DeviceOrientation | None = (  
 None  
)

)

Orientation used when the preview was paused.

**buildpreview\_size**class-attributeinstance-attribute

preview\_size: CameraPreviewSize | None = None

Preview dimensions, when available.

**buildrecording\_orientation**class-attributeinstance-attribute

recording\_orientation: DeviceOrientation | None = None

Orientation used for the current recording.

# ExposureMode

Exposure modes supported by the camera.

Inherits: `enum.Enum`

## Properties

- `AUTO` - Automatic exposure.
- `LOCKED` - Lock exposure to the current value.

## Properties

`buildAUTO`class-attributeinstance-attribute

Automatic exposure.

`buildLOCKED`class-attributeinstance-attribute

Lock exposure to the current value.

# FlashMode

Flash modes supported by the camera.

Inherits: `enum.Enum`

## Properties

- ALWAYS - Always fire flash.
- AUTO - Use flash automatically when required.
- OFF - Disable flash.
- TORCH - Continuous torch/flashlight mode.

## Properties

`buildALWAYS`class-attributeinstance-attribute

Always fire flash.

`buildAUTO`class-attributeinstance-attribute

Use flash automatically when required.

`buildOFF`class-attributeinstance-attribute

Disable flash.

`buildTORCH`class-attributeinstance-attribute

Continuous torch/flashlight mode.

# FocusMode

Focus modes supported by the camera.

Inherits: `enum.Enum`

## Properties

- `AUTO` - Automatic focus.
- `LOCKED` - Lock focus to the current value.

## Properties

`buildAUTO`class-attributeinstance-attribute

Automatic focus.

`buildLOCKED`class-attributeinstance-attribute

Lock focus to the current value.

# ImageFormatGroup

Describes the image output format.

Inherits: `enum.Enum`

## Properties

- `BGRA8888` - Raw BGRA 8888 format.
- `JPEG` - JPEG-compressed format.
- `NV21` - NV21 YUV format (Android).
- `UNKNOWN` - Unknown or unsupported format.
- `YUV420` - YUV420 planar format.

## Properties

`buildBGRA8888class-attributeinstance-attribute`

Raw BGRA 8888 format.

`buildJPEGclass-attributeinstance-attribute`

JPEG-compressed format.

`buildNV21class-attributeinstance-attribute`

NV21 YUV format (Android).

`buildUNKNOWNclass-attributeinstance-attribute`

Unknown or unsupported format.

`buildYUV420class-attributeinstance-attribute`

YUV420 planar format.

# ResolutionPreset

Represents a capture resolution preset.

Inherits: `enum.Enum`

## Properties

- `HIGH` - High resolution preset.
- `LOW` - Low resolution preset.
- `MAX` - Maximum available resolution preset.
- `MEDIUM` - Medium resolution preset.
- `ULTRA_HIGH` - Ultra high resolution preset.
- `VERY_HIGH` - Very high resolution preset.

## Properties

`buildHIGHclass-attributeinstance-attribute`

High resolution preset.

`buildLOWclass-attributeinstance-attribute`

Low resolution preset.

`buildMAXclass-attributeinstance-attribute`

Maximum available resolution preset.

`buildMEDIUMclass-attributeinstance-attribute`

Medium resolution preset.

`buildULTRA_HIGHclass-attributeinstance-attribute`

Ultra high resolution preset.

`buildVERY_HIGHclass-attributeinstance-attribute`

Very high resolution preset.



# BarChartEvent

Event payload emitted when the user hovers or clicks a bar chart.

Inherits: Event

## Properties

- `group_index` - Bar's index or -1 if chart is hovered or clicked outside of any bar.
- `rod_index` - Rod's index or -1 if chart is hovered or clicked outside of any bar.
- `stack_item_index` - Stack item's index or -1 if chart is hovered or clicked outside of any bar.
- `type` - The type of event that occurred on the chart.

## Properties

### `buildgroup_index`class-attributeinstance-attribute

`group_index: int | None = None`

Bar's index or -1 if chart is hovered or clicked outside of any bar.

### `buildrod_index`class-attributeinstance-attribute

`rod_index: int | None = None`

Rod's index or -1 if chart is hovered or clicked outside of any bar.

### `buildstack_item_index`class-attributeinstance-attribute

`stack_item_index: int | None = None`

Stack item's index or -1 if chart is hovered or clicked outside of any bar.

### `buildtype`instance-attribute

`type: ChartEventType`

The type of event that occurred on the chart.

# BarChartGroup

Groups one or more bar rods rendered at the same X-axis position.

Inherits: `BaseControl`

## Properties

- `group_vertically` - If set to `True` bar rods are drawn on top of each other; otherwise bar rods are drawn next to each other.
- `rods` - The list of `BarChartRod` objects to display in the group.
- `spacing` - The amount of space between bar rods.
- `x` - Group position on X axis.

## Properties

### `buildgroup_verticallyclass-attributeinstance-attribute`

```
group_vertically: bool = False
```

If set to `True` bar rods are drawn on top of each other; otherwise bar rods are drawn next to each other.

### `buildrodsclass-attributeinstance-attribute`

```
rods: list[BarChartRod] = field(default_factory=list)
```

The list of `BarChartRod` objects to display in the group.

### `buildspacingclass-attributeinstance-attribute`

```
spacing: Number | None = None
```

The amount of space between bar rods.

### `buildxclass-attributeinstance-attribute`

```
x: int = 0
```

Group position on X axis.

# BarChartRod

A bar rod in a BarChartGroup.

Inherits: BaseControl

## Properties

- `background_gradient` - An optional gradient to draw a background with.
- `bg_from_y` - An optional starting position of a background behind this rod.
- `bg_to_y` - An optional ending position of a background behind this rod.
- `bgcolor` - An optional color of a background behind this rod.
- `border_radius` - Border radius of a bar rod.
- `border_side` - Border to draw around rod.
- `color` - Rod color.
- `from_y` - Specifies a starting position of this rod on Y axis.
- `gradient` - Gradient to draw rod's background.
- `selected` - If set to True a tooltip is always shown on top of the bar when `BarChart.interactive` is set to False.
- `show_tooltip` - Whether a tooltip should be shown on top of hovered bar.
- `stack_items` - Optional list of `BarChartRodStackItem` objects to draw a stacked bar.
- `to_y` - Specifies an ending position of this rod on Y axis.
- `tooltip` - The rod's tooltip configuration for this rod.
- `width` - The width of this rod.

## Properties

### `buildbackground_gradientclass-attributeinstance-attribute`

`background_gradient: Gradient | None = None`

An optional gradient to draw a background with.

### `buildbg_from_yclass-attributeinstance-attribute`

`bg_from_y: Number | None = None`

An optional starting position of a background behind this rod.

### `buildbg_to_yclass-attributeinstance-attribute`

`bg_to_y: Number | None = None`

An optional ending position of a background behind this rod.

### `buildbgcolorclass-attributeinstance-attribute`

`bgcolor: ColorValue | None = None`

An optional color of a background behind this rod.

### `buildborder_radiusclass-attributeinstance-attribute`

`border_radius: BorderRadiusValue | None = None`

Border radius of a bar rod.

### `buildborder_sideclass-attributeinstance-attribute`

`border_side: BorderSide | None = None`

Border to draw around rod.

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

Rod color.

### **buildfrom\_yclass-attributeinstance-attribute**

```
from_y: Number = 0
```

Specifies a starting position of this rod on Y axis.

### **buildgradientclass-attributeinstance-attribute**

```
gradient: Gradient | None = None
```

Gradient to draw rod's background.

### **buildselectedclass-attributeinstance-attribute**

```
selected: bool = False
```

If set to True a tooltip is always shown on top of the bar when BarChart.interactive is set to False.

### **buildshow\_tooltipclass-attributeinstance-attribute**

```
show_tooltip: bool = True
```

Whether a tooltip should be shown on top of hovered bar.

### **buildstack\_itemsclass-attributeinstance-attribute**

```
stack_items: list[BarChartRodStackItem] = field(
 default_factory=list
)
```

Optional list of BarChartRodStackItem objects to draw a stacked bar.

### **buildto\_yclass-attributeinstance-attribute**

```
to_y: Number | None = None
```

Specifies an ending position of this rod on Y axis.

### **buildtooltipclass-attributeinstance-attribute**

```
tooltip: BarChartRodTooltip | str = field(
 default_factory=lambda: BarChartRodTooltip()
)
```

The rod's tooltip configuration for this rod.

### **buildwidthclass-attributeinstance-attribute**

```
width: Number | None = None
```

The width of this rod.

# BarChartRodStackItem

Colored segment used to build a stacked bar rod.

Inherits: `BaseControl`

## Properties

- `border_side` - A border around this item.
- `color` - The color of this item.
- `from_y` - The starting position of this item inside a bar rod.
- `to_y` - The ending position of this item inside a bar rod.

## Properties

**buildborder\_side**class-attributeinstance-attribute

```
border_side: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

A border around this item.

**buildcolor**class-attributeinstance-attribute

```
color: ColorValue | None = None
```

The color of this item.

**buildfrom\_y**class-attributeinstance-attribute

```
from_y: Number | None = None
```

The starting position of this item inside a bar rod.

**buildto\_y**class-attributeinstance-attribute

```
to_y: Number = 0
```

The ending position of this item inside a bar rod.

# BarChartRodTooltip

Tooltip configuration for the BarChartRod.

Inherits: `ChartDataPointTooltip`

## Properties

- `text` - The text to display in the tooltip.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildtextclass-attributeinstance-attribute`

```
text: str | None = None
```

The text to display in the tooltip.

When None, defaults to `BarChartRod.to_y`.

## Methods

### `deployed_codecopy`

```
copy(
 text: str | None = None,
 text_style: TextStyle | None = None,
 text_align: TextAlign | None = None,
 text_spans: list[TextSpan] | None = None,
 rtl: bool | None = None,
) -> BarChartRodTooltip
```

Returns a copy of this object with the specified properties overridden.

# BarChartTooltip

Configuration of the tooltip for BarCharts.

## Properties

- `bgcolor` - Background color of tooltips.
- `border_radius` - The border radius of the tooltip.
- `border_side` - The tooltip border side.
- `direction` - Defines the direction of this tooltip.
- `fit_inside_horizontally` - Forces the tooltip to shift horizontally inside the chart, if overflow happens.
- `fit_inside_vertically` - Forces the tooltip to shift vertically inside the chart, if overflow happens.
- `horizontal_alignment` - Defines the horizontal alignment of this tooltip.
- `horizontal_offset` - The horizontal offset of this tooltip.
- `margin` - Applies a bottom margin for showing tooltip on top of rods.
- `max_width` - Restricts the tooltip's width.
- `padding` - Applies a padding for showing contents inside the tooltip.
- `rotation` - The rotation angle of the tooltip.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

```
bgcolor: ColorValue = Colors.SECONDARY
```

Background color of tooltips.

### `buildborder_radiusclass-attributeinstance-attribute`

```
border_radius: BorderRadiusValue | None = None
```

The border radius of the tooltip.

### `buildborder_sideclass-attributeinstance-attribute`

```
border_side: BorderSide | None = None
```

The tooltip border side.

### `bulddirectionclass-attributeinstance-attribute`

```
direction: BarChartTooltipDirection = (
 BarChartTooltipDirection.AUTO
)
```

Defines the direction of this tooltip.

### `buildfit_inside_horizontallyclass-attributeinstance-attribute`

```
fit_inside_horizontally: bool = False
```

Forces the tooltip to shift horizontally inside the chart, if overflow happens.

### `buildfit_inside_verticallyclass-attributeinstance-attribute`

```
fit_inside_vertically: bool = False
```

Forces the tooltip to shift vertically inside the chart, if overflow happens.

### `buildhorizontal_alignmentclass-attributeinstance-attribute`

```
horizontal_alignment: HorizontalAlignment = (
 HorizontalAlignment.CENTER
)
```

Defines the horizontal alignment of this tooltip.

### **buildhorizontal\_offsetclass-attributeinstance-attribute**

```
horizontal_offset: Number = 0.0
```

The horizontal offset of this tooltip.

### **buildmarginclass-attributeinstance-attribute**

```
margin: Number = 16
```

Applies a bottom margin for showing tooltip on top of rods.

### **buildmax\_widthclass-attributeinstance-attribute**

```
max_width: Number | None = None
```

Restricts the tooltip's width.

### **buildpaddingclass-attributeinstance-attribute**

```
padding: PaddingValue = field(
 default_factory=lambda: Padding.symmetric(
 vertical=8, horizontal=16
)
)
```

Applies a padding for showing contents inside the tooltip.

### **buildrotationclass-attributeinstance-attribute**

```
rotation: Number = 0.0
```

The rotation angle of the tooltip.

## **Methods**

### **deployed\_codecopy**

```
copy(
 bgcolor: ColorValue | None = None,
 border_radius: BorderRadiusValue | None = None,
 margin: Number | None = None,
 padding: PaddingValue | None = None,
 max_width: Number | None = None,
 rotation: Number | None = None,
 horizontal_offset: Number | None = None,
 border_side: BorderSide | None = None,
 fit_inside_horizontally: bool | None = None,
 fit_inside_vertically: bool | None = None,
 direction: BarChartTooltipDirection | None = None,
 horizontal_alignment: HorizontalAlignment | None = None,
) -> BarChartTooltip
```

Returns a copy of this object with the specified properties overridden.



# BarChartTooltipDirection

Controls showing tooltip on top or bottom.

Inherits: `enum.Enum`

## Properties

- **AUTO** - Tooltip shows on top if value is positive, on bottom if value is negative.
- **BOTTOM** - Tooltip always shows on bottom.
- **TOP** - Tooltip always shows on top.

## Properties

**buildAUTO**`class-attributeinstance-attribute`

Tooltip shows on top if value is positive, on bottom if value is negative.

**buildBOTTOM**`class-attributeinstance-attribute`

Tooltip always shows on bottom.

**buildTOP**`class-attributeinstance-attribute`

Tooltip always shows on top.

# CandlestickChartEvent

Event raised for interactions with a CandlestickChart.

Inherits: Event

## Properties

- `spot_index` - Index of the candlestick that was interacted with; None if none.
- `type` - Type of pointer gesture that triggered the event.

## Properties

**buildspot\_index**class-attributeinstance-attribute

`spot_index: int | None = None`

Index of the candlestick that was interacted with; None if none.

**buildtype**instance-attribute

`type: ChartEventType`

Type of pointer gesture that triggered the event.

# CandlestickChartSpot

Represents a candlestick rendered on a CandlestickChart.

Inherits: BaseControl

## Properties

- close - The close value of the candlestick.
- high - The high value of the candlestick.
- low - The low value of the candlestick.
- open - The open value of the candlestick.
- selected - Whether to treat this candlestick as selected.
- show\_tooltip - Whether the tooltip should be shown when this candlestick is highlighted.
- tooltip - Tooltip configuration for this candlestick.
- x - The position of the candlestick on the X axis.

## Properties

### buildcloseinstance-attribute

close: Number

The close value of the candlestick.

### buildhighinstance-attribute

high: Number

The high value of the candlestick.

### buildlowinstance-attribute

low: Number

The low value of the candlestick.

### buildopeninstance-attribute

open: Number

The open value of the candlestick.

### buildselectedclass-attributeinstance-attribute

selected: bool = False

Whether to treat this candlestick as selected.

### buildshow\_tooltipclass-attributeinstance-attribute

show\_tooltip: bool = True

Whether the tooltip should be shown when this candlestick is highlighted.

### buildtooltipclass-attributeinstance-attribute

```
tooltip: CandlestickChartSpotTooltip | str = field(
 default_factory=lambda: CandlestickChartSpotTooltip()
)
```

Tooltip configuration for this candlestick.

### buildxinstance-attribute

x: Number

The position of the candlestick on the X axis.

# CandlestickChartSpotTooltip

Tooltip configuration for the CandlestickChartSpot.

Inherits: `ChartDataPointTooltip`

## Properties

- `bottom_margin` - Space between the tooltip bubble and the candlestick.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

`buildbottom_marginclass-attributeinstance-attribute`

`bottom_margin: Number = 8`

Space between the tooltip bubble and the candlestick.

## Methods

`deployed_code`  
`copy`

```
copy(
 text: str | None = None,
 text_style: TextStyle | None = None,
 text_align: TextAlign | None = None,
 text_spans: list[TextSpan] | None = None,
 rtl: bool | None = None,
 bottom_margin: Number | None = None,
) -> CandlestickChartSpotTooltip
```

Returns a copy of this object with the specified properties overridden.

# CandlestickChartTooltip

Configuration of the tooltip for CandlestickCharts.

## Properties

- `bgcolor` - Background color applied to the tooltip bubble.
- `border_radius` - Corner radius of the tooltip bubble.
- `border_side` - The tooltip bubble border.
- `fit_inside_horizontally` - Forces the tooltip bubble to remain inside the chart horizontally.
- `fit_inside_vertically` - Forces the tooltip bubble to remain inside the chart vertically.
- `horizontal_alignment` - Horizontal alignment of the tooltip relative to the tapped candlestick.
- `horizontal_offset` - Horizontal offset applied to the tooltip bubble.
- `max_width` - Maximum width of the tooltip bubble.
- `padding` - Padding inside the tooltip bubble.
- `rotation` - Rotation angle (in degrees) applied to the tooltip bubble.
- `show_on_top_of_chart_box_area` - When set to True, the tooltip is drawn at the top of the chart box.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

```
bgcolor: ColorValue = '#FFFECEF'
```

Background color applied to the tooltip bubble.

### `buildborder_radiusclass-attributeinstance-attribute`

```
border_radius: BorderRadiusValue = field(
 default_factory=lambda: BorderRadius.all(4)
)
```

Corner radius of the tooltip bubble.

### `buildborder_sideclass-attributeinstance-attribute`

```
border_side: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

The tooltip bubble border.

### `buildfit_inside_horizontallyclass-attributeinstance-attribute`

```
fit_inside_horizontally: bool = False
```

Forces the tooltip bubble to remain inside the chart horizontally.

### `buildfit_inside_verticallyclass-attributeinstance-attribute`

```
fit_inside_vertically: bool = False
```

Forces the tooltip bubble to remain inside the chart vertically.

### `buildhorizontal_alignmentclass-attributeinstance-attribute`

```
horizontal_alignment: HorizontalAlignment = (
 HorizontalAlignment.CENTER
)
```

Horizontal alignment of the tooltip relative to the tapped candlestick.

### `buildhorizontal_offsetclass-attributeinstance-attribute`

```
horizontal_offset: Number = 0
```

Horizontal offset applied to the tooltip bubble.

### **buildmax\_widthclass-attributeinstance-attribute**

```
max_width: Number = 120
```

Maximum width of the tooltip bubble.

### **buildpaddingclass-attributeinstance-attribute**

```
padding: PaddingValue = field(
 default_factory=lambda: Padding.symmetric(
 vertical=8, horizontal=16
)
)
```

Padding inside the tooltip bubble.

### **buildrotationclass-attributeinstance-attribute**

```
rotation: Number = 0.0
```

Rotation angle (in degrees) applied to the tooltip bubble.

### **buildshow\_on\_top\_of\_chart\_box\_areaclass-attributeinstance-attribute**

```
show_on_top_of_chart_box_area: bool = False
```

When set to True, the tooltip is drawn at the top of the chart box.

## **Methods**

### **deployed\_codecopy**

```
copy(
 bgcolor: ColorValue | None = None,
 border_radius: BorderRadiusValue | None = None,
 padding: PaddingValue | None = None,
 max_width: Number | None = None,
 rotation: Number | None = None,
 horizontal_offset: Number | None = None,
 horizontal_alignment: HorizontalAlignment | None = None,
 border_side: BorderSide | None = None,
 fit_inside_horizontally: bool | None = None,
 fit_inside_vertically: bool | None = None,
 show_on_top_of_chart_box_area: bool | None = None,
) -> CandlestickChartTooltip
```

Returns a copy of this object with the specified properties overridden.

# ChartAxis

Configures chart axis.

Inherits: `BaseControl`

## Properties

- `label_size` - The maximum space for each label in labels.
- `label_spacing` - The spacing/interval between labels.
- `labels` - The list of `ChartAxisLabel` objects to set custom axis labels for only specific values.
- `show_labels` - Whether to display the labels along the axis.
- `show_max` - Whether to display a label for the maximum value independent of the sampling interval.
- `show_min` - Whether to display a label for the minimum value independent of the sampling interval.
- `title` - A `Control` to display as axis title.
- `title_size` - The size of title area.

## Properties

### `buildlabel_size`class-attributeinstance-attribute

```
label_size: Number = 22
```

The maximum space for each label in labels.

Each label will stretch to fit this space.

### `buildlabel_spacing`class-attributeinstance-attribute

```
label_spacing: Number | None = None
```

The spacing/interval between labels.

If a value is not set, a suitable value will be automatically calculated and used.

### `buildlabelsc`class-attributeinstance-attribute

```
labels: list[ChartAxisLabel] = field(default_factory=list)
```

The list of `ChartAxisLabel` objects to set custom axis labels for only specific values.

### `buildshow_labels`class-attributeinstance-attribute

```
show_labels: bool = True
```

Whether to display the labels along the axis. If labels is empty then automatic labels are displayed.

### `buildshow_max`class-attributeinstance-attribute

```
show_max: bool = True
```

Whether to display a label for the maximum value independent of the sampling interval.

### `buildshow_min`class-attributeinstance-attribute

```
show_min: bool = True
```

Whether to display a label for the minimum value independent of the sampling interval.

### `buildtitle`class-attributeinstance-attribute

```
title: Control | None = None
```

A `Control` to display as axis title.

### `buildtitle_size`class-attributeinstance-attribute

```
title_size: Number = 16
```

The size of title area.

# ChartAxisLabel

Configures a custom label for specific value.

Inherits: `BaseControl`

## Properties

- `label` - The label to display for the specified value.
- `value` - A value to draw label for.

## Properties

### `buildlabelclass-attributeinstance-attribute`

`label: StrOrControl | None = None`

The label to display for the specified value.

### `buildvalueclass-attributeinstance-attribute`

`value: Number | None = None`

A value to draw label for.



# ChartCirclePoint

Draws a circle.

Inherits: `ChartPointShape`

## Properties

- `color` - The fill color to use for the circle.
- `radius` - The radius of the circle.
- `stroke_color` - The stroke color to use for the circle
- `stroke_width` - The stroke width to use for the circle.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue | None = None
```

The fill color to use for the circle.

### `buildradiusclass-attributeinstance-attribute`

```
radius: Number | None = None
```

The radius of the circle.

### `buildstroke_colorclass-attributeinstance-attribute`

```
stroke_color: ColorValue | None = None
```

The stroke color to use for the circle

### `buildstroke_widthclass-attributeinstance-attribute`

```
stroke_width: Number = 0
```

The stroke width to use for the circle.

## Methods

### `deployed_code``copy`

```
copy(
 color: ColorValue | None = None,
 radius: Number | None = None,
 stroke_color: ColorValue | None = None,
 stroke_width: Number | None = None,
) -> ChartCirclePoint
```

Returns a copy of this object with the specified properties overridden.

# ChartCrossPoint

Draws a cross-mark (X).

Inherits: `ChartPointShape`

## Properties

- `color` - The fill color to use for the cross-mark(X).
- `size` - The size of the cross-mark.
- `width` - The thickness of the cross-mark.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The fill color to use for the cross-mark(X).

### `buildsizeclass-attributeinstance-attribute`

`size: Number = 8.0`

The size of the cross-mark.

### `buildwidthclass-attributeinstance-attribute`

`width: Number = 2.0`

The thickness of the cross-mark.

## Methods

### `deployed_code``copy`

```
copy(
 color: ColorValue | None = None,
 size: Number | None = None,
 width: Number | None = None,
) -> ChartCrossPoint
```

Returns a copy of this object with the specified properties overridden.

# ChartDataPointTooltip

Configuration of the tooltip for data points in charts.

## Properties

- `rtl` - Whether the text is right-to-left.
- `text` - The text to display in this tooltip.
- `text_align` - The text alignment of the tooltip.
- `text_spans` - Additional text spans to show on this tooltip.
- `text_style` - A text style to display tooltip with.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildrtlclass-attributeinstance-attribute`

```
rtl: bool = False
```

Whether the text is right-to-left.

### `buildtextclass-attributeinstance-attribute`

```
text: str | None = None
```

The text to display in this tooltip.

### `buildtext_alignclass-attributeinstance-attribute`

```
text_align: TextAlign = TextAlign.CENTER
```

The text alignment of the tooltip.

### `buildtext_spansclass-attributeinstance-attribute`

```
text_spans: list[TextSpan] | None = None
```

Additional text spans to show on this tooltip.

### `buildtext_styleclass-attributeinstance-attribute`

```
text_style: TextStyle = field(
 default_factory=lambda: TextStyle()
)
```

A text style to display tooltip with.

## Methods

### `deployed_codecopy`

```
copy(
 text: str | None = None,
 text_style: TextStyle | None = None,
 text_align: TextAlign | None = None,
 text_spans: list[TextSpan] | None = None,
 rtl: bool | None = None,
) -> ChartDataPointTooltip
```

Returns a copy of this object with the specified properties overridden.

# ChartEventType

The type of event that occurred on the chart.

Inherits: `enum.Enum`

## Properties

- `LONG_PRESS_END` - When a pointer stops contacting the screen after a long press gesture was detected.
- `LONG_PRESS_MOVE_UPDATE` - When a pointer is moving after being held in contact at the same location for a long period of time.
- `LONG_PRESS_START` - When a pointer has remained in contact with the screen at the same location for a long period of time.
- `PAN_CANCEL` - When the pointer that previously triggered a pan-start did not complete.
- `PAN_DOWN` - When a pointer has contacted the screen and might begin to move
- `PAN_END` - When a pointer that was previously in contact with the screen and moving is no longer in contact with the screen.
- `PAN_START` - When a pointer has contacted the screen and has begun to move.
- `PAN_UPDATE` - When a pointer that is in contact with the screen and moving has moved again.
- `POINTER_ENTER` - The pointer has moved with respect to the device while the pointer is or is not in contact with the device, and it has entered our chart.
- `POINTER_EXIT` - The pointer has moved with respect to the device while the pointer is or is not in contact with the device, and exited our chart.
- `POINTER_HOVER` - The pointer has moved with respect to the device while the pointer is not in contact with the device.
- `TAP_CANCEL` - When the pointer that previously triggered a tap-down will not end up causing a tap.
- `TAP_DOWN` - When a pointer that might cause a tap has contacted the screen.
- `TAP_UP` - When a pointer that will trigger a tap has stopped contacting the screen.
- `UNDEFINED` - An undefined event.

## Properties

### `buildLONG_PRESS_END`class-attributeinstance-attribute

When a pointer stops contacting the screen after a long press gesture was detected. Also reports the position where the pointer stopped contacting the screen.

### `buildLONG_PRESS_MOVE_UPDATE`class-attributeinstance-attribute

When a pointer is moving after being held in contact at the same location for a long period of time. Reports the new position and its offset from the original down position.

### `buildLONG_PRESS_START`class-attributeinstance-attribute

When a pointer has remained in contact with the screen at the same location for a long period of time.

### `buildPAN_CANCEL`class-attributeinstance-attribute

When the pointer that previously triggered a pan-start did not complete.

### `buildPAN_DOWN`class-attributeinstance-attribute

When a pointer has contacted the screen and might begin to move

### `buildPAN_END`class-attributeinstance-attribute

When a pointer that was previously in contact with the screen and moving is no longer in contact with the screen.

**buildPAN\_START**class-attributeinstance-attribute

When a pointer has contacted the screen and has begun to move.

**buildPAN\_UPDATE**class-attributeinstance-attribute

When a pointer that is in contact with the screen and moving has moved again.

**buildPOINTER\_ENTER**class-attributeinstance-attribute

The pointer has moved with respect to the device while the pointer is or is not in contact with the device, and it has entered our chart.

**buildPOINTER\_EXIT**class-attributeinstance-attribute

The pointer has moved with respect to the device while the pointer is or is not in contact with the device, and exited our chart.

**buildPOINTER\_HOVER**class-attributeinstance-attribute

The pointer has moved with respect to the device while the pointer is not in contact with the device.

**buildTAP\_CANCEL**class-attributeinstance-attribute

When the pointer that previously triggered a tap-down will not end up causing a tap.

**buildTAP\_DOWN**class-attributeinstance-attribute

When a pointer that might cause a tap has contacted the screen.

**buildTAP\_UP**class-attributeinstance-attribute

When a pointer that will trigger a tap has stopped contacting the screen.

**buildUNDEFINED**class-attributeinstance-attribute

An undefined event.

# ChartGridLines

Configures the appearance of horizontal and vertical grid lines within the chart.

## Properties

- `color` - The color of a grid line.
- `dash_pattern` - Defines dash effect of the line.
- `interval` - The interval between grid lines.
- `width` - The width of a grid line.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildcolorclass-attributeinstance-attribute`

`color: ColorValue | None = None`

The color of a grid line.

### `bulddash_patternclass-attributeinstance-attribute`

`dash_pattern: list[int] | None = None`

Defines dash effect of the line. The value is a circular list of dash offsets and lengths. For example, the list [5, 10] would result in dashes 5 pixels long followed by blank spaces 10 pixels long. By default, a solid line is drawn.

### `buildintervalclass-attributeinstance-attribute`

`interval: Number | None = None`

The interval between grid lines.

### `buildwidthclass-attributeinstance-attribute`

`width: Number = 2.0`

The width of a grid line.

## Methods

### `deployed_codecopy`

```
copy(
 interval: Number | None = None,
 color: ColorValue | None = None,
 width: Number | None = None,
 dash_pattern: list[int] | None = None,
) -> ChartGridLines
```

Returns a copy of this object with the specified properties overridden.

# ChartPointLine

Defines style of a line.

## Properties

- `color` - The line's color.
- `dash_pattern` - The line's dash pattern.
- `gradient` - The line's gradient.
- `width` - The line's width.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue | None = None
```

The line's color.

### `bulddash_patternclass-attributeinstance-attribute`

```
dash_pattern: list[int] | None = None
```

The line's dash pattern.

### `buildgradientclass-attributeinstance-attribute`

```
gradient: Gradient | None = None
```

The line's gradient.

### `buildwidthclass-attributeinstance-attribute`

```
width: Number = 2
```

The line's width.

## Methods

### `deployed_code``copy`

```
copy(
 color: ColorValue | None = None,
 width: Number | None = None,
 dash_pattern: list[int] | None = None,
 gradient: Gradient | None = None,
) -> ChartPointLine
```

Returns a copy of this object with the specified properties overridden.

# ChartPointShape

Base class for chart point shapes.

See usable subclasses:

- `ChartCirclePoint`
- `ChartCrossPoint`
- `ChartSquarePoint`



# ChartSquarePoint

Draws a square.

Inherits: [ChartPointShape](#)

## Properties

- `color` - The fill color to use for the square.
- `size` - The size of the square.
- `stroke_color` - The stroke color to use for the square.
- `stroke_width` - The stroke width to use for the square.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue | None = None
```

The fill color to use for the square.

### `buildsizeclass-attributeinstance-attribute`

```
size: Number = 4.0
```

The size of the square.

### `buildstroke_colorclass-attributeinstance-attribute`

```
stroke_color: ColorValue | None = None
```

The stroke color to use for the square.

### `buildstroke_widthclass-attributeinstance-attribute`

```
stroke_width: Number = 1.0
```

The stroke width to use for the square.

## Methods

### `deployed_code``copy`

```
copy(
 color: ColorValue | None = None,
 size: Number | None = None,
 stroke_color: ColorValue | None = None,
 stroke_width: Number | None = None,
) -> ChartSquarePoint
```

Returns a copy of this object with the specified properties overridden.

# HorizontalAlignment

Defines an element's horizontal alignment to given point.

Inherits: `enum.Enum`

## Properties

- **CENTER** - Element shown horizontally center aligned to a given point.
- **LEFT** - Element shown on the left side of the given point.
- **RIGHT** - Element shown on the right side of the given point.

## Properties

**buildCENTER**`class-attributeinstance-attribute`

Element shown horizontally center aligned to a given point.

**buildLEFT**`class-attributeinstance-attribute`

Element shown on the left side of the given point.

**buildRIGHT**`class-attributeinstance-attribute`

Element shown on the right side of the given point.

# LineChartData

Styling and geometry configuration for a single line series.

Inherits: `BaseControl`

## Properties

- `above_line` - A vertical line drawn between a line point and the top edge of the chart.
- `above_line_bgcolor` - Fill the area above chart line with the specified color.
- `above_line_cutoff_y` - Cut off filled area above line chart at specific Y value.
- `above_line_gradient` - Fill the area above chart line with the specified gradient.
- `below_line` - A vertical line drawn between a line point and the bottom edge of the chart.
- `below_line_bgcolor` - Fill the area below chart line with the specified color.
- `below_line_cutoff_y` - Cut off filled area below line chart at specific Y value.
- `below_line_gradient` - Fill the area below chart line with the specified gradient.
- `color` - A color of chart line.
- `curve_smoothness` - Defines the smoothness of a curve line, when `curved` is set to `True`.
- `curved` - Whether to draw this chart line as a curve.
- `dash_pattern` - Defines dash effect of the line.
- `gradient` - Gradient to draw line's background.
- `point` - Defines the appearance and shape of a line point (dot).
- `points` - A list of points (dots) of `LineChartDataPoint` type representing a single chart line.
- `prevent_curve_over_shooting` - Whether to prevent overshooting when draw curve line on linear sequence spots.
- `prevent_curve_over_shooting_threshold` - Threshold for `prevent_curve_over_shooting` algorithm.
- `rounded_stroke_cap` - Whether to draw rounded line caps.
- `rounded_stroke_join` - Whether to draw rounded line joins.
- `selected_below_line` - A vertical line drawn between selected line point and the bottom edge of the chart.
- `selected_point` - Defines the appearance and shape of a selected line point.
- `shadow` - Shadow to drop by a chart line.
- `step_direction` - Determines the direction of each step.
- `stroke_width` - The width of a chart line.

## Properties

### `buildabove_lineclass-attributeinstance-attribute`

`above_line: ChartPointLine | None = None`

A vertical line drawn between a line point and the top edge of the chart.

### `buildabove_line_bgcolorclass-attributeinstance-attribute`

`above_line_bgcolor: ColorValue | None = None`

Fill the area above chart line with the specified color.

### `buildabove_line_cutoff_yclass-attributeinstance-attribute`

`above_line_cutoff_y: Number | None = None`

Cut off filled area above line chart at specific Y value.

### `buildabove_line_gradientclass-attributeinstance-attribute`

```
above_line_gradient: Gradient | None = None
```

Fill the area above chart line with the specified gradient.

### **buildbelow\_lineclass-attributeinstance-attribute**

```
below_line: ChartPointLine | None = None
```

A vertical line drawn between a line point and the bottom edge of the chart.

### **buildbelow\_line\_bgcolorclass-attributeinstance-attribute**

```
below_line_bgcolor: ColorValue | None = None
```

Fill the area below chart line with the specified color.

### **buildbelow\_line\_cutoff\_yclass-attributeinstance-attribute**

```
below_line_cutoff_y: Number | None = None
```

Cut off filled area below line chart at specific Y value.

### **buildbelow\_line\_gradientclass-attributeinstance-attribute**

```
below_line_gradient: Gradient | None = None
```

Fill the area below chart line with the specified gradient.

### **buildcolorclass-attributeinstance-attribute**

```
color: ColorValue = Colors.CYAN
```

A color of chart line.

### **buildcurve\_smoothnessclass-attributeinstance-attribute**

```
curve_smoothness: Number = 0.35
```

Defines the smoothness of a curve line, when curved is set to True.

### **buildcurvedclass-attributeinstance-attribute**

```
curved: bool = False
```

Whether to draw this chart line as a curve.

### **bulddash\_patternclass-attributeinstance-attribute**

```
dash_pattern: list[int] | None = None
```

Defines dash effect of the line. The value is a circular list of dash offsets and lengths. For example, the list [5, 10] would result in dashes 5 pixels long followed by blank spaces 10 pixels long. By default, a solid line is drawn.

### **buildgradientclass-attributeinstance-attribute**

```
gradient: Gradient | None = None
```

Gradient to draw line's background.

### **buildpointclass-attributeinstance-attribute**

```
point: None | bool | ChartPointShape = None
```

Defines the appearance and shape of a line point (dot).

Setting this property to True will draw a point with default style.

### **buildpointsclass-attributeinstance-attribute**

```
points: list[LineChartDataPoint] = field(
 default_factory=list
)
```

A list of points (dots) of LineChartDataPoint type representing a single chart line.

### **buildprevent\_curve\_over\_shootingclass-attributeinstance-attribute**

```
prevent_curve_over_shooting: bool = False
```

Whether to prevent overshooting when draw curve line on linear sequence spots.

### **buildprevent\_curve\_over\_shooting\_thresholdclass-attributeinstance-attribute**

```
prevent_curve_over_shooting_threshold: Number = 10.0
```

Threshold for prevent\_curve\_over\_shooting algorithm.

### **buildrounded\_stroke\_capclass-attributeinstance-attribute**

```
rounded_stroke_cap: bool = False
```

Whether to draw rounded line caps.

### **buildrounded\_stroke\_joinclass-attributeinstance-attribute**

```
rounded_stroke_join: bool = False
```

Whether to draw rounded line joins.

### **buildselected\_below\_lineclass-attributeinstance-attribute**

```
selected_below_line: None | bool | ChartPointLine = None
```

A vertical line drawn between selected line point and the bottom edge of the chart.

Setting this property to True will draw a line with default style.

### **buildselected\_pointclass-attributeinstance-attribute**

```
selected_point: None | bool | ChartPointShape = None
```

Defines the appearance and shape of a selected line point.

### **buildshadowclass-attributeinstance-attribute**

```
shadow: BoxShadow = field(
 default_factory=lambda: BoxShadow(
 color=Colors.TRANSPARENT
)
)
```

Shadow to drop by a chart line.

### **buildstep\_directionclass-attributeinstance-attribute**

```
step_direction: Number | None = None
```

Determines the direction of each step.

If not None, this chart will be drawn as a Step Line Chart.

Below are some typical values:

- 0.0: Go to the next spot directly, with the current point's y value.
- 0.5: Go to the half with the current spot y, and with the next spot y for the rest.
- 1.0: Go to the next spot y and direct line to the next spot.

### **buildstroke\_widthclass-attributeinstance-attribute**

```
stroke_width: Number = 2.0
```

The width of a chart line.

# LineChartDataPoint

A LineChartData point.

Inherits: BaseControl

## Properties

- `point` - Defines the appearance and shape of a line point.
- `selected` - Draw the point as selected when `LineChart.interactive` is set to `False`.
- `selected_below_line` - A vertical line drawn between selected line point and the bottom edge of the chart.
- `selected_point` - Defines the appearance and shape of a selected line point.
- `show_above_line` - Whether to display a line above data point.
- `show_below_line` - Whether to display a line below data point.
- `show_tooltip` - Whether the tooltip should be shown when this data point is hovered over.
- `tooltip` - Configuration of the tooltip for this data point.
- `x` - The position of a point on X axis.
- `y` - The position of a point on Y axis.

## Properties

### `buildpointclass-attributeinstance-attribute`

`point: None | bool | ChartPointShape = None`

Defines the appearance and shape of a line point.

### `buildselectedclass-attributeinstance-attribute`

`selected: bool = False`

Draw the point as selected when `LineChart.interactive` is set to `False`.

### `buildselected_below_lineclass-attributeinstance-attribute`

`selected_below_line: None | bool | ChartPointLine = None`

A vertical line drawn between selected line point and the bottom edge of the chart.

The value is either `True` - draw a line with default style, `False` - do not draw a line under selected point, or an instance of `ChartPointLine` class to specify line style to draw.

### `buildselected_pointclass-attributeinstance-attribute`

`selected_point: None | bool | ChartPointShape = None`

Defines the appearance and shape of a selected line point.

### `buildshow_above_lineclass-attributeinstance-attribute`

`show_above_line: bool = True`

Whether to display a line above data point.

### `buildshow_below_lineclass-attributeinstance-attribute`

`show_below_line: bool = True`

Whether to display a line below data point.

### `buildshow_tooltipclass-attributeinstance-attribute`

`show_tooltip: bool = True`

Whether the tooltip should be shown when this data point is hovered over.

### `buildtooltipclass-attributeinstance-attribute`

`tooltip: LineChartDataPointTooltip | str = field(  
 default_factory=lambda: LineChartDataPointTooltip()`

)

Configuration of the tooltip for this data point.

### **buildxinstance-attribute**

x: Number

The position of a point on X axis.

### **buildyinstance-attribute**

y: Number

The position of a point on Y axis.

# LineChartDataPointTooltip

Tooltip configuration for the LineChartDataPoint.

Inherits: ChartDataPointTooltip

## Properties

- `text` - The text to display in the tooltip.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildtextclass-attributeinstance-attribute`

`text: str | None = None`

The text to display in the tooltip.

When None, defaults to LineChartDataPoint.y.

## Methods

### `deployed_codecopy`

```
copy(
 text: str | None = None,
 text_style: TextStyle | None = None,
 text_align: TextAlign | None = None,
 text_spans: list[TextSpan] | None = None,
 rtl: bool | None = None,
) -> LineChartDataPointTooltip
```

Returns a copy of this object with the specified properties overridden.



# LineChartEvent

Event payload emitted when the user interacts with a line chart.

Inherits: Event

## Properties

- spots - Spots on which the event occurred.
- type - The type of event that occurred.

## Properties

### buildspotsinstance-attribute

spots: list[LineChartEventSpot]

Spots on which the event occurred.

Note

This list is empty when the event does not target a concrete point, for example when the pointer hovers over or taps empty chart space.

### buildtypeinstance-attribute

type: ChartEventType

The type of event that occurred.

# LineChartEventSpot

Identifies a concrete line/point pair involved in a chart interaction event.

## Properties

- `bar_index` - The line's index or -1 if no line was hovered.
- `spot_index` - The line's point index or -1 if no point was hovered.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildbar_indexinstance-attribute`

`bar_index: int`

The line's index or -1 if no line was hovered.

### `buildspot_indexinstance-attribute`

`spot_index: int`

The line's point index or -1 if no point was hovered.

## Methods

### `deployed_code``copy`

```
copy(
 bar_index: int | None = None,
 spot_index: int | None = None,
) -> LineChartEventSpot
```

Returns a copy of this object with the specified properties overridden.

# LineChartTooltip

Configuration of the tooltip for LineCharts.

## Properties

- `bgcolor` - Background color of tooltip.
- `border_radius` - The tooltip's border radius.
- `border_side` - Defines the borders of this tooltip.
- `fit_inside_horizontally` - Forces the tooltip to shift horizontally inside the chart, if overflow happens.
- `fit_inside_vertically` - Forces the tooltip to shift vertically inside the chart, if overflow happens.
- `horizontal_alignment` - The horizontal alignment of this tooltip.
- `horizontal_offset` - Applies horizontal offset for showing tooltip.
- `margin` - Applies a bottom margin for showing tooltip on top of rods.
- `max_width` - Restricts the tooltip's width.
- `padding` - Applies a padding for showing contents inside the tooltip.
- `rotation` - The tooltip's rotation angle in degrees.
- `show_on_top_of_chart_box_area` - Whether to force the tooltip container to top of the line.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

```
bgcolor: ColorValue = '#FF607D8B'
```

Background color of tooltip.

### `buildborder_radiusclass-attributeinstance-attribute`

```
border_radius: BorderRadiusValue | None = None
```

The tooltip's border radius.

### `buildborder_sideclass-attributeinstance-attribute`

```
border_side: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

Defines the borders of this tooltip.

### `buildfit_inside_horizontallyclass-attributeinstance-attribute`

```
fit_inside_horizontally: bool = False
```

Forces the tooltip to shift horizontally inside the chart, if overflow happens.

### `buildfit_inside_verticallyclass-attributeinstance-attribute`

```
fit_inside_vertically: bool = False
```

Forces the tooltip to shift vertically inside the chart, if overflow happens.

### `buildhorizontal_alignmentclass-attributeinstance-attribute`

```
horizontal_alignment: HorizontalAlignment = (
 HorizontalAlignment.CENTER
)
```

The horizontal alignment of this tooltip.

### `buildhorizontal_offsetclass-attributeinstance-attribute`

```
horizontal_offset: Number = 0.0
```

Applies horizontal offset for showing tooltip.

### **buildmarginclass-attributeinstance-attribute**

```
margin: Number = 16
```

Applies a bottom margin for showing tooltip on top of rods.

### **buildmax\_widthclass-attributeinstance-attribute**

```
max_width: Number = 120
```

Restricts the tooltip's width.

### **buildpaddingclass-attributeinstance-attribute**

```
padding: PaddingValue = field(
 default_factory=lambda: Padding.symmetric(
 vertical=8, horizontal=16
)
)
```

Applies a padding for showing contents inside the tooltip.

### **buildrotationclass-attributeinstance-attribute**

```
rotation: Number = 0.0
```

The tooltip's rotation angle in degrees.

### **buildshow\_on\_top\_of\_chart\_box\_areaclass-attributeinstance-attribute**

```
show_on_top_of_chart_box_area: bool = False
```

Whether to force the tooltip container to top of the line.

## **Methods**

### **deployed\_codecopy**

```
copy(
 bgcolor: ColorValue | None = None,
 border_radius: BorderRadiusValue | None = None,
 margin: Number | None = None,
 padding: PaddingValue | None = None,
 max_width: Number | None = None,
 rotation: Number | None = None,
 horizontal_offset: Number | None = None,
 border_side: BorderSide | None = None,
 fit_inside_horizontally: bool | None = None,
 fit_inside_vertically: bool | None = None,
 show_on_top_of_chart_box_area: bool | None = None,
) -> LineChartTooltip
```

Returns a copy of this object with the specified properties overridden.

# PieChartEvent

Event payload emitted when the user interacts with a pie chart section.

Inherits: Event

## Properties

- `local_x` - X coordinate of the local position where the event occurred.
- `local_y` - Y coordinate of the local position where the event occurred.
- `section_index` - Section's index or -1 if no section was hovered.
- `type` - Type of the event.

## Properties

### `buildlocal_xclass-attributeinstance-attribute`

`local_x: float | None = None`

X coordinate of the local position where the event occurred.

### `buildlocal_yclass-attributeinstance-attribute`

`local_y: float | None = None`

Y coordinate of the local position where the event occurred.

### `buildsection_indexclass-attributeinstance-attribute`

`section_index: int | None = None`

Section's index or -1 if no section was hovered.

### `buildtypeinstance-attribute`

`type: ChartEventType`

Type of the event.

# PieChartSection

Configures a PieChart section.

Inherits: BaseControl

## Properties

- `badge` - An optional Control drawn in the middle of a section.
- `badge_position` - The position/offset of the badge relative to the section's center.
- `border_side` - The border around section shape.
- `color` - Background color of the section.
- `gradient` - Defines the gradient of section.
- `radius` - External radius of the section.
- `title` - A title drawn at the center of the section.
- `title_position` - The position/offset of the title relative to the section's center.
- `title_style` - The style to draw title with.
- `value` - Determines how much the section should occupy.

## Properties

### `buildbadgeclass-attributeinstance-attribute`

```
badge: Control | None = None
```

An optional Control drawn in the middle of a section.

### `buildbadge_positionclass-attributeinstance-attribute`

```
badge_position: Number | None = None
```

The position/offset of the badge relative to the section's center.

- 0.0: near the center
- 1.0: near the outside of the chart

By default the badge is drawn in the middle of the section.

### `buildborder_sideclass-attributeinstance-attribute`

```
border_side: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

The border around section shape.

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue | None = None
```

Background color of the section.

### `buildgradientclass-attributeinstance-attribute`

```
gradient: Gradient | None = None
```

Defines the gradient of section. If specified, overrides the color setting.

### `buildradiusclass-attributeinstance-attribute`

```
radius: Number | None = None
```

External radius of the section.

### `buildtitleclass-attributeinstance-attribute`

```
title: str | None = None
```

A title drawn at the center of the section.

### **buildtitle\_positionclass-attributeinstance-attribute**

title\_position: Number | None = None

The position/offset of the title relative to the section's center.

- 0.0: near the center
- 1.0: near the outside of the chart

By default the title is drawn in the middle of the section.

### **buildtitle\_styleclass-attributeinstance-attribute**

title\_style: TextStyle | None = None

The style to draw title with.

### **buildvalueinstance-attribute**

value: Number

Determines how much the section should occupy. This depends on sum of all sections, each section should occupy (value / sum of all values) \* 360 degrees.

# RadarChartEvent

Event raised for interactions with a RadarChart.

Inherits: Event

## Properties

- `data_set_index` - The index of the touched data set, if any.
- `entry_index` - The index of the touched radar entry, if any.
- `entry_value` - The value of the touched radar entry, if any.
- `type` - The touch or pointer event that occurred.

## Properties

**builddata\_set\_index**class-attributeinstance-attribute

`data_set_index: int | None = None`

The index of the touched data set, if any.

**buildentry\_index**class-attributeinstance-attribute

`entry_index: int | None = None`

The index of the touched radar entry, if any.

**buildentry\_value**class-attributeinstance-attribute

`entry_value: Number | None = None`

The value of the touched radar entry, if any.

**buildtype**instance-attribute

`type: ChartEventType`

The touch or pointer event that occurred.



# RadarChartTitle

Custom title configuration displayed around a RadarChart.

Inherits: BaseControl

## Properties

- `angle` - Rotation angle (in degrees) applied to the title.
- `position_percentage_offset` - Defines the relative distance of this title from the chart center.
- `text` - The text displayed for the title.
- `text_spans` - Inline spans appended to the title.

## Properties

### `buildangleclass-attributeinstance-attribute`

```
angle: Number = 0
```

Rotation angle (in degrees) applied to the title.

### `buildposition_percentage_offsetclass-attributeinstance-attribute`

```
position_percentage_offset: Number | None = None
```

Defines the relative distance of this title from the chart center.

- 0 draws this title near the inside edge of each section.
- 1 draws this title near the outside edge of each section.

Must be between 0 and 1 (inclusive), if set.

Note

If set, it takes precedence over the parent `RadarChart.title_position_percentage_offset` value.

### `buildtextclass-attributeinstance-attribute`

```
text: str = ''
```

The text displayed for the title.

### `buildtext_spansclass-attributeinstance-attribute`

```
text_spans: list[TextSpan] | None = None
```

Inline spans appended to the title.

# RadarDataSet

A collection of RadarDataSetEntry drawn as a filled radar shape.

Inherits: BaseControl

## Properties

- `border_color` - The color of the dataset outline.
- `border_width` - The width of the dataset outline.
- `entries` - The data points that compose this set.
- `entry_radius` - The radius of each entry.
- `fill_color` - The color used to fill this dataset.
- `fill_gradient` - The gradient used to fill this dataset.

## Properties

### `buildborder_color`class-attributeinstance-attribute

```
border_color: ColorValue = Colors.CYAN
```

The color of the dataset outline.

### `buildborder_width`class-attributeinstance-attribute

```
border_width: Number = 2.0
```

The width of the dataset outline.

### `buildentries`class-attributeinstance-attribute

```
entries: list[RadarDataSetEntry] = field(
 default_factory=list
)
```

The data points that compose this set.

### `buildentry_radius`class-attributeinstance-attribute

```
entry_radius: Number = 5.0
```

The radius of each entry.

### `buildfill_color`class-attributeinstance-attribute

```
fill_color: ColorValue = Colors.CYAN
```

The color used to fill this dataset.

### `buildfill_gradient`class-attributeinstance-attribute

```
fill_gradient: Gradient | None = None
```

The gradient used to fill this dataset.

Takes precedence over `fill_color`.

# RadarDataSetEntry

A single data point rendered on a RadarChart.

Inherits: BaseControl

## Properties

- `value` - The numeric value drawn for this entry.

## Properties

`buildvalueinstance-attribute`

`value`: Number

The numeric value drawn for this entry.

# RadarShape

Shape of the radar grid and data polygons.

Inherits: `enum.Enum`

## Properties

- `CIRCLE` - Draws radial circles for the grid and data outlines.
- `POLYGON` - Draws straight-edged polygons for the grid and data outlines.

## Properties

`buildCIRCLEclass-attributeinstance-attribute`

Draws radial circles for the grid and data outlines.

`buildPOLYGONclass-attributeinstance-attribute`

Draws straight-edged polygons for the grid and data outlines.

# ScatterChartEvent

Event payload emitted when the user interacts with a scatter chart.

Inherits: Event

## Properties

- `spot_index` - The index of the touched spot, if any.
- `type` - The type of the event that occurred.

## Properties

**buildspot\_index**class-attributeinstance-attribute

`spot_index: int | None = None`

The index of the touched spot, if any.

**buildtype**instance-attribute

`type: ChartEventType`

The type of the event that occurred.

# ScatterChartSpot

A spot on a scatter chart.

Inherits: `BaseControl`

## Properties

- `color` - Color of a spot.
- `label_text` - TBD
- `label_text_style` - TBD
- `point` - TBD
- `radius` - Radius of a spot.
- `render_priority` - Sort by this to manage overlap.
- `selected` - Whether to treat this spot as selected.
- `show_tooltip` - Whether to show the tooltip.
- `tooltip` - Tooltip configuration for this spot.
- `visible` - Determines whether to show or hide the spot.
- `x` - The position of a spot on X axis.
- `x_error` - Determines the error range of the data point using `FlErrorRange` (which contains `lowerBy` and `upperValue`) for the X axis.
- `y` - The position of a spot on Y axis.
- `y_error` - Determines the error range of the data point using `FlErrorRange` (which contains `lowerBy` and `upperValue`) for the Y axis.

## Properties

### `buildcolorclass-attributeinstance-attribute`

```
color: ColorValue | None = None
```

Color of a spot.

### `buildlabel_textclass-attributeinstance-attribute`

```
label_text: str = ''
```

TBD

### `buildlabel_text_styleclass-attributeinstance-attribute`

```
label_text_style: TextStyle = field(
 default_factory=lambda: TextStyle()
)
```

TBD

### `buildpointclass-attributeinstance-attribute`

```
point: None | bool | ChartPointShape = None
```

TBD

### `buildradiusclass-attributeinstance-attribute`

```
radius: Number | None = None
```

Radius of a spot.

### `buildrender_priorityclass-attributeinstance-attribute`

```
render_priority: Number = 0
```

Sort by this to manage overlap.

### **buildselectedclass-attributeinstance-attribute**

```
selected: bool = False
```

Whether to treat this spot as selected.

### **buildshow\_tooltipclass-attributeinstance-attribute**

```
show_tooltip: bool = True
```

Whether to show the tooltip.

### **buildtooltipclass-attributeinstance-attribute**

```
tooltip: ScatterChartSpotTooltip | str = field(
 default_factory=lambda: ScatterChartSpotTooltip()
)
```

Tooltip configuration for this spot.

### **buildvisibleclass-attributeinstance-attribute**

```
visible: bool = True
```

Determines whether to show or hide the spot.

### **buildxclass-attributeinstance-attribute**

```
x: Number | None = None
```

The position of a spot on X axis.

### **buildx\_errorclass-attributeinstance-attribute**

```
x_error: Any | None = None
```

Determines the error range of the data point using FLErrorRange (which contains lowerBy and upperValue) for the X axis.

### **buildyclass-attributeinstance-attribute**

```
y: Number | None = None
```

The position of a spot on Y axis.

### **buildy\_errorclass-attributeinstance-attribute**

```
y_error: Any | None = None
```

Determines the error range of the data point using FLErrorRange (which contains lowerBy and upperValue) for the Y axis.

# ScatterChartSpotTooltip

Tooltip configuration for the ScatterChartSpot.

Inherits: ChartDataPointTooltip

## Properties

- `bottom_margin` - The bottom space from the spot.
- `text` - The text to display in the tooltip.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

**buildbottom\_margin**class-attributeinstance-attribute

```
bottom_margin: Number = 8
```

The bottom space from the spot.

**buildtext**class-attributeinstance-attribute

```
text: str | None = None
```

The text to display in the tooltip.

When None, defaults to ScatterChartSpot.y.

## Methods

**deployed\_code**copy

```
copy(
 text: str | None = None,
 text_style: TextStyle | None = None,
 text_align: TextAlign | None = None,
 text_spans: list[TextSpan] | None = None,
 rtl: bool | None = None,
 bottom_margin: float | None = None,
) -> ScatterChartSpotTooltip
```

Returns a copy of this object with the specified properties overridden.



# ScatterChartTooltip

Configuration of the tooltip for ScatterCharts.

## Properties

- `bgcolor` - The tooltip's background color.
- `border_radius` - The tooltip's border radius.
- `border_side` - The tooltip's border side.
- `fit_inside_horizontally` - Forces the tooltip to shift horizontally inside the chart, if overflow happens.
- `fit_inside_vertically` - Forces the tooltip to shift vertically inside the chart, if overflow happens.
- `horizontal_alignment` - The tooltip's horizontal alignment.
- `horizontal_offset` - Applies horizontal offset for showing tooltip.
- `max_width` - Restricts the tooltip's width.
- `padding` - Applies a padding for showing contents inside the tooltip.
- `rotation` - The tooltip's rotation angle in degrees.

## Methods

- `copy` - Returns a copy of this object with the specified properties overridden.

## Properties

### `buildbgcolorclass-attributeinstance-attribute`

```
bgcolor: ColorValue = '#FF607D8B'
```

The tooltip's background color.

### `buildborder_radiusclass-attributeinstance-attribute`

```
border_radius: BorderRadiusValue | None = None
```

The tooltip's border radius.

### `buildborder_sideclass-attributeinstance-attribute`

```
border_side: BorderSide = field(
 default_factory=lambda: BorderSide.none()
)
```

The tooltip's border side.

### `buildfit_inside_horizontallyclass-attributeinstance-attribute`

```
fit_inside_horizontally: bool = False
```

Forces the tooltip to shift horizontally inside the chart, if overflow happens.

### `buildfit_inside_verticallyclass-attributeinstance-attribute`

```
fit_inside_vertically: bool = False
```

Forces the tooltip to shift vertically inside the chart, if overflow happens.

### `buildhorizontal_alignmentclass-attributeinstance-attribute`

```
horizontal_alignment: HorizontalAlignment = (
 HorizontalAlignment.CENTER
)
```

The tooltip's horizontal alignment.

### `buildhorizontal_offsetclass-attributeinstance-attribute`

```
horizontal_offset: Number = 0
```

Applies horizontal offset for showing tooltip.

### `buildmax_widthclass-attributeinstance-attribute`

```
max_width: Number = 120
```

Restricts the tooltip's width.

### buildpaddingclass-attributeinstance-attribute

```
padding: PaddingValue = field(
 default_factory=lambda: Padding.symmetric(
 vertical=8, horizontal=16
)
)
```

Applies a padding for showing contents inside the tooltip.

### buildrotationclass-attributeinstance-attribute

```
rotation: Number = 0.0
```

The tooltip's rotation angle in degrees.

## Methods

### deployed\_codecopy

```
copy(
 bgcolor: ColorValue | None = None,
 border_radius: BorderRadiusValue | None = None,
 padding: PaddingValue | None = None,
 max_width: Number | None = None,
 rotation: Number | None = None,
 horizontal_offset: Number | None = None,
 horizontal_alignment: HorizontalAlignment | None = None,
 border_side: BorderSide | None = None,
 fit_inside_horizontally: bool | None = None,
 fit_inside_vertically: bool | None = None,
) -> ScatterChartTooltip
```

Returns a copy of this object with the specified properties overridden.

# CodeLanguage

Inherits: `enum.Enum`

## Properties

- ABNF - ABNF
- ACCESSLOG - Accesslog
- ACTIONSCRIPT - ActionScript
- ADA - Ada
- ANGELSCRIPT - Angelscript
- APACHE - Apache
- APPLESCRIPT - AppleScript
- ARCADE - Arcade
- ARDUINO - Arduino
- ARMASM - ARM Assembly
- ASCIIDOC - AsciiDoc
- ASPECTJ - Aspectj
- AUTOHOTKEY - AutoHotkey
- AUTOIT - AutoIt
- AVRASM - AVR Assembly
- AWK - Awk
- AXAPTA - Axapta
- BASH - Bash
- BASIC - Basic
- BNF - BNF
- BRAINFUCK - Brainfuck
- CAL - Cal
- CAPNPROTO - Cap'n Proto
- CEYLON - Ceylon
- CLEAN - Clean
- CLOJURE - Clojure
- CLOJURE\_REPL - Clojure REPL
- CMAKE - CMake
- COFFEESCRIPT - CoffeeScript
- COQ - Coq
- COS - Cos
- CPP - C++
- CRMSH - Crmsh
- CRYSTAL - Crystal
- CS - C#
- CSP - Csp

- CSS - CSS
- D - D
- DART - Dart
- DELPHI - Delphi
- DIFF - Diff
- DJANGO - Django
- DNS - DNS
- DOCKERFILE - Dockerfile
- DOS - DOS
- DSCONFIG - DSConfig
- DTS - DTS
- DUST - Dust
- EBNF - EBNF
- ELIXIR - Elixir
- ELM - Elm
- ERB - Erb
- ERLANG - Erlang
- ERLANG\_REPL - Erlang REPL
- EXCEL - Excel
- FIX - Fix
- FLIX - Flix
- FORTRAN - Fortran
- FSHARP - F#
- GAMS - Gams
- GAUSS - Gauss
- GCODE - G-code
- GHERKIN - Gherkin
- GLSL - GLSL
- GML - GML
- GN - Gn
- GO - Go
- GOLO - Golo
- GRADLE - Gradle
- GRAPHQL - GraphQL
- GROOVY - Groovy
- HAML - Haml
- HANDLEBARS - Handlebars
- HASKELL - Haskell
- HAXE - Haxe
- HSP - Hsp
- HTMLBARS - HTMLBars

- HTTP - HTTP
- HY - Hy
- INFORM7 - INFORM7
- INI - INI
- IRPF90 - IRPF90
- ISBL - ISBL
- JAVA - Java
- JAVASCRIPT - Javascript
- JBOSS\_CLI - JBoss CLI
- JSON - JSON
- JULIA - Julia
- JULIA\_REPL - Julia REPL
- KOTLIN - Kotlin
- LASSO - Lasso
- LDIF - LDIF
- LEAF - Leaf
- LESS - Less
- LISP - Lisp
- LIVECODESERVER - LiveCode Server
- LIVESCRIPT - Livescript
- LLVM - LLVM
- LSL - LSL
- LUA - Lua
- MAKEFILE - Makefile
- MARKDOWN - Markdown
- MATHEMATICA - Mathematica
- MATLAB - Matlab
- MAXIMA - Maxima
- MEL - Mel
- MERCURY - Mercury
- MIPSASM - MIPS Assembly
- MIZAR - Mizar
- MOJOLICIOUS - Mojolicious
- MONKEY - Monkey
- MOONSCRIPT - MoonScript
- N1QL - N1QL
- NGINX - Nginx
- NIMROD - Nimrod
- NIX - Nix
- NSIS - NSIS
- OBJECTIVEC - Objective-C

- OCAML - OCaml
- ONE\_C - 1C
- OPENS CAD - OpenSCAD
- OXYGENE - Oxygene
- PARSER3 - PARSER3
- PERL - Perl
- PF - PF
- PGSQL - PostgreSQL
- PHP - PHP
- PLAINTEXT - Plain text
- PONY - Pony
- POWERSHELL - PowerShell
- PROCESSING - Processing
- PROFILE - Profile
- PROLOG - Prolog
- PROPERTIES - Properties
- PROTOBUF - Protocol Buffers
- PUPPET - Puppet
- PUREBASIC - PureBasic
- PYTHON - Python
- Q - Q
- QML - QML
- R - R
- REASONML - ReasonML
- RIB - Rib
- ROBOCONF - Roboconf
- ROUTEROS - RouterOS
- RSL - RSL
- RUBY - Ruby
- RULESLANGUAGE - Rules language
- RUST - Rust
- SAS - SAS
- SCALA - Scala
- SCHEME - Scheme
- SCILAB - Scilab
- SCSS - SCSS
- SHELL - Shell
- SMALI - Smali
- SMALLTALK - Smalltalk
- SML - SML
- SOLIDITY - Solidity

- SQF - SQF
- SQL - SQL
- STAN - Stan
- STATA - Stata
- STEP21 - STEP21
- STYLUS - Stylus
- SUBUNIT - SubUnit
- SWIFT - Swift
- TAGGERSCRIPT - Tagger Script
- TAP - Tap
- TCL - Tcl
- TEX - TeX
- THRIFT - Thrift
- TP - TP
- TWIG - Twig
- TYPESCRIPT - TypeScript
- VALA - Vala
- VBNET - VB.NET
- VBSCRIPT - VBScript
- VBSCRIPT\_HTML - VBScript HTML
- VERILOG - Verilog
- VHDL - VHDL
- VIM - Vim script
- VUE - Vue
- X86ASM - x86 Assembly
- XL - XI
- XML - XML
- XQUERY - XQuery
- YAML - YAML
- ZEPHIR - Zephir

## Properties

buildABNFclass-attributeinstance-attribute

ABNF

buildACCESSLOGclass-attributeinstance-attribute

Accesslog

buildACTIONSRIPTclass-attributeinstance-attribute

ActionScript

buildADAc class-attributeinstance-attribute

Ada

buildANGELSCRIPTclass-attributeinstance-attribute

Angelscript

buildAPACHEclass-attributeinstance-attribute

Apache

buildAPPLESCRIPTclass-attributeinstance-attribute

AppleScript

buildARCADEclass-attributeinstance-attribute

Arcade

buildARDUINOclass-attributeinstance-attribute

Arduino

buildARMASMclass-attributeinstance-attribute

ARM Assembly

buildASCIIDOCclass-attributeinstance-attribute

AsciiDoc

buildASPECTJclass-attributeinstance-attribute

Aspectj

buildAUTOHOTKEYclass-attributeinstance-attribute

AutoHotkey

buildAUTOITclass-attributeinstance-attribute

AutoIt

buildAVRASMclass-attributeinstance-attribute

AVR Assembly

buildAWKclass-attributeinstance-attribute

Awk

buildAXAPTAcass-attributeinstance-attribute

Axapta

buildBASHclass-attributeinstance-attribute

Bash

buildBASICclass-attributeinstance-attribute

Basic

buildBNFclass-attributeinstance-attribute

BNF

buildBRAINFUCKclass-attributeinstance-attribute

Brainfuck

buildCALclass-attributeinstance-attribute

Cal

buildCAPNPROTOclass-attributeinstance-attribute

Cap'n Proto

buildCEYLONclass-attributeinstance-attribute



Ceylon

buildCLEANclass-attributeinstance-attribute

Clean

buildCLOJUREclass-attributeinstance-attribute

Clojure

buildCLOJURE\_REPLclass-attributeinstance-attribute

Clojure REPL

buildCMAKEclass-attributeinstance-attribute

CMake

buildCOFFEESCRIPTclass-attributeinstance-attribute

CoffeeScript

buildCOQclass-attributeinstance-attribute

Coq

buildCOSclass-attributeinstance-attribute

Cos

buildCPPclass-attributeinstance-attribute

C++

buildCRMSHclass-attributeinstance-attribute

CrmsH

buildCRYSTALclass-attributeinstance-attribute

Crystal

buildCSclass-attributeinstance-attribute

C#

buildCSPclass-attributeinstance-attribute

Csp

buildCSSclass-attributeinstance-attribute

CSS

buildDclass-attributeinstance-attribute

D

buildDARTclass-attributeinstance-attribute

Dart

buildDELPHIclass-attributeinstance-attribute

Delphi

buildDIFFclass-attributeinstance-attribute

Diff

buildDJANGOclass-attributeinstance-attribute

Django

buildDNSclass-attributeinstance-attribute

DNS

buildDOCKERFILEclass-attributeinstance-attribute

Dockerfile

buildDOSclass-attributeinstance-attribute

DOS

buildDSCONFIGclass-attributeinstance-attribute

DSConfig

buildDTSclass-attributeinstance-attribute

DTS

buildDUSTclass-attributeinstance-attribute

Dust

buildEBNFclass-attributeinstance-attribute

EBNF

buildELIXIRclass-attributeinstance-attribute

Elixir

buildELMclass-attributeinstance-attribute

Elm

buildERBclass-attributeinstance-attribute

Erb

buildERLANGclass-attributeinstance-attribute

Erlang

buildERLANG\_REPLclass-attributeinstance-attribute

Erlang REPL

buildEXCELclass-attributeinstance-attribute

Excel

buildFIXclass-attributeinstance-attribute

Fix

buildFLIXclass-attributeinstance-attribute

Flix

buildFORTRANclass-attributeinstance-attribute

Fortran

buildFSHARPclass-attributeinstance-attribute

F#

buildGAMSclass-attributeinstance-attribute

Gams

buildGAUSSclass-attributeinstance-attribute

Gauss

buildGCODEclass-attributeinstance-attribute

G-code

buildGHERKINclass-attributeinstance-attribute

Gherkin

buildGLSLclass-attributeinstance-attribute

GLSL

buildGMLclass-attributeinstance-attribute

GML

buildGNclass-attributeinstance-attribute

Gn

buildGOclass-attributeinstance-attribute

Go

buildGOLOclass-attributeinstance-attribute

Golo

buildGRADLEclass-attributeinstance-attribute

Gradle

buildGRAPHQLclass-attributeinstance-attribute

GraphQL

buildGROOVYclass-attributeinstance-attribute

Groovy

buildHAMLclass-attributeinstance-attribute

Haml

buildHANDLEBARSclass-attributeinstance-attribute

Handlebars

buildHASKELLclass-attributeinstance-attribute

Haskell

buildHAXEclass-attributeinstance-attribute

Haxe

buildHSPclass-attributeinstance-attribute

Hsp

buildHTMLBARSclass-attributeinstance-attribute

HTMLBars

buildHTTPclass-attributeinstance-attribute

HTTP

buildHYclass-attributeinstance-attribute

Hy

buildINFORM7class-attributeinstance-attribute

INFORM7

buildINIclass-attributeinstance-attribute

INI

buildIRPF90class-attributeinstance-attribute

IRPF90

buildISBLclass-attributeinstance-attribute

ISBL

buildJAVAcass-attributeinstance-attribute

Java

buildJAVASCRIPTclass-attributeinstance-attribute

Javascript

buildJBOSS\_CLIclass-attributeinstance-attribute

JBoss CLI

buildJSONclass-attributeinstance-attribute

JSON

buildJULIAclass-attributeinstance-attribute

Julia

buildJULIA\_REPLclass-attributeinstance-attribute

Julia REPL

buildKOTLINclass-attributeinstance-attribute

Kotlin

buildLASSOclass-attributeinstance-attribute

Lasso

buildLDIFclass-attributeinstance-attribute

LDIF

buildLEAFclass-attributeinstance-attribute

Leaf

buildLESSclass-attributeinstance-attribute

Less

buildLISPclass-attributeinstance-attribute

Lisp

buildLIVECODESERVERclass-attributeinstance-attribute

LiveCode Server

buildLIVESCRIPTclass-attributeinstance-attribute

Livescript

buildLLVMclass-attributeinstance-attribute

LLVM

buildLSLclass-attributeinstance-attribute

LSL

buildLUAcass-attributeinstance-attribute

Lua

buildMAKEFILEclass-attributeinstance-attribute

Makefile

buildMARKDOWNclass-attributeinstance-attribute

Markdown

buildMATHEMATICAclass-attributeinstance-attribute

Mathematica

buildMATLABclass-attributeinstance-attribute

Matlab

buildMAXIMAcass-attributeinstance-attribute

Maxima

buildMELclass-attributeinstance-attribute

Mel

buildMERCURYclass-attributeinstance-attribute

Mercury

buildMIPSASMclass-attributeinstance-attribute

MIPS Assembly

buildMIZARclass-attributeinstance-attribute

Mizar

buildMOJOLICIOUSclass-attributeinstance-attribute

Mojolicious

buildMONKEYclass-attributeinstance-attribute

Monkey

buildMOONSCRIPTclass-attributeinstance-attribute

MoonScript

buildN1QLclass-attributeinstance-attribute

N1QL

buildNGINXclass-attributeinstance-attribute

Nginx

buildNIMRODclass-attributeinstance-attribute

Nimrod

buildNIXclass-attributeinstance-attribute

Nix

buildNSISclass-attributeinstance-attribute

NSIS

buildOBJECTIVECclass-attributeinstance-attribute

Objective-C

buildOCAMLclass-attributeinstance-attribute

OCaml

buildONE\_Cclass-attributeinstance-attribute

1C

buildOPENSCADclass-attributeinstance-attribute

OpenSCAD

buildOXYGENEclass-attributeinstance-attribute

Oxygene

buildPARSER3class-attributeinstance-attribute

PARSER3

buildPERLclass-attributeinstance-attribute

Perl

buildPFclass-attributeinstance-attribute

PF

buildPGSQLclass-attributeinstance-attribute

PostgreSQL

buildPHPclass-attributeinstance-attribute

PHP

buildPLAINTEXTclass-attributeinstance-attribute

Plain text

buildPONYclass-attributeinstance-attribute

Pony

buildPOWERSHELLclass-attributeinstance-attribute

PowerShell

buildPROCESSINGclass-attributeinstance-attribute

Processing

buildPROFILEclass-attributeinstance-attribute

Profile

buildPROLOGclass-attributeinstance-attribute

Prolog

buildPROPERTIESclass-attributeinstance-attribute

Properties

buildPROTOBUFclass-attributeinstance-attribute

Protocol Buffers

buildPUPPETclass-attributeinstance-attribute

Puppet

buildPUREBASICclass-attributeinstance-attribute

PureBasic

buildPYTHONclass-attributeinstance-attribute

Python

buildQclass-attributeinstance-attribute

Q

buildQMLclass-attributeinstance-attribute

QML

buildRclass-attributeinstance-attribute

R

buildREASONMLclass-attributeinstance-attribute

ReasonML

buildRIBclass-attributeinstance-attribute

Rib

buildROBOCONFclass-attributeinstance-attribute

Roboconf

buildROUTEROSclass-attributeinstance-attribute

RouterOS

buildRSLclass-attributeinstance-attribute

RSL

buildRUBYclass-attributeinstance-attribute

Ruby

buildRULESLANGUAGEclass-attributeinstance-attribute

Rules language

buildRUSTclass-attributeinstance-attribute

Rust

buildSASclass-attributeinstance-attribute

SAS

buildSCALAcass-attributeinstance-attribute

Scala

buildSCHEMEclass-attributeinstance-attribute

Scheme

buildSCILABclass-attributeinstance-attribute

Scilab

buildSCSSclass-attributeinstance-attribute

SCSS

buildSHELLclass-attributeinstance-attribute

Shell

buildSMALLclass-attributeinstance-attribute

Smali

buildSMALLTALKclass-attributeinstance-attribute

Smalltalk

buildSMLclass-attributeinstance-attribute

SML

buildSOLIDITYclass-attributeinstance-attribute

Solidity

buildSQFclass-attributeinstance-attribute

SQF

buildSQLclass-attributeinstance-attribute

SQL

buildSTANclass-attributeinstance-attribute

Stan

buildSTATAclass-attributeinstance-attribute

Stata

buildSTEP21class-attributeinstance-attribute

STEP21

buildSTYLUSclass-attributeinstance-attribute

Stylus

buildSUBUNITclass-attributeinstance-attribute

SubUnit

buildSWIFTclass-attributeinstance-attribute

Swift

buildTAGGERSCRIPTclass-attributeinstance-attribute

Tagger Script

buildTAPclass-attributeinstance-attribute

Tap

buildTCLclass-attributeinstance-attribute

Tcl

buildTEXclass-attributeinstance-attribute

TeX

buildTHRIFTclass-attributeinstance-attribute

Thrift

buildTPclass-attributeinstance-attribute

TP

buildTWIGclass-attributeinstance-attribute

Twig

buildTYPESCRIPTclass-attributeinstance-attribute

TypeScript

buildVALAclass-attributeinstance-attribute



Vala

buildVBNETclass-attributeinstance-attribute

VB.NET

buildVBSCRIPTclass-attributeinstance-attribute

VBScript

buildVBSCRIPT\_HTMLclass-attributeinstance-attribute

VBScript HTML

buildVERILOGclass-attributeinstance-attribute

Verilog

buildVHDLclass-attributeinstance-attribute

VHDL

buildVIMclass-attributeinstance-attribute

Vim script

buildVUEclass-attributeinstance-attribute

Vue

buildX86ASMclass-attributeinstance-attribute

x86 Assembly

buildXLclass-attributeinstance-attribute

XI

buildXMLclass-attributeinstance-attribute

XML

buildXQUERYclass-attributeinstance-attribute

XQuery

buildYAMLclass-attributeinstance-attribute

YAML

buildZEPHIRclass-attributeinstance-attribute

Zephir

# CodeTheme

Alias for `flet.MarkdownCodeTheme`.

`MarkdownCodeTheme`

# CustomCodeTheme

Alias for `flet.MarkdownCustomCodeTheme`.

`MarkdownCustomCodeTheme`

# GutterStyle

Defines gutter appearance (line numbers) for the code editor.

## Properties

- `background_color` - Background color for the gutter.
- `margin` - Margin outside the gutter.
- `show_errors` - Whether to show errors in the gutter.
- `show_folding_handles` - Whether to show folding handles in the gutter.
- `show_line_numbers` - Whether to show line numbers in the gutter.
- `text_style` - Text style for line numbers.
- `width` - Fixed width of the gutter.

## Properties

### `buildbackground_colorclass-attributeinstance-attribute`

`background_color: ColorValue | None = None`

Background color for the gutter.

### `buildmarginclass-attributeinstance-attribute`

`margin: Number | None = None`

Margin outside the gutter.

### `buildshow_errorsclass-attributeinstance-attribute`

`show_errors: bool = True`

Whether to show errors in the gutter.

### `buildshow_folding_handlesclass-attributeinstance-attribute`

`show_folding_handles: bool = True`

Whether to show folding handles in the gutter.

### `buildshow_line_numbersclass-attributeinstance-attribute`

`show_line_numbers: bool = True`

Whether to show line numbers in the gutter.

### `buildtext_styleclass-attributeinstance-attribute`

`text_style: TextStyle | None = None`

Text style for line numbers.

### `buildwidthclass-attributeinstance-attribute`

`width: Number | None = None`

Fixed width of the gutter.

# DataColumnSize

Relative size of a column determines the share of total table width allocated to each individual column.

When determining column widths, ratios between S, M and L columns are kept (i.e. Large columns are set to 1.2x width of Medium ones).

See DataTable2.sm\_ratio, DataTable2.lm\_ratio.

Inherits: enum.Enum

## Properties

- L
- M
- S

## Properties

buildLclass-attributeinstance-attribute

buildMclass-attributeinstance-attribute

buildSclass-attributeinstance-attribute

# AttributionAlignment

Position to anchor RichAttribution control relative to the map.

Inherits: `enum.Enum`

## Properties

- `BOTTOM_LEFT` - The bottom left corner.
- `BOTTOM_RIGHT` - The bottom right corner.

## Properties

`buildBOTTOM_LEFT`class-attributeinstance-attribute

The bottom left corner.

`buildBOTTOM_RIGHT`class-attributeinstance-attribute

The bottom right corner.

# Camera

Snapshot of the map camera state.

## Properties

- `center` - The center of this camera.
- `max_zoom` - The maximum allowed zoom level.
- `min_zoom` - The minimum allowed zoom level.
- `rotation` - The rotation (in degrees) of the camera.
- `zoom` - Defines how far this camera is zoomed.

## Properties

### `buildcenterinstance-attribute`

`center: MapLatitudeLongitude`

The center of this camera.

### `buildmax_zoominstance-attribute`

`max_zoom: Number`

The maximum allowed zoom level.

### `buildmin_zoominstance-attribute`

`min_zoom: Number`

The minimum allowed zoom level.

### `buildrotationinstance-attribute`

`rotation: Number`

The rotation (in degrees) of the camera.

### `buildzoominstance-attribute`

`zoom: Number`

Defines how far this camera is zoomed.

# CameraFit

Defines how the camera should fit the bounds or coordinates, depending on which one was provided.

Raises:

- `ValueError` - If both bounds and coordinates are `None` or not `None`.

## Properties

- `bounds` - The bounds which the camera should contain once it is fitted.
- `coordinates` - The coordinates which the camera should contain once it is fitted.
- `force_integer_zoom_level` - Whether the zoom level of the resulting fit should be rounded to the nearest integer level.
- `max_zoom` - The inclusive upper zoom limit used for the resulting fit.
- `min_zoom`
- `padding` - Adds a constant/pixel-based padding to the normal fit.

## Properties

### `buildboundsclass-attributeinstance-attribute`

```
bounds: MapLatitudeLongitudeBounds | None = None
```

The bounds which the camera should contain once it is fitted.

Note

If this is not `None`, coordinates should be `None`, and vice versa.

### `buildcoordinatesclass-attributeinstance-attribute`

```
coordinates: list[MapLatitudeLongitude] | None = None
```

The coordinates which the camera should contain once it is fitted.

Note

If this is not `None`, bounds should be `None`, and vice versa.

### `buildforce_integer_zoom_levelclass-attributeinstance-attribute`

```
force_integer_zoom_level: bool = False
```

Whether the zoom level of the resulting fit should be rounded to the nearest integer level.

### `buildmax_zoomclass-attributeinstance-attribute`

```
max_zoom: Number | None = None
```

The inclusive upper zoom limit used for the resulting fit.

If the zoom level calculated for the fit exceeds the `max_zoom` value, `max_zoom` will be used instead.

### `buildmin_zoomclass-attributeinstance-attribute`

```
min_zoom: Number = 0.0
```

### `buildpaddingclass-attributeinstance-attribute`

```
padding: PaddingValue = field(
 default_factory=lambda: Padding.zero()
)
```

Adds a constant/pixel-based padding to the normal fit.



# CursorKeyboardRotationConfiguration

Options to configure cursor/keyboard rotation.

Cursor/keyboard rotation is designed for desktop platforms, and allows the cursor to be used to set the rotation of the map whilst a keyboard key is held down (as triggered by `is_key_trigger`).

## Properties

- `behavior` - The behaviour of the cursor/keyboard rotation function in terms of the angle that the map is rotated to.
- `set_north_on_click` - Whether to set the North of the map to the clicked angle, when the user clicks their mouse without dragging (a `on_pointer_down` event followed by `on_pointer_up` without a change in rotation).
- `trigger_keys` - List of keys that will trigger cursor/keyboard rotation, when pressed.

## Methods

- `disabled` - A disabled `CursorKeyboardRotationConfiguration`.

## Properties

### `buildbehaviorclass-attributeinstance-attribute`

```
behavior: CursorRotationBehaviour = (
 CursorRotationBehaviour.OFFSET
)
```

The behaviour of the cursor/keyboard rotation function in terms of the angle that the map is rotated to.

Does not disable cursor/keyboard rotation, or adjust its triggers: see `is_key_trigger`.

### `buildset_north_on_clickclass-attributeinstance-attribute`

```
set_north_on_click: bool = True
```

Whether to set the North of the map to the clicked angle, when the user clicks their mouse without dragging (a `on_pointer_down` event followed by `on_pointer_up` without a change in rotation).

### `buildtrigger_keysclass-attributeinstance-attribute`

```
trigger_keys: list = field(default_factory=lambda: [])
```

List of keys that will trigger cursor/keyboard rotation, when pressed.

## Methods

### `deployed_codedisabledclassmethod`

```
disabled() -> CursorKeyboardRotationConfiguration
```

A disabled `CursorKeyboardRotationConfiguration`.

# CursorRotationBehaviour

The behaviour of the cursor/keyboard rotation function in terms of the angle that the map is rotated to.

Does not disable cursor/keyboard rotation, or adjust its triggers: see `CursorKeyboardRotationConfiguration.is_key_trigger`.

Inherits: `enum.Enum`

## Properties

- `OFFSET` - Offset the current rotation of the map to the angle at which the user drags their cursor.
- `SET_NORTH` - Set the North of the map to the angle at which the user drags their cursor.

## Properties

`buildOFFSET`class-attributeinstance-attribute

Offset the current rotation of the map to the angle at which the user drags their cursor.

`buildSET_NORTH`class-attributeinstance-attribute

Set the North of the map to the angle at which the user drags their cursor.

# DashedStrokePattern

A stroke pattern of alternating dashes and gaps, defined by segments.

Inherits: `StrokePattern`

## Properties

- `pattern_fit` - Determines how this stroke pattern should be fit to a line when their lengths are not equal or multiples.
- `segments` - A list of even length with a minimum of 2, in the form of `[a1, b1, (a2, b2, ...)]`, where a should be the length of segments in 'units', and b the length of the space after each segment in units.

## Properties

### `buildpattern_fit`class-attributeinstance-attribute

`pattern_fit: PatternFit = PatternFit.SCALE_UP`

Determines how this stroke pattern should be fit to a line when their lengths are not equal or multiples.

### `buildsegments`instance-attribute

`segments: List[Number]`

A list of even length with a minimum of 2, in the form of `[a1, b1, (a2, b2, ...)]`, where a should be the length of segments in 'units', and b the length of the space after each segment in units. Both values must be strictly positive.

'Units' refers to pixels, unless the pattern has been scaled due to the `pattern_fit` being `PatternFit.SCALE_UP`.

If more than two items are specified, then each segments will alternate/iterate through the values.

For example, `[50, 10, 10, 10]` will cause:

- a segment of length 50px
- followed by a space of 10px
- followed by a segment of length 10px
- followed by a space of 10px
- followed by a segment of length of 50px
- followed by a space of 10px
- and so on ...

Raises:

- `ValueError` - If the list does not contain at least two items, or if its length is not even.

# DottedStrokePattern

A stroke pattern of circular dots, spaced with `spacing_factor`.

Inherits: `StrokePattern`

## Properties

- `pattern_fit` - Determines how this stroke pattern should be fit to a line when their lengths are not equal or multiples.
- `spacing_factor` - The multiplier used to calculate the spacing between dots in a dotted polyline, with respect to `Polyline.stroke_width / Polygon.border_stroke_width`.

## Properties

### `buildpattern_fit`class-attributeinstance-attribute

`pattern_fit: PatternFit = PatternFit.SCALE_UP`

Determines how this stroke pattern should be fit to a line when their lengths are not equal or multiples.

### `buildspacing_factor`class-attributeinstance-attribute

`spacing_factor: Number = 1.5`

The multiplier used to calculate the spacing between dots in a dotted polyline, with respect to `Polyline.stroke_width / Polygon.border_stroke_width`. A value of 1.0 will result in spacing equal to the `stroke_width`. Increasing the value increases the spacing with the same scaling.

May also be scaled by the use of `PatternFit.SCALE_UP`.

Raises:

- `ValueError` - If it is less than or equal to zero.

# FadeInTileDisplay

A TileDisplay that should get faded in.

Inherits: TileDisplay

## Properties

- `duration` - The duration of the fade in animation.
- `reload_start_opacity` - Opacity start value when a tile is reloaded.
- `start_opacity` - Opacity start value when a tile is faded in.

## Properties

### `builddurationclass-attributeinstance-attribute`

```
duration: DurationValue = field(
 default_factory=lambda: Duration(milliseconds=100)
)
```

The duration of the fade in animation.

### `buildreload_start_opacityclass-attributeinstance-attribute`

```
reload_start_opacity: Number = 0.0
```

Opacity start value when a tile is reloaded.

Raises:

- `ValueError` - If its value is not between 0.0 and 1.0 inclusive.

### `buildstart_opacityclass-attributeinstance-attribute`

```
start_opacity: Number = 0.0
```

Opacity start value when a tile is faded in.

Raises:

- `ValueError` - If its value is not between 0.0 and 1.0 inclusive.

# InstantaneousTileDisplay

A TileDisplay that should get instantaneously displayed.

Inherits: TileDisplay

## Properties

- opacity - The optional opacity of the tile.

## Properties

buildopacityclass-attributeinstance-attribute

opacity: Number = 1.0

The optional opacity of the tile.

Raises:

- ValueError - If its value is not between 0.0 and 1.0 inclusive.

# InteractionConfiguration

Configures user interaction behavior.

## Properties

- `cursor_keyboard_rotation_configuration` - Options to control the keyboard and mouse cursor being used together to rotate the map.
- `enable_multi_finger_gesture_race` - If True, then `rotation_threshold` and `pinch_zoom_threshold` and `pinch_move_threshold` will race.
- `flags` - Defines the map events to be enabled/disabled.
- `keyboard_configuration` - Options to configure how keyboard keys may be used to control the map.
- `pinch_move_threshold` - Map starts to move when `pinch_move_threshold` has been achieved or another multi finger gesture wins which allows `MultiFingerGesture.PINCH_MOVE`.
- `pinch_move_win_gestures` - When `pinch_move_threshold` wins over `rotation_threshold` and `pinch_zoom_threshold` then `pinch_move_win_gestures` gestures will be used.
- `pinch_zoom_threshold` - Map starts to zoom when `pinch_zoom_threshold` has been achieved or another multi finger gesture wins which allows `MultiFingerGesture.PINCH_ZOOM`.
- `pinch_zoom_win_gestures` - When `pinch_zoom_threshold` wins over `rotation_threshold` and `pinch_move_threshold` then `pinch_zoom_win_gestures` gestures will be used.
- `rotation_threshold` - Map starts to rotate when `rotation_threshold` has been achieved or another multi finger gesture wins which allows `MultiFingerGesture.ROTATE`.
- `rotation_win_gestures` - When `rotation_threshold` wins over `pinch_zoom_threshold` and `pinch_move_threshold` then `rotation_win_gestures` gestures will be used.
- `scroll_wheel_velocity` - The used velocity how fast the map should zoom in or out by scrolling with the scroll wheel of a mouse.

## Properties

### `buildcursor_keyboard_rotation_configurationclass-attributeinstance-attribute`

```
cursor_keyboard_rotation_configuration: CursorKeyboardRotationConfiguration = field(
 default_factory=lambda: (
 CursorKeyboardRotationConfiguration()
)
)
```

Options to control the keyboard and mouse cursor being used together to rotate the map.

### `buildenable_multi_finger_gesture_raceclass-attributeinstance-attribute`

```
enable_multi_finger_gesture_race: bool = False
```

If True, then `rotation_threshold` and `pinch_zoom_threshold` and `pinch_move_threshold` will race. If multiple gestures win at the same time, then precedence: `pinch_zoom_win_gestures` > `rotation_win_gestures` > `pinch_move_win_gestures`

### `buildflagsclass-attributeinstance-attribute`

```
flags: InteractionFlag = InteractionFlag.ALL
```

Defines the map events to be enabled/disabled.

### `buildkeyboard_configurationclass-attributeinstance-attribute`

```
keyboard_configuration: KeyboardConfiguration = field(
 default_factory=lambda: KeyboardConfiguration()
)
```

Options to configure how keyboard keys may be used to control the map.

Keyboard movements using the arrow keys are enabled by default.

### `buildpinch_move_thresholdclass-attributeinstance-attribute`

```
pinch_move_threshold: Number = 40.0
```

Map starts to move when `pinch_move_threshold` has been achieved or another multi finger gesture wins which allows `MultiFingerGesture.PINCH_MOVE`.

#### Note

If `InteractionConfiguration.flags` doesn't contain `InteractionFlag.PINCH_MOVE` or `enable_multi_finger_gesture_race` is false then pinch move cannot win.

### `buildpinch_move_win_gestures`class-attributeinstance-attribute

```
pinch_move_win_gestures: MultiFingerGesture = (
 MultiFingerGesture.PINCH_ZOOM
 | MultiFingerGesture.PINCH_MOVE
)
```

When `pinch_move_threshold` wins over `rotation_threshold` and `pinch_zoom_threshold` then `pinch_move_win_gestures` gestures will be used.

By default `MultiFingerGesture.PINCH_MOVE` and `MultiFingerGesture.PINCH_ZOOM` gestures will take effect see `MultiFingerGesture` for custom settings.

### `buildpinch_zoom_threshold`class-attributeinstance-attribute

```
pinch_zoom_threshold: Number = 0.5
```

Map starts to zoom when `pinch_zoom_threshold` has been achieved or another multi finger gesture wins which allows `MultiFingerGesture.PINCH_ZOOM`.

#### Note

If `InteractionConfiguration.flags` doesn't contain `InteractionFlag.PINCH_ZOOM` or `enable_multi_finger_gesture_race` is false then zoom cannot win.

### `buildpinch_zoom_win_gestures`class-attributeinstance-attribute

```
pinch_zoom_win_gestures: MultiFingerGesture = (
 MultiFingerGesture.PINCH_ZOOM
 | MultiFingerGesture.PINCH_MOVE
)
```

When `pinch_zoom_threshold` wins over `rotation_threshold` and `pinch_move_threshold` then `pinch_zoom_win_gestures` gestures will be used.

By default `MultiFingerGesture.PINCH_ZOOM` and `MultiFingerGesture.PINCH_MOVE` gestures will take effect see `MultiFingerGesture` for custom settings.

### `buildrotation_threshold`class-attributeinstance-attribute

```
rotation_threshold: Number = 20.0
```

Map starts to rotate when `rotation_threshold` has been achieved or another multi finger gesture wins which allows `MultiFingerGesture.ROTATE`.

#### Note

If `InteractionConfiguration.flags` doesn't contain `InteractionFlag.ROTATE` or `enable_multi_finger_gesture_race` is false then rotate cannot win.

### `buildrotation_win_gestures`class-attributeinstance-attribute

```
rotation_win_gestures: MultiFingerGesture = (
 MultiFingerGesture.ROTATE
)
```

When `rotation_threshold` wins over `pinch_zoom_threshold` and `pinch_move_threshold` then `rotation_win_gestures` gestures will be used.

### `buildscroll_wheel_velocity`class-attributeinstance-attribute

```
scroll_wheel_velocity: Number = 0.005
```

The used velocity how fast the map should zoom in or out by scrolling with the scroll wheel of a mouse.



# InteractionFlag

Flags to enable/disable certain interaction events on the map.

Example

- `InteractionFlag.ALL` to enable all events
- `InteractionFlag.NONE` to disable all events

Inherits: `enum.IntFlag`

## Properties

- `ALL` - All available interactive flags.
- `DOUBLE_TAP_DRAG_ZOOM` - Zooming with a single-finger double-tap-drag gesture.
- `DOUBLE_TAP_ZOOM` - Zooming with a single-finger double tap gesture.
- `DRAG` - Panning with a single finger or cursor.
- `FLING_ANIMATION` - Fling animation after panning if velocity is great enough.
- `NONE` - No interaction.
- `PINCH_MOVE` - Panning with multiple fingers.
- `PINCH_ZOOM` - Zooming with a multi-finger pinch gesture.
- `ROTATE` - Rotation with two-finger twist gesture.
- `SCROLL_WHEEL_ZOOM` - Zooming with a mouse scroll wheel.

## Methods

- `has_double_tap_drag_zoom` - Returns: True if the `DOUBLE_TAP_DRAG_ZOOM` interaction flag is enabled.
- `has_double_tap_zoom` - Returns: True if the `DOUBLE_TAP_ZOOM` interaction flag is enabled.
- `has_drag` - Returns: True if the `DRAG` interaction flag is enabled.
- `has_flag` - Returns: True if `left_flags` has at least one member in `right_flags` (intersection).
- `has_fling_animation` - Returns: True if the `FLING_ANIMATION` interaction flag is enabled.
- `has_fling_pinch_zoom` - Returns: True if the `PINCH_ZOOM` interaction flag is enabled.
- `has_multi_finger` - Returns: True if any multi-finger gesture flags (`MultiFingerGesture.PINCH_MOVE`, `MultiFingerGesture.PINCH_ZOOM`, `MultiFingerGesture.ROTATE`) are enabled.
- `has_pinch_move` - Returns: True if the `PINCH_MOVE` interaction flag is enabled.
- `has_rotate` - Returns: True if the `ROTATE` interactive flag is enabled.
- `has_scroll_wheel_zoom` - Returns: True if the `SCROLL_WHEEL_ZOOM` interaction flag is enabled.

## Properties

**buildALL**class-attributeinstance-attribute

All available interactive flags.

**buildDOUBLE\_TAP\_DRAG\_ZOOM**class-attributeinstance-attribute

Zooming with a single-finger double-tap-drag gesture.

**buildDOUBLE\_TAP\_ZOOM**class-attributeinstance-attribute

Zooming with a single-finger double tap gesture.

**buildDRAG**class-attributeinstance-attribute

Panning with a single finger or cursor.

**buildFLING\_ANIMATION**class-attributeinstance-attribute

Fling animation after panning if velocity is great enough.

**buildNONE**class-attributeinstance-attribute

No interaction.

**buildPINCH\_MOVE**class-attributeinstance-attribute

Panning with multiple fingers.

**buildPINCH\_ZOOM**class-attributeinstance-attribute

Zooming with a multi-finger pinch gesture.

**buildROTATE**class-attributeinstance-attribute

Rotation with two-finger twist gesture.

**buildSCROLL\_WHEEL\_ZOOM**class-attributeinstance-attribute

Zooming with a mouse scroll wheel.

## Methods

**deployed\_code**has\_double\_tap\_drag\_zoomstaticmethod

has\_double\_tap\_drag\_zoom(flags: int) -> bool

Returns:

- bool - True if the DOUBLE\_TAP\_DRAG\_ZOOM interaction flag is enabled.

**deployed\_code**has\_double\_tap\_zoomstaticmethod

has\_double\_tap\_zoom(flags: int) -> bool

Returns:

- bool - True if the DOUBLE\_TAP\_ZOOM interaction flag is enabled.

**deployed\_code**has\_dragstaticmethod

has\_drag(flags: int) -> bool

Returns:

- bool - True if the DRAG interaction flag is enabled.

**deployed\_code**has\_flagstaticmethod

has\_flag(left\_flags: int, right\_flags: int) -> bool

Returns:

- bool - True if left\_flags has at least one member in right\_flags (intersection).

**deployed\_code**has\_fling\_animationstaticmethod

has\_fling\_animation(flags: int) -> bool

Returns:

- bool - True if the FLING\_ANIMATION interaction flag is enabled.

**deployed\_code**has\_fling\_pinch\_zoomstaticmethod

has\_fling\_pinch\_zoom(flags: int) -> bool

Returns:

- bool - True if the PINCH\_ZOOM interaction flag is enabled.

**deployed\_code**has\_multi\_fingerstaticmethod

has\_multi\_finger(flags: int) -> bool

Returns:

- bool - True if any multi-finger gesture flags (MultiFingerGesture.PINCH\_MOVE, MultiFingerGesture.PINCH\_ZOOM, MultiFingerGesture.ROTATE) are enabled.

#### deployed\_codehas\_pinch\_movestaticmethod

has\_pinch\_move(flags: int) -> bool

Returns:

- bool - True if the PINCH\_MOVE interaction flag is enabled.

#### deployed\_codehas\_rotatestaticmethod

has\_rotate(flags: int) -> bool

Returns:

- bool - True if the ROTATE interactive flag is enabled.

#### deployed\_codehas\_scroll\_wheel\_zoomstaticmethod

has\_scroll\_wheel\_zoom(flags: int) -> bool

Returns:

- bool - True if the SCROLL\_WHEEL\_ZOOM interaction flag is enabled.

# KeyboardConfiguration

Options to configure how keyboard keys may be used to control the map. When a key is pushed down, an animation starts, consisting of a curved portion which takes the animation to its maximum velocity, an indefinitely long animation at maximum velocity, then ended on the key up with another curved portion.

If a key is pressed and released quickly, it might trigger a short animation called a 'leap'. The leap consists of a part of the curved portion, and also scales the velocity of the concerned gesture.

Info

See `CursorKeyboardRotationConfiguration` for options to control the keyboard and mouse cursor being used together to rotate the map.

## Properties

- `animation_curve_curve` - Curve of the curved portion of the animation occurring after key down and key up events.
- `animation_curve_duration` - Duration of the curved (`AnimationCurve.EASE_IN`) portion of the animation occurring after a key down event (and after a key up event if `animation_curve_reverse_duration` is `None`)
- `animation_curve_reverse_duration` - Duration of the curved (reverse `AnimationCurve.EASE_IN`) portion of the animation occurring after a key up event.
- `autofocus` - Whether to request focus as soon as the map control appears (and to enable keyboard controls).
- `enable_arrow_keys_panning` - Whether to allow arrow keys to pan the map (in their respective directions).
- `enable_qe_rotating` - Whether to allow the Q & E keys (\*) to rotate the map (Q rotates anticlockwise, E rotates clockwise).
- `enable_rf_zooming` - Whether to allow the R & F keys to zoom the map (R zooms IN (increases zoom level), F zooms OUT (decreases zoom level)).
- `enable_wasd_panning` - Whether to allow the W, A, S, D keys (\*) to pan the map (in the directions UP, LEFT, DOWN, RIGHT respectively).
- `leap_max_of_curve_component` - The percentage (0.0 - 1.0) of the curve animation component that is driven to (from 0), then in reverse from (to 0).
- `max_rotate_velocity` - The maximum angular difference to apply per frame to the camera's rotation during a rotation animation.
- `max_zoom_velocity` - The maximum zoom level difference to apply per frame to the camera's zoom level during a zoom animation.
- `pan_leap_velocity_multiplier` - The amount to scale the panning offset velocity by during a leap animation.
- `perform_leap_trigger_duration` - Maximum duration between the key down and key up events of an animation which will trigger a 'leap'.
- `rotate_leap_velocity_multiplier` - The amount to scale the rotation velocity by during a leap animation
- `zoom_leap_velocity_multiplier` - The amount to scale the zooming velocity by during a leap animation.

## Methods

- `disabled` - Disable keyboard control of the map.

## Properties

### `buildanimation_curve_curveclass-attributeinstance-attribute`

```
animation_curve_curve: AnimationCurve = (
 AnimationCurve.EASE_IN_OUT
)
```

Curve of the curved portion of the animation occurring after key down and key up events.

### `buildanimation_curve_durationclass-attributeinstance-attribute`

```
animation_curve_duration: DurationValue = field(
 default_factory=lambda: Duration(milliseconds=450)
```

)

Duration of the curved (AnimationCurve.EASE\_IN) portion of the animation occurring after a key down event (and after a key up event if animation\_curve\_reverse\_duration is None)

### **buildanimation\_curve\_reverse\_durationclass-attributeinstance-attribute**

```
animation_curve_reverse_duration: (
 DurationValue | None
) = field(
 default_factory=lambda: Duration(milliseconds=600)
)
```

Duration of the curved (reverse AnimationCurve.EASE\_IN) portion of the animation occurring after a key up event.

Set to None to use animation\_curve\_duration.

### **buildautofocusclass-attributeinstance-attribute**

```
autofocus: bool = True
```

Whether to request focus as soon as the map control appears (and to enable keyboard controls).

### **buildenable\_arrow\_keys\_panningclass-attributeinstance-attribute**

```
enable_arrow_keys_panning: bool = True
```

Whether to allow arrow keys to pan the map (in their respective directions).

### **buildenable\_qe\_rotatingclass-attributeinstance-attribute**

```
enable_qe_rotating: bool = True
```

Whether to allow the Q & E keys (\*) to rotate the map (Q rotates anticlockwise, E rotates clockwise).

QE are only the physical and logical keys on QWERTY keyboards. On non- QWERTY keyboards, such as AZERTY, the keys in the same position as on the QWERTY keyboard is used (ie. AE on AZERTY).

### **buildenable\_rf\_zoomingclass-attributeinstance-attribute**

```
enable_rf_zooming: bool = True
```

Whether to allow the R & F keys to zoom the map (R zooms IN (increases zoom level), F zooms OUT (decreases zoom level)).

RF are only the physical and logical keys on QWERTY keyboards. On non- QWERTY keyboards, such as AZERTY, the keys in the same position as on the QWERTY keyboard is used (ie. RF on AZERTY).

### **buildenable\_wasd\_panningclass-attributeinstance-attribute**

```
enable_wasd_panning: bool = True
```

Whether to allow the W, A, S, D keys (\*) to pan the map (in the directions UP, LEFT, DOWN, RIGHT respectively).

WASD are only the physical and logical keys on QWERTY keyboards. On non- QWERTY keyboards, such as AZERTY, the keys in the same position as on the QWERTY keyboard is used (ie. ZQSD on AZERTY).

If enabled, it is recommended to enable enable\_arrow\_keys\_panning to provide panning functionality easily for left handed users.

### **buildleap\_max\_of\_curve\_componentclass-attributeinstance-attribute**

```
leap_max_of_curve_component: Number = 0.6
```

The percentage (0.0 - 1.0) of the curve animation component that is driven to (from 0), then in reverse from (to 0).

Reducing means the leap occurs quicker (assuming a consistent curve animation duration). Also see \*\_leap\_velocity\_multiplier properties to change the distance of the leap assuming a consistent leap duration.

For example, if set to 1, then the leap will take animation\_curve\_duration + animation\_curve\_reverse\_duration to complete.

Must be greater than 0 and less than or equal to 1. To disable leaping, or change the maximum length of the key press that will trigger a leap, see perform\_leap\_trigger\_duration.

### **buildmax\_rotate\_velocityclass-attributeinstance-attribute**

```
max_rotate_velocity: Number = 3
```

The maximum angular difference to apply per frame to the camera's rotation during a rotation animation.

Measured in degrees. Negative numbers will flip the standard rotation keys.

### buildmax\_zoom\_velocityclass-attributeinstance-attribute

```
max_zoom_velocity: Number = 0.03
```

The maximum zoom level difference to apply per frame to the camera's zoom level during a zoom animation. Measured in zoom levels. Negative numbers will flip the standard zoom keys.

### buildpan\_leap\_velocity\_multiplierclass-attributeinstance-attribute

```
pan_leap_velocity_multiplier: Number = 5
```

The amount to scale the panning offset velocity by during a leap animation.

The larger the number, the larger the movement during a leap. To change the duration of a leap, see `leap_max_of_curve_component`.

### buildperform\_leap\_trigger\_durationclass-attributeinstance-attribute

```
perform_leap_trigger_duration: DurationValue | None = (
 field(
 default_factory=lambda: Duration(
 milliseconds=100
)
)
)
```

Maximum duration between the key down and key up events of an animation which will trigger a 'leap'.

To customize the leap itself, see the `leap_max_of_curve_component` & `*leap_velocity_multiplier` (`zoom_leap_velocity_multiplier`, `pan_leap_velocity_multiplier` and `rotate_leap_velocity_multiplier`) properties.

Set to None to disable leaping.

### buildrotate\_leap\_velocity\_multiplierclass-attributeinstance-attribute

```
rotate_leap_velocity_multiplier: Number = 3
```

The amount to scale the rotation velocity by during a leap animation

The larger the number, the larger the rotation difference during a leap. To change the duration of a leap, see `leap_max_of_curve_component`.

This may cause the pan velocity to exceed `max_rotate_velocity`.

### buildzoom\_leap\_velocity\_multiplierclass-attributeinstance-attribute

```
zoom_leap_velocity_multiplier: Number = 3
```

The amount to scale the zooming velocity by during a leap animation.

The larger the number, the larger the zoom difference during a leap. To change the duration of a leap, see `leap_max_of_curve_component`.

This may cause the pan velocity to exceed `max_zoom_velocity`.

## Methods

### deployed\_codedisabledclassmethod

```
disabled() -> KeyboardConfiguration
```

Disable keyboard control of the map.

Info

CursorKeyboardRotationConfiguration may still be active, and is not disabled if this is disabled.

# MapEvent

Map event emitted for camera/interaction updates from various sources.

Inherits: Event

## Properties

- camera - The camera state after the event.
- coordinates - Tap/press coordinates associated with this event.
- event\_type - Concrete subtype of this map event.
- id - Optional custom identifier associated with this event.
- old\_camera - Camera state before the event.
- source - Who/what issued the event.

## Properties

### buildcamerainstance-attribute

camera: Camera

The camera state after the event.

### buildcoordinatesclass-attributeinstance-attribute

coordinates: MapLatitudeLongitude | None = None

Tap/press coordinates associated with this event.

Set only for MapEventType.TAP, MapEventType.SECONDARY\_TAP, and MapEventType.LONG\_PRESS.

### buildevent\_typeinstance-attribute

event\_type: MapEventType

Concrete subtype of this map event.

### buildidclass-attributeinstance-attribute

id: str | None = None

Optional custom identifier associated with this event.

Set only for MapEventType.MOVE and MapEventType.ROTATE.

### buildold\_cameraclass-attributeinstance-attribute

old\_camera: Camera | None = None

Camera state before the event.

Set only for MapEventType.MOVE, MapEventType.FLING\_ANIMATION, MapEventType.DOUBLE\_TAP\_ZOOM, MapEventType.SCROLL\_WHEEL\_ZOOM, MapEventType.ROTATE, and MapEventType.NON\_ROTATED\_SIZE\_CHANGE.

### buildsourceinstance-attribute

source: MapEventSource

Who/what issued the event.

# MapEventSource

Defines the source of a MapEvent.

Inherits: `enum.Enum`

## Properties

- `CURSOR_KEYBOARD_ROTATION` - The MapEvent is caused by a 'CTRL + drag' rotation gesture.
- `CUSTOM` - The MapEvent is caused by a custom source.
- `DOUBLE_TAP` - The MapEvent is caused by a double tap gesture.
- `DOUBLE_TAP_HOLD` - The MapEvent is caused by a double tap and hold gesture.
- `DOUBLE_TAP_ZOOM_ANIMATION_CONTROLLER` - The MapEvent is caused by the AnimationController while performing the double tap zoom in animation.
- `DRAG_END` - The MapEvent is caused by the end of a drag gesture.
- `DRAG_START` - The MapEvent is caused by the start of a drag gesture.
- `FIT_CAMERA` - The MapEvent is caused by calling `fit_camera`.
- `FLING_ANIMATION_CONTROLLER` - The MapEvent is caused by the AnimationController while performing the fling gesture.
- `INTERACTIVE_FLAGS_CHANGED` - The MapEvent is caused by a change of the interactive flags.
- `KEYBOARD` - The MapEvent is caused by a keyboard key.
- `LONG_PRESS` - The MapEvent is caused by a long press gesture.
- `MAP_CONTROLLER` - The MapEvent is caused programmatically by the MapController.
- `MULTI_FINGER_GESTURE_END` - The MapEvent is caused by a the end of a two finger gesture.
- `MULTI_FINGER_GESTURE_START` - The MapEvent is caused by the start of a two finger gesture.
- `NON_ROTATED_SIZE_CHANGE` - The MapEvent is caused by a size change of the Map constraints.
- `ON_DRAG` - The MapEvent is caused by a drag update gesture.
- `ON_MULTI_FINGER` - The MapEvent is caused by a two finger gesture update.
- `SCROLL_WHEEL` - The MapEvent is caused by a scroll wheel zoom gesture.
- `SECONDARY_TAP` - The MapEvent is caused by a secondary tap gesture.
- `TAP` - The MapEvent is caused by a tap gesture.

## Properties

`buildCURSOR_KEYBOARD_ROTATION`class-attributeinstance-attribute

The MapEvent is caused by a 'CTRL + drag' rotation gesture.

`buildCUSTOM`class-attributeinstance-attribute

The MapEvent is caused by a custom source.

`buildDOUBLE_TAP`class-attributeinstance-attribute

The MapEvent is caused by a double tap gesture.

`buildDOUBLE_TAP_HOLD`class-attributeinstance-attribute

The MapEvent is caused by a double tap and hold gesture.

`buildDOUBLE_TAP_ZOOM_ANIMATION_CONTROLLER`class-attributeinstance-attribute

The MapEvent is caused by the AnimationController while performing the double tap zoom in animation.



**buildDRAG\_END**class-attributeinstance-attribute

The MapEvent is caused by the end of a drag gesture.

**buildDRAG\_START**class-attributeinstance-attribute

The MapEvent is caused by the start of a drag gesture.

**buildFIT\_CAMERA**class-attributeinstance-attribute

The MapEvent is caused by calling fit\_camera.

**buildFLING\_ANIMATION\_CONTROLLER**class-attributeinstance-attribute

The MapEvent is caused by the AnimationController while performing the fling gesture.

**buildINTERACTIVE\_FLAGS\_CHANGED**class-attributeinstance-attribute

The MapEvent is caused by a change of the interactive flags.

**buildKEYBOARD**class-attributeinstance-attribute

The MapEvent is caused by a keyboard key. See KeyboardConfiguration for details.

**buildLONG\_PRESS**class-attributeinstance-attribute

The MapEvent is caused by a long press gesture.

**buildMAP\_CONTROLLER**class-attributeinstance-attribute

The MapEvent is caused programmatically by the MapController.

**buildMULTI\_FINGER\_GESTURE\_END**class-attributeinstance-attribute

The MapEvent is caused by a the end of a two finger gesture.

**buildMULTI\_FINGER\_GESTURE\_START**class-attributeinstance-attribute

The MapEvent is caused by the start of a two finger gesture.

**buildNON\_ROTATED\_SIZE\_CHANGE**class-attributeinstance-attribute

The MapEvent is caused by a size change of the Map constraints.

**buildON\_DRAG**class-attributeinstance-attribute

The MapEvent is caused by a drag update gesture.

**buildON\_MULTI\_FINGER**class-attributeinstance-attribute

The MapEvent is caused by a two finger gesture update.

**buildSCROLL\_WHEEL**class-attributeinstance-attribute

The MapEvent is caused by a scroll wheel zoom gesture.

**buildSECONDARY\_TAP**class-attributeinstance-attribute

The MapEvent is caused by a secondary tap gesture.

**buildTAP**class-attributeinstance-attribute

The MapEvent is caused by a tap gesture.

# MapEventType

Concrete subtype of a MapEvent.

Inherits: enum.Enum

## Properties

- `DOUBLE_TAP_ZOOM` - A double tap zoom update event.
- `DOUBLE_TAP_ZOOM_END` - The end of a double tap zoom animation.
- `DOUBLE_TAP_ZOOM_START` - The start of a double tap zoom animation.
- `FLING_ANIMATION` - A fling animation update event.
- `FLING_ANIMATION_END` - The end of a fling animation.
- `FLING_ANIMATION_NOT_STARTED` - A fling was evaluated but not started.
- `FLING_ANIMATION_START` - The start of a fling animation.
- `LONG_PRESS` - A long press event.
- `MOVE` - A camera move update event.
- `MOVE_END` - The end of a move gesture/interaction.
- `MOVE_START` - The start of a move gesture/interaction.
- `NON_ROTATED_SIZE_CHANGE` - A map constraint/size-change event.
- `ROTATE` - A map rotation update event.
- `ROTATE_END` - The end of a rotation interaction.
- `ROTATE_START` - The start of a rotation interaction.
- `SCROLL_WHEEL_ZOOM` - A scroll wheel zoom update event.
- `SECONDARY_TAP` - A secondary tap event.
- `TAP` - A map tap event.
- `UNKNOWN` - Fallback value for unrecognized map event types.

## Properties

`buildDOUBLE_TAP_ZOOM`class-attributeinstance-attribute

A double tap zoom update event.

`buildDOUBLE_TAP_ZOOM_END`class-attributeinstance-attribute

The end of a double tap zoom animation.

`buildDOUBLE_TAP_ZOOM_START`class-attributeinstance-attribute

The start of a double tap zoom animation.

`buildFLING_ANIMATION`class-attributeinstance-attribute

A fling animation update event.

`buildFLING_ANIMATION_END`class-attributeinstance-attribute

The end of a fling animation.

`buildFLING_ANIMATION_NOT_STARTED`class-attributeinstance-attribute

A fling was evaluated but not started.

`buildFLING_ANIMATION_START`class-attributeinstance-attribute

The start of a fling animation.

**buildLONG\_PRESS**class-attributeinstance-attribute

A long press event.

**buildMOVE**class-attributeinstance-attribute

A camera move update event.

**buildMOVE\_END**class-attributeinstance-attribute

The end of a move gesture/interaction.

**buildMOVE\_START**class-attributeinstance-attribute

The start of a move gesture/interaction.

**buildNON\_ROTATED\_SIZE\_CHANGE**class-attributeinstance-attribute

A map constraint/size-change event.

**buildROTATE**class-attributeinstance-attribute

A map rotation update event.

**buildROTATE\_END**class-attributeinstance-attribute

The end of a rotation interaction.

**buildROTATE\_START**class-attributeinstance-attribute

The start of a rotation interaction.

**buildSCROLL\_WHEEL\_ZOOM**class-attributeinstance-attribute

A scroll wheel zoom update event.

**buildSECONDARY\_TAP**class-attributeinstance-attribute

A secondary tap event.

**buildTAP**class-attributeinstance-attribute

A map tap event.

**buildUNKNOWN**class-attributeinstance-attribute

Fallback value for unrecognized map event types.

# MapHoverEvent

Map hover event that includes geographic coordinates of the pointer.

Inherits: `HoverEvent`

## Properties

- `coordinates` - Coordinates of the point currently hovered by the pointer.

## Properties

### `buildcoordinatesinstance-attribute`

`coordinates`: `MapLatitudeLongitude`

Coordinates of the point currently hovered by the pointer.

# MapLatitudeLongitude

Map coordinates in degrees.

## Properties

- latitude - The latitude point of this coordinate.
- longitude - The longitude point of this coordinate.

## Properties

### buildlatitudeinstance-attribute

latitude: Number

The latitude point of this coordinate.

### buildlongitudeinstance-attribute

longitude: Number

The longitude point of this coordinate.

# MapLatitudeLongitudeBounds

Both corners have to be on opposite sites, but it doesn't matter which opposite corners or in what order the corners are provided.

## Properties

- `corner_1` - The corner 1.
- `corner_2` - The corner 2.

## Properties

### `buildcorner_1instance-attribute`

`corner_1`: MapLatitudeLongitude

The corner 1.

### `buildcorner_2instance-attribute`

`corner_2`: MapLatitudeLongitude

The corner 2.

# MapPointerEvent

Map pointer event with translated geographic coordinates.

Inherits: PointerEvent

## Properties

- `coordinates` - Coordinates of the point targeted by this pointer event.

## Properties

### `buildcoordinatesinstance-attribute`

`coordinates`: `MapLatitudeLongitude`

Coordinates of the point targeted by this pointer event.

# MapPositionChangeEvent

Event emitted when the map camera position changes.

Inherits: Event

## Properties

- camera - Current camera snapshot after the change.
- coordinates - Current center coordinates after the position change.
- has\_gesture - Whether this change was caused by a user gesture.

## Properties

### buildcamerainstance-attribute

camera: Camera

Current camera snapshot after the change.

### buildcoordinatesinstance-attribute

coordinates: MapLatitudeLongitude

Current center coordinates after the position change.

### buildhas\_gestureinstance-attribute

has\_gesture: bool

Whether this change was caused by a user gesture.



# MapTapEvent

Map tap event that includes information on the tap.

Inherits: TapEvent

## Properties

- `coordinates` - Coordinates of the point at which the tap occurred.

## Properties

**buildcoordinatesinstance-attribute**

`coordinates: MapLatitudeLongitude`

Coordinates of the point at which the tap occurred.

# MultiFingerGesture

Flags to enable/disable certain multi-finger gestures on the map.

Inherits: `enum.IntFlag`

## Properties

- `ALL` - All multi-finger gestures defined in this enum.
- `NONE` - No multi-finger gesture.
- `PINCH_MOVE` - Pinch move gesture, which allows moving the map by dragging with two fingers.
- `PINCH_ZOOM` - Pinch zoom gesture, which allows zooming in and out by pinching with two fingers.
- `ROTATE` - Rotate gesture, which allows rotating the map by twisting two fingers.

## Properties

`buildALLclass-attributeinstance-attribute`

All multi-finger gestures defined in this enum.

`buildNONEclass-attributeinstance-attribute`

No multi-finger gesture.

`buildPINCH_MOVEclass-attributeinstance-attribute`

Pinch move gesture, which allows moving the map by dragging with two fingers.

`buildPINCH_ZOOMclass-attributeinstance-attribute`

Pinch zoom gesture, which allows zooming in and out by pinching with two fingers.

`buildROTATEclass-attributeinstance-attribute`

Rotate gesture, which allows rotating the map by twisting two fingers.

# PatternFit

Determines how a non-solid StrokePattern should be fit to a line when their lengths are not equal or multiples

Inherits: `enum.Enum`

## Properties

- **APPEND\_DOT** - Uses the pattern exactly, truncating the final dash if it does not fit, or adding a single dot at the last point if the final dash does not reach the last point (there is a gap at that location).
- **EXTEND\_FINAL\_DASH** - Uses the pattern exactly, truncating the final dash if it does not fit, or extending the final dash to the last point if it would not normally reach that point (there is a gap at that location).
- **NONE** - Don't apply any specific fit to the pattern - repeat exactly as specified, and stop when the last point is reached.
- **SCALE\_DOWN** - Scale the pattern to ensure it fits an integer number of times into the polyline (smaller version regarding rounding, cf.
- **SCALE\_UP** - Scale the pattern to ensure it fits an integer number of times into the polyline (bigger version regarding rounding, cf.

## Properties

### **buildAPPEND\_DOT**`class-attributeinstance-attribute`

Uses the pattern exactly, truncating the final dash if it does not fit, or adding a single dot at the last point if the final dash does not reach the last point (there is a gap at that location).

### **buildEXTEND\_FINAL\_DASH**`class-attributeinstance-attribute`

Uses the pattern exactly, truncating the final dash if it does not fit, or extending the final dash to the last point if it would not normally reach that point (there is a gap at that location).

Only useful when working with DashedStrokePattern. Similar to APPEND\_DOT for DottedStrokePattern.

### **buildNONE**`class-attributeinstance-attribute`

Don't apply any specific fit to the pattern - repeat exactly as specified, and stop when the last point is reached.

Not recommended, as it may leave a gap between the final segment and the last point, making it unclear where the line ends.

### **buildSCALE\_DOWN**`class-attributeinstance-attribute`

Scale the pattern to ensure it fits an integer number of times into the polyline (smaller version regarding rounding, cf. SCALE\_UP).

### **buildSCALE\_UP**`class-attributeinstance-attribute`

Scale the pattern to ensure it fits an integer number of times into the polyline (bigger version regarding rounding, cf. SCALE\_DOWN).

# SolidStrokePattern

A solid/unbroken stroke pattern.

Inherits: `StrokePattern`

# StrokePattern

Determines whether a stroke should be solid, dotted, or dashed, and the exact characteristics of each.

This is an abstract class and shouldn't be used directly.

See usable derivatives:

- SolidStrokePattern
- DashedStrokePattern
- DottedStrokePattern

# TileDisplay

Defines how the tile should get displayed on the map.

This is an abstract class and shouldn't be used directly.

See usable derivatives:

- `InstantaneousTileDisplay`
- `FadeInTileDisplay`

# TileLayerEvictErrorTileStrategy

Strategies on how to handle tile errors.

Inherits: `enum.Enum`

## Properties

- **DISPOSE** - Evict images for tiles which failed to load when they are pruned.
- **NONE** - Never evict images for tiles which failed to load.
- **NOT\_VISIBLE** - Evict images for tiles which failed to load and: - do not belong to the current zoom level AND/OR - are not visible
- **NOT\_VISIBLE\_RESPECT\_MARGIN** - Evict images for tiles which failed to load and: - do not belong to the current zoom level AND/OR - are not visible, respecting the pruning buffer (the maximum of the `keep_buffer` and `pan_buffer`).

## Properties

### `buildDISPOSEclass-attributeinstance-attribute`

Evict images for tiles which failed to load when they are pruned.

### `buildNONEclass-attributeinstance-attribute`

Never evict images for tiles which failed to load.

### `buildNOT_VISIBLEclass-attributeinstance-attribute`

Evict images for tiles which failed to load and:

- do not belong to the current zoom level AND/OR
- are not visible

### `buildNOT_VISIBLE_RESPECT_MARGINclass-attributeinstance-attribute`

Evict images for tiles which failed to load and:

- do not belong to the current zoom level AND/OR
- are not visible, respecting the pruning buffer (the maximum of the `keep_buffer` and `pan_buffer`).

# WMSTileLayerConfiguration

Configuration for a WMS TileLayer.

## Properties

- `additional_parameters` - Additional request parameters.
- `base_url` - WMS service's URL, for example `http://ows.mundialis.de/services/service?`
- `format` - WMS image format (use 'image/png' for layers with transparency).
- `layers` - List of WMS layers to show.
- `styles` - List of WMS styles.
- `transparent` - Whether to make tiles transparent.
- `uppercase_bool_value` - Encode boolean values as uppercase in request.
- `version` - Version of the WMS service to use.

## Properties

### `buildadditional_parametersclass-attributeinstance-attribute`

```
additional_parameters: dict[str, str] = field(
 default_factory=dict
)
```

Additional request parameters.

### `buildbase_urlinstance-attribute`

```
base_url: str
```

WMS service's URL, for example `http://ows.mundialis.de/services/service?`

### `buildformatclass-attributeinstance-attribute`

```
format: str = 'image/png'
```

WMS image format (use 'image/png' for layers with transparency).

### `buildlayersclass-attributeinstance-attribute`

```
layers: list[str] = field(default_factory=list)
```

List of WMS layers to show.

### `buildstylesclass-attributeinstance-attribute`

```
styles: list[str] = field(default_factory=list)
```

List of WMS styles.

### `buildtransparentclass-attributeinstance-attribute`

```
transparent: bool = True
```

Whether to make tiles transparent.

### `builduppercase_bool_valueclass-attributeinstance-attribute`

```
uppercase_bool_value: bool = False
```

Encode boolean values as uppercase in request.

### `buildversionclass-attributeinstance-attribute`

```
version: str = '1.1.1'
```

Version of the WMS service to use.



# AdaptiveVideoControls

Platform-adaptive video controls.

Adaptive controls select the controls implementation at runtime based on the current `Page.platform`:

- Android and iOS use Material controls.
- macOS, Windows, and Linux use Material desktop controls.
- Other platforms show no built-in controls.

Use `material` and `material_desktop` to configure the controls family that may be selected for each platform.

Inherits: `VideoControls`

## Properties

- `material` - Controls used when adaptive controls select Material controls.
- `material_desktop` - Controls used when adaptive controls select Material desktop controls.

## Properties

### `buildmaterialclass-attributeinstance-attribute`

`material: MaterialVideoControls | None = None`

Controls used when adaptive controls select Material controls.

If omitted, Material controls use the default `MaterialVideoControls` value.

### `buildmaterial_desktopclass-attributeinstance-attribute`

`material_desktop: MaterialDesktopVideoControls | None = None`

Controls used when adaptive controls select Material desktop controls.

If omitted, Material desktop controls use the default `MaterialDesktopVideoControls` value.

# MaterialDesktopVideoControls

Desktop-oriented Material video controls.

Inherits: VideoControls

## Properties

- `automatically_imply_skip_next_button` - Whether a skip next button should be displayed if there are more than one videos in the playlist.
- `automatically_imply_skip_previous_button` - Whether a skip previous button should be displayed if there are more than one videos in the playlist.
- `bottom_button_bar` - Controls displayed in the bottom button bar.
- `bottom_button_bar_margin` - Margin around the bottom button bar.
- `button_bar_button_color` - Color of the button bar buttons.
- `button_bar_button_size` - Size of the button bar buttons.
- `button_bar_height` - Height of the button bar.
- `controls_hover_duration` - Duration after which the controls will be hidden when there is no mouse movement.
- `controls_transition_duration` - Duration for which the controls will be animated when shown or hidden.
- `display_seek_bar` - Whether to display seek bar.
- `hide_mouse_on_controls_removal` - Whether to hide mouse on controls removal.
- `modify_volume_on_scroll` - Modify volume on mouse scroll.
- `padding` - Padding around the controls.
- `play_and_pause_on_tap` - Whether to toggle play and pause on tap.
- `primary_button_bar` - Controls displayed in the primary button bar.
- `seek_bar_buffer_color` - Color of the playback buffer section in the seek bar.
- `seek_bar_color` - Color of the seek bar.
- `seek_bar_container_height` - Height of the seek bar container.
- `seek_bar_height` - Height of the seek bar.
- `seek_bar_hover_color` - Color of the hovered section in the seek bar.
- `seek_bar_hover_height` - Height of the seek bar when hovered.
- `seek_bar_margin` - Margin around the seek bar.
- `seek_bar_position_color` - Color of the playback position section in the seek bar.
- `seek_bar_thumb_color` - Color of the seek bar thumb.
- `seek_bar_thumb_size` - Size of the seek bar thumb.
- `seek_bar_thumb_transition_duration` - Duration for which the seek bar thumb will be animated when the user seeks.
- `seek_bar_transition_duration` - Duration for which the seek bar will be animated when the user seeks.
- `shift_subtitles_on_controls_visibility_change` - Whether to shift the subtitles upwards when the controls are visible.
- `toggle_fullscreen_on_double_press` - Whether to toggle fullscreen on double press.
- `top_button_bar` - Controls displayed in the top button bar.
- `top_button_bar_margin` - Margin around the top button bar.
- `visible_on_mount` - Whether the controls are initially visible.
- `volume_bar_active_color` - Color of the active region in the volume bar.
- `volume_bar_color` - Color of the volume bar.

- `volume_bar_thumb_color` - Color of the volume bar thumb.
- `volume_bar_thumb_size` - Size of the volume bar thumb.
- `volume_bar_transition_duration` - Duration for which the volume bar will be animated when the user hovers.

## Properties

### `buildautomatically_imply_skip_next_button`class-attributeinstance-attribute

`automatically_imply_skip_next_button`: `bool` = `True`

Whether a skip next button should be displayed if there are more than one videos in the playlist.

### `buildautomatically_imply_skip_previous_button`class-attributeinstance-attribute

`automatically_imply_skip_previous_button`: `bool` = `True`

Whether a skip previous button should be displayed if there are more than one videos in the playlist.

### `buildbottom_button_bar`class-attributeinstance-attribute

`bottom_button_bar`: (  
    `list[Control | VideoBarItem] | None`  
) = `None`

Controls displayed in the bottom button bar.

Set to a list to replace the bottom button bar, or an empty list to hide it; if `None`, the native default is:

```
[
 ftv.VideoSkipPreviousButton(),
 ftv.VideoPlayOrPauseButton(),
 ftv.VideoSkipNextButton(),
 ftv.VideoSpacer(),
 ftv.VideoPositionIndicator(),
 ftv.VideoFullscreenButton(),
 ftv.VideoVolumeButton(),
]
```

### `buildbottom_button_bar_margin`class-attributeinstance-attribute

`bottom_button_bar_margin`: `PaddingValue` = `field(`  
    `default_factory=lambda: Padding.symmetric(`  
        `horizontal=16.0`  
    `)`  
`)`

Margin around the bottom button bar.

### `buildbutton_bar_button_color`class-attributeinstance-attribute

`button_bar_button_color`: `ColorValue` = `'#FFFFFF'`

Color of the button bar buttons.

### `buildbutton_bar_button_size`class-attributeinstance-attribute

`button_bar_button_size`: `Number` = `28.0`

Size of the button bar buttons.

### `buildbutton_bar_height`class-attributeinstance-attribute

`button_bar_height`: `Number` = `56.0`

Height of the button bar.

### `buildcontrols_hover_duration`class-attributeinstance-attribute

`controls_hover_duration`: `DurationValue` = `3000`

Duration after which the controls will be hidden when there is no mouse movement.

### `buildcontrols_transition_duration`class-attributeinstance-attribute

`controls_transition_duration`: `DurationValue` = `150`

Duration for which the controls will be animated when shown or hidden.

### `bulddisplay_seek_bar`class-attributeinstance-attribute

`display_seek_bar`: `bool` = `True`

Whether to display seek bar.

## buildhide\_mouse\_on\_controls\_removalclass-attributeinstance-attribute

```
hide_mouse_on_controls_removal: bool = False
```

Whether to hide mouse on controls removal.

Note

On most platforms, the mouse must move before it becomes hidden. It works on macOS without moving the mouse.

## buildmodify\_volume\_on\_scrollclass-attributeinstance-attribute

```
modify_volume_on_scroll: bool = True
```

Modify volume on mouse scroll.

## buildpaddingclass-attributeinstance-attribute

```
padding: PaddingValue | None = None
```

Padding around the controls.

## buildplay\_and\_pause\_on\_tapclass-attributeinstance-attribute

```
play_and_pause_on_tap: bool = False
```

Whether to toggle play and pause on tap.

## buildprimary\_button\_barclass-attributeinstance-attribute

```
primary_button_bar: (
 list[Control | VideoBarItem] | None
) = None
```

Controls displayed in the primary button bar.

Set to a list to replace the primary button bar, or an empty list to hide it; if None, the native default is an empty list.

## buildseek\_bar\_buffer\_colorclass-attributeinstance-attribute

```
seek_bar_buffer_color: ColorValue = '#3DFFFFFF'
```

Color of the playback buffer section in the seek bar.

## buildseek\_bar\_colorclass-attributeinstance-attribute

```
seek_bar_color: ColorValue = '#3DFFFFFF'
```

Color of the seek bar.

## buildseek\_bar\_container\_heightclass-attributeinstance-attribute

```
seek_bar_container_height: Number = 36.0
```

Height of the seek bar container.

## buildseek\_bar\_heightclass-attributeinstance-attribute

```
seek_bar_height: Number = 3.2
```

Height of the seek bar.

## buildseek\_bar\_hover\_colorclass-attributeinstance-attribute

```
seek_bar_hover_color: ColorValue = '#3DFFFFFF'
```

Color of the hovered section in the seek bar.

## buildseek\_bar\_hover\_heightclass-attributeinstance-attribute

```
seek_bar_hover_height: Number = 5.6
```

Height of the seek bar when hovered.

## buildseek\_bar\_marginclass-attributeinstance-attribute

```
seek_bar_margin: PaddingValue = field(
 default_factory=lambda: Padding.symmetric(
 horizontal=16.0
)
)
```

Margin around the seek bar.

## buildseek\_bar\_position\_colorclass-attributeinstance-attribute

```
seek_bar_position_color: ColorValue = '#FFFF0000'
```

Color of the playback position section in the seek bar.

#### **buildseek\_bar\_thumb\_colorclass-attributeinstance-attribute**

```
seek_bar_thumb_color: ColorValue = '#FFFF0000'
```

Color of the seek bar thumb.

#### **buildseek\_bar\_thumb\_sizeclass-attributeinstance-attribute**

```
seek_bar_thumb_size: Number = 12.0
```

Size of the seek bar thumb.

#### **buildseek\_bar\_thumb\_transition\_durationclass-attributeinstance-attribute**

```
seek_bar_thumb_transition_duration: DurationValue = 150
```

Duration for which the seek bar thumb will be animated when the user seeks.

#### **buildseek\_bar\_transition\_durationclass-attributeinstance-attribute**

```
seek_bar_transition_duration: DurationValue = 300
```

Duration for which the seek bar will be animated when the user seeks.

#### **buildshift\_subtitles\_on\_controls\_visibility\_changeclass-attributeinstance-attribute**

```
shift_subtitles_on_controls_visibility_change: bool = True
```

Whether to shift the subtitles upwards when the controls are visible.

#### **buildtoggle\_fullscreen\_on\_double\_pressclass-attributeinstance-attribute**

```
toggle_fullscreen_on_double_press: bool = True
```

Whether to toggle fullscreen on double press.

#### **buildtop\_button\_barclass-attributeinstance-attribute**

```
top_button_bar: list[Control | VideoBarItem] | None = (
 None
)
```

Controls displayed in the top button bar.

Set to a list to replace the top button bar, or an empty list to hide it; if None, the native default is an empty list.

#### **buildtop\_button\_bar\_marginclass-attributeinstance-attribute**

```
top_button_bar_margin: PaddingValue = field(
 default_factory=lambda: Padding.symmetric(
 horizontal=16.0
)
)
```

Margin around the top button bar.

#### **buildvisible\_on\_mountclass-attributeinstance-attribute**

```
visible_on_mount: bool = False
```

Whether the controls are initially visible.

#### **buildvolume\_bar\_active\_colorclass-attributeinstance-attribute**

```
volume_bar_active_color: ColorValue = '#FFFFFFF'
```

Color of the active region in the volume bar.

#### **buildvolume\_bar\_colorclass-attributeinstance-attribute**

```
volume_bar_color: ColorValue = '#3DFFFFFF'
```

Color of the volume bar.

#### **buildvolume\_bar\_thumb\_colorclass-attributeinstance-attribute**

```
volume_bar_thumb_color: ColorValue = '#FFFFFFF'
```

Color of the volume bar thumb.

#### **buildvolume\_bar\_thumb\_sizeclass-attributeinstance-attribute**

volume\_bar\_thumb\_size: Number = 12.0

Size of the volume bar thumb.

buildvolume\_bar\_transition\_durationclass-attributeinstance-attribute

volume\_bar\_transition\_duration: DurationValue = 150

Duration for which the volume bar will be animated when the user hovers.

# MaterialVideoControls

Touch-oriented Material video controls.

Inherits: VideoControls

## Properties

- `automatically_imply_skip_next_button` - Whether a skip next button should be displayed if there are more than one videos in the playlist.
- `automatically_imply_skip_previous_button` - Whether a skip previous button should be displayed if there are more than one videos in the playlist.
- `backdrop_color` - Color of backdrop that comes up when controls are visible.
- `bottom_button_bar` - Controls displayed in the bottom button bar.
- `bottom_button_bar_margin` - Margin around the button bar.
- `brightness_gesture` - Whether to modify screen brightness on vertical drag gesture on the left side of the screen.
- `button_bar_button_color` - Color of the button bar buttons.
- `button_bar_button_size` - Size of the button bar buttons.
- `button_bar_height` - Height of the button bar.
- `controls_hover_duration` - Duration after which the controls will be hidden when there is no mouse movement.
- `controls_transition_duration` - Duration for which the controls will be animated when shown or hidden.
- `display_seek_bar` - Whether to display seek bar.
- `gestures_enabled_while_controls_visible` - Whether to allow gesture controls to work while controls are visible.
- `horizontal_gesture_sensitivity` - Gesture sensitivity on horizontal drag gestures, the higher the value is the less sensitive the gesture.
- `initial_brightness` - Initial brightness value used by the brightness gesture indicator.
- `initial_volume` - Initial volume value used by the volume gesture indicator.
- `padding` - Padding around the controls.
- `primary_button_bar` - Controls displayed in the primary button bar.
- `seek_bar_alignment` - Alignment of seek bar inside the seek bar container.
- `seek_bar_buffer_color` - Color of the playback buffer section in the seek bar.
- `seek_bar_color` - Color of the seek bar.
- `seek_bar_container_height` - Height of the seek bar container.
- `seek_bar_height` - Height of the seek bar.
- `seek_bar_margin` - Margin around the seek bar.
- `seek_bar_position_color` - Color of the playback position section in the seek bar.
- `seek_bar_thumb_color` - Color of the seek bar thumb.
- `seek_bar_thumb_size` - Size of the seek bar thumb.
- `seek_gesture` - Whether to seek on horizontal drag gesture.
- `seek_on_double_tap` - Whether to enable double tap to seek on left or right side of the screen.
- `seek_on_double_tap_backward_duration` - Duration of seek on double tap backward.
- `seek_on_double_tap_enabled_while_controls_visible` - Whether to allow double tap to seek on left or right side of the screen to work while controls are visible.
- `seek_on_double_tap_forward_duration` - Duration of seek on double tap forward.

- `seek_on_double_tap_layout_taps_ratios` - Width proportions for the backward seek, instant tap, and forward seek areas when a double tap occurs on the video widget.
- `seek_on_double_tap_layout_widget_ratios` - Width proportions for the visual indicators shown during backward seek, instant tap, and forward seek actions.
- `shift_subtitles_on_controls_visibility_change` - Whether to shift the subtitles upwards when the controls are visible.
- `speed_up_factor` - Factor to speed up on long press.
- `speed_up_on_long_press` - Whether to speed up on long press.
- `top_button_bar` - Controls displayed in the top button bar.
- `top_button_bar_margin` - Margin around the top button bar.
- `vertical_gesture_sensitivity` - Gesture sensitivity on vertical drag gestures, the higher the value is the less sensitive the gesture.
- `visible_on_mount` - Whether the controls are initially visible.
- `volume_gesture` - Whether to modify volume on vertical drag gesture on the right side of the screen.

## Properties

### `buildautomatically_imply_skip_next_button`class-attributeinstance-attribute

`automatically_imply_skip_next_button`: `bool` = `True`

Whether a skip next button should be displayed if there are more than one videos in the playlist.

### `buildautomatically_imply_skip_previous_button`class-attributeinstance-attribute

`automatically_imply_skip_previous_button`: `bool` = `True`

Whether a skip previous button should be displayed if there are more than one videos in the playlist.

### `buildbackdrop_color`class-attributeinstance-attribute

`backdrop_color`: `ColorValue` | `None` = `'#66000000'`

Color of backdrop that comes up when controls are visible.

### `buildbottom_button_bar`class-attributeinstance-attribute

```
bottom_button_bar: (
 list[Control | VideoBarItem] | None
) = None
```

Controls displayed in the bottom button bar.

Set to a list to replace the bottom button bar, or an empty list to hide it; if `None`, the native default is:

```
[
 ftv.VideoPositionIndicator(),
 ftv.VideoSpacer(),
 ftv.VideoFullscreenButton(),
]
```

### `buildbottom_button_bar_margin`class-attributeinstance-attribute

```
bottom_button_bar_margin: PaddingValue = field(
 default_factory=lambda: Padding.only(
 left=16.0, right=8.0
)
)
```

Margin around the button bar.

### `buildbrightness_gesture`class-attributeinstance-attribute

`brightness_gesture`: `bool` = `False`

Whether to modify screen brightness on vertical drag gesture on the left side of the screen.

### `buildbutton_bar_button_color`class-attributeinstance-attribute

`button_bar_button_color`: `ColorValue` = `'#FFFFFF'`

Color of the button bar buttons.

### `buildbutton_bar_button_size`class-attributeinstance-attribute



```
button_bar_button_size: Number = 24.0
```

Size of the button bar buttons.

### **buildbutton\_bar\_heightclass-attributeinstance-attribute**

```
button_bar_height: Number = 56.0
```

Height of the button bar.

### **buildcontrols\_hover\_durationclass-attributeinstance-attribute**

```
controls_hover_duration: DurationValue = 3000
```

Duration after which the controls will be hidden when there is no mouse movement.

### **buildcontrols\_transition\_durationclass-attributeinstance-attribute**

```
controls_transition_duration: DurationValue = 300
```

Duration for which the controls will be animated when shown or hidden.

### **bulddisplay\_seek\_barclass-attributeinstance-attribute**

```
display_seek_bar: bool = True
```

Whether to display seek bar.

### **buildgestures\_enabled\_while\_controls\_visibleclass-attributeinstance-attribute**

```
gestures_enabled_while_controls_visible: bool = True
```

Whether to allow gesture controls to work while controls are visible.

#### **Note**

This option is ignored when gestures are disabled.

### **buildhorizontal\_gesture\_sensitivityclass-attributeinstance-attribute**

```
horizontal_gesture_sensitivity: Number = 1000
```

Gesture sensitivity on horizontal drag gestures, the higher the value is the less sensitive the gesture.

### **buildinitial\_brightnessclass-attributeinstance-attribute**

```
initial_brightness: Number = 0.5
```

Initial brightness value used by the brightness gesture indicator.

It ranges from 0.0 to 1.0.

### **buildinitial\_volumeclass-attributeinstance-attribute**

```
initial_volume: Number = 0.5
```

Initial volume value used by the volume gesture indicator.

It ranges from 0.0 to 1.0.

### **buildpaddingclass-attributeinstance-attribute**

```
padding: PaddingValue | None = None
```

Padding around the controls.

### **buildprimary\_button\_barclass-attributeinstance-attribute**

```
primary_button_bar: (
 list[Control | VideoBarItem] | None
) = None
```

Controls displayed in the primary button bar.

Set to a list to replace the primary button bar, or an empty list to hide it; if None, the native default is:

```
[
 ftv.VideoSpacer(flex=2),
 ftv.VideoSkipPreviousButton(),
 ftv.VideoSpacer(),
 ftv.VideoPlayOrPauseButton(icon_size=48.0),
 ftv.VideoSpacer(),
 ftv.VideoSkipNextButton(),
 ftv.VideoSpacer(flex=2),
]
```

### buildseek\_bar\_alignmentclass-attributeinstance-attribute

```
seek_bar_alignment: Alignment = field(
 default_factory=lambda: Alignment.BOTTOM_CENTER
)
```

Alignment of seek bar inside the seek bar container.

### buildseek\_bar\_buffer\_colorclass-attributeinstance-attribute

```
seek_bar_buffer_color: ColorValue = '#3DFFFFFF'
```

Color of the playback buffer section in the seek bar.

### buildseek\_bar\_colorclass-attributeinstance-attribute

```
seek_bar_color: ColorValue = '#3DFFFFFF'
```

Color of the seek bar.

### buildseek\_bar\_container\_heightclass-attributeinstance-attribute

```
seek_bar_container_height: Number = 36.0
```

Height of the seek bar container.

### buildseek\_bar\_heightclass-attributeinstance-attribute

```
seek_bar_height: Number = 2.4
```

Height of the seek bar.

### buildseek\_bar\_marginclass-attributeinstance-attribute

```
seek_bar_margin: PaddingValue = 0
```

Margin around the seek bar.

### buildseek\_bar\_position\_colorclass-attributeinstance-attribute

```
seek_bar_position_color: ColorValue = '#FFFF0000'
```

Color of the playback position section in the seek bar.

### buildseek\_bar\_thumb\_colorclass-attributeinstance-attribute

```
seek_bar_thumb_color: ColorValue = '#FFFF0000'
```

Color of the seek bar thumb.

### buildseek\_bar\_thumb\_sizeclass-attributeinstance-attribute

```
seek_bar_thumb_size: Number = 12.8
```

Size of the seek bar thumb.

### buildseek\_gestureclass-attributeinstance-attribute

```
seek_gesture: bool = False
```

Whether to seek on horizontal drag gesture.

### buildseek\_on\_double\_tapclass-attributeinstance-attribute

```
seek_on_double_tap: bool = False
```

Whether to enable double tap to seek on left or right side of the screen.

### buildseek\_on\_double\_tap\_backward\_durationclass-attributeinstance-attribute

```
seek_on_double_tap_backward_duration: DurationValue = (
 10000
)
```

Duration of seek on double tap backward.

### buildseek\_on\_double\_tap\_enabled\_while\_controls\_visibleclass-attributeinstance-attribute

```
seek_on_double_tap_enabled_while_controls_visible: bool = (
 True
)
```

Whether to allow double tap to seek on left or right side of the screen to work while controls are visible.

Note

This option is ignored when seek\_on\_double\_tap is False.

### **buildseek\_on\_double\_tap\_forward\_durationclass-attributeinstance-attribute**

```
seek_on_double_tap_forward_duration: DurationValue = (
 10000
)
```

Duration of seek on double tap forward.

### **buildseek\_on\_double\_tap\_layout\_taps\_ratiosclass-attributeinstance-attribute**

```
seek_on_double_tap_layout_taps_ratios: list[int] = field(
 default_factory=lambda: [1, 1, 1]
)
```

Width proportions for the backward seek, instant tap, and forward seek areas when a double tap occurs on the video widget.

### **buildseek\_on\_double\_tap\_layout\_widget\_ratiosclass-attributeinstance-attribute**

```
seek_on_double_tap_layout_widget_ratios: list[int] = field(
 default_factory=lambda: [1, 1, 1]
)
```

Width proportions for the visual indicators shown during backward seek, instant tap, and forward seek actions.

### **buildshift\_subtitles\_on\_controls\_visibility\_changeclass-attributeinstance-attribute**

```
shift_subtitles_on_controls_visibility_change: bool = False
```

Whether to shift the subtitles upwards when the controls are visible.

### **buildspeed\_up\_factorclass-attributeinstance-attribute**

```
speed_up_factor: Number = 2.0
```

Factor to speed up on long press.

### **buildspeed\_up\_on\_long\_pressclass-attributeinstance-attribute**

```
speed_up_on_long_press: bool = False
```

Whether to speed up on long press.

### **buildtop\_button\_barclass-attributeinstance-attribute**

```
top_button_bar: list[Control | VideoBarItem] | None = (
 None
)
```

Controls displayed in the top button bar.

Set to a list to replace the top button bar, or an empty list to hide it; if None, the native default is an empty list.

### **buildtop\_button\_bar\_marginclass-attributeinstance-attribute**

```
top_button_bar_margin: PaddingValue = field(
 default_factory=lambda: Padding.symmetric(
 horizontal=16.0
)
)
```

Margin around the top button bar.

### **buildvertical\_gesture\_sensitivityclass-attributeinstance-attribute**

```
vertical_gesture_sensitivity: Number = 100
```

Gesture sensitivity on vertical drag gestures, the higher the value is the less sensitive the gesture.

### **buildvisible\_on\_mountclass-attributeinstance-attribute**

```
visible_on_mount: bool = False
```

Whether the controls are initially visible.

### **buildvolume\_gestureclass-attributeinstance-attribute**

```
volume_gesture: bool = False
```

Whether to modify volume on vertical drag gesture on the right side of the screen.

# PlaylistMode

Defines the playback mode for the video playlist.

Inherits: `enum.Enum`

## Properties

- **LOOP** - Loop over the playlist & restart it from beginning once end is reached.
- **NONE** - End playback once end of the playlist is reached.
- **SINGLE** - Indefinitely loop over the currently playing file in the playlist.

## Properties

**buildLOOP**`class-attributeinstance-attribute`

Loop over the playlist & restart it from beginning once end is reached.

**buildNONE**`class-attributeinstance-attribute`

End playback once end of the playlist is reached.

**buildSINGLE**`class-attributeinstance-attribute`

Indefinitely loop over the currently playing file in the playlist.

# VideoBarItem

Base class for built-in video controls button bar items.

# VideoConfiguration

Additional configuration for video playback.

## Properties

- `enable_hardware_acceleration` - Whether to enable hardware acceleration.
- `hardware_decoding_api` - Sets the `--hwdec` property on native backend.
- `height` - The fixed height for the video output.
- `mpv_properties` - Extra mpv/libmpv properties to set on native backends (Windows/macOS/Linux/iOS/Android).
- `output_driver` - Sets the `--vo` property on native backend.
- `scale` - The scale for the video output.
- `width` - The fixed width for the video output.

## Properties

### `buildenable_hardware_accelerationclass-attributeinstance-attribute`

`enable_hardware_acceleration: bool = True`

Whether to enable hardware acceleration. When disabled, may cause battery drain, device heating, and high CPU usage.

### `buildhardware_decoding_apiclass-attributeinstance-attribute`

`hardware_decoding_api: str | None = None`

Sets the `--hwdec` property on native backend.

The default value is platform dependent:

- Windows, GNU/Linux, macOS & iOS : "auto"
- Android: "auto-safe"

### `buildheightclass-attributeinstance-attribute`

`height: Number | None = None`

The fixed height for the video output.

### `buildmpv_propertiesclass-attributeinstance-attribute`

```
mpv_properties: (
 dict[str, str | int | float | bool] | None
) = None
```

Extra mpv/libmpv properties to set on native backends (Windows/macOS/Linux/iOS/Android).

The keys are mpv option/property names without the leading `--`. Values can be str, int, float or bool. All values are converted to strings before being passed to mpv; boolean values are converted to "yes" / "no".

Full list of mpv options: <https://mpv.io/manual/stable/#options>

## Example

```
>>> VideoConfiguration(
 mpv_properties={
 "profile": "low-latency", # --profile=low-latency
 "untimed": True, # --untimed
 "volume": 80, # --volume=80
 }
)
```

### `buildoutput_driverclass-attributeinstance-attribute`

`output_driver: str | None = None`

Sets the `--vo` property on native backend.

The default value is platform dependent:

- Windows, GNU/Linux, macOS & iOS : "libmpv"

- Android: "gpu"

### **buildscaleclass-attributeinstance-attribute**

scale: Number = 1.0

The scale for the video output. Specifying this option will cause width & height to be ignored.

### **buildwidthclass-attributeinstance-attribute**

width: Number | None = None

The fixed width for the video output.

# VideoControls

Base class for built-in video controls.



# VideoControlsMode

Modes that can receive different Video.controls values.

Inherits: `enum.Enum`

## Properties

- **DEFAULT** - Fallback controls used when no applicable mode-specific value is provided.
- **FULLSCREEN** - Controls used when the video is fullscreen.
- **NORMAL** - Controls used when the video is not fullscreen.

## Properties

**buildDEFAULT**class-attributeinstance-attribute

Fallback controls used when no applicable mode-specific value is provided.

**buildFULLSCREEN**class-attributeinstance-attribute

Controls used when the video is fullscreen.

**buildNORMAL**class-attributeinstance-attribute

Controls used when the video is not fullscreen.

# VideoFullscreenButton

A built-in fullscreen button item.

Inherits: VideoBarItem

## Properties

- `icon` - Icon displayed by the button.
- `icon_color` - Overrides the icon color.
- `icon_size` - Overrides the icon size.

## Properties

### `buildIcon` class-attribute instance-attribute

`icon: Control | None = None`

Icon displayed by the button.

If omitted, the default fullscreen icon is used.

### `buildIcon_color` class-attribute instance-attribute

`icon_color: ColorValue | None = None`

Overrides the icon color.

### `buildIcon_size` class-attribute instance-attribute

`icon_size: Number | None = None`

Overrides the icon size.

# VideoMedia

Represents a media resource for video playback.

## Properties

- `extras` - Additional metadata for the media resource.
- `http_headers` - HTTP headers to be used for the media resource.
- `resource` - URI of the media resource.

## Properties

### `buildextrasclass-attributeinstance-attribute`

`extras: dict[str, str] | None = None`

Additional metadata for the media resource.

### `buildhttp_headersclass-attributeinstance-attribute`

`http_headers: dict[str, str] | None = None`

HTTP headers to be used for the media resource.

### `buildresourceinstance-attribute`

`resource: str`

URI of the media resource.

# VideoPlayOrPauseButton

A built-in play/pause button item.

Inherits: VideoBarItem

## Properties

- `icon_color` - Overrides the icon color.
- `icon_size` - Overrides the icon size.

## Properties

**buildicon\_color**class-attributeinstance-attribute

`icon_color: ColorValue | None = None`

Overrides the icon color.

**buildicon\_size**class-attributeinstance-attribute

`icon_size: Number | None = None`

Overrides the icon size.

# VideoPositionIndicator

A built-in playback position indicator item.

Inherits: VideoBarItem

## Properties

- `text_style` - Overrides the text style.

## Properties

`buildTextStyle` class-attribute instance-attribute

`text_style`: TextStyle | None = None

Overrides the text style.

# VideoSkipNextButton

A built-in skip next button item.

Inherits: VideoBarItem

## Properties

- `icon` - Icon displayed by the button.
- `icon_color` - Overrides the icon color.
- `icon_size` - Overrides the icon size.

## Properties

### `buildIcon` class-attribute instance-attribute

`icon: Control | None = None`

Icon displayed by the button.

If omitted, the default skip next icon is used.

### `buildIcon_color` class-attribute instance-attribute

`icon_color: ColorValue | None = None`

Overrides the icon color.

### `buildIcon_size` class-attribute instance-attribute

`icon_size: Number | None = None`

Overrides the icon size.

# VideoSkipPreviousButton

A built-in skip previous button item.

Inherits: VideoBarItem

## Properties

- `icon` - Icon displayed by the button.
- `icon_color` - Overrides the icon color.
- `icon_size` - Overrides the icon size.

## Properties

### `buildiconclass-attributeinstance-attribute`

`icon: Control | None = None`

Icon displayed by the button.

If omitted, the default skip previous icon is used.

### `buildicon_colorclass-attributeinstance-attribute`

`icon_color: ColorValue | None = None`

Overrides the icon color.

### `buildicon_sizeclass-attributeinstance-attribute`

`icon_size: Number | None = None`

Overrides the icon size.

# VideoSpacer

A built-in spacer item for video controls button bars.

Inherits: VideoBarItem

## Properties

- flex - The flex factor to use for the spacer.

## Properties

**buildflexclass-attributeinstance-attribute**

flex: int = 1

The flex factor to use for the spacer.



# VideoSubtitleConfiguration

Represents the configuration for video subtitles.

## Properties

- padding - The padding to be used for the subtitles.
- text\_align - The text alignment to be used for the subtitles.
- text\_scale\_factor - Defines the scale factor for the subtitle text.
- text\_style - The text style to be used for the subtitles.
- visible - Whether the subtitles should be visible or not.

## Properties

### buildpaddingclass-attributeinstance-attribute

```
padding: PaddingValue = field(
 default_factory=lambda: Padding(
 left=16.0, top=0.0, right=16.0, bottom=24.0
)
)
```

The padding to be used for the subtitles.

### buildtext\_alignclass-attributeinstance-attribute

```
text_align: TextAlign = TextAlign.CENTER
```

The text alignment to be used for the subtitles.

### buildtext\_scale\_factorclass-attributeinstance-attribute

```
text_scale_factor: Number = 1.0
```

Defines the scale factor for the subtitle text.

### buildtext\_styleclass-attributeinstance-attribute

```
text_style: TextStyle = field(
 default_factory=lambda: TextStyle(
 height=1.4,
 size=32.0,
 letter_spacing=0.0,
 word_spacing=0.0,
 color=Colors.WHITE,
 weight=FontWeight.NORMAL,
 bgcolor=Colors.BLACK_54,
)
)
```

The text style to be used for the subtitles.

### buildvisibleclass-attributeinstance-attribute

```
visible: bool = True
```

Whether the subtitles should be visible or not.

# VideoSubtitleTrack

Represents a subtitle track for a video.

## Properties

- `album_art` - Whether the track represents album art rather than timed media.
- `audio_channels` - Explicit audio channel count override.
- `bitrate` - Overall media bitrate in bits per second.
- `channels` - Channel layout string describing the spatial arrangement of channels.
- `channels_count` - The number of audio channels detected in the media.
- `codec` - Codec identifier for the media stream.
- `decoder` - Decoder name used to process the media stream.
- `fps` - Video frames per second.
- `language` - The language of the subtitle track, e.g.
- `par` - Pixel aspect ratio value.
- `rotate` - Rotation metadata in degrees to apply when rendering the video.
- `sample_rate` - Audio sampling rate in hertz.
- `src` - The subtitle source.
- `title` - The title of the subtitle track, e.g.

## Methods

- `auto` - Default subtitle track.
- `none` - No subtitle track.

## Properties

### `buildalbum_artclass-attributeinstance-attribute`

`album_art: bool | None = None`

Whether the track represents album art rather than timed media.

### `buildaudio_channelsclass-attributeinstance-attribute`

`audio_channels: int | None = None`

Explicit audio channel count override.

### `buildbitrateclass-attributeinstance-attribute`

`bitrate: int | None = None`

Overall media bitrate in bits per second.

### `buildchannelsclass-attributeinstance-attribute`

`channels: str | None = None`

Channel layout string describing the spatial arrangement of channels.

### `buildchannels_countclass-attributeinstance-attribute`

`channels_count: int | None = None`

The number of audio channels detected in the media.

### `buildcodecclass-attributeinstance-attribute`

`codec: str | None = None`

Codec identifier for the media stream.

### builddecoderclass-attributeinstance-attribute

decoder: str | None = None

Decoder name used to process the media stream.

### buildfpsclass-attributeinstance-attribute

fps: Number | None = None

Video frames per second.

### builanguageclass-attributeinstance-attribute

language: str | None = None

The language of the subtitle track, e.g. 'en'.

### buildparclass-attributeinstance-attribute

par: Number | None = None

Pixel aspect ratio value.

### buildrotateclass-attributeinstance-attribute

rotate: int | None = None

Rotation metadata in degrees to apply when rendering the video.

### buildsample\_rateclass-attributeinstance-attribute

sample\_rate: int | None = None

Audio sampling rate in hertz.

### buildsrcinstance-attribute

src: str

The subtitle source.

Supported values:

- A URL (e.g. "https://example.com/subs.srt" or "www.example.com/sub.vtt")
- An absolute local file path (not supported on the web platform)
- A raw subtitle text string (e.g. the full contents of an SRT/VTT file)

### buildtitleclass-attributeinstance-attribute

title: str | None = None

The title of the subtitle track, e.g. 'English'.

## Methods

### deployed\_codeautoclassmethod

auto() -> VideoSubtitleTrack

Default subtitle track. Selects the first subtitle track.

### deployed\_codenoneclassmethod

none() -> VideoSubtitleTrack

No subtitle track. Disables subtitle output.

# VideoVolumeButton

note

VideoVolumeButton is currently rendered only by MaterialDesktopVideoControls.

A built-in volume button and slider item.

Note

This item is currently rendered only by MaterialDesktopVideoControls.

Inherits: VideoBarItem

## Properties

- `icon_color` - Overrides the icon color.
- `icon_size` - Overrides the icon size.
- `slider_width` - Width of the volume slider.
- `volume_high_icon` - Icon displayed when volume is high.
- `volume_low_icon` - Icon displayed when volume is low.
- `volume_mute_icon` - Icon displayed when volume is muted.

## Properties

### `buildicon_colorclass-attributeinstance-attribute`

`icon_color: ColorValue | None = None`

Overrides the icon color.

### `buildicon_sizeclass-attributeinstance-attribute`

`icon_size: Number | None = None`

Overrides the icon size.

### `buildslider_widthclass-attributeinstance-attribute`

`slider_width: Number | None = None`

Width of the volume slider.

### `buildvolume_high_iconclass-attributeinstance-attribute`

`volume_high_icon: Control | None = None`

Icon displayed when volume is high.

If omitted, the default high volume icon is used.

### `buildvolume_low_iconclass-attributeinstance-attribute`

`volume_low_icon: Control | None = None`

Icon displayed when volume is low.

If omitted, the default low volume icon is used.

### `buildvolume_mute_iconclass-attributeinstance-attribute`

`volume_mute_icon: Control | None = None`

Icon displayed when volume is muted.

If omitted, the default muted volume icon is used.

# LogLevelSeverity

Represents the severity of a JavaScript log message.

Inherits: `enum.Enum`

## Properties

- **DEBUG** - Indicates a debug message was logged using the `console.debug` method.
- **ERROR** - Indicates an error message was logged via an "error" event of the `console.error` method.
- **INFO** - Indicates an informational message was logged using the `console.info` method.
- **LOG** - Indicates a log message was logged using the `console.log` method.
- **WARNING** - Indicates a warning message was logged using the `console.warning` method.

## Properties

### **buildDEBUGclass-attributeinstance-attribute**

Indicates a debug message was logged using the `console.debug` method.

### **buildERRORclass-attributeinstance-attribute**

Indicates an error message was logged via an "error" event of the `console.error` method.

### **buildINFOclass-attributeinstance-attribute**

Indicates an informational message was logged using the `console.info` method.

### **buildLOGclass-attributeinstance-attribute**

Indicates a log message was logged using the `console.log` method.

### **buildWARNINGclass-attributeinstance-attribute**

Indicates a warning message was logged using the `console.warning` method.

# RequestMethod

Defines the supported HTTP methods for loading a page in a WebView.

Inherits: `enum.Enum`

## Properties

- GET - HTTP GET method.
- POST - HTTP POST method.

## Properties

`buildGETclass-attributeinstance-attribute`

HTTP GET method.

`buildPOSTclass-attributeinstance-attribute`

HTTP POST method.

# WebViewConsoleMessageEvent

Event payload for `WebView.on_console_message`.

Inherits: `Event`

## Properties

- `message` - The message written to the console.
- `severity_level` - The severity of a JavaScript log message.

## Properties

### `buildmessageinstance-attribute`

`message: str`

The message written to the console.

### `buildseverity_levelinstance-attribute`

`severity_level: LogLevelSeverity`

The severity of a JavaScript log message.

# WebViewJavaScriptEvent

Event payload for `WebView.on_javascript_alert_dialog`.

Represents a JavaScript `alert()` dialog request initiated by the loaded page.

Inherits: `Event`

## Properties

- `message` - The message to be displayed in the window.
- `url` - The URL of the page requesting the dialog.

## Properties

### `buildmessageinstance-attribute`

`message: str`

The message to be displayed in the window.

### `buildurlinstance-attribute`

`url: str`

The URL of the page requesting the dialog.



# WebViewScrollEvent

Event payload for `WebView.on_scroll`.

Emitted when the page scroll position changes.

Inherits: `Event`

## Properties

- `x` - The value of the horizontal offset with the origin being at the leftmost of the `WebView`.
- `y` - The value of the vertical offset with the origin being at the topmost of the `WebView`.

## Properties

### `buildxinstance-attribute`

`x: float`

The value of the horizontal offset with the origin being at the leftmost of the `WebView`.

### `buildyinstance-attribute`

`y: float`

The value of the vertical offset with the origin being at the topmost of the `WebView`.

# Hosting

Flet apps use WebSockets for real-time partial updates of their UI and sending events back to your program. When choosing a hosting provider for your Flet app you should pay attention to their support of WebSockets. Sometimes WebSockets are not allowed or come as a part of more expensive offering, sometimes there is a proxy that periodically breaks WebSocket connection by a timeout (Flet implements re-connection logic, but it could be unpleasant behavior for users of your app anyway).

Another important factor while choosing a hosting provider for Flet app is latency. Every user action on UI sends a message to Flet app and the app sends updated UI back to user. Make sure your hosting provider has multiple data centers, so you can run your app closer to the majority of your users.

# AudioDurationChangeEvent

Event triggered when the audio duration changes.

Inherits: Event

## Properties

- duration - The duration of the audio.

## Properties

builddurationinstance-attribute

duration: Duration

The duration of the audio.

# AudioPositionChangeEvent

Event triggered when the audio playback position changes.

Inherits: Event

## Properties

- position - The current playback position in milliseconds.

## Properties

**buildpositioninstance-attribute**

position: int

The current playback position in milliseconds.

# AudioState

The state of the audio player.

Inherits: `enum.Enum`

## Properties

- **COMPLETED** - The audio player has successfully reached the end of the audio.
- **DISPOSED** - The audio player has been disposed of and should not be used anymore.
- **PAUSED** - The audio player is paused and can be resumed.
- **PLAYING** - The audio player is currently playing audio.
- **STOPPED** - The audio player is stopped.

## Properties

**buildCOMPLETED**`class-attributeinstance-attribute`

The audio player has successfully reached the end of the audio.

**buildDISPOSED**`class-attributeinstance-attribute`

The audio player has been disposed of and should not be used anymore.

**buildPAUSED**`class-attributeinstance-attribute`

The audio player is paused and can be resumed.

**buildPLAYING**`class-attributeinstance-attribute`

The audio player is currently playing audio.

**buildSTOPPED**`class-attributeinstance-attribute`

The audio player is stopped.

# AudioStateChangeEvent

Event triggered when the audio playback state changes.

Inherits: Event

## Properties

- state - The current state of the audio player.

## Properties

### buildstateinstance-attribute

state: AudioState

The current state of the audio player.

# ReleaseMode

Determines what happens to the player's resources (the buffered audio and the underlying native player) when playback finishes or is stopped, and therefore how quickly, and at what cost, the same audio can be played again.

Inherits: `enum.Enum`

## Properties

- **LOOP** - Automatically restarts playback from the beginning every time it completes, creating a continuous loop.
- **RELEASE** - Frees all resources once playback completes, as if `Audio.release` had been called automatically.
- **STOP** - Stops playback when the audio completes but keeps all resources (the buffered audio and the native player) intact.

## Properties

### `buildLOOPclass-attributeinstance-attribute`

Automatically restarts playback from the beginning every time it completes, creating a continuous loop. All resources are kept buffered.

Best for audio that should repeat indefinitely, such as background music.

Note

Resources are never released automatically in this mode. To free them, change the source or call `Audio.release`.

### `buildRELEASEclass-attributeinstance-attribute` ☐

Frees all resources once playback completes, as if `Audio.release` had been called automatically. This is the default mode.

The buffered audio and the native player are released, so nothing is kept in memory while the audio is idle. Playing again is still supported, but the source is loaded again from scratch first (and a remote file is re-downloaded), which adds a short delay before playback starts.

Best when replays are rare or may never happen and keeping memory usage to a minimum matters.

Note

- On Android, the native media player is resource-intensive; this mode lets it go and re-buffers the data only when needed.
- On iOS and macOS, behaves like `Audio.release`.

### `buildSTOPclass-attributeinstance-attribute`

Stops playback when the audio completes but keeps all resources (the buffered audio and the native player) intact.

Because nothing is released, playing again is immediate: the audio restarts straight from the retained buffer, with no reloading or re-downloading. The trade-off is that these resources stay in memory while the audio is idle.

Best when you intend to replay the same audio and want instant, network-free playback (for example, short sound effects or repeated tones).

# AndroidAudioSource

Android-specific audio source types.

Inherits: `enum.Enum`

## Properties

- **CAMCORDER** - Microphone audio source tuned for video recording, with the same orientation as the camera, if available.
- **DEFAULT\_SOURCE** - Default audio source.
- **MIC** - Microphone audio source.
- **REMOTE\_SUBMIX** - Audio source for a submix of audio streams to be presented remotely.
- **UNPROCESSED** - Microphone audio source tuned for unprocessed (raw) sound if available, behaves like **DEFAULT\_SOURCE** otherwise.
- **VOICE\_CALL** - Voice call uplink + downlink audio source.
- **VOICE\_COMMUNICATION** - Microphone audio source tuned for voice communications such as VoIP.
- **VOICE\_DOWNLINK** - Voice call downlink (Rx) audio source.
- **VOICE\_PERFORMANCE** - Source for capturing audio meant to be processed in real time and played back for live performance (e.g karaoke).
- **VOICE\_RECOGNITION** - Microphone audio source tuned for voice recognition.
- **VOICE\_UPLINK** - Voice call uplink (Tx) audio source.

## Properties

**buildCAMCORDER**`class-attributeinstance-attribute`

Microphone audio source tuned for video recording, with the same orientation as the camera, if available.

**buildDEFAULT\_SOURCE**`class-attributeinstance-attribute`

Default audio source.

**buildMIC**`class-attributeinstance-attribute`

Microphone audio source.

**buildREMOTE\_SUBMIX**`class-attributeinstance-attribute`

Audio source for a submix of audio streams to be presented remotely.

**buildUNPROCESSED**`class-attributeinstance-attribute`

Microphone audio source tuned for unprocessed (raw) sound if available, behaves like **DEFAULT\_SOURCE** otherwise.

**buildVOICE\_CALL**`class-attributeinstance-attribute`

Voice call uplink + downlink audio source.

**buildVOICE\_COMMUNICATION**`class-attributeinstance-attribute`

Microphone audio source tuned for voice communications such as VoIP.

**buildVOICE\_DOWNLINK**`class-attributeinstance-attribute`

Voice call downlink (Rx) audio source.

**buildVOICE\_PERFORMANCE**`class-attributeinstance-attribute`

Source for capturing audio meant to be processed in real time and played back for live performance (e.g karaoke).

**buildVOICE\_RECOGNITION**`class-attributeinstance-attribute`



Microphone audio source tuned for voice recognition.

**buildVOICE\_UPLINKclass-attributeinstance-attribute**

Voice call uplink (Tx) audio source.

# AndroidRecorderConfiguration

Android specific configuration for recording.

## Properties

- `audio_source` - Defines the audio source.
- `manage_bluetooth` - Try to start a bluetooth audio connection to a headset (Bluetooth SCO).
- `mute_audio` - Whether to mute all audio streams like alarms, music, ring, etc.
- `use_legacy` - Whether to use the Android MediaRecorder.

## Properties

### `buildaudio_sourceclass-attributeinstance-attribute`

```
audio_source: AndroidAudioSource = (
 AndroidAudioSource.DEFAULT_SOURCE
)
```

Defines the audio source.

An audio source defines both a default physical source of audio signal, and a recording configuration. Some effects are available or not depending on this source.

Most of the time, you should use `AndroidAudioSource.DEFAULT_SOURCE` or `AndroidAudioSource.MIC`.

### `buildmanage_bluetoothclass-attributeinstance-attribute`

```
manage_bluetooth: bool = True
```

Try to start a bluetooth audio connection to a headset (Bluetooth SCO).

### `buildmute_audioclass-attributeinstance-attribute`

```
mute_audio: bool = False
```

Whether to mute all audio streams like alarms, music, ring, etc.

This is useful when you want to record audio without any background noise. The streams are restored to their previous state after recording is stopped and will stay at current state on pause/resume.

### `builduse_legacyclass-attributeinstance-attribute`

```
use_legacy: bool = False
```

Whether to use the Android MediaRecorder.

While advanced recorder (the default) unlocks additional features, the legacy recorder is stability oriented.

# AudioEncoder

Represents the different audio encoders for audio recording.

Inherits: `enum.Enum`

## Properties

- AACELD - Advanced Audio Codec Enhanced Low Delay.
- AACHE - Advanced Audio Codec High Efficiency.
- AACLC - Advanced Audio Codec Low Complexity.
- AMRNB - Adaptive Multi-Rate Narrow Band.
- AMRWB - Adaptive Multi-Rate Wide Band.
- FLAC - Free Lossless Audio Codec.
- OPUS - A codec designed for both speech and audio applications, known for its versatility.
- PCM16BITS - Pulse Code Modulation with 16-bit depth, used for high-fidelity audio.
- WAV - Standard audio format used for raw, uncompressed audio data.

## Properties

### `buildAACELDclass-attributeinstance-attribute`

Advanced Audio Codec Enhanced Low Delay. Suitable for low-latency applications like VoIP.

### `buildAACHEclass-attributeinstance-attribute`

Advanced Audio Codec High Efficiency. Optimized for high-quality audio at lower bit rates.

### `buildAACLCclass-attributeinstance-attribute`

Advanced Audio Codec Low Complexity. A commonly used encoder for streaming and general audio recording.

### `buildAMRNBclass-attributeinstance-attribute`

Adaptive Multi-Rate Narrow Band. Used for speech audio in mobile communication.

### `buildAMRWBclass-attributeinstance-attribute`

Adaptive Multi-Rate Wide Band. Used for higher-quality speech audio.

### `buildFLACclass-attributeinstance-attribute`

Free Lossless Audio Codec. Provides high-quality lossless audio compression.

### `buildOPUSclass-attributeinstance-attribute`

A codec designed for both speech and audio applications, known for its versatility.

### `buildPCM16BITSclass-attributeinstance-attribute`

Pulse Code Modulation with 16-bit depth, used for high-fidelity audio.

### `buildWAVclass-attributeinstance-attribute`

Standard audio format used for raw, uncompressed audio data.

# AudioRecorderConfiguration

Recording configuration.

## Properties

- `android_configuration` - Android specific configuration.
- `auto_gain` - The recorder will try to auto adjust recording volume in a limited range (if available on the device).
- `bit_rate` - The audio encoding bit rate in bits per second if applicable.
- `cancel_echo` - The recorder will try to reduce echo (if available on the device).
- `channels` - The numbers of channels for the recording.
- `device` - The device to be used for recording.
- `encoder` - The requested output format through this given encoder.
- `ios_configuration` - iOS specific configuration.
- `sample_rate` - The sample rate for audio in samples per second if applicable.
- `suppress_noise` - The recorder will try to negate the input noise (if available on the device).

## Properties

### `buildandroid_configurationclass-attributeinstance-attribute`

```
android_configuration: AndroidRecorderConfiguration = field(
 default_factory=lambda: AndroidRecorderConfiguration()
)
```

Android specific configuration.

### `buildauto_gainclass-attributeinstance-attribute`

```
auto_gain: bool = False
```

The recorder will try to auto adjust recording volume in a limited range (if available on the device).

Recording volume may be lowered by using this.

### `buildbit_rateclass-attributeinstance-attribute`

```
bit_rate: Number = 128000
```

The audio encoding bit rate in bits per second if applicable.

### `buildcancel_echoclass-attributeinstance-attribute`

```
cancel_echo: bool = False
```

The recorder will try to reduce echo (if available on the device).

Recording volume may be lowered by using this.

### `buildchannelsclass-attributeinstance-attribute`

```
channels: int = 2
```

The numbers of channels for the recording.

- 1 for mono
- 2 for stereo

Most platforms only accept at most 2 channels.

### `builddeviceclass-attributeinstance-attribute`

```
device: InputDevice | None = None
```

The device to be used for recording.

If None, default device will be selected.

### `buildencoderclass-attributeinstance-attribute`

```
encoder: AudioEncoder = AudioEncoder.WAV
```

The requested output format through this given encoder.

### **buildios\_configurationclass-attributeinstance-attribute**

```
ios_configuration: IosRecorderConfiguration = field(
 default_factory=lambda: IosRecorderConfiguration()
)
```

iOS specific configuration.

### **buildsample\_rateclass-attributeinstance-attribute**

```
sample_rate: int = 44100
```

The sample rate for audio in samples per second if applicable.

### **buildsuppress\_noiseclass-attributeinstance-attribute**

```
suppress_noise: bool = False
```

The recorder will try to negate the input noise (if available on the device).

Recording volume may be lowered by using this.

# AudioRecorderState

State of the audio recorder.

Inherits: `enum.Enum`

## Properties

- **PAUSED** - The audio recorder is paused and can be resumed.
- **RECORDING** - The audio recorder is currently recording audio.
- **STOPPED** - The audio recorder is stopped and not recording.

## Properties

**buildPAUSED**`class-attributeinstance-attribute`

The audio recorder is paused and can be resumed.

**buildRECORDING**`class-attributeinstance-attribute`

The audio recorder is currently recording audio.

**buildSTOPPED**`class-attributeinstance-attribute`

The audio recorder is stopped and not recording.

# AudioRecorderStateChangeEvent

Event payload for recorder state transitions.

Delivered by `AudioRecorder.on_state_change`.

Inherits: `Event`

## Properties

- `state` - The new state of the audio recorder.

## Properties

### `buildstateinstance-attribute`

`state: AudioRecorderState`

The new state of the audio recorder.

# AudioRecorderStreamEvent

Event payload for raw recording stream chunks.

Delivered by `AudioRecorder.on_stream`.

Inherits: `Event`

## Properties

- `bytes_streamed` - Total number of bytes delivered through chunk so far.
- `chunk` - Raw PCM16BITS audio bytes emitted by `AudioRecorder`.
- `sequence` - Incremental chunk number.

## Properties

### `buildbytes_streamedinstance-attribute`

`bytes_streamed: int`

Total number of bytes delivered through chunk so far.

### `buildchunkinstance-attribute`

`chunk: bytes`

Raw PCM16BITS audio bytes emitted by `AudioRecorder`.

### `buildsequenceinstance-attribute`

`sequence: int`

Incremental chunk number.



# AudioRecorderUploadEvent

Event payload for streaming recording uploads.

Delivered by `AudioRecorder.on_upload` for uploads started with `AudioRecorder.start_recording`.

Inherits: `Event`

## Properties

- `bytes_uploaded` - Number of bytes uploaded so far.
- `error` - Error message if the upload failed.
- `file_name` - Name provided by `AudioRecorderUploadSettings.file_name`.
- `progress` - Upload progress from 0.0 to 1.0.

## Properties

### `buildbytes_uploadedclass-attributeinstance-attribute`

`bytes_uploaded: int | None = None`

Number of bytes uploaded so far.

### `builderrorclass-attributeinstance-attribute`

`error: str | None = None`

Error message if the upload failed.

### `buildfile_nameclass-attributeinstance-attribute`

`file_name: str | None = None`

Name provided by `AudioRecorderUploadSettings.file_name`.

### `buildprogressclass-attributeinstance-attribute`

`progress: float | None = None`

Upload progress from 0.0 to 1.0.

Streaming uploads do not know their total size until recording stops, so `bytes_uploaded` is usually the best progress indicator while recording is active.

# AudioRecorderUploadSettings

Upload settings for streaming recordings.

## Note

Uploads started by `AudioRecorder.start_recording` send raw PCM16BITS bytes. They do not add a playable audio container such as WAV.

## Properties

- `file_name` - Friendly name reported in upload events.
- `headers` - HTTP headers sent with the upload request.
- `method` - HTTP method to use when uploading the streamed bytes.
- `upload_url` - Destination URL, for example one returned by `Page.get_upload_url`.

## Properties

### `buildfile_name`class-attributeinstance-attribute

`file_name: str | None = None`

Friendly name reported in upload events.

### `buildheaders`class-attributeinstance-attribute

`headers: dict[str, str] | None = None`

HTTP headers sent with the upload request.

### `buildmethod`class-attributeinstance-attribute

`method: str = 'PUT'`

HTTP method to use when uploading the streamed bytes.

### `buildupload_url`instance-attribute

`upload_url: str`

Destination URL, for example one returned by `Page.get_upload_url`.

# InputDevice

An audio input device.

## Properties

- `id` - The ID used to select the device on the platform.
- `label` - The label text representation.

## Properties

### `buildidinstance-attribute`

`id: str`

The ID used to select the device on the platform.

### `buildlabelinstance-attribute`

`label: str`

The label text representation.

# IosAudioCategoryOption

Audio behaviors.

Each option is valid only for specific audio session categories.

Inherits: `enum.Enum`

## Properties

- `ALLOW_AIRPLAY` - Stream audio from this session to AirPlay devices.
- `ALLOW_BLUETOOTH` - Bluetooth hands-free devices appear as available input routes.
- `ALLOW_BLUETOOTH_A2DP` - Stream audio from this session to Bluetooth devices that support the Advanced Audio Distribution Profile (A2DP).
- `DEFAULT_TO_SPEAKER` - Audio from the session defaults to the built-in speaker instead of the receiver.
- `DUCK_OTHERS` - Reduces the volume of other audio sessions while audio from this session plays.
- `INTERRUPT_SPOKEN_AUDIO_AND_MIX_WITH_OTHERS` - Pause spoken audio content from other sessions when your app plays its audio.
- `MIX_WITH_OTHERS` - Whether audio from this session mixes with audio from active sessions in other audio apps.
- `OVERRIDE_MUTED_MICROPHONE_INTERRUPTION` - System interrupts the audio session when it mutes the built-in microphone.

## Properties

`buildALLOW_AIRPLAYclass-attributeinstance-attribute`

Stream audio from this session to AirPlay devices.

Note

Available from iOS 10.0.

`buildALLOW_BLUETOOTHclass-attributeinstance-attribute`

Bluetooth hands-free devices appear as available input routes.

`buildALLOW_BLUETOOTH_A2DPclass-attributeinstance-attribute`

Stream audio from this session to Bluetooth devices that support the Advanced Audio Distribution Profile (A2DP).

Note

Available from iOS 10.0.

`buildDEFAULT_TO_SPEAKERclass-attributeinstance-attribute`

Audio from the session defaults to the built-in speaker instead of the receiver.

`buildDUCK_OTHERSclass-attributeinstance-attribute`

Reduces the volume of other audio sessions while audio from this session plays.

`buildINTERRUPT_SPOKEN_AUDIO_AND_MIX_WITH_OTHERSclass-attributeinstance-attribute`

Pause spoken audio content from other sessions when your app plays its audio.

Available from iOS 9.0.

`buildMIX_WITH_OTHERSclass-attributeinstance-attribute`

Whether audio from this session mixes with audio from active sessions in other audio apps.

`buildOVERRIDE_MUTED_MICROPHONE_INTERRUPTIONclass-attributeinstance-attribute`

System interrupts the audio session when it mutes the built-in microphone.

#### Note

Available from iOS 14.5.

# IosRecorderConfiguration

iOS specific configuration for recording.

## Properties

- `manage_audio_session` - Whether to manage the shared AVAudioSession.
- `options` - Optional audio behaviors.

## Properties

### `buildmanage_audio_sessionclass-attributeinstance-attribute`

```
manage_audio_session: bool = True
```

Whether to manage the shared AVAudioSession.

Set this to False if another plugin is already managing the AVAudioSession.

### `buildoptionsclass-attributeinstance-attribute`

```
options: list[IosAudioCategoryOption] = field(
 default_factory=lambda: [
 IosAudioCategoryOption.DEFAULT_TO_SPEAKER,
 IosAudioCategoryOption.ALLOW_BLUETOOTH,
 IosAudioCategoryOption.ALLOW_BLUETOOTH_A2DP,
]
)
```

Optional audio behaviors.

# ForegroundNotificationConfiguration

Android foreground-service notification settings for background tracking.

Supply this configuration through `GeolocatorAndroidConfiguration.foreground_notification_config` to control the persistent notification shown while background location updates are active.

## Properties

- `notification_channel_name` - The user visible name of the notification channel.
- `notification_enable_wake_lock` - When enabled, a `Wakelock` is acquired when background execution is started.
- `notification_enable_wifi_lock` - When enabled, a `WifiLock` is acquired when background execution is started.
- `notification_set_ongoing` - When enabled, the displayed notification is persistent and the user cannot dismiss it.
- `notification_text` - The body used for the foreground service notification.
- `notification_title` - The title used for the foreground service notification.

## Properties

### `buildnotification_channel_name`class-attributeinstance-attribute

```
notification_channel_name: str = 'Background Location'
```

The user visible name of the notification channel.

The notification channel name will be displayed in the system settings. The maximum recommended length is 40 characters, the name might be truncated if it is too long. Default value: "Background Location".

### `buildnotification_enable_wake_lock`class-attributeinstance-attribute

```
notification_enable_wake_lock: bool = False
```

When enabled, a `Wakelock` is acquired when background execution is started.

If this is false then the system can still sleep and all location events will be received at once when the system wakes up again.

Wake lock permissions should be obtained first by using a permissions library.

### `buildnotification_enable_wifi_lock`class-attributeinstance-attribute

```
notification_enable_wifi_lock: bool = False
```

When enabled, a `WifiLock` is acquired when background execution is started. This allows the application to keep the Wi-Fi radio awake, even when the user has not used the device in a while (e.g. for background network communications).

Wifi lock permissions should be obtained first by using a permissions library.

### `buildnotification_set_ongoing`class-attributeinstance-attribute

```
notification_set_ongoing: bool = False
```

When enabled, the displayed notification is persistent and the user cannot dismiss it.

### `buildnotification_text`instance-attribute

```
notification_text: str
```

The body used for the foreground service notification.

### `buildnotification_title`instance-attribute

```
notification_title: str
```

The title used for the foreground service notification.

# GeolocatorAndroidConfiguration

Android specific settings.

Inherits: GeolocatorConfiguration

## Properties

- foreground\_notification\_config
- interval\_duration - The desired interval for active location updates.
- use\_msl\_altitude - Whether altitude should be calculated as MSL (EGM2008) from NMEA messages and reported as the altitude instead of using the geoidal height (WSG84).

## Properties

### buildforeground\_notification\_configclass-attributeinstance-attribute

```
foreground_notification_config: (
 ForegroundNotificationConfiguration | None
) = None
```

### buildinterval\_durationclass-attributeinstance-attribute

```
interval_duration: DurationValue = field(
 default_factory=lambda: Duration(milliseconds=5000)
)
```

The desired interval for active location updates.

### builduse\_msl\_altitudeclass-attributeinstance-attribute

```
use_msl_altitude: bool = False
```

Whether altitude should be calculated as MSL (EGM2008) from NMEA messages and reported as the altitude instead of using the geoidal height (WSG84). Setting this property true will help to align Android altitude to that of iOS which uses MSL.

If the NMEA message is empty then the altitude reported will still be the standard WSG84 altitude from the GPS receiver.

MSL Altitude is only available starting from Android N and not all devices support NMEA message returning \$GPGGA sequences.

This property only works with position stream updates and has no effect when getting the current position or last known position.



# GeolocatorConfiguration

Base configuration for position requests and position-stream updates.

Use this type when you want the same settings to apply across all platforms. Platform-specific configuration classes extend it with additional options.

## Properties

- `accuracy` - Defines the desired accuracy that should be used to determine the location data.
- `distance_filter` - The minimum distance (measured in meters) a device must move horizontally before an update event is generated.
- `time_limit` - Specifies a timeout interval.

## Properties

### `buildaccuracyclass-attributeinstance-attribute`

```
accuracy: GeolocatorPositionAccuracy = (
 GeolocatorPositionAccuracy.BEST
)
```

Defines the desired accuracy that should be used to determine the location data.

### `bulddistance_filterclass-attributeinstance-attribute`

```
distance_filter: int = 0
```

The minimum distance (measured in meters) a device must move horizontally before an update event is generated.

Set to 0 when you want to be notified of all movements.

### `buildtime_limitclass-attributeinstance-attribute`

```
time_limit: DurationValue = None
```

Specifies a timeout interval.

For no time limit, set to None.

# GeolocatorIosActivityType

Represents the possible iOS activity types.

Inherits: `enum.Enum`

## Properties

- **AIRBORNE** - The location manager is being used specifically during airborne activities.
- **AUTOMOTIVE\_NAVIGATION** - The location manager is being used specifically during vehicular navigation to track location changes to the automobile.
- **FITNESS** - The location manager is being used to track fitness activities such as walking, running, cycling, and so on.
- **OTHER** - The location manager is being used for an unknown activity.
- **OTHER\_NAVIGATION** - The location manager is being used to track movements for other types of vehicular navigation that are not automobile related.

## Properties

**buildAIRBORNE**class-attributeinstance-attribute

The location manager is being used specifically during airborne activities.

**buildAUTOMOTIVE\_NAVIGATION**class-attributeinstance-attribute

The location manager is being used specifically during vehicular navigation to track location changes to the automobile.

**buildFITNESS**class-attributeinstance-attribute

The location manager is being used to track fitness activities such as walking, running, cycling, and so on.

**buildOTHER**class-attributeinstance-attribute

The location manager is being used for an unknown activity.

**buildOTHER\_NAVIGATION**class-attributeinstance-attribute

The location manager is being used to track movements for other types of vehicular navigation that are not automobile related.

# GeolocatorIosConfiguration

iOS specific settings.

Inherits: `GeolocatorConfiguration`

## Properties

- `activity_type` - The location manager uses the information in this property as a cue to determine when location updates may be automatically paused.
- `allow_background_location_updates` - Flag to allow the app to receive location updates in the background (iOS only)
- `pause_location_updates_automatically` - Allows the location manager to pause updates to improve battery life on the target device without sacrificing location data.
- `show_background_location_indicator` - Flag to ask the Apple OS to show the background location indicator (iOS only) if app starts up and background and requests the users location.

## Properties

### `buildactivity_typeclass-attributeinstance-attribute`

```
activity_type: GeolocatorIosActivityType = (
 GeolocatorIosActivityType.OTHER
)
```

The location manager uses the information in this property as a cue to determine when location updates may be automatically paused.

### `buildallow_background_location_updatesclass-attributeinstance-attribute`

```
allow_background_location_updates: bool = True
```

Flag to allow the app to receive location updates in the background (iOS only)

Note

For this setting to work Info.plist should contain the following keys:

- `UIBackgroundModes` and the value should contain "location"
- `NSLocationAlwaysUsageDescription`

### `buildpause_location_updates_automaticallyclass-attributeinstance-attribute`

```
pause_location_updates_automatically: bool = False
```

Allows the location manager to pause updates to improve battery life on the target device without sacrificing location data. When this property is set to True, the location manager pauses updates (and powers down the appropriate hardware) at times when the location data is unlikely to change.

### `buildshow_background_location_indicatorclass-attributeinstance-attribute`

```
show_background_location_indicator: bool = False
```

Flag to ask the Apple OS to show the background location indicator (iOS only) if app starts up and background and requests the users location.

For this setting to work and for the location to be retrieved the user must have granted "always" permissions for location retrieval.

# GeolocatorPermissionStatus

Represent the possible location permissions.

Inherits: `enum.Enum`

## Properties

- **ALWAYS** - Permission to access the device's location is allowed even when the app is running in the background.
- **DENIED** - Permission to access the device's location is denied.
- **DENIED\_FOREVER** - Permission to access the device's location is permanently denied.
- **UNABLE\_TO\_DETERMINE** - Permission status cannot be determined.
- **WHILE\_IN\_USE** - Permission to access the device's location is allowed only while the app is in use.

## Properties

### **buildALWAYS**class-attributeinstance-attribute

Permission to access the device's location is allowed even when the app is running in the background.

### **buildDENIED**class-attributeinstance-attribute

Permission to access the device's location is denied.

The app should try to request permission using the `Geolocator.request_permission` method.

### **buildDENIED\_FOREVER**class-attributeinstance-attribute

Permission to access the device's location is permanently denied.

When requesting permissions, the permission dialog will not be shown until the user updates the permission in the app settings.

### **buildUNABLE\_TO\_DETERMINE**class-attributeinstance-attribute

Permission status cannot be determined.

This status is only returned by the `Geolocator.request_permission` method on the web platform for browsers that did not implement the Permissions API. See: [https://developer.mozilla.org/en-US/docs/Web/API/Permissions\\_API](https://developer.mozilla.org/en-US/docs/Web/API/Permissions_API)

### **buildWHILE\_IN\_USE**class-attributeinstance-attribute

Permission to access the device's location is allowed only while the app is in use.

# GeolocatorPosition

Detailed location information.

## Properties

- **accuracy** - The estimated horizontal accuracy of the position in meters.
- **altitude** - The altitude of the device in meters.
- **altitude\_accuracy** - The estimated vertical accuracy of the position in meters.
- **floor** - The floor specifies the floor of the building on which the device is located.
- **heading** - The heading in which the device is traveling in degrees.
- **heading\_accuracy** - The estimated heading accuracy of the position in degrees.
- **latitude** - The latitude of this position in degrees normalized to the interval -90.0 to +90.0 (both inclusive).
- **longitude** - The longitude of the position in degrees normalized to the interval -180 (exclusive) to +180 (inclusive).
- **mocked** - Will be True on Android (starting from API level 18) when the location came from the mocked provider.
- **speed** - The speed at which the device is traveling in meters per second over ground.
- **speed\_accuracy** - The estimated speed accuracy of this position, in meters per second.
- **timestamp** - The time at which this position was determined.

## Properties

### **buildaccuracyclass-attributeinstance-attribute**

`accuracy: Number | None = None`

The estimated horizontal accuracy of the position in meters.

The accuracy is not available on all devices. In these cases the value is 0.0.

### **buildaltitudeclass-attributeinstance-attribute**

`altitude: Number | None = None`

The altitude of the device in meters.

The altitude is not available on all devices. In these cases the returned value is 0.0.

### **buildaltitude\_accuracyclass-attributeinstance-attribute**

`altitude_accuracy: Number | None = None`

The estimated vertical accuracy of the position in meters.

The accuracy is not available on all devices. In these cases the value is 0.0.

### **buildfloorclass-attributeinstance-attribute**

`floor: int | None = None`

The floor specifies the floor of the building on which the device is located.

The floor property is only available on iOS and only when the information is available. In all other cases this value will be None.

### **buildheadingclass-attributeinstance-attribute**

`heading: Number | None = None`

The heading in which the device is traveling in degrees.

The heading is not available on all devices. In these cases the value is 0.0.

### **buildheading\_accuracyclass-attributeinstance-attribute**

`heading_accuracy: Number | None = None`

The estimated heading accuracy of the position in degrees.

The heading accuracy is not available on all devices. In these cases the value is 0.0.

### **buildlatitudeclass-attributeinstance-attribute**

latitude: Number | None = None

The latitude of this position in degrees normalized to the interval -90.0 to +90.0 (both inclusive).

### **buildlongitudeclass-attributeinstance-attribute**

longitude: Number | None = None

The longitude of the position in degrees normalized to the interval -180 (exclusive) to +180 (inclusive).

### **buildmockedclass-attributeinstance-attribute**

mocked: bool | None = None

Will be True on Android (starting from API level 18) when the location came from the mocked provider.

On iOS this value will always be False.

### **buildspeedclass-attributeinstance-attribute**

speed: Number | None = None

The speed at which the device is traveling in meters per second over ground.

The speed is not available on all devices. In these cases the value is 0.0.

### **buildspeed\_accuracyclass-attributeinstance-attribute**

speed\_accuracy: Number | None = None

The estimated speed accuracy of this position, in meters per second.

The speed accuracy is not available on all devices. In these cases the value is 0.0.

### **buildtimestampclass-attributeinstance-attribute**

timestamp: datetime.datetime = None

The time at which this position was determined.

# GeolocatorPositionAccuracy

Represent the possible location accuracy values.

Inherits: `enum.Enum`

## Properties

- **BEST** - Location is accurate within a distance of ~0m on iOS and between 0m and 100m on Android.
- **BEST\_FOR\_NAVIGATION** - Location accuracy is optimized for navigation on iOS and matches the `GeolocatorPositionAccuracy.BEST` on Android.
- **HIGH** - Location is accurate within a distance of 10m on iOS and between 0m and 100m on Android.
- **LOW** - Location is accurate within a distance of 1000m on iOS and 500m on Android.
- **LOWEST** - Location is accurate within a distance of 3000m on iOS and 500m on Android.
- **MEDIUM** - Location is accurate within a distance of 100m on iOS and between 100m and 500m on Android.
- **REDUCED** - Location accuracy is reduced for iOS 14+ devices.

## Properties

### **buildBEST**class-attributeinstance-attribute

Location is accurate within a distance of ~0m on iOS and between 0m and 100m on Android.

On Android, corresponds to `PRIORITY_HIGH_ACCURACY`.

### **buildBEST\_FOR\_NAVIGATION**class-attributeinstance-attribute

Location accuracy is optimized for navigation on iOS and matches the `GeolocatorPositionAccuracy.BEST` on Android.

On Android, corresponds to `PRIORITY_HIGH_ACCURACY`.

### **buildHIGH**class-attributeinstance-attribute

Location is accurate within a distance of 10m on iOS and between 0m and 100m on Android.

On Android, corresponds to `PRIORITY_HIGH_ACCURACY`.

### **buildLOW**class-attributeinstance-attribute

Location is accurate within a distance of 1000m on iOS and 500m on Android.

On Android, corresponds to `PRIORITY_LOW_POWER`.

### **buildLOWEST**class-attributeinstance-attribute

Location is accurate within a distance of 3000m on iOS and 500m on Android.

On Android, corresponds to `PRIORITY_PASSIVE`.

### **buildMEDIUM**class-attributeinstance-attribute

Location is accurate within a distance of 100m on iOS and between 100m and 500m on Android.

On Android, corresponds to `PRIORITY_BALANCED_POWER_ACCURACY`.

### **buildREDUCED**class-attributeinstance-attribute

Location accuracy is reduced for iOS 14+ devices. Matches `GeolocatorPositionAccuracy.LOWEST` on iOS 13 and below and all other platforms.

On Android, corresponds to `PRIORITY_PASSIVE`.

# GeolocatorPositionChangeEvent

Event data provided to `Geolocator.on_position_change` handlers.

Inherits: `Event`

## Properties

- `position` - The current/new position of the device.

## Properties

### `buildpositioninstance-attribute`

`position`: `GeolocatorPosition`

The current/new position of the device.



# GeolocatorWebConfiguration

Web specific settings.

Inherits: GeolocatorConfiguration

## Properties

- `maximum_age` - A value indicating the maximum age of a possible cached position that is acceptable to return.

## Properties

`buildmaximum_ageclass-attributeinstance-attribute`

```
maximum_age: DurationValue = field(
 default_factory=lambda: Duration()
)
```

A value indicating the maximum age of a possible cached position that is acceptable to return. If set to 0, it means that the device cannot use a cached position and must attempt to retrieve the real current position.

# Permission

Defines the permissions which can be checked and requested.

Inherits: `enum.Enum`

## Properties

- `ACCESS_MEDIA_LOCATION` - Permission for accessing the device's media library.
- `ACCESS_NOTIFICATION_POLICY` - Permission for accessing the device's notification policy.
- `ACTIVITY_RECOGNITION` - Permission for accessing the activity recognition.
- `APP_TRACKING_TRANSPARENCY` - Permission for accessing the device's tracking state.
- `ASSISTANT` - Info: - Android: Nothing - iOS: SiriKit
- `AUDIO` - Permission for accessing the device's audio files from external storage.
- `BACKGROUND_REFRESH` - Permission for reading the current background refresh status.
- `BLUETOOTH` - Permission for accessing the device's bluetooth adapter state.
- `BLUETOOTH_ADVERTISE` - Permission for advertising Bluetooth devices Allows the user to make this device discoverable to other Bluetooth devices.
- `BLUETOOTH_CONNECT` - Permission for connecting to Bluetooth devices.
- `BLUETOOTH_SCAN` - Permission for scanning for Bluetooth devices.
- `CALENDAR_FULL_ACCESS` - Permission for reading from and writing to the device's calendar.
- `CALENDAR_WRITE_ONLY` - Permission for writing to the device's calendar.
- `CAMERA` - Permission for accessing the device's camera.
- `CONTACTS` - Permission for accessing the device's contacts.
- `CRITICAL_ALERTS` - Permission for sending critical alerts.
- `IGNORE_BATTERY_OPTIMIZATIONS` - Permission for accessing ignore battery optimizations.
- `LOCATION` - Permission for accessing the device's location.
- `LOCATION_ALWAYS` - Info: iOS: CoreLocation (Always)
- `LOCATION_WHEN_IN_USE` - Permission for accessing the device's location when the app is running in the foreground.
- `MANAGE_EXTERNAL_STORAGE` - Permission for accessing the device's external storage.
- `MEDIA_LIBRARY` - Permission for accessing the device's media library.
- `MICROPHONE` - Permission for accessing the device's microphone.
- `NEARBY_WIFI_DEVICES` - Permission for connecting to nearby devices via Wi-Fi.
- `NOTIFICATION` - Permission for pushing notifications.
- `PHONE` - Permission for accessing the device's phone state.
- `PHOTOS` - Permission for accessing (read & write) the device's photos.
- `PHOTOS_ADD_ONLY` - Permission for adding photos to the device's photo library (iOS only).
- `REMINDERS` - Permission for accessing the device's reminders.
- `REQUEST_INSTALL_PACKAGES` - Permission for requesting installing packages.
- `SCHEDULE_EXACT_ALARM` - Permission for scheduling exact alarms.
- `SENSORS` - Permission for accessing the device's sensors.
- `SENSORS_ALWAYS` - Permission for accessing the device's sensors in background.
- `SMS` - Permission for sending and reading SMS messages (Android only).
- `SPEECH` - Permission for accessing speech recognition.

- STORAGE - Permission for accessing external storage.
- SYSTEM\_ALERT\_WINDOW - Permission for creating system alert window.
- UNKNOWN - The unknown only used for return type, never requested.
- VIDEOS - Permission for accessing the device's video files from external storage.

## Properties

### buildACCESS\_MEDIA\_LOCATIONclass-attributeinstance-attribute

Permission for accessing the device's media library.

Allows an application to access any geographic locations persisted in the user's shared collection.

Note

Only supported on Android 10+ (API 29+) only.

### buildACCESS\_NOTIFICATION\_POLICYclass-attributeinstance-attribute

Permission for accessing the device's notification policy.

Allows the user to access the notification policy of the phone. Example: Allows app to turn on and off do-not-disturb.

Note

Only supported on Android Marshmallow+ (API 23+) only.

### buildACTIVITY\_RECOGNITIONclass-attributeinstance-attribute

Permission for accessing the activity recognition.

Note

Only supported on Android 10+ (API 29+) only.

### buildAPP\_TRACKING\_TRANSPARENCYclass-attributeinstance-attribute

Permission for accessing the device's tracking state. Allows user to accept that your app collects data about end users and shares it with other companies for purposes of tracking across apps and websites.

Note

Only supported on iOS only.

### buildASSISTANTclass-attributeinstance-attribute

Info

- Android: Nothing
- iOS: SiriKit

### buildAUDIOclass-attributeinstance-attribute

Permission for accessing the device's audio files from external storage.

Note

Only supported on Android 13+ (API 33+) only.

### buildBACKGROUND\_REFRESHclass-attributeinstance-attribute

Permission for reading the current background refresh status.

Note

Only supported on iOS only.

### buildBLUETOOTHclass-attributeinstance-attribute

Permission for accessing the device's bluetooth adapter state.

Depending on the platform and version, the requirements are slightly different:

Info

- Android: always allowed.

iOS:

- 13 and above: The authorization state of Core Bluetooth manager.
- below 13: always allowed.

### **buildBLUETOOTH\_ADVERTISEclass-attributeinstance-attribute**

Permission for advertising Bluetooth devices Allows the user to make this device discoverable to other Bluetooth devices.

Note

Only supported on Android 12+ (API 31+) only.

### **buildBLUETOOTH\_CONNECTclass-attributeinstance-attribute**

Permission for connecting to Bluetooth devices. Allows the user to connect with already paired Bluetooth devices.

Note

Only supported on Android 12+ (API 31+) only.

### **buildBLUETOOTH\_SCANclass-attributeinstance-attribute**

Permission for scanning for Bluetooth devices.

Note

Only supported on Android 12+ (API 31+) only.

### **buildCALENDAR\_FULL\_ACCESSclass-attributeinstance-attribute**

Permission for reading from and writing to the device's calendar.

### **buildCALENDAR\_WRITE\_ONLYclass-attributeinstance-attribute**

Permission for writing to the device's calendar.

On iOS 16 and lower, this permission is identical to CALENDAR\_FULL\_ACCESS.

### **buildCAMERAcass-attributeinstance-attribute**

Permission for accessing the device's camera.

Info

- Android: Camera
- iOS: Photos (Camera Roll and Camera)

### **buildCONTACTScass-attributeinstance-attribute**

Permission for accessing the device's contacts.

Info

- Android: Contacts
- iOS: AddressBook

### **buildCRITICAL\_ALERTScass-attributeinstance-attribute**

Permission for sending critical alerts. Allow for sending notifications that override the ringer.

Note

Only supported on iOS only.

### **buildIGNORE\_BATTERY\_OPTIMIZATIONSclass-attributeinstance-attribute**

Permission for accessing ignore battery optimizations.

Note

Only supported on Android only.

### **buildLOCATIONclass-attributeinstance-attribute**

Permission for accessing the device's location.

Info

- Android: Fine and Coarse Location
- iOS: CoreLocation (Always and WhenInUse)

**buildLOCATION\_ALWAYS**class-attributeinstance-attribute

Info

iOS: CoreLocation (Always)

**buildLOCATION\_WHEN\_IN\_USE**class-attributeinstance-attribute

Permission for accessing the device's location when the app is running in the foreground.

Info

- Android: Fine and Coarse Location
- iOS: CoreLocation - WhenInUse

**buildMANAGE\_EXTERNAL\_STORAGE**class-attributeinstance-attribute

Permission for accessing the device's external storage. Allows an application a broad access to external storage in scoped storage.

You should request this permission only when your app cannot effectively make use of the more privacy-friendly APIs. For more information: <https://developer.android.com/training/data-storage/manage-all-files>

Info

When the privacy-friendly APIs (i.e. Storage Access Framework or theMediaStore APIs) is all your app needs, the [PermissionGroup.storage] are the only permissions you need to request.

If the usage of this permission is needed, you have to fill out the Permission Declaration Form upon submitting your app to the Google Play Store. More details: <https://support.google.com/googleplay/android-developer/answer/9214102#zippy=>

Note

Only supported on Android 11+ (API 30+) only.

**buildMEDIA\_LIBRARY**class-attributeinstance-attribute

Permission for accessing the device's media library.

Note

Only supported on iOS 9.3+ only

**buildMICROPHONE**class-attributeinstance-attribute

Permission for accessing the device's microphone.

**buildNEARBY\_WIFI\_DEVICES**class-attributeinstance-attribute

Permission for connecting to nearby devices via Wi-Fi.

Note

Only supported on Android 13+ (API 33+) only.

**buildNOTIFICATION**class-attributeinstance-attribute

Permission for pushing notifications.

**buildPHONE**class-attributeinstance-attribute

Permission for accessing the device's phone state.

Note

Only supported on Android only.

**buildPHOTO**class-attributeinstance-attribute

Permission for accessing (read & write) the device's photos.

If you only want to add photos, you can use the `PHOTOS_ADD_ONLY` permission instead (iOS only).

### **buildPHOTOS\_ADD\_ONLYclass-attributeinstance-attribute**

Permission for adding photos to the device's photo library (iOS only).

If you want to read them as well, use the `Permission.PHOTOS` permission instead.

Info

iOS: Photos (14+ read & write access level)

### **buildREMINDERSclass-attributeinstance-attribute**

Permission for accessing the device's reminders.

Note

Only supported on iOS only.

### **buildREQUEST\_INSTALL\_PACKAGESclass-attributeinstance-attribute**

Permission for requesting installing packages.

Note

Only supported on Android Marshmallow+ (API 23+) only.

### **buildSCHEDULE\_EXACT\_ALARMclass-attributeinstance-attribute**

Permission for scheduling exact alarms.

Note

Only supported on Android 12+ (API 31+) only.

### **buildSENSORSclass-attributeinstance-attribute**

Permission for accessing the device's sensors.

Info

- Android: Body Sensors
- iOS: CoreMotion

### **buildSENSORS\_ALWAYSclass-attributeinstance-attribute**

Permission for accessing the device's sensors in background.

Note

Only supported on Android 13+ (API 33+) only.

### **buildSMSclass-attributeinstance-attribute**

Permission for sending and reading SMS messages (Android only).

### **buildSPEECHclass-attributeinstance-attribute**

Permission for accessing speech recognition.

Info

- Android: Requests access to microphone (identical to requesting `MICROPHONE`).
- iOS: Requests speech access (different from requesting `MICROPHONE`).

### **buildSTORAGEclass-attributeinstance-attribute**

Permission for accessing external storage.

Depending on the platform and version, the requirements are slightly different:

Info

Android:

- On Android 13 (API 33) and above, this permission is deprecated and always returns `PermissionStatus.denied`. Instead use `Permission.PHOTOS`, `Permission.VIDEO`, `Permission.AUDIO` or `Permission.MANAGE_EXTERNAL_STORAGE`. For more information see this.
- Below Android 13 (API 33), the `READ_EXTERNAL_STORAGE` and `WRITE_EXTERNAL_STORAGE` permissions are requested (depending on the definitions in the `AndroidManifest.xml` file).
- iOS: Access to folders like Documents or Downloads. Implicitly granted.

### **buildSYSTEM\_ALERT\_WINDOWclass-attributeinstance-attribute**

Permission for creating system alert window. Allows an app to create windows shown on top of all other apps.

Note

Only supported on Android only.

### **buildUNKNOWNclass-attributeinstance-attribute**

The unknown only used for return type, never requested.

### **buildVIDEOSclass-attributeinstance-attribute**

Permission for accessing the device's video files from external storage.

Note

Only supported on Android 13+ (API 33+) only.

# PermissionStatus

Defines the state of a Permission.

Inherits: `enum.Enum`

## Properties

- **DENIED** - The user denied access to the requested feature, permission needs to be asked first.
- **GRANTED** - The user granted access to the requested feature.
- **LIMITED** - The user has authorized this application for limited access.
- **PERMANENTLY\_DENIED** - Permission to the requested feature is permanently denied, the permission dialog will not be shown when requesting this permission.
- **PROVISIONAL** - The application is provisionally authorized to post non-interruptive user notifications.
- **RESTRICTED** - The OS denied access to the requested feature.

## Properties

### `buildDENIEDclass-attributeinstance-attribute`

The user denied access to the requested feature, permission needs to be asked first.

### `buildGRANTEDclass-attributeinstance-attribute`

The user granted access to the requested feature.

### `buildLIMITEDclass-attributeinstance-attribute`

The user has authorized this application for limited access. So far this is only relevant for the Photo Library picker.

Note

Only supported on iOS (iOS14+) and Android (Android 14+).

### `buildPERMANENTLY_DENIEDclass-attributeinstance-attribute`

Permission to the requested feature is permanently denied, the permission dialog will not be shown when requesting this permission. The user may still change the permission status in the settings.

Note

On Android:

- Android 11+ (API 30+): whether the user denied the permission for a second time.
- Below Android 11 (API 30): whether the user denied access to the requested feature and selected to never again show a request.
- On iOS: If the user has denied access to the requested feature.

### `buildPROVISIONALclass-attributeinstance-attribute`

The application is provisionally authorized to post non-interruptive user notifications.

Note

Only supported on iOS (iOS 12+).

### `buildRESTRICTEDclass-attributeinstance-attribute`

The OS denied access to the requested feature. The user cannot change this app's status, possibly due to active restrictions such as parental controls being in place.

Note

Only supported on iOS.



# AccessControlFlag

Keychain access control flags that define security conditions for accessing items.

These flags can be combined to create complex access control policies using the `access_control_flags` parameter in `IOSOptions` or `MacOsOptions`.

Rules for combining flags:

- Use `AccessControlFlag.OR` to allow access if any condition is met
- Use `AccessControlFlag.AND` to require that all specified conditions are met
- Only one logical operator (OR/AND) can be used per combination

Inherits: `enum.Enum`

## Properties

- `AND` - Combine multiple constraints with an AND operation.
- `APPLICATION_PASSWORD` - Use an application-provided password for encryption.
- `BIOMETRY_ANY` - Constraint to access an item with biometrics (Touch ID/Face ID).
- `BIOMETRY_CURRENT_SET` - Constraint to access an item with the currently enrolled biometrics.
- `DEVICE_PASSCODE` - Constraint to access an item with a passcode.
- `OR` - Combine multiple constraints with an OR operation.
- `PRIVATE_KEY_USAGE` - Enable private key usage for signing operations.
- `USER_PRESENCE` - Constraint to access an item with either biometry or passcode.
- `WATCH` - Constraint to access an item with a paired watch.

## Properties

**buildANDclass-attributeinstance-attribute**

Combine multiple constraints with an AND operation.

**buildAPPLICATION\_PASSWORDclass-attributeinstance-attribute**

Use an application-provided password for encryption.

**buildBIOMETRY\_ANYclass-attributeinstance-attribute**

Constraint to access an item with biometrics (Touch ID/Face ID).

**buildBIOMETRY\_CURRENT\_SETclass-attributeinstance-attribute**

Constraint to access an item with the currently enrolled biometrics.

**buildDEVICE\_PASSCODEclass-attributeinstance-attribute**

Constraint to access an item with a passcode.

**buildORclass-attributeinstance-attribute**

Combine multiple constraints with an OR operation.

**buildPRIVATE\_KEY\_USAGEclass-attributeinstance-attribute**

Enable private key usage for signing operations.

**buildUSER\_PRESENCEclass-attributeinstance-attribute**

Constraint to access an item with either biometry or passcode.

**buildWATCHclass-attributeinstance-attribute**

Constraint to access an item with a paired watch.

## Usage example

Require biometrics OR device passcode:

```
options = IOSOptions(
 access_control_flags=[
 AccessControlFlag.BIOMETRY_ANY,
 AccessControlFlag.OR,
 AccessControlFlag.DEVICE_PASSCODE
]
)
```

# AndroidOptions

Specific options for Android platform for secure storage.

Provides configurable options for encryption, key wrapping, biometric enforcement, and shared preferences naming.

## Properties

- `biometric_prompt_subtitle` - Subtitle displayed in the biometric authentication prompt.
- `biometric_prompt_title` - Title displayed in the biometric authentication prompt.
- `enforce_biometrics` - Whether to enforce biometric or PIN authentication.
- `key_cipher_algorithm` - Algorithm used to encrypt the secret key.
- `migrate_on_algorithm_change` - When the encryption algorithm changes, automatically migrate existing data to the new algorithm.
- `preferences_key_prefix` - Prefix for shared preference keys.
- `reset_on_error` - When an error is detected, automatically reset all data to prevent fatal errors with unknown keys.
- `shared_preferences_name` - The name of the shared preferences database to use.
- `storage_cipher_algorithm` - Algorithm used to encrypt stored data.

## Properties

### `buildbiometric_prompt_subtitleclass-attributeinstance-attribute`

```
biometric_prompt_subtitle: str = (
 "Use biometrics or device credentials"
)
```

Subtitle displayed in the biometric authentication prompt.

### `buildbiometric_prompt_titleclass-attributeinstance-attribute`

```
biometric_prompt_title: str = 'Authenticate to access'
```

Title displayed in the biometric authentication prompt.

### `buildenforce_biometricsclass-attributeinstance-attribute`

```
enforce_biometrics: bool = False
```

Whether to enforce biometric or PIN authentication.

When True:

- The plugin throws an exception if no biometric/PIN is enrolled.
- The encryption key is generated with authentication required.

When False:

- The plugin gracefully degrades if biometrics are unavailable.
- The key is generated without authentication required.

### `buildkey_cipher_algorithmclass-attributeinstance-attribute`

```
key_cipher_algorithm: KeyCipherAlgorithm = KeyCipherAlgorithm.RSA_ECB_OAEP_WITH_SHA256_AND_MGF1_PADDING
```

Algorithm used to encrypt the secret key.

Legacy RSA/ECB/PKCS1Padding is available for backwards compatibility.

### `buildmigrate_on_algorithm_changeclass-attributeinstance-attribute`

```
migrate_on_algorithm_change: bool = True
```

When the encryption algorithm changes, automatically migrate existing data to the new algorithm. Preserves data across algorithm upgrades.

If False, data may be lost when algorithm changes unless `reset_on_error` is True.

## buildpreferences\_key\_prefixclass-attributeinstance-attribute

```
preferences_key_prefix: str | None = None
```

Prefix for shared preference keys. Ensures keys are unique to your app.

An underscore (\_) is added automatically.

Changing this prevents access to existing preferences.

## buildreset\_on\_errorclass-attributeinstance-attribute

```
reset_on_error: bool = True
```

When an error is detected, automatically reset all data to prevent fatal errors with unknown keys.

Be aware that data is PERMANENTLY erased when this occurs.

## buildshared\_preferences\_nameclass-attributeinstance-attribute

```
shared_preferences_name: str | None = None
```

The name of the shared preferences database to use.

Changing this will prevent access to already saved preferences.

## buildstorage\_cipher\_algorithmclass-attributeinstance-attribute

```
storage_cipher_algorithm: StorageCipherAlgorithm = (
 StorageCipherAlgorithm.AES_GCM_NO_PADDING
)
```

Algorithm used to encrypt stored data.

Legacy AES/CBC/PKCS7Padding is available for backwards compatibility.

# Disabling Auto Backup

By default Android backups data on Google Drive. It can cause exception `java.security.InvalidKeyException: Failed to unwrap key`. You need to:

- Disable autobackup, details
- Exclude sharedprefs used by SecureStorage

Add the following to your `pyproject.toml`:

```
[tool.flet.android.manifest_application]
"allowBackup" = "false"
"fullBackupContent" = "false"
```

# Encryption Options

## Default

```
AndroidOptions()
```

- Key Cipher: RSA/ECB/OAEPWithSHA-256AndMGF1Padding
- Storage Cipher: AES/GCM/NoPadding
- Biometric Support: No
- Description: Standard secure storage with RSA OAEP key wrapping. Strong authenticated encryption without biometrics. Recommended for most use cases.

## Optional Biometrics

```
AndroidOptions(
 enforce_biometrics=False,
 key_cipher_algorithm=KeyCipherAlgorithm.AES_GCM_NO_PADDING,
)
```

- Key Cipher: AES/GCM/NoPadding
- Storage Cipher: AES/GCM/NoPadding
- Biometric Support: Optional
- Description: KeyStore-based with optional biometric authentication. Gracefully degrades if biometrics unavailable.

## Required Biometrics

- Key Cipher: AES/GCM/NoPadding
- Storage Cipher: AES/GCM/NoPadding
- Biometric Support: Required (API 28+)
- Description: KeyStore-based requiring biometric/PIN authentication. Throws error if device security not available.

For advanced users, all combinations below are supported using the `AndroidOptions()` constructor with custom parameters:

RSA\_ECB\_PKCS1\_PADDING  
AES\_CBC\_PKCS7\_PADDING  
RSA\_ECB\_PKCS1\_PADDING  
AES\_GCM\_NO\_PADDING  
RSA\_ECB\_OAEP\_WITH\_SHA256\_AND\_MGF1\_PADDING  
AES\_CBC\_PKCS7\_PADDING  
RSA\_ECB\_OAEP\_WITH\_SHA256\_AND\_MGF1\_PADDING  
AES\_GCM\_NO\_PADDING  
AES\_GCM\_NO\_PADDING  
AES\_CBC\_PKCS7\_PADDING  
enforce\_biometrics  
AES\_GCM\_NO\_PADDING  
AES\_GCM\_NO\_PADDING  
enforce\_biometrics

Secure Storage supports biometric authentication (fingerprint, face recognition, etc.) on Android API 23+.

To use biometric authentication on Android, you need to grant the necessary permissions (USE\_BIOMETRIC and optionally USE\_FINGERPRINT) in your project.

See the official Flet documentation for details: [Android Permissions in Flet](#)

```
[tool.flet.android.permission]
"android.permission.USE_BIOMETRIC" = true
"android.permission.USE_FINGERPRINT" = true
```

```
flet build \
 --android-permissions android.permission.USE_BIOMETRIC=True \
 android.permission.USE_FINGERPRINT=True
```

## Using Biometric Authentication

```
Optional biometric authentication (graceful degradation)
storage = SecureStorage(
 android_options=AndroidOptions(
 enforce_biometrics=False, # Default - works without biometrics
 biometric_prompt_title='Unlock to access your data',
 biometric_prompt_subtitle='Use fingerprint or face unlock',
),
)

Strict biometric enforcement (requires device security)
storage = SecureStorage(
 android_options=AndroidOptions(
 enforce_biometrics=True, # Requires biometric/PIN/pattern
 biometric_prompt_title: 'Biometric authentication required',
),
)
```

),

## Requirements

- API Level: Android 6.0 (API 23) minimum for basic encryption
- API Level: Android 9.0 (API 28) minimum for enforced biometric authentication
- Device Security: Device must have a PIN, pattern, password, or biometric enrolled (when using `enforce_biometrics = True`)
- Permissions: `USE_BIOMETRIC` permission in `pyproject.toml`

# AppleOptions

Specific options for Apple platforms (iOS/macOS) for secure storage.

This class allows configuring keychain access and storage behavior. Use `IOSOptions` for iOS-specific configuration or `MacOsOptions` for macOS-specific configuration.

## Note

- Most options apply to both iOS and macOS
- Some options (like `group_id` on macOS) only apply when certain keychain flags are set
- See individual option documentation for platform-specific behavior

## Properties

- `access_control_flags` - Keychain access control flags that define security conditions for accessing items.
- `accessibility` - Defines the accessibility level of the keychain item.
- `account_name` - Represents the service or application name associated with the item.
- `auth_ui_behavior` - Controls how authentication UI is presented during secure operations.
- `comment` - A comment associated with the keychain item.
- `creation_date` - The creation date of the keychain item.
- `description` - A description of the keychain item.
- `group_id` - Specifies the app group for shared access.
- `invisible` - Indicates whether the keychain item is hidden from user-visible lists.
- `is_negative` - Indicates whether the item is a placeholder or a negative entry.
- `is_persistent` - Indicates whether to return a persistent reference to the keychain item.
- `label` - A user-visible label for the keychain item.
- `last_modified_date` - The last modification date of the keychain item.
- `result_limit` - Specifies the maximum number of results to return in a query.
- `synchronizable` - Indicates whether the keychain item should be synchronized with iCloud.

## Properties

### `buildaccess_control_flagsclass-attributeinstance-attribute`

```
access_control_flags: list[AccessControlFlag] = field(
 default_factory=list
)
```

Keychain access control flags that define security conditions for accessing items.

### `buildaccessibilityclass-attributeinstance-attribute`

```
accessibility: KeychainAccessibility | None = (
 KeychainAccessibility.UNLOCKED
)
```

Defines the accessibility level of the keychain item.

Controls when the item is accessible (e.g., when device is unlocked or after first unlock).

### `buildaccount_nameclass-attributeinstance-attribute`

```
account_name: str | None = 'flet_secure_storage_service'
```

Represents the service or application name associated with the item.

Typically used to group related keychain items.

### `buildauth_ui_behaviorclass-attributeinstance-attribute`

```
auth_ui_behavior: str | None = None
```

Controls how authentication UI is presented during secure operations. Determines whether authentication prompts are displayed to the user.

### **buildcommentclass-attributeinstance-attribute**

`comment: str | None = None`

A comment associated with the keychain item. Often used for metadata or debugging information.

### **buildcreation\_dateclass-attributeinstance-attribute**

`creation_date: datetime | None = None`

The creation date of the keychain item. Automatically set by the system when an item is created.

### **bulldescriptionclass-attributeinstance-attribute**

`description: str | None = None`

A description of the keychain item. Can describe a category of items (shared) or a specific item (unique).

### **buildgroup\_idclass-attributeinstance-attribute**

`group_id: str | None = None`

Specifies the app group for shared access. Allows multiple apps in the same app group to access the item.

### **buildinvisibleclass-attributeinstance-attribute**

`invisible: bool | None = None`

Indicates whether the keychain item is hidden from user-visible lists. Can apply to all items in a category (shared) or specific items (unique).

### **buildis\_negativeclass-attributeinstance-attribute**

`is_negative: bool | None = None`

Indicates whether the item is a placeholder or a negative entry. Typically unique to individual keychain items.

### **buildis\_persistentclass-attributeinstance-attribute**

`is_persistent: bool | None = None`

Indicates whether to return a persistent reference to the keychain item. Used for persistent access across app sessions.

### **buildlabelclass-attributeinstance-attribute**

`label: str | None = None`

A user-visible label for the keychain item. Helps identify the item in keychain management tools.

### **buildlast\_modified\_dateclass-attributeinstance-attribute**

`last_modified_date: datetime | None = None`

The last modification date of the keychain item. Automatically updated when an item is modified.

### **buildresult\_limitclass-attributeinstance-attribute**

`result_limit: int | None = None`

Specifies the maximum number of results to return in a query. For example, 1 for a single result, or None for all matching results.

### **buildsynchronizableclass-attributeinstance-attribute**

`synchronizable: bool = False`

Indicates whether the keychain item should be synchronized with iCloud.

- True: Enables synchronization across user's devices
- False: Item stays local to this device only



# IOOptions

iOS-specific configuration for secure storage.

All configurable options are inherited from AppleOptions. There are currently no iOS-only options.

Inherits: AppleOptions

## Usage Example

### Usage with accessibility control

```
from flet_secure_storage import SecureStorage
from flet_secure_storage.types import IOOptions, KeychainAccessibility

storage = SecureStorage(
 ios_options=IOOptions(
 accessibility=KeychainAccessibility.FIRST_UNLOCK
)
)

await storage.set(key="token", value="secret_value")
```

### Biometric authentication:

```
from flet_secure_storage.types import IOOptions, AccessControlFlag

options = IOOptions(
 access_control_flags=[
 AccessControlFlag.BIOMETRY_ANY,
 AccessControlFlag.OR,
 AccessControlFlag.DEVICE_PASSCODE
]
)

await storage.set(key="secure_key", value="secure_value", ios=options)
```

# KeychainAccessibility

KeyChain accessibility attributes for iOS/macOS platforms.

These attributes determine when the app can access secure values stored in the Keychain.

Inherits: `enum.Enum`

## Properties

- `FIRST_UNLOCK` - The data in the keychain item cannot be accessed after a restart until the device has been unlocked once by the user.
- `FIRST_UNLOCK_THIS_DEVICE` - The data in the keychain item cannot be accessed after a restart until the device has been unlocked once by the user.
- `PASSCODE` - The data in the keychain can only be accessed when the device is unlocked.
- `UNLOCKED` - The data in the keychain item can be accessed only while the device is unlocked by the user.
- `UNLOCKED_THIS_DEVICE` - The data in the keychain item can be accessed only while the device is unlocked by the user.

## Properties

### `buildFIRST_UNLOCKclass-attributeinstance-attribute`

The data in the keychain item cannot be accessed after a restart until the device has been unlocked once by the user.

Enables access to secure values after the device is unlocked for the first time after a reboot.

### `buildFIRST_UNLOCK_THIS_DEVICEclass-attributeinstance-attribute`

The data in the keychain item cannot be accessed after a restart until the device has been unlocked once by the user.

Items with this attribute do not migrate to a new device.

Allows access to secure values only after the device is unlocked for the first time since installation on this device.

### `buildPASSCODEclass-attributeinstance-attribute`

The data in the keychain can only be accessed when the device is unlocked.

Only available if a passcode is set on the device. Items with this attribute do not migrate to a new device.

### `buildUNLOCKEDclass-attributeinstance-attribute`

The data in the keychain item can be accessed only while the device is unlocked by the user.

### `buildUNLOCKED_THIS_DEVICEclass-attributeinstance-attribute`

The data in the keychain item can be accessed only while the device is unlocked by the user.

Items with this attribute do not migrate to a new device.

# KeyCipherAlgorithm

Algorithm used to encrypt/wrap the secret key in Android KeyStore.

Different algorithms provide different security guarantees and compatibility levels:

- RSA algorithms wrap the AES encryption key with RSA (no biometric support)
- AES algorithm stores the key directly in Android KeyStore (supports biometric authentication)

See the [AndroidOptions] class for usage examples and combinations.

Inherits: enum.Enum

## Properties

- AES\_GCM\_NO\_PADDING - AES/GCM/NoPadding for KeyStore-based key wrapping (supports biometrics).
- RSA\_ECB\_OAEP\_WITH\_SHA256\_AND\_MGF1\_PADDING - RSA/ECB/OAEPWithSHA-256AndMGF1Padding (API 23+).
- RSA\_ECB\_PKCS1\_PADDING - Legacy RSA/ECB/PKCS1Padding for backwards compatibility.

## Properties

### buildAES\_GCM\_NO\_PADDINGclass-attributeinstance-attribute

AES/GCM/NoPadding for KeyStore-based key wrapping (supports biometrics).

Use this algorithm when you need biometric authentication support. Requires API 23+ for basic use, API 28+ for enforced biometric authentication.

### buildRSA\_ECB\_OAEP\_WITH\_SHA256\_AND\_MGF1\_PADDINGclass-attributeinstance-attribute

RSA/ECB/OAEPWithSHA-256AndMGF1Padding (API 23+).

This is the default and recommended algorithm for most use cases. Provides strong authenticated encryption without biometrics.

### buildRSA\_ECB\_PKCS1\_PADDINGclass-attributeinstance-attribute

Legacy RSA/ECB/PKCS1Padding for backwards compatibility.

# MacOsOptions

Specific options for macOS platform. Extends AppleOptions and adds the usesDataProtectionKeychain parameter.

Inherits: AppleOptions

## Properties

- `uses_data_protection_keychain` - Indicates whether the macOS data protection keychain is used.

## Properties

`builduses_data_protection_keychainclass-attributeinstance-attribute`

`uses_data_protection_keychain: bool = True`

Indicates whether the macOS data protection keychain is used.

# SecureStorageEvent

The event fired by SecureStorage when availability changes.

Inherits: Event

## Properties

- available - The availability of secure storage.

## Properties

### buildavailableinstance-attribute

available: bool | None

The availability of secure storage. True if secure storage is available, False if not, None if unknown.

# StorageCipherAlgorithm

Algorithm used to encrypt stored data on Android.

Modern applications should use AES\_GCM\_NO\_PADDING for better security. The legacy AES\_CBC\_PKCS7\_PADDING is provided for backwards compatibility only.

Inherits: `enum.Enum`

## Properties

- `AES_CBC_PKCS7_PADDING` - Legacy AES/CBC/PKCS7Padding for backwards compatibility.
- `AES_GCM_NO_PADDING` - AES/GCM/NoPadding (API 23+).

## Properties

`buildAES_CBC_PKCS7_PADDING`class-attributeinstance-attribute

Legacy AES/CBC/PKCS7Padding for backwards compatibility.

`buildAES_GCM_NO_PADDING`class-attributeinstance-attribute

AES/GCM/NoPadding (API 23+).

This is the default and recommended storage cipher algorithm. Provides authenticated encryption with associated data (AEAD).

# WebOptions

Specific options for the Web platform for secure storage.

Configures database, encryption, and storage behavior on web platforms.

## Properties

- `db_name` - The name of the database used for secure storage.
- `public_key` - The public key used for encryption.
- `use_session_storage` - Whether to use session storage instead of local storage.
- `wrap_key` - The key used to wrap the encryption key.
- `wrap_key_iv` - The initialization vector (IV) used for the wrap key.

## Properties

### `builddb_name`class-attributeinstance-attribute

```
db_name: str = 'FletEncryptedStorage'
```

The name of the database used for secure storage.

### `buildpublic_key`class-attributeinstance-attribute

```
public_key: str = 'FletSecureStorage'
```

The public key used for encryption.

### `builduse_session_storage`class-attributeinstance-attribute

```
use_session_storage: bool = False
```

Whether to use session storage instead of local storage.

### `buildwrap_key`class-attributeinstance-attribute

```
wrap_key: str = ''
```

The key used to wrap the encryption key.

### `buildwrap_key_iv`class-attributeinstance-attribute

```
wrap_key_iv: str = ''
```

The initialization vector (IV) used for the wrap key.

## Important Security Considerations

SecureStorage uses an experimental implementation using WebCrypto API. Use at your own risk. The browser creates the private key, and encrypted strings in `localStorage` are not portable to other browsers or machines and will only work on the same domain.

You MUST have HTTP Strict Forward Secrecy enabled and proper headers applied to your responses, or you could be subject to JavaScript hijacking.

Required security measures:

- Enable HSTS (HTTP Strict Transport Security)
- Use proper security headers

References:

- <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security>
- <https://www.netsparker.com/blog/web-security/http-security-headers/>

## Application-Specific Key Wrapping

On web, all keys are stored in `LocalStorage`. You can wrap this stored key with an application-specific key to make it more difficult to analyze:

```
storage = SecureStorage(
 web_options=WebOptions(
 wrap_key='your_application_specific_key',
 wrap_key_iv='your_application_specific_iv',
),
)
```



# WindowsOptions

Specific options for Windows platform for secure storage.

Allows configuring backward compatibility when reading/writing values from previous versions of storage.

Note

You need the C++ ATL libraries installed along with Visual Studio Build Tools. Download from:

<https://visualstudio.microsoft.com/downloads/?q=build+tools> Make sure the C++ ATL under optional components is installed as well.

## Properties

- `use_backward_compatibility` - If True, attempts to read values written by previous versions of the storage.

## Properties

`builduse_backward_compatibilityclass-attributeinstance-attribute`

`use_backward_compatibility: bool = False`

If True, attempts to read values written by previous versions of the storage. When reading or writing old storage values, they will be automatically migrated to new storage.

Note

- May introduce performance overhead.
- May cause errors for keys with `"`, `<`, `>`, `|`, `:`, `*`, `?`, `/`, `\` or any ASCII control characters.
- May cause errors for keys containing `./`, `\\.\\`, or similar patterns.
- May cause errors for very long keys (length depends on app's product name, company name, and executing account).

# Fly.io

Fly.io works well for Flet apps: it supports WebSockets, deploys close to users via multiple regions, and has a free tier suitable for small projects.

## Prerequisites

- Install flyctl.
- Authenticate:

```
fly auth login
```

## Project files

Place the following files in your app directory.

### requirements.txt

At minimum:

requirements.txt

```
flet
```

### fly.toml

Fly app configuration:

fly.toml

```
app = "<your-app-name>"

kill_signal = "SIGINT"
kill_timeout = 5
processes = []

[env]
 FLET_SESSION_TIMEOUT = "60"

[http_service]
 internal_port = 8000
 force_https = true
 auto_stop_machines = true
 auto_start_machines = true
 min_machines_running = 0

[http_service.concurrency]
 type = "connections"
 soft_limit = 200
 hard_limit = 250
```

- Replace <your-app-name> with the app name you want. It will also be used in the final URL of your app, in the form, <https://<your-app-name>.fly.dev>.
- internal\_port must correspond to the FLET\_SERVER\_PORT, which is 8000 by default.
- FLET\_SESSION\_TIMEOUT controls session lifetime (seconds).
- For other variables, see environment variables reference.

## Dockerfile

Create a Dockerfile containing the commands to build your application container, for example:

Dockerfile

```
FROM python:3-alpine

WORKDIR /app

COPY requirements.txt ./
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 8000

CMD ["python", "main.py"]
```

main.py is the app's entry point. If your file name differs, update accordingly.

#### Info

Fly.io deploys apps as Docker containers and provides a free remote Docker builder, so you don't need Docker installed locally.

## Deploy

From your app directory:

Create the Fly app:

```
fly apps create --name <your-app-name>
```

Deploy:

```
fly deploy
```

Open app in browser:

```
fly apps open
```

# Replit

Replit is an online IDE and hosting platform for web apps written in any language. Their free tier allows running any number of apps with some performance limitations.

To run your app on Replit:

Sign up on Replit.

Click "New Repl" button, select "Python" template and type the name of your repl, e.g. my-flet-app. Alternatively, go to Flet template page and click Use Template button. Flet template has everything configured for you, so you can jump to main.py and update your program right away.

Open .replit file on the left and add these two options to the root:

```
Stops the packager from installing packages when running the Repl
disableInstallBeforeRun = true
Stops the packager from guessing and auto-installing packages, but it still runs to install packages when running the Repl
disableGuessImports = true
```

Image:  - <https://flet.dev/img/docs/hosting-replit/replit-disable-guess-imports.png>

- On "Tools" pane click "Dependencies" and search for flet package and click "Install" button.
- Open main.py on "Files" pane and copy-paste your app.
- Modify call to ft.run() and include view=ft.AppView.WEB\_BROWSER parameter:

```
ft.run(main, view=ft.AppView.WEB_BROWSER)
```

Run the app. Enjoy.

Image:  - <https://flet.dev/img/docs/hosting-replit/replit-running-app.png>

# Self-Hosting

Host a Flet app on your own server with NGINX.

There are free and inexpensive cloud server tiers available at AWS, Oracle, Linode, and more.

## Setup Flet Environment

### requirements.txt and virtualenv

Create requirements.txt with a list of application dependencies. At minimum it should contain flet module:

```
flet
```

Create a virtualenv and install requirements:

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

### Sample Flet app

```
import flet as ft

def main(page: ft.Page):
 page.title = "My Flet app"
 page.add(ft.Text("Hello, world!"))

if __name__ == "__main__":
 ft.run(main)
```

Flet app will be served from the root URL, but you can configure FLET\_WEB\_APP\_PATH environment variable to serve beneath the root e.g. /apps/myapp.

By default, Flet web app will be running on port 8000, but you can change that by setting up FLET\_SERVER\_PORT environment variable.

Complete list of environment variables supported by a web app.

## Automatically start Flet app

### Create systemd unit file

Automatically start the Flet app using a systemd service unit file flet.service.

Setup below assumes your Flet app script is defined in \$HOME/flet-app/main.py. Replace User, Group, WorkingDirectory, etc. as per your setup.

```
[Unit]
Description=Flet App
After=network.target

[Service]
User=ubuntu
Group=ubuntu
WorkingDirectory=/home/ubuntu/flet-app
Environment="PATH=/home/ubuntu/flet-app/.venv/bin"
ExecStart=/home/ubuntu/flet-app/.venv/bin/python /home/ubuntu/flet-app/main.py

[Install]
WantedBy=multi-user.target
```

### Enable Flet app service

```
cd /etc/systemd/system
sudo ln -s /home/ubuntu/flet-app/flet.service
sudo systemctl start flet
sudo systemctl enable flet
sudo systemctl status flet
```

## NGINX Proxy Setup

NGINX will proxy the Flet app and the websocket.

In your /etc/nginx/sites-available/\* config file, updating path and port as needed:

```
location / {
 proxy_pass http://127.0.0.1:8000/;
```

```

 proxy_http_version 1.1;
 proxy_set_header Upgrade $http_upgrade;
 proxy_set_header Connection keep-alive;
 proxy_set_header Host $host;
 proxy_cache_bypass $http_upgrade;
 proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
 proxy_set_header X-Forwarded-Proto $scheme;
}

location /ws {
 proxy_pass http://127.0.0.1:8000/ws;
 proxy_http_version 1.1;
 proxy_set_header Upgrade $http_upgrade;
 proxy_set_header Connection "upgrade";
 proxy_set_header Host $host;
 proxy_cache_bypass $http_upgrade;
 proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
 proxy_set_header X-Forwarded-Proto $scheme;
}

```

That's it! Restart NGINX, and open your app in a browser.

# Hosting Flet website on Cloudflare

This guide provides the instructions for hosting Flet static website on Cloudflare.

Before we get to the deployment, you will need an account. Get one from [here](#), or simply login if you already have one. After signing up, you will have to verify your email address by clicking the link you will receive in your email. Check the spams too, if you don't get it in your inbox.

In your account, from the side menu, select "Pages" as shown below:

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/pages-from-sidebar.png>

And from there, select the "Create a project" button:

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/project-creation.png>

Cloudflare proposes three ways to create a project. Only the first two will be exposed here:

- Connect to a git provider
- Direct upload

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/deployment-methods.png>

## Connecting to a Git provider

note

This method works for flet publish command only.

For this, you will need to have a GitHub or GitLab account. In this account should be the repository you plan to use. An example could be found in this [repo](#).

Click on the "Connect to Git" blue button:

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/git-account-selection.png>

From there, select the tab with the service containing your repository. Then, connect your account. Select one of the suggested options, then click on "Install & Authorize".

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/git-auth.png>

Choose the repository to be used, and press on the "Begin setup" button.

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/repo-selection.png>

Before moving on, add a runtime.txt file in your repo. It should contain the python version to be used. In the file enter 3.7 which is the latest python version Cloudflare uses at time of writing.

Here is an example from the repo above.

Having this done, we can now move to the next step which will be to configure some build and deployment settings for your site.

Set the name of your project, and the production branch to be used. The production branch is simply the branch of your repo to which any push of a commit will automatically trigger a deployment to your production environment. Pushes to your other branches will trigger deployment instead to your preview environment.

After setting these two, jump down to the "Build settings" section where we will be setting up the build instructions.

Skip the "Framework preset" (allow None) because Flet is neither in the list nor a JavaScript framework :)

The "Build command" depends on your application's structure. Follow the guide in the sections above to come up with your custom build command.

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/flet-publish-help.png>

When the build command is ran by flet, a folder named 'dist' is created which will contain all the web files required by Cloudflare pages. Set it as your output directory. Note that this file will not be added to your repository, because Cloudflare only has read access to your code.

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/build-settings.png>

You could optionally specify advanced parameters: the root directory (the directory in which Cloudflare runs the build command), and Environment variables (variables to be used during build time). Now, click on the "Save and Deploy" button and let Cloudflare do the remaining job for you.

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/successful-deployment.png>

Click on the URL that will be shown to move to your deployed site.

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/app-link-1.png>

If when opening the site you see a Cloudflare error, it means they haven't completely finished the setup. So, simply wait for a minute then refresh the page, and you will see your application running. Test the above deployed site here.

## Direct Upload

note

This method works for both `flet build web` and `flet publish` commands.

Click on "Upload assets". In step one, give your project a name. Remember that the name you will give will be used to determine the link to which your project will be deployed to.

The second step requires you to upload your project's assets, either as folder or a zip file (with all the assets inside). If you already have one of them, then use the drag-and-drop or select them using the folder picker.

If you don't yet have these assets but already have an app you've built, use the `flet publish` command in the directory containing your app files, and a `dist` folder will be created which you will then upload to Cloudflare pages as mentioned above.

After the upload press on "Deploy site" button at the bottom.

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/assets-upload.png>

You will then see a success message with a link to your deployed website. Test an example of a deployed site here.

Image: Image - <https://flet.dev/img/docs/cloudflare-pages-deploy/app-link-2.png>

If when opening the site you see a Cloudflare error, it means they haven't completely finished the setup. So, simply wait for a minute then refresh the page, and you will see your application running. You can now click on the "Continue to project" button to monitor your deployments, or create new ones following the same steps above.

## Troubleshooting

When Flet app is running in a web browser all its `print()` statements are displayed in "Console" tab of Developer Tools in a browser. `print()` can be used as a simple debugging tool.

You can also use logging module and output messages to Console with different severity.

To enable detailed Flet logging add this to your program:

```
import logging
logging.basicConfig(level=logging.DEBUG)
```



# Hosting Flet website on GitHub Pages

This guide shows how to build a Flet static web app and deploy it to GitHub Pages with GitHub Actions.

## Setup

- Open your repository root.
- Create this folder if it does not exist: `.github/workflows/`
- Create a workflow file in that folder, for example: `.github/workflows/build-deploy.yml` - must have the `.yml` or `.yaml` file extension.
- Copy the workflow configuration below into that file and adjust to your needs.
- Commit and push to GitHub.
- Open the Actions tab to monitor build/deploy progress.

### Repository settings

In GitHub, open Settings → Pages and make sure deployment source is GitHub Actions.

## Workflow

```
name: Web Build + Deploy to GitHub Pages

on:
 push:
 pull_request:
 workflow_dispatch:

concurrency:
 group: "pages"
 cancel-in-progress: false

env:
 UV_PYTHON: 3.12

 # https://flet.dev/docs/reference/environment-variables
 FLET_CLI_NO_RICH_OUTPUT: 1

jobs:
 build:
 name: Build web app
 runs-on: ubuntu-latest

 steps:
 - name: Checkout repository
 uses: actions/checkout@v4

 - name: Setup uv
 uses: astral-sh/setup-uv@v6

 - name: Build app
 shell: bash
 run: |
 uv run flet build web --yes --verbose --base-url ${GITHUB_REPOSITORY#*/} --route-url-strategy hash

 - name: Upload Pages Artifact
 uses: actions/upload-pages-
 with:
 path: build/web
 name: web-build-artifact
 retention-days: 20

 deploy:
 name: Deploy to GitHub Pages
 if: github.event_name == 'push' && github.ref == 'refs/heads/main'
 needs: build
 runs-on: ubuntu-latest

 permissions:
 pages: write
 id-token: write

 environment:
 name: github-pages
 url: ${ steps.deploy.outputs.page_url }
```

```
steps:
 - name: Setup Pages
 uses: actions/configure-pages@v5

 - name: Deploy to GitHub Pages
 id: deployment
 uses: actions/deploy-
 with:
 artifact_name: web-build-artifact
```

See it in action [here](#).